POLYMER HANDBOOK

FOURTH EDITION

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J. BRANDRUP, E. H. IMMERGUT, and E. A. GRULKE

Associate Editors

A. ABE
D. R. BLOCH

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Preface

The purpose of the *Polymer Handbook* is to bring together the data and constants needed in theoretical and experimental polymer research. All polymer researchers have experienced the frustration of searching for data in the ever-expanding polymer literature and know the difficulties involved in trying to locate a particular constant that is buried in a long journal article. The contributors to this Handbook have taken on the arduous task of searching the literature and compiling the data and constants that polymer chemists, polymer physicists, and polymer engineers are likely to need.

The 520 and odd tables in this Handbook are divided into eight sections. The first lists the IUPAC nomenclature rules for polymers and the International System of Units. Although several naming conventions exist in the technical literature, IUPAC names permit a consistent listing of all polymers. Section II contains data and constants needed for polymer synthesis, kinetic mechanisms, and thermodynamic studies of polymerization and depolymerization reactions. Sections III and IV contain physical constants of monomers, solvents, and oligomers. Section V lists the physical constants of many important commercial polymers. Section VI and VII cover the solid state properties of polymers and the properties of polymer solutions. Section VIII of the Handbook lists the commonly used abbreviations or acronyms for polymers and Chemical Abstract Registry Numbers, and gives suggestions for electronic data searching for polymer information. This section should also be consulted in the few cases where contributors have not used IUPAC nomenclature.

As in the previous editions, the *Polymer Handbook* concentrates on synthetic polymers, poly(saccharides) and derivatives, and oligomers. Few data on biopolymers are included. Spectroscopic data as well as data needed by engineers and designers, such as mechanical and rheological data, are minimized, since many excellent compilations exist elsewhere. Only fundamental constants and parameters that refer to the polymer molecule, that describe the solid state of polymer molecules, or that describe polymer solutions, were compiled. Constants that depend on processing conditions or on sample history were not emphasized, as they can be found in existing plastics handbooks and encyclopedias.

A critical evaluation of the values published in the literature was not attempted, since such a task would have required an inordinate amount of time and a sizable staff. Therefore, the users of this Handbook should consult the original literature for details when in doubt about the validity of any data. (The authors of the individual tables were nevertheless requested to eliminate obviously erroneous data from otherwise complete compilations.)

The Fourth Edition revisions have focused on data generated in the ten years since the publication of the Third Edition. Therefore, a completely revised *Polymer Handbook* has been prepared. We have added new tables and incorporated a large amount of new data into existing tables. As a result, the Fourth Edition contains approximately twenty-five percent more data, and the number of pages has increased from about 1850 in the Third Edition to about 2250.

We hope that this new edition will be as useful to the polymer research community as the three earlier editions and that many of the *Polymer Handbook*'s previous users will also obtain the Fourth Edition for their laboratory and library.

The publisher plans a CD-ROM for the *Polymer Handbook* in the near future. We would be grateful if our contributors and users send us any new data they accumulate in the course of their research, and any errors, misprints, omissions and other flaws. We will pass on such data to the publisher, for the polymer database, and for future editions of this Handbook.

We would like to thank all of the contributors to the *Polymer Handbook* for their help and continued patience. The staff at John Wiley, especially Carla Fjerstad, Shirley Thomas, and Jacqueline Kroschwitz, have provided excellent help and support in getting all the work done. We hope that the outstanding efforts of all these people will find due appreciation among the users of this Handbook.

July, 1998

J. Brandrup
E. H. Immergut
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SECTION I

NOMENCLATURE RULES — UNITS

Nomenclature

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A. INTRODUCTION

Macromolecular (polymer) nomenclature has an almost 50-year history. As early as 1949 there existed a Subcommission on Nomenclature within the International Union of Pure and Applied Chemistry (IUPAC) under the chairmanship of Maurice L. Huggins. The Subcommission was part of the IUPAC Commission on Macromolecules, chaired then by Herman F. Mark. Other notable pioneers in polymer science, Jan Joseph Hermans, Otto Kratky, Harry W. Melville, and George J. Smets, were members of the Commission.

The Subcommission produced its first official report in 1952. It recognized then, what is just as true today, that the practice in the field of small molecules of providing rigorous definitions is impractical for polymers. The latter consist of molecules not necessarily exactly of the same size, chemical composition, or structure. The differences result from the presence of end groups, branches, variation in orientation of monomeric units, and irregularity in the sequence of different types of units.

The American Chemical Society (ACS) Division of Polymer Chemistry established its Nomenclature Committee in 1963 to try to unify earlier attempts to name polymers in some coherent way. Their first major project resulted in a structure-based nomenclature for regular linear polymers, first published in ACS Polymer Preprints in 1967, adopted by Chemical Abstracts in 1968, and incorporated in IUPAC recommendations in 1975 [1].

When the IUPAC Commission on Macromolecules was elevated to become the IUPAC Division of Macromolecular Chemistry, the latter in turn established its Commission on Macromolecular Nomenclature in 1968.

The Commission in the course of its nearly 30 years of existence produced a series of major documents that have shaped modern nomenclature and terminology of polymer science. The recommendations are being published in the IUPAC official journal, Pure and Applied Chemistry, and occasionally are republished in the form of a "Compendium" which groups all the currently valid documents in a single volume. The Commission published its "Compendium of Macromolecular Nomenclature" in 1991 [2]. The book, which IUPAC has designated the "Purple Book", in analogy to other IUPAC books ("Blue" – organic, "Red" – inorganic, "Orange" – analytical, "Green"– physical, and "Gold" – overall chemistry), has the two most basic macromolecular nomenclature recommendations: "nomenclature of regular single-strand organic polymers" and "source-based nomenclature for copolymers". In more recent years, the Commission extended that basic nomenclature to double-strand (ladder and spiro) organic polymers, irregular and crosslinked polymers, and polymer blends and interpenetrating networks.

In this article, the basic structure-based and source-based nomenclature of polymers is explained and illustrated in some detail. The continuing use of common and semisystematic nomenclature with reference to specific tables in this Handbook is highlighted. The current Chemical Abstracts (CA) index names [3] are compared and contrasted with the IUPAC practice and typical names employed in the polymer journals and textbooks.

B. IUPAC RECOMMENDATIONS

Since polymers, unlike low-molecular-weight compounds, have no uniform structure and are mixtures of macromolecules of different length and different structural

arrangement, their graphical representation and their names require a special approach.

Often enough, the structure of the polymer has not been sufficiently characterized and the researcher cannot draw its chemical structure. Consequently, no name of the polymer reflecting its structure is possible.

A polymerization reaction for a polymer formed from a monomer such as vinyl chloride can schematically be represented by

$$n\mathrm{CH_2{=}CHCl} \rightarrow \{\mathrm{CH_2\text{-}CHCl}\}_n$$

That shows an idealized product. In fact, however, the polymer consists of long-chains of various lengths. The repeating units, $-CH_2-CHCl-$, are not necessarily all uniquely oriented and joined in a regular fashion as shown in the idealized structure above. In addition to "head-to-tail" links

$$\mathrm{-CH_2\text{-}CHCl\text{-}CH_2\text{-}CHCl-}$$

other links such as "head-to-head"

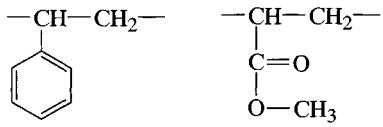
$$\mathrm{-CHCl\text{-}CH_2\text{-}CH_2\text{-}CHCl-}$$

and "tail-to-tail"

$$\mathrm{-CH_2\text{-}CHCl\text{-}CHCl\text{-}CH_2-}$$

can occur, and the exact sequence of all these repeating units usually is not known.

This becomes even more complicated when a copolymer is derived from more than one monomer, such as styrene and methyl acrylate, which contribute two constitutional or monomeric units:



They can combine into a polymeric chain, resulting in many types such as unspecified, statistical, random, alternating, periodic, block, or graft copolymers.

Because the exact structure of the polymer is not always known, two systems of naming polymers exist:

source-based nomenclature,
structure-based nomenclature.

1. Source-Based Nomenclature

1.1. Homopolymers Homopolymers are derived from only one species of monomer, which may be the actual starting reactant (or source), or be a hypothetical monomer if the homopolymer is formed by a modification of another homopolymer.

The name of the polymer is formed by attaching the prefix "poly" to the name of the actual or assumed monomer, or the starting reactant (source), from which the polymer is derived.

Examples: poly(acrylic acid)
polyacrylonitrile
polybutadiene
polyethylene
poly(methyl methacrylate)
poly(methyl vinyl ether)
polystyrene
poly(tetrafluoroethylene)
poly(vinyl alcohol)
poly(vinyl acetate)
poly(vinyl chloride)
poly(vinylidene dichloride)

Parentheses are used when the name of the monomer consists of two or more words, and when the monomer has substituents.

These are all olefinic type reactants, from which two carbon atoms originally linked by a double bond form a bivalent group:

$$\mathrm{R\text{-}CH{=}CH\text{-}R^1} \rightarrow \mathrm{-CHR\text{-}CHR^1-}$$

where R and R^1 represent a hydrogen atom or a substituent group. Many of these are joined, in turn, to each other but their exact orientation or sequence is seldom known.

On the other hand, different polymers derived from a single monomer and having identical constitutional units can still be differentiated by additional information, such as average molecular weight and other chemical and physical characteristics.

1.2. Copolymers For copolymers, the names of monomers are cited after the prefix "poly". In addition, an italicized connective (infix) is placed between the names of monomers to denote the kind of sequential arrangement by which the constitutional (monomeric) units, derived from each monomer, are related in the structure [4].

Seven types of sequence arrangements are listed with their corresponding connectives:

Type	Connective
unknown or unspecified	*-co-*
statistical (obeying known statistical laws)	*-stat-*
random (obeying Bernoullian distribution)	*-ran-*
alternating (for two monomeric units)	*-alt-*
periodic (ordered sequence of more than two)	*-per-*
block (linear arrangement of blocks)	*-block-*
graft (side blocks chains connected to main chain)	*-graft-*

Examples: poly[styrene-*co*-(methyl methacrylate)]
poly(styrene-*stat*-acrylonitrile-*stat*-butadiene)
poly[ethylene-*ran*-(vinyl acetate)]
poly[(ethylene glycol)-*alt*-(terephthalic acid)]
poly[formaldehyde-*per*-(ethylene oxide)-*per*-(ethylene oxide)]
polystyrene-*block*-polybutadiene
polybutadiene-*graft*-polystyrene

The names of the monomers are those common or semisystematic names that are encountered most often in the literature of polymer science. The order of citation of monomers in copolymer names is arbitrary.

An equally acceptable alternative scheme for naming copolymers utilizes the prefix "copoly", followed by citation of the names of the monomers, separated by an oblique stroke (a solidus). Parentheses are not needed to enclose monomer names consisting of two or more words.

Examples: copoly(styrene/methyl methacrylate)
stat-copoly(styrene/acrylonitrile/butadiene)
ran-copoly(ethylene/vinyl acetate)
alt-copoly(ethylene glycol/terephthalic acid)
per-copoly(formaldehyde/ethylene oxide/ethylene oxide)
block-copoly(styrene-butadiene)
graft-copoly(butadiene-styrene)

1.3. Nonlinear Macromolecules and Macromolecular Assemblies Most recently, the source-based nomenclature has been extended for non-linear macromolecules and macromolecular assemblies [5]. The non-linear macromolecules comprise branched, graft, comb, star, cyclic, and network macromolecules. The macromolecular assemblies comprise polymer blends, interpenetrating polymer networks, and polymer–polymer complexes.

The following italicized qualifiers can be used as both prefixes (e.g., *blend-*, *net-*) and infixes (connectives) (e.g., *-blend-*, *-net-*) to designate the skeletal structure of non-linear macromolecules or macromolecular assemblies:

Type	Connective
cyclic	*cyclo*
branched, unspecified	*branch*
short-chain-branched	*sh-branch*
long-chain-branched	*l-branch*
branched with branch point of functionality *f*	*f-branch*
comb	*comb*
star	*star*
star with *f* arms	*f-star*
network	*net*
crosslink	*ι* (Greek iota)
polymer blend	*blend*
interpenetrating polymer network	*ipn*
semi-interpenetrating polymer network	*sipn*
polymer–polymer complex	*compl*

In naming non-linear homopolymer molecules, the italicized prefix for the skeletal structure of the macromolecule is placed *before* the source-based name of the constituent linear chain.

Assemblies of macromolecules held together by non-covalent bonds are named by a combination of the names of the constituent macromolecules with an italicized connective between them.

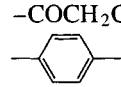
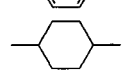
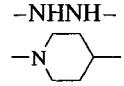
Examples: polystyrene-*comb*-polyacrylonitrile
comb-poly(styrene-*stat*-acrylonitrile)
polystyrene-*comb*-[polyacrylonitrile; poly(methyl methacrylate)]
4-*star*-polystyrene
star-(polyA-*block*-polyB-*block*-polyC)
star-(polyA; polyB; polyC)
star-(polyacrylonitrile; polystyrene) (M_r 100000 : 20000)
net-polystyrene-*ι*-divinylbenzene
net-poly[styrene-*alt*-(maleic anhydride)]-*ι*-(ethylene glycol)
polystyrene-*blend*-poly(2,6-dimethylphenol)
poly(methyl methacrylate)-*blend*-poly(*n*-butyl acrylate)
(*net*-polystyrene)-*sipn*-poly(vinyl chloride)
[*net*-poly(styrene-*stat*-butadiene]-*ipn*-[*net*-poly(ethyl acrylate)

2. Structure-Based Nomenclature

2.1. Regular Single-Strand Organic Polymers For regular organic polymers which have only one species of constitutional repeating unit (CRU) in a single sequential arrangement and consist of single strands only, the name is

poly(constitutional repeating unit),

wherein the repeating group is named as a bivalent organic group according to the IUPAC nomenclature rules for organic compounds [1]. Each such repeating group can consists of simple or substituted subunits such as:

$-CH_2-$	methylene
$-CH_2CH_2-$	ethylene
$-CH{=}CH-$	vinylene
$-CH_2CH_2CH_2CH_2CH_2CH_2-$	hexane-1,6-diyl
$-CHCl-CH_2-$	1-chloroethylene
$-COCH_2-CH_2-$	1-oxopropane-1,3-diyl
$-COCH_2CH_2CH_2CH_2CO-$	adipoyl
	1,4-phenylene
	cyclohexane-1,4-diyl
$-O-$	oxy
$-S-$	thio
$-SO_2-$	sulfonyl
$-NH-$	imino
$-N(CH_3)-$	methylimino
$-NHNH-$	hydrazo
	piperidine-1,4-diyl
$-SiH_2-$	silanediyl
$-Si(CH_3)_2-$	dimethylsilanediyl

Before the total constitutional repeating unit (CRU) can be given a unique name, a single preferred CRU must be selected.

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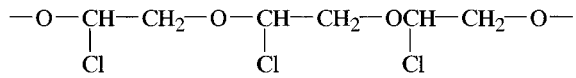
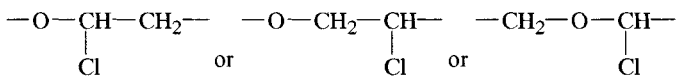
For instance, in a polymer such as

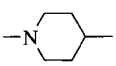


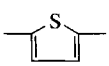
it is immediately evident that a three-atom group (oxygen, chlorine-substituted methylene group, and methylene group) are being regularly repeated. Yet, the CRU can be identified in at least three ways



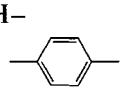
To obtain a unique name based on a preferred CRU, several rules have to be applied. Rules have been developed to specify both seniority among subunits, that is, the point at which to begin writing the CRU, and also the direction in which to move along the chain from left to right to reach the end of the chain.

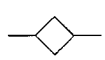
The order of seniority among the types of bivalent groups that are parts of the chain of a single-strand polymer is

a. heterocycles, e.g., piperidine-1,4-diyl: 

thiophene-2,5-diyl: 

b. heteroatomic acyclic groups, e.g., oxy: –O–, sulfonyl: $-SO_2-$, imino: –NH–

c. carbocycles, e.g., 1,4-phenylene: 

cyclobutane-1,3-diyl: 

d. carbon-containing acyclic groups,
e.g., 1-chloroethylene: $-CHCl-CH_2-$
propane-1,3-diyl: $-CH_2-CH_2-CH_2-$

Within each structural type, the seniority is established by further criteria:

a. for heterocycles, a ring system containing nitrogen is senior to a system containing a hetero atom other than nitrogen, with further descending order of seniority governed by the greatest number of rings in the ring system, the largest individual ring in the ring system, the largest number of hetero atoms, and the greatest variety of hetero atoms;

b. for heteroatomic acyclic chains, oxygen is senior to sulfur, sulfur to nitrogen, nitrogen to phosphorus, phosphorus to silicon, silicon to germanium, etc.;

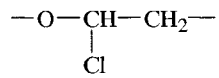
c. for carbocycles, a three-ring system is senior to a two-ring system, a two-ring system containing two 6-membered rings is senior to one containing a 5- and a 6-membered rings, a fused two-ring system (two atoms common to both rings) is senior to a spiro two-ring system (one atom in common), and an unsaturated ring is senior to a saturated ring of the same size;

d. for carbon-only acyclic chains, seniority is determined first by length, then by the number of substituents, by the ascending order of locants, and, finally, by the alphabetical order of the names of the substituent groups.

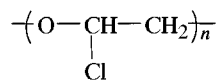
In all cases, the basic seniority rules apply only to those atoms or groups of atoms that are in the main chain. The kinds of substituents on the main chain (whether acyclic carbon-only groups, or hetero-atom-containing groups, or any ring systems) do not affect the selection of the CRU, unless identical basic subunits in the chain have to be further differentiated by the number of substituents and their alphabetical order, but not by the type of the substituent.

If completely identical subunits are separated by other subunits, the direction of citation is determined by the shorter part between them.

The preferred CRU is the one beginning with the subunit of highest seniority. To establish direction, one proceeds from this subunit to the neighboring subunit of the same or next in seniority. In the example of a poly(chloroethylene oxide), shown above, where a regularly repeating structure has been assumed, the subunit of the highest seniority is the oxygen atom and the subunit next in seniority is a substituted $-CH_2-CH_2-$. The substituted subunit, $-CHCl-CH_2-$, is oriented in such a way that the substituent, chlorine atom, is assigned the lowest locant (1 rather than 2). The CRU is written to read from left to right. Thus, the preferred CRU is



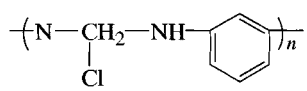
and the polymer



is named

poly[oxy(1-chloroethylene)]

The second example:

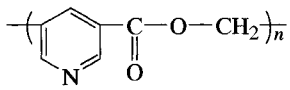


poly[(methylimino)methyleneimino-1,3-phenylene]

illustrates a polymer whose CRU starts with a substituted nitrogen atom and proceeds through the shortest path to the unsubstituted nitrogen atom and then through a carbocycle.

The chemical structure of the CRU is enclosed in parentheses or brackets. While dashes representing chemical bonds may be omitted within the formula unless necessary for clarity, at the ends of the CRU, dashes must be shown. They are drawn across the enclosing marks.

The third example:

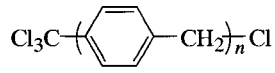


poly(pyridine-3,5-diylcarbonyloxymethylene)

shows a polymer, whose CRU starts with a heterocycle and then proceeds through a substituted carbon atom to a hetero atom.

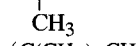
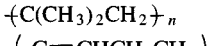
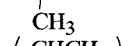
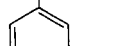
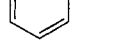
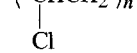
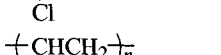
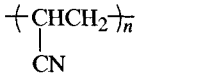
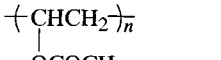
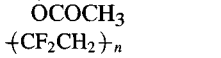
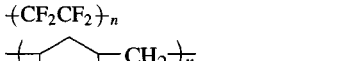
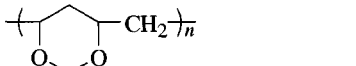
If the end groups of the chain are known, they may be specified by adding prefixes to the polymer name, with the symbol α denoting the left-hand end group and the symbol ω denoting the other end group.

Example:



α-(trichloromethyl)-ω-chloropoly(1,4-phenylene-methylene)

The following table compares and contrasts structure-based and source-based names of some common polymers.

Structure	Structure-based name	Source-based or trivial name
	Poly(methylene)	Polyethylene
	Poly(propylene)	Polypropene
	Poly(1,1-dimethylethylene)	Polyisobutylene
	Poly(1-methyl-1-butenylene)	Polyisoprene
	Poly(1-phenylethylene)	Polystyrene
	Poly(1-chloroethylene)	Poly(vinyl chloride)
	Poly(1-cyanoethylene)	Polyacrylonitrile
	Poly(1-acetoxyethylene)	Poly(vinyl acetate)
	Poly(1,1-difluoroethylene)	Poly(vinylidene fluoride)
	Poly(difluoromethylene)	Polytetrafluoroethylene
	Poly[(2-propyl-1,3-dioxane-4,6-diyl)methylene]	Poly(vinyl butyral)
	Poly[1-(methoxycarbonyl)-1-methylethylene]	Poly(methyl methacrylate)
	Poly(oxyethylene)	Poly(ethylene oxide)
	Poly(oxy-1,4-phenylene)	Poly(phenylene oxide)
	Poly[imino(1-oxohexane-1,6-diyl)]	Poly(ε-caprolactam)
	Poly(oxyethyleneoxyterephthaloyl)	Poly(ethylene terephthalate)
	Poly(iminoadipoyliminohexamethylene)	Poly(hexamethylenediamine-*alt*-adipic acid) *or* poly(hexamethylene adipamide)
	Poly[(2,5-dioxotetrahydrofuran-3,4-diyl)(1-phenylethylene)]	Poly(maleic anhydride-*alt*-styrene)

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2.2. Regular Double-Strand Organic Polymers In a double-strand polymer, the macromolecules consist of an uninterrupted sequence of rings with adjacent rings having two or more atoms in common (a ladder polymer) or one atom in common (a spiro polymer).

As for a single-strand polymer, a single preferred constitutional repeating unit (CRU) must be selected in order to obtain a unique name [6]. The CRU is usually a tetravalent group denoting attachment to four atoms and is named according to the usual rules of organic nomenclature. Again, the name of the polymer is in the form of

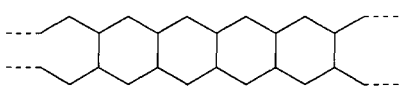
poly(constitutional repeating unit)

Since the polymer has a sequence of rings, in order to identify a preferred CRU, the rings must be broken by observing the following criteria in decreasing order of priority:

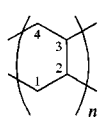
a. minimize the number of free valences in the CRU,
b. maximize the number of most preferred hetero atoms in the ring system,
c. retain the most preferred ring system, and
d. choose the longest chain for acyclic CRU.

Further decisions are based on the seniority of ring systems (indicated in the preceding section), on the orientation of the CRU to give the lowest free valence locant at the lower left of the structural diagram, and on placing the acyclic subunits, if any, on the right side of the ring system within the CRU.

For a polymer consisting of adjacent 6-membered saturated carbon rings:



the name of the preferred CRU, which in this case is an acyclic chain of four carbon atoms with four free valences at each atom (the lowest at the lower left), is



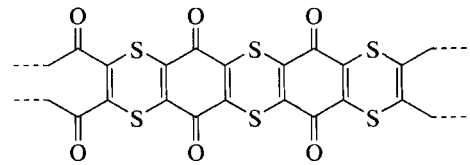
poly(butane-1,4 : 3,2-tetrayl)

The free valence locants are always placed just in front of the corresponding ending of the tetravalent unit and are cited in the order

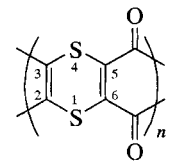
lower-left, upper-left : upper-right, lower-right

that is, in a clockwise direction, the left locants being separated from the right locants by a colon.

For a more complex ladder polymer consisting of an alternating sequence of 6-membered sulfur-containing rings and keto-group containing carbocycles, with two atoms in common at each junction:



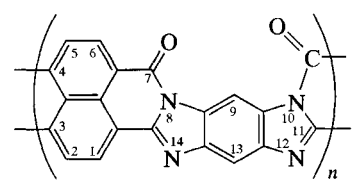
the name based on the preferred CRU is



poly(1,4-dithiin-2,3 : 5,6-tetrayl-5,6-dicarbonyl)

Here, by applying the seniority of the rings system, the heterocycle was left intact, but the carbocycle was broken. That left two bivalent acyclic subunits (carbonyl groups), which according to the rule, have been placed on the right side of the cyclic subunit, with attachments as in the original polymer.

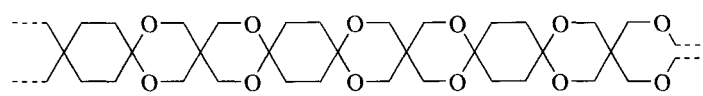
Another, still more complex ladder polymer, derived from the polycondensation of 1,4,5,8-naphthalenetetracarboxylic dianhydride with 1,2,4,5-benzenetetramine:



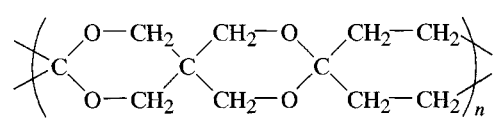
poly[(7-oxo-7*H*,10*H*-benz[*de*]imidazo[4′,5′ : 5,6]benzimidazo [2,1-*a*]isoquinoline-3,4:10,11-tetrayl)-10-carbonyl]

illustrates a six-ring system with four free valences oriented in such a way that the lowest free valence locant is at lower left. It is followed within the total CRU by an acyclic group at the upper right side of the diagram.

For a polymer consisting of regularly repeating adjacent cyclohexane and 1,3-dioxane rings in a spiro sequence (one atom in common at each junction):



the preferred CRU and the name are:



poly(2,4,8,10-tetraoxaspiro[5.5]undecane-3,3:9,9-tetrayl-9,9-diethylene)

2.3. Regular Single-Strand Inorganic and Coordination Polymers The names of inorganic and coordination polymers are based on the fundamentals developed for organic polymers [7]. As in the nomenclature of organic polymers, these rules apply to structural representations

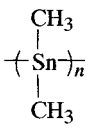
which may at times be idealized and do not take into account irregularities, chain imperfections, or random branching.

A constitutional repeating unit (CRU) is selected and named. However, because of the basic difference between the organic and inorganic nomenclature, the subunits of the CRU are not named as bivalent groups, but as inorganic or coordination groups indicating covalent and/or coordinate bonding.

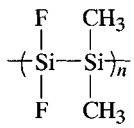
The name of the polymer is the name of the CRU prefixed by the term "poly", "*catena*", or other structural indicator. In order to arrive at the preferred CRU, seniorities of the constitutent subunits are considered as well as the preferred direction for the sequential citation.

The constituent subunit of the highest seniority must contain one or more central atoms; bridging groups between central atoms in the backbone of the polymer are of lower seniority. This is consistent with the principle of coordination nomenclature which puts the emphasis on the coordination center.

Examples of homoatomic backbones are

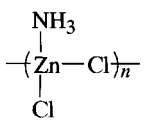


catena-poly[dimethyltin]

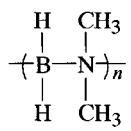


catena-poly[(difluorosilicon)(dimethylsilicon)]

More common coordination polymers consist of a mononuclear central atom with a bridging ligand. The CRU of such a polymer cites the central atom first, which is prefixed by its associated non-bridging ligands, followed in turn by the name of the bridging ligand prefixed by the Greek letter μ:

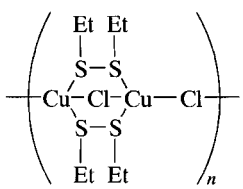


catena-poly[(ammine-chlorozinc)-μ-chloro]



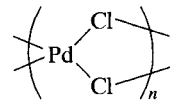
catena-poly[(dihydro-boron)-μ-(dimethylamido)]

Multiple bridging ligands between the pair of central atoms are cited in alphabetic order. Italicized element symbols indicating the coordinating atoms in bridging ligands are cited in the order of direction of the CRU and are separated by a colon:

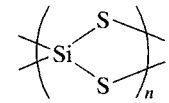


catena-poly[copper-[μ-chloro-bis-μ-(diethyl disulfide-*S*:*S*′)]-copper-μ-chloro]

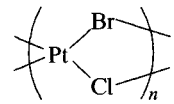
2.4. Regular Quasi-Single-Strand Coordination Polymers In a regular quasi-single-strand coordination polymer, the preferred CRU has one terminal constituent subunit connected through a single atom to the next CRU [7]. Such polymers are named as single-strand coordination polymers:



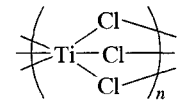
catena-poly[palladium-di-μ-chloro]



catena-poly[silicon-di-μ-thio]



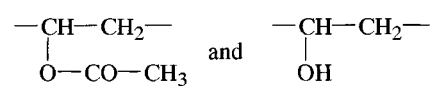
catena-poly[platinum(μ-bromo-μ-chloro)]



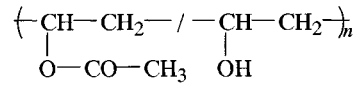
catena-poly[titanium-tri-μ-chloro]

2.5. Irregular Single-Strand Organic Polymers Irregular polymers are named by placing the prefix "poly" before the structure-based names of the constitutional units, collectively enclosed in parentheses or brackets, with the individual constitutional units separated by an oblique stroke (a solidus) [8]. The stroke indicates the irregular or unknown sequential arrangement of these units. The dashes at each end of the formula are drawn *fully inside* the enclosing marks to denote that these are not necessarily terminal bonds of the macromolecule.

For instance, a partially hydrolyzed poly(vinyl acetate) containing units:

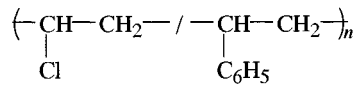


is represented graphically and named:



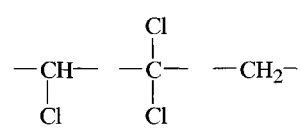
poly(1-acetoxyethylene/1-hydroxyethylene)

A copolymer of vinyl chloride and styrene joined head-to-tail is



poly(1-chloroethylene/1-phenylethylene)

A chlorinated polyethylene consisting of units



is shown and named as

poly(chloromethylene/dichloromethylene/methylene)

C. USE OF COMMON AND SEMISYSTEMATIC NAMES

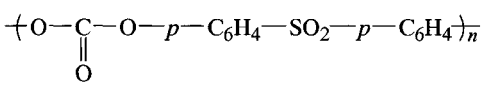
The main purpose of chemical nomenclature is to identify a chemical species by means of written or spoken words for a useful communication among chemists. A systematic nomenclature arose from the need to provide a relationship between the structure and the name. In other words, the reader should be able to deduce and identify the structure from the name.

Traditional names, however, not necessarily based on structures, have been widely used for many common compounds. These are the so-called common, trivial, or semisystematic names, which are satisfactory for communication within a given special chemical field.

It was earlier stated that in the macromolecular nomenclature IUPAC is recommending the use of common and semisystematic names for monomers in the source-based polymer names. The structure-based names, on the other hand, are more related to the structural characteristic of the polymers. But even there, IUPAC allows for a limited number of common names for such substituent groups as "allyl" and "vinyl", and for bivalent groups such as "adipoyl" and "terephthaloyl".

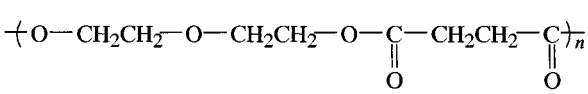
The authors in this *Handbook* for the most part are using the IUPAC-recommended names, or very close variants thereof. On the other hand, in some compilations such as "Crystallographic Data for Various Polymers" (in Section VI), the author provides an excellent introduction and explanation of the names used. For some structure-based names, reflecting constitutional repeating units (CRU), especially for those with a center of symmetry, the naming of linking bivalent groups begins with the central subunit and proceeds in both directions. The IUPAC method selects the most senior bivalent group and proceeds naming from left to right.

Examples:



Handbook name: poly(4,4′-sulfonyldiphenylene carbonate)

IUPAC name: poly(oxycarbonyloxy-1,4-phenylenesulfonyl-1,4-phenylene)

2. 

Handbook name: poly(oxydiethylene succinate)

IUPAC name: poly(oxyethyleneoxyethyleneoxysuccinyl)

In no case, there is any misunderstanding in either case as to which structure is associated with each name.

Abbreviations and acronyms are also extensively used in the chemical literature for monomers, polymers, as well as for additives, modifiers, and fillers. Their extensive list is provided in Section VIII of this Handbook.

As is clearly stated there, the same abbreviation is often used for different monomers and polymers, and the same polymer may have different abbreviations or acronyms.

Examples:

1.	DPP	diphenyl phthalate
		dipropyl phthalate
2.	PVA	poly(vinyl alcohol)
		poly(vinyl acetate)
3.	trioctyl phosphate	TOF
		TOP
4.	polyacrylonitrile	PAC
		PAN

The IUPAC policy [9] on the use of abbreviations in the chemical literature states that there are great advantages in defining all abbreviations in a single conspicuous place in each paper, preferably near the beginning of the paper in a single list. An alternative is to define each abbreviation fully the first time it appears in the text. No abbreviations should be used in the titles of publications. A chapter of this Handbook contains a set of abbreviations recognized by international organizations.

D. CHEMICAL ABSTRACTS (CA) INDEX NAMES

Chemical Abstracts Service (CAS), the publisher of printed Chemical Abstracts (CA) and corresponding products and services in microform, online databases, CD-ROM, and World Wide Web also names the polymers, selected for inclusion in the CAS Chemical Registry System and in the *CA Chemical Substance Indexes*, in two ways:

a. in terms of the component monomer(s) from which they are prepared
b. in terms of the final structure of the resulting polymer.

There are definitive rules that govern when each type of structure is recorded and named [3]. The *primary*, comprehensive representation of polymers by CAS is by citation of the component monomer(s) because, in many cases, the structure of the final polymer is either not known or is not described in sufficient detail by the author.

When the polymer structure is well documented by the author or can be confidently assumed, because one and only one structure is chemically possible, then a *supplementary* representation with a corresponding systematic name is added to the CAS Registry System and included in the printed indexes and online files.

The chemical nomenclature used by CAS has developed in parallel, and generally in accordance, with the rules and recommendations published by IUPAC. However, their rules of systematic nomenclature do not necessarily lead to a unique name for each compound, but do lead to an unambiguous one. This causes no difficulty in normal scientific communication, but is unacceptable in a formal, rigidly controlled, alphabetic listing such as the CA Chemical Substance Index.

The CA index names must not only be unambiguous, unique, and totally reproducible, but also selected so as to bring the names of structurally related substances into juxtaposition in the alphabetic index. CAS has always recognized that, while a unique name is needed for an index and for substance identification, the use of such invariant name in scientific papers is neither practicable nor desirable.

In this section, it is intended to highlight the characteristics of the CA index names for polymers and compare them with some of the typical names used in the scientific community.

a. IUPAC rules have been adapted to the specific needs of a highly ordered alphabetical index.
b. Most common names were eliminated and exceptional treatment for various classes of substances was discontinued.
c. A single preferred name is determined for each identifiable substance.
d. A strict order of precedence of chemical functions and compound classes is followed to determine the preferred "index heading parent".
e. A total name is "inverted" by citing first the *index heading parent* (usually, basic skeleton name with a locant and suffix denoting the principal function), followed by the *comma of inversion*, the *substituents*, and the *modification* (derivative of the principal function), e.g., 2-propenoic acid, 2-methyl-, methyl ester

As mentioned above, most common names, including those still sanctioned by IUPAC, have been replaced by fully systematic names. The following is a short list of most common monomers with both common names and CA Index names:

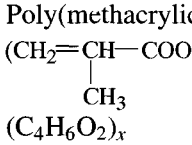
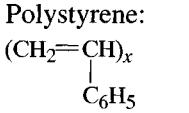
Common	CA Index names (inverted form)
acrylamide	2-propenamide
acrylic acid	2-propenoic acid
acrylonitrile	2-propenamide
adipic acid	hexanedioic acid
ε-caprolactam	2*H*-azepin-2-one, hexahydro-
ε-caprolactone	2-oxepanone
ethylene glycol	1,2-ethanediol
ethylene oxide	oxirane
fumaric acid	2-butenedioic acid, (*E*)-
glycolic acid	acetic acid, hydroxy-
hydroquinone	1,4-benzenediol
isophthalic acid	1,3-benzenedicarboxylic acid
lactic acid	propanoic acid, 2-hydroxy-
maleic acid	2-butenedioic acid, (*Z*)-
maleic anhydride	2,5-furandione
melamine	1,3,5-triazine-2,4,6-triamine
methacrylic acid	2-propenoic acid, 2-methyl-
methyl acrylate	2-propenoic acid, methyl ester
phthalic anhydride	1,3-isobenzofurandione
propylene oxide	oxirane, methyl-
styrene	benzene, ethenyl-
terephthalic acid	1,4-benzenedicarboxylic acid
vinyl alcohol	ethenol
vinyl chloride	ethene, chloro-
vinylidene chloride	ethene, 1,1-dichloro-

Similarly, commonly named substituent and multiplying groups have their systematic equivalents in CA Indexes:

adipoyl	1,6-dioxo-1,6-hexanediyl
allyl	2-propenyl
tert-butyl	1,1-dimethylethyl
ethylene	1,2-ethanediyl
hexamethylene	1,6-hexanediyl
isopropyl	1-methylethyl
succinyl	1,4-dioxo-1,4-butanediyl
vinyl	ethenyl
vinylene	1,2-ethenediyl

Homopolymers are described by the term "homopolymer" cited in the *modification* under the monomer name in the index. In the structural diagram derived from the CAS Registry System records, the structure of the monomer is enclosed in parentheses followed by a subscript "x". The corresponding empirical formula is also expressed in a similar way.

Examples:

1. Poly(methacrylic acid):

$(C_4H_6O_2)_x$
CA name: 2-Propenoic acid, 2-methyl-, homopolymer
2. Polystyrene:

$(C_8H_8)_x$
CA name: Benzene, ethenyl-, homopolymer
3. Poly(11-aminoundecanoic acid):
$(H_2N–(CH_2)_{10}–COOH)_x$
$(C_{11}H_{23}NO_2)_x$
CA name: Undecanoic acid, 11-amino-, homopolymer

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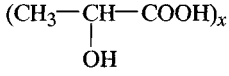
4. Poly(lactic acid):



$(C_3H_6O_3)_x$

CA name: Propanoic acid, 2-hydroxy-, homopolymer

Copolymers formed from two or more monomers are described by the term "polymer with" (followed by the other monomers in alphabetical order) cited in the *modification* under each monomer name. In the structural diagram derived from the CAS Registry System records, the structures of monomers, separated by a period or periods, are enclosed in parentheses followed by a subscript "*x*". The corresponding empirical formulas are similarly expressed. The descriptors *alternating, block,* and *graft* are cited whenever applicable.

Examples:

1. Butadiene-vinyl ethyl ether copolymer:

   $(CH_2{=}CH{-}CH{=}CH_2{\cdot}CH_2{=}CH{-}O{-}CH_2CH_3)_x$

   $(C_4H_8O{\cdot}C_4H_6)_x$ (in Formula Index order)

   CA names: 1,3-Butadiene, polymer with ethoxyethene
   Ethene, ethoxy-, polymer with 1,3-butadiene

2. Isophthalic acid-terephthalic acid-ethylene glycol polyester:

   $(HOOC\text{-}m\text{-}C_6H_4{-}COOH{\cdot}HOOC\text{-}p\text{-}C_6H_4{-}COOH{\cdot}OH{-}(CH_2)_2{-}OH)_x$

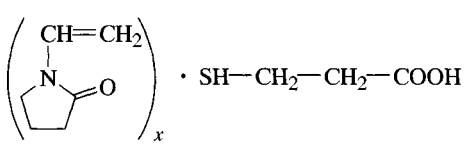
   $(C_8H_6O_4{\cdot}C_8H_6O_4{\cdot}C_2H_6O_2)_x$

   CA names: 1,3-Benzenedicarboxylic acid, polymer with 1,4-benzenedicarboxylic acid and 1,2-ethanediol
   1,4-Benzenedicarboxylic acid, polymer with 1,3-benzenedicarboxylic acid and 1,2-ethanediol
   1,2-Ethanediol, polymer with 1,3-benzenedicarboxylic acid and 1,4-benzenedicarboxylic acid

Telomers are named as copolymers with the term "telomer with" cited in the *modification* under both the monomer (taxogen) and the chain-transfer agent (telogen). In the structural representation, the monomer is enclosed in parentheses with a subscript "*x*", followed by the telogen structure. The corresponding empirical formula is expressed in the same way.

Examples:

1. Vinylpyrrolidone homopolymer terminated with mercaptopropionic acid:



$(C_6H_9NO)_x{\cdot}C_3H_6O_2S$

   CA names: 2-Pyrrolidinone, 1-ethenyl-, telomer with 3-mercaptopropanoic acid
   Propanoic acid, 3-mercapto-, telomer with 1-ethenyl-2-pyrrolidinone

2. Acrylic acid-vinyl choride copolymer terminated with carbon tetrachloride:

   $(CH_2{=}CH{-}COOH{\cdot}Cl{-}CH{=}CH_2)_x{\cdot}CCl_4$

   $(C_3H_4O_2{\cdot}C_2H_3Cl)_x{\cdot}CCl_4$

   CA names: 2-Propenoic acid, telomer with chloroethene and tetrachloromethane
   Ethene, chloro-, telomer with 2-propenoic acid and tetrachloromethane
   Methane, tetrachloro-, telomer with chloroethene and 2-propenoic acid

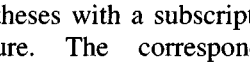
Structural repeating units (SRU), equivalent to IUPAC constitutional repeating units (CRU), are selected and named for polymers that have well documented regular structure, or can confidently be assumed. Assumptions are made for

a. polyamides formed from a dibasic acid (ester or halide) and a diamine; from an amino acid (ester or halide); from a lactam;
b. polyesters formed from a dibasic acid (anhydride, ester, or halide) and a dihydric alcohol (phenol); from a hydroxy acid (ester, halide); from a lactone;
c. polycarbonates formed from carbonic acid (ester, dihalide) and a dihydric alcohol (phenol);
d. polyurethanes formed from a diisocyanate and a dihydric alcohol (phenol).

The structural repeating unit is named by citation of one or more multivalent radicals of regular substitutive nomenclature. The selection of the preferred SRU, its orientation, and the construction of the name, proceeding from left to right, follows the same rules as those in the IUPAC recommendations for the CRU. In CA names, however, the names of the radicals are fully systematic, as explained and contrasted above. The empirical formula is enclosed in parentheses followed by a subscript "*n*".

Examples:

1. Nylon 11:



$(C_{11}H_{21}NO)_n$

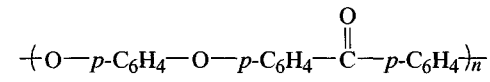
   CA name: Poly[imino(1-oxo-1,11-undecanediyl)]

2. Poly(ethylene adipate):

$(C_8H_{12}O_4)_n$

   CA name: Poly[oxy-1,2-ethanediyloxy(1,6-dioxo-1,6-hexanediyl)]

3. Poly(ether ether ketone) (PEEK):



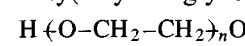
$(C_{19}H_{12}O_3)_n$

CA name: Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)

Poly(alkylene glycols) and their ethers and esters are indexed in CA as structural repeating units with the specified end groups, if aplicable.

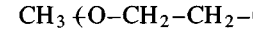
Examples:

1. Poly(ethylene glycol):



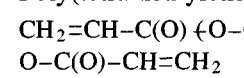
CA name: Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-

2. Poly(trimethylene glycol) dimethyl ether:



CA name: Poly(oxy-1, 3-propanediyl), α-methyl-ω-methoxy-

3. Poly(tetramethylene glycol) diacrylate:



CA name: Poly(oxy-1, 4-butanediyl), α-(1-oxo-2-propenyl)-ω-[(1-oxo-2-propenyl)oxy]-

E. POLYMER CLASS NAMES

In most textbooks, handbooks, encyclopedias, and indexes to polymer nomenclature, information on polymer is grouped under polymer class names. The following is a list of such terms extracted from these sources.

Acrylic Polymers
Alkyd resins
Aminoplasts
Coumarone-indene-resins
Epoxy resins
Fluoropolymers
Phenolic resins
Polyacetals
Polyacetylenes
Polyacrylics
Polyalkylenes
Polyalkenylenes
Polyalkynylenes
Polyamic acids
Polyamides
Polyamines
Polyanhydrides
Polyarylenealkenylenes
Polyarylenealkylenes
Polyarylenes
Polyazomethines
Polybenzimidazoles
Polybenzothiazoles
Polybenzoxazinones
Polybenzoxazoles
Polybenzyls
Polycarbodiimides
Polycarbonates
Polycarboranes
Polycarbosilanes
Polycyanurates
Polydienes
Polyester-polyurethanes
Polyesters
Polyetheretherketones
Polyether-polyurethanes
Polyethers
Polyhydrazides
Polyimidazoles
Polyimides
Polyimines
Polyisocyanurates
Polyketones
Polyolefins
Polyoxadiazoles
Polyoxides
Polyoxyalkylenes
Polyoxyarylenes
Polyoxymethylenes
Polyoxyphenylenes
Polyphenyls
Polyphosphazenes
Polypyrroles
Polypyrrones
Polyquinolines
Polyquinoxalines
Polysilanes
Polysilazanes
Polysiloxanes
Polysilsesquioxanes
Polysulfides
Polysulfonamides
Polysulfones
Polythiazoles
Polythioalkylenes
Polythioarylenes
Polythioethers
Polythiomethylenes
Polythiophenylenes
Polyureas
Polyurethanes
Polyvinyl acetals
Polyvinyl butyrals
Polyvinyl formals
Vinyl polymers

These classes are not mutually exclusive, some are relatively generic and some more specific. Some of them

could be grouped and arranged in a hierarchy, e.g.,

Polyacetals
  Polyvinyl acetals
    polyvinyl butyrals
    polyvinyl formals

Polyesters
  polycarbonates

Polyethers
  epoxy resins
  polyetheretherketones
  polyoxyalkylenes
    polyoxymethylenes
  polyoxyarylenes
    polyoxyphenylenes

Some are synonymous or near synonymous, e.g., polyalkenylenes and polydienes, polyalkynylenes and polyacetylenes, polyethers and polyoxides, polythioethers and polysulfides.

F. REFERENCES

1. IUPAC, "Nomenclature of Regular Single-Strand Organic Polymers (Recommendations 1975)", Pure Appl. Chem. **48**, 373 (1976). Reprinted as Chapter 5 in Ref. 2.
2. IUPAC, "Compendium of Macromolecular Nomenclature". W. V. Metanomski, (Ed.), Blackwell Scientific, Oxford, 1991.
3. Chemical Abstracts Service, "CA Index Guide 1997", Appendix IV, "Chemical Substance Index Names", Chemical Abstracts Service, Columbus, Ohio, 1997.
4. IUPAC, "Source-Based Nomenclature for Copolymers (Recommendations 1985)", Pure Appl. Chem. **57**, 1427 (1985). Reprinted as Chapter 7 in Ref. 2.
5. IUPAC, "Source-Based Nomenclature for Non-Linear Macromolecules and Macromolecular Assemblies (Recommendations 1997)", Pure Appl. Chem. **69**, 2511 (1997).
6. IUPAC, "Nomenclature of Regular Double-Strand (Ladder and Spiro) Organic Polymers (Recommendations 1993)", Pure Appl. Chem. **65**, 1561 (1993).
7. IUPAC, "Nomenclature for Regular Single-Strand and Quasi-Single-Strand Inorganic and Coordination Polymers (Recommendations 1984)", Pure Appl. Chem. **57**, 149 (1985). Reprinted as Chapter 6 in Ref. 2.
8. IUPAC, "Structure-Based Nomenclature for Irregular Single-Strand Polymers (Recommendations 1994)", Pure Appl. Chem. **66**, 873 (1994).
9. IUPAC, "Use of Abbreviations in the Chemical Literature (Recommendations 1979)", Pure Appl. Chem. **52**, 2229 (1980).

Units

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A. Introduction I-13
B. International Units I-13
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D. Conversion Factors I-14
E. Conversion Table for SI vs. English–American Units I-17

A. INTRODUCTION

The International System of Units (1) is used in this Handbook as far as possible, since this system has become obligatory in many European countries and since it is supported by the National Bureau of Standards (2) and the American Society for Testing and Materials (3). Only cursory information is given here for units needed in this Handbook. Detailed information may be found in the following References:

1. "SI-units and recommendations for the use of their multiples and of certain other units", ISO 1000–1973.
2. "The International System of Units (SI)", Eds. C. H. Page, P. Vigoreux, Natl. Bur. Std. Spec. Publ., 1981, 330.
3. "Standard Metric Practice Guide", ASTM E-380–85.
4. Manual of Symbols and Terminology for Physicochemical Quantities and Units; Pergamon Press, Oxford 1979 (Pure Appl. Chem., **51**, 1 (1979).
5. F. S. Conant, using the SI units, Polym. Eng. Sci., **17**, 222 (1977) (further references).

B. INTERNATIONAL UNITS

Quantity	SI-unit	Selection of multiples of the SI-unit	Accepted units used with SI	Units accepted temporarily	Non SI units that should not be used
Length	m (metre)	km, cm, mm, µm, nm		$1\,Å = 10^{-10}\,m$	$1\,µ = 10^{-6}\,m$
Area	m^2	km^2, dm^2, cm^2, mm^2			
Volume	m^3	dm^3, cm^3, mm^3	litre (l, L) ($1\,l = 1\,dm^3$)		
Time	s (second)	ks, ms, µs, ns	d, h, min		
Frequency	Hz (hertz)	THz, GHz, MHz, kHz			
Mass	kg (kilogram)	Mg, g, mg, µg	t		$1\gamma = 10^{-9}\,kg$
Density	kg/m^3	mg/m^3; kg/dm^3; g/cm^3	t/m^3; kg/l		
Force	N (newton)	MN, kN, mN, µN			$1\,dyn = 10^{-5}\,N$; pond, kilopond
Pressure	Pa (pascal) or N/m^2	GPa, MPa, kPa, mPa, µPa		$1\,bar = 10^5\,Pa$ $1\,atm = 101325\,Pa$	$1\,Torr = \frac{101325}{760}\,Pa = 133.32\,Pa$
Stress	Pa or N/m^2	GPa; MPa or N/mm^2; kPa			
Viscosity (dynamic)	Pa s	mPa s			poise; $1\,P = 0.1\,Pa\,s$
Viscosity (kinematic)	m^2/s	mm^2/s			stokes; $1\,St = 1\,cm^2/s$
Surface tension	N/m	mN/m			dyn/cm
Energy, work, heat	J (joule)	TJ, GJ, MJ, kJ, mJ	Electronvolt (eV) $1\,eV \approx 1.602 \times 10^{-19}\,J$		$1\,erg = 10^{-7}\,J$ $1\,cal = 4.1868\,J$
Power	W (watt)	GW, MW, kW, mW, µW			
Temperature	K (kelvin), °C				$1°K = 1\,K$
Thermal conductivity	W/mK				0.86 kcal/(m h grd)
Heat capacity	J/K	kJ/K			
Specific heat	J/(kg K)	kJ/(kg K)			
Entropy	J/K	kJ/K			
Amount of substance	mol (mole)				Mole
Electrical resistance	Ω				
Electrical conductivity	S (1/Ω)				

C. SI-PREFIXES

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{18}	exa	E	10^{-1}	deci	d
10^{15}	peta	P	10^{-2}	centi	c
10^{12}	tera	T	10^{-3}	milli	m
10^{9}	giga	G	10^{-6}	micro	μ
10^{6}	mega	M	10^{-9}	nano	n
10^{3}	kilo	k	10^{-12}	pico	p
10^{2}	hecto	h	10^{-15}	femto	f
10^{1}	deca	da	10^{-18}	atto	a

D. CONVERSION FACTORS

Length	m	in	ft	yd	thou or mil
1 m	1	3.937×10^{1}	3.281	1.094	3.937×10^{4}
in	2.540×10^{-2}	1	8.333×10^{-2}	2.778×10^{-2}	1.0×10^{3}
ft	3.048×10^{-1}	12	1	3.333×10^{-1}	1.2×10^{4}
yd	9.144×10^{-1}	36	3	1	3.6×10^{4}
thou or mil	2.540×10^{-5}	1.0×10^{-3}	8.334×10^{-5}	2.778×10^{-5}	1

Area	m^2	sq. in	sq. ft	sq. yd	ar
$1\,m^2$	1	1.550×10^{3}	1.076×10^{1}	1.196	1.0×10^{-2}
sq. in	6.452×10^{-4}	1	6.994×10^{-3}	7.716×10^{-4}	6.452×10^{-6}
sq. ft	9.290×10^{-2}	1.440×10^{2}	1	1.111×10^{-1}	9.290×10^{-4}
sq. yd	8.361×10^{-1}	1.296×10^{3}	9	1	8.361×10^{-3}
ar	1.0×10^{2}	1.550×10^{5}	1.076×10^{3}	1.196×10^{2}	1

Volume	m^3	1 (lit)	cu. in	cu. ft	cu. yd	gal (US)	gal (UK)
$1\,m^3$	1	10^{3}	6.102×10^{4}	3.531×10^{1}	1.308	2.642×10^{2}	2.20×10^{2}
1 (lit)	10^{-3}	1	6.102×10^{1}	3.531×10^{-2}	1.308×10^{-3}	2.642×10^{-1}	2.20×10^{-1}
cu. in	1.639×10^{-5}	1.639×10^{-2}	1	5.787×10^{-4}	2.143×10^{-5}	4.329×10^{-3}	3.605×10^{-3}
cu. ft	2.832×10^{-2}	2.832×10^{1}	1.728×10^{3}	1	3.703×10^{-2}	7.481	6.229
cu. yd	7.646×10^{-1}	7.645×10^{2}	4.666×10^{4}	2.7×10^{1}	1	2.020×10^{2}	1.682×10^{2}
gal (US)	3.785×10^{-3}	3.785	2.310×10^{2}	1.337×10^{-1}	4.951×10^{-3}	1	8.327×10^{-1}
gal (UK)	4.546×10^{-3}	4.546	2.774×10^{2}	1.605×10^{-1}	5.946×10^{-3}	1.201	1

Mass	kg	lbm	ton (UK) (long ton)	cwt (UK) (long cwt)	ton (US) (short ton)	ounce	grain
1 kg	1	2.205	9.842×10^{-4}	1.968×10^{-2}	1.102×10^{-3}	3.527×10^{1}	1.543×10^{4}
lbm	4.536×10^{-1}	1	4.464×10^{-4}	8.929×10^{-3}	5.0×10^{-4}	1.6×10^{1}	7.0×10^{3}
ton (metric)	1×10^{3}	2.205×10^{3}	9.842×10^{-1}	19.68	1.102	3.527×10^{4}	1.543×10^{7}
ton (UK)	1.016×10^{3}	2.240×10^{3}	1	2.0×10^{1}	1.120	3.584×10^{4}	1.568×10^{7}
cwt (UK)	5.080×10^{1}	1.120×10^{2}	5.0×10^{-2}	1	5.600×10^{-2}	1.792×10^{3}	7.840×10^{5}
ton (US)	9.072×10^{2}	2.0×10^{3}	8.929×10^{-1}	1.786×10^{1}	1	3.2×10^{4}	1.4×10^{7}
ounce	2.835×10^{-2}	6.250×10^{-2}	2.790×10^{-5}	5.580×10^{-4}	3.125×10^{-5}	1	4.375×10^{2}
grain	6.480×10^{-5}	1.429×10^{-8}	6.378×10^{-8}	1.276×10^{-6}	7.143×10^{-8}	2.286×10^{-3}	1

Density	kg/m^3	$Mg/m^3 = g/cm^3$	lbm/cu.ft	lbm/cu.in	lbm/gal (UK)	lbm/gal (US)
$1\,kg/m^3$	1	1.0×10^{-3}	6.243×10^{-2}	3.613×10^{-5}	1.002×10^{-2}	8.345×10^{-3}
$Mg/m^3 = g/cm^3$	1.0×10^{3}	1	6.243×10^{1}	3.613×10^{-2}	1.002×10^{1}	8.345
lbm/cu. ft	1.602×10^{1}	1.602×10^{-2}	1	5.789×10^{-4}	1.605×10^{-1}	1.337×10^{-1}
lbm/cu. in	2.768×10^{4}	2.768×10^{1}	1.728×10^{3}	1	2.774×10^{2}	2.310×10^{2}
lbm/gal (UK)	9.978×10^{1}	9.978×10^{-2}	6.229	3.605×10^{-3}	1	8.327×10^{-1}
lbm/gal (US)	1.198×10^{2}	1.198×10^{-1}	7.480	4.329×10^{-3}	1.201	1

Force	N (kg m/s^2)	kgf, kp	lbf	dyn	tonf (UK) (long ton)	tonf (US) (short ton)
1 N	1	1.020×10^{-1}	2.248×10^{-1}	1.0×10^{5}	1.004×10^{-4}	1.124×10^{-4}
kp	9.807	1	2.205	9.807×10^{5}	9.842×10^{-4}	1.102×10^{-3}
lbf	4.448	4.536×10^{-1}	1	4.448×10^{5}	4.464×10^{-4}	5.0×10^{-4}
dyn	1.0×10^{-5}	1.020×10^{-6}	2.248×10^{-6}	1	1.004×10^{-9}	1.124×10^{-9}
tonf (UK)	9.964×10^{3}	1.016×10^{3}	2.240×10^{3}	9.964×10^{8}	1	1.120
tonf (US)	8.896×10^{3}	9.072×10^{2}	2.0×10^{3}	8.896×10^{8}	8.929×10^{-1}	1

Pressure	Pa	kPa	MPa (N/mm^2)	bar	p/cm^2	kp/m^2	at†	atm†	Torr†	Psi (lbf/sq. in)†
1 Pa (= 1 N/m^2)	1	10^{-3}	10^{-6}	10^{-5}	1.02×10^{-2}	1.02×10^{-1}	1.02×10^{-5}	9.87×10^{-6}	7.50×10^{-6}	1.450×10^{-4}
1 kPa	10^{3}	1	10^{-3}	10^{-2}	1.02×10^{1}	1.02×10^{2}	1.02×10^{-2}	9.87×10^{-3}	7.50	1.450×10^{-1}
1 MPa (=1 N/mm^2	10^{6}	10^{3}	1	10^{1}	1.02×10^{4}	1.02×10^{5}	1.02×10^{1}	9.87	7.50×10^{3}	1.450×10^{2}
1 bar (=0.1 MPa)	10^{5}	10^{2}	10^{-1}	1	1.02×10^{3}	1.02×10^{4}	1.02	9.87×10^{-1}	7.50×10^{2}	1.450×10^{1}
1 p/cm^2 †	9.81×10^{1}	9.81×10^{-2}	9.81×10^{-5}	9.81×10^{-4}	1	10^{1}	10^{-3}	9.68×10^{-4}	7.36×10^{-1}	1.422×10^{-2}
1 kp/m^2 †	9.81	9.81×10^{-3}	9.81×10^{-6}	9.81×10^{-5}	10^{-1}	1	10^{-4}	9.68×10^{-5}	7.36×10^{-2}	1.422×10^{-3}
1 at (= 1 kp/cm^2)†	9.81×10^{4}	9.81×10^{1}	9.81×10^{-2}	9.81×10^{-1}	10^{3}	10^{4}	1	9.68×10^{-1}	7.36×10^{2}	1.422×10^{1}
1 atm (= 760 Torr)†	1.01325×10^{5}	1.01325×10^{2}	1.01325×10^{-1}	1.013	1.033×10^{3}	1.033×10^{4}	1.033	1	7.60×10^{2}	1.470×10^{1}
1 Torr (= 1/760 atm)†	1.33×10^{2}	1.33×10^{-1}	1.33×10^{-4}	1.33×10^{-3}	1.36	1.36×10^{1}	1.36×10^{-3}	1.32×10^{-3}	1	1.934×10^{-2}
Psi (lbf/sq. in)	6.895×10^{3}	6.895	6.895×10^{-3}	6.895×10^{-2}	7.031×10^{-5}	7.031×10^{-6}	7.031×10^{-2}	6.805×10^{-2}	5.171×10^{-1}	1

† Non SI units.

Stress	Pa	N/mm^2	kp/cm^2†	kp/mm^2†	Psi (lbf/sq. in)†
1 Pa (= 1 N/m^2)	1	10^{-6}	1.02×10^{-5}	1.02×10^{-7}	1.450×10^{-4}
1 N/mm^2 (= 1 MPa)	10^{6}	1	1.02×10^{1}	1.02×10^{-1}	1.450×10^{2}
1 kp/cm^2 (= 1 at)†	9.81×10^{4}	9.81×10^{-2}	1	10^{-2}	1.422×10^{1}
1 kp/mm^2 †	9.81×10^{6}	9.81	10^{2}	1	1.422×10^{3}
Psi lbf/sq. in	6.895×10^{3}	6.895×10^{-3}	7.031×10^{-2}	7.031×10^{-4}	1

Non SI units.

Viscosity (dynamic)	Pa s (N s/πm^2) (kg/(s m))	mPa s (mN s/m^2)	cP	kp s/m^2	kp h/m^2	lbm/(ft s)	lbm/(ft h)	lbf s/sq. ft
1 Pa s	1	1.0×10^{3}	1.0×10^{3}	1.020×10^{-1}	2.833×10^{-5}	6.720×10^{-1}	2.419×10^{3}	2.089×10^{-2}
mPa s	1.0×10^{-3}	1	1	1.020×10^{-4}	2.833×10^{-8}	6.720×10^{-4}	2.419	2.089×10^{-5}
cP	1.0×10^{-3}	1	1	1.020×10^{-4}	2.833×10^{-8}	6.720×10^{-4}	2.419	2.089×10^{-5}
kp s/m^2	9.807	9.807×10^{3}	9.807×10^{3}	1	2.778×10^{-4}	6.590	2.372×10^{4}	2.048×10^{-1}
kp h/m^2	3.530×10^{4}	3.530×10^{7}	3.530×10^{7}	3.60×10^{3}	1	2.372×10^{4}	8.540×10^{7}	7.373×10^{2}
lbm/(ft s)	1.488	1.488×10^{3}	1.488×10^{3}	1.518×10^{-1}	4.215×10^{-5}	1	3.60×10^{3}	3.103×10^{-2}
lbm/(ft h)	4.134×10^{-4}	4.134×10^{-1}	4.134×10^{-1}	4.215×10^{-5}	1.171×10^{-8}	2.778×10^{-4}	1	8.634×10^{-6}
lbf s/sq. ft	4.788×10^{1}	4.788×10^{4}	4.788×10^{4}	4.882	1.356×10^{-3}	3.217×10^{1}	1.158×10^{5}	1

Viscosity (kinematic)	m^2/s	mm^2/s	m^2/h	cSt	sq. ft/s	sq. ft/h
1 m^2/s	1	1.0×10^{6}	3.60×10^{3}	1.0×10^{6}	1.076×10^{1}	3.875×10^{4}
mm^2/s	1.0×10^{-6}	1	3.60×10^{-3}	1.0	1.076×10^{-5}	3.875×10^{-2}
m^2/h	2.778×10^{-4}	2.778×10^{2}	1	2.778×10^{2}	2.990×10^{-3}	1.076×10^{1}
cSt	1.0×10^{-6}	1.0	3.60×10^{-3}	1	1.076×10^{-5}	3.875×10^{-2}
sq. ft/s	9.290×10^{-2}	9.290×10^{4}	3.345×10^{2}	9.290×10^{4}	1	3.60×10^{3}
sq. ft/h	2.581×10^{-5}	2.581×10^{-1}	9.290×10^{-2}	2.581×10^{1}	2.778×10^{-4}	1

Surface tension	N/m (kg/s^2)	mN/m	kgf/m (kp/m)	dyn/cm
1 N/m	1	1.0×10^3	1.020×10^{-1}	1.0×10^3
mN/m	1.0×10^{-3}	1	1.020×10^{-4}	1.0
kp/m	9.807	9.807×10^3	1	9.807×10^3
dyn/cm	1.0×10^{-3}	1.0	1.020×10^{-4}	1

Energy	J (N m) ($kg\,m^2/s^2$)	kW h	kgf m kp m	Ps h	lbf ft	erg	HP h	kcal	BTU[a]
1 J	1	2.778×10^{-7}	1.020×10^{-1}	3.777×10^{-7}	7.376×10^{-1}	1.0×10^7	3.725×10^{-7}	2.388×10^{-4}	9.478×10^{-4}
kW h	3.60×10^6	1	3.671×10^5	1.360	2.655×10^6	3.60×10^{13}	1.341	8.598×10^2	3.412×10^3
kp m	9.807	2.724×10^{-6}	1	3.704×10^{-6}	7.233	9.807×10^7	3.653×10^{-6}	2.342×10^{-3}	9.295×10^{-3}
PS h	2.648×10^6	7.355×10^{-1}	2.70×10^5	1	1.953×10^6	2.648×10^{13}	9.863×10^{-1}	6.324×10^2	2.510×10^3
lbf ft	1.356	3.766×10^{-7}	1.363×10^{-1}	5.120×10^{-7}	1	1.356×10^7	5.051×10^{-7}	3.238×10^{-4}	1.285×10^{-3}
erg	1.0×10^{-7}	2.778×10^{-14}	1.020×10^{-8}	3.777×10^{-14}	7.375×10^{-8}	1	3.725×10^{-14}	2.388×10^{-11}	9.478×10^{-11}
HP h	2.685×10^6	7.457×10^{-1}	2.737×10^5	1.014	1.980×10^6	2.685×10^{13}	1	6.412×10^2	2.544×10^3
kcal	4.187×10^3	1.163×10^{-3}	4.269×10^2	1.581×10^{-3}	3.088×10^3	4.187×10^{10}	1.560×10^{-3}	1	3.968
BTU	1.055×10^3	2.931×10^{-4}	1.076×10^2	3.985×10^{-4}	7.782×10^2	1.055×10^{10}	3.930×10^{-4}	2.520×10^{-1}	1

[a] British thermal unit.

Heat	J (N m) ($kg\,m^2/s^2$)	kcal	BTU	CHU[a]
1 J	1	2.388×10^{-4}	9.478×10^{-4}	5.262×10^{-4}
kcal	4.187×10^3	1	3.968	2.203
BTU	1.055×10^3	2.520×10^{-1}	1	5.552×10^{-1}
CHU	1.900×10^3	4.539×10^{-1}	1.80	1
W h	3.60×10^3	8.598×10^{-1}	3.412	1.894

[a] Centigrade heat unit.

Power	W (J/s) ($kg\,m^2/s^3$)	kgf m/s kp m/s	PS HP (metr)	HP	erg/s	ft lbf/s	kcal/h	BTU/h
1 W	1	1.020×10^{-1}	1.360×10^{-3}	1.341×10^{-3}	1.0×10^7	7.376×10^{-1}	8.598×10^{-1}	3.412
kp m/s	9.807	1	1.333×10^{-2}	1.315×10^{-2}	9.807×10^7	7.233	8.432	3.346×10^1
PS	7.355×10^2	7.5×10^1	1	9.863×10^{-1}	7.355×10^9	5.425×10^2	6.324×10^2	2.510×10^3
HP	7.457×10^2	7.604×10^1	1.014	1	7.457×10^9	5.50×10^2	6.412×10^2	2.544×10^3
erg/s	1.0×10^{-7}	1.020×10^{-8}	1.360×10^{-10}	1.341×10^{-10}	1×10^{-8}	7.375	8.598×10^{-8}	3.412×10^{-7}
ft lbf/s	1.356	1.383×10^{-1}	1.843×10^{-3}	1.818×10^{-3}	1.356×10^7	1	1.166	4.626
kcal/h	1.163	1.186×10^{-1}	1.581×10^{-3}	1.560×10^{-3}	1.163×10^7	8.578×10^{-1}	1	3.968
BTU/h	2.931×10^{-1}	2.988×10^{-2}	3.985×10^{-4}	3.930×10^{-4}	2.931×10^6	2.162×10^{-1}	2.520×10^{-1}	1

Thermal conductivity	W/m K ($kg\,m/(s^3\,K)$)	kcal/(m h °C)	BTU/(ft h °F)	BTU/(in h °F)	BTU in/ (sq. ft h °F)
1 W/(m K)	1	8.598×10^{-1}	5.778×10^{-1}	4.815×10^{-2}	6.933
kcal/(m h °C)	1.163	1	6.720×10^{-1}	5.60×10^{-2}	8.064
BTU/(ft h °F)	1.731	1.488	1	8.333×10^{-2}	1.2×10^1
BTU/(in h °F)	2.077×10^1	1.786×10^1	1.2×10^1	1	1.44×10^2
BTU in/(sq. ft h °F)	1.442×10^{-1}	1.24×10^{-1}	8.333×10^{-2}	6.944×10^{-3}	1

Specific heat, heat capacity	J/kg K ($m^2/(s^2\,K)$)	kJ/(kg K)	kcal/(kg °C)	BTU/(lbm °F)
1 J/(kg K)	1	1.0×10^{-3}	2.389×10^{-4}	2.389×10^{-4}
kJ/(kg K)	1.0×10^3	1	2.389×10^{-1}	2.389×10^{-1}
kcal/(kg °C)	4.187×10^3	4.187	1	1.0
BTU/(lbm °F)	4.187×10^3	4.187	1.0	1
Wh/(kg K)	3.60×10^3	3.60	8.598×10^{-1}	0.8598

E. CONVERSION TABLE FOR SI vs. ENGLISH–AMERICAN UNITS

SI to English–American unit	English–American unit to SI
Length	
$1\,m = 1.0936\,yd = 3.28\,ft$ $= 39.37\,in$ $1\,mm = 0.03937\,in$	$1\,yd = 3\,ft = 0.9144\,m$ $1\,ft = 12\,in = 0.3048\,m$ $1\,in = 25.4\,mm$
Mass, weight	
$1\,kg = 2.2046\,lb$	$1\,lb = 0.45359\,kg$
Force	
$1\,N = 1(kgm)/s^2 = 2.248 \times 10^{-1}\,lbf$	$1\,lbf = 4.448\,N$
Pressure	
$1\,Pa = 1\,N/m^2 = 1.450 \times 10^{-4}\,lbf/in^2$ $= 2.953 \times 10^{-4}\,in\,Hg$ $= 4.015 \times 10^{-3}\,in\,H_2O$	$1\,lbf/in^2 = 1\,psi = 6.89475 \times 10^3\,Pa$ $= 6.89475 \times 10^{-2}\,bar$ $1\,in\,Hg = 3.38638 \times 10^3\,Pa$ $= 3.38638 \times 10^{-2}\,bar$
$1\,bar = 10^5\,Pa = 1.450 \times 10^1\,lbf/in^2$ $= 2.953 \times 10^1\,in\,Hg$ $= 4.015 \times 10^2\,in\,H_2O$	$1\,in\,H_2O = 2.49089 \times 10^2\,Pa$ $= 2.49089 \times 10^{-3}\,bar$
Tension, stress	
$1\,N/mm^2 = 1.450 \times 10^2\,lbf/in^2$	$1\,lbf/in^2 = 6.89475 \times 10^{-3}\,N/mm^2$
Dynamic viscosity	
$1\,Pa\,s = (N\,s)/m^2 = 2.089 \times 10^{-2}\,(lbf\,s)/ft^2$ $= 1\,kg/(m\,s) = 6.72 \times 10^{-1}\,lbf/(ft\,s)$	$1\,(lbf\,s)/ft^2 = 4.78802 \times 10^1\,Pa\,s$ $1\,lb/(ft\,s) = 1.488\,Pa\,s$
Kinematic viscosity	
$1\,m^2/s = 1.076 \times 10^1\,ft^2/s$	$1\,ft^2/s = 9.29 \times 10^{-2}\,m^2/s$
Energy, quantity of heat	
$1\,J = 1\,W\,s = 1\,N\,m = 7.376 \times 10^{-1}\,ft\,lbf$ $1\,kJ = 9.478 \times 10^{-1}\,BTU$	$1\,ft\,lbf = 1.35582\,J$ $1\,BTU = 1.05506\,kJ$
Power, heat flow rate	
$1\,W = 1\,Nm/s = 7.367 \times 10^{-1}\,(ft\,lbf)/s$ $= 1\,J/s = 4.425 \times 10^1\,(ft\,lbf)/min$ $= 3.412\,BTU/h$	$1\,(ft\,lbf)/s = 1.35582\,W$ $1\,(ft\,lbf)/min = 2.25969 \times 10^{-2}\,W$ $1\,BTU/h = 2.93072 \times 10^{-1}\,W$
Specific heat capacity	
$1\,J/(kg\,K) = 2.388 \times 10^{-4}\,BTU/(lb/°F)$ $1\,J/(m^3\,K) = 1.491 \times 10^{-5}\,BTU/(ft^3\,°F)$	$1\,BTU/(lb\,°F) = 4.1868 \times 10^3\,J/(kg\,K)$ $1\,BTU/(ft^3\,°F) = 6.71 \times 10^4\,J/(m^3\,K)$
Thermal conductivity	
$1\,W/(m\,K) = 6.933\,(BTU\,in)/(ft^2\,h\,°F)$ $= 5.778 \times 10^{-1}\,BTU/(ft\,h\,°F)$ $= 4.815 \times 10^{-2}\,BTU/(in\,h\,°F)$	$1\,(BTU\,in)/(ft^2\,h\,°F) = 1.442 \times 10^{-1}\,W/(m\,K)$ $1\,BTU/(ft\,h\,°F) = 1.7307\,W/(m\,K)$ $1\,BTU/(in\,h\,°F) = 2.07689 \times 10^1\,W/(m\,K)$

	SI to English–American unit	English–American unit to SI
Heat flux density		
	$1\,W/m^2 = 3.17 \times 10^{-1}\,BTU/(ft^2\,h)$	$1\,BTU/(ft^2\,h) = 3.1546\,W/m^2$
	$= 2.201 \times 10^{-3}\,BTU/(in^2\,h)$	$1\,BTU/(in^2\,h) = 4.54263 \times 10^2\,W/m^2$
Heat transfer coefficient		
	$1\,W/(m^2\,K) = 1.761 \times 10^{-1}\,BTU/(ft^2\,h\,^\circ F)$	$1\,BTU/(ft^2\,h\,^\circ F) = 5.678\,W/(m^2\,K)$
Thermal resistance		
	$1\,(m^2\,K)/W = 5.678\,(ft^2\,h\,^\circ F)/BTU$	$1\,(ft^2\,h\,^\circ F)/BTU = 1.761 \times 10^{-1}\,(m^2/K)/W$
Temperature		
	$1\,K = 1\,^\circ C = 1.8\,^\circ F$	$1\,^\circ F = 5.55\bar{5} \times 10^{-1}\,K = 5.55\bar{5} \times 10^{-1}\,^\circ C$
	Conversion:	
	$T_C = \frac{5}{9}(T_F - 32)$	$T_F = \frac{9}{5}T_C + 32$
	$T_K = \frac{5}{9}T_F + 255.372$	$T_F = \frac{9}{5}(T_K - 255.372)$

T_C – Temperature in celsius (°C); T_F – Temperature in fahrenheit (°F); T_K – Temperature in kelvin (K).

SECTION II

POLYMERIZATION AND DEPOLYMERIZATION

Decomposition Rates of Organic Free Radical Initiators

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A. INTRODUCTION

The decomposition of most organic free radical initiators follows first order kinetics. With certain peroxides, however, higher order decompositions are observed. Generally, the higher order reaction is caused by a reaction of radicals with the initiator (induced decomposition). The value of the rate for unimolecular decomposition may be determined either by extrapolation of the rate back to zero initiator concentration or by use of a monomer or other radical "trap". Some of the peroxides may also decompose by non-radical routes. Acids, bases, and polar solvents favor ionic intermediates. Koenig (296) presents an excellent discussion of azo and peroxide decomposition pathways.

Decomposition rate (k_d) data in these tables are reported for first order kinetics:

$$dL/dt = k_d L$$

where I is the initiator concentration (mol/l) and t is the time (s). The decomposition rate constant k_d is related to half-life ($t_{1/2}$) by the following equation:

$$t_{1/2} = 0.693/k_d$$

Fig. 1 relates k_d in s^{-1} to half-life for the range of k_d found in the tables.

For some of the initiators listed, the enthalpy (ΔH^*) is given (Note h) rather than the Arrhenius activation energy (E_a). The two quantities are related by the equation $E_a = \Delta H^* + RT$, where R is the gas constant (in kJ/mol-deg.) and T is the absolute temperature (94). Assuming that k_d is linear with respect to $1/T$ and that the activation energy, E_a, and the decomposition rate constant, k_d, for one temperature are known, k_d for any temperature can be calculated from the following expression:

$$\log k_2 = \log k_1 - \frac{E_a(T_2 - T_1)}{2.303R(T_2T_1)}$$

Where given by the author, the overall equation for k_d in terms of the frequency factor (A) and activation energy (E_a) has been included. Thus for any temperature (converted to K) the k_d may be calculated:

$$k_d = A \exp(-E_a/RT)$$

Although a wide range of initiators is reported in the tables, the author admits that the compilation is far from complete. There are several purposeful omissions: (a) azo compounds, where the azo group is part of a ring structure – these recombine almost exclusively; (b) compounds that decompose at appreciable rates only above 200 degrees. Neither class would be expected to be useful initiators. The data have been arranged into seven tables. Within each table the individual initiators are listed according to the following criteria:

1. Initiators:
   (a) according to increasing number of carbon atoms;
   (b) alphabetically (neglecting trivial prefixes), for compounds containing equal number of carbons;
   (c) miscellaneous initiators are listed alphabetically in Table 7.
2. For each initiator, solvents are listed alphabetically.
3. For a given solvent, all measurements reported by one investigator are listed in a series, with the activation energy listed opposite the lowest temperature.

Abbreviations: DMSO – dimethyl sulfoxide; DMAC – dimethylacetamide; DMF – dimethylformamide; THF – tetrahydrofuran; DCB – 1,2-dichlorobenzene.

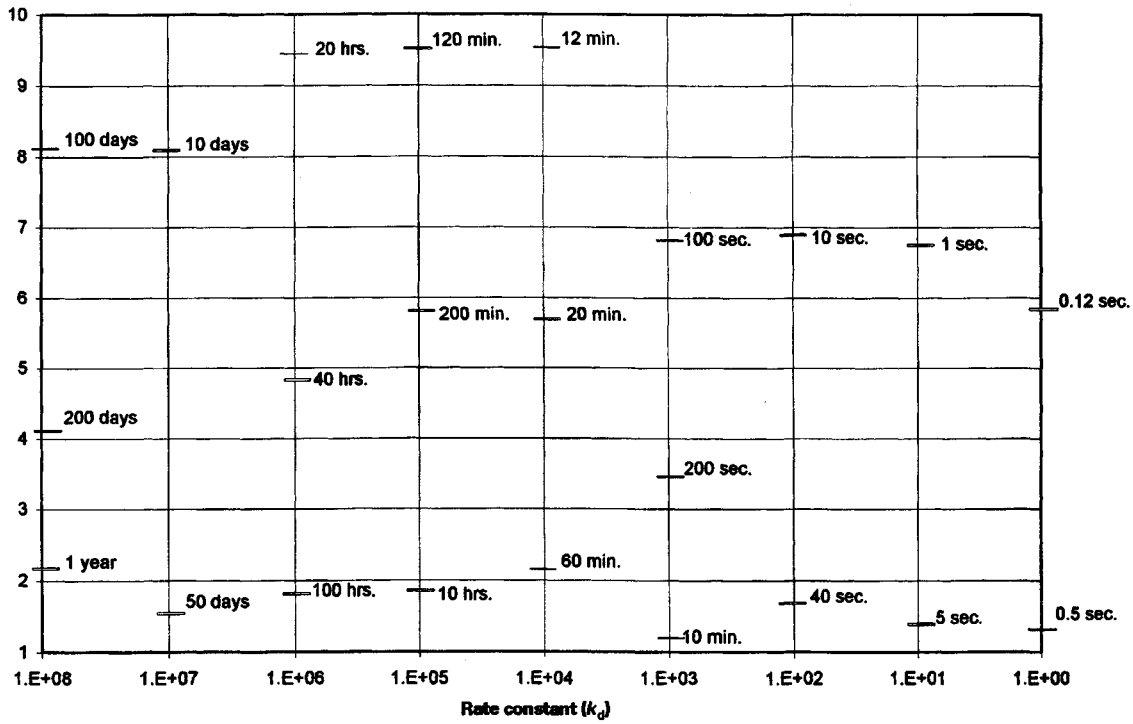

Figure 1. Relationship of half life to rate constant (k_d) (half lives are to the right of each vertical line)

B. TABLES OF DECOMPOSITION RATES OF ORGANIC FREE RADICAL INITIATORS

TABLE 1. AZONITRILES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
5	*N*-Acetyl *N'*-α-cyanoethyl diimide	Toluene	60.8	2.46×10^{-5}	110.2		322
			68.6	5.45×10^{-5}			322
			82.0	2.49×10^{-4}			322
	2-Cyano-2-propyl-azo-formamide	Chlorobenzene	100	1.5×10^{-5}			93
		Toluene	100	2.1×10^{-5}	144.3		93
			110	6.8×10^{-5}			93
			104	1.9×10^{-5}		a	339
		Xylene	100	2.1×10^{-5}	144.3		93
			120	2.4×10^{-4}			93
	2-(Carbamoylazo)isobutyronitrile	Toluene	104	1.93×10^{-5}		a	340
8	*N*-Acetyl *N'*-α-cyanocyclopentyl-diimide	Toluene	65.7	1.82×10^{-5}	120.3		322
			76.8	7.29×10^{-5}			322
			91.5	3.65×10^{-4}			322
	2,2'-Azo-bis-isobutyronitrile (2,2'-azo-bis-2-methylpropionitrile)	Acetic acid	79.9	1.43×10^{-4}			146
			79.9	1.48×10^{-4}		v_2	146
			79.9	1.62×10^{-4}		v_3	146
			80	1.52×10^{-4}		a	62
			82	1.50×10^{-4}		a	2
			82	1.49×10^{-4}		a	175
		Acetonitrile	79.9	1.25×10^{-4}			146
			79.9	1.24×10^{-4}		v_2	146
		tert-Amyl alcohol	80.2	1.40×10^{-4}			61
		Aniline	80.2	1.68×10^{-4}			61
		Benzene	40.0	5.44×10^{-7}	128.4	a,t_2	69
			45.2	1.12×10^{-6}		a,t_2	69
			50.0	2.64×10^{-6}		a,t_2	69
			55.0	5.19×10^{-6}		a,t_2	69
			60.5	1.16×10^{-5}		a,t_2	69
			69.5	3.78×10^{-5}		a,t_2	69

TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	2,2'-Azo-bis-isobutyronitrile	Benzene	40	4.83×10^{-7}	123.4		66
			50	2.085×10^{-6}			66
			60	8.45×10^{-6}			66
			70	3.166×10^{-5}			66
			78	8.023×10^{-5}			66
		Benzene or Toluene	37	2.83×10^{-7}	128.9		39
			43	7.35×10^{-7}			39
			50	2.16×10^{-6}			39
			60	9.15×10^{-6}			19
			100	1.52×10^{-3}			39
			T (K)	1.58×10^{15}	exp[− 128.9/RT]		39
		n-Butanol	82	1.54×10^{-4}		a	175
			82	1.55×10^{-4}		a	2
		Isobutanol	82	1.66×10^{-4}		a	2,175
			80.2	$(1.67–1.76) \times 10^{-4}$			61
		Di-*n*-butyl phthalate	80	2.64×10^{-4}	122.2	y_4	236
			90	6.47×10^{-4}		y_4	236
			100	1.78×10^{-3}		y_4	236
			110	4.88×10^{-3}		y_4	236
			120	1.43×10^{-2}		y_4	236
			127	2.48×10^{-2}		y_{16}	236
			137	5.43×10^{-2}		y_{16}	236
			145	1.24×10^{-1}		y_{32}	236
			80	2.22×10^{-4}			375
			90	4.23×10^{-4}			375
			100	1.99×10^{-3}			375
			110	5.3×10^{-3}			375
			120	1.48×10^{-2}			375
		Carbon tetrachloride	40	2.15×10^{-7}	128.4	a,t_2	69
			60	4.00×10^{-6}		a,t_2	69
			77	1.21×10^{-4}		a	62
		Chlorobenzene	64	1.93×10^{-5}			344
			82	1.93×10^{-4}			344
			101	1.93×10^{-3}			344
			T (K)	2.89×10^{15}	exp[− 130.23/RT]		344
		Cyclohexanone	82	1.43×10^{-4}		a	2,175
		Dichloroethane	70	2.01×10^{-5}			293
				1.64×10^{-5}		m_8	293
		Dichloroethane: propionitrile (1 : 1)	70	1.94×10^{-5}			293
		Diethylene glycol monobutylether	66.82	2.442×10^{-5}			26
		DMF	71.2	6.2096×10^{-5}			294
		DMF	60	6.45×10^{-4}			355
		DMF/methyl methacrylate (9/1)	60	5.55×10^{-4}			355
		(8/2)	60	5.10×10^{-4}			355
		(7/3)	60	5.10×10^{-4}			355
		(6/4)	60	5.30×10^{-4}			355
		(5/5)	60	5.65×10^{-4}			355
		(4/6)	60	5.35×10^{-4}			355
		(3/7)	60	5.65×10^{-4}			355
		(2/8)	60	6.25×10^{-4}			355
		(1/9)	60	6.50×10^{-4}			355
		N,*N*-Dimethylaniline	66.82	3.483×10^{-5}			26
			72.27	6.914×10^{-5}			26
			80	1.83×10^{-4}		a	62
		Dioxane/Water (80/20) pH 7.0	65.3	2.53×10^{-5}	141.0		186
			70.0	3.20×10^{-5}			186
			75.0	8.5×10^{-5}			186
			80.0	1.62×10^{-4}			186
		Dioxane/Water (80/20) pH 10.7	75.0	1.72×10^{-4}			186
		Dodecanethiol	80	1.46×10^{-4}		a	62

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TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	2,2′-Azo-bis-isobutyronitrile	Ethyl acetate	40	4.7×10^{-7}	128.5	a,t_2	69
			60	9.36×10^{-6}		a,t_2	69
		Ethyl acetate/ $SnCl_4$ (1 : 1)	T (K)	8.56×10^{15}	exp[− 127.4/RT]		290
		Glycerol/DMF 95/5, (v/v)	71.2	2.6054×10^{-5}			294
		Maleimide (solid state)	65	4.97×10^{-5}		a	185
			72	2.07×10^{-4}		a	185
		Methyl methacrylate	50	9.7×10^{-7}		a	289
			50	7.5×10^{-6}		a,v_{20}	289
			60	7.20×10^{-4}			355
			70	3.1×10^{-5}			216
			70	1.27×10^{-4}		v_4	216
		Nitrobenzene	80	1.98×10^{-4}	132.2	a	226
			100	2.24×10^{-3}		a	226
		1-Nitrobutane	82	1.45×10^{-4}		a	2,175
		Propionitrile/$SnCl_4$ (1 : 1)	T (K)	9.10×10^{15}	exp[− 127.0/RT]		290
		Propylene carbonate	72.27	5.821×10^{-5}			26
		Styrene	50	2.97×10^{-6}	127.6	a	132
			70	4.72×10^{-5}		a	132
			T (K)	1.29×10^{15}	exp[− 127.6/RT]		132
			68.4	3.8×10^{-5}			234
		Toluene	60.00	9.03×10^{-6}		c,m_3	291
			60	9.8×10^{-6}			290
			65	1.9×10^{-5}		a	339
			70	4.0×10^{-5}	121.3		47
			80.4	1.55×10^{-4}			47
			90.0	4.86×10^{-4}			47
			100.0	1.60×10^{-3}			47
			105.0	2.61×10^{-3}			47
			69.8	3.8×10^{-5}	142.3		61
			80.2	$(1.72–1.60) \times 10^{-4}$			61
			80.3	1.30×10^{-4}		m_2	234
		Toluene/$SnCl_4$ (1 : 1)	60	4.0×10^{-5}			290
		Xylene	50	2.0×10^{-6}		a	175
			77	9.46×10^{-5}		a	175
			82	1.44×10^{-4}		a	175
			80	1.53×10^{-4}	131.0	a	62
			80	1.50×10^{-4}		a,t_4	62
			82	1.45×10^{-5}		a	2
		o-Xylene	85	2.751×10^{-4}	131.8		332
			95	9.581×10^{-4}			332
			105	2.782×10^{-3}			332
			T (K)	3.16×10^{15}	exp[− 131.8/RT]		332
		Water	50	1.56×10^{-6}			285
	3,6-Dicyano-3,6-dimethyl-1,2-diazacyclohexene-1	Dichloromethane	− 6.06	4.60×10^{-4}	88.8	h	252
			0.03	1.14×10^{-3}			252
			2.52	1.64×10^{-3}			252
	2,2′-Azo-bis[2-(hydroxymethyl)-propionitrile	Methyl cellosolve	77	1.93×10^{-5}		a	339
9	*N*-Acetyl *N*′-α-cyanocyclohexyl-diimide	Toluene	82.0	5.21×10^{-5}	114.8		322
			92.8	1.70×10^{-4}			322
			100.6	3.74×10^{-4}			322
10	*N*-Acetyl *N*′-α-cyanocycloheptyl-diimide	Toluene	59.4	4.44×10^{-5}	112.0		322
			66.1	9.82×10^{-5}			322
			79.6	4.51×10^{-4}			322
	4,4′-Azo-bis-(4-cyanopentanol)	Acetone	70	4.26×10^{-5}		m_7	305
	4,4′-Azo-bis-(4-cyanovaleric acid)	Acetone	70	4.58×10^{-5}		m_7	305
	1,1′-Azo-bis-1-cyclobutanenitrile	Mesitylene	130.4	5.20×10^{-5}	134.3		101
			141.6	1.60×10^{-4}			101
		Xylene	120.4	2.05×10^{-5}			101
	2,2′-Azo-bis-2-methylbutyronitrile	Toluene	69.8	2.3×10^{-5}	129.7		61
			80.2	$8.4–9.0 \times 10^{-5}$			61
		Xylene	80	9.97×10^{-5}	123.0	a	62

TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
10 (*cont'd*)	2,2′-Azo-bis-2-ethylpropionitrile	Nitrobenzene	80	8.3×10^{-5}	143.1	a	226
			100	1.08×10^{-3}		a	226
	2,2′-Azo-bis-2-methylbutyronitrile	Cyclohexanone	79.9	1.06×10^{-4}	131.6	h	336
			90.0	3.71×10^{-4}			336
			100.1	1.27×10^{-3}			336
		Chlorobenzene	66	1.93×10^{-5}			344
			84	1.93×10^{-4}			344
			104	1.93×10^{-3}			344
			T (K)	1.38×10^{15}	exp[− 129.93/RT]		344
		Ethanol	80.0	1.05×10^{-4}	119.8	h	336
			90.0	3.23×10^{-4}			336
			100.0	9.87×10^{-4}			336
		Ethylbenzene	80.0	8.22×10^{-5}	134.9	h	336
			90.0	2.94×10^{-4}			336
			100.0	1.02×10^{-3}			336
		Toluene	67	1.9×10^{-5}		a	339
	N-benzoyl *N*′-α-cyanoethyldiimide	Toluene	69.2	1.31×10^{-5}	127.9		322
			83.2	7.96×10^{-5}			322
			90.0	1.744×10^{-4}			322
	4-Bromophenyl-azo-2-methyl-2-propionitrile	DCB	160	2.85×10^{-5}			329
	4-Bromophenyl-azo-(methyl-propanedinitrile)	DCB	80	2.4×10^{-5}	145.0		329
			85	4.8×10^{-5}			329
			90	9.4×10^{-5}			329
			T (K)	7.23×10^{16}	exp[− 145.0/RT]		329
	4-Nitrophenyl-azo-2-methyl-2-propionitrile	DCB	160	2.3×10^{-5}			329
	4-Nitrophenyl-azo-2-(methyl-propanedinitrile)	DCB	80	3.1×10^{-5}	139.1		329
			85	6.0×10^{-5}			329
			90	1.14×10^{-4}			329
			T (K)	1.35×10^{16}	exp[− 139.1/RT]		329
	Phenyl-azo-2-methyl-2-propionitrile	DCB	160	5.65×10^{-5}			329
		DMF	160	6.9×10^{-5}			329
		1,2,4-Trimethylbenzene	160	5.35×10^{-5}			329
	Phenyl-azo-2-(methyl-propanedinitrile)	DCB	80	4.45×10^{-5}	123.2		329
			85	8.0×10^{-5}			329
			90	1.42×10^{-4}			329
			T (K)	8.48×10^{13}	exp[− 123.2/RT]		329
	Phenyl-azo-2-(methyl-propanedinitrile)	DCB	80	2.4×10^{-5}	145.0		329
			85	4.8×10^{-5}			329
			90	9.4×10^{-5}			329
			T (K)	7.23×10^{16}	exp[− 145.0/RT]		329
11	4-Bromophenyl-azo-2-(ethyl-propanedinitrile)	DCB	80	1.85×10^{-5}	142.9		329
			85	3.65×10^{-5}			329
			90	7.05×10^{-5}			329
			T (K)	3.00×10^{16}	exp[− 142.9/RT]		329
	4-Methoxyphenylazo-2-(methylpropanedinitrile)	Butanol	85	2.9×10^{-4}	110		331
		Chlorobenzene	85	9.5×10^{-5}	71		331
		DCB	85	8.35×10^{-5}	115		331
		Decane	85	3.8×10^{-5}	142		331
		DMSO	85	4.87×10^{-4}	114		331
		Heptane	85	5.15×10^{-5}	136		331
		Hexanol	85	1.6×10^{-4}	102		331
		Octane	85	4.4×10^{-5}	140		331
		Octanol	85	1.5×10^{-4}	112		331
		Propanol	85	3.1×10^{-4}	103		331
		Toluene	85	6.75×10^{-5}	137		331
	4-Methoxyphenyl-azo-2-(methylpropanedinitrile)	Chlorobenzene	78.1	5.76×10^{-5}	71.5		330
			85.3	9.52×10^{-5}			330
			90.2	1.307×10^{-4}			330
		DCB	75.0	2.73×10^{-5}	116		330
			85.1	8.35×10^{-5}			330
			95.1	2.236×10^{-4}			330
			104.9	6.397×10^{-4}			330

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TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
11 (*cont'd*)	4-Methoxyphenyl-azo-2-(methylpropanedinitrile)	DCB	T (K)	5.67×10^{12}	exp[−116/RT]		330
			80	4.7×10^{-5}	114.8		329
			85	8.1×10^{-5}			329
			90	1.38×10^{-4}			329
			T (K)	2.94×10^{12}	exp[−114.8/RT]		329
	Phenyl-azo-2-(2-methylbutyronitrile)	DCB	160	5.0×10^{-5}			329
		DMF	160	5.2×10^{-5}			329
		1,2,4-Trimethylbenzene	160	5.05×10^{-5}			329
	4-Tolyl-azo-2-(methylpropanedinitrile)	DCB	80	4.5×10^{-5}	117.7		329
			85	7.9×10^{-5}			329
			90	1.365×10^{-5}			329
			T (K)	1.21×10^{13}	exp[−117.7/RT]		329
12	4,4′-Azo-bis-(1-bromo-4-cyanopentane)	Toluene	T (K)	1.0×10^{11}	exp[−104.7/RT]		307
	4,4′-Azo-bis-4-cyanopentanoic acid	Water	69	1.9×10^{-5}		a	339
			80	8.97×10^{-5}	142.3	a	62
	4,4′-Azo-bis-(4-cyanopentanol)	Di-*n*-butyl phthalate	80	2.55×10^{-4}			375
			90	7.99×10^{-4}			375
			100	2.81×10^{-3}			375
			110	8.99×10^{-3}			375
			120	2.6×10^{-2}			375
			130	5.2×10^{-2}			375
	1,1′-Azo-bis-1-cyclopentanenitrile	Toluene	80.3	7.45×10^{-5}	141.4		101
			89.2	2.43×10^{-4}			101
			95.1	5.18×10^{-4}			101
	2,2′-Azo-bis-2-cyclopropylpropionitrile (mp 64–65)	Toluene	44.2	3.50×10^{-5}	117.2	w	57
			49.5	7.53×10^{-5}		w	57
			59.2	2.68×10^{-4}		w	57
	2,2′-Azo-bis-2-cyclopropylpropionitrile (mp 76–77)	Toluene	44.2	3.90×10^{-5}	108.8	w	57
			49.5	8.17×10^{-5}		w	57
			59.3	2.46×10^{-4}		w	57
	2,2′-Azo-bis-2,3-dimethylbutyronitrile	Toluene	69.8	2.6×10^{-5}	133.9		61
			80.2	1.02×10^{-4}			61
	2,2′-Azo-bis-2-methylvaleronitrile	Toluene	69.8	4.2×10^{-5}	138.1		61
			80.2	$(1.65–1.74) \times 10^{-4}$			61
	4-Bromophenyl-azo-2-(isopropylpropanedinitrile)	DCB	80	1.4×10^{-5}	142.0		329
			85	2.80×10^{-5}			329
			90	5.4×10^{-5}			329
			T (K)	1.73×10^{16}	exp[−142.0/RT]		329
	2,6-Dimethylphenyl-azo-2-methylpropanedinitrile	DCB	60.1	3.06×10^{-5}	130.5		330
			70.0	1.177×10^{-4}			330
			80.6	4.594×10^{-4}			330
			84.5	7.320×10^{-4}			330
			T (K)	2.06×10^{17}	exp[−130.5/RT]		330
13	*N*-Benzoyl *N*′-α-cyanocyclopentyl diimide	Toluene	76.7	1.62×10^{-5}	131.9		322
			82.0	3.17×10^{-5}			322
			98.4	2.252×10^{-4}			322
	4-Bromophenyl-azo-isobutylpropanedinitrile	DCB	80	1.12×10^{-4}	141.6		329
			85	2.21×10^{-4}			329
			90	4.26×10^{-4}			329
			T (K)	1.19×10^{17}	exp[−141.6 kJ/RT]		329
	4-Bromophenyl-azo-2-(2-*tert*-butylpropionitrile)	DCB	160	4.85×10^{-5}			329
	4-Nitrophenyl-azo-2-(2-*tert*-butylpropionitrile)	DCB	160	1.8×10^{-5}			329
	Phenyl-azo-2-(2-*tert*-butylpropionitrile)	DCB	160	9.35×10^{-5}			329
		1,2,4-Trimethylbenzene	160	9.65×10^{-5}			329
		Hexamethylphosphoramide	160	9.5×10^{-5}			329
14	2,2′-Azo-bis-2-cyclobutylpropionitrile (mp 38–42)	Toluene	80.5	1.51×10^{-4}		w	104
	2,2′-Azo-bis-2-cyclobutylpropionitrile (mp 81.5–82.5)	Toluene	80.5	1.51×10^{-4}		w	104

TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	1,1′-Azo-bis-1-cyclohexane nitrile	Chlorobenzene	79.97	8.42×10^{-6}	140.2		179
			100.12	1.107×10^{-4}			179
		DMSO	80	1.01×10^{-5}	135.6	h	135
			85	2.01×10^{-5}			135
			90	3.89×10^{-5}			135
			95	6.83×10^{-5}			135
		Nitrobenzene	100	1.14×10^{-4}			226
		Toluene	80.3	6.5×10^{-6}	121.3		101
			95.2	5.44×10^{-5}			101
			102.4	1.26×10^{-4}			101
			88	1.9×10^{-5}		a	339
		Xylene	77	5.31×10^{-6}		a	175
	2,2′-Azo-bis-2,4-dimethyl-valeronitrile	Toluene	51	1.9×10^{-5}	121	a	343
			68	1.9×10^{-4}		a	343
			T (K)	6.98×10^{14}	exp[− 121/RT]		343
		Xylene	77	5.77×10^{-4}		a	175
	Dimethyl-4,4′-azo-bis-cyano-pentanoate (meso)	DMAC	77.9	1.43×10^{-4}	133.9	a	205
			85.0	3.76×10^{-4}		a	205
			90.2	6.80×10^{-4}		a	205
			99.7	2.05×10^{-3}		a	205
	(n_{D+})		77.6	1.46×10^{-4}	133.5	a	205
			85.9	4.03×10^{-4}		a	205
			90.0	6.90×10^{-4}		a	205
			99.2	2.04×10^{-3}		a	205
	(n_{D-})		77.9	1.49×10^{-4}	133.5	a	205
			85.4	3.90×10^{-4}		a	205
			90.2	6.95×10^{-4}		a	205
			99.0	1.97×10^{-3}		a	205
	4,4′-Azo-bis-4-cyano-l-methyl-piperidine	DMSO	80	9.7×10^{-6}	136.4	h	135
			85	1.75×10^{-5}			135
			90	3.64×10^{-5}			135
			95	6.47×10^{-5}			135
	2,2′-Azo-bis-2-propylbutyronitrile	Nitrobenzene	80	2.55×10^{-4}	128.9	a	226
			100	2.72×10^{-5}		a	226
	2,2′-Azo-bis-2,3,3-trimethylbutyro-nitrile (mp 114–116)	Toluene	79.9	7.42×10^{-5}	146.4	w	57
			89.0	2.59×10^{-4}		w	57
	2,2′-Azo-bis-2,3,3-trimethylbutyro-nitrile (mp 116–118)	Toluene	79.9	1.05×10^{-4}	125.5	w	57
			88.9	3.09×10^{-4}		w	57
	2,2′-Azo-bis-2-methylhexylnitrile	Toluene	80.2	1.58×10^{-4}			61
	2,2′-Azo-bis-2,4-dimethylvaleronitrile	Toluene	69.8	1.98×10^{-4}	121.3		61
			80.2	7.1×10^{-4}			61
	2,2′-Azo-bis-2,4-dimethylvaleronitrile (mp 56–57)	Toluene	59.7	8.05×10^{-5}	121.3	w	57
			69.9	2.89×10^{-4}		w	57
	2,2′-Azo-bis-2,4-dimethylvaleronitrile (mp 74–76)	Toluene	69.8	1.98×10^{-4}	121.3	w	57
			80.2	7.1×10^{-4}		w	57
	2,2′-Azo-bis-2-isopropylbutyronitrile	Toluene	80.5	1.01×10^{-4}			50
	N-(4-Chlorobenzoyl) *N*′-α-cyanocyclohexyl diimide	Toluene	79.1	8.6×10^{-6}	132.7		322
			86.4	2.01×10^{-5}			322
			99.7	1.044×10^{-4}			322
	N-(l-Cyanocyclohexyl)-pentamethyleneketenimine	Chlorobenzene	100	3.85×10^{-5}		p,z (l)	299
			100	2.07×10^{-5}		p,z (3950)	299
		Cumene	100	3.93×10^{-5}		p,z (1)	299
			100	1.42×10^{-5}		p,z (3950)	299
		Toluene	100	3.10×10^{-5}		p,z (1)	299
			100	1.83×10^{-5}		p,z (3950)	299
		tert-Butylbenzene	100	2.39×10^{-5}		p,z (l)	299
			100	1.22×10^{-5}		p,z (3950)	299
	2-Phenyl-azo-2,4-dimethyl-4-methoxyvaleronitrile	Xylene	122	1.9×10^{-5}			339
15	*N*-Benzoyl *N*′-α-cyanocyclo-heptyl diimide	Toluene	74.6	6.43×10^{-5}	122.8		322
			82.8	1.781×10^{-4}			322
			89.2	3.519×10^{-4}			322

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TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
15 (*cont'd*)	*N*-(4-Methoxybenzoyl) *N'*-α-cyanocyclohexyldiimide	Toluene	80.8	7.1×10^{-6}	142.8		322
			87.9	1.71×10^{-5}			322
			101.2	9.73×10^{-5}			322
	N-(4-Methylbenzoyl) *N'*-α-cyanocyclohexyl diimide	Toluene	81.4	1.06×10^{-5}	134.1		322
			87.3	2.20×10^{-5}			322
			101.9	1.255×10^{-4}			322
	N-Phenylacetyl *N'*-α-cyano-cyclohexyl diimide	Toluene	74.3	3.08×10^{-5}	133.9		322
			82.9	9.76×10^{-5}			322
			91.2	2.77×10^{-4}			322
	4-Cyano-1-methylpiperidine-4,4'-azo-4'-cyano-1,1'-dimethylpiperidinium nitrate	DMSO	80	1.76×10^{-5}	132.6	h	135
			85	3.31×10^{-5}			135
			90	6.23×10^{-5}			135
			95	1.156×10^{-4}			135
16	4,4'-Azo-bis-4-cyano-1,1'-dimethyl-piperidinium nitrate	DMSO	80	2.84×10^{-5}	124.7	h	135
			85	5.11×10^{-5}			135
			90	9.87×10^{-5}			135
			95	1.626×10^{-4}			135
	4,4'-Azo-bis(4-cyanopentanoyl)bis-2-aminoethanol	DMF	65	1.76×10^{-5}			357
			69	2.85×10^{-5}			357
			74	4.66×10^{-5}			357
		DMAC	65	1.73×10^{-5}			357
			69	2.73×10^{-5}			357
			74	4.53×10^{-5}			357
	1,1'-Azo-bis-1-cycloheptane-nitrile	Acetic acid	69.5	3.28×10^{-4}			101
		Toluene	48.9	2.69×10^{-5}	115.1		101
			58.9	9.72×10^{-5}			101
			67.3	2.69×10^{-4}			101
	2,2'-Azo-bis-(4-methoxy-2,4-dimethylvaleronitrile)	Toluene	30	1.9×10^{-5}		a	343
			47	1.9×10^{-4}		a	343
			T (K)	1.03×10^{15}	$\exp[-115/RT]$		343
	2,2'-Azo-bis-2-methylheptylonitrile	Toluene	80.0	1.63×10^{-4}			50
		Xylene	80.0	1.78×10^{-4}	126.4	a	62
	1,1'-Azo-bis-1-(2-methylcyclohexane)-nitrile	Toluene	80.2	7.43×10^{-6}			101
	1,1'-Azo-bis-1-cyclohexane-carbonitrile	Xylene	80	4.7×10^{-6}	166.9	a	62
	2,2'-Azo-bis-cyclopentylpropio-nitrile (mp 72.2–74.5)	Toluene	80.5	1.31×10^{-4}		w	104
	2,2'-Azo-bis-2-cyclopentyl-propionitrile (mp 96.3–97.6)	Toluene	80.5	1.30×10^{-4}		w	104
	2,2'-Azo-bis-2,4,4-trimethyl-valeronitrile (mp 67.5–69)	Toluene	40	1.175×10^{-4}	113.5	w	50
			50	4.45×10^{-4}		w	50
	2,2'-Azo-bis-2,4,4-trimethylvalero-nitrile (mp 94.5–95.5)	Toluene	40	6.95×10^{-5}	121.3	w	50
			50	2.89×10^{-4}		w	50
	2,2'-Azo-bis-2-isopropyl-3-methyl-butyronitrile	Toluene	80.5	$(1.325 \pm 0.35) \times 10^{-4}$			50
18	1,1'-Azo-bis-1-cyano-4,4' -dimethyl cyclohexane	DMSO	80	1.51×10^{-5}	132.2	h	135
			85	2.79×10^{-5}			135
			90	5.40×10^{-5}			135
			95	9.77×10^{-5}			135
	2,2'-Azo-bis-2-cyclohexyl-propionitrile	Toluene	80.3	8.3×10^{-6}			61
			80.5	2.27×10^{-4}			104
	1,1'-Azo-bis-1-cyclooctanenitrile	Toluene	36.6	5.35×10^{-5}	108.4		101
			45.4	1.45×10^{-4}			101
			49.7	2.60×10^{-4}			101
20	4,4'-Azo-bis(4-cyanopentanoyl)bis-2-(2-aminoethoxy)ethanol	DMF	65	1.77×10^{-5}			357
			69	2.90×10^{-5}			357
			74	4.90×10^{-5}			357
		DMAC	65	1.74×10^{-5}			357
			69	2.83×10^{-5}			357
			74	4.65×10^{-5}			357
	2,2'-Azo-bis-2-benzylpropionitrile	Toluene	80	1.16×10^{-4}			54

TABLE 1. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
20 (*cont'd*)	2,2′-Azo-bis-2-isobutyl-4-methyl-valeronitrile	Toluene	60.1	3.78×10^{-4}			50
			80.5	5.28×10^{-3}			50
	2,2′-Azo-bis-2-(4-chlorobenzyl)-propionitrile	Toluene	80	8.8×10^{-5}			54
	2,2′-Azo-bis-2-(4-nitrobenzyl)-propionitrile	Toluene	80	1.00×10^{-4}			54
22	1,1′-Azo-bis-1-cyclodecanenitrile	Toluene	50.8	5.40×10^{-5}			101
			60.2	1.70×10^{-4}			101
			69.5	5.69×10^{-4}			101
34	4,4′-Azo-bis-(4-cyanopentanyl)-bis(heptadecafluorododecanoate)	Di-*n*-butyl phthalate	90	9.86×10^{-4}			375
			100	2.11×10^{-3}			375
			110	4.4×10^{-3}			375
			120	1.1×10^{-2}			375
			130	1.42×10^{-2}			375

TABLE 2. MISCELLANEOUS AZO-DERIVATIVES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
2	1,1′-Azo-bis-formamide	DMF	86.0	1.56×10^{-5}		c_2	268
			100.3	5.73×10^{-5}		c_2	268
			115.3	1.14×10^{-4}		c_2	268
		DMSO	86.0	5.48×10^{-6}		c	268
			100.3	2.72×10^{-5}		c	268
			115.3	1.01×10^{-4}		c	268
		Formamide	86.0	9.25×10^{-5}			268
			115.3	$\sim 2.0 \times 10^{-3}$			268
		Hexamethyl-phosphoramide	115.3	3.83×10^{-5}			268
4	Methyl-azo-3-propene	Vapor	163.3	5.88×10^{-4}	148	h	298
			169.8	1.110×10^{-3}			298
			182.8	3.36×10^{-3}			298
			T (K)	3.2×10^{14}	exp[−148/RT]		298
5	2,3-Diazobicyclo[2,2,1]hept-2-ene	Isooctane	131.5	5.44×10^{-6}	157.7		180
			142.3	1.82×10^{-5}			180
			164.1	1.66×10^{-4}			180
			180.8	8.35×10^{-4}			180
		Toluene	164.1	1.64×10^{-4}			180
6	2,2′-Azo-bis-propane	Vapor	250	7.67×10^{-3}	171.1	a	110
			260	1.67×10^{-2}		a	110
			270	3.35×10^{-2}		a	110
			280	6.52×10^{-2}		a	110
			290	1.28×10^{-1}		a	110
	3,3′-Azo-bis-1-propene	Vapor	142.65	3.88×10^{-4}	151.3		319
			152.68	1.10×10^{-3}			319
			167.12	4.45×10^{-3}			319
	2,2′-Dichloro-2,2′-azo-bis-propane	Diphenyl ether	158.5	1.03×10^{-4}	138.3	h,u_4	297
			167.9	4.54×10^{-4}		u_4	297
			178.5	8.08×10^{-4}		u_4	297
		Silicone oil	185	4.8×10^{-4}			240
			199	2.1×10^{-3}			240
			221	1.6×10^{-2}			240
	Methyl-azo-1,1-dimethyl-2-propene (*cis*)	Acetone	−13.89	3.62×10^{-5}	96.8	h	252
			−5.04	1.70×10^{-4}			252
			4.00	6.82×10^{-4}			252
	(*trans*)	*p*-Diisopropyl-Benzene	114.55	1.02×10^{-4}	147.9	h	252
			123.83	2.68×10^{-4}			252
			133.33	8.44×10^{-4}			252
	Perfluoroazo-2-propane	Vapor	T (K)	5.01×10^{16}	exp[−189.0/RT]		360

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
6 (*cont'd*)	1-Propyl-azo-3′-propene	Vapor	101.32	1.10×10^{-6}	149.2	v_{21}	298
			114.10	5.60×10^{-6}		v_{21}	298
			126.02	2.11×10^{-5}		v_{21}	298
			T (K)	6.3×10^{14}	exp[− 149.2/RT]		298
	Triazobenzene	Nitrobenzene	105	3.5×10^{-7}	140.2		45
			115	1.20×10^{-6}			45
			130	6.25×10^{-6}			45
			145	2.50×10^{-5}			45
		Tetralin	105	4.0×10^{-7}	136.0		45
			115	1.34×10^{-6}			45
			130	6.01×10^{-6}			45
			145	2.47×10^{-5}			45
7	*tert*-Butyl-azo-3-propene	Vapor	80.88	2.13×10^{-6}	124.9	v_{21}	298
			90.90	7.27×10^{-6}		v_{21}	298
			106.81	3.89×10^{-5}		v_{21}	298
			T (K)	5.4×10^{12}	exp[− 124.9/RT]	v_{21}	298
8	2,2′-Azo-isobutyramide	Water	88	1.9×10^{-5}		a	342
	Azo-bis-isobutyramidine	Water	60	3.15×10^{-5}		q_2	288
			60	2.80×10^{-5}		q_3	288
		Chloroform	60	7.1×10^{-6}			184
		DMSO	70	3.68×10^{-6}			21
		Methanol	60	1.45×10^{-6}			184
	Azo-bis-isobutyramidine 2HCl (AIBA)	Water (pH 1.9)	50	8.3×10^{-6}	126.5		341
			60	3.42×10^{-5}			341
		Water (pH 5.8)	50	8.0×10^{-6}	124.4		341
			60	3.22×10^{-5}			341
		Water (pH 9.5)	50	4.7×10^{-6}	151.7		341
			60	2.53×10^{-5}			341
		Water	60.0	3.70×10^{-5}	128.9	h	233
			70.0	1.33×10^{-4}			233
			75.0	2.58×10^{-4}			233
			80.0	5.13×10^{-4}			233
	AIBA	Water	40.15	2.49×10^{-6}	122.6	h	237
			50.05	9.79×10^{-6}			237
			60.20	4.03×10^{-5}			237
			70.10	1.52×10^{-4}			237
			100.10	4.96×10^{-3}			237
			70	1.52×10^{-4}			21
	AIBA-kaolin adduct	Water	50.0	1.37×10^{-5}	133.9	h	233
			60.0	6.16×10^{-5}			233
			70.0	2.28×10^{-4}			233
			80.0	9.53×10^{-4}			233
	AIBA-bentonite adduct	Water	50.0	1.80×10^{-5}	126.4	h	233
			55.0	3.97×10^{-5}			233
			65.0	1.53×10^{-4}			233
			70.0	3.21×10^{-4}			233
			80.0	1.04×10^{-3}			233
	Azo-bis(2-amidinopropane) 2HCl	Methanol	50	1.45×10^{-5}			356
		Water	50	8.1×10^{-6}			356
	2,2′-Azo-bis(2-amidinopropane) 2HCl	Water	56	1.93×10^{-5}		a	340
	Azo-bis-isobutyramidine $2HNO_3$	DMSO–cumene	60	4.86×10^{-5}			21
			70	1.53×10^{-4}			21
	2,2′-Azo-bis-isobutane	Vapor	160	1.9×10^{-5}		a	339
			180	5.01×10^{-5}		m_4	221
			190	1.53×10^{-4}		m_4	221
			210	1.05×10^{-3}		m_4	221
			174.8	6.8×10^{-5}			243
			188.3	1.75×10^{-4}			243
			188.8	1.14×10^{-4}		m_4	243
	2,2′-Azo-bis-isobutane	Diphenyl ether/iso-quinoline (90/10)	165	2.782×10^{-5}	176.6	h	171
			175	8.74×10^{-5}			171
			185	2.513×10^{-4}			171
			190	4.143×10^{-4}			171

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	2,2′-Azo-bis-isobutane	Diphenyl ether/iso-quinoline (90/10)	195	6.731×10^{-4}			171
			200	1.093×10^{-3}			171
	2,2′-Dimethoxy-2,2′- azopropane	Diphenyl ether	150.1	4.02×10^{-5}	171.8		297
			159.9	1.50×10^{-4}			297
			174.6	6.67×10^{-4}			297
	2,2′-Dimethylmercapto-2,2′-azopropane	Diphenyl ether	125.0	3.40×10^{-4}	84.6	h,u_5	297
			135.0	7.10×10^{-4}		u_5	297
			145.0	1.15×10^{-3}		u_5	297
	Isopropyl-azo-l,1-dimethyl-2-propene (*cis*)	Acetone	− 25.09	6.30×10^{-5}	91.3	h	252
			− 16.65	2.87×10^{-4}			252
			− 8.20	1.10×10^{-3}			252
	(*trans*)	Diisopropylbenzene	97.41	1.15×10^{-4}	137.8	h	252
			106.06	3.19×10^{-4}			252
			115.06	9.22×10^{-4}			252
	2,2′-Azo-bis(2-methylpropane)	Vapor	160	1.93×10^{-5}		a	340
9	2-(4-Bromophenyl-azo)-2-nitropropane	DCB	130	1.62×10^{-4}	130		325
			135	2.59×10^{-4}			325
			140	4.30×10^{-4}			325
			T (K)	1.1×10^{13}	exp[− 130/RT]		325
	tert-Butyl-azo-l,l-dimethyl-2-propene (trans)	Diisopropylbenzene	73.72	8.69×10^{-5}	112.3	h	252
			80.31	1.83×10^{-4}			252
			87.08	3.80×10^{-4}			252
	2-(4-Nitrophenylazo)-2-nitro-propane	DCB	130	1.00×10^{-4}	134		325
			140	2.44×10^{-4}			325
			145	4.06×10^{-4}			325
			T (K)	1.1×10^{13}	exp[− 134/RT]		325
	α-Phenylethyl-azo-methane	Diphenyl ether	151	4.35×10^{-5}	161.5		149
			161	1.16×10^{-4}			149
			171	3.48×10^{-4}			149
		Hexadecane	161	1.20×10^{-4}			149
10	2,2′-Diacetylthio-2,2′-azopropane	Diphenyl ether	130.0	1.33×10^{-4}	151.3	h	297
			140.0	4.40×10^{-4}			297
			149.4	1.19×10^{-5}			297
	2,2′-Diacetoxy-2,2′-azopropane	Diphenyl ether	190.0	8.23×10^{-5}	171.4	h	297
			200.1	2.61×10^{-4}			297
			210.3	6.38×10^{-4}			297
		Silicone oil	258	1.46×10^{-2}			240
	Azo-bis(1,1-dimethyl-2-propene) (*trans*)	Xylene	42.03	5.95×10^{-5}	109.4	h	252
			49.98	1.73×10^{-4}			252
			57.39	4.54×10^{-4}			252
	Azo-bis-(1,1-dimethyl)-2-propyne (*trans*)	Xylene	41.32	5.37×10^{-5}	111.9	h	252
			48.20	1.30×10^{-4}			252
			57.19	4.39×10^{-4}			252
	2,2′-Azo-bis-2-methylbutane	Ethylbenzene	180.0	1.36×10^{-4}	176.4	h	336
			190.0	3.89×10^{-4}			336
			200.0	1.03×10^{-3}			336
		Isoamyl alcohol	180.0	1.14×10^{-4}	176.0	h	336
			190.0	3.51×10^{-4}			336
			200.0	9.26×10^{-4}			336
	2,2′-Azo-bis-2-methylbut-3-ene	*n*-Decane	44.90	6.73×10^{-5}	107.3	h	278
			55.44	2.631×10^{-4}			278
			63.74	7.045×10^{-4}			278
	2,2′-Azo-bis-2-(methylcarboxy)-propane	Water	50	3.51×10^{-6}			285
	2,2′-Azo-bis-methyl-2-methyl-propionate	Collidine	80	1.67×10^{-4}		a	226
		1,2-Dichlorobenzene	80	1.44×10^{-4}		a	226
		Diethyl oxalate	80	1.52×10^{-4}		a	226
		Ethylene glycol	80	1.24×10^{-4}		a	226
		Nitrobenzene	80	1.61×10^{-4}	129.3	a	226
			100	1.73×10^{-3}		a	226
	2,2′-Azo-bis-methyl-2-methyl-propionate	Undecane	80	1.04×10^{-4}		a	226
		Xylene	80	1.09×10^{-4}	149.8	a	62

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
10 (*cont'd*)	*N,N'*-Azo-piperidine	Silicone oil	181	1.7×10^{-3}			240
			228	4.3×10^{-2}			240
	2-(4-Methoxyphenylazo)-2-nitropropane	DCB	135	7.4×10^{-5}	153		325
			140	1.30×10^{-4}			325
			150	3.67×10^{-4}			325
			T (K)	2.5×10^{15}	exp[−153/RT]		325
	Dimethyl 2,2'-azo-isobutyrate	Acetic acid	40	8.61×10^{-7}		a	326
			50	3.16×10^{-6}		a	326
			60	1.56×10^{-5}		a	326
			70	5.49×10^{-5}		a	326
		Acetonitrile	40	5.44×10^{-7}		a	326
			50	2.68×10^{-6}		a	326
			60	1.01×10^{-5}		a	326
			70	3.57×10^{-5}		a	326
		Benzene	50	2.22×10^{-6}		a	326
			60	8.85×10^{-6}		a	326
			70	3.27×10^{-5}		a	326
		Cyclohexane	40	2.89×10^{-7}		a	326
			50	1.38×10^{-6}		a	326
			60	5.85×10^{-6}		a	326
			80	2.22×10^{-5}		a	326
	Dimethyl-2,2'-azo-isobutyrate	Methanol	40	6.86×10^{-7}		a	326
			50	3.12×10^{-6}		a	326
			60	1.44×10^{-5}		a	326
		Methanol/$SnCl_4$	40	1.20×10^{-6}		a	326
			50	4.46×10^{-6}			326
			60	1.98×10^{-5}		a	326
		Methanol/$ZnCl_2$	60	1.51×10^{-5}		a	326
		Paraffin	99.2	1.02×10^{-3}			240
		Toluene	60	8.9×10^{-6}			290
			60	8.11×10^{-6}	131.6		340
			66	1.93×10^{-5}		a	340
			70	3.17×10^{-5}			340
			80	1.19×10^{-4}			340
		Toluene/$SnCl_4$	60	1.63×10^{-4}			290
	2,3,7,8-Tetraazahexacyclo-[7.4.1.0^{4,12}.0^{6,11}.0^{10,13}]tetradeca-2,7-diene	Acetonitrile-d_3	215	1.18×10^{-5}	188.7	x_1	174
			230	6.65×10^{-5}		x_1	174
			250	3.40×10^{-4}		x_1	174
11	2-(2,6-Dimethylphenylazo)-2-nitropropane	DCB	135	9.0×10^{-5}			325
			140	1.44×10^{-4}			325
			150	3.86×10^{-4}			325
			T (K)	4.8×10^{13}	exp[−138/RT]		325
	α-Phenylethyl-azo-2-propane	Diphenyl ether/benzoquinone	143.2	1.50×10^{-4}			220
		Diphenyl ether	143.0	1.54×10^{-4}		t_{13}	310
			152.0	4.24×10^{-4}		t_{13}	310
			165.5	1.570×10^{-3}		t_{13}	310
12	Azo-bis-isobutanoldiacetate	Cyclohexane	170.0	4.86×10^{-5}			189
			180.0	1.27×10^{-4}			189
			189.0	3.47×10^{-4}			189
			199.0	1.01×10^{-3}			189
	Azo-bis-(1-carbomethoxy-3-methyl-propane)	Benzene	36	3.05×10^{-7}			68
			45	1.31×10^{-6}			68
			55	4.54×10^{-6}			68
			65	1.82×10^{-5}			68
		Carbon tetrachloride	45	7.18×10^{-7}			68
			55	3.79×10^{-6}			68
			65	1.02×10^{-5}			68
	4,4'-Azo-bis-(4-cyanopentanoic acid) (meso)	DMAC	77.6	1.33×10^{-4}	133.5	a	205
			85.3	3.70×10^{-4}		a	205
			90.9	7.17×10^{-4}		a	205
			99.8	2.02×10^{-3}		a	205

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	4,4′-Azo-bis-(4-cyanopentanoic acid)						
	($n_{D\pm}$)		77.9	1.53×10^{-4}	134.3	a	205
			85.8	4.11×10^{-4}		a	205
			90.1	7.00×10^{-4}		a	205
			99.1	2.09×10^{-3}		a	205
	(n_{D+})		78.0	1.55×10^{-4}	134.3	a	205
			85.4	4.05×10^{-4}		a	205
			90.0	6.98×10^{-4}		a	205
			99.0	2.00×10^{-3}		a	205
	(n_{D-})		77.7	1.51×10^{-4}	134.3	a	205
			86.0	4.34×10^{-4}		a	205
			90.5	7.33×10^{-4}		a	205
			99.7	2.24×10^{-3}		a	205
	2,2-Azo-bis-(ethyl-2-methyl-propionate)	Chlorobenzene	70	4.13×10^{-5}	123.0	a	139
			92.6	5.93×10^{-4}		a	139
		Nitrobenzene	80	1.56×10^{-4}		a	226
	2,2-Azo-bis-[2-methyl-*N*-(2-hydroxyethyl) propionamide]	Water	86	1.9×10^{-5}		a	342
	Azo-bis-(*N*,*N*′-dimethyleneisobutyr-amidine) $2HNO_3$	DMSO	60	2.06×10^{-4}			21
			70	6.64×10^{-4}			21
			75	1.08×10^{-3}			21
		DMSO–cumene	70	6.97×10^{-4}			21
		DMSO–tetralin	60	2.13×10^{-4}			21
	Azo-bis-(*N*,*N*′-dimethylene isobutyramidine) 2HCl	Water	44	1.9×10^{-5}		a	339
	Azo-bis-(*N*,*N*′-dimethyleneisobutyr-amidine)	DMSO	80	1.79×10^{-4}			21
		DMSO–cumene	75	8.04×10^{-5}			21
			80	1.39×10^{-4}			21
			85	2.6×10^{-4}			21
		DMSO–tetralin	60	1.09×10^{-5}			21
		Methanol	61	1.9×10^{-5}		a	339
	tert-Butyl(2,2,4-trimethyl-2-pentyl)diazene	Diphenyl ether	155	1.61×10^{-4}	159.6	h	314
			165.7	4.75×10^{-4}			314
			180.0	1.98×10^{-3}			314
	α-Cumyl-azo-isopropane	Diphenyl ether	120.0	1.06×10^{-4}			310
			135.0	5.94×10^{-4}			310
			143.0	1.421×10^{-3}			310
	2,2′-Dicyclopropyl-2,2′-azopropane	Diphenyl ether/ isoquinoline (90/10)	145.0	7.56×10^{-5}	158.2	h	171
			150.0	1.273×10^{-4}			171
			155.0	2.147×10^{-4}			171
			160.0	3.771×10^{-4}			171
			165.0	6.07×10^{-4}			171
			170.0	1.033×10^{-3}			171
	Di-(2-hydroxyethyl)-2,2′-azo-bis-isobutyrate	Dioxane	60	1.61×10^{-5}	125.7		292
			70	5.88×10^{-5}			292
			80	2.02×10^{-4}			292
		DMF	60	1.53×10^{-5}	129.9		292
			70	5.62×10^{-5}			292
			80	2.0×10^{-4}			292
		Toluene	60	1.62×10^{-5}	129.9		292
			70	5.83×10^{-5}			292
			80	2.066×10^{-4}			292
	2,2′-Azo-bis[*N*-(2-hydroxyethyl)-2-methylpropionamidine] 2HCl	Water	58	1.93×10^{-5}		a	339
	2,2′-Azo-bis[2-(imidazoline-2-yl)-propane 2HCl	Methanol	50	4.83×10^{-5}			356
		Water	50	4.02×10^{-5}			356
	2,2′-Azo-bis[2-(imidazoline-2-yl)-propane	Methanol	50	4.7×10^{-6}			356
14	2,2′-Azo-bis-(2-cyclobut-1-enylpropane)	*n*-Decane	39.21	1.286×10^{-4}	112.3	h	278
			50.71	5.678×10^{-4}			278
			58.95	1.762×10^{-3}			278

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	2,2′-Azo-bis-(2-methylhex-3-ene)	*n*-Decane	39.25	5.33×10^{-5}	115.2	h	278
			49.52	2.256×10^{-4}			278
			58.95	7.584×10^{-4}			278
	Di-(2-chloroethyl)-2,2′-azo-bis-isobutyrate	Dioxane	60	1.56×10^{-5}	129.9		292
			70	5.55×10^{-5}			292
			80	1.9672×10^{-4}			292
		DMF	60	1.25×10^{-5}	134.1		292
			70	4.902×10^{-5}			292
	Di-(2-chloroethyl)-2,2′-azo-bis-isobutyrate		80	1.9125×10^{-4}			292
		Toluene	60	1.666×10^{-5}	129.9		292
			70	5.018×10^{-5}			292
			80	1.99×10^{-4}			292
	1,1′-bis-(4-Chlorophenyl) azomethane	Diphenyl ether	145.56	2.30×10^{-4}	146.6	h	297
			155.56	7.60×10^{-4}			297
			165.56	1.49×10^{-3}			297
			175.56	4.40×10^{-3}			297
	Di-(2-hydroxypropyl)-2,2′-azo-bis-isobutyrate	Dioxane	60	1.80×10^{-5}	121.5		292
			70	7.6×10^{-5}			292
			80	2.22×10^{-4}			292
		DMF	60	1.97×10^{-5}	121.5		292
			70	6.55×10^{-5}			292
			80	2.30×10^{-4}			292
		Toluene	60	1.90×10^{-5}	121.5		292
			70	7.72×10^{-5}			292
			80	2.365×10^{-4}			292
	2,2′,3,3,3′,3′-Hexamethyl-2,2′-azo-butane	Diphenyl ether	170.0	2.37×10^{-4}	156.7	h	297
			180.0	5.29×10^{-4}			297
			190.0	1.00×10^{-3}			297
	1,1′-Diphenyl-azo-methane	Diphenyl ether	145.0	1.72×10^{-4}	119.0	h	297
			155.0	4.93×10^{-4}			297
			165.0	1.12×10^{-3}			297
			175.0	2.34×10^{-3}			297
	2,2′-Azo-bis[2-methyl-*N*-(phenyl-methyl)propionamidine] 2HCl	Water	53	1.93×10^{-5}		a	339
	2,2′-Azo-bis[2-(4-methyl-imidazoline-2-yl)-propane	Methanol	50	1.23×10^{-5}			356
	2,2′-Azo-bis[2-(4-methylimidazoline-2-yl)-propane 2HCl	Methanol	50	7.48×10^{-5}			356
		Water	50	6.73×10^{-5}			356
	2,2′-Azo-bis[2-(3,4,5-trihydro-pyrimidine-2-yl)propane	Methanol	50	1.1×10^{-6}			356
	2,2′-Azo-bis[2-(3,4,5-trihydro-pyrimidine-2-yl)propane 2HCl	Methanol	50	7.8×10^{-6}			356
		Water	50	5.9×10^{-6}			356
15	1-Methylcyclopentyl-azo-α-cumene	*tert*-Butyl benzene	88.3	5.00×10^{-5}	132.0	h	327
			107.3	4.95×10^{-4}			327
			117.6	1.43×10^{-3}			327
16	1,1′-Azo-bis-1-chloro-1-phenylethane	Acetophenone	74.8	8.6×10^{-4}			240
			83	1.95×10^{-3}			240
		Benzophenone	67.8	2.8×10^{-4}			240
			86.2	3.3×10^{-3}			240
			98.2	1.40×10^{-2}			240
		Paraffin	94.5	2.4×10^{-3}			240
			112.3	5.2×10^{-2}			240
			120.5	1.03×10^{-1}			240
			129.3	1.6×10^{-1}			240
			137	5.0×10^{-1}			240
		Toluene	64	1.8×10^{-4}			240
			74	7.2×10^{-4}			240
			64	1.79×10^{-4}	135.1	a	239
			74	7.20×10^{-4}		a	239
	1,1′-Azo-bis-1-chloro-1-(3-bromophenyl)ethane	Toluene	59	1.64×10^{-4}	131.8	a	239
			69	6.64×10^{-4}		a	239
	1,1′-Azo-bis-1-chloro-1-(4-bromophenyl)ethane	Toluene	59	1.77×10^{-4}	127.2	a	239
			69	6.75×10^{-4}		a	239

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	1,1′-Azo-bis-1-chloro-1-(4-chlorophenyl)ethane	Toluene	59	1.97×10^{-4}	118.8	a	239
			69	6.93×10^{-4}		a	239
	2,2′-Azo-bis-(2-cyclopent-1-enylpropane)	*n*-Decane	39.21	9.98×10^{-5}	116.9	h	278
			49.52	4.593×10^{-4}			278
			58.95	1.528×10^{-3}			278
	1,1′-Azo-bis-1,1-dicyclopropylethane	Diphenyl ether/ isoquinoline (90/10)	120	6.33×10^{-5}	149.0	h	171
			130	2.03×10^{-4}			171
			140	6.07×10^{-4}			171
			150	1.695×10^{-3}			171
	2,2′-Azo-bis-{2-methyl-*N*-[1,1-(bis-hydroxymethyl)-2-hydroxyethyl] propionamide}	Water	80	1.9×10^{-5}		a	342
	2,2′-Azo-bis-{2-methyl-*N*-[1,1-(bis-hydroxymethyl)ethyl] propionamide}	Water	82	1.9×10^{-5}		a	342
	2,2′-Azo-bis[2-(4, 4-dimethylimidazoline-2-yl) propane	Methanol	50	9.3×10^{-6}			356
	2,2′-Azo-bis[2-(4,4-dimethyl-imidazoline-2-yl)propane 2HCl	Methanol	50	9.55×10^{-5}			356
		Water	50	9.30×10^{-5}			356
	2,2′-Azo-bis[2-(4,5-dimethyl-imidazoline-2-yl)propane 2HCl	Methanol	50	9.21×10^{-5}			356
		Water	50	8.94×10^{-5}			356
	2,2′-Azo-bis[2-(4,5,6,7-tetrahydro-1H-1,3-diazepine-2-yl)propane 2HCl	Methanol	50	1.23×10^{-5}			356
		Water	50	1.04×10^{-5}			356
	2,2′-Azo-bis-(2,2,4-trimethyl-pentane)	Diphenyl ether	110	1.9×10^{-5}		a	339
	1,1′-Azo-bis-1-phenylethane	Dodecane	97.3	3.175×10^{-5}			26
	1,1′-Azo-bis-1-phenylethane	Ethylbenzene	100.4	$(5.45 \pm 0.05) \times 10^{-5}$	136.4		107,98
		Toluene	110.3	$(1.69 \pm 0.01) \times 10^{-4}$			107,98
		Ethylbenzene	105.02	8.473×10^{-5}			152
			105.28	9.02×10^{-5}			183
	($1,1'\text{-}d_2$)		105.28	7.62×10^{-5}			183
	($1,1,1,1',1',1'\text{-}d_6$)		105.02	7.623×10^{-5}			152
		N-Methyl-*N*-benzyl aniline	97.3	4.135×10^{-5}			26
		N-Methyl propionamide	97.3	3.688×10^{-5}			26
		Diphenylmethane	97.3	3.995×10^{-5}			26
		Propylene carbonate	97.3	3.294×10^{-5}			26
	tert-Butyl-(2,2,4,6,6-penta-methyl-4-heptyl)-diazine	Diphenyl ether	140.0	1.93×10^{-4}	141.6	h	314
			150.1	5.29×10^{-4}			314
			164.8	2.18×10^{-3}			314
	1,1′-Di-(3-chlorophenyl)-1,1′-azo-ethane	*p*-Cymene	95	4.13×10^{-5}	122.3	h	266,313
			105	1.218×10^{-4}			266,313
	1,1′-Di-(4-chlorophenyl)-1,1′-azo-ethane	*p*-Cymene	85	1.45×10^{-5}	134.1	h	266.313
			95	4.82×10^{-5}			266,313
			105	1.650×10^{-4}			266,313
	α,α-Diethylbenzyl-azo-3-pentane	Diphenyl ether	120.05	7.8×10^{-5}			310
			130.0	2.63×10^{-4}			310
			139.0	7.38×10^{-4}			310
			147.0	1.906×10^{-3}			310
	1,1'-Di-(3-fluorophenyl)-1,1′-azo-ethane	*p*-Cymene	105	1.121×10^{-4}			266,313
	1,1′-Di-(4-fluorophenyl)-1,1′-azo-ethane	*p*-Cymene	85	5.5×10^{-6}	147.0	h	266,313
			95	2.35×10^{-5}			266,313
			105	7.85×10^{-5}			266,313
	Di-(3-hydroxybutyl)-2,2′-azo-bis-isobutyrate	Dioxane	60	2.22×10^{-5}	117.3		292
			70	7.84×10^{-5}			292
			80	2.413×10^{-4}			292
		DMF	60	1.83×10^{-5}	121.5		292
			70	7.69×10^{-5}			292
			80	2.424×10^{-4}			292

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	Di-(3-hydroxybutyl)-2,2′-azo-bis-isobutyrate	Toluene	60	2.43×10^{-5}	117.3		292
			70	8.18×10^{-5}			292
			80	2.461×10^{-4}			292
	Di-(4-hydroxybutyl)-azo-bis-isobutyrate	Dioxane	60	2.12×10^{-5}	125.7		292
			70	6.57×10^{-5}			292
			80	2.539×10^{-4}			292
		DMF	60	1.97×10^{-5}	125.7		292
			70	6.19×10^{-5}			292
			80	2.50×10^{-4}			292
		Toluene	60	2.59×10^{-5}	121.5		292
			70	6.66×10^{-5}			292
			80	2.628×10^{-4}			292
	1,1′-Bis-(4-methoxyphenyl)-azo-methane	Diphenyl ether	145.0	2.23×10^{-4}	145.0	h	297
			155.0	5.67×10^{-4}			297
			165.0	1.61×10^{-3}			297
			175.0	3.79×10^{-3}			297
	2,2′,4,4,4′,4′-Hexamethyl-2,2′-azo-pentane	Diphenyl ether	130.2	1.82×10^{-4}	132.8	h	297
			140.0	4.51×10^{-4}			297
			150.0	1.28×10^{-3}			297
	1-Methylcyclohexyl-azo-α-cumene	*tert*-Butylbenzene	102.2	2.58×10^{-4}	131.1	h	327
			113.1	9.81×10^{-4}			327
			122.0	2.20×10^{-3}			327
	1,1′-Diphenyl-1,1′-azo-ethane	*p*-Cymene	95	2.74×10^{-5}	124.0	h	266,313
			105	8.21×10^{-5}			266,313
	2-(Phenylazo)-2-propyl perbenzoate	Chlorobenzene	10	4.5×10^{-4}			372
			35	1.9×10^{-3}			372
		Chloroform-d	10	5.1×10^{-4}			372
		Chlorotoluene	10	6.4×10^{-4}			372
	(1-Phenylethyl)-azo-diphenylmethane	Ethylbenzene	58	1.9×10^{-5}		a	339
	1,1′-Bis(*p*-tolyl)-azo-methane	Diphenyl ether	145.0	2.00×10^{-4}	145.4	h	297
			155.0	4.69×10^{-4}			297
			165.0	1.22×10^{-3}			297
			175.0	3.45×10^{-3}			297
17	3,7-Diphenyl-1,2-diaza-1-cyclo-heptene	Xylene	61.0	3.80×10^{-5}			138
			70.0	1.34×10^{-4}			138
			80.0	4.27×10^{-4}			138
			89.3	1.59×10^{-3}			138
			100.2	4.04×10^{-3}			138
			80	4.26×10^{-4}		a	44
			100.2	3.94×10^{-3}		a	44
18	1,1′-Azo-bis-cumene	Benzene	58.91	1.94×10^{-4}		v_5	217
		tert-Butylbenzene	60	1.03×10^{-4}			333
			60	4.03×10^{-5}		z (1256)	333
			60	4.44×10^{-6}		z (3890)	333
		Chlorobenzene	55	8.50×10^{-5}			280
			55	6.88×10^{-5}		z (1013)	280
			55	4.42×10^{-5}		z (4154)	280
		Cumene	55	8.10×10^{-5}			280
			55	6.75×10^{-5}		z (1013)	280
			55	3.83×10^{-5}		z (4154)	280
			60	1.16×10^{-4}			333
			60	4.73×10^{-5}		z (1256)	333
			60	7.23×10^{-6}		z (3890)	333
		Dodecane	60.23	1.47×10^{-4}			217
		n-Octane	60	1.14×10^{-4}			333
			60	4.44×10^{-5}		z (1256)	333
			60	8.32×10^{-6}		z (3890)	333
		Toluene	40.04	9.46×10^{-6}		h	217
			49.52	4.33×10^{-5}			217
			59.42	1.62×10^{-4}			217
			69.23	5.80×10^{-4}			217
			41.0	9.6×10^{-6}	122.8		309
			44.8	1.98×10^{-5}			309
			58.5	1.280×10^{-4}			309

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
18 (*cont'd*)	1,1′-Azo-bis-4-bromocumene	Toluene	42.8	2.72×10^{-5}	117.7		309
			48.6	6.28×10^{-5}			309
			58.0	2.189×10^{-4}			309
	1,1′-Azo-bis-3-chlorocumene	Toluene	36.0	8.8×10^{-6}	115.9		200
			42.8	2.46×10^{-5}			200
			48.2	5.55×10^{-5}			200
			61.1	2.796×10^{-4}			200
	1,1′-Azo-bis-4-chlorocumene	Toluene	36.0	1.11×10^{-5}	112.5		200
			42.8	2.90×10^{-5}			200
			48.2	6.52×10^{-5}			200
	1,1′-Azo-bis-1-chloro-1-(4-tolyl)-ethane	Acetophenone	74.8	2.1×10^{-3}			240
			79.5	3.0×10^{-3}			240
			86	7.4×10^{-3}			240
		Paraffin	105	1.66×10^{-3}			240
			109.5	2.5×10^{-2}			240
			126	6.4×10^{-2}			240
			135.5	1.7×10^{-1}			240
		Toluene	59	1.03×10^{-4}			240
			69	4.0×10^{-4}			240
			59	1.03×10^{-4}	126.8	a	239
			69	3.93×10^{-4}			239
	2,2′-Azo-bis[2-(4-ethyl-4-methyl-imidazoline-2-yl)propane 2HCl	Methanol	50	6.80×10^{-5}			356
		Water	50	7.38×10^{-5}			356
	2,2′-Azo-bis[2-(4-ethyl-5-methyl-imidazoline-2-yl)propane 2HCl	Methanol	50	9.48×10^{-5}			356
		Water	50	1.1×10^{-4}			356
	2,2′-Azo-bis[2-(5,5-dimethyl-3,4,5-tri-hydropyrimidine-2-yl)propane	Methanol	50	9×10^{-7}			356
	2,2′-Azo-bis[2-(5,5-dimethyl-3,4,5-tri-hydropyrimidine-2-yl)propane 2HCl	Methanol	50	8.0×10^{-6}			356
		Water	50	7.9×10^{-6}			356
	1,1′-Azo-bis-4-fluorocumene	Toluene	36.0	5.7×10^{-6}	115.9		200
			42.8	1.31×10^{-5}			200
			48.2	2.94×10^{-5}			200
			61.1	1.880×10^{-4}			200
	1,1′-Di-3-trifluoromethyl-phenyl)-1,1′-azo-ethane	*p*-Cymene	95	4.13×10^{-5}	134.9	h	266,313
			105	1.357×10^{-4}			266,313
	1,1′-Azo-bis-(2-methoxyphenyl)-ethane	*p*-Cymene	84.6	1.69×10^{-5}	135.6	h	266
			95	6.11×10^{-5}			266,313
			105	1.975×10^{-4}			266,313
	1,1′-Azo-bis-(2-methoxyphenyl)-ethane	Ethylbenzene	106.06	2.29×10^{-5}		a	300
			113.81	5.52×10^{-5}		a	300
			133.66	4.87×10^{-4}		a	300
	1,1′-Azo-bis-(3-methoxyphenyl)-ethane	*p*-Cymene	95	1.53×10^{-5}	171.8	h	266,313
			105	6.91×10^{-5}			266,313
	1,1′-Azo-bis-(4-methoxyphenyl)-ethane	*p*-Cymene	95	3.56×10^{-5}	126.5	h	266,313
			105	1.090×10^{-4}			266,313
	1,1′-Azo-bis-1-(4-methoxyphenyl)-ethane	Ethylbenzene	100.4	$(7.15 \pm 0.1) \times 10^{-5}$	149.8		98
			110.3	$(2.48 \pm 0.02) \times 10^{-4}$			98
	1,1′-Azo-bis-1-phenylpropane	Ethylbenzene	100.4	$(2.35 \pm 0.1) \times 10^{-5}$	135.1		98
			110.3	$(7.2 \pm 0.2) \times 10^{-5}$			98
	2,2′-Azo-bis-2-phenyl-hexafluoro-propane	Toluene	45.0	8.36×10^{-6}	137.2	h	173
			50.0	2.16×10^{-5}			173
			55.0	4.92×10^{-5}			173
			60.0	8.81×10^{-5}			173
			67.0	2.46×10^{-4}			173
			43.8	6.3×10^{-5}	115.5		42
			53.8	2.4×10^{-4}			42
	1,1′-Dimethyl-azo-cyclooctane	Benzene	130.0	1.09×10^{-4}	145.8	h	327
			140.0	3.05×10^{-4}			327
			150.0	8.96×10^{-4}			327
	1,1′-Diphenyl-azo-cyclopropane	Benzene	165.0	9.6×10^{-5}	135.3	h	323
			175.0	2.21×10^{-4}			323
			185.0	5.09×10^{-4}			323
		Ethylbenzene	170.0	1.67×10^{-4}			323
			180.0	3.48×10^{-4}			323

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
18 (*cont'd*)	3,8-Diphenyl-1,2-diaza-1-cyclooctene	Tetralin	143	9.52×10^{-5}	153.6		137
			151	2.09×10^{-4}			137
			165	8.77×10^{-4}			137
			173	1.82×10^{-3}			137
	1,1′-Di-(3-tolyl)-1,1′-azo-ethane	*p*-Cymene	105	8.17×10^{-5}			266,313
	1,1′-Di-(4-tolyl)-1,1′-azo-ethane	*p*-Cymene	95	3.06×10^{-5}	126.5	h	266,313
			105	9.35×10^{-5}			266,313
19	Phenyl-azo-diphenylmethane	Decalin	124.5	3.44×10^{-5}	142.3		46
			144.5	2.69×10^{-4}			46
	2-(2-Phenyl-4-methylpentyl)-azo-2-(4-methylpentane)	Diphenyl ether	103.0	1.17×10^{-4}			310
			110.0	2.52×10^{-4}			310
			120.0	8.51×10^{-4}			310
			130.0	2.611×10^{-3}			310
20	1,1′-Azo-bis-1-acetoxy-1-phenylethane	Paraffin	131	3.2×10^{-2}			240
			142	4.3×10^{-2}			240
			149.5	8.0×10^{-2}			240
	1,1′-Azo-bis-1,1-dicyclopropyl-butane	Diphenyl ether/ isoquinoline (90/10)	120.0	4.26×10^{-5}	159.0	h	171
			125.0	7.94×10^{-5}			171
			130.0	1.49×10^{-4}			171
			135.0	2.697×10^{-4}			171
			140.0	4.774×10^{-4}			171
			147.0	1.055×10^{-3}			171
	9-Azo-bis-decalin (*cis*)	Ethylbenzene	180.0	1.41×10^{-4}	175.1	h	335
			190.0	3.95×10^{-4}			335
			200.0	1.05×10^{-3}			335
	(*trans*)		180.0	1.90×10^{-4}	175.1	h	335
			190.0	5.35×10^{-4}			335
			200.0	1.41×10^{-3}			335
	α,α′-Azo-bis(α-methylbenzyl) diacetate	Dichlorobenzene	65	1.86×10^{-5}			366
			70	4.00×10^{-5}			366
			80	1.592×10^{-4}			366
			90	5.525×10^{-4}			366
	Azo-bis-4-methylcumene	Cumene	55	8.86×10^{-5}			280,333
			55	7.54×10^{-5}		z (1013)	280,333
			55	4.47×10^{-5}		z (4154)	280,333
		Toluene	41.0	1.33×10^{-5}	120.7		309
			49.5	5.02×10^{-5}			309
			58.5	1.690×10^{-4}			309
	2,2′-Azo-bis[2-(4-ethyl-5-isopropyl-imidazoline-2-yl)propane 2HCl	Methanol	50	7.72×10^{-5}			356
		Water	50	1.01×10^{-4}			356
	2,2′-Azo-bis-2-(4-tolyl)-propane	Toluene	36.0	8.4×10^{-6}	110.0		200
			42.8	1.62×10^{-5}			200
			61.1	2.08×10^{-4}			200
	1,1′-Azo-bis-1,1,1-tricyclopropyl-methane	Diphenyl ether/ isoquinoline (90/10)	105.0	7.89×10^{-5}	143.5	h	171
			110.0	1.374×10^{-4}			171
			115.0	2.36×10^{-4}			171
			120.0	4.58×10^{-4}			171
			125.0	7.76×10^{-4}			171
			130.0	1.38×10^{-3}			171
			135.0	2.39×10^{-3}			171
		Decalin/ isoquinoline (90/10)	118.5	3.60×10^{-4}			171
		Isoquinoline	118.5	3.30×10^{-4}			171
		Cumene/ isoquinoline (90/10)	118.5	445×10^{-4}		h	171
	1,1′-Di-(4-ethylphenyl)-1,1-azo-ethane	*p*-Cymene	95	3.02×10^{-5}	125.3		266,313
			105	9.14×10^{-5}		a	266,313
	1,1′-Diphenyl-azo-butane	Ethylbenzene	93.88	1.67×10^{-5}		a	300
			105.68	6.96×10^{-5}		a	300
			119.55	3.39×10^{-4}		h	300
	1,1′-Diphenyl-azo-cyclobutane	Ethylbenzene	63.2	8.2×10^{-6}	122.3		323
			73.8	3.5×10^{-5}			323
			79.1	6.9×10^{-5}			323
			87.7	1.69×10^{-4}		b,h	323

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
20 (*cont'd*)	(−)-(*s*)-1,1′-Diphenyl-1-methyl-azo-methane	Cumene	106.0	8.26×10^{-6}	149.2	b	318
			127.9	1.18×10^{-4}			318
	2,2′-Azo-bis(2-methyl-*N*-phenyl-propionamidine) 2HCl	Water	45	1.93×10^{-5}		a	339
	2,2′-Azo-bis[*N*-(4-chlorophenyl)-2-methylpropionamidine] 2HCl	Water	45.5	1.93×10^{-5}		a	339
	2,2′-Azo-bis[*N*-(4-hydroxyphenyl)-2-methylpropionamidine] 2HCl	Water	47.5	1.93×10^{-5}		a	339
22	1,1′-Azo-bis-1-phenyl-3-methylbutane	Ethylbenzene	100.4	7.6×10^{-5}	139.3		98
			110.3	2.42×10^{-4}			98
	1,1′-Azo-bis-4-ethylcumene	Toluene	42.8	1.29×10^{-5}	108.9		309
			48.6	3.29×10^{-5}			309
			58.0	1.240×10^{-4}			309
	1,1′-Azo-bis-4-isopropylcumene	Toluene	42.8	1.13×10^{-5}	124.4		309
			51.2	4.26×10^{-5}			309
			58.0	1.078×10^{-4}			309
	1,1′-Diisopropyl-1,1′-diphenyl-azo-ethane	Ethylbenzene	50	3.52×10^{-5}	113.6	h	324
	1,1′-Diphenyl-azo-cyclopentane	Benzene	29.5	3.16×10^{-4}	103.1		323
			40.0	1.24×10^{-3}			323
			50.0	4.68×10^{-3}			323
	2,2′-Azo-bis[2-methyl-*N*-(phenyl-methyl)propionamidine] 2HCl	Water	52	1.93×10^{-5}		a	339
	2,2′-Azo-bis[2-(4-methyl-4-isobutyl-imidazoline-2-yl)propane 2HCl	Methanol	50	5.54×10^{-5}			356
		Water	50	7.08×10^{-5}			356
	2,2′-Azo-bis[2-(4-methyl-5-isobutyl-imidazoline-2-yl)propane 2HCl	Methanol	50	7.72×10^{-5}			356
		Water	50	9.78×10^{-5}			356
	2,2′-Azo-bis[2-(4-ethyl-5-propyl-imidazoline-2-yl)propane 2HCl	Methanol	50	1.06×10^{-4}			356
		Water	50	1.41×10^{-4}			356
24	1,1′-Azo-bis-4-*tert*-butylcumene	Toluene	42.8	1.16×10^{-5}	123.6		309
			48.6	2.58×10^{-5}			309
			58.0	1.080×10^{-4}		h	309
	α,α′-Azo-bis(α-methylbenzyl)-dimethacrylate	Dichlorobenzene	65	3.65×10^{-5}			366
			70	7.98×10^{-5}			366
			80	2.313×10^{-4}			366
			90	1.344×10^{-3}			366
	Bis(2,2,4,6,6-pentamethyl-4-heptyl)diazine	Diphenyl ether	100.1	3.03×10^{-4}	125.7		314
			109.2	7.62×10^{-4}			314
			125.0	3.90×10^{-3}		h	314
	1,1′-Diphenyl-azo-cyclohexane	Benzene	45.5	2.52×10^{-5}	125.7		323
			54.0	8.74×10^{-5}			323
			69.7	7.12×10^{-4}			323
25	3-Bromophenyl-azo-triphenyl-methane	Toluene	53.8	1.14×10^{-4}	125.1		42
			64.0	4.58×10^{-4}			42
	4-Bromophenyl-azo-triphenyl-methane	Toluene	53.35	1.05×10^{-4}	117.2		52
			64.30	4.28×10^{-4}			52
	4-Hydroxyphenyl-azo-triphenyl-methane	Acetic acid	54.00	1.42×10^{-4}	133.9		52
			64.00	6.2×10^{-4}			52
		Pyridine	54.00	1.52×10^{-4}	133.9		52
			64.00	6.7×10^{-4}			52
		Toluene	54.00	1.70×10^{-4}	121.3		52
			64.00	6.4×10^{-4}			52
	2-Nitrophenyl-azo-9-phenylfluorene	Toluene	45.45	1.01×10^{-4}	119.7		75
			55.55	3.71×10^{-4}			75
	4-Nitrophenyl-azo-9-phenylfluorene	Toluene	45.45	2.06×10^{-4}	111.7		75
			55.55	7.53×10^{-4}			75
	2,4-Dinitrophenyl-azo-9-phenyl-fluorene	Toluene	55.55	2.06×10^{-4}	119.2		75
			64.94	6.92×10^{-4}			75
	2-Nitrophenyl-azo-triphenylmethane	Toluene	64.94	1.46×10^{-4}	123.8		75
			75.06	5.26×10^{-4}			75
	3-Nitrophenyl-azo-triphenylmethane	Toluene	53.8	5.8×10^{-5}	110.9		42
			64.0	1.99×10^{-4}			42
		Benzene	60	9.92×10^{-5}			263
		tert-Butylbenzene	60	9.18×10^{-5}			263

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
25 (*cont'd*)	3-Nitrophenyl-azo-triphenylmethane	Cumene	60	1.04×10^{-4}			263
		Ethylbenzene	60	1.10×10^{-4}			263
		Octane	60	1.01×10^{-4}			263
		Toluene	45	1.25×10^{-5}			263
			60	1.22×10^{-4}			263
	4-Nitrophenyl-azo-triphenylmethane	Benzene	60	1.09×10^{-4}			263
		tert-Butylbenzene	60	9.7×10^{-5}			263
			60	4.03×10^{-5}		z (1256)	263
			60	4.44×10^{-6}		z (3890)	263
		Chlorobenzene	55	8.50×10^{-5}			333
			55	6.88×10^{-5}		z (1013)	333
			55	4.42×10^{-5}		z (4154)	333
		Cumene	55	8.10×10^{-5}			333
			55	6.75×10^{-5}		z (1013)	333
			55	2.62×10^{-5}		z (6180)	333
			60	1.15×10^{-4}			263
			60	4.73×10^{-5}		z (1256)	263
			60	7.23×10^{-6}		z (3890)	263
		Decane	50	2.017×10^{-5}			168,263
			60	9.988×10^{-4}			168,263
			70	3.350×10^{-4}			168
			77.5	1.16×10^{-3}			168
		1-Decene	60	1.038×10^{-4}			168
		Dodecane	50	1.933×10^{-5}			168,263
			60	9.166×10^{-5}			168,263
			70	3.250×10^{-4}			168
		1-Eicosene	60	7.60×10^{-5}			168
		Heptane	50	2.716×10^{-5}			168
			60	1.043×10^{-4}			168,263
			70	3.900×10^{-4}			168
		Hexadecane	50	1.716×10^{-5}			168,263
			60	8.133×10^{-5}			168,263
			70	2.966×10^{-4}			168
			77.5	1.09×10^{-3}			168
		1-Hexadecene	60	8.25×10^{-5}			168
		Hexane	50	2.750×10^{-5}			168,263
			60	1.195×10^{-4}			168,263
		1-Hexene	60	1.143×10^{-4}			168
		Nonane	50	2.200×10^{-5}			168,263
			60	9.900×10^{-5}			168,263
			70	3.710×10^{-4}			168
		Octadecane	60	7.780×10^{-5}			168,263
			70	2.783×10^{-4}			168
		1-Octadecene	60	7.58×10^{-5}			168
		Octane	60	1.015×10^{-4}			168,263
			70	3.650×10^{-4}			168
			77.5	1.78×10^{-3}			168
		1-Octene	60	1.123×10^{-4}			168
		Pentane	50	3.600×10^{-5}			168,263
			60	1.280×10^{-4}			168,263
		Tetradecane	50	1.866×10^{-5}			168,263
			60	8.800×10^{-5}			168,263
			70	3.116×10^{-4}			168
	4-Nitrophenyl-azo-triphenylmethane	1-Tetradecene	60	9.02×10^{-5}			168
		Toluene	64.94	2.58×10^{-4}	123.0		75
			75.06	9.19×10^{-4}			75
			53.35	5.7×10^{-5}	113.0		52
			64.30	2.25×10^{-4}			52
	2,4-Dinitrophenyl-azo-triphenyl-methane	Toluene	75.06	1.90×10^{-4}	122.6		75
			84.98	6.13×10^{-4}			75
	Phenyl-azo-triphenylmethane	Acetic acid	43.30	5.7×10^{-5}	117.2		52
			64.00	8.4×10^{-4}			52
		Anisole	25.0	2.58×10^{-6}	120.5	b,h	103

TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
25 (*cont'd*)	Phenyl-azo-triphenylmethane	Anisole	50.2	1.31×10^{-4}		b	103
			74.7	3.03×10^{-3}		b	103
			25.0	3.0×10^{-6}	118.3		114
			74.5	2.9×10^{-3}			114
			80.1	6.8×10^{-3}			114
			85.9	1.1×10^{-2}			114
		Benzene	25.0	4.29×10^{-6}	112.1	b,h	103
			49.6	1.24×10^{-4}			103
			74.7	3.12×10^{-3}		b	103
			50	1.37×10^{-4}			263
		Benzonitrile	25.0	2.62×10^{-6}	121.3	b,h	103
			50.3	1.56×10^{-4}		b	103
			74.7	3.14×10^{-3}			103
		tert-Butylbenzene	50	1.33×10^{-4}			263
			50	3.83×10^{-5}		z (1140)	263
			50	9.53×10^{-6}		z (4265)	263
		Chlorobenzene	25.0	3.77×10^{-6}	118.4	b,h	103
			49.6	1.67×10^{-4}		b	103
			74.7	3.93×10^{-3}		b	103
		Cumene	50	1.50×10^{-4}			263
			50	6.82×10^{-5}		z (1140)	263
			50	7.65×10^{-6}		z (4265)	263
		Cyclohexane	25.0	4.22×10^{-6}	102.5	h	103
			49.6	9.90×10^{-5}		b	103
			74.7	1.75×10^{-3}		b	103
		Decane	60	5.72×10^{-4}			263
		Dodecane	60	5.35×10^{-4}			263
		Diethyl malonate	25.1	3.1×10^{-6}	116.6		114
			74.5	2.8×10^{-3}			114
			80.1	5.9×10^{-3}			114
			85.9	1.0×10^{-2}			114
		Ethylbenzene	50	1.58×10^{-4}			263
		Heptane	60	6.60×10^{-4}			263
		Hexadecane	60	4.76×10^{-4}			263
		Hexane	60	7.60×10^{-4}			263
		Nitrobenzene	25.0	2.6×10^{-6}	118.7		114
			74.5	3.0×10^{-3}			114
			80.1	5.8×10^{-3}			114
			85.9	9.8×10^{-3}			114
		Octane	50	1.59×10^{-4}			263
			50	7.53×10^{-5}		z (1140)	263
			50	1.17×10^{-5}		z (4265)	263
			60	6.41×10^{-4}			263
		Pyridine	53.35	1.74×10^{-4}	129.7		52
			64.00	8.0×10^{-4}			52
		Tetradecane	60	4.99×10^{-4}			263
		Toluene	43.8	6.3×10^{-5}	115.5		42
			53.8	2.4×10^{-4}			42
			45	9.23×10^{-5}			263
			50	1.73×10^{-4}			263
			45.45	8.48×10^{-5}	122.6		75
			55.55	3.51×10^{-4}			75
			43.30	6.0×10^{-5}	113.0		52
			53.35	2.25×10^{-4}			52
			53.3	2.25×10^{-4}	113.0		46
26	2,2′-Azo-bis[2-(4-ethyl-5-propyl-imidazoline-2-yl)propane 2HCl	Methanol	50	9.75×10^{-5}			356
		Water	50	1.33×10^{-4}			356
	Azo-bis-diphenylmethane	Toluene	54.0	1.01×10^{-4}			107,312
			64.0	3.40×10^{-4}	111.3		46,107,312
	1,1′-Bis-biphenyl-azo-methane	Diphenyl ether	135.0	1.23×10^{-4}	134.1	h	297
			145.0	3.58×10^{-4}			297
			155.0	7.76×10^{-4}			297
			165.0	2.09×10^{-3}			297
	1,1′-Azo-bis-1-(4-tolyl)-cyclohexane	Toluene	36.0	8.7×10^{-6}	107.1		200

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TABLE 2. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
26 (*cont'd*)	1,1′-Azo-bis-1-(4-tolyl)-cyclohexane	Toluene	42.8	2.01×10^{-5}			200
			61.1	2.009×10^{-4}			200
	1,1′-Diisopropyl-2,2′-dimethyl-1,1′-Diphenyl-azo-propane	Ethylbenzene	50.0	6.24×10^{-5}			324
			60.0	2.49×10^{-4}			324
			70.0	7.77×10^{-4}			324
	4-Methoxyphenyl-azo-triphenyl-methane	Toluene	54.00	2.13×10^{-4}	117.2		52
			64.00	7.6×10^{-4}			52
	1,1′-Diphenyl-azo-cycloheptane	Benzene	16.2	1.65×10^{-4}			323
			24.8	5.52×10^{-4}			323
			33.4	1.66×10^{-3}			323
			41.1	4.52×10^{-3}			323
	3-Tolyl-azo-triphenylmethane	Benzene	50	1.30×10^{-4}			263
		tert-Butylbenzene	50	1.30×10^{-4}			263
		Cumene	50	1.51×10^{-4}			263
		Ethylbenzene	50	1.72×10^{-4}			263
		Octane	45	8.9×10^{-5}			263
		Toluene	43.8	7.4×10^{-5}	113.8		42
			53.8	2.77×10^{-4}			42
			45	9.74×10^{-5}			263
			50	1.81×10^{-4}			263
	4-Tolyl-azo-triphenylmethane	Benzene	50	1.26×10^{-4}			263
		tert-Butylbenzene	50	1.35×10^{-4}			263
		Cumene	50	1.40×10^{-4}			263
		Ethylbenzene	50	1.62×10^{-4}			263
		Octane	45	8.6×10^{-5}			263
		Toluene	43.30	6.9×10^{-5}	100.4		52
			53.35	2.25×10^{-4}			52
			45	9.3×10^{-5}			263
			50	1.52×10^{-4}			263
27	4-Acetaminophenyl-azo-triphenyl-methane	Toluene	54.00	1.46×10^{-4}	125.5		52
			64.00	5.9×10^{-4}			52
28	1,1′-Azo-bis-1,2-diphenyl-ethane (meso)	Ethylbenzene	96.56	2.75×10^{-5}	138.5	a,h	162
			106.47	9.03×10^{-5}		a	162
			115.28	2.42×10^{-4}		a	162
	(D,L)		96.56	3.22×10^{-5}	138.9	a,h	162
			106.47	1.04×10^{-4}		a	162
			115.28	2.84×10^{-4}		a	162
30	1,1′-Dibutyl-1,1′-diphenyl-azo-pentane	Ethylbenzene	50.0	5.05×10^{-5}	108.9	h	324
			60.0	2.03×10^{-4}			324
			70.0	6.24×10^{-4}			324
34	1,1′-Azo-bis-(1,3-diphenyl-pentane)	Bromobenzene	89.7	1.5×10^{-5}			311
		Xylene	118.8	5.6×10^{-4}			311
40	3,10,13,20-Tetraphenyl-1,2,11,12-tetra-aza-l,11-cycloeicosadiene	Xylene	110	7.20×10^{-5}	145.6		38
			120	2.30×10^{-4}			38
			130	6.90×10^{-4}			38
44	3,12,15,24-Tetraphenyl-1,2,13,14-tetra-aza-1,13-cyclotetracosadiene	Ethylbenzene	112.95	8.761×10^{-5}			158
		Xylene	110.8	8.9×10^{-5}	126.4		48
			119.8	2.2×10^{-4}			48
			110	7.63×10^{-5}	143.9		38
			120	2.46×10^{-4}			38
			130	7.12×10^{-4}			38
	Poly[oxyethylenexoxy-(1-oxo-2,2-dimethylethylene)-azo-(2-oxo-1,1-dimethyl-ethylene)	*o*-Xylene	85	2.106×10^{-4}	124.0		332
			95	7.312×10^{-4}			332
			105	1.995×10^{-3}			332
			T (K)	1.99×10^{14}	$\exp[-124.0/RT]$		332

TABLE 3. ALKYL PEROXIDES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
2	Methyl peroxide	Vapor	T (K)	1.6×10^{15}	exp[− 147.7/RT]		72
		Methanol	T (K)	4.1×10^{15}	exp[− 153.9/RT]		225
4	Ethyl peroxide	Styrene	60	1.2×10^{-9}	147.3		92
		Vapor	140.2	1.75×10^{-4}	131.8	a	127
			147.8	3.60×10^{-4}		a	127
			160.0	1.08×10^{-3}		a	127
			176.5	3.78×10^{-3}		a	127
			184.5	7.16×10^{-3}		a	127
			145.9	6.69×10^{-4}			187
			145.9	5.02×10^{-4}		t_8	187
		Vapor (toluene)	200	3.58×10^{-2}	132.6		87
			210	6.76×10^{-2}			87
			218	1.47×10^{-1}			87
			226	2.23×10^{-1}			87
			234	3.86×10^{-1}			87
			245	6.43×10^{-1}			87
5	2,4-Pentanedione peroxide	Triethyl phosphate	100	1.9×10^{-6}	92	a	337
			115	6.1×10^{-6}		a	337
			125	1.4×10^{-5}		a	337
6	Propyl peroxide	Vapor	146.5	2.50×10^{-4}	132.2	a	128
			155.3	6.00×10^{-4}		a	128
			166.8	1.95×10^{-3}		a	128
			175.4	4.10×10^{-3}		a	128
	Isopropyl peroxide	Styrene	60	$6. \times 10^{-10}$	154.8		92
7	Allyl *tert*-butylperoxide	Toluene	130	5.76×10^{-5}	139.1	h,m_6	321
			140	1.785×10^{-4}			321
			150	4.800×10^{-4}			321
			160	1.270×10^{-3}			321
	Dimethylaminomethyl *tert*-butyl peroxide		110	5.5×10^{-5}	134.5		303
			120	1.9×10^{-4}			303
			130	4.76×10^{-4}			303
8	*tert*-Butyl peroxide	Acetic acid	115	1.2×10^{-5}	139.7	h	192
			120	2.19×10^{-5}			192
			125	2.98×10^{-5}			192
			130	6.29×10^{-5}			192
		Acetonitrile	95	9.53×10^{-7}			208
			125	3.89×10^{-5}		d	208
			115	1.19×10^{-5}	129.7	h	192
			120	2.21×10^{-5}			192
			125	3.47×10^{-5}			192
			130	5.63×10^{-5}			192
		Benzene	80	7.81×10^{-8}	142.3	m_2,u_2	129
			130	$(2.48–3.04) \times 10^{-5}$		m_2,u_2	129
			100	8.8×10^{-7}	146.9	a	126,337
			100	8.75×10^{-7}			197
			115	5.66×10^{-6}		a	126,337
			130	3.00×10^{-5}		a	126,337
			120	1.39×10^{-5}		z (0.98)	33
			120	7.6×10^{-6}		z (864)	33
			120	7.5×10^{-6}		z (1620)	33
			120	3.7×10^{-6}		z (3480)	33
			120	1.10×10^{-5}	147.7	h	192
			125	1.99×10^{-5}			192
			130	3.22×10^{-5}			192
			135	6.19×10^{-5}			192
		Benzhydrol	125	8.7×10^{-5}			140
		tert-Butanol	120	1.41×10^{-5}	143.5	h	192
			125	2.49×10^{-5}			192
			130	4.30×10^{-5}			192
			135	7.32×10^{-5}			192
		2-Butanol	125	4.8×10^{-5}			140
		n-Butyl mercaptan	125	1.5×10^{-5}			140
		Tri-*n*-butylamine	125	$(1.7 \pm 0.3) \times 10^{-5}$			64
			135	$(4.2 \pm 0.4) \times 10^{-5}$			64

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TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	*tert*-Butyl peroxide	Tri-*n*-butylamine	145	$(1.60 \pm 0.21) \times 10^{-4}$			64
			T (K)	2.8×10^{14}	exp[− 146.4/RT]		123
		tert-Butylbenzene	125	$(1.5 \pm 0.2) \times 10^{-5}$			64
			135	$(5.0 \pm 0.3) \times 10^{-5}$			64
			145	$(1.51 \pm 0.22) \times 10^{-4}$			64
			T (K)	2.8×10^{14}	exp[− 146.4/RT]		123
			125	2.10×10^{-6}		u_1	301
			125	2.49×10^{-6}		s	301
			125	2.03×10^{-6}		s,t_6	301
		Carbon tetrachloride	120	9×10^{-6}		z (0.98)	33
			120	2.4×10^{-6}		z (1930)	33
			120	2.3×10^{-6}		z (2890)	33
			120	8.6×10^{-7}		z (5525)	33
			125	1.40×10^{-6}		u_1,t_6	301
		Chlorobenzene	150	4.45×10^{4}			346
			121	1.93×10^{-5}			344
			141	1.93×10^{-4}			344
			164	1.93×10^{-3}			344
			T (K)	4.20×10^{15}	exp[− 153.46/RT]		344
		Cumene	125	$(1.6 \pm 0.1) \times 10^{-5}$			64
			135	$(5.2 \pm 0.3) \times 10^{-5}$			64
			145	$(1.56 \pm 0.13) \times 10^{-4}$			64
			T (K)	2.8×10^{14}	exp[− 146.4/RT]		123
		Cyclohexane	95	2.48×10^{-7}	170.7	h	208
			120	6.3×10^{-6}			192
			125	1.52×10^{-5}			192
			130	2.59×10^{-5}			192
			135	4.64×10^{-5}			192
		Cyclohexanol	125	2.4×10^{-5}			140
		Cyclohexene	120	7.6×10^{-6}	156.1	h	192
			125	1.38×10^{-5}			192
			130	2.81×10^{-5}			192
			135	4.41×10^{-5}			192
			120	8.3×10^{-6}		z (0.98)	33
			120	6.2×10^{-6}		z (1275)	33
			120	3.77×10^{-6}		z (2890)	33
			120	2.65×10^{-6}		z (5725)	33
		Cyclohexylamine	125	5.50×10^{-5}	157.3	h	143
		Decane	80	1.39×10^{-8}			270
			110	2.01×10^{-6}			270
			130	2.48×10^{-5}			270
			115	3.00×10^{-6}	163	a	337
			130	2.16×10^{-5}		a	337
			140	6.66×10^{-5}		a	337
		Dodecane	80	1.29×10^{-8}			270
			110	1.87×10^{-6}			270
			130	2.44×10^{-5}			270
		Dimethylaniline	120	9.6×10^{-6}			192
			125	1.89×10^{-5}			192
			130	3.41×10^{-5}			192
			135	5.84×10^{-5}			192
		Ethyl benzoate	120	1.07×10^{-5}	148.5	h	192
			125	1.92×10^{-5}			192
			130	3.39×10^{-5}			192
			135	5.90×10^{-5}			192
		N-Ethylcyclohexyl amine	125	4.01×10^{-5}			143
		Heptane	80	1.44×10^{-8}			270
			110	2.19×10^{-6}			270
		Hexadecane	80	1.07×10^{-8}			270
			110	1.82×10^{-6}			270
			130	2.33×10^{-5}			270
		Hexane	80	1.64×10^{-8}			270
			110	2.17×10^{-6}			270

TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	*tert*-Butyl peroxide	2-Methyl-2-butanol	120	1.26×10^{-5}	149.4	h	192
			125	2.34×10^{-5}			192
			130	4.47×10^{-5}			192
			135	6.80×10^{-5}			192
		Methyl methacrylate	T (K)	2.8×10^{14}	exp[− 146.4/RT]		123
		N-Methylpipendine	125	1.54×10^{-5}			143
		Mineral oil	T (K)	1.31×10^{15}	exp[− 154.6/RT]		376
		Nonane	80	1.36×10^{-8}			270
			110	2.01×10^{-6}			270
			130	2.54×10^{-5}			270
		Norbornanol	125	4.4×10^{-5}			140
		Nujol	125	7.8×10^{-7}		u	301
		Octane	80	1.48×10^{-8}			270
			110	2.19×10^{-6}			270
		2-Octanol	125	5.5×10^{-5}			140
		Piperidine	125	3.49×10^{-5}			143
		1-Propanol	125	2.8×10^{-5}			140
		Styrene	T (K)	2.8×10^{14}	exp[− 146.4/RT]		123
		Tetradecane	80	1.12×10^{-8}			270
			110	1.82×10^{-6}			270
			130	2.38×10^{-5}			270
		Tetrahydrofuran	120	9.7×10^{-6}	155.2	h	192
			125	1.84×10^{-5}			192
			130	3.39×10^{-5}			192
			135	5.76×10^{-5}			192
			125	1.5×10^{-5}			140
		Toluene	100	6.82×10^{-7}			197
			120	1.34×10^{-5}		z (0.98)	33
			120	9.5×10^{-6}		z (2000)	33
			120	8.0×10^{-6}		z (2850)	33
			120	5.7×10^{-6}		z (5170)	33
			125	1.6×10^{-5}			140
			125	1.62×10^{-5}			143
			T (K)	1.30×10^{16}	exp[− 158.3/RT]		376
		Triethylamine	120	7.9×10^{-6}	169.9	h	192
			125	1.69×10^{-5}			192
			130	3.15×10^{-5}			192
			135	5.55×10^{-5}			192
		Vapor (acetone)	127.5	7.4×10^{-6}			238
			131	1.13×10^{-5}			238
			146	6.0×10^{-5}			238
			151.5	1.03×10^{-4}			238
			162	3.6×10^{-4}			238
			167	5.2×10^{-4}			238
			145	1.3×10^{-4}	165.3		125
			T (K)	5.9×10^{16}	exp[− 165.3/RT]		125
		Vapor (carbon tetrachloride)	150	8.58×10^{-4}		l	130
		Vapor (chloroform)	150	1.167×10^{-3}		l	130
		Vapor (dichloro-difluoromethane)	150	$>(2.00 \times 10^{-4})$			130
		Vapor (trichloro ethylene)	150	2.35×10^{-3}		l	130
		Vapor (methylene chloride)	150	1.017×10^{-3}		l	130
		Vapor (3-pentanone)	145	1.5×10^{-4}	165.3		125
			T (K)	6.8×10^{16}	exp[− 165.3/RT]		125
		Vapor (silicon tetrafluoride)	160	1.05×10^{-3}	113.0	a,i(2.6)	117
			160	1.25×10^{-3}		a,i(33)	117
			160	1.27×10^{-3}		a,i(100)	117
		Vapor (toluene)	148	9.0×10^{-5}			88
			158	2.5×10^{-4}			88
		Vapor	103.2	5.6×10^{-7}			227
			111.9	2.03×10^{-6}			227

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TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	*tert*-Butyl peroxide	Vapor	120.2	6.39×10^{-6}			227
			129.5	1.98×10^{-5}			227
			138.5	6.00×10^{-5}			227
			145.4	1.24×10^{-4}			227
			125	1.1×10^{-5}			64
			135	3.6×10^{-5}			64
			145	1.15×10^{-4}			64
			129.6	1.64×10^{-5}			86
			141.0	6.28×10^{-5}			86
			152.5	2.25×10^{-4}			86
			166.8	8.92×10^{-4}			86
			130	1.82×10^{-5}	159.0	a	96
			140	5.75×10^{-5}		a	96
			150	1.75×10^{-4}		a	96
			160	4.88×10^{-4}		a	96
			170	1.35×10^{-3}		a	96
			129	1.97×10^{-5}			88
			138	4.3×10^{-5}			88
			149	1.30×10^{-4}			88
			152	1.62×10^{-4}			88
			139.7	6.0×10^{-5}	163.6		63
			147.2	1.43×10^{-4}			63
			154.6	3.22×10^{-4}			63
			159.8	5.53×10^{-4}			63
			T (K)	3.2×10^{16}	exp[−163.6/RT]		63
			145	1.3×10^{-4}	161.5		125
			T (K)	1.9×10^{16}	exp[−161.5/RT]		175,125
			149.5	1.79×10^{-4}	156.5	i(37-132)	83
			160	4.00×10^{-4}	154.8	a,i(2.6)	117
			160	4.53×10^{-4}		a,i(33)	117
			160	4.83×10^{-4}		a,i(100)	117
			130	1.91×10^{-5}	156.7	u_3	277
			140	5.95×10^{-5}		u_3	277
			150	1.78×10^{-4}		u_3	277
			160	4.96×10^{-4}		u_3	277
		Vapor (He)	280	7.7	154.8	i(10)	111
			290	1.51×10^{1}		i(10)	111
			300	2.77×10^{1}		i(10)	111
			310	4.87×10^{1}		i(10)	111
			320	8.34×10^{1}		i(10)	111
			330	1.38×10^{2}		i(10)	111
			340	2.13×10^{2}		i(10)	111
			350	3.22×10^{2}		i(10)	111
		KBr pellets	109	3.8×10^{-7}	159.0	h	241
			127	2.96×10^{-5}			241
			149	4.4×10^{-4}			241
	tert-Butyl 2-chloro-1,1-dimethylethyl-peroxide	Cumene	120	8.82×10^{-6}			363
			130	2.80×10^{-5}			363
			140	9.26×10^{-5}			363
			150	2.78×10^{-4}			363
	sec-Butyl peroxide	Toluene	100	2.7×10^{-6}			197
		Vapor	100	1.5×10^{-6}			197
	Butyl peroxide	Decane	80	1.43×10^{-8}		t_6	270
		Dodecane	80	1.17×10^{-8}		t_6	270
		Heptane	80	2.04×10^{-8}		t_6	270
		Hexadecane	80	8.8×10^{-9}		t_6	270
		Octane	80	1.58×10^{-8}		t_6	270
		Styrene	60	3.3×10^{-9}	142.3		92
		Tetradecane	80	9.6×10^{-9}		t_6	270
	Bromo-*tert*-butyl peroxide	Chlorobenzene	150	3.63×10^{-4}			346
	Chloro-*tert*-butyl peroxide	Chlorobenzene	150	4.15×10^{-4}			346
	Bis(chloro-*tert*-butyl) peroxide	Chlorobenzene	150	3.11×10^{-4}			346
	2-*tert*-Butyl peroxy-2-methyl-1-propanol	Chlorobenzene	150	1.16×10^{-3}			346

TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	2-*tert*-Butyl peroxy-2-methyl-1-propanol-d_1	Chlorobenzene	150	1.23×10^{-3}			346
	1-Hydroxybutyl-*n*-butyl peroxide	α-Methylstyrene	79.4	1.7×10^{-5}	102.9		32
			99.4	1.06×10^{-4}			32
			109.9	2.9×10^{-4}			32
	1-Hydroxyisobutyl-isobutyl peroxide	α-Methylstyrene	79.4	3.7×10^{-5}	83.7	c	32
			99.4	2.0×10^{-4}		c	32
			109.9	4.8×10^{-4}		c	32
	1-Hydroxyisobutyl-1-di-isobutyl-1,1-d_2 peroxide	α-Methylstyrene	99.5	6.9×10^{-5}	89.5		25
			109.8	1.8×10^{-4}			25
			122.0	4.2×10^{-4}			25
	Dimethylaminomethyl-*tert*-amyl peroxide	Styrene	110	4.9×10^{-5}	130.7	t_{11}	303
			120	1.19×10^{-4}		t_{11}	303
			130	3.76×10^{-4}		t_{11}	303
	Diethylaminomethyl-*tert*-butyl peroxide	Styrene	110	4.1×10^{-5}	132.4	t_{11}	303
			120	1.35×10^{-4}		t_{11}	303
			130	3.25×10^{-4}		t_{11}	303
	N-Piperidinomethyl-*tert*-butyl peroxide	Styrene	110	5.5×10^{-5}	130.7	t_{11}	303
			120	2.66×10^{-4}		t_{11}	303
			130	4.16×10^{-4}		t_{11}	303
9	*tert*-Butyl *tert*-amyl peroxide	Cumene	110	3.86×10^{-6}			363
			120	1.287×10^{-5}			363
			130	4.30×10^{-5}			363
			140	1.406×10^{-4}			363
10	*tert*-Amyl peroxide	Bulk	125	5.7×10^{-5}			202
			132.2	1.15×10^{-4}			202
		Chlorobenzene	108	1.93×10^{-5}			344
			128	1.93×10^{-4}			344
			150	1.93×10^{-3}			344
			T (K)	4.02×10^{15}	exp[−148.35/RT]		344
		Decalin	125	2.8×10^{-5}			202
		Octane	125	3.0×10^{-5}			202
		Triethylamine	125	3.5×10^{-5}			202
		Vapor	132.2	7.2×10^{-5}	154.8–171.5	i(200–225)	63
			136.7	1.15×10^{-4}		i(200–225)	63
			142.2	2.16×10^{-4}		i(200–225)	63
			149.2	4.8×10^{-4}		i(100–200)	63
			136.7	1.34×10^{-4}		i(440–610)	63
			142.2	2.41×10^{-4}		i(440–610)	63
			149.7	5.61×10^{-4}		i(440–610)	63
	tert-Butyl 1,1-dimethylbutyl peroxide	Cumene	110	4.38×10^{-6}			363
			120	1.537×10^{-5}			363
			130	4.99×10^{-5}			363
			140	1.544×10^{-4}			363
	tert-Butyl 1,1,2-trimethylpropyl peroxide	Cumene	110	7.07×10^{-6}			363
			120	2.374×10^{-5}			363
			130	7.62×10^{-5}			363
			140	2.308×10^{-4}			363
	2-Methyl-2-*tert*-butyl-peroxyacetate	Chlorobenzene	150	4.12×10^{-4}			346
	Ethyl 2-*tert*-butylperoxymethyl-propenoate	Triisopropylbenzene	127.5	1.63×10^{-4}		c_2	353
			130	1.95×10^{-4}		c_2	353
			132.5	2.16×10^{-4}		c_2	353
			135	2.56×10^{-4}		c_2	353
			137.5	3.47×10^{-4}		c_2	353
			140	3.92×10^{-4}		c_2	353
			142.5	4.68×10^{-4}		c_2	353
			145	5.37×10^{-4}		c_2	353
			147.5	6.30×10^{-4}		c_2	353
			150	7.76×10^{-4}		c_2	353
11	Apocamphane-1- formyl peroxide	Benzene	80	2.3×10^{-4}			244
12	2,2-Bis(*tert*-butylperoxybutane)	Benzene	85	3.0×10^{-6}	151	a	337
			100	1.9×10^{-5}		a	337
			115	9.36×10^{-5}		a	337

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TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	*tert*-Butyl 1,1,3,3-tetramethylbutyl peroxide	Cumene	110	1.085×10^{-5}			363
			120	3.73×10^{-5}			363
			130	1.141×10^{-4}			363
			140	3.40×10^{-4}			363
	1-Hydroxy-1-hydroperoxy-dicyclohexyl peroxide	Benzene	85	9.6×10^{-6}		a,c_2	337
			100	5.1×10^{-5}		a,c_2	337
			115	1.9×10^{-4}		a,c_2	337
			130	5.2×10^{-4}		a,c_2	337
	Di-(1-hydroxycyclohexyl)-peroxide	Chlorobenzene	105	$\sim 1.9 \times 10^{-5}$			344
			120	$\sim 1.9 \times 10^{-4}$			344
	Diisopropylaminomethyl-*tert*-amyl peroxide	Styrene	110	3.9×10^{-5}	132.4	t_{11}	303
			120	1.21×10^{-4}		t_{11}	303
			130	3.0×10^{-4}		t_{11}	303
	1-Phenylethyl *tert*-butyl peroxide	Chlorobenzene	129.2	3.09×10^{-5}			282
13	*tert*-Butylα-cumyl peroxide	*tert*-Butylbenzene	138	1.48×10^{-4}		a	80
			158	9.62×10^{-4}		a	80
		Chlorobenzene	115	1.93×10^{-5}			344
			136	1.93×10^{-4}			344
			159	1.93×10^{-3}			344
			T (K)	1.17×10^{15}	exp[− 146.98/RT]		344
		Cumene	138	1.44×10^{-4}		a	80
			158	8.88×10^{-4}		a	80
			125	3.04×10^{-5}			352
		Dodecane	128	4.44×10^{-5}	146.4	a	80
			138	1.39×10^{-4}		a	80
			148	3.21×10^{-4}		a	80
			158	8.88×10^{-4}		a	80
	tert-Butyl *p*-chlorocumyl peroxide	Cumene	125	2.42×10^{-5}			352
	tert-Butyl *p*-nitrocumyl peroxide	Cumene	125	2.20×10^{-5}			352
14	*tert*-Butyl 1,1-dimethyl-2-phenyl-ethyl peroxide	Cumene	110	4.35×10^{-6}			363
			120	1.458×10^{-5}			363
			130	4.70×10^{-5}			363
			140	1.463×10^{-4}			363
	1,1-Di-(*tert*-butylperoxy)-cyclohexane	Benzene	93	1.9×10^{-5}		a	338
	Ethyl-3,3-di-*tert*-butylperoxy)-butyrate	Benzene	111	1.9×10^{-5}		a	338
	1-[4-(Dimethylamino)phenyl]-ethyl *tert*-butyl peroxide	Chlorobenzene	100.4	1.82×10^{-6}	151.7		282
			110.8	6.97×10^{-6}			282
			120.9	2.18×10^{-5}			282
			129.2	6.18×10^{-5}			282
			129.2	5.22×10^{-5}		m_2	282
	tert-Butyl *p*-methoxycumyl peroxide	Cumene	125	3.72×10^{-5}			352
	tert-Butyl *p*-methylcumyl peroxide	Cumene	125	3.22×10^{-5}			352
	tert-Butyl *m*-methoxycumyl peroxide	Cumene	125	2.84×10^{-5}			352
	Bis(1,1,2,2-tetramethylpropyl) peroxide	Cumene	125	4.84×10^{-4}			352
15	2-[4-(Dimethylamino)phenyl]-propyl *tert*-butyl peroxide	Chlorobenzene	129.2	1.51×10^{-4}		m_2	282
16	1,1-Di-(*tert*-amylperoxy)-cyclohexane	Benzene	90	1.9×10^{-5}		a	338
	tert-Butyl 3-isopropenyl-cumyl peroxide	Chlorobenzene	104	1.93×10^{-5}			344
			126	1.93×10^{-4}			344
			152	1.93×10^{-3}			344
			T (K)	1.161×10^{13}	exp[− 128.33/RT]		344
	2,5-Dimethyl-2,5-di(*tert*-butyl-peroxy)-hexane	Benzene	115	1.15×10^{-5}	166.9	a	126,337
			130	6.86×10^{-5}		a	126,337
			145	4.75×10^{-4}		a	126,337
		Chlorobenzene	115	1.93×10^{-5}			344
			134	1.93×10^{-4}			344
			156	1.93×10^{-3}			344
			T (K)	1.68×10^{16}	exp[− 155.49/RT]		344
	2,5-Dimethyl-2,5-di(*tert*-butyl-peroxy)-3-hexyne	Benzene	115	3.91×10^{-6}	156.9	a	126,337
			130	2.35×10^{-5}		a	126,337
			145	1.14×10^{-4}		a	126,337
			160	6.17×10^{-4}		a	126,337

TABLE 3. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	2,5-Dimethyl-2,5-di(*tert*-butyl-peroxy)-3-hexyne	Chlorobenzene	120	1.93×10^{-5}			344
			141	1.93×10^{-4}			344
			164	1.93×10^{-3}			344
			T (K)	1.90×10^{15}	exp[−150.67/RT]		344
17	*n*-Butyl-4,4-bis(*tert*-butylperoxy)-valerate	Dodecane	100	5.83×10^{-6}		a	8
			115	3.53×10^{-5}		a	8
			130	2.91×10^{-4}		a	8
	1,1-Bis-(*tert*-butylperoxy)-3,3,5-trimethylcyclohexane	Benzene	85	6.9×10^{-6}	138	a	337
			100	5.05×10^{-5}		a	337
			115	2.71×10^{-4}		a	337
18	Cumyl peroxide	Benzene	115	2.05×10^{-5}	159	a	337
			130	1.05×10^{-4}		a	126,337
			145	6.86×10^{-4}		a	126,337
		tert-Butylbenzene	158	1.72×10^{-3}		a	80
		Chlorobenzene	112	1.93×10^{-5}			344
			132	1.93×10^{-4}			344
			154	1.93×10^{-3}			344
			T (K)	9.24×10^{15}	exp[−152.67/RT]		344
		Cumene	T (K)	4.31×10^{14}	exp[−144.3/RT]		67
			138	2.57×10^{-4}		a	80
			158	1.52×10^{-3}		a	80
		Dodecane	128	8.75×10^{-5}		a	80
			138	2.31×10^{-4}		a	80
			148	5.37×10^{-4}		a	80
			158	1.83×10^{-3}		a	80
		Diisopropylcarbinol	138	3.16×10^{-4}		a	80
		Neat	150	3.98×10^{-3}		a	371
			155	4.81×10^{-3}		a	371
			160	9.63×10^{-3}		a	371
			165	1.16×10^{-2}		a	371
	Bicyclo[2.2.2]octane-1-formyl peroxide	Carbon tetrachloride	30	1.71×10^{-4}			244
				1.73×10^{-3}		n	244
				1.68×10^{-4}		v_1	244
		Chlorobenzene	30	6.20×10^{-4}			244
		Cyclohexane	30	6.1×10^{-5}			244
		Isooctane	30	5.0×10^{-5}			244
		Toluene	30	3.66×10^{-4}			244
20	α,α′-Bis(*tert*-butylperoxy)-diisopropyl benzene	Benzene	115	1.9×10^{-5}	147	a	337
			125	4.91×10^{-5}		a	337
			130	9.61×10^{-5}		a	337
		Chlorobenzene	114	1.93×10^{-5}			344
			134	1.93×10^{-4}			344
			156	1.93×10^{-3}			344
			T (K)	7.65×10^{15}	exp[−152.69 kJ/RT]		344
		Neat	155	3.61×10^{-3}		a	371
			160	4.44×10^{-3}		a	371
			165	4.62×10^{-3}		a	371
24	2,5-Dimethyl-2,5-di-(2-ethyl-hexanoylperoxy) hexane	Benzene	68	1.9×10^{-5}		a	338

TABLE 4. ACYL PEROXIDES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4	Acetyl peroxide	Acetic acid	55.2	2.8×10^{-6}	126.4	d,e	106,176
			64.9	9.9×10^{-6}		d,e	106,176
			75.2	3.75×10^{-5}		d,e	106,176
			85.2	1.30×10^{-4}		d,e	106,176
			73.2	2.62×10^{-5}		a	100

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4 (*cont'd*)	Acetyl peroxide	Benzene	35	9.5×10^{-7}			68
			55	3.14×10^{-6}			68
			65	1.27×10^{-5}			68
			50	1.1×10^{-6}	134	a	337
			70	2.39×10^{-5}		a	126,337
			85	1.73×10^{-4}		a	126,337
			70	2.38×10^{-5}		a	16
			60.3	5.0×10^{-6}			4
			80	8.7×10^{-5}			20
			55.2	2.6×10^{-6}	135.1	c,e	106,176
			64.9	1.07×10^{-5}		c,e	106,176
			75.2	4.65×10^{-5}		c,e	106,176
			85.2	1.62×10^{-4}		c,e	106,176
		n-Butanol	60.3	3.4×10^{-5}			4
		sec-Butanol	60.3	3×10^{-5}			4
		tert-Butanol	60.3	3.1×10^{-6}	133.9		4
			80.3	4.9×10^{-5}			4
		Carbon tetrachloride	26	1.08×10^{-7}			68
			46	4.84×10^{-7}			68
			65	2.11×10^{-6}			68
			80	5.5×10^{-5}			20
		Chloroform	80.3	~ 5			4
		Cyclohexane	55.2	2.1×10^{-6}	131.4	c,e	106,176
			64.9	8.3×10^{-6}		c,e	106,176
			75.2	3.60×10^{-5}		c,e	106,176
			85.2	1.27×10^{-4}		c,e	106,176
		Cyclohexene	60	4.5×10^{-6}	133.5		20
			70	1.77×10^{-5}			20
			80	7.0×10^{-5}			20
			90	2.28×10^{-4}			20
			100	7.61×10^{-4}			20
		Cyclopentene	70	1.60×10^{-5}	137.2		20
			80	7.0×10^{-5}			20
			90	2.55×10^{-4}			20
			100	7.25×10^{-4}			20
		Cumene	80	7.6×10^{-5}			20
		Decane	80	6.85×10^{-5}			168
		n-Dodecane	60	2.3×10^{-6}			70
			80	6.15×10^{-5}			168
		Ethanol	60.3	1.01×10^{-4}	129.7		4
			80.3	1.40×10^{-3}			4
		Heptane	80	7.72×10^{-5}			168
		Hexadecane	80	5.39×10^{-5}			168
		n-Hexane	60	3.4×10^{-6}			70
		1-Hexene	70	2.35×10^{-5}	132.6		20
			80	8.7×10^{-5}			20
			90	3.05×10^{-4}			20
			100	9.83×10^{-4}			20
		2-Methyl-1-pentene	80	9.0×10^{-5}	126.8		20
			90	3.12×10^{-4}			20
			100	9.81×10^{-4}			20
		n-Octadecane	60	1.9×10^{-6}			70
		n-Octane	60	2.9×10^{-6}			70
			80	7.34×10^{-5}			168
		Isooctane	60	2.9×10^{-6}			70
			55.2	2.35×10^{-6}	134.7	c,e	106,176
			64.9	9.4×10^{-6}		c,e	106,176
			75.2	4.03×10^{-5}		c,e	106,176
			85.2	1.49×10^{-4}		c,e	106,176
		1-Pentene	70	2.45×10^{-5}			20
			80	9.4×10^{-5}			20
			90	3.22×10^{-4}			20
		Propionic acid	64.9	1.4×10^{-5}			106,176
			85.2	1.66×10^{-4}		d,e	106,176

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4 (*cont'd*)	Acetyl peroxide	*n*-Tetradecane	60	2.0×10^{-6}			70
			80	5.90×10^{-5}			168
		Toluene	60.3	5×10^{-6}	129.7		4
			55.2	2.7×10^{-6}	133.9	c,e	106,176
			64.9	1.14×10^{-5}		c,e	106,176
			75.2	4.70×10^{-5}		c,e	106,176
			85.2	1.59×10^{-4}		c,e	106,176
			73.2	3.06×10^{-5}	138.1	a	100
			85.5	1.72×10^{-4}		a	100
			90	7.33×10^{-5}	129.7	a	113
		Vapor (toluene)	88.0	3.12×10^{-4}	123.4		105
			134.7	3.1×10^{-2}			105
			150.7	1.18×10^{-1}			105
			161.7	2.77×10^{-1}			105
			170.7	6.10×10^{-1}			105
			184.2	1.76			105
6	Propionyl peroxide	Acetic acid	65.0	3.8×10^{-5}	123.0	c,e	51
			85.0	4.3×10^{-4}		c,e	51
		Acetic anhydride	65.0	3.5×10^{-5}	128.9	d,e	51
			85.0	4.5×10^{-4}		c,e	51
		Benzene	65.0	1.88×10^{-5}	129.3	c,e	51
			85.0	2.40×10^{-4}		c,e	51
			50	2.72×10^{-6}	127.6	a	126,337
			70	4.30×10^{-5}		a	126,337
			85	2.89×10^{-4}		a	126,337
		Benzonitrile	65.0	3.9×10^{-5}	130.5	d,e	51
			85.0	5.1×10^{-4}		d,e	51
		Dioxane	65.0	4.5×10^{-5}	116.7	c,e	51
			85.0	4.5×10^{-4}		c,e	51
		n-Hexane	65.0	1.50×10^{-5}	123.8		51
			85.0	1.72×10^{-4}			51
		Isooctane	65.0	9.8×10^{-6}	130.5	c,e	51
			86.5	1.44×10^{-4}		c,e	51
		Nitrobenzene	65.0	3.7×10^{-5}	120.9	c,e	51
			85.0	4.1×10^{-4}			51
		Toluene	65.0	1.87×10^{-5}	130.1		51
			85.0	2.54×10^{-4}			51
		Vapor	65.0	1.0×10^{-5}	125.5		51
			85.0	1.6×10^{-4}			51
			99.4	8×10^{-4}	125.5		85
			134.4	2.6×10^{-2}			85
			152.2	1.22×10^{-1}			85
			176.4	8.0×10^{-1}			85
			190.9	2.33			85
			T (K)	2.5×10^{-4}	$\exp[-125.5/RT]$		85
	2-Iodopropionyl peroxide	Acetone	56	2.19×10^{-4}		s	77
		Benzene	62.5	$(2.40–2.81) \times 10^{-4}$		s	77
			62.5	7.12×10^{-4}		n,s	77
			62.5	$(2.36 \pm 0.07) \times 10^{-4}$	108.8	p	77
		n-Butyl vinyl ether	62.5	2.47×10^{-4}		s	77
		Cyclohexene	62.5	2.7×10^{-4}		s	77
		95% Ethanol	62.5	4.0×10^{-4}		s	77
	Perfluoropropionyl peroxide	Freon 113	20	2.19×10^{-5}	100.1	h	283
			30	8.50×10^{-5}			283
			40	3.26×10^{-4}			283
			T (K)	1.8×10^{9}	$\exp[-83.8/RT]$		304
		Isooctane	T (K)	1.5×10^{17}	$\exp[-121.5/RT]$		304
	2,2,3,3-Tetrafluoropropionyl peroxide	Freon 113	25	2.26×10^{-4}	86.3	h	283
			35	7.17×10^{-4}			283
8	*tert*-Butyl permaleic acid	Acetone	87	1.9×10^{-5}		a	338
	Butyryl peroxide	Acetic acid	65.0	4.7×10^{-5}	125.1	c,e	51
			85.0	5.6×10^{-4}		c,e	51
		Acetic anhydride	65.0	4.3×10^{-5}			51
			85.0	5.5×10^{-4}		c,e	51

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	Butyryl peroxide	Benzene	65.0	2.24×10^{-5}	131.4	d,e	51
			85.0	3.02×10^{-4}		c,e	51
		Benzonitrile	65.0	4.3×10^{-5}	131.4	d,e	51
			85.0	5.8×10^{-4}			51
		Dioxane	65.0	4.6×10^{-5}	116.3	c,e	51
			85.0	4.6×10^{-4}		c,e	51
		Hexane	65.0	1.14×10^{-5}	131.4	c,e	51
			85.0	1.53×10^{-4}		c,e	51
		Isooctane	65.0	1.11×10^{-5}	133.5	c,e	51
			85.0	1.56×10^{-4}		c,e	51
		Toluene	65.0	2.14×10^{-5}	130.5	d,e	51
			85.0	2.87×10^{-4}		c,e	51
		Vapor	65.0	1.6×10^{-5}	123.8		51
			85.0	2.0×10^{-4}			51
			96.7	8.6×10^{-4}	123.8		85
			127.4	1.5×10^{-2}			85
			158.9	3.0×10^{-1}			85
			178.9	1.27			85
			T (K)	1.9×10^{-4}	$\exp[-123.8/RT]$		85
	Isobutyryl peroxide	Acetonitrile	40	6.81×10^{-4}		m_3	167
		Benzene	20	1.6×10^{-5}	109	a	337
			30	6.75×10^{-5}		a	337
			40	2.59×10^{-4}		a	337
			40	2.38×10^{-4}		m_3	167
			40	2.40×10^{-4}			148,244
		Benzonitrile	40	4.2×10^{-4}			148
		tert-Butanol	40	2.51×10^{-4}			148
		Carbon tetrachloride	40	7.8×10^{-5}			148
			45	1.58×10^{-4}			148
			50	3.05×10^{-4}			148
			55	5.61×10^{-4}			148
			60	7.67×10^{-4}			148
		Chlorobenzene	40	1.73×10^{-4}			148
			23	1.93×10^{-5}			344
			39	1.93×10^{-4}			344
			57	1.93×10^{-3}			344
			T (K)	3.37×10^{14}	$\exp[-109.06/RT]$		344
		Chloroform	40	7.5×10^{-5}			148
		Cyclohexane	40	4.5×10^{-5}			148
			40	4.70×10^{-5}			167
		Fluorobenzene	40	1.23×10^{-4}			148
		Isooctane	25	3.35×10^{-6}	114.2		84
			35	1.54×10^{-5}			84
			45	6.14×10^{-5}			84
			55	2.26×10^{-4}			84
			T (K)	2.8×10^{14}	$\exp[-114.2/RT]$		84
			40	3.2×10^{-5}			148
		Isopropanol	40	3.05×10^{-4}			148
		Nitrobenzene	40	5.80×10^{-4}			148
		Nujol	40	4.63×10^{-5}			167
		Tetralin	40	1.75×10^{-4}			148
		Toluene	40	1.43×10^{-4}			148
		Vapor	40	$\sim 1 \times 10^{-5}$			167
		p-Xylene	40	1.40×10^{-4}			148
	Cyclopropane formyl peroxide	Carbon tetrachloride	64.5	4.4×10^{-6}			34
			70.4	9.3×10^{-6}			34
			77.8	2.31×10^{-5}			34
	Diacetyl succinoyl diperoxide	Styrene	60	5.2×10^{-6}	125.5		215
			73.5	2.3×10^{-5}			215
			85	9.3×10^{-5}			215
	Succinoyl peroxide	Acetone	70	2.80×10^{-5}	99.6	a	126,337
			85	1.21×10^{-4}		a	126,337
			100	4.36×10^{-4}		a	126,337

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
9	Acetyl benzoyl peroxide	Chlorobenzene	70	2×10^{-5}			155
			96	5.12×10^{-4}		z(25),v_{13}	284
			96	4.08×10^{-4}		z(19),v_{13}	284
10	5-Bromo-2-thenoyl peroxide	Carbon tetrachloride	75	1.53×10^{-5}		a,m_1	232
	4-Bromo-2-thenoyl peroxide	Carbon tetrachloride	75	1.14×10^{-5}		a,m_1	232
	5-Chloro-2-thenoyl peroxide	Carbon tetrachloride	75	1.58×10^{-5}		a,m_1	232
	α-Chloropropionyl m-chlorobenzoyl-peroxide	Acetonitrile	41	3.05×10^{-5}		m_3	167
		Cyclohexane	41	1.51×10^{-5}		m_3	167
	Cyclobutane formyl peroxide	Carbon tetrachloride	65	5.15×10^{-5}			34
			70	$8.95,6.63 \times 10^{-5}$			34
			75	1.41×10^{-4}			34
	Cyclopropane acetyl peroxide	Carbon tetrachloride	14	9.45×10^{-5}			34
			25	$9.75,10.57 \times 10^{-4}$			34
			44.5	5.01×10^{-5}	101.7		24
			56.5	2.64×10^{-4}			24
			44.5	$6.5\text{-}8.0 \times 10^{-4}$		n	24
	Diacetyladipoyl diperoxide	Styrene	60	6.6×10^{-6}			215
			73.5	4.73×10^{-5}			215
			85	1.84×10^{-4}			215
	Difuroyl peroxide	styrene	50	7.0×10^{-7}	125.7	h	287
			60	2.9×10^{-6}			287
			70	1.03×10^{-5}			287
	2,2,3,3,4,4,5,5-Octafluoropentanoyl peroxide	Freon 113	20	3.29×10^{-5}	100.6	h	283
			30	1.27×10^{-4}			283
			40	5.01×10^{-4}			283
	Perfluoro-2-(2-ethoxysulfinic acid) propionyl peroxide	Freon 113	T (K)	2.8×10^{14}	exp[−95.1/RT]		304
	Pivaloyl peroxide	Carbon tetrachloride	10	1.9×10^{-4}			244
	2-Thenoyl peroxide	Carbon tetrachloride	75	2.21×10^{-5}		a,m_1	232
	3-Thenoyl peroxide	Carbon tetrachloride	75	2.14×10^{-5}		a,m_1	232
11	Benzoyl isobutyryl peroxide	Acetonitrile	41	4.06×10^{-4}			167
		Cyclohexane	41	1.63×10^{-5}			167
			70	3.05×10^{-4}			167
			40	1.45×10^{-5}	112.5	h,j	164
			50	5.398×10^{-5}		j	164
			60	1.924×10^{-4}		j	164
			70	6.872×10^{-4}		j	164
	m-Chlorobenzoyl isobutyryl peroxide	Acetonitrile	41	1.03×10^{-3}			167
		Cyclohexane	40	3.486×10^{-5}	111.3	h,j	164
			50	1.362×10^{-4}		j	164
			41	4.40×10^{-5}			167
	p-Chlorobenzoyl isobutyryl peroxide	Cyclohexane	50	1.029×10^{-4}		j	164
			55	1.651×10^{-4}			164
			60	3.21×10^{-4}			164
			65	5.638×10^{-4}			164
	p-Fluorobenzoyl isobutyryl peroxide	Cyclohexane	55	1.117×10^{-4}			164
			60	2.038×10^{-4}			164
			65	4.893×10^{-4}			164
			70	9.603×10^{-4}			164
	5-Methyl-bis-2-thenoyl peroxide	Carbon tetrachloride	75	2.92×10^{-5}		a	232
	p-Nitrobenzoyl isobutyryl peroxide	Cyclohexane	40	8.1×10^{-5}			164
			45	1.336×10^{-4}			164
			50	2.889×10^{-4}			164
			55	4.725×10^{-4}			164
			60	8.921×10^{-4}			164
12	β-Allyloxypropionyl peroxide	Toluene	70	2.01×10^{-5}		a	191
			80	8.62×10^{-5}		a	191
			90	2.53×10^{-4}		a	191
		p-Xylene	70	2.32×10^{-5}		a	191
			80	8.88×10^{-5}		a	191
			90	2.95×10^{-4}		a	191
	m-Chlorobenzoyl 2-methyl-butanoyl peroxide	Acetic acid	40	1.60×10^{-3}			256
		Acetonitrile	40	1.72×10^{-3}			256

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	*m*-Chlorobenzoyl 2-methyl-butanoyl peroxide	2-Butanol	40	9.0×10^{-4}			256
		Carbon tetrachloride	40	1.32×10^{-4}			256
		Cyclohexane	40	6.0×10^{-5}			256
	Cyclobutane acetyl peroxide	Carbon tetrachloride	65	1.37×10^{-5}			34
			70	$2.13,3.08 \times 10^{-5}$			34
			75	3.83×10^{-5}			34
	Cyclopentane formyl peroxide	Carbon tetrachloride	40	1.50×10^{-5}			34
			45	2.55×10^{-5}			34
			50	4.96×10^{-5}			34
			55	$8.17,7.85 \times 10^{-5}$			34
	Hexanoyl peroxide	Toluene	77	1.186×10^{-4}			209
	5-Hexenoyl peroxide	Toluene	60.1	1.06×10^{-5}	129.3	c	209
			70.4	4.15×10^{-5}		c	209
			76.4	8.59×10^{-5}		c	209
			85.0	2.668×10^{-4}		c	209
	4-Methoxybenzoyl isobutyryl-peroxide	Cyclohexane	55	5.465×10^{-5}			164
			60	1.024×10^{-4}			164
			65	1.876×10^{-4}			164
			70	3.208×10^{-4}			164
	4-Methylbenzoyl isobutyryl-peroxide	Cyclohexane	40	9.61×10^{-6}		j	164
			50	3.619×10^{-5}		j	164
			60	1.305×10^{-4}		j	164
			70	4.772×10^{-4}		j	164
	4-Methyl-2-thenoyl peroxide	Carbon tetrachloride	75	2.92×10^{-5}		a,m_1	232
	5-Methyl-2-thenoyl peroxide	Carbon tetrachloride	75	4.21×10^{-5}		a,m_1	232
	Perfluoro-2-furnanacetyl peroxide	Freon 113	T (K)	1.2×10^{11}	exp[−77.1/RT]		304
	Perfluoro-2-propoxypropionyl-peroxide	Freon 113	T (K)	4.4×10^{14}	exp[−98.5/RT]		304
		perF–ether mix.	T (K)	8.9×10^{16}	exp[−110.2/RT]		304
	Perfluoro-2-*n*-propoxypropionyl-peroxide	Freon 113	20	4.81×10^{-5}	100.1	h	283
			30	1.96×10^{-4}			283
			40	7.16×10^{-4}			283
	Perfluoro-2-iso-propoxypropionyl-peroxide	Freon 113	20	8.32×10^{-5}	102.7	h	283
			30	3.43×10^{-4}			283
			40	1.34×10^{-3}			283
14	2-Azidobenzoyl peroxide	Benzene	50	2.5×10^{-5}		d_2	161
			80	9.4×10^{-4}		d_2	161
	Benzoyl peroxide	Acetic acid	75	7.53×10^{-5}		a,r	74
		Acetone	50	2.25×10^{-6}	111.3	a	126
			70	2.63×10^{-5}		a	126
			85	1.34×10^{-4}		a	126
			100	5.83×10^{-4}		a	126
		Acetonitrile	70	1.76×10^{-5}		t_1	124
		Acetophenone	70	1.15×10^{-5}	126.4	a	5
			80	4.32×10^{-5}		a	5
			94.5	2.30×10^{-4}		a	5
		Allyl alcohol	80	3.80×10^{-4}		a,r	74
		Anisole	30	1.42×10^{-7}		a	109
		Benzaldehyde	80	5.50×10^{-5}		a,r	74
			90	1.71×10^{-4}		a,r	74
		Benzene	25	6.4×10^{-5}		p_2	279
			30	4.80×10^{-8}	116.3	a	109
			55	1.14×10^{-6}		a,r	74
			60	2.76×10^{-6}		a,r	74
			60	2.0×10^{-6}	124.3	m_2	14
			80	2.5×10^{-5}		m_2	14
			70	1.38×10^{-5}		a	213
			78	2.30×10^{-5}		c	230
			78	1.67×10^{-5}		t_9	230
			79.8	3.48×10^{-5}			170
			80	4.8×10^{-5}			20
			66	7.72×10^{-6}	129.7	a	131
			72.5	1.87×10^{-5}		a	131
			78	3.77×10^{-5}		a	131
			70	1.17×10^{-5}	133.9	a	73

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	Benzoyl peroxide	Benzene	75	2.62×10^{-5}		a	73
			80	4.39×10^{-5}		a	73
			80	3.27×10^{-5}		a,b_2	347
			50.8	4.28×10^{-7}	123.8	a,t_2	69
			54.9	8.53×10^{-7}		a,t_2	69
			60.9	1.66×10^{-6}		a,t_2	69
			65.6	3.22×10^{-6}		a,t_2	69
			71.0	5.94×10^{-6}		a,t_2	69
			75.8	1.19×10^{-5}		a,t_2	69
			70	1.27×10^{-5}	130	a	337
			85	8.94×10^{-5}		a	126,337
			100	4.96×10^{-4}		a	126,337
			70	1.03×10^{-5}		a,m_2	16
			70	1.18×10^{-5}		t_1	124
			75	1.48×10^{-5}	128.0		12
			75	1.66×10^{-5}	124.3	m_3	12
			85	4.7×10^{-5}			12
			85	5.5×10^{-5}		m_3	12
			100	2.28×10^{-4}			12
			100	2.56×10^{-4}		m_3	12
			79	2.58×10^{-5}		a,t_6	120
			80	3.35×10^{-5}		a	7
		Benzyl alcohol	80	4.44×10^{-4}		a,r	74
		Bromobenzene	80.2	8.15×10^{-5}			231
				2.19×10^{-5}		t_9	231
				3.84×10^{-5}	m_1		231
				3.55×10^{-5}	m_3		231
				4.34×10^{-5}	m_5		231
		Butanol	80	6.06×10^{-4}		a,r	74
		Butanone	80	4.64×10^{-5}		a,r	74
		Di-*n*-butyl phthalate	117	2.78×10^{-3}	120.1	y_{16}	236
			127	7.44×10^{-3}			236
			137	1.72×10^{-2}			236
			147	3.89×10^{-2}			236
		Carbon tetrachloride	75	1.07×10^{-5}		a,r	74
			79	1.69×10^{-5}		t_6,a	121
		Chlorobenzene	70	1.35×10^{-5}		t_1	124
			80	4.64×10^{-5}		a,r	74
			80.2	2.85×10^{-5}			231
			80.2	2.36×10^{-5}		t_9	231
			80.2	3.52×10^{-5}		c,m_1	231
			80.2	2.62×10^{-5}		c,m_3	231
			71	1.93×10^{-5}			344
			91	1.93×10^{-4}			344
			113	1.93×10^{-3}			344
			T (K)	6.94×10^{13}	exp[$-122.35/RT$]		344
		Chloroform	30	5.47×10^{-8}		a	109
		Cumene	80	3.69×10^{-5}		a,r	74
			85	6.39×10^{-5}		a,r	74
			90	1.19×10^{-4}		a,r	74
			30	7.30×10^{-8}		a	109
			45	1.85×10^{7}	120.5		67
			60	1.45×10^{-6}			67
			80	1.70×10^{-5}			67
			T (K)	1.20×10^{13}	exp[$-120.5/RT$]		67
			100	2.5×10^{-4}		a,t_7	122
		Cyclohexane	80	7.72×10^{-5}		a,r	74
		Decane	80	2.53×10^{-5}			270
		Decalin	80	2.26×10^{-4}		a,r	74
		Dioxane	70	1.30×10^{-5}		t_1	74
			80	6.72×10^{-4}		a,r	74
			80	4.20×10^{-4}		a,r	6
			80	4.18×10^{-5}		a,m_1	232

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	Benzoyl peroxide	Ethanol	25	3.02×10^{-8}		a	351
			40	2.77×10^{-7}		a	351
			50	4.72×10^{-7}		a	351
		Ethylbenzene	30	3.61×10^{-8}		a	109
			75	1.81×10^{-5}		a,r	74
			80	3.33×10^{-5}		a,r	74
			85	5.56×10^{-5}		a,r	74
			90	1.01×10^{-4}		a,r	74
			80	3.15×10^{-5}		c	239
		90% Formic acid	80	6.94×10^{-4}		a,r	74
		n-Heptane	80	3.11×10^{-5}		a,r	74
			80	2.71×10^{-5}			270
			130	5.73×10^{-3}		z (1500)	377
			135	1.02×10^{-2}		z (1500)	377
			140	1.54×10^{-2}		z (1500)	377
			145	2.25×10^{-2}		z (500)	377
			145	2.22×10^{-2}		z (1000)	377
			145	2.21×10^{-2}		z (1500)	377
			145	2.12×10^{-2}		z (2000)	377
			150	3.47×10^{-2}		z (1500)	377
			155	5.21×10^{-2}		z (1500)	377
			160	7.87×10^{-2}		z (1500)	377
		Hexadecane	80	2.51×10^{-5}			270
		Hexane	80	2.85×10^{-5}			270
		Isooctane	80	2.79×10^{-5}			270
			80	2.70×10^{-5}		m_2	308
		Isopropylbenzene	80	3.34×10^{-5}		c	229
		Methyl acetate	49.2	6.28×10^{-7}	123.8	a,t_2	69
			53.9	1.0×10^{-6}		a,t_2	69
		Methylcyclohexane	80	5.25×10^{-5}		a,r	74
		4-Methyl-2pentanone	80	4.28×10^{-5}		a,r	74
		α-Methylstyrene	70	3.02×10^{-5}		a	213
		Mineral oil	80	2.89×10^{-5}		m_2	308
		Neat	105	7.22×10^{-3}		a	371
		Nitrobenzene	80	4.58×10^{-5}		a,r	74
			30	6.61×10^{-8}	117.6	a	109
		PEG 400	25	5.01×10^{-7}		a	351
			40	5.73×10^{-6}		a	351
			50	1.15×10^{-5}		a	351
			70	1.15×10^{-4}		a	351
		90% aq. PEG 400	25	3.65×10^{-7}		a	351
			40	5.35×10^{-6}		a	351
			50	1.15×10^{-5}		a	351
			70	1.15×10^{-4}		a	351
		80% aq. PEG 400	25	2.77×10^{-7}		a	351
			40	3.09×10^{-6}		a	351
			50	7.29×10^{-6}		a	351
			70	1.15×10^{-4}		a	352
		70% aq. PEG 400	25	1.71×10^{-7}		a	351
			40	2.11×10^{-6}		a	351
			50	5.01×10^{-6}		a	351
			70	1.15×10^{-4}		a	351
		n-Pentanol	80	1.48×10^{-4}		a,r	74
		Phenol	80	6.25×10^{-4}		a,r	74
		Propionic acid	80	3.19×10^{-5}		a,r	74
		Propylene glycol	25	3.25×10^{-8}		a	351
			40	1.51×10^{-7}		a	351
			50	3.16×10^{-7}		a	351
		Styrene	34.8	3.89×10^{-8}		a	10
			49.4	5.28×10^{-7}		a	10
			61.0	2.58×10^{-6}		a	10
			74.8	1.83×10^{-5}		a	10
			100.0	4.58×10^{-4}		a	10

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	Benzoyl peroxide	Poly(styrene)	56.4	3.8×10^{-7}		a	17
			64.6	1.47×10^{-6}		a	17
			76.7	9.27×10^{-6}		a	17
			83.4	2.50×10^{-5}		a	17
			98.5	1.41×10^{-4}		a	17
			70.9	2.86×10^{-6}		a	15
			80.1	1.11×10^{-5}		a	15
			89.5	3.33×10^{-5}		a	15
		Tetradecane	80	2.64×10^{-5}			270
		Tetralin	80	3.72×10^{-5}		a,r	74
		Toluene	30	4.94×10^{-8}	120.5	a	109
			49.0	6.0×10^{-7}	123.8	a,t_2	69
			55.1	1.31×10^{-6}		a,t_2	69
			60.2	2.83×10^{-6}		a,t_2	69
			65.1	5.69×10^{-6}		a,t_2	69
			70.3	1.10×10^{-5}		a,t_2	69
			60.00	2.24×10^{-6}		e,m_3	291
		Poly(vinylchloride)	64.6	6.3×10^{-7}		a	17
			76.7	5.11×10^{-6}		a	17
			83.4	1.44×10^{-5}		a	17
			98.5	9.33×10^{-5}		a	17
		Styrene	50	7×10^{-7}	125.7		287
			60	7.7×10^{-6}			287
			70	9.9×10^{-6}			287
		p-Xylene	80	3.10×10^{-5}		c	229
	3-Bromobenzoyl peroxide	Benzene	60	1.1×10^{-6}			116
			80	1.22×10^{-5}			116
			80	2.60×10^{-5}		a	7
		Dioxane	80	2.57×10^{-5}		a,m_1	6,232
	4-Bromobenzoyl peroxide	Dioxane	80	3.23×10^{-5}		a,m_1	6,232
	4-*tert*-Butylbenzoyl peroxide	Dioxane	80	6.06×10^{-5}		a,m	232
	2-Chlorobenzoyl peroxide	Acetophenone	80	3.88×10^{-4}	123.0	a	5
		Benzene	80	3.12×10^{-4}		a	7
	3-Chlorobenzoyl peroxide	Acetophenone	80	2.85×10^{-5}	128.4	a	5
		Dioxane	80	2.63×10^{-5}		a,m_1	6
	4-Chlorobenzoyl peroxide	Acetophenone	80	3.83×10^{-5}	127.2	a	5
		Benzene	80	2.17×10^{-5}		a	7
			50	6.2×10^{-7}	128.9	a	126
			85	6.64×10^{-5}		a	126
			100	3.86×10^{-4}		a	126
		Dioxane	80	3.62×10^{-5}		a,m_1	6
		Styrene	34.8	8.3×10^{-8}		a	10
			49.4	8.3×10^{-7}		a	10
			61.0	3.33×10^{-6}		a	10
			74.8	2.22×10^{-5}		a	10
			100.0	4.17×10^{-4}		a	10
	Cyclohexane formyl peroxide	Benzene	30	9.64×10^{-5}	84.5	a,h	219
			35	1.46×10^{-4}			219
			40	3.10×10^{-4}			219
			45	5.11×10^{-4}			219
			50	7.77×10^{-4}			219
		Carbon tetrachloride	35	6.6×10^{-5}			148
			45	2.11×10^{-4}			148
			50	4.45×10^{-4}			148
			60	1.30×10^{-3}			148
			35	2.87×10^{-5}			34
			40	$5.22,5.29 \times 10^{-5}$			34
			45	9.67×10^{-5}			34
	Cyclopentane acetyl peroxide	Carbon tetrachloride	65	1.48×10^{-5}			34
			70	3.20×10^{-5}			34
			75	4.97×10^{-5}			34
	Diacetylsebacoyl diperoxide	Carbon tetrachloride	60	1.04×10^{-5}		x_1	215
			73.5	5.20×10^{-5}		x_1	215
			85	2.30×10^{-4}		x_1	215

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	2,4-Dichlorobenzoyl peroxide	Benzene	70	9.70×10^{-5}		a,j	16
			70	1.24×10^{-4}		a,k	16
			50	1.08×10^{-5}	117.6	a	126,337
			70	1.37×10^{-4}		a	126,337
			85	7.69×10^{-4}		a	126,337
		Styrene	34.8	3.88×10^{-6}		a	10
			49.4	2.39×10^{-5}		a	10
			61.0	7.78×10^{-5}		a	10
			74.8	2.78×10^{-4}		a	10
			100.0	4.17×10^{-3}		a	10
	2,2,3,3,4,4,5,5,6,6,7,7-Dodeca-fluoroheptanoyl peroxide	Freon 113	30	1.14×10^{-4}	100.6	h	283
			40	4.06×10^{-4}			283
	Heptanoyl peroxide	Toluene	77	1.24×10^{-4}		a	19
	6-Heptenoyl peroxide	Toluene	70	5.33×10^{-5}		a,k	19
			70	5.01×10^{-5}		a,j	19
			77	1.07×10^{-4}		a,j	19
			85	2.88×10^{-4}		a,k	19
	2-Iodobenzoyl peroxide	Chloroform	22	1.86×10^{-3}			82
	2-Iodobenzoyl 4-nitrobenzoyl peroxide	Acetone	25	3.0×10^{-4}			78
		Acetonitrile	25	2.1×10^{-4}			78
		Benzene	25	5.7×10^{-5}			78
		Carbon tetrachloride	25	3.4×10^{-5}			78
		Chloroform	25	2.8×10^{-4}			78
		Nitrobenzene	25	6.2×10^{-4}			78
	3-Methylbenzoyl peroxide	Dioxane	80	4.38×10^{-5}		a,m_1	232
	4-Methylbenzoyl peroxide	Dioxane	80	6.11×10^{-5}		a,m_1	232
	2-Nitrobenzoyl peroxide	Acetophenone	59.3	5.80×10^{-5}	119.7	a	5
			80	1.34×10^{-3}		a	5
		Methyl iodide	24.95	1.78×10^{-5}	81.2	h	82
			45.05	1.50×10^{-4}			82
	3-Nitrobenzoyl peroxide	Acetophenone	80	3.80×10^{-5}	126.4	a	5
	4-Nitrobenzoyl peroxide	Acetophenone	80	4.33×10^{-5}	126.8	a	5
	3,5-Dinitrobenzoyl peroxide	Acetophenone	80	1.87×10^{-5}	130.5	a	5
	Perfluoroheptanoyl peroxide	Freon 113	20	3.01×10^{-5}	98.0		283
			30	1.11×10^{-4}			283
			40	4.20×10^{-4}			283
15	Benzoyl phenylacetyl peroxide	Benzene	20	1.10×10^{-4}	90.8		222
			25	2.05×10^{-4}			222
			25	3.15×10^{-4}		v_1	222
			25	6.38×10^{-4}		n	222
			30	3.88×10^{-4}			222
			35	6.67×10^{-4}			222
	4-*tert*-Butylbenzoyl isobutyryl peroxide	Cyclohexane	55	6.543×10^{-5}		t_{10}	164
			60	1.171×10^{-4}		t_{10}	164
			65	2.663×10^{-4}		t_{10}	164
			70	5.016×10^{-4}		t_{10}	164
	3-Cyanobenzoyl benzoyl peroxide	Dioxane	80	2.73×10^{-2}		a,m_1	6
	3-Methoxybenzoyl benzoyl peroxide	Dioxane	80	4.82×10^{-2}		a,m_1	6
	4-Methoxybenzoyl benzoyl peroxide	Dioxane	80	7.57×10^{-5}		a,m_1	6
	4-Methoxybenzoyl 3-bromobenzoyl peroxide	Dioxane	80	4.43×10^{-2}		a,m_1	6
	4-Methoxybenzoyl-3,5-dinitro-benzoyl peroxide	Benzene	51	1.02×10^{-5}			108
		Nitrobenzene	51	9.61×10^{-4}			108
			51	1.67×10^{-3}		t_5	108
	4-Methoxybenzoyl 4-nitrobenzoyl peroxide	Benzene	70	2.08×10^{-5}		a	89
			70	8.03×10^{-5}		a	89
16	3,5-Dibromo-4-methoxybenzoyl peroxide	Benzene	60	9.5×10^{-7}			95
			60	6.1×10^{-7}			116
			80	9.4×10^{-6}			116
	Caprylyl peroxide	Benzene	50	3.44×10^{-6}	128.9	a	126
			70	5.78×10^{-5}		a	126
			85	3.78×10^{-4}		a	126

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	Caprylyl peroxide	Mineral oil	T (K)	9.8×10^{15}	exp[− 140.1/RT]		1
	p-(Chloromethyl)benzoyl peroxide	Acetone	70	2.07×10^{-5}			307
			70	1.82×10^{-5}		m_7	307
		THF	70	1.15×10^{-5}			307
		Toluene	70	8.8×10^{-6}			307
	3-Cyanobenzoyl peroxide	Dioxane	80	1.70×10^{-2}		a,m_1	6
	4-Cyanobenzoyl peroxide	Acetophenone	80	2.43×10^{-5}		a	5
		Dioxane	80	2.03×10^{-5}		a,m_1	6
	Cycloheptane formyl peroxide	Carbon tetrachloride	35	7.85×10^{-5}			34
			40	$1.63,1.34 \times 10^{-5}$			34
			45	2.02×10^{-4}			34
	Cyclohexane acetyl peroxide	Carbon tetrachloride	65	1.27×10^{-5}			34
			70	2.76×10^{-5}			34
			75	3.61×10^{-5}			34
			54.4	3.1×10^{-6}			24
			64.3	1.19×10^{-5}			24
			71.8	2.95×10^{-5}			24
	2-Ethyl-4-methyl-2-pentenoyl peroxide	Mineral oil	T (K)	7.1×10^{16}	exp[− 138.4/RT]		1
	2-Ethylhexanoyl peroxide	Mineral oil	T (K)	1.2×10^{14}	exp[− 106.4/RT]		1
	2-Ethyl-2-hexenoyl peroxide	Mineral oil	T (K)	1.6×10^{16}	exp[− 136.3/RT]		1
	2-Iodophenylacetyl peroxide	Acetone	0	2.60×10^{-5}			79
		Chloroform	0	3.98×10^{-5}		c,e,q	79
		Toluene	0	1.3×10^{-5}		q	79
	2-Methoxybenzoyl peroxide	Acetophenone	50	6.0×10^{-5}	113.8	a	5
			80	2.15×10^{-3}		a	5
	3-Methoxybenzoyl peroxide	Acetophenone	80	6.42×10^{-5}	120.9	a	5
		Dioxane	80	5.75×10^{-5}		a,m_1	6
	4-Methoxybenzoyl peroxide	Acetophenone	80	1.56×10^{-4}	120.1	a	5
		Dioxane	80	1.18×10^{-4}		a,m_1	6
	2-Methylbenzoyl peroxide	Acetophenone	70	9.02×10^{-5}	126.4	a	5
	3-Methylbenzoyl peroxide	Acetophenone	80	4.70×10^{-5}	126.4	a	5
		Dioxane	80	4.40×10^{-5}		a,m_1	6
	4-Methylbenzoyl peroxide	Acetophenone	80	5.92×10^{-5}	125.1	a	5
		Dioxane	80	6.13×10^{-5}		a,m_1	6
	Endo-norbornane-2-carbonyl-peroxide	Carbon tetrachloride	44.5	6.1×10^{-6}		j	23
			53.9	2.83×10^{-5}		j	23
			65.9	1.25×10^{-4}		j	23
			44.5	9.1×10^{-6}		j,m_2	23
			53.9	4.33×10^{-5}		j, m_2	23
			65.9	1.28×10^{-4}		j,m_2	23
	Exo-norbornane-2-carbonyl-peroxide	Carbon tetrachloride	44.5	4.68×10^{-5}		j	23
			53.9	2.05×10^{-4}		j	23
			65.9	8.18×10^{-4}		j	23
			44.5	7.2×10^{-5}		j, m_2	23
			53.9	1.60×10^{-4}		j,m_2	23
			65.9	8.48×10^{-4}		j, m_2	23
	Endo-norbornene-5-carbonyl-peroxide	Carbon tetrachloride	44.5	6.30×10^{-5}		j	23
			53.9	1.21×10^{-4}		j	23
			65.9	7.18×10^{-4}		j	23
			445	2.22×10^{-5}		j,m_2	23
			53.9	4.52×10^{-5}		j,m_2	23
			65.9	2.37×10^{-4}		j,m_2	23
	Exo-norbornene-5-carbonyl-peroxide	Carbon tetrachloride	44.5	6.58×10^{-5}		j	23
			53.9	1.21×10^{-4}		j	23
			65.9	8.42×10^{-4}		j	23
			44.5	2.58×10^{-5}		j,m_2	23
			53.9	1.20×10^{-4}		j,m_2	23
			65.9	7.10×10^{-4}		j,m_2	23
	Dioctanoyl peroxide	*n*-Heptane	70	1.70×10^{-5}		z (1500)	377
			80	7.01×10^{-5}		z (500)	377
			80	6.80×10^{-5}		z (1000)	377

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	Dioctanoyl peroxide	*n*-Heptane	80	6.27×10^{-5}		z (1500)	377
			80	6.01×10^{-5}		z (2000)	377
			80	5.77×10^{-5}		z (2500)	377
			90	2.13×10^{-4}		z (1500)	377
			95	3.80×10^{-4}		z (1500)	377
			100	6.79×10^{-4}		z (1500)	377
	Dioctanoyl peroxydicarbonate	Chlorobenzene	63	1.93×10^{-5}			344
			80	1.93×10^{-4}			344
			100	1.93×10^{-3}			344
			T (K)	2.30×10^{15}	exp[−129.05/RT]		344
	Phenylacetyl peroxide	Acetonitrile	20	1.02×10^{-3}			167
		Benzene	20	4.76×10^{-4}			167
		Carbon tetrachloride	20	1.86×10^{-4}			167
		Cyclohexane	20	1.60×10^{-4}			167
		Styrene	0	7.8×10^{-5}			316
		Toluene	0	2.50×10^{-5}	96.2	f,u_1	59
			18	3.34×10^{-4}		f,u_1	59
	Triptoyl peroxide	Benzene	80	1.42×10^{-4}			102
			80	1.4×10^{-4}			244
18	Apocamphoyl peroxide	Benzene	80	2.3×10^{-4}			102
	cis-5-*tert*-Butylcyclohexylformyl-*m*-chlorobenzoyl peroxide	Acetonitrile	40	5.9×10^{-3}			256
		2-Butanol	40	4.4×10^{-3}			256
		Carbon tetrachloride	40	6.7×10^{-4}			256
		Cyclohexane	40	4.4×10^{-4}			256
	trans-4-*tert*-Butylcyclohexylformyl-*m*-chlorobenzoyl peroxide	Acetonitrile	40	4.2×10^{-3}			256
		2-Butanol	40	2.8×10^{-3}			256
		Carbon tetrachloride	40	6.5×10^{-4}			256
		Cyclohexane	40	3.9×10^{-4}			256
	5-*tert*-Butylthenoyl peroxide	Carbon tetrachloride	75	4.03×10^{-5}		a,m_1	232
	Cinnamoyl peroxide	Styrene	50	1.20×10^{-5}			316
			70	1.67×10^{-4}			316
	Dibenzoyl succinoyl diperoxide	Unknown	70	7.7×10^{-6}	125.5	x_1	214
			75	1.42×10^{-5}		x_1	214
			85	5.4×10^{-5}		x_1	214
	Nonanoyl peroxide	Mineral oil	T (K)	8.4×10^{14}	exp[−127.1/RT]		1
	Isononanoyl peroxide (3,5,5-trimethylhexanoyl peroxide)	Benzene	50	6.1×10^{-6}	126	a	337
			70	8.02×10^{-5}		a	337
			85	5.30×10^{-4}		a	337
			60	1.9×10^{-5}		a	338
		Chlorobenzene	59	1.93×10^{-5}			344
			77	1.93×10^{-4}			344
			96	1.93×10^{-3}			344
			T (K)	2.84×10^{15}	exp[−128.34/RT]		344
		n-Heptane	60	5.28×10^{-6}		z (1500)	377
			70	2.06×10^{-5}		z (1500)	377
			80	9.53×10^{-5}		z (1)	377
			80	9.10×10^{-5}		z (500)	377
			80	8.62×10^{-5}		z (1000)	377
			80	8.24×10^{-5}		z (1500)	377
			80	7.86×10^{-5}		z (2000)	377
			80	7.46×10^{-5}		z (2500)	377
			90	2.52×10^{-4}		z (1500)	377
			100	8.81×10^{-4}		z (1500)	377
			120	6.98×10^{-3}		z (1500)	377
			125	1.27×10^{-2}		z (1500)	377
			130	2.15×10^{-2}		z (1500)	377
			135	2.60×10^{-2}		z (1500)	377
			140	5.71×10^{-2}		z (1500)	377
			145	8.62×10^{-2}		z (1500)	377
			150	1.17×10^{-1}		z (1500)	377
	2-Nonenoyl peroxide	Mineral oil	T (K)	1.6×10^{15}	exp[−128.8/RT]		1

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
18 (*cont'd*)	3-Nonenoyl peroxide	Mineral oil	T (K)	3.7×10^{14}	exp[− 108.0/RT]		1
	2-Phenylpropionyl peroxide	Cyclohexane	55.8	6.8×10^{-6}			258
		On silica	49.4	7.5×10^{-5}			258
		Styrene	70	5.5×10^{-5}			316
19	Dibenzoyl itaconyl diperoxide	Unknown	70	5.63×10^{-4}	83.7	x_2	214
				1.87×10^{-5}	115.1	x_2	214
			75	8.63×10^{-4}		x_2	214
				3.62×10^{-5}		x_2	214
			85	1.69×10^{-3}		x_2	214
				1.00×10^{-4}		x_2	214
	Dibenzoyl α-methylsuccinoyl-diperoxide	Unknown	70	2.74×10^{-4}	82.8	x_2	214
				1.26×10^{-5}	116.3	x_2	214
			75	4.26×10^{-4}		x_2	214
				2.40×10^{-5}		x_2	214
			85	9.60×10^{-4}		x_2	214
				8.7×10^{-5}		x_2	214
20	Decanoyl peroxide	Benzene	60	1.53×10^{-5}	127.2	a	126,337
			70	5.67×10^{-5}		a	126,337
			85	3.80×10^{-4}		a	126,337
		Chlorobenzene	63	1.93×10^{-5}			344
			80	1.93×10^{-4}			344
			100	1.93×10^{-3}			344
			T (K)	3.64×10^{15}	exp[− 130.48/RT]		344
		Mineral oil	T (K)	2.7×10^{15}	exp[− 131.9/RT]		1
	Dioctanoyl α-bromosuccinoyl-diperoxide	Unknown	70	1.52×10^{-4}	96.2	x_2	214
				1.82×10^{-5}	126.4	x_2	214
			75	2.74×10^{-4}		x_2	214
				3.83×10^{-5}		x_2	214
			85	6.97×10^{-4}		x_2	214
				1.15×10^{-4}		x_2	214
	Dioctanoyl α-chlorosuccinoyl-diperoxide	Unknown	70	2.02×10^{-4}	95.4	x_2	214
				2.42×10^{-5}	126.4	x_2	214
			75	3.23×10^{-4}		x_2	214
				4.83×10^{-5}		x_2	214
			85	8.50×10^{-4}		x_2	214
	Dioctanoyl α-chlorosuccinoyl-diperoxide	Unknown	85	1.58×10^{-4}		x_2	214
	4-Ethyl-2-octenoyl peroxide	Mineral oil	T (K)	8.2×10^{14}	exp[− 127.4/RT]		1
21	Dioctanoyl itaconoyl diperoxide	Unknown	70	3.23×10^{-4}	92.9	x_2	214
				2.80×10^{-5}	127.6	x_2	214
			75	5.47×10^{-4}		x_2	214
				5.60×10^{-5}		x_2	214
			85	1.28×10^{-3}		x_2	214
				1.92×10^{-4}		x_2	214
	Dioctanoyl α-methyl succinoyl-diperoxide	Unknown	70	5.48×10^{-4}	96.2	x_2	214
				5.88×10^{-5}	131.0	x_2	214
			75	9.58×10^{-4}		x_2	214
				1.06×10^{-4}		x_2	214
			85	2.61×10^{-3}		x_2	214
				3.84×10^{-4}		x_2	214
22	Benzoyl 2-[*trans*-2-(3-nitrophenyl)-vinyl]benzoyl peroxide	Chlorobenzene	70	1.87×10^{-4}		t_9	175
	Benzoyl 2-[*trans*-2-(4-nitrophenyl)-vinyl]benzoyl peroxide	Chlorobenzene	70	1.18×10^{-4}		t_9	175
		THF	70	6.2×10^{-5}		t_9	175
	Benzoyl 2-[*trans*-2-(4-nitrophenyl)-vinyl]-4-nitrobenzoyl peroxide	Chlorobenzene	70	1.54×10^{-5}		t_9	175
	Benzoyl 2-[*trans*-2-(phenyl)vinyl]-benzoyl peroxide	Chlorobenzene	35	9.42×10^{-5}	84.5	h,t_9	175
			70	3.02×10^{-3}		t_9	175
		Methanol	35	9.77×10^{-4}		t_9	175
	4-Benzylidenebutyryl peroxide	Acetophenone	50	7.92×10^{-5}	99.6		133
			55	1.45×10^{-4}			133
			65	4.13×10^{-4}			133
			T (K)	1.07×10^{12}	exp[− 99.6/RT]		133
			55	1.38×10^{-4}	98.7	m_2	133

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
22 (*cont'd*)	4-Benzylidenebutyryl peroxide	Acetophenone	65	4.02×10^{-4}		m_2	133
			T (K)	7.14×10^{11}	exp[−98.7/RT]	m_2	133
		Benzene	50	2.40×10^{-5}	91.2	h	194
			60	6.80×10^{-5}			194
			70	1.845×10^{-4}			194
		Carbon tetrachloride	60	3.47×10^{-5}	112.5	m_2	133
			70	1.06×10^{-4}		m_2	133
			T (K)	1.42×10^{13}	exp[−112.5/RT]	m_2	133
		Nitrobenzene	60	3.38×10^{-4}			133
		Propylene carbonate	40	7.36×10^{-5}	89.5		133
			50	2.13×10^{-4}			133
			T (K)	5.93×10^{10}	exp[−89.5/RT]		133
			50	2.08×10^{-4}		m_2	133
			40	7.23×10^{-5}	82.0	h	194
			50	2.093×10^{-4}			194
			60	5.117×10^{-4}			194
		Toluene	70	1.64×10^{-4}			133
	4-*tert*-Butylbenzoyl peroxide	Dioxane	80	6.08×10^{-2}		a,m_1	6
	cis-4-*tert*-Butylcyclohexane formyl-peroxide	Butane	40	1.25×10^{-5}		z (0.98)	145
			40	1.69×10^{-5}		z (1010)	145
		Carbon tetrachloride	40.0	$(8.65 \pm 0.35) \times 10^{-5}$	76.1	h	35
			45.45	$(1.32 \pm 0.04) \times 10^{-4}$			35
			50.7	$(2.35 \pm 0.05) \times 10^{-4}$			35
	trans-4-*tert*-Butylcyclohexane formyl-peroxide	Carbon tetrachloride	40.0	$(4.25 \pm 0.23) \times 10^{-5}$	81.6	h	35
			44.7	$(7.10 \pm 0.38) \times 10^{-5}$			35
			48.9	$(1.14 \pm 0.04) \times 10^{-4}$			35
	trans-4-(4-Chlorobenzylidene)-butyryl peroxide	Benzene	50	2.10×10^{-5}	95.0	h	194
			60	6.56×10^{-5}			194
			70	1.75×10^{-4}			194
		Propylene carbonate	40	5.33×10^{-5}			194
			50	1.448×10^{-4}			194
			60	2.791×10^{-4}			194
	trans-4-(4-Fluorobenzylidene)-butyryl peroxide	Benzene	50	2.67×10^{-5}	94.1	h	194
			60	8.44×10^{-5}			194
			70	2.119×10^{-4}			194
		Propylene carbonate	40	6.53×10^{-5}			194
			50	1.685×10^{-4}			194
			60	4.540×10^{-4}			194
	1-Naphthoyl peroxide	Benzene	54.6	1.01×10^{-4}			170
			59.9	1.86×10^{-4}			170
			64.5	3.0×10^{-4}			170
	4-Nitrobenzoyl-2-[*trans*-2-(4-nitro-phenyl)vinyl] benzoyl peroxide	Chlorobenzene	70	6.06×10^{-5}		t_9	175
	2-Phenylisovaleryl peroxide	Acetonitrile	55.0	2.27×10^{-5}			258
		Cyclohexane	55.0	6.4×10^{-6}			258
		On silica	50	2.56×10^{-4}		n_2	258
	5-Phenylpenta-2,4-dienoyl-peroxide	Styrene	70	2.18×10^{-4}			316
	5-Phenylpentanoyl peroxide	Acetophenone	77	2.37×10^{-4}			133
		Benzene	77	1.054×10^{-5}	127.6	h,m_2	194
		Carbon tetrachloride	70	2.76×10^{-5}	130.1	m_2	133
			77	7.19×10^{-5}		m_2	133
			85	1.87×10^{-4}		m_2	133
			T (K)	1.76×10^{15}	exp[−130.1/RT]	m_2	133
		Propylene carbonate	60	4.41×10^{-5}			133
			60	2.80×10^{-5}		m_2	133
24	Dibenzoyl 2-bromosebacoyl-diperoxide	Unknown	70	4.80×10^{-4}	87.9	x_2	214
				3.87×10^{-5}	110.8	x_2	214
			75	7.64×10^{-4}		x_2	214
				7.22×10^{-5}		x_2	214
			85	2.00×10^{-3}		x_2	214
				2.02×10^{-4}		x_2	214
	Dioctanoyl 2-bromosebacoyl-diperoxide	Unknown	70	6.46×10^{-4}	99.6	x_2	214
				5.23×10^{-5}	129.7	x_2	214
			75	1.171×10^{-3}		x_2	214

Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
24 (*cont'd*)	Dioctanoyl 2-bromosebacoyl-diperoxide	Unknown		1.08×10^{-4}		x_2	214
			85	3.00×10^{-3}		x_2	214
				3.72×10^{-4}		x_2	214
	Lauroyl peroxide	Benzene	30	2.56×10^{-7}		a	99
			40	4.91×10^{-7}			68
			50	2.19×10^{-6}			68
			60	9.17×10^{-6}			68
			70	2.86×10^{-5}			68
			60	1.51×10^{-5}	127.2	a	126,337
			70	5.58×10^{-5}		a	126,337
			85	3.75×10^{-4}		a	126,337
			70	4.33×10^{-5}		a,j	16
		Carbon tetrachloride	40	2.91×10^{-7}			68
			50	1.15×10^{-6}			68
			60	4.75×10^{-6}			68
			70	1.87×10^{-5}			68
		Chlorobenzene	61	1.93×10^{-5}			344
			79	1.93×10^{-4}			344
			99	1.93×10^{-3}			344
			T (K)	3.92×10^{14}	exp[− 123.37/RT]		344
		Ethyl acetate	40	6.03×10^{-7}			68
			50	2.70×10^{-6}			68
			61	1.05×10^{-5}			68
			70	3.99×10^{-5}			68
		Ethyl ether	30	1.97×10^{-6}		a	99
		Mineral oil	T (K)	2.2×10^{16}	exp[− 137.9/RT]		1
		Styrene	34.8	2.06×10^{-7}		a	10
			49.4	2.25×10^{-6}		a	10
			61.0	1.42×10^{-5}		a	10
			74.8	1.00×10^{-4}		a	10
			100.0	2.39×10^{-3}		a	10
			50	3.1×10^{-6}	121.1		287
			60	1.20×10^{-5}			287
			65	2.22×10^{-5}			287
			70	4.32×10^{-5}			287
	trans-4-(4-Methoxybenzylidene)-butyryl peroxide	Benzene	50	2.03×10^{-4}			194
	trans-4-(4-Methylbenzylidene)-butyryl peroxide	Benzene	50	6.58×10^{-5}	90.4	c,h	194
			60	1.815×10^{-4}		c	194
			70	4.729×10^{-4}		c	194
26	2-Phenoxybenzoyl peroxide	Acetophenone	65	8.18×10^{-5}	121.3	a	5
28	Myristoyl peroxide	Benzene	70	3.38×10^{-5}		a	16
36	Menthylphthaloyl peroxide	Dioxane	55	1.15×10^{-4}			65
x	Aliphatic polymeric diacyl peroxide	Chlorobenzene	75	9.04×10^{-5}		v_{14}	302
			75	9.00×10^{-5}		v_{15}	302
			80	1.720×10^{-4}		v_{15}	302

TABLE 5. HYDROPEROXIDES AND KETONE PEROXIDES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4	*sec*-Butyl hydroperoxide	Toluene	172.0	2.65×10^{-5}			199
			182.3	4.9×10^{-5}			199
	tert-Butyl hydroperoxide	Benzene	130	3×10^{-7}	138	a	337
			145	1.6×10^{-6}		a	337
			160	6.6×10^{-6}		a	337
			154.5	4.29×10^{-6}	170.7	h	76
			161.7	9.27×10^{-6}			76
			169.3	2.0×10^{-5}			76
			174.6	4.0×10^{-5}			76

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Table 4. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
24 (*cont'd*)	Dioctanoyl 2-bromosebacoyl-diperoxide	Unknown		1.08×10^{-4}		x_2	214
			85	3.00×10^{-3}		x_2	214
				3.72×10^{-4}		x_2	214
	Lauroyl peroxide	Benzene	30	2.56×10^{-7}		a	99
			40	4.91×10^{-7}			68
			50	2.19×10^{-6}			68
			60	9.17×10^{-6}			68
			70	2.86×10^{-5}			68
			60	1.51×10^{-5}	127.2	a	126,337
			70	5.58×10^{-5}		a	126,337
			85	3.75×10^{-4}		a	126,337
			70	4.33×10^{-5}		a,j	16
		Carbon tetrachloride	40	2.91×10^{-7}			68
			50	1.15×10^{-6}			68
			60	4.75×10^{-6}			68
			70	1.87×10^{-5}			68
		Chlorobenzene	61	1.93×10^{-5}			344
			79	1.93×10^{-4}			344
			99	1.93×10^{-3}			344
			T (K)	3.92×10^{14}	exp[− 123.37/RT]		344
		Ethyl acetate	40	6.03×10^{-7}			68
			50	2.70×10^{-6}			68
			61	1.05×10^{-5}			68
			70	3.99×10^{-5}			68
		Ethyl ether	30	1.97×10^{-6}		a	99
		Mineral oil	T (K)	2.2×10^{16}	exp[− 137.9/RT]		1
		Styrene	34.8	2.06×10^{-7}		a	10
			49.4	2.25×10^{-6}		a	10
			61.0	1.42×10^{-5}		a	10
			74.8	1.00×10^{-4}		a	10
			100.0	2.39×10^{-3}		a	10
			50	3.1×10^{-6}	121.1		287
			60	1.20×10^{-5}			287
			65	2.22×10^{-5}			287
			70	4.32×10^{-5}			287
	trans-4-(4-Methoxybenzylidene)-butyryl peroxide	Benzene	50	2.03×10^{-4}			194
	trans-4-(4-Methylbenzylidene)-butyryl peroxide	Benzene	50	6.58×10^{-5}	90.4	c,h	194
			60	1.815×10^{-4}		c	194
			70	4.729×10^{-4}		c	194
26	2-Phenoxybenzoyl peroxide	Acetophenone	65	8.18×10^{-5}	121.3	a	5
28	Myristoyl peroxide	Benzene	70	3.38×10^{-5}		a	16
36	Menthylphthaloyl peroxide	Dioxane	55	1.15×10^{-4}			65
x	Aliphatic polymeric diacyl peroxide	Chlorobenzene	75	9.04×10^{-5}		v_{14}	302
			75	9.00×10^{-5}		v_{15}	302
			80	1.720×10^{-4}		v_{15}	302

TABLE 5. HYDROPEROXIDES AND KETONE PEROXIDES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4	*sec*-Butyl hydroperoxide	Toluene	172.0	2.65×10^{-5}			199
			182.3	4.9×10^{-5}			199
	tert-Butyl hydroperoxide	Benzene	130	3×10^{-7}	138	a	337
			145	1.6×10^{-6}		a	337
			160	6.6×10^{-6}		a	337
			154.5	4.29×10^{-6}	170.7	h	76
			161.7	9.27×10^{-6}			76
			169.3	2.0×10^{-5}			76
			174.6	4.0×10^{-5}			76

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TABLE 5. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4 (*cont'd*)	*tert*-Butyl hydroperoxide	Benzene	172.3	1.09×10^{-5}			199
			182.6	3.1×10^{-5}			199
		Chlorobenzene	171	1.93×10^{-5}			344
			199	1.93×10^{-4}			344
			231	1.93×10^{-3}			344
		Cumene	182.6	8.1×10^{-5}		c	199
		Cyclohexane	100	1.2×10^{-7}			199
			172	1.41×10^{-4}		c	199
		Dodecane	86.1	1.32×10^{-6}	128.4	b	41
			98.5	5.55×10^{-6}		b	41
		Heptane	172	1.41×10^{-4}			199
		n-Octane	149.8	8×10^{-6}	163.2	f	118
			159.9	2.5×10^{-5}		f	118
			169.6	6.9×10^{-5}		f	118
			179.6	1.82×10^{-5}			118
		Toluene	100	5.7×10^{-8}			199
			172.5	9.2×10^{-6}			199
			181.5	2.69×10^{-5}			199
			192.6	8.3×10^{-5}			199
			204.5	1.52×10^{-4}		c	199
			214.9	3.24×10^{-4}			199
		Vapor	570	3.4×10^{1}			188
			670	4.95			188
			773	8.4×10^{1}			188
			873	5.66×10^{2}			188
			973	2.58×10^{3}			188
	tert-Butyl hydroperoxide/cobalt 2-Ethyl hexanoate	Chlorohenzene	0	1.01×10^{-3}			190
			0	8.29×10^{-5}		v_{10}	190
			0	5.40×10^{-4}		v_7	190
			25	2.3×10^{-3}			198
			25	1.1×10^{-3}		v_1	198
	tert-Butyl hydroperoxide/cobalt stearate		45	1.4×10^{-3}			198
			45	1.0×10^{-3}		v_1	198
	Methyl ethyl ketone peroxides	Benzene	100	1.3×10^{-5}	101	a	337
			115	4.80×10^{-5}		a	337
			130	1.60×10^{-4}		a	337
		Ethyl acetate	70	1.28×10^{-6}			13
5	*tert*-Amyl hydroperoxide	Chlorobenzene	153	1.93×10^{-5}			344
			190	1.93×10^{-4}			344
			228	1.93×10^{-3}			344
6	Acetone cyclic diperoxide	Benzene	135.5	6.25×10^{-7}	147.1	h	286
			145.5	2.09×10^{-6}			286
			165.0	1.41×10^{-5}			286
		Toluene	165.0	1.37×10^{-5}			286
	3,3,6,6-Tetramethyl-1,2,4,5-tetroxane	Benzene	135.5	6.25×10^{-7}			373
			140.5	1.19×10^{-6}			373
			145.5	2.09×10^{-6}			373
			150.5	2.83×10^{-6}			373
			165.0	1.41×10^{-5}			373
		Toluene	165.0	1.37×10^{-5}			373
		Vapor	130.1	1.8×10^{-6}			365
			150.1	3.4×10^{-6}			365
			170.1	3.5×10^{-6}			365
			177.1	4.6×10^{-6}			365
			185.1	1.3×10^{-5}			365
			194.1	2.4×10^{-5}			365
			207.1	8.5×10^{-5}			365
			215.1	1.66×10^{-4}			365
			250.1	4.88×10^{-4}			365
	Cyclohexyl hydroperoxide	Benzene	70	0		a	91
		Benzene/styrene (50/50)	70	1.27×10^{-3}		a	91
		Cyclohexane	130	2.38×10^{-3}		a	91
			140	1.16×10^{-2}		a	91
			150	3.20×10^{-2}		a	91

TABLE 5. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
6 (*cont'd*)	Cyclohexyl hydroperoxide	Benzene	80	Reaction order varies from 1.5 to 2.0			119
		Cyclohexane	80	Reaction order varies from 1.5 to 2.0			119
		Cyclohexene	80	Reaction order varies from 1.5 to 2.0			119
		Dimethylheptadiene	80	Reaction order varies from 1.5 to 2.0			119
		1-Octene	80	Reaction order varies from 1.5 to 2.0			119
7	Benzyl hydroperoxide	Styrene	60	9×10^{-6}		c_3	276
			60	1.8×10^{-5}		c_4	276
8	1,4-Dimethylcyclohexane-hydroperoxide	1,4-Dimethylcyclohexane	120	1.4×10^{-5}	137.2	c	224
	1,1,3,3-Tetramethylbutyl-hydroperoxide	Benzene	130	1.7×10^{-5}	134	a	337
			145	7.41×10^{-5}		a	337
			160	2.83×10^{-4}		a	337
		Chlorobenzene	135	1.93×10^{-5}			344
			153	1.93×10^{-4}			344
			173	1.93×10^{-3}			344
	2,5-Dimethyl-2,5-di(hydro-peroxy)hexane	Benzene	130	2.7×10^{-6}	113	a	337
			145	1.0×10^{-5}		a	337
			160	3.16×10^{-5}		a	337
	n-Octyl hydroperoxide	White oil	150	9.29×10^{-5}	112.5	a	177
	2,4,4-Trimethylpentyl-2-hydroperoxide	White oil	150	9.29×10^{-5}	112.5	a	177
$(8)_x$	Poly(phenyleneethyl hydro-peroxide)/manganese resinate	Chlorobenzene	26	2.00×10^{-4}	46.0	t_3	212
			30	2.75×10^{-4}		t_3	212
			35	3.67×10^{-4}		t_3	212
9	Cumene hydroperoxide	Benzene	115	4×10^{-7}	122	a	337
			130	1.7×10^{-6}		a	337
			145	6.6×10^{-6}		a	337
		Chlorobenzene	159	1.93×10^{-5}			344
			193	1.93×10^{-4}			344
			253	1.93×10^{-3}			344
		Cumene	110	2.742×10^{-6}			367
			120	5.194×10^{-6}			367
			132	9.305×10^{-6}			367
			T (K)	2.10×10^{11}	$\exp[-80.0/RT]$		367
		Mesitylene			98.7		58
		Styrene			101.3		58
		Toluene	125	9×10^{-6}			199
			139	3×10^{-5}			199
			182.3	6.45×10^{-5}			199
		White oil	150	1.34×10^{-4}	121.3	a	177
	Cumyl hydroperoxide	Acetone-d_6	−33.0	7.0×10^{-5}	62.0		264
			−23.0	2.28×10^{-4}			264
			−12.0	1.21×10^{-3}			264
			−18.0	8.3×10^{-5}	110.2	t_{14}	264
			−8.0	4.23×10^{-4}		t_{14}	264
			−3.0	1.65×10^{-3}		t_{14}	264
10	*p*-Cymene hydroperoxide	*p*-Cymene	110	5.500×10^{-6}			367
			120	1.200×10^{-5}			367
			132	2.089×10^{-5}			367
			T (K)	1.23×10^{12}	$\exp[-85.48/RT]$		367

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TABLE 5. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
10 (*cont'd*)	Decalin hydroperoxide	Acetic acid	130	5.5×10^{-4}			228
		Chlorobenzene	130	10% in 3 h			228
		Decalin	130	1.41×10^{-5}	124.3		228
		1,2-Dichlorobenzene	130	4.48×10^{-5}	176.4		228
		Ethylene glycol	130	1.65×10^{-4}	117.2		228
		Nitrobenzene	130	4.73×10^{-5}			228
		Pyridine	130	$(2.31-2.89) \times 10^{-4}$			228
	Pinane hydroperoxide	Benzene	130	7.08×10^{-6}	123.8	a	18
			145	2.72×10^{-5}		a	18
			160	9.17×10^{-5}		a	18
	2-Phenylbutyl 2-hydroperoxide	*sec*-Butylbenzene	110	2.944×10^{-6}			367
			120	6.611×10^{-6}			367
			132	1.089×10^{-5}			367
			T (K)	4.06×10^{12}	exp[−87.15/RT]		367
	1-Phenyl-2-methylpropyl-hydroperoxide	Benzene	133.8	3.18×10^{-6}	122.2	h	76
			143.9	8.95×10^{-6}			76
			153.9	2.0×10^{-5}			76
			163.7	4.03×10^{-5}			76
			174	9.77×10^{-5}			76
	1-Phenyl-2-methylpropyl-2-hydroperoxide	Benzene	144.2	5.04×10^{-6}	125.5	h	76
			154.5	1.21×10^{-5}			76
			165.6	2.92×10^{-5}			76
			176.0	6.97×10^{-5}			76
	Tetralin hydroperoxide	Poly(butene)	170	2.17×10^{-3}	78.2	t_3	71
		n-Butyl stearate	170	1.47×10^{-4}	125.5	t_3	71
		2-Ethyl-1-hexene	130	1.08×10^{-4}	82.8	e,t_3	71
			170	1.26×10^{-3}		t_3	71
		1-Hexadecene	170	7.92×10^{-4}	117.2	t_3	71
		Mineral oil	135.6	4.2×10^{-5}	131.4	t_3	71
			150.6	1.00×10^{-4}		e,t_3	71
			170	4.82×10^{-4}		t_3	71
		n-Octadecane	170	2.54×10^{-4}	119.2	t_3	71
		Isooctane	170	$1\ 31 \times 10^{-4}$		t_3	71
		Octyl ether	170	1.45×10^{-3}	121.3	t_3	71
		Poly(propylene)	170	2.50×10^{-3}		t_3	71
		n-Tetradecane	170	2.32×10^{-4}		t_3	71
		Tetralin	T (K)	2.27×10^{9}	exp[−102.1/RT]		112
		2,2,4-Trimethyl-1-pentene	170	1.67×10^{-3}		t_3	71
		white oil	150	1.34×10^{-4}	121.3		177
$(10)_n$	Poly(cumyleneethylhydroperoxide) (MW 1100)	Toluene	130	2.79×10^{-5}			210
			140	4.0×10^{-5}			210
			150	5.78×10^{-5}			210
11	2,2-Bis(*tert*-butyldioxy)propane	Cumene	90	2.16×10^{-6}			361
			100	9.14×10^{-6}			361
			110	2.97×10^{-5}			361
			120	9.56×10^{-5}			361
	1-Methoxy-1-(*tert*-butylperoxy) cyclohexane	Cumene	80	4.79×10^{-6}			348
			90	1.898×10^{-5}			348
			100	6.57×10^{-5}			348
			110	2.139×10^{-4}			348
	sec-Butyltoluene hydroperoxide	*sec*-Butyl toluene	100	4.111×10^{-6}			367
			110	8.194×10^{-6}			367
			120	1.9555×10^{-5}			367
			T (K)	1.02×10^{16}	exp[−110.2/RT]		367
12	2,2-Di-(*tert*-butylperoxy)butane	Chlorobenzene	98	1.93×10^{-5}			344
			116	1.93×10^{-4}			344
			136	1.93×10^{-3}			344
			T (K)	9.30×10^{16}	exp[−154.08/RT]		344
	2,2-Bis(*tert*-butyldioxy)butane	Cumene	100	1.84×10^{-5}			361
	Diisopropylbenzene monohydroperoxide	Chlorobenzene	150	1.93×10^{-5}			344
			182	1.93×10^{-4}			344
			218	1.93×10^{-3}			344
	1-Methoxy-1-(*tert*-amylperoxy) cyclohexane	Cumene	80	7.82×10^{-6}			348
			90	3.07×10^{-5}			348

TABLE 5. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	1-Methoxy-1-(*tert*-amylperoxy) cyclohexane	Cumene	100	1.058×10^{-4}			348
			110	3.450×10^{-4}			348
13	1-Methoxy-1-(*tert*-hexylperoxy) cyclohexane	Cumene	80	8.34×10^{-6}			348
			90	3.17×10^{-5}			348
			100	1.120×10^{-4}			348
			110	3.653×10^{-4}			348
	1,1-Bis(*tert*-butylperoxy)-cyclopentane	Cumene	80	4.98×10^{-6}			349,364
			90	1.80×10^{-5}			349,364
			100	6.67×10^{-5}			349,364
			110	2.17×10^{-4}			349,364
	2,2-Bis(*tert*-butyldioxy)-3-methyl-butane	Diphenylmethane	90	1.08×10^{-5}			358
			100	3.75×10^{-5}			358
			110	1.29×10^{-4}			358
			120	4.11×10^{-4}			358
		Cumene	100	3.17×10^{-5}			361
14	1,1-Bis(*tert*-butylperoxy)-cyclohexane	Cumene	80	3.47×10^{-6}			349,350
			90	1.45×10^{-5}			349,350
			100	5.06×10^{-5}			349,350
			110	1.62×10^{-4}			348,349,350
		Chlorobenzene	94	1.93×10^{-5}			344
			113	1.93×10^{-4}			344
			134	1.93×10^{-3}			344
			T (K)	3.47×10^{15}	exp[− 142.4/RT]		344
		n-Dodecane	100	3.51×10^{-5}			364
		n-Hexadecane	100	3.15×10^{-5}			364
		n-Hexane	100	3.82×10^{-5}			364
		n-Nonane	100	3.62×10^{-5}			364
	1,1-Bis(*tert*-butylperoxy)-2-methyl-cyclopentane	Cumene	80	1.55×10^{-5}			349
			90	5.82×10^{-5}			349
			100	1.85×10^{-4}			349
			110	6.00×10^{-4}			349
	1,1-Diphenylethyl hydroperoxide	Chlorobenzene	90	2.0×10^{-9}			374
			110	1.007×10^{-6}			374
			130	1.034×10^{-6}			374
			150	1.223×10^{-6}			374
			T (K)	1.38×10^{8}	exp[− 105.4/RT]		374
15	1-Methoxy-1-(*tert*-octylperoxy)-cyclohexane	Cumene	70	5.9×10^{-6}			348
			80	2.30×10^{-5}			348
			90	8.18×10^{-5}			348
			100	2.840×10^{-4}			348
	1,1-Bis(*tert*-butylperoxy)-2-methyl-cyclohexane	Cumene	80	1.29×10^{-5}			349
			90	4.51×10^{-5}			349
			100	1.43×10^{-4}			349
			110	4.58×10^{-4}			349
16	1,1-Bis(*tert*-amylperoxy)-cyclohexane	Cumene	80	5.34×10^{-6}			348
			90	2.02×10^{-5}			348
			100	7.32×10^{-5}			348
			110	2.44×10^{-4}			348
	1,1-Bis(*tert*-butylperoxy)-cyclooctane	Cumene	90	1.11×10^{-5}			364
			100	4.15×10^{-5}			364
			110	1.36×10^{-4}			364
			120	4.24×10^{-4}			364
17	1,1-Bis(*tert*-butylperoxy)-2-*n*-propyl-cyclohexane	Cumene	80	1.20×10^{-5}			349
			90	4.19×10^{-5}			349
			100	1.38×10^{-4}			349
	1,1-Bis(*tert*-butylperoxy)-2-isopropylcyclohexane	Cumene	70	2.59×10^{-5}			349
			80	4.86×10^{-5}			349
			90	1.98×10^{-4}			349
			100	7.46×10^{-4}			349
	1,1-Bis(*tert*-butylperoxy)-3,3,5-trimethylcyclohexane	Cumene	80	3.82×10^{-6}			349,364
			90	1.50×10^{-5}			349,364
			100	5.31×10^{-5}			349,364
			110	1.75×10^{-4}			349,364

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TABLE 5. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
17 (*cont'd*)	1,1-Bis(*tert*-butylperoxy)-3,3,5-trimethylcyclohexane	Chlorobenzene	85	1.93×10^{-5}			344
			105	1.93×10^{-4}			344
			128	1.93×10^{-3}			344
			T (K)	7.59×10^{13}	exp[−127.52/RT]		344
	n-Butyl 4,4-di-(*tert*-butylperoxy) valerate	Chlorobenzene	102	1.93×10^{-5}			344
			121	1.93×10^{-4}			344
			143	1.93×10^{-3}			344
			T (K)	1.91×10^{15}	exp[−143.51/RT]		344
18	1,1-Bis(*tert*-hexylperoxy)-cyclohexane	Cumene	80	5.38×10^{-6}			348
			90	2.06×10^{-5}			348
			100	7.48×10^{-5}			348
			110	2.48×10^{-4}			348
20	1,1-Bis(*tert*-butylperoxy)-cyclododecane	Cumene	90	8.14×10^{-6}			364
			100	3.22×10^{-5}			364
			110	9.93×10^{-5}			364
22	1,1-Bis(*tert*-octylperoxy)-cyclohexane	Cumene	80	1.513×10^{-5}			348
			90	5.62×10^{-5}			348
			100	1.983×10^{-4}			348
			110	6.26×10^{-4}			348
31	2,2-Bis[4,4-di-(*tert*-butylperoxycyclohexyl)propane	Chlorobenzene	87	1.93×10^{-5}			344
			106	1.93×10^{-4}			344
			128	1.93×10^{-3}			344
			T (K)	8.79×10^{14}	exp[−135.56/RT]		344

TABLE 6. PERESTERS AND PEROXYCARBONATES

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
4	Dimethyl peroxalate	Pentane	25	1.7×10^{-5}			203
5	*tert*-Butyl percarbamate	Chlorobenzene	90	6.6×10^{-6}			141
	tert-Butyl performate	Chlorobenzene	130.8	5.43×10^{-5}	159.0	h	218
			140.8	1.70×10^{-4}			218
			140.6	1.80×10^{-4}		m_2	218
			140	2.12×10^{-3}	64.0	v_8	218
			140	5.06×10^{-4}		v_1	218
		4-Chlorotoluene	140.6	1.61×10^{-4}			218
			140	1.62×10^{-3}		v_8	218
		Cumene	140.6	1.77×10^{-4}		b	218
			140	1.02×10^{-3}		b,v_6	218
6	Isobutyl peracetate	α-Methylstyrene	79.5	1.36×10^{-5}	115.2		253
			88.4	3.61×10^{-5}			253
			92.8	5.58×10^{-5}			253
	n-Butyl peracetate	α-Methylstyrene	64	$\sim 9.6 \times 10^{-3}$		g_2,v_{16}	253
			64	8.2×10^{-3}		g_2,v_{17}	253
			64	6.5×10^{-4}		g_2,v_{18}	253
			64	1.2×10^{-3}		g_2,v_{19}	253
			79.6	1.57×10^{-5}	101.0		253
			87.8	3.27×10^{-5}			253
			92.6	5.38×10^{-5}			253
	sec-Butyl peracetate	α-Methylstyrene	88.2	1.79×10^{-5}	104.3		253
			92.8	3.19×10^{-5}			253
			99.4	5.37×10^{-5}			253
	tert-Butyl peracetate	*tert*-Amyl alcohol	75	2.8×10^{-6}			193
		Benzene	85	1.2×10^{-6}			193
			85	2.18×10^{-6}	151.9	a	126,337
			100	1.54×10^{-5}		a	126,337
			115	1.02×10^{-4}		a	126,337
			130	5.69×10^{-4}		a	126,337
		n-Butanol	75	2.65×10^{-5}			193

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
6 (*cont'd*)	*tert*-Butyl peracetate	2-Butanol	75	1.13×10^{-4}		c	193
			75	5.3×10^{-5}		t_7	193
			75	4×10^{-7}		m_3	193
		Chlorobenzene	60	2.31×10^{-8}	159.0	a,h	81
			100	1.93×10^{-5}			344
			119	1.93×10^{-4}			344
			139	1.93×10^{-3}			344
			T (K)	1.57×10^{16}	exp[−149.36/RT]		344
		Decane	100	1.5×10^{-5}	134		337
			115	7.91×10^{-5}			337
			125	2.45×10^{-4}			337
			100	1.95×10^{-5}			270
		Dodecane	100	1.78×10^{-5}			270
		Hexadecane	100	1.55×10^{-5}			270
		Hexane	130.1	5.08×10^{-4}			169
		Octane	100	2.07×10^{-5}			270
		Paraffin	130.1	3.13×10^{-4}			169
		Tetradecane	100	1.55×10^{-5}			270
	tert-Butyl trichloroperacetate	Chlorobenzene	60	1.19×10^{-5}	125.9	a,h	81
			66.8	2.75×10^{-5}	126.8	h	28
			77.0	1.00×10^{-4}			28
	Diethyl peroxydicarbonate	*tert*-Butanol	45	1.25×10^{-5}	133.9–138.1	c	18
			55	5.7×10^{-5}		c	18
		2,2′-Oxydiethylene bis(allyl carbonate)	40	6.94×10^{-6}	127.2	a	90
			50	2.86×10^{-5}		a	90
			60	1.28×10^{-4}		a	90
	Diethyl peroxalate	Pentane	25	2.6×10^{-5}			203
7	*tert*-Amyl peracetate	Chlorobenzene	90	1.93×10^{-5}			344
			108	1.93×10^{-4}			344
			129	1.93×10^{-3}			344
			T (K)	4.13×10^{15}	exp[−141.1/RT]		344
	tert-Butyl methoxyperacetate	Ethylbenzene	40.0	2.2×10^{-4}	104.7	a,h	315
			39.6	1.94×10^{-4}	103.9	h	328
			50.2	8.24×10^{-4}			328
			60.1	2.35×10^{-3}			328
	tert-Butyl peracrylate	Benzene	90	1.12×10^{-5}	86.2	a	211
			100	1.51×10^{-5}		a	211
			110	4.85×10^{-5}		a	211
	tert-Butylperoxy isopropyl	Chlorobenzene	98	1.93×10^{-5}			344
			117	1.93×10^{-4}			344
			137	1.93×10^{-3}			344
			T (K)	2.49×10^{16}	exp[−150.15/RT]		344
	tert-Butyl perpropionate carborate	Benzene	80	1.53×10^{-6}		a	213
			90	6.04×10^{-6}		a	213
		Decane	100	1.82×10^{-5}			270
		Dodecane	100	1.78×10^{-5}			270
		Hexadecane	100	1.58×10^{-5}			270
		α-Methylstyrene	70	3.02×10^{-7}		a	213
			90	4.85×10^{-6}		a	213
		Octane	100	2.03×10^{-5}			270
		Tetradecane	100	1.68×10^{-5}			270
8	*tert*-Butyl 2-acetoxyperacetate	Ethylbenzene	91.8	2.73×10^{-5}		b	328
	tert-Butyl cyclopropane-percarboxylate	*n*-Dodecane	113.0	5.48×10^{-5}			255
		n-Hexadecane	102.6	1.28×10^{-5}	153.8	h	255
			113.0	4.92×10^{-5}			255
		2,2,4-Trimethyl-pentane	102.6	1.71×10^{-5}	146.2	h	255
			113.0	6.19×10^{-5}			255
			120.4	1.505×10^{-4}			255
	tert-Butyl ethoxyperacetate	Ethylbenzene	40.0	2.96×10^{-4}	105	a	315
			40.6	2.89×10^{-4}	102.7	m_2,h	328
			46.0	5.58×10^{-4}			328
			55.6	1.94×10^{-3}			328

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	*tert*-Butyl heptafluoroperoxy-butyrate	Benzene	65.0	1.56×10^{-5}		v_{24}	370
			70.0	3.09×10^{-5}		v_{24}	370
			75.0	5.27×10^{-5}		v_{24}	370
			80.0	9.08×10^{-5}		v_{24}	370
			60.0	9.5×10^{-6}		v_{25}	370
			65.0	1.61×10^{-5}		v_{25}	370
			70.0	2.70×10^{-5}		v_{25}	370
			75.0	5.23×10^{-5}		v_{25}	370
			80.0	9.18×10^{-5}		v_{25}	370
			60.0	9.6×10^{-6}		v_{26}	370
			65.0	1.93×10^{-5}		v_{26}	370
			70.0	3.57×10^{-5}		v_{26}	370
			75.0	6.35×10^{-5}		v_{26}	370
		Methanol	10.0	7.19×10^{-5}			369
			15.0	9.37×10^{-5}			369
			19.6	1.304×10^{-4}			369
			30.0	2.604×10^{-4}			369
		Methoxybenzene	55.0	5.01×10^{-5}			370
			60.2	8.56×10^{-5}			370
			65.0	1.378×10^{-4}			370
			70.0	2.444×10^{-4}			370
		Nitrobenzene	40.0	3.26×10^{-5}			370
			45.0	6.36×10^{-5}			370
			50.0	9.31×10^{-5}			370
		Octane	85.0	5.1×10^{-6}			370
			90.0	9.5×10^{-6}			370
			95.0	1.75×10^{-5}			370
			100.0	3.41×10^{-5}			370
		Pyridine	30.0	2.74×10^{-5}			370
			40.0	7.66×10^{-5}			370
			50.0	1.868×10^{-4}			370
		Toluene	65.0	1.88×10^{-5}			369
			75.0	4.66×10^{-5}			369
			80.0	7.06×10^{-5}			369
			85.0	1.203×10^{-4}			369
	tert-Butyl methoxyperpropionate	Ethylbenzene	40.3	4.30×10^{-3}			328
	tert-Butyl perisobutyrate	Benzene	78	3.77×10^{-5}		a	37
			70	6.69×10^{-5}	140.6	a	126,337
			85	5.33×10^{-5}		a	126,337
			100	3.50×10^{-4}		a	126,337
		Bulk	70	4.12×10^{-5}		a	37
		Chlorobenzene	90.6	8.13×10^{-5}	140.6	h	136
			100.7	2.75×10^{-4}			136
			110.0	8.92×10^{-4}			136
			79	1.93×10^{-5}			344
			98	1.93×10^{-4}			344
			118	1.93×10^{-3}			344
			T (K)	2.07×10^{15}	$\exp[-135.16/RT]$		344
		Cumene	90.6	6.9×10^{-5}	133.1	h	136
			100.7	2.35×10^{-4}			136
			110.0	6.57×10^{-4}			136
		Decane	70	4.7×10^{-6}	122	a	337
			85	2.6×10^{-5}		a	337
			100	1.55×10^{-4}		a	337
			100	1.46×10^{-4}			270
		Dodecane	100	1.56×10^{-4}			270
		Hexadecane	100	1.47×10^{-4}			270
		Iso-octane	103.4	2.16×10^{-4}			251
		Nujol	103.4	2.23×10^{-4}			251
		Octane	100	1.59×10^{-4}			270
		Tetradecane	100	1.52×10^{-4}			270
	tert-Butyl percrotonate	Benzene	100	2.5×10^{-5}	138	a	337
			115	1.50×10^{-4}		a	337
			130	7.39×10^{-4}		a	337

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
8 (*cont'd*)	*tert*-Butyl permaleic acid	Acetone	70	2.7×10^{-6}	113	a	337
			85	1.6×10^{-5}		a	337
			101	6.77×10^{-5}		a	337
	tert-Butyl permethacrylate	Benzene	90	1.92×10^{-5}	137.0	a	211
			100	7.66×10^{-5}		a	211
			110	1.92×10^{-4}		a	211
	tert-Butyl peroxyisopropyl carbonate	Benzene	90	6.64×10^{-6}		a	9
			100	2.21×10^{-5}		a	9
			110	6.87×10^{-5}		a	9
	Cyclohexyl peracetate	α-Methylstyrene	88.6	1.63×10^{-5}	109.4		253
			93.0	2.30×10^{-5}			253
			99.3	4.75×10^{-5}			253
	Diisopropyl peroxalate	Pentane	25	6.0×10^{-5}			203
	Diisopropyl peroxydicarbonate	Benzene	54.0	5.0×10^{-5}		m_2	60
		Di-*n*-butyl phthalate	77	1.39×10^{-3}		y_{16}	236
			87	4.09×10^{-3}		y_{16}	236
			97	1.25×10^{-2}		y_{16}	236
			107	3.54×10^{-2}		y_{16}	236
			117	7.98×10^{-2}		y_{16}	236
		Decane	35	6.3×10^{-6}	113	a	337
			45	2.7×10^{-5}		a	337
			60	1.85×10^{-4}		a	337
		Ethylbenzene	54.3	4.5×10^{-5}			60
			54.3	5.2×10^{-5}		m_2	60
		Mineral spirits	30	2×10^{-6}	113	a	337
			45	1.8×10^{-5}		a	337
			60	1.28×10^{-4}		a	337
			75	7.55×10^{-4}		a	337
		2,2′-Oxydiethylene-bis(allyl carbonate)	40	6.39×10^{-6}	117.6	a	90
			50	2.28×10^{-5}		a	90
			60	9.44×10^{-5}		a	90
		Toluene	50	3.03×10^{-5}		a	90
	Ethyl *tert*-butyl peroxalate	Benzene	45	4.48×10^{-5}	112.5	c,h	31
			55	1.63×10^{-4}			31
			65	5.93×10^{-4}			31
9	*tert*-Butyl 2-acetoxyperpropionate	Ethylbenzene	91.4	3.60×10^{-4}			328
	tert-Butyl 5-bromo-2-perthenoate	Carbon tetrachloride	99.2	2.24×10^{-6}	143.5	a	176
			112.0	1.18×10^{-5}		a	176
			124.5	4.30×10^{-5}		a	176
	tert-Butyl 5-chloro-2-perthenoate	Carbon tetrachloride	99.2	2.29×10^{-6}	143.5	a	176
			112.0	9.56×10^{-6}		a	176
			124.5	4.42×10^{-5}		a	176
	tert-Butyl cyclobutane-percarboxylate	*n*-Hexadecane	102.6	4.33×10^{-5}			255
		2,2,4-Trimethyl-pentane	92.9	1.21×10^{-5}	150.4	h	255
			102.6	4.84×10^{-5}			255
			113.0	1.668×10^{-4}			255
	tert-Butyl 2,2-dimethyl-3-bromoperpropionate	Cyclohexane	55	9.50×10^{-6}	118.2	h,t_6	320
			65	3.58×10^{-5}		t_6	320
			75	1.23×10^{-4}		t_6	320
	tert-Butyl 2,2-dimethyl-3-chloroperpropionate	Cyclohexane	55	3.55×10^{-6}	128.6	h,t_6	320
			65	1.48×10^{-5}		t_6	320
			75	5.65×10^{-5}		t_6	320
	tert-Butyl 2,2-dimethyl-3-iodoperpropionate	Cyclohexane	55	2.91×10^{-5}	116.9	h,t_6	320
			75	3.76×10^{-4}		t_6	320
	tert-Butyl isopropoxyperacetate	Ethylbenzene	40.0	4.8×10^{-4}	100.6	a,h	315
			39.9	4.22×10^{-4}	98.9	h,m_2	328
			49.9	1.39×10^{-3}			328
			55.3	2.68×10^{-3}			328

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
9 (*cont'd*)	*tert*-Butyl 1-methylcyclopropyl-percarboxylate	2,2,4-Trimethylpentane	102.6	2.30×10^{-5}	133.7	h	255
			113.0	7.47×10^{-5}			255
	Di-(*tert*-butylperoxy)-carbonate	Chlorobenzene	99.95	6.72×10^{-5}	133.1		27
			110.1	2.13×10^{-4}			27
			120.1	6.05×10^{-4}			27
		1,2-Dichlorobenzene	120.1	5.98×10^{-4}			27
		Cumene	99.95	6.72×10^{-5}			27
		Isopropyl ether	99.95	2.76×10^{-4}			27
	tert-Butylperoxy isopropyl carbonate	Benzene	85	3.3×10^{-6}	138	a	337
			100	2.2×10^{-5}		a	337
			115	1.36×10^{-4}		a	337
	tert-Butyl perpivalate	Benzene	50	9.77×10^{-6}	119.7	a	126,337
			70	1.24×10^{-4}		a	126,337
			85	7.64×10^{-4}		a	126,337
			50	1.03×10^{-5}			359
			55	1.94×10^{-5}			359
			60	3.92×10^{-5}			359
			65	7.31×10^{-5}			359
			70	1.30×10^{-4}			359
		Chlorobenzene	48.9	8.4×10^{-6}		k	249
			58.6	3.35×10^{-5}	125.5	h	28
			64.3	7.01×10^{-5}			28
			57	1.93×10^{-5}			344
			75	1.93×10^{-4}			344
			94	1.93×10^{-3}			344
			T (K)	7.09×10^{14}	$\exp[-123.59/RT]$		344
			74.8	2.79×10^{-4}			28
			60	3.85×10^{-5}	128.0	a,h	81
			60.6	4.00×10^{-5}		t_9	156
			74	1.93×10^{-4}		t_9	156
		Cumene	45	3.53×10^{-6}	106.7	h	160
			55	1.73×10^{-5}			160
			60	2.95×10^{-5}			362
			65	4.53×10^{-5}			160
			64.6	5.81×10^{-5}	115.5		159
			75.6	2.10×10^{-4}			159
			84.6	5.94×10^{-4}			159
			70.3	1.11×10^{-4}			267
		Cyclohexane	55	1.03×10^{-5}	122.8	h	320
			65	4.02×10^{-5}			320
			75	1.45×10^{-4}			320
		Decane	50	4.1×10^{-6}	117	a	337
			70	5.52×10^{-5}		a	337
			80	2.18×10^{-4}			270
		Dioxane/water (90/10)	60.6	6.26×10^{-5}		t_9	156
		Dodecane	80	2.38×10^{-4}			270
		Hexadecane	80	2.39×10^{-4}			270
		Hexane	55	8.42×10^{-6}			359
			60	1.59×10^{-5}			359
			65	3.39×10^{-5}			359
			70	6.19×10^{-5}			358
		Heptane	80	2.33×10^{-4}			270
		Isooctane	60.6	1.97×10^{-5}		t_9	165
			73.9	8.82×10^{-5}		t_9	165
		Mineral spirits	55	1.01×10^{-5}			359
			60	2.15×10^{-5}			359
			65	4.14×10^{-5}			359
			70	7.58×10^{-5}			359
			75	1.47×10^{-4}			359
			T (K)	3.56×10^{14}	$\exp[-123/RT]$		376
		Octane	80	2.34×10^{-4}			270
		Tetradecane	80	2.27×10^{-4}			270
	tert-Butyl perpivalate-d_6	Chlorobenzene	60.6	3.24×10^{-5}		t_9	156
			74.0	1.57×10^{-4}		t_9	156

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
9 (*cont'd*)	*tert*-Butyl perpivalate-d_6	Dioxane/water (90/10)	60.6	5.05×10^{-5}		t_9	156
		Isooctane	60.6	1.62×10^{-5}		t_9	165
			73.9	8.82×10^{-5}		t_9	165
	tert-Butyl 2-perthenoate	Carbon tetrachloride	99.2	3.3×10^{-6}	147.3	m_2	176
			112.0	1.66×10^{-5}		m_2	176
			124.5	6.87×10^{-5}		m_2	176
	tert-Butyl 1-pyrollidine-percarboxylate	Chlorobenzene	90	7.59×10^{-5}			141
	tert-Butyl *N*-succinimido-percarboxylate	Acetonitrile	100	9.9×10^{-4}			142
		Benzene	100	3.79×10^{-5}	113.4	h,m_2	142
		Chlorobenzene	90	1.32×10^{-5}			141
			100	4.91×10^{-5}		m_2	142
		Cumene	90	3.3×10^{-6}			141
		Cyclohexane	100	1.10×10^{-5}		t_9	142
		Cyclohexene	100	9.0×10^{-6}			142
		Methanol	100	6.00×10^{-3}			142
		Methylene chloride	90	9.24×10^{-5}			141
		Nitrobenzene	100	3.41×10^{-4}		m_2	142
10	*tert*-Amyl perpivalate	Benzene	51	1.9×10^{-5}		a	338
		Chlorobenzene	55	1.93×10^{-5}			344
			72	1.93×10^{-4}			344
			91	1.93×10^{-3}			344
			T (K)	4.12×10^{15}	exp[−127.76/RT]		344
		Cumene	50	1.05×10^{-5}			362
			60	3.51×10^{-5}			362
			70	1.309×10^{-4}			362
			80	4.239×10^{-4}			362
	Di-(*sec*-butyl)peroxy-dicarbonate	Decane	35	6.9×10^{-6}	113	a	337
			45	2.6×10^{-5}		a	337
			60	1.80×10^{-4}		a	337
		Chlorobenzene	47	1.93×10^{-5}			344
			63	1.93×10^{-4}			344
			82	1.93×10^{-3}			344
			T (K)	3.19×10^{15}	exp[−123.85/RT]		344
		Mineral spirits	30	2×10^{-6}	113	a	337
			40	1.0×10^{-5}		a	337
			60	1.34×10^{-4}		a	337
	Di-*n*-butyl peroxydicarbonate	Chlorobenzene	49	1.93×10^{-5}			344
			65	1.93×10^{-4}			344
			83	1.93×10^{-3}			344
			T (K)	7.00×10^{15}	exp[−126.65/RT]		344
	tert-Butyl 2-acetoxy-2-methyl perpropionate	Ethylbenzene	74.7	1.98×10^{-4}		m_2	328
			80.7	4.37×10^{-4}			328
			90.8	1.25×10^{-3}			328
	tert-Butyl *tert*-butoxyperacetate	Ethylbenzene	40.3	2.91×10^{-4}			328
	tert-Butyl 4-carbomethoxy-perbutyrate	Cumene	99.8	1.65×10^{-5}	151.3	h	257
			116.6	1.38×10^{-4}			257
			129.9	6.83×10^{-4}			257
	tert-Butyl cyclopentyl-percarboxylate	*n*-Hexadecane	113.0	3.02×10^{-4}			255
		2,2,4-Trimethyl-pentane	92.9	2.47×10^{-5}	140.8	h	255
			102.6	8.87×10^{-5}			255
			113.0	2.89×10^{-4}			255
	tert-Butyl peroxydiethyl acetate	Chlorobenzene	75	1.93×10^{-5}			344
			93	1.93×10^{-4}			344
			113	1.93×10^{-3}			344
			T (K)	2.45×10^{15}	exp[−134.00/RT]		344
	tert-Butyl 2,2-dimethylperbutenoate	Cumene	42.42	9.04×10^{-5}	100.7	h	274
			50.30	2.349×10^{-5}			274
			57.95	5.660×10^{-5}			274
	tert-Butyl 2,2-dimethylperbutyrate	Cumene	44.85	1.874×10^{-5}			274
			50.10	3.609×10^{-5}			274
			60.20	1.306×10^{-4}			274

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
10 (*cont'd*)	Di(*tert*-butylperoxy)-oxalate	Benzene	35.0	6.77×10^{-5}	106.7	c,h	30
			45.0	2.61×10^{-4}		c	30
			55.0	9.3×10^{-4}		c	30
		tert-Butanol	25.0	1.83×10^{-5}			197
		Decane	41	1.75×10^{-4}			270
		Dodecane	41	1.81×10^{-4}			270
		Octane	41	1.79×10^{-4}			270
		n-Pentane	37.8	1.01×10^{-4}			197
		Tetradecane	41	1.71×10^{-4}			270
		Styrene	35	5.5×10^{-5}	75.3		215
			45	1.71×10^{-4}			215
			55	6.01×10^{-4}			215
	tert-Butyl 5-methyl-7-perthenoate	Carbon tetrachloride	99.2	4.6×10^{-6}	137.7	a,m_2	176
			112.0	1.94×10^{-5}		a,m_2	176
			124.5	7.95×10^{-5}		a,m_2	176
	β-Methyl-β-phenyl-β-peroxy-propiolactone	Carbon tetrachloride	106.8	5.65×10^{-6}	131.8	h,v_9	195
			126	4.59×10^{-5}			195
			134	9.71×10^{-5}		c,v_9	195
	Bis(2-nitro-2-methylpropyl)-peroxy-dicarbonate	Toluene	50	2.22×10^{-5}		a	90
	1-Phenylethylperacetate	Benzene	100	1.14×10^{-4}	138.7	h	259
11	*tert*-Butyl 4-chloroperbenzoate	Phenyl ether	100.0	3.89×10^{-6}	164.4	a	53
			110.1	1.85×10^{-5}		a	53
			120.2	6.39×10^{-5}		a	53
			130.9	2.42×10^{-4}		a	53
		Carbon tetrachloride	80	4.2×10^{-7}			354
	tert-Butyl-*N*-(2-chlorophenylperoxy)-carbamate	Toluene	87.0	3.5×10^{-5}	157.3	h	36
			95.5	1.48×10^{-4}			36
			103.3	3.3×10^{-4}			36
	tert-Butyl-*N*-(3-chlorophenylperoxy)-carbamate	Toluene	78.0	2.6×10^{-5}	115.5	h	36
			87.0	7.8×10^{-5}			36
			96.6	2.03×10^{-4}			36
			102.7	4.62×10^{-4}			36
	tert-Butyl-*N*-(4-chlorophenylperoxy)-carbamate	Toluene	73.0	5.57×10^{-5}	119.7	h	36
			78.5	1.42×10^{-4}			36
			87.0	2.75×10^{-4}			36
			92.5	5.37×10^{-4}			36
	tert-Butyl cyclohexylperoxy-carboxylate	*n*-Hexadecane	92.9	6.70×10^{-5}	136.6	h	255
			102.6	2.046×10^{-4}			255
		2,2,4-Trimethyl-pentane	82.2	1.57×10^{-5}	135.8	h	255
			92.9	6.13×10^{-5}			255
			102.6	2.046×10^{-4}			255
			113.0	6.58×10^{-4}			255
	tert-Butyl-*N*-(2,5-dichlorophenyl-peroxy) carbamate	Toluene	88.0	2.75×10^{-5}	128.9	h	36
			95.0	5.58×10^{-5}			36
			103.3	1.48×10^{-4}			36
			114.5	4.82×10^{-4}			36
	tert-Butyl 2,2-dimethylperpentanoate	Cyclohexane	55	1.66×10^{-5}	120.7	h	320
			65	6.31×10^{-5}			320
			75	2.23×10^{-4}			320
		Cumene	45	6.2×10^{-6}	105.4	h	160
			55	2.15×10^{-5}			160
	tert-Butyl 5-ethyl-2-perthenoate	Carbon tetrachloride	99.2	5.39×10^{-6}	130.5		176
			112.0	2.12×10^{-5}			176
			124.5	7.91×10^{-5}			176
	tert-Butyl 2-iodoperbenzoate	Chlorobenzene	85.0	4.02×10^{-5}			134
			102.4	2.58×10^{-4}			134
			118.8	1.32×10^{-3}			134
	tert-butyl 4-nitroperbenzoate	Phenyl ether	110.1	7.56×10^{-6}	172.8		53
			120.2	3.19×10^{-5}			53
			130.9	1.11×10^{-4}			53
			141.5	3.92×10^{-4}			53
		Carbon tetrachloride	80	2.9×10^{-7}			354
	tert-Butyl 3,5-dinitroperbenzoate	Carbon tetrachloride	80	1.1×10^{-7}			354

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
11 (*cont'd*)	*tert*-Butyl-*N*-(3-nitrophenylperoxy)-carbamate	Toluene	78.0	8.4×10^{-6}	133.1	h	36
			88.7	2.43×10^{-5}			36
			98.0	1.01×10^{-4}			36
			106.0	2.36×10^{-4}			36
	tert-Butyl-*N*-(4-nitrophenylperoxy)-carbamate	Toluene	73.0	6.4×10^{-6}	113.8	h	36
			87.0	3.11×10^{-5}			36
			98.0	8.75×10^{-5}			36
			106.0	2.38×10^{-4}			36
	tert-Butyl perbenzoate	Acetic acid	100.0	3.83×10^{-5}	130.1	a	56
			110.0	1.14×10^{-4}		a	56
		Benzene	100	1.07×10^{-5}	145.2	a	126,337
			115	6.22×10^{-5}		a	126,337
			130	3.50×10^{-4}		a	126,337
			110.0	3.50×10^{-5}	144.3	a	56
			119.4	1.04×10^{-4}		a	56
			130.0	3.30×10^{-4}		a	56
		Bromobenzene	119.4	1.37×10^{-4}		a	56
		n-Butanol	90.0	9.27×10^{-5}	120.5	a	56
			100.0	2.70×10^{-4}		a	56
		n-Butyl acetate	110.0	1.06×10^{-4}	123.4	a	56
			119.4	2.67×10^{-4}		a	56
			110.0	3.61×10^{-5}	148.5	a,m_2	56
			1194	1.10×10^{-4}		a,m_2	56
		tert-Butylbenzene	119.4	1.03×10^{-4}		a	56
		n-Butyl ether	100.0	7.80×10^{-5}	99.2	a	56
			110.0	1.80×10^{-4}		a	56
		Carbon tetrachloride	80	8.8×10^{-7}			354
		Chlorobenzene	110.0	3.83×10^{-5}	141.8	a	56
			119.4	1.11×10^{-4}		a	56
			120	1.31×10^{-4}		m_2	22
			135	6.74×10^{-4}		m_2	22
			150	3.12×10^{-3}		m_2	22
			103	1.93×10^{-5}			344
			122	1.93×10^{-4}			344
			142	1.93×10^{-3}			344
			T (K)	2.23×10^{16}	exp[$-151.59/RT$]		344
		4-Chlorotoluene	110.0	3.25×10^{-5}	144.3	c	55
			119.4	9.80×10^{-5}		c	55
			130.0	3.06×10^{-4}		c	55
			60	3.85×10^{-7}	140.2	a,h	81
		Cumene	79.6	7.21×10^{-5}			272
			85.0	1.320×10^{-6}			345
			99.3	9.37×10^{-6}			345
			115.0	5.81×10^{-5}			345
			130.2	3.24×10^{-4}			345
		Decane	100	1.4×10^{-5}	134	a	337
			115	8.05×10^{-5}		a	337
			130	3.56×10^{-4}		a	337
			115	6.77×10^{-5}			270
		Dodecane	115	6.44×10^{-5}			270
		Ethylbenzene	119.4	1.07×10^{-4}		a	56
		Heptane	115	7.21×10^{-5}			270
		Hexadecane	115	5.97×10^{-5}			270
		Methyl benzoate	119.4	7.80×10^{-5}		a	56
		Octane	115	7.06×10^{-5}			270
		Phenyl ether	100.0	6.94×10^{-6}	156.9	a	53
			110.1	2.28×10^{-5}		a	53
			120.2	9.00×10^{-5}		a	53
			130.9	2.92×10^{-4}		a	53
		Tetradecane	115	6.03×10^{-5}			270
		Trimethylbenzene	T (K)	2.62×10^{16}	exp[$-152.0/RT$]		376
		Xylene	119.4	1.09×10^{-4}	141.4	a	56
			130.0	3.42×10^{-4}		a	56

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
11 (*cont'd*)	*tert*-Butyl-*N*-(4-bromophenylperoxy)-carbamate	Toluene	70.8	3.32×10^{-5}	177.6	h	36
			79.8	1.04×10^{-4}			36
			84.0	1.75×10^{-4}			36
			96.0	7.70×10^{-4}			36
	tert-Butyl percarboxycyclohexane	Chlorobenzene	100.1	2.75×10^{-4}	131.0	h	181
			111.4	9.65×10^{-4}			181
			120.0	2.39×10^{-3}			181
		Cumene	79.6	1.86×10^{-5}		z (l)	163
			79.6	1.42×10^{-5}		z (2030)	163
			79.6	1.15×10^{-5}		z (4050)	163
	tert-Butyl per-2-methylphenylacetate	Isooctane	60.6	6.60×10^{-5}		m_2	165
			73.9	3.233×10^{-4}		m_2	165
	tert-Butyl-*N*-(phenylperoxy) carbamate	Toluene	51.2	3.4×10^{-6}	139.7	h	36
			67.7	2.31×10^{-5}			36
			77.7	1.15×10^{-4}			36
			90.7	6.41×10^{-4}			36
		Chlorobenzene	T (K)	1.51×10^{16}	exp[−136.0/RT]		40
	tert-Hexyl peroxypivalate	Cumene	40	2.39×10^{-6}			362
			50	8.83×10^{-6}			362
			60	3.37×10^{-5}			362
			70	1.226×10^{-4}			362
			80	4.166×10^{-4}			362
	1,1,2-Trimethylpropyl perpivalate	Cumene	40	3.6×10^{-6}			362
			50	1.59×10^{-5}			362
			60	5.14×10^{-5}			362
			70	1.921×10^{-4}			362
	Cumyl peracetate	Carbon tetrachloride	100	4.68×10^{-5}		c	271
		Toluene	100	3.43×10^{-5}		p	271
12	*tert*-Amyl peroxybenzoate	Chlorobenzene	99	1.93×10^{-5}			344
			118	1.93×10^{-4}			344
			139	1.93×10^{-3}			344
			T (K)	8.38×10^{15}	exp[−147.02/RT]		344
	4-Bromocumyl perpropionate	Benzene	70	1.48×10^{-6}	129.7	a	213
			80	5.73×10^{-6}		a	213
			90	1.81×10^{-5}		a	213
		α-Methylstyrene	70	1.99×10^{-6}	121.3	a	213
			80	4.93×10^{-6}		a	213
			90	1.55×10^{-5}			213
	tert-Butyl bicyclo[2.2.1]lheptane-2-percarboxylate	Cumene	85	1.16×10^{-6}	155.2	h	160
			100	1.02×10^{-5}	160		
			110	3.76×10^{-5}			160
			110	4.6×10^{-5}	150.2	h	223
			120	1.56×10^{-4}			223
			130	4.99×10^{-4}			223
	tert-Butyl endo-bicyclo[2.2.1]-heptane percarboxylate	Chlorobenzene	94.0	1.75×10^{-5}	137.2	h	147
			101.9	6.20×10^{-5}			147
			109.6	1.7×10^{-4}			147
			120.5	5.72×10^{-4}			147
			94.5	9.69×10^{-5}			147
			101.9	1.96×10^{-5}			182
			111.9	6.53×10^{-5}			182
			100.1	4.61×10^{-5}	149.0	h	181
			111.4	1.96×10^{-4}			181
			120.0	5.40×10^{-4}			181
		Cumene	94.0	1.87×10^{-5}	154.0	h	147
			101.9	5.30×10^{-5}			147
			109.6	1.10×10^{-4}			147
			120.3	4.3×10^{-4}			147
			94.5	5.27×10^{-5}	130.1	h	182
			101.9	1.12×10^{-4}			182
			111.9	3.57×10^{-4}			182
	tert-Butyl exo-bicyclo[2.2.11]-heptane-2-percarboxylate	Chlorobenzene	94.5	8.86×10^{-5}			182
			101.9	2.28×10^{-4}			182
			112.1	8.85×10^{-4}			182

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)			96.0	1.33×10^{-4}			147
			100.1	1.90×10^{-4}	129.3	h	181
			111.4	6.79×10^{-4}			181
			120.0	1.64×10^{-3}			181
		Cumene	84.4	2.19×10^{-5}	129.3	h	147
			94.7	7.29×10^{-5}			147
			100.1	1.37×10^{-4}			147
			108.6	3.6×10^{-4}			147
			113.5	6.19×10^{-4}			147
			94.5	6.05×10^{-5}	131.4	h	182
			101.9	1.53×10^{-4}			182
			112.1	4.72×10^{-4}			182
	tert-Butyl endo-bicyclo[2.2.1]-hept-5-ene-2-percarboxylate	Cumene	94.5	3.11×10^{-5}	136.4	h	182
			101.9	7.35×10^{-5}			182
			112.0	2.48×10^{-4}			182
	tert-Butyl exo-bicyclo[2.2.1]-hept-5-ene-2-percarboxylate	Chlorobenzene	94.5	5.59×10^{-5}			182
			101.9	$1\ 25 \times 10^{-4}$			182
			112.1	5.00×10^{-4}			182
		Cumene	94.5	4.48×10^{-5}	138.1	h	182
			101.9	1.18×10^{-4}			182
			112.1	3.97×10^{-4}			182
	tert-Butyl 4-bromophenoxyperacetate	Ethylbenzene	70.5	2.6×10^{-4}		a	315
	tert-Butyl 4-chlorophenoxyperacetate	Ethylbenzene	70.5	2.9×10^{-4}		a	315
	tert-Butyl 3-chlorophenylperacetate	Chlorobenzene	79.6	4.05×10^{-5}	123.0	h	178
			90.7	1.44×10^{-4}			178
			100.5	4.38×10^{-4}			178
		Cumene	79.6	2.98×10^{-5}		z (0.98)	207
			79.6	1.99×10^{-5}		z (6090)	207
	tert-Butyl 4-chlorophenyl peracetate	Chlorobenzene	79.6	8.44×10^{-5}	117.2	h	178
			90.7	2.95×10^{-4}			178
			100.5	8.19×10^{-4}			178
		Cumene	79.6	6.54×10^{-5}		z (0.98)	207
			79.6	4.45×10^{-5}		z (6090)	207
	tert-Butylcycloheptyl percarboxylate	*n*-Hexadecane	92.9	1.801×10^{-4}	121.5	h	255
			102.6	5.17×10^{-4}			255
		2,2,4-Trimethylpentane	92.9	1.703×10^{-4}	125.7	h	255
			102.6	5.07×10^{-4}			255
	tert-Butyl 2,4-dichlorophenoxy-peracetate	Ethylbenzene	70.5	9.4×10^{-5}		a	315
	tert-Butyl 2,2-diethyl perbutyrate	Cumene	45	1.23×10^{-5}	100.3	h	160
			55	4.92×10^{-5}			160
			65	1.46×10^{-4}			160
	tert-Butyl per-2-ethylhexanoate	Benzene	72	1.9×10^{-5}		a	338
		Chlorobenzene	72	1.93×10^{-5}			344
			91	1.93×10^{-4}			344
			113	1.93×10^{-3}			344
			T (K)	1.54×10^{14}	exp[−124.90/RT]		344
	tert-Butyl 2-methoxy-2-methyl-perpropionate	Chlorobenzene	−14.0	2.05×10^{-5}	86.3	m_2	275
			−7.0	6.73×10^{-5}		m_2	275
			−1.0	1.67×10^{-4}		m_2	275
			15.0	1.28×10^{-3}		m_2	275
	tert-Butyl 4-methoxyperbenzoate	Phenyl ether	100.0	1.07×10^{-5}	149.8	a	53
			110.1	4.17×10^{-5}		a	53
			120.2	1.28×10^{-4}		a	53
			130.9	4.28×10^{-4}		a	53
	tert-Butyl peroxy-2-methyl-benzoate	Chlorobenzene	96	1.93×10^{-5}			344
			115	1.93×10^{-4}			344
			136	1.93×10^{-3}			344
			T (K)	8.56×10^{15}	expr[−145.97/RT]		344
	tert-Butyl 4-methoxyperbenzoate	Carbon tetrachloride	80	1.0×10^{-6}			354
	tert-Butyl 4-methylperbenzoate	Phenyl ether	100.0	9.42×10^{-6}	151.0	a	53
			110.1	3.19×10^{-5}		a	53
			120.2	1.06×10^{-4}		a	53

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	*tert*-Butyl 4-methylperbenzoate	Phenyl ether	130.9	3.25×10^{-4}		a	53
	tert-Butyl 2-methyl-2-*tert*-butylperoxyperpropionate	Benzene	9.0	6.66×10^{-5}	78.8	m_2	275
			19.0	1.99×10^{-4}			275
			30.0	6.94×10^{-4}			275
			42.0	2.43×10^{-3}			275
	tert-Butyl 2-methylsulfonyl perbenzoate	Chlorobenzene	105	6.68×10^{-6}			22
			120	5.57×10^{-5}			22
			135	2.76×10^{-4}			22
			150.6	2.05×10^{-3}			22
	tert-Butyl 2-(methylthio)perbenzoate	Chlorobenzene	60	8.08×10^{-4}	95.0	a,h	97
			39.4	2.59×10^{-5}		b	134
			50.1	2.42×10^{-4}		c	134
			50.2	1.88×10^{-4}		m_2	134
			69.8	1.96×10^{-3}			134
	tert-Butyl 4-(methylthio)perbenzoate	Chlorobenzene	120.4	1.75×10^{-4}			134
	tert-Butyl 4-nitrophenoxyperacetate	Ethylbenzene	70.5	6.7×10^{-5}	129.9	a,h	315
	tert-Butyl 4-nitrophenylperacetate	Chlorobenzene	79.6	2.5×10^{-5}	124.7	h	178
			90.7	8.9×10^{-5}			178
			100.5	2.83×10^{-4}			178
			80	3.77×10^{-5}			169
		Decane	77.5	6.30×10^{-6}			168
	tert-Butyl 4-nitrophenylperacetate	Dodecane	77.5	5.81×10^{-6}			168
			100	1.31×10^{-4}			168
		Hexadecane	77.5	5.11×10^{-6}			168
		Octane	77.5	6.42×10^{-6}			168
			100	1.60×10^{-4}			168
		Tetradecane	77.5	5.56×10^{-6}			168
	Di-*tert*-butyl per-2-chlorosuccinoate	Unknown	105	2.74×10^{-4}	102.9	x_2	214
				1.22×10^{-5}	143.5	x_2	214
			115	5.90×10^{-4}		x_2	214
				3.83×10^{-5}		x_2	214
			125	1.52×10^{-3}		x_2	214
				1.00×10^{-4}		x_2	214
	tert-Butyl peroctoate	Benzene	70	1.4×10^{-5}	130	a	337
			85	8.55×10^{-5}		a	337
			100	4.55×10^{-4}		a	337
		Decane	70	6.9×10^{-6}	130	a	337
			85	4.39×10^{-5}		a	337
			100	2.64×10^{-4}		a	337
	Di-*tert*-butyl persuccinoate	Styrene	105	1.93×10^{-5}	154.8	x_1	215
			115	6.7×10^{-5}		x_1	215
			125	2.53×10^{-4}		x_1	215
	tert-Butyl phenoxyperacetate	Ethylbenzene	70.5	4.4×10^{-4}	113.1	a,h	315
	tert-Butyl phenylperacetate	Chlorobenzene	60	6.79×10^{-6}	120.1	a,h	81
			77.0	6.85×10^{-5}	117.6	h	28
			88.6	2.45×10^{-4}		m_2	28
			79.6	1.05×10^{-4}	116.7	h	178
			90.7	3.53×10^{-4}			178
			100.5	1.003×10^{-3}			178
			79.6	1.07×10^{-4}		z (0.98)	157
			79.6	9.1×10^{-5}		z (4050)	157
			79.6	1.02×10^{-4}		z (0.98)	163
			79.6	9.0×10^{-4}		z (4050)	163
			85.0	1.945×10^{-4}		m_2	165
		Cumene	79.6	6.6×10^{-5}		z (0.98)	157
			79.6	5.6×10^{-5}		z (4050)	157
			79.6	6.78×10^{-5}		z (0.98)	163,207
			79.6	4.73×10^{-5}		z (6090)	163,207
			79.61	1.1×10^{-5}			274
		Decane	77.5	3.00×10^{-5}			168
		Dodecane	77.5	2.75×10^{-5}			168
		Hexadecane	77.5	2.60×10^{-5}			168
		Isooctane	85.0	1.08×10^{-4}			165
			95.2	3.527×10^{-4}			165

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
12 (*cont'd*)	*tert*-Butyl phenylperacetate	Octane	77.5	3.11×10^{-5}			168
		Paraffin oil	85.0	1.150×10^{-4}			165
		Tetradecane	77.3	2.66×10^{-5}			168
	tert-Butyl 2-propylperpenten-2-oate	Cumene	94.9	2.78×10^{-5}	137.2	h	172
	(*cis*)		100.1	5.4×10^{-5}		z (0.98)	172
			100.1	4.26×10^{-5}		z (1100)	172
			100.1	2.31×10^{-5}		z (4000)	172
			110.1	1.72×10^{-4}			172
	(*trans*)	Cumene	94.9	2.47×10^{-5}	143.9	h	172
			100.1	4.7×10^{-5}		z (0.98)	172
			100.1	3.13×10^{-5}		z (1100)	172
			100.1	1.42×10^{-5}		z (3830)	172
			110.1	1.64×10^{-4}			172
	tert-Butyl 2,2,3,3-tetramethyl-perpropionate	Cumene	42.2	8.76×10^{-6}	114	h	267
			54.9	5.84×10^{-5}			267
			70.3	3.36×10^{-4}			267
	tert-Butyl thiophenylperacetate	Ethylbenzene	70.5	3.8×10^{-4}		a	315
	tert-Butyl per-*o*-toluate	Benzene	97	1.9×10^{-5}		a	338
	tert-Butyl per-paratoluate	Cumene	84.0	1.660×10^{-6}			345
			99.3	1.139×10^{-5}			345
			115.0	6.77×10^{-5}			345
	tert-Butyl-*N*-(3-tolylperoxy)-carbamate	Toluene	64.0	4.58×10^{-5}	102.9	h	36
			70.7	9.17×10^{-5}			36
			78.0	2.03×10^{-4}			36
			88.5	5.78×10^{-4}			36
	tert-Butyl 2,4,5-trichloro-phenoxyperacetate	Ethylbenzene	70.5	5.8×10^{-5}		a	315
	4-Chlorocumyl perpropionate	Benzene	70	1.46×10^{-6}	129.3	a	213
			80	5.68×10^{-6}		a	213
			90	1.79×10^{-5}		a	213
		α-Methylstyrene	70	1.50×10^{-6}	118.8	a	213
			80	4.54×10^{-6}		a	213
			90	1.47×10^{-5}		a	213
	Cumyl perpropionate	Benzene	70	2.60×10^{-6}	101.7	a	213
			80	6.15×10^{-6}		a	213
			90	1.90×10^{-5}		a	213
		α-Methylstyrene	70	6.31×10^{-6}	106.7	a	213
			80	1.71×10^{-5}		a	213
			90	4.94×10^{-5}		a	213
	4-Iodocumyl perpropionate	Benzene	70	1.37×10^{-6}	133.1	a	213
			80	5.76×10^{-6}		a	213
			90	1.79×10^{-5}		a	213
		α-Methylstyrene	70	1.54×10^{-6}	125.5	a	213
			80	4.96×10^{-6}		a	213
			90	1.62×10^{-5}		a	213
	4-Nitrocumyl perpropionate	Benzene	70	1.36×10^{-6}	133.5	a	213
			80	5.32×10^{-6}		a	213
			90	1.78×10^{-5}		a	213
		α-Methylstyrene	70	1.35×10^{-6}	124.7	a	213
			80	4.07×10^{-6}		a	213
			90	1.43×10^{-5}		a	213
	1,1,3-Trimethylbutyl perpivalate	Cumene	40	4.80×10^{-6}			362
			50	1.77×10^{-5}			362
			60	6.18×10^{-5}			362
			70	2.344×10^{-4}			362
13	*tert*-Amyl per-2-ethylhexanoate	Benzene	70	1.9×10^{-5}		a	338
		Chlorobenzene	73	1.93×10^{-5}			344
			91	1.93×10^{-4}			344
			111	1.93×10^{-3}			344
			T (K)	1.77×10^{15}	$\exp[-132.11/RT]$		344
	Benzyl(*tert*-butylperoxy)oxalate	Benzene	45	3.65×10^{-5}	111.3	c,h	31
			55	1.33×10^{-4}		c	31
			65	4.69×10^{-4}		c	31

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
13 (*cont'd*)	*tert*-Butyl benzyloxyperacetate	Ethylbenzene	40.0	1.3×10^{-4}	100.6	a,h	315
			40.0	1.45×10^{-4}	101.8	h,m_2	328
			51.0	5.03×10^{-4}			328
			60.7	1.62×10^{-3}			328
	tert-Butyl benzylthioperacetate	Ethylbenzene	40.0	2.60×10^{-5}	108.1	h,m_2	328
			60.0	3.31×10^{-4}			328
			70.0	1.14×10^{-3}			328
	tert-Butyl bicyclo[2.2.2]octane-1-percarboxylate	Cumene	65	7.5×10^{-6}	119.2	h	160
			75	2.82×10^{-5}			160
			85	8.47×10^{-5}			160
			80	5.1×10^{-5}	120.1	h	223
			90	1.67×10^{-4}			223
			100	4.86×10^{-4}			223
	tert-Butyl endo-bicyclo[2.2.1]-2-methylheptane-2-percarboxylate	Cumene	75	7.35×10^{-5}	126.8	h	206
			80	1.39×10^{-4}			206
			85	2.64×10^{-4}			206
			90	4.81×10^{-4}			206
			95	8.41×10^{-4}			206
	tert-Butyl exo-bicyclo[2.2.1]-2-methylheptane-2-percarboxylate	Cumene	60	7.78×10^{-5}	116.7	h	206
			65	1.44×10^{-4}			206
			70	2.75×10^{-4}			206
			75	4.89×10^{-4}			206
			80	8.93×10^{-4}		c	206
	tert-Butyl 1,4-dimethylcyclohexane-1-percarboxylate (*cis*)	Cumene	60.0	7.52×10^{-5}	115.2	h	206
			70.0	2.66×10^{-4}			206
			80.0	8.49×10^{-4}		c	206
	(*trans*)	Cumene	60.0	6.93×10^{-5}	116.3	c,h	206
			70.0	2.52×10^{-4}		c	206
			80.0	7.79×10^{-4}			206
	tert-Butyl 3-methoxyphenyl-peracetate	Chlorobenzene	79.6	9.9×10^{-5}	122.2	h	178
			90.7	3.45×10^{-4}			178
			100.5	1.051×10^{-3}			178
	tert-Butyl 4- methoxyphenyl-peracetate	Chlorobenzene	56.0	4.57×10^{-5}	105.4	h	178
			60.3	9.9×10^{-5}			178
			70.2	3.06×10^{-4}			178
			79.3	7.99×10^{-4}			178
			48.9	1.93×10^{-5}		k	249
		Cumene	60	6.85×10^{-5}		z (0.98)	207
			60	5.31×10^{-5}		z (4050)	207
			79.6	5.942×10^{-4}	207		
		Decane	77.5	2.52×10^{-4}			168
		Dodecane	77.5	2.60×10^{-4}			168
		Ethylbenzene	70.5	1.4×10^{-3}	104.8	a,h	315
		Isooctane	60.5	4.72×10^{-5}			169
		Octane	77.5	2.53×10^{-4}			168
		Paraffin oil	60.5	4.20×10^{-5}			169
		Tetradecane	77.5	2.68×10^{-4}			168
	tert-Butyl 4-methylphenyl peracetate	Chlorobenzene	70.4	8.67×10^{-5}	110.9	h	178
			79.6	2.37×10^{-4}			178
			90.7	7.95×10^{-4}			178
		Cumene	79.6	1.649×10^{-4}		z (0.98)	207
			79.6	1.180×10^{-4}		z (6080)	207
		Decane	77.5	9.08×10^{-5}			168
		Dodecane	77.5	8.57×10^{-5}			168
		Ethylbenzene	70.5	7.2×10^{-4}		a	315
		Hexadecane	77.5	1.032×10^{-4}			168
		Octane	77.5	8.83×10^{-5}			168
		Tetradecane	77.5	7.75×10^{-5}			168
	Di-*tert*-butyl perglutarate	Styrene	105	2.65×10^{-5}		x_1	215
			115	1.06×10^{-4}		x_1	215
			125	3.00×10^{-4}			215
	tert-Butylperoxy 2-ethylhexyl-carbonate	Chlorobenzene	99	1.93×10^{-5}			344
			117	1.93×10^{-4}			344
			137	1.93×10^{-3}			344

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
13 (*cont'd*)	*tert*-Butylperoxy 2-ethylhexyl-carbonate	Chlorobenzene	T (K)	4.07×10^{16}	exp[−151.72/RT]		344
	tert-Butyl 2-phenylperpropionate	Cumene	40.7	7.88×10^{-6}			273
			60.1	8.44×10^{-5}			273
			80.1	8.10×10^{-4}			273
	tert-Butyl 2,2,4,4-tetra-methylperbutyrate	Cumene	40.8	6.15×10^{-6}	111	h	267
			54.7	3.82×10^{-5}			267
			70.7	2.73×10^{-4}			267
	tert-Butylthiobenzylperacetate	Ethylbenzene	40.0	2.62×10^{-5}	100.6	a,h	315
	tert-Butyl peroxy-3,5,5-tri-methylhexanoate	Chlorobenzene	95	1.93×10^{-5}			344
			114	1.93×10^{-4}			344
			135	1.93×10^{-3}			344
			T (K)	1.94×10^{15}	exp[−140.78/RT]		344
	4-Nitrobenzyl(*tert*-butylperoxy)-oxalate	Benzene	45	1.30×10^{-5}	116.7	c,h	31
			55	4.89×10^{-5}		c	31
			65	1.89×10^{-4}		c	31
14	*tert*-Amylperoxy 2-ethylhexyl-carbonate	Chlorobenzene	95	1.93×10^{-5}			344
			113	1.93×10^{-4}			344
			134	1.93×10^{-3}			344
			T (K)	2.22×10^{16}	exp[−148.41/RT]		344
	tert-Butyl di-*tert*-butylperacetate	Cumene	54.7	4.38×10^{-6}	125.3	h	245
			70.2	3.56×10^{-5}			245
			85.4	2.44×10^{-4}			245
	tert-Butyl 2-carbomethoxyphenyl-peracetate	Cumene	70.8	1.48×10^{-5}	116.9	h	257
			85.4	8.07×10^{-5}			257
			101.0	4.38×10^{-4}			257
	tert-Butyl 2-(4-chlorophenoxy)-2-methylperpropionate	Ethylbenzene	40.0	7.2×10^{-4}		a	315
	tert-Butyl 2-(4-chlorophenyl)-2-methylperpropionate	Acetonitrile	50.0	2.55×10^{-4}			265
		Benzene	50.0	2.03×10^{-4}			265
		Dodecane	50.0	7.98×10^{-5}	108.1	h	265
			60.0	2.68×10^{-4}			265
			67.0	6.36×10^{-4}			265
		Hexadecane	50.0	8.00×10^{-5}			265
			60.0	2.56×10^{-4}			265
			67.0	5.96×10^{-4}			265
		Octane	50.0	8.10×10^{-5}			265
			60.0	2.65×10^{-4}			265
			67.0	5.95×10^{-4}			265
	tert-Butyl α-methylpercinnamate (*cis*)	Cumene	99.6	1.23×10^{-4}	128.0	f,h	150
			109.8	3.70×10^{-4}		f	150
	(*trans*)	Cumene	99.6	3.5×10^{-5}	125.5	f,h	150
			109.8	1.03×10^{-4}		f	150
	tert-Butyl 2-methyl-2-phenylper-propionate	Chlorobenzene	34.1	3.26×10^{-5}			249
		Cumene	39.7	5.63×10^{-5}			249
	tert-Butyl perneodecanoate	Benzene	47	1.9×10^{-5}			338
		Chlorobenzene	46	1.93×10^{-5}			344
			64	1.93×10^{-4}			344
			84	1.93×10^{-3}			344
			T (K)	1.52×10^{14}	exp[−115.47/RT]		344
	tert-Butyl 2-(4-nitrophenyl)-2-methylperpropionate	Acetonitrile	50.0	7.50×10^{-5}	105.2	h	265
		Benzene	50.0	6.40×10^{-5}			265
		Dodecane	50.0	2.63×10^{-5}			265
			60.0	9.27×10^{-5}			265
			67.0	2.03×10^{-4}			265
		Octane	67.0	1.98×10^{-4}			265
	tert-Butyl permesitoate	Cumene	84.0	1.464×10^{-5}			345
			99.3	9.94×10^{-5}			345
			115.0	6.7×10^{-5}			345
	tert-Butyl 1-phenylcyclo-propylpercarboxylate	*n*-Dodecane	113.0	8.67×10^{-5}			255
		Ethylbenzene	120.0	2.49×10^{-4}	147.5	h,t_1	323
			130.0	7.49×10^{-4}		t_1	323
			140.0	2.22×10^{-3}		t_1	323

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
14 (*cont'd*)	*tert*-Butyl 1-phenylcyclo-propylpercarboxylate	2,2,4-Trimethyl pentane	102.6	2.53×10^{-5}	141.2	h	255
			113.0	8.96×10^{-5}			255
			120.4	2.04×10^{-4}			255
	tert-Butyl phenyldimethylperacetate	Chlorobenzene	60	9.6×10^{-4}	109.2	c,h	81
		Isooctane	40.6	2.95×10^{-5}		t_9	165
			60.6	3.059×10^{-4}		t_9	165
			60.6	4.064×10^{-4}		m_2	165
	tert-Butyl 2-phenyl-3-perbutenoate	Chlorohenzene	60	2.9×10^{-3}	96.2	c,h	81
	tert-Butyl 4-phenyl-3-perbutenoate	Chlorobenzene	60	1.15×10^{-4}	98.3	c,h,g	81
	tert-Butyl 2-phenyl-2-methyl-perpropionate	Acetonitrile	50.0	3.08×10^{-4}			265
		Benzene	50.0	2.18×10^{-4}			265
		Benzophenone	60	7.73×10^{-4}			334
		Benzyl benzoate	60	7.92×10^{-4}			334
		Butyl phthalate	60	6.5×10^{-4}			334
		Chlorobenzene	60	7.00×10^{-4}			334
		Cumene	60	5.25×10^{-4}			334
		Dodecane	50.0	8.77×10^{-5}	109.3	h	265
			60.0	3.00×10^{-4}			265
			67.0	7.13×10^{-4}			265
			60	2.88×10^{-4}			334
		Hexadecane	60	3.53×10^{-4}			334
		Methyl benzoate	60	7.55×10^{-4}		t_{12}	334
		Octyl phthalate	60	5.70×10^{-4}			334
		Toluene	60	5.66×10^{-4}			334
	tert-Butyl 4-vinylphenylperacetate	Cumene	70.9	5.54×10^{-5}	116.5	h,k	254
			85.4	3.00×10^{-4}		k	254
			100.3	1.48×10^{-3}			254
	Dibenzyl peroxalate	Pentane	25	6.7×10^{-5}			203
	Di-*tert*-butyl adipate	Styrene	105	3.50×10^{-5}	151.0	x_1	215
			115	1.23×10^{-4}		x_1	215
			125	3.05×10^{-4}		x_1	215
	Dicyclohexyl peroxydicarbonate	Benzene	50	5.4×10^{-5}		c	201
		Chlorobenzene	44	1.93×10^{-5}			344
			60	1.93×10^{-4}			344
			77	1.93×10^{-3}			344
			T (K)	3.3×10^{16}	exp[−129.0/RT]		344
		α-Methylstyrene	50	5.9×10^{-5}			201
		Mineral spirits	45	2.78×10^{-5}	117	a	337
			50	5.58×10^{-4}		a	337
			60	2.07×10^{-4}		a	337
	4-Methoxybenzyl (*tert*-butylperoxy)-oxalate	Benzene	45	6.69×10^{-5}	109.6	c,h	31
			55	2.48×10^{-4}		c	31
			65	8.27×10^{-4}		c	31
	1-Methyl-1-phenylethyl 2,2-dimethyl-peroxyporpionate	Cumene	40	6.91×10^{-6}			368
			50	2.80×10^{-5}			368
			60	1.032×10^{-4}			368
			70	3.57×10^{-4}			368
15	*tert*-Amyl perneodecanoate	Benzene	45	1.9×10^{-5}		a	338
		Chlorobenzene	43	1.93×10^{-5}			344
			61	1.93×10^{-4}			344
			81	1.93×10^{-3}			344
			T (K)	1.47×10^{14}	exp[−114.38/RT]		344
	Dibenzoyl monopercarbonate	Benzene	60.00	8.05×10^{-5}	116.5	h	280
	tert-Butyl 1-adamantyl-percarboxylate	Cumene	45	5.15×10^{-5}	115.5	h	160
			55	2.05×10^{-5}			160
			65	7.40×10^{-5}			160
			64.6	6.87×10^{-5}	116.7	h	159
			74.6	2.44×10^{-4}			159
			84.6	7.78×10^{-4}			159
			60	5.4×10^{-5}	124.7	h	223
			70	2.12×10^{-4}			223
			80	7.20×10^{-4}			223

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
15 (*cont'd*)	*tert*-Butyl 4-(carboethoxy)-phenoxyperacetate	Ethylbenzene	70.5	1.6×10^{-4}		a	315
	tert-Butyl α,β-dimethyl-percinnamate (*cis*)		95.0	1.02×10^{-4}	118.8	a,h	151
			103.2	2.29×10^{-4}		a	151
			110.1	5.15×10^{-4}		a	151
	(*trans*)		85.1	3.9×10^{-5}	122.6	a,h	151
			95.0	1.17×10^{-4}		a	151
			103.7	2.84×10^{-4}		a	151
			110.0	6.05×10^{-4}		a	151
			110.0	7.90×10^{-4}		a,v_6	151
	tert-Butyl 1-pernaphthoate	Chlorobenzene	110.1	8.69×10^{-5}			154
	tert-Butyl 2,2-dimethyl-3-phenylthio perpropionate	Cyclohexane	55	1.63×10^{-5}	116.9	h,t_6	320
			65	1.61×10^{-5}			320
			75	1.98×10^{-4}		t_6	320
	tert-Butyl 2-(4-methoxyphenyl)-2-methylperpropionate	Acetonitrile	50	1.675×10^{-3}			265
		Benzene	50	9.96×10^{-4}			265
		Dodecane	40	1.49×10^{-4}	93.0	h	265
			50	4.21×10^{-4}			265
			57	1.028×10^{-3}			265
		Octane	50	4.18×10^{-4}			265
	tert-Butyl 1-phenylcyclo-*n*-Dodecane butylpercarboxylate	52.3	7.89×10^{-5}			255	
		Ethylbenzene	60.0	3.62×10^{-4}		t_1	323
		n-Hexadecane	52.3	8.80×10^{-5}			255
		2,2,4-trimethyl pentane	52.3	7.69×10^{-5}	110.6	h	255
			61.6	2.46×10^{-4}			255
	tert-Butyl 4-*tert*-butylperbenzoate	Chlorobenzene	100.1	3.81×10^{-5}			134
			119.8	4.50×10^{-4}			134
			135.9	2.38×10^{-3}			134
	Di-*tert*-butyl perpimelate	Styrene	105	4.45×10^{-5}	150.6	x_1	215
			115	1.05×10^{-4}		x_1	215
			125	5.06×10^{-4}		x_1	215
	tert-Butyl 2-(4-tolyl)-2-methylperpropionate	Acetonitrile	50.0	6.00×10^{-4}			265
		Benzene	50.0	3.85×10^{-4}			265
		Dodecane	50.0	1.90×10^{-4}	100.6	h	265
			60.0	5.58×10^{-4}			265
			67.0	1.320×10^{-3}			265
		Octane	60.0	5.43×10^{-4}			265
	tert-Butyl triisopropylperacetate	Cumene	24.8	5.32×10^{-6}	111.5	h	267
			41.6	6.84×10^{-5}			267
			55.2	3.78×10^{-4}			267
	1,1-Dimethyl-2-phenylethyl-perpivalate	Cumene	40	4.56×10^{-6}			362
			50	1.833×10^{-5}			362
			60	6.1×10^{-5}			362
			70	2.51×10^{-4}			362
	1-Phenylethyl 4-methoxyperacetate	Benzene	99.7	3.50×10^{-4}	134.1	h	259
	1-Phenylethyl 3-nitroperbenzoate	Benzene	100	5.97×10^{-4}	136.2	h	259
	1-Phenylethyl 4-nitroperbenzoate	Benzene	100	4.10×10^{-4}	146.2	h	259
	1-Phenylethyl perbenzoate	Benzene	100	4.03×10^{-4}	123.6	h	259
16	Dibenzyl peroxydicarbonate	Benzene	40	2.0×10^{-5}	117	a	337
			50	6.25×10^{-5}		a	337
			60	2.67×10^{-4}		a	337
	Di-*tert*-butyl 2-iodoperisophthalate	Chlorobenzene	68.4	8.24×10^{-5}	93.9	h	248
			85.4	4.45×10^{-4}			248
			101.6	1.68×10^{-3}			248
	Di-*tert*-butyl perphthalate	Benzene	100	1.1×10^{-5}	159	a	337
			115	7.81×10^{-5}		a	337
			130	4.81×10^{-4}		a	337
		Di-*n*-butyl phthalate	100	1.08×10^{-5}	155.0	h	269
			115	7.8×10^{-5}			269
			130	4.8×10^{-4}			269
		Chlorobenzene	107	1.93×10^{-5}			344
			125	1.93×10^{-4}			344
			146	1.93×10^{-3}			344
			T (K)	9.39×10^{16}	exp[− 157.85/RT]		344

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
16 (*cont'd*)	Di-*tert*-butyl perphthalate	Cumene	99.9	8.34×10^{-6}	140.8	h	269
			115.0	4.68×10^{-5}			269
			130.1	2.60×10^{-4}			269
	tert-Butyl 1-phenyl-1-cyclopentane percarboxylate	Ethylbenzene	60.0	1.52×10^{-3}		t_1	323
	tert-Butyl 2,3-dimethyl-2-phenylperbutyrate	Ethylbenzene	50	4.30×10^{-5}			324
	Cumyl *N*-phenylperoxy carbamate	Xylene	T (K)	1.26×10^{14}	exp[−117.6/RT]		40
	Dibenzyl peroxydicarbonate	Toluene	50	2.92×10^{-5}		a	90
	Di-*tert*-butyl perphthalate	Benzene	100	1.08×10^{-5}	157.7	a	126
			115	7.83×10^{-5}		a	126
			130	4.80×10^{-4}		a	126
	Di-*tert*-butyl persuberate	Styrene	115	7.86×10^{-5}	159.0	x_1	215
			125	2.81×10^{-4}		x_1	215
	tert-Butyl 1-phenylcyclopentyl-percarboxylate	2,2,4-Trimethyl-pentane	43.5	1.184×10^{-4}	103.5	h	255
			52.3	3.51×10^{-4}			255
	1,1,3,3-Tetramethylbutyl-per-2-ethylhexanoate	Benzene	60	1.9×10^{-5}	101	a	337
			70	5.30×10^{-5}		a	337
			80	1.38×10^{-4}		a	337
	2-Phenylazo-2-propylperbenzoate	Chlorobenzene	10	4.5×10^{-4}			260
			35	1.9×10^{-3}			260
		p-Chlorotoluene	10	6.4×10^{-4}			260
		Chloroform-d_1	10	5.1×10^{-4}			260
	1-Phenylethyl 4-dimethyl-aminoperbenzoate	Benzene	100	2.16×10^{-3}	111.5	h	259
17	Di-*tert*-butylperhomoterephthalate	Cumene	79.6	2.38×10^{-5}			272
		Toluene	70.6	9.44×10^{-6}	122.3	h	272
			80.3	2.80×10^{-5}			272
			100.5	3.09×10^{-4}			272
	tert-Butyl 1-phenylcyclo-hexanepercarboxylate	Ethylbenzene	60.0	1.90×10^{-3}			323
		2,2,4-Trimethyl-pentane	43.5	1.27×10^{-4}	107.7	h	255
			52.3	3.91×10^{-4}			255
			61.6	1.218×10^{-3}			255
	tert-Butyl 2-(phenylthio)perbenzoate	Acetone	25	1.89×10^{-5}		m_2	22
			40	1.22×10^{-4}		m_2	22
		Acetonitrile	25	1.01×10^{-4}		m_2	22
			40	5.44×10^{-4}		m_2	22
		tert-Butanol	25	5.26×10^{-5}		m_2	22
			40	3.24×10^{-4}		m_2	22
		Chlorobenzene	39.3	4.98×10^{-5}		b	134
			53.8	5.23×10^{-4}			134
			53.8	4.58×10^{-4}		m_2	134
			70.0	2.62×10^{-3}			134
			25	1.76×10^{-5}		m_2	22
			40	1.03×10^{-4}		m_2	22
		Cyclohexane	25	9.8×10^{-7}		m_2	22
			40	6.9×10^{-6}		m_2	22
		DMSO	25	1.11×10^{-4}		m_2	22
			40	6.02×10^{-4}		m_2	22
		Ethanol	25	2.31×10^{-4}		m_2	22
			40	1.65×10^{-3}		m_2	22
		Isopropanol	25	1.33×10^{-4}		m_2	22
			40	7.25×10^{-4}		m_2	22
		Methanol	25	8.21×10^{-4}		m_2	22
			40	4.75×10^{-3}		m_2	22
	tert-Butyl dibenzothiophene-4-per-carboxylate	Chlorobenzene	105	2.79×10^{-5}	128.0	h	22
			119.4	1.29×10^{-4}			22
			135.1	6.07×10^{-4}			22
	Di-*tert*-butyl perazelate	Styrene	105	5.13×10^{-5}	154.8		215
			115	1.20×10^{-4}			215
			125	5.10×10^{-4}			215
		Chlorobenzene	98	1.93×10^{-5}			344
			116	1.93×10^{-4}			344
			137	1.93×10^{-3}			344
			T (K)	1.35×10^{16}	exp[−147.95/RT]		344

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
18	*tert*-Butyl 4-benzoylperbenzoate	Benzene	25	1.84×10^{-4}		p_2	279
		Chlorobenzene	110	2.88×10^{-5}			279
	tert-Butyl diphenylperacetate	Chlorobenzene	60	4.44×10^{-4}	101.7	a,h	81
		Cumene	40.3	2.75×10^{-5}	104.6	h	136
			49.6	9.81×10^{-5}			136
			59.9	3.10×10^{-4}			136
			70.4	1.04×10^{-3}			136
		Decane	77.5	1.66×10^{-3}			168
		Dodecane	77.5	1.688×10^{-3}			168
		Hexadecane	77.5	1.632×10^{-3}			168
		Octane	77.5	1.651×10^{-3}			168
		tetradecane	77.5	1.627×10^{-3}			168
	2,4,4-Trimethylpentyl-2-perneodecanoate	Chlorobenzene	40	1.93×10^{-5}			344
			57	1.93×10^{-4}			344
			76	1.93×10^{-3}			344
			T (K)	3.97×10^{14}	exp[−115.79/RT]		344
	α,ω-Di-*O*,*O*-*tert*-butyl-(bis-persuccinyl)ethylene glycol	Ethylbenzene	95	2.68×10^{-6}	147.5		306
			105	9.10×10^{-6}			306
			115	2.97×10^{-5}			306
	tert-Butyl l-phenyl-1-cycloheptanepercarboxylate	Ethylbenzene	60.0	2.75×10^{-3}			323
	Di-*tert*-butyl *p*-phenylenediperacetate	Cumene	70.9	3.69×10^{-5}	112.7	h,k	254
			85.4	2.08×10^{-4}		k	254
			100.3	8.83×10^{-4}		k	254
	tert-Butyl 2-(phenylthiomethyl)-perbenzoate	Chlorobenzene	98.7	4.64×10^{-5}	134.7	h	154
			120.1	4.20×10^{-4}			154
	tert-Butyl thioxanthone-4-percarboxylate	Chlorobenzene	120	1.89×10^{-4}	156.5	h,m_2	22
			135	1.15×10^{-3}		m_2	22
	Di-*tert*-butyl persebacate	Styrene	105	5.10×10^{-5}	154.8	x_1	215
			115	1.20×10^{-4}		x_1	215
			125	5.00×10^{-4}		x_1	215
	Di-(2-ethylhexyl)peroxydicarbonate	Benzene	40	1.9×10^{-5}		a	338
		Chlorobenzene	47	1.93×10^{-5}			344
			64	1.93×10^{-4}			344
			83	1.93×10^{-3}			344
			T (K)	1.83×10^{15}	exp[−122.45/RT]		344
		Decane	30	5.3×10^{-6}	105	a	337
			45	3.31×10^{-5}		a	337
			60	2.32×10^{-4}		a	337
		Mineral spirits	40	1.3×10^{-5}	126	a	337
			50	6.14×10^{-5}		a	337
			60	2.31×10^{-4}		a	337
	Di-*n*-heptyl persuccinate	Styrene	60	6.1×10^{-6}	131.0	x_1	215
			73.5	3.85×10^{-5}		x_1	215
			85	1.70×10^{-4}		x_1	215
19	*tert*-Butyl diphenylmethylperacetate	Chlorobenzene	60	1.9×10^{-3}	103.3	a,h	81
	tert-Butyl diphenylperglycidate (*cis*)	Cumene	60.0	2.8×10^{-5}	112.1	h	204
			70.0	5.5×10^{-5}			204
			80.0	1.68×10^{-4}			204
	(*trans*)	Cumene	60.0	1.6×10^{-5}	117.2	h	204
			70.0	3.9×10^{-5}			204
			80.0	1.20×10^{-4}			204
	tert-Butyl 2,2-diphenylperpropionate	Chlorobenzene	34.1	7.76×10^{-5}			249
	tert-Butyl 4-hydroxy-3,5-di-(*tert*-butyl)perbenzoate	Benzene	100	3.18×10^{-5}		c,e	246
		Carbon tetrachloride	100	3.60×10^{-5}		c,e	246
	tert-Butyl 4-(4-methylbenzoyl)-perbenzoate	Benzene	25	1.45×10^{-4}		p_2	279
		Chlorobenzene	110	3.03×10^{-5}			279

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TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
19 (*cont'd*)	*tert*-Butyl α-phenylpercinnamate	Cumene	99.6	1.52×10^{-4}	136.8	h	150
	(*cis*)		109.8	4.41×10^{-4}			150
	(*trans*)	Cumene	99.6	8.2×10^{-5}	133.9	h	150
			109.8	2.59×10^{-4}			150
	Cumyl perneodecanoate	Benzene	39	1.9×10^{-5}		a	338
		Chlorobenzene	39	1.93×10^{-5}			344
			56	1.93×10^{-4}			344
			75	1.93×10^{-3}			344
			T (K)	3.12×10^{14}	exp[− 114.59/RT]		344
20	Di-*n*-heptyl peradipate	Styrene	60	1.02×10^{-5}	133.1	x_1	215
			73.5	6.57×10^{-5}		x_1	215
			85	2.98×10^{-4}		x_1	215
	tert-Butyl 2-carbobenzoxy-phenylperacetate	Cumene	71.0	8.84×10^{-6}	129.5	h	257
			85.4	5.46×10^{-5}			257
			102.4	4.23×10^{-4}			257
	tert-Butyl 2-methyl-2-phenyl-perhexanoate	Ethylbenzene	50	5.01×10^{-5}			324
	α,ω-Di-*O*,*O*-*tert*-butyl-(bis-persuccinyl)diethylene glycol	Ethylbenzene	95	2.34×10^{-6}	155.0	h	306
			105	8.88×10^{-6}			306
			115	3.27×10^{-5}			306
21	*tert*-Butyl 8-(phenylthio)-1-pernaphthoate	Chlorobenzene	50	3.83×10^{-5}	99.2	h,m_2	154
			70	3.44×10^{-4}		m_2	154
			70	3.12×10^{-4}			154
			80	9.82×10^{-4}		m_2	154
	tert-Butyl 4-(2,4,6-trimethyl-benzoyl)perbenzoate	Benzene	25	1.34×10^{-4}		p_2	279
		Chlorobenzene	110	1.80×10^{-5}			279
22	*tert*-Butyl 2,2-diphenyl-3,3-dimethylperbutanoate	Toluene (?)	30	2.76×10^{-5}	114.8	j_2	250
			39.7	1.15×10^{-4}		j_2	250
			50	5.45×10^{-4}		k	250
	Di-*tert*-butyl 2-thiophenyl-perisophthalate	Chlorobenzene	12.1	1.68×10^{-4}	73.7	h	248
			25.0	6.45×10^{-4}			248
			40.0	3.09×10^{-3}			248
	2,5-Dimethylhexyl 2,5-di(peroxybenzoate)	Benzene	100	1.87×10^{-5}	154.0	a	126,337
			115	1.25×10^{-4}		a	126,337
			130	7.14×10^{-4}		a	126,337
	Di-(4-*tert*-butylcyclohexyl)-peroxydicarbonate	Chlorobenzene	48	1.93×10^{-5}			344
			64	1.93×10^{-4}			344
			82	1.93×10^{-3}			344
			T (K)	7.44×10^{15}	exp[− 126.39/RT]		344
23	*tert*-Butyl 2,6-di(phenylthio)-perbenzoate	Chlorobenzene	40.1	7.07×10^{-5}			154
	tert-Butylperoxy stearyl carbonate	Chlorobenzene	98	1.93×10^{-5}			344
			117	1.93×10^{-4}			344
			137	1.93×10^{-3}			344
			T (K)	3.94×10^{16}	exp[− 151.59/RT]		344
	tert-Butyl 2,4,6-tri-*tert*-butyl-perbenzoate	Cumene	70.4	5.84×10^{-6}			345
			84.0	3.54×10^{-5}			345
			99.3	2.67×10^{-4}			345
			115.0	1.656×10^{-3}			345
24	*tert*-Butyl triphenylperacetate	Chlorobenzene	34.9	7.8×10^{-4}			153
		Cumene	25.7	1.7×10^{-4}	100.8	h	153
			34.9	5.8×10^{-4}			153
			45.5	2.3×10^{-3}			153
	Di-*tert*-butyl 2,3-diphenyl-persuccinate	Cumene	70.1	9.83×10^{-5}	125.5	c,h	196
			79.9	3.29×10^{-4}		c	196
			90.0	1.20×10^{-3}		c	196
	Di-*n*-heptyl persebacate	Styrene	60	1.17×10^{-5}	131.0	x_1	215
			73.5	7.13×10^{-5}		x_1	215
			85	3.03×10^{-5}		x_1	215
	2,5-Dimethyl-2,5-bis(2-ethyl-hexanoylperoxy)hexane	Benzene	60	6.7×10^{-6}	142	a	337
			70	2.92×10^{-5}		a	337
			80	1.14×10^{-4}		a	337
		Chlorobenzene	68	1.93×10^{-5}			344
			86	1.93×10^{-4}			344

TABLE 6. *cont'd*

Number of C atoms	Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
24 (*cont'd*)	2,5-Dimethyl-2,5-bis(2-ethyl-hexanoylperoxy)hexane	Chlorobenzene	106	1.93×10^{-3}			344
			T (K)	2.19×10^{15}	exp[−130.88/RT]		344
25	*tert*-Butyl 2-(2,2-diphenylvinyl)-perbenzoate	Chlorobenzene	90.0	1.32×10^{-4}			175
			90.3	7.45×10^{-5}		t_9	175
			100.0	3.27×10^{-4}			175
			105.5	4.64×10^{-4}		t_9	175
			119.3	1.54×10^{-3}		t_9	175
		Cyclohexane	90.0	3.5×10^{-5}			175
		Methanol	90.4	2.3×10^{-3}		t_9	175
	tert-Butyl 2-percarboxybenzal-fluorene	Chlorobenzene	90	3.75×10^{-4}			175
		Methanol	90	2.47×10^{-3}			175
26	α,ω-Di-*O*,*O*-caprylyl-(bis succinyl)ethylene glycol	Ethylbenzene	65	1.48×10^{-5}	120.7	h	306
			75	4.10×10^{-5}			306
			85	2.19×10^{-4}			306
			95	6.92×10^{-4}			306
30	α,ω-Di-*O*,*O*-caprylyl-(bis-persuccinyl)triethylene glycol	Ethylbenzene	65	2.74×10^{-5}	116.9		306
			75	8.47×10^{-5}			306
			85	2.95×10^{-4}			306
			95	8.92×10^{-4}			306
	Dimyristyl peroxydicarbonate	Chlorobenzene	48	1.93×10^{-5}			344
			65	1.93×10^{-4}			344
			84	1.93×10^{-3}			344
			T (K)	2.82×10^{15}	exp[−124.10/RT]		344
34	Dicetyl peroxydicarbonate	Chlorobenzene	48	1.93×10^{-5}			344
			65	1.93×10^{-4}			344
			84	1.93×10^{-3}			344
			T (K)	3.02×10^{15}	exp[−124.30/RT]		344
36	α,ω-Di-*O*,*O*-heptanoyl-(bis-perphthaloyl)triethylene glycol	Ethylbenzene	65	5.02×10^{-5}	124.4		306
			75	1.420×10^{-4}			306
			85	5.19×10^{-4}			306
			95	1.410×10^{-3}			306
42	α,ω-Di-*O*,*O*-caprylyl-(bis-persuccinyl)nonaethylene glycol	Ethylbenzene	65	1.66×10^{-5}	116.9		306
			75	6.57×10^{-5}			306
			85	1.78×10^{-4}			306
			95	9.72×10^{-4}			306
46	α,ω-Di-*O*,*O*-heptanoyl-(bis-perphthaloyl)tridecaethylene glycol	Ethylbenzene	75	1.130×10^{-4}	122.3		306
			85	3.50×10^{-4}			306
			95	8.80×10^{-4}			306
50	α,ω-Di-*O*,*O*-caprylyl-(bis-persuccinyl)tridecaethylene glycol	Ethylbenzene	65	1.28×10^{-5}	128.2		306
			75	4.46×10^{-5}			306
			85	1.91×10^{-4}			306

TABLE 7. MISCELLANEOUS INITIATORS

Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
Acetyl cyclohexyl sulfonyl peroxide	Benzene	30	1.4×10^{-5}	147	a	337
		40	9.42×10^{-5}		a	337
		45	2.10×10^{-4}		a	337
N-(1-Cyanocyclohexyl)pentamethylene keteneimine	Chlorobenzene	80.0	3.25×10^{-6}			179
		89.2	1.007×10^{-5}			179
		100.1	4.025×10^{-5}			179
Dibenzyl hyponitrite	Paraffin	61.5	6.5×10^{-4}			240
		68.5	3.7×10^{-3}			240
		75.5	8.3×10^{-3}			240
		80.5	1.45×10^{-2}			240
		132	8.7×10^{-1}			240

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TABLE 7. *cont'd*

Initiator	Solvent	T (°C)	k_d (s^{-1})	E_a (kJ/mol)	Notes	Refs.
Di-*tert*-butyl hyponitrite	*tert*-Butanol	65	4.68×10^{-4}			317
	n-Butyl ether	65	4.50×10^{-4}			317
	Ethanol (95%)	65	3.93×10^{-4}			317
	Isooctane	45	2.72×10^{-5}	116.9	h	317
		55	1.07×10^{-4}			317
		65	4.00×10^{-4}			317
		75	1.330×10^{-3}			317
	Nujol	65	5.17×10^{-4}			317
Peroxybenzoylnitrate	Vapor	7.6	2.79×10^{-6}	69.1	h	261
		19.9	1.05×10^{-5}			261
		30.4	2.02×10^{-5}			261
		40.6	3.64×10^{-5}			261
		50.1	1.21×10^{-4}			261
		60.5	2.74×10^{-4}			261
		T (K)	1×10^{7}	$\exp[-69.1/RT]$		261
	Vapor + NO	18.1	8.18×10^{-5}	108.5	h	261
		30.4	3.62×10^{-4}			261
		40.6	1.42×10^{-3}			261
		T (K)	1.6×10^{15}	$\exp[-108.5/RT]$		261
	Vapor + NO	30.4	1.34×10^{-5}	79.2	h	261
		50.2	7.22×10^{-5}			261
		68.5	3.41×10^{-4}			261
		T (K)	6.3×10^{-8}	$\exp[-79.2/RT]$		261
2,3-Dimethyl-2,3-diphenylbutane	Chlorobenzene	237	1.93×10^{-5}			344
		259	1.93×10^{-4}			344
		284	1.93×10^{-3}			344
		T (K)	7.34×10^{18}	$\exp[-230.19/RT]$		344
3,4-Dimethyl-3,4-diphenylhexane	Chlorobenzene	201	1.93×10^{-5}			344
		226	1.93×10^{-4}			344
		254	1.93×10^{-3}			344
		T (K)	1.45×10^{15}	$\exp[-180.32/RT]$		344
1,4-Dimethyl-1,4-diphenyl-tetrazene-2	Benzophenone	121	3.3×10^{-4}			240
		149	3.45×10^{-3}			240
	Cumene	120	2.3×10^{-4}			240
		130	6.5×10^{-4}			240
		140	1.67×10^{-3}			240
	Paraffin	164.2	2.6×10^{-2}			240
		174.3	5.6×10^{-2}			240
		186	1.43×10^{-1}			240
		194	2.5×10^{-1}			240
	Silicone oil	126	3.7×10^{-4}			240
		139	1.92×10^{-3}			240
Potassium persulfate	0.1 M NaOH	50	9.5×10^{-7}	140.2	a	242
		60	3.16×10^{-6}		a	242
		70	2.33×10^{-5}		a	242
		80	9.16×10^{-5}		a	242
		90	3.5×10^{-4}		a	242
	Water (pH 3)	50	1.66×10^{-6}		a	242
	Water	80	6.89×10^{-5}		a,ν_{11}	115
		80	5.78×10^{-5}		a,ν_{11}	115
		40	1.65×10^{-2}	83.4	d_3,ν_{22}	262
		50	4.02×10^{-2}		d_3,ν_{22}	262
		60	1.08×10^{-1}		d_3,ν_{22}	262
		50	3.78×10^{-3}	121.5	d_3,ν_{23}	262
		60	2.18×10^{-2}		d_3,ν_{23}	262
		70	5.01×10^{-2}		d_3,ν_{23}	262
		50	1.1×10^{-6}			262
1-Pentanesulfonylazide	Diphenyl ether	166	4.46×10^{-4}			166
1,4-Butanedisulfonyl azide	Diphenyl ether	163	5.02×10^{-4}			166
1,6-Hexanedisulfonyl azide	Diphenyl ether	163	5.02×10^{-4}			166
1,9-Nonanedisulfonyl azide	Diphenyl ether	150	8.84×10^{-5}			166
		160	2.25×10^{-4}			166
		170	4.45×10^{-4}			166
1,10-Decanedisulfonyl azide	Diphenyl ether	163	4.45×10^{-4}			166
1,4-Dimethylcyclohexane-α,α′-disulfonyl azide	Diphenyl ether	163	4.82×10^{-4}			166

m-Xylene-α,α′-disulfonyl azide	Diphenyl ether	163	6.09×10^{-4}			166
p-Xylene-α,α′-disulfonyl azide	Diphenyl ether	163	5.78×10^{-4}			166
Benzenesulfonyl azide	Naphthalene	110	3.6×10^{-6}	152.3		235
		120	1.07×10^{-5}			235
		125	1.97×10^{-5}			235
		130	3.41×10^{-5}			235
		175	6.08×10^{-5}			235
p-Bromobenzenesulfonyl azide	Naphthalene	120	1.36×10^{-5}			235
p-Chlorobenzenesulfonyl azide	Naphthalene	120	1.15×10^{-5}			235
p-Methoxybenzenesulfonylazide	Naphthalene	120	1.31×10^{-5}			235
p-Nitrobenzenesulfonyl azide	Naphthalene	120	1.60×10^{-5}			235
p-Toluenesulfonyl azide	1,4-Dichlorobutane	145	1.70×10^{-4}			166
	Dimethyl terephthalate	155	3.23×10^{-4}			166
	Diphenyl ether	130	3.30×10^{-5}			166
		145	1.44×10^{-4}			166
		155	3.43×10^{-4}			166
	Hexanoic acid	155	2.97×10^{-4}			166
	Naphthalene	120	1.12×10^{-5}			235
	Nitrobenzene	155	3.97×10^{-4}			166
	1-Octanol	155	3.63×10^{-4}			166
	Tetradecane	155	3.80×10^{-4}			166
p-Toluenesulfonyl-*p*-tolylsulfone	Acetonitrile	29.3	3.9×10^{-5}	103.8	h	144
		39.1	1.45×10^{-4}			144
		49.3	5.4×10^{-4}			144
	Dioxane	29.3	2.1×10^{-5}	115.5	h	144
		39.5	1.01×10^{-4}			144
		49.3	3.9×10^{-4}			144

C. NOTES

a k_d converted to s^{-1} from author's units
b k_d values for several concentrations averaged
b_2 analyzed from nonisothermal conditions
c k_d increases with increasing initiator concentration
c_2 rate non-linear
c_3 hydroperoxide concentration 7.83×10^{-3} mol/l
c_4 hydroperoxide concentration 2.50×10^{-3} mol/l
d k_d decreases with increasing initiator concentration
d_2 after 1st half life; rate slower initially
d_3 rate is second order: units are 1/mol sec.
e k_d listed is for lowest initiator concentration
f k_d is extrapolated value for zero initiator concentration
g $k_d \neq$ has been corrected for induced decomposition
g_2 corrected for uncatalysed reaction ($7 \times 10^{-5}\ s^{-1}$)
h ΔH (not E_a)
i pressure (number gives mbar)
j iodometric analysis
j_2 manometer measurement
k infrared analysis
l k_d is limiting value with respect to additive concentration
m_1 3,4-dichlorostyrene added to minimize induced decomposition
m_2 styrene added to minimize induced decomposition
m_3 methyl methacrylate added to minimize induced decomposition
m_4 isobutene added to minimize induced decomposition
m_5 acenaphthalene added to minimize induced decomposition
m_6 1 mol/l α-methylstyrene added to minimize induced decomposition
m_7 butadiene added to minimize induced decomposition
m_8 acrylonitrile added to minimize induced decomposition
n trichloroacetic acid added
n_2 3.2 wt.% or about 0.1 monolayer
o addition of trichloroacetic acid did not affect k_d
p degassed
p_2 photochemical benzophenone added: 6.56×10^{17} quants/min @ 366 nm
q addition of trichloroacetic acid increased k_d several fold
q_2 pH 2.90
q_3 pH 7.05
r not inhibited, but initiator concentration low enough (0.01–0.09 M) so that higher order decomposition is unimportant
s solvent not degassed
t_1 2,6-di-*tert*-butylphenol added to inhibit induced decomposition
t_2 α,α-diphenyl-β-picrylhydrazyl added to inhibit induced decomposition
t_3 phenyl-α-naphthylamine added to inhibit induced decomposition
t_4 tetrachloroquinone added to inhibit induced decomposition
t_5 1,3,5-trinitrobenzene added to inhibit induced decomposition
t_6 I_2 added to inhibit induced decomposition
t_7 O_2 added to inhibit induced decomposition
t_8 5–20% NO_2 added to inhibit decomposition

t_9 galvanoxyl added to inhibit induced decomposition
t_{10} α,γ-bisdiphenylene-β-phenylallyl added to inhibit induced decomposition
t_{11} α-naphthylamine added to minimize induced decomposition
t_{12} 2,3′,5′,6′-tetra-*tert*-butyl indophenoxy added
t_{13} benzoquinone added
t_{14} 2,6-di-*tert*-butyl-4-methylphenol added
u_1 in absence of oxygen
u_2 from initiation data
u_3 in multisurface reaction vessel
u_4 rate reduced when nujol layer used to exclude Hg vapor
u_5 rate increased when nujol layer used to exclude Hg vapor
v_1 acetic acid added
v_2 $CuCl_2$ added
v_3 CuCl added
v_4 0.1 mol $AgClO_4$/mol AIBN added
v_5 3.9 mol thiophenol added
v_6 *tert*-butyl mercaptan added
v_7 2.5 mol cyclohexane added
v_8 $\sim 4 \times 10^{-2}$ mol pyridine added
v_9 2×10^{-2} mol pyridine added
v_{10} 2.5 mol cumene added
v_{11} buffered with sodium pyrophosphate
v_{12} saturated with ethyl acetate
v_{13} 3.11 mol CCl_4 present
v_{14} peroxide in P(MMA)
v_{15} peroxide in P(sty.)
v_{16} 0.26 mol piperidine
v_{17} 0.24 mol triethylamine
v_{18} 0.21 mol *N,N*-diethylaniline
v_{19} 0.22 mol pyridine
v_{20} 0.9 mol $ZnCl_2$
v_{21} in presence of ^{15}NO
v_{22} 0.12 mol 18 crown 6 added
v_{23} 0.24 mol diglyme added
v_{24} 0.02 M pyridine added
v_{25} 0.10 M pyridine added
v_{26} 0.10 M 2,6-lutidine added
w stereoisomers
x_1 actual rate divided by 2 because of two identical peroxide groups
x_2 each peroxide group has different k_d
y measured in differential scanning calorimeter: subscript is heating rate in deg/min
z pressure (in bar)

D. REFERENCES

1. J. E. Guillet, T. R. Walker, M. F. Meyer, J. P. Hawk, E. B. Towne, Ind. Eng. Chem., Prod. Res. Develop., **3**, 257 (1964).
2. L. M. Arnett, J. Am. Chem. Soc., **74**, 2027 (1952).
3. L. M. Arnett, J. H. Peterson, J. Am. Chem. Soc., **74**, 2031 (1952).
4. W. M. Thomas, M. T. O'Shaughnessy, J. Polym. Sci., **11**, 455 (1953).
5. A. T. Blomquist, A. J. Buselli, J. Am. Chem. Soc., **73**, 3883 (1951).
6. C. G. Swain, J. T. Parke, W. H. Stockmeyer, J. Am. Chem. Soc., **72**, 5426 (1950).
7. D. J. Brown, J. Am. Chem. Soc., **70**, 1208 (1948).
8. S. W. Butaka, L. L. Zabrocki, M. F. McLaughlin, J. R. Kolcznski, O. L. Mageli, Ind. Env. Chem. Prod. Res. Develop., **3**, 261 (1964).
9. W. A. Strong, Ind. Eng. Chem. Prod. Res. Develop., **3**, 264 (1964).
10. L. E. Redington, J. Polym. Sci., **3**, 503 (1948).
11. B. Baysal, A. V. Tobolsky, J. Polym. Sci., **8**, 529 (1952).
12. A. Conix, G. Smets, J. Polym. Sci., **10**, 525 (1953).
13. M. R. Gopalan, M. Santhappa, J. Polym. Sci., **25**, 333 (1957).
14. J. C. Bevington, J. Toole, J. Polym. Sci., **28**, 413 (1958).
15. H. C. Haas, J. Polym. Sci., **39**, 493 (1959).
16. A. I. Lowell, J. R. Price, J. Polym. Sci., **43**, 1 (1960).
17. H. C. Haas, J. Polym. Sci., **55**, 33 (1961).
18. D. F. Doehnert, O. L. Mageli, Mod. Plastics., **36**, (6), 142 (1959).
19. R. C. Lamb, P. W. Ayers, M. K. Toney, J. Am. Chem. Soc., **85**, 3483 (1963).
20. H. J. Shine, J. A. Waters, D. M. Hoffman, J. Am. Chem. Soc., **85**, 3613 (1963).
21. G. S. Hammond, R. C. Neuman, Jr., J. Am. Chem. Soc., **85**, 1501 (1963).
22. D. L. Tuleen, W. G. Bentrude, J. C. Martin, J. Am. Chem. Soc., **85**, 1938 (1963).
23. H. Hart, F. J. Chloupek, J. Am. Chem. Soc., **85**, 1155 (1963).
24. H. Hart, R. A. Cipriani, J. Am. Chem. Soc., **84**, 3697 (1962).
25. L. J. Durham, H. S. Mosher, J. Am. Chem. Soc., **84**, 2811 (1962).
26. R. C. Petersen, J. H. Markgraf, S. D. Ross, J. Am. Chem. Soc., **83**, 3819 (1961).
27. M. M. Martin, J. Am. Chem. Soc., **83**, 2869 (1961).
28. P. D. Bartlett, D. M. Simons, J. Am. Chem. Soc., **82**, 1753 (1960).
30. P. D. Bartlett, E. P. Benzing, R. E. Pincock, J. Am. Chem. Soc., **82**, 1762 (1960).
31. P. D. Bartlett, R. E. Pincock, J. Am. Chem. Soc., **82**, 1769 (1960).
32. L. J. Durham, H. S. Mosher, J. Am. Chem. Soc., **82**, 4537 (1960).
33. C. Walling, G. Metzger, J. Am. Chem. Soc., **81**, 5365 (1959).
34. H. Hart, D. P. Woman, J. Am. Chem. Soc., **81**, 4891 (1959).
35. H. H. Lau, H. Hart, J. Am. Chem. Soc., **81**, 4897 (1959).
36. E. L. O'Brien, F. M. Bennger, R. B. Mesrobian, J. Am. Chem. Soc., **81**, 1506 (1959).
37. N. A. Milas, A. Golubovic, J. Am. Chem. Soc., **80**, 5994 (1958).
38. C. G. Overberger, I. Tashlick, M. Vernstein, R. G. Hiskey, J. Am. Chem. Soc., **80**, 6556 (1958).

39. J. P. Van Hook, A. V. Tobolsky, J. Am. Chem. Soc., **80**, 779 (1958).
40. E. L. O'Brien, F. M. Beringer, R. B. Mesrobian, J. Am. Chem. Soc., **79**, 6238 (1957).
41. B. K. Morse, J. Am. Chem. Soc., **79**, 3375 (1957).
42. S. Solomon, C. H. Wang, S. G. Cohen, J. Am. Chem. Soc., **79**, 4104 (1957).
43. C. Walling, J. Pellon, J. Am. Chem. Soc., **79**, 4786 (1957).
44. C. G. Overberger, J. G. Lombardino, I. Tashlich, R. G. Hiskey, J. Am. Chem. Soc., **79**, 2662 (1957).
45. K. E. Russel, J. Am. Chem. Soc., **77**, 3487 (1955).
46. S. G. Cohen, C. H. Wang, J. Am. Chem. Soc., **77**, 3628 (1955).
47. M. Talat-Erben, S. Bywater, J. Am. Chem. Soc., **77**, 3712 (1955).
48. C. G. Overberger, M. Lapkin, J. Am. Chem. Soc., **77**, 4651 (1955).
49. G. S. Hammond, J. N. Sen, C. E. Boozer, J. Am. Chem. Soc., **77**, 3244 (1955).
50. C. G. Overberger, W. F. Hale, M. B. Berenbaum, A. B. Finestone, J. Am. Chem. Soc., **76**, 6185 (1954).
51. J. Smid, A. Rembaum, M. Szwarc, J. Am. Chem. Soc., **78**, 3315 (1956).
52. S. G. Cohen, C. H. Wang, J. Am. Chem. Soc., **75**, 5504 (1953).
53. A. T. Blomquist, I. A. Berstein, J. Am. Chem. Soc., **73**, 5546 (1951).
54. C. G. Overberger, H. Biletch, J. Am. Chem. Soc., **73**, 4880 (1951).
55. A. T. Blomquist, A. F. Ferris, J. Am. Chem. Soc., **73**, 3408 (1951).
56. A. T. Blomquist, A. F. Ferris, J. Am. Chem. Soc., **73**, 3412 (1951).
57. C. G. Overberger, M. B. Berenbaum, J. Am. Chem. Soc., **73**, 2618 (1951).
58. V. Stannett, R. B. Mesrobian, J. Am. Chem. Soc., **72**, 4125 (1950).
59. P. D. Bartlett, J. E. Leffler, J. Am. Chem. Soc., **72**, 3030 (1950).
60. S. G. Cohen, D. B. Sparrow, J. Am. Chem. Soc., **72**, 611 (1950).
61. C. G. Overberger, M. T. O'Shaughnessy, H. Shalit, J. Am. Chem. Soc., **71**, 2661 (1949).
62. F. M. Lewis, M. S. Matheson, J. Am. Chem. Soc., **71**, 747 (1949).
63. J. H. Raley, F. F. Rust, W. E. Vaughan, J. Am. Chem. Soc., **70**, 88 (1948).
64. J. H. Raley, F. F. Rust, W. E. Vaughan, J. Am. Chem. Soc., **70**, 1336 (1948).
65. C. S. Marvel, R. L. Frank, E. Prill, J. Am. Chem. Soc., **65**, 1647 (1943).
66. C. E. H. Bawn, D. Verdin, Trans. Faraday Soc., **56**, 815 (1960).
67. H. C. Bailey, G. W. Godin, Trans. Faraday Soc., **52**, 68 (1956).
68. C. E. H. Bawn, R. G. Halford, Trans. Faraday Soc., **51**, 780 (1955).
69. C. E. H. Bawn, S. F. Mellish, Trans. Faraday Soc., **47**, 1216 (1951).
70. W. Braun, L. Rajbenbach, F. R. Eirich, J. Phys. Chem., **66**, 1591 (1962).
71. J. R. Thomas, O. L. Harle, J. Phys. Chem., **63**, 1027 (1959).
72. P. L. Hanst, J. G. Calvert, J. Phys. Chem,. **63**, 104 (1959).
73. B. Barnett, W. E. Vaughan, J. Phys. Chem., **51**, 926 (1947).
74. B. Barnett, W. E. Vaughan, J. Phys. Chem., **51**, 942 (1947).
75. S. G. Cohen, F. Cohen, C. H. Wang, J. Org. Chem., **28**, 1479 (1963).
76. R. R. Hiatt, W. M. J. Strachan, J. Org. Chem., **28**, 1893 (1963).
77. J. E. Leffler, J. S. West, J. Org. Chem., **27**, 4191 (1962).
78. W. Honsberg, J. E. Leffler, J. Org. Chem., **26**, 733 (1961).
79. J. E. Leffler, A. F. Wilson, J. Org. Chem., **25**, 424 (1960).
80. M. S. Kharasch, A. Fono, W. Nudenberg, J. Org. Chem., **16**, 105 (1951).
81. P. D. Bartlett, R. R. Hiatt, J. Am. Chem. Soc., **80**, 1398 (1958).
82. J. E. Leffler, R. D. Faulkner, C. C. Petropoulos, J. Am. Chem. Soc., **80**, 5435 (1958).
83. L. Batt, S. W. Benson, J. Chem. Phys., **36**, 895 (1962).
84. J. Smid, M. Szwarc, J. Chem. Phys., **29**, 432 (1958).
85. A. Rembaum, M. Szwarc, J. Chem. Phys., **23**, 909 (1955).
86. R. K. Brinton, D. H. Volman, J. Chem. Phys., **20**, 25 (1952).
87. R. E. Rebbert, K. J. Laidler, J. Chem. Phys., **20**, 574 (1952).
88. J. Murawski, J. S. Roberts, M. Szwarc, J. Chem. Phys., **19**, 698 (1951).
89. J. E. Leffler, J. Am. Chem. Soc., **72**, 67 (1950).
90. F. Strain, W. E. Bissinger, W. R. Dial, H. Rudolf, B. J. DeWitt, H. C. Stevens, J. H. Langston, J. Am. Chem. Soc., **72**, 1254 (1950).
91. A. Farkas, E. Passaglia, J. Am. Chem. Soc., **72**, 3333 (1950).
92. W. A. Pryor, D. M. Huston, T. R. Fiske, T. L. Pickering, E. Ciuffarin, J. Am. Chem. Soc., **86**, 4237 (1964).
93. J. C. Bevington, A. Wahid, Polymer, **4**, 129 (1963).
94. A. A. Frost, R. G. Pearson, "Kinetics and Mechanism," Wiley, New York, 1953, p. 97.
95. J. C. Bevington, T. D. Levis, Polymer, **1**, 1 (1960).
96. F. W. Birss, C. J. Danby, C. Hinshelwood, Proc. Roy Soc. (London). A, **239**, 154 (1957).
97. J. C. Martin, W. G. Bentrude, Chem. Ind. (London), 192 (1959).
98. S. G. Cohen, S. J. Groszos, D. B. Sparrow, J. Am. Chem. Soc., **72**, 3947 (1950).
99. W. E. Cass, J. Am. Chem. Soc., **72**, 4915 (1950).
100. S. D. Ross, M. A. Fineman, J. Am. Chem. Soc., **73**, 2176 (1951).
101. C. G. Overberger, H. Biletch, A B. Finestone, J. Lilker, J. Herbert, J. Am. Chem. Soc., **75**, 2078 (1953).
102. P. D. Bartlett, F. D. Greene, J. Am. Chem. Soc., **76**, 1088 (1954).
103. M. G. Alder, J. E. Leffler, J. Am. Chem. Soc., **76**.,1425 (1954).

104. C. G. Overberger, A. Lebovits, J. Am. Chem. Soc., **76**, 2722 (1954).
105. A. Rembaum, M. Szwarc, J. Am. Chem. Soc., **76**, 5975 (1954).
106. M. Levy, M. Steinberg, M. Szwarc, J. Am. Chem. Soc., **76**, 5978 (1954).
107. S. G. Cohen, C. H. Wang, J. Am. Chem. Soc., **77**, 2457 (1955).
108. J. E. Leffler, C. C. Petropoulos, J. Am. Chem. Soc., **79**, 3068 (1957).
109. W. E. Cass, J. Am. Chem. Soc., **68**, 1976 (1946).
110. H. C. Ramsperger, J. Am. Chem. Soc., **50**, 714 (1928).
111. F. P. Lossing, A. W. Tickner, J. Chem. Phys., **20**, 907 (1952).
112. A. Robertson, W. A. Waters, J. Chem. Soc., 1578 (1948).
113. O. J. Walker, G. L. E. Wild, J. Chem. Soc., 1132 (1937).
114. J. E. Leffler, R. A. Hubbard, II, J. Org. Chem., **19**, 1089 (1954).
115. P. D. Bartlett, K. Nozaki, J. Polym. Sci., **3**, 216 (1948).
116. J. C. Bevington, J. Toole, L. Trossarelli, Makromol. Chem., **32**, 57 (1959).
117. A. N. Bose, C. Hinshelwood, Proc. Roy. Soc., (London) A, **249**, 173 (1959).
118. E. R. Bell, J. H. Raley, F. F. Rust, F. H. Seubold, W. E. Vaughan, Disc. Faraday Soc., **10**, 242 (1951).
119. L. Bateman, H. Hughes, A. L. Morns, Disc. Faraday Soc., **14**, 190 (1953).
120. G. S. Hammond, J. Am. Chem. Soc., **72**, 3737 (1950).
121. G. S. Hammond, L. M. Softer, J. Am. Chem. Soc., **72**, 4711 (1950).
122. G. A. Russell, J. Am. Chem. Soc., **78**, 1044 (l956).
123. J. A. Off'enbach, A. V. Tobolsky, J. Am. Chem. Soc., **79**, 278(1957).
124. G. S. Hammond, U. S. Nandi, J. Am. Chem. Soc., **83**, 1213 (1961).
125. G. O. Pritchard, H. O. Pritchard, A. F. Trotman-Dickenson, J. Chem. Soc., 1425 (1954).
126. O. L. Mageli, S. D. Butaka, D. J. Bolton, Wallace and Tiernan, Lucidol Division, Bulletin 30.30, "Evaluation of Organic Peroxides from Half-Life Data".
127. E. J. Harris, A. C. Egenon, Proc. Roy. Soc., (London) A, **168**, 1 (1938).
128. E. J. Harry, Proc. Roy. Soc., (London) A, **173**, 126 (1939).
129. J. K. Allen, J. C. Bevington, Proc. Roy. Soc., (London) A, **262**, 271 (1961).
130. G. Archer, C. Hinshelwood, Proc. Roy. Soc., (London) A, **261**, 293 (1961).
131. J. H. McClure, R. E. Robertson, A. C. Cuthbertson, Can. J. Res. B, **20**, 103 (1942).
132. J. W. Breitenbach, A. Schindler, Monatsh. Chem., **83**, 724 (1952).
133. R. C. Lamb, F. F. Rogers, G D. Dean, F. W Voight, J. Am. Chem. Soc., **84**, 2635 (1962).
134. W. G. Bentrude, J. C. Martin, J. Am. Chem,. **84**, 1561 (1962).
135. R. C. Neuman, R. P. Pankratz, J. Org. Chem., **36**, 4046 (1971).
136. P. D. Bartlett, L. B. Gonler, J. Am. Chem. Soc., **85**, 1864 (1963).
137. C. G. Overberger, I. Tashlick, J. Am. Chem. Soc., **81**, 217 (1959).
138. C. G. Overberger, J. G. Lambardino, J. Am. Chem. Soc., **80**, 2317 (1958).
139. G. S. Hammond, J. R. Fox, J. Am. Chem. Soc., **86**, 1918 (1964).
140. E. S. Huyser, C. J. Bredeweg, J. Am. Chem. Soc., **84**, 2401 (1964).
141. E. Hedaya, R. L. Hinman, L. M. Kibler, S. Theodoropulos, J. Am. Chem. Soc., **86**, 2727 (1964).
142. T. Koenig, W. Brewer, J. Am. Chem. Soc., **86**, 2728 (1964).
143. E. S. Huyser, C. J. Bredeweg, R. M. Van Scoy, J. Am. Chem. Soc., **86**, 4148 (1964).
144. J. L. Kice, N. E. Pawlowski, J. Am. Chem. Soc., **86**, 4898 (1964).
145. C. Walling, H. N. Moulden, J. H. Waters, R. C. Newman, J. Am. Chem. Soc., **87**, 518 (1965).
146. J. K. Kochi, D. M. Mog, J. Am. Chem. Soc., **87**, 522 (1965).
147. P. D. Bartlett, M. McBride, J. Am. Chem. Soc., **87**, 1727 (1965).
148. R. C. Lamb, J. G. Pacifici, P. W. Ayers, J. Am. Chem. Soc., **87**, 3928 (1965).
149. S. Seltzer, F. T. Dunne, J. Am. Chem. Soc., **87**, 2628 (1965).
150. L. A. Singer, N. P. Kong, J. Am. Chem. Soc., **88**, 5213 (1966).
151. R. M. Fantazier, J. A. Kampmeier, J. Am. Chem. Soc., **88**, 5219 (1966).
152. S. Seltzer, E. J. Hamilton, Jr., J. Am. Chem. Soc., **88**, 3775 (1966).
153. J. P. Lorand, P. D. Bartlett, J. Am. Chem. Soc., **88**, 3294 (1966).
154. T. H. Fischer, J. C. Martin, J. Am. Chem. Soc., **88**, 3382 (1966).
155. C. Walling, Z. Cekovic, J. Am. Chem. Soc., **89**, 6681 (1967).
156. T. Koenig, R. Wolf, J. Am. Chem. Soc., **89**, 2948 (1967).
157. R. C. Neuman, J. V. Behar, J. Am. Chem. Soc., **89**, 4549 (1967).
158. S. G. Mylonakis, S. Seltzer, J. Am. Chem. Soc., **90**, 5487 (1968).
159. J. P. Lorand, S. D. Cnodroff, R. W. Wallace, J. Am. Chem. Soc., **90**, 5266 (1968).
160. R. C. Fort Jr., R. E. Franklin, J. Am. Chem. Soc., **90**, 5267 (1968).
161. J. E. Leffler, H. H. Gibson. Jr., J. Am. Chem. Soc., **90**, 4117 (1968).
162. S. E. Scheppelle, S. Seltzer, J. Am. Chem. Soc., **90**, 358 (1968).
163. R. C. Neuman, J. V. Behar, J. Am. Chem. Soc., **91**, 6024 (1969).
164. R. C. Lamb, J. R. Sanderson, J. Am. Chem. Soc., **91**, 5034 (1969).
165. T. Koenig, R. Wolf, J. Am. Chem. Soc., **91**, 2574 (1969).
166. D. S. Breslow, M. F. Sloan, N. R. Newburg, W. B. Renfrow, J. Am. Chem. Soc., **91**, 2273 (1969).
167. C. Walling, H. P. Warts, J. Milovanovic, C. G. Pappiaonnou, J. Am. Chem. Soc., **92**, 4927 (1970).

168. W. A. Pryor, K. Smith, J. Am. Chem. Soc., **92**, 5403 (1970).
169. T. Koenig, J. Huntington, R. Cruthoff, J. Am. Chem. Soc., **92**, 5413 (1970).
170. J. E. Leffler, R. G. Zepp, J. Am. Chem. Soc., **92**, 3713 (1970).
171. J. C. Martin, J. W. Timberlake, J. Am. Chem. Soc., **92**, 978 (1970).
172. R. C. Newman Jr., G. D. Holmes, J. Am. Chem. Soc., **93**, 4242 (1971).
173. J. B. Levy, E. J. Lehmann, J. Am. Chem. Soc., **93**, 5790 (1971).
174. K. Shen, J. Am. Chem. Soc., **93**, 3064 (1971).
175. T. W. Koenig, J. C. Martin, J. Org. Chem., **29**, 1520 (1964).
176. R. D. Schuetz, J. L. Shea, J. Org. Chem., **30**, 844 (1965).
177. J. R. Thomas, J. Am. Chem. Soc., **77**, 246 (1955).
178. P. D. Bartlett, C. Ruchardt, J. Am. Chem. Soc., **82**, 1756 (1960).
179. C. S. Wu, G. S. Hammond, J. M. Wright, J. Am. Chem. Soc., **82**, 5386 (1960).
180. S. G. Cohen, R. Zand, C. Steel, J. Am. Chem. Soc., **83**, 2895 (1961).
181. P. D. Bartlett, R. E. Pincock, J. Am. Chem. Soc., **84**, 2445 (1962).
182. M. M. Martin, D. C. De Jongh, J. Am. Chem. Soc., **84**, 3526 (1962).
183. S. Selzer, J. Am. Chem. Soc., **83**, 2625 (1961).
184. S. N. Gupta, U. S. Nandi, J. Polym. Sci. A-1, **8**, 3019 (1970).
185. T. Kagiya, M. Izu, S. Kauai, K. Fukni, J. Polym. Sci. A-1, **6**, 1719 (1968).
186. K-P. S. Kwei, J. Polym. Sci. A, **3**, 2387 (1965).
187. C. Leggett, J. C. J. Thynne, Trans. Faraday Soc., **63**, 2504 (1967).
188. S. W. Benson, G. N. Spokes, J. Phys. Chem., **72**, 1182 (1968).
189. G. A. Mortimer, J. Org. Chem., **30**, 1632 (1965).
190. W. H. Richardson, J. Org. Chem., **30**, 2804 (1965).
191. R. C. Lamb, J. G. Pacifici, P. W. Ayers, J. Org. Chem., **30**, 3099 (1965).
192. E. S. Huyser, R. M. Van Scoy, J. Org. Chem., **33**, 3524 (1968).
193. C. Walling, J. C. Azar, J. Org. Chem., **33**, 3888 (1968).
194. R. C. Lamb, L. P. Spadafino, R. G. Webb, E. B. Smith, W. E. McNew, J. G. Pacifici, J Org. Chem., **31**, 147 (1966).
195. F. D. Greene, W. Adam, G. A. Knudsen Jr., J. Org. Chem., **31**, 2087 (1966).
196. L. M. Bobroff, L. B. Gortler, D. J. Sahn, H. Wiland, J. Org. Chem., **31**, 2678 (1966).
197. R. Hiatt, T. Mill, K. C. Irwin, J. K. Castleman, J. Org. Chem., **33**, 1421 (1968).
198. R. Hiatt, K. C. Irwin, C. W. Gould, J. Org. Chem., **33**, 1430 (1968).
199. R. Hiatt, K. C. Irwin, J. Org. Chem., **33**, 1436 (1968).
200. P. Kovacic, R. R. Flynn, J. F. Gormish, A. H. Kappelman, J. R. Shelton, J. Org. Chem., **34**, 3312 (1969).
201. D. E. Van Sickle, J. Org. Chem., **34**, 3446 (1969).
202. E. S. Huyser, K. J. Jankauskas, J. Org. Chem., **35**, 3196 (1970).
203. R. A. Sheldon, J. K. Kochi, J. Org. Chem., **35**, 1223 (1970).
204. A. Padwa, N. C. Das, J. Org. Chem., **34**, 816 (1969).
205. C. G. Overberger, D. A. Labianca, J. Org. Chem., **35**, 1762 (1970).
206. W. G. Schindel, R. E. Pincock, J. Org. Chem., **35**, 1789 (1970).
207. R. C. Newman Jr., J. V. Behar, J. Org. Chem., **36**, 654 (1971).
208. C. Walling, D. Bristol, J. Org. Chem., **36**, 733 (1971).
209. R. C. Lamb, W. E. McNew Jr., J. R. Sanderson, D. C. Lunney, J. Org. Chem., **36**, 174 (1971).
210. G. S. Kolesnikov, A. Y. Chuchin, Polym. Sci. USSR, **7**, 1931 (1965).
211. G. A. Nosayev, O. N. Romanlsova, Polym. Sci. USSR, **8**, 14 (1966).
212. Y. A. Chuchin, V. A. Lazarev, M. B. Fromberg, Polym. Sci. USSR, **10**, 2968 (1968).
213. L. M. Aparovich, T. I. Yurzhenko, Polym. Sci. USSR, **10**, 1313 (1968).
214. A. I. Prisyazhnyuk, S. S. Ivanchev, Polym. Sci. USSR, **12**, 514 (1970).
215. S. G. Yerigova, S. S. Ivanchev. Polym. Sci. USSR, **11**, 2377 (1969).
216. C. H. Bamford, R. Denyer, J. Hobbs, Polymer, **8**, 493 (1967).
217. S. F. Nelsen, P. D. Bartlett, J. Am. Chem. Soc., **88**, 137 (1966).
218. R. E. Pincock, J. Am. Chem. Soc., **86**, 1820 (1964).
219. R. C. Lamb, J. G. Pacifici, J. Am. Chem. Soc., **86**, 914 (1964).
220. S. Seltzer, J. Am. Chem. Soc., **85**, 14 (1963).
221. J. B. Levy, B. K. W. Copeland, J. Am. Chem. Soc., **82**, 5314 (1960).
222. T. Suchiro, H. Tsuruta, S. Hibino, Bull. Chem. Soc. Japan, **40**, 674 (1967).
223. L. B. Humphrey, B. Hodgson, R. E. Pincock, Can. J. Chem., **46**, 3099 (1968).
224. V. Stannett, R. B. Mesrobian, Disc. Faraday Soc., **14**, 242 (1953).
225. Y. Takezaki, C. Takeuchi, J. Chem. Phys., **22**, 1527 (1954).
226. K. Ziegler, W. Deparade, W. Meye, Annalen, **567**, 141 (1950).
227. A. R. Blake, K. O. Kutschke, Can. J. Chem., **37**, 1462 (1959).
228. C. F. H. Tipper, J. Chem. Soc., 1675 (1953).
229. W. R. Foster, G. H. Williams, J. Chem. Soc., 2862 (1962).
230. G. B. Gill, G. H. Williams, J. Chem. Soc., 995 (1965).
231. G. B. Gill, G. H. Williams, J. Chem. Soc., 7127 (1965).
232. R. D. Schuetz, D. M. Teller, J. Org. Chem., **27**, 110 (1962).
233. H. G. G. Dekking, J. Appl. Polym. Sci., **9**, 1641 (1965).
234. C. G. Overberger, P. Fram, T. Alfrey Jr., J. Polym. Sci., **6**, 539 (1951).
235. K. Takemoto, R. Fujita, M. Imoto, Makromol. Chem., **112**, 116 (1968).
236. K. E. J. Barrett, J. Appl. Polym. Sci., **11**, 1617 (1967).
237. T. J. Dougherty, J. Am. Chem. Soc., **83**, 4849 (1961).

238. M. T. Jaquiss, J. S. Roberts, M. Szwarc, J. Am. Chem. Soc., **74**, 6005 (1952).
239. S. Goldschmidt, B. Acksteiner, Annalen, **618,** 173 (1958).
240. J. C. McGowen, T. Powell, Rec. Trav. Chim., **81**, 1061 (1962).
241. H. A. Bent, B. Crawford Jr., J. Am. Chem. Soc., **79**, 1793 (1957).
242. I. M. Kolthoff, I. K. Miller, J. Am. Chem. Soc., **73**, 3055 (1951).
243. D. F. McMillen, D. M. Golden, S. W. Benson, J. Am. Chem. Soc., **94**, 4403 (1972).
244. J. E. Leffler, A. A. More, J. Am. Chem. Soc., **94**, 2483 (1972).
245. G. J. Abruscato, T. T. Tidwell, J. Am. Chem. Soc., **94**, 672 (1972).
246. A. M. Hucek, J. T. Barbas, J. E. Leffler, J. Am. Chem. Soc., **95**, 4698 (1973).
247. E. L. Allred, K. J. Voorhees, J. Am. Chem. Soc., **95**, 620 (1973).
248. J. C. Martin, M. M. Chau, J. Am. Chem. Soc., **96**, 3319 (1974).
249. J. P. Lorand, J. Am. Chem. Soc., **96**, 2867 (1974).
250. J. P. Lorand, R. W. Wallace, J. Am. Chem. Soc., **96**, 1402 (1974).
251. T. Koenig, J. G. Huntington, J. Am. Chem. Soc., **96**, 592 (1974).
252. P. S. Engel, D. J. Bishop, J. Am. Chem. Soc., **97**, 6754 (1975).
253. R. R. Hiatt, L. C. Glover, H. S. Mosher, J. Am. Chem. Soc., **97**, 1556 (1975).
254. L-F. Lai, T. T. Tidwell, J. Am. Chem. Soc., **99**, 1465 (1977).
255. R. A. Wolf, M. J. Migliore, P. H. Fuery, P. R. Gagnier, I. C. Saheta, R. J. Trocino, J. Am. Chem. Soc., **100**, 7967 (1978).
256. C. Walling, J. P. Sloan, J. Am. Chem. Soc., **101**, 7679 (1979).
257. C. M. Rynard, C. Thankachan, T. T. Tidwell, J. Am. Chem. Soc., **101**, 1196 (1979).
258. J. E. Leffler, J. T. Barbas, J. Am. Chem. Soc., **103**, 768 (1981).
259. B. G. Dixon, G. B. Schuster, J. Am. Chem. Soc., **103**, 3068 (1981).
260. A. S. Nazran, J. Warkentin, J. Am. Chem. Soc., **104**, 6405 (1982).
261. R. A. Kenley, D. G. Hendry, J. Am. Chem. Soc., **104**, 220 (1982).
262. J. K. Rasmussen, S. M. Heilmann, P. E. Toren, A. V. Pocius, T. A. Kotnour, J. Am. Chem. Soc., **105**, 6845 (1983).
263. R. C. Neuman, G. D. Lockyer, J. Am. Chem. Soc., **105**, 3982 (1983).
264. W. A. Pryor, N. Ohto, D. F. Church, J. Am. Chem. Soc., **105**, 3614 (1983).
265. J. P. Engstrom, J. C. DuBose, J. Org. Chem., **38**, 3817 (1973).
266. J. R. Shelton, C. K. Liang, J. Org. Chem., **38**, 2301 (1973).
267. J. A. Ernst, C. Thankachan, T. T. Tidwell, J. Org. Chem., **39**, 3614 (1974).
268. J. E. Herweh, R. M. Fantazier, J. Org. Chem., **39**, 786 (1974).
269. D. A. Cable, J. A. Ernst, T. T. Tidwell, J. Org. Chem., **37**, 3420 (1972).
270. W. A. Pryor, E. H. Morkved, H. T. Bickley, J. Org. Chem., **37**, 1999 (1972).
271. J. E. Leffler, F. E. Scrivener, J. Org. Chem., **37**, 1794 (1972).
272. A. I. Dalton, T. T. Tidwell, J. Org. Chem., **37**, 1504 (1972)
273. J. P. Engstrom, F. D. Greene, J. Org. Chem., **37**, 968 (1972).
274. P. S. Engel, A. I. Dalton, L. Shen, J. Org. Chem., **39**, 384 (1974).
275. W. B. Richardson, W. C. Koskinen, J. Org. Chem., **41**, 3182 (1976).
276. W. A. Pryor, D. Graham, J. Org. Chem., **43,** 770 (1978).
277. L. Batt, S. W. Benson, J. Chem. Phys., **36**, 895 (1962).
278. P. S. Engel, R. L. Allgren, W. K. Chae, R. A. Leckonby, N. A. Marron, J. Org. Chem., **44**, 4233 (1979).
279. L. Thijs, S. N. Gupta, D. C. Neckers, J. Org. Chem., **44**, 4123 (1979).
280. R. C. Neuman, M. J. Amrich, J. Org. Chem., **45**, 4629 (1980).
281. J. R. Hurst, G. B. Shuster, J. Org. Chem., **45**, 1053 (1980).
282. W. H. Richardson, S. A. Thomson, J. Org. Chem., **47**, 4515 (1982).
283. Z. Chengxue, Z. Renmo, P. Hegi, J. Xiangshan, Q. Yangling, W. Chengjiu, J. Xikui, J. Org. Chem., **47**, 2009 (1982).
284. E. M. Shulman, A. E. Merbach, W. J. le Noble, J. Org. Chem., **47**, 431 (1982).
285. A. Pohlman, T. Mill, J. Org. Chem., **48**, 2133 (1983).
286. L. F. R. CaRerata, G. N. Eyler, M. V. Mirifico, J. Org. Chem., **49**, 2107 (1984).
287. S. Molnar, J. Polym. Sci. A-1, **10**, 2245 (1972).
288. K. Ito, J. Polym. Sci., Polym. Chem. Ed., **11**, 1673 (1973).
289. M. B. Lachinov, V P. Zubov, V. A. Kabanov, J. Polym. Sci., Polym. Chem. Ed., **15**, 1777 (1977).
290. B. Yamada, H. Kamei, T. Otsu, J. Polym. Sci., Polym. Chem. Ed., **18**, 1917 (1980).
291. S. G. Ng, K. K. Che, J. Polym. Sci., Polym. Chem. Ed., **20**, 409 (1982).
292. B. B. Idage, S. P. Vernckar, N. D. Ghatge, J. Polym. Sci., Polym. Chem. Ed., **21**, 2145 (1983).
293. B. L. Funt, G. Pawel Chak, J. Polym. Sci., Lett. Ed., **13**, 451 (1975).
294. M. G. Kulkarni, R. A. Mashelkar, L. K. Doraiswam, J. Polym. Sci., Lett. Ed., **17**, 713 (1979).
295. P. S. Engel, Chem. Rev., **80**, 99 (1980).
296. T. Koenig, in: "Free Radicals", Wiley, New York, 1973, p. 113.
297. B. K. Bandlish, A. W. Garner, M. L. Hodges, J. W. Timberlake, J. Am. Chem. Soc., **97**, 5856 (1975).
298. R. J. Crawford, K. Takagi, J. Am. Chem. Soc., **94**, 7406 (1972).
299. R. C. Neuman., J. J. Amrich, J. Am. Chem. Soc., **94**, 2730 (1972).
300. S. E. Scheppele, W. H. Rapp, D. W. Miller, D. Wright, T. Marriott, J. Am. Chem. Soc., **94**, 539 (1972).
301. C. Walling, H. P. Waits, J. Phys. Chem., **71**, 2361 (1967).
302. Y. L. Zherebin, S. S. Ivanchev, N. M. Domareva, Polym. Sci., USSR, **16**, 1033 (1974).

303. R. V. Kucher, A. A. Turovskii, N. V. Dzumedzei, V. A. Tamko, Polym. Sci., USSR, **16**, 1216 (1974).
304. V. A. Novikov, V. P. Sass, S. L. Ivanova, L. F. Sokolov, S. V. Sokolov, Polym. Sci., USSR, **17**, 1414 (1975).
305. V. P. Kartavykh, V. A. Drach, Y. N. Barantsevich, Y. L. Abramenko, Polym. Sci., USSR, **19**, 1413 (1977).
306. V. I. Galibei. L. V. Dudnik, T. A. Tolpygina, A. B. Petrova, V. I. Sokolova, Polym. Sci., USSR, **19**, 1519 (1977).
307. B. N. Pronin, Y. N. Barantsevich, L. V. Shumnyi, S. S. Ivanchev, Polym. Sci., USSR, **19**, 1850 (1977).
308. J. C. Martin, J. H. Hargis, J. Am. Chem. Soc., **91**, 5399 (1969).
309. J. R. Shelton, C. K. Liang, P. Kovacic, J. Am. Chem. Soc., **90**, 354 (1968).
310. C. G. Overberger. A. V. DiGiulio, J. Am. Chem. Soc., **81**, 2154 (1959).
311. C. G. Overberger, A. B. Finestone, J. Am. Chem. Soc., **78**, 1638 (1956).
312. S. G. Cohen, C. H. Wang, J. Am. Chem. Soc., **77**, 2457 (1955).
313. J. R. Shelton, C. K. Liang. J. Org. Chem., **38**, 2301 (1973).
314. J. W. Timberlake. A. W. Garner, J. Org. Chem., **41**, 1666 (1976).
315. C. Ruchardt, H. Bock, I. Ruthardt, Angew. Chem. Int. Ed., **5**, 253 (1966).
316. W. Cooper, J. Chem. Soc., 2408 (1952).
317. H. Kiefer, T. G. Traylor, Tetrahedron Lett., 6163 (1966).
318. K. R. Kopecky, T. Gillan, Can. J. Chem., **47**, 2371 (1969).
319. B. H. Al-Sader, R. J. Crawford, Can. J. Chem., **48**, 2745 (1970).
320. D. D. Tanner, H. Yabuuchi, H, Lutzer, Can. J. Chem., **55**, 612 (1977).
321. R. Hiatt, V. G. K. Nair, Can. J. Chem., **58**, 450 (1980).
322. T. R. Lynch, F. N. MacLachlan, J. L. Suschitzky, Can. J. Chem., **51**, 1378 (1973).
323. J. Bonnekessel, C. Ruchardt, Chem. Ber., **106**, 2890 (1973).
324. W. Duismann, C. Ruchardt, Chem. Ber., **106**, 1083 (1973).
325. R. Kerber, O. Nuyken, L, Weithmann, Chem. Ber., **108**, 1533 (1975).
326. T. Otsu, B. Yamada, J. Macromol. Sci.-Chem., **A3**, 187 (1969).
327. I. Hinz, C. Ruchardt, Ann. Chem., **765**, 94 (1972).
328. C. Ruchardt, I. Mayer-Ruthardt, Chem. Ber., **104**, 593 (1971).
329. R. Kerber, O. Nuyken, Makromol. Chem., **164**, 183 (1973).
330. R. Kerber, O. Nuyken, V. Pasupathy, Makromol. Chem., **170**, 155 (1973).
331. R. Kerber, O. Nuyken, R. Steinhausen, Makromol. Chem., **175**, 3225 (1974).
332. R. Walz, B. Bomer, W. Heitz, Makromol. Chem., **178**, 2527 (1977).
333. R. C. Neuman, G. D. Lockyer, M. J. Amrich, Tetrahedron Lett., 1221 (1972).
334. F. E. Herkes, J. Friedman, P. D. Bartlett, Int. J. Chem. Kinet., **1**, 193 (1969).
335. N.-T.-Giac, C. Ruchardt, Chem. Ber., **110**, 1095 (1977).
336. W. Duismann, C. Ruchardt, Chem. Ber., **111**, 596 (1978).
337. Anon., "Evaluation of Organic Peroxides from Half-Life Data," Technical Bulletin, Lucidol Division, Pennwalt (no date).
338. Anon., Product Guide US Peroxygen Division, Witco, 1984.
339. Anon., "Azo Polymerization Initiators", Wako Chemicals, USA, 1983.
340. Anon., Product Bulletin "V-601", Wako Chemicals, USA (no date).
341. Anon., Product Bulletin " V-50", Wako Chemicals, USA, 1985.
342. Anon., "Azo Amide Compounds", Wako Chemicals, USA (no date).
343. Anon., Product Bulletin "V-65" Wako Chemicals, USA, 1985.
344. Anon., "Initiators for Polymer Production", Akzo Nobel Chemicals Inc., pp. 94–116.
345. S. Icli, K. A. Kandil, C. Thankachan, T. T. Tidwell, Can. J. Chem., **53**, 979 (1975).
346. W. H. Richardson, M. B. Yelvington, A. H. Andrist, E. W. Ertley, R. S. Smith, T. D. Johnson, J. Org. Chem., **38**, 4219 (1973).
347. D. J. Robbins, A. J. Almquist, D. C. Timm, J. I. Brand, R. E. Gilbert, Macromol., **28,** 8729 (1995).
348. K. Matsuyama, T. Sugiura, Y. Minoshima, J. Org. Chem., **60**, 5520 (1995).
349. S. Suyama, H. Ishigaki, T. Nakamura, Y. Sugihara, H. Kumura, Y. Watanabe, Polym. J., **26**, 273 (1994).
350. Y. Sugihara, Y. Watanabe, H. Kumura, T. Nakamura, S. Suyama, Y. Sawaki, Bull. Chem. Soc. Japan, **65**, 664 (1992).
351. E. M. Chellquist, W. G. Gorman, Pharmaceutical Research, **9**, 1341 (1992).
352. W. H. Hendrickson, C. C. Nguyen, J. T. Nguyen, K. T. Simons, Tetrahedron Lett., **36**, 7217 (1995).
353. C. Navarro, B. Maillard, Thermochimica Acta, **220**, 103 (1993).
354. W. A. Pryor, W. H. Hendrickson Jr., J. Am. Chem. Soc., **105**, 7114 (1983).
355. J. Szafko, W. Feist, J. Polym. Sci., Polym. Chem., **33**, 1637 (1995).
356. H. Fujie, K. Shiraki, T. Miyagawa, N. Minamii, J. Macromol. Sci., Pure Appl. Chem. A, **29**, 741 (1992).
357. S. Idage, B. Idage, S. Vernekar, J. Appl. Polym. Sci., **45**, 931 (1992).
358. T. Komai, S. Suyama, Bull. Chem. Soc. Japan, **58**, 3045 (1985).
359. T. Komai, H. Ishigaki, K. Matsuyama, Bull. Chem. Soc. Japan, **58**, 2431 (1985).
360. K. V. Scherer Jr., L. Batt, P. H. Stewart, Int. J. Chem. Kinet., **26**, 73 (1994).
361. S. Suyama, Y. Watanabe, Y. Sawaki, Bull. Chem. Soc. Japan, **63**, 716 (1990).
362. T. Komai, K. Matsuyama, M. Matsushima, Bull. Chem. Soc. Japan, **61**, 1641 (1988).
363. K. Matsuyama, Y. Higuchi, Bull. Chem. Soc. Japan, **64**, 259 (1991).
364. K. Matsuyama, H. Kumura, J. Org. Chem., **58**, 1766 (1993).

365. L. F. R. Cafferata, J. D. Lombardo, Int. J. Chem. Kinet., **26**, 503 (1994).
366. B. Yamada, T. Otsu, Makromol. Chem., **190**, 915 (1989).
367. U. Shanker, G. N. Kulsrestha, J. S. Sharma, B. S. Pathania, Indian J. Tech., **23**, 318 (1985).
368. T. Komai, K. Kato, K. Matsuyama, Bull. Chem. Soc. Japan, **61**, 2641 (1988).
369. H. Sawada, H. Hagii, K. Aoshima, T. Arai, Bull. Chem. Soc. Japan, **57**, 1161 (1984).
370. H. Sawada, H. Hagii, K. Aoshima, M. Yoshida, M. Kobayashi, Bull. Chem. Soc. Japan., **58**, 3448 (1985).
371. F. Severini, R. Gallo, J. Thermal Anal., **30**, 841 (1985).
372. A. S. Nazran, J. Warkentin, J. Am. Chem. Soc., **104**, 6405 (1982).
373. L. F. R. Cafferata, G. N. Eyler, M. V. Mirifico, J. Org. Chem., **49**, 2107 (1984).
374. A. F. Shushunova, L. Y. Prokhorova, J. Chrom., **283**, 365 (1984).
375. J. M. Bessiere, B. Boutevin, O. Loubet, Polym. Bull., **30**, 545 (1993).
376. J. C. M. Torfs, L. Deij, A. J. Dorrepaal, J. C. Heijens, Anal. Chem., **56**, 2863 (1984).
377. M. Buback, C. Hinton, Zeit. Phys. Chem., **193**, 61 (1996).

Propagation and Termination Constants in Free Radical Polymerization

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A. INTRODUCTION

In free radical polymerization the propagation and termination rate constants describe the reactions

$$P_n^{\bullet} + M \xrightarrow{k_p} P_{n+1}^{\bullet} \qquad (A1)$$

$$2P_n^{\bullet} \xrightarrow{k_t} \text{dead polymer} \qquad (A2)$$

where $P_n^{\bullet}$ is a propagating chain of any length n and M is the monomer. The rate constants are defined by the following equations:

$$R_p = -\frac{d[M]}{dt} = k_p[P^{\bullet}][M] \qquad (A3)$$

$$R_t = -\frac{d[P^{\bullet}]}{dt} = k_t[P^{\bullet}]^2 \qquad (A4)$$

where

$$[P^{\bullet}] = \sum_{n=1}^{\infty} [P_n^{\bullet}] \qquad (A5)$$

The rate constants k_p and k_t have usually been assumed to be independent of chain length. In this chapter termination constants depending on the length of the polymer radicals are also reported. In American literature the right hand side of (A4) is written as $2k_t[P^{\bullet}]^2$. In this chapter the definition given in Eq. (A4) has been used.

Simultaneous determination of absolute values of both k_p and k_t from a single experiment has not been reported. In practice, the ratio k_p^2/k_t is determined from measurements of molecular weight as a function of rate of polymerization for a low conversion polymerization or from measurements of initiation rate and polymerization rate in a low conversion. The ratio k_p/k_t is determined from nonsteady-state measurements of the average lifetime, τ, of the growing polymer chain in a photochemically initiated polymerization. This lifetime may be defined by noting that the concentration of chains present must be related to their average lifetime and rate of disappearance by

$$[P^{\bullet}]/\tau = R_t \qquad (A6)$$

which, from (A3) and (A4), yields

$$\frac{k_p}{k_t} = \frac{\tau R_p}{[M]} \qquad (A7)$$

By combining the separately determined ratios, k_p^2/k_t and k_p/k_t, the individual propagation and termination rate constants may be calculated. Alternatively, the rate of initiation, R_i, may be measured as the rate of initiator disappearance and equated to R_t. This gives (from (A6) and (A3))

$$k_p = \frac{R_p}{R_i \tau [M]} \qquad (A8)$$

There is a large degree of imprecision inherent in measuring τ and in combining data from different experiments, which helps to explain the scatter in the data tabulated here.

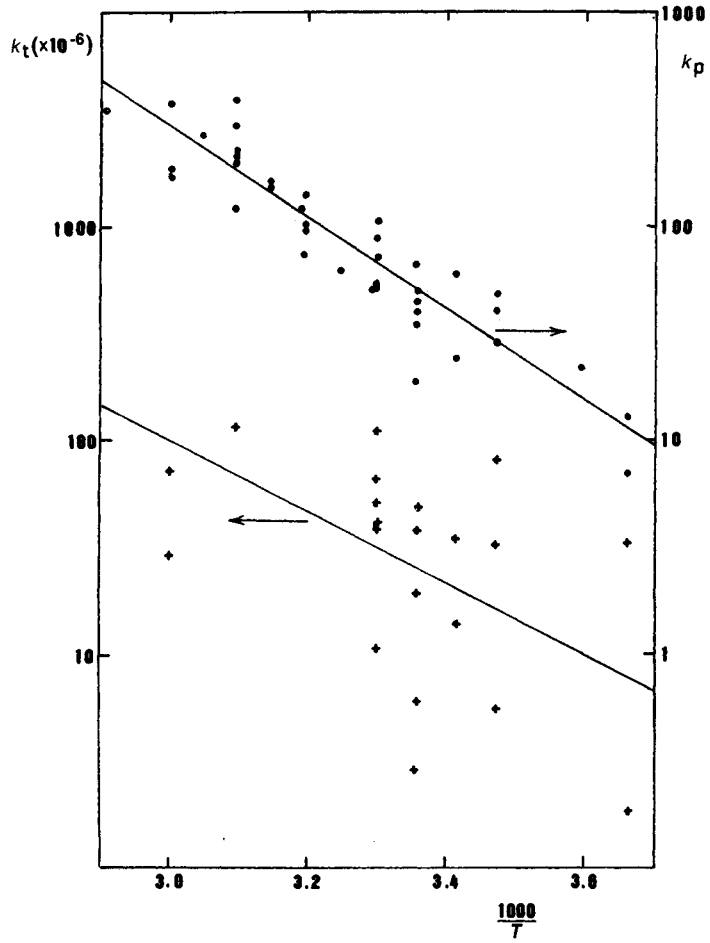

Figure 1. Arrhenius plots of all bulk polymerization data for styrene for k_p (•) and k_t (+). Solid lines are least squares obtained by assuming all points to be of equal value.

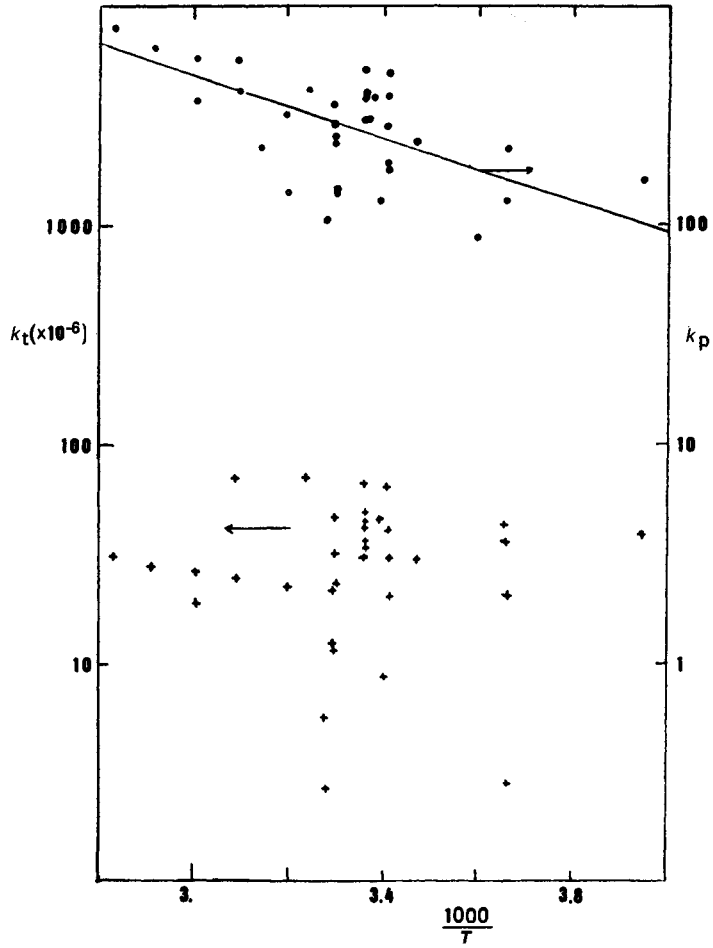

Figure 2. Arrhenius plots of all bulk polymerization data for methyl methacrylate for k_p (•) and k_t (+). Solid lines are least squares obtained by assuming all points to be of equal value. The line for k_t is not shown since a horizontal line for a temperature independent k_t could be a better representation.

The two monomers styrene and methyl methacrylate have been so extensively studied that their data are presented as Arrhenius plots (Figs. 1 and 2). Because of the influence of solvents on the rates, only data of bulk polymerizations are shown. It remains a wide scatter, which should serve as a warning against casual acceptance of any single number. Solid lines are the least squares that have been calculated assuming all points to be of equal value.

For the termination rate of methyl methacrylate the least-squares line results in a positive slope and is not shown. In this case a temperature-independent k_t could be a better representation of the experiments.

The pulse laser polymerization (PLP) method has been developed as a new method of obtaining k_p. In the PLP method, the value of k_p can be estimated from the degree of polymerization of polymer formed, ν_p, a knowledge of the monomer concentration, [M], and t_f the time between pulses:

$$\nu_p = k_p[M]t_f \qquad \text{(A9)}$$

The reproducibility of the PLP method has been recognized by an IUPAC Working Party on Modeling of Kinetic and Process of Polymerization, which is establishing a critical review of the literature values of k_p.

Classification of the methods for the determination of rate constants

The tabulated data refer to seven different methods.

Method A uses the following four methods for the measurement of the lifetime τ:

A1 – rotating sector or a "flashing" laser
A2 – flow through a tube with spatially separated light and dark sections
A3 – spacial intermittent polymerization (SIP)
A4 – intermittent illumination method

Method B effectively measures only a single decrease (increase), in the radical concentration by the following methods:

B1 – dilatometry
B2 – dielectric constant
B3 – interferomety
B4 – temperature change
B5 – viscosity
B6 – light scattering
B7 – monomer pressure

B8 – cathetometer
B9 – inhibitor
B10 – scavenger

Method C uses electron spin resonance (ESR) for the determination of the radical concentration

Method D refers to values obtained in emulsion polymerization by application of the Smith–Ewart theory

Method E refers to recalculated values

Method F uses pulse laser photopolymerization (PLP) method

F1 – PLP
F2 – PLP-GPC (MWD)
F3 – PLP-MALDI TOF (-GPC)

Method G refers to the method of measuring molecular weight (MW) and molecular weight distribution (MWD) by

G1 – gel permeation chromatography (GPC)
G2 – high performance liquid chromatography (HPLC)

B. TABLES OF PROPAGATION AND TERMINATION CONSTANTS

TABLE 1. DIENES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
1-Acetoxybutadiene	18.0	281	25	A1		152
Butadiene	$k_p = 1.2 \times 10^8 \times \exp(-39000/RT)$		10	D		23
	150 ± 40	–	5	C		223
		$k_t = 1.13 \times 10^4 \times \exp(-711/T)$	45–60	G1		202
	$\ln k_p = \ln (3.873 \times 10^{10}) - 53400/RT$		50	F1		221
	111.6	–	50	F1	Solvent: chlorobenzene	221
	$k_p = 8.05 \times 10^7 \times \exp(-35710/RT)$			F2	Solvent: chlorobenzene	187
Chloroprene	220	–	40	D	Quoted in Ref. 86	59
	$k_p = 2.9 \times 10^9 \times \exp(-41000/RT)$			D		84
	$k_p = 1.95 \times 10^7 \times \exp(-26630/RT)$			F1		198
2,3-Dimethyl-1,3-butadiene	$k_p = 8.9 \times 10^7 \times \exp(-38000/RT)$			D		71
Ethyl 4-ethoxy-2,4-pentadienoate	9.9	9.3	25	A1		152
Ethyl 4-methyl-2,4-pentadienoate	29.7	23	25	A1		152
Ethyl pentadienoate	30.9	19	25	A1		152
1,3-Hexadiene	20 ± 10	–	5	C		223
2,4-Hexadiene	16 ± 12	–	5	C		223
Isoprene	2.8	–	5	D	Calalyst system: diisopropylbenzene, monohydroperoxide-tetraethylenepentamine	24
	125 ± 30	–	5	C		223
2-Methyl-1,3-pentadiene	35 ± 10	–	5	C		223

TABLE 2. OLEFINS

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Ethylene	470 ± 30	1050 ± 50	83	A1	Solvent: benzene	53
	18.6 ± 2	455 ± 50	−20.01	A1		66
	5400	200	130	A1	Pressure 1.8×10^8 Pa	139
	$k_p = 4.8 \times 10^7 \times \exp[(-4450 + 3.1 \times 10^{-6} p)/T]$				Pressure $(5–17.5) \times 10^7$ Pa, using results of Ref. 141	142
	1.2×10^4	$3.9\ (7.8) \times 10^2$	130	A1	Pressure 1.9×10^8 Pa	135
	1.09×10^4	1.99×10^2	190	C	Initial pressure 2.27×10^8 Pa	151
	3.08×10^4	–	230	F1	Pressure 2.55×10^8 Pa, at low or moderate conversion	167
	–	3.2×10^3	190	F1	Pressure 2.50×10^8 Pa, at conversion 0	167
Propylene	$k_p = 2 \times 10^6 \times \exp(-32000/RT)$		50–150		Radical telomerization	129

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TABLE 3. ACRYLIC DERIVATIVES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Acrylamide	6000 ± 1000	3.3 ± 0.6		A1, B1	Solvent: water, pH 5.5	79
	18000 ± 1500	14.5 ± 2.0	25	A1	Solvent: water	41
	8200	5.5	19	A	Solvent: water	89
	220	1.0	26	A	Solvent: DMSO	89
	$(7.9 \pm 0.5) \times 10^4$	660 ± 40	30	A1	Solvent: water 0.38 M	126
	$(4.3 \pm 0.2) \times 10^4$	350 ± 20	30	A1	H_2O : DMSO (90 : 10)	119
	$(3.3 \pm 0.2) \times 10^4$	230 ± 20		A1	H_2O : dioxane (90 : 10)	119
	$(2.3 \pm 0.1) \times 10^4$	160 ± 10		A1	H_2O : THF (92 : 8)	119
	$k_p = 7 \times 10^6$ $\times \exp[(-21000 \pm 2000)/RT]$			F1		197
Acrylic acid						
	650	2.6	23	A1	Solvent: Water, pH 7.9, (with 1.2 M NaOH)	99
	3150	2.6		A1	Solvent: water, pH 7.9 (with 1.5 N NaCl)	99
	6600	2.7		A1	Solvent: water, pH 11	99
–, butyl ester	13	0.018	25	A1		10
	2100	330	25	B4		57
	1977	3.84		A1	Pressure 5.0×10^7 Pa	107
	679 ± 66	6.4 ± 0.6	30	A1	Solvent: benzene 1.76 M	137
	$\log_{10} k_p = 6.0123 - 748.4/T$		25–80	F1		201
		$\ln[(k_t[p]/(1.00 + 0.06p)] = \ln k_t\,[1] - 0.60p$	30	B5	$k_t\,[p]$; k_t for pressure p (bar), $k_t\,[1] = 3.5 \times 10^6$	122
	1360	1.28	30	A1	Solvent: anisole 2.00 M	138
	963	1.18			Solvent: benzene 2.00 M	138
	1320	1.07			Solvent: chlorobenzene 2.00 M	138
	360	0.36			Solvent: benzonitrile 2.00 M	138
–, cyclohexyl ester	2722 ± 248	3.2 ± 0.3	30	A1	Solvent: benzene 1.76 M	137
–, ethyl ester	800	1.76	50	A1	Solvent: toluene 1.84 M	108
	12.39×10^3	139.2	50	A1	Solvent: benzene x; molar fractions of monomer, $x = 0.401$, k_p and k_t also given by other [M]	171
–, 2-ethylhexyl ester	155	0.233	50	A1	Solvent: toluene 1.94 M	108
–, methyl ester	1300	75	15	A1	From unpublished results of Ross and Melville	22
	880	260	15	A1	From unpublished results of Matheson	22
	1580	55	25	A1		19
	580	6.5	25	B1		48
	$k_p = 1.0 \times 10^8$ $\times \exp(-30000/RT)$	$k_t = 2.8 \times 10^{11}$ $\times \exp(-22000/RT)$		A1		21
	1000	3.5550		A1		50
	11680	194		A1	Solvent: benzene 2.69 M	144
	21300	–		A1		144
	$(1.5 \pm 0.2) \times 10^4$	–	60	G2	Primary propagating step (k_{p1}), Ph-MA•	190
	$(3.7 \pm 0.6) \times 10^4$	–	60	G2	$PhCO_2$-MA•	190
	6.3×10^4	–	60	G2	*tert*-BuO-MA•	190
	$(1.4 \pm 0.2) \times 10^4$	–	60	G2	Secondary propagating step (k_{p2}), Ph-MA-MA•, $PhCO_2$-MA-MA•, *tert*-BuO-MA-MA•	
–, 2-(acetoacetoxy-methyl)-, ethyl ester	300	1.0	60	C	Solvent: benseme	170
–, 2-cyano-, ethyl ester	1622	411	30	A1	AcOH[7] wt.%	143
	1610	404	30	A1	1,3-propanesulfone 0.5 wt.%	143
	1613	411	30	A1	Optimum value, solvent: AcOH	123
	1607	404	30	A1	Optimum value Solvent: propanesulfone	123
–, 2-chloro-, ethyl ester	1660	333	30	A1	AIBN 4.88×10^{-2} M	140
	1408	244	30	A1	Optimum value	123
–, 2-fluoro-, ethyl ester	1120	4.8×10^2	30	A1		146
	978	435	30	A1	Optimum value	123
–, 2-acetoxymethyl-, methyl ester	350	2.1	60	C	Solvent: benzene 1.00 M, MAIB 0.050 M	225

TABLE 3. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
–, 2-acetyloxy-, methyl ester	430	–	60	C	Solvent: 1,1,2-trichloroethane 2.0 M 2,2′-azobis (4-methoxy-2,4-dimethylvaleronitrile) 0.030 M, k_p also given by other temp.	224
–, 2-benzoyloxy-, ethyl ester	990	2.9	60	C	Solvent: benzene 1.56 M	172
–, 2-(benzyloxymethyl)-, methyl ester	182	1.6	60	C	Solvent: benzene 2.0 M, AIBN 5.0×10^{-3} M	206
–, 2-[2,2-bis(carbomethoxy) ethyl], methyl ester	4.0	0.038	60	C	Bulk AVN 0.05 M, k_p and k_t also given by other temp.	228
–, 2-butoxy-, methyl ester	298	8	60	C	Solvent: benzene 2 M	176
	184	–	60	C	Solvent: 1,1,2-trichloroethane 2.0 M 2,2′-azobis (4-methoxy-2,4-dimethylvaleronitrile) 0.030 M, k_p also given by other temp.	224
–, 2-butyroxymethyl-, methyl ester	360	1.4	60	C	Solvent: benzene 1.00 M, MAIB 0.050 M	225
–, 2-(2-carbomethoxy)- ethyl-, methyl ester	19	0.51	60	C	Bulk MAIB 0.05 M, k_p and k_t also given by other temp.	229
–, 2-ethyl-, methyl ester	8.6	21	60	C	Bulk AIBN 0.10 M	230
–, 2-ethyl-, cyclohexyl ester	1.6	1.8	60	C	Bulk AIBN 0.10 M	230
–, 2-isobutyroxymethyl-, methyl ester	300	1.1	60	C	Solvent: benzene 1.00 M, MAIB 0.050 M	225
–, -2-(methoxycarbonylmethyl)-, phenetyl ester	11	0.08	50	C	Solvent: benzene 2.22 M, k_p and k_t also given by other temp. and [M]	157
–, 3-methyl, adamantyl ester	0.57	0.11	60	C	Bulk 4.8 M MAIB 0.05 M	232
	0.76	1.4	60	C	Solvent: benzene 2.4 M, MAIB 0.05 M, k_p also given by other [M]	232
–, 3-methyl, dimethyladamantyl ester	0.41	0.081	60	C	Bulk 4.0 M, MAIB 0.05 M	232
	0.71	1.1	60	C	Solvent: benzene 2.0 M, MAIB 0.05 M	232
–, 3-methyl, *tert*-butyl ester	0.90	2.3	60	C	Bulk 5.9 M, MAIB 0.05 M	232
	1.0	4.7	60	C	Solvent: benzene 2.0 M, MAIB 0.05 M	232
–, 2-naphthoyloxymethyl-, ethyl ester	320	0.37	60	C	Solvent: benzene 1.00 M, MAIB 0.01 M	226
–, 2-pivaroxymethyl-, methyl ester	230	0.59	60	C	Solvent: benzene 1.00 M, MAIB 0.005 M	225
–, *trans*-2-vinyl-, methyl ester	125	–	60	C	Solvent: benzene 1.0 M, AIBN 0.25 M	227
Acrylonitrile	3000–5000	–	0	D		42
	127	12.2	25	B4		49
	52	5	25	B4		63
	51	1.8	25	B4	Solvent: DMF	75
	20000		40	D		42
	24	23	50		Primary radical termination in a precipitating medium	35
	1960	782	60	A1	Solvent: DMF	43
	15400	2700	25	A1	Solvent: water	54
	28000	3700	25	A1	Solvent: water	55
	1910	290	25	A1	Solvent: DMSO	70
	382 ± 230	47.6 ± 22	25	A1	Solvent: DMF	68
	3300 ± 300	1200 ± 170	50	A1	Solvent: DMF 3.8 M	88
	3200 ± 400	300 ± 50	50	A1	Solvent: DMSO 3.8 M	88
	3000 ± 600	240 ± 70	50	A1	Solvent: Ethylene carbonate 3.8 M	88
	$(6.5 \pm 1.3) \times 10^3$	4700 ± 1300	50	A1	Solvent: $Mg(ClO_4)_2 \cdot 12H_2O$ 3.04 M	88
N-Acryloyl-2,2-dimethyl-5 (R)-phenyl-1,3-dioxazoline	28.2	0.20	60	C	Solvent: benzene 1.08 M, MAIB 0.05 M, k_p and k_t also given by other temp.	231
N-Acryloylpiperidine	273	17.9	30	A1		145
	273	11.9	30	A1	Optimum value	123
N-Acryloylpyrrolidine	8267	2200	30	A1	Optimum value	123

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TABLE 3. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
N,N'-Dimethylacrylamide	11000	38	50	A1		74
	27200	3540	30	A1		121
	29198	4080	30	A1	Optimum value	123
	15×10^3	–	25	F1	Solvent: water, pH 1.4	197
3-Dimethyl-(acryloyloxyethyl)-ammonium propane sulfonate	2330 ± 230	0.540 ± 0.053	30	A4	Solvent: formamide : 1,4-dioxane (3 : 1 (v/v))	192
	2760 ± 342	0.452 ± 0.056	30		Solvent: formamide: 1,4-dioxane (3 : 1 (v/v)), NaCl 0.1 M	192
N-Methylacrylamide	9.9×10^3	–	25	F1	Solvent: water, pH 1.9	197

TABLE 4. METHACRYLIC DERIVATIVES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Methacrylamide	79.6 ± 5.6	16.5 ± 1.4	25	A1	Solvent: water	69
	1.1×10^3	–	25	F1	Solvent: water, pH 1	197
Methacrylonitrile	~21	~27	30			26
	26	21	25	A1		39
	$k_p = 10^{6.43 \pm 0.26} \times \exp[(-29700 \pm 1500)/RT]$					
Methacrylic acid						
	670	2.1	23	A1	Solvent: water, pH 8.0 (with 0.22 M NaOH)	99
	1950	2.25		A1	pH = 13.6	99
–, benzyl ester	1410	41.9	30	A1		101
	110	80	32.5	C		153
	895	40	32.5	C		153
	1250	41.9	30	A1		87
	510	2.87	20	A1	k_p and k_t also given as a function of viscosity	103
–, bornyl ester	580	3.3	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M	193
–, *n*-butyl ester	369	10.2	30	A1		27
	573	18.0	30	A1	Pressure: 5.0×10^7 Pa	106
	360	10	30	A1		109
	$\ln k_p = 6.13 + 6.33 \times 10^{-9} p$		30	F2		216
	$\ln k_p = 15.8 - 2751/T$		12–93	F2	Pressure 1.0×10^8 Pa	216
	–	2.6	70	B5	$\eta = 1.92$ cp	212
		$\ln[(k_t[p]/(1.00 + 0.01p)] = \ln k_t[1] - 0.73p$	30	B5	$k_t[p]$; k_t for pressure p (bar), $k_t[1] = 13.4 \times 10^6$	122
	$\ln k_p = (14.41 \pm 0.09) - (2472 \pm 29)/T$		10–90	F1		219
	$k_p = 3.44 \times 10^6 \exp(-23300/RT)$			F1		169
	1576	9.74	30	A3		225
–, *tert*-butyl ester	350	14	25	A1		52
	$k_p = 10^{7.4 \pm 0.4} \times \exp[-(27.7 \pm 2.5) \times 10^3/RT]$		9–66	F1		203
–, *trans*-4-*tert*-butyl-cyclohexyl ester	550	1.9	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M, *trans* 100%	194
	510	1.9	60	C	*trans* 28.9%	194
	570 ± 10	0.5–2.3	60	C	Solvent: benzene	210
–, cetyl ester	300 ± 90	0.16 ± 0.04	30	A1		109
–, *p*-[*p*-(cetyloxy)-benzoyloxy]-phenyl ester	300	0.25	50	A1	Solvent: dioxane 0.2 M	97
	170	0.30	50	A1	Solvent: toluene 0.2 M	97
–, 2-chloroethyl ester	254	6.71	30	A1		87
–, 2-cyclohexyl-ethyl ester	1190	32.8	30	A1		87
–, cyclohexyl ester	510	5.4	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M	193
–, 2-decahydro-naphthyl ester	570	3.1	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M	193

TABLE 4. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
–, 2,6-dimethyl-phenyl ester	68	2.1	30	A1		120
	72	2.4	30	A1	Optimum value	123
–, dodecyl ester	–	0.6	70	B5	$\eta = 3.96$ cp	212
–, ethyl ester	126	7.35	30	A1		87
	–	10	70	B5	$\eta = 0.91$ cp	212
	$k_p = 1.50 \times 10^6 \times \exp(-20460/RT)$			F1		169
	$\ln k_p = (15.11 \pm 0.17) - (2753 \pm 55)/T$		10–90	F1		219
–, hexadecyl ester	–	0.16	70	B5	$\eta = 7.52$ cp	212
–, isobornyl ester	390	3.5	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M	193
–, isobutyl ester	$\ln k_p = (14.72 \pm 0.13) - (2590 \pm 42)/T$		10–90	F1		219
–, isopropyl ester	121	4.52	30	A1		87
–, lauryl ester	460 ± 140	0.6 ± 0.2	30	A1		109
	$k_p = 2.93 \times 10^5 \times \exp(-16190/RT)$			F1		169
–, 2-methoxyethyl ester	249	9.30	30	A1		87
–, methyl ester	310 ± 20	66 ± 4	23.6	A1		9
	$k_p = 5.13 \times 10^6 \times \exp(-26400/RT)$	$k_t = 1.36 \times 10^3 \times \exp(-11900/RT)$		A1		7
	187	72.1	32.5	C		153
	404	17.6	40	B1	Solvent: ethyl acetate	51
	410	24	50	E	Using results of Ref. 7	62
	13.2	0.488	–30	B1	Solvent: ethyl acetate	72
	390	–	20	A1	k_p also given as a function of temp. and of viscosity for other solvents	92
	517	37	10	B1	Solvent: methanol	94
	527	23	10	B1	Solvent: pyridine	94
	28	4.4	10	B1	Solvent: DMF	94
	260	21	25	A1	Solvent: benzene 4.69 M	85
	270	21	25	A1	Solvent: fluorobenzene 4.69 M	85
	280	19.5	25	A1	Solvent: chlorobenzene 4.69 M	85
	285	17.5	25	A1	Solvent: anisole 4.69 M	85
	310	17	25	A1	Solvent: bromobenzene 4.69 M	85
	330	17	25	A1	Solvent: benzonitrile 4.69 M	85
	340	17	25	B5	Solvent: methyl benzoate 4.69 M	90
	240	11.5	25	B5	Solvent: methylphenyl acetate 4.69 M	90
	335	9	25	B5	Solvent: dimethyl phthalate 4.69 M	90
	270	22	25	B5	Solvent: dimethyl carbonate 4.69 M	90
	330	16	25	B5	Solvent: diethyl oxalate 4.69 M	90
	250	29	25	B5	Solvent: methyl formate 4.69 M	90
	240	25	25	B5	Solvent: methyl propionate 4.69 M	90
	280	14	25	B5	Solvent: diethyl succinate 4.69 M	90
	320	28	25	B5	Solvent: acetonitrile 4.69 M	90
	390 ± 40	42 ± 4	25	A1	Solvent: methanol 50% by vol.	104
	410 ± 40	29 ± 3	25	A1	Solvent: *n*-octane 5% by vol.	104
	410 ± 40	26 ± 3	25	A1	Solvent: *n*-nonane 20% by vol.	104
	180 ± 50	20 ± 6	20	E	$\overline{DP} = (2-8) \times 10^3$ using results of Ref. 86	96
	500 ± 250	63 ± 32	20	B6	$\overline{DP} = (3-5) \times 10^4$	96
	41.6	2.69	0	B5	Assuming biradical initiation	11
	62	–	5	D		65
	200	–	20	B6	k_p found as a linear function of M_w	91
	128	8.44	22	B1		29
	364	41.8	22.5	B4	Rate of initiation; 1.20×10^{-8} (moles/l/s)	60
	512.6	46.6	25	A1		30
	410	42.7	25	A1	k_p and k_t also given as a function of viscosity for other solvents	93
	248	22.7	30	A1		14
	141	11.6	30	A		87
	106	5.7	32	B4		58
	140	–	40	D		65

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TABLE 4. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
	0.224	–	45	D		67
	573	2.0	60	E	Termination by combinat. using results of Refs. 7 and 31	33
	573	11.9	60	E	Termination by disprop. using results of Refs. 7 and 31	33
	995 ± 83	43.6 ± 4.9	80		Solvent: benzene 4.7 M	111
	506 ± 48	35.6 ± 3.9	30	A1	Solvent: anisole 2.0 M	133
	450 ± 33	42.0 ± 3.0		A1	Solvent: benzene 2.0 M	133
	456 ± 29	43.8 ± 2.0		A1	Solvent: C_6D_6 2.0 M	133
	448 ± 47	47.2 ± 6.6		A1	Solvent: fluorobenzene 2.0 M	133
	498 ± 39	42.9 ± 4.7		A1	Solvent: chlorobenzene 2.0 M	133
	614 ± 43	39.9 ± 3.3		A1	Solvent: benzonitrile 2.0 M	133
	427 ± 38	30.9 ± 3.7	30	A1	Solvent: benzene 1.0 M	133
	290	21	60	A1		118
	1020	292		A1	Addition of 1×10^{-3} mol pyridine	118
	530	55		A1	Addition of 1×10^{-3} mol acetone	118
	550	68		A1	Addition of 1×10^{-3} mol triethanolamine	118
	$k_p = 4.92 \times 10^5 \times \exp(-18210/RT)$	$k_t = 98 \times \exp(-2930/RT)$	15–30	A2	$\bar{P}_n = 10^4$	114
	315	33.9	25	A2	k_p and k_t also given as a function of $\bar{P}_n$	110
	336 ± 90	28 ± 12	25	A1	Pressure 5.0×10^6 Pa	127
	441	19.7	30	A1	Pressure 5.0×10^7 Pa	105
	$k_p = 2.5 \times 10^6 \times \exp[(-23000 \pm 2500)/RT]$	720	25–60	C		45
	$k_p = k_p^0$ $(0.33 \leq w_p \leq 0.84)$ $= k_p^0 \exp[-29.8\ (w_p - 0.84)]$ $(0.84 \leq w_p \leq 0.99)$ $(k_p^0 = 790 \pm 300)$		50	C	w_p (weight fraction of polymer)	150
	130	20.7 ± 1.0	0	A1	At high conversions (~65%)	148
	705.6	25	60		Special evaluation from steady-state and $\bar{M}_n$	134
	170	–	0.2	F3	Solvent: toluene	222
	306	–	25	F1		196
	292	–	25	F1	Solvent: ethanol (50%) 4.67 M, k_p also given by other [M] and solvents	
	670 ± 30	22	60	C		175
	$\ln k_p = 14.69 - 2670/T + 0.201/T \times (1.0 \times 10^{-5} p - 1)$		20–90	F1		207
		$\ln k_t[p] = \ln k_t[1] - 0.57p$	30	B5	$k_t[p]$; k_t for pressure p (bar), $k_t[1] = 15.4 \times 10^6$	122
	26.6	1.4	5	B10	Scavenger DPPH, solvent: DMF, blank polymn.	115
	5.8	0.017	5	B10	Scavenger DPPH, solvent: DMF, template polymn., template; it-MMA	115
		$k_t(m,n) = 1.22 \times 10^2 (n \times m)^{-0.075}$	25	A3	n: n-mer; m: m-mer, coupling with n-mer and m-mer	147
	–	21	70	B5	$\eta = 0.53$ cP	212
	–	0.034	50	D	No addition of chain-transfer agent, Time 7.5–20.5 min.	149
	–	0.16	50	D	Addition of 2% CBr_4, Time 7.5–20.5 min.	149
	$k_p = 10^{6.694} \exp(-23940/RT)$		−1 to −70	F1		217
	$\ln k_p$ (l/mol/min) $= \ln k_{p,0} - (3.86\ [\text{AIBME}]_0 + 0.8) \times (1 - \phi_m) - 1.6\ (1 - \phi_m)^{6.0}$		60	C	AIBME (dimethyl 2,2′-azodiisobutyrate) 0.0434 M, ϕ_m; monomer volume fraction $k_{p,0} = 700$ (l/mol/min)	158
	131	–	−8	F3	Benzoin 7×10^{-4} M	199
	680–820	–	30	F1	Solvent: toluene, pressure 1.0×10^8 Pa, k_p also given by other pressures	208
	690–810	–		F1	Solvent/2-butanone, pressure 1.0×10^8 Pa	208

TABLE 4. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
	313 ± 8.1	39 ± 4	25	F1		161
	794.0	62.4	60	B5	Solvent: benzene 3 M	204
	815.0	51.6			Solvent: chlorobenzene 3 M	204
	957.0	38.2			Solvent: benzonitrile 3 M	204
	–	$25(\pm 3)$	60	C		166
	510 ± 100	21 ± 2	60	C		191
	299	45	25	A1		160
	431	50	40			160
	$k_p = 2.39 \times 10^6 \times \exp(-22000/RT)$			F1		198
	359	–	25	F1		220
	12	–	50	C	Emulsion polymn., particle size 260 nm, k_p also given by other particle sizes	184
	135	–	0	A1	k_p; average value	159
	292	–	25	F1		164
	200	25.6	28	B1	Values from Chem. Abstr.	83
	135 ± 50	–	0	A1		165
	170	–	50	C	Emulsion polymn., *n*-butyl acrylate: MMA: methacrylic acid (8 : 91 : 1), particle size (p.s.) 50 nm, k_p also given by other temp. and particle sizes	155
	35	–	50		p.s. 500 nm	155
	2960	29.1	30	A3		225
	$k_p = 1.9068 \times 10^6 \times \exp(-21181.07/RT)$			F1		227
	$k_p = 1.2169 \times 10^6 \times \exp(-25203.59/RT)$			F1	30% MMA in benzyl alcohol	227
	$k_p = 3.0598 \times 10^7 \times \exp(-28008.18/\mathrm{RT})$			F1	30% MMA in NMP	227
	35	–	50		p.s. 500 nm	155
–, octyl ester	1710	62.6		A1	Pressure 5.0×10^7 Pa	113
	240 ± 80	2.6 ± 0.9	30	A1		109
		$\ln[(k_t[p]/(1.00-0.19)] = \ln k_t[1] - 1.01p$	30	B5	$k_t[p]$; k_t for pressure p (bar), $k_t[1] = 40.0 \times 10^6$	122
–, phenyl ester	411	11.9	30	A1		87
	230 ± 15	2.02 ± 0.22	30	A1	Solvent: anisole 2.00 M	116
	176 ± 8	2.06 ± 0.21		A1	Solvent: benzene 2.00 M	116
	180 ± 9	2.16 ± 0.19		A1	Solvent: fluorobenzene 2.00 M	116
	223 ± 11	2.26 ± 0.22		A1	Solvent: chlorobenzene 2.00 M	116
	235 ± 8	1.72 ± 0.11		A1	Solvent: bromobenzene 2.00 M	116
	273 ± 8	1.96 ± 0.12		A1	Solvent: benzonitrile 2.00 M	116
–, 2-phenylethyl ester	228	1.88	30	A1		87
–, 3-phenylpropyl ester	149	0.813	30	A1		87
–, *n*-propyl ester	467	45.1	30	A1		28
–, 3-tetracyclo-[4.4.0^{2,5}.1^{7,10}]-dodecyl ester	590	2.6	60	C	Solvent: benzene 1.0 M, AIBN 0.05 M	193
–, triphenylmethyl ester	26	0.301	32.5	C		153
3-Dimethyl-(methacryloyloxy-ethyl) ammonium propane sulfonate	1470 ± 170	0.499 ± 0.058	30	A4	Solvent: formamide: 1,4-dioxane (3 : 1 (v/v))	192
	2760 ± 342	0.452 ± 0.056	30		Solvent: formamide: 1,4-dioxane (3 : 1 (v/v)), NaCl 0.1 M	192
Polyhydroxytetramethylene-α, ω-methacrylate of MW = 700	185	0.034	20	B8	Calcd. from nonstationary state kinetics	125
	230	0.053	20	B8	Calcd. from stationary state kinetics	125

TABLE 5. ITACONIC DERIVATIVES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Itaconic acid						
–, bis (4-*tert*-butyl-cyclohexyl) ester	1.7	2.5×10^{-3}	60	C	Solvent: benzene 0.5 M	213
–, bis (cyclohexyl-methyl) ester	3.9	3.4×10^{-3}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182

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TABLE 5. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
–, bis (3,5-dimethyl-1-adamantyl) ester	0.92	0.32×10^{-3}	60	C	Solvent: benzene 1.5 M	213
–, bis (2-ethylhexyl) ester	6.8	–	50	C	Solvent: hexane 1.03 M	185
	3.4	–	50	C	Solvent: cyclohexane 1.03 M	185
	2.0	–	50	C	Solvent: toluene 1.03 M	185
	2.1	–	50	C	Solvent: benzene 1.03 M	185
	2.2	–	50	C	Solvent: chlorobenzene 1.03 M	185
	2.1	–	50	C	Solvent: ethyl acetate 1.03 M	185
	2.1	–	50	C	Solvent: THF 1.03 M	185
	1.4	–	50	C	Solvent: acetone 1.03 M	185
	3.0	0.001	50	C	Solvent: benzene 2.0 M, MAIB 0.05 M	177
–, diadamantyl ester	0.56	0.30×10^{-3}	60	C	Solvent: benzene 1.5 M, k_p and k_t also given by other [M]	213
–, di-*n*-butyl ester	3.5	1.59×10^{-2}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
	13	0.11	50	C	Solvent: cyclohexane 1.33 M, MAIB 5.00×10^{-2} M, k_p and k_t also given by other temp.	162
	11	0.13	50		Solvent: *n*-hexane 1.33 M	162
	6.8	0.07	50		Solvent: chlorobenzene 1.33 M	162
	9.4	0.11	50		Solvent: methyl benzoate 1.33 M	162
	6.7	0.083	50		Solvent: benzene 1.33 M	162
	3.4	0.068	50		Solvent: acetone 1.33 M	162
	3.2	0.13	50		Solvent: acetonitrile 1.33 M	162
	–	1.0×10^{-3}	50	C	Solvent: benzene 0.794 M, AIBN 0.05 M k_t also given by other [M]	154
	5.3	8.5×10^{-2}	50	C	Solvent: benzene 0.794 M, MAIB 0.05 M	154
	5.9	0.053	50	C	Solvent: benzene 2.0 M, MAIB 0.05 M	177
–, di-*sec*-butyl ester	0.6	4.0×10^{-3}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, di-*tert*-butyl ester	0.2	5.6×10^{-3}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, dicyclohexyl ester	2.3	1.4×10^{-3}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, diethyl ester	7.7	0.24	50	C	Solvent: benzene 2.0 M, MAIB 0.05 M	177
	3.7	6.41×10^{-2}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, diisobutyl ester	2.7	1.56×10^{-2}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, diisopropyl ester	1.1	5.0×10^{-3}	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	182
–, dimethyl ester	5.2	0.36	60	C	Solvent: benzene 1.5 M, MAIB 0.02 M	195
	10	0.59	50	C	Solvent: benzene 2.0 M, MAIB 0.05 M	177
–, α-ethyl-β-hexafluoropropyl ester	5.0	0.28	60	C	Solvent: benzene 2.0 M, MAIB 0.05 M, k_p and k_t also given by other [M] and temp.	233
–, α-methyl-β-isopropyl ester	3.4	0.06	60	C		195
–, α-methyl-β-*tert*-butyl ester	4.2	0.051	60	C		195
–, α-isopropyl-β-methyl ester	1.9	0.07	60	C		195
–, α-*tert*-butyl-β-methyl ester	0.91	0.056	60	C		195
N(2,6-dimethylphenyl) itaconimide	26	0.082	50	C	Solvent: THF 0.44 M, AIBN 0.05 M, k_p and k_t also given by other [M]	234
Methyl *N*-phenyl-itaconamate	15	0.29	60	C	Solvent: DMF 1.4 M, AIBN 0.05 M, k_p and k_t also given by other [M] and temp.	235

TABLE 6. FUMARIC DERIVATIVES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Fumaric acid						
–, bis (2-methoxy-ethyl) ester	0.19	330×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, bis (2-chloro-ethyl) ester	0.26	320×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, *tert*-butyl methyl ester	0.23	150×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, *tert*-butyl isopropyl ester	0.51	26×10^{-6}	60	C	Solvent: benzene 1.0 M	211
	0.35		60	C	AIBN	181
	0.39	1.75×10^{-5}			MAIB	181
–, di-*tert*-butyl ester	0.61	30×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, dicyclohexyl ester	0.60	40×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, diethyl ester	0.21	$(510-560) \times 10^{-6}$	60	C	Solvent: benzene 1.0 M	211
	0.015	1.64×10^{-4}	30	A1	AVN (2,2′-azobis-2,4-dimethylvaleronitrile), ACN (1.1′-azobiscyclo-hexane-1-carbonitrile)	173
	0.029 ± 0.003	8×10^{-6}	30	B10	ACN, scavenger TPV (1,3,5-triphenylverdazyl), sovent: benzene	179
–, diisopropyl ester	0.46	$(80-100) \times 10^{-6}$	60	C	Solvent: benzene 1.0 M	211
	0.31 ± 0.07	0.84×10^{-6}	30	B10	ACN, scavenger TPV, solvent: benzene	180
–, dimethyl ester	0.058	430×10^{-6}	60	C	Solvent: benzene 1.0 M	211
–, dineopentyl ester	0.028	44×10^{-6}	60	C	Solvent: benzene 1.0 M	211
Ethyl *o*-formylphenyl fumarate	4.0	1.4	60	C	Solvent: benzene 1.16 M, MAIB 0.100 M, k_p and k_t also given by other [M]	236

TABLE 7. VINYL HALIDES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Tetrafluoroethylene	7400	7.4×10^{-5}	40		Large active chain end concentration measured by addition of inhibitor in aqueous solution polymn.	81
Vinyl bromide	22.7	92	−30	A1		80
	570	385	20		Quoted in Ref. 80	44
Vinyl chloride	$k_p = 3.3 \times 10^6 \times \exp(-15000/RT)$	$k_t = 1.3 \times 10^6 \times \exp(-4200/RT)$		A1		32
	11000	2100	25	A1		56
	3130	2300	25	A1		77
Vinylidene chloride	8.6	0.175	25	A1		18
	$k_p = 10^9 \times \exp(-4540/T)$	$k_t = 10^6 \times \exp(-2670/T)$	22–75	A1	Solvent: *N*-methylpyrrolidone 2.0 M	117

TABLE 8. VINYL ESTERS

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Vinyl acetate	4600	220	25	B5		13
	795	46	15	B1		47
	556–586	2860–3040	15.9	A1		1
	670–770	2500–3100	15.9	A1		3
	700	2600	15.9	A1	Solvent: *n*-hexane	4
	559	51.8	20	B4		37
	1100	80	25	A1		2
	1000	59	25	A1		15
	895	24	25	B4	At 4% conversion	36
	$k_p = 2.43 \times 10^8 \times \exp(-30600/RT)$	$k_t = 4.16 \times 10^5 \times \exp(-21900/RT)$		A1		8
	9500–19000	380–760	60	E	Using results of Refs. 8 and 15	64
	$(15-26) \times 10^2$	25	50	A1		128

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TABLE 8. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
	117 ± 12	319 ± 61	30	A1	Solvent: benzene 2.00 M	130
	113 ± 10	351 ± 54		A1	Solvent: C_6D_6 2.00 M	130
	48 ± 5	239 ± 45		A1	Solvent: anisole 2.00 M	130
	97 ± 10	311 ± 59		A1	Solvent: fluorobenzene 2.00 M	130
	61 ± 9	266 ± 75		A1	Solvent: chlorobenzene 2.00 M	130
	37 ± 5	412 ± 104		A1	Solvent: ethyl benzoate 2.00 M	130
	8 ± 1	258 ± 76		A1	Solvent: benzonitrile 2.00 M	130
	637 ± 101	94.2 ± 23.7		A1	Solvent: ethyl acetate 2.00 M	130
	• 78	–			Solvent: ethyl acetate: ethyl benzoate (1 : 1 (v/v))	130
	120	310	32.5	A1	Solvent: benzene	153
	530	630	32.5	C	Solvent: benzene 5.41 M	153
	1900	280	32.5	C	Solvent: ethyl acetate 5.43 M	153
	1400	–	32.5	C		153
	4435	64.4	30	A1	Pressure 5.0×10^7 Pa	112
	$k_p = 2.0 \times 10^6 \times \exp[(-19000 \pm 2900)/RT]$	27.5	15–60	C		45
	$k_p = 2.7 \times 10^8 \times \exp(-27820/RT)$			F1		209
		$\ln[(k_t[p]/(1.00-0.07p)] = \ln k_t[1] - 0.50p$	30	B5	$k_t[p]$; k_t for pressure p (bar) $k_t[1] = 52.0 \times 10^6$	122
Vinyl benzoate	106 ± 14	253 ± 63	30	A1	Solvent: anisole 1.01 M	116
	185 ± 8	402 ± 32		A1	Solvent: benzene 1.01 M	116
	245 ± 13	522 ± 48		A1	Solvent: fluorobenzene 1.01 M	116
	168 ± 18	383 ± 76		A1	Solvent: chlorobenzene 1.01 M	116
	69 ± 13	257 ± 72		A1	Solvent: ethyl benzoate 1.01 M	116
	33 ± 3	299 ± 44		A1	Solvent: benzonitrile 1.01 M	116
	267 ± 33	451 ± 65		A1	Solvent: ethyl acetate 1.01 M	116

TABLE 9. VINYL ETHERS

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Refs.
Monovinyl ether of ethylene glycol	5.0 ± 1.0	1.5 ± 0.3	50	A1	136
Monovinyl ether of diethylene glycol	3.0 ± 0.8	1.4 ± 0.3	50	A1	136

TABLE 10. STYRENE DERIVATIVES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Styrene	40 ± 20	80 ± 40	15	A1	Dimension of k_p and k_t: kg/mol/s	46
	24	14	20	A1	From copolymn. data with sulfur dioxide	25
	51.9	10.5	30	A1		16
	108	32.5	30	A1		38
	106	108	30	A1		78
	102	–	40	D	Quoted in Ref. 86	59
	390	–	50	D		5
	209	115	50	E	Using results of Ref. 20	61
	18.7	2.79	25	B5		6
	39.5	5.96	25	B1		17
	$k_p = 2.16 \times 10^7 \times \exp(-32500/RT)$	$2k_t = 2.59 \times 10^3 \times \exp(-9920/RT)$		A1		20
	22	–	5	D	Catalyst: cumene hydroperoxide–triethylenetetramine	24
	139	–	40	D	Catalyst: persulfate	24
	223	–	50	D		40
	206	–	50	D		67

TABLE 10. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
	$k_p = 2.24 \times 10^{14} \times \exp(-73510/RT)$			B1		73
	50	–	25	D		65
	80	64	60	A1		118
	370	915			Addition of 1×10^{-3} mol pyridine	118
	25 ± 3	9 ± 1.5	25	A1	Solvent: dodecane 40% by vol	104
	35 ± 7	19 ± 4	25	A1		100
	35 ± 5	19 ± 3	25	A1	Solvent: bromobenzene 20% by vol	100
	29 ± 7	15 ± 4	25	A1	Solvent: diethyl malonate 20% by vol	100
	23 ± 8	11 ± 2	25	A1	Solvent: diethyl phthalate 20% by vol	100
	27 ± 4	31 ± 0.4	25	A1	Solvent: dinonyl phthalate 60% by vol	100
	$k_p = 1.09 \times 10^7 \times \exp(-31380/RT)$	$k_t = 1.703 \times 10^3 \times \exp(-9489/RT)$	15–30	A2	$\overline{DP} = 2500$	114
	66.6	22.4	30	A1	Pressure 5.0×10^7 Pa	102
	66.5	–	25	C		98
	$k_p = 2.4 \times 10^8 \times \exp(-38000 \pm 1700)/RT$		50–90			98
	187.1	29.4	60		Special evaluation from steady-state kinetics and $\bar{M}_n$	134
		$k_p = 10^{7.630} \times \exp(-32510/RT)$	–12–93	F2		215
	84	–	25	F1	k_p also given by other [M] and solvents	196
	76	–	25	F1	Solvent/ethanol (25%) 6.52 M	196
	70	–	25	F1	Solvent: methanol (25%) 6.52 M	196
	380 ± 110	–	40	F1	Pressure 2.0×10^8 Pa	186
	480 ± 10	–	70	C	Over a range of degree of polymn. of the polymer radical from 40 to 410	188
	–	16.6–17.0	100	C	Conversion 0–20% TBP (*tert*-butyl peroxide)	174
	256	190	50	C	k_p and k_t also given by other temp.	189
	107	–	25	G1	Laser-flash-initiated polymn. by benzoin	156
		$k_t(m,n) = 1.97 \times 10^2 (n \times m)^{-0.12}$	30	A3	*n*: *n*-mer, *m*: *m*-mer, coupling with *n*-mer and *m*-mer	147
	180 ± 10	–	5	C		223
	$k_p = 1.99 \times 10^7 \exp(-30780/RT)$			F1		169
	$\ln k_p = 17.14 - 1.873 \times 10^{-9} p - 3748/T + 2.02 \times 10^{-6} p/T$		30–90	F2	Pressure 1×10^5–2.8×10^8 Pa	214
	77 ± 4.1	78 ± 12	25	F1		161
	79 ± 5.1	96 ± 9	25	F1	Solvent: toluene 50% (v/v)	161
	187.0	63.2	60	B5	Solvent: benzene 3 M	204
	201.0	51.1			Solvent: chlorobenzene 3 M	204
	167.0	37.9			Solvent: benzonitrile 3 M	204
	641 ± 48	–	91.2	G1	Solvent: toluene	178
	89	132	25	A1		160
	107	–	30	F1	k_p also given by other temp.	198
	92	–	25	F1		220
	78	–	25	F1		164
	23	0.6	28	B1		83
	$\ln k_p = 16.09 - 28950/RT$ (overall) $= 16.47 - 3003/RT$ (chain length ≥ 4)			F1	Solvent: benzene 4.33 M	200
	$k_p = 10^{7.1 \pm 0.5} \times \exp[-(29 \pm 3) \times 10^3/RT]$			B1		131
	$k_p = 1.8861 \times 10^7 \times \exp(-30737.52/RT)$			F1		227
	$k_p = 4.2060 \times 10^8 \times \exp(-37468.93/RT)$			F1	30% styrene in benzylalcohol	227
	$k_p = 4.2458 \times 10^8 \times \exp(-39014.90/RT)$			F1	30% styrene in NMP	227
–, *p*-bromo-	186	46	30	A1		78
–, *p*-chloro-	150	77	30	A1		78
–, *p*-cyano-	219	35	30	A1	Solvent: dimethylacetamide	78
–, *p*-fluoro-	112	127	30	A1		78
–, *p*-methoxy-	2.92	1.06	0	B5		12
	71	33	30	A1		78

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TABLE 10. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
–, *o*-methyl-	$k_p = 1.67 \times 10^{11} \times \exp(-58240/RT)$			B1		73
–, *p*-methyl-	84	66	30	A1		78
	$k_p = 2.27 \times 10^7 \times \exp(-32060/RT)$			B1		73
	$k_p = 10^{5.77} \times \exp(-23.0/RT)$		24–55	F1	Solvent: toluene 2 M	226
Styrene-d_8	$k_p = 3.63 \times 10^7 \times \exp(-31500/RT)$	$k_t = 2.55 \times 10^4 \times \exp(-14200/RT)$		A1		168

TABLE 11. VINYL HETEROAROMATICS

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
N-Vinylcarbazole	6.0	0.306	10	A1		80
2-Vinylpyridine	186	33	25	A1		76
	96.6	8.9	25	B4		76
4-Vinylpyridine	12	3	25	B5		34
5-Vinylpyridine	47	3.5	25	B4		75
–, 2-methyl-	122	66	20	B1	Solvent: methanol 1.95 M, k_p also given as a function of [M]	94
	209	43	20	B1	Solvent: 50% (molar) aqueous methanol	94
	17.3	1.2	20	B1	Solvent: acetic acid	94
N-Vinylpyrrolidone	953	65	20	B4		82
	22000 ± 4000	600 ± 200	25	A1	Solvent: water	132

TABLE 12. ALDEHYDES

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Formaldehyde	–	$(7-10) \times 10^{-3}$	– 190	B4	Solid monomer, gamma irradiation	95

TABLE 13. OTHERS

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
Maleimide						
–, *N-tert*-amyl-	69	0.048[a],0.017[b]	60	C	Solvent: benzene 1 M	205
–, *N-tert*-butyl-	100	0.021[a],0.023[b]	60	C	(a) Determined from steady-state equation $k_t = (2k_d f[I])/[P^\bullet]^2$, AIBN 0.005 M; (b) determined from second order plot of the decay curve of the radical concentration, ATMP (2,2′-azobis-(2,4,4-trimethyl-pentane)) 0.02 M, for *tert*-BMI, ATMP 0.01 M	205
–, *N-tert*-butyl-dimethylsilyl-	120	0.20[a]	60	C		205
–, *N-tert*-octyl-	23	0.027[a],0.015[b]	60	C		205
–, *N*-trimethylsilyl-	120	0.15[a],0.23[b]	60	C		205
–, *N*-cyclohexyl-	49	0.054	50	C	Solvent: benzene 0.559 M, AIBN 1.39×10^{-2} M	163
	55	0.069			Solvent: benzene 0.698 M, AIBN 1.00×10^{-2} M, k_p and k_t also given by other [M] and [I]	163

TABLE 13. *cont'd*

Monomer	k_p (l/mol/s)	k_t ($\times 10^{-6}$) (l/mol/s)	Temp. (°C)	Method	Remarks	Refs.
–, *N*-(2,6-diethylphenyl-	2.0	0.0078	60	C	Solvent: benzene 1 M, AIBN 5×10^{-3} M	183
–, *N*-(2,6-dimethylphenyl)-	14	0.034	60	C	Solvent: benzene 1 M, AIBN 5×10^{-3} M	183
–, *N*-(4-ethylphenyl)-	1200	3.9	60	C	Solvent: benzene 1 M, AIBN 5×10^{-3} M	183
–, *N*-(2-methylphenyl)-	190	0.59	60	C	Solvent: benzene 1 M, AIBN 5×10^{-3} M	183
–, *N*-(2-carboethoxyphenyl)-	96	0.10	60	C	Solvent: benzene MAIB 0.005 M	237
–, *N*-dodecyl-	54	0.12	50	C	Solvent: benzene MAIB 0.01 M	238
	104	0.45	50	C	Solvent: toluene MAIB 0.01 M	238
	175	1.0	50	C	Solvent: anisole MAIB 0.01 M	238
	80	0.40	50	C	Solvent: chlorobenzene MAIB 0.01 M	238
	128	0.064	50	C	Solvent: bromobenzene MAIB 0.01 M	238
	149	0.17	50	C	Solvent: methyl benzoate MAIB 0.01 M	238
	386	–	50	C	Solvent: THF, MAIB 0.01 M	238
	250	–	50	C	Solvent: dioxane, MAIB 0.01 M	238
	235	0.55	50	C	Solvent: ethyl acetate MAIB 0.01 M	238
	196	–	50	C	Solvent: methyl ethyl ketone MAIB 0.01 M	238
–, *N*-octadecyl-	33	0.10	50	C	Solvent: benzene AIBN 0.20 M, k_p and k_t also given by other [I]	239
Oligotetramethylene glycol dimethacrylic ester (MW 600)	400	0.07	50	B9		124

C. REFERENCES

1. G. M. Burnett, H. W. Melville, Nature, **156**, 661 (1945).
2. C. G. Swain, P. D. Bartlett, J. Am. Chem. Soc., **68**, 2381 (1946).
3. G. M. Burnett, H. W. Melville, Proc. R. Soc. London, Ser. A, **189**, 456 (1947).
4. G. M. Burnett, H. W. Melville, Proc. R. Soc. London, Ser. A, **189**, 494 (1947).
5. W. V. Smith, J. Am. Chem. Soc., **70**, 3695 (1948).
6. C. H. Bamford, M. J. S. Dewar, Proc. R. Soc. London, Ser. A, **192**, 308 (1948).
7. M. S. Matheson, E. E. Auer, E. B. Bevilacque, E. J. Hart, J. Am. Chem. Soc., **71**, 497 (1949).
8. M. S. Matheson, E. E. Auer, E. B. Bevilacque, E. J. Hart, J. Am. Chem. Soc., **71**, 2610 (1949).
9. M. H. Mackay, H. W. Melville, Trans Faraday Soc., **45**, 323 (1949).
10. H. W. Melville, A. F. Bickel, Trans. Faraday Soc., **45**, 1049 (1949).
11. C. H. Bamford, M. J. S. Dewar, Proc. R. Soc. London, Ser. A, **197**, 356 (1949).
12. D. W. E. Axford, Proc. R. Soc. London, Ser. A, **197**, 374 (1949).
13. G. Dixon-Lewis, Proc. R. Soc. London, Ser. A, **198**, 510 (1949).
14. L. Valentine, Thesis, Aberdeen (1949).
15. H. Kwart, H. S. Broadbent, P. D. Bartlett, J. Am. Chem. Soc., **72**, 1060 (1950).
16. H. W. Melville, L. Valentine, Trans. Faraday Soc., **46**, 210 (1950).
17. G. M. Burnett, Trans. Faraday Soc., **46**, 772 (1950).
18. J. D. Burnett, H. W. Melville, Trans. Faraday Soc., **46**, 976 (1950).
19. Ross, Thesis, Aberdeen (1950).
20. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **73**, 1700 (1951).
21. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **73**, 5395 (1951).
22. T. G. Majury, H. W. Melville, Proc. R. Soc. London, Ser. A, **205**, 496 (1951).
23. M. Morton, P. P. Salatiello, H. Landfield, J. Polym. Sci., **8**, 215 (1952).
24. M. Morton, P. P. Salatiello, H. Landfield, J. Polym. Sci., **8**, 279 (1952).
25. W. G. Barb, Proc. R. Soc. London, Ser. A., **212**, 177 (1952).
26. P. J. Flory, in: P. J. Flory (Ed.), "Principles of Polymer Chemistry", Cornell University Press, New York, 1953, p. 158.
27. G. M. Burnett, P. Evans, H. W. Melville, Trans. Faraday Soc., **49**, 1096 (1953).
28. G. M. Burnett, P. Evans, H. W. Melville, Trans. Faraday Soc., **49**, 1105 (1953).
29. G. M. Burnett, in: G. M. Burnett (Ed.), "Mechanism of Polymer Reactions", Interscience, New York, 1954, p. 230, p. 233.
30. B. R. Chinmayanandam, H. W. Melville, Trans. Faraday Soc., **50**, 73 (1954).
31. J. C. Bevington, H. W. Melville, R. P. Taylor, J. Polym. Sci., **14**, 463 (1954).

32. G. M. Burnett, W. W. Wright, Proc. R. Soc. London, Ser. A, **211**, 41 (1954).
33. J. L. O'Brien, F. Gormick, J. Am. Chem. Soc., **77**, 4757 (1955).
34. P. F. Onyon, Trans. Faraday Soc., **51**, 400 (1955).
35. J. Durup, M. Magat, J. Polym. Sci., **18**, 586 (1955).
36. W. J. Bengough, H. W. Melville, Proc. R. Soc. London, Ser. A, **230**, 429 (1955).
37. H. Miyama, Bull. Chem. Soc. Japan, **29**, 720 (1956); *ibid.* **30**, 10 (1957).
38. E. A. Nicholson, R. G. W. Norrish, Disc. Faraday Soc., **22**, 104 (1956).
39. N. Grassie, E. Vance, Trans. Faraday Soc., **52**, 727 (1956).
40. E. Bartholome, H. Gerrens, R. Herbeck, H. Weitz, Z. Elektrochem., **60**, 334 (1956).
41. F. S. Dainton, M. Tordoff, Trans. Faraday Soc., **53**, 499 (1957).
42. W. M. Thomas, R. L. Webb, J. Polym. Sci., **25**, 124 (1957).
43. C. H. Bamford, A. D. Jenkins, R. Johnston, Proc. R. Soc. London, Ser. A, **241**, 364 (1957).
44. M. Kryszewski, Roczniki Chemii, **31**, 893 (1957).
45. S. E. Bresler, E. N. Kasbezov, V. N. Shandrin, Vysokomol. Soedin., Ser. A, **17**, 507 (1957).
46. K. Ueberreiter, G. Sorge, Z. Phys. Chem. (Frankfurt), **13**, 158 (1957).
47. W. J. Bengough, Trans. Faraday Soc., **54**, 868 (1958).
48. W. J. Bengough, A. C. K. Smith, Trans. Faraday Soc., **54**, 1553 (1958).
49. W. J. Bengough, J. Polym. Sci., **28**, 475 (1958).
50. Z. A. Sinitsyna, Kh. S. Bagdasaryan, Zh. Fiz. Khim., **32**, 1319 (1958).
51. S. W. Benson, A. M. North, J. Am. Chem. Soc., **81**, 1339 (1959).
52. D. H. Grant, N. Grassie, Trans. Faraday Soc., **55**, 1042 (1959).
53. Z. Litia, Z. Machacek, J. Polym. Sci., **38**, 459 (1959).
54. F. S. Dainton, D. G. L. James, J. Polym. Sci., **39**, 299 (1959).
55. F. S. Dainton, R. S. Eaton, J. Polym. Sci., **39**, 313 (1959).
56. W. J. Bengough, S. A. McIntosh, R. A. M. Thomson, Nature, **184**, 266 (1959).
57. W. J. Bengough, H. W. Melville, Proc. R. Soc. London, Ser. A, **249**, 445 (1959).
58. W. J. Bengough, H. W. Melville, Proc. R. Soc. London, Ser. A, **249**, 455 (1959).
59. Z. Manásek, A. Rezábek, Int. Symp. Macromol. Chem. (Moscow), Section 2 (Russian version) (1960).
60. P. Hayden, H. W. Melville, J. Polym. Sci., **43**, 201 (1960).
61. G. Henrici-Olive, S. Olive, Makromol. Chem., **37**, 71 (1960).
62. G. V. Shulz, G. Henrici-Olive, S. Olive, Z. Phys. Chem. (Frankfurt), **27**, 1 (1960).
63. W. J. Bengough, Proc. R. Soc. London, Ser. A, **260**, 205 (1961).
64. G. V. Schulz, D. J. Stein, Makromol. Chem., **52**, 1 (1962).
65. P. E. M. Allen, G. M. Burnett, J. M. Downer, J. R. Majer, Makromol. Chem., **58**, 169 (1962).
66. W. Rabel, K. Ueberreiter, Ber. Bunsenges. Phys. Chem., **67**, 710 (1963).
67. H. Gerrens, Ber. Bunsenges. Phys. Chem., **67**, 741 (1963).
68. N. Coleboume, E. Collinson, D. J. Currie, F. S. Dainton, Trans. Faraday Soc., **59**, 1357 (1963).
69. F. S. Dainton, W. D. Sisley, Trans. Faraday Soc., **59**, 1369 (1963).
70. E. F. T. White, M. J. Zissell, J. Polym. Sci., Part A: Gen. Pap., **1**, 2189 (1963).
71. M. Morton, W. E. Gibbs, J. Polym. Sci., Part A: Gen. Pap., **1**, 2679 (1963).
72. J. Hughes, A. M. North, Trans. Faraday Soc., **60**, 960 (1964).
73. K. P. Paoletti, Jr. F. W. Billmeyer, J. Polym. Sci., Part A: Gen. Pap., **2**, 2049 (1964).
74. A. M. North, A. M. Scallan, Polymer, **5**, 447 (1964).
75. A. F. Revzin, Kh. S. Bagdasaryan, Zh. Fiz. Khim., **38**, 1020 (1964).
76. W. I. Bengough, W. Henderson, Trans. Faraday Soc., **61**, 141 (1965).
77. W. I. Bengough, R. A. M. Thomson, Trans. Faraday Soc., **61**, 1735 (1965).
78. M. Imoto, M. Kinoshita, M. Nishigaki, Makromol. Chem., **86**, 217 (1965).
79. D. J. Currie, F. S. Dainton, W. S. Watt, Polymer, **6**, 451 (1965).
80. J. Hughes, A. M. North, Trans. Faraday Soc., **62**, 1866 (1966).
81. A. N. Plyusnin, N. M. Chirkor, Teor. Eksp. Khimiya, **2**, 777 (1966).
82. V. A. Agasandyan, E. A. Trosman, Kh. S. Bagdasaryan, A. D. Litmonovich, V. Ya. Shtern, Vysokomol. Soedin., **8**, 1580 (1966).
83. K. M. Gibov, G. P. Gladyshev, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **17**, 53 (1967).
84. F. Hrabák, M. Bezdêk, V. Hynková, Z. Pelzbauer, J. Polym. Sci., Part C: Polym. Symp., **16**, 1345 (1967).
85. C. H. Bamford, S. Brumby, Makromol. Chem., **105**, 122 (1967).
86. Kh. S., in: Kh. S. Bagdasaryan (Ed.), "Bagdasaryan Theory of Free Radical Polymerization", translated from the Russian Second Edition (1966) by J. Schmorak, Israel Program for Scientific Translations, 1968.
87. K. Yokota, M. Kani, Y. Ishii, J. Polym. Sci., Part A-1: Polym. Chem., **6**, 1325 (1968).
88. J. Ulbricht, J. Polym. Sci., Part C: Polym. Symp., **16**, 3747 (1968).
89. W. F. Gromov, P. M. Chomikovski, A. D. Abkin, N. A. Rosanova, Vysokomol. Soedin., Ser. B, **10**, 754 (1968).
90. C. H. Bamford, S. Brumby, Chem. Ind., 1020 (1969).
91. G. N. Kornienko, A. Chervenka, I. M. Belugovskii, N. S. Yenikolopyan, Vysokomol. Soedin., Ser. A, **11**, 2703 (1969).
92. J. P. Fischer, G. V. Schulz, Ber. Bunsenges. Phys. Chem., **74**, 1077 (1970).
93. G. M. Burnett, G. G. Cameron, M. M. Zafar, Eur. Polym. J., **6**, 823 (1970).

94. A. V. Angelova, Yu. L. Spirin, R. Ye. Koval'chuk, Vysokomol. Soedin., Ser. A, **12**, 2703 (1970).
95. D. P. Kiryukin, A. M. Kaplan, I. M. Barkalov, V. I. Gol'danskii, Dokl. Akad. Nauk SSSR, **199**, 857 (1971).
96. S. V. Koslov, I. M. Bulugovskii, N. S. Enikolopyan, Vysokomol. Soedin., Ser. B, **13**, 46 (1971).
97. A. A. Baturin, Yu. B. Amerik, B. A. Krentsel, V. N. Tsvetkov, I. N. Shtennikova, E. I. Ryumtsev, Dokl. Akad. Nauk SSSR, **202**, 586 (1972).
98. S. B. Bresler, E. N. Kasbekov, V. N. Formichev, V. N. Shadrin, Makromol. Chem., **157**, 167 (1972).
99. T. M. Karaputadse, A. I. Kurilova, D. A. Topchiev, V. A. Kabanov, Vysokomol. Soedin., Ser. B, **14**, 323 (1972).
100. G. M. Burnett, G. G. Cameron, S. N. Joiner, J. Chem. Soc., Faraday Trans. I, **69**, 322 (1973).
101. K. Yokota, A. Konda, Makromol. Chem., **171**, 113 (1973).
102. Y. Ogo, M. Yokawa, T. Imoto, Makromol. Chem., **171**, 123 (1973).
103. G. Mayer, G. V. Schulz, Makromol. Chem., **173**, 101 (1973).
104. G. G. Cameron, J. Cameron, Polymer, **14**, 107 (1973).
105. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 179 (1974).
106. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 2903 (1974).
107. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 2913 (1974).
108. M. Ratzsch, J. Zschach, Plaste und Kautschuk, **21**, 245 (1974).
109. N. A. Plate, A. G. Ponomarenko, Polym. Sci. USSR (Engl. Transl.), **16**, 3067 (1974).
110. K. F. O'Driscoll, H. K. Mahabadi, J. Polym. Sci., Polym. Chem. Ed., **14**, 869 (1976).
111. P. Penchev, Makromol. Chem., **177**, 413 (1976).
112. M. Yokawa, Y. Ogo, Makromol. Chem., **177**, 429 (1976).
113. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **178**, 443 (1976).
114. H. K. Mahabadi, K. F. O'Driscoll, J. Macromol. Sci. A: Chem., **11**, 967 (1977).
115. J. Gons, E. J. Vorenkamp, G. Challa, J. Polym. Sci., Polym. Chem. Ed., **15**, 3031 (1977).
116. M. Kamachi, J. Satoh, D. J. Liaw, S. Nozakura, Macromolecules, **10**, 501 (1977).
117. K. Matsuo, G. W. Nelb, W. H. Stockmayer, Macromolecules, **10**, 654 (1977).
118. B. M. Soghomonyan, N. M. Beilerian, Armen. Chim. Zh., **31**, 567 (1978).
119. T. O. Osmanov, V. F. Gromov, P. M. Khomikovskii, A. D. Abkin, Dokl. Akad. Nauk SSSR, **240**, 910 (1978).
120. T. Otsu, B. Yamada, S. Sugiyama, Kobunshi Ronbunsyu, **35**, 705 (1978).
121. B. Yamada, M. Yoshioka, T. Otsu, Kobunshi Ronbunsyu, **35**, 795 (1978).
122. Y. Ogo, T. Kyotani, Makromol. Chem., **179**, 2407 (1978).
123. B. Yamada, H. Ohnishi, T. Otsu, Mem. Fac. Eng., Osaka City Univ., **19**, 189 (1978).
124. N. N. Tvorogov, A. G. Kondrat'eva, Vysokomol. Soedin., Ser. A, **20**, 230 (1978).
125. N. N. Tvorogov, A. G. Kondrat'eva, Vysokomol. Soedin., Ser. A, **20**, 1550 (1978).
126. P. M. Komikovskii, A. D. Abin, Vysokomol. Soedin., Ser. B, **20**, 263 (1978).
127. N. N. Bashenova, B. A. Egorov, Yu. D. Semchikov, Dokl. Akad. Nauk SSSR, **245**, 621 (1979).
128. H. A. Bruk, S. A. Pavlov, A. D. Abkin, Dokl. Akad. Nauk SSSR, **245**, 626 (1979).
129. A. G. Shostenko, V. E. Myshkin, Kinet. Katal., **20**, 781 (1979).
130. M. Kamachi, D. J. Liaw, S. Nozakura, Polym. J., **11**, 921 (1979).
131. S. W. Lansdowne, R. G. Gilbert, D. H. Napper, J. Chem. Soc., Faraday Trans. I, **76**, 1344 (1980).
132. E. V. Shtamm, Yu. I. Skurlatav, I. M. Karaputadse, Yu. E. Kirsh, A. P. Purmal, Vysokomol. Soedin., Ser. B, **22**, 420 (1980).
133. M. Kamachi, D. J. Liaw, S. Nozakura, Polym. J., **13**, 41 (1981).
134. K. Jen-Feng, C. Chu-Yung, Polym. J., **13**, 453 (1981).
135. T. Takahashi, P. Ehrlich, Polym. Prepr., **22**, 203 (1981).
136. Z. S. Nurkeeva, E. M. Shaidutinov, A. S. Seitov, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **1982**, 47 (1982).
137. D. J. Liaw, K. C. Chung, J. Chinese Inst. Chem. Eng., **13**, 145 (1982).
138. M. Kamachi, M. Fujii, S. Nozakura, J. Polym. Sci., Polym. Chem. Ed., **20**, 1489 (1982).
139. T. Takahashi, P. Ehrlich, Macromolecules, **15**, 714 (1982).
140. B. Yamada, T. Hayashi, T. Otsu, J. Macromol. Sci. A: Chem., **19**, 1023 (1983).
141. P. C. Lim, G. Luft, Makromol. Chem., **184**, 207 (1983).
142. P. C. Lim, G. Luft, Makromol. Chem., **184**, 849 (1983).
143. B. Yamada, M. Yoshioka, T. Otsu, Makromol. Chem., **184**, 1025 (1983).
144. A. Nagy, T. Foldes-Beresznich, T. Tudos, Eur. Polym. J., **20**, 25 (1984).
145. B. Yamada, M. Yoshioka, T. Otsu, J. Polym. Sci., Polym. Chem. Ed., **22**, 463 (1984).
146. B. Yamada, T. Kontani, M. Yoshioka, T. Otsu, J. Polym. Sci., Polym. Chem. Ed., **22**, 2381 (1984).
147. H. K. Mahabadi, Macromolecules, **18**, 1319 (1985).
148. G. Meyerhoff, R. Sack, M. Kouloumbris, Polym. Prepr., **26** (1), 293 (1985).
149. M. J. Ballard, D. H. Napper, R. G. Gilbert, D. F. Sangster, J. Polym. Sci., Part A: Polym. Chem., **24**, 1027 (1986).
150. M. J. Ballard, R. G. Gilbert, D. H. Napper, P. J. Pomery, P. W. O'Sullivan, J. H. O'Donnell, Macromolecules, **19**, 1303 (1986).
151. M. Buback, H. Hippler, J. Schweer, H. -P. Vögele, Makromol. Chem., Rapid Commun., **7**, 261 (1986).
152. M. Kamachi, H. Umetani, S. Nozakura, Polym. J., **18**, 211 (1986).
153. M. Kamachi, Adv. Polym. Sci., **82**, 209 (1987).
154. T. Sato, S. Inui, H. Tanaka, T. Ota, M. Kamachi, K. Tanaka, J. Polym. Sci., Part A: Polym. Chem., **25**, 637 (1987).
155. W. Lau, D. G. Westmoreland, R. W. Novak, Macromolecules, **20**, 457 (1987).

156. O. F. Olaj, I. Schnöll-Bitai, F. Hinkelmann, Makromol. Chem., **188**, 1689 (1987).
157. T. Sato, K. Morino, H. Tanaka, T. Ota, Makromol. Chem., **188**, 2951 (1987).
158. J. Shen, Y. Tian, Y. Zeng, Z. Qiu, Makromol. Chem., Rapid Commun., **8**, 615 (1987).
159. R. Sack, G. V. Schulz, G. Meyerhoff, Macromolecules, **21** 3345 (1988).
160. O. F. Olaj, P. Kremminger, I. Schnöll-Bitai, Makromol. Chem., Rapid Commun., **9**, 771 (1988).
161. O. F. Olaj, I. Schnöll-Bitai, Eur. Polym. J., **25**, 635 (1989).
162. T. Sato, N. Morita, H. Tanaka, T. Ota, J. Polym. Sci., Part A: Polym. Chem., **27**, 2497 (1989).
163. T. Sato, K. Arimoto, H. Tanaka, T. Ota, K. Kato, K. Doiuchi, Macromolecules, **22**, 2219 (1989).
164. T. P. Davis, K. F. O'Driscoll, M. C. Piton, M. A. Winnik, Macromolecules, **22**, 2785 (1989).
165. R. Sack-Kouloumbris, G. Meyerhoff, Makromol. Chem., **190**, 1133 (1989).
166. R. W. Garrett, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, C. L. Winzor, Polym. Bull., **22**, 611 (1989).
167. M. Buback, J. Schweer, Z. Phys. Chem. (Munich), **161**, 153 (1989).
168. G. Clouet, P. Chaffanjon, J. Macromol. Sci. A: Chem., **27**, 193 (1990).
169. T. P. Davis, K. F. O'Driscoll, M. C. Piton, M. A. Winnik, Macromolecules, **23**, 2113 (1990).
170. T. Sato, N. Morita, H. Tanaka, T. Ota, Makromol. Chem., **191**, 2559 (1990).
171. F. Tudos, A. Nagy, T. Foldes-Berezsnich, Angew. Makromol. Chem., **185/186**, 303 (1991).
172. T. Sato, I. Kamiya, H. Tanaka, T. Ota, Eur. Polym. J., **27**, 1087 (1991).
173. T. Otsu, B. Yamada, T. Ishikawa, Macromolecules, **24**, 415 (1991).
174. B. Yamada, M. Kageoka, T. Otsu, Macromolecules, **24**, 5234 (1991).
175. J. Shen, Y. Tian, G. Wang, M. Yang, Makromol. Chem., **192**, 2669 (1991).
176. B. Yamada, M. Satake, T. Otsu, Makromol. Chem., **192**, 2713 (1991).
177. T. Sato, Y. Takahashi, M. Seno, H. Nakamura, H. Tanaka, T. Ota, Makromol. Chem., **192**, 2909 (1991).
178. O. F. Olaj, I. Schnöll-Bitai, Makromol. Chem., Rapid Commun., **12**, 373 (1991).
179. B. Yamada, E. Yoshikawa, H. Miura, T. Otsu, Polym. Bull., **26**, 543 (1991).
180. B. Yamada, E. Yoshikawa, K. Shiraishi, H. Miura, T. Otsu, Polymer, **32**, 1892 (1991).
181. M. Yoshioka, T. Otsu, Macromolecules, **25**, 559 (1992).
182. T. Otsu, K. Yamagishi, M. Yoshioka, Macromolecules, **25**, 2713 (1992).
183. A. Matsumoto, Y. Oki, T. Otsu, Eur. Polym. J., **29**, 1225 (1993).
184. W. Lau, D. G. Westmoreland, Macromolecules, **25**, 4448 (1992).
185. T. Sato, T. Shimizu, M. Seno, H. Tanaka, T. Ota, Makromol. Chem., **193**, 1439 (1992).
186. M. Buback, B. Huckestein, B. Lugwig, Makromol. Chem., Rapid Commun., **13**, 1 (1992).
187. S. Deibert, F. Bandermann, J. Schweer, J. Sarnecki, Makromol. Chem., Rapid Commun., **13**, 351 (1992).
188. B. Yamada, M. Kageoka, T. Otsu, Polym. Bull., **28**, 75 (1992).
189. B. Yamada, M. Kageoka, T. Otsu, Polym. Bull., **29**, 385 (1992).
190. G. Moad, E. Rizzardo, D. H. Solomon, A. L. J. Beckwith, Polym. Bull., **29**, 647 (1992).
191. T. G. Carswell, D. J. T. Hill, D. I. Londero, J. H. O'Donnell, P. J. Pomery, C. L. Winzor, Polymer, **33**, 137 (1992).
192. D. J. Liaw, J. R. Lin, K. C. Chung, J. Macromol. Sci. A: Pure Appl. Chem., **30**, 51 (1993).
193. A. Matsumoto, K. Mizuta, T. Otsu, J. Polym. Sci., Part A: Polym. Chem., **31**, 2531 (1993).
194. A. Matsumoto, K. Mizuta, T. Otsu, Macromolecules, **26**, 1659 (1993).
195. T. Otsu, K. Yamagishi, A. Matsumoto, M. Yoshioka, H. Watanabe, Macromolecules, **26**, 3026 (1993).
196. B. R. Morrison, M. C. Piton, M. A. Winnik, R. G. Gilbert, D. H. Napper, Macromolecules, **26**, 4368 (1993).
197. P. Pascal, M. A. Winnik, D. H. Napper, R. G. Gilbert, Macromolecules, **26**, 4572 (1993).
198. R. A. Hutchinson, M. T. Aronson, J. R. Richards, Macromolecules, **26**, 6410 (1993).
199. P. O. Danis, D. E. Karr, D. G. Westmoreland, M. C. Piton, D. I. Christie, P. A. Clay, S. H. Kable, R. G. Gilbert, Macromolecules, **26**, 6684 (1993).
200. M. Deady, A. W. H. Mau, G. Moad, T. H. Spurling, Makromol. Chem., **194**, 1691 (1993).
201. M. Buback, B. Degener, Makromol. Chem., **194**, 2875 (1993).
202. S. Deibert, F. Bandermann, Makromol. Chem., **194**, 3287 (1993).
203. P. Pascal, M. A. Winnik, D. H. Napper, R. G. Gilbert, Makromol. Chem., Rapid Commun., **14**, 213 (1993).
204. E. L. Madruga, Makromol. Chem., Rapid Commun., **14**, 581 (1993).
205. A. Matsumoto, Y. Oki, T. Otsu, Polym. J., **25**, 237 (1993).
206. B. Yamada, S. Kobatake, S. Aoki, Macromol. Chem. Phys., **195**, 933 (1994).
207. S. Beuermann, M. Buback, G. T. Russell, Macromol. Rapid Commun., **15**, 351 (1994).
208. S. Beuermann, M. Buback, G. T. Russell, Macromol. Rapid Commun., **15**, 647 (1994).
209. R. A. Hutchinson, J. R. Richards, M. T. Aronson, Macromolecules, **27**, 4530 (1994).
210. A. Matsumoto, K. Mizuta, Macromolecules, **27**, 5863 (1994).
211. A. Matsumoto, T. Otsu, Proc. Jpn. Acad., Ser. B, **70**, 43 (1994).
212. V. P. Budtov, B. V. Revnov, Vysokomol. Soedin., Ser. A Ser. B, **36**, 1061 (1994).
213. A. Matsumoto, K. Yamagishi, T. Otsu, Eur. Polym. J., **31**, 121 (1995).

214. M. Buback, F. D. Kuchta, Macromol. Chem. Phys., **196**, 1887 (1995).
215. M. Buback, R. G. Gilbert, R. A. Hutchison, B. Klumperman, F. Kuchta, B. G. Manders, K. F. O'Driscoll, G. T. Russell, J. Schweer, Macromol. Chem. Phys., **196**, 3267 (1995).
216. U. Bergert, M. Buback, Macromol. Rapid Commun., **16**, 275 (1995).
217. U. Bergert, S. Beuermann, M. Buback, C. H. Kurz, G. T. Russell, C. Schmaltz, Macromol. Rapid Commun., **16**, 425 (1995).
218. D. A. Shipp, T. A. Smith, D. H. Solomon, H. David, G. Moad, Macromol. Rapid Commun., **16**, 837 (1995).
219. R. A. Hutchinson, D. A. Paquet, J. H. McMinn, R. E. Fuller, Macromolecules, **28**, 4023 (1995).
220. J. Sarnecki, J. Schweer, Macromolecules, **28**, 4080 (1995).
221. F. Bandrmann, C. Guenther, J. Schweer, Macromol. Chem. Phys., **197**, 1055 (1996).
222. M. D. Zammit, T. P. Davis, D. M. Haddleton, Macromolecules, **29**, 492 (1996).
223. M. Kamachi, A. Kajiwara, Macromolecules, **29**, 2378 (1996).
224. H. Tanaka, S. Yoshida, Macromolecules, **28**, 8117 (1995).
225. S. Kobatake, B. Yamada, S. Aoki, Polymer, **36**, 413 (1995).
226. T. Sato, I. Kamiya, M. Seno, H. Tanaka, J. Macromol. Sci. A: Pure Appl. Chem., **32**, 415 (1995).
227. M. Ueda, S. Shimada, T. Ogata, K. Ookawa, H. Ito, B. Yamada, J. Polym. Sci., Part A: Polym. Chem., **33**, 1059 (1995).
228. S. Kobatake, B. Yamada, Macromolecules, **28**, 4047 (1995).
229. S. Kobatake, B. Yamada, J. Polym. Sci., Part A: Polym. Chem., **34**, 95 (1996).
230. S. Kobatake, B. Yamada, Polym. J., **28**, 535 (1996).
231. M. Seno, T. Takikawa, H. Tanaka, T. Sato, Macromolecules, **28**, 4795 (1995).
232. A. Matsumoto, K. Shimizu, K. Mizuta, T. Otsu, J. Polym. Sci., Part A: Polym. Chem., **32**, 1957 (1994).
233. T. Sato, Y. Hirose, M. Seno, H. Tanaka, J. Polym. Sci., Part A: Polym. Chem., **33**, 797 (1995).
234. T. Sato, A. Takarada, H. Tanaka, T. Ota, Makromol. Chem., **192**, 2231 (1991).
235. T. Sato, M. Okazaki, M. Seno, H. Tanaka, Makromol. Chem., **194**, 637 (1993).
236. T. Sato, S. Shimooka, M. Seno, H. Tanaka, J. Polym. Sci., Part A: Polym. Chem., **33**, 2865 (1995).
237. A. Matsumoto, T. Kimura, J. Macromol. Sci. A: Pure Appl. Chem., **33**, 1049 (1996).
238. T. Sato, K. Masaki, M. Seno, H. Tanaka, Polym. Bull., **35**, 345 (1995).
239. T. Sato, K. Masaki, M. Seno, H. Tanaka, Makromol Chem., **194**, 849 (1993).

Transfer Constants to Monomers, Polymers, Catalysts and Initiators, Solvents and Additives, and Sulfur Compounds in Free Radical Polymerization†

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A. INTRODUCTION

The transfer reaction in radical polymerization describes a process in which the polymer radical reacts with another molecule (monomer, polymer, catalyst, solvent, modifier, etc.) forming a dead polymer and new radical. This new radical can continue the kinetic chain:

$$P_n^{\bullet} + RX \xrightarrow{k_{tr,X}} P_nR + X^{\bullet}$$

$$X^{\bullet} + M \xrightarrow{k_{i,X}} P_1^{\bullet}$$

The dimensionless transfer constant is defined by the ratio

$$C_X = k_{tr,X}/k_p$$

where k_p is the rate constant of propagation.

Based on the tables in the third edition, by K. C. Berger and G. Brandrup.

If it is assumed that

(a) all new radicals $X^{\bullet}$ react only by formation of growing polymer radicals;
(b) all polymer radicals have equal reactivities independent of their size;
(c) all rate constants are independent of solvent;
(d) the consumption of monomer by initiation and transfer is negligible compared with propagation;
(e) a steady-state concentration of polymer radicals is quickly established ($d[P^{\bullet}]/dt = 0$).

The application of these assumptions to the common reaction scheme results in the following relationship of ideal kinetics (Ref. 145):

$$\frac{1}{P_n} = \frac{1}{P_{n0}} + \sum C_X \frac{[X]}{[M]}$$

P_n, P_{n0} = number-average degree of polymerization obtained in the presence or absence, respectively, of transfer agent X.
[M], [X] = concentration of monomer or transfer agent, respectively.

The left-hand side of this equation is obtained by measuring the number average molecular weight of polymer at different concentrations of the transfer agent X, all other variables being kept constant. In many cases the degree of polymerization was determined by measurement of viscosity (Remark R). It should be noted that the ratio of

viscosimetric to number average molecular weight shows a considerable dependence on the molecular weight distribution.

The first term P_{n0} of the right-hand side of this equation is the number average molecular weight measured in the absence of transfer agent X. This is limited to cases where the transfer agent is a modifier. In all other cases, P_{n0} is calculated by combination of several kinetic expressions and used with its numerical value:

Initiation	$R_i = 2fk_d[I]$ or therm. $\quad R_i = R_t$
Termination	$R_t = k_t[P^\bullet]^2 = (k_{t,d} + k_{t,c})[P^\bullet]^2$
Propagation	$R_p = k_p[M][P]^\bullet]$
Number-average molecular weight	$P_{n0} = \dfrac{2(1+\delta)}{1+2\delta}\dfrac{R_p}{R_t}$

where k_t, $k_{t,d}$, and $k_{t,c}$ are the rate constants of termination, termination by disproportionation, and termination by combination, respectively.

$$\delta = k_{t,d}/k_{t,c}$$

The second term of the right-hand side of the above equation is given by the following expression:

$$C_X = C_M + C_I\frac{[I]}{[M]} + C_S\frac{[S]}{[M]} + C_p\frac{[P]}{[M]} + \cdots$$

where C_p is the constant of chain transfer with added polymer before starting polymerization.

The transfer constant of very reactive molecules can be determined from the rate of disappearance of both transfer agent and monomer:

$$C_X = \frac{d\ln([X])}{d\ln([M])}$$

This decrease of the transfer agent concentration with increasing monomer conversion is important in practice because very reactive transfer agents are used up preferentially, giving a broad molecular weight distribution if they cannot be replenished.

In analogy to quickly decomposing initiators with their criterion of "half-life", one can define, in addition to the transfer constant C_X, a "half-conversion" $U_{1/2}$ as that monomer conversion where the transfer agent is half consumed:

$$U_{1/2} = 100(1 - 0.5^{1/C_X})$$

The following table demonstrates (with some examples for a calibration curve) that the "half-conversion" $U_{1/2}$ decreases with increasing transfer constant C_X:

The "Half-Conversion" of Transfer Agents

C_X	$U_{1/2}$(%)	C_X	$U_{1/2}$(%)
0.1	99.9	5	13.0
0.2	96.8	10	6.7
0.5	75.0	20	3.4
1	50.0	50	1.4
2	29.3	100	0.7

The transfer agent must be replenished at this monomer conversion at the latest, if polymerization is to proceed, still yielding a narrow molecular weight distribution.

Numerous examples are known of polymerization systems that possibly do not satisfy basic assumptions: retardation (Remark J), induced decomposition and primary radical termination by peroxide or hydroperoxide initiation (Remarks C and B), diffusion control of termination (Remark X), electron donor/acceptor complexes (Remarks JJ), etc. Corresponding data in this table have to be used with caution.

B. TABLES OF TRANSFER CONSTANTS

TABLE 1. TRANSFER CONSTANTS TO MONOMERS

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Acetic acid, allyl ester	80	170		29
		700		27
Acrylamide	25	0.12	D	92
		0.16		74
		0.2		75
		0.2	E	103
	50	0.558	C2, F13, LL	560
	60	0.651	C2, F13, LL	560
	60	0.6	E	103
Acrylamide, *N,N*-dimethyl-	50	1.5		272
Acrylic acid, benzyl ester	55	0.883		291
	60	0.905		291
–, butyl ester	60	0.333	C1, H, KK	561
	60	1.05	C1, H, KK	561

TABLE 1. *cont'd*

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Acetic acid, butyl ester (*cont'd*)	60	1.25	C1, H, KK	561
	65	0.955		291
–, ethyl ester	50	0.193		306
		0.90	F7	510
		0.41	C, F17, R	512
	60	0.579	C, F17, R	512
	65	0.332		306
	70	0.351		306
		0.789	C, F17, R	512
–, -2 ethylhexyl ester	70	6.28	C, F17, R	512
	80	3.79	C, F17, R	512
–, methyl ester	50	0.80	F7	510
	55	0.275	C, AA	221
	60	0.036	B, AA	221
		0.325	C, AA	221
	65	0.10	C, AA	221
		0.11	C, F10	123
		0.11	C, AA	221
		0.37	C, AA	221
	70	0.01	C, F10	322
		0.072	B, AA	221
		0.16	C, AA	221
		0.18	C, AA	221
		0.405	C, AA	221
	75	0.224	C, AA	221
		0.25	C, F10	123
		0.25	C, AA	221
–, tetrahydrofurfuryl ester	40	4.0	C, F2	319
–, 2-chloro-, ethyl ester	60	3.0		342
–, thio-, methyl ester	60	560		140
Acrylonitrile	20	0.18	E	299
	25	0.105		284
	30	1.5	F8	196
	40	0.17		284
	50	0.050	F18	410
		0.27	F17	376
		8.2	F16	362
	60	0.26		95
		0.3		23
	60	0.333	C1, H, KK	561
	60	0.57	C1, H, KK	561
	60	1.02	C1, H, KK	561
Allyl bromide	60	30	C, R, HH	505
Ammonium chloride, *N,N*-diallyl-*N,N*-diethyl-	60	18	C1, F30	538
–, *N,N*-diallyl-*N,N*,-dimethyl-	60	7	C1, F30	538
Anisole, *p*-vinyl	0	0.198		13
	60	0.74		53
Anthranilic acid, vinyl ester	70	80		139
Benzoic acid, vinyl ester	60	6.0		321,393
	80	7.0		26
	95	2.0		321
		4.0		213
1,3-Butadiene, 2-chloro-	50	2.32	F2	153
–, 1,1,2-trichloro-	25	16	E	230
1-Butene	40	3.1		334
	50	5.1		334
	60	7.3		334
2-Butene, *cis*-	40	3.2		334
	50	4.9		334
	60	11.2		334
–, *trans*-	40	3.0		334
	50	5.2		334
	60	10.8		334
3-Butene-2-one, 3-methyl-	80	4.00	C	84

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TABLE 1. *cont'd*

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Butyric acid, vinyl ester	50	26.7		71
	80	22.3		200
Carbamic acid, vinyl-, ethyl ester	60	0.25		117
Carbazole, *N*-vinyl-7H-benzo(2)-	70	11	R, F2	522
Decanoic acid, vinyl ester	50	45.5		71
Ethylene	60	0.4	W5	102
		1.4	W4, W5	102
		4.2	W4	102
	83	5.0	D, W	204
		5.32	D, W	205
	110	1.1	W4, W5	102
		3.5	W4, W5	102
		9.0	W4	102
	130	0.0	C, F21, W5	259
		1.6	W4, W5	102
		4.7	W4, W5	102
		11.2	W4	102
	50–70	5	W3	218
–, chloro-	20	3.2		94
	25	3.2		315
		7.0		425
	30	6.25	Y	58
	40	19.4	F5	463
		50		246
	50	6.4	F3	379
		7.8	F4	379
		8.5		379
		11	C, F26	392
		13.5	Y	58
	60	10.8	F3	379
		12.3		379
		12.8	F4	379
	70	23.8	Y	58
Glutaramic acid, *N,N*-diisobutyl-, vinyl ester	60	42	C, F10	106
–, *N,N*-diisopentyl-, vinyl ester	60	51	C, F10	107
	70	51	C, F10	107
	80	51	C, F10	107
Hexanoic acid, vinyl ester	80	36		200
5-Hexen-3-yn-2-ol, 2-methyl-	60	5.0		226
Isobutyric acid, vinyl ester	80	46		200
Lauric acid, vinyl ester	50	45.5		71
Maleic anylhydride	75	750	C	175
Maleic anhydride/methyl methacrylate	60	100		367
Maleimide, *N*-(3-dimethylamino-6-methyl-phenyl)	50	720	F2	542
Methacrylamide	60	10×10^5	JJ1	515
Methacrylic acid, bornyl ester	60	2.85		162
–, butyl ester	50	0.54		233
	60	120×10^5	JJ1	515
		0.14		264
–, tert-butyl ester	70	0.35		9
–, *o-t*-butylphenyl ester	60	30.41	C	550
–, *p-t*-butylphenyl ester	60	2.01	C	550
–, *n*-dodecyl ester	60	80×10^5	JJ1	515
–, 2,3-epoxypropyl ester	60	0.59		198
–, ethyl ester	45	0.248	CC	193
	60	0.259	CC	193
	80	0.456	CC	193
	90	0.442	CC	193
–, *o*-ethylphenyl ester	60	3.63	C	550
–, *p*-ethylphenyl ester	60	1.36	C	550
–, glycidyl ester	60	45×10^5	JJ1	515
–, hexadecyl ester	70	0.14		138
–, isobornyl ester	60	1.85		162

TABLE 1. *cont'd*

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Methacrylic acid, isobutyl ester	35	0.189	DD	193
	50	0.179	DD	193
	60	0.14		264
		0.165	F2	264
		0.224	DD	193
	80	0.301	DD	193
	100	0.382	DD	193
–, *n*-nonyl ester	60	105×10^5	JJ1	515
–, phenyl ester	27	400	R	534
	60	0.61	C	550
–, *o*-tolyl ester	60	1.13	C	550
–, *p*-tolyl ester	60	1.06	C	550
Methacrylonitrile	25	2.08	Z	125
	60	5.81	Z	125
	70	8.00	Z	125
	80	10.05	Z	125
Methyl methacrylate	0	0.128		69
		0.148		18
	30	0.117		69
		0.260	D	224
		0.15		508
		30×10^5	JJ2	525
	40	242×10^5	JJ1	528
	50	0.10	F2	147,327
		0.15		145
		0.477	D	224
		241×10^5	JJ1	528
		0.85		69
	60	0.07		274
		0.10		34,290,317
		0.103	C	316
		0.18		145
		240×10^5	JJ0	556
	65	0.20		123
	70	0.17	C, F10	123
		0.20		267
		0.23		145
		0.265		440
		0.29	C, F17	123
		0.30		123
		0.45		104
		0.807	D	224
		1.37		459
		240×10^5	JJ1	528
	75	0.27	C, F10	123
		0.33	C	123
		0.60	C, F11	123
		0.70	C, F2	123
	80	0.25		145
		0.40		123
	90	0.10		147
	100	0.38		145
	120	0.58		145
Naphthalene, 1-vinyl-	50	290		216
	60	310		216
	70	300		216
1,4-Pentadiene, 1,1,2,3,3,4,5,5-octafluoro-	110	500	W10	68
Phernol, *o*-vinyl-	70	130		186
2-Picoline, 5-vinyl-	70	6.7		11
1-Propene, 2-chloro-	80	1600		27
–, 2-methyl-	40	2.5		334
	50	4.4		334
	60	6.9		334
Propionic acid, vinyl ester	–	3.6		158
	50	48.9		71

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TABLE 1. *cont'd*

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Pyridine, 2-vinyl-	15–35	0	D	38
–, 4-vinyl-	25	6.7		283
2-Pyrrolidinone, 1-vinyl-	20	4.0		59
Stearic acid, vinyl ester	50	69.8		71
Styrene	0	0	BB	360
		0.108		17,69
	25	0.279		69,245
		0.358		17
	27	0.31	A	145
	30	0.2	BB	318,360
		0.32		245
	45	0.3	BB	360
	50	0.35	H	354,355
		0.40	F2	318,327
		0.50	C, H	388
		0.6	C	145
		0.62	A, K	145
		0.65	B, C, H	388
		0.78	A	146
	60	0.07	C	467
		0.6		34,171,223
		0.6	C	145,240
		0.6	BB	318,360
		0.75		467
		0.79	A	145
		0.85	A, F	145
		0.92		447
		1.1		53
		1.37	C	234
		20×10^5	JJ1	515
	67.8	1.0		34
	70	0	C, F10	123
		0.6		104
		0.6	BB	360
		0.8	B, H	388
		0.96		67
		1.16	A	145
		1.35	A	145
		2.0	C, F2	123
	75	0	C, F10	123
		1.6	C	123
		5.0	C, F2	123
		5.00	C, F11	123
	80	0.7		98
		0.75	BB	360
		1.00		318
	80.3	4.0		34
	90	0.85	BB	360
		1.25	C, F	145
		1.47	A	145
		1.79	A	145
	99	1.5		98
	100	1.72	A	145
		1.8	A, F	145
		1.83	A	145
	110	2.80	A	145
	117	1.40		318
	132	2.45	A	145
		3.0	A, F	145
		3.4	A	145
		5.33	A	145
Styrene, *p*-boromo-	50	23		192
–, *o*-chloro-	30	0.25		62
	50	0.25	C	60
		0.28	C, H	60

TABLE 1. *cont'd*

Monomer	T (°C)	C_M ($\times 10^4$)	Remarks	Refs.
Styrene, *p*-iodo-	50	1.9	F1	56
Succinimide, *N*-vinyl-	55	0.55	C	93
Valeric acid, 4-methyl-, vinyl ester	80	24.8		200
Vinyl acetate	–	1.9		158
	−60	0.40	D	464
	−40	0.60	D	464
	−20	0.70	D	464
	−20	0.30	D	241
	0	0.50	D	241
		0.90	I4	86
		0.96	D	464
	20	0.94		328
	25	0.90	D	241
		1.3	I7	86
		1.45	I7	86
		2.4	D	202
		10.7		69
	40	1.29		12
		1.32		328
	45	2.0		51
	50	0.25		466
		0.41		265
		0.54	W14	265
		0.61	W15	265
		1.29		12
		2		552
		4.55	C	71
		20		225
	60	1.75		12
		1.8		466
		1.9		124
		1.91		328
		1.93		348
		2.0		231
		2.1	I12	145
		2.4		294
		2.5	C	86
		2.5		167
		2.6		167
		2.8	C	80
	65	2.1	C	466
	70	2.4		403
		2.9	I13	86
Vinylidene chloride	50	22	R	541
	60	38	R	541

TABLE 2. TRANSFER CONSTANTS TO POLYMERS

Polymer	T (°C)	C_P ($\times 10^4$)	Remarks	Refs.
Acrylamide, *N,N*-dimethyl-				
Poly(*N,N*-dimethylacrylamide)	50	0.61		272
Acrylic acid, ethyl ester				
Poly(methyl methacrylate)	60	12800	M6	120
–, methyl ester				
Poly(isoprene)				
–, chlorinated	75	12.6	C, F2	308
Poly(methyl acrylate)	60	0.5	L	212
		1.0	L	105
Poly(methyl methacrylate)	60	18000	M6	120
Acrylonitrile				
Cellulose	60	1.0	L, N	361
		11	L, M4	361
		20	L, M5	361

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TABLE 2. *cont'd*

Polymer	T (°C)	C_P ($\times 10^4$)	Remarks	Refs.
Acrylonitrile, (*cont'd*)				
Poly(acrylonitrile)	50	4.7	F14	376
	60	3.5	L	135
Poly(methyl methacrylate)	60	0.2	N	25
		240	M2	25
		900	M3	25
		1270	M	25
Poly(sarcosine)	60	400		23
1,3-Butadiene				
Poly(1,3-butadiene)	50	11		142
Ethylene	174.8	108.40	C, W20, F32	540
	201.6	150.70	C, W20, F32	540
	215.0	194.81	C, W20, F32	540
	231.2	214.46	C, W20, F32	540
	230.3	337.31	C, W22, F32	540
	229.2	256.85	C, W21, F32	540
	230.3	195.81	C, W20, F32	540
	230.4	199.89	C, W19, F32	540
	230.5	155.78	C, W18, F32	540
	256.0	348.46	C, W20, F32	540
–, chloro-				
Poly(vinyl chloride)	50	5	L	211
Hexanoic acid, vinyl ester				
Poly(oxyethylene), dodecyl ether	60	780		278
Methacrylic acid, butyl ester				
Poly(methyl methacrylate)	60	7700	M6	120
–, dodecyl ester				
Poly(methyl methacrylate)	60	12800	M6	120
–, ethyl ester				
Poly(isoprene), chlorinated	70	18.3	C, F2	308
Poly(vinyl chloride)	70	21.0	C, F28	308
Methyl methacrylate				
Poly(ethylene)	50	0.6		209
Poly(isoprene), chlorinated	80	23.4	C, F17	307
Poly(methyl methacrylate)	40	1.5	N	326
		360	M	326
	50	0.22	N	147
		1.5	N	326
		1.5	M	327
		360	M	326
		360	N	327
		1000	M	147
	60	0.1		105
		1.5	N	326
		2.1		262
		360	M	326
	80	2.48		262
	90	0.22	N	147
		1000	M	147
Poly(propylene)	50	0.04	C	292
		1.0		209
	130	0.42		292
Poly(styrene)	50	0.75		148
	60	2.20		262
	80	2.95		262
Poly(vinyl acetate)	60	2.0		263
	80	2.8		263
Poly(vinyl chloride)	–	11		250
	70	10.0	C, F28	308
Poly(vinyl urethane)	50	17	C, F8	157
Rubber, natural	50	10.9		258
		11.0	C	258
2-Pyrrolidinone, 1-vinyl-dextran				
Dextran	50	5		333
		5.87		203

TABLE 2. *cont'd*

Polymer	T (°C)	C_P ($\times 10^4$)	Remarks	Refs.
Styrene				
Poly(oxyadipoyloxy-2,2-dibromomethyl trimethylene)	60	4.05	L	325
Poly(oxyethylene)	70	19	M10	562
	70	14	M10	562
–, dodecyl ether	60	20		278
Poly(oxythexamethyleneoxy sebacoyl)	80	13	C, N	394
Poly(methyl methacrylate)	50	0.4	N	327
		<0.3	M	326
		1110	N	326
		1140	M	327
	60	16.4	M1	64
		17.5	M1	64
		5.7×10^4	M7	120
		32×10^4	M6	120
	80	3.74		262
	100	6.04		262
Poly(propylene)	60	0.025	C	292
	66	26		486
	130	0.3		292
		0.30		207
Poly(styrene)	50	1.9		144,148
		4.5		327
		14.0		327
		16.6		176
	55	15		177
	60	0.8	L	105
		1.9		73
		3.1		105
		15.4	M1	64
		15.8	M1	64
		16.6		176
	73.5	1.0	C	8
	85	1.4	C	8
	90	5.8		327
	100	2.0	N	276
	110	9.2		177
		10.8		176
	130	1.8		43
	154	1.5		43
Poly(vinyl acetate)	100	6.6		263
	130	9.2		263
Poly(vinyl chloride)	–	160		250
Poly(2-vinylpyridine)	50	8–10		279
Vinyl acetate				
Poly(oxyethylene)	60	17	N	277
		40	M	277
–, dodecyl ether	60	10	M8	277
		40	M9	277
		750		271
Poly(methyl methacrylate)	60	21		263
	75	26		263
Poly(styrene)	40	12		263
	60	15		263
	75	19		263
Poly(vinyl acetate)	–15	0.36		97
	0	0.5		97
		1.7		350
	11	2	O	145
	21	4	O	145
	31	16	O	145
	40	11.2		12
		30.9		43
		32.0		263
	50	0.06	W15	265

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TABLE 2. *cont'd*

Polymer	T (°C)	C_P ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Poly(vinyl acetate) (*cont'd*)		0.11	W14	265
		0.15		265
		3		43
		10.2		12
	60	1.2		124
		1.4		168,169
		1.8		348
		1.9		328
		2.5		329
		3.0		165
		4.0		105
		6.8		12
		8.0		105
		47.0		263
	60–70	3.5	P	350
	70	2	O	403
		4		403
Poly(vinyl acetate-*co*-vinyl chloride)	60	0.21	VCL-Part	166
		3.0	VOAC-Part	166

TABLE 3. TRANSFER CONSTANTS TO CATALYSTS AND INITIATORS

Catalyst/Initiator	T (°C)	C_I	Remarks	Refs.
Acrylamide				
Bisulfite ion	75	0.17	F13	352
Hydrogen peroxide	25	0.0005	D	92
Potassium persulfate	50	0.0258	F13, LL	560
	60	0.028	F13, LL	560
Acrylic acid, benzyl ester				
Isobutyronitrile, 2,2′-azobis-	55	0		291
	60	0		291
	65	0		291
–, ethyl ester				
Isobutyronitrile, 2,2′-azobis-	65	0		306
–, methyl ester				
Benzoyl peroxide	55	0.0143		221
	60	0.0246		221
	65	0.0375		221
	70	0.01	C, F10	322
		0.05		221
2-Butanone, peroxide	65	0.05	F10	123
		0.05		221
	70	0.077		221
	75	0.113	F10	123
		0.113		221
tert-Butyl hydroperoxide	60	0.01		221
	70	0.0266		221
tert-Butyl peroxide	65	0.00047		221
	70	0.00082		221
	75	0.00111		221
Chloroform/$CuCl_2$	120	600	Q(5–19), F1	551
Acrylonitrile				
Isobutyronitrile, 2,2′-azobis-	50	0	F16	362
	60	0		95
Azodiphenylmethane, benzyl-	50	0.07	R	536
	60	0.07	R	536
	80	0.07	R	536
–, cyclohexyl-	50	0.02	R	536
	60	0.04	R	536
	80	0.05	R	536

TABLE 3. *cont'd*

Catalyst/Initiator	T (°C)	C_I	Remarks	Refs.
Azodiphenylmethane (*cont'd*)				
–, 1-phenylethyl-	50	0.1	R	536
	60	0.1	R	536
	80	0.1	R	536
–, 2-propyl-	50	0.02	R	536
	60	0.04	R	536
	80	0.05	R	536
Benzoic acid, vinyl ester				
Benzoyl peroxide	80	0.0527		26
Isobutyronitrile, 2,2′-azobis-	80	0		26
3-Buten-2-one, 3-methyl-				
Benzoyl peroxide	80	0.0509		84
Ethylene				
Azoethane, 1,1′-dimethyl-	83	0.5	D, W	204
		0.51	D, W	205
Ethylene, chloro-				
Valeronitrile, 2,2′-azobis[2,4,4-trimethyl-	25	0.85		425
Maleic anhydride				
Benzoyl peroxide	75	2.63		175
Methacrylonitrile				
Isobutyronitrile, 2,2′-azobis-	60	0		125
Methyl methacrylate				
Acetophenone, 2-diazo-2-phenyl-	70	0.0		459
p-Anisoyl peroxide	60	0.037		316
Benzoyl peroxide	50	0.01		145
	60	0		34
		0.02	12	145
2-Butanone peroxide	65	0.0025	F2	123
		0.00698		123
	70	0.0033	F17	123
		0.0033	F10	123
		0.0040	F11	123
		0.0092		123
	75	0.00553	F11	123
		0.00667	F2	123
		0.0071	F10	123
		0.0089		123
	80	0.0111	F17	123
		0.0128		123
tert-Butyl hydroperoxide	60	1.27 × [cat]		34
tert-Butyl peroxide	20	< 0.0001	C, D	35
Butyronitrile, 2-ethyl-,2,2′-azobis-	60	0		317
–, 2-methyl-, 2,2′-azobis-	60	0		317
–, 2,3,3-trimethyl-, 2,2′-azobis-	60	0	317	
Cinnamoyl peroxide	60	0.009		316
Cobalt, [bis-[μ-[(2,3-butane-dionedioximato)(2-)-*O*,*O*′]] tetrafluorodiborato(2-)-*N*,*N*′,*N*″,*N*″]-		36000		563
		2500	F17	563
Cyclohexanecarbonitrile, 1,1′-azodi-	60	0	317	
Cyclopropanecarboxylic acid, 1-methyl- 2-(9′-anthryl)-, methyl ester	70	0.002	440	
Hydrogen peroxide	60	0.046	F11	290
Hydroperoxide, α,α-dimethylbenzyl	60	0.33		34
Isobutyronitrile, 2,2′-azobis-	50	0	F2	327
	60	0	F10	34,290,317
Methane, diazodiphenyl-	70	0.0		267
Palmitoyl peroxide	60	0		316
		0.16	I11	145
Peroxide, bis(*m*-chlorobenzoyl)	60	0.003		316
–, bis(*o*-chlorobenzoyl)	60	0.019		316
		0.35	F10	290
		0.8	I10	145
–, bis(*p*-chlorobenzoyl)	60	0.009		316
–, bis(*m*-nitrobenzoyl)	60	0.012		316

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TABLE 3. *cont'd*

Catalyst/Initiator	T (°C)	C_I	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
–, bis(*p*-nitrobenzoyl)	60	0.144		316
–, bis(*m*-phenylazobenzoyl)	50	0.00001	C	282
2-Tetrazene, 1,1,4,4-tetramethyl-	30	0.038	D	483
o-Toluoyl peroxide	60	0.046		316
		0.06	I11	145
Valeronitrile, 2-methyl, 2,2′-azobis-	60	0		317
Styrene				
Acetyl peroxide	70	0		87,217
p-Anisoyl peroxide	70	0.074		87
Benzoyl peroxide	22	0.1	D	65
	50	0.13		58
	60	0.048		171
		0.055		240
		0.101	C	234
	70	0		217
		0.075		87
		0.12		67
		0.18		58,65
	80	0.13		58,65
		0.813	F17	530
Cobalt, [Bis-[μ-[(2,3-butane-dionedioximato)(2-)-*O,O′*]] tetrafluorodiborato(2-)-*N,N′,N″,N″*]-		1500		563
Di-3-phenylazo benzoyl peroxide	50	6.7	R	527
	70	5.5	R	527
	90	4.8	R	527
2-Butanone, peroxide	50	0.46	H	388
	70	0.0667	F2	123
		0.1250	F10	123
		0.1670	F11	123
	75	0.1250	F2	123
		0.1670	F10	123
		0.2000	F11	123
		0.243		123
tert-Butyl hydroperoxide	60	0.035		171
	70	0.051	H	388
		0.060		399
		0.063	F6	400
		0.064		400
		0.066	F2, F5	400
	80	0.003		385
Butyl peroxide	60	0.00076	F9	302
		0.00092	F2	302
	80	0.0029	F9	302
sec-Butyl peroxide	60	0.0004	F2	301
	80	0.0021	F2	301
tert-Butyl peroxide	60	0.00023	F9	303
		0.0003		303
		0.0006	F2, F12	303
		0.00086		300
		0.0013		304
	70	0.039	F2	400
	80	0.0022	F2	303
		0.0033		385
Butyryl peroxide	70	0.018		87
Cinnamoyl peroxide	70	1.10		87
Crotonyl peroxide	70	0.146		87
Cyclohexanone peroxide	60	0.062		33
Ethyl peroxide	60	0.00066	F2	301
	80	0.0024	F2	301
Formamide, 2-cyano-2-propylazo-	100	0.17		47
Furoyl peroxide	70	0.23		87
Hexanoyl peroxide	70	0.166		87
Hydroperoxide, α,α-dimethylbenzyl	40	0.052	H	388

TABLE 3. *cont'd*

Catalyst/Initiator	T (°C)	C_I	Remarks	Refs.
Styrene (*cont'd*)				
Hydroperoxide, α,α-dimethylbenzyl (*cont'd*)	50	0.069	H	388
	60	0.063		171
	70	0.082	H	388
		0.10		399
–, α,α-dimethylbenzyl, *p*-isopropyl-	70	0.033	H	388
Isobutyronitrile,2,2′-azobis-	50	0	F2	327
		0	H	388
	60	0		171
		0		318
		0.012		468
		0.16		447
Isopropyl peroxide	60	0.0003	F2	301
	80	0.0015	F2	301
Lauroyl peroxide	70	0		217
		0.024		87
	84	0		217
p-Menth-8-yl hydroperoxide	50	0.048	H	388
Myristoyl peroxide	70	0		217
		0.116		87
2-Naphthoyl peroxide	70	0.178		87
Nickel peroxide	60	0.00265	C, F2	266
Octanoyl peroxide	70	0		217
		0.098		87
Oleoyl peroxide	70	0.154		87
Palmitoyl peroxide	70	0.142		87
Peroxide, bis(*p*-acetoxybenzoyl)	70	0.187		87
–, bis(*m*-bromobenzoyl)	70	0.465		87
–, bis(*o*-[bromobenzoyl)	50	1.0		58,65
	70	2.17		87
–, bis(*p*-bromobenzoyl)	70	0.193		87
–, bis(*p*-*tert*-butylbenzoyl)	70	0		87
–, bis(*m*-chlorobenzoyl)	70	0.346		87
Peroxide, bis(*o*-chlorobenzoyl)	22	2.0	D	65
	70	1.91		87
–, bis(*p*-chlorobenzoyl)	70	0.216		217
–, bis(*p*-cyanobenzoyl)	70	0.804		87
–, bis(2,4-dichlorobenzoyl)	60	2.9		217
	70	2.6		217
–, bis(α,α-dimethylbenzyl)	50	0.01	H	388
–, bis(*m*-fluorobenzoyl)	70	0.246		87
–, bis(*o*-fluorobenzoyl)	70	0.40		87
–, bis(*p*-fluorobenzoyl)	70	0.219		87
–, bis(hydroxyheptyl)	50	<0.005	H	388
–, bis(*m*-iodobenzoyl)	70	0.262		87
–, bis(*p*-methoxycarbonyloxy-benzoyl)	70	0.208		87
–, bis(*m*-nitrobenzoyl)	70	6.2		87
–, bis(*p*-nitrobenzoyl)	70	7.4		87
–, bis(*m*-phenylazobenzoyl)	70	5.5	C	281
	90	4.8	C	281
–, bis(5-phenyl-2,4-pentadienoyl)	70	5.24		87
–, bis(2-thiophenecarbonyl)	50	0.23		65
	70	0.38		65
–, *tert*-butoxymaleoyl	70	1.52		87
–, *tert*-butoxyphthaloyl	70	0.018		87
–, butylidenebis[*tert*-butyl	80	0.00077	C	411
	90	0.00140	C	411
–, *sec*-butylidenebis[*tert*-butyl	80	0.00072	C	411
	90	0.00115	C	411
–, isobutylidenebis[*tert*-butyl	80	0.00083	C	411
	90	0.00155	C	411
–, isopropylidenebis[*tert*-butyl	80	0.00040	C	411
	90	0.00105	C	411
Pinanyl hydroperoxide	70	0.026	H	388

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TABLE 3. *cont'd*

Catalyst/Initiator	T (°C)	C_I	Remarks	Refs.
Styrene (*cont'd*)				
Pivalonitrile	60	0.000038		468
Propylene, oxidized	60	1.01		396
	70	1.14		396
Propyl peroxide	60	0.00084		305
Sorboyl peroxide	70	1.19		87
Stearoyl peroxide	70	0.154		87
Succinonitrile, tetramethyl-	60	0.000037		468
2-Tetrazene, 1,1,4,4-tetramethyl-	30	0.038	D	483
o-Toluoyl peroxide	70	0.175		87
p-Toluoyl peroxide	50	0.17		65
	70	0.003		87
		0.19		65
9-Undecenoyl peroxide	70	0.065		87
Valeronitrile, 2,2′-azobis[2,44-trimethyl-	25	0.59		426
Vinyl acetate				
Benzoyl peroxide	60	0.032		465
		0.09		231
		0.15		80
	65	0.040		466
Isobutyronitrile, 2,2′-azobis-	50	0.025		466
	60	0.055		466
Lauroyl peroxide	60	0.10		80
Palmitoyl peroxide	60	0.10		80
		0.17	I3	145
Peroxide, bis(*m*-bromobenzoyl)	60	0.24		80
		0.6	I3	145
–, bis(*o*-bromobenzoyl)	60	0.25		80
		3.5	I3	145
–, bis(*p*-bromobenzoyl)	60	0.17		80
–, bis(*o*-chlorobenzoyl)	60	0.17	I3	145

TABLE 4. TRANSFER CONSTANTS TO SOLVENTS AND ADDITIVES

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acetic acid, allyl ester				
Benzene	80	21.0		273
p-Benzoquinone	80	5.2×10^5	J	27
–, 2,3,5,6-tetrachloro-	80	16×10^5	J	27
–, 2,3,5,6-tetramethyl-	80	4.14×10^4	J	27
–, 2,3,5-trichloro-	80	5.5×10^5	J	27
Carbon tetrachloride	100	2.0×10^4	C	210
Phosphorus trichloride	40	1.0×10^4	C, D	214
Acrylamide				
Acetonitrile		5.5	F13	564
tris(2-Carbamoylethyl)amine		85	F13	564
Iron(III) chloride	25	4.26×10^4		75
Isopropyl alcohol	50	19	F13	353
	80	7.2	F13, S	353
Methanol	30	0.13	C, F13	130
Propionamide	25	220	E	103
	60	64	E	103
Water	25	0.204	C, F13	437
	40	5.8	F13	444
Acrylic acid, butyl ester				
Aniline, *N,N*-dimethyl-	50	380		161
Ethanol	80	4.28	A, R	548
Methanol	80	0.47	F31, A, R	548
Propanol	80	3.78	A, R	548
Iso-propanol	80	14.12	A, R	548

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylic acid, ethyl ester				
Acetic acid	50	0.176		472
	80	0.537	A	471
	100	1.05	A	472
Acetone	40	0.207		472
	60	0.27	C	134
	80	1.10	A	471
	100	2.30	A	472
Acetonitrile	50	0.158		472
	60	0.245		472
	80	0.55	A	471
	100	1.43	A	472
Aniline, *N,N*-dimethyl-	50	2300	F16	228
Benzene	50	0.016		472
	60	0.22		472
		0.27	C	133
		0.45	C	134
	80	0.525	A	471
	100	2.2	A	472
–, bromo-	60	0.163		472
	80	0.685	A	471
	100	3.34	A	472
–, chloro-	40	0.054		472
	80	0.37	A	471
	100	1.68	A	472
–, ethyl-	40	0.668		472
	60	11.6		472
	80	16.80	A	471
	100	28.7	A	472
2-Butanone	50	1.44		472
	60	0.151	C	134
	80	1.92		472
		3.29	A	471
	90	4.45	A	472
Butyl alcohol	40	1.28		472
	60	2.91		472
	80	5.85	A	471
	100	12.6	A	472
sec-Butyl alcohol	45	10.6		472
	63	18.5		472
	80	22.20	A	471
	100	31.5	A	472
tert-Butyl alcohol	40	0.068		472
	60	0.17		472
	80	0.712	A	471
	100	1.64	A	472
Butyric acid	80	0.855	A	471
Carbon tetrachloride	40	0.332		472
	60	0.90		472
		3.2		416
	70	1.13	C	134
	80	1.55	A	471
	100	2.80	A	472
Chloroform	40	0.195		472
	60	0.89	C	134
	70	1.57	C	134
	80	1.49	A	471
	100	4.74	A	472
Cumene	50	11.7		472
	60	13.8		472
	80	22.2	A	471
	90	26.0	A	472
Cyclohexane	50	28.9		472
	60	0.48		472
		0.61	C	134

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylic acid, ethyl ester (*cont'd*)				
cyclohexane (*cont'd*)	80	1.22	A	471
	100	3.08	A	472
Dimethylformamide	50	0.1	F7	510
Ethanol	80	4.38	A, R	548
Ethyl acetate	50	0.298		472
	60	0.448		472
		0.69	C	134
	80	0.89	A	471
	100	1.82	A	472
Formic acid	80	0.046	A	471
Heaxane	50	0.524		472
	60	0.593		472
	80	0.97	A	471
	100	1.46	A	472
Isobutyl alcohol	45	2.1		472
	63	3.31		472
	80	4.65	A	471
	100	8.06	A	472
Isopropyl alcohol	–	21	C	445
	80	28.70	A, R	548
Methanol	60	0.32	C	134
		1.70	4	472
	80	0.36	A, R	548
Propanol	80	4.73	A,R	548
Toluene	40	0.88		472
	50	0.611	C, F17, R	512
	60	0.929	C, F17, R	512
	70	1.37	C, F17, R	512
		1.84	C	134
	80	2.60	A	471
	100	6.80	A	471
–, 2-ethylhexylester				
Toluene	70	28.70	C, F17, R	512
	80	2.13	C, F17, R	512
–, methyl ester				
Acetone	60	0.23	C	470
	80	0.622		332
		1.1	C	111
Acetophenone	50	<2.5		15
Aluminum, triethyl	60	480	F2, S	156
Aniline	50	<1.0		15
–, *N,N*-dimethyl-	50	60	F16	228
–, *m*-nitro-	50	42.1		15
–, *p*-nitro-	50	26.7		15
Anisole, *m*-nitro-	50	72.7		15
Anthracene	50	1000		15
Benzene	80	0.326		332
		0.45	C	111
–, chloro-	80	0.52	C	111
		0.986		332
–, *o*-dichloro-	80	0.71	C	111
–, *m*-dinitro-	50	309		15
–, ethyl-	80	6.056		332
–, nitro-	50	46.4		15
–, 1,3,5-trinitro-	50	2040		15
Benzoic acid, *p*-nitro-	50	107		15
Benzonitrile	50	<1.0		15
Copper(II) chloride	120	600	F34	565
Phosphine, tributyl-	60	1890	F2	156
Phosphorus trichloride	40	1000	C, D	214
Propanol	80	3.92	A, R	548
Silane, dimethyphenyl-	60	400	A, R	539
–, methyldichloro-	60	1200	A, R	539
–, tetraethyl-	60	33.1	F2	156
–, trichloro-	60	1600	A, R	539

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylic acid, methyl ester(*cont'd*)				
–, triethyl-	60	300	A, R	539
–, triphenyl-	60	800	A, R	539
Stearic acid, methyl ester	60	0.751		441
Toluene	60	2.7		21,24
	80	1.775		332
		2.7	C	111
–, *m*-nitro-	50	41.2		15
–, *p*-nitro-	50	48.6		15
Triethylamine	60	400		24
Tripropylamine	60	470	F2	156
–, tetrahydrofurfuryl ester				
Carbon tetrachloride	40	1.0	C, F2	319
Acrylonitrile				
Acetaldehyde	–	14	F	436
	50	47	F7	377
Acetamide, *N,N*-dimethyl-	50	4.945		375
		5.05		197
Acetic acid	50	0.81	F14	377
Acetone	50	1.7	F14	377
	60	1.13	F	95
Acetonitrile	50	0.7	F14	377
	60	2.0	F15	86
Acrolein, bis(2-ethoxyethyl)acetal	60	120	F7	499
–, bis(2-butoxyethyl)acetal	60	90.5	F7	156
Aluminum, hydrodiisobutyl	60	3940	F7	156
–, triethyl	60	590	F7	156
	100	17.0×10^4	F7	247
–, triisobutyl	100	28.0×10^4	F7	247
Aniline	40	32.0	F2, G	229
	50	44.0	F2, J	229
		9600	F16, G, J	227,229
	60	1.22×10^4	F16, G, J	229
–, *N,N*-diethyl-	40	215	F2, J	229
		5.81×10^4	F16, G, J	229
	50	359	F2, J	229
		9.38×10^4	F16, G, J	227,229
	60	547	F2, J	229
		14.3×10^4	F16, G, J	229
Aniline, *N,N*-dimethyl-	40	605	F2, J	229
		1.19×10^4	F16, G, J	229
	50	708	F2, J	229
		1040		161
		1.5×10^4	F16, J	228
		1.54×10^4	F16, G, J	227,229
	60	870	J	24
		964	F2, J	229
		2.18×10^4	F16, G, J	229
Anthracene	50	1.8×10^4		15
Arabinose	60	13.0	F8	361
Benzene	60	2.46	G	95
–, bromo-	60	1.36	G, S	95
–, *tert*-butyl-	60	1.93	G	95
–, chloro-	60	0.79	G, S	95
–, ethyl-	60	35.73	G	95
–, iodo-	60	5.19	G, S	95
Benzoic acid, vinyl ester	65	1400		136
p-Benzoquinone	50	1.3×10^4		15
Borane, tributyl	60	6470	F7	156
2-Butanone	60	6.43	G	95
–, 3-methyl-	60	21.08	G	95
1-Buten-3-yne	50	3800	F7	377
Butyl alcohol	60	15.42	G	95
sec-Butyl alcohol	60	97.55	G	95
tert-Butyl alcohol	60	0.44	G	95

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylonitrile (*cont'd*)				
Butyric acid, 4-hydroxy-γ-lactone	50	0.658		375
		0.74		374
Cadmium, dibutyl	60	5.5×10^4	F7	156
Carbonic acid, cyclic ethylene ester	50	0.073		293
		0.33		197
		0.39		374
		0.474		375
		0.5		404
		0.511	E	458
		1.0		356
	60	0.128		293
Carbon tetrabromide	50	400		493
	60	500		20
		1900	F7, J	24
Carbon tetrachloride	60	0.85	G	96
	80	1.13	C	336
Chloroform	60	5.64	G	95
	80	5.90	C	336
Copper(II) chloride	35	1.8×10^5	U(0.01)	402
		1.9×10^5	U(0.1)	402
		3.2×10^5	U(1.0)	402
	60	10×10^5	F7	37
Copper(II) sulfate	35	1900	F13, J	402
		2800	J, V(0.0001)	402
		3000	J, U(0.01)	402
		1.07×10^4	J, V(0.001)	402
		1.35×10^4	J, U(0.1)	402
		3.93×10^4	J, V(0.01)	402
		13.6×10^4	J, U(1.0)	402
		21.0×10^4	J, V(0.1)	402
Crotonaldehyde	50	47	F7	377
Crotononitrile, *trans*-	50	19	F7	377
Cumene	60	41.41	G	95
Cyclohexane	60	2.06	G	95
–, methyl-	60	2.31	G	95
Dimethylamine	50	175	F7	377
m-Dioxane, 5,5-dimethyl-2-vinyl-	60	2.20	F7	498
m-Dioxane, 5,5-dimethyl-2-vinyl-				
–, 4-methyl-2-vinyl-	60	4.40	F7	498
–, 2-vinyl-	60	2.71	F7	498
1,3,-Dioxolane, 2-vinyl-	60	16.60	F7	498
1,3-Dioxolane-4-methanol, 2-vinyl-	60	2.40	F7	498
Diphenylamine-T	60	700	F7	61
Erythritol	60	12.8	F8	361
Ethane, 1,2-dichloro-	60	1.47	G	95
–, 1,1,2,2-tetrachloro-	60	3.11	G	95
–, 1,1,1-trichloro-	60	1.25	G	95
–, 1,1,2-trichloro-	60	1.68	G	95
Ethanol, 2,2′-iminodi-	30	10.1	F13	476
–, 2,2′,2″-nitrilotri-	30	76.0	F13	476
Ether, dodecyl vinyl	50	4.95	C	4
Ethyl acetate	60	2.54	G	95
Formamide, *N,N*-dimethyl-	20	1.4	C, J, F7	248
	25	4.97	D, J	91
	40	3.24	E	458
	50	1.0		347
		2.70		374
		2.78		197
		2.8	F7	377
		2.83		375
		10		356
	60	2.412		347
		4.494		22
		5.0	F15	86

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylonitrile (*cont'd*)				
Formamide, *N,N*-dimethyl- (*cont'd*)				
–, *N*-methyl-	50	1.8	F14	377
Formic acid	50	0.5	F14	377
Glucose	60	6.9	F8	361
α,D-Glucoside, methyl-	60	20	F7	220
–, –, 6-deoxy-6-iodo-	60	50	F7	220
–, –, 6-deoxy-6-mercapto-	60	1300	F7	220
–, –, 6-deoxy-6-phthalimido-	60	60	F7	220
–, –, 2,3-di-*O*-benzyl-	60	90	F7	220
–, –, 2,3,4,6-tetra-*O*-acetyl-	60	30	F7	220
–, –, 6,-*O*-(*p*-toluenesulfonyl)-	60	10	F7	220
–, –, 6-*O*-triphenylmethyl-	60	80	F7	220
β,D-Glucoside, methyl-	60	20	F7	220
–, –, 6-deoxy-6-dipropylamino-	60	1100	F7	220
Glutaronitrile, 2,4-dimethyl-	50	0.6	F14	376
Glyceraldehyde	60	32.1	F8	361
Glycerol	60	23.5	F8	361
2,4,6-Heptanetricarbonitrile	50	1.0	F14	376
1-Heptanol, 2,2,3,3,4,4,5,5,6,6,7,7-dodecafluoro-	60	1.9	F2	442
1,5-Hexadien-3-yne	50	1.17×10^4	F7	377
Hydrocyanic acid	50	0.81	F14	377
	–	6.2	F	436
Indium, triethyl	60	2220	F7	156
Iron(III) chloride	60	3.33×10^4	F7	19
Isobutyl alcohol	60	24.06	G	95
Isobutylene	60	70		553
Isobutyronitrile	50	1.3	F14	376
	60	1.8	I6	376
		3.5		135
		4.0	F15	86
Lactonitrile	50	1	F14	377
Lead, tetraethyl	60	243	F7	156
Magnesium perchlorate	50	<0.05		376
Mercury, diethyl	60	72.2	F7	156
Methane, dichloro-	60	3.06	G	95
–, nitro-	60	6.0	F15	86
Methanol	50	0.5	F14	377
Methylamine	50	175	F7	377
2,6-Octadiene, 2,6-dimethyl-	60	450		6
Oxime, acrolein	60	6.26×10^5		555
–, crotonaldehyde-	60	2.05×10^4		555
–, ethylisopropenylketone-	60	1.03×10^4		555
–, methacrolein-	60	9.41×10^4		555
–, methylacrolein-	60	5.35×10^4		555
–, methylvinylketone-	60	1.2×10^4		555
–, methylisopropenylketone-	60	1.71×10^4		555
–, methylisobutenylketone-	60	2.65×10^4		555
1-Pentanol, 2,2,3,3,4,4,5,5-octa-fluoro-	60	1.9	F2	442
1-Pentanol, 4-methyl-	60	11.79	G	95
Piperidine, 1-ethyl-	60	3300	J	24
–, 1-methyl-	60	2300	J	24
1-Propanol, 2,2,3,3-tetrafluoro-	60	1.5	F2	442
Silane, tetraethyl	60	21.0	F7	156
Sorbitol	60	6.1	F8	361
Stibine, tributyl	60	11.1×10^4	F7	156
Succinonitrile	60	0.6		86
Sulfur dioxide	50	0		362
Tin, tetrabutyl	60	80.8	F7	156
Toluene	50	1.153	G	347
	60	2.632	G	347
		3.2		21
		5.83	G	24,95
o-Toluidine, *N,N*-dimethyl-	40	272	F2, J	229
	50	334	F2, J	229

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Acrylonitrile (*cont'd*)				
o-Toluidine, *N,N*-dimethyl- (*cont'd*)		30200	F16, G, J	227,229
	60	463	F2, J	229
Tributylamine	60	6700	J	24
Triethylamine	60	1700	T	20
		1900		20
		3800		22
		5900	F7, J	24
		6600	J	24
Trimethylamine	50	175	F7	377
	60	790	J	24
Tripropylamine	60	4820	F7	156
		1.05×10^4	J	24
Zinc, diethyl	60	1.06×10^4	F7	156
Zinc chloride	50	0.006	F13	410
Acrylonitrile/styrene (38.5–61.5 mol%)				
Methanol	65	1.4		116
Anisole, *p*-vinyl-				
Anisole, *p*-isopropyl-	60	3.40		407
Benzonitrile, *p*-isopropyl-	60	40.9		407
Cumene	60	4.28		407
–, *p*-bromo-	60	11.8		407
–, *p*-tert-butyl-	60	3.64		407
–, *p*-chloro-	60	8.52		407
Benzoic acid, vinyl ester				
Benzene	60	1.5		393
	80	0.4		26
Benzoic acid, isopropyl ester	60	1.0		393
2-Butanone	80	29.0		26
Carbon tetrachloride	80	730		26
Chloroform	80	105		26
Cyclohexane	80	2.3		26
1,3-Butadiene, 2-chloro-				
1-Butene, 3-chloro-3-methyl-	50	5.47	F1	152
2-Butene, 1-chloro-3-methyl-	50	4.57	F2	152
2,6-Octadiene, 2,7-dichloro-	50	5.24	F2	152
3-Buten-2-one, 3-methyl-				
Benzene	80	2.489	C	84
–, ethyl-	80	6.934	C	84
Cyclohexane, methyl-	80	0.500	C	84
Toluene	80	3.282	C	84
Butyraldehyde, divinyl acetal				
Aniline, *N,N*-dimethyl-	50	1060		430
Benzene	50	27.7		430
Butyl alcohol	50	3500		430
tert-Butyl alcohol	50	37.2		430
1,3-Dioxolane, 2-propyl-	50	708		430
Butyric acid, vinyl ester				
Benzene	80	3.28		201
Carbamic acid, vinyl-, ethyl ester				
Acetone	60	8.5		117
Benzene	60	1.25		117
Chloroprene				
3-Chloro-(methyl-^{14}C)-1-butene(4-^{14}C)	50	5.37	F2	509
3-Chloro-2-pentene-(2,4-^{14}C)	50	3.94	F2	509
p-Dioxin, 2,3-dihydro-/maleic anhydride				
Benzene	60	11500	R	169
Carbon tetrabromide	60	100×10^5	R	169
Carbon tetrachloride	60	1.7×10^5	R	169
Chloroform	60	3.6×10^4	R	169
p-Dioxane	60	1.07×10^4	R	169
Toluene	60	6.0×10^4	R	169
m-Xylene	60	9.9×10^4	R	169

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene				
Acetamide, *N,N*-diethyl-	130	125	C, W5	454
–, *N,N*-diisopropyl-	130	57	C, W5	454
–, *N,N*-dimethyl-	130	182	C, W5	454
–, *N*-ethyl-	130	115	C, W5	454
–, *N*-methyl-	130	61	C, W5	454
Acetic acid, butyl ester	130	89	C, W5	454
–, *tert*-butyl ester	130	40	C, W5	454
–, methyl ester	130	40	C, W5	454
Acetic acid, chloro-				
–, –, methyl ester	90	610	C, Q1	434
		1210	C, Q2	434
		2170	C, Q3	434
		2150	C, Q4	434
		2580	C, Q5	434
	120	630	C, Q1	434
		1160	C, Q2	434
		1880	C, Q3	434
		1990	C, Q4	434
		2090	C, Q5	434
	130	1120	C, W5	454
Acetic acid, cyano-				
–, –, methyl ester	130	6700	C, W5	454
Acetic acid, dichloro-				
–, –, methyl ester	55	1800	C, Q1	435
		7900	C, Q2	435
		11100	C, Q3	435
		13000	C	435
	90	1900	C, Q1	435
		6900	C, Q2	435
		9400	C, Q3	435
		10500	C	435
	120	2000	C, Q1	435
		5900	C, Q2	435
		8100	C, Q3	435
		11100	C	435
Acetic acid, trichloro-				
–, –, methyl ester	55	700	C, Q1	484
		16100	C, Q2	484
		27500	C, Q3	484
		38000	C, Q4	484
		65700	C	484
	90	1000	C, Q1	484
		17000	C, Q2	484
		27700	C, Q3	484
		39000	C, Q4	484
		65200	C	484
	120	1200	C, Q1	484
		15400	C, Q2	484
		23600	C, Q3	484
		33000	C, Q4	484
		54100	C	484
Acetic anhydride	130	130	C, W5	454
Acetone	130	160	C, W17	456
		165	C, F21, W5	259
		168	C, W5	456
	200	282	C, W5	455
Acetonitrile	130	100	C, F21, W5	159
Benzaldehyde	130	1970	C, W5	457
Benzene	20	0.629	E, W	143
	83	20	D	204
		21	D, W	205
	130	9.4	C, W5	260
	50–70	18	W3	218

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene *(cont'd)*				
–, ethyl-	130	430	C, W17	456
		480	C, W5	52
		520	C, W5	454
		560	C, W5	52
	200	500	C, W5	455
Benzene-D6	130	5.6	C, W5	260
Benzoic acid, butyl ester	130	140	C, W5	454
–, ethyl ester	130	55	C, W5	454
–, methyl ester	130	53	C, W5	454
Butane	130	40	C, F21, W5	259
		45	C, W17	456
–, 1-iodo-	100	9.45×10^4	Q1, W14	187
		11.3×10^4	Q2, W14	187
		12.6×10^4	Q3, W14	187
2-Butanone	130	550	C, W17	456
		600	C, F21, W5	259
	200	750	C, W5	455
–, 3-methyl-	130	840	C, F21, W5	259
1-Butene	130	330	C, W5	52
		470	C, W5	52
		560	C, F21, W5	259
	189	900	B, W5	443
	200	570	C, W5	455
–, 2-methyl-	130	210	C, F21, W5	259
		530	C, F21, W5	259
–, 3-methyl-	130	1200	C, F21, W5	259
2-Butene	130	250	C, F21, W5	259
		380	C, W5	454
–, 1,4-dichloro-	130	4100	C, W5	454
–, 2-methyl	130	470	C, F21, W5	259
tert-Butyl alcohol	30	0	E, W4	482
	130	2	C, F21, W5	259
Butylamine	130	220	C-W5	454
Butyraldehyde	130	3250	C, W5	457
Butyric acid, methyl ester	130	220	C, W5	454
Butyronitrile	130	520	C, F21, W5	259
Carbon tetrachloride	20	215	E, W	143
	50	772	C, Q1,	391
	70	974	C, Q1	391
		7000	C, W4	215
		32000	C	210
	90	740	Q1, W12	453
		1210	C, Q1	391
		20200	Q2, W12	453
		32500	Q3, W12	453
		34000	W12	453
	95	1250 ± 200	W23, Q(1)	518
		22000 ± 1000	W23, Q(2)	518
		43000 ± 4000	W23, Q(3)	518
		1130 ± 700	W20, Q(1)	518
		2000 ± 1000	W20, Q(2)	518
		37000 ± 3000	W20, Q(3)	518
	130	9800	C, F21, W5	259
	140	1600	C, Q1	170
		1700	C, Q1, F19	170
		1800	C, Q1, W13	170
		2200	C, Q1, F20	170
		22000	C, Q2	170
		23000	C, Q2, F19	170
		30000	C, Q1, W1	170
		36000	C, Q3	170
		38000	C, Q2, F20	170
		39000	C, Q3, F19	170
		60000	C, Q2, W13	170

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene (*cont'd*)				
Carbon tetrachloride (*cont'd*)		60000	C, Q3	170
		61000	C, Q3, F20	170
		70000	C, Q4, F20	170
		90000	C, Q4, F20	170
		100000	C, Q2, W1	170
		110000	C, Q3, W1	170
		130000	C, Q3, W13	170
		140000	C, Q4, W1	170
		180000	C	170
Chloroacetic acid, methyl ester	90	590, 550, 1200	C, Q(1), Q(2), Q(3)	557
		1680, 2240, 2340	C, Q(4), Q(5), Q(6)	557
Chloroform	28	2100	E, Q1, W	244
		13000	E, Q2, W	244
		15000	E, Q3, W	244
	70	8000	C, W4	215
		30000	C	210
	80	2470	C, Q1, W12	101
		15500	C, Q2, W12	101
		24900	C, Q3, W12	101
		31200	C, Q4, W12	101
		43900	C, W12	101
	95	2000 ± 120	W23, Q(1)	518
		10000 ± 500	W23, Q(2)	518
		16000 ± 3000	W23, Q(3)	518
		1150 ± 50	W20, Q(1)	518
		5000 ± 300	W20, Q(3)	518
		8000 ± 1000	W20, Q(3)	518
	101	1500	E, Q1, W	224
		4500	E, Q3, W	244
		5400	E, Q2, W	244
	103	2890	C, Q1, W12	101
		15400	C, Q2, W12	101
		23800	C, Q3, W12	101
		29800	C, Q4, W12	101
		41100	C, W12	101
	130	2900	C, F21, W5	259
	140	3210	C, Q1, W12	101
		15200	C, Q2, W12	101
		22200	C, Q3, W12	101
		28000	C, Q4, W12	101
		37600	C, W12	101
Cumene	130	500	C, F21, W5	259
Cyclohexane	130	80	C, F21, W5	259
		90	C, W17	456
		91	C, F21, W5	259
	200	190	C, W5	455
–, methyl-	130	110	C, F21, W5	259
Cyclopentane	130	109	C, W17	456
		127	C, F21, W5	259
	200	228	C, W5	455
Cyclopropane	130	0	C, F21, W5	259
Decane	130	120	C, F21, W5	259
	189	425	B, W5	443
1-Decene	189	1090	B, W5	443
Dibutylamine	130	1070	C, W5	454
Dichloroacetic acid, methyl ester	90	1900, 6900, 9400	C, Q(1), Q(2), Q(3)	558
Dimethylamine	130	1900	C, F21, W5	259
p-Dioxane	130	320	C, F21, W5	259
Ethane	130	6	C, F21, W5	259
–, 1,1-bis(dimethylamino)-	130	1070	C, W5	457
–, 1-bromo-2-chloro-	130	390	C, W5	454
–, chloro-	70	120	C, W4	215
–, 1,2-dibromo-	130	1250	C, W5	454
–, 1,1-dichloro-	70	1500	C, W4	215

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene (*cont'd*)				
Ethane, 1,2-dichloro-	130	110	C, F21, W5	259
–, iodo-	65	1.51×10^5	Q1	189
		1.59×10^5	Q2	189
		1.65×10^5	Q3	189
	100	1.1×10^5	Q1	452
		1.36×10^5	Q1, W14	188
		1.43×10^5	Q2, W14	188
		1.48×10^5	Q3, W14	188
		1.49×10^5	Q3	452
–, 1,1,1-trichloro-	70	500	C, W4	215
Ethanol	20	5.71	E, W	143
	30	110, 210, 270	E, Q(1), Q(3), Q(5)	523
	60	76, 240, 280	E, Q(1), Q(3), Q(5)	523
	100	89, 270, 340	E, Q(1), Q(3), Q(5)	523
	125	100, 310, 320	E, Q(1), Q(3), Q(5)	523
	130	68	C, W17	456
		69	C, F21, W5	259
		75	C, W5	454
	125–135	190	C, W2	381
	150	92, 260	E, Q(1), Q(3)	523
	180	73, 350	E, Q(1), Q(3)	523
	200	135	C, W5	455
Ethyl acetate	130	45	C, W5	454
	200	121	C, W17	455
Ethylene oxide	130	7	C, F21, W5	259
Formaldehyde	130	560	C, W5	457
Fomamide, *N,N*-dimethyl-	130	260	C, F21, W5	259
Formic acid, methyl ester	130	42	C, W5	454
Furan, tetrahydro-	130	288	C, W5	454
		289	C, F21, W5	259
	200	401	C, W5	455
Heptaldehyde	130	2600	C, W17	457
		3900	C, W5	457
	200	4800	C, W5	457
Heptane	50–70	90	W3	218
	130	80	C, F21, W5	259
Hexane	130	68	C, F21, W5	259
	189	225	B, W5	443
1-Hexene	189	900	B, W5	443
–, 2-ethyl-	130	3300	C, F21, W5	259
Hydrogen	130	159	C, W5	454
		160	C, F21, W5	259
	200	400	C, W5	455
Isobutyronitrile	130	1070	C, F21, W5	259
Isocyanic acid, butyl ester	130	212	C, W5	454
Isopropanol	30	96, 410, 520	E, Q(1), Q(3), Q(5)	523
	60	65, 310, 380	E, Q(1), Q(3), Q(5)	523
	100	65, 300, 350	E, Q(1), Q(3), Q(5)	523
	125	75, 380, 380	E, Q(1), Q(3), Q(5)	523
	130	130	C, W17	456
		140	C, F21, W5	259
		144	C, W5	454
	125–135	570	C, W1	381
	150	98, 470, 460	E, Q(1), Q(3), Q(5)	523
	180	86, 440, 500	E, Q(1), Q(3), Q(5)	523
	200	234	C, W5	455
Isothiocyanic acid, butyl ester	130	0	C, W5	454
Methane	130	0	C, W5	259
–, bromochloro-	100	1600	C, Q1, W11	2
		4500	C, Q2, W11	2
		7500	C, Q3, W11	2
		8000	C, Q4, W11	2
	130	10000	C, W5	454
	140	16.0×10^5	Q1, W11, II	1

TABLE 4. *cont'd*

Solvent/additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene *(cont'd)*				
Methane, bromochloro- *(cont'd)*		47.0×10^5	Q2, W11, II	1
–, chloro-	70	4	C, W4	215
–, dichloro-	70	700	C, W4	215
	130	360	C, F21, W5	259
–, dimethoxy-	130	73	C, W5	454
–, iodo-	65	41000	Q1	189
		45000	Q2	189
		45000	Q3	189
	100	41000	Q1	189
		45000	Q2	189
		45000	Q3	189
Methanol	30	36, 84, 100	E, Q(1), Q(2), Q(3)	523
	60	46, 100, 140	E, Q(1), Q(2), Q(3)	523
	100	34, 90, 120	E, Q(1), Q(2), Q(3)	523
	125	56, 100, 140	E, Q(1), Q(2), Q(3)	523
	130	21	C, F21, W5	259
	150	53, 120, 160	E, Q(1), Q(2), Q(3)	523
	180	37, 120, 160	E, Q(1), Q(2), Q(3)	523
Methylamine	130	53	C, W5	454
1-Octene	130	360	C, W5	52
		360	C, F21, W5	259
Pentane, 2,2,4-trimethyl-	130	64	C, F21, W5	259
Pentene	189	900	B, W5	443
–, 4,4-dimethyl-	130	175	C, W5	454
–, 4-methyl-	130	310	C, W5	454
Phosphine	130	207000	C, W5	457
–, dibutyl	130	36000	C, W5	454
–, tributyl	130	4500	C, W5	454
–, triphenyl	130	50	C, W5	454
Phosphorous acid, dimethyl ester	130	5100	C, F21, W5	259
Propane	130	27	C, F21, W5	259
		27.6	C, W17	456
		31	C, F21, W5	259
	200	65.2	C, W5	455
–, 2-chloro-	70	250	C, W4	215
Propane, 2-chloro-2-methyl-	70	40	C, W4	215
–, 2,2-dimethyl-	130	8	C, F21, W5	259
–, 2-iodo-	65	5.7×10^5	Q1	189
		5.9×10^5	Q2	189
		6.07×10^5	Q3	189
	100	4.55×10^5	Q1	189
		4.70×10^5	Q2	189
		4.83×10^5	Q3	189
–, 2-methyl-	130	50	C, F21, W5	259
		72	C, W5	454
	200	136	C, W5	455
–, 1,1,1,2,2,3,3,3-octafluoro-	130	4	C, F21, W5	259
1-Propene	130	110	C, W5	52
		122	C, W5	454
		150	C, W5	52
	200	200	C, W5	455
Propionaldehyde	130	2300	C, W17	457
		3300	C, W5	457
	200	2830	C, W5	457
Propionic acid, methyl ester	90	63000	C, Q1	485
		92000	C, Q3	485
		1.08×10^5	C, Q5	485
		630, 420, 780	C, Q(1), Q(2), Q(3)	516
		890, 1030, 1230	C, Q(4), Q(5), Q(6)	516
	120	78000	C, Q1	485
		1.12×10^5	C, Q3	485
		1.39×10^5	C, Q5	485
		780, 430, 1060	C, Q(1), Q(2), Q(3)	516
		1220, 1300, 1410	C, Q(4), Q(5), Q(6)	516

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Ethylene (*cont'd*)				
–, 3-cyano-, methyl ester	130	790	C, W5	454
Silane, tetramethyl	130	0	C, F21, W5	259
Tetradecane	189	580	B, W5	443
1-Tetradecene	189	1760	B, W5	443
Toluene	130	130	C, W5	52
		154	C, W5	454
		180	C, W17	456
	200	220	C, W5	455
Tributylamine	130	820	C, W5	454
Trichloroacetic acid, methyl ester	90	1000, 1700, 2770	C, Q(1), Q(2), Q(3)	559
Tridecane	130	140	C, F21, W5	259
Trimethylamine	130	180	C, F21, W5	259
		330	C, W5	454
Water	20	1.71	E, W	143
p-Xylene	130	300	C, F21, W5	259
		317	C, W5	454
		400	C, W17	456
	200	434	C, W5	455
Ethylene, bromo-				
Carbon tetrachloride	60	50		417
Ethylene, chloro-				
Acetaldehyde	50	110		164
Aniline, *N,N*-dimethyl-	50	2700		161
Benzene	35	1500		219
Butyraldehyde	50	350	W8	500
		420	W7	500
		500	W15	500
		580		500
Carbon tetrabromide	50	4.7×10^4	F26	492
		50.0×10^4		66
	60	3300	Q1	418
		1.85×10^4	Q2	418
		7.45×10^4	Q3	418
		12.15×10^4	Q4	418
Cyclohexane, 1,2-epoxy-4-vinyl-	–	264		428
Ethane, 1,2-dichloro-	25	4.0		425
	50	4.5	C, F26	392
Ether, dodecyl vinyl	50	156		4
Furan, tetrahydro-	25	16		425
	40	30		246
	50	24		164
Heptane, 2,4,6-trichloro-	50	5		211
Oxalic acid, diethyl ester	25	1.4		315
Pentane, 2,4-dichloro-	25	9.0	G	315
	50	5		211
Ethylene, 1,1-dichloro-				
Phosphorus trichloride	60	14	D	214
Ethylene, tetrafluoro-				
Ethanol	100	800	C, Q5, W2	3
		820	C, Q6, W2	3
		860	C, Q4, W2	3
		880	C, Q3, W2	3
Isopropanol	100	1540	C, Q3, W2	3
		1660	C, Q4, W2	3
		1700	C, Q5, W2	3
Methanol	100	350	C, Q3, W2	3
		390	C, Q4, W2	3
Hexanoic acid, vinyl ester				
Benzene	80	4.9		201
1-Hexene				
Carbon tetrachloride	140	4.1×10^4	C, Q1, F19	170
Cyclohexanol	125–135	390	C	381
Ethyl alcohol	30–35	170	D	381

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Isobutene				
Carbon tetrachloride	100	$(7.1 \pm 0.2) \times 10^4$	E, Q(1)	537
		$(8.1 \pm 0.5) \times 10^5$	E, Q(2)	
Isobutyric acid, vinyl ester				
Benzene	80	4.49		201
Maleic acid, diethyl ester				
Stearamide, *N*-allyl-	90	22.4	J	173
Maleic anhydride/methyl methacrylate				
Acetone	60	1.6	HH	367
Maleic anhydride/styrene				
Aniline, *N,N*-dimethyl-	60	930		489
Carbon tetrachloride	70	1.04	HH	365
Methacrylic acid				
Bromotrichloromethane	70	300		566
–, butyl ester				
Benzene	60	0.158		264
–, 2-(diethylamino)ethyl ester/styrene				
Carbon tetrachloride	80	23.6		335
Toluene	80	13.3		335
–, ethyl ester				
Acetic acid	80	0.095	S	83
Acetone	80	0.102		83
Acetophenone	80	0.281		83
Benzene	80	0.081		83
–, chloro-	80	0.436		83
–, ethyl-	80	1.428		83
2-Butanone	80	0.252		83
Butyl alcohol	80	0.454		83
sec-Butyl alcohol	80	1.604		83
tert-Butyl alcohol	80	0.417		83
Carbon tetrachloride	80	0.901		83
		5.640	C	183
Chloroform	80	0.703		83
		2.360	C	183
Cumene	80	2.067		83
Cyclohexane	80	0.928		83
Ethane, 1,2-dichloro-	80	1.821	C	183
Ethane, 1,1,2,2-tetrachloro-	80	0.311		83
		1.820	C	183
–, 1,1,1-trichloro-	80	0.536		83
Ethyl acetate	80	0.919		83
Ethyl alcohol	80	0.429		83
Heptane	80	0.865		83
2-Heptanone	80	0.702		83
Isobutyl alcohol	80	0.445		83
2,4-Pentanedione	80	0.236		83
Toluene	80	0.436		83
–, hexadecyl ester				
Carbon tetrachloride	70	0.983	–	137
Cumene	70	2.05	–	137
–, isobutyl ester				
Benzene	60	0.165		264
Carbon tetrachloride	80	1.971	C	183
Chloroform	80	1.110	C	183
Ethane, 1,2-dichloro-	80	0.510	C	183
–, 1,1,2,2-tetrachloro-	80	0.510	C	183
Methacrylonitrile				
Carbon tetrabromide	100	900	A, F7	109
	120	1000	A, F7	109
Iron(III) chloride	60	30800	F7	19
Isobutyraldehyde	60	0.001		70
Toluene	60	0.996		21
Methyl methacrylate				
Acetaldehyde	60	6.5		86
Acetic acid	80	0.24	A	31

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TABLE 4. *cont'd*

Solvent/Additive	T (°)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
Acetic acid, monochloro-	60	0.64	R	529
–, dichloro-	60	0.80	R	529
–, 1,1-dimethyl-2,2,2-trinitro-ethyl ester	45	520		110
–, trichloro-	60	1.10	R	529
Acetone	60	0.195		81
	80	0.225	A	31
		0.275		81
Acetonitrile, (*m*-bromophenyl)-	60	3.89		462
–, (*p*-bromophenyl)-	60	2.78		461
–, (*m*-chlorophenyl)-	60	4.28		462
–, (*p*-chlorophenyl)-	60	3.21		461
–, (*p*-methoxyphenyl)-	60	19.0		461
–, phenyl-	60	5.18		461
–, *m*-tolyl-	60	5.75		462
–, *p*-tolyl-	60	7.83		461
Acetylene, *p*-bromophenyl-	60	41.3	K	151
–, *p*-chlorophenyl-	60	38.9	K	151
–, *p*-nitrophenyl-	60	127.5	K	151
Aluminum, hydrodiisobutyl	60	3600	S	156
–, triethyl	50	1550	R, F2	547
	60	1240		156
Aniline	60	4.2	F2	424
	80	6.3	F2	424
	100	9.0	F2	424
–, *N,N*-dimethyl-	30	2.45		508
	50	18	F16	228
		30.4	H	409
		430		161
	60	11.3	A, F2	424
	70	10.8		270
	80	17.0	A, F2	424
	100	20.0	A, F2	424
–, *N,N*-divinyl-	60	340		82
–, *N*-methyl-	60	7.0	F2	424
	80	10.0	F2	424
	100	13.3	F2	424
p-Anisaldehyde	60	1.11		495
Anisole	25	0	D	16
–, *p*-ethynyl-	60	40.3	K	151
–, *p*-isopropyl-	60	3.46		406
–, *p*-methyl-	60	0.57		496
Anthracene	50	0		15
Azobenzene	50	100	C	282
Benzaldehyde	60	2.5		86
		0.86		495
–, *p*-bromo-	60	1.43		495
–, *m*-chloro-	60	0.96		495
–, *p*-chloro-	60	1.03		495
–, *p*-cyano-	60	2.06		495
Benzene	25	0	D	16
	30	0.01		508
	50	0.036		147,327
	52	0.027		81
	60	0.040		81
		0.83	C	181
	75	0.33	C	123
	80	0.075	A	31
		0.080		81
		0.24	I1	145
	90	0.036		147
–, allyl-	60	400		502
–, bromo-	25	0	D	16
–, *tert*-butyl-	80	0.260	A	31

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
–, chloro-	25	0	D	16
	60	0.074		81
	80	0.200	A	31
		0.207		81
–, *p*-diisopropyl-	60	5.72		406
–, *m*-dinitro-	50	52		15
–, ethyl-	52	0.501		81
	60	0.766		81
	80	1.311		81
		1.350	A	31
		2.1	I1	145
–, ethynyl-	60	21.9	K	151
		22.3	J	150
–, fluoro-	25	0	D	16
Benzene sulfonylchloride	60	5.16	R	543
–, *p*-chloro-	60	8.56	R	543
–, *p*-dimethyl-	60	4.06	R	543
Benzonitrile	25	0	D	16
	60	0.162		150
–, *p*-hydroxy-	50	6.0		479
–, *p*-isopropyl-	60	4.99		406
p-Benzoquinone	44.1	5.5×10^4		194
	50	5.7×10^4		15
	60	4.5×10^4		42
–, 2,3,5,6-tetrachloro-	44.1	2600		194
Benzothiazolethion, 3-allyl-	60	153		502
Benzyl ether	25	10.4	D	368
	60	8.0		368
Bibenzyl	50	0.0	F2	354
Biphenyl, 2,2-methylene-	60	41.8		567
	60	53.5		567
Borane, tributyl-	60	7.45		156
Bromoform	77	23	D, C, R	506
Butane, 1-chloro-	80	1.20	A	31
–, 1,1,1-trinitro-	45	8300		110
1,4-Butanediol	60	0.61	F24	421
	80	1.07	F24	421
2-Butanone	60	0.45	C	290
	70	0.56	C	123
	75	0.83	C	123
	80	0.70	A	31
1-Butene	40	3.1		287
	50	5.1		287
–, 3,4-epoxy-2-methyl-	80	29.6		269
2-Butene, *cis*-	40	3.2		287
	50	4.9		287
–, *trans*-	40	3.0		287
	50	5.2		287
3-Buten-2-ol, 1-chloro-3-methyl-	80	18.8		269
Butyl alcohol	60	0.394		479
	80	0.25	A	31
sec-Butyl alcohol	60	0.259		479
	80	0.85	A	31
tert-Butyl alcohol	60	0.085		81
	80	0.100	A	31
		0.152		81
Butylamine, *N*-nitro-	45	0		110
Butyl ether	60	0.8		86
1-Butyne, 1-phenyl-	60	10.8		150
Butyraldehyde	50	1.47	W8	500
		2.25	W7	500
		3.40		500
Carbon tetrabromide	30	2800 ± 600	D, R, JJ3a	521
		1000 ± 100	D, R, JJ3b	521

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
Carbon tetrabromide (*cont'd*)		1000 ± 500	D, R, JJ3c	521
		500 ± 200	D, R, JJ3d	521
	60	2700	F2	109
	60	1500		568
	60	1900		568
	80	3300	F2	109
	100	4600	F2	109
Carbon tetrachloride	20	0.2	C, D	35
	30	20	D, F2	451
		0.20		514
	50	0.82		514
	60	0.925		81
		2.40	C	181
		5		86
		20.11	R	529
		18.52	R, H	529
		0.42	R	568
		0.5	R, H	568
	70	1.74		514
	80	2.393	A	31
		2.421		81
		3.3	I1	145
	–	24.4	D	427
Chloroform	60	0.454		81
		1.77	C	181
	80	1.129		81
		1.400	A	31
		1.9	I1	145
	–	1.9	D	427
Copper(II) chloride	60	105×10^5	F7	36
Cumene	60	1.9		7
		2.56		406
	80	1.9	A	31
		2.4	I1	145
–, *p*-bromo-	60	3.71		406
–, *p*-*tert*-butyl-	60	2.74		406
–, *p*-chloro-	60	3.07		406
Cyclohexane	60	12		86
	80	0.10	A	31
–, methyl-	80	0.195	A	31
–, *trans*-1,4-, diacetate	60	8.46	C	569
–, *cis*/*trans*-1,4-, dicarboxylic acid dimethylester	60	1.16	C	569
Cyclotetrasiloxane, octamethyl-	50	2.5	C	450
	79.5	0.080		474
p-Dioxane	80	0.222	A	31
Diphenylamine	44.1	0		194
Diphenylamine-T	60	0.3	F2	46
Disiloxane, 1,1-dimethoxy-3,3,3-trimethyl-1-phenyl-	79.5	0.032		474
–, hexamethyl-	79.5	0.104		474
Epibromohydrin	60	16.53	R	529
Epichlorohydrin	60	9.3	R	529
Ethane, 1,2-dichloro-	60	0.35		81
	80	0.756		81
–, nitro-	45	2.0		110
–, 1,1,2,2-tetrachloro-	60	0.155		81
	80	0.200	A	31
		0.235		81
–, 1,1,1-trichloro-	80	0.600	A	31
–, 1,1,1-trinitro-	45	1400		110
Ethyl acetate	60	0.100	C	316
		0.13	A	290,316
		0.132	C	216
		0.134		290
		0.155		290

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
Ethyl acetate (*cont'd*)		0.156	C	290
		0.46	I10	146
	70	0.55	C	123
	75	0.83	C	123
	80	0.240	A	31
Ethyl alcohol	60	0.40	A, F2	423
	80	0.625	A, F2	423
	100	0.80	A	49
Ethylene glycol	60	0.28	F24	421
	80	0.60	F24	421
Germane, diethylchloro-	60	22000	A, R	539
–, ethyldichloro-	60	39300	A, R	539
–, triethyl-	60	5800	A, R	539
–, triphenyl-	60	18500	A, R	539
Glycerol	60	0.152		479
Heptane	50	1.8		206
1-Heptanol, 2,2,3,3,4,4,5,5,6,6,7,7-dodecafluoro-	60	2.8	F2	442
Hydrazyl, 2,2-diphenyl-1-picryl	44.1	200×10^5		194
Hydroquinone	45	7.0		30
	131	100	A	398
Indium, triethyl-	60	332		156
Isobutyl alcohol	60	0.10		81
		0.505		479
	80	0.229		81
		0.250	A	31
Isobutyric acid	80	0.900	A	31
–, methyl ester	60	0.26		275
Isopropyl alcohol	60	0.583		81
	80	1.907		81
	100	3.00	A	49
		8	C	337
	130	43	C	337
		45	C	480
Lead, tetraethyl	60	3.14	S	156
Malononitrile, furfurylidene-	44.1	1.2×10^4		194
Mercury, diethyl	60	0.898		156
Methane, bromotrichloro-	30	830	D, Q3	311
		1.2×10^4	D	311
		4.5×10^4	D, Q4	311
–, bromotrinitro-	45	1.0×10^4		110
Methane, dichloro-	60	0.100		81
	80	0.217		81
–, nitro-	60	2.0		86
–, trinitro-	45	5400		110
–, triphenyl-	60	4.0		44
Methanol	60	0.2	A, F2	423
	80	0.33	A, F2	423
	100	0.45	A	49
1-Naphthol	45	<5.0		30
2,6-Octadiene, 2,6-dimethyl-	60	6.7		324
Oleic acid, methyl ester	60	1.68		441
		8.0		7
Oxime, acrolein-	60	25500		555
–, crotonaldehyde-	60	500		555
–, ethylisopropenylketone-	60	3000		555
–, methacrolein-	60	17000		555
–, methylacrolein-	60	5200		555
–, methylisobutenylketone-	60	4700		555
–, methylisopropenylketone-	60	2900		555
–, methylvinylketone-	60	1300		555
Oxolane, *cis*-3,4-diacetoxy-	60	1.39	C	569
Pentane, 2,2,4-trimethyl-	50	1.2		206
1-Pentanol, 2,2,3,3,4,4,5,5-octa-fluoro-	60	2.36	F2	442
2-Pentanone, 4,4-dimethyl-5,5,5-trinitro-	45	400		110

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
–, 4-methyl-	80	0.700	A	31
3-Pentanone	60	0.833		81
	80	1.729		81
		1.775	A	31
		2.7	I1	145
Pentasiloxane, dodecamethyl-	79.5	0.145		474
Phenol	50	2.5		479
–, *o*-bromo-	50	5.0		479
–, *p*-bromo-	50	5.0		479
–, *o*-chloro-	50	3.5		479
–, *o*-ethyl-	50	7.2		479
–, *p*-ethyl-	50	7.5		479
–, *o*-isopropyl-	50	13.3		479
–, *p*-isopropyl-	50	13.3		479
–, *p*-methoxy-	45	<5.0		30
–, 2,3,4,6-tetramethyl-	45	11.0		30
Phenyl ether	25	6.42	D	368
	60	9.13		368
Phosphine, octyl-	60	2.3×10^4		295
–, phenyl-	60	16.1×10^4	S	295
–, tributyl-	60	30.6		156
Phthalic acid, dimethylester	30	0.2		508
Piperidine, 1-nitroso-	45	8.2		110
Propane, 1,2-dichloro-	80	0.675	A	31
–, 1,1-dinitro-	45	68		110
–, 2,2-dinitro-	45	15		110
–, 1-nitro-	45	5		110
1,2-Propanediol	60	0.48	F24	421
	80	0.86	F24	421
1-Propanol, 2,2,3,3-tetrafluoro-	60	1.86	F2	442
Propene, 1,3-diphenyl-	60	15.6		567
1-Propene, 2-methyl-	40	2.5		287
	50	4.4		287
Propionic acid, 2,2,2-trinitroethyl ester	45	3000		110
Propionitrile, 3-phosphino-	60	1.4×10^4		295
–, 3,3′-phosphinylidenedi-	60	1.3×10^4		295
Propyl alcohol	60	0.69	A, F2	423
	80	0.84	A, F2	423
	100	1.25	A	49
Pyridine	70	0.176		514
Pyrocatechol	45	<5		30
–, *p-tert*-butyl-	45	9		30
Pyrogallol	45	26		30
Silane, (α-bromotolyl)trimethoxy-	79.5	29.8		474
–, bromomethylphenyltrimethoxy-	79.5	29.8	R	554
–, chlorotrimethyl-	30	144	D, F2	451
	50	2.2		249
–, dichlorodimethyl-	30	175	D, F2	451
	50	2.45		249
–, dichloromethyl-	79.5	15.2		474
–, diethylchloro-	60	1700	A, R	539
–, dimethoxymethylphenyl-	79.5	0.20		474
		0.2	R	554
–, dimethylphenyl-	60	300	A, R	539
–, hexamethoxy-(3,3′-thiodipropyl)bis-	79.5	25.8	R	554
–, methyldichloro-	79.5	15.2	R	554
	60	3100	A, R	539
–, (3-mercaptopropyl)trimethoxy-	79.5	6930	R	554
–, tetraethyl-	60	5.75		156
–, tetramethyl-	30	5.0	D, F2	451
	50	1.3		249
–, trichloro	60	5000	A, R	539
–, trichloromethyl-	30	185	D, F2	451
	50	2.7		249

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
–, triethyl	60	200	A, R	539
–, triisopropoxymethyl-	79.5	0.697		474
		0.697	R	554
–, trimethoxymethyl-	79.5	0.331		474
		0.331	R	554
–, trimethoxyphenyl-	79.5	0.053		474
		0.053	R	554
–, triphenyl-	60	1200	A, R	539
Silicon chloride	30	256	D, F2	451
	50	3.0		249
Siloxane, 1,1-dimethoxy-trimethyl-1-phenyl-di-3,3,3-	79.5	0.032	R	554
–, dodecamethyl penta-	79.5	0.145	R	554
–, 1,1,1,3,5,5,5-heptamethyl-3-phenyltri-	79.5	0.232	R	554
–, 1,1,1,3,5,5,5-heptamethyl-tri-	79.5	0.284	R	554
–, 1,1,1,3,5,5,5-heptamethyl-tri-	79.5	0.189	R	554
–, 1,1,1,3,5,5,5-heptamethyl-3-(3,3,3,3-tri-fluoropropyl)-tri-	79.5	0.205	R	554
–, hexamethyl-di-	79.5	0.104	R	554
–, hexamethyl-phenyl-tri-	79.5	1.84	R	554
–, 1,1,1,5,5,5-bexamethyl-3-(2-phenylpropyl)-tri-	79.5	2.08	R	554
–, 3-(3-mercaptopropyl)-1,1,1,3,5,5,5-heptamethyl-tri-	79.5	1284.5	R	554
–, octamethyl cyclotetra-	79.5	0.08	R	554
–, octamethyl tri-	79.5	0.0316	R	554
Sorbit-2,5-diacetate, 1,4:3,6-, dianhydro-	60	29	C	569
Stearamide, *N*-allyl-	90	3.01	C, J	173
Stearic acid, methyl ester	60	0.282		441
Stibine, tributyl	60	< 1.0		156
Tin, tetrabutyl	60	1.32		156
Toluene	20	0.04	C, D	35
	52	0.084		81
	60	0.170		21,81
		0.190	C	316
		0.202	C	32,316,317
		0.250	C	316
		0.26		275
		0.400	C	316
		0.45		496
	70	0.567	C	123
	80	0.292	C	32
		0.303		81
		0.52	I1	145
		0.525	A	31
		0.91	C	123
–, *p*-bromo-	60	0.58		496
–, *m*-chloro-	60	0.48		496
–, *p*-chloro-	60	0.49		496
–, α-chloro	60	4.17	C	181
–, *p*-ethynyl-	60	64.4	K	151
–, *p*-nitro-	44.1	0		194
–, 2,4,6-trinitro-	44.1	500		194
m-Tolunitrile, α-cyano-	60	4.55		462
p-Tolunitrile	60	0.73		496
–, α-cyano-	60	1.74		461
Tributylamine	70	25.7		270
Triethylamine	20	1.5	C, D	35
	60	8.3	J	24
		1900		25
Tripropylamine	60	14.6		156
Trisiloxane, 1,1,1,3,5,5,5-heptamethyl-	79.5	0.284		474
–, –, 3-(β-methylphenethyl)-	79.5	2.08		474
–, –, 3-phenyl-	79.5	0.232		474
–, –, 3-propyl-	79.5	0.189		474
–, –, 3-(3,3,3-trifluoropropyl)-	79.5	0.205		474

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
–, 1,1,1,5,5,5-hexamethyl-3-phenyl-	79.5	1.84		474
–, octamethyl-	79.5	0.032		474
Urea, thio-	42	1.21		503
Water	60	0		268
		0.002		479
		0.03	A, R	529
	80.5	0	F7, F24	48
m-Xylene	60	0.43		496
p-Xylene	60	0.50		496
2,4-Xylenol	50	6.5		479
Naphthalene, 1-vinyl-				
Naphthalene	60	30–50		216
Nonanoic acid, vinyl ester				
Toluene	50	11.1	R	357
	60	13.9	R	357
	70	15.2	R	357
1-Octene				
Butyl alcohol	117–118	270	C	381
sec-Butyl alcohol	117–118	520	C	381
Ethyl alcohol	115–118	230	C	381
Isopropyl alcohol	30–35	630	D	381
	120–125	520	C	381
Methanol	116–120	110	C	381
Phthalimide, *N*-vinyl-				
Chloroform	55	3.4		313
Ethane, 1,2-dichloro-	55	0.35		313
Methane, dichloro-	55	0.97		313
2-Picoline, 5-vinyl-				
Allyl ether	70	21.4		11
1-Propene				
Carbon tetrachloride	100	$(5-10)\times 10^4$	C	210
Chloroform	100	1.03×10^4	C	210
1-Propene, 3-chloro-				
Carbon tetrachloride	100	4800	C	210
Hydrochloric acid	80	1.8×10^4	Q1	238
		5.4×10^4	Q2	238
1-Propene, 2-methyl-				
Carbon tetrachloride	100	17.0×10^4	C	210
–, 3,3,3-trifluoro-				
Carbon tetrabromide		640000	C, Q(1–3)	570
Propionic acid, vinyl ester				
Toluene	–	1.4		158
Propylene				
Bromoethane	50	1200	Q(1–4)	517
	75	1400	Q(1)	517
		1500	Q(2)	517
		1800	Q(3–4)	517
	100	1700	Q(1)	517
		1900	Q(2)	517
		1800	Q(3–4)	517
	125	2200	Q(1–2)	517
		2100	Q(3–4)	517
	150	2700	Q(1–4)	517
Chloroform	95	17200 ± 1100	W23, Q(1)	518
		33000 ± 15000	W23, Q(2)	518
		17000 ± 800	W24, Q(1)	518
		299000 ± 8000	W24, Q(2)	518
		13800 ± 800	W20, Q(1)	518
		266000 ± 10000	W20, Q(2)	518
Stearic acid, vinyl ester				
Toluene	50	20.7	R	357
	70	21.6	R	357
Styrene				
Acetaldehyde	60	8.5		86

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Acetamide, *N,N*-dimethyl-	50	0.743		375
	60	4.6		113
Acetic acid	40	2.22	C	385
–, methyl ester	100	0.33		72
–, bromo-	60	430		112
	68	300		149
–, chloro-	68	200		149
–, –, methyl ester	60	0.3	A	128
	100	0.75	A	128
–, dibromo-, ethyl ester	90	2700	C	64
–, monochloro-	60	28.6	R	529
		0.76	R, H	529
–, dichloro-	60	35.0	R	529
		1.0	R, H	529
–, dichloro-, ethyl ester	60	1.3	A	128
–, iodo-	68	8000		149
–, phenyl-	60	6.0	A	128
–, –, ethyl ester	100	8.8		72
–, tribromo-	–	3.0×10^4		178
	90	2.4×10^4	C	64
–, –, ethyl ester	90	10.5×10^4	C	64
–, trichloro-	–	270		178
	60	66.0	A	128
		100	R	529
		3.75	R, H	529
–, –, ethyl ester	60	65.0	A	128
	90	90.0	C	64
	100	145.0	A	128
Acetic anhydride	60	0.7	A	128
Acetoacetic acid, ethyl ester	100	1.3		72
–, 2-acetyl, ethyl ester	100	3.0		72
Acetone	60	4.1	C	182
		<0.5	A	128
		0.32	A	501
–, oxime	60	2.2	A	128
Acetonitrile	60	0.44	A	128
–, *m*-bromophenyl-	60	66.7		462
Acetonitrile, *p*-bromophenyl-	60	68.5		461
–, *m*-chlorophenyl-	60	65.2		462
–, *p*-chlorophenyl-	60	66.0		461
–, *p*-methoxyphenyl-	60	51.0		461
–, phenyl-	60	45.1		461
–, *m*-tolyl-	60	48.5		462
–, *p*-tolyl-	60	49.2		461
Acetyl bromide	60	8600	S	112
Acetyl chloride, chloro-	60	3300		112
Acetylene, *p*-bromophenyl-	60	188.5	K	151
–, *p*-chlorophenyl-	60	161	K	151
–, *p*-nitrophenyl-	60	3130	K	151
Adipic acid, diallyl ester	60	6.0		446
–, di-(3,3,5,5-tetramethyl-4-nitrosocyclohexyl) ester	60	1.1×10^4	F9, J	314
Allyl alcohol	60	1.5	A	128
Aluminum, hydrodiisobutyl	100	26.9×10^4		156
		28.0×10^4	A	155
–, triethoxy-	100	<0.1	A	155,156
–, triethyl	100	8.05×10^4		156
		17.0×10^4	A	155
–, triisobutyoxy-	100	<1.0	A	155
–, triisobutyl	110	28.5×10^4	A	155
Aniline	50	20	F2	154
	60	2.0	A	128
–, *N,N*-dimethyl-	50	12	F2	154
		53		161
–, *N,N*-divinyl-	60	130		82

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TABLE 4. *cont'd*

Solvent/Additive	T (°)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Aniline, *N*-methyl-	50	13	F2	154
–, 2,4,6-trinitro-	50	11.8×10^4		370,372
p-Anisaldehyde	60	2.86		495
Anisole, *p*-ethynyl-	60	60.0	K	151
–, *m*-isopropyl-	60	5.23		405
–, *p*-isopropyl-	60	3.23		406
–, *p*-methyl-	60	0.78		496
–, 2,4,6-trinitro-	50	20.3×10^4		370,372
Anthracene	44.1	2.0×10^4	J	159
–, dihydro-	50	610	D	271
2H-Azepin-2-one, hexahydro-	130	0.61	C, I	108
Azobenzene	70	750	C	281
Benzaldehyde	60	4.54		495
		5.5		86
	100	2.6	A	298
–, *p*-bromo-	60	12.2		495
–, *m*-chloro-	60	13.7		495
–, *p*-chloro-	60	8.63		495
	100	5.6	A	298
–, *p*-cyano-	60	76.7		495
Benzene	35	3.9	F2	119
	40	5.8	F2	119
	50	0.01		327
	60	0.018	A	127
		0.023		79
		0.028	C	255
		0.04	A, I5	145
		1.92	H	415
	70	5.50	C	123
	75	6.67	C	123
	80	0.061	I5	185
		0.121	C	255
		0.156		273
	100	0.184	A	127
		0.23	A, I5	145
		0.31	A	235
		0.42	A	235
	132	0.81	A	235
		0.89	A	127
		1.5	A, I5	145
–, allyl-	100	36	A, R	524
–, bromo-	60	1.78	C	182
	155	3	A	237
–, *sec*-butyl-	60	6.22	H	415
–, *tert*-butyl-	60	0.04	A, I5	145
		0.06	A	127
	80	0.193	I5	185
	100	0.55	A	127
–, chloro-	60	0.133	C	255
		1.50	C	182
	80	0.235	C	255
		0.874	I5	185
	100	0.54	A	235
	140	0.6		58
–, –, 2,4,6-trinitro-	50	58.5×10^4		370,372
–, 1,2-dibromoethyl-	60	1950		112
–, *p*-dibutyl	60	7.02	H	415
–, *p*-di-*sec*-butyl-	60	10.70	H	415
–, *p*-di-*tert*-butyl-	60	0.87	H	415
–, *m*-dichloro-	140	0.2		58
		1.4		67
–, *o*-dichloro-	60	3.4	C	182
–, *p*-dichloro-	60	2.6	C	182

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Benzene, diethyl-, (mixture)	100	3.35	A	235
		6.33	A	235
	132	5.13	A	235
–, *p*-diisopropyl-	60	3.30		497
		6.60		406
–, ethyl-	60	0.67	A	127
		0.70		79
		0.710	C	255
		0.83	A, I5	145
		2.7	C	363
	80	1.07	I5	185
		1.113	C	255
	100	1.38	A	235
		1.62	A	127
		2.2	A, I5	145
		2.33	A	235
	132	2.31	A	235
		2.9	A	127
		4.9	A, I5	145
–, ethynyl-	60	96.3	K	151
		98.3	J	150
–, *sec*-hexyl-	60	12.76	H	415
–, *sec*-pentyl-	60	9.43	H	415
–, tri-*sec*-butyl-	60	13.30	H	415
–, 1,3,5-trinitro-	40	9.48×10^5		372
	50	6.43×10^5		370,372,373
	60	3.51×10^5		372
Benzo[B] chrysene	44.1	1.3×10^5	J	159
Benzoic acid				
–, *m*-(phenylazo)-, methyl ester	70	700	C	281
–, 2,4,6-trinitro-, ethyl ester	50	5.72×10^5		370,372
Benzoin	60	40	A	128
Benzonitrile	60	5.3		150
–, *p*-isopropyl	60	18.6		406
Benzo[A]pyrene	44.1	1.4×10^5	J	159
p-Benzoquinone	60	22.7×10^5		42
	80	56.6×10^5	J	27
p-Benzoquinone, 2-anilino-	45	53.2×10^5	J	199
–, 2,5-dimethyl-	80	4.3×10^5	J	27
–, 2-methyl-	80	21.0×10^5	J	27
–, 2,3,5,6-tetrachloro-	80	95.0×10^5	J	27
–, 2,3,5,6-tetramethyl-	80	6700	J	27
–, 2,3,5-trimethyl-	80	2.6×10^5	J	27
Benzyl ether	25	71.6	D	368
	60	62.4		368
Bibenzyl, α,α′-dibromo-	60	3020		112
Bicyclohexyl, 3,3,3′,3′,5,5,5′,5′-octamethyl-4,4′-dinitroso-	60	1.12×10^4	F9, J	314
	80	1.2×10^4	F9, J	314
Biphenyl, 2,2′-methylene-	60	142		567
	60	97.9		567
Borane, tributoxy-	100	<0.1		156
–, tributyl	100	34.8		156
Butane, 1-bromo-	60	0.06	A	128
	100	0.35	A	128
–, 1-chloro-	60	0.04	A	128
	100	0.37	A	128
–, 2-chloro-	60	1.2	A, S	128
	100	0.3	A	128
–, 2,2-dimethyl-	60	0.43		304
–, 1-iodo-	60	1.85	A	128
	100	5.5	A	128
1,4-Butanediol	60	3.2	F24	422
	80	5.6	F24	422
	100	9.3	F24	422

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
2-Butanone	60	4.98	C	182
	70	8.60	C	123
	75	12.00	C	123
1-Butene	100	2.6	A	358
–, 3,3-dimethyl-2-phenyl-	110	10	A	433
–, 2,4-diphenyl-	110	70	A	433
–, 3,4-epoxy-2-methyl-	80	17.4		269
–, 2-methyl-	100	3.1	A	358
–, 3-methyl-	100	6.9	A	358
		6.1	A, R	524
2-Butene	100	2.0	A	358
–, 1,4-dichloro-	80	51	C	112
–, 2,3-dimethyl-	100	5.4	A	358
		5.1	A, R	524
–, 2-methyl-	100	2.9	A	358
3-Buten-2-ol, 1-chloro-3-methyl-	80	8.2		269
Butyl alcohol	40	11.2	F24	119
	50	6.5		430
	60	0.06	A	128
		1.6	A, C	323,385
sec-Butyl alcohol	60	0.562		479
tert-Butyl alcohol	50	6.6		430
	60	0.22		422
		6.65	A	385
	80	0.345		422
	100	0.55	A	261
	130	1.0	A	261
Butylamine	60	0.5	A	128
tert-Butyl ether	60	2.6	C, F2	301
	80	1.0	C, F2	301
tert-Butyl isocyanide	100	33.0		4,75
1-Butyne, 1-phenyl-	60	34.3		150
Butyraldehyde	–	2.7	W8	412
	–	14.3		412
	50	2.7	F29, W8	414
		3.7	F29, W16	414
		4.0	F2, W7	413
		4.7	F29, W7	414
		8.0	F2, W15	413
		8.0	F29, W15	414
		11.7	F2	413
		14.3	F29	414
	60	5.7	A	128
	100	11.0	A	128
–, diallyl acetal	60	20.2		446
Butyric acid, 4-hydroxy-, γ-lactone	50	0.409		375
Cadmium, dibutyl	100	1170		156
Carobonic acid, cyclic ethylene ester	50	0.235	E	458
–, diallyl ester	60	6.2		446
–, di-(3,3,5,5-tetramethyl-4-nitrosocyclohexyl) ester	60	1.3×10^4	F27, J	314
	80	1.4×10^4	F9, J	314
Carbon tetrabromide	40	1.8×10^4	F2	109
	60	1.78×10^4	C	64
		2.2×10^4	F2	109
		2.5×10^4	F1	109
		13.6×10^4	A	128
		4.2×10^4		487
		$(5.09 \pm 0.18) \times 10^5$	II2	520
	60	88300		568
	60	40600		568
	70	1.8×10^4		61
	80	2.3×10^4	F2	109
	90	2.51×10^4	C	64
	100	2.35×10^4	A	128

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Carbon tetrachloride	–	80	W8	412
	–	100		412
	50	17		513
		248		513
	60	69	Q36	63
		84	C, W8	401
		87	C	113,183
		90	C, W7	401
		92	A	128,129,210
		98	C	401
		100		366
		109	Q142	63
		110		223
		120		416
		122	A, X	114
		144.1	R	529
		130.9	R, H	529
		148	X	115
	60	92		568
	60	110		568
	76	18	C	236
		117	A	236
	80	133	I5	185
	85	34.1		163
	95	148	Q38	63
		186	Q142	63
	100	185	A	128,129
	132	304	A	129
	140	300		58
Chloroform	60	0.41	A	501
		0.5	A	128,210
		0.566	C	255
		3.40	C	182
	68	4.0		149
	80	0.50	I5	185
		0.916	C	255
Chloroform/acetone	60	3.3		501
m-Cresol	50	11		247,248
o-Cresol	50	33		247,248
	60	43		121
p-Cresol	50	22.1		247,248
	60	39		121
–, α-phenyl-	60	<5		121
Cumene	60	0.8		7
		0.82	A	127
		1.04	A, I5	145
		3.88		406
	80	1.31	I5	185
	100	2.00	A	127
		2.90	A, I5	145
–, *m*-bromo-	60	8.29		405
–, *p*-bromo-	60	7.57		406
–, *p*-tert-butyl-	60	3.46		406
–, *p*-chloro-	60	6.90		406
Cyclohexane	60	0.024	A	127
		0.031		79
		0.04	A, I5	145
		0.063	C	255
		8.5	F23	119
	80	0.066	I5	185
		0.083	C	255
		0.156		273
	100	0.16	A	127
		0.23	A, I5	145

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Cyclohexane (*cont'd*)		0.31	A	235
		12.3	A, R	524
	132	0.81	A	235
		0.87	A	127
		1.5	A, I5	145
Cyclohexanone	60	7.90	C	182
–, *trans*-1,4-, diacetate	60	16.9	C	569
–, *cis/trans*-1,4-, dicarboxylic acid dimethylester	60	4.2	C	569
Cyclopentanone	60	3.30	C	182
1-Cyclopentene-1-carboxylic acid, 3,3,5,5-tetramethyl-4-nitroso-, diester with pyrocatechol	60	1.3×10^4	F27, J	314
	80	1.0×10^4	F9, J	314
Cyclotetrasiloxane, octamethyl-	50	4.0	C	450
1,2,3,4-Dibenzpyrene	44.4	6.0×10^4	J	159
1,2,7,8-Dibenztetracene	44.4	13.0×10^4	J	159
1,2,9,10-Dibenztetracene	44.4	13.0×10^4	J	159
p-Dioxane	60	0	A, S	128
		2.28	C	4
		2.75	C	182
	100	0.8		220
Diphenylamine-T	60	0.9	F2	46
Disiloxane, hexamethyl-	79.5	0.387		474
Epibromohydrine	60	67.48	R	529
Epichlorohydrine	60	37.87	R	529
Ethane, 1,2-dibromo-	60	0.988	C	255
	80	1.914	C	255
–,1,2-dichloro-	60	0.333	C	255
		4.12	C	182
	70	1.1		67
	80	1.137	C	255
		9.8	C	112
	100	3.84	A	235
–, pentaphenyl-	60	20000	A	127
–, 1,1,2,2-tetrachloro-	100	10.8	A	235
–, 1,1,2,2-tetrachloro-1,2-difluoro-	90	1.13	A	14
–, 1,1,2-trichloro-1,2,2-trifluoro-	90	0.84	A	14
Ethanehexacarboxylic acid, hexaethyl ester	50	< 50		89
1,1,2-Ethanetricarboxyic acid	100	0.94		72
Ether, benzyl methyl	68	6.0		149
–, *p*-bromobenzyl methyl	68	6.0		149
–, *p*-chlorobenzyl methyl	68	4.0		149
–, *p*-cyanobenzyl methyl	68	20.0		149
–, dodecyl vinyl	60	3.32	C	4
		4.11	C	4
Ethyl acetate	60	15.5	F10	119
	70	5.5	C	123
	75	6.67	C	123
	100	0.39		72
Ethyl alcohol	60	1.32	F2	423
		1.611		479
	80	2.60	F2	423
Ethylene, 1,1-diphenyl-	110	450	A	432
Ethylene glycol	60	1.36	F24	422
	80	2.70	F24	422
	100	4.70	F24	422
–, bis(*m*-phenylazobenzoate)	70	900	C	281
Ethyl ether	60	5.64	C	4
Fluorene	60	75.0	A	127
	100	124.0	A	127
Formamide, *N,N*-dimethyl-	40	1.09	E	458
	50	0.869		375
	60	4.0		113
	100	1.08		256

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Furan, tetrahydro-	50	0.50	C	449
α, D-Glucoside, methyl-				
–, –, 6-deoxy-6-mercapto-	100	55000		220
–, –, 2,3-di-*O*-benzyl-	100	62		220
–, –, 2,3,4,6-tetra-*O*-acetyl-	100	2.0		220
–, –,6-*O*-(*p*-toluenesulfonyl)-	100	2.0		220
–, –, 2,3,4-tri-*O*-acetyl-6-deoxy-6-iodo-	100	50		220
–, –, 6-*O*-triphenylmethyl-	100	21		220
β, D-Glucoside, methyl-				
–, –, 6-deoxy-6-dipropylamino-	100	22		220
Germane, ethyldichloro-	60	56900	A, R	539
–, diethylchloro-	60	31600	A, R	539
–, triethyl-	60	2400	A, R	539
–, dimethylchloro-	60	33500	A, R	539
–, triphenyl-	60	23000	A, R	539
Heptane	60	0.42	A	127
	100	0.95	A	127
1-Heptanol, 2,2,3,3,4,4,5,5,6,6,7,7-dodecafluoro-	60	13.33	F2	442
1-Heptene	100	2.7	A	358
2-Heptene	100	3.2	A	358
Hexane	100	0.9	A	358
1-Hexene	100	2.5	A	358
		2.5	A, R	524
2-Hexene	100	3.6	A	358
Hydrochloric acid	100	0		238
Hydroquinone	60	3.6	A	128
Hydroxylamine, *N,N*-dibenzyl-	60	5000	R	533
–, *N*-benzyl-*N*-phenyl-	60	3.8×10^4	R	533
–, *N,N*-diethylene-	60	2.5×10^4	R	533
–, *N*-ethyl-*N*-phenyl-	60	20.5×10^4	R	533
Indium triethyl	100	1.7600×10^4		156
Iron(III) chloride	60	536.0×10^4		19
Isobutyl alcohol	60	0.17	A	385
		0.497		479
Isobutyl alcohol-D	100	2.9	A	261
	130	7.8	A	261
Isobutyraldehyde		8.2	W8	412
		21.0		412
Isobutyric acid	60	2.5	C	385
		4.6	A	385
Isobutyronitrile	100	2.7	A	261
	130	3.5	A	261
Isophthalic acid, diallyl ester	60	3.5		446
Isopropyl alcohol	60	3.05	F2	423
	80	4.00	F2	423
	100	1.7	A	261
		6.0	F2	423
	130	2.7	A	261
Isopropyl alcohol-D	100	1.6	A	261
Isopropyl-1-D alcohol-D	100	0.78	A	261
Lead, tetraethyl	100	1.24		156
Malonic acid, diallyl ester	60	5.2		446
–, diethyl ester	60	0.47	A	128
	100	0.46		72
–, dimethyl ester	100	0.42		72
–, acetyl-, diethyl ester	100	1.2		72
–, bromo-, diethyl ester	60	700	A	128
	100	1200	A	128
–, butyl-, diethyl ester	100	0.82		72
–, dibromo-, diethyl ester	60	1.2×10^4	A	128
–, dichloro-, diethyl ester	60	30.0	A	128
	100	62.0	A	128
–, diethyl-, diethyl ester	100	0.88		72
–, ethyl-, diethyl ester	100	0.72		72

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TABLE 4. *cont'd*

Solvent/Addition	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Malonic acid, phenyl-, diethyl ester	100	3.5		72
Mercury, diethyl	100	0.335		156
Methane, bis(2-chloroethoxy)-	80	6.0	C	112
–, bromotrichloro-	80	7.6×10^4	D, Q2	312
		7.7×10^4	D, Q2	195
		2.4×10^6	D, Q3	312
		2.78×10^6	D, Q3	195
	60–80	6.5×10^5		439
		$(6.5 \pm 0.4) \times 10^5$		504
	80	9450	I5	185
–, dibromo-	60	110		112
–, dichloro-	60	0.15	A	128
	80	9.5	C	112
	100	11.8	A	128
–, diiodo-	60	710		112
–, diphenyl-	60	2.3	A	127
	100	4.2	A	127
		35	A, R	524
–, nitro-	60	10		86
–, triphenyl-	60	3.5	A	127
	80	6.0		44
	100	8.0	A	127
Methanetricarboxylic acid	100	0.91		72
Methanol	60	0.296		479
		0.74	F2	423
	80	1.10	F2	423
	100	1.22	F2	423
–, (4-biphenylyl)phenyl(*p*-vinyl-phenyl)-	50	3.5		57
–, bis(4-biphenylyl)(*p*-vinyl-phenyl)-	50	3.5		57
Naphthalene	60	11		216
–, decahydro-	60	0.4	A	127
–, 2-isopropenyl-	80	56		98
		70		98
	99	69		98
		86		98
–, 2-methoxy-	60	<5.0		121
–, 1,2,3,9-tetahydro-1-phenyl-	60	10000	C	467
1-Naphthol	60	480	J	50,121
–, 2,4-dichloro-	60	490	J	121
1-Naphthol-D	60	75		50,121
2-Naphthol	60	77		122
Naphtho[2,3-a]pyrene	44.4	24.0×10^5	J	159
2,6-Octadiene, 2,6-dimethyl-	60	2.0	C	324
2-Octene	100	2.8	A	358
Oleic acid, methyl ester	60	3.15	C	441
		3.52		441
		455		239
	70	420	C	141
	90	6.64	A	441
Oxalic acid, diallyl ester	60	4.2		446
–, diethyl ester	60	13.5	F22	119
Oxetane, 3,3-bis(chloromethyl)-	50	8.0	C	449
Oxime, acrolein	60	10800		555
–, crotonaldehyde-	60	1500		555
–, ethylisopropenylketone-	60	4300		555
–, methacrolein-	60	13000		555
–, methylacrolein-	60	400		555
–, methylisobutylketone-	60	2300		555
–,methylisopropenylketone-	60	1100		555
–, methylvinylketone-	60	2700		555
Oxolane, *cis*-3,4-diacetoxy-	60	1.89	C	569
Pentane, 1-chloro-	60	0.49	A	128
–, 2,2,4-trimethyl-	100	<10	A	155
2,4-Pentanedione	100	2.0		72

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
1-Pentanol, 2,2,3,3,4,4,5,5-octa-fluoro-	60	11.36	F2	442
3-Pentanone	60	2.6	A	385
Pentasiloxane, dodecamethyl-	79.5	0.285		474
1-Pentene	100	2.3	A	358
–, 4,4-dimethyl-2-phenyl-	110	10	A	433
–, 2,4-diphenyl-	110	170	A	433
–, 4-methyl-2,4-diphenyl-	110	2900	A	433
–, 4-methyl-2-phenyl-	110	10	A	433
–, 2,4,4-triphenyl-	110	2600	A	433
2-Pentene	100	4.2	A	358
–, 2-methyl-	100	6.3	A	358
–, 4-methyl-	100	6.9	A	358
Phenol	50	8.1		247,248
	60	14		122
–, *p*-benzyloxy	60	290	J	50,121
–, *p*-*tert*-butyl-	60	26		121
–, *o*-chloro-	60	6.0		122
–, *p*-chloro-	60	−11		121
–, 2,6-di-*tert*-butyl-	60	49		121
–, 2,6-diisopropyl-	60	310	J	121
–, *p*-fluoro-	60	54		121
–, *m*-methoxy-	60	<5		121
–, *o*-methoxy-	60	43		122
–, *p*-methoxy-	60	260	J	121
–, *o*-phenyl-	60	<5		121
–, 2,3,4,6-tetramethyl-	60	580	J	50,121
–, 2,4,6-trinitro-	50	2.11×10^5		371
Phenol-D, *p*-benzyloxy-	60	10		50
–, 2,3,4,6-tetramethyl-	60	20		50
Phenyl ether	25	7.94	D	368
	60	7.86		368
Phosphine, dibutyl	100	2.08×10^4		296
–, diethyl	100	1.35×10^4		296
–, octyl	60	3.6×10^4	A	295
Phosphine, phenyl	60	43.9×10^4	A, S	295
–, tributyl	100	24.4		156
Phosphoric acid, tributyl ester	100	<0.1		156
Phosphorus, white (P4)	25	400	E	99
Phosphorus trichloride	57	250	C, D	214
	75	800		494
o-Phthalic acid				
–, bis(2-methylallyl) ester	60	6.3		446
–, diallyl ester	60	6.3		446
–, di-(3,3,5,5-tetramethyl-4-nitrosocyclohexyl) ester	80	1.2×10^4	F9, J	314
		1.32×10^4	F27, J	314
Piperidine	60	1.0	A	128
Propane, 1-chloro-2,3-epoxy-	50	7.5	C	449
–, 1-chloro-2-methyl	60	1.4	A	128
	100	3.0	A	128
–, 1,1,1,3-tetrabromo-3-phenyl-	90	3.65×10^4	C	64
–, 1,1,1-tribromo-	90	2.41×10^4	C	64
1,2-Propanediol	60	2.08	F24	422
	80	3.90	F24	422
	100	6.80	F24	422
1,3-Propanediol, 2,2-bis(bromomethyl)-, diacetate	60	4.05		325
1,3-Propanedione, 1,3-diphenyl-	60	7.0	A	128
1-Propene, 3-chloro-2-methyl-	60	24.0	A	128
–, 2-methyl-	100	1.7	A	358
–, 1,3-diphenyl-	60	87.3		567
Propionaldehyde, diallyl acetal	60	12.3		446
Propionic acid	60	0.05	A	128
		4.3	C	385
		4.5		323
		4.65	A	385

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Propionitrile, 3-phosphino-	60	50000	A	295
–, 3,3′-phophinylidenedi-	60	50000	A	295
Propyl alcohol	60	2.00	F2	423
	80	3.14	F2	423
	100	3.60	F2	423
Propylene oxide	50	1.6	C	449
2-Propyn-1-ol	60	7.0	A	128
Pyridine	60	0.6	A	128
Pyrocatechol	60	1340	J	50,121
–, *p-tert*-butyl-	60	3600	J	50,121
Pyrocatechol-D	60	260		50
–, *p-tert*-butyl-	60	370		50
Pyrogallol	60	10400	J	50,121
Pyrogallol-D	60	1600		50
Sebacic acid, diallyl ester	60	4.8		446
Silane, chlorotrimethyl-	50	12.5		249
–, dichlorodimethyl-	50	17.8		249
–, diethylchloro-	60	5700	A, R	539
–, dimethylphenyl-	60	2200	A, R	539
–, hexamethoxy-(3,3′-thiodipropyl) bis-	79.5	118	R	554
–, methyldichloro-	60	9800	A, R	539
–, 3-mercaptopropyl-trimethoxy-	79.5	5.9×10^4	R	554
–, trichloro-	60	1.4×10^4	A, R	539
–, tetraethyl-	100	8.12		156
–, tetramethyl-	50	3.1		249
–, trichloromethyl-	50	19.2		249
–, triethyl-	60	1200	A, R	539
	70	33.4		90
	80	36.8		90
–, trimethoxymethyl-	79.5	0.23		474
		0.230	R	554
–, triphenyl-	60	3700	A, R	539
	70	2.44		90
	80	1.3		90
Silicon chloride	50	20.0		249
Siloxane, hexamethyl-di-	79.5	0.387	R	554
–, dodecamethyl-penta-	79.5	0.285	R	554
–, octamethyltri-	79.5	0.069	R	554
Sorbit-2,5-diacetate, 1,4:3,6-, dianhydro-	60	77	C	569
Stannane, triphenyl-	60	3.3×10^4	A, R	539
Stearamide, *N*-allyl-	90	5.82	A, J	173
		8.30	C, J	173
Stearic acid, methyl ester	60	1.06		441
		526		239
	70	15.6	C	141
	90	0.676	A	441
Stibine, tributyl	100	58.0		156
Stibene, dibromo-	60	3020	A	112
Styrene, α-bromo-	70	10000		172
–, β-bromo-	70	2000		172
–, dibromo-	60	1950		112
–, α-ethyl-	110	10		439
–, α-methyl-	60	0.86		431
	74	95		343
	80	3.2		98
		4.9		98
	99	5.6		98
		8.5		98
	110	1.67		431
Succinic acid, diallyl ester	60	5.4		446
Succinonitrile, tetraphenyl-	50	28000		89
Sulfonyl chloride, benzene-	60	4330	R	543
–, *p*-chloro-benzene-	60	7650	R	543
–, *p*-methyl-benzene-	60	3180	R	543

TABLE 4. *cont'd*

Solvent/Additive	T (°)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Sulfonyl chloride, *p*-methoxy-benzene-	60	3110	R	543
–, methane-	60	1180	R	543
–, phenylmethane-	60	3190	R	543
–, α-phenyl-β-(methanesulfonyl)-ethane-	60	27600	R	543
Terephthalamide, *N,N'*-dimethyl-*N,N'*-dinitroso-	40	1400		160
	60	2000		160
Terephthalic acid, diallyl ester	60	4.5		446
Tin, tetrabutyl	100	3.71		156
Toluene	60	0.105	C	255
		0.121		21,24,79
		0.125	A	127
		0.134		78
		0.16	A, I5	145
		0.21		318
		0.82		496
		1.10	C	363
		2.05		363
	80	0.15	C	385
		0.298	I5	185
		0.3	C	385
		0.308	C	255
		0.310		76,77
		0.313		78,273
		0.813	C, F17	530
	100	0.53	A	235
		0.55	C	385
		0.645	A	127
		0.72	A	235
		0.8	A, I5	145
	132	1.12	A	235
–, *p*-bromo-	60	1.30		496
–, *m*-chloro-	60	1.25		496
	70	0.62		67
	140	1.8		67
–, *p*-chloro-	60	1.07		496
–, α-chloro-	60	1.56	A	128
–, α,α-dichloro-	60	50.0	A	128
–, *p*-ethynyl-	60	72.0	K	151
–, α-$^2H_1$, toluene	80	0.825	C, F17	530
–, ββ-$^2H_2$, toluene	80	0.820	C, F17	530
–, αββ-$^2H_3$, toluene	80	0.992	C, F17	530
–, ααα-trichloro-	60	57.5	A	128
–, 2,4,6-trinitro-	50	14.6×10^4		370,372
p-Toluidine	50	78	F2	154
–, *N,N*-dimethyl-	50	16	F2	154
–, *N*-methyl-	50	11	F2	154
m-Tolunitrile, α-cyano-	60	91.4		462
p-Tolunitrile	60	2.07		496
–, α-cyano-	60	112		461
s-Triazine, trimethyl-	60	0.468		88
Triethylamine	60	1.4	W9	364
		3.0	W6	364
		7.1		24
		7.5		363,364
Tripropylamine	100	24.2		156
Trisiloxane, octamethyl-	79.5	0.069		474
Urea, 1,3-bis(3,3,5,5-tetramethyl-4-nitrosocyclohexyl)-	60	14500	F9, J	314
	80	15000	F9, J	314
Water	60	0.006		479
		0.31	A, R	529
m-Xylene	60	0.78		496
p-Xylene	60	0.84		496
–, α,α'-dibromo-	60	150		112
2,6-Xylenol	60	110		121

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Styrene (*cont'd*)				
Zinc, diethyl	100	3660		156
–, *p*-chloro-				
Anisole, *p*-isopropyl-	60	3.86		406
Benzaldehyde	100	1.9	A	298
–, *p*-chloro-	100	9.7	A	298
Benzene, *p*-diisopropyl-	60	3.62		497
		7.24		406
Benzonitrile, *p*-isopropyl-	60	8.84		406
Carbon tetrabromide	60	52000	F2	109
Carbon tetrachloride	60	45		366
Cumene	60	3.44		406
–, *p*-bromo-	60	5.71		406
–, *p*-*tert*-butyl-	60	3.52		406
–, *p*-chloro-	60	4.97		406
–, *p*-iodo-				
Benzene	50	0.2		56
–, *p*-methyl				
Anisole, *p*-isopropyl-	60	3.27		408
Benzene, *p*-diisopropyl-	60	3.67		408
		7.34		406
Benzonitrile, *p*-isopropyl-	60	26.0		408
Cumene	60	4.12		408
–, *p*-bromo-	60	9.23		408
–, *p*-*tert*-butyl-	60	3.59		408
–, *p*-chloro-	60	7.67		408
–, pentafluoro-				
Benzene, bromo-	60	0		469
–, chloro-	60	0		469
–, ethyl-	60	5.35		469
–, fluoro-	60	0.117		469
2-Butanone, 3-methyl-	60	1.09		469
Cumene	60	6.76		469
Furan, tetrahydro-	60	1.53	A	469
		1.61		469
2-Pentanone, 4-methyl-	60	0.033		469
Toluene	60	1.20		469
Styrene/Styrene, *p*-chloro-				
Carbon tetrachloride	60	15	EF	366
		115	EE	366
Succinimide, *N*-vinyl-				
Acetic acid	55	0.077	C, F25	93
Ethane, 1,2-dichloro-	55	0.127	C, F26	93
Valeric acid, 4-methyl-, vinyl ester				
Benzene	80	6.2		201
Vinyl acetate				
Acetaldehyde	30	400		346
	45	530	J	86
	50	390		460
	60	0.72		257
		200	I8	145
		220		232
		500		460
		570	F2	86
		660	J	86
	70	610		460
	75	700	J	86
Acetamide, *N*-butyl-	60	40		86
Acetic acid	50	0.180		466
	60	0.166	C	465
		0.200		466
		1.0	I9	145
		1.13		289
		10	F2	86
	65	0.170	C	466

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Acetic acid, allyl ester	60	85	F2	86
		94	J	86
–, benzyl ester	60	80	F2, J	86
–, butyl ester	50	13.2		208
–, *sec*-butyl ester	50	4.4		208
	60	8.0		86
–, *tert*-butyl ester	50	1.5		208
	60	6.2		232
–, isobutyl ester	50	9.1		208
–, isopropyl ester	50	3.5		208
	60	3.1		145,232
		8.0		86
	67.5	9.0	J	86
	75	10	J	86
–, methyl ester	60	1.56		145,231
		1.6		145,232
		2.5		86
–, propyl ester	50	6.2		208
	60	3.4		145,232
–, 1,3,3,3-tetrachloropropyl ester	70	423.2	C	389
–, bromo-	60	489	C, J	386
	70	4450	C	389
–, chloro-	70	2550	C	389
–, cyano-, methyl ester	60	5000	J	86
–, dichloro-, ethyl ester	60	210		86
–, phenyl-	60	400	J	86
–, trichloro-	70	1445	C	389
–, ethyl ester	60	4400		86
–, trifluoro-, ethyl ester	50	30	C	71
Acetic anhydride	60	8.0		86
Acetoacetic acid, ethyl ester	70	80.4	C	390
Acetone	60	1.5		86
		11.70		289
		12.0		145
	70	25.6	C	330
	75	42	C	184
Acetonitrile	60	10		86
–, phenyl-	60	2100	J	86
Acetophenone	50	91.5		15
	60	100	J	86
	70	62.0	C	390
–, 3′-hydroxy-	45	1405	J	51
–, 3′-hydroxy-D-	45	1290	J	51
Aniline	50	149		15
		210	F2	154
–, *N,N*-dimethyl-	50	260	F2	154
–, *N*-methyl-	50	360	F2	154
–, *p*-nitro-	50	48600		15
m-Anisaldehyde	60	2500	J	86
o-Anisaldehyde	60	420	J	86
p-Anisaldehyde	60	370	J	86
Anisole	60	10	F2, J	86
Anthracene	20	6.68×10^5	J	41
	30	6.03×10^5	J	41
	40	4.55×10^5	J	41
	50	2.057×10^5		15
		3.64×10^5	J	41
	60	2.78×10^5	J	41
–, 9-phenyl-	50	2.73×10^5	J	41
Benzaldehyde	60	230	J	145,355
		460	F2	86
		540	J	86
	70	421	C, J	390
	75	600	J	86

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Benzaldehyde, *m*-chloro-	60	860	J	86
–, *o*-chloro-	60	390	J	86
–, *p*-chloro-	60	340	J	86
–, *m*-cyano-	60	1070	J	86
–, *p*-cyano-	60	610	J	86
–, *p*-isopropyl-	60	540	J	86
Benz[A]anthracene	50	1.17×10^5	J	41
Benzene	60	1.07		263
		1.2	J	86
		1.2		294
		2.4	C	80
		2.96		289
		20		294
	70	5.27	C	390
	75	1.4	J	86
		1.40		263
		3.6	C	184
–, bromo-	50	18.9		15
	60	8.0	W7	265
		9.2	W15	265
		10.0		265
	70	134.2	C	389
	75	25.2	C	184
–, *tert*-butyl-	60	3.61		289
–, chloro-	60	5.6	W7	265
		6.8	W15	265
		8.0		265
		8.35		289
		80	J	86
	70	2.61	C	389
	75	12.7	C	184
Benzene, *o*-dichloro-	75	42	C	184
–, *p*-dichloro-	70	11.8	C	389
–, *m*-dinitro-	45	1.05×10^6	C, J	28
	50	645700		15
–, *o*-dinitro-	45	9.6×10^5	C, J	28
–, *p*-dinitro-	45	26.7×10^5	C, J	28
	50	662800		15
–, ethyl-	60	55.15		289
		100	J	86
–, nitro-	50	110300		15
–, 1,3,5-trinitro-	45	8.9×10^6	C, J	28
	50	434280		15
Benzo[B]chrysene	50	2.83×10^5	J	41
Benzoic acid	60	50	J	86
–, ethyl ester	60	26	J	86
–, *p*-nitro-	50	245700		15
Benzoic anhydride	60	130	J	86
Benzoin	60	800	J	86
Benzonitrile	50	40.6		15
–, *m*-hydroxy-	45	820	J	51
–, *m*-hydroxy-D-	45	805	J	51
Benzo[GHI]perylene	50	1.82×10^4	J	370
Benzophenone	50	286		15
Benzo[A]pyrene	50	3.06×10^5	J	41
Benzo[E]pyrene	50	8400	J	370
p-Benzoquinone, 2,3,5,6-tetramethyl-	45	9.5×10^5	J	27
Benzoyl chloride	50	366		15
	70	300	C	389
Benzyl alcohol	70	556	C	390
Biphenyl	50	263		15
	60	6.4		132
Borane, tributyl	10	9000		191

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Butane, 1-bromo-	60	50		86
	70	1100	C	389
–, 1-chloro-	60	10		86
–, 1-iodo-	60	800		86
2,3-Butanedione	60	670	F2, J	86
2-Butanone	60	73.80		289
	75	165	C	184
–, 3-methyl-	60	118.16		289
1-Butene, 3-chloro-3-methyl-	50	28900	F2	152
2-Butene, 1-chloro-3-methyl-	50	4300	F2	152
1-Buten-3-yne	60	0	J	232
Butyl alcohol	60	20.0	I9	145
		20.39		289
	70	29.1	C	390
sec-Butyl alcohol	60	31.74		289
	70	6.21	C	390
	75	95.0	C	184
tert-Butyl alcohol	60	0.46		289
		0.5	I9	145
		1.3		86
	70	12.1	C, J	390
Butyl ether	60	76		86
3-Butyn-2-ol, 2-methyl-	60	400	J	86
Butyraldehyde	40	440	W8	500
		510	W7	500
		590	W15	500
		670		500
	60	650		232,355
		1000		86
	70	388	C	390
Butyric acid, ethyl ester	50	45	C	71
–, methyl ester	60	18		145,232
		19		86
Carbon tetrabromide	60	7.39×10^6	F2	109,145
	70	28740	C	389
Carbon tetrachloride	0	1500		346
	20	4700	E	309
	45	7600	J	86
	60	800	Q1	10
		1300	Q2	10
		4700	Q3	10
		6700	Q4	10
		7300	F2	86
		8000	Q5	10
		9000	Q6	10
		9600	J	86
		10000		210
		10500	Q7	10
		10700	Q8	10
	70	2023	C	389
	75	10500	J	86
Chloral	60	5000		86
	70	4927	C	389
Chloral hydrate	70	4312	C	389
Chloroform	–	100		345
	30	140		346
	60	125.18		289
		130	I9	145
		150	J	86
		160		210
		170	F2	86
	70	554	C	389
2-Chloro-3-(methyl-^{14}C)-1butene(4-^{14}C)	50	3980	F2	509
3-Chloro-2-pentene-(2,4-^{14}C)	50	628	F2	509

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Chrysene	50	3360	J	370
m-Cresol	45	375	J	51
m-Cresol-D	45	85	J	61
p-Cresol	45	710	J	51
p-Cresol-D	45	130	J	51
Crotonaldehyde	60	1800	J	355
Cumene	60	89.9		289
		100		7,86
	70	139	C	390
	75	356	C	184
Cyclohexane	60	6.59		289
		7.0	I9	145
		100	F2	86
–, methyl-	60	11.75		289
		24		86
1,3-Cyclohexanedione, 5,5-dimethyl-	70	5580	C	390
Cyclohexanol	70	127	C	390
Cyclohexanone	60	180	F2	86
	75	670	C	184
Cyclohexene	60	620	J	86
		1600	F2	86
	75	770	J	86
Dibenz[a,h]anthracene	50	8.7×10^4	J	370
Dibenzo[def,mno]chrysene	40	156.5×10^4	J	41
	50	129.2×10^4	J	41
	60	105.4×10^4	J	41
1,2,3,4-Dibenzpyrene	50	73.6×10^4	J	41
Diethylene glycol	70	85.3	C	390
Diethylphosphonate	60	350	F1	571
	60	910	F2	571
	60	830	F2	571
	60	730	F2	571
p-Dioxane	60	20	F2	86
	70	49.1	C	390
Diphenylamine	50	138		15
	60	240		46
Diphenylamine-D	60	170		46
Diphenylamine-T	60	230		46
Ethane, 1,1-dibromo-	60	1100		86
–, 1,2-dibromo-	70	134	C	389
–, 1,1-dichloro-	60	65		86
–, 1,2-dichloro-	60	5		86
		7	I9	145
		7.18		289
–, 1,2-dichloro-	70	10.2	C	389
–, hexachloro-	70	1210	C	389
–, pentachloro-	70	1348	C	389
–, 1,1,2,2-tetrabromo-	60	6000		86
–, 1,1,2,2-tetrachloro-	60	107.03		289
		160		86
	70	67.72	C	389
–, 1,1,1-trichloro-	60	71.11		289
–, 1,1,2-trichloro-	60	35.98		289
1,1-Ethanediol, diacetate	60	40		86
Ether, benzyl methyl	60	280	F2, J	86
–, bis(2-chloroethyl)	70	245	C	389
–, dodecyl vinyl	60	57.2	C	4
		73.5	C	4
–, ethyl	60	45.3	C	4
Ethyl acetate	20	1.52		328
	40	2.11		328
	50	2.9		208
		12	C	71
	60	1.07		289

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Ethyl acetate (*cont'd*)		1.25	C	80
		2.6		145,232
		2.96		328
		3.3		86
		3.4		420
	70	7.8	C	390
Ethyl alcohol	60	25	F2	86
	70	26.3	C	390
Ethylene, tetrabromo-	70	2800	C	389
–, tetrachloro-	70	465	C	389
–, tribromo-	70	34720	C	389
–, trichloro-	70	3810	C	389
Ethylene glycol	70	83.0	C, J	390
Fluorene	60	4700	J	86
	70	3610	C, J	390
–, 9-phenyl-	70	16240	C	390
Formamide, *N,N,*-dimethyl-	60	50		86
Formic acid, ethyl ester	50	22	C	71
–, methyl ester	60	3		86
2-Furaldehyde	60	15000	J	86
Furfuryl alcohol	70	2520	C, J	390
Germane, diethylchloro-	60	7800	A, R	539
–, dimethylchloro-	60	86600	A, R	539
–, triethyl-	60	4500	A, R	539
–,triphenyl-	60	56000	A, R	539
Glycolic acid, methyl ester	60	300		86
Heptane	50	17.0		208
1-Heptanol, 2,2,3,3,4,4,5,5,6,6,7,7-dodecafluoro-	60	33.3	F2	442
1,5-Hexadien-3-yne	60	0		232
Hexanoic acid, 2-ethyl-, ethyl ester	50	65	C	71
Hydroquinone	50	7000	J	338
–, 2,5-di-*tert*-butyl-	50	38000	J	338
–, 2,6-dimethyl-	50	29000	J	338
–, tetramethyl-	50	74000	J	338
Isobutyl alcohol	60	21.75		289
	70	32.4	C	390
Isobutyric acid	60	5.02		289
Isobutyric acid, ethyl ester	50	160	C	71
–, methyl ester	60	58		145,232
		71		420
		86		86
Isobutyronitrile	60	100		86
Isopropyl alcohol	70	44.6	C	390
Lactic acid, ethyl ester	60	700		86
–, methyl ester	60	640		86
Lauric acid, ethyl ester	50	105	C	71
Malonic acid, dimethyl ester	60	17		86
p-Mentha-1,8-dene	60	1900	J	86
Mesitol	50	5000	J	338
Methane, bromotrichloro-	25	$\gg 10^4$	D, S	39
	60	6000	C	39
		$>4 \times 10^5$	C	39
	70	6303	C	389
–, dichloro-	60	4		86
–, iodo-	70	1230	C	389
–, nitro-	60	2300	F2, J	86
	75	2600	J	86
–, tribromo-	70	34760	C	389
–, triphenyl-	60	700		45
Methanol	–	9.0		5
		10.0		5
		10.5		5
	10	3.4		190
	60	2.26		145,231

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TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Methanol (*cont'd*)		4.3		320
		6.0		86
	70	5.5		320
Naphthacene	50	8.62×10^6	J	370
Naphthalene	50	1150	J	370
		1457		15
	70	1715	C, J	390
–, decahydro-	60	48		86
Naphtho[1,2,3,4-def]chrysene	50	1.35×10^5	J	41
Nonanoic acid, ethyl ester	50	80	C	71
2,6-Octadiene, 2,7-dichloro-	50	3200	F2	152
–, 2,6-dimethyl-	60	430		324
		700		7
Octanoic acid, ethyl ester	50	70	C	71
Oleic acid, methyl ester	60	217		441
		1000	J	86
	90	358		441
Orthoformic acid, trimethyl ester	–	7.6		345
	30	8.0		346
Oxalic acid, diethyl ester	60	4.0		86
–, dimethyl ester	60	1.0		86
		2.0		145,232
Paraldehyde	70	136	C	390
Pentane, 2,2,4-trimethyl-	50	8.0		208
2,4-Pentanedione	60	10		86
1-Pentanol, *n* 2,2,3,3,4,4,5,5-octafluoro-	60	26.1	F2	442
2-Pentanone, 4-methyl-	60	34.52		289
3-Pentanone	60	10.0	I9	145
		114.39		289
Pentyl acetate	70	7.2	C	390
	75	87.0	C	184
Pentyl alcohol	75	56	C	184
Perylene	50	2.88×10^5	J	41
Phenanthrene	50	870	J	370
		5600		15
	70	3380	C, J	390
Phenol	45	215	J	51
	50	120		15
	60	600	J	86
–, *m*-chloro-	45	205	J	51
–, *p*-chloro-	45	400	J	51
–, *p*-nitro-	50	8.86×10^4		15
–, 2,3,4,6-tetramethyl-	45	1.13×10^4	J	51
Phenol-D	45	35	J	51
–, *m*-chloro-	45	70	J	51
–, *p*-chloro-	45	80	J	51
–, 2,3,4,6-tetramethyl-	45	800	J	51
Phosphorus trichloride	60	1.5×10^4	C, D	214
Propane, 2-bromo-2-methyl-	60	150		86
–, 2-chloro-2-methyl	60	26		86
1-Propanol, 2,2,3,3-tetrafluoro-	60	7.11	F2	442
1-Propene, 3-chloro-	60	3100		86
–, 3-chloro-2-methyl-	60	400	J	86
Propionaldehyde	60	950	J	86
		1000	F2	86
	70	457	C	390
Propionic acid, ethyl ester	50	40	C	71
–, methyl ester	60	23		232
		26		420
Pyrene	50	11500	J	370
Pyrocatechol, 4-(1,1,3,3,-tetramethylbutyl)-	50	11000	J	338
Pyrogallol	50	50000	J	338
Salicylic acid	70	296	C	390
Silane, dimethylphenyl-	60	2200	A, R	539

TABLE 4. *cont'd*

Solvent/Additive	T (°C)	C_S ($\times 10^4$)	Remarks	Refs.
Vinyl acetate (*cont'd*)				
Silane, methyldichloro-	60	5800	A, R	539
–, trichloro-	60	8800	A, R	539
–, triethyl-	60	1200	A, R	539
–, triphenyl-	60	2700	A, R	539
Stearamide, *N*-allyl-	90	620.1	C, J	173
Stearic acid, ehtyl ester	50	140	C	71
–, methyl ester	60	38.2		441
Succinic acid, diethyl ester	60	41		420
–, dihexyl ester	60	40		420
–, dimethyl ester	60	31		420
m-Tolualdehyde	60	570	J	86
p-Tolualdehyde	60	440	J	86
Toluene		3.3		158
	50	12.0		15
		14.9	R	357
		100	C	71
		123		71
	60	17.8	R	357
		20.75	C	80
		20.89		21,59,289
		21.6		357
		34	J	86
	60	35	F2	86
		49.0	W15	265
		69.0		265
	70	21.1	R	357
		21.8	C	390
		29.2		357
	75	66	C	184
	80	91.6		273
–, bromo-	50	20.6		15
–, *o*-chloro-	70	92.86	C	389
–, *p*-chloro-	70	195	C	389
–, α-chloro-	60	450	J	86
	70	584	C	389
	75	118	C	184
Toluene, *p*-nitro-	50	106300		15
p-Toluidine	50	750	F2	154
–, *N,N*-dimethyl-	50	380	F2	154
–, *N*-methyl-	50	830	F2	154
Toluonitrile	50	38.3		15
Triethylamine	50	360	F2	154
	60	370	J	24,145
Triphenylene	50	160	J	370
Veratraldeyhyde	60	550	J	86
Xylene	50	14.9		15
	70	278	C	390
	75	166	C	184
Vinyl chloride				
Carbon tetrachloride	60	280	Q(3)	511
Chloroform	60	290	Q(3)	511
Bromotrichloromethane	60	400000	Q(3)	511
Vinylidene chloride				
N-Methylpyrolidone	50	20	R	541
	60	20	R	541
2-Vinylpyridine				
Silane, diethylchloro-	60	5400	A, R	539
–, Methyldichloro-	60	8600	A, R	539
–, trichloro-	60	12000	A, R	539
–, triethyl-	60	2000	A, R	539
–, triphenyl-	60	4000	A, R	539
Vinyltrimethylsilane				
Bromoform		1200000	C, Q(1–2)	572

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TABLE 5. TRANSFER CONSTANTS TO SULFUR COMPOUNDS

Modifier	T (°C)	C_X	Remarks	Refs.
Acrylamide, *N*-octadecyl-/acrylonitrile (1:1 mol ratio)				
1-Dodecanethiol	60	0.539	C, F24	174
Acrylic acid				
1-Dodecanethiol	45	1.9	C, F31	549
–, methyl ester				
1-Butanethiol	30	1.53	A	397
	60	1.69	A	69,397
Ethanethiol	50	0.78	Q1	330
		0.94 ± 0.07	Q1	478,507
		1.65 ± 0.12	Q2	478,507
		1.57 ± 0.09	Q3	508
		1.57 ± 0.18	Q4	330,509
		1.79	Q2	330
		5.00	Q3	330
2-Propanethiol	60	0.544	Q1	477
		0.668	Q2	477
		0.700	Q3	477
		0.656	Q4	477
Acrylonitrile				
1-Dodecanethiol	50	0.73		378
Hydrogen sulfide	50	0.13		378
Methyl sulfoxide	20	0.1	C, F8	248
	30	0.47		196
	40	0.0000812	E	458
	50	0.11		404
		0.29		197
		0.795		375
Methyl tetrasulfide	80	0.69	J	85
p-Toluenethiol	50	0.97		378
Acrylonitrile/1,3-butadiene (10–90%)				
1-Hexanethiol,				
–, 1,1,3,3,5,5-hexamethyl-	5	1.5	C, H	380
1-Pentanethiol,				
–, 1,1,3,3-tetramethyl-		53.3	C, H	380
Acrylonitrile/1,3-butadiene (20–80%)				
1-Hexanethiol				
–, 1,1,3,3,5,5-hexamethyl-	5	1.2	C, H	380
1-Pentanethiol				
–, 1,1,3,3-tetramethyl-	5	2.1	C, H	380
Acrylonitrile/1,3-butadiene (30–70%)				
1-Hexanethiol				
–, 1,1,3,3,5,5-hexamethyl-	5	1.1	C, H	380
1-Pentanethiol				
–, 1,1,3,3-tetramethyl-	5	1.8	C, H	380
Acrylonitrile/styrene (38.5–61.5 mol%)				
1-Pentanethiol				
–, 1,1,3,3,4-pentamethyl-	50	1.02	C, H	380
Butadiene				
Xanthogendisulfide, dibutyl-	60	6.9	J	532
–, diethyl-	60	8.1	J	532
–, di-2-ethylhexyl-	60	7.5	J	532
–, di-isopropyl-	60	8.3	J	532
1,3-Butadiene				
1-Butanethiol, 1,1,3,3-tetramethyl-	5	5.3	H	242
	50	3.7	H	242
1-Octanethiol	5	21.8	H	242
	50	16.0	H	242
		19.0	H	243
–, 1,1,3,3,5,5,7,7-octamethyl-	50	3.0	H	242
1-Tetradecanethiol	50	19.5	H	242
1,3-Butadiene/styrene				
1-Dodecanethiol	5	0.66	B, H, J	384
Formic acid, thio-, dithiobis-				
–, –, –, *O,O'*-dibutyl ester	5	2.42	B, H, J	383

TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
1,3-Butadiene/styrene (*cont'd*)	−5	1.75	H	180
	50	5.34	C, H, J	382
−, −, −, *O,O'*-di-*sec*-butyl ester	5	1.65	B, H, J	
	−5	1.565	H	180
−, −, − *O,O'*-dicyclohexyl ester	5	1.78	B, H, J	383
	50	3.71	C, H, J	383
−, −, −, *O,O'*-diethyl ester	5	8.43	B, H, J	383
	−5	4.23	H	180
	50	16.04	C, H, J	382
−, −, −, *O,O'*-diheptyl ester	5	0.41	B, H, J	383
	50	1.12	C, H, J	382
−, −, −, *O,O'*-dihexyl ester	5	0.74	B, H, J	383
	50	1.78	C, H, J	382
−, −, −, *O,O'*-diisobutyl ester	5	1.87	B, H, J	383
	−5	1.573	H	180
	50	6.43	C, H, J	382
−, −, *O,O'*-diisopentyl ester	5	1.21	B, H, J	383
		7.0	B, H, J	341
	−5	1.85	H	180
	50	4.38	C, H, J	382
−, −, −, *O,O'*-diisopropyl ester	5	2.83	B, H, J	383
		3.4	B, H, J	341
		12.5	B, H	100
		0.40–1.31	B, H, J, GG4	100
		0.80–3.01	B, H, J, GG3	100
		1.40–3.70	B, H, J, GG2	100
		1.80–3.50	B, H, J, GG1	100
		1.80–6.0	B, H, J, GG5	100
	–	9.78		179
	−5	1.185	H	180
	50	9.78	C, H, J	382
−, −, −, *O,O'*-dimethyl ester	5	16.43	B, H, J	383
	−5	15.86	H	180
	50	27.04	C, H, J	382
−, −, −, *O,O'*-dineopentyl ester	5	1.01	B, H, J	383
	50	3.40	C, H, J	382
−, −, −, *O,O'*-dioctyl ester	5	0.23	B, H, J	383
	50	0.64	C, H, J	382
−, −, −, *O,O'*-dipentyl ester	5	1.45	B, H, J	383
	−5	1.70	H	180
	50	3.20	C, H, J	382
−, −, −, *O,O'*-dipropyl ester	5	4.42	B, H, J	383
	−5	1.815	H	180
	50	9.20	C, H, J	382
1-Heptanethiol				
−, 1,1,3,3,5,5,6-heptamethyl-	5	0.64	B, H, J	384
1-Hexanethiol				
−, 1,1,3,3,5,5-hexamethyl-	5	1.22	B, H, J	384
		4.1	H	54,55
		2.7–4.7	H	54
1-Octanethiol				
−, 1,1,3,5,5,7,7-octamethyl-	5	0.34	B, H, J	384
p-Dioxin, 2,3-dihydro/maleic anhydride				
1-Butanethil	60	6300	R	169
Ethylene				
1-Butanethiol	130	5.8	C, W5	457
Ethyl sulfide	130	0.027	C, W5	454
Methyl sulfoxide	130	12	C, F21, W5	259
2-Propanethiol, 2-methyl-	130	15.0	C, W5	457
Sulfur hexafluoride	130	0.0000	C, F21, W5	259
Methyl methacrylate				
Acetic acid, dithiodi-, diethyl ester	60	0.00065		86
−, mercapto-, methyl ester	60	0.63		274
Acetophenone, 3′-mercapto-	45	4.2		30
−, 4′-mercapto-	45	2.6		30

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TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
Anisole, 4,4′-dithiodi-	50	0.0044		285
	60	0.0052		285
p-Anisoyl disulfide	60	14.6		369
Benzenesulfonic acid				
–, compound with pyridine	60	0.00365		490
–, thio-, *S*-phenyl ester	60	0.0196	A	491
Benzenethiol	45	4.7		30
	60	2.7		274
	70	15		566
–, *m*-bromo-	45	3.8		30
–, *p*-bromo-	45	5.6		30
–, *m*-chloro-	45	3.5		30
–, *p*-chloro-	45	4.6		30
Benzoyl disulfide	60	10.0		369
Benzyl disulfide	25	0.134		368
	50	0.016	F2	354
	60	0.00627		368
Benzyl sulfide	25	0.0183	D	368
	50	0.0098		252
		0.0098	F2	354
	60	0.00154		368
Benzyl sulfone	50	0.0040		252
Benzyl sulfoxide	50	0.0039		252
Benzyl tetrasulfide	50	0.084	F2, J	354
Benzyl trisulfide	50	0.021	F2, J	354
1-Butanethiol	60	0.66		274
		0.67	A	69,397
Carbanilide, thio-	50	0.008	F2	251
		0.22	C, F2	251
–, –, 4,4′-bis(dimethylamino)-	50	0.09	C, F2	251
–, –, 4,4′-dichloro-	50	0.35	C, F2	251
–, –, 3,3′-dimethoxy-	50	0.33	C, F2	251
–, –, 3,3′-dimethyl-	50	0.21	C, F2	251
–, –, 4,4′-dimethyl-	50	0.13	C, F2	251
–, –, 3,3′-dinitro-	50	0.55	C, F2	251
Carbon disulfide	60	0.00007	A	288
2,13-Dioxa-7,8-dithia-3,12-disila-tetradecane, 3,3,12,12-tetrameth oxy-	79.5	0.00258		474
Disulfide, bis(*p*-bromobenzoyl)-	60	16.7		369
–, bis (*p*-bromophenyl)-	50	0.0046		285
	60	0.0098		285
–, bis(*p*-chlorobenzoyl)-	60	10.0		369
–, bis(*p*-chlorophenyl)-	50	0.0072		285
	60	0.0117		285
–, bis(*p*-cyanobenzoyl)-	60	290		369
–, bis(dimethylthiocarbamoyl)-	70	0.0115		104
–, bis(*p*-nitrobenzoyl)-	60	694		369
–, bis(*o*-nitrophenyl)-	50	0.0176		285
	60	0.0508		285
–, bis(*p*-nitrophenyl)-	50	0.0127		285
	60	0.0193		285
1-Dodecanethiol	60	0.7		573
	70	0.55		573
	80	0.42		573
	90	0.33		573
Ethanethiol, 2-amino-, hydrochloride	50	0.17	F7	473
	70	0.15	F7	473
Ethanol, 2-mercapto-	50	0.43	F2	473
	60	0.62		274
	70	0.40	F2	473
Ethyl disulfide	60	0.00013		310
Formamidine, *N,N′*-diphenyl-				
–, –, 1,1′-dithiobis-	50	0.22	C, F2	251
Methyl sufoxide	60	0.0000071	A	438

TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Methyl methacrylate (*cont'd*)				
2-Naphthalenethiol	45	1.3		30
	60	3.1		274
1-Penthanethiol	50	0.8	H	344
Phenyl disulfide	25	0.0174	D	368
	50	0.0038		285
	60	0.0007		310
		0.00176		368
		0.0085		285
		0.0110	A	491
Phenyl disufone	60	0.0003	A	491
Phenyl sulfide	25	0.00554	D	368
	50	0.0064		252
	60	0.00132		368
Phenyl sulfone	50	0.0006		252
Phenyl sulfoxide	50	0.0013		252
1-Propanethiol, 3-[1,3,3,3-tetramethyl-1-(trimethylsiloxy)disiloxanyl]-	79.5	0.128		474
–, 3-(trimethoxy)-	79.5	0.693		474
2-Propanethiol	60	0.38		274
–, 2-methyl-	60	0.18		274
Propionic acid, 3-mercapto-	50	0.39	F2	473
	70	0.38	F2	473
Sulfide, benzoyl dimethylthiocarbamoyl	20	5.5	D	280
	60	0.08	A	280
Sulfur	–	1.3	C	118
	44.1	0.075		194
	50	0.106	F2, J	354
Thiosulfuric acid, *S*-butyl ester				
–, –, sodium salt	60	0.00226		490
–, *S*-isopropyl ester				
–, –, sodium salt	60	0.00167		490
–, *S*-phenyl ester				
–, –, compound with pyridine	60	0.374		490
–, –, potassium salt	60	0.00537		490
–, *S*-propyl ester				
–, –, sodium salt	60	0.00190		490
m-Toluenethiol	45	4.7		30
p-Toluenethiol	45	7.4		30
α-Toluenethiol	50	0.027	F2, J	354
p-Toluoyl disulfide	60	11.0		369
p-Tolyl disulfide	50	0.0031		285
	60	0.0044		285
Styrene				
Acetic acid, dithiodi-	50	<0.005		297
	99	0.2		96
–, –, diethyl ester	60	0.015		86
–, –, dimethyl ester	99	0.1		96
–, –, mercapto-	99	>14.0	J, R	96
–, –, butyl ester	50	26.0	A	146
–, –, ethyl ester	60	58.0	A	126
–, –, methyl ester	60	0.63		323
	99	27.6		96
–, phenyl-				
–, –, *p,p'*-dithiodi-	50	0.24		89
–, thio	99	>14.7	J	96
Aniline, 2,2′ dithiodi-	25	3.8		297
	50	3.0		297
		3.4		297
	75	3.0		297
–, 4,4′-dithiodi-	50	0.24		89
–, *N*-dodecyl-, 4,4′-dithiodi-	50	0.21		89
Anisole, 2,2′-dithiodi-	50	3.0		89
–, 4,4′-dithiodi-	50	0.18		89
–, *p*-methyl-, α,α′-dithiobis-	99	0.021		96

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TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Styrene (*cont'd*)				
p-Anisoyl disulfide	60	96.0		369
Anthranilic acid, 4,4′-dithiobis-	50	3.0		89
Arylbenzene thiolsulfonate	60	0.95	R, F2	544
–, *p*-chloro-	60	2.22	R, F2	544
–, *p*-methoxy-	60	1.872	R, F2	544
–, *p*-methyl-	60	1.237	R, F2	544
Benzenesulfonic acid				
–, compound with pyridine	60	0.00617		490
–, thio-, *S*-phenyl ester	50	1.60	A	491
	60	1.67	A	491
Benzenethiol	99	0.08		96
–, *o*-ethoxy-	99	25.1		96
–, *p*-ethoxy-	99	0.13		96
Benzene thiosulfonate				
–, aryl-*p*-chloro-	60	2.597	R, F2	544
–, aryl-*p*-methoxy-	60	1.475	R, F2	544
–, aryl-*p*-methyl	60	1.099	R, F2	544
2-Benzimidazolethiol	99	0.21		96
Benzoic acid, 2,2′-dithiodi-	50	0.01		89
–, –, diethyl ester	50	<0.005		89
–, –,dimethyl ester	99	0.2		96
–, 4,4′-dithiodi-	50	0.11		297
		0.17		89
–, –, diethyl ester	50	0.11		89,297
–, *o*-mercapto-	99	>14.7		96
–, –, methyl ester	99	17.0		96
–, thio-	99	6.23	J	96
Benzothiazole, 2,2′-dithiobis-	25	2.3		297
	50	2.1		297
		2.3		297
	75	2.4		297
	99	2.73		96
–, 2,2′-thiobis-	50	<0.005		297
2-Benzothiazolethiol	50	0.03		297
	99	0.26		96
Benzo[B]thiophene, 4,5,6,7-tetrahydro-4(2-thienyl)-	60	0.011		419
Benzoyl disulfide	50	<0.005		89
	60	0.010		545
		0.0107		304
		36.0		369
	99	0.11		96
Benzyl alcohol, *o,o*′-dithiobis-	50	0.58		89
–, *p,p*′-dithiobis-	50	0.09		297
Benzyl diselenide	60	2.0	A	128
Benzyl disulfide	50	0.02		89
		0.03		297
	60	0.00878		368
		0.01	A	128
	99	0.011		96
Benzyl sulfide	25	0.548	D	368
	50	0.069		252
	60	0.00335		368
Benzyl sulfone	50	0.052		252
Benzyl sulfoxide	50	0.051		252
1-Butanethiol	25	5.4	D, Q1	340
		13.2	D	340
	60	21.0	A	395
		22.0	A	69,397
		25		131
	70	15.0	A	395
	80	17.0	A	395
	99	15.4		96

TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Styrene (*cont'd*)				
1-Butanethiol, 1,1,3,3-tetramethyl-	5	5.5	H	242
		6.4	H	222
	50	4.3	H	242
1-Butanethiol-D	60	5.2	A	395
	70	4.0	A	395
	80	7.0	A	395
Butyl disulfide	25	0.00079		448
	55	0.00154		448
	60	0.0024		304
	99	0.0068		96
	150	0.022	A	359
sec-Butyl disulfide	50	<0.005		89
tert-Butyl disulfide	50	<0.005		89
	60	0.00014		304
Butyl sulfide	60	0.0022		304
tert-Butyl sulfide	60	0.025		304
Carbanilic acid, *p,p'*-dithiodi-diethyl ester	50	0.24		89
Carbon disulfide	60	0.00066	A	288
Carbonic acid, dithio-				
–, –, *S,S'*-bis(carboxymethyl)ester	50	0.36		297
–, trithio-				
–, –, *S,S'*-bis(carboxymethyl) ester	50	0.21		297
1-Decanethiol	80	4.3	C3, H	574
	80	4.2	C3, H	574
2,13-Dioxa-7,8-dithia-3,12-disilatetradecane, 3,3,12,12-tetramethoxy-	79.5	0.0118		474
Diphenyldisulfide	60	0.147		546
Disulfide, bis(*p*-bromobenzoyl)	60	745		369
–, bis(α-bromo-*o*-tolyl)	50	1.0		297
–, bis(*p*-chlorobenzoyl)	60	196		369
–, bis(chlorobenzyl)	50	<0.005		297
–, bis(2-chloroethyl)	50	0.01		297
–, bis(α-chloro-*o*-tolyl)	50	1.3		297
–, bis(*p*-cyanobenzoyl)	60	3190		369
–, bis(diethylthiocarbamoyl)	60	0.724		286
–, bis(dimethylthiocarbamoyl)	60	1.11		286
	70	0.0136		104
	80	0.568	J	40
		0.620		481
	95	0.780		481
		0.860	J	40
	115	0.939	J	40
		1.035		481
	130	0.984	J	40
		1.150		481
–, bis(2-ethylhexyl)	50	<0.005		297
–, bis(*N*-ethyl-*N*-phenylthio-carbamoyl)	60	1.75		286
–, bis(1-methylheptyl)	99	0.0104		96
–, bis(morpholinothiocarbonyl)	60	6.1		286
–, bis(1-naphthylmethyl)	99	0.033		96
–, bis(*p*-nitrobenzoyl	60	6650		369
–, bis(*o*-nitrophenyl)	50	–		89
–, bis(1-phenylethyl)	50	<0.005		89
–, bis(β-(2-pyridyl)ethyl)	50	0.03		297
–, bis(2,3,5,6-tetramethylphenyl)	50	0.73		297
–, bis(2,4,6-triisopropylphenyl)	50	0.12		89
1-Dodecanethiol	60	14.8	A	69,126
		19.0		59
	70	19		562
	70	0.69	C3, H	574
	90	1.3	C3, H	574
	100	13.0		156
	110	26.0		156

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TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Styrene (*cont'd*)				
tert-Dodecanethiol	70	1.5	C3, H	574
	90	1.5	C3, H	574
	90	1.6	C3, H	574
Ethanethiol	50	17.1		331
Ethanol, 2,2′-dithiodi-	50	<0.005		297
–, –, di(chloroacetate)	50	<0.005		297
Ether, ethyl 3-mercaptopropyl	60	14.1	A	126
		21.0		69
	100	13.7		69
Ethyl disulfide	99	0.0045		96
Formamidine, *N,N′*-diphenyl-, 1,1′-dithiobis-	50	6.72	C, F2	251
Formic acid, thio-, dithiobis, *O,O′*–diisopropyl ester	50	5.3		297
	99	7.5		96
1-Heptanethiol	99	15.1		96
1-Hexanethiol	99	15.3		96
–, 1,1,3,3,5,5-hexamethyl-	25	3.2		297
	50	2.9		297
	75	2.4		297
Hexyl disulfide	99	0.0104		96
Hydrogen sulfide	70	5.0		145
Isobutyl disulfide	60	0.0020		304
Isopropyl disulfide	60	0.00066		304
Lauryl disulfide	60	0.00023	A	128
Lepidine, 2,2-dithiodi-	50	0.04		89
Mesityl disulfide	50	0.69		89
Methyl disulfide	60	0.0094		304
Methyl sulfoxide	40	0.0000693	E	458
	50	0.548		375
	60	0.0000242		429
		0.000048	A	438
Morpholine, 4,4′-dithiodi-	50	<0.005		297
1-Naphthalenemethanethiol	25	12.7		297
	50	18.3		297
	75	15.7		297
	99	24.6		96
1-Naphthalenethiol	99	0.15		96
2-Naphthalenethiol	99	0.18		96
1-Naphthoyl disulfide	50	0.34		89
1-Naphthyl disulfide	99	1.57		96
2-Naphthyl disulfide	25	0.17		297
	50	0.19		297
	75	0.29		297
	99	0.36		96
1-Octadecanethiol	99	14.7		96
Octadecyl disulfide	99	0.024		96
1-Octanethiol	5	19.3	H	339
		23.0	H	242
	50	19.0	H	242
–, 1,1,3,3,5,5,7,7-octamethyl-	50	4.7	H	242
2-Octanethiol	99	3.2		96
1,4,5-Oxadithiepane	150	0.057	A	359
1-Pentanethiol	50	20.0	H	344
Phenetole, 2,2′-dithiodi-	99	0.075		96
–, 4,4′-dithiodi-	99	0.33		96
Phenyl disulfide	50	0.06		297
		0.102	A	491
	60	0.0103		368
		0.111	A	491
		0.147		285
Phenyl disulfone	50	0.022	A	491
	60	0.025	A	491
Phenyl sulfide	25	0.0325	D	368
	50	0.056		252
	60	0.00548		368

TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Styrene (*cont'd*)				
Phenyl sulfone	50	0.021		252
Phenyl sulfoxide	50	0.024		252
1-Propanethiol, 3-(trimethoxysilyl)-	79.5	5.90		474
2-Propanethiol, 2-methyl-	50	4.0	H	344
	60	3.1	A	261
		3.7	A	69,126
	60	4.6		59
	100	1.8	A	261
		2.3	A	69,126
Propionic acid, 3,3-dithiodi-				
–, –, dipropyl ester	50	<0.005		297
–, 2-mercapto-	–	7.7		178
–, 3-mercapto-	–	9.4		178
	50	6.0		297
Propyl disulfide	60	0.00234		304
Pyridine, 2,2′-dithiodi-	50	0.01		89
Quinoline, 2,2′-dithiodi-	50	0.05		89
Sulfide, ethyl 2,4-diphenylbutyl	50	30.0		331
–, ethyl phenethyl	50	7.15		331
1-Tetradecanethiol	50	19.0	H	242
Thiosulfuric acid, *S*-butyl ester				
–, –, sodium salt	60	0.173		490
–, *S*-isopropyl ester, sodium salt	60	0.407		490
–, *S*-phenyl ester				
–, – potassium salt	60	0.763		490
–, *S*-propyl ester, sodium salt	60	0.150		490
p-Toluenethiol	99	0.07		96
α-Toluenethiol	99	25.5		96
–, *p*-methoxy-	99	26.0		96
p-Toluoyl disulfide	60	46.3		369
o-Tolyl disulfide	25	0.22		297
	50	0.23		297
		0.28		297
	75	0.32		297
p-Tolyl disulfide	50	0.11		297
	99	0.15		96
2,6-Xylyl disulfide	50	0.69		297
Vinyl acetate				
Acetic acid, dithiodi-				
–, –, diethyl ester	60	1.41		86
		1.5		351
Acetic acid, ithio-				
–, –, *S*-(2-hydroxyethyl)ester, acetate	60	0.0132	J	140
Acetyl disulfide	60	0.29	J	86
1-Butanethiol	60	48.0		140,397
Butyl disulfide	60	1.0	J	86, 351
Butyl sulfide	60	0.026	J	351
1,4,5-Oxadithiepane	60	0.25–2.5		351
Sulfur	45	470.0	J	28
α-Toluenethiol	70	0.885	C	390
Thiol acetate	60	0.186	F20	575

C. REMARKS

A. Thermal initiation
B. Hydroperoxide initiation
C. Peroxide initiation
  C1. Ammonium persulfate
  C2. Potassium persulfate
  C3. Sodium persulfate
D. Photoinitiation
E. γ-Ray initiation
F. Solution polymerization in:
  F1. Acetonitrile
  F2. Benzene
  F3. Butyl acetate
  F4. Butyl chloride
  F5. Chlorobenzene

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TABLE 5. *cont'd*

Modifier	T (°C)	C_X	Remarks	Refs.
Styrene (*cont'd*)				
Phenyl sulfone	50	0.021		252
Phenyl sulfoxide	50	0.024		252
1-Propanethiol, 3-(trimethoxysilyl)-	79.5	5.90		474
2-Propanethiol, 2-methyl-	50	4.0	H	344
	60	3.1	A	261
		3.7	A	69,126
	60	4.6		59
	100	1.8	A	261
		2.3	A	69,126
Propionic acid, 3,3-dithiodi-				
–, –, dipropyl ester	50	<0.005		297
–, 2-mercapto-	–	7.7		178
–, 3-mercapto-	–	9.4		178
	50	6.0		297
Propyl disulfide	60	0.00234		304
Pyridine, 2,2′-dithiodi-	50	0.01		89
Quinoline, 2,2′-dithiodi-	50	0.05		89
Sulfide, ethyl 2,4-diphenylbutyl	50	30.0		331
–, ethyl phenethyl	50	7.15		331
1-Tetradecanethiol	50	19.0	H	242
Thiosulfuric acid, *S*-butyl ester				
–, –, sodium salt	60	0.173		490
–, *S*-isopropyl ester, sodium salt	60	0.407		490
–, *S*-phenyl ester				
–, – potassium salt	60	0.763		490
–, *S*-propyl ester, sodium salt	60	0.150		490
p-Toluenethiol	99	0.07		96
α-Toluenethiol	99	25.5		96
–, *p*-methoxy-	99	26.0		96
p-Toluoyl disulfide	60	46.3		369
o-Tolyl disulfide	25	0.22		297
	50	0.23		297
		0.28		297
	75	0.32		297
p-Tolyl disulfide	50	0.11		297
	99	0.15		96
2,6-Xylyl disulfide	50	0.69		297
Vinyl acetate				
Acetic acid, dithiodi-				
–, –, diethyl ester	60	1.41		86
		1.5		351
Acetic acid, ithio-				
–, –, *S*-(2-hydroxyethyl)ester, acetate	60	0.0132	J	140
Acetyl disulfide	60	0.29	J	86
1-Butanethiol	60	48.0		140,397
Butyl disulfide	60	1.0	J	86, 351
Butyl sulfide	60	0.026	J	351
1,4,5-Oxadithiepane	60	0.25–2.5		351
Sulfur	45	470.0	J	28
α-Toluenethiol	70	0.885	C	390
Thiol acetate	60	0.186	F20	575

C. REMARKS

A. Thermal initiation
B. Hydroperoxide initiation
C. Peroxide initiation
  C1. Ammonium persulfate
  C2. Potassium persulfate
  C3. Sodium persulfate
D. Photoinitiation
E. γ-Ray initiation
F. Solution polymerization in:
  F1. Acetonitrile
  F2. Benzene
  F3. Butyl acetate
  F4. Butyl chloride
  F5. Chlorobenzene

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F6. *o*-Dichlorobenzene
F7. Dimethylformamide
F8. Dimethyl sulfoxide
F9. *p*-Dioxane
F10. Ethyl acetate
F11. Ethyl methyl ketone
F12. Heptane
F13. Water
F14. Magnesium perchlorate
F15. Succinonitrile
F16. Sulfur dioxide
F17. Toluene
F18. Zinc chloride
F19. Octane
F20. Methanol
F21. Propane
F22. Diethyl oxalate
F23. Cyclohexane
F24. Butyl alcohol
F25. Acetic acid
F26. 1,2-Dichloroethane
F27. Ethylbenzene
F28. Cylohexanone
F29. Parabutyr aldehyde
F30. Water–methanol mixture
F31. *tert*-Butyl alcohol
F32. Hexane
F33. 1-Propanol
F34. Carbon tetrachloride

G. Heterogeneous polymerization
H. Emulsion polymerization
I. Recalculated from data of:
I1. Basu, Sen, and Palit (1950)
I2. Baysal and Tobolsky (1952)
I3. Chadha and Misra (1958)
I4. Dixon-Lewis (1949)
I5. Gregg and Mayo (1953)
I6. Ham (1956)
I7. Kwart, Broadbent, and Bartlett (1950)
I8. Matsumoto, et. al. (1959)
I9. Palit and Das (1954)
I10. Palit, Nandi, and Saha (1954)
I11. Saha, Nandi, and Palit (1956)
I12. Stein and Schulz (1960)
I13. Wheeler, Lavin, and Crozier (1952)

J. Apparent transfer constant; retardation occurred
K. Average value
L. Estimated from model compounds
M. For end groups:
M1. Tribromomethyl
M2. Butylamino
M3. Diethylamino
M4. Hydroxymethyl
M5. Aldehyde
M6. Thioglycolate
M7. $-CH(OH)CH_2$-SH
M8. Dodecyl
M9. 2-Hydroxyethyl
M10. $-(CH_2)_{11}SH$
N. For middle groups
O. For side chain
P. For main chain
Q. Telomerization (number of monomer units in transferring chain)
R. Calculated from viscosity average molecular weight
S. Value uncertain
T. In presence of 0.4 mol/l lithium nitrate
U. In presence of ( ) mol/l sodium chloride
V. In presence of ( ) mol/l sodium bromide
W. Under pressure of:
W1. 200 psi (13.8 bar)
W2. 250 psi (17.2 bar)
W3. 1470 psi (101.4 bar)
W4. 5000 psi (344.8 bar)
W5. 20000 psi (1379.0 bar)
W6. 26600 psi (1834.1 bar)
W7. 28400 psi (1958.2 bar)
W8. 56000 psi (3861.2 bar)
W9. 64600 psi (4454.2 bar)
W10. 168000 psi (11583.6 bar)
W11. 600 psi (41.4 bar)
W12. 2500 psi (172.4 bar)
W13. 18 psi (1.2 bar)
W14. 735 psi (50.7 bar)
W15. 14200 psi (979.1 bar)
W16. 42600 psi (2937.3 bar)
W17. 34500 psi (2378.8 bar)
W18. 3000 bar
W19. 2700 bar
W20. 2400 bar
W21. 2000 bar
W22. 1800 bar
W23. 150 bar
W24. 600 bar
X. Corrected for loss of low molecular weight polymer
Y. $CM = 125 \times \exp(-7300/RT)$
Z. $CM = 3.0 \times \exp(-5673/RT)$
AA. $CM = 0.4 \times \exp(-6219/RT)$
BB. $CM = 0.2 \times \exp(-5400/RT)$
CC. $CM = 6.4 \times \exp(-4120/RT)$
DD. $CM = 4.77 \times \exp(-3540/RT)$
EE. Radical is styrene
FF. Radical is *p*-chlorostyrene
GG. 1/C varies with polym. rate and emulsifier of:
GG1. K caprate
GG2. K laurate
GG3. K myristate
GG4. K palmitate
GG5. K rosinate
HH. *C* is combined constant for copolymerization,
$= (C_{S1} \times R_1 \times M_1 + C_{S2} \times R_2 \times M_2)/$
$(R_1 \times M_1 \times M_1 + 2 \times M_1 \times M_2$
$+R_2 \times M_2 \times M_2)$
II. In presence of $FeBr_2$

II1. In presence of $FE(CO)_5$
II2. In presence of methacrylic acid methyl ester
JJ0. Catalyzed by Co(II)-tetramethylhematoporphyrine
JJ1. Catalyzed by Co(II)-hematoporphyrine-tetramethyl ether
JJ2. Catalyzed by Co(II)-tetra-4-*tert*-butylbenzoporphyrine
JJ3. In presence of $ZnCl_2$; $[ZnCl_2]/[MMA] =$
(a) 0
(b) 0.1
(c) 0.23
(d) 0.40
KK. Copolymerization of acrylonitrile and butyl acrylate
LL. Controlled to pH 3–4 with acetic acid

D. REFERENCES

1. I. B. Afanas'ev, T. N. Eremina, Zh. Org. Khim., **2**, 1832 (1966).
2. I. B. Afanas'ev, T. N. Eremina, E. D. Safronenko, Zh. Org. Khim., **1**, 844 (1965); J. Org. Chem. USSR, **1**, 849 (1965).
3. I. B. Afanas'ev, E. D. Safronenko, A. A. Beer, Vysokomol. Soedin., Ser. B, **9**, 802 (1967).
4. G. Akazome, S. Sakai, K. Maurai, Kogyo Kagaku Zasshi, **63**, 592 (1960); from CA, **56**, 4924F (1962).
5. L. Alexandru, M. Oprish, Polym. Sci. USSR, **3**, 99 (1962).
6. P. W. Allen, G. P. Mcsweeney, Trans. Faraday Soc., **54**, 715 (1958).
7. P. W. Allen, F. M. Merrett, J. Scanlan, Trans. Faraday Soc., **51**, 95 (1955).
8. U. N. Anistmov, S. S. Ivanchev, A. I. Yurzhenko, Vysokomol, Soedin., Ser. A, **9**, 692 (1967).
9. I. A. Arbusova, L. I. Medvedeva, Izv. Akad. Nauk SSSR, Ser. Khim., 1957, 1349.
10. T. Asahara, T. Makishima, Kogyo Kagaku Zasshi, **69**, 2173 (1966).
11. C. Aso, M. Sugabe, Kogyo Kagaku Zasshi, **68**, 1970 (1965).
12. R. Autrata, J. Muller, Collect. Czech. Chem. Commun., **24**, 3442 (1959).
13. D. W. E. Axford, Proc. Roy. Soc., Ser. A, **197**, 137 (1949).
14. T. M. Babchinitser, K. K. Mozgova, V. V. Korshak, Dokl. Akad. Nauk SSSR, **173**, 575 (1967); from CA, **67**, 117314E (1967).
15. Kh. S. Bagdasar'ian, Z. A. Sinitsina, J. Polym. Sci., **52**, 31 (1961).
16. C. H. Bamford, S. Brumby, Makromol. Chem., **105**, 122 (1967).
17. C. H. Bamford, M. J. S. Dewar, Proc. Roy. Soc., Ser. A, **192**, 309 (1948).
18. C. H. Bamford, M. J. S. Dewar, Proc. Roy. Soc., Ser. A, **197**, 356 (1949).
19. C. H. Bamford, A. D. Jenkins, R. Johnston, Proc. Roy. Soc., Ser. A, **239**, 214 (1957).
20. C. H. Bamford, A. D. Jenkins, R. Johnston, Proc. Roy. Soc., Ser. A, **241**, 364 (1957).
21. C. H. Bamford, A. D. Jenkins, R. Johnston, Trans. Faraday Soc., **55**, 418 (1959).
22. C. H. Bamford, A. D. Jenkins, R. Johnston, E. F. T. White, Trans. Faraday Soc., **55**, 168 (1959).
23. C. H. Bamford, A. D. Jenkins, E. F. T. White, J. Polym. Sci., **34**, 271 (1959).
24. C. H. Bamford, E. F. T. White, Trans. Faraday Soc., **52**, 716 (1956).
25. C. H. Bamford, E. F. T. White, Trans. Faraday Soc., **54**, 268 (1958).
26. S. Banerjee, M. S. Muthana, J. Polym. Sci., **37**, 469 (1959).
27. P. D. Bartlett, G. S. Hammond, H. Kwart, Dis. Faraday Soc., **2**, 342 (1947).
28. P. D. Bartlett, H. Kwart, J. Am. Chem. Soc., **74**, 3969 (1952).
29. P. D. Bartlett, F. A. Tate, J. Am. Chem. Soc., **75**, 91 (1953).
30. S. C. Barton, R. A. Bird, K. E. Russell, Can. J. Chem., **41**, 2737 (1963).
31. S. Basu, J. N. Sen, S. R. Palit, Proc. Roy. Soc., Ser. A, **202**, 485 (1950).
32. S. Basu, J. N. Sen, S. R. Palit, Proc. Roy. Soc., Ser. A, **214**, 247 (1952).
33. B. Baysal, J. Polym. Sci., **33**, 381 (1958).
34. B. Baysal, A. V. Tobolsky, J. Polym. Sci., **8**, 529 (1952).
35. I. M. Bel'govskii, L. S. Sakhonenko, N. S. Yenikolopyan, Vysokomol. Soedin., **8**, 369 (1966).
36. W. I. Bengough, W. H. Fairservice, Trans. Faraday Soc., **61**, 1206 (1965).
37. W. I. Bengough, W. H. Fairservice, Trans. Faraday Soc., **63**, 382 (1967).
38. W. I. Bengough, W. Henderson, Trans. Faraday Soc., **61**, 141 (1965).
39. W. I. Bengough, R. A. M. Thomson, Trans. Faraday Soc., **56**, 407 (1960).
40. J. Beniska, E. Staudner, J. Polym. Sci. C, **16**, 1301 (1967).
41. T. Berezhnykh-Földes, F. Tudös, Eur. Polym. J., **2**, 219 (1996).
42. J. C. Bevington, N. A. Ghanem, H. W. Melville, J. Chem. Soc., 1995, 2822.
43. J. C. Bevington, G. M. Guzman, H. W. Melville, Proc. Roy. Soc. Ser. A, **221**, 437 (1954).
44. J. C. Bevington, H. G. Troth, Trans. Faraday Soc., **58**, 2005 (1962).
45. J. C. Bevington, H. G. Troth, Trans. Faraday Soc., **59**, 127 (1963).
46. J. C. Bevington, H. G. Troth, Trans. Faraday Soc., **59**, 1348 (1963).
47. J. C. Bevington, A. Wahid, Polymer, **3**, 585 (1962).
48. B. R. Bhattacharyya, U. S. Nandi, J. Polym. Sci. A-1, **4**, 2675 (1966).
49. B. R. Bhattacharyya, U. S. Nandi, Makromol. Chem. **116**, 8 (1968).
50. R. A. Bird, G. A. Harpell, K. E. Russell, Can. J. Chem., **40**, 701 (1962).
51. R. A. Bird, K. E. Russell, Can. J. Chem., **43**, 2123 (1965).

52. L. Bogetich, G. A. Mortimer, G. W. Daues, J. Polym. Sci., **61**, 3 (1962).
53. E. P. Bonsall, L. Valentine, H. W. Melville, J. Polym. Sci., **7**, 39 (1951).
54. C. Booth, L. R. Beason, J. Polym. Sci., **42**, 93 (1960).
55. C. Booth, L. R. Beason, J. T. Bailey, J. Appl. Polym. Sci., **5**, 116 (1961).
56. D. Braun, T.-O. O. Ahn., W. Kern, Makromol. Chem., **53**, 154 (1962).
57. D. Braun, G. Arcache, R. J. Faust, W. Neumann, Makromol. Chem., **114**, 51 (1968).
58. J. W. Breitenbach, Makromol Chem., **8**, 147 (1952).
59. J. W. Breitenbach, Z. Elektrochem., **60**, 286 (1956).
60. J. W. Breitenbach, H. Edelhauser, R. Hochrainer, Monatsh. Chem., **97**, 217 (1966).
61. J. W. Breitenbach, H. Karlinger, Monatsh. Chem., **82**, 245 (1951).
62. J. W. Breitenbach, O. F. Olaj, Makromol. Chem., **96**, 83 (1966).
63. J. W. Breitenbach, O. F. Olaj, K. Kuchner, H. Horacek, Makromol. Chem., **87**, 295 (1965).
64. J. W. Breitenbach, O. F. Olaj, A. Schindler, Monatsh. Chem., **91**, 205 (1960).
65. J. W. Breintenbach, A. Schindler, Monatsh. Chem., **83**, 724 (1952).
66. J. W. Breintenbach, A. Schindler, Monatsh. Chem., **86**, 437 (1955).
67. J. W. Breintenbach, A. Schindler, Monatsh. Chem., **88**, 53 (1957).
68. D. W. Brown, J. E. Fearn, R. E. Lowry, J. Polym. Sci. A, **3**, 16 (1965).
69. G. M. Burnett, Quart. Rev. (London), **4**, 292 (1950).
70. G. M. Burnett, F. L. Ross, J. N. Hay, Makromol. Chem., **105**, 1 (1965).
71. A. J. Buselli, M. K. Lindemann, C. E. Blades, J. Polym. Sci., **28**, 485 (1958).
72. J. I. G. Cadogan, D. H. Hey, J. T. Sharp, J. Chem. Soc., B, 1966, 933.
73. M. Cantow, G. Meyerhoff, G. V. Schulz, Makromol. Chem., **49**, 1 (1961).
74. E. A. S. Cavell, Makromol. Chem., **54**, 70 (1962).
75. E. A. S. Cavell, I. T. Gilson, J. Polym. Sci. A-1, **4**, 541 (1966).
76. R. N. Chadha, G. S. Misra, Indian J. Phys., **28**, 37 (1954).
77. R. N. Chadha, G. S. Misra, Current Sci. (India), **23**, 186 (1954).
78. R. N. Chadha, G. S. Misra, Makromol. Chem., **14**, 97 (1954).
79. R. N. Chadha, G. S. Misra, Current Sci. (India), **24**, 26 (1955).
80. R. N. Chadha, G. S. Misra, Trans. Faraday Soc., **54**, 1227 (1958).
81. R. N. Chadha, J. S. Shukla, G. S. Misra, Trans. Faraday Soc., **53**, 240 (1957).
82. E. Y. C. Chang, C. C. Price, J. Am. Chem. Soc., **83**, 4650 (1961).
83. S. R. Chatterjee, S. N. Khanna, S. R. Palit, J. Indian Chem. Soc., **41**, 622 (1964).
84. A. K. Chaudhuri, Makromol. Chem., **31**, 214 (1959).
85. S. Chubachi, P. K. Chatterjee, A. V. Tobolsky, J. Org. Chem., **32**, 1511 (1967).
86. J. T. Clarke, R. O. Howard, W. H. Stockmayer, Makromol. Chem., **44–46**, 427 (1961).
87. W. Cooper, J. Chem. Soc. **1952**, 2408.
88. A. T. Coscia, R. L. Kugel, J. Pellon, J. Polym. Sci., **55**, 303 (1961).
89. A. J. Costanza, R. J. Coleman, R. M. Pierson, C. S. Marvel, C. King, J. Polym. Sci., **17**, 319 (1955).
90. J. Curtice, H. Gilman, G. S. Hammond, J. Am. Chem. Soc., **79**, 4754 (1957).
91. F. S. Dainton, R. G. Jones, Trans. Faraday Soc., **63**, 1512 (1967).
92. F. S. Dainton, M. Tordoff, Trans. Faraday Soc., **53**, 499 (1957).
93. N. V. Daniel, A. F. Nikolaev, Vysokomol. Soedin., **8**, 465 (1966).
94. F. Danusso, D. Sianesi, Chim. Ind. (Milan), **37**, 695 (1955).
95. S. K. Das, S. R. Chatterjee, S. R. Palit, Proc. Roy. Soc., Ser. A, **227**, 252 (1955).
96. V. A. Dinaburg, A. A. Vansheidt, Zh. Obshch. Khim., **24**, 840 (1954).
97. G. Dixon-Lewis, Proc. Roy. Soc., Ser. A, **198**, 510 (1949).
98. K. W. Doak, M. A. Deahl, I. H. Christmas, 137th ACS Meeting, Cleveland, Ohio, Abstr. Papers, Vol. 1, No. 1, 151 (April 1960).
99. H. Drawe, A. Henglein, Makromol. Chem., **84**, 203 (1965).
100. E. Dvorak, F. Hrabak, J. Polym. Sci. C, **16**, 1051 (1967).
101. B. A. Englin, R. Kh. Freidlina, Izv. Akas. Nauk SSSR, Ser. Khim., 1965 (13), 425; from CA, **63**, 4122E (1965).
102. B. Erussalimsky, N. Tumarkin, F. Duntoff, S. Lyubetzky, A. Goldenberg, Makromol. Chem., **104**, 288 (1967).
103. T. A. Fadner, H. Morawetz, J. Polym. Sci., **45**, 475 (1960).
104. T. E. Ferington, A. V. Tobolsky, J. Am. Chem. Soc., **77**, 4510 (1955).
105. T. G. Fox, S. Gratch, Ann. N. Y. Acad. Sci., **57**, 367 (1953).
106. G. N. Freidlin, K. A. Solop, Vysokomol. Soedin., **7**, 1060 (1965).
107. G. N. Freidlin, K. A. Solop, Vysokomol. Soedin., **8**, 1151 (1966).
108. T. M. Frunze, V. V. Korshak, E. L. Baranov, B. V. Lokshin, Vysokomol. Soedin, **8**, 455 (1966).
109. N. Fuhrma, R. B. Mesrobian, J. Am. Chem. Soc., **76**, 3281 (1954).
110. T. R. Fukuto, J. P. Kispersky, US Dept. Com., Office Tech. Serv., Pb Rept. **147**, 271 (1953).
111. S. Gadkary, S. Kapur, Makromol. Chem., **17**, 29 (1955).
112. J. A. Gannon, E. M. Fettes, A. V. Tobolsky, J. Am. Chem. Soc., **74**, 1854 (1952).
113. M. H. George, J. Polym. Sci. A, **2**, 3169 (1964).
114. M. H. George, P. F. Onyon, Trans. Faraday Soc., **59**, 134 (1963).
115. M. H. George, P. F. Onyon, Trans. Faraday Soc., **59**, 1390 (1963).
116. H. Gerrens, H. Ohlinger, R. Fricker, Markromol. Chem., **87**, 209 (1965).

117. L. Ghosez, G. Smets, J. Polym. Sci., **37**, 445 (1959).
118. G. P. Gladyshev, G. V. Leplyanin, Vysokomol. Soedin., Ser. A, **9**, 2438 (1967).
119. J. E. Glass, N. L. Zutty, J. Polym. Sci. A-1, **4**, 1223 (1966).
120. M. S. Gluckman, M. J. Kampf, J. L. O'Brien, T. G. Fox, R. K. Graham, J. Polym. Sci., **37**, 411 (1959).
121. M. P. Godsay, G. A. Harpell, K. E. Russell, J. Polym. Sci., **57**, 641 (1962).
122. M. P. Godsay, D. H. Lohmann, K. E. Russell, Chem. Ind. (London), **1959**, 1603.
123. M. R. Gopalan, M. Santhappa, J. Polym. Sci., **25**, 333 (1957).
124. W. W. Graessley, H. Mittelhauser, R. Maramba, Makromol. Chem., **86**, 129 (1965).
125. N. Grassie, E. Vance, Trans. Faraday Soc., **52** 727 (1956).
126. R. A. Gregg, D. M. Alderman, F. R. Mayo, J. Am. Chem. Soc., **70**, 3740 (1948).
127. R. A. Gregg, F. R. Mayo, Disc. Faraday Soc., **2**, 328 (1947).
128. R. A. Gregg, F. R. Mayo, J. Am. Chem. Soc., **70**, 2373 (1948).
129. R. A. Gregg, F. R. Mayo, J. Am. Chem. Soc., **75**, 3530 (1953).
130. V. F. Gromov, A. V. Matveeva, P. M. Khomikovskii, A. D. Abkin, Vysokomol. Soedin., Ser. A, **9**, 1444 (1967).
131. J. Guillot, A. Guyot, Compt. Rend., Ser. C, **266**, 1209 (1968).
132. H. C. Haas, H. Husek, J. Polym. Sci. A, **2**, 2297 (1964).
133. Y. Hachihama, H. Sumitomo, Technol. Repts. Osaka Univ., **5**, 491 (1955); from CA, **51**, 8474B (1957).
134. Y. Hachihama, H. Sumitomo, Technol. Repts. Osaka Univ., **5**, 497 (1955); from CA, **51**, 8474C (1957).
135. G. E. Ham, J. Polym. Sci., **21**, 337 (1956).
136. G. E. Ham, E. L. Ringwald, J. Polym. Sci., **8**, 91 (1952).
137. D. Hardy, K. Nytrai, N. Fedorova, G. Kovacs, Polym. Sci. USSR, **4**, 584 (1963).
138. D. Hardy, K. Nytrai, N. Fedorova, G. Kovacs, Vysokomol. Soedin, **4**, 1872 (1962).
139. D. Hardy, V. Spiegel, K. Nytrai, Polym. Sci. USSR, **2**, 528 (1961).
140. G. Hardy, J. Varga, K. Nytrai, I. Tsajlik, L. Zubonyai, Vysokomol. Soedin., **6**, 758 (1964).
141. S. A. Harrison, W. E. Tolberg, J. Am. Oil Chem. Soc., **30**, 114 (1953); calc. by F. R. Mayo and C. W. Gould (1964).
142. R. A. Hayes, J. Polym. Sci., **13**, 583 (1954).
143. E. J. Henley, C. Chong, J. Polym. Sci., **36**, 511 (1959).
144. G. Henrici-Olive, S. Olive, Makromol. Chem., **37**, 71 (1960).
145. G. Henrici-Olive, S. Olive, Fortschr. Hochpolymer. Forsch., **2**, 496 (1961).
146. G. Henrici-Olive, S. Olive, Makromol. Chem., **53**, 122 (1962).
147. G. Henrici-Olive, S. Olive, G. V. Schulz, Makromol. Chem., **23**, 207 (1957).
148. G. Henrici-Olive, S. Olive, G. V. Schulz, Z. Phys. Chem. (Frankfurt), **20**, 176 (1959).
149. R. Hiddema, Proefschrift Rijks Universitat Groningen, 1953, Ref. 16 in Breitenbach (1956).
150. K. Higashiura, Kogyo Kagaku Zasshi, **69**, 349 (1966).
151. K. Higaashiura, M. Oiwa, J. Polym. Sci., A-1, **6**, 1857 (1968).
152. F. Hrabak, M. Bezdek, Makromol. Chem., **115**, 43 (1968).
153. F. Hrabak, M. Bezdek, Collect. Czech. Chem. Commun., **33**, 278 (1968).
154. F. Hrabak, L. Jiresova, Collect. Czech. Chem. Commun., **26**, 1283 (1961).
155. T. Huff, E. Perry, J. Am. Chem. Soc., **82**, 4277 (1960).
156. T. Huff, E. Perry, J. Polym. Sci. A, **1**, 1553 (1963).
157. F. Ide, K. Nakatsuka, H. Tamura, Kobunshi Kagaku, **23**, 45 (1965); from CA, **64**, 17716G (1966).
158. F. Ide, K. Okano, S. Nakano, K. Nakstsuka, Shikizai Kyokaishi, **40**, 571 (1967); from CA, **69**, 3162Q (1968).
159. J. L. Ihrig, S. P. Sood, J. Polym. Sci. A-3, 1573 (1965).
160. M. Imoto, K. Higashiura, Kobunshi Kagaku, **17**, 468 (1960); from CA, **55**, 22900G (1961).
161. M. Imoto, T. Otsu, T. Oda, H. Takatsugi, M. Matsuda, J. Polym. Sci., **22**, 137 (1956).
162. M. Imoto, T. Otsu, K. Tsuda, T. Ito, J. Polym. Sci. A-2, 1407 (1964).
163. M. Imoto, K. Takemoto, H. Azuma, Makromol. Chem., **114**, 210 (1968).
164. M. Imoto, K. Takemoto, Y. Nakai, Makromol. Chem., **48**, 80 (1961).
165. S. Imoto, T. Kominami, Kobunshi Kagaku, **15**, 60 (1958); from CA, **53**, 8690G (1959).
166. S. Imoto, T. Kominami, Kobunshi Kagaku, **15**, 279 (1958); from CA, **54**, 2803A (1960).
167. S. Imoto, J. Ukida, T. Kominami, Kobunshi Kagaku, **14**, 127 (1957); from CA, **52**, 1670A (1958).
168. S. Imoto, J. Ukida, T. Kominami, Kobunshi Kagaku, **14**, 384 (1957); from CA, **52**, 5024D (1958).
169. S. Iwatsuki, K. Nishio, Y. Yamashita, Kogyo Kagaku Zasshi, **70**, 384 (1967).
170. V. Jaacks, F. R. Mayo, J. Am. Chem. Soc., **87**, 3371 (1965).
171. D. H. Johnson, A. V. Tobolsky, J. Am. Chem. Soc., **74**, 938 (1952).
172. M. H. Jones, Can. J. Chem., **34**, 108 (1956).
173. E. F. Jordan, Jr., B. Artymyshyn, A. N. Wrigley, J. Polym. Sci. A-1, **6**, 575 (1968).
174. E. F. Jordan, Jr., G. R. Riser, W. E. Parker, A. N. Wrigley, J. Polym. Sci. A-2, **4**, 975 (1966).
175. R. M. Joshi, Makromol. Chem., **53**, 33 (1962).
176. H. Kaemmerer, F. Rocaboy, Compt. Rend. Ser. AB, **256**, 4440 (1963).
177. H. Kaemmerer, F. Rocaboy, Makromol. Chem., **72**, 76 (1964).
178. K. Kaeriyama, Nippon Kagaku Zasshi, **88**, 783 (1967); from CA, **69**, 19607Z (1968).
179. M. Kalfus, J. Kopytowski, S. Lesniak, Z. Skupinska, Polimery, **9**, (2) 54 (1964); from CA, **62**, 9325E (1965).
180. S. Kamenar, Chem. Zvesti, **14**, 525 (1960); from Chem. Zentr., **133**, 493 (1961).
181. S. L. Kapur, J. Sci. Ind. Res., **108**, 186 (1951).
182. S. L. Kapur, J. Polym. Sci., **11**, 399 (1953).
183. S. L. Kapur, S. D. Gadkary, J. Sci. Ind. Res., **17B**, 152 (1958).

184. S. L. Kapur, R. M. Joshi, J. Polym. Sci., **14**, 489 (1954).
185. K. Katagiri, K. Uno, S. Okamura, J. Polym. Sci., **17**, 142 (1955).
186. M. Kato, H. Kamogawa, J. Polym. Sci. A-1, **4**, 2771 (1966).
187. V. Ya. Katsobashvili, R. Ya. Chernaya, I. B. Afanas'ev, Vysokomol. Soedin., Ser. B, **9**, 342 (1967).
188. V. Ya. Katsobashvili, E. D. Safronenko, I. B. Afanas'ev, Vysokomol. Soedin., **7**, 823 (1965).
189. V. Ya. Katsobashvili, E. D. Safronenko, I. B. Afanas'ev, Vysokomol. Soedin., **8**, 282 (1966).
190. H. Kawakami, N. Mori, K. Kawashima, M. Sumi, Kogyo Kagaku Zasshi, **66**, 88 (1963); from CA, **59**, 4042B (1963).
191. H. Kawakami, N. Mori, M. Sumi, Kobunshi Kagaku, **20**, 408 (1963); from CA, **61**, 13422D (1964).
192. W. Kern, D. Braun, Makromol. Chem., **27**, 23 (1958).
193. S. N. Khanna, S. R. Chatterjee, U. S. Nandi, S. R. Palit, Trans. Faraday Soc., **58**, 1827 (1962).
194. J. L. Kice, J. Am. Chem. Soc., **76**, 6274 (1954).
195. W. J. Kirkham, J. C. Robb, Trans. Faraday Soc., **57**, 1757 (1961).
196. H. Kitagama, Kobunshi Kagaku, **20**, 5 (1963); from Makromol. Chem., **64**, 229 (1963).
197. H. Kiuchi, M. Watanabe, Kobunshi Kagaku, **21**, 37 (1964) from CA, **61**, 7107F (1964).
198. I. M. Kochnov, M. F. Sorokin, Polym. Sci. USSR, **6**, 869 (1964).
199. M. Kubin, L. Zikmund, Collect. Czech. Chem. Commun., **32**, 535 (1967).
200. C. J. Kurian, M. S. Muthana, Makromol. Chem., **29**, 1 (1959).
201. C. J. Kurian, M. S. Muthana, Makromol. Chem., **29**, 19 (1959).
202. H. Kwart, H. S. Broadbent, P. D. Bartlett, J. Am. Chem. Soc., **72**, 1060 (1950).
203. K. S. Kwei, F. R. Eirich, J. Phys. Chem., **66**, 828 (1962).
204. Z. Laita, J. Polym. Sci., **38**, 247 (1959).
205. Z. Laita, Z. Machacek, J. Polym. Sci., **38**, 459 (1959).
206. M. Lazar, J. Pavlinec, Chem. Zvesti, **15**, 428 (1961); from CA, **55**, 22896C (1961).
207. M. Lazar, J. Pavlinec, J. Polym. Sci. A, **2**, 3197 (1964).
208. M. Lazar, J. Pavlinec, Z. Manasek, Collect. Czech. Chem. Commun., **26**, 1380 (1961).
209. M. Lazar, R. Rado, J. Pavlinec, J. Polym. Sci., **53**, 163 (1961).
210. F. M. Lewis, F. R. Mayo, J. Am. Chem. Soc., **76**, 457 (1954).
211. D. Lim, M. Kolinsky, J. Polym. Sci., **53**, 173 (1961).
212. D. Lim, O. Wichterle, J. Polym. Sci., **29**, 579 (1958).
213. M. Litt, V. Stannett, Makromol. Chem., **37**, 19 (1960).
214. J. R. Little, P. F. Hartman, J. Am. Chem. Soc., **88**, 96 (1966).
215. J. R. Little, L. W. Hartzel, F. O. Guenther, F. R. Mayo, private communication to C. Walling, "Free radicals in solution", Wiley, NY, 1957, p. 257.
216. S. Loshaek, E. Broderick, P. Bernstein, J. Polym. Sci., **39**, 223 (1959).
217. A. I. Lowell, J. R. Price, J. Polym. Sci., **43**, 1 (1960).
218. S. G. Lyubetskii, B. A. Dolgoplosk, B. L. Erusalimskii, Polym. Sci. USSR, **3**, 164 (1962).
219. Z. Machacek, F. Cermak, Chem, Prum., **16**, 604 (1966); from CA, **66**, 18907Y (1967).
220. G. Machell, G. N. Richards, J. Chem. Soc., **1961**, 3308.
221. V. Mahadevan, M. Santhappa, Makromol. Chem., **16**, 119 (1955).
222. K. L. Malik, Naturwissenschaften, **45**, 385 (1958).
223. T. Manabe, T. Utsumi, S. Okamura, J. Polym. Sci., **58**, 121 (1962).
224. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **71**, 497 (1949).
225. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **71**, 2610 (1949).
226. S. G. Matsoyan, N. N. Morlyan, F. S. Kinoyan, Vysokomol. Soedin., **7**, 1159 (1965).
227. M. Matsuda, S. Abe, N. Tokura, J. Polym. Sci. A, **2**, 3877 (1964).
228. M. Matsuda, K. Matsumoto, N. Tokura, Kogyo Kagaku Zasshi, **68**, 1269 (1965).
229. M. Matsuda. N. Tokura, J. Polym. Sci. A, **2**, 4281 (1964).
230. M. Matsuda, S. Fujii, J. Polym. Sci. A-1, **5**, 2617 (1967).
231. M. Matsumoto, M. Maeda, J. Polym. Sci., **17**, 438 (1955).
232. M. Matsumoto, J. Ukida, G. Takayama, T. Eguchi, K. Mukumoto, K. Imai, Y. Kazusa, M. Maeda, Makromol. Chem. **32**, 13 (1959).
233. B. Matuska, J. Kossler, V. Srajer, Collect, Czech. Chem. Commun., **23**, 1456 (1958).
234. J. A. May, Jr., W. B. Smith, J. Phys. Chem., **72**, 216 (1968).
235. F. R. Mayo, J. Am. Chem. Soc., **65**, 2324 (1943).
236. F. R. Mayo, J. Am. Chem. Soc., **70**, 3689 (1948).
237. F. R. Mayo, J. Am. Chem. Soc., **75**, 6133 (1953).
238. F. R. Mayo. J. Am. Chem. Soc., **76**, 5392 (1954).
239. F. R. Mayo, C. W. Gould, J. Am. Oil Chem. Soc., **41**, 25 (1964).
240. F. R. Mayo, R. A. Gregg, M. S. Matheson, J. Am. Chem. Soc., **73**, 1691 (1951).
241. V. V. Mazurek, V. G. Gasan-Zade, G. T. Nesterchuk, Polym. Sci. USSR, **6**, 1587 (1964).
242. E. J. Meehan, I. M. Kolthoff, H. R. Sinha, J. Polym. Sci., **16**, 471 (1955).
243. E. J. Meehan, I. M. Kolthoff, P. R. Sinha, J. Polym. Sci. A-2, 4911 (1964).
244. F. W. Mellows, M. Burton, J. Phys. Chem., **66**, 2164 (1962).
245. H. W. Melville, L. Valentine, Trans. Faraday Soc., **46**, 210 (1950).
246. H. S. Mickley, A. S. Michaels, A. L. Moore, J. Polym. Sci., **60**, 121 (1962).
247. E. B. Milovskaya, T. G. Zhuravleva, Vysokomol. Soedin., Ser. A, **9**, 1128 (1967).
248. E. B. Milovskaya, T. G. Zhuravleva, L. V. Zamoyskaya, J. Polym. Sci., C, **16**, 899 (1967).
249. Y. Minoura, Y. Enomoto, Kogyo Kagaku Zasshi, **70**, 1021 (1967).
250. Y. Minoura, Y. Hayashi, M. Imoto, Kobunshi Kagaku, **15**, 260 (1958); from CA, **54** 2803F (1960).
251. Y. Minoura, T. Sugimura, J. Polym. Sci. A-1, **4**, 2721 (1966).

252. Y. Minoura, T. Sugimura, T. Hirahara, Kogyo Kagaku Zasshi, **70**, 357 (1967).
253. Y. Minoura, N. Yasumoto, T. Ishii, Kogyo Kagaku Zasshi, **65**, 1299 (1962); from CA, **58**, 1538D (1963).
254. Y. Minoura, N. Yasumoto, T. Ishii, Makromol. Chem., **71**, 159 (1964).
255. G. S. Misra, R. N. Chadha, Makromol. Chem., **23**, 134 (1957).
256. G. S. Misra, R. C. Rastogi, V. P. Gupta, Makromol. Chem., **50**, 72 (1961).
257. T. Miyake, M. Matsumoto, Kogyo Kagaku Zasshi, **62**, 1101 (1959); from CA, **57**, 15342A (1962).
258. Y. Mori, K. Sato, Y. Minoura, Kogyo Kagaku Zasshi, **61**, 462 (1958); from CA, **55**, 4021F (1961).
259. G. A. Mortimer, J. Polym. Sci. A-1, **4**, 881 and 1895 (1966).
260. G. A. Mortimer, L. C. Arnold, J. Am. Chem. Soc., **84**, 4986 (1962).
261. M. Morton, J. A. Cala, I. Piirma, J. Am. Chem. Soc., **78**, 5394 (1956).
262. M. Morton, I. Piirma, J. Am. Chem. Soc., **80**, 5596 (1958).
263. M. Morton, I. Piirma, J. Polym. Sci. A, **1**, 3043 (1963).
264. A. S. Nair, M. S. Muthana, Makromol. Chem., **47**, 114, 128 (1961).
265. H. Nakamoto, Y. Ogo, T. Imoto, Makromol. Chem., **111**, 93 (1968).
266. T. Nakata, T. Otsu, M. Imoto, J. Polym. Sci. A, **3**, 3383 (1965).
267. T. Nakaya, K. Ohashi, M. Imoto, Makromol. Chem., **114**, 201 (1968).
268. U. S. Nandi, P. Ghosh, S. R. Palit, Nature, **195**, 1197 (1962).
269. A. F. Nikolaev, N. V. Meiya, G. A. Balaev, Vysokomol. Soedin., **7**, 2122 (1965).
270. K. Noma, Y. Tajima, M. Niwa, Sci. Eng. Rev. Doshisha Univ., **3**, 91 (1962); from CA, **59**, 2955C (1963).
271. R. G. W. Norrish, J. P. Simons, Proc. Roy. Soc., Ser. A, **251**, 4 (1959).
272. A. M. North, A. M. Scallan, Polymer, **5**, 447 (1964).
273. K. Nozaki, Disc. Faraday Soc., **2**, 337 (1947).
274. J. L. O'Brien, F. Gornick, J. Am. Chem. Soc., **77**, 4757 (1955).
275. J. L. O'Brien, J. R. Panchak, T. G. Fox, Abstract of 124th ACS Meeting, Chicago, 1953.
276. S. Okamura, K. Katagiri, Makromol. Chem., **28**, 177 (1958).
277. S. Okamura, K. Katagiri, T. Motoyama, J. Polym. Sci., **43**, 509 (1960).
278. S. Okamura, T. Motoyama, J. Polym. Sci., **58**, 221 (1962).
279. S. Okamura, K. Takeya, Kobunshi Kagaku, **15**, 353 (1958); from CA, **54**, 8143 (1960).
280. M. Okawara, T. Nakai, E. Imoto, Kogyo Kagaku Zasshi, **69**, 973 (1966).
281. O. F. Olaj, J. W. Breitenbach, I. Hofreitere, Makromol. Chem., **91**, 264 (1966).
282. O. F. Olaj, J. W. Breitenbach, I. Hofreitere, Makromol. Chem., **110**, 72 (1967).
283. P. F. Onyon, Trans. Faraday Soc., **51**, 400 (1955).
284. P. F. Onyon, J. Polym. Sci., **22**, 19 (1956).
285. T. Otsu, Y. Kinoshita, M. Imoto, Makromol. Chem., **73**, 225 (1964).
286. T. Otsu, K. Nayatani, Makromol. Chem., **27**, 149 (1955).
287. T. Otsu, A. Shimizu, M. Imoto, J. Polym. Sci. B, **2**, 973 (1964).
288. T. Otsu, K. Tsuda, N. Kita, Mem. Fac. Eng., Osaka City Univ., **7**, 95 (1965); from CA, **66**, 95434U (1967).
289. S. R. Palit, S. K. Das, Proc. Roy. Soc., Ser. A, **226**, 82 (1954).
290. S. R. Palit, U. S. Nandi, N. G. Saha, J. Polym. Sci., **14**, 295 (1954).
291. S. K. Patra, D. Mangaraj, Makromol. Chem., **111**, 168 (1968).
292. J. Pavlinec, M. Lazar, J. Polym. Sci. C, **22**, 297 (1968).
293. L. H. Peebles, J. Polym. Sci. A, **3**, 341 (1965).
294. L. H. Peebles, J. T. Clarke, W. H. Stockmayer, J. Am. Chem. Soc., **82**, 4780 (1960).
295. J. J. Pellon, J. Polym. Sci., **43**, 537 (1960).
296. E. Perry, J. Polym. Sci., **54**, S-46 (1961).
297. R. Pierson, A. Costanza, A. Weinstein, J. Polym. Sci., **17**, 221 (1955).
298. G. Platau, F. R. Eirich, R. B. Mesrobian, A. E. Woodward, J. Polym. Sci., **39**, 357 (1959).
299. A. Prevot-Bernas, J. Sebban-Danon, J. Chem. Phys., **53**, 418 (1956).
300. W. A. Pryor, J. Phys. Chem., **67**, 519 (1963).
301. W. A. Pryor, D. M. Huston, T. R. Fiske, T. L. Pickering, E. Ciuffarin, J. Am. Chem. Soc., **86**, 4237 (1964).
302. W. A. Pryor, G. L. Kaplan, J. Am. Chem. Soc., **86**, 4234 (1964).
303. W. A. Pryor, A. Lee, C. E. Witt, J. Am. Chem. Soc., **86**, 4229 (1964).
304. W. A. Pryor, T. L. Pickering, J. Am. Chem. Soc., **84**, 2705 (1962).
305. W. A. Pryor, E. P. Pultinas, Jr., J. Am. Chem. Soc., **85**, 133 (1963).
306. P. V. T. Raghuram, U. S. Nandi, J. Polym. Sci. A-1, **5**, 2005 (1967).
307. S. P. Rao, M. Santhappa, Curr. Sci. (India), **34**, (6) 174 (1965).
308. S. P. Rao, M. Santhappa, J. Polym. Sci. A-1, **5**, 2681 (1967).
309. A. I. Restaino, W. N. Reed, J. Polym. Sci., **36**, 499 (1959).
310. E. H. Riddle, "Monomeric Acrylic Ester", Reinhold, NY, 1954, Ref. 14, p. 64.
311. J. C. Robb, E. Senogles, Trans. Faraday Soc., **58**, 708 (1962).
312. J. C. Robb, D. Vofsi, Trans, Faraday Soc., **55**, 558 (1959).
313. M. E. Rozenberg, A. F. Nikolaev, A. V. Pustovalova, Vysokomol. Soedin., **8**, 1155 (1966).
314. L. V. Ruban, A. L. Buchachenko, M. B. Neiman, Yu. V. Koknanov, Vysokomol. Soedin., **8**, 1642 (1966).
315. M. Ryska, M. Kolinsky, D. Lim, J. Polym. Sci. C, **16**, 621 (1967).
316. N. G. Saha, U. S. Nandi, S. R. Palit, J. Chem. Soc., 427 (1956).
317. N. G. Saha, U. S. Nandi, S. R. Palit, Chem. Soc., 7 (1958).
318. N. G. Saha, U. S. Nandi, S. R. Palit, Chem. Soc., 12 (1958).
319. I. Sakurada, K. Noma, Y. Ofuji, Kobunshi Kagaku, **20**, 481 (1963); from CA, **63**, 8487C (1965).

320. I. Sakurada, Y. Sakaguchi, K. Hashimoto, Kobunshi Kagaku, **19**, 593 (1962); from CA, **61**, 16159D (1964).
321. G. F. Santee, R. H. Marchessault, H. G. Clark, J. J. Kearny, V. Stannett, Makromol. Chem., **73**, 177 (1964).
322. M. Santhappa, V. M. Iyer, Current Sci. (India), **24**, 173 (1955).
323. M. Santhappa, V. S. Vaidhyanathan, Current Sci. (India) **23**, 259 (1954).
324. J. Scanian, Trans. Faraday Soc., **50**, 756 (1954).
325. E. Schonfeld, I. Waltcher, J. Polym. Sci., **35**, 536 (1959).
326. G. V. Schulz, G. Henrici, S. Olive, J. Polym. Sci., **17**, 45 (1955).
327. G. V. Schulz, G. Henrici, S. Olive, Z. Elektrochem., **60**, 296 (1956).
328. G. V. Schulz, L. Roberts-Nowakowska, Makromol. Chem., **80**, 36 (1964).
329. G. V. Schulz, D. J. Stein, Makromol. Chem., **52**, 1 (1962).
330. G. P. Scott, C. C. Soong, W.-S. Huang, J. L. Reynolds, J. Org. Chem., **29**, 83 (1964).
331. G. P. Scott, J. C. Wang, J. Org. Chem., **28**, 1314 (1963).
332. J. N. Sen, U. S. Nandi, S. R. Palit, J. Indian Chem. Soc., **40**, 729 (1963).
333. K. P. Shen, F. R. Eirich, J. Polym. Sci., **53**, 81 (1961).
334. A. Shimzu, T. Otsu, M. Imoto, Bull. Chem. Soc. Japan, **38**, 1535 (1965).
335. T. Shimomura, Y. Kuwabara, E. Tsuchida, I. Shinohara, Kogyo Kagaku Zasshi, **71**, 283 (1968).
336. T. Shimomura, E. Tsuchida, I. Shinohara, Mem. School Sci. Eng., Waseda Univ., **8**, 37 (1964); from CA, **63**, 14985A (1965).
337. T. Shimomura, E. Tsuchida, I. Shinohara, Mem. School Sci. Eng., Waseda Univ., **29**, 1 (1965); from CA, **65**, 10675A (1966).
338. M. Simonyi, F. Tudös, J. Pospisil, Eur. Polym. J., **3**, 101 (1967).
339. P. R. Sinha, K. L. Mallik, J. Indian Chem. Soc., **34**, 424 (1957).
340. C. Sivertz, J. Phys. Chem., **63**, 34 (1959).
341. B. Skrabal, L. Rosik, Chem. Prumysl, **8**, 46 (1958).
342. G. Smets, L. Convent, X. van der Borght, Makromol. Chem., **23**, 162 (1957).
343. G. Smets, L. Dehaes, Bull. Soc. Chim. Belges, **59**, 13 (1950).
344. W. V. Smith, J. Am. Chem. Soc., **68**, 2059 (1964).
345. R. D. Spencer, M. B. Fulton, U.S. Dept. Com., Office Tech. Serv., PB Rept., **144**, 990 (1961).
346. R. D. Spencer, M. B. Fulton, B. H. Beggs, Abstracts of 137th ACS Meeting, Cleveland, 1960.
347. N. T. Srinivasan, M. Santhappa, Makromol. Chem., **26**, 80 (1958).
348. D. J. Stein, Makromol. Chem., **76**, 170 (1964).
349. D. J. Stein, G. V. Schulz, Makromol. Chem., **38**, 248 (1960).
350. D. J. Stein, G. V. Schulz Makromol. Chem., **52**, 249 (1962).
351. W. H. Stockmayer, R. O. Howard, J. T. Clarke, J. Am. Chem. Soc., **75**, 1756 (1953).
352. T. J. Suen, Y. Jen, J. V. Lockwood, J. Polym. Sci., **31**, 481 (1958).
353. T. J. Suen, A. M. Schiller, W. N. Russell, Adv. Chem. Series No. 34, "Polymerization and Polycondensation Processes," ACS, Washington, DC, 1962, pp. 217–24.
354. T. Sugimura, Y. Ogata, Y. Minoura, J. Polym. Sci. A-1, **4**, 2747 (1966).
355. G. Takayama, Kobunshi Kagaku, **15**, 89 (1958); from CA, **53**, 868D (1959).
356. W. M. Thomas, E. H. Gleason, J. J. Pellon, J. Polym. Sci., **17**, 275 (1955).
357. C. F. Thompson, W. S. Port, L. P. Witnauer, J. Am. Chem. Soc., **81**, 2552 (1959).
358. A. P. Titov, I. A. Livshits, Zh. Obshch. Khim., **29**, 1605 (1959).
359. A. V. Tobolsky, B. Baysal, J. Am. Chem. Soc., **75**, 1757 (1953).
360. A. V. Tobolsky, J. Offenbach, J. Polym. Sci., **16**, 311 (1955).
361. T. Toda, J. Polym. Sci., **58**, 411 (1962).
362. N. Tokura, M. Matsuda, F. Yazaki, Makromol. Chem., **42**, 108 (1960).
363. A. C. Toohey, K. E. Weale, Trans. Faraday Soc., **58**, 2439 (1962).
364. A. C. Toohey, K. E. Weale, Trans. Faraday Soc., **58**, 2446 (1962).
365. E. Tsuchida, Y. Ohtani, H. Nakadai, I. Shinohara, Kogyo Kagaku Zasshi, **70**, 573 (1967).
366. E. Tsuchida, Y. Ohtani, H. Nakadai, I. Shinohara, Kogyo Kagaku Zasshi, **69**, 1230 (1966).
367. E. Tsuchida, T. Shimomura, K. Fujimori, Y. Ohtani, I. Shinohara, Kogyo Kagaku Zasshi, **70**, 566 (1967).
368. K. Tsuda, S. Kobayashi, T. Otsu, Bull, Chem. Soc. Japan, **38**, 1517 (1965).
369. K. Tsuda, T. Otsu, Bull. Chem. Soc. Japan, **39**, 2206 (1966).
370. F. Tudös, T. Berezhnykh-Földes, Eur. Polym. J., **2**, 229 (1966).
371. F. Tudös, I. Kende, M. Azori, J. Polym. Sci., **53**, 17 (1961).
372. F. Tudös, I. Kende, M. Azori, J. Polym. Sci. A, **1**, 1353 (1963).
373. F. Tudös, I. Kende, M. Azori, J. Polym. Sci. A, **1**, 1369 (1963).
374. J. Ulbricht, Faserforsch. Textiltech., **10**, 166 (1959)
375. J. Ulbricht, Faserforsch. Textiltech., **11**, 62 (1960).
376. J. Ulbricht, Z. Phys. Chem. (Frankfurt), **221**, 346 (1962).
377. J. Ulbricht, B. Sandner, Faserforsch. Textiltech., **17**, 208 (1966).
378. J. Ulbricht, R. Sourisseau, Faserforsch. Textiltech., **16**, 213 (1965).
379. T. Uno, K. Yoshida, Kobunshi Kagaku, **15**, 819 (1958); from CA, **54**, 20298D (1960).
380. C. A. Uraneck, J. E. Bulreigh, J. Appl. Polym. Sci., **12**, 1075 (1968).
381. W. H. Urry, F. W. Stacey, E. S. Huyser, O. O. Juveland, J. Am. Chem. Soc., **76**, 450 (1954).
382. V. Vaclavek, Chem. Prumysl, **10**, 103 (1960).
383. V. Vaclavek, J. Appl. Polym. Sci., **11**, 1881 (1967).
384. V. Vaclavek, J. Appl. Polym. Sci., **11**, 1903 (1967).

385. V. S. Vaidhyanathan, M. Santhappa, Makromol. Chem., **16**, 140 (1955).
386. R. L. Vale, W. G. P. Robertson, J. Polym. Sci., **33**, 518 (1958).
387. B. M. E. van der Hoff, J. Polym. Sci., **44**, 241 (1960).
388. B. M. E. van der Hoff, J. Polym. Sci., **48**, 175 (1960).
389. A. A. Vansheidt, G. Khardi, Acta Chim. Acad. Sci. Hung., **20**, 261 (1959); from CA, **54**, 6180B (1960).
390. A. A. Vansheidt, G. Khardi, Acta Chim. Acad. Sci. Hung., **20**, 381 (1959); from CA, **54**, 1152F (1960).
391. F. K. Velichko, I. P. Lavrent'ev, Yu. P. Chizhov, Izv. Akad. Nauk SSSR, Ser, Khim., 1966, 172; from CA, **64**, 12485C (1966).
392. G. Vidotto, A. Crosato-Amaldi, G. Talmini, Makromol. Chem., **114**, 217 (1968).
393. M. Vrancken, G. Smets, Makromol. Chem., **30**, 197 (1959).
394. J. Vuillemenot, B. Barbier, G. Riess, A. Banderet, J. Polym. Sci. A, **3**, 1969 (1965).
395. L. A. Wall, D. W. Brown, J. Polym. Sci., **14**, 513 (1954).
396. R. A. Wallace, K. L. Hadley, J. Polym. Sci. A-1, **4**, 71 (1966).
397. C. Walling, J. Am. Chem. Soc., **70**, 2561 (1948).
398. C. Walling, E. R. Briggs, J. Am. Chem. Soc., **68**, 1141 (1946).
399. C. Walling, Y. Chang, J. Am. Chem. Soc., **76**, 4878 (1954).
400. C. Walling, L. Heaton, J. Am. Chem. Soc., **87**, 38 (1965).
401. C. Walling, J. J. Pellon, J. Am. Chem. Soc., **79**, 4776 (1957).
402. M. Watanabe, H. Kiuchi, J. Polym. Sci., **58**, 103 (1962).
403. O. L. Wheeler, E. Lavin, R. N. Crozier, J. Polym. Sci., **9**, 157 (1952).
404. E. F. T. White, M. J. Zissell, J. Polym. Sci. A, **1**, 2189 (1963).
405. T. Yamamoto, Bull. Chem. Soc. Japan, **40**, 642 (1967).
406. T. Yamamoto, T. Otsu, J. Polym. Sci. B, **4**, 1039 (1966); see also Bull. Chem. Soc. Japan, **40**, 2449 (1967).
407. T. Yamamoto, T. Otsu, Koguo Kagaku Zasshi, **70**, 2403 (1967).
408. T. Yamamoto, T. Otsu, M. Imoto, Kogyo Kagaku Zasshi, **69**, 990 (1966).
409. S. D. Yevstratova, M. F. Margaritova, S. S. Medvedev, Polym. Sci. USSR, **5**, 681 (1964).
410. M. Yoshida, K. Tanouchi, Kobunshi Kangaku, **20**, 545 (1963); from Makromol. Chem., **71**, 216 (1964).
411. V. V. Zaitseva, V. D. Enal'ev, A. I. Yurzhenko, Vysokomol. Soedin., Ser. A, **9**, 1958 (1967).
412. V. M. Zhulin, M. G. Gonikberg, V. N. Zagorbinina, Dokl. Akad. Nauk SSSR, **163**, 106 (1965); from CA, **63**, 11706G (1965).
413. V. M. Zhulin, M. G. Gonikberg, V. N. Zagorbinina, Izv. Akad. Nauk SSSR, Ser. Khim., 827 (1966).
414. V. M. Zhulin, M. G. Bonikberg, V. N. Zagorbinina, Izv. Akad. Nauk SSSR, Ser. Khim., 997 (1966).
415. M. G. Zimina, N. P. Apukhtina, Kolloid Zh., **21**, 181 (1959).
416. J. Aoyagi, K. Kitamura, I. Shinohara, Kogyo Kagaku Zasshi, **73**, 2045 (1970).
417. J. Aoyagi, I. Shinohara, Kogyo Kagaku Zasshi, **74**, 1191 (1971).
418. T. Asahara, M. C. Chou, Bull. Chem. Soc. Japan, **42**, 1373 (1969).
419. C. Aso, T. Kunitake, M. Shinsenji, H. Miyazaki, J. Polym. Sci. A-1, **7**, 1497 (1969).
420. J. C. Bevington, M. Johnson, J. P. Sheen, Eur. Polym. J., **8**, 209 (1972).
421. G. C. Bhaduri, U. S. Nandi, Makroml. Chem., **128**, 176 (1969).
422. G. C. Bhaduri, U. S. Nandi, Makromol. Chem., **128**, 183 (1969).
423. B. R. Bhattacharyya, U. S. Nandi, Makromol. Chem., **149**, 231 (1971).
424. B. R. Bhattacharyya, U. S. Nandi, Makromol. Chem., **149**, 243 (1971).
425. D. Braun, F. Weiss, Makromol. Chem., **138**, 83 (1970).
426. D. Braun, F. Weiss, Angew. Makromol. Chem., **15**, 127 (1971).
427. C-T. Chen, W-D. Huang, J. Chin. Chem. Soc., **16**, 46 (1969); from CA, **71**, 102247U (1969).
428. Y. Choshi, A. Tsuji, G. Akazome, K. Murai, Kogyo Kagaku Zasshi, **71**, 908 (1968); from CA, **69**, 67745Q (1968).
429. N. N. Dass, M. H. George, Eur. Polym. J., **6**, 897 (1970).
430. H. J. Dietrich, M. A. Raymond, J. Macromol. Sci. Chem. A, **6**, 191 (1972).
431. J. P. Fischer, Makromol. Chem., **155**, 211 (1972).
432. J. P. Fischer, Makromol. Chem., **155**, 227 (1972).
433. J. P. Fischer, W. Lüders, Makromol. Chem. **155**, 239 (1972).
434. R. Kh. Freidlina, A. B. Terent'ev, N. S. Ikonnikov, Dokl. Akad. Nauk SSSR, **193**, 605 (1970).
435. R. Kh. Freidlina, A. V. Terent'ev, N. S. Ikonnikov, Izv. Akad. Nauk SSSR, Ser. Khim., 1970, 554.
436. S. S. Frolov, T. M. Slivchenko, Izv. Vyssh. Ucheb. Zaved., Khim. Khim. Tekhnol., **14**, 1264 (1971); from CA, **76**, 25660X (1972).
437. I. Geczy, H. I. Nasr, Acta Chim. (Budapest), **70**, 319 (1971); from CA, **76**, 100150M (1972).
438. S. N. Gupta, U. S. Nandi, J. Polym. Sci. A-1, **8**, 1493 (1970).
439. D. A. J. Harker, R. A. M. Thomson, I. R. Walters, Trans. Faraday Soc., **67**, 3057 (1971).
440. M. Imoto, K. Ohashi, Makromol. Chem., **117**, 117 (1968).
441. E. F. Jordan, Jr., B. Artymyshyn, A. N. Wrigley, J. Polym. Sci. A-1, **7**, 2605 (1969).
442. I. Kar, B. M. Mandal, S. R. Palit, J. Polym. Sci. A-1, **7**, 2829 (1969).
443. S. Kobayashi, Kogyo Kagaku Zasshi, **72**, 2511 (1969).
444. C. Kwang-Fu, Kobunshi Kagaku, **29**, 233 (1972).
445. I. M. Likhterova, E. M. Lukina, Zh. Obshch. Khim., **42**, 194 (1972); from CA, **76**, 127504S (1972).
446. A. Matsumoto, M. Oiwa, J. Polym. Sci. A-1, **10**, 103 (1972).
447. J. A. May, Jr., W. B. Smith, J. Phys. Chem., **72**, 2993 (1968).
448. L. A. Miller, V. Stannett, J. Polym. Sci. A-1, **7**, 3159 (1969).
449. Y. Minoura, M. Mitoh, Kogyo Kagaku Zasshi, **74**, 747 (1971).
450. Y. Minoura, A. Tabuse, Kogyo Kagaku Zasshi, **74**, 990 (1971).
451. Y. Minoura, H. Toshima, J. Polym. Sci. A-1, **7**, 2837 (1969).
452. M. Modena, P. Piccardi, Eur. Polym. J., **7**, 1 (1971).
453. N. Mogi, M. Shindo, Kogyo Kagaku Zasshi, **73**, 786 (1970).

454. G. A. Mortimer, J. Polym. Sci. A-1, **8**, 1513 (1970).
455. G. A. Mortimer, J. Polym. Sci. A-1, **8**, 1535 (1970).
456. G. A. Mortimer, J. Polym. Sci. A-1, **8**, 1543 (1970).
457. G. A. Mortimer, J. Polym. Sci. A-1, **10**, 163 (1972).
458. I. G. Murgulescu, I. Vlagiu, Rev. Roum. Chim., **14**, 411 (1969); from CA, **71**, 71773F (1969).
459. T. Nakaya, M. Tanaka, M. Imoto, Makromol. Chem., **149**, 221 (1971).
460. J. H. Ok, S. B. Pak, Hwahak Kwa Hwahak Kongop, 1971, 239 (Korean); from CA, **77**, 20147U (1972).
461. T. Ota, S. Masuda, Kogyo Kagaku Zasshi, **73**, 2020 (1970).
462. T. Ota, S. Masuda, C. Aoyama, M. Ebisudani, Kogyo Kagaku Zasshi, **74**, 994 (1971).
463. G. S. Park, D. G. Smith, Trans. Faraday Soc., **65**, 1854 (1969).
464. F. Patat, P. Mehnert, Monatsh. Chem., **98**, 538 (1967).
465. S. P. Potnis, A. M. Deshpande, Makromol. Chem., **125**, 48 (1969).
466. S. P. Potnis, A. M. Deshpande, Makromol. Chem., **153**, 139 (1972).
467. W. A. Pryor, J. H. Coco, Macromolecules, **3**, 500 (1970).
468. W. A. Pryor, T. R. Fiske, Macromolecules, **2**, 62 (1969).
469. W. A. Pryor, T.-L. Huang, Macromolecules, **2**, 70 (1969).
470. E. Pulat, C. B. Senvar, Commun. Fac. Sci. Univ. Ankara, Ser. B, **15**, (3) 25 (1968); from CA, **72**, 55934Z (1970).
471. P. V. T. Raghuram, U. S. Nandi, J. Polym. Sci. A-1, **7**, 2379 (1969).
472. P. V. T. Raghuram, U. S. Nandi, J. Polym. Sci. A-1, **8**, 3079 (1970).
473. K. K. Roy, D. Pramanick, S. R. Palit, Makromol. Chem., **153**, 71 (1972).
474. J. C. Saam, D. J. Gordon, J. Polym. Sci. A-1, **8**, 2509 (1970).
475. T. Saegusa, Y. Ito, N. Yasuda, Polym. J., **1**, 591 (1970).
476. S. K. Saha, A. K. Chaudhuri, J. Polym. Sci. A-1, **10**, 797 (1972).
477. G. P. Scott, A. M. R. Elghoul, J. Polym. Sci. A-1, **8**, 2255 (1970).
478. G. P. Scott, F. J. Foster, Macromolecules, **2**, 428 (1969).
479. R. B. Seymour, J. M. Sosa, V. J. Patel, J. Paint Technol., **43**, (563), 45 (1971).
480. T. Shimomura, E. Tsuchida, I. Shinohara, Kogyo Kagaku Zasshi, **71**, 1074 (1968); from CA, **69**, 97249Y (1968).
481. E. Staudner, J. Beniska, Eur. Polym. J. Suppl., 1969, 537.
482. F. Suganuma, H. Mitsui, S. Machi, M. Hagiwara, T. Kagiya, J. Polym. Sci. A-1, **6**, 3127 (1968).
483. K. Sugiyama, T. Nakaya, M. Imoto, J. Polym. Sci. A-1, **10**, 205 (1972).
484. A. B. Terent'ev, N. S. Ikonnikov, R. Kh. Freidlina, Izv. Akad. Nauk SSSR, Ser. Khim, 73 (1971).
485. A. B. Terent'ev, N. S. Ikonnikov, R. Kh. Freidlina, Dokl. Akad. Nauk USSR, **196**, 1373 (1971).
486. A. S. Tevlina, H. S. Kolesnikov, S. N. Sividova, V. V. Ryltsev, Vysokomol. Soedin., Ser. A, **9**, 2473 (1967); Polym. Sci. USSR, **9**, 2797 (1967).
487. R. A. M. Thomson, I. R. Walters, Trans. Faraday Soc., **67**, 3046 (1971).
488. R. A. M. Thomson, I. R. Walters, J. R. King, J. Polym. Sci., B, **10**, 63 (1972).
489. E. Tsuchida, T. Tomono, Kogyo Kagaku Zasshi, **73**, 2040 (1970).
490. M. Tsunooka, M. Fujii, N. Ando, M. Tanaka, N. Murata, Kogyo Kagaku Zasshi, **73**, 805 (1970).
491. M. Tsunooka, T. Higuchi, M. Fujii, M. Tanaka, N. Murata, Kogyo Kagaku Zasshi, **73**, 596 (1970).
492. G. Vidotto, S. Brugnaro, G. Talamini, Makromol. Chem., **140**, 249 (1970).
493. G. Vidotto, S. Brugnaro, G. Talamini, Makromol. Chem., **140**, 263 (1970).
494. F. Yamada, M. Kanbe, I. Shinohara, Memo. Sch. Sci. Eng. Waseda Univ. 1969, No. 33, 67: from CA, **73**, 88255C (1970).
495. T. Yamamoto, M. Hasegawa, T. Otsu, Bull. Chem. Soc. Japan, **42**, 1364 (1969).
496. T. Yamamoto, S. Nakamura, M. Hasegawa, T. Otsu, Kogyo Kagaku Zasshi, **72**, 727 (1969).
497. T. Yamamoto, T. Otsu, J. Polym. Sci. A-1, **7**, 1279 (1969).
498. N. Yamashita, T. Seita, M. Yoshihara, T. Maeshima, Kogyo Kagaku Zasshi, **74**, 2157 (1971).
499. N. Yamashita, T. Seita, M. Yoshihara, T. Maeshima, J. Polym. Sci. B, **9**, 641 (1971).
500. V. M. Zhulin, M. G. Gonikberg, A. L. Goff, V. N. Zagorbinina, Vysokomol. Soedin., Ser. A, **11**, 777 (1969).
501. N. Ya. Kaloforov, E. Borsig, J. Polym. Sci., Polym. Chem. Ed., **11**, (10) 2665 (1973).
502. O. A. Aminova, B. D. Saidov, Dokl. Akad. Nauk Uzb. SSSR, **4**, 45 (1979).
503. D. Pramanick, A. K. Chaterjee, Eur. Polym. J., **16**, (9) 895 (1980).
504. D. A. J. Harker, R. A. M. Thomson, I. R. Walters, Trans. Faraday Soc., **67**, (Pt. 10) 3057 (1971).
505. U. A. Safaev, Kh. Rakhmatullaev, A. T. Dzhalilov, Uzb. Khim. Zh., **3**, 42 (1983).
506. E. Abuin, E. A. Lissi, J. Macromol. Sci., Chem., **13** (8), 1147 (1979).
507. G. P. Scott, F. J. Foster, Macromolecules, **2** (4), 428 (1969).
508. K. Yokota, H. Tomioka, A. Tazumi, J. Macromol. Sci., Chem., **6** (7), 1337 (1972).
509. M. Bezdek, J. Exner, J. Kriz, F. Hrabak, Makromol. Chem., **147**, 1 (1971).
510. A. Fehervari, E. Boros Gyevi, T. Foldes-Bereznish, F. Tudös, J. Macromol. Sci. A: Chem., **18** (3), 431 (1982).
511. B. A. Englin, T. A. Onishchenko, R. Kh. Freidlina, Izv. Akad. Nauk SSSR, Ser, Khim. **12**, 2542 (1971).
512. M. Raetzsch, I. Zschach, Plaste Kaut., **21** (5), 345 (1974).
513. T. Foldes-Bereznich, M. Szesztay, F. Tudös, J. Macromol. Sci. A: Chem., **16** (5), 977 (1981).
514. J. Pavlinec, E. Borsig, J. Polym. Sci., Polym. Chem. Ed., **19** (9), 2305 (1981).
515. N. S. Enikolopyan, B. R. Smirnov, G. V. Ponomarev, I. M. Belgovskii, J. Polym. Sci., Polym. Chem. Ed., **19** (4), 879 (1981).
516. A. B. Terentev, N. S. Ikonnikov, R. Kh. Freidlina, Dokl. Akad. Nauk SSSR, **196**, (6) 1373 (1971).

517. A. G. Shostenko, V. E. Myshkin, Vysokomol. Soedin., Ser. B, **20**, 163 (1978).
518. Sl. A. Karapetyan, G. P. Shakhovskoi, N. A. Grigorév, V. M. Zhulin, B. A. Englin, Izv. Akad. Nauk SSSR, Ser. Khim., **6**, 1303 (1970).
519. B. A. Englin, B. N. Ospipov, Izv. Akad. Nauk SSSR, Ser. Khim., **1**, 65 (1971).
520. C. H. Bamford, J. Chem. Soc., Faraday Trans., **1**, 72 (12), 2805 (1976).
521. M. R. Lachinov, T. R. Aslamazova, V. P. Zubov, V. A. Kabanov, Vysokomol. Soedin., Ser. A, **17** (5), 1146 (1975).
522. V. P. Naidenov, V. G. Syromyatnikov, Ukr. Khim. Zh. (Russ. Ed.), **45** (10), 982 (1979).
523. P. A. Zagorets, A. G. Shostenko, A. M. Dodonov, N. P. Tarsova, Zh. Org. Khim., **10** (10), 2093 (1974).
524. A. P. Titov, V. V. Kotov, Vysokomol. Soedin., Ser. B, **10** (7), 495 (1968).
525. B. R. Smirnov, V. E. Mironychev, I. V. Golikov, M. M. Mogilevich, I. V. Enikolopov, Depos. Doc., SPSTL 598 Khp-D82, 9 pp. Avail. SPSTL (1982).
526. M. Niwa, N. Higashi, Sci. Eng. Rev. Doshisha Univ., **24** (3), 129 (1983).
527. O. F. Olaj, J. W. Breitenbach, I. Hofreiter, Rec. Chem. Progr., **30** (2), 87 (1969).
528. B. R. Smirnov, A. P. Marchenko, G. V. Korolev, I. M. Belgovskii, N. S. Enikolopyan, Vysokomol. Soedin., Ser. A, **23** (5), 1042 (1981).
529. R. B. Seymour, V. Patel, J. Paint Technol., **44** (569), 53 (1972).
530. N. M. Kozyreva, A. I. Kirilin, V. V. Korshak, Plast. Massy, **2**, 24 (1983).
531. N. G. Podosenova, E. G. Zotilov, V. P. Budtov, Vysokomol. Soedin., Ser. A, **25** (1), 43 (1983).
532. S. K. Verma, K. Wendler, M. Fedtke, Acta Polym., **35** (1), 58 (1984).
533. A. Hrivik, J. Beniska, E. Hudecova, Chem. Zvesti, **37** (4), 503 (1983).
534. S. Raghunath, M. H. Rao, K. N. Rao, Radiat, Phys. Chem., **22** (6), 1011 (1983).
535. K. Kodaira, K. Ito, Nagoya Kogyo Gijutsu Shikensho Hokoku, **31** (1/2), 22 (1982).
536. K. Kodaira, T. Ito, M. Omi, Y. Onishi, K. Ito, Nagoya Kogyo Gijutsu Shik, Hokoku, **31** (6/7), 186 (1982).
537. V. Kim, A. G. Shostenko, M. D. Gasparyan, React. Kinet. Catal. Lett., **12** (4), 479 (1979).
538. V. A. Kabanov, D. A. Topchiev, G. T. Nazhmethdinova, Izv. Akad. Nauk SSSR, Ser. Khim., **9**, 2146 (1983).
539. L. A. Smirnova, Yu. D. Semchikov, L. I. Kamyshenkova, T. G. Sveshnikova, A. N. Egorochkin, G. S. Kalinina, B. A. Egorov, Vysokomol. Soedin., Ser. A, **24** (5), 999 (1982).
540. K. Yamamoto, M. Sugimoto, J. Makromol. Sci. A: Chem., **13** (8), 1067 (1979).
541. K. Matsuo, G. W. Nelb, R. G. Nelb, W. H. Stockmayer, Macromolecules, **10** (3), 654 (1977).
542. J. Lokaj, F. Hrabak, Eur. Polym. J., **14** (12), 1039 (1978).
543. M. Iino, M. Igarashi, M. Matsuda, Macromolecules, **12** (4), 697 (1979).
544. M. Yoshihara, N. Miryamoto, M. Fukumura, T. Maeshima, J. Polym. Sci., Polym. Lett. Ed., **14** (2), 97 (1976).
545. W. A. Pryor, Tetrahedron Lett., 1201 (1963).
546. T. Otsu, Radical Polymerization – I, Kapakudojin, Koyoto, 1971, p. 143.
547. I. Grotewold, M. M. Hirschler, J. Polym. Sci., Polym. Chem. Ed., **15**, 383 (1977).
548. U. S. Nandi, M. Singh, P. V. T. Raghuram, Makromol. Chem., **183** (8), 1467 (1982).
549. B. Cellard, Ch. Pichot, A. Revillon, Makromol. Chem., **183** (8), 1935 (1982).
550. L. Vrhovac, J. Velickovic, D. Filipovic, Makromol. Chem., **185** (8), 1637 (1984).
551. B. Boutevin, Y. Piétrasanta, Makromol. Chem., **186** (4), 831 (1985).
552. M. A. Bruk, S. A. Pavlov, A. D. Abkin, Dokl. Akad. Nauk SSSR, **245**, 626 (1979).
553. W. Berger, H. J. Schneider, H. Schmidt, Faserforsch. Textiltech., **27** (8), 437 (1976).
554. J. C. Saam, D. J. Gordon, J. Polym. Sci., Part. A-1, **8** (9), 2509 (1970).
555. T. Ota, Kenkyu Hokoku–Asyhi Garasu Kogyo Gijutsu Shoreikai, **42**, 65 (1983).
556. B. R. Smirnov, I. M. Bel'Govskii, G. V. Ponomarev, N. S. Enikolopyan, Dokl. Akad. Nauk SSSR, **254** (1), 127 (Chem.).
557. R. Kh. Freidlina, A. B. Terentjev, N. S. Ikonnikov, Dokl. Akad. Nauk, **3**, 193, 605 (1970).
558. R. Kh. Freidlina, A. B. Terentjev, N. S. Ikonnikov, Isv. Akad. Nauk, Ser. Chem., 554 (1970).
559. A. B. Terentijev, N. S. Ikonnikov, R. Kh. Freidlina, Isv. Akad. Nauk, Ser. Chem., **1**, 73 (1971).
560. C. F. Jasso, E. Mendizabal, M. E. Hernandez, Rev. Plast. Mod., **62**, 823 (1991); from CA, **117**, 131596f (1992).
561. I. Capek, Collect. Czech. Chem. Commun., **51**, 2546 (1986).
562. F. Vidal, R. G. Gilbert, Macromol. Chem. Phys., **197**, 1835 (1996).
563. K. G. Suddaaby, D. R. Maloney, D. M. Haddleton, Macromolecules, **30**, 702 (1997).
564. L. I. Abramova, E. N. Zilberman, V. V. Leshin, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **30**, 117 (1987); from CA, **107**, 237343e (1987).
565. B. Boutevin, Y. Pietrasanta, Makromol. Chem., **186**, 831 (1985).
566. G. Bauduin, B. Boutevin, J. P. Mistral, L. Sarraf, Makromol. Chem., **186**, 1445 (1985).
567. C. A. Barson, J. C. Bevington, B. J. Hunt, J. Polym. Sci., Part A, Polym. Chem., **34**, 227 (1996).
568. C. A. Barson, J. C. Bevington, B. J. Hunt, J. Polym. Sci., Part A, Polym. Chem., **33**, 863 (1995).
569. D. Braun, P. Hempler, Polym. Bull. (Berlin), **30**, 55 (1993).
570. T. T. Vasileva, V. A. Kochetkova, B. V. Nelyubin, V. I. Dostovalova, R. K. Freidlina, Izv. Akad. Nauk SSSR, Ser. Khim., 808b (1987); from CA, **108**, 111753r (1988).
571. G. Bauduin, B. Boutevin, B. Pucci, J. P. Rigaud, Makromol. Chem., **188**, 2339 (1987).

572. T. T. Vasileva, V. A. Kochetkova, B. V. Nelyubin, R. K. Freidlina, Izv. Akad. Nauk. SSSR, Ser. Khim., 1397 (1986); from CA, **106**, 11993e (1987).

573. F. Jahanzad, M. Kazemi, S. Sajjadi, F. A. Taromi, Polymer, **34**, 3542 (1993).

574. B. K. Dietrich, W. A. Pryor, S. J. Wu, J. Appl. Polym. Sci., **36**, 1129 (1988).

575. S. Y. Lee, Y. J. Shin, Pollimo, **20**, 233 (1966); from CA, **124**, 318076y (1996).

Photopolymerization Reactions

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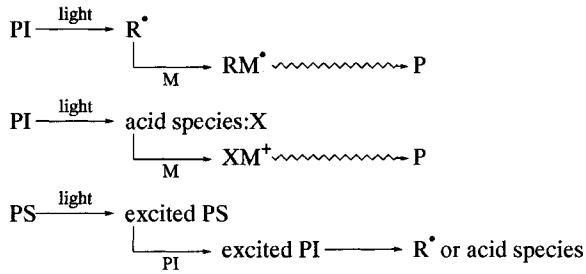
A. INTRODUCTION

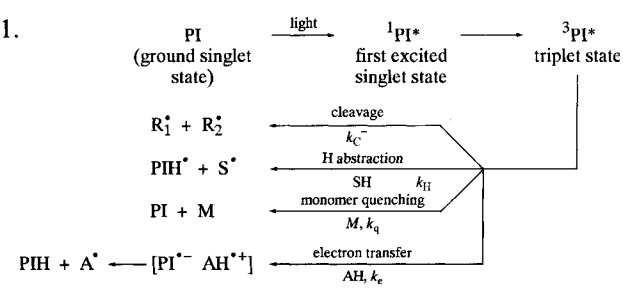
UV curing technologies use light beams to start photochemical and chemical reactions in organic materials (monomers, oligomers, prepolymers, polymers), mostly through a Photo-Induced Polymerization (PIP) reaction. This leads to the formation of a new polymeric material whose applications lie in various industrial sectors, such as coatings, graphic arts, imaging, microelectronics, etc. Specific advantages of these technologies over the usual thermal operations are rapid through-cure, solvent-free formulation, room temperature treatment and low energy requirements.

This PIP process is concerned with the creation of a polymer P through a chain radical or cationic reaction initiated by light in the presence of a photoinitiator (PI) and a coupled PI/photosensitizer (PS):



The reactivities of PI and PS govern, for a large part, the practical efficiency of the PIP reaction. The present chapter reports typical data obtained (through time-resolved laser spectroscopy experiments) on the excited state processes in PI and PS occurring after the absorption of the photon. Rate constants reported in the following tables correspond to the following processes:

1.



2. $R^{\bullet}$ (or $A^{\bullet}$ or $S^{\bullet}\ldots$) + M $\xrightarrow{k_{RM}}$ $RM^{\bullet}$

3. $^3PI^*$ + light stabilizer (LS) $\xrightarrow{k_a}$

4. $^3PS^*$ + PI $\xrightarrow{k_T}$ excitation transfer

5. τ_T°: Triplet state lifetime under the given conditions (equal to the reciprocal value of the sum of the first-order rate constants of the different processes)
   τ_T: Triplet state lifetime in the presence of a given additive

6. $^3PS^*$ + cationic photoinitiator C^+ $\xrightarrow{k_e}$ $PS^{\bullet+}$ $C^{\bullet}$
   (a) Energy transfer can sometimes occur:
   $^3PS^*$ + C^+ ⟶ PS + C^{+*}
   (b) Energy and electron transfer can also occur in the first excited singlet state $^1PS^*$.

Detailed data are available, especially in several chapters of two edited books (1) and in a recent monograph (2). Few data are known on the photopolymerization itself and largely depend on the practical formulation used as well as the experimental conditions. Typically, one photon absorbed can lead to ~10000 polymerized double bonds (3). Rate constants of propagation k_p and termination k_t for a polyurethane acrylate resin containing an acrylate monomer (weight ratio, 1 : 1) as reactive diluent are $\sim 10^4$ l/mol/s and 3×10^4 l/mol/s respectively (when half of the double bonds have been polymerized) (3).

B. TABLES

TABLE 1. RATE CONSTANTS OF CLEAVAGE, ELECTRON TRANSFER AND MONOMER QUENCHING IN RADICAL PHOTOINITIATORS[a]

Compound	$10^{-8}k_c$ (s^{-1})	$10^{-9}k_e$ (l/mol/s)	$10^{-6}k_q$ (l/mol/s)	Monomers[b]/amines[c]/ solvents[d]	Refs.
IIa		1.3	66	M_1, AH_1, S_1	4
IIb		2.0	150		
IIc		1.2	2.5		
IId		1.2	2		
IIe		0.2	1		
IIf		0.07	13		
IIg		1.5	180		
IIh		0.6	8		
IIi		0.27	0.05		
IIa			3200	M_2, S_2	21
			360	M_5, S_2	21
			5.4	M_3, S_2	21
			34	M_4, S_2	21
IIa		0.003[e]			22
		0.001[f]			
IIa		7		S_2, AH_3	26
		2		S_2, AH_4	
		1.9		S_2, AH_2	
		2		S_2, AH_5	
		0.56		S_2, AH_6	
IIj	10			S_3	23
IIk		0.5	1500	M_7, AH_2, S_4	25
III		0.9	1550		
IIm		0.0064[e]		S_2	26
IIn		0.0025[e]			
IIp		0.0056[e]			
IIq		0.0043[e]			
IIr			56	S_6, M_1	26
IIs			110		
IIr			5500	S_6, M_2	26
IIs			7400		
IIr			5100	S_6, M_5	26
IIs			6500		
IIr			49	S_6, M_3	26
IIs			140		
IIr			9.4	S_6, M_4	26
IIs			50		
I2a	7.14		250	M_1, AH_1, S_1	5
I2b	0.87	0.26	20		
I2c	0.003	<0.001	4.5		
I2d	0.00025	0.08	8		
I2e	0.83	0.20	22		
I2f	0.003	0.08	10		
I2g	0.67	0.17	29		
I2h	1.18	0.15	20		
I2i	0.80	0.05	21		

TABLE 1. *cont'd*

Compound	$10^{-8}k_c$ (s^{-1})	$10^{-9}k_e$ (l/mol/s)	$10^{-6}k_q$ (l/mol/s)	Monomers[b]/amines[c]/ solvents[d]	Refs.
I3a	10		≥200	M_1, AH_1, S_1	6
I3b	1		2		
I3c	25				
I3d	0.005	$<10^{-4}$			
I3e	10				
I3f	0.006	$\approx 10^{-4}$			
I3g	0.007	$<10^{-4}$			
I3h	0.13	$<10^{-4}$			
I4a	≥≥100			S_1	7
I4b	≥≥100				
I4c	≥≥100				
I4d	≥100				
I4e			8100	M_2, S_2	7
I4f			4800		
I4g			6500		
I4h			1500		
I4i			190		
I4e			480	M_1, S_2	7
I4f			800		
I4g			1000		
I4h			200		
I4i			13		
I4f			60	M_3, S_2	7
I4g			17		
I4h			<1		
I4i			5.7		
I4f			1300	M_4, S_2	7
I4g			700		
I4h			380		
I4i			180		
I5a	≥10				8
I6a	7×10^{-3}		0.7	M_1, S_1	9
I7a		0.7	0.05	M_1, AH_1, S_1	9
I7b		6×10^{-5}	440		
I8a		0.85	0.35	M_1, AH_1, S_1	9
I9a		2.5	11	M_1, AH_1, S_1	9
I10a	≥10				10
I11a	0.8			M_1, AH_1, S_1	10
I11b	≥1				
I11c	0.12	0.9	700		
I11d	0.01	0.4	1200		
I12a			480	M_1, S_2	11
I12b			800		
I12c			1000		
I12d			200		
I12e			13		
I12a			8100	M_2, S_2	11
I12b			4800		
I12c			6500		
I12d			1500		
I12e			190		
I12b			6	M_3, S_2	11
I12c			17		
I12d			<10		
I12e			5.7		
I12b			1300	M_4, S_2	11
I12c			700		
I12d			380		
I12e			180		
I13a	1.4	1.6		AH_2, S_2	12
I14a			11000	M_1, AH_1, S_1	13
I14b		6.5	2900		
I14c		6.0	15		
I14d		3.5	1200		
I15a	>5			S_2	14
I15b	4				
I15c	>5				

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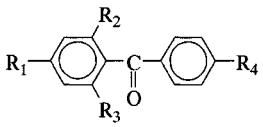
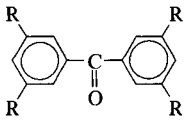
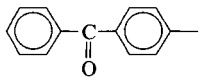
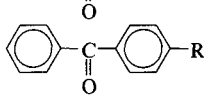
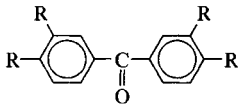
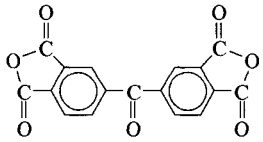
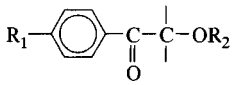
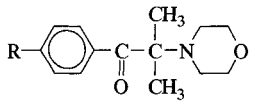
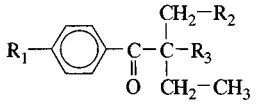
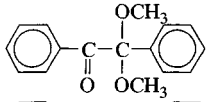
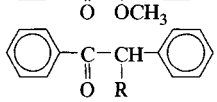
TABLE 1. *cont'd*

Compound	$10^{-8}k_c$ (s^{-1})	$10^{-9}k_e$ (l/mol/s)	$10^{-6}k_q$ (l/mol/s)	Monomers[b]/amines[c]/ solvents[d]	Refs.
I15d	3.3				
I16a		1.5	180	AH_1, S_1	15
I16b	>3				
I17a	0.05		300	S_1	16
I18a	0.16			S_1	17
I19a	0.3	0.7	100	S_1	18
I20a	29			S_2	18
I21a	60			S_2	18
I22a	~10			S_3	19
I22b	~5		93		
I22c	0.02		2.3		
I22d	0.014		2.0		
I23a		6	15	M_1, AH_3, S_2	20
I23b		5	2		
I23c		4	3		
I23d		4	3		
I23e		0.01		AH_2, S_5	24
I23f		0.017			
I23a		2.5	3000	M_2, S_2, AH_2	20
I23b			6000		
I23c			6000		
I23d			6000		
I23a			0.2	M_3, S_2	20
I23b			0.02		
I23c			0.03		
I23d			0.04		
I23a			4	M_4, S_2	20
I23b			0.4		
I23c			1		
I23d			1		
I23a			40	M_5, S_2	20
I23b			30		
I23c			5		
I23d			6		
I23a			1	M_6, S_2	20
I23b			1		
I23c			0.3		
I23d			0.3		
I23a		8		S_2, AH_4	26
I23b		6			
I23c		6			
I23d		6			
I23a			26	S_1, M_8	13
I23a			23	S_1, M_9	13
I23b		8		S_2, AH_7	26
I23b		0.85		S_2, AH_8	26
I23g		$<10^{-5}$	65	M_7, S_4	25
I23h		$<10^{-5}$	37		
I24			0.016	S_2, M_1	37
I25			0.006	S_2, M_{10}	37
I26			0.007	S_2, M_{10}	37
I27		4.4[f]		S_2	39
I28		0.3			
I29		0.015		S_7, AH_9	41

[a]Compound chemistries given in Annex to Table 1.
[b]*Monomers:* M_1 – methyl methacrylate; M_2 – styrene; M_3 – vinyl acetate; M_4 – acrylonitrile; M_5 – vinyl pyrrolidone; M_6 – butylvinylether; M_7 – acrylamide; M_8 – PETA; M_9 – TMPTA; M_{10} – butylmethacrylate.
[c]*Amines:* AH_1 – methyl diethanolamine; AH_2 – triethylamine; AH_3 – ethyl-4-(dimethylamino)-benzoate; AH_4 – bis-(2-hydroxy ethyl)-methylamine; AH_5 – methyl-4-amino benzoate; AH_6 – 1,4-diazabicyclo [2.2.2] octane (DABCO); AH_7 – dimethylamine; AH_8 – 2-(dimethyl aminoethyl)-benzoate; AH_9 – diethylaniline.
[d]*Solvents:* S_1 – toluene; S_2 – benzene; S_3 – acetonitrile; S_4 – water; S_5 – 2-propanol; S_6 – acetone; S_7 – ethanol.
[e]Rate constant of H abstraction by THF.
[f]Rate constant of H abstraction by isopropanol.

ANNEX TO TABLE 1. PHOTOINITIATOR COMPOUND CHEMISTRIES

	Compound	R	R_1	R_2	R_3	R_4
	I1a		H	H	H	H
	I1b		H	H	H	OCH_3
	I1c		H	H	H	SCH_3
	I1d		H	H	H	S-ϕ
	I1e		H	H	H	$N(CH_3)_2$
	I1f		$N(CH_3)_2$	H	H	$N(CH_3)_2$
	I1g		O-ϕ	H	H	O-ϕ
	I1h		CH_3	CH_3	CH_3	H
	I1i		H	H	H	ϕ
	I1j	COOOR				
	I1k	$-CH_2N(CH_3)_3^+Cl^-$				
	I1l	$-CH_2SO_3^-Na^+$				
	I1m	Cl				
	I1n	OCH_3				
	I1p	F				
	I1q	COOH				
	I1r	$COOCH_3$				
	I1s					
	I2a		H	H		
	I2b		OCH_3	H		
	I2c		$N(CH_3)_2$	H		
	I2d		SCH_3	H		
	I2e		$O(CH_2)_2OH$	H		
	I2f		$S(CH_2)_2OH$	H		
	I2g		$O(CH_2)_2Si(C_2H_5O)_3$	H		
	I2h		$O(CH_2)_2OCOCH{=}CH_2$	H		
	I2i		$O(CH_2)_2OCOCH{=}CH_2$	$COCH{=}CH_2$		
	I3a	H				
	I3b	SCH_3				
	I3c	OCH_3				
	I3d	$N(CH_3)_2$				
	I3e		H	C_6H_5	$N(CH_3)_2$	
	I3f			C_6H_5	$N(CH_3)_2$	
	I3g			$CH{=}CH_2$	$N(CH_3)_2$	
	I3h			$CH{=}CH_2$		
	I4a					
	I4b	OCH_3				
	I4c	OC_4H_9				
	I4d	$OCH(CH_3)_2$				
	I4e	OH				
	I4f	$OCOCH_3$				
	I4g	H				
	I4h	CH_3				
	I4i					

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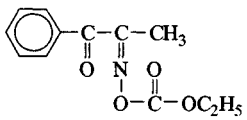
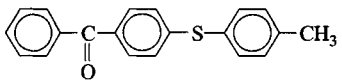
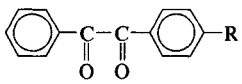
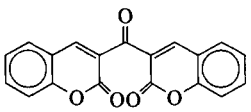
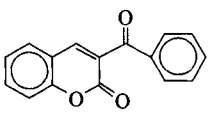
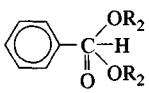
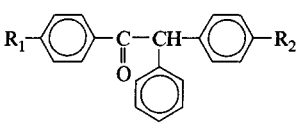
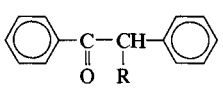
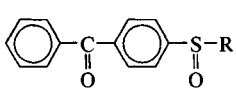
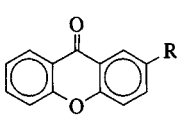
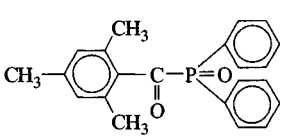
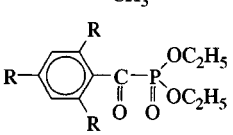
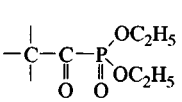
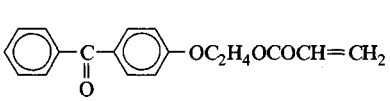
ANNEX TO TABLE 1. *cont'd*

Compound		R	R_1	R_2	R_3	R_4
	I5a					
	I6a					
	I7a	H				
	I7b	$CH_2SO_3^-Na^+$				
Fluorenone	I8a					
	I9a					
	I9b					
	I10a			C_2H_5		
	I11a		H	H		
	I11b		H	CH_3		
	I11c		CH_3	H		
	I11d		CH_3O	H		
	I12a	OH				
	I12b	$OCOCH_3$				
	I12c	H				
	I12d	CH_3				
	I12e					
	I13a	CH_3				
	I14a	H				
	I14b	i-C_3H_7				
	I14c	OC_2H_5				
	I14d	t-C_4H_9				
	I15a					
	I15b	H				
	I15c	CH_3				
	I15d					
	I16a					

ANNEX TO TABLE 1. *cont'd*

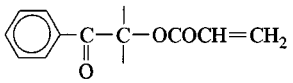
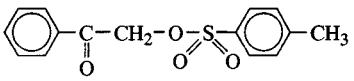
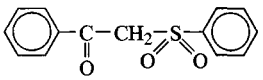
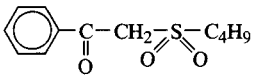
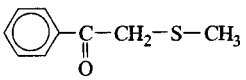
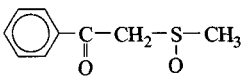
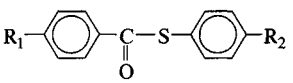
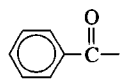
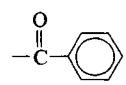
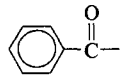
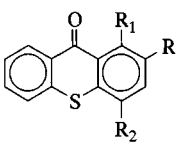
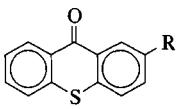
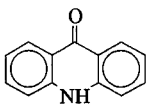
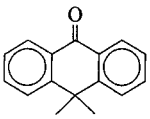
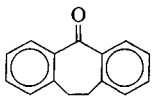
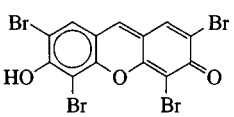
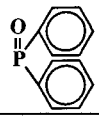
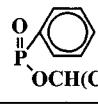
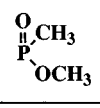
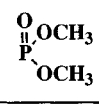
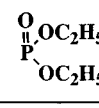
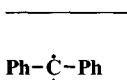
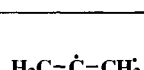
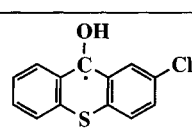
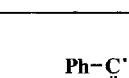
	Compound	R	R_1	R_2	R_3	R_4
	I16b					
	I17a					
	I18a					
	I19a					
	I20a					
	I21a					
	I22a		H	C_8H_{17}		
	I22b		NO_2	H		
	I22c					
	I22d			$C_{12}H_{25}$		
	I23a	H	H	H		
	I23b	Cl	H	H		
	I23c	CH_3	H	H		
	I23d	$CH_3(CH_3)_2$	H	H		
	I23e	H	Br	OC_3H_7		
	I23f	H	Cl	OC_3H_7		
	I23g	$-OCH_2COOH$				
	I23h	$OCH_2CH_2CH_2\overset{+}{N}(CH_3)_3SO_3Me^-$				
	I24					
Acridine	I25					
Phenazine	I26					
	I27					
	I28					
	I29					

TABLE 2. BIMOLECULAR RATE CONSTANTS[a] FOR THE REACTION OF PHOSPHONYL RADICALS WITH VARIOUS MONOMERS IN CYCLOHEXANE AT ROOM TEMPERATURE[b]

Monomer	Radical				
					
Methacrylonitrile	5.0	4.6	4.5	9.2	11
Styrene	6.0	4.5	8.0	22	25
Methylmethacrylate	8.0	5.0	5.8	5.8	5.3
Acrylonitrile	2.0	2.0	0.18	0.58	0.26
Methyl acrylate	3.5	2.1	1.3	1.7	1.6
n-Butyl vinyl ether	0.4	0.3	0.23	2.1	1.4
Vinyl acetate	0.16	0.13	0.082	0.29	0.18

[a] k_{RM} in 10^7 l/mol/s.
[b] Ref. 26.

TABLE 3. BIMOLECULAR RATE CONSTANTS[a] FOR THE REACTION OF VARIOUS RADICALS WITH VARIOUS OLEFINIC MONOMERS AT ROOM TEMPERATURE[b]

Monomer	Radical							
					$Ph_2\dot{P}{=}O$	$Ph^{\cdot}$	$PhS\dot{O}_2$	$PhS\dot{O}_3$
Styrene		3.5×10^{-2}	7×10^{-4}	1.2×10^{-2}	6			
Methylmethacrylate	9×10^{-4}	5.4×10^{-2}	4×10^{-5}	0.9×10^{-2}	6	18[c]	10^{-3}[†]	10^{-1}[†]
Acrylonitrile	4×10^{-4}	1.6×10^{-2}		2×10^{-3}	2			
Methyl acrylate					2			
n-Butyl vinyl ether			$< 10^{-6}$		0.5			
Vinyl acetate	6×10^{-4}			2×10^{-2}	0.2			
N-Vinylpyrrolidone			4×10^{-5}					

[a] k_{RM} in 10^7 l/mol/s.
[b] Ref. 26.
[c] From Ref. 40.
[†] From Ref. 16.

TABLE 4. BIMOLECULAR RATE CONSTANTS[a] FOR THE REACTION OF $Ph_2\dot{P}{=}O$ AND $Ph_2\dot{P}{=}S$ WITH VARIOUS MONOMERS[b,c]

Monomer	Radical	
	$Ph_2\dot{P}{=}O$	$Ph_2\dot{P}{=}S$
Methacrylonitrile	1.9	0.9
Styrene	4.6	0.4
Methylmethacrylate	4.1	0.19
Acrylonitrile	1.3	0.52
Methyl acrylate	1.7	0.62
n-Butyl vinyl ether	5.0	0.15
Vinyl acetate	1.4	0.042

[a] k_{RM} in 10^7 l/mol/s
[b] Ref. 27.
[c] Experiments were carried out in CH_2Cl_2 solution at room temperature.

TABLE 5. ELECTRON TRANSFER REACTION OF RADICALS WITH DIPHENYLIODONIUM SALTS

Radicals	$10^{-9}k_e$ (l/mol/s)	Refs.
$Ph_2P\dot{O}$	$\leq 10^{-3}$	42
$Ph_2\dot{C}OH$	0.03	43
$(CH_3)_2\dot{C}OH$	0.06	

TABLE 6. ELECTRON TRANSFER RATE CONSTANTS (k_e) BETWEEN PHOTOSENSITIZERS AND CATIONIC PHOTOINITIATORS AND QUENCHING RATE CONSTANTS (k_q) FOR CYCLOHEXENE OXIDE IN METHANOL (M) AND ACETONITRILE (AN)

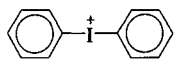
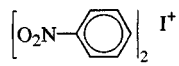
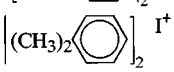
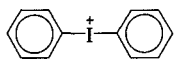
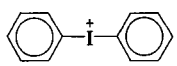
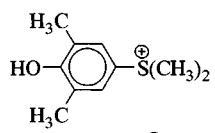
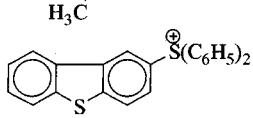
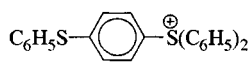
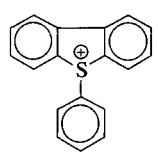
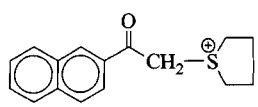
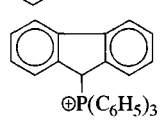
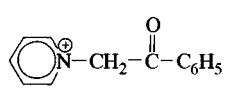
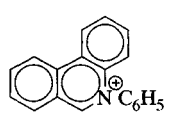
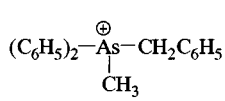
Photosensitizer	Cationic photoinitiator	Anion	Solvent	$10^{-7}k_e$ (l/mol/s)	$10^{-7}k_q$ (l/mol/s)	Refs.
Chlorothioxanthone		AsF_6^-	M	24	0.3	28
		Cl^-		24		
		PF_6^-		26		
		BF_4^-		24		
		AsF_6^-	AN	290		
Chlorothioxanthone		AsF_6^-	M	400		
				36		
Benzophenone		AsF_6^-	M	15	0.33	
		Cl^-		1		
		BF_4^-		47		
Ketocoumarin		BF_4^-	M	6.4	0.04	
Chlorothioxanthone	$(C_6H_5)_3S^+ PF_6^-$		M	14		29
		PF_6^-		35		
		PF_6^-		28		
		SbF_6^-		4		
		AsF_6^-		8		
		BF_4^-		570		
		BF_4^-		110		
		PF_6^-		14		
		AsF_6^-		100		
		BF_4^-		38		
	$(C_6H_5)_3Se^+AsF_6^-$			15		

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TABLE 6. *cont'd*

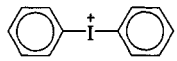
Photosensitizer	Cationic photoinitiator	Anion	Solvent	$10^{-7}k_e$ (l/mol/s)	$10^{-7}k_q$ (l/mol/s)	Refs.
Xanthone		Cl^-	M	140		30
Michler's ketone				0.7		
Thioxanthone				200		
10-Methylacridone				350		
Acridone				740		
Anthracene			M	1		31
Chlorothioxanthone	$(NO_2\varphi)_2I^+$	Cl^-		400		28
Pyrene				130		
Benzophenone				500		
Isopropylthioxanthone				250		

TABLE 7. EXCITATION TRANSFER RATE CONSTANTS (k_T) FOR THIOXANTHONES AND PHOTOINITIATORS[a]

Thioxanthone	Initiator	Solvent	$10^{-6}k_T$ (l/mol/s)
TXI	HMK	Toluene	100
		Methanol	290
	OMK	Toluene	75
		Methanol	380
	TPMK	Toluene	240
		Methanol	630
	NMK	Toluene	4700
		Methanol	7900
ETX	TPMK	Toluene	12
		Methanol	110
ITX	TPMK	Toluene	60
		Methanol	55

[a]Ref. 32.

ANNEX TO TABLE 7. COMPOUND CHEMISTRIES

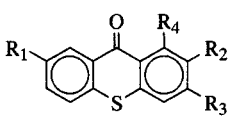
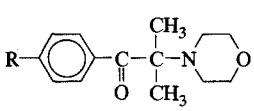
TX's	R	R_1	R_2	R_3	R_4	Abbrev.	Triplet state energy level (kcal/mol)
		H	Cl	H	H	CTX	62
		H	$CH(CH_3)_2$	H	H	ITX	61
		CH_3	H	COOET	H	ETX	58,5
		H	H	H	COOET	TXI	63
Mok's							
	H					HMK	65
	SCH_3					TPMK	61
	OCH_3					OMK	65
	$N(CH_3)_2$					NMK	63

TABLE 8. TRIPLET STATE LIFETIMES (τ_T) OF THE SENSITIZER (TXI) IN DIFFERENT MEDIA[a], AND RATE CONSTANT (k_T) OF THE INTERACTION BETWEEN TXI AND TPMK[b]

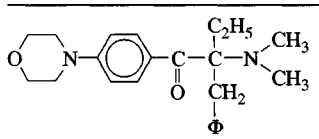
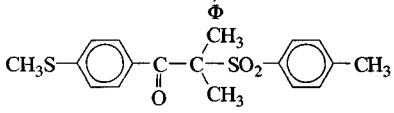
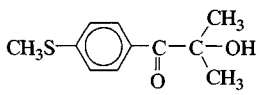
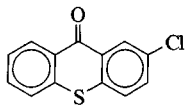
Medium	τ_T (ns)	η (cp at 28°C)	$10^{-6}k_T$ (l/mol/s)
TMPTA[c] (100%)	650	40	80
Acrylate (66/33)	2200	185	31
Toluene–Ep. acrylate (50/50)	1500	9.5	35
Toluene–TMPTA (25/75)	270	5.8	135
Toluene–TMPTA (50/50)	200	1.7	<40
Toluene–TMPTA (75/25)	280	0.9	150
HDDA[d] (100%)	650	5.2	95
HDDA–Ep. acrylate (50/50)	2500	58	22
Toluene–HDDA (50/50)	1000	1.2	140
Toluene–HDDA–Ep. acrylate (25/50/25)	850	5.8	68
Toluene–PETA[e] (50/50)	220	3.1	40
Toluene–PETA–Ep. acrylate (43/43/14)	360	6.9	69
Toluene	50	0.53	240
Methanol	140	0.52	630

[a]Viscosity: η.
[b]Ref. 33; for formula, see Annex to Table 7.
[c]Trimethylolpropane triacrylate.
[d]Hexanedioldiacrylate.
[e]Pentaerythritol triacrylate.

TABLE 9. SOME VALUES OF THE TRIPLET STATE ENERGY LEVELS OF PHOTOINITIATORS AND MONOMERS

Compound	Triplet state energy (ΔE) (kcal/mol)	
I1a	69	Cited in Ref. 13
I4a	73	
I7a	53	
I8a	53	
I9b	59	
M_2	61	
I14a	75	Cited in Ref. 38
I24	57	

TABLE 10. VALUES OF τ_T^0, τ_T, AND $k_a^{b\dagger}$ IN SOLUTION

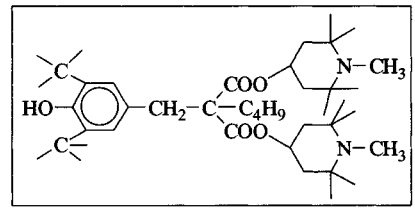
Photoinitiators	τ_T^0 (ns)	τ_T (ns)	$10^{-6}k_a^b$ (l/mol/s)
	4200	4000	0.85
	3300	2000	14
	470	470	≤5
	4000	1750	24

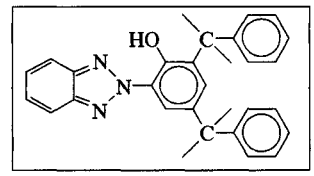
†Determined in bulk epoxyacrylate–HDDA (60:40 w/w); a in solution b in the presence of LS1 (2%).; Ref. 34.

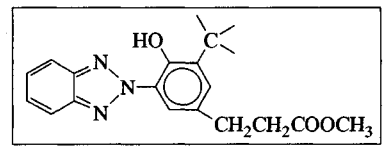
TABLE 11. RATE CONSTANT OF INTERACTION OF KETONES AND LIGHT STABILIZERS IN SOLUTION

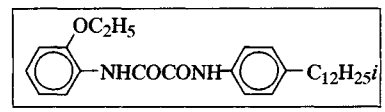
Photoinitiator	Light stabilizer	$10^{-6}k_a$ (l/mol/s)	Refs.
Benzophenone	LS1	6500	34
	LS2	8550	
	LS3	10300	
	LS4	15000	
	LS5	7050	
	SL6	5900	35
Acetone	LS1	500	36
	LS7	700	

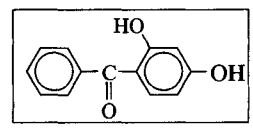
ANNEX TO TABLE 11.

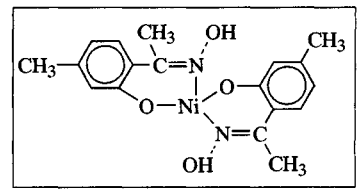
LS1:


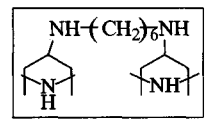
LS2:


LS3:


LS4:


LS5:


LS6:


LS7:


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C. REFERENCES

1. (a) "Lasers in Polymer Science and Technology: Applications", J. P. Fouassier, J. F. Rabek (Eds.), CRC Press, Boca Raton, 1990.
   (b) "Radiation Curing in Polymer Science and Technology", J. P. Fouassier, J. F. Rabek (Eds.), Chapman & Hall, London, (1993).
2. J. P. Fouassier, "Photoinitiation, Photopolymerization, Photocuring", Hanser, Münich, (1995).
3. (a) C. Decker, in: S. P. Pappas, (Ed.), "Radiation Curing: Science and Technology", Plenum Press, New York, 1992.
   (b) C. Decker, B. Elzaouk in "Laser Curing of Photopolymers", Proc. Polymer Photochemistry Symposium, Gênes, 112, (1993).
   (c) C. Decker, B. Elzaouk, Eur. Polym. J., **31** (12), 1155 (1995).
4. D. Ruhlmann, J. P. Fouassier, Eur. Polym. J., **27** (9), 991 (1991).
5. D. Ruhlmann, J. P. Fouassier, W. Schnabel, Eur. Polym. J., **28** (3), 287 (1992).
6. D. Ruhlmann, F. Wieder, J. P. Fouassier, Eur. Polym. J., **28** (6), 591 (1992).
7. R. Kuhlmann, W. Schnabel, Polymer, **18**, 1163 (1977).
8. J. P. Fouassier, in: J. P. Fouassier, J. F. Rabek (Eds.) "Lasers in Polymer Science and Technology: Applications", CRC Press, Boca Raton, 1990.
9. J. P. Fouassier, D. J. Lougnot, Polymer Comm., **31**, 418 (1990).
10. J. P. Fouassier, D. J. Lougnot, J. Chem. Soc., Faraday Trans., 1, **83** (9), 2935 (1987).
11. G. Amirzadeh, R. Kuhlmann, W. Schnabel, J. Photochem., **10**, 133 (1979).
12. J. P. Fouassier, D. J. Lougnot, L. Avar, Polymer, **36** (26), 5005 (1995).
13. J. P. Fouassier, in: J. P. Fouassier, J. F. Rabek (Eds.) "Radiation Curing in Polymer Science and Technology", Chapman & Hall, London, vol. II, 1993.
14. T. Sumiyoshi, W. Schnabel, A. Henne, J. Photochem., **32**, 191 (1986).
15. D. Ruhlmann, K. Zahouily, J. P. Fouassier, Eur. Polym. J., **28** (9), 1063 (1992).
16. J. P. Fouassier, D. Burr, Macromolecules, **23**, 3615 (1990).
17. J. P. Fouassier, D. J. Lougnot, J. C. Scaiano, Chem. Phys. Lett., **160**, 335 (1989).
18. P. J. Wagner, M. J. Lindstrom, Jacs, **109**, 3062 (1987).
19. F. Morlet-Savary, J. P. Fouassier, H. Tomioka, Polymer, **33**, 4202 (1992).
20. G. Amirzadeh, W. Schnabel, Makromol. Chem., **182**, 2821 (1981).
21. R. Kuhlmann, W. Schnabel, Angew. Makromol. Chem., **57**, 195 (1977).
22. R. Kuhlmann, W. Schnabel, Polymer, **17**, 419 (1976).
23. F. Morlet-Savary, J. P. Fouassier, T. Matsumoto, K. Inomata, Polymers for Adv. Techn., **5**, 56 (1994).
24. N. S. Allen, A. W. Timms, W. A. Green, F. Catalina, T. Corrales, S. Navaratnam, B. J. Parsons, J. Chem. Soc. Faraday, **90**, 83 (1994).
25. D. J. Lougnot, C. Turck, J. P. Fouassier, Macromolecules, **22**, 108 (1989).
26. W. Schnabel, in: J. P. Fouassier, J. F. Rabek (Eds.), "Lasers in Polymer Science and Technology: Applications", vol. II, CRC Press, Boca Raton, 1990.
27. T. Sumiyoshi, W. Weber, W. Schnabel, Z. Naturforsch, **40a**, 541 (1985).
28. J. P. Fouassier, D. Burr, J. V. Crivello, J. Photochem. Photobiol., A: Chem., **49**, 318 (1989).
29. G. Mannivannan, J. P. Fouassier, J. V. Crivello, J. Polym. Sci. Part A: Polym. Chem., **30**, 1999 (1992).
30. H. J. Timpe, K. P. Kronfeld, U. Lammel, J. P. Fouassier, D. J. Lougnot, J. Photochem., **52**, 111 (1990).
31. M. R. V. Sahyun, R. J. DeVoe, P. M. Olofson, in: J. P. Fouassier, J. F. Rabek, (Eds.), "Radiation Curing in Polymer Science and Technology", vol. II, Chapman & Hall, London, 1993.
32. A. Bohrer, G. Rist, K. Dietliker, V. Desobry, J. P. Fouassier, D. Ruhlmann, Macromolecules, **25**, 4182 (1992).
33. J. P. Fouassier, D. Ruhlmann, Eur. Polym. J., **29** (4), 505 (1993).
34. J. P. Fouassier, D. Ruhlmann, A. Erddalane, Macromolecules, **26**, 721 (1993).
35. J. P. Guillory, C. F. Cook, Jacs, **95**, 4885 (1973).
36. P. Bortolus, in: J. P. Fouassier, J. F. Rabek (Eds.), "Radiation Curing in Polymer Science and Technology", vol. II, Chapman & Hall, London, 1993.
37. H-J. Timpe, Topics in Current Chemistry, vol. **156**, 167, (1990).
38. H-J. Timpe, K-P. Kronfeld, J. Photochem. & Photobiol., A: Chem., **46**, 253 (1989).
39. J. C. Netto-Ferreira, D. Weir, J. C. Scaiano, J. Photochem. & Photobiol., A: Chem., **48**, 345 (1989).
40. J. C. Scaiano, L. C. Stewart, J. Am. Chem. Soc., **105** (11), 3609 (1983).
41. E. Klimtchuk, M. A. J. Rodgers, D. C. Neckers, J. Phys. Chem., **96** (24), 9817 (1992).
42. Y. Yagci, W. Schnabel, Makromol. Chem., Rapid Commun., **8**, 209 (1987).
43. Y. Yagci, S. P. Pappas, W. Schnabel, Z. Naturforsch, **42a**, 1425 (1987).

Free Radical Copolymerization Reactivity Ratios

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A. Introduction II-181
B. Tables II-182
Table 1. Copolymer Reactivity Ratios II-182
Table 2. Listing of Quick Basic (Microsoft) Program for Calculating Reactivity Ratios II-288
C. References II-290

A. INTRODUCTION

When a vinyl monomer is copolymerized with a second monomer, the relationship between the composition of the initially formed copolymer and the initial monomer mixture is given by

$$\frac{dm_1}{dm_2} = \frac{M_1(r_1M_1 + M_2)}{M_2(r_2M_2 + M_1)}$$

where m_1 is the number of moles of monomer 1 entering the copolymer, m_2, the number of moles of monomer 2 entering the copolymer, M_1, the number of moles of monomer 1 in the monomeric mixture, M_2, the number of the moles of monomer 2 in the monomer mixture, and r_1 and r_2 are the monomer reactivity ratios.

The monomer reactivity ratios, r_1 and r_2, for any monomer pair are the ratios of the rate constants of the different propagation reactions:

$$\sim M_1^{\bullet} + M_1 \longrightarrow \sim M_1M_1^{\bullet} \quad k_{11}$$
$$\sim M_1^{\bullet} + M_2 \longrightarrow \sim M_1M_2^{\bullet} \quad k_{12}$$
$$\sim M_2^{\bullet} + M_2 \longrightarrow \sim M_2M_2^{\bullet} \quad k_{22}$$
$$\sim M_2^{\bullet} + M_1 \longrightarrow \sim M_2M_1^{\bullet} \quad k_{21}$$

with $r_1 = k_{11}/k_{12}$, $r_2 = k_{22}/k_{21}$. $\sim M^{\bullet}$ represents a polymer chain ending in a radical derived from monomer M.

The original compilations of reactivity ratios and their references were published by L. J. Young in the first two editions of this Handbook. In the third edition, the original listings and those through 1986 were re-evaluated. This fourth edition contains an additional 548 evaluated monomer pairs giving a total of 3,265 such pairs. Only free radical reactivity ratios are listed. When the original references were available, they were read. If the experimental data was published, the reactivity ratios were recalculated according to the methods of Kelen and Tűdös (803,804). The 95% confidence limits for the reactivity ratios were also calculated (805). If the authors used this method for r-value evaluation, including adjustment of the monomer feeds for total conversion, then the reported values were utilized. There are a number of reported reactivity ratios that are not reported here. This is due to the absence of these values in the abstracts of papers which were published in journals unavailable to the author.

All of the monomer pairs in Table 1 are cross-referenced. The full spelling of each monomer was included in its Monomer 1 citation. Sometimes truncated but distinguishable names have been used in the Monomer 2 lists.

If only the reactivity ratios and reference are given, then, either no experimental data was given or the journal was not available and the Chemical Abstracts summary was the source of the data. If a Y or N (yes or no) appears in the conversion (Conv.) column then the reactivity ratios were recalculated. If a recalculation was performed but the 95% confidence columns (95%) are still left blank, it indicates that only two feed/polymer data pairs were available. In general, if there is a citation (Y or N) in the Conv. column but no reactivity ratios are shown in the reactivity ratio columns, the copolymerization did not follow the copolymerization equation (ionic or penultimate effects were prevalent). In a few cases, the data were too scattered to allow a reactivity ratio calculation.

Recalculations yield negative reactivity ratios in several cases. We are aware that this is a physically unrealistic artifact. When the 95% confidence limits are applied to these negative numbers, a value of zero usually falls within these limits. It may be noted that a single reference sometimes contains a variety of reactivity ratios for one monomer pair. The reader may assume that these result from a change of polymerization conditions; e.g., different polymerization temperatures or polymerization media of varying polarity.

For those who are interested in using the Kelen–Tűdös calculations, a listing of a simplistic program written in QuickBasic (Microsoft) is also given (Table 2).

*Retired.

B. TABLES

TABLE 1. COPOLYMER REACTIVITY RATIOS

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acenaphthalene	Acrylate, (−)-menthyl					Y	645
Acenaphthalene	Acrylonitrile	2.56		0.02			575
Acenaphthalene	Carbazole, *N*-vinyl-	6.62	1.77	−0.03	0.12	N	331
Acenaphthalene	Fumarate, diisopropyl	2.57		0.06			1039
Acenaphthalene	Maleic anhydride	0.46		0.02			1039
Acenaphthalene	Maleic anhydride	0.46	0.13	−0.04	0.05	N	26
Acenaphthalene	Maleimide, *N*-cyclohexyl-	0.57		0.11			1039
Acenaphthalene	Methacrylate, (−)-menthyl	2.48	0.55	0.316	0.085	Y	645
Acenaphthalene	Methacrylate, methyl	1.05	0.14	0.36	0.04	N	326
Acenaphthalene	Methacrylate, methyl	0.99	0.15	0.38	0.04	N	326
Acenaphthalene	Methacrylate, methyl	2.25		0.44			575
Acenaphthalene	Methacrylonitrile	2.38		0.15			575
Acenaphthalene	Pyrrolidone, *N*-vinyl-					N	326
Acenaphthalene	Pyrrolidone, *N*-vinyl-					N	326
Acenaphthalene	Stilbene	5.65	0.23	−0.01	0.04	N	725
Acenaphthalene	Styrene	0.33		3.81			525
Acetamide, *N*-(4-methacryloyloxyphenyl)-2-(4-methoxy)	Methacrylate, 2-hydroxyethyl	1.49		0.61			1051
Acetamide, *N*-vinyl-	Acrylamide	0.3		1.4			984
Acetamide, *N*-vinyl-	Methacrylate, methyl	0.19		2.65			984
Acetamide, *N*-vinyl-	Methacrylate, methyl	0.71		1.18			984
Acetamide, *N*-vinyl-	Methacrylate, methyl	0.01		2.1			984
Acetamide, *N*-vinyl-	Vinyl acetate	5.5		0.6			984
Acetamide, *N*-vinyl-	Vinyl acetate	21		0			984
Acetanilide, 4-(2-methacryloyloxy)-ethyloxy)	Methacrylate, 2-hydroxyethyl	0.67		1.39			1096
Acetanilide, 4-(2-methacryloyloxy)-hexyloxy)	Methacrylate, 2-hydroxyethyl	0.87		1.19			1096
Acetanilide, 4-(methacryloyloxy)-	Methacrylate, 2-hydroxyethyl	1.94		0.78			1096
Acetate, 2-chloro-, vinyl	Vinyl acetate	1.18		0.8			892
Acetate, allyl	Vinyl acetate	0.43		1.8			1045
Acetate, chloro-, allyl	Vinyl acetate	0.69		0.77			1045
Acetate, dichloro-, allyl	Vinyl acetate	0.54		0.57			1045
Acetate, isopropenyl	Fumarate, diisopropyl	0.011		0.67			1038
Acetate, trichloro-, allyl	Vinyl acetate	0.28		0.41			1045
Acetylene, phenyl-	Acrylate, methyl	0.272	0.072	0.622	0.062	Y	60
Acetylene, phenyl-	Acrylate, methyl	0.093	0.01	0.664	0.045	Y	731
Acetylene, phenyl-	Acrylonitrile	0.325	0.083	0.266	0.044	Y	60
Acetylene, phenyl-	Isoprene	0.1		3.01			648
Acetylene, phenyl-	Maleic anhydride	0.08		0.06			797
Acetylene, phenyl-	Methacrylate, methyl	0.2		1.5			443
Acetylene, phenyl-	Methacrylate, butyl	0.21		1.7			565
Acetylene, phenyl-	Methacrylate, isobutyl	0.27		1.9			565
Acetylene, phenyl-	Methacrylate, methyl	1.69	26.56	−0.06	0.16	Y	357
Acetylene, phenyl-	Methacrylate, methyl	0.07	0.016	1.111	0.095	Y	732
Acetylene, phenyl-	Methacrylonitrile	0.04		0.78			648
Acetylene, phenyl-	Pyridine, 2-vinyl-					Y	192
Acetylene, phenyl-	Styrene	0.33	0.12	0.324	0.02	Y	357
Aconitate, trimethyl	Acrylonitrile	−0.48	0.11	4.24	0.92	Y	147
Aconitate, trimethyl	Butadiene	0	0.22	0.4	0.4	Y	147
Aconitate, trimethyl	Styrene	−0.013	0.009	1.026	0.091	Y	147
Aconitate, trimethyl	Vinyl chloride	0.19		0.04		Y	147
Aconitate, trimethyl	Vinylidene chloride	0.15	0.94	64.6	23.74	N	149
Acrolein	Acrylamide	1.95	0.58	0.8	0.23	N	207
Acrolein	Acrylamide	1.59	0.1	0.18	0.02	N	208
Acrolein	Acrylate, butyl	1.86	0.19	0.638	0.032	Y	291
Acrolein	Acrylate, butyl	2.29	0.58	1.12	0.41	N	292
Acrolein	Acrylate, ethyl	1.2		0.6			291
Acrolein	Acrylate, ethyl	1.98	0.44	1.09	0.25	N	292
Acrolein	Acrylate, methyl	−0.07	0.03	7.86	3.58	N	207
Acrolein	Acrylate, methyl	1.41	0.4	0.83	0.12	Y	291

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrolein	Acrylate, methyl	2.54	0.93	1.08	0.38	N	292
Acrolein	Acrylic acid	2.48	1.24	0.08	0.28	Y	57
Acrolein	Acrylic acid	0.5		1.15			57
Acrolein	Acrylonitrile	1.16	0.13	0.88	0.06	N	207
Acrolein	Acrylonitrile	1.07	0.08	0.71	0.03	N	207
Acrolein	Acrylonitrile	1.52	0.07	0.48	0.01	N	208
Acrolein	Acrylonitrile	1.28		0.6			504
Acrolein	Methacrylate, methyl	0.76	0.12	1.136	0.05	Y	291
Acrolein	Methacrylate, methyl	0.59	0.3	1.33	0.4	N	292
Acrolein	Methacrylonitrile	0.68	0.12	1.17	0.05	N	208
Acrolein	Methacrylonitrile	0.72		1.2			446
Acrolein	Pyridine, 2-vinyl-	2.64	0.14	−0.12	0.03	N	208
Acrolein	Styrene	0.216	0.036	0.257	0.017	Y	291
Acrolein	Styrene	0.02	0.01	0.22	0.02	N	292
Acrolein	Styrene	0.32	0.002	0.205	0.006	Y	307
Acrolein	Styrenesulfonate, *p*-, sodium	0.395	0.092	0.327	0.033	Y	57
Acrolein	Styrenesulfonate, *p*-, sodium	0.26	0.16	0.047	0.071	Y	57
Acrolein	Styrenesulfonate, *p*-, sodium	0.113	0.018	0.01	0.009	Y	57
Acrolein	Vinyl acetate	3.04	0.43	−0.02	0.12	N	207
Acrolein	Vinyl chloride	5.22	0.13	0.03	0.11	N	271
Acrolein diethylacetal	Acrylonitrile	0.02		11.19			767
Acrolein diethylacetal	Maleic anhydride	0.18		0.07			767
Acrolein, methyl-	Acrylate, butyl	2.5		0.02			589
Acrolein, methyl-	Acrylonitrile	1.7		0.15			455
Acrolein, methyl-	Acrylonitrile	3.8		0.15			587
Acrolein, methyl-	Acrylonitrile	3.8		0.15			589
Acrolein, methyl-	Acrylonitrile	1.72	0.42	0.03	0.19	Y	91
Acrolein, methyl-	Methacrylate, methyl	0.4		0.1			589
Acrolein, methyl-	Methacrylonitrile	1.78	0.07	0.41	0.01	N	208
Acrolein, methyl-	Methacrylonitrile	1.25		0.45			587
Acrolein, methyl-	Methacrylonitrile	1.25		0.45			589
Acrolein, methyl-	Styrene	0.833	0.092	0.173	0.017	N	223
Acrolein, methyl-	Styrene	0.3		0.41			589
Acrolein, methyl-	Vinyl acetate	0.98		0.15			589
Acrolein, 2-chloro-	Styrene	0.15	0.07	0.02	0.05		806
Acryl-2′-ethyl-4,5,7-trinitro-9-flurenone-2-carboxylate	Acrylate, *N*-(2-hydroxyethyl)-carbazolyl	0.585	0.028	0.912	0.079	Y	715
Acryl-2′-ethyl-4,5,7-trinitro-9-flurenone-2-carboxylate	Methacrylate, *N*-(2-hydroxyethyl)-carbazolyl	0.102	0.022	0.773	0.096	Y	703
Acrylamide	Acetamide, *N*-vinyl-	1.4		0.3			984
Acrylamide	Acrolein	0.8	0.23	1.95	0.58	N	207
Acrylamide	Acrolein	0.18	0.02	1.59	0.1	N	208
Acrylamide	Acrylic acid	1.08	0.17	0.288	0.029	Y	221
Acrylamide	Acrylic acid	0.445	0.075	1.644	0.069	Y	278
Acrylamide	Acrylic acid	1.346	0.044	0.341	0.012	Y	278
Acrylamide	Acrylic acid	0.598	0.047	1.38	0.14	N	37
Acrylamide	Acrylic acid	0.58		3.8			869
Acrylamide	Acrylic acid	1.06		0.29			869
Acrylamide	Acrylic acid, *cis*-3-bromo-	7.64	1.67	−0.2	0.11	Y	298
Acrylamide	Acrylic acid, *cis*-3-ethyl-	6.32	0.68	−0.16	0.1	Y	298
Acrylamide	Acrylic acid, *trans*-3-bromo-	4.13	0.76	−0.33	0.16	Y	298
Acrylamide	Acrylic acid, *trans*-3-ethyl-	8.19	1.09	−0.21	0.11	Y	298
Acrylamide	Acrylonitrile	0.81	0.11	0.863	0.033	Y	228
Acrylamide	Acrylonitrile	1.08	0.51	0.97	0.1	Y	228
Acrylamide	Acrylonitrile	0.5	0.26	1.8	0.75	N	330
Acrylamide	Acrylonitrile	1.36		0.88			421
Acrylamide	Acrylonitrile	1.3		0.8			461
Acrylamide	Cinnamic acid, *cis*-	3.84	0.68		0.28	Y	298
Acrylamide	Cinnamic acid, *trans*-	3.4	0.24	−0.24	0.15	Y	298
Acrylamide	Crotonic acid	3.76	0.89	−0.18	0.14	Y	298
Acrylamide	Crotonic acid	4.23	1.37	−0.19	0.19	Y	298
Acrylamide	Crotonic acid	4.72		0.11			569
Acrylamide	Crotonic acid	5.32		0.12			569
Acrylamide	Crotonic acid, *cis*-					Y	341
Acrylamide	Crotonic acid, *trans*-	7.2	0.33	−0.085	0.024	Y	341

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylamide	Dioxolane, 1,3,4-methylene-2-trichloro-	8.6		0.037			616
Acrylamide	Maleic anhydride	0.56		0			655
Acrylamide	Methacrylate, 2-hydroxyethyl	0.05		1.89			689
Acrylamide	Methacrylate, 2-hydroxyethyl	0.14		0.98			928
Acrylamide	Methacrylate, 3-methoxy-2-hydroxypropyl	0.04		3.98			689
Acrylamide	Methacrylate, dimethylaminoethyl sulfate	0.52	0.05	1.9	0.2		1052
Acrylamide	Methacrylate, methyl	2.29	0.44	2.34	0.17	N	314
Acrylamide	Methacrylate, methyl	0.53	0.09	3	0.09	N	314
Acrylamide	Methacrylate, methyl	0.82		2.53			314
Acrylamide	Methacrylate, methyl	0.9	1.07	3	0.45	N	691
Acrylamide	Methacrylic acid	0.57		1.63			1028
Acrylamide	Methacrylic acid	0.58		4.4			852
Acrylamide	Methacrylic acid	0.56		0.15			852
Acrylamide	Methylenebutyrolactone	0.3		2.85			746
Acrylamide	Styrene	0.58	0.12	1.17	0.06	N	295
Acrylamide	Styrene	1.32	0.13	1.21	0.04	N	314
Acrylamide	Styrene	0.33	0.14	1.49	0.16	N	314
Acrylamide	Styrene	0.59		1.13			314
Acrylamide	Styrene	8.97	0.43	0.65	0.03	N	343
Acrylamide	Styrene	0.2		1.05			555
Acrylamide	Succinimide, *N*-vinyl-	1.86		0.17			600
Acrylamide	Vinyl chloride					Y	156
Acrylamide	Vinyl methyl ketone	3.99	0.7	0.47	0.44	Y	609
Acrylamide	Vinyl methyl ketone	1.02	0.13	0.75	0.23	Y	609
Acrylamide	Vinylene carbonate	13.8		0.05			387
Acrylamide	Vinylmethylphenylsulfonium tetrafluoro	3.4		0.1			793
Acrylamide	Vinylsulfonic acid	3.5	0.15	0.3	0.1		1066
Acrylamide, 2-chloro-*N,N*-dimethyl-	Methacrylate, methyl	−0.159	0.502	2.71	0.398	Y	917
Acrylamide, 2-chloro-*N,N*-dimethyl-	Styrene	−0.036	0.152	1.29	0.065	Y	917
Acrylamide, *N,N*-dibutyl-	Styrene	0.294	0.058	1.6	0.058		806
Acrylamide, *N,N*-diethyl-	Acrylic acid	0.35	0.12	0.36	0.04	N	205
Acrylamide, *N,N*-diethyl-	Butanoate, 3-acrylamido-3-methyl-, sodium	0.987	0.153	0.22	0.138	Y	857
Acrylamide, *N,N*-diethyl-	Methacrylate, methyl	0.41	0.15	1.65	0.16	N	170
Acrylamide, *N,N*-diethyl-	Styrene	0.39		1.23			739
Acrylamide, *N,N*-diethyl-	Styrene	0.34		1.36			739
Acrylamide, *N,N*-diethyl-	Styrene	0.18		1.54			739
Acrylamide, *N,N*-dimethyl-	Methacrylate, methyl	0.57	0.08	2.15	0.08	N	315
Acrylamide, *N,N*-dimethyl-	Methacrylate, methyl	0.59	0.09	2.8	0.1	N	315
Acrylamide, *N,N*-dimethyl-	Methacrylate, methyl	0.21	0.21	2.64	0.15	N	691
Acrylamide, *N,N*-dimethyl-	Styrene	0.12	0.3	1.15	0.35	N	170
Acrylamide, *N,N*-dimethyl-	Styrene	0.49	0.15	1.37	0.1	N	315
Acrylamide, *N,N*-dimethyl-	Styrene	0.47	0.2	1.32	0.12	N	315
Acrylamide, *N,N*-dimethyl-	Sulfonate, 2-acrylamido-2-methylpropane	1.108	0.59	0.162	0.949	Y	857
Acrylamide, *N,N*-dimethyl-	Vinyl methyl ketone	0.82	0.36	0.1	0.43	Y	609
Acrylamide, *N*-(2-(4-hydroxyphenyl)ethyl)-	Methacrylamide, *N*-(2-hydroxypropyl)-	0.95	0.09	1.01	0.09		965
Acrylamide, *N*-(2-(propionamide)-	Styrene	0.86	0.3	1.84	0.21	N	295
Acrylamide, *N*-methyl-	Methacrylate, methyl	0.05	0.72	1.14	0.09	N	691
Acrylamide, *N*-methylol	Acrylate, butyl	0.61		0.87			441
Acrylamide, *N*-methylol	Acrylate, ethyl	1.4		1.4			442
Acrylamide, *N*-methylol	Acrylate, methyl	1.9		1.3			441
Acrylamide, *N*-methylol	Acrylonitrile	1.2		0.7			442
Acrylamide, *N*-methylol	Acrylonitrile	2.33		0.98			577
Acrylamide, *N*-methylol	Acrylonitrile	2.56		0.22			596
Acrylamide, *N*-methylol	Methacrylate, methyl	0.7		1.62			593
Acrylamide, *N*-methylol	Methacrylate, methyl	0.7		1.62			596
Acrylamide, *N*-methylol	Styrene	0.48		0.03			596
Acrylamide, *N*-methylol	Vinyl chloride					Y	157
Acrylamide, *N*-methylol-	Styrene	0.4	0.03	1.55	0.05		891
Acrylamide, *N*-octadecyl-	Acrylonitrile	1.4	0.17	1.034	0.02	Y	119
Acrylamide, *N*-octadecyl-	Methacrylate, methyl	0.42	0.11	4.119	0.07	Y	290

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylamide, *N*-octadecyl-	Styrene	0.54	0.14	2.08	0.12	Y	290
Acrylamide, *N*-octadecyl-	Vinyl acetate	8.25	1.46	0.004	0.011	Y	119
Acrylamide, *N*-octadecyl-	Vinylidene chloride	1.4	0.11	0.432	0.008	Y	119
Acrylamide, *N*-octyl-	Styrene	0.258	0.119	2.715	0.244		806
Acrylamide, *N*-propyl-	Acrylate, butyl	0.4		0.8			730
Acrylamide, *N*-propyl-	Acrylate, methyl	0.26		1.2			730
Acrylamide, *N-tert*-butyl-	Acrylonitrile	1.14		0.2			709
Acrylamide, *N-tert*-butyl-	Methacrylate, methyl	2.83		0.25			709
Acrylamide, α-fluoro-	Methacrylate, methyl	0.148	0.081	2.13	0.136	Y	823
Acrylamido, 1-,-1-deoxy-D-glucitol	Methacrylate, methyl	0.05	0.09	3.75	0.24	N	258
Acrylamido, 1-,-1-deoxy-D-glucitol	Styrene	0.03	0.02	2.42	0.08	N	258
Acrylamido, 1-,-1-deoxy-D-glucitol	Vinyl acetate	0.98	0.87	0.03	0.17	N	258
Acrylamido, 2-,2-methylpropanesulfonate, sodium	Vinyl acetate	11.6		0.05			700
Acrylamido, 2-,-2-methylpropanesulfonic acid	Methacrylate, 2-hydroxyethyl	0.9		0.86			618
Acrylamido, 2-,-2-methylpropanesulfonic acid	Methacrylate, 2-hydroxypropyl	1.03		0.89			618
Acrylamido-2-methylpropane sulfonate sodium salt	*N*-Vinylpyrrolidone	0.66		0.13			939
Acrylamidomethylamino, *p*-, azobenzene	Styrene	8.5		0.083			858
Acrylamidomethylamino, *p*-, azobenzene	Vinyl acetate	12.98		0.126			858
Acrylate, (−)-menthyl	Acenaphthalene					Y	645
Acrylate, (−)-menthyl	Pyridine, 4-vinyl-	0.291	0.017	2.32	0.45	Y	667
Acrylate, 1,1,5-trihydroperfluoroamyl	Methacrylonitrile	0.14		1.03			809
Acrylate, 2,3-dibromopropyl	Styrene	0.16		0.42			643
Acrylate, 2,4,5-trichlorophenyl	Pyrrolidone, *N*-vinyl-	0.163	0.02	0.012	0.008	Y	840
Acrylate, 2,4,5-trichlorophenyl	Styrene	0.23	0.021	0.277	0.048	Y	622
Acrylate, 2,4,6-tribromophenyl	Styrene	0.1		0.2			643
Acrylate, 2,4-dinitrophenyl	Acrylate, *N*-(2-hydroxyethyl)-carbazolyl	0.158	0.015	0.109	0.026	Y	716
Acrylate, 2,4-dinitrophenyl	Methacrylate, *N*-(2-hydroxyethyl)-carbazolyl	0.186	0.009	0.243	0.034		716
Acrylate, 2(*O*-ethyl methylphosphonoxy)-, methyl	Methacrylate, methyl	0.44	0.25	0.73	0.04		882
Acrylate, 2-chloroethyl	Acrylate, ethyl	1.03		0.9			726
Acrylate, 2-chloroethyl	Acrylate, methyl	1.07		0.9			726
Acrylate, 2-chloroethyl	Acrylonitrile	0.87		1.03			790
Acrylate, 2-chloroethyl	Itaconate, bis(tri-*n*-butyltin)	0.683		0.391			940
Acrylate, 2-chloroethyl	Itaconic anhydride	−0.016	0.022	2.46	0.38	Y	365
Acrylate, 2-chloroethyl	Itaconic anhydride	0.251	0.021	2.61	0.32	Y	365
Acrylate, 2-chloroethyl	Maleic anhydride	7.15	0.31	0.027	0.004	Y	117
Acrylate, 2-chloroethyl	Methacrylate, methyl	0.37		2.15			790
Acrylate, 2-chloroethyl	Pyrrolidone, *N*-vinyl-	0.52		0.03			790
Acrylate, 2-chloroethyl	Styrene	0.1	0.17	0.55	0.47	Y	140
Acrylate, 2-chloroethyl	Styrene	0.139	0.083	0.494	0.097	Y	178
Acrylate, 2-chloroethyl	Styrene	0.12		0.54			790
Acrylate, 2-chloroethyl	Vinyl butyl ether	2.01		0			790
Acrylate, 2-chloroethyl	Vinyl isobutyl ether	2.24		0			790
Acrylate, 2-chloroethylglucitol	Styrene	0.14	0.09	0.49	0.2		806
Acrylate, 2-cyano-, methyl	Styrene, α-methyl-	0.001		0.05			452
Acrylate, 2-cyano-, methyl	Vinyl acetate	0.5		0.005			452
Acrylate, 2-cyanoethyl	Styrene	0.128	0.074	0.399	0.032	Y	178
Acrylate, 2-ethylhexyl	Acrylate, glycidyl	1.12	0.07	1.18	0.06	N	126
Acrylate, 2-ethylhexyl	Methacrylate, glycidyl	0.09	0.13	2.41	0.46	Y	674
Acrylate, 2-ethylhexyl	Styrene	0.31	0.6	0.96	0.3	N	126
Acrylate, 2-ethylhexyl	Styrene	0.26		0.94			463
Acrylate, 2-ethylhexyl	Vinyl acetate	7.5		0.04			570
Acrylate, 2-ethylhexyl	Vinyl chloride	4.15		0.16			510
Acrylate, 2-hydroxyethyl	Acrylate, butyl	0.9	0.23	0.3	0.06	N	868
Acrylate, 2-hydroxyethyl	Acrylate, ethyl	0.97	0.24	0.5	0.03	N	868
Acrylate, 2-hydroxyethyl	Acrylate, methyl	0.9	0.13	0.94	0.04	N	868
Acrylate, 2-nitrobutyl	Acrylonitrile	1.76	0.69	0.67	0.25	N	234
Acrylate, 2-nitrobutyl	Methacrylate, methyl	0.27	0.04	1.19	0.12	N	234
Acrylate, 2-nitrobutyl	Styrene	0.12	0.01	0.35	0.03	N	234
Acrylate, 3,4-epoxyhexahydrobenzyl	Acrylonitrile	0.388	0.074	0.25	0.13	Y	680

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, 3,4-epoxyhexahydrobenzyl	Styrene	1.97	0.35	0.27	0.3	N	680
Acrylate, 4-acetylphenyl	Methacrylate, glycidyl	0.292	0.141	0.644	0.13	Y	1012
Acrylate, β-acetoxymethyl-, methyl	Methacrylate, methyl	0.24		0.91			1091
Acrylate, β-acetoxymethyl-, methyl	Styrene	0.19		0.34			1091
Acrylate, β-chloro-2-hydroxypropyl	Acrylonitrile	0.7		0.24			654
Acrylate, β-ethoxy-, ethyl	Acrylonitrile	−0.26	0.56	2.42	1.8	N	194
Acrylate, β-ethoxy-, ethyl	Styrene	−0.21	0.44	46.98	27.03	Y	194
Acrylate, *N*-(2-hydroxyethyl)-3,6-dichlorocarbazolyl	Methacrylate, 2,4-dinitrophenyl	0.115	0.029	1.207	0.032	Y	736
Acrylate, *N*-(2-hydroxyethyl)carbazolyl	Acryl-2′-ethyl-4,5,7-trinitro-9-fluorene	0.912	0.079	0.585	0.028	Y	715
Acrylate, *N*-(2-hydroxyethyl)carbazolyl	Acrylate, 2,4-dinitrophenyl	0.109	0.026	0.158	0.015	Y	716
Acrylate, *N*-(2-hydroxyethyl)carbazolyl	Acryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	0.62	0.044	0.361	0.016	Y	702
Acrylate, *N*-(2-hydroxyethyl)carbazolyl	Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	1.04	0.14	0.08	0.026	Y	702
Acrylate, *N*-(2-hydroxyethyl)carbazolyl	Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	0.334	0.022	1.364	0.018	Y	719
Acrylate, α-(4-chlorobenzyl)-, ethyl	Styrene	1.03	0.13	0.04	0.09		806
Acrylate, α-(4-cyanobenzyl)-, ethyl	Styrene	0.65	0.09	0.06	0.05		806
Acrylate, α-(4-methoxybenzyl)-, ethyl	Styrene	0.71	0.24	0.33	0.3		806
Acrylate, α-(*O*-ethyl methylphosphonoxy)-methyl	Methacrylate, methyl	0.44		0.73			851
Acrylate, α-(hydroxymethyl)-, methyl	Styrene	0.34		0.36			932
Acrylate, α-[2,2-bis(carbomethoxy)ethyl]-, methyl	Methacrylate, methyl	0.04		3.7			1092
Acrylate, α-[2,2-bis(carbomethoxy)ethyl]-, methyl	Styrene	0.09		0.58			1092
Acrylate, α-acetoxy-, ethyl	Acrylate, α-chloro-, ethyl	0.3		1.71			638
Acrylate, α-acetoxy-, ethyl	Acrylate, ethyl	0.968	0.033	0.943	0.075	Y	244
Acrylate, α-acetoxy-, ethyl	Methacrylate, methyl	0.608	0.02	1.65	0.12	Y	244
Acrylate, α-acetoxy-, ethyl	Styrene	0.173	0.01	0.557	0.031	Y	244
Acrylate, α-acetoxy-, ethyl	Vinyl acetate	5.621	0.094	0.08	0.021	Y	244
Acrylate, α-benzyl-, methyl	Styrene	0.175	0.053	0.57	0.028	Y	49
Acrylate, α-benzyl-, ethyl	Styrene	0.9	0.2	0.14	0.17		806
Acrylate, α-benzyl-, methyl	Methacrylate methyl	0.173	0.022	2.36	0.23	Y	610
Acrylate, α-benzyl-, methyl	Methacrylate methyl	0.09	0.01	3.9	0.19	N	610
Acrylate, α-bromo-, ethyl	Styrene	0.44	0.08	0.02	0.04		806
Acrylate, α-butyl-, methyl	Styrene	0.205	0.053	0.812	0.043	Y	49
Acrylate, α-chloro-, methyl	Methacrylate, methyl	1.13	0.19	0.31	0.24	Y	6
Acrylate, α-chloro-, ethyl	Acrylate, α-methoxy-, methyl	0.58		0.11			638
Acrylate, α-chloro-, ethyl	Acrylate, α-acetoxy-, ethyl	1.71		0.3			638
Acrylate, α-chloro-, ethyl	Acrylate, α-fluoro-, ethyl	2.9		0.21			638
Acrylate, α-chloro-, ethyl	Acrylate, methyl	3.22		0.09			638
Acrylate, α-chloro-, ethyl	Acrylonitrile, α-methoxy-	0.9		0.3			638
Acrylate, α-chloro-, ethyl	Methacrylonitrile	2		0.45			638
Acrylate, α-chloro-, ethyl	Styrene	0.32	0.11	0.08	0.07		806
Acrylate, α-chloro-, ethyl	Vinyl acetate	30		0.03			638
Acrylate, α-chloro-, methyl	Acrylonitrile	1.76	0.18	0.122	0.081	Y	6
Acrylate, α-chloro-, methyl	Methacrylate, methyl	1.92	0.08	0.11	0.09	N	31
Acrylate, α-chloro-, methyl	Styrene	0.3		0.25			74
Acrylate, α-chloro-, methyl	Vinylidene cyanide	0.313	0.021	0.066	0.07	Y	82
Acrylate, α-chloro-, sodium	Styrenesulfonate, *p*-, sodium	0.27	0.06	1.44	0.31	Y	834
Acrylate, α-cyano-, methyl	Acrylonitrile	0.68	0.09	0.01	0.02	N	141
Acrylate, α-cyano-, methyl	Acrylate, methyl	0.34	0.07	0.02	0.04	N	141
Acrylate, α-cyano-, methyl	Methacrylate, methyl	0.03	0.01	0.21	0.02	N	141
Acrylate, α-cyano-, methyl	Methacrylate, methyl	0.135	0.016	0.068	0.014	Y	182
Acrylate, α-cyano-, methyl	Methacrylonitrile	0.15	0.01	0.18	0.01	N	143
Acrylate, α-cyano-, methyl	Styrene	0.61	0.05	0.05	0.01	N	141
Acrylate, α-cyclohexyl-, methyl	Styrene	−0.001	0.034	1.615	0.048	Y	49
Acrylate, α-ethyl-, methyl	Styrene	0.197	0.014	0.758	0.014	Y	49
Acrylate, α-ethyloxymethyl-, methyl	Styrene	0.24		0.37			1047
Acrylate, α-fluoro-, ethyl	Acrylate, α-chloro-, ethyl	0.21		2.9			638
Acrylate, α-fluoro-, methyl	Styrene	0.16	0.11	0.66	0.17		806
Acrylate, α-hexafluoropropyloxymethyl-, methyl	Styrene	0.07		0.23			1047

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, α-hydroxymethyl-, ethyl	Azlactone, 2-vinyl-4,4-dimethyl-	0.668	0.12	0.389	0.229	Y	1011
Acrylate, α-hydroxymethyl-, ethyl	Methacrylate, methyl	1.34	0.062	1.26	0.17	N	1027
Acrylate, α-hydroxymethyl-, ethyl	Styrene	0.546	0.127	0.468	0.03	Y	982
Acrylate, α-isobutyl-, methyl	Styrene	0.202	0.042	0.974	0.022	Y	49
Acrylate, α-isopropyl-, methyl	Styrene	0.036	0.034	1.872	0.052	Y	49
Acrylate, α-isopropyloxymethyl-, methyl	Styrene	0.26		0.41			1047
Acrylate, α-methoxy, methyl	Acrylate, α-chloro-, ethyl	0.11		0.58			638
Acrylate, α-methoxy-, methyl	Styrene	0.513	0.058	1.131	0.078		806
Acrylate, α-*p*-chlorobenzyl-, methyl	Methacrylate, methyl	0.255	0.034	1.87	0.222	Y	905
Acrylate, α-*p*-methoxybenzyl-, methyl	Methacrylate, methyl	0.13	0.008	1.71	0.109	Y	905
Acrylate, α-phenoxymethyl-, methyl	Methacrylate, methyl	0.755	0.061	1.13	0.038	Y	1048
Acrylate, α- phenoxymethyl-, methyl	Styrene	0.256	0.102	0.176	0.05	Y	1048
Acrylate, α-phenyl-, methyl	Acrylate, methyl	1		0.06			141
Acrylate, α-phenyl-, methyl	Acrylonitrile	6.7		0.08			141
Acrylate, α-phenyl-, methyl	Methacrylate, methyl	0		0.21			141
Acrylate, α-phenyl-, methyl	Methacrylate, methyl	0		0.3			49
Acrylate, α-phenyl-, methyl	Methacrylonitrile	0.25		0.19			143
Acrylate, α-phenyl-, methyl	Styrene	1		0.06			141
Acrylate, α-phenyl-, methyl	Styrene	0.4		0.03			399
Acrylate, α-phenyl-, methyl	Styrene	0.45		0.06			488
Acrylate, α-phenyl-, methyl	Styrene	1.275	0.008	1.176	0.015	Y	49
Acrylate, α-phenyl-, butyl	Styrene	0.107	0.055	0.04	0.022	Y	488
Acrylate, α-phenyl-, chloroethyl	Styrene	0.201	0.09	0.025	0.015	Y	488
Acrylate, α-phenyl-, propyl	Styrene	0.126	0.063	0.03	0.015	Y	488
Acrylate, α-propyl-, methyl	Styrene	0.208	0.07	0.821	0.029	Y	49
Acrylate, α-propyloxymethyl-, methyl	Styrene	0.29		0.47			1047
Acrylate, α-*sec*-butyl-, methyl	Styrene	0.004	0.024	2.29	0.068	Y	49
Acrylate, α-tetrafluoropropyloxymethyl-, methyl	Methacrylate, methyl	0.61		1.43			1047
Acrylate, α-tetrafluoropropyloxymethyl-, methyl	Styrene	0.17		0.37			1047
Acrylate, α-trifluoroethyloxymethyl-, methyl	Styrene	0.2		0.26			1047
Acrylate, α-trifluoromethyl-, methyl	Methacrylate, methyl	−0.17	0.18	1.8	0.23	Y	681
Acrylate, α-trifluoromethyl-, methyl	Methacrylate, methyl	0	0.1	3.2	0.5	N	825
Acrylate, α-trifluoromethyl-, methyl	Methacrylate, methyl	0.1	0.1	2.3	0.2	N	825
Acrylate, α-trifluoromethyl-, methyl	Styrene, *p*-chloro-	0	0.03	0.24	0.02	N	825
Acrylate, α-trimethylsiloxy-, methyl	Styrene	1.42		0.46			1101
Acrylate, benzyl	Acrylonitrile	0.72	0.16	0.28	0.19	N	23
Acrylate, benzyl	Acrylonitrile	0.63		1.49			385
Acrylate, benzyl	Allyl chloride	9.9		0.06			438
Acrylate, benzyl	Methacrylate, methyl	0.19	0.2	2.23	0.14	Y	277
Acrylate, benzyl	Styrene	0.2	0.12	0.494	0.05	Y	178
Acrylate, benzyl	Styrene	0.248	0.041	0.534	0.028	Y	277
Acrylate, butyl	2-Oxazoline, 2-isopropenyl-	0.24	0.01	1.4	0.08		894
Acrylate, butyl	Acrolein	0.638	0.032	1.86	0.19	Y	291
Acrylate, butyl	Acrolein	1.12	0.41	2.29	0.58	N	292
Acrylate, butyl	Acrolein, methyl-	0.02		2.5			589
Acrylate, butyl	Acrylamide, *N*-propyl-	0.8		0.4			730
Acrylate, butyl	Acrylamide, *N*-methylol-	0.87		0.61			441
Acrylate, butyl	Acrylate, 2-hydroxyethyl	0.3	0.06	0.9	0.23	N	868
Acrylate, butyl	Acrylic acid	1.08	0.18	0.59	0.27	N	187
Acrylate, butyl	Acrylic acid	0.91	0.13	1.31	0.3	N	252
Acrylate, butyl	Acrylic acid, α-bromo-	0.19		0.09			733
Acrylate, butyl	Acrylonitrile	1.06	0.15	1.679	0.058	Y	118
Acrylate, butyl	Acrylonitrile	0.894	0.019	0.781	0.083	Y	162
Acrylate, butyl	Acrylonitrile	0.82	0.13	1.08	0.08	N	233
Acrylate, butyl	Acrylonitrile	0.9		1			513
Acrylate, butyl	Acrylonitrile	1.2		1			513
Acrylate, butyl	Allyl acetate	10.2	42.2	0.039	0.066	N	1053
Acrylate, butyl	Allyl chloride	5.83		0.1			438
Acrylate, butyl	Benzofuran, 2-vinyl-	0.041	0.043	7.8	1.36	N	1020
Acrylate, butyl	Butadiene	0.074	0.03	1.04	0.11	Y	257
Acrylate, butyl	Ethylene	13.94	0.38	0.01	0.003	N	42
Acrylate, butyl	Furan, 2-vinyl-5-methyl	0.118	0.019	0.392	0.051	Y	1105

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, butyl	Itaconate, bis(tri-*n*-butyltin)	0.91		0.422			940
Acrylate, butyl	Itaconate, dimethyl	0.4		0.94			398
Acrylate, butyl	Maleimide, *N*-(2,4-dimethylphenyl)-	0.62		0.2			788
Acrylate, butyl	Methacrylate, 2-(*N*,*N*-dimethylcarbamoyloxy)ethyl	0.47		1.89			600
Acrylate, butyl	Methacrylate, 2-chloro-2,3,3,3-fluoropropyl	0.079	0.304	0.221	0.636	N	927
Acrylate, butyl	Methacrylate, 2-hydroxyethyl	0.09	0.11	4.75	4.2	Y	358
Acrylate, butyl	Methacrylate, 2-hydroxypropyl	0.171	0.041	5.35	0.32	Y	333
Acrylate, butyl	Methacrylate, 2-hydroxypropyl	0.24	0.16	3.3	6.82	Y	662
Acrylate, butyl	Methacrylate, butyl	0.3		2.2			187
Acrylate, butyl	Methacrylate, glycidyl	0.083	0.052	2.16	0.57	Y	674
Acrylate, butyl	Methacrylate, methyl	0.13	0.15	0.92	0.2	N	30
Acrylate, butyl	Methacrylate, methyl	0.11	0.018	2.86	0.32	Y	639
Acrylate, butyl	Methacrylate, methyl	0.43	0.07	1.88	0.4	N	87
Acrylate, butyl	Methacrylic acid	0.31	0.1	1.25	0.31	N	187
Acrylate, butyl	Oxazoline, 2,2-isopropenyl-	0.24		1.4			862
Acrylate, butyl	Pyridine, 2-vinyl-	0.1		2.51			236
Acrylate, butyl	Pyridine, 2-vinyl-	0.11	0.01	2.59	0.14	N	78
Acrylate, butyl	Pyridine, 4-vinyl-	0.23	0.04	4.3	0.34	N	78
Acrylate, butyl	Styrene	0.25	0.05	0.79	0.08		1032
Acrylate, butyl	Styrene	0.34	0.06	0.79	0.11		1032
Acrylate, butyl	Styrene	0.38	0.05	0.75	0.1		1032
Acrylate, butyl	Styrene	−0.106	0.099	1.23	0.21	N	19
Acrylate, butyl	Styrene	0.29	0.05	0.44	0.03	N	20
Acrylate, butyl	Styrene	0.18	0.03	0.84	0.2	N	38
Acrylate, butyl	Styrene	0.19		0.66			38
Acrylate, butyl	Styrene	0.21		0.82			463
Acrylate, butyl	Styrene	0.15		0.8			549
Acrylate, butyl	Styrene	0.164		0.698			612
Acrylate, butyl	Styrene	0.164		0.698			632
Acrylate, butyl	Styrene, *o*-chloro-	0.2		2.25			558
Acrylate, butyl	Styrene, *p*-1-(2-hydroxybutyl)-	0.17		0.4			548
Acrylate, butyl	Styrene, *p*-octylamine sulfonate	0.3		2.3			817
Acrylate, butyl	Succinimide, *N*-vinyl-	1.54		0.15			789
Acrylate, butyl	Thiophene, 3-vinyl	0.386		0.499			1000
Acrylate, butyl	Vinyl acetate	3.48	3.39	0.018	0.072	Y	301
Acrylate, butyl	Vinyl bromide	4.07	0.44	0.18	0.16	N	268
Acrylate, butyl	Vinyl bromide	3.7		0.19			268
Acrylate, butyl	Vinyl chloride	4.4		0.07			518
Acrylate, butyl	Vinyl fluoride	19		0.01			604
Acrylate, butyl	Vinyl fluoride	19		0.01			606
Acrylate, butyl	Vinyl methyl ketone	0.8	0.06	1.81	0.24	N	53
Acrylate, butyl	Vinyl, *p*-, benzylethylcarbinol	0.17		0.4			591
Acrylate, butyl	Vinylanthracene, 9-	3.717	0.046	0.126	0.054	Y	123
Acrylate, butyl	Vinylidene chloride	0.873	0.022	0.934	0.006	Y	118
Acrylate, butyl	Vinylidene chloride	0.46		0.84			464
Acrylate, butyl	Vinylidene chloride	0.58		0.87			464
Acrylate, *cis*-B-cyano-, methyl	Acrylonitrile					N	1034
Acrylate, *cis*-B-cyano-, methyl	Styrene					N	1034
Acrylate, cresyl	Acrylonitrile	0.773	0.067	0.72	0.15	Y	363
Acrylate, cyclododecyl	Styrene	0.34	0.15	0.47	0.14		806
Acrylate, cyclododecyl	Styrene	0.31	0.08	0.54	0.08		806
Acrylate, cyclododecyl	Styrene	0.301	0.072	0.461	0.07		806
Acrylate, cyclohexyl	*N*-Vinylpyrrolidone	1.297	0.088	0.122	0.014		1061
Acrylate, cyclohexyl	Styrene	0.272	0.048	0.913	0.107		1061
Acrylate, cyclohexyl	Styrene	0.254	0.141	0.885	0.102		806
Acrylate, cyclohexyl	Styrene	0.16	0.13	0.53	0.11		806
Acrylate, cyclohexyl	Styrene	0.17	0.09	0.7	0.12		806
Acrylate, cyclohexyl	Styrene	0.18	0.07	0.58	0.1		806
Acrylate, di-, copper	Methacrylate, di-, dicyclopentadienyl-titanium	1.09		0.89			876
Acrylate, di-, copper	Styrene	0.12	0.08	5.94	0.05	N	829
Acrylate, di-, copper	Styrene	0.56	0.09	1.74	0.03	N	829

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, di-, nickel	Methacrylate, di-, dicyclopentadienyltitanium	0.95		0.65			876
Acrylate, di-, nickel	Styrene	0.53	0.06	1.83	0.02	N	829
Acrylate, di-, zinc	Acrylonitrile	0.24		0.41			646
Acrylate, di-, zinc	Styrene	0.9	0.07	1.1	0.02	N	829
Acrylate, dibutylchlorotin	Acrylate, methyl	0.09		0.81			659
Acrylate, diethylaluminum	Styrene	0.046	0.035	−0.002	0.016	Y	652
Acrylate, diethylaluminum	Styrene					Y	652
Acrylate, dodecyl	Acrylonitrile	0.88	0.16	2.37	0.07	N	233
Acrylate, ethyl	2-Oxazoline, 2-isopropenyl-	0.19	0.02	1.39	0.06		894
Acrylate, ethyl	Acrolein	0.6		1.2			291
Acrylate, ethyl	Acrolein	1.09	0.25	1.98	0.44	N	292
Acrylate, ethyl	Acrylamide, *N*-methylol-	1.4		1.4			442
Acrylate, ethyl	Acrylate, 2-chloroethyl	0.9		1.03			726
Acrylate, ethyl	Acrylate, 2-hydroxyethyl	0.5	0.03	0.97	0.24	N	868
Acrylate, ethyl	Acrylate, α-acetoxy-, ethyl	0.943	0.075	0.968	0.033	Y	244
Acrylate, ethyl	Acrylate, hydroxyethyl	0.5	0.05	0.88	0.09		890
Acrylate, ethyl	Acrylic acid	1.02		0.91			711
Acrylate, ethyl	Acrylonitrile	1.2	0.07	0.92	0.08		1023
Acrylate, ethyl	Acrylonitrile	0.81	0.27	1.16	0.78	N	198
Acrylate, ethyl	Acrylonitrile	4		0.66			389
Acrylate, ethyl	Acrylonitrile	0.95		0.44			542
Acrylate, ethyl	Allyl chloride	7.73		0.08			438
Acrylate, ethyl	Benzofuran, 2-vinyl-	0.015	0.01	5.76	0.334	N	1020
Acrylate, ethyl	Carbazole, *N*-vinyl-	1.1		0.27			757
Acrylate, ethyl	Ethylene, 1,1-diphenyl-	0.8		0.5			65
Acrylate, ethyl	Imidazolid-2-one, 1,3-divinyl-	0.41	0.13	−0.08	0.03	N	800
Acrylate, ethyl	Imidazolid-2-one, 1-ethyl-3-vinyl-	0.47	0.06	0.01	0.01	N	800
Acrylate, ethyl	Isopropenylisocyanate	0.79		0.15			434
Acrylate, ethyl	Maleimide, 2,3-dimethyl-*N*-(2-methacryloyloxy)ethyl	0.28	1.15	1.55	3.06	Y	619
Acrylate, ethyl	Methacrylate, 2-(sulfonic acid)ethyl	0.3		3.2			564
Acrylate, ethyl	Methacrylate, 2-bromoethyl	0.37		2.7			657
Acrylate, ethyl	Methacrylate, 2-hydroxyethyl	0.189	0.008	11.21	0.43	Y	358
Acrylate, ethyl	Methacrylate, 2-hydroxypropyl	0.273	0.054	13.32	1.05	Y	333
Acrylate, ethyl	Methacrylate, 2-hydroxypropyl	0.207	0.045	9.08	3.95	Y	662
Acrylate, ethyl	Methacrylate, butyl	0.22	0.17	2.43	0.39	Y	690
Acrylate, ethyl	Methacrylate, methyl	0.47		1.83			30
Acrylate, ethyl	Methacrylate, methyl	0.28		2			469
Acrylate, ethyl	Methacrylate, methyl	0.22	0.05	2.04	0.28	N	87
Acrylate, ethyl	Methacrylic acid					N	1065
Acrylate, ethyl	Norbornadiene	2.39	0.44	−0.01	0.03	N	267
Acrylate, ethyl	Oxazoline, 2-, 2-isopropenyl-	0.19		1.39			862
Acrylate, ethyl	Propanesulfonate, 3-[diethyl-2-(2-methacroyloxyethoxy)]ethyl	3.97	1.03	0.19	0.02	Y	962
Acrylate, ethyl	Pyridine, 2-vinyl-	0.21	0.06	0.28	0.1	N	250
Acrylate, ethyl	Pyridine, 2-vinyl-	0.21		2.25			832
Acrylate, ethyl	Pyridine, 4-vinyl-	0.29		2.58			832
Acrylate, ethyl	Pyrimid-2-one, 1,3-divinylhexahydro-	0.59	0.13	−0.05	0.03	N	800
Acrylate, ethyl	Styrene	0.139	0.085	0.699	0.072	Y	178
Acrylate, ethyl	Styrene	0.16		1.01			463
Acrylate, ethyl	Styrene	0.19		0.79			530
Acrylate, ethyl	Styrene	0.2		0.8			530
Acrylate, ethyl	Styrene	0.48		0.8			542
Acrylate, ethyl	Styrene	0.152		0.787			673
Acrylate, ethyl	Styrene	0.152		0.787			677
Acrylate, ethyl	Styrene 3-tri-*n*-butylstannyl-	10.4		0.01			985
Acrylate, ethyl	Tropone, 2-methacryloyloxy-	0.387	0.027	3.13	0.528	Y	1104
Acrylate, ethyl	Vinyl 2-chloroethyl ether	5		0.15			439
Acrylate, ethyl	Vinyl 2-chloroethyl ether	4.65		0			726
Acrylate, ethyl	Vinyl isobutyl sulfide	0.36	0.02	0.05	0.04	N	285
Acrylate, ethyl	Vinyl isobutyl sulfide	0.32	0.02	0.05	0.05	N	336
Acrylate, ethyl	Vinylanthracene, 9-	3.498	0.041	0.295	0.056	Y	123
Acrylate, ethyl	Vinylidene chloride	0.72	0.39	0.58	0.15	Y	65
Acrylate, ferrocenylmethyl	Acrylate, methyl	0.14		4.46			293

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, ferrocenylmethyl	Maleic anhydride	−0.3	2.96	0.09	0.14	Y	325
Acrylate, ferrocenylmethyl	Methacrylate, methyl	0.024	0.064	2.8	0.19	Y	293
Acrylate, ferrocenylmethyl	Methacrylate, methyl	0.08		2.9			293
Acrylate, ferrocenylmethyl	Styrene	−0.016	0.06	1.87	0.17	Y	293
Acrylate, ferrocenylmethyl	Styrene	0.02		2.3			293
Acrylate, ferrocenylmethyl	Styrene	0.02		2.5			576
Acrylate, furfuryl	Methacrylate, 2-hydroxyethyl	0.88	0.19	1.35	0.08		1063
Acrylate, glycidyl	Acrylate, 2-ethylhexyl	1.18	0.06	1.12	0.07	N	126
Acrylate, glycidyl	Acrylonitrile	1		1.01			114
Acrylate, glycidyl	Styrene	0.17		0.6			114
Acrylate, glycidyl	Styrene	0.24		0.73	0.34	N	126
Acrylate, glycidyl	Vinyl acetate	7.6		0.003			553
Acrylate, heptafluorobutyl	Butadiene	0.073	0.031	0.359	0.039	Y	206
Acrylate, heptafluorobutyl	Methacrylate, methyl	0.177		0.084		Y	206
Acrylate, heptafluorobutyl	Styrene	0.049	0.013	0.31	0.01	Y	206
Acrylate, heptyl	Maleimide, *N*-(2,4-dimethylphenyl)-	0.14		0.2			788
Acrylate, heptyl	Methacrylate, methyl	0.46	0.02	2.51	0.13		980
Acrylate, hydroxyethyl	Acrylate, ethyl	0.88	0.09	0.5	0.05		890
Acrylate, hydroxyethyl	Acrylate, methyl	1	0.1	1	0.1		890
Acrylate, hydroxyethyl	Acrylate, methyl	0.94	0.1	0.23	0.03		890
Acrylate, isobutyl	Methacrylate, methyl	0.135	0.02	4.1	0.5	Y	639
Acrylate, isobutyl	Methacrylate, glycidyl	0.282	0.026	1.24	0.12	Y	674
Acrylate, isobutyl	Methacrylate, methyl	0.29	0.04	1.04	0.08	N	30
Acrylate, isopropyl	Styrene	0.195	0.027	0.755	0.024	Y	178
Acrylate, *m*-chlorophenyl	Acrylonitrile	0.85	0.036	1.3	0.073	Y	363
Acrylate, *m*-nitrophenyl	Methacrylate methyl	1.04	0.135	0.928	0.248	Y	1074
Acrylate, methyl	2-Oxazoline, 2-isopropenyl-	0.16	0.04	1.9	0.08		894
Acrylate, methyl	2-Oxazolinium BF4, 3-methyl-2-isopropenyl-	0.096	0.057	0.251	0.483	Y	908
Acrylate, methyl	2-Oxazolinium, 2-isopropenyl-	0.191	0.031	2.08	0.327	Y	908
Acrylate, methyl	Acetylene, phenyl-	0.622	0.062	0.272	0.072	Y	60
Acrylate, methyl	Acetylene, phenyl-	0.664	0.045	0.093	0.01	Y	731
Acrylate, methyl	Acrolein	7.86	3.58	−0.07	0.03	N	207
Acrylate, methyl	Acrolein	0.83	0.12	1.41	0.4	Y	291
Acrylate, methyl	Acrolein	1.08	0.38	2.54	0.93	N	292
Acrylate, methyl	Acrylamide, *N*-methylol-	1.3		1.9			441
Acrylate, methyl	Acrylamide, *N*-propyl-	1.2		0.26			730
Acrylate, methyl	Acrylate, 2-chloroethyl	0.9		1.07			726
Acrylate, methyl	Acrylate, 2-hydroxyethyl	0.94	0.04	0.9	0.13	N	868
Acrylate, methyl	Acrylate, α-chloro-, ethyl	0.09		3.22			638
Acrylate, methyl	Acrylate, α-cyano-, methyl	0.02	0.04	0.34	0.07	N	141
Acrylate, methyl	Acrylate, α-phenyl-, methyl	0.06		1			141
Acrylate, methyl	Acrylate, dibutylchlorotin	0.81		0.09			659
Acrylate, methyl	Acrylate, ferrocenylmethyl	4.46		0.14			293
Acrylate, methyl	Acrylate, hydroxyethyl	1	0.1	1	0.1		890
Acrylate, methyl	Acrylate, hydroxyethyl	0.23	0.03	0.94	0.1		890
Acrylate, methyl	Acrylate, tributyltin	0.82		0.03			659
Acrylate, methyl	Acrylic acid, α-bromo-	0.24		0.1			733
Acrylate, methyl	Acrylonitrile	0.51	0.02	0.56	0.03		1024
Acrylate, methyl	Acrylonitrile	0.844	0.037	1.54	0.044	Y	151
Acrylate, methyl	Acrylonitrile	0.85	0.2	1.31	0.08	N	233
Acrylate, methyl	Acryloyl chloride	0.34		2.3			508
Acrylate, methyl	Allyl acetate	5		0			470
Acrylate, methyl	Allyl acrylate	0.52		0.33			578
Acrylate, methyl	Allyl chloride					Y	204
Acrylate, methyl	Allyl chloride	8.45		0.05			437
Acrylate, methyl	Allyl chloride	5.45		0.07			437
Acrylate, methyl	Allyl chloride	6.46		0.4			437
Acrylate, methyl	Allyl chloride	9.1		−0.02			437
Acrylate, methyl	Allylbenzene	11.35	0.66	−0.036	0.016	Y	351
Acrylate, methyl	Benzophenone, *p*-vinyl-	0.09	0.02	3.6	1.06	Y	866
Acrylate, methyl	Benzophenone, *p*-vinyl-	1.1	0.02	3.54	0.077	Y	911
Acrylate, methyl	Butadiene	0.07	0.87	1.09	4.53	Y	257
Acrylate, methyl	Butadiene, 2-chloro-	0.06		10.4		Y	61
Acrylate, methyl	Carbamate, *N,N*-diethyl-, vinyl	4.78	0.22	−0.027	0.049	Y	92

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, methyl	Carbazole, *N*-vinyl-	0.53	0.13	0.049	0.029	Y	13
Acrylate, methyl	Carbazole, *N*-vinyl-	0.43		0.11			551
Acrylate, methyl	Carbazole, *N*-vinyl-	0.51	0.073	0.028	0.045	Y	98
Acrylate, methyl	Cinnamate, dibutylchlorotin	1.65		−0.2			659
Acrylate, methyl	Diallyl phthalate	6.208	0.069	0.028	0.005	Y	297
Acrylate, methyl	Diallyl phthalate	6.18	0.16	0.038	0.008	Y	297
Acrylate, methyl	Diallylcyanamide	6.7		0.05			603
Acrylate, methyl	Diallylcyanamide	6.7		0.05			604
Acrylate, methyl	Ethylene					N	41
Acrylate, methyl	Ethylene	19.4		0.02			768
Acrylate, methyl	Ethylene 1,1-diphenyl-	0.09	0.01	−0.24	0.15	N	334
Acrylate, methyl	Ethylene, 1,1-diphenyl-	0.12	0.01	0.21	0.05	N	334
Acrylate, methyl	Ethylene tetrachloro-	830		0			402
Acrylate, methyl	Ethylene, trichloro-	33		0			470
Acrylate, methyl	Fumarate, diisopropyl	1.9		0.091			1038
Acrylate, methyl	Hexatriene, tetrachloro-	0.262	0.087	3.19	1.36	Y	5
Acrylate, methyl	Hexene-1	−0.13	0.6	−0.67	0.03	Y	60
Acrylate, methyl	Imidazole, 1-vinyl-2-methyl-	1.28		0.05			465
Acrylate, methyl	Indene	0.63		0.1			606
Acrylate, methyl	Indene	0.62		0.12			606
Acrylate, methyl	Isoprene	0.12		0.75			424
Acrylate, methyl	Isoprene, 3-acetoxy-	0.27		5.71			770
Acrylate, methyl	Isopropenyl, 3-1-cyclohexenyl, acetate	0.37		0.57			770
Acrylate, methyl	Isopropenylisocyanate	0.6		0.11			434
Acrylate, methyl	Isopropenylisocyanate	0.799	0.049	0.079	0.047	Y	99
Acrylate, methyl	Maleic anhydride	2.788	0.051	0.012	0.013	Y	260
Acrylate, methyl	Maleic anhydride	2.5		0			470
Acrylate, methyl	Maleimide, *N*-(4-bromophenyl)-	0.429	0.099	0.39	0.17	Y	628
Acrylate, methyl	Maleimide, *N*-(4-chlorophenyl)-	0.684	0.052	0.155	0.052	Y	628
Acrylate, methyl	Maleimide, *N*-(4-tolyl)-	0.64	0.16	0.14	0.16	Y	644
Acrylate, methyl	Maleimide, *N*-phenyl-	0.554	0.048	0.103	0.052	Y	644
Acrylate, methyl	Methacrylate, 2,3-epithiopropyl	0.34		2.81			787
Acrylate, methyl	Methacrylate, 2-(*N*,*N*-dimethyl carbamoyloxy)ethyl	0.22		1.32			600
Acrylate, methyl	Methacrylate, 2-chloroethyl	0.29		2.19			726
Acrylate, methyl	Methacrylate, 2-hydroxyethyl	−0.005	0.036	8.67	8.28	Y	358
Acrylate, methyl	Methacrylate, 2-hydroxypropyl	0.013	0.007	7.335	0.083	Y	333
Acrylate, methyl	Methacrylate, ferrocenylmethyl	1.4	0.35	0.042	0.085	Y	293
Acrylate, methyl	Methacrylate, methyl	0.4	0.12	2.15	0.04	N	324
Acrylate, methyl	Methacrylate, *t*-butyl dimethylsilyl	0.288	0.057	1.07	0.21	Y	1069
Acrylate, methyl	Methacrylic anhydride	0.16		4.75			275
Acrylate, methyl	Oxazoline, 2-,2-isopropenyl-	0.16		1.9			862
Acrylate, methyl	Oxazoline, 2-,2-isopropenyl-	0.328	0.032	2.07	0.191	Y	877
Acrylate, methyl	Oxazoline, 2-,2-isopropenyl-4,4-dimethyl	0.46		1.3			596
Acrylate, methyl	Oxazoline, 2-,4-acryloxy-methyl-2,4-dimethyl	0.29		0.11			596
Acrylate, methyl	Oxazoline, 2-,4-methacryl-oxy-2,4-dimethyl	0.4		0.56			596
Acrylate, methyl	Oxazolinium, 2-, tetrafluoro-borate, 3-methyl	0.552	0.043	0.344	0.105	Y	877
Acrylate, methyl	Phthalimide, *N*-(4-vinylphenyl)-	1.14		1.56			636
Acrylate, methyl	Phthalimide, *N*-(methacryloyloxy)-	0.193	0.028	1.23	0.168	Y	979
Acrylate, methyl	Propenyl, 2-chloro-, acetate	0.7		0			445
	Pyridine, 2-methyl-5-vinyl-	0.18	0.05	1	0.3	N	232
Acrylate, methyl	Pyridine, 2-vinyl-	0.37	0.4	2.13	0.54	Y	2
Acrylate, methyl	Pyridine, 2-vinyl-	0.17	0.03	1.72	0.26	N	231
Acrylate, methyl	Pyridine, 2-vinyl-	0.2	0.03	2.12	0.1	N	749
Acrylate, methyl	Pyridine, 2-vinyl-5-ethyl-	0.18	0.004	1.32	0.03	N	231
Acrylate, methyl	Pyridine, 4-vinyl-	0.2	0.03	1.78	0.32	N	231
Acrylate, methyl	Pyridine, 4-vinyl-	0.15		2.14			424
Acrylate, methyl	Pyrrolidone, 1-benzyl-3-methylene-5-methyl	0.125	0.025	1.58	0.24	Y	633
Acrylate, methyl	Styrene	0.24		0.82			139
Acrylate, methyl	Styrene	0.8	0.21	0.192	0.057	Y	14

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, methyl	Styrene	0.168	0.067	0.722	0.056	Y	151
Acrylate, methyl	Styrene	0.18		0.75			393
Acrylate, methyl	Styrene	0.18		0.75			397
Acrylate, methyl	Styrene	0.14		0.68			463
Acrylate, methyl	Styrene	0.07		0.9			495
Acrylate, methyl	Styrene	0.3		0.65			496
Acrylate, methyl	Styrene	0.16		0.4			496
Acrylate, methyl	Styrene	0.148	0.026	0.871	0.022	Y	865
Acrylate, methyl	Styrene, 2,5-dichloro-	0.142	0.054	3.93	1.38	Y	136
Acrylate, methyl	Styrene, 2,5-dichloro-	0.288	0.069	4.47	0.15	Y	2
Acrylate, methyl	Styrene, α-methoxy-	0.17	0.02	0.003	0.01	N	142
Acrylate, methyl	Styrene, *m*-methyl-	0.14		1.65			558
Acrylate, methyl	Styrene, *p*-1-(2-hydroxybutyl)-	0.18		0.48			548
Acrylate, methyl	Styrene, *p*-*N*,*N*-dimethylamino-	0.06	0.01	0.38	0.16	Y	866
Acrylate, methyl	Styrene, *p*-dimethylamino-	0.192	0.01	0.372	0.055	Y	911
Acrylate, methyl	Styrene, *p*-methoxy-	0.07		2			558
Acrylate, methyl	Styrene, *p*-methyl-	0.17		1.54			558
Acrylate, methyl	Succinimide, *N*-vinyl-	0.357	0.063	0.9	0.17	Y	80
Acrylate, methyl	Vinyl 2-chloroethyl ether	3.27		0			726
Acrylate, methyl	Vinyl acetate	7.28	0.37	0.04	0.01		1067
Acrylate, methyl	Vinyl acetate	6.38		0.03		Y	158
Acrylate, methyl	Vinyl acetate	2.58	0.3	0.405	0.043	Y	25
Acrylate, methyl	Vinyl acetate	6.7		0.03			489
Acrylate, methyl	Vinyl acetate	6.3		0.03			568
Acrylate, methyl	Vinyl butyl ether	3.65	0.07	0.01	0.01	N	345
Acrylate, methyl	Vinyl butyl ether	3.6		0			574
Acrylate, methyl	Vinyl butylsulfonate	4.14	0.58	0	0.4	Y	185
Acrylate, methyl	Vinyl chloride	4.4	0.22	0.093	0.002	Y	151
Acrylate, methyl	Vinyl chloride	7.66	0.5	0.002	0.01	Y	46
Acrylate, methyl	Vinyl chloride	5		0			470
Acrylate, methyl	Vinyl chloride	4		0.06			518
Acrylate, methyl	Vinyl cymantrene	0.46	0.14	0.211	0.067	Y	354
Acrylate, methyl	Vinyl ethyl ether	3.3		0			470
Acrylate, methyl	Vinyl fluoride	43		0.01			604
Acrylate, methyl	Vinyl fluoride	43		0.01			606
Acrylate, methyl	Vinyl hendecanoate	3.615	0.012	0.033	0.008	Y	145
Acrylate, methyl	Vinyl isobutyl ether	3.63	0.04	0	0.002	N	347
Acrylate, methyl	Vinyl isobutyl sulfide	0.3	0.01	0.05	0.02	N	285
Acrylate, methyl	Vinyl isobutyl sulfide	0.29	0.01	0.06	0.02	N	336
Acrylate, methyl	Vinyl methyl sulfide	0.209	0.034	0.084	0.086	Y	195
Acrylate, methyl	Vinyl phenyl sulfide	0.57	0.1	−0.02	0.22	Y	193
Acrylate, methyl	Vinyl stearate	5.8		0.03			537
Acrylate, methyl	Vinyl thiolacetate	0.8		0.23			566
Acrylate, methyl	Vinyl, 2-bromo-, ethyl ether	16.83	2.34	−0.79	0.32	Y	194
Acrylate, methyl	Vinyl, *p*-, benzylethylcarbinol	0.18		0.48			591
Acrylate, methyl	Vinylanthracene, 9-	2.998	0.067	0.08	0.092	Y	123
Acrylate, methyl	Vinylhydroquinone dibenzoate	0.46		0.75			440
Acrylate, methyl	Vinylidene chloride	0.95	0.21	0.9	0.11	Y	107
Acrylate, methyl	Vinylidene chloride	0.8		0.7			107
Acrylate, methyl	Vinylidene chloride	1		1			377
Acrylate, methyl	Vinylisocyanate	1.38		0.14			435
Acrylate, mono-, ethylene glycol	Acrylonitrile	1.2		0.8			682
Acrylate, *o*-chlorophenyl	Acrylonitrile	1.171	0.023	1.123	0.037	Y	363
Acrylate, *o*-chlorophenyl	Methacrylate, methyl	0.4		1.25			621
Acrylate, octadecyl	Acrylonitrile	1.04	0.38	1.91	0.1	Y	118
Acrylate, octadecyl	Acrylonitrile	1.35	0.4	4.39	0.64	N	233
Acrylate, octadecyl	Acrylonitrile	0.32	0.35	1.76	0.2	Y	290
Acrylate, octadecyl	Methacrylate, methyl	0.464	0.07	2.477	0.024	Y	290
Acrylate, octadecyl	Styrene	0.37	0.12	0.777	0.018	Y	290
Acrylate, octadecyl	Styrene	0.18		0.44			290
Acrylate, octadecyl	Styrene	0.31		0.79			290
Acrylate, octadecyl	Styrene	0.18		0.44			290
Acrylate, octadecyl	Vinylidene chloride	1	0.14	0.9	0.037	Y	118
Acrylate, octyl	Acrylonitrile	0.84	0.74	1.98	0.3	Y	118
Acrylate, octyl	Styrene	0.01	0.53	0.39	0.11		806

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylate, octyl	Vinyl chloride	4.8		0.12			518
Acrylate, octyl	Vinylidene chloride	0.679	0.057	0.851	0.018	Y	118
Acrylate, *p*-(cinnamoylamino)phenyl	Styrene	0.35		1.08			685
Acrylate, *p*-bromophenyl	Acrylonitrile	1.026	0.062	0.8	0.13	Y	363
Acrylate, *p*-bromophenyl	Methacrylate, methyl	0.191	0.014	0.926	0.056	Y	1074
Acrylate, *p*-chlorophenyl	Acrylonitrile	0.951	0.063	0.32	0.056	Y	63
Acrylate, *p*-chlorophenyl	Methacrylate, methyl	0.235	0.023	1.01	0.132	Y	1074
Acrylate, *p*-chlorophenyl	Methacrylate, methyl	0.3		1.09			621
Acrylate, *p*-methylphenyl	Methacrylate, methyl	0.223	0.036	1.45	0.275	Y	1074
Acrylate, pentabromophenyl	Styrene	0.1		0.16			643
Acrylate, pentachlorophenyl	Acrylic acid	1.12		0.82			900
Acrylate, pentachlorophenyl	Styrene	0.9		2.11			900
Acrylate, phenyl	Acrylonitrile	0.327	0.045	0.43	0.13	Y	363
Acrylate, phenyl	Fumarate, diisopropyl	3.4		0.07			1038
Acrylate, phenyl	Methacrylate, methyl	0.45		1.56			621
Acrylate, phenyl	Methacrylate, methyl	0.416	0.041	1.54	0.14	Y	672
Acrylate, sodium	Vinyl acetate					Y	37
Acrylate, *tert*-butyl	Methacrylate, glycidyl	0.463	0.079	2.096	0.443	Y	814
Acrylate, tetrahydrofurfuryl	Styrene	0.489	0.196	0.475	0.186		806
Acrylate, thio- butyl	Styrene	0.421	0.17	0.439	0.14		806
Acrylate, thio-, ethyl	Styrene	0.291	0.075	0.18	0.038		806
Acrylate, thio-, *tert*-butyl	Styrene	0.368	0.055	0.157	0.034		806
Acrylate, tributyltin	Acrylate, methyl	0.03		0.82			659
Acrylate, trifluoro-, methyl	Styrene	0		1.4			595
Acrylate, trifluoro-, methyl	Styrene	0		1.4			598
Acrylate, trifluoro-, methyl	Vinyl acetate	0		0.24			595
Acrylate, trifluoro-, methyl	Vinyl acetate	0		0.24			598
Acrylate, trifluoro-, methyl	Vinyl chloride	0		1.5			595
Acrylate, trifluoro-, methyl	Vinyl chloride	0		1.5			598
Acrylic acid	Acrolein	0.08	0.28	2.48	1.24	Y	57
Acrylic acid	Acrolein	1.15		0.5			57
Acrylic acid	Acrylamide	0.288	0.029	1.08	0.17	Y	221
Acrylic acid	Acrylamide	1.644	0.069	0.445	0.075	Y	278
Acrylic acid	Acrylamide	0.341	0.012	1.346	0.044	Y	278
Acrylic acid	Acrylamide	1.38	0.14	0.598	0.047	Y	37
Acrylic acid	Acrylamide	3.8		0.58			869
Acrylic acid	Acrylamide	0.29		1.06			869
Acrylic acid	Acrylamide, *N,N*-dimethyl-	0.36	0.04	0.35	0.12	N	205
Acrylic acid	Acrylate, pentachlorophenyl	0.82		1.12			900
Acrylic acid	Acrylate, butyl	0.59	0.27	1.08	0.18	N	187
Acrylic acid	Acrylate, butyl	1.31	0.3	0.91	0.13	N	252
Acrylic acid	Acrylate, ethyl	0.91		1.02			711
Acrylic acid	Acrylic acid, 2-benzamido-	0.48		2.08			1019
Acrylic acid	Acrylic acid, *N*-acetyl-α-amino-	1.41		0.49			661
Acrylic acid	Carbamate, *N,N*-diethyl-, vinyl	5.58	0.32	−0.097	0.068	Y	92
Acrylic acid	Carbamate, *N*-vinyl-, ethyl	4.69		0.26		Y	81
Acrylic acid	Imidazole, 1-vinyl-	0.0025		0.32			955
Acrylic acid	Methacrylate, isopropyl	1.03		0.68			847
Acrylic acid	Methacrylate, methyl	1.73	0.096	0.418	0.02	N	964
Acrylic acid	Methacrylate, butyl	0.24	0.21	3.53	0.42	N	187
Acrylic acid	*N*-Vinylimidazole	1.932	0.077	0.187	0.025	Y	879
Acrylic acid	Phosphonic acid, α-phenylvinyl-	0.7	0.13	0.04	0.83	Y	132
Acrylic acid	Propenyl, 2-chloro-, acetate	1		0			445
Acrylic acid	Propylene					Y	183
Acrylic acid	Pyrrolidone, *N*-vinyl-	0.88	0.69	0.1	0.21	N	250
Acrylic acid	Styrene	0.05		0.25			312
Acrylic acid	Styrene	0.08		1.1			312
Acrylic acid	Styrene	0.25		0.15			397
Acrylic acid	Styrene	0.35		0.22			414
Acrylic acid	Styrene	0.136	0.029	0.253	0.069	Y	47
Acrylic acid	Styrene	0.15		0.25			490
Acrylic acid	Styrene	0.45		0.25			542
Acrylic acid	Styrene	0.25		0.15			561
Acrylic acid	Vinyl acetate	8.66	6.63	0.021	0.059	Y	37
Acrylic acid	Vinyl chloride	6.69	0.88	0.107	0.011	Y	135

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylic acid	Vinyl chloride	9.28	0.51	0.03	0.09	N	229
Acrylic acid	Vinyl chloride	7.04	0.33	0.024	0.069	Y	288
Acrylic acid	Vinyl chloride	6.8		0.11			379
Acrylic acid	Vinylhydroquinone	0.29	0.11	0.026	0.092	Y	337
Acrylic acid	Vinylhydroquinone dibenzoate	0.44		0.95			444
Acrylic acid	Vinylidene chloride	1.246	0.042	0.437	0.065	Y	288
Acrylic acid	Vinylidene cyanide	0.212	0.016	0.291	0.042	Y	82
Acrylic acid, 2-benzamido-	Acrylic acid	2.08		0.48			1019
Acrylic acid, 2-benzamido-	Styrene	0.78		0.13			1019
Acrylic acid, 2-ethyl	Methacrylic acid	0.194	0.027	1.12	0.132	Y	1004
Acrylic acid, 2-ethyl	Methacrylic acid	0.069	0.04	2.29	0.417	Y	1004
Acrylic acid, 2-ethyl	Methacrylic acid	0.23		1.14		Y	961
Acrylic acid, *N*-acetyl-α-amino	Acrylic acid	0.49		1.41			661
Acrylic acid, *N*-acetyl-α-amino-	Styrene	0.44		0.91			661
Acrylic acid, α-bromo	Methacrylate, methyl	0.11		0.37			733
Acrylic acid, α-bromo	Acrylate, butyl	0.09		0.19			733
Acrylic acid, α-bromo	Acrylate, methyl	0.1		0.24			733
Acrylic acid, α-bromo	Methacrylate, butyl	0.13		0.85			733
Acrylic acid, α-bromo	Pyrrolidone, *N*-vinyl	0.07		0.25			694
Acrylic acid, *cis*-3-bromo-	Acrylamide	−0.2	0.11	7.64	1.67	Y	298
Acrylic acid, *cis*-3-ethyl	Acrylamide	−0.16	0.1	6.32	0.68	Y	298
Acrylic acid, *trans*-3-bromo-	Acrylamide	−0.33	0.16	4.13	0.76	Y	298
Acrylic acid, *trans*-3-ethyl-	Acrylamide	−0.21	0.11	8.19	1.09	Y	298
Acrylic anhydride	Allyl chloride	11.66	0.56	0.02	0.02	N	219
Acrylic anhydride	Methacrylonitrile	1	0.13	0.42	0.04	N	219
Acrylic anhydride	Styrene	0.1	0.004	0.16	0.01	N	219
Acrylonitrile	1,3-pentadiene, *cis*-	0.026	0.173	0.017	0.012	Y	912
Acrylonitrile	1,3-pentadiene, *cis*-	0.06	0.242	0.071	0.029	Y	912
Acrylonitrile	1,3-pentadiene, *trans*-	0.069	0.104	0.018	0.01	Y	912
Acrylonitrile	1,3-pentadiene, *trans*-	0.068	0.096	0.064	0.02	Y	912
Acrylonitrile	2-Oxazoline, 2-isopropenyl-	0.13	0.04	0.52	0.06		894
Acrylonitrile	Acenaphthalene	0.02		2.56			575
Acrylonitrile	Acetylene, phenyl-	0.266	0.044	0.325	0.083	Y	60
Acrylonitrile	Aconitate, trimethyl	4.24	0.92	−0.48	0.11	Y	147
Acrylonitrile	Acrolein	0.88	0.06	1.16	0.13	N	207
Acrylonitrile	Acrolein	0.71	0.03	1.07	0.08	N	207
Acrylonitrile	Acrolein	0.48	0.01	1.52	0.07	N	208
Acrylonitrile	Acrolein	0.6		1.28			504
Acrylonitrile	Acrolein diethylacetal	11.19		0.02			767
Acrylonitrile	Acrolein, methyl-	0.15		1.7			455
Acrylonitrile	Acrolein, methyl-	0.15		3.8			587
Acrylonitrile	Acrolein, methyl-	0.03	0.19	1.72	0.42	Y	91
Acrylonitrile	Acrylamide	0.863	0.033	0.81	0.11	Y	228
Acrylonitrile	Acrylamide	0.97	0.1	1.08	0.51	Y	228
Acrylonitrile	Acrylamide	1.8	0.75	0.5	0.26	N	330
Acrylonitrile	Acrylamide	0.88		1.36			421
Acrylonitrile	Acrylamide	0.8		1.3			461
Acrylonitrile	Acrylamide, *N*-methylol-	0.7		1.2			442
Acrylonitrile	Acrylamide, *N*-methylol-	0.98		2.33			577
Acrylonitrile	Acrylamide, *N*-methylol-	0.22		2.56			596
Acrylonitrile	Acrylamide, *N*-octadecyl-	1.034	0.02	1.4	0.17	Y	119
Acrylonitrile	Acrylamide, *N-tert*-butyl-	0.2		1.14			709
Acrylonitrile	Acrylate, 2-chloroethyl	1.03		0.87			790
Acrylonitrile	Acrylate, 2-nitrobutyl	0.67	0.25	1.76	0.69	N	234
Acrylonitrile	Acrylate, 3,4-epoxyhexahydrobenzyl	0.25	0.13	0.388	0.074	Y	680
Acrylonitrile	Acrylate, β-chloro-2-hydroxypropyl	0.24		0.7			654
Acrylonitrile	Acrylate, β-ethoxy-, ethyl	2.42	1.8	−0.26	0.56	N	194
Acrylonitrile	Acrylate, α-chloro-, methyl	0.122	0.081	1.76	0.18	Y	6
Acrylonitrile	Acrylate, α-cyano-, methyl	0.01	0.02	0.68	0.09	N	141
Acrylonitrile	Acrylate, α-phenyl-, methyl	0.08		6.7			141
Acrylonitrile	Acrylate, benzyl	0.28	0.19	0.72	0.16	N	23
Acrylonitrile	Acrylate, benzyl	1.49		0.63			385
Acrylonitrile	Acrylate, butyl	1.679	0.058	1.06	0.15	Y	118
Acrylonitrile	Acrylate, butyl	0.781	0.083	0.894	0.019	Y	162
Acrylonitrile	Acrylate, butyl	1.08	0.08	0.82	0.13	N	233

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Acrylate, butyl	1		0.9			513
Acrylonitrile	Acrylate, butyl	1		1.2			513
Acrylonitrile	Acrylate, *cis*-β-cyano-, methyl					N	1034
Acrylonitrile	Acrylate, cresyl	0.72	0.15	0.773	0.067	Y	363
Acrylonitrile	Acrylate, di-, zinc	0.41		0.24			646
Acrylonitrile	Acrylate, dodecyl	2.37	0.07	0.88	0.16	N	233
Acrylonitrile	Acrylate, ethyl	0.92	0.08	1.2	0.07		1023
Acrylonitrile	Acrylate, ethyl	1.16	0.78	0.81	0.27	N	198
Acrylonitrile	Acrylate, ethyl	0.66		4			389
Acrylonitrile	Acrylate, ethyl	0.44		0.95			542
Acrylonitrile	Acrylate, glycidyl	1.01		1			114
Acrylonitrile	Acrylate, *m*-chlorophenyl	1.3	0.073	0.85	0.036	Y	363
Acrylonitrile	Acrylate, methyl	0.56	0.03	0.51	0.02		1024
Acrylonitrile	Acrylate, methyl	1.54	0.044	0.844	0.037	Y	151
Acrylonitrile	Acrylate, methyl	1.31	0.08	0.85	0.2	N	233
Acrylonitrile	Acrylate, mono-, ethylene glycol	0.8		1.2			682
Acrylonitrile	Acrylate, *o*-chlorophenyl	1.123	0.037	1.171	0.023	Y	363
Acrylonitrile	Acrylate, octadecyl	1.91	0.1	1.04	0.38	Y	118
Acrylonitrile	Acrylate, octadecyl	4.39	0.64	1.35	0.4	N	233
Acrylonitrile	Acrylate, octadecyl	1.76	0.2	0.32	0.35	Y	290
Acrylonitrile	Acrylate, octyl	1.98	0.3	0.84	0.74	Y	118
Acrylonitrile	Acrylate, *p*-bromophenyl	0.8	0.13	1.026	0.062	Y	363
Acrylonitrile	Acrylate, *p*-chlorophenyl	0.32	0.056	0.951	0.063	Y	363
Acrylonitrile	Acrylate, phenyl	0.43	0.13	0.327	0.045	Y	363
Acrylonitrile	Acryloyl chloride	1.2		1			580
Acrylonitrile	Acryloylpyrrolidone	1.56		0.43			666
Acrylonitrile	Allyl acetate	6.57	2.11	−0.09	0.12	Y	108
Acrylonitrile	Allyl alcohol					N	108
Acrylonitrile	Allyl alcohol	2.6		0.05			108
Acrylonitrile	Allyl alcohol	3.9	1.06	0.1	0.16	N	176
Acrylonitrile	Allyl alcohol	1.79	0.11	0.12	0.37	Y	300
Acrylonitrile	Allyl chloride	2.145	0.049	−0.152	0.008	Y	108
Acrylonitrile	Allyl chloride	3.45	0.65	0.08	0.03	Y	47
Acrylonitrile	Allylbenzene	3.82	1.7	−0.05	0.15	Y	24
Acrylonitrile	Allylbenzene	5.35	0.3	−0.026	0.022	Y	351
Acrylonitrile	Allylcyclohexane	3.81	11.42	−0.18	1.01	Y	24
Acrylonitrile	Allylstearamide, *N*-	3.79	0.28	0.03	0.27	Y	119
Acrylonitrile	Benzothiazole, vinylmercapto-	0.191	0.014	0.178	0.081	Y	763
Acrylonitrile	Butadiene	0.046	0.006	0.358	0.046	Y	21
Acrylonitrile	Butadiene	0.1	0.059	0.452	0.023	Y	255
Acrylonitrile	Butadiene	0.05		0.35			418
Acrylonitrile	Butadiene	0.06		0.1			501
Acrylonitrile	Butadiene	0.03	0.02	0.2	0.01	N	66
Acrylonitrile	Butadiene, 1,4-dicarboxylate, diethyl	0.29	0.12	2.79	0.9	N	327
Acrylonitrile	Butadiene, 1-(diethylamino)-	0.481	0.049	0.426	0.208	Y	864
Acrylonitrile	Butadiene, 1-(diethylamino)-	0.208	0.049	0.425	0.48	Y	889
Acrylonitrile	Butadiene, 1-acetoxy-	−0.012	0.043	0.82	0.68	Y	91
Acrylonitrile	Butadiene, 1-ethoxy	0.02	0.008	0.065	0.009	Y	864
Acrylonitrile	Butadiene, 1-ethoxy	0.01	0.008	0.065	0.02	Y	889
Acrylonitrile	Butadiene, 2-chloro-	0.04		5.35			402
Acrylonitrile	Butadiene, 2-chloro-	0.05		4.8			504
Acrylonitrile	Butadiene, 2-chloro-	0.056	0.023	5.38	0.45	Y	61
Acrylonitrile	Butadiene, 2-fluoro-	0.077	0.021	0.6	0.18	Y	174
Acrylonitrile	Butadiene, 2-trimethylsilyloxy-	0.036		0.065			993
Acrylonitrile	Butadiene-1-carboxylate, ethyl	0.3		3.2			791
Acrylonitrile	Butadiene-1-carboxylic acid	0.2		4			791
Acrylonitrile	Butadiene-1-carboxylic acid	0.03		11			799
Acrylonitrile	Butene-1	13.46	3.61	0.24	0.81	Y	211
Acrylonitrile	Butene-1, 4-(*p*-chlorophenyl)-	4.68	0.63	0.066	0.044	Y	24
Acrylonitrile	Butene-1, 4-(*p*-methoxy-phenyl)-	2.56	1.11	−0.085	0.088	Y	24
Acrylonitrile	Butene-1, 4-cyclohexyl-	5.03	0.76	0.014	0.055	Y	24
Acrylonitrile	Butene-1, 4-phenyl-	4.51	0.86	0.024	0.069	Y	24
Acrylonitrile	Butene-2, *cis*	22.32		0.11		Y	211
Acrylonitrile	Butene-2, *trans*	62.99	10.67	0.84	0.6	Y	211
Acrylonitrile	Carbazole, *N*-vinyl-	0.39	0.04	0.09	0.04	N	269

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Cinnamate, α-cyano-, ethyl	12.6	2.3	−0.18	0.15	Y	310
Acrylonitrile	Cinnamonitrile	8.46	0.37	0.36	0.1	Y	310
Acrylonitrile	Cinnamonitrile, α-cyano-					Y	310
Acrylonitrile	Citraconimide, *N*-methyl-	0.53		0.6			460
Acrylonitrile	Crotonaldehyde	8		0			429
Acrylonitrile	Crotonaldehyde	25		0.01			504
Acrylonitrile	Crotonate, α-chloro-, ethyl	9.53		0			785
Acrylonitrile	Crotonate, α-cyano-, ethyl	11.4		0.06			785
Acrylonitrile	Crotonate, α-methoxy-, methyl	1		0			785
Acrylonitrile	Crotonate, α-methyl-, methyl	2.97		0			785
Acrylonitrile	Crotonate, α-acetyl-, methyl	8.68		0			785
Acrylonitrile	Crotonate, α-carboethoxy-, ethyl	18.7		0			785
Acrylonitrile	Crotonate, ethyl	25.2		0			785
Acrylonitrile	Crotonic acid	21		0			402
Acrylonitrile	Cyclopentene, 4-, -1,3-dione	3.02	0.71	0.04	0.058	Y	95
Acrylonitrile	Cyclopropene, 3,3-dimethoxy-	1.424	0.038	0.245	0.037	Y	838
Acrylonitrile	Diallyl phthalate	4.23	0.15	0.04	0.012	Y	153
Acrylonitrile	Diallyl phthalate	2.78	0.52	0.037	0.031	Y	153
Acrylonitrile	Ethylene	7		0			415
Acrylonitrile	Ethylene,1,1-diphenyl-	0.05	0.01	−0.02	0.01	N	143
Acrylonitrile	Ethylene, tetrachloro-	456		0			571
Acrylonitrile	Ethylene, tetrachloro-	470		0			59
Acrylonitrile	Ethylene, trichloro-	67		0			470
Acrylonitrile	Ethylene, trichloro-	62.1		0			572
Acrylonitrile	Fumarate, diisopropyl	16		0.2			1038
Acrylonitrile	Fumarate, diethyl	8		0			470
Acrylonitrile	Fumarate, diethyl	10		0.1			640
Acrylonitrile	Hexatriene, tetrachloro-	0.234	0.05	4.01	1.25	Y	4
Acrylonitrile	Hexene-1	2.07		−1.08		Y	60
Acrylonitrile	Indene	0.281	0.05	0.09	0.25	Y	360
Acrylonitrile	Indene	0.29	0.12	0.07	0.24	Y	360
Acrylonitrile	Isobutylene	1.295	0.008	0	0.002	Y	21
Acrylonitrile	Isobutylene	0.98	0.25	0.02	0.38	Y	211
Acrylonitrile	Isobutylene	1.02		0			484
Acrylonitrile	Isoprene	0.03		0.45			417
Acrylonitrile	Isopropenyl methyl ketone	0.3		1.2			490
Acrylonitrile	Isopropenyl methyl ketone	0.36		0.7			542
Acrylonitrile	Isopropenylisocyanate	0.24		0.1			434
Acrylonitrile	Itaconic acid	0.59	0.34	0.86	0.74	Y	164
Acrylonitrile	Itaconic anhydride	0.034	0.014	4.83	0.33	Y	365
Acrylonitrile	Maleate, diethyl	12		0			470
Acrylonitrile	Maleate, diethyl	20		0			640
Acrylonitrile	Maleic anhydride	6		0			470
Acrylonitrile	Maleimide, *N*-(2-bromophenyl)-	2.029	0.214	0.338	0.045	Y	863
Acrylonitrile	Maleimide, *N*-(2-chlorophenyl)-	0.956	0.095	1.078	0.048	Y	863
Acrylonitrile	Maleimide, *N*-(4-chlorophenyl)-	0.972	0.112	0.743	0.048	Y	863
Acrylonitrile	Maleimide, *N*-2-bromophenyl-	2.02	0.205	0.348	0.044	Y	915
Acrylonitrile	Maleimide, *N*-2-chlorophenyl-	0.955	0.095	1.08	0.048	Y	915
Acrylonitrile	Maleimide, *N*-4-chlorophenyl-	0.972	0.112	0.743	0.048	Y	915
Acrylonitrile	Maleimide, *N*-octadecyl-	2.482	0.119	0.389	0.023	Y	863
Acrylonitrile	Maleimide, *N*-octadecyl-	2.46	0.115	0.39	0.023	Y	915
Acrylonitrile	Methacrylamide, *N*-(*p*-chlorophenyl)-	0.399	0.019	0.627	0.051	Y	352
Acrylonitrile	Methacrylamide, *N*-(*p*-methoxyphenyl)-	0.337	0.019	0.559	0.048	Y	352
Acrylonitrile	Methacrylamide, *N*-(*p*-methylphenyl)-	0.336	0.026	0.549	0.06	Y	352
Acrylonitrile	Methacrylamide, *N*-(*p*-nitrophenyl)-	0.37	0.14	2.34	1.07	Y	352
Acrylonitrile	Methacrylamide, *N*-phenyl-	0.381	0.085	0.71	0.19	Y	352
Acrylonitrile	Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	0.02		14			664
Acrylonitrile	Methacrylate, 2-(*N*,*N*-dimethylcarbamoyloxy)ethyl	0.38		1.89			600
Acrylonitrile	Methacrylate, 2-bromoethyl	0.31		2.38			657
Acrylonitrile	Methacrylate, 2-chloroethyl	0.14		1.3			382
Acrylonitrile	Methacrylate, 2-hydroxyethyl	0.2		1			689
Acrylonitrile	Methacrylate, 2-naphthyl	0.15	0.2	1.17	0.2	Y	611
Acrylonitrile	Methacrylate, 2-naphthyl	0.094	0.099	1.02	0.19	Y	611

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Methacrylate, 2-naphthyl	0.06	0.1	0.76	0.18	Y	611
Acrylonitrile	Methacrylate, 2-naphthyl	−0.01	0.16	0.64	0.24	N	611
Acrylonitrile	Methacrylate, 3,5-dimethyladamantyl	0.19	0.01	1.3	0.23	N	335
Acrylonitrile	Methacrylate, benzyl	0.2	0.1	0.96	0.1	N	23
Acrylonitrile	Methacrylate, butyl	0.291	0.039	0.98	0.2	Y	165
Acrylonitrile	Methacrylate, ferrocenylmethyl	0.15	0.04	0.82	0.42	Y	325
Acrylonitrile	Methacrylate, glycidyl	0.95	0.06	0.85	0.05	N	114
Acrylonitrile	Methacrylate, glycidyl	0.14		1.32			433
Acrylonitrile	Methacrylate, isobutyl	0.217	0.037	1.05	0.22	Y	165
Acrylonitrile	Methacrylate, methyl	0.138	0.037	1.322	0.05	Y	216
Acrylonitrile	Methacrylate, methyl	−0.016	0.048	1.11	0.12	Y	681
Acrylonitrile	Methacrylate, mono-, ethylene glycol	0.19		1.5			682
Acrylonitrile	Methacrylate, phenyl	0.36		0.46			363
Acrylonitrile	Methacrylate, potassium	0.226	0.04	0.203	0.086	Y	808
Acrylonitrile	Methacrylate, potassium 18-crown-6-ether	0.07	0.006	0.175	0.018	Y	808
Acrylonitrile	Methacrylic acid	0.04	0.13	0.2	0.26	Y	16
Acrylonitrile	Methacrylic acid	0.092	0.06	2.38	0.29	Y	216
Acrylonitrile	Methacrylonitrile	0.43	0.04	1.67	0.12	N	198
Acrylonitrile	Methacryloyl chloride	0.35		2.8			580
Acrylonitrile	Methacryloyl acetone	0.01	0.004	3.74	0.22	Y	339
Acrylonitrile	Methylenebutyrolactone	0.09		1.1			745
Acrylonitrile	*N,N*-Divinylaniline	0.246	0.014	−0.006	0.07	Y	45
Acrylonitrile	*N*-Vinylimidazole	3.393	0.008	0.832	0.044	Y	879
Acrylonitrile	Naphthalene, 1-vinyl-	0.107	0.028	0.451	0.088		967
Acrylonitrile	Norbornadiene	0.48	0.27	0.05	0.23	N	267
Acrylonitrile	Norbornadiene	0.66	0.01	0.44	0.05	N	69
Acrylonitrile	Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro	0.985	0.675	1.484	1.078	Y	838
Acrylonitrile	Oxazoline, 2-,2-isopropenyl-	0.13		0.52			862
Acrylonitrile	Oxazoline, 2-,2-isopropenyl-, 4-dimethyl	0.24		1.83			596
Acrylonitrile	Oxazoline, 2-,4-methacryloxy-2,4-dimethyl	0.11		0.89			596
Acrylonitrile	Pentadiene, *trans*-1,3-	0.06	0.064	0.095	0.021	Y	867
Acrylonitrile	Pentadiene, *cis*-1,3-	0.026	0.018	0.172	0.012	Y	867
Acrylonitrile	Pentadiene, *cis*-1,3-	0.056		0.198			952
Acrylonitrile	Pentadiene, *cis*-1,3-	0.033	0.017	0.188	0.011	N	997
Acrylonitrile	Pentadiene, *trans*-1,3-	0.07	0.017	0.104	0.01	Y	867
Acrylonitrile	Pentadiene, *trans*-1,3-	0.085		0.132			952
Acrylonitrile	Pentadiene, *trans*-1,3-	0.079	0.018	0.114	0.009	N	997
Acrylonitrile	Pentadiene, *cis*-1,3-	0.06	0.071	0.241	0.029	Y	867
Acrylonitrile	Pentadienoate, *trans*-4-ethoxy-2,4-, ethyl	0.01		5.4			723
Acrylonitrile	Pentene-1, 5-cyclohexyl-	4.22	0.45	−0.043	0.044	N	24
Acrylonitrile	Pentene-1, 5-phenyl-	3.84	0.48	−0.026	0.046	Y	24
Acrylonitrile	Phosphate, diethyl isopropenyl	15.2		0.03			650
Acrylonitrile	Phosphine oxide, diphenylvinyl-	3.8		0.16			843
Acrylonitrile	Phosphonic acid, α-phenylvinyl-	0.34	0.12	−0.45	0.41	Y	132
Acrylonitrile	Phosphonic acid, α-phenylvinyl-	2.36		0			590
Acrylonitrile	Phthalide, 3-(4-vinylphenoxy)-	0.084	0.053	0.43	0.19	Y	721
Acrylonitrile	Phthalimide, *N*-vinyl-	0.43	0.01	0.24	0.02		1046
Acrylonitrile	Phthalimide, *N*-(methacryloyloxy)-	0.204	0.016	1.45	0.072	Y	979
Acrylonitrile	Pinene	0.07	0.01	13.5	0.1		936
Acrylonitrile	Propene, 2-chloro-	1.164	0.053	−0.023	0.062	Y	181
Acrylonitrile	Propene, 3,3,3-trichloro-	12.2		0.1			402
Acrylonitrile	Propene, 1-chloro-, *cis*-	23.37	2.99	−0.12	0.19	Y	181
Acrylonitrile	Propene, 1-chloro-, *trans*-	19.61	1.72	0.01	0.17	Y	181
Acrylonitrile	Pyridazinone, 3-(2-vinyl)-6-methyl-	0.32	0.02	0.19	0.04	N	340
Acrylonitrile	Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-	0.74	0.02	0.02	0.02	N	340
Acrylonitrile	Pyridine, 2-methyl-5-vinyl-	0.16	0.01	0.31	0.02	N	232
Acrylonitrile	Pyridine, 2-vinyl-	0.05		21.9			456
Acrylonitrile	Pyridine, 2-vinyl-	0.1	0.03	0.44	0.08	N	50
Acrylonitrile	Pyridine, 2-vinyl-5-ethyl-	0.02		0.43			115
Acrylonitrile	Pyridine, 2-vinyl-5-ethyl-	0.06	0.07	0.43	0.18	N	50
Acrylonitrile	Pyridine, 4-vinyl-	0.11		0.41			115
Acrylonitrile	Pyridine, 4-vinyl-	0.09	0.04	0.34	0.13	N	50

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Pyrrolidone, *N*-vinyl-					Y	213
Acrylonitrile	Silane, γ-methacryloxypropyltrimethoxy-	0.094	0.034	3.79	0.594	Y	977
Acrylonitrile	Styrene	0		0.33			140
Acrylonitrile	Styrene	0.02	0.06	0.29	0.04	N	198
Acrylonitrile	Styrene	0.05		0.4			269
Acrylonitrile	Styrene	0.03		0.36			396
Acrylonitrile	Styrene	0.06		0.39			403
Acrylonitrile	Styrene	0.05		0.37			417
Acrylonitrile	Styrene	0.07		0.37			543
Acrylonitrile	Styrene	0.04		0.4			60
Acrylonitrile	Styrene	0.17		0.3			727
Acrylonitrile	Styrene	0.17		0.33			727
Acrylonitrile	Styrene	0.15		0.33			727
Acrylonitrile	Styrene	0.17		0.36			727
Acrylonitrile	Styrene	0.07		0.46			727
Acrylonitrile	Styrene	0.06		0.41			727
Acrylonitrile	Styrene	0.03		0.43			727
Acrylonitrile	Styrene	0.06		0.55			727
Acrylonitrile	Styrene	0.01		0.35			812
Acrylonitrile	Styrene, 2,4,6-trimethyl-	0.874	0.051	0.065	0.048	Y	58
Acrylonitrile	Styrene, 2,5-dichloro-	0.22	0.01	0.08	0.03	N	84
Acrylonitrile	Styrene, 2,5-dichloro-	0.25		0.07			84
Acrylonitrile	Styrene, 2,5-dichloro-	0.26		0.09			84
Acrylonitrile	Styrene, 3-tri-*n*-butylstannyl-	4.88		0.001			985
Acrylonitrile	Styrene, 4-methyl-	0.079	0.005	0.227	0.021	Y	929
Acrylonitrile	Styrene, α-(trimethylsilyloxy)-	0.05		0			970
Acrylonitrile	Styrene, α-methoxy-	0.06	0.01	−0.002	0.01	N	142
Acrylonitrile	Styrene, α-methyl-	0.07	0.04	0.08	0.03	N	171
Acrylonitrile	Styrene, α-methyl-	0.04		0.17			265
Acrylonitrile	Styrene, α-methyl-	0.04		0.24			265
Acrylonitrile	Styrene, α-methyl-	0.04	0.005	0.14	0.03	N	65
Acrylonitrile	Styrene, α-methyl-	0.063	0.041	0.09	0.024	Y	75
Acrylonitrile	Styrene, α-methyl-	0.34	0.15	0.103	0.036	Y	75
Acrylonitrile	Styrene, α-methyl-	0.03		0.14			835
Acrylonitrile	Styrene, α-methyl-	−0.02	0.27	0.08	0.07	N	94
Acrylonitrile	Styrene, *m*-methyl-	0.07		0.43			378
Acrylonitrile	Styrene, *p*-1-(2-hydroxy-butyl)-	0.04		0.31			548
Acrylonitrile	Styrene, *p*-1-(2-hydroxy-propyl)-	0.1		0.53			380
Acrylonitrile	Styrene, *p*-2-(2-hydroxy-propyl)-	0.05		0.41			547
Acrylonitrile	Styrene, *p*-acetoxy-	0.07	0.01	0.4	0.03	N	47
Acrylonitrile	Styrene, *p*-chloro-	−0.09	0.06	0.92	0.06	N	825
Acrylonitrile	Styrene, *p*-chloromethyl-	0.067		0.56			1035
Acrylonitrile	Styrene *p*-methyl-	0.05		0.33			378
Acrylonitrile	Succinimide, *N*-vinyl-	0.116	0.024	0.516	0.081		80
Acrylonitrile	Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	0.42		1.1			583
Acrylonitrile	Tetrazole, 5-phenyl-2-(4′-vinyl)phenyl-	0.32		1.4			583
Acrylonitrile	Tetrazole, 1-vinyl-	0.54	0.032	0.314	0.066	Y	925
Acrylonitrile	Toluenesulfonamide, *N*,*N*-methyl-vinyl-	0.42	0.12	0.04	1.04	Y	80
Acrylonitrile	Triallyl citrate	1.76	0.082	−0.08	0.1	Y	367
Acrylonitrile	Tricyclo[4.2.2.0{2,5}]dec-7-ene-3,4,9,10-tetracarboxylic acid	2.75		0.25			963
Acrylonitrile	Vinyl 12-ketostearate	3.3	0.5	−0.18	0.36	Y	146
Acrylonitrile	Vinyl 2-chloroethyl ether	1.09		0			597
Acrylonitrile	Vinyl 2-ethylhexanoate	9.24	1.96	−0.047	0.05	Y	47
Acrylonitrile	Vinyl 3,3-bis(ethoxycarbonyl)propyl ether	1.02		0.06			1015
Acrylonitrile	Vinyl acetate	4.05	0.82	0.04	0.067	Y	158
Acrylonitrile	Vinyl acetate					N	235
Acrylonitrile	Vinyl acetate	5.51	0.91	0.06	0.22	Y	6
Acrylonitrile	Vinyl acetate	5.29	1.13	−0.06	0.08	Y	75
Acrylonitrile	Vinyl acetate	9.2	6.39	−0.03	0.09	Y	75
Acrylonitrile	Vinyl benzoate	5.03	1.08	0.019	0.051	Y	47
Acrylonitrile	Vinyl bromide	2.79	0.2	0.06	0.04	N	268
Acrylonitrile	Vinyl bromide	2.25		0.06			268
Acrylonitrile	Vinyl butyl ether					N	345

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Vinyl butyl ether	0.98	0.13	−0.03	0.01	N	347
Acrylonitrile	Vinyl butyl sulfide	0.086	0.01	0.041	0.028	Y	303
Acrylonitrile	Vinyl chloride	3.26	0.34	−0.01	0.037	Y	140
Acrylonitrile	Vinyl chloride	2.55	0.12	0.07	0.03	N	202
Acrylonitrile	Vinyl chloride	2.62	0.25	0.02	0.08	N	22
Acrylonitrile	Vinyl chloride	3.62	0.21	0.044	0.002	Y	238
Acrylonitrile	Vinyl chloride	3.65	0.22	0.052	0.01	Y	46
Acrylonitrile	Vinyl chloride	4		0.04			529
Acrylonitrile	Vinyl chloroacetate	4.18	0.33	−0.007	0.016	Y	362
Acrylonitrile	Vinyl chloroacetate	0.34		0.09			542
Acrylonitrile	Vinyl chloroethyl ether	1.04	0.05	−0.02	0.02	N	116
Acrylonitrile	Vinyl chloromethyl ketone	0.064	0.019	0.88	0.12	Y	356
Acrylonitrile	Vinyl cymantrene	0.446	0.048	−0.065	0.093	Y	354
Acrylonitrile	Vinyl dichloroacetate	7.45	0.32	0.044	0.015	N	362
Acrylonitrile	Vinyl dichloroacetate	0.25		0.18			542
Acrylonitrile	Vinyl dodecyl ether	0.82		0			373
Acrylonitrile	Vinyl ether	0.94		0.02			386
Acrylonitrile	Vinyl ethyl ether	0.69	0.12	0.06	0.12	N	194
Acrylonitrile	Vinyl ethyl oxalate	1.34	0.45	0.06	0.03	N	285
Acrylonitrile	Vinyl ethyl sulfide	0.09	0.01	0.06	0.03	N	285
Acrylonitrile	Vinyl ethyl sulfide	0.06	0.01	0.05	0.03	N	336
Acrylonitrile	Vinyl ethyl sulfoxide	1.63	0.16	−0.08	0.14	Y	286
Acrylonitrile	Vinyl fluoride	44		0.005			528
Acrylonitrile	Vinyl fluoride	24		1			590
Acrylonitrile	Vinyl formate	2.54	1.79	−0.004	0.12	Y	47
Acrylonitrile	Vinyl hendecanoate	1.88	0.44	0.09	0.91	Y	145
Acrylonitrile	Vinyl isobutyl ether	−0.24	0.26	0.7	0.09	N	322
Acrylonitrile	Vinyl isobutyl ether	−0.05	0.07	0.99	0.06	N	322
Acrylonitrile	Vinyl isobutyl ether	0.98	0.15	−0.04	0.01	N	347
Acrylonitrile	Vinyl isobutyl ether	1.26	0.19	−0.04	0.02	N	347
Acrylonitrile	Vinyl isobutyl sulfide	0.08	0.005	0.03	0.01	N	285
Acrylonitrile	Vinyl isobutyl sulfide	0.062	0.004	0.022	0.038	Y	303
Acrylonitrile	Vinyl isobutyl sulfide	0.08	0.004	0.03	0.01	N	336
Acrylonitrile	Vinyl isothiocyanate	0.36		1.4			542
Acrylonitrile	Vinyl laurate	4		0.04			381
Acrylonitrile	Vinyl methyl ketone	0.59	0.13	1.75	0.11	Y	140
Acrylonitrile	Vinyl methyl ketone	0.63		1.39			504
Acrylonitrile	Vinyl octadecyl ether	0.85		0			373
Acrylonitrile	Vinyl octyl ether	0.81		0			373
Acrylonitrile	Vinyl phenyl ether	2.5		0.23			556
Acrylonitrile	Vinyl phenyl sulfide	0.11	0.01	0.03	0.02	N	285
Acrylonitrile	Vinyl phenyl sulfide	0.11	0.01	0.03	0.01	N	336
Acrylonitrile	Vinyl stearate	4.639	0.015	0.078	0.007	Y	145
Acrylonitrile	Vinyl *tert*-butyl sulfide	0.09	0.024	0	0.013	Y	303
Acrylonitrile	Vinyl trichloroacetate	7.34	0.9	0.047	0.039	Y	362
Acrylonitrile	Vinyl, α-chloro-, triethoxysilane					Y	
Acrylonitrile	Vinyl, *p*-, benzylethylcarbinol	0.04		0.31			591
Acrylonitrile	Vinyl, *p*-, benzylmethylcarbinol	0.11	0.17	0.54	0.2	N	48
Acrylonitrile	Vinyl-tris(trimethoxysiloxy)-silane	3.9		0.07			209
Acrylonitrile	Vinyl-tris(trimethoxysiloxy)-silane	3.85		0.08			573
Acrylonitrile	Vinylbenzoate, *p*-, sodium	0.174	0.04	0.23	0.34	Y	175
Acrylonitrile	Vinylbenzoic acid, *p*-	0.06	0.009	1.63	0.42	Y	175
Acrylonitrile	Vinylene carbonate	14.9		0.08			579
Acrylonitrile	Vinylethynyl-4-piperidinol, 1,2,5-trimethyl	0.08		0.38			783
Acrylonitrile	Vinylferrocene	0.186	0.025	0.165	0.092	Y	294
Acrylonitrile	Vinylferrocene	0.16		0.15			294
Acrylonitrile	Vinylhydroquinone	0.56	0.36	0.04	0.03	N	338
Acrylonitrile	Vinylidene chloride	0.92	0.11	0.32	0.14	Y	138
Acrylonitrile	Vinylidene chloride	1.04	0.2	0.28	0.034	Y	238
Acrylonitrile	Vinylidene chloride	0.44		0.4			404
Acrylonitrile	Vinylidene chloride	0.58		0.42			404
Acrylonitrile	Vinylidene chloride	0.65		0.49			404
Acrylonitrile	Vinylidene chloride	0.57		0.69			404
Acrylonitrile	Vinylidene chloride	0.5		0.76			404

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Acrylonitrile	Vinylidene chloride	0.48		0.95			404
Acrylonitrile	Vinylidene chloride	0.65		0.95			404
Acrylonitrile	Vinylidene chloride	0.65		1.3			404
Acrylonitrile	Vinylidene chloride	0.66		1.74			404
Acrylonitrile	Vinylidene chloride	0.63		1.43			404
Acrylonitrile	Vinylidene chloride	0.63		1.8			404
Acrylonitrile	Vinylidene chloride	0.58		0.39			468
Acrylonitrile	Vinylisocyanate	0.19		0.16			435
Acrylonitrile	Vinylmethyldiacetoxysilane	2.246	0.054	−0.11	0.23	Y	754
Acrylonitrile	Vinylmethylphenylsulfonium tetrafluoroborate	2.2		0.1			793
Acrylonitrile	Vinyltriethoxysilane	6.59	0.6	0.41	0.38	Y	209
Acrylonitrile	Vinyltriethoxysilane	26.83	29.59	2.69	2.5	Y	237
Acrylonitrile	Vinyltrimethoxysilane	9.09		0.45		Y	237
Acrylonitrile	Vinyltrimethylsilane	4.08	0.76	0.1	0.27	Y	209
Acrylonitrile, 2-chloro-	Styrene	0.087	0.04	0.014	0.004	Y	884
Acrylonitrile, 2-fluoro-	Styrene					Y	884
Acrylonitrile, α-chloro-	Styrene	0.03	0.02	0.13	0.04		806
Acrylonitrile, α-fluoro-	Styrene	0.03		0.44			850
Acrylonitrile, α-hydroxymethyl-	Styrene	0.528	0.086	0.205	0.029	Y	981
Acrylonitrile, α-methoxy	Acrylate, α-chloro-, ethyl	0.3		0.9			638
Acrylonitrile, α-methoxy	Styrene	0.312	0.054	0.534	0.069		806
Acrylonitrile, α-perfluoropropyl-	Methacrylate, methyl	0	0.05	1.9	0.3		926
Acryloxymethylpentamethyldisiloxane	Methacrylate, 2-hydroxyethyl	0.52		0.86			822
Acryloyl chloride	Acrylate, methyl	2.3		0.34			508
Acryloyl chloride	Acrylonitrile	1		1.2			580
Acryloyl chloride	Methacrylate, 2-hydroxypropyl	0.29		1.79			649
Acryloyl chloride	Methacrylate, methyl	1.51		0.48			508
Acryloyl chloride	Methacrylate, methyl	0.05		0.42			776
Acryloyl chloride	Pyrrolidone, *N*-vinyl	0.15		0.04			599
Acryloyl chloride	Styrene	0.02	0.02	0.1	0.05	N	220
Acryloyl chloride	Vinyl chloride	3.03	0.22	0.02	0.029	Y	288
Acryloyl chloride	Vinylidene chloride	1.12		0.5			288
Acryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	Acrylate, *N*-(2-hydroxyethyl)-carbazolyl	0.361	0.016	0.62	0.044	Y	702
Acryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	Methacrylate, *N*-ethyl-3-hydroxymethylcarbazolyl	0.103	0.015	1.425	0.09	Y	701
Acryloyloxy, 2-, benzoic acid	Methacrylate, *N*-methyl-*N*-phenyl-2-amino	1.82	0.2	0.22	0.03	N	651
Acryloyloxy, 2-, benzoic acid	Styrene	0.64	0.02	0.06	0.02	N	651
Acryloylpyrrolidone	Acrylonitrile	0.43		1.56			666
Adipate, divinyl	Styrene	0.011	0.011	20.36	0.26	Y	815
Allyl acetate	Acrylate, butyl	0.039	0.066	10.2	42.2	N	1053
Allyl acetate	Acrylate, methyl	0		5			470
Allyl acetate	Acrylonitrile	−0.09	0.12	6.57	2.11	Y	108
Allyl acetate	Isobutylene, 3-chloro-	0.15		4.5			412
Allyl acetate	Maleic anhydride	−0.001	0.007	0.028	0.019	Y	27
Allyl acetate	Methacrylate, ethyl	−0.57	0.82	81.27	1.93	N	210
Allyl acetate	Methacrylate, methyl	−2.42	37.66	99.24	24.6	N	210
Allyl acetate	Methacrylate, methyl	0		23			470
Allyl acetate	Styrene	0.01	0.05	62.8	82.7	N	1053
Allyl acetate	Styrene	0		90			470
Allyl acetate	Vinyl acetate	0.45		0.6			140
Allyl acetate	Vinyl acetate	0.7		1			502
Allyl acetate	Vinyl acetate	0.7		1			998
Allyl acetate	Vinyl chloride	1.2					470
Allyl acetate	Vinyl chloride	0		1.16			475
Allyl acetate	Vinylidene chloride	0		6.6			470
Allyl acrylate	Acrylate, methyl	0.33		0.52			578
Allyl acrylate	Allyl chloride	10.4		0.08			578
Allyl acrylate	Vinyl acetate	17.35		0.11			578
Allyl alcohol	Acrylonitrile					N	108
Allyl alcohol	Acrylonitrile	0.05		2.6			108
Allyl alcohol	Acrylonitrile	0.1	0.16	3.9	1.06	N	176
Allyl alcohol	Acrylonitrile	0.12	0.37	1.79	0.11	Y	300

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Allyl alcohol	Methacrylate, ethyl	−0.02	1.92	107.4	5.3	N	210
Allyl alcohol	Methacrylate, methyl	−1.62	2.9	78.79	6.5	N	210
Allyl butyrate	Vinyl acetate	0.64		0.97			998
Allyl chloride	Acrylate, benzyl	0.06		9.9			438
Allyl chloride	Acrylate, butyl	0.1		5.83			438
Allyl chloride	Acrylate, ethyl	0.08		7.73			438
Allyl chloride	Acrylate, methyl					Y	204
Allyl chloride	Acrylate, methyl	0.05		8.45			437
Allyl chloride	Acrylate, methyl	0.07		5.45			437
Allyl chloride	Acrylate, methyl	0.4		6.46			437
Allyl chloride	Acrylate, methyl	−0.02		9.1			437
Allyl chloride	Acrylic anhydride	0.02	0.02	11.66	0.56	N	219
Allyl chloride	Acrylonitrile	−0.152	0.008	2.145	0.049	Y	108
Allyl chloride	Acrylonitrile	0.08	0.03	3.45	0.65	Y	47
Allyl chloride	Allyl acrylate	0.08		10.4			578
Allyl chloride	Methacrylate, benzyl	0.02		58.7			438
Allyl chloride	Methacrylate, methyl					Y	204
Allyl chloride	Styrene	0.05	0.16	35.55	15.82	Y	10
Allyl chloride	Styrene						204
Allyl chloride	Styrene	0.03		36.8			438
Allyl chloride	Vinyl acetate	−2.4	3.21	1.355	0.089	Y	204
Allyl chloride	Vinyl acetate	0.61	0.15	0.66	0.15	Y	3
Allyl chloride	Vinyl acetate	0.75		0.34			438
Allyl chloride	Vinyl benzoate	0.88		0.46			438
Allyl chloride	Vinyl butyrate	1.15		0.31			438
Allyl chloride	Vinyl formate	0.78		0.57			438
Allyl chloride	Vinyl propionate	0.68		0.62			438
Allyl chloride	Vinylidene chloride	0.24	0.24	3.49	1.4	Y	3
Allyl chloroacetate	Methacrylic anhydride	−0.01	0.13	29.76	13	Y	106
Allyl chloroacetate	Styrene	0.06		34.3			782
Allyl glycidyl hexahydrophthalate	Styrene	0.06		34.3			782
Allyl glycidyl phthalate	Styrene	0.14		34.8			782
Allyl isobutyrate	Vinyl acetate	0.51		1.04			998
Allyl propionate	Vinyl acetate	0.42		1.29			998
Allyl trichloroacetate	Styrene	0.81	0.02	2.32	0.15		913
Allyl trimethylacetate	Vinyl acetate	0.34		1.15			998
Allyl valerate	Vinyl acetate	0.58		1.07			998
Allylbenzene	Acrylate, methyl	−0.036	0.016	11.35	0.66	Y	351
Allylbenzene	Acrylonitrile	−0.05	0.15	3.82	1.7	Y	24
Allylbenzene	Acrylonitrile	−0.026	0.022	5.35	0.3	Y	351
Allylbenzene	Maleic anhydride					Y	351
Allylbenzene	Methacrylonitrile	−0.061	0.03	16.24	1.36	Y	351
Allylbenzene	Triallyl citrate	0.339	0.039	2.01	0.5	Y	367
Allylcyclohexane	Acrylonitrile	−0.18	1.01	3.81	11.42	Y	24
Allylstearamide, *N*-	Acrylonitrile	0.03	0.27	3.79	0.28	Y	119
Allylstearamide, *N*-	Vinyl acetate	0.68	0.26	0.923	0.092	Y	119
Allylstearamide, *N*-	Vinylidene chloride	−0.24	0.19	5.22	0.15	Y	119
Allyltriethoxysilane	Vinyl chloride	−0.94	0.48	1.68	0.38	Y	237
Allylurea	Methacrylic anhydride	−0.06	0.17	29.01	18.48	Y	106
Azlactone, 2-vinyl-4,4-dimethyl-	Acrylate, α-hydroxymethyl-, ethyl	0.389	0.229	0.668	0.12	Y	1011
Azlactone, 2-vinyl-4,4-dimethyl-	Styrene	0.237	0.129	0.175	0.054	Y	1011
Azobenzene, 4-(acrylamidomethylamino)-	Styrene	8.5		0.083			880
Azobenzene, 4-(acrylamidomethylamino)-	Vinyl acetate	12.98		0.126			880
Benzenesulfamide, *p*-methacrylamide-	Vinyl butyl ether	3.4	0.1	0			1005
Benzimidazole, vinylmercapto-	Styrene	0.56	0.41	6.51	0.51	Y	764
Benzoate, *p*-(acryloylethoxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	0.36		0.91			881
Benzoate, *p*-(acryloylethoxy)-, phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	0.61		0.32			881
Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	0.91		0.36			881
Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy)-, phenyl	0.32		0.61			881
Benzoate, *p*-(acryloyloxy)-, *p*-methoxy phenyl	Benzoate, *p*-(acryloyloxy)-, phenyl *p*-butoxyphenyl	0.49		1.5			881

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Benzoate, *p*-(acryloyloxy)-, phenyl *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-methoxyphenyl	1.5		0.49			881
Benzoate, *p*-(acryloyloxy)-, phenyl *p*-butoxy phenyl	Benzoate, *p*-(methacryloyloxy)-, (*p*-octyloxy)phenyl	0.23		1.26			881
Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy-ethoxy)-, *p*-butoxy phenyl	0.91		0.36			856
Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy-ethoxy)-, phenyl	0.32		0.61			856
Benzoate, *p*-(acryloyloxy)-, phenyl	Benzoate, *p*-(acryloyloxy-ethoxy)-, *p*-methoxy phenyl	1.5		0.49			856
Benzoate, *p*-(acryloyloxy)-, phenyl	Benzoate, *p*-(methacryloyloxy-ethoxy)-, *p*-methoxy phenyl	0.23		1.26			856
Benzoate, *p*-(acryloyloxy-ethoxy)-, *p*-butoxy phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	0.36		0.91			856
Benzoate, *p*-(acryloyloxy-ethoxy)-, *p*-methoxy phenyl	Benzoate, *p*-(acryloyloxy)-, phenyl	0.49		1.5			856
Benzoate, *p*-(acryloyloxy-ethoxy)-,phenyl	Benzoate, *p*-(acryloyloxy)-, *p*-butoxy phenyl	0.61		0.32			856
Benzoate, *p*-(methacryloyloxy)-, *p*-octyloxy phenyl	Benzoate, *p*-(acryloyloxy)-, phenyl butoxy phenyl	1.26		0.23			881
Benzoate, *p*-(methacryloyloxyethoxy)-, *p*-octyloxy phenyl	Benzoate, *p*-(acryloyloxy)-, phenyl	1.26		0.23			856
Benzocyclobutane, α-methylene-	Maleic anhydride	0.002		0.025			1083
Benzocyclobutane, α-methylene-	Methacrylate, methyl	0.46		0.32			1083
Benzocyclobutene, 4-vinyl-	Methacrylate, methyl	1.47		0.6			1085
Benzofuran, 2-vinyl-	Acrylate, butyl	7.8	1.36	0.041	0.043	N	1020
Benzofuran, 2-vinyl-	Acrylate, ethyl	5.76	0.334	0.015	0.01	N	1020
Benzofuran, 2-vinyl-	Methacrylate, methyl	3.89	0.617	0.006	0.02	N	1020
Benzophenone, 2-hydroxy-4-(methacryloxy)-	Methacrylate, methyl	1.65	0.114	0.315	0.036	Y	934
Benzophenone, *p*-vinyl-	Acrylate, methyl	3.6	1.06	0.09	0.02	Y	866
Benzophenone, *p*-vinyl-	Acrylate, methyl	3.54	0.077	1.1	0.02	Y	911
Benzophenone, *p*-vinyl-	Styrene, *p*-*N*,*N*-dimethyl-aminomethyl-	2.54	0.136	0.26	0.017	Y	86
Benzophenone, *p*-vinyl-	Styrene, *p*-*N*,*N*-dimethylamino-	0.84	0.18	0.07	0.05	Y	866
Benzophenone, *p*-vinyl-	Styrene, *p*-dimethylamino-	0.842	0.072	0.176	0.046	Y	911
Benzophenone, *p*-vinyl-	Styrene, *p*-dimethylaminomethyl-	2.54	0.259	0.135	0.017	Y	911
Benzothiazole, vinylmercapto-	Acrylonitrile	0.178	0.081	0.191	0.014	Y	763
Benzothiazole, vinylmercapto-	Fumaronitrile					Y	763
Benzothiazole, vinylmercapto-	Maleic anhydride					Y	763
Benzothiazole, vinylmercapto-	Methacrylate, methyl	0.43	0.14	1.25	0.083	Y	763
Benzothiazole, vinylmercapto-	Styrene	0.349	0.087	2.171	0.07	Y	764
Benzothiazole, vinylmercapto-	Styrene	0.5	0.095	3.02	0.14	Y	764
Benzothiazole, vinylmercapto-	Vinyl acetate	2.53	0.47	0.013	0.037	Y	763
Benzothiazole, vinylmercapto-	Vinyl phenyl sulfide	0.95	0.61	0.24	0.12	Y	763
Benzothiazolone, *N*-[(methacroyloxy)-methyl]	Styrene	3.1	0.01	0.35	0.02		960
Benzoxazole, vinylmercapto-	Styrene	0.24	0.12	2.87	0.19	Y	764
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Methacrylate, methyl	0		30.5			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Methacrylate, methyl	0		108			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Styrene	0		18.2			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Styrene	0		89.2			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Vinyl acetate	0.35		1.26			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Vinyl acetate	0.15		1.46			994
Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	Vinyl acetate	0.014		1.99			994
Borazine, β-vinyl	Styrene	0.001	0.016	4.53	0.62		1018
Butadiene	Aconitate, trimethyl	0.4	0.4	0	0.22	Y	147
Butadiene	Acrylate, butyl	1.04	0.11	0.074	0.03	Y	257
Butadiene	Acrylate, heptafluorobutyl	0.359	0.039	0.073	0.031	Y	206

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Butadiene	Acrylate, methyl	1.09	4.53	0.07	0.87	Y	257
Butadiene	Acrylonitrile	0.358	0.046	0.046	0.006	Y	21
Butadiene	Acrylonitrile	0.452	0.23	0.1	0.059	Y	255
Butadiene	Acrylonitrile	0.35		0.05			418
Butadiene	Acrylonitrile	0.1		0.06			501
Butadiene	Acrylonitrile	0.2	0.01	0.03	0.02	N	66
Butadiene	Butadiene, 2,3-dimethyl-	1.26		0.78			173
Butadiene	Butadiene, 2,3-dimethyl-	0.3	0.92	0.86	0.24	Y	83
Butadiene	Butadiene, 2-chloro-	0.06		3.41			418
Butadiene	Butadiene, 2-phthalidomethyl	1.014	0.049	1.35	0.45	Y	735
Butadiene	Cyclobutane, 1,2-dimethylene-	0.1		5.76			854
Butadiene	Cyclobutane, 1,2-dimethylene-	0.1	0.05	5.76	0.36		885
Butadiene	Ethylene, trichloro-	9.64	0.21	−0.03	0.07	N	89
Butadiene	Ferrocene, ethyl-α-cylopentadienylide	0.02		1.51			699
Butadiene	Fumarate, diethyl	2.13		0.25			466
Butadiene	Isoprene	0.94		1.06			173
Butadiene	Isoprene	0.14	0.37	0.608	0.062	Y	83
Butadiene	Maleamic acid, *N*-3-dimethylamino-6-methyl phenyl	0.37	0.04	−0.08	0.03	N	679
Butadiene	Maleate, diethyl	8.08		0.11			466
Butadiene	Methacrylate, 2-(*N*,*N*-dimethylcarbamoyloxy)ethyl	1.06		0.35			600
Butadiene	Methacrylate, methyl	0.75		0.25			140
Butadiene	Methacrylate, methyl	0.504	0.024	0.027	0.004	Y	257
Butadiene	Methacrylate, methyl	0.7		0.32			467
Butadiene	Methacrylonitrile	0.39	0.15	0.053	0.016	Y	257
Butadiene	Methyleneglutaronitrile, 2-					N	196
Butadiene	Pentadiene, *cis*-1,3-	0.63		0.26			743
Butadiene	Pentadiene, *trans*-1,3-	1.22		0.09			743
Butadiene	Pyridine, 2-methyl-5-vinyl-	0.61		0.47			462
Butadiene	Pyridine, 2-methyl-5-vinyl-	1.32		0.72			520
Butadiene	Pyridine, 2-methyl-5-vinyl-	1.29	0.02	0.41	0.02	N	56
Butadiene	Styrene	1.44	0.56	0.84	0.4	Y	140
Butadiene	Styrene	1.55	0.14	0.37	0.25	N	159
Butadiene	Styrene	1.37		0.38			173
Butadiene	Styrene					Y	21
Butadiene	Styrene	1.35		0.58			418
Butadiene	Styrene	1.59		0.44			428
Butadiene	Styrene	1.8		0.6			473
Butadiene	Styrene	1.4		0.44			486
Butadiene	Styrene	1.4		0.38			486
Butadiene	Styrene	0.5		0.68			491
Butadiene	Styrene	1.38	0.11	0.82	0.13	N	56
Butadiene	Styrene	1.83	0.41	0.83	0.21	Y	83
Butadiene	Styrene	1.38		0.64			83
Butadiene	Styrene, 2,5-dichloro-	0.449	0.043	0.45	0.17	Y	255
Butadiene	Styrene, 2,5-dichloro-	0.662	0.017	0.187	0.049	Y	8
Butadiene	Styrene, *p*-chloromethyl	0.87		0.42			1075
Butadiene	Styrene, α-methyl-	1.5	0.13	0.139	0.087	Y	70
Butadiene	Styrene, *p*-chloro-	1.22	2.16	1.6	5.7	Y	255
Butadiene	Vinyl chloride	8.8		0.04			517
Butadiene	Vinyl chloride	5.27	3.35	−0.11	0.31	Y	717
Butadiene	Vinylhendecanoate	22.56	24.54	−0.22	1.26	Y	145
Butadiene	Vinyl pelargonate	26.3		0.02			145
Butadiene	Vinyl stearate	42.5	4.73	0.02	0.27	Y	145
Butadiene	Vinylferrocene	4.34	0.18	0.16	0.15	Y	802
Butadiene	Vinylidene chloride	1.97	0.25	−0.044	0.038	Y	257
Butadiene monoxide	Maleic anhydride	0.05	0.18	0.64	0.06		1064
Butadiene, 1,4-dicarboxylate-,diethyl	Acrylonitrile	2.79	0.9	0.29	0.12	N	327
Butadiene, 1,4-dicarboxylate-, diethyl	Pyridine, 2-vinyl-	0.69	0.23	0.41	0.07	N	327
Butadiene, 1,4-dicarboxylate-,diethyl	Styrene	0.55	0.1	0.09	0.014	Y	327
Butadiene, 1-(2-hydroxyethylthio)-	Styrene, *p*-chloro-	3.27		0.24			810
Butadiene, 1-(diethylamino)-	Acrylonitrile	0.426	0.208	0.481	0.049	Y	864
Butadiene, 1-(diethylamino)-	Acrylonitrile	0.425	0.48	0.208	0.049	Y	889
Butadiene, 1-acetoxy-	Acrylonitrile	0.82	0.68	−0.012	0.043	Y	91

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Butadiene, 1-chloro-	Styrene	1.44	3.1	0.27	0.27	Y	695
Butadiene, 1-ethoxy-	Acrylonitrile	0.065	0.009	0.02	0.008	Y	864
Butadiene, 1-ethoxy-	Acrylonitrile	0.065	0.02	0.01	0.008	Y	889
Butadiene, 1-phthalimido-	Styrene	1.56	0.63	0.299	0.049	Y	163
Butadiene, 1-succinimido-	Styrene	1.68	0.16	0.286	0.018	Y	163
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide	0.14		0.43			759
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide, *N*-(2-hydroxyethyl)-	0.12		0.49			759
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide, *N*-(4-chlorophenyl)-	0.03		0.41			759
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide, *N*-(4-methoxy-phenyl)-	0.04		0.47			759
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide, *N*-chloromethyl-	0.05		0.3			759
Butadiene, 2,3-bis(diethylphosphono)-	Maleimide, *N*-phenyl-	0.05		0.42			759
Butadiene, 2,3-dichloro-	Butadiene, 2-chloro-	1.98	0.25	0.31	0.12	Y	122
Butadiene, 2,3-dichloro-	Ethylene, 1,1-diphenyl-	3.58		−0.08		Y	61
Butadiene, 2,3-dichloro-	Methacrylate, methyl	11.04		0.08		Y	61
Butadiene, 2,3-dichloro-	Phosphonic acid, α-phenylvinyl-	7.75		0.07			588
Butadiene, 2,3-dichloro-	Phosphonic acid, α-phenylvinyl-	7.75		0.07			597
Butadiene, 2,3-dichloro-	Styrene	8.65		0.25			587
Butadiene, 2,3-dichloro-	Styrene	10.6		0.05		Y	61
Butadiene, 2,3-dimethyl-	Butadiene	0.78		1.26			173
Butadiene, 2,3-dimethyl-	Butadiene	0.86	0.24	0.3	0.92	Y	83
Butadiene, 2,3-dimethyl-	Isoprene	0.84		1.18			173
Butadiene, 2,3-dimethyl-	Styrene	1.048	0.066	0.534	0.035	Y	173
Butadiene, 2,3-chloro-		5.19		0.06			457
Butadiene, 2,3-chloro-	Acrylate, methyl	10.4		0.06		Y	61
Butadiene, 2,3-chloro-	Acrylonitrile	5.35		0.04			402
Butadiene, 2,3-chloro-	Acrylonitrile	4.8		0.05			504
Butadiene, 2,3-chloro-	Acrylonitrile	5.38	0.45	0.056	0.023	Y	61
Butadiene, 2,3-chloro-	Butadiene	3.41		0.06			418
Butadiene, 2,3-chloro-	Butadiene, 1-(2-hydroxyethyl-thio)-	1		0.2			810
Butadiene, 2,3-chloro-	Butadiene, 2,3-dichloro-	0.31	0.12	1.98	0.25	Y	122
Butadiene, 2,3-chloro-	Butadiene, 2-fluoro-	3.7		0.22			493
Butadiene, 2,3-chloro-	Butadiene, hexafluoro-	5.47		0.1			127
Butadiene, 2,3-chloro-	Butadiene, hexafluoro-	5.52		0.1			453
Butadiene, 2,3-chloro-	Ethylene, 1,1-diphenyl-	3.4		0.03		Y	61
Butadiene, 2,3-chloro-	Fumarate, diethyl	6.51		0.02		Y	61
Butadiene, 2,3-chloro-	Hexatriene, tetrachloro-	3.6		0.2			459
Butadiene, 2,3-chloro-	Isoprene	2.82		0.06			418
Butadiene, 2,3-chloro-	Isopropenyl methyl ketone	4.035	0.087	0.14	0.01	Y	259
Butadiene, 2,3-dichloro-	Methacrylate, methyl	11.04	0.08			Y	61
Butadiene, 2-chloro-	Maleic anhydride	0.026	0.03	0.329	0.107	N	1068
Butadiene, 2-chloro-	Methacrylate, isobutyl	1.8	0.06	0.9	0.088	N	1057
Butadiene, 2-chloro-	Methacrylate, methyl	6.33		0.08		Y	61
Butadiene, 2-chloro-	Methacrylic acid	2.52	0.2	−0.035	0.058	Y	259
Butadiene, 2-chloro-	Phosphonate, vinyl-,dimethyl	7.5		0.1			594
Butadiene, 2-chloro-	Phosphonic acid, α-phenylvinyl-	7.5		0.1			546
Butadiene, 2-chloro-	Phosphonic acid, α-phenylvinyl-	7.5		0.1			597
Butadiene, 2-chloro-	Quinoline, 2-vinyl-	2.1		0.38			456
Butadiene, 2-chloro-	Styrene	7.84		0.05		Y	61
Butadiene, 2-chloro-	Styrene	5.98	0.5	0.025	0.047	Y	8
Butadiene, 2-chloro-	Vinyl acetate	33.52	8.98	−0.02	0.09	N	247
Butadiene, 2-chloro-	Vinyl butyrate					N	247
Butadiene, 2-chloro-	Vinyl formate	15		−0.04			127
Butadiene, 2-chloro-	Vinyl formate	11.91	3.89	−0.05	0.01	N	247
Butadiene, 2-chloro-	Vinyl propionate					N	247
Butadiene, 2-chloro-	Vinylidene cyanide					Y	82
Butadiene, 2-fluoro-	Acrylonitrile	0.6	0.18	0.077	0.021	Y	174
Butadiene, 2-fluoro-	Butadiene, 2-chloro-	0.22		3.7			493
Butadiene, 2-fluoro-	Butadiene, hexafluoro-	2.93		0.24			453
Butadiene, 2-fluoro-	Isoprene	2.63	0.96	0.25	0.12	Y	174
Butadiene, 2-fluoro-	Methacrylate, methyl	1.23	1.77	0.76	0.77	Y	174
Butadiene, 2-fluoro-	Styrene	2.1	0.12	0.214	0.03	Y	174
Butadiene, 2-fluoro-	Styrene	1.24	0.61	0.22	0.15	Y	174
Butadiene, 2-fluoro-	Styrene, α-methyl-	2.22	0.7	0.72	0.24	Y	174
Butadiene, 2-phthalidomethyl-	Butadiene	1.35	0.45	1.014	0.049	Y	735
Butadiene, 2-trimethylsilyloxy-	Acrylonitrile	0.065		0.036			993
Butadiene, 2-trimethylsilyloxy-	Styrene	1.2		0.64			993

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Butadiene, hexafluoro-	Butadiene, 2-chloro-	0.1		5.47			127
Butadiene, hexafluoro-	Butadiene, 2-chloro-	0.1		5.52			453
Butadiene, hexafluoro-	Butadiene, 2-fluoro-	0.24		2.93			453
Butadiene, hexafluoro-	Isoprene	0.78		1.19			453
Butadiene,1-(2-hydroxyethylthio)-	Butadiene, 2-chloro-	0.2		1			810
Butadiene-1carboxylate, ethyl	Styrene	0.56		0.12			791
Butadiene-1carboxylate, *S*-(-)α-phenethyl-ammonium	Styrene	6.81	1.68	0.45	0.13	Y	748
Butadiene-1-carboxylate, ethyl	Acrylonitrile	3.2		0.3			791
Butadiene-1-carboxylate, ethyl	Pyridine, 2-vinyl-	0.8		0.4			791
Butadiene-1-carboxylic acid	Acrylonitrile	4		0.2			791
Butadiene-1-carboxylic acid	Acrylonitrile	11		0.03			799
Butadiene-1-carboxylic acid	Styrene	0.52		0.15			791
Butadiene-1-carboxylic acid	Styrene	5.55		0.08			799
Butanoate, 3-acrylamido-3-methyl-, sodium	Acrylamide, *N,N*-dimethyl-	0.22	0.138	0.987	0.153	Y	857
Butene, *cis*-2-	Maleimide, *N*-cyclohexyl-	3		0			995
Butene-1	Acrylonitrile	0.24	0.81	13.46	3.61	Y	211
Butene-1	Ethylene	3.25					388
Butene-1	Ethylene	0.16		3.6			407
Butene-1	Vinyl acetate	0.11	0.42	1.55	0.24	Y	211
Butene-1	Vinyl chloride	0.12	0.12	3.34	0.58	Y	30
Butene-1, 2-ethyl-	Vinyl chloride	−1.05	0.89	1.56	0.26	N	329
Butene-1, 2-methyl-	Vinyl chloride	−1.18	1.2	1.01	0.31	N	329
Butene-1, 3,3-dimethyl-	Vinyl chloride	−0.44	0.28	5.64	0.2	N	329
Butene-1, 3,3-dimethyl-	Vinyl chloride	−0.07	2.5	6.18	13.57	Y	6
Butene-1, 4-(*p*-chlorophenyl)-	Acrylonitrile	−0.066	0.044	4.68	0.63	Y	24
Butene-1, 4-(*p*-methoxyphenyl)-	Acrylonitrile	−0.085	0.088	2.56	1.11	Y	24
Butene-1, 4-cyclohexyl-	Acrylonitrile	0.014	0.055	5.03	0.76	Y	24
Butene-1,4-phenyl-	Acrylonitrile	0.024	0.069	4.51	0.86	Y	24
Butene-2	Ethylene			7.5			477
Butene-2	Vinyl chloride	−1.79	5.22	3.3	2.18	N	329
Butene-2, 2-methyl-	Vinyl chloride	0.3	5.31	8.52	10.54	N	329
Butene-2, *cis*-	Acrylonitrile	0.11		22.32		Y	211
Butene-2, *cis*-	Vinyl acetate	0.03	0.25	8.27	1.05	Y	211
Butene-2, *cis*-	Vinyl chloride	0		8.8			230
Butene-2, *trans*-	Acrylonitrile	0.84	0.6	62.99	10.67	Y	211
Butene-2, *trans*-	Vinyl acetate	−0.03	0.44	7.2	1.69	Y	211
Butenoate, 3-phenyl-3-, methyl	Styrene	−0.532	0.512	1.97	0.364	Y	1008
Butyrolactone, α-methylene-	Styrene	0.85	0.06	0.14	0.01	N	663
Caprolactam, *N*-vinyl-	Crotonaldehyde	0.31		0			875
Caprolactam, *N*-vinyl-	Crotonaldehyde	0.31	0.005	0			907
Caprolactam, *N*-vinyl-	*N*-Vinyl pyrrolidone	1.7	0.2	2.8	0.2		883
Caprolactam, *N*-vinyl-	Pyridine, 2-methyl-5-vinyl-	0.04		11.58			589
Caprolactam, *N*-vinyl-	Pyridine, 2-methyl-5-vinyl-	0.04		11.58			592
Caprolactam, *N*-vinyl-	Pyridine, 2-vinyl-	0.12		4.35			589
Caprolactam, *N*-vinyl-	Pyridine, 2-vinyl-	0.12		4.35			592
Caprolactam, *N*-vinyl-	Pyridine, 4-vinyl-	0.1		4.43			589
Caprolactam, *N*-vinyl-	Pyridine, 4-vinyl-	0.1		4.43			592
Caprolactam, *N*-vinyl-	Pyrrolidone, *N*-vinyl-	1.7		2.8			859
Caprolactam, *N*-vinyl-	Styrene	1.02	0.01	0.2	0.01		916
Caprolactam, *N*-vinyl-	Vinyl acetate	0.31		0.63			557
Caprolactam, *N*-vinyl-	Vinyl acetate	2.5		0.35			859
Caprolactam, *N*-vinyl-	Vinyl acetate	2.5		0.35			883
Caprolactam, *N*-vinyl-	Vinyl butyl ether	3.1	5.3	0.03	0.73	Y	215
Caprolactam, *N*-vinyl-	Vinyl isopropyl ether	0.22	1.95	0.01	0.96	Y	215
Caprolactam, *N*-vinyl-	Vinyl phenyl ether	1.3	2.43	0.26	0.37	Y	214
Carbamate, *N,N*-diethyl-, vinyl	Acrylate, methyl	−0.027	0.049	4.78	0.22	Y	92
Carbamate, *N,N*-diethyl-, vinyl	Acrylic acid	−0.097	0.068	5.58	0.32	Y	92
Carbamate, *N,N*-diethyl-, vinyl	Maleic anhydride	0.024	0.066	0.025	0.034	Y	92
Carbamate, *N,N*-diethyl-, vinyl	Styrene	−0.08	0.18	29.56	3.59	Y	197
Carbamate, *N,N*-diethyl-, vinyl	Vinyl acetate	0.13	0.19	1.7	0.04	Y	197
Carbamate, *N*-phenyl-, vinyl	Methacrylate, butyl	0.329		0.418			833
Carbamate, *N*-phenyl-, vinyl	Methacrylate, di-, triethylene-glycol	0.034		0.504			833
Carbamate, *N*-vinyl-, ethyl	Acrylic acid	0.26		4.69		Y	81

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Carbamate, *N*-vinyl-, ethyl	Pyrrolidone, *N*-vinyl-	0.371	0.05	1.77	0.12	Y	81
Carbamate, *N*-vinyl-, ethyl	Vinyl acetate	2.09		0.39			629
Carbamate, *N*-vinyl-, ethyl	Vinyl acetate	0.4	0.26	0.418	0.063	Y	81
Carbamate, *N*-vinyl-, *tert*-butyl	Vinyl acetate	0.46	0.03	1.12	0.07	N	827
Carbamate, *N*-vinyl-, *tert*-butyl	Methacrylate, methyl	0.17	0.1	1.87	0.2		991
Carbamate, *N*-vinyl-, *tert*-butyl	*N*-Vinylpyrrolidone	2.4	0.5	0.4	0.06		991
Carbazole, 2-butenyl-	Carbazole, 9-vinyl-	0.02	0.003	2.6	0.5		904
Carbazole, 9-propenyl-	Carbazole, 9-vinyl-	0.02	0.003	2.8	0.5		904
Carbazole, 9-vinyl-	Carbazole, 2-butenyl-	2.6	0.5	0.02	0.003		904
Carbazole, 9-vinyl-	Carbazole, 9-propenyl-	2.8	0.5	0.02	0.003		904
Carbazole, *N*-butenyl-	Carbazole, *N*-vinyl-	0.02		2.6			871
Carbazole, *N*-methyl-3-vinyl	Styrene	1.404	0.146	0.735	0.067		806
Carbazole, *N*-propenyl-	Carbazole, *N*-vinyl-	0.02		2.8			871
Carbazole, *N*-vinyl-	Acenaphthalene	−0.03	0.12	6.62	1.77	N	331
Carbazole, *N*-vinyl-	Acrylate, ethyl	0.27		1.1			757
Carbazole, *N*-vinyl-	Acrylate, methyl	0.049	0.029	0.53	0.13	Y	13
Carbazole, *N*-vinyl-	Acrylate, methyl	0.11		0.43			551
Carbazole, *N*-vinyl-	Acrylate, methyl	0.028	0.045	0.51	0.073	Y	98
Carbazole, *N*-vinyl-	Acrylonitrile	0.09	0.04	0.39	0.04	N	269
Carbazole, *N*-vinyl-	Carbazole, *N*-butenyl-	2.6		0.02			871
Carbazole, *N*-vinyl-	Carbazole, *N*-propenyl-	2.8		0.02			871
Carbazole, *N*-vinyl-	Fumarate, diisopropyl	0		0			1038
Carbazole, *N*-vinyl-	Maleate, diethyl	0.21	0.05	0	0.01	N	361
Carbazole, *N*-vinyl-	Maleate, dimethyl	0.51	0.11	0	0.01	N	361
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.14	0.03	1.55	0.16	N	11
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.033	0.015	2.12	0.14	Y	13
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.06		1.8			738
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.75		0.57			738
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.08	0.05	1.84	0.26	N	758
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.33	0.18	0.44	0.22	N	758
Carbazole, *N*-vinyl-	Methacrylate, methyl	0.031	0.022	2.123	0.048	Y	98
Carbazole, *N*-vinyl-	Propene, 2,3-dichloro-	1		0			376
Carbazole, *N*-vinyl-	Styrene	−0.02	0.06	6	1.34	N	11
Carbazole, *N*-vinyl-	Styrene	0.026	0.034	5.93	1.05	Y	13
Carbazole, *N*-vinyl-	Styrene	0.032	0.057	5.72	0.4	Y	98
Carbazole, *N*-vinyl-	Styrene, *p*-chloro-	0.02		7			551
Carbazole, *N*-vinyl-	Succinimide, *N*-vinyl-	0.04	0.09	0.11	0.83	Y	284
Carbazole, *N*-vinyl-	Vinyl acetate	2.67	0.13	0.126	0.023	Y	246
Carbazole, *N*-vinyl-	Vinyl acetate	3.02		0.15			246
Carbazole, *N*-vinyl-	Vinyl acetate	3.9		0.13			551
Carbazole, *N*-vinyl-	Vinyl butyrate	1.33	0.34	0.07	0.12	Y	246
Carbazole, *N*-vinyl-	Vinyl chloride	4.77		0.17			427
Carbazole, *N*-vinyl-	Vinyl formate	4.142	0.011	0.192	0.001	Y	246
Carbazole, *N*-vinyl-	Vinyl propionate	1.69	0.21	0.081	0.073	Y	246
Carbazole, *N*-vinyl-	Vinyl thiolacetate	0.44	0.28	0.45	0.15	N	97
Carbazole, *N*-vinyl-	Vinylidene chloride	3.6	0.12	−0.023	0.068	Y	13
Carbazole, *N*-vinyl-	Vinylidene chloride	3.65	0.3	−0.008	0.018	Y	98
Carbon monoxide	Ethylene	0.32	0.65	0.068	0.012	Y	51
Carbon monoxide	Ethylene	0.004	0.012	0.025	0.006	Y	51
Carbon monoxide	Ethylene	−0.15	0.07	0.04	0.01	N	52
Carbon monoxide	Ethylene	0		0.5			52
Carbon monoxide	Vinyl acetate	0.33		0.24			474
Carbon monoxide	Vinyl chloride	0.121	0.011	13.52	0.25	Y	184
Cinnamat, methyl	Styrene	0.04	0.13	1.799	0.073	Y	488
Cinnamate, 2-chloroethyl	Styrene	0.01	0.11	1.394	0.068	Y	488
Cinnamate, α-cyano-, 2-ethylhexyl	Styrene	0.06	0.01	0.58	0.004	N	282
Cinnamate, α-cyano-, benzyl	Styrene	−0.16	0.17	0.4	0.03	N	282
Cinnamate, α-cyano-, butyl	Styrene	−0.09	0.08	0.47	0.03	N	282
Cinnamate, α-cyano-, cyclohexyl	Styrene	−0.24	0.22	0.37	0.04	N	282
Cinnamate, α-cyano-, ethyl	Acrylonitrile	−0.18	0.15	12.6	2.3	Y	310
Cinnamate, α-cyano-, ethyl	Styrene					Y	134
Cinnamate, α-cyano-, hexyl	Styrene	−0.17	0.26	0.39	0.04	N	282
Cinnamate, α-cyano-, methyl	Pyrrolidone, *N*-vinyl-	−0.001	0.001	0.044	0.008	Y	816
Cinnamate, α-cyano-, methyl	Styrene	−0.21	0.35	0.4	0.06	N	282
Cinnamate, benzyl	Styrene	0.14	0.13	1.33	0.12	Y	488

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Cinnamate, butyl	Styrene	0.04	0.15	1.757	0.082	Y	488
Cinnamate, dibutylchlorotin	Acrylate, methyl	−0.2		1.65			659
Cinnamate, ethyl	Styrene	−0.076	0.054	1.52	0.31	Y	200
Cinnamate, ethyl	Styrene	0.082	0.045	1.566	0.035	Y	488
Cinnamate, methyl	Styrene	0.02		1.27			653
Cinnamate, *p*-chloro-, methyl	Styrene	0.04		2.24			653
Cinnamate, *p*-methyl-, methyl	Styrene	0		2.84			653
Cinnamate, phenyl	Styrene	0.029	0.055	1.66	0.12	Y	488
Cinnamate, *tert*-butyl	Styrene	0.1	0.12	1.692	0.087	Y	488
Cinnamic acid, *cis*-	Acrylamide	−0.48	0.28	3.84	0.68	Y	298
Cinnamic acid, *trans*-	Acrylamide	−0.24	0.15	3.4	0.24	Y	298
Cinnamide, *N,N*-diethyl-	Styrene	0.01		4.17			737
Cinnamide, *p*-chloro-, *N,N*-diethyl-	Styrene	0.01		2.99			737
Cinnamide, *p*-methyl-, *N,N*-diethyl-	Styrene	0.01		4.95			737
Cinnamonitrile	Acrylonitrile	0.36	0.1	8.46	0.37	Y	310
Cinnamonitrile	Styrene	−0.008	0.021	2.549	0.092	Y	134
Cinnamonitrile, α-cyano-	Acrylonitrile					Y	310
Cinnamonitrile, α-cyano-	Styrene					Y	134
Citraconate, di-*sec*-butyl	Styrene	0.13		2			841
Citraconate, dibenzyl	Styrene	0.047		1.55			841
Citraconate, dibutyl	Styrene	0.054		2.74			841
Citraconate, dicyclohexyl	Styrene	0.03		1.88			841
Citraconate, diethyl	Styrene	0.029		2.48			841
Citraconate, diisobuty	Styrene	0.03		1.8			841
Citraconate, diisopropyl	Styrene	0.076		1.62			841
Citraconate, dimethyl	Styrene	0.049		1.58			841
Citraconate, dipropyl	Styrene	0.027		2.19			841
Citraconic acid	Styrene	0		1.36			1021
Citraconic anhydride	Styrene	0.021	0.058	0.16	0.15	Y	47
Citraconimide, β-iso-, *N*-(4-acetylphenyl)-	Styrene	0		0.42			608
Citraconimide, β-iso-, *N*-(4-acetylphenyl)-	Styrene	0		0.28			608
Citraconimide, β-iso-, *N*-(4-ethoxycarbonylphenyl)-	Styrene	0		0.3			608
Citraconimide, β-iso-, *N*-(4-methoxyphenyl)-	Styrene	0		0.27			608
Citraconimide, β-iso-, *N*-(*p*-tolyl)-	Styrene	0		0.4			608
Citraconimide, β-iso-, *N*-phenyl-	Styrene	0		0.39			608
Citraconimde, *N*-(1-menthylcarboxylatomethyl)-	Methacrylate, methyl	0.109	0.115	3.47	0.335	Y	950
Citraconimide, *N*-(4-acetoxyphenyl)-	Methacrylate, methyl	0.18	0.13	1.88	0.16	Y	728
Citraconimide, *N*-(4-acetoxyphenyl)-	Styrene	0.015	0.086	0.151	0.054	Y	728
Citraconimide, *N*-(4-chlorophenyl)-	Methacrylate, methyl	−0.01	0.18	2.93	0.4	Y	728
Citraconimide, *N*-(4-chlorophenyl)-	Styrene	0.075	0.093	0.181	0.057	Y	728
Citraconimide, *N*-(4-ethoxycarbonylphenyl)-	Methacrylate, methyl	−0.027	0.084	2.72	0.18	Y	728
Citraconimide, *N*-(4-ethoxycarbonylphenyl)-	Styrene	0.004	0.033	0.148	0.018	Y	728
Citraconimide, *N*-(4-methoxyphenyl)-	Methacrylate, methyl	−0.1	0.19	3.16	0.73	Y	728
Citraconimide, *N*-(4-methoxyphenyl)-	Styrene	0.04	0.11	0.63	0.19	Y	728
Citraconimide, *N*-allyl	Styrene	0.004		0.142			959
Citraconimide, *N*-benzyl-	Methacrylate, methyl	0.02	0.14	2.77	0.28	Y	714
Citraconimide, *N*-benzyl	Styrene	−0.066	0.082	0.19	0.087	Y	714
Citraconimide, *N*-butyl-	Methacrylate, methyl	−0.12	0.36	2.17	0.56	Y	714
Citraconimide, *N*-butyl-	Styrene	0.03	0.045	0.169	0.028	Y	714
Citraconimide, *N*-ethyl-	Methacrylate, methyl	−0.04	0.33	3.25	0.79	Y	714
Citraconimide, *N*-ethyl-	Styrene	−0.002	0.05	0.231	0.044	Y	714
Citraconimide, *N*-hexyl-	Methacrylate, methyl	−0.08	0.28	2.38	0.45	Y	714
Citraconimide, *N*-hexyl-	Styrene	−0.03	0.11	0.316	0.086	Y	714
Citraconimide, *N*-isobutyl-	Methacrylate, methyl	−0.18	0.42	1.78	0.85	Y	714
Citraconimide, *N*-isobutyl-	Styrene	0.07	0.14	0.326	0.094	Y	714
Citraconimide, *N*-isopropyl-	Methacrylate, methyl	0.09	0.27	2.7	0.51	Y	714
Citraconimide, *N*-isopropyl-	Styrene	−0.014	0.063	0.244	0.055	Y	714
Citraconimide, *N*-methyl-	Acrylonitrile	0.6		0.53			460
Citraconimide, *N*-methyl-	Methacrylate, methyl	0.15		3.24			460
Citraconimide, *N*-methyl-	Styrene	0.24		0.15			460

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Citraconimide, *N*-methyl-	Styrene	0.24		0.14			506
Citraconimide, *N*-phenyl-	Methacrylate, methyl	0.001	0.033	3.081	0.074	Y	728
Citraconimide, *N*-phenyl-	Styrene	−0.001	0.023	0.175	0.016	Y	728
Citraconimide, *N*-propyl-	Methacrylate, methyl	−0.03	0.43	3.23	0.98	Y	714
Citraconimide, *N*-propyl-	Styrene	−0.014	0.087	0.196	0.062	Y	714
Citraconimide, *N*-*tert*-butyl-	Methacrylate, methyl	0.038	0.072	2.88	0.19	Y	714
Citraconimide, *N*-*tert*-butyl-	Styrene	0.045	0.05	0.465	0.039	Y	714
Citraconimide, *N*-tolyl-	Methacrylate, methyl	0.003	0.088	2.75	0.19	Y	728
Citraconimide, *N*-tolyl-	Styrene	−0.02	0.18	0.31	0.13	Y	728
Citraconimide, α-iso-, *N*-(4-acetylphenyl)-	Styrene	0		0.81			608
Citraconimide, α-iso-, *N*-(4-chlorophenyl)-	Styrene	0		1.03			608
Citraconimide, α-iso-, *N*-(4-ethoxycarbonylphenyl)-	Styrene	0		0.96			608
Citraconimide, α-iso-, *N*-(4-methoxyphenyl)-	Styrene	0		0.39			608
Citraconimide, α-iso-, *N*-(p-tolyl)-	Styrene	0		1.69			608
Citraconimide, α-iso-, *N*-phenyl-	Styrene	0		1.76			608
Copper diacrylate	Titanium, dicylopentadienyl-, dimethacrylate	1.09		0.89			895
Crotonaldehyde	Acrylonitrile	0		8			429
Crotonaldehyde	Acrylonitrile	0.01		25			504
Crotonaldehyde	Caprolactam, *N*-vinyl-	0		0.31			875
Crotonaldehyde	Caprolactam, *N*-vinyl-	0		0.31	0.005		907
Crotonaldehyde	Pyrrolidone, *N*-vinyl-	−0.15	0.01	0.42	0.02	N	248
Crotonaldehyde	Styrene	0.03		14.7			428
Crotonaldehyde	Styrene	0.07		14.7			428
Crotonaldehyde	Styrene	0.12		14.7			428
Crotonaldehyde	Vinyl chloride	−0.56	0.25	1.55	0.26	N	271
Crotonaldehyde	Vinylidene chloride	0					405
Crotonate, hydrocortisone 21-	*N*-Vinyl pyrrolidone	0.05	0.05	0.75	0.02		1022
Crotonate, α-acetyl-, methyl	Acrylonitrile	0		8.68			785
Crotonate, α-acetyl-, methyl	Styrene	0		2.7			785
Crotonate, α-bromo-, ethyl	Styrene	0		4.42			785
Crotonate, α-carboethoxy-, ethyl	Acrylonitrile	0		18.7			785
Crotonate, α-carboethoxy-, ethyl	Styrene	0		8.24			785
Crotonate, α-chloro-, ethyl	Acrylonitrile	0		9.53			785
Crotonate, α-chloro-, ethyl	Styrene	0		5.13			785
Crotonate, α-cyano-, ethyl	Acrylonitrile	0.06		11.4			785
Crotonate, α-cyano-, ethyl	Styrene	0.02		0.26			785
Crotonate, α-methoxy-, methyl	Acrylonitrile	0		1			785
Crotonate, α-methoxy-, methyl	Styrene	0.04		18.4			785
Crotonate, α-methyl-, methyl	Acrylonitrile	0		2.97			785
Crotonate, α-methyl-, methyl	Styrene	0.02		30.7			785
Crotonate, ethyl	Acrylonitrile	0		25.2			785
Crotonate, ethyl	Styrene	0		27			785
Crotonate, methyl	Styrene	−0.27	0.22	9.68	1.86	Y	112
Crotonic acid	Acrylamide	−0.18	0.14	3.76	0.89	Y	298
Crotonic acid	Acrylamide	−0.19	0.19	4.23	1.37	Y	298
Crotonic acid	Acrylamide	0.11		4.72			569
Crotonic acid	Acrylamide	0.12		5.32			569
Crotonic acid	Acrylonitrile	0		21			402
Crotonic acid	Pyrrolidone, *N*-vinyl-	0	0.03	0.84	0.07	N	245
Crotonic acid	Styrene	0		20			402
Crotonic acid	Vinyl acetate	0.04	0.14	0.317	0.081	Y	47
Crotonic acid	Vinyl acetate	0.01		0.3			526
Crotonic acid	Vinyl acetate	0.001	0.1	0.31	0.07	N	94
Crotonic acid	Vinylidene chloride	−0.2	1.03	18.1	43.7	Y	47
Crotonic acid, *cis*-	Acrylamide					Y	341
Crotonic acid, *trans*-	Acrylamide	−0.085	0.024	7.2	0.33	Y	341
Crotononitrile	Styrene	−0.012	0.06	23.29	1.7	Y	112
Crotononitrile	Styrene	0		23.8			785
Crotononitrile, α-phenyl-	Styrene	0		0.43			785
Cyclobutane, 1,2-dimethylene-	Butadiene	5.76		0.1			854
Cyclobutane, 1,2-dimethylene-	Butadiene	5.76	0.36	0.1	0.05		885
Cyclobutane, 1,2-dimethylene-	Styrene	3.6		0.26			854

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Cyclobutane, 1,2-dimethylene-	Styrene	3.6	0.16	0.28	0.05		885
Cyclopentene, 4-,-1,3-dione	Acrylonitrile	0.04	0.058	3.02	0.71	Y	95
Cyclopentene, 4-,-1,3-dione	Methacrylate, methyl	0.094	0.017	8.11	0.38	Y	95
Cyclopentene, 4-,-1,3-dione	Styrene					Y	263
Cyclopentene, 4-,-1,3-dione	Styrene, *p*-chloro-	0.014	0.006	0.294	0.081	Y	264
Cyclopentene, 4-,-1,3-dione	Vinylidene chloride	0.113	0.027	1.88	0.35	Y	264
Cyclopropane, 1,1-bis(ethoxycarbonyl)-2-vinyl-	Methacrylate, methyl	0.223	0.194	4.21	0.905	Y	1076
Cyclopropane, 1,1-bis(ethoxycarbonyl)-2-vinyl-	Methacrylate, methyl	0.111	0.194	18.3	2.5	Y	1076
Cyclopropene, 3,3-dimethoxy-	Acrylonitrile	0.245	0.037	1.424	0.038	Y	838
Diallyl β-cyanoethylisocyanurate	Vinyl acetate	4.6		0.06			778
Diallyl melamine	Methacrylate, methyl					Y	201
Diallyl melamine	Styrene	−0.32	0.2	36.96	4.02	Y	201
Diallyl melamine	Vinyl acetate	0.318	0.073	0.68	0.026	Y	201
Diallyl phthalate	Acrylate, methyl	0.028	0.005	6.208	0.069	Y	297
Diallyl phthalate	Acrylate methyl	0.038	0.008	6.18	0.16	Y	297
Diallyl phthalate	Acrylonitrile	0.04	0.012	4.23	0.15	Y	153
Diallyl phthalate	Acrylonitrile	0.037	0.031	2.78	0.52	Y	153
Diallyl phthalate	Fumarate, dibutyl	0.02		1.01			665
Diallyl phthalate	Fumarate, diethyl	0.01		1.25			665
Diallyl phthalate	Fumarate, dioctyl	0.02		0.96			665
Diallyl phthalate	Methacrylate, methyl	−0.001	0.011	26.56	1.98	Y	153
Diallyl phthalate	Methacrylate, methyl	0.029	0.026	22.73	1.48	Y	153
Diallyl phthalate	Methacrylate, methyl	0.076	0.006	21.52	0.39	Y	153
Diallyl phthalate	Styrene	0.041	0.038	36.41	1.23	Y	152
Diallyl phthalate	Styrene	0.026	0.071	25.59	2.19	Y	152
Diallyl phthalate	Styrene	0.038	0.04	23.67	1.2	Y	152
Diallyl phthalate	Styrene	0.08		18.8			152
Diallyl phthalate	Styrene	0.09		27.5			152
Diallyl phthalate	Styrene	0.1		16.9			152
Diallyl phthalate	Styrene	0.1		23.8			152
Diallyl phthalate	Styrene	0.13		15.1			152
Diallyl phthalate	Vinyl acetate	2.19	0.34	0.336	0.049	Y	297
Diallyl phthalate	Vinyl acetate	1.66	0.16	0.414	0.029	Y	297
Diallyl phthalate	Vinyl acetate	2		0.72			509
Diallyl phthalate	Vinyl chloride	0.7	0.046	0.833	0.012	Y	296
Diallyl phthalate	Vinylidene chloride	0.2		5			376
Diallyl, 1,3-,5-(2-hydroxy-3-phenoxypropyl)isocyanate	Vinyl acetate	5.5		0.15			671
Diallyl, 1,3-,5-(2-hydroxy-3-phenoxypropyl)isocyanate	Vinyl acetate	3.65		0.16			676
Diallylcyanamide	Acrylate, methyl	0.05		6.7			603
Diallylcyanamide	Acrylate, methyl	0.05		6.7			604
Diallylcyanamide	Methacrylate, methyl	0.04		25			603
Diallylcyanamide	Methacrylate, methyl	0.04		25			604
Diallylcyanamide	Vinyl acetate	1.62		0.01			581
Diallylcyanamide	Vinyl chloride	0.68		0.44			602
Diallylcyanamide	Vinyl chloride	0.68		0.44			603
Diallyldimethylammonium chloride	Vinyl acetate	1.95		0.35			1029
Dicyclopentadiene	Maleic anhydride	0.037	0.112	0.071	0.045	Y	914
Dihydroxydiphenylsulfone diallyl ether	Styrene	0.92	0.16	0.975	0.015		1016
Dimethyleneheptanedioate, 2,6-,dimethyl	Methacrylate, methyl	0.53	0.02	2.36	0.25	N	795
Dioxolane,1,3-,4,4-dimethyl-5-methylene-2-trichloro-	Pyrrolidone, *N*-vinyl-	0.01		3.87			647
Dioxolane,1,3-,4-methylene-2-trichloromethyl-	Acrylamide	0.037		8.6			616
Dioxolane,1,3-,4-methylene-2-trichloromethyl-	Pyrrolidone, *N*-vinyl-	0.04		2.98			647
Diphenyl sulfone,3,3′-bisitaconimido-*m,m′*-	Methacrylic acid	0.015	0.058	0.564	0.032	Y	1042
Dithiane, 2-methylene-1,3-	Methacrylate, methyl	4.12		0.08			1031
Dithiane, 2-methylene-1,3-	Styrene	3.35		0.03			1031
Divinylbenzene, *m*-	Styrene	0.58	0.04	0.58	0.15	N	262
Divinylbenzene, *p*-	Maleic anhydride	0.01	0.01	0.09	0.01		943

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Divinylbenzene, *p*-	Styrene	1.18	0.17	0.26	0.25	N	262
Ethene, 2-phenyl-1,1-dicyano-	*N*-Vinylpyrrolidone					Y	947
Ethylene	Acrylate, butyl	0.01	0.003	13.94	0.38	N	42
Ethylene	Acrylate, methyl					N	41
Ethylene	Acrylate, methyl	0.02		19.4			768
Ethylene	Acrylonitrile	0		7			415
Ethylene	Butene-1	3.25					388
Ethylene	Butene-1	3.6		0.16			407
Ethylene	Butene-2	7.5					477
Ethylene	Carbon monoxide	0.068	0.012	0.32	0.65	Y	51
Ethylene	Carbon monoxide	0.025	0.006	0.004	0.012	Y	51
Ethylene	Carbon monoxide	0.04	0.01	−0.15	0.07	N	52
Ethylene	Carbon monoxide	0.5		0			52
Ethylene	Ethylene, tetrafluoro-	0.52	0.18	0.067	0.12		1093
Ethylene	Ethylene, tetrafluoro-	0.575	0.09	0.065	0.027		1094
Ethylene	Ethylene, tetrafluoro-	0.61		0.02			563
Ethylene	Ethylene, tetrafluoro-	0.38		0.1			563
Ethylene	Fumarate, diethyl	0.11	1.14	1.91	3.4	N	41
Ethylene	Maleate, diethyl	0.25		10			41
Ethylene	Maleic anhydride	0.04		0			620
Ethylene	Methacrylate, di-, zinc	0.19		40			935
Ethylene	Methacrylate, methyl					N	41
Ethylene	Phosphonate, vinyl-, diphenyl	0.088	0.01	1.66	2.22	Y	801
Ethylene	Phosphonate, vinyl-, bis(2-chloroethyl)	0.077	0.003	7.73	2.19	Y	801
Ethylene	Phosphonate, vinyl- diisobutyl					Y	801
Ethylene	Propylene	3.43					388
Ethylene	Propylene	3.2		0.62			477
Ethylene	Styrene	0.05		14.88			615
Ethylene	Vinyl acetate	0.16	0.03	0.72	1.2	N	309
Ethylene	Vinyl acetate					N	41
Ethylene	Vinyl acetate	0.28		1.14			413
Ethylene	Vinyl acetate	0.88	0.52	1.03	0.77	N	42
Ethylene	Vinyl acetate	0.74		1.52			559
Ethylene	Vinyl acetate	0.67	0.03	3.74	0.64	N	68
Ethylene	Vinyl acetate	0.13	0.01	1.23	0.21	N	68
Ethylene	Vinyl acetate	0.79		1.4			766
Ethylene	Vinyl butyrate	0.7		1.5			605
Ethylene	Vinyl chloride	0.19	0.17	1.99	0.42	N	180
Ethylene	Vinyl chloride	0.24	0.13	1.74	0.58	N	270
Ethylene	Vinyl chloride	0.34	0.13	4.38	0.44	N	42
Ethylene	Vinyl chloride	0.2		2			472
Ethylene	Vinyl chloride	0.21		3.21			536
Ethylene	Vinyl chloride	0.21	0.01	1.85	0.1	N	63
Ethylene	Vinyl chloride	0.14	0.12	0.96	0.22	N	63
Ethylene	Vinyl chloride	0.23	0.01	1.63	0.22	N	64
Ethylene	Vinyl chloride	0.16	0.01	1.55	0.18	N	68
Ethylene	Vinyl diethylphosphinate	0.15	0.16	−11.99	80.64	Y	311
Ethylene	Vinyl diphenylphosphinate	0.05	0.01	−62.84	34.66	Y	311
Ethylene	Vinyl ethyl ether	2.7		0			584
Ethylene	Vinyl fluoride	2.52	0.32	0.01	0.15	N	42
Ethylene	Vinyl fluoride	1.7		0.3			590
Ethylene	Vinyl propionate	0.67		1.5			605
Ethylene	Vinylidene chloride	0.018	0.005	15.71	1.96	Y	299
Ethylene, 1,1-dichloro-2,2-difluoro	Vinyl acetate	−0.13	0.36	0.33	0.58	Y	6
Ethylene, 1,1-diphenyl-	Acrylate, ethyl	0.5		0.8			65
Ethylene, 1,1-diphenyl-	Acrylate, methyl	−0.24	0.15	0.09	0.01	N	334
Ethylene, 1,1-diphenyl-	Acrylate, methyl	0.21	0.05	0.12	0.01	N	334
Ethylene, 1,1-diphenyl-	Acrylonitrile	−0.02	0.01	0.05	0.01	N	143
Ethylene, 1,1-diphenyl-	Butadiene, 2,3-dichloro-	−0.08		3.58		Y	61
Ethylene, 1,1-diphenyl-	Butadiene, 2-chloro-	0.03		3.4		Y	61
Ethylene, 1,1-diphenyl-	Methacrylate, methyl	0.09		2.1			65
Ethylene, 1,1-diphenyl-	Methacrylate, methyl	0		8.7			901
Ethylene, 1,1-diphenyl-	Methacrylonitrile	−0.45	0.11	0.48	0.1	N	143
Ethylene, 1,1-diphenyl-	Styrene	0.01	0.02	0.33	0.02	N	189
Ethylene, 1,1-diphenyl-	Styrene	−0.02	0.01	0.34	0.01	N	281

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Ethylene, 1,1-diphenyl-	Styrene	1.15		0.87			901
Ethylene, chlorotrifluoro-	Ethylene, tetrafluoro-	1		1			392
Ethylene, chlorotrifluoro-	Isobutylene	0.01	0.02	0.06	0.03	N	318
Ethylene, chlorotrifluoro-	Methacrylate, methyl	−0.074	0.062	51.36	3.31	Y	236
Ethylene, chlorotrifluoro-	Propionate, vinyl	0.08	0.01	0.63	0.04		1097
Ethylene, chlorotrifluoro-	Propylene	0.02	0.08	0.08	0.07	N	318
Ethylene, chlorotrifluoro-	Pyrrolidone, *N*-vinyl-	0.3		0.38			830
Ethylene, chlorotrifluoro-	Styrene	−0.1	0.21	4.38	0.99	Y	236
Ethylene, chlorotrifluoro-	Styrene	0.01		5.2			567
Ethylene, chlorotrifluoro-	Vinyl acetate	0.04	0.02	0.68	0.11		1097
Ethylene, chlorotrifluoro-	Vinyl acetate	−0.02		0.47		Y	236
Ethylene, chlorotrifluoro-	Vinyl chloride	0.01		2.53			454
Ethylene, chlorotrifluoro-	Vinyl fluoride	1.2		0.8			406
Ethylene, chlorotrifluoro-	Vinylidene chloride	0.02		17.14			454
Ethylene, dichloro-, *cis*	Styrene	−3.1		3.1		Y	137
Ethylene, dichloro-, *cis*	Styrene	−2.2		63.1		Y	137
Ethylene, dichloro-, *cis*	Styrene	−0.28	0.12	21.4	4.9	Y	40
Ethylene, dichloro-, *cis*	Styrene	−0.02	0.062	161.3	30.3	Y	9
Ethylene, dichloro-, *cis*	Styrene	0.067	0.053	24.75	2.64	Y	9
Ethylene, dichloro-, *cis*	Vinyl acetate	0.055	0.024	1.02	0.27	Y	137
Ethylene, dichloro-, *cis*	Vinyl acetate	0.01		6.4		Y	137
Ethylene, dichloro-, *cis*	Vinyl acetate	0.058	0.03	0.898	0.06	Y	9
Ethylene, dichloro-, *cis*	Vinyl acetate	0.1	0.14	4.13	1.94	Y	9
Ethylene, dichloro-, *cis*	Vinylidene cyanide					Y	82
Ethylene, dichloro-, *cis*	Vinylidene cyanide					Y	82
Ethylene, *p*-vinyl-1,1-diphenyl-	Styrene	1.31		0.46			742
Ethylene, tetrachloro-	Acrylate, methyl	0		830			402
Ethylene, tetrachloro-	Acrylonitrile	0		456			571
Ethylene, tetrachloro-	Acrylonitrile	0		470			59
Ethylene, tetrachloro-	Methacrylate, methyl					Y	204
Ethylene, tetrachloro-	Styrene					Y	204
Ethylene, tetrachloro-	Styrene	200					204
Ethylene, tetrachloro-	Styrene					Y	40
Ethylene, tetrachloro-	Styrene	0		187			40
Ethylene, tetrachloro-	Styrene	0		185			59
Ethylene, tetrachloro-	Vinyl acetate	−0.07	0.15	4.45	1.87	Y	3
Ethylene, tetrachloro-	Vinyl acetate					Y	59
Ethylene, tetrachloro-	Ethylene	0.067	0.12	0.52	0.18		1093
Ethylene, tetrafluoro-	Ethylene	0.065	0.027	0.575	0.09		1094
Ethylene, tetrafluoro-	Ethylene	0.02		0.61			563
Ethylene, tetrafluoro-	Ethylene	0.1		0.38			563
Ethylene, tetrafluoro-	Ethylene, chlorotrifluoro-	1		1			392
Ethylene, trichloro-	Acrylate, methyl	0		33			470
Ethylene, trichloro-	Acrylonitrile	0		67			470
Ethylene, trichloro-	Acrylonitrile	0		62.1			572
Ethylene, trichloro-	Butadiene	−0.03	0.07	9.64	0.21	N	89
Ethylene, trichloro-	Methacrylate, methyl	0		100			470
Ethylene, trichloro-	Styrene	0		17.1			40
Ethylene, trichloro-	Styrene	0		16.5			40
Ethylene, trichloro-	Styrene	0		12.1			40
Ethylene, trichloro-	Styrene	0		12.7			40
Ethylene, trichloro-	Styrene	−0.006	0.37	16	11	Y	59
Ethylene, trichloro-	Vinyl acetate	−0.06	0.34	0.607	0.076	Y	158
Ethylene, trichloro-	Vinyl acetate	−0.006	0.019	0.7	0.1	Y	9
Ethylene, trichloro-	Vinyl isobutyl ether	0.014	0.003	0.05	0.009	Y	332
Ferrocene, ethyl-α-cyclopentadienylidene	Butadiene	1.51		0.02			699
Ferrocene, ethyl-α-cyclopentadienylidene	Isoprene	1.91		0.01			699
Formaldehyde-*S*-vinyl-*O*-ethylthioacetal	Styrene	0.32	0.24	7.75	0.68	Y	273
Fumarate, allyl methyl	Styrene	0.03	0.02	0.25	0.01	N	713
Fumarate, bis(2-butyl)	Styrene	0	0.01	0.51	0.04	N	249
Fumarate, bis(3-[tris(trimethylsiloxy)-silyl)propyl]	Fumarate, di-*tert*-butyl	0.21		0.67			1040
Fumarate, bis(3-[tris(trimethylsiloxy)-silyl)propyl]	Fumarate, diisopropyl	0.07		0.87			1040
Fumarate, bis(3-chloro-2-butyl)	Styrene	−0.06	0.03	0.26	0.04	N	249

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Fumarate, di-(2-chloroethyl)	Styrene	0		0.18			1038
Fumarate, di-(2-cyanoethyl)	Styrene	0.09		0.07			1038
Fumarate, di-(2-(methoxyethyl)	Styrene	0.03		0.35			1038
Fumarate, di-(2-phenylethyl)	Styrene	0		0.27			1038
Fumarate, di-*sec*-butyl	Styrene	0.04		0.55			1038
Fumarate, di-*tert*-amyl	Styrene	0.08		0.69			1038
Fumarate, di-*tert*-butyl	Fumarate, bis(3-[tris(trimethylsiloxy) silyl)propyl	0.67		0.21			1040
Fumarate, di-*tert*-butyl	Fumarate, dimethyl	1.57	0.365	0.156	0.227	Y	1049
Fumarate, di-*tert*-butyl	Maleate, dimethyl	18.2	6.7	0.011	0.037	Y	1049
Fumarate, di-*tert*-butyl	Styrene	0.04		0.57			1038
Fumarate, dibenzyl	Styrene	0		0.32			1038
Fumarate, dibutyl	Diallyl phthalate	1.01		0.02			665
Fumarate, dibutyl	Imidazole, 1-vinyl-	0.182	0.054	0.03	0.01	Y	614
Fumarate, dibutyl	Styrene	0.04		0.35			1038
Fumarate, dibutyl	Styrene, *tert*-butoxy	0.01		0.18			1077
Fumarate, dicyclohexyl	Styrene	0.06		0.49			1038
Fumarate, dicyclopentyl	Styrene	0.06		0.3			1038
Fumarate, diethyl	Acrylonitrile	0		8			470
Fumarate, diethyl	Acrylonitrile	0.1		10			640
Fumarate, diethyl	Butadiene	0.25		2.13			466
Fumarate, diethyl	Butadiene, 2-chloro-	0.02		6.51		Y	61
Fumarate, diethyl	Diallyl phthalate	1.25		0.01			665
Fumarate, diethyl	Ethylene	1.91	3.4	0.11	1.14	N	41
Fumarate, diethyl	Methacrylate, methyl	0.04	0.004	40.3	0.6	N	28
Fumarate, diethyl	Methacrylate, methyl	0.05		2.1			394
Fumarate, diethyl	*N*-Vinylimidazole	0.191	0.022	0.113	0.019	Y	879
Fumarate, diethyl	Styrene	0.02		0.29			1038
Fumarate, diethyl	Styrene	0.081	0.005	6.3	0.26	Y	139
Fumarate, diethyl	Styrene	0.11	0.022	0.394	0.06	Y	139
Fumarate, diethyl	Styrene	0.07		0.3			139
Fumarate, diethyl	Styrene	0	0.01	0.31	0.02	N	249
Fumarate, diethyl	Styrene	0.1		0.36			562
Fumarate, diethyl	Styrene	0.02		0.29			640
Fumarate, diethyl	Vinyl acetate	0.443	0.007	0.011	0.004	Y	137
Fumarate, diethyl	Vinyl acetate	0.33		0.09			718
Fumarate, diethyl	Vinyl chloride	0.48	0.1	0.13	0.1	Y	137
Fumarate, diethyl	Vinyl ethyl ether	1.63	0.16	−0.124	0.059	Y	194
Fumarate, diethyl	Vinyl isobutyl ether	0.52		0			640
Fumarate, diethyl	Vinylidene chloride	0.012	0.038	12.19	1.58	Y	59
Fumarate, diisobutyl	Styrene	0.05		0.37			1038
Fumarate, diisopropyl	Acenaphthalene	0.06		2.57			1039
Fumarate, diisopropyl	Acetate, isopropenyl	0.67		0.011			1038
Fumarate, diisopropyl	Acrylate, methyl	0.091		1.9			1038
Fumarate, diisopropyl	Acrylate, phenyl	0.07		3.4			1038
Fumarate, diisopropyl	Acrylonitrile	0.2		16			1038
Fumarate, diisopropyl	Carbazole, *N*-vinyl-	0		0			1038
Fumarate, diisopropyl	Fumarate, bis(3-[tris(trimethylsiloxy) silyl)propyl	0.87		0.07			1040
Fumarate, diisopropyl	Isobutylene	0.42		0			1038
Fumarate, diisopropyl	Methacrylate, methyl	0.17		23			1038
Fumarate, diisopropyl	Methacrylonitrile	0		85			1038
Fumarate, diisopropyl	Styrene	0.055		0.4			1038
Fumarate, diisopropyl	Vinyl acetate	0.9		0.012			1038
Fumarate, diisopropyl	Vinyl chloride	0.49		0.29			1038
Fumarate, diisopropyl	Vinylidene chloride	0.095		48			1038
Fumarate, dimethyl	Fumarate, di-*tert*-butyl	0.156	0.227	1.57	0.365	Y	1049
Fumarate, dimethyl	*N*-Vinylimidazole	0.252	0.015	0.133	0.021	Y	879
Fumarate, dimethyl	Styrene	0.03		0.39			1038
Fumarate, dioctyl	Diallyl phthalate	0.96		0.02			665
Fumarate, dioctyl	Pyrrolidone, *N*-vinyl-	0.04	0.04	0.02	0.01	N	121
Fumarate, dipropyl	Imidazole, 1-vinyl-	0.163	0.69	0.009	0.012	Y	614
Fumarate, dipropyl	Styrene	0.02		0.32			1038
Fumarate, isopropyl butyl	Styrene	0.01		0.42			1056
Fumarate, isopropyl butyl	Styrene	0.01		0.3			1056

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Fumarate, isopropyl isobutyl	Styrene	0.01		0.46			1056
Fumarate, isopropyl methyl	Styrene	0.16		0.3			1056
Fumarate, isopropyl *sec*-butyl	Styrene	0.02		0.54			1056
Fumarate, methyl butyl	Styrene	0.02		0.36			1056
Fumarate, methyl ethyl	Styrene	0.02		0.32			1056
Fumarate, methyl isobutyl	Styrene	0.03		0.32			1056
Fumarate, methyl isopropyl	Styrene	0.16		0.3			1056
Fumarate, methyl propyl	Styrene	0.05		0.28			1056
Fumarate, methyl *sec*-butyl	Styrene	0.02		0.45			1056
Fumarate, methyl *tert*-amyl	Styrene	0.06		0.53			1056
Fumarate, methyl *tert*-butyl	Styrene	0.03		0.45			1056
Fumarate, mono-2-butyl	Styrene	−0.04	0.03	0.33	0.03	N	249
Fumarate, *tert*-butyl 2-ethylhexyl	Styrene	0.01		0.54			1056
Fumarate, *tert*-butyl 4-methyl-2-pentyl	Styrene	0.01		0.57			1056
Fumarate, *tert*-butyl ethyl	Styrene	0.04		0.25			1056
Fumarate, *tert*-Butyl isoamyl	Styrene	0.06		0.45			1056
Fumarate, *tert*-butyl methyl	Styrene	0.03		0.45			1056
Fumarate, *tert*-butyl *sec*-butyl	Styrene	0.07		0.55			1056
Fumaric acid	*N*-vinylimidazole	0.313	0.012	0.121	0.01	Y	879
Fumaronitrile	Benzothiazole, vinylmercapto-					Y	763
Fumaronitrile	Methacrylate, methyl	−0.064	0.096	6.74	0.79	Y	191
Fumaronitrile	Styrene	−0.016	0.021	0.202	0.058	Y	137
Fumaronitrile	Styrene	0.006	0.031	0.294	0.044	Y	199
Fumaronitrile	Styrene	0		0.3			478
Fumaronitrile	Styrene					Y	77
Fumaronitrile	Styrene, α-methyl-	−0.01	0.078	0.094	0.02	Y	77
Fumaryl chloride	Styrene					Y	91
Fumaryl chloride	Vinyl acetate					Y	91
Furan, 2,3-dihydro-	Styrene	0		20			806
Furan, 2-vinyl-5-methyl	Acrylate, butyl	0.392	0.051	0.118	0.019	Y	1105
Furan, 2-vinyl-5-methyl	Methacrylate, isobutyl	0.716	0.169	0.105	0.044	Y	1105
Furan, 2-vinyl-5-methyl	Methacrylate, methyl	0.867	0.292	0.183	0.146	Y	1105
Hex-3-yne-5-ene, 2-methyl-*tert*-butyl peroxide	Maleic anhydride	0.36		0.01			637
Hexatriene, tetrachloro-	Acrylate, methyl	3.19	1.36	0.262	0.087	Y	5
Hexatriene, tetrachloro-	Acrylonitrile	4.01	1.25	0.234	0.05	Y	4
Hexatriene, tetrachloro-	Butadiene, 2-chloro-	0.2		3.6			459
Hexatriene, tetrachloro-	Isoprene	1.58		0.58			459
Hexatriene, tetrachloro-	Methacrylate, methyl	1.76	0.28	0.504	0.057	Y	5
Hexatriene, tetrachloro-	Styrene	0.85	0.22	0.123	0.042	Y	4
Hexatriene, tetrachloro-	Vinyl acetate					Y	4
Hexene, 1-, 3,4-dioic anhydride	Styrene	−0.052	1	8.57	8.4	Y	933
Hexene, 2-, 3,4-dioic anhydride	Styrene	−0.08	1.17	3.63	5.6	Y	933
Hexene, 3-, 3,4-dioic anhydride	Styrene	−0.54	5.83	17.17	436	Y	933
Hexene, 5-, 2,4-dione	Styrene	1.22		0.031			931
Hexene, 5-, 2,4-dione	Styrene	1.38		0.049			931
Hexene, 5-, 2,4-dione	Styrene	1.52		0.066			931
Hexene, 5-, 2,4-dione	Styrene	1.73		0.07			931
Hexene, 5-, 2,4-dione	Styrene	1.91		0.171			931
Hexene, 5-, 2,4-dione	Styrene	1.91		0.179			931
Hexene, 5-, 2,4-dione	Styrene	2.07		0.173			931
Hexene, 5-, 2,4-dione	Styrene	2.46		0.121			931
Hexene, 5-, 2,4-dione	Styrene	2.69		0.24			931
Hexene, 5-, 2,4-dione	Styrene	2.93		0.014			931
Hexene, 5-, 2,4-dione	Styrene	3.19		0.044			931
Hexene, 5-, 2,4-dione	Styrene	4.11		0.028			931
Hexene-1	Acrylate, methyl	−0.67	0.03	−0.13	0.6	Y	60
Hexene-1	Acrylonitrile	−1.08		2.07		Y	60
Hexene-1	Vinyl chloride	−1.72	9.23	1.54	0.98	N	329
Hexene-1, 2,5-dicarboxylate, dimethyl	Styrene	0.196	0.043	0.678	0.031	Y	828
Imidazole, 1-vinyl	Triazole, 1-*N*-vinyl-1,2,4-					N	972
Imidazole, 1-vinyl-	Acrylic acid	0.32		0.002			955
Imidazole, 1-vinyl-	Fumarate, dibutyl	0.03	0.01	0.182	0.054	Y	614
Imidazole, 1-vinyl-	Fumarate, dipropyl	0.009	0.012	0.163	0.069	Y	614

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Imidazole, 1-vinyl-	Maleate, dibutyl	0.486	0.057	0.093	0.08	Y	614
Imidazole, 1-vinyl-	Maleate, dipropyl	0.314	0.028	0.053	0.035	Y	614
Imidazole, 1-vinyl-	Maleate, monomethyl	0.4		0.6			724
Imidazole, 1-vinyl-	Maleic acid	0.3		0.6			635
Imidazole, 1-vinyl-	Methacrylate, methyl	0.014	0.03	4.36	1.86	Y	364
Imidazole, 1-vinyl-	Pyrrolidone, *N*-vinyl-	0.96	0.19	0.16	0.074	Y	364
Imidazole, 1-vinyl-	Styrene	0.071	0.029	9.94	0.24	Y	364
Imidazole, 1-vinyl-	Vinyl acetate	1.9	6.99	−0.11	1.17	Y	364
Imidazole, 1-vinyl-2-methyl-	Acrylate, methyl	0.05		1.28			465
Imidazole, 1-vinyl-2-methyl-	Methacrylate, methyl	0.047	0.088	4	0.54	Y	364
Imidazole, 1-vinyl-2-methyl-	Methacrylate, methyl	0.003		3.48			479
Imidazole, 1-vinyl-2-methyl-	Styrene	0.09	0.064	8.87	0.62	Y	364
Imidazole, 1-vinyl-2-methyl-	Styrene	0.07		8.97			479
Imidazole, 1-vinyl-2-methyl-	Vinyl chloride	2.15	0.26	0.188	0.077	Y	154
Imidazole, *N*-vinyl-	Silane, methacryloxypropyldimethoxy methyl-	0.079		1.39			1095
Imidazole, *N*-vinyl-	Silane, methacryloxypropyltrimethoxy-	0.158	0.04	2.27	1.12	Y	1088
Imidazole, *N*-vinyl-	Styrene, *p*-trimethyoxysilyl-	0.002		6.9			1095
Imidazolid-2-one, 1,3-divinyl-	Acrylate, ethyl	−0.08	0.03	0.41	0.13	N	800
Imidazolid-2-one, 1-ethyl-3-vinyl-	Acrylate, ethyl	0.01	0.01	0.47	0.06	N	800
Indene	Acrylate, methyl	0.1		0.63			606
Indene	Acrylate, methyl	0.12		0.62			606
Indene	Acrylonitrile	0.09	0.25	0.281	0.05	Y	360
Indene	Acrylonitrile	0.07	0.24	0.29	0.12	Y	360
Indene	Styrene, *p*-chloro-	−0.069	0.066	2.86	0.83	Y	6
Indene	Vinyl acetate	8.64	2.8	0.39	0.2	Y	360
Indene	Vinyl 1-menthyl ether	12.64	12.59	0.67	2.82	Y	750
Indene	Vinyl methyl ketone	0.2		2.63			606
Indene	Vinyl methyl ketone	0.25		2.9			606
Indene	Vinylidene chloride	0.293	0.095	0.49	0.15	Y	6
Indene	Vinylidene chloride	0.34	0.85	−0.02	1.15	Y	6
Isobutene	Maleimide, *N*-cyclohexyl-	0.41		0			995
Isobutylene	Acrylonitrile	0	0.002	1.295	0.008	Y	21
Isobutylene	Acrylonitrile	0.02	0.38	0.98	0.25	Y	211
Isobutylene	Acrylonitrile	0		1.02			484
Isobutylene	Ethylene, chlorotrifluoro-	0.06	0.03	0.01	0.02	N	318
Isobutylene	Fumarate, diisopropyl	0		0.42			1038
Isobutylene	Maleic anhydride	0.012		0.065			988
Isobutylene	Vinyl acetate	0.17	0.19	1.66	0.18	Y	211
Isobutylene	Vinyl chloride	0.06	0.8	2.04	1.82	Y	140
Isobutylene	Vinyl chloride	0.268	0.043	2.12	0.2	Y	155
Isobutylene	Vinyl chloride	−0.54	5.8	1.2	2.05	N	329
Isobutylene	Vinyl chloride	0.08		1.54			560
Isobutylene, 3-chloro-	Allyl acetate	4.5		0.15			412
Isobutylene, 3-chloro-	Vinyl acetate	0		0.13			475
Isobutylene, 3-chloro-	Vinyl chloride	0		0.31			475
Isobutylene, 3-chloro-	Vinylidene chloride	0		1.1			470
Isooctene	Maleimide, *N*-cyclohexyl-	0.72		0			995
Isoprene	Acetylene, phenyl-	3.01		0.1			648
Isoprene	Acrylate, methyl	0.75		0.12			424
Isoprene	Acrylonitrile	0.45		0.03			417
Isoprene	Butadiene	1.06		0.94			173
Isoprene	Butadiene	0.608	0.062	0.14	0.37	Y	83
Isoprene	Butadiene, 2,3-dimethyl-	1.18		0.84			173
Isoprene	Butadiene, 2-chloro-	0.06		2.82			418
Isoprene	Butadiene, 2-fluoro-	0.25	0.12	2.63	0.96	Y	174
Isoprene	Butadiene, hexafluoro-	1.19		0.78			453
Isoprene	Ferrocene, ethyl-α-cyclo-pentadienylide	0.01		1.91			699
Isoprene	Hexatriene, tetrachloro-	0.58		1.58			459
Isoprene	Methacrylate, methyl	0.65	0.23	0.26	0.21	Y	302
Isoprene	Methacrylate, methyl	0.64		0.25			698
Isoprene	Methacrylate, methyl	0.64		0.25			704
Isoprene	Pyridine, 2-vinyl-	0.59		0.47			456
Isoprene	Pyridine, 2-vinyl-	0.58		0.46			458
Isoprene	Pyridine, 4-vinyl-	0.32		2.49			424

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Isoprene	Quinoline, 2-vinyl-	1.88		0.53			456
Isoprene	Styrene	1.922	0.016	0.513	0.02	Y	366
Isoprene	Styrene	1.957	0.011	0.476	0.018	Y	366
Isoprene	Styrene	1.98		0.44			544
Isoprene	Styrene	2.02		0.42			544
Isoprene	Styrene	1.3		0.48			83
Isoprene, 3-acetoxy-	Acrylate, methyl	5.71		0.27			770
Isoprene, 3-acetoxy-	Methacrylate, methyl	2.81		0.16			770
Isoprene, 3-acetoxy-	Styrene	4.47		0.34			770
Isopropenyl acetate	Maleic anhydride	0.034	0.022	0.002	0.011	Y	260
Isopropenyl acetate	Methacrylate, methyl	−0.09	0.12	17.36	1.95	Y	100
Isopropenyl acetate	Vinyl acetate	1.082	0.074	1.171	0.038	Y	100
Isopropenyl acetate	Vinyl chloride	0.23	0.16	2.26	0.13	Y	100
Isopropenyl methyl ketone	Acrylonitrile	1.2		0.3			490
Isopropenyl methyl ketone	Acrylonitirle	0.7		0.36			542
Isopropenyl methyl ketone	Butadiene, 2-chloro-	0.14	0.01	4.035	0.087	Y	259
Isopropenyl methyl ketone	Styrene	0.29		0.44			542
Isopropenyl methyl ketone	Styrene	0.66		0.32			542
Isopropenyl methyl ketone	Vinylidene chloride	4.06	0.66	0.134	0.099	Y	47
Isopropenyl, 3-(1-cyclohexenyl), acetate	Acrylate, methyl	0.57		0.37			770
Isopropenyl, 3-(1-cyclohexenyl), acetate	Methacrylate, methyl	0.4		0.81			770
Isopropenyl, 3-(1-cyclohexenyl), acetate	Styrene	0.56		1.59			770
Isopropenyl-, *p*-, phenol glycidyl ether	Maleic anhydride	0.003		0.002			937
Isopropenylacetophenone, *p*-	Styrene	0.84	0.59	0.3	0.18	Y	266
Isopropenylisocyanate	Acrylate, ethyl	0.15		0.79			434
Isopropenylisocyanate	Acrylate, methyl	0.11		0.6			434
Isopropenylisocyanate	Acrylate, methyl	0.079	0.047	0.799	0.049	Y	99
Isopropenylisocyanate	Acrylonitrile	0.1		0.24			434
Isopropenylisocyanate	Methacrylate, methyl	0.14		3.1			434
Isopropenylisocyanate	Styrene	0.07		8.12			434
Isopropenylisocyanate	Styrene	0.122	0.047	6.84	0.18	Y	99
Isopropenylisocyanate	Vinyl chloride	3		0.39			777
Isopropenylisocyanate	Vinylidene chloride	0.31		0.85			434
Isopropenyltoluene	Methacrylate, methyl	0.04		0.68			688
Isopropenyltoluene	Methacrylate, methyl	0.04		0.68			722
Isopropylidene, 2,3:4,5-di-*O*-, -1-vinylbenzyl-β-D-fructose	Styrene	3.26	1.12	0.14	0.055	Y	772
Itaconamate, *N*-(4-carboethoxyphenyl)-, butyl	Styrene	0.61		0.2			678
Itaconamate, *N*-(4-carboethoxyphenyl)-, ethyl	Methacrylate, methyl	0.48		1.21			678
Itaconamate, *N*-(4-carboethoxyphenyl)-, ethyl	Styrene	0.2		0.23			678
Itaconamate, *N*-(4-carboethoxyphenyl)-, methyl	Styrene	0.58		0.15			678
Itaconamate, *N*-(4-carboethoxyphenyl)-, propyl	Methacrylate, methyl	0.53		1			678
Itaconamate, *N*-(4-carboethoxyphenyl)-, propyl	Styrene	0.4		0.26			678
Itaconamate, *N*-(4-chlorophenyl)-, ethyl	Methacrylate, methyl	0.36		1.24			678
Itaconamate, *N*-(4-chlorophenyl)-, ethyl	Styrene	0.16		0.35			678
Itaconamate, *N*-(4-chlorophenyl)-, methyl	Styrene	0.18		0.38			678
Itaconamate, *N*-(4-chlorophenyl)-, propyl	Methacrylate, methyl	0.31		1.26			678
Itaconamate, *N*-(4-chlorophenyl)-, propyl	Styrene	0.31		0.39			678
Itaconamate, *N*-(4-methoxyphenyl)-, butyl	Styrene	0.69		0.12			678
Itaconamate, *N*-(4-methoxyphenyl)-, ethyl	Methacrylate, methyl	0.43		1.31			678
Itaconamate, *N*-(4-methoxyphenyl)-, ethyl	Styrene	0.34		0.33			678
Itaconamate, *N*-(4-methoxyphenyl)-, methyl	Styrene	0.41		0.26			678
Itaconamate, *N*-(4-methoxyphenyl)-, propyl	Methacrylate, methyl	0.4		1.03			678
Itaconamate, *N*-(4-methoxyphenyl)-, propyl	Styrene	0.4		0.38			678
Itaconamate, *N*-(4-methylphenyl)-, propyl	Methacrylate, methyl	0.28		1.11			678
Itaconamate, *N*-(4-methylphenyl)-, propyl	Styrene	0.19		0.54			678

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Itaconamate, *N*-(4-methylphenyl)-, butyl	Styrene	0.31		0.19			678
Itaconamate, *N*-(4-methylphenyl)-, ethyl	Methacrylate, methyl	0.63		1.43			678
Itaconamate, *N*-(4-methylphenyl)-, ethyl	Styrene	0.26		0.38			678
Itaconamate, *N*-(4-methylphenyl)-, methyl	Styrene	0.16		0.3			678
Itaconamate, *N*-phenyl-, butyl	Styrene	0.49		0.12			678
Itaconamate, *N*-phenyl-, ethyl	Methacrylate, methyl	0.47		1.32			678
Itaconamate, *N*-phenyl-, ethyl	Styrene	0.21		0.45			678
Itaconamate, *N*-phenyl-, methyl	Styrene	0.52		0.31			1059
Itaconamate, *N*-phenyl-, methyl	Styrene	0.18		0.39			678
Itaconamate, *N*-phenyl-, propyl	Methacrylate, methyl	0.15		1.13			678
Itaconamate, *N*-phenyl-, propyl	Styrene	0.18		0.5			678
Itaconate, bis(2-ethylhexyl)	Maleimide, *N*-cyclohexyl-	0.38		0.34			1060
Itaconate, bis(tri-*n*-butyltin)	Acrylate, 2-chloroethyl	0.391		0.683			940
Itaconate, bis(tri-*n*-butyltin)	Acrylate, butyl	0.422		0.91			940
Itaconate, bis(tri-*n*-butyltin)	Methacrylate,allyl	0.038		1.08			940
Itaconate, bis(tri-*n*-butyltin)	*N*-Vinylpyrrolidone	0.065	0.04	0.21	0.013	Y	941
Itaconate, di-2[-2(2-methoxyethoxy)-ethoxy]ethyl	Styrene	0.2	0.015	0.53	0.022		1098
Itaconate, di-*p*-butyl	Maleimide, *N*-cyclohexyl	0.34	0.04	0.21	0.04		1014
Itaconate, diamyl	Styrene	0.5	0.09	0.36	0.05	N	39
Itaconate, dibutyl	Itaconate, dimethyl	1.1		1.1			398
Itaconate, dibutyl	Methacrylate, methyl	0.717	0.11	1.33	0.09		1073
Itaconate, dibutyl	Methacrylate, methyl	0.4		0.8			398
Itaconate, dibutyl	Styrene	0.34	0.05	0.42	0.06	N	39
Itaconate, diethyl	Pyridine, 2-methyl-5-vinyl-	0.17		0.51			545
Itaconate, diethyl	Styrene	0.34	0.12	0.23	0.11	N	39
Itaconate, diethyl	Vinyl chloride	5.65		0.06			480
Itaconate, dimenthyl	Maleimide, *N*-cyclohexyl-	0.318	0.213	0.601	0.193	Y	975
Itaconate, dimenthyl	Methacrylate, methyl	−0.156	0.02	3.34	0.557	Y	975
Itaconate, dimenthyl	Styrene	0.483	0.364	0.502	0.188	Y	975
Itaconate, dimenthyl	Vinyl acetate					Y	975
Itaconate, dimethyl	Acrylate, butyl	0.94		0.4			398
Itaconate, dimethyl	Itaconate, dibutyl	1.1		1.1			398
Itaconate, dimethyl	Methacrylate, methyl	0.3		1.2			398
Itaconate, dimethyl	Methacrylonitrile	0.28		1.26			398
Itaconate, dimethyl	Pyridine, 2-methyl-5-vinyl-	0.22		0.88			545
Itaconate, dimethyl	Styrene					N	39
Itaconate, dimethyl	Styrene	0.14		0.48			398
Itaconate, dimethyl	Styrene	0.06		0.5			586
Itaconate, dimethyl	Styrene, *p*-chloro-	0.15		0.69			398
Itaconate, dimethyl	Vinyl chloride	5		0.05			479
Itaconate, dioctyl	Styrene	0.14	0.12	0.33	0.16	N	39
Itaconate, dipropy	Styrene	0.28	0.06	0.16	0.07	N	39
Itaconate, ethyl hexafluoroisopropyl	Styrene	0.039	0.035	0.435	0.025	N	1103
Itaconate, glycidyl methyl	Styrene	0.041	0.079	0.376	0.036		806
Itaconate, mono-methyl	Methacrylate, 2-naphthyl	0.31	0.42	0.52	0.16	Y	613
Itaconate, mono-methyl	Methacrylate, 2-naphthyl	0.36	0.21	0.59	0.14	Y	613
Itaconate, mono-methyl	Methacrylate, 2-naphthyl	0.135	0.028	0.223	0.02	Y	613
Itaconate, mono-methyl	Methacrylate, 2-naphthyl	0.04	0.11	0.189	0.055	Y	613
Itaconate, mono-methyl	Styrene	0.111	0.052	0.029	0.039	Y	656
Itaconate, mono-methyl	Styrene	0.33	0.15	0.34	0.077	Y	656
Itaconate, mono-methyl	Styrene	0.33	0.11	0.52	0.085	Y	656
Itaconate, mono-methyl	Styrene	0.218	0.086	0.404	0.06	Y	656
Itaconate, monomethyl	Methacrylate, methyl	0.139	0.103	1.1	0.071	Y	1010
Itaconic acid	Acrylonitrile	0.86	0.74	0.59	0.34	Y	164
Itaconic acid	Methacrylate, methyl	0		1.23			260
Itaconic acid	Methacrylate, methyl	0		1.14			408
Itaconic acid	Styrene	0		0.34			420
Itaconic acid	Styrene	0.12	0.67	0.26	0.08	Y	76
Itaconic acid mono(*p*-sulfamidoanilide)	Methacrylate, propargyl	0.58	0.05	1.65	0.02		957
Itaconic anhydride	Acrylate, 2-chloroethyl	2.46	0.38	−0.016	0.022	Y	365
Itaconic anhydride	Acrylate, 2-chloroethyl	2.61	0.32	0.251	0.021	Y	365
Itaconic anhydride	Acrylonitrile	4.83	0.33	0.034	0.014	Y	365
Itaconic anhydride	Methacrylate, methyl	1.17	0.33	0.155	0.024	Y	1010
Itaconic anhydride	Styrene	0.405	0.09	0.004	0.01	Y	365

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Itaconic anhydride	Styrene	0.456	0.09	0.086	0.015	Y	62
Itaconic anhydride	Styrene	0.785	0.05	0.003	0.005	Y	62
Itaconic anhydride	Vinyl acetate	1.57	0.3	−0.019	0.045	Y	365
Itaconic anhydride	Vinyl acetate	2.87	0.32	−0.04	0.024	Y	365
Itaconimide, *N*-(2-chloroethyl)-	Methacrylate, methyl	0.54	0.39	0	0.11	Y	642
Itaconimide, *N*-(2-chloroethyl)-	Styrene	0.162	0.092	0.069	0.045	Y	642
Itaconimide, *N*-(4-acetoxyphenyl)-	Styrene	0.21	0.21	0.35	0.12	Y	729
Itaconimide, *N*-(4-chlorophenyl)-	Styrene	0.2	0.18	0.159	0.084	Y	729
Itaconimide, *N*-(4-ethoxycarbonylphenyl)-	Styrene	0.112	0.054	0.232	0.032	Y	729
Itaconimide, *N*-(*p*-chlorophenyl)-	Styrene	0.14		0.15			740
Itaconimide, *N*-(*p*-tolyl)-	Styrene	0.24		0.16			740
Itaconimide, *N*-benzyl-	Methacrylate, methyl	1.06	0.75	0.46	0.18	Y	642
Itaconimide, *N*-benzyl-	Styrene	0.2	0.11	0.056	0.048	Y	642
Itaconimide, *N*-butyl-	Styrene	−0.019	0.04	0.22	0.03	Y	642
Itaconimide, *N*-ethyl-	Methacrylate, methyl	0.42	0.13	0.486	0.066	Y	642
Itaconimide, *N*-ethyl-	Styrene	0.172	0.06	0.232	0.029	Y	642
Itaconimide, *N*-isobutyl-	Styrene	−0.03	0.11	0.44	0.11	Y	642
Itaconimide, *N*-isopropyl-	Styrene	0.045	0.064	0.308	0.043	Y	642
Itaconimide, *N*-methyl-	Methacrylate, methyl	1.12	0.73	0.38	0.22	Y	642
Itaconimide, *N*-methyl-	Styrene	0.12	0.12	0.153	0.069	Y	642
Itaconimide, *N*-naphthyl-	Styrene	0.35		0.08			740
Itaconimide, *N*-phenyl-	Styrene	0.109	0.082	0.116	0.043	Y	729
Itaconimide, *N*-phenyl-	Styrene	0.08		0.05			740
Itaconimide, *N*-phenyl-	Styrene	0.998	0.123	0.05	0.248	N	978
Itaconimide, *N*-propyl-	Styrene	0.04	0.18	0.28	0.12	Y	642
Itaconimide, *N*-tolyl-	Styrene	0.219	0.079	0.134	0.031	Y	729
Malemic acid, *N*-3-dimethylamino-6-methylphenyl-	Butadiene	−0.08	0.03	0.37	0.04	N	679
Malemic acid, *N*-3-dimethylamino-6-methylphenyl-	Styrene	−0.19	0.03	0.55	0.03	N	679
Maleate, chloro-, diethyl	Styrene	−0.043	0.021	2.07	0.14	Y	14
Maleate, dibutyl	Imidazole, 1-vinyl-	0.093	0.08	0.486	0.057	Y	614
Maleate, allyl methyl	Styrene	0.02	0.01	8.29	0.14	N	713
Maleate, butyl stannylallyl	Styrene	0.06	0.03	0.1	0.02	N	811
Maleate, di-*tert*-butyl	Vinyl acetate	0.199	0.02	0.005	0.024	Y	1049
Maleate, dibutyl tin	Methacrylate, methyl	−0.095	0.199	72.7	5.93	Y	987
Maleate, dibutyl tin	Styrene	0.194	0.121	2.08	0.118	Y	987
Maleate, diethyl	Acrylonitrile	0		12			470
Maleate, diethyl	Acrylonitrile	0		20			640
Maleate, diethyl	Butadiene	0.11		8.08			466
Maleate, diethyl	Carbazole, *N*-vinyl-	0	0.01	0.21	0.05	N	361
Maleate, diethyl	Ethylene	10		0.25			41
Maleate, diethyl	Methacrylate, methyl	−0.1	0.05	341	46	N	28
Maleate, diethyl	Methacrylate, methyl	0		20			470
Maleate, diethyl	Methacrylate, methyl	−0.02	0.015	285.8	35.9	Y	873
Maleate, diethyl	*N*-Vinylimidazole	0.106	0.147	0.057	0.028	Y	879
Maleate, diethyl	Styrene					Y	14
Maleate, diethyl	Styrene	−0.013	0.022	6.59	0.46	Y	139
Maleate, diethyl	Styrene	0.07		8			640
Maleate, diethyl	Styrene	0.01	0.01	6.07	0.07	N	874
Maleate, diethyl	Styrene	0.001		6.592			888
Maleate, diethyl	Vinyl acetate	0.04	0.012	0.171	0.025	Y	137
Maleate, diethyl	Vinyl chloride	−0.003	0.044	0.768	0.078	Y	137
Maleate, diethyl	Vinyl chloride	0		0.8			449
Maleate, diethyl	Vinyl chloride	0.046	0.07	0.9	0.11	Y	6
Maleate, diethyl	Vinyl isobutyl ether	0.17		0			640
Maleate, diethyl	Vinylidene chloride	12.5					3
Maleate, dimethyl	Carbzole, *N*-vinyl-	0	0.01	0.51	0.11	N	361
Maleate, dimethyl	Fumarate, di-*tert*-butyl	0.011	0.037	18.2	6.7	Y	1049
Maleate, dimethyl	*N*-Vinylimidazole	0.046	0.438	0.046	0.046	Y	879
Maleate, dimethyl	Styrene	0.15	0.12	10.65	2.81	Y	137
Maleate, dimethyl	Styrene	0.105	0.058	0.083	0.015	Y	317
Maleate, dimethyl	Succinimide, *N*-vinyl-	−0.012	0.021	1.25	0.16	Y	284
Maleate, dimethyl	Vinyl acetate	0.257	0.086	−0.006	0.084	Y	1049
Maleate, dioctyl	Vinyl chloride	0.608	0.099	0.05	0.14	Y	3

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Maleate, dipropyl	Imidazole, 1-vinyl-	0.053	0.035	0.314	0.028	Y	614
Maleate, monoallyl	Maleic anhydride	0.027	0.014	0.051	0.054	N	906
Maleate, monoallyl	Styrene	0.07	0.02	0.14	0.01	N	811
Maleate, monomethyl	Imidazole, 1-vinyl-	0.6		0.4			724
Maleate, tributylstannyl allyl	Maleic anhydride	0.095	0.104	0.032	0.017	N	906
Maleic Anhydride	Isopropenyl-, *p*–, phenol glycidyl ether	0.002		0.003			937
Maleic acid	Imidazole, 1-vinyl-	0.6		0.3			635
Maleic acid	*N*-Vinylimidazole	0.094	0.027	0.018	0.007	Y	879
Maleic anhydride	Acenaphthalene	0.02		0.46			1039
Maleic anhydride	Acenaphthalene	−0.04	0.05	0.46	0.13	N	326
Maleic anhydride	Acetylene, phenyl-	0.06		0.08			797
Maleic anhydride	Acrolein diethylacetal	0.07		0.18			767
Maleic anhydride	Acrylamide	0		0.56		Y	655
Maleic anhydride	Acrylate, 2-chloroethyl	0.027	0.004	7.15	0.31	Y	117
Maleic anhydride	Acrylate, ferrocenylmethyl				2.96	Y	325
Maleic anhydride	Acrylate, methyl	0.012	0.013	2.788	0.051	Y	260
Maleic anhydride	Acrylate, methyl	0		2.5			470
Maleic anhydride	Acrylonitrile	0		6			470
Maleic anhydride	Allyl acetate	0.028	0.019	−0.001	0.007	Y	27
Maleic anhydride	Allylbenzene					Y	351
Maleic anhydride	Benzocyclobutane, α-methylene-	0.025		0.002			1083
Maleic anhydride	Benzothiazole, vinylmercapto-					Y	763
Maleic anhydride	Butadiene monoxide	0.64	0.06	0.05	0.18		1064
Maleic anhydride	Butadiene, 2-chloro-	0.329	0.107	0.026	0.03	N	1068
Maleic anhydride	Carbamate, *N*,*N*-diethyl-, vinyl	0.025	0.034	0.024	0.066	Y	92
Maleic anhydride	Dicyclopentadiene	0.071	0.045	0.037	0.112	Y	914
Maleic anhydride	Divinylbenzene, *p*-	0.09	0.01	0.01	0.01		943
Maleic anhydride	Ethylene	0		0.04			620
Maleic anhydride	Hex-3-yne-5-ene, 2-methyl-*tert*-butyl-peroxide	0.01		0.36			637
Maleic anhydride	Isobutylene	0.065		0.012			988
Maleic anhydride	Isopropenyl acetate	0.002	0.011	0.034	0.022	Y	260
Maleic anhydride	Maleate, monoallyl	0.051	0.054	0.027	0.014	N	906
Maleic anhydride	Maleate, tributylstannyl allyl	0.032	0.017	0.095	0.104	N	906
Maleic anhydride	Maleimide, *N*-phenyl-	−0.095	0.106	1.08	0.13	Y	990
Maleic anhydride	Maleimide, *N*-phenyl-	0.02	0.65	1.92	11.83	Y	341
Maleic anhydride	Methacrylate, ferrocenylmethyl	0.24	0.28	0.33	0.65	Y	325
Maleic anhydride	Methacrylate, methyl	−0.163	0.027	4.18	0.31	Y	169
Maleic anhydride	Methacrylate, methyl	−0.019	0.026	6.36	0.64	Y	260
Maleic anhydride	Methacrylate, methyl	0.08	1.77	0.46	0.69	N	320
Maleic anhydride	Methacrylate, methyl	0.02	0.004	5.2	0.47	N	33
Maleic anhydride	Methacrylate, methyl	0.01		3.1			524
Maleic anhydride	Methacrylate, methyl	0.01		3.85			524
Maleic anhydride	Methacrylate, methyl	0.01		3.4			524
Maleic anhydride	Methacrylate, methyl	0.02		3.85			524
Maleic anhydride	Methacrylate, di-, di-*n*-butylstannyl	0.024	0.008	0.012	0.016	Y	954
Maleic anhydride	Methacrylate, benzyl					Y	720
Maleic anhydride	Methacrylate, benzyl	0.372	0.093	−0.075	0.088	Y	720
Maleic anhydride	Phthalimide, *N*-vinyl-	0.008	0.012	0.344	0.061	Y	284
Maleic anhydride	Propene, 1-chloro-, *cis*-	0.347	0.082	−0.022	0.08	Y	181
Maleic anhydride	Propene, 1-chloro-, *trans*-	0.28	0.012	0.05	0.015	Y	181
Maleic anhydride	Styrene	0.001	0.008	0.097	0.002	N	105
Maleic anhydride	Styrene	0	0.13	0.04	0.12	Y	12
Maleic anhydride	Styrene	0.003	0.03	0.02	0.003	N	17
Maleic anhydride	Styrene	0.02	0.01	0	0.01	N	241
Maleic anhydride	Styrene	0		0.02			384
Maleic anhydride	Styrene	0		0.02			397
Maleic anhydride	Styrene	0.02		0.04			483
Maleic anhydride	Styrene	0.01		0.04			506
Maleic anhydride	Styrene	0.001		0.04			522
Maleic anhydride	Styrene	0.005		0.05			522
Maleic anhydride	Styrene	0.01		0.02			522
Maleic anhydride	Styrene	0.02		0.04			522
Maleic anhydride	Styrene	0.05		0.04			522
Maleic anhydride	Styrene					N	923
Maleic anhydride	Styrene, α-methyl-	0.08		0.04			492

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Maleic anhydride	Succinimide, *N*-vinyl-	0.021	0.027	0.148	0.073	Y	284
Maleic anhydride	Vinyl 2-chlorocyclohexyl ketone	0.83		1.7			836
Maleic anhydride	Vinyl 4-chlorocyclohexyl ketone	0.08		0.12			836
Maleic anhydride	Vinyl acetate	−0.058	0.074	0.019	0.008	Y	260
Maleic anhydride	Vinyl chloride	−0.22	0.38	0.098	0.042	Y	260
Maleic anhydride	Vinyl chloride	0.4		0.04			521
Maleic anhydride	Vinyl chloride	0.67		0.1			521
Maleic anhydride	Vinylidene cyanide					Y	82
Maleic anhydride, α-chloro-	Styrene	0		0.07			845
Maleimide	Butadiene, 2,3-bis(diethyl-phosphono)-	0.43		0.14			759
Maleimide	Methacrylate, methyl	0.166	0.047	2.576	0.061	Y	251
Maleimide	Methacrylate, methyl	0.17		2.5			992
Maleimide	Piperidinol, 4-, *N*-methyl-4-(vinylethynyl)-	0.49	0.01	2.05	0.03		948
Maleimide	Styrene	0.088	0.036	0.057	0.012	Y	251
Maleimide	Vinylidene chloride	0.464	0.083	0.666	0.056	Y	251
Maleimide, *N*-hydroxymethyl	Methacrylate, methyl	0.263	0.026	1.745	0.03	Y	323
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	Acrylate, ethyl	1.55	3.06	0.28	1.15	Y	619
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	Methacrylate, methyl	1.1	3.58	0.51	1.66	Y	619
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	Methacrylic acid	1.33	0.1	0.746	0.059	Y	619
Maleimide, *N*-(2,4-dimethylphenyl)-	Acrylate, butyl	0.2		0.62			788
Maleimide, *N*-(2,4-dimethylphenyl)-	Acrylate, heptyl	0.2		0.14			788
Maleimide, *N*-(2,6-diethylphenyl)-	Methacrylate, methyl	0.05		4.78			992
Maleimide, *N*-(2,6-diethylphenyl)-	Styrene	0.02		0.19			992
Maleimide, *N*-(2,6-diisopropylphenyl)-	Methacrylate, methyl	0.02		10.9			992
Maleimide, *N*(2,6-dimethylphenyl)-	Methacrylate, methyl	0.07		3.62			992
Maleimide, *N*-(2,6-dimethylphenyl)-	Styrene	0.08		0.14			992
Maleimide, *N*-(2-bromophenyl)-	Acrylonitrile	0.338	0.045	2.029	0.214	Y	863
Maleimide, *N*-(2-chlorophenyl)-	Acrylonitrile	1.078	0.048	0.956	0.095	Y	863
Maleimide, *N*-(2-chlorophenyl)-	Methacrylate, methyl	0.1		2.1			992
Maleimide, *N*-(2-chlorophenyl)-	Styrene	0.013	0.002	0.011	0.004	Y	853
Maleimide, *N*-(2-chlorophenyl)-	Styrene	0.028	0.003	0.038	0.006	Y	853
Maleimide, *N*-(2-chlorophenyl)-	Styrene	0.01	0.005	0.013	0.012	Y	910
Maleimide, *N*-(2-chlorophenyl)-	Styrene	0.006	0.003	0.038	0.28	Y	910
Maleimide, *N*-(2-hydroxyethyl)-	Butadiene, 2,3-bis(diethyl-phosphono)-	0.49		0.12			759
Maleimide, *N*-(2-hydroxyethyl)-	Methacrylate, methyl	0.201	0.083	1.396	0.089	Y	323
Maleimide, *N*-(2-hydroxyethyl)-	Pyrrolidone, *N*-vinyl-	0.038	0.01	0.002	0.002	Y	323
Maleimide, *N*-(2-hydroxyethyl)-	Styrene	0.011	0.027	0.046	0.017	Y	323
Maleimide, *N*-(2-hydroxyethyl)-	Vinyl acetate	0.61	0.12	−0.014	0.037	Y	323
Maleimide, *N*-(2-hydroxyethyl)-	Vinyl acetate	0.88	0.17	0.01	0.041	Y	323
Maleimide, *N*-(2-methylphenyl)-	Methacrylate, methyl	0.09		2.22			992
Maleimide, *N*-(3,5-dimethylphenyl)-	Methacrylate, methyl	0.28		1.63			992
Maleimide, *N*-(3-dimethylaminophenyl)-	Styrene	0.19		0.03			747
Maleimide, *N*-(3-methylphenyl)-	Methacrylate, methyl	0.22		1.18			992
Maleimide, *N*-(3-trifluoromethylphenyl)-	Methacrylate, methyl	0.1		1.57			992
Maleimide, *N*-(4-bromophenyl)-	Acrylate, methyl	0.39	0.17	0.429	0.099		628
Maleimide, *N*-(4-bromophenyl)-	Styrene	0.018	0.007	0.02	0.006	Y	1030
Maleimide, *N*-(4-carboxyethylphenyl)-	Vinyl chloride	3.29	0.18	0.04	0.03	N	306
Maleimide, *N*-(4-carboxyethylphenyl)-α-chloro-	Methacrylate, methyl	−0.004	0.168	0.472	0.114	Y	973
Maleimide, *N*-(4-carboxyethylphenyl)-α-chloro-	Styrene	0.022	0.041	0.059	0.021	Y	973
Maleimide, *N*-(4-chlorophenyl)-	Acrylate, methyl	0.155	0.052	0.684	0.052	Y	628
Maleimide, *N*-(4-chlorophenyl)-	Acrylonitrile	0.743	0.048	0.972	0.112	Y	863
Maleimide, *N*-(4-chlorophenyl)-	Butadiene, 2,3-bis(diethyl-phosphono)-	0.41		0.03			759
Maleimide, *N*-(4-chlorophenyl)-	Vinyl chloride	3.65	0.39	0.026	0.044	Y	306
Maleimide, *N*-(4-chlorophenyl)-α-chloro-	Methacrylate, methyl	0.02	0.03	0.562	0.025	Y	973
Maleimide, *N*-(4-chlorophenyl)-α-chloro-	Styrene	0.019	0.038	0.149	0.048	Y	973
Maleimide, *N*-(4-hydroxyphenyl)-	Methacrylate, methyl	0.345	0.001	1.392	0.001	Y	323
Maleimide, *N*-(4-hydroxyphenyl)-	Styrene	0.13	0.1	−0.059	0.041	Y	323
Maleimide, *N*-(4-hydroxyphenyl)-	Vinyl acetate	1.75	0.17	−0.011	0.027	Y	323
Maleimide, *N*-(4-hydroxyphenyl)-	Butadiene, 2,3-bis(diethyl-phosphono)-	0.47		0.04			75

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Maleimide, *N*-(4-methoxyphenyl)-α-chloro-	Methacrylate, methyl	0.064	0.189	0.556	0.145	Y	973
Maleimide, *N*-(4-methoxyphenyl)-α-chloro-	Styrene	0.049	0.061	0.112	0.031	Y	973
Maleimide, *N*-(4-methylphenyl)-	Methacrylate, methyl	0.34		0.83			992
Maleimide, *N*-(4-methylphenyl)-	Vinyl chloride	4.49	0.34	0.022	0.048	Y	306
Maleimide, *N*-(4-methylphenyl)-α-chloro-	Methacrylate, methyl	0.029	0.158	0.564	0.117	Y	973
Maleimide, *N*-(4-methylphneyl)-α-chloro-	Styrene	0.019	0.078	0.107	0.048	Y	973
Maleimide, *N*-(4-tolyl)-	Acrylate, methyl	0.14	0.16	0.64	0.16	Y	644
Maleimide, *N*-(β-chloroethyl)-α-chloro-	Methacrylate, methyl	0.05	0.04	0.94	0.06		974
Maleimide, *N*-(β-chloroethyl)-α-chloro-	Methacrylate, methyl	0.11	0.05	0.59	0.06		974
Maleimide, *N*-(β-chloroethyl)-α-chloro-	Styrene	0	0.06	0.29	0.05		974
Maleimide, *N*-(L-menthoxycarbonylmethyl)-	Maleimide, *N*-benzyl	0.848	0.221	0.474	0.087	Y	1089
Maleimide, *N*-(L-menthoxycarbonyl-methyl)-	Maleimide, *N*-phenyl-	0.79	0.261	0.798	0.141	Y	1089
Maleimide, *N*-2-bromophenyl-	Acrylonitrile	0.348	0.044	2.02	0.205	Y	915
Maleimide, *N*-2-chlorophenyl-	Acrylonitrile	1.08	0.048	0.955	0.095	Y	915
Maleimide, *N*-4-chlorophenyl-	Acrylonitrile	0.743	0.048	0.972	0.112	Y	915
Maleimide, *N*-[*N'*-α-methylbenzyl) aminocarbonylpentyl]-	Methacrylate, methyl	0.582	0.359	1.37	0.291	Y	1099
Maleimide, *N*-[*N'*-α-methylbenzyl) aminocarbonylpentyl]-	Styrene	0.258	0.25	0.198	0.143	Y	1099
Maleimide, *N*-[*N'*-α-methylbenzyl) aminocarbonylpentyl]-	Styrene					Y	1099
Maleimide, *N*-α-methylbenzyl-	Maleimide, *N*-cyclohexyl-	1.07	0.177	0.392	0.214	Y	1055
Maleimide, *N*-α-methylbenzyl-	Maleimide, *N*-phenyl-	0.683	0.177	0.788	0.5	Y	1055
Maleimide, *N*-allyl-	Styrene	0.035		0.056			1003
Maleimide, *N*-benzyl-	Maleimide, *N*-(L-menthoxycarbonyl-methyl)-	0.474	0.087	0.848	0.221	Y	1089
Maleimide, *N*-benzyl-	Methacrylate, methyl	0.03	0.23	1.28	0.23	Y	348
Maleimide, *N*-benzyl-	Styrene	0.013	0.031	0.058	0.021	Y	348
Maleimide, *N*-benzyl-α-chloro-	Styrene	0	0.06	0.42	0.06		974
Maleimide, *N*-butyl-	Methacrylate, methyl	−0.1	0.16	1.41	0.19	Y	115
Maleimide, *N*-butyl-	Styrene					Y	115
Maleimide, *N*-butyl-	Styrene	60		0.02			50
Maleimide, *N*-butyl-	Vinyl chloride	−12.7	106.2	0.24	1.13	Y	305
Maleimide, *N*-chloromethyl-	Butadiene, 2,3-bis(diethyl-phosphono)-	0.3		0.05			759
Maleimide, *N*-cyclohexyl-	Methacrylate, methyl	0.24		1.35			992
Maleimide, *N*-cyclohexyl-	Acenaphthalene	0.11		0.57			1039
Maleimide, *N*-cyclohexyl-	Butene, *cis*-2-	0		3			995
Maleimide, *N*-cyclohexyl-	Isobutene	0		0.41			995
Maleimide, *N*-cyclohexyl-	Itaconate, bis(2-ethylhexyl)	0.34		0.38			1060
Maleimide, *N*-cyclohexyl-	Itaconate, di-*n*-butyl	0.21	0.04	0.34	0.04		1014
Maleimide, *N*-cyclohexyl-	Itaconate, dimenthyl	0.601	0.193	0.318	0.213	Y	975
Maleimide, *N*-cyclohexyl-	Maleimide, *N*-α-methylbenzyl-	0.392	0.214	1.07	0.177	Y	1055
Maleimide, *N*-cyclohexyl-	Methacrylate, methyl	0.19		1.31			1100
Maleimide, *N*-cyclohexyl-	Methacrylate, methyl	0.106	0.048	2.26	0.066	Y	348
Maleimide, *N*-cyclohexyl-	Styrene	0.1		0.102			1100
Maleimide, *N*-cyclohexyl-	Styrene	0.162	0.054	−0.01	0.017	Y	348
Maleimide, *N*-cyclohexyl-	Isooctene	0		0.72			995
Maleimide, *N*-ethyl-α-chloro-	Methacrylate, methyl	0.13	0.03	0.53	0.04		974
Maleimide, *N*-ethyl-α-chloro-	Styrene	0	0.04	0.9	0.05		974
Maleimide, *N*-hexyl-	Methacrylate, methyl	0.1	0.15	1.9	0.2	Y	348
Maleimide, *N*-hexyl-	Styrene	0.09	0.16	−0.041	0.069	Y	348
Maleimide, *N*-hydroxymethyl-	Styrene	0.06	0.11	−0.035	0.053	Y	323
Maleimide, *N*-hydroxymethyl-	Vinyl acetate	1.45	0.21	0.017	0.037	Y	323
Maleimide, *N*-methyl-α-chloro-	Methacrylate, methyl	0.11	0.03	0.63	0.34		974
Maleimide, *N*-methyl-α-chloro-	Styrene	0	0.03	0.63	0.04		974
Maleimide, *N*-octadecyl-	Acrylonitrile	0.389	0.023	2.482	0.119	Y	863
Maleimide, *N*-octadecyl-	Acrylonitrile	0.39	0.023	2.46	0.115	Y	915
Maleimide, *N*-phenyl-	Acrylate, methyl	0.103	0.052	0.554	0.048	Y	644
Maleimide, *N*-phenyl-	Butadiene, 2,3-bis(diethyl-phosphono)-	0.42		0.05			759
Maleimide, *N*-phenyl-	Maleic anhydride	1.92	11.83	0.02	0.65	Y	341
Maleimide, *N*-phenyl-	Maleic anhydride	1.08	0.13	−0.095	0.106	Y	990
Maleimide, *N*-phenyl-	Maleimide, *N*-(L-menthoxycarbonyl-methyl)-	0.798	0.141	0.79	0.261	Y	1089
Maleimide, *N*-phenyl-	Maleimide, *N*-α-methylbenzyl-	0.788	0.5	0.683	0.177	Y	1055

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Maleimide, *N*-phenyl-	Methacrylate, methyl	0.3		0.98			323
Maleimide, *N*-phenyl-	Methacrylate, methyl	0.164	0.065	1.014	0.06	Y	328
Maleimide, *N*-phenyl-	Methacrylate, methyl	0.3		0.91			992
Maleimide, *N*-phenyl-	Styrene					N	1026
Maleimide, *N*-phenyl-	Styrene	0.044	0.017	0.016	0.004	Y	328
Maleimide, *N*-phenyl-	Styrene	0.01		0.07			992
Maleimide, *N*-phenyl-	Vinyl acetate	1.284	0.046	−0.003	0.006	Y	328
Maleimide, *N*-phenyl-	Vinyl chloride	4.01	0.18	0.023	0.026	Y	306
Maleimide, *N*-phenyl-α-chloro-	Methacrylate, methyl	0.077	0.098	0.657	0.083	Y	973
Maleimide, *N*-phenyl-α-chloro-	Styrene	−0.002	0.073	0.087	0.04	Y	973
Maleimide, *N*-propyl-	Vinyl chloride	2.38	0.16	0.06	0.05	N	305
Maleimido-*N*,*N*-dimethylaniline, 4-(2-chlorophenyl)azo	Styrene	0.04		0.05			747
Mesaconate, dibenzyl	Styrene	0.08		0.73			658
Mesaconate, di-*sec*-butyldithiol	Styrene	0.02		0.45			624
Mesaconate, diamyl	Styrene	0.02		1.27			658
Mesaconate, dibenzyldithiol	Styrene	0.06		0.28			624
Mesaconate, dibutyl	Styrene	0.05		1.28			658
Mesaconate, dibutyldithiol	Styrene	0.05		0.41			624
Mesaconate, dicyclohexyl	Styrene	0.1		1.25			658
Mesaconate, diethyl	Styrene	0.06		1.24			658
Mesaconate, diethyldithiol	Styrene	0.03		0.4			624
Mesaconate, dihexyl	Styrene	0.1		1.51			658
Mesaconate, diisoamyl	Styrene	0.07		1.34			658
Mesaconate, diisobutyl	Styrene	0.1		1.29			658
Mesaconate, diisobutyldithiol	Styrene	0.03		0.41			624
Mesaconate, diisopropyl	Styrene	0.04		1.31			658
Mesaconate, diisopropyldithiol	Styrene	0.05		0.44			624
Mesaconate, dimethyl	Styrene	0.04		0.98			658
Mesaconate, dimethyldithiol	Styrene	0.07		0.35			624
Mesaconate, dioctyl	Styrene	0.01		1.23			658
Mesaconate, diphenyl	Styrene	0.07		0.45			658
Mesaconate, diphenyldithiol	Styrene	0.15		0.16			624
Mesaconate, dipropyl	Styrene	0.05		1.22			658
Mesaconate, dipropyldithiol	Styrene	0.06		0.39			624
Mesaconic acid	Styrene	0		1.6			1021
Methacryl-2′-ethyl-4,5,7-trinitro-9-fluorenone-2-carboxylate	Methacrylate, *N*-(2-hydroxyethyl)carbazolyl	0.382	0.014	0.63	0.041	Y	703
Methacrylamide	Methacrylate, methyl	1.27	0.24	1.57	0.11	N	316
Methacrylamide	Methacrylate, methyl	0.425	0.085	1.39	0.074	Y	54
Methacrylamide	Methacrylate, methyl	0.49		1.65			55
Methacrylamide	Methacrylic acid	0.33	0.25	2.73	0.28	Y	190
Methacrylamide	Methacrylic acid	1.28		1.55			527
Methacrylamide	Methacrylic acid	0.2	0.014	2.006	0.09	Y	55
Methacrylamide	Styrene	0.48	0.2	1.4	0.13	N	316
Methacrylamide	Styrene	1.24	0.38	1.42	0.17	N	316
Methacrylamide	Vinyl methyl ketone	3	0.18	0.25	0.12	Y	609
Methacrylamide	Vinyl methyl ketone	1.72	0.26	0.28	0.24	Y	609
Methacrylamide, 1-deoxy-D-glucitol	Vinyl acetate	0.56		0.16			258
Methacrylamide, *N*-(2-hydroxypropyl)-	Acrylamide, *N*-(2-(4-hydroxyphenyl)-ethyl)-	1.01	0.09	0.95	0.09		965
Methacrylamide, *N*-(4′-nitro-4-stilbenyl)-	Methacrylate, 2-hydroxyethyl	0.593	0.047	1.474	0.05	Y	342
Methacrylamide, *N*-(4-stilbenyl)-	Methacrylate, 2-hydroxyethyl	0.7	0.23	1.842	0.062	Y	342
Methacrylamide, *N*-(*p*-chlorophenyl)-	Acrylonitrile	0.627	0.051	0.399	0.019	Y	352
Methacrylamide, *N*-(*p*-methoxyphenyl)-	Acrylonitrile	0.559	0.048	0.337	0.019	Y	352
Methacrylamide, *N*-(*p*-methylphenyl)-	Acrylonitrile	0.549	0.06	0.336	0.026	Y	352
Methacrylamide, *N*-(*p*-nitrophenyl)-	Acrylonitrile	2.34	1.07	0.37	0.14	Y	352
Methacrylamide, *N*-[4-(4-methoxyphenylacetyloxy)phenyl]-	Methacrylate, 2-hydroxyethyl	0.29		1.47			1051
Methacrylamide, *N*-methoxymethyl-	Methacrylate, methyl	0.265	0.009	1.678	0.048	Y	227
Methacrylamide, *N*-*p*-bromophenyl-	Methacrylate, methyl	0.38	0.2	3.48	0.2		968
Methacrylamide, *N*-*p*-tolyl-	Methacrylate, methyl	0.78	0.2	1.43	0.2		968
Methacrylamide, *N*-phenyl	Methacrylate, methyl	0.28	0.2	1.95	0.2		968
Methacrylamide, *N*-phenyl-	Acrylonitrile	0.71	0.19	0.381	0.085	Y	352
Methacrylamide, *N*-phenyl-	Methacrylate, methyl	0.56	0.12	1.42	0.096	Y	352

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylamide, *N*-phenyl-	Styrene	0.88	0.17	1.24	0.06	N	352
Methacrylamide, 1-deoxy-D-glucitol	Methacrylate, methyl	0.04		4.2			258
Methacrylamide, 1-deoxy-D-glucitol	Styrene	0.005		2.09			258
Methacrylamide, *N*-*p*-nitrophenyl-	Methacrylate, methyl	0.29	0.2	5.05	0.2		968
Methacrylate, 2-ethylhexyl	Methacrylate, 2-hydroxypropyl	0.083	0.006	4.56	0.18	Y	708
Methacrylate, ferrocenylmethyl	Maleic anhydride	0.33	0.65	0.24	0.28	Y	325
Methacrylate, ferrocenylmethyl	Pyrrolidone, *N*-vinyl-	2.57	2.77	0.073	0.096	Y	325
Methacrylate, octyl	Methacrylonitrile	0.61	0.05	0.75	0.04	N	44
Methacrylate, potassium 18-crown-6-ether	Acrylonitrile	0.175	0.018	0.07	0.006	Y	808
Methacrylate, tert-butyl	Methacrylate, 2-chloroethyl	0.98	0.22	1.18	0.11	Y	179
Methacrylate, (−)-menthyl	Acenaphthalene	0.316	0.085	2.48	0.55	Y	645
Methacrylate, (2,2-dimethyl-1,3-dioxolane-4-yl)methyl	Methacrylate, methyl	1.1		0.9			807
Methacrylate, 1,1-diphenylethyl	Methacrylate, methyl	0.77		0.72			761
Methacrylate, 1-naphthyl	Methacrylate, methyl	1.8	0.73	0.77	0.14	Y	752
Methacrylate, 1-naphthyl	Methacrylate, methyl	1.68	0.38	0.809	0.078	Y	752
Methacrylate, 1-naphthyl	Methacrylate, methyl	2.62	0.47	0.851	0.084	Y	752
Methacrylate, 1-naphthyl	Methacrylate, methyl	1.52		0.61			761
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	Acrylonitrile	14		0.02			664
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	Styrene	0.3		0.63			664
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	Vinyl acetate	13.7		0.2			664
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	Vinyl methyl ketone	0.53		0.41			664
Methacrylate, 2,2-dimethylaminoethyl	Methacrylate, methyl	1.01	0.54	0.42	0.02	N	691
Methacrylate, 2,3-epithiopropyl	Acrylate, methyl	2.81		0.34			787
Methacrylate, 2,3-epithiopropyl	Methacrylate glycidyl	0.78		0.81			787
Methacrylate, 2,3-epithiopropyl	Methacrylate, methyl	0.97		0.83			787
Methacrylate, 2,3-epithiopropyl	Styrene	0.28		0.41			787
Methacrylate, 2,4,6-trinitrophenyl	Methacrylate, 2-naphthyl					Y	744
Methacrylate, 2,4-dinitrophenyl ethyl	Acrylate, *N*-(2-hydroxyethyl)-oxy) 3,6-dichloro	1.207	0.032	0.115	0.029	Y	736
Methacrylate, 2,4-dinitrophenyl	Methyacrylate, *N*-(2-hydroxyethyl)-3,6-dichloro	1.112	0.026	0.126	0.026	Y	
Methacrylate, 2,6-(diphenyl)phenyl	Methacrylate, methyl	0.071	0.004	1.42	0.06		1079
Methacrylate, 2-(2-carboxybenzoyl-oxy)ethyl	Methacrylate, butyl	0.85		0.75			1009
Methacrylate, 2-(2-carboxybenzoyl-oxy)ethyl	Methacrylate, butyl	0.85		0.75			1041
Methacrylate, 2-(*N*,*N*-dimethyl-carbamoyloxy)ethyl	Acrylate, butyl	1.89		0.47			600
Methacrylate, 2-(*N*,*N*-dimethyl-carbamoyloxy)ethyl	Acrylate, methyl	1.32		0.22			600
Methacrylate, 2-(*N*,*N*-dimethyl-carbamoyloxy)ethyl	Acrylonitrile	1.89		0.38			600
Methacrylate, 2-(*N*,*N*-dimethyl carbamoyloxy)ethyl	Butadiene	0.35		1.06			600
Methacrylate, 2-(*N*,*N*-dimethyl-carbamoyloxy)ethyl	Methacrylate, methyl	0.42		0.59			600
Methacrylate, 2-acetoxyethyl	Methacrylate, 2-hydroxyethyl	0.99	0.02	1.02	0.01	N	346
Methacrylate, 2-acetoxyethyl	Methacrylate, methyl	1.12	0.06	0.94	0.02	N	346
Methacrylate, 2-acetoxyethyl	Styrene	0.58	0.04	0.44	0.02	N	346
Methacrylate, 2-acetylsalicylicoyloxy-3-hydroxypropyl	*N*-Vinylpyrrolidone	0.85		0.34			1080
Methacrylate, 2-bromoethyl	Acrylate, ethyl	2.7		0.37			657
Methacrylate, 2-bromoethyl	Acrylonitrile	2.38		0.31			657
Methacrylate, 2-bromoethyl	Methacrylate, methyl	1.18		0.79			657
Methacrylate, 2-bromoethyl	Pyrrolidone, *N*-vinyl-	2.44		0.02			657
Methacrylate, 2-bromoethyl	Styrene	0.41		0.32			657
Methacrylate, 2-bromoethyl	Vinyl butyl ether	13.7		0			657
Methacrylate, 2-chloro-2,3,3,3-fluoro-propyl	Acrylate, butyl	0.221	0.636	0.079	0.304	N	927

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, 2-chloro-2,3,3,3-fluoro-propyl	Methacrylate, methyl	0.583	1.17	0.984	0.395	N	927
Methacrylate, 2-chloro-2,3,3,3-fluoro-propyl	Styrene	0.243	0.15	0.121	0.068	N	927
Methacrylate, 2-chloroethyl	Acrylate, methyl	2.19		0.29			726
Methacrylate, 2-chloroethyl	Acrylonitrile	1.3		0.14			382
Methacrylate, 2-chloroethyl	Methacrylate, benzyl	0.989	0.098	1.068	0.042	Y	179
Methacrylate, 2-chloroethyl	Methacrylate, butyl	0.82	0.041	1.086	0.02	Y	179
Methacrylate, 2-chloroethyl	Methacrylate, isobutyl	0.85	0.14	1.014	0.064	Y	179
Methacrylate, 2-chloroethyl	Methacrylate, methyl	0.97	0.18	1.156	0.082	Y	179
Methacrylate, 2-chloroethyl	Methacrylate, methyl	0.48		0.55			382
Methacrylate, 2-chloroethyl	Methacrylate, phenyl	1.02	0.21	0.785	0.083	Y	179
Methacrylate, 2-chloroethyl	Methacrylate, *tert*-butyl	1.18	0.11	0.98	0.22	Y	179
Methacrylate, 2-chloroethyl	Styrene	0.341	0.075	0.35	0.046	Y	179
Methacrylate, 2-chloroethyl	Styrene	0.34	0.08	0.345	0.048	Y	179
Methacrylate, 2-chloroethyl	Styrene	0.23		0.42			382
Methacrylate, 2-chloroethyl	Methacrylate, tributyltin	0.53	0.16	0.2	0.444	N	945
Methacrylate, 2-ethylhexyl	Methacrylate, 2-hydroxypropyl	0.083	0.006	4.56	0.18	Y	692
Methacrylate, 2-hydroxyethyl	Acetamide, *N*-(4-methacryloyloxy-phenyl)-2-(4-methoxy)	0.61		1.49			1051
Methacrylate, 2-hydroxyethyl	Acetanilide, 4-(2-(methacryloyloxy) ethyloxy)	1.39		0.67			1096
Methacrylate, 2-hydroxyethyl	Acetanilide, 4-(2-(methacryloyloxy) hexyloxy)	1.19		0.87			1096
Methacrylate, 2-hydroxyethyl	Acetanilide, 4-(methacryloyloxy)-	0.78		1.94			1096
Methacrylate, 2-hydroxyethyl	Acrylamide	1.89		0.05			689
Methacrylate, 2-hydroxyethyl	Acrylamide	0.98		0.14			928
Methacrylate, 2-hydroxyethyl	Acrylamido, 2-,2-methylpropanesulfonic	0.86		0.9			618
Methacrylate, 2-hydroxyethyl	Acrylate, butyl	4.75	4.2	0.09	0.11	Y	358
Methacrylate, 2-hydroxyethyl	Acrylate, ethyl	11.21	0.43	0.189	0.008	Y	358
Methacrylate, 2-hydroxyethyl	Acrylate, furfuryl	1.35	0.08	0.88	0.19		1063
Methacrylate, 2-hydroxyethyl	Acrylate, methyl	8.67	8.28	−0.005	0.036	Y	358
Methacrylate, 2-hydroxyethyl	Acrylonitrile	1		0.2			689
Methacrylate, 2-hydroxyethyl	Acryloxymethylpentamethyl disiloxane	0.86		0.52			822
Methacrylate, 2-hydroxyethyl	Methacrylate, 2-hydroxypropyl	0.55		1.82			686
Methacrylate, 2-hydroxyethyl	Methacrylamide, *N*-(4′-nitro-4-stilbenyl)-	1.474	0.05	0.593	0.047	Y	342
Methacrylate, 2-hydroxyethyl	Methacrylamide, *N*-(4-stilbenyl)-	1.842	0.062	0.7	0.23	Y	342
Methacrylate, 2-hydroxyethyl	Methacrylamide, *N*-[4-(4-methoxy-phenylacetyloxy)phenyl]-	1.47		0.29			1051
Methacrylate, 2-hydroxyethyl	Methacrylate, 2-acetoxyethyl	1.02	0.01	0.99	0.02	N	346
Methacrylate, 2-hydroxyethyl	Methacrylate, dodecyl	11.2		0.7			837
Methacrylate, 2-hydroxyethyl	Methacrylate, dodecyl	1.6		0.5			837
Methacrylate, 2-hydroxyethyl	Methacrylate, dodecyl	0.8		0.8			837
Methacrylate, 2-hydroxyethyl	Methacrylate, methyl	0.63	0.076	0.824	0.01	Y	350
Methacrylate, 2-hydroxyethyl	Methacrylate, methyl	0.81	1.02	0.192	0.09	Y	358
Methacrylate, 2-hydroxyethyl	Methacrylate, methyl	1.5		0.75			710
Methacrylate, 2-hydroxyethyl	Methacryloxymethylpentamethyl disiloxane	0.97		0.33			822
Methacrylate, 2-hydroxyethyl	Pyridine, 2-vinyl-	0.56	0.04	0.64	0.16	N	342
Methacrylate, 2-hydroxyethyl	Pyridine, 3-vinyl-	0.69	0.01	0.73	0.04	N	342
Methacrylate, 2-hydroxyethyl	Pyridine, 4-vinyl-	0.62	0.04	0.9	0.09	N	342
Methacrylate, 2-hydroxyethyl	Pyrrolidone, *N*-vinyl-	4.841	0.133	−0.019	0.046	Y	840
Methacrylate, 2-hydroxyethyl	Styrene	0.54	0.03	0.44	0.02	N	346
Methacrylate, 2-hydroxyethyl	Styrene	0.856	0.018	0.332	0.006	Y	350
Methacrylate, 2-hydroxyethyl	Styrene	0.65		0.57			463
Methacrylate, 2-hydroxyethyl	Styrene	1.65		0.5			878
Methacrylate, 2-hydroxyethyl	Styrene	0.53		0.59			878
Methacrylate, 2-hydroxyethyl	Styrene	0.59		0.53			903
Methacrylate, 2-hydroxyethyl	Styrene	1.65		0.5			903
Methacrylate, 2-hydroxypropyl	Acrylamido, 2-, 2-methylpropane-sulfonic	0.89		1.03			618
Methacrylate, 2-hydroxypropyl	Acrylate, butyl	5.35	0.32	0.171	0.041	Y	333
Methacrylate, 2-hydroxypropyl	Acrylate, butyl	3.3	6.82	0.24	0.16	Y	662
Methacrylate, 2-hydroxypropyl	Acrylate, ethyl	13.32	1.05	0.273	0.054	Y	333
Methacrylate, 2-hydroxypropyl	Acrylate, ethyl	9.08	3.95	0.207	0.045	Y	662

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, 2-hydroxypropyl	Acrylate, methyl	7.335	0.083	0.013	0.007	Y	333
Methacrylate, 2-hydroxypropyl	Acryloyl chloride	1.79		0.29			649
Methacrylate, 2-hydroxypropyl	Methacrylate, 2-ethylhexyl	4.56	0.18	0.083	0.006	Y	692
Methacrylate, 2-hydroxypropyl	Methacrylate, 2-ethylhexyl	4.56	0.18	0.083	0.006	Y	708
Methacrylate, 2-hydroxypropyl	Methacrylate, 2-hydroxyethyl	1.82		0.55			686
Methacrylate, 2-hydroxypropyl	Methacrylate, butyl	2.35	0.13	0.158	0.007	Y	692
Methacrylate, 2-hydroxypropyl	Methacrylate, butyl	2.35	0.13	0.158	0.007	Y	708
Methacrylate, 2-hydroxypropyl	Methacrylate, ethyl	1.878	0.056	0.245	0.007	Y	692
Methacrylate, 2-hydroxypropyl	Methacrylate, ethyl	1.844	0.039	0.267	0.005	Y	708
Methacrylate, 2-hydroxypropyl	Methacrylate, methyl	1.055	0.04	0.402	0.017	Y	333
Methacrylate, 2-hydroxypropyl	Methacrylate, tributyltin	1.81	0.275	0.728	0.517	N	945
Methacrylate, 2-hydroxypropyl	Methacrylic acid	0.31		0.99			886
Methacrylate, 2-naphthyl	Acrylonitrile	1.17	0.2	0.15	0.2	Y	611
Methacrylate, 2-naphthyl	Acrylonitrile	1.02	0.19	0.094	0.099	Y	611
Methacrylate, 2-naphthyl	Acrylonitrile	0.76	0.18	0.06	0.1	Y	611
Methacrylate, 2-naphthyl	Acrylonitrile	0.64	0.24	−0.01	0.16	N	611
Methacrylate, 2-naphthyl	Itaconate, mono-methyl	0.52	0.16	0.31	0.42	Y	613
Methacrylate, 2-naphthyl	Itaconate, mono-methyl	0.59	0.14	0.36	0.21	Y	613
Methacrylate, 2-naphthyl	Itaconate, mono-methyl	0.223	0.02	0.135	0.028	Y	613
Methacrylate, 2-naphthyl	Itaconate, mono-methyl	0.189	0.055	0.04	0.11	Y	613
Methacrylate, 2-naphthyl	Methacrylate, 2,4,6-trinitrophenyl					Y	744
Methacrylate, 2-naphthyl	Methacrylate, methyl	0.85	0.34	0.75	0.16	Y	540
Methacrylate, 2-naphthyl	Methacrylate, methyl	2.57	0.48	0.602	0.059	Y	760
Methacrylate, 2-naphthyl	Methacrylate, methyl	2.53	0.27	0.581	0.023	Y	760
Methacrylate, 2-naphthyl	Methacrylate, methyl	1.91	0.6	0.532	0.055	Y	760
Methacrylate, 2-naphthyl	Methacrylate, methyl	4.41	0.76	0.614	0.051	Y	760
Methacrylate, 2-naphthyl	Methacrylate, methyl	3.83	0.57	0.623	0.048	Y	760
Methacrylate, 2-naphthyl	Styrene	0.534	0.099	0.56	0.025	Y	540
Methacrylate, 2-naphthyl-, methyl	Styrene	0.113	0.126	0.786	0.16	Y	924
Methacrylate, 2-naphthyl-, methyl	Styrene	0.321	0.098	0.486	0.117	Y	924
Methacrylate, 2-naphthyl-, methyl	Styrene	0.576	0.477	0.61	0.195	Y	924
Methacrylate, 2-naphthyl-, methyl	Styrene	0.216	0.422	0.384	0.225	Y	924
Methacrylate, 2-sulfonic acid)ethyl	Acrylate, ethyl	3.2		0.3			564
Methacrylate, 2-sulfonic acid)ethyl	Methacrylate, ethyl	2		1			564
Methacrylate, 2-sulfonic acid)ethyl	Styrene	0.6		0.37			564
Methacrylate, 2-sulfonic acid)ethyl	Vinylidene chloride	3.6		0.22			564
Methacrylate, 2-methoxyethyl	Styrene	0.48		0.46			918
Methacrylate, 3,5-dimethyladamantyl	Acrylonitrile	1.3	0.23	0.19	0.01	N	335
Methacrylate, 3,5-dimethyladamantyl	Methacrylate, methyl	0.45	0.16	0.68	0.04	N	335
Methacrylate, 3,5-dimethyladamantyl	Styrene	0.63	0.21	0.89	0.14	N	335
Methacrylate, 3,5-dimethyladamantyl	Vinyl chloride	1.85	8.87	0.21	0.04	N	335
Methacrylate, 3-methoxy-2-hydroxypropyl	Acrylamide	3.98		0.04			689
Methacrylate, 5-oxo-pyrrolidinylmethyl	Methacrylate, methyl	0.694	0.087	0.421	0.033	Y	633
Methacrylate, 5-oxo-pyrrolidinylmethyl	Styrene	0.411	0.033	0.342	0.013	Y	633
Methacrylate, β-(2-benzothiazolylthio) ethyl	Vinyl chloride	0.26		0.64			741
Methacrylate, *N*-(2-hydroxyethyl)-3,6-dichlorocarbazolyl	Methacrylate, 2,4-dinitrophenyl	0.126	0.026	1.112	0.026	Y	736
Methacrylate, *N*-(2-hydroxyethyl)-carbazolyl	Acryl-2′-ethyl-4,5,7-trinitroflourenone	0.773	0.096	0.102	0.022	Y	703
Methacrylate, *N*-(2-hydroxyethyl)-carbazolyl	Acrylate, 2,4-dinitrophenyl	0.243	0.034	0.186	0.009	Y	716
Methacrylate, *N*-(2-hydroxyethyl)-carbazolyl	Methacryl-2′-ethyl-4,5,7-trinitro-9-fluorenone	0.63	0.041	0.382	0.014	Y	703
Methacrylate, *N*-α-benzothiazolonylmethyl	Methacrylate, methyl	0.6	0.02	0.5	0.01		887
Methacrylate, *N*-ethyl-3-hydroxymethylcarbazolyl	Acryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	1.425	0.09	0.103	0.015	Y	701
Methacrylate, *N*-ethyl-3-hydroxymethylcarbazolyl	Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	0.729	0.051	0.554	0.019	Y	701
Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	Acryloyloxy, 2-, benzoic acid	0.22	0.03	1.82	0.2	N	651
Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	Methacrylic acid	0.2	0.01	2.16	0.18	N	651
Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	Methacryloyloxy, 2-,benzoate, methyl	0.55	0.04	1.11	0.18	N	651

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	Methacryloyloxy, 2-,benzoic acid	0.2	0.01	2.6	0.08	N	651
Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	Styrene	0.58	0.02	0.44	0.04	N	651
Methacrylate, α,α-dimethylbenzyl	Methacrylate, methyl	0.92		0.81			761
Methacrylate, α-naphthoyloxy-, ethyl	Styrene	0.242	0.018	0.216	0.043		1102
Methacrylate, acetonyl	Styrene	0.505	0.115	0.285	0.041	Y	958
Methacrylate, allyl	Itaconate, bis(tri-*n*-butyltin)	1.08		0.038			940
Methacrylate, amyl	Styrene	0.46	0.2	0.52	0.11	Y	179
Methacrylate, benzyl	Acrylonitrile	0.96	0.1	0.2	0.1	N	23
Methacrylate, benzyl	Allyl chloride	58.7		0.02			438
Methacrylate, benzyl	Maleic anhydride					Y	720
Methacrylate, benzyl	Maleic anhydride	−0.075	0.088	0.372	0.093	Y	720
Methacrylate, benzyl	Methacrylate, 2-chloroethyl	1.068	0.042	0.989	0.098	Y	179
Methacrylate, benzyl	Methacrylate, methyl	1.112	0.093	0.808	0.03	Y	111
Methacrylate, benzyl	Methacrylate, methyl	1.05		0.93			32
Methacrylate, benzyl	Methacrylate, methyl	1.38		0.78			540
Methacrylate, benzyl	Methacrylate, phenyl	0.67	0.06	1.42	0.09	N	32
Methacrylate, benzyl	Styrene	0.658	0.084	0.463	0.041	Y	111
Methacrylate, benzyl	Styrene	0.467	0.048	0.435	0.026	Y	179
Methacrylate, benzyl	Styrene	0.51		0.45			487
Methacrylate, benzyl	Styrene	0.42		0.48			540
Methacrylate, benzyl	Styrene	0.3		0.8			602
Methacrylate, benzyl	Styrene	0.5	0.38	0.45	0.32		806
Methacrylate, benzyl	Vinylidene chloride	3.3		0.34			432
Methacrylate, benzylmethylaminoethyl	Methacrylate, methyl	0.296	0.123	0.523	0.134	Y	1081
Methacrylate, butyl	Acetylene, phenyl-	1.7		0.21			565
Methacrylate, butyl	Acrylate, butyl	2.2		0.3			187
Methacrylate, butyl	Acrylate, ethyl	2.43	0.39	0.22	0.17	Y	690
Methacrylate, butyl	Acrylic acid	3.53	0.42	0.24	0.21	N	187
Methacrylate, butyl	Acrylic acid, α-bromo-	0.85		0.13			733
Methacrylate, butyl	Acrylonitrile	0.98	0.2	0.291	0.039	Y	165
Methacrylate, butyl	Carbamate, *N*-phenyl-, allyl	0.418		0.329			833
Methacrylate, butyl	Methacrylate, 2-(2-carboxy-benzoyloxy)ethyl	0.75		0.85			1009
Methacrylate, butyl	Methacrylate, 2-(2-carboxy-benzoyloxy)ethyl	0.75		0.85			1041
Methacrylate, butyl	Methacrylate, 2-chloroethyl	1.086	0.02	0.82	0.041	Y	179
Methacrylate, butyl	Methacrylate, 2-hydroxypropyl	0.158	0.007	2.35	0.13	Y	692
Methacrylate, butyl	Methacrylate, 2-hydroxypropyl	0.158	0.007	2.35	0.13	Y	708
Methacrylate, butyl	Methacrylate, ethyleneglycol-*N*-phenylcarbamate	0.395		0.299			833
Methacrylate, butyl	Methacrylate, glycidyl	0.85	0.03	0.94	0.06	N	224
Methacrylate, butyl	Methacrylate, methyl	2.11	0.08	0.52	0.07	N	30
Methacrylate, butyl	Methacrylate, methyl	1.2		1.27			844
Methacrylate, butyl	Methacrylic acid	1.26	0.12	0.8	0.16	N	187
Methacrylate, butyl	Methacrylic acid	1.15	0.12	0.73	0.2	N	224
Methacrylate, butyl	Methacrylonitrile	0.75	0.07	0.56	0.15	N	44
Methacrylate, butyl	Styrene	0.31	0.2	0.56	0.12	Y	179
Methacrylate, butyl	Styrene	2.52	1.1	−0.05	1.09	N	43
Methacrylate, butyl	Styrene	0.47		0.52			463
Methacrylate, butyl	Styrene	0.64		0.63			530
Methacrylate, butyl	Styrene	0.64		0.54			530
Methacrylate, butyl	Styrene	0.59		0.74			835
Methacrylate, butyl	Styrene, *o*-chloro-	0.32		1.24			558
Methacrylate, butyl	Styrene, *p*-chloro-		0.1		0.1	N	872
Methacrylate, butyl	Styrene *p*-chloro-	0.564	0.131	1.025	0.124		966
Methacrylate, butyl	Styrene chloromethyl-	0.38		1.46			1035
Methacrylate, butyl	Styrene *p*-octylamine sulfonate	0.7		1.5			817
Methacrylate, butyl	Thioindigoid	11.5		0.5			1007
Methacrylate, butyl	Vinyl acetate	30.18	14.47	−0.003	0.15	Y	165
Methacrylate, butyl	Vinyl chloride	13.5		0.05			519
Methacrylate, butyl	Vinylidene chloride	2.07	0.18	0.35	0.03	Y	3
Methacrylate, chloromethyl	Methacrylate, methyl	1.02	0.16	0.477	0.029	Y	712
Methacrylate, chloromethyl	Styrene	0.298	0.03	0.208	0.006	Y	712

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, cyanomethyl	Methacrylate, methyl	0.73	0.11	0.399	0.019	Y	625
Methacrylate, cyanomethyl	Styrene	0.28	0.041	0.144	0.011	Y	625
Methacrylate, cyclohexyl	Methacrylate, tributyltin	0.891	0.136	0.395	0.428	N	945
Methacrylate, cyclohexyl	Styrene	0.57	0.28	0.586	0.088	Y	179
Methacrylate, decyl	Oxazolidone, *N*-vinyl-	12.19	1.35	−0.052	0.049	Y	36
Methacrylate, di-, di-*n*-butylstannyl	Maleic anhydride	0.012	0.016	0.024	0.008	Y	954
Methacrylate, di-, di-*n*-butylstannyl	Vinyl acetate	0.013	0.036	0.013	0.018	Y	949
Methacrylate, di-, zinc	Ethylene	40		0.19			935
Methacrylate, di-, dicyclopentadienyl-titanate	Acrylate, di-, copper	0.89		1.09			876
Methacrylate, di-, dicyclopentadienyl-titanate	Acrylate, di-, nickel	0.65		0.95			876
Methacrylate, di-, ethylene glycol	Styrene	0.67	0.08	0.39	0.15	N	262
Methacrylate, di-, triethyleneglycol	Carbamate, *N*-phenyl-, allyl	0.504		0.034			833
Methacrylate, di-, triethyleneglycol	Methacrylate, ethylene-glycol *N*-phenylcarbamate	0.59		0.24			833
Methacrylate, diethylaminoethyl	Methacrylate, methyl	1.44	0.085	0.843	0.128	Y	1081
Methacrylate, diethylaminoethyl	Methacrylic acid	2.34	1.59	0.63	1.47	Y	16
Methacrylate, dimethylaminoethyl	Methacrylate, methyl	1.2	0.154	0.699	0.051	Y	1081
Methacrylate, dimethylaminoethyl sulfate	Acrylamide	1.9	0.2	0.52	0.05		1052
Methacrylate, diphenylmethyl	Methacrylate, 2-hydroxyethyl	0.5		1.6			837
Methacrylate, diphenylmethyl	Methacrylate, 2-hydroxyethyl	0.8		0.8			837
Methacrylate, diphenylmethyl	Methacrylate, dodecyl	0.7		11.2			837
Methacrylate, diphenylmethyl	Methacrylate, methyl	1.07		0.81			761
Methacrylate, diphenylmethyl	Styrene	0.3	0.1	0.528	0.05	Y	179
Methacrylate, ethyl	Allyl acetate	81.27	1.93	−0.57	0.82	N	210
Methacrylate, ethyl	Allyl alcohol	107.4	5.3	−0.02	1.92	N	210
Methacrylate, ethyl	Methacrylate, 2-(sulfonic acid)ethyl	1		2			564
Methacrylate, ethyl	Methacrylate, 2-hydroxypropyl	0.245	0.007	1.878	0.056	Y	692
Methacrylate, ethyl	Methacrylate, 2-hydroxypropyl	0.267	0.005	1.844	0.039	Y	708
Methacrylate, ethyl	Methacrylate, methyl					N	32
Methacrylate, ethyl	Methacrylate, methyl	1.08		1.08			844
Methacrylate, ethyl	Methacrylate, methyl	1	0.09	1.16	0.19	N	87
Methacrylate, ethyl	Methacrylate, tributyltin	0.601	0.123	0.299	0.607	N	945
Methacrylate, ethyl	Methacrylic acid	0.71		0.57			820
Methacrylate, ethyl	Methacrylonitrile	0.83	0.05	0.46	0.04	N	44
Methacrylate, ethyl	Styrene	0.36	0.12	0.55	0.062	Y	179
Methacrylate, ethyl	Styrene	0.26		0.67			530
Methacrylate, ethyl	Styrene	0.29		0.65			530
Methacrylate, ethyl	Styrene	0.33		0.55			570
Methacrylate, ethyl	Styrene, *o*-chloro-	0.45		1.34			558
Methacrylate, ethyl	Styrene, *p*-chloromethyl-	0.36		1.29			1035
Methacrylate, ethyl	Vinyl acetate	131.8	29.8	−0.77	5.29	N	210
Methacrylate, ethyl	Vinylidene chloride	2.18	0.1	0.353	0.02	Y	3
Methacrylate, ethyleneglycol-*N*-phenylcarbamate	Methacrylate, butyl	0.299		0.395			833
Methacrylate, ethyleneglycol-*N*-phenylcarbamate	Methacrylate, di-, triethyleneglycol	0.24		0.59			833
Methacrylate, ferrocenylmethyl	Acrylate, methyl	0.042	0.085	1.4	0.35	Y	293
Methacrylate, ferrocenylmethyl	Acrylonitrile	0.82	0.42	0.15	0.04	Y	325
Methacrylate, ferrocenylmethyl	Styrene	−0.01	0.14		3.63	Y	293
Methacrylate, ferrocenylmethyl	Styrene	0.01		3.6			576
Methacrylate, ferrocenylmethyl	Vinyl acetate	1.52		0.2			293
Methacrylate, furfuryl	*N*-Vinylpyrrolidone	5.16	1.13	0.0036	0.02	Y	1050
Methacrylate, glycidyl	Acrylate, 2-ethylhexyl	2.41	0.46	0.09	0.13	Y	674
Methacrylate, glycidyl	Acrylate, 4-acetylphenyl	0.644	0.13	0.292	0.141	Y	1012
Methacrylate, glycidyl	Acrylate, butyl	2.16	0.57	0.083	0.052	Y	674
Methacrylate, glycidyl	Acrylate, isobutyl	1.24	0.12	0.282	0.026	Y	674
Methacrylate, glycidyl	Acrylate, *tert*-butyl	2.096	0.443	0.463	0.079	Y	814
Methacrylate, glycidyl	Acrylonitrile	0.85	0.05	0.95	0.06	N	114
Methacrylate, glycidyl	Acrylonitrile	1.32		0.14			433
Methacrylate, glycidyl	Methacrylate, 2,3-epithiopropyl	0.81		0.78			787
Methacrylate, glycidyl	Methacrylate, butyl	0.94	0.06	0.85	0.03	N	224
Methacrylate, glycidyl	Methacrylate, methyl	0.52	0.06	0.71	0.08	Y	113
Methacrylate, glycidyl	Methacrylate, methyl	0.934	0.085	0.726	0.042	Y	226

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, glycidyl	Methacrylate, methyl	1.05		0.8			463
Methacrylate, glycidyl	Methacrylate, methyl	0.501	0.027	0.688	0.076	Y	639
Methacrylate, glycidyl	Methacrylic acid	1.2	0.1	0.98	0.16	N	224
Methacrylate, glycidyl	*N*-Vinylpyrrolidone	4.29	0.498	0.003	0.018		1017
Methacrylate, glycidyl	Styrene	0.16	0.03	0.54	0.07	N	114
Methacrylate, glycidyl	Styrene	0.539	0.04	0.278	0.086	Y	217
Methacrylate, glycidyl	Styrene	0.46	0.08	0.47	0.02	N	225
Methacrylate, glycidyl	Styrene	0.54		0.44			225
Methacrylate, glycidyl	Styrene	0.55		0.45			463
Methacrylate, glycidyl	Styrene	0.74	0.1	0.31	0.04	N	849
Methacrylate, glycidyl	Styrene	0.73	0.284	0.11	0.045	N	893
Methacrylate, glycidyl	Styrene	0.514	0.098	0.435	0.033	Y	999
Methacrylate, glycidyl	Vinyl phenyl ketone	0.719		0.697			946
Methacrylate, glycidyl	Vinyl, mono-, ethylene glycol ether	4.41		0			780
Methacrylate, hexyl	Styrene	0.44	0.13	0.58	0.07	Y	179
Methacrylate, hydroxyethyl	Methacrylate, tetra(2-hydroxypropyl) ethylenediamine	0.606	0.416	1.44	1	Y	996
Methacrylate, hydroxyethyl	Methacrylate, tributyltin	2.33	0.185	0.282	0.424	N	945
Methacrylate, iso-butyl	Butadiene, 2-chloro-	0.9	0.088	1.8	0.06	N	1057
Methacrylte, isobornyl	Styrene	0.24	0.3	0.61	0.45		806
Methacrylate, isobutyl	Acetylene, phenyl-	1.9		0.27			565
Methacrylate, isobutyl	Acrylic acid	0.68		1.03			847
Methacrylate, isobutyl	Acrylonitrile	1.05	0.22	0.217	0.037	Y	165
Methacrylate, isobutyl	Furan, 2-vinyl-5-methyl	0.105	0.044	0.716	0.169	Y	1105
Methacrylate, isobutyl	Methacrylate, 2-chloroethyl	1.014	0.064	0.85	0.14	Y	179
Methacrylate, isobutyl	Methacrylate, methyl	1.88	0.07	0.62	0.08	N	30
Methacrylate, isobutyl	Methacrylate, methyl	1.2		0.89			541
Methacrylate, isobutyl	Methacrylate, methyl	0.488	0.029	0.92	0.15	Y	639
Methacrylate, isobutyl	Methacrylic acid	0.47		2.01			847
Methacrylate, isobutyl	Methacrylonitrile	0.68	0.05	0.74	0.07	N	44
Methacrylate, isobutyl	Methacrylonitrile	0.94	0.12	0.43	0.08	N	44
Methacrylate, isobutyl	Styrene	0.271	0.036	0.509	0.014	Y	179
Methacrylate, isobutyl	Styrene	0.58		0.56			463
Methacrylate, isobutyl	Styrene	0.42		0.5			463
Methacrylate, isobutyl	Styrene	0.4		0.55			487
Methacrylate, isobutyl	Styrene	0.74		0.47			541
Methacrylate, isobutyl	Vinyl acetate	29.7	24.16	−0.06	0.22	Y	165
Methacrylate, *m*-chlorophenyl	Styrene	0.209	0.013	0.375	0.05	Y	177
Methacrylate, *m*-nitrophenyl	Styrene	0.185	0.054	0.44	0.16	Y	177
Methacrylate, methoxymethyl	Styrene	0.586	0.069	0.395	0.014	Y	821
Methacrylate, methyl	2-Oxazoline, 2-isopropenyl-	0.69	0.02	0.99	0.07		894
Methacrylate, methyl	3-Pyridazone, 2-vinyl	0.63		1.56			897
Methacrylate, methyl	3-Pyridazone, 2-vinyl-6-methyl-	0.16		0.3			897
Methacrylate, methyl	3-Pyridazone, 2-vinyl-6-methyl-4,5-benz-3-	1.65		0.27			897
Methacrylate, methyl	3-Pyridazone, 2-vinyl-6-phenyl-	0.58		0.49			897
Methacrylate, methyl	4H-1,3-Oxazinium F_3CSO_3, 3-methyl-2-vinyl-5,6-dihydro-	0.37		0.31			909
Methacrylate, methyl	Acenaphthalene	0.36	0.04	1.05	0.14	N	326
Methacrylate, methyl	Acenaphthalene	0.38	0.04	0.99	0.15	N	326
Methacrylate, methyl	Acenaphthalene	0.44		2.25			575
Methacrylate, methyl	Acetamide, *N*-vinyl-	2.65		0.19			984
Methacrylate, methyl	Acetamide, *N*-vinyl-	1.18		0.71			984
Methacrylate, methyl	Acetamide, *N*-vinyl-	2.1		0.01			984
Methacrylate, methyl	Acetylene, phenyl-	−0.06	0.16	1.69	26.56	Y	357
Methacrylate, methyl	Acetylene, phenyl-	1.5		0.2			443
Methacrylate, methyl	Acetylene, phenyl-	1.111	0.095	0.07	0.016	Y	732
Methacrylate, methyl	Acrolein	1.136	0.05	0.76	0.12	Y	291
Methacrylate, methyl	Acrolein	1.33	0.4	0.59	0.3	N	292
Methacrylate, methyl	Acrolein, methyl-	0.1		0.4			589
Methacrylate, methyl	Acrylamide	2.34	0.17	2.29	0.44	N	314
Methacrylate, methyl	Acrylamide	3	0.09	0.53	0.09	N	314
Methacrylate, methyl	Acrylamide	2.53		0.82			314
Methacrylate, methyl	Acrylamide	3	0.45	0.9	1.07	N	691
Methacrylate, methyl	Acrylamide, 2-chloro-*N*,*N*-dimethyl-	2.71	0.398	−0.159	0.502	Y	917

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Acrylamide, *N,N*-dimethyl-	1.65	0.16	0.41	0.15	N	170
Methacrylate, methyl	Acrylamide, *N,N*-dimethyl-	2.15	0.08	0.57	0.08	N	315
Methacrylate, methyl	Acrylamide, *N,N*-dimethyl-	2.8	0.1	0.59	0.09	N	315
Methacrylate, methyl	Acrylamide, *N,N*-dimethyl-	2.64	0.15	0.21	0.21	N	691
Methacrylate, methyl	Acrylamide, *N*-methyl-	1.14	0.09	0.05	0.72	N	691
Methacrylate, methyl	Acrylamide, *N*-methylol-	1.62		0.7			593
Methacrylate, methyl	Acrylamide, *N*-methylol	1.62		0.7			596
Methacrylate, methyl	Acrylamide, *N*-octadecyl-	4.119	0.07	0.42	0.11	Y	290
Methacrylate, methyl	Acrylamide, *N-tert*-butyl-	0.25		2.83			709
Methacrylate, methyl	Acrylamide, α-fluoro-	2.13	0.136	0.148	0.081	Y	823
Methacrylate, methyl	Acrylamido, 1-,-1-deoxy-D-glucitol	3.75	0.24	0.05	0.09	N	258
Methacrylate, methyl	Acrylate, α-benzyl-, methyl	3.9	0.19	0.09	0.01	N	610
Methacrylate, methyl	Acrylate, α-chloro-methyl	0.11	0.09	1.92	0.08	N	31
Methacrylate, methyl	Acrylate, ferrocenylmethyl	2.9		0.08			293
Methacrylate, methyl	Acrylate, 2-(*O*-ethyl methyl-phosphonoxy)-, methyl	0.73	0.04	0.44	0.25		882
Methacrylate, methyl	Acrylate, 2-chloroethyl	2.15		0.37			790
Methacrylate, methyl	Acrylate, 2-nitrobutyl	1.19	0.12	0.27	0.04	N	234
Methacrylate, methyl	Acrylate, β-acetoxymethyl-, methyl	0.91		0.24			1091
Methacrylate, methyl	Acrylate, α-(*O*-ethyl methylphos-phenoxy)	0.73		0.44			851
Methacrylate, methyl	Acrylate, α-[2,2-bis(carbomethoxy)-ethyl]-, methyl	3.7		0.04			1092
Methacrylate, methyl	Acrylate, α-acetoxy-, ethyl	1.65	0.12	0.608	0.02	Y	244
Methacrylate, methyl	Acrylate, α-benzyl-, methyl	2.36	0.23	0.173	0.022	Y	610
Methacrylate, methyl	Acrylate, α-chloro-, methyl	0.31	0.24	1.13	0.19	Y	6
Methacrylate, methyl	Acrylate, α-cyano-, methyl	0.21	0.02	0.03	0.01	N	141
Methacrylate, methyl	Acrylate, α-cyano-, methyl	0.068	0.014	0.135	0.016	Y	182
Methacrylate, methyl	Acrylate, α-hydroxymethyl-, ethyl	1.26	0.17	1.34	0.062	N	1027
Methacrylate, methyl	Acrylate, α-phenoxymethyl-, methyl	1.13	0.038	0.755	0.061	Y	1048
Methacrylate, methyl	Acrylate, α-phenyl-, methyl	0.21		0			141
Methacrylate, methyl	Acrylate, α-phenyl-, methyl	0.3		0			49
Methacrylate, methyl	Acrylate, α-tetrafluoropropyloxymethyl-, methyl	1.43		0.61			1047
Methacrylate, methyl	Acrylate, α-trifluoromethyl-, methyl	1.8	0.23	−0.17	0.18	Y	681
Methacrylate, methyl	Acrylate, α-trifluoromethyl-, methyl	3.2	0.5	0	0.1	N	825
Methacrylate, methyl	Acrylate, α-trifluoromethyl-, methyl	2.3	0.2	0.1	0.1	N	825
Methacrylate, methyl	Acrylate, benzyl	2.23	0.14	0.19	0.2	Y	277
Methacrylate, methyl	Acrylate, butyl	0.92	0.2	0.13	0.15	N	30
Methacrylate, methyl	Acrylate, butyl	2.86	0.32	0.11	0.018	Y	639
Methacrylate, methyl	Acrylate, butyl	1.88	0.4	0.43	0.07	N	87
Methacrylate, methyl	Acrylate, ethyl	1.83		0.47			30
Methacrylate, methyl	Acrylate, ethyl	2		0.28			469
Methacrylate, methyl	Acrylate, ethyl	2.04	0.28	0.22	0.05	N	87
Methacrylate, methyl	Acrylate, ferrocenylmethyl	2.8	0.19	0.024	0.064	Y	293
Methacrylate, methyl	Acrylate, heptafluorobutyl	1.387	0.072	0.177	0.084	Y	206
Methacrylate, methyl	Acrylate, heptyl	2.51	0.13	0.46	0.02		980
Methacrylate, methyl	Acrylate, isobutyl	1.04	0.08	0.29	0.04	N	30
Methacrylate, methyl	Acrylate, isobutyl	4.1	0.5	0.135	0.02	Y	639
Methacrylate, methyl	Acrylate, *m*-nitrophenyl	0.928	0.248	1.04	0.135	Y	1074
Methacrylate, methyl	Acrylate, methyl	2.15	0.04	0.4	0.12	N	324
Methacrylate, methyl	Acrylate, *o*-chlorophenyl	1.25		0.4			621
Methacrylate, methyl	Acrylate, octadecyl	2.477	0.024	0.464	0.07	Y	290
Methacrylate, methyl	Acrylate, *p*-bromophenyl	0.926	0.056	0.191	0.014	Y	1074
Methacrylate, methyl	Acrylate, *p*-chlorophenyl	1.01	0.132	0.235	0.023	Y	1074
Methacrylate, methyl	Acrylate, *p*-chlorophenyl	1.09		0.3			621
Methacrylate, methyl	Acrylate, methylphenyl	1.45	0.275	0.223	0.036	Y	1074
Methacrylate, methyl	Acrylate, phenyl	1.56		0.45			621
Methacrylate, methyl	Acrylate, phenyl	1.54	0.14	0.416	0.041	Y	672
Methacrylate, methyl	Acrylate, α-*p*-chlorobenzyl-, methyl	1.87	0.222	0.255	0.034	Y	905
Methacrylate, methyl	Acrylate, α-*p*-methoxybenzyl-, methyl	1.71	0.109	0.13	0.008	Y	905
Methacrylate, methyl	Acrylic acid	0.418	0.02	1.73	0.096	N	964
Methacrylate, methyl	Acrylic acid, α-bromo-	0.37		0.11			733
Methacrylate, methyl	Acrylonitrile	1.322	0.05	0.138	0.037	Y	216
Methacrylate, methyl	Acrylonitrile, α-trifluoromethyl-	1.11	0.12	−0.016	0.048	Y	681

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Acrylonitrile, α-perfluoropropyl-	1.9	0.3	0	0.05		926
Methacrylate, methyl	Acryloyl chloride	0.48		1.51			508
Methacrylate, methyl	Acryloyl chloride	0.42		0.05			776
Methacrylate, methyl	Allyl acetate	99.24	24.6	−2.42	37.66	N	210
Methacrylate, methyl	Allyl acetate	23		0			470
Methacrylate, methyl	Allyl alcohol	78.79	6.5	−1.62	2.9	N	210
Methacrylate, methyl	Allyl chloride					Y	204
Methacrylate, methyl	Benzocyclobutane, α-methylene-	0.32		0.46			1083
Methacrylate, methyl	Benzocyclobutene, 4-vinyl-	0.6		1.47			1085
Methacrylate, methyl	Benzofuran, 2-vinyl-	0.006	0.02	3.89	0.617	N	1020
Methacrylate, methyl	Benzophenone, 2-hydroxy-4-(methacryloxy)-	0.315	0.036	1.65	0.114	Y	934
Methacrylate, methyl	Benzothiazole, vinylmercapto-	1.25	0.083	0.43	0.14	Y	763
Methacrylate, methyl	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	30.5		0			994
Methacrylate, methyl	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	108		0			994
Methacrylate, methyl	Butadiene	0.25		0.75			140
Methacrylate, methyl	Butadiene	0.027	0.004	0.504	0.024	Y	257
Methacrylate, methyl	Butadiene	0.32		0.7			467
Methacrylate, methyl	Butadiene, 2,3-dichloro-	0.08		11.04		Y	61
Methacrylate, methyl	Butadiene, 2-chloro-	0.08		6.33		Y	61
Methacrylate, methyl	Butadiene, 2-fluoro-	0.76	0.77	1.23	1.77	Y	174
Methacrylate, methyl	Carbamate, *N*-vinyl-*t*-butyl-	1.87	0.2	0.17	0.1		991
Methacrylate, methyl	Carbazole, *N*-vinyl-	1.55	0.16	0.14	0.03	N	11
Methacrylate, methyl	Carbazole, *N*-vinyl-	2.12	0.14	0.033	0.015	Y	13
Methacrylate, methyl	Carbazole, *N*-vinyl-	1.8		0.06			738
Methacrylate, methyl	Carbazole, *N*-vinyl-	0.57		0.75			738
Methacrylate, methyl	Carbazole, *N*-vinyl-	1.84	0.26	0.08	0.05	N	758
Methacrylate, methyl	Carbazole, *N*-vinyl-	0.44	0.22	0.33	0.18	N	758
Methacrylate, methyl	Carbazole, *N*-vinyl-	2.123	0.048	0.031	0.022	Y	98
Methacrylate, methyl	Citraconimide, *N*-(4-acetoxyphenyl)-	1.88	0.16	0.18	0.13	Y	728
Methacrylate, methyl	Citraconimide, *N*-(4-chlorophenyl)-	2.93	0.4	−0.01	0.18	Y	728
Methacrylate, methyl	Citraconimide, *N*-(4-ethoxycarbonyl-phenyl)-	2.72	0.18	−0.027	0.084	Y	728
Methacrylate, methyl	Citraconimide, *N*-(4-methoxyphenyl)-	3.16	0.73	−0.1	0.19	Y	728
Methacrylate, methyl	Citraconimide, *N*-(1-menthylcarboxylatomethyl)-	3.47	0.335	0.109	0.115	Y	950
Methacrylate, methyl	Citraconimide, *N*-benzyl-	2.77	0.28	0.02	0.14	Y	714
Methacrylate, methyl	Citraconimide, *N*-butyl-	2.17	0.56	−0.12	0.36	Y	714
Methacrylate, methyl	Citraconimide, *N*-ethyl-	3.25	0.79	−0.04	0.33	Y	714
Methacrylate, methyl	Citraconimide, *N*-hexyl-	2.38	0.45	−0.18	0.28	Y	714
Methacrylate, methyl	Citraconimide, *N*-isobutyl-	1.78	0.85	−0.18	0.42	Y	714
Methacrylate, methyl	Citraconimide, *N*-isopropyl-	2.7	0.51	0.09	0.27	Y	714
Methacrylate, methyl	Citraconimide, *N*-methyl	3.24		0.15			460
Methacrylate, methyl	Citraconimide, *N*-phenyl	3.081	0.074	0.001	0.033	Y	728
Methacrylate, methyl	Citraconimide, *N*-propyl	3.23	0.98	−0.03	0.43	Y	714
Methacrylate, methyl	Citraconimide, *N*-*tert*-butyl-	2.88	0.19	0.038	0.072	Y	714
Methacrylate, methyl	Citraconimide, *N*-tolyl-	2.75	0.19	0.003	0.088	Y	728
Methacrylate, methyl	Cyclopentene, 4-,-1,3-dione	8.11	0.38	0.094	0.017	Y	95
Methacrylate, methyl	Cyclopropane, 1,1-bis(ethoxy-carbonyl)-2-vinyl-	4.21	0.905	0.223	0.194	Y	1076
Methacrylate, methyl	Cyclopropane, 1,1-bis(ethoxy-carbonyl)-2-vinyl-	18.3	2.5	0.111	0.194	Y	1076
Methacrylate, methyl	Diallyl melamine					Y	201
Methacrylate, methyl	Diallyl phthalate	26.56	1.98	−0.001	0.011	Y	153
Methacrylate, methyl	Diallyl phthalate	22.73	1.48	0.029	0.026	Y	153
Methacrylate, methyl	Diallyl phthalate	21.52	0.39	0.076	0.006	Y	153
Methacrylate, methyl	Diallylcyanamide	25		0.04			603
Methacrylate, methyl	Diallylcyanamide	25		0.04			604
Methacrylate, methyl	Dimethyleneheptanedioate, 2,6-dimethyl	2.36	0.25	0.53	0.02	N	795
Methacrylate, methyl	Ditiane, 2-methylene-1,3-	0.08		4.12			1031
Methacrylate, methyl	Ethylene					N	41

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Ethylene, 1,1-diphenyl-	2.1		0.09			65
Methacrylate, methyl	Ethylene, 1,1-diphenyl-	8.7		0			901
Methacrylate, methyl	Ethylene, chlorotrifluoro-	51.36	3.31	−0.074	0.062	Y	236
Methacrylate, methyl	Ethylene, tetrachloro-					Y	204
Methacrylate, methyl	Ethylene, trichloro-	100		0			470
Methacrylate, methyl	Fumarate, diethyl	40.3	0.6	0.04	0.004	N	28
Methacrylate, methyl	Fumarate, diethyl	2.1		0.05			394
Methacrylate, methyl	Fumarate, diisopropyl	23		0.17			1038
Methacrylate, methyl	Fumaronitrile	6.74	0.79	−0.064	0.096	Y	191
Methacrylate, methyl	Furan, 2-vinyl-5-methyl	0.183	0.146	0.867	0.292	Y	1105
Methacrylate, methyl	Hexatriene, tetrachloro-	0.504	0.057	1.76	0.28	Y	5
Methacrylate, methyl	Imidazole, 1-vinyl-	4.36	1.86	0.014	0.03	Y	364
Methacrylate, methyl	Imidazole, 1-vinyl-2-methyl-	4	0.54	0.047	0.088	Y	364
Methacrylate, methyl	Imidazole, 1-vinyl-2-methyl-	3.48		0.003			479
Methacrylate, methyl	Isoprene	0.26	0.21	0.65	0.23	Y	302
Methacrylate, methyl	Isoprene	0.25		0.64			698
Methacrylate, methyl	Isoprene	0.25		0.64			704
Methacrylate, methyl	Isoprene, 3-acetoxy-	0.16		2.81			770
Methacrylate, methyl	Isopropenyl acetate	17.36	1.95	−0.09	0.12	Y	100
Methacrylate, methyl	Isopropenyl, 3-(1-cyclohexenyl), acetate	0.81		0.4			770
Methacrylate, methyl	Isopropenylisocyanate	3.1		0.14			434
Methacrylate, methyl	Isopropenyltoluene	0.68		0.04			688
Methacrylate, methyl	Isopropenyltoluene	0.68		0.04			722
Methacrylate, methyl	Itaconamate, *N*-(4-chlorophenyl)-, ethyl	1.24		0.36			678
Methacrylate, methyl	Itaconamate, *N*-(4-chlorophenyl)-, propyl	1.26		0.31			678
Methacrylate, methyl	Itaconamate, *N*-(4-methoxyphenyl)-, ethyl	1.31		0.43			678
Methacrylate, methyl	Itaconamate, *N*-(4-carboethoxyphenyl)-ethyl	1.21		0.48			678
Methacrylate, methyl	Itaconamate, *N*-(4-methylphenyl)-, ethyl	1.43		0.63			678
Methacrylate, methyl	Itaconamate, *N*-(4-methoxyphenyl)-, propyl	1.03		0.4			678
Methacrylate, methyl	Itaconamate, *N*-(4-methylphenyl)-, propyl	1.11		0.28			678
Methacrylate, methyl	Itaconamate, *N*-phenyl-, ethyl	1.32		0.47			678
Methacrylate, methyl	Itaconamate, *N*-phenyl-, propyl	1.13		0.15			678
Methacrylate, methyl	Itaconamate, *N*-(4-carboethoxyphenyl)-propyl	1		0.53			678
Methacrylate, methyl	Itaconate, dibutyl	1.33	0.09	0.717	0.11		1073
Methacrylate, methyl	Itaconate, dibutyl	0.8		0.4			398
Methacrylate, methyl	Itaconate, dimethyl	3.34	0.557	−0.156	0.02	Y	975
Methacrylate, methyl	Itaconate, dimethyl	1.2		0.3			398
Methacrylate, methyl	Itaconate, monomethyl	1.1	0.071	0.139	0.103	Y	1010
Methacrylate, methyl	Itaconic acid	1.23		0			260
Methacrylate, methyl	Itaconic acid	1.14		0			408
Methacrylate, methyl	Itaconic anhydride	0.155	0.024	1.17	0.33	Y	1010
Methacrylate, methyl	Itaconimide, *N*-(2-chloroethyl)-	0	0.11	0.54	0.39	Y	642
Methacrylate, methyl	Itaconimide, *N*-benzyl-	0.46	0.18	1.06	0.75	Y	642
Methacrylate, methyl	Itaconimide, *N*-ethyl-	0.486	0.066	0.42	0.13	Y	642
Methacrylate, methyl	Itaconimide, *N*-methyl-	0.38	0.22	1.12	0.73	Y	642
Methacrylate, methyl	Maleate, dibutyl tin	72.7	5.93	−0.095	0.199	Y	987
Methacrylate, methyl	Maleate, diethyl	341	46	−0.1	0.05	N	28
Methacrylate, methyl	Maleate, diethyl	20		0			470
Methacrylate, methyl	Maleate, diethyl	285.8	35.9	−0.02	0.015	Y	873
Methacrylate, methyl	Maleic anhydride	4.18	0.31	−0.163	0.027	Y	169
Methacrylate, methyl	Maleic anhydride	6.36	0.64	−0.019	0.026	Y	260
Methacrylate, methyl	Maleic anhydride	0.46	0.69	0.08	1.77	N	320
Methacrylate, methyl	Maleic anhydride	5.2	0.47	0.02	0.004	N	33
Methacrylate, methyl	Maleic anhydride	3.1		0.01			524
Methacrylate, methyl	Maleic anhydride	3.85		0.01			524
Methacrylate, methyl	Maleic anhydride	3.4		0.01			524
Methacrylate, methyl	Maleic anhydride	3.85		0.02			524
Methacrylate, methyl	Maleimide	2.576	0.061	0.166	0.047	Y	251
Methacrylate, methyl	Maleimide	2.5		0.17			992

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Maleimide, 2,3-dimethyl-*N*-(2-methacryloyloxyethyl)	0.51	1.66	1.1	3.58	Y	619
Methacrylate, methyl	Maleimide, *N*-(4-hydroxyphenyl)-	1.392	0.001	0.345	0.001	Y	323
Methacrylate, methyl	Maleimide, *N*-(2-hydroxyethyl)-	1.396	0.089	0.201	0.083	Y	323
Methacrylate, methyl	Maleimide, *N*-(2,6-diethylphenyl)-	4.78		0.05			992
Methacrylate, methyl	Maleimide, *N*-(2,6-diisopropylphenyl)-	10.9		0.02			992
Methacrylate, methyl	Maleimide, *N*-(2,6-dimethylphenyl)-	3.62		0.07			992
Methacrylate, methyl	Maleimide, *N*-(2-chlorophenyl)-	2.1		0.1			992
Methacrylate, methyl	Maleimide, *N*-(2-methylphenyl)-	2.22		0.09			992
Methacrylate, methyl	Maleimide, *N*-(3,5-dimethylphenyl)-	1.63		0.28			992
Methacrylate, methyl	Maleimide, *N*-(3-methylphenyl)-	1.18		0.22			992
Methacrylate, methyl	Maleimide, *N*-(3-trifluoromethylphenyl)-	1.57		0.1			992
Methacrylate, methyl	Maleimide, *N*-(4-carboxyethylphenyl)-α-chloro-	0.472	0.114	−0.004	0.168	Y	973
Methacrylate, methyl	Maleimide, *N*-(4-chlorophenyl)-α-chloro-	0.562	0.025	0.02	0.03	Y	973
Methacrylate, methyl	Maleimide, *N*-4-methoxyphenyl)-α-chloro-	0.556	0.145	0.064	0.189	Y	973
Methacrylate, methyl	Maleimide, *N*-(4-methylphenyl)-	0.83		0.34			992
Methacrylate, methyl	Maleimide, *N*-(4-methylphenyl)-α-chloro-	0.564	0.117	0.029	0.158	Y	973
Methacrylate, methyl	Maleimide, *N*-(β-chloroethyl)-α-chloro-	0.94	0.06	0.05	0.04		974
Methacrylate, methyl	Maleimide, *N*-(β-chloroethyl)-α-chloro-	0.59	0.06	0.11	0.05		974
Methacrylate, methyl	Maleimide, *N*-[*N'*-(α-methylbenzyl) aminocarbonylpentyl]-	1.37	0.291	0.582	0.359	Y	1009
Methacrylate, methyl	Maleimide, *N*-benzyl	1.28	0.23	0.03	0.23	Y	348
Methacrylate, methyl	Maleimide, *N*-butyl	1.41	0.19	−0.1	0.16	Y	115
Methacrylate, methyl	Maleimide, *N*-cyclohexyl	1.35		0.24			992
Methacrylate, methyl	Maleimide, *N*-cyclohexyl	1.31		0.19			1100
Methacrylate, methyl	Maleimide, *N*-cyclohexyl	2.26	0.066	0.106	0.048	Y	348
Methacrylate, methyl	Maleimide, *N*-ethyl-α-chloro-	0.53	0.04	0.13	0.03		974
Methacrylate, methyl	Maleimide, *N*-hexyl-	1.9	0.2	0.1	0.15	Y	348
Methacrylate, methyl	Maleimide, *N*-hydroxymethyl-	1.745	0.03	0.263	0.026	Y	323
Methacrylate, methyl	Maleimide, *N*-methyl-α-chloro-	0.63	0.34	0.11	0.03		974
Methacrylate, methyl	Maleimide, *N*-phenyl-	0.98		0.3			323
Methacrylate, methyl	Maleimide, *N*-phenyl-	0.91		0.3			992
Methacrylate, methyl	Maleimide, *N*-phenyl-α-chloro-	0.657	0.083	0.077	0.098	Y	973
Methacrylate, methyl	Maleimide, *N*-phenyl-	1.014	0.06	0.164	0.065	Y	328
Methacrylate, methyl	Methacrylate, 2-acetoxyethyl	0.94	0.02	1.12	0.06	N	346
Methacrylate, methyl	Methacrylate, cyanomethyl	0.399	0.019	0.73	0.11	Y	625
Methacrylate, methyl	Methacrylate, diphenylmethyl	0.81		1.07			761
Methacrylate, methyl	Methacrylamide	1.57	0.11	1.27	0.24	N	316
Methacrylate, methyl	Methacrylamide	1.7	0.04	0.44	0.06	N	316
Methacrylate, methyl	Methacrylamide	1.39	0.074	0.425	0.085	Y	54
Methacrylate, methyl	Methacrylamide	1.65		0.49			55
Methacrylate, methyl	Methacrylamide, *N*-methoxymethyl-	1.678	0.048	0.265	0.009	Y	227
Methacrylate, methyl	Methacrylamide, *N*-*p*-bromophenyl-	3.48	0.2	0.38	0.2		968
Methacrylate, methyl	Methacrylamide, *N*-*p*-tolyl-	1.43	0.2	0.78	0.2		968
Methacrylate, methyl	Methacrylamide, *N*-phenyl	1.95	0.2	0.28	0.2		968
Methacrylate, methyl	Methacrylamide, *N*-phenyl-	1.42	0.096	0.56	0.12	Y	352
Methacrylate, methyl	Methacrylamide, 1-deoxy-D-glucitol	4.2		0.04			258
Methacrylate, methyl	Methacrylamide, *N*-*p*-nitrophenyl-	5.05	0.2	0.29	0.2		968
Methacrylate, methyl	Methacrylate, *tert*-butyl	0.96		1.35			761
Methacrylate, methyl	Methacrylate, α,α-dimethylbenzyl	0.81		0.92			761
Methacrylate, methyl	Methacrylate, (2,2-dimethyl-1,3-dioxolane)	0.9		1.1			807
Methacrylate, methyl	Methacrylate, 1,1-diphenylethyl	0.72		0.77			761
Methacrylate, methyl	Methacrylate, 1-naphthyl	0.77	0.14	1.8	0.73	Y	752
Methacrylate, methyl	Methacrylate, 1-naphthyl	0.809	0.078	1.68	0.38	Y	752
Methacrylate, methyl	Methacrylate, 1-naphthyl	0.851	0.084	2.62	0.47	Y	752
Methacrylate, methyl	Methacrylate, 1-naphthyl	0.61		1.52			761
Methacrylate, methyl	Methacrylate, 2,2-dimethylaminoethyl	0.42	0.02	1.01	0.54	N	691
Methacrylate, methyl	Methacrylate, 2,3-epithiopropyl	0.83		0.97			787
Methacrylate, methyl	Methacrylate, 2,6-(diphenyl)phenyl	1.42	0.06	0.071	0.004		1079
Methacrylate, methyl	Methacrylate, 2-(*N*,*N*-dimethylcarbamoyloxy)ethyl	0.59		0.42			600

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Methacrylate, 2-bromoethyl	0.79		1.18			657
Methacrylate, methyl	Methacrylate, 2-chloro-2,3,3,3-fluoropropyl	0.984	0.395	0.583	1.17	N	927
Methacrylate, methyl	Methacrylate, 2-chloroethyl	1.156	0.082	0.97	0.18	Y	179
Methacrylate, methyl	Methacrylate, 2-chloroethyl	0.55		0.48			382
Methacrylate, methyl	Methacrylate, 2-hydroxyethyl	0.824	0.01	0.63	0.076	Y	350
Methacrylate, methyl	Methacrylate, 2-hydroxyethyl	0.192	0.09	0.81	1.02	Y	358
Methacrylate, methyl	Methacrylate, 2-hydroxyethyl	0.75		1.5			710
Methacrylate, methyl	Methacrylate, 2-hydroxypropyl	0.402	0.017	1.055	0.04	Y	333
Methacrylate, methyl	Methacrylate, 2-naphthyl	0.602	0.059	2.57	0.48	Y	760
Methacrylate, methyl	Methacrylate, 2-naphthyl	0.581	0.023	2.53	0.27	Y	760
Methacrylate, methyl	Methacrylate, 2-naphthyl	0.532	0.055	1.91	0.6	Y	760
Methacrylate, methyl	Methacrylate, 2-naphthyl	0.614	0.051	4.41	0.76	Y	760
Methacrylate, methyl	Methacrylate, 2-naphthyl	0.623	0.048	3.83	0.57	Y	760
Methacrylate, methyl	Methacrylate, 2-phenylethyl	0.75	0.16	0.85	0.34	Y	540
Methacrylate, methyl	Methacrylate, 3,5-dimethyladamantyl	0.68	0.04	0.45	0.16	N	335
Methacrylate, methyl	Methacrylate, 5-oxo-pyrrolidinylmethyl	0.421	0.033	0.694	0.087	Y	633
Methacrylate, methyl	Methacrylate, *N*-α-benzothiazolonyl-methyl	0.5	0.01	0.6	0.02		887
Methacrylate, methyl	Methacrylate, benzyl	0.808	0.03	1.112	0.093	Y	111
Methacrylate, methyl	Methacrylate, benzyl	0.93		1.05			32
Methacrylate, methyl	Methacrylate, benzyl	0.78		1.38			540
Methacrylate, methyl	Methacrylate, benzylmethylaminoethyl	0.523	0.134	0.296	0.123	Y	1081
Methacrylate, methyl	Methacrylate, butyl	0.52	0.07	2.11	0.08	N	30
Methacrylate, methyl	Methacrylate, butyl	1.27		1.2			844
Methacrylate, methyl	Methacrylate, chloromethyl	0.477	0.029	1.02	0.16	Y	712
Methacrylate, methyl	Methacrylate, diethylaminoethyl	0.843	0.128	1.44	0.085	Y	1081
Methacrylate, methyl	Methacrylate, dimethylaminoethyl	0.699	0.051	1.2	0.154	Y	1081
Methacrylate, methyl	Methacrylate, ethyl	0.81	0.22	0.86	0.12	N	32
Methacrylate, methyl	Methacrylate, ethyl	1.08		1.08			844
Methacrylate, methyl	Methacrylate, ethyl	1.16	0.19	1	0.09	N	87
Methacrylate, methyl	Methacrylate, glycidyl	0.71	0.08	0.52	0.06	Y	113
Methacrylate, methyl	Methacrylate, glycidyl	0.726	0.042	0.934	0.085	Y	226
Methacrylate, methyl	Methacrylate, glycidyl	0.8		1.05			463
Methacrylate, methyl	Methacrylate, glycidyl	0.688	0.076	0.501	0.027	Y	639
Methacrylate, methyl	Methacrylate, isobutyl	0.62	0.08	1.88	0.07	N	30
Methacrylate, methyl	Methacrylate, isobutyl	0.92	0.15	0.488	0.029	Y	639
Methacrylate, methyl	Methacrylate, isopropyl	0.89		1.2			541
Methacrylate, methyl	Methacrylate, methyl diphenylsilyl	0.438	0.075	0.38	0.178	Y	1069
Methacrylate, methyl	Methacrylate, methylthiomethyl	0.639	0.057	1.1	0.356	Y	920
Methacrylate, methyl	Methacrylate, *p*-acetylphenyl	0.47		0.67			769
Methacrylate, methyl	Methacrylate, *p*-chlorophenyl	0.54		1.08			769
Methacrylate, methyl	Methacrylate, *p*-methoxyphenyl	0.72		1.19			769
Methacrylate, methyl	Methacrylate, *p*-tolyl	0.59		1.28			769
Methacrylate, methyl	Methacrylate, phenyl	0.53		1.67			32
Methacrylate, methyl	Methacrylate, phenyl	0.58		1.69			32
Methacrylate, methyl	Methacrylate, phenyl	0.56		1.72			32
Methacrylate, methyl	Methacrylate, phenyl	0.66		1.55			761
Methacrylate, methyl	Methacrylate, phenyl	0.55		0.99			769
Methacrylate, methyl	Methacrylate, propyl	1.21		1.24			844
Methacrylate, methyl	Methacrylate, sulfolanyl	1.07	2.48	7.16	23.7	N	359
Methacrylate, methyl	Methacrylate, *t*-butyl dimethylsilyl	0.897	0.05	0.725	0.455	Y	1069
Methacrylate, methyl	Methacrylate, *t*-butyl diphenylsilyl	0.636	0.014	0.396	0.257	Y	1069
Methacrylate, methyl	Methacrylate, tributyltin	0.58		0.62			1082
Methacrylate, methyl	Methacrylate, tributyltin	0.62		0.87			1082
Methacrylate, methyl	Methacrylate, trimethyltin	0.63		1.07			1082
Methacrylate, methyl	Methacrylate, trimethyltin	0.6		0.68			1082
Methacrylate, methyl	Methacrylic acid	0.63	0.03	0.25	0.05		1071
Methacrylate, methyl	Methacrylic acid	0.1	0.01	1.06	0.02		1072
Methacrylate, methyl	Methacrylic acid	0.78	0.03	0.33	0.02		1072
Methacrylate, methyl	Methacrylic acid	0.31	0.02	0.63	0.02		1072
Methacrylate, methyl	Methacrylic acid	0.27	0.03	0.46	0.03		1072
Methacrylate, methyl	Methacrylic acid	0.1	0.01	1.06	0.02		1087
Methacrylate, methyl	Methacrylic acid	0.78	0.03	0.33	0.02		1087
Methacrylate, methyl	Methacrylic acid	0.32	0.02	0.63	0.02		1087

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Methacrylic acid	0.87	0.05	2.16	0.28	N	203
Methacrylate, methyl	Methacrylic acid	1.81	0.16	1.63	6.25	N	210
Methacrylate, methyl	Methacrylic acid	0.56	0.23	0.99	0.55	Y	216
Methacrylate, methyl	Methacrylic acid	0.29	0.04	1.18	0.19	N	276
Methacrylate, methyl	Methacrylic acid	1.28	0.1	0.48	0.1	N	276
Methacrylate, methyl	Methacrylic acid	0.77	0.05	1.84	0.24	N	312
Methacrylate, methyl	Methacrylic acid	0.36	0.08	1.38	0.27	N	313
Methacrylate, methyl	Methacrylic acid	1.18		0.63			313
Methacrylate, methyl	Methacrylic acid	0.55		1.55			424
Methacrylate, methyl	Methacrylic acid	0.209	0.05	1.26	0.2	Y	54
Methacrylate, methyl	Methacrylic acid	1.25		0.78			630
Methacrylate, methyl	Methacrylic acid	0.48		0.68			819
Methacrylate, methyl	Methacrylic anhydride	0.22	0.02	1.58	0.06	N	275
Methacrylate, methyl	Methacrylophenone, α-	1.79	0.07	0.11	0.02	N	798
Methacrylate, methyl	Methacrylophenone, α-	1.74	0.22	0.119	0.009	Y	798
Methacrylate, methyl	Methacryloyl chloride	0.45		0.2			591
Methacrylate, methyl	Methacryloyl chloride	0.45		0.2			593
Methacrylate, methyl	Methacryloyl fluoride	0.405	0.037	0.43	0.12	Y	617
Methacrylate, methyl	Methacryloyl fluoride	0.53	0.06	0.9	0.16	N	660
Methacrylate, methyl	Methacryloyl isocyanate	0.75		0.35			944
Methacrylate, methyl	Methacryloyloxy-, *o*-, benzoic acid	0.908	0.252	1.37	0.64	Y	938
Methacrylate, methyl	Methylenebutyrolactone	0.6		1.67			746
Methacrylate, methyl	Methyleneglutaronitrile, 2-	1.36	0.28	0.12	0.74	N	196
Methacrylate, methyl	Methyleneglutaronitrile, 2-	1.27		0.12			196
Methacrylate, methyl	Myrcene	0.256	0.013	0.448	0.06	Y	1062
Methacrylate, methyl	*N,N*-divinylaniline	1.06	0.18	0.005	0.018	Y	45
Methacrylate, methyl	*N*-Vinylimidazole	4.306	0.185	0.663	0.104	Y	879
Methacrylate, methyl	*N*-Vinylpyrrolidone	4.78	0.236	0.006	2.29		1084
Methacrylate, methyl	*N*-Vinylpyrrolidone	5.93	0.198	0.014	0.899		1084
Methacrylate, methyl	*N*-Vinylpyrrolidone	2.07	0.247	0.027	0.062		1084
Methacrylate, methyl	*N*-Vinylpyrrolidone	4.04	0.956	0.066	3.96		1084
Methacrylate, methyl	Naphthalene, 1-vinyl-	0.321	0.096	0.574	0.073		967
Methacrylate, methyl	Norbornadiene	14.68	16.78	−0.5	0.52	N	267
Methacrylate, methyl	Oxazolidone, *N*-vinyl-	6.75	0.37	0.04	0.13	Y	102
Methacrylate, methyl	Oxazolidone, *N*-vinyl-	9.46	0.39	0.027	0.008	Y	36
Methacrylate, methyl	Oxazoline, 2-,2-isopropenyl 4,4-dimethyl	0.53		0.82			596
Methacrylate, methyl	Oxazoline, 2-,2-isopropenyl	0.69		0.99			862
Methacrylate, methyl	Oxazoline, 2-,4-acryloxymethyl 2,4-dimethyl	0.41		0.11			596
Methacrylate, methyl	Oxazoline, 2-,4-methacryloxy- 2,4-dimethyl	1.17		0.19			596
Methacrylate, methyl	Pentadiene, *trans*-1,3-	0.404	0.038	0.379	0.013	Y	1058
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.555	0.032	3.39	0.263	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.424	0.022	2.02	0.141	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.591	0.046	3.06	0.349	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.443	0.032	1.67	0.183	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.34	0.031	1.24	0.168	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.57	0.023	1.72	0.123	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.606	0.045	1.69	0.236	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.269	0.027	0.72	0.12	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.715	0.048	1.55	0.224	N	1033
Methacrylate, methyl	Pentenoate, 3-oxo-4-, ethyl	0.528	0.02	1.09	0.09	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.597	0.03	2.56	0.218	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.38	0.034	0.748	0.155	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.513	0.022	1.14	0.109	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.43	0.034	0.952	0.164	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.474	0.023	1.67	0.143	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.581	0.025	1.04	0.111	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.616	0.024	1.05	0.108	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.663	0.023	0.725	0.08	N	1033
Methacrylate, methyl	Pentenoate, 4-methyl-3-oxo-4-, ethyl	0.528	0.02	1.09	0.09	N	1033
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	2.37	0.046	0.486	0.088	N	971
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	2.23	0.047	0.521	0.138	N	971
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	1.87	0.086	0.946	0.436	N	971

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	1.97	0.029	1.16	0.248	N	971
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	2.1	0.045	0.42	0.097	N	971
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	2.56	0.094	0.258	0.226	N	971
Methacrylate, methyl	Phosphate, 2-(acryloyloxy)ethyl diethyl	2.04	0.012	0.0701	0.052	N	971
Methacrylate, methyl	Phosphonate, α-bromovinyl-,diethyl	1.25	0.04				684
Methacrylate, methyl	Phosphonate, α-carboethoxyvinyl-, diethyl	1		0.05			684
Methacrylate, methyl	Phosphonate, α-carbomethoxyvinyl-, diethyl	4.7		0.01			684
Methacrylate, methyl	Phosphonate, isopropenyl-, dimethyl	30		0.2			684
Methacrylate, methyl	Phosphonate, vinyl-, bis(2-chloroethyl)	29.9		0.26			410
Methacrylate, methyl	Phosphonate, vinyl-, diethyl	11.1	0.04				684
Methacrylate, methyl	Phosphonic acid, α-phenylvinyl-	1.362	0.084	−0.063	0.077	Y	131
Methacrylate, methyl	Phthalimide,-*N*-)methacryloyloxy)-	0.69	0.048	1.45	0.151	Y	979
Methacrylate, methyl	Pinene, α-	10		0.08			48
Methacrylate, methyl	Propene, 2,3-dichloro-	6.02	1.66	0.18	0.1	N	11
Methacrylate, methyl	Propene, 2-chloro-3-hydroxy-	4.4		0			445
Methacrylate, methyl	Propenyl, 2-chloro-, acetate	1		0			445
Methacrylate, methyl	Propenyl, 2-methyl, acetate	10		0			470
Methacrylate, methyl	Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-	1.19	0.04	0.08	0.03	N	340
Methacrylate, methyl	Pyridazinone 3-(2-vinyl)-6-methyl-	0.86	0.02	0.56	0.04	N	340
Methacrylate, methyl	Pyridine, 2-methyl-5-vinyl-	0.42	0.09	0.54	0.19	N	232
Methacrylate, methyl	Pyridine, 2-vinyl-	0.42	0.02	0.73	0.06	N	231
Methacrylate, methyl	Pyridine, 2-vinyl-	0.4		0.86			532
Methacrylate, methyl	Pyridine, 2-vinyl-	0.35	0.19	0.76	0.12	Y	6
Methacrylate, methyl	Pyridine, 2-vinyl-	0.31		0.64			623
Methacrylate, methyl	Pyridine, 2-vinyl-	0.27	0.04	1.1	0.08	N	749
Methacrylate, methyl	Pyridine, 2-vinyl-5-ethyl-	0.39	0.02	0.68	0.06	N	231
Methacrylate, methyl	Pyridine, 4-vinyl-	0.57	0.02	0.77	0.07	N	231
Methacrylate, methyl	Pyridine, 4-vinyl-	0.54	0.01	0.99	0.18	N	691
Methacrylate, methyl	Pyrrolidone, 1-benzyl-3-methylene-5-methyl	0.387	0.07	2.28	0.47	Y	33
Methacrylate, methyl	Pyrrolidone, *N*-vinyl-					Y	213
Methacrylate, methyl	Pyrrolidone, *N*-vinyl-	5.62	0.3	−0.05	0.08	N	320
Methacrylate, methyl	Pyrrolidone, *N*-vinyl-	4.63	0.34	−0.032	0.068	Y	321
Methacrylate, methyl	Pyrrolidone, *N*-vinyl-	4.36	0.61	−0.052	0.028	Y	36
Methacrylate, methyl	Pyrrolidone, *N*-vinyl-	4.04	0.61	0.01	0.15	N	691
Methacrylate, methyl	Pyrrolidone, α-methylene-*N*-methyl-	0.577	0.058	0.24	0.14	Y	641
Methacrylate, methyl	Silane, γ-methacryloxypropyltrimethoxy-	0.788	0.082	2.06	0.309	Y	977
Methacrylate, methyl	Styrene	0.5		0.5			138
Methacrylate, methyl	Styrene	0.46		0.52			139
Methacrylate, methyl	Styrene	0.46		0.52			179
Methacrylate, methyl	Styrene	0.42		0.48			20
Methacrylate, methyl	Styrene	0.45		0.52			21
Methacrylate, methyl	Styrene	0.46		0.49			21
Methacrylate, methyl	Styrene	0.47		0.45			21
Methacrylate, methyl	Styrene	0.48		0.58			21
Methacrylate, methyl	Styrene	0.5		0.44			21
Methacrylate, methyl	Styrene	0.22	0.13	0.396	0.048	Y	241
Methacrylate, methyl	Styrene	0.32		0.42			241
Methacrylate, methyl	Styrene	0.418	0.085	0.49	0.14	Y	254
Methacrylate, methyl	Styrene	0.54	0.029	0.564	0.047	Y	254
Methacrylate, methyl	Styrene	0.478	0.011	0.585	0.007	Y	287
Methacrylate, methyl	Styrene	0.45	0.01	0.44	0.08	N	324
Methacrylate, methyl	Styrene	0.45		0.47			324
Methacrylate, methyl	Styrene	0.49		0.54			403
Methacrylate, methyl	Styrene	0.47		0.52			416
Methacrylate, methyl	Styrene	0.48		0.38			451
Methacrylate, methyl	Styrene	0.46		0.52			463
Methacrylate, methyl	Styrene	0.42		0.54			492
Methacrylate, methyl	Styrene	0.5		0.44			492
Methacrylate, methyl	Styrene	0.46		0.52			505
Methacrylate, methyl	Styrene	0.44		0.5			531
Methacrylate, methyl	Styrene	0.49		0.56			531
Methacrylate, methyl	Styrene	0.49		0.53			534

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Styrene	0.409	0.02	0.485	0.052	Y	751
Methacrylate, methyl	Styrene	0.504	0.026	0.48	0.059	Y	751
Methacrylate, methyl	Styrene	0.611	0.028	0.371	0.055	Y	751
Methacrylate, methyl	Styrene	0.59		0.62			753
Methacrylate, methyl	Styrene	0.6		0.52			753
Methacrylate, methyl	Styrene	0.63		0.5			753
Methacrylate, methyl	Styrene	0.58		0.55			753
Methacrylate, methyl	Styrene	0.64		0.55			753
Methacrylate, methyl	Styrene	0.41		0.41			753
Methacrylate, methyl	Styrene	0.454		0.472			824
Methacrylate, methyl	Styrene	0.464		0.497			824
Methacrylate, methyl	Styrene	0.422		0.432			824
Methacrylate, methyl	Styrene	0.41		0.57			835
Methacrylate, methyl	Styrene	0.314		0.275			860
Methacrylate, methyl	Styrene, 2,4,6-trimethyl-	1.511	0.072	0.083	0.045	Y	58
Methacrylate, methyl	Styrene, 2,4,6-trimethyl-	1.45	0.16	−0.01	0.11	Y	58
Methacrylate, methyl	Styrene, 2,5-dichloro-	0.44	0.13	2.66	0.38	Y	3
Methacrylate, methyl	Styrene, 2,6-dichloro-	1.98	0.16	−0.15	0.27	N	324
Methacrylate, methyl	Styrene, 3-methyl	0.48	0.04	0.53	0.04		1087
Methacrylate, methyl	Styrene, 3-methyl	0.38	0.03	0.36	0.06		1087
Methacrylate, methyl	Styrene, 3-methyl	0.41	0.04	0.51	0.05		1087
Methacrylate, methyl	Styrene, 3-tri-*n*-butylstannyl-	0.69		0.03			985
Methacrylate, methyl	Styrene, 4-methyl-	0.345	0.008	0.46	0.022	Y	929
Methacrylate, methyl	Styrene, α-methoxy-	2.5	0.39	−0.02	0.04	N	142
Methacrylate, methyl	Styrene, α-methyl-	0.48	0.02	0.27	0.07	N	265
Methacrylate, methyl	Styrene, α-methyl-	0.55		0.51			265
Methacrylate, methyl	Styrene, α-methyl-	0.55		0.6			265
Methacrylate, methyl	Styrene, α-methyl-	0.5		0.14			532
Methacrylate, methyl	Styrene, α-methyl-	0.42		0.22			835
Methacrylate, methyl	Styrene, α-methyl-	0.47		0.48			899
Methacrylate, methyl	Styrene, *m*-bromo-	0.48		1.17			256
Methacrylate, methyl	Styrene, *m*-chloro-					Y	150
Methacrylate, methyl	Styrene, *m*-methyl-	0.512	0.036	0.486	0.02	Y	256
Methacrylate, methyl	Styrene, *m*-nitro-	0.35	0.11	0.8	0.32	Y	222
Methacrylate, methyl	Styrene, *p*-chloro-	0.5		1.37			532
Methacrylate, methyl	Styrene, *o*-chloro-	0.46		1.34			558
Methacrylate, methyl	Styrene, *p*-1-(2-hydroxypropyl)-	0.26		1			380
Methacrylate, methyl	Styrene, *p*-*N*,*N*-dimethylamino-	0.204	0.022	0.101	0.016	Y	256
Methacrylate, methyl	Styrene, *p*-bromo-	0.4		1.1			256
Methacrylate, methyl	Styrene, *p*-chloro-	0.334	0.016	0.77	0.12	Y	150
Methacrylate, methyl	Styrene, *p*-chloro-	0.42		0.89			256
Methacrylate, methyl	Styrene, *p*-chloro-	0.43		0.47			476
Methacrylate, methyl	Styrene, *p*-chloromethyl-	0.408	0.038	0.91	0.15	Y	756
Methacrylate, methyl	Styrene, *p*-cyano-	0.22		1.41			256
Methacrylate, methyl	Styrene, *p*-methoxy-	0.29		0.32			35
Methacrylate, methyl	Styrene, *p*-methyl-	0.4		0.44			256
Methacrylate, methyl	Styrene, *p*-octylamine sulfonate	0.6		2.6			817
Methacrylate, methyl	Styrene, *p*-*tert*-butyl-	0.44		0.5			634
Methacrylate, methyl	Styrene, pentachloro-	3.65	0.85	0.3	0.05	N	7
Methacrylate, methyl	Succinamete, potassium *N*-vinyl-	1.1	0.17	−0.02	0.12	Y	771
Methacrylate, methyl	Succinimide, *N*-vinyl-	9.94	0.27	0.048	0.026	Y	104
Methacrylate, methyl	Succinimide, *N*-vinyl-	0.01	0.031	−0.059	0.096	Y	693
Methacrylate, methyl	Tetrazole, 1-Vinyl-	7.13	0.093	0.262	0.026	Y	925
Methacrylate, methyl	Tetrazole, 2-methyl-5-vinyl-	1.026	0.029	0.486	0.075	Y	670
Methacrylate, methyl	Tetrazole, 2-phenyl-5(4′-vinyl) phenyl-	0.7		0.4			583
Methacrylate, methyl	Thiophene, 3-vinyl	0.563		0.25			1000
Methacrylate, methyl	Thioxanthone, 2-(acryloyloxy)-	1.36		0.5			986
Methacrylate, methyl	Toluenesulfonamide, *N*,*N*-methyl-vinyl-	1.87	0.19	−1.83	0.26	Y	80
Methacrylate, methyl	Triallyl cyanurate	16.12	0.56	−0.005	0.05	Y	201
Methacrylate, methyl	Triallyl isocyanurate	16.88	6.64	−0.49	0.94	Y	201
Methacrylate, methyl	Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vinyl	0.832	0.022	0.266	0.058	Y	670
Methacrylate, methyl	Triazole, 1,2,3-, 4(5)-vinyl-	0.839	0.055	0.27	0.13	Y	670
Methacrylate, methyl	Urea, *N*-vinyl-*N*′-ethyl-	1.85	0.23	−0.008	0.029	Y	101

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Vinyl 2-chloroethyl ether	18.8		0			597
Methacrylate, methyl	Vinyl 4-chlorocyclohexyl ketone	0.77		0.51			668
Methacrylate, methyl	Vinyl acetate	2.71	0.25	−0.2	0.01		158
Methacrylate, methyl	Vinyl acetate	137.5	5.9	−0.81	1.17	N	210
Methacrylate, methyl	Vinyl acetate	26		0.03			272
Methacrylate, methyl	Vinyl acetate	28.6		0.04			383
Methacrylate, methyl	Vinyl acetate	22.21		0.07			471
Methacrylate, methyl	Vinyl benzoate	20.3		0.07			272
Methacrylate, methyl	Vinyl bromide	14.11	0.67	0.12	0.06	N	268
Methacrylate, methyl	Vinyl bromide	17.1		0.33			268
Methacrylate, methyl	Vinyl bromide	25.1	0.88	0.052	0.026	Y	34
Methacrylate, methyl	Vinyl bromide	19.84	2.58	0.05	0.074	Y	34
Methacrylate, methyl	Vinyl butyl ether					N	127
Methacrylate, methyl	Vinyl butyrate	25		0.03			272
Methacrylate, methyl	Vinyl chloride	8.99	2.97	0.07	0.047	Y	3
Methacrylate, methyl	Vinyl chloromethyl ketone	0.121	0.021	0.593	0.092	Y	356
Methacrylate, methyl	Vinyl ether	4.9	0.99	−0.15	0.16	Y	45
Methacrylate, methyl	Vinyl ethyl ether	37		0.01			106
Methacrylate, methyl	Vinyl ethyl ketone	0.42	0.17	0.65	0.42	Y	794
Methacrylate, methyl	Vinyl ethyl oxalate	4.74	2.95	−0.05	0.44	Y	124
Methacrylate, methyl	Vinyl ethyl sulfide	1.03	0.11	0.036	0.017	Y	303
Methacrylate, methyl	Vinyl ethyl sulfide	2.7		0.3			507
Methacrylate, methyl	Vinyl ethyl sulfoxide	11.17	0.48	0.37	0.06	Y	286
Methacrylate, methyl	Vinyl isobutyl sulfide	0.89	0.01	0.05	0.03	N	285
Methacrylate, methyl	Vinyl isobutyl sulfide	0.985	0.077	0.021	0.034	Y	303
Methacrylate, methyl	Vinyl isobutyl sulfide	0.88	0.02	0.03	0.02	N	336
Methacrylate, methyl	Vinyl isopropyl ketone	0.81		1.44			601
Methacrylate, methyl	Vinyl isopropyl ketone	0.54	0.22	0.6	0.47	Y	794
Methacrylate, methyl	Vinyl isopropyl sulfide	0.822	0.064	0.046	0.025	Y	303
Methacrylate, methyl	Vinyl isothiocyanate	0.85		0.6			422
Methacrylate, methyl	Vinyl *m*-cresyl ether	3.3		0.2			605
Methacrylate, methyl	Vinyl methyl sulfide	1		0.03			106
Methacrylate, methyl	Vinyl *o*-cresyl ether	1.8		0.1			605
Methacrylate, methyl	Vinyl *p*-cresyl ether	2.6		0			605
Methacrylate, methyl	Vinyl phenyl ether	11.34	5.76	0.01	0.32	Y	172
Methacrylate, methyl	Vinyl phenyl ether	2.8		0			605
Methacrylate, methyl	Vinyl phenyl ketone	0.71	0.6	0.59	1.9	Y	794
Methacrylate, methyl	Vinyl phenyl sulfide	0.79	0.06	0.06	0.09	N	239
Methacrylate, methyl	Vinyl phenyl sulfide	0.85		0.08			239
Methacrylate, methyl	Vinyl propionate	24		0.03			272
Methacrylate, methyl	Vinyl sulfone	4.5	0.49	0.1	0.16	Y	45
Methacrylate, methyl	Vinyl *tert*-butyl ketone	1.21	0.11	0.65	0.15	Y	794
Methacrylate, methyl	Vinyl, 4-, dithiobenzoate, methyl	0.151	0.092	0.3	0.29	Y	697
Methacrylate, methyl	Vinyl, *p*-,benzylmethylcarbinol	0.24	0.26	0.98	0.28	Y	48
Methacrylate, methyl	Vinyl-tris(trimethoxysiloxy)silane	34		0.02			482
Methacrylate, methyl	Vinylacetophenone, *p*-	0.33	0.36	1.65	3.81	Y	266
Methacrylate, methyl	Vinylanthracene, 9-	4.068	0.04	0.117	0.043	Y	123
Methacrylate, methyl	Vinylbenzoic acid, *p*-	0.398	0.097	1.16	0.3	Y	266
Methacrylate, methyl	Vinylbenzyl chloride	0.46		1.02			792
Methacrylate, methyl	Vinylene carbonate	126	298	−0.04	0.85	Y	103
Methacrylate, methyl	Vinylferrocene	1.22		0.52			294
Methacrylate, methyl	Vinylhydroquinone	0.49	0.21	0.02	0.13	Y	337
Methacrylate, methyl	Vinylhydroquinone dibenzoate	0.34		0.41			440
Methacrylate, methyl	Vinylidene chloride	2.6		0.3		Y	138
Methacrylate, methyl	Vinylidene chloride	2.38	0.02	0.36	0.03	N	324
Methacrylate, methyl	Vinylidene chloride	2.16	0.06	0.18	0.18	N	324
Methacrylate, methyl	Vinylidene chloride	2.5		0.5			426
Methacrylate, methyl	Vinylidene chloride	2.5		0.34			432
Methacrylate, methyl	Vinylidene chloride	2.1		0.09			538
Methacrylate, methyl	Vinylidene chloride	1.8	0.95	0.094	0.032	Y	65
Methacrylate, methyl	Vinylidene chloride	2.83		0.19			762
Methacrylate, methyl	Vinylidene chloride	1.23		0.02			762
Methacrylate, methyl	Vinylidene chloride	2.77	0.06	0.27	0.01	N	786
Methacrylate, methyl	Vinylidene cyanide		0.012		0.013	Y	82
Methacrylate, methyl	Vinylisocyanate	3.3		0.01			422

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylate, methyl	Vinylphenyldimethylsilane	29.79	1.89	−0.009	0.034	Y	482
Methacrylate, methyl	Vinyltrimethylsilane	35.66	0.85	0.028	0.016	Y	482
Methacrylate, methyl	Xanthate, *S*-methacryloyl *O*-ethyl	0.76	0.04	0.6	0.08		1086
Methacrylate, methyl diphenylsilyl	Methacrylate, methyl	0.38	0.178	0.438	0.075	Y	1069
Methacrylate, methylthiomethyl	Methacrylate, methyl	1.1	0.356	0.639	0.057	Y	920
Methacrylate, methylthiomethyl	Styrene	0.512	0.05	0.338	0.01	Y	920
Methacrylate, mono-,ethylene glycol	Acrylonitrile	1.5		0.19			682
Methacrylate, octyl	Styrene	0.56	0.16	0.65	0.1	Y	179
Methacrylate, octyl	Styrene	0.68		0.56			463
Methacrylate, octyl	Vinyl chloride	14		0.04			511
Methacrylate, *p*-(cinnamoylamino)phenyl	Styrene	0.31		0.62			685
Methacrylate, *p*-acetylphenyl	Methacrylate, methyl	0.67		0.47			769
Methacrylate, *p*-carbomethoxyphenyl	Styrene	0.178	0.027	0.482	0.084	Y	177
Methacrylate, *p*-chlorophenyl	Methacrylate, methyl	1.08		0.54			769
Methacrylate, *p*-chlorophenyl	Styrene	0.243	0.022	0.34	0.11	Y	177
Methacrylate, *p*-methoxyphenyl	Methacrylate, methyl	1.19		0.72			769
Methacrylate, *p*-methoxyphenyl	Styrene	0.244	0.026	0.499	0.089	Y	177
Methacrylate, *p*-methylphenyl	Styrene	0.225	0.04	0.49	0.25	Y	177
Methacrylate, *p*-methylphenyl	Styrene	0.185	0.039	0.223	0.09	Y	177
Methacrylate, *p*-tolyl	Methacrylate, methyl	1.28		0.59			769
Methacrylate, phenyl	Acrylonitrile	0.46		0.36			363
Methacrylate, phenyl	Methacrylate, 2-chloroethyl	0.785	0.083	1.02	0.21	Y	179
Methacrylate, phenyl	Methacrylate, benzyl	1.42	0.09	0.67	0.06	N	32
Methacrylate, phenyl	Methacrylate, methyl	1.67		0.53			32
Methacrylate, phenyl	Methacrylate, methyl	1.69		0.58			32
Methacrylate, phenyl	Methacrylate, methyl	1.72		0.56			32
Methacrylate, phenyl	Methacrylate, methyl	1.55		0.66			761
Methacrylate, phenyl	Methacrylate, methyl	0.99		0.55			769
Methacrylate, phenyl	Methacrylic, acid	1.89	0.17	0.211	0.018	Y	88
Methacrylate, phenyl	Methacrylic, acid	0.589	0.081	0.398	0.019	Y	88
Methacrylate, phenyl	*N*-Vinylpyrrolidone	4.17	0.756	−0.026	0.032	Y	1013
Methacrylate, phenyl	Styrene	0.48	0.23	0.246	0.049	Y	177
Methacrylate, phenyl	Styrene	0.55	0.33	0.247	0.064	Y	179
Methacrylate, phenyl	Styrene	0.51		0.26			463
Methacrylate, potassium	Acrylonitrile	0.203	0.086	0.226	0.04	Y	808
Methacrylate, propargyl	Itaconic acid mono(*p*-sulfamidoanilide)	1.65	0.02	0.58	0.05		957
Methacrylate, propyl	Methacrylate, methyl	1.24		1.21			844
Methacrylate, propyl	Styrene	0.38	0.11	0.565	0.081	Y	179
Methacrylate, sodium (2-sulphoxyethyl)-	Methacrylic, acid	1.2	0.19	0.78	0.08	N	745
Methacrylate, sulfolanyl	Methacrylate methyl	7.16	23.7	1.07	2.48	N	359
Methacrylate, sulfolanyl	Methacrylic, acid	0.81	0.24	0.81	0.03	N	359
Methacrylate, sufolanyl	Styrene	−0.2	2.46	0.2	0.1	N	359
Methacrylate, *t*-butyl dimethylsilyl	Acrylate, methyl	1.07	0.21	0.288	0.057	Y	1069
Methacrylate, *t*-butyl dimethylsilyl	Methacrylate, methyl	0.725	0.455	0.897	0.05	Y	1069
Methacrylate, *t*-butyl diphenylsilyl	Methacrylate, methyl	0.396	0.257	0.636	0.014	Y	1069
Methacrylate, *tert*-butyl	Methacrylate, methyl	1.35		0.96			761
Methacrylate, *tert*-butyl	Styrene	0.61	0.51	0.545	0.054	Y	179
Methacrylate, tetra(2-hydroxypropyl)-ethylenediamine	Methacrylate, hydroxyethyl	1.44	1	0.606	0.416	Y	996
Methacrylate, tetra(2-hydroxypropyl)-ethylenediamine	Methacrylic acid	0.328	0.181	1.56	0.237	Y	996
Methacrylate, tributyltin	Methacrylate, 2-ethoxyethyl	0.2	0.444	0.53	0.16	N	945
Methacrylate, tributyltin	Methacrylate, 2-hydroxypropyl	0.728	0.517	1.81	0.275	N	945
Methacrylate, tributyltin	Methacrylate, cyclohexyl	0.395	0.428	0.891	0.136	N	945
Methacrylate, tributyltin	Methacrylate, ethyl	0.299	0.607	0.601	0.123	N	945
Methacrylate, tributyltin	Methacrylate, hydroxyethyl	0.282	0.424	2.33	0.185	N	945
Methacrylate, tributyltin	Methacrylate, methyl	0.62		0.58			1082
Methacrylate, triethyltin	Methacrylate, methyl	0.87		0.62			1082
Methacrylate, trimethyltin	Methacrylate, methyl	1.07		0.63			1082
Methacrylate, triphenyltin	Methacrylate, methyl	0.68		0.6			1082
Methacrylic acid	Acrylamide	1.63		0.57			1028
Methacrylic acid	Acrylamide	4.4		0.58			852
Methacrylic acid	Acrylamide	0.15		0.56			852
Methacrylic acid	Acrylate, butyl	1.25	0.31	0.31	0.1	N	187

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylic acid	Acrylate, ethyl					N	1065
Methacrylic acid	Acrylic acid, 2-ethyl	1.12	0.132	0.194	0.027	Y	1004
Methacrylic acid	Acrylic acid, 2-ethyl	2.29	0.417	0.069	0.04	Y	1004
Methacrylic acid	Acrylic acid, 2-ethyl	1.14		0.23		Y	961
Methacrylic acid	Acrylonitrile	0.2	0.26	0.04	0.13	Y	16
Methacrylic acid	Acrylonitrile	2.38	0.29	0.092	0.06	Y	216
Methacrylic acid	Butadiene, 2-chloro-	−0.035	0.058	2.52	0.2	Y	259
Methacrylic acid	Diphenyl sulfone, 3,3′-bisitaconimido-*m,m′*-	0.564	0.032	0.015	0.058	Y	1042
Methacrylic acid	Maleimide, 2,3-dimethyl-*N*-(2-methacryl)	0.746	0.059	1.33	0.1	Y	619
Methacrylic acid	Methacrylamide	2.73	0.28	0.33	0.25	Y	190
Methacrylic acid	Methacrylamide	1.55		1.28			527
Methacrylic acid	Methacrylamide	2.006	0.09	0.2	0.014	Y	55
Methacrylic acid	Methacrylate, 2-hydroxypropyl	0.99		0.31			886
Methacrylic acid	Methacrylate, *N*-methyl-*N*-phenyl-2-amino	2.16	0.18	0.2	0.01	N	651
Methacrylic acid	Methacrylate, butyl	0.8	0.16	1.26	0.12	N	187
Methacrylic acid	Methacrylate, butyl	0.73	0.2	1.15	0.12	N	224
Methacrylic acid	Methacrylate, diethylaminoethyl	0.63	1.47	2.34	1.59	Y	16
Methacrylic acid	Methacrylate, ethyl	0.57		0.71			820
Methacrylic acid	Methacrylate, glycidyl	0.98	0.16	1.2	0.1	N	224
Methacrylic acid	Methacrylate, isopropyl	2.01		0.47			847
Methacrylic acid	Methacrylate, methyl	0.25	0.05	0.63	0.03		1071
Methacrylic acid	Methacrylate, methyl	1.06	0.02	0.1	0.01		1072
Methacrylic acid	Methacrylate, methyl	0.33	0.02	0.78	0.03		1072
Methacrylic acid	Methacrylate, methyl	0.63	0.02	0.31	0.02		1072
Methacrylic acid	Methacrylate, methyl	0.46	0.03	0.27	0.03		1072
Methacrylic acid	Methacrylate, methyl	1.06	0.02	0.1	0.01		1087
Methacrylic acid	Methacrylate, methyl	0.33	0.02	0.78	0.03		1087
Methacrylic acid	Methacrylate, methyl	0.63	0.02	0.32	0.02		1087
Methacrylic acid	Methacrylate, methyl	2.16	0.28	0.87	0.05	N	203
Methacrylic acid	Methacrylate, methyl	1.63	6.25	1.81	0.16	N	210
Methacrylic acid	Methacrylate, methyl	0.99	0.55	0.56	0.23	Y	216
Methacrylic acid	Methacrylate, methyl	1.18	0.19	0.29	0.04	N	276
Methacrylic acid	Methacrylate, methyl	0.48	0.1	1.28	0.1	N	276
Methacrylic acid	Methacrylate, methyl	1.84	0.24	0.77	0.05	N	312
Methacrylic acid	Methacrylate, methyl	1.38	0.27	0.36	0.08	N	313
Methacrylic acid	Methacrylate, methyl	0.63		1.18			313
Methacrylic acid	Methacrylate, methyl	1.55		0.55			424
Methacrylic acid	Methacrylate, methyl	1.26	0.2	0.209	0.05	Y	54
Methacrylic acid	Methacrylate, methyl	0.78		1.25			630
Methacrylic acid	Methacrylate, methyl	0.68		0.48			819
Methacrylic acid	Methacrylate, phenyl	0.211	0.018	1.89	0.17	Y	88
Methacrylic acid	Methacrylate, phenyl	0.398	0.019	0.589	0.081	Y	88
Methacrylic acid	Methacrylate, sodium (2-sulphoxyethyl)-	0.78	0.08	1.2	0.19	N	745
Methacrylic acid	Methacrylate, sulfolanyl	0.81	0.03	0.81	0.24	N	359
Methacrylic acid	Methacrylate, tetra(2-hydroxypropyl)ethylenediamine	1.56	0.237	0.328	0.181	Y	996
Methacrylic acid	Methacrylonitrile	1.74	0.1	0.64	0.07	N	44
Methacrylic acid	Methacrylonitrile	0.54	3.71	0.52	0.32	N	85
Methacrylic acid	*N*-Vinylimidazole	1.384	0.034	0.145	0.022	Y	879
Methacrylic acid	Phosphonate, vinyl-, diethyl	1.9		0.15			431
Methacrylic acid	Phosphonic acid, α-phenylvinyl-	1.49	0.16	0.17	0.3	Y	130
Methacrylic acid	Piperylene, *cis*-	0.4		0.38			773
Methacrylic acid	Piperylene, *trans*-	0.38		0.33			773
Methacrylic acid	Propene, 2,3-dichloro-	4		0			445
Methacrylic acid	Propene, 2-chloro-3-hydroxy-	4.5		0			445
Methacrylic acid	Pyridine, 2-methyl-5-vinyl-	0.43		0.85			552
Methacrylic acid	Pyridine, 2-vinyl-	0.58	0.21	1.69	0.71	Y	15
Methacrylic acid	Pyridine, 2-vinyl-	0.44		1.38			498
Methacrylic acid	Styrene	0.55	0.66	0.21	0.02	N	166
Methacrylic acid	Styrene	0.64		0.22			166
Methacrylic acid	Styrene	0.66		0.2			203
Methacrylic acid	Styrene	0.49	0.08	0.15	0.04	N	312

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylic acid	Styrene	0.56	0.19	0.2	0.04	N	313
Methacrylic acid	Styrene	0.28		0.38			313
Methacrylic acid	Styrene	0.602	0.088	0.124	0.092	Y	47
Methacrylic acid	Styrene	0.85		0.17			812
Methacrylic acid	Styrene	0.631	0.146	0.041	0.03	Y	842
Methacrylic acid	Styrene	0.6	0.249	0.067	0.046	Y	842
Methacrylic acid	Styrene	0.44	0.089	0.627	0.031	Y	842
Methacrylic acid	Styrene	0.39	0.199	0.55	0.06	Y	842
Methacrylic acid	Styrene	0.12	0.118	0.221	0.048	Y	842
Methacrylic acid	Styrene, 3-methyl	0.35	0.08	0.06	0.03		1087
Methacrylic acid	Styrene, 3-methyl	0.18	0.02	0.33	0.02		1087
Methacrylic acid	Styrene, 3-methyl	0.51	0.01	0.42	0.01		1087
Methacrylic acid	Styrene, *o*-chloro-	0.622	0.058	0.101	0.034	Y	6
Methacrylic acid	Sulfonic acid, *o*-methacryloylamino-benzene-	0.203	0.067	0.217	0.122	Y	1006
Methacrylic acid	Sulfonic acid, *o*-methacryloylamino-benzene-	0.188	0.076	0.523	0.195	Y	1006
Methacrylic acid	Sulfonic acid, *p*-methacryloylamino-benzene-	0.156	0.081	1.08	0.338	Y	1006
Methacrylic acid	Sulfonic acid, *p*-methacryloylamino-benzene-	0.155	0.052	1.65	0.27	Y	1006
Methacrylic acid	Vinyl acetate	0.2		0.01			377
Methacrylic acid	Vinyl chloride	23.52	0.98	0.064	0.008	Y	135
Methacrylic acid	Vinyl chloride	23.26	0.96	−0.022	0.02	Y	229
Methacrylic acid	Vinyl chloride	9.05		0.07			46
Methacrylic acid	Vinyl, mono-, diethyleneglycol ether	0		2.2			831
Methacrylic acid	Vinylethynyl-4-piperidinol, 1,2,5-trimethyl	0.4		0.05			734
Methacrylic acid	Vinylidene chloride	3.368	0.098	0.154	0.047	Y	6
Methacrylic anhydride	Acrylate, methyl	4.75		0.16			275
Methacrylic anhydride	Allyl chloroacetate	29.76	13	−0.01	0.13	Y	106
Methacrylic anhydride	Allylurea	29.01	18.48	−0.06	0.17	Y	106
Methacrylic anhydride	Methacrylate, methyl	1.58	0.06	0.22	0.02	N	275
Methacrylic anhydride	Methacrylonitrile	1.45	0.18	0.26	0.06	N	219
Methacrylic anhydride	Styrene	0.27	0.02	0.13	0.01	N	219
Methacrylic anhydride	Styrene	0.33		0.1			275
Methacrylic anhydride	Vinyl benzyl sulfide	0.97	0.27	−0.048	0.033	Y	106
Methacrylic anhydride	Vinyl chloroethyl ether					Y	106
Methacrylonitrile	Acenaphthalene	0.15		2.38			575
Methacrylonitrile	Acetylene, phenyl-	0.78		0.04			648
Methacrylonitrile	Acrolein	1.17	0.05	0.68	0.12	N	208
Methacrylonitrile	Acrolein	1.2		0.72			446
Methacrylonitrile	Acrolein, methyl-	0.41	0.01	1.78	0.07	N	208
Methacrylonitrile	Acrolein, methyl-	0.45		1.25			587
Methacrylonitrile	Acrolein, methyl-	0.45		1.25			589
Methacrylonitrile	Acrylate, 1,1,5-trihydroperfluoroamyl	1.03		0.14			809
Methacrylonitrile	Acrylate, α-chloro-, ethyl	0.45		2			638
Methacrylonitrile	Acrylate, α-cyano-, methyl	0.18	0.01	0.15	0.01	N	143
Methacrylonitrile	Acrylate, α-phenyl-, methyl	0.19		0.25			143
Methacrylonitrile	Acrylic anhydride	0.42	0.04	1	0.13	N	219
Methacrylonitrile	Acrylonitrile	1.67	0.12	0.43	0.04	N	198
Methacrylonitrile	Allylbenzene	16.24	1.36	−0.061	0.03	Y	351
Methacrylonitrile	Butadiene	0.053	0.016	0.39	0.15	Y	257
Methacrylonitrile	Ethylene 1,1-diphenyl-	0.48	0.1	−0.45	0.11	N	143
Methacrylonitrile	Fumarate, diisopropyl	85		0			1038
Methacrylonitrile	Itaconate, dimethyl	1.26		0.28			398
Methacrylonitrile	Methacrylate, butyl	0.56	0.15	0.75	0.07	N	44
Methacrylonitrile	Methacrylate, ethyl	0.46	0.04	0.83	0.05	N	44
Methacrylonitrile	Methacrylate, isobutyl	0.74	0.07	0.68	0.05	N	44
Methacrylonitrile	Methacrylate, isopropyl	0.43	0.08	0.94	0.12	N	44
Methacrylonitrile	Methacrylate, octyl	0.75	0.04	0.61	0.05	N	44
Methacrylonitrile	Methacrylic acid	0.64	0.07	1.74	0.1	N	44
Methacrylonitrile	Methacrylic acid	0.52	0.32	0.54	3.71	N	85
Methacrylonitrile	Methacrylic anhydride	0.26	0.06	1.45	0.18	N	219
Methacrylonitrile	Phosphonic acid, α-phenylvinyl-	2.24		0			590

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Methacrylonitrile	Styrene	0.41		0.37			311
Methacrylonitrile	Styrene	0.42		0.38			311
Methacrylonitrile	Styrene	0.28		0.43			542
Methacrylonitrile	Styrene	0.21		0.34			707
Methacrylonitrile	Styrene	0.25		0.25			75
Methacrylonitrile	Styrene	0.21		0.34			835
Methacrylonitrile	Styrene, α-methoxy-	0.83	0.04	−0.02	0.005	N	143
Methacrylonitrile	Styrene, α-methyl-	0.21		0.15			542
Methacrylonitrile	Styrene, *o*-chloro-	0.86		0.78			44
Methacrylonitrile	Vinyl acetate	12		0.01			75
Methacrylonitrile	Vinyl cinnamate	4.6		0.15			200
Methacrylonitrile	Vinylidene chloride	2.43	0.1	0.3	0.1	N	781
Methacrylonitrile	Vinylidene chloride	2.38	0.05	0.36	0.05	N	781
Methacrylophenone, α-	Methacrylate, methyl	0.11	0.02	1.79	0.07	N	798
Methacrylophenone, α-	Methacrylate, methyl	0.119	0.009	1.74	0.22	Y	798
Methacryloxymethylpentamethyldisiloxane	Methacrylate, 2-hydroxyethyl	0.33		0.97			822
Methacryloxypropyltrimethoxysilane	Styrene	0.58		0.36			951
Methacryloyl chloride	Acrylonitrile	2.8		0.35			580
Methacryloyl chloride	Methacrylate, methyl	0.2		0.45			591
Methacryloyl chloride	Methacrylate, methyl	0.2		0.45			593
Methacryloyl chloride	Pyrrolidone, *N*-vinyl-	0.58		0.05			599
Methacryloyl chloride	Vinyl chloride	2		0.3			432
Methacryloyl fluoride	Methacrylate, methyl	0.43	0.12	0.405	0.037	Y	617
Methacryloyl fluoride	Methacrylate, methyl	0.9	0.16	0.53	0.06	N	660
Methacryloyl fluoride	Styrene	0.164	0.045	0.069	0.014	Y	660
Methacryloyl fluoride, perfluoro-	Vinylidene fluoride	0.02		7.6			669
Methacryloyl isocyanate	Methacrylate, methyl	0.35		0.75			944
Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	Acrylate, *N*-(2-hydroxyethyl)carbazolyl	0.08	0.026	1.04	0.14	Y	702
Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	Acrylate, *N*-(2-hydroxyethyl)carbazolyl	1.364	0.018	0.334	0.022	Y	719
Methacryloyl-β-hydroxyethyl-3,5-dinitrobenzoate	Methacrylate, *N*-ethyl-3-hydroxymethyl-carbazolyl	0.554	0.019	0.729	0.051	Y	701
Methacryloylacetone	Acrylonitrile	3.74	0.22	0.01	0.004	Y	339
Methacryloylacetone	Pyrrolidone, *N*-vinyl-	5.66	0.4	0.008	0.004	Y	339
Methacryloylacetone	Styrene	1.66	0.16	0.067	0.007	Y	339
Methacryloyloxy, 2-benzoate methyl	Methacrylate, *N*-methyl-*N*-phenyl-2-amino ethyl	1.11	0.18	0.55	0.04	N	651
Methacryloyloxy, 2-benzoate methyl	Styrene	0.48	0.02	0.27	0.04	N	651
Methacryloyloxy, 2-benzoic acid methyl	Methacrylate, *N*-methyl-*N*-phenyl-2-amino ethyl	2.6	0.08	0.2	0.01	N	651
Methacryloyloxy, 2-benzoic acid	Styrene	1.17	0.03	0.15	0.04	N	651
Methacryloyloxy, 4-,2,2,6,6-tetramethylpiperidine	Styrene	0.32		0.6			687
Methacryloyloxy-, *o*-, benzoic acid	Methacrylate, methyl	1.37	0.64	0.908	0.252	Y	938
Methacryloyloxy-, *o*-, benzoic acid	Vinylchloride	11.26	3.5	0.004	0.052	Y	938
Methylenebutyrolactone	Acrylamide	2.85		0.3			746
Methylenebutyrolactone	Acrylonitrile	1.1		0.09			745
Methylenebutyrolactone	Methacrylate, methyl	1.67		0.6			746
Methylenebutyrolactone	Styrene	0.7		0.09			746
Methyleneglutaronitrile, 2-	Butadiene					N	196
Methyleneglutaronitrile, 2-	Methacrylate, methyl	0.12	0.74	1.36	0.28	N	196
Methyleneglutaronitrile, 2-	Methacrylate, methyl	0.12		1.27			196
Methyleneglutaronitrile, 2-	Styrene	−0.02	1.3	0.42	0.52	N	196
Methyleneglutaronitrile, 2-	Styrene	0.02		0.5			196
Methyleneglutaronitrile, 2-	Styrene	0.25		0.85			512
Methyleneglutaronitrile, 2-	Vinyl methyl ketone	1.24		5.05			512
Myrcene	Methacrylate, methyl	0.448	0.06	0.256	0.013	Y	1062
Myrcene	Styrene	1.38	0.081	0.891	0.015	Y	1062
Myrcene	Styrene *p*-fluoro-	0.87	0.143	0.362	0.022	Y	1062
N-N-Divinylaniline	Acrylonitrile	−0.006	0.07	0.246	0.014	Y	45
N-N-Divinylaniline	Methacrylate, methyl	0.005	0.018	1.06	0.18	Y	45
N-N-Divinylaniline	Styrene	0.033	0.067	5.38	0.36	Y	45
N-N-Divinylaniline	Styrene, *p*-methyl-	0.06	0.03	6.15	0.16	Y	45
N-N-Divinylaniline	Vinyl acetate	3.99	0.12	0.058	0.009	Y	45

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
N-Vinylimidazole	Acrylic acid	0.187	0.025	1.932	0.077	Y	879
N-Vinylimidazole	Acrylonitrile	0.832	0.044	3.393	0.008	Y	879
N-Vinylimidazole	Fumarate, diethyl	0.113	0.019	0.191	0.022	Y	879
N-Vinylimidazole	Fumarate, dimethyl	0.133	0.021	0.252	0.015	Y	879
N-Vinylimidazole	Fumaric acid	0.121	0.01	0.313	0.012	Y	879
N-Vinylimidazole	Maleate, diethyl	0.057	0.028	0.106	0.147	Y	879
N-Vinylimidazole	Maleate, dimethyl	0.046	0.046	0.046	0.438	Y	879
N-Vinylimidazole	Maleic acid	0.018	0.007	0.094	0.027	Y	879
N-Vinylimidazole	Methacrylate, methyl	0.663	0.104	4.306	0.185	Y	879
N-Vinylimidazole	Methacrylic acid	0.145	0.022	1.384	0.034	Y	879
N-Vinylimidazole	Styrene	0.68	0.02	8.377	0.149	Y	879
N-Vinylpyrolidone	Caprolactam, *N*-vinyl-	2.8	0.2	1.7	0.2		883
N-Vinylpyrolidone	Crotonate, hydrocortisone 21-	0.75	0.02	0.05	0.05		1022
N-Vinylpyrrolidone	2-Oxazoline, 2-isopropenyl-	0.01	0.02	3.5	1.2		894
N-Vinylpyrrolidone	Acrylamido-2-methylpropane sulfonate sodium salt	0.13		0.66			939
N-Vinylpyrrolidone	Acrylate, cyclohexyl	0.122	0.014	1.297	0.088		1061
N-Vinylpyrrolidone	Carbamate, *N*-vinyl-*t*-butyl-	0.4	0.06	2.4	0.5		991
N-Vinylpyrrolidone	Ethene, 2-phenyl-1,1-dicyano-					Y	947
N-Vinylpyrrolidone	Itaconate, bis(tri-*n*-butyltin)	0.21	0.013	0.065	0.04	Y	941
N-Vinylpyrrolidone	Methacrylate, 2-acetylsalicylicoyloxy-3-hydroxypropyl	0.34		0.85			1080
N-Vinylpyrrolidone	Methacrylate, furfuryl	0.0036	0.02	5.16	1.13	Y	1050
N-Vinylpyrrolidone	Methacrylate, glycidyl	0.003	0.018	4.29	0.498		1017
N-Vinylpyrrolidone	Methacrylate, methyl	0.006	2.29	4.78	0.236		1084
N-Vinylpyrrolidone	Methacrylate, methyl	0.014	0.899	5.93	0.198		1084
N-Vinylpyrrolidone	Methacrylate, methyl	0.027	0.062	2.07	0.247		1084
N-Vinylpyrrolidone	Methacrylate, methyl	0.066	3.96	4.04	0.956		1084
N-Vinylpyrrolidone	Methacrylate, phenyl	−0.026	0.032	4.17	0.756	Y	1013
N-Vinylpyrrolidone	Pyridine, 2-methyl-5-vinyl-	0.039	0.006	13	2		1070
N-Vinylpyrrolidone	Pyridine, 2-vinyl-	0.014	0.002	12.4	2.3		1070
N-Vinylpyrrolidone	Pyridine, 4-vinyl-	0.01	0.002	9.8	1.5		1070
N-Vinylpyrrolidone	Quinine	2.2		0.37			922
N-Vinylpyrrolidone	Styrene sulfonate, sodium salt	0.084		7.19			939
N-Vinylpyrrolidone	Vinyl trimethylammonioethyl ether iodide	11.6	0.03	0	0.02		1037
Naphthalene, 1-vinyl-	Acrylonitrile	0.451	0.088	0.107	0.028		967
Naphthalene, 1-vinyl-	Methacrylate, methyl	0.574	0.073	0.321	0.096		967
Naphthalene, 1-vinyl-	Styrene	2.02	0.401	0.699	0.233		967
Naphthalene, 1-vinyl-4-chloro-	Styrene	0.75	0.45	0.76	0.68		806
Naphthalene, 2-vinyl-	Styrene	1.345	0.093	0.395	0.049		806
Naphthalene, 2-vinyl-6-chloro-	Styrene	2.44	1.38	0.39	0.6		806
Naphthalene, 6-vinyl-2-*tert*-butyl	Styrene	1.014	0.244	0.616	0.069		806
Nickel diacrylate	Titanium, dicylopentadienyl-, dimethacrylate	0.95		0.65			895
Norbornadiene	Acrylate, ethyl	−0.01	0.03	2.39	0.44	N	267
Norbornadiene	Acrylonitrile	0.05	0.23	0.48	0.27	N	267
Norbornadiene	Acrylonitrile	0.44	0.05	0.66	0.01	N	69
Norbornadiene	Methacrylate, methyl	−0.5	0.52	14.68	16.78	N	267
Norbornadiene	Styrene, *p*-chloro-	0	0.12	69.93	24.19	Y	188
Norbornadiene	Vinyl acetate	1.354	0.07	0.818	0.019	Y	188
Norbornadiene	Vinyl chloride	0.41	0.01	0.67	0.02	N	267
Norbornadiene	Vinylidene chloride	0.09	0.17	1.36	0.63	N	267
Oct-1-ene, 4,8-dioxaspiro(2.5)-	Styrene	0.246	0.024	1.831	0.027	Y	838
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	Acrylonitrile	1.484	1.078	0.985	0.675	Y	838
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	Pyridine, 2-vinyl-	0.073	0.04	2.216	0.062	Y	838
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	Pyrrolidone, *N*-vinyl-	0.552	0.075	−0.009	0.018	Y	838
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	Styrene	0.271	0.109	1.93	0.098	Y	838
Octene-1	Vinyl chloride	−0.57	0.85	3.26	0.45	N	329
Oxazinium F3CSO3, 3-methyl-2-vinyl-5,6-dihydro-4H-1,3-	Methacrylate, methyl	0.31		0.37			909

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Oxazolidone, *N*-vinyl-	Methacrylate, decyl	−0.052	0.049	12.19	1.35	Y	36
Oxazolidone, *N*-vinyl-	Methacrylate, methyl	0.04	0.13	6.75	0.37	Y	102
Oxazolidone, *N*-vinyl-	Methacrylate, methyl	0.027	0.008	9.46	0.39	Y	36
Oxazolidone, *N*-vinyl-	Styrene	−0.029	0.014	29.42	2.19	Y	36
Oxazolidone, *N*-vinyl-	Vinyl acetate	1.55	0.28	0.602	0.038	Y	102
Oxazolidone, *N*-vinyl-	Vinyl acetate	2.04	0.13	0.55	0.18	Y	36
Oxazolidone, *N*-vinyl-	Vinyl chloride	0.822	0.048	0.313	0.095	Y	36
Oxazolidone, *N*-vinyl-	Vinylidene chloride	0.091	0.077	1.42	0.11	Y	102
Oxazoline, 2-, 2-isopropenyl-	Acrylate, butyl	1.4		0.24			862
Oxazoline, 2-, 2-isopropenyl-	Acrylate, ethyl	1.39		0.19			862
Oxazoline, 2-, 2-isopropenyl-	Acrylate, methyl	1.9		0.16			862
Oxazoline, 2-, 2-isopropenyl-	Acrylate, methyl	2.07	0.191	0.328	0.032	Y	877
Oxazoline, 2-, 2-isopropenyl-	Acrylonitrile	0.52		0.13			862
Oxazoline, 2-, 2-isopropenyl-	Methacrylate, methyl	0.99		0.69			862
Oxazoline, 2-, 2-isopropenyl-	Pyrrolidone, *N*-vinyl-	3.5		0.01			862
Oxazoline, 2-, 2-isopropenyl-	Styrene	0.64		0.67			862
Oxazoline, 2-, 2-isopropenyl-	Vinyl acetate	7.1		0.25			862
Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl-	Acrylate, methyl	1.3		0.46			596
Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl-	Acrylonitrile	1.83		0.24			596
Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl-	Methacrylate, methyl	0.82		0.53			596
Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl-	Styrene	0.68		0.55			596
Oxazoline, 2-, 4-acryloxymethyl-2,4-dimethyl-	Acrylate, methyl	0.11		0.29			596
Oxazoline, 2-, 4-acryloxymethyl-2,4-dimethyl-	Methacrylate, methyl	0.11		0.41			596
Oxazoline, 2-, 4-acryloxymethyl-2,4-dimethyl-	Styrene	0.14		0.56			596
Oxazoline, 2-, 4-acryloxymethyl-2,4-dimethyl-	Vinyl acetate	1.41		0.03			596
Oxazoline, 2-, 4-methacryloxy-2,4-dimethyl-	Acrylate, methyl	0.56		0.4			596
Oxazoline, 2-, 4-methacryloxy-2,4-dimethyl-	Acrylonitrile	0.89		0.11			596
Oxazoline, 2-, 4-methacryloxy-2,4-dimethyl-	Methacrylate, methyl	0.19		1.17			596
Oxazoline, 2-, 4-methacryloxy-2,4-dimethyl-	Vinyl acetate	6.2		0.05			596
Oxazoline, 2-isopropenyl-2-	Acrylate, butyl	1.4	0.08	0.24	0.01		894
Oxazoline, 2-isopropenyl-2-	Acrylate, ethyl	1.39	0.06	0.19	0.02		894
Oxazoline, 2-isopropenyl-2-	Acrylate, methyl	1.9	0.08	0.16	0.04		894
Oxazoline, 2-isopropenyl-2-	Acrylonitrile	0.52	0.06	0.13	0.04		894
Oxazoline, 2-isopropenyl-2-	Methacrylate, methyl	0.99	0.07	0.69	0.02		894
Oxazoline, 2-isopropenyl-2-	*N*-Vinylpyrrolidone	3.5	1.2	0.01	0.02		894
Oxazoline, 2-isopropenyl-2-	Styrene	0.64	0.03	0.67	0.02		894
Oxazoline, 2-isopropenyl-2-	Vinyl acetate	7.1	2	0.25	0.04		894
Oxazolinium BF_4, 3-methyl-2-isopropenyl-2-	Acrylate, methyl	0.251	0.483	0.096	0.057	Y	908
Oxazolinium BF_4, 3-methyl-2-isopropenyl-2-	Vinyl acetate	0.413	0.005	0.114	0.021	Y	908
Oxazolinium, 2-, tetrafluoroborate 3-methyl-2-isopropenyl-	Acrylate, methyl	0.344	0.105	0.552	0.043	Y	877
Oxazolinium, 2-, tetrafluoroborate 3-methyl-2-isopropenyl-	Vinyl acetate	0.366	0.151	0.43	0.057	Y	877
Oxazolinium, 2-isopropenyl-2-	Acrylate methyl	2.08	0.327	0.191	0.031	Y	908
Oxirane, 4-vinylphenyl-	Styrene	1.06		0.78			1001
Pentadiene, *cis*-1,3-	Acrylonitrile	0.172	0.012	0.026	0.018	Y	867
Pentadiene, *cis*-1,3-	Acrylonitrile	0.241	0.029	0.06	0.071	Y	867
Pentadiene, *cis*-1,3-	Acrylonitrile	0.017	0.012	0.026	0.173	Y	912
Pentadiene, *cis*-1,3-	Acrylonitrile	0.071	0.029	0.06	0.242	Y	912
Pentadiene, *cis*-1,3-	Acrylonitrile	0.198		0.056			952
Pentadiene, *cis*-1,3-	Acrylonitrile	0.188	0.011	0.033	0.017	N	997

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Pentadiene, *cis*-1,3-	Butadiene	0.26		0.63			743
Pentadiene, *cis*-1,3-	Pentadiene, *trans*-1,3-	0.86		0.37			743
Pentadiene, *trans*-1,3-	Acrylonitrile	0.104	0.01	0.07	0.017	Y	867
Pentadiene, *trans*-1,3-	Acrylonitrile	0.095	0.021	0.06	0.064	Y	867
Pentadiene, *trans*-1,3-	Acrylonitrile	0.018	0.01	0.069	0.104	Y	912
Pentadiene, *trans*-1,3-	Acrylonitrile	0.064	0.02	0.068	0.096	Y	912
Pentadiene, *trans*-1,3-	Acrylonitrile	0.132		0.085			952
Pentadiene, *trans*-1,3-	Acrylonitrile	0.114	0.009	0.079	0.018	N	997
Pentadiene, *trans*-1,3-	Butadiene	0.09		1.22			743
Pentadiene, *trans*-1,3-	Methacrylate, methyl	0.379	0.013	0.404	0.038	Y	1058
Pentadiene, *trans*-1,3-	Pentadiene, *cis*-1,3-	0.37		0.86			743
Pentadienoate, *trans*-4-ethoxy-2,4-,ethyl	Styrene	12.2		0.04			723
Pentadienoate, *trans*-4-ethoxy-2,4-,ethyl	Acrylonitrile	5.4		0.01			723
Pentadienoate, *trans*-4-ethoxy-2,4-	Styrene	11.8		0.08			723
Penten-2,4-one	Styrene	−0.033	0.055	13.84	0.57	Y	112
Pentene-1	Vinyl chloride	0.001		4			3
Pentene-1	Vinyl chloride	0.81	5.2	2.08	1.78	N	329
Pentene-1, 2-methyl-	Vinyl chloride	−1.16	2.31	1.59	0.68	N	329
Pentene-1, 5-cyclohexyl-	Acrylonitrile	−0.043	0.044	4.22	0.45	N	24
Pentene-1, 5-phenyl-	Acrylonitrile	−0.026	0.046	3.84	0.48	Y	24
Pentenoate, 3-oxo-4-, ethyl	Styrene	3.29	0.712	0.233	0.069	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	1.57	0.193	0.161	0.028	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	1.29	0.309	0.215	0.051	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	1.4	0.322	0.189	0.05	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	0.923	0.075	0.179	0.014	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	0.747	0.073	0.164	0.015	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	0.635	0.068	0.162	0.015	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	1.12	0.112	0.178	0.02	N	930
Pentenoate, 3-oxo-4-, ethyl	Styrene	0.568	0.035	0.217	0.009	N	930
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	3.39	0.263	0.555	0.032	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	2.02	0.141	0.424	0.022	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	3.06	0.349	0.591	0.046	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.67	0.183	0.443	0.032	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.24	0.168	0.34	0.031	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.72	0.123	0.57	0.023	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.69	0.236	0.606	0.045	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	0.72	0.12	0.269	0.027	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.55	0.224	0.715	0.048	N	1033
Pentenoate, 3-oxo-4-, ethyl	Methacrylate, methyl	1.09	0.09	0.528	0.02	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	2.56	0.218	0.597	0.03	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	0.748	0.155	0.38	0.034	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	1.14	0.109	0.513	0.022	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	0.952	0.164	0.43	0.034	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	1.67	0.143	0.474	0.023	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	1.04	0.111	0.581	0.025	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	1.05	0.108	0.616	0.024	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	0.725	0.08	0.663	0.023	N	1033
Pentenoate, 4-methyl-3-oxo-4-, ethyl	Methacrylate, methyl	1.09	0.09	0.528	0.02	N	1033
Phenol, 3-(methacryloylamino)-	Styrene	0.99	0.04	0.5	0.06		896
Phenol, 4-(methacryloylamino)-	Styrene	0.86	0.03	0.71	0.04		896
Phosphate diethyl isopropenyl	Acrylonitrile	0.03		15.2			650
Phosphate diethyl isopropenyl	Vinyl acetate	0.44		1.56			650
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.486	0.088	2.37	0.046	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.521	0.138	2.23	0.047	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.946	0.436	1.87	0.086	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	1.16	0.248	1.97	0.029	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.42	0.097	2.1	0.045	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.258	0.226	2.56	0.094	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Methacrylate, methyl	0.0701	0.052	2.04	0.012	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.171	0.01	0.629	0.003	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.15	0.05	0.612	0.004	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.146	0.073	0.595	0.005	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.168	0.355	0.597	0.044	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.094	0.02	0.633	0.002	N	971
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.23	0.187	0.61	0.018	N	971

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Phosphate, 2-(acryloyloxy)ethyl diethyl	Styrene	0.108	0.034	0.599	0.002	N	971
Phosphate, vinyl bis(chloroethyl)	Styrene	0.03	0.23	2.47	0.78	Y	133
Phosphate, vinyl diethyl	Vinyl chloride	0.15	0.047	3.4	0.21	Y	779
Phosphate, vinyl diethyl	Vinyl chloroacetate	0.374	0.046	2.01	0.2	Y	779
Phosphate, vinyl diphenyl	Vinyl chloride	0.327	0.023	3.009	0.077	Y	779
Phosphate, vinyl diphenyl	Vinyl chloroacetate	0.53	0.12	1.42	0.15	Y	779
Phosphazene, isopropenylpentafluoro-	Styrene	−0.121	0.088	1.6	0.27	Y	765
Phosphazene, isopropenylpentafluoro-	Styrene, *p*-chloromethyl-	−0.048	0.067	1.64	0.58	Y	765
Phosphine oxide, diphenylvinyl-	Acrylonitrile	0.16		3.8			843
Phosphine oxide, diphenylvinyl-	Vinyl acetate	1.7		0.38			843
Phosphonate, 2-(4-vinylphenyl)ethyl, diethyl	Styrene	1.36		0.55			796
Phosphonate, 4-vinylphenyl, diethyl	Styrene	1.4		0.7			796
Phosphonate, α-bromovinyl-, diethyl	Methacrylate, methyl	0.04		1.25			684
Phosphonate, α-carboethoxyvinyl-, diethyl	Methacrylate, methyl	0.05		1			684
Phosphonate, α-carboethoxyvinyl-, diethyl	Phosphonate, vinyl-, diethyl	0.55		0.62			684
Phosphonate, α-carboethoxyvinyl-, diethyl	Methacrylate, methyl	0.01		4.7			684
Phosphonate, isopropenyl-, dimethyl	Methacrylate, methyl	0.2		30			684
Phosphonate, isopropenyl-, dimethyl	Styrene	0.23		8.6			684
Phosphonate, isopropenyl-, dimethyl	Vinyl acetate	0.99		0.14			684
Phosphonate, vinyl-, bis(2-chloroethyl)	Ethylene	7.73	2.19	0.077	0.003	Y	801
Phosphonate, vinyl-, bis(2-chloroethyl)	Methacrylate, methyl	0.26		29.9			410
Phosphonate, vinyl-, bis(2-chloroethyl)	Styrene	0.02	0.39	2.46	0.43	Y	133
Phosphonate, vinyl-, bis(2-chloroethyl)	Styrene	0.16		2.3			410
Phosphonate, vinyl-, bis(2-chloroethyl)	Styrene	0.2		2.2			485
Phosphonate, vinyl-, bis(2-chloroethyl)	Styrene	0.18		6.32			592
Phosphonate, vinyl-, bis(2-chloroethyl)	Styrene	0.18		6.32			594
Phosphonate, vinyl-, bis(2-chloroethyl)	Vinyl acetate	0.85		0.09			592
Phosphonate, vinyl-, bis(2-chloroethyl)	Vinyl acetate	0.85		0.09			594
Phosphonate, vinyl-, bis(2-chloroethyl)	Vinyl acetate	0.47		0.33			595
Phosphonate, vinyl-, dibutyl	Styrene	−0.12	2.33	4.97	3.44	Y	129
Phosphonate, vinyl-, diethyl	Methacrylate, methyl	0.04		11.1			684
Phosphonate, vinyl-, diethyl	Methacrylic acid	0.15		1.9			431
Phosphonate, vinyl-, diethyl	Phosphonate, α-carbomethoxyvinyl-, diethyl	0.62		0.55			684
Phosphonate, vinyl-, diethyl	Styrene	−0.15	2.18	4.36	4.27	Y	129
Phosphonate, vinyl-, diethyl	Styrene	−0.52	0.22	1.59	0.24	Y	18
Phosphonate, vinyl-, diethyl	Styrene	0.02		2.45			592
Phosphonate, vinyl-, diethyl	Styrene	0		3.88			684
Phosphonate, vinyl-, diethyl	Vinyl acetate	0.74		0.04			592
Phosphonate, vinyl-, diethyl	Vinyl acetate	0.6		0.09			684
Phosphonate, vinyl-, diisobutyl	Ethylene					Y	801
Phosphonate, vinyl-, diisobutyl	Styrene	−0.32	1.46	3.87	1.59	Y	129
Phosphonate, vinyl-, diisopropyl	Styrene	−0.52	2.52	2.57	2.13	Y	129
Phosphonate, vinyl-, dimethyl	Butadiene, 2-chloro-	0.1		7.5			594
Phosphonate, vinyl-, dimethyl	Styrene					Y	129
Phosphonate, vinyl-, dimethyl	Styrene	0.4		4.61			129
Phosphonate, vinyl-, dimethyl	Styrene	0.15		2.45			592
Phosphonate, vinyl-, dimethyl	Styrene	0.02		2.45			594
Phosphonate, vinyl-, dimethyl	Vinyl acetate	0.74		0.04			592
Phosphonate, vinyl-, dimethyl	Vinyl acetate	0.74		0.04			594
Phosphonate, vinyl-, diphenyl	Ethylene	1.66	2.22	0.088	0.01	Y	801
Phosphonate, vinyl-, diphenyl	Styrene	−0.38	0.29	1.74	0.16	Y	129
Phosphonate, vinyl-, dipropyl	Styrene	−0.41	0.72	3.81	0.68	Y	129
Phosphonic acid, α-phenylvinyl	Vinylidene cyanide	0.1		4.58			921
Phosphonic acid, α-phenylvinyl-	Acrylic acid	0.04	0.83	0.7	0.13	Y	132
Phosphonic acid, α-phenylvinyl-	Acrylonitrile	−0.45	0.41	0.34	0.12	Y	132
Phosphonic acid, α-phenylvinyl-	Acrylonitrile	0		2.36			590
Phosphonic acid, α-phenylvinyl-	Butadiene, 2,3-dichloro-	0.07		7.75			588
Phosphonic acid, α-phenylvinyl-	Butadiene, 2,3-dichloro-	0.07		7.75			597
Phosphonic acid, α-phenylvinyl-	Butadiene, 2-chloro-	0.1		7.5			546
Phosphonic acid, α-phenylvinyl-	Butadiene, 2-chloro-	0.1		7.5			597
Phosphonic acid, α-phenylvinyl-	Methacrylate, methyl	−0.063	0.077	1.362	0.084	Y	131
Phosphonic acid, α-phenylvinyl-	Methacrylic acid	0.17	0.3	1.49	0.16	Y	130
Phosphonic acid, α-phenylvinyl-	Methacrylonitrile	0		2.24			590

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Phthalide, 3-(4-vinylphenyl)-	Acrylonitrile	0.43	0.19	0.084	0.053	Y	721
Phthalimide, *N*-(4-vinylphenyl)-	Acrylate, methyl	1.56		1.14			636
Phthalimide, *N*-(methacryloyloxy)-	Acrylate, methyl	1.23	0.168	0.193	0.028	Y	979
Phthalimide, *N*-(methacryloyloxy)-	Acrylonitrile	1.45	0.072	0.204	0.016	Y	979
Phthalimide, *N*-(methacryloyloxy)-	Methacrylate, methyl	1.45	0.151	0.69	0.048	Y	979
Phthalimide, *N*-vinyl-	Acrylonitrile	0.24	0.02	0.43	0.01		1046
Phthalimide, *N*-vinyl-	Maleic anhydride	0.344	0.061	0.008	0.012	Y	284
Phthalimide, *N*-vinyl-	Styrene	0.07	0.11	6.27	0.66	Y	163
Pinene	Acrylonitrile	13.5	0.1	0.07	0.01		936
Pinene, α-	Methacrylate, methyl	0.08		10			848
Piperidinol, 4-, *N*–methyl-4-(vinylethynyl)-	Maleimide	2.05	0.03	0.49	0.01		948
Piperylene, *cis*-	Methacrylic acid	0.38		0.4			773
Piperylene, *trans*-	Methacrylic acid	0.33		0.38			773
Propanesulfonate, 3-[diethyl-[2-(2-methacroyloxyethoxy)ethyl	Acrylate, ethyl	0.19	0.02	3.97	1.03	Y	962
Propene, 3,3,3-trichloro-	Acrylonitrile	0.1		12.2			402
Propene, 3,3,3-trichloro-	Styrene	0		6.9			402
Propene, 3,3,3-trichloro-	Vinyl acetate	0.19		0.19			402
Propene, 1-chloro-*cis*-	Acrylonitrile	−0.12	0.19	23.37	2.99	Y	181
Propene, 1-chloro-*cis*-	Maleic anhydride	−0.022	0.08	0.347	0.082	Y	181
Propene, 1-chloro-*cis*-	Vinyl acetate	−0.012	0.061	7.08	0.35	Y	181
Propene, 1-chloro-*cis*-	Vinyl chloride	0.1	0.63	11.52	2.14	Y	181
Propene, 1-chloro-*trans*-	Acrylonitrile	0.01	0.17	19.61	1.72	Y	181
Propene, 1-chloro-*trans*-	Maleic anhydride	0.05	0.015	0.28	0.012	Y	181
Propene, 1-chloro-*trans*-	Vinyl acetate	0.011	0.074	3.13	0.15	Y	181
Propene, 1-chloro-*trans*-	Vinyl chloride	0.33	0.4	4.45	0.81	Y	181
Propene, 2,3-dichloro-	Carbazole, *N*-vinyl-	0		1			376
Propene, 2,3-dichloro-	Methacrylate, methyl	0.18	0.1	6.02	1.66	N	11
Propene, 2,3-dichloro-	Methacrylic acid	0		4			445
Propene, 2,3-dichloro-	Styrene	0.05	0.01	3.51	0.18	N	11
Propene, 2,3-dichloro-	Styrene	0.036		3			983
Propene, 2-chloro-	Acrylonitrile	−0.023	0.062	1.164	0.053	Y	181
Propene, 2-chloro-	Vinyl acetate	1.65	0.36	0.186	0.091	Y	181
Propene, 2-chloro-	Vinyl chloride	4.75	5.12	0.095	0.085	Y	181
Propene, 2-chloro-	Vinyl chloride	0.58		0.75			517
Propene, 2-chloro-	Vinylidene cyanide	0.001	0.003	0.204	0.079	Y	82
Propene, 2-chloro-3-hydroxy-	Methacrylate, methyl	0		4.4			445
Propene, 2-chloro-3-hydroxy-	Methacrylic acid	0		4.5			445
Propene, 2-chloro-3-hydroxy-	Styrene	0		12.5			445
Propene, 3-chloro-2-chloromethyl-	Styrene	0.01	0.1	3.37	0.73		806
Propenyl methyl ketone	Styrene	0.08	0.3	18.12	6.09		806
Propenyl, 2-, acetate	Vinyl chloride	0.25		2.2			189
Propenyl, 2-chloro-, acetate	Acrylate, methyl	0		0.7			445
Propenyl, 2-chloro-, acetate	Acrylic acid	0		1			445
Propenyl, 2-chloro-, acetate	Methacrylate, methyl	0		1			445
Propenyl, 2-chloro-, acetate	Styrene	0		4.1			445
Propenyl, 2-chloro-, acetate	Vinyl chloride	0		0.7			445
Propenyl, 2-methyl-, acetate	Methacrylate, methyl	0		10			470
Propenyl, 2-chloro-, acetate	Styrene	0		71			470
Propenyl, 2-methyl-, acetate	Vinylidene chloride	0		2.4			470
Propenyltriethoxysilane	Vinyl chloride					Y	237
Propiolactam, *N*-phenyl-α-methylene-	Styrene	0.406	0.106	0.626	0.025	Y	1002
Propionate, vinyl	Ethylene, chlorotrifluoro-	0.63	0.04	0.08	0.01		1097
Propylene	Acrylic acid					Y	183
Propylene	Ethylene			3.43			388
Propylene	Ethylene	0.62		3.2			477
Propylene	Ethylene, chlorotrifluoro-	0.08	0.07	0.02	0.08	N	318
Propylene	Vinyl chloride	0.196	0.023	2.35	0.14	Y	155
Propylene	Vinyl chloride	3.1	0.21	−0.05	0.1	N	329
Propylene	Vinyl chloride	−0.02	0.04	5.16	0.06	N	355
Propylene	Vinyl chloride	0.09		2.45			560
Propylene	Vinyl chloride			2.9			588
Pyrazole, 3-methyl-1-vinyl-	Pyrazole, 5-methyl-1-vinyl-	0.42		1.35			898
Pyrazole, 5-methyl-1-vinyl-	Pyrazole, 3-methyl-1-vinyl-	1.35		0.42			898
Pyrazole, *N*-vinyl-3,5-dimethyl-	Styrene	0.049	0.011	8.663	0.196		806

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Pyridazinone, 3-(2-vinyl)-6-methyl-	Acrylonitrile	0.19	0.04	0.32	0.02	N	340
Pyridazinone, 3-(2-vinyl)-6-methyl-	Methacrylate, methyl	0.56	0.04	0.86	0.02	N	340
Pyridazinone, 3-(2-vinyl)-6-methyl-	Styrene	0.85	0.05	0.9	0.03	N	340
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	Acrylonitrile	0.02	0.02	0.74	0.02	N	340
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	Methacrylate, methyl	0.08	0.03	1.19	0.04	N	340
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	Styrene	0.13	0.03	5.92	0.1	N	340
Pyridazone, 2-vinyl-3-	Methacrylate, methyl	1.56		0.63			897
Pyridazone, 2-vinyl-3-	Styrene	0.43		0.53			897
Pyridazone, 2-vinyl-6-methyl-3-	Methacrylate, methyl	0.3		0.16			897
Pyridazone, 2-vinyl-6-methyl-3-	Styrene	1.2		0.9			897
Pyridazone, 2-vinyl-6-methyl-4,5-benz-3-	Methacrylate, methyl	0.27		1.65			897
Pyridazone, 2-vinyl-6-methyl-4,5-benz-3-	Styrene	0.24		0.7			897
Pyridazone, 2-vinyl-6-phenyl-3-	Methacrylate, methyl	0.49		0.58			897
Pyridazone, 2-vinyl-6-phenyl-3-	Styrene	1.45		0.51			897
Pyridine, 2-methyl-5-vinyl-	Acrylate, methyl	1	0.3	0.18	0.05	N	232
Pyridine, 2-methyl-5-vinyl-	Acrylonitrile	0.31	0.02	0.16	0.01	N	232
Pyridine, 2-methyl-5-vinyl-	Butadiene	0.47		0.61			462
Pyridine, 2-methyl-5-vinyl-	Butadiene	0.72		1.32			520
Pyridine, 2-methyl-5-vinyl-	Butadiene	0.41	0.02	1.29	0.02	N	56
Pyridine, 2-methyl-5-vinyl-	Caprolactam, *N*-vinyl-	11.58		0.04			589
Pyridine, 2-methyl-5-vinyl-	Caprolactam, *N*-vinyl-	11.58		0.04			592
Pyridine, 2-methyl-5-vinyl-	Itaconate, diethyl	0.51		0.17			545
Pyridine, 2-methyl-5-vinyl-	Itaconate, dimethyl	0.88		0.22			545
Pyridine, 2-methyl-5-vinyl-	Methacrylate, methyl	0.54	0.19	0.42	0.09	N	232
Pyridine, 2-methyl-5-vinyl-	Methacrylic acid	0.85		0.43			552
Pyridine, 2-methyl-5-vinyl-	*N*-Vinylpyrrolidone	13	2	0.039	0.006		1070
Pyridine, 2-methyl-5-vinyl-	Styrene	0.75	0.08	0.74	0.03	N	232
Pyridine, 2-methyl-5-vinyl-	Styrene	0.68		0.6			500
Pyridine, 2-methyl-5-vinyl-	Styrene	1.2		0.72			514
Pyridine, 2-methyl-5-vinyl-	Styrene	0.78	0.03	0.73	0.03	N	56
Pyridine, 2-methyl-5-vinyl-	Sulfonic acid, *o*-methacryloyl-aminobenzene-	−0.002	0.031	1.83	0.218	Y	1006
Pyridine, 2-methyl-5-vinyl-	Sulfonic acid, *p*-methacryloyl-aminobenzene-	−0.054	0.048	2.72	0.525	Y	1006
Pyridine, 2-methyl-5-vinyl-	Vinyl acetate	9		0.11			812
Pyridine, 2-vinyl-	Acetylene, phenyl-					Y	192
Pyridine, 2-vinyl-	Acrolein	−0.12	0.03	2.64	0.14	N	208
Pyridine, 2-vinyl-	Acrylate, butyl	2.51		0.1			236
Pyridine, 2-vinyl-	Acrylate, butyl	2.59	0.14	0.11	0.01	N	78
Pyridine, 2-vinyl-	Acrylate, ethyl	0.28	0.1	0.21	0.06	N	250
Pyridine, 2-vinyl-	Acrylate, ethyl	2.25		0.21			832
Pyridine, 2-vinyl-	Acrylate, methyl	2.13	0.54	0.37	0.4	Y	2
Pyridine, 2-vinyl-	Acrylate, methyl	1.72	0.26	0.17	0.03	N	231
Pyridine, 2-vinyl-	Acrylate, methyl	2.12	0.1	0.2	0.03	N	749
Pyridine, 2-vinyl-	Acrylonitrile	21.9		0.05			456
Pyridine, 2-vinyl-	Acrylonitrile	0.44	0.08	0.1	0.03		50
Pyridine, 2-vinyl-	Butadiene, 1,4-dicarboxylate-, diethyl	0.41	0.07	0.69	0.23	N	327
Pyridine, 2-vinyl-	Butadiene, 1-carboxylate, ethyl	0.4		0.8			791
Pyridine, 2-vinyl-	Butadiene, 2-chloro-	0.06		5.19			457
Pyridine, 2-vinyl-	Caprolactam, *N*-vinyl-	4.35		0.12			589
Pyridine, 2-vinyl-	Caprolactam, *N*-vinyl-	4.35		0.12			592
Pyridine, 2-vinyl-	Isoprene	0.47		0.59			456
Pyridine, 2-vinyl-	Isoprene	0.46		0.58			458
Pyridine, 2-vinyl-	Methacrylate, 2-hydroxyethyl	0.64	0.16	0.56	0.04	N	342
Pyridine, 2-vinyl-	Methacrylate, methyl	0.73	0.06	0.42	0.02	N	231
Pyridine, 2-vinyl-	Methacrylate, methyl	0.86		0.4			532
Pyridine, 2-vinyl-	Methacrylate, methyl	0.76	0.12	0.35	0.19	Y	6
Pyridine, 2-vinyl-	Methacrylate, methyl	0.64		0.31			623
Pyridine, 2-vinyl-	Methacrylate, methyl	1.1	0.08	0.27	0.04	N	749
Pyridine, 2-vinyl-	Methacrylic acid	1.69	0.71	0.58	0.21	Y	15
Pyridine, 2-vinyl-	Methacrylic acid	1.38		0.44			498
Pyridine, 2-vinyl-	*N*-Vinylpyrrolidone	12.4	2.3	0.014	0.002		1070

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Pyridine, 2-vinyl-	Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro	2.216	0.062	0.073	0.04	Y	838
Pyridine, 2-vinyl-	Styrene	0.75	0.32	0.46	0.15	N	231
Pyridine, 2-vinyl-	Styrene	1.27		0.5			41
Pyridine, 2-vinyl-	Styrene	1.81		0.55			456
Pyridine, 2-vinyl-	Styrene	1.33		0.57			499
Pyridine, 2-vinyl-	Styrene	1.14		0.55			532
Pyridine, 2-vinyl-	Styrene, 2,5-dichloro-	0.62	0.12	0.14	0.18	N	2
Pyridine, 2-vinyl-	Vinyl acetate	13.65	3.64	−0.11	0.17	Y	6
Pyridine, 2-vinyl-	Vinyl acetate	30		0			6
Pyridine, 2-vinyl-5-ethyl-	Acrylate, methyl	1.32	0.03	0.18	0.004	N	231
Pyridine, 2-vinyl-5-ethyl-	Acrylonitrile	0.43		0.02			115
Pyridine, 2-vinyl-5-ethyl-	Acrylonitrile	0.43	0.18	0.06	0.07	N	50
Pyridine, 2-vinyl-5-ethyl-	Methacrylate, methyl	0.68	0.06	0.39	0.02	N	231
Pyridine, 2-vinyl-5-ethyl-	Styrene	1.09	0.12	0.74	0.05	N	231
Pyridine, 3-ethynyl-6-methyl-	Styrene	0.15		2.3			784
Pyridine, 3-vinyl-	Methacrylate, 2-hydroxyethyl	0.73	0.04	0.69	0.01	N	342
Pyridine, 4-vinyl-	Acrylate, (−)-menthyl	2.32	0.45	0.291	0.017	Y	667
Pyridine, 4-vinyl-	Acrylate, butyl	4.3	0.34	0.23	0.04	N	78
Pyridine, 4-vinyl-	Acrylate, ethyl	2.58		0.29			832
Pyridine, 4-vinyl-	Acrylate, methyl	1.78	0.32	0.2	0.03	N	231
Pyridine, 4-vinyl-	Acrylate, methyl	2.14		0.15			424
Pyridine, 4-vinyl-	Acrylonitrile	0.41		0.11			115
Pyridine, 4-vinyl-	Acrylonitrile	0.34	0.13	0.09	0.04	N	50
Pyridine, 4-vinyl-	Caprolactam, *N*-vinyl-	4.43		0.1			589
Pyridine, 4-vinyl-	Caprolactam, *N*-vinyl-	4.43		0.1			592
Pyridine, 4-vinyl-	Isoprene	2.49		0.32			424
Pyridine, 4-vinyl-	Methacrylate, 2-hydroxyethyl	0.9	0.09	0.62	0.04	N	342
Pyridine, 4-vinyl-	Methacrylate, methyl	0.77	0.07	0.57	0.02	N	231
Pyridine, 4-vinyl-	Methacrylate, methyl	0.99	0.18	0.54	0.01	N	691
Pyridine, 4-vinyl-	*N*-Vinylpyrrolidone	9.8	1.5	0.01	0.002		1070
Pyridine, 4-vinyl-	Styrene	0.69	0.16	0.52	0.07	N	231
Pyridine, 4-vinyl-	Styrene	1.04	0.22	−0.73	1.09	Y	79
Pyridine, 4-vinyl-	Vinyl chloride					Y	154
Pyridine, 4-vinyl-	Vinyl ether	32		0.03			395
Pyridine, 4-vinyl-	Vinylhydroquinone	0.53	0.4	−0.23	0.36	Y	337
Pyridine, *N*-oxide-, 2-vinyl-5-ethyl-	Styrene	2.27	0.88	0.06	0.11		806
Pyrimidine-2-one, 1,3-divinylhexahydro-	Acrylate, ethyl	−0.05	0.03	0.59	0.13	N	800
Pyrimidine, 2-*N*-*N*-dimethylamino-4-vinyl-	Styrene	1.404	0.071	0.338	0.017	Y	243
Pyrimidine, 4-vinyl-	Styrene	2.38	1.06	0.219	0.095	Y	243
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	Acrylate, methyl	1.58	0.24	0.125	0.025	Y	633
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	Methacrylate, methyl	2.28	0.47	0.387	0.07	Y	633
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	Styrene	0.71	0.17	0.555	0.053	Y	633
Pyrrolidone, *N*-vinyl-	Acenaphthalene					N	326
Pyrrolidone, *N*-vinyl-	Acrylate, 2,4,5-trichlorophenyl	0.012	0.008	0.163	0.02	Y	840
Pyrrolidone, *N*-vinyl-	Acrylate, 2-chloroethyl	0.03		0.52			790
Pyrrolidone, *N*-vinyl-	Acrylic acid	0.1	0.21	0.88	0.69	N	250
Pyrrolidone, *N*-vinyl-	Acrylic acid, α-bromo-	0.25		0.07			694
Pyrrolidone, *N*-vinyl-	Acrylonitrile					Y	213
Pyrrolidone, *N*-vinyl-	Acryloyl chloride	0.04		0.15			599
Pyrrolidone, *N*-vinyl-	Caprolactam, *N*-vinyl-	2.8		1.7			859
Pyrrolidone, *N*-vinyl-	Carbamate, *N*-vinyl-, ethyl	1.77	0.12	0.371	0.05	Y	81
Pyrrolidone, *N*-vinyl-	Cinnamate, α-cyano-, methyl	0.044	0.008	−0.001	0.001	Y	816
Pyrrolidone, *N*-vinyl-	Crotonaldehyde	0.42	0.02	−0.15	0.01	N	248
Pyrrolidone, *N*-vinyl-	Crotonic acid	0.84	0.07	0	0.03	N	245
Pyrrolidone, *N*-vinyl-	Dioxolane, 1,3,4-methylene-2-trichlor	2.98		0.04			647
Pyrrolidone, *N*-vinyl-	Dioxolane, 1,3-, 4,4-dimethyl-5-methyl	3.87		0.01			647
Pyrrolidone, *N*-vinyl-	Ethylene, chlorotrifluoro-	0.38		0.3			830
Pyrrolidone, *N*-vinyl-	Fumarate, dioctyl	0.02	0.01	0.04	0.04	N	121
Pyrrolidone, *N*-vinyl-	Imidazole, 1-vinyl-	0.16	0.074	0.96	0.19	Y	364
Pyrrolidone, *N*-vinyl-	Maleimide, *N*-(2-hydroxyethyl)-	0.002	0.002	0.038	0.01	Y	323
Pyrrolidone, *N*-vinyl-	Methacrylate, 2-bromoethyl	0.02		2.44			657
Pyrrolidone, *N*-vinyl-	Methacrylate, 2-hydroxyethyl	−0.019	0.046	4.841	0.133	Y	840

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Pyrrolidone, *N*-vinyl-	Methacrylate, ferrocenylmethyl	0.073	0.096	2.57	2.77	Y	325
Pyrrolidone, *N*-vinyl-	Methacrylate, methyl					Y	213
Pyrrolidone, *N*-vinyl-	Methacrylate, methyl	−0.05	0.08	5.62	0.3	N	320
Pyrrolidone, *N*-vinyl-	Methacrylate, methyl	−0.032	0.068	4.63	0.34	Y	321
Pyrrolidone, *N*-vinyl-	Methacrylate, methyl	−0.052	0.028	4.36	0.61	Y	36
Pyrrolidone, *N*-vinyl-	Methacrylate, methyl	0.01	0.15	4.04	0.61	N	691
Pyrrolidone, *N*-vinyl-	Methacryloyl chloride	0.05		0.58			599
Pyrrolidone, *N*-vinyl-	Methacryloylacetone	0.008	0.004	5.66	0.4	Y	339
Pyrrolidone, *N*-vinyl-	Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro	−0.009	0.018	0.552	0.075	Y	838
Pyrrolidone, *N*-vinyl-	Oxazoline, 2-, 2-isopropenyl-	0.01		3.5			862
Pyrrolidone, *N*-vinyl-	Styrene	0.057	0.028	17.2	1.55	Y	36
Pyrrolidone, *N*-vinyl-	Succinimide, *N*-vinyl-	0.97	0.42	3.67	0.89	N	96
Pyrrolidone, *N*-vinyl-	Triazine, 2-allyloxy-4,6-dichloro-	0.98		0.5			826
Pyrrolidone, *N*-vinyl-	Vinyl acetate	1.06	0.25	0.26	0.25	Y	103
Pyrrolidone, *N*-vinyl-	Vinyl acetate	1.75	2.79	0.19	1.77	N	121
Pyrrolidone, *N*-vinyl-	Vinyl acetate	3.4	0.21	0.195	0.057	Y	36
Pyrrolidone, *N*-vinyl-	Vinyl benzoate	2.507	0.039	0.408	0.03	Y	253
Pyrrolidone, *N*-vinyl-	Vinyl butyl ether	1.49	0.95	−0.027	0.063	Y	215
Pyrrolidone, *N*-vinyl-	Vinyl chloride	0.38		0.53			40
Pyrrolidone, *N*-vinyl-	Vinyl chloride	0.34		0.55			582
Pyrrolidone, *N*-vinyl-	Vinyl chloride	0.73		0.74			582
Pyrrolidone, *N*-vinyl-	Vinyl cinnamate	0.01		1.2			200
Pyrrolidone, *N*-vinyl-	Vinyl cyclohexyl ether	1.91	9.39	−0.11	0.59	Y	214
Pyrrolidone, *N*-vinyl-	Vinyl cymantrene	0.037	0.03	0.093	0.053	Y	308
Pyrrolidone, *N*-vinyl-	Vinyl ethylene glycol glycidyl ether	1.4		0.06			369
Pyrrolidone, *N*-vinyl-	Vinyl ethyleneglycol glycidyl ether	1.4		0.06			818
Pyrrolidone, *N*-vinyl-	Vinyl isopropyl ether	0.87	9.3	0.03	2.09	Y	215
Pyrrolidone, *N*-vinyl-	Vinyl phenyl ether	4.48	0.1	0.253	0.007	Y	214
Pyrrolidone, *N*-vinyl-	Vinylene carbonate	1.08	0.11	0.056	0.059	Y	103
Pyrrolidone, *N*-vinyl-	Vinylferrocene	0.364	0.068	0.71	0.16	Y	308
Pyrrolidone, α-methylene-*N*-methyl-	Methacrylate, methyl	0.24	0.14	0.577	0.058	Y	641
Quinine	*N*-Vinylpyrrolidone	0.37		2.2			922
Quinoline, 2-vinyl-	Butadiene, 2-chloro-	0.38		2.1			456
Quinoline, 2-vinyl-	Isoprene	0.53		1.88			456
Quinoline, 2-vinyl-	Styrene	2.09		0.49			456
Quinoline, 2-vinyl-	Styrene	2.69		0.49			458
S-Ethyl *N*-vinyl-*N*-methylthiolcarbamate	Styrene	−0.11	0.26	10	1.72	Y	197
S-Ethyl *N*-vinyl-*N*-methylthiolcarbamate	Vinyl acetate	1.22	1.17	0.67	0.1	Y	197
S-Vinyl *N*, *N*-diethylthiolcarbamate	Styrene	0.1	0.34	4.45	0.48	Y	197
S-Vinyl *N*, *N*-diethylthiolcarbamate	Vinyl acetate	1.89	0.77	0.133	0.025	Y	197
Silane, γ-methacryloxypropyltrimethoxy-	Acrylonitrile	3.79	0.594	0.094	0.034	Y	977
Silane, γ-methacryloxypropyltrimethoxy-	Methacrylate, methyl	2.06	0.309	0.788	0.082	Y	977
Silane, γ-methacryloxypropyltrimethoxy-	Styrene	0.868	0.096	0.425	0.048	Y	977
Silane, methacryloxypropyldimethoxymethyl-	Imidazole, *N*-Vinyl-	1.39		0.079			1095
Silane, methacryloxypropyltrimethoxy-	Imidazole, *N*-Vinyl-	2.27	1.12	0.158	0.04	Y	1088
Stilbene	Acenaphthalene	−0.01	0.04	5.65	0.23	N	725
Stilbene	Styrene	0	0.09	3.92	0.17	N	725
Styrene	2-Oxazoline, 2-isopropenyl-	0.67	0.02	0.64	0.03		894
Styrene	3-Pyridazone, 2-vinyl-	0.53		0.43			897
Styrene	3-Pyridazone, 2-vinyl-6-methyl-	0.9		1.2			897
Styrene	3-Pyridazone, 2-vinyl-6-methyl-4,5-benz-3-	0.7		0.24			897
Styrene	3-Pyridazone, 2-vinyl-6-phenyl-	0.51		1.45			897
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.233	0.069	3.29	0.712	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.161	0.028	1.57	0.193	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.215	0.051	1.29	0.309	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.189	0.05	1.4	0.322	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.179	0.014	0.923	0.075	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.164	0.015	0.747	0.073	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.162	0.015	0.635	0.068	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.178	0.02	1.12	0.112	N	930
Styrene	4-Pentenoate, 3-oxo-, ethyl	0.217	0.009	0.568	0.035	N	930
Styrene	Acenaphthalene	3.81		0.33			525
Styrene	Acetylene, phenyl-	0.324	0.02	0.33	0.12	Y	357

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Aconitate, trimethyl	1.026	0.091	−0.013	0.009	Y	147
Styrene	Acrolein	0.257	0.017	0.216	0.036	Y	291
Styrene	Acrolein	0.22	0.02	0.02	0.01	N	292
Styrene	Acrolein	0.205	0.006	0.32	0.002	Y	307
Styrene	Acrolein, 2-chloro-	0.02	0.05	0.15	0.07		806
Styrene	Acrolein, methyl-	0.173	0.017	0.833	0.092	Y	223
Styrene	Acrolein, methyl-	0.41		0.3			589
Styrene	Acrylamide	1.17	0.06	0.58	0.12	N	295
Styrene	Acrylamide	1.21	0.04	1.32	0.13	N	314
Styrene	Acrylamide	1.49	0.16	0.33	0.14	N	314
Styrene	Acrylamide	1.13		0.59			314
Styrene	Acrylamide	0.65	0.03	8.97	0.43	N	343
Styrene	Acrylamide	1.05		0.2			555
Styrene	Acrylamide, 2-chloro-*N*,*N*-dimethyl-	1.29	0.065	−0.036	0.152	Y	917
Styrene	Acrylamide, *N*,*N*-dibutyl-	1.6	0.058	0.294	0.058		806
Styrene	Acrylamide, *N*,*N*-diethyl-	1.23		0.39			739
Styrene	Acrylamide, *N*,*N*-diethyl-	1.36		0.34			739
Styrene	Acrylamide, *N*,*N*-diethyl	1.54		0.18			739
Styrene	Acrylamide, *N*,*N*-dimethyl-	1.15	0.35	0.12	0.3	N	170
Styrene	Acrylamide, *N*,*N*-dimethyl-	1.37	0.1	0.49	0.15	N	315
Styrene	Acrylamide, *N*,*N*-dimethyl-	1.32	0.12	0.47	0.2	N	315
Styrene	Acrylamide, *N*-(2-propionamido)-	1.84	0.21	0.86	0.3	N	295
Styrene	Acrylamide, *N*-methylol-	0.03		0.48			596
Styrene	Acrylamide, *N*-methylol-	1.55	0.05	0.4	0.03		891
Styrene	Acrylamide, *N*-octadecyl-	2.08	0.12	0.54	0.14	Y	290
Styrene	Acrylamide, *N*-octyl-	2.715	0.244	0.258	0.119		806
Styrene	Acrylamido, 1- , -1-deoxy-D-glucitol	2.42	0.08	0.03	0.02	N	258
Styrene	Acrylamidomethylamino, *p*-, azobenzene	0.083		8.5			858
Styrene	Acrylate, 2,3-dibromopropyl	0.42		0.16			643
Styrene	Acrylate, 2,4,5-trichlorophenyl	0.277	0.048	0.23	0.021	Y	622
Styrene	Acrylate, 2,4,6-tribromophenyl	0.2		0.1			643
Styrene	Acrylate, 2-chloroethyl	0.55	0.47	0.1	0.17	Y	140
Styrene	Acrylate, 2-chloroethyl	0.494	0.097	0.139	0.083	Y	178
Styrene	Acrylate, 2-chloroethyl	0.54		0.12			790
Styrene	Acrylate, 2-chloroethylglucitol	0.49	0.2	0.14	0.09		806
Styrene	Acrylate, 2-cyanoethyl	0.399	0.032	0.128	0.074	Y	178
Styrene	Acrylate, 2-ethylhexyl	0.96	0.3	0.31	0.6	N	126
Styrene	Acrylate, 2-ethylhexyl	0.94		0.26			463
Styrene	Acrylate, 2-nitrobutyl	0.35	0.03	0.12	0.01	N	234
Styrene	Acrylate, 3,4-epoxyhexahydrobenzyl	0.27	0.3	1.97	0.35	N	680
Styrene	Acrylate, β-acetoxymethyl-, methyl	0.34		0.19			1091
Styrene	Acrylate, β-ethoxy-, ethyl	46.98	27.03	−0.21	0.44	Y	194
Styrene	Acrylate, α-(4-chlorobenzyl)-, ethyl	0.04	0.09	1.03	0.13		806
Styrene	Acrylate, α-(4-cyanobenzyl)-, ethyl	0.06	0.05	0.65	0.09		806
Styrene	Acrylate, α-(4-methoxybenzyl)-, ethyl	0.33	0.3	0.71	0.24		806
Styrene	Acrylate, α-(hydroxymethyl)-, methyl	0.36		0.34			932
Styrene	Acrylate, α-[2,2-bis(carbomethoxy)-ethyl]-, methyl	0.58		0.09			1092
Styrene	Acrylate, α-acetoxy-, ethyl	0.557	0.031	0.173	0.01	Y	244
Styrene	Acrylate, α-benzyl-, ethyl	0.14	0.17	0.9	0.2		806
Styrene	Acrylate, α-benzyl-, methyl	0.57	0.028	0.175	0.053	Y	49
Styrene	Acrylate, α-bromo-, ethyl	0.02	0.04	0.44	0.08		806
Styrene	Acrylate, α-butyl-, methyl	0.812	0.043	0.205	0.053	Y	49
Styrene	Acrylate, α-chloro-, ethyl	0.08	0.07	0.32	0.11		806
Styrene	Acrylate, α-chloro-, methyl	0.25		0.3			74
Styrene	Acrylate, α-cyano-, methyl	0.05	0.01	0.61	0.05	N	141
Styrene	Acrylate, α-cyclohexyl-, methyl	1.615	0.048	−0.001	0.034	Y	49
Styrene	Acrylate, α-ethyl-, methyl	0.758	0.014	0.197	0.014	Y	49
Styrene	Acrylate, α-ethyloxymethyl-, methyl	0.37		0.24			1047
Styrene	Acrylate, α-fluoro-, methyl	0.66	0.17	0.16	0.11		806
Styrene	Acrylate, α-hexafluoropropyloxymethyl-, methyl	0.23		0.07			1047
Styrene	Acrylate, α-hydroxymethyl-, ethyl	0.468	0.03	0.546	0.127	Y	982
Styrene	Acrylate, α-isobutyl-, methyl	0.974	0.022	0.202	0.042	Y	49

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Acrylate, α-isopropyl-, methyl	1.872	0.052	0.036	0.034	Y	49
Styrene	Acrylate, α-isopropyloxymethyl-, methyl	0.41		0.26			1047
Styrene	Acrylate, α-methoxy-, methyl	1.131	0.078	0.513	0.058		806
Styrene	Acrylate, α-phenoxymethyl-, methyl	0.176	0.05	0.256	0.102	Y	1048
Styrene	Acrylate, α-phenyl-, chloroethyl	0.025	0.015	0.201	0.09	Y	488
Styrene	Acrylate, α-phenyl-, methyl	0.06		1			141
Styrene	Acrylate, α-phenyl-, methyl	0.03		0.4			399
Styrene	Acrylate, α-phenyl-, methyl	0.06		0.45			488
Styrene	Acrylate, α-phenyl-, methyl	1.176	0.015	1.275	0.008	Y	49
Styrene	Acrylate, α-phenyl-, propyl	0.03	0.015	0.126	0.063	Y	488
Styrene	Acrylate, α-phenyl-, butyl	0.04	0.022	0.107	0.055	Y	488
Styrene	Acrylate, α-propyl-, methyl	0.821	0.029	0.208	0.07	Y	49
Styrene	Acrylate, α-propyloxymethyl-, methyl	0.47		0.29			1047
Styrene	Acrylate, α-*sec*-butyl-, methyl	2.29	0.068	0.004	0.024	Y	49
Styrene	Acrylate, α-tetrafluoropropyloxymethyl-, methyl	0.37		0.17			1047
Styrene	Acrylate, α-trifluoroethyloxymethyl-, methyl	0.26		0.2			1047
Styrene	Acrylate, α-trimethylsiloxy-, methyl	0.46		1.42			1101
Styrene	Acrylate, benzyl	0.494	0.05	0.2	0.12	Y	178
Styrene	Acrylate, benzyl	0.534	0.028	0.248	0.041	Y	277
Styrene	Acrylate, butyl	0.79	0.08	0.25	0.05		1032
Styrene	Acrylate, butyl	0.79	0.11	0.34	0.06		1032
Styrene	Acrylate, butyl	0.75	0.1	0.38	0.05		1032
Styrene	Acrylate, butyl	1.23	0.21	−0.106	0.099	N	19
Styrene	Acrylate, butyl	0.44	0.03	0.29	0.05	N	20
Styrene	Acrylate, butyl	0.84	0.2	0.18	0.03	N	38
Styrene	Acrylate, butyl	0.66		0.19			38
Styrene	Acrylate, butyl	0.82		0.21			463
Styrene	Acrylate, butyl	0.8		0.15			549
Styrene	Acrylate, butyl	0.698		0.164			612
Styrene	Acrylate, butyl	0.698		0.164			632
Styrene	Acrylate, *cis*-β-cyano-, methyl					N	1034
Styrene	Acrylate, cyclododecyl	0.47	0.14	0.34	0.15		806
Styrene	Acrylate, cyclododecyl	0.54	0.08	0.31	0.08		806
Styrene	Acrylate, cyclododecyl	0.461	0.07	0.301	0.072		806
Styrene	Acrylate, cyclohexyl	0.913	0.107	0.272	0.048		1061
Styrene	Acrylate, cyclohexyl	0.885	0.102	0.254	0.141		806
Styrene	Actylate, cyclohexyl	0.53	0.11	0.16	0.13		806
Styrene	Acrylate, cyclohexyl	0.7	0.12	0.17	0.09		806
Styrene	Acrylate, cyclohexyl	0.58	0.1	0.18	0.07		806
Styrene	Acrylate, di-, copper	1.74	0.03	0.56	0.09	N	829
Styrene	Acrylate, di-, copper	5.94	0.05	0.12	0.08	N	829
Styrene	Acrylate, di-, nickel	1.83	0.02	0.53	0.06	N	829
Styrene	Acrylate, di-, zinc	1.1	0.02	0.9	0.07	N	829
Styrene	Acrylate, diethylaluminum	−0.002	0.016	0.046	0.035	Y	652
Styrene	Acrylate, diethylaluminum					Y	652
Styrene	Acrylate, ethyl	0.699	0.072	0.139	0.085	Y	178
Styrene	Acrylate, ethyl	1.01		0.16			463
Styrene	Acrylate, ethyl	0.79		0.19			530
Styrene	Acrylate, ethyl	0.8		0.2			530
Styrene	Acrylate, ethyl	0.8		0.48			542
Styrene	Acrylate, ethyl	0.787		0.152			673
Styrene	Acrylate, ethyl	0.787		0.152			677
Styrene	Acrylate, ferrocenylmethyl	1.87	0.17	−0.016	0.06	Y	293
Styrene	Acrylate, ferrocenylmethyl	2.3		0.02			293
Styrene	Acrylate, ferrocenylmethyl	2.5		0.02			576
Styrene	Acrylate, glycidyl	0.6		0.17			114
Styrene	Acrylate, glycidyl	0.73	0.34	0.24	0.58	N	126
Styrene	Acrylate, heptafluorobutyl	0.31	0.01	0.049	0.013	Y	206
Styrene	Acrylate, isopropyl	0.755	0.024	0.195	0.027	Y	178
Styrene	Acrylate, methyl	0.82		0.24			139
Styrene	Acrylate, methyl	0.192	0.057	0.8	0.21	Y	14
Styrene	Acrylate, methyl	0.722	0.056	0.168	0.067	Y	151

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Acrylate, methyl	0.75		0.18			393
Styrene	Acrylate, methyl	0.75		0.18			397
Styrene	Acrylate, methyl	0.68		0.14			463
Styrene	Acrylate, methyl	0.9		0.07			495
Styrene	Acrylate, methyl	0.65		0.3			496
Styrene	Acrylate, methyl	0.4		0.16			496
Styrene	Acrylate, methyl	0.871	0.022	0.148	0.026	Y	865
Styrene	Acrylate, octadecyl	0.777	0.018	0.37	0.12	Y	290
Styrene	Acrylate, octadecyl	0.44		0.18			290
Styrene	Acrylate, octadecyl	0.79		0.31			290
Styrene	Acrylate, octadecyl	0.44		0.18			290
Styrene	Acrylate, octyl	0.39	0.11	0.01	0.53		806
Styrene	Acrylate, *p*-(cinnamoylamino)phenyl	1.08		0.35			685
Styrene	Acrylate, pentabromophenyl	0.16		0.1			643
Styrene	Acrylate, pentachlorophenyl	2.11		0.9			900
Styrene	Acrylate, tetrahydrofurfuryl	0.475	0.186	0.489	0.196		806
Styrene	Acrylate, thio-, butyl	0.439	0.14	0.421	0.17		806
Styrene	Acrylate, thio-, ethyl	0.18	0.038	0.291	0.075		806
Styrene	Acrylate, thio-, *tert*-butyl	0.157	0.034	0.368	0.055		806
Styrene	Acrylate, trifluoro-, methyl	1.4		0			595
Styrene	Acrylate, trifluoro-, methyl	1.4		0			598
Styrene	Acrylic acid	0.25		0.05			312
Styrene	Acrylic acid	1.1		0.08			312
Styrene	Acrylic acid	0.15		0.25			397
Styrene	Acrylic acid	0.22		0.35			414
Styrene	Acrylic acid	0.253	0.069	0.136	0.029	Y	47
Styrene	Acrylic acid	0.25		0.15			490
Styrene	Acrylic acid	0.25		0.45			542
Styrene	Acrylic acid	0.15		0.25			561
Styrene	Acrylic acid, 2-benzamido-	0.13		0.78			1019
Styrene	Acrylic acid, *N*-acetyl-α-amino-	0.91		0.44			661
Styrene	Acrylic anhydride	0.16	0.01	0.1	0.004	N	219
Styrene	Acrylonitrile	0.33		0			140
Styrene	Acrylonitrile	0.29	0.04	0.02	0.06	N	198
Styrene	Acrylonitrile	0.4		0.05			269
Styrene	Acrylonitrile	0.36		0.03			396
Styrene	Acrylonitrile	0.39		0.06			403
Styrene	Acrylonitrile	0.37		0.05			417
Styrene	Acrylonitrile	0.37		0.07			543
Styrene	Acrylonitrile	0.4		0.04			60
Styrene	Acrylonitrile	0.3		0.17			727
Styrene	Acrylonitrile	0.33		0.17			727
Styrene	Acrylonitrile	0.33		0.15			727
Styrene	Acrylonitrile	0.36		0.17			727
Styrene	Acrylonitrile	0.46		0.07			727
Styrene	Acrylonitrile	0.41		0.06			727
Styrene	Acrylonitrile	0.43		0.03			727
Styrene	Acrylonitrile	0.55		0.06			727
Styrene	Acrylonitrile	0.35		0.01			812
Styrene	Acrylonitrile- 2-chloro-	0.014	0.004	0.087	0.04	Y	884
Styrene	Acrylonitrile, 2-fluoro-					Y	884
Styrene	Acrylonitrile, α-chloro-	0.13	0.04	0.03	0.02		806
Styrene	Acrylonitrile, α-fluoro-	0.44		0.03			850
Styrene	Acrylonitrile, α-hydroxymethyl-	0.205	0.029	0.528	0.086	Y	981
Styrene	Acrylonitrile, α-methoxy-	0.534	0.069	0.312	0.054		806
Styrene	Acryloyl chloride	0.1	0.05	0.02	0.02	N	220
Styrene	Acryloyloxy, 2-, benzoic acid	0.06	0.02	0.64	0.02	N	651
Styrene	Adipate, divinyl	20.36	0.26	0.011	0.011	Y	815
Styrene	Allyl acetate	62.8	82.7	0.01	0.05	N	1053
Styrene	Allyl acetate	90		0			470
Styrene	Allyl chloride	35.55	15.82	0.05	0.16	Y	10
Styrene	Allyl chloride					Y	204
Styrene	Allyl chloride	36.8		0.03			438
Styrene	Allyl glycidyl hexahydrophthalate	34.3		0.06			782
Styrene	Allyl glycidyl phthalate	34.8		0.14			782

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Allyl trichloroacetate	2.32	0.15	0.81	0.02		913
Styrene	Azlactone, 2-vinyl-4,4-dimethyl-	0.175	0.054	0.237	0.129	Y	1011
Styrene	Azobenzene, 4-(acrylamidomethylamino)-	0.083	8.5				880
Styrene	Benzimidazole, vinylmercapto-	6.51	0.51	0.56	0.41	Y	764
Styrene	Benzothiazole, vinylmercapto-	2.171	0.07	0.349	0.087	Y	764
Styrene	Benzothiazole, vinylmercapto-	3.02	0.14	0.5	0.095	Y	764
Styrene	Benzothiazolone, *N*-[(methacroyloxy)methyl]	0.35	0.02	3.1	0.01		960
Styrene	Benzoxazole, vinylmercapto-	2.87	0.19	0.24	0.12	Y	764
Styrene	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	18.2		0			994
Styrene	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	89.2		0			994
Styrene	Borazine, β-vinyl	4.53	0.62	0.001	0.016		1018
Styrene	Butadiene	0.84	0.4	1.44	0.56	Y	140
Styrene	Butadiene	0.37	0.25	1.55	0.14	N	159
Styrene	Butadiene	0.38		1.37			173
Styrene	Butadiene					Y	21
Styrene	Butadiene	0.58		1.35			418
Styrene	Butadiene	0.44		1.59			428
Styrene	Butadiene	0.6		1.8			473
Styrene	Butadiene	0.44		1.4			486
Styrene	Butadiene	0.38		1.4			486
Styrene	Butadiene	0.68		0.5			491
Styrene	Butadiene	0.82	0.13	1.38	0.11	N	56
Styrene	Butadiene	0.83	0.21	1.83	0.41	Y	83
Styrene	Butadiene	0.64		1.38			83
Styrene	Butadiene, 1,4-dicarboxylate-, diethyl	0.09	0.014	0.55	0.1	Y	327
Styrene	Butadiene, 1-chloro-	0.27	0.27	1.44	3.1	Y	695
Styrene	Butadiene, 1-phthalimido-	0.299	0.049	1.56	0.63	Y	163
Styrene	Butadiene, 1-succinimido-	0.286	0.018	1.68	0.16	Y	163
Styrene	Butadiene, 2,3-dichloro-	0.25		8.65			587
Styrene	Butadiene, 2,3-dichloro-	0.05		10.6		Y	61
Styrene	Butadiene, 2,3-dimethyl-	0.534	0.035	1.048	0.066	Y	173
Styrene	Butadiene, 2-chloro-	0.05		7.84		Y	61
Styrene	Butadiene, 2-chloro-	0.025	0.047	5.98	0.5	Y	8
Styrene	Butadiene, 2-fluoro-	0.214	0.03	2.1	0.12	Y	174
Styrene	Butadiene, 2-fluoro-	0.22	0.15	1.24	0.61	Y	174
Styrene	Butadiene, 2-trimethylsilyloxy-	0.64		1.2			993
Styrene	Butadiene-1-carboxylate, *S*(-)-α-phenethyl	0.45	0.13	6.81	1.68	Y	748
Styrene	Butadiene-1-carboxylate, ethyl	0.12		0.56			791
Styrene	Butadiene-1-carboxylic acid	0.15		0.52			791
Styrene	Butadiene-1-carboxylic acid	0.08		5.55			799
Styrene	Butenoate, 3-phenyl-3-, methyl	1.97	0.364	−0.532	0.512	Y	1008
Styrene	Bntyrolactone, α-methylene-	0.14	0.01	0.85	0.06	N	663
Styrene	Caprolactam, *N*-vinyl-	0.2	0.01	1.02	0.01		916
Styrene	Carbamate, *N*, *N*-diethyl-, vinyl	29.56	3.59	−0.08	0.18	Y	197
Styrene	Carbazole, *N*-methyl-3-vinyl-	0.735	0.067	1.404	0.146		806
Styrene	Carbazole, *N*-vinyl-	6	1.34	−0.02	0.06	N	11
Styrene	Carbazole, *N*-vinyl-	5.93	1.05	0.026	0.034	Y	13
Styrene	Carbazole, *N*-vinyl-	5.72	0.4	0.032	0.057	Y	98
Styrene	Cinnamate, 2-chloroethyl	1.394	0.068	0.01	0.11	Y	488
Styrene	Cinnamate, α-cyano-, 2-ethylhexyl	0.58	0.004	−0.06	0.01	N	282
Styrene	Cinnamate, α-cyano-, benzyl	0.4	0.03	−0.16	0.17	N	282
Styrene	Cinnamate, α-cyano-, butyl	0.47	0.03	−0.09	0.08	N	282
Styrene	Cinnamate, α-cyano-, cyclohexyl	0.37	0.04	−0.24	0.22	N	282
Styrene	Cinnamate, α-cyano-, ethyl					Y	134
Styrene	Cinnamate, α-cyano-, hexyl	0.39	0.04	−0.17	0.26	N	282
Styrene	Cinnamate, α-cyano-, methyl	0.4	0.06	−0.21	0.35	N	282
Styrene	Cinnamate, benzyl	1.33	0.12	0.14	0.13	Y	488
Styrene	Cinnamate, butyl	1.757	0.082	0.04	0.15	Y	488
Styrene	Cinnamate, ethyl	1.52	0.31	−0.076	0.054	Y	200
Styrene	Cinnamate, ethyl	1.566	0.035	0.082	0.045	Y	488

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Cinnamate, methyl	1.799	0.073	0.04	0.13	Y	488
Styrene	Cinnamate, methyl	1.27		0.02			653
Styrene	Cinnamate, *p*-chloro-, methyl	2.24		0.04			653
Styrene	Cinnamate, *p*-methyl-, methyl	2.84		0			653
Styrene	Cinnamate, phenyl	1.66	0.12	0.029	0.055	Y	488
Styrene	Cinnamate, *tert*-butyl	1.692	0.087	0.1	0.12	Y	488
Styrene	Cinnamide, *N,N*-diethyl-	4.17		0.01			737
Styrene	Cinnamide, *p*-chloro-, *N,N*-diethyl-	2.99		0.01			737
Styrene	Cinnamide, *p*-methyl-, *N,N*-diethyl-	4.95		0.01			737
Styrene	Cinnamonitrile	2.549	0.092	−0.008	0.021	Y	134
Styrene	Cinnamonitrile, α-cyano-					Y	134
Styrene	Citraconate, di-*sec*-butyl	2		0.13			841
Styrene	Citraconate, dibenzyl	1.55		0.047			841
Styrene	Citraconate, dibutyl	2.74		0.054			841
Styrene	Citraconate, dicyclohexyl	1.88		0.03			841
Styrene	Citraconate, diethyl	2.48		0.029			841
Styrene	Citraconate, diisobutyl	1.8		0.03			841
Styrene	Citraconate, diisopropyl	1.62		0.076			841
Styrene	Citraconate, dimethyl	1.58		0.049			841
Styrene	Citraconate, dipropyl	2.19		0.027			841
Styrene	Citraconic acid	1.36		0			1021
Styrene	Citraconic anhydride	0.16	0.15	0.021	0.058	Y	47
Styrene	Citraconimide, β-iso-, *N*-(4-acetylphenyl)-	0.42		0			608
Styrene	Citraconimide, β-iso-, *N*-(4-chlorophenyl)-	0.28		0			608
Styrene	Citraconimide, β-iso-, *N*-(4-ethoxycarbonylphenyl)-	0.3		0			608
Styrene	Citraconimide, β-iso-, *N*(4-methoxyphenyl)-	0.27		0			608
Styrene	Citraconimide, β-iso-, *N*-(*p*-tolyl)-	0.4		0			608
Styrene	Citraconimide, β-iso-, *N*-phenyl-	0.39		0			608
Styrene	Citraconimide, *N*-(4-acetoxyphenyl)-	0.151	0.054	0.015	0.086	Y	728
Styrene	Citraconimide, *N*-(4-chlorophenyl)-	0.181	0.057	0.075	0.093	Y	728
Styrene	Citraconimide, *N*-(4-ethoxycarbonylphenyl)-	0.148	0.018	0.004	0.033	Y	728
Styrene	Citraconimide, *N*-(4-methoxyphenyl)-	0.63	0.19	0.04	0.11	Y	728
Styrene	Citraconimide, *N*-allyl	0.142		0.004			959
Styrene	Citraconimide, *N*-benzyl-	0.19	0.087	−0.066	0.082	Y	714
Styrene	Citraconimide, *N*-butyl-	0.169	0.028	0.03	0.045	Y	714
Styrene	Citraconimide, *N*-ethyl-	0.231	0.044	−0.002	0.05	Y	714
Styrene	Citraconimide, *N*-hexyl-	0.316	0.086	−0.03	0.11	Y	714
Styrene	Citraconimide, *N*-isobutyl-	0.326	0.094	0.07	0.14	Y	714
Styrene	Citraconimide, *N*-isopropyl-	0.244	0.055	−0.014	0.063	Y	714
Styrene	Citraconimide, *N*-methyl-	0.15		0.24			460
Styrene	Citraconimide, *N*-methyl-	0.14		0.24			506
Styrene	Citraconimide, *N*-phenyl-	0.175	0.016	−0.001	0.023	Y	728
Styrene	Citraconimide, *N*-propyl-	0.196	0.062	−0.014	0.087	Y	714
Styrene	Citraconimide, *N*-*tert*-butyl-	0.465	0.039	0.045	0.05	Y	714
Styrene	Citraconimide, *N*-tolyl-	0.31	0.13	−0.02	0.18	Y	728
Styrene	Citraconimide, α-iso-, *N*-(4-acetylphenyl)-	0.81		0			608
Styrene	Citraconimide, α-iso-, *N*-(4-chlorophenyl)-	1.03		0			608
Styrene	Citraconimide, α-iso-, *N*-(4-ethoxycarbonylphenyl)-	0.96		0			608
Styrene	Citraconimide, α-iso-, *N*-(4-methoxyphenyl)-	0.39		0			608
Styrene	Citraconimide, α-iso-, *N*-(*p*-tolyl)-	1.69		0			608
Styrene	Citraconimide, α-iso-, *N*-phenyl-	1.76		0			608
Styrene	Crotonaldehyde	14.7		0.03			428
Styrene	Crotonaldehyde	14.7		0.07			428
Styrene	Crotonaldehyde	14.7		0.12			428
Styrene	Crotonate, α-acetyl-, methyl	2.7		0			785
Styrene	Crotonate, α-bromo-, ethyl	4.42		0			785

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Crotonate, α-carboethoxy-, ethyl	8.24		0			785
Styrene	Crotonate, α-chloro-, ethyl	5.13		0			785
Styrene	Crotonate, α-cyano-, ethyl	0.26		0.02			785
Styrene	Crotonate, α-methoxy-, methyl	18.4		0.04			785
Styrene	Crotonate, α-methyl-, methyl	30.7		0.02			785
Styrene	Crotonate, ethyl	27		0			785
Styrene	Crotonate, methyl	9.68	1.86	−0.27	0.22	Y	112
Styrene	Crotonic acid	20		0			402
Styrene	Crotononitrile	23.29	1.7	−0.012	0.06	Y	112
Styrene	Crotononitrile	23.8		0			785
Styrene	Crotononitrile, α-phenyl-	0.43		0			785
Styrene	Cyclobutane, 1,2-dimethylene-	0.26		3.6			854
Styrene	Cyclobutane, 1,2-dimethylene-	0.28	0.05	3.6	0.16		885
Styrene	Cyclopentene, 4-, -1,3-dione					Y	263
Styrene	Diallyl melamine	36.96	4.02	−0.32	0.2	Y	201
Styrene	Diallyl phthalate	36.41	1.23	0.041	0.038	Y	152
Styrene	Diallyl phthalate	25.59	2.19	0.026	0.071	Y	152
Styrene	Diallyl phthalate	23.67	1.2	0.038	0.04	Y	152
Styrene	Diallyl phthalate	18.8		0.08			152
Styrene	Diallyl phthalate	27.5		0.09			152
Styrene	Diallyl phthalate	16.9		0.1			152
Styrene	Diallyl phthalate	23.8		0.1			152
Styrene	Diallyl phthalate	15.1		0.13			152
Styrene	Dihydroxydiphenylsulfone diallyl ether	0.975	0.015	0.92	0.16		1016
Styrene	Dithiane, 2-methylene-1,3-	0.03		3.35			1031
Styrene	Divinylbenzene, *m*-	0.58	0.15	0.58	0.04	N	262
Styrene	Divinylbenzene, *p*-	0.26	0.25	1.18	0.17	N	262
Styrene	Ethylene	14.88		0.05			615
Styrene	Ethylene, 1,1-diphenyl-	0.34	0.01	−0.02	0.01	N	281
Styrene	Ethylene, 1,1-diphenyl-	0.87		1.15			901
Styrene	Ethylene, 1-diphenyl-	0.33	0.02	0.01	0.02	N	189
Styrene	Ethylene, chlorotrifluoro-	4.38	0.99	−0.1	0.21	Y	236
Styrene	Ethylene, dichloro-, *cis*-	3.1		−3.1		Y	137
Styrene	Ethylene, dichloro-, *cis*-	63.1		−2.2		Y	137
Styrene	Ethylene, dichloro-, *cis*-	21.4	4.9	−0.28	0.12	Y	40
Styrene	Ethylene, dichloro-, *cis*-	161.3	30.3	−0.02	0.062	Y	9
Styrene	Ethylene, dichloro-, *cis*-	24.75	2.64	0.067	0.053	Y	9
Styrene	Ethylene, *p*-vinyl-1,1-diphenyl-	0.46		1.31			742
Styrene	Ethylene, tetrachloro-					Y	204
Styrene	Ethylene, tetrachloro-	200					204
Styrene	Ethylene, tetrachloro-					Y	40
Styrene	Ethylene, tetrachloro-	187		0			40
Styrene	Ethylene, tetrachloro-	185		0			59
Styrene	Ethylene, tetrafluoro-	5.2		0.01			567
Styrene	Ethylene, trichloro-	17.1		0			40
Styrene	Ethylene, trichloro-	16.5		0			40
Styrene	Ethylene, trichloro-	12.1		0			40
Styrene	Ethylene, trichloro-	12.7		0			40
Styrene	Ethylene, trichloro-	16	11	−0.006	0.37	Y	59
Styrene	Formaldehyde,-*S*-vinyl-*O*-ethylthioacetal	7.75	0.68	0.32	0.24	Y	273
Styrene	Fumarate, allyl methyl	0.25	0.01	0.03	0.02	N	713
Styrene	Fumarate, bis(2-butyl)	0.51	0.04	0	0.01	N	249
Styrene	Fumarate, bis(3-chloro-2-butyl)	0.26	0.04	−0.06	0.03	N	249
Styrene	Fumarate, di-(2-chloroethyl)	0.18		0			1038
Styrene	Fumarate, di-(2-cyanoethyl)	0.07		0.09			1038
Styrene	Fumarate, di-(2-methoxyethyl)	0.35		0.03			1038
Styrene	Fumarate, di-(2-phenylethyl)	0.27		0			1038
Styrene	Fumarate, di-*sec*-butyl	0.55		0.04			1038
Styrene	Fumarate, di-*tert*-amyl	0.69		0.08			1038
Styrene	Fumarate, di-*tert*-butyl	0.57		0.04			1038
Styrene	Fumarate, dibenzyl	0.32		0			1038
Styrene	Fumarate, dibutyl	0.35		0.04			1038
Styrene	Fumarate, dicyclohexyl	0.49		0.06			1038
Styrene	Fumarate, dicyclopentyl	0.3		0.06			1038

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Fumarate, diethyl	0.29		0.02			1038
Styrene	Fumarate, diethyl	6.3	0.26	0.081	0.005	Y	139
Styrene	Fumarate, diethyl	0.394	0.06	0.11	0.022	Y	139
Styrene	Fumarate, diethyl	0.3		0.07			139
Styrene	Fumarate, diethyl	0.31	0.02	0	0.01	N	249
Styrene	Fumarate, diethyl	0.36		0.1			562
Styrene	Fumarate, diethyl	0.29		0.02			640
Styrene	Fumarate, diisobutyl	0.37		0.05			1038
Styrene	Fumarate, diisopropyl	0.4		0.055			1038
Styrene	Fumarate, dimethyl	0.39		0.03			1038
Styrene	Fumarate, dipropyl	0.32		0.02			1038
Styrene	Fumarate, isopropyl butyl	0.42		0.01			1056
Styrene	Fumarate, isopropyl ethyl	0.3		0.01			1056
Styrene	Fumarate, isopropyl isobutyl	0.46		0.01			1056
Styrene	Fumarate, isopropyl methyl	0.3		0.16			1056
Styrene	Fumarate, isopropyl *sec*-butyl	0.54		0.02			1056
Styrene	Fumarate, methyl butyl	0.36		0.02			1056
Styrene	Fumarate, methyl ethyl	0.32		0.02			1056
Styrene	Fumarate, methyl isobutyl	0.32		0.03			1056
Styrene	Fumarate, methyl isopropyl	0.3		0.16			1056
Styrene	Fumarate, methyl propyl	0.28		0.05			1056
Styrene	Fumarate, methyl *sec*-butyl	0.45		0.02			1056
Styrene	Fumarate, methyl *tert*-amyl	0.53		0.06			1056
Styrene	Fumarate, methyl tert-butyl	0.45		0.03			1056
Styrene	Fumarate, mono-2-butyl	0.33	0.03	−0.04	0.03	N	249
Styrene	Fumarate, *tert*-butyl 2-ethylhexyl	0.54		0.01			1056
Styrene	Fumarate, *tert*-butyl 4-methyl-2-pentyl	0.57		0.01			1056
Styrene	Fumarate, *tert*-butyl ethyl	0.25		0.04			1056
Styrene	Fumarate, *tert*-butyl isoamyl	0.45		0.06			1056
Styrene	Fumarate, *tert*-butyl methyl	0.45		0.03			1056
Styrene	Fumarate, *tert*-butyl *sec*-butyl	0.55		0.07			1056
Styrene	Fumaronitrile	0.202	0.058	−0.016	0.021	Y	137
Styrene	Fumaronitrile					Y	191
Styrene	Fumaronitrile	0.294	0.044	0.006	0.031	Y	199
Styrene	Fumaronitrile	0.3		0			478
Styrene	Fumaronitrile					Y	77
Styrene	Fumaryl chloride					Y	91
Styrene	Furan, 2,3-dihydro-	20		0			806
Styrene	Hexatriene, tetrachloro-	0.123	0.0428	0.85	0.22	Y	4
Styrene	Hexene, 1-, 3,4-dioic anhydride	8.57	8.4	−0.052	1	Y	933
Styrene	Hexene, 2-, 3,4-dioic anhydride	3.63	5.6	−0.08	1.17	Y	933
Styrene	Hexene, 3-, 3,4-dioic anhydride	17.17	436	−0.54	5.83	Y	933
Styrene	Hexene, 5-, 2,4-dione	0.031		1.22			931
Styrene	Hexene, 5-, 2,4-dione	0.049		1.38			931
Styrene	Hexene, 5-, 2,4-dione	0.066		1.52			931
Styrene	Hexene, 5-, 2,4-dione	0.07		1.73			931
Styrene	Hexene, 5-, 2,4-dione	0.171		1.91			931
Styrene	Hexene, 5-, 2,4-dione	0.179		1.91			931
Styrene	Hexene, 5-, 2,4-dione	0.173		2.07			931
Styrene	Hexene, 5-, 2,4-dione	0.121		2.46			931
Styrene	Hexene, 5-, 2,4-dione	0.24		2.69			931
Styrene	Hexene, 5-, 2,4-dione	0.014		2.93			931
Styrene	Hexene, 5-, 2,4-dione	0.044		3.19			931
Styrene	Hexene, 5-, 2,4-dione	0.028		4.11			931
Styrene	Hexene-1,5-dicarboxylate, dimethyl	0.678	0.031	0.196	0.043	Y	828
Styrene	Imidazole, 1-vinyl-	9.94	0.24	0.071	0.029	Y	364
Styrene	Imidazole, 1-vinyl-2-methyl-	8.87	0.62	0.09	0.064	Y	364
Styrene	Imidazole, 1-vinyl-2-methyl-	8.97		0.07			479
Styrene	Isoprene	0.513	0.02	1.922	0.016	Y	366
Styrene	Isoprene	0.476	0.018	1.957	0.011	Y	366
Styrene	Isoprene	0.44		1.98			544
Styrene	Isoprene	0.42		2.02			544
Styrene	Isoprene	0.48		1.3			83
Styrene	Isoprene, 3-acetoxy-	0.34		4.47			770
Styrene	Isopropenyl methyl ketone	0.44		0.29			542

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Isopropenyl methyl ketone	0.32		0.66			542
Styrene	Isopropenyl, 3-(1-cyclohexenyl), acetate	1.59		0.56			770
Styrene	Isopropenylacetophenone, *p*-	0.3	0.18	0.84	0.59	Y	266
Styrene	Isopropenylisocyanate	8.12		0.07			434
Styrene	Isopropenylisocyanate	6.84	0.18	0.122	0.047	Y	99
Styrene	Isopropylidene, 2,3:4,5-di-*O*-, -1-vinylbenzyl-	0.14	0.055	3.26	1.12	Y	772
Styrene	Itaconamate, *N*-(4-carboethoxyphenyl)-, butyl	0.2		0.61			678
Styrene	Itaconamate, *N*-(4-carboethoxyphenyl)-, ethyl	0.23		0.2			678
Styrene	Itaconamate, *N*-(4-carboethoxyphenyl)-, methyl	0.15		0.58			678
Styrene	Itaconamate, *N*-(4-carboethoxyphenyl)-, propyl	0.26		0.4			678
Styrene	Itaconamate, *N*-(4-chlorophenyl)-, ethyl	0.35		0.16			678
Styrene	Itaconamate, *N*-(4-chlorophenyl)-, methyl	0.38		0.18			678
Styrene	Itaconamate, *N*-(-chlorophenyl)-, propyl	0.39		0.31			678
Styrene	Itaconamate, *N*-(4-methoxyphenyl-, butyl	0.12		0.69			678
Styrene	Itaconamate, *N*-(4-methoxyphenyl)-, ethyl	0.33		0.34			678
Styrene	Itaconamate, *N*-(4-methoxyphenyl)-, methyl	0.26		0.41			678
Styrene	Itaconamate, *N*-(4-methoxyphenyl)-, propyl	0.38		0.4			678
Styrene	Itaconamate, *N*-(4-methylphenyl)-, butyl	0.19		0.31			678
Styrene	Itaconamate, *N*-(4-methlyphenyl)-, ethyl	0.38		0.26			678
Styrene	Itaconamate, *N*-(4-methylphenyl)-, methyl	0.3		0.16			678
Styrene	Itaconamate, *N*-(4-methylphenyl)-, propyl	0.54		0.19			678
Styrene	Itaconamate, *N*-phenyl, methyl	0.39		0.18			678
Styrene	Itaconamate, *N*-phenyl-, butyl	0.12		0.49			678
Styrene	Itaconamate, *N*-phenyl, ethyl	0.45		0.21			678
Styrene	Itaconamate, *N*-phenyl-, methyl	0.31		0.52			1059
Styrene	Itaconamate, *N*-phenyl-, propyl	0.5		0.18			678
Styrene	Itaconate, di-2[-2(2-methoxyethoxy)-ethoxy]ethyl	0.53	0.022	0.2	0.015		1098
Styrene	Itaconate, diamyl	0.36	0.05	0.5	0.09	N	39
Styrene	Itaconate, dibutyl	0.42	0.06	0.34	0.05	N	39
Styrene	Itaconate, diethyl	0.23	0.11	0.34	0.12	N	39
Styrene	Itaconate, dimenthyl	0.502	0.188	0.483	0.364	Y	975
Styrene	Itaconate, dimethyl					N	39
Styrene	Itaconate, dimethyl	0.48		0.14			398
Styrene	Itaconate, dimethyl	0.5		0.06			586
Styrene	Itaconate, dioctyl	0.33	0.16	0.14	0.12	N	39
Styrene	Itaconate, dipropyl	0.16	0.07	0.28	0.06	N	39
Styrene	Itaconate, ethyl hexafluoroisopropyl	0.435	0.025	0.039	0.035	N	1103
Styrene	Itaconate, glycidyl methyl	0.376	0.036	0.041	0.079		806
Styrene	Itaconate, mono-methyl	0.029	0.039	0.111	0.052	Y	656
Styrene	Itaconate, mono-methyl	0.34	0.077	0.33	0.15	Y	656
Styrene	Itaconate, mono-methyl	0.52	0.085	0.33	0.11	Y	656
Styrene	Itaconate, mono-methyl	0.404	0.06	0.218	0.086	Y	656
Styrene	Itaconic acid	0.34		0			420
Styrene	Itaconic acid	0.26	0.08	0.12	0.67	Y	76
Styrene	Itaconic anhydride	0.004	0.01	0.405	0.09	Y	365
Styrene	Itaconic anhydride	0.086	0.015	0.456	0.09	Y	62
Styrene	Itaconic anhydride	0.003	0.005	0.785	0.05	Y	62
Styrene	Itaconimide, *N*-(2-chloroethyl)-	0.069	0.045	0.162	0.092	Y	642
Styrene	Itaconimide, *N*-(4-acetoxyphenyl)-	0.35	0.12	0.21	0.21	Y	729
Styrene	Itaconimide, *N*-(4-chlorophenyl)-	0.159	0.084	0.2	0.18	Y	729

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Itaconimide, *N*-(4-ethoxycarbonylphenyl)	0.232	0.032	0.112	0.054	Y	729
Styrene	Itaconimide, *N*-(*p*-chlorophenyl)-	0.15		0.14			740
Styrene	Itaconimide, *N*-(*p*-tolyl)-	0.16		0.24			740
Styrene	Itaconimide, *N*-benzyl-	0.056	0.048	0.2	0.11	Y	642
Styrene	Itaconimide, *N*-butyl-	0.22	0.03	−0.019	0.04	Y	642
Styrene	Itaconimide, *N*-ethyl	0.232	0.029	0.172	0.06	Y	642
Styrene	Itaconimide, *N*-isobutyl-	0.44	0.11	−0.03	0.11	Y	642
Styrene	Itaconimide, *N*-Isopropyl-	0.308	0.043	0.045	0.064	Y	642
Styrene	Itaconimide, *N*-methyl-	0.153	0.069	0.12	0.12	Y	642
Styrene	Itaconimide, *N*-naphthyl-	0.08		0.35			740
Styrene	Itaconimide, *N*-phenyl-	0.116	0.043	0.109	0.082	Y	729
Styrene	Itaconimide, *N*-phenyl-	0.05		0.08			740
Styrene	Itaconimide, *N*-phenyl-	0.05	0.248	0.998	0.123	N	978
Styrene	Itaconimide, *N*-propyl-	0.28	0.12	0.04	0.18	Y	642
Styrene	Itaconimide, *N*-tolyl-	0.134	0.031	0.219	0.079	Y	729
Styrene	Maleamic acid, *N*-3-dimethylamino-6-methyl	0.55	0.03	−0.19	0.03	N	679
Styrene	Maleate, allyl methyl	8.29	0.14	0.02	0.01	N	713
Styrene	Maleate, butyl stannyl, allyl	0.1	0.02	0.06	0.03	N	811
Styrene	Maleate, chloro-, diethyl	2.07	0.14	−0.043	0.021	Y	14
Styrene	Maleate, dibutyl tin	2.08	0.118	0.194	0.121	Y	987
Styrene	Maleate, diethyl	6.59	0.46	−0.013	0.022	Y	139
Styrene	Maleate, diethyl					Y	14
Styrene	Maleate, diethyl	8		0.07			640
Styrene	Maleate, diethyl	6.07	0.07	0.01	0.01	N	874
Styrene	Maleate, diethyl	6.592		0.001			888
Styrene	Maleate, dimethyl	10.65	2.81	0.15	0.12	Y	137
Styrene	Maleate, dimethyl	0.083	0.015	0.105	0.058	Y	317
Styrene	Maleate, monoallyl	0.14	0.01	0.07	0.02	N	811
Styrene	Maleic anhydride	0.097	0.002	0.001	0.008	N	105
Styrene	Maleic anhydride	0.04	0.12	0	0.13	Y	12
Styrene	Maleic anhydride	0.02	0.003	0.003	0.03	N	17
Styrene	Maleic anhydride	0	0.01	0.02	0.01	N	241
Styrene	Maleic anhydride	0.02		0			384
Styrene	Maleic anhydride	0.02		0			397
Styrene	Maleic anhydride	0.04		0.02			483
Styrene	Maleic anhydride	0.04		0.01			506
Styrene	Maleic anhydride	0.04		0.001			522
Styrene	Maleic anhydride	0.05		0.005			522
Styrene	Maleic anhydride	0.02		0.01			522
Styrene	Maleic anhydride	0.04		0.02			522
Styrene	Maleic anhydride	0.04		0.05			522
Styrene	Maleic anhydride					N	923
Styrene	Maleic anhydride, α-chloro-	0.07		0			845
Styrene	Maleimide	0.057	0.012	0.088	0.036	Y	251
Styrene	Maleimide, *N*-(2,6-diethylphenyl)-	0.19		0.02			992
Styrene	Maleimide, *N*-(2,6-dimethylphenyl)-	0.14		0.08			992
Styrene	Maleimide, *N*-(2-chlorophenyl)-	0.011	0.004	0.013	0.002	Y	853
Styrene	Maleimide, *N*-(2-chlorophenyl)-	0.038	0.006	0.028	0.003	Y	853
Styrene	Maleimide, *N*-(2-chlorophenyl)-	0.013	0.012	0.01	0.005	Y	910
Styrene	Maleimide, *N*-(2-chlorophenyl)-	0.038	0.028	0.006	0.003	Y	910
Styrene	Maleimide, *N*-(2-hydroxyethyl)-	0.046	0.017	0.011	0.027	Y	323
Styrene	Maleimide, *N*-(3-dimethylaminophenyl)-	0.03		0.19			747
Styrene	Maleimide, *N*-(4-bromophenyl)-	0.02	0.006	0.018	0.007	Y	1030
Styrene	Maleimide, *N*-(4-carboxyethylphenyl)-α-chloro-	0.059	0.021	0.022	0.041	Y	973
Styrene	Maleimide, *N*-(4-chlorophenyl)-α-chloro-	0.149	0.048	0.019	0.038	Y	973
Styrene	Maleimide, *N*-(4-hydroxyphenyl)-	−0.059	0.041	0.13	0.1	Y	323
Styrene	Maleimide, *N*-(4-methoxyphenyl)-α-chloro-	0.112	0.031	0.049	0.061	Y	973
Styrene	Maleimide, *N*-(4-methylphenyl)-α-chloro-	0.107	0.048	0.019	0.078	Y	973
Styrene	Maleimide, *N*-(β-chloroethyl)-α-chloro-	0.29	0.05	0	0.06		974

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Maleimide, *N*-[*N'*-(α-methylbenzyl)-aminocarbonylpentyl]-	0.198	0.143	0.258	0.25	Y	1099
Styrene	Maleimide, *N*-[*N'*-(α-methylbenzyl)-aminocarbonylundecyl]-					Y	1099
Styrene	Maleimide, *N*-allyl	0.056		0.035			1003
Styrene	Maleimide, *N*-benzyl-	0.058	0.021	0.013	0.031	Y	348
Styrene	Maleimide, *N*-benzyl-α-chloro-	0.42	0.06	0	0.06		974
Styrene	Maleimide, *N*-butyl-					Y	115
Styrene	Maleimide, *N*-butyl-	0.02		0.06			50
Styrene	Maleimide, *N*-cyclohexyl-	0.102		0.1			1100
Styrene	Maleimide, *N*-cyclohexyl-	−0.01	0.017	0.162	0.054	Y	348
Styrene	Maleimide, *N*-ethyl-α-chloro-	0.9	0.05	0	0.04		974
Styrene	Maleimide, *N*-hexyl-	−0.041	0.069	0.09	0.16	Y	348
Styrene	Maleimide, *N*-hydroxymethyl-	−0.035	0.053	0.06	0.11	Y	323
Styrene	Maleimide, *N*-methyl-α-chloro-	0.63	0.04	0	0.03		974
Styrene	Maleimide, *N*-phenyl-					N	1026
Styrene	Maleimide, *N*-phenyl-	0.016	0.004	0.044	0.017	Y	328
Styrene	Maleimide, *N*-phenyl-	0.07		0.01			992
Styrene	Maleimide, *N*-phenyl-α-chloro-	0.087	0.04	−0.002	0.073	Y	973
Styrene	Maleimide-*N*,*N*-dimethylaniline, 4-(2-chlorophenyl)-	0.05		0.04			747
Styrene	Mesaconate, di-*sec*-butyldithiol	0.45		0.02			624
Styrene	Mesaconate, diamyl	1.27		0.02			658
Styrene	Mesaconate, dibenzyl	0.73		0.08			658
Styrene	Mesaconate, dibenzyldithiol	0.28		0.06			624
Styrene	Mesaconate, dibutyl	1.28		0.05			658
Styrene	Mesaconate, dibutyldithiol	0.41		0.05			624
Styrene	Mesaconate, dicyclohexyl	1.25		0.1			658
Styrene	Mesaconate, diethyl	1.24		0.06			658
Styrene	Mesaconate, diethyldithiol	0.4		0.03			624
Styrene	Mesaconate, dihexyl	1.51		0.1			658
Styrene	Mesaconate, diisoamyl	1.34		0.07			658
Styrene	Mesaconate, diisobutyl	1.29		0.1			658
Styrene	Mesaconate, diisobutyldithiol	0.41		0.03			624
Styrene	Mesaconate, diisopropyl	1.31		0.04			658
Styrene	Mesaconate, diisopropyldithiol	0.44		0.05			624
Styrene	Mesaconate, dimethyl	0.98		0.04			658
Styrene	Mesaconate, dimethyldithiol	0.35		0.07			624
Styrene	Mesaconate, dioctyl	1.23		0.01			658
Styrene	Mesaconate, diphenyl	0.45		0.07			658
Styrene	Mesaconate, diphenyldithiol	0.16		0.15			624
Styrene	Mesaconate, dipropyl	1.22		0.05			658
Styrene	Mesaconate, dipropyldithiol	0.39		0.06			624
Styrene	Mesaconic acid	1.6		0			1021
Styrene	Methacrylamide	1.4	0.13	0.48	0.2	N	316
Styrene	Methacrylamide	1.42	0.17	1.24	0.38	N	316
Styrene	Methacrylamide, 1-deoxy-D-glucitol	2.09		0.005			258
Styrene	Methacrylamide, *N*-phenyl-	1.24	0.06	0.88	0.17	N	352
Styrene	Methacrylate, 2,2,6,6,-tetramethyl-4-piperidinyl	0.63		0.3			664
Styrene	Methacrylate, 2,3-epithiopropyl	0.41		0.28			787
Styrene	Methacrylate, 2-(sulfonic acid)ethyl	0.37		0.6			564
Styrene	Methacrylate, 2-acetoxyethyl	0.44	0.02	0.58	0.04	N	346
Styrene	Methacrylate, 2-bromoethyl	0.32		0.41			657
Styrene	Methacrylate, 2-chloro-2,3,3,3-fluoropropyl	0.121	0.068	0.243	0.15	N	927
Styrene	Methacrylate, 2-chloroethyl	0.35	0.046	0.341	0.075	Y	179
Styrene	Methacrylate, 2-chloroethyl	0.345	0.048	0.34	0.08	Y	179
Styrene	Methacrylate, 2-chloroethyl	0.42		0.23			382
Styrene	Methacrylate, 2-hydroxyethyl	0.44	0.02	0.54	0.03	N	346
Styrene	Methacrylate, 2-hydroxyethyl	0.332	0.006	0.856	0.018	Y	350
Styrene	Methacrylate, 2-hydroxyethyl	0.57		0.65			463
Styrene	Methacrylate, 2-hydroxyethyl	0.5		1.65			878
Styrene	Methacrylate, 2-hydroxyethyl	0.59		0.53			878
Styrene	Methacrylate, 2-hydroxyethyl	0.53		0.59			903

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Methacrylate, 2-hydroxyethyl	0.5		1.65			903
Styrene	Methacrylate, 2-methoxyethyl	0.46		0.48			918
Styrene	Methacrylate, 2-naphthyl-, methyl	0.786	0.16	0.113	0.126	Y	924
Styrene	Methacrylate, 2-naphthyl-, methyl	0.486	0.117	0.321	0.098	Y	924
Styrene	Methacrylate, 2-naphthyl-, methyl	0.61	0.195	0.576	0.477	Y	924
Styrene	Methacrylate, 2-naphthyl-, methyl	0.384	0.225	0.216	0.422	Y	924
Styrene	Methacrylate, 2-phenethyl	0.56	0.025	0.534	0.099	Y	540
Styrene	Methacrylate, 3,5-dimethyladamantyl	0.89	0.14	0.63	0.21	N	335
Styrene	Methacrylate, 5-oxo-pyrrolidinylmethyl	0.342	0.013	0.411	0.033	Y	633
Styrene	Methacrylate, *N*-methyl-*N*-phenyl-2-aminoethyl	0.44	0.04	0.58	0.02	N	651
Styrene	Methacrylate, α-naphthoyloxy-, ethyl	0.216	0.043	0.242	0.018		1102
Styrene	Methacrylate, acetonyl	0.285	0.041	0.505	0.115	Y	958
Styrene	Methacrylate, amyl	0.52	0.11	0.46	0.2	Y	179
Styrene	Methacrylate, benzyl	0.463	0.041	0.658	0.084	Y	111
Styrene	Methacrylate, benzyl	0.435	0.026	0.467	0.048	Y	179
Styrene	Methacrylate, benzyl	0.45		0.51			487
Styrene	Methacrylate, benzyl	0.48		0.42			540
Styrene	Methacrylate, benzyl	0.8		0.3			602
Styrene	Methacrylate, bornyl	0.45	0.32	0.5	0.38		806
Styrene	Methacrylate, butyl	0.56	0.12	0.31	0.2	Y	179
Styrene	Methacrylate, butyl	−0.05	1.09	2.52	1.1	N	43
Styrene	Methacrylate, butyl	0.52		0.47			463
Styrene	Methacrylate, butyl	0.63		0.64			530
Styrene	Methacrylate, butyl	0.54		0.64			530
Styrene	Methacrylate, butyl	0.74		0.59			835
Styrene	Methacrylate, chloromethyl	0.208	0.006	0.298	0.03	Y	712
Styrene	Methacrylate, cyanomethyl	0.144	0.011	0.28	0.041	Y	625
Styrene	Methacrylate, cyclohexyl	0.586	0.088	0.57	0.28	Y	179
Styrene	Methacrylate, di-ethylene glycol	0.39	0.15	0.67	0.08	N	262
Styrene	Methacrylate, dodecyl	0.528	0.05	0.3	0.1	Y	179
Styrene	Methacrylate, ethyl	0.55	0.062	0.36	0.12	Y	179
Styrene	Methacrylate, ethyl	0.67		0.26			530
Styrene	Methacrylate, ethyl	0.65		0.29			530
Styrene	Methacrylate, ethyl	0.55		0.33			570
Styrene	Methacrylate, ferrocenylmethyl	3.63	0.14	−0.019	0.022	Y	293
Styrene	Methacrylate, ferrocenylmethyl	3.6		0.01			576
Styrene	Methacrylate, glycidyl	0.54	0.07	0.16	0.03	N	114
Styrene	Methacrylate, glycidyl	0.278	0.086	0.539	0.04	Y	217
Styrene	Methacrylate, glycidyl	0.47	0.02	0.46	0.08	N	225
Styrene	Methacrylate, glycidyl	0.44		0.54			225
Styrene	Methacrylate, glycidyl	0.45		0.55			463
Styrene	Methacrylate, glycidyl	0.31	0.04	0.74	0.1	N	849
Styrene	Methacrylate, glycidyl	0.11	0.045	0.73	0.284	N	893
Styrene	Methacrylate, glycidyl	0.435	0.033	0.514	0.098	Y	999
Styrene	Methacrylate, hexyl	0.58	0.07	0.44	0.13	Y	179
Styrene	Methacrylate, isobornyl	0.61	0.45	0.24	0.3		806
Styrene	Methacrylate, isobutyl	0.509	0.014	0.271	0.036	Y	179
Styrene	Methacrylate, isobutyl	0.56		0.58			463
Styrene	Methacrylate, isobutyl	0.55		0.4			487
Styrene	Methacrylate, isopropyl	0.5		0.42			463
Styrene	Methacrylate, isopropyl	0.47		0.74			541
Styrene	Methacrylate, *m*-chlorophenyl	0.375	0.05	0.209	0.013	Y	177
Styrene	Methacrylate, *m*-nitrophenyl	0.44	0.16	0.185	0.054	Y	177
Styrene	Methacrylate, methoxymethyl	0.395	0.014	0.586	0.069	Y	821
Styrene	Methacrylate, methyl	0.5		0.5			138
Styrene	Methacrylate, methyl	0.52		0.46			139
Styrene	Methacrylate, methyl	0.52		0.46			179
Styrene	Methacrylate, methyl	0.48		0.42			20
Styrene	Methacrylate, methyl	0.52		0.45			21
Styrene	Methacrylate, methyl	0.49		0.46			21
Styrene	Methacrylate, methyl	0.45		0.47			21
Styrene	Methacrylate, methyl	0.58		0.48			21
Styrene	Methacrylate, methyl	0.44		0.5			21
Styrene	Methacrylate, methyl	0.396	0.048	0.22	0.13	Y	241

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Methacrylate, methyl	0.42		0.32			241
Styrene	Methacrylate, methyl	0.49	0.14	0.418	0.085	Y	254
Styrene	Methacrylate, methyl	0.564	0.047	0.54	0.029	Y	254
Styrene	Methacrylate, methyl	0.585	0.007	0.478	0.011	Y	287
Styrene	Methacrylate, methyl	0.44	0.08	0.45	0.01	N	324
Styrene	Methacrylate, methyl	0.47		0.45			324
Styrene	Methacrylate, methyl	0.54		0.49			403
Styrene	Methacrylate, methyl	0.52		0.47			416
Styrene	Methacrylate, methyl	0.38		0.48			451
Styrene	Methacrylate, methyl	0.52		0.46			463
Styrene	Methacrylate, methyl	0.54		0.42			492
Styrene	Methacrylate, methyl	0.44		0.5			492
Styrene	Methacrylate, methyl	0.52		0.46			505
Styrene	Methacrylate, methyl	0.5		0.44			531
Styrene	Methacrylate, methyl	0.56		0.49			531
Styrene	Methacrylate, methyl	0.53		0.49			534
Styrene	Methacrylate, methyl	0.485	0.052	0.409	0.02	Y	751
Styrene	Methacrylate, methyl	0.48	0.059	0.504	0.026	Y	751
Styrene	Methacrylate, methyl	0.371	0.055	0.611	0.028	Y	751
Styrene	Methacrylate, methyl	0.62		0.59			753
Styrene	Methacrylate, methyl	0.52		0.6			753
Styrene	Methacrylate, methyl	0.5		0.63			753
Styrene	Methacrylate, methyl	0.55		0.58			753
Styrene	Methacrylate, methyl	0.55		0.64			753
Styrene	Methacrylate, methyl	0.41		0.41			753
Styrene	Methacrylate, methyl	0.472		0.454			824
Styrene	Methacrylate, methyl	0.497		0.464			824
Styrene	Methacrylate, methyl	0.432		0.422			824
Styrene	Methacrylate, methyl	0.57		0.41			835
Styrene	Methacrylate, methyl	0.275		0.314			860
Styrene	Methacrylate, methylthiomethyl	0.338		0.512	0.05	Y	920
Styrene	Methacrylate, octyl	0.65	0.1	0.56	0.16	Y	179
Styrene	Methacrylate, octyl	0.56		0.68			463
Styrene	Methacrylate, *p*-(cinnamoylamino)-phenyl	0.62		0.31			685
Styrene	Methacrylate, *p*-carbomethoxyphenyl	0.482	0.084	0.178	0.027	Y	177
Styrene	Methacrylate, *p*-chlorophenyl	0.34	0.11	0.243	0.022	Y	177
Styrene	Methacrylate, *p*-methoxyphenyl	0.499	0.089	0.244	0.026	Y	177
Styrene	Methacrylate, *p*-methylphenyl	0.49	0.25	0.225	0.04	Y	177
Styrene	Methacrylate, *p*-nitrophenyl	0.223	0.09	0.185	0.039	Y	177
Styrene	Methacrylate, phenyl	0.246	0.049	0.48	0.23	Y	177
Styrene	Methacrylate, phenyl	0.247	0.064	0.55	0.33	Y	179
Styrene	Methacrylate, phenyl	0.26		0.51			463
Styrene	Methacrylate, propyl	0.565	0.081	0.38	0.11	Y	179
Styrene	Methacrylate, sulfolanyl	0.2	0.1	−0.2	2.46	N	359
Styrene	Methacrylate, *tert*-butyl	0.545	0.054	0.61	0.51	Y	179
Styrene	Methacrylic acid	0.21	0.02	0.55	0.66	N	166
Styrene	Methacrylic acid	0.22		0.64			166
Styrene	Methacrylic acid	0.2		0.66			203
Styrene	Methacrylic acid	0.15	0.04	0.49	0.08	N	312
Styrene	Methacrylic acid	0.2	0.04	0.56	0.19	N	313
Styrene	Methacrylic acid	0.38		0.28			313
Styrene	Methacrylic acid	0.124	0.092	0.602	0.088	Y	47
Styrene	Methacrylic acid	0.17		0.85			812
Styrene	Methacrylic acid	0.041	0.03	0.631	0.146	Y	842
Styrene	Methacrylic acid	0.067	0.046	0.6	0.249	Y	842
Styrene	Methacrylic acid	0.627	0.031	0.44	0.089	Y	842
Styrene	Methacrylic acid	0.55	0.06	0.39	0.199	Y	842
Styrene	Methacrylic acid	0.221	0.048	0.12	0.118	Y	842
Styrene	Methacrylic anhydride	0.13	0.01	0.27	0.02	N	219
Styrene	Methacrylic anhydride	0.1		0.33			275
Styrene	Methacrylonitrile	0.37		0.41			311
Styrene	Methacrylonitrile	0.38		0.42			311
Styrene	Methacrylonitrile	0.43		0.28			542
Styrene	Methacrylonitrile	0.34		0.21			707

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Methacrylonitrile	0.25		0.25			75
Styrene	Methacrylonitrile	0.34		0.21			835
Styrene	Methacryloxypropyltrimethoxysilane	0.36		0.58			951
Styrene	Methacryloyl fluoride	0.069	0.014	0.164	0.045	Y	660
Styrene	Methacryloylacetone	0.067	0.007	1.66	0.16	Y	339
Styrene	Methacryloyloxy, 2-, benzoate, methyl	0.27	0.04	0.48	0.02	N	651
Styrene	Methacryloyloxy, 2-, benzoic acid	0.15	0.04	1.17	0.03	N	651
Styrene	Methacryloyloxy, 4-, 2,2,6,6-tetramethyl	0.6		0.32			687
Styrene	Methylenebutyrolactone	0.09		0.7			746
Styrene	Methyleneglutaronitrile, 2-	0.42	0.52	−0.02	1.3	N	196
Styrene	Methyleneglutaronitrile, 2-	0.5		0.02			196
Styrene	Methyleneglutaronitrile, 2-	0.85		0.25			512
Styrene	Myrcene	0.891	0.015	1.38	0.081	Y	1062
Styrene	*N,N*-Divinylaniline	5.38	0.36	0.033	0.067	Y	45
Styrene	*N*-Vinylimidazole	8.377	0.149	0.68	0.02	Y	879
Styrene	Naphthalene, 1-vinyl-	0.699	0.233	2.02	0.401		967
Styrene	Naphthalene, 1-vinyl-4-chloro-	0.76	0.68	0.75	0.45		806
Styrene	Naphthalene, 2-vinyl-	0.395	0.049	1.345	0.093		806
Styrene	Naphthalene, 2-vinyl-6-chloro-	0.39	0.6	2.44	1.38		806
Styrene	Naphthalene, 6-vinyl-2-*tert*-butyl	0.616	0.069	1.014	0.244		806
Styrene	Oct-1-ene, 4,8-dioxaspiro(2,5)-	1.831	0.027	0.246	0.024	Y	838
Styrene	Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro	1.93	0.098	0.271	0.109	Y	838
Styrene	Oxazolidone, *N*-vinyl-	29.42	2.19	−0.029	0.014	Y	36
Styrene	Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl	0.55		0.68			596
Styrene	Oxazoline, 2-, 4-acryloxymethyl-2,4-dimethyl	0.56		0.14			596
Styrene	Oxazoline, 2-isopropenyl-	0.67		0.64			862
Styrene	Oxirane, 4-vinylphenyl-	0.78		1.06			1001
Styrene	Pentadienoate, *trans*-4-ethoxy-2,4-, ethyl	0.04		12.2			723
Styrene	Pentadienonitrile, *trans*-4-ethoxy-2,4-	0.08		11.8			723
Styrene	Penten, 2-, -4-one	13.84	0.57	−0.033	0.055	Y	112
Styrene	Phenol, 3-(methacryloylamino)-	0.5	0.06	0.99	0.04		896
Styrene	Phenol, 4-(methacryloylamino)-	0.71	0.04	0.86	0.03		896
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.629	0.003	0.171	0.01	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.612	0.004	0.15	0.05	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.595	0.005	0.146	0.073	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.597	0.044	0.168	0.355	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.633	0.002	0.094	0.02	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.61	0.018	0.23	0.187	N	971
Styrene	Phosphate, 2-(acryloyloxy)ethyl diethyl	0.599	0.002	0.108	0.034	N	971
Styrene	Phosphate, vinyl bis(chloroethyl)	2.47	0.78	0.03	0.23	Y	133
Styrene	Phosphazene, isopropenylpentafluoro-	1.6	0.27	−0.121	0.088	Y	765
Styrene	Phosphonate, 2-(4-vinylphenyl)ethyl, diethyl	0.55		1.36			796
Styrene	Phosphonate, 4-vinylphenyl, diethyl	0.7		1.4			796
Styrene	Phosphonate, α-carbomethoxyvinyl-, diethyl	3.88		0			684
Styrene	Phosphonate, isopropenyl-, dimethyl	8.6		0.23			684
Styrene	Phosphonate, vinyl-, bis(2-chloroethyl)	2.46	0.43	0.02	0.39	Y	133
Styrene	Phosphonate, vinyl-, bis(2-chloroethyl)	2.3		0.16			410
Styrene	Phosphonate, vinyl-, bis(2-chloroethyl)	2.2		0.2			485
Styrene	Phosphonate, vinyl-, bis(2-chloroethyl)	6.32		0.18			592
Styrene	Phosphonate, vinyl-, bis(2-chloroethyl)	6.32		0.18			594
Styrene	Phosphonate, vinyl-, dibutyl	4.97	3.44	−0.12	2.33	Y	129
Styrene	Phosphonate, vinyl-, diethyl	4.36	4.27	−0.15	2.18	Y	129
Styrene	Phosphonate, vinyl-, diethyl	1.59	0.24	−0.52	0.22	Y	18
Styrene	Phosphonate, vinyl-, diethyl	2.45		0.02			592
Styrene	Phosphonate, vinyl-, diisobutyl	3.87	1.59	−0.32	1.46	Y	129
Styrene	Phosphonate, vinyl-, diispropyl	2.57	2.13	−0.52	2.52	Y	129
Styrene	Phosphonate, vinyl-, dimethyl					Y	129
Styrene	Phosphonate, vinyl-, dimethyl	4.61		0.4			129
Styrene	Phosphonate, vinyl-, dimethyl	2.45		0.15			592
Styrene	Phosphonate, vinyl-, dimethyl	2.45		0.02			594

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Phosphonate, vinyl-, diphenyl	1.74	0.16	−0.38	0.29	Y	129
Styrene	Phosphonate, vinyl-, dipropyl	3.81	0.68	−0.41	0.72	Y	129
Styrene	Phthalimide, *N*-vinyl-	6.27	0.66	0.07	0.11	Y	163
Styrene	Propene, 2,3-dichloro-	3.51	0.18	0.05	0.01	N	11
Styrene	Propene, 2,3-dichloro-	3		0.036			983
Styrene	Propene, 2-chloro-3-hydroxy-	12.5		0			445
Styrene	Propene 3,3,3-trichloro-	6.9		0			402
Styrene	Propene, 3-chloro-2-chloromethyl-	3.37	0.73	0.01	0.1		806
Styrene	Propneyl methyl ketone	18.12	6.09	0.08	0.3		806
Styrene	Propenyl, 2-chloro-, acetate	4.1		0			445
Styrene	Propenyl, 2-methyl, acetate	71		0			470
Styrene	Propiolactam, *N*-phenyl-α-methylene-	0.626	0.025	0.406	0.106	Y	1002
Styrene	Pyrazole, *N*-vinyl-3,5-dimethyl-	8.663	0.196	0.049	0.011		806
Styrene	Pyridazinone, 3-(2-vinyl)-6-methyl-	0.9	0.03	0.85	0.05	N	340
Styrene	Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro	5.92	0.1	0.13	0.03	N	340
Styrene	Pyridine, 2-methyl-5-vinyl-	0.74	0.03	0.75	0.08	N	232
Styrene	Pyridine, 2-methyl-5-vinyl-	0.6		0.68			500
Styrene	Pyridine, 2-methyl-5-vinyl-	0.72		1.2			514
Styrene	Pyridine, 2-methyl-5-vinyl-	0.73	0.03	0.78	0.23	N	56
Styrene	Pyridine, 2-vinyl-	0.46	0.15	0.75	0.32	N	231
Styrene	Pyridine, 2-vinyl-	0.5		1.27			41
Styrene	Pyridine, 2-vinyl-	0.55		1.81			456
Styrene	Pyridine, 2-vinyl-	0.57		1.33			499
Styrene	Pyridine, 2-vinyl-	0.55		1.14			532
Styrene	Pyridine, 2-vinyl-5-ethyl-	0.74	0.05	1.09	0.12	N	231
Styrene	Pyridine, 3-ethynyl-6-methyl-	2.3		0.15			784
Styrene	Pyridine, 4-vinyl-	0.52	0.07	0.69	0.16	N	231
Styrene	Pyridine, 4-vinyl-	−0.73	1.09	1.04	0.22	Y	79
Styrene	Pyridine, *N*-oxide-, 2-vinyl-5-ethyl-	0.06	0.11	2.27	0.88		806
Styrene	Pyrimidine, 2-*N*,*N*-dimethylamino-4-vinyl	0.338	0.017	1.404	0.071	Y	243
Styrene	Pyrimidine, 4-vinyl-	0.219	0.095	2.38	1.06	Y	243
Styrene	Pyrrolidone, 1-benzyl-3-methylene-5-methyl	0.555	0.053	0.71	0.17	Y	633
Styrene	Pyrrolidone, *N*-vinyl-	17.2	1.55	0.057	0.028	Y	36
Styrene	Quinoline, 2-vinyl-	0.49		2.09			456
Styrene	Quinoline, 2-vinyl-	0.49		2.69			458
Styrene	*S*-Ethyl *N*-vinyl-*N*-methylthiol-carbamate	10	1.72	−0.11	0.26	Y	197
Styrene	*S*-Vinyl *N*,*N*-diethylthiolcarbamate	4.45	0.48	0.1	0.34	Y	197
Styrene	Silane, γ-methacryloxypropyl-trimethoxy-	0.425	0.048	0.868	0.096	Y	977
Styrene	Stilbene	3.92	0.17	0	0.09	N	725
Styrene	Styrene, 2,4-dibromo-	0.14	1.61	0.91	2.88		806
Styrene	Styrene, 2,5-dichloro-	0.268	0.08	0.81	0.14	Y	14
Styrene	Styrene, 2,5-dichloro-	0.18		0.25			411
Styrene	Styrene, 2,5-dichloro-	0.23		2.2			419
Styrene	Styrene, 2,5-dichloro-	0.26	0.17	0.14	0.04	N	84
Styrene	Styrene, 2,4,5-tribromo-	0.23		2.78			1054
Styrene	Styrene, 3-*tri*-*n*-butylstannyl-	18.5		0.03			985
Styrene	Styrene, 4-(*t*-butyldimethylsilyloxy)-	0.877	0.351	0.863	0.193	Y	976
Styrene	Styrene, 4-(trimethylsilyl)-	0.48		0.28			1036
Styrene	Styrene, 4-phenoxy-	1.38		0.94			602
Styrene	Styrene, α-(trimethylsilyloxy)-	1.48		0			970
Styrene	Styrene, α-methoxy-	2.51	0.34	0.07	0.05	N	142
Styrene	Styrene, α-methyl-	1.21	0.41	0.14	0.06	N	171
Styrene	Styrene, α-methyl-	0.96	0.03	0.05	0.13	N	281
Styrene	Styrene, α-methyl-	1.09		0.3			281
Styrene	Styrene, α-methyl-	1.13		0.4			281
Styrene	Styrene, α-methyl-	1.08	0.11	0.521	0.084	Y	289
Styrene	Styrene, α-methyl-	1.12		0.63			289
Styrene	Styrene, *m*-amino-	1.04		1.16			775
Styrene	Styrene, *m*-amino-	0.786	0.01	1.18	0.007	Y	989
Styrene	Styrene, *m*-bromo-	0.5		1.6			168
Styrene	Styrene, *m*-bromo-	0.55		1.05			256

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Styrene, *m*-chloro-	0.57		2.3			168
Styrene	Styrene, *m*-chloro-	0.64		1.09			256
Styrene	Styrene, *m*-cyano-	0.851	0.534	0.363	0.08		806
Styrene	Styrene, *m*-diethylaminoethyl-	1.2	0.28	1.21	0.26		806
Styrene	Styrene, *m*-dimethylamino-	1.027	0.248	0.617	0.264		806
Styrene	Styrene, *m*-hydroxy-	0.78	0.07	1.33	0.06	N	35
Styrene	Styrene, *m*-methoxy-	1.742	0.456	0.661	0.232		806
Styrene	Styrene, *m*-nitro-	0.279	0.055	0.5	0.14	Y	222
Styrene	Styrene, *m*-sulfonic acid fluoride	0.788	0.183	1.268	0.236		806
Styrene	Styrene, *p*-2-methoxyphenyl ketone	0.23	0.07	1.52	0.58		806
Styrene	Styrene, *p*-4-methoxyphenyl ketone	0.423	0.071	1.85	0.93		806
Styrene	Styrene, *p*-phenyl ketone	0.11	0.15	2.58	0.83		806
Styrene	Styrene, *p*-phenyl ketone	0.13	0.02	0.55	9.44		806
Styrene	Styrene, *p*-1-(2-hydroxypropyl)-	0.97		0.91			380
Styrene	Styrene, *p*-2-(2-hydroxypropyl)-	0.5		0.56			390
Styrene	Styrene, *p*-2-(2-hydroxypropyl)-	0.79		1.25			547
Styrene	Styrene, *p*-acetoxy-	0.835	0.029	1.305	0.05		806
Styrene	Styrene, *p*-acetoxy-	0.887	0.034	1.218	0.069		806
Styrene	Styrene, *p*-amino-	0.41		0.22			775
Styrene	Styrene, *p*-amino-	0.876	0.007	1.07	0.004	Y	989
Styrene	Styrene, *p*-bromo-	0.7	0.03	1.05	0.07	N	125
Styrene	Styrene, *p*-bromo-	0.67	0.09	1.1	0.05	N	128
Styrene	Styrene, *p*-bromo-	0.71	0.14	0.99	0.11	Y	256
Styrene	Styrene, *p*-bromo-	0.71		1.05			447
Styrene	Styrene, *p*-bormo-	0.69		1			554
Styrene	Styrene, *p*-chloro-	0.62	0.1	1.08	0.07	N	128
Styrene	Styrene, *p*-chloro-	0.744	0.08	1.029	0.056	Y	139
Styrene	Styrene, *p*-chloro-	0.816	0.029	1.062	0.014	Y	139
Styrene	Styrene, *p*-chloro-	0.74		1.03			140
Styrene	Styrene, *p*-chloro-	0.39	0.043	0.76	0.12	Y	150
Styrene	Styrene, *p*-chloro-	0.76		1.76			523
Styrene	Styrene, *p*-chloro-	0.74		1.02			554
Styrene	Styrene, *p*-chloromethyl-	0.622	0.047	1.12	0.17	Y	756
Styrene	Styrene, *p*-chloromethyl-	0.62	0.15	1.12	0.2		806
Styrene	Styrene, *p*-chloromethyl-	0.6	0.3	1.66	0.46		806
Styrene	Styrene, *p*-cyano-	0.24	0.07	1.22	0.11	N	128
Styrene	Styrene, *p*-cyano-	0.28		1.16			256
Styrene	Styrene, *p*-decyl-	0.88	0.22	0.26	1.32		806
Styrene	Styrene, *p*-diethylaminoethyl-	1.12	0.19	0.59	0.19		806
Styrene	Styrene, *p*-ethoxy	0.98	0.19	0.71	0.14		806
Styrene	Styrene, *p*-fluoromethyl-	0.474	0.151	1.399	0.236		806
Styrene	Styrene, *p*-formyl-	0.2		3			1025
Styrene	Styrene, *p*-formyl-	0.16		1.43			956
Styrene	Styrene, *p*-iodo-	0.485	0.093	1.044	0.159		806
Styrene	Styrene, *p*-iodo-	0.629	0.039	0.965	0.124	Y	839
Styrene	Styrene, *p*-mercapto-	0.607	0.188	0.907	0.223		806
Styrene	Styrene, *p*-methoxy-	0.85	0.19	0.69	0.1	N	128
Styrene	Styrene, *p*-methoxy-	1.14	0.6	1.01	1.27	N	160
Styrene	Styrene, *p*-methoxy-	1.16		0.82			256
Styrene	Styrene, *p*-methoxy-	1.05		0.79			535
Styrene	Styrene, *p*-methyl-	0.891	0.086	0.993	0.028	Y	261
Styrene	Styrene, *p*-nitro-	0.193	0.092	1.13	0.13	Y	256
Styrene	Styrene, *p*-octylamine sulfonate	0.7		2.7			817
Styrene	Styrene, *p*-sulfonic acid fluoride	0.171	0.045	1.372	0.188		806
Styrene	Styrene, *p*-trimethoxysilyl-	0.774	0.126	1.581	0.342		806
Styrene	Styrene, *p*-trimethylsilyl-	1.26	0.47	1.15	0.72		806
Styrene	Styrene, pentachloro-	1.33	0.19	0.07	0.1	N	7
Styrene	Succinimide, *N*-vinyl-	10.74	0.49	0.045	0.036	Y	104
Styrene	Succinimide, *N*-vinyl-	10.09	0.46	0.045	0.072	Y	163
Styrene	Succinimide, *N*-vinyl-	2.31	0.98	0.01	0.23	Y	693
Styrene	Succinimide, *N*-vinyl-	0.052	0.032	7	1.6	Y	80
Styrene	Terpyridinyl, 4′-vinyl-2,2′:6′,2″-	0.53	0.384	1.26	0.33	N	953
Styrene	Tetrazole, 1-vinyl-	3.85	0.045	0.184	0.031	Y	925
Styrene	Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	0.65		1.3			583
Styrene	Tetrazole, 2-methyl-5-vinyl-	1.124	0.081	0.62	0.21	Y	670

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	0.95		1			583
Styrene	Tetrazole, 5-phenyl-2-(4′vinyl)phenyl-	0.48		2.2			583
Styrene	Tetrazole, 5-vinyl-	0.78		0.31			870
Styrene	Tetrazole, 5-vinyl-	0.78	0.3	0.31	0.02		902
Styrene	Toluenesulfonamide, *N,N*-methyl-vinyl-	5.6	1.13	−0.58	0.44	Y	80
Styrene	Triallyl citrate	20.02	0.81	0.076	0.041	Y	367
Styrene	Triallyl cyanurate	32.45	1.27	0.22	0.12	Y	201
Styrene	Triallyl isocyanurate	28.6	1.39	−0.05	0.11	Y	201
Styrene	Triazine, 1,3,5-, 2-amino-4-(*N*-methyl-*p*-aminoanilino)-	0.306	0.054	0.44	0.12	Y	364
Styrene	Triazine, 1,3,5-, 2-amino-4-(*N*-methyl-*p*-nitroanilino)-	0.169	0.013	1.676	0.065	Y	364
Styrene	Triazine, 1,3,5-, 2-amino-4-(*p*-nitroanil)-	0.257	0.062	0.91	0.18	Y	364
Styrene	Triazine, 4,6-diamino-2-vinyl-	1.104	0.157	0.538	0.141		806
Styrene	Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vi	1.32	0.15	0.73	0.4	Y	670
Styrene	Triazole, 1,2,3-, 4(5)-vinyl-	1.48	0.12	0.52	0.26	Y	670
Styrene	Urea, 1,3-divinyl-1,3-diphenyl-	3.62	0.98	−1.76	1.36	Y	800
Styrene	Urea, *N*-acryloyl-*N′*-benzoyl-	0.404		0.053			1090
Styrene	Urea, *N*-vinyl-*N′*-ethyl-	10.78	2.07	−0.33	0.33	Y	101
Styrene	Valerolactone, α-methylen-	0.097		1.42			919
Styrene	Valerolactone, α-methylene-	0.096		1.44			855
Styrene	Vinyl 2-, 1,1-dichlorocyclopropane	17.98	0.59	0	0.034		806
Styrene	Vinyl 2-, 2-methyl-4,4,6-6-tetraphenyl cyclotrisiloxane	0.11		0.8			626
Styrene	Vinyl 2-chloroethyl ether	160		0.07			597
Styrene	Vinyl 4, dithiobenzoate methyl	0.111	0.009	2.08	0.81	Y	697
Styrene	Vinyl 4-chlorocyclohexyl ketone	0.53		0.58			668
Styrene	Vinyl acetate	42.48	26.85	−0.04	0.072	Y	158
Styrene	Vinyl acetate	18.8	2.87	0.02	0.12	Y	241
Styrene	Vinyl acetate	56		0.01			481
Styrene	Vinyl acetate	48		0.05			481
Styrene	Vinyl acetate	60		0.16			481
Styrene	Vinyl acetate	57.8		0.08			812
Styrene	Vinyl acetoxymethyl ketone	0.103	0.011	0.368	0.018		806
Styrene	Vinyl benzoate	34.8	29.3	−0.02	1.62	Y	26
Styrene	Vinyl benzoate	31.56	4.33	0.06	0.15	N	696
Styrene	Vinyl benzyl sulfide	2.32	0.25	0.04	0.17	Y	303
Styrene	Vinyl bromide	16.88	0.94	0.012	0.049	Y	34
Styrene	Vinyl bromide	18		0.06			34
Styrene	Vinyl bromide	15	8.26	0.09	0.2	N	368
Styrene	Vinyl butyl ether	21.25	16.53	−0.08	0.78	Y	241
Styrene	Vinyl butyl sulfide	2.69	0.25	−0.06	0.18	Y	303
Styrene	Vinyl butylsulfonate	1.74	0.27	0	0.16	Y	185
Styrene	Vinyl chloride	27.54	7.35	−0.06	0.08	Y	109
Styrene	Vinyl chloride	17.24	1.12	0.058	0.003	Y	238
Styrene	Vinyl chloride	23.92	3.6	0.16	0.08	N	242
Styrene	Vinyl chloride	25	5	0.005	0.031	Y	46
Styrene	Vinyl chloride	12.4		0.04			516
Styrene	Vinyl chloride	14.7	29.2	0.01	0.87	Y	59
Styrene	Vinyl chloroacetate	14.57	62.41	−0.04	0.24	Y	212
Styrene	Vinyl chloroacetate	45		0.03			542
Styrene	Vinyl chloromethyl ketone	0.127	0.047	0.507	0.039	Y	353
Styrene	Vinyl cymantrene	2.324	0.06	0.096	0.014	Y	354
Styrene	Vinyl dichloroacetate	20		0.28			542
Styrene	Vinyl dodecyl ether	27		0			374
Styrene	Vinyl dodecyl ether	56		0			374
Styrene	Vinyl ether	152	185	1.2	5.3	Y	45
Styrene	Vinyl ethyl ether	90		0			140
Styrene	Vinyl ethyl ether	111	15.5	−0.122	0.066	Y	194
Styrene	Vinyl ethyl ketone	13.83	1.44	−0.033	0.023	Y	112
Styrene	Vinyl ethyl ketone	0.2	0.08	0.43	0.06	N	344
Styrene	Vinyl ethyl ketone	0.29	0.16	0.332	0.043	Y	353
Styrene	Vinyl ethyl ketone	0.665	0.021	0.122	0.017	Y	550
Styrene	Vinyl ethyl oxalate	5.62	1.06	−0.12	0.19	Y	124
Styrene	Vinyl ethyl sulfide	4.76	0.2	0.113	0.032	Y	303

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Vinyl ethyl sulfide	6		0.25			507
Styrene	Vinyl ethyl sulfioxde	9.64	0.54	0.101	0.044	Y	286
Styrene	Vinyl ethyl sulfoxide	6		0.1			286
Styrene	Vinyl hendecanoate	24.89	1.22	−0.17	0.18	Y	145
Styrene	Vinyl iodide	7.401	0.315	0.126	0.035		806
Styrene	Vinyl isobutyl ether	11.58	1.91	0.65	0.39	N	349
Styrene	Vinyl isobutyl ether	11.96	2.32	0.29	0.33	N	349
Styrene	Vinyl isobutyl ether	50		0.01			490
Styrene	Vinyl isobutyl sulfide	2.38	0.41	−0.11	0.2	Y	303
Styrene	Vinyl isopropyl ketone	0.46	0.05	0.26	0.07	N	344
Styrene	Vinyl isopropyl ketone	0.36	0.1	0.272	0.046	Y	353
Styrene	Vinyl isopropyl ketone	0.4		0.3			599
Styrene	Vinyl isopropyl sulfide	4.61	0.18	0.16	0.068	Y	303
Styrene	Vinyl isothiocyanate	0.65		0.37			422
Styrene	Vinyl isothiocyanate	0.8		0.5			542
Styrene	Vinyl methyl ketone	0.288	0.092	0.349	0.068	Y	140
Styrene	Vinyl methyl ketone	0.237	0.023	0.348	0.009	Y	353
Styrene	Vinyl methyl ketone	0.35		0.27			550
Styrene	Vinyl methyl sulfide	4.5		0.15			193
Styrene	Vinyl methyl sulfide	4.46	0.11	0.066	0.056	Y	195
Styrene	Vinyl methyl sulfide	5	0.24	0.046	0.084	Y	303
Styrene	Vinyl methyl sulfone	1.146	0.087	−0.22	0.17	Y	195
Styrene	Vinyl methyl sulfoxide	3.57		0.01			737
Styrene	Vinyl octyl ether	65		0			374
Styrene	Vinyl pelargonate	49.5		0.01			145
Styrene	Vinyl phenyl ether	1.7		0.01			423
Styrene	Vinyl phenyl ketone	0.24	0.04	0.3	0.02	N	344
Styrene	Vinyl phenyl ketone	0.21	0.042	0.465	0.03	Y	353
Styrene	Vinyl phenyl sulfide	3.63	0.12	0.019	0.054	Y	193
Styrene	Vinyl phenyl sulfide	3.96	0.27	0.26	0.18	N	239
Styrene	Vinyl phenyl sulfone	1.562	0.091	−0.31	0.16	Y	193
Styrene	Vinyl stearate	15.96	13.32	−0.39	0.71	Y	145
Styrene	Vinyl sulfone	0.731	0.058	−0.027	0.047	Y	45
Styrene	Vinyl *tert*-butyl ketone	0.3	0.31	3.2	0.46	N	344
Styrene	Vinyl *tert*-butyl ketone	0.36	0.11	0.248	0.05	Y	353
Styrene	Vinyl *tert*-butyl sulfide	4.36	0.76	0.158	0.073	Y	303
Styrene	Vinyl thiolacetate	4		0.25			566
Styrene	Vinyl, 1-cyano, acetate	0.181	0.409	0.147	0.033		806
Styrene	Vinyl, 2-bromo-, ethyl ether					Y	194
Styrene	Vinyl, *p*-benzylmethylcarbinol	0.98	0.32	0.94	0.15	Y	48
Styrene	Vinyl-*tris*(trimethoxysiloxy)silane	26		0			209
Styrene	Vinyl-*tris*(trimethoxysiloxy)silane	24		0.01			482
Styrene	Vinylacetophenone, *p*-	0.246	0.072	1.12	0.42	Y	266
Styrene	Vinylanthracene, 9-	2.12		0.25			123
Styrene	Vinylanthracene, 9-	2.12		0.3			48
Styrene	Vinylbenzoic acid, *p*-	0.282	0.011	1.029	0.06	Y	266
Styrene	Vinylene carbonate	411		3.16		Y	103
Styrene	Vinylene carbonate	21.03	23.45	−0.73	0.98	Y	120
Styrene	Vinylferrocene	4.37	1.44	0.2	0.1	Y	274
Styrene	Vinylferrocene	2.76	0.22	0.14	0.14	Y	294
Styrene	Vinylhydroquinone dibenzoate	0.22		0.43			440
Styrene	Vinylidene chloride	1.81	0.35	0.134	0.48	Y	138
Styrene	Vinylidene chloride	1.79	0.14	0.098	0.01	Y	238
Styrene	Vinylidene chloride	1.8		1.5			393
Styrene	Vinylidene chloride	1.839	0.024	0.87	0.007	Y	59
Styrene	Vinylidene chloride	1.81		0.15			812
Styrene	Vinylidene chloride	1.7	0.12	0.11	0.005	N	93
Styrene	Vinylidene cyanide	0.015	0.007	0.72	0.16	Y	134
Styrene	Vinylidene cyanide	0.003	0.001	0.001	0.001	Y	82
Styrene	Vinylisocyanate	8.13		0.08			435
Styrene	Vinylisocyanate	6.9		0.1			533
Styrene	Vinylmethyldiacetoxysilane	4.08	0.14	0	0.15	Y	754
Styrene	Vinylmethyldiethoxysilane	5.68	0.1	0.08	0.14	Y	754
Styrene	Vinylmethyldiethoxysilane	11.29	0.43	−0.08	0.2	Y	754
Styrene	Vinylphenyldimethylsilane	32.53	0.49	0.024	0.016	Y	482

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene	Vinylphenylpropanol, 2-*p*-	0.48	0.3	0.41	0.4	Y	72
Styrene	Vinyltriethoxysilane	20.86	1.99	−0.09	0.14	Y	209
Styrene	Vinyltrimethylsilane	14.66	5.15	−0.19	0.36	Y	209
Styrene	Vinyltrimethylsilane					Y	482
Styrene	Vinyltrimethylsilane	5.98	0.18	0.049	0.025	Y	73
Styrene	Xanathate, *S*-methacryloyl *O*-ethyl	0.44	0.06	0.36	0.08		1086
Styrene sulfonate, sodium salt	*N*-Vinylpyrrolidone	7.19		0.084			939
Styrene, 2,4,6-trimethyl-	Acrylonitrile	0.065	0.048	0.874	0.051	Y	58
Styrene, 2,4,6-trimethyl-	Methacrylate, methyl	0.083	0.045	1.511	0.072	Y	58
Styrene, 2,4,6-trimethyl-	Methacrylate, methyl	−0.01	0.11	1.45	0.16	Y	58
Styrene, 2,4,6-trimethyl-	Styrene, *p*-chloro-	0.34	1.05	17.83	5.24	Y	58
Styrene, 2,4-dibromo-	Styrene	0.91	2.88	0.14	1.61		806
Styrene, 2,5-dichloro-	Acrylate, methyl	3.93	1.38	0.142	0.054	Y	136
Styrene, 2,5-dichloro-	Acrylate, methyl	4.47	0.15	0.288	0.069	Y	2
Styrene, 2,5-dichloro-	Acrylonitrile	0.08	0.03	0.22	0.01	N	84
Styrene, 2,5-dichloro-	Acrylonitrile	0.07		0.25			84
Styrene, 2,5-dichloro-	Acrylonitrile	0.09		0.26			84
Styrene, 2,5-dichloro-	Butadiene	0.45	0.17	0.449	0.043	Y	255
Styrene, 2,5-dichloro-	Butadiene	0.187	0.049	0.662	0.017	Y	8
Styrene, 2,5-dichloro-	Methacrylate, methyl	2.66	0.38	0.44	0.13	Y	3
Styrene, 2,5-dichloro-	Pyridine, 2-vinyl-	0.14	0.18	0.62	0.12	N	2
Styrene, 2,5-dichloro-	Styrene	0.81	0.14	0.268	0.08	Y	14
Styrene, 2,5-dichloro-	Styrene	0.25		0.18			411
Styrene, 2,5-dichloro-	Styrene	2.2		0.23			419
Styrene, 2,5-dichloro-	Styrene	0.14	0.04	0.26	0.17	N	84
Styrene, 2,5-dichloro-	Styrene, 2,5-dimethyl-	1.543	0.068	0.263	0.022	Y	91
Styrene, 2,5-dichloro-	Styrene, α-methyl-	3		0.14			377
Styrene, 2,5-dichloro-	Vinyl methyl ketone	2		0.5			375
Styrene, 2,5-dichloro-	Vinylidene cyanide	0.026	0.006	0.005	0.009	Y	82
Styrene, 2,5-dimethyl-	Styrene, 2,5-dichloro-	0.263	0.022	1.543	0.068	Y	91
Styrene, 2,6-dichloro-	Methacrylate, methyl	−0.15	0.27	1.98	0.16	N	324
Styrene, 2,4,5-tribromo-	Styrene	2.78		0.23			1054
Styrene, 3-tri-*n*-butylstannyl-	Acrylate, ethyl	0.01		10.4			985
Styrene, 3-tri-*n*-butylstannyl-	Acrylonitrile	0.001		4.88			985
Styrene, 3-tri-*n*-butylstannyl-	Methacrylate, methyl	0.03		0.69			985
Styrene, 3-tri-*n*-butylstannyl-	Styrene	0.03		18.5			985
Styrene, 3-tri-*n*-butylstannyl-	Vinyl acetate	0.38		5.33			985
Styrene, 4-(*tert*-butyldimethylsilyloxy)-	Styrene	0.863	0.193	0.877	0.351	Y	976
Styrene, 4-(trimethylsilyl)-	Styrene	0.28		0.48			1036
Styrene, 4-methyl-	Acrylonitrile	0.227	0.021	0.079	0.005	Y	929
Styrene, 4-methyl-	Methacrylate, methyl	0.46	0.022	0.345	0.008	Y	929
Styrene, 4-phenoxy-	Styrene	0.94		1.38			602
Styrene, α-(trimethylsilyloxy)-	Acrylonitrile	0		0.05			970
Styrene, α-(trimethylsilyloxy)-	Styrene	0		1.48			970
Styrene, α-methoxy-	Acrylate, methyl	0.003	0.01	0.17	0.02	N	142
Styrene, α-methoxy-	Acrylonitrile	−0.002	0.01	0.06	0.01	N	142
Styrene, α-methoxy-	Methacrylate, methyl	−0.02	0.04	2.5	0.39	N	142
Styrene, α-methoxy-	Methacrylonitrile	−0.02	0.005	0.83	0.04	N	143
Styrene, α-methoxy-	Styrene	0.07	0.05	2.51	0.34	N	142
Styrene, α-methyl-	Acrylate, 2-cyano-, methyl	0.05		0.001			452
Styrene, α-methyl-	Acrylonitrile	0.08	0.03	0.07	0.04	N	171
Styrene, α-methyl-	Acrylonitrile	0.14	0.03	0.04	0.005	N	265
Styrene, α-methyl-	Acrylonitrile	0.17		0.04			265
Styrene, α-methyl-	Acrylonitrile	0.24		0.04			265
Styrene, α-methyl-	Acrylonitrile	0.09	0.024	0.063	0.041	Y	75
Styrene, α-methyl-	Acrylonitrile	0.103	0.036	0.34	0.15	Y	75
Styrene, α-methyl-	Acrylonitrile	0.14		0.03			835
Styrene, α-methyl-	Acrylonitrile	0.08	0.07	−0.02	0.27	N	94
Styrene, α-methyl-	Butadiene	0.139	0.087	1.5	0.13	Y	70
Styrene, α-methyl-	Butadiene, 2-fluoro-	0.72	0.24	2.22	0.7	Y	174
Styrene, α-methyl-	Fumaronitrile	0.094	0.02	−0.01	0.078	Y	77
Styrene, α-methyl-	Maleic anhydride	0.04		0.08			492
Styrene, α-methyl-	Methacrylate, methyl	0.27	0.07	0.48	0.02	N	265
Styrene, α-methyl-	Methacrylate, methyl	0.51		0.55			265
Styrene, α-methyl-	Methacrylate, methyl	0.6		0.55			265

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene, α-methyl-	Methacrylate, methyl	0.14		0.5			532
Styrene, α-methyl-	Methacrylate, methyl	0.22		0.42			835
Styrene, α-methyl-	Methacrylate, methyl	0.48		0.47			899
Styrene, α-methyl-	Methacrylonitrile	0.15		0.21			542
Styrene, α-methyl-	Styrene	0.14	0.06	1.21	0.41	N	171
Styrene, α-methyl-	Styrene	0.05	0.13	0.96	0.03	N	281
Styrene, α-methyl-	Styrene	0.3		1.09			281
Styrene, α-methyl-	Styrene	0.4		1.13			281
Styrene, α-methyl-	Styrene	0.521	0.084	1.08	0.11	Y	289
Styrene, α-methyl-	Styrene	0.63		1.12			289
Styrene, α-methyl-	Styrene, 2,5-dichloro-	0.14		3			377
Styrene, α-methyl-	Styrene, *p*-chloro-	0.21	0.2	1.56	0.42	Y	218
Styrene, α-methyl-	Vinyl chloromethyl ketone	0.009	0.01	0.386	0.034	Y	356
Styrene, α-methyl-	Vinylhydroquinone dibenzoate	0.11		0.3			440
Styrene, *m*-amino-	Styrene	1.16		1.04			775
Styrene, *m*-amino-	Styrene	1.18	0.007	0.786	0.01	Y	989
Styrene, *m*-amino-	Styrene, *p*-amino-	0.63		0.49			774
Styrene, *m*-bromo-	Methacrylate, methyl	1.17		0.48			256
Styrene, *m*-bromo-	Styrene	1.6		0.5			168
Styrene, *m*-bromo-	Styrene	1.05		0.55			256
Styrene, *m*-bromo-	Styrene, *p*-methoxy-	1.4		0.25			168
Styrene, *m*-bromo-	Methacrylate, methyl					Y	150
Styrene, *m*-chloro-	Styrene	2.3		0.57			168
Styrene, *m*-chloro-	Styrene	1.09		0.64			256
Styrene, *m*-chloro-	Styrene, *p*-methoxy-	1.9		0.2			168
Styrene, *m*-cyano-	Styrene	0.363	0.08	0.851	0.534		806
Styrene, *m*-diethylaminoethyl-	Styrene	1.21	0.26	1.2	0.28		806
Styrene, *m*-dimethylamino-	Styrene	0.617	0.264	1.027	0.248		806
Styrene, *m*-hydroxy-	Styrene	1.33	0.06	0.78	0.07	N	35
Styrene, *m*-methoxy-	Styrene	0.661	0.232	1.742	0.456		806
Styrene, *m*-methyl	Methacrylate, methyl	0.53	0.04	0.48	0.04		1087
Styrene, *m*-methyl	Methacrylate, methyl	0.36	0.06	0.38	0.03		1087
Styrene, *m*-methyl	Methacrylate, methyl	0.51	0.05	0.41	0.04		1087
Styrene, *m*-methyl	Methacrylic acid	0.06	0.03	0.35	0.08		1087
Styrene, *m*-methyl	Methacrylic acid	0.33	0.02	0.18	0.02		1087
Styrene, *m*-methyl	Methacrylic acid	0.42	0.01	0.51	0.01		1087
Styrene, *m*-methyl	Acrylate, methyl	1.65		0.14			558
Styrene, *m*-methyl-	Acrylonitrile	0.43		0.07			378
Styrene, *m*-methyl-	Methacrylate, methyl	0.486	0.02	0.512	0.036	Y	256
Styrene, *m*-nitro-	Methacrylate, methyl	0.8	0.32	0.35	0.11	Y	222
Styrene, *m*-nitro-	Styrene	0.5	0.14	0.279	0.055	Y	222
Styrene, *m*-nitro-	Styrene, *p*-chloro-	1.73	0.47	0.3	0.14	Y	222
Styrene, *m*-sulfonic acid fluoride	Styrene	1.268	0.236	0.788	0.183		806
Styrene, *o*-chloro-	Acrylate, butyl	2.25		0.2			558
Styrene, *o*-chloro-	Indene	2.86	0.83	−0.069	0.066	Y	6
Styrene, *o*-chloro-	Methacrylate, butyl	1.24		0.32			558
Styrene, *o*-chloro-	Methacrylate, ethyl	1.34		0.45			558
Styrene, *o*-chloro-	Methacrylate, methyl	1.37		0.5			532
Styrene, *o*-chloro-	Methacrylate, methyl	1.34		0.46			558
Styrene, *o*-chloro-	Methacrylic acid	0.101	0.034	0.622	0.058	Y	6
Styrene, *o*-chloro-	Methacrylonitrile	0.78		0.86			44
Styrene, *p*-2-methoxyphenyl ketone	Styrene	1.52	0.58	0.23	0.07		806
Styrene, *p*-phenyl ketone	Styrene	2.58	0.83	0.11	0.15		806
Styrene, *p*-phenyl ketone	Styrene	0.55	9.44	0.13	0.02		806
Styrene, *p*-1-(2-hydroxybutyl)-	Acrylate, butyl	0.4		0.17			548
Styrene, *p*-1-(2-hydroxybutyl)-	Acrylate, methyl	0.48		0.18			548
Styrene, *p*-1-(2-hydroxybutyl)-	Acrylonitrile	0.31		0.04			548
Styrene, *p*-1-(2-hydroxypropyl)-	Acrylonitrile	0.53		0.1			380
Styrene, *p*-1-(2-hydroxypropyl)-	Methacrylate, methyl	1		0.26			380
Styrene, *p*-1-(2-hydroxypropyl)-	Styrene	0.91		0.97			380
Styrene, *p*-1-(2-hydroxypropyl)-	Styrene, *p*-chloro-	0.63		0.91			380
Styrene, *p*-2-(2-hydroxypropyl)-	Acrylonitrile	0.41		0.05			547
Styrene, *p*-2-(2-hydroxypropyl)-	Styrene	0.56		0.5			390
Styrene, *p*-2-(2-hydroxypropyl)-	Styrene	1.25		0.79			547
Styrene, *p*-2-(2-hydroxypropyl)-	Styrene, *p*-chloro-	0.53		1.24			547

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene, *p*-4-methoxyphenyl ketone	Styrene	1.85	0.93	0.423	0.071		806
Styrene, *p*–*N*,*N*-dimethylamino-	Acrylate, methyl	0.38	0.16	0.06	0.01	Y	866
Styrene, *p*-*N*,*N*-dimethylamino-	Benzophenone, *p*-vinyl-	0.07	0.05	0.84	0.18	Y	866
Styrene, *p*-*N*,*N*-dimethylamino-	Methacrylate, methyl	0.101	0.016	0.204	0.022	Y	256
Styrene, *p*-*N*,*N*-dimethylaminomethyl-	Benzophenone, *p*-vinyl	0.26	0.017	2.54	0.136	Y	86
Styrene, *p*-acetoxy-	Acrylonitrile	0.4	0.03	0.07	0.01	N	47
Styrene, *p*-acetoxy-	Styrene	1.305	0.05	0.835	0.029		806
Styrene, *p*-acetoxy-	Styrene	1.218	0.069	0.887	0.034		806
Styrene, *p*-amino-	Styrene	0.22		0.41			775
Styrene, *p*-amino-	Styrene	1.07	0.004	0.876	0.007	Y	989
Styrene, *p*-amino-	Styrene, *m*-amino-	0.49		0.63			774
Styrene, *p*-bromo-	Methacrylate, methyl	1.1		0.4			256
Styrene, *p*-bromo-	Styrene	1.05	0.07	0.7	0.03	N	125
Styrene, *p*-bromo-	Styrene	1.1	0.05	0.67	0.09	N	128
Styrene, *p*-bromo-	Styrene	0.99	0.11	0.71	0.14	Y	256
Styrene, *p*-bromo-	Styrene	1.05		0.71			447
Styrene, *p*-bromo-	Styrene	1		0.69			554
Styrene, *p*-bromo-	Styrene, *p*-methoxy-	1.15	0.08	0.44	0.02	N	128
Styrene, *p*-chloro-	Acrylate, α-trifluoromethyl-, methyl	0.24	0.02	0	0.03	N	825
Styrene, *p*-chloro-	Acrylonitrile, α-trifluoromethyl-	0.92	0.06	−0.09	0.06	N	825
Styrene, *p*-chloro-	Butadiene	1.6	5.7	1.22	2.16	Y	255
Styrene, *p*-chloro-	Butadiene, 1-(2-hydroxyethylthio)-	0.24		3.27			810
Styrene, *p*-chloro-	Carbazole, *N*-vinyl-	7		0.02			551
Styrene, *p*-chloro-	Cyclopentene, 4-, -1,3-dione	0.294	0.081	0.014	0.006	Y	264
Styrene, *p*-chloro-	Itaconate, dimethyl	0.69		0.15			398
Styrene, *p*-chloro-	Methacrylate, butyl	1.036	0.1	0.517	0.1	N	872
Styrene, *p*-chloro-	Methacrylate, butyl	1.025	0.124	0.564	0.131		966
Styrene, *p*-chloro-	Methacrylate, methyl	0.77	0.12	0.334	0.016	Y	150
Styrene, *p*-chloro-	Methacrylate, methyl	0.89		0.42			256
Styrene, *p*-chloro-	Methacrylate, methyl	0.47		0.43			476
Styrene, *p*-chloro-	Norbornadiene	69.93	24.19	0	0.12	Y	188
Styrene, *p*-chloro-	Styrene	1.08	0.07	0.62	0.1	N	128
Styrene, *p*-chloro-	Styrene	1.029	0.056	0.744	0.08	Y	139
Styrene, *p*-chloro-	Styrene	1.062	0.014	0.816	0.029	Y	139
Styrene, *p*-chloro-	Styrene	1.03		0.74			140
Styrene, *p*-chloro-	Styrene	0.76	0.12	0.39	0.043	Y	150
Styrene, *p*-chloro-	Styrene	1.76		0.76			523
Styrene, *p*-chloro-	Styrene	1.02		0.74			554
Styrene, *p*-chloro-	Styrene, 2,4,6-trimethyl-	17.83	5.24	0.34	1.05	Y	58
Styrene, *p*-chloro-	Styrene, α-methyl-	1.56	0.42	0.21	0.2	Y	218
Styrene, *p*-chloro-	Styrene, *m*-nitro-	0.3	0.14	1.73	0.47	Y	222
Styrene, *p*-chloro-	Styrene, *p*-1-(2-hydroxypropyl)-	0.91		0.63			380
Styrene, *p*-chloro-	Styrene, *p*-2-(2-hydroxypropyl)-	1.24		0.53			547
Styrene, *p*-chloro-	Styrene, *p*-cyano-	0.26	0.08	1.28	0.13	N	128
Styrene, *p*-chloro-	Styrene, *p*-methoxy-	0.47	0.16	0.41	0.08	N	128
Styrene, *p*-chloro-	Styrene, *p*-methoxy-	0.86		0.56			256
Styrene, *p*-chloro-	Styrene, *p*-methyl-	1.15		0.61			256
Styrene, *p*-chloro-	Vinyl methyl sulfoxide	6.46		0.01			737
Styrene, *p*-chloro-	Vinyl, *p*-, benzylmethylcarbinol	1		1		Y	48
Styrene, *p*-chloromethyl-	Acrylonitrile	0.56		0.067			1035
Styrene, *p*-chloromethyl-	Butadiene	0.42		0.87			1075
Styrene, *p*-chloromethyl-	Methacrylate, butyl	1.46		0.38			1035
Styrene, *p*-chloromethyl-	Methacrylate, ethyl	1.29		0.36			1035
Styrene, *p*-chloromethyl-	Methacrylate, methyl	0.91	0.15	0.408	0.038	Y	756
Styrene, *p*-chloromethyl-	Phosphazene, isopropenylpentafluoro-	1.64	0.58	−0.048	0.067	Y	765
Styrene, *p*-chloromethyl-	Styrene	1.12	0.17	0.622	0.047	Y	756
Styrene, *p*-chloromethyl-	Styrene	1.12	0.2	0.62	0.15		806
Styrene, *p*-chloromethyl-	Styrene	1.66	0.46	0.6	0.3		806
Styrene, *p*-cyano-	Methacrylate, methyl	1.41		0.22			256
Styrene, *p*-cyano-	Styrene	1.22	0.11	0.24	0.07	N	128
Styrene, *p*-cyano-	Styrene	1.16		0.28			256
Styrene, *p*-cyano-	Styrene, *p*-chloro	1.28	0.13	0.26	0.08	N	128
Styrene, *p*-cyano-	Styrene, *p*-methoxy-	0.89	0.03	0.14	0.05	N	128
Styrene, *p*-decyl-	Styrene	0.26	1.32	0.88	0.22		806
Styrene, *p*-diethylaminoethyl-	Styrene	0.59	0.19	1.12	0.19		806

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Styrene, *p*-dimethylamino-	Acrylate, methyl	0.372	0.055	0.192	0.01	Y	911
Styrene, *p*-dimethylamino-	Benzophenone, *p*-vinyl-	0.176	0.046	0.842	0.072	Y	911
Styrene, *p*-dimethylaminomethyl-	Benzophenone, *p*-vinyl-	0.135	0.017	2.54	0.259	Y	911
Styrene, *p*-ethoxy-	Styrene	0.71	0.14	0.98	0.19		806
Styrene, *p*-fluoro-	Myrcene	0.362	0.022	0.87	0.143	Y	1062
Styrene, *p*-fluoromethyl-	Styrene	1.399	0.236	0.474	0.151		806
Styrene, *p*-formyl-	Styrene	3		0.2			1025
Styrene, *p*-formyl-	Styrene	1.43		0.16			956
Styrene, *p*-iodo-	Styrene	1.044	0.159	0.485	0.093		806
Styrene, *p*-iodo-	Styrene	0.965	0.124	0.629	0.039	Y	839
Styrene, *p*-mercapto-	Styrene	0.907	0.223	0.607	0.188		806
Styrene, *p*-methoxy-	Acrylate, methyl	2		0.07			558
Styrene, *p*-methoxy	Methacrylate, methyl	0.32		0.29			35
Styrene, *p*-methoxy	Styrene	0.69	0.1	0.85	0.19	N	128
Styrene, *p*-methoxy	Styrene	1.01	1.27	1.14	0.6	N	160
Styrene, *p*-methoxy	Styrene	0.82		1.16			256
Styrene, *p*-methoxy	Styrene	0.79		1.05			535
Styrene, *p*-methoxy	Styrene, *m*-bromo-	0.25		1.4			168
Styrene, *p*-methoxy	Styrene, *m*-chloro-	0.2		1.9			168
Styrene, *p*-methoxy	Styrene, *p*-bromo-	0.44	0.02	1.15	0.08	N	128
Styrene, *p*-methoxy	Styrene, *p*-chloro-	0.41	0.08	0.47	0.16	N	128
Styrene, *p*-methoxy	Styrene, *p*-chloro-	0.56		0.86			256
Styrene, *p*-methoxy	Styrene, *p*-cyano-	0.14	0.05	0.89	0.03	N	128
Styrene, *p*-methyl	Acrylate, methyl	1.54		0.17			558
Styrene, *p*-methyl	Acrylonitrile	0.33		0.05			378
Styrene, *p*-methyl	Methacrylate, methyl	0.44		0.4			256
Styrene, *p*-methyl	*N,N*-Divinylaniline	6.15	0.16	0.06	0.03	Y	45
Styrene, *p*-methyl	Styrene	0.993	0.028	0.891	0.086	Y	261
Styrene, *p*-methyl	Styrene, *p*-chloro-	0.61		1.15			256
Styrene, *p*-methyl	Vinyl methyl sulfoxide	2.73		0.01			737
Styrene, *p*-nitro	Styrene	1.13	0.13	0.193	0.092	Y	256
Styrene, *p*-octylamine sulfonate	Acrylate, butyl	2.3		0.3			817
Styrene, *p*-octylamine sulfonate	Methacrylate, butyl	1.5		0.7			817
Styrene, *p*-octylamine sulfonate	Methacrylate, methyl	2.6		0.6			817
Styrene, *p*-octylamine sulfonate	Styrene	2.7		0.7			817
Styrene, *p*-sulfonic acid fluoride	Styrene	1.372	0.188	0.171	0.045		806
Styrene, *p-tert*-butyl-	Methacrylate, methyl	0.5		0.44			634
Styrene, *p-tert*-butyl-	Styrene	1.581	0.342	0.774	0.126		806
Styrene, *p*-trimethoxysilyl-	Imidazole, *N*-vinyl-	6.9		0.002			1095
Styrene, *p*-trimethylsilyl-	Styrene	1.15	0.72	1.26	0.47		806
Styrene, pentachloro-	Methacrylate, methyl	0.3	0.05	3.65	0.85	N	7
Styrene, pentachloro-	Styrene	0.07	0.1	1.33	0.19	N	7
Styrene, pentachloro-	Vinyl chloride	5.3		0.43			516
Styrene, *tert*-butoxy-	Fumarate, dibutyl	0.18		0.01			1077
Styrenesulfonate, *p*-, sodium	Acrolein	0.047	0.071	0.26	0.16	Y	57
Styrenesulfonate, *p*-, sodium	Acrolein	0.327	0.033	0.395	0.092	Y	57
Styrenesulfonate, *p*-, sodium	Acrolein	0.01	0.009	0.113	0.018	Y	57
Styrenesulfonate, *p*-, sodium	Acrylate, α-chloro-, sodium	1.44	0.31	0.27	0.06	Y	834
Succinamate, potassium *N*-vinyl-	Methacrylate, methyl	−0.02	0.12	1.1	0.17	Y	771
Succinimide, *N*-vinyl-	Acrylamide	0.17		1.86			600
Succinimide, *N*-vinyl-	Acrylate, butyl	0.15		1.54			789
Succinimide, *N*-vinyl-	Acrylate, methyl	0.9	0.17	0.357	0.063	Y	80
Succinimide, *N*-vinyl-	Acrylonitrile	0.516	0.081	0.116	0.024	Y	80
Succinimide, *N*-vinyl-	Carbazole, *N*-vinyl-	0.11	0.83	0.04	0.09	Y	284
Succinimide, *N*-vinyl-	Maleate, dimethyl	1.25	0.16	−0.012	0.021	Y	284
Succinimide, *N*-vinyl-	Maleic anhydride	0.148	0.073	0.021	0.027	Y	284
Succinimide, *N*-vinyl-	Methacrylate, methyl	0.048	0.026	9.94	0.27	Y	104
Succinimide, *N*-vinyl-	Methacrylate, methyl	−0.059	0.096	0.01	0.031	Y	693
Succinimide, *N*-vinyl-	Pyrrolidone, *N*-vinyl-	3.67	0.89	0.97	0.42	N	96
Succinimide, *N*-vinyl-	Styrene	0.045	0.036	10.74	0.49	Y	104
Succinimide, *N*-vinyl-	Styrene	0.045	0.072	10.09	0.46	Y	163
Succinimide, *N*-vinyl-	Styrene	0.01	0.23	2.31	0.98	Y	693
Succinimide, *N*-vinyl-	Styrene	7	1.6	0.052	0.32	Y	80
Succinimide, *N*-vinyl-	Vinyl acetate	5.68	0.12	0.072	0.009	Y	104
Succinimide, *N*-vinyl-	Vinyl acetate	4.49	9.49	0.18	0.54	Y	167

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Succinimide, *N*-vinyl-	Vinyl acetate	1.99	0.43	0.229	0.06	Y	80
Succinimide, *N*-vinyl-	Vinyl thiolacetate	0.17	0.15	2.47	0.37	N	97
Succinimide, *N*-vinyl-	Vinylidene chloride	0.311	0.058	1.441	0.061	Y	104
Sulfonate, 2-acrylamido-2-methylpropane-, sodium	Acrylamide, *N*,*N*-dimethyl-	0.162	0.949	1.108	0.59	Y	857
Sulfonic acid, *o*-methacryloylamino-benzene-	Methacrylic acid	0.217	0.122	0.203	0.067	Y	1006
Sulfonic acid, *o*-methacryloylamino-benzene-	Methacrylic acid	0.523	0.195	0.188	0.076	Y	1006
Sulfonic acid, *o*-methacryloylamino-benzene-	Pyridine, 2-methyl-5-vinyl-	1.83	0.218	−0.002	0.031	Y	1006
Sulfonic acid, *p*-methacryloylamino-benzene-	Methacrylic acid	1.08	0.338	0.156	0.081	Y	1006
Sulfonic acid, *p*-methacryloylamino benzene-	Methacrylic acid	1.65	0.27	0.155	0.052	Y	1006
Sulfonic acid, *p*-methacryloylamino benzene-	Pyridine, 2-methyl-5-vinyl-	2.72	0.525	−0.054	0.048	Y	1006
Terpyridinyl 4′-vinyl-2,2′:6′,2′′-	Styrene	1.26	0.33	0.53	0.384	N	953
Tetrazole, 1-vinyl-	Acrylonitrile	0.314	0.066	0.54	0.032	Y	925
Tetrazole, 1-vinyl-	Methacrylate, methyl	0.262	0.026	7.13	0.093	Y	925
Tetrazole, 1-vinyl-	Styrene	0.184	0.031	3.85	0.045	Y	925
Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	Acrylonitrile	1.1		0.42			583
Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	Styrene	1.3		0.65			583
Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	Vinylidene chloride	1.9		0.13			583
Tetrazole, 2-methyl-5-vinyl-	Methacrylate, methyl	0.486	0.075	1.026	0.029	Y	670
Tetrazole, 2-methyl-5-vinyl-	Styrene	0.62	0.21	1.124	0.081	Y	670
Tetrazole, 2-methyl-5-vinyl-	Vinyl acetate	28.51	31.34	0.04	0.16	Y	670
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Methacrylate, methyl	0.4		0.7			583
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Styrene	1		0.95			583
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Vinylidene chloride	1.8		0.25			583
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Acrylonitrile	1.4		0.32			583
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Styrene	2.2		0.48			583
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	Vinylidene chloride	1.9		0.11			583
Tetrazole, 5-vinyl-	Styrene	0.31		0.78			870
Tetrazole, 5-vinyl-	Styrene	0.31	0.02	0.78	0.3		902
Thioindigoid	Methacrylate, butyl	0.5		11.5			1007
Thiophene, 3-vinyl	Acrylate, butyl	0.499		0.386			1000
Thiophene, 3-vinyl	Methacrylate, methyl	0.25		0.563			1000
Thioxanthone, 2-(acryloyloxy)-	Methacrylate, methyl	0.5		1.36			986
Titanium, dicylopentadienyl-, dimethacrylate	Copper diacrylate	0.89		1.09			895
Titanium, dicylopentadienyl-, dimethacrylate	Nickel diacrylate	0.65		0.95			895
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	Acrylonitrile	0.04	1.04	0.42	0.12	Y	80
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	Methacrylate methyl	−1.83	0.26	1.87	0.19	Y	80
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	Styrene	−0.58	0.44	5.6	1.13	Y	80
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	Vinyl butyl ether	−0.31	0.29	3.37	1.65	Y	80
Triallyl citrate	Acrylonitrile	−0.08	0.1	1.76	0.082	Y	367
Triallyl citrate	Allylbenzene	2.01	0.5	0.339	0.039	Y	367
Triallyl citrate	Styrene	0.076	0.041	20.02	0.81	Y	367
Triallyl citrate	Vinyl acetate	2.97	0.38	0.222	0.025	Y	367
Triallyl citrate	Vinyl chloride	1.03	0.17	0.605	0.033	Y	367
Triallyl cyanurate	Methacrylate, methyl	−0.005	0.05	16.12	0.56	Y	201
Triallyl cyanurate	Styrene	0.22	0.12	32.45	1.27	Y	201
Triallyl cyanurate	Vinyl acetate	1.29	0.42	0.197	0.053	Y	201
Triallyl isocyanurate	Methacrylate, methyl	−0.49	0.94	16.88	6.64	Y	201
Triallyl isocyanurate	Styrene	−0.05	0.11	28.6	1.39	Y	201
Triallyl isocyanurate	Vinyl acetate	1.95	0.42	0.278	0.043	Y	201
Triazine, 4,6-diamino-2-vinyl-	Styrene	0.538	0.141	1.104	0.157		806
Triazine, 1,3,5-2-amino-4-(*N*-methyl-*p*-aminoanilino)-6-isopropenyl-	Styrene	0.44	0.12	0.306	0.054	Y	364
Triazine, 1,3,5-2-amino-4-(*N*-methyl-*p*-aminoanilino)-6-isopropenyl-	Styrene	1.676	0.065	0.169	0.013	Y	364
Triazine, 1,3,5-2-amino-4-(*p*-nitroanilino)-6-isopropenyl-	Styrene	0.91	0.18	0.257	0.062	Y	364

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Triazine, 2-allyloxy-4,6-dichloro-	Pyrrolidone, *N*-vinyl-	0.5		0.98			826
Triazine, 4,6-diamino-2-vinyl-	Vinyl methyl ketone	1.35	0.68	0.17	0.11	Y	243
Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vinyl-	Methacrylate, methyl	0.266	0.058	0.832	0.022	Y	670
Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vinyl-	Styrene	0.73	0.4	1.32	0.15	Y	670
Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vinyl-	Vinyl acetate	42.47	9.84	−0.011	0.008	Y	670
Triazole, 1,2,3-, 4(5)-vinyl-	Methacrylate, methyl	0.27	0.13	0.839	0.055	Y	670
Triazole, 1,2,3-, 4(5)-vinyl-	Styrene	0.52	0.26	1.48	0.12	Y	670
Triazole, 1-*N*-vinyl-1,2,4-	Imidazole, 1-*N*-vinyl					N	972
Tricyclo[4,2,2,0{2,5}]dec-7-ene-3,4,9,10-tetracarboxylicacid	Acrylonitrile	0.25		2.75			963
Tropone, 2-methacryloyloxy-	Acrylate, ethyl	3.13	0.528	0.387	0.027	Y	1104
Urea, 1,3-divinyl-1,3-diphenyl-	Styrene	−1.76	1.36	3.62	0.98	Y	800
Urea, *N*-acryloyl-*N'*-benzoyl-	Styrene	0.053		0.404			1090
Urea, *N*-vinyl-*N'*-ethyl-	Methacrylate, methyl	−0.008	0.029	1.85	0.23	Y	101
Urea, *N*-vinyl-*N'*-ethyl-	Styrene	−0.33	0.33	10.78	2.07	Y	101
Urea, *N*-vinyl-*N'*-ethyl-	Vinyl acetate	0.621	0.081	0.463	0.029	Y	101
Urea, *N*-vinyl-*N'*-ethyl-	Vinyl acetate	3.61		0.16			683
Urea, *N*-vinyl-*N'*-ethyl-	Vinyl acetate	1.45		0.38			683
Valerolactone, α-methylene-	Styrene	1.42		0.097			919
Valerolactone, α-methylene-	Styrene	1.44		0.096			855
Vinyl 1-cyano- acetate	Styrene	0.147	0.033	0.181	0.409		806
Vinyl 12-ketostearate	Acrylonitrile	−0.18	0.36	3.3	0.5	Y	146
Vinyl 12-ketostearate	Vinyl acetate	−0.28		0.18		Y	146
Vinyl 12-ketostearate	Vinyl chloride	−0.01		0.03		Y	146
Vinyl 12-ketostearate	Vinylidene chloride	0.26	2.61	4.4	4.8	Y	146
Vinyl 2-, 1,1-dichlorocyclopropane	Styrene	0	0.034	17.98	0.59		806
Vinyl 2-bromo-ethyl ether	Acrylate, methyl	−0.79	0.32	16.83	2.34	Y	194
Vinyl 2-bromo-ethyl ether	Styrene					Y	194
Vinyl 2-chlorocyclohexyl ketone	Maleic anhydride	1.7		0.83			836
Vinyl 2-chloroethyl ether	Acrylate, ethyl	0.15		5			439
Vinyl 2-chloroethyl ether	Acrylate, ethyl	0		4.65			726
Vinyl 2-chloroethyl ether	Acrylate, methyl	0		3.27			726
Vinyl 2-chloroethyl ether	Acrylonitrile	0		1.09			597
Vinyl 2-chloroethyl ether	Methacrylate, methyl	0		18.8			597
Vinyl 2-chloroethyl ether	Styrene	0.07		160			597
Vinyl 2-chloroethyl ether	Vinyl acetate	0.16		2.36			597
Vinyl 2-chloroethyl ether	Vinyl chloride	−0.07	0.11	2.46	0.52	Y	304
Vinyl 2-ethylhexanoate	Acrylonitrile	−0.047	0.05	9.24	1.96	Y	47
Vinyl 3,3-bis(ethoxycarbonyl)propyl ether	Acrylonitrile	0.06		1.02			1015
Vinyl 4-dithiobenzoate methyl	Methacrylate, methyl	0.3	0.29	0.151	0.092	Y	697
Vinyl 4-dithiobenzoate methyl	Styrene	2.08	0.81	0.111	0.009	Y	697
Vinyl 4-chlorocyclohexyl ketone	Maleic anhydride	0.12		0.08			836
Vinyl 4-chlorocyclohexyl ketone	Methacrylate, methyl	0.51		0.77			668
Vinyl 4-chlorocyclohexyl ketone	Styrene	0.58		0.53			668
Vinyl 4-chlorocyclohexyl ketone	Vinyl acetate	0.15		0.1			668
Vinyl 4-chlorocyclohexyl ketone	Vinyl cyclohexyl ketone	0.5		1.6			668
Vinyl *N,N*-methylacetamide	Vinyl acetate	0.93		0.49			631
Vinyl α-chloro-, triethoxysilane	Acrylonitrile					Y	237
Vinyl acetate	2-Oxazoline, 2-isopropenyl-	0.25	0.04	7.1	2		894
Vinyl acetate	2-Oxazolinium BF4, 3-methyl-2-isopropenyl-	0.114	0.021	0.413	0.005	Y	908
Vinyl acetate	Acetamide, *N*-vinyl-	0.6		5.5			984
Vinyl acetate	Acetamide, *N*-vinyl-	0		21			984
Vinyl acetate	Acetate, 2-chloro-, vinyl	0.8		1.18			892
Vinyl acetate	Acetate, allyl	1.8		0.43			1045
Vinyl acetate	Acetate, chloro-, allyl	0.77		0.69			1045
Vinyl acetate	Acetate, dichloro-, allyl	0.57		0.54			1045
Vinyl acetate	Acetate, trichloro-, allyl	0.41		0.28			1045
Vinyl acetate	Acrolein	−0.02	0.12	3.04	0.43	N	207
Vinyl acetate	Acrolein, methyl	0.15		0.98			589
Vinyl acetate	Acrylamide, *N*-octadecyl-	0.004	0.011	8.25	1.46	Y	119
Vinyl acetate	Acrylamido, 1-, -1-deoxy-D-glucitol	0.03	0.17	0.98	0.87	N	258
Vinyl acetate	Acrylamido, 2-, -2-methylpropane-sulfonate	0.05		11.6			700

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl acetate	Acrylamidomethylamino, *p*-, azobenzene	0.126		12.98			858
Vinyl acetate	Acrylate, 2-cyano-, methyl	0.005		0.05			452
Vinyl acetate	Acrylate, 2-ethylhexyl	0.04		7.5			570
Vinyl acetate	Acrylate, α-acetoxy-, ethyl	0.08	0.021	5.621	0.094	Y	244
Vinyl acetate	Acrylate, α-chloro-, ethyl	0.03		30			638
Vinyl acetate	Acrylate, butyl	0.018	0.072	3.48	3.39	Y	301
Vinyl acetate	Acrylate, glycidyl	0.003		7.6			553
Vinyl acetate	Acrylate, methyl	0.04	0.01	7.28	0.37		1067
Vinyl acetate	Acrylate, methyl	0.03		6.38		Y	158
Vinyl acetate	Acrylate, methyl	0.405	0.043	2.58	0.3	Y	25
Vinyl acetate	Acrylate, methyl	0.03		6.7			489
Vinyl acetate	Acrylate, methyl	0.03		6.3			568
Vinyl acetate	Acrylate, sodium					Y	37
Vinyl acetate	Acrylate, trifluoro-, methyl	0.24		0			595
Vinyl acetate	Actylate, trifluoro-, methyl	0.24		0			598
Vinyl acetate	Acrylic acid	0.021	0.059	8.66	6.63	Y	37
Vinyl acetate	Acrylonitrile	0.04	0.067	4.05	0.82	Y	158
Vinyl acetate	Acrylonitrile					N	235
Vinyl acetate	Acrylonitrile	0.06	0.22	5.51	0.91	Y	6
Vinyl acetate	Acrylonitrile	−0.06	0.08	5.29	1.13	Y	75
Vinyl acetate	Acrylonitrile	−0.03	0.09	9.2	6.39	Y	75
Vinyl acetate	Allyl acetate	0.6		0.45			140
Vinyl acetate	Allyl acetate	1		0.7			502
Vinyl acetate	Allyl acetate	1		0.7			998
Vinyl acetate	Allyl acrylate	0.11		17.35			578
Vinyl acetate	Allyl butyrate	0.97		0.64			998
Vinyl acetate	Allyl chloride	1.355	0.089	−2.4	3.21	Y	204
Vinyl acetate	Allyl chloride	0.66	0.15	0.61		Y	3
Vinyl acetate	Allyl chloride	0.34		0.75	0.15		438
Vinyl acetate	Allyl isobutyrate	1.04		0.51			998
Vinyl acetate	Allyl propionate	1.29		0.42			998
Vinyl acetate	Allyl trimethylacetate	1.15		0.34			998
Vinyl acetate	Allyl valerate	1.07		0.58			998
Vinyl acetate	Allystearamide, *N*-	0.923	0.092	0.68	0.26	Y	119
Vinyl acetate	Azobenzene, 4-(acrylamidomethyl-amino)-	0.126		12.98			880
Vinyl acetate	Benzothiazole, vinylmercapto-	0.013	0.037	2.53	0.47	Y	763
Vinyl acetate	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	1.26		0.35			994
Vinyl acetate	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	1.46			0.15		994
Vinyl acetate	Bicyclo[2,2,1]hept-2-ene-5,6-dicarboximide, *N*-benzyl	1.99		0.014			994
Vinyl acetate	Butadiene, 2-chloro-	−0.02	0.09	33.52	8.98	N	247
Vinyl acetate	Butene-1	1.55	0.24	0.11	0.42	Y	211
Vinyl acetate	Butene-2, *cis*-	8.27	1.05	0.03	0.25	Y	211
Vinyl acetate	Butene-2, *trans*-	7.2	1.69	−0.03	0.44	Y	211
Vinyl acetate	Caprolactam, *N*-vinyl-	0.63		0.31			557
Vinyl acetate	Caprolactam, *N*-vinyl-	0.35		2.5			859
Vinyl acetate	Caprolactam, *N*-vinyl-	0.35		2.5			883
Vinyl acetate	Carbamate, *N,N*-diethyl-, vinyl	1.7	0.04	0.13	0.19	Y	197
Vinyl acetate	Carbamate, *N*-vinyl-, ethyl	0.39		2.09			629
Vinyl acetate	Carbamate, *N*-vinyl-, ethyl	0.418	0.063	0.4	0.26	Y	81
Vinyl acetate	Carbamate, *N*-vinyl-, *tert*-butyl	0.07	1.12	0.46	0.03	N	827
Vinyl acetate	Carbazole, *N*-vinyl-	0.126	0.023	2.67	0.13	Y	246
Vinyl acetate	Carbazole, *N*-vinyl-	0.15		3.02			246
Vinyl acetate	Carbazole, *N*-vinyl-	0.13		3.9			551
Vinyl acetate	Carbon monoxide	0.24		0.33			474
Vinyl acetate	Crotonic acid	0.317	0.081	0.04	0.14	Y	47
Vinyl acetate	Crotonic acid	0.3		0.01			526
Vinyl acetate	Crotonic acid	0.31	0.07	0.001	0.1	N	94
Vinyl acetate	Diallyl β-cyanoethylisocyanurate	0.06		4.6			778
Vinyl acetate	Diallyl melamine	0.68	0.026	0.318	0.073	Y	201
Vinyl acetate	Diallyl phthalate	0.336	0.049	2.19	0.34	Y	297

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl acetate	Diallyl phthalate	0.414	0.029	1.66	0.16	Y	297
Vinyl acetate	Diallyl phthalate	0.72		2			509
Vinyl acetate	Diallyl, 1,3-, 5-(2-hydroxy-3-phenoxypropyl)	0.15		5.5			671
Vinyl acetate	Diallyl, 1,3-, 5-(2-hydroxy-3-phenoxypropyl)	0.16		3.65			676
Vinyl acetate	Diallylcyanamide	0.01		1.62			581
Vinyl acetate	Diallyldimethylammonium chloride	0.35		1.95			1029
Vinyl acetate	Ethylene	0.72	1.2	0.16	0.03	N	309
Vinyl acetate	Ethylene					N	41
Vinyl acetate	Ethylene	1.14		0.28			413
Vinyl acetate	Ethylene	1.03	0.77	0.88	0.52	N	42
Vinyl acetate	Ethylene	1.52		0.74			559
Vinyl acetate	Ethylene	3.74	0.64	0.67	0.03	N	68
Vinyl acetate	Ethylene	1.23	0.21	0.13	0.01	N	68
Vinyl acetate	Ethylene	1.4		0.79			766
Vinyl acetate	Ethylene	0.33	0.58	−0.13	0.36	Y	6
Vinyl acetate	Ethylene, chlorotrifluoro-	0.68	0.11	0.04	0.02		1097
Vinyl acetate	Ethylene, chlorotrifluoro-	0.47		−.0.02		Y	236
Vinyl acetate	Ethylene, dichloro-, *cis*-	1.02	0.27	0.055	0.024	Y	137
Vinyl acetate	Ethylene, dichloro-, *cis*-	6.4		0.01		Y	137
Vinyl acetate	Ethylene, dichloro-, *cis*-	0.898	0.06	0.058	0.03	Y	9
Vinyl acetate	Ethylene, dichloro-, *cis*-	4.13	1.94	0.1	0.14	Y	9
Vinyl acetate	Ethylene, tetrachloro-	4.45	1.87	−0.07	0.015	Y	3
Vinyl acetate	Ethylene, tetrachloro-					Y	59
Vinyl acetate	Ethylene, trichloro-	0.607	0.076	−0.06	0.34	Y	158
Vinyl acetate	Ethylene, trichloro-	0.7	0.1	−0.006	0.019	Y	9
Vinyl acetate	Fumarate, diethyl	0.011	0.004	0.443	0.007	Y	137
Vinyl acetate	Fumarate, diethyl	0.09		0.33			718
Vinyl acetate	Fumarate, diisopropyl	0.012		0.9			1038
Vinyl acetate	Fumaryl chloride					Y	91
Vinyl acetate	Hexatriene, tetrachloro-					Y	4
Vinyl acetate	Imidazole, 1-vinyl-	−0.11	1.17	1.9	6.99	Y	364
Vinyl acetate	Indene	0.39	0.2	8.64	2.8	Y	360
Vinyl acetate	Isobutylene	1.66	0.18	0.17	0.19	Y	211
Vinyl acetate	Isobutylene, 3-chloro-	0.13		0			475
Vinyl acetate	Isopropenyl acetate	1.171	0.038	1.082	0.074	Y	100
Vinyl acetate	Itaconate, dimenthyl					Y	975
Vinyl acetate	Itaconic anhydride	−0.019	0.045	1.57	0.36	Y	365
Vinyl acetate	Itaconic anhydride	−0.04	0.024	2.87	0.32	Y	365
Vinyl acetate	Maleate, di-*tert*-butyl	0.005	0.024	0.199	0.02	Y	1049
Vinyl acetate	Maleate, diethyl	0.171	0.025	0.04	0.012	Y	137
Vinyl acetate	Maleate, dimethyl	−0.006	0.084	0.257	0.086	Y	1049
Vinyl acetate	Maleic anhydride	0.019	0.008	−0.058	0.074	Y	260
Vinyl acetate	Maleimide, *N*-(2-hydroxyethyl)-	−0.014	0.037	0.61	0.12	Y	323
Vinyl acetate	Maleimide, *N*-(2-hydroxyethyl)-	0.01	0.041	0.88	0.17	Y	323
Vinyl acetate	Maleimide, *N*-(2-hydroxyphenyl)-	−0.011	0.027	1.75	0.17	Y	323
Vinyl acetate	Maleimide, *N*-hydroxymethyl-	0.017	0.037	1.45	0.21	Y	323
Vinyl acetate	Maleimide, *N*-phenyl-	−0.003	0.006	1.284	0.046	Y	328
Vinyl acetate	Methacrylamide, 1-deoxy-D-glucitol	0.16		0.56			258
Vinyl acetate	Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	0.2		13.7			664
Vinyl acetate	Methacrylate, butyl	−0.003	0.15	30.18	14.47	Y	165
Vinyl acetate	Methacrylate, di-, di-*n*-butylstannyl	0.013	0.018	0.013	0.036	Y	949
Vinyl acetate	Methacrylate, ethyl	−0.77	5.29	131.8	29.8	N	210
Vinyl acetate	Methacrylate, ferrocenylmethyl	0.2		1.52			293
Vinyl acetate	Methacrylate, isobutyl	−0.06	0.22	29.7	24.16	Y	165
Vinyl acetate	Methacrylate, methyl	−0.2	0.01	2.71	0.25	Y	158
Vinyl acetate	Methacrylate, methyl	−0.81	1.17	137.5	5.9	N	210
Vinyl acetate	Methacrylate, methyl	0.03		26			272
Vinyl acetate	Methacrylate, methyl	0.04		28.6			383
Vinyl acetate	Methacrylate, methyl	0.07		22.21			471
Vinyl acetate	Methacrylate acid	0.01		0.2			377
Vinyl acetate	Methacrylonitrile	0.01		12			75
Vinyl acetate	*N,N*-Divinylaniline	0.058	0.009	3.99	0.12	Y	45

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl acetate	Norbornadiene	0.818	0.019	1.354	0.07	Y	188
Vinyl acetate	Oxazolidone, *N*-vinyl-	0.602	0.038	1.55	0.28	Y	102
Vinyl acetate	Oxazolidone, *N*-vinyl-	0.55	0.18	2.04	0.13	Y	36
Vinyl acetate	Oxazoline, 2-, 2-isopropenyl-	0.25		7.1			862
Vinyl acetate	Oxazoline, 2-, 4-acryloxymethyl-dimethyl-2,4-dimethyl	0.03		1.41			596
Vinyl acetate	Oxazoline, 2-, 4-methacryloxy-2,4-dimethyl	0.05		6.2			596
Vinyl acetate	Oxazolinium, 2- tetrafluoroborate, 3-methyl	0.43	0.057	0.366	0.151	Y	877
Vinyl acetate	Phosphate, diethyl isopropenyl	1.56		0.44			650
Vinyl acetate	Phosphine oxide, diphenylvinyl-	0.38		1.7			843
Vinyl acetate	Phosphonate, α-carbomethoxyvinyl-, diethyl	0.09		0.6			684
Vinyl acetate	Phosphonate, isopropenyl-, dimethyl	0.14		0.99			684
Vinyl acetate	Phosphonate, vinyl-, bis(2-chloroethyl)	0.09		0.85			592
Vinyl acetate	Phosphonate, vinyl-, bis(2-chloroethyl)	0.09		0.85			594
Vinyl acetate	Phosphonate, vinyl-, bis(2-chloroethyl)	0.33		0.47			595
Vinyl acetate	Phosphonate, vinyl-, diethyl	0.04		0.74			592
Vinyl acetate	Phosphonate, vinyl-, dimethyl	0.04		0.74			592
Vinyl acetate	Phosphonate, vinyl-, dimethyl	0.04		0.74			594
Vinyl acetate	Propene, 1-chloro-, *cis*-	7.08	0.35	−0.012	0.061	Y	181
Vinyl acetate	Propene, 1-chloro-, *trans*-	3.13	0.15	0.011	0.074	Y	181
Vinyl acetate	Propene, 2-chloro-	0.186	0.091	1.65	0.36	Y	181
Vinyl acetate	Propene, 3,3,3-trichloro-	0.19		0.19			402
Vinyl acetate	Pyridine, 2-methyl-5-vinyl-	0.11		9			812
Vinyl acetate	Pyridine, 2-vinyl-	−0.11	0.17	13.65	3.64	Y	6
Vinyl acetate	Pyridine, 2-vinyl-	0		30			6
Vinyl acetate	Pyrrolidone, *N*-vinyl-	0.26	0.25	1.06	0.25	Y	103
Vinyl acetate	Pyrrolidone, *N*-vinyl-	0.19	1.77	1.75	2.79	N	121
Vinyl acetate	Pyrrolidone, *N*-vinyl-	0.195	0.057	3.4	0.21	Y	36
Vinyl acetate	*S*-Ethyl *N*-vinyl-*N*-methylthiolcarbamate	0.67	0.1	1.22	1.17	Y	197
Vinyl acetate	*S*-Vinyl *N*,*N*-diethylthiolcarbamate	0.133	0.025	1.89	0.77	Y	197
Vinyl acetate	Styrene	−0.04	0.072	42.48	26.85	Y	158
Vinyl acetate	Styrene	0.02	0.12	18.8	2.87	Y	241
Vinyl acetate	Styrene	0.01		56			481
Vinyl acetate	Styrene	0.05		48			481
Vinyl acetate	Styrene	0.16		60			481
Vinyl acetate	Styrene	0.08		57.8			812
Vinyl acetate	Styrene, 3-*tri*-*n*-butylstannyl-	5.33		0.38			985
Vinyl acetate	Succinimide, *N*-vinyl-	0.072	0.009	5.68	0.12	Y	104
Vinyl acetate	Succinimide, *N*-vinyl-	0.18	0.54	4.49	9.49	Y	167
Vinyl acetate	Succinimide, *N*-vinyl-	0.229	0.06	1.99	0.43	Y	80
Vinyl acetate	Tetrazole, 2-methyl-5-vinyl-	0.04	0.16	28.51	31.34	Y	670
Vinyl acetate	Triallyl citrate	0.222	0.025	2.97	0.38	Y	367
Vinyl acetate	Triallyl cyanurate	0.197	0.053	1.29	0.42	Y	201
Vinyl acetate	Triallyl isocyanurate	0.278	0.043	1.95	0.42	Y	201
Vinyl acetate	Triazole, 1,2,3-, 1-*p*-bromophenyl-4-vinyl-	−0.011	0.008	42.47	9.84	Y	670
Vinyl acetate	Urea, *N*-vinyl-*N'*-ethyl-	0.463	0.029	0.621	0.081	Y	101
Vinyl acetate	Urea, *N*-vinyl-*N'*-ethyl-	0.16		3.61			683
Vinyl acetate	Urea, *N*-vinyl-*N'*-ethyl-	0.38		1.45			683
Vinyl acetate	Vinyl 12-ketostearate	0.18		−0.28		Y	146
Vinyl acetate	Vinyl 2-chloroethyl ether	2.36		0.16			597
Vinyl acetate	Vinyl 4-chlorocyclohexyl ketone	0.1		0.15			668
Vinyl acetate	Vinyl benzenesulfonate	0.628	0.036	−1.34	1.7	Y	942
Vinyl acetate	Vinyl benzoate	0.455	0.069	1.07	0.27	Y	253
Vinyl acetate	Vinyl benzoate	0.35		0.99			253
Vinyl acetate	Vinyl benzoate	0.78	0.52	1.64	1.93	Y	26
Vinyl acetate	Vinyl benzoate	0.7		1.13			272
Vinyl acetate	Vinyl benzoate	0.66		1.74			430
Vinyl acetate	Vinyl benzoate	0.24	0.2	1.98	0.98	N	696
Vinyl acetate	Vinyl bromide	0.41	0.13	5.26	1.23	Y	158
Vinyl acetate	Vinyl bromide	0.6	0.09	1.92	0.35	N	268
Vinyl acetate	Vinyl bromide	0.38		3			268

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl acetate	Vinyl butyl ether	3.72	0.09	0.01	0.02	N	345
Vinyl acetate	Vinyl butyl ether	0.71		0.2			448
Vinyl acetate	Vinyl butyl ether	2.5		0.16			553
Vinyl acetate	Vinyl butylsulfonate	0.025	0.006	0.13	0.33	Y	185
Vinyl acetate	Vinyl butyrate	1		0.97			272
Vinyl acetate	Vinyl butyrate	1.35		1.25			515
Vinyl acetate	Vinyl chloride	0.43	0.14	1.63	0.32	Y	109
Vinyl acetate	Vinyl chloride	0.981	0.036	1.032	0.097	Y	148
Vinyl acetate	Vinyl chloride	0.24	0.15	1.84	0.68	Y	158
Vinyl acetate	Vinyl chloride	0.263	0.028	2.13	0.2	Y	283
Vinyl acetate	Vinyl chloride	0.25	0.15	1.64	0.12	Y	3
Vinyl acetate	Vinyl chloride	0.65		1.35			450
Vinyl acetate	Vinyl chloride	0.26	0.14	2.3	0.64	N	86
Vinyl acetate	Vinyl chloroacetate	0.83	0.12	1.39	0.3	N	71
Vinyl acetate	Vinyl chloroacetate	0.803	0.089	0.88	0.19	Y	779
Vinyl acetate	Vinyl chloroacetate	0.8		1.18		Y	861
Vinyl acetate	Vinyl chloroformate	0.67	0.16	0.579	0.076	Y	675
Vinyl acetate	Vinyl chloroformate	0.66	0.12	0.586	0.052	Y	706
Vinyl acetate	Vinyl chloromethyl ketone	0.044	0.054	49.52	15.79		356
Vinyl acetate	Vinyl cinnamate	0.04		1.2			200
Vinyl acetate	Vinyl cyclohexyl ketone	0.08		13.76			846
Vinyl acetate	Vinyl cymantrene	0.072	0.003	7.21	1.01	Y	354
Vinyl acetate	Vinyl dodecyl ether	3.67		0			372
Vinyl acetate	Vinyl ethyl ether	3.42		0.26		Y	158
Vinyl acetate	Vinyl ethyl ketone	−0.13	0.18	10.18	6.62	Y	794
Vinyl acetate	Vinyl ethyl oxalate	1.35	0.84	9.89	8.44	Y	124
Vinyl acetate	Vinyl fluoride	3.5		0.25			755
Vinyl acetate	Vinyl formate	0.94	0.14	0.98	0.06	N	272
Vinyl acetate	Vinyl formate	1.41		0.68			401
Vinyl acetate	Vinyl isopropyl ketone	0.3		7.87			601
Vinyl acetate	Vinyl isopropyl ketone	−0.154	0.073	6.47	2.18	Y	794
Vinyl acetate	Vinyl laurate	1.4		0.7			503
Vinyl acetate	Vinyl methanesulfonate	0.639	0.014	0.206	0.427	Y	942
Vinyl acetate	Vinyl methyl ketone					Y	91
Vinyl acetate	Vinyl methyl sulfone	0	0.02	0.355	0.21	Y	195
Vinyl acetate	Vinyl octadecyl ether	4.5		0			372
Vinyl acetate	Vinyl octyl ether	3.47		0			372
Vinyl acetate	Vinyl phenyl carbonate	0.85	0.23	0.384	0.083	Y	675
Vinyl acetate	Vinyl phenyl carbonate	0.85	0.23	0.384	0.083	Y	705
Vinyl acetate	Vinyl phenyl ketone	−0.06	0.057	5.81	0.96	Y	794
Vinyl acetate	Vinyl phenyl sulfone	0.171	0.064	−0.35	0.37	Y	193
Vinyl acetate	Vinyl propionate	0.98		0.98			272
Vinyl acetate	Vinyl propionate	0.9		1			425
Vinyl acetate	Vinyl stearate	0.953	0.015	0.96	0.17	Y	1
Vinyl acetate	Vinyl stearate	0.9		0.73			537
Vinyl acetate	Vinyl *tert*-butyl sulfide	13		0.07			598
Vinyl acetate	Vinyl *tert*-butyl sulfide	13		0.07			601
Vinyl acetate	Vinyl thiolacetate	0.05		5.5			566
Vinyl acetate	Vinyl, *N*-, *N*-methylacetamide	0.49		0.93			631
Vinyl acetate	Vinylene carbonate	3.2	0.62	0.3	0.3	Y	103
Vinyl acetate	Vinylene carbonate	3		0.27			103
Vinyl acetate	Vinylene carbonate	7.16	0.64	0.05	0.16	Y	120
Vinyl acetate	Vinylene carbonate	7.3		0.13			120
Vinyl acetate	Vinylene carbonate	3.83	0.18	0.044	0.039	Y	144
Vinyl acetate	Vinylene carbonate	3.92	0.32	0.08	0.11	Y	90
Vinyl acetate	Vinylidene chloride	0.03	0.1	4.66	0.26	Y	3
Vinyl acetate	Vinylidene chloride	0.05		6.7			432
Vinyl acetate	Vinylidene chloride	0.05		5			436
Vinyl acetate	Vinylidene chloride	−0.002	0.026	3.49	0.17	Y	59
Vinyl acetate	Vinylidene chloride	0.12		4.7			812
Vinyl acetate	Vinylidene cyanide	0.003	0.002	0.1	0.11	Y	82
Vinyl acetate	Vinylidene fluoride	6		0.08			627
Vinyl acetoxymethyl ketone	Styrene	0.368	0.018	0.103	0.011		806
Vinyl benzenesulfonate	Vinyl acetate	−1.34	1.7	0.628	0.036	Y	942
Vinyl benzoate	Acrylonitrile	0.019	0.051	5.03	1.08	Y	47

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl benzoate	Allyl chloride	0.46		0.88			438
Vinyl benzoate	Methacrylate, methyl	0.07		20.3			272
Vinyl benzoate	Pyrrolidone, *N*-vinyl-	0.408	0.03	2.507	0.039	Y	253
Vinyl benzoate	Styrene	−0.02	1.62	34.8	29.3	Y	26
Vinyl benzoate	Styrene	0.06	0.15	31.56	4.33	N	696
Vinyl benzoate	Vinyl acetate	1.07	0.27	0.455	0.069	Y	253
Vinyl benzoate	Vinyl acetate	0.99		0.35			253
Vinyl benzoate	Vinyl acetate	1.64	1.93	0.78	0.52	Y	26
Vinyl benzoate	Vinyl acetate	1.13		0.7			272
Vinyl benzoate	Vinyl acetate	1.74		0.66			430
Vinyl benzoate	Vinyl acetate	1.98	0.98	0.24	0.2	N	696
Vinyl benzoate	Vinyl chloride	0.343	0.036	1.66	0.16	Y	283
Vinyl benzoate	Vinyl chloride	0.5		1.7			450
Vinyl benzoate	Vinyl chloride	0.28		0.72			519
Vinyl benzoate	Vinyl *p*-bromobenzoate	0.716	0.031	1.17	0.13	Y	110
Vinyl benzoate	Vinyl *p*-chlorobenzoate	0.832	0.028	0.81	0.072	Y	110
Vinyl benzoate	Vinyl *p*-cyanobenzoate	0.659	0.033	0.878	0.094	Y	110
Vinyl benzoate	Vinylidene chloride	0.065	0.093	5.58	1.76	Y	47
Vinyl benzoate	Vinylidene cyanide	0	0.017	0.061	0.047	Y	82
Vinyl benzyl sulfide	Methacrylic anhydride	−0.048	0.033	0.97	0.27	Y	106
Vinyl benzyl sulfide	Styrene	0.04	0.17	2.32	0.25	Y	303
Vinyl bromide	Acrylate, butyl	0.18	0.16	4.07	0.44	N	268
Vinyl bromide	Acrylate, butyl	0.19		3.7			268
Vinyl bromide	Acrylonitrile	0.06		2.25			268
Vinyl bromide	Acrylonitrile	0.06	0.04	2.79	0.2	N	286
Vinyl bromide	Methacrylate, methyl	0.12	0.06	14.11	0.67	N	268
Vinyl bromide	Methacrylate, methyl	0.33		17.1			268
Vinyl bromide	Methacrylate, methyl	0.052	0.026	25.1	0.88	Y	34
Vinyl bromide	Methacrylate, methyl	0.05	0.074	19.84	2.58	Y	34
Vinyl bromide	Styrene	0.012	0.049	16.88	0.94	Y	34
Vinyl bromide	Styrene	0.06		18			34
Vinyl bromide	Styrene	0.09	0.2	15	8.26	N	368
Vinyl bromide	Vinyl acetate	5.26	1.23	0.41	0.13	Y	158
Vinyl bromide	Vinyl acetate	1.92	0.35	0.6	0.09	N	268
Vinyl bromide	Vinyl acetate	3		0.38			268
Vinyl butyl ether	Acrylate, 2-chloroethyl	0		2.01			790
Vinyl butyl ether	Acrylate, methyl	0.01	0.01	3.65	0.07	N	345
Vinyl butyl ether	Acrylate, methyl	0		3.6			574
Vinyl butyl ether	Acrylonitrile					N	345
Vinyl butyl ether	Acrylonitrile	−0.03	0.01	0.98	0.13	N	347
Vinyl butyl ether	Benzenesulfamide, *p*-methacrylamide-	0		3.4	0.1		1005
Vinyl butyl ether	Caprolactam, *N*-vinyl-	0.03	0.73	3.1	5.3	Y	215
Vinyl butyl ether	Methacrylate, 2-bromoethyl	0		13.7			657
Vinyl butyl ether	Methacrylate, methyl					N	127
Vinyl butyl ether	Pyrrolidone, *N*-vinyl	−0.027	0.063	1.49	0.95	Y	215
Vinyl butyl ether	Styrene	−0.08	0.78	21.25	16.53	Y	241
Vinyl butyl ether	Toluenesulfonamide, *N,N*-methyl-vinyl-	3.37	1.65	−0.31	0.29	Y	80
Vinyl butyl ether	Vinyl acetate	0.01	0.02	3.72	0.09	N	345
Vinyl butyl ether	Vinyl acetate	0.2		0.71			448
Vinyl butyl ether	Vinyl acetate	0.16		2.5			553
Vinyl butyl ether	Vinyl chloride	0.023	0.001	2.084	0.016	Y	279
Vinyl butyl ether	Vinyl chloroacetate	0.12	4.39	1.49	34.5	Y	212
Vinyl butyl ether	Vinylidene chloride	0.004	0.001	1.75	0.11	Y	279
Vinyl butyl sulfide	Acrylonitrile	0.041	0.028	0.086	0.01	Y	303
Vinyl butyl sulfide	Styrene	−0.06	0.18	2.69	0.25	Y	303
Vinyl butylsulfonate	Acrylate, methyl	0	0.14	4.14	0.58	Y	185
Vinyl butylsulfonate	Styrene	0	0.16	1.74	0.27	Y	185
Vinyl butylsulfonate	Vinyl acetate	0.13	0.33	0.025	0.006	Y	185
Vinyl butylsulfonate	Vinyl chloride	0.267	0.046	0.325	0.013	Y	185
Vinyl butylsulfonate	Vinyl chloride	0.083	0.096	8.47	0.86	Y	185
Vinyl butyrate	Allyl chloride	0.31		1.15			438
Vinyl butyrate	Butadiene, 2-chloro-					N	247
Vinyl butyrate	Carbazole, *N*-vinyl-	0.07	0.12	1.33	0.34	Y	246
Vinyl butyrate	Ethylene	1.5		0.7			605
Vinyl butyrate	Methacrylate, methyl	0.03		25			272

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl butyrate	Vinyl acetate	0.97		1			272
Vinyl butyrate	Vinyl acetate	1.25		1.35			515
Vinyl butyrate	Vinyl chloride	0.302	0.013	2.092	0.074	Y	283
Vinyl butyrate	Vinyl chloride	0.55		1.75			400
Vinyl butyrate	Vinyl chloride	0.65		1.35			494
Vinyl caproate	Vinyl chloride	0.348	0.05	2.26	0.22	Y	283
Vinyl chloride	Aconitate, trimethyl	0.04		0.19		Y	147
Vinyl chloride	Acrolein	0.03	0.11	5.22	0.13	N	271
Vinyl chloride	Acrylamide					Y	156
Vinyl chloride	Acrylamide, *N*-methylol-					Y	157
Vinyl chloride	Acrylate, 2-ethylhexyl	0.16		4.15			510
Vinyl chloride	Acrylate, butyl	0.07		4.4			518
Vinyl chloride	Acrylate, methyl	0.093	0.002	4.4	0.22	Y	151
Vinyl chloride	Acrylate, methyl	0.002	0.01	7.66	0.5	Y	46
Vinyl chloride	Acrylate, methyl	0		5			470
Vinyl chloride	Acrylate, methyl	0.06		4			518
Vinyl chloride	Acrylate, octyl	0.12		4.8			518
Vinyl chloride	Acrylate, trifluoro-, methyl	1.5		0			595
Vinyl chloride	Acrylate, trifluoro-, methyl	1.5		0			598
Vinyl chloride	Acrylic acid	0.107	0.011	6.69	0.88	Y	135
Vinyl chloride	Acrylic acid	0.03	0.09	9.28	0.51	N	229
Vinyl chloride	Acrylic acid	0.024	0.069	7.04	0.33	Y	288
Vinyl chloride	Acrylic acid	0.11		6.8			379
Vinyl chloride	Acrylonitrile	−0.01	0.037	3.26	0.34	Y	140
Vinyl chloride	Acrylonitrile	0.07	0.03	2.55	0.12	N	202
Vinyl chloride	Acrylonitrile	0.02	0.08	2.62	0.25	N	22
Vinyl chloride	Acrylonitrile	0.044	0.002	3.62	0.21	Y	238
Vinyl chloride	Acrylonitrile	0.052	0.01	3.65	0.22	Y	46
Vinyl chloride	Acrylonitrile	0.04		4			529
Vinyl chloride	Acryloyl chloride	0.02	0.029	3.03	0.22	Y	288
Vinyl chloride	Allyl acetate	1.2					470
Vinyl chloride	Allyl acetate	1.16		0			475
Vinyl chloride	Allyltriethoxysilane	1.68	0.38	−0.94	0.48	Y	237
Vinyl chloride	Butadiene	0.04		8.8			517
Vinyl chloride	Butadiene	−0.11	0.31	5.27	3.35	Y	717
Vinyl chloride	Butene-1	3.34	0.58	0.12	0.12	Y	230
Vinyl chloride	Butene-1, 2-ethyl-	1.56	0.26	−1.05	0.89	N	329
Vinyl chloride	Butene-1, 2-methyl-	1.01	0.31	−1.18	1.2	N	329
Vinyl chloride	Butene-1,3, 3-dimethyl-	5.64	0.2	−0.44	0.28	N	329
Vinyl chloride	Butene-1,3, 3-dimethyl-	6.18	13.57	−0.07	2.5	Y	6
Vinyl chloride	Butene-2	3.3	2.18	−1.79	5.22	N	329
Vinyl chloride	Butene-2, 2-methyl-	8.52	10.54	0.3	5.31	N	329
Vinyl chloride	Butene-2, *cis*-	8.8		0			230
Vinyl chloride	Carbazole, *N*-vinyl-	0.17		4.77			427
Vinyl chloride	Carbon monoxide	13.52	0.25	0.121	0.011	Y	184
Vinyl chloride	Crotonaldehyde	1.55	0.26	−0.56	0.25	N	271
Vinyl chloride	Diallyl phthalate	0.833	0.012	0.7	0.046	Y	296
Vinyl chloride	Diallylcyanamide	0.44		0.68			602
Vinyl chloride	Diallylcyanamide	0.44		0.68			603
Vinyl chloride	Ethylene	3.82	0.59	0.02	0.06	N	161
Vinyl chloride	Ethylene	1.99	0.42	0.19	0.17	N	180
Vinyl chloride	Ethylene	1.74	0.58	0.24	0.13	N	270
Vinyl chloride	Ethylene	4.38	0.44	0.34	0.13	N	42
Vinyl chloride	Ethylene	2		0.2			472
Vinyl chloride	Ethylene	3.21		0.21			536
Vinyl chloride	Ethylene	1.85	0.1	0.21	0.01	N	63
Vinyl chloride	Ethylene	0.96	0.22	0.14	0.12	N	63
Vinyl chloride	Ethylene	1.63	0.22	0.23	0.01	N	64
Vinyl chloride	Ethylene	1.55	0.18	0.16	0.01	N	68
Vinyl chloride	Ethylene, chlorotrifluoro-	2.53		0.01			454
Vinyl chloride	Fumarate, diethyl	0.13	0.1	0.48	0.1	Y	137
Vinyl chloride	Fumarate, diisopropyl	0.29		0.49			1038
Vinyl chloride	Hexene-1	1.54	0.98	−1.72	9.23	N	329
Vinyl chloride	Imidazole, 1-vinyl-2-methyl-	0.188	0.077	2.15	0.26	Y	154
Vinyl chloride	Isobutylene	2.04	1.82	0.06	0.8	Y	140

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl chloride	Isobutylene	2.12	0.2	0.268	0.043	Y	155
Vinyl chloride	Isobutylene	1.2	2.05	−0.54	5.8	N	329
Vinyl chloride	Isobutylene	1.54		0.08			560
Vinyl chloride	Isobutylene, 3-chloro-	0.31		0			475
Vinyl chloride	Isopropenyl acetate	2.26	0.13	0.23	0.16	Y	100
Vinyl chloride	Isopropenylisocyanate	0.39		3			777
Vinyl chloride	Itaconate, diethyl	0.06		5.65			480
Vinyl chloride	Itaconate, dimethyl	0.05		5			479
Vinyl chloride	Maleate, diethyl	0.768	0.078	−0.003	0.044	Y	137
Vinyl chloride	Maleate, diethyl	0.8		0			449
Vinyl chloride	Maleate, diethyl	0.9	0.11	0.046	0.07	Y	6
Vinyl chloride	Maleate, dioctyl	0.05	0.14	0.608	0.099	Y	3
Vinyl chloride	Maleic anhydride	0.098	0.042	−0.22	0.38	Y	260
Vinyl chloride	Maleic anhydride	0.04		0.4			521
Vinyl chloride	Maleic anhydride	0.1		0.67			521
Vinyl chloride	Maleimide, *N*-(4-carboxyethylphenyl)-	0.04	0.03	3.29	0.18	N	306
Vinyl chloride	Maleimide, *N*-(4-chlorophenyl)-	0.026	0.044	3.65	0.39	Y	306
Vinyl chloride	Maleimide, *N*-(4-methylphenyl)-	0.022	0.048	4.49	0.34	Y	306
Vinyl chloride	Maleimide, *N*-butyl-	0.24	1.13	−12.7	106.2	Y	305
Vinyl chloride	Maleimide, *N*-phenyl-	0.023	0.026	4.01	0.18	Y	306
Vinyl chloride	Maleimide, *N*-propyl-	0.06	0.05	2.38	0.16	N	305
Vinyl chloride	Methacrylate, 3,5-dimethyladamantyl	0.21	0.04	1.85	8.87	N	335
Vinyl chloride	Methacrylate, β-(2-benzothiazolylthio)-ethyl	0.64		0.26			741
Vinyl chloride	Methacrylate, butyl	0.05		13.5			519
Vinyl chloride	Methacrylate, methyl	0.07	0.047	8.99	2.97	Y	3
Vinyl chloride	Methacrylate, octyl	0.04		14			511
Vinyl chloride	Methacrylic acid	0.064	0.008	23.52	0.98	Y	135
Vinyl chloride	Methacrylic acid	−0.022	0.02	23.26	0.96	Y	229
Vinyl chloride	Methacrylic acid	0.07		9.05			46
Vinyl chloride	Methacryloyl chloride	0.3		2			432
Vinyl chloride	Methacryloyloxy-, *o*-, benzoic acid	0.004	0.052	11.26	3.5	Y	938
Vinyl chloride	Norbornadiene	0.67	0.02	0.41	0.01	N	267
Vinyl chloride	Octene-1	3.26	0.45	−0.57	0.85	N	329
Vinyl chloride	Oxazolidone, *N*-vinyl-	0.313	0.095	0.822	0.048	Y	36
Vinyl chloride	Pentene-1	4		0.001		Y	3
Vinyl chloride	Pentene-1	2.08	1.78	0.81	5.2	N	329
Vinyl chloride	Pentene-1, 2-methyl-	1.59	0.68	−1.16	2.31	N	329
Vinyl chloride	Phosphate, vinyl diethyl	3.4	0.21	0.15	0.047	Y	779
Vinyl chloride	Phosphate, vinyl diphenyl	3.009	0.077	0.327	0.023	Y	779
Vinyl chloride	Propene, 1-chloro-, *cis*-	11.52	2.14	0.1	0.63	Y	181
Vinyl chloride	Propene, 1-chloro-, *trans*-	4.45	0.81	0.33	0.4	Y	181
Vinyl chloride	Propene, 2-chloro-	0.095	0.085	4.75	5.12	Y	181
Vinyl chloride	Propene, 2-chloro-	0.75		0.58			517
Vinyl chloride	Propenyl, 2-, acetate	2.2		0.25			189
Vinyl chloride	Propenyl, 2-chloro-, acetate	0.7		0			445
Vinyl chloride	Propenyltriethoxysilane					Y	237
Vinyl chloride	Propylene	2.35	0.14	0.196	0.023	Y	155
Vinyl chloride	Propylene	−0.05	0.1	3.1	0.21	N	329
Vinyl chloride	Propylene	5.16	0.06	−0.02	0.04	N	355
Vinyl chloride	Propylene	2.45		0.09			560
Vinyl chloride	Propylene	2.9					588
Vinyl chloride	Pyridine, 4-vinyl-					Y	154
Vinyl chloride	Pyrrolidone, *N*-vinyl-	0.53		0.38			40
Vinyl chloride	Pyrrolidone, *N*-vinyl-	0.55		0.34			582
Vinyl chloride	Pyrrolidone, *N*-vinyl-	0.74		0.73			582
Vinyl chloride	Styrene	−0.06	0.08	27.54	7.35	Y	109
Vinyl chloride	Styrene	0.058	0.003	17.24	1.12	Y	238
Vinyl chloride	Styrene	0.16	0.08	23.92	3.6	N	242
Vinyl chloride	Styrene	0.005	0.031	25	5	Y	46
Vinyl chloride	Styrene	0.04		12.4			516
Vinyl chloride	Styrene	0.01	0.87	14.7	29.2	Y	59
Vinyl chloride	Styrene, pentachloro-	0.43		5.3			516
Vinyl chloride	Triallyl citrate	0.605	0.033	1.03	0.17	Y	367
Vinyl chloride	Vinyl 12-ketostearate	0.03		−0.01		Y	146

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl chloride	Vinyl 2-chloroethyl ether	2.46	0.52	−0.07	0.11	Y	304
Vinyl chloride	Vinyl acetate	1.63	0.32	0.43	0.14	Y	109
Vinyl chloride	Vinyl acetate	1.032	0.097	0.981	0.036	Y	148
Vinyl chloride	Vinyl acetate	1.84	0.68	0.24	0.15	Y	158
Vinyl chloride	Vinyl acetate	2.13	0.2	0.263	0.028	Y	283
Vinyl chloride	Vinyl acetate	1.64	0.12	0.25	0.15	Y	3
Vinyl chloride	Vinyl acetate	1.35		0.65			450
Vinyl chloride	Vinyl acetate	2.3	0.64	0.26	0.14	N	86
Vinyl chloride	Vinyl benzoate	1.66	0.16	0.343	0.036	Y	283
Vinyl chloride	Vinyl benzoate	1.7		0.5			450
Vinyl chloride	Vinyl benzoate	0.72		0.28			519
Vinyl chloride	Vinyl butyl ether	2.084	0.016	0.023	0.001	Y	279
Vinyl chloride	Vinyl butylsulfonate	0.325	0.013	0.267	0.046	Y	185
Vinyl chloride	Vinyl butylsulfonate	8.47	0.86	0.083	0.096	Y	185
Vinyl chloride	Vinyl butyrate	2.092	0.074	0.302	0.013	Y	283
Vinyl chloride	Vinyl butyrate	1.75		0.55			400
Vinyl chloride	Vinyl butyrate	1.35		0.65			494
Vinyl chloride	Vinyl caproate	2.26	0.22	0.348	0.05	Y	283
Vinyl chloride	Vinyl chloroacetate	1.64	0.34	0.421	0.055	Y	283
Vinyl chloride	Vinyl dichloroacetate	1.3	0.55	0.91	0.49	Y	283
Vinyl chloride	Vinyl dodecyl ether	1.93		0.15			370
Vinyl chloride	Vinyl fluoride	11		0.05			590
Vinyl chloride	Vinyl fluoride	11.6		0.11			755
Vinyl chloride	Vinyl hendecanoate	1.081	0.016	0.38	0.022	Y	145
Vinyl chloride	Vinyl isobutyl ether	2.19	1.43	0.04	0.37	Y	47
Vinyl chloride	Vinyl laurate	7.4		0.2			450
Vinyl chloride	Vinyl *m*-cresyl ether	1.23	0.04	0.13	0.02	Y	279
Vinyl chloride	Vinyl methyl ketone	0.1		8.3			375
Vinyl chloride	Vinyl *o*-cresyl ether	1.33	0.02	0.14	0.01	Y	279
Vinyl chloride	Vinyl octadecyl ether	2.1		−0.1			370
Vinyl chloride	Vinyl octyl ether	1.9		0.1			370
Vinyl chloride	Vinyl *p*-chlorophenyl ether	1	0.04	0.63	0.05	Y	279
Vinyl chloride	Vinyl *p*-cresyl ether	1.17	0.02	0.13	0.01	Y	279
Vinyl chloride	Vinyl *p-tert*-butylphenyl ether	1.08	0.06	0.14	0.02	Y	279
Vinyl chloride	Vinyl *p*-vinyloxybenzoate	0.58	0.06	0.78	0.07	N	497
Vinyl chloride	Vinyl pelargonate	1.16		0.28			145
Vinyl chloride	Vinyl phenyl ether	1.28	0.14	0.01	0.59	Y	172
Vinyl chloride	Vinyl phenyl ether	1.06		0.16			279
Vinyl chloride	Vinyl phenyl ether	1.21	0.03	0.17	0.03	N	280
Vinyl chloride	Vinyl propionate	1.6		0.6			400
Vinyl chloride	Vinyl propionate	1.35		0.65			450
Vinyl chloride	Vinyl stearate	0.71	0.097	0.24	0.64	Y	145
Vinyl chloride	Vinyl *tert*-butyl sulfide	3		0.15			598
Vinyl chloride	Vinyl *tert*-butyl sulfide	3		0.15			601
Vinyl chloride	Vinyl valerate	2.006	0.071	0.277	0.014	Y	283
Vinyl chloride	Vinylene carbonate	4.82	0.49	0.03	0.12	Y	103
Vinyl chloride	Vinylene carbonate	15.38	1.44	0.069	0.068	Y	319
Vinyl chloride	Vinylene carbonate	17.81	7.84	0	0.22	Y	319
Vinyl chloride	Vinylethyldiethoxysilane	0.94	1.53	−0.44	2.97	Y	237
Vinyl chloride	Vinylidene chloride	0.071	0.025	2.06	0.32	Y	238
Vinyl chloride	Vinylidene chloride	0.14	0.32	3.39	1.16	N	29
Vinyl chloride	Vinylidene chloride	0.25		8.92		Y	3
Vinyl chloride	Vinylidene chloride	0.3		3.2			470
Vinyl chloride	Vinylidene chloride	0.2		4.5			517
Vinyl chloride	Vinylidene chloride	0.2		1.8			539
Vinyl chloride	Vinylidene chloride	0.16		4.5			585
Vinyl chloride	Vinylidene chloride	0.205	0.003	3.068	0.076	Y	67
Vinyl chloride	Vinylidene cyanide	0.01		0.72			409
Vinyl chloride	Vinylidene cyanide	0.016	0.002	0.6	0.068	Y	82
Vinyl chloride	Vinylisocyanate	0.46		3.7			777
Vinyl chloride	Vinylphenylisocyanate, *p*-	0.01		11.9			777
Vinyl chloride	Vinyltriethoxysilane	0.82	0.16	−0.04	0.44	Y	237
Vinyl chloride	Vinyltrimethoxysilane	0.93		0.09		Y	237
Vinyl chloroacetate	Acrylonitrile	−0.007	0.016	4.18	0.33	Y	362
Vinyl chloroacetate	Acrylonitrile	0.09		0.34			542

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl chloroacetate	Phosphate, vinyl diethyl	2.01	0.2	0.374	0.046	Y	779
Vinyl chloroacetate	Phosphate, vinyl diphenyl	1.42	0.15	0.53	0.12	Y	779
Vinyl chloroacetate	Styrene	−0.04	0.24	14.57	62.41	Y	212
Vinyl chloroacetate	Styrene	0.03		45			542
Vinyl chloroacetate	Vinyl acetate	1.39	0.3	0.83	0.12	N	71
Vinyl chloroacetate	Vinyl acetate	0.88	0.19	0.803	0.089	Y	779
Vinyl chloroacetate	Vinyl acetate	1.18		0.8			861
Vinyl chloroacetate	Vinyl butyl ether	1.49	34.5	0.12	4.39	Y	212
Vinyl chloroacetate	Vinyl chloride	0.421	0.055	1.64	0.34	Y	283
Vinyl chloroacetate	Vinyl ethyl ether	1.92	17.19	0.43	3.9	Y	212
Vinyl chloroacetate	Vinyl phenyl ether	0.84	1.67	0.55	0.61	Y	212
Vinyl chloroacetate	Vinylidene cyanide	0	0.006	0.137	0.037	Y	82
Vinyl chloroethyl ether	Acrylonitrile	−0.02	0.02	1.04	0.05	N	116
Vinyl chloroethyl ether	Methacrylic anhydride					Y	106
Vinyl chloroformate	Vinyl acetate	0.579	0.076	0.67	0.16	Y	675
Vinyl chloroformate	Vinyl acetate	0.586	0.052	0.66	0.12	Y	706
Vinyl chloromethyl ketone	Acrylonitrile	0.88	0.12	0.064	0.019	Y	356
Vinyl chloromethyl ketone	Methacrylate, methyl	0.593	0.092	0.121	0.021	Y	356
Vinyl chloromethyl ketone	Styrene	0.507	0.039	0.127	0.047	Y	353
Vinyl chloromethyl ketone	Styrene, α-methyl-	0.386	0.034	0.009	0.01	Y	356
Vinyl chloromethyl ketone	Vinyl acetate	49.52	15.79	0.044	0.054	Y	356
Vinyl chloromethyl ketone	Vinylidene chloride	6.8	0.3	0.018	0.009	Y	356
Vinyl cinnamate	Methacrylonitrile	0.15		4.6			200
Vinyl cinnamate	Pyrrolidone, *N*-vinyl-	1.2		0.01			200
Vinyl cinnamate	Vinyl acetate	1.2		0.04			200
Vinyl cyclohexyl ether	Pyrrolidone, *N*-vinyl-	−0.11	0.59	1.91	9.39	Y	214
Vinyl cyclohexyl ketone	Vinyl 4-chlorocyclohexyl ketone	1.6		0.5			668
Vinyl cyclohexyl ketone	Vinyl acetate	13.76		0.08			846
Vinyl cymantrene	Acrylate, methyl	0.211	0.067	0.46	0.14	Y	354
Vinyl cymantrene	Acrylonitrile	−0.065	0.093	0.446	0.048	Y	354
Vinyl cymantrene	Pyrrolidone, *N*-vinyl-	0.093	0.053	0.037	0.03	Y	308
Vinyl cymantrene	Styrene	0.096	0.014	2.324	0.06	Y	354
Vinyl cymantrene	Vinyl acetate	7.21	1.01	0.072	0.003	Y	354
Vinyl dichloroacetate	Acrylonitrile	0.044	0.015	7.45	0.32	Y	362
Vinyl dichloroacetate	Acrylonitrile	0.18		0.25			542
Vinyl dichloroacetate	Styrene	0.28		20			542
Vinyl dichloroacetate	Vinyl chloride	0.91	0.49	1.3	0.55	Y	283
Vinyl diethylphosphinate	Ethylene	−11.99	80.64	0.15	0.16	Y	311
Vinyl diethylphosphinate	Ethylene	−62.84	34.66	0.05	0.01	Y	311
Vinyl dodecyl ether	Acrylonitrile	0		0.82			373
Vinyl dodecyl ether	Styrene	0		27			374
Vinyl dodecyl ether	Styrene	0		56			374
Vinyl dodecyl ether	Vinyl acetate	0		3.67			372
Vinyl dodecyl ether	Vinyl chloride	0.15		1.93			370
Vinyl dodecyl ether	Vinylidene chloride	0		1.3			371
Vinyl ether	Acrylonitrile	0.02		0.94			386
Vinyl ether	Methacrylate, methyl	−0.15	0.16	4.9	0.99	Y	45
Vinyl ether	Pyridine, 4-vinyl-	0.03		32			395
Vinyl ether	Styrene	1.2	5.3	152	185	Y	45
Vinyl ethyl ether	Acrylate, methyl	0		3.3			470
Vinyl ethyl ether	Acrylonitrile	0.06	0.12	0.69	0.12	N	194
Vinyl ethyl ether	Ethylene	0		2.7			584
Vinyl ethyl ether	Fumarate, diethyl	−0.124	0.059	1.63	0.16	Y	194
Vinyl ethyl ether	Methacrylate methyl	0.01		37			106
Vinyl ethyl ether	Styrene	0		90			140
Vinyl ethyl ether	Styrene	−0.122	0.066	111	15.5	Y	194
Vinyl ethyl ether	Vinyl acetate	0.26		3.42		Y	158
Vinyl ethyl ether	Vinyl chloroacetate	0.43	3.9	1.92	17.19	Y	212
Vinyl ethyl ether	Vinylidene chloride	0		3.2			470
Vinyl ethyl ketone	Methacrylate, methyl	0.65	0.42	0.42	0.17	Y	794
Vinyl ethyl ketone	Styrene	−0.033	0.023	13.83	1.44	Y	112
Vinyl ethyl ketone	Styrene	0.43	0.06	0.2	0.08	N	344
Vinyl ethyl ketone	Styrene	0.332	0.043	0.29	0.16	Y	353
Vinyl ethyl ketone	Styrene	0.122	0.017	0.665	0.021	Y	550
Vinyl ethyl ketone	Vinyl acetate	10.18	6.62	−0.13	0.18	Y	794

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl ethyl oxalate	Acrylonitrile	0	0.28	1.34	0.45	Y	124
Vinyl ethyl oxalate	Methacrylate, methyl	−0.05	0.44	4.74	2.95	Y	124
Vinyl ethyl oxalate	Styrene	−0.12	0.19	5.62	1.06	Y	124
Vinyl ethyl oxalate	Vinyl acetate	9.89	8.44	1.35	0.84	Y	124
Vinyl ethyl sulfide	Acrylonitrile	0.06	0.03	0.09	0.01	N	285
Vinyl ethyl sulfide	Acrylonitrile	0.05	0.03	0.06	0.01	N	336
Vinyl ethyl sulfide	Methacrylate, methyl	0.036	0.017	1.03	0.11	Y	303
Vinyl ethyl sulfide	Methacrylate, methyl	0.3		2.7			507
Vinyl ethyl sulfide	Styrene	0.113	0.032	4.76	0.2	Y	303
Vinyl ethyl sulfide	Styrene	0.25		6			507
Vinyl ethyl sulfoxide	Acrylonitrile	−0.08	0.14	1.63	0.16	Y	286
Vinyl ethyl sulfoxide	Methacrylate, methyl	−0.037	0.06	11.17	0.48	Y	286
Vinyl ethyl sulfoxide	Styrene	0.101	0.044	9.64	0.54	Y	286
Vinyl ethyl sulfoxide	Styrene	0.1		6			286
Vinyl ethylene glycol glycidyl ether	Pyrrolidone, *N*-vinyl-	0.06		1.4			369
Vinyl ethylene glycol glycidyl ether	Pyrrolidone, *N*-vinyl-	0.06		1.4			818
Vinyl fluoride	Acrylate, butyl	0.01		19			604
Vinyl fluoride	Acrylate, butyl	0.01		19			606
Vinyl fluoride	Acrylate, methyl	0.01		43			604
Vinyl fluoride	Acrylate, methyl	0.01		43			606
Vinyl fluoride	Acrylonitrile	0.005		44			528
Vinyl fluoride	Acrylonitrile	1		24			590
Vinyl fluoride	Ethylene	0.01	0.15	2.52	0.32	N	42
Vinyl fluoride	Ethylene	0.3		1.7			590
Vinyl fluoride	Ethylene, chlorotrifluoro-	0.8		1.2			406
Vinyl fluoride	Vinyl acetate	0.25		3.5			755
Vinyl fluoride	Vinyl chloride	0.05		11			590
Vinyl fluoride	Vinyl chloride	0.11		11.6			755
Vinyl fluoride	Vinylidene chloride	0.16		6			607
Vinyl formate	Acrylonitrile	−0.004	0.12	2.54	1.79	Y	47
Vinyl formate	Allyl chloride	0.57		0.78			438
Vinyl formate	Butadiene, 2-chloro-	−0.04		15			127
Vinyl formate	Butadiene, 2-chloro-	−0.05	0.01	11.91	3.89	N	247
Vinyl formate	Carbazole, *N*-vinyl-	0.192	0.001	4.142	0.011	Y	246
Vinyl formate	Vinyl acetate	0.98	0.06	0.94	0.14	N	272
Vinyl formate	Vinyl acetate	0.68		1.41			401
Vinyl hendecanoate	Acrylate, methyl	0.033	0.008	3.615	0.012	Y	145
Vinyl hendecanoate	Acrylonitrile	0.09	0.91	1.88	0.44	Y	145
Vinyl hendecanoate	Butadiene	−0.22	1.26	22.56	24.54	Y	145
Vinyl hendecanoate	Styrene	−0.17	0.18	24.89	1.22	Y	145
Vinyl hendecanoate	Vinyl chloride	0.38	0.022	1.081	0.016	Y	145
Vinyl hendecanoate	Vinylidene chloride	0.04	0.16	2.72	0.37	Y	145
Vinyl iodide	Styrene	0.126	0.035	7.401	0.315		806
Vinyl isobutyl ether	Acrylate, 2-chloroethyl	0		2.24			790
Vinyl isobutyl ether	Acrylate, methyl	0	0.002	3.63	0.04	N	347
Vinyl isobutyl ether	Acrylonitrile	0.7	0.09	−0.24	0.26	N	322
Vinyl isobutyl ether	Acrylonitrile	0.99	0.06	−0.05	0.07	N	322
Vinyl isobutyl ether	Acrylonitrile	−0.04	0.01	0.98	0.15	N	347
Vinyl isobutyl ether	Acrylonitrile	−0.04	0.02	1.26	0.19	N	347
Vinyl isobutyl ether	Ethylene, trichloro-	0.05	0.009	0.014	0.003	Y	332
Vinyl isobutyl ether	Fumarate, diethyl	0		0.52			640
Vinyl isobutyl ether	Maleate, diethyl	0		0.17			640
Vinyl isobutyl ether	Styrene	0.65	0.39	11.58	1.91	N	349
Vinyl isobutyl ether	Styrene	0.29	0.33	11.96	2.32	N	349
Vinyl isobutyl ether	Styrene	0.01		50			490
Vinyl isobutyl ether	Vinyl chloride	0.04	0.37	2.19	1.43	Y	47
Vinyl isobutyl ether	Vinylene carbonate	0.18		0.16			322
Vinyl isobutyl sulfide	Acrylate, ethyl	0.05	0.04	0.36	0.02	N	285
Vinyl isobutyl sulfide	Acrylate, ethyl	0.05	0.05	0.32	0.02	N	336
Vinyl isobutyl sulfide	Acrylate, methyl	0.05	0.02	0.3	0.01	N	285
Vinyl isobutyl sulfide	Acrylate, methyl	0.06	0.02	0.29	0.01	N	336
Vinyl isobutyl sulfide	Acrylonitrile	0.03	0.01	0.08	0.005	N	285
Vinyl isobutyl sulfide	Acrylonitrile	0.022	0.038	0.062	0.004	Y	303
Vinyl isobutyl sulfide	Acrylonitrile	0.03	0.01	0.08	0.004	N	336
Vinyl isobutyl sulfide	Methacrylate, methyl	0.05	0.03	0.89	0.01	N	285

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl isobutyl sulfide	Methacrylate, methyl	0.021	0.034	0.985	0.077	Y	303
Vinyl isobutyl sulfide	Methacrylate, methyl	0.03	0.02	0.88	0.02	N	336
Vinyl isobutyl sulfide	Styrene	−0.11	0.2	2.38	0.41	Y	303
Vinyl isopropyl ether	Caprolactam, *N*-vinyl-	0.01	0.96	0.22	1.95	Y	215
Vinyl isopropyl ether	Pyrrolidone, *N*-vinyl-	0.03	2.09	0.87	9.3	Y	215
Vinyl isopropyl ketone	Methacrylate, methyl	1.44		0.81			601
Vinyl isopropyl ketone	Methacrylate, methyl	0.6	0.47	0.54	0.22	Y	794
Vinyl isopropyl ketone	Styrene	0.26	0.07	0.46	0.05	N	344
Vinyl isopropyl ketone	Styrene	0.272	0.046	0.36	0.1	Y	353
Vinyl isopropyl ketone	Styrene	0.3		0.4			599
Vinyl isopropyl ketone	Vinyl acetate	7.87		0.3			601
Vinyl isopropyl ketone	Vinyl acetate	6.47	2.18	−0.154	0.073	Y	794
Vinyl isopropyl sulfide	Methacrylate, methyl	0.046	0.025	0.822	0.064	Y	303
Vinyl isopropyl sulfide	Styrene	0.16	0.068	4.61	0.18	Y	303
Vinyl isothiocyanate	Acrylonitrile	1.4		0.36			542
Vinyl isothiocyanate	Methacrylate, methyl	0.6		0.85			422
Vinyl isothiocyanate	Styrene	0.37		0.65			422
Vinyl isothiocyanate	Styrene	0.5		0.8			542
Vinyl 1-methyl ether	Indene	0.67	2.82	12.64	12.59	Y	750
Vinyl laurate	Acrylonitrile	0.04		4			381
Vinyl laurate	Vinyl acetate	0.7		1.4			503
Vinyl laurate	Vinyl chloride	0.2		7.4			450
Vinyl *m*-cresyl ether	Methacrylate, methyl	0.2		3.3			605
Vinyl *m*-cresyl ether	Vinyl chloride	0.13	0.02	1.23	0.04	Y	279
Vinyl *m*-cresyl ether	Vinylidene chloride	0.04	0.004	1.95	0.07	Y	279
Vinyl methanesulfonate	Vinyl acetate	0.206	0.427	0.639	0.014	Y	942
Vinyl methyl ketone	Acrylamide	0.47	0.44	3.99	0.7	Y	609
Vinyl methyl ketone	Acrylamide	0.75	0.23	1.02	0.13	Y	609
Vinyl methyl ketone	Acrylamide, *N,N*-dimethyl	0.1	0.43	0.82	0.36	Y	609
Vinyl methyl ketone	Acrylate, butyl	1.81	0.24	0.8	0.06	N	53
Vinyl methyl ketone	Acrylonitrile	1.75	0.11	0.59	0.13	Y	140
Vinyl methyl ketone	Acrylonitrile	1.39		0.63			504
Vinyl methyl ketone	Indene	2.63		0.2			606
Vinyl methyl ketone	Indene	2.9		0.25			606
Vinyl methyl ketone	Methacrylamide	0.25	0.12	3	0.18	Y	609
Vinyl methyl ketone	Methacrylamide	0.28	0.24	1.72	0.26	Y	609
Vinyl methyl ketone	Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	0.41		0.53			664
Vinyl methyl ketone	Methyleneglutaronitrile, 2-	5.05		1.24			512
Vinyl methyl ketone	Styrene	0.349	0.068	0.288	0.092	Y	140
Vinyl methyl ketone	Styrene	0.348	0.009	0.237	0.023	Y	353
Vinyl methyl ketone	Styrene	0.27		0.35			550
Vinyl methyl ketone	Styrene, 2,5-dichloro-	0.5		2			375
Vinyl methyl ketone	Triazine, 4,6-diamino-2-vinyl-	0.17	0.11	1.35	0.68	Y	243
Vinyl methyl ketone	Vinyl acetate					Y	91
Vinyl methyl ketone	Vinyl chloride	8.3		0.1			375
Vinyl methyl ketone	Vinyl phenyl ether	4.4		0.01			423
Vinyl methyl ketone	Vinyl phenyl ketone	0.74		0.671			946
Vinyl methyl ketone	Vinylidene chloride	1.8		0.55			375
Vinyl methyl sulfide	Acrylate, methyl	0.084	0.086	0.209	0.034	Y	195
Vinyl methyl sulfide	Methacrylate, methyl	0.03		1			106
Vinyl methyl sulfide	Styrene	0.15		4.5			193
Vinyl methyl sulfide	Styrene	0.066	0.056	4.46	0.11	Y	195
Vinyl methyl sulfide	Styrene	0.046	0.084	5	0.24	Y	303
Vinyl methyl sulfide	Vinylene carbonate	7.72	1.68	−0.08	0.17	Y	120
Vinyl methyl sulfone	Styrene	−0.22	0.17	1.146	0.087	Y	195
Vinyl methyl sulfone	Vinyl acetate	0.55	0.21	0	0.02	Y	195
Vinyl methyl sulfoxide	Styrene	0.01		3.57			737
Vinyl methyl sulfoxide	Styrene, *p*-chloro-	0.01		6.46			737
Vinyl methyl sulfoxide	Styrene, *p*-methyl-	0.01		2.73			737
Vinyl mono-diethyleneglycol ether	Methacrylic acid	2.2		0			831
Vinyl mono-ethyleneglycol ether	Methacrylate, glycidyl	0		4.41			780
Vinyl *o*-cresyl ether	Methacrylate, methyl	0.1		1.8			605
Vinyl *o*-cresyl ether	Vinyl chloride	0.14	0.01	1.33	0.02	Y	279
Vinyl *o*-cresyl ether	Vinylidene chloride	0.04	0.01	2.1	0.1	Y	279

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl octadecyl ether	Acrylonitrile	0		0.85			373
Vinyl octadecyl ether	Vinyl acetate	0		4.5			372
Vinyl octadecyl ether	Vinyl chloride	−0.1		2.1			370
Vinyl octadecyl ether	Vinylidene chloride	0		1.5			371
Vinyl octyl ether	Acrylonitrile	0		0.81			373
Vinyl octyl ether	Styrene	0		65			374
Vinyl octyl ether	Vinyl acetate	0		3.47			372
Vinyl octyl ether	Vinyl chloride	0.1		1.9			370
Vinyl *p*-benzylethylcarbinol	Acrylate, butyl	0.4		0.17			591
Vinyl *p*-benzylethylcarbinol	Acrylate, methyl	0.48		0.18			591
Vinyl *p*-benzylethylcarbinol	Acrylonitrile	0.31		0.04			591
Vinyl *p*-benzylmethylcarbinol	Acrylonitrile	0.54	0.2	0.11	0.17	N	48
Vinyl *p*-benzylmethylcarbinol	Methacrylate, methyl	0.98	0.28	0.24	0.26	Y	48
Vinyl *p*-benzylmethylcarbinol	Styrene	0.94	0.15	0.98	0.32	Y	48
Vinyl *p*-benzylmethylcarbinol	Styrene, *p*-chloro-	1		1		Y	48
Vinyl *p*-bromobenzoate	Vinyl benzoate	1.17	0.13	0.716	0.031	Y	110
Vinyl *p*-bromobenzoate	Vinyl *p*-cyanobenzoate	0.83	0.021	0.901	0.034	Y	110
Vinyl *p*-bromobenzoate	Vinyl *p*-methoxybenzoate	1.08	0.29	0.72	0.1	Y	110
Vinyl *p*-bromobenzoate	Vinyl *p*-methylbenzoate	1.05	0.17	0.594	0.045	Y	110
Vinyl *p*-chlorobenzoate	Vinyl benzoate	0.81	0.072	0.832	0.028	Y	110
Vinyl *p*-chlorobenzoate	Vinyl *p*-cyanobenzoate	0.813	0.014	1.031	0.028	Y	110
Vinyl *p*-chlorobenzoate	Vinyl *p*-methoxybenzoate	1	0.14	0.699	0.057	Y	110
Vinyl *p*-chlorobenzoate	Vinyl *p*-methylbenzoate	0.853	0.054	0.682	0.021	Y	110
Vinyl *p*-chlorophenyl ether	Vinyl chloride	0.63	0.05	1	0.04	Y	279
Vinyl *p*-chlorophenyl ether	Vinylidene chloride	0.14	0.01	2.44	0.04	Y	279
Vinyl *p*-cresyl ether	Methacrylate, methyl	0		2.6			605
Vinyl *p*-cresyl ether	Vinyl chloride	0.13	0.01	1.17	0.02	Y	279
Vinyl *p*-cresyl ether	Vinylidene chloride	0.03	0.004	1.97	0.06	Y	279
Vinyl *p*-cyanobenzoate	Vinyl benzoate	0.878	0.094	0.659	0.033	Y	110
Vinyl *p*-cyanobenzoate	Vinyl *p*-bromobenzoate	0.901	0.034	0.83	0.021	Y	110
Vinyl *p*-cyanobenzoate	Vinyl *p*-chlorobenzoate	1.031	0.028	0.813	0.014	Y	110
Vinyl *p*-cyanobenzoate	Vinyl *p*-methoxybenzoate	0.63	0.22	0.685	0.075	Y	110
Vinyl *p*-cyanobenzoate	Vinyl *p*-methylbenzoate	0.81	0.1	0.624	0.04	Y	110
Vinyl *p*-methoxybenzoate	Vinyl *p*-bromobenzoate	0.72	0.1	1.08	0.29	Y	110
Vinyl *p*-methoxybenzoate	Vinyl *p*-chlorobenzoate	0.699	0.057	1	0.14	Y	110
Vinyl *p*-methoxybenzoate	Vinyl *p*-cyanobenzoate	0.685	0.075	0.63	0.22	Y	110
Vinyl *p*-methylbenzoate	Vinyl *p*-bromobenzoate	0.594	0.045	1.05	0.17	Y	110
Vinyl *p*-methylbenzoate	Vinyl *p*-chlorobenzoate	0.682	0.021	0.853	0.054	Y	110
Vinyl *p*-methylbenzoate	Vinyl *p*-cyanobenzoate	0.624	0.04	0.81	0.1	Y	110
Vinyl *p*-*tert*-butylphenyl ether	Vinyl chloride	0.14	0.02	1.08	0.06	Y	279
Vinyl *p*-*tert*-butylphenyl ether	Vinylidene chloride	0.05	0.004	2.04	0.06	Y	279
Vinyl *p*-vinyloxybenzoate	Vinyl chloride	0.78	0.07	0.58	0.06	N	497
Vinyl pelargonate	Butadiene	0.02		26.3			145
Vinyl pelargonate	Styrene	0.01		49.5			145
Vinyl pelargonate	Vinyl chloride	0.28		1.16			145
Vinyl pelargonate	Vinylidene chloride	0		4.08			145
Vinyl phenyl carbonate	Vinyl acetate	0.384	0.083	0.85	0.23	Y	675
Vinyl phenyl carbonate	Vinyl acetate	0.384	0.083	0.85	0.23	Y	705
Vinyl phenyl ether	Acrylonitrile	0.23		2.5			556
Vinyl phenyl ether	Caprolactam, *N*-vinyl-	0.26	0.37	1.3	2.43	Y	214
Vinyl phenyl ether	Methacrylate, methyl	0.01	0.32	11.34	5.76	Y	172
Vinyl phenyl ether	Methacrylate, methyl	0		2.8			605
Vinyl phenyl ether	Pyrrolidone, *N*-vinyl-	0.253	0.007	4.48	0.1	Y	214
Vinyl phenyl ether	Styrene	0.01		1.7			423
Vinyl phenyl ether	Vinyl chloride	0.01	0.59	1.28	0.14	Y	172
Vinyl phenyl ether	Vinyl chloride	0.16		1.06			279
Vinyl phenyl ether	Vinyl chloride	0.17	0.03	1.21	0.03	N	280
Vinyl phenyl ether	Vinyl chloroacetate	0.55	0.61	0.84	1.67	Y	212
Vinyl phenyl ether	Vinyl methyl ketone	0.01		4.4			423
Vinyl phenyl ether	Vinylidene chloride	0.433	0.051	2.45	0.34	Y	240
Vinyl phenyl ether	Vinylidene chloride	0.04	0.03	2.11	0.11	N	280
Vinyl phenyl ketone	Methacrylate, glycidyl	0.697		0.719			946
Vinyl phenyl ketone	Methacrylate, methyl	0.59	1.9	0.71	0.6	Y	794
Vinyl phenyl ketone	Styrene	0.3	0.02	0.24	0.04	N	344
Vinyl phenyl ketone	Styrene	0.465	0.03	0.21	0.042	Y	353

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinyl phenyl ketone	Vinyl acetate	5.81	0.96	−0.06	0.057	Y	794
Vinyl phenyl ketone	Vinyl methyl ketone	0.671		0.74			946
Vinyl phenyl sulfide	Acrylate, methyl	−0.02	0.22	0.57	0.1	Y	193
Vinyl phenyl sulfide	Acrylonitrile	0.03	0.02	0.11	0.01	N	285
Vinyl phenyl sulfide	Acrylonitrile	0.03	0.01	0.11	0.01	N	336
Vinyl phenyl sulfide	Benzothiazole, vinylmercapto-	0.24	0.12	0.95	0.61	Y	763
Vinyl phenyl sulfide	Methacrylate, methyl	0.06	0.09	0.79	0.06	N	239
Vinyl phenyl sulfide	Methacrylate, methyl	0.08		0.85			239
Vinyl phenyl sulfide	Styrene	0.019	0.054	3.63	0.12	Y	193
Vinyl phenyl sulfide	Styrene	0.26	0.18	3.96	0.27	N	239
Vinyl phenyl sulfone	Styrene	−0.31	0.16	1.562	0.091	Y	193
Vinyl phenyl sulfone	Vinyl acetate	−0.35	0.37	0.171	0.064	Y	193
Vinyl propionate	Allyl chloride	0.62		0.68			438
Vinyl propionate	Butadiene, 2-chloro-					N	247
Vinyl propionate	Carbazole, *N*-vinyl-	0.081	0.073	1.69	0.21	Y	246
Vinyl propionate	Ethylene	1.5		0.67			605
Vinyl propionate	Methacrylate, methyl	0.03		24			272
Vinyl propionate	Vinyl acetate	0.98		0.98			272
Vinyl propionate	Vinyl acetate	1		0.9			425
Vinyl propionate	Vinyl chloride	0.6		1.6			400
Vinyl propionate	Vinyl chloride	0.65		1.35			450
Vinyl stearate	Acrylate, methyl	0.03		5.8			537
Vinyl stearate	Acrylonitrile	0.078	0.007	4.639	0.015	Y	145
Vinyl stearate	Butadiene	0.02	0.27	42.5	4.73	Y	145
Vinyl stearate	Styrene	−0.39	0.71	15.96	13.32	Y	145
Vinyl stearate	Vinyl acetate	0.96	0.17	0.953	0.015	Y	1
Vinyl stearate	Vinyl acetate	0.73		0.9			537
Vinyl stearate	Vinyl chloride	0.24	0.64	0.71	0.097	Y	145
Vinyl stearate	Vinylidene chloride	0.08	0.17	3.92	0.14	Y	145
Vinyl sulfone	Methacrylate, methyl	0.1	0.16	4.5	0.49	Y	45
Vinyl sulfone	Styrene	−0.027	0.047	0.731	0.058	Y	45
Vinyl *tert*-butyl ketone	Methacrylate, methyl	0.65	0.15	1.21	0.11	Y	794
Vinyl *tert*-butyl ketone	Styrene	3.2	0.46	0.3	0.31	N	344
Vinyl *tert*-butyl sulfide	Styrene	0.248	0.05	0.36	0.11	Y	353
Vinyl *tert*-butyl sulfide	Acrylonitrile	0	0.013	0.09	0.024	Y	303
Vinyl *tert*-butyl sulfide	Styrene	0.158	0.073	4.36	0.76	Y	303
Vinyl *tert*-butyl sulfide	Vinyl acetate	0.07		13			598
Vinyl *tert*-butyl sulfide	Vinyl acetate	0.07		13			601
Vinyl *tert*-butyl sulfide	Vinyl chloride	0.15		3			598
Vinyl *tert*-butyl sulfide	Vinyl chloride	0.15		3			601
Vinyl thiolacetate	Acrylate, methyl	0.23		0.8			566
Vinyl thiolacetate	Carbazole, *N*-vinyl-	0.45	0.15	0.44	0.28	N	97
Vinyl thiolacetate	Styrene	0.25		4			566
Vinyl thiolacetate	Succinimide, *N*-vinyl-	2.47	0.37	0.17	0.15	N	97
Vinyl thiolacetate	Vinyl acetate	5.5		0.05			566
Vinyl thiolacetate	Vinylene carbonate	13.57	1.04	0.041	0.01	Y	186
Vinyl trichloroacetate	Acrylonitrile	0.047	0.039	7.34	0.9	Y	362
Vinyl trimethylammonioethyl ether iodide	*N*-Vinylpyrrolidone	0	0.02	11.6	0.03		1037
Vinyl valerate	Vinyl chloride	0.277	0.014	2.006	0.071	Y	283
Vinyl, 2-, 2-methyl-4,4,6,6-tetraphenyl-cyclotrisiloxane	Styrene	0.8		0.11			626
Vinyl-tris(trimethoxysiloxy)silane	Acrylonitrile	0.07		3.9			209
Vinyl-tris(trimethoxysiloxy)silane	Acrylonitrile	0.08		3.85			573
Vinyl-tris(trimethoxysiloxy)silane	Methacrylate, methyl	0.02		34			482
Vinyl-tris(trimethoxysiloxy)silane	Styrene	0		26			209
Vinyl-tris(trimethoxysiloxy)silane	Styrene	0.01		24			482
Vinylacetophenone	Methacrylate, methyl	1.65	3.81	0.33	0.36	Y	266
Vinylacetophenone	Styrene	1.12	0.42	0.246	0.072	Y	266
Vinylanthracene, 9-	Acrylate, butyl	0.126	0.054	3.717	0.046	Y	123
Vinylanthracene, 9-	Acrylate, ethyl	0.295	0.056	3.498	0.041	Y	123
Vinylanthracene, 9-	Acrylate, methyl	0.08	0.092	2.998	0.067	Y	123
Vinylanthracene, 9-	Methacrylate, methyl	0.117	0.043	4.068	0.04	Y	123
Vinylanthracene, 9-	Styrene	0.25		2.12			123
Vinylanthracene, 9-	Styrene	0.3		2.12			48
Vinylbenzoate, *p*-, sodium	Acrylonitrile	0.23	0.34	0.174	0.04	Y	175

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinylbenzoic acid, *p*-	Acrylonitrile	1.63	0.42	0.06	0.009	Y	175
Vinylbenzoic acid, *p*-	Methacrylate, methyl	1.16	0.3	0.398	0.097	Y	266
Vinylbenzoic acid, *p*-	Styrene	1.029	0.06	0.282	0.011	Y	266
Vinylbenzyl chloride	Methacrylate, methyl	1.02		0.46			792
Vinylene carbonate	Acrylamide	0.05		13.8			387
Vinylene carbonate	Acrylonitrile	0.08		14.9			579
Vinylene carbonate	Methacrylate, methyl	−0.04	0.85	126	298	Y	103
Vinylene carbonate	Pyrrolidone, *N*-vinyl-	0.056	0.059	1.08	0.11	Y	103
Vinylene carbonate	Styrene	3.16		411		Y	103
Vinylene carbonate	Styrene	−0.73	0.98	21.03	23.45	Y	120
Vinylene carbonate	Vinyl acetate	0.3	0.3	3.2	0.62	Y	103
Vinylene carbonate	Vinyl acetate	0.27		3			103
Vinylene carbonate	Vinyl acetate	0.05	0.16	7.16	0.64	Y	120
Vinylene carbonate	Vinyl acetate	0.13		7.3			120
Vinylene carbonate	Vinyl acetate	0.044	0.039	3.83	0.18	Y	144
Vinylene carbonate	Vinyl acetate	0.08	0.11	3.92	0.32	Y	90
Vinylene carbonate	Vinyl chloride	0.03	0.12	4.82	0.49	Y	103
Vinylene carbonate	Vinyl chloride	0.069	0.068	15.38	1.44	Y	319
Vinylene carbonate	Vinyl chloride	0	0.22	17.81	7.84	Y	319
Vinylene carbonate	Vinyl isobutyl ether	0.16		0.18			322
Vinylene carbonate	Vinyl 1-menthyl ether	0.14	1.91	0.25	3.32	Y	750
Vinylene carbonate	Vinyl methyl sulfide	−0.08	0.17	7.72	1.68	Y	120
Vinylene carbonate	Vinyl thiolacetate	0.041	0.01	13.57	1.04	Y	186
Vinylethyldiethoxysilane	Vinyl chloride	−0.44	2.97	0.94	1.53	Y	237
Vinylethynyl-4-piperidinol, 1,2,5-trimethyl-4-	Acrylonitrile	0.38		0.08			783
Vinylethynyl-4-piperidinol, 1,2,5-trimethyl-4-	Methacrylic acid	0.05		0.4			734
Vinylferrocene	Acrylonitrile	0.165	0.092	0.186	0.025	Y	294
Vinylferrocene	Acrylonitrile	0.15		0.16			294
Vinylferrocene	Butadiene	0.16	0.15	4.34	0.18	Y	802
Vinylferrocene	Methacrylate, methyl	0.52		1.22			294
Vinylferrocene	Pyrrolidone, *N*-vinyl-	0.71	0.16	0.364	0.068	Y	308
Vinylferrocene	Styrene	0.2	0.1	4.37	1.44	Y	274
Vinylferrocene	Styrene	0.14	0.14	2.76	0.22	Y	294
Vinylhydroquinone	Acrylic acid	0.026	0.092	0.29	0.11	Y	337
Vinylhydroquinone	Acrylonitrile	0.04	0.03	0.56	0.36	N	338
Vinylhydroquinone	Methacrylate, methyl	0.02	0.13	0.49	0.21	Y	337
Vinylhydroquinone	Pyridine, 4-vinyl-	−0.23	0.36	0.53	0.4	Y	337
Vinylhydroquinone dibenzoate	Acrylate, methyl	0.75		0.46			440
Vinylhydroquinone dibenzoate	Acrylic acid	0.95		0.44			444
Vinylhydroquinone dibenzoate	Methacrylate, methyl	0.41		0.34			440
Vinylhydroquinone dibenzoate	Styrene	0.43		0.22			440
Vinylhydroquinone dibenzoate	Styrene, α-methyl-	0.3		0.11			440
Vinylidene chloride	Aconitate, trimethyl	64.6	23.74	0.15	0.94	N	149
Vinylidene chloride	Acrylamide, *N*-octadecyl-	0.432	0.008	1.4	0.11	Y	119
Vinylidene chloride	Acrylate, butyl	0.934	0.006	0.873	0.022	Y	118
Vinylidene chloride	Acrylate, butyl	0.84		0.46			464
Vinylidene chloride	Acrylate, butyl	0.87		0.58			464
Vinylidene chloride	Acrylate, ethyl	0.58	0.15	0.72	0.39	Y	65
Vinylidene chloride	Acrylate, methyl	0.9	0.11	0.95	0.21	Y	107
Vinylidene chloride	Acrylate, methyl	0.7		0.8			107
Vinylidene chloride	Acrylate, methyl	1		1			377
Vinylidene chloride	Acrylate, octadecyl	0.9	0.037	1	0.14	Y	118
Vinylidene chloride	Acrylate, octyl	0.851	0.018	0.679	0.057	Y	118
Vinylidene chloride	Acrylic acid	0.437	0.065	1.246	0.042	Y	288
Vinylidene chloride	Acrylonitrile	0.32	0.14	0.92	0.11	Y	138
Vinylidene chloride	Acrylonitrile	0.28	0.034	1.04	0.2	Y	238
Vinylidene chloride	Acrylonitrile	0.4		0.44			404
Vinylidene chloride	Acrylonitrile	0.42		0.58			404
Vinylidene chloride	Acrylonitrile	0.49		0.65			404
Vinylidene chloride	Acrylonitrile	0.69		0.57			404
Vinylidene chloride	Acrylonitrile	0.76		0.5			404
Vinylidene chloride	Acrylonitrile	0.95		0.48			404
Vinylidene chloride	Acrylonitrile	0.95		0.65			404

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinylidene chloride	Acrylonitrile	1.3		0.65			404
Vinylidene chloride	Acrylonitrile	1.74		0.66			404
Vinylidene chloride	Acrylonitrile	1.43		0.63			404
Vinylidene chloride	Acrylonitrile	1.8		0.63			404
Vinylidene chloride	Acrylonitrile	0.39		0.58			468
Vinylidene chloride	Acryloyl chloride	0.5		1.12			288
Vinylidene chloride	Allyl acetate	6.6		0			470
Vinylidene chloride	Allyl chloride	3.49	1.4	0.24	0.24	Y	3
Vinylidene chloride	Allylstearamide, *N*-	5.22	0.15	−0.24	0.19	Y	119
Vinylidene chloride	Butadiene	−0.044	0.038	1.97	0.25	Y	257
Vinylidene chloride	Carbazole, *N*-vinyl-	−0.023	0.068	3.6	0.12	Y	13
Vinylidene chloride	Carbazole, *N*-vinyl-	−0.008	0.018	3.65	0.3	Y	98
Vinylidene chloride	Crotonaldehyde			0			405
Vinylidene chloride	Crotonic acid	18.1	43.7	−0.2	1.03	Y	47
Vinylidene chloride	Cyclopentene, 4-, 1,3-dione	1.88	0.35	0.113	0.027	Y	264
Vinylidene chloride	Diallyl phthalate	5		0.2			376
Vinylidene chloride	Ethylene	15.71	1.96	0.018	0.005	Y	299
Vinylidene chloride	Ethylene chlorotrifluoro-	17.14		0.02			454
Vinylidene chloride	Fumarate, diethyl	12.19	1.58	0.012	0.038	Y	59
Vinylidene chloride	Fumarate, diisopropyl	48		0.095			1038
Vinylidene chloride	Indene	0.49	0.15	0.293	0.095	Y	6
Vinylidene chloride	Indene	−0.02	1.15	0.34	0.85	Y	6
Vinylidene chloride	Isobutylene, 3-chloro-	1.1		0			470
Vinylidene chloride	Isopropenyl methyl ketone	0.134	0.099	4.06	0.66	Y	47
Vinylidene chloride	Isopropenylisocyanate	0.85		0.31			434
Vinylidene chloride	Maleate, diethyl	12.5					3
Vinylidene chloride	Maleimide	0.666	0.056	0.464	0.083	Y	251
Vinylidene chloride	Methacrylate, 2-(sulfonic acid)ethyl	0.22		3.6			564
Vinylidene chloride	Methacrylate, benzyl	0.34		3.3			432
Vinylidene chloride	Methacrylate, butyl	0.35	0.03	2.07	0.18	Y	3
Vinylidene chloride	Methacrylate, ethyl	0.353	0.02	2.18	0.1	Y	3
Vinylidene chloride	Methacrylate, methyl	0.3		2.6		Y	138
Vinylidene chloride	Methacrylate, methyl	0.36	0.03	2.38	0.02	N	324
Vinylidene chloride	Methacrylate, methyl	0.18	0.18	2.16	0.06	N	324
Vinylidene chloride	Methacrylate, methyl	0.5		2.5			426
Vinylidene chloride	Methacrylate, methyl	0.34		2.5			432
Vinylidene chloride	Methacrylate, methyl	0.09		2.1			538
Vinylidene chloride	Methacrylate, methyl	0.094	0.032	1.8	0.95	Y	65
Vinylidene chloride	Methacrylate, methyl	0.19		2.83			762
Vinylidene chloride	Methacrylate, methyl	0.02		1.23			762
Vinylidene chloride	Methacrylate, methyl	0.27	0.01	2.77	0.06	N	786
Vinylidene chloride	Methacrylic acid	0.154	0.047	3.368	0.098	Y	6
Vinylidene chloride	Methacrylonitrile	0.3	0.1	2.43	0.1	N	781
Vinylidene chloride	Methacrylonitrile	0.036	0.05	2.38	0.05	N	781
Vinylidene chloride	Norbornadiene	1.36	0.63	0.09	0.17	N	267
Vinylidene chloride	Oxazolidone, *N*-vinyl-	1.42	0.11	0.091	0.077	Y	102
Vinylidene chloride	Propenyl 2-methyl, acetate	2.4		0			470
Vinylidene chloride	Styrene	0.134	0.048	1.81	0.35	Y	138
Vinylidene chloride	Styrene	0.098	0.01	1.79	0.14	Y	238
Vinylidene chloride	Styrene	1.5		1.8			393
Vinylidene chloride	Styrene	0.087	0.007	1.839	0.024	Y	59
Vinylidene chloride	Styrene	0.15		1.81			812
Vinylidene chloride	Styrene	0.11	0.005	1.7	0.12	N	93
Vinylidene chloride	Succinimide, *N*-vinyl-	1.441	0.061	0.311	0.058	Y	104
Vinylidene chloride	Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	0.13		1.9			583
Vinylidene chloride	Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	0.25		1.8			583
Vinylidene chloride	Tetrazole, 5-phenyl-2-(4′-vinyl)phenyl-	0.11		1.9			583
Vinylidene chloride	Vinyl 12-ketostearate	4.4	4.8	0.26	2.61	Y	146
Vinylidene chloride	Vinyl acetate	4.66	0.26	0.03	0.1	Y	3
Vinylidene chloride	Vinyl acetate	6.7		0.05			432
Vinylidene chloride	Vinyl acetate	5		0.05			436
Vinylidene chloride	Vinyl acetate	3.49	0.17	−0.002	0.026	Y	59
Vinylidene chloride	Vinyl acetate	4.7		0.12			812
Vinylidene chloride	Vinyl benzoate	5.58	1.76	0.065	0.093	Y	47
Vinylidene chloride	Vinyl butyl ether	1.75	0.11	0.004	0.001	Y	279

TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinylidene chloride	Vinyl chloride	2.06	0.32	0.071	0.025	Y	238
Vinylidene chloride	Vinyl chloride	3.39	1.16	0.14	0.32	N	29
Vinylidene chloride	Vinyl chloride	8.92		0.25		Y	3
Vinylidene chloride	Vinyl chloride	3.2		0.3			470
Vinylidene chloride	Vinyl chloride	4.5		0.2			517
Vinylidene chloride	Vinyl chloride	1.8		0.2			539
Vinylidene chloride	Vinyl chloride	4.5		0.16			585
Vinylidene chloride	Vinyl chloride	3.068	0.076	0.205	0.003	Y	67
Vinylidene chloride	Vinyl chloromethyl ketone	0.018	0.009	6.8	0.3	Y	356
Vinylidene chloride	Vinyl dodecyl ether	1.3		0			371
Vinylidene chloride	Vinyl ethyl ether	3.2		0			470
Vinylidene chloride	Vinyl fluoride	6		0.16			607
Vinylidene chloride	Vinyl hendecanoate	2.72	0.37	0.04	0.16	Y	145
Vinylidene chloride	Vinyl *m*-cresyl ether	1.95	0.07	0.04	0.004	Y	279
Vinylidene chloride	Vinyl methyl ketone	0.55		1.8			375
Vinylidene chloride	Vinyl *o*-cresyl ether	2.1	0.1	0.04	0.01	Y	279
Vinylidene chloride	Vinyl octadecyl ether	1.5		0			371
Vinylidene chloride	Vinyl *p*-chlorophenyl ether	2.44	0.04	0.14	0.01	Y	279
Vinylidene chloride	Vinyl *p*-cresyl ether	1.97	0.06	0.03	0.004	Y	279
Vinylidene chloride	Vinyl *p-tert*-butyl phenyl ether	2.04	0.06	0.05	0.004	Y	279
Vinylidene chloride	Vinyl pelargonate	4.08		0			145
Vinylidene chloride	Vinyl phenyl ether	2.45	0.34	0.433	0.051	Y	240
Vinylidene chloride	Vinyl phenyl ether	2.11	0.11	0.04	0.03	N	280
Vinylidene chloride	Vinyl stearate	3.92	0.14	0.08	0.17	Y	145
Vinylidene chloride	Vinylidene cyanide	0.009	0.001	0.049	0.02	Y	82
Vinylidene chloride	Vinylisocyanate	1.46		0.33			435
Vinylidene cyanide	Acrylate, α-chloro-, methyl	0.066	0.07	0.313	0.021	Y	82
Vinylidene cyanide	Acrylic acid	0.291	0.042	0.212	0.016	Y	82
Vinylidene cyanide	Butadiene, 2-chloro-					Y	82
Vinylidene cyanide	Ethylene, dichloro-, *cis*-					Y	82
Vinylidene cyanide	Ethylene, dichloro-, *cis*-					Y	82
Vinylidene cyanide	Maleic anhydride					Y	82
Vinylidene cyanide	Methacrylate, methyl	0.028	0.013	0.057	0.012	Y	82
Vinylidene cyanide	Phosphonic acid, 1-phenylvinyl	4.58		0.1			921
Vinylidene cyanide	Propene, 2-chloro-	0.204	0.079	0.001	0.03	Y	82
Vinylidene cyanide	Styrene	0.72	0.16	0.015	0.007	Y	134
Vinylidene cyanide	Styrene	0.001	0.001	0.003	0.001	Y	82
Vinylidene cyanide	Styrene, 2,5-dichloro-	0.005	0.009	0.026	0.006	Y	82
Vinylidene cyanide	Vinyl acetate	0.1	0.11	0.003	0.002	Y	82
Vinylidene cyanide	Vinyl benzoate	0.061	0.047	0	0.017	Y	82
Vinylidene cyanide	Vinyl chloride	0.72		0.01			409
Vinylidene cyanide	Vinyl chloride	0.6	0.068	0.016	0.002	Y	82
Vinylidene cyanide	Vinyl chloroacetate	0.137	0.037	0	0.006	Y	82
Vinylidene cyanide	Vinyl chloride	0.049	0.02	0.009	0.001	Y	82
Vinylidene fluoride	Methacryloyl fluoride, perfluoro-	7.6		0.02			669
Vinylidene fluoride	Vinyl acetate	0.08		6			627
Vinylisocyanate	Acrylate, methyl	0.14		1.38			435
Vinylisocyanate	Acrylonitrile	0.16		0.19			435
Vinylisocyanate	Methacrylate, methyl	0.01		3.3			422
Vinylisocyanate	Styrene	0.08		8.13			435
Vinylisocyanate	Styrene	0.1		6.9			533
Vinylisocyanate	Vinyl chloride	3.7		0.46			777
Vinylisocyanate	Vinylidene chloride	0.33		1.46			435
Vinylmethyldiacetoxysilane	Acrylonitrile	−0.11	0.23	2.246	0.054	Y	754
Vinylmethyldiacetoxysilane	Styrene	0	0.15	4.08	0.14	Y	754
Vinylmethyldiethoxysilane	Styrene	0.08	0.14	5.68	0.1	Y	754
Vinylmethyldiethoxysilane	Styrene	−0.08	0.2	11.29	0.43	Y	754
Vinylmethylphenylsulfonium tetrafluoroborate	Acrylamide	0.1		3.4			793
Vinylmethylphenylsulfonium tetrafluoroborate	Acrylonitrile	0.1		2.2			793
Vinylphenyldimethylsilane	Methacrylate, methyl	−0.009	0.034	29.79	1.89	Y	482
Vinylphenyldimethylsilane	Styrene	0.024	0.016	32.53	0.49	Y	482
Vinylphenylisocyanate, *p*-	Vinyl chloride	11.9		0.1			777
Vinylphenylpropanol, 2-*p*-	Styrene	0.41	0.4	0.48	0.3	Y	72

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	r_1	±95%	r_2	±95%	Conv.	Refs.
Vinylsulfonic acid	Acrylamide	0.3	0.1	3.5	0.15		1066
Vinyltriethoxysilane	Acrylonitrile	0.41	0.38	6.59	0.6	Y	209
Vinyltriethoxysilane	Acrylonitrile	2.69	2.5	26.83	29.59	Y	237
Vinyltriethoxysilane	Styrene	−0.09	0.14	20.86	1.99	Y	209
Vinyltriethoxysilane	Vinyl chloride	−0.04	0.44	0.82	0.16	Y	237
Vinyltrimethoxysilane	Acrylonitrile	0.45		9.09		Y	2.37
Vinyltrimethoxysilane	Vinyl chloride	0.09		0.93		Y	237
Vinyltrimethylsilane	Acrylonitrile	0.1	0.27	4.08	0.76	Y	209
Vinyltrimethylsilane	Methacrylate, methyl	0.028	0.016	35.66	0.85	Y	482
Vinyltrimethylsilane	Styrene	−0.19	0.36	14.66	5.15	Y	209
Vinyltrimethylsilane	Styrene					Y	482
Vinyltrimethylsilane	Styrene	0.049	0.025	5.98	0.18	Y	73
Xanthate, *S*-methacryloyl *O*-ethyl	Methacrylate, methyl	0.6	0.08	0.76	0.04		1086
Xanthate, *S*-methacryloyl *O*-ethyl	Styrene	0.36	0.08	0.44	0.06		1086

TABLE 2. LISTING OF QUICK BASIC (MICROSOFT) PROGRAM FOR CALCULATING REACTIVITY RATIOS*

```
100  OPTION BASE 1
110  DIM eta (20)
120  DIM ksi (20)
130  DIM gee (20)
140  DIM ef (20)
150  DIM t (20)
151  FOR b = 1 TO 20
152  READ t(b)
153  NEXT b
170  DATA 12.71,4.3,3.18,2.78,2.57,2.45,2.37,2.31,2.26,2.23,2.2,2.18,2.16,2.14,2.13,2.12,2.11,2.1,2.09,2.09
180  DEF FNLog10(X) = LOG(X)/LOG(10#)
190  PRINT "Calculation of reactivity ratios by the method of Kelen and Tudos"
210  PRINT "Enter 0 for monomer 1 in the feed to start the calculation"
220  REM initialize all of the summing addresses
230  n = 0
231  m = 0
232  var = 0
233  sumasi = 0
234  sumi = 0
235  suma = 0
236  sumsqi = 0
237  sumnegi = 0
238  sum2i = 0
250  INPUT "Do you wish to stop (Y/N)?";f$
260  IF f$ = "y" THEN GO TO 1510
270  REM enter the names and molecular weights of the monomers
280  INPUT "Monomer 1 is:"; mone$
290  INPUT "Monomer 2 is:"; mtwo$
340  REM enter the feed, polymer and conversion values
350  PRINT "Are the feed and polymer values in mol fractions (mol),
mol ratios (ratio) or weight fractions (wt) ?"
370  INPUT type$
380  IF type$ = "ratio" THEN GOTO 1210
390  IF type$ = "wt" THEN GOTO 1360
400  PRINT "Mol fraction of"; mone$; "in the feed = "
410  INPUT m10
420  IF m10 = 0 THEN GOTO 800
430  n = n + 1
440  m20 = 1-m10
450  PRINT "Mol fraction of"; mone$; "in the polymer = "
460  INPUT p1
470  p2 = 1-p1
480  PRINT "Conversion, weight fraction = ? (If none given enter 0)"
490  INPUT w
500  IF w = 0 THEN GOTO 680
```

* Method of Kelen and Tűdös (803,804).

```
510 REM calculate G and F
520 X = m10/m20
530 y = p1/p2
540 mu = mw2/mw1
550 zeta2 = w*(mu + X)/(mu + y)
560 zeta1 = (y/X)*zeta2
570 z = (FNLog10(1-zeta1))/(FNLog10(1-zeta2))
580 g = (y-1)/z
590 f = y/(z^2)
600 IF n = 1 THEN fmin = f
610 IF n = 1 THEN fmax = f
620 IF fmin>f THEN fmin = f
630 IF fmax<f THEN fmax = f
640 gee(n) = g
650 ef(n) = f
660 GOTO 380
670 REM calculate G and f without conversion data
680 X = m10/m20
690 y = p1/p2
700 g = ((y-1)*X)/y
710 f = (X^2)/Y
720 IF n = 1 THEN fmin = f
730 IF n = 1 THEN fmax = f
740 IF fmin>f THEN fmin = f
750 IF fmax<f THEN fmax = f
760 gee(n) = g
770 ef(n) = f
780 GOTO 380
790 REM calculate eta and ksi
800 alpha = SQR(fmin*fmax)
810 m = m + 1
820 a = gee (m)/(alpha + ef(m))
830 si = ef(m)/(alpha + ef(m))
840 REM sum variables for least squares
850 sumasi = (a*si) + sumasi
860 sumi = si + sumi
870 suma = a + suma
880 sumsqi = (si^2) + sumsqi
890 sumnegi = (1-si) + sumnegi
900 sum2i = si*(1-si) + sum2i
910 eta(m) = a
920 ksi(m) = si
930 IF m = n THEN GOTO 960
940 GOTO 810
950 REM calculate the reactivity ratios
960 r1 = (sumasi*(n-sumi)-suma*(sumi-sumsqi))/((n*sumsqi)-(sumi^2))
970 r2 = (alpha*((sumasi*sumi)-(suma*sumsqi)))/((n*sumsqi)-(sumi^2))
980 REM calculate S squared
990 m = 0
1000 m = m + 1
1010 smin = ((eta(m)-(r1*ksi(m))) + ((r2/alpha)*(1-ksi(m))))^2
1020 var = smin + var
1030 IF m = n THEN GOTO 1050
1040 GOTO 1000
1050 d = sumsqi*(sumnegi^2)-(sum2i^2)
1060 REM calculate the delta r's
1070 deltr1 = t(m-2)*SQR((var/(n-2))*((sumnegi^2)/d))
1080 deltr2 = alpha*t(m-2)*SQR((var/(n-2))*(sumsqi/d))
1090 REM print out the answers
1100 PRINT "Reactivity ratios for"; mone$; "(M1) and"; mtwo$; "(M2)"
1110 PRINT "r1 = "; r1; "delta r1 = "; deltr1
1120 PRINT "r2 = "; r2; "delta r2 = "; deltr2
1130 PRINT "alpha = "; alpha
1140 PRINT "ksi"; "eta"
1150 m = 0
1160 m = m + 1
1170 PRINT ksi(m), eta (m)
1180 IF m = n THEN GOTO 220
1190 GOTO 1160
```

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```
1200  REM calculation of M1 and M2 from mol ratios
1210  PRINT "Mol ratio of"; mone$; "in the feed =";
1220  INPUT qf
1230  IF qf=0 THEN GOTO 800
1240  n=n+1
1250  m10=qf/(qf+1)
1260  m20=1-m10
1270  PRINT "Mole ratio of"; mone$ "in the polymer="
1280  INPUT qp
1290  p1=qp/(qp+1)
1300  p2=1-p1
1310  PRINT "Weight fraction conversion="
1320  INPUT w
1330  IF w=0 THEN GOTO 680
1340  GOTO 510
1350  REM calculation of M1 and M2 from weight fractions
1351  PRINT "The molecular weight of"; mone$; "is"
1352  INPUT mw1
1353  PRINT "The molecular weight of"; mtwo$; "is"
1354  INPUT mw2
1360  PRINT "Weight fraction of"; mone$ "in the feed="
1370  INPUT rf1
1380  If rf1=0 THEN GOTO 800
1390  n=n+1
1400  rf2=1-rf1
1410  m10=(rf1/mw1)/((rf1/mw1)+(rf2/mw2))
1420  m20=1-m10
1430  PRINT "Weight fraction of"; mone$; "in the polymer="
1440  INPUT rp1
1450  rp2=1-rp1
1460  p1=(rp1/mw1)/((rp1/mw1)+(rp2/mw2))
1470  PRINT "Weight fraction conversion="
1480  INPUT w
1490  IF w=0 THEN GOTO 680
1500  GOTO 510
1510  PRINT "FINI"
1520  END
```

C. REFERENCES

1. A. Adicoff, A. Buselli, J. Polym. Sci., **21**, 340 (1956).
2. S. L. Aggarwal, F. A. Long, J. Polym. Sci., **11**, 127 (1953).
3. P. Argon, T. Alfrey, J. Bohrer, H. Hass, H. Wechsler, J. Polym. Sci., **3**, 157 (1948).
4. A. N. Akopyan, G. Ye. Krbekyan, Vysokomol. Soedin., **5**, 201 (1963).
5. A. N. Akopyan, G. Ye. Krbekyan, E. G. Sinanyan, Vysokomol. Soedin., **5**, 681 (1963).
6. T. Alfrey, J. Bohrer, H. Haas, C. Lewis, J. Polym. Sci., **5**, 719 (1950).
7. T. Alfrey, W. H. Ebelke, J. Am. Chem. Soc., **71**, 3235 (1949).
8. T. Alfrey, A. I. Goldberg, W. P. Hohenstein, J. Am. Chem. Soc., **68**, 2464 (1946).
9. T. Alfrey, S. Greenberg, J. Polym. Sci., **3**, 297 (1948).
10. T. Alfrey, J. G. Harrison, J. Am. Chem. Soc., **68**, 299 (1946).
11. T. Alfrey, S. L. Kapur, J. Polym. Sci., **4**, 215 (1949).
12. T. Alfrey, E. Lavin, J. Am. Chem. Soc., **67**, 2044 (1945).
13. T. Alfrey, B. Magel, quoted by Hart, Makromol. Chem., **47**, 143 (1961).
14. T. Alfrey, E. Mertz, H. Mark, J. Polym. Sci., **1**, 37 (1946).
15. T. Alfrey, H. Morawetz, J. Am. Chem. Soc., **74**, 436 (1952).
16. T. Alfrey, C. G. Overberger, S. H. Pinner, J. Am. Chem. Soc., **75**, 4221 (1953).
17. T. L. Ang, H. J. Hardwood, Am. Chem. Soc., Div. Polymer Chem. Preprints, **5**, 306 (1964).
18. C. L. Arcus, R. J. S. Mathews, J. Chem. Soc., 4607, (1956).
19. E. J. Arlman, H. W. Melville, Proc. Roy. Soc. (London) A, **203**, 301 (1950).
20. E. J. Arlman, H. W. Melville, L. Valentine, Rec. Trav. Chim., **68**, 945 (1949).
21. H. Asai, T. Imoto, J. Polym. Sci. B, **2**, 553 (1964).
22. N. Ashikari, A. Nishimura, J. Polym. Sci., **31**, 250 (1958).
23. M. A. Askarov, A. S. Bank, Vysokomol. Soedin., **8**, 1012 (1966).
24. C. Aso, T. Kunitake, K. Watanabe, Kogyo Kagaku Zasshi, **70**, 1001 (1967).
25. I. S. Avetisyan, V. I. Eliseeva, O. G. Larinovo, Vysokomol Soedin. A, **9**, 570 (1967).
26. S. Banarjee, M. S. Muthana, J. Polym. Sci., **35**, 292 (1959).
27. P. D. Bartlett, K. Nozaki, J. Am. Chem. Soc., **68**, 1495 (1946).

28. W. I. Bengough, D. Goldrich, R. A. Young, Eur. Polym. J., **3**, 117 (1967).
29. W. I. Bengough, R. G. W. Norrish, Proc. Roy. Soc. (London) A, **218**, 155 (1953).
30. J. C. Bevington, D. O. Harris, J. Polym. Sci. B, **5**, 799 (1967).
31. J. C. Bevington, M. Johnson, Makromol. Chem., **87**, 66 (1965).
32. J. C. Bevington, B. W. Malpass, Eur. Polym. J., 1, **85** (1965).
33. D. C. Blackley, H. W. Melville, Makromol. Chem., **18**, 16 (1956).
34. G. Blauer, L. Goldstein, J. Polym. Sci., **25**, 19 (1957).
35. E. P. Bonsall, L. Valentine, H. W. Melville, Trans Faraday Soc., **48**, 763 (1952).
36. J. F. Bork, L. E. Coleman, J. Polym. Sci., **43**, 413 (1960).
37. J. Bourdais, Bull. Soc. Chim. France, 485 (1955).
38. J. H. Bradbury, H. W. Melville, Proc. Roy. Soc. (London) A, **222**, 456 (1954).
39. D. Braun, T.-O. Ahn, Kolloid Z., **188**, 1 (1963).
40. J. W. Breitenbach, A. Schindler, C. Pflug, Monatsh. Chem., **81**, 21 (1950).
41. F. E. Brown, G. E. Ham, J. Polym. Sci. A, **2**, 3623 (1964).
42. R. D. Burkhart, N. L. Zutty, J. Polym. Sci. A, **1**, 1137 (1963).
43. G. M. Burnett, P. Evans, H. W. Melville, Trans. Faraday Soc., **49**, 1096 (1953).
44. G. G. Cameron, D. H. Grant, N. Grassie, J. E. Lamb, I. C. McNeill, J. Polym. Sci., **36**, 173 (1959).
45. E. Y. C. Chang, C. C. Price, J. Am. Chem. Soc., **83**, 4650 (1961).
46. E. C. Chapin, G. E. Ham, R. G. Fordyce, J. Am. Chem. Soc., **70**, 538 (1948).
47. E. C. Chapin, G. E. Ham, C. L. Mills, J. Polymer Sci., **4**, 597 (1949); errata, J. Polymer Sci., **55**, S-6 (1961).
48. K. Anda, S. Iwai, Kogyo Kagaku Zasshi, **70**, 557 (1967).
49. K. Chikanishi, T. Tsuruta, Makromol. Chem., **81**, 211 (1965).
50. L. E. Coleman, J. A. Conrady, J. Polym. Sci., **38**, 241 (1959).
51. P. Colombo, L. E. Kukacka, J. Fontana, R. N. Chapman, M. Steinberg, J. Polym. Sci. A-1, **4**, 29 (1966).
52. P. Colombo, M. Steinberg, J. Fontana, J. Polym. Sci. B, **1**, 447 (1963).
53. W. Cooper, E. Catterall, Can. J. Chem., **34**, 387 (1956).
54. K. Crauwels, G. Smets, Bull. Soc. Chim. Belg., **59**, 443 (1950).
55. K. Crauwels, G. Smets, Bull. Soc. Chim. Belg., **59**, 182 (1950).
56. I. Crescentini, G. B. Getchele, A. Zanella, J. Appl. Polym. Sci., **9**, 1323 (1965).
57. G. F. D'Alelio, T. Huemmer, J. Polym. Sci. A-1, **5**, 77 (1967).
58. A. DePauw, G. Smets, Bull. Soc. Chim. Belg., **59**, 629 (1950).
59. K. W. Doak, J. Am. Chem. Soc., **70**, 1525 (1948).
60. K. W. Doak, J. Am. Chem. Soc., **72**, 4681 (1950).
61. K. W. Doak, D. L. Dineen, J. Am. Chem. Soc., **73**, 1084 (1951).
62. J. Drougas, R. L. Guile, J. Polym. Sci., **55**, 297 (1961).
63. F. I. Duntov, B. L. Erusalimskii, Vyskomol. Soedin., **7**, 1075 (1965).
64. F. I. Duntov, A. L. Gol'denberg, M. A. Litvinova, B. L. Erusalimskii, Vysokomol. Soedin. A, **9**, 1920 (1967).
65. B. G. Elgood, B. J. Sauntson, Chem. Ind. (London), 1558 (1965).
66. W. H. Embree, J. M. Mitchell, H. L. Williams, Can. J. Chem., **29**, 253 (1951).
67. S. Enomoto, J. Polym. Sci., **55**, 95 (1961).
68. B. Erussalimsky, N. Tumarkin, F. Duntoff. S. Lyubetzky, A. Goldenberg, Makromol. Chem., **104**, 288 (1967).
69. E. S. Ferdinandi, W. P. Garby, D. G. L. James, Can. J. Chem., **42**, 2568 (1964).
70. R. E. Florin, J. Am. Chem. Soc., **71**, 1867 (1949).
71. J. W. L. Fordham, G. H. McCain, L. E. Alexander, J. Polym. Sci., **39**, 335 (1959).
72. D. Braun, H.-G. Keppler, Makromol. Chem., **82**, 132 (1965).
73. V. G. Filippova, N. S. Nametkin, S. G. Durgaryan, , Izv. Akad. Nauk SSR, Ser. Khim., 1727 (1966).
74. R. G. Fordyce, E. C. Chapin, J. Am. Chem. Soc., **69**, 581 (1947).
75. R. G. Fordyce, E. C. Chapin, G. E. Ham, J. Am. Chem. Soc., **70**, 2489 (1948).
76. R. G. Fordyce, G. E. Ham, J. Am. Chem. Soc., **69**, 695 (1947).
77. R. G. Fordyce, G. E. Ham, J. Am. Chem. Soc., **73**, 1186 (1951).
78. B. L. Funt, E. A. Ogryzlo, J. Polym. Sci., **25**, 279 (1957).
79. R. M. Fuoss, G. I. Cathers, J. Polym. Sci., **4**, 97 (1949).
80. J. Furukawa, T. Tsuruta, N. Yamamoto, H. Fukutani, J. Polym. Sci., **37**, 215 (1959).
81. L. Ghosez, G. Smets, J. Polym. Sci., **35**, 215 (1959).
82. H. Gilbert, F. F. Miller, S. J. Averill, E. J. Carlson, V. L. Folt, H. J. Heller, F. D. Stewart, R. F. Schmidt, H. L. Trumbull, J. Am. Chem. Soc., **78**, 1669 (1956).
83. R. D. Gilbert, H. L. Williams, J. Am. Chem. Soc., **74**, 4114 (1952).
84. G. Goldfinger, M. Steidlitz, J. Polym. Sci., **3**, 786 (1948).
85. N. Grassie, I. C. McNeill, J. Polym. Sci., **27**, 207 (1958).
86. N. Grassie, I. C. McNeill, I. F. McLaren, J. Polym. Sci. B, **3**, 897 (1965).
87. N. Grassie, B. J. D. Torrance, J. D. Fortune, J. D. Gemmell, Polymer, **6**, 653 (1965).
88. E. Yun, L. V. Lobanova, A. D. Litmanovich, N. A. Plate, M. V. Shishkina, T. A. Polikarpova, Vysokomol. Soedin., **12**, 2488 (1970).
89. E. Grzyma, Z. Jedlinski, Vysokomol. Soedin., **8**, 1653 (1966).
90. H. C. Haas, N. W. Schuler, J. Polym. Sci., **31**, 237 (1958).
91. H. C. Hass, M. S. Simon, J. Polym. Sci., **9**, 309 (1952).
92. G. Hagele, H. Frolich, D. Bischoff, K. Hamann, Makromol. Chem., **75**, 98 (1964).
93. R. A. Haldron, J. N. Hay, J. Polym. Sci. A-1, **6**, 951 (1968).

94. G. E. Ham, J. Polym. Sci., **14**, 87 (1954).
95. F. L. Hamb, A. Winston, J. Polym. Sci. A, **2**, 4475 (1964).
96. G. Hardy, J. Varga, G. Nagy, Makromol. Chem., **85**, 58 (1965).
97. G. Hardy, J. Varga, K. Nytrai, I. Tsajlik, L. Zubonyai, Vysokomol. Soedin., **6**, 758 (1964).
98. R. Hart, Makromol. Chem., **47**, 143 (1961).
99. R. Hart, A. E. vanDormael, Bull. Soc. Chim. Belg., **65**, 571 (1956).
100. R. Hart, G. Smets, J. Polym. Sci., **5**, 55 (1950).
101. R. Hart, D. Timmerman, Bull. Soc. Chim. Belg., **67**, 123 (1958).
102. R. Hart, D. Timmerman, Makromol. Chem., **31**, 223 (1959).
103. K. Hayashi, G. Smets, J. Polym. Sci., **27**, 275 (1958).
104. H. Hopff, P. Schlumbom, Makromol. Chem., **43**, 173 (1961).
105. M. B. Huglin, Polymer, **3**, 335 (1962).
106. J. C. H. Hwa, L. Miller, J. Polym. Sci., **55**, 197 (1961).
107. M. Ibonai, T. Kato, Kogyo Kagaku Zasshi, **70**, 2078 (1967).
108. M. Imoto, T. Otsu, B. Yamada, Kogyo Kagaku Zasshi, **68**, 1132 (1965).
109. T. Imoto, Y. Ogo, H. Nakamoto, Bull. Chem. Soc. Japan, **41**, 543 (1968).
110. K. Ito, Y. Yamashita, J. Polym. Sci. B, **3**, 637 (1965).
111. K. Ito, Y. Yamashita, Kogyo Kagaku Zasshi, **68**, 1469 (1965).
112. Y. Minoura, T. Tadokoro, Y. Suzuki, J. Polym. Sci. A-1, **5**, 2641 (1967).
113. Y. Iwakura, T. Kurosaki, N. Ariga, T. Ito, Makromol. Chem., **97**, 128 (1966).
114. Y. Iwakura, T. Kurosaki, N. Nakabayashi, Makromol. Chem., **46**, 570 (1961).
115. Y. Iwakura, T. Tamikado, M. Yamaguchi, K. Takei, J. Polym. Sci., **39**, 203 (1959).
116. S. Iwatsuki, M. Murakami, Y. Yamashita, Kogyo Kagaku Zasshi, **68**, 1967 (1965).
117. S. Iwatsuki, Y. Yamashita, Kogyo Kagaku Zasshi, **68**, 1963 (1965).
118. E. F. Jordan, K. M. Doughty, W. S. Port, J. Appl. Polym. Sci., **4**, 203, (1960).
119. E. F. Jordan, A. N. Wrigley, J. Appl. Polym. Sci., **8**, 527 (1964).
120. J. M. Judge, C. C. Price, J. Polym. Sci., **41**, 435 (1959).
121. D. J. Kahn, H. H. Horowitz, J. Polym. Sci., **54**, 363 (1961).
122. N. G. Karapetyan, I. S. Boshnyakov, A. S. Magaryan, Vysokomol. Soedin., **7**, 1993 (1965).
123. D. Katz, J. Relis, J. Polym. Sci. A-1, **6**, 2079 (1968).
124. N. Kawabata, T. Tsuruta, J. Furukawa, Makromol. Chem., **48**, 106 (1961).
125. R. Kerber, Makromol. Chem., **96**, 30 (1966).
126. J. T. Khamis, Polymer, **6**, 98 (1965).
127. A. M. Khomutov, Izv. Akad. Nauk SSSR Otd. Khim. Nauk, 352 (1961).
128. M. Kinoshita, M. Imoto, Kogyo Kagaku Zasshi, **68**, 2454 (1965).
129. G. S. Kolesnikov, E. F. Rodionova, I. G. Safaralieva, Izv. Akad. Nauk SSSR, Ser. Khim., 2028, (1963).
130. G. S. Kolesnikov, A. S. Tevlina, A. B. Alovitdinov, Vysokomol. Soedin., **7**, 1913 (1965).
131. G. S. Kolesnikov, A. S. Tevlina, A. B. Alovitdinov, Plasticheskie Massy, 12 (1966).
132. G. S. Kolesnikov, A. S. Tevlina, S. P. Novikova, S. N. Sividova, Vysokomol. Soedin., **7**, 2160 (1965).
133. S. Konya, M. Yokoyama, Kogyo Kagaku Zasshi, **68**, 1080 (1965).
134. M. Kreisel, U. Garbatski, D. H. Kohn, J. Polym. Sci. A, **2**, 105 (1964).
135. A. G. Kronman, I. V. Pasmanik, B. I. Fedoseev, V. A. Kargin, Vysokomol. Soedin. A, **9**, 2503 (1967).
136. F. Leonard, W. P. Hohenstein, E. Merz, J. Am. Chem. Soc., **70**, 1283 (1948).
137. F. M. Lewis, F. R. Mayo, J. Am. Chem. Soc., **70**, 1533 (1948).
138. F. M. Lewis, F. R. Mayo, W. F. Hulse, J. Am. Chem. Soc., **67**, 1701 (1945).
139. F. M. Lewis, C. Walling, W. Cummings, E. R. Briggs, F. R. Mayo, J. Am. Chem. Soc., **70**, 1519 (1948).
140. F. M. Lewis, C. Walling, W. Cummings, E. R. Briggs, W. J. Wenisch, J. Am. Chem. Soc., **70**, 1527 (1948).
141. H. Lussi, Makromol. Chem., **103**, 62 (1967).
142. H. Lussi, Makromol. Chem., **103**, 68 (1967).
143. H. Lussi, Makromol. Chem., **110**, 100 (1967).
144. H. L. Marder, C. Schuerch, J. Polym. Sci., **44**, 129 (1960).
145. C. S. Marvel, W. G. Dipierri, J. Polym. Sci., **27**, 39 (1958).
146. C. S. Marvel, T. K. Dykstra, F. C. Magne, J. Polym. Sci., **62**, 369 (1962).
147. C. S. Marvel, J. W. Johnson, J. P. Economy, G. P. Scott, W. K. Taft, B. G. Labbe, J. Polym. Sci., **20**, 437 (1956).
148. C. S. Marvel, G. D. Jones, T. W. Mastin, G. L. Schertz, J. Am. Chem. Soc., **64**, 2356 (1942).
149. C. S. Marvel, E. B. Mano, J. Polym. Sci., **31**, 165 (1958).
150. C. S. Marvel, G. L. Schertz, J. Am. Chem. Soc., **65**, 2045 (1943).
151. C. S. Marvel, R. Schwen, J. Am. Chem. Soc., **79**, 6003 (1957).
152. A. Matsumoto, M. Oiwa, Kogyo Kagaku Zasshi, **70**, 360 (1967).
153. A. Matsumoto, S. Shoda, T. Harada, M. Oiwa, Kogyo Kagaku Zasshi, **70**, 1007 (1967).
154. K. Matsuoka, M. Otsuka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **69**, 137 (1966).
155. K. Matsuoka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **68**, 1941 (1965).
156. K. Matsuoka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **69**, 134 (1966).
157. K. Matsuoka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **69**, 142 (1966).
158. F. R. Mayo, C. Walling, F. M. Lewis, W. F. Hulse, J. Am. Chem. Soc., **70**, 1523 (1948).
159. E. J. Meehan, J. Polym. Sci., **1**, 318 (1946).
160. H. W. Melville, L. Valentine, Trans. Faraday Soc., **51**, 1474 (1955).

161. A. Misono, Y. Uchida, Bull. Chem. Soc. Japan, **39**, 2458 (1966).
162. J. Muller, Collection Czech. Chem. Commun., **20**, 241 (1955).
163. K. Murata, A. Terada, Bull. Chem. Soc. Japan, **39**, 2494 (1966).
164. S. Nagai, Bull. Chem. Soc. Japan, **36**, 1459 (1963).
165. A. S. Nair, M. S. Muthana, Makromol. Chem., **47**, 138 (1961).
166. A. F. Nikolaev, V. M. Gaperin, Vysokomol. Soedin. A, **9**, 2469 (1967).
167. A. F. Nikolaev, S. N. Ushakov, L. S. Mishkileeva, Vysokomol. Soedin., **6**, 287 (1964).
168. M. Nishigaki, M. Kinoshita, M. Imoto, Kogyo Kagaku Zasshi, **70**, 1938 (1967).
169. A. M. North, D. Postlethwaite, Polymer, **5**, 237 (1964).
170. A. M. North, A. M. Scallan, Polymer, **5**, 447 (1964).
171. K. F. O'Driscoll, F. P. Gasparro, J. Macromol. Sci. A: Chem., **1**, 643 (1967).
172. E. Ohmori, Y. Ohi, T. Otsu, M. Imoto, Kogyo Kagaku Zasshi, **68**, 1600 (1965).
173. R. J. Orr, H. L. Williams, Can. J. Chem., **30**, 108 (1952).
174. R. J. Orr, H. L. Williams, Can. J. Chem., **33**, 1328 (1955).
175. E. Osawa, K. Wang, O. Kurihara, Makromol. Chem., **83**, 100 (1965).
176. G. Oster, Y. Mizutani, J. Polym. Sci., **22**, 173 (1956).
177. T. Otsu, T. Ito, Y. Fujii, M. Imoto, Bull. Soc. Chem. Japan, **41**, 204 (1968).
178. T. Otsu, T. Ito, T. Fukumizu, M. Imoto, Bull. Soc. Chem. Japan, **39**, 2257 (1966).
179. T. Otsu, T. Ito, M. Imoto, Kogyo Kagaku Zasshi, **69**, 986 (1966).
180. T.Otsu, Y. Kinoshita, A. Nakamachi, Makromol. Chem., **115**, 27 (1968).
181. T. Otsu, A. Shimizu, M. Imoto, J. Polym. Sci. A, **3**, 615 (1965).
182. T. Otsu, B. Yamada, Makromol. Chem., **110**, 297 (1967).
183. M. Otsuka, K. Takemoto, M. Imoto, Kobunshi Kagaku, **23**, 765 (1966).
184. M. Otsuka, Y. Yashuhara, K. Takemoto, M. Imoto, Makromol. Chem., **103**, 291 (1967).
185. C. G. Overberger, D. E. Baldwin, H. P. Gregor, J. Am. Chem. Soc., **72**, 4864 (1950).
186. C. G. Overberger, H. Biletch, R. G. Nickerson, J. Polym. Sci., **27**, 381 (1958).
187. T. R. Paxton, J. Polym. Sci. B, **1**, 73 (1963).
188. J. Pellon, R. L. Kugel, R. Marcus, R. Rabinowitz, J. Polym. Sci., A-2, 4105 (1964).
189. J. P. Fischer, Makromol. Chem., **156**, 227 (1972).
190. S. H. Pinner, J. Polym. Sci., **10**, 379 (1953).
191. C. C. Price, R. D. Gilbert, J. Polym. Sci., **8**, 580 (1952).
192. C. C. Price, C. E. Greene, J. Polym. Sci., **6**, 111 (1951).
193. C. C. Price, H. Morita, J. Am. Chem. Soc., **75**, 4747 (1953).
194. C. C. Price, T. C. Schwan, J. Polym. Sci., **16**, 577 (1955).
195. C. C. Price, J. Zomlefer, J. Am. Chem. Soc., **72**, 14 (1950).
196. E. G. Pritchett, P. M. Kamath, J. Polym. Sci. B, **4**, 849 (1966).
197. H. Ringsdorf, N. Weinshenker, C. G. Overberger, Makromol. Chem., **64**, 126 (1963).
198. W. M. Ritchey, L. E. Ball, J. Polym. Sci. B, **4**, 557 (1966).
199. L. Rodriguez, Makromol. Chem., **12**, 110 (1954).
200. J. Roovers, G. Smets, Makromol. Chem., **60**, 89 (1963).
201. R. W. Roth, R. F. Church, J. Polym. Sci., **55**, 41 (1961).
202. E. W. Rugeley, T. A. Field, G. H. Fremon, Ind. Eng. Chem., **40**, 1724 (1948).
203. A. V. Ryabov, Yu. D. Semchikov, N. N. Slavnitskaya, Dokl. Akad. Nauk SSSR, **145**, 822 (1962).
204. M. K. Saha, P. Ghosh, S. R. Palit, J. Polym. Sci. A, **2**, 1365 (1964).
205. G. Saini, G. Polla-Mattiot, M. Meirone, J. Polym. Sci., **50**, S-13 (1961).
206. C. L. Sandberg, F. A. Bovey, J. Polym. Sci., **15**, 553 (1955).
207. R. C. Schulz, H. Cherdron, W. Kern, Makromol. Chem., **28**, 197 (1958).
208. R. C. Schulz, E. Kaiser, W. Kern, Makromol. Chem., **58**, 160 (1962).
209. C. E. Scott, C. C. Price, J. Am. Chem. Soc., **81**, 2670 (1959).
210. P. K. Sengupta, A. R. Mukherjee, P. Ghosh, J. Macromol. Chem., **1**, 481 (1966).
211. A. Shimizu, T. Otsu, M. Imoto, Bull. Chem. Soc. Japan, **38**, 1535 (1965).
212. M. F. Shostakovskii, A. M. Khomutov, A. P. Alimov, Izv. Akad. Nauk SSSR, Ser. Khim., 1839, (1963).
213. M. F. Shostakovskii, A. M. Khomutov, F. P. Sidel'kovskaya, Izv. Akad. Nauk SSSR, Ser. Khim., 2222, (1961).
214. F. P. Sidel'kovskaya, M. A. Askarov, F. Ibragimov, Vysokomol. Soedin., **6**, 1810 (1964).
215. F. P. Sidel'kovskaya, M. F. Shostakovskii, F. Ibragimov, M. A. Askarov, Vysokomol. Soedin., **6**, 1585 (1964).
216. C. Simionescu, N. Asandei, A. Liga, Makromol. Chem., **110**, 278 (1967).
217. J. A. Simms, J. Appl. Polym. Sci., **5**, 58 (1961).
218. G. Smets, L. Dehaes, Bull. Soc. Chim. Belg., **59**, 13 (1950).
219. G. Smets, N. Deval, P. Hous, J. Polym. Sci. A, **2**, 4835 (1964).
220. G. Smets, E. Dysseleer, Makromol. Chem., **91**, 160 (1966).
221. G. Smets, A. M. Hesbain, J. Polym. Sci., **40**, 217 (1959).
222. G. Smets, A. Reckers, Rec. Trav. Chim., **68**, 983 (1949).
223. W. E. Smith, G. E. Ham, H. D. Anspon, S. E. Gebura, D. W. Alwani, J. Polym. Sci. A-1, **6**, 2001 (1968).
224. M. F. Sorokin, M. M. Babkina, Vysokomol. Soedin., **7**, 734 (1965).
225. M. F. Sorokin, I. M. Kochnov, Vysokomol. Soedin., **6**, 798 (1964).
226. M. F. Sorokin, K. A. Lyalyushko, R. A. Dudakova, V. S. Vasil'ev, A. N. Shuvalova, Plasticheskie Massy, 3 (1963).
227. E. Muccer, K. Dinges, W. Graulich, Makromol. Chem., **57**, 27 (1962).
228. A. Takahashi, H. Tanaka, I. Kagawa, Kogyo Kagaku Zasshi, **70**, 988 (1967).
229. K. Takemoto, Y. Kikuchi, M. Imoto, Kogyo Kagaku Zasshi, **69**, 1367 (1966).

230. K. Takemoto, K. Matsuoka, M. Otsuka, M. Imoto, Kogyo Kagaku Zasshi, **69**, 2026 (1966).
231. T. Tamikado, J. Polym. Sci., **43**, 489 (1960).
232. T. Tamikado, Makromol. Chem., **38**, 85 (1960).
233. T. Tamikado, Y. Iwakura, J. Polym. Sci., **36**, 529 (1959).
234. A. Tanaka, K. Sasaki, Y. Hozumi, O. Hashimoto, J. Appl. Polym. Sci., **8**, 1787 (1964).
235. M. Taniyama, G. Oster, Bull. Chem. Soc. Japan, **30**, 856 (1957).
236. W. M. Thomas, M. T. O'Shaughnessy, J. Polym. Sci., **11**, 455 (1953).
237. B. R. Thompson, J. Polym. Sci., **19**, 373 (1956).
238. B. R. Thompson, R. H. Raines, J. Polym. Sci., **41**, 265 (1959).
239. K. Tsuda, S. Kobayashi, T. Otsu, J. Polym. Sci. A-1, **6**, 41 (1968).
240. N. A. Tyukavkina, A. V. Kalabina, G. I. Deryabina, G. T. Zhikarev, A. D. Biryukova, Vysokomol. Soedin., **6**, 1573 (1964).
241. K. Uehara, T. Nishi, T. Tsuyuri, F. Tamura, N. Murata, Kogyo Kagaku Zasshi, **70**, 750 (1967).
242. J. Ulbricht, J. Giesemann, M. Gebauer, Angew. Makromol. Chem., **3**, 69 (1968).
243. C. G. Overberger, F. W. Michelotti, J. Am. Chem. Soc., **80**, 988 (1958).
244. C. C. Unruh, T. M. Laasko, J. Polym. Sci., **33**, 87 (1958).
245. S. N. Ushakov, V. A. Kropachev, L. B. Trukhmanova, R. I. Gruz, T. M. Markelova, Vysokomol. Soedin. A, **9**, 1807 (1967).
246. S. N. Ushakov, A. F. Nikolaev, Izv. Akad. Nauk SSSR, Ser. Khim., 83 (1956).
247. S. N. Ushakov, L. B. Trukhmanova, Izv. Akad. Nauk SSSR, Ser. Khim., 980 (1957).
248. S. N. Ushakov, L. B. Trukhmanova, T. M. Markelova, V. A. Kropachev, Vysokomol. Soedin. A, **9**, 999 (1967).
249. L. F. Vandenburgh, C. E. Brockway, J. Polym. Sci. A, **3**, 575 (1965).
250. G. vanPaesschen, G. Smets, Bull. Soc. Chim. Belg., **64**, 173 (1955).
251. G. vanPaesschen, D. Timmerman, Makromol. Chem., **78**, 112 (1964).
252. B. Vollmert, Angew. Makromol. Chem., **3**, 1 (1968).
253. M. Vrancken, G. Smets, Makromol. Chem., **30**, 197 (1969).
254. F. T. Wall, R. E. Florin, C. J. Delbecq. J. Am. Chem. Soc., **72**, 4769 (1950).
255. F. T. Wall, R. W. Powers, G. D. Sands, G. S. Stent, J. Am. Chem. Soc., **70**, 1031 (1948).
256. C. Walling, F. R. Briggs, K. B. Wolfstirn, F. R. Mayo, J. Am. Chem. Soc., **70**, 1537 (1948).
257. C. Walling, J. A. Davison, J. Am. Chem. Soc., **73**, 5736 (1951).
258. R. L. Whistler, J. L. Goatley, J. Polym. Sci., **50**, 127 (1961).
259. G. S. Wich, N. Brodoway, J. Polym. Sci. A, **1**, 2163 (1963).
260. M. C. DeWilde, G. Smets, J. Polym. Sci., **5**, 253 (1950).
261. R. H. Wiley, B. Davis, J. Polym. Sci., **46**, 423 (1960).
262. R. H. Wiley, E. E. Sale, J. Polym. Sci., **42**, 497 (1960).
263. A. Winston, F. L. Hamb, J. Polym. Sci. A, **3**, 583 (1965).
264. A. Winston, G. T. C. Li., J. Polym. Sci. A-1, **5**, 1223 (1967).
265. P. Wittmer, Makromol. Chem., **103**, 188 (1967).
266. B. A. Zaitsev, G. A. Shtraikhman, Vysokomol. Soedin. A, **10**, 438 (1968).
267. N. L. Zutty, J. Polym. Sci. A, **1**, 2231 (1963).
268. J. Aoyagi, I. Shinohara, Nippon Kagaku Kaishi, 788, 1972.
269. S. Tazuke, S. Okamura, J. Polym. Sci. A-1, **6**, 2907 (1968).
270. T. Otsu, T.-C. Lai, Y. Kinoshita, A. Nakamachi, M. Imoto, Kogyo Kagaku Zasshi, **71**, 904 (1968).
271. K. Takemoto, S. Takahashi, M. Imoto, Kogyo Kagaku Zasshi, **71**, 742 (1968).
272. J. C. Bevington, M. Johnson, Eur. Polym. J., **4**, 669 (1968).
273. R. Kroker, H. Ringsdorf, Makromol. Chem., **121**, 240 (1969).
274. C. Aso, T. Kunitake, T. Nakashima, Kogyo Kagaku Zasshi, **72**, 1411 (1969).
275. F. C. Baines, J. C. Bevington, Polymer, **11**, 647 (1970).
276. V. D. Bezuglyi, I. B. Voskresenskaya, T. A. Alekseeva, M. M. Gemer, Vysokomol. Soedin. A, **14**, 540 (1972).
277. D. Braun, G. Mott, Angew. Makromol. Chem., **18**, 183 (1971).
278. W. R. Cabaness, T. Y.-C. Lin, C. Parkanyi, J. Polym. Sci. A-1, **9**, 2155 (1971).
279. G. I. Deryabina, Yu. L. Frolov, A. V. Kalabina, Vysokomol. Soedin., **A13**, 1162 (1971).
280. G. I. Deryabina, G. I. Gershengorn, Vysokomol. Soedin. A, **11**, 941 (1969).
281. J. P. Fischer, Makromol. Chem., **155**, 211 (1972).
282. A. Gilath, S. H. Ronel, M. Shmueli, D. H. Kohn, J. Appl. Polym. Sci., **14**, 1491 (1970).
283. K. Hayashi, T. Otsu, Makromol. Chem., **127**, 54 (1969).
284. H. Hopff, G. Becker, Makromol. Chem., **133**, 1 (1970).
285. H. Inoue, T. Otsu, Makromol. Chem., **153**, 21 (1972).
286. H. Inoue, I. Umeda, T. Otsu, Makromol. Chem., **147**, 271 (1971).
287. T. Ito, T. Otsu, J. Macromol. Sci. Chem., **3**, 197 (1969).
288. Z. Janovic, C. S. Marvel, M. J. Diamond, D. J. Kertesz, G. Fuller, J. Polymer Sci. A-1, **9**, 2639 (1971).
289. H. K. Johnson, A. Rudin, J. Paint Technol., **42**, 435 (1970).
290. E. F. Jordan, R. Bennett, A. C. Shuman, A. N. Wrigley, J. Polym. Sci. A-1, **8**, 3113 (1970).
291. Y. Kinoshita, J. Kobayashi, F. Ide, K. Nakatsuka, Kobunshi Kagaku, **27**, 469 (1970).
292. Y. Kinoshita, J. Kobayashi, F. Ide, K. Nakatsuka, Kobunshi Kagaku, **28**, 430 (1971).
293. J. C. Lai, T. D. Rounsefell, C. U. Pittman, Macromolecules, **4**, 155 (1971).
294. J. C. Lai, T. Rounsefell, C. U. Pittman, J. Polym. Sci. A-1, **9**, 651 (1971).
295. A. Leoni, S. Franco, Macromolecules, **4**, 355 (1971).
296. A. Matsumoto, T. Ise, M. Oiwa, Nippon Kagaku Kaishi, 209 (1972).
297. A. Matsumoto, H. Muraoka, M. Taniguchi, M. Oiwa, Kogyo Kagaku Zasshi, **74**, 1913 (1971).
298. K. Matsuo, S. Machida, J. Polym. Sci. A-1, **10**, 187 (1972).

299. H. Mitsui, K. Tsuneta, T. Kagiya, Kogyo Kagaku Zasshi, **74**, 1918 (1971).
300. Z. Moniyama, A. Yamamoto, M. Kagiyama, R. Chujo, J. Macromol. Sci. A: Chem., **4**, 1649 (1970).
301. A. F. Nikolaev, L. P. Vishnevetskaya, O. A. Gromova, M. M. Grigor'eva, M. S. Kleshcheva, Vysokomol. Soedin. A, **11**, 2418 (1969).
302. E. Oikawa, K. Yamamoto, Polym. J., **1**, 669 (1970).
303. T. Otsu, H. Inoue, J. Macromol. Sci. A-1, **4**, 35 (1970).
304. T. Otsu, O. Nakatsuka, Kobunshi Kagaku, **29**, 168 (1972).
305. M. Otsuka, K. Matsuoka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **72**, 2505 (1969).
306. M. Otsuka, K. Matsuoka, K. Takemoto, M. Imoto, Kogyo Kagaku Zasshi, **73**, 1062 (1970).
307. T. Ouchi, M. Oiwa, Kogyo Kagaku Zasshi, **72**, 1587 (1969).
308. C. U. Pittman, P. L. Grube, J. Polym. Sci. A-1, **9**, 3175 (1971).
309. M. Ratzch, W. Schneider, D. Musche, J. Polym. Sci. A-1, **9**, 785 (1971).
310. S. H. Ronel, D. H. Kohn, J. Polym. Sci. A-1, **7**, 2209 (1969).
311. R. A. Terteryan, Vysokomol. Soedin., **11**, 1798 (1969).
312. A. V. Ryabov, Yu. D. Semchikov, N. N. Sivanitskaya, Vysokomol. Soedin. A, **12**, 553 (1970).
313. A. V. Ryabov, L. A. Smirnova, V. A. Soldatov, Dokl. Akad. Nauk SSSR, **194**, 1338 (1970).
314. G. Saini, A. Leoni, S. Franco, Makromol. Chem., **144**, 235 (1971).
315. G. Saini, A. Leoni, S. Franco, Makromol. Chem., **146**, 165 (1971).
316. G. Saini, A. Leoni, S. Franco, Makromol. Chem., **147**, 213 (1971).
317. F. F. Shcherbina, O. I. Bakumenko, Vysokomol. Soedin. A, **11**, 772 (1969).
318. Y. Tabata, K. Ishigure, H. Higaki, K. Oshima, J. Macromol. Sci. A: Chem., **4**, 801 (1970).
319. S. Takahashi, K. Takemoto, M. Imoto, Kobunshi Kagaku, **27**, 156 (1970).
320. H. Tamura, M. Tanaka, N. Murata, Kobunshi Kagaku, **27**, 652 (1970).
321. H. Tamura, M. Tanaka, N. Murata, Kobunshi Kagaku, **27**, 736 (1970).
322. S. Tazuke, S. Okamura, J. Polym. Sci. A-1, **7**, 715 (1969).
323. M. Yamada, I. Takase, T. Tsukano, Y. Ueda, N. Koutou, Kobunshi Kagaku, **26**, 593 (1969).
324. V. P. Zubov, L. I. Valuev, V. A. Kabanov, V. A. Kargin, J. Polym. Sci. A-1, **9**, 833 (1971).
325. O. E. Ayers, S. P. McManus, C. U. Pittman, J. Polym. Sci., Polym. Chem. Ed., **11**, 1201 (1973).
326. J. Ballesteros, G. J. Howard, L. Teasdale, J. Macromol. Sci. A: Chem., **11**, 29 (1977).
327. Y. Bando, T. Dodou, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **15**, 1917 (1977).
328. J. M. Barrales-Rienda, J. I. Gonzalez de la Campa, R. J. Gonzalez, J. Macromol. Sci. A: Chem., **11**, 267 (1977).
329. A. R. Cain, Am. Chem. Soc., Div. Polym. Chem. Preprints, **11**, 312 (1970).
330. A. Chapiro, L. Perec-Spitzer, Eur. Polym. J., **11**, 59 (1975).
331. A. Chapiro, P. Lessulnier, Eur. Polym. J., **12**, 761 (1976).
332. T. A. DuPlessis, A. C. Thomas, J. Polym. Sci., Polym. Chem. Ed., **11**, 2681 (1973).
333. N. B. Gaddam, S. F. Xavioir, T. C. Goel, J. Polym. Sci., Polym. Chem. Ed., **15**, 1473 (1977).
334. C. V. Hinton, H. G. Spencer, Macromolecules, **9**, 864 (1976).
335. M. E. Hoaglund, I. N. Duling, Am. Chem. Soc., Petrol. Chem. Preprints, **15**, B85 (1970).
336. H. Inoue, T. Otsu, Makromol. Chem., **153**, 21 (1972).
337. S. Iwabuchi, K. Kojima, T. Nakahira, M. Ueda, M. Kobayashi, M. Iwakura, J. Polym. Sci., Polym. Chem. Ed., **12**, 2801 (1974).
338. S. Iwabuchi, M. Ueda, M. Kobayashi, K. Kojima, Polym. J., **6**, 185 (1974).
339. V. V. Kopeikin, E. F. Panarin, I. S. Milevskaya, N. S. Redi, Vysokomol. Soedin. A, **19**, 861 (1977).
340. Y. Matsubara, M. Yoshihara, T. Maeshima, Nippon Kagaku Kaishi, 2186 (1974).
341. D. R. Abayasekara, R. M. Ottenbrite, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **25**, 164 (1984).
342. F. Mikes, P. Strop, O. Seycek, J. Roda, J. Kalal, Eur. Polym. J., **10**, 1029 (1974).
343. L. M. Minsk, C. Kotlarchik, G. N. Meyer, J. Polym. Sci., Polym. Chem. Ed., **11**, 3037 (1973).
344. J. C. Salmone, P. Taylor, B. Snider, S. C. Israel, Am. Chem. Soc., Polym. Chem. Preprints, **15**, 462 (1974).
345. M. Niwa, Kobunshi Kagaku, **29**, 52 (1972).
346. M. Niwa, S. Iida, Y. Nakazato, Kobunshi Robunshu, **32**, 189 (1975).
347. K. Noma, M. Niwa, K. Morita, Kobunshi Kagaku, **29**, 907 (1972).
348. T. Oishi, T. Kimura, Kobunshi Robunshu, **33**, 685 (1976).
349. S. Okamura, K. Hayashi, Polym. J., **4**, 495 (1973).
350. T. Okano, J. Aoyagi, I. Shinohara, Nippon Kagaku Kaishi, 161 (1976).
351. T. Masuda, C. Aoyama, Kobunshi Kagaku, **30**, 223 (1973).
352. T. Otsu, M. Inoue, Kobunshi Robunshu, **31**, 250 (1974).
353. T. Otsu, H. Tanaka, J. Polym. Sci. Polym. Chem. Ed., **13**, 2605 (1975).
354. C. U. Pittman, G. Marlin, T. D. Rounsefell, Macromolecules, **6**, 1 (1973).
355. G. Sielaff, D. O. Hummel, Makromol. Chem., **175**, 1579 (1974).
356. H. Tanaka, T. Otsu, J. Macromol. Sci. A: Chem., **11**, 231 (1977).
357. A. Kh. Usbekova, V. V. Razumovskii, Vysokomol. Soedin. A, **14**, 1681 (1972).
358. I. K. Varma, S. Patnaik, Eur. Polym. J., **12**, 259 (1976).
359. A. I. Vorob'eva, E. I. Ablyakimov, G. V. Leplyanin, Vysokomol. Soedin. A, **16**, 349 (1974).
360. T. Yala-Lutokanu, G. Jenner, A. Deluzarche, Bull. Soc. Chim. France, 609 (1974).
361. M. Yoshimura, Y. Shirota, H. Mikawa, Macromolecules, **8**, 713 (1975).
362. A. Miller, J. Szafko, E. Turska, J. Polym. Sci., Polym. Chem. Ed., **15**, 51 (1977).

363. A. Miller, J. Szafko, J. Polym. Sci., Polym. Chem. Ed., **15**, 1595 (1977).
364. K. L. Petrak, J. Polym. Sci., Polym. Letters Ed., **16**, 393 (1978).
365. M. Sharabash, R. L. Guile, J. Macromol. Sci. A: Chem., **10**, 1039 (1976).
366. R. H. Wiley, B. Davis, J. Polym. Sci. A, **1**, 2819 (1963).
367. T. Ohata, A. Matsumoto, M. Oiwa, Bull. Chem. Soc. Japan, **47**, 928 (1974).
368. J. Aoyagi, I. Shinohara, Kogyo Kagaku Zasshi, **74**, 1191 (1971).
369. L. V. Morozova, T. T. Minakova, B. A. Trofimov, Deposited Doc., VINITI 1228 (1983).
370. G. Akasome, S. Sakai, Y. Choshi, K. Murai, Kobunshi Kagaku, **17**, 478 (1960).
371. G. Akasome, S. Sakai, Y. Choshi, K. Murai, Kobunshi Kagaku, **17**, 558 (1960).
372. G. Akasome, S. Sakai, K. Murai, Kobunshi Kagaku, **17**, 449 (1960).
373. G. Akasome, S. Sakai, K. Murai, Kobunshi Kagaku, **17**, 452 (1960).
374. G. Akasome, S. Sakai, K. Murai, Kobunshi Kagaku, **17**, 482 (1960).
375. T. Alfrey, L. Arond, C. G. Overberger, quoted by Alfrey, "Copolymerization", p. 35.
376. T. Alfrey, jr., J. Bohrer, H. Mark, "Copolymerization", High Polymer Series, Vol. VIII, Interscience, New York, 1952, p. 40.
377. T. Alfrey, B. Magel, quoted by Alfrey, "Copolymerization", Wiley, New York.
378. American Cyanamid Co., "Chemistry of Acrylonitrile", 2nd Ed., 1960, p. 34.
379. American Cyanamid Co., private communication to L. J. Young.
380. K. Anda, S. Iwai, Kogyo Kagaku Zasshi, **70**, 557 (1967).
381. T. Asahara, K. Mitsuhashi, Yukagaku, **6**, 331 (1957).
382. M. A. Askarov, S. R. Pinkhasov, A. S. Bank, Vysokomol. Soedin. B, **9**, 601 (1967).
383. J. N. Atherton, A. M. North, Trans. Faraday Soc., **58**, 2049 (1962).
384. C. H. Bamford, W. G. Barb, Discussions Faraday Soc., **14**, 208 (1953).
385. A. S. Bank, S. D. Savranskaya, M. A. Askarov, Khim. iFiz.-Khim. Prirodn. iSintetich. Polimerov, Akad. Nauk Uz. SSR, Inst. Khim. Polimerov No. 2, 110 (1964).
386. I. M. Barton, G. B. Butler, E. C. Chapin, J. Polym. Sci. A, **3**, 501 (1965).
387. T. F. Baskova, O. M. Klimova, Vysokomol. Soedin. B, **10**, 63 (1968).
388. L. Bogetich, G. A. Mortimer, G. W. Daues, J. Polym. Sci., **61**, 3, (1962).
389. J. Brandrup, Faserforsch. Textiltech., **12**, 135 (1961).
390. D. Braun, H.-G. Keppler, Makromol. Chem., **82**, 132 (1965).
391. J. W. Breitenbach, H. Edelhauser, Ric. Sci., **25**, 242 (1955).
392. M. A. Bruk, A. D. Abkin, P. M. Khomikovskii, G. A. Gold'er, H.-L. Chu, Dokl. Akad. Nauk SSSR, **157**, 1399 (1964).
393. W. J. Burlant, D. H. Green, J. Polym. Sci., **31**, 227 (1958).
394. G. M. Burnett, J. M. Pearson, J. D. B. Smith, J. Polym. Sci. A-1, **4**, 2024 (1966).
395. G. B. Butler, G. Vanhaeren, M.-F. Ramadier, J. Polym. Sci. A-1, **5**, 1265 (1967).
396. A. Chapiro, A.-M. Jendrychowska-Bonamour, J. Polym. Sci. C, **1**, 1211 (1963).
397. C. B. Chapman, L. Valentine, J. Polym. Sci., **34**, 319 (1959).
398. Chas. Pfizer, Co., Product News, quoted by L. J. Young (1961).
399. K. Chikanishi, T. Tsuruta, Makromol. Chem., **73**, 231 (1964).
400. M. A. Chilingaryan, A. E. Akopyan, M. G. Barkudaryan, Arm. Khim. Zh., **20**, 428 (1967).
401. L. E. DeMillo, G. A. Shtraikhman, E. N. Rostovskii, Zh. Prikl. Khim., **39**, 1360 (1966).
402. K. W. Doak, quoted by Mayo, Walling, 1950.
403. K. W. Doak, M. A. Deahl, I. H. Christmas, 137th ACS Meeting, Cleveland, Ohio, Abstr. Papers, Vol. 1, No. 1, 151 (April, 1960).
404. T. Doi, M. Matsuki, A. Sugiyama, Kogyo Kagaku Zasshi, **70**, 1792 (1967).
405. G. Dolgin, P. Gordon, quoted by Alfrey, "Copolymerization", Wiley, New York, p. 40.
406. DuPont Co., British Patent, **593**, 605 (1947).
407. F. H. C. Edgecombe, Nature, **198**, 1085 (1963).
408. J. Exner, M. Bohdanecky, Chem. Listy, **48**, 483 (1954).
409. V. L. Folt, Canadian Patent, **509**, 259 (1955).
410. S. Fujii, Radiation Res., **33**, 249 (1968).
411. J. Furukawa, T. Tsuruta, N. Yameda, Kogyo Kagaku Zasshi, **61**, 734 (1958).
412. L. H. Gale, J. Org. Chem., **31**, 2475 (1966).
413. A. L. Gol'denberg, S. G. Lubetskii, Dokl. Nauk SSSR, **179**, 900 (1968).
414. M. Goodstein, quoted in Alfrey, "Copolymerization", Wiley, New York, p. 35.
415. V. F. Gromov, P. M. Khomikovskii, A. D. Abkin, Vysokomol. Soedin., **3**, 1015 (1961).
416. A. Guyot, J. Guillot, J. Chem. Phys., **61**, 1434 (1964).
417. K. R. Henery-Logan, R. V. V. Nichlos, quoted by Simha and Wall.
418. K. R. Henery-Logan, R. V. V. Nichlos, quoted by Mayo and Walling.
419. R. Hess, quoted in Alfrey, "Copolymerization", Wiley, New York, p. 36, p. 37.
420. T. Higuchi, H. Imoto, Kogyo Kagaku Zasshi, **61**, 1053 (1958).
421. A. Hunyar, H. Reichert, Faserforsch. Textiltech., **5**, 204 (1954).
422. E. Husemann, Univ. Freiberg i. Br. Germany, private communication to L. J. Young.
423. F. Ida, K. Uemura, S. Abe, Kagaku to Kogyo (Osaka), **38**, 215 (1964).

424. F. Ida, K. Uemura, S. Abe, Kagaku to Kogyo (Osaka), **39**, 565 (1965).
425. F. Ide, K. Okano, S. Nakano, K. Nakstuka, Shikizai Kyokaishi, **40**, 571 (1967).
426. M. Imoto, T. Otsu, Y. Harada, Makromol. Chem., **65**, 180 (1963).
427. M. Imoto, S. Shimizu, Kobunshi Kagaku, **18**, 747 (1961).
428. T. Imoto, Y. Ogo, S. Goto, T. Mitani, Kogyo Kagaku Zasshi, **69**, 1371 (1966).
429. T. Imoto, Y. Ogo, T. Mitani, Kogyo Kagaku Zasshi, **70**, 1217 (1967).
430. T. Irie, M. Kinoshita, M. Imoto, Kogyo Kagaku Zasshi, **69**, 980 (1966).
431. O. A. Iskhakov, E. V. Kuznetzov, G. M. Eliseeva, Vysokomol. Soedin. B, **10**, 32 (1968).
432. K. Ito, S. Iwase, Y. Yamashita, Makromol. Chem., **110**, 233 (1967).
433. Y. Iwakura, K. Matsuzaki, Kobunshi Kagaku, **17**, 187 (1960).
434. Y. Iwakura, M. Sato, T. Tamikado, S. Mimashi, Kobunshi Kagaku, **13**, 125 (1956).
435. Y. Iwakura, M. Sato, T. Tamikado, T. Mizoguchi, Kobunshi Kagaku, **13**, 390 (1956).
436. U. Johnsen, K. Kolbe, Kolloid-Z. Z. Polymer, **216**, 97 (1967).
437. R. M. Joshi, S. L. Kapur, J. Sci. Ind. Res., **16B**, 379 (1957).
438. R. M. Joshi, S. L. Kapur, J. Sci. Ind. Res., **16B**, 441 (1957).
439. N. Kamiya, H. Matsuura, Nippon Gomu Kyokaishi, **40**, 64 (1967).
440. H. Kamogawa, H. G. Cassidy, J. Polym. Sci. A, **1**, 1971 (1963).
441. H. Kamogowa, R. Murase, T. Sekiya, Kogyo Kagaku Zasshi, **62**, 1749 (1959).
442. H. Kamogawa, T. Sekiya, Kogyo Kagaku Zasshi, **62**, 1117 (1959).
443. R. G. Karzhaubaeva, G. P. Gladyshev, S. R. Rafikov, Vysomokol. Soedin. B, **9**, 453 (1967).
444. V. F. Kazanskaya, O. M. Klimova, B. M. Khlebnikov, Vysomokol. Soedin., **6**, 1799 (1964).
445. W. O. Kenyon, J. H. vanCampen, U.S. Patent, 2,419,221 (1947).
446. W. Kern, Univ. Mainz, private communication to L. J. Young.
447. W. Kern, D. Braun, Makromol. Chem., **27**, 23 (1958).
448. A. M. Khomutov, Vysokomol. Soedin., **5**, 1121 (1963).
449. T. Kimura, K. Yoshida, Kagaku to Kogyo (Osaka), **27**, 288 (1953).
450. T. Kimura, K. Yoshida, Kagaku to Kogyo (Osaka), **32**, 223, 341 (1958).
451. J. B. Kinsinger, J. S. Bartlett, N. H. Rauscher, J. Appl. Polym. Sci., **6**, 529 (1962).
452. J. B. Kinsinger, J. R. Panchak, R. L. Kelso, J. S. Bartlett, R. K. Graham, J. Appl. Polym. Sci., **9**, 429 (1965).
453. A. L. Klebanskii, O. A. Timofeev, Zh. Prikl. Khim., **32**, 2294 (1959).
454. N. Kliman, M. Lazar, Chem. Prum., **9**, 668 (1959).
455. A. R. Kol'k, A. A. Konkin, Z. A. Rogovin, Khim. Volokna, 12 (1963).
456. M. M. Koton, J. Polym. Sci., **30**, 331 (1958).
457. M. M. Koton, Inst. Akad. Nauk Latv. SSR Riga, 119 (1957).
458. M. M. Koton, O. K. Sumina, Dokl. Akad. Nauk SSSR, **113**, 1063 (1957).
459. G. E. Krybekyan, E. G. Sinanyan, A. N. Akopyan, Izv. Akad. Nauk Arm. SSR, Khim. Nauki, **15**, 527 (1962).
460. G. N. Larina, Z. V. Borisova, T. V. Sheremeteva, Vysokomol. Soedin., **3**, 1664 (1961).
461. F.-M. Li, Y.-L. Ch'en, L.-C. Wang, H.-C. Wei, L.-H. Yeh, Ko Fen Tzu T'ung Hsun, **8**, 1 (1966).
462. P. T. Li, K.-K. Wu, K'O Hsueh Ch'u Pan She, 151 (1963).
463. L. S. Luskin, R. J. Myers, "Encyclopedia of Polymer Science and Technology", Vol. 1, Interscience Publishers, 1964, p. 246.
464. Z. Machacek, Chem. Listy, **48**, 477 (1954).
465. S. Machida, Y. Shimizu, T. Makita, Kamipa Gikyoshi, **21**, 352 (1967).
466. M. F. Margaritova, G. D. Berezhnov, Tr. Mosk. Khim.-Tekhnol. Inst., **4**, 46 (1953).
467. M. F. Margaritova, V. A. Raiskaya, Tr. Mosk. Khim.-Tekhnol. Inst., **4**, 37 (1953).
468. L. Marker, O. J. Sweeting, J. G. Wepsic, J. Polym. Sci., **57**, 855 (1962).
469. G. Markert, Makromol. Chem., **103**, 109 (1967).
470. F. R. Mayo, F. M. Lewis, C. Walling, J. Am. Chem. Soc., **70**, 1529 (1948).
471. Min. (Szu-Kwei), Chen Ho Chu, Hua Hsueh Hsueh Pao, **23**, 262 (1957).
472. A. Misono, Y. Uchida, K. Yamada, Bull. Chem. Soc. Japan, **40**, 2366 (1967).
473. J. M. Mitchell, H. L. Williams, Can. J. Res., **27**, 35 (1949).
474. A. Mitsutani, M. Yono, Kogyo Kagaku Zasshi, **67**, 935 (1964).
475. E. W. Moffett, R. E. Smith, U.S. Patent 2,356,871 (1944).
476. R. B. Mohite, S. Gundiah, S. L. Kapur, Makromol. Chem., **108**, 52 (1967).
477. G. A. Mortimer, J. Polymer Sci. B, **3**, 343 (1965).
478. D. T. Mowry, U.S. Patent 2,417,607 (1947).
479. S. Murahashi, S. Nozakura, A. Umehara, K. Obata, Kobunshi Kagaku, **21**, 625 (1964).
480. S. Nagai, K. Yoshida, Kobunshi Kagaku, **17**, 77 (1960).
481. T. Nakata, T. Otsu, M. Imoto, J. Polym. Sci. A, **3**, 3383 (1965).
482. N. S. Nametkin, S. G. Durgar'yan, V. G. Filippova, Dokl. Akad. Nauk SSSR, **177**, 853 (1967).
483. K. Noma, M. Niwa, K. Iwasaki, Kobunshi Kagaku, **20**, 646 (1963).
484. S. Okamura, T. Yamashita, J. Soc. Textile Cellulose Ind. Japan, **9**, 446 (1953).
485. V. A. Orlov, O. G. Tarakanov, Plasticheskie Massy, 6 (1964).
486. R. J. Orr, Polymer, **2**, 79 (1961).
487. T. Otsu, T. Ito, M. Imoto, J. Polym. Sci. B, **3**, 113 (1965).
488. T. Otsu, B. Yamada, T. Nozaki, Kogyo Kagaku Zasshi, **70**, 1941 (1967).

489. G. S. Park, private communication to L. J. Young, (1966).
490. N. Platzer, Monsanto Co., private communication to L. J. Young.
491. Polymer Corp. Ltd., Sarnia, Can., private communication to L. J. Young.
492. C. C Price, J. G. Walsh, J. Polym. Sci., **6**, 239 (1951).
493. N. V. Rakityanskii, R. L. Rabinovich, Rept. Allunion Sci. Inst. Synthetic Rubber, 1951.
494. R. C. Reinhardt, Ind. Eng. Chem., **35**, 422 (1943).
495. N. N. Rozovskaya, A. P. Sheinker, A. D. Abkin, Vysokomol. Soedin., **7**, 1383 (1965).
496. N. N. Rozovskaya, A. P. Sheinker, A. D. Abkin, Vysokomol. Soedin., **7**, 1500 (1965).
497. M. Kinoshita, S. Kataoka, M. Imoto, Kogyo Kagaku Zasshi, **72**, 969 (1969).
498. A. V. Ryabov, Yu. D. Semchikov, N. N. Slavnitskaya, Tr. Khim. Khim. Tekhnol., 334 (1963).
499. A. V. Ryabov, Yu. D. Semchikov, V. N. Vakhrusheva, Tr. Khim. Khim. Tekhnol., 188 (1963).
500. I. Sakurada, Kobunshi Kagaku, **18**, 496 (1961).
501. I. Sakurada, T. Okada, S. Hatakeyama, F. Kimura, J. Polym. Sci. C, **1**, 1233 (1963).
502. I. Sakurada, G. Takahashi, Kobunshi Kagaku, **11**, 286 (1954).
503. I. Sakurada, G. Takahashi, H. Mata, Kobunshi Kagaku, **12**, 362 (1955).
504. B. Sandner, J. Ulbricht, Faserforsch. Textiltech., **17**, 286 (1966).
505. A. P. Sheinker, A. D. Abkin, Tr. Tashkentsk Konf. po Mimomu Ispol Z. at. Energii Akad. Nauk Uz. SSR, **1**, 395 (1961).
506. T. V. Sheremeteva, G. N. Larina, Dokl. Akad. Nauk SSSR, **162**, 1323 (1965).
507. M. F. Shostakovskii, E. N. Prilezhaeva, J. M. Karavaeva, Vysokomol. Soedin., **1**, 781 (1959).
508. G. Smets, A. Poot, G. L. Duncan, J. Polym. Sci., **54**, 65 (1961).
509. G. Takashi, Kobunshi Kagaku, **14**, 151 (1957).
510. G. Talamini, G. Vidotto, C. Garbuglio, Chim. Ind. (Milan), **47**, 955 (1965).
511. A. Tanaka, Y. Hozumi, K. Hatada, Kobunshi Kagaku, **22**, 216 (1965).
512. M. Tanaka, A. Asai, S. Takeya, Kogyo Kagaku Zasshi, **62**, 1786 (1959).
513. H. Tatemichi, S. Suzuki, Kogyo Kagaku Zasshi, **63**, 1843 (1960).
514. S. Tazuke, N. Sato, S. Okamura, J. Polym. Sci. A-1, **4**, 2461 (1966).
515. R. W. Tess, W. T. Tsatsos, A. Chem. Soc., Div. Org. Coatings Plast. Chem., Preprints, **26**, 276 (1966).
516. G. V. Tkachenko, V. S. Etlis, L. V. Stupen, L. P. Kofman, Zh. Fiz. Khim, **33**, 25 (1959).
517. G. V. Tkachenko, P. H. Khomikovskii, A. D. Abkin, S. S. Medvedev, Zh. Fiz. Khim., **31**, 242 (1957).
518. G. V. Tkachenko, L. V. Stupen, L. P. Kofman, L. Z. Frolova, Zh. Fiz. Khim., **31**, 2676 (1957).
519. G. V. Tkachenko, L. V. Stupen, L. P. Kofman, L. A. Karacheva, Zh. Fiz. Khim., **32**, 2492 (1958).
520. V. L. Tsailingol'd, M. I. Farberov, G. A. Burgova, Vysokomol. Soedin., **1**, 415 (1959).
521. E. Tsuchida, T. Kawagoe, Enka Biniiru to Porima, **8**, 21 (1968).
522. E. Tsuchida, Y. Ohtani, H. Nakadai, I. Shinohara, Kogyo Kagaku Zasshi, **70**, 573 (1967).
523. E. Tsuchida, Z. Okuno, T. Yao, I. Shinohara, Kogyo Kagaku Zasshi, **69**, 1230 (1966).
524. E. Tsuchida, T. Shimomura, K. Fujimora, Y. Ohtani, I. Shinohara, Kogyo Kagaku Zasshi, **70**, 1230 (1966).
525. K. Ueberreiter, W. Krull, Z. Physik. Chem. (Frankfort), **12**, 303 (1957).
526. S. N. Ushakov, E. M. Lavrent'eva, Zh. Prikl. Khim., **31**, 1686 (1958).
527. Kh. U. Usmanov, R. S. Tillaev, U. N. Musaev, M. M. Nauchn. Tr., Tashkenysk. Gos. Univ. No. 257, 30 (1964).
528. Kh. U. Usmanov, A. A. Yul'chibaev, Kh. Yuldasheva, Uzb. Khim. Zh., **11**, 27 (1967).
529. Kh. U. Usmanov, A. A. Yul'chibaev, Kh. Yuldasheva, Uzb. Khim. Zh., **11**, 41 (1967).
530. J. W. Vanderhoff in E. H. Riddle, "Monomeric Acrylic Esters", Reinhold, New York, 1954, p. 94, Ref. 154.
531. F. T. Wall, quoted by F. R. Mayo and C. Walling.
532. C. Walling, E. R. Briggs, K. B. Wolfstirn, J. Am. Chem. Soc., **70**, 1543 (1948).
533. G. Welzel, G. Greber, Makromol. Chem., **31**, 230 (1959).
534. R. H. Wiley, B. Davis, J. Polym. Sci., **62**, S-132 (1962).
535. R. H. Wiley, L. K. Heidemann, B. Davis, J. Polym. Sci. B, **1**, 521 (1963).
536. C. E. Wilkes, J. C. Westfahl, R. H. Backderf, Am. Chem. Soc., Div. Polym. Chem., Preprints, **8**, 386 (1967).
537. L. P. Witnauer, E. Watkins, W. S. Port, J. Polym. Sci., **20**, 213 (1956).
538. D. M. Woodford, Chem. Ind. (London), 1966, 316.
539. N. Yamazaki, S. Kambara, J. Polym. Sci. C, **22**, 75 (1968).
540. K. Yokota, Y. Ishii, Kogyo Kagaku Zasshi, **69**, 1057 (1966).
541. K. Yokota, M. Kani, Y. Ishii, J. Polym. Sci. A-1, **6**, 1325 (1968).
542. L. J. Young, unpublished data, quoted in J. Polym. Sci., **54**, 411 (1961).
543. N. L. Zutty, R. D. Burkhart, Advances in Chem. Series No, **34**, "Polymerization and Condensation Processes", ACS, Washington D. C., 1962, p. 52.
544. M. P. Zverev, M. F. Margaritova, Ukr. Khim. Zh., **24**, 626 (1958).
545. D. D. Al-Khashimi, T. R. Abdurashidov, M. A. Askarov, Uzb. Khim. Zh., **14**, 87 (1970).
546. A. B. Alovitdinov, A. B. Kuchkarov, A. I. Kurbanov, Vysokomol. Soedin. B, **13**, 657 (1971).
547. K. Anda, S. Iwai, J. Polym. Sci. A-1, **7**, 2414 (1969).
548. K. Anda, S. Iwai, Kobunshi Kagaku, **29**, 212 (1972).
549. J. Aoyagi, K. Kitamura, I. Shinohara, Kogyo Kagaku Zasshi, **73**, 2045 (1970).
550. J. Ushirone, T. Otsu, Kogyo Kagaku Zasshi, **71**, 772 (1968).
551. A. M. North, K. E. Whitelock, Polymer, **9**, 590 (1968).

552. B. K. Basov, V. L. Tsailingol'd, E. G. Lazaryants, Prom. Sin. Kauch., 17 (1967).
553. J. M. Carcamo, F. Arranz, Rev. Plast. Mod., **22**, (182), 1169 (1971).
554. A. V. Chernobai, Zh. Kh. Zelichenko, Vysokomol. Soedin. A, **11**, 1470 (1969).
555. K. H. Chung, Daehan Hwahak Hwoejee, **14**, 333 (1970).
556. G. I. Deryabina, A. K. Khaliullin, S. Yu. Federova, L. A. Kron, Izv. Nauch.-Issled. Inst. Nefte-Uglekhim. Sin. Irkutsk. Univ., **12**, 57 (1970).
557. M. Z. El-Sabban, S. I. Dmitrieva, A. I. Meos, Vysokomol. Soedin. B, **12**, 243 (1970).
558. J. W. H. Faber, W. F. Fowler, J. Polym. Sci. A-1, **8**, 1777 (1970).
559. A. L. German, D. Heikens, J. Polym. Sci. A-1, **9**, 2225 (1971).
560. K. Hamanoue, Rev. Phys. Chem. Japan, **38**, 120 (1968).
561. G. W. Hastings, J. Chem. Soc. D, 1039, (1969).
562. K. Horie, I. Mita, H. Kambe, J. Polym. Sci. A-1, **7**, 2561 (1969).
563. A. S. Kabankin, S. A. Balabanova, A. M. Markevich, Vysokomol. Soedin. A, **12**, 267 (1970).
564. D. A. Kangas, R. R. Pelletier, J. Polym. Sci. A-1, **8**, 3543 (1970).
565. R. G. Karzhaubaeva, G. M. L'dokova, G. P. Gladyshev, S. R. Rafikov, Tr. Inst. Khim. Nauk Akad. Nauk Kaz. SSR, **28**, 115 (1970).
566. M. Kinoshita, T. Irie, M. Imoto, Kogyo Kagaku Zasshi, **72**, 1210 (1969).
567. K. Kitanaka, Y. Tabata, Kobunshi Kagaku, **28**, 206 (1971).
568. N. G. Kulkami, N. Krishnamurti, P. C. Chatterjee, M. A. Sivasamban, Makromol. Chem., **139**, 165 (1970).
569. S. Machida, K. Matsuo, Yukei Gosei Kagaku Kyokoi Shi, **27**, 759 (1969).
570. K. Moser, R. Signer, H. U. Stuber, Chimia, **23**, 393 (1969).
571. S. U. Mullik, M. A. Quddus, Pak. J. Sci., Ind. Res., **12**, 186 (1970).
572. S. U. Mullik, M. A. Quddus, Pak. J. Sci., Ind. Res., **12**, 181 (1970).
573. N. S. Nametkin, I. N. Kozhukhova, S. G. Durgar'yan, V. G. Filippova, Vysokomol. Soedin. A, **11**, 2523 (1969).
574. K. Noma, M. Niwa, Kobunshi Kagaku, **29**, 52 (1972).
575. K. Noma, M. Niwa, H. Norisada, Doshisha Daigaku Rikogaku Kenkyu Hokoku, **10**, 349 (1970).
576. C. U. Pittman, N. C. Lai, D. P. Vanderpool, Macromolecules, **3**, 105 (1970).
577. G. P. Popova, T. R. Kirpichenko, K. L. Glazomitskii, B. E. Glo'tsin, E. S. Roskin, E. N. Rostovskii, Izv. Vyssh. Ucheb. Zaved., Khim. Khim. Tekhnol., **13**, 259 (1970).
578. M. Ratzsch, L. Stephan, Plaste Kautschuk, **18**, 572 (1971).
579. E. A. Rassolova, M. A. Zharkova, G. I. Kudryavtsev, V. S. Klimenkov, Khim. Volokna, 50 (1969).
580. S. A. Rostovtseva, Vysokomol. Soedin. B, **12**, 588 (1970).
581. A. G. Sayadyan, D. A. Simonyan, Arm. Khim. Zh., **22**, 528 (1969).
582. Yu. L. Spirin, T. S. Yatsmirskaya, Vysokomol. Soedin. B, **11**, 515 (1969).
583. J. K. Stille, L. D. Gotter, Macromolecules, **2**, 468 (1969).
584. R. A. Tterteryan, N. V. Fomicheva, V. N. Monastyrskii, Vysokomol. Soedin. B, **13**, 485 (1971).
585. Y. Yamashita, K. Ito, H. Ishii, S. Hoshino, M. Kai, Macromolecules, **1**, 529 (1968).
586. K. Yokota, J. Macromol. Sci. A: Chem., **4**, 65 (1970).
587. A. B. Alovitdinov, A. I. Kurbanov, A. B. Kuchkarov, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **17**, 1443, (1974).
588. A. B. Alovitdinov, A. I. Kurbanov, A. B. Kuchkarov, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **18**, 160 (1975).
589. L. Ambroz, J. Caskova, M. Jelinek, J. Majer, Chem. Prum., **26**, 27 (1976).
590. A. B. Alovitdinov, Kh. U. Kochkarova, D. K. Khamdamova, A. B. Kuchkarov, Tr. Tashk. Politekh. Inst., **135**, 12 (1974).
591. K. Anda, S. Iwai, Kobunshi Kagaku, **29**, 212 (1972).
592. N. Cobianu, C. Boghina, B. Marculescu, C. Cincu, I. Amalinei, Rev. Roum. Chim., **19**, 1251 (1974).
593. A. T. Dzhalilov, O. M. Yariev, N. N. Yadgarov, Uzb. Khim. Zh., 29, (1976).
594. G. B. Fridman, Ya. A. Levin, B. E. Ivanov, Sb. Nekot. Probl. Org. Khim., Mater. Nauch. Sess., Inst. Org. Fiz. Khim., Akad. Nauk SSSR, 89, (1972).
595. J.-I. Jin. H.-S. Byun, Y.-M. Park, Taehan Hwahak Hoechi, **21**, 219 (1977).
596. T. G. Kulagina, M. A. Askarov, A. S. Bank, Sin. Vysokomol. Soedin., 9, (1972).
597. A. I. Kurbanov, A. B. Kucharov, A. B. Alovitdinov, Tr. Tashk. Politekh. Inst., **90**, 133 (1972).
598. Z. Laita, O. Paleta, A. Posta, F. Liska, Collect. Czech. Chem. Commun., **40**, 2059 (1975).
599. Sh. Nadzhimutdinov, A. S. Turnev, A. Bazarbaev, Kh. U. Usmanov, Vysokomol. Soedin. B, **15**, 574 (1973).
600. A. F. Nikolaev, V. Bondarenko, L. E. Klubikova, S. A. Golenishcheva, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **19**, 1141 (1976).
601. T. Otsu, I. Umeda, H. Inoue, Mem. Fac. Eng., Osaka City Univ., **16**, 107 (1975).
602. V. S. Romanova, M. A. Bulatov, V. V. Sinitsyn, Deposited Doc., 1975, VINITI 2407.
603. A. G. Savadyan, O. A. Dzhanikyan, V. A. Mirzoyan, Arm. Khim. Zh., **29**, 708 (1976).
604. A. G. Sayadyan, O. A. Dzhanikyan, M. G. Ketikyan, Arm. Khim. Zh., **30**, 433 (1977).
605. E. M. Shaikhutdinov, B. A. Zhubanov, N. Kadyrsizov, A. B. Sagyndykova, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **25**, 55 (1974).
606. Kh. U. Usmanov, A. A. Yul'chibaev, A. A. Mat'yakubov, Kh. Kuzieva, S. Kazakov, Dokl. Akad. Nauk Uzb. SSR, **29**, 37 (1972).
607. Kh. U. Usmanov, Kh. Kuzieva, A. A. Yul'chibaev, M. Zh. Akimdzhanova, Izv. Vyssh. Ucheb. Zaved., Khim. Khim. Tekhnol., **16**, 948 (1973).
608. T. Oishi, S. Maruyama, M. Momoi, M. Fujimoto, T. Kimura, Makromol. Chem., **185**, 479 (1984).
609. N. Yamashita, K. Ikezawa, S. Ayukawa, T. Toshihisa, J. Macromol. Sci. A: Chem., **21**, 621 (1984).

610. E. L. Madruga, J. San Roman, M. A. Lavia, J. Polym. Sci., Poly. Chem. Ed., **22**, 801 (1984).
611. Kh. Budevska, V. Sinigerski, Makromol. Chem., **185**, 531 (1984).
612. G. Kaszas, T. Foldes-Berezsnich, F. Tudos, Eur. Polym. J., **20**, 395 (1984).
613. H. Boudevska, O. Todorova, Makromol. Chem., **185**, 353 (1984).
614. J. Tanaka, A. Yamada, J. Macromol. Sci. A: Chem., **21**, 253 (1984).
615. R. A. Terteryan, S. D. Livshits, Sb. Nauch. Tr.-Vses. Nauchno-Issled. Inst. Pererab. Nefti, **41**, 118 (1982).
616. V. A. Kruglova, V. V. Annenkov, I. V. Zaitseva, A. V. Kalabina, A. N. Mirskova, Vysokomol. Soedin. B, **25**, 852 (1983).
617. M. Ueda, T. Kumakura, Y. Imai, C. U. Pittman, E. Wallace, J. Polym. Sci., Polym. Chem. Ed., **22**, 85 (1984).
618. Y. K. Sung, S. Y. Lee, J. O. Pollimo, **7**, 392 (1983).
619. J. Berger, H. Zweifel, Angew. Makromol. Chem., **115**, 163 (1983).
620. R. A. Terteryan, V. S. Khrapov, Vysokomol. Soedin. A, **25**, 1850 (1983).
621. J. San Roman, E. L. Madruga, M. A. Del Puerto, J. Polym. Sci., Polym. Chem. Ed., **21**, 3303 (1983).
622. B. S. R. Reddy, R. Arshady, M. H. George, Macromolecules, **16**, 1813 (1983).
623. G. Yuan, Huaxue Xuebao, **41**, 746 (1983).
624. T. Oishi, M. Momoi, M. Fujimoto, T. Kimura, J. Macromol. Sci. A: Chem., **20**, 763 (1983).
625. M. Ueda, T. Suzuki, Y. Imai, C. U. Pittman, E. Wallace, J. Polym. Sci., Polym. Chem. Ed., **21**, 2997 (1983).
626. A. A. Zhdanov, B. G. Zavin, O. G. Blokhina, Vysokomol. Soedin. B, **25**, 582 (1983).
627. T. L. Latypov, A. A. Yul'chibaev, Kh. Yu. Kuzieva, Uzb. Khim. Zh., 62 (1983).
628. J. D. Patel, M. R. Patel, J. Polym. Sci., Polym. Chem. Ed., **21**, 3027 (1983).
629. W. S. Kim, O. R. Kwan, I. K. Kang, U. R. Cho, Pollimo, **7**, 242 (1983).
630. Y. Kihira, K. Moriwaki, H. Nishio, H. Yamamura, Kinki Daigaku Kogakubu Kenkyu Hokoku, **16**, 1 (1982).
631. J. P. Fishcer, S. Roesinger, Makromol. Chem., **184**, 1247 (1983).
632. G. Kaszas, T. Foldes-Berezsnich, F. Tudos, Magy. Kem. Foly., **89**, 154 (1983).
633. C. Samyn, G. Smets, Makromol. Chem., **184**, 949 (1983).
634. J. J. Uebel, F. J. Dinan, J. Polym. Sci., Polym. Chem. Ed., **21**, 1773 (1983).
635. V. Konsulov, Yu. Piruleva, Vysokomol. Soedin. B, **25**, 307 (1983).
636. R. N. Majumdar, S. L. Yang, H. J. Harwood, J. Polym. Sci., Polym. Chem. Ed., **21**, 1717 (1983).
637. V. S. Kurganskii, V. A. Puchin, S. A. Voronov, V. S. Tokarev, Vysokomol. Soedin. A, **25**, 997 (1983).
638. B. Yamada, T. Hayashi, T. Otsu, J. Macromol. Sci. A: Chem., **19**, 1023 (1983).
639. J. C. Brosse, J. M. Gauthier, J. C. Lenain, Makromol. Chem., **184**, 505 (1983).
640. N. Toyoda, M. Yoshida and T. Otsu, Polym. J. (Tokyo), **15**, 255 (1983).
641. M. Ueda, M. Takahashi, T. Suzuki, Y. Imai, C. U. Pittman, J. Polym. Sci., Polym. Chem. Ed., **21**, 1139 (1983).
642. T. Oishi, M. Momoi, M. Fujimoto, T. Kimura, J. Polym. Sci., Polym. Chem. Ed., **21**, 1139 (1983).
643. K. Saric, Z. Janovic, O. Vogl, J. Macromol. Sci. A: Chem., **19**, 837 (1983).
644. J. D. Patel, M. R. Patel, J. Macromol. Sci. A: Chem., **19**, 801 (1983).
645. G. Galli, R. solaro, E. Chiellini, A. Ledwith, Macromolecules, **16**, 497 (1983).
646. Z. Wojtczak, T. Czerniawski, B. Rozwadowska, Acta Polym., **34**, 125 (1983).
647. V. A. Kruglova, N. I. Skobeeva, A. A. Antonovich, L. M. Dobrynina, A. V. Kalabina, A. N. Mirskova, Deposited Doc., SPSTL 744, Khp-D81 (1981).
648. C. I. Simionescu, B. C. Simionescu, C. Mihailescu, Acta Polym., **34**, 62 (1983).
649. A. C. Constantinescu, S. Dumitriu, C. I. Simionescu, Acta Polym., **34**, 60 (1983).
650. J. I. Jin, H. K. Shim, S. M. Lee, Taehan Hwahakhoe Chi, **26**, 421 (1982).
651. F. Hrabak, M. Bezdek, V. Hynkova, J. Svobodova, Makromol. Chem., **183**, 2675 (1982).
652. L. Lukasik, W. Kuran, J. Kiryjow, J. Polym. Sci., Polym. Lett. Ed., **21**, 17 (1983).
653. J. Asakura, M. Yoshihara, H. Fujihara, Y. Matsubara, T. Maeshima, J. Macromol. Sci. A: Chem., **19**, 311 (1983).
654. E. Dumitriu, S. Oprea, G. Datcu, Mater. Plast. (Bucharest), **19**, 141 (1982).
655. M. Tomescu, S. F. Cretu, Bul. Inst. Politeh. "Gheorghe Gheorghiu-Dej" Bucuresti, Ser. Chim.-Metal., **44**, 67 (1982).
656. Kh. Budevska, S. Plachkova, O. Todorova, Makromol. Chem., **183**, 2583 (1982).
657. M. Niwa, N. Higashi, K. Noma, Sci. Eng. Rev. Doshisha Univ., **23**, 87 (1982).
658. T. Oishi, M. Fujimoto, T. Kimura, J. Polym. Sci., Polym. Chem. Ed., **20**, 2999 (1982).
659. G. Ayrey, M. J. Humphrey, R. C. Poller, Eur. Polym. J., **18**, 693 (1982).
660. M. Ueda, T. Kumakura, Y. Imai, C. U. Pittman, J. Polym. Sci., Polym. Chem. Ed., **20**, 2829 (1982).
661. H. I. Park, C. K. Lee, J. H. Choi, J. I. Jin, Taehan Hwahakhoe Chi, **26**, 235 (1982).
662. G. N. Babu, A. Deshpande, D. D. Deshpande, Angew. Makromol. Chem., **105**, 83 (1982).
663. M. Ueda, M. Takahashi, Y. Imai, C. U. Pittman, J. Polym. Sci., Polym. Chem. Ed., **20**, 2819 (1982).
664. H. Feng, X. Zhou, W. Liang, Q. Lu, R. Zhao, Gaofenzi Tongxun, 126, (1982).
665. A. Matsumoto, H. Kamigaki, M. Oiwa, J. Polym. Sci., Polym. Chem. Ed., **20**, 2611 (1982).
666. H. G. Woo, S. K. Choi, Taehan Hwahakhoe, **26**, 179 (1982).

667. R. N. Majumdar, C. Carlini, C. Bertucci, Makromol. Chem., **183**, 2047 (1982).
668. A. G. Ismailov, S. G. Karaeva, N. Sh. Rasulov, Azerb. Khim. Zh., 66, (1982).
669. S. G. Khodzhaev, F. Z. Yusupbekova, A. A. Yul'chibaev, Sb. Nauchn. Tr.-Tashk. Gos. Univ. im V. I. Lenina, **667**, 34 (1981).
670. G. Wouters, G. Smets, Makromol. Chem., **183**, 1861 (1982).
671. A. V. Agasaryan, M. L. Eritsyan, A. O. Marukyan, Arm. Khim. Zh., **35**, 329 (1982).
672. J. San Roman, E. L. Madruga, Eur. Polym. J., **18**, 481 (1982).
673. A. Fehervari, T. Foldes-Berezsnich, F. Tudos, J. Macromol. Sci. A: Chem., **18**, 337 (1982).
674. P. K. Dhal, M. S. Ramakrishna, G. N. Babu, J. Polym. Sci., Polym. Chem. Ed., **20**, 1581 (1982).
675. G. Meunier, P. Hemery, S. Boileau, Polymer, **23**, 855 (1982).
676. M. L. Eritsyan, A. V. Agasaryan, Arm. Khim. Zh., **35**, 265 (1982).
677. A. Fehervari, T. Foldes-Berezsnich, F. Tudos, Magy. Kem. Foly., **88**, 226 (1982).
678. T. Oishi, M. Fujimoto, T. Kimura, Polym. J. (Tokyo), **14**, 323 (1982).
679. O. Sosanwo, J. Lokaj, F. Hrabak, Eur. Polym. J., **18**, 341 (1982).
680. S. Yu Zaitsev, A. E. Batog, A. V. Bondarenko, V. V. Zaitseva, Vysokomol. Soedin. A, **24**, 778 (1982).
681. H. Ito, D. C. Miller, C. G. Willson, Macromolecules, **15**, 915 (1982).
682. O. I. Sorokina, M. L. Syrkina, G. A. Kiselev, Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol., **25**, 227 (1982).
683. W. S. Kim, D. H. Oh, O. R. Kwon, I. K. Kang, K. H. Seo, Pollimo, **6**, 113 (1982).
684. Ya. A. Levin, G. B. Fridman, V. Sh. Gurskaya, L. Kh. Gazizova, S. V. Shulyndin, B. E. Ivanov, Vysokomol. Soedin. A, **24**, 601 (1982).
685. L. P. Paskal, V. G. Syromyatnikov, Ukr. Khim. Zh. (Russ. Ed.), **48**, 204 (1982).
686. M. Macret, G. Hild, Polymer, **23**, 81 (1982).
687. C. Hu, H. Feng, D. Jiang, Gaofenzi Tongxun, 275, (1981).
688. A. K. Askerov, A. D. Aliev, G. R. Slamanovva, A. A. Ibragimov, Mater. Sumgaitskoi Gor. Nauchno-Tekh. Konf. Probl. "Org. Khlororg. Sint.", 1980, p. 45.
689. K. Qiu, Y. Sun, X. Feng, Gaofenzi Tongxun, 1981, 373.
690. S. Pitchumani, R. C. Rami, S. Rajadurai, J. Polym. Sci., Polym. Chem. Ed., **20**, 277 (1982).
691. M. Orbay, R. Laible, L. Dulog, Makromol. Chem., **183**, 47 (1982).
692. G. N. Babu, A. Deshpande, J. Macromol. Sci., Chem., **A17**, 717 (1982).
693. S. G. Bondarenko, A. F. Nikolaev, S. A. Baranova, N. I. Plyashechnik, G. A. Smirnova, S. V. Obukhova, I. V. Baidenok, E. M. Stepanov, I. N. Gluschenok, E. D. Andreeva, Vysokomol. Soedin. A, **23**, 2639 (1981).
694. M. K. Abdullaeva, K. M. Muminov, M. A. Askarov, A. T. Dzhalilov, Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol., **24**, 1022 (1981).
695. S. Kohjiya, H. Takeuchi, K. Kawamoto, S. Yamashita, Bull. Chem. Soc. Jpn., **54**, 3245 (1981).
696. B. Plavljanic, Z. Janovic, J. Polym. Sci., Polym. Chem. Ed., **19**, 1795 (1981).
697. R. Haraoubia, C. Bonnans-Plaisance, G. Levesque, Makromol. Chem., **182**, 2409 (1981).
698. D. Jiang, C. Hu, G. Yuan, Sci. Sin. (Engl. Ed.), **24**, 1248 (1981).
699. V. I. Gulimov, T. P. Vishnyakova, I. D. Vlasova, T. Z. Tabasaranskaya, Vysokomol. Soedin. B, **23**, 633 (1981).
700. K. Arai, Y. Ogiwara, Makromol. Chem., Rapid Commun., **2**, 363 (1981).
701. V. Percec, A. Natansohn, C. I. Simionescu, Polym. Bull. (Berlin), **4**, 255 (1981).
702. V. Percec, A. Natansohn, C. I. Simionescu, Polym. Bull. (Berlin), **4**, 247 (1981).
703. C. I. Simionescu, V. Percec, A. Natansohn, Polym. Bull., **4**, 623 (1981).
704. D.-Z. Jiang, C. Hu, G.-Q. Yuan, Proc. China-U.S. Bilateral Symp. Polym. Chem. Phys., 1979, 138. Sci. Press: Peking, Peop. Rep. China.
705. G. Meunier, P. Hemery, J. P. Senet, S. Boileau, Polym. Bull. (Berlin), **4**, 705 (1981).
706. G. Meunier, P. Hemery, J. P. Senet, S. Boileau, Polym. Bull. (Berlin), **4**, 699 (1981).
707. B. C. Simionescu, A. Natansohn, M. Leanca, C. Ananiescu, C. I. Simionescu, Polym. Bull. (Berlin), **4**, 569 (1981).
708. G. N. Babu, A. Deshpande, J. Macromol. Sci. A: Chem., **16**, 1299 (1981).
709. T.-C. Chiang, K.-H. Yu, Polym. Bull. (Berlin), **4**, 425 (1981).
710. J. K. Fink, Makromol. Chem., **182**, 2105 (1981).
711. M. Niwa, M. Kobayashi, T. Matsumoto, Kobunshi Robunshu, **38**, 413 (1981).
712. M. Ueda, K. Iri, Y. Imai, C. U. Pittman, Macromolecules, **14**, 1046 (1981).
713. K. Urushido, A. Matsumoto, M. Oiwa, J. Polym. Sci., Polym. Chem. Ed., **19**, 245 (1981).
714. T. Oishi, Polym. J. (Tokyo), **13**, 65 (1981).
715. C. I. Simionescu, V. Percec, A. Natansohn, Polym. Bull. (Berlin), **3**, 551 (1980).
716. C. I. Simionescu, V. Percec, A. Natansohn, Polym. Bull. (Berlin), **3**, 529 (1980).
717. K. Hayashi, M. Tachibana, Y. Tanaka, S. Okamura, J. Polym. Sci., Polym. Chem. Ed., **19**, 1571 (1981).
718. J. L. De la Pena, J. Guzman, An. Quim. C, **76**, 245 (1980).
719. C. I. Simionescu, V. Percec, A. Natansohn, Polym. Bull. (Berlin), **3**, 535 (1980).
720. Kh. Budevska, S. Plachkova, Makromol. Chem., **182**, 1119 (1981).
721. T. Shimidzu, M. Yoshikawa, B. Ohtani, Macromolecules, **14**, 506 (1981).
722. A. K. Askerov, A. D. Aliev, G. R. Salmanova, A. A. Ibragimov, Azerb. Khim. Zh., 65, (1980).

723. M. Kamachi, R. Miwa, S. Nozakura, Polym. J. (Tokyo), **12**, 899 (1980).
724. V. Konsulov, Yu. Piruleva, V. Mircheva, I. Mladenov, Dokl. bolg. Akad. Nauk, **33**, 807 (1980).
725. A. I. Shepeleva, V. D. Bezuglyi, Vysokomol. Soedin. A, **23**, 121 (1981).
726. K. Noma, N. Higashi, M. Kobayashi, M. Niwa, Sci. Eng. Rev. Doshisha Univ., **21**, 141 (1980).
727. J. Asakura, M. Yoshihara, Y. Matsubara, T. Maeshima, J. Macromol. Sci. A: Chem., **15**, 1473 (1981).
728. T. Oishi, Polym. J. (Tokyo), **12**, 799 (1980).
729. T. Oishi, Polym. J. (Tokyo), **12**, 719 (1980).
730. S. Oprea, E. Dumitriu, Bul. Inst. Politeh. Iasi, Sec. 2: Chim. Ing. Chim., **26**, 71 (1980).
731. C. I. Simionescu, A. Natansohn, V. Percec, J. Macromol. Sci. A: Chem., **15**, 659 (1981).
732. C. I. Simionescu, A. Natansohn, V. Percec, J. Macromol. Sci. A: Chem., **15**, 643 (1981).
733. M. K. Shaikhova, K. M. Muminov, A. T. Dzhalilov, M. A. Askarov, Uzb. Khim. Zh., 48, (1980).
734. O. Sh. Kurmanaliev, E. M. Shaikhutdinov, Sh. S. Tul'baev, T. M. Mukhametkaliev, Vysokomol. Soedin. B, **22**, 526 (1980).
735. F. Borg-Visse, F. Dawans, E. Marechal, J. Polym. Sci., Polym. Chem. Ed., **18**, 2481 (1980).
736. V. Percec, A. Natansohn, V. Barboiu, B. C. Simionescu, D. Galea, Polym. Bull. (Berlin), **2**, 505 (1980).
737. H. Fujihara, T. Shindo, M. Yoshihara, T. Maeshima, J. Macromol. Sci. A: Chem., **14**, 1029 (1980).
738. E. Chiellini, G. Galli, R. Solaro, A. Ledwith, Conv. Ital. Sci. Macromol., {Atti}, 4th 85. Univ. Genova, 1st Chim. Ind.: Genoa, Italy.
739. H. Fujihara, M. Yoshihara, T. Maeshima, J. Macromol. Sci. A: Chem., **14**, 867 (1980).
740. T. Oishi, I. Seo, T. Kimura, Technol. Rep. Yamaguchi Univ., **2**, 199 (1979).
741. M. A. Askarov, E. N. Shakirova, S. Masharipov, A. S. Bank, Kh. D. Islomov, Vysokomol. Soedin. B, **22**, 68 (1980).
742. Sh. S. Vezirov, S. M. Aliev, N. F. Shakhmamedova, E. S. Zeinalova, Vysokomol. Soedin. B, **22**, 22 (1980).
743. V. V. Rabotnov, L. I. Batik, B. K. Basov, V. A. Lysanov, V. A. Kotov, Prom-st. Sint. Kauch., 1979, 5.
744. C. I. Simionescu, V. Percec, A. Natansohn, Polym. Bull. (Berlin), **2**, 57 (1980).
745. H. Boudevska, C. Brutchkov, V. Veltchev, Eur. Polym. J., **15**, 907 (1979).
746. M. K. Akkapeddi, Polymer, **20**, 1215 (1979).
747. F. Hrabak, J. Lokaj, Eur. Polym. J., **15**, 701 (1979).
748. Y. Bando, Y. Minoura, Eur. Polym. J., **15**, 333 (1979).
749. A. Natansohn, S. Maxim, D. Feldman, Polymer, **20**, 629 (1979).
750. M. Kurokawa, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **17**, 3297 (1979).
751. E. L. Madruga, J. San Roman, M. A. Del Puerto, J. Macromol. Sci., Chem., **13** 1105 (1979).
752. E. L. Madruga, J. San Roman, J. Guzman, J. Macromol. Sci., Chem., **13** 1089 (1979).
753. H. Fujihara, K. Yamazaki, Y. Matsubara, M. Yoshihara, T. Maeshima, J. Macromol. Sci., Chem., **13**, 1081 (1979).
754. P. Bajaj, D. C. Gupta, Eur. Polym. J., **15**, 271 (1979).
755. D. Raucher, M. Levy, J. Polym. Sci., Polym. Chem. Ed., **17**, 2663 (1979).
756. S. Kondo, T. Ohtsuka, K. Tsuda, J. Macromol. Sci., Chem., **13**, 767 (1979).
757. J. L. Acosta, R. Sastre, J. A. Penaranda, Rev. Plast. Mod., **37**, 709 (1979).
758. A. Ledwith, G. Galli, E. Chiellini, R. Solaro, Polym. Bull. (Berlin), **1**, 491 (1979).
759. S. V. Shulyndin, V. A. Byl'ev, E. F. Gubanov, Kh. G. Sanatullin, E. P. Dikolenko, T. G. Valeeva, B. E. Ivanov, Vysokomol. Soedin. A, **21**, 985 (1979).
760. M. Sorm, S. Nespurek, Eur. Polym. J., **14**, 977 (1978).
761. H. Yuki, Y. Okamoto, Y. Shimada, K. Ohta, K. Hatada, J. Polym. Sci., Polym. Chem. Ed., **17**, 1215 (1979).
762. N. N. Bazhenova, B. A. Egorov, Yu. D. Semchikov, Dokl. Akad. Nauk SSSR, **245**, 621 (1979).
763. H. Ohnishi, T. Otsu, J. Macromol. Sci. A: Chem., **12**, 1491 (1978).
764. T. Otsu, H. Ohnishi, J. Macromol. Sci. A: Chem., **12**, 1477 (1978).
765. J. G. DuPont, C. W. Allen, Macromolecules, **12**, 169 (1979).
766. R. Van der Meer, A. L. German, J. Polym. Sci., Polym. Chem. Ed., **17**, 571 (1979).
767. R. Sastre, J. L. Mateo, J. Acosta, Angew. Makromol. Chem., **73**, 25 (1978).
768. M. Raetzsch, H. Lange, Plaste Kautsch., **26**, 6 (1979).
769. M. Kamachi, D. J. Liaw, S. Nozakura, Polym. J., **10**, 641 (1978).
770. C. Bonnans-Plaisance, P. F. Casals, G. Levesque, Makromol. Chem., **180**, 79 (1979).
771. S. G. Bondarenko, A. F. Nikolaev, N. I. Dubkova, V. P. Kochanenkov, Vysokomol. Soedin. A, **20**, 2433 (1978).
772. W. K. Busfield, F. P. Franke, R. D. Guthrie, Aust. J. Chem., **31**, 2559 (1978).
773. A. I. Vorob'eva, G. V. Leplyanin, S. R. Rafikov, Vysokomol. Soedin. B, **20**, 763 (1978).
774. A. P. Donya, A. M. Shur, Deposited Doc., VINITI 2672 (1972).
775. A. P. Donya, A. M. Shur, G. N. Agafanova, Z. Z. Malinina, S. I. Sokhina, Deposited Doc., VINITI 2675 (1976).
776. T. Tajima, T. Otsu, Mem. Fac. Eng., Osaka City Univ., **18**, 107 (1977).
777. S. Hoering, D. Kleine, J. Ulbricht, J. Prakt. Chem., **320**, 473 (1978).
778. M. L. Eritsyan, V. V. Zolotukhin, G. I. Zolotukhina, Deposited Doc., VINITI 3792 (1976).
779. K. Hayashi, Makromol. Chem., **179**, 1753 (1978).
780. E. F. Osadchaya, R. G. Karzhaubaeva, B. A. Zhubanov, E. M. Shaikhutdinov, Prikl. Teor. Khim., **6**, 135 (1975).
781. J. R. Suggate, Makromol. Chem., **179**, 1219 (1978).
782. Y. Yamamoto, A. Fukami, Nippon Kagaku Kaishi, 1978, 288.
783. O. Sh. Kurmanaliev, E. M. Shaikhutdinov, Sh. S. Tul'baev, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **28**, 53 (1978).

784. Yu. M. Rodionov, M. G. Chauser, M. I. Cherkashin, Tezisy Dokl.-Vses. Konf. Khim. Atsetilena, 5th, 404. "Metsniereba": Tiflis, USSR (1975).
785. B. Yamada, M. Itahashi, T. Otsu, J. Macromol. Sci. A: Chem., **11**, 1773 (1977).
786. T. C. Chiang, C. Graillat, J. Guillot, Q. T. Pham, J. Polym. Sci., Polym. Chem. Ed., **15**, 2961 (1977).
787. H. Egawa, T. Nonaka, Kobunshi Robunshu, **35**, 21 (1978).
788. E. N. Konstantinova, V. K. Smirnova, V. S. Ivanov, Vestn. Leningr. Univ., Fiz. Khim., 155, (1977).
789. A. F. Nikolaev, V. M. Bondarenko, L. M. Lopukhinskii, Deposited Doc., VINITI 3271 (1975).
790. K. Noma, S. Emi, N. Niwa, Doshisha Daigaku Rikogaku Kenkyu Hokoku, **18**, 28 (1977).
791. Y. Bando, T. Dodou, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **15**, 1917 (1977).
792. M. A. Askarov, A. T. Dzhalilov, G. A. Babakhanov, Deposited Doc., VINITI 1693 (1975).
793. S. Kondo, M. Tsumadori, K. Tsuda, J. Polym. Sci., Polym. Lett. Ed., **15**, 333 (1977).
794. R. Sastre, J. L. Acosta, R. Garrido, J. Fontan, Angew. Makromol. Chem., **62**, 85 (1977).
795. E. Casorati, A. Martina, M. Guaita, Makromol. Chem., **178**, 765 (1977).
796. M. Hartmann, U. C. Hipler, K. Huebig, Z. Chem., **16**, 487 (1976).
797. S. N. Novikov, D. V. Pebalk, L. K. Vasyanina, A. N. Pravednikov, Vysokomol. Soedin. A, **18**, 2333 (1976).
798. R. Roussel, M. Galin, J. C. Galin, J. Macromol. Sci. A: Chem., **10**, 1479 (1976).
799. Y. Bando, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **14**, 693 (1976).
800. G. C. Corfield, H. H. Monks, L. P. Ellinger, Polymer, **16**, 770 (1975).
801. S. M. Samoilov, S. D. Yankova, L. S. Sememova, V. N. Monastyrskii, Vysokomol. Soedin. A, **18**, 340 (1976).
802. M. H. George, G. F. Hayes, Makromol. Chem., **177**, 399 (1976).
803. T. Kelen, F. Tudos, J. Macromol. Sci. A: Chem., **9**, 1 (1975).
804. F. Tudos, T. Kelen, T. Foldes-Berezsnich, B. Turcsanyi, J. Macromol. Sci. A: Chem., **10**, 1513 (1976).
805. T. Kelen, F. Tudos, B. Turcsanyi, Polym. Bull., **2**, 71 (1980).
806. D. Braun, W. Czerwinski, G. Disselhoff, F. Tudos, T. Kelen, B. Turcsanyi, Angew. Makromol. Chem., **125**, 161 (1984).
807. S. Hudecek, J. Otoupalova, M. Ryska, J. Svetlik, P. Cefelin, Collect. Czech. Chem. Commun., **48**, 2015 (1983).
808. L. Lukasik, W. Kuran, B. Laguna, J. Gosk, J. Polym. Sci., Polym. Chem. Ed., **21**, 3333 (1983).
809. G. Bajoras, R. Gedraitis, A. V. Kashkin, S. Slaucka, Vysokomol. Soedin. A, **25**, 2626 (1983).
810. G. D. Jones, H. E. Doorenbos, J. Macromol. Sci. A: Chem., **21**, 155 (1984).
811. Z. Rzaev, K. I. Gurbanov, S. G. Mamedova, M. M. Guseinov, G. S. Sharifov, Vysokomol. Soedin. A, **26**, 736 (1984).
812. Yu. D. Semchikov, L. A. Smirnova, T. E. Knyazeva, S. A. Bulgakova, G. A. Voskoboinik, V. I. Sherstyanykh, Vysokomol. Soedin. A, **26**, 704 (1984).
813. M. F. Sorokin, E. L. Gershanova, T. P. Skopina, Deposited Doc., VINITI 1642 (1983).
814. P. K. Dahl, G. N. Babu, J. Polym. Sci., Polym. Chem. Ed., **22**, 1817 (1984).
815. B. G. Zadontsev, V. G. Neroznik, O. I. Vasilova, T. V. Novikova, S. I. Kuznetsova, Yu. S. Zaitsev, Vysokomol. Soedin. A, **26**, 1591 (1984).
816. G. B. Kharas, M. H. Litt, Polym. Bull. (Berlin), **12**, 65 (1984).
817. Y. Oda, T. Shintani, N. Emura, Toyo Soda Kenkyu Hokoku, **28**, 71 (1984).
818. T. T. Minakova, L. V. Morozova, B. A. Trofimov, Zh. Prokl. Khim. (Leningrad), **57**, 1667 (1984).
819. E. A. Gonyukh, E. V. Kuznetsov, L. Kh. Khazryatova, V. P. Prokop'ev, M. A. Akhmerov, Isv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol., **27**, 1070 (1984).
820. Y. Kihira, H. Yamamura, Kinki Daigaku Kogakubu Kenkyu Hokoku, **17**, 1 (1983).
821. M. Ueda, S. Ishibashi, T. Suzuki, T. Masuko, C. U. Pittman, J. Polym. Sci., Polym. Chem. Ed., **22**, 2305 (1984).
822. M. A. Al-Issa, T. P. Davis, M. B. Huglin, I. B. Yahya, D. C. F. Yip, Eur. Polym. J., **20**, 947 (1984).
823. M. Ueda, S. Shouji, T. Ogata, M. Kamachi, C. U. Pittman, Macromolecules, **17**, 2800 (1984).
824. K. F. O'Driscoll, L. T. Kale, L. H. Rubio, P. M. Reilly, J. Polym. Sci., Polym. Chem. Ed., **22**, 2777 (1984).
825. S. Iwatsuki, A. Kondo, H. Harashina, Macromolecules, **17**, 2473 (1984).
826. O. P. Kosharnaya, Yu. V. Korshak, M. I. Shtil'man, T. M. Yarmizina, Deposited Doc., VINITI 5295 (1983).
827. W. M. Brouwer, P. Piet, A. L. German, J. Polym. Sci., Polym. Chem. Ed., **22**, 2353 (1984).
828. M. Miura, F. Akutsu, M. Yamamoto, K. Naruchi, K. Nagakubo, Makromol. Chem., Rapid Commun., **5**, 745 (1984).
829. Z. Wojtczak, A. Gronowski, Makromol. Chem., **186**, 139 (1985).
830. C. Jin, K. Kazushige, Y. Tabata, J. Macromol. Sci. A: Chem., **22**, 379 (1985).
831. O. Sh. Kurmanaliev, E. M. Shaikhutdinov, K. A. Akhmetkarimov, Deposited Doc., VINITI 1114 (1984).
832. M. Niwa, T. Matsumoto, M. Kagami, K. Kajiyama, Sci. Eng. Rev. Doshisha Univ., **25**, 192 (1984).
833. M. F. Sorokin, L. A. Onosova, L. G. Shode, O. F. Guseva, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **28**, 91 (1985).
834. G. B. Kharas, S. Ponrathnam, S. Jiarui, Y. Ozcayir, A. Blumstein, Polym. Bull. (Berlin), **13**, 357 (1985).
835. C. Simionescu, B. C. Simionescu, S. Ioan, J. Macromol. Sci. A: Chem., **22**, 765 (1985).
836. N. Sh. Rasulov, L. V. Medyakova, N. Yu. Lezgiev, Z. M. Rzaev, Vysokomol. Soedin. B, **27**, 247 (1985).
837. K. Ito, K. Uchida, T. Kitano, E. Yamada, T. Matsumoto, Polym. J. (Tokyo), **17**, 761 (1985).
838. G. A. Cook, G. B. Butler, J. Macromol. Sci. A: Chem., **22**, 1049 (1985).
839. D. Braun, D. Chaudhari, W. Czerwinski, Makromol. Chem., **186**, 1435 (1985).

840. B. S. R. Reddy, R. Arshady, M. H. George, Eur. Polym. J., **21**, 511 (1985).
841. T. Oishi, M. Fujimoto, J. Macromol. Sci. A: Chem., **22**, 1201 (1985).
842. H. Boudevska, O. Todorova, Makromol. Chem., **186**, 1711 (1985).
843. J. I. Jin, H. K. Shim, Pollimo, **9**, 301 (1985).
844. Y. Musha, Y. Hori, Y. Sato, M. Katayama, Nihon Daigaku Kogakubu Kiyo, Bunrui A, **26**, 175 (1985).
845. K. Nagai, K. Akiyama, N. Kuramoto, Makromol. Chem., **186**, 1855 (1985).
846. N. Sh. Rasulov, Sh. Z. Mekhtieva, N. Yu. Lezgiev, S. Z. Rizaeva, Z. M. Rzaev, Azerb. Khim. Zh., 60 (1985).
847. Y. Kihira, H. Yamamura, Kinki Daigaku Kogakubu Kenkyu Hokoku, **18**, 1 (1984).
848. A. H. K. Yousafzai, A. R. Khan, T. Akhtar, Pak. J. Sci. Ind. Res., **28**, 135 (1985).
849. P. K. Dhal, J. Macromol. Sci. A: Chem., **23**, 181 (1986).
850. M. Ueda, M. Yazawa, Nippon Kagaku Kaishi, 1862 (1985).
851. V. Likhterov, V. S. Etlis, Vysokomol. Soedin. B, **27**, 664 (1985).
852. C. Pichot, C. Graillat, V. Glukhikh, Makromol. Chem., Suppl., **10/11**, 199 (1985).
853. A. A. Mohamed, F. H. Jebrael, M. Z. Maher, Macromolecules, **19**, 32 (1986).
854. A. V. Yurkovetskii, V. L. Kofman, G. N. Bondarenko, A. M. Taber, K. L. Makovetskii, Vysokomol. Soedin. B, **27**, 786 (1985).
855. M. Ueda, T. Suzuki, M. Takahashi, Z. B. Li, K. Koyama, C. U. Pittman, Macromolecules, **19**, 558 (1986).
856. J. Horvath, F. Cser, G. Hardy, Magy. Kem. Foly., **91**, 529 (1985).
857. C. L. McCormick, D. L. Elliott, Macromolecules, **19**, 542 (1986).
858. K. R. Cho, K. H. Kim, Han'guk Somyu Konghakhoechi, **22**, 320 (1985).
859. E. E. Skorikova, T. M. Karaputadze, A. M. Ovespyan, A. I. Aksenov, Yu. E. Kirsh, Vysokomol. Soedin. B, **27**, 869 (1985).
860. S. A. Chen, L. C. Tsai, Makromol. Chem., **187**, 653 (1986).
861. S. Das, F. Rodriguez, Polym. Mater. Sci. Eng., **54**, 32 (1986).
862. D. M. Dibona, R. F. Fibiger, E. F. Gurnee, J. E. Shuetz, J. Appl. Polym. Sci., **31**, 1509 (1986).
863. M. W. Sabaa, M. G. Mikhael, A. Yassin, M. Z. Elsabee, Angew. Makromol. Chem., **139**, 95 (1986).
864. G. B. Butler, J. C. Chen, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **27**, 94 (1986).
865. C. Simionescu, V. Barboiu, B. C. Simionescu, V. Talmaciu, C. Sava, J. Polym. Sci., Polym. Chem. Ed., **24**, 851 (1986).
866. T. J. Oh, G. Smets, J. Polym. Sci., Polym. Lett., **24**, 229 (1986).
867. A. Petti, J. Neel, J. Polym. Sci., Polym. Chem. Ed., **24**, 883 (1986).
868. J. M. Catala, A. Nonn, J. M. Pujol, J. Brossas, Polym. Bull. (Berlin), **15**, 311 (1986).
869. F. Canadau, Z. Zekhnini, F. Heatley, Macromolecules, **19**, 1895 (1986).
870. V. A. Kruglova, V. V. Annenkov, S. R. Buzilova, Vysokomol. Soedin. B, **28**, 257 (1986).
871. L. I. Leonenko, S. I. Kudinova, N. N. Chernyshova, E. E. Sirotkina, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **29**, 92 (1986).
872. D. Braun, E. Manger, Colloid Polym. Sci., **264**, 494 (1986).
873. D. Braun, G. Cei, Makromol. Chem., **187**, 1699 (1986).
874. D. Braun, G. Cei, Makromol. Chem., **187**, 1713 (1986).
875. U. M. Mirzaev, M. D. Inoyatov, I. F. Shadrin, Uzb. Khim. Zh., 39 (1986).
876. G. I. Dzhardimalieva, V. A. Zhorin. I. N. Ivleva, A. D. Pompgailo, N. S. Enikolopyan, Dokl. Akad. Nauk SSSR, **287**, 654 (1986).
877. M. Miyamoto, Y. Sano, Y. Kimura, T. Saegusa, Makromol. Chem., **187**, 1807 (1986).
878. J. Lebduska, J. Snuparek, L. Pryskyric, Chem. Prum., **36**, 472 (1986).
879. J. Tanaka, A. Yamada, J. Macromol. Sci. A: Chem., **19**, 453 (1983).
880. K. R. Cho, K. H. Kim, H. Kyung, Han'guk Somyu Konghakhoechi, **22**, 320 (1985).
881. J. Horvath, F. Cser, G. Hardy, Magy. Kem. Foly., **91**, 529 (1095).
882. V. R. Likhterov, V. S. Etlis, Vysokomol. Soedin., Ser. B, **27**, 664 (1985).
883. E. E. Skorikova, T. M. Karaputadze, A. M. Ovsepyan, A. I. Aksenov, Yu. E. Kirsh, Vysokomol. Soedin., Ser. B, **27**, 869 (1985).
884. M. Ueda, M. Yazawa, Nippon Kagaku Kaishi, **10**, 1862 (1985).
885. A. V. Yurkovetskii, V. I. Kofman, G. N. Bondarenko, A. M. Taber, K. L. Makovetskii, Vysokomol. Soedin., Ser. B, **27**, 786 (1985).
886. V. S. Aliev, Z. G. Asadov, A. Yu. Yusubov, A. D. Agazade, D. I. Allakhverdiev, A. I. Gasanov, Ch. K. Salmanova, Azerb. Khim. Zh., **2**, 55 (1986).
887. B. M. Beshimov, O. M. Yariev, A. T. Dzhalilov, Uzb. Khim. Zh., **5**, 38 (1986).
888. D. Bruan, G. Sei, Makromol. Chem., **187**, 1713 (1986).
889. G. B. Butler, J. C. Chen, Polym. Propr., (Am. Chem. Soc., Div. Polym. Chem.), **27**, 94 (1986).
890. J. M. Catala, A. Nonn, J. M. Pujol, J. Brossas, Polym. Bull. (Berlin), **15**, 311 (1986).
891. K. H. Chung, J. K. Park, Nonmumjip-Sanop Kwahak Kisul Yoneuso, **14**, 391 (1986).
892. S. Das, F. Rodriguez, Polym. Mater. Sci. Eng., **54**, 32 (1986).
893. P. K. Dhal, J. Macromol. Sci. A: Chem., **23**, 181 (1986).
894. D. M. Dibona, R. F. Fibiger, E. F. Gurnee, J. E. Shuetz, J. Appl. Polym. Sci., **31**, 1509 (1986).
895. G. I. Dzhardimalieva, V. A. Zhorin, I. N. Ivleva, A. D. Pomogailo, N. S. Enikolopyan, Dokl. Akad. Nauk SSSR, **287**, 654 (1986).
896. E. E. Ergozhin, N. I. Chugunova, B. R. Tausarova, Z. S. Birimzhanova, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **6**, 58 (1986).

897. S. A. Gridchin, M. B. Lachinov, A. V. Kisin, G. V. Shatalov, V. P. Zubov, Vysokomol. Soedin., Ser. A, **28**, 2191 (1986).
898. A. G. Gzyryan, R. V. Egoyan, O. S. Attaryan, V. A. Danielyan, E. G. Darbinyan, Arm. Khim. Zh., **39**, 369 (1986).
899. K. Ito, K. Kodaira, Polym. J. (Tokyo), **18**, 667 (1986).
900. A. P. Kiseleva, V. T. Kolesnikov, B. I. Budzan, Vestn. L'vov. Politekh. Inst., **201**, 54 (1986).
901. K. Kodaira, K. Ito, Nagoya Kogyo Gijutsu Shikensho Hokoku, **35**, 291 (1986).
902. V. A. Kruglova, V. V. Annenkov, S. R. Buzilova, Vysokomol. Soedin., Ser. B, **28**, 257 (1986).
903. J. Lebduska, J. Snuparek, K. Kaspar, Chem. Prum., **36**, 472 (1986).
904. L. I. Leonenko, S. I. Kudinova, N. N. Chernyshova, E. E. Sirotkina, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **29**, 92 (1986).
905. E. L. Madruga, J. San Roman, M. A. Lavia, J. Polym. Sci., Polym. Chem. Ed., **24**, 1379 (1986).
906. S. G. Mamedova, K. I. Gurbanov, Z. M. Rzaev, Vysokomol. Soedin., Ser. A, **28**, 758 (1986).
907. U. M. Mirzaev, M. D. Inoyatov, I. F. Shadrin, Uzb. Khim. Zh., **1**, 39 (1986).
908. M. Miyamoto, Y. Sano, Y. Kimura, T. Saegusa, Makromol. Chem., **187**, 1807 (1986).
909. M. Miyamoto, Y. Sano, T. Saegusa, Makromol. Chem., **187**, 2747 (1986).
910. A. A. Mohamed, F. H. Jebrael, M. Z. Elsabee, Macromolecules, **19**, 32 (1986).
911. T. J. Oh, G. Smets, J. Polym. Sci., Polym. Lett., **24**, 229 (1986).
912. A. Petit, J. Neel, J. Polym. Sci., Polym. Chem., **24**, 883 (1986).
913. Z. M. Rzaev, S. G. Mamedova, N. Sh. Rasulov, U. Kh. Agaev, Azerb. Khim. Zh., **3**, 58 (1986).
914. Z. I. Rzaeva, L. V. Medyakova, U. Kh. Agaev, Vysokomol. Soedin., Ser. A, **28**, 1089 (1986).
915. M. W. Sabaa, M. G. Mikhael, A. A. Yassin, M. Z. Elsabee, Angew. Makromol. Chem., **139**, 95 (1986).
916. B. B. Sadriddinov, M. A. Askarov, I. I. Ismailov, Uzb. Khim. Zh., **5**, 32 (1986).
917. G. Smets, C. Samyn, K. Swaelen, Makromol. Chem., **187**, 2853 (1986).
918. J. Stejskal, P. Kratochvil, D. Strakova, O. Prochazka, Macromolecules, **19**, 1575 (1986).
919. M. Ueda, T. Suzuki, M. Takahashi, Z. B. Li, K. Koyama, C. U. Pittman, Macromolecules, **19**, 558 (1986).
920. M. Ueda, M. Yazawa, T. Suzuki, C. U. Pittman, J. Polym. Sci., Polym. Chem., **24**, 3177 (1986).
921. A. Akmalov, A. B. Alovitdinov, Zh. M. Niyazova, Uzb. Khim. Zh., **1**, 33 (1987).
922. S. M. A. Borban, I. V. Zadneprovskaya, T. M. Babaev, U. N. Musaev, Vysokomol. Soedin., Ser. A, **29**, 39 (1987).
923. A. S. Brown, K. Fujimori, Makromol. Chem., **188**, 2177 (1987).
924. K. Budevska, K. Brachkov, O. Todorova, D. Braun, W. K. Czerwinski, Makromol. Chem., **188**, 1157 (1987).
925. P. N. Gaponik, O. A. Ivashkevich, T. B. Kovaleva, T. N. Andreeva, J. Appl. Polym. Sci., **33**, 769 (1987).
926. S. Iwatsuki, M. Yamaxaki, Kenkyu Hokoku-Asahi Garasu Kogyo Gijutsu Shoreikai, **51**, 113 (1987).
927. A. Karpatyova, J. Barton, O. Paleta, Makromol. Chem., Rapid Commun., **8**, 621 (1987).
928. V. Konsulov, D. Kirova, Z. Grozeva, N. Kirov, Khim. Ind. (Sofia), **59**, 216 (1987).
929. D. J. Lin, A. Petit, J. Neel, Makromol. Chim., **188**, 1163 (1987).
930. S. Masuda, M. Tanaka, T. Ota, Makromol. Chem., **188**, 371 (1987).
931. S. Masuda, M. Tanaka, T. Ota, Polymer, **28**, 1945 (1987).
932. L. J. Mathias, S. H. Kusefoglu, A. O. Kress, Macromolecules, **20**, 2326 (1987).
933. K. Naruchi, F. Akutsu, M. Miura, Makromol. Chem., Rapid Commun., **8**, 281 (1987).
934. M. Patel, J. S. Parmar, M. R. Patel, M. M. Patel, J. Macromol. Sci.-Chem., **24**, 1085 (1987).
935. U. Kredel, G. Luft, Angew. Makromol. Chem., **185/186**, 16 (1991).
936. K. A. Rasheed, A. H. K. Yousufzai, T. Akhtar, Pak. J. Sci. Ind. Res., **30**, 230 (1987).
937. Z. M. Rzaev, M. R. Bairamov, S. M. Aliev, S. G. Mamedova, M. D. Ibragimova, R. V. Dzhafarov, S. G. Aliev, Vysokomol. Soedin., Ser. B, **29**, 89 (1987).
938. J. San Roman, M. Lopez, E. Madruga, L. Pargada, Polymer, **28**, 315 (1987).
939. D. N. Shulz, K. Kitano, J. A. Danik, J. J. Kaladas, Polym. Mater. Sci. Eng., **57**, 149 (1987).
940. A. F. Shaaban, M. M. Arief, A. A. Mahmoud, N. N. Messiha, Acta Polym., **38**, 492 (1987).
941. A. F. Shaaban, M. M. Arief, A. A. Mahmoud, N. N. Messiha, Polymer, **28**, 1423 (1987).
942. Y. Tezuka, Y. Horie, K. Imai, Polymer, **28**, 1025 (1987).
943. S. V. Zlev, L. Ya. Tsarik, G. V. Ratovskii, N. A. Ivanova, D. D. Chuvashev, Vysokomol. Soedin., Ser. A, **29**, 2026 (1987).
944. K. Aoki, S. Urano, H. Umemoto, R. Mizuguchi, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **29**, 423 (1988).
945. J. R. Dharia, C. P. Pthak, G. N. Babu, S. K. Gupta, J. Polym. Sci., Part A.: Polym. Chem., **26**, 595 (1988).
946. Y. Ikeda, Y. Takeda, Y. Yasuaki, Mem. Konan Univ., Sci. Ser., **35**, 17 (1988).
947. G. B. Kharas, J. Appl. Polym. Sci., **35**, 733 (1988).
948. O. Sh. Kurmanaliev, L. M. Sugralina, E. M. Shaikhutdinov, A. V. Omasheva, Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol., **31** 98 (1988).
949. S. G. Mamedova, G. A. Yusifov, Z. M. Rzaev, F. B. Rustamov, Vysokomol. Soedin., Ser. A, **30**, 316 (1988).
950. T. Oishi, N. Okamoto, M. Fujimoto, J. Macromol. Sci. A: Chem., **25**, 1039 (1988).
951. T. Otsu, K. Endo, M. Tanaka, Mem. Fac. Eng., Osaka City Univ., **29**, 151 (1988).
952. A. Petit, M. T. Cung, J. Neel, J. Chim. Phys. Phys.-Chim., Biol., **85**, 319 (1988).
953. K. T. Potts, D. A. Usifer, Macromolecules, **21**, 1985 (1988).

954. Z. M. Rzaev, S. G. Mamedova, G. A. Yusifov, F. B. Rustamov, J. Polym. Sci., part A: Polym. Chem., **26**, 849 (1988).
955. A. I. Skushnikova, E. S. Domnina, A. I. Pavlova, I. M. Korotaeva, Vysokomol. Soedin., Ser. B, **30**, 537 (1988).
956. K. Takeda, M. Akiyama, T. Yamamizu, Kobunshi Ronbunshu, **45**, 485 (1988).
957. D. Tulyaganova, F. Kh. Khadzhaeva, D. D. Il'yasova, Uzb. Khim. Zh., **3**, 28 (1988).
958. M. Ueda, M. Mano, M. Yazawa, J. Polym. Sci., Part A: Polym. Chem., **26**, 2295 (1988).
959. K. Urushido, M. Yokoyama, Y. Shiratori, A. Matsumoto, Kobunshi Ronbunshu, **45**, 435 (1988).
960. O. M. Yariev, B. M. Beshimov, A. T. Dzhalilov, B. A. Mavlonov, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **31**, 93 (1988).
961. H. You, D. A. Tirrell, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **29**, 330 (1988).
962. Y.-L. Zheng, M. Galin, J. C. Galin, Polymer, **29**, 724 (1988).
963. B. A. Zhubanov, O. V. Grinina, N. P. Lyubchenko, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **1**, 48 (1988).
964. A. S. Brar, E. Arunan, G. S. Kapur, Polym. J. (Tokyo), **21**, 689 (1989).
965. J. Brown, P. Goddard, K. Petrak, J. Polym. Sci., Part C: Polym. Lett., **27**, 515 (1989).
966. D. Braun, E. Manger, Colloid Polym. Sci., **264**, 494 (1989).
967. D. Braun, A. E. Pour, W. K. Czerwinski, Colloid Polym. Sci., **267**, 1096 (1989).
968. Y. Ding, D. Qi, Beijing Daxue Xuebao, Ziran Kexueban, **25**, 648 (1989).
969. O. Sh. Kurmanaliev, N. U. Aliev, M. Zh. Burkeev, E. M. Shaikhutdinov, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **5**, 75 (1989).
970. K. Nagai, K. Asada, K. Chiba, N. Kuramoto, J. Polym. Sci., Part A: Polym. Chem., **27**, 3779 (1989).
971. C. P. R. Nair, G. Clouet, Eur. Polym. J., **25**, 251 (1989).
972. V. V. Nikolaenko, A. V. Nekrasov, V. V. Smolyaninov, G. A. Bodun, Vysokomol. Soedin., Ser. A, **31**, 780 (1989).
973. T. Oishi, M. Moriwaki, M. Momoi, M. Fujimoto, J. Macromol. Sci. A: Chem, **26**, 861 (1989).
974. T. Oishi, K. Saeki, M. Fujimoto, J. Polym. Sci., Part A: Polym. Chem., **27**, 1429 (1989).
975. T. Oishi, M. Fujimoto, J. Macromol. Sci. A: Chem., **26**, 1611 (1989).
976. S. Packirisamy, A. Hirao, S. Nakahama, J. Polym. Sci., Part A: Polym. Chem., **27**, 2811 (1989).
977. V. L. Rao, G. N. Babu, Eur. Polym. J., **25**, 605 (1989).
978. T. Sato, K. Morina, H. Tanaka, T. Ota, Eur. Polym. J., **25**, 1281 (1989).
979. A. F. Shaaban, A. A. Khalil, N. N. Messiha, J. Appl. Polym. Sci., **37**, 2051 (1989).
980. Yu. E. Shapiro, N. A. Budanov, Z. V. Orlova, S. I. Kuchanov, Vysokomol. Soedin., Ser. B, **31**, 123 (1989).
981. M. Ueda, K. Kawaguchi, M. Mano, J. Polym. Sci., Part A: Polym. Chem., **27**, 737 (1989).
982. M. Ueda, T. Koyama, M. Mano, M. Yazawa, J. Polym. Sci., Part A: Polym. Chem., **27**, 751 (1989).
983. F. D. Agaev, Ch. A. Chalbiev, Vysokomol. Soedin., Ser. B, **32**, 725 (1990).
984. M. Akashi, E. Yashima, T. Yamashita, N. Miyauchi, S. Sugita, K. Marumo, J. Polym. Sci., Part A: Polym. Chem., **28**, 3487 (1990).
985. S. S. S. Al-Diab, H. K. Suh, J. E. Mark, H. Zimmer, J. Polym. Sci., Part A: Polym. Chem., **28**, 299 (1990).
986. F. Catalina, C. Peinado, E. L. Madruga, R. Sastre, J. L. Mateo, N. S. Allen, J. Polym. Sci., Part A: Polym. Chem., **28**, 967 (1990).
987. H. Chen, Y. Wu, Z. Jin, J. Appl. Polym. Sci., **41**, 427 (1990).
988. C. Deng, Y. Qi, Shiyou Huagong, **19**, 739 (1990).
989. A. P. Donya, O. I. Kachurin, Yu. B. Vysotskii, V. M. Murav'eva, Vysokomol. Soedin., Ser. A, **32**, 1309 (1990).
990. Z. Florjanczyk, W. Krawiec, K. Such, J. Polym. Sci., Part A: Polym. Chem., **28**, 795 (1990).
991. L. Janus, H. Tbal, M. Delporte, J. Morcellet, M. Morcellet, Polym. Bull. (Berlin), **23**, 13 (1990).
992. A. Matsumoto, T. Kubota, T. Otsu, Macromolecules, **23**, 4508 (1990).
993. K. Nagai, K. Asada, N. Kuramoto, J. Polym. Sci., Part A: Polym. Chem., **b**, 2845 (1990).
994. T. Oishi, M. Iwahara, M. Fujimoto, J. Macromol. Sci. A: Chem., **27**, 843 (1990).
995. T. Otsu, A. Matsumoto, H. Ito, Chem. Express, **b**, 901 (1990).
996. D. R. Patil, D. J. Smith, J. Polym. Sci., Part A: Polym. Chem., **28**, 949 (1990).
997. A. Petit, J. Neel, J. Appl. Polym. Sci., **41**, 267 (1990).
998. Y. Shigetomi, T. Kojima, N. Ono, J. Polym. Sci., Part A: Polym. Chem., **28**, 3317 (1990).
999. S. Soundararaan, B. S. Reddy, S. Rajadurai, Polymer, **31**, 366 (1990).
1000. D. L. Trumbo, Polym. Bull. (Berlin), **24**, 215 (1990).
1001. R. Truxa, Makromol. Chem., **191**, 1941 (1990).
1002. M. Ueda, H. Mori, H. Ito, J. polym. Sci., Part A: Polym. Chem., **28**, 2597 (1990).
1003. K. Urushido, K. Koike, K. M. Hiroharu, S. Kuribayashi, Kobunshi Ronbunshu, **47**, 79 (1990).
1004. H. You, D. A. Tirrell, J. Polym. Sci., Part A: Polym. Chem., **28**, 3155 (1990).
1005. S. Z. Zhakupova, R. G. Karzhaubaeva, E. E. Ergozhin, Sh. M. Nurymbetova, Dokl. Akad. Nauk SSSR, **314**, 885 (1990).
1006. E. E. Ergozhin, B. R. Tausarova, S. A. Shitybaev, Dokl. Akad. Nauk SSSR, **321**, 108 (1991).
1007. L. F. Gorbas, P. P. Kisilita, Plast. Massy, **2**, 55 (1991).
1008. K. Itoh, T. Harada, H. Nagashima, Bull. Chem. Soc. Jpn., **64**, 3746 (1991).
1009. Z. Sedlakova, K. Bouchal, M. Ilavsky, Angew. Makromol. Chem., **201**, 33 (1992).
1010. A. F. Miles, J. M. G. Cowie, Eur. Polym. J., **b**, 165 (1991).
1011. J. Muthiah, L. J. Mathias, J. Polym. Sci., Part A:, **b**, 29 (1991).
1012. T. Narasimhaswamy, S. C. Sumathi, B. S. R. Reddy, Eur. Polym. J., **27**, 255 (1991).

1013. T. Narasimhaswamy, S. C. Sumathi, B. S. R. Reddy, J. Macromol. Sci. A: Chem., **28**, 517 (1991).
1014. T. Sato, K. Takahashi, H. Tanaka, T. Ota, Macromolecules, **24**, 2330 (1991).
1015. T. Sato, D. Ito, M. Kuki, H. Tanaka, T. Ota, Macromolecules, **24**, 2963 (1991).
1016. R. V. Sharipov, E. E. Ergozhin, L. N. Prodius, K. Kh. Tastanov, Izv. Akad. Nauk Kaz. SSR, Ser. Khim., **2**, 46 (1991).
1017. S. Soundarrarajan, B. S. R. Reddy, J. Appl. Polym. Sci., **43**, 251 (1991).
1018. K. Su, E. E. Remsen, H. M. Thompson, L. G. Sneddon, Macromolecules, **24**, 3760 (1991).
1019. P. S. Chung, C. K. Lee, Pollimo, **15**, 749 (1991).
1020. D. I. Trumbo, J. Polym. Sci., Part A: Polym. Chem., **29**, 357 (1991).
1021. J. Z. Yang, T. Otsu, Chem. Express, **6**, 41 (1991).
1022. V. E. Baikov, E. F. Panarin, S. L. Timofeevskii, Vysokomol. Soedin., Ser. B, **34**, 3 (1992).
1023. A. S. Brar, Eur. Polym. J., **28**, 803 (1992).
1024. A. S. Brar, J. Polym. Sci., Part A: Polym. Chem., **30**, 2549 (1992).
1025. B. Charleux, C. Pichot, M. F. Llauro, Makromol. Chem., **193**, 187 (1992).
1026. H. Cheng, G. Zhao, D. Yan, J. Polym. Sci., Part A: Polym. Chem., **30**, 2181 (1992).
1027. M. C. Fernandez-Monreal, R. Cuervo, E. L. Madruga, J. Polym. Sci., Part A: Polym. Chem., **30**, 2313 (1992).
1028. M. Georgieva, T. Arabadzhieva, Dokl. Blug. Akad. Nauk, **45**, 81 (1992).
1029. S. Janietz, M. Hahn, W. Jaeger, Acta Polym., **43**, 230 (1992).
1030. Z. Janovi, T. T. Matusinovitch, J. Macromol. Sci. A: Pure Appl. Chem., **29**, 801 (1992).
1031. S. Kobayashi, J. Kadokawa, Y. Matsumura, I. F. Yen, H. Uyama, Macromol. Rep. A, **29**, (Suppl. 3), 243 (1992).
1032. L. K. Kostanski, A. E. Hamielec, Polymer, **33**, 3706 (1992).
1033. S. Masuda, J. Macromol. Sci. A: Pure Appl. Chem., **29**, 821 (1992).
1034. K. Nagai, K. Okada, J. Kido, J. Polym. Sci., Part A: Polym. Chem., **30**, 1187 (1992).
1035. Y. Lu, J. Wu, S. Lin, Gaofenzi Xuebao, **1**, 112 (1992).
1036. Y. Nagasaki, S. B. Han, M. Kato, T. Tsuruta, Makromol. Chem., **193**, 1633 (1992).
1037. Z. S. Nurkeeva, G. G. Khazrenova, G. A. Mun, Vysokomol. Soedin. Ser. B, **34**, 34 (1992).
1038. T. Otsu, A. Matsumoto, K. Shiraishi, N. Amaya, Y. Koiunuma, J. Polym. Sci., Part A: Polym. Chem., **30**, 1559 (1992).
1039. T. Otsu, A. Matsumoto, K. Nakamura, J. Appl. Polym. Sci., **45**, 1889 (1992).
1040. T. Otsu, M. Yoshioka, Makromol. Chem., **193**, 2283 (1992).
1041. Z. Sedlakova, K. Bouchal, M. Ilavsky, Angew. Makromol. Chem., **201**, 33 (1992).
1042. K. V. C. Rao, S. K. Nema, V. N. G. Kumar, S. Mathew, Angew. Makromol. Chem., **152**, 21 (1987).
1043. K. Takenaka, A. Hirao, S. Nakahama, Makromol. Chem., **193**, 1943 (1992).
1044. S. Wen, X. Yin, W. T. K. Stevenson, Polym. Int., **27**, 81 (1992).
1045. Y. Shigetomi, N. Ono, H. Kato, M. Oki, Polym. J. (Tokyo), **24**, 87 (1992).
1046. G. A. Valieva, A. S. Maksumova, I. I. Ismailov, M. A. Askarov, Izv. Vyssh. Uchebn. Zaved., Khim. Khim. Tekhnol., **35**, 109 (1992).
1047. B. Yamada, M. Satake, T. Otsu, J. Macromol. Sci. A: Pure Appl. Chem., **29**, 533 (1992).
1048. B. Yamada, M. Satake, T. Otsu, Polym. J. (Tokyo), **24**, 563 (1992).
1049. M. Yoshioka, A. Matsumoto, T. Otsu, Macromolecules, **25**, 2837 (1992).
1050. D. Zaldivar, C. Peniche, A. Bulay, J. San Romano, Polymer, **33**, 4625 (1992).
1051. A. Gallardo, J. San Roman, Polymer, **34**, 567 (1993).
1052. V. F. Gromov, Yu. S. Bogachev, E. V. Bune, I. L. Zhuravleva, E. N. Teleshov, Vysokomol. Soedin., Ser. A, **35**, 7 (1993).
1053. F. Heatley, P. A. Lovell, J. McDonald, Eur. Polym. J., **29**, 255 (1993).
1054. T. Oishi, H. Tsotomu, S. Haruyuki, Polym. J. (Tokyo), **25**, 781 (1993).
1055. T. Oishi, K. Kagawa, M. Fujimoto, Polymer, **34**, 2644 (1993).
1056. T. Otsu, K. Shiraishi, A. Matsumoto, J. Polym. Sci., Part A: Polym. Chem., **31**, 2523 (1993).
1057. C. K. Park, C. S. Ha, J. K. Lee, J. Won, J. Appl. Polym. Sci., **50**, 1239 (1993).
1058. A. Petit, Eur. Polym. J., **29**, 1419 (1993).
1059. T. Sato, M. Ikazaki, M. Seno, H. Tanaka, Makromol. Chem., **194**, 637 (1993).
1060. T. Sato, S. Kawasaki, M. Seno, H. Tanaka, K. Kato, Makromol. Chem., **194**, 2247 (1993).
1061. S. Soundrarajan, B. S. R. Reddy, Polymer, **34**, 2224 (1993).
1062. D. L. Trumbo, Polym. Bull. (Berlin), **31**, 629 (1993).
1063. D. Zaldivar, C. Peniche, A. Bulay, J. San Roman, Polym. Sci., Part A: Polym. Chem., **31**, 625 (1993).
1064. R. A. Akhmed'yanova, A. G. Liakumovich, S. V. Shulyndin, P. A. Kirpichnikov, G. B. Kamardin, S. V. Mokeeva, Zh. Prikl. Khim. (S.-Peterburg), **67**, 514 (1994).
1065. P. Bajaj, M. Goyal, R. B. Chavan, J. Appl. Polym. Sci., **53**, 1771 (1994).
1066. A. I. Barabanova, V. F. Gromov, E. V. Bune, Yu. S. Bogachev, N. V. Kozlova, E. N. Teleshov, Vysokomol. Soedin., Ser. A. Ser. B, **36**, 901 (1994).
1067. A. S. Brar, S. Charan, J. Appl. Polym. Sci., **53**, 1813 (1994).
1068. W. -J. Cho, C. -H. Choi, C. -S. Ha, Polym. Sci., Part A: Polym. Chem., **32**, 2301 (1994).
1069. P. Durand, A. Margaillan, M. Camail, J. L. Vernet, Polymer, **35**, 4392 (1994).
1070. E. K. Fedorov, O. E. Labanov, L. F. Mosalova, V. I. Svergun, S. A. Kedik, Yu. E. Kirsh, Vysokomol. Soedin., Ser. A Ser. B, **36**, 1446 (1994).
1071. G. S. Georgiev, I. G. Dakova, Eur. Polym. J., **30**, 1417 (1994).

1072. G. S. Georgiev, I. G. Dakova, Macromol. Chem. Phys., **195**, 1695 (1994).
1073. E. L. Madruga, M. Fernandez-Garcia, Polymer, **35**, 4437 (1994).
1074. A. Miller, Eur. Polym. J., **30**, 185 (1994).
1075. U. Rehn, W. Frank, M. Arnold, Plaste Kautsch., **41**, 69 (1994).
1076. F. Sanda, T. Takata, T. Endo, Macromolecules, **27**, 3982 (1994).
1077. T. Sato, S. Shimooka, M. Seno, H. Tanaka, Macromol. Chem. Phys., **195**, 833 (1994).
1078. H. Tanaka, M. Sakaguchi, Y. Kikukawa, T. Sato, Macromol. Chem. Phys., **195**, 2083 (1994).
1079. Y. Y. Tan, G. O. R. A. van Ekenstein, Eur. Polym. J., **30**, 1363 (1994).
1080. G. Bauduin, B. Boutevin, M. Belbachir, R. Meghabar, Macromolecules, **28**, 1750 (1995).
1081. M. Camail, H. Essaoudi, A. Margaillan, J. L. Vernet, Eur. Polym. J., **31**, 1119 (1995).
1082. Y. Chen, Z. Han, P. Wu, Qingdao Daxue Xuebao Ziran Kexueban, **8**, 86 (1995).
1083. K. Chino, T. Takata, T. Endo, Macromolecules, **28**, 5947 (1995).
1084. W. K. Czerwinski, Macromolecules, **28**, 5411 (1995).
1085. T. Endo, T. Koizumi, T. Takata, K. Chino, Polym. Sci., Part A: Polym. Chem., **33**, 707 (1995).
1086. R. Francis, A. Ajayaghosh, Polymer, **36**, 1091 (1995).
1087. G. S. Georgiev, I. G. Dakova, S. J. Simpson, Macromol. Sci. A: Pure Appl. Chem., **32**, 497 (1995).
1088. J. Jang, H. Kim, J. Appl. Polym. Sci., **56**, 1495 (1995).
1089. K. Kagawa, T. Oishi, K. Matsusaki, M. Fujimoto, Polymer, **36**, 941 (1995).
1090. M. Kanamaru, T. Takata, T. Endo, Polym. Sci., Part A: Polym. Chem., **33**, 1361 (1995).
1091. S. Kobatake, B. Tamada, S. Aoki, Polymer, **36**, 413 (1995).
1092. S. Kobatake, B. Yamada, Macromolecules, **28**, 4047 (1995).
1093. G. K. Kostov, Al. T. Nikolov, J. Appl. Polym. Sci., **55**, 1529 (1995).
1094. G. K. Kostov, Al. T. Nikolov, J. Appl. Polym. Sci., **55**, 1545 (1995).
1095. M. Kumagai, K. Tsuchida, Y. Ogino, J. Hansen, H. Ishida, Polymer, **36**, 535 (1995).
1096. B. Levenfeld, J. San Roman, Macromolecules, **28**, 3650 (1995).
1097. D. L. Murray, H. J. Harwood, M. M. Samy, I. Piirma, Polymer, **36**, 3841 (1995).
1098. H. Nakamura, M. Seno, H. Tanaka, T. Sato, Colloid Polym. Sci., **273**, 122 (1995).
1099. T. Oishi, K. Kagawa, M. Fujimoto, Polym. Sci., Part A: Polym. Chem., **33**, 1341 (1995).
1100. T. Oishi, K. Sase, K. Saeki, S. Yao, K. Ohdan, Polymer, **36**, 3935 (1995).
1101. J. Penelle, S. verraver, P. Raucq, J. Marchand-Brynaert, Macromol. Chem. Phys., **196**, 857 (1995).
1102. T. Sato, I. Kamiya, M. Seno, H. Tanaka, J. Macromol. Sci. A: Pure Appl. Chem., **32**, 415 (1995).
1103. T. Sato, Y. Hirose, M. Seno, H. Tanaka, Polym. Sci., Part A: Polym. Chem., **33**, 797 (1995).
1104. D. L. Trumbo, Polym. Bull. (Berlin), **34**, 27 (1995).
1105. D. L. Trumbo, Polym. Bull. (Berlin), **34**, 399 (1995).

Q and e Values for Free Radical Copolymerizations of Vinyl Monomers and Telogens

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A. INTRODUCTION

The Alfrey and Price Q and e equation (A1) is well known:

$$Q_2 = (Q_1/r_1)\exp\{-e_1(e_1 - e_2)\} \qquad (A1)$$

Q and e are measures of the reactivity and polarity, respectively, of a vinyl monomer and $r_1 = k_{11}/k_{12}$, a reactivity ratio. This equation may be arranged to a logarithmic form (A2) which provides a more convenient linear solution:

$$\{[\ln(Q_1/r_1)] - e_1^2\} = -e_2e_1 + \ln Q_2. \qquad (A2)$$

When monomer 2 is the one to be characterized, its Q and e values may be determined from the slope $(-e_2)$ and intercept $(\ln Q_2)$ of the equation.

The first requirement for the use of Eq. (A2) is the selection of a small group of common monomers for which Q and e could be independently established. Since, by definition the Q and e scheme is based on styrene ($Q = 1.0$ and $e = -0.8$), the average Q and e values of these primary monomers were calculated using Eqs. (A1) and (A3) (Ref. 1).

$$e_2 = e_1 \pm (-\ln r_1 r_2)^{1/2} \qquad (A3)$$

utilizing only their reactivity ratios with styrene. The six primary monomers (Group I) – acrylic acid, acrylonitrile, butadiene, methyl acrylate, methacrylonitrile and methyl methacrylate – were chosen because their reactivity ratios had been narrowly defined by several investigators. These primary reactivity ratios were taken from the compilation in the second edition of this Handbook.

The Q and e values of a secondary group (Group II) of ten common vinyl monomers were determined using Eq. (A2), with the primary monomer group supplying the Q_1 and e_1 values. The Q and e values were then calculated for a tertiary set (Group III) of reactivity ratios for which a set of reactivity ratios with four or more of the primary and secondary monomers was available. For all groups except Group I, recalculated reactivity ratios (see the preceding chapter) were used when available. Otherwise, reported values were included.

The Q and e values developed for the three groups during the first pass through the data were subsequently used to recalculate the Q and e values for Groups II and III. Group I was not redetermined so that the styrene bias of the system could be better maintained. After four iterations, no significant changes were observed. During the iterative process, a number of reactivity ratio citations were dropped due to a lack of fit with the majority of data points.

The final Q and e values for these three groups were then used to determine the Q and e values of other monomers with four or more reactivity ratio citations with at least three comonomers (Group IV), and monomers with three citations with three different comonomers (Group V).

By replacing r_1 in Eq. (A2) with the reciprocal of the chain transfer constant, $1/C_s$ (3), Q and e values of telogens may also be estimated (4) from their coreactivity with vinyl monomers.

The Monomer Table lists the monomer group designation, the Q and e values, and the correlation coefficient (r) for the regression. The Number column (n) represents the number of r_1 values actually used in the regression. If the monomer is in Group V, the number is understood to be 3. The data are listed alphabetically as well as in increasing values of Q and e.

*Retired.

Recent work by Jenkins (5) has uncovered another approach to characterizing the copolymerization character of vinyl monomers. By deriving data from r_1 and r_2 values, he was able to calculate three coefficients which appear to allow a more precise prediction of the actual reactivity ratios.

B. *Q* AND *e* VALUES FOR FREE RADICAL COPOLYMERIZATIONS

TABLE 1. MONOMERS

Monomer	Group	*Q*	*e*	*n*	*r*
Acenaphthalene	IV	0.72	−1.88	5	0.98
Acetylene, phenyl-	IV	0.45	0.10	6	0.75
Aconitate, trimethyl	III	0.25	2.27	5	0.63
Acrolein	IV	0.80	1.31	16	0.97
Acrolein, methyl-	IV	1.83	0.71	5	0.98
Acrylamide	IV	0.23	0.54	17	0.64
Acrylamide, *N*-methylol-	IV	0.52	1.15	6	0.91
Acrylamide, *N*-octadecyl-	IV	0.66	1.64	4	0.93
Acrylamide, *N,N*-diethyl	IV	0.48	−0.31	5	0.56
Acrylamide, *N,N*-dimethyl-glucitol	V	0.22	0.61		0.68
Acrylate, α-acetoxy-, ethyl	V	0.52	0.77		0.99
Acrylate, α-chloro-, ethyl	V	1.00	−1.03		0.87
Acrylate, α-chloro-, methyl	IV	2.43	0.35	4	0.38
Acrylate, α-cyano-, methyl	IV	4.91	0.91	4	0.99
Acrylate, α-phenyl-, methyl	IV	5.19	0.96	8	0.90
Acrylate, benzyl	IV	0.33	1.13	5	0.92
Acrylate, butyl	II	0.38	0.85	20	0.94
Acrylate, ethyl	III	0.41	0.55	12	0.91
Acrylate, ferrocenylmethyl	IV	0.15	0.51	4	0.65
Acrylate, glycidyl	IV	0.48	1.28	4	0.87
Acrylate, heptafluorobutyl	V	0.96	1.34		0.81
Acrylate, methyl	I	0.45	0.64		
Acrylate, octadecyl	IV	0.33	1.26	7	0.97
Acrylate, octyl	V	0.63	2.01		0.99
Acrylate, trifluoro-, methyl	V	0.048	1.20		0.55
Acrylate, 2-chloroethyl	V	0.49	1.03	5	0.95
Acrylate, 2-ethylhexyl	IV	0.37	0.24	4	0.37
Acrylate, 2-nitrobutyl	V	0.69	1.09		0.98
Acrylic acid	I	0.83	0.88		
Acrylic acid, α-bromo-	V	3.07	1.37		0.99
Acrylic anhydride	V	1.46	0.31		0.37
Acrylonitrile	I	0.48	1.23		
Acryloyl chloride	IV	1.82	1.92	6	0.90
Allyl acetate	III	0.24	−1.07	8	0.85
Allyl acrylate	V	0.32	−0.99		0.96
Allyl alcohol	IV	0.005	−1.48	5	0.98
Allyl chloride	III	0.026	−0.60	12	0.93
Allylbenzene	IV	0.038	0.40	4	0.60
N-Allylstearamide	V	0.024	−0.41		0.62
Benzothiazole, vinylmercapto-	IV	0.36	−0.44	4	0.95
Bicyclo[2.2.1]hept-2-ene-5,6-dicarboxamine, *N*-benzyl-	IV	0.01	−0.36	6	0.47
Butadiene	I	1.70	−0.50	24	0.82
Butadiene-1-carboxylate, ethyl	V	1.67	1.26		0.99
Butadiene, 1,4-dicarboxylate, diethyl	V	1.94	1.39		0.99
Butadiene, 2,3-dichloro-	V	9.08	0.14		0.93
Butadiene, 2,3-dimethyl-	V	1.42	−0.43		0.58
Butadiene, hexafluoro-	IV	0.82	0.58	4	0.93
Butadiene, 2-chloro-	II	10.52	1.20	10	0.98
Butadiene, 2-fluoro-	III	1.88	0.63	6	0.77
Butene-1	IV	0.007	−0.06	5	0.08
Butene-2	IV	0.002	−0.29	6	0.44
Carbamate, *N*-vinyl-, ethyl	V	0.037	−1.12		0.91
Carbamate, *N,N*-diethyl-, vinyl	IV	0.028	−1.11	4	0.95
Carbazole, *N*-vinyl-	III	0.26	−1.29	12	0.99
Carbon monoxide	V	0.013	1.68		0.70

TABLE 1. *cont'd*

Monomer	Group	*Q*	*e*	*n*	*r*
Citraconimide, *N*-methyl-	IV	0.87	1.58	4	0.94
Crotonaldehyde	IV	0.023	0.84	5	0.82
Crotonic acid	III	0.017	0.89	6	0.98
4-Cyclopentene-1,3-dione	IV	0.42	2.43	4	0.96
Diallyl melamine	V	0.059	−1.57		0.99
Diallyl phthalate	IV	0.031	−0.26	20	0.50
Diallycyanamide	IV	0.14	2.41	4	0.96
N,N-Divinylaniline	II	0.26	−0.68	5	0.78
Ethylene	III	0.016	0.05	6	0.52
Ethylene, 1,1-diphenyl-	IV	0.17	−1.71	5	0.84
Ethylene, chlorotrifluoro-	III	0.026	1.56	5	0.99
Ethylene, tetrachloro-	III	0.001	1.24	7	0.99
Ethylene, tetrafluoro-	IV	0.032	1.63	4	0.99
Ethylene, trichloro-	III	0.010	1.29	11	0.94
Fumarate, diethyl	IV	0.25	2.26	10	0.91
Fumarate, diisopropyl	IV	0.11	2.58	8	0.84
Fumaronitrile	IV	0.29	2.73	5	0.96
Hexatriene, tetrachloro-	IV	1.83	0.94	6	0.98
Hexene-1	V	0.035	0.92		0.69
Imidazole, 1-vinyl-	V	0.11	−0.68		0.98
Imidazole, 1-vinyl-2-methyl-	IV	0.14	−0.98	6	0.93
Indene	IV	0.13	−0.71	8	0.45
Isobutylene	III	0.023	−1.20	7	0.99
Isobutylene, 3-chloro-	IV	0.17	−0.86	4	0.98
Isoprene	II	1.99	−0.55	11	0.86
Isoprene, 3-acetoxy-	V	1.91	−0.11		0.10
Isopropenyl acetate	IV	0.023	−0.94	4	0.98
Isopropenyl methyl ketone	III	1.03	0.64	6	0.78
Isopropenyl, 3-(1-cyclohexenyl), acetate	V	0.57	−0.66		0.98
Isopropenylisocyanate	III	0.18	−1.05	8	0.94
Itaconate, dibutyl	V	0.82	0.57		0.88
Itaconate, diethyl	V	1.04	0.88		0.98
Itaconate, dimethyl	III	0.73	0.57	8	0.90
Itaconic acid	IV	0.78	1.07	5	0.99
Maleate, diethyl	IV	0.053	1.08	12	0.87
Maleic anhydride	IV	0.86	3.69	25	0.98
Maleimide	V	0.94	2.86		0.99
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	V	2.54	1.64		0.98
Maleimide, *N*-(2-hydroxyethyl)-	V	1.26	1.07		0.68
Maleimide, *N*-(2-chlorophenyl)-	IV	2.29	2.87	5	0.96
Maleimide, *N*-(4-chlorophenyl)-	V	2.98	2.75		0.99
Maleimide, *N*-butyl-	V	0.88	3.70		0.92
Maleimide, *N*-phenyl-	IV	2.81	3.24	4	0.99
Methacrylamide	IV	0.40	−0.05	8	0.29
Methacrylamide, *N*-phenyl-	V	0.40	0.19		0.70
Methacrylamide, 1-deoxy-D-glucital	V	0.15	−0.16		0.19
Methacrylate, 2-chloro-2,3,3,3-tetrafluoropropyl	V	2.09	0.62		0.56
Methacrylate, 2,2,6,6-tetra-methyl-4-piperidinyl	IV	0.52	−1.09	4	0.68
Methacrylate, 2,3-epithio-propyl	IV	1.04	0.29	4	0.84
Methacrylate, benzyl	IV	0.88	0.35	10	0.97
Methacrylate, butyl	III	0.82	0.28	10	0.68
Methacrylate, ethyl	III	0.76	0.17	7	0.62
Methacrylate, ferrocenylmethyl	IV	0.22	0.65	7	0.85
Methacrylate, glycidyl	III	0.96	0.20	9	0.92
Methacrylate, isobutyl	III	0.82	0.27	5	0.76
Methacrylate, isopropyl	IV	0.97	0.10	4	0.46
Methacrylate, methyl	I	0.78	0.40		
Methacrylate, phenyl	IV	1.25	0.79	7	0.91
Methacrylate, sulfolanyl	V	1.18	0.95		0.95

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TABLE 1. *cont'd*

Monomer	Group	*Q*	*e*	*n*	*r*
Methacrylate, 2-naphthyl	IV	1.04	−0.09	8	0.46
Methacrylate, 2-(*N*,*N*-dimethyl-carbamoyloxy)ethyl	IV	1.14	0.84	5	0.74
Methacrylate, 2-(sulfonic acid)ethyl	IV	1.09	0.25	4	0.52
Methacrylate, 2-acetoxyethyl	V	0.60	0.51		0.80
Methacrylate, 2-bromoethyl	IV	1.18	0.74	4	0.96
Methacrylate, 2-chloroethyl	IV	1.04	0.31	7	0.79
Methacrylate, 2-hydroxyethyl	IV	1.78	−0.39	7	0.67
Methacrylate, 2-hydroxypropyl	IV	4.38	1.86	7	0.81
Methacrylic acid	III	0.98	0.62	23	0.73
Methacrylic anhydride	IV	3.00	0.56	5	0.93
Methacrylonitrile	I	0.86	0.68		
Methacryloyl chloride	V	2.04	1.54		0.94
Methacryloylacetone	V	5.47	−0.76		0.79
Methylenebutyrolactone	V	2.48	0.83		0.90
2-Methyleneglutaronitrile	IV	0.41	1.25	6	0.83
Norbornadiene	III	0.051	−1.48	7	0.90
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	V	0.25	0.61		0.82
Oxazolidone, *N*-vinyl-	III	0.087	−1.70	6	0.95
Oxazoline, 2,-2-isopropenyl-4,4-dimethyl-	IV	0.87	0.34	4	0.65
Oxazoline, 2-isopropenyl-2-	IV	0.59	−0.64	7	0.54
Oxazoline, 2-,4-acryloxymethyl-2,4-dimethyl-	IV	0.97	−0.51	4	0.71
Oxazoline, 2-,4-methacryloxymethyl-2,4-dimethyl-	IV	0.44	−0.70	4	0.99
Pentadiene, *trans*-1,3-	IV	1.28	−0.13	8	0.54
Phosphonate, α-carbomethoxyvinyl-, diethyl	V	0.14	−0.04		0.96
Phosphonate, isopropenyl-, dimethyl	V	0.032	0.96		0.98
Phosphonate, vinyl-, bis(2-chloroethyl)	V	0.044	1.64		0.89
Phosphonate, vinyl-, diethyl	V	0.27	−0.40		0.54
Phosphonate, vinyl-, dimethyl	V	0.24	−0.25		0.53
Phosphonic acid, α-phenyl-vinyl-	IV	0.65	0.52	5	0.76
Phthalimide, *N*-(methacryloyloxy)-	V	1.89	0.96		0.74
Propene, 3,3,3-trichloro-	V	0.030	1.00		0.99
Propene, 1-chloro-, *cis*-	V	0.003	−0.46		0.84
Propene, 1-chloro-, *trans*-	V	0.006	−0.16		0.28
Propene, 2-chloro-	V	0.073	−0.16		0.99
Propene, 2-chloro-3-hydroxy-	V	0.091	−0.95		0.99
Propene, 2,3-dichloro-	IV	0.12	−0.40	4	0.67
Propenyl, 2-chloro-, acetate	IV	0.20	−0.85	5	0.61
Propenyl, 2-methyl-, acetate	V	0.040	−2.08		0.97
Propylene	V	0.009	−1.69	7	0.84
Pyridazinone, 3-(2-vinyl)-6-methyl-	V	0.57	0.24		0.58
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	V	0.18	−0.32		0.35
Pyridine, 2-methyl-5-vinyl-	III	1.32	−0.66	9	0.95
Pyridine, 2-vinyl-	II	1.41	−0.42	12	0.97
Pyridine, 2-vinyl-5-ethyl-	IV	1.29	−0.91	4	0.94
Pyridine, 4-vinyl-	III	2.47	0.84	7	0.86
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	V	1.50	−0.59		0.98
Quinoline, 2-vinyl-	IV	1.04	−0.09	4	0.46
Silane, γ-methacryloxypropyl-trimethoxy	V	1.08	0.07		0.35
Styrene	I	1.00	−0.80		
Styrene, 2,4,6-trimethyl-	IV	0.15	−0.58	4	0.41
Styrene, 2,5-dichloro-	II	1.50	0.94	12	0.96
Styrene, 3-tri-*n*-butylstannyl-	IV	0.014	−0.61	4	0.51
Styrene, α-methoxy-	IV	0.39	−1.02	5	0.77
Styrene, α-methyl-	III	0.97	−0.81	15	0.98
Styrene, *m*-bromo-	IV	1.25	−0.27	4	0.98
Styrene, *m*-chloro-	IV	2.46	−0.90	4	0.87
Styrene, *m*-methyl-	V	1.57	−0.03		0.06
Styrene, *m*-nitro-	IV	2.66	1.57	7	0.85
Styrene, *p*-bromo-	IV	1.30	−0.68	7	0.99
Styrene, *p*-chloro-	III	1.33	−0.64	12	0.92

TABLE 1. *cont'd*

Monomer	Group	*Q*	*e*	*n*	*r*
Styrene, *p*-chloromethyl-	IV	1.39	−0.38	7	0.74
Styrene, *p*-cyano-	IV	2.93	−0.38	5	0.75
Styrene, *p*-methoxy-	III	1.53	−1.40	6	0.99
Styrene, *p*-methyl-	III	1.10	−0.63	5	0.98
Styrene, *p*-1-(2-hydroxybutyl)-	V	0.70	−0.97		0.65
Styrene, *p*-1-(2-hydroxypropyl)-	IV	1.08	−0.35	4	0.51
Styrene, *p*-2-(2-hydroxypropyl)-	IV	1.15	−0.49	4	0.91
Styrene, pentachloro-	V	0.20	0.79		0.87
Succinimide, *N*-vinyl-	III	0.19	−1.42	8	0.96
Tetrazole, 1-vinyl-	V	0.13	−0.14		0.39
Tetrazole, 2-methyl-5-(4′-vinyl)phenyl-	V	0.86	0.51		0.49
Tetrazole, 2-methyl-5-vinyl-	V	0.55	−0.46		0.85
Tetrazole, 2-phenyl-5-(4′-vinyl)phenyl-	V	0.84	−0.52		0.97
Tetrazole, 5-phenyl-2-(4′-vinyl)phenyl-	V	1.11	0.53		0.53
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	V	0.18	−0.53		0.78
Triallyl citrate	IV	0.054	0.26	4	0.50
Triallyl cyanurate	V	0.14	2.41		0.17
Triallyl isocyanurate	V	0.035	−0.23		0.35
Urea, *N*-vinyl-*N*′-ethyl-	V	0.17	−1.88		0.98
Vinyl acetate	II	0.026	−0.88	49	0.87
Vinyl benzoate	III	0.030	−0.89	9	0.95
Vinyl bromide	IV	0.038	−0.23	10	0.94
Vinyl butyl ether	IV	0.038	−1.50	10	0.87
Vinyl butylsulfonate	IV	0.16	1.06	4	0.97
Vinyl butyrate	IV	0.024	−0.89	8	0.91
Vinyl chloride	II	0.056	0.16	38	0.37
Vinyl chloroacetate	IV	0.039	−1.61	5	0.98
Vinyl chloromethyl ketone	IV	16.00	1.78	4	0.97
Vinyl cinnamate	IV	0.18	0.76	4	0.99
Vinyl cymantrene	IV	0.39	−0.57	4	0.88
Vinyl dichloroacetate	V	0.059	−1.38		0.95
Vinyl dodecyl ether	IV	0.041	−1.69	6	0.83
Vinyl ether	IV	0.029	−1.16	4	0.80
Vinyl ethyl ether	III	0.018	−1.80	8	0.90
Vinyl ethyl oxalate	IV	0.056	−0.65	4	0.53
Vinyl ethyl sulfide	IV	0.27	−1.31	6	0.96
Vinyl ethyl sulfoxide	IV	0.065	0.05	4	0.21
Vinyl fluoride	IV	0.008	0.72	8	0.71
Vinyl formate	III	0.043	−1.19	5	0.99
Vinyl hendecanoate	III	0.056	−0.84	5	0.82
Vinyl isobutyl ether	IV	0.030	−1.27	7	0.98
Vinyl isobutyl sulfide	IV	0.49	−0.95	7	0.99
Vinyl isopropyl ketone	V	0.58	−1.16		0.44
Vinyl isothiocyanate	IV	0.59	0.37	4	0.77
Vinyl laurate	V	0.011	−0.54		0.81
Vinyl-*m*-cresyl ether	V	0.016	−6.33		0.99
Vinyl methyl ketone	II	0.66	1.05	8	0.98
Vinyl methyl sulfide	IV	0.42	−1.66	5	0.99
Vinyl methyl sulfoxide	V	0.60	−1.79		0.46
Vinyl *o*-cresyl ether	V	0.010	−8.53		0.97
Vinyl octadecyl ether	IV	0.024	−1.93	4	0.87
Vinyl octyl ether	IV	0.020	−1.57	4	0.98
Vinyl *p*-cresyl ether	V	0.015	−6.86		0.99
Vinyl pelargonate	IV	0.046	−1.22	4	0.79
Vinyl phenyl ether	IV	0.046	−2.16	9	0.95
Vinyl phenyl ketone	IV	1.16	1.02	5	0.95
Vinyl phenyl sulfide	IV	0.33	−0.99	5	0.96
Vinyl propionate	IV	0.027	−0.68	7	0.82
Vinyl stearate	III	0.043	−0.97	7	0.85
Vinyl *tert*-butyl sulfide	IV	0.046	−2.20	4	0.72
Vinyl thiolacetate	IV	0.27	−0.52	5	0.85

TABLE 1. *cont'd*

Monomer	Group	Q	e	n	r
Vinyl 4-chlorocyclohexyl ketone	IV	0.66	−0.82		0.50
Vinyl 2-chloroethyl ether	IV	0.019	−1.64	9	0.96
Vinyl-tris(trimethoxysiloxy)silane	IV	0.022	−0.12	5	0.75
p-Vinylbenzylethylcarbinol	V	0.69	−0.98		0.67
p-Vinylbenzylmethylcarbinol	V	1.70	−1.27		0.98
9-Vinylanthracene	IV	0.14	0.82	5	0.90
Vinyl *N,N*-diethylcarbamate	IV	0.028	−1.10	4	0.95
p-Vinylbenzoic acid	V	5.17	1.08		0.95
N-Vinylcaprolactam	IV	0.14	−1.18	4	0.72
Vinylene carbonate	III	0.004	−0.49	9	0.73
Vinylferrocene	IV	0.31	−1.34	6	0.91
Vinylhydroquinone	V	4.30	2.26		0.87
Vinylhydroquinone dibenzoate	IV	1.73	0.84	5	0.87
Vinylidene chloride	II	0.31	0.34	40	0.68
Vinylidene cyanide	IV	14.22	1.92	9	0.82
Vinylisocyanate	III	0.14	−0.95	6	0.98
N-Vinylpyrrolidone	II	0.088	−1.62	12	0.95
Vinyltriethoxysilane	V	0.021	0.82		0.61
Vinyltrimethylsilane	V	0.027	0.19		0.58

TABLE 2. TELOGENS

Telogen	Temp. (°C)	Q ($\times 10^4$)	e	No.	r
Acetone	60	0.11	0.35	5	0.80
	80	0.32	0.45	5	0.68
Benzene	60	0.05	−1.21	16	0.94
	80	0.07	−0.61	9	0.72
–, chloro-	60	0.07	0.08	7	0.30
	80	0.13	−0.06	5	0.30
–, ethyl-	60	0.95	−1.02	8	0.84
	80	0.91	−0.61	7	0.86
–, methyl-	60	0.16	−0.95	12	0.93
	80	0.34	−0.87	11	0.94
Benzoyl peroxide	60	36	−1.89	4	0.93
Butanol	60	0.53	−0.57	6	0.63
Butanone	60	0.82	0.53	4	0.60
	80	1.40	1.00	7	0.80
Cyclohexane	60	0.11	−0.64	6	0.86
	80	0.14	−1.41	7	0.77
1,2-Dichloroethane	60	0.62	1.34	3	0.96
	80	1.1	2.07	6	0.96
Ethyl acetate	60	0.07	−0.87	11	0.97
Mercaptan, alkyl-	60	14500	2.44	11	0.99
Mercaptoacetate, alkyl-	60	15100	3.27	4	0.99
Methane, dichloro-	60	0.10	−0.68	4	0.82
–, nitro-	60	3.26	−1.50	4	0.92
–, tetrabromo-	60	7300	2.90	8	0.99
Methane, tetrachloro-	60	3.64	3.21	13	0.99
	80	5.15	3.41	9	0.99
–, trichloro-	60	1.18	0.47	8	0.95
	80	1.24	0.70	10	0.82
Methanol	60	0.18	−0.93	5	0.76
Triethylamine	60	28.8	−2.39	7	0.99

TABLE 3. MONOMERS ARRANGED BY Q VALUES

Monomer	Q	e	Monomer	Q	e
Ethylene, tetrachloro-	0.001	1.24	Vinylene carbonate	0.004	−0.49
Butene-2	0.002	−0.29	Allyl alcohol	0.005	−1.48
Propene, 1-chloro-, *cis-*	0.003	−0.46	Propene, 1-chloro-, *trans-*	0.006	−0.16

TABLE 3. *cont'd*

Monomer	*Q*	*e*
Butene-1	0.007	−0.06
Vinyl fluoride	0.008	0.72
Propylene	0.009	−1.69
Bicyclo[2.2.1]hept-2-ene-5,6 dicarboxamine, *N*-benzyl-	0.010	−0.36
Ethylene, trichloro-	0.010	1.29
Vinyl *o*-cresyl ether	0.010	−8.53
Vinyl laurate	0.011	−0.54
Carbon monoxide	0.013	1.68
Styrene, 3-tri-*n*-butylstannyl-	0.014	−0.61
Vinyl *p*-cresyl ether	0.015	−6.86
Ethylene	0.016	0.05
Vinyl *m*-cresyl ether	0.016	−6.33
Crotonic acid	0.017	0.89
Vinyl ethyl ether	0.018	−1.80
Vinyl 2-chloroethyl ether	0.019	−1.64
Vinyl octyl ether	0.020	−1.57
Vinyltriethoxysilane	0.021	0.82
Vinyl-tris(trimethoxysiloxy)silane	0.022	−0.12
Crotonaldehyde	0.023	0.84
Isobutylene	0.023	−1.20
Isopropenyl acetate	0.023	−0.94
N-Allylstearamide	0.024	−0.41
Vinyl butyrate	0.024	−0.89
Vinyl octadecyl ether	0.024	−1.93
Allyl chloride	0.026	−0.60
Ethylene, chlorotrifluoro-	0.026	1.56
Vinyl acetate	0.026	−0.88
Vinyl propionate	0.027	−0.68
Vinyltrimethylsilane	0.027	0.19
Carbamate, *N,N*-diethyl-, vinyl	0.028	−1.11
Vinyl *N,N*-diethylcarbamate	0.028	−1.10
Vinyl ether	0.029	−1.16
Propene, 3,3,3-trichloro-	0.030	1.00
Vinyl benzoate	0.030	−0.89
Vinyl isobutyl ether	0.030	−1.27
Diallyl phthalate	0.031	−0.26
Ethylene, tetrafluoro-	0.032	1.63
Phosphonate, isopropenyl-, dimethyl	0.032	0.96
Hexene-1	0.035	0.92
Triallyl isocyanurate	0.035	−0.23
Carbamate, *N*-vinyl-, ethyl	0.037	−1.12
Allylbenzene	0.038	0.40
Vinyl bromide	0.038	−0.23
Vinyl butyl ether	0.038	−1.50
Vinyl chloroacetate	0.039	−1.61
Propenyl, 2-methyl-, acetate	0.040	−2.08
Vinyl dodecyl ether	0.041	−1.69
Vinyl formate	0.043	−1.19
Vinyl stearate	0.043	−0.97
Phosphonate, vinyl-, bis(2-chloroethyl)	0.044	1.64
Vinyl pelargonate	0.046	−1.22
Vinyl phenyl ether	0.046	−2.16
Vinyl *tert*-butyl sulfide	0.046	−2.20
Acrylate, trifluoro-, methyl	0.048	1.20
Norbornadiene	0.051	−1.48
Maleate, diethyl	0.053	1.08
Triallyl citrate	0.054	−0.26
Vinyl chloride	0.056	0.16
Vinyl ethyl oxalate	0.056	−0.65
Vinyl hendecanoate	0.056	−0.84
Diallyl melamine	0.059	−1.57
Vinyl dichloroacetate	0.059	−1.38
Vinyl ethyl sulfoxide	0.065	0.05
Propene, 2-chloro-	0.073	−0.16
Oxazolidone, *N*-vinyl-	0.087	−1.70
N-Vinylpyrrolidone	0.088	−1.62
Propene, 2-chloro-3-hydroxy-	0.091	−0.95
Fumarate, diisopropyl	0.11	2.58
Imidazole, 1-vinyl-	0.11	−0.68
Propene, 2,3-dichloro-	0.12	−0.40
Indene	0.13	−0.71
Tetrazole, 1-vinyl-	0.13	−0.14
N-Vinylcaprolactam	0.14	−1.18
Diallylcyanamide	0.14	2.41
Imidazole, 1-vinyl-2-methyl-	0.14	−0.98
Phosphonate, α-carbomethoxyvinyl-diethyl	0.14	−0.04
Triallyl cyanurate	0.14	2.41
9-Vinylanthracene	0.14	0.82
Vinylisocyanate	0.14	−0.95
Acrylate, ferrocenylmethyl	0.15	0.51
Methacrylamide, 1-deoxy-D-glucitol	0.15	−0.16
Styrene, 2,4,6-trimethyl-	0.15	−0.58
Vinyl butylsulfonate	0.16	1.06
Ethylene, 1,1-diphenyl-	0.17	−1.71
Isobutylene, 3-chloro-	0.17	−0.86
Urea, *N*-vinyl-*N'*ethyl-	0.17	−1.88
Isopropenylisocyanate	0.18	−1.05
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	0.18	−0.32
Toluenesulfonamide, *N,N*-methyl-vinyl-	0.18	−0.53
Vinyl cinnamate	0.18	0.76
Succinimide, *N*-vinyl-	0.19	−1.42
Propenyl, 2-chloro-, acetate	0.20	−0.85
Styrene, pentachloro-	0.20	0.79
1-Acrylamido-1-deoxy-D-glucitol	0.22	0.61
Methacrylate, ferrocenylmethyl	0.22	−0.65
Acrylamide	0.23	0.54
Allyl acetate	0.24	−1.07
Phosphonate, vinyl-, dimethyl	0.24	−0.25
Aconitate, trimethyl	0.25	2.27
Fumarate, diethyl	0.25	2.26
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2,5)-	0.25	0.61
Carbazole, *N*-vinyl-	0.26	−1.29
N,N-Divinylaniline	0.26	−0.68
Phosphonate, vinyl-, diethyl	0.27	−0.40
Vinyl ethyl sulfide	0.27	−1.31
Vinyl thiolacetate	0.27	−0.52
Fumaronitrile	0.29	2.73
Vinylferrocene	0.31	−1.34
Vinylidene chloride	0.31	0.34
Allyl acrylate	0.32	−0.99
Acrylate, benzyl	0.33	1.13
Acrylate, octadecyl	0.33	1.26
Vinyl phenyl sulfide	0.33	−0.99
Benzothiazole, vinylmercapto-	0.36	−0.44
Acrylate, 2-ethylhexyl	0.37	0.24
Acrylate, butyl	0.38	0.85
Styrene, α-methoxy-	0.39	−1.02
Vinyl cymantrene	0.39	−0.57
Methacrylamide	0.40	−0.05
Methacrylamide, *N*-phenyl-	0.40	0.19
Acrylamide, *N,N*-dimethyl-	0.41	−0.26
Acrylate, ethyl	0.41	0.55
2-Methyleneglutaronitrile	0.41	1.25
4-Cyclopentene-1,3-dione	0.42	2.43
Vinyl methyl sulfide	0.42	−1.66
Oxazoline, 2-,4-methacryloxymethyl-2,4-dimethyl-	0.44	−0.70
Acetylene, phenyl-	0.45	0.10

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TABLE 3. *cont'd*

Monomer	Q	e
Acrylate, methyl	0.45	0.64
Acrylamide, *N,N*-diethyl	0.48	−0.31
Acrylate, glycidyl	0.48	1.28
Acrylonitrile	0.48	1.23
Acrylate, 2-chloroethyl	0.49	1.03
Vinyl isobutyl sulfide	0.49	−0.95
Acrylamide, *N*-methylol-	0.52	1.15
Acrylate, α-acetoxy-, ethyl	0.52	0.77
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl	0.52	−1.09
Tetrazole, 2-methyl-5-vinyl-	0.55	−0.46
Isopropenyl, 3-(1-cyclohexenyl)acetate	0.57	−0.66
Pyridazinone, 3-(2-vinyl)-6-methyl-	0.57	0.24
Vinyl isopropyl ketone	0.58	−1.16
Oxazoline, 2-isopropenyl-2-	0.59	−0.64
Vinyl isothiocyanate	0.59	0.37
Methacrylate, 2-acetoxyethyl	0.60	0.51
Vinyl methyl sulfoxide	0.60	−1.79
Acrylate, octyl	0.63	2.01
Phosphonic acid, α-phenyl-	0.65	0.52
Acrylamide, *N*-octadecyl-	0.66	1.64
Vinyl methyl ketone	0.66	1.05
Vinyl 4-chlorocyclohexyl ketone	0.66	−0.82
Acrylate, 2-nitrobutyl	0.69	1.09
p-Vinylbenzylethylcarbinol	0.69	−0.98
Styrene, *p*-1-(2-hydroxybutyl)-	0.70	−0.97
Acenaphthalene	0.72	−1.88
Itaconate, dimethyl	0.73	0.57
Methacrylate, ethyl	0.76	0.17
Itaconic acid	0.78	1.07
Methacrylate, methyl	0.78	0.40
Acrolein	0.80	1.31
Butadiene, hexafluoro-	0.82	0.58
Itaconate, dibutyl	0.82	0.57
Methacrylate, butyl	0.82	0.28
Methacrylate, isobutyl	0.82	0.27
Acrylic acid	0.83	0.88
Tetrazole, 2-phenyl-5-(4′-vinyl)-phenyl-	0.84	−0.52
Maleic anhydride	0.86	3.69
Methacrylonitrile	0.86	0.68
Tetrazole, 2-methyl-5-(4′-vinyl)-phenyl-	0.86	0.51
Citraconimide, *N*-methyl-	0.87	1.58
Oxazoline, 2-, 2-isopropenyl-4,4-dimethyl-	0.87	0.34
Maleimide, *N*-butyl-	0.88	3.70
Methacrylate, benzyl	0.88	0.35
Maleimide	0.94	2.86
Acrylate, heptafluorobutyl	0.96	0.20
Methacrylate, isopropyl	0.97	0.10
Oxazoline, 2-,4-acryloxymethyl-2,4-dimethyl-	0.97	−0.51
Styrene, α-methyl-	0.97	−0.81
Methacrylic acid	0.98	0.62
Acrylate, α-chloro-, ethyl	1.00	−1.03
Styrene	1.00	−0.80
Isopropenyl methyl ketone	1.03	0.64
Itaconate, diethyl	1.04	0.88
Methacrylate, 2-naphthyl	1.04	−0.09
Methacrylate, 2,3-epithiopropyl	1.04	0.29
Methacrylate, 2-chloroethyl	1.04	0.31
Quinoline, 2-vinyl-	1.04	−0.09
Silane, γ-methacryloxypropyl-trimethoxy	1.08	0.07
Styrene, *p*-1-(2-hydroxypropyl)-	1.08	−0.35
Methacrylate, 2-(sulfonic acid)ethyl	1.09	0.25
Styrene, *p*-methyl-	1.10	−0.63
Tetrazole, 5-phenyl-2-(4′-vinyl)-phenyl-	1.11	0.53
Methacrylate, 2-(*N,N*-dimethylcarbamoyloxy)ethyl	1.14	0.84
Styrene, *p*-2-(2-hydroxypropyl)-	1.15	−0.49
Vinyl phenyl ketone	1.16	1.02
Methacrylate, sulfolanyl	1.18	0.95
Methacrylate, 2-bromoethyl	1.18	0.74
Methacrylate, phenyl	1.25	0.79
Styrene, *m*-bromo-	1.25	−0.27
Maleimide, *N*-(2-hydroxyethyl)-	1.26	1.07
Pentadiene, *trans*-1,3-	1.28	−0.13
Pyridine, 2-vinyl-5-ethyl-	1.29	−0.91
Styrene, *p*-bromo-	1.30	−0.68
Pyridine, 2-methyl-5-vinyl-	1.32	−0.66
Styrene, *p*-chloro-	1.33	−0.64
Styrene, *p*-chloromethyl-	1.39	−0.38
Pyridine, 2-vinyl-	1.41	−0.42
Butadiene, 2,3-dimethyl-	1.42	−0.43
Acrylic anhydride	1.46	0.31
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	1.50	−0.59
Styrene, 2,5-dichloro-	1.50	0.94
Styrene, *p*-methoxy-	1.53	−1.40
Styrene, *m*-methyl-	1.57	−0.03
Butadiene-1-carboxylate, ethyl	1.67	1.26
Butadiene	1.70	−0.50
p-Vinylbenzylmethylcarbinol	1.70	−1.27
Vinylhydroquinone dibenzoate	1.73	0.84
Methacrylate, 2-hydroxyethyl	1.78	−0.39
Acryloyl chloride	1.82	1.92
Acrolein, methyl-	1.83	0.71
Hexatriene, tetrachloro-	1.83	0.94
Butadiene, 2-fluoro-	1.88	0.63
Phthalimide, *N*-(methacryloyloxy)-	1.89	0.96
Isoprene, 3-acetoxy-	1.91	−0.11
Butadiene, 1,4-dicarboxylate, diethyl	1.94	1.39
Isoprene	1.99	−0.55
Methacryloyl chloride	2.04	1.54
Methacrylate, 2-chloro-2,3,3,3-tetrafluoropropyl	2.09	0.62
Styrene, *m*-nitro-	2.19	0.20
Maleimide, *N*-(2-chlorophenyl)-	2.29	2.87
Acrylate, α-chloro-, methyl	2.43	0.35
Styrene, *m*-chloro-	2.46	−0.90
Pyridine, 4-vinyl-	2.47	0.84
Methylenebutyrolactone	2.48	0.83
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	2.54	1.64
Styrene, *o*-chloro-	2.66	1.57
Maleimide, *N*-phenyl-	2.81	3.24
Styrene, *p*-cyano-	2.93	−0.38
Maleimide, *N*-(4-chlorophenyl)-	2.98	2.75
Methacrylic anhydride	3.00	0.56
Acrylic acid, α-bromo-	3.07	1.37
Vinylhydroquinone	4.30	2.26
Methacrylate, 2-hydroxypropyl	4.38	1.86
Acrylate, α-cyano-, methyl	4.91	0.91
p-Vinylbenzoic acid	5.17	1.08
Acrylate, α-phenyl-, methyl	5.19	0.96
Methacryloylacetone	5.47	−0.76
Butadiene, 2,3-dichloro-	9.08	0.14
Butadiene, 2-chloro-	10.52	1.20
Vinylidene cyanide	14.22	1.92
Vinyl chloromethyl ketone	16.00	1.78

TABLE 4. MONOMERS ARRANGED BY *e* VALUES

Monomer	*Q*	*e*
Vinyl ethyl sulfoxide	0.065	0.05
Ethylene	0.016	0.05
Silane, γ-methacryloxypropyl-trimethoxy	1.08	0.07
Acetylene, phenyl-	0.45	0.10
Methacrylate, isopropyl	0.97	0.10
Butadiene, 2,3-dichloro-	9.08	0.14
Vinyl chloride	0.056	0.16
Methacrylate, ethyl	0.76	0.17
Vinyltrimethylsilane	0.027	0.19
Methacrylamide, *N*-phenyl-	0.40	0.19
Methacrylate, glycidyl	0.96	0.20
Styrene, *m*-nitro-	2.19	0.20
Acrylate, 2-ethylhexyl	0.37	0.24
Pyridazinone, 3-(2-vinyl)-6-methyl-	0.57	0.24
Methacrylate, 2-(sulfonic acid)ethyl	1.09	0.25
Methacrylate, isobutyl	0.82	0.27
Methacrylate, butyl	0.82	0.28
Methacrylate, 2,3-epithiopropyl	1.04	0.29
Methacrylate, 2-chloroethyl	1.04	0.31
Acrylic anhydride	1.46	0.31
Vinylidene chloride	0.31	0.34
Oxazoline, 2-,2-isopropenyl-4,4-dimethyl-	0.87	0.34
Methacrylate, benzyl	0.88	0.35
Acrylate, α-chloro-, methyl	2.43	0.35
Vinyl isothiocyanate	0.59	0.37
Allylbenzene	0.038	0.40
Methacrylate, methyl	0.78	0.40
Acrylate, ferrocenylmethyl	0.15	0.51
Methacrylate, 2-acetoxyethyl	0.60	0.51
Tetrazole, 2-methyl-5-(4′-vinyl)-phenyl-	0.86	0.51
Phosphonic acid, α-phenylvinyl-	0.65	0.52
Tetrazole, 5-phenyl-2-(4′-vinyl)-phenyl-	1.11	0.53
Acrylamide	0.23	0.54
Acrylate, ethyl	0.41	0.55
Methacrylic anhydride	3.00	0.56
Itaconate, dimethyl	0.73	0.57
Itaconate, dibutyl	0.82	0.57
Butadiene, hexafluoro-	0.82	0.58
1-Acrylamideo-1-deoxy-D-glucitol	0.22	0.61
Oct-1-ene, 6,6-dimethyl-4,8-dioxaspiro(2.5)-	0.25	0.61
Methacrylic acid	0.98	0.62
Butadiene, 2-fluoro-	1.88	0.63
Acrylate, methyl	0.45	0.64
Isopropenyl methyl ketone	1.03	0.64
Methacrylonitrile	0.86	0.68
Acrolein, methyl-	1.83	0.71
Vinyl fluoride	0.008	0.72
Methacrylate, 2-bromoethyl	1.18	0.74
Vinyl cinnamate	0.18	0.76
Acrylate, α-acetoxy-, ethyl	0.52	0.77
Styrene, pentachloro-	0.20	0.79
Methacrylate, phenyl	1.25	0.79
Vinyltriethoxysilane	0.021	0.82
9-Vinylanthracene	0.14	0.82
Methylenebutyrolactone	2.48	0.83
Vinylhydroquinone dibenzoate	1.73	0.84
Crotonaldehyde	0.023	0.84
Methacrylate, 2-(*N*,*N*-dimethylcarbamoyloxy)ethyl	1.14	0.84
Pyridine, 4-vinyl-	2.47	0.84
Acrylate, butyl	0.38	0.85
Acrylic acid	0.83	0.88
Itaconate, diethyl	1.04	0.88
Crotonic acid	0.017	0.89
Acrylate, α-cyano-, methyl	4.91	0.91
Hexene-1	0.035	0.92
Styrene, 2,5-dichloro-	1.50	0.94
Hexatriene, tetrachloro-	1.83	0.94
Methacrylate, sulfolanyl	1.18	0.95
Phosphonate, isopropenyl-, dimethyl	0.032	0.96
Phthalimide, *N*-(methacryloyloxy)-	1.89	0.96
Acrylate, α-phenyl-, methyl	5.19	0.96
Propene, 3,3,3-trichloro-	0.030	1.00
Vinyl phenyl ketone	1.16	1.02
Acrylate, 2-chloroethyl	0.49	1.03
Vinyl methyl ketone	0.66	1.05
Vinyl butylsulfonate	0.16	1.06
Itaconic acid	0.78	1.07
Maleimide, *N*-(2-hydroxyethyl)-	1.26	1.07
p-Vinylbenzoic acid	5.17	1.08
Maleate, diethyl	0.053	1.08
Acrylate, 2-nitrobutyl	0.69	1.09
Acrylate, benzyl	0.33	1.13
Acrylamide, *N*-methylol-	0.52	1.15
Acrylate, trifluoro-, methyl	0.048	1.20
Butadiene, 2-chloro-	10.52	1.20
Acrylonitrile	0.48	1.23
Ethylene, tetrachloro-	0.001	1.24
2-Methyleneglutaronitrile	0.41	1.25
Acrylate, octadecyl	0.33	1.26
Butadiene-1-carboxylate, ethyl	1.67	1.26
Acrylate, glycidyl	0.48	1.28
Ethylene, trichloro-	0.010	1.29
Acrolein	0.80	1.31
Acrylate, heptafluorobutyl	0.96	1.34
Acrylic acid, α-bromo-	3.07	1.37
Butadiene, 1,4-dicarboxylate, diethyl	1.94	1.39
Methacryloyl chloride	2.04	1.54
Ethylene, chlorotrifluoro-	0.026	1.56
Styrene, *o*-chloro-	2.66	1.57
Citraconimide, *N*-methyl-	0.87	1.58
Ethylene, tetrafluoro-	0.032	1.63
Phosphonate, vinyl-, bis(2-chloro-ethyl)	0.044	1.64
Acrylamide, *N*-octadecyl-	0.66	1.64
Maleimide, 2,3-dimethyl-*N*-(2-methacryloxyethyl)-	2.54	1.64
Carbon monoxide	0.013	1.68
Vinyl chloromethyl ketone	16.00	1.78
Methacrylate, 2-hydroxypropyl	4.38	1.86
Vinylidene cyanide	14.22	1.92
Acryloyl chloride	1.82	1.92
Acrylate, octyl	0.63	2.01
Methacrylate, 2-chloro-2,3,3,3-tetrafluoropropyl	0.62	2.09
Vinylhydroquinone	4.30	2.26
Fumarate, diethyl	0.25	2.26
Aconitate, trimethyl	0.25	2.27
Fumarate, diisopropyl	0.11	2.58
Triallyl cyanurate	0.14	2.41
Diallylcyanamide	0.14	2.41
4-Cyclopentene-1,3-dione	0.42	2.43
Fumaronitrile	0.29	2.73
Maleimide, *N*-(4-chlorophenyl)-	2.98	2.75
Maleimide	0.94	2.86
Maleimide, *N*-(2-chlorophenyl)-	2.29	2.87
Maleimide, *N*-phenyl-	2.81	3.24
Maleic anhydride	0.86	3.69
Maleimide, *N*-butyl-	0.88	3.70
Styrene, *m*-methyl-	1.57	−0.03
Phosphonate, α-carbomethoxyvinyl-, diethyl	0.14	−0.04

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TABLE 4. *cont'd*

Monomer	Q	e
Methacrylamide	0.40	−0.05
Butene-1	0.007	−0.06
Methacrylate, 2-naphthyl	1.04	−0.09
Quinoline, 2-vinyl-	1.04	−0.09
Isoprene, 3-acetoxy-	1.91	−0.11
Vinyl-tris(trimethoxysiloxy)silane	0.022	−0.12
Pentadiene, *trans*-1,3-	1.28	−0.13
Tetrazole, 1-vinyl-	0.13	−0.14
Propene, 1-chloro-, *trans*-	0.006	−0.16
Propene, 2-chloro-	0.073	−0.16
Methacrylamide,1-deoxy-D-glucitol	0.15	−0.16
Triallyl isocyanurate	0.035	−0.23
Vinyl bromide	0.038	−0.23
Phosphonate, vinyl-, dimethyl	0.24	−0.25
Triallyl citrate	0.054	−0.26
Diallyl phthalate	0.031	−0.26
Acrylamide, *N*,*N*-dimethyl-	0.41	−0.26
Styrene, *m*-bromo-	1.25	−0.27
Butene-2	0.002	−0.29
Acrylamide, *N*,*N*-diethyl	0.48	−0.31
Pyridazinone, 3-(2-vinyl)-6-methyl-	0.18	−0.32
Styrene, *p*-1-(2-hydroxypropyl)-	1.08	−0.35
Bicyclo[2.2.1]hept-2-ene-5,6 dicarboxamine, *N*-benzyl-	0.01	−0.36
Styrene, *p*-chloromethyl-	1.39	−0.38
Styrene, *p*-cyano-	2.93	−0.38
Methacrylate, 2-hydroxyethyl	1.78	−0.39
Propene, 2,3-dichloro-	0.12	−0.40
Phosphonate, vinyl-, diethyl	0.27	−0.40
N-Allylstearamide	0.024	−0.41
Pyridine, 2-vinyl-	1.41	−0.42
Butadiene, 2,3-dimethyl-	1.42	−0.43
Benzothiazole, vinylmercapto-	0.36	−0.44
Tetrazole, 2-methyl-5-vinyl-	0.55	−0.46
Propene, 1-chloro-, *cis*-	0.003	−0.46
Vinylene carbonate	0.004	−0.49
Styrene, *p*-2-(2-hydroxypropyl)-	1.15	−0.49
Butadiene	1.70	−0.50
Oxazoline, 2,4-acryloxymethyl-2,4-dimethyl-	0.97	−0.51
Vinyl thiolacetate	0.27	−0.52
Tetrazole, 2-phenyl-5-(4′-vinyl)-phenyl-	0.84	−0.52
Toluenesulfonamide, *N*,*N*-methyl-vinyl-	0.18	−0.53
Vinyl laurate	0.011	−0.54
Isoprene	1.99	−0.55
Vinyl cymantrene	0.39	−0.57
Styrene, 2,4,6-trimethyl-	0.15	−0.58
Pyrrolidone, 1-benzyl-3-methylene-5-methyl-	1.50	−0.59
Allyl chloride	0.026	−0.60
Styrene, 3-tri-*n*-butylstannyl-	0.014	−0.61
Styrene, *p*-methyl-	1.10	−0.63
Oxazoline, 2-isopropenyl-2-	0.59	−0.64
Styrene, *p*-chloro-	1.33	−0.64
Vinyl ethyl oxalate	0.056	−0.65
Methacrylate, ferrocenylmethyl	0.22	−0.65
Isopropenyl, 3-(1-cyclohexenyl)-acetate	0.57	−0.66
Pyridine, 2-methyl-5-vinyl-	1.32	−0.66
Vinyl propionate	0.027	−0.68
Imidazole, 1-vinyl-	0.11	−0.68
N,*N*-Divinylaniline	0.26	−0.68
Styrene, *p*-bromo-	1.30	−0.68
Oxazoline, 2-,4-methacryloxymethyl-2,4-dimethyl-	0.44	−0.70
Indene	0.13	−0.71
Methacryloylacetone	5.47	−0.76
Styrene	1.00	−0.80
Styrene, α-methyl-	0.97	−0.81
Vinyl 4-chlorocyclohexyl ketone	0.66	−0.82
Vinyl hendecanoate	0.056	−0.84
Propenyl, 2-chloro-, acetate	0.20	−0.85
Isobutylene, 3-chloro-	0.17	−0.86
Vinyl acetate	0.026	−0.88
Vinyl butyrate	0.024	−0.89
Vinyl benzoate	0.030	−0.89
Styrene, *m*-chloro-	2.46	−0.90
Pyridine, 2-vinyl-5-ethyl-	1.29	−0.91
Isopropenyl acetate	0.023	−0.94
Vinylisocyanate	0.14	−0.95
Vinyl isobutyl sulfide	0.49	−0.95
Propene, 2-chloro-3-hydroxy-	0.091	−0.95
Vinyl stearate	0.043	−0.97
Styrene, *p*-1(2-hydroxybutyl)-	0.70	−0.97
p-Vinylbenzylethylcarbinol	0.69	−0.98
Imidazole, 1-vinyl-2-methyl-	0.14	−0.98
Vinyl phenyl sulfide	0.33	−0.99
Allyl acrylate	0.32	−0.99
Styrene, α-methoxy-	0.39	−1.02
Acrylate, α-chloro-, ethyl	1.00	−1.03
Isopropenylisocyanate	0.18	−1.05
Allyl acetate	0.24	−1.07
Methacrylate, 2,2,6,6-tetramethyl-piperidinyl	0.52	−1.09
Vinyl *N*,*N*-diethylcarbamate	0.028	−1.10
Carbamate, *N*,*N*-diethyl-, vinyl	0.028	−1.11
Carbamate, *N*-vinyl-, ethyl	0.037	−1.12
Vinyl ether	0.029	−1.16
Vinyl isopropyl ketone	0.58	−1.16
N-Vinylcaprolactam	0.14	−1.18
Vinyl formate	0.043	−1.19
Isobutylene	0.023	−1.20
Vinyl pelargonate	0.046	−1.22
Vinyl isobutyl ether	0.030	−1.27
p-Vinylbenzylmethylcarbinol	1.70	−1.27
Carbazole, *N*-vinyl-	0.26	−1.29
Vinyl ethyl sulfide	0.27	−1.31
Vinylferrocene	0.31	−1.34
Vinyl dichloroacetate	0.059	−1.38
Styrene, *p*-methoxy-	1.53	−1.40
Succinimide, *N*-vinyl-	0.19	−1.42
Allyl alcohol	0.005	−1.48
Norbornadiene	0.051	−1.48
Vinyl butyl ether	0.038	−1.50
Vinyl octyl ether	0.020	−1.57
Diallyl melamine	0.059	−1.57
Vinyl chloroacetate	0.039	−1.61
N-Vinylpyrrolidone	0.088	−1.62
Vinyl 2-chloroethyl ether	0.019	−1.64
Vinyl methyl sulfide	0.42	−1.66
Vinyl dodecyl ether	0.041	−1.69
Propylene	0.009	−1.69
Oxazolidone, *N*-vinyl-	0.087	−1.70
Ethylene, 1,1-diphenyl-	0.17	−1.71
Vinyl methyl sulfoxide	0.60	−1.79
Vinyl ethyl ether	0.018	−1.80
Urea, *N*-vinyl-*N*′ethyl-	0.17	−1.88
Acenaphthalene	0.72	−1.88
Vinyl octadecyl ether	0.024	−1.93
Propenyl, 2-methyl-, acetate	0.040	−2.08
Vinyl phenyl ether	0.046	−2.16
Vinyl *tert*-butyl sulfide	0.046	−2.20
Vinyl *m*-cresyl ether	0.016	−6.33
Vinyl *p*-cresyl ether	0.015	−6.86
Vinyl *o*-cresyl ether	0.010	−8.53

C. REFERENCES

1. T. Alfrey, C. C. Price, J. Polym. Sci., **2**, 101 (1947).
2. R. Z. Greenley, J. Macromol. Sci. A: Chem., **9**, 505 (1975).
3. L. J. Young, in: J. Brandrup, E. H. Immergut (Eds.), "Polymer Handbook", 2nd ed., Wiley, New York, 1975.
4. R. Z. Greenley, J. Macromol. Sci. A: Chem., **11**, 933 (1977).
5. A. D. Jenkins, J. Polym. Sci.: Part A: Polym. Chem., **34**, 3495 (1996).

Patterns of Reactivity (U,V) Parameters for the Prediction of Monomer Reactivity Ratios and Transfer Constants in Radical Polymerization

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A. INTRODUCTION

The relation between monomer reactivity ratios and the Alfrey–Price Q–e parameters is explained in the introduction to the tables of monomer reactivity ratios and Q–e values, compiled by Robert Z. Greenley and published in the present volume (1,2). Although very widely used, the Q–e scheme is well known to have serious limitations (3), which have prompted several attempts to improve upon it. One such endeavour was the "Patterns of Reactivity" scheme, first described as long ago as 1959 (4–7), when the Q–e scheme was only about ten years old; despite the indisputably more satisfactory basis of this procedure, it did not achieve popularity but recent revisions have greatly improved both its accessibility and its accuracy (3,8,9).

In the Q–e scheme, four parameters (Q_1, Q_2, e_1, e_2; two for each monomer) are necessary for the prediction of a monomer reactivity ratio (e.g. $r_{12} = k_{11}/k_{12}$, where subscript 1 refers to the radical and subscript 2 to the monomer) but the same four parameters also facilitate the prediction of the partner monomer reactivity ratio ($r_{21} = k_{22}/k_{21}$) because identical Q and e parameters are used to characterise both a given monomer and its derived radical. This device is certainly economical on input data but it introduces an approximation of very doubtful validity.

In the "Patterns" scheme (to use the abbreviated title), different parameters are designated for the monomer and its conjugate radical. Hence, while four input parameters are again necessary for the calculation of a single monomer reactivity ratio, eight are required to calculate both r_{12} and r_{21}. The reward for the greater input of data is a substantial increase in the precision of the result; furthermore, all four parameters for each monomer/radical conjugate pair are experimentally determined from polymerization data, specifically monomer reactivity ratios from Greenley's tabulation, so no arbitrary assignment is involved, as it is with the Q–e scheme.

The fundamental equation for the calculation of a reactivity ratio, r_{12}, is given below.

$$\log r_{12} = \log r_{1S} - u_2\pi_1 - v_2. \qquad \text{(A1)}$$

Here, the symbol S denotes the monomer styrene, and $\log r_{1S}$ is the counterpart of Q_1 in the Q–e scheme. The counterpart of e_1 is the polarity parameter π_1; this is usually almost exactly equal to the Hammett σ parameter for the substituent(s) present on the α-carbon atom of the radical derived from monomer 1 but it is best calculated (8) from monomer reactivity ratio data according to Eq. (A2).

$$\pi_1 = 0.385 \log[(r_{1A})/0.377(r_{1S})] \qquad \text{(A2)}$$

Here, the symbol A denotes the monomer acrylonitrile. It is truly an astonishing fact that the Hammett σ parameter, derived originally from studies of the dissociation of substituted benzoic acids, is equal in value to a simple quantity derived from a knowledge of the monomer reactivity ratios for the reactions of a monomer 1 with (separately) acrylonitrile and styrene.

The values of u_2 and v_2, the respective counterparts of the Q–e scheme's e_2 and Q_2, are determined by reference to data for the (separate) copolymerizations of monomer 2 with the members of a Basic Monomer Set. These are five monomers for which reliable data exist in the literature: styrene (S); methyl methacrylate (MM); methyl acrylate

(MA); methacrylonitrile (MAN); and acrylonitrile (A). Ideally, monomer reactivity ratios should be known for the copolymerization of the monomer of interest (labelled 2 in this case) with each of these five reference monomers. If the reference monomer is monomer 1 in this context, a plot is made of $[\log r_{12} - \log r_{1S}]$ against π_1; the slope of the resulting straight line is $-u_2$ and the intercept on the ordinate axis is $-v_2$, in conformity with Eq. (A1).

If data for reactions with all five members of the Basic Monomer Set are not available, use can be made of such data as exists, always provided that styrene and acrylonitrile are among the monomers included. This condition ensures that the data are spread over a wide range of radical polarity, represented by π_1, because styrene has a very low value (zero), while acrylonitrile has one of the highest values known (0.701).

In Table 1, each monomer is designated either as Basic or assigned to a category according to the number of members of the Basic Monomer Set for which data have been employed in the determination of its u_2 and v_2; for category 5, all five Basic monomers were involved, and so on down to category 2, where data for only styrene and acrylonitrile are available. Clearly, the higher the category number, the more reliable the u_2 and v_2 values. The procedure described thus far is known as the *Patterns U,V Scheme* (3).

For Group 2 monomers, there is no need to make a plot because the use of just two data points permits algebraic solutions to be found. In fact, it can be shown (3) that the following relations hold.

$$u_2 = [\log r_{AS} - \log r_{A2} + \log r_{S2}]/\pi_A \quad \text{and} \quad v_2 = -\log r_{S2} \tag{A3}$$

Substitution in Eq. (A1) leads to the following relationships (A4, A5), giving the two monomer reactivity ratios for the copolymerization of any two monomers, 1 and 2.

$$\log(r_{12}) = \log[(r_{1S})(r_{S2})] - \frac{\{\log[(r_{AS})(r_{S2})/(r_{A2})]\}\{\log[(r_{SA})(r_{1S})/(r_{1A})]\}}{\log[(r_{AS})(r_{SA})]} \tag{A4}$$

$$\log(r_{21}) = \log[(r_{2S})(r_{S1})] - \frac{\{\log[(r_{AS})(r_{S1})/(r_{A1})]\}\{\log[(r_{SA})(r_{2S})/(r_{2A})]\}}{\log[(r_{AS})(r_{SA})]} \tag{A5}$$

Application of Eqs. (A3–A5) corresponds to the *Patterns A,S Scheme* (4), so-called, because data for reaction with only acrylonitrile and styrene are involved. According to the Patterns A,S Scheme, it is possible to calculate the two monomer reactivity ratios for copolymerization of monomers 1 and 2, if each of them has separately been copolymerized with acrylonitrile and styrene. (The two monomer reactivity ratios for copolymerization of acrylonitrile and styrene are also required. These are taken to be: $r_{AS} = 0.04$, $r_{SA} = 0.38$ (3).)

The π, u, and v values listed in the table below have been derived as explained above, with the Patterns U,V Scheme being used for the monomers in categories 3–5 and the Patterns A,S Scheme for those in category 2. The r_{1S} values are mean values from the figures supplied in Greenley's table, making due allowance for the consistency of the data.

B. EXAMPLE

Suppose one wants to evaluate the monomer reactivity ratios for the copolymerization of 2-chlorobutadiene (CB) and 2-vinylpyridine (VP). To use the Patterns U,V Scheme for this purpose, it is first necessary to consult Greenley's table of monomer reactivity ratios for data characterizing the copolymerizations of each of these monomers with as many members of the Basic Monomer Set as possible. In fact, in both cases, values exist for reactions with styrene, methyl methacrylate, methyl acrylate and acrylonitrile, but not methacrylonitrile. The π values for the four useful basic monomers are, respectively, 0.000, 0.339, 0.423 and 0.701. The relevant monomer reactivity ratios are listed in the table below.

MONOMER REACTIVITY RATIOS FOR 2-CHLOROBUTADIENE AND 2-VINYLPYRIDINE AND MEMBERS OF THE BASIC MONOMER SET

Monomer reactivity ratio	2-Chlorobutadiene	2-Vinylpyridine
$r_{X,S}$	6.91	1.26
$r_{X,A}$	5.18	0.44
$r_{S,X}$	0.038	0.53
$r_{A,X}$	0.05	0.10
$r_{MM,X}$	0.08	0.35
$r_{MA,X}$	0.06	0.25

X = 2-chlorobutadiene or 2-vinylpyridine.

Plots of $[\log r_{12} - \log r_{1S}]$ against π_1, where 2 = CB or VP, are linear with slopes and intercepts that provide the following u and v values (9). For CB, $u = -2.18$, $v = 1.44$, and for VP, $u = -0.982$ and $v = 0.323$. All the data are now available for substitution in Eq. (A1), first for CB = monomer 1 and VP = monomer 2, and secondly with the monomers' roles reversed, to obtain values of r_{12} and r_{21}. The results are $r_{12} = 4.26$ and $r_{21} = 0.04$. The Patterns A,S Scheme can be applied by simply substituting the appropriate monomer reactivity ratio data in Eqs. (A4) and (A5), giving $r_{12} = 4.71$ and $r_{21} = 0.05$. The corresponding experimental results are $r_{12} = 5.19$ and $r_{21} = 0.06$, while the $Q = e$ scheme predicts that $r_{12} = 1.07$ and $r_{21} = 0.07$.

C. TRANSFER CONSTANTS

Transfer constants can be predicted by exactly parallel reasoning (3). It is necessary only to realise that whereas the rate constant for the propagation step of a radical with its own parent monomer appears in the *numerator* of a monomer reactivity ratio ($r_{12} = k_{11}/k_{12}$), where both rate

constants refer to propagation reactions, it figures in the *denominator* of a transfer constant because $(C_2)_1 = k_{12}/k_{11}$, where k_{12} represents a transfer reaction between the radical derived from a monomer (species 1) and a transfer agent (species 2); the equations for calculating transfer constants are thus easily obtained from the equations above by replacing $\log r_{12}$ by $-\log (C_2)_1$. The only formal difference between the two cases is that, for transfer, there is no symmetrical counterpart of the equations representing reaction of a radical of type 2 with a transfer agent of type 1. Thus, the equivalent of Eq. (A1), for transfer is Eq. (A6), and this enables the Patterns U,V Scheme to be applied to chain transfer.

$$-\log(C_2)_1 = \log r_{1S} - u_2\pi_1 - v_2 \qquad \text{(A6)}$$

For use of the Patterns A,S Scheme in transfer, a parallel derivation to that used in copolymerization leads to the following equation, where $(C_2)_S$ and $(C_2)_A$ are, respectively, the (known) transfer constants for reaction of the same transfer agent (species 2) with radicals derived from styrene and acrylonitrile (species 1).

$$\log(C_2)_1 = \log[(C_2)_S/r_{1S}] + (\pi_1/\pi_A)\log[r_{AS}(C_2)_A/(C_2)_S] \qquad \text{(A7)}$$

Since $\pi_A = 0.701$ and $r_{AS} = 0.04$, this equation can be reduced to Eq. (A8) for general use.

$$\log(C_2)_1 = \log[(C_2)_S/r_{1S}] + (1.43\pi_1)\log[0.04(C_2)_A/(C_2)_S] \qquad \text{(A8)}$$

The transfer constants used here are taken from two tabulations, one due to Eastmond (10) and one in the *Polymer Handbook* (11).

D. TABLES OF PARAMETERS

As explained above, each monomer or transfer agent is assigned to a category, designated by Basic or the number 2, 3, 4, or 5, according to the number of members of the Basic Monomer Set with which this reagent has been reacted to obtain data for the evaluation of u, v and (in the case of monomers) π. Although it must be true that the higher the category, the more reliable the derived parameters, the difference between Categories 2 and 5 is not so great as might appear at first sight. In *all* cases, two of the basic monomers involved are styrene, S, (the least polar) and acrylonitrile, A, (the most polar), so the role played by other basic monomers is merely to add intermediate points to what should be a straight line joining the data points for S and A; if the data for these two latter monomers are accurate, intermediate points add little or no value beyond confirmation of the slope and intercept. They do contribute valuable weighting when there are some discrepancies in the data, but parameters have not been recorded here in cases where the discrepancies are large.

Where *either* r_{1S} *or* r_{1A} was reported as zero, the value 0.05 has been assigned arbitrarily in order to make it possible to calculate an approximate value for the π parameter but the values of the u and v parameters are not influenced by this device; entries of this type are printed in italics in Table 1. Sometimes both r_{1S} *and* r_{1A} are reported to be zero; when this happens, π cannot usefully be estimated but u and v can still be determined, if positive values of r_{S1} and r_{A1} are available.

Nomenclature In the tables below, compounds are listed in alphabetical sequence in accordance with the following rules:

The primary listing is based on the root name of the compound and the secondary sequence on the name(s) of the substituent(s) or the esterifying moiety. For example, the compounds, *p*-methyl styrene, *N*-vinyl carbazole, and ethyl acrylate, will be found listed under Styrene, *p*-methyl, Carbazole, *N*-vinyl, and Acrylate, ethyl, respectively.

In deciding on the appropriate order of names, *all prefixes (alphanumeric, Greek or whatever) and spaces* are ignored, and reliance is placed solely on the alphabetical priority of the strictly chemical part of the name. In styrene, *p*-methyl, for example, the prefix *p*- plays no part in determining the place of this name in the list; this is governed by the *m* in methyl.

TABLE 1. MONOMERS

Monomer	Category	r_{1S}	r_{1A}	r_{A1}	r_{S1}	$\log(r_{1S})$	π	u	v
Acenaphthalene	4	0.33	2.56	0.02	3.81	−0.4815	0.506	1.33	−0.46
Acetylene, phenyl	5	0.33	0.33	0.27	0.32	−0.4815	0.159	−1.88	0.37
Aconitate, trimethyl	2			4.24	1.026			−2.88	−0.01
Acrolein	5	0.27	1.11	0.78	0.23	−0.5686	0.397	−2.75	0.59
Acrolein, methyl	4	0.60	3.10	0.15	0.26	−0.2219	0.436	−1.73	0.77
Acrylamide	3	0.70	1.10	0.90	1.20	−0.1549	0.237	−1.82	−0.07
Acrylamide, *N*-methylol	4	0.48	2.43	0.60	0.03	−0.3188	0.432	−3.94	1.18
Acrylamide, *N*-octadecyl	3	0.54	1.40	1.03	2.08	−0.2676	0.321	−1.56	−0.35
Acrylate, benzyl	3	0.20	0.72	1.49	0.49	−0.6990	0.376	−2.68	0.28
Acrylate, butyl	3	0.18	0.97	1.11	0.77	−0.7447	0.443	−2.22	0.12
Acrylate, 2-chloroethyl	4	0.12	0.87	1.03	0.53	−0.9208	0.492	−2.38	0.25
Acrylate, α-chloro-, methyl	3	0.30	1.76	0.12	0.25	−0.5229	0.457	−1.55	0.64

TABLE 1. *cont'd*

Monomer	Category	r_{1S}	r_{1A}	r_{A1}	r_{S1}	$\log(r_{1S})$	π	u	v
Acrylate, α-cyano-, methyl	5	0.61	0.68	0.01	0.05	−0.2147	0.180	−1.09	1.20
Acrylate, 3,4-epoxyhexahydrobenzyl	2	1.97	0.39	0.25	0.27	0.2945	−0.110	−1.95	0.57
Acrylate, β-ethoxy-, ethyl	2			2.42	46.98			−0.16	−1.67
Acrylate, ethyl	3	0.17	0.87	0.80	0.81	−0.7696	0.436	−1.99	0.08
Acrylate, methyl	B	0.18	0.85	1.42	0.75	−0.7447	0.421	−2.34	0.16
Acrylate, octadecyl	3	0.26	1.20	3.15	0.61	−0.5850	0.417	−3.01	0.24
Acrylate, octyl	2	0.125	0.84	1.98	0.39	−2.0000	0.482	−3.01	0.41
Acrylate, 2-nitrobutyl	3	0.12	1.76	0.67	0.35	−0.9208	0.610	−2.39	0.44
Acrylate, α-phenyl-, methyl	5	1.28	6.70	0.08	0.06	0.1072	0.438	−2.13	1.20
Acrylate, di-zinc	2	0.90	0.24	0.41	1.10	−0.0458	−0.060	−1.39	−0.04
Acrylonitrile	B	0.04	1.00	1.00	0.38	−1.3980	0.701	−2.60	0.42
Acryloyl chloride	4	0.02	1.00	1.20	0.10	−1.6990	0.815	−3.50	1.09
Allyl acetate	3			6.57	90			−0.37	−1.97
Allyl chloride	3	0.04	0.04	2.80	36	−1.3980	0.162	−0.39	−1.53
Aniline, N,N-divinyl	*2*	*0.033*	*0.05*	*0.246*	*5.38*	*−1.482*	*0.231*	*−0.08*	*−0.73*
Benzothiazole, vinylmercapto-	3	0.42	0.18	0.19	2.6	−0.3768	0.020	−0.38	−0.38
Butadiene	5	1.40	0.29	0.06	0.57	0.1461	−0.100	−0.30	0.41
Butadiene-1-carboxylic acid	2	5.55	7.5	0.12	0.115	0.7443	0.213	−2.00	0.94
Butadiene-1-carboxylate, ethyl	2	0.3	3.2	0.3	0.12	−0.5229	0.559	−2.57	0.92
Butadiene-1,4-dicarboxylic acid	2	0.52	4.00	0.20	0.15	−0.2840	0.502	−2.18	0.82
Butadiene, 1,4-dicarboxylate-, diethyl	2	0.55	2.79	0.29	0.09	−0.2596	0.433	−2.73	1.05
Butadiene, 2-chloro-	4	6.91	5.18	0.05	0.038	0.8395	0.113	−2.18	1.44
Butadiene, 2-fluoro-	3	1.67	0.60	0.077	0.22	0.2227	−0.0095	−1.32	0.51
Butadiene, 2-trimethylsilyloxy-	2	1.20	0.07	0.036	0.64	0.0792	−0.324	−0.21	0.19
Carbazole, *N*-vinyl	4			0.39	5.82			−0.25	−0.58
Cinnamonitrile	*2*	*0.05*	*0.36*	*8.46*	*2.55*	*−1.3010*	*0.491*	*−2.74*	*−0.41*
Citraconimide, *N*-methyl-	3	0.24	0.60	0.53	0.145	−0.6198	0.315	−2.77	0.59
Crotonaldehyde	2	0.07	0.01	25	14.7	−1.1549	−0.164	−2.33	−1.17
Crotonate, α-acetyl-, methyl	2			8.68	2.7			−2.72	−0.43
Crotonate, α-carboethoxy-, ethyl	2			18.7	8.24			−2.51	−0.92
Crotonate, α-chloro-, ethyl	2			9.53	5.13			−2.38	−0.71
Crotonate, α-cyano-, ethyl	2	0.02	0.06	11.4	0.26	−1.6990	0.345	−4.35	0.59
Crotonate, ethyl	2			25.2	27			−1.96	−1.43
Crotonate, α-methoxy-, methyl	*2*	*0.04*	*0.05*	*1.00*	*18.4*	*−1.3980*	*0.199*	*−0.19*	*−1.26*
Crotonate, α-methyl-, methyl	*2*	*0.02*	*0.05*	*2.97*	*30.7*	*−1.699*	*0.315*	*−0.55*	*−1.49*
Crotonic acid	2			21	20	−1.301		−2.03	−1.30
Diallyl phthalate	4	0.076	0.0385	3.5	23.5	−1.12	0.048	−0.77	−1.35
Ethylene	*3*	*0.05*	*0.05*	*7*	*14.88*	*−1.301*	*0.162*	*−1.57*	*−1.23*
Ethylene, tetrachloro-	3			463	195			−2.60	−2.36
Ethylene, trichloro-	4			64.5	14.6			−2.87	−1.20
Ethylene, diphenyl-	5			0.05	0.335			−0.74	0.23
Fumarate, diethyl	3	0.06	0.05	9	0.33	−1.222	0.131	−4.05	0.56
Fumarate, diisopropyl	5	0.055	0.20	16	0.4	−1.26	0.377	−4.31	0.17
Hexatriene, tetrachloro-	4	0.85	4.01	0.234	0.123	−0.0706	0.421	−2.39	0.86
Imidazole, *N*-vinyl	3	0.68	0.832	3.39	8.38	−0.1675	0.195	−1.44	−0.92
Isoprene	4	1.84	0.45	0.03	0.458	0.2648	−0.074	−0.32	0.34
Isopropenyl isocyanate	3	0.096	0.1	0.24	7.48	−1.0177	0.170	0.14	−0.87
Isopropenyl methyl ketone	2	0.48	0.95	0.33	0.38	−0.3188	0.276	−1.91	0.42
Itaconic acid	3	0.12	0.86	0.59	0.26	−0.9208	0.491	−2.50	0.54
Itaconic anhydride	2	0.55	4.83	0.034	0.031	−0.2596	0.526	−2.06	1.51
Maleate, diethyl	*3*	*0.04*	*0.05*	*16*	*7.03*	*−1.398*	*0.199*	*−2.51*	*−0.83*
Maleic anhydride	*4*	*0.011*	*0.05*	*6*	*0.36*	*−1.959*	*0.416*	*−5.20*	*1.22*
Maleimide, *N*-(2-chlorophenyl)-	3	0.021	1.078	0.956	0.0245	−1.688	0.826	−4.23	1.32
Methacrylamide, *N*-phenyl-	3	0.88	0.71	0.381	1.24	−0.0555	0.126	−1.26	−0.08
Methacrylate, benzyl	3	0.47	0.96	0.2	0.52	−0.3279	0.281	−1.40	0.26
Methacrylate, 2-bromoethyl	3	0.41	2.38	0.31	0.32	−0.3872	0.455	−1.97	0.47
Methacrylate, butyl	4	0.53	0.98	0.291	0.6	−0.2757	0.267	−1.49	0.26
Methacrylate, isobutyl	4	0.42	1.05	0.217	0.54	−0.3768	0.315	−1.43	0.27

TABLE 1. *cont'd*

Monomer	Category	r_{1S}	r_{1A}	r_{A1}	r_{S1}	log (r_{1S})	π	u	v
Methacrylate, 2-chloroethyl	3	0.3	1.30	0.14	0.37	−0.5229	0.406	−1.39	0.42
Methacrylate, ferrocenylmethyl	2	0.01	0.82	0.15	3.6	−2	0.898	−0.03	−0.56
Methacrylate, glycidyl	3	0.5	1.32	0.14	0.415	−0.301	0.325	−1.32	0.34
Methacrylate, 2-hydroxyethyl	3	0.64	1	0.2	0.48	−0.1938	0.238	−1.45	0.34
Methacrylate, methyl	B	0.46	1.32	0.138	0.5	−0.3372	0.339	−1.18	0.23
Methacrylate, 3,5-dimethyladamantyl	3	0.63	1.30	0.19	0.89	−0.2007	0.283	−1.04	0.10
Methacrylate, 2,2,6,6-tetramethyl-4-piperidinyl-	2	0.30	14	0.02	0.63	−0.5229	0.806	0.14	0.20
Methacrylate, phenyl	3	0.51	0.46	0.36	0.25	−0.2924	0.146	−2.22	0.62
Methacrylic acid	4	0.524	0.2	0.04	0.24	−0.2807	0.002	−0.95	0.39
Methacrylonitrile	B	0.33	1.67	0.43	0.38	−0.4815	0.432	−2.08	0.44
Methacryloylacetone	2	1.66	3.74	0.01	0.067	0.2202	0.297	−0.82	1.17
Methylenebutyrolactone	3	0.7	1.10	0.09	0.09	−0.1549	0.237	−1.98	0.88
Naphthalene, 1-vinyl-	2	2.02	0.451	0.107	0.699	0.3054	−0.089	−0.83	0.16
Oct-1-ene,6,6-dimethyl-4,8-dioxaspiro(2,5)-	2	0.271	1.484	0.985	1.93	−0.567	0.446	−1.58	−0.29
Oxazoline, 2,2-isopropenyl-	4	0.64	0.52	0.13	0.67	−0.1938	0.127	−0.95	0.19
Oxazoline, 2,2-isopropenyl-4,4-dimethyl-	4	0.68	1.83	0.24	0.55	−0.1675	0.327	−1.51	0.31
Pentadienoate, *trans*-4-ethoxy-2,4-,ethyl	2	12.2	5.40	0.01	0.04	1.0864	0.025	−1.14	1.40
Phthalimide, *N*-vinyl-	2	0.07	0.24	0.43	6.27	−1.155	0.367	−0.33	−0.80
Propene, 3,3,3-trichloro-	*2*	*0.05*	*0.10*	*12.2*	*6.9*	*−1.301*	*0.277*	*−2.35*	*−0.84*
Pyridazinone, 3-(2-vinyl)-6-methyl-	3	0.85	0.19	0.32	0.90	−0.0706	−0.089	−1.36	0.09
Pyridazinone, 3-(2-vinyl)-6-methyl-4,5-dihydro-	3	0.13	0.02	0.74	5.92	−0.8861	−0.151	−0.73	−0.57
Pyridine, 2-methyl-5-vinyl-	4	0.85	0.31	0.16	0.70	−0.0706	−0.007	−1.04	0.28
Pyridine, 2-vinyl-	4	1.26	0.44	0.10	0.53	0.1004	−0.014	−0.98	0.32
Pyridine, 2-vinyl-5-ethyl-	4	1.09	0.43	0.04	0.74	0.03743	0.006	−0.20	0.12
Pyridine, 4-vinyl-	4	0.69	0.375	0.10	0.52	−0.1612	0.06	−0.94	0.30
Silane, 3-methacryloxypropyl, trimethoxy-	2	0.868	3.79	0.094	0.425	−0.0615	0.408	−1.06	0.37
Styrene	B	1.00	0.38	0.04	1.00	0	0	0.00	0.00
Styrene, *p*-acetoxy-	2	1.26	0.40	0.07	0.86	0.1004	−0.030	−0.44	0.07
Styrene, 3-tri-*n*-butylstannyl-	2	0.03	0.001	4.88	18.5	−1.5229	−0.407	−1.17	−1.27
Styrene, 2,5-dichloro-	4	0.40	0.08	0.24	0.236	−0.3979	−0.107	−1.98	0.67
Styrene, *p*-chloromethyl-	2	1.12	0.56	0.067	0.62	0.0492	0.046	−0.62	0.21
Styrene, *p*-1-(2-hydroxypropyl)-	3	0.91	0.53	0.10	0.97	−0.041	0.071	−0.60	0.16
Styrene, α-methoxy	5	0.07	0.05	0.06	2.51	−1.1549	0.105	0.41	−0.49
Styrene, α-methyl	3	0.60	0.143	0.047	1.1	−0.2219	−0.077	−0.04	−0.03
Styrene, *p*-methyl	4	0.993	0.33	0.05	0.891	−0.0031	−0.021	−0.20	0.08
Succinimide, *N*-vinyl	4	0.033	0.516	0.116	7.71	−1.4815	0.623	0.75	−1.02
Tetrazole, 1-vinyl-	2	0.184	0.314	0.54	3.85	−0.7352	0.251	−0.78	−0.59
Tetrazole, 5-phenyl-2-(4′-vinyl)-phenyl-	2	2.2	1.4	0.32	0.48	0.3424	0.088	−1.75	0.32
Toluenesulfonamide, N,N-methyl-vinyl-	*3*	*0.05*	*0.04*	*0.42*	*5.6*	*−1.301*	*0.126*	*−0.40*	*−0.66*
Triallyl citrate	*2*	*0.076*	*0.05*	*1.76*	*20*	*−1.12*	*0.092*	*−0.49*	*−1.30*
Vinyl acetate	5	0.02	0.05	4.78	48	−1.699	0.315	−0.44	−1.56
Vinyl benzoate	3	0.06	0.019	5.03	31.56	−1.222	−0.031	−0.86	−1.45
Vinylbenzoic acid, *p*-	3	1.03	1.63	0.06	0.282	0.0124	0.238	−1.03	0.50
Vinyl benzyl methyl carbinol	3	0.94	0.54	0.11	0.98	−0.027	0.069	−0.66	0.18
Vinyl bromide	3	0.054	0.06	2.52	16.6	−1.268	0.179	−0.82	−1.27
Vinyl isobutyl ether	*3*	*0.32*	*0.40*	*0.49*	*24.5*	*−0.495*	*0.200*	*0.41*	*−1.41*
Vinyl butyl sulfide	2	0.05	0.041	0.086	2.69	−1.301	0.128	0.14	−0.43
Vinyl isobutyl sulfide	*4*	*0.05*	*0.027*	*0.074*	*2.38*	*−1.301*	*0.059*	*0.18*	*−0.36*
Vinyl tert-butyl sulfide	*2*	*0.158*	*0.05*	*0.09*	*4.36*	*−0.8013*	*−0.029*	*0.41*	*−0.64*
Vinyl chloride	4	0.055	0.045	3.29	18.7	−1.26	0.128	−0.90	−1.16
Vinyl chloroacetate	2	0.03	0.09	0.34	45	−1.523	0.345	1.04	−1.65
Vinyl dichloroacetate	2	0.28	0.04	3.85	20	−0.553	−0.162	−0.98	−1.30

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TABLE 1. *cont'd*

Monomer	Category	r_{1S}	r_{1A}	r_{A1}	r_{S1}	$\log(r_{1S})$	π	u	v
Vinyl 2-chloroethyl ether	4	0.07	0.05	1.09	160	−1.155	0.105	1.18	−2.06
Vinyl chloromethyl ketone	3	0.507	0.88	0.064	0.127	−0.295	0.255	−1.58	0.97
Vinyl cymantrene	*2*	*0.096*	*0.05*	*0.446*	*2.32*	*−1.018*	*0.053*	*−0.97*	*−0.37*
Vinyl dodecyl ether	2			0.82	41.5			0.44	−1.62
Vinylene carbonate	3	0.05	0.08	14.9	70	0.29	0.252	−1.03	−1.95
Vinyl ethyl ether	4	0.05	0.06	0.69	100	−1.301	0.192	1.11	−2.00
Vinyl ethyl oxalate	3			1.34	5.62			−1.11	−0.71
Vinyl ethyl sulfide	3	0.182	0.055	0.075	5.38	−0.740	−0.038	0.66	−0.77
Vinyl ethyl sulfoxide	*3*	*0.10*	*0.05*	*1.63*	*7.82*	*−1.000*	*0.046*	*−1.02*	*−0.94*
Vinylferrocene	3	0.17	0.158	0.173	3.57	−0.7696	0.149	−0.12	−0.50
Vinyl hendecanoate	*3*	*0.05*	*0.09*	*1.88*	*24.89*	*−1.301*	*0.260*	*−0.34*	*−1.33*
Vinylidene chloride	5	0.1075	0.32	0.64	1.79	−0.9686	0.346	−1.34	−0.24
Vinyl isocyanate	3	0.08	0.16	0.19	8.13	−1.097	0.277	0.29	−0.90
Vinyl isothiocyanate	3	0.435	1.40	0.36	0.725	−0.3615	0.357	−1.56	0.18
Vinyl methyl ketone	2	0.32	1.57	0.61	0.29	−0.495	0.427	−2.46	0.54
Vinyl phenyl ether	3	0.01	0.23	2.50	1.70	−2.000	0.685	−2.23	−0.30
Vinyl phenyl sulfide	3	0.14	0.03	0.11	3.8	−0.854	−0.090	0.20	−0.58
Vinyl stearate	*3*	*0.05*	*0.078*	*4.64*	*15.96*	*−1.301*	*0.236*	*−1.19*	*−1.15*
Vinyltriethoxysilane	*2*	*0.05*	*0.41*	*6.59*	*20.86*	*−1.301*	*0.513*	*−1.28*	*−1.32*
Vinyl-tris(trimethoxysiloxy) silane	3	0.005	0.075	3.90	25	−2.301	0.616	−0.84	−1.46
Vinyltrimethylsilane	3	0.05	0.10	4.08	10.32	−1.301	0.277	−1.41	−1.15

TABLE 2. TRANSFER AGENT

Transfer agent	Category	C_S	C_A	u	v
Acetaldehyde	3	0.00085	0.0047	−0.93	−3.12
Acetamide, *N,N*-dimethyl-	2	0.00046	0.0005	−1.94	−3.34
Acetic acid	3	0.000222	0.000081	−2.61	−3.80
Acetone	4	0.000032	0.000113	−1.28	−4.59
Acetonitrile	2	0.000044	0.0002	−1.06	−4.36
Allyl chloride	2	0.00151	0.000595	−2.57	−2.82
Aluminium, hydrodiisobutyl	3	27.5	0.394	−4.60	1.21
Aluminium, triethyl	4	12.5	0.059	−5.46	0.79
Aluminium, triisobutyl	2	28.5	28	−2.01	1.45
Aniline	4	0.0020	0.0050	−1.43	−2.70
Aniline, *N,N*-dimethyl-	4	0.0053	0.0547	−0.55	−2.20
Anthracene	3	2	1.8	−2.29	−0.01
Benzene	4	0.000003	0.000246	0.80	−5.65
Benzene, bromo-	3	0.000178	0.000136	−2.07	−4.73
Benzene, *tert*-butyl-	3	0.000005	0.000193	0.26	−5.20
Benzene, chloro-	4	0.000041	0.000079	−1.49	−4.51
Benzene, ethyl-	4	0.00007	0.003573	0.49	−4.28
p-Benzoquinone	3	227	1.3	−5.19	2.29
Borane, tributyl	3	0.00348	0.647	1.29	−2.95
Butanone	3	0.0005	0.000643	−1.82	−3.55
Butyl alcohol	3	0.00016	0.001542	−0.57	−4.05
sec-Butyl alcohol	3	0.000056	0.009755	1.23	−4.62
tert-Butyl alcohol	3	0.00003	0.000044	−1.75	−4.62
Butyl mercaptan	4	21.99	0.409	−4.42	1.23
Butyric acid, 4-hydroxy-γ-lactone	2	0.00004	0.00007	−1.66	−4.39
Cadmium, dibutyl	2	0.117	5.5	0.39	−0.93
Carbonic acid, cyclic ethylene ester	2	0.000024	0.000013	−2.36	−4.63
Carbon tetrabromide,		See methane, tetrabromo-			
Carbon tetrachloride,		See methane, tetrachloro-			
Chloroform	3	0.00005	0.000577	−0.49	−4.17
Copper(II) chloride	5	10300	67.19	−4.61	4.33
Cumene	3	0.00009	0.004141	0.38	−4.07
Cyclohexane	3	0.000005	0.000206	0.25	−4.63

TABLE 2. *cont'd*

Transfer agent	Category	C_S	C_A	u	v
Dimethyl sulfoxide	3	0.00005	0.000029	−2.32	−4.44
Diphenylamine-T	3	0.00009	0.07	2.18	−4.57
Ethane, 1,2-dichloro-	3	0.0002	0.000147	−2.17	−3.82
Ethane, 1,1,2,2-tetrachloro-	3	0.00108	0.000311	−2.73	−3.39
Ether, dodecyl vinyl	2	0.000372	0.000495	−1.82	−3.43
Ethyl acetate	3	0.00155	0.000254	−3.07	−3.28
Formamide, *N,N*-dimethyl-	2	0.0001	0.000278	−1.36	−4.00
α-D-Glucoside, methyl, 6-deoxy-6-mercapto-	2	5.5	0.13	−4.31	0.74
α-D-Glucoside, methyl-, di-*O*-benzyl-	2	0.0062	0.009	−1.76	−2.21
α-D-Glucoside, methyl-, 2,3,4,6-tetra-*O*-acetyl-	2	0.0002	0.003	−0.32	−3.70
α-D-Glucoside, methyl-, 6-(*p*-toluene sulfonyl)-	2	0.0002	0.001	−1.00	−3.70
α-D-Glucoside, methyl-, 6-*O*-triphenylmethyl-	2	0.0021	0.008	−1.17	−2.68
β-D-Glucoside, methyl-, 6-deoxy-6-dipropylamino-	2	0.0022	0.11	0.43	−2.66
Glycerol	2	0.002864	0.00235	−2.12	−2.54
Heptanol, dodecafluoro-	3	0.001333	0.00019	−3.20	−2.85
Indium, triethyl	3	1.76	0.222	−3.25	−0.08
Iron(III) chloride	3	536	3.33	−5.44	2.43
Isobutyl alcohol	3	0.00005	0.002406	0.41	−4.54
Isobutyronitrile	2	0.00027	0.00029	−1.95	−3.57
Lead, tetraethyl	2	0.000124	0.0243	1.28	−3.91
Mercury, diethyl	3	0.000034	0.00722	1.35	−4.60
Methane, dichloro-	2	0.000015	0.000306	−0.13	−4.82
Methane, tetrabromo-	4	2.2	0.085	−4.04	0.37
Methane, tetrachloro-	3	0.01	0.000099	−4.85	−2.09
Methane, nitro-	3	0.001	0.0006	−2.30	−3.09
Methanol	3	0.000074	0.00005	−2.23	−4.18
Octadiene, 2,6-dimethyl-	3	0.0002	0.045	1.37	−3.79
Oxime, acrolein-	3	1.08	62.6	0.53	−0.02
Oxime, crotonaldehyde-	3	0.15	2.05	−0.35	−1.06
Oxime, ethyl isopropenyl ketone-	3	0.43	1.03	−1.45	−0.37
Oxime, methacrolein-	3	1.3	9.41	−0.77	0.13
Oxime, methylacrolein-	3	0.04	5.35	1.03	−1.25
Oxime, methyl isopropenyl ketone-	3	0.11	1.71	−0.30	−0.90
Oxime, methyl vinyl ketone-	3	0.27	1.20	−1.06	−0.67
Pentanol, octafluoro-	3	0.001136	0.00019	−3.10	−2.93
Silane, tetraethyl	4	0.000812	0.0021	−1.30	−3.02
Stibine, tributyl	3	0.0058	11.1	2.78	−3.27
Tin, tetrabutyl	3	0.000371	0.00808	−0.06	−3.69
Toluene	5	0.000012	0.00032	0.26	−4.84
Triethylamine	4	0.00071	0.304	1.85	−3.04
Tripropylamine	4	0.00242	0.428	1.32	−2.89
Zinc, diethyl	2	0.366	1.6	−1.08	−0.44

E. REFERENCES

1. R. Z. Greenley, "Polymer Handbook", this volume, p. 181.
2. R. Z. Greenley, "Polymer Handbook", this volume, p. 309.
3. A. D. Jenkins, J. Polymer Sci., **34**, 3495 (1996).
4. C. H. Bamford, A. D. Jenkins, R. Johnston, Transactions. Farad. Soc., **55**, 418 (1959).
5. C. H. Bamford, A. D. Jenkins, J. Polymer Sci., **53**, 149 (1961).
6. C. H. Bamford, A. D. Jenkins, Transactions. Farad. Soc., **58**, 530 (1962).
7. A. D. Jenkins, Eur. Polymer J., **1**, 177 (1965).
8. A. D. Jenkins, Macromol. Chem. Phys. Rapid Commun., **17**, 275 (1996).
9. A. D. Jenkins, J. Jenkins, Macromol. Chem. Phys., Macromol. Symp., **111**, 159 (1996).
10. G. C. Eastmond, Comprehensive Chemical Kinetics, (C. H. Bamford, C. F. H. Tipper (Eds.)), **14A**, 105 (1976).
11. A. Ueda, S. Nagai, "Polymer Handbook", this volume, p. 97.

Copolymerization Parameters of Metallocene-Catalyzed Copolymerizations

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A. INTRODUCTION

In contrast to radical and ionic polymerization reactions in which the incoming monomer always adds to the *end* of the growing chain

$$R\text{-}M_n\text{-}M_{n+1}\text{-}M_{n+2}\text{-}M_{n+3}\text{-}M_{n+4}^{\bullet}$$

the most distinct feature of Ziegler–Natta-catalysis is the attachment of a monomer molecule to a highly structured catalyst complex (which may be homogeneously solvated or heterogeneously fixed to a surface) and its insertion *between* the catalyst complex and the growing chain:

$$R\text{-}M_{n+4}\text{-}M_{n+3}\text{-}M_{n+2}\text{-}M_{n+1}\text{-}M_n$$

Since the structural features of the catalyst – in particular the steric bulk of ligands, the bite angle of metallocenes, the configuration and conformation on the one hand, and the increasing space-filling demand of the growing chain and of the various monomers on the other – do influence the addition of the monomers, elucidation of the copolymerization behavior, and, in particular, determination of copolymerization parameters requires correct modeling, i.e. adequate Markov-statistics.

B. A BRIEF THEORETICAL OUTLINE OF COPOLYMERIZATION REACTIONS

Various mathematical models have been applied to copolymerization reactions induced by Ziegler–catalysts. Wall (1) assumed the velocity of the monomer addition M_1 and M_2 to be independent of the previously integrated monomeric unit (zero-order Markov model):

$$R\text{-Kat} + M_1 \xrightarrow{k_1} R\text{-}M_1\text{-Kat}$$

$$R\text{-Kat} + M_2 \xrightarrow{k_2} R\text{-}M_2\text{-Kat}$$

$$R = \text{Polymer chain}$$

This scheme gives rise to the following equation for a copolymerization reaction:

$$\frac{d[M_1]}{d[M_2]} = r\frac{[M_1]}{[M_2]} \qquad r = \frac{k_1}{k_2}$$

Mayo and Lewis (2) extended this model by including the influence of the last built-in monomer into the chain during the subsequent step (first-order Markov model):

$$R\text{-}M_1\text{-Kat} + M_1 \xrightarrow{k_{11}} R\text{-}M_1\text{-}M_1\text{-Kat}$$

$$R\text{-}M_1\text{-Kat} + M_2 \xrightarrow{k_{12}} R\text{-}M_1\text{-}M_2\text{-Kat}$$

$$R\text{-}M_2\text{-Kat} + M_1 \xrightarrow{k_{21}} R\text{-}M_2\text{-}M_1\text{-Kat}$$

$$R\text{-}M_2\text{-Kat} + M_2 \xrightarrow{k_{22}} R\text{-}M_2\text{-}M_2\text{-Kat}$$

This results in the so-called Mayo–Lewis equation,

$$\frac{d[M_2]}{d[M_1]} = \frac{[M_2]\,r_2([M_2]/[M_1]) + 1}{[M_1]\,([M_2]/[M_1]) + r_1}$$

with $r_1 = k_{11}/k_{12}$ and $r_2 = k_{22}/k_{21}$. Taking the effect of

the last *two* built-in monomers into account, Merz, *et al.* (3) developed a second-order Markov model. This 'penultimate model' necessarily results in a more complex kinetic scheme:

$$\text{R-M}_1\text{-M}_1\text{-Kat} + \text{M}_1 \xrightarrow{k_{111}} \text{R-M}_1\text{-M}_1\text{-M}_1\text{-Kat}$$
$$\text{R-M}_1\text{-M}_1\text{-Kat} + \text{M}_2 \xrightarrow{k_{112}} \text{R-M}_1\text{-M}_1\text{-M}_2\text{-Kat}$$
$$\text{R-M}_2\text{-M}_1\text{-Kat} + \text{M}_1 \xrightarrow{k_{211}} \text{R-M}_2\text{-M}_1\text{-M}_1\text{-Kat}$$
$$\text{R-M}_2\text{-M}_1\text{-Kat} + \text{M}_2 \xrightarrow{k_{212}} \text{R-M}_2\text{-M}_1\text{-M}_2\text{-Kat}$$
$$\text{R-M}_1\text{-M}_2\text{-Kat} + \text{M}_1 \xrightarrow{k_{121}} \text{R-M}_1\text{-M}_2\text{-M}_1\text{-Kat}$$
$$\text{R-M}_1\text{-M}_2\text{-Kat} + \text{M}_2 \xrightarrow{k_{122}} \text{R-M}_1\text{-M}_2\text{-M}_2\text{-Kat}$$
$$\text{R-M}_2\text{-M}_2\text{-Kat} + \text{M}_1 \xrightarrow{k_{221}} \text{R-M}_2\text{-M}_2\text{-M}_1\text{-Kat}$$
$$\text{R-M}_2\text{-M}_2\text{-Kat} + \text{M}_2 \xrightarrow{k_{222}} \text{R-M}_2\text{-M}_2\text{-M}_2\text{-Kat}$$

With $r_{11} = k_{111}/k_{112}$, $r_{21} = k_{211}/k_{212}$, $r_{22} = k_{222}/k_{221}$ and $r_{12} = k_{122}/k_{121}$, the following copolymerization equation is obtained:

$$\frac{\text{d}[\text{M}_1]}{\text{d}[M_2]} = \frac{1 + r_{21}\,x\,(1 + r_{11}\,x)/(1 + r_{21}\,x)}{1 + (r_{12}/x)\,(1 + r_{22}/x)/(1 + r_{12}/x)}$$

with $x = [\text{M}_1]/[\text{M}_2]$.

Expanding this model even further to the influence of the last *three* monomeric units, Ham (4) proposed a third-order Markov model. However, due to the many parameters involved, it has been of little practical use.

Thus, most authors restrict themselves to first- or second-order Markov models when dealing with Ziegler–Natta-catalysis for copolymerization, and this is elaborated below.

1. First-Order Markov Model

The Mayo–Lewis equation describes the composition of a copolymer as a function of the composition of the monomer mixture and their respective copolymerization parameters. These parameters not only determine the composition of the resulting copolymer, but also its microstructure, e.g. the distribution of the sequence length.

A knowledge of the ratio of the starting monomers and their copolymerization parameters enables the sequence length distribution to be calculated. The reactivity probabilities P_{ij} are defined (5,6) that give the probability of the monomer j adding to the polymer chain with the terminating monomer i:

$$P_{11} = \frac{k_{11}[\text{R-M}_1\text{-Kat}][\text{M}_1]}{k_{11}[\text{R-M}_1\text{-Kat}][\text{M}_1] + k_{12}[\text{R-M}_1\text{-Kat}][M_2]} = \frac{r_1}{r_1 + [\text{M}_2]/[\text{M}_1]}$$

$$P_{12} = \frac{k_{12}[\text{R-M}_1\text{-Kat}][\text{M}_2]}{k_{11}[\text{R-M}_1\text{-Kat}][\text{M}_1] + k_{12}[\text{R-M}_1\text{-Kat}][M_2]} = \frac{1}{r_1[\text{M}_1]/[\text{M}_2] + 1}$$

$$P_{21} = \frac{k_{21}[\text{R-M}_2\text{-Kat}][\text{M}_1]}{k_{21}[\text{R-M}_2\text{-Kat}][\text{M}_1] + k_{22}[\text{R-M}_2\text{-Kat}][M_2]} = \frac{1}{1 + r_2[\text{M}_2]/[\text{M}_1]}$$

$$P_{22} = \frac{k_{22}[\text{R-M}_2\text{-Kat}][\text{M}_2]}{k_{21}[\text{R-M}_2\text{-Kat}][\text{M}_1] + k_{22}[\text{R-M}_2\text{-Kat}][M_2]} = \frac{r_2}{[\text{M}_1]/[\text{M}_2] + r_2}$$

with $P_{11} + P_{12} = 1$ and $P_{21} + P_{22} = 1$

The proportion of the M_1 and M_2 units ($^1P_1$ or $^1P_2$) in the copolymer is also determined by reactivity probabilities:

$$^1P_1 = {}^1P_1P_{11} + {}^1P_2P_{21}$$
$$^1P_2 = {}^1P_1P_{12} + {}^1P_2P_{22}$$

giving $^1P_1 = P_{21}/(P_{12} + P_{21})$ and $^1P_2 = P_{12}/(P_{12} + P_{21})$.

All the potential sequence distributions can thus be calculated. Consider, for example, the triad distributions:

$$\begin{aligned}
\text{M}_1\text{M}_1\text{M}_1 &= {}^1P_1P_{11}P_{11} \\
&= [P_{21}(1 - P_{12})^2]/[P_{12} + P_{21}] \\
\text{M}_1\text{M}_1\text{M}_2 + \text{M}_2\text{M}_1\text{M}_1 &= 2\,{}^1P_1P_{11}P_{12} \\
&= 2[P_{21}(1 - P_{12})P_{12}]/[P_{12} + P_{21}] \\
\text{M}_2\text{M}_1\text{M}_2 &= {}^1P_2P_{21}P_{12} \\
&= [P_{12}^2P_{21}]/[P_{12} + P_{21}] \\
\text{M}_2\text{M}_2\text{M}_2 &= {}^1P_2P_{22}P_{22} \\
&= [P_{12}(1 - P_{21})^2]/[P_{12} + P_{21}] \\
\text{M}_2\text{M}_2\text{M}_1 + \text{M}_1\text{M}_2\text{M}_2 &= 2\,{}^1P_1P_{12}P_{22} \\
&= 2[P_{21}P_{12}(1 - P_{21})]/[P_{12} + P_{21}] \\
\text{M}_1\text{M}_2\text{M}_1 &= {}^1P_1P_{12}P_{21} \\
&= [P_{21}^2P_{12}]/[P_{12} + P_{21}]
\end{aligned}$$

The triad distribution allows the calculation of the true copolymerization parameters. For a given polymer, the triad distribution is deduced from experimental ^{13}C-NMR spectra (7) (see below).

2. Second-Order Markov Model

The kinetic scheme for this model gives rise to eight reaction probabilities P_{ijk} that result from the probability of the monomer k adding to the polymer chain with the terminating monomers ij. The reactivity probability P_{121} is

given for instance as

$$P_{121} = \frac{k_{121}[\text{R-M}_1\text{-M}_2\text{-Kat}][\text{M}_1]}{k_{121}[\text{R-M}_1\text{-M}_2\text{-Kat}][\text{M}_1] + k_{122}[\text{R-M}_1\text{-M}_2\text{-Kat}][\text{M}_2]}$$

$$= \frac{1}{1 + r_{12}([\text{M}_2]/[\text{M}_1])}$$

with $P_{111} + P_{112} = 1$, $P_{222} + P_{221} = 1$, $P_{122} + P_{121} = 1$, $P_{211} + P_{212} = 1$.

The triad distribution is calculated in a manner analogous to the first-order Markov model:

$$\begin{aligned}
\text{M}_1\text{M}_1\text{M}_1 &= {}^1P_{11}P_{111} \\
\text{M}_1\text{M}_1\text{M}_2 + \text{M}_2\text{M}_1\text{M}_1 &= 2\,{}^1P_{11}P_{112} \\
\text{M}_2\text{M}_1\text{M}_2 &= {}^1P_{21}P_{212} \\
\text{M}_2\text{M}_2\text{M}_2 &= {}^1P_{22}P_{222} \\
\text{M}_2\text{M}_2\text{M}_1 + \text{M}_1\text{M}_2\text{M}_2 &= 2\,{}^1P_{22}P_{221} \\
\text{M}_1\text{M}_2\text{M}_1 &= {}^1P_{12}P_{121}
\end{aligned}$$

${}^1P_{ij}$ is the frequency of the diad *ij*, and is calculated as follows:

$$\begin{aligned}
{}^1P_{11} &= {}^1P_{11}P_{111} + {}^1P_{21}P_{211} \\
{}^1P_{22} &= {}^1P_{22}P_{222} + {}^1P_{12}P_{122} \\
{}^1P_{12} &= {}^1P_{11}P_{112} + {}^1P_{21}P_{212} \\
{}^1P_{21} &= {}^1P_{22}P_{221} + {}^1P_{12}P_{121}
\end{aligned}$$

It results in

$${}^1P_{21} = \frac{(1 - P_{111})(1 - P_{222})}{2(1 - P_{111})(1 - P_{222}) + P_{211}(1 - P_{222}) + P_{122}(1 - P_{111})}$$

$${}^1P_{11} = \frac{(1 - P_{222})P_{211}}{2(1 - P_{111})(1 - P_{222}) + P_{211}(1 - P_{222}) + P_{122}(1 - P_{111})}$$

$${}^1P_{22} = \frac{(1 - P_{111})P_{122}}{2(1 - P_{111})(1 - P_{222}) + P_{211}(1 - P_{222}) + P_{122}(1 - P_{111})}$$

The diad frequency ${}^1P_{ij}$ is ${}^1P_{12} = {}^1P_{21}$ and ${}^1P_{11} + 2\,{}^1P_{12} + {}^1P_{22} = 1$

C. CALCULATION OF THE COPOLYMERIZATION PARAMETERS

Once the copolymerization parameters of a catalyst system with a given ratio of monomers are known, the composition of the copolymer and all its sequence distributions result, e.g., the complete microstructure of the copolymer is revealed. These parameters characterize any catalyst system and enable different systems to be compared. The major practical problem is the choice of the adequate statistical model.

1. First-Order Markov Model

1.1. Copolymerization Parameters Deduced from the Mayo–Lewis Equation The Mayo–Lewis equation calculates the composition of the copolymer formed as a function of the ratio of the monomer concentration at a given time. At arbitrary intervals these values are generally inaccessible; thus a sufficiently small interval (a small percent of the total conversion) is taken and the relative change of the monomer concentration $\text{d}[\text{M}_2]/\text{d}[\text{M}_1]$ is assumed to be the composition of the copolymer ($[\text{m}_1], [\text{m}_2]$) itself according to

$$\lim(\text{d}[\text{M}_2]/\text{d}[\text{M}_1] = [m_2]/[m_1]$$
$$\text{d}U \longrightarrow 0$$

It is advisable to keep the monomer ratio $[\text{M}_2]/[\text{M}_1]$ constant during polymerization by constantly feeding both monomers, or at least the most reactive monomer. To determine the copolymerization parameters the experiments are run with varying ratios of the starting monomers, and their compositions are subsequently measured. The data obtained can be analyzed by linear or nonlinear methods.

Linear Methods: Mayo and Lewis (2), Fineman and Ross (8), Kelen and Tüdös (9)
Nonlinear Methods: Behnken (10), Tidwell and Mortimer (11), Braun et al. (12)

Independent of the analytical methods the copolymerization parameters derived from the Mayo–Lewis equation have some major drawbacks, but do enable a part of the information contained in each copolymerization experiment to be extracted, namely the integral composition of the copolymer. A series of experiments is necessary to obtain the copolymerization parameters. Furthermore, using this approach it cannot be decided whether the application of the first-order Markov model is valid or not. One may even obtain a nearly perfect Fineman–Ross plot, where other methods show that a first-order Markov model cannot be applied.

1.2. Determination of Copolymerization Parameters from the Sequence Distribution (Triad Distribution) As mentioned above, a knowledge of the copolymerization parameters enables all the sequence distributions to be calculated. Or inversely, experimentally determined sequence distributions yield the copolymerization parameters. In the simplest case, intensities of appropriate single sequences can be compared. The results are even more reliable, if the determination of the copolymerization parameters is based on the maximal accessible information on the microstructure of the copolymer. To derive the *r*-parameters from the overall triad distribution is the safest way (13). This may be done by calculating a triad distribution based on a first-order Markov model and optimizing by variation of the reaction probabilities P_{ij} until the best fit between calculated and experimental data

is reached. Based on the optimized reaction probabilities the copolymerization parameters r_1 and r_2 are calculated according to the following equations:

$$r_1 = ([1/P_{12}] - 1)([M_2]/[M_1])$$
$$r_2 = ([1/P_{21}] - 1)([M_1]/[M_2])$$

This, procedure for determining the copolymerization parameters is superior to the Mayo–Lewis approach. While the latter requires several experiments, using the above approach, all the information can be obtained from one experiment.

To determine the copolymerization parameters from the triad distribution, six variables are available, five of which are independent. The system is over-determined and thus allows critical evaluation.

2. Second-Order Markov Model

The second-order Markov model gives four copolymerization parameters, which are gained from the sequence distribution (full triad distribution) with a high degree of reliability. The calculated triad distribution from the second-order Markov model is optimized by varying the reaction probabilities P_{ijk} until the best fit is reached. The four copolymerization parameters are calculated from reaction probabilities P_{ijk} as follows (13–15):

$$r_{11} = ([1/P_{112}] - 1)([M_2]/[M_1])$$
$$r_{21} = ([1/P_{212}] - 1)([M_2]/[M_1])$$
$$r_{22} = ([1/P_{221}] - 1)([M_1]/[M_2])$$
$$r_{12} = ([1/P_{121}] - 1)([M_1]/[M_2])$$

The second-order Markov model results in four copolymerization parameters from six variables based on triad distribution. A higher degree of determination clearly is desirable to justify this copolymerization model. However, the complete distribution is not obtained from ^{13}C-NMR spectra of ethene/α-olefin copolymers. The full triad distribution does not exhaust the full information of the spectra, since some peaks exhibit tetrad or even pentad sensitivity. Cheng (16,17) suggested that all recorded peaks of a ^{13}C-NMR spectrum should be included to calculate the copolymerization parameters. Arbitrarily chosen reaction probabilities give rise to a polymer chain built at random, and its ^{13}C-NMR spectrum is calculated. This spectrum is superimposed on the experimental one and optimized by varying the starting values, until the minimal difference is reached. Since the ^{13}C-NMR spectra of ethene/α-olefin copolymers display many overlapping peaks, the quality of the spectra simulation very much influences the success of this method.

Randall (7) found a way to avoid this problem by creating a "collective assignment method". Here the complete triad distribution results from combining the experimental spectral information with the relation between different *n*-ades. In the first step the ^{13}C-NMR spectrum of an ethene/α-olefin copolymer is split into several spectral areas given by signal overlap. The ^{13}C-NMR spectra of ethene/propene copolymers, for example, are divided into eight appropriate spectral areas A to H, as shown in Fig. 1. The integrals T_x of different spectral areas are expressed by the number of contributing triads ending up in the complete triad distribution by appropriately combining several T_x-data. Randall exemplified this method in detail for ethene/propene-, ethene/1-butene-, and ethene/1-hexene-copolymerizations (7).

The resulting triad distribution allows the composition of the copolymer

$$[m_1] = M_1M_1M_1 + M_2M_1M_2 + M_2M_1M_1 + M_1M_1M_2$$
$$[m_2] = M_2M_2M_2 + M_1M_2M_1 + M_2M_2M_1 + M_1M_2M_2$$

and the average sequence length:

$$n(M_1) = [m_1]/[M_2M_1M_2 + 0.5\,(M_2M_1M_1 + M_1M_1M_2)]$$
$$n(M_2) = [m_2]/[M_1M_2M_1 + 0.5\,(M_2M_2M_2 + M_2M_2M_1)]$$

to be calculated.

3. Example

To demonstrate which model describes the experimental results best, the following table from a recent publication by Fink is reproduced (15). It is based on ethene/1-octene copolymerization experiments using a homogeneous metallocene/MAO catalyst.

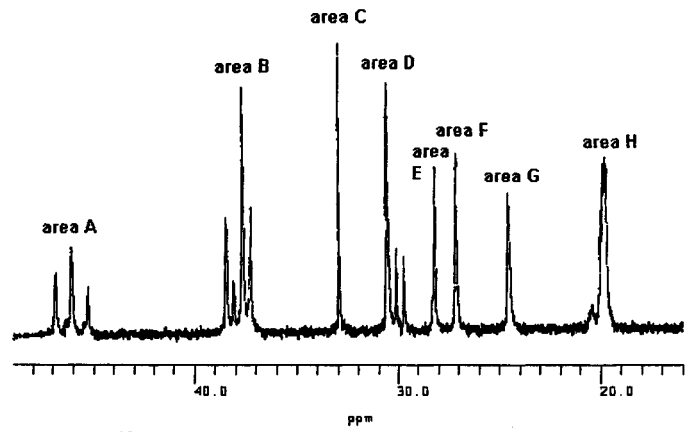

Figure 1. ^{13}C-NMR spectrum of an ethene/propene copolymer.

COMPARISON BETWEEN THE EXPERIMENTAL MEASURED TRIAD DISTRIBUTION AND THE CALCULATED TRIAD DISTRIBUTION ACCORDING TO THE FIRST- (M1) AND THE SECOND-ORDER (M2) MARKOVIAN-STATISTIC OF THE ETHYLENE (E)/1-OCTADECENE (O) COPOLYMERIZATION[a]

[O]/[E] in solution	Model	EEE	EEO + OEE	OEO	OOO	EOO + OOE	EOE	R^b ($\times 10^6$)
0.28	Exp.	0.976	0.014	0.000	0.000	0.001	0.008	–
	M1	0.980	0.004	0.000	0.011	0.004	0.000	50
	M2	0.976	0.014	0.001	0.000	0.001	0.008	0.22
1.39	Exp.	0.878	0.077	0.004	0.000	0.004	0.038	–
	M1	0.878	0.078	0.002	0.000	0.004	0.038	0
	M2	0.878	0.078	0.003	0.000	0.004	0.039	0.3
1.41	Exp.	0.883	0.074	0.005	0.000	0.005	0.033	–
	M1	0.883	0.074	0.002	0.000	0.006	0.035	0
	M2	0.882	0.072	0.003	0.000	0.006	0.036	6.27
2.84	Exp.	0.795	0.118	0.011	0.006	0.008	0.063	–
	M1	0.798	0.124	0.005	0.000	0.010	0.062	20
	M2	0.795	0.117	0.010	0.006	0.008	0.064	0.64
7.14	Exp.	0.878	0.143	0.020	0.011	0.011	0.084	–
	M1	0.738	0.158	0.008	0.001	0.014	0.080	90
	M2	0.731	0.143	0.019	0.011	0.011	0.085	0.27
14.18	Exp.	0.718	0.112	0.046	0.010	0.030	0.084	–
	M1	0.733	0.153	0.008	0.005	0.035	0.066	610
	M2	0.718	0.111	0.045	0.010	0.031	0.085	0.60

[a] $T_p = 60\,°C$, Kat: iPr(CpFlu)$ZrCl_2$ and MAO, $[Zr] = 7.56 \times 10^{-6}$ mol/l, Al/Zr = 9800.
[b] R is the sum of the least squares divided by the number of measured values.

If the experimentally determined triad distribution is compared with the calculated values based on a first-order or alternatively on a second-order Markov model (M_1 or M_2 of the table), both models give a reasonable good fit up to a monomer/comonomer ratio in solution of 1.5. Beyond a ratio of 3, however, the second-order Markov model proves to be the better approximation as indicated by the sum of the error squares divided by the number of experimental values (R of the table).

Second-order Markov statistics is only applicable when there is a sufficiently high comonomer concentration in solution, and enables the formation of α-olefin/α-olefin dual-blocks. Now the insertion of another α-olefin unit into this dual-block sequence is preferred to that into in a single-block. The complete triad distribution is gained by this procedure and it has to be performed over a wide range of the monomer/comonomer ratio in solution.

D. TABLE OF COPOLYMERIZATION PARAMETERS

Monomer/ comonomer	r_1	r_2	r_{11}	r_{21}	r_{22}	r_{12}	Cat./ cocat.	Remarks	T_{Poly} (°C)	Model tested	Analytical method*	Refs.
Ethene/propene			3.4	2.2	0.270	0.153	19		25	M1; M2	C.T.A.	13
			4.1	3.9	0.153	0.065	28		25	M1; M2	C.T.A.	13
	6.26	0.11					24		25	M1	F.-R.	18
	6.61	0.06					24		50	M1	F.-R.	18
	18.6	0.032					2		36	M1	F.-R.	19
	2.57	0.39					24		50	–	See Ref. 20	20
	2.90	0.28					25		50	–	See Ref. 20	20
	15.7	0.009					1		50	–	See Ref. 20	20
	16.1	0.025					7		50	–	See Ref. 20	20
	20.6	0.074					35		50	–	See Ref. 20	20
	12.43	0.08					29	A[a]	120–220	M1	F.-R.	21
	5.1	0.14					24		0	M1	F.-R.	22
	1.3	0.2					19		25	M1	F.-R.	22
	5.60	0.13					24	B[b]	40	M1	See Ref. 24	23
	4.23	0.12					28	B	40	M1	See Ref. 24	23
	25.42	–					13	B	40	M1	See Ref. 24	23
	13.75	0.18					12	B	40	M1	See Ref. 24	23
	16.53	–					7	B	40	M1	See Ref. 24	23
	19.61	–					11	B	40	M1	See Ref. 24	23
	24	0.0085					4		50	M1	F.-R.	25

*C.T.A. – Complete Triad Analysis (7); D.A. – Diad Analysis (28); F.-R. – Fineman–Ross (8); K.-T. – Kelen–Tüdös (9); M.-L. – Mayo–Lewis (2).

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Monomer/ comonomer	r_1	r_2	r_{11}	r_{21}	r_{22}	r_{12}	Cat./ cocat.	Remarks	T_{Poly} (°C)	Model tested	Analytical method	Refs.
	19.5	0.015					3		50	M1	F.-R.	25
	50	0.007					8		50	M1	F.-R.	25
	48	0.015					6		50	M1	F.-R.	25
	24	0.029					27		50	M1	F.-R.	25
	250	0.002					10		50	M1	F.-R.	25
	60	–					9		50	M1	F.-R.	25
	3.9	0.12					37	B	40	M1	see ref. 24	26
	4.4	0.12					37	B	60	M1	see ref. 24	26
	5.5	0.11					24	B	40	M1	see ref. 24	26
Ethene/ 1-butene			3.6	2.9	0.21	0.041	19		25	M1; M2	C.T.A.	13
			7.9	6.8	0.085	0.017	28		25	M1; M2	C.T.A.	13
	19.4	0.05					24		30	M1	F.-R.	27
	23.6	0.03					24		50	M1	F.-R.	27
	29.2	0.04					24		70	M1	F.-R.	27
	5.4	0.17					37		30	M1	F.-R.	27
	6.6	0.10					37		50	M1	F.-R.	27
	6.8	0.21					37		70	M1	F.-R.	27
Ethene/ 1-hexene			3.2	2.6	0.150	0.065	19		25	M1; M2	C.T.A.	13
			10.3	6.4	0.111	0.022	28		25	M1; M2	C.T.A.	13
	32.67	0.027					26		60	M1	F.-R.	29
	44.75	0					26		60	M1	D.A.	29
	62.31	0.003					6		60	M1	F.-R.	29
	74.60	0					6		60	M1	D.A.	29
	31.00	0.013					24		60	M1	F.-R.	29
	36.00	0					24		60	M1	D.A.	29
	88.00	0.005					12		60	M1	F.-R.	29
	86.70	0					12		60	M1	D.A.	29
	55	0.004					6		20	M1	F.-R.	30
	54	0.005					6		40	M1	F.-R.	30
	52	0.005					6		60	M1	F.-R.	30
	79	0.005					6		70	M1	F.-R.	30
			5.93	4.58	0.130	0.034	19		25	M1; M2	C.T.A.	15
			7.95	6.60	0.109	0.024	19		40	M1; M2	C.T.A.	15
			10.08	8.85	0.098	0.022	19		60	M1; M2	C.T.A.	15
Ethene/ 1-octene	20.2	–					6	B	85	M1	D.A.	31
	2.5	0.55					5	B	85	M1	D.A.	31
	42.28	0.035					26		60	M1	F.-R.	29
	43.35	0					26		60	M1	D.A.	29
	61.56	0.03					6		60	M1	F.-R.	29
	99.43	0					6		60	M1	D.A.	29
	4.71	0.23					32		0	M1	C.T.A.	32
	6.36	0.18					32		20	M1	C.T.A.	32
	8.16	0.14					32		40	M1	C.T.A.	32
	10.6	0.11					32		60	M1	C.T.A.	32
	10.1	0.03					24		40	M1	see ref. 24	26
	7.5	0.07					24		60	M1	see ref. 24	26
	10.1	0.118					32		40	M1	C.T.A.	33
	18.9	0.014					28		40	M1	C.T.A.	33
	19.5	0.013					30		40	M1	C.T.A.	33
	10.7	0.076					31		40	M1	C.T.A.	33
Ethene/ 1-dodecene			7.2	5.0	2.9	0.057	19		28	M1; M2	C.T.A.	34
			12.0	5.7	0.3	0.036	28		28	M1; M2	C.T.A.	34
Ethene/ 1-octadecene			11.2	6.3	1.6	0.044	19		28	M1; M2	C.T.A.	34
			14.7	8.4	0.4	0.035	28		28	M1; M2	C.T.A.	34
			38.45	18.58	0.085	0.041	19		60	M1; M2	C.T.A.	15
Ethene/styrene	23.4	0.015					5		40	M1	F.-R.	35,46
Ethene/2-allyl-norbornane	43.7	0.038					6		35	M1	F.-R.	36
	42.6	0.027					6		35	M1	K.-T.	36

Monomer/ comonomer	r_1	r_2	r_{11}	r_{21}	r_{22}	r_{12}	Cat./ cocat.	Remarks	T_{Poly} (°C)	Model tested	Analytical method	Refs.
Ethene/												
cyclopentene	80	<0.05					24		−10	M1	F.-R.	39
	120	<0.02					24		0	M1	F.-R.	39
	250	<0.02					24		10	M1	F.-R.	39
	300	<0.07					24		20	M1	F.-R.	39
	1.9	<1					25		0	M1	F.-R.	40
	2.2	<1					25		25	M1	F.-R.	40
Ethene/												
norbornene	2.66	n.d.[c]					28		30	M1	F.-R.	37
	3.44	n.d.					34		30	M1	F.-R.	37
	2.93	n.d.					19		30	M1	F.-R.	37
	2.61	n.d.					23		30	M1	F.-R.	37
	1.85	n.d.					23		0	M1	F.-R.	37
	1.5	n.d.					25		−25	M1	F.-R.	38
	1.9	n.d.					25		0	M1	F.-R.	38
	2.2	n.d.					25		25	M1	F.-R.	38
	3.2	n.d.					25		50	M1	F.-R.	38
	20	n.d.					6		25	M1	F.-R.	38
	6.6	n.d.					24		25	M1	F.-R.	38
	2.6	<2					28		30	M1	F.-R.	40
	3.4	0.06					19		30	M1	F.-R.	40
	2.0	0.05					23		0	M1	F.-R.	40
	3.0	0.05					23		30	M1	F.-R.	40
	3.1	0					22		30	M1	F.-R.	40
Ethene/norbornene												
	0.88	0.05					15		70	M1	K.-T.	41
	1.14	0.1					16		70	M1	K.-T.	41
	1.1	0.026					17		70	M1	K.-T.	41
	0.83	0.29					14		70	M1	K.-T.	41
	1.3	0.03					19		70	M1	K.-T.	41
			1.2	3.6	0	0	20	C[d]	70	M2	C.T.A.	41
			2.2	2.7	0	0	21	C	70	M2	C.T.A.	41
			2.2	8.9	0	0	22	C	70	M2	C.T.A.	41
			5.2	18.2	0	0	33	C	70	M2	C.T.A.	41
			2.2	54	0	0	5	C	70	M2	C.T.A.	41
Ethene/5-phenyl-2-norbornene												
	7.8	0.1					28		30	M1	M.-L.	43
	7.0	0.05					19		30	M1	M.-L.	43
Ethene/dimethanooctahydronaphthalene												
	7.8	0.1					23		50	M1	F.-R.	40,42
	7.0	0.05					18		50	M1	F.-R.	40,42
	7.1	0.04					38		50	M1	F.-R.	40,42
Ethene/phenyldimethanooctahydronaphthalene												
	11.5	0.05					28		30	M1	M.-L.	43
	7.1	0.03					19		30	M1	M.-L.	43
Ethene/trimethanododecahydroanthracene												
	15.6	0.06					23		50	M1	F.-R.	40
Propene/1-octene	3.3	0.39					28		40	M1	D.A.	44
	3.6	0.32					30		40	M1	D.A.	44
	2.6	0.37					31		40	M1	D.A.	44
	2.4	0.48					32		40	M1	D.A.	44
Propene/cyclopentene												
	40	0.001					28		60	M1	K.-T.	45

[a] High pressure copolymerization (100–150 MPa).
[b] [Monomer]/[Comonomer] ratio varied. The cited r-values hold only for a selected ratio of [Ethene]/[Propene].
[c] Not determined.
[d] Alternating copolymerization, hence, $r_{12} = r_{22} = 0$.

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E. LIST OF CATALYSTS/COCATALYSTS USED

1	$[Cp]_2TiCl_2$	MAO	
2	$[Cp]_2TiMe_2$	$AlMe_3$	
3	[Cp] 2$TiPh_2$	MAO : $AlMe_3$	(1 : 1 mixture)
4	$[Cp]_2Ti=CH_2$	$AlMe_2Cl$	(Tebbe reagent)
5	$Me_2Si[Me_4CpNt\text{-}Bu]TiCl_2$	MAO	
6	$[Cp]_2ZrCl_2$	MAO	
7	$[Cp]_2ZrCl_2$	MAO : $AlMe_3$	(1 : 1 mixture)
8	$([Cp]_2ZrCl_2)O$	MAO : $AlMe_3$	(1 : 1 mixture)
9	$[MeCp]_2ZrCl_2$	MAO : $AlMe_3$	(1 : 1 mixture)
10	$[C_5Me_4]_2ZrCl_2$	MAO : $AlMe_3$	(1 : 1 mixture)
11	$[n\text{-}Bu\text{-}Cp]_2ZrCl_2$	MAO	
12	$[Ind]_2ZrCl_2$	MAO	
13	$[Ind]_2ZrBenzyl_2$	MAO	
14	$MeCH[Cp]_2ZrCl_2$	MAO	
15	$Me_2C[CpInd]ZrCl_2$	MAO	
16	$Me_2C[MeCpInd]ZrCl_2$	MAO	
17	$Me_2C[3\text{-}t\text{-}BuCpInd]ZrCl_2$	MAO	
18	$Ph_2C[CpInd]ZrCl_2$	MAO	
19	$Me_2C[CpFlu]ZrCl_2$	MAO	
20	$Me_2C[3\text{-}MeCpFlu]ZrCl_2$	MAO	
21	$Me_2C[3\text{-}PropCpFlu]ZrCl_2$	MAO	
22	$Me_2C[3\text{-}t\text{-}BuCpFlu]ZrCl_2$	MAO	
23	$Ph_2C[CpFlu]ZrCl_2$	MAO	
24	$C_2H_4[Ind]_2ZrCl_2$	MAO	
25	$C_2H_4[IndH4]_2ZrCl_2$	MAO	
26	$Me_2Si[Cp]_2ZrCl_2$	MAO	
27	$Me_2Si[Cp]_2ZrCl_2$	MAO : $AlMe_3$	(1 : 1 mixture)
28	$Me_2Si[Ind]_2ZrCl_2$	MAO	
29	Modified silylbridged $[IndH4]_2ZrCl_2$	MAO	
30	$Me_2Si[2\text{-}MeInd]_2ZrCl_2$	MAO	
31	$Me_2Si[Benz(e)Ind]_2ZrCl_2$	MAO	
32	$Me_2Si[2\text{-}MeBenz(e)Ind]_2ZrCl_2$	MAO	
33	$Me_2Si[3\text{-}t\text{-}BuCpFlu]ZrCl_2$	MAO	
34	$Ph_2Si[Ind]_2ZrCl_2$	MAO	
35	$[Cp]_2HfCl_2$	MAO	
36	$Me_2C[CpFlu]HfCl_2$	MAO	
37	$C_2H_4[Ind]_2HfCl_2$	MAO	
38	$Ph_2C[CpFlu]HfCl_2$	MAO	

F. REFERENCES

1. F. T. Wall, J. Am. Chem. Soc., **63**, 1862 (1941).
2. F. R. Mayo, F. M. Lewis, J. Am. Chem. Soc., **66**, 1594 (1944).
3. E. Merz, T. Alfrey, and G. Goldfinger, J. Polym. Sci., **1**, 75 (1946).
4. G. E. Ham, J. Polym. Sci., **45**, 169 (1960).
5. F. P. Price, J. Chem. Phys., **36**, 209 (1962).
6. K. Ito, Y. Yamashita, J. Polym. Sci., Part A: Gen. Pap., **3**, 2165 (1965).
7. J. C. Randall, J. Macromol. Sci., C: Rev. Macromol. Chem. Phys., **29**, 201 (1989).
8. M. Fineman, S. D. Ross, J. Polym. Sci., **5**, 269 (1950).
9. T. Kelen, F. Tüdös, J. Macromol. Sci., A: Chem., **9**, 1 (1975).
10. D. W. Behnken, J. Polym. Sci., Part A: Gen. Pap., **2**, 645 (1964).
11. P. W. Tidwell, G. A. Mortimer, J. Polym. Sci., Part A: Gen. Pap., **3**, 369 (1965).
12. D. Braun, W. Brendlein, G. Mott, Eur. Polym. J., **9**, 1007 (1973).
13. N. Herfert, P. Montag, G. Fink, Makromol. Chem., **194**, 3167 (1993).
14. J. Koivumäki, G. Fink, Macromolecules, **27**, 6254 (1994).
15. P. Mühlenbrock, G. Fink, Z. Naturforsch., B: Chem. Sci., **50**, 423 (1995).
16. H. N. Cheng, J. Chem. Inf. Comput. Sci., **27**, 8 (1987).
17. H. N. Cheng, J. Appl. Polym. Sci.: Appl. Polym. Symp., **43**, 129 (1989).
18. H. Drögemüller, K. Heiland, W. Kaminsky, in: W. Kaminsky, H. Sinn (Eds.), "Transition Metals and Organometallics as Catalysts for Olefin Polymerization", Springer, Heidelberg, 1988, p. 302.
19. V. Busico, L. Mevo, G. Palumbo, A. Zambelli, Makromol. Chem., **184**, 2193 (1983).
20. J. C. W. Chien, D. He, J. Polym. Sci., Part A: Polym. Chem., **29**, (1991).
21. C. Bergemann, R. Cropp, G. Luft, J. Mol. Catal., **102**, 1 (1995).

22. A. Zambelli, A. Grassi, M. Galimberti, R. Mazzocchi, F. Piemontesi, Macromol. Chem., Rapid Commun., **12**, 523 (1991).
23. C. Lethinen, B. Löfgren, Eur. Polym. J., **33**, 115 (1997).
24. M. Kakugo, Y. Naito, K. Mizunuma, T. Miyatake, Macromolecules, **15**, 1150 (1982).
25. J. A. Ewen, in: T. Keii, K. Soga (Eds.), Cataytic Polymerization of Olefins, Stud. Surf. Sci. Catal., **25**, 271 Elsevier, Amsterdam (1986).
26. P. Starck, C. Lethinen, B. Löfgren, Angew. Makromol. Chem., **249**, 115 (1997).
27. K. Heiland, W. Kaminsky, Makromol. Chem., **193**, 601 (1992).
28. K. Soga, T. Uozumi, Makromol. Chem., **193**, 823 (1992).
29. R. Quijada, J. Dupont, M. S. L. Miranda, R. B. Scipioni, G. B. Galland, Macromol. Chem. Phys., **196**, 3991 (1995).
30. W. Kaminsky, in: T. Keii, K. Soga, Catalytic Polymerization of Olefins, Stud. Surf. Sci. Catal, **25**, 293, Elsevier, Amsterdam (1986).
31. K. Soga, T. Uozumi, S. Nakamura, T. Toneri, T. Teranishi, T. Sano, T. Arai, Macromol. Chem. Phys., **197**, 4237 (1996).
32. J. Suhm, M. J. Schneider, R. Mülhaupt, J. Polym. Sci., Part A: Polym. Phys., **35**, 735 (1997).
33. M. J. Schneider, J. Suhm, R. Mülhaupt, M.-H. Prosenc, H.-H. Brintzinger, Macromolecules, **30**, 3164 (1997).
34. J. Koivumäki, G. Fink, J. V. Seppälä, Macromolecules, **27**, 6254 (1994).
35. F. G. Sernetz, R. Mülhaupt, R. M. Waymouth, Macromol. Chem. Phys., **197**, 1071 (1996).
36. S. Marathe, T. P. Mohandas, S. Sivaram, Macromolecules, **28**, 7318 (1995).
37. W. Kaminsky, A. Noll, Polym. Bull. (Berlin), **31**, 175 (1993).
38. W. Kaminsky, A. Bark, M. Arndt, Makromol. Chem., Makromol. Symp., **47**, 83 (1991).
39. W. Kaminsky, R. Spiehl, Makromol. Chem., **190**, 515 (1989).
40. W. Kaminsky, Macromol. Chem. Phys., **197**, 3907 (1996).
41. D. Ruchatz, G. Fink, Macromolecules, **31**, 4681 (1998) and the two preceeding papers by the same authors on pages 4669 and 4674 (1998).
42. W. Kaminsky, R. Engehausen, J. Kopf, Angew. Chem., **107**, 2469 (1995), Angew. Chem. Int. Ed. Engl., **34**, 2273 (1995).
43. W. Kaminsky, A. Noll, in: G. Fink, R. Mülhaupt, H.-H. Brinzinger (Eds.), Ziegler Catalysts, Springer, Berlin, 1995, p. 149.
44. M. J. Schneider, R. Mülhaupt, Macromol. Chem. Phys., **198**, 1121 (1997).
45. M. Arnold, O. Henschke, F. Köller, Macromol. Rep., **A33**, 219 (1996).
46. L. Oliva, P. Longo, L. Izzo, M. DiSerio, Macromolecules, **30**, (1997).

Rates of Polymerization and Depolymerization, Average Molecular Weights, and Molecular Weight Distributions of Polymers

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A. INTRODUCTION

An attempt is made to present a systematic guide to the literature dealing with rates of polymerization, average molecular weights, and molecular weight distributions of polymers for various types of polymerization. This chapter is based on a review of molecular weight distributions (1) in which many of the equations are given in detail along with graphs showing the interrelationship among various distributions; here, we present only references to the literature. In addition, sections have been added on the effects of degradation and reactor design on the reaction rates and the molecular weight distributions. Literature references beyond (312) are those that have been added since the third edition of Polymer Handbook.

The theoretical description of the molecular weight distribution of a polymer and its rate of polymerization is dependent on the assumed mechanism of polymerization and on the mathematical simplifications used to obtain analytical expressions. As the number of distinct reactions is increased, such as the various transfer reactions, the mathematical expressions can become quite complex and unwieldy. In general, the equations for the rate of polymerization are the most difficult to describe, the distribution equations are somewhat easier, and the average molecular weights the simplest. In condensation polymerization, many of the distribution formulas are derived by considering the statistics or the probability of a given reaction instead of the kinetics of the reactions. Depending on which assumptions are made, quite different average molecular weights are derived, despite the rigor of the derivation. The emphasis in this section is, therefore, on the distribution functions and their averages; the rates of polymerization are given only if they have been explicitly derived.

This chapter is divided into several tables and sections, each treating various types of polymerization. The Stockmayer distribution function for condensation polymers is given in detail because of its general applicability and usefulness. Some general distribution functions are given in Section C. For all the other expressions, the reader must refer to the original literature. Many of the simpler functions are adequately described in textbooks of polymer chemistry. Flory (2), Bamford et al. (3), Odian (4), Billmeyer (5), and Kuechler (6) give extended descriptions of many systems. Bagdasarian (7) gives an extended discussion of methods of determining absolute values of propagation and termination constants, the influence of cage effects on the rate of initiation, and the influence

of retardation, inhibition, and diffusion-controlled termination on the rate of polymerization. A review of the various ways of deriving molecular weight distributions and the moments of a distribution is given by Chappelear and Simon (8).

Section B presents a series of tables describing the main assumption or conditions imposed on the theoretical models and references to the articles where the corresponding equations may be found. Tables 1 and 2 present rate equations and the distribution formulas for addition polymerization by a variety of mechanisms. No distinction is made among free radical, cationic, anionic, or coordination-type polymerization. While Table 1 treats those cases where termination reactions predominate and where steady-state assumptions are usually made, Table 2 treats those cases where termination reactions either do not exist, or may be considered as side reactions, having a minor to major control over the molecular weight distribution. The sequence length distributions for addition-type copolymers are omitted. However, see Kuechler (6) for an extended discussion of copolymerization distributions. Table 3 contains distribution formulas for linear condensation polymers in which the polymer is assumed to be perfectly linear and to contain no rings. Table 4 treats equilibrium polymerization. Table 5 describes nonlinear systems. Table 6 treats those cases where polymers are degraded (or altered) by the application of heat, light, or ionizing radiation. In the latter case, the polymer may undergo scission, crosslinking, or both reactions simultaneously. Several references on crosslinking are included here but not in Table 5. Finally, Table 7 is concerned with the influence of reactor design on molecular weight distribution; the kinetic equations for addition polymerization (with and without termination) and condensation polymerization are considered.

Section C lists a number of distribution functions and their properties. Among them is the generalized exponential function which is a good approximation to many real systems (Eq. C29).

Section D presents the Stockmayer distribution function for condensation polymerization wherein molecules of various types of kind A react with molecules of various types of kind B.

B. REFERENCE TABLES FOR THE CALCULATION OF RATES OF POLYMERIZATION, AVERAGE MOLECULAR WEIGHTS, AND MOLECULAR WEIGHT DISTRIBUTIONS OF POLYMERS FOR VARIOUS TYPES OF POLYMERIZATION

The following symbols are used in this chapter. R_p: rate of polymerization; R_d: rate of depolymerization; R_1^*: active, growing polymer containing one monomer unit; R_n^* and Q_n^*: active, growing polymer chains containing n monomer units; P_r: molecular weight distribution; k_i: rate constant for initiation; k_p: rate constant for propagation; I: initiator concentration; M: monomer concentration; M_n and M_w: number-average and weight-average molecular weights, respectively; 1° and 2° represent first and second order, respectively. No distinction is made among free radical, cationic, anionic, or coordination-type polymerizations.

The rate of initiation may be held constant (const) throughout the polymerization, or it may depend on some function of the catalyst and monomer concentrations, or it may be instantaneous (instant), in which case only the total number of initiating species need be known.

The rate of propagation is normally only the consumption of monomer; in some cases, the rate of propagation through terminal or pendant double bonds is also considered.

Transfer reactions may occur to initiator, momomer, solvent, or polymer.

Termination of active species may occur by a first-order deactivation, by second-order combination (comb) or disproportionation (disprop), or not at all ("living" polymers).

Confusion exists over the meaning of the transfer-to-catalyst reaction. In free radical systems, it means transfer of the active species to the initiator by a second-order mechanism. In ionic polymerization, it means the expulsion of an active fragment from a growing chain by a first-order mechanism to form dead polymer and an active initiator fragment. The first-order mechanism is called here the "catalyst expulsion reaction" (cat ex).

The nomenclature of distribution functions can be quite confusing. In this work, the Flory distribution (Eq. C41) is also known as the Schulz–Flory distribution, the "most probable" distribution, and the exponential distribution. The Schulz distribution (Eq. C36) is also known as the Schulz–Zimm distribution or the generalized Poisson distribution; at large values of k it approximates the Poisson distribution (Eq. C48). The Pearson Type III distribution is a variation of the Schulz distribution. If an addition polymer is made at constant monomer concentration, no transfer reactions occur, termination is only by second-order combination, and the distribution of the polymer is described by the Schulz distribution with $k = 2$. This distribution is sometimes called the self-convolution distribution or the convoluted exponential distribution. In a uniform distribution, all molcules have the same size – it is monodisperse. A rectangular or box distribution has no molecules below r_a, an equal number (or weight) of molecules between r_a and r_b, and no molecules whose size is above r_b.

In Table 5, systems are treated where branching reactions, ring formation, or gel formation may occur. See also Table 6 for crosslinking reactions during degradation. The symbol RA_f means a monomer containing f reactive A units. The problem of calculating the species distribution of polyfunctional condensation polymers where branched, cyclic, and crosslinked species can be formed is exceedingly complex. Most treatments apply the restriction that ring formation does not occur prior to the gel point, which obviously is incorrect. The restriction is invoked because of the transition from the pre-gel condition, where all unreacted functional groups can react without steric limitations to a condition containing rings in which some of the unreacted groups are sterically unable to react with all of the remaining unreacted groups. The problem is

further compounded because each different reacting monomer will have various degrees of steric hindrance. Therefore, a general treatment must ignore this aspect. An examination of the Stockmayer distribution function in Section D shows that in-depth calculations are required even for simple oligomer species. A number of attempts have been made to provide simpler expressions for the number or weight of individual species and the average molecular weights before and after the gel point. These include the theory of stochastic processes (271), stochastic graph theory (272), the theory of cascade processes which is based on probability generating functions (151), the use of link distribution functions (273), the recursive nature of branching processes and elementary probability laws (274), Monte Carlo methods (275), graph theory (276), and a kinetic approach (277).

TABLE 1. ADDITION POLYMERIZATION WITH TERMINATION

Set	Initiation	Monomer	Transfer	Termination	References		
					R_p	P_r	M_n, M_w
1.1. INVARIANT MONOMER CONCENTRATION							
1.1.1	Const	Const	None	2° Disprop or comb	2–4	1–4	1–4
1.1.2	Const	Const	Monomer, solvent	2° Disprop	2,3,10–12	1–3	1–3
1.1.3	Const	Const	Monomer, solvent	2° Disprop and comb	3	1,3	1,3
1.1.4	Const, redox	Const	Activator	2° Comb	13	13	13
1.1.5	Const, initiation by activator	Const	None	2° With catalyst; redox system	14	–	14
1.1.6	$k_i M^2$	Const	Dimer, Monomer Initiator	2° Term	15	–	15
1.1.7	Instant	Const	None	2° Term; 1° or 2° reactivation	16	–	16
1.2. VARYING MONOMER CONCENTRATION, NO TRANSFER-TO-MONOMER REACTION							
1.2.1	Const	Varies	None	2° Comb or disprop	3	1,3	1,3
1.2.2	Const	Varies	Solvent	2° Comb and disprop	3	1,3	1,3
1.2.3	Photosensitized	Varies	None	2° Disprop	39	39	–
1.2.4	Const	Varies	None	2° Disprop and comb pseudo 1° with scavenger	17 (rate of scavenger disappearance)	–	–
1.3. VARYING MONOMER CONCENTRATION, TRANSFER-TO-MONOMER REACTION OCCURS							
1.3.1	Const	Varies	Monomer, solvent	2° Disprop or comb	3,18	1,3,18	1
1.3.2	Const	Varies	Monomer, solvent	2° Disprop and comb	–	–	19
1.3.3	$k_i M^2$	Varies	Monomer	2° Comb and disprop	3,4	1,3,4	1,3,4
1.3.4	$k_i MI$	Varies	Monomer, initiator	2° Disprop; degradative chain transfer	20	–	20
.3.5	Instant	Varies	Monomer	Non-steady state conditions, term by comb or disprop	313	313	313
1.4. 1° TERMINATION OR DEACTIVATION, STEADY-STATE KINETICS							
1.4.1	Const	Const	Monomer	1° Deactivation	4	1,4	1,4
1.4.2	$k_i MI$	Const	Monomer, solvent	1° Deactivation	4	1.4,21,22	1,4,21,22
1.4.3	$k_i M^2$	Const	Monomer	1° Deactivation	4	1,4	1,4
1.4.4	$k_i MI$	Varies	Solvent	1° Plus solvent term	4,23	1,4	1,4
1.4.5	$k_i M^2$	Varies	None	1° Deactivation	4,23	1,4	1,4
1.4.6	$k_i MI$	Const	None	Deactivation by init. expulsion	21	1,21	1,21
1.5. TERMINATION BY 2° REACTION WITH MONOMER, STEADY-STATE KINETICS							
1.5.1	Const	Varies	None	2° With monomer	4,23	1,4	1,4
1.5.2	$k_i MI$	Varies	Monomer	2° With monomer	23	1	1
1.5.3	Const	Const	Monomer	2° With monomer	4	1,4	1,4
1.6. TWO ACTIVE ENDS PER CHAIN							
1.6.1	Const	Const	None	2° Disprop	3	1,3	1,3
1.6.2	Const	Const	Monomer	2° Disprop	–	18	18
1.6.3	Const	Const	Monomer	2° Comb; two active chains couple to form a chain with two active ends	3	1,3	1,3
1.7. SLOW EXHAUSTION OF INITIATOR, NONSTEADY-STATE KINETICS							
1.7.1	$k_i I$	Const	None	2° Disprop* or comb	25*,26	–	1,25*,26
1.7.2	$k_i I$	Varies	None	2° Comb	27	–	27

*Data belong together.

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TABLE 1. *cont'd*

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
1.8. DEAD-END POLYMERIZATION, RAPID DECAY OF INITIATOR, MONOMER CONCENTRATION INVARIANT							
1.8.1	$k_i I$	Const	Monomer, solvent	2° Disprop or comb	28	1,28	1,28
1.8.2	$k_i I^2$	Const	None	2° Disprop or comb	28	1,28	1,28
1.9. DEAD-END POLYMERIZATION, SLOW DECAY OF INITIATOR, MONOMER CONCENTRATION VARIES							
1.9.1	$k_i I$	Varies	Monomer, solvent	2° Disprop* or comb	30	–	1,19*
1.9.2	$k_i I$	Varies	Monomer, solvent	2° Comb	30	–	1
1.9.3	$k_i IM$	Varies	None	1° Term, bimolecular monomer addition. I^* and R_1^* do not terminate	31	–	–
1.9.4	Instant	Varies	Monomer	1° Term, or 2° with monomer	40	–	–
1.9.5	Redox	Varies	None	2° Term	41	–	–
1.10. COPOLYMERIZATION: TWO DIFFERENT MONOMERS PRESENT							
1.10.1	Const	Const	Monomer	2° Disprop or comb	1,2,3,4,10 32,33	34	1,35
1.10.2	Const	Varies	None	Diffusion controlled and inversely proportional to viscosity	239	239	239
1.10.3	Const	Const	None	Rate const for term is known as fn(conc). Comparison of cross term parameter Φ	314	–	–
1.10.4	Const	Const	None	Alternating copolymerization	315	–	–
1.10.5	Varies	Varies	Agent	Various; Monte Carlo calculations; initiation and term influences composition	–	316	316
1.11. DIFFUSION-CONTROLLED TERMINATION (see also 1.10.2 and 5.1.4)							
1.11.1	Const	Varies	None or solvent*	2° Disprop or comb* and diffusion controlled	37*	1,36	–
1.11.2	$k_i I$	Varies	None	2° Comb, $R_t = k_t M_n, n > 1$	27	–	27
1.11.3	Const	Varies	None	2° Process	42	–	–
1.11.4	$k_i I$	Varies	Monomer	2° Disprop and diffusion controlled	240	240	–
1.11.5	$k_i I$	Varies	None	2° Disprop and comb, diffusion controlled	241	241	–
1.11.6	k_i	Varies	None	2° Disprop, term rate for long chains depends on entanglement density	242	242	242
1.11.7	k_i	Varies	Monomer, solvent	2° Disprop, term rate for long chains depends on entanglement density	243	243	243
1.11.8	$k_i I$ or none	Varies	Monomer, solvent	2° Comb. and disprop term depends on free volume or free volume plus entanglement, Calc. of gel point	317	–	317
1.11.9	Const	Const	Solvent	$k_{t(n,m)} = k_o(nm)^{-a}$. Term by comb and disprop.	318	–	–
1.12. PRIMARY RADICAL TERMINATION							
1.12.1	Const	Varies	Monomer	2° Comb and disprop and primary radicals	38	–	1,38
1.12.2	Const	Const	None or init. and monomer*	2° Term and primary radicals	33,43–45	–	44,45,15*
1.13. MIXED SPECIES PROPAGATION							
1.13.1	Const	Varies	Monomer	Propagation, transfer and termination rates depend on species type, i.e. free radical and cationic occurring simultaneously	46 244	46 244	– 244
1.13.2	Const	Const	None	Zwitter-ion polymerization; distance between ions varies as $r^{3/2}$, equilibrium between free and solvated ions; 1° term	47	–	47
1.13.3	Const	Varies	None	Bifunctional initiator, term by comb	319	–	319

*Data belong together.

TABLE 1. *cont'd*

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
1.14. EMULSION POLYMERIZATION (see also Tables 2 and 7)							
1.14.1	Const	Const within particles	None	General theory; particles act as independent units; 2° term within particles; number of particles remain constant; number of radicals per particle	48–54	–	53
1.14.2	Const	Const within particles	None	General theory; Slow-term rate within particles	55	–	55
1.14.3	Const	Const within particles	None	Inst. term within particles; number of particles varies	56	–	–
1.14.4	Const	Const within particles	Monomer	Normal 2° term within particles; rapid interchange between phases of small-sized radicals; number of radicals per particle	57	–	–
1.14.5	Const	Const within particles varies outside	Monomer	Rapid exchange between phases of small-sized radicals; term in aq phase and in particles	58		
1.14.6	Slow and const	Const	None	Slow 2° comb; calcn. of number of radicals per particle	–	59	59
1.14.7	Const	Const within particles	None	Const number of particles; term by 2° comb	60	–	60
1.14.8	Const	Const	None	Inst term on entering a particle; const number of particles	–	–	61
1.14.9	Varies	Varies	Minimize	Adjust initiation and term rates while minimizing transfer to obtain "monodisperse" polymer	72	72	72
1.14.10	Const	Const within particles	Monomer, agent	Instant 2° term when a free radical enters a particle containing two growing chains and reacts with one of them. No term by disprop or comb within particles	245	245	245
1.14.11	Const	Const within particles	Monomer, agent	Same as case 1.14.10 but plus term by disprop or particle by comb	245	245	245
1.14.12	Varies	Varies	Monomer	Varying rate of entry, desorption of oligometic radicals, term by comb and disprop, Monte Carlo simulation model	–	320	–
1.14.13	Const	Const	None	Any number of chains per particle, 1° term, 2° term by disprop or by comb. Use of Markov chains	–	321	–
1.14.14	Const	Const	None	See 1.14.13. Copolymerization, chain composition distribution	–	322	–
1.15. HETEROGENEOUS POLYMERIZATION							
1.15.1	$k_i MI$, I varies	Varies	Monomer	Polymerization in monomer rich and monomer poor phases; 2° disprop	62	62	62
1.15.2	Instant	Varies	None	Term by precipitation onto growing solid particle	63	–	63
1.15.3	Const	Const	None	2° Comb, 1° occlusion onto particle surfaces, and primary radical term	64	–	–
1.15.4	Const	Const	None	2° Term in liquid phase, 1° radical precipitation, propagation and 2° term at solid–liquid interface	65	–	–
1.16. INHIBITION AND RETARDATION							
1.16.1	Const	Varies	None	2° Term, 2° addition to inhibitor	3,66,67,73	–	–
1.16.2	Const	Const	Inhibitor	2° Term, inhibitor term, inhibitor coupling	68	–	68
1.16.3	Const	Const	Retarder	2° Term, retarder reinitiation and term	69	–	69

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TABLE 1. *cont'd*

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
1.16.4	Const	Const	Retarder	2° Term, retarder reinitiation, term and coupling	3,12,70 71,74	–	–
1.16.5	Const	Const	Retarder	2° Term, rate of transfer equals rate of reinitiation	10	–	–
1.16.6	Const	Const	Retarder	2° Term, copolymerization of retarder	11	–	11
1.16.7	Const	Varies	None	Pseudo 1° with scavenger	17	–	–
1.16.8	Const	Const	Inhibitor	2° Disprop, 2° reaction with inhibitor	75	–	75
1.16.9	$k_i I$	Const	Inhibitor	2° with inhibitor or with monomer to assess the efficiency of initiation	246	–	–
1.17. POLYFUNCTIONAL INITIATOR WITH VARIOUS THERMAL STABILITIES							
1.17.1	$k_i MI$ On raising the temp., other peroxide groups can initiate	Const	None	2° disproportionation	–	–	247
1.18. PERIODIC MODULATION OF TERMINATION OR INITIATION							
1.18.1	Const	Varies	None	Periodic modulation of term by disprop by addition of an inhibitor or by magnetic or electric fields	–	323	–
1.18.2	Varies	Varies	None	Periodic modulation of initiation, term by comb and/or disprop, computer calculation	324	324	324
1.18.3	Varies	Varies	None	Periodic modulation of initiation, term by comb. Calcn. of MWD by generating functions	–	325	–

TABLE 2. ADDITION POLYMERIZATION – "LIVING" POLYMERS WITH PARTIAL DEACTIVATION

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
2.1. THE POISSON DISTRIBUTION: $k_i = k_p$							
2.1.1	$k_p MI$	Varies	None	None	2,101,102	1,2,101, 102	1,2,101, 102
2.1.2	$k_p MIt$	Varies	None	None; initiator added at a constant rate	103	–	103
2.2. THE GOLD DISTRIBUTION: $k_i \neq k_p$							
2.2.1	$k_i MI$	Varies	None	None	102, 104	1,102	1,102
2.3. PARTIAL DEACTIVATION BY A 1° PROCESS OR A 2° PROCESS WITH AN IMPURITY							
2.3.1	Instant	Varies	None	1° or 2°	105	1,106	1,106
2.3.2	Instant	Varies	None	Rate of term/R_p = const.	–	1,106,107	1,106,107
2.3.3	Instant initiator has two active sites	Varies	None	Rate of term/R_p = const.	–	1,107	1,107
2.3.4	Instant	Varies	None	Rate of term independent of R_p	1,106	–	1,106
2.3.5	Instant	Varies	None	1° at infinite time	108,109	–	108,109
2.3.6	Instant	Const	None	Probability that a chain will terminate rather than propagate is $k_t/k_p M$	–	–	248
2.4. MULTIPLE PROPAGATING SPECIES: R_n^* CAN TRANSFORM INTO Q_n^*, ETC.							
2.4.1	Instant	Const	None	None, two propagating species	–	76	76
2.4.2	$k_i MI$	Varies	None	None, two propagating species	1, 77–79	1,77,78 80	1,77–80

TABLE 2. *cont'd*

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
2.4.3	Instant	Varies	None	None, multiple propagating species	249	249	249
2.4.4	Activated monomer addition	Varies; volume also varies	None	None; two prop constants k_p can increase	– –	326 327	326 327
2.5. SIMULTANEOUS POLYMERIZATION AND DEPOLYMERIZATION							
2.5.1	$k_i MI$	Const	None	None	–	1,81	1,81
2.5.2	$k_i MI$	Varies	None	None	82,83	81	83
2.5.3	$k_p M^2$	Varies	None	None	84	–	–
2.6. THE k_p VARIES WITH CHAIN LENGTH (see also 2.4.4, 3.3, and 3.4)							
2.6.1	$k_i MI$	Varies	None	None			
	(a) all propagation constants (k_r) are different				–	1,85,86	–
	(b) $k_i : k_1 : k_2 : k_3 : \cdots k_n = m : (m-1) : (m-2) : (m-3) : \cdots$				–	1,85	–
	(c) $k_1 \neq k_2 \cdots \neq k_m = k_{m+1} = \cdots = k_n$			–	1,85	1,85	
	(d) $k_i \neq k_1 \neq k_2 = k_m$				–	1,85	–
2.6.2	Instant, $k_1 = k_2 \cdots = k_m, k_{m+1}$, etc.= 0			86	86	86	–
2.6.3	Instant, k_p is a linear decreasing function of r				86	86	86
2.6.4	Const	Varies	None	None; k_p for monomer, dimer, and trimer different from remaining molecules	–	328	328
2.6.5	Rate constants can vary with extent of conversion; determination of rate constants from P_r and the use of generating functions; termination by monomolecular decay, disprop and/or comb.				–	329	–
2.7. DEACTIVATION BY TRANSFER TO MONOMER							
2.7.1	$k_i MI$	Varies	Monomer	None	1,87	1,87	1,74,87,88
2.7.2	Instant	Varies	Monomer	None	1,74,90	1,89,90	1,74,90
2.7.3	Instant	Varies	Monomer	None, two active ends per initiator	1,91	1,91,92	1,91
2.8. DEACTIVATION BY SLOW 1° TERMINATION							
2.8.1	Instant	Varies	None	Slow 1°	1,74	–	1,74
2.8.2	Instant	Varies	Monomer	Slow 1°, at infinite time	1,74	–	1,74
2.9. DEACTIVATION BY INITIATOR EXPULSION REACTION							
2.9.1	Instant	Varies	None	None	–	–	1,93
2.9.2	$k_i MI$	Varies	Monomer	1° term	67	–	67
2.9.3	Instant	Varies	Monomer	None	250	250	250
2.10. DEACTIVATION BY DEGRADATIVE CHAIN TRANSFER							
2.10.1	$k_i MI$	Varies	Degradative transfer to polymer	None	94	–	95
2.11. COPOLYMERIZATION							
2.11.1	Instant	Varies	None	None	–	96	
2.12. HETEROGENEOUS POLYMERIZATION							
2.12.1	Instant	Varies	None	None; diffusion of monomer through solid polymer	97	–	97
2.12.2	Instant	Varies	None	2° Term with monomer; diffusion of monomer through solid polymer	98	–	98
2.12.3	Const	Const	None	Sorption and desorption of chains from the surface; slow 1° term which depends on chain length	–	99	99
2.13. SPONTANEOUS POLYMERIZATION							
2.13.1	Const	Const	None	Vinyl compound and activator form monomer; monomer both initiates and propagates; no term	100	–	100

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TABLE 2. *cont'd*

Set	Initiation	Monomer	Transfer	Termination	References R_p	P_r	M_n, M_w
2.14.	**REVERSIBLE INITIATION AND REACTIVATION**						
2.14.1	Rev 2° with monomer	Const	Monomer	No termination; dead species in equilibrium with living species, irreversible propagation	251	251	251
2.15.	**DEACTIVATION BY TRANSFER TO AGENT**						
2.15.1	$k_i MI$	Const or varies	Agent const varies	No termination. Transfer creates dead polymer and active initiator	–	–	252
2.15.2	$k_i MI$	Varies	Agent	Numerical method for calc. of MWD with mono- or polyfunctional transfer agent. Latter leads to branching.	–	330	–
2.16.	**SLOW EQUILIBRIA**						
2.16.1	Varies	Varies	None	3-State mechanism consisting of two successive equilibria	–	–	331
2.16.2	Varies	Varies	None	Equilibrium between dormant aggregated polymer and "living" polymer	332	–	–
2.17.	**POLYMER MICROSTRUCTURE**						
2.17.1	Varies	Varies	None	None; concentration of head-to-head and tail-to-tail, sequence length distributions, various dyad and triad fractions	–	–	334
2.17.2	NA	NA	NA	Block copolymers	–	–	335

TABLE 3. LINEAR CONDENSATION POLYMERIZATION WITHOUT RING FORMATION

Set		Conditions	References
3.1		Condensation of bifunctional monomer AB; the Flory distribution. See Eq. (C41)	1,2,110
3.2		Bifunctional monomer AA reacting with bifunctional monomer BB; the nylon case of hexamethylene diamine and adipic acid	1,24,111
3.3		Deviations from the principle of equal reactivity; the "substitution effect"	1,112,253,254
3.4		Rate of condensation varies	
	3.4.1	Rate of condensation proportional to chain size	1,113,114,134
	3.4.2	Condensation requires rotation into colinear orientation prior to reaction	255
	3.4.3	The order of reaction varies with conversion, or a catalytic influence of unreacted A or B groups, or both effects occurring simultaneously	256
3.5		Other simple linear condensation cases	
	3.5.1	AA reacting with BC. BC is an anhydride; within a given molecule, B must react before C	1,111
	3.5.2	AA reacting with BC. BC is an unsymmetrical acid or glycol; B reacts only with A at a different rate from that of C reacting with A	1,111
	3.5.3	AB reacting with C and itself. B and C react only with A; C is a terminator or capping material	1,111
	3.5.4	AA reacting with BB and C; the nylon case again with acetic acid as terminator	1,111
	3.5.5	AA reacting with BC; A and B react with C	1,111
	3.5.6	AB reacts with CC or DD kinetics	115
	3.5.7	AA and A reacting with BB and B	336
3.6		Further polymerization of polymers with an initial geometric distribution	
	3.6.1	Further polymerization of AB when the initial distribution is geometric	1,116
	3.6.2	Further polymerization of AB when the initial distribution is a superposition of two geometric distributions	1,116
	3.6.3	Further polymerization of AA with BB when the initial distribution of both is geometric	1,116
3.7		Copolymerization of condensation polymers	
	3.7.1	AB reacting with CD; AB and CD can be hydroxy acids or similar materials	1,111
	3.7.2	AA reacting with BB and CC. A reacts with B and C only and vice versa; BB and CC can be adipic and sebacic acids	111,257
	3.7.3	AA and BB reacting with CC and DD; A and B react only with C and D and vice versa	111
	3.7.4	AA reacting with BC and DD; A reacts only with B, C, and D	111
	3.7.5	AA and DD reacting with BC; A and B react only with C and D and vice versa	111
	3.7.6	AA reacting with BC–BC, the latter can be a dianhydride; B must react before C	258,259

TABLE 3. *cont'd*

Set	Conditions	References
3.8	Coupled polymers	
	3.8.1 AB polymerized to extent of conversion α, then coupled with CC	1,117,118
	3.8.2 AB polymerized to extent of conversion α, then coupled with CD. A and B can react with C or D	1,117
	3.8.3 AA and BB polymerized to extent of conversion α, then coupled with an excess of CC. A and B to react completely on coupling	1,117
	3.8.4 Poisson-distribution polymer of AA coupled with BC	1,117
	3.8.5 Monomer AB polymerized to extent of conversion α, then coupled with excess CC to extent of conversion γ, then recoupled with excess DD	1,117
	3.8.6 Particularly narrow distributions via coupling reaction	
	I. AA and BB (great excess)→BBAABB, then remove excess BB. CC and BBAABB (great excess)→ BBAABBCCBBAABB, then remove excess BBAABB, and continue in like manner	119
	II. AA and 2BC→CBAABC then CBAABC and 2DE→EDCBAABCDE, etc	119
	III. AB and CD→ABCD then ABCD and EF→ABCDEF, etc	119
	3.8.7 Blocks of polymers of known distribution are coupled together	
	I. A series of Poisson-type-polymers coupled together	
	II. Poisson-type-polymers coupled to "most probable"-type polymers	
	III. A series of "most probable"-type polymers coupled together	120,118
3.9	AB reacts with CC or CD; rate of reaction of A dependent upon whether or not B has reacted	115
3.10	Segmented block copolymers; distribution of block sizes	260–262
3.11	Reacted sequences in homopolymers	263, 264

TABLE 4. EQUILIBRIUM POLYMERIZATION

Set	Conditions	References
4.1	The "most probable" distribution of Flory has been derived for condensation polymerization when all reactions are assumed to have the same probability, regardless of chain length, and whether or not exchange reactions occur:	Eq. C41
	$$P_r + P_s = P_{r+s-i} + P_i, \quad i < r+s$$	
	Here, P_i can be a byproduct such as water, or a polymer molecule whose size is smaller than $r+s$. The same distribution results when random scission occurs to infinitely long chains. See also (265) for multifunctional monomers.	2,121
4.2	The theoretical equilibrium molecular weight distribution for the system	122
	$$rM = P_r$$	
	depends upon the thermodynamic definition of the final product. If the change in Gibbs free energy of formation and polymerization is a linear function of molecular size r, then two cases can be considered:	
	4.2.1 The monomer is polymerized to a pure perfectly ordered state (solid); this gives the "most probable" distribution	Eq. C41
	4.2.2 The monomer is polymerized to a pure randomly ordered state (liquid); a completely different distribution results	123
	If the free-energy change upon disordering of the pure polymer is independent of the molecular weight distribution, a new distribution arises	124
4.3	Thermodynamics of living equilibrium polymers	125
4.4	Given an initial distribution, find the distribution as a function of time when chain ends react at random with all monomer units	126
4.5	Initiation, propagation, and termination reactions are all equilibrium reactions; an initiator is required	127
4.6	Initiation, propagation, and termination reactions are all equilibrium reactions; the system is self-initiating	127
4.7	Equilibrium initiation and equilibrium propagation with a multifunctional initiator	128,266
4.8	Equilibrium initiation and equilibrium propagation; no termination; nonequilibrium solution.	129
4.9	The catalyst can form an active species with any polymer molecule, and the total number of *moles* of the system is kept constant (constant pressure and volume if all polymer molecules are ideal gases)	130
4.10	Addition polymerization, instantaneous initiation, no transfer, no termination, monomer concentration held constant, active species in equilibrium with inactive species	77
4.11	Addition polymerization, rate of initiation equals $k_i MC$, no transfer, no termination, two active species in equilibrium, monomer concentration varies	78
4.12	Addition polymerization, instantaneous initiation, no transfer, no termination, active species in equilibrium with inactive species, monomer concentration varies	77
4.13	Poly(phosphate) equilibria	131
4.14	Instantaneous initiation, polymerization and depolymerization, 1° termination	132
4.15	Instantaneous initiation, polymerization and depolymerization, degradative transfer to monomer	133
4.16	Copolymerization: bimolecular addition of one monomer, equilibrium addition of other monomer	230
4.17	Copolymerization: equilibrium growth with monomer A or B adding to a chain with a terminal group of either A or B. Three cases: 2° reaction with initiator; 1° formation of active radical; 2° reaction with another monomer	267

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TABLE 4. *cont'd*

Set	Conditions	References
4.18	Equilibrium copolymerization of two or more comonomers	268
4.19	Polycondensation where groups have different equilibrium constants	269,270
4.20	Equilibrium copolymerization initiated by polymer chains with active sites: average mol. wts, grafting ratio, grafting efficiency, grafting success, branching distribution, frequency of grafts	337

TABLE 5. NONLINEAR POLYMERIZATION SYSTEMS

Set	Conditions	References
5.1	Vinyl polymerization; self-grafting	
5.1.1	General case; molecular weight averages for grafted and ungrafted polymer	278
5.1.2	Constant rate of initiation, monomer concentration invariant, transfer to monomer and to polymer, termination by 2° disproportionation	1,3,18
5.1.3	Constant rate of initiation, monomer concentration varies, transfer to polymer, termination by 2° combination	135
5.1.4	Constant rate of initiation, monomer concentration can vary, transfer to monomer and to polymer, termination by 2° disproportionation and/or combination	279
5.1.5	Instantaneous initiation, monomer varies, either no transfer occurs or transfer to monomer, solvent, or polymer is allowed; termination by coupling and varies with chain length	280
5.2	Vinyl polymerization; grafting onto a performed polymer whose initial distribution is either (i) of constant length or (ii) a geometric distribution	18
5.3	Vinyl polymerization; all segments of the backbone chains have equal probability to be grafted	136
5.4	Vinyl polymerization; terminal double-bond polymerization, monomer concentration varies, transfer to monomer and to polymer, terminal double bond polymerization	
5.4.1	Termination predominantly by transfer, constant rate of initiation; batch or continuous polymerization	1,137–139
5.4.2	Termination by disproportionation, constant rate of initiation, includes transfer to solvent	338
5.4.3	Termination by combination, constant rate of initiation, amount of short and long chain branching, transfer to a tertiary site	338,339
5.4.4	Termination by combination, initiator and monomer concentration vary, transfer to monomer and to polymer	340
5.4.5	Termination by combination, initiator and monomer concentration vary, transfer to monomer and to polymer, characterization following saponification and reacetylation	341
5.5	Vinyl polymerization; long-chain branching.	
5.5.1	constant rate of initiation, monomer concentration invariant, transfer to monomer and to polymer, termination by 2° disproportionation	1,140
5.5.2	"Living polymer". Instantaneous initiation, β-hydride or β-methyl elimination to form a terminal double bond, transfer with agent to form a saturated end	355
5.6	Vinyl polymerizaton; "living" branched polymer; polyfunctional initiator RA_f; rate of initiation equals rate of propagation, no transfer, no termination	141
5.7	Vinyl polymerization; R_p, in a diffusion controlled gelling system; termination by disproportionation or combination, no transfer	142
5.8	Vinyl polymerization; branching density as a function of conversion; branches formed by polymerization through a vinyl group (diene polymers) or by transfer-to-polymer reaction	
5.8.1	Batch polymerization	1,2
5.8.2	Continuous polymerization	1,138
5.9	Vinyl polymerization	
5.9.1	Branching density as a function of conversion of (1) long-chain branching, (2) short-chain branching (backbiting), (3) very short chain branching (radical migration)	143
5.9.2	Copolymerization with divinyl, no ring formation prior to gelation	274,281
5.9.3	Copolymerization with divinyl; ring formation permitted; Monte Carlo calculations for crosslinking density, weight fraction of gel; chain-length distribution between crosslinks	342,343
5.9.4	Circuits in a network, elastically active chains	282,283
5.10a	Emulsion polymerization; branch density	144,344
5.10b	Emulsion polymerization; Monte Carlo calculations; can account for desorption of oligomers, chain-length dependence of kinetic parameters, chain transfer to monomer and solvent, polymerization through terminal double bonds and long-chain branching	345
5.11	Cyclopolymerization: the concentration of pendent double bonds per molecule	145–149
5.12	Gelation condition in a system when cyclopolymerization also occurs; transfer to monomer reaction occurs. Molecular weight given	150
5.13	General theory of gelation for polyaddition- and polycondensation-type polymers	151
5.14	The general distribution function for various molecules of type A reacting with various molecules of type B; A can only react with B	
5.14.1	All functional groups have equal reactivity, independent of position within the molecule or the size of the molecule. Ring formation excluded	Section D 276

TABLE 5. *cont'd*

Set	Conditions	References
	5.14.2 Monomers of type RA_f (f can vary) can only react with monomers of type RB_g (g can vary) where the various B's can have different reactivities. No ring formation	284
	5.14.3 Monomers of type RA_f and RB_g where f and g can vary, the reactivity of a given B can depend on whether or not an adjacent B on the same molecule has reacted. No ring formation	285
	5.14.4 Monomers of type RA_f and RB_g where f and g can vary; no ring formation; calculation of higher distribution moments	346
5.15	Crosslinking or coupling of a polymer with a known primary distribution (see also Table 6)	
	5.15.1 General case. No ring formation prior to gelation	1,152–155, 286
	5.15.2 All primary chains have the same size; no ring formation prior to gelation	1,2,152, 155–157
	5.15.3 Primary chains of the Poisson type, (Eq. C48); no ring formation prior to gelation	155,157
	5.15.4 Primary chains of the Schulz type, (Eq. C36); no ring formation prior to gelation	286
	5.15.5 Primary chains of the "most probable" type; ring formation allowed prior to gelation	287
	5.15.6 Formation of star-type polymers	288
5.16	Homopolymer of ARB_{f-1}. A can react only with B; the B's may have different reactivities. Formation of rings excluded prior to gelation	
	5.16.1 All B groups have the same reactivity	1,2,158,265, 289
	5.16.2 Let B groups have different reactivities	1,159,160
	5.16.3 A controversy over the statistical approach	161,162,163, 164
5.17	Copolymer of ARB_{f-1} and AB; A can react only with B; formation of rings excluded prior to gelation	1,2,158
5.18	Homopolymer of RA_f; formation of rings excluded prior to gelation	
	5.18.1 All A's equally reactive	1,2,121,158, 165,166, 265,274, 277,281, 290–292
	5.18.2 Reactivity of A depends on number of A's previously reacted	167,273,293
5.19	Various copolymers	
	5.19.1 Copolymer of RA_f and AA; formation of rings excluded prior to gelation	1,2,165
	5.19.2 Copolymer of RA_f, AA, and BB; A can react only with B; formation of rings excluded prior to gelation; all branch points completely reacted. See set 5.16.3 for a controversy	1,168,169
	5.19.3 A mixture of self-condensing monomers R_1A_g, R_2A_h, etc., no ring formation	277,294,295
	5.19.4 A stoichiometric mixture of RA_f and RB_f of the same functionality f, indistinguishable reactivity, no ring formation. Note that different authors get different equations for M_n and M_w	277
	5.19.5 A stoichiometric mixture of RA_f, RA_g, ... and RB_h, RB_i ..., f, g, h, i being different functionalities, indistinguishable reactivities, no ring formation	277,294
	5.19.6 A copolymer of RA_f and BB; no ring formation prior to gelation	274,281,290
	5.19.7 A copolymer of RA_f, AA, and B′B″ where B′ has a different reaction rate from B″; no rings prior to gelation	293
	5.19.8 Copolymer of RA_f, AA, and BB where the A's in RA_f can have different rates of reaction	293
5.20	Polymers of RA_fB_g	
	5.20.1 Homopolymer of RA_fB_g; A can only react with B; formation of rings excluded prior to gelation	170,265
	5.20.2 A mixture of monomers RA_fB_g, where $f, g = 0, 1, 2, \ldots$, stoichiometry not required	295
5.21	Branching without gelation; copolymer of RA_f and AB. A can react only with B	1,171
5.22	Gelation conditions; formation of rings excluded prior to gelation	
	5.22.1 AA reacting with B_fC_g. A can react with B and C, but with different velocities. B cannot react with C	1,172
	5.22.2 AA, BB, and C reacting with DDE and FF; A, B, and C individually can react with D, E, and F, but with different velocities	1,172
	5.22.3 AAB, RC_4, and GC reacting with DE and F. DE is an anhydride or similar material. D must react first and may have a different velocity from E. A, B, C. and G may react with different velocities	1,172
	5.22.4 AB and CD reacting with EEF and GG. AB and CD are anhydrides or similar materials, where A must react before B and C must react before D. A, B, C, and D may react with different velocities	1,172
	5.22.5 AA and BC reacting with DDE and FG. BC is an anhydride, B reacting first, and FG is like an unsymmetrical glycol	1,172
5.23	Ring formation in linear polymers	
	5.23.1 Homopolymer of type AA	1,173,174
	5.23.2 Homopolymer of type AA. Formation of catenanes	296
	5.23.3 Homopolymer of type AB	1,173,175
	5.23.4 Copolymer of type AABB	1,173,176
	5.23.5 Distribution for chain fraction	1,2
	5.23.6 Copolymer of AA with BC. BC can be an anhydride; within a given molecule, B must react before C; A single cyclic oligomer is permitted to form	297
	5.23.7 Ring-chain equilibrium constant, R_p of rings, cut-off point of monomer conc below which only cyclics formed	347
5.24	Crosslinking of a polymer with a known distribution; rings are permitted prior to gelation	153

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TABLE 5. *cont'd*

Set	Conditions	References
5.25	Gelation conditions for RA_f when rings are permitted prior to gelation	1,177
5.26	Homopolymer of RA_f. Ring formation permitted prior to gelation	1,178,179, 298
5.27	Copolymer of RA_f and BB when rings are permitted prior to gelation	180
5.28	Gelation condition for RA_f, AA, and BB when rings are permitted prior to gelation	1,181
5.29	Monte Carlo calculations on small amounts of crosslinking	299
5.30	Monte Carlo calculations for RA_3, the reactivity of an A group depends on the number of previously reacted groups on that monomer, no ring formation prior to gelation	275
5.31	Monte Carlo calculations for vinyl and divinyl copolymerization	300
5.32	Grafting of reactive polymers of the type AA, AAB, BB, and BBA where rings can form prior to gelation; M_w and the gel point	348
5.33	Star type molecules prepared by anionic polymerization and different precursors (see also 5.19.1, 5.21 and 5.27)	349

TABLE 6. DEGRADATION OF POLYMERS – MAY BE ACCOMPANIED BY CROSSLINKING

Set	Initiation	Mass	Initial distribution	Transfer	Termination	References		
						R_d	P_r	M_n, M_w
6.1. RANDOM SCISSION ONLY								
6.1.1	Proportional to time	Const	Arbitrary	None	None	–	183–186	182–184
6.1.2	Proportional to time	Const	Infinite or uniform	None	None	–	183–187	183,184,188
6.1.3	Random at all bonds	Const	Uniform	None	None	–	189	189
6.1.4	Random at chain ends	Const	Uniform	None	None	–	190	–
6.1.5	Random at all chains	Const	Uniform	None	None	–	191	191
6.1.6	Random	Const	Rectangular	None	None	–	184,187	184
6.1.7	Random	Const	Flory	None	None	–	184,187	184,188
6.1.8	Random	Const	Schulz	None	None	–	184,187, 301	187,301
6.1.9	Random	Const	Poisson	None	None	–	187	–
6.1.10	Random at normal and weak bonds	Const	Uniform	None	None	–	–	192
6.1.11	Gaussian about the midpoint	Const	Schulz	None	None	301	301	–
6.1.12	Central	Const	Schulz	None	None	301	301	–
6.2. UNZIPPING-TYPE REACTION								
6.2.1	1° end	Varies	Uniform	None	1°	193	–	193
6.2.2	1° end	Varies	Uniform	Polymer	2° Disprop	194	–	194
6.2.3	1° end	Varies	Schulz	None	1°	193	–	193
6.2.4	1° end	Varies	Schulz	None	2° Disprop	195	–	195
6.2.5	1° end	Varies	Schulz ($k = 2$)	None	1°	196	–	196
6.2.6	1° end	Varies	Schulz ($k = 2$)	None	2° Comb	196	–	196
6.2.7	1° end; various rates of initiation	Varies; monomer radicals may evaporate	Flory	None	1° or 2° Disprop	197	–	–
6.2.8	Random scission only	Varies	Uniform	Polymer	2° Disprop	194	–	194
6.2.9	Random scission only	Varies	Schulz	None	2° Disprop	198	–	198
6.2.10	1° end and random scission	Varies	Unspecified	None	1° and 2° Process	199 (all molecules must terminate) 200 (some molecules can completely disappear)		
6.2.11	1° end and random scission	Varies	Flory	Polymer	1° and 2° Disprop	201	–	201
6.2.12	1° end and random scission	Varies	Flory	Polymer	2° Comb	201	–	201

TABLE 6. *cont'd*

Set	Initiation	Mass	Initial Distribution	Transfer	Termination	References R_d	References P_r	References M_n, M_w
6.2.13	1° end and random scission	Varies	Flory	Polymer	1° and 2° Process; comparison of random scission, weak link, and transfer theories.	202 203	– –	– 203
6.2.14	1° end and random scission	Varies	Uniform	Polymer	Comparison of various term mechanisms	204	–	–
6.2.15	1° end and random scission	Varies; units other than monomer may evaporate	Uniform	Polymer	2° Disprop	205	–	205
6.2.16	1° end and random scission	Varies; units other than monomer may evaporate	Flory	polymer	1° and 2° Disprop or 2° comb	206	206	206
6.2.17	Unzipping reactions	Varies	Various	Various	Various, a review	207	–	–
6.3. RANDOM SCISSION AND CROSSLINKING								
6.3.1	Proportional to time	Const	Arbitrary	None	Unspecified term, with and without ring formation	153 208 – 302 303	153 208 282 302 303	153 208 – 302 –
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6.3.5	Proportional to time	Const	Schulz	None	Unspecified term, without ring formation	29	–	29
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[a]CFSTR: Continuous-flow, stirred tank reactor; BR: batch reactor; TR: tube reactor; MWD: molecular weight distribution.

C. SOME DISTRIBUTION FUNCTIONS AND THEIR PROPERTIES

The frequency function, $F(r)$, is the fraction of molecules of size r. Furthermore, $F(r)$ is normalized

$$\sum_1^\infty F(r) = \int_0^\infty F(r)\,dr = 1 \tag{C1}$$

The weight fraction of molecules of size r is

$$W(r) = rF(r) \Big/ \sum_1^\infty rF(r) = rF(r) \Big/ \int_0^\infty rF(r)\,dr \tag{C2}$$

Averages of any distribution are defined by

$$\sum_1^\infty r^i F(r) \Big/ \sum_1^\infty r^{i-1} F(r) = \int_0^\infty r^i F(r)\,dr \Big/ \int_0^\infty r^{i-1} F(r)\,dr \tag{C3}$$

The number-average degree of polymerization is defined by

$$\bar{r}_n = \sum_1^\infty rF(r) \Big/ \sum_1^\infty F(r) = \sum_1^\infty W(r) \Big/ \sum_1^\infty [W(r)/r] \tag{C4}$$

Thus, by the definitions of $F(r)$ and $W(r)$

$$W(r) = rF(r)/\bar{r}_n \tag{C5}$$

The weight-average degree of polymerization is

$$\bar{r}_w = \sum_1^\infty r^2 F(r) \Big/ \sum_1^\infty rF(r) = \sum_1^\infty rW(r) \Big/ \sum_1^\infty W(r) \tag{C6}$$

The z and $z+1$ averages are defined by Eq. (C3) with i set equal to 3 and 4. There is no need to restrict averages of a distribution to positive integers – any useful average can be defined; such as the $(-5/2)$ average. If the intrinsic viscosity of a polymer is related to the degree of polymerization through the equation

$$[\eta] = k\bar{r}_v^a, \tag{C7}$$

where $\bar{r}_v$ is the viscosity-average degree of polymerization, it is related to the frequency function by

$$\bar{r}_v = \left[\sum_1^\infty r^{1+a} F(r) \Big/ \sum_1^\infty rF(r) \right]^{1/a} \tag{C8}$$

As a approaches unity, $\bar{r}_v$ approaches $\bar{r}_w$. In principle, a distribution function can be determined if sufficient averages of the distribution can be determined. In practice, only the number, weight, and perhaps z averages can be found, which are insufficient to define any distribution without making further assumptions.

The degree of polymerization is a useful concept as long as one is describing polymers made with a single monomer or with monomers of equal molecular weight. When considering copolymers, one must work with the actual molecular weight of the reacted unit. This is done in condensation polymerization. The molecular weight distribution of addition copolymers is not included here because of the extreme complexity of these systems.

When performing the summation of a distribution equation to find an average of the distribution, use is frequently made of the following sums:

$$\sum_{x=1}^{\infty} p^{x-1} = 1/(1-p) \quad p < 1 \tag{C9}$$

$$\sum_{x=1}^{\infty} xp^{x-1} = 1/(1-p)^2 \quad p < 1$$

$$\sum_{x=1}^{\infty} x^2p^{x-1} = (1+p)/(1-p)^3 \quad p < 1$$

$$\sum_{x=1}^{\infty} x^n p^{x-1} = \frac{\mathrm{d}}{\mathrm{d}p}\left[p\sum_{x=1}^{\infty} x^{n-1}p^{x-1}\right] \quad p < 1$$

$$\sum_{i=0}^{\infty} a^{n-i}b^i n!/(n-i)!\,i! = (a+b)^n \quad n \text{ is an integer}$$

$$\sum_{i=0}^{\infty} a^i i n!/(n-i)!\,i! = na(1+a)^{n-1}$$

$$\sum_{i=0}^{\infty} a^i i^2 n!/(n-i)!\,i! = na(1+na)(1+a)^{n-2}$$

$$\sum_{n=1}^{\infty} A^{n-1}/(n-1)! = \exp A$$

$$\sum_{n=1}^{\infty} nA^{n-1}/(n-1)! = (1+A)\exp A$$

$$\sum_{n=1}^{\infty} n^2A^{n-1}/(n-1)! = (1+3A+A^2)\exp A$$

$$\sum_{n=1}^{\infty} n^pA^{n-1}/(n-1)! = \frac{\mathrm{d}}{\mathrm{d}A}\left[A\sum_{n=1}^{\infty} n^{p-1}A^{n-1}/(n-1)!\right]$$

1. Normal Distribution Function (Gaussian Distribution)

$$F(r) = \frac{\exp\{-(r-\bar{r})^2/2\sigma^2\}}{(2\pi)^{1/2}\sigma} \tag{C10}$$

$$W(r) = (r/\bar{r})F(r) \tag{C11}$$

$$\bar{r} = \int_{-\infty}^{+\infty} \frac{r\exp\{-(r-\bar{r})^2/2\sigma^2\}}{(2\pi)^{1/2}\sigma} = \bar{r}_n \tag{C12}$$

$$\sigma^2 = \int_{-\infty}^{+\infty} \frac{(r-\bar{r})^2\exp\{-(r-\bar{r})^2/2\sigma^2\}}{(2\pi)^{1/2}\sigma} \tag{C13}$$

$$\bar{r}_w = \frac{\sigma^2}{\bar{r}} + \bar{r} \tag{C14}$$

Values of $F(r)$ and $\int_0^t F(r)\,\mathrm{d}r$ are found in many statistical tables (223).

2. Logarithmic Normal Distribution Function

Under the assumption that the weight distribution of the logarithm of molecular size is normally distributed, we can replace r in the normal distribution function by $\ln r$ and $\bar{r}$ by $\ln \bar{r}_m$

$$W(\ln r) = \frac{\exp\{-(\ln r - \ln \bar{r}_m)^2/2\sigma^2\}}{(2\pi)^{1/2}\sigma} \tag{C15}$$

where

$$\int_1^{\infty} W(\ln r)\,\mathrm{d}(\ln r) = 1 \tag{C16}$$

or in the alternate form

$$W(r) = \frac{\exp\{-(\ln r - \ln \bar{r}_m)^2/2\sigma^2\}}{r(2\pi)^{1/2}\sigma} \tag{C17}$$

where

$$\int_0^{\infty} W(r)\,\mathrm{d}r = 1 \tag{C18}$$

and

$$\ln \bar{r}_m = \int_0^{\infty} (\ln r)W(r)\,\mathrm{d}r \tag{C19}$$

Here $\bar{r}_m$ is the median value of the distribution, that is, one-half of the values of r are less than $\bar{r}_m$.

$$\sigma^2 = \int_0^{\infty} (\ln r - \ln \bar{r}_m)^2 W(r)\,\mathrm{d}r \tag{C20}$$

$$\bar{r}_n = \bar{r}_m \exp(-\sigma^2/2) \tag{C21}$$

$$\bar{r}_w = \bar{r}_m \exp(+\sigma^2/2) \tag{C22}$$

$$\bar{r}_z = \bar{r}_m \exp(+3\sigma^2/2) \tag{C23}$$

$$\bar{r}_i = \bar{r}_m \exp\{(2i-3)\sigma^2/2\} \tag{C24}$$

Note that

$$\bar{r}_m = (\bar{r}_n\bar{r}_w)^{1/2} \tag{C25}$$

$$\bar{r}_w/\bar{r}_n = \bar{r}_z/\bar{r}_w = \bar{r}_{z+1}/\bar{r}_z = \exp\sigma^2 \tag{C26}$$

The logarithmic normal distribution sometimes is given in a "generalized" form (224)

$$W_s(r) = \frac{r^s\exp\left[-(\ln r/\bar{r}_s)^2/2\sigma^2\right]}{(2\pi)^{1/2}\sigma\bar{r}_s^{s+1}\exp\{(s+1)^2\sigma^2/2\}} \tag{C27}$$

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but Honig (225) has shown that

$$\ln \bar{r}_m = \ln \bar{r}_s + (s+1)\sigma^2 \tag{C28}$$

Thus, the normalized Lansing–Kramer function ($s = 0$) (226) is identical to the normalized Wesslau function ($s = -1$) (227).

The generalized logarithmic normal distribution is skewed to large values of r, but Kotliar (237, 238) shows that this distribution is not a good representation of a polymer after either low- or high-molecular weight material is removed or degradation has occurred.

3. Generalized Exponential Distribution (1,228)

$$F_{m,k,y}(r) = my^{k/m}r^{k-1}[\exp(-yr^m)]/\Gamma(k/m) \tag{C29}$$

$$W_{m,k,y}(r) = my^{(k+1)/m}r^k[\exp(-yr^m)]/\Gamma[(k+1)/m] \tag{C30}$$

$$\bar{r}_n = \Gamma[(k+1)/m]/y^{1/m}\Gamma(k/m) \tag{C31}$$

$$\bar{r}_w = \Gamma[(k+2)/m]/y^{1/m}\Gamma[(k+1)/m] \tag{C32}$$

$$\bar{r}_z = \Gamma[(k+3)/m]/y^{1/m}\Gamma[(k+2)/m] \tag{C33}$$

$$\bar{r}_i = \Gamma[(k+i)/m]/y^{1/m}\Gamma[(k+i-1)/m] \tag{C34}$$

and

$$r_v = \{\Gamma[(k+a+1)/m]/y^{a/m}\Gamma[(k+1)/m]\}^{1/a} \tag{C35}$$

where a is defined by equation (C7).

The variance of a distribution σ^2, the usual measure of distribution breadth for small populations, depends strongly on the average molecular weight of the distribution:

$$\sigma^2 = \bar{r}_n(\bar{r}_w - \bar{r}_n)$$

Hence, the preferred measure of the molecular weight distribution breadth is $\bar{r}_w/\bar{r}_n$.

(a) The distribution function of Schulz (9) and Zimm (229) is obtained by setting $m = 1$ and requiring that $k > 0$. This distribution is equivalent to the Pearson Type III distribution

$$F(r) = y^k r^{k-1}[\exp(-yr)]/\Gamma(k) \tag{C36}$$

$$W(r) = y^{(k+1)}r^k[\exp(-yr)]/\Gamma(k+1) \tag{C37}$$

The distribution reduces to the "most probable" distribution where $k = 1$.

The cumulative number or weight fraction of the Schulz distribution may be computed from

$$\int_0^r F_{1,k,y}(r) = \frac{ky^k}{\Gamma(1+k)} \sum_{i=0}^{\infty} \frac{(-1)^i y^i r^{k+i}}{i!\,(k+i)} \tag{C38}$$

$$\int_0^r W_{1,k,y}(r) = \frac{y^{k+1}}{\Gamma(1+k)} \sum_{i=0}^{\infty} \frac{(-1)^i y^i r^{k+i+1}}{i!\,(a+i+1)}. \tag{C39}$$

(b) The Weibull (231) or Tung (232) distribution is obtained by setting $m = 1 + k$ and $k > 0$. This distribution is usually seen in the form

$$\int_0^r W(r)\,dr = 1 - \exp[-yr^{(1+k)}] \tag{C40}$$

Kotliar (233) shows that evaluation of y and k by the Tung method can lead to erroneous values of $\bar{r}_w/\bar{r}_n$.

(c) The Flory distribution, as noted above, occurs when $k = m = 1$, provided that $\ln p$ can be replaced by $-(1-p)$, i.e. $p \approx 1$. The Flory distribution is also written as:

$$F(r) = p^{r-1}(1-p) \tag{C41}$$

$$W(r) = rp^{r-1}(1-p)^2 \tag{C42}$$

$$\bar{r}_n = 1/(1-p) \tag{C43}$$

$$\bar{r}_w = (1+p)/(1-p) \tag{C44}$$

$$\bar{r}_v = \bar{r}_n([\Gamma(2+a)]^{1/a} \tag{C45}$$

and with the approximation $\ln p = p - 1$

$$\int_0^r W(r)\,dr \cong (1/p) - [1 + (1-p)r]p^{r-1} \tag{C46}$$

For high molecular weights $p \approx 1$, hence

$$r_n : r_w : r_z = 1:2:3. \tag{C47}$$

4. Poisson Distribution

$$F(r) = \exp(-\nu)\,\nu^{r-1}/(r-1)! \tag{C48}$$

$$W(r) = [\nu/(\nu+1)]r\exp(-\nu)\,\nu^{r-2}/(r-1)! \tag{C49}$$

$$\bar{r}_n = 1 + \nu \tag{C50}$$

$$\bar{r}_w = 1 + \nu + \nu/(1+\nu) \tag{C51}$$

D. MOLECULAR WEIGHT DISTRIBUTION IN CONDENSATION POLYMERS: THE STOCKMAYER DISTRIBUTION FUNCTION

Stockmayer (234) has presented a generalized distribution formula for a variety of monomers containing end groups of type A which can only react with a variety of monomers containing end groups of type B. In the original mixture there are $A_1, A_2, A_3, \ldots, A_i, \ldots$ moles of reactants bearing respectively, $f_1, f_2, f_3, \ldots, f_i, \ldots$ functional groups of type A each, together with $B_1, B_2, \ldots, B_j, \ldots$ moles of reactants of functionalities $g_1, g_2, \ldots g_j, \ldots$ in groups of type B. all functional groups of a given type are equally reactive and ring formation does not occur appreciably; which obviously is not true near the gel point. The system reacts until a fraction α of the A groups and a fraction β of the B groups

have reacted. Further

$$\alpha \sum_i f_i A_i = \beta \sum_j g_j B_j$$

Now $N\{m_n, n_j\}$ represents the number of moles of that species which consists of $m_1, m_2, \ldots, m_i, \ldots$ monomer units of the A type combined with $n_1, n_2, \ldots, n_j$ units of the B type.

$$N\{m_i, n_j\} = \frac{K(\sum_i f_i m_i - \sum_i m_i)!\,(\sum_j g_j n_j - \sum_j n_j)!\,\Pi_i(x_i^{m_i}/m_i!)\Pi_j(y_j^{n_j}/n_j!)}{(\sum_i f_i m_i - \sum_i m_i - \sum_j n_j + 1)!\,(\sum_j g_j n_j - \sum_j n_j - \sum_i m_i + 1)!} \tag{D1}$$

where

$$\begin{aligned} x_i &= \left[\frac{f_i A_i}{(\sum_i f_i A_i)}\right]\left[\frac{\beta(1-\alpha)^{f_i-1}}{(1-\beta)}\right] \\ y_j &= \left[\frac{g_j B_j}{(\sum_j g_j B_j)}\right]\left[\frac{\alpha(1-\beta)^{g_j-1}}{(1-\alpha)}\right] \\ K &= \left(\sum_i f_i A_i\right)(1-\alpha)(1-\beta)/\beta \\ &= \left(\sum_j g_j B_j\right)(1-\alpha)(1-\beta)/\alpha \end{aligned} \tag{D2}$$

An example will illustrate the use of Eq. (D1).

Suppose we have the monomers acetic acid (CH_3COOH, $A_1, f_1 = 1$) and adipic acid ($HOOC(CH_2)_4COOH$, $A_2, f_2 = 2$) reacting with ethylene glycol ($HOCH_2CH_2OH$, $B_1, g_1 = 2$) and glycerol ($HOCH_2CHOHCH_2OH$, $B_2, g_2 = 3$) (all hydroxyl groups of the glycerol are considered equally reactive). What is the number of molecules which contain exactly 1 acetic acid unit, 4 adipic acid units, 3 glycol units, and 2 glycerol units? It is $N\{1,4,3,2\}$ and

$$\begin{aligned} N\{1,4,3,2\} = K&(1+8-1-4)!\,(6+6-3-2)! \\ &\div (1+8-1-4-3-2+1)! \\ &\div (6+6-3-2-1-4+1)! \\ &\times \left(\frac{x_1^1}{1!}\right)\left(\frac{x_2^4}{4!}\right)\left(\frac{y_1^3}{3!}\right)\left(\frac{y_2^2}{2!}\right) \end{aligned} \tag{D3}$$

where

$$\begin{aligned} x_1 &= \left[\frac{A_1}{A_1+2A_2}\right]\left[\frac{\beta}{1-\beta}\right], \\ x_2 &= \left[\frac{2A_2}{A_1+2A_2}\right]\left[\frac{\beta(1-\alpha)}{(1-\beta)}\right] \\ y_1 &= \left[\frac{2B_1}{2B_1+3B_2}\right]\left[\frac{\alpha(1-\beta)}{(1-\alpha)}\right], \\ y_2 &= \left[\frac{3B_2}{2B_1+3B_2}\right]\left[\frac{\alpha(1-\beta)^2}{(1-\alpha)}\right] \\ K &= (A_1+2A_2)(1-\alpha)(1-\beta)/\beta \end{aligned} \tag{D4}$$

If each species i has an effective molecular weight M_i, which is lower than the original molecular weight by the term $W_0 f_i/2$, where W_0 is the molecular weight of the by-product, then

$$W = \sum_i M_i A_i + \sum_j B_j M_j \tag{D5}$$

The number of molecules of the end of the reaction is, neglecting byproduct,

$$N = \sum_i A_i + \sum_j B_j - \alpha \sum_i f_i A_i \tag{D6}$$

The number-average molecular weight is

$$\bar{M}_n = W/N \tag{D7}$$

The weight-average molecular weight is

$$\begin{aligned} \bar{M}_w = &\left[\beta\frac{\sum_i M_i^2 A_i}{\sum_i f_i A_i} + \alpha\frac{\sum_j M_j^2 B_j}{\sum_j g_j B_j} \right. \\ &\left. + \frac{\alpha\beta\,[\alpha(f_c-1)M_b^2 + \beta(g_c-1)M_a^2 + 2M_a M_b]}{1-\alpha\beta(f_c-1)(g_c-1)}\right] \\ &\times \left[\beta\frac{\sum_i M_i A_i}{\sum_i f_i A_i} + \alpha\frac{\sum_j M_j B_j}{\sum_j g_j B_j}\right]^{-1} \end{aligned} \tag{D8}$$

where

$$\begin{aligned} f_c &= \left(\sum_i f_i^2 A_i\right)\Big/\left(\sum_i f_i A_i\right) \\ g_c &= \left(\sum_j g_j^2 B_j\right)\Big/\left(\sum_j g_j B_j\right) \\ M_a &= \left(\sum_i M_i f_i A_i\right)\Big/\left(\sum_i f_i A_i\right) \\ M_b &= \left(\sum_j M_j g_j B_j\right)\Big/\left(\sum_j g_j B_j\right) \end{aligned} \tag{D9}$$

Simpler expressions for M_n and M_w are given by Ziegel et al. (235) for the case when only bifunctional B monomers are present.

The gel point is

$$(\alpha\beta)_c = 1/(f_c-1)(g_c-1) \tag{D10}$$

Extensive computations of $N\{m_i, n_j\}$ for the case $f_1 = 1, f_2 = 2, f_4$, and $g_2 = 2$ have appeared (236).

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E. REFERENCES

1. L. H. Peebles, Jr., "Molecular Weight Distributions in Polymers", Interscience, New York, 1971.
2. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, N.Y., 1953.
3. C. H. Bamford, W. G. Barb, A. D. Jenkins, P. F. Onyon, "The Kinetics of Vinyl Polymerization by Radical Mechanisms", Academic Press, New York, 1958.
4. G. Odian, "Principles of Polymerization", McGraw-Hill, New York, 1970; 2nd ed., Wiley-Interscience, New York, 1981.
5. F. W. Billmeyer, Jr., "Textbook of Polymer Science", 2nd ed., Interscience, New York, 1971; 3rd ed., 1984.
6. L. Kuechler, "Polymerisationskinetik", Springer, Berlin, 1951.
7. Kh. S. Bagdasarian, "Theory of Free Radical Polymerization", Israel Program for Scientific Translations, Jerusalem, 1968.
8. D. C. Chappelear, R. H. M. Simon. "Addition and Condensation Polymerization Processes", in: N. Platzer (Ed.), Adv. in Chem., **91**, Am. Chem. Soc., Washington, DC, 1969, p. 1.
9. G. V. Schulz, Z. Physik, Chem. B, **43**, 25 (1939).
10. G. M. Burnett, L. D. Loan, Trans. Faraday Soc., **51**, 214 (1955).
11. L. H. Peebles, Jr., J. T. Clarke, W. H. Stockmayer, J. Am. Chem. Soc., **82**, 4780 (1960).
12. J. L. Kice, J. Am. Chem. Soc., **76**, 6274 (1954).
13. L. H. Peebles, Jr., J. Appl. Polym. Sci., **17**, 113 (1973).
14. S. Saccubai, M. Santappa, Makromol. Chem., **117**, 50 (1968).
15. W. A. Pryor, J. H. Coco, Macromolecules, **3**, 300 (1970).
16. L. S. Bresler, V. A. Grechanovsky, A. Muzsay, I. Ya. Poddubnyi, Makromol. Chem., **133**, 111 (1970).
17. C. H. Bamford, A. D. Jenkins, R. Johnson, Trans. Faraday Soc., **58**, 1212 (1962).
18. C. H. Bamford, H. Tompa, Trans. Faraday Soc., **50**, 1097 (1954).
19. M. Litt, J. Polym. Sci. A-2, **8**, 2105 (1970).
20. M. Litt, F. R. Eirich, J. Polym. Sci., **45**, 379 (1960).
21. P. H. Plesch, Ed., "The Chemistry of Cationic Polymerization", MacMillan Co., New York, 1963.
22. A. A. Korotkov, S. P. Mitsengendler, V. N. Krasulina, J. Polym. Sci., **53**, 217 (1961).
23. D. O. Jordan, A. R. Mathieson, J. Chem. Soc., **1952**, 2358. (Note: equation 13 et seq. contain errors owing to an integration error.)
24. H. E. Grethlein, Ind. Eng. Chem. Fund., **8**, 206 (1969).
25. J. C. W. Chien, J. Polym. Sci. A, **1**, 1839 (1963).
26. J. C. W. Chien, J. Am. Chem. Soc., **81**, 86 (1959).
27. S. Katz, G. M. Saidel, Am. Inst. Chem. Eng. J., **13**, 319 (1967).
28. C. H. Bamford, Polymer, **6**, 63 (1965).
29. M. Inokuti, J. Chem. Phys., **38**, 2999 (1963).
30. A. V. Tobolsky, J. Am. Chem. Soc., **80**, 5927 (1958); A. V. Tobolsky, C. E. Rogers, R. D. Brinkman, ibid., **82**, 1277 (1960).
31. C. B. Wooster, Macromolecules, **1**, 324 (1968).
32. K. F. O'Driscoll, R. Knorr, Macromolecules, **1**, 367 1968).
33. J. C. Bevington, "Radical Polymerization", Academic Press, New York, 1961.
34. W. H. Ray, T. L. Douglas, E. W. Goodsolve, Macromolecules, **4**, 166 (1971).
35. W. H. Stockmayer, J. Chem. Phys., **13**, 199 (1945).
36. K. Ito, J. Polym. Sci. A-2, **7**, 241 (1969).
37. S. W. Benson, A. M. North, J. Am. Chem. Soc., **84**, 935 (1962).
38. S. Okamura, T. Manabe, Polymer, **2**, 83 (1961).
39. K. Venkatarao, M. Santappa, J. Polym. Sci. A-1, **8**, 1785, 3429 (1970).
40. S. Kohjiya, Y. Imanishi, T. Higashimura, J. Polym. Sci. A-1, **9**, 747 (1971).
41. K. F. O'Driscoll, S. A. McArdle, J. Polym. Sci., **40**, 557 (1959).
42. K. Ito, J. Polym. Sci., A-1, **7**, 827, 2995 (1969).
43. C. H. Bamford, A. D. Jenkins, R. Johnson, Trans. Faraday Soc., **55**, 1451 (1959).
44. K. Ito, J. Polym. Sci. A-1, **7**, 2247, 3387 (1969); **8**, 1823 (1970); **10**, 1481 (1972); K. Ito, T. Matsuda, J. Appl. Polym. Sci., **14**, 311 (1970).
45. T. Nagabhushanam, M. Santappa, J. Polym. Sci. A-1, **10**, 1511 (1972).
46. J. F. Westlake, R. Y. Hunag, J. Polym. Sci. A-1, **10**, 1429, 1442 (1972).
47. M. A. Markevich, Ye. V. Kochetov, N. S. Yenikolopyan, Vysokomol. Soedin. A, **13**, 1013 (1971); Polym. Sci., USSR, **13**, 1153 (1971).
48. R. N. Haward, J. Polym. Sci., **4**, 273 (1949).
49. W. V. Smith, R. H. Ewart, J. Chem. Phys., **16**, 592 (1948).
50. W. H. Stockmayer, J. Polym. Sci., **24**, 314 (1957).
51. J. T. O'Toole, J. Appl. Polym. Sci., **9**, 1291 (1965).
52. A. G. Parts, D. E. Moore, J. G. Watterson, Makromol. Chem., **89**, 156 (1965).
53. J. L. Gardon, J. Polym. Sci. A-1, **6**, 623 (1968).
54. L. M. Pisman, S. I. Kuchanov, Polym. Sci., USSR, **13**, 1187 (1971); Vysokomol. Soedin. A, **13**, 1055 (1971).
55. J. L. Gardon, J. Polym. Sci. A-1, **6**, 665 (1968).
56. J. L. Gardon, J. Polym. Sci. A-1, **9**, 2763 (1971).
57. P. Harriot, J. Polym. Sci. A-1, **9**, 1153 (1971).
58. M. Litt, R. Patsiga, V. Stannett, J. Polym. Sci. A-1, **8**, 3607 (1970).
59. S. Katz, R. Shinnar, G. M. Saidel, "Addition and Condensation Polymerization Processes", N. Platzer, Ed., Adv. in Chem. Ser., **91**, Am. Chem. Soc., Washington, DC, 1969, p. 145.
60. G. M. Saidel, S. Katz, J. Polym. Sci. C, 149 (1969).
61. J. G. Watterson, J. G. Parts, Markromol. Chem., **146**, 1 (1971).
62. A. H. Abdeh-Alim, A. E. Hamielec, J. Appl. Polym. Sci., **16**, 783 (1972).
63. M, Iguchi, J. Polym. Sci. A-1, **8**, 1013 (1970).

64. O. G. Lewis, R, M. King, Jr., "Addition and Condensation Polymerization Processes", N. Platzer, Ed., in Adv. in Chem. Ser. 91, Am. Chem. Soc., Washington, DC, 1969, p. 25.
65. R. Wessling, I. R. Harrion, J. Polym. Sci. A-1, **9**, 3471 (1971).
66. P. D. Barlett, H. Kwart, J. Am. Chem. Soc., **72**, 1051 (1950).
67. Th. Deleanu, M. Dimonie, J. Polym. Sci. A-1, **8**, 95 (1970).
68. I. Kar, B. M. Mandal, S. R. Palit, Makromol. Chem., **127**, 196 (1969).
69. W. H. Atkinson, C. H. Bamford, G. C. Eastmond, Trans. Faraday Soc., **66**, 1446 (1970).
70. A. D. Jenkins, Trans. Faraday Soc., **54**, 1885 (1958).
71. A. D. Jenkins, Trans. Faraday Soc., **54**, 1895 (1958).
72. J. P. Bianchi, F. P. Price, B. H. Zimm, J. Polym. Sci., **25**, 27 (1957).
73. F. F. Williams, K. Hayashi, K. Ueno, K. Hayashi, S. Okamura, Trans. Faraday Soc., **63**, 1501 (1967).
74. P. W. Allen, F. M. Merrett, J. Scanlan, Trans. Faraday Soc., **51**, 95 (1955).
75. C. H. Bamford, A. D. Jenkins, R. Johnson, Proc. Roy. Soc., (London) **A239**, 214 (1957).
76. B. D. Coleman, T. G. Fox, J. Am. Chem. Soc., **85**, 1241 (1963).
77. M. Szwarc, J. J. Hermans, Polym. Lett. B, **2**, 815 (1964).
78. R. V. Figini, Makromol. Chem., **71**, 193 (1964).
79. R. V. Figini, Makromol. Chem., **107**, 170 (1967).
80. R. V. Figini, Makromol. Chem., **88**, 271 (1965).
81. A. Miyake, W. H. Stockmayer, Makromol. Chem., **88**, 90 (1965).
82. C. R. Huang, H. H. Wang, J. Polym. Sci. A-1, **10**, 791 (1972).
83. V. S. Nanda, S. C. Jain, Eur. Polym. J., **6**, 1517 (1970).
84. J. Leonard, Macromolecules, **2**, 661 (1969).
85. H. J. R. Maget, J. Polym. Sci. A-2, 1281 (1964).
86. I. Mita, J. Macromol. Sci. Chem. A-5, 883 (1971).
87. W. T. Kyner, J. R. M. Radok, M. Wales. J. Chem. Phys., **30**, 363 (1959).
88. V. S. Nanda, Trans. Faraday Soc., **60**, 949 (1964).
89. L. H. Peebles, Jr., Polym. Lett., **7**, 75 (1969). See also Ref. 91.
90. S. C. Jain, V. S. Nanda, Polym. Lett., **8**, 843 (1970).
91. L. H. Peebles, Jr., J. Polym. Sci. A-2, **8**, 1235 (1970) (Note: Eq. 13 of Ref. 89 should be written $x = (\tau - r)/r^{1/2}$; the figures are correct.)
92. V. S. Nanda, R. K. Jain, Trans. Faraday Soc., **64**, 1022 (1968).
93. J. C. W. Chien, J. Polym. Sci. A, **1**, 425 (1963).
94. S. Penzek, P. Kubisa, Makromol. Chem., **130**, 186 (1969).
95. V. I. Irzhak, N. S. Enikolopyan, Dokl. Akad. Nauk SSSR **185**, 862 (1969); Dokl. Phys. Chem., **185**, 226 (1969).
96. W. H. Ray, Macromolecules, **4**, 162 (1971).
97. W. R. Schmeal, J. R. Street, Am. Inst. Chem. Eng. J., **17**, 1188 (1971).
98. D. Singh, R. P. Merrill, Macromolecules, **4**, 599 (1971).
99. M. Gordon, R. J. Roe, Polymer, **2**, 41 (1961).
100. M. I. Mustafayev, K. V. Aliev, V. A. Kabanov, Vysokomol. Soedin. A, **12**, 855 (1970); Polym. Sci., USSR, **12**, 968 (1970).
101. P. J. Flory, J. Am. Chem. Soc., **62**, 1561 (1940).
102. L. Gold, J. Chem. Phys., **28**, 91 (1958).
103. M. Litt, D. H. Richards, Polym. Lett., **5**, 867 (1967).
104. L. F. Beste, H. K. Hall, Jr., J. Phys. Chem., **68**, 269 (1964).
105. H. N. Friedlander, J. Polym. Sci. A, **2**, 3885 (1964).
106. B. D. Coleman, F. Gornick, G. Weiss, J. Chem. Phys., **39**, 3233 (1963).
107. T. A. Orofino, F. Wenger, J. Chem. Phys., **35**, 532 (1961).
108. R. Chiang, J. J. Hermans, J. Polym. Sci. A-1, **4**, 2843 (1966).
109. A. Guyot, J. Chim. Phys., 548, 1964.
110. P. J. Flory, J. Am. Chem. Soc., **58**, 1877 (1936).
111. L. C. Case, J. Polym. Sci., **29**, 455 (1958).
112. G. Challa, Makromol. Chem., **38**, 105 (1960).
113. V. S. Nanda, S. C. Jain, J. Chem. Phys., **49**, 1318 (1968).
114. A. Ninagawa, I. Ijichi, M. Imoto, Makromol. Chem., **107**, 196 (1967).
115. L. C. Case, J. Polym. Sci., **48**, 27 (1960).
116. J. J. Hermans, Makromol. Chem., **87**, 21 (1965).
117. L. C. Case, J. Polym. Sci., **37**, 147 (1959).
118. C. H. Bamford, A. D. Jenkins, Trans. Faraday Soc., **56**, 907 (1960).
119. L. C. Case, J. Polym. Sci., **39**, 175 (1959).
120. L. C. Case, J. Polym. Sci., **39**, 183 (1959).
121. P. J. Flory, Chem. Rev., **39**, 137 (1946).
122. W. Kuhn, Chem. Ber., **63**, 1503 (1930).
123. J. L. Lundberg, J. Polym. Sci. A, **2**, 1121 (1964).
124. F. E. Harries, J. Polym. Sci., **18**, 351 (1955).
125. M. Szwarc, Adv. Polym. Sci., **4**, 457 (1967).
126. J. J. Hermans, J. Polym. Sci. C, **12**, 345 (1966).
127. A. V. Tobolsky, J. Polym. Sci., **25**, 220 (1957); J. Polym. Sci., **31**, 126 (1958); A. V. Tobolsky, A. Eisenberg, J. Am. Chem. Soc., **81**, 2302 (1959); J. Am. Chem. Soc., **82**, 289 (1960).
128. M. E. Baur, A. Eisenberg, J. Chem. Phys., **42**, 85 (1965).
129. F. P. Adams, J. B. Carmichael, R. J. Zeman, J. Polym. Sci. A-1, **5**, 741 (1967).
130. C. M. Fonatana in: The Chemistry of Cationic Polymerization, P. H. Plesch (Ed.), Macmillan, New York, 1963.
131. J. R. Van Wazer, "Phosphorus and Its Compounds", Vol. 1, Interscience Publishers, Inc., New York, 1958.
132. T. Saegusa, T. Shiota, S. Matsumoto, H. Fujii, Macromolecules, **5**, 34 (1972).
133. R. Z. Greenley, J. C. Stauffer, J. E. Kurz, Macromolecules, **2**, 561 (1969).
134. V. S. Nanda, S. C. Jain, J. Polym. Sci. A-1, **8**, 1871 (1970).
135. G. M Saidel, S. Katz, J. Polym. Sci. A-2, **6**, 1149 (1968).
136. L. H. Tung. T. M. Wiley, Polym. Preprints, **13**, 1060 (1972).
137. W. W. Graessley. H. Mittelhauser, R. Maramba, Markromol. Chem., **86**, 129 (1965).
138. W. W. Graessley, Am. Inst. Chem. Eng.-Inst. Chem. Eng. Joint Meeting, London, June, 1965, **3**, 16.
139. O. Saito, K. Najasubramanian, W. W. Graessley, J Polym. Sci. A-2, **7**, 1937 (1969).

140. J. K. Beasley, J. Am. Chem. Soc., **75**, 6123 (1953).
141. K. Fukii, T. Yamade, J. Polym. Sci. A, **2**, 3743 (1964).
142. M. Gordon, R. J. Roe, J. Polym. Sci., **21**, 57 (1956).
143. A. Rigo, G. Palma, G. Talamini, Markromol. Chem., **153**, 219 (1972).
144. L. E. Dannals. J. Polym. Sci. A-1, **8**, 2989 (1970).
145. M. Gordon, J. Chem. Phys., **22**, 610 (1954).
146. C. Aso, T. Nawata, H. Kamao, Makromol. Chem., **68**, 1 (1963).
147. Y. Minoura, M. Mitoh, J. Polym. Sci. A, **3**, 2149 (1965).
148. G. B. Butler, M. A. Raymond, J. Polym. Sci. A, **3**, 3413 (1965).
149. L. Trassarelli, M. Guaita, T. Macromol. Sci.-Chem. **1**, 471 (1966).
150. M. Gordon, R. J. Roe, J. Polym. Sci., **21**, 75 (1956).
151. M. Gordon, Proc. Roy. Soc. (London), **A268**, 240 (1962).
152. W. H. Stockmayer, J. Chem. Phys., **12**, 125 (1944).
153. O. Saito, J. Phys. Soc. (Japan), **13**, 198 (1958).
154. A. A. Miler, J. Polym. Sci., **42**, 441 (1960).
155. G. R. Dobson, M. Gordon, J. Chem. Phys., **43**, 705 (1965).
156. P. J. Flory, J. Am. Chem. Soc., **63**, 3096 (1941).
157. A. Amemiya, J. Phys. Soc. (Japan), **23**, 1394, 1402 (1967).
158. P. J. Flory, J. Am. Chem. Soc., **74**, 2718 (1952).
159. E. S. Allen, J. Polym. Sci., **21**, 349 (1956).
160. S. Erlander, D. French, J. Polym. Sci., **20**, 7 (1956).
161. M. Gordon, M. Judd, Nature, **234**, 96 (1971).
162. C. R. Masson, I. B. Smith, S. G. Whiteway, Nature, **234**, 97 (1971).
163. C. R. Masson, I. B. Smith, S. G. Whiteway, Can. J. Chem., **48**, 1456 (1970).
164. S. G. Whiteway, I. B. Smith, C. R. Masson, Can. J. Chem., **48**, 33 (1970).
165. P. J. Flory, J. Am. Chem. Soc., **63**, 3091 (1941).
166. W. H. Stockmayer, J. Chem. Phys., **11**, 45 (1943).
167. M. Gordon, G. R. Scantlebury, Trans. Faraday Soc., **60**, 604 (1964).
168. A. R. Shultz, J. Polym. Sci. A, **3**, 4211 (1965).
169. H. L. Berger, A. R. Shultz, J. Polym. Sci. A, **3**, 4227 (1965).
170. L. M. Pismen, S. I. Kuchanov, Polym. Sci., USSR **13**, 890 (1971); Vysokomol. Soedin. A, **13**, 791 (1971).
171. J. R. Schaefgen, P. J. Flory, J. Am. Chem. Soc., **70**, 2709 (1948).
172. L. C. Case, J. Polym. Sci., **26**, 333 (1957).
173. H. Jacobson, W. H. Stockmayer, J. Chem. Phys., **18**, 1600 (1950).
174. P. J. Flory, J. A. Semlyen, J. Am. Chem. Soc., **88**, 3209 (1966).
175. H. Morawetz, N. Goodman, Macromolecules, **3**, 699 (1970).
176. M. Gordon, W. B. Temple, Makromol. Chem., **152**, 277 (1972).
177. F. E. Harris, J. Chem. Phys., **23**, 1518 (1955).
178. C. A. J. Hoeve, J. Polym. Sci., **21**, 11 (1956).
179. M. Gordon, G. R. Scantlebury, J. Polym. Sci. C, **16**, 3933 (1968).
180. M. Gordon, G. R. Scantlebury, J. Chem. Soc. B, **1967**, 1 (1967).
181. R. W. Kilb, J. Phys. Chem., **62**, 969 (1958).
182. A. Charlesby, S. H. Pinner, Proc. Roy. Soc. (London), A, **249**, 367 (1959).
183. D. Mejzler, J. Schmorak, M. Lewin, J. Polym. Sci., **46**, 289 (1960).
184. V. S. Nanda, R. K. Pathria, Proc. Roy. Soc. (London), A, **270**, 14 (1962).
185. L. Monnerie, J. Neel, J. Chim. Phys., **62**, 53 (1965).
186. S. W. Lee, J. Polym. Sci. A-2, **7**, 77 (1969).
187. L. Monnerie, J. Neel, J. Chim. Phys., **62**, 510 (1965).
188. J. Durup. J. Chim. Phys., **51**, 64 (1954).
189. R. Simha, J. Appl. Phys., **12**, 569 (1941).
190. B. J. Coyne, J. Polym. Sci. A-2, **5**, 633 (1967).
191. E. W. Montroll, R. Simha, J. Chem. Phys., **8**, 721 (1940).
192. J. R. MacCallum, Markromol. Chem., **83**, 129 (1965).
193. M. Gordon, L. R. Shenton, J. Polym. Sci., **38**, 157 (1959).
194. R. Simba, L. A. Wall, J. Phys. Chem., **56**, 707 (1952).
195. R. H. Boyd, T. P. Lin, J. Chem. Phys., **45**, 773 (1966).
196. M. Gordon, L. R. Shenton, J. Polym. Sci., **38**, 179 (1959).
197. M. Gordon, J. Phys. Chem., **64**, 19 (1960).
198. R. H. Boyd, T. P. Lin, J. Chem. Phys., **45**, 778 (1966).
199. G. G. Cameron, Makromol. Chem., **100**, 225 (1967).
200. G. G. Cameron, G. P. Kerr, Makromol. Chem., **115**, 268 (1968).
201. R. H. Boyd, J. Chem. Phys., **31**, 321 (1959).
202. M. Gordon, Trans. Faraday Soc., **53**, 1662 (1957).
203. R. Simha, Trans. Faraday Soc., **54**, 1345 (1958).
204. L. A. Wall, S. Strauss, Polym. Preprints, **10**, 1472 (1969).
205. L. A. Walls, S. Strauss, J. H. Flynn, D. McIntyre, J. Phys. Chem., **70**, 53 (1966).
206. R. H. Boyd, J. Polym. Sci. A-1, **5**, 1573 (1967).
207. M. Gordon, Soc. Chem. Ind. (London) Monograph, **13**, 163 (1961).
208. O. Saito, J. Phys. Soc. (Japan), **13**, 1465 (1958).
209. A. R. Shultz, J. Appl. Polym. Sci., **10**, 353 (1966).
210. A. R. Shultz, J. Chem. Phys., **29**, 200 (1958); J. Phys. Chem., **65**, 967 (1961).
211. O. Satio, J. Phys. Soc. (Japan), **13**, 1451 (1958).
212. T. Kimura, J. Phys. Soc. (Japan), **17**, 1884 (1962). See also ref. 213.
213. D. I. C. Kells, J. E. Guillet, J. Polym. Sci. A-2, **7**, 1895 (1969).
214. K. G. Denbigh, Trans. Faraday Soc., **43**, 648 (1947).
215. A. W. DeGraft, G. W. Poehlin, J. Polym. Sci. A-2, **9**, 1955 (1971).
216. Z. Tadmor. J. A. Biesenberger, Ind. Eng. Chem. Fund., **5**, 336 (1966).
217. M. E. Sacks, S. Lee, J. A. Biesenberger, Chem. Eng. Sci., **28**, 241 (1973).
218. R. V. Figini, G. V. Schulz, Z. Physik, Chem. (Frankfurt), **23**, 233 (1960); Makromol. Chem., **41**, 1 (1960).
219. M. Litt, J. Polym. Sci., **58**, 429 (1962).

220. L. F. Beste, H. K. Hall, Jr., J. Macromol. Chem., **1**, 121 (1966).
221. A. Eisenberg, D. A. McQuarrie, J. Polym. Sci. A-1, **4**, 737 (1966).
222. T. E. Corrigan, M. J. Dean, J. Macromol. Sci. A: Chem., **2**, 645 (1968).
223. "Handbook of Chemistry and Physics", Vol. 34 th Edition et seq. Chemical Rubber Publishing Co., Cleveland.
224. W. F. Espenscheid, M. Kerker, E. Matijevic, J. Phys. Chem., **68**, 3093 (1964).
225. E. P. Honig, J. Phys. Chem., **69**, 4418 (1965).
226. W. D. Lansing, E. O. Kramer, J. Am. Chem. Soc., **57**, 1369 (1935).
227. H. Wesslau, Makromol. Chem., **20**, 111 (1956).
228. L. T. Muus, W. H. Stockmayer, quoted by F. W. Billmeyer, Jr., in "Textbook of Polymer Science", Interscience, New York, 1962.
229. B. H. Zimm, J. Chem. Phys., **16**, 1099 (1948).
230. M. Matsuda, M. Iino, T. Hirayama, T. Miyashita, Macromolecules, **5**, 240 (1972).
231. W. Weibull, J. Appl. Mech., **18**, 293 (1951).
232. L. H. Tung, J. Polym. Sci., **20**, 495 (1956).
233. A. M. Kotliar, J. Polym. Sci. A, **2**, 1373 (1964).
234. W. H. Stockmayer, J. Polym. Sci., **9**, 69 (1952); **11**, 424 (1953).
235. K. D. Siegel, A. W. Fogiel, R. Pariser, Macromolecules, **5**, 95 (1972).
236. I. Nakamura, R. Yokouchi, T. Ito, D. Miura, K. Fujii, Kobunshi Kagaku, **21**, 553 (1964).
237. A. M. Kotliar, J. Polym. Sci. A, **2**, 4304 (1964).
238. A. M. Kotliar, J. Polym. Sci. A, **2**, 4329 (1964).
239. O. Prochazka, P. Kratochvil, J. Polym. Sci., Polym. Phys., Ed. **22**, 501 (1984).
240. T. J. Tulig, M. Tirrell, Macromolecules, **14**, 1501 (1981).
241. W. Y. Chiu, G. M. Carratt, D. S. Soong, Macromolecules, **16**, 348 (1983).
242. J. N. Cardenas, K. F. O'Driscoll, J. Polym. Sci., Polym. Chem. Ed., **14**, 883 (1976).
243. J. N. Cardenas, K. F. O'Driscoll, J. Polym. Sci., Polym. Chem. Ed., **15**, 1883 (1977).
244. N. S. Yenikolopyan, F. S. D'yachkovskii, P. Ye. Matkovshii, Ye. L. Pechatnikov, Kh. M. A. Brikenshtein, G. A. Furman, S. P. Davtyan, Vysokomol. Soyedin. A, **25**, 265 (1983); Polym. Sci., USSR, **25**, 303 (1983).
245. G. Lichti, R. G. Gilbert, D. H. Napper, J. Polym. Sci., Polym. Chem. Ed., **18**, 1297 (1980).
246. K. Ito, J. Polym. Sci., A-1, 57 (1972).
247. V. A. Ivanov, S. I. Kuchanov, S. S. Ivanchev, Vysokomol. Soyedin. A, **19**, 1684 (1977); Polym. Sci., USSR, **19**, 1923 (1977).
248. H. Morawetz, Macromolecules, **12**, 532 (1979).
249. L. L. Boehm, Z. Phys. Chem. N. F., **88**, 297 (1974).
250. S. C. Jain, V. S. Nanda, Eur. Polym. J., **13**, 137 (1977).
251. M. Szwarc, B. H. Zimm, Macromolecules, **16**, 1918 (1983).
252. J. Largo-Cabrerizo, J. Guzman, Macromolecules, **12**, 526 (1979).
253. S. I. Kuchanov, Ye. S. Povolotskaya, Vysokomol. Soyedin. A, **24**, 2179 (1982); Polym. Sci., USSR, **24**, 2499 (1982).
254. S. K. Gupta, A. Kumar, A. Bhargava, Eur. Polym., J., **15**, 557 (1979).
255. L. H. Peebles, Jr., Macromolecules, **16**, 929 (1983).
256. J. W. Stafford, J. Polym. Sci., Polym. Chem. Ed., **21**, 1627 (1983).
257. E. Ozizmir, G. Odian, J. Polym. Sci., Polym. Chem. Ed., **18**, 2281 (1980).
258. V. I. Kolegov, S. Ya. Frenkel, Vysokomol. Soyedin. A, **18**, 1680 (1975); Polym. Sci., USSR, **18**, 8 (1975).
259. V. I. Kolegov, Vysokomol. Soyedin. A, **18**, 1689 (1976); Polym. Sci., USSR, **18**, 1929 (1976).
260. L. H. Peebles, Jr., Macromolecules, **7**, 872 (1974).
261. L. H. Peebles, Jr., Macromolecules, **9**, 58 (1976).
262. Ye. P. Piskareva, Ye. G. Erenburg, I. Ya Poddubnyi, Vysokomol. Soyedin. A, **20** (1978); Polym. Sci., USSR, **20**, 883 (1979).
263. J. J. Gonzalez, K. W. Kehr, Macromolecules, **11**, 996 (1978). See also Ref. 264.
264. N. A. Plate, A. D. Litmanovich, O. V. Noah, Macromolecules, **12**, 1231 (1979).
265. J. L. Spouge, Macromolecules, **16**, 121 (1983).
266. D. Y. Yan, G. F. Cai, Makromol. Chem., **186**, 583 (1985).
267. G. F. Cai, D. Y. Yan, Makromol. Chem., **186**, 597 (1985).
268. M. Szwarc, C. L. Perrin, Macromolecules, **18**, 528 (1985).
269. Ye. B. Brun, S. I. Kuchanov, Vysokomol. Soyedin. A, **21**, 691 (1979); Polym. Sci., USSR, **21**, 757 (1979).
270. S. I. Kuchanov, Ye. B. Brun, Vysokomol. Soyedin. A, **21**, 700 (1979); Polym. Sci., USSR, **21**, 767 (1979).
271. P. Whittle, Proc. Cambridge Phil. Soc., **61**, 475 (1965); Proc. Roy. Soc. London, Ser. A, **285**, 501 (1965).
272. D. W. Matula, L. C. D. Groenweghe, J. R. Van Wazer, J. Chem. Phys., **41**, 3105 (1964).
273. P. Luby, J. Phys. Chem., **78**, 1083 (1974).
274. C. W. Macosko, D. R. Miller, Macromolecules, **9**, 199 (1976).
275. J. Mikes, K. Dusek, Macromolecules, **15**, 93 (1982).
276. D. Durand, C. M. Bruneau, Macromolecules, **12**, 1216 (1979).
277. J. W. Stafford, J. Polym. Sci., Polym. Chem. Ed., **19**, 3219 (1981).
278. R. Collins, M. B. Huglin, R. W. Richards, Eur. Polym. J., **11**, 197 (1975).
279. B. L. Gutin, Vysokomol. Soyedin. A, **20**, 620 (1978); Polym. Sci., USSR, **20**, 699 (1979).
280. T. Yasukawa, Y. Sasaki, K. Murikami, J. Polym. Sci., Polym. Chem. Ed., **13**, 17 (1975).
281. D. R. Miller, C. W. Macosko, Macromolecules, **9**, 206 (1976).
282. D. S. Pearson, W. W. Graessley, Macromolecules, **11**, 528 (1978).
283. A. Ziabicki, J. Walasek, Macromolecules, **11**, 471 (1978).
284. D. Durand, C. M. Bruneau, Polymer, **24**, 587 (1983).
285. D. Durand, C. M. Bruneau, Polymer, **24**, 592 (1983).
286. V. Vojta, K. Dusek, Brit. Polym. J., **12**, 1 (1980).

287. K. Dusek, M. Gordon, S. B. Ross-Murphy, Macromolecules, **11**, 236 (1978).
288. T. Kotaka, Bull. Inst. Chem. Res., Kyoto Univ., **55**, 135 (1977).
289. V. S. Nanda, J. Macromol. Sci., Phys. B, **18**, 685 (1980).
290. M. A. Bibbo, E. M. Valles, Macromolecules, **15**, 1293 (1982).
291. D. R. Miller, E. M. Valles, C. W. Macosko, Polym. Eng. Sci., **19**, 272 (1979).
292. E. Donoghue, Macromolecules, **15**, 1634 (1982).
293. D. R. Miller, C. W. Macosko, Macromolecules, **11**, 656 (1978).
294. J. W. Stafford, J. Polym. Sci., Polym. Chem. Ed., **22**, 365 (1984).
295. D. Durand, C. M. Bruneau, Brit. Polym. J. **13**, 33 (1981).
296. H. Jacobson, Macromolecules, **17**, 705 (1984).
297. V. I. Kolegov, T. S. Kozlova, V. M. Artemichev, O. I. Rodionova, Vysokomol. Soyedin. A, **25**, 1005 (1983); Polym. Sci., USSR, **25**, 1162 (1983).
298. M. Gordon, G. R. Scantlebury, Proc. Roy. Soc. London, Ser. A, **292**, 380 (1966).
299. I. I. Romantsova, Yu. A. Taran, O. V. Noa, N. A. Plate, Vysokomol, Soyedin A, **21**, 1176 (1979); Polym. Sci., USSR, **21**, 1293 (1980).
300. R. Bansil, H. J. Herrmann, D. Stauffer, Macromolecules, **17**, 998 (1984).
301. M. Ballauff, B. A. Wolf, Macromolecules, **14**, 654 (1981).
302. O. Saito in: M. Dole (Ed.), "The Radiation Chemistry of Macromolecules", vol. 1, Academic Press, New York, 1972, p. 223.
303. J. H. O'Donnell, N. P. Rahman, C. A. Smith, D. J. Winzor Macromolecules, **12**, 113 (1979).
304. C. C. Lin, W. Y. Chiu, J. Appl. Polym. Sci., **23**, 2049 (1979).
305. A. Kumar, S. K. Gupta, B. Kumar, N. Somu, Polymer, **24**, 1180 (1983).
306. A. Ya. Malkin, Yu. B. Lavochnik, V. P. Begishev, Polym. Proc. Eng., **1**, 71 (1983).
307. A. Ya. Malkin, Yu. B. Lavochnik, V. P. Begishev, Vysokomol. Soyedin A, **26**, 775 (1984); Polym. Sci., USSR, **26**, 868 (1984).
308. H. T. Chen, J. Steenord, Polym. Eng. Sci., **15**, 357 (1975).
309. A. Kumar, S. K. Gupta, R. Saraf, Polymer, **21**, 1323 (1980).
310. A. Kumar, P. Rajora, N. L. Agarwalla, S. K. Gupta, Polymer, **23**, 222 (1982).
311. H. Kilkson, I & EC Fund., **3**, 281 (1964).
312. J. Ugelstad, H. R. Mfutakamba, P. C. Mørk, T. Ellington, A. Berge, R. Schmid, L. Holm, A. Jørgedal, F. K. Hansen, K. Justad, J. Polym. Sci., Polym, Symp., **72**, 225 (1985).
313. D. Yan, Macromolecules, **22**, 2926 (1989).
314. G. Kaszás, T. Földes-Berezsnich, F. Tüdös, J. Macromol. Sci.-Chem. A, **22**, 1571 (1985).
315. K. Fujimori, P. P. Organ, M. J. Costigan, I. A. Craven, J. Macromol. Sci.-Chem. A, **33**, 647 (1986).
316. M. N. Galbraith, G. Moad, D. H. Soloman, T. H. Spurling Macromolecules, **20**, 675 (1987).
317. K. Ito, Eur. Polym. J., **23**, 409 (1987).
318. H.-Kh. Mahabadi, Macromolecules, **24**, 606 (1991).
319. B. V. Revnov. E. G. Zotikov, N. G. Podosenova, V. P. Budtov, J. Appl. Chem., **58**, 766 (1985); Zh. Prikl. Khim. (Leningrad), **58**, 845 (1985).
320. H. Tobita, Y. Takada, M. Nomura, J. Polym. Sci., Polym. Chem. Ed., **33**, 441 (1995).
321. G. Storti, G. Polotti, M. Cociani, M. Morbidelli, J. Polym. Sci., Polym. Chem. Ed., **30**, 731 (1992).
322. G. Storti, G. Polotti, P. Canu, N. Morbidelli, J. Polym. Sci., Polym. Chem. Ed., **30**, 751 (1992).
323. G. Zifferer, O. F. Olaj, Macromol. Theory Simul., **6**, 53 (1997).
324. A. N. Nikitin, A. V. Evsseev, Macromol. Reports A, **33** (Suppls. **3–4**), 187 (1996).
325. K. W. McLaughlin, D. D. Latham, C. E. Hoyle, M. A. Trapp, J. Phys. Chem., **93**, 3643 (1989).
326. R. Szymański, Makromol. Chem., **191**, 933 (1990).
327. R. Szymański, N. Manolova, I. Rashkov, Makromol. Chem., **194**, 941 (1993).
328. D.-Y. Yan. G.-F. Cai, Z.-D. Lan, Makromol Chem., **187**, 2179 (1986).
329. N. G. Taganov, Vysokomol. Soyed. A, **29**, 1682 (1987); Polym. Sci., USSR, **29**, 1847 (1987).
330. I. Ermakov, R. DeClercq, E. Goethals, Macromol. Theory Simul., **3**, 427 (1994).
331. D. Yan, H. Dong, P. Ye, Macromolecules, **29**, 8057 (1996).
332. M. Fontanille, G. Hélary, M. Szwarc, Macromolecules, **21**, 1532 (1988).
333. D. Yan, X. Hu, Polymer, **29**, 1858 (1988).
334. X. Hu, D. Yan, Polymer, **31**, 1950 (1990).
335. D. Yan, C. Yuan, Makromol. Chem. Rapid Commun., **8**, 83 (1987).
336. E. T. Thachil, Polym.-Plast. Technol. Eng., **32**, 151 (1993).
337. G.-F. Cai, D.-Y. Yan, Makromol. Chem., **188**, 1885 (1987).
338. J. Villermaux, L. Blavier, M. Pons, in: K. H. Richert, W. Geisler, (Eds.), Influence React. Eng. Polym. Prop., [Workshop Proc.] Hanser, Munich, 1983, p. 1.
339. P. Lorenzini, M. Pons, J. Villermaux, Chem. Eng. Sci., **47**, 3969 (1992).
340. N. G. Taganov, Vysokomol. Soyed. A, **27** 732 (1985); Polym. Sci., USSR, **27**, 816 (1985).
341. N. G. Taganov, Vysokomol. Soyed. A, **27**, 741 (1985); Polym. Sci., USSR, **27**, 828 (1985).
342. H. Tobita, Macromolecules, **26**, 836; 5427 (1993).
343. H. Tobita, Makromol. Chem. Theory Simul., **2**, 761 (1993).
344. H. Tobita, Polymer, **35**, 3023 (1994).
345. H. Tobita, Y. Takada, M. Nomura, Macromolecules, **27**, 3804 (1994).
346. L.-D. Shiau, Macromolecules, **28**, 6273 (1995).
347. L. Rief, H. Höcker, Macromolecules, **17**, 952 (1984).
348. L. Nie, R. Narayan, Polym. Preprints, **34**, 632 (1993).
349. M. Kosmas, N. Hadjichristidis, Macromolecules, **27**, 5216 (1994).

350. J. H. O'Donnell, C. L. Winzor, D. J. Winzor, Macromolecules, **23**, 167 (1990).
351. V. S. Papkov, M. N. Il'ina, G. L. Slonimskii, Vysokomol. Soyed. A, **30**, 1487 (1988); Polym. Sci., USSR, **30**, 1566 (1988).
352. V. S. Papkov, M. N. Il'ina, G. L. Slonimskii, Vysokomol. Soyed. A, **30**, 1495 (1988); Polym. Sci., USSR, **30**, 1576 (1988).
353. A. Ya. Malkin, Yu. B. Lavochnik, V. P. Begishev, Vysokomol. Soyed. A, **26**, 775 (1984); Polym. Sci., USSR, **26**, 863 (1984).
354. A. Kumar, J. Appl. Polym. Sci., **34**, 1383 (1987).
355. J. B. P. Soares, A. E. Hamielec, Macromol. Theory Simul., **5**, 547 (1996).

Heats and Entropies of Polymerization, Ceiling Temperatures, Equilibrium Monomer Concentrations, and Polymerizability of Heterocyclic Compounds*

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*Based on the tables compiled for the third edition, by W.K. Buslield, School of Sciences Griffith University, Brisbane, Australia.

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A. HEATS OF POLYMERIZATION

Symbols: The subscripts xx to ΔH denote the state of the monomer (first letter) and the state of the polymer (second letter), as follows:

g Gaseous state (hypothetical in case of polymer)
c Condensed amorphous state
c′ Crystalline or partially crystalline state
l Liquid state
s In solution; solvent specified in the sixth column (ls denotes polymer dissolved in monomer).

For emulsion polymerization the subscript lc is used and a note added in the sixth column. In all cases where monomer or polymer is present in solution etc., the value of ΔH will depend on the composition to some extent. Where the polymer is crystalline, ΔH will depend on the degree of crystallinity. ΔH values are in kJ per mol of monomer and are generally the limiting values for high degree of polymerization n [1 kcal = 4.187 kJ]. Where more than one value is available in the literature for given states of monomer and polymer and a given method, only one value is shown in the tables. This corresponds to one of the references indicated N_i refers to the ith note given at the end of the table.

Example: $C_2H_4\,(g) \rightarrow (1/n)(C_2H_4)_n\,(g)$, $\Delta H_{gg} = -93.5\,kJ$

Precision: generally 0.5–4.0 kJ/mol (for details see Ref. 1).

Methods of determination: These are summarized below, using the numbering system in Ref. 1.

2 Combustion of monomer or polymer or both.
3 Reaction calorimetry.
4a Thermodynamic (van't Hoff Equation).
4b Semiempirical rules applied to evaluate heat of formation of polymer or monomer or both. Such rules may be found in Refs. 25,143,223,224,335.
5 Heat capacity and heat of transition (Kirchhoff's equation).

TABLE 1. MONOMERS GIVING POLYMERS CONTAINING CARBON ATOMS ONLY IN THE MAIN CHAIN

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
1.1. WITH ACYCLIC CARBONS ONLY IN THE MAIN CHAIN						
1.1.1. DIENES						
1,3-Butadiene	gg	73	25	4b	1:2 Polymerization	10
	gg	78	25	4b	1:4 Polymerization	10
	lc	73	25	2	N_1	45
Chloroprene	lc	68	61.3	3		41
Cyclopentadiene	ss	59	−70	3	Methylene chloride	190
Isoprene	gg	70.5	25	4b		10
	lc	75	25	2		46
	ls	71	74.5	3		15
	ls	65.5	34.6	3	N_2	47
1.1.2 MONOMERS GIVING POLYMERS WITH OR WITHOUT ALIPHATIC SIDE CHAINS THAT CONTAIN ONLY C, H						
1.1.2.1. Olefins (listed by increasing carbon number)						
Ethylene	gg	93.5	25	4b		2
	gc′	108.5	25	2		2,3
	gc′	107.5	25	3		183
	gc	101.5	25	2	ΔH_{fus} (Polymer) taken as 5.0	2,4
	c′c′	173	−173	5	Polymer is poly(1-butene), iso	385
	c′c	169	−173	5	Polymer is poly(1-butene), iso	385
	gc′	212	173	5	Polymer is poly(1-butene), iso	385
	gc	209	173	5	Polymer is poly(1-butene), iso	385
	gc′	212	25	4b	Polymer is poly(1-butene), iso	385
	gc	209	25	4b	Polymer is poly(1-butene), iso	385
	gc′	211	127	5	Polymer is poly(1-butene), iso	385
	gc	204	127	5	Polymer is poly(1-butene), iso	385
	gl	198	227	5	Polymer is poly(1-butene), iso	385
Propene	gg	86.5	25	4b		5
	gc′	104	25	2		3
	lc	84	25	4b		5
	sc	69	−78	3	*n*-Butane	6

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TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
1-Butene	gg	86.5	25	4b		5
	lc	83.5	25	4b		5
	lc′	78	−173	5	Isotactic polymer	385
	lc	74	−173	5	Isotactic polymer	385
	lc′	83	−73	5	Isotactic polymer	385
	lc	79	−73	5	Isotactic polymer	385
	gc′	108	25	4b	Isotactic polymer	385
	gc	103	25	4b	Isotactic polymer	385
	gc′	107	127	5	Isotactic polymer	385
	gc	99	127	5	Isotactic polymer	385
	gl	94	227	5	Isotactic polymer	385
2-Butene, *cis*	gg	80	25	4b		5
	lc	75	25	4b		5
	c′c′	63	−173	5	Isotactic polymer	385
	c′c	59	−173	5	Isotactic polymer	385
	lc′	75	−73	5	Isotactic polymer	385
	lc	71	−73	5	Isotactic polymer	385
	gc′	101	25	4b	Isotactic polymer	385
	gc	96	25	4b	Isotactic polymer	385
	gc′	99	127	5	Isotactic polymer	385
	gc	92	127	5	Isotactic polymer	385
	gl	86	227	5	Isotactic polymer	385
2-Butene, *trans*	gg	75.5	25	4b		5
	lc	71	25	4b		5
	c′c′	57	−173	5	Isotactic polymer	385
	c′c	54	−173	5	Isotactic polymer	385
	lc′	71	−73	5	Isotactic polymer	385
	lc	67	−73	5	Isotactic polymer	385
	gc′	96	25	4b	Isotactic polymer	385
	gc	91	25	4b	Isotactic polymer	385
	gc′	96	127	5	Isotactic polymer	385
	gc	89	127	5	Isotactic polymer	385
	gl	83	227	5	Isotactic polymer	385
Isobutene	gc	72	25	2		7
	lc	48	25	2	Low polymer	3,8
	ss	53.5	25	?	Solvent? low polymer	8
	ss	54	−35	3	CH_2Cl_2, polymer swollen, N_3	9
1-Pentene	c′c′	84	−173	5		386
	c′c	81	−173	5		386
	lc′	95	−73	5		386
	lc	93	−73	5		386
	lc′	99	25	2,4b		386
	lc	95	25	2,4b		386
	gl	117	127	5		386
	gl	113	227	5		386
–, 4-methyl	c′c′	82	−173	5		387
	c′c	79	−173	5		387
	lc′	93	−73	5		387
	lc	90	−73	5		387
	lc′	97	25	4b		387
	lc	95	25	4b		387
	gc′	124	127	5		387
	gc	117	127	5		387
	gc′	122	227	5		387
	gc	112	227	5		387
	gl	105	327	5		387
2-Pentene, *cis*	gg	80.5	25	4b		10
	c′c′	75	−173	5		386
	c′c	72	−173	5		386
	lc′	87	−73	5		386
	lc	85	−73	5		386
	lc′	91	25	2,4b		386
	lc	87	25	2,4b		386

TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	gl	109	127	5		386
	gl	105	227	5		386
–, 4-methyl	lc′	90	25	4b		387
	lc	88	25	4b		387
	gc′	118	127	5		387
	gc	111	127	5		387
	gc′	116	227	5		387
	gc	114	227	5		387
	gl	100	327	5		387
2-Pentene, *trans*	gg	75.5	25	4b		10
	c′c′	70	−173	5		386
	c′c	68	−173	5		386
	lc′	83	−73	5		386
	lc	81	−73	5		386
	lc′	87	25	2,4b		386
	lc	83	25	2,4b		386
	gl	106	127	5		386
	gl	103	227	5		386
–, 4–methyl	lc′	86	25	4b		387
	lc	83	25	4b		387
	gc′	115	127	5		387
	gc	107	127	5		387
	gc′	113	227	5		387
	gc	103	227	5		387
	gl	97	327	5		387
1-Hexene	lc	83	25	4b		5
1-Heptene	gg	86	25	4b		5
1.1.2.2. *Cycloalkanes* (listed by increasing ring size)						
Cyclopropane	lc	113	25	2		11
–, methyl-	lc	105	25	4b		11
–, 1,1-dimethyl-	lc	97.5	25	4b		11
Cyclobutane	lc	105	25	2		12
–, methyl-	lc	100	25	4b		11
–, 1,1-dimethyl-	lc	93.5	25	4b		11
Cyclopentane	lc	22	25	2		12
–, methyl-	lc	17	25	4b		11
– 1,1-dimethyl-	lc	13.5	25	4b		11
Cyclohexane	lc	−3	25	2		12
–, methyl-	lc	−9	25	4b		11
–, 1,1-dimethyl-	lc	−7.5	25	4b		11
Cycloheptane	lc	21.5	25	2		12
Cyclooctane	lc	34.5	25	2		12
Cyclononane	lc	47	25	2		13
Cyclodecane	lc	48	25	2		13
Cycloundecane	lc	45	25	2		13
Cyclododecane	lc	14	25	2		13
Cyclotridecane	lc	22	25	2		13
Cyclotetradecane	lc	7	25	2		13
Cyclopentadecane	lc	12	25	2		13
Cyclohexadecane	lc	8	25	2		13
Cycloheptadecane	lc	8.5	25	2		13
1.1.2.3. *Cycloalkenes*						
Cyclobutene	ll	121	25	2,4b	*cis*-Polybutadiene	388
	gl	145	25	2,4b	*cis*-Polybutadiene	388
Cyclopentene	lc	17.5	25	2	N_{66}	361
	c′(I)c	11	−173	5	*cis*-Polymer	388
	lc	16	−73	5	*cis*-Polymer	388
	lc	15	25	2,4b	*cis*-Polymer	388
	c′(I)c	23	−173	5	*trans*-Polymer	388
	lc	19	−73	5	*trans*-Polymer	388
	lc	18	25	2,4b	*trans*-Polymer	388
	lc	18	77	5	*trans*-Polymer	388

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TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
Cyclopentadiene	ss	59	−173	3	Butyl chloride, N_{70}	371
Cyclohexene	ss	1	−65	4a	Toluene, N_{78}	389
Cycloheptene	c′(III)c	12	−173	5	70% *trans*, 30% *cis*-Polymer, N_{54}	388
	c′(III)c′	18	−173	5	70% *trans*, 30% *cis*-Polymer, N_{54}	388
	c′(II)c	18	−73	5	70% *trans*, 30% *cis*-Polymer, N_{54}	388
	c′(II)c′	26	−73	5	70% *trans*, 30% *cis*-Polymer, N_{54}	388
	lc	18	25	2,4b	70% *trans*, 30% *cis*-Polymer	388
	lc′	29	25	2,4b	70% *trans*, 30% *cis*-Polymer	388
	ll	17	51	5	70% *trans*, 30% *cis*-Polymer	388
Cyclooctene	c′(II)c′	18	−173	5	48% *trans*, 52% *cis*-Polymer, N_{54}	388
	c′(II)c	6	−173	5	48% *trans*, 52% *cis*-Polymer, N_{54}	388
	c′(I)c′	27	−73	5	48% *trans*, 52% *cis*-Polymer, N_{54}	388
	c′(I)c	19	−73	5	48% *trans*, 52% *cis*-Polymer, N_{54}	388
	lc′	30	25	2,4b	48% *trans*, 52% *cis*-Polymer	388
	lc	13	25	2,4b	48% *trans*, 52% *cis*-Polymer	388
	ll	13	57	5	48% *trans*, 52% *cis*-Polymer	388
1.1.3. MONOMERS GIVING POLYMERS WITH ALIPHATIC SIDE CHAINS THAT CONTAIN HETEROATOMS						
1.1.3.1. Acrylic Derivatives (listed alphabetically)						
Acrolein	lc	80	74.5	3		15
	ss	57.5	74.5	3	Hexane	17
	ss	81.5	74.5	3	Water	17
Acrylamide	ss	81.5	74.5	3	Water	17,14
	lc	79.4	90	3		332
	ss?	70.5	74.5	3	Acetone	17
	sc	60	74.5	3	Benzene	17
	sc	57.5	74.5	3	Hexane	17
–, *N*-methylol-	lc	74.5	190	3		332
–, *N,N′*-methylene bis-	lc	82.7	190	3		332
Acrylic acid	lc	67	74.5	3		15
	ss	77.5	20	3	Water	36,37
	sc	73.5	74.5	3	Benzene	15
	sc	72	74.5	3	Carbon tetrachloride	15
	sc	74.5	74.5	3	Hexane	15
–, *n*-butyl ester	lc	78	74.5	3		17,37
	ss	77	74.5	3	Acetone	17
	ss	79	74.5	3	Butanone	17
	ss	76	74.5	3	Benzene	17
	ss	75.5	74.5	3	Carbon tetrachloride	17
	ss	78	74.5	3	Hexane	17
–, ethyl ester	lc	78	74.5	3		34,37
	lc	80.7	90	3		318
–, ethylhexyl ester	lc	78.2	90	3		318
–, lauryl ester	lc	80.5	30	3		352
–, methyl ester	lc	78	76.8	3		21,34,37
	lc	81.8	80	3		319
	ss	84.5	20	3	Ethyl alcohol	36
	ss	81	74.5	3	Hexane	34
Acrylonitrile	lc′	76.5	74.5	3		17,21,38,201
	sc′	77.5	74.5	3	Benzene	17
1.1.3.2. α-Substituted Acrylic Derivatives (listed alphabetically)						
α-Chloroacrylic acid, methyl ester	lc	52.7	80	3		319
α-Cyanoacrilic acid, methyl ether	lc	42	60	3		354
–, ethyl ester	c′c	28	−173	5		390
	c′c	29	−73	5		390
	lc	47	25	2		390
	lc	53	127	5		390
	lc	54	177	5		390
–, allyl ester	c′c	41	−173	5		391
	c′c	42	−73	5		391
	c′c	43	−8	5		391
	lc	62	−8	5		391

TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	lc	64	25	2,4b		391
	lc	68	127	5		391
–, *n*-decyl ester	c′c	29	−173	5		392
	c′c	29	−73	5		392
	lc	69	25	2,4b		392
	lc	74	127	5		392
–, allyloxyethyl	cc	41	−173	5		393
	lc	43	−73	5		393
	lc	49	25	2,4b		393
	lc	51	57	5		393
Ethacrylic acid, methyl ester	ls	31	87	4a		395
Methacrolein	lc	65.5	74.5	3		15
Methacrylamide	ss	56	74.5	3	Water	17
	ss	42.5	74.5	3	Chloroform	17
	ss	39.5	74.5	3	Acetone	17
	ss	35	74.5	3	Benzene	17
Methacrylic acid	lc	42.5	74.5	3		15
	lc	64.5	25	2		206
	ss	66	20	3	Water	36
	ss	56.5	25	3	Water	37
	ss	54	25	3	Dimethylformamide, N_7	203,316
	sc	57	74.5	3	Methanol, N_5	15
–, benzoic acid ester	c′c	55	−173	5		394
	c′c	56	−73	5		394
	c′c	59	25	5		394
	ss	57	77	2,3	1,4 Dioxane	394
	ll	90	227	5		394
–, benzyl ester	lc	56	76.8	3	N_6	39
–, *n*-butyl ester	lc	57.5	74.5	3		34,37,39
	ls	60	26.9	3		14
–, *tert*-butyl ester	ls	54.5	26.9	3		14
–, cyclohexyl ester	lc	51	76.8	3		39
	ls	53	26.9	3		14
–, 2-ethoxyethyl ester	lc	57.5	74.5	3		34
	ls	62	26.9	3		14
–, ethyl ester	lc	60	120	4a		40
	lc	59.5	74.5	3		34,37
	ls	57.5	26.9	3		14
–, β(*N*-piperidyl)ethyl ester	lc	57	40	3		188
–, *n*-hexyl ester	lc	58.5	25	3	Emulsion (aq.)	37
	ls	60	26.9	3		14
–, 2-hydroxyethyl ester	lc	50	25	3	Emulsion (aq.)	37
–, 2-hydroxypropyl ester	lc	50.5	25	3	Emulsion (aq.)	37
–, isobutyl ester	lc	60		3		34
–, isopropyl ester	lc	60	74.5	3		34
–, methyl ester	lc	56	130	4a		42,43,29
	lc	55.5	74.5	3		17,34,36,37, 41,201,235
	lc	55	25	2		206
	ls	57.5	26.9	3		14
	c′c	34	−173	5		414
	c′c	36	−73	5		414
	lc	55	25	5		414
	lc	56	77	3		414
	ss	54	130	4a	*o*-Dichlorobenzene	42,29
	ss	58.5	74.5	3	Acetonitrile	34
	ss	57.5	74.5	3	Tetrahydrofuran	34
	ss	58.5	74.5	3	Hexane	34
–, methyl ester deuterated	c′c	33	−173	5		414
	c′c	35	−73	5		414
	lc	55	25	5		414
	lc	57	77	3		414
–, *n*-octyl ester	lc	57	65	3		188

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TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
–, phenyl ester	lc	51.5	76.8	3	N_6	39
–, *n*-propyl ester	lc	57.5	74.5	3		34
Methacrylonitrile	lc	56.5	74.5	3		15
	ss	64	130	4a	Benzonitrile	44,29
1.1.3.3. Vinyl Ethers						
Vinyl *n*-butyl ether	lc?	60	50	3	N_6	216
1.1.3.4. Vinyl Alcohol, Ketones, Halogens						
Vinyl alcohol					see Acetaldehyde	195
Methyl vinyl ketone	ls	74	74.5	3		17
Vinyl chloride	gc	132	25	2		18.19,20
	lc	71	25	4b		10
	lc	111.5	25	2	ΔH_{fus} (Monomer) taken as 20	18
	lc	96	74.5	3		215
1.1.3.5. Vinyl Esters						
Vinyl acetate	lc	88	74.5	3		17,34,21
	ls	89.5	25	3		35
	ss	90	74.5	3	Acetone	17
	ss	86.5	74.5	3	Hexane	17
	ss	86	74.5	3	Benzene	17
Vinyl propionate	lc	86	74.5	3		17,34
Vinyl 2-ethylhexoate	lc	88	74.5	3		34
1.1.3.6. Others (listed alphabetically)						
Allyl chloride	ls	77.5	74.5	3		15
	lc	88	20	2		214
	ss	91	20	3	Dimethylformamide, tetrahydrofuran	214
Ethylene, nitro						
–, tetrafluoro-	gg	155	25	2		22
	gc′	172	25	2		22,23
	lc′	163 ± 17	25	2		1
Fumaric acid, diethyl ester	lc	65	100	3		235
Itaconic acid, dimethyl ester	ss	60.5	26.9	3	*o*-Dichlorobenzene	14
Maleic anhydride	ls	59	74.5	3		15
Maleimide	ss	67.5	74.5	3	Chlorobenzene	24
	ss	89.5	74.5	3	Dioxane	24
	ss	88.5	74.5	3	Acetonitrile	24
	ss	87.5	74.5	3	Dimethylformamide	24
–, phenyl	ss	37	55	4a	DMF, N_{78}, N_{75}	440
α-Methylene pentanolactone	ss	38.8	83	4a	DMF	315
Vinyl cyclohexane	c′c′	88	−173	5		396
	c′c	68	−173	5		396
	lc′	100	−73	5		396
	lc	79	−73	5		396
	lc′	105	25	2,4b		396
	lc	85	25	2,4b		396
	lc′	111	127	5		396
	lc	86	127	5		396
Vinylidene chloride	lc′	60	76.8	3		21
	lc′	75.5	25	2		18
	lc′	73	74.5	3		15
	c′c′	67	−173	5		397
	c′c	56	−173	5		397
	lc′	77	−73	5		397
	lc	66	−73	5		397
	lc′	81	25	2		397
	lc	70	25	2		397
	gc′	108	25	2		397
	gc	97	25	5		397
	gc′	107	127	5		397
	gc	95	127	5		397
N-Vinyl succinimide	ls	54.7	97	3		384

TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
1.1.4. MONOMERS GIVING POLYMERS WITH AROMATIC SIDE CHAINS THAT CONTAIN ONLY C, H (listed alphabetically)						
Acenaphthalene	ss	87.8	0	3	Dichloromethane	320
	ss	84.8	18	3	Dichloromethane	320
	ss	98.5	26.9	3	*o*-Dichlorobenzene	14
	ss	67	74.5	3	*o*-Dichlorobenzene	15
	ss	67	74.5	3	Benzene	15
	c'c	82	26.9	3		14
Acenaphthylene	ss	43	60	4a	Toluene, N_{78}, N_{75}	399
Biphenyl, *p*-isopropenyl-	ss	34	−15	4a	Tetrahydrofuran	16
Indene	ss	58	−30	3	Methylene chloride indep. of T [−70° to 10°]	190
−, 5-methyl-	ss	58	−50	3	Methylene chloride	175
−, 6-methyl-	ss	57.5	−60	3	Methylene chloride	175,169
−, 7-methyl-	ss	58.5	−70	3	Methylene chloride	175,169
−, 4,6-dimethyl-	ss	77.5	−40	3	Methylene chloride	175
−, 4,7-dimethyl-	ss	65	−40	3	Methylene chloride	175
−, 5,6-dimethyl-	ss	79	−40	3	Methylene chloride	175
−, 4,5,6,7-tetramethyl-	ss	75	−40	3	Methylene chloride	175
Naphthalene, 2-isopropenyl-	ss	36.5	−5	4a	Tetrahydrofuran	16
	ss	38	10	4a	Toluene, N_{78}	400
Styrene	gg	74.5	25	4b		25
	lc	70	25	2		26
	lc	68.5	26.9	3		14,17,27,201,235
	lc	73	127	4a		28,29
	ls	73	25	2		26
	ls	73	26.9	3		14
	ss	66.5	−60	3	Methylene chloride	30,190
− α-acetoxy	ss	26	64	4a	Toluene, N_{78}, N_{75}	401
−, *o*-chloro-	lc	68.5	76.8	3		27
−, *p*-chloro-	lc	67	76.8	3		27
−, 2,5-dichloro-	lc	69	76.8	3		27
−, ar-ethyl-	lc	68	76.8	3		27
−, α-methyl-	lc	35	25	2		31
	lc	34.5	−20	4a		32,29
	lc	39	25		N_9	123
	ss	33.5	−20	4a	Tetrahydrofuran	32,29,33,16
	ss	47.7	22	4a	Cyclohexane	341
	ss	15	−10	4a	*n*-Heptane, N_{78}	402
	ss	14	−10	4a	Toluene, N_{78}	402
−, α-methyl, *p-tert*-butyl-	ss	29.7		4a	Solvent and temp. not stated	381
−, 2,4,6-trimethyl-	lc	70	26.9	3		14
1.1.5. MONOMERS GIVING POLYMERS WITH AROMATIC SIDE CHAINS AND THAT CONTAIN HETEROATOMS						
2-Amino-4-alkyl/aryl-amino-6-isopropenyl-1,3,5-triazine,						
R_1 = H, R_2 = ethyl	ls	64.8	170	3	N_{65}	370
R_1 = H, R_2 = propyl	ls	63.5	160	3	N_{65}	370
R_1 = ethyl, R_2 = ethyl	ls	53.9	161	3	N_{65}	370
R_1 = propyl, R_2 = propyl	ls	47.7	153	3	N_{65}	370
R_1 = methyl, R_2 = phenyl	ls	50.2	140	3		379
R_1 = methyl, R_2 = phenyl	ls	53.1	150	4a		379
R_1 = H, R_2 = *p*-methoxyphenyl	ls	50.2	160	3		378
R_1 = H, R_2 = *p*-methoxyphenyl	ls	46.8	160	4a		378
2-Amino-4-(*N*-R. anilino)-6-isopropenyl-1,3,5-triazine						
R = *n*-alkyl = *n*-propyl	lc	46	60	3		403
= *n*-hexyl	lc	43	60	3		403
= *n*-octyl	lc	42	60	3		403
= *n*-decyl	lc	41	60	3		403
= *n*-dodecyl	lc	41	60	3		403
= *n*-tetradecyl	lc	40	60	3		403
= *n*-hexadecyl	lc	39	60	3		403
R = alkyl = isopropyl	lc	44	60	3		404
= isobutyl	lc	44	60	3		40
= *sec*-butyl	lc	43	60	3		404

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TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
= *n*-pentyl	lc	44	60	3		404
= isopentyl	lc	43	60	3		404
= 1-methylbutyl	lc	42	60	3		404
R = benzyl	lc	44	60	3		404
R = chloro	ls	11	170	3		405
R = – 4-methoxy-	ls	14	183	3		406
R = – 4-ethoxy-	ls	14	182	3		406
2-Anilo-4-R-6-isopropenyl-1,3,5-triazine						
R = methoxy	ls	12	174	3		406
R = ethoxy	ls	12	174	3		406
Atropic acid, methyl ester	ss	28	– 40	4b	Toluene, N_{10}	149
Atroponitrile	ss	39	50	4b	Toluene, N_{10}	149
Benzoic acid, *n*-methacryloyloxy-	lc	59	25	2		363
–, vinyl ester	lc	84.5	74.5	3		34
Carbazole, *N*-vinyl-	sc′	63.5	74.5	3	Hexane	15
	ss	94.9	0	3	Dichloromethane	321
2,4-Hexadiyne-1,6-di(R group)						
R = $-SO_3C_6H_4CH_3$(p)	c′c′	152.6	187	3		372
R = $-NHC_6H_3(NO_2)_2$(2,4 posn.)	c′c′	77.5	135	3		373
R = $-SO_3C_6H_4NNC_6H_5$(p)	c′c′	138	140	3		374
R = $-OCONHCH_2COOH$	c′c′	130	137	3		375
Oxazole, 2-isopropenyl-	ss	39.5	20	4a	Tetrahydrofuran	284
Pyridine						
–, 2-isopropenyl-	ss	26	20	4a	Tetrahydrofuran	284
	ss	40	– 15	4a	Toluene, N_{78}	407
–, 2-vinyl-	lc	71.5	74.5	3		15
	ls	75.5	74.5	3		15
	sc	73.5	74.5	3	Benzene	15
–, 4-vinyl-	lc	78	74.5	3		17
	ss	78	74.5	3	Benzene, hexane	17
Silane, benzyl dimethyl vinyl-	c′c′	48.2	– 73	2		355
	lc	66.7	25	3	N_{62}	366
	lc	67.1	77	3	N_{62}	366
–, dimethyl phenyl vinyl-	lc	56.5	25	3	N_{62}	362
–, trimethyl vinyl-	lc	65.8	25	3	N_{62}	367
N,N,N′,N′-Tetraethyl-4-isopropenyl-phenylphosphonous diamide	ss	27	– 55	4a	Tetrahydrofuran	408
Thiazole, 2-isopropenyl-	ss	28.5	20	4a	Tetrahydrofuran	284
p-[bis(Trimethylsilyl)methyl]isopropenylbenzene	ss	30	– 30	4a	Tetrahydrofuran	409
1.2. WITH AROMATIC OR CYCLIC CARBONS IN THE MAIN CHAIN						
– 11,12-Dicyano-2,6-naphtoquinodimethane						
11,12-bis (alkylthio)						
alkyl = butyl	ss	67	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= s-butyl	ss	59	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= isopropyl	ss	65	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= ethyl	ss	69	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= methyl	ss	70	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
– , 7,8-Dicyanoquinodimethane						
7,8-bis(butoxycarbonyl)-	ss	26	60	4a	Chloroform, N_{78}, N_{81}	411
7,8-diacetyl-	ss	25	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-dibenzoyl-	ss	22	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-dibutoxycarbonyl-	ss	26	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-diisobutyryl-	ss	23	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-dipropionyl-	ss	24	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-bis(ethylthio)-	ss	23	60	4a	Chloroform, N_{78}, N_{81}	411
7,8-bis(phenylacetyl)-	ss	25	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-bis(phenylthio)-	ss	23	50	4a	Chloroform, N_{78}, N_{81}	411
Norbornylene	c′(II)c	56	– 173	5	55% *cis*/45% *trans* Polymer, N_{54}	415
	c′(I)c	62	– 73	5	55% *cis*/45% *trans* Polymer, N_{54}	415
	c′(I)c	62	25	4b	55% *cis*/45% *trans* Polymer, N_{54}	415
	lc	65	57	5	55% *cis*/45% *trans* Polymer, N_{54}	415
5-Trimethylsilyl-2-	c′c	38	– 173	5		416
	cc	40	– 173	5	Overcooled monomer	416

TABLE 1. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	c′c	39	−73	5		416
	cc	46	−73	5	Overcooled monomer	416
	lc	52	25	4b		416
	lc	55	107	5		416
N-7,7-Triacyanobenzoquinonemethideimine	ss	22	55	4a	Toluene N_{78}, N_{82}	438

TABLE 2. MONOMERS GIVING POLYMERS CONTAINING HETEROATOMS IN THE MAIN CHAIN

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
2.1. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO CARBON ONLY						
2.1.1. ETHERS AND ACETALS (listed by increasing ring size of monomer)						
2-RINGS (aldehydes and ketones)						
Acetaldehyde	lc	0	25	4b	N_{11}	48
	lc′	64.5	25	2	To poly(vinyl alcohol)	195
	lc	62.5	25	2	To poly(vinyl alcohol)	195
	c′c	31	−173	5		417
	lc	34	−73	5		417
	lc	36	25	2,4b		417
–, bromochlorofluoro-	sc′	36.0		4a	Toluene, N_{74}	233
–, bromodichloro-	sc′	31.2		4a	Toluene, N_{74}	344
–, bromodifluoro-	sc′	43.4		4a	Toluene, N_{74}	344
–, dibromochloro-	sc′	26.0		4a	Toluene, N_{74}	344
–, dibromofluoro-	sc′	29.0		4a	Toluene, N_{74}	344
–, dichlorofluoro-	sc′	56.2		4a	Toluene, N_{74}	344
–, difluorochloro-	sc′	51.0		4a	Toluene, N_{74}	344
–, tribromo-	sc′	19.6		4a	Toluene, N_{74}	344
–, trichloro-	gc′	71 ± 8	50	4a		50
	gc′	65.5	−50	4a		166
	gc	51	−50	4a		166
	lc′	17	57	5		418
	lc′	38 ± 8	50	4a		50
	lc′	34	25	3		418
	lc′	34.5	−50	4a		166
	lc	20	−50	4a		166
	lc′	30	−73	5		418
	cc′	25	−173	5		418
	ss	14.5	−50	4a	Tetrahydrofuran	167
	sc′	33.5	13	4a	Pyridine	50
	sc′	39	10	4a	*n*-Heptane	166
	sc′	37.8		4a	Toluene, N_{74}	344
–, trifluoro-	gc	64.5	45	4a		129
	lc	47	25	4a	Dichloromethane	322
	sc	50.8	25	4a	Dichloromethane	322
	sc′	54.9		4a	Toluene, N_{74}	344
Acetone	gg	−12	25	4b		234
	gc	10	25	4b		144,148
	lc	−25	25	4b		48,234
Butanal	lc	20.5	20	3		161
	sc	35.5	18	4a	*n*-Hexane (9 kbar), N_{12}	161
	sc	21.5	20	3	*n*-Hexane (1 bar, 1M), N_{12}	161
4,7-Dioxaoctanal	lc	15.9	−80	4a	N_{69}	330
	ss	16.7	−80	4a	THF	330
Formaldehyde	gc′	66	25	4a	N_{13}	49,164, 165,132
	gc′	55	25	2		7,49
Heptanal	c′c′	20	−173	5		419
	c′c	10	−173	5		419

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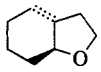
TABLE 2. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	c′c′	22	−73	5		419
	c′c	8	−73	5		419
	lc(I)	39	25	3	N_{77}	419
	lc	33	25	3		419
	ll	36	97	5		419
Hexanal	c′c′	8	−173	5		420
	c′c	7	−173	5		420
	c′c′	12	−73	5		420
	c′c	10	−73	5		420
	ll	27	25	3		420
	ll	28	67	5		420
Octanal	lc	14.7	−75	4a	N_{69}	330
	sc	14.2	−75	4a	THF	330
4-Oxapentanal	lc	19.7	−80	4a	N_{69}	329
	ls	20.1	−80	4a		329
Pentanal	lc	22.7	−75	4a	N_{69}	328
	sc	22.2	−75	4a	THF	328
o-Phthalaldehyde	ss	22	−70	4a	Dichloromethane, cyclopolymer	154
Propanal	c′c′	5	−173	5		421
	c′c	2	−173	5		421
	lc′	15	−73	5		421
	lc	12	−73	5		421
	lc′	19	25	3		421
	lc	14	25	3		421
	lc′	21	57	5		421
	lc	15	57	5		421
–, 2,2-dichloro-	gc	50	−65	4a		172
	lc	20.5	−65	4a		172
	ss	17	−65	4a	Tetrahydrofuran	172
–, 2-chloro, 2-methyl-	gc	47	−65	4a		172
	lc	19.5	−65	4a		172
	ss	19.5	−65	4a	Tetrahydrofuran	172
–, 2-methyl-	gc	46	−65	4a		166
	lc	16	−65	4a		166
	ss	15.5	−65	4a	Tetrahydrofuran	167,172
	ss	16.5	−65	4a	Diethyl ether	166
	ss	22	−65	4a	*n*-Pentane	166
3-RINGS						
Cyclohexene oxide	lc	97	25	2		422
Ethylene oxide	gg	104	25	4b		14
	gc′	140	25	3		168
	gc	127.3	25	2		306
	lc′	102.4	25	2		306
	lc	94.5	25	2		51
Propylene oxide	gg	75.5	25	4b		14,52
Styrene oxide	lc	101.5	26.9	3		14
–, 3-nitro-	lc	101	26.9	3		14
4-RINGS						
Oxetane	gc	81.8	25	4a		325
	lc	51.4	25	4a	N_{17}	325
	ss	81	−9	3	Chloromethane + chloroethane	53
–, 3,3-di(chloromethyl)-	lc	84.5	26.9	3		14
–, 3,3-dimethyl-	ss	67.5	−9	3	Chloromethane + chloroethane	53
– 3,3-di(phenoxymethyl)-	ss	83.5	26.9	3	*o*-Dichlorobenzene	14
5-RINGS						
1,3-Dioxolane	gg	26	20	4b		54
	gc	50	25	4a		292
	gc	52.0	57	4a	N_{60}	353
	lc	24	100	4b	N_{14}	198
	lc	17	90	4a	N_{69}, N_{78}	437
	lc	15	25	4a	N_{17}	292

TABLE 2. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	lc	23	40	4a	N_{15}	159
	ls	6.5	55	4a		219
	ss	22	0	4a/3	Dichloromethane	133,159,220
	ss	27	20	4a	Chloroethane	293
	ss	15	30	4a	Benzene	134
	ss	17.5	70	3	Dichloromethane	220
	ss	8	75	4a	Toluene, N_{78}	423
–, 4-ethyl-	lc	13.0	−24	4a		333
–, 4,4-dimethyl-	lc	9.6	80	4b		335
–, *cis*-4,5-dimethyl-	lc	12.1	80	4b		335
–, *trans*-4,5-dimethyl-	lc	7.5	80	4b		335
–, 4-isopropyl-	lc	12.1	80	4b		335
–, 4-methyl-	lc	13.4	−8	4a		334
trans-7-Oxabicyclo[4.3.0]nonane 	lc	14.0				357
7-Oxa-bicyclo[2.2.1]heptane, 2-alkyl-						
R = H	lc	44.3	25	2		308
R = exo-CH_3	lc	49.7	25	2		308
R = endo-CH_3	lc	45.4	25	2		308
Tetrahydrofuran	gg	21	20	4b		55
	gg	12	25	4b		56
	gc	39.5	25	4a		292
	gc	47.1	50	4a	N_{61}	353
	lc′	29.6	25	2		358
	lc′	38	25	2		56,57
	lc	18.6	25	2		358
	lc	7.5	25	4a	N_{17}	292
	lc	12.5	50	4a		130
	lc	15	30	4a	N_{16}	231
	lc	25.5	25	2		140,208
	ls	23	40	4a		57,58,211
	ss	23	40	4a	Diethyl ether	211
–, 3-methyl-	ls	22.6	−6	4a		342
6-RINGS						
m-Dioxane	gg	0	20	4b		54,59,60
p-Dioxane	ls	−4	20	4a	N_{78}	436
Tetrahydropyran	gg	1.5	20	4b		55,56,59
Trioxane	sc′	17.5	40	4a	1,2-Dichloroethane	178
	sc′	26.5	58	3	1,2-Dichloroethane	212
	sc′	19.5	30	3	Dichloromethane	182,202
	sc′	12.5	30	4a	Nitrobenzene	182
	sc′	21.5	30	3	Nitrobenzene	202
	c′c′	6	30	3	N_{18}	202
	c′c′	11.5	58	3	N_{19}	212
	c′c′	4.5	25	4a		164
	c′c′	5.5	25	2		see 202
7-RINGS						
1,3-Dioxepane	gg	19.5	20	4b		54
	gc	53.9	25	4a		323
	lc	13.0	25	4a	N_{17}	323
	ss	12	75	4a	Toluene, N_{78}	423
	ss	13.5	60	4a	Benzene	134,87
	ss	14	0	4a	1,2-Dichloroethane, N_{78}	424
	ss	15	−30	4a/3	Dichloromethane	136
– 2-butyl	ss	7	−5	4a	1,2-Dichloroethane, N_{78}	424
–, 4-methyl-	ss	9.3	−15	4a	1,2-Dichloroethane	339
–, 2-methyl-	ss	8.8	−15	4a	1,2-Dichloroethane	340
	ss	9	−10	4a	1,2-Dichloroethane, N_{78}	424
1-Oxa-4,5-dithiepane $\boxed{-O(CH_2)_2S_2(CH_2)_2-}$	lc	7.5	26.9	3		61
	ss	9	26.9	3	Dioxane	61
	ss	8	26.9	3	Benzene	61

Notes page II-406; References page II-407

TABLE 2. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
Oxepane	gc	38.6	25	4a		323
	lc	−1.7	25	4a	N_{17}	323
8- AND LARGER RINGS						
2-Butyl-1,3,6-trioxocane	ss	8	10	4a	Toluene, N_{78}	425
1,3-Dioxocane	gg	53.5	20	4b		54
	gc	62.1	25	4a		324
	lc	18.3	25	4a	N_{17}	324
Tetraoxane	sc′	26	30	3	Nitrobenzene	202
	c′c′	3.5	30	3		131,202
	c′c′	4	25	2		see 202
1,3,6-Trioxocane	gc	62.2	5	4a		324
	lc	12.0	25	4a	N_{17}	324
	ss	13	20	4a	Toluene, N_{78}	425
	ss	22	60	4a	Benzene	134
(1,3,6-Trioxocane)$_n$						
$n=1$	ss	16.7	0	4a	Dichloromethane	343
$n=2$	ss	4.6	0	4a	Dichloromethane	343
$n=3$–8	ss	0	0	4a	Dichloromethane	343
(1,3,6,9-Tetraoxacycloundecane)$_n$						
$n=1$	ss	13.4	0	4a	Dichloromethane	343
$n=2$	ss	3.8	0	4a	Dichloromethane	343
$n=3$	ss	3.8	0	4a	Dichloromethane	343
$n=4$–8	ss	0	0	4a	Dichloromethane	343
1,3,6,9,12-Pentaoxacyclotetradecane)$_n$						
$n=1$	ss	11.3	5	4a	Dichloromethane	343
$n=2$	ss	7.5	5	4a	Dichloromethane	343
$n=3$–7	ss	0	5	4a	Dichloromethane	343
1,3,6,9,12,15-Hexaoxacycloheptadecane)$_n$						
$n=1$	ss	10.5	−2	4a	Dichloromethane	343
$n=2$	ss	2.9	−2	4a	Dichloromethane	343
$n=3$–5	ss	0	−2	4a	Dichloromethane	343
2.1.2. CYCLIC ESTERS						
4-RINGS						
β-Propiolactone	lc	75	25	3		368
	lc′	82	25	3		368
	lc′	80.5	25	2/4b		137
	lc	74	127	3		368
5-RINGS						
γ-Butyrolactone	lc	−5	25	3		368
	lc′	7	25	3		368
	lc	−6	77	3		368
6-RINGS						
2,2-Dimethyltrimethylene carbonate	c′(II)c′(II)	7	−173	5	N_{54}, N_{63}, N_{77}	426
	c′(II)c	−9	−173	5	N_{54}, N_{63}	426
	c′(II)c′(II)	7	25	4b		426
	c′(II)c	−8	25	4b		426
	lc′(I)	18	117	5	N_{63}, N_{77}	426
	ll	12	147	5	N_{63}	426
p-Dioxanone	c′c′	21	−173	5		427
	c′c	6	−173	5		427
	c′c′	21	−73	5		427
	c′c	6	−73	5		427
	c′c′	21	25	2		427
	c′c	4	25	2		427
	ll	17	127	5		427
Ethylene oxalate	c′c	8.6	−173	5		312
	c′c	9.0	−73	5		312
	c′c	10.1	25	2		312
	c′c	8.3	87	5		312

TABLE 2. *cont'd*

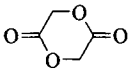
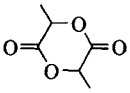
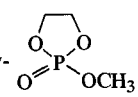
Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	lc	20.0	142	2		312
Glycollide	c′c′	34.0	25	2		364
	cc	10.8	25	2		364
	lc	31.4	277	2		364
	c′c	8.7	25	2		311
DL-Lactide	lc	27.0	127	2		311
L-Lactide	ss	25	40	4a	Nitrobenzene, N_{78}	428
	ss	23	105	4a	1,4-Dioxane, N_{78}	429
	c′c′	20.9	25	2	N_{53}	307
8-Oxo-7-oxabicyclo[2,2,2]octane	gc′	90.5	25	2	N_{53}	307
	lc′	34.5	25	2	N_{53}	307
Pentanolactone	lc′	27.5	25	2		314
	lc	10.5	25	2		314
	lc	9.9	77	3		314
7-RINGS						
ε-Caprolactone	lc	17	25	3		368
	lc′	31	25	3		368
	lc	13.7	77	3		368
	ss	15	40	4a	Nitrobenzene, N_{78}	428
Spiroorthoester						
2-R-1,4,6-trioxaspiro[4.6] undecane						
R = H	ss	11	−30 to 5	4b	Dichloromethane	430
$R = CH_3$	ss	8	−50 to 5	4b	Dichloromethane	430
$R = CH_3Br$	ss	5	−35 to −5	4b	Dichloromethane	430
R = Ph	ss	7	−50 to −10	4b	Dichloromethane	430
8-AND LARGER RINGS						
Heptanolactone	lc	16	127	4b		368
Octanolactone	lc	19	127	4b		368
Nonanolactone	lc	22	127	4b		368
Decanolactone	lc	24	127	4b		368
Undecanolactone	lc	28	25	3		368
	lc′	62	25	3		368
	lc	27	127	3		368
Dodecanolactone	lc	17	127	4b		368
Tridecanolactone	c′c	−2	25	3		368
	c′c′	38	25	3		368
	lc	6	127	3		368
Tetradecanolactone	lc	2	127	4b		368
Pentadecanolactone	c′c	−10	25	1		313
	c′c′	39	25	1		313
	lc	−3	97	1		313
	c′(II)c	−37	−73	1	N_{54}	313
Hexadecanolactone	lc	0	127	4b		368
2.2. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO OTHER HETEROATOMS (S, Si, P)						
Cyclotrisiloxane, 1,1,3,3,5,5-hexaethyl	c′(III)c′(II)	4.6	−173	3	N_{54}, N_{63}, N_{77}	431
	c′(II)c′(II)	5	−73	3	N_{54}, N_{63}, N_{77}	431
	c′(II)c	−10	−23	3	N_{54}, N_{63}	431
	c′(I)c′(I)	10	−23	3	N_{54}, N_{63}, N_{77}	431
	c′(I)c	1	25	3	N_{54}, N_{63}	431
	ll	14	25	3	N_{63}	431
	ls	15	170	3	95% Conversion	431
–, hexamethyl	lc	14.5	25	3		84
	lc	23.4	77	3	N_{62}	365
	c′c	2.79	25	3	N_{63}	365
Cyclotetrasiloxane, octamethyl-	lc	−13.4	77	3	N_{62}	365
	lc	−6.4	25	3	N_{63}	365
N,N-Diethylamino-1,3,2-dioxaphosphorinane	ls	−6.3	−20	4a	N_{58}	346
	ss	15		3	Dichloromethane	309
1,3,2-Dioxaphospholan, 2-methoxy-	ss	14		4a	1,1,2,2-Tetrachloroethane	309

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TABLE 2. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
2-Oxo-1,3,2-dioxaphosphorinane –, 2-alkoxy-	lc	−6.28	90	4a	N_{59}	347
R = methyl	ls	2.9	140	4a	Diglyme or DMSO	348
R = methyl	ss	0.8	100	4a	Dichloromethane	356
R = ethyl	ss	−1.0	100	4a	Dichloromethane	356
R = propyl	ss	−2.4	100	4a	Dichloromethane	356
R = trimethyl silyl	ss	−2.7	100	4a	Dichloromethane	356
Sulfur trioxide	gc′	56	13	4a		86
	lc′	12.5	25	4a		86

2.3. MONOMERS GIVING POLYMERS CONTAINING S IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY (listed by increasing ring size) (polysulfones listed under Table 3. Copolymers)

Carbon disulfide	lc′	23	25	2		179
Thiirane	gg	81	25	4b	N_{20}	69,70,71
	lc	73 ± 4	20	3		173
	sc′	80 ± 4	20	3	Various solvents	173
–, 2,2-dimethyl-	gg	70.5	25	4b	N_{20}	69,70,71
–, *cis*-2,3-dimethyl-	gg	63	25	4b	N_{20}	69,70,71
–, *trans*-2,3-dimethyl-	gg	5.5	25	4b	N_{20}	69,70,71
–, 2-methyl-	gg	71	25	4b	N_{20}	69,70,71
Thietane	gg	80	25	4b	N_{20}	69,70,71
Thioacetone	ls	10.8	50–90	4a	Monomer	327
Thiolane	gg	7.5	25	4b	N_{20}	69,70,71
1,2-Dithiolane $\overline{SSCH_2CH_2CH_2}$	ss	26.5	30	4a	Ethyl alcohol	72
Thiane	gg	−2.5	25	4b	N_{20}	69,70,71
o-Dithiane $\overline{SS(CH_2)_3CH_2}$	lc	2	26.9	3		14
exo-3,4,5-Trithiatricyclo-[5.2.2.0^{2,6}]decane	ss	5.8	40	4a	Toluene, N_{78}	377
exo-3,4,5-Trithiatetracyclo-[5.5.1.0^{2,6}.0^{8,12}]tridec-10-ene	ss	6.6	50	4a	Toluene, N_{78}	376
Thiepane	gg	14.5	25	4b	N_{20}	69,70,71
1,2-Dithiepane $\overline{SS(CH_2)_4CH_2}$	lc	10.5	26.9	3		14
	sc	11.5	26.9	3	Dioxane	14
1,2-Dithiocane $\overline{SS(CH_2)_5CH_2}$	lc	16	26.9	3		14

2.4. MONOMERS GIVING POLYMERS CONTAINING N IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY (listed by increasing ring size)

2-RINGS						
Butylisocyanate	ss	24	−55	4a	Dimethylformamide, N_{78}	432
3-RINGS						
Ethyleneimine	lc	96.2	27	2		360
–, *N*-(β-trimethylsilylethyl)(TEI)	lc	119	27	2		360
–, – $ZnCl_2 \cdot (TEI)_2$ (complex)	c′c	264	27	2		360
4-RINGS						
Trimethyleneimine,						
–, *N*-(β-trimethylsilylethyl)(TMI)	lc	115	27	2		360
–, –, $ZnCl_2 \cdot (TMI)_2$(complex)	c′c	258	27	2		360
5-RINGS						
8-Oxa-6-azabicyclo[3.2.1]octan-7-one	ss	24	38	4a	Dimethylsulfoxide, N_{78}	433
2-Pyrollidinone	lc	4.5	75	2/4b		62
	lc	3.5	43	3		135
	lc′	12	43	3		135
	c′c′	−5.5	25	2/4b		63
–, 1-methyl-	lc	3.5	25	4b		64
Pyrrole	ss	145	15	3	Acetonitrile	434
	ss	104	30	3	Acetonitrile	434
	ss	60	50	3	Acetonitrile	434

TABLE 2. *cont'd*

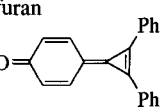
Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
6-RINGS						
2-Piperidone	lc	9	75	2/4b		62
	c′c′	4.5	25	2/4b		63
–, 1-methyl-	lc	−2	25	4b		64
7-RINGS						
ε-Caprolactam	lc	16.3	186	3		310
	lc	16	75	2		62
	lc	15.5	200	3		65,135,138
	ls	16.5	250	4a		68,67,66
	c′c′	12.5	25	2		63
–, 1-methyl	lc	9.5	25	4b		64
–, 3-methyl	ls	12.5	260	4a		194
–, 3-ethyl	ls	16	240	3		191
–, 3-propyl	ls	16.5	240	3		191
–, 4-ethyl	ls	14.5	240	3		171
–, 5-methyl	lc	16	75	4b		62
	ls	11	200	4a		225
–, 7-methyl	lc	16	75	4b		62
	ls	16.5	225	4a		226
–, 7-ethyl	ls	14.5	240	3		191
–, 7-propyl	ls	14.5	240	3		191
8-RINGS						
2-Oxo-hexamethyleneimine	lc	22	75	4b		62
	lc	22	230	3		138,207
	c′c′	24	25	2		63
–, 1-methyl-	lc	16.5	25	4b		64
–, 4-ethyl-	ls	22	240	3		171
–, 7-ethyl-	ls	21.5	240	3		171
9–13-RINGS						
Ω-Dodecalactam	lc	10.6	186	3		310
2-Oxo-heptamethyleneimine	lc	32.5	230	3		138,189
2-Oxo-octamethyleneimine	c′c′	23.5	25	2/4b		138
2-Oxo-nonamethyleneimine	c′c′	11.5	25	2/4b		138
2-Oxo-decamethyleneimine	c′c′	−2	25	2/4b		138
2-Oxo-undecamethyleneimine	lc	~0	230	3		138,139
	ls	13	290	4a		139
2.5. MONOMERS GIVING POLYMERS CONTAINING N IN THE MAIN CHAIN, BONDED TO OTHER HETEROATOMS (P)						
Phosphonitrile chloride						
–, cyclic trimer	gc	61	230	3		125
	lc	6	230	3		125
–, cyclic tetramer	gc	68	230	3		125
	lc	4	230	3		125
–, cyclic pentamer	gc	79.5	230	3		125
	lc	3.5	230	3		125
–, cyclic hexamer	gc	88	230	3		125
	lc	1.5	230	3		125
–, cyclic heptamer	gc	96.5	230	3		125
	lc	0	230	3		125
2.6. OTHER MONOMERS GIVING POLYMERS NOT LISTED ABOVE						
Carbon C (graphite)	c′c′	−39.5	25	2	Linear polymer (carbyne)	187
4,7-bis[Cyano(ethoxycarbonyl) methylene]-4,7-dihydrobenzofuran	ss	21.6	60		Chloroform, N_{78}, N_{79}	435
4-(2,3-Diphenyl cyclopropenyl-idene)cyclohexa-2,5-dienone 	ss	21.2	−15	4a	Dichloromethane, N_{67}	317
1,3-Disilacyclobutane, 1,1,3,3-tetramethyl-	lc	73.9	25	2		359
Selenium Se_8	ls	−9.5	400	4a		85

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TABLE 2. *cont'd*

Monomer	State of monomer and polymer (xx)	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method		Solvent/Notes	Refs.
1-Silacyclobutane, 1,1-dimethyl-	lc′	92.7	25	2			359
	lc	85.4	25	2			359
–, 1-phenyl, 1-methyl-	lc	91.5	25	2			359
Sulfur, S_6	ls	10	138	2	N_{72}		380
–, S_7	ls	4	138	2	N_{72}		380
–, S_8	ls	−19	160	4a	N_{71}		380
–, S_8	ls	−13.5	200	4a			73,74

TABLE 3. COPOLYMERS (listed alphabetically under Monomer A) N_{21}

Monomer A	Monomer B	State of monomer and polymer xx	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
Acrylonitrile	Methacrylic acid,	sc		25	3	Emulsion (aq.)	38
	Methyl ester	ss		30.5	3		76
	Styrene	ss		20	3		76,201
	Vinyl acetate	ss		20	3		76
	Vinylidene chloride	sc		25	3	Emulsion(aq.), N_{22}	75
Allyl chloride	Maleic anhydride	ss	74	74.5	3		34
bis-(*o*-Aminophenyl)-2,2′ dibenzymidazoleoxide	Terephthalic acid	c′c	24	−173	5	Polymer + H_2O (c′)	445
		c′c	18	−73	5	Polymer + H_2O (c′)	445
		c′c	6.5	25	2	Polymer + H_2O (l)	445
	–, dichloride	c′c	136	−173	5	Polymer + HCl (c′)	445
		c′c	96	−73	5	Polymer + HCl (g)	445
		c′c	83.5	25	2	Polymer + HCl (g)	445
	–, dimethylether	c′c	31.5	−173	5	Polymer + CH_3OH (c′)	445
		c′c	16.5	−73	5	Polymer + CH_3OH (l)	445
		c′c	5.5	25	2	Polymer + CH_3OH (l)	445
1,3-Butadiene	Styrene	lc		25	2		45
	Sulfur dioxide						197
1-Butene	Sulfur dioxide	ss	44.5	26.9	3	Excess B	5
		ss	43.5	55	4a	Excess B	77
–, 3-methyl-	Sulfur dioxide	lc′	46	35	4a	From B-rich mixture	199,200
		lc	22	25	4a	From A-rich mixture	199,200
2-Butene, *cis*	Sulfur dioxide	ss	42.5	26.9	3	Excess B	5
		ss	43.5	25	4a	Excess B	5
–, *trans*	Sulfur dioxide	ss	39.5	26.9	3	Excess B	5
		ss	40.5	25	4a	Excess B	5
Cyclohexene	Sulfur dioxide	ss	38	25	4a	Excess B	78
Cyclopentene	Sulfur dioxide	ss	45	26.9	3	Excess B	5
1,3-Dioxane	1,3-Dioxolane	ss	16	20	4a	Methylene dichloride, N_{78} $\Delta H_{BA} = -11$	446
Fumaric acid, diethyl ester	Vinyl acetate	ss?	78	76.8	3	Excess B, N_{22}	79
Fumaroyl chloride	Styrene	ss	80	74.5	3	Hexane	34
	Styrene, α-methyl-	ss	71.5	74.5	3	Excess B	34
Isobutene	Sulfur dioxide	lc	31	25	2		82
		lc	39.5	0	4a	N_{23}	83
1-Hexadecene	Sulfur dioxide	ss	42	26.9	3	Chloroform	5
		ss	40	30	4a	Chloroform	80
1-Hexene	Sulfur dioxide	ss	43.5	26.9	3	Excess B	5
Isopropenyl acetate	Maleic anhydride	sc	74.5	76.8	3		79
Maleic acid, diethyl ester	Vinyl acetate	ss?	83.5	76.8	3	Excess B, N_{22}	79
Maleic anhydride	Allyl chloride	ss	74	74.5	3		34
	Isopropenyl acetate	sc	74.5	76.8	3		79
	Styrene	ss	81	74.5	s	Benzene	34
		ss	82.5	74.5	3	Acetonitrile	34

TABLE 3. *cont'd*

Monomer A	Monomer B	State of monomer and polymer xx	$-\Delta H_{xx}$ (kJ/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	Styrene, α-methyl-	ss	72.5	74.5	3	Excess B	34
	Vinyl acetate	sc	84.5	76.8	3	Excess B	79
	Vinyl *n*-butyl ether	ss	90	74.5	3	Benzene	34
Maleimide	Styrene	ss	87.5	74.5	3	Acetonitrile	34
	Styrene, α-methyl-	ss	72	74.5	3	Acetonitrile	34
Methacrylic acid	Methacrylic acid, methyl ester	lc		25	2		206
–, methyl ester	Acrylonitrile	sc		25	3	Emulsion (aq.)	38
		ss		30.5	3		76
	Methacrylic acid	lc		25	2		206
	Styrene	ss		24	3		81
	Vinyl acetate	ss		24	3		81
Propene	Sulfur dioxide	sc	42.5	26.9	3	Excess B	5
Styrene	Acrylonitrile	ss		20	3		76,201
	1,3-Butadiene	lc		25	2		45,
	Fumaroyl chloride	ss	80	74.5	3	Hexane	34
	Maleic anhydride	ss	81	74.5	3	Benzene	34
		ss	82.5	74.5	3	Acetonitrile	34
	Maleimide	ss	87.5	74.5	3	Acetonitrile	34
	Methacrylic, acid, methyl ester	ss		24	3		81
	Vinyl acetate	ss		35	3		81
–, α-methyl-	Fumaroyl chloride	ss	71.5	74.5	3	Excess A	34
	Maleic anhydride	ss	72.5	74.5	3	Excess A	34
	Maleimide	ss	72	74.5	3	Acetonitrile	34
Sulfur dioxide	Butadiene						197
	1-Butene	ss	44.5	26.9	3	Excess A	5
		ss	43.5	55	4a	Excess A	77
	–, 3-methyl-	lc′	92	35	4a	From A-rich mixture	199
		lc	44.5	25	4a	From B-rich mixture	199
	2-Butene, *cis*	ss	42.5	26.9	3	Excess A	5
		ss	43.5	25	4a	Excess A	5
	–, *trans*	ss	39.5	26.9	3	Excess A	5
		ss	40.5	25	4a	Excess A	5
	Cyclohexene	ss	38	25	4a	Excess A	78
	Cyclopentene	ss	45	26.9	3	Excess A	5
	1-Hexadecene	ss	42	26.9	3	Chloroform	5
		ss	40	30	4a	Chloroform	80
	1-Hexene	ss	43.5	26.9	3	Excess A	5
	Isobutene	lc	31	25	2		82
		lc	39.5	0	4a	N_{24}	83
	Propene	sc	42.5	26.9	3	Excess A	5
Vinyl acetate	Acrylonitrile	ss		20	3		76
	Fumaric acid, diethyl ester	ss?	78	76.8	3	Excess A, N_{22}	79
	Maleic acid, diethyl ester	ss?	83.5	76.8	3	Excess A, N_{22}	79
	Maleic anhydride	sc	84.5	76.8	3	Excess A	79
	Methacrylic acid, methyl ester	ss		24	3		81
	Styrene	ss		35	3		81
Vinyl *n*-butyl ether	Maleic anhydride	ss	90	74.5	3	Benzene	34
Vinylidene chloride	Acrylonitrile	sc		25	3	Emulsion (aq.), N_{22}	75

B. ENTROPIES OF POLYMERIZATION

Symbols: The subscripts to $\Delta S°$ denote the state of the monomer (first letter) and the state of the polymer (second letter) as in Section A. The standard state of the monomer is 1 atm for the gaseous state and 1*M* in solution, unless otherwise stated. In all cases where monomer or polymer is present in solution, the value of $\Delta S°$ will depend to some extent on the composition. Where the polymer is crystalline, $\Delta S°$ will depend on the degree of crystallinity. ΔS values are in (J/K) per mol of monomer and are generally the limiting values for high degree of polymerization (1 kcal = 4.187 kJ).

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Precision: generally 0.5–8.0 (J/K/mol)

Methods of determination: (numbering conforms with that in Section A)
1 Third law or statistical.
4a Thermodynamic (van't Hoff Equation).
4b Semiempirical rules applied to evaluate entropy of monomer or polymer or both.

TABLE 4. MONOMERS GIVING POLYMERS CONTAINING MAIN CHAIN CARBON ONLY

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
4.1. MAIN CHAIN ACYCLIC CARBON ONLY						
4.1.1. DIENES						
1,3-Butadiene	lc	89	25	1		95,96
	lc	84	25	1	*cis*-1,4-Polymer	97
Isoprene	lc	101	25	1		100,101
4.1.2. MONOMERS GIVING POLYMERS WITHOUT OR WITH ALIPHATIC SIDE CHAINS WHICH CONTAIN ONLY C, H						
4.1.2.1. Olefins (listed by increasing carbon number)						
Ethylene	c′c′	66	−173	1	Isotactic poly(-1-butene)	385
	c′c	59	−173	1	Isotactic poly(-1-butene)	385
	gc′	338	−73	1	Isotactic poly(-1-butene)	385
	gc	331	−73	1	Isotactic poly(-1-butene)	385
	gg	142	25	4b		2,87
	gc	155	25	4b		11
	gc	158	25	1		88–91
	gc′	172	25	4b		11
	gc′	174	25	1	100% Cryst. polymer	88–91
	gc′	336	25	1	Isotactic poly(-1-butene)	385
	gc	325	25	1	Isotactic poly(-1-butene)	385
	gc′	334	127	1	Isotactic poly(-1-butene)	385
	gc	317	127	1	Isotactic poly(-1-butene)	385
	gl	311	227	1	Isotactic poly(-1-butene)	385
Propene	gg	167	25	4b		5
	gc′	191	25	1	Syndiotactic polymer	186
	gc′	205	25	1	Isotactic polymer	207
	lc	113	25	4b		5
	lc	116	25	1	Isotactic polymer	88
	lc	116	25	1	Atactic polymer	88,92
	lc′	136	25	1	100% Cryst. isotactic	88,92
1-Butene	lc′	75	−173	1	Isotactic polymer	385
	lc	69	−173	1	Isotactic polymer	385
	lc′	114	−73	1	Isotactic polymer	385
	lc	107	−73	1	Isotactic polymer	385
	gg	166	25	4b		5
	gc′	213	25	1	Isotactic polymer	88,385
	gc	193	25	1	Isotactic polymer	88,385
	lc	113	25	4b		5
	lc	112	25	1	Isotactic polymer	88
	lc′	141	25	1	100% Cryst. isotactic	88
	gc′	204	127	1	Isotactic polymer	385
	gc	187	127	1	Isotactic polymer	385
	gl	182	227	1	Isotactic polymer	385
2-Butene, *cis*	c′c′	21	−173	1	Isotactic polymer	385
	c′c	14	−173	1	Isotactic polymer	385
	lc′	106	−73	1	Isotactic polymer	385
	lc	99	−73	1	Isotactic polymer	385
	gg	163	25	4b		5
	lc	104	25	4b		5
	gc′	201	25	1	Isotactic polymer	385
	gc	190	25	1	Isotactic polymer	385
	gc′	196	127	1	Isotactic polymer	385
	gc	179	127	1	Isotactic polymer	385
	gl	172	227	1	Isotactic polymer	385

TABLE 4. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^{\circ}_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
–, *trans*	c′c′	17	−173	1	Isotactic polymer	385
	c′c	10	−173	1	Isotactic polymer	385
	lc′	100	−73	1	Isotactic polymer	385
	lc	93	−73	1	Isotactic polymer	385
	gg	159	25	4b		5
	lc	100	25	4b		5
	gc′	195	25	1	Isotactic polymer	385
	gc	194	25	1	Isotactic polymer	385
	gc′	193	127	1	Isotactic polymer	385
	gc	175	127	1	Isotactic polymer	385
	gl	170	227	1	Isotactic polymer	385
Isobutene	gg	172	25	4b		5
	lc	112	25	4b		5
	lc	121	25	1		11,93
1-Pentene	c′c′	24	−173	1		386
	c′c	20	−173	1		386
	lc′	117	−73	1		386
	lc	113	−73	1		386
	lc′	133	25	1		386
	lc	124	25	1		386
	gl	198	127	1		386
	gl	190	227	1		386
–, 4-methyl	c′c′	28	−173	1		387
	c′c	28	−173	1		387
	lc′	111	−73	1		387
	lc	111	−73	1		387
	gc′	216	25	1	Isotactic polymer	196
	lc′	131	25	1		387
	lc	131	25	1		387
	gc′	214	127	1		387
	gc	200	127	1		387
	gc′	209	227	1		387
	gc	188	227	1		387
	gl	176	327	1		387
2-Pentene, *cis*	c′c′	20	−173	1		386
	c′c	16	−173	1		386
	lc′	114	−73	1		386
	lc	111	−73	1		386
	lc′	129	25	1		386
	lc	120	25	1		386
	gl	195	127	1		386
	gl	185	227	1		386
–, 4-methyl	lc′	134	25	1		387
	lc	134	25	1		387
	gc′	220	127	1		387
	gc	205	127	1		387
	gc′	215	227	1		387
	gc	215	227	1		387
	gl	183	327	1		387
2-Pentene, *trans*	c′c′	18	−173	1		386
	c′c	14	−173	1		386
	lc′	111	−73	1		386
	lc	102	−73	1		386
	lc′	127	25	1		386
	lc	118	25	1		386
	gl	194	127	1		386
	gl	186	227	1		386
–, 4-methyl	lc′	130	25	1		387
	lc	130	25	1		387
	gc′	216	127	1		387
	gc	202	127	1		387
	gc′	214	227	1		387

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TABLE 4. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	gc	194	227	1		387
	gl	182	327	1		387
1-Hexene	lc	113	25	4b		5
1-Heptene	gg	168	25	4b		5
4.1.2.2. Cycloalkanes (listed by increasing ring size)						
Cyclopropane	lc	69	25	4b		11
–, methyl-	lc	85	25	4b		87
–, 1,1-dimethyl-	lc	93	25	4b		87
Cyclobutane	lc	55	25	4b		11
–, methyl-	lc	72	25	4b		87
–, 1,1-dimethyl-	lc	75	25	4b		87
Cyclopentane	lc	43	25	4b		11
–, methyl-	lc	64	25	4b		87
–, 1,1-dimethyl-	lc	66	25	4b		87
Cyclohexane	lc	11	25	4b		11,87
–, methyl-	lc	32	25	4b		87
–, 1,1-dimethyl-	lc	36	25	4b		87
Cycloheptane	lc	16	25	4b	N_{25}	11,94
Cyclooctane	lc	3	25	4b	N_{25}	11,94
Cycloalkenes						
Cyclobutene	ll	52	25	1,4b	*cis*-Polybutadiene	388
	gl	131	25	1,4b	*cis*-Polybutadiene	388
Cyclopentene	c′(I)c	17	−173	1	*cis*-Polymer	388
	lc	54	−73	1	*cis*-Polymer	388
	lc	52	25	1,4b	*cis*-Polymer	388
	lc	47.2	25	1	N_{66}	361
	c′(I)c	18	−173	1	*trans*-Polymer	388
	lc	56	−73	1	*trans*-Polymer	388
	lc	52	25	1,4b	*trans*-Polymer	388
	lc	51	77	1	*trans*-Polymer	388
Cyclohexene	ss	17	−65	4b	Toluene, N_{78}	389
Cycloheptene	c′(III)c	−7	−173	1	70% *trans*, 30% *cis* Polymer, N_{54}	388
	c′(III)c′	6	−173	1	70% *trans*, 30% *cis* Polymer, N_{54}	388
	c′(II)c	37	−73	1	70% *trans*, 30% *cis* Polymer, N_{54}	388
	c′(II)c′	56	−73	1	70% *trans*, 30% *cis* Polymer, N_{54}	388
	lc	37	25	4b	70% *trans*, 30% *cis* Polymer	388
	lc′	70	25	4b	70% *trans*, 30% *cis* Polymer	388
	ll	36	51	1	70% *trans*, 30% *cis* Polymer	388
Cyclooctene	c′(II)c′	10	−173	1	48% *trans*, 52% *cis* Polymer, N_{54}	388
	c′(II)c	−25	−173	1	48% *trans*, 52% *cis* Polymer, N_{54}	388
	c′(I)c′	49	−73	1	48% *trans*, 52% *cis* Polymer, N_{54}	388
	c′(I)c	8	−73	1	48% *trans*, 52% *cis* Polymer, N_{54}	388
	lc′	63	25	1,4b	48% *trans*, 52% *cis* Polymer	388
	lc	9	25	1,4b	48% *trans*, 52% *cis* Polymer	388
	ll	9	57	1	48% *trans*, 52% *cis* Polymer	388
4.1.3. MONOMERS GIVING POLYMERS WITH ALIPHATIC SIDE CHAINS THAT CONTAIN HETEROATOMS						
Acrylonitrile	lc′	109	25	1		193
–, allylester	c′c	−6	−173	1		391
	c′c	1	−73	1		391
	c′c	24	−8	1		391
	lc	97	−8	1		391
	lc	85	25	1		391
	lc	96	127	1		391
–, allylloxyethyl	cc	16	−173	1		393
	lc	29	−73	1		393
	lc	56	25	1		393
	lc	60	57	1		393
–, *n*-decylester	c′c	4	−173	1		392
	c′c	8	−73	1		392

TABLE 4. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	lc	141	25	1		392
	lc	155	127	1		392
–, ethylester	c′c	3	−173	1		390
	c′c	16	−73	1		390
	lc	88	25	1		390
	lc	104	127	1		390
	lc	107	177	1		390
Ethacrylic acid						
-methyl ester	ls	105	87	4b		395
Ethylene, tetrafluoro-	gc′	197	−75.7	1		98,99
	lc′	112	−75.7	1		98,99
Methacrylic acid	ss	146	80	4b		316
–, ethyl ester	lc	126	120	4a		40
–, methyl ester	lc	117	127	4a		42,43,102
	c′c	9	−173	1		414
	c′c	24.1	−73	1		414
	c′c′	40	−63	1		102,103
	lc	105	25	1		414
	lc	115	77	1		414
	ss	130	127	4a	*o*-Dichlorobenzene	42,49
–, methyl ester deuterated	c′c	10	−173	1		414
	c′c	26	−73	1		414
	lc	110	25	1		414
	lc	118	77	1		414
Methacrylonitrile	ss	151	127	4a	Benzonitrile	44,29
α-Methylene pentanolactone	ss	108.8	83	4a	DMF	315
Vinylidene chloride	c′c′	31	−173	1		397
	c′c	25	−173	1		397
	lc′	97	−73	1		213,397
	lc	91	−73	1		397
	lc′	112	25	1		397
	lc	106	25	1		397
	gc′	201	25	1		397
	gc	195	25	1		397
	gc′	198	127	1		397
	gc	191	127	1		397
4.1.4. MONOMERS GIVING POLYMERS WITH AROMATIC SIDE CHAINS THAT CONTAIN ONLY C, H						
Acenaphtylene	ss	98	60	4a	Toluene, N_{78}	399
Biphenyl, *p*-isopropenyl-	ss	118	−20	4a	Tetrahydrofuran	16
Naphthalene, 2-isopropenyl-	ss	122	−5	4a	Tetrahydrofuran	16
	ss	125	10	4a	Toluene, N_{78}	400
Styrene	gg	149	25	4b		25
	lc	90	−23	1		104
	lc	104	25	1		104
	lc	116	127	1		104
	lc	104	127	4a	N_{26}	28
	lc	105	25	1	Isotactic polymer	88
	lc	112	25	1		105
	lc′	111	25	1	100% Cryst. isotactic	88
–, α-methyl-	lc	110	−20	4a		32
	ss	130	−20	4a	Tetrahydrofuran	32,29,16
	ss	51	−10	4a	*n*-Heptane, N_{78}	402
	ss	47	−10	4a	Toluene, N_{78}	402
	ss	163.3	22	4a	Cyclohexane	341
–, α-methyl, *p-tert*-butyl-	ss	106.6		4a	Solvent and temp. not stated	381
4.1.5. MONOMERS GIVING POLYMERS WITH AROMATIC SIDE CHAINS AND THAT CONTAIN HETEROATOMS						
α-Acetoxystyrene	ss	83	64	4a	Toluene, N_{78}	401
2-Amino-4-(*N*-alkalino)-6-isopropenyl-1,3,5-triazine						
alkyl = *n*-propyl	ls	117	157	4a		402
isopropyl	ls	114	151	4a		403
isobutyl	ls	114	154	4a		403

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TABLE 4. *cont'd*

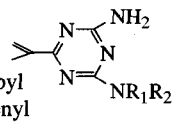
Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
sec-butyl	ls	112	148	4a		403
1-methylbutyl	ls	111	146	4a		403
n-pentyl	ls	114	150	4a		403
isopentyl	ls	113	151	4a		403
n-hexyl	ls	112	149	4a		402
n-octyl	ls	110	143	4a		402
n-decyl	ls	109	140	4a		402
n-dodecyl	ls	109	134	4a		402
n-tetradecyl	ls	105	132	4a		402
n-hexadecyl	ls	102	130	4a		402
benzyl	ls	113	157	4a		403
2-Amino-4-alkyl/aryl-amino-6-isopropenyl-1,3,5-triazine						
R_1 = H, R_2 = ethyl	ls	160.5	150	4a		370
R_1 = H, R_2 = propyl 	ls	153.8	145	4a		370
R_1 = ethyl, R_2 = ethyl	ls	137.5	130	4a		370
R_1 = propyl, R_2 = propyl	ls	124.6	125	4a		370
R_1 = methyl, R_2 = phenyl	ls	131.7	150	4a		379
R_1 = H, R_2 = *p*-methoxyphenyl	ls	117	160	4a		378
Benzoic acid, *n*-methacryloyloxy-	c′c	−2	−173	4a		394
	c′c	5	−73	4a		394
	c′c	8	25	1,4a		363,394
	ll	85	227	4a		394
Oxazole, 2-isopropenyl, 4,5-dimethyl-	ss	96	20	4a	Tetrahydrofuran	284
Pyridine, 2-isopropenyl-	ss	70	20	4a	Tetrahydrofuran	284
Silane, benzyl dimethyl vinyl-	c′c	0.7	−73	1		355
	lc	83.5	25	1		366
	lc	107.3	77	3		366
–, dimethyl phenyl vinyl-	lc	127.5	25	1		362
–, trimethyl vinyl-	lc	109.5	25	1		367
Thiazole, 2-isopropenyl-	ss	69	20	4a	Tetrahydrofuran	284
p-[bis(Trimethylsilyl)methyl] isopropenylbenzene	ss	110	−30	4a	Tetrahydrofuran	409
N,N,N′,N′-Tetraethyl-4-isopropenylphenyl-phosphonous diamide	ss	10	−55	4a	Tetrahydrofuran	408
4.1.6. OTHER MONOMERS GIVING POLYMERS NOT LISTED ABOVE						
Vinylcyclohexane	c′c	28	−173	1		396
	c′c	3	−173	1		396
	lc′	95	−73	1		396
	lc	75	−73	1		396
	lc′	120	25	1		396
	lc	85	25	1		396
	lc′	136	127	1		396
	lc	102	127	1		396
4.2. WITH AROMATIC OR CYCLIC CARBONS IN THE MAIN CHAIN						
–11,12-Dicyano-2,6-naphtoquinodimethane 11,12-bis(alkylthio)						
alkyl = butyl	ss	109	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= *sec*-butyl	ss	109	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= isopropyl	ss	116	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= ethyl	ss	115	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
= methyl	ss	114	30	4a	Oligomer, benzene, N_{78}, N_{80}	410
–7,8-Dicyanoquinodimethane						
7,8-bis(butoxycarbonyl)-	ss	37	60	4a	Chloroform, N_{78}, N_{81}	411
7,8-diacetyl-	ss	37	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-dibenzoyl-	ss	38	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-dibutoxycarbonyl-	ss	37	55	4a	Chloroform, N_{78}, N_{81}	412
7,8-diisobutyryl-	ss	39	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-dipropionyl-	ss	37	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-bis(ethylthio)-	ss	37	60	4a	Chloroform, N_{78}, N_{81}	411

TABLE 4. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
7,8-bis(phenylacetyl)-	ss	37	55	4a	Chloroform, N_{78}, N_{81}	413
7,8-bis(phenylthio)-	ss	34	50	4a	Chloroform, N_{78}, N_{81}	411
–4,7-Dihydrobenzofuran						
4,7-bis-[cyano(ethoxycarbonyl)methylene]	ss	41	60	4a	Chloroform, N_{78}	435
Norbornylene	c′(II)c	5	−173	3	55% *cis*/45% *trans* Polymer N_{54}	415
	c′(I)c	50	−73	3	55% *cis*/45% *trans* Polymer N_{54}	415
	c′(I)c	50	25	4b	55% *cis*/45% *trans* Polymer N_{54}	415
	lc	58	57	3	55% *cis*/45% *trans* Polymer N_{54}	415
5-trimethylsilyl–2-	c′c	4	−173	1		416
	cc	13	−173	1	Overcooled monomer	416
	c′c	14	−73	1		416
	cc	48	−73	1	Overcooled monomer	416
	lc	70	25	1		416
	lc	81	107	1		416
N,7,7-Tricyanobenzoquinonemethideimine	ss	38	55	4a	Toluene, N_{78}, N_{81}	438

TABLE 5. MONOMERS GIVING POLYMERS CONTAINING HETEROATOMS IN THE MAIN CHAIN

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
5.1. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO CARBON ONLY						
5.1.1. ETHERS AND ACETALS (listed by increasing ring size of monomer)						
2-RINGS (aldehydes and ketones)						
Acetaldehyde	c′c	−6	−173	1		417
	lc	14	−73	1		417
	lc	21	25	1		417
–, bromochlorofluoro-	sc′	122		4a	Toluene, N_{74}	233
–, bromodichloro-	sc′	122		4a	Toluene, N_{74}	344
–, bromodifluoro-	sc′	135		4a	Toluene, N_{74}	344
–, dibromochloro-	sc′	116		4a	Toluene, N_{74}	344
–, dibromofluoro-	sc′	114		4a	Toluene, N_{74}	344
–, dichlorofluoro-	sc′	179		4a	Toluene, N_{74}	344
–, difluorochloro-	sc′	152		4a	Toluene, N_{74}	344
–, tribromo-	sc′	100		4a	Toluene, N_{74}	344
–, trichloro-	gc′	190 ± 30	50	4a		50
	lc′	95 ± 30	50	4a		50
	sc′	134		4a	Toluene, N_{74}	344
	sc′	117	13	4a	Pyridine, N_{27}	50
	ss	52	−50	4a	Tetrahydrofuran	167
	sc′	142	10	4a	*n*-Heptane	166
–, trichloro	cc′	37	−173	1		418
	lc′	69	−73	1		418
	lc′	80	25	1		418
	lc′	30	57	1		418
–, trifluoro-	gc	187	45	4a		129
	lc	97	25	4a	Dichloromethane	322
	sc′	155		4a	Toluene, N_{74}	344
	sc	142.7	25	4a	Dichloromethane	322
Acetone	gc	188	25	4b		144,148
Butanal	sc	122	10	4a	*n*-Hexane [9 kbar], N_{12}	161
4,7-Dioxaoctanal	lc	67.4	−80	4a	N_{69}	330
	ss	77.0	−80	4a	THF	330
Formaldehyde	gg	124	25	1		141,142
	gc′	174	25	1		106
	gc′	169	25	4a	N_{28}	164,165
Heptanal	c′c′	25	−173	1		419
	c′c	1	−173	1		419
	c′c′	32	−73	1		419

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TABLE 5. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	c′c	4	−73	1		419
	lc′(I)	111	25	1	N_{77}	419
	lc	101	25	1		419
	ll	110	97	1		419
Hexanal	c′c′	19	−173	1		420
	c′c	16	−173	1		420
	c′c′	43	−73	1		420
	c′c	41	−73	1		420
	ll	117	25	1		420
	ll	121	67	1		420
Octanal	lc	57.8	−75	4a	N_{69}	330
	sc	65.7	−75	4a	THF	330
4-Oxapentanal	lc	81.6	−80	4a	N_{69}	329
	ls	93.6	−80	4a		329
Pentanal	lc	97.1	−75	4a	N_{69}	328
	sc	107.6	−75	4a	THF	328
o-Phthalaldehyde	ss	96	−70	4a	Dichloromethane	154
Propanal	c′c′	8	−173	1		421
	c′c	6	−173	1		421
	lc′	69	−73	1		421
	lc	67	−73	1		421
	lc′	85	25	1		421
	lc	76	25	1		421
	lc′	90	57	1		421
	lc	77	57	1		421
–, 2-chloro 2-methyl-	ss	90	−65	4a	Tetrahydrofuran	172
–, 2,2-dichloro-	ss	69	−65	4a	Tetrahydrofuran	172
–, 2-methyl-	ss	74	−65	4a	Tetrahydrofuran	172
	ss	78	−65	4a	Diethyl ether	166
	ss	94	−65	4a	*n*-Pentane	166
3-RINGS						
Ethylene oxide	gc′	174	25		100% Cryst. polymer	210
Propylene oxide	gc′	189	25		Estimated for 100% cryst. polymer	210
4-RINGS						
Oxetane	gc	162.7	25	4a		325
	lc	68.4	25	4a	N_{17}	325
–, 3,3-dichloromethyl	lc	83	25	1		107
5-RINGS						
1,3-Dioxolane	gc′	205	25	1		177
	gc	167	25	1		177
	gc	139	25	4a		292
	gc	147	57	4a	N_{60}	353
	lc′	100	25	1		177
	lc	37	25	4a	N_{17}	292
	lc	67	25	1		177
	lc	63	40	4a	N_{15}	159
	lc	76	100	4a	N_{14}	198
	lc	18	90	4a	N_{69}, N_{78}	398
	ss	78	0	4a	Dichloromethane	133,154
	ss	94	20	4a	Chloromethane	213
	ss	22	75	4a	Toluene, N_{78}	423
	ss	59	30	4a	Benzene	134
–, 4,4-dimethyl-	lc	54.8	80	b4		335
–, *cis* 4,5-dimethyl-	lc	55.3	80	4b		335
–, *trans* 4,5-dimethyl-	lc	55.3	80	4b		335
–, 4-ethyl-	lc	59.4	−24	4a		333
–, 4-isopropyl	lc	60.7	80	4b		335
–, 4-methyl-	lc	53.2	−8	4a		334
7-Oxa-bicyclo[2,2,1]heptane	lc	75.3	25	4b		308
–, 2-exo-methyl-	lc	96.2	25	4b		308
–, 2-endo-methyl-	lc	96.2	25	4b		308

TABLE 5. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
trans-7-Oxabicyclo[4.3.0]nonane	lc	55.7				357
Tetrahydrofuran	gc′	177	25	1		140
	gc	112	25	4a		292
	gc	139	25	1		140
	gc	143	50	4a	N_{61}	353
	lc′	100	25	1		140
	lc′	81.9	25	1		358
	lc	47.6	25	1		358
	lc	16	25	4a	N_{17}	292
	lc	62	25	1		140
	lc	41	50	4a		130
	lc	49	30	4a	N_{16}	231
	ls	67	40	4a		57,58
	ss	87	40	4a	Diethyl ether	211
–, 3-methyl-	ls	100.9	−6	4a		342
6-RINGS						
1,3-Dioxane	ls	−23	20	4a	N_{78}, Cyclic oligomer	436
Paraldehyde	gc′	201	25	1	Isotactic polymer	184
	gc′	159	25	1	Syndiotactic polymer	184
Trioxane	gg	64	25	1		141
	gc′	156	25	1		141
	ss	41	40	4a	1,2-Dichloroethane	178
	sc′	42	25	4a	Nitrobenzene	182
	c′c′	18 ± 16	25	4a		164
7-RINGS						
1,3-Dioxepane	gc′	181	25	1		170
	gc′	144	25	1		170
	gc	131.4	25	4a		323
	lc	25.1	25	4a	N_{17}	323
	lc′	77	25	1		170
	lc	39	25	1		170
	ss	48	−30	4a	Dichloromethane	136
	ss	39	60	4a	Benzene	134
	ss	26	75	4a	Toluene, N_{78}	423
	ss	43	0	4a	1,2-Dichloroethane, N_{78}	424
–, 2-butyl	ss	33	−5	4a	1,2-Dichloroethane, N_{78}	424
–, 2-methyl-	ss	37.2	−15	4a	1,2-Dichloroethane	340
	ss	39	−10	4a	1,1-Dichloromethene, N_{78}	424
–, 4-methyl-	ss	38.8	−15	4a	1,2-Dichloroethane	339
Oxepane	gc	91.8	25	4a		323
	lc	−12.4	25	4a	N_{17}	323
1-Oxa-4,5-dithiepane	ss	>12	25	4a	Benzene	61
8-AND LARGER RINGS						
2-Butyl-1,3,6-trioxocane	ss	16	10	4a	Toluene, N_{78}	425
1,3-Dioxocane	gc	121.7	25	4a		324
	lc	10.2	25	4a	N_{17}	324
Tetraoxane	gg	51	25	1		142
	c′c′	−3	27	1		131
1,3,6-Trioxocane	gc	129.1	25	4a		324
	lc	8.0	25	4a	N_{17}	324
	ss	39	60	4a	Benzene	134
	ss	13	20	4a	Toluene, N_{78}	425
$(1,3,6\text{-Trioxocane})_n$						
$n=1$	ss	34.3	0	4a	Dichloromethane	343
$n=2$	ss	12.6	0	4a	Dichloromethane	343
$n=3$	ss	36.4	0	4a	Dichloromethane	343
$n=4$	ss	43.1	0	4a	Dichloromethane	343
$n=5$	ss	47.7	0	4a	Dichloromethane	343
$n=6$	ss	51.5	0	4a	Dichloromethane	343
$n=7$	ss	54.8	0	4a	Dichloromethane	343
$n=8$	ss	57.8	0	4a	Dichloromethane	343

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TABLE 5. *cont'd*

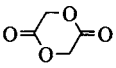
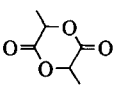
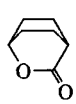
Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
(1,3,6,9-Tetraoxacycloundecane)$_n$						
$n=1$	ss	13.0	0	4a	Dichloromethane	343
$n=2$	ss	20.1	0	4a	Dichloromethane	343
$n=3$	ss	28.9	0	4a	Dichloromethane	343
$n=4$	ss	48.2	0	4a	Dichloromethane	343
$n=5$	ss	52.3	0	4a	Dichloromethane	343
$n=6$	ss	56.9	0	4a	Dichloromethane	343
$n=7$	ss	59.9	0	4a	Dichloromethane	343
$n=8$	ss	62.8	0	4a	Dichloromethane	343
(1,3,6,9,12-Pentaoxacyclotetradecane)$_n$						
$n=1$	ss	8.4	5	4a	Dichloromethane	343
$n=2$	ss	10.0	5	4a	Dichloromethane	343
$n=3$	ss	43.1	5	4a	Dichloromethane	343
$n=4$	ss	49.8	5	4a	Dichloromethane	343
$n=5$	ss	54.4	5	4a	Dichloromethane	343
$n=6$	ss	58.2	5	4a	Dichloromethane	343
$n=7$	ss	61.1	5	4a	Dichloromethane	343
(1,3,6,9,12,15-Hexaoxacycloheptadecane)$_n$						
$n=1$	ss	8.4	−2	4a	Dichloromethane	343
$n=2$	ss	28.1	−2	4a	Dichloromethane	343
$n=3$	ss	44.8	−2	4a	Dichloromethane	343
$n=4$	ss	50.7	−2	4a	Dichloromethane	343
$n=5$	ss	54.8	−2	4a	Dichloromethane	343
5.1.2. CYCLIC ESTERS						
4-RINGS						
β-Propiolactone	lc	54	25	1		368
	lc′	74	25	1		368
	lc	51	127	1		368
5-RINGS						
γ-Butyrolactone	lc	30	25	1		368
	lc′	65	25	1		368
	lc	29	77	1		368
6-RINGS						
2,2-Dimetyltrimethylene carbonate	c′(II)c′(II)	7.7	−173	1	N_{63}	426
	c′(II)c	−35.3	−173	1	N_{63}	426
	c′(II)c′(II)	5.8	25	1	N_{63}	426
	c′(II)c	−35.2	25	1	N_{63}	426
	lc′(I)	38.7	117	1	N_{63}	426
	ll	24.0	147	1	N_{63}	426
p-Dioxanone	c′c′	6	−173	1		427
	c′c	−28	−173	1		427
	c′c′	9	−73	1		427
	c′c	−25	−73	1		427
	c′c′	10	25	1		427
	c′c	−32	25	1		427
	ll	13	127	1		427
Ethylene oxalate	c′c	−12	−173	1		312
	c′c	−9	−73	1		312
	c′c	−5	25	1		312
	lc	15	142	1		312
Glycollide 	c′c′	0.3	25	1		364
	cc	−10.5	25	1		364
	lc	17.5	277	1		364
DL-Lactide 	c′c	−31.3	25	1		311
	lc	13.0	127	1		311
L-Lactide	ss	41	105	4a	1,4-Dioxane, N_{78}	429
	ss	25	40	4a	Nitrobenzene, N_{78}	428
8-Oxo-7-oxabicyclo[2,2,2]octane 	gg	127.7	25	4b	N_{53}	307
	lc′	76.2	25	4b	N_{53}	307

TABLE 5. *cont'd*

Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
Pentanolactone	lc′	65	25	1		314
	lc	15	25	1		314
	lc	13	77	1		314
7-RINGS						
ε-Caprolactone	lc	4	25	1		368
	lc′	54	25	1		368
	lc	8	77	1		368
Spiroorthoester 2-R-1,4,6-trioxaspiro[4.6] undecane						
R = H	ss	−39.6	−10	4a	Dichloromethane, N_{78}	430
= CH_3	ss	−30.7	−25	4a	Dichloromethane, N_{78}	430
= CH_2Br	ss	−20.8	−20	4a	Dichloromethane, N_{78}	430
= Ph	ss	−25.7	−30	4a	Dichloromethane, N_{78}	430
8-AND LARGER RINGS						
Heptanolactone	lc	1	127	4b		368
Octanolactone	lc	−5	127	4b		368
Nonanolactone	lc	−12	127	4b		368
Decanolactone	lc	−18	127	4b		368
Undecanolactone	lc	−24	25	1		368
	lc′	69	25	1		368
	lc	−25	127	1		368
Dodecanolactone	lc	−25	127	4b		368
Tridecanolactone	c′c	−57	25	1		368
	c′c′	50	25	1		368
	lc	−26	127	1		368
Tetradecanolactone	lc	−25	127	4b		368
Pentadecanolactone	c′c	−41	25	1		313
	c′c′	86	25	1		313
	lc	−23	97	1		313
	c′(II)c	−109	−73	1	N_{54}	313
Hexadecanolactone	lc	−21	127	4b		368
5.2. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO OTHER HETEROATOMS (S, Si, P)						
Cyclotrisiloxane, 1,1,3,3,5,5-hexaethyl	c′(III)c′(II)	7	−173	1	N_{54}, N_{63}, N_{77}	431
	c′(II)c′(II)	0	−73	1	N_{54}, N_{63}, N_{77}	431
	c′(II)c	−62	−23	1	N_{54}, N_{63}	431
	c′(I)c′(I)	9	−23	1	N_{54}, N_{63}, N_{77}	431
	c′(I)c	−22	25	1	N_{54}, N_{63}	431
	ll	25	25	1	N_{63}	431
hexamethyl-	lc	3.03	77	1		365
	c′c	−51.0	25	1		365
Cyclotetrasiloxane, octamethyl-	lc	−190	77	1		365
	lc	−194.4	25	1		365
N,N-Diethylamino-1,3,2-dioxaphosphorinane	ls	−20.9	−20	4a	N_{58}	346
1,3,2-Dioxaphospholan, 2-methoxy-	ss	13.5		4a	1,1,2,2-Tetrachloroethane	309
2-Oxo-1,3,2-dioxaphosphorinane	lc	−19.3	90	4a	N_{59}	347
−, 2-alkoxy-						
R = methyl	ls	11.7	140	4a	Diglyme or DMSO	348
R = methyl	ss	2.5	100	4a	Dichloromethane	356
R = ethyl	ss	−1.5	100	4a	Dichloromethane	356
R = propyl	ss	−4.5	100	4a	Dichloromethane	356
R = trimethyl silyl	ss	−6.0	100	4a	Dichloromethane	356
Sulfur trioxide	gc′	178	41	4a		86
5.3. MONOMERS GIVING POLYMERS CONTAINING N IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY (listed by increasing ring size)						
3-RINGS						
Ethyleneimine	lc	51.9	27	1		360
−, *N*(β-trimethylsilyl ethyl)-(TEI)	lc	113	27	1		360

Notes page II-406; References page II-407

TABLE 5. *cont'd*

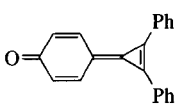
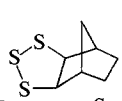
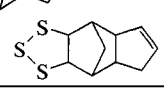
Monomer	State of monomer and polymer xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
–,–, $ZnCl_2(TEI)_2$(complex)	c′c	114	27	1		360
4-RINGS						
Trimethyleneimine, *N*-(β-trimethylsilyl ethyl)-(TMI)	lc	83.0	27	1		360
–, $ZnCl_2(TEI)_2$(complex)	c′c	37.2	27	1		360
5-RINGS						
Bicyclic oxalactam	ss	72	38	4a	Dimethylsulfoxide	433
2-Pyrrolidinone	c′c′	31	25	1/4b		63
6-RINGS						
2-Piperidone	c′c′	25	25	1/4b		63
7-RINGS						
ε-Caprolactam	ls	29	250	4a		68
	c′c′	−5	25	1		63
–, 3-methyl-	ls	11	260	4a		194
–, 5-methyl-	ls	16	200	4a		225
–, 7-methyl-	ls	21	225	4a		226
8-RINGS						
2-Oxo-hexamethyleneimine	c′c′	−17	25	1		63
N. B. Values for 9–13 rings by probably unreliable extrapolation of $\Delta S_{c'c'}$ for 5–8 rings are given in Ref. 138						
13-RINGS						
2-Oxo-undecamethyleneimine	ls	16	290	4a		139
5.4. OTHER MONOMERS GIVING POLYMERS NOT LISTED ABOVE						
4-(2,3-Diphenyl cyclopropenylidene)cyclohexa-2,5-dienone 	ss	69.4	−15	4a	Dichloromethane, N_{67}	317
1,3-Disilacyclobutane, 1,1,3,3-tetramethyl-	lc	18.1	25	1		359
N-Phenylmaleimide	ss	86	55	4a	DMF, N_{78}	440
Selenium Se_g	ls	−27	200	4a	From estimated T_e	88,111
1-Silacyclobutane, 1,1-dimethyl-	lc′	85.4	25	1		359
	lc	71.2	25	1		359
–, 1-phenyl 1-methyl-	lc	57.4	25	1		359
Sulfur, S_6	ls	2	80	1	N_{73}	380
–, S_7	ls	−11	80	1	N_{73}	380
–, S_8	ls	−44.7	160	4a	N_{71}	380
	ls	−31	159	4a		73,74,111
Thioacetone	ls	48.9	50–90	4a	Monomer	327
exo-3,4,5-Trithiatricyclo-[5.2.1.$0^{2,6}$]decane 	ss	31.4	40	4a	Toluene, N_{76}	377
exo-3,4,5-Trithiatetracyclo-[5.5.1.$0^{2,6}$.$0^{8,12}$]tridec-10-ene 	ss	29.3	50	4a	Toluene, N_{76}	376

TABLE 6. COPOLYMERS (listed alphabetically under Monomer A)

Monomer A	Monomer B	State of monomer and polymer, xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
Bis-(*o*-aminophenyl)-2,2′ dibenzimidazoleoxide	Terephthalic acid	c′c	19	−173	1	Polymer + H_2O (c′)	445
		c′c	−17	−73	1	Polymer + H_2O (c′)	445
		c′c	−112	25	1	Polymer + H_2O (l)	445

TABLE 6. *cont'd*

Monomer A	Monomer B	State of monomer and polymer, xx	$-\Delta S^\circ_{xx}$ (J/K/mol)	Temp. (°C)	Method	Solvent/Notes	Refs.
	–, dichloride	c′c	−33	−173	1	Polymer + Hcl (c′)	445
		c′c	−295	−73	1	Polymer + Hcl (g)	445
		c′c	−320	25	1	Polymer + HCl (g)	445
	–, dimethylester	c′c	−13	−173	1	Polymer + CH_3OH (c′)	445
		c′c	−108	−73	1	Polymer + CH_3OH (l)	445
		c′c	−160	25	1	Polymer + CH_3OH (l)	445
1,3-Butadiene	Styrene	lc		30	1		108
1-Butene	Sulfur dioxide	lc	116	25	1		109
		ss	145	64	4a	Excess B	110
–, 3-methyl-	Sulfur dioxide	lc	162	35	4a	Excess B	199,200
		lc	81	25	4a	Excess A	199,200
2-Butene, *cis*	Sulfur dioxide	ss	146	25	4a	Excess B	5
–, *trans*	Sulfur dioxide	ss	140	25	4a	Excess B	5
Cyclopentene	Sulfur dioxide	ss	134	102	4a	Excess B	110
1,3 Dioxane	1,3 Dioxolane	ss	57	20	4a	Methylene dichloride, N_{78}, $\Delta S_{BA} = -34$	446
1-Hexadecene	Sulfur dioxide	ss	139	30	4a	Chloroform	110
1-Hexene	Sulfur dioxide	lc	116	25	1		109
		ss	145	60	4a	Excess B	110
Isobutene	Sulfur dioxide	lc	134	0	4a	N_{23}	83
Methacrylic acid	Methacrylic acid, methyl ester						206
Propene	Sulfur dioxide	lc	117	25	1		109
		sc	130	90	4a	Excess B	110
	Ethylene	gc	172	25	1	31 mol% A in copolymer, N_{30}	222

C. CEILING TEMPERATURES AND EQUILIBRIUM MONOMER CONCENTRATIONS

Most addition polymerization reactions are exothermic and exentropic. The free energy of polymerization per monomer unit therefore becomes less negative as the temperature is raised. At the *ceiling temperature* T_c, the free energy of polymerization under the prevailing conditions is zero, and above this temperature, polymerization to long-chain polymer is impossible (just as in physical aggregation a liquid cannot form a solid when the temperature is above the melting point). The reverse phenomenon of a *floor* temperature is also known, e.g., for sulfur.

In general a pure liquid monomer which gives an *insoluble polymer* will have a single well-defined ceiling temperature, given by $T_c = \Delta H_{lc}/\Delta S_{lc}$. A pure liquid monomer which gives a *soluble* polymer will have a series of ceiling temperatures corresponding to different percentage conversions of monomer to polymer. The condition for equilibrium is then

$$-\Delta\bar{G}_1 + \Delta G_{lc} + \Delta\bar{G}_2 = 0$$

The partial molar free energy per mol of monomer, $\Delta\bar{G}_1$ and per base mol of polymer, $\Delta\bar{G}_2$ are then functions of composition and may be evaluated from an appropriate equation for mixing of monomer and polymer, e.g., the Flory–Huggins equation. For a monomer dissolved in a solvent the situation is more complex and the ceiling temperature at a given monomer concentration (or the equilibrium concentration of monomer at a given temperature) is dependent on the nature of the solvent and the composition of the medium (Refs. 28,29,112,113,114,160). For the case where both monomer and polymer are in solution the variation of T_c with concentration is given to a first approximation by

$$T_c = \Delta H^\circ_{ss}/(\Delta S_{ss} + R \ln[M])$$

where $[M]$ is the concentration of monomer and ΔH°_{ss} and ΔS°_{ss} refer to the heat and entropy changes in an appropriate standard state. A more general expression may be derived from the free energy condition by insertion of suitable expressions for $\Delta\bar{G}_1$ and $\Delta\bar{G}_2$. These will contain the various interaction parameters appropriate to the polymer–monomer–solvent system.

The values of T_c quoted in the table are mostly obtained from experimental values by interpolation or short extrapolation. Some unpolymerizable monomers are included where these are structurally closely related to monomers which do polymerize and where the cause of nonpolymerization appears to be thermodynamic. This is amplified for cyclic monomers in Section D.

Section C is divided into three sections

1 Equilibria involving pure liquid monomers (lc, lc′, ls)
2 Equilibria involving gaseous monomers (gc, gc′)
3 Equilibria involving monomers in solution (sc, sc′, ss) and copolymerizations

For meaning of symbols, see Section A.

Notes page II-406; References page II-407

1. Equilibria Involving Pure Liquid Monomers

TABLE 7. MONOMERS GIVING POLYMERS CONTAINING MAIN CHAIN ACYCLIC CARBON ONLY

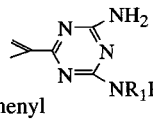
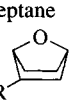
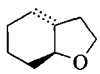
Monomer	State of monomer and polymer	T_c [°C]	Wt. fraction monomer at equilibrium	Notes	Refs.
2-Amino-4-(*N*-alkylanilino)-6-isopropenyl-1,3,5-triazine					
alkyl = *n*-propanyl	lc	157	1		402
= *n*-hexanyl	lc	149	1		402
= *n*-octanyl	lc	143	1		402
= *n*-decanyl	lc	140	1		402
= *n*-dodecanyl	lc	134	1		402
= *n*-tetradecanyl	lc	132	1		402
= *n*-hexadecanyl	lc	130	1		402
= isopropyl	lc	151	1		403
= isobutyl	lc	154	1		403
= *sec*-butyl	lc	148	1		403
= *n*-pentyl	lc	150	1		403
= isopentyl	lc	151	1		403
= 1-methylbutyl	lc	146	1		403
= benzyl	lc	157	1		403
2-Amino-4-alkyl/aryl-amino-6-isopropenyl-1,3,5-triazine					
R_1 = H, R_2 = ethyl	ls	181	1		370
R_1 = H, R_2 = propyl	ls	177	1		370
R_1 = ethyl, R_2 = ethyl 	ls	163	1		370
R_1 = propyl, R_2 = propyl	ls	161	1		370
R_1 = methyl, R_2 = phenyl	ls	169	1		379
R_1 = H, R_2 = *p*-methoxyphenyl	ls	169	1		378
2-Amino-4-(*p*-chloroanilino)-6-isopropenyl-1,3-5-triazine					
	lc	170	1		404
2-Amino-4-methoxy	ls	183	1		405
-4-ethoxy	ls	182	1		405
6-isopropenyl-1-3,5-triazine					
2-Anilo-4-methoxy	ls	174	1		405
-4-ethoxy	ls	174	1		405
6-isopropenyl-1,3,5-triazine					
Atropic acid, methyl ester	ls	−8	1		149
1-Butene	lc	247	1	Isotactic polymer, N_{57}	385
2-Butene, *cis*	lc	227	1	Isotactic polymer, N_{57}	385
, *trans*	lc	217	1	Isotactic polymer, N_{57}	385
–, 2-methyl-	ll	−29	1	Calculated, N_{31}	143
–, 2,3-dimethyl-	ll	−223	1	Calculated, N_{31}	143
α-Cyanoacrilic acid					
–, allyl ester	lc	307	1	N_{57}	391
–, *n*-decyl ester	lc	217	1	N_{57}	392
–, allyloxyethyl	lc	607	1	N_{57}	393
Cyclopentene	lc	97	1	N_{66}	361
Ethacrylic acid					
–, methyl ester	ls	87	1	N_{75}	395
Ethylene	lc	367	1	Polymer is polybutene-1-iso	385
Methacrylic acid					
–, benzoil acid ester	lc	787	1	N_{57}	394
–, methyl ester	lc	102	1		414
–, methyl ester deuterated	lc	93	1		414
Naphtalene, 2-isopropenyl	ls	69	1	N_{43}	400
Pentene-1	lc	317	1	N_{57}	386
Pentene-2, *cis*	lc	287	1	N_{57}	386
Pentene-2, *trans*	lc	277	1	N_{57}	386
–, 4-methyl, *cis*	lc	317	1	N_{57}	387
–, 4-methyl, *trans*	lc	267	1	N_{57}	387
Silane, benzyl dimethyl vinyl-	lc	397	1		366
–, dimethyl phenyl vinyl-	lc	197	1		362
Styrene, α-methyl-	ls	61	1	1 bar	119
	ls	54	1		341
	ls	170	1	6.57 kbar	119

TABLE 7. *cont'd*

Monomer	State of monomer and polymer	T_c [°C]	Wt. fraction monomer at equilibrium	Notes	Refs.
Vinylcyclohexane	lc	567	1	N_{57}	396
Vinylidenechloride	lc	387	1	N_{57}	397
Vinyldimethylbenzylsilane	lc	397	1	N_{57}	367
Vinyldimethylphenylsilane	lc	117	1	N_{57}	367
Vinyltrimethylsilane	lc	327	1	N_{57}	367

N.B. The following compounds are not polymerizable for thermodynamic reasons but can sometimes be copolymerized with other monomers (see Ref. 205): α-trifluoromethyl vinyl acetate, α-methoxystyrene, 1,1-diphenylethylene, *trans*-crotonitrile, *trans*-stilbene, *trans*-1,2-di(2-pyridyl)-ethylene, *trans*-1,2-dibenzoylethylene, *trans*-1,2-diacetylethylene, α-stillbazole, methyl 2-*tert*-butylacrylate, 1-isopropenylnaphthalene, 2,4-dimethyl-α-methyl-styrene.

TABLE 8. MONOMERS GIVING POLYMERS CONTAINING HETEROATOMS IN THE MAIN CHAIN

Monomer	State of monomer and polymer	T_c [°C]	Wt. fraction monomer at equilibrium	Notes	Refs.
8.1. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO CARBON ONLY					
Acetaldehyde	gg	−31	1	Atactic polymer (1 bar)	115
	ls	−39	1	Isotactic polymer (1 bar)	115
	ls	20	1	< 10 kbar	158
–, trichloro-	lc′	58	1		233
	lc	177	1	N_{57}	418
–, trifluoro-	lc	212	1	N_{68}	322
2,2-Dimethyltrimethylene carbonate	lc	227	1	N_{57}	426
4,7-Dioxaoctanal	lc	−38	1	N_{69}	330
1,3-Dioxepane	ls	100	0.1	N_{22}	116
	ls	192	1	N_{43}	423
	ls	144	1		398
–, 2-phenyl-	ls	20	0.36	N_{22}	116
$(1,3\text{-Dioxolane})_n$ $n=1$	ls	100	0.3		116,219
	ls	165	1		198
	ls	100	1	N_{43}	423
	lc	98	1		353
$n=1$–8	ls	60		K_n determined for cyclic monomers	232
1,3-Dioxolane, 4,4-dimethyl-	lc	−98	1	N_{57}	335
–, *cis* 4,5-dimethyl-	lc	−54	1	N_{57}	335
–, *trans* 4,5-dimethyl-	lc	−137	1	N_{57}	335
–, 4-ethyl-	ls	−48	4.74 Md/l	N_{56}	333
	lc	−54	1	N_{57}	333
–, 4-isopropyl-	lc	−74	1	N_{57}	335
–, 4-methyl-	ls	−20	4.47 Md/l	N_{56}	334
	lc	−21	1	N_{57}	334
–, 4-phenyl-	ls	20	1		285
Heptanal	lc	74	1	N_{57}	419
Hexanal	lc′	−43	1	N_{57}	420
	lc	−25	1	N_{57}	420
L-Lactide	lc	640	1	N_{57}, calculated	429
Octanal	lc	−19	1	N_{69}	330
7-Oxabicyclo[2,2,1]heptane	lc	320	1	N_{57}	308
–, 2-exo-methyl- 	lc	240	1	N_{57}	308
–, 2-endo-methyl-	lc	200	1	N_{57}	308
trans-7-Oxabicyclo[4,3,0]nonane 	lc	12	1		357
4-Oxapentanal	lc	−35	1	N_{69}	329
	ls	−59	1		329
1,2-Oxathiolane-2,2-dioxide	ls	95	0.74	N_{22}	152
Oxepane	ls	41.5	1		147
	lc	−136	1	N_{32}, N_{57}	323
Oxetane	lc	478	1	N_{31}	325
8-Oxo-7-oxabicyclo[2,2,2]octane	lc′	180	1	N_{53}	307

Notes page II-406; References page II-407

TABLE 8. *cont'd*

Monomer	State of monomer and polymer	T_c [°C]	Wt. fraction monomer at equilibrium	Notes	Refs.
Pentanal	lc	−42	1	N_{69}	328
Propanal	ls	−31	1	Atactic polymer	115
	ls	−39	1	Isotactic polymer	115
	lc′	−48	1	N_{57}	421
	lc	−83	1	N_{57}	421
Spiroorthoester					
2-R-1,4,6-trioxaspiro[4,6]undecane					
R = H	lc	172	1	N_{68}	430
= CH_3	lc	254	1	N_{68}	430
= CH_2Br	lc	498	1	N_{68}	430
= Ph	lc	258	1	N_{68}	430
Tetrahydrofuran	ls	80	1	1 bar	58,130
	ls	129	1	12.5 kbar	146
–, 3-methyl	ls	4	1		342
Tri(ethylene terephthalate)	ls	297	0.014		see 174

N.B. The following compounds are not polymerizable for thermodynamic reasons but can sometimes be copolymerized with other monomers (see Ref. 205); benzaldehyde (Refs. 297–299), hexafluoroacetone, 1,1-dimethylpropanal; also numerous cyclic ethers – see Section D.

8.2. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO OTHER HETEROATOMS (S, Si, P)

Sulfur trioxide		lc′	30.4	1		86
Sodium metaphosphate $Na_n(OPO_2)_n$						
$n = 3$		ls	800	0.043	K_n determined for cyclic monomers, $n = 3$–7	300,301
$n = 4$		ls	800	0.026		300,301
$n = 5$		ls	800	0.008		300,301
Siloxanes $(R_1R_2SiO)_n$						
R_1, R_2						
Me, H	$n = 3$–5	ls	0	0.045	K_n determined for cyclic monomers, $n = 4$–15	176
	$n = 6$–18	ls	0	0.034		176
	$n = 19$–∞	ls	0	0.046		176
	$n = 3$–∞	ls	0	0.125		176
Me, Me	$n = 3$–5	ls	110	0.100	K_n determined for cyclic monomers, $n = 4$–40	176
	$n = 6$–18	ls	110	0.036		176
	$n = 19$–∞	ls	110	0.047		176
	$n = 3$–∞	ls	110	0.183		176
	$n = 4$–40	ls	110		K_n determined for values of n indicated	180
	$n = 4$–30	ls	145		K_n determined for values of n indicated	180
Me, Et	$n = 3$–5	ls	110	0.170	K_n determined for cyclic monomers, $n = 4$–20	176
	$n = 6$–18	ls	110	0.049		176
	$n = 19$–∞	ls	110	0.039		176
	$n = 3$–∞	ls	110	0.258		176
Me, Pr	$n = 3$–5	ls	110	0.270	K_n determined for cyclic monomers, $n = 4$–8	176
	$n = 3$–8	ls	110	0.310		176
Me, $CF_3(CH_2)_2$	$n = 3$–5	ls	110	0.711	K_n determined for cyclic monomers, $n = 4$–20	176
	$n = 6$–18	ls	110	0.089		176
	$n = 19$–∞	ls	110	0.027		176
	$n = 3$–∞	ls	110	0.827		176
Et, Et	$n = 3$	lc	247	1	Calculated	431

8.3. MONOMERS GIVING POLYMERS CONTAINING S IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY

Thiacetone	lc	95	1		148
	ls	−52	1	N_{55}	327
Thioacetophenone	ls	approx. 40	1		288

N.B. Isothiocyanates (Ref. 295) are not polymerizable for thermodynamic reasons, but can be copolymerized with thiiranes; carbon disulfide (Ref. 296) can also be copolymerized with thianes

TABLE 8. *cont'd*

Monomer	State of monomer and polymer	T_c [°C]	Wt. fraction monomer at equilibrium	Notes	Refs.
8.4. MONOMERS GIVING POLYMERS CONTAINING N IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY					
(ε-Caprolactam)$_n$ $n=1$	ls	220	0.055		68
	ls	225	0.067		192
	ls	277	0.084		See 181
$n=2$	ls	277	0.0078	Cyclic monomer	See 181
$n=3$	ls	277	0.0052	Cyclic monomer	See 181
$n=4$	ls	277	0.0056	Cyclic monomer	See 181
$n=5$	ls	277	0.0048	Cyclic monomer	See 181
$n=\geq 6$	ls	277	0.0076	Cyclic monomer	See 181
–, 3-methyl-	ls	225	0.086		192,194
–, 3-ethyl-	ls	240	0.1		191
–, 3-propyl-	ls	240	0.1		191
–, 4-ethyl-	ls	240	0.35		171
–, 5-methyl-	ls	172	0.18		225,302
–, 7-methyl-	ls	225	0.107		226,302
–, 7-ethyl-	ls	240	0.1		191,302
–, 7-propyl-	ls	240	0.1		191,302
2,5-Dioxopiperazine	ls	171	1	From calorimetric data	437
2-Oxo-hexamethylenimine	ls	240			302
–, 4-ethyl-	ls	240	0.03		171
–, 7-ethyl-	ls	240	0.06		171
–, 8-propyl-	ls	240			302
2-Oxo-undecamethylenimine	ls	290	0.02		139
2-Piperidone	ls	60	0.32		118
2-Pyrrolidone	lc	76	1	N_{75}	383
	ls	70	1	T_c varies with the accelerator, N_{75}	439
8.5. OTHER MONOMERS GIVING POLYMERS NOT LISTED ABOVE					
Norbornene	lc	777	1	55% *cis*/45% *trans* polymer, N_{57}	415
5-trimethylsilyl-2-	lc	417	1		416
Selenium Se_8	ls	83	1	N_{32}	85
Sulfur S_8	ls	159	1	N_{33}	73,,111
				Effect of pressure	145
	ls	157	1	N_{33}	380

2. Equilibria Involving Gaseous Monomers (listed alphabetically)

Monomer	State of monomer and polymer xx	T_c (°C)	Equilibrium pressure (mbar)	Notes	Refs.
Acetaldehyde, trichloro-	gc′	98	1013		50
–, trifluoro-	gc	73	1013		129
Acetone	gc	−220		Estimated, N_{36}	144
1,3-Dioxepane	gc	149	1013		323
1,3-Dioxocane	gc	288	1013		324
1,3-Dioxolane	gc	87	1013		292
Ethylene, tetrafluoro-	gc	560	128	N_{34}	23,120
Formaldehyde	gc′	119	1013	N_{28}	49,164,165
Methacrylic acid					
–, ethyl ester	gs	173	1013	N_{35}	40
–, methyl ester	gs	164	1013	N_{35}	43
Oxepane	gc	135	1013		323
Oxetane	gc	319	1013		325
Sulfur trioxide	gc′	27.0	372		86
		0.0	40.8		86
Tetrahydrofuran	gc	83	1013		292
1,3,6-Trioxocane	gc	243	1013		324

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3. Equilibria Involving Monomers in Solution

TABLE 9. MONOMERS GIVING POLYMERS CONTAINING MAIN CHAIN CARBON ONLY

Monomer	State of monomer and polymer xx	T_c (°C)	M (mol/l)	Solvent/Notes	Refs.
9.1. MONOMERS GIVING POLYMERS CONTAINING MAIN CHAIN ACYCLIC CARBON ONLY					
Acrylonitrile		25	3×1^{-8}	Estimated from $\Delta G_{lc'}$	193
Atropic acid					
–, methyl ester	ss	−40	1.0	Toluene	149
–, ethyl ester	ss	−67	1.0	Toluene	150
–, *n*-propyl ester	ss	−72	1.0	Toluene	150
–, *n*-butyl ester	ss	−80	1.0	Toluene	150
–, *p*-methyl-, methyl ester	ss	−37	1.0	Toluene	150
Atroponitrile	ss	50	1.0	Toluene	149
Biphenyl, *p*-isopropenyl-	ss	0	0.515	Tetrahydrofuran	16
Cyclohexene	ss	−23	5	Toluene	389
Cyclooctene	ss	20		Benzene, N_{38}	229,230
Isobutylene	ss	88	1.2	Supercritical CO_2	441
Methacrylamide, *N*-butyl-	ss	122	0.651	Ethyl benzoate	326
–, *N*-phenyl-	ss	125	0.611	Ethyl benzoate	326
Methacrylic acid, methyl ester	ss	155.5	0.82	*o*-Dichlorobenzene	42
	ss	135	0.611	Ethyl benzoate	326
–, *o*-methyl phenyl ester	ss	140	0.64	*p*-Xylene	345
–, phenyl ester	ss	146	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-*tert*-butyl phenyl ester	ss	81	0.64	*p*-Xylene	345
–, *o*-*sec*-butylphenyl ester	ss	100	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-chlorophenyl ester	ss	139	0.64	Benzene or *p*-xylene, N_{75}	442
–, 2,6-diisopropylphenyl ester	ss	33	0.64	Benzene or *p*-xylene, N_{75}	442
–, 2,6-dimethyl phenyl ester	ss	77	0.64	*p*-Xylene	345
–, *o*-ethylphenyl ester	ss	118	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-isopropylphenyl ester	ss	112	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-methoxyphenyl ester	ss	145	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-methoxycarbonylphenyl ester	ss	112	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-nitrophenyl ester	ss	108	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-phenylphenyl ester	ss	100	0.64	Benzene or *p*-xylene, N_{75}	442
–, *o*-propylphenyl ester	ss	118	0.64	Benzene or *p*-xylene, N_{75}	442
2,4,6-tribromophenyl ester	ss	92	0.64	Benzene or *p*-xylene, N_{75}	442
2,4,6-trichlorophenyl ester	ss	104	0.64	Benzene or *p*-xylene, N_{75}	442
2,4,6-trimethylphenyl ester	ss	69	0.64	Benzene or *p*-xylene, N_{75}	442
Methacrylonitrile	ss	145	0.27	Benzonitrile	44
Methyl-α-*p*-chlorobenzylacrylate	ss	85	5	Toluene	443
α-Methylene pentanolactone	ss	83	1	DMF	315
	ss	55	0.324	DMF	315
Methyl-α-*p*-methoxybenzylacrylate	ss	95	5	Toluene	443
Methylphenylitaconate	ss	120	2,22	Dichlorobenzene	444
Naphthalene, 2-isopropenyl-	ss	0	0.284	Tetrahydrofuran	16
Oxazole,					
–, 2-isopropenyl-4,5-dimethyl-	ss	0	0.0025	Tetrahydrofuran	284
Pyridine, 2-isopropenyl-	ss	0	0.043	Tetrahydrofuran	284
Styrene	ss	150	9.1×10^{-4}	Benzene	28
	ss	150	6.5×10^{-4}	Cyclohexane	28
–, α, acetoxy	ss	47	1.0	Toluene	401
–, α-methyl-	ss	0	0.76	Tetrahydrofuran, N_{37}	16,32,33
	ss	53	1.8	*n*-Heptane	402
	ss	58	1.8	Toluene	402
–,–, *o*-methoxy-	ss	−25	~2	Methylene chloride	155
N,*N*,*N*′,*N*′-Tetraethyl-4-isopropenylphenylphosphorous diamide	ss	10	1.0	Tetrahydrofuran	408
Thiazole, 2-isopropenyl-	ss	0	0.010	Tetrahydrofuran	284
p-[Bis(Trimethylsilyl)methyl] isopropenylbenzene	ss	1.2	1.0	Tetrahydrofuran	409

TABLE 9. *cont'd*

Monomer	State of monomer and polymer xx	T_c (°C)	M (mol/l)	Solvent/Notes	Refs.
9.2. MONOMERS GIVING POLYMERS CONTAINING MAIN CHAIN CYCLIC CARBON					
–, 7,8-Dycyanoquinodimethane					
7,8-bis(butoxycarbonyl)	ss	93	0.014	Chloroform, N_{75}	411
	ss	442	1.0	Chloroform, N_{43}	411
7,8-Diacetyl-	ss	87	0.018	Chloroform, N_{75}	412
	ss	415	1.0	Chloroform, N_{43}	412
7,8-Dibenzoyl-	ss	88	0.069	Chloroform, N_{75}	412
	ss	301	1.0	Chloroform, N_{43}	412
7,8-Dibutoxycarbonyl	ss	93	0.014	Chloroform, N_{75}	412
	ss	442	1.0	Chloroform, N_{43}	411
	ss	79	0.014	Chloroform, N_{75}	411
	ss	422	1.0	Chloroform, N_{43}	411
	ss	67	0.014	Chloroform, N_{75}	411
	ss	481	1.1	Chloroform, N_{43}	411

TABLE 10. MONOMERS GIVING POLYMERS CONTAINING HETEROATOMS IN THE MAIN CHAIN

Monomer	State of monomer and polymer xx	T_c (°C)	M (mol/l)	Solvent/Notes	Refs.
10.1. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO CARBON ONLY					
Acetaldehyde	sc	~ –80	6	Toluene, N_{22}	228
–, bromochlorofluoro-	sc′	23	1	Toluene	233
–, bromodichloro-	sc′	–17	1	Toluene	344
–, bromodifluoro-	sc′	48	1	Toluene	344
–, dibromochloro-	sc	–40	1	Toluene	337
–, dibromofluoro-	sc′	–19	1	Toluene	344
–, dichlorofluoro-	sc	52	1	Methyl cyclohexane	336
	sc′	41	1	Toluene	344
–, chlorodifluoro-	sc	70	1	Methyl cyclohexane	336
	sc′	63	1	Toluene	344
–, tribromo-	sc′	–75	1	Toluene	338
–, trichloro-	ss	11	1.0	Tetrahydrofuran	167
	sc′	12.5	(0.1)	Pyridine, N_{39}, N_{40}	50
	sc′	35	1.0	*n*-Heptane	166
	sc′	18	1	Methyl cyclohexane	336
	sc′	9	1	Toluene	344
–, trifluoro-	sc′	81	1	Toluene	344
	sc	9	1	Dichloromethane	322
Butanal	sc	–18	~2	Toluene, isotactic polymer	227
	sc	12	1.0	*n*-Hexane [8 kbar], N_{12}	161
	sc	27	1.0	*n*-Hexane [10 kbar], N_{12}	161
4,7-Dioxaoctanal	ss	–56	1	THF	330
1,3-Dioxepane	ss	78	1.0	Benzene	116,134
	ss	27	1.0	Dichloromethane	136
	ss	32	1.0	1,2-Dichloroethane, N_{57}	424
–, 2-butyl	ss	–51	1.0	1,2-Dichloroethane, N_{57}	424
–, 2-methyl-	ss	–37	1	1,2-Dichloroethane	340
	ss	–40	1.0	1,2-Dichloroethane, N_{57}	424
–, 4-methyl-	ss	–33	1	1,2-Dichloroethane	339
$(1,3\text{-Dioxolane})_n$ $n=1$	ss	1	1.0	Dichloromethane	133,159
	ss	–8	1.0	Benzene, N_{41}	134,159
	ss	20	2.0	1,3-Dioxane, N_{41}	159
	ss	20	1.0	Chloroethane	293
	ls	60	2.0	A few percent dichloromethane present	198
	ls	145	10.0	A few percent dichloromethane present	198
$n=1$–8	ss	60		Dichloroethane, N_{42}	232

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TABLE 10. *cont'd*

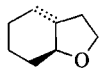
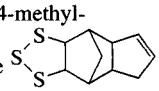
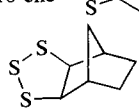
Monomer	State of monomer and polymer xx	T_c (°C)	M (mol/l)	Solvent/Notes	Refs.
Formaldehyde	sc′	30	0.06	Dichloromethane	122
	sc′	25	0.004	Nitrobenzene	182
L-Lactide	ss	284	1.0	1,4-Dioxane	429
Octanal	sc	−59	1	THF	330
trans-8-Oxabicyclo[4,3,0]nonane 	ss	−51	1	Dichloromethane	357
trans-2-Oxabicyclo[3,3,0]octane	ss	0	0.63	Dichloromethane	357
Oxa-4,5-dithiepane	ss	25	<0.005	Benzene	61
Oxepane	ss	30	0.08	Dichloromethane	236
Pentanal	sc	−68	1	THF	328
o-Phthalaldehyde	ss	−43	1.0	Dichloromethane, cyclopolymer	154
Propanal, 2-chloro-, 2-methyl-	ss	−54	1.0	Tetrahydrofuran	172
−, 2,2-dichloro-	ss	−24	1.0	Tetrahydrofuran	172
	ss	approx. 5	(0.25)	Dichloromethane, N_{22}, N_{40}	153
−, 2-methyl-	ss	−36.6	1.0	*n*-Pentane	166
	ss	−63	1.0	Tetrahydrofuran	172
	ss	−61.6	1.0	Diethyl ether	166
Tetrahydrofuran	ss	25	4.45	Benzene, N_{37}, N_{44}	231
	ss	20	2.8	Diethyl ether	211
Tetraoxane	sc′	50	<0.008	1,2-Dichloroethane	178
	sc′	25	0.12	Dichloromethane	182
	sc′	25	0.10	Nitrobenzene	182
Trioxane	sc′	50	0.20	1,2-Dichloroethane	178
	sc′	30	0.13	1,2-Dichloroethane	178
	sc′	30	0.05	Benzene	178
	sc′	30	0.19	Nitrobenzene, N_{45}	178
	sc′	25	1.5	Nitrobenzene, N_{45}	182
	sc′	25	2.5	Dichloromethane	182
1,3,6-Trioxocane	ss	297	1.0	Benzene, N_{45}	134
	ss	330	1.0	Toluene, N_{57}	425
−, 2-Butyl-	ss	220	1.0	Toluene, N_{57}	425
10.2. MONOMERS GIVING POLYMERS CONTAINING O IN THE MAIN CHAIN, BONDED TO OTHER HETEROATOMS (Si, P)					
1,3,2-Dioxaphospholan, 2-methoxy-	ss	157	0.1	1,1,2,2-Tetrachloroethane	309
1,3-Dioxepane, 4-methyl-	ss	−33	1	1,2-Dichloroethane	339
−, 2-methyl-	ss	−37	1	1,2-Dichloroethane	340
Siloxanes $(R_1R_2SiO)_n$					
R_1, R_2					
H, Me	ss	0	(0.2)	Toluene, N_{46}	176
Me, Me	ss	110	0.3	Toluene, N_{47}	217,218
Me, Et	ss	110	(0.4)	Toluene, N_{46}	176
Me, $CF_3(CH_2)_2$	ss	110	(0.9)	Cyclohexanone, N_{46}	176
10.3. MONOMERS GIVING POLYMERS CONTAINING S IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY					
1,3-Dithiolane, 2-phenylimino-4-methyl-	ss	65	2.7	Benzene	290
exo-3,4,5-Trithiatetracyclo-[5,5,1,$0^{2,6}$.$0^{8,12}$]tridec-10-ene 	ss	25	2.37	Toluene, N_{76}	376
	ss	90	3.72	Toluene, N_{76}	376
exo-3,4,5-Trithiatricyclo-5.2.1.$0^{2,6}$]decane 	ss	10	3.8	Toluene, N_{76}	377
	ss	70	6.0	Toluene, N_{76}	377
10.4. MONOMERS GIVING POLYMERS CONTAINING N IN THE MAIN CHAIN, BONDED IN THE CHAIN TO CARBON ONLY					
Bicyclic oxalactam	ss	60	1.0	Dimethylsulfoxide	433
n-Hexyl isocyanate	sc	−22	2	Dimethylformamide	127
Pyrrole	ss	77	0.017	Acetonitrile	434
N,7,7-Tricyanobenzoquinone methide imine	ss	74	0.14	Toluene, N_{75}, N_{81}	438

TABLE 11. COPOLYMERS

Monomer A	Monomer B	State of monomer and polymer	T^a (°C)	[A][B] (mol^2/l^2)	Solvent/Notes	Refs.
11.1. 1:1 COPOLYMERS						
Allyl acetate	Sulfur dioxide	ss	45	27	Excess B	110
Allyl alcohol	Sulfur dioxide	sc	76	27	Excess B	110
Allyl ethyl ether	Sulfur dioxide	ss	68	27	Excess B	110
Allyl formate	Sulfur dioxide	sc	45	27	Excess B	110
1-Butene	Sulfur dioxide	ss	64	27	Excess B	110
–, 2-ethyl-	Sulfur dioxide		<– 80		All compositions	110
–, 3-methyl-	Sulfur dioxide	sc	36	27	Excess B	110,199
2-Butene, *cis*	Sulfur dioxide	ss	46	27	Excess B, N_{48}	110
–, *trans*	Sulfur dioxide	ss	38	27	Excess B, N_{48}	110
2-Butene, (50% *cis*)	Sulfur dioxide	ss	34.6	27	Excess B, N_{49}	110
–, 2-methyl-	Sulfur dioxide		<– 80		All compositions	110
Cycloheptene	Sulfur dioxide	ss	11	27	Excess B	124
Cyclohexene	Sulfur dioxide	ss	24	27	Excess B	110
1,3-Cyclooctadiene	Sulfur dioxide	ss	37	10	Excess B (1,2 addition)	156
Cyclopentene	Sulfur dioxide	ss	103	27	Excess B	110
Ethene	Carbon monoxide	gg	150 (– 35)		N_{64}	369
	Carbon dioxide	gg	– (50)		N_{64}	369
	Sulfur dioxide	sc′	>135	27	Excess B	110
Ethene oxide	Carbon monoxide	gg	350 (– 127)		N_{64}	369
	Carbon dioxide	gg	220 (– 58)		N_{64}	369
Formaldehyde	Carbon monoxide	gg	190 (– 47)		N_{64}	369
	Carbon dioxide	gg	– 150 (46)		N_{64}	369
2-Heptene (88% *cis*)	Sulfur dioxide	ss	– 38	33	Excess B, N_{49}	110
1-Hexadecene	Sulfur dioxide	sc	69	27	Excess B	110
		ss	30	1.15	Chloroform	80
1-Hexene	Sulfur dioxide	ss	60	27	Excess B	110
–, 2-ethyl-	Sulfur dioxide		<– 80		All compositions	110
Isobutene	Sulfur dioxide	sc	5	27	Excess B	110
1-Pentene	Sulfur dioxide	ss	63	27	Excess B	110
–, 2-methyl-	Sulfur dioxide	ss	– 34	27	Excess B	110
–, 4,4-dimethyl-	Sulfur dioxide	sc	14	27	Excess B	110
–, 2,4,4-trimethyl-	Sulfur dioxide		<– 80		All compositions	110
2-Pentene (50% *cis*)	Sulfur dioxide	ss	8.5	32	N_{49}	110
–, 4-methyl-			<– 80		All compositions	110
4-Pentenoic acid	Sulfur dioxide	sc	66	27	Excess B	110
Propene	Sulfur dioxide	sc	90	27	Excess B	110
	Carbon monoxide	gg	140 (– 39)		N_{64}	369
	Carbon dioxide	gg	– 120 (39)		N_{64}	369
Propene oxide	Carbon monoxide	gg	490 (– 129)		N_{64}	369
	Carbon dioxide	gg	180 (– 49)		N_{64}	369
3-Propenyl phenyl ether	Carbon monoxide	gg	590 (– 136)		N_{64}	369
	Carbon dioixde	gg	190 (– 36)		N_{64}	369
Styrene	Carbon monoxide	gg	– 190 (53)		N_{64}	369
Tetrafluoroethene	Carbon monoxide	gg	280 (– 74)		N_{64}	369
	Carbon dioxide	gg	30 (– 2)		N_{64}	369
Vinyl chloride	Carbon monoxide	gg	230 (– 55)		N_{64}	369
	Carbon dioxide	gg	– 45 (18)		N_{64}	369
11.2. GENERAL COPOLYMERS						
ε-Caprolactam	ε-Caprolactam, 3-methyl-	ls	225		N_{50}	192,194

[a] Values in parentheses are for ΔG°_{298} in (kJ/mol).

D. POLYMERIZABILITY OF 5-, 6- AND 7-MEMBERED HETEROCYCLIC RING COMPOUNDS

The free energy of polymerization is generally negative for 3-, 4-, 8-, and higher-membered ring compounds. With heterocyclic rings there is generally an ionic mechanism available which allows these compounds to polymerize. With 5-, 6- and 7-membered ring compounds the sign of ΔG, and hence the polymerizability, is critically dependent on the nature of the ring and on the extent and position of substitution. Substitution generally makes ΔG less negative (more positive) and reduces polymerizability in the thermodynamic sense.

Tables 12–14 summarize the polymerizability of 5-, 6- and 7-membered rings, respectively, as a function of the nature of the ring and extent of substitution. Table 15 provides a comparison of polymerizability of unsubstituted 5-, 6-, and 7-membered rings.

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TABLE 11. COPOLYMERS

Monomer A	Monomer B	State of monomer and polymer	T^a (°C)	[A][B] (mol^2/l^2)	Solvent/Notes	Refs.
11.1. 1:1 COPOLYMERS						
Allyl acetate	Sulfur dioxide	ss	45	27	Excess B	110
Allyl alcohol	Sulfur dioxide	sc	76	27	Excess B	110
Allyl ethyl ether	Sulfur dioxide	ss	68	27	Excess B	110
Allyl formate	Sulfur dioxide	sc	45	27	Excess B	110
1-Butene	Sulfur dioxide	ss	64	27	Excess B	110
–, 2-ethyl-	Sulfur dioxide		<– 80		All compositions	110
–, 3-methyl-	Sulfur dioxide	sc	36	27	Excess B	110,199
2-Butene, *cis*	Sulfur dioxide	ss	46	27	Excess B, N_{48}	110
–, *trans*	Sulfur dioxide	ss	38	27	Excess B, N_{48}	110
2-Butene, (50% *cis*)	Sulfur dioxide	ss	34.6	27	Excess B, N_{49}	110
–, 2-methyl-	Sulfur dioxide		<– 80		All compositions	110
Cycloheptene	Sulfur dioxide	ss	11	27	Excess B	124
Cyclohexene	Sulfur dioxide	ss	24	27	Excess B	110
1,3-Cyclooctadiene	Sulfur dioxide	ss	37	10	Excess B (1,2 addition)	156
Cyclopentene	Sulfur dioxide	ss	103	27	Excess B	110
Ethene	Carbon monoxide	gg	150 (– 35)		N_{64}	369
	Carbon dioxide	gg	– (50)		N_{64}	369
	Sulfur dioxide	sc′	>135	27	Excess B	110
Ethene oxide	Carbon monoxide	gg	350 (– 127)		N_{64}	369
	Carbon dioxide	gg	220 (– 58)		N_{64}	369
Formaldehyde	Carbon monoxide	gg	190 (– 47)		N_{64}	369
	Carbon dioxide	gg	– 150 (46)		N_{64}	369
2-Heptene (88% *cis*)	Sulfur dioxide	ss	– 38	33	Excess B, N_{49}	110
1-Hexadecene	Sulfur dioxide	sc	69	27	Excess B	110
		ss	30	1.15	Chloroform	80
1-Hexene	Sulfur dioxide	ss	60	27	Excess B	110
–, 2-ethyl-	Sulfur dioxide		<– 80		All compositions	110
Isobutene	Sulfur dioxide	sc	5	27	Excess B	110
1-Pentene	Sulfur dioxide	ss	63	27	Excess B	110
–, 2-methyl-	Sulfur dioxide	ss	– 34	27	Excess B	110
–, 4,4-dimethyl-	Sulfur dioxide	sc	14	27	Excess B	110
–, 2,4,4-trimethyl-	Sulfur dioxide		<– 80		All compositions	110
2-Pentene (50% *cis*)	Sulfur dioxide	ss	8.5	32	N_{49}	110
–, 4-methyl-			<– 80		All compositions	110
4-Pentenoic acid	Sulfur dioxide	sc	66	27	Excess B	110
Propene	Sulfur dioxide	sc	90	27	Excess B	110
	Carbon monoxide	gg	140 (– 39)		N_{64}	369
	Carbon dioxide	gg	– 120 (39)		N_{64}	369
Propene oxide	Carbon monoxide	gg	490 (– 129)		N_{64}	369
	Carbon dioxide	gg	180 (– 49)		N_{64}	369
3-Propenyl phenyl ether	Carbon monoxide	gg	590 (– 136)		N_{64}	369
	Carbon dioixde	gg	190 (– 36)		N_{64}	369
Styrene	Carbon monoxide	gg	– 190 (53)		N_{64}	369
Tetrafluoroethene	Carbon monoxide	gg	280 (– 74)		N_{64}	369
	Carbon dioxide	gg	30 (– 2)		N_{64}	369
Vinyl chloride	Carbon monoxide	gg	230 (– 55)		N_{64}	369
	Carbon dioxide	gg	– 45 (18)		N_{64}	369
11.2. GENERAL COPOLYMERS						
ε-Caprolactam	ε-Caprolactam, 3-methyl-	ls	225		N_{50}	192,194

[a] Values in parentheses are for ΔG°_{298} in (kJ/mol).

D. POLYMERIZABILITY OF 5-, 6- AND 7-MEMBERED HETEROCYCLIC RING COMPOUNDS

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Tables 12–14 summarize the polymerizability of 5-, 6- and 7-membered rings, respectively, as a function of the nature of the ring and extent of substitution. Table 15 provides a comparison of polymerizability of unsubstituted 5-, 6-, and 7-membered rings.

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TABLE 12. 5-MEMBERED RING COMPOUNDS

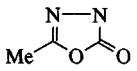
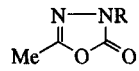
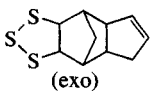
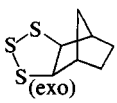
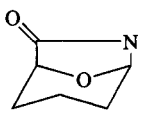
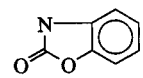
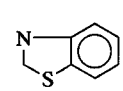
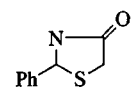
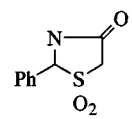
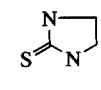
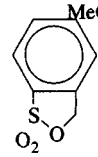
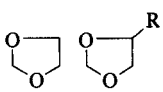
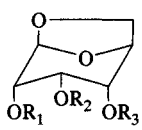
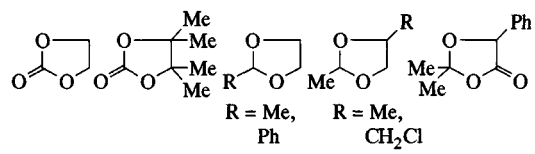
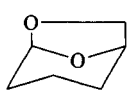
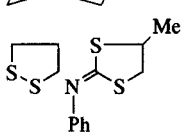
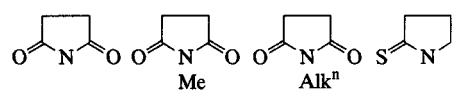
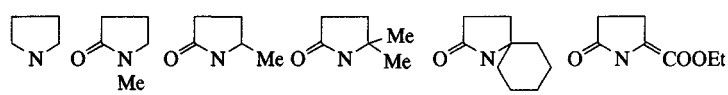
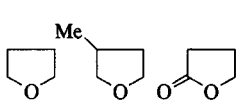
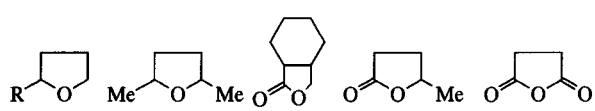
Ring	Polymerized	Not polymerized	Refs.
C_2N_2O	 N_{51}	 R = Me, Ph	237
C_2S_3	 (exo)		376
	 (exo)		377
C_3NO	R = Me, Ph etc. N_{52} 		238,239,433
C_3NS		  	239,240
C_3N_2		 R = Me, MeCO	239
C_3OS			152
C_3O_2	 R = Me, CH_2Cl, Ph  $R_1 = R_2 = R_3 = H$ $R_1 = R_2 = R_3 = Me$ $R_1 = R_2 = R_3 = CH_2Ph$ $R_1 = R_3 = CH_2Ph, R_2 = H$ $R_1 = R_2 = CH_2Ph, R_3 = H$	 R = Me, Ph R = Me, CH_2Cl	116,133,134, 159,177. 239,241–244, 285–287, 350,351,349, 398,423
			331
C_3S_2			245,246,290
C_4N			239,240,243
			240,247–249
C_4O			51,58,117, 130,140,146, 239,243,250–256, 291,303,304
	R′ = H, R = H R′ = H, R = endo – Me R′ = H, R = exo – Me R′ = endo – Me, R = exo – Me	R′ = exo-Me R = exo-Me	

TABLE 12. *cont'd*

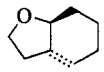
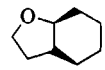
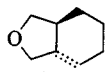
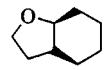
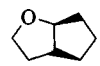
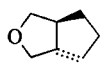
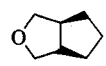
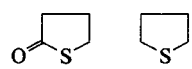
Ring	Polymerized	Not polymerized	Refs.
	 (*trans*)	 (*cis*)	357
	 (*trans*)	 (*cis*)	357
	(*trans*)	 (*cis*)	357
	 (*trans*)	 (*cis*)	357
C_4S			185,256

TABLE 13. 6-MEMBERED RING COMPOUNDS

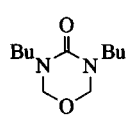
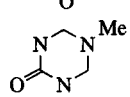
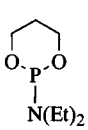
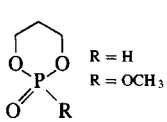
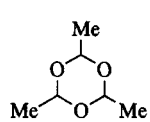
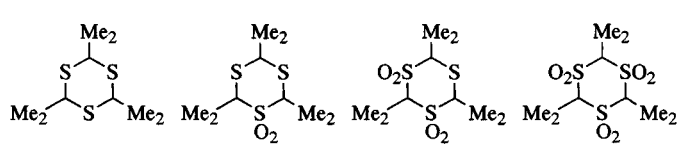
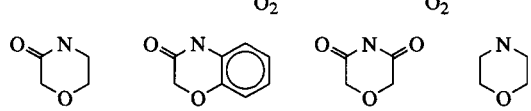
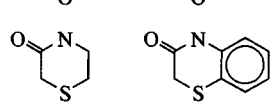
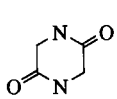
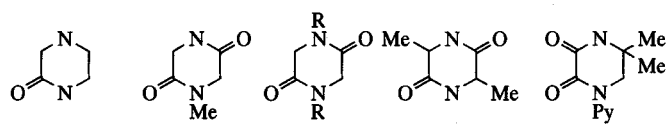
Ring	Polymerized	Not polymerized	Refs.
C_3N_2O			239
C_3N_3			239
C_3O_2P			346
	 R = H, R = OCH_3		347, 348
C_3O_3			141,178,184,257
C_3S_3			148,258,259,260
C_4NO			185,239,240
C_4NS			240
C_4N_2		 R = Me, Ph	239,240,261

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TABLE 13. *cont'd*

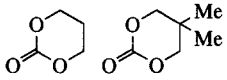
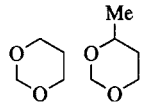
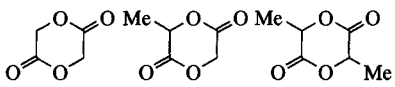
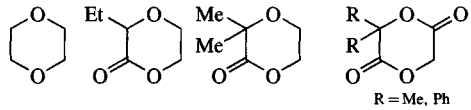
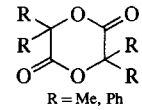
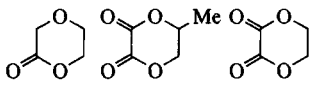
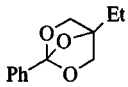
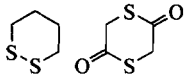
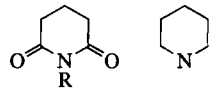
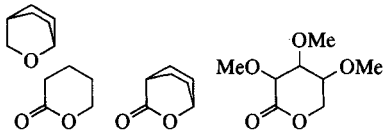
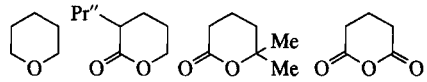
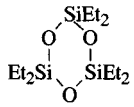
Ring	Polymerized	Not polymerized	Refs.
C_4OS			152
C_4O_2			116,136,239,243,262,426,436
		 R = Me, Ph  R = Me, Ph	239,243,262,263,428,429
			427
			382
C_4S_2			246,265
C_5N		 R = H, Et, Ph	118,209,239,240,243,266
C_5O			153,239,243,264,226–270,283
			331
C_5S			185,256
Si_3O_3			431

TABLE 14. 7-MEMBERED RING COMPOUNDS

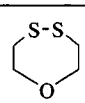
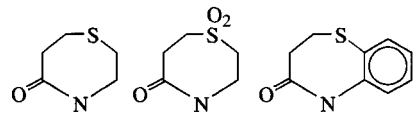
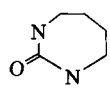
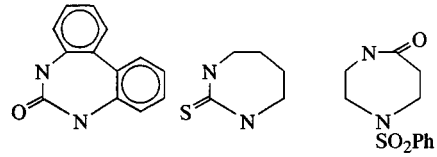
Ring	Polymerized	Not polymerized	Refs.
C_4OS_2			61
C_5NS			240
C_5N_2			239,240

TABLE 14. *cont'd*

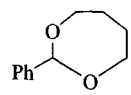
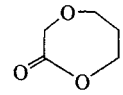
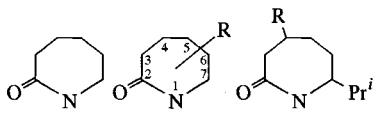
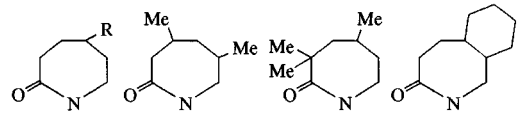
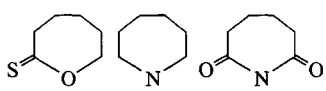
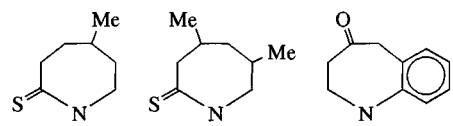
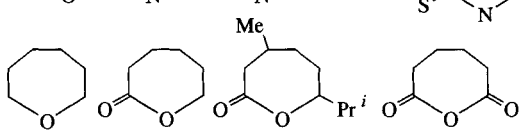
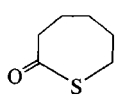
Ring	Polymerized	Not polymerized	Refs.
C_5O_2		 	116,136,170,241, 262,271,423,424
C_5S_2			246
C_6N	 R = 3 – Me, 4 – Me, R = H, Me 5 – Me, 5 – Et, 5 – n – propyl, 5–*i*–propyl 5 – Cyclohexyl 6 – Me, 7 – Me	 R = Ph, $PhCH_2$, *n* – Heptyl	63,68,240, 249,272–279
			240,243,276, 277,280,305
C_6O			153,239,243, 281,282
C_6S			256

TABLE 15. COMPARISON OF POLYMERIZABILITY (+ OR –) OF UNSUBSTITUTED 5-, 6- AND 7-MEMBERED RING COMPOUNDS[a]

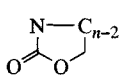
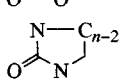
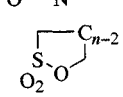
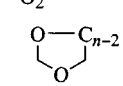
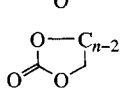
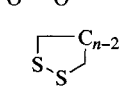
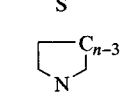
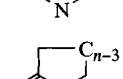
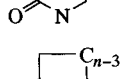
Type of ring		Name of 5-ring	5-Ring	6-Ring	7-Ring
1,3-C_nNO		2-Oxazolidinone	–	+	
1,3-C_nN_2		2-Imidazolidinone	+	–	+
1,2-C_nOS		1,2-Oxathiolane-2,2-dioxide	+	–	
1,3-C_nO_2		1,3-Dioxolane	+	–	+
		Ethylene carbonate	–	+	
1,2-C_nS_2		1,2-Dithiolane	+	+	+
C_nN		Pyrrolidine	–	–	+
		2-Pyrrolidinone	+	+	+
		Succinimide	–	–	+

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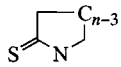
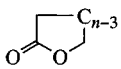
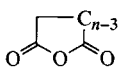
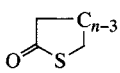
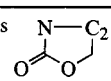
TABLE 15. *cont'd*

Type of ring		Name of 5-ring	5-Ring	6-Ring	7-Ring
		2-Pyrrolidinethione	–		+
C_nO		Tetrahydrofuran (oxolane)	+	–	+
		Oxolane-2-one (γ-butyrolactone)	+	+	+
		Oxolane-2,5-dione(succinic anhydride)	–	–	+
C_nS		Thiolan-2-one (γ–thiobutyrolactone)	–	+	+

[a] The formulae denote rings of any size. Thus in the first line $1,3C_4NO$ is drawn as  and represents

E. NOTES

N_1 Corrected for end-group effects.

N_2 Is assumed.

N_3 Partial allowance for unreacted monomer.

N_4 Semi-empirical values for a number of other olefins are given in Ref. 143.

N_5 Value given is for dilute solution. Values determined for complete range of composition; maximum (14.4) at 50 mol% monomer.

N_6 No allowance for unreacted monomer.

N_7 Strongly dependent on both monomer concentration and temperature.

N_9 Corrected for enthalpy of glassy state.

N_{10} $-\Delta S_{ss}$ assumed to be 120.3 J/K mol.

N_{11} Zero heat not necessarily in conflict with observed polymerizability below $-40°C$. Additional loss of free energy may be provided by the crystallization of the polymer.

N_{12} Polymer largely insoluble and remains in suspension.

N_{13} After correction for species in vapour other than formaldehyde.

N_{14} In the presence of a few per cent methylene chloride.

N_{15} From measurements in methylene chloride, benzene and 1,4-dioxane.

N_{16} From measurements in benzene.

N_{17} From measurements on vapor–solid equilibrium.

N_{18} From measurements in both methylene chloride and nitrobenzene.

N_{19} From measurements in 1,2-dichloroethane.

N_{20} Value calculated by the compiler of this table from data in Ref. cited.

N_{21} Numerical values for heats of copolymerization are listed only for those systems yielding 1 : 1 copolymers. The values refer to the copolymerization of 0.5 mol of each monomer. In all other cases listed the copolymers have a range of composition; details of the corresponding heats of copolymerization are given in the Ref. cited. Where no solvent is specified in the seventh column, the symbol ΔH_{ss} denotes that the measured heat is for a liquid mixture of monomers going to a solution of copolymers. ΔH_{lc} denotes the heat change for pure liquid monomers going to condensed amorphous polymer. (This symbolism differs from that in Ref. 1.)

N_{22} The states of monomer and/or polymer are not stated in Ref. but are likely to be the ones given.

N_{23} Value calculated from measurements on mixtures containing excess B.

N_{24} Value calculated from measurements on mixtures containing excess A.

N_{25} Value in Ref. 11 corrected using entropy of monomer in Ref. 94.

N_{26} From measurements in both benzene and cyclohexane.

N_{27} Standard state: mol fraction of monomer = 0.1.

N_{28} After correction for species in vapor other than formaldehyde.

N_{29} Numerical values for entropies of polymerizaiton listed are for systems yielding 1:1 copolymers. The values refer to the copolymerization of 0.5 mol of each monomer. The symbolism and standard states are the same as used for the heats of copolymerization (see N_{21}).

N_{30} Value of $\Delta S°$ for the composition of polymer indicated.

N_{31} No allowance for free energy of mixing of polymer and monomer.

N_{32} Floor temperature (hypothetical for supercooled liquid).

N_{33} Floor temperature.

N_{34} Calculated value. Not measurable experimentally because of side reactions.

N_{35} Small amount of vapor dissolved in the polymer: approximately gc.

N_{36} Hypothetical value for acetone as a gas at 1013 mbar pressure.

N_{37} Decreases with increasing polymer concentration (Ref. 112,160,289).

N_{38} Recognised that polymerization over $WCl_6/EtAlCl_2/EtOH$ yields an equilibrium mixture of unsaturated cyclic rings $(C_8H_{14})_n$. Species up to $n = 15$ have been identified and higher cyclic polymers are also likely to be present. No firm equilibrium data exist as yet.

N_{39} $dT_c/dP = 19$ deg/kbar.

N_{40} Value in [*M*] column is the mol fraction of monomer.

N_{41} Decreases with increasing polymer concentration in methylene chloride, in benzene and in 1,4-dioxane (Ref. 159).

N_{42} K_n values determined for cyclic monomers, $n = 1-8$.

N_{43} Extrapolated value.

N_{44} For equilibrium concentrations of linear oligomeric species see Ref. 294.

N_{45} Note the substantial discrepancy between the two sets of results with nitrobenzene as solvent.

N_{46} Value in [*M*] column is the volume fraction of cyclic monomer (rings of all sizes).

N_{47} Concentration of rings, $n = 13-\infty$, taking mol. wt. = 74 (value for $n = 1$); K_n determined for cyclic monomers, $n = 3-400$.

N_{48} Corrected for isomerization effect.

N_{49} Uncorrected for isomerization effect.

N_{50} Equilibrium concentrations of A and B lower than expected from values for homopolymerizations.

N_{51} Polymer is $\dashv$CONHN(COMe)$\vdash$.

N_{52} Polymer is $\dashv CH_2CH_2N(COR)\vdash$.

N_{53} *cis* and *trans* forms of polymer.

N_{54} c′(I),c′(II),c(III) refer to crystal forms I, II, III of monomer.

N_{55} Standard state is 1M, calculated from data in Ref. 148.

N_{56} Polymer dissolved in monomer; equilibrium concentration in mol/L.

N_{57} T_c calculated from $\Delta H^\circ_{lc}/\Delta S^\circ_{lc}$ or $\Delta H^\circ_{ss}/\Delta S^\circ_{ss}$.

N_{58} Valid in concentration range 2.3 to 4.6 M.

N_{59} Corrected for monomer–polymer interactions; required temperature variable interaction parameter.

N_{60} Data from Ref. 292, with corrections for monomer adsorbed in polymer.

N_{61} lc data from Ref. 231 combined with vaporization data.

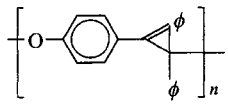
N_{62} Calorimetry in solution with adjustments for solution enthalpies.

N_{63} With heat capacity adjustment for T change: may apply to N_{62}.

N_{64} From group method calculations; ΔG° (298 K) value given in brackets in kJ/mol; T_c refers to standard state conditions, $p_A = p_B = 1$ atm.

N_{65} ΔH_p by method 4a in close agreement.

N_{66} Ring opening polymerization giving 78% *trans* polymer.

N_{67} Polymer is 

N_{68} Calculated from solution values with experimentally determined data.

N_{69} Calculated from solution data utilizing monomer-polymer interaction parameters, see Ref. 130.

N_{70} Mixture of 1,2 and 1,4 polymerization.

N_{71} Includes adjustment for the presence of small concentrations of other cyclics; $S_n \neq S_8$.

N_{72} Calculations based on standard enthalpies of formation of S_n and ΔH_p (S_8).

N_{73} Calculation based on standard entropies of S_n and ΔS_p (S_8).

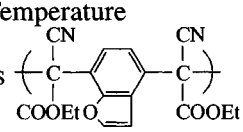
N_{74} Values apply in temperature range below the appropriate ceiling temperature, see Section C, Table 10, under "Monomers Giving Polymers Containing O in the Main Chain, Bonded to C Only."

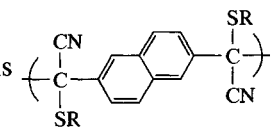
N_{75} Obtained by extrapolation of kinetic data.

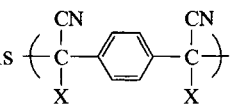
N_{76} Ring opening at S–C bond.

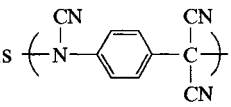
N_{77} c′(I), c′(II) refer to crystal forms I, II of polymer.

N_{78} Average Temperature

N_{79} Polymer is 

N_{80} Polymer is 

N_{81} Polymer is 

N_{82} Polymer is 

F. REFERENCES

1. F. S. Dainton, K. J. Ivin, in: H. A. Skinner (Ed.) "Experimental Thermochemistry", vol. II, Interscience, New York, 1962, p. 251.
2. R. S. Jessup, J. Chem. Phys., **16**, 661 (1948).
3. J. W. Richardson, G. S. Parks, J. Am. Chem. Soc., **61**, 3545 (1939).
4. F. A. Quinn, L. Mandelkern, J. Am. Chem. Soc., **80**, 3178 (1958).
5. F. S. Dainton, J. Diaper, K. J. Ivin, D. R. Sheard, Trans. Faraday Soc., **53**, 1269 (1957).,
6. C. M. Fontana, G. A. Kidder, J. Am. Chem. Soc., **70**, 3745 (1948).
7. G. S. Parks, H. P. Mosher, J. Polym. Sci. A, **1**, 1979 (1963).
8. A. G. Evans, M. Polanyi, Nature, **152**, 738 (1943).

9. R. H. Biddulph, P. H. Plesch, P. P. Rutherford, Polymer, **1**, 521 (1960).
10. D. E. Roberts, J. Res. Natl. Bur. Std. **44**, 221 (1950).
11. F. S. Dainton, T. R. E. Devlin, P. A. Small, Trans. Faraday Soc., **51**, 1710 (1955).
12. S. Kaarsemaker, J. Coops, Rec. Trav. Chim, **71**, 261 (1952).
13. H. van Kamp, J. Coops, W. A. Lambregts, B. J. Visser, H. Dekker, Rec. Trav. Chim, **79**, 1226 (1960).
14. F. S. Dainton, K. J. Ivin, D. A. G. Walmsley, Trans. Faraday Soc., **56**, 1784 (1960).
15. R. M. Joshi, Makromol. Chem., **55**, 35 (1962).
16. H. Hopff, H. Luessi, Makromol. Chem., **62**, 31 (1963).
17. R. M. Joshi, J. Polym. Sci., **56**, 313 (1962).
18. G. C. Sinke, D. R. Stull, J. Phys. Chem., **62**, 397 (1958).
19. J. R. Lacher, E. E. Merz, E. Bohmfalk, J. D. Park, J. Phys. Chem., **60**, 492 (1956).
20. J. R. Lacher, H. B. Gottlieb, J. D. Park, Trans. Faraday Soc., **58**, 2348 (1962).
21. L. K. J. Tong, W. O. Kenyon, J. Am. Chem. Soc., **69**, 2245 (1947).
22. W. M. D. Bryant, J. Polym. Sci., **56**, 277 (1962).
23. C. R. Patrick, Tetrahedron, **4**, 26 (1958).
24. R. M. Joshi, Makromol. Chem., **62**, 140 (1963).
25. F. S. Dainton, K. J. Ivin, Trans. Faraday Soc., **46**, 331 (1950).
26. D. E. Roberts, W. W. Walton, R. S. Jessup, J. Res. Natl. Bur. Std., **38**, 627 (1947).
27. L. K. J. Tong, W. O. Kenyon, J. Am. Chem. Soc., **69**, 1402 (1947).
28. S. Bywater, D. J. Worsfold, J. Polym. Sci., **58**, 571 (1962).
29. S. Bywater, Makromol. Chem., **52**, 120 (1962).
30. R. H. Biddulph, W. R. Longworth, J. Penfold, P. H. Plesch, P. P. Rutherford, Polymer, **1**, 521 (1960).
31. D. E. Roberts, R. S. Jessup, J. Res. Natl. Bur. Std., **46**, 11 (1951).
32. D. J. Worsfold, S. Bywater, J. Polym. Sci., **26**, 299 (1957).
33. H. W. McCormick, J. Polym. Sci., **25**, 488 (1957).
34. R. M. Joshi, Makromol. Chem., **66**, 114 (1963).
35. W. I. Bengough, Trans. Faraday Soc., **54**, 1560 (1958).
36. A. G. Evans, E. Tyrrall, J. Polym. Sci., **2**, 387 (1947).
37. K. G. McCurdy, K. J. Laidler, Can. J. Chem., **42**, 818 (1964).
38. J. H. Baxendale, G. W. Madaras, J. Polym. Sci., **19**, 171 (1956).
39. L. K. J. Tong, W. O. Kenyon, J. Am. Chem. Soc., **68**, 1355 (1946).
40. R. E. Cook, K. J. Ivin, Trans. Faraday Soc., **53**, 1132 (1957).
41. S. Ekegren, S. Oehrn, K. Granath, P. Kinell, Acta Chem. Scand., **4**, 126 (1950).
42. S. Bywater, Trans. Faraday Soc., **51**, 1267 (1955).
43. K. J. Ivin, Trans. Faraday Soc., **51**, 1273 (1955).
44. S. Bywater, Can. J. Chem., **35**, 552 (1957).
45. R. A. Nelson, R. S. Jessup, D. E. Roberts, J. Res. Natl. Bur. Std., **48**, 275 (1952).
46. R. S. Jessup, A. D. Cummings, J. Res. Natl. Bur. Std., **13**, 357 (1934).
47. A. A. Korotkov, E. N. Marandzheva, Russ. J. Phys. Chem. (English Transl.), **37**, 135 (1963).
48. V. A. Kargin, V. A. Kabanov, V. P. Zubov, I. M. Papisov, Dokl. Akad. Nauk SSSR, **134**, 1098 (1960). Chem. Abst., **55**, 8282 (1961).
49. F. S. Dainton, K. J. Ivin, D. A. G. Walmsley, Trans. Faraday Soc., **55**, 61 (1959).
50. W. K. Busfield, E. Whalley, Trans. Faraday Soc., **59**, 679 (1963).
51. H. C. Raine, R. B. Richards, H. Ryder, Trans. Faraday Soc., **41**, 56 (1945).
52. P. Gray, A. Williams. Trans. Faraday Soc., **55**, 760 (1959).
53. J. B. Rose, J. Chem. Soc., 546 (1946).
54. S. M. Skuratov, A. A. Strepikheev, S. M. Shtekher, A. V. Volokhina, Dokl. Akad. Nauk. SSSR, **117**, 263 (1957).
55. S. M. Skuratov, A. A. Strepikheev, M. P. Kozina, Dokl. Akad. Nauk. SSSR, **117**, 452 (1957).
56. R. C. Cass, S. E. Fletcher, C. T. Mortimer, H. D. Springall, T. R. White, J. Chem. Soc., 1406, (1958).
57. D. Sims, J. Chem. Soc., 864, (1964).
58. C. E. H. Bawn, R. M. Bell, A. Ledwith, Polymer, **6**, 95 (1965).
59. A. Snelson, H. A. Skinner, Trans. Faraday Soc., **57**, 2125 (1961).
60. S. E. Fletcher, C. T. Mortimer, H. D. Springall, J. Chem. Soc., 580 (1959).
61. F. S. Dainton, J. A. Davies, P. P. Manning, S. A. Zahir, Trans. Faraday Soc., **53**, 813 (1957).
62. A. A. Strepikheev, S. M. Skuratov, O. N. Kachinskaya, R. S. Muramova, E. P. Brykina, S. M. Shtekher, Dokl. Akad. Nauk. SSSR, **102**, 105 (1955).
63. V. P. Kolesov, I. E. Paukov, S. M. Skuratov, Zh. Fiz. Khim., **36**, 770 (1962); Russ. J. Phys. Chem., **36**, 400 (1962).
64. M. P. Kozina, S. M. Skuratov, Dokl. Akad. Nauk. SSSR, **127**, 561 (1959).
65. S. M. Skuratov, A. A. Strepikheev, Y. N. Kanarskaya, Kolloidn. Zh., **14**, 185 (1952).
66. A. B. Meggy, J. Chem. Soc., 796, (1953).
67. P. F. van Velden, G. M. van der Want, D. Heikens, C. A. Kruissink, P. H. Hermans, A. J. Staverman, Rec. Trav. Chim., **74**, 1376 (1955).
68. A. V. Tobolsky, A. Eisenberg, J. Am. Chem. Soc., **81**, 2302 (1959).
69. H. Mackle, P. A. G. O'Hare, Tetrahedron, **19**, 961 (1963).
70. S. Sunner, Acta Chem. Scand., **17**, 728 (1963).
71. H. Mackle, R. G. Mayrick, Trans. Faraday Soc., **58**, 230 (1962).
72. J. A. Barltrop, P. M. Hayes, M. Calvin, J. Am. Chem. Soc., **76**, 4348 (1954).
73. F. Fairbrother, G. Gee, G. T. Merrall, J. Polym. Sci.,**16**, 459 (1955).
74. A. V. Tobolsky, A. Eisenberg, J. Am. Chem. Soc., **81**, 780 (1959).
75. H. Nagao, T. Yamaguchi, J. Chem. Soc. Japan, Ind. Eng. Chem. Sect., **59**, 1363 (1956).
76. H. Miyama, S. Fujimoto, J. Polym. Sci., **54**, S 32 (1961).
77. F. S. Dainton, K. J. Ivin, Proc. Roy. Soc. (London) A, **212**, 207 (1952).
78. J. E. Hazell, K. J. Ivin, Trans. Faraday Soc., **58**, 342 (1962).

79. L. K. J. Tong, W. O. Kenyon, J. Am. Chem. Soc., **71**, 1925 (1949).
80. F. S. Dainton, K. J. Ivin, D. R. Sheard, Trans. Faraday Soc., **52**, 414 (1956).
81. M. Suzuki, H Miyama, S. Fujimoto, J. Polym. Sci., **31**, 212 (1958).
82. K. J. Ivin, W. A. Keith, H. Mackle, Trans. Faraday Soc., **55**, 262 (1959).
83. R. E. Cook, K. J. Ivin, J. H. O'Donnell, Trans. Faraday Soc., **61**, 1881 (1965).
84. W. A. Piccoli, G. G. Haberland, R. L. Merker, J. Am. Chem. Soc., **82**, 1883 (1960).
85. A. Eisenberg, A. V. Tobolsky, J. Polym. Sci., **46**, 19 (1960).
86. D. C. Abercromby, R. A. Hyne, P. F. Tiley, J. Chem. Soc., 5832 (1963).
87. F. S. Dainton, K. J. Ivin, Quart. Rev., **12**, 61 (1958).
88. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer **3**, 277, 286 (1962).
89. E. Passaglia, H. K. Kevorkian, J. Appl. Polym. Sci., **7**, 119 (1963).
90. R. W. Warfield, M. C. Petree, Makromol. Chem., **51**, 113 (1962).
91. B. Wunderlich, J. Chem. Phys. **37**, 1203 (1962).
92. E. Passaglia, H. K. Kevorkian, J. Appl. Phys., **34**, 90 (1963).
93. G. T. Furukawa, M. L. Reilly, J. Res. Natl. Bur. Std., **56**, 285 (1956).
94. H. L. Finke, D. W. Scott, M. E. Gross, J. F. Messerly, G. Waddington, J. Am. Chem. Soc., **78**, 5469 (1956).
95. R. B. Scott, C. H. Meyers, R. D. Rands, F. G. Brickwedde, N. Bekkedahl, J. Res. Natl. Bur. Std., **35**, 39 (1945).
96. G. T. Furukawa, R. E. McCoskey, J. Res. Natl. Bur. Std., **51**, 321 (1953).
97. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 297 (1962).
98. G. T. Furukawa, R. E. McCoskey, M. L. Reilly, J. Res. Natl. Bur. Std., **51**, 69 (1953).
99. G. T. Furukawa, R. E. McCoskey, G. J. King, J. Res. Natl. Bur. Std., **49**, 273 (1952).
100. N. Bekkedahl, L. A. Wood, J. Res. Natl. Bur. Std., **19**, 551 (1937).
101. N. Bekkedahl, H. Matheson, J. Res. Natl. Bur. Std., **15**, 503 (1953).
102. T. P. Melia, Polymer, **3**, 317 (1962).
103. R. W. Warfield, M. C. Petree, J. Polym. Sci. A, **1**, 1701 (1963).
104. R. H. Boundy, R. F. Boyer, "Styrene", Reinhold, New York, 1952, p. 67.
105. R. W. Warfield, M. C. Petree, J. Polym. Sci., **55**, 497 (1961).
106. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 263 (1962)
107. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 271 (1962).
108. R. J. Orr, Polymer, **2**, 74 (1961).
109. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 310 (1962).
110. R. E. Cook, F. S. Dainton, K. J. Ivin, J. Polym. Sci., **26**, 351 (1957).
111. G. Gee, Chemical Society (London) Special Publication, **15**, 67 (1961).
112. A. Vrancken, J. Smid, M. Szwarc, Trans. Faraday Soc., **58**, 2036 (1962).
113. A. V. Tobolsky, A. Rembaum, A. Eisenberg, J. Polym. Sci., **45**, 347 (1960).
114. K. J. Ivin, Pure Appl. Chem., **4**, 271 (1962).
115. A. M. North, D. Richardson, Polymer, **6**, 333 (1965).
116. A. A. Strepikheev, A. V. Volokhina, Dokl. Akad. Nauk. SSSR, **99**, 407 (1954).
117. C. L. Hamermesh, V. E. Haury, J. Org. Chem., **26**, 4748 (1961).
118. N. Yoda, A. Miyake, J. Polym. Sci., **43**, 117 (1960).
119. J. G. Kilroe, K. E. Weale, J. Chem. Soc., 3849, (1960).
120. H. H. G. Jellinek, H. Kachi, Makromol. Chem., **85**, 1 (1965).
121. R. B. Whitney, M. Calvin, J. Chem. Phys., **23**, 1750 (1955).
122. W. Kern, V. Jaacks, J. Polym. Sci., **48**, 399 (1960).
123. B. J. Cottam, J. M. G. Cowie, S. Bywater, Makromol. Chem., **86**, 116 (1965).
124. J. E. Hazell, K. J. Ivin, Trans. Faraday Soc., **58**, 176 (1962).
125. J. K. Jacques, M. F. Mole, N. L. Paddock, J. Chem. Soc., 2112, (1965).
126. J. W. C. Crawford, J. Chem. Soc., 2658, (1953).
127. V. E. Shashoua, W. Sweeny, R. E. Tietz, J. Am. Chem. Soc., **82**, 866 (1960).
128. R. C. P. Cubbon, Makromol. Chem., **80**, 44 (1964).
129. W. K. Busfield, Polymer, **7**, 541 (1966).
130. K. J. Ivin, J. Leonard, Polymer, **6**, 621 (1965).
131. K. Nakatsuka, H. Suga, S. Seki, J. Polym. Sci. B, **7**, 361 (1969).
132. J. B. Thompson, W. M. D. Bryant, Am. Chem. Soc. Polym. Preprints, **11**, 204 (1970).
133. P. H. Plesch, P. H. Westermann, J. Polym. Sci. C, **16**, 3837 (1968).
134. Y. Yamashita, M. Okada, K. Suyama, H. Kasahara, Makromol. Chem., **144**, 146 (1968).
135. O. Riedel, P. Wittmer, Makromol. Chem., **97**, 1 (1966).
136. P. H. Plesch, P. H. Westermann, Polymer, **10**, 105 (1969).
137. B. Boerjesson, Y. Nakase, S. Sunner, Acta Chem. Scand., **20**, 803 (1966).
138. A. K. Bonetskaya, S. M. Skuratov, Vysokomol. Soedin. A, **11**, 532 (1969).
139. H. G. Elias, A. Fritz, Makromol. Chem., **114**, 31 (1968); **120**, 238 (1968).
140. G. A. Clegg, D. R. Gee, T. P. Melia, A. Tyson, Polymer, **9**, 501 (1968).
141. G. A. Clegg, T. P. Melia, A. Tyson, Polymer, **9**, 75 (1968).
142. G. A. Clegg, T. P. Melia, Makromol. Chem., **123**, 184 (1969).
143. R. M. Joshi, B. J. Zwolinski, C. W. Hayes, Macromolecules, **1**, 30 (1968).
144. D. R. Waywell, J. Polym. Sci. B, **8**, 327 (1970).
145. G. C. Vezzoli, F. Dachille, R. Roy. J. Polym. Sci. A-1, **7**, 1557 (1969).
146. M. Rahman, K. E. Weale, Polymer, **11**, 122 (1970).
147. F. P. Jones, Ph.D. Thesis, Keele University, 1970.

148. V. C. E. Burnop, K. G. Latham, Polymer, **8**, 589 (1967).
149. H. Hopff, H. Luessi, L. Borla, Makromol. Chem., **81**, 268 (1965).
150. L. Borla, Promotionsarbeit Nr. 3668, E. T. H. Zuerich (1965).
151. H. Luessi, Makromol. Chem., **103**, 68 (1967).
152. S. Hashimoto, T. Yamashita, Kobunshi Kagaku, **27**, 400 (1970).
153. H. Sumimoto, T. Nakagawa, J. Polym. Sci. B, **7**, 739 (1969).
154. C. Aso, S. Tagami, T. Kunitake, J. Polym. Sci. A-1, **7**, 497 (1969).
155. Y.Okamoto, H. Takano, H. Yuki, Polym. J., **1**, 403 (1970).
156. T. Yamaguchi, K. Nagi, Kobunshi Kagaku, **26**, 809 (1969).
157. J. Stein, P. Wittmer, J. Toelle, Angew. Makromol. Chem., **8**, 61 (1969).
158. A. Novak, E. Whalley, Can. J. Chem., **37**,1710 (1959).
159. L. I. Kuzub, M. A. Markevich, A. A. Berlin, N. S. Yenikolopyan, Vysokomol. Soedin. A, **10**, 2007 (1968); Polym. Sci. SSSR, **10**, 2332 (1968).
160. K. J. Ivin J. Leonard, Eur. Polym. J., **6**, 331 (1970).
161. Y. Ohtsuka, C. Walling, J. Am. Chem. Soc., **88**, 4167 (1966).
162. C. Walling, T. A. Augurt, J. Am. Chem. Soc., **88**, 4163 (1966).
163. G. Natta, G. Mazzanti, P. Corradini, I. W. Bassi, Makromol. Chem., **37**, 156 (1960).
164. W. K. Busfield, D. Merigold, Makromol. Chem., **138**, 65 (1970).
165. Y. Iwasa, T. Imoto, J. Chem. Soc. Japan, Pure Chem. Sect. (Nippon Kagaku Zasshi), **84**, (29 (1963).
166. I. Mita, I. Imai, H. Kambe, Makromol. Chem., **137**, 169 (1970).
167. I. Mita, I. Imai, H. Kambe, Makromol. Chem., **137**, 143 (1970).
168. V. E. Ostrovskii, V. A. Khodzhemirov, A. P. Barkova, Dokl. Akad. Nauk. SSSR, **191**, 1095 (1970).
169. P. Caillaud, J. M. Huet, E. Maréchal, Bull. Soc. Chim. France, 1473 (1970).
170. G. A. Clegg, T. P. Melia, Polymer, **11**, 245 (1970).
171. A. K. Bonetskaya, T. V. Sopova, O. B. Salamatina, S. M. Skuratov, B. P. Fabrichuyi, I. F. Shalavina, V. L. Gol'dfarb, Vysokomol. Soedin. B, **11**, 894 (1969).
172. I. Mita, I. Imai, H. Kambe, Makromol. Chem., **137**, 155 (1970).
173. A. Nicco, B. Boucheron, Europ. Polym. J., **6**, 1477 (1970).
174. G. R. Walker, J. A. Semlyen, Polymer, **11**, 472 (1970).
175. E. Maréchal, J. Polym. Sci. A-1, **8**, 2867 (1970).
176. P. V. Wright, J. A. Semlyen, Polymer, **11**, 462 (1970).
177. G. A. Cleg, T. P. Melia, Polymer, **10**, 912 (1969).
178. T. Miki, T. Higashimura, S. Okamura, J. Polym. Sci. A-1, **8**, 157 (1970).
179. E. G. Butcher, H. Mackle, D. V. McNally, Trans. Faraday Soc., **65**, 2331 (1969).
180. J. A. Semlyen, P. V. Wright, Polymer, **10**, 543 (1969).
181. J. A. Semlyen, G. R. Walker, Polymer, **10**, 597 (1969).
182. A. A. Berlin, K. A. Bogdanova, I. P. Kravchuk, G. P. Rakova, N. S. Yenikolopyan, Dokl. Akad. Nauk. SSSR, **184**, 1128 (1969).
183. V. E. Ostrovskii, V. A. Khodzhemirov, S. P. Kostareva, Dokl. Akad. Nauk. SSSR, **184**, 103 (1969).
184. G. A. Clegg, T. P. Melia, Makromol. Chem., **123**, 194 (1969).
185. G. Nabi, Pak. J. Sci., **20**, 29 (1968).
186. D. R. Gee, T. P. Melia, Makromol. Chem., **116**, 122 (1968).
187. V. I. Kasatochkin, M. E. Kazakov, Y. P. Kudryavtsev, A. M. Sladkov, V. V. Korshak, Dokl. Akad. Nauk. SSSR, **183**, 109 (1968).
188. E. M. Morazova, V. I. Yeliseyeva, M. A. Korshunov, Vysokomol. Soedin. A, **10**, 2354 (1968); Polym. Sci. A, SSSR, **10**, 2736 (1968).
189. A. K. Bonetskaya, S. M. Skuratov, N. A. Lukina, A. A. Strel'tsova, K. E. Kuznetsova, M. P. Lazareva, Vysokomol. Soedin. B, **10**, 75 (1968).
190. H. Cheradame, J. P. Vairon, P. Sigwalt, Europ. Polym. J., **4**, 13 (1968).
191. O. B. Salamatina, A. K. Bonetskaya, S. M. Skuratov, B. P. Fabrichnyi, I. F. Shalavina, Y. L. Gol'dfarb, Vysokomol. Soedin. B, **10**, 10 (1968).
192. P., Čefelín, P. Schmidt, J. Šebenda, J. Polym. Sci. C, **23**, 175 (1966).
193. B. V. Lebedev, I. B. Rabonovich, L. Y. Martynenko, Vysokomol. Soedin. A, **9**, 1640 (1967).
194. P. Čefelín, A. Frydrychová, J. Labský, P. Schmidt, J. Šebenda, Coll. Czech Chem. Comm., **32**, 2787 (1967).
195. M. Furue, S. Nozakura, S. Murahashi, T. Tanaka, Bull. Chem. Soc. Japan, **40**, 2700 (1967).
196. T. P. Melia, A. Tyson, Makromol. Chem., **109**, 87 (1967).
197. S. Iwatsuki, S. Amano, Y. Yamashita, Kogyo Kagaku Zassi, **70**, 2027 (1967).
198. L. A. Kharitonova, G. V. Rakova, A. A. Shaginyan, N. S. Yenikolopyan, Vysokomol. Soedin. A, **9**, 2586 (1967).
199. B. H. G. Brady, J. H. O'Donnell, Trans. Faraday Soc., **64**, 29 (1968).
200. B. H. G. Brady, J. H. O'Donnell, Europ. Polym. J., **4**, 537 (1968).
201. J. Chiu, Proc. 2nd Symp. on Anal. Calorimetry 171, (1970).
202. K. A. Bogdanova, A. K. Bonetskaya, A. A. Berlin, G. V. Rakova, N. S. Yenikolopyan, Dokl. Akad. Nauk. SSSR, **197**, 618 (1971).
203. F. Delben, V. Crescenzi, Ann. Chim. (Rome), **60**, 782 (1970).
204. G. Natta, G. Mazzanti, P. Corradini, I. W. Bassi, Makromol. Chem., **37**, 156 (1968).
205. K. J. Ivin, Thermodynamics of Addition Polymerization Processes, Ch. 16 in: A. D. Jenkins, A. Ledwith, (Eds.), "Reactivity, Mechanism and Structure in Polymer Chemistry", Wiley, New York, 1973.
206. I. B. Rabinovich, L. I. Pavlinov, Vysokomol. Soedin. A, **10**, 416 (1968).
207. S. M. Skuratov, V. V. Voevodskii, A. A. Strepikheev, Y. N. Kanarskaya, R. S. Muramova, N. V. Fok, Dokl. Akad. Nauk. SSSR, **95**, 591 (1954).
208. J. H. S. Green, Quart. Rev. Chem. Soc., **15**, 125 (1961).
209. F. Korte, W. Glet, J. Polym. Sci. B, **4**, 685 (1966).
210. R. H. Beaumont, B. Clegg, G. Gee, J. B. M. Herbert, D. J. Marks, R. C. Roberts, D. Sims, Polymer, **7**, 401 (1966).

211. B. A. Rozenberg, O. M. Chekhuta, Y. B., Lyudvig, A. R. Gantmakher, S. S. Medvedev, Vysokomol. Soedin., **6**, 2030 (1964); Polym. Sci. SSSR, **6**, 2246 (1964).
212. L. Lesse, M. W. Baumber, Polymer, **6**, 269 (1965).
213. R. W. Warfield, M. C. Petree, J. Polym. Sci. A-2, **4**, 532 (1966).
214. J. Grodzinsky, A. Katchalsky, D. Vofsi, Makromol. Chem., **44/46**, 591 (1961).
215. R. M. Joshi, Indian. J. Chem., **2**, 215 (1964).
216. M. F. Shostakovskii, I. F. Bogdanov, J. Appl. Chem. Russ., **15**, 249 (1942).
217. J. F. Brown, G. M. J. Slusarczuk, J. Am. Chem. Soc., **87**, 931 (1965).
218. P. J. Flory, J. A. Semlyen, J. Am. Chem. Soc., **88**, 3209 (1966).
219. M. Kučera, Y. Pichler, Vysokomol. Soedin., **7**, 10 (1965); Polym. Sci. SSSR, **7**, 9 (1965).
220. G. M. Chil'-Gevorgyan, A. K. Bonetskaya, S. M. Skuratov, N. S. Yenikolopyan, Vysokomol. Soedin. A, **9**, 1363 (1967).
221. I. Mita, S. Yabe, I. Imai, H. Kambe, Makromol. Chem., **137**, 133 (1970).
222. T. P. Melia, G. A. Clegg, A. Tyson, Makromol. Chem., **112**, 84 (1968).
223. O. E. Grikina, V. M. Tatevskii, N. F. Stepanov, S. S. Yarovoi, Vestn. Mosk. Univ. Ser. II, **22**, 8 (1967).
224. O. E. Grikina, N. F. Stepanov, V. M. Tatevskii, S. S. Yarovoi, Vysokomol. Soedin. A, **13**, 575 (1971); Polym. Sci. SSSR, **13**, 653 (1971).
225. P. Čefelín, D. Doskočilova, A. Frydrychová, J. Šebenda, Coll. Czech. Chem. Comm., **29**, 485 (1964).
226. P. Čefelín, A. Frydrychová, P. Schmidt, J. Šebenda, Coll. Czech. Chem. Comm., **32**, 1006 (1967).
227. O. Vogl. J. Polym. Sci. A, **2**, 4607 (1964).
228. H. Takida, K. Noro, Chemistry of High Polymers, Japan, **21**, 452 (1964).
229. E. Wasserman, D. A. Ben-Efraim, R. Wolovsky, J. Am. Chem. Soc., **90**, 3286 (1968).
230. N. Calderon, J. Macromol. Sci. Rev. Macromol. Chem. **C7**, 105 (1972).
231. J. Leonard, D. Maheux, Am. Chem. Soc. Polymer Preprints, **13**, 78 (1972).
232. J. M. Andrews, J. A. Semlyen, Polymer, **13**, 142 (1972).
233. O. Vogl, personal communication.
234. O. Vogl, W. M. D. Bryant, J. Polym. Sci. A, **2**, 4633 (1964).
235. H. Kambe, I. Mita, K. Horie, Proc. 2nd Int. Conf. on Thermal Analysis (1969) 1071.
236. T. Saegusa, T. Shiota, S. Matsumoto, H. Fujii, quoted in Macromolecules, **5**, 34 (1972).
237. T. Endo, Bull. Chem. Soc. Japan, **44**, 870 (1971).
238. T. Kagiya, S. Narisawa, T. Maeda, K. Fukui, Kogyo Kagaku Zasshi, **69**, 732 (1969); J. Polym. Sci. B, **4**, 441 (1966).
239. H. K. Hall, A. K. Schneider, J. Am. Chem. Soc., **80**, 6409 (1958).
240. H. K. Hall, J. Am. Chem. Soc., **80**, 6404 (1958).
241. J. W. Hill W. H. Carothers, J. Am. Chem. Soc., **57**, 925 (1935).
242. M. Okada, Y. Yamashita, Y. Ishii, Makromol. Chem., **80**, 196 (1964).
243. W. H. Carothers, Collected papers of, G. S. Whitby, H. Mark, (Ed.), Interscience, New York, 1940.
244. I. Maruyama, M. Nakaniwa, T. Saegusa, J. Furukawa, J. Chem. Soc. Japan Ind. Chem. Sect., **68**, 1149 (1965).
245. R. B. Whitney, M. Calvin, J. Chem. Phys., **23**, 1750 (1955).
246. J. G. Affleck, G. Dougherty, J. Org. Chem., **15**, 865 (1950).
247. W. O. Ney, W. R. Nummy, C. E. Bames, USP. 2638463 (1953).
248. W. O. Ney, M. Crowther, USP. 2739959 (1956).
249. M. P. Kozina, S. M. Skuratov, Dokl. Akad. Nauk. SSSR, **127**, 561 (1959).
250. H. Meerwein, Angew. Chem., **59**, 168 (1947).
251. H. Meerwein, D. Delfs, H. Morschel, Angew. Chem., **72**, 927 (1960).
252. M. P. Dreyfuss, P. Dreyfuss, J. Polym. Sci. A, **4**, 2179 (1966).
253. T. Saegusa, S. Matsumoto, J. Macromol. Sci. A: Chem., **4**, 873 (1970).
254. J. B. Rose, J. Stuart-Webb, unpublished results quoted in: P. H. Plesch, (Ed.), "The Chemistry of Cationic Polymerization", Pergamon, Oxford 1963.
255. R. Chiang, J. H. Rhodes, J. Polym. Sci. B, **7**, 643 (1969).
256. C. G. Overberger, J. K. Weise, J. Am. Chem. Soc., **90**, 3533 (1968).
257. W. Kern, H. Cherdron, V. Jaacks, Angew. Chem., **73**, 177 (1961).
258. E. Gipstein, E. Wellisch, O. L. Sweeting, J. Polym. Sci. B, **1**, 237 (1963).
259. J. B. Lando, V. Stannett, J. Polym. Sci. B, **2**, 375 (1964).
260. O. G. von Ettinghausen, E. Kendrick, Polymer, **7**, 469 (1966).
261. A. B. Meggy, J. Chem. Soc., 1444, (1956).
262. H. K. Hall, Am. Chem. Soc. Polymer Preprints, **6**, 535 (1965).
263. W. Dittrich, R. C. Schulz, Angew. Makromol. Chem., **15**, 109 (1971).
264. E. Hollo, Bericht, **61**, 895 (1928).
265. A. Schoeberl, G. Wiehler, Ann., **595**, 101 (1955).
266. P. A. Small, Trans. Faraday Soc., **51**, 1717 (1955).
267. F. Fichter, A. Beisswenger, Ber., **36**, 1200 (1903).
268. K. Saotome, Y. Kodaira, Makromol. Chem., **82**, 41 (1965).
269. H. Batzer, G. Fritz, Makromol. Chem., **14**, 179 (1954).
270. H. D. K. Drew, W. N. Haworth, J. Chem. Soc., 775, (1927).
271. M. H. Palomaa, V. Toukola, Ber. **66B**, 1629 (1933).
272. D. D. Coffman, N. L. Cox, E. L. Martin, W. E. Mochel, F. J. van Natta, J. Polym. Sci., **3**, 85 (1948).
273. W. E. Hanford, R. M. Joyce, J. Polym. Sci., **3**, 167 (1948).
274. H. R. Mighton, USP. 2647105 (1953).
275. F. N. S. Carver, B. L. Hollingsworth, Makromol. Chem., **95**, 135 (1966).
276. A. Schäffler, W. Ziegenbein, Ber., **88**, 1374 (1955).
277. W. Ziegenbein, A. Schäffler, R. Kaufhold, Ber., **88**, 1906 (1955).
278. L. E. Wolinski, H. R. Mighton, J. Polym. Sci., **49**, 217 (1961).

279. M. Imoto, H. Sakurai, T. Kono, J. Polym. Sci., **50**, 467 (1961).
280. N. L. Coxx, W. E. Hanford, USP. 2276164 (1942); Swiss P. 270546 (1951), 276924 (1952).
281. F. P. Jones, Ph.D. thesis, Univers. of Keele, 1970.
282. R. Gehm, Angew. Makromol. Chem., **18**, 159 (1971).
283. T. Saegusa, T. Hodaka, H. Fujii, Polym. J., **2**, 670 (1971).
284. K. Yagi, T. Miyazaki, H. Okitsu, F. Toda, Y. Iwakura, J. Polym. Sci. A-1, **10**, 1149 (1972).
285. B. Krummenacher, H. G. Elias, Makromol. Chem., **150**, 271 (1971).
286. P. C. Wollwage, P. A. Seib, J. Polym. Sci. A-1, **9**, 2877 (1971).
287. C. C. Tu, C. Schuerch, J. Polym. Sci. B, **1**, 163 (1963).
288. T. Kunitake, M. Yasumatsu, C. Aso, J. Polym. Sci. A-1, **9**, 3678 (1971).
289. I. Mita, H. Okuyama, J. Polym. Sci. A-1, **9**, 3437 (1971).
290. G. Belonovskaya, Z. Tchernova, B. Dolgoplosk, Europ. Polym. J., **8**, 35 (1972).
291. J. Kops, H. Spanggaard, Makromol. Chem., **151**, 21 (1972).
292. W. K. Busfield, R. M. Lee, D. Merigold, Makromol. Chem., **156**, 183 (1972).
293. Y. B. Lyudvig, Y. E. Berman, Z. N. Nysenko, V. A. Ponomarenko, S. S. Medvedev, Vysokomol. Soedin. A, **13**, 1375 (1971); Polym. Sci. SSSR, **13**, 1546 (1971).
294. J. M. Andrews, J. A. Semlyen, Polymer, **12**, 642 (1971).
295. V. S. Etlis, G. A. Razuvaev, A. P. Sineokof, Khim. Geterosicl. Soedin., **2**, 223 (1967).
296. G. A. Razuvaev, V. S. Etlis, L. N. Gribov, Zh. Obshch. Khim., **33**, 1366 (1963).
297. C. Aso, S. Tagami, T. Kunitake, Kobunshi Kagaku, **23**, 63 (1966).
298. A. A. Dugaryan, A. V. Agumyan, Vysokomol. Soedin., **5**, 1755 (1963).
299. R. Raff, J. L. Cook, B. V. Etting, J. Polym. Sci. A, **3**, 3511 (1965).
300. E. Thilo, U. Schuelke, Z. Anorg. Allgem. Chem., **341**, 293 (1965).
301. D. R. Cooper, J. A. Semlyen, Polymer, **13**, 414 (1972).
302. O. B. Salamatina, A. K. Bonetskaya, S. M. Skuratov, B. P. Fabrichnyi, I. F. Shalavina, Y. L. Gol'dfarb, Vysokomol. Soedin., **7**, 485 (1965).
303. K. Weissermel, E. Noelken, Makromol. Chem., **68**, 140 (1963).
304. T. Saegusa, M. Matoi, S. Matsumoto, H. Fujii, Macromolecules, **5**, 233, 236 (1972).
305. T. Kodaira, J. Stehlicek, J. Šebenda, Eur. Polym. J., **6**, 1451 (1970).
306. F. Andruzzi, S. Suradi, G. Pilcher, Makromol. Chem., **183**, 2183 (1982).
307. F. Andruzzi, G. Pilcher, J. M. Hacking, S. Cavell, Makromol. Chem., **181**, 923 (1980).
308. F. Andruzzi, G. Pilcher, Y. Virmani, P. H. Plesch, Makromol. Chem., **178**, 2367 (1977).
309. S. Sosnowski, J. Libiszowski, S. Slomkowski, S. Penczek, Macromol. Chem., Rapid Commun., **5**, 239 (1984).
310. T. M. Frunze, V. A. Kolelnikov, M. P. Ivanov, V. V. Kurashev, L. B. Danilevskaya, S. P. Davtyan, V. V. Korshak, Dokl. Akad. Nauk. SSSR (Chem.), **260**, 1379 (1981).
311. T. G. Kulagina, B. V. Lebedev, E. G. Kiparisova, E. B. Lyudvig, I. G. Barskaya, Polym. Sci. USSR, **24**, 1702 (1982).
312. B. V. Lebedev, T. G. Kulagina, E. B. Lyudvig, T. N. Ovchinnikova; Polym. Sci. USSR, **24**, 1695 (1982).
313. A. A. Yevstropov, B. V. Lebedev, E. Kiparisova, Polym. Sci., USSR, **25**, 1951 (1983).
314. A. A. Yevstropov, B. V. Lebedev, T. G. Kulagina, N. K. Lebedev, Polym. Sci., USSR, **24**, 628 (1982).
315. M. Ueda, M. Takahashi, Y. Imai, C. U. Pitman, Macromolecules, **16**, 1300 (1983).
316. G. O. R. Alberda von Ekenstein, Y. Y. Tan, Eur. Polym. J., **18**, 1061 (1982).
317. K. Takahashi, K. Ohnishi, K. Takase, Tetrahedron Lett., **25**, 73 (1984).
318. T. Malavasic, U. Osredkar, I. Anzur, I. Vizovisk, J. Macromol. Sci. A, Chem., **19**, 987 (1983).
319. T. Malavasik, I. Visorisek, U. Osredkar, I. Anzur, J. Polym. Sci., Polym. Symp., **69**, 73 (1981).
320. A. M. Soledad, L. M. Leon, J. Polym. Sci., Polym. Lett. Ed., **21**, 979 (1983).
321. M. Rodriguez, L. M. Leon, J. Polym. Sci., Polym. Lett. Ed., **21**, 881 (1983).
322. W. K. Busfield, I. J. McEwen, Eur. Polym. J., **9**, 1143 (1973).
323. W. K. Busfield, R. M. Lee, Makromol. Chem., **169**, 199 (1973).
324. W. K. Busfield, R. M. Lee, Makromol. Chem., **176**, 2017 (1975).
325. R. M. Lee, Ph.D. Thesis, University of Dundee, 1972.
326. T. Otsu, B. Yamada, T. Mari, M. Inoue, J. Polym. Sci., Polym. Lett. Ed., **14**, 283 (1976).
327. R. M. Joshi, Polymer, **20**, 394 (1979).
328. K. Hashimoto, H. Sumitomo, S. Ohsawa, J. Polym. Sci., Polym. Chem. Ed., **14**, 1221 (1976).
329. K. Hashimoto, H. Sumitomo, S. Ohsawa, J. Polym. Sci., Polym. Chem. Ed., **13**, 2775 (1975).
330. K. Hashimoto, H. Sumitomo, S. Ohsawa, J. Polym. Sci., Polym. Chem. Ed., **16**, 435 (1978).
331. M. Okada, H. Sumitomo, M. Atsumi, Macromolecules, **17**, 1840 (1984).
332. V. R. Pai Verneker, K. N. Santhanalakshimi, J. Polym. Sci., Polym. Chem. Ed., **22**, 3217 (1984).
333. M. Okada, K. Mita, H. Sumitomo, Makromol. Chem., **177**, 895 (1976).
334. M. Okada, K. Mita, H. Sumitomo, Makromol. Chem., **176**, 859 (1975).
335. M. Okada, K. Mita, H. Sumitomo, Makromol. Chem., **177**, 2055 (1976).
336. B. Yamada, R. W. Campbell, O. Vogl, J. Polym. Sci., Polym. Chem. Ed., **15**, 1123 (1977).
337. D. W. Lipp, O. Vogl, Polym. J., **9**, 499 (1977).
338. D. W. Lipp, O. Vogl, Polymer, **18**, 1051 (1977).
339. M. Okada, T. Hisada, H. Sumitomo, Makromol. Chem., **179**, 959 (1978).

340. M. Okada, K. Yagi, H. Sumitomo, Makromol. Chem., **163**, 225 (1973).
341. R. E. Cunningham, Polymer, **19**, 729 (1978).
342. L. Garrido, J. Guzman, E. Riande, Macromol., **14**, 1132 (1981).
343. Y. Yamashita, J. Mayumi, Y. Kawakami, K. Ito, Macromolecules, **13**, 1075 (1980).
344. P. Kubisa, O. Vogl, Polymer, **21**, 525 (1980).
345. T. Otsu, B. Yamada, S. Sugiyama, S. Mori, J. Polym. Sci., Polym. Chem. Ed., **18**, 2197 (1980).
346. J. Pretula, K. Kaluzynski, S. Penczek, J. Polym. Sci., Polym. Chem. Ed., **22**, 1251 (1984).
347. K. Kaluzynski, S. Penczek, Makromol. Chem., **180**, 2289 (1979).
348. G. Lapienis, S. Penczek, J. Polym. Sci., J. Polym. Sci., Polym. Chem. Ed., **15**, 371 (1977).
349. K. Hatanaka, S. Kanazawa, T. Uryu; J. Polym. Sci., Polym. Chem. Ed., **22**, 1987 (1984).
350. K. Kobayashi, H. Sumitomo, Macromolecules, **16**, 710 (1983).
351. T. Uryu, Y. Sakamoto, K. Hatanaka, K. Matsuzaki, Macromolecules, **17**, 1307 (1984).
352. J. E. Moore, S. H. Schroeter, A.R. Shultz, L. D. Stang, Am. Chem. Soc. Symp. Ser., **25**, 90 (1976).
353. J. C. Favier, J. Leonard, Macromolecules, **11**, 424 (1978).
354. J. C. Bevington, J. A. L. Jemmett, P. F. Onyon, Polymer, **18**, 73 (1977).
355. N. K. Lebedev, B. V. Lebedev, V. S. Khotimskii, S. G. Durgar'yan, Polym. Sci., USSR, **21**, 1131 (1980).
356. G. Lapienis, S. Penczek, Macromolecules, **10**, 1301 (1977).
357. J. Kops, S. Hvilsted, H. Spanggaard, Pure Appl. Chem., **53**, 1777 (1981).
358. B. V. Lebedev, V. Ya Lityagov, Vysokomol. Soedin. A, **19**, 2283 (1977).
359. B. V. Lebedev, I. B. Rabinovich, N. K. Lebedev, N. V. Ushakov, Dokl. Phys. Chem., **239**, 376 (1978).
360. B. V. Lebedev, I. B. Rabinovich, Dokl. Phys. Chem., **235**, 762 (1977).
361. B. V. Lebedev, I. B. Rabinovich, V. Ya Lityagov, Yu. V. Kobshak, V. M. Kuteinikov, Vysokomol. Soedin. A, **18**, 2444 (1976).
362. B. V. Lebedev, I. B. Rabinovich, N. K. Lebedev, V. G. Tsvetkov, S. G. Durgar'yan, V. S. Khotimskii, Vysokomol. Soedin. A, **19**, 513 (1977).
363. B. V. Lebedev, N. K. Lebedev, Ye. G. Kiparisova, K. A. Yesayan, L. K. Golova, Yu. B. Amerik, Vysokomol. Soedin., **26**, 909 (1984).
364. B. V. Lebedev, A. A. Yevstropov, Ye. G. Kiparisova, V. I. Belov, Vysokomol. Soedin. A, **20**, 29 (1978).
365. B. V. Lebedev, N. N. Mukhina, T. G. Kulagina, Vysokomol. Soedin. A, **20**, 1297 (1978).
366. N. K. Lebedev, B. V. Lebedev, V. S. Khotimiskii, S. G. Durgar'yan, Vysokomol. Soedin. A, **21**, 1031 (1979).
367. B. V. Lebedev, N. K. Lebedev, V. S. Khotimskii, S. G. Durgar'yan, N. S. Nametkin, Dokl. Phys. Chem., **259**, 666 (1981).
368. B. V. Lebedev, A. A. Evstropov, Dokl. Phys. Chem., **264**, 334 (1982).
369. M. A. Markevich, L. I. Pavlinov, Yu. A. Lebedev, Dokl. Phys. Chem., **262**, 64 (1982).
370. Y. Yuki, H. Kunisada, T. Seo, T. Kakurai, Kobunshu Ronbunshu, **40**, 733 (1983).
371. I. M. Barkalov, D. P. Kirukhin, G. A. Kichigina, M. P. Muidinov, Eur. Polym. J., **20**, 1013 (1984).
372. G. M. Patel, R. R. Chance, E. A. Turi, Y. P. Khanna, J. Am. Chem. Soc., **100**, 6644 (1978).
373. A. F. Garito, C. J. Horner, P. S. Kalyanaraman, K. N. Desai, Makromol. Chem., **181**, 1605 (1980).
374. G. N. Patel, Macromolecules, **14**, 1170 (1981).
375. G. N. Patel A. F. Preziosi, H. R. Bhattacharjee, J. Polym. Sci., Polym. Symp., **71**, 247 (1984).
376. T. Baran, A. Duda, S. Penczek, Makromol. Chem., **185**, 2337 (1984).
377. T. Baran, A. Duda, S. Penczek, J. Polym. Sci., Polym. Chem. Ed., **22**, 1085 (1984).
378. Y. Yuki, T. Ouchi, Kobunshu Ronbunshu, **33**, 303 (1976).
379. Y. Yuki, T. Ouchi, Kobunshu Ronbunshu, **32**, 97 (1975).
380. R. Steudal, S. Passlack-Stephan, G. Holdt, Z. Anorg. Allg. Chem., **517**, 7 (1984).
381. H. Sawada, "Thermodynamics of Polymerization", Marcel Dekker, New York, 1976.
382. H. Uno, T. Endo, M. Okawara, J. Polym. Sci., Polym. Chem. Ed., **23**, 63 (1985).
383. T. Fries, J. Belohlávková, J. Roda, J. Králicek, Polym. Bull., **12**, 87 (1984).
384. M. Morcellet, Thermochim. Acta, **80**, 361 (1984).
385. B. V. Lebedev, L. Ya. Tsvetkova, V. G. Vasil'ev, V. I. Kleiner, B. A. Krentsel, Vysokomol. Soedin. A, **35**, 1941 (1993).
386. B. V. Lebedev, N. N. Smirnova, E. G. Kiparisova, D. G. Faminskii, V. G. Vasil'ev, V. F. Vasilenko, Vysokomol. Soedin. A, **35**, 1951 (1993).
387. B. V. Lebedev, N. N. Smirnova, V. G. Vasil'ev, E. G. Kiparisova, V. I. Kleiner, Vysokomol. Soedin. A, **36**, 1413 (1994).
388. N. N. Smirnova, B. V. Lebedev, Vysokomol. Soedin. A, **32**, 2536 (1990).
389. P. A. Patton, T. J. McCarthy, Polym. Prep., **26** (1), 160 (1985).
390. T. A. Bykova, Ye. G. Kiparisova, B. V., Lebedev, K. A. Mager, Yu. G. Gololobov, Vysokomol. Soedin. A, **33**, 614 (1991).
391. E. G. Kiparisova, T. A. Bykova, B. V. Lebedev, T. I. Guseva, K. A. Mager, Yu. G. Gololobov, Vysokomol. Soedin. A, **35**, 615 (1993).
392. T. A. Bykova, E. G. Kiparisova, B. V. Lebedev, N. G. Senchenya, K. A. Mager, Yu. G. Gololobov, Vysokomol. Soedin. A, **35**, 13 (1993).
393. T. A. Bykova, Ye. G. Kiparisova, B. V. Lebedev, T. I. Guseva, K. A. Mager, Yu. G. Gololobov, Vysokomol. Soedin. A, **33**, 2602 (1991).
394. B. V. Lebedev, N. K. Lebedev, Ye. G. Kiparosiva, K. A. Yesayan, L. G. Golova, Yu. B. Amerik, Polym. Sci. USSR, **26**, 1013 (1984).
395. J. Penelle, J. Collot, G. Rufflard, J. Polym. Sci. A: Polym. Chem., **31**, 2407 (1993).

396. B. V. Lebedev, N. N. Smirnova, E. G. Kiparisova, V. G. Vasil'ev, N. Yu. Surkova, V. I. Kleiner, B. A. Krentsel, Vysokomol. Soedin. A, **35**, 767 (1993).
397. B. V. Lebedev, T. G. Kulagina, N. N. Smirnova, T. A. Bykova, V. G. Visil'ev, D. F. Grishin, Vysokomol. Soedin. A, **37**, 1896 (1995).
398. R. Binet, J. Leonard, Polymer, **14**, 355 (1973).
399. S. Iwatsuki, M. Kubo, H. Iwayama, Macromolecules, **26**, 7309 (1993).
400. R. E. Cunningham, H. A. Colvin, Polymer, **33**, 5073 (1992).
401. M. Ueda, T. Ito, H. Ito, Macromolecules, **23**, 2895 (1990).
402. V. Gebauer, K. Burghardt, S. Jaehrig, K. Gehrke, Plaste Kautsch., **38**, 372 (1991).
403. H. Kunisada, Y. Yuki, S. Kondo, Macromolecules, **24**, 4733 (1991).
404. H. Kunisada, Y. Yuki, S. Kondo, K. Adachi, N. Takahashi, Polymer, **33**, 3512 (1992).
405. H. Kunisada, Y. Yuki, Kobunshi Ronbunshu, **43**, 895 (1986).
406. Y. Yuki, H. Kunisada, Nippon Kagaku Kaishi, **2**, 242 (1985).
407. A. Aboudalle, A. Soum, M. Fontanille, T. E. Hogen-Esch, Polym. Prepr., **27**(1), 143 (1986).
408. T. Kase, M. Imahori, T. Kazama, Y. Isono, T. Fujimoto, Macromolecules, **24**, 1714 (1991).
409. Y. Nagasaki, N. Yamazaki, K. Takeda, N. Kato, M. Kato, Macromolecules, **27**, 3702 (1994).
410. T. Itoh, T. Ario, M. Kido, S. Iwatsuki, Macromolecules, **27**, 7035 (1994).
411. S. Iwatsuki, T. Itoh, N. Kusaka, H. Maeno, Macromolecules, **25**, 6395 (1992).
412. S. Iwatsuki, T. Itoh, T. Higuchi, K. Enomoto, Macromolecules, **21**, 1571 (1988).
413. S. Iwatsuki, T. Itoh, H. Yabunouchi, M. Kubo, Macromolecules, **23**, 3450 (1990).
414. N. K. Lebedev, B. V. Lebedev, N. N. Smirnova, V. G. Vasil'ev, N. M. Kozyreva, S. B. Skubina, V. V. Korshak, Vysokomol. Soedin. B, **28**, 357 (1986).
415. B. V. Lebedev, N. Smirnova, Y. Kiparisova, K. Makovetsky, Makromol. Chem., **193**, 1399 (1992).
416. B. Lebedev, N. Smirnova, N. Novosyolova, K. Makovetsky, I. Ostrovskaya, Macromol. Chem. Phys., **195**, 1807 (1994).
417. B. V. Lebedev, V. G. Vasil'ev, Ye. G. Kiparisiova, Vysokomol. Soedin. A, **32**, 1070 (1990).
418. V. G. Vasil'ev, B. V. Lebedev, Vysokomol. Soedin. A, **33**, 818 (1991).
419. V. G. Vasil'ev, B. V. Lebedev, N. V. Novoselova, Vysokomol. Soedin. A, **35**, 621 (1993).
420. B. V. Lebedev, V. G. Vasil'ev, N. Novosyolova, Makromol. Chem., **194**, 739 (1993).
421. B. V. Lebedev, V. G. Vasil'ev, N. Novosyolova, J. Therm. Anal., **38**, 1299 (1992).
422. F. Andruzzi, S. Li, G. Pilcher, F. Heatley, Makromol. Chem., **188**, 2643 (1987).
423. H. Krüger, J. Rübner, W. Wittke, G. Gnauk, Acta Polym., **37**, 601 (1986).
424. J. C. W. Chien, Y. G. Cheun, P. Lillya, Macromolecules, **21**, 870 (1988).
425. B. Xu, C. P. Lillya, J. C. W. Chien, Macromolecules, **20**, 1445 (1987).
426. B. V. Lebedev, T. G. Kulagina, V. I. Telnoy, V. G. Vasil'ev, Macromol. Chem. Phys., **196**, 3487 (1995).
427. B. V. Lebedev, T. A. Bykova, E. G. Kiparisova, B. G. Belen'kaya, V. N. Filatova, Vysokomol, Soedin. A, **37**, 187 (1995).
428. I. Rangel, M. Richard, A. Ricaard, Macromol. Chem. Phys., **195**, 3095 (1994).
429. A. Duda, S. Penczek, Macromolecules, **23**, 1636 (1990).
430. S. Chikaoka, T. Takata, T. Endo, Macromolecules, **24**, 6557 (1991).
431. B. V. Lebedev, T. G. Kulagina, V. S. Svistunov, V. S. Papkov, A. A. Zhdanov, Polym. Sci. USSR, **26**, 2773 (1984).
432. I. C. Eromosele, D. C. Pepper, J. Polym. Sci., Part A: Polym. Chem., **25**, 3499 (1987).
433. K. Hashimoto, H. Sumitomo, H. Shinoda, A. Washio, J. Polym. Sci., Part C, Polymer Letters, **27**, 307 (1989).
434. S. Cavallero, A. Colligiani, G. Cum, J. Therm. Anal., **38**, 2649 (1992).
435. S. Iwatsuki, T. Itoh, K. Enomoto, T. Itoh, E. Yamamoto, T. Myamoto, Macromolecules, **22**, 1014 (1989).
436. R. Szymanski, Makromol. Chem., **192**, 2961 (1991).
437. T. G. Kulagina, B. V. Lebedev, Fiz, -Chim. Osn. Sint. Perebab. Polim., **99**, (1983).
438. S. Iwatsuki, T. Itoh, H. Ito, M. Kubo, Macromolecules, **23**, 2423 (1990).
439. T. Fries, J. Bělohlávková, O. Nováková, J. Roda, J. Králíček, Makromol. Chem., **188**, 239 (1987).
440. S. Iwatsuki, M. Kubo, M. Wakita, Y. Matsui, H. Hanoh, Macromolecules, **24**, 5009 (1991).
441. G. Deák, T. Pernecker, J. P. Kennedy, Polym. Bull., **33**, 259 (1994).
442. B. Yamada, T. Tanaka, T. Otsu, Eur. Polym. J., **25**, 117 (1989).
443. E. L. Madruza, J. S. Roman, M. A. Lavia, M. C. Fernandez-Monreal, Eur. Polym. J., **22**, 357 (1986).
444. T. Sato, K. Morino, H. Tanaka, T. Ota, Eur. Polym. J., **26**, 1279 (1990).
445. N. V. Karyakin, N. G. Shvetsova, G. P. Kamelova, A. L. Rusanov, V. V. Korshak, L. Kh. Plieva, Vysokomol. Soedin. A, **26**, 36 (1984).
446. R. Szymanski, Makromol. Chem., **192**, 2943 (1991).

Activation Energies of Propagation and Termination in Free Radical Polymerization

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A. INTRODUCTION

The conversion of a free radical homopolymerization as a function of time is usually expressed by the following equation:

$$\frac{1}{1-x}\frac{\partial x}{\partial t} = k_p\left(\frac{2fk_dI}{k_t}\right)^{0.5}$$

where x is the conversion, t the time, k_p, k_t, and k_d the propagation, termination and initiator decomposition rate constants, respectively, f the initiator efficiency and I the initiator concentration.

It is generally assumed that the temperature dependence of free radical polymerization rate constants can be expressed using the well-known Arrhenius equation:

$$k = A_0 \exp\left(\frac{-E}{RT}\right)$$

where A_0 is the frequency factor, R the universal gas constant, T the absolute temperature, and E the activation of the reaction in question. Measurement of k_p and k_t at different temperatures allows one to determine values of the activation energies of the same two reactions, E_p and E_t, respectively.

If we accept this representation, then the overall activation energy of a homopolymerization can be written as

$$E_{OV} = E_p - \frac{1}{2}E_t + \frac{1}{2}E_d$$

The value of the activation energy for the initiator decomposition reaction, E_d, can often be estimated separately. When data are not available for E_p and E_t as separate quantities, the term $\left(E_p - \frac{1}{2}E_t\right)$ can be used to model the polymerization reaction.

In the following tables, all energies are reported in terms of kJ/mol, and any appropriate remark(s) associated with the data are presented.

B. TABLES

TABLE 1. DIENES

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Butadiene	10.9			In gas phase	13
	38.9			In emulsion	14
	24.3 & 20.9			In gas phase	16
	20.5			In gas phase	21
	35.7	5.9		Solution polymerization in CCl_4; temperature range: 303–333 K	90
	38.9			For emulsion polymerization, supposing $\bar{n} = 0.5$	91
1,3-Hexadiene	8.42			Thermal and photoinitiation; temperature range: 278–333 K; k_p measured by ESR	92
Chloroprene	40.6			In emulsion	35
Dimethyl butadiene	37.7			In emulsion	45
Isoprene	46.0			In emulsion	15

TABLE 2. OLEFINS

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Acetylene	21.4			In gas phase	21
Ethylene	34.3			In gas phase	21
	18.4	1.3	17.8	Calculated using results of Ref. 24	35
	23.9			In gas phase	44
	46.4	33.5	29.7	High pressure in gas phase (up to 300 NPa); initiated by oxygen; temperature range: 60–250 °C; conversion found by gravimetry	73
			31.0	High pressure polymerization (up to 300 MPa); initiated by DTBPO; temperature range: 130–165 °C	82
	36.8	6.22			93
	29.7	10.0			94
	40.0	15.2		Activation volumes (cal/mol/atm): propagation: − 19.7, termination: 13.0	95
	25.1	− 1.59			96
	20.8	3.13			97
	29.7	10.0			98
	32.5	4.19			99
	36.9	1.05			100
	31.4			Estimation for simulation of emulsion copolymerization with vinyl acetate	101
		14.2			102
			31.6	Activation volume $= -26.2E \times 10^{-6}$ m^3/gmol	103
	30.3	4.2		Activation volume for propagation = − 0.5 cal/mol/atm	104
	17.6 ± 0.31	15.3 ± 0.26		Activation volumes (cal/mol/atm): propagation: − 16.8 ± 0.1; termination: 9.2.	105
	30.2	6.44		Optimized from values of industrial runs	106
Propene	23.4			In gas phase	21
Isoprene	41.0			In emulsion	15

TABLE 3. ACRYLIC DERIVATIVES

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Acrylamide					
A	11.7 ± 1.2	11.7 ± 0.8	5.8 ± 1.6	Solution polymerization in H_2O/DMSO mixture with potassium persulfate; initiators: H_2O : DMSO : (A) 100 : 0	68
B	16.3 ± 0.8	10.9 ± 0.4	10.9 ± 1.3	(B) 88 : 12	
C	18.8 ± 1.9	10.9 ± 0.4	13.4 ± 0.3	(C) 65 : 35	
D	20.5 ± 0.8	7.9 ± 0.4	16.9 ± 1.0	(D) 38 : 62	
E	25.1 ± 0.9	0 ± 0.8	25.1 ± 1.3	(E) 0 : 100	
	22 ± 0.2			pH 1	
	20 ± 0.2			pH 4	
	11.5	6.2		Compilation of data, validated for inverse suspension polymerization	111
	57.8	10.46		Compilation from various sources, validated for solution polymerization	112
N,N-methylene-bis-acrylamide			9.3	Aqueous polymerization with temperature range: 293–313 K	113
Acrylic acid					
			23.4	Initiation with APS; temperature range: 309.5–324.5 K; solution polymerization at pH_0 4.38	107
			21.7	Precipitation polymerization; temperature range: 313–333 K	108
			14.1		109
				By PLP; temperature range: 288–328 K	110
–, sodium salt	(1) 67.5 (2) 72.1			Thermally induced solid state polymerization; two methods to find E_p: (1) ESR, (2) DSC; temperature range: 60–160 °C	87
–, chromium salt			33.5	Polymerized with styrene–arsenic sulfide complex initiator in DMF	114
–, zinc salt			13.0	Initiated with an As2S3–styrene complex in DMSO at 363 K	115
–, butyl ester	8.8	0			8
	52.3	73.7		At 20% conversion	26
	15.5			From PLP–GPC; temperature range: (208–266 K); bulk polymerization	116
		38.1			109
	14 ± 6			From PLP; temperature range: (298–353 K); conversion range: (10–80%); bulk polymerization; $P = 1$ bar.	117
			$E_p - E_T = 0$	From PLP; conversion = 30% in bulk polymerization	118
	9.6 ± 0.56			By PLP–GPC; bulk polymerization with photoinitiators: benzoin and 2,2-dimethoxy-2-phenylacetophenone; temperature range: 298–303 K	119
–, ethyl ester	41	12.3			109
–, methyl ester	29.7	22.2			
	19.7	~ 0			
	31.8 ± 2.5	20.1 ± 0.8		Gamma ray induced solid state polymerization; temperature range: 243–283 K	
			40.7		109
–, 1,3-hydroxyneopentyl ester			31.8	Measured by dilatometry in both benzene and 1,4 dioxane ($k_p/k_t^{0.5}$ higher in dioxane)	120
–, methyl α-acetoxy ester	35.5	46.8		By ESR; temperature range: 303–333 K; solution polymerization in 1,1,2-trichloroethane; initiator: 2,2′-azobis(4-methoxy-2,4-dimethylvaleronitrile) (V-70)	121
–, methyl α-butoxy ester	30.5	25.1		By ESR; assuming $E_d = 109.5$; solution polymerization in 1,1,2-trichloroethane; initiator: 2,2′-azobis(4-methoxy-2, 4-dimethylvaleronitrile (V-70)	121
Acrylonitrile	17.2	22.6	5.9	In water solvent	25
	16.2	15.5	8.5	In DMF solvent	32

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TABLE 3. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Acrylonitrile (*cont'd*)	26.8	~0	6.8	In DMF solvent	37
			21.4		109
	29.6	7.0		Estimated from data on solution copolymerization with styrene as azeotrope	122
	42.7				123
			23.9	Polymerization in NaSCN solution with $Mn(C_5H_7O_2)_2$ as free radical initiator; temperature range: 295–313 K; value estimated from overall activation energy with $E_d = 58.6$ kJ/mol	124
D-Fluoroacrylic acid					
–, methyl ester			9.6	Solution: acrylonitrile; initiator: AIBN; (assume $E_d = 128$ kJ/mol)	71

TABLE 4. METHACRYLIC DERIVATIVES

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Methacrylamide	15.5	16.7	7.1	Solvent: water	34
	22.1			By PLP; temperature range: (288–328 K); pH 1	110
Methacrylic acid	17.7			By PLP–GPC; 33% monomer in methanol; temperature range: 293–333 K	125
	17.4				109
–, butyl ester	18.4	4.6	16.1	Gamma ray initiation	27
	20.4	12.7	14.1	Kinetics from calorimetry	55
	20.9	8.8	16.7	Temperature range: 35–27°C	Quoted in 58
–, butyl ester	21.8 ± 0.27			By PLP	131
	22.9 ± 1.1			By PLP–GPC; $P = 1000$ bars; temperature range: −10 to 60°C	141
	23.3			Temperature range: 25–55°C; by PLP bulk polymerization; photoinitiator: benzoin	135
	20.4				126
	20.5 ± 0.24			By PLP–MWD; bulk polymerization; pressure: 1 bar; photoinitiator: benzoin	127
–, *i*-butyl ester	22.5 ± 0.42				131
	21.5 ± 0.35			By PLP–MWD; bulk polymerization; pressure: 1 bar; photoinitiator: benzoin	127
–, *t*-butyl ester	27.7 ± 2.5			Temperature range: 9–66°C; by PLP with benzoinethyl ether as free rapid initiator	128
–, chloromethyl ester			5.5	Ampoule and emulsion polymerization; AIBN initiator	78
–, cyanomethyl ester			~8.4	Solution: acetonitrile AIBN initiator calculated, assuming E_d of AIBN = 128 kJ/mol from overall activation energy; temperature range: 50–70 °C	87
–, dodecyl ester (lauryl ester)		15.5			Quoted in 58
			29.8	Bulk polymerization with BPO initiator; temperature range 60–80 °C; maximum conversion 1%	66
	21.5 ± 0.34			By PLP	131
	13.7				135
	16.2			By PLP–MWD; bulk polymerization; photoinitiator: benzoin	135
–, ethyl ester	23.2	8.9	18.0	Gamma ray initiation; temperature range: −30 to 55 °C; kinetics from calorimetry	52
	21.3	7.1	17.7		Quoted in 58
			16.5	Data from bulk and solution (50% DMF); AIBN initiator; kinetics from GLC and gravimetry	63

TABLE 4. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Methacrylic acid, ethyl ester (*cont'd*)	27.0	−6.6	30.4		Quoted in 79
	22.8 ± 0.35			By PLP	131
	20.46			By PLP; temperature range: 298–328 °C; bulk polymerization; photoinitiator: benzoin	135
	27.0			Evaluated for emulsion polymerization	129
	22.9 ± 0.46			By PLP–GPC; bulk polymerization, pressure: 1 bar; photoinitiator: benzoin	127
			22.2	Bulk polymerization; initiator: AIBN	130
			16.3	Solution polymerization in benzene; initiator: AIBN	130
–, hexadecyl ester	20.1	18.8	10.5		Quoted in 58
–, hexyl ester	18.0		~8.4		Quoted in 58
–, methyl ester	26.4	11.9	20.5		4
	18.4	0	18.4		7
	24.3	2.1	23.2		28
	20.9	0–16.7	20.9–12.6	Termination by combination or disproportionation	30
	18.8				47
	22.4			In emulsion	31
	22.8	13.2	11.2	Gamma ray induced; temperature range: −25 to 55 °C; used a combination of DCS and inhibition with DPPH to find kinetic parameters	48
				Polymerization in benzene; DMDM initiator; temperature range: 50–80 °C	49
	22.6 ± 2.5	Tends to 0	22.6 ± 2.5	Bulk polymerization; DBPO initiator; maximum conversion 25%; ESR used to find kinetic parameters	53
			13.0	Isothermal bulk; polymerization with AIBN initiator; DCS used to follow reaction; value found assuming E_d AIBN = 128 kJ/mol	56
	19.7	3.3	18.0		Quoted in 58
	18.2 ± 0.2	2.9 ± 0.0	16.7 ± 0.2	Bulk polymerization; AIBN initiator temperature range 15–30 °C; reaction followed by spatially intermittent polymerization; DP 10000–15000	64
			23.4	Bulk polymerization; AIBN initiator; 1-butanethiol used as CTA; value assumes E_d AIBN = 128 kJ/mol	65
	20.4	~10.5	15.1	Solution: benzene; AIBN initiator; kinetic data from molecular weight data; maximum conversion 5%	68
	29.1			Emulsion data; particle sizes from turbidity; potassium persulfate initiator	72
	20.9 ± 3.3	1.6 ± 0.4	20.1 ± 3.6	Solution: benzene; AIBN initiator; kinetic parameters from intermittent illumination	77
	31.3			Bulk polymerization	85
	21.3 ± 0.25			By PLP	131
	18.2				132
	26.4	11.7			133
			3.55		109
	22.6	0 ± 2.5		Temperature range: 293–353 K	156
	18.9	2.93		Suspension polymerization	134
	15.7			Spatially intermittent polymerization/ emulsion polymerization	164
	18.0				135
	18.2	2.9			136
	24.2	26.9			137
	18.3	2.94		Solution polymerization in toluene	138
	31.3	21.1			139
		36		1 Bar; temperature range: 323–353 K, low conversion	140
		−14 ± 8		100 bars	
		−16 ± 15		1000 bars	
	22.2 ± 0.3				141

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TABLE 4. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Methcrylic acid, methyl ester (*cont'd*)	22.4			PLP plus GPC for bulk polymerization; temperature range: 272–363 K	142
	22.2				143
		2.9			144
	23.9			By PLP–GPC; bulk polymerization; pressure: 1 bar; photoinitiator: 2-hydroxy-2,2-dimethylacetophenone; temperature range: 272–363 K	145
	22.2			By PLP–GPC; bulk polymerization; pressure: 1 bar; photoinitiator: benzoin; temperature range: 272–363 K	146
–, nonyl ester	17.2	13.0	10.9		Quoted in 58
–, octyl ester	15.9	12.1	9.7		Quoted in 58
–, phenyl ester			9.2		147
–, *o*-methylphenyl ester			9.6		147
–, *o*-ethylphenyl ester			12.2		147
–, *o*-iso-propylphenyl ester			14.4		147
–, *p*-methylphenyl ester			10.6		147
–, *p*-ethylphenyl ester			9.3		147
–, *p*-iso-propylphenyl ester			13.2		147
–, *p-tert*-butylphenyl ester			9.5		147
–, ethyl α-Benzoyloxymethyl ester			11.5	By ESR, assuming $E_d = 129$; solution polymerization in benzene; initiator: AIBN	148
–, ethyl α-acetacetoxymethyl ester			24.55		149
–, 2-(3-chloro-3-methylcyclobutyl)-2-hydroxyethyl ester			30.68	Polymerization in dioxane using Bz_2O_2 as initiator	150
Methacryloyl fluoride			10.4	Solution: MEK; AIBN initiator; calculated using E_d AIBN = 128 kJ/mol; temperature range: 50–70 °C	81
Methacrylonitrile	48.1	20.9	37.7		20
	29.7 ± 1.5			By PLP; solution polymerization in benzylactone; temperature range: 283–333 K	151
–, *n*-Phenyl-α-methylene-β-lactam			22.8	Solution polymerization in *n*-methyl-2-pyrrolidone with AIBN (calculated from E_{OV} assuming E_d of AIBN = 129 kJ/mol)	152

TABLE 5. VINYL HALOGENS

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Tetrafluoroethylene	17.4	13.6	10.6		46
	5.4	0.8	5.0		43
Vinyl bromide	30.0	16.3	21.8		42
Vinyl chloride	16.0	17.6	7.4		17
			12	Validated for model of suspension polymerization	153
	27.6	18.2		For suspension polymerization	154
Vinylidene chloride	104	167	20.5	Solvent: hexane	9
	37.7 ± 5.9	20.9 ± 10.5	27.2 ± 11.1	Solution: *N*-methylpyrrolidone; AIBN used as photoinitiator; most runs $\leq 1\%$ conversion	67

TABLE 6. VINYL ETHERS AND VINYL ESTERS

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Butyl vinyl ether			46.4	Bulk polymerization; initiator DTBPO; assume E_d = 51.5 kJ/mol (Ref. 82) at given conditions; GPC data used to find kinetic parameters	84
Vinyl acetate	18.0	0	18.0		1
	30.6	21.9	19.6		5
	13	0	13		6
	18	<4	16–18		18
	18	0	0		22
	39 ± 3				38
	19.6 ± 2.9	0	19.6 ± 2.9	Bulk polymerization; DBPO and cyclohexyloxyperoxide initiators; ESR used to find kinetics	53
			~26.4	Bulk polymerization; DBPO and cyclohexyl-oxyperoxide initiators; temperature range: 55–70 °C; DSC used for kinetics; maximum conversion 20%	57
	20.5 ± 0.39				131
	26.4	11.7			133
	23.5			Validated for emulsion copolymerization with ethylene; temperature range: 293–323 K; pressure range: 0–34 bar	102
		14.2			155
			28.1		109
	19.7 ± 2.9	0		Temperature range: 293–353 K	156
	23.7			For emulsion polymerization	157
	23.6			Emulsion polymerization	158
	24.2	4.0			159
	26.4	13.4			160
	35.2	14.2		Rotating-sector method; bulk polymerization; photoinitiator: benzoin	155
	27.8			By PLP–MWD; bulk polymerization; photoinitiator: benzoin	161

TABLE 7. STYRENE AND DERIVATIVES

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Styrene	49.0			In emulsion	2
	27.2 ± 4.2	11.7 ± 4.2	21.3 ± 6.3		3
	26.0	8.0	22.0		10
	32.5	9.9	27.5		11
	31.0 & 35.0			In emulsions	15
	54.0 & 59.0			In emulsions	19
	25.0	2.0	24.0		29
	73.6				36
	29.0				42
			25.1	Gamma ray induced; bulk liquid phase polymerization; maximum conversion 10%; temperature range: 19–74 °C	50
	37.7			Low conversion; bulk polymerization; ESR spectroscopy used for kinetics	51
	29.5 ± 0.1	9.2 ± 0.5	24.9 ± 0.4	Bulk photopolymerization; kinetics from spatially intermittent polymerization technique; temperature range: 15–36 °C DP 2400–2700	64
			34.1	AIBN initiator; assume E_d = 128 kJ/mol; bulk polymerization 1-butanethiol as CTA; maximum conversion 0.4%	65
	29.0 ± 3.0			In emulsion; reaction followed by dilatometry	70

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Table 7. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
	32.2			Emulsion data; assume $N = 0.5$, sodium persulfate and sodium bisulfite initiators	76
	32.5	14.5	25.2	Bulk polymerization; AIBN initiator; temperature range: 45–100 °C	79
	17.7			In emulsion; sodium persulfate initiator	89
	29.6				132
			25.2		109
	37.8 ± 1.7				156
	39.8	15.6		By ESR; temperature range: 273–403 K	162
	36.1	18.7		Suspension polymerization; conversion less than 0.8; temperature range: 283–353 K	163
	29.5			Spatially intermittent polymerization/emulsion polymerization	164
	44.4	36.4		High conversion suspension polymerization	165
	29			Emulsion polymerization; temperature range: 318–338 K	166
	37.5 ± 1.6	10.91		ESR; temperature range: 323–363 K	167
	30.8			By PLP; temperature range: 298–328 K	168
				By PLP; temperature range: 293–353	169
	30.0			For d_p (degree of polymerization) > 4	
	29.0			For all chain lengths together	
			32.3	Bulk and diluted systems initiated by (*p*-acetylbenzylidene)triphenylarsonium	170
				Measurements by rotating sector method; initiators: AIBN (30–50 °C), BPO (70–90 °C)	171
	31.5	9.5		Styrene	
	31.5	14.2		Perdeuterated styrene	
	31.5			PLP; temperature range: 303–363 K; in bulk	172
	18			Measurments by ESR; photo and thermal initiation; temperature range: 293–363 °C	173
	32.5			PLP; temperature range: 261–366 K; bulk polymerization with photoinitiators: benzoin and 2,2-dimethoxy-2-phenylacetophenone (DMPA)	174
p-Bromostyrene	32.0				39
o-Methylstyrene	58.2			In emulsion	36
p-Methylstyrene	32.0			In emulsion	36
			25.4		109
	29.6	30.5			175
	29.6	8.27			176
α-Methyl styrene	54.3	8.4			177
Divinylbenzene	35.5	29.8		Searched by EVM	178

TABLE 8. VINYL HETEROAROMATICS

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
3-Methylene-5,5-dimethyl-2-pyrrolidone			8.2	Solution polymerization in DMSO with AIBN (calculated from E_{OV} assuming E_d of AIBN = 129 kJ/mol)	179
Methylene-*N*-methylpyrrolidone			20.1	In bulk and in solution with acetonitrile; AIBN initiator (assume $E_d = 128$ kJ/mol)	86
N-Vinylcarbazole	28.9				42
2-Vinylpyridine	33.5	20.9	23.0		40
N-Vinylpyrrolidone	29.7	6.7	26.4	Low conversion	41
(A)	68.0	19.0	58.5	In various solvents: (A) Water	61
(B)	40.0 ± 2.0	13.0	33.5 ± 2.0	(B) L-Propanol	
(C)	31.6 ± 3.0	9.0	27.5 ± 3.0	(C) Methanol	
(D)	23.0	8.0	19.0	(D) Ethyl acetate; rates followed by dilatometry	

Table 8. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
			4.8	Template polymerization in DMF; AIBN initiator; DSC used to follow reaction	74
			(1) 12.8	Solution polymerization	75
			(2) 10.8	In DMF; assume E_d of AIBN in DMF = 142 kJ/mol; reaction followed by DSC in 2 modes: (1) dynamic (2) isothermal	
			65.8 ± 0.1	Aqueous phase polymerization with potassium peroxydiphosphate Ag^+; temperature range: 293–323 K	180
	29.7	6.7			181
4-Methylene-4H-1,3-benzodioxin-2-one			11.2	Solution polymerization in *n*-methyl-2-pyrrolidone with AIBN (calculated from E_{OV} assuming E_d of AIBN = 129 kJ/mol)	182

TABLE 9. MISCELLANEOUS COMPOUNDS

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
Acetaldehyde	46.0 ± 6.3			Radiation induced in solid phase; calorimetry used to follow reaction; temperature range: 130–150 °K	54
Diethyl fumarate			25.0 ± 2.0	Bulk polymerization; AIBN initiator; (assume E_d = 128 kJ/mol) dilatometry used to follow the reaction; temperature range: 60–80 °C	59
			26.5	Assuming $E_d = 129$; by dilatometry; bulk polymerization; initiator: AIBN	183
Diisopropyl fumarate	16.8	16.3			184
Formaldehyde	8.4				Quoted in 54
D-Methylene-γ-butyrolactone			12.1	Solution: DMF; AIBN initiator (assume $E_d = 128$ kJ/mol); $E_p - E_t \approx 12.1$	80
Vinylferrocene	20.0	7.9		Note nonclassical kinetics; overall activation energy does not equal $E_d + E_p - E_t/2$; can approximate E_t = that of styrene	62
N-Vinyl-methylacetamide			26.9	Solution: methanol; AIBN initiator; maximum conversion 10%; temperature range: 25–75 °C	83
α-Ethyl, β-hexafluoroisopropyl itaconate	18.0	57.4		Polymerization in benzene catalyzed by dimethyl 2,2-azobisisobutyrate	185
Diethyl itaconate				Polymerization in benzene catalyzed by dimethyl 2,2-azobisisobutyrate	186
	29.0	30.0		(no $SnCl_4$)	
	21.0	24.0		(0.342 mol/l $SnCl_4$)	
Diisopropyl itaconate	28.5	66.1		Polymerization in benzene catalysed by dimethyl 2,2-azobisisobutyrate	187
Di-*n*-Butyl itaconate (DBI)			0.5	Assuming $E_d = 129$; by ESR; solution polymerization in benzene; initiator: AIBN	188
			4.35	Assuming $E_d = 129.1$; By ESR; solution polymerization in benzene; initiator: 2,2′-azobis(isobutyrate): MAIB	188
1,1-bis[(1-adamantyloxy)carbonyl]-2-vinyl-cyclopropane			87.1	Ring-opening solution homopolymerization in benzene; temperature range: 323–343 K; with AIBN (assume $E_{D,AIBN} = 123.5$ kJ/mol)	189
Diethylene glycol bis-allyl carbonate (DEAC)	53.6			Rotating sector method; Initiator: BPO	190
N-Acryloyl-2,2-dimethyl-5®-phenyl-1, 3-oxazolidine (ADPO)	29	21		Assuming $E_d = 124.3$; temperature range: 323–343 °C; solution polymerization in benzene; initiator: 2,2′-azobis(isobutyrate): MAIB	191

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Table 9. *cont'd*

Monomer	E_p	E_t	$E_p - \frac{1}{2}E_t$	Remarks	Refs.
N-Cyclohexylmaleimide (CHMIm)	40.9	20.1		By ESR; temperature range: 313–333 K; solution polymerization in benzene; initiator: AIBN	192
N-Acryloyl-*p*-aminobenzoic acid			48.3	Assuming $E_d = 129$; solution polymerization in dioxane; initiator: AIBN	193

C. REFERENCES

1. G. M. Burnett, H. W. Burnett, Nature, **156**, 661 (1945).
2. W. V. Smith, J. Am. Chem. Soc., **70**, 3695 (1948).
3. C. H. Bamford and M. J. S. Dewar, Proc. Roy. Soc. (London) A, **192**, 369 (1948).
4. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **71**, 497 (1949).
5. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **71**, 2610 (1949).
6. G. Dixon-Lewis, Proc. Roy. Soc. (London) A, **198**, 501 (1949).
7. M. H. Mackay, H. W. Melville, Trans. Faraday Soc., **45**, 323 (1949).
8. H. W. Melville, A. F. Bickel, Trans. Faraday Soc., **45**, 1049 (1949).
9. J. D. Burnett, H. W. Melville, Trans. Faraday Soc., **46**, 976 (1950).
10. G. M. Burnett, Trans. Faraday Soc., **47**, 772 (1950).
11. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **73**, 1700 (1951).
12. M. S. Matheson, E. E. Auer, E. B. Bevilacqua, E. J. Hart, J. Am. Chem. Soc., **73**, 5395 (1951).
13. D. H. Volman, J. Chem. Phys., **19**, 68 (1951).
14. M. Morton, P. P. Salatiello, H. Landfield, J. Polym. Sci., **8**, 215 (1952).
15. M. Morton, P. P. Salatiello, H. Landfield, J. Polym. Sci., **8**, 279 (1952).
16. D. H. Volman, W. M. Graven, J. Am. Chem. Soc., **75**, 3111 (1953).
17. G. M. Burnett, W. W. Wright, Proc. Roy. Soc. (London) A, **221**, 41 (1954).
18. W. I. Bengough, H. W. Melville, Proc. Roy. Soc. (London) A, **230**, 429 (1955).
19. E. Bartholme, H. Gerrens, R. Herbeck, H. Weitz, Z. Electrochem., **60**, 334 (1956).
20. N. Grassie, E. Vance, Trans. Faraday Soc., **52**, 727 (1956).
21. L. C. Landers, D. H. Volman, J. Am. Chem. Soc., **79**, 2996 (1957).
22. W. I. Bengough, Trans. Faraday Soc., **54**, 868 (1958).
23. Z. A. Sinitsyna, Kh. S. Bagdasaryan, Zh. Fiz. Khim., **32**, 1319 (1958).
24. Z. Laita, Z. Machacek, J. Polym. Sci., **38**, 459 (1959).
25. F. S. Dainton, R. S. Eaton, J. Polym. Sci., **39**, 313 (1959).
26. W. I. Bengough, H. W. Melville, Proc. Roy. Soc. (London) A, **249**, 445 (1959).
27. D. H. Grant, N. Grassie, Trans. Faraday Soc., **55**, 1042 (1959).
28. P. Heyden, H. W. Melville, J. Polym. Sci., **43**, 201 (1960).
29. G. Henrici-Olive, S. Olive, Makromol. Chem., **37**, 71 (1960).
30. G. V. Schulz, G. Henrici-Olive, S. Olive, Z. Physik. Chem. (Frankfurt), **27**, 1 (1960).
31. P. E. M. Allen, G. M. Burnett, J. M. Downer, J. R. Majer, Makromol. Chem., **58**, 169 (1962).
32. W. Rabel, K. Ueberreiter, Ber. Bunsenges., **67**, 710 (1963).
33. N. Coleboume, E. Collinson, D. J. Currie, and F. S. Dainton, Trans. Faraday Soc., **59**, 1357 (1963).
34. F. S. Dainton, W. D. Sisley, Trans. Faraday Soc., **59**, 1369 (1963).
35. M. Morton, W. E. Gibbs, J. Polym. Sci. A, **1**, 2679 (1963).
36. K-P. Paoletti, F. W. Billmeyer, J. Polym. Sci. A, **2**, 2049 (1964).
37. A. F. Revzin, Kh. S. Bagdasaryan, Zh. Fiz. Khim., **38**, 1020 (1964).
38. T. A. Berezsnich-Foldes, F. Tudos, Vysokomol. Soedin., **6**, 1529 (1964).
39. M. Imoto, M. Kinoshita, M. Nishigaki, Makromol. Chem., **86**, 2167 (1965).
40. W. I. Bengough, W. Henderson, Trans. Faraday Soc., **61**, 141 (1965).
41. V. A. Agasandyan, E. A. Trosman, Kh. S. Bagdasarayan, A. D. Litmanovich, V. Ya. Shtern, Vysokomol. Soedin., **8**, 1580 (1966).
42. J. Hughes, A. M. North, Trans. Faraday Soc., **62**, 1866 (1966).
43. A. N. Plyusnin, N. M. Chirkov, Teoret. Eksp. Khim., **2**, 777 (1966).
44. S. Machi, S. Kise, M. Hagiwara, T. Kagiya, J. Polym. Sci., A-1, **5**, 3115 (1967).
45. F. Khrabk, M. Bezdek, V. Hynkova, Z. Pelzbauer, J. Polym. Sci. C, **16**, 1345 (1967).
46. E. F. Nosov, Kinet. Katal. **8**, 680 (1967).
47. J. P. Fischer, G. V. Schulz, Ber. Bunsenges. Phys. Chem., **74**, 1077 (1970).
48. G. Palma, L. Busolini, S. Lora, Eur. Polym. J., **6**, 453 (1970).

49. T. Nakaya, H. Wada, M. Tanaka, J. Macromol. Sci., Chem., A: **5**, 1043 (1971).
50. S. P. Sood, J. W. Hodgins, J. Polym. Sci. A-1, **9**, 3383 (1971).
51. S. E. Bresler, E. N. Kazebov, V. N. Fomichev, V. N. Shadrin, Makromol. Chem., **157**, 167 (1972).
52. L. Busulini, S. Lora, G. Palma, G. Lunardon, Eur. Polym. J., **8**, 455 (1972).
53. S. E. Bresler, E. N. Kazebov, V. N. Shadrin, Makromol. Chem., **175**, 2875 (1974).
54. P. P. Kuryukhin, I. M. Barkalov, V. I. Gol'danski, Eur. Polym. J., **10**, 309 (1974).
55. S. Lora, G. Palma, L. Busulini, B. Castiletti, Eur. Polym. J., **10**, 1223 (1974).
56. T. Malavasic, I. Vizovisek, S. Lapanje, A. Moze, Makromol. Chem., **175**, 873 (1974).
57. A. Moze, I. Vizovisek, T. Malavasic, F. Cernec, S. Lapanje, Makromol. Chem., **175**, 1507 (1974).
58. N. A. Plate, A. G. Ponomarenko, Polym. Sci. U.S.S.R., **16**, 3067 (1974).
59. W. I. Bengough, G. B. Park, R. A. Young, Eur. Polym. J., **11**, 305 (1975).
60. C. Chachaty, M. Latimer, A. Forchioni, J. Polym. Sci.: Polym. Chem. Ed., **13**, 189 (1975).
61. E. Senogles, R. Thomas, J. Polym. Sci.: Polym. Symp. Ed., **49**, 203 (1975).
62. M. H. George, G. E. Hayes, J. Polym. Sci.: Polym. Chem. Ed., **13**, 1059 (1975).
63. J. N. Cardenas, K. F. O'Driscoll, J. Polym. Sci.: Polym. Chem. Ed., **15**, 2097 (1977).
64. H. K. Mahabadabi, K. F. O'Driscoll, J. Macromol. Sci. Chem. A: **11**, 967 (1977).
65. H. K. Mahabadabi, K. F. O'Driscoll, Makromol. Chem., **178**, 2629 (1977).
66. H. K. Mahabadabi, K. F. O'Driscoll, Makromol. Chem., **179**, 1921 (1978).
67. K. Matsuo, G. W. Nells, R. G. Nelb, W. H. Stockmayer, Macromolecules, **10**, 654 (1977).
68. P. C. Deb, S. K. Kapoor, Eur. Polym. J., **15**, 477 (1979).
69. V. F. Kurenkov, V. A. Myagchenkov, Eur. Polym. J., **16**, 1229 (1980).
70. S. W. Landsdowne, R. G. Gilbert, D. H. Napper, J. Chem. Soc., Faraday Trans. I, **76**, 1344 (1980).
71. C. U. Pittman, M. Ueda, K. Iri, Y. Imai, Macromolecules, **13**, 1031 (1980).
72. S. K. Soh, J. Appl. Polym. Sci., **25**, 2993 (1980).
73. U. Tatsuikami, T. Takahashi, H. Yoshioka, Makromol. Chem., **181**, 1097 (1980).
74. G. O. R. Alberda van Ekenstein, Y. Y. Tan. Eur. Polym. J., **17**, 838 (1981).
75. G. O. R. Alberda van Ekenstein, D. W. Koetsier, Y. Y. Tan, Eur. Polym. J., **17**, 845 (1981).
76. B. Harris, A. E. Hamielec, L. Marten, A. C. S. Symp. Series, **165**, 315 (1981).
77. M. Kamachi, D. J. Liaw, S. Nozakura, Polym. J., **13**, 41 (1981).
78. M. Ueda, K. Iri. Y. Imai, C. U. Pittman, Macromolecules, **14**, 1046 (1981).
79. S. K. Soh, D. C. Sundberg, J. Polym. Sci.: Polym. Chem. Ed., **20**, 1345 (1982).
80. M. Ueda, M. Takahashi, Y. Imai, C. U. Pittman, J. Polym. Sci.: Polym. Chem. Ed., **20**, 3819 (1982).
81. M. Ueda, M. Takahashi, Y. Imai, C. U. Pittman, J. Polym. Sci.: Polym. Chem. Ed., **20**, 3829 (1982).
82. M. Buback, H. Lendle, Makromol. Chem., **184**, 193 (1983).
83. J. P. Fischer, S. Rosinger, Makromol. Chem. **184**, 1247 (1983).
84. A. Matsumoto, T. Nakana, M. Oiwa, Makromol. Chem.: Rapid Commun., **4**, 277 (1983).
85. M. Stickler, Makromol. Chem., **184**, 2563 (1983).
86. M. Ueda, M. Takahashi, T. Suzuki, Y. Imai, C. U. Pittman, J. Polym. Sci.: Polym. Chem. Ed., **20**, 1139 (1983).
87. M. Ueda, T. Suzuki, Y. Imai, C. U. Pittman, E. Wallace, J. Polym. Sci.: Polym. Chem. Ed., **21**, 2997 (1983).
88. V. R. P. Verneker, R. Vasanthakumari, J. Polym. Chem. Ed., **21**, 1657, 1664 (1983).
89. C. I. Kao, D. B. Gundlach, R. T. Nelson, J. Polym. Sci. Polym. Chem. Ed., **22**, 3499 (1984).
90. S. Deibert, F. Banderman, Makromol. Chem. Rapid. Commun., **13**, 351 (1992).
91. M. Morton, P. P. Salatillo, H. Landfield, J. Polym. Sci., **8**, 215 (1952).
92. M. Kamachi, Polym. Pre., **38**, 649 (1997).
93. S. Agrawal, C. D. Han, AlChE J., **21** (3), 449 (1975).
94. C. H. Chen, J. G. Vermeychuk, S. A. Howell, P. Elrich, AlChE J, **22**, 463 (1976).
95. S. Goto, S. Yamamoto, S. Furui, M. Sugimoto, J. Polym. Sci.: Appl. Polym. Symp., **36**, 21 (1981).
96. K. H. Lee, J. P. Marano, in: J. N. Henderson, T. C. Bouton (Eds.), "Polymerization Reactors and Processes", ACS, Symp. Series 104, 1979, p. 221.
97. G. M. Donati, M. Marini, L. Marziano, C. Mazzaferri, M. Spaminaot, E. Langianni, in: J. Wei, C. Georgakis (Eds.), "Chemical Reaction Engineering", ACS Symp. Ser. 196, 1982, p. 579.
98. W. Hollard, P. Elrich, Chem. Eng. Commun., **24**, 57 (1983).
99. H. Mavridis, C. Kiparissides, Polym. Proc. Engng., **3** (3), 263 (1985).
100. P. P. Shirodkar, G. O. Tsien, Chem. Engng. Sci., **41** (4), p. 1031 (1986).
101. P. J. Scott, A. Penlidis, G. L. Rempel, Polym. Rxn Engng., **3** (2), p. 93 (1995).
102. A. Penlidis, Ph. D. Thesis, McMaster University, 1986.
103. C. Kiparissides, G. Verros, G. Kalfas, M. Koutoudi, C. Kantzia, Chem. Eng. Commun., **121**, 193 (1993).
104. L. Marini, C. Georgakis, AlChE J., **30** (3), 410 (1984).
105. A. Brandolin, M. H. Lacunza, P. E. Ugrin, N. J. Capiati, Polym React. Engng. **4** (4), 193 (1996).
106. P. Lorenzini, M. Pons, J. Villermaux, Chem. Engng. Sci., **47** (15), 3981 (1992).
107. R. A. Scott, N. Peppas, AlChE J., **43** (1), 135 (1997).

108. H. G. Poersch-Panke, A. Avela, K. H. Reichert, Angew. Chemie, **206**, 157 (1993).
109. J. Gao, A. Penlidis, J. Macromol Si., Rev. Macromol. Chem. C, **36** (2), 199 (1996).
110. P. Pascal, D. H. Napper, R. G. Gilbert, M. C. Piton, M. A. Winnik, Macromolecules, **23** (24), 5161 (1990).
111. J. Hernadez-Barajak, D. J. Hunkler, Polym. Adv. Tech., **6**, 509 (1995).
112. N. F. Umoh, P. Harriott, R. Hughes, Chem. Eng. J., **21**, 85 (1981).
113. K. C. Gupta, J. Macromol. Sci., Pure & Appl. Chem. A, **31**, 805 (1994).
114. B. Chaturvedi, A. K. Srivastava, Indian J. Technol., **31** (12), 851 (1993).
115. B. P. Agrawal, A. K. Srivastava, Macromol. Rep. A, **29** (Suppl. 3), 325 (1992).
116. R. A. Lyons, J. Hutovic, M. C. Piton, D. Christie, P. Clay, B. G. Mandrs, S. H. Kable, R. G. Gilbert, Macromolecules **29**, 1918 (1996).
117. M. Buback, B. Degener, Makromol Chemie, **194**, 2875 (1993).
118. M. Buback, B. Degener, B. Huckstein, Makromol Chemie, Rapid Commun., **10**, 311 (1989).
119. S. Beuermann, D. A. Paquet Jr., R. A. Hutchinson, Macromolecules, **29** (12), 4206 (1996).
120. M. T. Iglesias, G. M. Guzman, J. Riande, E., J. Polym. Sci., Part A: Polym. Chem. **32** (13), 2565 (1994).
121. H. Tanaka, S. Yoshida, Macromolecules, **28** (24), 8117 (1995).
122. C. C. Lin, W. Y. Chiu, C. T. Wang, J. Appl. Polym. Sci., **23**, 1203 (1979).
123. C. Lin, W. Chiu, C. Wang, J. Appl. Polym. Sci. **23**, p. 1203 (1979).
124. N. Jocheska, G. Petrov, A. Jenevski, A., Sedenik, Eur. Polym. J., **32** (2), 209 (1996).
125. S. Beuermann, D. A. Paquet, J. McMinn, R. A. Hutchinson, Macromolecules, **30** (2), 194 (1997).
126. S. Lora, G. Palma, L. Busilini, B. Castelleti, Eur. Polym. J., **10**, 1223 (1974).
127. R. A. Hutchinson, D. A. Paquet Jr., J. H. McMinn, R. E. Fuller, Macromolecules, **28**, p. 4023 (1995).
128. P. Pascal, M. A. Winnik, D. H. Napper, R. G. Gilbert, Makromol. Chem., Rapid Commun., **14** (3), 213 (1993).
129. J. Cardenas, K. F. O'Driscoll, J. Polym. Sci.: Polym. Chem., **15**, 2097 (1977).
130. H. H. Catalgil, B. M. Baysal, Eur. Polym. J., **27** (1), 7 (1991).
131. R. A. Hutchinson, D. A. Paquet, J. H. McMinn, S. Beuermann, in 5th International Workshop on Polymer Reaction Engineering, Dechema Monographs, vol. 131, 1995, p. 467.
132. N. Tefera, G. Weickert, K. R. Westerterp, J. Schweer, in 5th International Workshop on Polymer Reaction Engineering, Dechema Monographs, vol. 131, 1995, p. 557.
133. K. Y. Choi, D. N. Butala, Polym. Eng. Sci., **31** (5), 353 (1991).
134. G. Maschio, T. Bello, C. Scali, Chem. Engng. Sci., **47** (9), 2609 (1992).
135. T. P. Davis, K. F. O'Driscoll, M. C. Piton, M. A. Winnick, Macromol., **23**, 2113 (1990).
136. G. Polacco, D. Semino, M. Palla, Polym. Eng. Sci., **36** (11), 2088 (1996).
137. J. Congalides, J. Richards, W. H. Ray, AlChE J., **35** (6), 891 (1989).
138. P. E. Baillagou, D. S. Soong, Chem. Engng. Sci., **40** (1), 87 (1985).
139. D. Panke, M. Stickler, W. Wunderlich, Makromol. Chem., **184**, 175 (1983).
140. S. Beuermann, Ph.D. Thesis, University of Göttingen 1993.
141. U. Begert, M. Buback, J. Heyne, Macromol. Chem. Rapid Commun., **16**, 275 (1995).
142. S. Beuermann, M. Buback, T. P. Davis, R. G. Gilbert, R. A. Hutchinson, O. F. Olaj, G. T. Russel, J. Schweer, A. M. van Herk, Macromol. Chem. Phys., **198**, 1545 (1997).
143. G. T. Russel, Macromol. Theory Simul., **4**, 549 (1995).
144. R. A. Hutchinson, Polym. React. Eng., **1**, 521 (1993).
145. U. Berger, S. Beuermann, M. Buback, H. K. Caroline, G. T. Russell, C. Schmaltz, Macromol. Rapid Commun., **16**, 425 (1995).
146. R. A. Hutchinson, M. T. Aronson, J. R. Richards, Macromolecules, **26**, 6410 (1993).
147. L. Vrhovac, D. Filipovic, M. Misic-Vukovic, J. Vellekovic, Int. J. Chem. Kin., **24** (10), 861 (1992).
148. T. Sato, I. Kamiya, H. Tanaka, T. Ota, Eur. Polym. J., **27** (10), 1087 (1991).
149. T. Sato, N. Morita, H. Tanaka, T. Ota, Macromol. Chem. Phys., **191** (11), 2599 (1990).
150. M. Coskun, K. Demirelli, M. A. Ahmetzade, J. Macromol. Sci., Pure Appl. Chem. A, **34** (3), 429 (1997).
151. D. A. Shipp, T. A. Smith, D. H. Soloman, G. Moad, Macromol. Chem. Rapid Commun., **16**, 837 (1995).
152. M. Ueda, H. Mori, H. Ito, J. Polym. Sci., Polym. Chem. A, **28** (10), 2597 (1990).
153. J. C. Pinto, Polym. Eng. Sci., **30** (5), 291 (1990).
154. C. Sidiropoulou, C. Kiparissides, J. Macromol. Sci. A: Chem., **27** (3), 257 (1990).
155. T. Pernecker, T. Foldes-Berezsnich, F. Tudos, J. Macromol. Sci. A: Chem., **27** (8), 1029 (1990).
156. M. Kamachi, Adv. Polym. Sci., **82**, p. 207 (1987).
157. N. Friis, D. Goosney, J. D. Wright, A. E. Hamielec, J. Polym. Sci., **18**, 1247 (1974).
158. S. S. Jang, P. H. Liu, Chem. Engng. Sci., **46** (12), 3153 (1991).
159. J. Congalides, J. Richards, W. H. Ray AlChE J., **35** (6), 891 (1989).
160. F. Teymour, Ph.D. Thesis, University of Wisconsin (Madison) (1989).
161. R. A. Hutchinson, J. R. Richards, M. T. Aronson, Macromolecules, **27** (16), 4530 (1994).
162. B. Yamada, M. Kageoka, T. Otsu, Polym. Bull. **29** (3–4), 385 (1992).
163. N. Tefera, G. Weickert, R. Bloodworth, J. Schweer, Macromol. Chem. Phys., **195**, 3067.
164. M. Buback, L. H. Garcia-Rubio, R. G. Gilbert, D. H. Napper, J. Guillot, A. E. Hamielec, D. Hill, K. F. O'Driscoll, F. Olaj, J. Shen, D. Soloman, G. Moad, M. Stickler, M. Tirrell,

M. A. Winnick, J. Polym. Sci.: Polym. Let., **26**, 293 (1988).
165. D. K. Sharma, D. S. Soong, Macromol., **21** (3), (1988)
166. W. Lansdowne, R. G. Gilbert, D. H. Napper, D. F. Sangster, J. Chem. Soc., Faraday Trans., **76**, 1344 (1980).
167. S. E. Bresler, E. N. Kazbekov, U. N. Shardin, Makromol. Chem., **157**, 167 (1972).
168. T. P. Davis, K. F. O'Driscoll, M. C. Piton, M. A. Winnick, Macromol., **22**, 2785 (1990).
169. M. Deady, A. W. H. Mau, G. Moad, T. H. Spurly, Macromol Chem., **194**, 1691 (1993).
170. R. Vasishtha, A. K. Srivastava, Polymer, **31** (1), 150 (1990).
171. G. Clouet, P. Chaffanjon, J. Macromol. Sci. A: Chem., **27** (2), 193 (1990).
172. M. Buback, F. D. Kuchta, Macromol. Chem. Phys., **196**, 1887 (1995).
173. M. Kamachi, A. Kajiwara, Macromol. Chem. Phys., **198**, 787 (1997).
174. M. Buback, R. G. Gilbert, R. A. Hutchinson, B. Klumpermann, F. D. Kuchta, B. A. Manders, K. F. O'Driscoll, G. T. Russell, J. Schweer, Macromol. Chem. Phys., **196**, 3267 (1995).
175. S. K. Soh, D. C. Sundberg, J. Polym. Sci.: Polym. Chem., **20**, 1345 (1982).
176. O. Chiantore, A. E. Hamielec, Polymer, **26**, 608 (1985).
177. J. P. Fisher, Makromol. Chem., **155**, 211 (1972).
178. E. Vivaldo-Lima, A. E. Hamielec, P. E. Wood, Polym. React. Engng., **2** (2–1), 87 (1994).
179. K. C. Gupta, J. Appl. Polym. Sci., **53** (1), 71 (1994).
180. K. C. Gupta, J. Appl. Polym. Sci., **53** (1), 71 (1994).
181. V. A. Agasandyan, E. A. Trosman, Kh. S. Bagdasar'yan, A. D. Litmovich, V. Ya. Shtern, Vysokomol. Soyed., **8**, 1580 (1966).
182. M. Ueda, H. Toyota, A. Sane, H. Ito, J. Polym. Sci., Polym. Chem., **131**, 1895 (1993).
183. O. Takayuki, Y. Bunichiro, I. Toshihiro, Macromolecules, **24/2**, 415 (1991).
184. B. Yamada, E. Yoshikawa, H. Miura, T. Otso, Macromol. Chem. Rapid Commun., **13**, 531 (1992).
185. T. Sato, Y. Hirose, M. Seno, H. Tanaka, J. Polym. Sci., Part A: Polym. Chem., **33** (5), 797 (1995).
186. T. Sato, H. Nakamura, H. Tanaka, T. Ota, Makromol. Chem., **192** (11), 2659 (1991).
187. T. Sato, Y. Hirose, M. Seno, H. Tanaka, N. Uchiumi, M. Matsumoto, Eur. Polym. J., **30** (3), 347 (1994).
188. T. Sato, S. Inui, H. Tanaka, T. Ota, M. Kamachi, T. Tanaka, J. Polym. Sci., Part A: Polym. Chem., **25**, 637 (1987).
189. J. Sugiyama, N. Kanamori, S. Shimada, Macromol., **29**, 1943 (1996).
190. V. S. Nikiforenko, Yu. S. Zaitsev, V. V. Zaitseva, Academician R. V. Kucher, S. V. Muravchenko, A. V. Bulavin, L. N. Mikhailova, Doklady Physical Chemistry, **300** (1–3), 455 (1988).
191. M. Seno, T. Takikawa, H. Tanaka, T. Sato, Macromolecules, **28**, 4795 (1995).
192. T. Sato, K. Arimoto, H. Tanaka, T. Ota, K. Kato, K. Doiuchi, Macromolecules, **22**, 2219 (1989).
193. B. Patel, B. Suthar, R. Patel, Macromolecular Reports, **29** (1), 55 (1992).

Activation Volumes of Polymerization Reactions

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A. INTRODUCTION

The activation volume is included in the pressure dependence of the reaction rate constant (1–4):

$$\left(\frac{\partial \ln k}{\partial p}\right)_T = -\frac{\Delta v^*}{RT} + \frac{1}{V}\left(\frac{\partial v}{\partial p}\right)_T \qquad \text{(A1)}$$

where k is the reaction rate constant, p is the pressure, T, the temperature V, the volume, and Δv^*, the activation volume (analogous to the activation energy). It is the difference between the partial molar volume of the activated complex, as formed in the transition state theory by the efficient collision of molecules, and those of the initial reactants. The sign of the activation volume depends on the type of the chemical reaction. If new bonds are formed in the transition state, the activated complex is larger than the initial species, and hence the activation volume is positive. Inversely the activation volume is negative if bonds are stretched and broken.

For a rough calculation of the activation volume, we can assume that the formation of new bonds as well as stretching occurs along the axis of a cylinder, whose constant cross section is determined by the van der Waal's radii $d_{mA/2}$, $d_{mB/2}$, and $d_{mC/2}$ of the atoms A, B, C:

$$\Delta v^* = \frac{\pi}{8} N\left[(d_{mA}^2 + d_{mB}^2)(\Delta_{AB}) + (d_{mB}^2 + d_{mC}^2)(\Delta_{B,C})\right] \qquad \text{(A2)}$$

with $\Delta_{AB} = (0.10\text{–}0.35)\, d_{mAB}$, $\Delta_{B,C} = (1.10\text{–}1.35) d_{mBC} - d_{mB,C}$ (see Refs. 2,5,6), N is the Avogadro number, d_{mAB}, d_{mBC} = bond length (values, see Ref. 7), and d_{mBC} is the interatomic distance at the minimum of potential (sum of van der Waal's radii), $d_{mA/2}$, $d_{mB/2}$ (see Ref. 8).

The stretching of the bonds in the transition state is about 0.01–0.1 nm, whereas the change in the distance $d_{mB,C}$ between the unbonded atoms is usually larger (up to 0.25 nm).

If the decomposition of the activated complex causes only small changes in the bond length or in the distance between the atoms, its partial molar volume differs from that of the activated complex only slightly. Hence we can assume

$$\Delta v^* \cong \Delta v$$

where Δv is the excess of the partial molar volume of the reaction products over the partial molar volumes of the initial species.

This simplification is valid for reactions in which products of cyclic structure are formed, as in the dimerization of cyclopentadiene, with $\Delta v^* \approx \Delta v = -30\,\text{cm}^3/\text{mol}$ (9, 10).

*Table 3. Activation Volumes.

A second term is added to the volume change of the reacting molecules in the transition state if a polar solvent is involved. This term takes into account the change in the packing density of the surrounding solvent molecules due to the arising or disappearing of electrostatic charges between the solvent and the reactant species. This volume change is negative in bimolecular association and positive in unimolecular dissociation and tends therefore to counteract the effect of volume change during the reaction. The activation volume can be determined according to Eq. (A1) from rate measurements at different pressures and constant temperature using a semilog plot of reaction rate constant versus pressure. In this evaluation the value of $(\partial V/\partial p)_T/V$ can often be neglected, because the compressibility of fluid reactants and compressed gases is generally small at pressures above 100–200 MPa.

At very high pressures the increase of the reaction rate constant at negative activation volume is retarded by the lower mobility of the molecules due to the increased viscosity. This effect was first measured by Hamann (11) in an examination of the alkaline etherification of ethyl bromide at pressures up to 4000 MPa. It was also found in the polymerization of styrene by Nicholson and Norrish (12).

In order to obtain the reaction rate as an explicite function of pressure, Eq. (A1) can be integrated, neglecting the compressibility and assuming that the activation volume does not change greatly with pressure (13):

$$k_p = k_0 \times \exp -\frac{\Delta v*}{RT}(p - p_0) \qquad \text{(A3)}$$

where k_p is the reaction rate constant at pressure p and k_0 is the reaction rate constant at reference pressure p_0.

According to Eq. (A3) the reaction rate constant increases exponentially with pressure if the activation volume is negative and decreases if Δv^* is positive. Eq. (A3) is valid only in the high pressure range ($p > 50$ MPa). At low pressures the compressibility cannot be neglected anymore, and the activation volume changes noticeably with pressure, especially in the neighborhood of the critical point. Simmons and Mason (14) studied the dimerization of chlorotrifluoroethylene at pressures up to 10 MPa. They found that the value of the negative activation volume decreases first with pressure and then increases rapidly. It has a maximum at the critical point and after decreasing again it approaches asymptotically a constant value.

The authors describe the pressure dependence of the rate constant, taking account of the partial molar volume of the initial reactants as well as that of the activated complex by the use of suitable equations of state (e.g., Redlich–Kwong or virial coefficient equations).

In order to appreciate the influence of pressure on the reaction rate constant one can compare it with the influence of the temperature: an increase in pressure (activation volume $\Delta v^* = -25\,\text{cm}^3/\text{mol}$) at a temperature of 50°C from 0.1 to 450 MPa corresponds to a temperature increase (activation energy $E = 84$ kJ/mol) from 50 to 105°C.

Experimentally determined overall activation volumes of some polymerization reactions are listed in Section B. The data were determined from the dependence of the overall rate constant of the polymerization reactions on pressure. The values are more or less negative, hence the rate of polymerization increases with pressure.

The overall activation volume is composed of the activation volumes of the different polymerization steps, initiation, or initiator decomposition, chain propagation and chain termination:

$$\Delta v^* = \Delta v_p^* + \Delta v_d^*/2 - \Delta v_t^*/2 \qquad \text{(A4)}$$

where Δv_p^* is the activation volume of chain propagation, Δv_d^* is the activation volume of initiation with respect to initiator decomposition, and Δv_t^* the activation volume of chain termination.

The activation volumes of initiator decomposition (values in Section C) are always positive because this reaction is a unimolecular dissociation, in which a bond (e.g., the O–O bond in peroxides and peresters or the N–N bond in azo-compounds) is stretched in the transition state and finally broken. As mentioned before, Δv_d^* depends on the solvent. It can significantly be influenced by a change of electrostatic charges, viscosity, and cage effects. The activation volume of initiator decomposition is different when induced decomposition is involved at high concentration of the initiator or when the initiator decomposes in a concerted mechanism. Δv_d^* is small for two-bond scission initiators (15). The activation volume is higher for one-bond scission initiator, bulky molecules, and when more bonds brake simultaneously. The data collected in Section C were obtained in separate measurements, observing the decrease of the initiator concentration with time at different pressures. The disappearance of the initiator can also be measured during polymerization by observing the light absorption at a characteristic waver length or using scavenger techniques.

In the chain propagation reaction, the decrease in the distance between the radical and the monomer molecule is greater than the increase in length of the double bond of the monomer. Hence the activation volumes Δv_p^* listed in Section D are always negative. The data for Δv_p^* are measured at high pressures by the method of the rotating sector together with Δv_t^*, the activation volume of chain termination.

The activation volumes Δv_t^* are also negative as shown in Section E. Mostly the termination reactions are diffusion-controlled. The reactant molecules undergo translational diffusion at first to form a collision pair. It follows the movement of the reactive sites on these molecules into a position favorable to chemical reaction. This process has been considered to be the rate-determining step. The activation volume involved in this so-called segment diffusion is observed. When chain termination is governed by translational diffusion, the activation volume, Δv_{vis}^* for viscous flow, is the key parameter for the evaluation of the influence of pressure.

The chain transfer of a radical to the monomer, to a "dead" polymer, or to a modifier molecule is mostly favored by the pressure which means that the activation volumes should be negative, whereas the activation volume of the intramolecular transfer (by "back biting"), which initiates the formation of short side chains, can be negative or positive. The activation volumes of these reactions are determined from the change in short- and long-chain branching or from the change of the average molar mass or degree of polymerization of the formed polymers with pressure.

Under the assumption that the monomer and the polymer concentration do not change over a wide range and that the temperature is kept constant, the following relation can be derived from kinetic considerations:

$$\Delta v_p^* - \Delta v_{tr}^* = RT\left(\frac{\partial \ln \mathrm{VZG}}{\partial p}\right)_T \qquad \text{(A5)}$$

where VZG is the number of branches and Δv_{tr}^* is the activation volume of the chain transfer reaction.

Analogously, the influence of the pressure on the chain transfer to a modifier can be described by the expression

$$\Delta v_p^* - \Delta v_{tr}^* = RT\left(\frac{\partial \ln(k_{tr}/k_p)}{\partial p}\right)_T = RT\left(\frac{\partial \ln(1/\bar{P} - 1/\bar{P}_{no})}{\partial p}\right)_T \qquad \text{(A6)}$$

where P is the average degree of polymerization obtained in the presence of a transfer agent and $\bar{P}_{no}$ is the average degree of polymerization obtained in the absence of a transfer agent.

Because of the mostly negative sign of the activation volumes (listed in Section F), the radical transfer to the modifier molecule is often slightly decreased at high pressures.

The activation volume of degradation reactions which can occur at high temperatures should be positive because bond cleavage increases the volume of the activated complex more than the formation of double bonds from single bonds decreases it.

The difference in the activation volumes of the various component reactions results in a different influence of the pressure on the rate constants and leads to modified polymers. Because of its larger negative activation volume the chain propagation reaction is more favored by high pressures than the termination. The chain length and consequently the tensile strength as well as the tensile modulus increase with pressure:

$$\left(\frac{\partial \ln \nu}{\partial p}\right)_T = -\frac{1}{RT}\left[\Delta v_p^* - \frac{1}{4}(3\Delta v_p^* + \Delta v_{td}^*\right] \qquad \text{(A7)}$$

where ν = chain length.

The small negative activation volume of the intramolecular chain transfer shows that this reaction step, which determines the short chain branching, is less favored by high pressures, which means that polymers with high density and crystallinity can be synthesized at high pressures.

Similar considerations also show that the composition of copolymers and thus their molecular structure is influenced by the synthesis pressure. The pressure dependence of the copolymerization parameters r_1 and r_2, which determine the composition of a copolymer, is expressed by the relationship

$$\left(\frac{\partial \ln(r_1/r_2)}{\partial p}\right)_T = -\frac{1}{RT}(\Delta v_{p11}^* - \Delta v_{p12}^* - \Delta v_{p22}^* + \Delta v_{p21}^*) \approx -\frac{1}{RT}(\Delta v_{p11}^* - \Delta v_{p22}^*) \qquad \text{(A8)}$$

where Δv_{p11}^*, Δv_{p22}^* are the activation volumes of the propagation reaction between a monomer molecule and a radical formed from the same monomer, and Δv_{p12}^*, Δv_{p21}^* are the activation volumes of the propagation reaction between a monomer molecule and a radical formed from the comonomer.

Equation (A8) is valid if the composition of the initial reaction mixture does not change appreciably with pressure. According to Eq. (A8), an increase in pressure favors the inclusion of monomers, which in homopolymerization show a large negative activation volume (e.g., substituted olefins). Values of copolymerization parameters obtained at different pressures and activation volumes of some copolymerization reactions are listed in Section G.

When r_1 and r_2 should be calculated for high-pressure copolymerization, following Alfrey and Price, the pressure dependence of Q, e data must be taken into account. Some values of Q and e for high pressures are collected in Section G. The dependence of the e data of two monomers on pressure is given by the equation

$$\left(\frac{\partial\left[-(e_1 - e_2)^2\right]}{\partial p}\right)_T = -\frac{1}{RT}[(\Delta v_{p11}^* - \Delta v_{p12}^*) + (\Delta v_{22}^* - \Delta v_{21})] \qquad \text{(A9)}$$

B. ACTIVATION VOLUMES (cm^3/mol) OF SOME POLYMERIZATION REACTIONS

Monomer	Maximum pressure (MPa)	Temp. (°C)	Solvent	Initiator[a]	Δv^*	Refs.
Acenaphthylene	390	60		AIBN	−5.8	16
Acetaldehyde	150	100–200		Diethyamine	−12.5	127
	844	−60			−11.1	142
	844	−70			−8.0	
	844	−78			−6.2	
Acrylic acid anhydride	400	50	DMF	AIBN	−14	51
Acrylonitrile	200	50	DMF	AIBN	−22[b]	52
Allyl acetate	850	80		BPO	−13.4	17
N-(4-Anilinophenyl)acrylamide	100	60	THF	AIBN	−36.1	150
	200	60			−31.2	
	300	60			−22.3	
	500	60			−15.4	
	1500	60			−6.4	
3,3-Bis(chloromethyl)oxetane	500	60	Toluene	BF_3/diethyl ether	−16	55
Butadiene	500	50			−8.8	141
	500	70			−13.3	
	500	95			−17.2	
	100	30		ACN	−26.3	91
Butyl acrylate	100	30		ACN	−26.3	154
Butyl acrylate	100	20		Radiation	−21.0	147
	300	20			−21.2	
	460	20			−5.7	
	700	20			−11.4	
Butyl methacrylate	100	60	Benzene	AIBN	−25.6	128
	100	30		ACN	−17.4	90
	100	20		Radiation	−21	
	300	20			−21.2	94,95
	460	20			−5.7	
	>460	20			−14.4	
2-Cyclopropylpropene-1	140	70		AIBN	−23	53
	140	20		Stannic chloirde	−15	53
Diallyl cyanamide	600	80	Toluene	TBPB	−14	54
Diallyl phthalate	500	70		AIBN	−9.9 to −11.8	130
	500	70	Benzene	AIBN	−14.3	
3,3-Dichloropropene-1	1000	40		DCPC	0 to 8[b]	57
1,1-Dichloro-2-vinylcyclopropane	1000	80		AIBN	−14[b]	56
	1000	20		Stannic chlorde	−30 to −70	
Diethyl fumarate	300	60			−14	18
3,3,3-Difluorochloropropene-1	1400	30		DCPC	−15	58
1,1-Difluoro-2-vinylcyclopropane	1000	50		AIBN	−15	59
	1000	20		Stannic chloride	−45	59,60
2,3-Dimethyl-1,3-butadiene			Heptane	BuLi	−36	133
			DEE	BuLi	−48	
Ethylene	250	129		DTBP	−20 to −23	46,20
	760	50–70		AIBN	−3 to −6	19
	200	225–235		Peroxide	−20 to −26	73
	190	80		DCPC	−10	74
				Enanthylperoxide	−58 to −68	151
	100–210	200–300		Methylinobutylketone	−18	114
	110–220	190		TBPO	−18 to −22	61
	250	130–165		TBPP	−22.6	79
	110–550	170–240		Oxygen	−12 to −18	21
	175	127–197		Diphenyl disulfide	−29	85
	300	135–165		DTBP	−17.5	79
1,2-Epoxycyclohexane	840–1250	60		BPO	−13	55
n-Hexyl methacrylate	450	20		Radiation	−24	94
	800	20			−17.4	
Indene	800	64	Heptane		−21.0	22
Isobutyl methacrylate	100	60	Benzene	AIBN	−23.9	122
Isoprene	1400	20–30		Oxygen	−11.0	23
	1400	20–30		BPO	−7.9 to 18.0	
	1400	20	Toluene	HI/H_2O	−7.9	24
	1400	20	Ethyl acetate	HI/H_2O	−9.8[c]	

Monomer	Maximum pressure (MPa)	Temp. (°C)	Solvent	Initiator[a]	Δv^*	Refs.
	1400	20	Dichloromethane	HI/H_2O	−10.8	
Isopropenylcyclopropane	1400	70		AIBN	−23.0	158
	1400	20		Stannic chloride	−15.0	
Itaconic acid	500	40	DMF	AIBN	−23.2	140
	500	50	DMF	AIBN	−27.4	
	500	60	DMF	AIBN	−30.8	
	500	70	DMF	AIBN	−32.3	
Maleic anhydride					−4.6	132
Maleimide	500	20–40	Methanol	BPO	−15 to −25	122
Methyl acrylate	100	50	Benzene	AIBN	−17.6	129
	180	20		Radiation	−21.4	147
	270	20			−22.9	
	400	20			−3.6	
	700	20			−12.8	
Methyl methacrylate	200	40		Peroxide	−19	25
	300	40		Peroxide	−19	26
	100	30		ACN	−15.6	87,153
					−21.8[c]	
	300	20		Radiation	−23.6	93,95
	550	20			−13.7	
	100	50	Acetone	Peroxide	−26.0	27
	100	50	Benzene	AIBN	−21.6	129
	100	40	Benzene	Peroxide	−17	27
	100	50	Benzene	Peroxide	−19	27
	100	60	Benzene	Peroxide	−21	27
	100	70	Benzene	Peroxide	−23	27
	400	50	Butyraldehyde	AIBN	−25	62
	100	50	Carbon tetrachloride	AIBN	−19	27
	100	50	Dichloroethane	AIBN	−21	27
	100	50	Ethanol	AIBN	−20	27
	100	50	Ethyl acetate	Peroxide	−24	27
	100	50	*n*-Hexane	Peroxide	−19	27
	100	50	Trichloromethane	Peroxide	−22	27
α-Methylstyrene	350	60			−18	18
	500	50		AIBN	−14.5	135
	500	60		AIBN	−13.8	
	500	70		AIBN	−11.8	
	500	80		AIBN	−10.7	
Octyl methacrylate	100	30		ACN	−26.3	99
					−25.6[c]	
Propylene	1600	21		Radiation	−9.6	28
	1600	48		Radiation	−9.6	
	1600	83		Radiation	−12.2	
	800	80–100		AIBN	−14	144
	400	80		BPO	−9.0	
	400	90		BPO	−10.8	
	400	100		BPO	−12.7	
Sodium allylsulfonate	900		H_2O	Radiation	−5 to −7	145
Styrene	208	22		Radiation	−20.5	137
					−18.6[c]	
					+5.8	
	273	25		Radiation	−20.9	136
	400	30		Radiation	−18	126
	100	30		ACN	−17.1	89
					−17.6[c]	
	200	40		Radiation	−15.8	146
	280	40		Radiation	−50.5	146
	400	40		Radiation	−4.3	146
	600	40		Radiation	−29.5	146
	300	50		AIBN	−21	64
	100	50		AIBN	−20.2	27
	200	60		AIBN	−15	149

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Monomer	Maximum pressure (MPa)	Temp. (°C)	Solvent	Initiator[a]	Δv^*	Refs.
	293	60		AIBN	−18	29
	265	80		BPO	−17	125
	265	80		Thermal	−25.8	125
	274	25	Emulsion	Radiation	138	
			Emulsion with SDS		−18.7[c]	138
			SDS		−17.9[c]	138
			Emulsion with aerosol OT		−18.3[c]	138
	274	25	Emulsion	Radiation		139
			Emulsion with nonionic emulsifier		−21.6	139
	200–500	60	Emulsion		−11.1	35
	100	50	Acetone	AIBN	−25.5	27
	100	50	Benzene	AIBN	−19.2	27
			Benzene	AIBN	−23.0	63
	100	50	Benzene	Peroxide	−18	30
	400	50	Butyraldehyde	AIBN	−25	65
	400	50	Butyraldehyde	AIBN	−22	63
	100	50	Carbon tetrachloride	AIBN	−19.1	27
	300	25–30	Chlorobenzene	$HClO_4$	−20 to −25	155
	100	50	Chloroform	AIBN	−21.9	27
	100	50	Dichloroethene	AIBN	−21.2	27
	100	50	Ethanol	AIBN	−19.6	27
	100	50	Ethyl acetate	AIBN	−24.3	27
	400	60	Isopropyl aldehyde	AIBN	−6.8	157
	100	50	Hexane	AIBN	−19.1	27
	400	60	Propionaldehyde	AIBN	−15.5	157
	268	60	Tetrachloroethylene	Peroxide	−16	31
	293	60	Toluene	AIBN	−17	29
	440	60	Triethylamine	AIBN	−17	29
	300	30	Triethylamine	BPO	−18	32
	300	40	Triethylamine	BPO	> −11.5	33
	100–600	72	Triethylamine	Thermal	−17.5	34
1,1,2,2-Tetrachloro-1,3-butadiene	1400	25–80			−36	159
Tetrahydrofuran	250–540	60		BF_3/diethyl ether	−22	55
Vinyl acetate	100	30		ACN	−17.2	86
	600	20		Radiation	−9	148
	400	40	Acetone	AIBN	−19	62
	400	40	Butyraldehyde	AIBN	−21	62
	300	40	Carbon tetrachloride	AIBN	−13	36
	400	65	Trichloroethylene	AIBN	−10.5	65
Vinyl chloride	500	60			−16.7	131
	390	40		BPO	−4.6	143
	390	50		BPO	−6.8	
	390	60		BPO	−8.1	
	390	70		BPO	−10.4	
	250	50		LPO	−18	124
	200	60	Benzaldehyde	AIBN	−4.1	98
	300	60	Benzene	AIBN	−20.5	131
	200	60	Benzonitrile	AIBN	−12.0	98
	400	60	Toluene	AIBN	−8.2	98

[a]Abbreviation: ACN: 1,1′-azobiscyclohexane-1-carbonitrile; AIBN: 2,2′-azobisisobutyronitrile; BPO: benzoyl peroxide; DCPC: dicyclohexyl peroxidicarbonate; DEE: diethyl ehter; DMF: dimethylformamide; DTBP: di-*tert*-butyl peroxide; LPO: lauroyl peroxide; TBPB: *tert*-butyl perbenzoate; TBPO: *tert*-butyl peroctoate; THF: tetrahydrofuran; SDS: sodium dodecylsulfate; Aerosol OT: sodium dioctyl sulfosuccinate; TBPP: *tert*-butyl pervivalate.
[b]Calculated by other authors.
[c]Cationic polymerization.

C. ACTIVATION VOLUMES OF INITIATOR DECOMPOSITION

Initiator	Solvent	Absolute pressure (MPa)	Temperature (°C)	Activation volume Δv_d^* (cm^3/mol)	Refs.
AIBN[a]	Toluene	0.1–150	70	13.1	37
	Toluene	0.1–1000	62.5	3.8[b] to 9.4[c]	37
Azobis-*tert*-butanolate	Octane	0.1–630	55	4.3	104
Benzoyl peroxide	Acetophenone	0.1–650	80	4.8	38
	Allyl acetate	0.1–550	80	4.7	17
	Carbon tetrachloride	0.1–300	60	9.7	39
	Carbon tetrachloride	0.1–300	70	8.6	39
2,2-Bis(*tert*-butyl peroxy)butane	Iso-dodecane	0.1–200	135–212	22.3	81
Decanoyl peroxide	Benzene	0.1–300	<100	3	78
Dicyclohexyl peroxydicarbonate	Iso-dodecane	0.1–300	95–200	4.3	75
Di-*n*-propyl peroxydicarbonate	Benzene	0.1–300	<100	4	78
Dioctanoyl peroxide	Iso-dodecane	0.1–300	80–200	5.9	75
Di-*tert*-butyl peroxide	Benzene	0.1–300	<100	15	78
		0.1–730	120	12.6	40
	Iso-dodecane	0.1–300	143–230	13.4	75
	Carbon tetrachloride	0.1–730	120	13.3	40
	n-Heptane	10–230	140–200	10.1	77
	Cyclohexane	0.1–730	120	6.7	40
	Toluene	0.1–730	120	5.4	40
Pentaphenylethane	Toluene	0.1–150	60	7	37
	Toluene	100–270	50	5	37
tert-Amyl perneodecanoate	Iso-dodecane	0.1–300	100–200	7.0	75
tert-Amyl perpivalate	Iso-dodecane	0.1–300	120–200	9.4	75
tert-Butyl cyclohexanecarboxylate	Cumene	0.1–410	80	3.9	101
tert-Butyl perbenzoate	Cumene	0.1–410	80	10	102
	Benzene	0.1–300	<100	7	78
tert-Butyl perneodecanoate	Iso-dodecane	0.1–300	100–200	6.1	75
tert-Butyl perpivalate	Iso-dodecane	0.1–300	120–200	6.1	88
	n-Heptane	0.1–200	65–105	1.6	78
tert-Butyl peroxy-2-ethylhexanoate	Benzene	0.1–300	<100	6	78
tert-Butyl phenylperacetate	Cumene	0.1–410	80	1	103
1-Methyl-1 propylpentyl perpivalate	Iso-dodecane	0.1–300	120–200	13	76
1,1,4-Trimethylpentyl perpivalate	Iso-dodecane	0.1–300	120–200	14	76
1,1,4-Trimethylpentyl perneodecanoate	Iso-dodecane	0.1–300	100–200	9	76

[a] Azobisisobutyronitrile.
[b] Photometric analysis.
[c] By I_2 scavenger technique.

D. ACTIVATION VOLUMES OF CHAIN PROPAGATION

Monomer	Solvent	Initiator	Absolute pressure (MPa)	Temperature (°C)	Activation volume Δv^* (cm^3/mol)	Refs.
Butyl acrylate	–	ACN[a]	0.1–100	30	−22.5	91
Butyl methacrylate	–	ACN	0.1–100	30	−23.2	90
Ethylene	*t*-Butanol	Radiation	−40	24	−39	152
Ethylene	–	Diphenyl disulfide	50–175	130–200	−25.5	85
Methyl methacrylate	–	ACN	0.1–100	30	−19	87
Octyl methacrylate	–	ACN	0.1–100	30	−24.7	99
Styrene	–	ACN	0.1–100	30	−17.9	89
Vinyl acetate	–	ACN	0.1–100	30	−23.3	86

[a] ACN: 1,1-azocyclohexane-1,1′-dicarbonitrile.

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E. ACTIVATION VOLUMES OF CHAIN TERMINATION

Monomer	Solvent	Initiator	Absolute pressure (MPa)	Temperature (°C)	Activation volume Δv^* (cm^3/mol)	Refs.
Butyl acrylate	–	ACN[a]	0.1–100	30	20.8	91
Butyl methacrylate	–	ACN	0.1–100	30	17.8	90
Ethylene	–	Diphenyl disulfide	50–175	130–200	7	85
Methyl methacrylate	–	ACN	0.1–100	30	25.0	87
Octyl methacrylate	–	ACN	0.1–100	30	20.8	99
Styrene	–	ACN	0.1–100	30	13.1	89
Vinyl acetate	–	ACN	0.1–100	30	16.3	86

[a]ACN: 1,1-azocyclohexane-1,1′-dicarbonitrile.

F. ACTIVATION VOLUMES OF CHAIN TRANSFER REACTIONS

Monomer	Transfer agent	Absolute pressure (MPa)	Temperature (°C)	Activation volume (cm^3/mol)		Refs.
				Δv^*_{tr}	$\Delta v^*_p - \Delta v^*_{tr}$	
Acrylonitrile	Butyraldehyde	0.1–400	50		−6.6	62
3,3-Dichloropropene-1	3,3-Dichloropropene-1	0.1	50		−8	62
Ethylene	Acetone	136–238	130		−1.1	41
		138–240	130–200	−24.9	−1.1	83
	Butane	136–238	130		−3.2	41
	Butanone	136–238	130		−2.9	41
	Cyclohexane	136–238	130		−1.5	41
		138–240	130–200	−24.5	−1.5	83
	Cyclopentane	198–240	130–200	−21.3	−4.7	83
		136–238	130		−4.7	41
	Ethanol	138–240	130–200	−22.9	−3.1	83
		136–238	130		−3.1	41
	Ethylbenzene	138–240	130–200	−19.3	−6.7	83
		136–238	130	–	−6.7	41
	Ethylene	−250	130–255	−20.1	–	96
	Isopropanol	138–240	130–200	−19.9	−6.1	83
		136–238	130	–	−6.1	41
	Polyethylene – intermolecular	110–200	100–300	−23	3	82,85
	intramolecular	110–200	100–300	−21	5	82,85
		180–300	175–275		20–24	83,84
	Propane	138–240	130–200	−22.9	−3.1	83
		136–238	130		−3.1	41
	Toluene	138–240	130–200	−30.7	4.7	83
		136–238	130	4.7	4.7	41
	Xylene	138–240	130–200	−33.8	7.8	83
		138–238	130		7.8	83
Methyl methacrylate	Butyraldehyde	0.1–400	50		−6.6	67
Styrene	Butyraldehyde	0.1–400	50		−15	63,65,68
	Carbon tetrachloride	600	60	−11		33
	Isobutyraldehyde	0.1–400	50		−6.6	68
	Triethylamine	181–440	60	−2 to −3		42
Vinyl acetate	Butyraldehyde	0.1–400	50		−3.2	67
Vinyl chloride	Butyraldehyde	0.1–400	50		−3.8	67

G. INFLUENCE OF PRESSURE ON COPOLYMERIZATION

TABLE 1. COPOLYMERIZATION PARAMETERS

Monomer 1	Monomer 2	Solvent	Absolute pressure (MPa)	Temperature (°C)	Copolymerization parameters r_1	r_2	Refs.
Acrylonitrile	Ethyl cinnamate	–	0.1	60	2.4	0.17	69
			80		2.8	0.21	69
			280		3.9	0.36	69
	Ethyl fumarate		0.1		5.5	0.15	105
			300		9.20	0.10	105
	Methyl acrylate		0.1		0.65	0.46	105
			300		0.76	0.43	105
	α-Methylstyrene		0.1		0.05	0.15	105
			300		0.04	0.10	105
	Styrene		0.1		0.02	0.65	105
			300		0.05	0.53	105
	Vinyl acetate		0.1		3.10	0.10	105
			300		3.50	0.05	105
Butyl acrylate	Styrene		0.1		0.20	0.62	105
			300		0.17	0.50	105
Cyclopentene	Maleic anhydride	Benzene	0.1	60	0.49	0.049	100
			200		0.21	0.041	100
			400		0.10	0.030	100
Ethylene	Vinyl acetate	–	3.5	62	0.74	1.51	97
			60	62	0.94	1.20	97
		–	51	120	0.6	0.95	41
			102–204		0.82	0.99	41
		Benzene	3.5	62	0.80	1.39	116
		Isopropyl alcohol	3.5	62	0.77	1.485	116
		N,*N*-Dimethylformamide	3.5	62	0.92	1.13	116
		tert-Butyl alcohol	3.5	62	0.74	1.50	116
			60	62	0.79	1.40	117
			120	62	0.80	1.37	118
	Vinyl butyrate	*tert*-Butyl alcohol	3.5	62	0.696	1.505	120
	Vinyl chloride	–	34	74	0.16	2.7	41
			102		0.16	2.1	41
			204		0.21	2.0	41
	Vinyl formate	*tert*-Butyl alcohol	3.5	62	0.586	1.29	120
	Vinyl isobutyrate	*tert*-Butyl alcohol	3.5	62	0.609	1.49	120
	Vinyl pivalate	*tert*-Butyl alcohol	3.5	62	0.645	1.49	120
	Vinyl propionate	*tert*-Butyl alcohol	3.5	62	0.674	1.50	120
			118	62	0.68	1.26	115
Maleic anhydride	Acenaphthylene	Acetic anhydride	0.1	60	0.07	0.43	43
			288		0.10	0.32	43
	2-Ethyl-1-hexene		0.1	70	0.15	0.015	106
			300		0.11	0.011	106
	1-Hexene		0.1	70	0.15	0.015	106
			200		0.10	0.010	106
	2-Methyl-2-butene		0.1	70	0.10	0.035	106
			300		0.005	0.025	106
	α-Methylstyrene		0.1	70	0.005	0.04	106
			300		0.03	0.03	106
	Norbornene		0.1	70	0.03	0.05	106
			300		0.02	0.03	106
	Thiophene		0.1	70	0.20	0	106
			320		0.15	0	106
Methyl acrylate	α-Methylstyrene		0.1	50	0.10	0.27	105
			300		0.08	0.23	105
	Croton aldehyde		0.1	50	20	0.02	105
			600		20	0.02	105
	Methyl vinyl ketone		0.1	50	0.50	1.80	105
			300		0.60	1.20	105
	Vinyl-2-pyridine		0.1	50	0.20	1.10	105
			300		0.30	1.05	105

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TABLE 1. *cont'd*

Monomer 1	Monomer 2	Solvent	Absolute pressure (MPa)	Temperature (°C)	Copolymerization parameters r_1	r_2	Refs.
Methyl methacrylate	Acenaphthylene	Toluene	0.1	60	0.57	3.1	43
			190		0.60	1.9	43
			390		0.58	1.6	43
	Acrylonitrile	Toluene	0.1	70	1.34	0.12	44
			10		1.46	0.37	44
			100		2.01	0.45	44
	Methyl acrylate	Toluene	3.4	62	2.42	0.31	119
			590	62	2.36	0.31	119
			118	62	2.40	0.30	119
Styrene	Acrylic acid	–	0.1	60	0.25	0.07	69
			100		0.27	0.08	69
			250		0.31	0.095	69
	Acrylonitrile	Toluene	0.1	70	0.37	0.07	44
			10		0.43	0.13	44
			100		0.55	0.14	44
	Diethyl fumarate	Benzene	0.1	60	0.26	0.06	70
			10		0.29	0.09	70
			100		0.32	0.15	70
	Diethyl maleate	Benzene	0.1	60	6.53	0.01	70
			10		7.08	0.02	70
			100		9.50	0.02	70
	Ethyl methacrylate	Benzene	0.1	50	0.56	0.50	45
			10		0.62	0.55	45
			100		0.72	0.67	45
	Methyl methacrylate	–	0.1	60	0.704	0.159	69
			100		0.710	0.163	69
			300		0.718	0.171	69
	Methyl acrylate	Toluene	3.4	62	1.20	0.09	119
			59	62	1.12	0.11	119
			118	62	1.10	0.13	119
	Methylacrylate	Toluene	3.4	62	0.86	0.38	119
			59	62	0.82	0.39	119
			118	62	0.77	0.41	119
	Vinyl acetate	Benzene	0.1	60	44	0.01	46
			10		46	0.01	46
			100		47	0.01	46
	Vinylidene chloride	–	0.1	60	2.6	0.04	47
			150		2.5	0.04	47
			200		2.5	0.01	47
	Vinylidene chloride	–	0.1	60	0.54–0.55	0.80–1.135	69,105
			200		0.560	1.145	69
			300		0.605–0.70	0.80–1.165	69,105
Vinyl acetate	Isopropyl acetate		0.1		0.90	0.30	105
			300		0.60	0.35	105
	Vinylidene chloride	–	0.1	60	0.025	3.2	47
			150		0.025	3.1	47
			250		0.03	3.1	47
Vinylidene chloride	Indene	–	0.1	60	0.475	0.205	69
			200		0.500	0.215	69
			400		0.525	0.225	69

TABLE 2. REACTIVITY RATIOS OF TERPOLYMERIZATION[a]

Pressure (MPa)	r_{12}	r_{21}	r_{13}	r_{31}	r_{23}	r_{32}	Refs.
0.1	5.5	0.15	0.02	0.65	0.06	0.15	107,108
300	3.2	0.10	0.05	0.53	0.15	0.25	

[a]Monomers: (i) acrylonitrile; (ii) diethyl fumarate; (iii) styrene.

TABLE 3. ACTIVATION VOLUMES

Monomer 1	Monomer 2	Solvent	Absolute pressure (MPa)	Temperature (°C)	Activation volume (cm^3/mol) Δv^*	$\Delta v_{11}^* - \Delta v_{12}^*$	$\Delta v_{22}^* - \Delta v_{21}^*$	Refs.
Acrylonitrile	Ethylfumarate		0.1–300	50		−5.0	3.7	107
	Methyl acrylate		0.1–300	50		−1.5	~0	105
	α-Methylstyrene		0.1–300	50		~0	3.7	105
	Styrene		0.1–300	50		−8.5	1.9	105
	Vinyl acetate		0.1–300	50		−1.1	6.3	105
Butyl acrylate	Styrene		0.1–300	50		0	1.0	105
Cyclopentene	Maleic anhydride	Benzene	0.1	60	−26.4 to −37.4			100
Ethylene	Vinyl acetate	–	−0.1–600	60		9.3	2.7	100
			2.5–60	62		−11.3	11.9	97
Ethylene	Vinyl acetate	–	110–190	240	−36			72
Indene	Acrylonitrile	Heptane		50	−17.5	0	−4.3	48
		Dioxane		50	−18.0	0	−3.6	48
	Methyl vinyl ketone	Heptane		50	−24.0	2.5	−1.0	49
	Methyl methacrylate	Heptane		50	−20.1	0	0	48
	Vinylidiene chloride				−17.8 to −19.1			71
Isoprene	Acrylonitrile	Heptane		50	−27.0	–	–	50
Maleic anhydride	2-Ethyl-1-hexene		0.1–300	70	−36			106
	2,3-Dimethyl-2-butene		0.1–300	70	−25			106
	1,1-Diphenyl ethylene		0.1–300	70	−50 to −60			106
	1-Hexane		0.1–200	70	−18			106
	Cyclohexene		0.1–300	70	−23			121
	2-Methyl-2-butene		0.1–300	70	−25			106
	α-Methylstyrene		0.1–300	70	−35			106
	Norbornene		0.1–300	70	−28			106
	Thiophene		0.1–320	70	−26			106
	trans-Stilbene		0.1–300	60	−25			106

COPOLYMERIZATION OF MALEIC ANHYDRIDE WITH MONO- AND POLY-SUBSTITUTED OLEFINS 160

Substitution	Comonomer	Activation volume (cm^3/mol)
mono-	=CH(Pr)	−16.6
	=CH(Bu)	−18.0
	=CH(*i*-Pr)	−17.0
	=CH(*i*-Bu)	−17.5
	=CH(*t*-Bu)	−31
	=CH(Ph)	−18.5
di-(1,2)	CH(Me)=CH(Et)	−19.0
	CH(Me)=CH(*i*-Pr)	−20.0
	CH(Me)=CH(*t*-Bu)	−35
	CH(*i*-Pr)=CH(*i*-Pr)	−18.5
	CH(Ph)=CH(Ph)	−25.0
gem-	=C(Me)(Pr)	−31
	=C(Me)(*i*-Pr)	−33
	=(Me)(*t*-Bu)	−60
	=C(Me)(CH_2*t*-Bu)	−46
	=C(Me)(Ph)	−35
	=C(Et)(Et)	−34
	=C(Et)(Bu)	−36
	=(Ph)(Ph)	−50
tri-	CH(Me)=C(Me)(Me)	−25
	CH(Et)=C(Me)(Me)	−31
tetra-	C(Me)(Me)=C(Me)(Me)	−25

Methyl acrylate	α-Methylstyrene	0.1–300	50		1.0	0.7	105
	Croton aldehyde	0.1–300	50		0	0	105
	Maleic anhydride			−18 to −25			71
	Methyl vinyl ketone	0.1–300	50		−1.9	1.9	105
	Vinyl-2- pyridine	0.1–300	50		−3.7	~0	105

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TABLE 3. *cont'd*

Monomer 1	Monomer 2	Solvent	Absolute pressure (MPa)	Temperature (°C)	Activation volume (cm^3/mol) Δv^*	$\Delta v_{11}^* - \Delta v_{12}^*$	$\Delta v_{22}^* - \Delta v_{21}^*$	Refs.
Methyl methylacrylate	Methyl acrylate		3.4–118	62		0.9	1.8	119
Styrene	Diethyl fumarate				−18 to −25			71,105
	Methyl acrylate				−13.8 to −14.5			71
	Methyl methacrylate				−18 to −25			71
	Vinyl-2-pyridine				−23.5 to −26	−2.2	0	71

TABLE 4. *Q, e* **VALUES**

Monomer	Q^a At 0.1 MPa	Q^a At 300 MPa	e^a At 0.1 MPa	e^a At 300 MPa	Refs.
Acrylic acid	0.53 (0.83)[b]	0.45	+0.91 (0.88)[b]	+0.64	108
Acrylonitrile	0.53 (0.50)	0.62	+0.90 (1.20)	+0.95	105
n-Butyl acrylate	0.40 (0.30)	0.40	+0.35 (0.74)	+0.35	105
n-Butyl methacrylate	0.62 (0.66)	0.70	−0.08 (0.34)	−0.54	92
Crotonaldehyde	0.015 (0.02)	0.015	+0.80 (0.84)	+0.80	105
Diethyl fumarate	0.20 (0.03)	0.25	+1.35 (2.17)	+1.25	105,109,25
Diethyl maleate	0.03 (0.04)	0.03	+0.62 (1.40)	+0.40	109
Ethylene	0.015 (0.013)	0.05	−0.15 (−0.15)	−0.02	110
Ethyl methacrylate	0.62 (0.57)	0.66	0.00 (0.35)	−0.35	92
Indene	0.30 (0.36)	0.28	−1.10 (−1.03)	−1.05	111,108
Isobutylene	0.025 (0.029)	0.022	−1.72 (−1.06)	−1.25	112
Isopropenyl acetate	0.08 (0.02)	0.12	−1.30 (−0.34)	−1.60	105
Methyl acrylate	0.44 (0.44)	0.44	+0.60 (0.60)	+0.60	105
Methyl methacrylate	0.63 (0.74)	0.63	−0.20 (0.38)	−0.05	108,25
α-Methylstyrene	1.40 (1.00)	1.65	−1.30 (−1.27)	−1.40	105
Methyl vinyl ketone	1.07 (0.65)	1.03	+0.95 (1.11)	+1.15	105
Propylene	0.010 (0.008)	0.015	−1.50 (−2.20)	−1.05	112
Styrene	1.20 (1.00)	1.35	−1.10 (−0.80)	−1.22	105,108,92
Vinyl acetate	0.06 (0.03)	0.05	−0.20 (−0.22)	−0.35	105,108,110
Vinyl chloride	0.0450 (0.056)	0.045	+0.16 (0.20)	+0.19	112,41
Vinyl pivalate	0.023 (–)	0.046	−0.19 (–)	−0.02	113
2-Vinylpyridine	1.05 (1.03)	0.95	−0.60 (−0.50)	−0.50	105
Vinylidene chloride	0.27 (0.26)	0.24	+0.42 (0.40)	+0.43	108

[a] Reference monomer is methyl acrylate (105).
[b] Values in parentheses refer to the values of styrene.

H. REFERENCES

1. M. G. Evans, M. Polanyi, Trans. Faraday Soc., **32**, 1333 (1936).
2. S. D. Hamann, in: R. S. Bradley (Ed.), "High Pressure Physics and Chemistry", London, 1963.
3. M. G. Gonikberg., "Chemical Equilibrium and Reaction Rates at High Pressures", 2nd. ed., Khmiya, Moscow, 1969, Td. Y. Ogo., "Kouatsu Kouatsu Kagaku", Nikkan Kogyo Shinbunsha, Tokyo, 1972.
4. G. Luft, Dissertation, Darmstadt 1967.
5. A. E. Steam, H. Eyring, Chem. Rev., **29**, 509 (1941).
6. G. Luft, Chem. Ing. Tech., **41**, 712 (1969).
7. Landolt–Boernstein, "Zahlenwerte und Funktionen aus Physik und Chemie", Berlin, 1951.
8. W. Kleinpaul, Z. Physik. Chem. (Frankfurt), **29**, 201 (1961); **30**, 262 (1961).
9. B. Raistrick, R. H. Sapiro, D. M. Newitt, J. Chem. Soc. (London), 1761 (1939).
10. M. G. Gonikberg, L. F. Vereshchagin, Zh. Fiz. Khim., **23**, 1447 (1949).
11. S. D. Hamann, Trans. Faraday Soc., **54**, 507 (1958).
12. A. E. Nicholson, R. G. W. Norrish, Disc. Faraday Soc., **22**, 104 (1956).
13. R. Steiner, G. Luft, Chem. Eng. Sci., **22**, 537 (1967).
14. G. M. Simmons, D. M. Mason, Chem. Eng. Sci., **27**, 89 (1972).
15. R. C. Neumann, Jr., Acc. Chem. Res., **5**, 381 (1972).
16. M. N. Romanii, K. E. Weale, Trans. Faraday Soc., **62**, 2264 (1966).
17. C. Walling, J. Pellon, J. Am. Chem. Soc., **79**, 4782 (1957).

18. R. L. Hemmings, K. E. Weale, "Chemical Reactions at High Pressures", Spon, London, 1967.
19. A. L. Shriers, B. F. Dodge, R. H. Bretton, "Free Radical Polymerization of Ethylene at High Pressures", paper presented on the A.I.Ch.E./London, 1965, Chem. Eng. Symp. Ser., **2**, 10 (1965).
20. R. O. Symcox, P. Ehrilich, J. Am. Chem. Soc., **84**, 531 (1962).
21. D. Lierse, G. Luft, unpublished.
22. T. Yala, G. Jenner, in: H. Kelm (Ed.), "High Pressure Chemistry", Reidel, Dordrecht, 1977, p. 376.
23. G. Jenner, Thesis, Strasbourg, 1966 and Bull. Soc. Chim. France, **1**, 344 (1967).
24. H. Abdi, G. Jenner, C. Brun, Makromol. Chem., **164**, 149 (1973).
25. J. A. Lamb, K. E. Weale, Symp. Phys. Chem. High Pressures, 229 (1963) (Soc. Chem. Ind., London).
26. A. A. Zharov, N. S. Enikolopyan, Zh. Fiz. Khim., **38**, 2727 (1964).
27. H. Asai, T. Imoto, J. Chem. Soc. (Japan), **66**, 863 (1963); Nirrou Kagaku Zasshi, **84**, 863 (1963).
28. D. W. Brown, L. A. Wall, J. Phys. Chem., **67**, 1016 (1963).
29. A. C. Toohey, K. E. Weale, Trans. Faraday Soc., **58**, 2439, 2446 (1962).
30. H. Asai, T. Imoto, J. Chem. Soc. (Japan), **66**, 871 (1963).
31. K. Salahuddin, K. E. Weale, in: K. E. Weale (Ed.), "Chemical Reactions at High Pressures", Spon, London, 1967.
32. A. E. Nicholson, R. G. W. Norrish, Disc. Faraday Soc., **22**, 104 (1956).
33. C. Walling, J. Pellon, J. Am. Chem. Soc., **79**, 4776 (1957).
34. P. P. Kobeko, E. U. Kuvshinskii, A. S. Semonova, Zh. Fiz. Khim., **24**, 345 (1950).
35. F. M. Merrett, R. G. W. Norrish, Proc. Roy. Soc. (London) A, **206**, 309 (1951).
36. V. M. Zhulin, M. G. Gonikberg, R. F. Barkova, Izv. Akad. Nauk. SSSR, 432 (1965).
37. A. H. Ewald, Disc. Faraday Soc., **22**, 138 (1956).
38. C. Walling, J. Pellon, J. Am. Chem. Soc., **79**, 4786 (1957).
39. A. H. Ewald, Disc. Faraday Soc., **22**, 146 (1956).
40. C. Walling, G. Metzger, J. Am. Chem. Soc., **81**, 5365 (1959).
41. P. Ehrlich, G. Mortimer in Adv. Polymer Sci., **7** (3), 420 (1970).
42. A. C. Toohey, K. E. Weale, Trans. Faraday Soc., **58**, 2446 (1962).
43. M. N. Romani, K. E. Weale, Thesis, London University 1964.
44. R. D. Burkhart, N. L. Zutti, J. Polym. Sci., **57**, 793 (1962).
45. H. Asai, T. Imoto, J. Chem. Soc. (Japan), **85**, 155 (1964).
46. Ibid., **85**, 152 (1964).
47. B. S. Maulik, K. E. Weale, in: K. E. Weale (Ed.), "Chemical Reactions at High Pressures", Spon, London, 1967.
48. T. Yala, G. Jenner, Communication 10, Meeting EHPRG, Strasbourg, 1972.
49. T. Yala, Thesis 3, Cycle, Strasbourg, 1972.
50. G. Jenner, J. Rimmelin, unpublished.
51. J. P. J. Higgins, K. E. Weale, J. Polym. Sci. A-1, **8**, 1705 (1970).
52. R. I. Baikova, V. M. Zhulin, M. G. Gonikberg, Izv. Akad. Nauk SSSR, Ser. Khim., 154 (1966).
53. V. M. Zhulin, A. R. Volchek, M. G. Gonikberg, A. S. Shashkow, S. V. Zotova, Vysokomol. Soedin., **7A**, 1513 (1972).
54. J. P. J. Higgins, K. E. Weale, unpublished.
55. S. A. Mehdi, K. E. Weale, unpublished.
56. A. R. Volchek, V. M. Zhulin, A. S. Shashkov, O. M. Nefedov, A. A. Ivashenko, Vysokomol. Soedin., **A14**, 2258 (1973).
57. V. M. Zhulin, N. V. Klimentova, M. G. Gonikberg, V. V. Korshak, A. P. Suprun, Vysokomol. Soedin., **11A**, 101 (1969).
58. V. M. Zhulin, A. P. Suprun, G. P. Lopatina, T. A. Soboleva, A. S. Shashkov, M. G. Gonikberg, G. P. Shakhovskoi, Vysokomol. Soedin., **13A**, 2518 (1971).
59. V. M. Zhulin, M. G. Gonikberg, A. R. Volchok, O. M. Nefedov, A. S. Shashkov, Vysokomol. Soedin., **9A**, 2153 (1971).
60. A. R. Volchek, V. M. Zhulin, A. S. Shashkov, O. M. Nefedov, A. A. Ivashenko, Vysokomol. Soedin., **A14**, 2258 (1973).
61. G. Luft, H. Bitsch, unpublished.
62. V. M. Zhulin, Thesis for Doctor of Science degree, Moscow (1971).
63. V. M. Zhulin, M. G. Gonikberg, V. N. Zagorbinina, Izv. Akad. Nauk, SSSR, Ser. Khim., 827 (1966).
64. V. M. Zhulin, R. I. Baikova, M. G. Gonikberg, G. P. Shakhovskoi, Vysokomol. Soedin., **13**, 1071 (1971).
65. V. M. Zhulin, M. G. Gonikberg, V. N. Zagorbinina, Izv. Akad. Nauk, SSSR, Ser. Khim., 997 (1966).
66. M. G. Gonikberg, R. I. Baikova, V. M. Zhulin, Izv. Akad. Nauk, SSSR, 1164 (1962).
67. V. M. Zhulin, M. G. Gonikberg, A. L. Goff, V. N. Zagorbinina, Vysokomol. Soedin., **11A**, 777 (1969).
68. V. M. Zhulin, M. G. Gonikberg, V. N. Zagorbinina, Dokl. Akad. Nauk SSSR, **163**, 106 (1965).
69. W. F. Dellsperger, K. E. Weale, ACS Polym. Preprints, **11**, 645 (1970).
70. H. Asai, J. Chem. Soc. (Japan), **85**, 247 (1964).
71. J. A. Lamb, Thesis, London University, 1961.
72. H. Bitsch, G. Luft, Angew. Makromol. Chem., **32**, 17 (1973).
73. Th. J. van der Molen, Preprint IUPAC Symp. on Macromolecular Chemistry, Budapest, 1969.
74. Th. J. van der Molen, International Symp. on Macromolecules, Helsinki, 1972.
75. G. Luft, P. Mehrling, H. Seidl, Angew. Makromol. Chem., **73**, 85 (1978).
76. P. Mehrling, G. Luft, B. Giese, H. Seidl, Chem. Ber., **118**, 240 (1985).
77. M. Buback, H. Lendle, Z. Naturforsch., **34A**, 1482 (1979).
78. R. Halle, Plastic Compound., 73 (1980).
79. M. Buback, H. Lendle, Makromol. Chem., **184**, 193 (1983).
80. M. Buback, Z. Naturforsch., **39A**, 399 (1984).
81. G. Luft, P. -Ch. Lim, S. Pavlakis, J. Macromol. Sci. Chem. A, **22** (9), 1183 (1985).

82. G. Luft, R. Kämpf, H. Seidl, Angew. Makromol. Chem., **108**, 203 (1982).
83. K. Yamamoto, J. Macromol. Sci. Chem. A, **17** (3), 415 (1982).
84. K. Yamamoto, M. Sugimoto, J. Macromol. Sci. Chem. A, **13** (8), 1067 (1979).
85. P.-Ch. Lim, G. Luft, Makromol. Chem., **184**, 849 (1983).
86. M. Yokawa, Y. Ogo, Makromol. Chem., **177**, 429 (1976).
87. P. Ehrlich, G. A. Mortimer, Adv. Polym. Sci., **7**, 386 (1970).
88. M. Buback, H. Lendle, Z. Naturforsch, **36A**, 1371 (1981).
89. Y. Ogo, M. Yokawa, T. Imoto, Makromol. Chem., **171**, 123 (1973).
90. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 2903 (1974).
91. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 2913 (1974).
92. H. Asai, T. Imoto, Nippon Kagaku Zasshi, **85**, 149 (1964).
93. T. Sasuga, M. Takehisa, J. Makromol. Sci. Chem. A, **12** (9), 1307 (1978).
94. T. Sasuga, M. Takehisa, J. Makromol. Sci. Chem. A, **12** (9), 1333 (1978).
95. T. Sasuga, S. Kawanishi, M. Takehisa, J. Makromol. Sci. Chem. A, **14** (4), 605 (1980).
96. M. Buback, Ch.-R. Choe, E.-U. Franck, Makromol. Chem., **185**, 1685 (1984).
97. F. de Kok, D. Heikens, A. L. German, Proc. 4th Int. Conf. High Pressure, Kyoto 1974, Phys. Soc. Japan.
98. Y. Ogo, M. Yoshikawa, S. Ontani, T. Imoto, J. Polym. Sci. Chem., **16**, 1413 (1978).
99. M. Yokawa, I. Yoshida, Y. Ogo, Makromol. Chem., **178**, 443 (1977).
100. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **163**, 135 (1973).
101. R. C. Neumann, J. V. Behar, J. Am. Chem. Soc., **91**, 6024 (1969).
102. R. C. Neumann, J. V. Behar, Tetrahedron Lett., 3281 (1968).
103. R. C. Neumann, J. V. Behar, J. Am. Chem. Soc., **89**, 4549 (1967).
104. R. C. Neumann, R. J. Bussey, J. Am. Chem. Soc., **92**, 2440 (1970).
105. G. Jenner, S. Aieche, J. Polym. Sci. Chem., **16**, 1017 (1978).
106. M. Kellou, G. Jenner, Makromol. Chem., **180**, 1687 (1979).
107. S. Aieche, G. Jenner, Polymer, **19**, 1236 (1978).
108. W. Dellsperger, K. Weale, Am. Chem. Soc., Polym. Prepr., **11**, 645 (1970).
109. H. Asai, Nippon Kagaku Zasshi, **85**, 247 (1964).
110. R. van der Meer, E. H. M. van Gorp, A. German, J. Polym. Sci., Polym. Chem. Ed., **15**, 1489 (1977).
111. T. Yala-Lutokanu, G. Jenner, A. Deluzarche, Bull. Soc. Chim. France, 609 (1974).
112. K. Hamanoue, Rev. Phys. Chem. Japan, **38**, 120 (1968).
113. R. van der Meer, A. German, 15th Meeting EHPRG, Darmstadt, Germany (1977).
114. G. Luft, W. Henning, M. Dorn, Angew. Makromol. Chem. (in press).
115. J. Schrijver, J. L. Ammerdorffer, A. L. German, J. Polym. Sci. Chem., **20**, 2693 (1982).
116. R. van der Meer, M. W. A. M. Aarts, A. L. German, J> Polym. Sci. Chem., **18**, 1347 (1980).
117. R. van der Meer, A. L. German, J. Polym. Sci. Chem., **17**, 571 (1979).
118. R. van der Meer, A. L. German, D. Heikens, J. Polym. Sci. Chem., **15**, 1765 (1977).
119. J. Schrijver, A. L. German, J. Polym. Sci. Chem., **21**, 341 (1983).
120. R. van der Meer, E. H. M. van Gorp, A. L. German, J. Polym. Sci. Chem., **15**, 1498 (1977).
121. G. Jenner, M. Kellou, M. Papadopoulos, Angew. Chem. Suppl., 929 (1982).
122. T. Bartnik, B. Baranowski, Polym. J. Chem., **53** (3), 741 (1979)
124. A. Crosato-Arnaldi, G. B. Guarise, G. Talamini, Polymer, **10**, 385 (1969).
125. G. B. Gualise, Polymer, **7**, 497 (1966).
126. G. B. Gualise, G. Palma, E. Siviero, G. Talamini, ibid., **11**, 613 (1970).
127. T. Imoto, Mem. Fac. Eng., Osaka City Univ., **3**, 217 (1961).
128. T. Imoto, H. Asai, Nippon Kagaku Zasshi, **85**, 149 (1964).
129. T. Imoto, H. Asai, ibid., **85**, 347 (1964).
130. T. Imoto, T. Nakajima, Kogyo Kagaku Zasshi, **69**, 520 (1966).
131. T. Imoto, Y. Ogo, Y. Hashimoto, ibid., **70**, 1952 (1967).
132. G. Jenner, M. Kellou, Abstracts, 15th Annual Meeting of European High Pressure Research Group, Darmstadt, 1977, p. 12.
133. A. R. Khalilpour, G. Jenner, A. Deluzarche, Bull. Soc. Chim. Fr., **3–4** (2), 583 (1976).
134. G. Luft, P. Lim, M. Yokawa, Makromol. Chem., **184**, 207 (1983).
135. T. Mitani, Y. Ogo, T. Imoto, Rev. Phys. Chem. Soc. Japan, **42**, 25 (1972).
136. P. W. Moore, F. W. Ayscough, J. G. Clouston, J. Polym. Sci., Polym. Chem. Ed., **15**, 1291 (1977).
137. P. W. Moore, J. G. Clouston, R. P. Chaplin, ibid., **19**, 1659 (1981).
138. P. W. Moore, J. G. Clouston, R. P. Chaplin, ibid., **21**, 2491 (1983).
139. P. W. Moore, J. G. Clouston, R. P. Chaplin, ibid., **21**, 2503 (1983).
140. H. Nakomoto, Y. Ogo, T. Imoto, Makromol. Chem., **111**, 104 (1968).
141. J. Osugi, K. Hamanoue, E. Hirayama, Nippon Kagaku Zasshi, **87**, 1032 (1966).
142. J. Osugi, T. Mizukami, ibid., **87**, 1157 (1966).
143. J. Osugi, K. Hamanoue, ibid., **90**, 544 (1969).
144. J. Osugi, K. Hamanoue, ibid., **90**, 549 (1969).
145. T. Sasuga, N. Morishita, A. Udagawa, Y. Kusama, M. Takehisa, J. Polym. Sci., Polym. Chem. Ed., **14**, 2527 (1976).
146. T. Sasuga, S. Kawanishi, J. Phys. Chem., **83**, 3290 (1979).
147. T. Sasuga, S. Kawanishi, M. Takehisa, J. Macromol. Sci.-Chem. A, **14** (4), 605 (1980).
148. T. Sasuga, S. Kawanishi, M. Takehisa, Macromolecules, **16**, 545 (1983).

149. T. Sano, Y. Ogo, T. Imoto, Kobunshi Ronbunshu., **32**, 338 (1975).
150. Y. Tanaka, K. Noguchi, T. Shibata, A. Okada, Macromolecules, **11**, 1017 (1978).
151. O. N. Tsvetkov, T. N. Tsvetkova, V. N. Monastyrskii, Plast. Massy, **9**, 9 (1975).
152. T. Wada, T. Watanabe, M. Takehisa, J. Polym. Sci., Polym. Chem. Ed., **10**, 2639 (1972).
153. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 179 (1974).
154. M. Yokawa, Y. Ogo, T. Imoto, Makromol. Chem., **175**, 2913 (1974).
155. A. A. Zharov, Al. Al. Berlin, N. S. Enikolopyan, J. Polym. Sci., Part C, **16**, 2313 (1967).
156. V. A. Zhorin, A. A. Zharov, Yu. V. Kissin, N. S. Enikolopyan, Dokl. Akad. Nauk SSSR, **219**, 674 (1974).
157. V. M. Zhulin, M. G. Gonikberg, Dokl. Akad. Nauk, SSSR, **163**, 106 (1965).
158. V. M. Zhulin, A. R. Volchek, M. G. Gonikberg, A. S. Shashkov, S. V. Zotova, ibid., **14A**, 1484 (1972).
159. V. M. Zhulin, Z. G. Makarova, N. V. Klimentova, A. P. Suprun, A. N. Akopyan, G. P. Shakhovskoi, ibid., **24A**, 2621 (1982).
160. G. Jenner, M. Kellou, Tetrahedron, **37**, 1153–60 (1981).

Activation Enthalpies and Entropies of Stereocontrol in Free Radical Polymerizations

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A. INTRODUCTION

In free radical polymerization, the tacticity of the resulting polymer depends on the propagation step, i.e., on stereocontrol by the propagating end. The following simple cases exist:

(a) The last monomeric unit does not control the stereospecificity of the propagation step. Isotactic (i) and syndiotactic (s) additions thus exhibit the same conditional probabilities ($p_i = p_s$). The mole fractions x of the resulting isotactic and syndiotactic diads are equal ($x_i = x_s$). If all steps occur at random, the four conditional probabilities for the formation of triads are equal ($p_{i/i} = p_{i/s} = p_{s/i} = p_{s/s}$), and so are the conditional probabilities for the formation of tetrads, pentads, etc. The resulting polymer is a true atactic polymer. It follows that

$$p_i = x_i = 1/2 \qquad p_s = x_s = 1/2 \tag{A1}$$

$$x_{ii} = x_{is} = x_{si} = x_{ss} = 1/4 \tag{A2}$$

(b) The last monomeric unit of the chain, but not the last diad, controls the propagation step. The probabilities for the formation of isotactic and syndiotactic diads are thus different ($p_i \neq p_s$). The process is Bernoullian (zeroth order Markov statistics) with respect to the formation of diads. Consequently, the probability of forming an isotactic diad at an existing isotactic diad equals the probability of forming an isotactic diad at an existing syndiotactic diad ($p_{i/i} = p_{s/i}$), and, by analogy, $p_{s/s} = p_{i/s}$. It follows for the diads (6,8)

$$p_i = x_i \neq 1/2 \qquad p_s = x_s \neq 1/2 \tag{A3}$$

and for the triads

$$x_{ii} = p_i^2 = x_i^2 \qquad x_{ss} = p_s^2 = x_s^2 \tag{A4}$$

$$x_{ht} = p_i p_s + p_s p_i = 2\,x_i x_s = 2x_i(1 - x_i) \tag{A5}$$

where x_{ii} is the mole fraction of isotactic triads and x_{ss} is the mole fraction of syndiotactic triads. x_{ht} is the mole fraction of heterotactic triads, regardless of whether formed by the formation of an isotactic unit at a syndiotactic one (conditional probability $p_{s/i}$) or vice versa (conditional probability $p_{i/s}$).

(c) The last two monomeric units of the chain, i.e., the last monomeric diad, regulate the stereocontrol of the propagation step. The four conditional probabilities do not equal each other ($p_{i/i} \neq p_{i/s} \neq p_{s/i} \neq p_{s/s}$). The mole fractions of isotactic (x_{ii}), syndiotactic (x_{ss}) and heterotactic triads (x_{ht}) are thus given by (8,11)

$$x_{ii} = x_i\,p_{i/i} \qquad x_{ss} = x_s\,p_{s/s} \tag{A6}$$

$$x_{ht} = x_i\,p_{i/s} + x_s p_{s/i} \tag{A7}$$

This process corresponds to a Markov trial of first order.

(d) Second order Markov statistics are present if the last two diads (i.e., the last triad or the last three monomeric

units) control the propagation step. In this case

$$x_{iii} = x_{ii}p_{i/i} \qquad x_{sss} = x_{ss}p_{s/s} \tag{A8}$$

$$x_{iis} = x_{ii}p_{i/s} \qquad x_{ssi} = x_{ss}p_{s/i} \tag{A9}$$

$$x_{iss} = x_{is}p_{s/s} \qquad x_{sii} = x_{si}p_{i/i} \tag{A10}$$

$$x_{isi} = x_{is}p_{s/i} \qquad x_{sis} = x_{si}p_{i/s} \tag{A11}$$

In order to distinguish between Markov second order and Markov first order mechanisms, at least tetrads must be known.

The probability of the addition of a 'b' diad (b = i, s) to an existing end 'a' (a = i, s) is given by the corresponding rates R:

$$P_{a/b} = \frac{R_{a/b}}{R_{a/a} + R_{a/b}} \tag{A12}$$

with a = i, s or b = i, s, in the Markov first order case, or a = ii, is, si, ss b = ii, is, si, ss, in the Markov second order case.

Using these definitions and assuming steady states and the equality of instantaneous and final diad and triad fractions, one can express ratios of all diads, triads, and tetrads in terms of rate constants (Table 1) or ratios of rate constants in terms of diad, triad, etc. fractions (Table 2). The rate constants themselves can be calculated if the rate constants of propagation are known for the corresponding experimental conditions (1).

From the temperature dependence of the expressions given in Table 2, differences in activation enthalpies ($\Delta HA^{\ddagger} - \Delta H_B^{\ddagger}$) and differences in activation entropies ($\Delta S_A^{\ddagger} - \Delta S_B^{\ddagger}$) can be calculated:

$$\frac{k_A}{k_B} = \exp\left(\frac{\Delta S_A^{\ddagger} - \Delta S_B^{\ddagger}}{R}\right)\exp\left(\frac{\Delta H_A^{\ddagger} - \Delta H_B^{\ddagger}}{RT}\right) \tag{A13}$$

The quantities ($\Delta H_A^{\ddagger} - \Delta H_B^{\ddagger}$) and ($\Delta S_A^{\ddagger} - \Delta S^{\ddagger}$) may reflect activation parameters (2,3), conformation parameters (4,5), or combinations thereof.

Table 3 gives the difference $\Delta\Delta H^{\ddagger} = \Delta H_A^{\ddagger} - \Delta H_B^{\ddagger}$ and $\Delta\Delta S^{\ddagger} = \Delta S_A^{\ddagger} - \Delta S_B^{\ddagger}$ calculated from the temperature dependence of diad ratios x_i/x_s. In the case of Bernoulli trials, they represent $\Delta H_i^{\ddagger} - \Delta H_s^{\ddagger}$ and $\Delta S_i^{\ddagger} - \Delta S_s^{\ddagger}$, respectively. For first order Markov trials, they stand for $\Delta H_{s/i}^{\ddagger} - \Delta H_{i/s}^{\ddagger}$ and $\Delta S_{s/i}^{\ddagger} - \Delta S_{i/s}^{\ddagger}$, respectively. In general, both $\Delta\Delta H^{\ddagger}$ and $\Delta\Delta S^{\ddagger}$ depend on the solvent (9,10), and consequently on the monomer/solvent ratio (11,12), but they are nearly independent of monomer conversion.

Table 4 gives the differences in activation enthalpies for modes of addition other than s/i vs. i/s, and Table 5 gives the corresponding differences in activation entropies. Only a few of these activation parameters could be calculated from literature data as very few triad fractions have been reported.

For any two modes of addition, the compensation effect between $\Delta\Delta H^{\ddagger}$ and $\Delta\Delta S^{\ddagger}$ (10,12,43) for the polymerization of a given monomer, in different solvents and/or at different monomer/solvent ratios, can be described by

$$\Delta H_A^{\ddagger} - \Delta H_B^{\ddagger} = \Delta\Delta H_0^{\ddagger} - (\Delta S_A^{\ddagger} - \Delta S_B^{\ddagger})T_0 \tag{A14}$$

The compensation temperature T_0 has been found to be independent, of the mode of addition (e.g., A = i/i vs. B = i/s) within the limits of error (12,43). However, the compensation enthalpy $\Delta\Delta H_0^{\ddagger}$ depends on the mode of addition.

B. TABLES

TABLE 1. RATIOS OF SOME *i*-ADS FOR DIFFERENT MECHANISMS[a]

	Mechanism		
***i*-ad Ratio**	**Bernoulli**	**Markov 1st order**	**Markov 2nd order**
$\frac{x_i}{x_s}$	$\frac{k_i}{k_s}$	$\frac{k_{s/i}}{k_{i/s}}$	$\frac{k_{ss/i}(k_{si/i} + k_{ii/s})}{k_{ii/s}(k_{is/s} + k_{ss/i})}$
$\frac{x_{ii}}{x_{ss}}$	$\frac{k_i^2}{k_s^2}$	$\frac{k_{i/i}k_{s/i}}{k_{s/s}k_{i/s}}$	$\frac{k_{ss/i}k_{si/i}}{k_{ii/s}k_{is/s}}$
$\frac{x_{iii}}{x_{sss}}$	$\frac{k_i^3}{k_s^3}$	$\frac{k_{i/i}^2k_{s/i}}{k_{s/s}^2k_{i/s}}$	$\frac{k_{ss/i}k_{si/i}k_{ii/i}}{k_{ii/s}k_{is/s}k_{ss/s}}$
$\frac{x_{is}}{x_{ss}}$	$\frac{k_i}{k_s}$	$\frac{2k_{s/i}}{k_{s/s}}$	

[a] From Ref. 42.

TABLE 2. RATIOS OF RATE CONSTANT FOR MARKOV FIRST ORDER MECHANISMS[a]

Ratios of rate constants	**Expressions giving ratios of rate constants**
$\frac{k_{i/i}}{k_{i/s}}$	$\frac{2x_{ii}}{x_{is}}$
$\frac{k_{i/i}}{k_{s/i}}$	$\frac{2x_sx_{ii}}{x_ix_{is}} = \frac{1 + (2x_{ss}/x_{is})}{1 + (x_{is}/(2x_{ii}))}$
$\frac{k_{i/i}}{k_{s/s}}$	$\frac{x_sx_{ii}}{x_ix_{is}} = \frac{1 + (x_{is}/(2x_{ss}))}{1 + (x_{is}/(2x_{ii}))}$
$\frac{k_{i/s}}{k_{s/i}}$	$\frac{x_s}{x_i} = \frac{x_{ss} - (x_{is}/2)}{x_{ii} - (x_{is}/2)}$
$\frac{k_{i/s}}{k_{s/s}}$	$\frac{x_sx_{is}}{2x_ix_{ss}} = \frac{1 + (x_{is}/(2x_{ss}))}{1 + (2x_{ii}/(x_{is}))}$
$\frac{k_{s/i}}{k_{s/s}}$	$\frac{x_{is}}{2x_{ss}}$

[a] From Ref. 43.

TABLE 3. $\Delta H^{\ddagger}_{s/i} - \Delta H^{\ddagger}_{i/s}$ AND $\Delta S^{\ddagger}_{s/i} - \Delta S^{\ddagger}_{i/s}$ OF FREE RADICAL POLYMERIZATIONS IN DIFFERENT SOLVENTS*

Monomer (+additive)	Vol.%	Solvent	$\Delta H^{\ddagger}_{s/i} - \Delta H^{\ddagger}_{i/s}$ (J/mol)	$\Delta S^{\ddagger}_{s/i} - \Delta S^{\ddagger}_{i/s}$ (J/mol/K)	Remarks	Refs.
Acrylonitrile	30	Toluene	3770 ± 59	2.7 ± 0.19	–	10,14
	100	Bulk	0	0	–	13
Acrylic acid, isopropyl ester	?	?	0	−5.9	[a]	15
–, methyl ester	?	?	0	−5.9	[a]	15
α-Chloroacrylic acid, methyl ester	?	Toluene	3800 ± 190	2.5 ± 0.59	–	16
Methacrylonitrile	100	Bulk	250 ± 50	3.0 ± 0.14	–	17
	100	Bulk	−4100 ± 71	−10.8 ± 0.21	–	18
–, (+1 mol $SnCl_4$/2 mol monomer)	3:1[b]	Toluene	−630 ± 54	−0.7 ± 0.18	–	18
–, (+1 mol $SnCl_4$/1 mol monomer)	2:1[b]	Toluene	170 ± 63	−0.1 ± 0.18	–	18
–, (+1 mol $SnCl_4$/1 mol monomer)	1:1[b]	Benzene	251	0.75 ± 0.88	–	18
–, (+1 mol $SnCl_4$/2 mol monomer)	1:1[b]	Benzene	2350 ± 38	9.76 ± 0.096	–	18
Methacrylic acid	10	1-Propanol	3980 ± 92	−2.2 ± 0.32	–	19
	20	1-Propanol	3850 ± 80	−1.8 ± 0.28	–	19
	20	2-Propanol	7800 ± 180	10.5 ± 0.63	–	19
	20	Methanol	3140 ± 42	−4.1 ± 0.13	–	19
–, anhydride	?	Benzene	13300 ± 210	44.4 ± 0.63	–	20
–, benzyl ester	50	Toluene	2300 ± 130	−1.7 ± 3.8	[a]	21
–, *s*-butyl ester	50	Toluene	3430 ± 63	2.0 ± 0.22	–	10,22
–, *t*-butyl ester	50	Toluene	2090 ± 80	−2.6 ± 0.29	–	10,22
–, *p*-carboxyphenyl ester	?	Dimethylformamide	4310	2.09	[c]	23
	?	*p*-Cetyloxybenzoic acid	0	−6.87	–	10,23
–, cyclohexyl ester	100	Bulk	1210 ± 80	−5.0 ± 0.20	–	10,24
–, glycidyl ester	45	Acetone	700 ± 100	9.5 ± 0.36	–	10,25
	50	Dimethylformamide	5780 ± 42	5.0 ± 0.15	–	10,25
	57	Toluene	8600 ± 130	14.2 ± 0.042	–	10,25
	100	Bulk	9800 ± 110	17.5 ± 0.28	–	10,25
–, α-methylbenzyl ester	50	Toluene	3640 ± 59	2.2 ± 0.22	–	10,22
–, methyl ester	9.1[d]	Acetone	6280 ± 75	10.2 ± 0.23	–	12
	9.1[d]	Acetonitrile	5360 ± 75	7.3 ± 0.26	–	12
	9.1[d]	Benzene	3480 ± 71	2.3 ± 0.23	–	12
	9.1[d]	Butanone	3900 ± 160	4.2 ± 0.46	–	12
	9.1[d]	γ-Butyrolactone	3140 ± 88	1.4 ± 0.27	–	12
	9.1[d]	Chloroform	5100 ± 71	7.5 ± 0.23	–	12
	9.1[d]	Dimethylformamide	3500 ± 150	3.4 ± 0.46	–	12
	9.1[d]	Dimethyl sulfoxide	3350 ± 96	2.5 ± 0.28	–	12
	9.1[d]	1,4-Dioxane	5317 ± 8	7.4 ± 0.30	–	12
	9.1[d]	Methylene chloride	3100 ± 67	2.0 ± 0.22	–	12
	9.1[d]	Pyridine	4860 ± 75	7.7 ± 0.22	–	12
	9.1[d]	Tetrachloroethane	5200 ± 100	7.2 ± 0.29	–	12
	9.1[d]	Tetrahydrofuran	3900 ± 110	4.4 ± 0.33	–	12
	9.1[d]	Toluene	2810 ± 96	0.8 ± 0.26	–	12
	20	Paraffin	2720 ± 34	1.6 ± 0.13	–	27
	25	Acetone	7030 ± 75	9.0 ± 0.23	IR	9
	25	Carbon tetrachloride	5860 ± 21	5.36 ± 0.067	IR	9
	25	Chloroform	9380 ± 71	16.0 ± 0.21	IR	9
	25	Dichloroethylene	6700 ± 46	8.4 ± 0.13	IR	9
	25	Ethanol	4980 ± 50	3.6 ± 0.14	IR	9
	25	*n*-Heptane	5200 ± 210	3.69 ± 0.059	IR	9
	25	*n*-Hexane	6700 ± 29	8.50 ± 0.092	IR	9
	28.5	Toluene	5650 ± 80	3.9 ± 0.26	–	10,26
	50	Toluene	2900 ± 420	1.3 ± 1.3	[a]	21
	100	Bulk	6200 ± 110	9 ± 3.8	–	27
	100	Bulk	4480	4.2	[a]	7
	100	Bulk	3800 ± 230	−0.04 ± 0.63	[a]	25
	100	Bulk	2300 ± 54	−1.8 ± 0.16	–	28
	100	Bulk	3560 ± 84	3.3 ± 0.24	–	12
–, –, (+8.7 mol $ZnCl_2$/1 mol MMA)	1.1[e]	Water	2850 ± 71	2.2 ± 0.20	–	10,29
–, –, (+1 mol $ZnCl_2$/1 mol MMA)	3.2[e]	Ethyl acetate	4100 ± 71	5.2 ± 0.2	–	10,29
–, –, (+0.04 mol $ZnCl_2$/1 mol MMA)	100	Bulk	2700 ± 190	1.13 ± 0.054	–	10,29
–, –, (+1 mol $ZnCl_2$/2 mol MMA)	100	Bulk	−130 ± 80	−9.80 ± 0.67	–	27
–, –, (+1 mol $ZnCl_2$/1 mol MMA)	100	Bulk	−40 ± 130	−6.7 ± 0.46	–	27
–, –, (+1 mol $SnCl_4$/2 mol MMA)	100	Bulk	800 ± 100	−8.2 ± 0.33	–	27

*In all cases, except those marked by IR (infrared) (see Remarks column), diad fractions were determined via NMR.

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TABLE 3. *cont'd*

Monomer (+ additive)	Vol.%	Solvent	$\Delta H^{\ddagger}_{s/i} - \Delta H^{\ddagger}_{i/s}$ (J/mol)	$\Delta S^{\ddagger}_{s/i} - \Delta S^{\ddagger}_{i/s}$ (J/mol/K)	Remarks	Refs.
–, –, (+1 mol $SnCl_4$/1 mol MMA)	100	Bulk	420 ± 13	− 14.07 ± 0.054	–	27
–, L-menthyl ester	50	Toluene	960 ± 46	0.3 ± 0.18	–	30
–, phenetyl ester	50	Toluene	5600 ± 1700	4.6 ± 4.6	[a]	21
Trifluorochloroethylene	0.4[f]	Carbon tetrachloride	0	− 5.8	[a]	31
Vinyl bromide	100	Bulk	710	2.5	[a]	32
	?	*n*-Butyl acetate	250	2.1	[a]	32
Vinyl bromide	6.5	*o*-Dichlorobenzene	1930 ± 54	5.3 ± 0.17	–	34
	100	Bulk	2140 ± 17	5.99 ± 0.080	–	34
	100	Bulk	1270 ± 30	2.72	–	44
N-Vinyl carbazole	?	Dimethylacetamide	− 4600 ± 300	− 20 ± 1	–	45
	?	Dimethylformamide	− 2850 ± 100	− 13.5 ± 0.4	–	45
	?	Dimethylsulfoxide	− 3300 ± 500	− 15.4 ± 1.4	–	45
	?	Hexamethylphosphoric triamide	− 3400 ± 200	− 14.7 ± 0.7	–	45
	?	Hexane/nonane	− 1500 ± 200	− 10.5 ± 0.7	–	45
	?	Methylene dichloride	− 2650 ± 100	− 11.7 ± 0.4	–	45
	?	Toluene	− 3650 ± 1000	− 16.0 ± 0.8	–	45
Vinyl chloride	10	Cyclohexane	1300 ± 84	2.5 ± 0.42	[a]	36
	33.3	Water/ethanol	5000 ± 1000	18 ± 4.2	–	35
	33.3	Water	1500 ± 630	2.9 ± 2.0	–	35
	100	Bulk	2900 ± 160	6.78 ± 0.067	IR	37
	100	Bulk	2260 ± 50	5.7 ± 0.14	–	34
	100	Bulk	1090 ± 20	1.9	–	44
– β, β-d_2	100	Bulk	− 2650 ± 4	− 6.28 ± 0.04	–	34
Vinyl formate	50[d]	Acetone	340 ± 25	0.21 ± 0.80	–	41
	50[d]	Chloroform	290 ± 50	0.08 ± 0.16	–	41
	50[d]	Dimethylformamide	250 ± 38	0.5 ± 0.12	–	41
	100	Bulk	1050 ± 54	3.3 ± 0.17	–	38
	100	Bulk	880 ± 25	3.4 ± 0.080	–	41
Vinyl trichloroacetate	100	Bulk	2830	84	[a]	32
	?	*n*-Butyl acetate	1800	4.6	[a]	32
	100	Bulk	1590	4.2	[a]	32
	?	*n*-Butyl acetate	880	2.1	[a]	32

[a] Data directly from literature, mechanistic assumptions unknown.
[b] Solvent/monomer molar ratio.
[c] Literature data, calculated under the assumption of Bernoulli statistics.
[d] mol%.
[e] mol/l.
[f] g/ml.

TABLE 4. ACTIVATION ENTHALPY DIFFERENCE ($\Delta H^{\ddagger}_A - \Delta H^{\ddagger}_B$) CALCULATED FROM VARIOUS LITERATURE DATA

		$\Delta H^{\ddagger}_A - \Delta H^{\ddagger}_B$ (J/mol)					
Monomer	Solvent	A = i/i B = i/s	A = s/i B = i/i	A = i/i B = s/s	A = i/s B = s/s	A = s/i B = s/s	Refs.
Glycidyl methacrylate	Acetone	33000 ± 2300	–	–	–	7100 ± 110	25
	Bulk	–	–	–	–	9300 ± 140	25
	Dimethylformamide	–	–	–	–	580 ± 33	25
	Toluene	–	–	–	–	10000 ± 1000	25
Methacrylic acid (20 vol.%)	Methanol	25000 ± 1400	38000 ± 2500	6500 ± 750	420 ± 80	3300 ± 110	19
	1-Propanol	14000 ± 1500	20000 ± 1800	300 ± 160	− 1240 ± 75	4010 ± 96	19
	2-Propanol	27000 ± 1800	33000 ± 2600	5556 ± 8	150 ± 29	8000 ± 220	19
Methyl methacrylate/$ZnCl_2$							
9.34/0.369 (mol/mol)	Bulk	2800 ± 120	− 130 ± 110	2430 ± 59	− 340 ± 59	2310 ± 54	29
2.00/1.00 (mol/mol)	Bulk	− 13 ± 110	− 100 ± 75	− 42 ± 88	− 29 ± 29	− 140 ± 75	27
1.00/1.00 (mol/mol)	Bulk	− 20 ± 160	− 21 ± 110	− 29 ± 130	− 8 ± 17	− 46 ± 130	27
3.19/3.19 (mol/mol)	Ethyl acetate	2300 ± 230	1800 ± 180	2600 ± 170	370 ± 63	4480 ± 54	29
1.10/9.6 (mol/mol)	70% $ZnCl_2/H_2O$	700 ± 240	2200 ± 180	1400 ± 150	700 ± 92	3550 ± 38	29
Vinyl formate	Acetone	670 ± 25	− 1060 ± 88	− 419 ± 4	− 1060 ± 92	− 2150 ± 92	41
	Bulk	2850 ± 84	− 1700 ± 100	650 ± 25	− 1900 ± 130	− 900 ± 130	41
	Chloroform	1700 ± 180	− 3200 ± 190	− 520 ± 46	− 300 ± 190	− 3700 ± 1250	41
	Dimethylformamide	0 ± 150	− 250 ± 190	− 180 ± 200	− 180 ± 200	− 440 ± 230	41
Vinyl chloride	Water	423	1210	170	980	2500	35
	Water/methanol	3600 ± 63	− 1230 ± 54	4420 ± 29	830 ± 84	5600 ± 140	35
Vinyl chloride, β, β-d_2	Bulk	3180 ± 46	540 ± 46	2860 ± 21	− 330 ± 29	2320 ± 25	39

TABLE 5. ACTIVATION ENTROPY DIFFERENCES ($\Delta S_A^{\ddagger} - \Delta S_B^{\ddagger}$) CALCULATED FROM VARIOUS LITERATURE DATA

Monomer	Solvent	$\Delta S_A^{\ddagger} - \Delta S_B^{\ddagger}$ (J/mol/K) A = i/i B = i/s	A = s/i B = i/i	A = i/i B = s/s	A = i/s B = s/s	A = s/i B = s/s	Refs.
Glycidyl methacrylate	Acetone	85 ± 8.8	–	–	–	10.3 ± 0.42	25
	Bulk	–	–	–	–	16.1 ± 0.54	25
	Dimethylformamide	–	–	–	–	5.8 ± 0.13	25
	Toluene	–	–	–	–	17 ± 3.9	25
Methacrylic acid (20 vol.%)	Methanol	54 ± 4.2	125 ± 8.4	− 1.3 ± 2.3	2.3 ± 0.25	− 1.1 ± 0.13	19
	1-Propanol	32 ± 7.1	72 ± 8.0	− 15.8 ± 0.54	− 3.4 ± 0.25	− 0.9 ± 0.38	19
	2-Propanol	75 ± 7.5	100 ± 11	− 1.84 ± 0.04	0.7 ± 0.13	11.1 ± 0.75	19
Methyl methacrylate/$ZnCl_2$							
9.34/0.369 (mol/mol)	Bulk	4.0 ± 0.34	− 2.9 ± 0.29	1.8 ± 0.17	− 2.2 ± 0.17	− 1.1 ± 0.13	29
2.00/1.00 (mol/mol)	Bulk	− 7.0 ± 0.46	− 2.8 ± 0.29	− 8.0 ± 0.38	− 1.5 ± 0.13	− 10.8 ± 0.29	27
1.00/1.00 (mol/mol)	Bulk	− 5.2 ± 0.63	− 1.7 ± 0.38	− 6.0 ± 0.46	− 0.8 ± 0.21	− 7.7 ± 0.42	27
3.19/3.19 (mol/mol)	Ethyl acetate	1.4 ± 0.88	3.9 ± 0.71	1.8 ± 0.63	0.4 ± 0.25	5.7 ± 0.13	29
1.10/9.6 (mol/mol)	70% $ZnCl_2$ in water	3.1 ± 0.71	5.2 ± 0.54	− 1.5 ± 0.42	1.6 ± 0.25	3.8 ± 0.13	29
Vinyl formate	Acetone	5.1 ± 0.25	− 5.6 ± 0.29	− 0.75 ± 0.04	− 5.8 ± 0.29	− 9.3 ± 0.29	41
	Bulk	11.1 ± 0.25	− 7.6 ± 0.25	2.55 ± 0.080	− 8.6 ± 0.34	− 5.0 ± 0.38	41
	Chloroform	7.7 ± 0.54	− 13.0 ± 0.63	− 1.1 ± 0.13	− 13.9 ± 0.59	− 13.9 ± 0.75	41
	Dimethylformamide	4.6 ± 0.46	− 3.9 ± 0.59	− 0.3 ± 0.13	− 4.3 ± 0.39	− 3.9 ± 0.71	41
Vinyl chloride	Water	− 1.3	4.6	4.5	3.8	6.8	35
	Water/methanol	11.7 ± 0.25	− 4.1 ± 0.21	15.3 ± 0.54	3.6 ± 0.34	20.6 ± 0.54	35
Vinyl chloride, β, β-d_2	Bulk	8.3 ± 0.17	2.1 ± 0.17	7.03 ± 0.080	− 1.30 ± 0.08	4.98 ± 0.08	39

TABLE 6. CALCULATED COMPENSATION TEMPERATURES T_0 AND COMPENSATION ENTHALPIES $\Delta\Delta H_0^{\ddagger}$ FOR VARIOUS MONOMERS AND MODES OF ADDITION, ASSUMING MARKOV FIRST ORDER TRIALS

Monomer	Modes of addition A	B	T_0 (K)	$\Delta\Delta H_0^{\ddagger}$ (J/mol)
Acrylonitrile	s/i	i/s	1400	0
Methacrylic acid, in alcohols	s/i	i/s	455	2780
	i/i	i/i	294 ± 26	6300 ± 1260
	s/i	i/i	323 ± 5	− 2300 ± 510
	i/i	s/s	403 ± 140	6600 ± 200
	s/i	i/s	323 ± 10	4480 ± 21
	i/s	s/s	302 ± 74	− 180 ± 63
	s/i	s/s	333 ± 23	4250 ± 25
−, *i*-butyl ester	s/i	i/s	294 ± 3	3304 ± 4
−, *n*-butyl ester	s/i	i/s	236 ± 4	3576 ± 8
−, *p*-carboxyphenyl ester	s/i	i/s	481	3308
−, glycidyl ester	s/i	i/s	345 ± 15	3800 ± 200
	s/i	s/s	373 ± 11	3500 ± 120
−, hexyl ester	s/i	i/s	303 ± 11	3260 ± 29
−, methyl ester	i/i	i/s	315 ± 49	120 ± 160
	s/i	i/i	310 ± 20	2600 ± 130
	i/i	s/s	323 ± 124	1440 ± 84
	s/i	i/s	362 ± 15	1470 ± 50
	i/s	s/s	338 ± 51	1400 ± 92
	s/i	s/s	334 ± 54	4010 ± 75
−, −, (+$ZnCl_2$)	i/i	i/s	314 ± 39	1650 ± 67
	s/i	i/i	321 ± 9	540 ± 33
	i/i	s/s	331 ± 24	1930 ± 42
	s/i	i/s	336 ± 21	2250 ± 50
	i/s	s/s	304 ± 10	243 ± 4
	s/i	s/s	328 ± 42	2530 ± 80
−, *i*-propyl ester	s/i	i/s	230 ± 54	3420 ± 67
−, *n*-propyl ester	s/i	i/s	248 ± 41	3300 ± 176
Vinyl acetate	s/i	i/s	1100	− 2050
Vinyl chloride	i/i	i/s	283 ± 25	820 ± 130
	s/i	i/i	282 ± 6	− 75 ± 17
	i/i	s/s	235 ± 28	900 ± 160
	s/i	i/s	305 ± 124	− 220 ± 420

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TABLE 6. *cont'd*

Monomer	Modes of addition A	B	T_0 (K)	$\Delta\Delta H_0^{\ddagger}$ (J/mol)
	i/s	s/s	248 ± 9	− 13 ± 33
	s/i	s/s	219 ± 10	1130 ± 59
Vinyl formate	i/i	i/s	362 ± 23	− 1400 ± 130
	s/i	i/i	315 ± 16	850 ± 80
	i/i	s/s	323 ± 27	− 200 ± 25
	s/i	i/s	330 ± 95	− 290 ± 42
	i/s	s/s	320 ± 18	970 ± 96
	s/i	s/s	316 ± 5	740 ± 21
N-Vinyl carbazole	s/i	i/s	− 243	288
Vinyl chloroacetate	s/i	i/s	300	419
Vinyl trifluoroacetate	s/i	i/s	340	167

[a]Based on data in Tables 3–5; see also Refs. 40,43.

REFERENCES

1. H. -G. Elias, Makromol. Chem. **137**, 277 (1970).
2. J. W. L. Fordham, J. Polym. Sci., **39**, 321 (1959).
3. C. E. H. Bawn, W. H. Janes, A. M. North, J. Polym. Sci. C **4**, 427 (1963).
4. H. Fischer, Kolloid-Z., **206**, 131 (1965).
5. P. L. Luisi, R. M. Mazo, J. Polym. Sci. A-2, **7**, 775 (1969).
6. F. A. Bovey, G. V. D. Tiers, J. Polym. Sci., **44**, 173 (1960).
7. T. G. Fox, H. W. Schnecko, Polymer **3**, 575 (1962).
8. F. A. Bovey, "Polymer Conformation and Configuration", Academic Press, New York, 1969.
9. H. Watanabe, Y. Sono, Kogyo Kagaku Zasshi, **65**, 273 (1962).
10. H. -G. Elias, P. Goeldi, V. S. Kamat, Makromol. Chem., **117**, 269 (1968).
11. E. L. Miller, SPE Trans., **3**, 1 (1963).
12. P. Goeldi, H. -G. Elias, Makromol. Chem., **153**, 81 (1972).
13. G. Svegliado, G. Talamini, G. Vidotto, J. Polym. Sci. A-1, **5**, 2875 (1967).
14. H. Murano, R. Yamadera, J. Polym. Sci. B, **5**, 333 (1967).
15. C. Schuerch, W. Fowells, F. P. Hood, F. A. Bovey, cited in Ref. 36.
16. K. Matsuzaki, T. Uryu, K. Ito, Makromol. Chem., **126**, 292 (1969).
17. K. Matsuzaki, T. Uryu, J. Polym. Sci. B, **4**, 255 (1966).
18. H. Hirai, T. Ikegami, S. Makishima, J. Polym. Sci. A-1, **7**, 2059 (1969).
19. J. B. Lando, J. Semen, B. Farmer, Polym. Preprints, **10**, 586 (1969).
20. W. L. Miller, W. S. Brey, G. B. Butler, J. Polym. Sci., **54**, 329 (1961).
21. K. Yokota, Y. Ishi, Kogyo Kagaku Zasshi, **69**, 1966 (1966).
22. K. Matsuzaki, A. Ishida, N. Tateno, T. Asakura, A. Hasegawa, T. Tameda, Kogyo Kagaku Zasshi, **68**, 852 (1965).
23. Y. B. Amerik, I. I. Konstantinov, B. A. Krentsel, IUPAC, International Symposium, Macromolecular Chemistry, Tokyo, 1966, Preprint 1.1.09.
24. K. Matsuzaki, A. Ishida, N. Tateno, IUPAC, International Symposium, Macromolecular Chemistry, Prague, 1965, Preprint 265.
25. Y. Iwakura, F. Toda, T. Ito, K. Aoshima, Makromol. Chem., **104**, 26 (1967).
26. Y. Kato, A. Nishioka, Bull. Chem. Soc. Japan, **37**, 1614 (1964).
27. S. Okuzawa, H. Hirai, S. Makishima, J. Polym. Sci. A-1, **7**, 1039 (1969).
28. F. A. Bovey, J. Polym. Sci., **46**, 59 (1960).
29. T. Otsu, B. Yamada, M. Imoto, J. Macromol. Chem., **1**, 61 (1966).
30. H. Sobue, K. Matsuzaki, S. Nakano, J. Polym. Sci. A, **2**, 3339 (1964).
31. F. A. Bovey, G. V. D. Tiers, Fortschr. Hochpolym. -Forschg., **3**, 139 (1963).
32. M. Uoi, M. Sumi, S. Nozakura, S. Murahashi, cited in Ref. 33.
33. S. Murahashi, in: IUPAC, International Symposium, Makromol. Chem., **3**, 435 (1967) (= Pure Appl. Chem., **15**, nos. 3–4 (1967)).
34. G. Talamini, G. Vidotto, Markromol. Chem., **100**, 48 (1967).
35. J. Bargon, K. H. Hellwege, U. Johnsen, Makromol. Chem., **95**, 187 (1966).
36. F. A. Bovey, F. D. Hood, E. W. Anderson, R. L. Kornegay, J. Phys. Chem., **71**, 312 (1967).
37. H. U. Pohl, D. O. Hummel, Makromol. Chem., **113**, 203 (1968).
38. K. C. Ramey, D. C. Lini, G. Statton, J. Polym. Sci. A-1, **5**, 257 (1967).
39. L. Cavalli, G. C. Borsini, G. Carraro, G. Confalonieri, J. Polym. Sci. A-1, **8**, 801 (1970).
40. H. -G. Elias, P. Goeldi, Makromol. Chem., **144**, 85 (1971).
41. H. -G. Elias, M. Riva, P. Goeldi, Makromol. Chem., **145**, 163 (1971).
42. B. L. Johnson, H. -G. Elias, Makromol. Chem., **155**, 121 (1972).
43. H. -G. Elias, P. Geoldi, B. L. Johnson, Adv. Chem. Ser., **128**, 21 (1973).
44. R. E. Cais, W. L. Brown, Macromolecules, **13**, 801 (1980).
45. D. R. Torrell, F. Evers, Makromol. Chem., **183**, 863 (1982).

Products of Thermal Degradation of Polymers

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A. INTRODUCTION

Polymer degradation is a complex phenomenon, and the nature of the chemistry occurring during thermal decomposition can be difficult to determine precisely. The observed products of degradation can be the result of a mixture of primary, secondary or even tertiary decomposition processes, and as a consequence, the distribution of products will depend on such factors as sample thickness and melt viscosity, in addition to the more obvious factors such as temperature, atmosphere and the chemical structure of the polymer itself. It is often overlooked that abnormal structures (terminal unsaturation, head-to-head structures, peroxide links etc.) can also markedly effect polymer stability and product distribution.

Products are of three types. Firstly, there are the volatile small molecules (typically of the size of monomer or smaller) which are fairly readily separated and characterized. Secondly, there are the larger molecules that represent fragments of the chain. These species are volatile at the degradation temperature but condense outwith the degradation zone. Generally produced as a complex spectrum of oligomers, these fragments are more difficult to separate and characterize in detail, particularly as internal rearrangements of the backbone often occur before and during fragmentation. In many cases, identification is limited to key functionalities but the advent of sophisticated chromatographic and mass spectrometric techniques is leading to increasingly detailed analyses. In particular, the smaller fragments such as dimer and trimer are in principle more readily fully characterized than the larger fragments; nevertheless there often remains some uncertainty, particularly in relation to end groups. The most intractable product is the third type, the residue, which is often (but not always) insoluble, and characterization may again be limited to identification of key functionalities.

Naturally, most of the information in the literature relates to the easily identified low molecular weight species, and the data in this section reflects this. In addition, most thermal degradation studies focus on inert rather than oxidizing environments. Following the lead of the previous author, Norman Grassie, the vast majority of the entries included here are concerned with products of the first two types from purely thermal degradation processes under inert atmospheres.

B. TABLES

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Temperature range (°C)	Degradation products	Refs.
1.1. UNSATURATED POLYMERS			
Poly(acetylene)	650	Methane (0.87 moles relative to benzene = 1), ethylene (0.47), ethane (0.26), propylene (0.27), propane (0.047), 1,3-butadiene (0.19), cyclopentadiene (0.086), cyclopentene (0.063), benzene (1.0), toluene (0.18), *m*-xylene (0.091), *p*-xylene (0.015), styrene (0.041), *o*-xylene (0.013), methylstyrene (0.010), *trans*-β-methylstyrene (0.011), indane (0.0074), indene (0.036), methylindene (0.0072), naphthalene (0.052), 2-methylnaphthalene (0.015), 1-methylnaphthalene (0.018), biphenyl (0.0091), fluorene (1.0), anthracene (0.005)	82
Poly(butadiene)	325–475	14.1% of products are volatile at 25°C including 1.5% of monomer among other saturated and unsaturated hydrocarbons; 85.9% of products are larger fragments involatile at 25°C	16
	20°C/min to 15% weight loss	Effect of *cis*/*trans* ratio; increasing *trans* content gives more cyclopentene and 1,3-cyclohexadiene and less 4-vinyl-1-cyclohexene and 1,3-butadiene	83

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(butadiene-*co*-acrylonitrile) (70/30)	310–400	14.5% of products are volatile at 25°C, consisting of saturated hydrocarbons	10
25–100 mol% butadiene	350–400	Main product is structurally modified chain fragments; ammonia and hydrogen cyanide increasing with acrylonitrile content to approximately 4% each by weight of copolymer containing mole 75% acrylonitrile; traces of hydrogen and hydrocarbons up to C_7	98
Poly(butadiene)-*blend*-poly(styrene) (0–100 wt.%)	Ambient to 500	Volatile products as for individual polymers	84
Poly(butadiene-*co*-sytrene) (75/25)	327–430	11.8% of products are volatile at 25°C; 1.9% butadiene with other saturated and unsaturated hydrocarbons	10,16
Poly(chloroprene)	377	Hydrogen chloride	19
	170–400	HCl (90% of theoretical), ethylene, chloroprene (trace), chloroprene dimers and other modified chain fragments	85
	400–500	HCl (trace), methane, ethylene, propylene (trace)	85
Poly(chloroprene)-*blend* poly(methyl methacrylate)	Ambient to 500	Methyl methacrylate, HCl, CO_2, CO, methyl chloride, methanol.	86
Poly(isoprene), synthetic	287–400	3.4% isoprene, 8.8% dipentene, small amounts of *p*-menthene.	2,16,31
97% *cis*-1,4; 3%, 3,4	384	In helium; methane (0.04 wt.%), ethylene (0.08), ethane (0.03), propylene (0.15), propane (0.06), isobutylene (0.09), butenes (0.07), methyl butene (0.09), isoprene (29), toluene (0.02), 2,3-dimethyl cyclopentene (1.2), octene (0.05), 2,4-dimethyl cyclohexene (0.16), *m*-xylene (0.33), 1-methyl-4-ethyl cyclohexene (0.18), 1,5-dimethyl-5-vinyl cyclohexene (2.5), dipentene (60), $C_{11}H_{18}$ (0.2), $C_{12}H_{18}$ (0.2), $C_{15}H_{24}$ (3.5), $C_{16}H_{26}$ (1.8)	87
natural rubber	287–400	3.9% isoprene, 13.2% dipentene, small amounts of *p*-menthene	2,16,31
	450–800	Dipentene main product at 450°C, optimum yields of isoprene in range 675–800°C, e.g., 58% at 750°C and 13 mbar	32
	500	Monomer, dimers (1,7,7-trimethyl-2,2,1-bicyclohept-2-ene, 2,5,6-trimethyl 1,3,6-heptatriene, dipentene/d,1-limonene), trimers	259
gutta percha	287–400	3.0% Isoprene, 15.6% dipentene, small amounts of *p*-menthene	2,16,31
Poly(isoprene)-*blend*-poly(styrene) (4/1 to 1/4, w/w)	Ambient to 500	Styrene, isoprene, dipentene larger chain fragments of both polymers, traces of ethylene, ethane, and propane	88
chlorinated rubber	Ambient to 500	HCl (almost quantitative), CH_4, C_2H_4, and H_2 are minor products formed in the later stages of reaction	89
chlorinated rubber-*blend*-poly(vinyl chloride)	160–350	HCl	90
Poly(perfluoro-4-chloro-1,6-heptadiene)	320–400	Completely volatilized – products unknown.	49
1.2. POLY(ALKENES)			
Poly(ethylene)	335–450	Continuous spectrum of saturated and unsaturated hydrocarbons from C_2–C_{90}; lower temperature favors larger fragments	13,16
–, low density	Ambient to 500	Continuous spectrum of saturated and unsaturated hydrocarbons C_1–C_{15}; trace of hydrogen	243
	264–289	In air; CO_2, H_2O, ethene, propene, propane, cyclopropane, butene, butane, pentene, pentane, hexene, hexane, heptene, heptane, octene, octane, methanol, ethanol, furan, tetrahydrofuran, formaldehyde, acetaldehyde, propanal, acrolein, butanal, isobutanal, pentanal, acetone, methyl vinyl ketone, methyl ethyl ketone, 2-pentanone, 2-hexanone, 2-heptanone, formic acid, acetic acid, propionic acid, acrylic acid, butyric acid, isovaleric acid, hydroxyvaleric acid, crotonic acid, caproic acid, butyrolactone, valerolactone	91
Poly(ethylene-*co*-carbon monoxide) Alternating copolymer	500	CO, H_2O, ethene, series of ketones with general structure $R-[-CO-CH_2-CH_2-]_n-CO-R'$ where R, $R' = -CH_3$, $-CH_2-CH_3$ or $-CH=CH_2$	270
Poly(ethylene-*co*-ethyl acrylate) 18 : 1 mole ratio	Ambient to 500	Broad spectrum of saturated and unsaturated hydrocarbons, ethyl esters and carboxylic acids; ethanol, ethylene, CO_2, CO and small amounts of ketene and ethyl acrylate	243
Poly(ethylene-*co*-vinyl acetate) (12–33 mol% vinyl acetate)	Ambient to 500	Acetic acid, ketene, CO_2, ethylene, saturated and usaturated hydrocarbons, chain fragment fraction containing some vinyl acetate units	92
(38.5% (wt.%) of ester groups)	260–290	Acetic acid > 99%	93
Poly(isobutene)	288–425	18.1% monomer together with methane, isobutane, and C_5 and higher saturated and unsaturated hydrocarbons	16
	Up to 1200	As temperature is increased the yields of fragments smaller than monomer increase at the expense of larger fragments	28

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
	325	At 4.9% weight loss, products are isobutene (64.3%), CH_4 (13.6%), neopentane (10.3%); remainder C_2–C_{12} hydrocarbons	29
	345	At 15.7% weight loss, products are isobutene (78.9%), CH_4 (5.9%), neopentane (4.7%); remainder C_2–C_{12} hydrocarbons	29
	365	At 46.8% weight loss, products are isobutene (81.6%), CH_4 (3.9%), neopentane (3.1%); remainder C_2–C_{12} hydrocarbons	29
	300–320	Two types of *t*-butyl ended and two types of isopropyl ended terminal monoolefins in range 2-mers to 12-mers	249
Poly(isopropenylcyclohexane)	400–900	Methylenecyclohexane and/or methyl-1-cyclohexene, ethylcyclohexane, toluene, isopropenylcyclohexane, isomer of isopropenylcyclohexane, ethyl benzene, isopropylcyclohexadiene, pentadienylcyclohexane (5 isomers), dicyclohexylpropane, 3-cyclohexenyl-1-cyclohexylpentadiene, dicyclohexylhexadiene	94
Poly(isopropenylcyclohexane-*co*-α-methylstyrene) (92/8 and 33/67)	400–900	Methylenecyclohexane and/or methyl-1-cyclohexane, benzene, toluene, isopropenylcyclohexane, ethylbenzene (92/8 only), decalin (92/8 only), cumene (33/67 only), α-methylstyrene, styrene, (33/67 only), pentadienylcyclohexane (92/8 only), indene and methyleneindene (33/67 only), naphthalene, dicyclohexylhexadiene (92/8 only). The following from the 33/67 copolymer only: 2-methylnaphthalene, 1-methylnaphthalene, biphenyl, methylstilbene, α-methylstilbene, anthracene/phenanthrene, dihydropyrene	94
Poly(4-methyl-1-pentene)	291–341	After 22 hours at 341°C, 20% residue, 71% saturated and chain fragments; 9% of products volatile at 25°C, comprising isobutene (56%), propane (34%), traces of monomer, and hydrocarbons	43
Poly(propylene)	328–410	Saturated and hydrocarbons from C_2 upwards, monomer yield 0.17%	13
	400–1200	As temperature is raised, yield of small fragments increases at expense of large fragments	28
	360–400	At 400°C, chain fragments with 7.05 wt.% of volatile products comprising principally, methane 0.057 wt.%, ethane 0.13, propane 0.078, propene 0.83, isobutene 0.16, butane 0.004, 2-methyl-1-butene 0.010, 1-pentene 0.008, pentane 1.62, 2-pentene (*cis* and *trans*) 0.015, 2-methyl pentane 0.074, 2-methyl-1-pentene 1.19, 2,4-dimethylpentane 0.032, 2,4-di-methyl-1-pentene 0.089, 4-methylheptane 0.24, 2,4-dimethyl-1-heptene 2.20, 4,6-dimethylnonane (threo form) 0.042 (erythro form) 0.052, 2,4,6-trimethyl-1-nonene (threo form) 0.055 (erythro form) 0.066	95
	120–280	In air; ethene, ethane, propene, propane, isobutene, butane, isobutane, pentadiene, 2-methyl-1-pentene, 2,4-dimethyl-1-pentene, 5-methyl-1-heptene, dimethylbenzene, methanol, ethanol, 2-methyl-2-propene-1-ol, 2-methyl furan, 2,5-dimethylfuran, formaldehyde, acetaldehyde, acrolein, propanal, methacrolein, 2-methyl propanal, butanal, 2-vinyl crotonaldehyde, 3-methyl pentanal, 3-methyl hexanal, octanal, nonanal, decanal, ethenone, acetone, 3-buten-2-one, 2-butanone, 1-hydroxy-2-propanone, 1-cyclopropylethanone, 3-methyl-2-buten-2-one, 3-penten-2-one, 2-pentanone, 2,3-butanedione	96
	200–356	Under 254 nm radiation; additional methane and ethylene are formed compared with the thermal reaction	97
Poly(propylene)-*blend*-poly(methyl methacrylate)	Ambient to 500	Poly(methyl methacrylate) stabilized by blending with polypropylene; pre-irradiation with 2537 Å radiation at ambient temperature reduces amount of methyl methacrylate formed on subsequent thermal degradation; missing methyl methacrylate units appear in a chain fragment fraction	99
Poly(vinylcyclohexane)	335–391	Small amounts of cyclohexane, cyclohexane, methylcyclohexene, methyl cyclohexane, vinylcyclohexene, vinylcyclohexane, ethylcyclohexane with larger chain fragments	2,3
	360–380	Cyclohexane (main product), cyclohexene, vinylcyclohexane, vinylcyclo hexane oligomers, methane, ethane, ethylene, propane, propylene, butane, butylene, hydrogen	100
Poly(chlorotrifluoro-ethylene)	Ambient to 500	Monomer, traces of dichlorodifluoroethene, chloropentafluoropropene, trichlorotrifluoroethane	261

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
1.3. POLY(ACRYLICS), POLY(METHACRYLICS)			
Poly(acrylic acid)	Ambient to 500	CO, CO_2, H_2O, CH_4, acetone, ketene, ethene, propylene, 1-butene, methyl vinyl ketone, benzene, acrylic acid, toluene, xylene, short chain fragments such as dimer to octamer with unsaturated and anhydride functionalities	297
–, methyl ester	292–399	26% of products are volatile at 25°C, mainly methyl alcohol and carbon dioxide with traces of monomer and methyl methacrylate and C_4–C_6 oxygenated compounds (74% of products are large chain fragments involatile at 25°C)	2,3
	Up to 980	Methanol (1.9%), methyl formate (trace), methyl acetate (1.4%), methyl acrylate (19.0%), methyl methacrylate (1.9%), saturated ester (2.1%), dimethyl glutarate (7.5%), methyl 4-methoxycarbonyl-pent-4-enoate (11.6%), trimethyl pentane 1,3,5-tricarboxylate (0.6%) methyl 4,6-di-methoxycarbonylhept-6-enoate (53%), saturated and tetramers (1.0%)	101
	50	Irradiated with high-pressure mercury lamp; methanol, formaldehyde, methyl acetate, methyl formate, CO_2, CO, H_2, CH_4	102
–, ethyl ester	300–500	Major products, carbon dioxide, ethylene, ethanol, and chain fragments; minor products, ethane, ethyl acrylate, and ethyl methacrylate	103
	50	Irradiated with high-pressure mercury lamp; CO, CO_2, ethanol, acetaldehyde, ethane, ethyl formate, acetal, ethyl propionate	104
	Ambient to 500	Ethylene, carbon dioxide, trace of ketene, ethanol (major product), various ethyl esters (see Ref.)	243
–, *n*-propyl ester	300–500	Major products carbon dioxide, propene, propanol, and chain fragments; minor products, propane, carbon monoxide, hydrogen, *n*-propyl acrylate, and *n*-propyl methacrylate	103
–, isopropyl ester	250–500	Major products carbon dioxide, propene; minor products isopropanol with trace carbon monoxide	105
–, *n*-butyl ester	300–500	Major products carbon dioxide, 1-butene, butanol, and chain fragments; minor products, butane, *n*-butyl acrylate, and *n*-butyl methacrylate	103
	50	Irradiated with high-pressure mercury lamp. CO, CO_2 *n*-butanol, butyraldehyde, *n*-butene, *n*-butyl formate, *n*-butyl valerate (after prolonged exposure)	104
–, *tert*-butyl ester	>160	86% isobutylene, 11% water, 3% carbon dioxide	4
–, 2-ethyl hexyl ester	300–500	Carbon dioxide, 2-ethyl-1-hexene, 2-ethyl-1-hexanol	103
–, benzyl ester	260–300	Major products are benzyl alcohol, chain fragments, partially crosslinked residue; minor products are CO_2, toluene, CO, CH_4, H_2	1
	–	CO_2, benzyl alcohol, and low polymers	106
–, 1,1,5-trihydroperfluoroamyl ester	300–315	Oligomers, 1,1,5-trihydroperfluoroamyl alcohol, monomer, 1,1,5-trihydroperfluoroamyl methacrylate	107
–, 2,2,2-trichloroethyl ester	319	2,2,2-Trichloroethanol (80%), monomer (9%), 1,1-dichloroethene, 2-chloroethanal, 2,2-dichloroethanal, 1,1-dichloropropane, 1,1,1-trichloroethane, 2,2,2-trichloroethylethanoate, 2,2,2-trichloroethyl chloromethanoate, 2,2,2-trichloroethylpropanoate, 2,2,2-trichloroethylmethacrylate, 2,2,2-trichloroethyl chloroethanoate, 2,2,2-trichloroethyl-2-chloroacrylate	263
–, α-bromo-, methyl ester	110–150	Methylbromide, hydrogen bromide	5
	600	CO_2, propene, methyl bromide, vinyl bromide	305
–, α-chloro-, methyl ester	600	CO_2, propene, methyl chloride, monomer	306
–, α-chloro-, *sec*-butyl ester	190	*sec*-Butyl chloride, butylene, hydrogen chloride	7
–, α-cyano-, methyl ester	>180	Yellows and some monomer formed	6
–, α-phenyl-, methyl ester	210–280	Monomer is the sole product	8
Poly(acrylic acid, magnesium salt)	Ambient to 500	CO, CH_4, CO_2, acetone, ketene, ethene, propene, 1-butene, benzene, toluene, xylene, cyclopentene, methyl ethyl ketone, diethyl ketone, methyl-*n*-propyl ketone, di-*n*-propyl ketone, methyl vinyl ketone, methyl isopropenyl ketone, methyl isopropyl ketone, ethyl vinyl ketone, trace amounts of methyl-*n*-butyl ketone, cyclopentanone, cyclohexanone, acrolein, ethanal and butanal, chain fragments, some monomer	312
Poly(acrylic acid, calcium salt)	Ambient to 500	CO, CH_4, CO_2, acetone, ketene, ethene, propene, 1-butene, benzene, toluene, mesitylene, xylene, methyl ethyl ketone, diethyl ketone, methyl-*n*-propyl ketone, methyl-*n*-butyl ketone, ethyl vinyl ketone, methyl propenyl ketone (trace), ethyl propyl ketone (trace), 3-methyl-cyclopentanone, cyclohexanone (trace), cyclohexa-2-enone,	312

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
		2-methyl-cyclohexanone, 1-methyl-cyclohexa-1-ene-3-one (trace), acrolein, mesityl oxide, ethanal, propanal, butanal, chain fragments, some monomer	
Poly(acrylic acid, sodium salt)	Ambient to 500	CO, CH_4, CO_2, ketene, 1-butene, propene, acetone, methyl ethyl ketone, methyl isopropyl ketone, methyl-*n*-propyl ketone, diethyl ketone, methyl propenyl ketone, 3-hexanone, toluene, 2-hexanone, 1,3-cyclopentadiene, cyclopentanone, 2-methyl cyclopentanone, mesityl oxide, xylenes, benzene, propionaldehyde, acrolein, acetaldehyde, ethene, short chain fragments, traces of monomer	303
Poly(acrylic acid, potassium salt)	Ambient to 500	CO, CH_4, CO_2, ketene, 1-butene, propene, acetone, methyl ethyl ketone, methyl *n*-propyl ketone, 1,4-cyclohexadiene, toluene, 1-methyl-1,3-cyclohexadiene, 2-hexanone, cyclopentene, 1-methyl cyclopentene, mesityl oxide, xylenes, benzene, ethene, cyclopentanone, 1,3-cyclopentadiene, diethyl ketone, short chain fragments, traces of monomer	303
Poly(methyl acrylate-*co*-methyl methacrylate) (mole ratios 112/1–2/1)	Ambient to 500	Monomers, CO_2, chain fragments larger than monomer, H_2; more stable than poly(methyl methacrylate); rapid decrease in molecular weight; products of phtodegradation at 170° by 254 nm radiation are qualitatively similar	108,109
Poly(methyl acrylate-*co*-acryloyl chloride) 6, 15, 41, 83 mol% AC	Ambient to 500	Acrylic acid, HCl, H_2O, CO, CO_2, acryloyl chloride, methyl acrylate, methyl acrylate chain fragments, anhydride and γ-lactone structures, chloro-methane, methyl methacrylate, methanol, 3,5-(6-chloro-2-pyrone) and fragments thereof, C_6, C_9, C_{12}, C_{15} fragments. Relative yields depend upon copolymer composition	262
Poly(methyl acrylate-*co*-2-bromoethyl methyacrylate) (10–82 mol% methyl acrylate)	Ambient to 500	CO_2, ketene, methyl bromide, acetaldehyde, methyl acrylate, methanol, 1,2-dibromoethane, methyl methacrylate, 2-bromoethanol, ethyl methacrylate, CH_4, CO, H_2 modified chain fragments	110
Poly(methyl acrylate)-*blend*-poly((2-bromoethyl methacrylate) (50/50, w/w)	Ambient to 500	CO_2, ketene (minor), methyl bromide, vinyl bromide (minor), acetaldehyde, methanol, methyl methacrylate, methyl acrylate, 1,2-dibromoethane, 2-bromoethyl methacrylate	111
Poly(*n*-butyl acrylate-*co*-methyl methacrylate) (3.9–93.4 mol% butyl acrylate)	313–332	1-Butene, carbon dioxide, butanol methyl methacrylate, butyl acrylate, and butyl methacrylate; small quantities of methane, ethane, propane, butane, ethylene, propene, *cis*, and *trans* 2-butene, hydrogen, and carbon monoxide increase with butyl acrylate content of the copolymer	112
	165	Under 2537 Å radiation; methyl methacrylate with smaller quantities of *n*-butyl acrylate, *n*-butanol, *n*-butyraldehyde, hydrogen, carbon monoxide, and methane; a chain fragment fraction increases with *n*-butyl acrylate content	113
Poly(*tert*-butyl acrylate-*co*-glycidyl methacrylate) (34/66)	0–700	CO, CO_2, propene, ethanal, isobutene, acrolein, allyl alcohol	114
Poly(methyl-α-chloroacrylate-*co*-2,2,2-trifluoroethyl methacrylate) (70/30)	Ambient	Gamma Radiolysis; CH_4, CO, ethylene, ethane, CH_3OH, CHF_3, HCl, C_3H_6, CO_2, C_2H_3F, CH_3Cl, methyl formate, CH_3CH_2Cl, CHF_3, CH_3CF_3, $CH_2=C(CH_3)COOCH_2CF_3$	115
Poly(methyl-α-chloroacrylate-*co*-2,2,2-trifluoroethyl methacrylate) (70/30)	Ambient	Gamma Radiolysis; CH_4, CO, ethylene, ethane, CH_3OH, HCl, C_3H_6, CO_2, CH_3Cl, methyl formate, CH_3CH_2Cl, CH_2Cl_2, $CHCl_3$	115
Poly(acrylonitrile)	<200	Colours through yellow, orange, red, and black	9
	250–280	12% of products are volatile at 25°C, consisting of hydrogen cyanide, acrylonitrile, and vinyl acetonitrile; 88% of products are involatile at room temperature	10,11
	280–450	Five major volatile products are cyanogen, HCN, acrylonitrile, acetonitrile, vinyl acetonitrile; involatile residue remains	12
	250–350	NH_3, HCN	116
	235	HCN, hydrogenated acrylonitrile, dimer and trimer, acetonitrile, methacrylonitrile	117
–, α-chloro	150–200	Hydrogen chloride	118
	Ambient to 500	Hydrogen chloride and small amount of HCN at higher temperatures	119
–, α-phenyl	Ambient to 500	High yields of monomer	119
Poly(acrylonitrile-*co*-2-bromoethyl methacrylate (50/50, mol/mol)	Below 310	Ethylene, CO_2, vinyl bromide acetaldehyde, 1,2-dibromoethane	120
	310–500	Propane, isobutene, CO_2, HCN, isocyanic acid, yellow-chain fragment fraction	121
Poly(acrylonitrile-*co*-methyl methacrylate)(1/410–1/8)	220–280	Only methyl methacrylate from 410/1 and 40/1 (molar ratio) copolymers; methyl methacrylate and small amounts of acrylonitrile from 16/1 and 8/1 copolymers; rapid decrease in the molecular weight of all copolymers	121

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TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
	160	Photodegradation by 254 nm radiation gives methyl methacrylate from 410/1 to 8/1 copolymers with a 8/1 copolymer; rapid decrease in the molecular weight of all copolymers	121
Poly(acrylonitrile-*co*-methyl vinyl ketone)	–	Production of water from poly(methyl vinyl ketone) accelerated by copolymerization with acrylonitrile	122
Poly(acrylonitrile-*co*-styrene) (0/100–50/50, mol/mol)	292	Acrylonitrile causes increase in rate of chain scission; proportion of acrylonitrile among degradation products greater than from pure polyacrylonitrile; chain fragments increase with acrylonitrile content of copolymer; styrene, acrylonitrile, toluene, benzene, four dimers, and four trimers	123
(1.86–96.62 mol% acrylonitrile)	500	Hydrocyanic acid, acetonitrile, acrylonitrile, propionitrile, toluene, ethyl benzene, styrene, methane, ethylene, ethane, C_3 hydrocarbons, but-1-ene, prop-1-ene-3-nitrile, isopropylnitrile, benzene	124
Poly(acrylonitrile-*co*-vinyl acetate) (7.5 wt.% vinyl acetate)	260	HCN, hydrogenated acrylonitrile, dimer and trimer, acetonitrile, methacrylonitrile, acetic acid	117
Poly(acrylonitrile-*co*-methyl acrylate-*co*-vinyl bromide) (methyl acrylate, 10 wt.%; vinyl bromide, 5 wt.%)	180–320	CH_3Br, HBr, H_2O, CO_2, CH_3OH, NH_3, HCN, hydrogenated acrylonitrile, dimer and trimer, acetonitrile, methacrylonitrile, naphthyridine fragments	117
Poly(α-chloroacrylonitrile-*co*-methyl methacrylate)	>150	Hydrogen chloride, methyl methacrylate, α-chloroacrylonitrile	125
(0–50 mol% α-chloroacrylonitrile)	Ambient	Under 254 nm radiation. HCl, CO_2, methyl chloride, methyl formate, methanol, methyl methacrylate, CO, CH_4, H_2	126
Poly(acrylonitrile)-*blend*-poly(methyl methacrylate) (50/50, w/w)	Up to 320	NH_3, HCN, MMA	127
	320–500	Chain-fragment fraction and small amounts of CO, CO_2, methanol, CH_4	
Poly(acryloyl chloride)	Ambient to 500	Acrylic acid, HCl, H_2O, CO, acryloyl chloride, 3,5-(6-chloro-2-pyrone) and fragments thereof, C_6, C_9, C_{12}, C_{15} fragments	262
Poly(methacrylic acid)	200	Almost quantitative yields of H_2O, traces of monomer, residue of poly(methacrylic anhydride)	33
	900	CO_2, propene, butenes, pentenes, pentadienes, hexene, hexadienes, hexatrienes, benzene, toluene, heptene, heptatriene, methacrylic acid, xylene, octatriene, octadienyne, nonatriene, hexadienoic acid, decatriene, decatetraene, decapentaene, undecatriene, methylbenzoic acids, trimethylbenzoic acid	128
–, methyl ester	170–300	100% monomer	34, 35
	246–400	As temperature is raised fragmentation increases to give complex series of products and monomer yields correspondingly decreases	2
	160	100% monomer under 253.7 nm radiation	36
	100–400	Under 254 nm radiation; 100% monomer	129
–, ethyl ester	250	Monomer	37
	245	Monomer, ethanol, ethanoic acid	300
–, *n*-propyl ester	250	Monomer	37
–, *iso*-propyl ester	250	Monomer	37
–, *n*-butyl ester	250	40% monomer and traces of 1-butene	38
	170	100% monomer under 253.7 nm radiation	38
–, *iso*-butyl ester	250	Monomer	37
–, *sec*-butyl ester	250	Monomer and small amount of olefin by cracking of side chain	37
–, *tert*-butyl ester	180–200	High yields of isobutylene and water, 1% monomer, trace of methacrylic acid, residue of poly(methacrylic anhydride)	39
	<180	100% monomer under 253.7 nm radiation	
–, *n*-amyl ester	250	Monomer	37
–, *iso*-amyl ester	250	Monomer	37
–, 1,2-dimethylpropyl ester	250	Monomer and small amount of olefin by side chain cracking	37
–, neopentyl ester	250	Monomer	37
–, 3,3-dimethylbutyl ester	250	Monomer	37
–, 1,3-dimethylbutyl ester	250	Monomer and small amount of olefin by side chain cracking	37
–, 2-bromoethyl ester	Ambient to 500	2-Bromoethyl methacrylate (>95%), CO_2 (<1%), vinyl bromide (<1%)	130
–, 2-sulfoethyl ester	50–600	CO, CO_2, H_2O, SO_2, ethene	290
	600	CO, CO_2, propene, acetaldehyde, methyl bromide, butene, vinyl bromide, ethyl bromide, methyl methacrylate	305
–, 2-chloroethyl ester	600	CO_2, propene, acetaldehyde, vinyl chloride, ethyl chloride, monomer	306
	268	Monomer (99%), traces of acetaldehyde, methyl methacrylate, 1,2-dichloroethane, ethylene methacrylate	268

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(methacrylic acid), 2-fluoroethyl ester	600	CO_2, vinyl fluoride, acetaldehyde, fluoroacetaldehyde	310
–, 2,2-dichloroethyl ester	242	Monomer (90.7%), chloroacetaldehyde (8%), 1,1,2-trichloroethane (0.5%), 2,2-dichloroethanol, methacrylic acid (0.1%), 2-chloroethylene methacrylate (0.1%)	268
–, 2-methoxyethyl ester	Ambient to 300	Monomer, CO_2	309
	Ambient to 500	Monomer, CO_2, CO, CH_4, ethene, formaldehyde, isobutene, methyl vinyl ether, methanol	309
–, 2,3-dibromopropyl ester	600	CO_2, propene, acetaldehyde, methyl bromide, butene, ethyl bromide, bromocyclopropane or 2-bromopropane	305
–, 2-hydroxyethyl ester	375–500	2-Hydroxyethyl methacrylate (20% at 500°C), ethylene dimethacrylate (6%at 500°C)(main products), ethylene glycol, 3-methyl-5-hydroxy-δ-valerolactone (probably), 1,2-di-iospropenyl-oxyethane (probably), CO, CO_2, methane, ethylene, ethane, propane, H_2O	131
–, 2,2,2-trifluoroethyl ester	Ambient	Gamma radiolysis; CH_4, CO, ethylene, ethane, CH_3OH, CH_3F, C_3H_6, CO_2, C_2H_3F, methyl formate, CH_3CHFOH, CHF_3, CF_3CH_3, CF_3OH, $CH_2=C(CH_3)-COOCH_2CF_3$	115
	600	CO_2, vinylidene fluoride, trifluoroacetaldehyde, trifluorethanol, monomer	310
–, 2,2,2-trichloroethyl ester	600	CO_2, propene, vinyl chloride, vinylidene chloride	306
	240	1,1-Dichloroethene, chloroform, methyl methacrylate, methacrylic acid, 2,2,2-trichloroethanol, 2,2,2-trichloroethyl acrylate, 2,2,2-trichloroethyl methacrylate, 2,2-dichloroethyl methacrylate, $CH_2=C(CH_3)-COOCH=CCl_2$ or $CH_2=C(CH_3)-COOCCl=CHCl$	300
–, hexafluoroisopropyl ester	600	CO_2, trifluoracetaldehyde, acetaldehyde, 1,1,1,3,3,3-hexafluoropropane, hexafluoro-isopropylformate	310
–, bisphenol S ester	464	CO_2, SO_2, phenol, bisphenol S, hydroquinone, fragments of backbone, benzene sulfonic acid	248
–, allyl ester	Ambient to 500	CO_2, CH_4, CO, H_2, propene, isobutene, dimethyl ketene, acrolein, diallyl ether, allyl methacrylate monomer and dimer, chain fragments including anhydride structures from side-group cyclization and an aliphatic ketone in the vicinity of unsaturation	283
–, phenyl ester	Ambient to 500	Monomer, some CO_2	314
–, glycidyl ester	Ambient to 500	CO, CO_2, dimethyl ketene, isobutene, propene, acrolein, allyl alcohol, glycidyl methacrylate, glycidol, short chain fragments with some unsaturation	302
–, ammonium salt	Ambient to 500	Isocyanic acid, NH_3, HCN, isobutene, CO_2 (trace), methacrylonitrile (trace)	132
–, Li, Na, K, and Co salts	Ambient to 500	Ethylene, butene-1, isobutene, acetylene, CO_2, CH_4, CO, acetone, isopropylaldehyde, methacrolein, butyraldehyde, methyl ethyl ketone, benzene (trace), methyl isopropyl ketones, methyl isopropenyl ketone, diethyl ketone, methyl *n*-propenyl ketone, cyclopentanone, methyl cyclopentanone, dimethyl cyclopentanone, dimethyl cyclopentenone and traces of toluene, di-isopropyl ketone, mesitylene, mesityl oxide	133
–, Mg, Ca, Sr, and Ba salts	Ambient to 500	CO_2, dimethyl ketene trace amounts of CO, CH_4, isobutene, butene-1, ethylene, acetylene, benzene toluene isobutyraldehyde, methyl ethyl ketone, methyl isopropenyl ketone, diethyl ketone, methyl isopropenyl ketone, di-isopropyl ketone, cyclopentanone, 2-methyl cyclopentanone, 2,5-dimethyl-cyclopentanone, 2,5-dimethylcyclopent-3-ene-1-one, methacrolein, butyraldehyde	134
Poly[α-ω-(diethylene glycol phthalate)dimethacrylate]	300	Monomer	194
	400	Monomer, phthalic anhydride, diethylene glycol, C_2H_4, propylene, CO_2	194
Poly(methyl methacrylate-*co*-alkali (metal methacrylates) (Li, Na, K) (0–100 mol%)	Ambient to 500	CO_2, alkenes, CH_4, CO, propanal methanol (major), acetone, 2-methyl propanal, methacrolein, butanal, methyl ethyl ketone, methyl isopropyl ketone, methyl isopropenyl ketone, methyl methacrylate (major), methyl prop-2-enyl ketone, toluene, di-isopropyl ketone, cyclopentanone, 2-methyl cyclopentanone, 2,5-dimethyl cyclopentanone, 2,5-di-methyl cyclopent-3-enone. Relative yields depend upon salt and copolymer composition	135
Poly(methyl methacrylate-*co*-2-bromoethyl methacrylate) (50/50, mol/mol)	Ambient to 500	Methyl methacrylate, 2-bromoethyl methacrylate, traces of vinyl bromide and CO_2	130
Poly(methyl methacrylate-*co*-methacrylamide)	Up to 340	> 35 mol% methacrylamide. MMA, methanol, NH_3, H_2O	146

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
	>340	Methacrylamide; chain fragments and small amounts of CH_4, CO_2, CO, HCN, methacrylonitrile and isobutene	146
	–	<10 mol% methacrylamide. Degradation becomes progressively more like that of PMMA	146
Poly(methyl methacrylate-*co*-methacrylic acid)	<300	Water and methanol	136
	>300	Methyl methacrylate, CO_2, CO, CH_4	
	900	(50/50 and 75/25); CO_2, propene, butenes, pentenes, pentadienes, hexene, hexadienes, hexatrienes, benzene, methyl isobutyrate, methyl methacrylate, toluene, heptene, heptatriene, methyl pentenoate, methacrylic acid, xylene, octatriene, octadienyne, nonatriene, hexadienoic acid, decatriene, decatetraene, decapentaene, undecatriene, methylbenzoic acids	128
Poly(methyl methacrylate-*co*-2-chloroethylmethacrylate) 22 mol% CEMA	600	CO_2, propene, acetaldehyde, vinyl chloride, ethyl chloride, methyl methacrylate	306
Poly(methyl methacrylate-*co*-2,2,2-trichloroethyl methacrylate) 24 mol% TCEMA	600	CO_2, propene, vinyl chloride, vinylidene chloride, dichloroacetaldehyde, methyl methacrylate	306
Poly(methyl methacrylate-*co*-methyl-α-chloroacrylate) 38 mol% MCA	600	CO_2, propene, methyl chloride, methyl methacrylate	306
Poly(methyl methacrylate-*co*-2-bromoethyl methacrylate) 22 mol% BEMA	600	CO_2, propene, acetaldehyde, methyl bromide, butene, vinyl bromide, methyl methacrylate	305
Poly(methyl methacrylate-*co*-2,3-dibromopropyl methacrylate) 16 mol% DBPMA	600	CO, CO_2, ethene, propene, methyl bromide, butene, vinyl bromide, ethyl bromide, 1-bromopropene, 2-bromopropene, cyclopropyl bromide, methyl methacrylate	305
Poly(methyl methacrylate-*co*-methyl-α-bromoacrylate) 22 mol% MBA	600	CO_2, acetaldehyde, methyl bromide, butene, vinyl bromide, unidentified alkenyl bromides, methyl methacrylate	305
Poly(methyl methacrylate-*co*-chlorotrifluoroethylene) 7.7 mol% MMA	Ambient to 500	HCl, CO_2, CH_3Cl, C_2F_3Cl (monomer), C_3F_5Cl, $C_2F_3Cl_3$, methyl methacrylate, chain fragments possibly with some lactonization	281
77.7 and 85.7 mol% MMA	Ambient to 500	HCl, CO_2, CH_3Cl, C_2F_3Cl (monomer), methyl methacrylate	281
Poly(methyl methacrylate-*co*-2-sulfoethyl methacrylate) 20, 40, 60 mol% MMA	50–600	CO, CO_2, H_2O, SO_2, ethene, methanol, methyl methacrylate (only 60 and 80 mol% methyl methacrylate copolymers)	290
Poly(methyl methacrylate-*co*-4-vinylpyridine) 22, 35, 46, 56 mol% 4-vinyl pyridine	Ambient to 500	4-vinyl pyridine, methyl methacrylate, traces of pyridine and 4-methyl pyridine	260
Poly(methyl methacrylate-*co*-phenyl methacrylate) 9, 18, 47, 88 mol% PMA	Ambient to 500	CO_2, isobutene, ketene, formaldehyde, methanol, methyl methacrylate, phenyl methacrylate, chain fragments including six-membered anhydride ring structures. Product distribution varies with copolymer composition with non-monomer products most pronounced at low PMA contents	314
Poly(methyl methacrylate-*co*-glycidyl methacrylate)	Ambient to 500	CH_4, CO, CO_2, propene, isobutene, dimethyl ketene, acrolein, allyl alcohol, glycidol, glycidyl methyl ether, methyl methacrylate, glycidyl methacrylate, chain fragments. Distribution varies with copolymer composition	286
Poly(methyl methacrylate *-block-t-*butylaziridine) ABA type copolymer with MW 12000 butylaziridine (B) and MW 16000 MMA (A)	Ambient to 600	CO, CH_4, CO_2, ethene, propene, isobutene, aziridine, methanol, methyl methacrylate, α-methylstyrene, 2-phenyl-2-butene, 1-*t*-butylamino-2-isopropenylamino-ethane, 2-*t*-butylaminoethyl-ethyl-isopropenylamine, 2-aminoethyl-2-*t*-butylaminoethyl-isopropenylamine, oligomers – principally butylaziridine based	287
Poly(methyl methacrylate-*co*-styrene)	260–340	Methyl methacrylate, styrene and oligomers of styrene; formation of oligomers of styrene strongly inhibited by presence of methyl methacrylate units	137
Poly (methyl methacrylate-*co*-vinyl acetate) (5–40 mol% vinyl acetate)	Ambient to 500	Acetic acid, methyl methacrylate, methyl acetate, chain fragment fraction, ketene, CO_2	92

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(methyl methacrylate-*co*-vinyl bromide) (6–75 mol% vinyl bromide)	Ambient to 500	Methyl methacrylate, methyl bromide, methanol, CO_2, methanol, HBr, CO, H_2 and other unidentified minor products. Relative amounts depend upon copolymer composition	138
Poly(2-bromoethyl methacrylate-*co*-styrene) (50/50, mol/mol)	Ambient to 500	Styrene, 2-bromoethyl methacrylate, modified chain fragement fraction. Minor products; ethylene, CO_2, HBr, vinyl bromide, acetaldehyde	139
Poly(methyl methacrylate)-*blend*-ammonium polyphosphate	Ambient to 450	H_2O and NH_3 from ammonium polyphosphate. Methyl methacrylate (major product), CO, CO_2, methanol, dimethyl ether, hydrocarbons, high boiling chain fragments	140
Poly(methyl methacrylate)-*blend*-poly(styrene)	Ambient to 500	Polymers degrade separately without interaction to give same products as polymers degraded separately; copolymers and mixtures of the same molar compositions can be distinguished by degradation behavior	141
Poly(methyl methacrylate)-*blend*-poly(vinyl acetate) (10/1–1/1, w/w)	Ambient to 500	Methyl methacrylate, acetic acid, methyl acetate, methanol, CO_2, butene, benzene, ketene	142
Poly(methyl methacrylate)-*blend*-poly(vinyl bromide) (50/50, mol/mol)	Ambient to 500	Methyl methacrylate, HBr, CO, CO_2 and other unidentified products	143
Poly(methyl methacrylate)-*blend*-poly(vinyl chloride)	Ambient to 500	Major products: HCl, methyl methacrylate; minor products: CO_2, methyl chloride, benzene	144,145
Poly(methacrylamide)	Up to 340	NH_3 and H_2O	146
	Above 340	Chain fragments (50%) in which a high proportion of amide groups have been converted to cyclic imides; small amounts of CO_2, CH_4, HCN	146
	Ambient to 500	NH_3, HCN, isobutene, CO_2(trace), methacrylonitrile (trace)	132
Poly(methacrylonitrile)	<200	Non-volatile material, coloration through yellow, orange, and red	40
	220–270	50–100% monomer depending upon pretreatment and purity of polymer	40
	Ambient to 500	Ethene, propene, isobutene, HCN, CO_2(from benzoyl peroxide end groups), methacrylonitrile (major product), 2-cyanobutane, 2-cyanopentene, CH_4	277
Poly(methacrylonitrile-*co*-styrene) 10 : 1 mole ratio	Ambient to 500	Ethene, isobutene, HCN, methacrylonitrile, styrene, toluene, α-methylstyrene, β-methylstyrene, 2-phenylbutene, 4-cyano-2,4-dimethylpentene, 4-phenyl-2-methylbutene, 3-phenyl-2-propenenitrile, chain fragments	279
Poly(methacrylonitrile-*co*-styrene) 1 : 1 mole ratio	Ambient to 500	Ethene, isobutene, HCN, methacrylonitrile, styrene, toluene, α-methylstyrene, 4-phenylbutene, 4-phenyl-2-methylbutene, 3-phenyl-2-propenenitrile, chain fragments	279
Poly(methacrylonitrile-*co*-styrene) 1 : 10 mole ratio	Ambient to 500	Ethene, isobutene, HCN, methacrylonitrile, styrene, toluene, α-methylstyrene, β-methylstyrene, 4-phenylbutene, allylbenzene, 4-phenyl-2-methylbutene, 3-phenyl-2-propenenitrile, chain fragments	279
1.4. POLY(VINYLS)			
Poly(tert-butyl-*N*-vinyl carbamate)	185–220	Equal amounts of CO_2 and isobutylene, in yields ranging from 43 to 60%	196
Poly(chlorotrifluoroethylene) (KEL-F)	347–418	25% of products volatile at 25°C–monomer with traces of C_3F_5Cl and $C_3F_4Cl_2$; 72.1% of larger chain fragments involatile at 25°C	14
Poly(1,1-dichloro-2,2-difluoroethylene)	240	20% black involatile residue; 80% monomer	22
Poly(perfluoroheptene)	210–270	100% monomer	49
Poly(perfluoropropylene)	280–400	100% monomer	49
Poly(tetrafluoroethylene)	504–538	>95% monomer, 2–3% C_3F_6, no larger fragments (in vacuum)	14
	1200	Monomer yield drops, larger fragments appear (in vacuum)	23
	600–700	At 6.66 mbar pressure; pure monomer; at 1013 mbar pressure, 15.9% monomer, 25.7% C_3F_6, 58.4% C_4F_8	24
Poly(trifluoroethylene)	380–800	High yields of HF and products involatile at 25°C	23
Poly(vinyl acetate)	213–235	Quantitative yields of acetic acid	71
	300	Small amounts of aromatics including benzene	71
	280–350	Acetic acid, benzene, naphthalene, toluene, styrene, ethyl benzene, propenyl benzene, propylbenzene, methylnaphthalene, ethylnaphthalene, propenylnaphthalene, propylnaphthalene, anthracene, methylanthracene, ethylanthracene, propylanthracene	147
Poly(vinyl acetate) in presence of zinc bromide	Ambient to 500	Acetic acid produced as low as 100°C	148
Poly(vinyl acetate-*co*-maleic anhydride) alternating copolymer	Ambient to 500	Acetic acid, CO_2, H_2O, maleic anhydride, CO	240
Poly(vinyl acetate-*co*-crotonic acid) alternating copolymer	Ambient to 500	Ethene, propene, acetic acid, CO_2, propanoic acid, 2-methylpropanoic acid, butanoic acid, cyclopentanone, some unsaturated cyclic and acyclic aliphatic ketones, some chain fragments containing lactones	241

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TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(vinyl acetate-*co*-methylvinylidene cyanide) Alternating with 10-15 mol% VA homosequence	285–350	Vinyl acetate, acetonitrile, HCN, acetic acid, acetamide, 2-methyl-1-aminonaphthalene, 3-cyano-3-pentene	266
Poly(vinyl acetate-*co*-styrene) (5–20 mol% vinyl acetate)	Ambient to 500	Acetic acid, styrene, chain fragment fraction, ketene, CO_2	92
Poly(vinyl acetate-*co*-vinyl alcohol) (50–99 mol% vinyl alcohol)	0–600	Acetaldehyde, acetone, benzene, crotonaldehyde, acetic acid	154
Poly(vinyl acetate-*co*-vinyl chloride)	Ambient to 500	Bulk polymer; hydrogen chloride and acetic acid in the proportions of the monomers in the copolymer; at each end of the composition range, incorporation of the comonomer results in a copolymer less stable than the homopolymer; minimum stability at 40–50 mol% vinyl acetate	155
	180	In tritolyl phosphate solution; similar to bulk polymer with minimum in stability at 30–40 mol% vinyl acetate	156
Poly(vinyl acetate-*co*-vinyl fluoride) (91 : 9 to 1 : 89)	200–500	Hydrogen fluoride, acetic acid, hydrocarbons	150
(80 : 20 to 23 : 77)	200–500	Hydrogen fluoride, HCl, hydrocarbons	150
Poly(vinyl acetate)-*blend*-poly(styrene) (1/1, w/w)	Ambient to 500	Products as for individual polymers	158
Poly(vinyl acetate-*blend*-poly(vinyl chloride)	Ambient to 500	Acetic acid, HCl, and traces of CO_2, ketene, acetyl chloride, CO and CH_4	157
Poly(vinyl alcohol)	250	Quantitative yields of H_2O	72
	240	Main products H_2O and C_2H_5OH, with aldehydes CH_3—$(CH{=}CH)_n$—CHO and ketones CH_3—$(CH{=}CH)_n$—$COCH_3$ where $n = 0,1,2$, etc.	73
Poly(vinyl bromide)		HBr, ethylene, benzene, and other unidentified minor products	143
Poly(vinyl butyrate)	300–325	Butyric acid	74
Poly(*N*-vinylcarbazole)	230–410	Below 350°, monomer; above 350°, monomer and low-molecular weight oligomers	175
	300–500	Monomer (49% of total volatiles at 300°C, 39% at 500°C) ethane, ethene, propene, *n*-butane, isobutene, *cis* and *trans* but-2-ene, 1,3-butadiene, 2-methylbutadiene, trace amounts of C_5 and C_6 hydrocarbons	244
Poly(3,6-dichloro *N*-vinylcarbazole)	300–500	Monomer (83% of total volatiles at 300°C, 67% at 500°C), ethane, ethene, propene, *n*-butane, isobutene, *cis* and *trans* but-2-ene, 1,3- butadiene, 2-methylbutadiene, trace amounts of C_5 and C_6 hydrocarbons	244
Poly(3,6-dibromo *N*-vinylcarbazole)	350–500	Monomer (48% of total volatiles at 350°C, 40% at 500°C), ethane, ethene, propene, *n*-butane, isobutene, *cis* and *trans* but-2-ene, 1,3-butadiene, 2-methylbutadiene, trace amounts of C_5 and C_6 hydrocarbons	244
Poly(vinyl chloride)	200–300	Quantitative yields of HCl	75
	400	Saturated and unsaturated, aliphatic and aromatic hydrocarbons are produced with benzene and toluene in high yield	75
	600	In helium; quantitative yield of HCl, remainder residue and hydrocarbons; benzene is major volatile hydrocarbons product	76
	200–800	Aliphatic hydrocarbons, benzene (major product), toluene, ethylbenzene, *o*-xylene, monochlorobenzene, styrene, vinyl toluene, *p*-dichlorobenzene, *o*-dichlorobenzene, indene, 1,3,5-trichlorobenzene, 1,2,4-trichlorobenzene, naphthalene, α-methylnaphthalene, β-methylnaphthalene; effect of ZnO, SnO_2, and Al_2O_3 on the yields of products is also recorded	149
	Ambient to 1000	HCl, CO_2, ethene, ethane, propane, 1-butene, 2-butene, 1-pentene, cyclopentene, *n*-pentane, 2-methylbutane, 1,3-pentadiene, 2-methyl-1,3-pentadiene, complex series (60 identified) of aromatic and polyaromatic species, including benzene, styrene, methylstyrenes, toluene, *o*-xylene, *m*-xylene, *p*-xylene, biphenyl, naphthalene, anthracene, phenanthrene, pyrene, etc. (see Ref.)	267
Poly(vinyl chloride)-*blend*-poly(α-methylstyrene)	Ambient to 500	HCl, α-methyl styrene	159
Poly(vinyl chloride)-*blend*-poly(styrene)	Ambient to 500	HCl, styrene	159
(10/1 to 1/10, w/w)	Ambient to 500	Products as for individual polymers	160
PVC 1 : 1 blend with bisphenol A polycarbonate	Ambient to 500	CO_2, HCl, benzene, traces of toluene, phenol and other phenolic compounds, bisphenol A, polycarbonate chain fragments	291

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
PVC 1 : 1 blend with poly(dimethylsiloxane)	Ambient to 500	HCl, benzene, traces of toluene, octamethylcyclotetrasiloxane, oligomeric siloxanes	292
Poly(vinyl fluoride)	372–480	High yields of HF and products involatile at 25°C – little carbonization	23
	200–500	Hydrogen fluoride, C_2–C9 fractions	150
	450	85% wt. loss, 33 ± 5% volatile products, remainder was oily green liquid; Volatile products: HF (82 mol%), CH_4 (5.2), C_2H_6 (0.6), ethylene (0.8), acetylene (0.03), fluoroethylene (0.07), propene (0.5), C_3H_5F (0.4), butane (0.06), butene (1.1), 1,4-butadiene (0.2), 1,3-butadiene (0.7), C_4H_5F (0.06), Cyclopentadiene (0.05), benzene (4.5), fluorobenzene (0.3), toluene (1.0), $C_6H_4(CH_3)F$ (0.04), $C_6H_5C_2H_5$ or $C_6H_4(CH_3)_2$, and styrene (0.9), *n*-propylbenzene (0.2), iso-propylbenzene (0.3), C_6H_5–C_3H_5 (0.2), indene (0.3), naphthalene (0.8), fluoronaphthalene (0.3)	151
Poly(vinylidene chloride)	225–275	High yields of HCl	77
	170	Polymer prepared in presence of oxygen; phosgene, formaldehyde, HCl	152
Poly(vinylidene cyanide)	>160	High yields of monomer	78
Poly(vinylidene fluoride)	400–530	35% HF and high yields of products involatile at 25°C – some carbonization	23
	Ambient to 500	Monomer, dimer ($C_4F_3H_3$), SiF_4, (from reaction of HF with glass)	265
Poly(vinylidene fluoride-*co*-chlorotrifluoroethylene) 80 mol% vinylidene fluoride	Ambient to 500	HCl, HF, vinylidene fluoride, chlorotrifluoroethylene, dimer ($C_4F_3H_3$)	265
3, 10 mol% vinylidene fluoride	Ambient to 500	HCl, chlorotrifluoroethylene, dichlorodifluoroethene, chloropentafluoropropene, trichlorotrifluoroethane, dimer ($C_4F_3H_3$)	265
Poly(vinyltrimethylsilane)	300–600 in argon	Tetramethylsilane, trimethylvinylsilane, 2-trimethylsilylpropene, 1,3-tri-methylsilylpropane, 1,3-trimethylsilylpropene, 2,4-trimethylsilylbutene-1,2,4,6-trimethylsilylhexene-1	153
Poly(4-vinyl pyridine)	Ambient to 500	Monomer, traces of pyridine and 4-methyl pyridine	260
Poly(vinylacetophenone)	380	CO, CH_4, ethane, 4-acetyl styrene, 4-acetyl toluene, styrene, toluene, 4-vinyl acetaldehyde, α-methylstyrene, α-methyl-4-acetylstyrene, 4-methylstyrene	307
Poly(*N*-acryloyl-*N'*-cyanoacetohydrazide)	600	Cyanoactetamide	289
1.5. POLY(VINYL KETONES), POLY(VINYL ETHERS)			
Poly(methyl isopropenyl ketone)	270–360	H_2O	41
	Ambient to 500	Methyl isopropenyl ketone (50%), modified chain fragments, water, small amounts of CH_4 and CO	161
	150–190	Monomer under 313 nm radiation	42
Poly(methyl vinyl ketone)	270–360	H_2O, 3-methyl-2-cyclohexene-1-one and other six-membered ring ketones	44
	Ambient to 500	Water, modified chain fragments, small amounts of CH_4 and CO	161
Poly(trifluorovinyl phenyl ether)	275–500	Maximum of 75% volatilization–products unknown	49
1.6. POLY(STYRENES)			
Poly(styrene)	300–400	40.6% monomer, 2.0% toluene, 0.1% CO, remainder dimer, trimer, and tetramer – monomer yield increases with pressure of nitrogen – 62% at 1013 mbar	62,63
	500–1200	Small hydrocarbon fragments appear (C_1–C_6) – fragmentation is greater the higher the temperature and the greater the pressure of inert gas	28
	300–570	Pulsed pyrolysis and 40 µg of polymer. Styrene (92.4–99.8% depending upon temperature), benzene, toluene, and ethyl benzene	163
	Ambient to 500	Thin films (1000 Å or less), monomer yield about 55%	164
–, (crosslinked)	346–450	Crosslinking with increasing quantities of divinyl or trivinyl benzene progressively decreases the styrene yield–the yield of larger chain fragments and the amount of carbonization also increase	11,64
–, (head-to-head)	320	Monomer, dimer, trimer, tetramer, pentamer, stilbene	165
	500 under N_2	Hydrocarbons (C_1–C_4) (2.2 relative GC peak intensity), benzene (0.4), toluene (3.3), ethylbenzene (2.0), styrene (20.4), 3-phenyl-1-propene (1.5), 1-phenylpropane (2.3), α-methylstyrene (0.3), 1-phenylbutadiene (1.0), diphenylmethane (0.3), 1,1-diphenylethylene (2.6), 1,2-diphenylethane (1.3), 1,2-diphenylpropane, (0.7), 1,2-diphenyl-1-propene (0.5), 2,3-diphenyl-1-propene (2.2), 3,4-diphenyl-1-butene (0.5), 2,3-diphenylbutadiene (2.0), 1,2-diphenylethylene (6.5), 1,4-diphenylbutane (3.5), 1,4-diphenyl-1-butene (3.8), 1,4-diphenyl-1-pentene (1.3),	166

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TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
		1,3-diphenylpropadiene (2.1), 1,2-diphenylbutadiene (1.8), 1,4-diphenylbutadiene (1.9), 1,3,4-triphenylbutene (0.6), 1,2,5-triphenylpentane (1.5), 1,2,5-triphenyl-1-pentene (4.6), 2,3,6-triphenyl-1-hexene and 2,5,6-triphenyl-1-hexene (8.9), 1,4,5-triphenyl-1-pentene (4.4)	
	Ambient to 600	Styrene, 1,2-diphenylethene, 1,2-diphenylpropane, 1,4-diphenylbutane, 1,4-diphenyl-1-butene, 1,4-diphenylpentane, 4,5-diphenyl-1,8-octadiene, 2,3,6-triphenyl-1-hexene, 1,4,5-triphenyl-1-pentene	271
–, (poly(divinyl benzene))	385–450	Volatile products include toluene, benzene, styrene, and xylene	11,64
–, (poly(trivinyl benzene))	470–500	Mixture of aliphatic and aromatic hydrocarbons	11,64
–, α-acetoxy-	270	Acetic acid (99%), mixture of α-acetoxystyrene and acetophenone (1%)	167
–, *m*-amino-	340–500	Gaseous fraction CO_2, CH_4, C_2H_6, C_3H_6, C_3H_8; liquid fraction *m*-aminostyrene, *m*-toluidine; gum fraction mainly dimer and trimer; crosslinked residue remains	65
	320–500	Monomer yield rises from 41% (320°C) to 61%(500°C); minor product is *m*-toluidine, with traces of aniline, *m*-ethylaniline, α-methyl-*m*-aminostyrene, H_2, CO_2, CH_4; remainder comprises chain fragments and residue	66
–, *p*-chloro-α-methyl-	Ambient	Gamma radiolysis: CH_4 (12.9 relative mass spec. ion current), C_2H_6 (21.8), C_3H_6 (13.6), C_3H_8 (10.5), C_4H_{10} (61), HCl (4.2), CH_3Cl (17.4), C_3H_4Cl (20), C_2H_5Cl (28.1), C_3H_7Cl (3.3), CH_2Cl_2 (9.9), C_6H_5Cl (3.0), *p*-chlorostyrene (1.5), *p*-chloro-α-methyl styrene (12.5)	168
–, α-deutero-	334–387	68.4% monomer, 1.5% α-deuterostyrene, 0.6% α-methylstyrene, 29.5% larger chain fragments	2,13
–, β-deutero-	345–384	39.7% monomer, 1,2% toluene, 0.1% deuterotoluene, 59% larger chain fragments	2,13
–, *p*-fluoro-α-methyl-	Ambient	Gamma radiolysis: CH_4 (0.4 relative mass spec. ion current), C_3H_4 (3.0), C_3H_6 (0.6), C_3H_8 (0.8), C_4H_4 (3.5), C_4H_6 (6.0), C_3H_7F (3.2), C_4H_3F (6.0), C_4H_9F (14.5), C_5H_5F (9.37), C_6H_5F (29), *p*-fluorotoluene (23), *p*-fluorostyrene (6.45), *p*-fluoro-α-methyl styrene (175.5), *p*-fluoro-isopropyl benzene (11.5)	168
–, *p*-methoxy-α-methyl-	Ambient	Gamma radiolysis: CH_4 (32 relative mass spec. ion current), H_2O (26), C_2H_4 (35), C_3H_4 (7.3), C_3H_6 (15.2), C_3H_8 (68.7), C_4H_{10} (8.2), CH_3OH (24.2), phenol (63.8), methoxybenzene (38.6), *p*-methoxytoluene (19.6)	168
–, 3-methyl-	309–399	44.4% monomer, 7.3% xylene, 48.3% larger chain fragments	3
–, 4-methyl-	200–350	Ratio of monomer to oligomer rises from 40% at 200°C to 95% at 350°C	79
	560	C_1–C_4 hydrocarbons, benzene, toluene, ethylbenzene, styrene, ethyltoluene, vinyltoluene, naphthalene, ethylstyrene	278
Poly(*o*-propionylstyrene)	385	CO, CH_4, ethane, ethene, butane, 2-propionylstyrene, 2-propionyltoluene, styrene, toluene, α-methylstyrene, α-methyl-2-propionylstyrene, various dimeric and trimeric species not fully characterized	313
–, α-methyl-	200–500	95–100% monomer	28
	520–1200	Fragments both larger and smaller than monomer appear in increasing amounts the higher the temperature, particularly CH_4, C_2H_4, and C_6H_6 until at 1200°C the monomer yield is only 33.9%	28
	Ambient	Gamma radiolysis: CH_4 (1.0 relative mass spec. ion current) C_2H_6 (1.2), C_3H_4 (6.7), C_3H_6 (1.8), C_3H_8 (2.7), C_4H_4 (22), C_4H_{10} (9.9), cyclopentadiene (6.1), benzene (72.8), toluene (20), styrene (50.5), α-methyl styrene (184), isopropyl benzene (10)	168
	400–900	Benzene, toluene, ethylbenzene, cumene, styrene, α-methyl styrene, indene/phenylpropyne, methyl-1,2-dihydronaphthalene, methylene indene, naphthalene, 2-methylnaphthalene, 1-methylnaphthalene, biphenyl, anthracene/phenanthrene, dihydropyrene	169
–, *p*-*N*,*N*-diethylamino-	350–500	Toluene, styrene, *p*-ethylstyrene, *N*,*N*-diethyl-*p*-toluidine, *N*,*N*-diethyl-*p*-ethylaniline, *p*-*N*,*N*-diethylaminostyrene, α-methyl-*N*,*N*-diethylaminostyrene, 2-(*p*-*N*,*N*-diethylaminophenyl) butane, α-propyl-*p*-*N*,*N*-diethylaminostyrene, *N*-ethyl-*p*-toluidine, *N*-ethyl-*p*-toluidine, *N*-ethyl-*p*-ethylaniline, *N*-ethyl-*p*-aminostyrene and closely related compounds	170
–, *m*-*N*,*N*-dimethylamino-	350–500	*m*-*N*,*N*-Dimethylaminostyrene, *N*,*N*-dimethyl-*m*-toluidine, *m*-xylene, styrene, *N*,*N*-dimethyl-*m*-ethylaniline, toluene, *m*-methylstyrene, α-methyl-*m*-*N*,*N*-dimethylaminostyrene, *trans*-β-methyl-*m*-*N*,*N*-dimethylaminostyrene, α-ethyl-*m*-*N*,*N*-dimethylaminostyrene, *N*-methyl-*m*-toluidine, *N*-methyl-*m*-ethylaniline, *N*-methyl-*m*-aminostyrene, oligomers of *m*-*N*,*N*-dimethylaminostyrene and closely related compounds	171

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
–, *p*-*N*,*N*-dimethylamino-	500 for 30 min	*N*,*N*-dimethylaniline (0.2% of original polymer weight), *N*,*N*-dimethyl-*p*-toluidine (2.8%), *N*-methylaniline (0.4%), *N*,*N*-dimethyl-*p*-ethylaniline (0.4%), *N*-methyl-*p*-toluidine (0.4%), *N*-methyl-*o*-toluidine (0.4%), *p*-toluidine (0.4%), *N*-methyl-*p*-ethylaniline (0.4%), *p*-*N*,*N*-dimethylaminostyrene (36.3%), α-methyl-*p*-*N*,*N*-dimethylaminostyrene (0.7%)	172
Poly(*p*-(2,4-dichlorobenzyl)styrene)	Ambient to 500	Benzene, toluene, ethylbenzene, styrene, α-methylstyrene, 1,3-dichlorobenzene, 2,4-dichlorotoluene, monomer (major product), complex series of halogenated and dehalogenated chain fragments (see Ref.)	275
–, *p*-isopropyl-α-methyl-	Ambient	Gamma radiolysis: CH_4 (8, relative mass spec. ion current), C_2H_4 (4.6), C_3H_4 (7.2), C_3H_6 (21.2), C_3H_8 (108), C_4H_{10}, (57.4)	168
–, 2,3,4,5,6-pentafluoro-	390–446	63% of the products are volatile at 25°C – contains some monomer	67
–, α,β,β-trifluoro-	333–382	72% monomer, 28% larger chain fragments	14
Polystyrene (chain brominated on α-position)	150–300	HBr, trace styrene	271
	300–500	Styrene, HBr, toluene, benzene	271
	Ambient to 500	HBr, styrene, traces of toluene and benzene, *p*-bromostyrene	271
Poly(styrene-*co*-methylene) $\sim\sim CH(\phi)-CH_2-CH_2-CH(\phi)-(CH_2)_n\sim\sim$ n = 0,1,3,4 and 5	350 for 4 hr	Benzene, toluene, ethylbenzene, styrene (major volatile product), *n*-propylbenzene (except with $n=2$), phenylpropene, butylbenzene (except with $n=0$), phenylbutene (with $n=4$ and 5 only), amylbenzene (with $n=4$ and 5 only) hexylbenzene (with $n=4$ and 5 only), chain fragment fraction (major product)	173
Poly(styrene-*co*-SO_2) (1.85 : 1)	200	At 26.7 mbar pressure; 40% weight loss; main product is 2,4-diphenyl thiophene; at least 11 unidentified minor products	68
Poly(styrene-*co*-methacrylic acid) 18, 44 and 72 mol% MAA	Ambient to 500	CO_2, H_2O, butene, isobutene, dimethyl ketene, styrene, methacrylic acid, succinic-type 5-membered cyclic anhydrides	294
Poly(styrene-*co*-chlorotrifluoroethylene) 7.7, 14, 20 mol% CTFE	Ambient to 500	Chlorotrifluoroethylene, styrene, HCl, chloropentafluoropropene, ethene, chloroethene, toluene, α-methylstyrene, dimer and trimer structures with some unsaturation; SiF_4 (from reaction of HF with glass). Distribution of products varies with polymer composition	261
Poly(styrene-*co*-glycidyl methacrylate) 10, 23, 51, 62, 78, 86 mol% GMA	Ambient to 500	CO, CO_2, propene, isobutene, dimethyl ketene, acrolein, allyl alcohol, toluene, styrene, α-methylstyrene, ethylbenzene, glycidol, glycidylmethacrylate; product distribution depends on copolymer composition	264
Poly(styrene-*co*-phenyl methacrylate) 10–90 mol% PMA	Ambient to 500	CH_4, CO, CO_2, propene, dimethyl ketene, toluene, styrene, 4-phenylbutene, 4-phenyl-2-methylbutene, phenyl methacrylate, α-methylstyrene, short chain fragments	280
Poly(styrene-*co*-1,2,2,2,-tetrachloroethyl acrylate) (4.7/1)	310 and 500	Styrene, benzene, toluene, ethylbenzene, α-methyl styrene, diphenyl, 5,6-benzo-2-methyl-3,5-cyclohexadienone, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene, chloroacetaldehyde, dichloroacetaldehyde, trichloroacetaldehyde, trichloropropanal	174
Poly(styrene-*co*-1,2,2,2-tetrachloroethyl methacrylate) (3.8/1)	310 and 500	Styrene, benzene, toluene, allylbenzene, α-methylstyrene, indane, isopropenylstyrene, 5,6-benzo-2-methyl-3,5-cyclo-hexadienone, propyl-propenylbenzene, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene chloroacetaldehyde, dichloroacetaldehyde, trichloroacetaldehyde, trichloropropanal, benzyl chloride, 2-chloroethylstyrene, 2,2-dichloroethenyl methacrylate, 1,2,2,2-tetrachloroethyl methacrylate	174
Poly(styrene-*co*-bis-(1,2,2,2-tetrachloroethyl)fumarate) (8.9/1)	310 and 500	Styrene, benzene, toluene, ethylbenzene, allylbenzene, α-methylstyrene, indane, diphenyl propylpropenylbenzene, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene, chloroacetaldehyde, dichloroacetaldehyde, trichloroacetaldehyde, benzyl chloride	174
Poly(styrene-*co*-bis-(1,2,2,2-tetrachloroethyl) muconate) (7.72/1)	310 and 500	Styrene, benzene, toluene, ethylbenzene, α-methylstyrene, diphenyl, propylpropenylbenzene, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene, chloroacetaldehyde, dichloroacetaldehyde, trichloroacetaldehyde, trichloropropanal, benzyl chloride	174
Poly(styrene-*co*-*N*-(2,4,6-tribromophenyl)-maleimide) (6.4/1)	310 and 500	Styrene, benzene, toluene, ethylbenzene, allylbenzene, α-methylstyrene, indane, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene, *N*-(2,4,6-tribromophenyl) maleimide, *N*-dibromophenyl maleinimide	174

TABLE 1. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(styrene-*co*-*N*-(pentachlorophenyl)-maleimide) (4.9/1)	310 and 500	Styrene, benzene, toluene, ethylbenzene, allylbenzene, α-methylstyrene, indane, diphenylethane, 1,3-diphenylpropane, dimers and trimers of styrene, pentachlorophenyl isocyanate, *N*-pentachlorophenyl formamide, *N*-tetrachlorophenyl maleinimide, *N*-(pentachlorophenyl) maleimide, methyl maleic acid pentachlorophenylimide	174
Poly(styrene) 1 : 1 (w/w) *blend* with bisphenol A polycarbonate	Ambient to 500	CO_2 styrene, α-methylstyrene, *p*-cresol, phenol, *p*-ethylphenol, *p*-vinylphenol, *p*-isopropylphenol, polystyrene short chain fragments with vinylidene and saturated ends, polycarbonate cyclic dimer and chain fragments, bisphenol A	272
Poly(vinyl toluene-*co*-divinyl benzene) 10-50% DVB	560	C_1–C_4 hydrocarbons, benzene, toluene, ethylbenzene, styrene, ethyltoluene, α-methylstyrene, vinyltoluene, divinylbenzene, naphthalene, ethylstyrene; distribution varies with copolymer composition	278
Poly(4-methoxystyrene) 1 : 1 (w/w) *blend* with bisphenol A polycarbonate	Ambient to 500	CO_2, 4-methoxystyrene, α-methyl-4-methoxystyrene, *p*-cresol and other phenols in small amounts, poly(4-methoxystyrene) dimer and trimer with vinylidene and saturated ends, polycarbonate cyclic dimer and chain fragments, bisphenol A	272

TABLE 2. MAIN-CHAIN CARBOCYCLIC POLYMERS

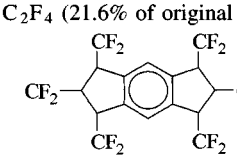
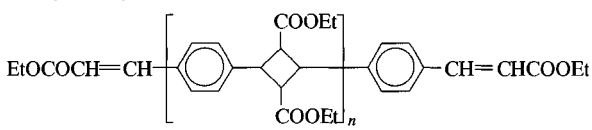
Polymer	Temperature range (°C)	Degradation products	Refs.
Poly((E,E)-[6.2]paracyclophane-1,5-diene)	460, 515, 620	Primary product is monomer; in addition, a series of secondary products based on ring-opened monomer are formed by H-abstraction (see Ref.); also detected: toluene (not 460°C) and *p*-xylene	251
Poly(perfluoro-*m*-phenylene)	700	Volatiles comprise SiF_4 (from silica vessel), CO_2, with traces of C_6F_6 and C_6F_5H; white sublimate on vessel walls; residue contains 1% fluorine	50
Poly(perfluoro-*p*-phenylene) high molecular weight (15000)	700	Volatiles comprise SiF_4 (from silica vessel), CO_2, with traces of C_6F_6 and C_6F_5H; white sublimate on vessel walls; residue contains 1% fluorine	50
low molecular weight (3700)	700	Volatiles comprise SiF_4 (from silica vessel), CO_2, with traces of C_6F_6 and C_6F_5H; no white sublimate; residue contains 33–41% fluorine	50
Poly(1,4-phenylene)	250–620	79% residue, 10% chain fragments, 11% volatiles comprising H_2, CH_4, H_2O, HCl (from catalyst)	55
–, 2-hydroxy-	300–620	60% residue, 24% chain fragments, 16% volatiles comprising CO, CO_2 H_2, CH_4	55
Poly(1,4-phenylene ethylene)	420–465	3.6% of products are volatile at 25°C and consist of 2.83% xylene, 0.29% toluene, 0.28% methylethyl benzene, 0.14% methylstyrene, 0.06% benzene, products involatile at 25°C consist of dimeric–octameric fragments	13,14,80
	408–515	Mainly chain fragments (dimer–pentamer); traces of H_2, monomer	81
	436–475	H_2, CH_4, C_2H_4, C_2H_6, toluene, *p*-xylylene, *p*-ethyltoluene, *p*-methylbibenzyl, 1,2-di-*p*-tolylethane, 4,4′-dimethylstilbene	176
Poy(1,3-phenylene-hexafluoro-trimethylene	660 for 30 s	C_2F_4 (21.6% of original weight of polymer), HF (17),  CF_2 (13.9%), $C_6H_5(CF_2)_5C_6H_5$ (10.2%), C_6H_6 (8.5%), $C_6H_5(CF_2)_3C_6H_5$ (7.9%)	177
Poly(1,4-phenylenemethylene)	386–416	7.4% of products are volatile at 25°C; 5.9% toluene, 1.4% benzene, 0.1% xylene	13,14
Poly α-(ethoxycarbonyl-vinyl)-ω-[4-(ethoxycarbonyl-vinyl)-phenyl] (1,4-phenylene-2,4-bis-ethoxycarbonyl-1,3-cyclobutylene) 	320–360 for 1 h under N_2	Low-molecular-weight polymer and small amount of monomer	195

TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

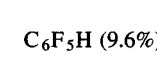
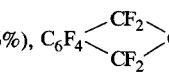
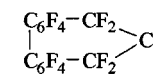
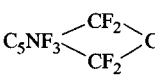
Polymer	Temperature range (°C)	Degradation products	Refs.
3.1. POLY(OXIDES) AND POLY(KETONES)			
Poly(fluoro ethers)	510–540		178
a. $-\{O-CH_2-(CF_2)_3CH_2O-CH_2\}_n-$		$CH_3-(CF_2)_3CH_2-$ OCH_3 (46.9%) relative areas of glc traces, $CH_3(CF_2)_3CH_3$(14%), $CH_3CF_2CF_2H$ (6.7%), $CF_2=CH_2$ (6.7%), CF_2H-CF_2H (2.3%)	
b. $-\{OCH_2(CF_2)_3CH_2O-p-C_6F_4\}_n-$		$CF_2=CH_2$ (17.0%), $CH_3(CF_2)_3H$ (16.5%), CF_2HCF_2H (6.4%), $CH_3CF_2CF_2H$ (4.1%),  C_6F_5H (9.6%),  (9.4%), C_6F_5CHO (7.2%), C_6F_4HCHO (4.9%), $C_6F_4H_2$ (3.3%)	
c. $-\{OCH_2(CF_2)_3CH_2O-p-C_6F_4-p-C_6F_4\}_n-$		HF (31%), $CF_2=CH_2$ (22.0%), $CH_3(CF_2)_3H$ (19.7%), CF_2HCF_2H (10.1%), $C_6F_5C_6F_4H$ (5.6%),  (8.1%), $C_6F_4HC_6F_4H$ (9.3%)	
d. $-\{OCH_2(CF_2)_3CH_2O-m-C_5NF_5\}_n-$		HF (50%), $CH_3(CF_2)_3H$ (27.6%), $CF_2=CH_2$ (7.8%), CF_2HCF_2H (4.3%),  (5.6%), $C_5NF_3H_2$ (3.1%)	
Poly(oxybutylethylene)	321–365	Mixture of saturated and hydrocarbons (C_1-C_6), aldehydes (C_1-C_6), H_2, H_2O	20
Poly(oxy-2,5-dimethoxy-1,4-phenylene)	100–550	41% residue, 29% fragments, 30% volatiles comprising H_2, CH_4, CO, CO_2, CH_3OH, traces of H_2O, C_2H_6, other hydrocarbons	55
Poly(oxy-2,6-dimethyl-1,4-phenylene)	100–550	26% residue, 66% chain fragments, 8% volatiles comprising H_2, CH_4, H_2O, CO, CO_2	55
	400–550	*o*-Cresol, 2,6-dimethyl phenol, 2,4-dimethyl phenol, 2,4,6-trimethyl phenol, various ketones, dimers, and more complicated structures (see Ref.), various combinations of the monomeric skeletons	179
Poly(oxyethylene)	324–363	9.7% of products volatile at 25°C–3.9% monomer with smaller amounts of CO_2, formaldehyde, ethanol, and saturated and C_1-C_7 compounds	25
	280	Diethyl ether, ethyl methyl ether, acetaldehyde, formaldehyde, CO_2, CO, ethene, cyclohex-3-ene-1-one	252
Poly(oxymethylene)	222	100% Monomer	27
	100–180	100% Formaldehyde; inverse relationship between molecular weight and rate for hydroxyl terminated polymers; acetylation of terminal hydroxy groups inhibits the reaction	181
Poly(oxymethyleneoxyethylene)	314–338	Major volatile products CH_4, C_2H_6, C_2H_4, C_2H_2, $HCO_2C_2H_5$	20
Poly(oxy-2-methylphenylene)	250–500	*p*-Xylene, phenol, 2-methylphenol, 4-methylphenol, 2,4-dimethylphenol, and a variety of more complex structures (see Ref.)	180
Poly(oxy-1,3-phenylene)	300–620	34% Residue, 53% chain fragments, 13% volatiles comprising H_2O, CO, H_2, CH_4, CO_2, trace of C_6H_6	55
Poly(oxy-1,4-phenylene)	550	Hydroquinone, phenol, benzene, diphenylether, 4,4′-dihydroxydiphenylether, 4-phenoxyphenol. In addition, a broad spectrum of phenyl and phenolic terminated oligomers with up to 10 aromatic units	255
	585	As 550°C plus dibenzofuran and dibenzofuran-containing oligomers	255
Poly(oxy-1,4-phenylene)-*blend*-polystyrene (60 : 40)	300–360	Ethylbenzene, styrene, α-methylstyrene, 2-methylphenol, 2,6-dimethylphenol, 2,4-dimethylphenol, 2,4,6-trimethylphenol, 1,3-diphenylpropane, 2,4-diphenyl-1-butene, 2,4,6-triphenyl-1-hexene, PPO dimers	318
Poly(oxypropylene) atactic	270–330	12.8% of products volatile at 25°C, including 4.00% acetaldehyde, 2.22% acetone, 1.43% dipropyl ether and 0.75% propylene	25
isotactic	275–355	20% of products volatile at 25°C, including 6.34% acetaldehyde, 2.39% acetone, 2.19% dipropyl ether, and 2.22% propylene	25
Poly(oxytetramethylene)	347	Volatiles comprise a mixture of CH_4, C_2H_6, C_3H_8, C_4H_{10}, C_2H_4, C_3H_6, CH_3CHO, C_2H_5CHO, C_3H_7CHO	20
Poly(oxytrichloromethyl methylene) (Poly(trichloro acetaldehyde))	Ambient	Monomer	182

TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(oxy-1,4-phenylene-carbonyl-1,4-phenylene) (Poly(ether ketone), PEK)	390, 490, 530	Low MW polymer at 490°C produces benzene, phenol, benzaldehyde, 4-hydroxybenzaldehyde, diphenyl ether, biphenylene, benzophenone, series of –H, –OH and –CHO terminated chain fragments containing <8 aromatic units and fragments containing dibenzofuran units. High MW polymer at 530°C as above. Low MW polymer at 390°C produces significant quantities of 4-hydroxy benzaldehyde and benzophenone-4,4′-diol	250
	506	Phenol, dibenzofuran, benzofuran derivatives, mixtures of oligomers mostly hydroxy terminated	315
Poly(oxy-1,4-phenylene-oxy-1,4-phenylene -carbonyl-1,4-phenylene) (Poly(ether ether ketone), PEEK)	506	Phenol, dibenzofuran, benzofuran derivatives, mixtures of oligomers mostly hydroxy terminated	315
Poly(etherketone-*co*-ethersulphone) Mole ratio 1 : 3, 3 : 1, 1 : 1 (random), 1 : 1 (alternating)		All polymers have same products with distribution varying with composition. Biphenyl, hydroquinone, 4-phenoxyphenol, benzaldehyde, 4-hydroxybenzaldehyde, diphenyl sulphone, diphenylether, phenol, benzophenone-4,4′-diol. Also two series of chain fragments containing PEK and PES units with –H, –OH and –CHO ends. Analogous fragments with biphenyl units	250
3.2. POLY(ESTERS)			
Poly(3-oxybenzoyl)	330	Monomer, cyclic dimer–heptamer	187
Poly(4-oxybenzoyl)	505–565	CO, CO_2, phenol, *p*-hydroxy phenyl benzoate	197
Poly(oxycarbonyl-1,3-phenylene-hexafluoroisopropylidene-1,3-phenylene carbonyloxy-1,3-phenylene-hexafluoro-isopropylidene-1,3-phenylene)		HF (12%), cyclic structures (see Ref.)	177
Poly(oxycarbonyl-1-chloroisopropylidene)	253–384	HCl and a compound with a glycollidic structure	185
Poly(oxycarbonylcyclohexylidene)	200–500	Cyclohexen-1-carboxylic acid, cyclohexanone	198
Poly(oxycarbonylcyclopentylidene)	200–500	Cyclopenten-1-carboxylic acid, cyclopentanone, dicyclopentyl glycollide	198
Poly(oxycarbonyl-3-pentylidene)	200–300	*Cis*- and *trans*-2-ethyl crotonic acid (major products)	188
Poly(oxyethyleneoxyterephthaloyl) (Poly(ethylene terephthalate))	283–306	Acetaldehyde major gaseous product with CO_2, CO, C_2H_4, H_2O, CH_4, benzene, 2-methyl-dioxolane, terephthalic acid, and more complex chain fragments	26
	900	CO, CH_4, CO_2, ethylene, acetylene, ethane, H_2O, propylene, ethanal, acetone, propanal, ethanol, benzene, toluene, ethyl benzene, styrene, *p*-vinyl toluene, benzaldehyde, *p*-ethyl toluene, acetophenone, methyl benzoate, vinyl benzoate, ethyl benzoate, *p*-methyl acetophenone, benzoic acid, *p*-methyl vinyl benzoate, *p*-vinyl acetophenone, propyl benzoate, *p*-ethyl vinyl benzoate, *p*-vinyl vinyl benzoate, biphenyl, 1-hydroxy ethyl benzoate, diacetyl benzene, *p*-acetyl vinyl benzoate, divinyl terephthalate, ethyl vinyl terephthalate, ethyl vinyl terephthalate, *p*-acetyl benzoic acid, methyl 1-hydroxyethyl terephthalate, ethylene dibenzoate	192
	Ambient to 500	CO, CH_4, CO_2, ethene, ketene, acetaldehyde, 1,4-dioxane, toluene, benzaldehyde, vinylbenzoate, divinylterephthalate, benzoic acid, terephthaldehydic acid, terephthalic acid, hydroxyethyl methyl terephthalate, short chain fragments	298
Polyester from terephthaloyl chloride and $n = 4$ polyethylene glycol	Ambient to 500	CO, CH_4 (trace), CO_2, ethene (trace), formaldehyde, acetaldehyde, 1,3-dioxalane, ethoxyacetaldehyde, 1,4-dioxane, glycidol, diethylether, 1,2-dimethoxyethane, 2-methoxyethanol, 2-ethoxyethanol, $-CH_2CH_2COCH_2-$(ring), $HOCH_2CH_2OCH_2CHO$, $CH_3OCH_2CH_2OC_2H_5$, terephthalic acid, triethylene glycol, tetraethylene glycol, chain fragments	274
Polyester from terephthaloyl chloride and $n = 22$ polyethylene glycol		CO, CH_4 (trace), CO_2, ethene (trace), formaldehyde, acetaldehyde, 1,3-dioxalane, methoxyacetaldehyde, 1,4-dioxane, 7-membered cyclic ether, methyl benzoate, ethyl benzoate, benzaldehyde, $-CH_2CH_2COCH_2-$(ring), $HOCH_2CH_2OCH_2CHO$, $CH_3OCH_2CH_2OC_2H_5$, $-(CH_2CH_2O)_2CH_2CO-$(ring), $CH_2=CHOCH_2CH_2OCH_2CHO$, $HO(CH_2CH_2O)_2C_2H_5$, $C_2H_5(OCH_2CH_2)_3H$, terephthalic acid, triethylene glycol, tetraethylene glycol, chain fragments	274

TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(oxyisophthaloyl-oxy-1,3-phenylene)	330	Cyclic monomer (10 relative abundance), cyclic dimer (83.5), cyclic trimer (24.5), benzoic acid, isophthalic acid, resorcinol and various fragments of the main chain (see Ref.)	187
Poly(oxyisophthaloyl-oxy-1,4-phenylene)	330	Cyclic dimer (17), cyclic trimer (4), *m*-phthalic acid (26), benzoic acid (7.5), hydroquinone (46), and various fragments of the main chain (see Ref.)	187
Poly(oxy-1-oxohexamethylene)	220	Monomer (ε-caprolactone)	184
Poly((D)-oxy-1-oxo-3-methyl-trimethylene) (Poly(3-hydroxybutyrate))	Ambient to 338	Dimer (41.2 wt.%), crotonic acid (35.3), trimer (12.5), tetramer (2.9), iso-crotonic acid (0.9). Small amounts of CO_2, propene, ketene, acetaldehyde, and β-butyrolactone are formed at higher temperatures due to decomposition of primary products	186
	250	Crotonic acid (64.4% of original polymer), dehydrated dimer (7.2%), dehydrated trimer (1.2%)	245
	400	Crotonic acid (65.4% of original polymer), dehydrated dimer (8.2%), dehydrated trimer (2.1%)	245
	600	Crotonic acid (58.3% of original polymer), dehydrated dimer (9.0%), dehydrated trimer (2.7%)	245
	350	Crotonic acid (*trans*), dehydrated dimer (*trans*), dehydrated trimer and tetramer (*cis*/*trans* not resolved)	246
Poly((D)-oxy-1-oxotetramethylene) (Poly(4-hydroxybutyrate))	320	Cyclic oligomers (γ-butyrolactone and higher homologues)	257
Poly(oxy-1-oxo-2-methylethylene) (Poly(l-lactic acid))	500	CO, CO_2, acetaldehyde, acetone, acrylic acid, acetic acid, DL-lactide, meso-lactide, cyclic oligomers up to nonamer, 2,3 pentanedione (?), lactoyl lactic acid (?)	282
Poly(oxy-1-oxo-2,2-dimethyltrimethylene (Poly(α,α-dimethyl-β-propiolactone))	250–600	Isobutylene and CO_2	183
Poly(oxy-1-oxo-trimethylene) (Poly(β-propiolactone))	180–220	Acrylic acid	189
Poly(oxyterephthaloyloxy-1,2-phenylene)	Ambient to 500	CO_2, benzene, phenol, benzaldehyde, catechol, cyclic dimer	285
Poly(oxyterephthaloyloxy-1,3-phenylene)	345	Cyclic dimer (2), cyclic trimer (0.3), terephthalic acid (0.4), benzoic acid (1.5), resorcinol (20) and various fragments of the main chain (see Ref.)	187
	Ambient to 500	CO_2, benzene, phenol, benzaldehyde, biphenyl, benzoic acid, 3-hydroxyphenylbenzoate, resorcinol, mono and di(3-hydroxyphenyl) terephthalate, terephthalic acid, terephthaldehydic acid, short chain fragments	285
Poly(oxyterephthaloyloxy-1,4-phenylene)	350	Cyclic dimer (0.1), terephthalic acid (0.5), benzoic acid (6), hydroquinone (40.5), and various fragments of the main chain (see Ref.)	187
	Ambient to 500	Benzene, phenol, benzaldehyde, biphenyl, benzoquinone, 4-hydroxyphenylbenzoate, benzoic acid, mono and di(4-hydroxyphenyl) terephthalate, terephthalic acid, terephthaldehydic acid, short chain fragments	285
Poly(oxyterephthaloyloxy-4,4′-diphenylene)	Ambient to 500	CO_2, benzene, phenol, benzaldehyde, biphenyl, 4-phenylphenol, *p*,*p*′-dihydroxybiphenyl, 4-hydroxybiphenylbenzoate, di(4-hydroxybiphenyl) terephthalate	285
Poly(oxytetramethyleneoxyterephthaloyl) (Poly(butylene terephthalate))	240–280	Butadiene, terephthalic acid, mono-3-butenyl terephthalate, and di-3-butenyl terephthalate	190
	200–400	Tetrahydrofuran, 1,3-butadiene, CO_2, CO, H_2O, benzoic acid, terephthalic acid, terephthalic acid mono-3-butenyl, toluene, phenol, benzene	191
	Ambient to 500	CO, CO_2, butadiene, tetrahydrofuran, toluene, benzene, 1,5-hexadiene, dihydrofuran, 4-vinyl cyclohexene, 1,4-butane diol, benzaldehyde, benzoic acid, terephthaldehydic acid, terephthalic acid, mono-3-butenyl terephthalate, cyclic dimer, short chain fragments	298
Poly(oxydecamethylene-oxyterephthaloyl)	Ambient to 500	CO, CO_2, 1,9-decadiene, 1,10-decane diol, 1-decene-10-ol, benzoic acid, terephthalic acid, mono-decenyl terephthalate	298
Poly(oxyethyleneoxyadipoyl)	Ambient to 500	Cyclic oligomers, CO, CO_2, acetaldehyde, 2-ethylacrolein	269
Poly(oxytetramethylene-oxysebacoyl)	Ambient to 500	CO_2, H_2O, butadiene, tetrahydrofuran	299
Polyester from maleic anhydride, hexolic acid and butanediol; mole ratio 1 : 1 : 2.2	100–440	Tetrahydrofuran, 1-butanol, toluene, maleic anhydride, chlorosuccinic acid isomers, chlorovalerolactone isomers, hydroxypyran, allylpyran, trichlorocylopentadiene, tetrachlorocylopentadiene, pentachlorocylopentadiene, hexachlorocylopentadiene, phthalic anhydride, 1,4-butanediol, 1,4-butanediol dibenzoate, benzoic acid, 4-cyclohexene-1,2-dicarboxylic acid anhydride, benzoic acid butenyl ester, cyclohexadiene carboxylic acid propyl ester, hexolic anhydride,	311

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TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
		unidentified butenyl ester, compound containing C_5HCl_5 unit, $CH_3-CH=CH-CH_2-O-CH_3$, $CH_3-CH=CH-CH_2-O-CH=CH_2$, $Cl-CH=CH-O-CH_2-CH_3$	
Polyester from maleic anhydride, phthalic anhydride and butanediol; mol ratio 1 : 1 : 2.2	100–440	Tetrahydrofuran, 1-butanol, 3-butene-1-ol, toluene, maleic anhydride, phthalic anhydride, 1,4-butanediol, 1,4-butanediol dibenzoate, phthalic acid butenyl ester, various esters containing butenyl, maleate and phthalate units, $CH_3-CH=CH-CH_2-O-CH_3$, $CH_3-CH=CH-CH_2-O-CH=CH_2$	311
Unsaturated polyester resins	300–800		
a. Maleic acid–phthalic acid–propylene glycol–styrene		Ethane (above 600°C), propionaldehyde (above 400°C), styrene (major product), phthalic anhydride (major product), benzene (above (450°C), toluene (above 500°C), xylene (above 400°C)	193
b. Maleic acid–tetrabromophthalic acid–phthalic acid–propylene glycol–styrene		Styrene (major product), phthalic anhydride (major product), benzene (above 550°C), toluene (above 450°C), xylene (above 500°C)	193
c. Maleic acid–phthalic acid–dibromoneopentyl glycol–styrene		Ethane (above 500°C), propionaldehyde (above 400°C), styrene (major product), phthalic anhydride (major product), benzene (above 500°C), toluene (above 500°C), xylene (above 500°C)	193
3.3. POLY(CARBONATES)			
Poly(oxycarbonyloxy-1,3-phenylene)	270	Cyclic trimer, tetramer, and pentamer, and various chain fragments	226
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	300–389	Major products: CO_2, bisphenol A; minor products: CO, CH_4, 4-alkyl phenols	60
	200–400	CO, CO_2, CH_4, phenol, diphenyl carbonate, 4-(4-hydroxyphenyl)-2-phenyl propane	225
	290	Cyclic dimer, cyclic trimer, bisphenol A, and other chain fragments	226
	Ambient to 500	CO, CH_4, CO_2, H_2O, phenol, *p*-cresol, *p*-ethyl phenol, *p*-isopropyl phenol, *p*-vinyl phenol, bisphenol A, cyclic dimer, chain fragments including some ester structures	304
	300	Phenol, *p*-cresol	304
	375	CO_2, phenol, *p*-cresol	304
Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene-oxy-carbonyl-tetramethylene)	240	Cyclic monomer and dimer, and higher cyclic fragments	226
Poly(oxycarbonyloxy-1,3-phenylene hexafluorotrimethylene-1,3-phenylene)		HF (3%), cyclic dimer	177
Poly(oxycarbonyloxy-1,4-phenylene-oxycarbonyloxy-tetramethylene)	270	Cyclic dimer and trimer, and higher cyclic fragments	226
Poly(oxycarbonyloxy-1,4-phenylene-phthalidylidene-1,4-phenylene)	350	CO_2, CO, O_2, H_2O, phenol, fluorenone, diphenyl carbonate, xanthone, anthraquinone, 2-hydroxyanthraquinone, 2-benzoxyanthraquinone, phenolphthalein, traces of benzoxyphenol and hydroquinone	227
Poly(trimethylene carbonate)	*Ambient to 500*	CO_2, H_2O (possibly desorbed), trimethylene carbonate, cyclic oligomers	295
Poly(neopentylene carbonate)	Ambient to 500	CO_2, monomer, oligomer	296
Poly(2-phenyl trimethylene carbonate)	Ambient to 500	CO, CO_2, H_2O (possibly desorbed), α-methylstyrene, chain fragments (85%, w/w)	296
3.4. POLY(ANHYDRIDES)			
Poly(oxycarbonyl-1,3-phenylene-hexafluorotrimethylene-1,3-phenylene-carbonyl)	350	Cyclic dimer	177
Poly(maleic anhydride-*co-trans* stilbene) (alternating copolymer)	Ambient to 500	CO_2, styrene, maleic anhydride; small amounts benzene, cyclohexadiene, dimethylbutene, ethylcyclobutanol and 4-methoxystyrene; major products (79%) are chain fragments, including aromatic, ketonic and unsaturated structures	242
3.5. POLY(SULFIDES) AND POLY(SULFONES)			
Poly(thioacetone)	145	Main product is cyclic trimer [$(CH_3)_2CS$]	69
Poly(thioethylene)	220–260	Ethylene, H_2S, ethanethiol	199
Poly(thio-1,4-perfluorophenylene)	500	73% Residue; remainder, chain fragments, H_2, SiF_4 (from silica vessel), CO_2	51
Poly(thiomethylene-1,4-phenylene-methylene)	250–350	Sulphur, styrene, thiobenzaldehyde, 4-methyl styrene, *p*-xylene, 4-ethyl toluene, H_2S, 4-methyl thiobenzaldehyde, stilbene and chain	254

TABLE 3. *cont'd*

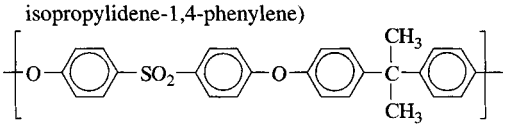
Polymer	Temperature range (°C)	Degradation products	Refs.
		fragments containing stilbene units, chain fragments containing polymer repeat units with $-CH_2SH$, $-CH{=}S$ and $-CH_3$ terminal units	
	650	Toluene, ethylbenzene, *p*-xylene, 1,4-diethylbenzene, 4-ethyltoluene; series of polycyclic aromatics containing phenanthrene and dihydrophenanthrene	254
Poly(dithiomethylene-1,4-phenylene-methylene)	250–350	Sulphur, styrene, thiobenzaldehyde, 4-methyl styrene, *p*-xylene, 4-ethyl toluene, H_2S, 4-methyl thiobenzaldehyde, stilbene and chain fragments containing stilbene units, chain fragments containing polymer repeat units with $-CH_2SH$, $-CH{=}S$ and $-CH_3$ terminal units	254
	650	Toluene, ethylbenzene, *p*-xylene, 1,4-diethylbenzene, 4-ethyltoluene; series of polycyclic aromatics containing phenanthrene and dihydrophenanthrene	254
Poly(thio-1,2-phenylene)	430	Cyclic dimer (thianthrene), some cyclic trimer	258
Poly(thio-1,3-phenylene)	540	Cyclic oligomers (trimer to heptamer) in decreasing yield	258
Poly(thio-1,4-phenylene)	460	72% Residue; remainder chain fragments, H_2, dibenzthiophene	51
	300–620	38% Residue, 47% chain fragments, 15% volatiles comprising mainly H_2S and H_2	59
	325–625	Benzene, thiophenol, diphenyl, diphenylsulfide, dibenzothiophene, chlorobenzene (from polymers with terminal chlorine atoms)	200
	560	Cyclic oligomers tetramer to heptamer (pentamer dominates)	258
Poly(oxy-1,4-phenylene-sulfonyl-1,4-phenylene-oxy-1,4-phenylene-isopropylidene-1,4-phenylene) 	380	Phenol with smaller amounts of *p*-cresol, *p*-ethyl phenol, *p*-isopropyl-phenyl phenol, bisphenol A, diphenyl ether, *p*-tolyl phenyl ether, *p*-ethyl diphenyl ether, *p*-isopropenyl diphenyl ether, isomers of tolyl (ethyl–phenyl)ether	201
	350–450	In 90 min at 450°C, 15% (of initial weight) of gaseous products are formed; SO_2 (76% of gaseous products), CO_2 (15%), CO (4.5%), CH_4 (4.5%)	202
	470 for 2 h	Benzene, toluene, *p*-xylene, isopropylbenzene, phenol (main product), diphenyl oxide, diphenyl sulfide	203
	800	SO_2, benzene, toluene, styrene, phenol, methylphenol, diphenylether, methyl diphenylether	288
Poly(oxy-1,4-phenylene-sulfonyl-1,4-phenylene)	470 for 2 h	Phenol (main product), diphenyl oxide	203
	550–600	SO_2, diphenylsulfone, phenol, diphenylether, hydroquinone, benzenesulfonic acid, 4-hydroxybenzenesulfonic acid, 4,4′-dihydroxydiphenylether, 4-phenoxyphenol; series of $-H$ and $-OH$ terminated fragments up to trimer, series of compounds containing consecutive diphenylether and dibenzofuran units, fragments containing biphenyl units as a result of SO_2 elimination; complex ring structures with sulfonate links	255
	800	SO_2, benzene, phenol, diphenyl ether, dibenzofuran	288
Poy(tris-(oxy-1,4-phenylene)-sulfonyl-1,4-phenylene)		Benzene, phenol (major product), diphenyl oxide, *p*-hydroxydiphenyl, or *p*-hydroxydiphenyl oxide	203
Poly(oxy-4,4′-diphenylene-oxy-1,4-phenylene sulfonyl-1,4-phenylene)		Phenol (main product), diphenyl oxide	203
Poly(oxy-1,4-phenylene-sulfonyl-1,4-phenylene-oxy-1,4-phenylenemethylene-1,4-phenylene)		Benzene, toluene, *p*-xylene, phenol (main product), diphenyl oxide	203
Poly(sulfur dioxide-*co*-1-butene)	700–800	Monomers	204
Poly(sulfur dioxide-*co*-1-hexene)	700–800	Monomers	204
Poly(sulfur dioxide-*co*-1-octadecene)	700–800	Monomers	204
Poly(sulfur dioxide-*co*-bicycloheptene)	700–800	SO_2, bicycloheptene, cyclopentadiene, ethylene, methyl cyclohexadiene, benzene, indene, naphthalene, dihydronaphthalenes	204
Poly(sulfur dioxide-*co*-cyclopentene)	700–800	SO_2, cyclopentene, xylenes, ethylbenzene, styrene, indene, decadienes, dihydronaphthalenes, tetralin, napthalene	204
Poly(sulfur dioxide-*co*-*cis*-2-butene)	700–800	SO_2, butenes, benzene, toluene, octadienes, xylenes, styrene, indene, naphthalene	204
Poly(sulfur dioxide-*co*-methyl methacrylate-*co*-1-hexene) (24.2/1)	700–800	SO_2, hexenes, hexadienes, methyl methacrylate, toluene, naphthalene, dodecadienes	204
Poly(sulfur dioxide-*co*-methyl methacrylate-*co*-bicycloheptene) (2.5/1)	700–800	SO_2, ethylene cyclopentadiene, benzene, methyl methacrylate, bicyclo-heptene, methylcyclohexa-2,4-diene, ethylbenzene, xylenes, styrene, indene, naphthalene, dihydronaphthalenes	204

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TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
3.6. POLY(AMIDES)			
Poly(2,5-dimethyl-1,4-piperazinediyl-isophthaloyl)	475	After 1 h, 5.1% residue, 73.8% chain fragments, 20.5% volatiles comprising mainly CO, H_2O, CH_4, CO_2, H_2, with traces of hydrocarbons, pyrazines, pyrroles	30
Poly(2,5-dimethyl-1,4-piperazinediyloxaloyl)	475	After 1 h, 1% residue, 73.5% chain fragments, comprising mainly CO, with traces of H_2, CO_2, H_2O, hydrocarbons, pyrazines, pyrroles	47
Poly(2,5-dimethyl-1,4-piperazinediyl-terephthaloyl)	475	After 1 h, 9.4% residue, 3.2% chain fragments 87.4% volatiles comprising mainly CO, H_2O, CO_2, CH_4, NH_3, with traces of hydrocarbons, pyrazines, and pyrroles	47
Poly(iminoisophthaloylimino-1,3-phenylene) (Nomex)	300–500	Mixture of H_2, CO, CO_2, H_2O, HCN, benzene, toluene, benzonitrile	57
	300–500	CO_2, H_2O, CO	205
	450–550	Benzene, HCN, toluene, benzonitrile, H_2	205
		CO, CO_2, H_2O, benzene, benzonitrile, 1,3-dicyanobenzene, 3-cyanobenzoic acid, other trace compounds	319
Poly(iminoisophthaloylimino-1,3-perchlorophenylene)	550	Water (46 mol%), CO (18), CO_2 (31), CH_4 (1.2), HCN (1.3), and smaller quantities of, N_2O, NO, CH_3Cl, ethylene, acetylene, cyanogen, acetonitrile, acetone, acetic acid, cyanogen chloride, propenenitrile, benzene, toluene, benzonitrile, phenol, chlorobenzene	206
Poly(iminoisophthalylimino-1,4-phenylene)	300–500	Mixture of H_2, CO, CO_2, H_2O, HCN, benzene, toluene, benzonitrile	57
Poly(iminoisophthaloyl-*co*-terephthaloyl-imino-1,4-phenylene)	250–620	36% residue, 24% chain fragments, 40% volatiles comprising H_2, CO, CO_2, CH_4	58
Poly(imino(1-oxohexamethylene) (Nylon 6)	327	Cyclic monomer, acetonitrile, HCN, NH_3, acrylonitrile, 3-cyanopropene, 4-cyanobutadiene, 4-cyano-1-butene, 5-cyano-1,3-pentadiene, 4-pentenal, 5-cyano-1-pentene, 3,5-hexadienal, 5-isocyanato-1-pentene, linear dimer	293
Poly(iminohexamethyleneiminoadipoyl) (Nylon 66)	310–380	H_2O, CO_2, cyclopentanone, traces of saturated and hydrocarbons; purification from water and acid polymerization catalysts increases stability and decreases yield of CO_2	45,46
	305	H_2O, CO_2, NH_3, cyclic monomer, cyclopentanone, cyclopentylidine-cyclopentanone, cyclopentylcyclopentanone, hexylamine, hexamethyleneimine, hexamethylenediamine	207
	305	CO_2, NH_3, H_2O, cyclic monomer, cyclopentanone, 2-cyclopentylidenecyclo pentanone, 2-cyclopentylcyclopentanone, hexylamine, hexamethyleneimine, hexamethylenediamine, 1,2,3,5,6,7-hexahydro dicyclopenta[b,e]pyridine	207
	400	Hexamethylenediamine, aminohexamethylene isocyanate, hexamethylenediamine-monocyclopentanimine, hexamethylenediamine-dicyclopentanimine and larger oligomers related to above; cyclopentanone	247
Poly(imino(1-oxoundecamethylene) (Nylon 11)	800	CO_2, HCN, saturated and olefinic hydrocarbons C_5–C_{11}, saturated and olefinic mononitriles $C_nH_{(2n+1)}CN$ and $C_nH_{(2n-1)}CN$, undecalactam, hydrocarbons and nitriles containing amide units	308
Poly(imino(1-oxododecamethylene) (Nylon 12)	800	CO_2, HCN, saturated and olefinic hydrocarbons C_5–C_{11}, saturated and olefinic mononitriles $C_nH_{(2n+1)}CN$ and $C_nH_{(2n-1)}CN$, dodecalactam, hydrocarbons and nitriles containing amide units	308
Poly(iminoterephthaloylimino-1,3-phenylene)	300–500	Mixture of H_2, CO, CO_2, H_2O, HCN, benzene, toluene, benzonitrile	57
Poly(iminoterephthaloylimino-1,4-phenylene) (Kevlar)	300–500	Mixture of H_2, CO, CO_2, H_2O, HCN, benzene, toluene, benzonitrile	57
	370–450	CO_2, H_2O, CO	205
	450–550	Benzene, HCN, toluene, benzonitrile, H_2	
	527	Cyanobenzene, dicyanobenzene, aniline, 1,4-diaminobenzene, 4-cyanoaniline, benzaldehyde, benzene, CO, NH_3, CO_2, H_2O, benzamide, benzoic acid, phenylisocyanate, terephthaldehyde, benzanilide, NH_2–Ph–NH–CO–Ph, NC–Ph–NH–CO–Ph, NH_2–Ph–NH–CO–Ph–CN, Ph–NH–CO–Ph–CHO	293
Poly(2-methylpiperazinediylterephthaloyl)	475	After 1 h, 11.3% residue, 46.2% chain fragments, 42.5% volatiles comprising mainly CO, H_2O, CO_2, CH_4, NH_3, H_2, with traces of hydrocarbons, pyrazines, and pyrroles	30
Poly(1,4-piperazinediylterephthaloyl)	475	After 1 h, 21.1% residue, 45.6% chain fragments, 33.3% volatiles comprising mainly H_2O, CO, H_2, CO_2, NH_3, with traces of hydrocarbons, pyrazines, and pyrroles	30

TABLE 3. *cont'd*

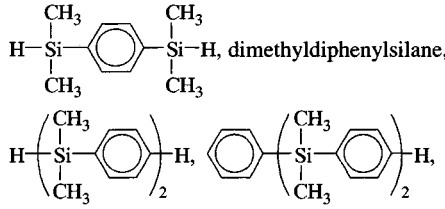
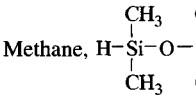
Polymer	Temperature range (°C)	Degradation products	Refs.
3.7. POLY(SILOXANES)			
Poly(oxy-dimethylsilylene) (Poly(dimethyl siloxane))	Ambient to 500	Predominantly cyclic trimer with smaller amounts of cyclic tetramer and higher cyclics; threshold degradation temperature of about 300°C reduced to 100°C by 5% KOH and CH_4 formed as additional product	208
	470	Distribution of cyclic oligomers, $n = 3$ (243.9 relative mass spec. intensity), $n = 4$ (100), $n = 5$ (35.6), $n = 6$ (37.6), $n = 7$ (13.1), $n = 8$ (1.1), $n = 9$ (0.2)	317
Poly(oxy-methylphenylsilylene)	Ambient to 500	Mixture of all possible stereoisomeric cyclic trimers and tetramers with small amounts of pentamer, benzene, and two more complex oligomers	209
Poly(oxy-methyl-3,3,3-trifluoro-propylsilylene)	400	Low-molecular-weight cyclic oligomers, $CF_2{=}CH{-}CH_3$, CHF_3	70
	615 for 30 s	Cyclic methyl-(3,3,3-trifluoropropyl)siloxanes, CH_4, CHF_3, $CF_2{=}CH{-}CH_3$, $CH_2{=}CH{-}CF_3$	210
Poly(oxy-phenylbutoxysilylene)	430–500	CO, H_2 and hydrocarbons	211
Poly(dimethyl siloxane-*co*-diphenyl siloxane)	Ambient to 500	Complex mixture of cyclic oligomers, which have been individually identified, whose composition varies with the dimethyl/diphenyl ratio; small amounts of benzene	212
Poly(dimethyl siloxane-*co*-phenylmethyl siloxane)	Ambient to 500	Complex mixture of cyclic oligomers, which have been individually identified, whose composition varies with the dimethyl/methyl phenyl ratio; small amounts of benzene	213
Poly(dimethyl-1,4-silphenylene)	400 in argon	Methane, dimethylsilane, benzene, dimethylphenylsilane,  dimethyldiphenylsilane,	214
Poly(oxy-methylphenyl-1,4-silphenylene-silylene)	Ambient to 500	Benzene, diphenylmethylsilane, triphenylmethylsilane, and higher linear oligomers	215
Poly(oxy-tetramethyl-1,4-silphenylene-silylene)	500 in argon	Methane,  benzene, cyclic trimer of dimethylsiloxane, and various higher chain fragments (see Ref.)	214
	Ambient to 500	Cyclic trimer and tetramer of dimethylsiloxane and a complex mixture of higher linear and cyclic oligomers	216
Poly(oxy-tetramethyl-1,3-silphenylene-silylene-hexafluoropropylenephenylene)	615 for 30 s	HF (15–18%), methane, methyltrifluorosilane, dimethyldifluorosilane, benzene, toluene, phenyldifluoromethane	210
Poly(oxy-tetramethyl-1,3-silphenylene-silylene-tetrafluoroethylenephenylene)	615 for 30 s	HF (15–18%), methane, methyltrifluorosilane, dimethyldifluorosilane, benzene, toluene, phenyldifluoromethane	210
Poly(oxy-tetramethyl-1,3-silphenylene-siloxane-*co*-dimethylsiloxane)	Ambient to 500	Cyclic trimer and tetramer of dimethylsiloxane and a complex mixture of higher linear and cyclic oligomers	216
Poly(boron tri(dimethylsiloxane)), nonlinear	250–350	Molecular weight increases; traces of cyclic products $[(CH_3)_2SiO]_3$ and $[(CH_3)_2SiO]_4$	15
3.8. POLY(URETHANES)			
Poly(oxy-2,2-dimethyltrimethylene) oxycarbonyl-*N*-imino-1,4-phenylenemethylene-1,4-phenylene-*N*-iminocarbonyl)	350	Various fragments of the main chain (see Ref.)	231
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxycarbonylimino-1,4-phenylene methylene-1,4-phenylene-iminocarbonyl)	190	4,4′-Diphenylmethane diisocyanate, bisphenol A	231
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenylene-oxycarbonyl-*N*-methylimino-1,4-phenylenemethylene-1,4-phenylene-*N*-methyliminocarbonyl)		Bisphenol A, 4,4′-*N*-methyldiphenyl methane and linear and cyclic fragments of the main chain (see Ref.)	231
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenylene-oxycarbonyl-*N*-methyliminohexa-methylene-*N*-methyliminocarbonyl)	330	Cyclic monomer and dimer, and various chain fragments	232
Poly((oxy-1,4-phenyleneisopropylidene-1,4-phenylene-oxycarbonyl-piperazine-1,4-diyl-carbonyl)	330	Cyclic dimer, bisphenol A and various chain fragments	232

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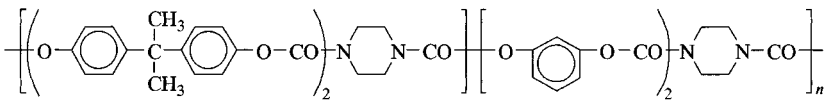
TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(oxy-1,3-phenylene-oxycarbonyl-*N*-methylimino-hexamethylene-*N*-methyliminocarbonyl)	320	Cyclic monomer, dimer and trimer, resorcinol	232
Poly(oxy-1,4-phenyleneoxycarbonyl-*N*-methylimino-1,4-phenylenemethylene-1,4-phenylene-*N*-methyliminocarbonyl)	325	Cyclic dimer, 4,4′-*N*-methyldiphenylmethane, hydroquinone, and higher chain fragments (see Ref.)	232
Poly(oxy-1,3-phenylene-oxycarbonyl-piperazine-1,4-diyl-carbonyl)	360	Cyclic dimer and trimer, resorcinol, and various chain fragments (see Ref.)	232
Poly(oxytetramethylene oxycarbonyl-imino-1,4-phenylenemethylene-1,4-phenylene-iminocarbonyl)	Ambient to 500	1,4-Butanediol, methylene bis-(4-phenyl isocyanate), tetrahydrofuran, CO_2, H_2O, butadiene, HCN, CO	228
with 10% (w/w) ammonium polyphosphate	Ambient to 500	Methylene bis-(4-phenyl isocyanate), tetrahydrofuran, H_2O, aniline, formaldehyde, CO_2	229
Poly(oxy-xylylene-oxycarbonylimino-xylylene-iminocarbonyl)	300	CO_2, *trans*-1,4-cyclohexane-dimethanol, 4-methylene-cyclohexane-methanol, xylylene diamine	230
Polyurethane from methylene bis-(4-phenyl isocyanate) (105 parts) and propoxylated trimethyl propane (100 parts)	600	Methane, CO_2, ethylene ethane, H_2O, propene, propane, ethanol, isobutene, acrolein, propanal, 2-methyl-1-butene, 2-methylpropenal, 2-butanone, benzene, dihydropyran, 2,2,4-trimethyl-1,3-dioxalane	234
	250–350	CO_2, propene, some methane and propanal	256
	350–500	Methane, ethane, ethene, propane, propene, ethanal, propanal, diminishing quantities of CO_2	256
	500–650	Methane (max. rate of production at 600°C), ethane, ethene, propene, ethanal, propanal	256
	650–800	Methane, ethane, ethene, propene, ethanal, propanal	256
Polyurethane from butane diol, phenylphosphonic dichloride, and methylene bis-(4-phenyl isocyanate)	Ambient to 500	CO_2 tetrahydrofuran, dihydrofuran, aniline, *p*-toluidine, *N*-phenyl pyrrolidine, *N*-(*p*-tolyl) pyrrolidine, cyclobutylene phenylphosphonate, butane diol, methylene bis-(4-phenyl isocyanate), 4,4′-methylene dianiline	235
Polyester urethane from an ethylene/propylene glycol/adipic acid polyester, butane diol and methyl bis-(4-phenyl isocyanate)	Ambient to 500	Modified chain fragments, ethylene, propylene, CO_2, ethylene oxide, propylene oxide, acetaldehyde, tetrahydrofuran, water, cyclopentanone, ethylene, glycol, propylene glycol, adipic anhydride (trace), aniline (trace)	236
Poly(urethane-*co*-carbonates)	330	Cyclic monomer, bisphenol A and higher linear and cyclic fragments of the chain (see Ref.)	237



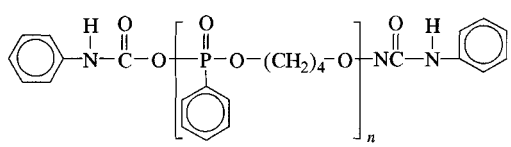
	290	Cyclic monomer, resorcinol, and various fragments of the chain (see Ref.)	237

3.9. OTHER NITROGEN CONTAINING POLYMERS

Poly(*t*-butyl aziridine)	Ambient to 500	Ethene, propane, propene, isobutane, isobutene, ethylamine, aziridine, 1-*t*-butyl-perhydro-1,4-diazine, *N*-*t*-butyl ethylamine, 1-*t*-butyl-4-isopropenyl-1,2,3,4-tetrahydro-1,4-diazine, 1,4-di-(*t*-butyl)-perhydro-1,4-diazine, oligomers	276

3.10. PHOSPHOROUS CONTAINING POLYMERS

Poly(oxytetramethylene oxyphenyl phosphono)	Ambient to 550	Butadiene, tetrahydrofuran, dihydrofuran, H_2O, cyclic ester of phenylphosphonic acid and 1,4-butanediol, phenylphosphonic acid, butane diol	233
Poly(oxytetramethylene oxyphenyl-phosphono) capped with phenylisocyanate)	Ambient to 500	Butadiene, tetrahydrofuran, dihydrofuran, H_2O, cyclic ester of phenylphosphonic acid and butanediol, phenylphosphonic acid, butane diol, CO_2, aniline	233



Poly(bis(trifluoroethoxy)phosphazene)	150–400	Cyclic trimer, tetramer, pentamer, hexamer, and higher oligomers, 1,1,1-trifluoro-2-chloroethane (due to residual chlorine in the polymer)	238
Poly(diphenoxyphosphazene)	100–400	Cyclic tetramer, trimer and higher oligomers, chlorobenzene (due to residual chlorine in the polymer), phenol, triphenylphosphate	239

TABLE 3. *cont'd*

Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(diphenoxy-phosphazene) (residual chlorine content: <0.05%)	Ambient to 480	Traces of phenol, phenoxy cyclic trimer and monochlorophenoxy cyclic trimer	316
Poly(diphenoxy-phosphazene) (residual chlorine content: 2.4%)	Ambient to 480	Phenoxy cyclic trimer and tetramer, monochloropentaphenoxy cyclic trimer, monochloroheptaphenoxy cyclic tetramer, dichlorohexaphenoxy cyclic tetramers	316
Poly(diphenoxy-phosphazene) (residual chlorine content: 9.9%)	Ambient to 480	Phenoxy cyclic trimer and tetramer, monochloropentaphenoxy cyclic trimer, monochloroheptaphenoxy cyclic tetramer, dichlorohexaphenoxy cyclic tetramers, dichlorotetraphenoxy cyclic trimers, trichloropentaphenoxy cyclic tetramers, HCl	316

TABLE 4. MAIN-CHAIN HETEROCYCLIC POLYMERS

Polymer	Temperature range (°C)	Degradation products	Refs.
Phenol-formaldehyde resin	250–400	Volatiles comprise xylene (76%), traces of phenol, cresol, benzene	53
Poly(1-methylene-2-imidazolidinone)	240	Formaldehyde (trace)	253
(Poly(ethyleneurea formaldehyde))	Ambient to 600	Low boiling fraction consists of NH_3, CO_2, HCN, CH_4, CO (trace); IR data also suggests presence of isocyanate, ketamine and carbodiimide or cyanogen compounds; high boiling fraction (77% of total volatiles) consists of chain fragments up to pentamer, including *N*-methylethylene urea, corresponding unsaturated compounds and higher homologues; similarly, *N,N'*-dimethylethylene urea and homologues	253
Poly(*N*-phenylmaleimide)	100–400	Aniline, Ph–N=CH–CH_2–CH_3, *N*-phenylsuccinimide, *N*-phenylmaleimide (trace), Ph–N=CH–CH=CH–CO–NH–Ph	301
Poly(5,5'-bibenzimidazole-2,2'-diyl-1,4-phenylene)	400–700	Phenol, terephthalodinitrile, benzonitrile, 4,4'-diaminodiphenyl, H_2, CO, NH_3, HCN, traces of aniline, carbonaceous residue	56
	570, 700, 1000	H_2 (700 and 1000°C), NH_3 (570 and 700), N_2O, HCN, cyanogen (570 only), CH_4, C_2H_6, ethylene, acetylene (700 and 1000), acetonitrile (700 and 1000), propene (570 and 700), benzene, toluene, benzonitrile (700 only), ethylbenzene, methylbenzonitrile (700 only)	217
Poly(5,5'-bibenzimidazole-2,2'-diyl-1,3-phenylene)	599–667	Gaseous products; H_2, NH_3, HCN, and CH_4; products involatile at ambient temperature after approx. 50% weight loss; benzimidazole (29.8%), benzoxazole (4.9%), 2-methylbenzimidazole (7.3%), 2-phenyl-benzimidazole (24.3%), 2-(3-methylphenyl)benzimidazole (5.3%), 5-(3-aminophenyl)benzimidazole (9.6%), 2-(3-cyanophenyl)benzimidazole (7.8%), 5,5'-bibenzimidazole (4.4%), 2-phenyl-5-(3-aminophenyl)-benzimidazole (4.4%), 2-phenyl-5,5'-bibenzimidazole (1.9%), 2-phenyl-5-(benzoxazolo)-benzimidazole (0.3%)	218
Poly(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-(1H,5H)-2,6-benzo[1,2-c : 4,5-c']-dipyrroldiyl-1,4-phenylene-carbonyl-1,4-phenylene) (Poly(pyromellitimide))	700	H_2, CO, CH_4, CO_2, H_2O, benzene, benzonitrile, chain fragments, 60% residue	61
Poly(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-(1H,5H)-2,6-benzo[1,2-c : 4,5-c']-dipyrroldiyl-1,4-phenylene-methylene-1,4-phenylene) (Poly(pyromellitimide))	700	H_2, CO, CH_4, CO_2, H_2O, benzene, toluene, benzonitrile, chain fragments, 60% residue	61
Poly(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-(1H,5H)-2,6-benzo[1,2-c : 4,5-c']-dipyrroldiyl-1,4-phenylene-oxy-1,4-phenylene) (Poly(pyromellitimide))	700	H_2, CO, CH_4, CO_2, H_2O, benzene, benzonitrile, phenol, chain fragments, 60% residue	61
Poly(1,5-diimino-3,7-dioxo-2,3,6,7-tetrahydro-(1H,5H)-2,6-benzo [1,2-c : 4,5-c']dipyrroldiyl-1,3-phenylene)	450–500	CO, CO_2, HCN, NH_3, H_2 and traces of CH_4 and C_6H_6; NH_3 is the main product at 440°C, CO increases at higher temperatures	219
Poly(1,5-diimino-3,7-dioxo-2,3,6,7-tetrahydro-(1H,5H)-2,6-benzo [1,2-c : 4,5-c']dipyrroldiyl-1,4-phenyleneoxy-1,4-phenylene)	450–500	CO, CO_2 HCN, NH_3, H_2 and traces of CH_4 and C_6H_6; NH_3 is the main product at 440°C, CO increases at higher temperatures	

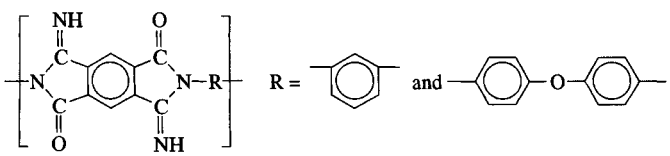


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TABLE 4. *cont'd*

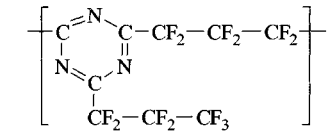
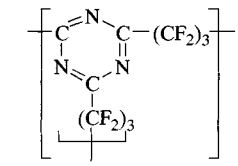
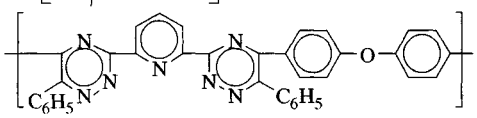
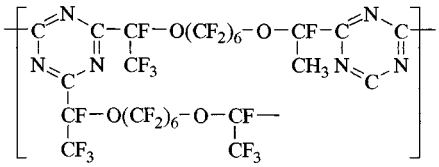
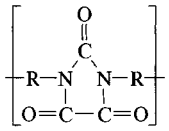
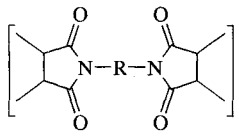
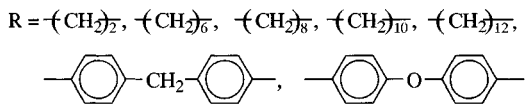
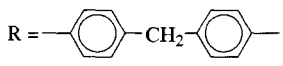
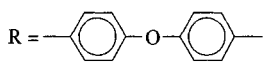
Polymer	Temperature range (°C)	Degradation products	Refs.
Poly(triazines)			
	415–505	C_2F_4, C_2F_6, and smaller amounts of C_3H_6 and C_3H_8 and larger chain fragments	48
	430–503	C_2F_4 with traces of CF_4, C_3H_6, and larger chain fragments	48
	305–465	Only volatile product is benzonitrile	54
	480–660	HCN and NH_3	54
	690–760	H_2 and carbonaceous residue	54
	424–550	Perfluoroethane, perfluoromethane, perfluoroethylene, perfluoropropane, perfluoropropylene, acetal fluoride, carbonyl fluoride	220
Poly(perfluoropyridine)	600	Volatiles comprise CO_2, CO, SiF_4 (all from silica vessel); carbonaceous residue remains	52
Poly(parabanic acid)			
	300	CO_2, CO, NO	221
Poly(bismaleimides)			
			
R = 	350–500	Aliphatic: C_2–C_6 fractions, maleimide (MI), succinimide (SI) MI–CH_3, SI–CH_3, MI–C_2H_5, SI–C_2H_5, MI–C_3H_7, SI–C_3H_7, MI–C_4H_9, SI–C_4H_9, MI–C_5H_{11}, SI–C_5H_{11}, MI–$(CH_2)_2$–MI, SI–$(CH_2)_2$–SI, MI–$(CH_2)_6$–MI, MI–$(CH_2)_8$–MI, MI–$(CH_2)_{10}$–MI, MI–$(CH_2)_{12}$–MI	222
R = 	350–500	Benzene, toluene, aniline, SI, methylaniline, phenylisocyanate, tolylisocyanate, and various other products (see Ref.)	222
R = 	350–500	Aniline, phenol, SI, anisidine, phenylisocyanate, and various other products (see Ref.)	222
Poly(3-(4-octylphenyl)-2,2′-bithiophene)	800	H_2S, COS, SO_2, CH_3SH, CS_2, S_8, C_1–C_8 alkanes and alkenes, benzene, ethylbenzene, toluene, styrene, octylbenzene, thiophene, methylthiophene, bithiophene	284
Poly(3-octylthiophene)	550–1400	C_1–C_8 alkanes and alkenes, H_2S, CS_2, benzene, toluene	284
Poly(3-octyl-2,2′-bithiophene)	550–1400	C_1–C_8 alkanes and alkenes, H_2S, CS_2, benzene, toluene	284
Poly(3-(4-octylphenyl)thiophene)	550–1400	C_1–C_8 alkanes and alkenes, H_2S, CS_2, benzene, toluene	284

TABLE 5. CELLULOSE AND ITS DERIVATIVES

Polymer	Temperature range (°C)	Degradation products	Refs.
Cellulose	250–397	H_2O with smaller amounts of CO_2 and CO and a tar containing principally levoglucosan	17,18
	Ambient to 500	H_2, CO, CO_2, H_2O, furan, furancarboxyaldehyde, methylfurans, acetone, hydroxypropanone, butenal, butanone, butanedione, levoglucosan	273
–, α form	330–440	Fixed gases (not identified), pentene, acetaldehyde, furan, propionaldehyde, acetone, acrolein, 2-methylfuran, butyraldehyde, methyl ethyl ketone, benzene, 2,5-dimethylfuran, methyl vinyl ketone, 2,3-butanedione, toluene, crotonaldehyde, H_2O, cyclopentanone, cyclooctatetraene, acetol, pyruvaldehyde, acetic acid, furfural, formic acid, 5-methyl-2-furfural, furfuryl alcohol, butyrolactone, 2-hydroxy-3-methyl-3-cyclopenten-2-one, phenol, *o*-cresol, *m*-cresol, *p*-cresol, 2,5-dimethylphenol, 3,4-dimethylphenol, 5-hydroxymethylfurfural	162
Cellulose (oxidized)	180–331	Mainly H_2O and CO_2, smaller amounts of CO, formaldehyde, methanol, acetic acid, ethanol and acetaldehyde, and very little tar	
Cellulose triacetate	250–310	Product fraction volatile at 25°C contains acetic acid, CO_2, CO, CH_4, H_2, acetaldehyde and acetone; heavier fractions do not contain levoglucosan acetate	18
	250–900	Main products: acetic acid, water, CO, CH_4, H_2, CO_2; minor products: ketene, acetaldehyde, acetone, acetylene, ethylene, ethane, propylene, and C_4, C_5 and C_6 hydrocarbons	223
	230–320	Acetic acid, acetyl derivatives of D-glucose	224
Ethyl cellulose	306	H_2O, CO, CO_2, C_2H_4, C_2H_6, C_2H_5OH, CH_3CHO, aliphatic compounds, furan derivatives	224

C. REFERENCES

1. G. G. Cameron, D. R. Kane, Polymer, **9**, 461 (1968).
2. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **50**, 165 (1953).
3. S. L. Madorsky, J. Polym. Sci., **11**, 491 (1953).
4. J. R. Schaefgen, I. M. Sarasohn, J. Polym. Sci., **58**, 1049 (1962).
5. C. S. Marvel, J. C. Cowan, J. Am. Chem. Soc., **61**, 3156 (1939).
6. A. J. Canale, W. E. Goode, J. B. Kinsinger, J. R. Panchak, R. L. Kelso, R. K. Graham, J. Appl. Polym. Sci., **4**, 231 (1960).
7. J. W. C. Crawford, D. Plant, J. Chem. Soc., 4492, 1952.
8. G. G. Cameron, G. P. Kerr, J. Polym. Sci. A-1, **7**, 3067 (1969).
9. W. J. Burlant, J. L. Parsons, J. Polym. Sci., **22**, 249 (1956).
10. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **61**, 77 (1958).
11. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **63A**, 261 (1959).
12. A. R. Monahan, J. Polym. Sci. A-1, **4**, 2391 (1966).
13. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **53**, 361 (1954).
14. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **55**, 223 (1955).
15. M. A. Verkhotin, K. A. Andrianov, M. N. Yermakova, S. R. Rafikov, V. V. Rode, Polym. Sci. USSR, **8**, 2369 (1966).
16. S. L. Madorsky, S. Straus, D. Thompson, L. Williamson, J. Res. Natl. Bur. Std., **42**, 499 (1949).
17. S. L. Madorsky, V. E. Hart, S. Straus, J. Res. Natl. Bur. Std., **56**, 343 (1956).
18. S. L. Madorsky, V. E. Hart, S. Straus, J. Res. Natl. Bur. Std., **60**, 343 (1958).
19. R. F. Schwenker, Jr., L. R. Beck, Textile Res. J., **624**, (August, 1960).
20. A. B. Blyumenfel'd, B. M. Kovarskaya, Polym. Sci. USSR, **12**, 710 (1970).
21. L. G. Kaufman, P. T. Funke, A. A. Volpe, Macromolecules, **3**, 358 (1970).
22. J. L. Cotter, G. J. Knight, W. W. Wright, J. Polym. Sci. B, **6**, 763 (1968).
23. S. L. Madorsky, V. E. Hart, S. Straus, V. A. Sedlek, J. Res. Natl. Bur. Std., **51**, 327 (1953).
24. E. E. Lewis, M. A. Naylor, J. Am. Chem. Soc., **69**, 1968 (1947).
25. S. L. Madorksy, S. Straus, J. Polym. Sci., **36**, 183 (1959).
26. E. P. Goodings, Soc. Chem. Ind. (London), Monograph, **13**, 211 (1961).
27. C. E. Schweitzer, R. N. MacDonald, J. O. Punderson, J. Appl. Polym. Sci., **1**, 158 (1959).
28. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **66A**, 401 (1962).
29. Y. Tsuchiya, K. Sumi, J. Polym. Sci. A-1, **7**, 813 (1969).
30. S. D. Bruck, Polymer, **7**, 231 (1966).
31. S. Straus, S. L. Madorsky, Ind. Eng. Chem., **48**, 1212 (1956).
32. B. S. T. Boonstra, G. J. Van Amerongen, Ind. Eng. Chem., **41**, 161 (1949).
33. D. H. Grant, N. Grassie, Polymer, **1**, 125 (1960).
34. N. Grassie, H. W. Melville, Proc. Roy. Soc. A, **190**, 1 (1949).
35. P. D. Zemany, Nature, **171**, 391 (1953).

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–, α form	330–440	Fixed gases (not identified), pentene, acetaldehyde, furan, propionaldehyde, acetone, acrolein, 2-methylfuran, butyraldehyde, methyl ethyl ketone, benzene, 2,5-dimethylfuran, methyl vinyl ketone, 2,3-butanedione, toluene, crotonaldehyde, H_2O, cyclopentanone, cyclooctatetraene, acetol, pyruvaldehyde, acetic acid, furfural, formic acid, 5-methyl-2-furfural, furfuryl alcohol, butyrolactone, 2-hydroxy-3-methyl-3-cyclopenten-2-one, phenol, *o*-cresol, *m*-cresol, *p*-cresol, 2,5-dimethylphenol, 3,4-dimethylphenol, 5-hydroxymethylfurfural	162
Cellulose (oxidized)	180–331	Mainly H_2O and CO_2, smaller amounts of CO, formaldehyde, methanol, acetic acid, ethanol and acetaldehyde, and very little tar	
Cellulose triacetate	250–310	Product fraction volatile at 25°C contains acetic acid, CO_2, CO, CH_4, H_2, acetaldehyde and acetone; heavier fractions do not contain levoglucosan acetate	18
	250–900	Main products: acetic acid, water, CO, CH_4, H_2, CO_2; minor products: ketene, acetaldehyde, acetone, acetylene, ethylene, ethane, propylene, and C_4, C_5 and C_6 hydrocarbons	223
	230–320	Acetic acid, acetyl derivatives of D-glucose	224
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C. REFERENCES

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3. S. L. Madorsky, J. Polym. Sci., **11**, 491 (1953).
4. J. R. Schaefgen, I. M. Sarasohn, J. Polym. Sci., **58**, 1049 (1962).
5. C. S. Marvel, J. C. Cowan, J. Am. Chem. Soc., **61**, 3156 (1939).
6. A. J. Canale, W. E. Goode, J. B. Kinsinger, J. R. Panchak, R. L. Kelso, R. K. Graham, J. Appl. Polym. Sci., **4**, 231 (1960).
7. J. W. C. Crawford, D. Plant, J. Chem. Soc., 4492, 1952.
8. G. G. Cameron, G. P. Kerr, J. Polym. Sci. A-1, **7**, 3067 (1969).
9. W. J. Burlant, J. L. Parsons, J. Polym. Sci., **22**, 249 (1956).
10. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **61**, 77 (1958).
11. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **63A**, 261 (1959).
12. A. R. Monahan, J. Polym. Sci. A-1, **4**, 2391 (1966).
13. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **53**, 361 (1954).
14. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **55**, 223 (1955).
15. M. A. Verkhotin, K. A. Andrianov, M. N. Yermakova, S. R. Rafikov, V. V. Rode, Polym. Sci. USSR, **8**, 2369 (1966).
16. S. L. Madorsky, S. Straus, D. Thompson, L. Williamson, J. Res. Natl. Bur. Std., **42**, 499 (1949).
17. S. L. Madorsky, V. E. Hart, S. Straus, J. Res. Natl. Bur. Std., **56**, 343 (1956).
18. S. L. Madorsky, V. E. Hart, S. Straus, J. Res. Natl. Bur. Std., **60**, 343 (1958).
19. R. F. Schwenker, Jr., L. R. Beck, Textile Res. J., **624**, (August, 1960).
20. A. B. Blyumenfel'd, B. M. Kovarskaya, Polym. Sci. USSR, **12**, 710 (1970).
21. L. G. Kaufman, P. T. Funke, A. A. Volpe, Macromolecules, **3**, 358 (1970).
22. J. L. Cotter, G. J. Knight, W. W. Wright, J. Polym. Sci. B, **6**, 763 (1968).
23. S. L. Madorsky, V. E. Hart, S. Straus, V. A. Sedlek, J. Res. Natl. Bur. Std., **51**, 327 (1953).
24. E. E. Lewis, M. A. Naylor, J. Am. Chem. Soc., **69**, 1968 (1947).
25. S. L. Madorksy, S. Straus, J. Polym. Sci., **36**, 183 (1959).
26. E. P. Goodings, Soc. Chem. Ind. (London), Monograph, **13**, 211 (1961).
27. C. E. Schweitzer, R. N. MacDonald, J. O. Punderson, J. Appl. Polym. Sci., **1**, 158 (1959).
28. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **66A**, 401 (1962).
29. Y. Tsuchiya, K. Sumi, J. Polym. Sci. A-1, **7**, 813 (1969).
30. S. D. Bruck, Polymer, **7**, 231 (1966).
31. S. Straus, S. L. Madorsky, Ind. Eng. Chem., **48**, 1212 (1956).
32. B. S. T. Boonstra, G. J. Van Amerongen, Ind. Eng. Chem., **41**, 161 (1949).
33. D. H. Grant, N. Grassie, Polymer, **1**, 125 (1960).
34. N. Grassie, H. W. Melville, Proc. Roy. Soc. A, **190**, 1 (1949).
35. P. D. Zemany, Nature, **171**, 391 (1953).

36. P. R. E. J. Cowley, H. W. Melville, Proc. Roy. Soc. A, **210**, 461 (1952).
37. J. W. C. Crawford, J. Soc. Chem. Ind., **68**, 201 (1949).
38. N. Grassie, J. R. MacCallum, J. Polym. Sci., **2**, 983 (1964).
39. D. H. Grant, N. Grassie, Polymer, **1**, 445 (1960).
40. N. Grassie, I. C. McNeill, J. Chem. Soc., 3929, 1956.
41. C. S. Marvel, E. H. Riddle, J. O. Corner, J. Am. Chem. Soc., **64**, 92 (1942).
42. K. F. Wissbrun, J. Am. Chem. Soc., **81**, 58 (1959).
43. L. Reginato, Makromol. Chem., **132**, 113 (1970).
44. C. S. Marvel, C. L. Levesque, J. Am. Chem. Soc., **60**, 280 (1938).
45. S. Straus, L. A. Wall, J. Res. Natl. Bur. Std., **60**, 39 (1958).
46. S. Straus, L. A. Wall, J. Res. Natl. Bur. Std., **63A**, 269 (1959).
47. S. D. Bruck, Polymer, **6**, 483 (1965).
48. L. A. Wall, S. Straus, J. Res. Natl. Bur,. Std., **65A**, 227 (1961).
49. S. Straus, L. A. Wall, Soc. Plastics Eng. J., **56**, (January 1964).
50. J. L. Cotter, G. J. Knight, J. M. Lancaster, W. W. Wright, J. Appl. Polym. Sci., **12**, 2481 (1968).
51. N. S. J. Christopher, J. L. Cotter, G. J. Knight, W. W. Wright, J. Appl. Polym. Sci., **12**, 863 (1968).
52. J. L. Cotter, H. Dickinson, G. J. Knight, W. W. Wright, J. Appl. Polym. Sci., **15**, 317 (1971).
53. A. A. Berlin, V. V. Yarkina, A. P. Firsov, Polym. Sci. USSR, **10**, 2219 (1968).
54. W. Wrasidlo, P. M. Hergenrother, Macromolecules, **3**, 548 (1970).
55. G. F. L. Ehlers, K. R. Fisch, W. R. Powell, J. Polym. Sci. A-1, **7**, 2931 (1969).
56. V. V. Rode, N. M. Kotsoyeva, G. M. Cherkasova, D. S. Tugushi, G. M. Tseitlin, A. L. Rusanov, V. V. Korshak, Polym. Sci. USSR, **12**, 2103 (1970).
57. Ye. P. Krasnov, V. M. Savinov, L. B. Sokolov, V. I. Logunova, V. K. Belyakov, T. A. Polyakova, Polym. Sci. USSR, **8**, 413 (1966).
58. G. F. L. Ehlers, K. R. Fisch, W. R. Powell, J. Polym. Sci. A-1, **7**, 2969 (1969).
59. G. F. L. Ehlers, K. R. Fisch, W. R. Powell, J. Polym. Sci. A-1, **7**, 2955 (1969).
60. A. Davis, J. H. Golden, J. Chem. Soc. B, **1**, 45 (1968).
61. D. P. Bishop, D. A. Smith, J. Appl. Polym. Sci., **14**, 345 (1970).
62. S. L. Madorsky, S. Straus, J. Res. Natl. Bur. Std., **40**, 417 (1948).
63. S. L. Madorsky, "Thermal Degradation of Organic Polymers", Wiley, New York, 1964.
64. S. Straus, S. L. Madorsky, J. Res. Natl. Bur. Std., **65A**, 243 (1961).
65. R. H. Still, P. B. Jones, A. L. Mansell, J. Appl. Polym. Sci., **13**, 401 (1969).
66. R. H. Still, P. B. Jones, J. Appl. Polym. Sci., **13**, 2033 (1969).
67. L. A. Wall, J. M. Antonucci, S. Straus, M. Tryon, Soc. Chem. Ind. (London), Monograph, **13**, 295 (1961).
68. D. C. Allport, Polymer, **8**, 492 (1967).
69. V. C. E. Burnop, K. G. Latham, Polymer, **8**, 589 (1967).
70. S. N. Novikov, Ye. G. Kagan, A. N. Pravednikov, Polym. Sci. USSR, **8**, 1114 (1966).
71. N. Grassie, Trans. Faraday Soc., **48**, 379 (1952).
72. J. B. Gilbert, J. J. Kipling, Fuel, **12**, 249 (1962).
73. Y. Tsuchiya, K. Sumi, J. Polym. Sci. A-1, **7**, 3151 (1969).
74. J. B. Gilbert, J. J. Kipling, B. McEnaney, J. N. Sherwood, Polymer, **3**, 1 (1962).
75. R. R. Stromberg, S. Straus, B. G. Achhammer, J. Polym. Sci., **35**, 355 (1959).
76. M. M. O'Mara, J. Polym. Sci. A-1, **8**, 1887 (1970).
77. F. H. Winslow, W. O. Baker, W. A. Yager, quoted in Ch. V of Ref. 63.
78. H. Gilbert, F. F. Miller, S. J. Averill, R. F. Schmidt, F. D. Stewart, H. L. Trumbull, J. Am. Chem. Soc., **76**, 1074 (1954).
79. Y. A. Glagoleva, V. R. Regel, Polym. Sci. USSR, **12**, 1078 (1970).
80. P. Brandt, V. H. Dibeler, F. L. Mohler, J. Res. Natl. Bur. Std., **50**, 201 (1953).
81. H. H. G. Jellinek, S. N. Lipovac, J. Polym. Sci. A-1, **8**, 2517 (1970).
82. J. C. W. Chien, P. C. Uden, Ju-Li Fan, J. Polym. Sci., Chem. Ed., **20**, 2159 (1982).
83. S. Tamura, J. K. Gillham, J. Appl. Polym. Sci., **22**, 1867 (1978).
84. I. C. McNeill, L. Ackerman, S. N. Gupta, J. Polym. Sci., Chem. Ed., **16**, 2169 (1978).
85. D. L. Gardner, I. C. McNeill, Eur. Polym. J., **7**, 593 (1971).
86. D. L. Gardner, I. C. McNeill, Eur. Polym. J., **7**, 603 (1971).
87. J. C. W. Chien, J. K. Y. Kiang, Eur. Polym. J., **15**, 1059 (1979).
88. I. C. McNeil, S. N. Gupta, Polym. Degrad. Stab., **2**, 95 (1980).
89. B. Dodson, I. C. McNeill, J. Polym. Sci., Chem. Ed., **12**, 2305 (1974).
90. B. Dodson, I. C. McNeill, T. Straiton, J. Polym. Sci., Chem. Ed., **12**, 2369 (1974).
91. A. Hoff, S. Jacobson, J. Appl. Polym. Sci., **26**, 3409 (1981).
92. I. C. McNeill, A. Jamieson, D. J. Tosh, J. J. McClune, Eur. Polym. J., **12**, 305 (1976).
93. G. A. Razuvaev, B. B. Troitskii, L. V. Chochlova, Z. B. Dubova, J. Polym. Sci. (Letters), **11**, 521 (1973).
94. R. J. Gritter, E. Gipstein, G. E. Adams, J. Polym. Sci., Chem. Ed., **17**, 3959 (1979).
95. Y. Tsuchiya, K. Sumi, J. Polym. Sci. A-1, **7**, 1599 (1969).
96. A. Hoff, S. Jacobson, J. Appl. Polym. Sci., **29**, 465 (1984).
97. N. Grassie, W. B. H. Leeming, Eur. Polym. J., **11**, 809 (1975).
98. N. Grassie, A. Heaney, Eur. Polym. J., **10**, 415 (1974).
99. N. Grassie, W. B. H. Leeming, Eur. Polym. J., **11**, 819 (1975).
100. V. V. Konovalov, A. B. Blyumenfel'd, Ye. F. Zinin, M. S. Akutin, B. M. Kovarskaya, B. V. Andrianov, V. I. Kleiner, Vysokomolkul Soedin. A, **16**, 323 (1974).
101. J. K. Haken, D. K. M. Ho, E. Houghton, J. Polym. Sci., Chem. Ed., **12**, 1163 (1974).

102. L. Ackerman, W. J. McGill, J. S. African Chem. Inst., **26**, 82 (1973).
103. N. Grassie, J. G. Speakman, T. I. Davis, J. Polym. Sci. A-1, **9**, 931 (1971).
104. W. J. McGill, L. Ackerman, J. Polym. Sci., Chem. Ed., **12**, 1541 (1974).
105. N. Grassie, J. G. Speakman, J. Polym. Sci. A-1, **9**, 949 (1971).
106. G. G. Cameron, D. R. Kane, Polymer, **9**, 461 (1968).
107. Yu. P. Gorelov, R. Ya Khvilivitskii, R. P. Chernovskaya, L. M. Terman, Polym. Sci., USSR, **18**, 2309 (1976).
108. N. Grassie, B. J. D. Torrance, J. Polym. Sci. A-1, **6**, 3303 (1968).
109. N. Grassie, B. J. D. Torrance, J. B. Colford, J. Polym. Sci. A-1, **7**, 1425 (1969).
110. N. Grassie, A. Johnston, A. Scotney, Polym. Degrad. Stab., **3**, 349 (1980–81).
111. N. Grassie, A. Johnston, A. Scotney, Polym. Degrad. Stab., **3**, 365 (1980–81).
112. N. Grassie, J. D. Fortune, Makromol. Chem., **168**, 1 (1973).
113. N. Grassie, R. M. Jenkins, Eur. Polym. J., **9**, 697 (1973).
114. P. K. Dhal, G. N. Babu, J. Polym. Sci., Chem. Ed., **22**, 1817 (1984).
115. G. N. Babu, P. H. Lu, S. L. Hsu, J. C. W. Chien, J. Polym. Sci., Chem. Ed., **22**, 195 (1984).
116. N. Grassie, R. McGuchan, Eur. Polym. J., **7**, 1091, 1357 (1971).
117. A. Ballistreri, S. Foti, G. Montaudo, S. Pappalardo, E. Scamporrino, Makromol. Chem., **180**, 2835 (1979).
118. N. Grassie, E. M. Grant, J. Polym. Sci., C, **16**, 591 (1967).
119. N. Grassie, R. McGuchan, Eur. Polym. J., **8**, 243 (1972).
120. N. Grassie, A. Johnston, A. Scotney, Polym. Degrad. Stab., **4**, 173 (1982).
121. N. Grassie, E. Farish, Eur. Polym. J., **3**, 619 (1967).
122. N. Grassie, J. N. Hay, Makromol. Chem., **64**, 82 (1963).
123. N. Grassie, D. R. Bain, J. Polym. Sci. A-1, **8**, 2653, 2665 (1970).
124. D. Braun, R. Disselhoff, Angew. Makromol. Chem., **23**, 103 (1972).
125. N. Grassie, E. M. Grant, Eur. Polym. J., **2**, 255 (1966).
126. N. Grassie, A. S. Holmes, Polym. Degrad. Stab., **3**, 145 (1981).
127. N. Grassie, I. C. McNeill, J. N. R. Samson, Polym. Degrad. Stab., **1**, 17 (1979).
128. R. J. Gritter, M. Seeger, D. E. Johnson, J. Polym. Sci., Chem. Ed., **16**, 169 (1978).
129. N. Grassie, A. Scotney, L. Mackinnon, J. Polym. Sci., Chem. Ed., **15**, 251 (1977).
130. N. Grassie, A. Johnston, A. Scotney, Eur. Polym. J., **17**, 589 (1981).
131. J. Razga, J. Petranek, Eur. Polym. J., **11**, 805 (1975).
132. I. C. McNeill, M. Zulfiqar, J. Polym. Sci., Chem. Ed., **16**, 2465 (1978).
133. I. C. McNeill, M. Zulfiqar, J. Polym. Sci., Chem. Ed., **16**, 3201 (1978).
134. I. C. McNeill, M. Zulfiqar, Polym. Degrad. Stab., **1**, 89 (1979).
135. A. Hamoudi, I. C. McNeill, Eur. Polym. J., **14**, 525 (1978).
136. A. Jamieson, I. C. McNeill, Eur. Polym. J., **10**, 217 (1974).
137. N. Grassie, E. Farish, Eur. Polym. J., **3**, 305 (1967).
138. I. C. McNeill, T. Straiton, P. Anderson, J. Polym. Sci., Chem. Ed., **18**, 2085 (1980).
139. N. Grassie, A. Johnston, A. Scotney, Polym. Degrad. Stab., **4**, 123 (1982).
140. G. Camino, N. Grassie, I. C. McNeill, J. Polym. Sci., Chem. Ed., **16**, 95 (1978).
141. N. Grassie, I. C. McNeill, I. Cooke, J. Polym. Sci., **12**, 831 (1968).
142. A. Jamieson, I. C. McNeill, J. Polym. Sci., Chem. Ed., **14**, 1839 (1976).
143. I. C. McNeill, T. Straiton, P. Anderson, J. Polym. Sci., Chem. Ed., **18**, 2085 (1980).
144. I. C. McNeill, D. Neil, Eur. Polym. J., **6**, 569 (1970).
145. A. Guyot, M. Bert, A. Michel, I. C. McNeill, Eur. Polym. J., **7**, 471 (1971).
146. N. Grassie, I. C. McNeill, J. N. R. Samson, Eur. Polym. J., **14**, 931 (1978).
147. A. Ballistreri, S. Foti, G. Montaudo, E. Scamporrino, J. Polym. Sci., Chem. Ed., **18**, 1147 (1980).
148. I. C. McNeill, R. C. McGuiness, Polym. Degrad. Stab., **5**, 303 (1983).
149. T. Iida, M. Nakanishi, K. Goto, J. Polym. Sci., Chem. Ed., **12**, 737 (1974).
150. D. Raucher, M. Levy, J. Polym. Sci., Chem. Ed., **17**, 2675 (1979).
151. D. A. Chatfield, J. Polym. Sci., Chem. Ed., **21**, 1681 (1983).
152. D. R. Roberts, A. L. Gatzke, J. Polym. Sci., Chem. Ed., **16**, 1211 (1978).
153. M. Blazso, T. Szekely, Eur. Polym. J., **10**, 115 (1974).
154. C. Vasile, C. N. Cascaval, P. Barbu, J. Polym. Sci., Chem. Ed., **19**, 907 (1981).
155. N. Grassie, I. F. McLaren, I. C. McNeill, Eur. Polym. J., **6**, 679 (1970).
156. N. Grassie, I. F. McLaren, I. C. McNeill, Eur. Polym. J., **6**, 865 (1970).
157. A. Jamieson, I. C. McNeill, J. Polym. Sci., Chem. Ed., **12**, 387 (1974).
158. A. Jamieson, I. C. McNeill, J. Polym. Sci., Chem. Ed., **14**, 603 (1976).
159. I. C. McNeill, D. Neil, A. Guyot, M. Bert, A. Michel, Eur. Polym. J., **7**, 453 (1971).
160. B. Dodson, I. C. McNeill, J. Polym. Sci., Chem. Ed., **14**, 353 (1976).
161. I. C. McNeill, D. Neil, Eur. Polym. J., **7**, 115 (1971).
162. F. A. Wodley, J. Appl. Polym. Sci., **15**, 835 (1971).
163. N. B. Zaitsev, I. Ya Poddubnyi, Polym. Sci. USSR, **17**, 1299 (1975).
164. I. C. McNeill, M. A. J. Mohammed, Eur. Polym. J., **8**, 975 (1972).
165. I. Luderwald, O. Vogl, Makromol. Chem., **180**, 2295 (1979).
166. Y. Sugimura, T. Nagaya, S. Tsuge, Macromolecules, **14**, 520 (1981).
167. J. P. Montheard, M. Camps, S. Kawayi, Makromol. Chem., **183**, 1191 (1982).

168. G. N. Babu, P. H. Lu, S. L. Hsu, J. C. W. Chien, J. Polym. Sci., Chem. Ed., **22**, 213 (1984).
169. R. J. Gritter, E. Gipstein, G. E. Adams, J. Polym. Sci., Chem. Ed., **17**, 3959 (1979).
170. T. S. Ellis, R. H. Still, J. Appl. Polym. Sci., **23**, 2871, 2881 (1979).
171. T. S. Ellis, R. H. Still, J. Appl. Polym. Sci., **23**, 2837, 2855 (1979).
172. R. H. Still, M. B. Evans, A. Whitehead, J. Appl. Polym. Sci., **16**, 3207 (1972).
173. J. R. MacCallum, K. Paterson, Eur. Polym. J., **10**, 477 (1974).
174. F. Hrabak, J. Mitera, V. Kubelka, M. Bezdek, Eur. Polym. J., **14**, 219 (1978).
175. J. Y. C. Chu, M. Stolka, J. Polym. Sci., Chem. Ed., **13**, 2867 (1975).
176. H. H. G. Jellinek, S. H. Ronel, J. Polym. Sci. A-1, **9**, 2605 (1971).
177. J. L. Cotter, G. J. Knight, W. W. Wright, Br. Polym. J., **7**, 389 (1975).
178. J. L. Cotter, G. J. Knight, W. W. Wright, Br. Polym. J., **7**, 375 (1975).
179. J. Jachowicz, M. Kryszewski, P. Kowalski, J. Appl. Polym. Sci., **22**, 2891 (1978).
180. J. Jachowicz, M. Kryszewski, A. Sobol, Polymer, **20**, 995 (1979).
181. N. Grassie, R. S. Roche, Makromol. Chem., **112**, 16 (1968).
182. G. H. McCain, J. M. Sanders, J. Appl. Polym. Sci., **18**, 3077 (1974).
183. N. A. Nikolayeva, B. G. Belenkii, N. A. Glukhov, Yu. V. Zhuravlev, Yu N. Sazanov, Polym. Sci. USSR, **12**, 2978 (1970).
184. S. Iwabuchi, V. Jaacks, W. Kern, Makromol. Chem., **177**, 2675 (1976).
185. A. Patterson, G. J. Sutton, B. J. Tighe, J. Polym. Sci., Chem. Ed., **11**, 2343 (1973).
186. N. Grassie, E. J. Murray, P. A. Holmes, Polym. Degrad. Stab., **6**, 47 (1984).
187. S. Foti, M. Giuffrida, P. Maravigna, G. Montaudo, J. Polym. Sci., Chem. Ed., **22**, 1201 (1984).
188. D. R. Cooper, R. Molloy, G. J. Sutton, B. J. Tighe, J. Polym. Sci., Chem. Ed., **18**, 123 (1980).
189. S. Iwabuchi, V. Jaacks, F. Galil, W. Kern, Makromol. Chem., **165**, 59 (1973).
190. V. Passalacqua, F. Pilati, V. Zamboni, B. Fortunato, P. Manarese, Polymer, **17**, 1044 (1976).
191. R. M. Lum, J. Polym. Sci., Chem. Ed., **17**, 203 (1979).
192. M. E. Bednas, M. Day, K. Ho, R. Sander, D. M. Wiles, J. Appl. Polym. Sci., **26**, 277 (1981).
193. M. Ravey, J. Polym. Sci., Chem. Ed., **21**, 1 (1983).
194. R. M. Aseyeva, T. V. Zelenetskaya, O. G. Sel'skaya, A. A. Berlin, Polym. Sci. USSR, **14**, 1766 (1972).
195. F. Nakanishi, M. Hasegawa, T. Tasai, Polymer, **16**, 218 (1975).
196. G. O. Schulz, H. J. Harwood, J. Polym. Sci., Chem. Ed., **12**, 1451 (1974).
197. H. H. G. Jellinek, H. Fujiwara, J. Polym. Sci. A-1, **10**, 1719 (1972).
198. G. P. Norton, B. J. Tighe, R. Molloy, J. Polym. Sci., Chem. Ed., **16**, 283 (1978).
199. E. H. Catsiff, M. N. Gillis, R. H. Gobran, J. Polym. Sci. A-1, **9**, 1271 (1971).
200. V. B. Gavalyan, I. V. Zhieravleva, S-S. A. Pavlova, V. I. Nedel'hin, V. A. Sergeyev, Vysokomol. Soedin., **A22**, 2121 (1980).
201. A. Davis, Makromol. Chem., **128**, 242 (1969).
202. I. I. Levantovaskaya, G. V. Dralyuk, O. A. Mochalova, I. A. Yurkova, M. S. Akutin, B. M. Kovarskaya, Polym. Sci. USSR, **13**, 7 (1971).
203. L. I. Danilina, E. N. Teleshov, A. N. Pravednikov, Polym. Sci. USSR, **16**, 672 (1974).
204. R. J. Gritter, M. Seeger, E. Gipstein, J. Polym. Sci., Chem. Ed., **16**, 353 (1978).
205. Yl. P. Krasnov, V. M. Savinov, L. B. Sokolov, V. I. Logunova, V. K. Belyakov, T. A. Polyakova, Vysokomol. Soedin., **8**, 380 (1970).
206. D. A. Chatfield, I. N. Einhorn, R. W. Michelson, J. H. Futrell, J. Polym. Sci., Chem. Ed., **17**, 1353 (1979).
207. L. H. Peebles, Jr., M. W. Huffman, J. Polym. Sci. A-1, **9**, 1807 (1971).
208. N. Grassie, I. G. Macfarlane, Eur. Polym. J., **14**, 875 (1978).
209. N. Grassie, I. G. Macfarlane, K. F. Francey, Eur. Polym. J., **15**, 415 (1979).
210. J. L. Cotter, G. J. Knight, W. W. Wright, Br. Polym. J., **7**, 381 (1975).
211. A. I. Sidnev, Yu. V. Khvashchevskaya, A. F. Moiseyev, I. A. Zubkov, A. N. Pravednikov, Polym. Sci. USSR, **12**, 2905 (1970).
212. N. Grassie, K. F. Francey, I. G. Macfarlane, Polym. Degrad. Stab., **2**, 67 (1980).
213. N. Grassie, K. F. Francey, Polym. Degrad. Stab., **2**, 53 (1980).
214. B. Zelei, M. Blazso, S. Dobos, Eur. Polym. J., **17**, 503 (1981).
215. N. Grassie, S. R. Beattie, Polym. Degrad. Stab., **9**, 23 (1984).
216. N. Grassie, S. R. Beattie, Polym. Degrad. Stab., **7**, 231 (1984).
217. D. A. Chatfield, I. N. Einhorn, J. Polym. Sci., Chem. Ed., **19**, 601 (1981).
218. N. R. Lerner, J. Polym. Sci., Chem. Ed., **15**, 1145 (1977).
219. R. M. Kromaite, S. V. Lavrov, L. A. Oksentevich, I. V. Vasileva, E. N. Teleshov, A. N. Pravednikov, Polym. Sci. USSR, **16**, 1006 (1974).
220. V. A. Ponomarenko, V. N. Shilgayev, A. G. Kechina, A. A. Yarosh, S. P. Krukovskii, Polym. Sci. USSR, **16**, 637 (1974).
221. J. P. Luongo, H. Schonhorn, J. Polym. Sci., Chem. Ed., **13**, 1363 (1975).
222. H. D. Stenzenberger, K. U. Heinen, D. O. Hummel, J. Polym. Sci., Chem. Ed., **14**, 2911 (1976).
223. A. Scotney, Eur. Polym. J., **8**, 163 (1972).
224. W. P. Brown, C. F. H. Tipper, J. Appl. Polym. Sci., **22**, 1459 (1978).
225. A. Davis, J. H. Golden, J. Macromol. Sci. Rev. Macromol. Chem. C, **3**, 49 (1969).

226. S. Foti, M. Giuffrida, P. Maravigna, G. Montaudo, J. Polym. Sci., Chem. Ed., **21**, 1567 (1983).
227. M. S. Lim, B. J. Bulken, E. M. Pearce, J. Polym. Sci., **19**, 2773 (1981).
228. N. Grassie, M. Zulfiqar, J. Polym. Sci., Chem. Ed., **16**, 1563 (1978).
229. N. Grassie, M. Zulfiqar, in: G. Scott (Ed.), "Developments in Polymer Stabilisation", vol. 1, Applied Science Publishers, London, 1979, p. 197.
230. L. P. Rumao, K. C. Frisch, J. Polym. Sci. A-1, **10**, 1499 (1972).
231. S. Foti, P. Maravigna, G. Montaudo, J. Polym. Sci., **19**, 1679 (1981).
232. S. Foti, M. Giuffrida, P. Maravigna, G. Montaudo, J. Polym. Sci., Chem. Ed., **21**, 1583 (1983).
233. N. Grassie, D. H. Mackerron, Polym. Degrad. Stab., **5**, 43 (1983).
234. F. Gaboriaud, J. P. Vantelon, J. Polym. Sci., Chem. Ed., **19**, 139 (1981).
235. N. Grassie, D. H. Mackerron, Polym. Degrad. Stab., **5**, 89 (1983).
236. N. Grassie, M. Zulfiqar, M. I. Guy, J. Polym. Sci., Chem. Ed., **18**, 265 (1980).
237. S. Foti, M. Giuffrida, P. Maravigna, G. Montaudo, J. Polym. Sci., Chem. Ed., **21**, 1599 (1983).
238. H. R. Allcock, W. J. Cook, Macromolecules, **7**, 284 (1974).
239. H. R. Allcock, G. Y. Moore, W. J. Cook, Macromolecules, **7**, 571 (1974).
240. I. C. McNeill, A. Ya. Polishchuk, G. E. Zaikov, Polym. Degrad. Stab., **37**, 223 (1992).
241. I. C. McNeill, S. Ahmed, L Memetea, Polym. Degrad. Stab., **46**, 303 (1994).
242. I. C. McNeill, A. Ya. Polishchuk, G. E. Zaikov, Polym. Degrad. Stab., **47**, 319 (1995).
243. I. C. McNeill, M. H. Mohammed, Polym. Degrad. Stab., **48**, 175 (1995).
244. E. Trebacz, Thermochimica Acta, **143**, 51 (1989).
245. H. Morikawa, R. H. Marchessault, Can. J. Chem., **59**, 2306 (1981).
246. R. S. Lehrle, R. J. Williams, Macromolecules, **27**, 3782 (1994).
247. A. Ballistreri, D. Garozzo, M. Giufrida, G. Montaudo, Macromolecules, **20**, 2991 (1987).
248. D.-J. Liaw, K.-F. Teng, Polym. Int., **37**, 77 (1995).
249. T. Sawaguchi, T. Takesue, T. Ikemura, M. Seno, Macromol. Chem. Phys., **196**, 4139 (1995).
250. G. Montaudo, C. Puglosi, F. Samperi, Macromol. Chem. Phys., **195**, 1241 (1994).
251. R. C. White, E. C. Sikabwe, J. Polym. Sci., Chem. Ed., **30**, 2781 (1992).
252. M. M. Fares, J. Hacaloglu, S. Suzer, Eur. Polym. J., **30**, 845 (1994)
253. G. Camino, M. P. Luda, L. Costa, M. Guaita, Macromol. Chem. Phys., **197**, 41 (1996).
254. G. Montaudo, C. Puglisi, J. W. de Leeuw, W. Hartgers, K. Kishore, K. Ganesh, Macromolecules, **29**, 6466 (1996).
255. G. Montaudo, C. Puglisi, R. Rapisardi, F. Samperi, Macromol. Chem. Phys., **195**, 1225 (1994).
256. F. Gaboriaud, J. P. Vantelon, J. Polym. Sci., Chem. Ed., **20**, 2063 (1982).
257. R. Abate, A. Ballistreri, G. Montaudo, G. Impallomeni, Macromolecules, **27**, 332 (1994).
258. G. Montaudo, C. Puglosi, E. Scamporrino, D. Vitalini, Macromolecules, **19**, 2157 (1986).
259. S. Groves, R. Lehrle, Eur. Polym. J., **28**, 373 (1992).
260. K. S. Khairou, M. A. Diab, Polym. Degrad. Stab., **44**, 17 (1994).
261. K. S. Khairou, M. A. Diab, Polym. Degrad. Stab., **44**, 21 (1994).
262. K. S. Khairou, M. A. Diab, Polym. Degrad. Stab., **43**, 329 (1993).
263. J. Song, W. Schnabel, Polym. Degrad. Stab., **43**, 335 (1994).
264. M. Zulfiqar, A. Paracha, S. Zulfiqar, Polym. Degrad. Stab., **43**, 403 (1994).
265. S. Zulfiqar, M. Zulfiqar, M. Rizvi, A. Munir, I. C. McNeill, Polym. Degrad. Stab., **43**, 423 (1994).
266. B. Boinon, M. Raihane, J. P. Montheard, Polym. Degrad. Stab., **43**, 27 (1994).
267. I. C. McNeill, L. Memetea, Polym. Degrad. Stab., **43**, 9 (1994).
268. J. Song, Ch.-H. Fischer, W. Schnabel, Polym. Degrad. Stab., **41**, 141 (1993).
269. I. C. McNeill and S. Basan, Polym. Degrad. Stab., **41**, 311 (1993).
270. J. C. W. Chien, A. X. Zhao, Polym. Degrad. Stab., **40**, 257 (1993).
271. M. Kryszewski, J. Jachowicz, M. Malanga, O. Vogl, Polymer, **23**, 271 (1982).
272. A. Rincon, I. C. McNeill, Polym. Degrad. Stab., **40**, 343 (1993).
273. S. Soares, G. Camino, S. Levchik, Polym. Degrad. Stab., **49**, 275 (1995).
274. M. Bounekhel, I. C. McNeill, Polym. Degrad. Stab., **49**, 347 (1995).
275. M. Coskun, K. Demirelli, E. Odzemir, Polym. Degrad. Stab., **47**, 251 (1995).
276. S. Zulfiqar, M. Zafar-uz-Zaman, A. Munir, I. C. McNeill, Polym. Degrad. Stab., **47**, 59 (1995).
277. I. C. McNeill, T. Mahmood, Polym. Degrad. Stab., **45**, 285 (1994).
278. C. N. Cascaval, V. Neagu, Polym. Degrad. Stab., **45**, 423 (1994).
279. I. C. McNeill, T. Mahmood, Polym. Degrad. Stab., **46**, 195 (1994).
280. S. Zulfiqar, K. Masud, B. Siddique, A. Piracha, Polym. Degrad. Stab., **52**, 293 (1996).
281. S. Zulfiqar, M. Rizvi, A. Munir, A. Ghaffar, I. C. McNeill, Polym. Degrad. Stab., **52**, 341 (1996).
282. F.-D. Kopinke, M. Remmler, K. Mackenzie, M. Moder, O. Wachsen, Polym. Degrad. Stab., **53**, 329 (1996).
283. S. Zulfiqar, A. Piracha, K. Masud, Polym. Degrad. Stab., **52**, 89 (1996).
284. P. Selsbo, I. Ericsson, Polym. Degrad. Stab., **51**, 83 (1996).

285. M. Bounekhel, I. C. McNeill, Polym. Degrad. Stab., **51**, 35 (1996).
286. A. Piracha, S. Zulfiqar, Polym. Degrad. Stab., **51**, 27 (1996).
287. S. Zulfiqar, M. Zafar-uz-Zaman, A. Munir, I. C. McNeill, Polym. Degrad. Stab., **50**, 33 (1996).
288. P. Almen, I. Ericsson, Polym. Degrad. Stab., **50**, 223 (1995).
289. N. A. Al-Awadi, M. Z. Elsabee, Polym. Degrad. Stab., **50**, 319 (1995).
290. S. L. Hurley, M. L. Mittleman, C. A. Wilkie, Polym. Degrad. Stab., **39**, 345 (1993).
291. I. C. McNeill, S. Basan, Polym. Degrad. Stab., **39**, 145 (1993).
292. I. C. McNeill, S. Basan, Polym. Degrad. Stab., **39**, 139 (1993).
293. H.-J. Dussel, H. Rosen, D. O. Hummel, Makromol. Chem., **177**, 2343 (1976).
294. I. C. McNeill, J. J. Liggat, Polym. Degrad. Stab., **36**, 291 (1992).
295. I. C. McNeill, A. Rincon, Polym. Degrad. Stab., **24**, 59 (1989).
296. I. C. McNeill, A. Rincon, Polym. Degrad. Stab., **24**, 171 (1989).
297. I. C. McNeill, S. M. T. Sadeghi, Polym. Degrad. Stab., **29**, 233 (1990).
298. I. C. McNeill, M. Bounekhel, Polym. Degrad. Stab., **34**, 187 (1991).
299. I. C. McNeill, J. G. Gorman, S. Basan, Polym. Degrad. Stab., **33**, 263 (1991).
300. I. Popovic, J. Song, Ch.-H. Fischer, L. Katsikas, G. Hohne, J. Velickovic, W. Schnabel, Polym. Degrad. Stab., **32**, 265 (1991).
301. C. T. Vijayakumar, K. Lederer, A. Kramer, Polym. Degrad. Stab., **32**, 9 (1991).
302. S. Zulfiqar, M. Zulfiqar, M. Nawaz, I. C. McNeill, J. G. Gorman, Polym. Degrad. Stab., **30**, 195 (1990).
303. I. C. McNeill, S. M. T. Sadeghi, Polym. Degrad. Stab., **30**, 213 (1990).
304. I. C. McNeill, A. Rincon, Polym. Degrad. Stab., **31**, 163 (1990).
305. P. K. Dhal, G. N. Babu, J. C. W. Chien, Polym. Degrad. Stab., **18**, 1 (1987).
306. V. L. Rao, P. K. Dhal, G. N. Babu, Polym. Degrad. Stab., **18**, 19 (1987).
307. N. A. Weir, K. Whiting, J. Arct, G. McCulloch, Polym. Degrad. Stab., **18**, 293 (1987).
308. V. Mailhos-Lefievre, D. Sallot, B. Martel, Polym. Degrad. Stab., **23**, 327 (1989).
309. S. Zulfiqar, N. Akthar, M. Zulfiqar, I. C. McNeill, Polym. Degrad. Stab., **23**, 299 (1989).
310. P. K. Dhal, G. N. Babu, J. C. W. Chien, Polym. Degrad. Stab., **16**, 135 (1986).
311. G. Sivasubramanian, P. Sivasamy, C. T. Vijayakumar, J. K. Fink, Polym. Degrad. Stab., **21**, 151 (1988).
312. I. C. McNeill, S. M. T. Sadeghi, Polym. Degrad. Stab., **30**, 267 (1990).
313. N. A. Weir, K. Whiting, J. Arct, Polym. Degrad. Stab., **22**, 17 (1988).
314. S. Zulfiqar, M. Zulfiqar, T. Kausar, I. C. McNeill, Polym. Degrad. Stab., **17**, 347 (1987).
315. J. N. Hay, D. J. Kemmish, Polymer, **28**, 2047 (1987).
316. S. J. Maynard, T. R. Sharp, J. F. Haw, Macromolecules, **24**, 2794 (1991).
317. A. Ballistreri, D. Garozzo, G. Montaudo, Macromolecules, **17**, 1312 (1984).
318. J. Jachowicz, M. Kryszewski, M. Mucha, Macromolecules, **17**, 1315 (1984).
319. D. A. Chatfield, I. N. Einhorn, R. W. Mickelson, J. H. Futrell, J. Polym. Sci., Chem. Ed., **17**, 1367 (1979).

Radiation Chemical Yields: *G* Values

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A. INTRODUCTION

The *G* values quoted in the following tables are defined as the radiation chemical yields of individual atomic or molecular events for 100 eV of energy absorbed by the system. *G*(R) represents the yield of free radicals, *G*(X) the yield of crosslinks, *G*(S) the yield of main chain scission, and *G*(products) the yield of product molecules per 100 eV of absorbed radiation.

This table has been compiled from the polymer literature that has been published since the third edition of the "Polymer Handbook" (i.e. 1986–1997). In many cases, the tabulated data are only representative of the total data in the paper, and the original publication should be consulted.

B. TABLES OF *G* VALUES

Notes to the Tables

General: Unless otherwise stated, the conditions applying are γ-rays, vacuum, room temperature (ambient).

Symbols/Abbreviations

+	cation radical	d.b.	double bond
−	anion radical	*t*-v	*trans*-vinylene
vac	vacuum	Y	crosslinks
RT	room temperature	X	crosslinks
e	electron beam	n	neutron

TABLE 1. HOMOPOLYMERS

Polymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
1.1. POLYDIENES							
Polybutadiene				38.7 (d.b.)	e, 263 K	Excitation scavenger	64
			610 (1,2)			1, 2-PBD	138
	0.0054				77 K	Radical pairs, isotactic	74
		9.3		112 (Vinyl)		Syndiotactic	97
		105		Various	e	*cis* 1,4	84
	2.1	3.36/9.4	0.34/0.19		77 K/RT		98
		5.3	90		Air, 295 K	Effect of pressure	104
	Various	Various		Various	Various	Review	186
						G(Cy) > *G*(X)	261
						G(X) depend on MW	
						G (initiation),	262
						G(X) increases *G*(Cy), *G*(db) decreases with Crosslinker	
		41			Vac, 303 K		180
Poly(2-phenyl butadiene)		7.2			Vac/303 K, ethylene dichloride		178
Polyisoprene	0.009				77 K	Radical pairs	74
	0.5–0.8	1.6–2.1		94–168 (d.b.), CH_4, H_2		Var *cis* content	15
	Various	Various		Various	Various	Review	186
NR latex		Various				+ acrylate	248
Polychloroprene	3.0	3.9		2.8 (–Cl)	Vac/77, 303 K		179
1.2. POLYOLEFINS							
Squalene				3.0 (H_2)			235
Poly(butene-1)		1.0	3.2	13.9 (–isotactic pentad)	RT, vac		226
Poly(ethylene)		1.09–1.39			e, air		89
		0.63	4	7 (COOH)	Vac/O_2		101
			4.1	9.1 (O_2)			107
	4			–2 (*t*-v)	300 K		172
		1.4	0.48				116
		15–30	8–15	Carries	Air, 300/450 K	Pressure effects	169
		0.25					30
		1.50–2.40	1.08–1.69	Various	303–373 K	e pre-irradiated	63
	2.8, 5.8 (C_2D_4)				He, 77 K		52
			2.4		O_2, CCl_4		147
						Fibers, *G*(S)/*G*(X) = 1	151
		1.1–2.8	0.05–2.7			e, N_2	155
				Various gases	Air	Various compounds	70
		1.1	< 0.05			e, N_2	87
		1.5	0.48			Gamma/n	174
	2.6/0.56 (alkyl)				Gamma/alpha, 77 K		163
		0.74 G (H)			Vac, molten state		183
		0.27 G (Y)					
		1.0	0.2	3.9 (H_2)			228
				3.5 (damage)		Crystalline region units	270

TABLE 1. *cont'd*

Polymer	G(R)	G(X)	G(S)	G(products)	Radiation conditions	Comments	Refs.
–, LDPE	2.5 (alkyl)				77 K		177
				12.4–15.4 (O_2)	O_2		96
					air	$G(S)/G(X) = 1.6$	117
				4.45 (butanes)	303 K		10
				Various	298–423 K	$+I_2$	27
				−1.9 (*t*-v)		$+CH_3SH$ antioxidants	71
				Alkanes			129
	3.0 (total)				77 K		164
	2.8 (alkyl)						
				Various $G(t\text{-v})$			231
–, LLDPE		2.4	0.4				105
				Various	Air	Butene-1, hexene-1	184
–, HDPE		1.3			Air		88
	2.9–3.6				77 K		92
	2.3–4.5(alkyl)				77 K	Morphology	177
				0.16 (butanes)	303 K		10
	0.3*					Allylic	28
	0.21						32
				Alkanes			125
		0.3, 0.43		Y, C=C, (*t*-v)		Monolayer crystals	45
		3.7			303 K		47
		0.76 (bulk)		−9.6 (*t*-v)	Air, 77 K/RT	Warming, morphology	134
		1.6*				*G(X/unsat.)	75
–, HDLPE	5.5	0.76				77 K	79
				2.4 (*trans*)			240
				0.7 (*cis*)			
–, Chlorosulfonated polyisobutylene				4–5 (gas)			119
			4.0		e		131
	2.5				77 K		133
			3.8–11.1		Various temperatures		78,90,100
	Various		Various	Various	Various		186
	2.1		3.6	Various	Vac/77, 303 K		
Poly(4-methylpentene-1)				0.3,0.6*	N2/air	*G(S)/G(X)	43
polypropylene	4.8, 2.7(dPP)				He, e, 77 K		52
			3.1/0.64		Air/vac	Also 373–623 K	129
			9.4		$O_2 + CCl_4$		147
				2.9 (H_2)		Also O_2	158
			2.19*		$+CCl_4$	Decreases with dose *G(S)− G(X)	159
				Various	Air		118
				Various	Air		40
i-Polypropylene				94 (-*i*-pentad)	Gamma		244
				Various (pentad sequences)			
				220 (-*i*-pentad)	e		245
1.3. POLYACRYLATES							
Poly(acrylic acid)				Various gases	Vac/air		170
	3.6						2
	3.49				293–308 K		4
		0.0	0.44		Vac		185
				Various	77, 300 K		189
		0.47	0.62				190
				Various	Gamma, aq.		233
Poly(acrylic anhydride)			1.1*			*G(S)− S(X)	59
Poly(methyl acrylate)				H_2, CO, CO_2, CH_4 etc.	195–423		126

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TABLE 1. *cont'd*

Polymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
Poly(methyl bromoacrylate)	6.98						139
Poly(methyl α-chloroacrylate)	5.1				77 K		133
		0.3–0.9	3.2–7.4			Dose dependent	24
		0.8	6.0				62
			6.7*			*G(S)− G(X)	17
Poly(methyl α-cyanoacrylate)	1.6				77 K		133
Poly(methyl α-fluoroacrylate)	1.0		0.05		77 K		65
1.4. POLY (METHACRYLATES)							
Poly(benzyl methacrylate)			0.29*		e	*G(S)− G(X)	22
		0	0		300 K	Syndiotactic	192
			0.14				238
Poly(*t*-butyl methacrylate)			1.28*		e	*G(S)− G(X)	22
Poly(cyclohexyl methacrylate)			0.44*		e	*G(S)− G(X)	22
			2.4*	Various		*G(S)− G(X)	120
Poly(glycidyl methacrylate)		9.2					246
Poly(hexafluoroisopropyl methacrylate)			2.6		7 K		65
Poly(hexyl methacrylate)			0.38*		e	*G(S)− G(X)	22
Poly(isobutyl methacrylate)			1.1*		e	*G(S)− G(X)	22
		0.1	0.8		300 K	Syndiotactic	192
Poly(2-methyl heptyl methacrylate)		0.5	0.3		300 K	Syndiotactic	192
Poly(methacrylic acid)			3.4		N_2	Aq. solution	91
				Various	Gamma/e		176
			0.6/2.9		77/273 K		123
			4.8		77 K		133
		0.2	6.0				185
				Various	77, 303 K		189
		0.04	6				195
Poly(methacrylic anhydride)				Various	Gamma/e		176
			1.8–2.9				59
Poly(methyl methacrylate)			1.9 (DMF) 1.3 (THF)			Soln.	59
	1.1–1.5(+)		1.5				73
	0.7–0.9(−)				77 K	Aromatic additives	57
	0.5, 1.5(+)				77 K	Aromatic additives	51
	2.8–4.1				303, 343 K	Absorbed on zeolite	53
			1.3		77 K		65
			1.31				50
			1.9		e		131
	0.58				77 K		132
			1.63, 1.09		Gamma/n, vac/air 206 K		135
	1.42		1.4				139
			1.5/0.33		Gamma/X	Si wafers	152
			1.2				161
				Various	e/gamma, 303–383 K		176
			0.71–1.63		Gamma/n, vac/air		86
	1.9/4.1				77/273 K	Various radicals	173
	1.04/1.11				Vac/air, 273 K		146
	1.9, 4.4				77 K	Bulk, alcohol	171
	1.5						162
			1.5*			*G(S)− G(X)	17
				Various		+ ethane diol	14
				15.8 (units)	Air	Isotactic	23
				0.34 (H_2)			33
			18.6		Vac, 363 K	Isotactic	187
			0.22,0.46,1.08	Various	Deep UV, e, X-ray		188

TABLE 1. *cont'd*

Polymer	G(R)	G(X)	G(S)	G(products)	Radiation conditions	Comments	Refs.
			1.5			Isotactic	190
			1.1		300 K	Syndiotactic	192
			1.7		N_2, RT		222
			1.7		Vac	Protons, gamma	223
			1.6		N_2	e, gamma	224
			0.99, 0.4		Vac	C_5+, O_5+ ions	225
			0.77				238
			5.1 (soln.)			*i*-PMMA	249
			0.89 (amorphous solid)				
			0.48 (crystalline solid)				
			1.21 (333 K)				
			0.89 (303 K)				
			0.46 (195 K)				
				G(damage) = 8.8		Morphology effects, racemization	258
		1.0			Air, RT		264
Poly(1-naphthyl methacrylate)			0.14			Benzene solution	1
			0.14				238
Poly(2-naphthyl methacrylate)			0.19				238
Poly(2,2,2-trichloroethyl methacrylate)		0.03	3.7		Gamma/e, 298–416 K		175
			3.7			Dioxane solution	11
Poly(ethyl methacrylate)		0	1.7		300 K	Syndiotactic	192
Poly(*n*-butyl methacrylate)		0.4	1.3		300 K	Syndiotactic	192
Poly(2-hydroxyethyl methacrylate)	1.2, 1.7				77, 300 K		193
Poly(phenyl methacrylate)			0.44				238
1.5. POLY(STYRENES)							
Poly(α-methyl styrene)			0.3		e		131
			0.33		Gamma/e, 77–373 K		38
			0.78		273–319 K	Benzene solution	25
		0	0.29		303, 353, 373,	+ Depolymeriza-	194
			0.48,0.89, 0.97, 1.13		383, 393 K	tion	
	0.105, 0.05				77, 298 K		197
Poly(*p*-methyl styrene)		0.1			Gamma/e, 77–373 K		38
Poly(styrene)		0.45–7.9			303 K	Emulsion polymerization	48
		0.022	0.022		Air	Sheet	56
		0.17	1.09, 2.12			absorbed on silica gel	58
		0.035–0.042	0.010–0.051	0.032 (H_2)	303–373 K		60
		0.043	0.0009		303 K	Includes lit. review	61
		0.043–0.027	0.086–0.074		303–423 K		67
			0.2*		77 K	*G(S)− G(X)	73
		0.048, 0.021	0.27, 0.07		Air	Various doses	130
	0.05				77 K		132
		0.03/0.2	0.02/0.2		Gamma/n, vac/air, 206 K		135
		0.043	0.00086			Sheet	136
			3.9		$O_2 + CCl_4$ 373–623 K		147
		0.02	0.1		Air		157

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TABLE 1. *cont'd*

Polymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
		0.077	0.055				161
	0.0009				77 K	Radical pairs	74
		0.094	0.0027		313 K		81
		Various	Various		Gamma/n, vac/air		85
	1.8				77 K		98
		0.018, 0.15	0.06, 0.07		Gamma/n/alpha, air		112
	0.5				77 K	Warming	19
		0.0174	0.0012				6
		0.0425					7
				0.033 (H_2)			33
		0.014–0.05	0.105		Gamma/e, 77–373 K		38
		0.03–0.05			Abs of air	Various sources	195
		6.7–15.2	24.7–30.4 *G*(degradation)		Various dose rates	+ Crosslinking agents	196
		0.05	0.01	0.03 (H_2)			228
				H_2, CH_4, C_2H_2, C_3H_4, C_6H_6	Ion beam	Deuterium subs	234
				Various ($-O_2$)		Variable temp. dose rate	250
		Various	Various	Various hydrogen	Vac, various temp.	Tacticity, crystallinity effect	257
		0.03–0.05				Gamma, X-ray, Ne, Ar	255

1.6. POLY(VINYLS)

1.6.1. ACRYLAMIDES AND NITRILES

Poly(acrylamide)		2.5			Air/H_2O		149
			0.04		e, O_2	Aq. soln.	103
		5			Aq. soln, 304 K	Hydrogel	108
	0.88	1.33/0.62	0.45/0.52	Various	Vac/air, 77–393 K		167
Poly(methacrylamide)			1.2*			*G(S)–G(X)	141
Poly(acrylonitrile)	3.7–4.1				303 K	Absorbed on zeolite	53
	0.15–3.0						5
	2.8, 4.2	0.59	0.0		77, 300 K		198
Poly(methacrylonitrile)	2.2, 2.8				77, 300 K		200
			0.4–3.3		Absence of O_2	Various sources	195
			3.3 gamma				255
			3.6 X				
			0.5 Ne				
			0.4 Ar				
Poly(α-chloroacrylonitrile)	1.4				77 K		133
			2.7*			*G(S)– G(X)	59
			1.1–7.2		Vac/air	Solution + bulk	99
			3.2				145
			2.1–3.3*			*G(S)– G(X)	59
			3.1–3.3				62
			3.3*			*G(S)– G(X)	17

1.6.2. VINYL MONOMERS

Poly(*t*-butyl crotonate)		0.66	0.59				54
Poly(*t*-butyl vinyl ether)			3.6, 10.3		273, 348 K		122
Poly(ethyl vinyl ether)			0.36, 0.50		273, 348 K	Liquid	122
Poly(isobutyl vinyl ether)			0.39, 1.73		273, 348 K		122
Poly(isopropyl vinyl ether)			0.86, 1.50		273, 348 K		122

TABLE 1. *cont'd*

Polymer	G(R)	G(X)	G(S)	G(products)	Radiation conditions	Comments	Refs.
Poly(methyl vinyl ether)			0.31, 0.64		273, 348 K		122
Poly(tetrafluoroethylene)			0.5		$O_2 + CCl_4$ K	Also 373–623 K	147
Poly(vinyl acetate)	2.4–4.1				243–273 K		21
				0.9–2.4 gel dose	Absence of O_2		195
Poly(vinyl alcohol)			1.08		O_2		236
			1.02		Air		
			0.71		N_2		
Poly(vinyl bromide)				39–55(HBr)		Solution	9
Poly(vinyl butyral)		0.35	0.6		e, N_2		241
Poly(vinyl carbazole)	0.7 (holes)				e		168
Poly(vinyl chloride)	1.7				77 K		165
		2.36			Air	Under pressure	113
				Various	Vac/O_2		115
				7.7 (HCl)	Vac/O_2		119
		10–11	20–115	(d.b.)		THF soln. + additives	8,44
				17(HCl)	e, N_2/vac		154
				0.24 (H_2)	Vac/O_2		160
				8 (HCl)			
				2.8–3.5 (Py +)		G(cation)	247
				0–1.2 (Phenyl +)		273 K	
						G(X) Increases with crosslinker	263
				3.0 (–Cl)			269
Poly(vinyl imidazole)		10–54			Aq	+ Crosslinker	256
Poly(vinyl pyrrolidone)		0.41–0.75				Aq. soln.	12
	3.3				77 K	+ Warming	13
		0.01–1.32				Various solutions	205

1.7. MISCELLANEOUS POLYMERS

1.7.1. CELLULOSE AND DERIVATIVES

Cellulose				CO, CHO, COOH	Vac/air		111
			13–186		Vac/air	Post irrad.	26
Cellulose acetate			7.1		Air	H_2O incr. G(S)	156

1.7.2. POLY(SILOXANES)

Poly(oxydimethyl diphenylsilylene)		Various				Liq.	94
Poly(oxydimethyl silylene)		2.48			N_2, 303 K	Also blends with PS	143
		18.5/120			Air	Effect of pressure	104
		2.7					76
		3.0					80
		2.8				Liquid	82
	Various	Various	Various	Various			186
						G(X) increases with crosslinker	265
Poly(methyl phenyl silane)		0.01–0.21				LET dependence	230
Poly(cyclohexyl methyl silane)	1		1.8–17.4			Dose dependence	232
Poly(dimethyl siloxane)		1.7					243
Silicon containing poly(acetylene)		0.3–2.3				Effect of various pendent groups	252

1.7.3. POLY (AMINO ACIDS)

Poly(alanine)	2.9		3.8–7.0	Various			95
	3.6			Various			128
Poly(glycine)	2.2		3.0–3.3	Various			95
Poly(tyrosine)	< 0.4		0.12	Various			95
				0.04–0.12 (NH_3)	$Air/N_2/N_2O$	Aq. soln.	102
Poly(valine)				Various			95

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TABLE 1. *cont'd*

Polymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
1.7.4. POLYESTERS							
Poly(hydroxybutyrate)			1.65		Air		191
Poly(L-lactic acid)	2.0, 1.5				77, 300 K		201
	2.37–1.23				77, 300 K		203
Poly(D, L lactic acid)	2.4, 1.2				77, 300 K		201
						Pressure and temp. effect on *G*(S) and *G*(X)	274
Poly(glycolic acid)	1.72, 1.47				77, 300 K		203
						Pressure and temp. effects on *G*(S) and *G*(X)	274
1.7.5. POLYSULFONES							
Bis-A-PSF	0.13	0.13–1.34	0.11–0.67	Various gases	Various temp		207
	0.13			0.14 (SO_2)			208
	0.57			0.145 (SO_2)			211
Hydroquinone PSF	0.19			0.136 (SO_2)			207
	0.14			0.14 (SO_2)			208
	0.56			0.14 (SO_2)			211
Biphenol PSF	0.084			0.063 (SO_2)			207
	0.084			0.06 (SO_2)			208
	0.51			0.6 (SO_2)			211
Bis-S-PSF	0.128			0.198 (SO_2)			207
PES (Polyethersulfone)	0.24			0.243 (SO_2)			207
1.7.6. POLYKETONES							
PEEK	0.11				Vac, 77 K	209	
	0.004				Vac, 300 K		
	0.001				Air, 300 K		
	1.0e20 spin/kG				77 K	211	
	0.11			0.00054 gases			212
PEEKK	0.12				Vac, 77 K		209
	0.004				Vac, 300 K		
	0.001				Air, 300 K		
	1.1e20 spin/kG				77 K		211
PEEK-Me	0.28				Vac, 77 K		209
	0.009				Vac, 300 K		
	0.001				Air, 300 K		
PEEKK-Me	0.37				Vac, 77 K		209
	0.012				Vac, 300 K		
	0.001				Air, 300 K		
PEK-Ar	0.42				Vac, 77 K		209
	0.039				Vac, 300 K		
	0.005				Air, 300 K		
1.7.7. FLUOROPOLYMERS							
Fluoro elastomers	Various	Various	Various	Various	Various	Review	186
Poly(perfluoroethers)				Various gases	e, argon 9 K		202
				G(decomposition) = 0.2–1.2	e	Structure, temp. effects	259
			G(S)	*G*(wt. loss), *G*(gas)			260
Poly(tetrafluoroethylene)			2.15			As polymerized sintered, low crystalline	237
			6.0			Dose rate effects on *G*(S)	267

TABLE 1. *cont'd*

Polymer	G(R)	G(X)	G(S)	G(products)	Radiation conditions	Comments	Refs.
			0.3 at 373 0.57 at 473 0.89 at 573 1.33 at 673 2.07 at 773			+ Dose effects	271
Poly(vinyl fluoride)	0.22			0.07 (Polyvinyl)			46
Poly(vinylidene fluoride)	0.37			0.11 (Polyvinyl)			46
		0.46–0.78	0.38–0.86	1–4 (HF)	293 K		49
		0.6–0.78	0.29			+ Crosslinker	254
1.7.8. OTHERS							
Aromatic polymers	Various			Various gases			204,211
Alkyl aromatic poly amides		Various	Various			G(S)/G(X) Reduces with increase in methylene	251
Aryloxyphosphazenes		0.22 (R = Me)		0.26			110
–[(R-Ph-O)$_2$-PN]$_n$–		2.12 (R = Et)		0.26			110
–biphenol	0.6, 0.19				77, 300 K		210
–bis phenol	0.6, 0.22				77, 300 K		210
–hydroquinone	0.6, 0.35				77, 300 K		210
Poly(bis(phenoxy)phosphazene)		0.9	0.7				273
Poly(bis(4-methyl-phenoxy)phosphazene)		3.5	2.0				273
Poly(bis(4-*t*-butyl-phenoxy)phosphazene)		0.1	0.1				273
Poly(bis(4-cumyl-phenoxy)phosphazene)		0.2	0.2				273
Poly(1-(*t*-butoxycarbonylthio)ethylene)	3.39				77 K	Warming	13
Poly(*t*-butyl isopropenyl ketone)			1.5				142
Poly(butylene), 1,4-disubstituted(urethane side chain)		0.33					93
Poly(epoxide)(Araldite)	0.59				77 K		36
Poly(ether imide) (Ultem)	0.01	0.139	0.00528		77 K		114
Hyaluronic acid				5.0–6.7 (O_2) 2.0 (H_2O_2) 0–0.1 (peroxide)		pH dependence	253
Poly(1-mercapto ethylene)	2.65				77 K	Warming	19
Poly(oxadiazole-2,5-diyl-1,4-phenylene-phthalidylidene-1,4-phenylene)			Various	Vac/air		39	
Poly(oxycarbonyl-1,1-dimethyl-2-isopropylidene-ethylene)			1.9				50
Poly(oxycarboxyethylene)		0.4–11	14.5, 40.5		N_2 air		35
Poly(oxyethylene)			2.6		e, O_2 air	Aq. soln.	103
		36			200 K		83
	5		57–144	Various		Dose rate dependence	55
Poly(oxyethyleneoxyterephthaloyl)		0.023				Also e	29
Poly(oxymethylene)			8	320 (HCHO)	423 K		37
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenelene oxy-1,4-phenylene-isopropylidene-1,4-phenylene)		0.05	0.012	0.04 gas	300–423 K		153
Poly(oxy-1,4-phenylenesulfony-1,4-phenylene)				0.04 gas	300–423 K		153
Poly(oxysebacoyloxy-1,4-phenylene) isopropylidene-1,4-phenylene)			9		Air		68
Poly(phenylene oxide)	5.7e20 spin/kG				77 K		211
	0.6, 0.33			0.08 H_2 0.015 CH_4	77, 300 K		206
Poly(phenylene sulfide)	0.16, 0.04			0.03H_2	77, 300 K		206
Poly(thiophenylene)		0.015/0.067	0.030/0.15		Gamma/e, vac/air		109
Poly(thiopropylene)	0.0014				77 K	Radical pairs	74

TABLE 2. COPOLYMERS

Copolymers	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
2.1. COPOLYMERS WITH ETHYLENE							
Poly(ethylene-*co*-1-butene)				Various	303–423 K	+ Additives	10
				Various			125
Poly(ethylene-*co*-ethyl acrylate)		2.53–3.2			e, air		89
Poly(ethylene-*co*-hexene)				Various			125
				Various	303–423 K	+ Additives	10
Poly(ethylene-*co*-propylene)		0.43, 0.83	0.11, 0.35		303 K	23%, 36% Propylene	127
				Various	Air	+ Additives	70
				Various			125
				Various	303–423 K	+ Additives	10
				8.6–11.0 (O_2)	Air		96
		0.69	1.2	5(COOH)	Air		101
		1.1	0.33				116
		0.9/3.2	3.2		Air	2.0 GPa + variation	104
	2.92	0.26	0.16			In terpolymer	16
				Various gases, O_2	293–298		182
	Various	Various	Various	Various 0.07–0.62 gel dose	Various	Review various sources	186 195
Butyl rubber	2.3 (total) 0.19 (allyl)	0–0.6	3.9–6.09	Various chain ends	77–398 K		219
Halogenated butyl rubber	3.7–4.0	Dose dependent	Dose dependent	Various gases	77 K	Chloro and bromo	220
2.2. COPOLYMER WITH METHYL METHACRYLATE							
Poly(methyl methacrylate-*co*-bromomethyl methacrylate)	0.2–1.4	0.06–0.1	0.7–1			139	
Poly(methyl methacrylate-*co*-chlorotrifluoroethylene)			2.1		77 K		65
			2.1				137
Poly(methyl methacrylate-*co*-dichlorostyrene)		1.3				50	
Poly(methyl methacrylate-*co*-dimethyl itaconate)			3.2–3.5*			*G(S)− G(X)	17
			1.0*			*G(S)− G(X)	59
Poly(methyl methacrylate-*co*-hexafluoroisopropyl methacrylate)			2.0–3.1		77 K		65
		0.4	3.6				137
Poly(methyl methacrylate-*co*-hexyl methacrylate)		0.53–0.87				22	
Poly(methyl methacrylate-*co*-methacrylic acid)	1.8, 2.06				273 K		146
	2.9				77 K	20% MAA	133
Poly(methyl methacrylate-*co*-methacrylic acid-styrene)		1.2				238	
Poly(methyl methacrylate-*co*-methacrylic acid-*co*-cumyl phenyl methacrylate)		1.6				238	
Poly(methyl methacrylate-*co*-methacrylic acid-*co*-2-naphthyl methacrylate)			1.09				238
Poly(methyl methacrylate-*co*-methacrylic acid)			2.0				238
Poly(methyl methacrylate-*co*-2 naphthyl methacrylate)			1.06				238
Poly(methyl methacrylate-*co*-methacrylonitrile, vinylidene chloride, methyl-α-chloroacrylate, α-chloronitrile ter polymers)		0.06	1.9–3.1				145
Poly(methyl methacrylate-*co*-methyl bromoacrylate)	2.0	0.41	4.16			139	

TABLE 2. *cont'd*

Copolymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
Poly(methyl methacrylate-*co*-methyl-α-chloroacrylate)	3.1				77 K	14% MCA	133
Poly(methyl methacrylate-*co*-methyl-α-chloroacrylate)		0.06	3.1				24
Poly(methyl methacrylate-*co*-methyl-2-chloroacrylate)			2.8*			*G(S)− G(X)	17
Poly(methyl methacrylate-*co*-methyl-α-chloroacrylonitrile)		6.6			77 K	14% CAN	133
Poly(methyl methacrylate-*co*-methyl-α-cyanoacrylate)	5.9				77 K	20% MCA	133
Poly(methyl methacrylate-*co*-methyl-α-fluoroacrylate)		0-0.44	1.0–2.2		77 K		65
Poly(methyl methacrylate-*co*-methyl fluoroacrylate)			2.4				137
Poly(methyl methacrylate-*co*-3-oximino-2-butanone methacrylate)			> 0.55	1.4–2.4	Various		69
Poly(methyl methacrylate-*co*-tribromophenyl methacrylate)		0.07–0.19	0.38–2.2		Air		72
Poly(methyl methacrylate-*co*-trichloroethyl-alpha-chloroacrylate)		3.8				Also TFEM	140
Poly(methyl methacrylate-*co*-trichlorophenyl methacrylate)		0.05–0.7	0.4–4.4				72
Poly(methyl methacrylate-*co*-trifluoroethylene methacrylate)			1.9		77 K		65
Poly(methyl methacrylate-*co*-vinylidene chloride)			2.8–3.4*			*G(S)− G(X)	17
Poly(methyl methacrylate-*co*-vinylidene fluoride)			1.6				137
			1.6		77 K		65
Poly(glycidyl methacrylate-*co*-ethyl methacrylate)		1.35–8.5	1.51–5.5			Various compositions	246
Poly(glycidyl methacrylate-*co*-ethyl acrylate)	6.5					246	

2.3. COPOLYMERS WITH STYRENE

Poly(styrene-*co*-acrylates)			Various				73
Poly(styrene-*co*-acrylonitrile)		0.077	0.055				34
	0.4–2.8 77 K 0.14–4.2 300 K	0.04–0.59	0.0–0.09		Vac	Various compositions	215,221
Poly(styrene-*co*-butadiene)		4.3/33	2.8/17		Air	Effect of pressure	104
		0.3–3.0			313 K		3
		Various	Various		77 K	Block copolymers	98
Poly(styrene-*co*-cyanoacrylates)		0.04–0.38	2.18–2.47				73
Poly(styrene-*co*-1-mercaptoethylene)	Various				77 K		19
Poly(styrene-*co*-methyl acrylate)				CO, CH_4, CO_2 etc.	Various temperatures	12–80% sty	126
	0.40–2.76 77 K 0.14–1.39 300 K				Vac	Various compositions	216
Poly(styrene-*co*-methyl methacrylate)	0.18			0.83 (gases)	77 K/(R), 300 K/(gas)	51% sty	132
Poly(styrene-*co*-methacrylonitrile)	0.31–2.16					Various compositions	199
Poly(styrene-*co*-butyl vinyl ketone)					e		239
Poly(styrene-*co*-methyl isopropyl ketone)					e		239

2.4. COPOYMERS WITH SULFUR DIOXIDE

Poly(1-butene-*co*-sulfur dioxide)				Various gases	273–423 K		31,124,150
			12		e		131
Poly(2-butene-*co*-sulfur dioxide)				Various gases	273–423 K		31,124
Poly(cyclohexene-*co*-sulfur dioxide)				Various gases	273–423 K		31,124
Poly(4,4-dimethylpentene-*co*-sulfur dioxide)				Various gases	273–423 K		31,124

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TABLE 2. *cont'd*

Copolymer	*G*(R)	*G*(X)	*G*(S)	*G*(products)	Radiation conditions	Comments	Refs.
Poly(ethylene-*co*-sulfur dioxide)				Various gases	273–423 K		124
Poly(hexene-1-*co*-sulfur dioxide)				Various gases	273–423 K		31,124
Poly(isobutene-*co*-sulfur dioxide)				Various gases	273–423 K		31,124
Poly(3-methyl-1-butene-*co*-sulfur dioxide)	1.8			Various gases	273–423 K		Poly(31,124, 150
Poly(propylene-*co*-sulfur dioxide)	2.2			Various gases	273–423 K		31, 124, 150
2.5. OTHER COPOLYMERS							
Poly(butyl methacrylate-*co*-1-naphthyl methacrylate)			0.14			Benzene soln.	1
Poly(glutamic acid-*co*-tyrosine)		0.03–0.5	0–0.5		300 K	Various compositions	214
Poly(glycolic acid-*co*-1-lactic acid)	2.23–2.36				77 K	Various	203
	1.49–1.53				300 K	compositions	
Poly(glycidyl methacrylate-*co*-3-chlorostyrene)	1.0	0.42					144
Poly(hydroxy butyrate-*co*-valerate)	Various				77, 263, 273, 303 K	Different compositions	217
		3.0–0.5	2.7–3.4	Various gases 1,4-ester	e, gamma		218
Poly(methacrylonitrile-*co*-methyl-2-chloroacrylate)			3.2–43.*			*G(S)– G(X)	17
Poly(*N*-vinylpyrrolidone-*co*-*t*-butyl-*s*-vinyl thiocarbonate)	3.01				77 K	+ Warming	13
Poly(acrylonitrile-*co*-butadiene)		20/150			Air		104
Poly(acrylonitrile-*co*-chloroprene)				3.5 (Gas)			119
Poly(tetrafluoroethylene-*co*-hexafluoropropylene)		3.3					106
Poly(tetrafluoroethylene-*co*-propylene)		1.3	0.42			Emulsion polymer + H_2O	121
				0.6–5.2 (HF)	N_2		116
		1.16			240 C		266
Poly(vinyl chloride-*co*-ethylglycodi-methacrylate)				30 (d.b.)			8
Poly(vinylidene chloride-*co*-dimethyl itaconate)			2.3–2.7			*G(S)– G(X)	17
Poly(vinylidene chloride-*co*-hexafluoro-propylene)		1.2			Vac/air/H_2O	+ Additives	166
Tetrafluoroethylene, perfluoro methyl vinyl ether copolymer 2:1				0.93 (CF_4) 0.31 (COF_2) 0.055 (CO_2) 0.14 (CF_3 OCF_3) 1.43 (total gas) 0.34 (acid fluoride)	e		229
		0.9		New chain ends 1 ($-CF_2-CF_3$) 0.09 ($-CF_2-O-CF_3$) 0.5 (CF_2-COOH) 0.2($-CF=CF_2$)			268
Polysulphone–polydimethyl siloxane *block* copolymer	Various					Dose and molecular weight effects	272

TABLE 3. POLYMERS BLENDS

Polymer Blends	$G(X)$	$G(S)$	Comments	Refs.
Poly(methyl methacrylate)/ Poly(1-naphthyl(methacrylate)		0.14	Benzene	1
Poly(methyl methacrylate)/ Poly(styrene-*co*-acrylonitrile)	0.025	0.051	50% PMMA	161
Poly(methyl methacrylate)/styrene-acrylonitrile	0.041	0.027	SAN 100	242
	0.034	0.15	85	
	0.02	0.21	50	
	0.008	0.23	15	
Poly(methyl methacrylate)/Acrylonitrile	0.042	0.14	ACN 100	242
	0.036	0.25	85	
	0.009	0.34	50	
	0.01	0.88	15	

TABLE 4. COMPOSITES

Polymer Composites	G(products)	Radiation conditions	Refs.
Glass-DGEBA-DDM	H_2, CO, CO_2	77 K, RT	227
Color-DGEBA-DDM	H_2, CO, CO_2	77 K, RT	227

C. REFERENCES

1. S. Nishimoto, K. Yamamoti, T. Kagiya, Polym. Photochem., **5**, 231 (1984).
2. M. R. Clay, A. Charlesby, Europ. Polym. J., **11**, 187 (1975).
3. M. A. El-Azmirly, A. A. Yehia, W. M. Khalifa, N. A. Ghanem, Europ. Polym. J., **11**, 499 (1975).
4. A. Chapiro, D. Goldfield-Freilich, J. Perichon, Europ. Polym. J., **11**, 515 (1975).
5. A. Chapiro, Z. Mankowski, Europ. Polym. J., **14**, 15, (1978).
6. C. David, D. Breyens-Volant, Europ. Polym. J., **14**, 29 (1978).
7. S. Warty, J. A. Sauer, A. Charlesby, Europ. Polym. J., **15**, 445, (1979).
8. G. Palma, M. Gnrenza, C. Pollini, Europ. Polym. J., **16**, 333, (1980).
9. A. Chapiro, Z. Mankowski, Europ. Polym. J., **18**, 707 (1982).
10. S. Y. Ho, J. H. O'Donnell, Europ. Polym. J., **20**, 421 (1984).
11. J. Rosiak, W. Schnabel, Europ. Polym. J., **20**, 1159 (1984).
12. A. Chapiro, C. Legris, Europ. Polym. J., **21**, 49 (1985).
13. H. Monig, H. Ringsdorf, S. Wessel, Makromol. Chem., **176**, 1323 (1975).
14. C. David, F. de Bie, G. Geuskens, Makromol. Chem., **176**, 3493 (1975).
15. R. Kaufman, H. Heusinger, Makromol. Chem., **177**, 871 (1976).
16. W. Geissler, H. Zott, H. Heusinger, Makromol. Chem., **179**, 697 (1978).
17. C-Y. Chen, M. Igbal, C. U. Pitman Jr., Makromol. Chem., **179**, 2109 (1978).
18. M. B. Huglin, J. Smith, Makromol. Chem., **180**, 2775 (1979).
19. H. Monig, H. Ringsdorf, E. Spirk, Makromol. Chem., **181**, 2293 (1980).
20. R. Basheer, M. Dole, Makromol. Chem., **183**, 2141 (1982).
21. J. G. Braks, R. Y. M. Huang, Makromol. Chem., **185**, 317 (1984).
22. J. H. Lai, J. H. Helbert, Macromolecules, **11**, 617 (1978).
23. R. P. Kusy, A. R. Greenburg, Macromolecules, **11**, 1051 (1978).
24. J. N. Helbert, C-Y. Chen, C. U. Pittman Jr., G. L. Hagnauer, Macromolecules, **11**, 1104 (1978).
25. T. Ikeda, H. S. von Euler-Chelpin, S. Okamura, H. Yamaoka, Macromolecules, **17**, 1655 (1984).
26. J. T. Guthrie, Polymer, **16**, 134 (1975).
27. T. N. Bowmer, J. H. O'Donnell, Polymer, **18**, 1032 (1977).
28. Y. Hori, Z. Fukunaga, S. Shimada, H. Kashiwabara, Polymers, **20**, 181 (1979).
29. T. Memetea, V. Stannett, Polymer, **20**, 469 (1979).
30. G. Ungar, Polymer, **21**, 1278 (1980).
31. T. N. Bowmer, J. H. O'Donnell, Polymer, **22**, 71 (1981).
32. M. Dole, Polymer **22**, 1458 (1981).
33. W. K. Busfield, J. H. O'Donnell, C. A. Smith, Polymer, **23**, 431 (1982).
34. T. Q. Nguyen, H. H. Kausch, K. Jud, M. Dettenmaier, Polymer, **23**, 1305 (1982).
35. M. C. Gupta, V. G. Deshmukh, Polymer, **24**, 827 (1983).
36. K. R. Schaeffer, R. E. Fornes, R. D. Gilbert, J. D. Memory, Polymer, **25**, 54 (1984).

37. L. P. Burlak, Y. S. Deyev, G. V. Levit, V. P. Pshenitsyna, V. I. Serennkov, N. A. Slovokhotova, F. F. Sukhov, V. S. Tikhomirov, Polymer Science USSR, **21**, 2326 (1979).
38. I. K. Chernova, S. S. Leshchenko, V. P. Golikov, V. L. Karpov, Polymer Science USSR, **22**, 2382 (1980).
39. V. V. Lyashevich, V. V. Korshak, D. R. Tur, V. V. Rode, Polym. Sci. USSR, **22**, 2422 (1980).
40. S. G. Kiryushkin, Y. A. Shlyapnikov, Polym. Sci. USSR, **23**, 617 (1981).
41. A. P. Tyutnev, V. S. Sayenko, Y. D. Pozhidayev, Polym. Sci. USSR, **24**, 120 (1982).
42. A. P. Tyutnev, V. S. Sayenko, V. S. Tikhomirov, Y. D. Pozhidayev, Polym. Sci. USSR, **25**, 113 (1983).
43. N. S. Soboleva, S. S. Leshchenko, V. L. Karpov, Polym. Sci. USSR, **25**, 446 (1983).
44. V. I. Dakin, A. V. Danchenko, V. L. Karpov, Polym. Sci. USSR, **26**, 2473 (1984).
45. G. N. Patel, J. Polym. Sci. Polym. Phys. Ed., **13**, 339 (1975).
46. J. N. Helbert, E. W. Burkhard, E. H. Poindexter, L. Kevan, J. Polym. Sci., Polym. Phys. Ed., **13**, 825 (1975).
47. J. C. Randall, Makromol. Chem., Rapid Commun., **4**, 149 (1983).
48. K. Sato, A. Rudin, R. Y. M. Huang, J. Polym. Sci., Polym. Chem. Ed., **14**, 37 (1976).
49. K. Makuuchi, M. Asano, T. Abe, J. Polym. Sci., Polym. Chem. Ed., **14**, 617 (1976).
50. A. C. Ouano, D. E. Johnson, B. Dawson, L. A. Pederson, J. Polym. Sci., Polym. Chem. Ed., **14**, 701 (1976).
51. A. Torikai, H. Kato, Z-I. Kuri, J. Polym. Sci., Polym. Chem. Ed., **14**, 1065 (1976).
52. V. K. Micinchuk, E. R. Klinshpont, V. V. Vasilenko, J. Polym. Sci., Polym. Chem. Ed., **14**, 1419 (1976).
53. J. P. Quaegebeur, T. Teguchi, H. LeBail, C. Chachaty, J. Polym. Sci., Polym. Chem. Ed., **14**, 2703 (1976).
54. J. H. O'Donnell, N. P. Rahman, D. J. Winzor, J. Polym. Sci., Polym. Chem. Ed., **15**, 131 (1977).
55. C. Decker, J. Polym. Sci., Polym. Chem. Ed., **15**, 781 (1977).
56. J. M. Nichol, J. H. O'Donnell, N. P. Rahman, D. J. Winzor, J. Polym. Sci., Polym. Chem. Ed., **15**, 2919 (1977).
57. A. Torikai, R. Kato, J. Polym. Sci., Polym. Chem. Ed., **16**, 1487 (1978).
58. Y. Kasama, A. Udagawa, M. Takehisa, J. Polym. Sci., Polym. Chem. Ed., **16**, 2607 (1978).
59. J. Helbert, E. Poindexter, G. Stahl, C-Y. Chen, C. Pittman, J. Polym. Sci., Polym. Chem. Ed., **17**, 49 (1979).
60. Y. Shimizu, H. Mitsui, J. Polym. Sci., Polym. Chem. Ed., **17**, 2307 (1979).
61. J. H. O'Donnell, N. P. Rahman, C. A. Smith D. J. Winzor, J. Polym. Sci., Polym. Chem. Ed., **17**, 4081 (1979).
62. C-Y. Chen, C. U. Pittman, Jr., J. N. Helbert, J. Polym. Sci., Polym. Chem. Ed., **18**, 169 (1980).
63. H, Mitsui, Y. Shimizu J. Polym. Sci., Polym. Chem. Ed., **18**, 1115 (1980).
64. K. Hayashi, M. Tachibana, S. Okamura, J. Polym. Sci., Polym. Chem. Ed., **18**, 2785 (1980).
65. C. Pittman, Jr., C-Y. Chen, M. Ueda, J. N. Helbert, J. Kwiatkowski, J. Polym. Sci., Polym. Chem. Ed., **18**, 3413 (1980).
66. T. N. Bowmer, J. H. O'Donnell, J. Polym. Sci., Polym. Chem. Ed., **19**, 45 (1981).
67. T. N. Bowmer, J. H. O'Donnell, D. J. Winzor, J. Polym. Sci., Polym. Chem. Ed., **19**, 1167 (1981).
68. M. C. Gupta, A. K. Srivastava, J. Polym. Sci., Polym. Chem. Ed., **20**, 541 (1982).
69. T. N. Bowmer, E. Reichmans, C. W. Wilkins, Jr., M. Y. Hellman, J. Polym. Sci., Polym. Chem. Ed., **20**, 2661 (1982).
70. K. Arakawa, T. Seguchi, N. Hayakawa, S. Machi, J. Polym. Sci., Polym. Chem. Ed., **21**, 1173 (1983).
71. F. J. Zoepfl, J. Silverman, J. Polym. Sci., Polym. Chem. Ed., **21**, 1763 (1983).
72. P. H. Lu, G. Babu, J. C. W. Chien, J. Polym. Sci., Polym. Chem. Ed., **23**, 1421 (1985).
73. H. Ito, C. Hrusa, H. K. Hall, Jr. A. B. Padias, J. Polym. Sci., Polym. Chem. Ed., **24**, 955 (1986).
74. P. Hesse, H. Heusinger, Radiat. Phys. Chem., **7**, 1 (1975).
75. S. Rafi Ahmad, A. Charlesby, Radiat. Phys. Chem., **8**, 497 (1976).
76. R. Folland, A. Charlesby, Radiat. Phys. Chem., **8**, 555 (1976).
77. A. Rafi Ahmad, A. Charlesby, Radiat. Phys. Chem., **8**, 585 (1976).
78. A. Charlesby, J. Steven, Radiat. Phys. Chem., **8**, 719 (1976).
79. M. Dole, V. M. Patel, Radiat. Phys. Chem., **9**, 433 (1977).
80. R. Folland, A. Charlesby, Radiat. Phys. Chem., **10**, 61 (1977).
81. A. Charlesby, Radiat. Phys. Chem., **10**, 177 (1977).
82. C. G. Delides, I. W. Shepherd, Radiat. Phys. Chem., **10**, 379 (1977).
83. A. Charlesby, M. Byrne, Radiat. Phys. Chem., **12**, 191 (1978).
84. S. G. Burnay, Radiat. Phys. Chem., **13**, 171 (1979).
85. Y. Tabata, Radiat. Phys. Chem., **14**, 235 (1979).
86. K. Ishigure, S. Egusa, S. Tagawa, Y. Tabata, Radiat. Phys. Chem., **14**, 585 (1979).
87. V. Handlos, Radiat. Phys. Chem., **14**, 721 (1979).
88. G. N. Patel. Radiat. Phys. Chem., **14**, 729 (1979).
89. N. M. Burns, Radiat. Phys. Cehm., **14**, 97 (1979).
90. A. Charlesby, R. Folland, Radiat. Phys. Chem., **15**, 393 (1980).
91. T. S. Chien, J. K. Thomas, Radiat. Phys. Chem., **15**, 429 (1980).
92. N. Gvozdic, M. Dole, Radiat. Phys. Chem., **15**, 435 (1980).
93. G. N. Patel, Radiat. Phys. Chem. **15**, 637 (1980).
94. G. G. Delides, Radiat. Phys. Chem., **16**, 345 (1980).
95. D. J. T. Hill, R. W. Garrett, S. Y. Ho, J. H. O'Donnell, P. W. O'Sullivan, P. J. Pomery, Radiat. Phys. Chem., **17**, 163 (1981).
96. T. Seguchi, S. Hashimoto, K. Arakawa, N. Hayakawa, W. Kawakami, I. Kuriyama, Radiat. Phys. Chem., **17**, 195 (1981).
97. H. Okamoto, T. Iwai, Radiat. Phys. Chem., **18**, 407 (1981).
98. R. Basheer, M. Dole, Radiat. Phys. Chem., **18**, 1053 (1981).
99. H. Yamaoka, K. Nagta, T. Ikeda, S. Okamura, Radiat. Phys. Chem., **18**, 1073, (1981).

100. A. Charlesby, B. J. Bridges, Radiat. Phys. Chem., **19**, 155 (1982).
101. T. Seguchi, K. Arakawa, N. Hayakawa, Y. Watanabe, I. Kuriyama, Radiat. Phys. Chem., **19**, 321 (1982).
102. G. Boguta, A. M. Dancewicz, Radiat. Phys. Chem., **20**, 359 (1982).
103. W. Schnabel, O. Denk, V. Grollmann, I. A. Raapand, K. Washino, Radiat. Phys. Chem., **21**, 225 (1983).
104. V. K. Milinchuk, E. R. Klinshpont, V. P. Kiryukhin, Radiat. Phys. Chem., **21**, 519 (1983).
105. M. Dole, Radiat. Phys. Chem., **22**, 11 (1983).
106. N. Lin, Radiat. Phys. Chem., **22**, 259 (1983).
107. K. Hayakawa, K. Kawase, H. Yamikita, Radiat. Phys. Chem., **22**, 929 (1983).
108. Z. -C. Zhang, Q. Li, D. -H. Li, Radiat. Phys. Chem., **22**, 1064 (1983).
109. G. -L Huang, Z. -F. Chang, Y. -X. Han, Radiat. Phys. Chem., **22**, 1067, (1983).
110. S. Yanai, V. T. Stannett, D. R. Squire, G. L. Hagnauer, R. L. Singler, Radiat. Phys. Chem., **23**, 489 (1984).
111. W. M. Dziedziela, D. J. Kotynska, Radiat. Phys. Chem., **23**, 723 (1984).
112. M. C. Gupta, V. P. Bansod, Radiat. Phys. Chem., **24**, 499 (1984).
113. G. Urillo, Radiat. Phys. Chem., **25**, 383 (1985).
114. R. Basheer, M. Dole, Radiat. Phys. Chem., **25**, 389 (1985).
115. A. H. Zahran, E. A. Hegazy, F. M. Ezz Eldin, Radiat. Phys. Chem., **26**, 25 (1985).
116. T. Seguchi, N. Hayakawa, K. Yoshida, N. Tamura, Y. Katsumura, Y. Tabata, Radiat. Phys. Chem., **26**, 221 (1985).
117. O. S. Gal, V. M. Markovic, L. J. R. Novakovic, V. T. Stannett, Radiat. Phys. Chem., **26**, 325 (1985).
118. E-S. A. Hegazy, A. H. Zahran, Radiat. Phys. Chem., **27**, 139 (1986).
119. K. Arakawa, T. Seguchi, K. Yoshida, Radiat. Phys. Chem., **27**, 157 (1986).
120. M. Kumar, M. H. Rao, K. N. Rao, Radiat. Phys. Chem., **27**, 219 (1986).
121. N. Suzuki, J. Okamoto, J. Macromol. Sci. A: Chem., **9** (2), 285 (1975).
122. Y. Suzuki, J. M. Rooney, V. Stannett, J. Macromol Sci. A: Chem., **12** (7), 1055 (1978).
123. J. H. O'Donnell, R. D. Sothman, J. Macromol. Sci. A: Chem., **14** (46), 879 (1980).
124. T. N. Bowmer, J. H. O'Donnell, J. Macromol. Sci. A: Chem., **17** (2), 243 (1982).
125. T. N. Bowmer, S. Ho, J. H. O' Donnell, P. W. O'Sullivan, J. Macromol. Sci. A: Chem., **21** (6&7), 745 (1984).
126. W. H. Busfield, J. H. O'Donnell, C. A. Smith, J. Macromol. Sci. A: Chem., **17** (8), 1263 (1982).
127. J. H. O'Donnell, A. K. Whittaker, Brit. Polym. J., **17**, 51 (1985).
128. D. J. T. Hill, S. Y. Ho, J. H. O'Donnell, P. W. O'Sullivan, P. J. Pomery, Polym. Degrad. Stability, **3**, 83 (1980–81).
129. T. Kayiga S. Nishimoto, Polym. Degrad. Stability, **12**, 261 (1985).
130. M. C. Gupta, B. S. Shaprio, B. P. Kanphade, V. P. Bansoo, Polym. Comm., **25**, 335 (1984).
131. M. J. Bowden, J. Polym. Sci., Polym. Symp., **49**, 221, (1975).
132. J. H. O'Donnell, P. J. Pomery, J. Polym. Sci. Polym. Symp., **55**, 269 (1976).
133. J. Helbert, B. Wagner, P. Caplan, E. Poindexter, Polym. Preprints, **16** (1), 708 (1975).
134. N. Gvozdic, M. Dole, Polym. Preprints, **19** (1), 584 (1978).
135. K. Ishigure, S. Egusa, M. Ogawa, S. Tagawa, Y. Tabata, Polym. Preprints, **20** (1), 372 (1979).
136. T. Bowmer, L. Cowen, J. Nichol, J. H. O'Donnell, N. Rahman, C. Smith, D. Winzor, Polym. Preprints, **20** (1), 376 (1979).
137. C. Pittman, C. Chen, M. Ueda, J. Helbert, Polym. Preprints, **20** (2), 602, (1979).
138. M. Golub, R. Cormia, Polym. Preprints, **22** (1), 40 (1981).
139. C. Babu, J. Chien, Polym. Preprints, **23** (2), 85 (1982).
140. E. Wallace, C. Pittman, J. Kwiatkowski, C. Cook, J. Helbert, Polym. Preprints, **24** (1), 157 (1983).
141. D. O'Connor, N. Yang, A. Woodward, Polym. Preprints, **24** (1), 172 (1983).
142. S. MacDonald, H. Ito, C. Willson, J. Moore, H. Gharapetian, J. Guillet, Polym. Preprints, **25** (1), 298 (1984).
143. G. Corfield, D. Astill, D. Clegg, Polym. Preprints, **25** (1), 311 (1984).
144. A. Novembre, T. Bowmer, Polym. Preprints, **25** (1), 324 (1984).
145. E. Wallace, C. Chen, C. Pittman, J. Helbert, J. Kwiatkowski, C. Cook, Polym. Preprints, **25** (1), 326 (1984).
146. P. R. Wunsche, K. Pfeiffer, H-J. Lorkowski, J. Reinhardt, Acta Polymerica, **36**, 265 (1985).
147. T. Tagawa, E. Tsuchida, I. Shinohara, M. Hagiwara, T. Kagiya, J. Polym. Sci., Polym. Letters Ed., **13**, 287 (1975).
148. G. Gasparrini, M. Carenza, G. Palma, J. Polym. Sci., Polym. Letters Ed., **18**, 29 (1980)
149. G. Burillo, T. Ogave, J. Polym. Sci., Polym. Letters Ed., **21**, 615 (1983).
150. T. N. Bowmer, J. H. O'Donnell, P. R. Wells, Polym. Bull., **2**, 103 (1980).
151. J. de Boer, A. J. Pennings, Polym. Bull., **5**, 317 (1981).
152. H. Sotobayashi, F. Asmussen, K. Thimm, W. Schnabel, H. Betz, D. Einfeld, Polym. Bull., **7**, 95 (1982).
153. J. R. Brown, J. H. O'Donnell, J. Appl. Polym. Sci., **19**, 405 (1975).
154. W. Szymanski, G. Smietanska, J. Appl. Polym. Sci., **19**, 1955 (1975).
155. V. N. Handlos, K. A. Singer, J. Appl. Polym. Sci., **20**, 3375 (1976).
156. S. M. Ali, P. G. Clay, J. Appl. Polym. Sci., **23**, 2893 (1979).
157. T. N. Bowmer, L. K. Cowen, J. H. O'Donnell, D. J. Winzor, J. Appl. Polym. Sci., **24**, 425 (1979).
158. E. A. Hegazy, T. Seguchi, K. Arakawa, S. Machi, J. Appl. Polym. Sci., **26**, 1361 (1981).
159. G. Ramanan, M. H. Rao, J. Appl. Polym. Sci., **26**, 1439 (1981).

160. E. A. Hegazy, T. Seguchi, S. Machi, J. Appl. Polym. Sci., **26**, 2947 (1981).
161. T. Q. Nguyen, H. H. Kausch, J. Appl. Polym. Sci., **29**, 455 (1984).
162. P. A. Zagorets, A. G. Shostenko, A. I. Krivonosov, A. M. Dodonov, B. A. Gorelik, M. V. Postrigan, High Energy Chem., **10**, 254 (1976).
163. B. K. Pasal'skii, Ya. I. Lavrentovich, V. A. Vonsyatskii, A. M. Kabakchi, High Energy Chem., **10**, 316 (1976).
164. A. P. Mal'tseva, V. P. Golikov, S. S. Leshchenko, V. L. Karpov, V. I. Muromtsev, High Energy Chem., **11**, 189 (1977).
165. V. I. Dakin, Z. S. Egorova, V. L. Karpov, High Energy Chem., **11**, 317 (1977).
166. T. G. Samoilenko, High Energy Chem., **12**, 72 (1978).
167. V. N. Khabarov, L. L. Kozlov, G. Panchenkov, High Energy Chem., **15**, 382 (1981).
168. A. P. Tyutnev, A. V. Vannikov, V. S. Saenko, S. Likhovidov, E. D. Pozhidaev, High Energy Chem., **16**, 386 (1982).
169. P. I. Dubenskov, A. P. Tyutnev, V. S. Shenko, A. V. Vannikov, High Energy Chem., **19**, 92, (1985).
170. A. M. Afanas'ev, V. A. Barakova, V. N. Demishev, V. A. Novozhilov, High Energy Chem., **19**, 342 (1985).
171. G. S. Zhadanov, A. S. Smolyanskii, V. R. Milinchuk, High Energy Chem., **20**, 27 (1986).
172. J. Silverman, F. J. Zoepfl, J. Randall, V. Markovic, in: J. Dobo, P. Hedvig, R. Schiller (Eds.), Proc. Fifth Tihany Symp. Radiation Chem. (Siofok, Hungary, 19–24 Sept, 1982), vol. 2, Akad. Kiado, Budapest, **2**, 1983, 895.
173. J. Dobo, P. Hedvig, R. Schiller (Eds.), Proc. Fifth Tihany Symp. Radiation Chem. (Siofok, Hungary, 19–24 Sept, 1982), vol. 2, Akad. Kiado, Budapest, 1983, p. 979.
174. Y. Tabata, Y. Katsumuka, T. Seguchi, N. Hayakawa, K. Yoshida, N. Tamura, Radiat. Process. Plastics and Rubber, 2nd Int. Conf., Canterbury, UK, 28–30 March, 1984, p. 1.
175. J. Rosiak, W. Schnabel, Radiat. Process. Plastics and Rubber, 2nd Int. Conf., Canterbury, UK, 28–30 March, 1984, p. 37.
176. H. Hiroka, IBM J. Res. Develop., **21**, 121 (1977)
177. G. Ungar, J. Mater. Sci., **16**, 2635 (1981).
178. H. Yamaoka, K. Kato, S. Okamura, Polym. J., **19**, 667 (1987).
179. D. J. T. Hill, J. H. O'Donnell, M. C. S. Perera, P. J. Pomery, ACS Symposium Series, **527**, 74 (1993)
180. P. F. Barron, J. H. O'Donnell, A. K. Whittaker, Polym. Bull., **14**, 339 (1985).
181. A. J. Whittaker, Ph.D. Thesis, University of Queensland, (1986).
182. K. Arakawa, J. Polym. Sci., Polym. Chem. Ed., **25**, 1713 (1987).
183. F. Horii, Q. Zhu, R. Kitamaru, H. Yamaoka, Macromolecules, **23**, 977 (1990).
184. A. Tidjani, Y. Watanbe J. Polym., Sci., Polym. Chem. Ed., **33**, 1455 (1995).
185. D. J. T. Hill, J. H. O'Donnell, C. L. Winzor, D. J. Winzor, Polymer, **31**, 538 (1990).
186. G. G. A. Bohm, J. O, Tveekrem, Rubb. Chem. Technol., **55**, 575 (1982).
187. L. Dong, D. J. T. Hill, J. H. O'Donnell, T. G. Carswell-Pomerantz, P. J. Pomery, A. K. Whittaker, K. Hatada, Macromolecules, **28**, 3681 (1995).
188. J. A. Moore, J. O. Choi, ACS Symposium series, **475**, 156 (1991)
189. D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, C. L. Winzor, Radiat. Phys. Chem., **39**, 237 (1992).
190. D. J. T. Hill, K. A. Milne, J. H. O'Donnell, P. J. Pomery, ACS Symposium Series, **620**, 130 (1996).
191. H. Mitomo, Y. Watanabe, I. Ishigaki, T. Saito, Polym. Degrad. Stab., **5**, 11 (1994).
192. L. Dong, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, K. Hatada, J. Appl. Polym. Sci., **59**, 589, (1996).
193. D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, G. Saadat, Radiat. Phys. Chem., **48**, 605 (1996).
194. R. W. Garrett, T. Le, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, in: T. Tabata, I. Mira, T. Nonogaki, (Eds.), "Polymers for Microelectronics Science and Technology", VCH, New York, 1990, p. 47.
195. W. Schnabel, Q. Q. Zin, S. Klaumunzer, ACS Symposium Series, **475**, 44 (1991).
196. D. Lopez, G. Burillo, ACS Symposium Series, **475**, 262 (1991).
197. R. W. Garrett. D. J. T. Hill, T. T. Le, J. H. O'Donnell, P. J. Pomery, Radiat. Phys. Chem., **39**, 215 (1992).
198. D. J. T. Hill, A. P. Lang, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **38**, 193 (1992).
199. L. Dong, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **48**, 71 (1995).
200. L. Dong, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polymer, **36**, 2873 (1995).
201. A. Babanalbandi, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **50**, 297 (1995).
202. J. Pacansky, R. J. Waltman, J. Phys. Chem., **95**, 1512 (1991).
203. A. Babanalbandi, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **52**, 59 (1996).
204. E. S. A. Hegazy, T. Sasuga, M. Nishii, T. Seguchi, Polymer, **33**, 2897 (1992).
205. J. M. Rosiak, ACS Symposium Series, **475**, 271 (1991).
206. D. J. T. Hill, D. S. Hunter, D. A. Lewis, J. H. O'Donnell, P. J. Pomery, Radiat. Phys. Chem., **36**, 550 (1990).
207. D. J. T. Hill, D. A. Lewis, J. H. O'Donnell, P. J. Pomery, J. L. Hedrick, J. E. McGrath, Polym. Adv. Tech., **2**, 15 (1991).
208. D. J. T. Hill, D. A. Lewis, J. H. O'Donnell, P. J. Pomery, Polym. International, **28**, 233 (1992).
209. K. Heiland, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polym. Adv., **8**, 116 (1994).
210. J. L. Hopewell, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **293**, (1994).
211. K. Heiland, D. J. T. Hill, J. L. Hopewell, D. A. Lewis, J. H. O'Donnell, P. J. Pomery, A. K. Whittaker, Adv. Chem. ser., **249**, 637 (1996).
212. L. Dong, K. Heiland, D. J. T. Hill, M. C. S. Perera, in: M. Kitajima, H. Haneda, K. Ishioka, K. Hirata, (Eds.), "Materials Chemistry in Nuclear Environment", National Research Institute of Metals, Tsukuba, Japan, 1996, 61.

213. E. A. Hegazy, T. Sasuga, M. Nishi, T. Seguchi, Polymer, **33**, 2897 (1992).
214. D. J. T. Hill, S. Y. Ho, J. H. O'Donnell, P. J. Pomery, Radiat. Phys. Chem., **36**, 467 (1990),
215. D. J. T. Hill, A. P. Lang, J. H. O'Donnell, P. J. Pomery, Polym. Degrad. Stab., **38**, 205 (1992).
216. R. Kellman, D. J. T. Hill, D. S. Hunter, J. H. O'Donnell, P. J. Pomery, ACS Symposium Series, **475**, 119 (1991).
217. T. G. Carswell-Pomerantz, D. J. T. Hill, J. H. O'Donnell, P. J. Pomery, Radiat. Phys. Chem., **45**, 737 (1995).
218. T. Bremner, D. J. T. Hill, J. H. O'Donnell, M. C. S. Perera, P. J. Pomery, J. Polym. Sci., Polym. Chem. Ed., **34**, 971 (1996).
219. D. J. T. Hill, J. H. O'Donnell, M. C. S. Perera, P. J. Pomery, Radiat. Phys. Chem., **40**, 127 (1992).
220. D. J. T. Hill, J. H. O'Donnell, M. C. S. Perera, P. J. Pomery, Polymer, **36**, 4185 (1995).
221. T. Q. Nguyen, H. H. Kausch, K. Jud, M. Dettnmaier, Polymer, **23**, 1305 (1982).
222. H. Kudoh, T. Sasuga, T. Seguchi, ACS Symposium Series, **620**, 2 (1996).
223. H. Kudoh, T. Sasuga, T. Seguchi, Y. Katsumura, Polymer, 4663 (1996),
224. H. Kudoh, M. Celina, G. M. Malone, R. J. Kaye, K. T. Gillen, R. L. Clough, Radiat. Phys. Chem., **48**, 553 (1996).
225. H. Kudoh, T. Sasuga, T. Seguchi, Y. Katsumura, Polymer, **37**, 2903 (1996).
226. W. K. Busfield, G. S. Watson, ACS Symposium Series, **620**, 74 (1996).
227. H. Kudoh, N. Kasai, T. Sasuga, T. Seguchi, Radiat. Phys. Chem., **48**, 695 (1996).
228. M. Dole, Radiat. Phys. Chem. **37**, 65 (1991).
229. J. Pacansky, R. J. Waltman, D. Jebens, Macromolecules, **29**, 7699 (1996).
230. S. Seik, H. Shibata, H. Ban, K. Ishigure, S. Tagawa, Radiat. Phys. Chem., **48**, 539 (1996).
231. Y. Hama, K. Hamanaka, H. Matsumoto, T. Takano, H. Kudoh, M. Sugimoto, T. Seguchi, Radiat. Phys. Chem., **48**, 549 (1996).
232. J. Kumagai, K. Oyama, H. Yoshida, T. Ichikawa, Radiat. Phys. Chem., **47**, 631 (1996).
233. P. Ulanski, E. Bothe, J. M. Rosiak, C. Von Sonntag, J. Chem. Soc., Perkin Trans., **2**, 1, 5 (1996).
234. M. B. Lewis, E. H. Lee, J. Nucl. Mater., **203**, 224 (1993).
235. Y. S. Soebianto, Y. Katsumura, K. Ishigure, J. Kubo, T. Koizumi, H. Shigekuni, K. Azami, Polym. Degrad. Stab., **42**, 29 (1993).
236. G. Lu, H. Chen, D. Liu, Radiat. Phys. Chem., **42**, 229 (1993).
237. X. Zhong, L. Yu, W. Zhao, Y. Zhang, J. Sun, J. Appl. Polym. Sci., **48**, 741 (1993).
238. I. Gitsov, O. Todorova, J. Appl. Polym. Sci., **46**, 1631 (1992).
239. S. A. M. Hesp, J. E. Guillet, Polym. Microelectron. Proc. Int. Symp., **113**, CA 118: 39547, (1989).
240. Q. Zhu, F. Horii, R. Kitamaru, H. Yamaoka, J. Polym. Sci. Polym. Chem. ed., **28**, 2741 (1990).
241. J. Pacansky, R. J. Waltman, Proc. Water-Borne Higher Solids Coat. Symp., **175**, CA: 111: 195538 (1989).
242. A. Daro, F. Zabeau, C. David, Eur. Polym. J., **25**, 71 (1989).
243. J. Rosiak, J. Olejniczak, A. Charlesby, Radiat. Phys. Chem., **32**, 691 (1988).
244. P. F. Barron, W. K. Busfield, J. V. Hanna, J. Polym. Sci., Polym. Lett. ed., **26**, 225 (1988).
245. P. F. Barron, W. K. Busfield, J. V. Hanna, Polym. Commun., **29**, 70 (1988).
246. D. N. Payne, F. Williams, Polym. Prepr., **28**, 303 (1987).
247. A. Torikai, T. Adachi, K. Fueki, J. Polym. Sci., Polym. Chem. Ed., **24**, 1413 (1986).
248. P. Peele, Song, Hu, Radiat. Phys. Chem., **42**, 121 (1993).
249. L. Yunxia, Y. Hang, L. Shuhua, Y. Bin, D. Mengxian, J. Bingzhen, Radiat. Phys. Chem., **42**, 233 (1993).
250. K. T. Gillen, J. S. Wallace, R. L. Clough, Radiat. Phys. Chem., **41**, 101 (1993).
251. B. J. Lyons, L. C. Glover, Radiat. Phys. Chem., **37**, 93 (1991).
252. H. Yamaoka, T. Matsuyama, T. Masuda, T. Higashimura, Radiat. Phys. Chem., **37**, 111 (1991).
253. D. J. Deeble, G. O. Phillips, E. Bothe, H. P. Schuchmann, C. von Sonntag, Radiat, Phys. Chem., **37** 115 (1991).
254. V. S. Ivanov, I. I. Migunova, A. J. Mikhailov, Radiat. Phys. Chem., **37**, 119 (1991).
255. W. Schnabel, S. Klaumunzer, Radiat. Phys. Chem., **37**, 131 (1991).
256. A. Chapiro, S. Galant, Radiat. Phys. Chem., **37**, 505 (1991).
257. Y. Tabata, A. Oshima, K. Takashika, T. Seguchi, Radiat. Phys. Chem., **48**, 563 (1996).
258. Y. Lau, J. Guan, H. Yang, S. Li, D. Bin, B. Jiang, CA: 121: 36389, (1992).
259. J. Pacansky, R. J. Waltman, G. Pacansky, Chem. Mater, **5**, 1526 (1993).
260. J. Pacansky, R. J. Waltman,, Chem. Mater., **5**, 486 (1993).
261. L. Zhang, L. Feng, S. Li, S. Wang, Z. Zhang, CA, 115:184090, (1990).
262. L. Zhang, L. Feng, S. Li, Z. Zhang, CA, **114**, 102961 (1990).
263. J Shou, Z. Zhu, S. Shi, Y Liu, CA, **111**, 195542, (1988).
264. L. Ye, P. Li, Z. Mei, Y. Ye, CA, 107: 78430, (1986).
265. X. Chen, X. Tang, H. Guan, Z. Zhang, L. Zhang, S. Li, J. Zhang, Y. Zhang, CA. 107: 78428, (1985).
266. J Sun, S. Zhu, Y. Zhang, CA, 107: 78425, (1986).
267. M. Wang, M. Xu, X. Mao, Y. Liu, CA, 108: 78424 (1986).
268. J. S. Forsythe, D. J. T. Hill, A. L. L. Logothetis, A. K. Whittaker, "Radiothon 96", Australian Institute of Nuclear Science and Engineering, Sydney, 1996.
269. J. Rickards, R. Trejo-Luna, E. Andrade, Radiat. Phys. Chem., **45**, 629 (1995).
270. Y. Luo, K. Yang, Y. Chui, CA, 122:82249 (1993).
271. K. Lunkwitz, H. J. Brink, D. Handle, A. Ferse, Radiat. Phys. Chem., **33**, 523 (1989).
272. C. Xinfang, Z. Chunshan, Radiat. Phys. Chem., **31**, 623 (1988).
273. S. Lora, F. Minto, M. Carenza, G. Palma, A. Faucitanu, Radiat. Phys. Chem., **31**, 629 (1988).
274. J. H. Collet, L. Y. Lim, P. L. Gould, Polym. Prep., **30**, 468 (1989).

SECTION III

PHYSICAL PROPERTIES OF MONOMERS AND SOLVENTS

Physical Properties of Monomers

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A. INTRODUCTION

This chapter contains some of the principal properties of the more common monomers. The monomers are arranged alphabetically, in groups, with derivatives listed alphabetically within the groups. Compounds are listed under their most commonly used names in polymer chemistry. Therefore, some names are not in agreement with the nomenclature rules of the International Union of Pure and Applied Chemistry.

Data are taken from a variety of published literature, including the 3rd edition of the "Polymer Handbook", "The Brandon Worldwide Monomer Reference Guide and Sourcebook", the "Aldrich Catalog Handbook of Fine Chemicals", and "Polysciences Monomers/Polymers, Scientific Polymer Products", "TCI America", "Lancaster and Janssen Chemica" catalogs. Where a range is reported in the literature, the average value is used here for melting points and boiling points. No attempt was made to verify the reported values experimentally.

Linear structures are given for most monomers. Molecular formulas are provided when linear structures cannot be unambiguously drawn.

Molecular weight is calculated based on the International Atomic Weight values.
Boiling point is given in degrees Celsius (°C), at 760 mmHg, unless indicated otherwise by the value in mmHg following a slash (/).
Melting point is given in degrees Celsius (°C), at 760 mmHg, unless indicated otherwise by the value in mmHg following a slash (/).
Refractive index is given at 20°C, unless indicated otherwise by the temperature in Celsius following a slash (/).
Density is relative to water at 4°C and measured at 20°C, unless otherwise indicated by the value following a slash (/).
Flash point was determined by the closed-cup method, using a Setaflash apparatus.

RTECS# is the reference to the Registry of Toxic Effects of Chemical Substances.

R&S is the reference found in the "Sigma–Aldrich Library of Regulatory & Safety Data", available from Aldrich Chemical Company.

FT-NMR is the reference to spectra in the "Aldrich Library of ^{13}C and ^{1}H FT-NMR Spectra", available from the Aldrich Chemical Company.

FT-IR is the reference to spectra in the "Aldrich Library of FT-IR Spectra", available from the Aldrich Chemical Company.

Merck Index is the reference to "The Merck Index", Merck & Co., Inc., Whitehouse Station, NJ.

Beilstein is the reference to "Beilsteins Handbuch der Organischen Chemie", Springer–Verlag, New York, NY.

Fieser is the reference to L. F. Fieser and M. Fieser, "Reagents for Organic Synthesis", John Wiley, New York, NY.

Abbreviations: (d) decomposes; (s) sublimes; (aq) aqueous.

B. TABLES OF PHYSICAL PROPERTIES

TABLE 1. ACETYLENES

Name	Formula RC≡CR′ R	R′	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Acetylene	–H	–H	74-86-2	26.04	–82	–84(s)
–, Chloro	–Cl	–H		60.48	–32	–126
–, Dichloro-	–Cl	–Cl		94.93		–66
–, Diphenyl-	$-C_6H_5$	$-C_6H_5$	501-65-5	178.23	60	170/19
–, Phenyl-	$-C_6H_5$	–H		102.14	143	

TABLE 2. ACID DICHLORIDES

Name	Formula ClC(O)RC(O)Cl R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Adipoyl	$-(CH_2)_4-$	111-50-2	183.03	106/2	
Azelaoyl	$-(CH_2)_7-$	123-98-8	255.12	166/18	
Diethylmanoyl	$-C(CH_2CH_3)_2-$	54505-72-5	197.06	198	
Dodecanedioyl	$-(CH_2)_{10}-$	4834-98-4	267.20	140/0.5	
Glutaryl	$-(CH_2)_3-$	2873-74-7	169.01	217	
Isophthaloyl	$-C_6H_4-$	99-63-8	203.02	276	43
Malonyl	$-CH_2-$	1663-67-8	140.95	54/19	
Methylpimeloyl	$-(CH_2)_2CH(CH_3)(CH_2)_2-$	44987-62-4	197.06	118/10	
Oxalyl	–CO–CO–	79-37-8	126.93	61	–8
Phthaloyl	$-C_6H_4-$	88-95-9	203.02	270	12
Pimeloyl	$-(CH_2)_5-$	142-79-0	197.06	113/5	
Sebacoyl	$-(CH_2)_8-$	111-19-3	239.14	168/12	–2.5
Suberoyl	$-(CH_2)_6-$	10027-07-3	211.09	162/15	
Succinyl	$-(CH_2)_2-$	543-20-4	154.98	190	16
Terephthaloyl	$-C_6H_4-$	100-20-9	203.02	266	80
Thionyl	ClC(S)Cl	7719-09-7	118.97	79	–105

TABLE 3. ACROLEINS

Name	Formula HC(O)CR=CHR′ R	R′	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Acrolein	–H	–H	107-02-8	56.06	53	–87
–, 2-Chloro-	–Cl	–H		90.51	40/30	
–, 2-Methyl-	$-CH_3$	–H	78-85-3	70.09	68	–81
–, 3-Methyl-, *trans*-(Crotonaldehyde)	–H	$-CH_3$	123-73-9	70.09		104
–, 2-Chloro-3-methyl-	–Cl	$-CH_3$		104.54	148	

TABLE 4. ACRYLAMIDES/METHACRYLAMIDES

Name	R	R′	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
4.1. ACRYLAMIDES H_2C=CHC(O)NRR′						
Acrylamide	–H	–H	79-06-1	71.08	125/25	
–, Acetyl-, *N*-	$-C(O)CH_3$	–H	1432-45-7	113.12		129
–, Adamantyl-, *N*-	$-C_{10}H_5$	–H	19026-83-6	205.30		155
–, Allyl-, *N*-	$-CH_2CH=CH_2$	–H	2555-13-7	111.14	120/19	
–, Benzyl-, *N*-	$-CH_2C_6H_5$	–H	13304-62-6	161.20		70
–, Butoxymethyl-, *N*-	$-CH_2O(CH_2)_3CH_3$	–H	1852-16-0	157.21	125/0.3	–9
–, *tert*-Butyl-, *N*-	$-C(CH_3)_3$	–H	107-58-4	127.19		128
–, Cyclohexyl-, *N*-	$-C_6H_{11}$	–H	3066-72-6	153.23		113
–, Cystamine-bis-, *N,N′*-	$-CH_2CH_2SSCH_2CH_2-$	–H	60984-57-8	260.38		123
–, Diacetone-, *N*-	$-C(CH_3)_2CH_2C(O)CH_3$	–H	2873-97-4	169.23	93/0.1	55
–, Diallyl-, *N,N*-	$-CH_2CH=CH_2$	$-CH_2CH=CH_2$	3085-68-5	151.21	108/3	
–, Di-*n*-butyl-, *N,N*-	$-(CH_2)_3CH_3$	$-(CH_2)_3CH_3$	2274-13-7	183.29	97/1	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.000			AO9600000						1,11
0.990				1(2),2711I	1(3), 533A	1(2),946B	11,9428	5,656	1,335
0.929	1.548								

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.259	1.4700	>110		1(1), 867J		1(1),741A		2,653	
1.143	1.4680	>110		1(1),869C				2,709	
1.145	1.4590	74		1(1),867G	1(1),1211B	1(1),740B		2,687	
1.069	1.4680	>110		1(1),869E	1(1)1212C			2(4),2127	
1.324	1.4720	107		1(1),867I	1(1),1212A	1(1),740D		2,634	
		180	NT2625000	1(2),1981H		1(2),352A		9,834	
1.449	1.4650	47		1(1),867F	1(1),1211A	1(1),740A		2(1),252	12,291
1.217	1.4720	>110		1(1),867K	1(1),1212B	1(1),741B			
1.455	1.4290	None	KI2950000	1(1),867D		1(1),739C	11,6867	2,542	17,241
1.409	1.5684	>110		1(2),1981G		1(2),351D	11,7439	9,805	14,263
1.205	1.4690	>110		1(1),869A		1(1),741C		2,671	
1.121	1.4680	>110		1(1),869D		1(1),742B		2,719	
1.172	1.4680	>110		1(1),869B		1(1),741D		2,694	
1.407	1.4680	77	WN4900000	1(1),867H	1(1),1211C	1(1),740C	11,8844	2,613	
		180	WZ1797000	1(2),1981I		1(2),352B		9,844	
1.631	1.5190	None	XM5150000	1(2),3197F		1(2),1227B	11,9278		13,297

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.839	1.4025	−19	AS105000	1(1),515A	1(1),733B	1(1),470D	11,122	1,725	10,2
1.199	1.463								
0.847	1.4160	−16	OZ2625000	1(1),515B	1(1),733C	1(1),471A		1(3),2981	
0.846	1.4365	9	GP9625000	1(1),515G		1(1),472C	11,2599	1,728	
1.478									

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.122			AS3325000	1(1),877K	1(1),1221A	1(1),749B	11,123	2,400	
0.9790	1.4620	>110	AS3450000						
			AS3460000					4(4),664	
		>110	AS3475000	1(1),887I	1(1),1236A	1(1),750C	11,2943		

TABLE 4. *cont'd*

Name	R	R′	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
–, Diethyl-, *N,N*-	$-CH_2CH_3$	$-CH_2CH_3$	2675-94-7	127.19	93/19	
–, 1,2-Dihydroxyethylene-bis-,*N,N′*	$-CH(OH)CH(OH)-$	–H	868-63-3	200.19		156(d)
–, Diisopropyl-, *N,N*-	$-CH(CH_3)_2$	$-CH(CH_3)_2$	44975-46-4	155.24	54/0.4	
–, 4,4-Dimethoxybutyl-, *N*-	$-(CH_2)_3CH(OCH_3)_2$	–H	38486-53-2	187.24	135/0.7	18
–, Dimethyl-, *N,N*-	$-CH_3$	$-CH_3$	2680-03-7	99.13	82/21	
–, 3-(Dimethylamino)-propyl-, *N*-	$-(CH_2)_3N(CH_3)_2$	–H	3845-76-9	156.23	117/2	
–, 1,1-Dimethyl-2-sulfoethyl-, *N*-(AMPS)	$-C(CH_3)_2CH_2SO_3H$	–H	15214-89-8	207.25		195(d)
–, Diphenylmethyl-,*N*-	$-CH(C_6H_5)_2$	–H	10254-08-7	237.30	178	
–, Ethylenebis-,*N,N′*-	$-CH_2CH_2-$	–H	2956-58-3	168.20		140
–, Ethylhexyl-,*N*-	$-CH_2CH(CH_2CH_3)(CH_2)_3CH_3$	–H	91625-16-0	183.30	123/0.5	
–, 2-Glycolic acid, *N*-	$-CH(OH)C(O)OH$	–H	6737-24-2	163.13	95	
–, 2-Glycolic acid methyl ester, *N*-	$-CH(OH)C(O)OCH_3$	–H	77402-03-0	173.17		74
–, Hexamethylenebis-, -*N,N′*	$-(CH_2)_6-$	–H	7150-41-6	224.28		135
–, *tert*-Hexyl-, *N*-	$-CH_2CH_2C(CH_3)_3$	–H	51330-07-5	155.24		100
–, Hydroxymethyl-, *N*- (solution)	$-CH_2OH$	–H	924-42-5	101.11		
–, Isobornyl-, *N*-	$-C_{10}H_{17}$	–H	17159-04-5	207.31		135
–, Isobutoxymethyl-, *N*-	$-CH_2OCH_2CH(CH_3)_2$	–H	16669-59-3	157.21	108	
–, Isopropyl-, *N*-	$-CH(CH_3)_2$	–H	2210-25-5	113.16	90/2	61
–, Isopropylidenebis-, *N,N′*-	$-C(CH_3)_2-$	–H		182.22		210
–, 3-Methoxypropyl-, *N*-	$-(CH_2)_3OCH_3$	–H	107374-86-7	143.19	99/0.3	
–, Methyl-, *N*-	$-CH_3$	–H	1187-59-3	85.11	106/4	
–, Methylenebis-, *N,N′*-	$-CH_2-$	–H	110-26-9	154.17		>300
–, (1-Methoxy)						
–, (Methoxy)-acetoxymethyl-, *N*-	$-C(OCH_3)C(O)OCH_3$	–H	77402-03-0	173.17		73
–, Morpholinoethyl-, *N*-	$-CH_2CH_2NC_4H_8O$	–H	5117-12-4	141.17	158/50	>110
–, 1-Naphthyl-, *N*-	$-C_{10}H_7$	–H	22302-62-1	197.24		138
–, Octadecyl-, *N*-	$-(CH_2)_{17}CH_3$	–H	1506-54-3	323.56		75
–, Octamethylenebis-, *N,N′*-	$-(CH_2)_8-$	–H	2945-02-0	252.35		138
–, *tert*-Octyl-, *N*-	$-(CH_2)_4C(CH_3)_3$	–H	4223-03-4	183.29		58
–, Phthalamidomethyl-, *N*-	$-CH_2NC_8H_4O_2$	–H	80500-94-3	230.22		190
–, 2,2,2-Trichloro-1-hydroxyethyl-, *N*-	$-CH(OH)CCl_3$	–H		218.47		170
–, Tri(hydroxymethyl)-methyl-, *N*	$-C(CH_2OH)_3$	–H	13880-05-2	175.18		139
–, 1,1,3-Trimethylbutyl-, *N*-	$-C(CH_3)_2CH_2CH(CH_3)_2$	–H	25269-93-6	169.27	125/25	55
–, Trimethylenebis-, *N,N′*-	$-(CH_2)_3-$	–H	4887-13-2	182.22		112
–, Triphenylmethyl-, *N*-	$-C(C_6H_5)_3$	–H	102548-89-0	313.40		204
4.2. METHACRYLAMIDES $H_2C=C(CH_3)C(O)NRR'$						
Methacrylamide	–H	–H	79-39-0	85.11	215	109
–, Acetyl-, *N*-	$-C(O)CH_3$	–H	44810-87-9	127.14	76/1.2	138
–, Acetylphenyl-, *N*-	$-C_6H_4C(O)CH_3$	–H	58813-69-7	203.24		138
–, Allyl-, *N*-	$-CH_2CH=CH_2$	–H	2186-33-6	125.17	90/2.2	
–, 3-Aminopropyl-, *N*-, hydrochloride	$-(CH_2)_3NH_2\cdot HCl$	–H	72607-53-5	178.66		124
–, Benzyl-, *N*-	$-CH_2C_6H_5$	–H	3219-55-4	175.23		82
–, Bis(diethylaminoethyl)-, *N,N*-	$-CH_2CH_2N(CH_2CH_3)_2$	$-CH_2CH_2N-(CH_2CH_3)_2$	76392-05-7	183.28	122/0.4	
–, Butoxymethyl-, *N*-	$-CH_2O(CH_2)_3CH_3$	–H	5153-77-5 28384-61-4	171.24	105/0.3	
–, *tert*-Butyl-, *N*-	$-C(CH_3)_3$	–H	6554-73-0	142.20	163	60
–, 2-Cyanoethyl-, *N*-	$-CH_2CH_2CN$	–H	24854-94-2	138.17		46
–, 2-Cyanoethyl-, *N*-, methyl-, *N*	$-CH_2CH_2CN$	$-CH_3$		152.20	113/1	46

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
				1(1),885K	**1**(1),1233C				
0.962	1.4730	71	AU3230000	**1**(1),893I	**1**(1),1244C	**1**(3),784B		**4**(3),130	
0.949	1.4820	>110						**4**(4),1268	
		160		**1**(1),1065K	**1**(1),1438C				
				1(1),883K				**4**(3),526	
				1(1),915E	**1**(1),1276B	**1**(1),781A			
			AI8967330	**1**(1),913D	**1**(1),1272B				
1.074	1.4130	None	AS3600000	**1**(1),885J		**1**(1),750A		**2**(4),1472	
0.970	1.4610	79	AS3630000						
			AS3675000					**4**(4),517	
			AS3678000	**1**(1),883J	**1**(1),1230C	**1**(1),749D			
			AI8967330	**1**(1),913D	**1**(1),1272B				
1.122	1.5120	>110						**27**(4),278	
				1(1),883M	**1**(1),1231B				
1.10			UC6475000	**1**(1),877L	**1**(1),1221B	**1**(1),749C		**2**(2),399	
	1.4702								
0.96	1.46	122							
	1.4755								

TABLE 4. *cont'd*

Name	R	R′	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
–, Cyclohexyl-, *N*-	$-C_6H_{11}$	–H	2918-67-4	167.25		102
–, Diallyl-, *N,N*-	$-CH_2CH{=}CH_2$	$-CH_2CH{=}CH_2$	13169-99-8	197.32	50/1.1	
–, Dibenzyl-, *N,N*-	$-CH_2C_6H_5$	$-CH_2C_6H_5$	57625-30-6	265.36		41
–, Dibutyl-, *N,N*-	$-(CH_2)_3CH_3$	$-(CH_2)_3CH_3$	22240-86-4	197.32	81/0.8	
–, 3-Di(butyl)aminopropyl-, *N*-	$-(CH_2)_3N((CH_2)_3CH_3)_2$	–H	76392-03-5	253.41	125/0.13	
–, Diethyl-, *N,N*-	$-CH_2CH_3$	$-CH_2CH_3$	5441-99-6	141.21	82/7	
–, 3-(Diethylamino)propyl-, *N*-	$-(CH_2)_3N(CH_2CH_3)_2$	–H		197.30	134/10	
–, Diisopropyl-, *N,N*-	$-CH(CH_3)_2$	$-CH(CH_3)_2$	51745-62-1	169.27		47
–, 2,2-Dimethoxyethyl-, *N*-	$-CH_2CH(OCH_3)_2$	–H	95984-11-5	173.21		27
–, Dimethyl-, *N,N*-	$-CH_3$	$-CH_3$	6976-91-6	113.16	66/10	
–, 2-(Dimethylamino)ethyl-, *N*-	$-CH_2CH_2N(CH_3)_2$	–H	13081-44-2	156.34	87/1.8	
–, 3-(Dimethylamino)propyl-, *N*-	$-(CH_2)_3N(CH_3)_2$	–H	5205-93-6	170.26	134/2	
–, Dodecyl-, *N*-	$-(CH_2)_{11}CH_3$	–H	1191-39-5	253.43	145/0.08	
–, Ethylenebis-, *N,N′*-	$-CH_2CH_2-$	–H	6117-25-5	196.25		170
–, 2-Ethylhexyl-, *N*-	$-CH_2CH(CH_2CH_3)(CH_2)_3CH_3$	–H		197.32	110/0.3	
–, 2-Hydroxyethyl-, *N*-	$-CH_2CH_2OH$	–H	5238-56-2	129.16	147/1.2	
–, Hydroxymethyl-, *N*-	$-CH_2OH$	–H	923-02-4	115.13	*ca.* 100	– 37
–, 4-Hydroxyphenyl-, *N*-	$-C_6H_4OH$	–H	19243-95-9	177.20		154
–, Isobutoxymethyl-, *N*-	$-CH_2OCH(CH_3)_2$	–H	4548-27-0	171.24	127/5	
–, Isopropyl-, *N*-	$-CH(CH_3)_2$	–H	13749-61-6	127.19	112/115	90
–, Methoxymethyl-, *N*-	$-CH_2OCH_3$	–H	3644-12-0	129.16	80.03	
–, Methyl-, *N*-	$-CH_3$	–H	3887-02-3	99.13	88/3.5	
–, Methylenebis-, *N,N′*-	$-CH_2-$	–H	2359-15-1	182.22		164
–, Methyl-, *N*-, Phenyl-, *N*-	$-CH_3$	$-C_6H_5$	2918-73-2	238.31	88/2	50
–, Methyl-, *N*-, 2,2-dimethoxyethyl-, *N*-	$-CH_3$	$-CH_2CH(OCH_3)_2$	95984-81-9	187.24	72/2	
–, 3-(*N*-morpholino)propyl-, *N*-	$-(CH_2)_3NC_4H_4O$	–H	55937-58-1	212.30	150/0.08	
–, 2-(2-Oxo-1-imidazolidinyl)-ethyl-, *N*-	$-CH_2CH_2C_3H_5N_2O$	–H	3089-19-8	197.24	100	
–, Phenyl-, *N*-	$-C_6H_5$	–H	1611-83-2	161.20		84
–, Phenyl-, *N*-, Phenylethyl-, *N*-	$-C_6H_5$	$-CH_2CH_2C_6H_5$	76392-02-4	266.37		63
–, 3-(Trifluoromethyl)phenyl-, *N*-	$-C_6H_4CF_3$	–H	783-05-1	229.20		95
–, 3-(Trimethylammonium)-propyl-, chloride, *N*- (eq. solution)	$-(CH_2)_3N(CH_3)_3 \cdot HCl$	–H	51410-72-1	220.74		– 22.5

TABLE 5. ACRYLATES/METHACRYLATES

Name	R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
5.1. ACRYLATE, ACIDS/ESTERS $H_2C=CHC(O)OR$					
Acid	–H	79-10-7	72.06	139	13
Allyl ester	$-CH_2CH{=}CH_2$	999-55-3	112.13	123	
Anhydride	$-C(O)CH{=}CH_2$	2051-76-5	126.11	97/35	
Benzyl ester	$-CH_2C_6H_5$	2495-35-4	162.19	110/8	
4-Biphenylyl ester	$-C_6H_4C_6H_5$	13026-23-8	226.27		224
Bisphenol A ethoxylate diester	$(-(CH_2CH_2OC_6H_4)_2C(CH_3)_2$	64401-02-1	424		
Bisphenol A diglycidyl ether diester	$(-CH_2CH(OH)CH_2OC_6H_4)_2C(CH_3)_2$	4687-94-9	484.55		
2-Bromo-	$H_2C=CBrC(O)OH$	10443-65-9	150.97		63
3-Bromo-, *cis*-	$BrCH=CHC(O)OH$	1609-92-3	150.96		63
2-Bromo-, ethyl ester	$H_2C=CBrC(O)OCH_2CH_3$	4519-46-4	164.99	72/78	
2-Bromoethyl ester	$-CH_2CH_2Br$	4823-47-6	179.03	52/5	
2-Bromomethyl-	$H_2C=C(CH_2Br)C(O)OH$	72707-66-5	164.99		70
2-Bromomethyl-, ethyl ester	$H_2C=C(CH_2Br)C(O)OCH_2CH_3$	17435-72-2	193.05	86/20	
2-Bromomethyl-, methyl ester	$H_2C=C(CH_2Br)C(O)OCH_3$	4224-69-5	179.02	36/1.3	
1,3-Butylene diester	$-CH_2CH_2CH(CH_3)-$	19485-03-1	198.22	73/0.1	
1,4-Butylene diester	$-(CH_2)_4-$	1070-70-8	198.22	83/0.3	
2-Butylene-1,4 diester	$-CH_2CH{=}CHCH_2-$		127.16	60/23	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
	1.4731								
	1.4600								
	1.4744								
0.940	1.4790	140							
	1.5002								
1.100	1.4300	>110	UC6380000						
0.97	1.46	107							
			SI1225000					**4**(3),276	
	1.4707								
0.97	1.4740								
1.110	1.4560	>110							
1.053	1.427								

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.051	1.4202	54	AS4375000	**1**(1),547A	**1**(1),776A	**1**(1),498B	**11**,124	**2**,397	
0.880	1.4320	8					**2**,400		
	1.4487								
1.0573	1.5143	101							
1.146	1.5450	>110							
1.180	1.5570	>110							
				1(1),563H				**2**,402	
				1(1),563I				**2**(4),1486	
1.4774	1.4770								
				1(1),563K	**1**(1),806C	**1**(1),516C			
1.388	1.478								
1.489	1.490	78	AS4900000	**1**(1),745J	**1**(1),1031C	**1**(3),696D			
1.030	1.4500	93	AS5250000						
1.051	1.4560	>110	UD3130000					**2**(4),170	
0.9344	1.4422								

TABLE 5. *cont'd*

Name	R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
2-(2-Butoxyethoxy)ethyl ester	$-CH_2CH_2OCH_2CH_2O(CH_2)_3CH_3$		232.28	103/23	
2-Butoxyethyl ester	$-CH_2CH_2O(CH_2)_3CH_3$	7251-90-3	172.22	63/2	
n-Butyl ester	$-(CH_2)_3CH_3$	141-32-2	128.17	145	−64
s-Butyl ester	$-CH(CH_3)CH_2CH_3$	2998-08-5	128.17	59/25	
t-Butyl ester	$-C(CH_3)_3$	1663-39-4	128.17	30/25	
2-Chloro-	$H_2C{=}CClC(O)OH$	26952-44-3	106.51	176(d)	60
2-Chloro-, butyl ester	$H_2C{=}CCl(O)O(CH_2)_3CH_3$	13401-85-9	162.62	100/42	
2-Chloro-, ethyl ester	$H_2C{=}CClC(O)OCH_2CH_3$	687-46-7	134.56	41/20	
2-Chloro-, methyl ester	$H_2C{=}CClC(O)OCH_3$	80-63-7	120.54	52/50	
3-Chloro-, *cis*-	$ClCH{=}CHC(O)OH$	1609-93-4	106.51		61
3-Chloro-, *trans*-	$ClCH{=}CHC(O)OH$	2345-61-1	106.51		84
2-Chloroethyl ester	$-CH_2CH_2Cl$	2206-89-5	134.57	64/20	
Cinnamyl ester	$-CH_2CH{=}CHC_6H_5$		188.23	105/2	
Crotyl ester	$-CH_2CH{=}CHCH_3$	23916-33-8	126.16	38/10	
2-Cyano-, butyl ester	$H_2C{=}C(CN)C(O)O(CH_2)_3CH_3$	6606-65-1	153.18	92/6	
2-Cyano-, ethyl ester	$H_2C{=}C(CN)C(O)OCH_2CH_3$	7085-85-0	125.13	90/5	−17
2-Cyano-, isobutyl ester	$H_2C{=}C(CN)C(O)OCH_2CH(CH_3)_2$	1069-55-2	153.18	81/4	
2-Cyanoethyl ester	$-CH_2CH_2CN$	106-71-8	125.13	103/8	−17
Cyclohexyl ester	$-C_6H_{11}$	3066-71-5	154.21	183	
Cyclopentyl ester	$-C_5H_9$	16868-13-6	282.38	145/0.08	
n-Decyl ester	$-(CH_2)_9CH_3$	2156-96-9	212.34	185/50	
2,3-Dibromopropyl ester	$-CH_2CHBrCH_2Br$	19660-16-3	271.94	105/6	
2,3-Dichloropropyl ester	$-CH_2CHClCH_2Cl$	24910-84-7	183.04	58/0.5	
Dicyclopentenyl ester	$-C_{10}H_{13}$	33791-58-1	202.25	105/4	
Dicyclopentenyloxyethyl ester	$-CH_2CH_2OC_{10}H_{13}$	65983-31-5	248.32	100/0.5	
2-(Diethylamino)ethyl ester	$-CH_2CH_2N(CH_2CH_3)_2$	2426-54-2	171.27	171.27	< −60
3-(Diethylamino)propyl ester	$-(CH_2)_3N(CH_2CH_3)_2$		185.27	44/0.1	
Di(ethylene glycol) diester	$-CH_2CH_2OCH_2CH_2-$	4074-88-8	214.22	162	
Dihydrodicyclopentadienyl ester	$-C_{10}H_{15}$	12542-30-2	204.27	119/5	−40
2,3-Dihydroxypropyl ester	$-CH_2CH(OH)CH_2OH$	10095-20-2	146.15	115/0.1	
2-(Dimethylamino) ethyl ester	$-CH_2CH_2N(CH_3)_2$	2439-35-2	143.19	64/12	< −60
3-(Dimethylamino) neopentyl ester	$-CH_2C(CH_3)_2CH_2N(CH_3)_2$	20166-73-8	181.24	53/1.4	
3-(Dimethylamino) propyl ester	$-(CH_2)_3N(CH_3)_2$	18526-07-3	157.22	167	
Dipentaerythritol pentaester	$(-CH_2)_3CCH_2OCH_2C(CH_2OH)(CH_2-)_2$	60506-81-2	524.53		
Di(propylene glycol) diester	$-CH_2CH(CH_3)OCH_2CH(CH_3)-$	85136-76-1	242.27	95/0.5	
Di(trimethylolpropane) tetraester	$((-CH_2)_2C(CH_2CH_3)CH_2)_2O$	94108-97-1	466.53		
Dodecyl ester	$-(CH_2)_{11}CH_3$	2156-97-0	240.39	120/1	
1H,1H,11H-Eicosafluoro-undecylester	$-CH_2(CF_2)_9CHF_2$	4998-38-3	586.17	115/1	49
2-(2-Ethoxyethoxy)ethyl ester	$-(CH_2CH_2O)_2CH_2CH_3$	7328-17-8	188.23	250	
2-Ethoxyethyl ester	$-CH_2CH_2OCH_2CH_3$	106-74-1	144.17	65/16	
Ethyl ester	$-CH_2CH_3$	140-88-5	100.12	99	−72
Ethylene diester	$-CH_2CH_2-$	2274-11-5	170.16	67/2	
2-Ethylhexyl ester	$-CH_2CH(CH_2CH_3)(CH_2)_3CH_3$	103-11-7	184.28	217	−90
Furfuryl ester	$-CH_2C_4H_3O$	10525-17-4	152.15	92/16	
Glycidyl ester	$-CH_2CH(-O-)CH_2$	106-90-1	128.13	115/78	
Glycerol propoxylate triester	$-CH(CH_3)CH_2OCH(CH_2OCH_2CH(CH_3)-)_2$	52408-84-1	428		
1H,1H,2H,2H-Heptadecafluoro-decyl ester	$-CH_2CH_2(CF_2)_7CF_3$	27905-45-9	518.10	100/4	
1H,1H-Heptafluorobutyl ester	$-CH_2CF_2CF_2CF_3$	424-64-6	254.11	121/743	
Heptyl ester	$-(CH_2)_6CH_3$	2499-58-3	170.25	56/1	
Hexadecyl ester	$-(CH_2)_{15}CH_3$	13402-02-3	296.50	170/15	
2,2,3,4,4,4-Hexafluorobutyl ester	$-CH_2CF_2CHFCF_3$	54052-90-3	236.11	48/12	
1H-Hexafluoroisoporpyl ester	$-CH(CF_3)_2$	2160-89-6	222.09	84	
Hexanediol diester	$-(CH_2)_6-$	13048-33-4	226.28	295	
n-Hexyl ester	$-(CH_2)_5CH_3$	2499-95-8	156.23	190	−45
4-Hydroxybutyl ester	$-(CH_2)_4OH$	2478-10-6	144.17	95/0.1	
2-Hydroxyethyl ester	$-CH_2CH_2OH$	818-61-1	116.11	90/12	−101
2-Hydroxy-3-phenoxypropyl ester	$-CH_2CH(OH)CH_2OC_6H_5$	16969-10-1	222.24		
2-Hydroxypropyl ester	$-CH_2CH(OH)CH_3$	999-61-1	130.15	77/15	−92
Isobornyl ester	$-C_{10}H_{18}$	5888-33-5	208.30	275	−15
Isobutyl ester	$-CH_2CH(CH_3)_2$	106-63-8	128.19	132	−61

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.9821	1.4394								
0.9497	1.4323	90							
0.894	1.4180	39	UD3150000	**1**(1),719E	**1**(1),974A	**1**(1),642A	**11**,1539	**2**(2),388	
0.8114	1.4140								
0.8850	1.4108	17		**1**(1),719F	**1**(1),974B	**1**(3),674D		**2**(3),1228	
1.1404	1.4384								
1.189	1.4420	33							
				1(1),563F	**1**(1),805C	**1**(1),516A		**2**,400	
			AS5960000	**1**(1),563G	**1**(1),806A	**1**(1),516B		**2**,400	
1.1404	1.4384								
1.0495	1.5660								
0.986	1.4420	99							
1.069		108							
1.0690	1.4433	124	AT1500000					**3**(3),543	
0.975	1.4673	29	AS7350000						
0.8781	1.440	227							
1.7803	1.5520								
1.2603	1.4765								
1.085	1.5010	99							
0.9250	1.443	68	AS8225000						
0.9180	1.441								
1.118	1.4630	> 110	AS9450000						
1.08	1.509								
0.943	1.4380	58	AS8578000	**1**(1),771G	**1**(1),1073C	**1**(3),734A		**4**(3),649	
0.90	1.439	75							
0.928	1.4400	68							
1.1915		115							
	1.4488								
1.101	1.4790	> 110							
0.884	1.4450	> 110						**2**(3),1230	
1.016	1.4390	102						**2**(3),1232	
0.9819	1.4282	65	AS9800000					**2**(3),1232	
0.924	1.4060	15	AT0700000	**1**(1),719D	**1**(1),973C	**1**(1),638C	**11**,3715	**2**,399	**6**,251
1.090	1.4610	100	AT0350000	**1**(1),727M		**1**(3),675B		**2**(4),1496	
0.8859	1.4360	79	AT0855000	**1**(1),719G	**1**(1),974C	**1**(3),675A		**2**(3),1229	
1.1125	1.4800								
1.099	1.4490	76	AS9275000		**1**(1),1088C			**17**(3),1005	
1.064	1.4610								
	1.3380								
1.418	1.3301	31						**2**(4),1464	
0.8846	1.4311								
0.8620	1.4470								
1.398	1.352	59							
1.330	1.3190	10		**1**(1),745C	**1**(1),1029B				
1.010	1.4562	> 110	AT1430000	**1**(1),729E		**1**(1),639B			
0.8882	1.4280	68	AT1450000					**2**(3),1228	
1.039	1.4520	> 110		**1**(1),753L	**1**(1),1044C	**1**(3),720C		**2**(4),1470	
1.011	1.4502	98	AT1750000	**1**(1),753J	**1**(1),1044A	**1**(3),720B		**2**(4),1496	
1.160	1.5280	89							
1.05	1.4448	65							
0.986	1.4760	93		**1**(1),719H					
0.8896	1.4140	33	AT2100000					**2**(3),1227	

TABLE 5. *cont'd*

Name	R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Isodecyl ester	$-(CH_2)_7CH(CH_3)_2$	1330-61-6	212.34	121/10	−100
Isooctyl ester	$-(CH_2)_5CH(CH_3)_2$	29590-42-9	184.25	125/20	
Isopropoxyethyl ester	$-CH_2CH_2OCH(CH_3)_2$		159.21	82/19	
Isopropyl ester	$-CH(CH_3)_2$	689-12-3	114.14	51/103	
Methallyl ester	$-CH_2C(CH_3)=CH_2$	818-67-7	126.16	70/50	
2-(2-Methoxyethoxy) ethyl ester	$-CH_2CH_2OCH_2CH_2OCH_3$	7328-18-9	174.20	86/4.6	
2-Methoxyethyl ester	$-CH_2CH_2OCH_3$	3121-61-7	130.14	56/12	
Methyl ester	$-CH_3$	96-33-3	86.09	80	−75
2-Methylbutyl ester	$-CH_2CH(CH_3)CH_2CH_3$		142.20	63/27	
2-(*N*-Morpholino)ethyl ester	$-CH_2CH_2NC_4H_8O$	19727-38-9	185.2	67/0.2	
1-Naphthyl ester	$-C_{10}H_7$	20069-66-3	198.22	212/1	
2-Naphthyl ester	$-C_{10}H_7$	52684-34-1	198.22	138/0.4	
Neopentyl ester	$-CH_2C(CH_3)_3$	4513-36-4	142.24	31/3	
Neopentyl glycol diester	$-CH_2C(CH_3)_2CH_2-$	2223-82-7	212.25	105/5	
Nonyl ester	$-(CH_2)_8CH_3$		198.31	76/0.2	
Octadecyl ester	$-(CH_2)_{17}CH_3$	4813-57-4	324.55	160/3	32
1H,1H,5H-Octafluoropentyl ester	$-CH_2(CF_2)_3CF_2H$	376-84-1	286.12	88/41	
n-Octyl ester	$-(CH_2)_7CH_3$	2499-59-4	184.28	57/0.05	
1H,1H-Pentadecafluorooctyl ester	$-CH_2(CF_2)_6CF_3$	307-98-2	454.13	65/5	
Pentaerythritol tetraester	$(-CH_2)_4C$	4986-89-4	352.34		18
Penterythritol triester	$(-CH_2)_3CCH_2OH$	3524-68-3	298.30	>315	15
Pentaerythritol stearate diester	$(-CH_2)_2C(CH_2OH)CH_2OC(O)(CH_2)_{16}CH_3$	92092-01-8	510.72		30
2,2,3,3,3-Pentafluoropropyl ester	$-CH_2CF_2CF_3$	356-86-5	204.10	50/100	
1,5-Pentanediol diester	$-(CH_2)_5-$	36840-85-4	212.25	94/0.3	
n-Pentyl ester	$-(CH_2)_4CH_3$	2998-23-4	142.20	47/7	
2-Phenoxyethyl ester	$-CH_2CH_2OC_6H_5$	48145-04-6	192.21	103/0.6	
Phenyl ester	$-C_6H_5$	937-41-7	148.16	87/12	
1,4-Phenylene diester	$-C_6H_4-$	6729-79-9	218.21		88
1,4-Phenylene di(acrylic acid)	$C_6H_4(CH=CHC(O)OH)_2$	16323-43-6	218.21		>300
2-Phenylethyl ester	$-CH_2CH_2C_6H_5$	3530-36-7	176.22	104/5	
Trimethyl 2-phosphonoacrylate	$(CH_3O)_2P(O)C(C(O)OCH_3)=CH_2$	55168-74-6	194.13	91/0.1	
Propargyl ester	$-CH_2C\equiv CH$	10477-47-1	110.11		138
n-Propyl ester	$-CH_2CH_2CH_3$	925-60-0	114.14	43/40	
1,2-Propylene glycol diester	$-CH_2CH(CH_3)-$		184.19	63/0.3	
1,3-Propylene glycol diester	$-(CH_2)_3-$	25151-33-1	184.19	122/0.5	
Tetradecyl ester	$-(CH_2)_{13}CH_3$	21643-42-5	268.44	138/0.4	
Tetra(ethylene glycol) diester	$-(CH_2CH_2O)_3CH_2CH_2-$	17831-71-9	302.33	>120/0.3	
2,2,3,3-Tetrafluoropropyl ester	$-CH_2CF_2CF_2H$	7383-71-3	186.11	132	
Tetrahydrofurfuryl ester	$-C_5H_9O$	2399-48-6	156.18	87/9	<−60
S,S'-Thiodi-1,4-phenylene dithiol diester	$-SC_6H_4SSC_6H_4S-$	129283-82-5	386.56		64
2,3,3-Trichloro-	$ClCH_2=CClC(O)OH$	2257-35-4	175.40		72
Tridecyl ester	$-(CH_2)_{12}CH_3$	3076-04-8	254.41	150/10	
Tri(ethylene glycol) diester	$-(CH_2CH_2O)_2CH_2CH_2-$	1680-21-3	258.30	125/0.2	
2,2,2-Trifluoroethyl ester	$-CH_2CF_3$	407-47-6	154.09	92	
1,1,1-Tri(2-hydroxyethoxy-methyl)propane triester	$(-CH_2CH_2OCH_2)_3CCH_2CH_3$	28961-43-5	428.48	157	
Tri(2-hydroxyethyl) isocyanurate triester	$(-CH_2CH_2)_3C_3N_3O_3$	40220-08-4	423.38		97
3,5,5-Trimethylcyclohexyl ester	$-C_6H_8(CH_3)_3$	86178-38-3	196.29	40/0.1	
3,5,5-Trimethylhexyl ester	$-(CH_2)_2CH(CH_3)CH_2C(CH_3)_3$	2664-55-3	198.31	58/0.8	−34
Trimethylolpropane triester	$(-CH_2)_3CCH_2CH_3$	15625-89-5	296.32	316	
Trimethylolpropane ethoxylate triester	$((-CH_2CH_2O)_xCH_2)_3CCH_2CH_3$	28961-43-5	ca. 428	157	
Tri(propylene glycol) diester	$(-CH_2(CH_3)CH_2O)_2CH_2CH(CH_3)-$	42978-66-5	300	>120/1	<−20
Vinyl ester	$-CH=CH_2$	2177-18-6	98.10	91	

5.2. METHACRYLATE, ACIDS/ESTERS ($H_2C=C(CH_3)C(O)OR$)

Acid	$-H$	79-41-4	86.09	163	16
2-(Acetoacetoxy)ethyl ester	$-CH_2CH_2C(O)OCH_2C(O)CH_3$	21282-97-3	214.22	274	
Allyl ester	$-CH_2CH=CH_2$	96-05-9	126.16	60/43	
Anhydride	$-C(O)C(CH_3)=CH_2$	760-93-0	154.17	200	−20
2-(1-Aziridinyl)ethyl ester	$-CH_2CH(-NH-)CH_2$	6498-81-3	155.20	190	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.875	1.4420	106	AT2190000						
0.880	1.4370	80	UD3391000						
0.9549	1.4258								
0.8932	1.4060								
0.9285	1.4372								
1.0421	1.4392								
1.012	1.4272	60	KL6000000					**2**(3),1232	
0.956	1.4020	7	AT2800000	**1**(1),719C	**1**(1),973B	**1**(1),638B	**11**,5935	**3**,638B	**17**,183
0.8883	1.4800								
1.0711	1.4728								
0.859		> 37							
1.030	1.4530	123	AS8925000						
0.875	1.4375								
0.800		190						**2**(4),1468	
	1.3467	70							
0.8810	1.4350								
1.63	1.3279								
1.190	1.4870	170							
1.180	1.4840	> 110	UD3370000	**1**(1),753I		**1**(1),639C			
1.018									
1.32	1.3363								
	1.4551								
0.8920	1.4240								
1.104	1.5180	> 110	KM0700000					**6**(3),572	
				1(2),1775F	**1**(2),1050A	**1**(2),179B		**9**,914	
1.249	1.4540	105		**1**(1),1109B	**1**(1),1489B			**4**(4),3638	
0.9078	1.4130								
	1.4470								
	1.4529								
0.8700	1.4468								
1.110	1.4638	> 110	AS8100000	**1**(1),759E	**1**(1),1053B	**1**(1),639A			
1.317	1.3629	45		**1**(1),745A				**17**(3),1104	
1.063	1.4580	> 110						**17**(3),1104	
1.099	1.4609	63							
1.216	1.3506	12		**1**(1),743M	**1**(1),1028C	**1**(3),696C		**2**(4),1462	
1.110	1.4710	> 110	KM2890000		**1**(1),999C				
0.9292	1.455	88							
0.875	1.4370	89	UD3643625					**2**(3),1230	
1.100	1.4736	> 110	AT4810000	**1**(1),729F		**1**(1),640A			
1.110	1.4720	86	KM2890000		**1**(1),999C				
1.030	1.4500		AT4690000	**1**(1),729A		**1**(1),638D			
0.942	1.4320	2							
1.015	1.432	76	OZ2975000	**1**(1),547B	**1**(1),776B	**1**(1),499C	**11**,5849	**2**,421	
1.122	1.4560	106		**1**(1),791E	**1**(1),1107C				
0.934	1.4360	34	UD3483000	**1**(1),721E	**1**(1),977A	**1**(1),641B		**2**(3),1290	
1.035	1.454	83	OZ5700000	**1**(1),831C	**1**(1),1170C			**2**(3),1293	
								6(3),1481	

TABLE 5. *cont'd*

Name	R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Benzyl ester	$-CH_2C_6H_5$	2495-37-6	176.22	232	
Bisphenol A diester	$-C_6H_4C(CH_3)_2C_6H_4-$	3253-39-2	364.44		73
Bisphenol A tetraethoxylate diester	$((-CH_2CH_2O)_2C_6H_4)_2C(CH_3)_2$	41637-38-1	452.55	ca. 540	
2-Bromoethyl ester	$-CH_2CH_2Br$	4513-56-8	193.05	47/2.7	
1,3-Butylene diester	$-CH_2CH_2CH(CH_3)-$	1189-08-8	226.28	290	
1,4-Butylene diester	$-(CH_2)_4-$	2082-81-7	226.28	133/4	
2-Butoxyethyl ester	$-CH_2CH_2O(CH_2)_3CH_3$	13532-94-0	186.25	90/3	
n-Butyl ester	$-(CH_2)_3CH_3$	97-88-1	142.20	161	
s-Butyl ester	$-CH(CH_3)CH_2CH_3$	2998-18-7	142.20	146	
tert-Butyl ester	$-C(CH_3)_3$	585-07-9	142.20	132	
N-tert-Butyl-2-aminoethyl ester	$-CH_2CH_2NHC(CH_3)_3$	3775-90-4	185.27	100/12	
2-Chloro-2-hydroxypropyl ester	$-CH_2C(OH)ClCH_3$	13159-52-9	178.62	95/2	
2-Chloroethyl ester	$-CH_2CH_2Cl$	1888-94-4	148.60	59/7.5	
Chloromethyl ester	$-CH_2Cl$	27550-73-8	146.57	54/20	
Cinnamyl ester	$-CH_2CH=CHC_6H_5$	31736-34-2	202.25	141/3	
Chloride	$CH_2=C(CH_3)C(O)Cl$	920-46-7	104.54	95	
2-Cyanoethyl ester	$-CH_2CH_2CN$	4513-53-5	139.16	99/2	
1,4-Cyclohexanediol diester	$-C_6H_{10}-$	38479-34-4	252.31		50
Cyclohexyl ester	$-C_6H_{11}$	101-43-9	168.24	69/4	
Decanediol diester	$-(CH_2)_{10}-$	6701-13-9	310.44	170/2	
Decyl ester	$-(CH_2)_9CH_3$	3179-47-3	226.36	327	−44
2,3-Dibromopropyl ester	$-CH_2CHBrCH_2Br$	3066-70-4	285.96	72/0.03	
2-(Dibutylamino)ethyl ester	$-(CH_2)_2N(CH_2CH_2CH_2CH_3)_2$	2397-75-3	241.38	100/1	
Dicyclopentenyl ester	$-C_{10}H_{13}$	31621-69-9	218.30	137/13	
Dicyclopentenyloxyethyl ester	$-C_{12}H_{17}O$	68586-19-6	262.35	150/10	
2-(Diethylamino) ethyl ester	$-(CH_2)_2N(CH_2CH_3)_2$	105-16-8	185.28	80/10	< −60
3-(Dimethylamino) propyl ester	$-(CH_2)_3N(CH_3)_2$	17577-32-1	199.30	105/1.5	
Di(ethylene glycol) diester	$-CH_2CH_2OCH_2CH_2-$	2358-84-1	242.27	134/2	
3,4-Dihydroxybutyl ester	$-CH_2CH_2CH(OH)CH_2OH$	62180-57-8	174.20	110/0.25	
2,3-Dihydroxypropyl ester	$-CH_2CH(OH)CH_2OH$	5919-74-4	160.17	140/0.6	
2-(Dimethylamino) ethyl ester	$-CH_2CH_2N(CH_3)_2$	2867-47-2	157.22	187	−30
Diurethane diester (isomers)	$C_{23}H_{38}N_2O_8$	72869-86-4	470.57	200	
1H,1H,7H-Dodecafluoroheptyl ester	$-CH_2(CF_2)_5CF_2H$	2261-99-6	400.17	107/23	
Dodecanediol diester	$-(CH_2)_{12}-$	72829-09-5	338.49		0.95
Dodecyl ester	$-(CH_2)_{11}CH_3$	142-90-5	252.42	142/4	−7
2,3-Epithiopropyl ester	$-CH_2CH(-S-)CH_2$	3139-91-1	158.22	59/1.4	
2,3-Epoxybutyl ester	$-CH_2CH(-O-)CHCH_3$	68212-07-7	156.18	45/0.25	
3,4-Epoxybutyl ester	$-CH_2CH_2CH(-O-)CH_2$	55750-22-6	156.18	55/0.8	
2,3-Epoxyopropyl ester	$-CH_2CH(-O-)CH_2$	106-91-2	142.16	189	
4-Ethoxybutyl ester	$-(CH_2)_4OCH_2CH_3$	76392-24-0	186.25	85/23	
2-Ethoxyethyl ester	$-CH_2CH_2OCH_2CH_3$	2370-63-0	158.20	92/35	
Ethyl ester	$-CH_2CH_3$	97-63-2	114.15	118	< −75
Ethyl 2-bromomethyl- ester	$CH_2=C(CH_2Br)C(O)OCH_2CH_3$	17435-72-2	193.05	38/0.8	
2-Ethylbutyl ester	$-CH_2CH(CH_2CH_3)CH_2CH_3$	5138-86-3	170.25	74/16	
1,2-Ethylene diester	$-CH_2CH_2-$	97-90-5	198.22	240	−40
2-Ethylhexyl ester	$-CH_2CH(CH_2CH_3)(CH_2)_3CH_3$	688-84-6	198.31	218	
2-(Ethylthio)ethyl ester	$-CH_2CH_2SCH_2CH_3$	14216-25-2	174.26	102/15	
Ethyl 2-(trimethoxysilylmethyl-) ester	$CH_2=C(CH_2Si(OCH_3)_3)C(O)OCH_2CH_3$	74976-84-4	186.33	71/10	
Furfuryl ester	$-CH_2C_4H_3O$	3454-28-2	166.18	81/5	
Glycerol diester	$-CH_2CH(OH)CH_2-$	1830-78-0	228.25	120/1	<−18
Glycerol triester	$-CH(CH_2-)_2$	52408-84-1	296.3	150/0.1	
Glycidyl ester	$-CH(-O-)CH_2$	106-91-2	142.16	189	
1H,1H,2H,2H-Heptadeca-fluorodecyl ester	$-(CH_2)_2(CF_2)_7CF_3$	1996-88-9	532.20	120/4	
1H,1H-Heptafluorobutyl ester	$-CH_2(CF_2)_2CF_3$	13695-31-3	268.13	134	
Heptyl ester	$-(CH_2)_6CH_3$	5459-37-0	184.28	104/14	
1,6-Hexanediol diester	$-(CH_2)_6-$	6606-59-3	254.33	>315	
2,2,3,4,4,4-Hexafluorobutyl ester	$-CH_2CF_2CHFCF_3$	36405-47-7	250.14	158	
1H-Hexafluoroisopropyl ester	$-CH(CF_3)_2$	3063-94-3	236.11	99	
Hexyl ester	$-(CH_2)_5CH_3$	142-09-6	170.25	88/14	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.040	1.5120	107							
				1(2),1931F	**1**(2),1296C	**1**(2),316B			
1.120	1.5320	>110							
1.010	1.4520	130							
1.023	1.4565	113		**1**(1),729D	**1**(1),999A	**1**(1),639D		**2**(4),1534	
0.939	1.4335	93						**2**(3),1292	
0.894	1.4230	50	OZ3675000	**1**(1),721A	**1**(1),975C	**1**(1),641A		**2**(3),1286	
	1.4195								
0.875	1.4150	50	OZ3675500					**2**(4),1582	
0.914	1.4420	71	OZ3500000					**4**(4),1509	
1.190	1.4750	150						**2**(4),1533	
	1.4434								
1.070	1.4420	3	OZ5791000	**1**(1),857B		**1**(1),732C		**2**(3),1293	
	1.4459								
		82							
0.9626	1.459	82						**6**(3),25	
	1.4577								
0.875	1.443	101							
	1.4474								
1.050	1.4990	>110						**6**(3),1942	
1.064	1.4970	>110							
0.922	1.4442	85						**4**(3),676	
1.082	1.4580	66						**2**(3),1292	
0.933	1.4400	70	OZ4200000	**1**(1),771H	**1**(1),1074A	**1**(1),675C		**4**(3),649	
1.110	1.485	>150							
1.46									
0.868	1.4450	>110	OZ4300000	**1**(1),721D	**1**(1),976C	**1**(3),678B		**2**(3),1290	
	1.4422								
1.038	1.4472								
1.042	1.4490	76	OZ4375000	**1**(1),781F	**1**(1),1089A	**1**(1),668D			
	1.4223								
0.964	1.4290	71		**1**(1),759C	**1**(1),1052C	**1**(3),723C		**2**(3),1291	
0.917	1.4130	15	OZ4550000	**1**(1),719J	**1**(1),975B	**1**(1),640D		**2**,423	
1.398	1.4790	87							
1.051	1.4540	86	OZ4400000	**1**(1),729B	**1**(1),998B	**1**(1),642D		**2**(3),1292	
0.885	1.4380	92	OZ4630000	**1**(1),721C	**1**(1),976B			**2**(3),1289	
0.897	1.4380	59		**1**(2),2985A	**1**(3),686B				
1.078	1.4820	90						**17**(3),1248	
1.120	1.4720	>115							
1.042	1.4490	76	OZ4375000	**1**(1),781F	**1**(1),1089A	**1**(1),668D			
1.345	1.3410	37							
0.995	1.4580	150							
1.348	1.3610	57		**1**(1),745F	**1**(1),1030B				
1.302	1.3310	14		**1**(1),745E	**1**(1),1030A				
0.885	1.432	63							

TABLE 5. *cont'd*

Name	R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
4-Hydroxybutyl ester	$-(CH_2)_4OH$	997-46-6	158.20	105/0.01	
2-Hydroxyethyl ester	$-CH_2CH_2OH$	868-77-9	130.14	250	−12
3-(5-Hydroxypentyloxy)-3-oxopropyl ester	$-CH_2CH_2C(O)O(CH_2)_5OH$	85099-10-1	244.29		−48
3-Hydroxypropyl ester	$-(CH_2)_3OH$	276-09-3	144.17	90/9	
Isobornyl ester	$-C_{10}H_8$	7534-94-3	222.33	245	−51
Isobutyl ester	$-CH_2CH(CH_3)_2$	97-86-9	142.20	155	
2-Isocyanatoethyl ester	$-CH_2CH_2NCO$	30674-80-7	155.17	211	−45
Isodecyl ester	$-(CH_2)_7CH(CH_3)_2$	29964-84-9	226.36	126/10	−41
Isopropyl ester	$-CH(CH_3)_2$	4655-34-9	128.17	125	
Methallyl ester	$-CH_2C(CH_3)=CH_2$	816-74-0	140.18	157	
2-(2-Methoxyethoxy) ethyl ester	$-(CH_2CH_2O)_2CH_3$	45103-58-0	188.22	67/1	
2-Methoxyethyl ester	$-CH_2CH_2OCH_3$	6976-93-8	144.17	66/16	
Methyl ester	$-CH_3$	80-62-6	100.12	100	−48
2-Methyl-2-nitropropyl ester	$-CH_2C(CH_3)(NO_2)CH_3$	2177-42-6	187.20	102/4	
2-(Methylthio) ethyl ester	$-CH_2CH_2SCH_3$	14216-23-0	160.24	38/0.06	
Methyl 2-bromomethyl ester	$CH_2=C(CH_2Br)C(O)OCH_3$	4224-69-5	179.02	36/1.3	
Methyl 2-(1-hydroxyethyl-)ester	$CH_2=C(CH(OH)CH_3)C(O)OCH_3$	18020-65-0	130.14	91/20	
2-*N*-Morpholinoethyl ester	$-CH_2CH_2NC_4H_8O$	2997-88-8	199.25	80/0.08	
Neopentylglycol diester	$-CH_2C(CH_3)_2CH_2-$	1985-51-9	240.30	112/1.2	
Nona(ethylene glycol) diester	$-(CH_2CH_2O)_8CH_2CH_2-$	25852-47-5	550		
Nona(propylene glycol) diester	$-(CH(CH_3)CH_2O)_8CH(CH_3)CH_2-$	25852-49-7	560	>200/1	
Nonyl ester	$-(CH_2)_8CH_3$	2696-43-7	212.33		
4-Nonylphenyl ester	$-C_6H_4(CH_2)_8CH_3$	76391-98-5	288.43	125/0.03	
Octadecyl ester	$-(CH_2)_{17}CH_3$	32360-05-7	338.58	195/6	19
n-Octyl ester	$-(CH_2)_7CH_3$	2157-01-9	198.31	105/4	
Pentabromophenyl ester	$-C_6Br_5$	18967-31-2	556.69		132
Pentachlorophenyl ester	$-C_6Cl_5$	16184-61-5	334.41		87
1H,1H-Pentafluorooctyl ester	$-CH_2(CF_2)_6CF_3$	3934-23-4	468.16	67/5	
Pentaerythritol tetraester	$C(CH_2-)_4$	3253-41-6	408.45		66
2,2,3,3,3-Pentafloropropyl ester	$-CH_2CF_2CF_3$	45115-53-5	218.12	55/110	
Pentyl ester	$-(CH_2)_4CH_3$	2849-98-1	156.23	73/17	
2-Phenoxyethyl ester	$-CH_2CH_2OC_6H_5$	10595-06-9	206.24	183	
Phenyl ester	$-C_6H_5$	2177-70-0	162.19	115/10	
2-Phenylethyl ester	$-CH_2CH_2C_6H_5$	3683-12-3	190.24	119/11	
n-Propyl ester	$-CH_2CH_2CH_3$	2210-28-8	128.17	140	
1,2-Propylene diester	$-CH_2CH(CH_3)-$	1188-09-6	212.25	68/1	
1,3-Propylene diester	$-(CH_2)_3-$		212.25	68/1	
2-Sulfoethyl ester	$-CH_2CH_2SO_3H$	10595-80-9	194.21	(d)	
3-Sulfopropyl ester, potassium salt	$-(CH_2)_3SO_3K$	31098-21-2	246.33		295(d)
Tetra(ethylene glycol) diester	$-CH_2CH_2(OCH_2CH_2)_3-$	109-17-1	300.37	220	
2,2,3,3-Tetrafluoropropyl ester	$-CH_2CF_2CF_2H$	45102-52-1	200.14	124	
Trimethylsilyl ester	$-Si(CH_3)_3$	13688-56-7	158.28	51/20	
2-(Trimethylsilyloxy)ethyl ester	$-CH_2CH_2OSi(CH_3)_3$	17407-09-9	202.33	65/0.9	
3-(Trimethylsilyloxy)propyl ester	$-(CH_2)_3OSi(CH_3)_3$	2530-85-0	248.35	190	
3-(Tris(trimethylsilyloxy)silyl) propyl ester	$-(CH_2)_3Si(OSi(CH_3)_3)_3$	17096-07-0	422.82	113/0.2	
Vinyl ester	$-CH=CH_2$	4245-37-8	112.13	111	

TABLE 6. ALCOHOLS

Name	Formula HOROH: R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
6.1. ALKANEDIOLS					
1,2-Butane	$-CH(C_2H_5)CH_2-$	584-03-2	90.12	191/747	
1,3-Butane	$-CH(CH_3)CH_2CH_2-$	107-88-0	90.12	203	
1,4-Butane	$-(CH_2)_4-$	110-63-4	90.12	230	16
1,10-Decane	$-(CH_2)_{10}-$	112-47-0	174.28	170/8	73
1,12-Dodecane	$-(CH_2)_{12}-$	5675-51-4	202.34	189/12	82
1,2-Ethylene	$-(CH_2)_2-$	107-21-1	62.07	197	−13

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.073	1.4520	97	OZ4725000	**1**(1),753M	**1**(1),1045A	**1**(1),665B			
1.080	1.4360	>110							
1.066	1.4470	96							
0.9830	1.4770	107							
0.886	1.420	41	OZ4900000	**1**(1),721B	**1**(1),976A	**1**(3),677C		**2**(3),1287	
1.096		97							
0.878	1.4430	>110	OZ5000000						
0.8847	1.4122								
1.020	1.4397	>110							
0.993	1.4310	65							
0.936	1.4140	10	OZ5075000	**1**(1),719I	**1**(1),975A	**1**(1),640C	**11**,5849	**2**(2),398	
1.087	1.450	>110						**2**(3),1288	
1.040	1.4800	88							
1.489	1.4900	78	AS4900000	**1**(1),745J	**1**(1),1031C				
1.071	1.4520	43							
1.003	1.4530	117						**2**(4),1535	
1.099	1.4660	>110							
1.010	1.4520	>110							
1.099	1.4660	>110							
	1.5020	>110							
0.864	1.4510	196						**2**(3),1290	
	1.4373								
		96							
		>100							
1.5732	1.3482	6							
1.0687	1.5130	100							
	1.5184								
0.98	1.508	122							
0.9022	1.4450								
	1.4450								
1.3245	1.4772								
				1(1),1065F	**1**(1),1438B	**1**(1),890D			
1.080	1.4630	82	OZ4000000					**2**(4),1531	
1.250	1.3730	50		**1**(1),745B	**1**(1),1029A				
0.890	1.4147	32		**1**(2),2985B	**1**(3),686C				
0.928	1.4280	76		**1**(2),2985C	**1**(3),687A				
1.045	1.4310	92	UC0230000	**1**(2),2985D	**1**(3),687B	**1**(2),1116C			
0.918	1.4190	>110							
0.933	1.4360	13						**2**(3),1290	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.006	1.4380	93	EK0380000	**1**(1),141A	**1**(1),196C	**1**(1),130B		**1**,477	
1.005	1.4400	121	EK0440000	**1**(1),139L	**1**(1),195C	**1**(1),130D	**11**,1566	**1**,477	
1.017	1.4450	>110	EK0525000	**1**(1),139K	**1**(1),195B	**1**(1),130A		**1**,478	
			HD8433713	**1**(1),143N	**1**(1),205C	**1**(1),136D	**11**,2842	**1**(2),560	
				1(1),145A	**1**(1),206B	**1**(1),137A		**1**(2),562	
1.113	1.4310	>110	KW2975000	**1**(1),139A		**1**(1),127D	**11**,3755	**1**,465	**15**,156

TABLE 6. *cont'd*

Name	Formula HOROH R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
1,7-Heptane	$-(CH_2)_7-$	629-30-1	132.20	259	18
1,16-Hexadiene	$-(CH_2)_{16}-$	23079-20-1	258.45	198/3	92
1,6-Hexane	$-(CH_2)_6-$	629-11-8	118.18	250	42
1,7-Heptane	$-(CH_2)_7-$	629-30-1	132.20	259	18
1,16-Hexadiene	$-(CH_2)_{16}-$	23079-20-1	258.45	198/3	92
1,9-Nonane	$-(CH_2)_9-$	3937-56-2	160.26	177/15	48
1,8-Octane	$-(CH_2)_8-$	629-41-4	146.23	172/20	60
1,5-Pentane	$-(CH_2)_5-$	111-29-5	104.15	242	
1,2-Propane	$-CH(CH_3)CH_2-$	57-55-6	76.10	187	−60
1,3-Propane	$-(CH_2)_3-$	504-63-2	76.10	214	−27
1,14-Tetradecane	$-(CH_2)_{14}-$	19812-64-7	230.39		88
6.2. ETHER DIOLS					
Di(ethylene glycol)	$-CH_2CH_2OCH_2CH_2-$	111-46-6	106.12	245	−10
Di(propylene glycol)	$-CH(CH_3)CH_2OCH(CH_3)CH_2-$	110-98-5	134.18		
Hexa(ethylene glycol)	$-CH_2CH_2(OCH_2CH_2)_5-$	2615-15-8	282.34	217/4	6
Penta(ethylene glycol)	$-CH_2CH_2(OCH_2CH_2)_4-$	4792-15-8	238.28	184/2	
Tetra(ethylene glycol)	$-CH_2CH_2(OCH_2CH_2)_3-$	112-60-7	194.23	314	−6
Tri(ethylene glycol)	$-CH_2CH_2(OCH_2CH_2)_2-$	112-27-6	150.17	285	−7
Tri(propylene glycol)	$-CH(CH_3)CH_2(OCH(CH_3)CH_2)_2-$	24800-44-0	192.26	273	

TABLE 7. ALLYL FUNCTIONAL

Name	Formula $CH_2=CHCH_2R$ R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Acetate	$-OC(O)CH_3$	591-87-7	100.12	104	
−Acetic acid	$-CH_2C(O)OH$	591-80-0	100.12	84/12	−22
Acetoacetate	$-OC(O)CH_2C(O)CH_3$	1118-84-9	142.16	194/737	
Alcohol	$-OH$	107-18-6	58.08	97	−129
−, 2-Bromo-	$H_2C=CBrCH_2OH$		136.98	152	
−, 2-Chloro-	$H_2C=CClCH_2OH$	5976-47-6	92.53	134	
−, 3-Methyl(crotyl alcohol)	$CH_3CH=CHCH_2OH$	6117-91-5	72.12	122	
−Amine	$-NH_2$	107-11-9	57.10	53	−88
−*N*-Aniline	$-NC_6H_5$	589-09-3	133.19	219	
−4-Anisole	$-C_6H_4OCH_3$	140-67-0	148.21	215	
−Benzene	$-C_6H_5$	300-57-2	118.18	156	
Benzyl ether	$-OCH_2C_6H_5$	14593-43-2	148.21	204	
Bromide	$-Br$	106-95-6	120.98	70	−119
Butyl ether	$-O(CH_2)_3CH_3$	3739-64-8	114.19		
Butyrate	$-OC(O)CH_2CH_2CH_3$	2051-78-7	128.17	44/15	
Chloride	$-Cl$	107-05-1	76.09	45	−134
Chloroacetate	$-OC(O)CH_2Cl$	2916-14-5	134.56	164/265	
Chloroformate	$-OC(O)Cl$	2937-50-0	120.54	109	
Cyanide	$-CN$	109-75-1	67.09	118	
Cyanoacetate	$-OC(O)CH_2CN$	13361-32-5	125.13	110/20	
Diallyl amine	$-NH-$	124-02-7	97.16	112	−88
2,3-Dibromide	$H_2=CBrCH_2Br$	513-31-5	199.88	141	
2,3-Dichloride	$H_2C=CClCH_2Cl$		110.97	94	
Ether	$-O-$	557-40-4	98.15	95	
Ethyl ether	$-OCH_2CH_3$	557-31-3	86.13	66	
Fluoride	$-F$	818-92-8	60.07	−10	
Formate	$-OC(O)H$		86.09	84	
Glycidyl ether	$-O-CH_2CH(-O-)CH_2$	106-92-3	114.14	154	
Iodide	$-I$	556-569	167.98	103	
Isocyanate	$-NCO$	1476-23-9	83.09	88	
Isopropyl ether	$-OCH(CH_3)_2$		100.16	84	
4-(2-Methoxyphenol)	$-C_6H_4(OCH_3)OH$	97-53-0	164.20	154	−11
Methyl ether	$-OCH_3$	627-40-7	72.11	46	
−2-Phenol	$-C_6H_4OH$	1745-81-9	134.18	220	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.951	1.4550	>110	MI9804000	**1**(1),143J	**1**(1),204B	**1**(1),135C		**1**,489	
				1(1),145E		**1**(1),137D			
		102	MO2100000	**1**(1),143D	**1**(1),202B	**1**(1),134B	**11**,4610	**1**,484	
0.951	1.4550	>110	MI9804000	**1**(1),143J	**1**(1),204B	**1**(1),135C		**1**,489	
				1(1),145E		**1**(1),137D		**1**,489	
		>110		**1**(1),143M	**1**(1),205B	**1**(1),136B		**1**,493	
				1(1),143K	**1**(1),204C	**1**(1),135D		**1**,490	
0.994	1.4500	129	SA0480000	**1**(1),141I	**1**(1),199A	**1**(1),132B	**11**,7073	**1**,481	
1.036	1.4320	107	TY2000000	**1**(1),139C	**1**(1),192C	**1**(1),128B	**11**,7868	**1**,472	
1.053	1.4400	79	TY2010000	**1**(1),139B	**1**(1),192B	**1**(1),128A	**11**,9629	**1**,475	
				1(1),145C	**1**(1),207A		**1**(2),564		
1.118	1.4460	143	ID5950000	**1**(1),225D	**1**(1),341B	**1**(1),223B	**11**,3109	**1**,468	
1.023	1.4410	138	UB8785000	**1**(1),225E	**1**(1),341C	**1**(1),226A		**1**(2),537	
1.127	1.4640	>110	MM3670000	**1**(1),229B	**1**(1),347A	**1**(1),227A		**1**,468	
1.126	1.4620	>110	RZ2670000	**1**(1),227K	**1**(1),346B	**1**(1),226D		**1**,468	
1.125	1.4590	177	XC2100000	**1**(1),227H	**1**(1),346A	**1**(1),226C		**1**,468	
1.125	1.4550	166	YE4550000	**1**(1),227D	**1**(1),345A	**1**(1),226B	**11**,9585	**1**,468	
1.021	1.4440	>110	YK6825000	**1**(1),227F	**1**(1),345B				

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.928	1.4040	7	AF1760000	**1**(1),715C	**1**(1),963C	**1**(1),633B		**2**,136	
0.981	1.4283	89	SB2800000	**1**(1),547H	**1**(1),778A	**1**(1),499B		**2**,425	
1.037	1.4390	75		**1**(1),787A	**1**(1),687C	**1**(1),1098A	**1**(1),687C	**19**(3),1203	
0.854	1.4120	22	BA5075000	**1**(1),147A	**1**(1),207B	**1**(1),138B	**11**,284	**1**,436	
1.6	1.5								
1.162	1.4590	54	UD4725000	**1**(1),183D				**1**,439	
0.845	1.4270	37	EM9275000	**1**(1),147C	**1**(1),208A	**1**(1),138C	**11**,2604	**1**,442	
0.761	1.4205	−29	BA5425000	**1**(1),347A	**1**(1),518C	**1**(1),322B	**11**,285	**4**,205	
0.982	1.5630	89		**1**(1),1383K	**1**(2),454B	**1**(1),1191A		**12**,170	
0.965	1.5210	81	BZ8225000	**1**(1),1243C	**1**(2),208B	**1**(1),1053B	**11**,3657	**6**,751	
0.892	1.5100	33	CY2275000	**1**(1),1139E	**1**(2),24B	**1**(1),946B		**5**,484	
0.959	1.5070	76						**6**(3),1459	
1.398	1.4690	−2	UC7090000	**1**(1),95I	**1**(1),135A	**1**(1),93D	**11**,286	**1**,201	
0.783	1.4060	14		**1**(1),219G	**1**(1),333B			**1**(3),1882	
0.902	1.4140	41	ES5775000	**1**(1),717N	**1**(1),970B	**1**(1),637B		**2**,272	
0.939	1.4135	−29	UC7350000	**1**(1),95E	**1**(1),133C	**1**(1),93C	**11**,287	**1**,198	
1.159	1.4460	61						**2**,198	
1.136	1.4220	31	LQ5775000	**1**(1),863B	**1**(1),1206B	**1**(1),737A		**3**,12	**13**,2
0.834	1.4050	23	EM8050000	**1**(1),991F	**1**(1),1358C	**1**(1),843A	**11**,288	**2**,408	
1.065	1.4430	>110	AG3690000	**1**(1),1013F	**1**(1),1391B				
0.787	1.4405	16	UC6650000	**1**(1),347K	**1**(1),520A	**1**(1),322C	**11**,2951	**4**,208	
1.934	1.5470	81	UC8200000	**1**(1),101K		**1**(1),99C	**11**,3009	**1**,201	
1.211	1.4603								
0.803	1.4160	−7	KN7525000	**1**(1),219H	**1**(1),333C	**1**(1),212D	**11**,290	**1**,438	
0.760	1.3880	−21		**1**(1),219E	**1**(1),332C	**1**(1),212C	**11**,291	**1**,438	
0.9460									
0.962	1.4330	57	RR0875000	**1**(1),251C	**1**(1),374A	**1**(1),233D			
1.837	1.5540	18	VD0450000	**1**(1),95J		**1**(1),94A	**11**,292	**1**,202	
0.940	1.4170	43	NQ8175000	**1**(1),1023E	**1**(1),1404A	**1**(1),872A		**4**,214	
0.7764	1.3946								
1.066	1.5410	>110	SJ4375000	**1**(1),1283E	**1**(2),278A	**1**(1),1091A	**11**,3855	**6**,961	
0.978	1.5200	63							
1.028	1.5450	88	SJ3850000	**1**(1),1265H	**1**(2),245B	**1**(1),1070D		**6**,572	

TABLE 7. *cont'd*

Name	Formula $CH_2=CHCH_2R$ R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Phenyl ether	$-OC_6H_5$	1746-13-0	134.18	192	
4-Chloro-	$-OC_6H_4Cl$		168.63	107/12	
2,4,6-Tribromo-	$-OC_6H_2Br_3$	3278-89-5	370.88		75
Phenyl sulfone	$-S(O)(O)C_6H_5$	16212-05-8	182.24	111/0.5	
Propyl	$-CH_2CH_2CH_3$	1471-03-0	100.16	91	
-Succinic anhydride	$-C_4H_3O_3$	7539-12-0	140.1	260	6
Sulfide	$-S-$	592-88-1	114.21	138	
2-Tolyl ether	$-OC_6H_4CH_3$		148.21	81/2	
3-Tolyl ether	$-OC_6H_4CH_3$		148.21	213	
4-Tolyl ether	$-OC_6H_4CH_3$		148.21	91/10	
-Urea	$-NHC(O)NH_2$	557-11-9	100.12		85

TABLE 8. AMINES, DIFUNCTIONAL

Name	Formula H_2NRNH_2 R	CAS Registry Number	Mol. wt.	Boiling Point(°C)	Melting Point(°C)
Butane-	$-(CH_2)_4-$	110-60-1	88.15	159	27
Decane-	$-(CH_2)_{10}-$	646-25-3	172.32	140/12	62
Dodecane-	$-(CH_2)_{12}-$	2783-17-7	200.37		70
Ethylene-	$-(CH_2)_2-$	107-15-3	60.10	118	8.5
Heptane-	$-(CH_2)_7-$	646-19-5	130.24	224	28
Hexane-	$-(CH_2)_6-$	124-09-4	116.21	204	43
Nonane-	$-(CH_2)_9-$	646-24-2	158.29	258/756	37
Octane-	$-(CH_2)_8-$	373-44-4	144.26	225	51
Pentane-	$-(CH_2)_5-$	462-94-2	102.18	179	
1,2-Phenylene-	$-C_6H_4-$	95-54-5	108.14	257	104
1,3-Phenylene-	$-C_6H_4-$	108-45-2	108.14	283	65
1,4-Phenylene-	$-C_6H_4-$	106-50-3	108.14	267	144
Propane-	$-(CH_2)_3-$	109-76-2	74.13	140	-12

TABLE 9. ANHYDRIDES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
9.1. MONOANHYDRIDES					
Acetoxysuccinic	$C_6H_6O_5$	79814-40-7	158.11		56
S-Acetylmercaptosuccinic	$C_6H_6O_4S$	6953-60-2	174.18		83
Aconitic, *cis*-	$C_6H_4O_5$	6318-55-4	156.10		75
Acrylic (see Acrylates)					
Allylsuccinic	$C_7H_8O_3$	7539-12-0	140.1	260	6
4-Amino-1,8-naphthalic	$C_{12}H_7NO_3$	6492-86-0	213.20		
1,2,4-Benzenetricarboxylic (trimellitic)	$C_9H_4O_5$	552-30-7	192.13		167
Bicyclo[2.2.2.]oct-5-ene-2,3-dicarboxylic, *endo*-	$C_{10}H_{10}O_3$	24327-08-0	178.19		145
Bromomaleic	C_4HBrO_3	5926-51-2	176.96	215	
4-Bromo,1,8-naphthalic	$C_{12}H_5BrO_3$	21563-29-1	277.08		218
3-(*tert*-Butyldimethylsilyloxy)-glutaric	$C_{11}H_{20}O_4Si$	91424-40-7	244.37		80
4-*tert*-Butylphthalic	$C_{12}H_{12}O_3$	32703-79-0	204.23		73
N-Carboxybenzyloxy-L-aspartic	$C_{12}H_{11}NO_5$	4515-23-5	249.23		123
Cantharidin	$C_{10}H_{12}O_4$	56-25-7	196.20		216
5-Chloroisotoic	$C_8H_4ClNO_3$	4743-17-3	197.58		300(d)
4-Chloro-1,8-naphthalic	$C_{12}H_5ClO_3$	4053-08-1	232.63		208
Citraconic	$C_5H_4O_3$	616-02-4	112.08	213	7
Cyclohexane-1,2-dicarboxylic, *cis*-	$C_8H_{10}O_3$	13149-00-3	154.17	158/17	33

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.978	1.5200	63	DA8575000	1(1),1241M	1(2),207A	1(1),1052B	6,144	11,14	
	1.5348								
								6,205	
1.189	1.5480	>110	WR2400000	1(2),2185B	1(2),1580A			6,299	
0.767	1.3990	−5		1(1),219F	1(1),333A	1(3),2780		1,438	
0.887	1.4889	46	BC4900000	1(1),287J	1(1),432C	1(1),269C	11,295	1,440	
0.950	1.5179								
0.9564	1.5179								
0.970	1.5168								
			YR7875000	1(1),939M	1(1),1308B	1(1),800D	11,296	4,209	

Density (at 20°C)	Refractive Index(At 20°C)	Flash Point(°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.877	1.4569	52	EJ6800000	1(1),309G	1(1),465A	1(1),291D	11,7964	4,264	
			HD7175000	1(1),311I	1(1),469B	1(1),293D		4,273	
		155	JR2200000	1(1),311J	1(1),469C	1(1),294A		4,273	
0.899	1.4565	34	KH8575000	1(1),309A	1(1),463B	1(1),289C	11,3752	4,230	4,231
		88		1(1),311E	1(1),468A	1(1),293A		4,271	
		81	MO1180000	1(1),311B	1(1),467A	1(1),292C	11,4614	4,269	
		>110		1(1),311H	1(1),469A	1(1),293C		4,272	
		165	RG8841500	1(1),311F	1(1),468B	1(1),293B		4,271	
0.873	1.4582	63	SA0200000	1(1,309K		1(1,292B	11,1608	4,266	
			SS7875000	1(1),1429A	1(2),536B	1(1),1235C	11,7255	13,6	8,393
		>110	SS7700000	1(1),1431C	1(2),539A	1(1),1239A	11,7254	3,33	
			SS8050000	1(1),1433D	1(2),542C	1(1),1242B	11,7256	13,61	
0.888	1.4565	34	TX6825000	1(1),309D	1(1),464A	1(1),290A		4,261	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
								18(2),1128	
				1(1),837E	1(1),1179B	1(1),715C			
				1(1),837D				18,463	
1.163									
				1(2),1961B		1(2),336A		18(2),469	
			DC2050000	1(2),1959G	1(2),1329C	1,(2),334A	11,9617	18,468	
				1(1),835N	1(1),1178B	1(1),717D			
1.905		>110		1(1),839A	1(1),1182B	1(1),719C		17,435	
				1(2),1961A	1(2),1330B			17,523	
				1(2),2987F	1(3),692A				
			RN8575000					19,161	
				1(2),1957H		1(2),332C		27,265	
			QL6127295	1(2),1959N		1(2),335D		17,522	
1.247	1.4710	101	GE6825000	1(1),837J	1(1),1181A	1(1),718D		17,440	
		>110		1(1),835G	1(1),1175C	1(1),715D			

TABLE 9. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Cyclohexane-1,2-dicarboxylic, *trans-*	$C_8H_{10}O_3$	14166-21-3	154.17		146
Diacetyl-L-tartaric	$C_8H_8O_7$	6283-74-5	216.15		131
3,5-Diacetyltetrahydropyran-2,4,6-trione	$C_9H_8O_6$	33524-89-9	212.16		154
Dichloromaleic	$C_4Cl_2O_3$	1122-17-4	166.95		
3,6-Dichlorophthalic	$C_8H_2Cl_2O_3$	4466-59-5	217.01		188
4,5-Dichlorophthalic	$C_8H_2Cl_2O_3$	942-06-3	217.01		186
3,6-Difluorophthalic	$C_8H_2F_2O_3$	652-40-4	184.10		219
Diglycolic	$C_4H_4O_4$	4480-83-5	116.07	240	92
2,2-Dimethylglutaric	$C_7H_{10}O_3$	2938-48-9	142.16	175/60	36
3,3-Dimethylglutaric	$C_7H_{10}O_3$	4160-82-1	142.16	181/25	124
2,3-Dimethylmaleic	$C_6H_6O_3$	766-39-2	126.11	223	94
2,2-Dimethylsuccinic	$C_6H_8O_3$	17347-61-4	128.13	219	30
Diphenic	$C_{14}H_8O_3$	6050-13-1	224.22		225
2,3-Diphenylmaleic	$C_{16}H_{10}O_3$	4808-48-4	250.26		160
2-Dodecen-1-ylsuccinic	$C_{16}H_{26}O_3$	19780-11-1	266.38	180/5	42
3,6-Epoxy-1,2,3,6-tetrahydrophthalic, *exo-*	$C_8H_6O_4$	6118-51-0	166.13		118(d)
3-Ethyl-3-methylglutaric	$C_8H_{12}O_3$	6970-57-6	156.18	185/20	
3-Fluorophthalic	$C_8H_3FO_3$	652-39-1	166.11		159
4-Fluorophthalic	$C_8H_3FO_3$	319-03-9	116.11		79
2-Formamidosuccinic	$C_5H_5NO_4$	33605-73-1	143.10		145
Glutaric	$C_5H_6C_3$	108-55-4	114.10		56
1,4,5,6,7,7-Hexachloro-5-norbornene-2,3-dicarboxylic	$C_9H_2Cl_6O_3$	115-27-5	370.83		237
Hexafluoroglutaric	$C_5F_6O_3$	376-68-1	222.04	72	
Hexahydro-4-methylphthalic	$C_9H_{12}O_3$	19438-60-9	168.19		
Homophthalic	$C_9H_6O_3$	703-59-3	162.14		141
3-Hydroxyphthalic	$C_8H_4O_4$	37418-88-5	164.12		200
Isatoic	$C_8H_5NO_3$	118-48-9	163.13		233(d)
Isobutenylsuccinic	$C_8H_{10}O_3$	18908-20-8	157.17	146/14	64
Itaconic	$C_5H_4O_3$	2170-03-8	112.09	114/12	71
Maleic	$C_4H_2O_3$	108-31-6	98.06	202	52
3-Methylglutaric	$C_6H_8O_3$	4166-53-4	128.13	181/25	45
N-Methylisatoic	$C_9H_7NO_3$	10328-92-4	117.16		
Methyl-5-norbornene-2,3-dicarboxylic	$C_{10}H_{10}O_3$	25134-21-8	178.19		
4-Methylphthalic	$C_9H_6O_3$	19438-61-0	162.14	295	91
Methylsuccinic	$C_5H_6O_3$	4100-80-5	114.10	239	34
1,8-Naphthalic	$C_{12}H_6O_3$	81-84-5	198.18		268
3-Nitro-1,8-naphthalic	$C_{12}H_5NO_5$	3027-38-1	243.17		248(d)
4-Nitro-1,8-naphthalic	$C_{12}H_5NO_5$	34087-02-0	343.17		228
3-Nitrophthalic	$C_8H_3NO_5$	641-70-3	193.11		164
4-Nitrophthalic	$C_8H_3NO_5$	5466-84-2	193.11		118
5-Norbornene-*endo*-2,3-dicarboxylic, *cis-*	$C_9H_8O_3$	129-64-6	164.16		166
Octadecylsuccinic	$C_{22}H_{38}O_3$		350.55	251/4	62
2-Octen-1-ylsuccinic	$C_{12}H_{18}O_3$	26680-54-6	210.27	168/10	10
3-Oxabicyclo[3.1.0] hexane-2,4-dione	$C_5H_4O_3$	5617-74-3	112.08	100/5	60
2,5-Oxazolidenedione	$C_3H_3NO_3$	2185-00-4	101.06		120(d)
2-Phenylglutaric	$C_{11}H_{10}O_3$	2959-96-8	190.20	218/13	97
Phenylmaleic	$C_{10}H_6O_3$	36122-35-7	176.16		121
1-Phenyl-2,3-naphthalenedicarboxylic	$C_{18}H_{10}O_3$	1985-37-1	247.28		259
Phenylsuccinic	$C_{10}H_8O_3$	1131-15-3	176.17	191/12	54
Phthalic	$C_8H_4O_3$	85-44-9	148.12	284	132
N-Phthaloylglutamic	$C_{13}H_9NO_5$	3343-28-0	259.22		199
1-Propenyloxymethylmaleic	$C_7H_{10}O_4$	130221-78-2	158.16	251	
Pyrazinedicarboxylic	$C_6H_2N_2O_3$	4744-50-7	150.09		210(d)

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
				1(1),835H	1(1),1176A	1(1),716A		17,452	
				1(1),837F	1(1),1179C			18,162	
				1(1),839L	1(1),1174A	1(1),457D		17(4),6841	
				1(1),839B		1(1),719D		17,434	5,195
				1(2),1959B	1(2),1328B			17,483	
				1(2),1959C	1(2),1328C			17,483	
				1(2),1959A	1(2),1328A			17(5),11,259	
				1(1),841C		1(1),722A		19,153	
				1(1),839G	1(1),1183C	1(1),720D		17,418	
				1(1),839H	1(1),1184A	1(1),721A		17,419	
			ON4025000	1(1),837K	1(1),1181B	1(1),719A		17,445	
1.135		108		1(1),835C	1(1),1175A			17,417	
				1(2),1959L		1(2),335B		17,526	
				1(2),1957C				17,532	
		177		1(1),835K	1(1),1177A	1(1),717A			
				1(1),837B	1(1),1179A			19(2),181	
				1(1),839I	1(1),1184B	1(1),721B		17(1),231	
								17(5),11,259	
								18(11),5,548	
		>110	MA3850000	1(1),839E	1(1),1183A	1(1),720B		17,411	
			RB9080000	1(1),839C	1(1),1182C	1(1),720A	11,2084		
1.654	1.3240	None		1(1),839K	1(1),1184C			17(4),5814	
1.162	1.4770	>110		1(1),835I	1(1),1176B	1(1),716B			
				1(2),1957E		1(2),331D		17,489	11,254
			TI3300000	1(2),1959F	1(2),1329B			18,94	
			DM3100000	1(2),1957F		1(2),332A		27,264	
				1(1),835E	1(1),1175B			17(4),5927	
				1(1),835J	1(1),1176C	1(1),716D		17,442	
		103	ON3675000	1(1),837I	1(1),1180C	1(1),718C	11,5586	17,432	5,422
		>110		1(1),839F	1(1),1183B	1(1),720C		17,415	
			DM3140000	1(2),1957G		1(2),332B		27,265	
1.232	1.5060	>110	RB9100000	1(1),835O	1(1),1178C	1(1),718C		17(2),461	
				1(2),1957M	1(2),1327B			17,492	
1.220		>110		1(1),835B	1(1),1174C	1(1),715B		17,414	
			QK5350000	1(2),1959M		1(2),335C		17,521	1,713
			QK5370000	1(2),1961C				17,523	
				1(2),1961D	1(2),1330C			17,524	
				1(2),1959H		1(2),334B		17,486	
			TI3328000	1(2),1959I	1(2),1330A	1(2),334C		17,486	
			DT5600000	1(1),835M	1(1),1177C	1(1),717C	11,1801	17(2),461	
1.055									
1.000	1.4694	>110							
				1(1),835F				17,442	
				1(1),907M				27,245	
				1(2),1957D		1(2),331C		17,494	
								17,510	
								17,492	
			TI3150000	1(2),1957J	1(2),1327A	1(2),332D	11,7346	17,469	1,882
			MA3900000	1(2),2017M	1(2)1391A			21(3),5370	
1.100	1.4610	>110							
								27,681	

TABLE 9. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
2,3-Pyridenedicarboxylic	$C_7H_3NO_3$	699-98-9	149.11		138
3,4-Pyridinedicarboxylic	$C_7H_3NO_3$	4664-08-8	149.11		76
Succinic	$C_4H_4O_3$	108-30-5	100.07	126	119
4-Sulfo-1,8-naphthalic	$C_{12}H_6O_6S$	71501-16-1	316.34		
Tetrabromophthalic	$C_8Br_4O_3$	632-79-1	463.72		275
Tetrachlorophthalic	$C_8Cl_4O_3$	117-08-8	285.90	371	256
Tetrafluorophthalic	$C_8F_4O_3$	652-12-0	220.08		95
1,2,3,6-Tetrahydrophthalic, *cis*-	$C_8H_8O_3$	935-79-5	152.15		98
3,4,5,6-Tetrahydrophthalic	$C_8H_8O_3$	2426-02-0	152.15		72
3,3-Tetramethyleneglutaric	$C_9H_{12}O_3$	5662-95-3	168.19	186/15	65
Tetrapropenylsuccinic (isomers)	$C_{16}H_{26}O_3$	26544-38-7	266.38	150/3	
Trimellitic	$C_9H_3ClO_4$	1204-28-0	210.57		67
2-(Triphenylphosphoranylidene)-succinic	$C_{22}H_{17}O_3P$	906-65-0	360.35		167(d)
9.2. DIANHYDRIDES					
Benzene-1,2,4,5-tetracarboxylic	$C_{10}H_2O_6$	89-32-7	218.12	397	283
Benzophenone-3,3′,4,4′-tetracarboxylic	$C_{17}H_6O_7$	2421-28-5	322.23		220
Bicyclo[2.2.2.]oct-7-ene-2,3,5,6-tetracarboxylic	$C_{12}H_8O_6$	1719-83-1	248.19		>300
Biphenylene-3,3′,4,4′-tetracarboxylic	$C_{16}H_6O_6$	2420-87-3	294.22		300
Cyclobutanetetracarboxylic	$C_8H_4O_6$	4415-87-6	196.11		>300
Cyclopentane-1,2,3,4-tetracarboxylic, *cis,cis*-	$C_9H_6O_6$	6053-68-5	210.14		225(d)
Diethylenetriaminepentaacetic	$C_{14}H_{19}N_3O_8$	23911-26-4	357.32		183
Ethylenediaminetetraacetic	$C_{10}H_{12}N_2O_6$	23911-25-3	256.22		190(d)
4,4′-(Hexafluoroisopropylidene) diphthalic	$C_{19}H_6F_6O_6$	1107-00-2	444.25		244
Naphthalene-1,4,5,8-tetra-carboxylic	$C_{14}H_4O_6$	81-30-1	268.18		>300
Perylene-3,4,9,10-tetracarboxylic	$C_{24}H_8O_6$	128-69-8	392.32		>300
Tetrahydrofuran-2,3,4,5-tetra-carboxylic	$C_8H_4O_7$	25774-69-0	212.11		223

TABLE 10. BUTADIENES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
10.1. 1,2-BUTADIENES					
1,2-Butadiene	$H_2C{=}C{=}CHCH_3$	590-19-2	54.09	11	− 136
−, 4-Bromo-	$H_2C{=}C{=}CHCH_2Br$		133.00	110	
−, 4-Chloro-	$H_2C{=}C{=}CHCH_2Cl$		88.54	88	
−, 4-Hydroxy-	$H_2C{=}C{=}CHCH_2OH$		70.09	127	
−, 4-Iodo-	$H_2C{=}C{=}CHCH_2I$		179.99	130	
−, 3-Methyl-	$H_2C{=}C{=}C(CH_3)_2$	598-25-4	68.12	41	− 148
10.2. 1,3-BUTADIENES					
1,3-Butadiene	$H_2C{=}CHCH{=}CH_2$	106-99-0	54.09	− 4	− 109
−, 2-Bromo-	$H_2C{=}CBrCH{=}CH_2$		133.00	42/165	
−, 1-Chloro-	$HCCl{=}CHCH{=}CH_2$		88.54	68	
−, 1-Chloro-2-methyl-	$HCCl{=}C(CH_3)CH{=}CH_2$		102.57	107	
−, 1-Chloro-3-Methyl-	$HCCl{=}CHC(CH_3){=}CH_2$		102.57	100	
−, 2-Chloro-	$H_2C{=}CClCH{=}CH_2$	126-99-8	88.54	59	
−, 2-Chloro-3-methyl-	$H_2C{=}CClC(CH_3){=}CH_2$		102.57	93	
−, 1,2-Dichloro-	$HCCl{=}CClCH{=}CH_2$		122.98	63/105	
−, 2,3-Dichloro-	$H_2C{=}CClCCl{=}CH_2$		122.98	98	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
				1(2),2547K	1(3),336A	1(2),795B		27,261	
				1(2),2547L	1(3),336B			27,261	
			WN0875000	1(1),835A	1(1),1174B	1(1),715A	11,8841	17,407	4,468
				1(2),2203F		1(2),501D			
			TI3350000	1(2),1959E		1(2),333D		17,485	
			TI3450000	1(2),1959D		1(2),333C		17,484	15,300
					1(2),1329A			17(5),11,260	
			GW5775000	1(1),835L	1(1),1177B	1(1),717B		17,462	
				1(1),837M	1(1),1182A	1(1),719B		17,461	
				1(1),839J		1(1),721C		17,237	
1.005	1.4790	>110	WN1300000					17(5),11,105	
				1(2),1981E	1(2),1353B	1(2),351B			
					1(2),1680C	1(2),550D			
			DB9300000	1(2),1959K		1(2),335A		19,196	
		162		1(21),1957L		1(2),333B			
				1(1),837H	1(1),1180B	1(1),716C			
				1(1),837G	1(1),1180A				
				1(1),841B					
				1(1),841A	1(1),1185A				
				1(2),1957N	1(2),1327C				
			QK3695000	1(2),1961F		1(2),336B		19,196	
				1(2),1961G		1(2),336C			

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.652	1.4205/1.3							1(4),975	
1.4255	1.5248								
0.9891	1.4775								
0.9164	1.4759								
1.7129	1.5709								
0.694	1.4190	−23		1(1),27B	1(1),35C	1(1),30C		1,252	
0.650/−6	1.4292/ −25		EI9275000	1(1),27D	1(1),36B		11,1500	1,249	
1.397	1.4988								
0.961	1.4709								
0.9710	1.4792								
0.9543	1.4719								
0.9583	1.4583		EI9625000						
0.9593	1.4686								
1.1991	1.4960								
1.1829	1.4890								

TABLE 10. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
–, 2,3-Dimethyl-	$H_2C{=}C(CH_3)C(CH_3){=}CH_2$	513-81-5	82.15	68	– 76
–, 2-Fluoro-	$H_2C{=}CFCH{=}CH_2$		72.08	12	
–, Hexachloro-	$CCl_2{=}CClCCl{=}CCl_2$	87-68-3	260.76	215	– 20
–, Hexafluoro-	$CF_2{=}CFCF{=}CF_2$	685-63-2	162.03	7	– 132
–, 2-Iodo-	$H_2C{=}CICH{=}CH_2$		179.99	112	
–, 2-Methyl-	$H_2C{=}C(CH_3)CH{=}CH_2$	78-79-5	68.12	34	– 146

TABLE 11. BUTENES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
11.1. 1-BUTENES					
1-Butene	$H_2C{=}CHCH_2CH_3$	106-98-9	56.11	– 6	– 185
–, 4-Bromo-	$H_2C{=}CHCH_2CH_2Br$	5162-44-7	135.01	99	
–, 2,3-Dimethyl-	$H_2C{=}C(CH_3)CH(CH_3)_2$	563-78-0	84.16	56	– 158
–, 3,3-Dimethyl-	$H_2C{=}CHC(CH_3)_2CH_3$	558-37-2	84.16	41	– 115
–, 2-Methyl-	$H_2C{=}C(CH_3)CH_2CH_3$	563-46-2	70.14	31	– 137
–, 3-Methyl-	$H_2C{=}CHCH_2(CH_3)_2$	563-45-1	70.14	20	– 168
11.2. 2-BUTENES					
2-Butene, *cis*-	$CH_3CH{=}CHCH_3$	590-18-1	56.11	4	– 139
2-Butene, *trans*-	$CH_3CH{=}CHCH_3$	624-64-6	56.11	1	– 105
1,4-Dihydroxy-, *cis*-	$H_2C(OH)CH{=}CHCH_2(OH)$	6117-80-2	88.11	131/12	7
2,3-Dimethyl-	$(CH_3)_2C{=}C(CH_3)_2$	563-79-1	84.16	73	– 75
2-Methyl-	$(CH_3)_2C{=}CHCH_3$	513-35-9	70.14	36	– 134

TABLE 12. EPOXIDES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
12.1. MONOEPOXIDES $CH_2(-O-)CHR$					
Allylglycidyl ether	$-CH_2OCH_2CH{=}CH_2$	106-92-3	114.14	154	
2-Biphenylyl glycidyl ether	$-CH_2OC_6H_4C_6H_5$	7144-65-2	266.28	120/0.1	31
n-Butyl glycidyl ether	$-CH_2O(CH_2)_3CH_3$	2426-08-6	130.19	165	
t-Butyl glycidyl ether	$-CH_2OC(CH_3)_3$	7665-72-7	130.19	151	
4-*t*-Butylphenyl glycidyl ether	$-CH_2OC_6H_4C(CH_3)_3$	3101-60-8	206.29	166/4	
4-Chlorophenyl glycidyl ether	$-CH_2OC_6H_4Cl$	2212-05-7	184.62		32
Cyclohexene oxide	$C_6H_{10}O$	286-20-4	98.15	129	
Cyclooctene oxide	$C_8H_{14}O$	286-62-4	126.20	55/5	54
Cyclopentene oxide	C_5H_8O	285-67-6	84.12	102	
Epibromohydrin	$-CH_2Br$	3132-64-7	136.98	135	– 40
Epichlorohydrin	$-CH_2Cl$	106-89-8	92.53	116	– 57
Epifluorohydrin	$-CH_2F$	503-09-3	76.07	85	
1,2-Epoxybutane, *trans*-	$CH_3CH(-O-)CHCH_3$	21490-63-1	72.11	54	
1,2-Epoxydecane	$-(CH_2)_7CH_3$	2404-44-6	156.27	94/15	
1,2-Epoxydodecane	$-(CH_2)_9CH_3$	2855-19-8	184.32	124/15	
1,2-Epoxyhexadecane	$-(CH_2)_{13}CH_3$	7320-37-8	240.43	176/12	21
1,2-Epoxyhexane	$-(CH_2)_3CH_3$	1436-34-6	100.16	119	
2,3-Epoxynorbornane, *exo*-	$C_7H_{10}O$	3146-39-2	110.16		123
1,2-Epoxyoctadecane	$-(CH_2)_{15}CH_3$	7390-81-0	268.49	137/0.5	34
2,3-Epoxypropyl benzene	$-CH_2C_6H_5$	4436-24-2	134.18	99/17	
N-(2,3-Epoxypropyl)phthalimide	$-CH_2NC_8H_4O_2$	5455-98-1	203.20		99
1,2-Epoxytetradecane	$-(CH_2)_{11}CH_3$	3234-28-4	212.38	95/10.4	
Ethylene oxide	$-H$	75-21-8	44.05	10.7	– 111
2-Ethylhexyl glycidyl ether	$-CH_2OCH_2CH(CH_2CH_3)(CH_2)_3CH_3$	2461-15-6	186.30	61/0.3	
Furfuryl glycidyl ether	$-CH_2OCH_2C_4H_3O$	5380-87-0	154.17	103/11	
Glycidol	$-CH_2OH$	556-52-5	74.08	61/15	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.726	1.4390	−22		1(1),27F	1(1),37A	1(1),32A	11,3228	1(3),991	1,276
0.843	1.400								
1.665	1.5550	None	EJ0700000	1(1),105H	1(1),150A	1(1),102A		1,250	10,195
1.553/−20	1.378/−20								
	1.5616								
1.4220	1.4220	−54	NT4037000	1(1),27E	1(1),36C	1(1),30D	11,5087	1,252	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
	1.3962			1(1),15C			11,1513	1,203	
1.330	1.4625	9		1(1),97E	1(1),136C	1(1),94B		1(1),84	
0.684	1.3890	−18		1(1),23I	1(1),31B	1(1),22A		1(3),816	
0.653	1.3760	−29		1(1),23B	1(1),29B	1(1),25B		1,217	
0.650	1.3780	<−35		1(1),23G	1(1),30C	1(1),21B		1,211	
0.627/0	1.3640	−57		1(1),23A	1(1),29A	1(1),21C		1,213	
	1.3931/−25	37		1(1),19A				11,1514	1,205
	1.3848/−25			1(1),19B				11,1514	1,205
1.070	1.4780	126	EM4970000	1(1),153A	1(1),221A	1(1),146C		1(2),567	
0.708	1.4120	−17		1(1),25H	1(1),35B	1(1),25A	11,644	1,218	
0.662	1.3870	−46		1(1),25D	1(1),34A	1(1),21D	11,644	1,211	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.962	1.4330	57	RR0875000	1(1),251C	1(1),374A	1(1),233D			
				1(1),1253D	1(2),225A	1(1),1065C			
0.910	1.4190	56	TX4200000	1(1),249K	1(1),373A			17(3),988	
0.917	1.4170	43	RR0475000	1(1),251A	1(1),373B	1(1),232C		17(3),988	
1.038	1.5150	102	TX4250000	1(1),1253A	1(2),224A	1(1),1065B			
	1.5450	107	TX5600000	1(1),1253B	1(2),224B	1(1),1064D			
0.970	1.4520	27	RN7175000	1(1),251L	1(1),376C	1(1),235A		17,21	
		56		1(1),251M	1(1),377A	1(1),235B			
0.964	1.4340	10	RN8935000	1(1),251K	1(1),376B	1(1),234D		17,21	
1.601	1.4820	56	TX4115000	1(1),247L	1(1),370B	1(1),232A		17,21	
1.183	1.4380	34	TX4900000	1(1),247I	1(1),369B	1(1),231D	11,3563	17,6	5,290
1.067	1.3680	4	TZ3325000	1(1),247H	1(1),369A	1(1),231C			
0.804	1.3730	−27	EK3855000	1(1),245F	1(1),364C	1(1),229B		17(3),49	
0.840	1.4290	78		1(1),245J	1(1),365C	1(1),230A		17,18	
0.844	1.4360	105	JR2450000	1(1),245K	1(1),366A	1(1),230B		17(3),136	
0.846	1.4460	93	ML9450000	1(1),247A	1(1),366C	1(1),230D		17,20	
0.831	1.4060	16	MO3630000	1(1),245H		1(1),229A		17,86	
		10	RB7176000	1(1),253D	1(1),378B	1(1),235D			
		>110		1(1),247B	1(1),367A	1(1),231B		17(3),140	
1.020	1.5230	82	DA0178000	1(1),1251D	1(2),220C			17,51	
			TI4950000	1(2),2083B	1(2),1461C	1(2),408A			
0.845	1.4408	>110		1(1),245L	1(1),366B	1(2),230C		17(3),139	
0.891/4	1.3597/7		KX2450000	1(1),245A			11,3758	17,4	8,150
0.891	1.4340	97	TZ3300000	1(1), 251B	1(1),373C	1(1),232D			
1.122	1.4810	102	LU1423000	1 (2),2319J	1(3),17C			17(5),347	
1.117	1.4330	81	UB4375000	1(1),249B	1(1),370C	1(1),232B	11,4385	17,104	

TABLE 12. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
3-Glycidoxypropyl trimethoxysilane	$-CH_2O(CH_2)_3Si(OCH_3)_3$	2530-83-8	236.34	120/2	
(R)-(−)-Glycidyl Butyrate	$-CH_2OC(O)CH_2CH_2CH_3$	60456-26-0	144.17	90/19	
Glycidyl isopropyl ether	$-CH_2OCH(CH_3)_2$	4016-14-2	116.16	131	
Glycidyl methacrylate (see Methacrylates)					
Glycidyltrimethylammonium chloride, (70% aq. solution)	$-CH_2N(CH_3)_3Cl$	3033-77-0	151.54		
Hexafluoropropylene oxide	$F_2C(-O-)CFCF_3$	428-59-1	166.02	−42	
4-Methoxyphenyl glycidyl ether	$-CH_2OC_6H_4OCH_3$	2211-94-1	180.20		46
α-Pinene oxide	$C_{10}H_{16}O$	1686-14-2	152.24	102/50	
Propylene oxide	$-CH_3$	75-56-9	58.08	34	−112
Styrene oxide	$-C_6H_5$	96-09-3	120.15	194	−37
Tetracyanoethylene oxide	$(NC)_2C(-O-)C(CN)_2$	3189-43-3	144.09		
1,1,2,2-Tetrafluoroethyl glycidyl ether	$-CH_2OCF_2CHF_2$	85567-21-1	174.10	143	
Vinylethylene oxide	$-CH=CH_2$	930-22-3	70.09	65	
12.2. DIEPOXIDES $CH_2(-O-)CHRCH(-O-)CH_2$					
Bis(3,4-epoxycyclohexylmethyl) adipate	$C_6H_9(-O-)CH_2OC(O)(CH_2)_4C(O)OCH_2C_6H_9(-O-)$	3130-19-6	366.46		
1,3-Butadiene diepoxide	$CH_2(-O-)CHCH(-O-)CH_2$	1464-53-5	86.09	57/25	3
1,4-Butanediol diglycidol ether	$-CH_2O(CH_2)_4OCH_2-$	2425-79-8	202.25	158/11	
1,2,5,6-Diepoxycyclooctane	$C_8H_{12}O_2$	27035-39-8	140.18	70/0.4	
1,2,7,8-Diepoxyoctane	$-(CH_2)_4-$	2426-07-5	142.20	240	
3,4-Epoxycyclohexylmethyl 3,4-epoxycyclohexanecarboxylate	$C_6H_9(-O-)CH_2OC(O)C_6H_9(-O-)$	2386-87-0	252.31		−37
Ethylene glycol digycidyl ether	$-CH_2O(CH_2)_2OCH_2-$	2224-15-9	174.20	112/4.5	

TABLE 13. ETHYLENE HALIDES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Ethylene	$H_2C=CH_2$	74-85-1	28.05	−104	−169
−, Bromo- (vinyl bromide)	$BrCH=CH_2$	593-60-2	106.96	16/750	−139
−, Chloro- (vinyl chloride)	$ClCH=CH_2$	75-01-4	62.50	−13.4	−154
−, 1-Chloro-1-fluoro-	$ClCF=CH_2$		84.48	−24	−169
−, Chlorotrifluoro-	$ClCF=CF_2$	79-38-9	116.47	−29	−157
−, 1,1-Dibromo- (vinylidene bromide)	$Br_2C=CH_2$		185.86	92	
−, 1,2-Dibromo-, *cis*-	$BrCH=CHBr$		185.86	112	−53
1,2-Dibromo-, *trans*-	$BrCH=CHBr$		185.86	108	−6
−, 1,1-Dichloro- (vinylidene chloride)	$Cl_2C=CH_2$	75-35-4	96.94	31	−122
−, 1,2-Dichloro-, *cis*-	$ClCH=CHCl$	156-59-2	96.94	60	−80
−, 1,2-Dichloro-, *trans*-	$ClCH=CHCl$	156-60-5	96.94	48	−50
−, Tetrabromo-	$Br_2C=CBr_2$		343.66	226	56
−, Tetrachloro-	$Cl_2C=CCl_2$	127-18-4	165.83	121	−22
−, Tetrafluoro-	$F_2C=CF_2$	116-14-3	100.02	−76	−142
−, Tetraiodo-	$I_2C=CI_2$	513-92-8	531.64		192
−, Tribromo-	$Br_2C=CHBr$		264.76	163	
−, Trichloro-	$Cl_2C=CHCl$	79-01-6	131.39	87	−85

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.070	1.4290	>110	VV4025000	**1**(2),2973B	**1**(3),667C	**1**(2),1113D			
1.018	1.4280	85		**1**(1),781E	**1**(1),1088B			**17**(5),3,34	
0.924	1.4100	33	TZ3500000	**1**(1),249J	**1**(1),372C	**1**(3),306A		**17**(3),988	
1.129	1.4780	>110	BQ3480000		**1**(1),616B			**18**,583	
			TZ3400000	**1**(1),249A				**17**(3),20	
		>110	DA0176000	**1**(1),1253C	**1**(2),224C	**1**(1),1065A			
0.964	1.4690	66	TK4565000	**1**(1),253B	**1**(1),377C	**1**(1),237A		**5**,152	
0.830	1.3660	−35	TZ2975000	**1**(1),245B	**1**(1),363B	**1**(1),228B	**11**,7869	**17**,6	
1.054	1.5350	79	CZ9625000	**1**(1),1251A	**1**(2),219C	**1**(1),1063C		**17**,49	
				1(1),1003F	**1**(1),1376C		**18**(5),7,215		
1.380	1.3490	51							
0.870	1.4170	−50	EM7350000	**1**(1),247D	**1**(1),367C	**1**(1),233A		**17**(1),13	**4**,53
1.149	1.4930	>110	MO1880550						
1.113	1.4340	46	EJ8400000	**1**(1),251E	**1**(1),374B	**1**(1),234A	**11**,3621	**19**,14	
1.100	1.4610	>110	EJ5100000	**1**(1),251I	**1**(1),375C	**1**(1),234C			
1.138	1.4690	105	GX9627000	**1**(1),253E	**1**(1),378C	**1**(1),235C			
0.997	1.4450	98	RG9450000	**1**(1),251F	**1**(1),374C	**1**(1),234B			
1.170	1.4980	118	RN7750000						
1.118	1.4630	>110	KH5780000	**1**(1),251G	**1**(1),375A			**17**(3),997	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
			KU5340000	**1**(1),15A	**1**(3),17C		**11**,3748	**1**,180	
1.517	1.4350	None	KU8400000	**1**(1),95B	**1**(1),133A	**1**(1),93A		**1**,188	
0.911	1.3700	−61	KU9625000	**1**(1),95A			**11**,9898	**1**,186	**1**,1274
			KV0525000	**1**(1),103I				**1**(3),646	
2.178									
2.2464	1.5428								
2.2308	1.550								
1.213	1.4254	−9	KV9275000	**1**(1),101B	**1**(1),142B	**1**(1),98B	**11**,9900	**1**,186	
1.284	1.4481	6	KV9420000	**1**(1),101D	**1**(1),143A	**1**(1),98A	**11**,86	**1**,188	**14**,127
1.257	1.4456	6	KV9400000	**1**(1),101E	**1**(1),134B	**1**(1),79D	**11**,86	**1**,188	**12**,175
1.623	1.5060	None	KX3850000	**1**(1),103J		**1**(1),101C	**11**,9126	**1**,187	
1.1507/−40									
2.983			KX4125000	**1**(1),103K			**11**,9151	**1**,195	
2.708	1.6045/16								
1.463	1.4760	None	KX4550000	**1**(1),103G	**1**(1),147C	**1**(1),101B	**11**,9552	**1**,187	**11**,552

TABLE 14. FUMARATE ACIDS/ESTERS

Name	Formula ROC(O)CH=CHC(O)OR R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Acid	–H	110-17-8	116.07		299(s)
2-Bromo-	HOC(O)CBr=CHC(O)OH		194.98	200(d)	186
2-Chloro-	HOC(O)CCl=CHC(O)OH		150.52		193
2-Chloro-, diethyl ester	$CH_3CH_2OC(O)CCl=CHC(O)OCH_2CH_3$		206.63	136/19	
2-Chloro-, dimethyl ester	$CH_3OC(O)CCl=CHC(O)OCH_3$		178.57	224	
Di-*n*-amyl ester	$-(CH_2)_4CH_3$		256.33	162/7	
Diethyl ester	$-CH_2CH_3$	623-91-6	172.18	217	2
Diisoamyl ester	$-CH_2CH_2CH(CH_3)_2$		256.33	166/11	
Diisobutyl ester	$-CH_2CH(CH_3)_2$		228.29	170/160	
Diisopropyl ester	$-CH(CH_3)_2$		200.24	226	
2,3-Dimethyl-	$HOC(O)C(CH_3)=C(CH_3)C(O)OH$		144.13		241
Dimethyl ester	$-CH_3$	624-49-7	144.13	192	103
Dinitrile- (fumaronitrile)	NCCH=CHCN	764-42-1	78.07	186	96
Diphenyl ester	$-C_6H_5$		268.27	219/14	162
Di-*n*-propyl ester	$-CH_2CH_2CH_3$		200.24	110/5	
2-Methyl- (mesaconic acid)	$HOC(O)C(CH_3)=CHC(O)OH$	498-24-8	130.10	205(d)	204
2-Methyl-, diethyl ester	$CH_3CH_2OC(O)C(CH_3)=CHC(O)CH_2CH_3$			186.21	229
2-Methyl-, dimethyl ester	$CH_3OC(O)C(CH_3)=CHC(O)OCH_3$			158.16	204

TABLE 15. ISOCYANATES

Name	Formula OCNRNCO R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
1,3-Benzene diisocyanate	$-C_6H_4-$	123-61-5	160.13	121/25	50
1,4-Benzene diisocyanate	$-C_6H_4-$	104-49-4	160.13	260	98
1,3-Bis(isopropyl)benzene α,α′-diisocyanate	$-C(CH_3)_2C_6H_4C(CH_3)_2-$	2778-42-9	244.30	157/10	–10
1,4-Bis(isopropyl)benzene α,α′-diisocyanate	$-C(CH_3)_2C_6H_4C(CH_3)_2-$	2778-41-8	244.30	150/3	72
1,4-Butene diisocyanate	$-(CH_2)_4-$	4538-37-8	140.14	102/14	
α-Chlorotoluene 2,4-diisocyanate	$-C_6H_3(CH_2Cl)-$	51979-57-8	208.60	153/11	
1,4-Cyclohexylene diisocyanate, *trans*-	$-C_6H_{10}-$	7517-76-2	166.18		60
1,12-Dodecane diisocyanate	$-(CH_2)_{12}-$	13879-35-1	252.36	168/3	
1,6-Hexane diisocyanate	$-(CH_2)_6-$	822-06-0	168.20	255	
Isopherone diisocyanate (isomers)	$-C_{10}H_{18}-$	4098-71-9	222.29	158/15	
4,4′-Methylenebis(cyclohexyl isocyanate) (isomers)	$-(C_6H_{10})CH_2(C_6H_{10})-$	5124-30-1	262.35		
4,4′-Methylenebis(2,6-diethylphenyl isocyanate)	$-C_6H_2(CH_2CH_3)_2CH_2C_6H_2(CH_2CH_3)_2-$	105442-35-1	362.48		56
4,4′-Methylenebis(phenylisocyanate)	$-C_6H_4CH_2C_6H_4-$	101-68-8	250.26	200/5	43
2-Methylpentane 1,5-diisocyanate	$-CH_2CH(CH_3)(CH_2)_3-$	34813-62-2	168.20		
1,8-Octane diisocyanate	$-(CH_2)_8-$	10124-86-4	196.25	156/15	
2,4-Toluene diisocyanate	$-C_6H_3(CH_3)-$	584-84-9	174.16	120/10	21

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
			LS9625000	**1**(1),553A	**1**(1),787B	**1**(1),502D	**11**,4200	**2**,737	**5**,319
1.1880	1.4571								
1.290									
0.9681	1.4496								
1.052	1.4410	94	EM5950000	**1**(1),725M	**1**(1),993C	**1**(1),645A		**2**,742	
0.9655	1.4479								
0.9760	1.4432								
			EM6125000	**1**(1),725L	**1**(1),993B	**1**(1),644D		**2**,741	
0.9416	1.4349		LT2300000	**1**(1),993A	**1**(1),1360B	**1**(1),844A		**2**(1),302	
1.0129	1.4439								
1.466			OX5075000	**1**(1),553C	**1**(1),788A	**1**(1),503B	**11**,5806	**2**,763	
	1.0453	1.4488							
	1.0914	1.4512							

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
		>110	NR0150000	**1**(2),2165G		**1**(3),1432D		**13**,50	
		>110	CZ6150000	**1**(2),2165H				**13**,105	
1.05		153	CY8480000						
		>93							
1.105		106		**1**(1),1023F	**1**(1),1404B			**4**(3),578	
1.362	1.5960	>110		**1**(2),2169H	**1**(2),1572A			**13**(4),245	
			GU9642500	**1**(1),1025A	**1**(1),1406A			**13**(3),12	
0.940	1.4590	>110		**1**(1),1023H	**1**(1),1405A			**4**(3),624	
1.040	1.4520	140	MO1740000	**1**(1),1023G	**1**(1),1404C	**1**(1),871A		**4**(2),711	
1.049	1.4840	>110	NQ9370000	**1**(1),1023J	**1**(1),1405C				
1.066	1.4970	>110	NQ9250000	**1**(1),1025B				**13**(4),33	
				1(2),2169K	**1**(2),1572C				
1.180		>110	NQ9350000	**1**(2),2163D				**13**(3),461	
1.049	1.4550	>110		**1**(1),1023I	**1**(1),1405B				
1.007	1.4550	>110						**4**(3),614	
1.214	1.5680	>110	CZ630000	**1**(2),2169F		**1**(2),478B	**11**,9456	**13**,138	**1**,117

TABLE 16. LACTAMS

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
γ-Butyrolactam (2-pyrrolidinone)	$-C(O)(CH_2)_3NH-$	616-45-5	85.11	245	24
ε-Caprolactam	$-C(O)(CH_2)_5NH-$	105-60-2	113.16	137/10	69
12-Dodecanolactam	$-C(O)(CH_2)_{11}NH-$	947-04-6	197.32		151
7-Heptanolactam	$-C(O)(CH_2)_6NH-$	673-66-5	127.19	149/10	36
β-Propiolactam	$-C(O)CH_2CH_2NH-$	930-21-2	71.08	106/15	75
δ-Valerolactam	$-C(O)(CH_2)_4NH-$	675-20-7	99.13	256	39

TABLE 17. LACTONES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
γ-Butyrolactone	$-C(O)(CH_2)_3O-$	96-48-0	86.09	204	−45
ε-Caprolactone	$-C(O)(CH_2)_5O-$	502-44-3	114.14	97/15	
β-Propiolactone	$-C(O)(CH_2)_2O-$	57-57-8	72.06	162	−33
δ-Valerolactone	$-C(O)(CH_2)_4O-$	542-28-9	100.12	59/0.5	

TABLE 18. MALEATE ACIDS/ESTERS

Name	Formula ROC(O)CH=CHC(O)OR R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Acid	$-H$	110-16-7	116.07		141
2-Chloro-	$HOC(O)CCl=CHC(O)OH$		150.52		110
2-Chloro-, diethyl ester	$CH_3CH_2OC(O)CCl=CHC(O)OCH_2CH_3$		206.63	235	
2-Chloro-, dimethyl ester	$CH_3OC(O)CCl=CHC(O)OCH_3$		178.57	107	
Diallyl ester	$-CH_2CH=CH_2$	999-21-3	196.20	110/4	−47
Di-*n*-amyl ester	$-(CH_2)_4CH_3$		256.33	161/10	
Di-*n*-butyl ester	$-(CH_2)_3CH_3$	105-76-0	228.29	281	
2,3-Dichloro-	$HOC(O)CCl=CClC(O)OH$		184.97		120
Diethyl ester	$-CH_2CH_3$	141-05-9	172.18	225	−10
2,3-Dihydroxy-	$HOC(O)C(OH)=C(OH)C(O)OH$		148.07		155
Diisoamyl ester	$-CH_2CH_2CH(CH_3)_2$		156.33	157/13	
Dimethyl ester	$-CH_3$	624-48-6	144.13	204	
Diphenyl ester	$-C_6H_5$		268.27	226/15	93
Di-*n*-propyl ester	$-CH_2CH_2CH_3$		200.24	126/12	
2-Methyl-, *cis-* (citraconic acid)	$HOC(O)C(CH_3)=CHC(O)OH$	498-23-7	130.10		90
2-Methyl-, diethyl ester, *cis-*	$CH_3CH_2OC(O)C(CH_3)=CHC(O)OCH_2CH_3$		186.21	230	
2-Methyl-, dimethyl ester, *cis-*	$CH_3OC(O)C(CH_3)=CHC(O)OCH_3$			158.16	211

TABLE 19. PROPENES

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Propene	$H_2C=CHCH_3$	115-07-1	42.08	−48	−185
–, 1-Bromo-, *cis-*	$BrCH=CHCH_3$	590-13-6	120.98	58	
–, 1-Bromo-, *trans-*	$BrCH=CHCH_3$	590-15-8	120.98	64	
–, 2-Bromo-	$H_2C=CBrCH_3$	557-93-7	120.98	48	−125
–, 1-Chloro-, *cis-*	$ClCH=CHCH_3$		76.53	33	−135
–, 1-Chloro-, *trans-*	$ClCH=CHCH_3$		76.53	37	−99
–, 2-Chloro-	$H_2C=CClCH_3$	557-98-2	76.53	22.5	−139
–, 1-Chloro-2-methyl- (isocrotyl chloride)	$ClCH=C(CH_3)_2$	513-37-1	90.55	68	
–, 3-Chloro-2-methyl- (methallyl chloride)	$H_2C=C(CH_3)CH_2Cl$	563-47-3	90.55	71	−80
–, 1,1-Dichloro-	$Cl_2C=CHCH_3$	563-58-6	110.97	76	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.120	1.4870	>110	UY5715000	**1**(1),923B	**1**(1),1285B	**1**(1),788D	**11**,8027	**21**,236	
			CM3675000	**1**(1),927G	**1**(1),1293B	**1**(1),792B	**11**,1762	**21**(2),216	
			CL6940000	**1**(1),929A	**1**(1),1296A	**1**(1),794A			
		>110	CN4810000	**1**(1),927M	**1**(1),1295B	**1**(1),793C		**21**,242	
				1(1),923A	**1**(1),1285A			**21**(3),3141	
		>110	TO0110000	**1**(1),927E	**1**(1),1292C	**1**(1),791D		**21**,238	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.120	1.4360	98	LU3500000	**1**(1),803J	**1**(1),1127A	**1**(1),697D	**11**,1596	**17**,234	**1**,101
1.030	1.4630	109	MO8400000	**1**(1),819E	**1**(1),1161C	**1**(1),699B		**17**(2),290	
1.146	1.4120	70	RQ7350000	**1**(1),803A			**11**,7832	**17**(1),130	**1**,957
1.079	1.4580	100		**1**(1),815A	**1**(1),1152A	**1**(1),704D		**17**,235	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.590			OM9625000	**1**(1),5510	**1**(1),787A	**1**(1),503A	**11**,5585	**2**,748	
1.1741									
1.277									
1.073	1.4690	>110	ON0700000	**1**(1),725K	**1**(1),993A			**2**(3),1926	
0.9741	1.4475								
0.988	1.4452	>110	ON0875000	**1**(1),725J	**1**(1),992C			**2**(3),1925	
1.064	1.4410	93	ON1225000	**1**(1),725I	**1**(1),992B	**1**(1),644C	**11**,3113	**2**,751	
0.9714	1.4459								
1.152	1.4410	91	EM6300000	**1**(1),725H	**1**(1),992A	**1**(1),644B		**2**,751	
1.0245	1.4433								
			GE6650000	**1**(1),553B	**1**(1),787C	**1**(1),503C	**11**,2323	**2**,768	
1.0491	1.4468								
	1.9491	1.4473							

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
			UC6740000	**1**(1),15B			**11**,7862	**1**,196	
1.423	1.4545	−34		**1**(1),95F	**1**(1),134A			**1**(3),710	
1.408	1.4530	−15		**1**(1),95C				**1**(4),754	**4**,236
1.362	1.436	4	UC7085000	**1**(1),95H	**1**(1),134C	**1**(1),96B		**1**,200	
0.9347	1.4055								
0.9351	1.4054								
0.899	1.3939	−34	UC7200000	**1**(1),95D	**1**(1),133B	**1**(1),96A		**1**,198	
0.920	1.4225	−1	UC8045000	**1**(1),97I	**1**(1),137C	**1**(1),97A	**11**,2147	**1**,209	
0.917	1.4278	−12	UC8050000	**1**(1),97J	**1**(1),138A	**1**(1),96D	**11**,2148	**1**,209	
1.169	1.4450	0	UC8290000	**1**(1),101G	**1**(1),144A	**1**(3),138D		**1**,199	

TABLE 19. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
–, 1,2-Dichloro-	$ClCH=CClCH_3$		110.97	77	
–, Hexafluoro-	$F_2C=CFCF_3$	116-15-4	150.02	–28	–153
–, 2-Methyl- (isobutylene)	$H_2C=C(CH_3)_2$	115-11-7	56.11	–7	–140
–, 1,1,2-Trichloro-	$Cl_2C=CClCH_3$		145.42	118	
–, 1,1,3-Trichloro-	$Cl_2C=CHCH_2Cl$		145.42	131	
–, 1,2,3-Trichloro-	$ClCH=CClCH_2Cl$		145.42	142	
–, 3,3,3-Trichloro-	$H_2C=CHCCl_3$		145.42	114	

TABLE 20. STYRENES

Name	Formula $CH_2=CHR$ R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Styrene	$-C_6H_5$	100-42-5	104.15	145	–30
–, 4-Acetoxy-	$-C_6H_4OC(O)CH_3$	2628-16-2	162.19	260	7
–, 2-Amino-	$-C_6H_4NH_2$	3867-18-3	119.18	104/8	
–, 4-Amino-	$-C_6H_4NH_2$	1520-21-4	119.18	249	23
–, 4-Benzyloxy-3-methoxy-	$-C_6H_3(OCH_3)OCH_2C_6H_5$	55708-65-1	240.30	51	
–, 3,5-Bis(trifluoromethyl)-	$-C_6H_4(CF_3)_2$	349-59-7	240.15	60/20	7
α-Bromo-	$H_2C=CBrC_6H_5$	98-81-7	183.05	69/4	–44
β-Bromo-	$HCBr=CHC_6H_5$	103-64-0	183.05	111/20	7
–, 2-Bromo-	$-C_6H_4Br$	2039-88-5	183.05	210	–53
–, 3-Bromo-	$-C_6H_4Br$	2039-86-3	183.05	74/3	
–, 4-Bromo-	$-C_6H_4Br$	2039-82-9	183.05	212	4.5
–, 4-*tert*-Butyl-	$-C_6H_4C(CH_3)_3$	1746-23-2	160.26	219	–37
–, 4-Carboxy-	$-C_6H_4CO_2H$	1075-49-6	148.16		143
α-Chloro-	$H_2C=CClC_6H_5$		138.60	199	–23
β-Chloro-	$HCCl=CHC_6H_5$		138.60	199	
–, 2-Chloro-	$-C_6H_4Cl$	2039-87-4	138.60	188	–63
–, 3-Chloro-	$-C_6H_4Cl$	2039-85-2	138.60	62/6	
–, 3-Chloromethyl (3-vinylbenzyl chloride)	$-C_6H_4CH_2Cl$	57458-41-0	152.62	229	
–, 4-Chloro-	$-C_6H_4Cl$	1073-67-2	138.60	192	–16
–, 4-Chloromethyl-	$-C_6H_4CH_2Cl$	1592-20-7	152.62	229	
–, 4-Chloro-α-Methyl-	$H_2C=C(CH_3)C_6H_4Cl$	1712-70-5	152.62		
–, 2-Cyano-	$-C_6H_4CN$		129.16	53/0.15	
–, 3-Cyano-	$-C_6H_4CN$	5338-96-5	129.16	83/2.6	
–, 4-Cyano-	$-C_6H_4CN$	3435-51-6	129.16	92/3	–15
–, 2,5-Dichloro-	$-C_6H_3Cl_2$	1123-84-8	173.04	74/3	
–, 2,6-Dichloro-	$-C_6H_3Cl_2$	28469-92-3	173.04	88/8	
–, 3,4-Dichloro-	$-C_6H_3Cl_2$	2039-83-0	173.04	76/3	
–, α,β-Difluoro-	$HCF=CFC_6H_5$		140.03	88/60	
–, 1,2-Difluoro-	$-C_6H_3F_2$	366-37-0	140.03	90/45	
–, 2,6-Difluoro-	$-C_6H_3F_2$		140.03	59/51	
–, 1,3-Diisopropenyl benzene	$C_6H_4(C(CH_3)=CH_2)_2$	3748-13-8	158.25	231	
–, 3,4-Dimethoxy-	$-C_6H_3(OCH_3)_2$	6380-23-0	164.20	120/10	
–, α,2-dimethyl-	$H_2C=C(CH_3)C_6H_4CH_3$	26444-18-8	132.21	54/11	
–, 2,4-Dimethyl-	$-C_6H_3(CH_3)_2$	2234-20-0	132.21	70/11	–64
–, 2,5-Dimethyl-	$-C_6H_3(CH_3)_2$	2039-89-6	132.21	71/10	–35
–, Divinyl-, (isomers)	$C_6H_4(CH=CH_2)_2$	1321-74-0	130.19		
–, 4-Ethoxy-	$-C_6H_4OCH_2CH_3$	5459-40-5	148.21	59/1	
–, 2-Ethyl-	$-C_6H_4CH_2CH_3$	7564-63-8	132.21	191	–127
–, 4-Ethyl-	$-C_6H_4CH_2CH_3$	3454-07-7	132.21	193	–50
–, 3-Iodo-	$-C_6H_4I$		233.05		73
–, 2-Fluoro-	$-C_6H_4F$	394-46-7	122.14	67/15	
–, 3-Fluoro-	$-C_6H_4F$	350-51-6	122.14	30/4	
–, 4-Fluoro-	$-C_6H_4F$	405-99-2	122.14	30/14	
–, 2-Hydroxy-	$-C_6H_4OH$		120.15	108/15	29
–, 3-Hydroxy-	$-C_6H_4OH$		120.15	115	
–, 4-Hydroxy-	$-C_6H_4OH$		120.15		73.5
–, 4-Isopropyl-	$-C_6H_4CH(CH_3)_2$	2055-40-5	146.22	204	–45

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.1818	1.4471								
			UD0350000	1(1),105A		1(3),143C		1(3),697	
			UD0890000	1(1),23F		1(3),28B	11,5024	1,207	
1.382	1.4827								
1.403	1.4950	48	UD1928500	1(1),103H	1(1),148A	1(3),142A		1(3),707	
1.414	1.5020								
1.369	1.4827								

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
0.9059	1.5470	31	WL3675000	1(1),1139A	1(2),23A	1(1),945B	11,8830	5,474	
1.060	1.5380	87	SL3784000	1(2),1927F	1(2),1287C			6(3),2387	
1.608									
1.014	1.6267							12,1187	
		>110		1(1),1237M	1(2),197C	1(1),1053A			
1.334	1.4250	122		1(1),1213A	1(2),155A			5(4),1386	
1.410	1.5880	87	WL3840000	1(1),1203F				5,477	
1.427	1.6070	102	WL3850000	1(1),1203H	1(2),136C	1(1),1017B		5,477	
1.460	1.5927	85		1(1),1203L	1(2),138A	1(1),1018A		5(3),1176	
1.406	1.5891	67		1(1),1205D	1(2),139C	1(3),195B		5(3),1176	
1.400	1.5940	75		1(1),1205H	1(2),141A	1(1),1019C		5(2),367	
0.875	1.5260	80						5(3),1254	
				1(2),1789D	1(2),1076A	1(2),194D		9(3),2755	
1.1016/18	1.5612								
1.1095/15	1.5648								
1.080	1.5648	58	WL4160000	1(1),1203K	1(2),137C	1(1),1017D			
1.090	1.5613	62		1(1),1205C	1(2),139B	1(1),1018C			
1.074		105							
1.155	1.5662	60		1(1),1205F	1(2),140B	1(1),1019B		5(2),367	
1.083	1.5740	104							
1.065	1.5550	74		1(1),1205G	1(2),140C	1(1),1018D			
	1.5756								
1.083	1.5630	104							
	1.575								
1.267	1.5740	71		1(1),1205J	1(2),141C	1(1),1019D			
	1.5061								
1.130	1.4990	30		1(1),1205I	1(2),141B				
0.925	1.5570	91	CY8535000	1(1),1139L	1(2),26B	1(1),947C			
1.109	1.5710	>110		1(1),1243D	1(2),208C	1(1),1053D		6,954	
0.894	1.5150	43							
0.906	1.5390	60	WL4450000	1(1),1141D	1(2),27C			5,491	
0.904	1.5391	63	WL4460000	1(1),1141E	1(2),28A			5,491	
0.914	1.5470	64	CZ9370000	1(1),1141B					
0.990	1.5498	87		1(1),1243A		1(3),984B		6,561	
0.8955	1.5351								
0.8925	1.5376								
	1.6390								
1.025	1.5200	35		1(1),1203J	1(2),137B			5(3),1171	
1.025	1.5175	29		1(1),1205B	1(2),139A	1(1),1081B		5(3),1171	
1.024	1.5156	26		1(1),1205E	1(2),140A	1(1),1019A			
1.0609/18	1.5783/27								
1.0468/35	1.5804/31								
0.885	1.5289								

TABLE 20. *cont'd*

Name	Formula CH_2=CHR: R	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
–, 2-Methoxy-	$-C_6H_4OCH_3$	612-15-7	134.18	61.3	29
–, 3-Methoxy-	$-C_6H_4OCH_3$	626-20-0	134.18	144/2	
–, 4-Methoxy-	$-C_6H_4OCH_3$	637-69-4	134.18	204	
–, 2-Methoxy-4-hydroxy-	$-C_6H_3OCH_3(OH)$	7786-61-0	150.18	224	
α-Methyl-	$H_2C=C(CH_3)C_6H_5$	98-83-9	118.18	167	– 24
–, 2-Methyl-	$-C_6H_4CH_3$	611-15-4	118.18	171	– 69
–, 3-Methyl-	$-C_6H_4CH_3$	100-80-1	118.18	170	– 82
–, 4-Methyl-	$-C_6H_4CH_3$	622-97-9	118.18	172	– 34
–, β-Nitro-	$HC(NO_2)=CHC_6H_5$		149.15	250	58
–, 2-Nitro-	$-C_6H_4NO_2$		149.15		13
–, 3-Nitro-	$-C_6H_4NO_2$	586-39-0	149.15	90/3.5	– 5
–, 4-Nitro-	$-C_6H_4NO_2$	100-13-0	149.15	120/10	25
–, Pentabromo-	$-C_6Br_5$	53097-59-9	498.66		188
–, Pentafluoro-	$-C_6F_5$	653-34-9	194.10	139	
–, 4-Phenoxy-	$-C_6H_4OC_6H_5$	4973-29-9	196.25		71
–, 4-Phenyl-	$-C_6H_4C_6H_5$	2350-89-2	180.25		120
–, 4-Sulfonic acid, sodium salt	$-C_6H_4SO_3Na$	2695-37-6	206.20		225
–, 2-Trifluoromethyl-	$-C_6H_4CF_3$	395-45-9	172.15	61/40	
–, 3-Trifluoromethyl-	$-C_6H_4CF_3$	402-24-4	172.15	55/12	
–, 4-Trifluoromethyl-	$-C_6H_4CF_3$	402-50-6	172.15	65/40	
–, 2,4,6-Trimethyl-	$-C_6H_2(CH_3)_3$	769-25-5	146.24	209	

TABLE 21. VINYL FUNCTIONAL

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
21.1. ARYL H_2C=CHR					
9-Anthracene	$-C_{14}H_9$	2444-68-0	204.27	62/10	65
Benzene (see Styrene)					
2-Naphthalene	$-C_{10}H_7$	827-54-3	154.21		61
2-Pyridine	$-C_5H_4N$	100-69-6	105.14	80/29	
3-Pyridine	$-C_5H_4N$	1121-55-7	105.14	68/15	
4-Pyridine	$-C_5H_4N$	100-43-6	105.14	63/15	
2-Quinoline	$-C_9H_6N$	772-03-2	155.20	71/0.5	
21.2. ESTERS H_2C=CHOC(O)R					
Acetate	$-CH_3$	108-05-4	86.09	72	– 93
Acrylate (see Acrylates)					
Allyl adipate	$-(CH_2)_4C(O)OCH_2CH=CH_2$	56860-96-9	212.25	135/9	
Benzoate	$-C_6H_5$	769-78-8	148.16	203	
Bromoacetate	$-CH_2Br$	5309-70-6	165.0	50/10	
4-*tert*-Butyl benzoate	$-C_6H_4C(CH_3)_3$	15484-80-7	204.27	111/2	
Butyrate	$-(CH_2)_2CH_3$	123-20-6	114.14	116	
Carbonate	$-OCH=CH_2$	872-36-6	86.05	162	22
Chloroacetate	$-CH_2Cl$	2549-51-1	120.54	134	
2-Chlorobenzoate	$-C_6H_4Cl$	15721-27-4	182.61		
Chloroformate	$-Cl$	5130-24-5	106.51	68/735	
Cinnamate	$-CH=CHC_6H_5$	3098-92-8	174.20	125/7	
Crotonate	$-CH_2CH=CHCH_3$	14861-06-4	112.13	133	
Decanoate	$-(CH_2)_8CH_3$	4704-31-8	198.31	146/50	
Dodecanoate	$-(CH_2)_{10}CH_3$	2146-71-6	226.36	140/10	
2-Ethylhexanoate	$-CH(CH_2CH_3)(CH_2)_3CH_3$	94-04-2	170.25	128/20	– 90
Formate	$-H$	692-45-5	72.06	46	
2-Furoate	$-C_4H_3O$	1917-10-8	138.12	182/10	14
Hexadecanoate (palmitate)	$-(CH_2)_{14}CH_3$	693-38-9	282.47	165/2	24
Hexanoate (caproate)	$-(CH_2)_4CH_3$	3050-69-9	142.20	160/10	
Maleate	$-CH=CHC(O)OCH=CH_2$		168.15	94/10	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.0049	1.5388								
0.9999	1.5540								
0.994	1.5620	72		1(1),1241O	1(2),207C	1(1),1052C		6,561	
1.110	1.5820	>110	SL8205000					6(3),4981	
0.909	1.5380	45	WL5075300	1(1),1139B	1(2),2437C	1(1),945C		5,484	
0.917	1.544	58	WL5075900	1(1),1139K		1(1),947A		5(1),233	
0.890	1.5408	52	WL5075800	1(1),1139M	1(2),26C	1(3),869B		5,485	
0.897	1.5412	45	WL5076000	1(1),1139N	1(2),27A	1(1),947B		5,485	
1.070	1.5840	107		1(1),1555D	1(2),687C	1(1),1334D		5,478	
	1.6016		WL5470000						
				1(1),1205N		1(1),1020B			
1.406	1.4455	34		1(1),1205K	1(2),142A	1(1),1020A			
				1(1),1145G	1(2),35B	1(1),951D			
				1(2),2207I	1(2),1598C				
1.175	1.4700	42		1(1),1207C	1(2),143A			5(4),1367	
1.161	1.4655			1(1),1207K	1(2),138C			5(3),1197	
1.165	1.4660			1(1),1209F	1(2),148A			5(4),1369	
0.906	1.5320	75		1(1),1141G		1(3),871A		5,500	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
				1(1),1153I	1(2),51A	1(1),963B			
				1(1),1151G	1(2),46A	1(1),959C			
0.975	1.5490	47	UU1040000	1(2),2495G	1(3),242C	1(2),734A		20,256	
	1.5530								
0.975	1.5500	52	UU1045000	1(2),2499B	1(3),248B	1(2),738D		20(2),170	
1.340		128				1(2),853D		20,425	
0.934	1.3950	−6	AK0875000	1(1),715A	1(1),963A	1(1),632D	11,9896	2(1),63	12,565
1.070	1.5290	82	DI1050000					9(1),65	
0.999	1.5180	>110		1(2),1903L	1(2),1248C			9(4),1885	
0.9022	1.411	20						2(4),792	
1.355	1.4210	72	FG3325000	1(1),697E	1(1),933A	1(1),644A			13,342
1.19	1.4440								
1.22									
1.253	1.4100	−4		1(1),863A	1(1),1206A	1(1),736D		3(3),28	8,530
1.07									
0.940	1.4480	27	GQ5850000	1(1),721H	1(1),978C	1(1),642B		2(3),1263	
0.886	1.4350	104						2(3),843	
0.8639	1.4387	136							
0.875	1.4256	65	MO7875000	1(1),719A	1(1),793A	1(1),637C		2(4),1005	
0.9651	1.4757		LR0525000		1(1),1206A	1(1),736C		3(3),28	8,530
0.8602	1.4444								
0.8837	1.4159								

TABLE 21. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
Maleate, monoester	$-CH=CHC(O)OH$	19896-47-0	142.11	80/1	
Methacrylate (see Methacrylates)					
Neodecanoate	$-C_9H_{19}$	51000-52-3	198.31	216	
Nonate	$-(CH_2)_7CH_3$	6280-03-1	184.28	133/50	
Octadecanoate (stearate)	$-(CH_2)_{16}CH_3$	111-63-7	310.51	181/2	32
9-Octadecenoate (oleate)	$-(CH_2)_7CH=CH(CH_2)_7CH_3$	3896-58-0	308.49	145	36
Octanoate	$-(CH_2)_6CH_3$	818-44-0	170.25	79/7	
Pentafluoropropionate	$-CF_2CF_3$		190.08	58/560	
Pentanoate	$-(CH_2)_3CH_3$	5873-43-8	128.17	132	
Pivalate	$-C(CH_3)_3$	3377-92-2	128.17	112	
Propionate	$-CH_2CH_3$	105-38-4	100.12	95	−81
Sebacate	$-(CH_2)_8C(O)OCH=CH_2$	10355-50-7	254.33	142/3	
Succinate	$-CH_2CH_2C(O)OCH=CH_2$	13416-90-5	170.16	50/3	
Tetradecanoate (myristate)	$-(CH_2)_{12}CH_3$	5809-91-6	254.41	152/3	15
Thioacetate	$H_2C=CHSC(O)CH_3$	10340-63-3	102.15	115	
Trichloroacetate	$-CCl_3$	7062-87-5	189.43	38/5	
Trifluoroacetate	$-CF_3$	433-28-3	140.06	42	
1-Trifluoromethyl acetate	$H_2C=C(CF_3)OC(O)CH_3$	2247-91-8	154.09	86	
Trithiocarbonate	$H_2C=CHSC(S)SCH=CH_2$	930-35-8	134.24		48
Undecanoate	$-(CH_2)_9CH_3$		212.33	125/8	
Versatate	$-C(CH_3)_2(CH_2)_5CH_3$	49863-74-3	198.31	60/1	
21.3. ETHERS $H_2C=CHOR$					
Allyl	$-CH_2CH=CH_2$	3917-15-5	84.12	67	
Butoxyethyl	$-CH_2CH_2O(CH_2)_3CH_3$	4223-11-4	144.21	70/20	
n-Butyl	$-(CH_2)_3CH_3$	111-34-2	100.16	94	−92
2-Butyl	$-CH(CH_2CH_3)CH_3$	4181-12-8	100.16	87	
tert-Butyl	$-C(CH_3)_3$	926-02-3	100.16	75	
1-Chloroethyl	$-CHClCH_3$	110-75-8	106.55	109	
2-Chloroethyl	$-CH_2CH_2Cl$	110-75-8	106.55	109	−70
Cyclohexyl	$-C_6H_{11}$	2182-55-0	126.20	147	−109
1,2-Dichloroethyl	$-CHClCH_2Cl$		141.00	128	
Di(ethylene glycol)	$-CH_2CH_2OCH_2CH_2OH$	929-37-3	132.16	196	
Divinyl	$-CH=CH_2$	109-93-3	70.09	28	−101
Dodecyl	$-(CH_2)_{11}CH_3$	765-14-0	212.38	118	−12
Ethyl	$-CH_2CH_3$	109-92-2	72.11	33	−116
2-Ethylhexyl	$-CH_2CH(CH_2CH_3)(CH_2)_3CH_3$	103-44-6	156.27	177	−85
Glycidyl	$-CH_2CH(-O-)CH_2$	3678-15-7	100.12	139	
Hexachlorodivinyl	$(Cl_2C=CCl)_2O$		276.76	210	
Hexadecyl	$-(CH_2)_{15}CH_3$	822-28-6	268.47	173/5	16
4-Hydroxybutyl	$-(CH_2)_4OH$	17832-28-9	116.16	189	−33
2-Hydroxyethyl	$-CH_2CH_2OH$	764-48-7	88.11	143	
Isoamyl	$-CH_2CH_2CH(CH_3)_2$		114.19	112	
Isobutyl	$-CH_2CH(CH_3)_2$	109-53-5	100.16	82	−112
Isooctyl	$-(CH_2)_5CH(CH_3)_2$	37769-62-3	156.27	175	
Isopropyl	$-CH(CH_3)_2$	926-65-8	86.13	83	−140
2-(2-Methoxyethoxy)ethyl	$-(CH_2CH_2O)_2CH_3$	26256-87-1	190.24	242	
2-Methoxyethyl	$-CH_2CH_2OCH_3$	1663-35-0	102.13	108	−83
Methyl	$-CH_3$	107-25-5	58.08	5	−123
Octadecyl	$-(CH_2)_{17}CH_3$	930-02-9	296.54	178/5	27
Perfluoropropyl	$F_2C=CFOCF_2CF_2CF_3$	1623-05-8	266.04	36	
Phenyl	$-C_6H_5$	766-94-9	120.15	154	
1-Phenylethyl	$-CH(C_6H_5)CH_3$	6230-62-2	148.21	211	
Propyl	$-CH_2CH_2CH_3$	764-47-6	86.13	65	−50
21.4. *N*-SUBSTITUTED $H_2C=CHR$					
–Caprolactam	$-NC_6H_{10}O$	2235-00-9	139.20	128/21	36
–Carbazole	$-NC_{12}H_8$	1484-13-5	193.25	154/3	65
–Formamide	$-NHC(O)H$	13162-05-5	71.08	210	−16
–Phthalimide	$-NC_8H_4O_2$	3485-84-5	173.17		85
–Pyrolidone	$-NC_4H_6O$	88-12-0	111.14		
–*N*-Methylacetamide	$-N(CH_3)C(O)CH_3$	3195-78-6	99.10	166	−36

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
								2(3),1290	
0.882	1.4360	83		**1**(1),719B				**2**(4),1045	
0.8689	1.4291								
0.904		>110							
		>110							
0.8719	1.4256								
0.866	1.4050	10							
0.9173	1.4030	6	UF8575000					**2**(3),532	
0.87									
1.203	1.3170	−31		**1**(1),745G	**1**(1),1030C	**1**(3),704B		**2**(4),466	
1.212	1.3410	2							
		>110		**1**(1),697F	**1**(1),933B			**19**(3),1598	
0.8	1.4109	<20							
0.866	1.4220	46	KH7175000					**1**(4),2387	
0.774	1.410	−9	KN5950000	**1**(1),219C	**1**(1),332B	**1**(1),213D			
	1.4558								
0.762	1.3980	−17						**1**(3),1863	
1.048	1.4380	16	KN6300000	**1**(1),221C	**1**(1),336A	**1**(1),213B	**11**,2139	**1**(2),473	
1.048	1.4380	16	KN6300000	**1**(1),221C	**1**(1),336A	**1**(1),213B	**11**,2139	**1**(2),473	
0.891	1.4540	35						**6**(3),18	
1.197	1.4558/17							**1**(4),2398	
0.986	1.4480	83	KM5495500					**1**(4),2398	
0.773	1.3989	−30							
0.817	1.4382	>110							
0.754	1.3770	−45	KO0710000	**1**(1),219B	**1**(1),332A	**1**(1),212B		1,433	**11**,235
0.816	1.4280	52	KO0175000					**1**(3),1864	
1.006	1.4326	35						**17**(5),8,12	
1.654									
0.939	1.4440	85						**1**(4),2518	
0.982	1.4360	48	KM5495000					**1**(2),520	
0.7826	1.4072								
0.7645	1.3950	−13	KO1300000	**1**(1),219D		**1**(3),278A		**1**(3),1862	
0.7534	1.3840	−32							
0.990	1.4390								
0.8967	1.4072	17							
0.7511/0	1.3730/0	−56	KO2300000	**1**(1),219A		**1**(3),277C		**1**(3),1857	1,697
0.821	1.440	177	RG0300000					**1**(4),2057	
1.53									
	1.5226								
	1.3908	−26						**1**(3),1859	
1.029		101						**2**(3),3207	
			FE6350000	**1**(2),2425E	**1**(3),161C			**20**(2),282	
1.014	1.4940	102							
								21(1),363	
1.040	1.5120	94	UY6107000	**1**(1),923K	**1**(1),1288A	**1**(1),790B			
0.9600	1.4835	60	AC6475000	**1**(1),891K	**1**(1),1242C	**1**(1),759C		**4**(3),442	

TABLE 21. *cont'd*

Name	Formula	CAS Registry Number	Mol. wt.	Boiling point (°C)	Melting point (°C)
21.5. SULFONATES $H_2C=CHSO_3R$					
Ethylene sulfonic acid	$-H$		108.10	100/0.5	
n-Amyl ester	$-(CH_2)_4CH_3$		178.23	131/17	
n-Butyl ester	$-(CH_2)_3CH_3$		164.21	117/15	
Ethyl ester	$-CH_2CH_3$		136.16	76/5	
n-Hexyl ester	$-(CH_2)_5CH_3$		192.26	146/15	
Isoamyl ester	$-(CH_2)_2CH(CH_3)_2$		178.23	124/15	
Isobutyl ester	$-CH_2CH(CH_3)_2$		164.21	78/5	
Isopropyl ester	$-CH(CH_3)_2$		152.20	70/4	
Methyl ester	$-CH_3$		122.13	91/15	
Phenyl ester	$-C_6H_5$	1562-34-1	184.21	45/2	
n-Propyl ester	$-CH_2CH_2CH_3$		152.20	110/18	

Density (at 20°C)	Refractive index (at 20°C)	Flash point (°C)	RTECS Number	R and S	FT-NMR	FT-IR	Merck	Beilstein	Fieser
1.4003	1.4493								
1.087	1.4412								
1.122	1.4416								
1.183	1.431								
1.050	1.4430								
1.082	1.4415								
1.190	1.426								
1.132	1.4321								
1.248	1.4316								
1.165	1.426	>110		**1**(2),2219A	**1**(2),1605A	**1**(3),1437A		**6**(3),651	**13**,241
1.156	1.4368								

Isorefractive and Isopycnic Solvent Pairs

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A. INTRODUCTION

Isorefractive solvents are solvents having the same refractive index, and isopycnic solvents are those that have the same density.

The determination of molar masses by non-colligative methods such as light scattering and ultracentrifugation will lead only to *apparent* molar masses if solvent mixtures are used whose components are not isorefractive. The observed increases or decreases of these apparent molar masses depend in sign and magnitude on both the preferential solvation and the refractive index increment of solvent 2 in solvent 1 at a fixed polymer concentration. The effect disappears if isorefractive solvent pairs are used.

A similar effect may be observed in ultracentrifugal experiments with non-isopycnic solvent pairs. To suppress these effects, isopycnic-isorefractive solvent pairs should be used. A table of isorefractive and isopycnic solvent pairs was prepared (for 25°C), starting with 392 commonly used solvents. A solvent pair was classified as isorefractive and isopycnic if the components had differences no greater than ± 0.002 in refractive index and ± 0.015 g/ml in density. The miscibility of the components was not checked.

B. TABLE OF ISOREFRACTIVE AND ISOPYCNIC SOLVENT PAIRS

		Refractive index		Density (g/ml)	
Solvent 1	**Solvent 2**	**1**	**2**	**1**	**2**
Acetone	Ethanol	1.357	1.359	0.788	0.786
Ethyl formate	Methyl acetate	1.358	1.360	0.916	0.935
	Propionitrile	1.359	1.363	0.786	0.777
2,2-Dimethylbutane	2-Methylpentane	1.366	1.369	0.644	0.649
2-Methylpentane	*n*-Hexane	1.369	1.372	0.649	0.655
	2,3-Dimethylbutane	1.369	1.372	0.649	0.657
	3-Methylpentane	1.369	1.374	0.649	0.660
2,3-Dimethylbutane	*n*-Hexane	1.372	1.372	0.657	0.655
	3-Methylpentane	1.372	1.374	0.657	0.660
n-Hexane	3-Methylpentane	1.372	1.374	0.655	0.660
Isopropyl acetate	2-Chloropropane	1.375	1.376	0.868	0.865
2-Butanone	Butyraldehyde	1.377	1.378	0.801	0.799
Butyraldehyde	Butyronitrile	1.378	1.382	0.799	0.786
n-Propyl ether	*n*-Butyl ethyl ether	1.379	1.380	0.753	0.746
2,4-Dimethylpentane	2-Methylhexane	1.379	1.382	0.799	0.786
Acetaldehyde diethyl acetal	*n*-Butyl ethyl ether	1.379	1.380	0.753	0.746
n-Propyl acetate	Ethyl propionate	1.382	1.382	0.883	0.888
	Isobutyl formate	1.382	1.383	0.883	0.881
	1-Chloropropane	1.382	1.386	0.883	0.890
Butyronitrile	*t*-Butanol	1.382	1.385	0.786	0.781
Ethyl propionate	Isobutyl formate	1.382	1.383	0.888	0.881
	1-Chloropropane	1.382	1.386	0.888	0.890
2-Methylhexane	*n*-Heptane	1.382	1.385	0.674	0.680
	3-Methylhexane	1.382	1.386	0.674	0.683

		Refractive index		Density (g/ml)	
Solvent 1	**Solvent 2**	**1**	**2**	**1**	**2**
n-Propanol	3-Methyl-2-butanone	1.383	1.386	0.806	0.807
	2-Pentanone	1.383	1.387	0.806	0.804
Isobutyl formate	1-Chloropropane	1.383	1.386	0.881	0.890
	s-Butyl acetate	1.383	1.387	0.881	0.868
	n-Butylformate	1.383	1.387	0.881	0.888
Diethylamine	*n*-Propylamine	1.384	1.386	0.702	0.713
n-Heptane	3-Methylhexane	1.385	1.386	0.680	0.683
	2,3,3-Trimethylbutane	1.385	1.387	0.680	0.686
	2,2,4-Trimethylpentane	1.385	1.389	0.680	0.687
	2,3-Dimethylpentane	1.385	1.389	0.680	0.691
3-Methylhexane	2,3,3-Trimethylbutane	1.386	1.387	0.683	0.686
	2,4-Trimethylpentane	1.386	1.389	0.683	0.687
	2,3-Dimethylpentane	1.386	1.389	0.683	0.691
1-Chloropropane	Butyl formate	1.386	1.387	0.890	0.888
3-Methyl-2-butanone	2-Pentanone	1.386	1.390	0.807	0.802
	3-Pentanone	1.386	1.390	0.807	0.810
n-Propylamine	Diisopropylamine	1.386	1.390	0.713	0.712
	s-Butylamine	1.386	1.390	0.713	0.720
2,3,3-Trimethylbutane	2,2,4-Trimethylpentane	1.387	1.389	0.686	0.683
	2,3-Dimethylpentane	1.387	1.389	0.686	0.691
s-Butyl acetate	Methyl butyrate	1.387	1.391	0.868	0.875
n-Butyl formate	*n*-Dodecane	1.387	1.391	0.888	0.775
Isobutyl acetate	Methyl butyrate	1.388	1.391	0.871	0.875
	n-Butyl acetate	1.388	1.392	0.871	0.877
2,2,4-Trimethylpentane	2,3-Dimethylpentane	1.389	1.389	0.687	0.691
Diisopropylamine	*s*-Butylamine	1.390	1.390	0.712	0.720
2-Pentanone	3-Pentanone	1.390	1.390	0.802	0.810
	4-Methyl-2-pentanone	1.390	1.394	0.802	0.797
	2-Methyl-1-propanol	1.390	1.394	0.802	0.798
3-Pentanone	4-Methyl-2-pentanone	1.390	1.394	0.810	0.797
	2-Methyl-1-propanol	1.390	1.394	0.810	0.798
Methyl butyrate	*n*-Butyl acetate	1.391	1.392	0.875	0.877
	2-Chlorobutane	1.391	1.395	0.875	0.868
2-Chloro-2-methyl-propane	2-Chlorobutane	1.392	1.395	0.872	0.868
n-Butyl acetate	2-Chlorobutane	1.392	1.395	0.877	0.868
4-Methyl-2-pentanone	2-Methyl-1-propanol	1.394	1.394	0.797	0.798
	Valeronitrile	1.394	1.395	0.797	0.795
	2-Butanol	1.394	1.395	0.797	0.803
	2-Hexanone	1.394	1.395	0.797	0.810
	1-Butanol	1.394	1.397	0.797	0.812
	Methacrylonitrile	1.394	1.398	0.797	0.795
	3-Methyl-2-pentanone	1.394	1.398	0.797	0.808
2-Methyl-1-propanol	Valeronitrile	1.394	1.395	0.798	0.795
	2-Butanol	1.394	1.395	0.798	0.803
	2-Hexanone	1.394	1.395	0.798	0.810
	Butanol	1.394	1.397	0.798	0.812
	Methacrylonitrile	1.394	1.398	0.798	0.795
	3-Methyl-2-pentanone	1.394	1.398	0.798	0.808
Octane	2,2,5-Trimethylhexane	1.395	1.397	0.698	0.703
2-Butanol	Butanol	1.395	1.397	0.803	0.812
	Methacrylonitrile	1.395	1.398	0.803	0.795
	3-Methyl-2-pentanone	1.395	1.398	0.803	0.808
	2,4-Dimethyl-3-pentanone	1.395	1.399	0.803	0.805
2-Hexanone	Butanol	1.395	1.397	0.810	0.812

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	Methacrylonitrile	1.395	1.398	0.810	0.795
	3-Methyl-2-pentanone	1.395	1.398	0.810	0.808
	2,4-Dimethyl-3-pentanone	1.395	1.399	0.810	0.805
Valeronitrile	Methacrylonitrile	1.395	1.398	0.795	0.795
	3-Methyl-2-pentanone	1.395	1.398	0.795	0.808
	2,4-Dimethyl-3-pentanone	1.395	1.399	0.795	0.805
2-Hexanone	3-Methyl-2-pentanone	1.395	1.398	0.810	0.808
Isobutylamine	Triethylamine	1.395	1.399	0.729	0.723
	n-Butylamine	1.395	1.399	0.729	0.736
2-Chlorobutane	Isobutyl *n*-butyrate	1.395	1.399	0.868	0.860
Butyric acid	2-Methoxyethanol	1.396	1.400	0.955	0.960
n-Butanol	3-Methyl-2-pentanone	1.397	1.398	0.812	0.808
	2,4-Dimethyl-3-pentanone	1.397	1.399	0.812	0.805
1-Chloro-2-methyl-propane	Isobutyl *n*-Butyrate	1.397	1.399	0.872	0.860
	Amyl acetate	1.397	1.400	0.872	0.871
	1-Chlorobutane	1.397	1.400	0.872	0.881
2,5,5-Trimethylhexane	2,2,3-Trimethylpentane	1.397	1.401	0.703	0.712
Methyl methacrylate	3-Methyl-2-pentanone	1.398	1.398	0.795	0.808
Methacrylonitrile	2,4-Dimethyl-3-pentanone	1.398	1.399	0.795	0.805
	2-Methyl-2-butanol	1.398	1.404	0.795	0.805
3-Methyl-2-pentanone	2,4-Dimethyl-4-pentanone	1.398	1.399	0.808	0.805
Triethylamine	*n*-Butylamine	1.399	1.399	0.723	0.736
	2,2,3-Trimethylpentane	1.399	1.401	0.723	0.712
	n-Nonane	1.399	1.401	0.723	0.714
	Dipropylamine	1.399	1.401	0.723	0.736
n-Butylamine	*n*-Dodecane	1.399	1.400	0.736	0.746
Isobutyl *n*-butyrate	*n*-Amyl acetate	1.399	1.400	0.860	0.871
	Isoamyl acetate	1.399	1.403	0.860	0.868
	1-Chlorobutane	1.399	1.401	8.860	0.875
1-Nitropropane	Propionic anhydride	1.399	1.400	0.995	1.007
Amyl acetate	1-Chlorobutane	1.400	1.400	0.871	0.881
	Tetrahydrofuran	1.400	1.404	0.871	0.885
n-Dodecane	Dipropylamine	1.400	1.403	0.746	0.736
	Cyclopentane	1.400	1.404	0.746	0.740
1-Chlorobutane	Tetrahydrofuran	1.400	1.404	0.871	0.885
2,2,3-Trimethylpentane	*n*-Nonane	1.401	1.403	0.712	0.714
Isovaleric acid	2-Ethoxyethanol	1.402	1.405	0.923	0.926
	Valeric acid	1.402	1.406	0.923	0.936
Dipropylamine	Cyclopentane	1.403	1.404	0.736	0.740
	Methylcyclopentane	1.403	1.407	0.736	0.744
n-Nonane	2,2,4-Trimethyl-1-pentene	1.403	1.407	0.714	0.712
Isoamylacetate	Tributyl borate	1.403	1.407	0.868	0.854
2-Pentanol	2-Methyl-1-butanol	1.404	1.404	0.804	0.805
	3-Methyl-1-butanol	1.404	1.404	0.804	0.805
	4-Heptanone	1.404	1.405	0.804	0.813
	2-Heptanone	1.404	1.406	0.804	0.811
2-Methoxybutanol	3-Methyl-1-butanol	1.404	1.404	0.805	0.805
	Capronitrile	1.404	1.405	0.805	0.801
	4-Heptanone	1.404	1.405	0.805	0.813
	2-Heptanone	1.404	1.406	0.805	0.811
	Pentanol	1.404	1.408	0.805	0.810
	3-Methyl-2-butanol	1.404	1.408	0.805	0.815
3-Methyl-1-butanol	Capronitrile	1.404	1.405	0.805	0.801
	4-Heptanone	1.404	1.405	0.805	0.813

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	2-Heptanone	1.404	1.406	0.805	0.811
	Pentanol	1.404	1.408	0.805	0.810
	3-Methyl-2-butanol	1.404	1.408	0.805	0.815
Cyclopentane	Methylcyclopentane	1.404	1.407	0.740	0.744
Capronitrile	4-Heptanone	1.405	1.405	0.801	0.813
	2-Heptanone	1.405	1.406	0.801	0.811
	2-Pentanol	1.405	1.407	0.801	0.804
	1-Pentanol	1.405	1.408	0.801	0.810
	3-Methyl-2-butanol	1.405	1.408	0.801	0.815
	4-Methyl-2-pentanol	1.405	1.409	0.801	0.802
	3-Isopropyl-2-pentanone	1.405	1.409	0.801	0.808
	2-Methyl-1-butanol	1.405	1.409	0.801	0.815
4-Heptanone	2-Heptanone	1.405	1.406	0.813	0.811
	1-Pentanol	1.405	1.408	0.813	0.810
	3-Methyl-2-butanol	1.405	1.408	0.813	0.815
	4-Methyl-2-pentanol	1.405	1.409	0.813	0.802
	3-Isopropyl-2-pentanone	1.405	1.409	0.813	0.808
	2-Methyl-1-butanol	1.405	1.409	0.813	0.815
2-Ethoxyethanol	Valeric acid	1.405	1.406	0.926	0.936
2-Heptanone	1-Pentanol	1.406	1.408	0.811	0.810
	3-Methyl-2-butanol	1.406	1.408	0.811	0.802
	4-Methyl-2-pentanol	1.406	1.409	0.811	0.802
	3-Isopropyl-2-pentanone	1.406	1.409	0.811	0.808
	2-Ethoxyethanol	1.406	1.409	0.811	0.815
	2-Methyl-1-butanol	1.406	1.409	0.811	0.815
	Amyl ether	1.406	1.410	0.811	0.799
2-Pentanol	1-Pentanol	1.407	1.408	0.804	0.810
	3-Methyl-2-butanol	1.407	1.408	0.804	0.815
	4-Methyl-2-pentanol	1.407	1.409	0.804	0.802
	3-Isopropyl-2-pentanone	1.407	1.409	0.804	0.808
	2-Methyl-1-butanol	1.407	1.409	0.804	0.815
	Amyl ether	1.407	1.410	0.804	0.799
2,2,4-Trimethyl-1-pentene	*n*-Decane	1.407	1.409	0.712	0.726
Tributyl borate	Isoamyl isovalerate	1.407	1.410	0.854	0.853
	Allyl alcohol	1.407	1.411	0.854	0.847
1-Pentanol	3-Methyl-2-butanol	1.408	1.408	0.810	0.815
	4-Methyl-2-pentanol	1.408	1.409	0.810	0.802
	3-Isopropyl-2-pentanone	1.408	1.409	0.810	0.808
	2-Methyl-1-butanol	1.408	1.409	0.810	0.815
	Amyl ether	1.408	1.410	0.810	0.799
3-Methyl-2-butanol	4-Methyl-2-pentanol	1.408	1.409	0.815	0.802
	3-Isopropyl-2-pentanone	1.408	1.409	0.815	0.808
	2-Methyl-1-butanol	1.408	1.409	0.815	0.815
	Amyl ether	1.408	1.410	0.815	0.799
4-Methyl-2-pentanol	3-Isopropyl-2-pentanone	1.409	1.409	0.802	0.808
	2-Methyl-1-butanol	1.409	1.409	0.802	0.815
	Amyl ether	1.409	1.410	0.802	0.799
3-Isopropyl-2-pentanone	2-Methyl-1-butanol	1.409	1.409	0.808	0.815
	Amyl ether	1.409	1.410	0.808	0.799
2-Methyl-1-butanol	Amyl ether	1.409	1.410	0.815	0.799
Isoamyl isovalerate	Allyl alcohol	1.410	1.411	0.853	0.847
Amyl ether	2-Octanone	1.410	1.414	0.799	0.814
2,4-Dimethyldioxane	Allyl chloride	1.412	1.413	0.935	0.932
	Caproic acid	1.412	1.415	0.935	0.923

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
Diethyl malonate	Ethyl cyanoacetate	1.412	1.415	1.051	1.056
Allyl chloride	Capric acid	1.413	1.415	0.932	0.923
2-Octanone	3-Methyl-2-heptanone	1.414	1.415	0.814	0.818
	1-Hexanol	1.414	1.416	0.814	0.814
	2-Pentanol	1.414	1.416	0.814	0.826
	Caprylnitrile	1.414	1.418	0.814	0.810
	2-Heptanol	1.414	1.418	0.814	0.818
3-Octanone	3-Methyl-2-heptanone	1.414	1.415	0.830	0.818
	2-Pentanol	1.414	1.416	0.830	0.826
3-Methyl-2-heptanone	1-Hexanol	1.415	1.416	0.818	0.814
	2-Pentanol	1.415	1.416	0.818	0.826
	Caprylonitrile	1.415	1.418	0.818	0.810
	2-Heptanol	1.415	1.418	0.818	0.818
1-Hexanol	2-Pentanol	1.416	1.416	0.814	0.826
	Caprylonitrile	1.416	1.418	0.814	0.810
	2-Heptanol	1.416	1.418	0.814	0.818
	3-Methyl-2-pentanol	1.416	1.420	0.814	0.823
	2-Ethyl-1-butanol	1.416	1.420	0.814	0.829
2-Pentanol	2-Heptanol	1.416	1.418	0.826	0.818
	3-Methyl-2-pentanol	1.416	1.418	0.816	0.818
	2-Ethyl-1-butanol	1.416	1.420	0.826	0.829
Dibutylamine	Allylamine	1.416	1.419	0.756	0.758
Caprylonitrile	2-Heptanol	1.418	1.418	0.810	0.818
	3-Methyl-2-pentanol	1.418	1.420	0.810	0.823
	1-Heptanol	1.418	1.422	0.810	0.818
2-Heptanol	2-Methyl-2-pentanol	1.418	1.420	0.818	0.823
	2-Ethyl-1-butanol	1.418	1.420	0.818	0.829
	1-Heptanol	1.418	1.422	0.818	0.818
	3-Isopropyl-2-heptanone	1.418	1.423	0.818	0.815
Allylamine	Methylcyclohexane	1.419	1.421	0.758	0.765
3-Methyl-2-pentanol	2-Ethyl-1-butanol	1.420	1.420	0.823	0.829
	1-Heptanol	1.420	1.422	0.823	0.818
	3-Isopropyl-2-heptanone	1.420	1.423	0.815	0.815
2-Ethyl-1-butanol	1-Heptanol	1.420	1.420	0.829	0.818
	3-Isopropyl-2-heptanone	1.420	1.423	0.829	0.815
Methylcyclohexane	Cyclohexane	1.421	1.424	0.765	0.774
1-Heptanol	3-Isopropyl-2-heptanone	1.422	1.423	0.818	0.815
3-Isopropyl-2-heptanone	1-Octanol	1.423	1.427	0.815	0.821
	3-Methyl-2-pentanone	1.423	1.427	0.815	0.824
3-Chloro-2-methyl-1-propene	Caprylic acid	1.425	1.426	0.917	0.905
Caprylic acid	*N*-Methylalaninenitrile	1.426	1.429	0.905	0.895
1-Octanol	3-Methyl-2-pentanol	1.427	1.427	0.821	0.824
1-Chlorooctane	1-Chloro-2-ethylhexane	1.428	1.430	0.867	0.872
2-Methyl-7-ethylnonane	2-Methyl-7-ethyl-4-undecanone	1.433	1.435	0.830	0.832
Butyrolactone	Chloro-*t*-butanol	1.434	1.436	1.051	1.059
	1,3-Propanediol	1.434	1.438	1.051	1.049
	Diethyl maleate	1.434	1.438	1.051	1.064
4-*n*-Propyl-5-ethyldioxane	*N*-methylmorpholine	1.435	1.436	0.927	0.924
2-Methyl-7-ethyl-4-undecanone	2-Methyl-7-ethyl-4-nonanol	1.435	1.438	0.832	0.829
	6-Ethyl-2-nonanol	1.435	1.438	0.832	0.836
6-Ethyl-3-octanol	5-Ethyl-2-nonanol	1.435	1.438	0.832	0.830
Chloro-*t*-butanol	1,3-Propanediol	1.436	1.438	1.059	1.049
	Diethyl maleate	1.436	1.438	1.059	1.064
N-Methylmorpholine	Dibutyl sebacate	1.436	1.440	0.924	0.932

		Refractive index		Density (g/ml)	
Solvent 1	**Solvent 2**	**1**	**2**	**1**	**2**
2-Methyl-7-ethyl-4-nonanol	5-Ethyl-2-nonanol	1.438	1.438	0.829	0.830
	6-Ethyl-3-octanol	1.438	1.438	0.829	0.836
	Butanethiol	1.438	1.440	0.829	0.837
	7-Methyl-7-ethyl-4-undecanol	1.438	1.442	0.829	0.829
	Ethyl sulfide	1.438	1.442	0.829	0.831
	6-Ethyl-3-decanol	1.438	1.441	0.829	0.838
5-Ethyl-2-nonanol	6-Ethyl-3-octanol	1.438	1.438	0.830	0.836
	Butanethiol	1.438	1.440	0.830	0.837
	2-Methyl-7-ethyl-4-undecanol	1.438	1.442	0.830	0.829
	Ethyl sulfide	1.438	1.442	0.830	0.831
	6-Ethyl-3-decanol	1.438	1.441	0.830	0.838
1,3-Propanediol	Diethyl maleate	1.438	1.438	1.049	1.064
Methyl salicylate	2-Methyl-7-ethyl-1-undecanol	1.438	1.442	0.836	0.829
	Ethyl sulfide	1.438	1.442	0.836	0.831
	Butanethiol	1.438	1.442	0.836	0.837
6-Ethyl-3-octanol	6-Ethyl-3-decanol	1.438	1.441	0.836	0.838
Butanethiol	6-Ethyl-3-decanol	1.440	1.441	0.837	0.838
	2-Methyl-7-ethyl-4-undecanol	1.440	1.442	0.837	0.829
	Ethyl sulfide	1.440	1.442	0.837	0.831
	Mesityl oxide	1.440	1.442	0.837	0.850
6-Ethyl-3-decanol	2-Methyl-7-ethyl-4-undecanol	1.441	1.442	0.838	0.829
	Ethyl sulfide	1.441	1.442	0.838	0.831
	Mesityl oxide	1.441	1.442	0.838	0.850
1-Chlorododecane (technical)	Mesityl oxide	1.441	1.442	0.862	0.850
	Butyl stearate	1.441	1.442	0.862	0.854
	1-Chlorotetradecane	1.441	1.445	0.862	0.858
2-Methyl-7-ethyl-4-undecanol	Ethyl sulfide	1.442	1.442	0.829	0.831
	2-Butyloctyl-3-aminopropyl ether	1.442	1.446	0.829	0.842
Mesityl oxide	Butyl stearate	1.442	1.442	0.850	0.854
	1-Chlorotetradecane	1.442	1.445	0.850	0.858
	2-Butyloctyl-3-aminopropyl ether	1.442	1.446	0.850	0.842
Butyl stearate	1-Chlorotetradecane	1.442	1.445	0.850	0.858
	2-Butyloctyl-3-aminopropyl ether	1.442	1.446	0.854	0.842
Ethyl sulfide	2-Butyloctyl-3-aminopropyl ether	1.442	1.446	0.831	0.842
1,3-Butanediol sulfite	1,2-Dichloroethane	1.444	1.444	1.231	1.245
	trans-1,2-Dichloroethylene	1.444	1.444	1.231	1.257
1,2-Dichloroethane	*trans*-1,2-Dichloroethylene	1.444	1.444	1.231	1.257
1-Chlorotetradecane	2-Butyloctyl-3-aminopropyl ether	1.445	1.446	0.857	0.842
	1-Chlorohexadecane	1.445	1.448	0.857	0.859
Diethylene glycol	Formamide	1.445	1.446	1.128	1.129
	Ethylene glycol diglycidyl ether	1.445	1.447	1.128	1.134
2-Butyloctyl-3-aminopropyl ether	3-Lauroxy-1-propylamine	1.446	1.447	0.842	0.840
Formamide	Ethylene glycol diglycidyl ether	1.446	1.447	1.129	1.134
2-Methylmorpholine	Cyclohexanone	1.446	1.448	0.951	0.943
	1-Amino-2-propanol	1.446	1.448	0.951	0.961
Dipropylene glycol monoethyl ether	Tetrahydrofurfuryl alcohol	1.446	1.450	1.043	1.050
1-Amino-2-methyl-2-pentanol	2-Butylcyclohexanone	1.449	1.453	1.050	1.047
3-Methyl-5-ethyl-2,4-heptanediol	2-Propylcyclohexanone	1.452	1.452	0.922	0.923
	4-Methylcyclohexanone	1.452	1.454	0.922	0.908
	2,2′-Dimethyl-2,2′-dipropyldiethanolamine	1.452	1.456	0.922	0.922
2-Propylcyclohexanone	4-Methylcyclohexanol	1.452	1.454	0.923	0.908
	3-Methylcyclohexanol	1.452	1.455	0.923	0.913
	2,2′-Dimethyl-2,2′-dipropyldiethanolamine	1.452	1.456	0.923	0.922
	1,8-Cineole	1.452	1.456	0.923	0.921

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
4-Methylcyclohexanol	2,2′-Dimethyl-2,2′-dipropyldiethanolamine	1.454	1.456	0.908	0.922
3-Methylcyclohexanol	2,2′-Dimethyl-2,2′-dipropyldiethanolamine	1.455	1.456	0.913	0.922
Cyclohexylamine	1-Chloroeicosane	1.456	1.459	0.862	0.872
1-Chloroeicosane (technical)	Oleic acid	1.459	1.459	0.872	0.887
Oleic acid	2-(β-Ethyl)butylcyclohexanone	1.459	1.461	0.887	0.892
	2-Butylcyclohexanol	1.459	1.462	0.887	0.898
	2-(β-Ethyl)hexylcyclohexanone	1.459	1.463	0.887	0.892
1,1′,2,2′-Tetramethyldiethanolamine	1-Aminopropanol	1.459	1.459	0.973	0.965
	N-(*n*-Butyl)diethanolamine	1.459	1.461	0.973	0.965
Carbon tetrachloride	4,5-Dichloro-1,3-dioxolane-2-one	1.459	1.461	1.584	1.591
2-(β-Ethyl)butylcyclohexanone	2,4-(bis)α-Phenylethyl)phenylmethyl ether	1.461	1.462	0.892	0.898
	2-(β-Ethyl)hexylcyclohexanone	1.461	1.463	0.892	0.892
N-(*n*-Butyl)diethanolamine	Cyclohexanol	1.461	1.465	0.965	0.968
2-Butylcyclohexanol	2-(β-Ethyl)hexylcyclohexanone	1.462	1.463	0.898	0.892
	2-Ethylcyclohexanol	1.462	1.463	0.898	0.908
N-β-Oxypropylmorpholine	Fluorobenzene	1.462	1.463	1.013	1.020
Fluorobenzene	*N*-(2-Hydroxyethyl)-2-hydroxybutyl amine	1.463	1.467	1.020	1.027
D-α-Pinene	1-α-Pinene	1.464	1.465	0.855	0.855
	trans-Decahydronaphthalene	1.464	1.468	0.855	0.867
m-Fluorotoluene	*p*-Fluorotoluene	1.465	1.467	0.994	0.995
1-α-Pinene	*trans*-Decahydronaphthalene	1.465	1.468	0.855	0.867
p-Fluorotoluene	*o*-Fluorotoluene	1.467	1.468	0.994	0.995
N-(2-Hydroxyethyl)-2-hydrobutylamine	*N*-(2-Hydroxyethyl)-2-hydropropylamine	1.467	1.468	1.027	1.042
	2-Allyloxy-2-hydroxypropylamine	1.467	1.469	1.027	1.017
	Di(2-Hydroxybutyl)ethanolamine	1.467	1.469	1.027	1.018
	Di(2-Hydroxypropyl)ethanolamine	1.467	1.469	1.027	1.042
2-Allyloxy-2-hydroxypropylamine	Di(2-Hydroxybutyl)ethanolamine	1.469	1.469	1.017	1.018
cis-Decahydronaphthalene	1-Methoxy-1-butene-3-yn	1.479	1.480	0.893	0.902
	n-Dodecyl-4-*t*-butylphenyl ether	1.479	1.482	0.893	0.881
	n-Dodecylphenyl ether	1.479	1.482	0.893	0.891
	n-Dodecyl-4-methylphenyl ether	1.479	1.483	0.893	0.889
1-Methoxy-1-butene-3-yn	*n*-Dodecylphenyl ether	1.480	1.482	0.902	0.891
	n-Dodecyl-4-methylphenyl ether	1.480	1.483	0.902	0.889
n-Dodecyl-4-*t*-butylphenyl ether	*n*-Dodecylphenyl ether	1.482	1.482	0.881	0.891
Butylbenzene	Dioctylbenzene (90% *p*; 10% *m*)	1.487	1.487	0.856	0.856
	p-Cymene	1.487	1.488	0.856	0.853
	Isopropylbenzene	1.487	1.489	0.856	0.857
	t-Butylbenzene (80% *p*, 15% *m*, 5% *o*)	1.487	1.490	0.856	0.856
	n-Propylbenzene	1.487	1.490	0.856	0.856
	s-Butylbenzene	1.487	1.490	0.856	0.856
	Hexyl-*m*-xylene	1.487	1.490	0.856	0.860
	t-Butylbenzene	1.487	1.490	0.856	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.487	1.491	0.856	0.856
p-Cymene	Isopropylbenzene	1.488	1.489	0.853	0.857
	t-Butylcumene (80% *p*, 15% *m*, 5% *o*)	1.488	1.490	0.853	0.856
	n-Propylbenzene	1.488	1.490	0.853	0.858
	s-Butylbenzene	1.488	1.490	0.853	0.858
	Hexyl-*m*-xylene (mainly 1,3,5)	1.488	1.490	0.853	0.860
	t-Butylbenzene	1.488	1.490	0.853	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.488	1.491	0.853	0.856
	t-Butyltoluene	1.488	1.491	0.853	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.488	1.492	0.853	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.488	1.492	0.853	0.866
Isopropylbenzene	*t*-Butylcumene (80% *p*, 15% *m*, 5% *o*)	1.489	1.490	0.857	0.856

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	n-Propylbenzene	1.489	1.490	0.857	0.858
	s-Butylbenzene	1.489	1.490	0.857	0.858
	Hexyl-*m*-xylene (mainly 1,3,5)	1.489	1.490	0.857	0.860
	t-Butylbenzene	1.489	1.490	0.857	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.489	1.491	0.857	0.856
	t-Butylbenzene (80% *p*, 15% *m*, 5% *o*)	1.489	1.491	0.857	0.858
t-Butylcumene (80% *p*, 15% *m*, 5% *o*)	*n*-Propylbenzene	1.490	1.490	0.856	0.858
	s-Butylbenzene	1.490	1.490	0.856	0.858
	Hexyl-*m*-xylene (mainly 1,3,5)	1.490	1.490	0.856	0.860
	t-Butylbenzene	1.490	1.490	0.856	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.490	1.491	0.856	0.856
	t-Butyltoluene (80% *p*, 15% *m*, 5% *o*)	1.490	1.491	0.856	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.490	1.492	0.856	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.490	1.492	0.856	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.490	1.492	0.856	0.869
	Dihexylbenzene	1.490	1.492	0.856	0.870
	p-Xylene	1.490	1.493	0.856	0.857
	1,3-Diethylbenzene	1.490	1.493	0.856	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.490	1.493	0.856	0.862
	Ethylbenzene	1.490	1.493	0.856	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.490	1.493	0.856	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.490	1.494	0.856	0.860
	Toluene	1.490	1.494	0.856	0.862
n-Propylbenzene	*s*-Butylbenzene	1.490	1.490	0.858	0.858
	Hexyl-*m*-xylene (mainly 1,3,5)	1.490	1.490	0.858	0.860
	t-Butylbenzene	1.490	1.490	0.858	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.490	1.491	0.858	0.856
	t-Butyltoluene (80% *p*, 15% *m*, 5% *p*)	1.490	1.491	0.858	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.490	1.492	0.858	0.863
	Octyltoluene (96% *p*, 2% *m* 2% *o*)	1.490	1.492	0.858	0.866
	Octylcumene (90% *p*, 4% *m* 6% *o*)	1.490	1.492	0.858	0.869
	Dihexylbenzene	1.490	1.492	0.858	0.870
	p-Xylene	1.490	1.493	0.858	0.857
	1,3-Diethylbenzene	1.490	1.493	0.858	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.490	1.493	0.858	0.862
	Ethylbenzene	1.490	1.493	0.858	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.490	1.493	0.858	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.490	1.494	0.858	0.860
	Toluene	1.490	1.494	0.858	0.862
s-Butylbenzene	Hexyl-*m*-xylene (mainly 1,3,5)	1.490	1.490	0.858	0.860
	t-Butylbenzene	1.490	1.490	0.858	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.490	1.491	0.858	0.856
	t-Butyltoluene (80% *p*, 15% *m*, 5% *o*)	1.490	1.491	0.858	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.490	1.492	0.858	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.490	1.492	0.858	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.490	1.492	0.858	0.869
	Dihexylbenzene	1.490	1.492	0.858	0.870
	p-Xylene	1.490	1.493	0.858	0.857
	1,3-Diethylbenzene	1.490	1.493	0.858	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.490	1.493	0.858	0.862
	Ethylbenzene	1.490	1.493	0.858	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.490	1.493	0.858	0.866

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.490	1.494	0.858	0.860
	Toluene	1.490	1.494	0.858	0.862
Hexyl-*m*-xylene (mainly 1,3,5)	*t*-Butylbenzene	1.490	1.490	0.860	0.862
	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.490	1.491	0.860	0.856
	t-Butyltoluene (80% *p*, 15% *m*, 5% *o*)	1.490	1.491	0.860	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.490	1.492	0.860	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.490	1.492	0.860	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.490	1.492	0.860	0.869
	Dihexylbenzene	1.490	1.492	0.860	0.870
	p-Xylene	1.490	1.493	0.860	0.857
	1,3-Diethylbenzene	1.490	1.493	0.860	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.490	1.493	0.860	0.862
	Ethylbenzene	1.490	1.493	0.860	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.490	1.493	0.860	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.490	1.494	0.860	0.860
	Toluene	1.490	1.494	0.860	0.862
t-Butylbenzene	Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.490	1.491	0.862	0.856
	t-Butyltoluene (80% *p*, 15% *m*, 5% *o*)	1.490	1.491	0.862	0.858
	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.490	1.492	0.862	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.490	1.492	0.862	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.490	1.492	0.862	0.869
	Dihexylbenzene	1.490	1.492	0.862	0.870
	p-Xylene	1.490	1.493	0.862	0.857
	1,3-Diethylbenzene	1.490	1.493	0.862	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.490	1.493	0.862	0.862
	Ethylbenzene	1.490	1.493	0.862	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.490	1.493	0.862	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.490	1.494	0.862	0.860
	Toluene	1.490	1.494	0.862	0.862
Isopropylethylbenzene (35% *p*, 60% *m*)	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.491	1.492	0.858	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.491	1.492	0.858	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.491	1.492	0.858	0.869
	Dihexylbenzene	1.491	1.492	0.858	0.870
	p-Xylene	1.491	1.493	0.858	0.857
	1,3-Diethylbenzene	1.491	1.493	0.858	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.491	1.493	0.858	0.862
	Ethylbenzene	1.491	1.493	0.858	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.491	1.493	0.858	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.491	1.494	0.858	0.860
	Toluene	1.491	1.494	0.858	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.491	1.495	0.858	0.854
	m-Xylene	1.491	1.495	0.858	0.860
	Hexylethylbenzene	1.491	1.495	0.858	0.868
t-Butyltoluene (85% *p*, 10% *m*)	Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.491	1.492	0.858	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.491	1.492	0.856	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.491	1.492	0.856	0.869
	Dihexylbenzene	1.491	1.492	0.856	0.870
	p-Xylene	1.491	1.493	0.856	0.857
	1,3-Diethylbenzene	1.491	1.493	0.856	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.491	1.493	0.856	0.862
	Ethylbenzene	1.491	1.493	0.856	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.491	1.493	0.856	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.491	1.494	0.856	0.860
	Toluene	1.491	1.494	0.856	0.862

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.491	1.495	0.856	0.854
	m-Xylene	1.491	1.495	0.856	0.860
	Hexylethylbenzene(70% *p*, 25% *m*, 5% *o*)	1.491	1.495	0.856	0.868
1-Phenyl-1-hydroxyphenyl ether	1,3-Dimorpholyl-2-propanol	1.491	1.493	1.081	1.094
Hexylcumene (90% *p*, 5% *m*)	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.492	1.492	0.863	0.866
	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.492	1.492	0.863	0.869
	Dihexylbenzene	1.492	1.492	0.863	0.870
	p-Xylene	1.492	1.493	0.863	0.857
	1,3-Diethylbenzene	1.492	1.493	0.863	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.492	1.493	0.863	0.862
	Ethylbenzene	1.492	1.493	0.863	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.492	1.493	0.863	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.492	1.494	0.863	0.860
	Toluene	1.492	1.494	0.863	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.863	0.854
	m-Xylene	1.492	1.495	0.863	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.863	0.868
	1,4-Diethylbenzene	1.492	1.496	0.863	0.858
	Isopropylbenzene	1.492	1.498	0.863	0.857
Dihexylbenzene (85% *p*, 10% *m*)	Octylcumene (90% *p*, 4% *m*, 6% *0*)	1.492	1.492	0.870	0.869
	p-Xylene	1.492	1.493	0.870	0.857
	1,3-Diethylbenzene	1.492	1.493	0.870	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.492	1.493	0.870	0.862
	Ethylbenzene	1.492	1.493	0.870	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.492	1.493	0.870	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.492	1.494	0.870	0.860
	Toluene	1.492	1.494	0.870	0.862
	m-Xylene	1.492	1.495	0.870	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.870	0.868
	1,4-Diethylbenzene	1.492	1.496	0.870	0.858
	Isopropylbenzene	1.492	1.498	0.870	0.857
Octyltoluene (96% *p*, 2% *m*)	Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.492	1.492	0.866	0.869
	Dihexylbenzene (85% *p*, 10% *m*, 5% *o*)	1.492	1.492	0.866	0.870
	p-Xylene	1.492	1.493	0.866	0.857
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.492	1.493	0.866	0.862
	Ethylbenzene	1.492	1.493	0.866	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.492	1.493	0.866	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.492	1.494	0.866	0.860
	Toluene	1.492	1.494	0.866	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.866	0.854
	m-Xylene	1.492	1.495	0.866	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.866	0.868
	1,4-Diethylbenzene	1.492	1.496	0.866	0.858
	Isopropylbenzene	1.492	1.498	0.866	0.857
Octylcumene (90% *p*, 4% *m*, 6% *o*)	*p*-Xylene	1.492	1.493	0.866	0.857
	1,3-Diethylbenzene	1.492	1.493	0.866	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.492	1.493	0.866	0.862
	Ethylbenzene	1.492	1.493	0.866	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.492	1.493	0.866	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.492	1.494	0.866	0.860
	Toluene	1.492	1.494	0.866	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.866	0.854
	m-Xylene	1.492	1.495	0.866	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.492	1.495	0.866	0.868

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	1,4-Diethylbenzene	1.492	1.496	0.866	0.858
	Isopropylbenzene	1.492	1.498	0.866	0.857
p-Xylene	1,3-Diethylbenzene	1.493	1.493	0.857	0.860
	t-Butyl-*m*-xylene (mainly 1,3,5)	1.493	1.493	0.857	0.862
	Ethylbenzene	1.493	1.493	0.857	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.493	1.493	0.857	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.493	1.494	0.857	0.860
	Toluene	1.493	1.494	0.857	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.857	0.854
	m-Xylene	1.493	1.495	0.857	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.857	0.868
	1,4-Diethylbenzene	1.493	1.496	0.857	0.858
	Mesitylene	1.493	1.497	0.857	0.861
	Isopropylbenzene	1.493	1.498	0.857	0.857
1,3-Diethylbenzene	*t*-Butyl-*m*-xylene (mainly 1,3,5)	1.493	1.493	0.860	0.862
	Ethylbenzene	1.493	1.493	0.860	0.863
	Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.493	1.493	0.860	0.866
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.493	1.493	0.860	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.493	1.494	0.860	0.860
	Toluene	1.493	1.494	0.860	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.860	0.854
	m-Xylene	1.493	1.495	0.860	0.860
	1,4-Diethylbenzene	1.493	1.496	0.860	0.858
	Mesitylene	1.493	1.497	0.860	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.493	1.497	0.860	0.870
	Isopropylbenzene	1.493	1.498	0.860	0.860
t-Butyl-*m*-xylene (mainly 1,3,5)	Ethylbenzene	1.493	1.493	0.862	0.863
	Octylethylbenzene (80–90% *p*, 10% *m*)	1.493	1.493	0.862	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.493	1.494	0.862	0.860
	Toluene	1.493	1.494	0.862	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.862	0.854
	m-Xylene	1.493	1.495	0.862	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.862	0.868
	1,4-Diethylbenzene	1.493	1.496	0.862	0.858
	Mesitylene	1.493	1.497	0.862	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.493	1.497	0.862	0.870
	Isopropylbenzene	1.493	1.498	0.862	0.857
Ethylbenzene	Octylethylbenzene (80–90% *p*, 10% *m*)	1.493	1.493	0.863	0.866
	Isopropyl-*m*-xylene (mainly 1,3,5)	1.493	1.494	0.863	0.860
	Toluene	1.493	1.494	0.863	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.863	0.854
	m-Xylene	1.493	1.495	0.863	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.863	0.868
	1,4-Diethylbenzene	1.493	1.496	0.863	0.858
	Mesitylene	1.493	1.497	0.863	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.493	1.497	0.863	0.870
	Isopropylbenzene	1.493	1.498	0.863	0.857
Octylethylbenzene (80–90% *p*, 10% *m*)	Isopropyl-*m*-xylene (mainly 1,3,5)	1.493	1.494	0.866	0.860
	Toluene	1.493	1.494	0.866	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.866	0.854
	m-Xylene	1.493	1.495	0.866	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.493	1.495	0.866	0.868
	1,4-Diethylbenzene	1.493	1.496	0.866	0.858
	Mesitylene	1.493	1.497	0.866	0.861

Solvent 1	Solvent 2	Refractive index		Density (g/ml)	
		1	2	1	2
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.493	1.497	0.866	0.870
	Isopropylbenzene	1.493	1.498	0.866	0.857
Isopropyl-*m*-xylene (mainly 1,3,5)	Toluene	1.494	1.494	0.860	0.862
	t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.494	1.495	0.860	0.854
	m-Xylene	1.494	1.495	0.860	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.494	1.495	0.860	0.868
	1,4-Diethylbenzene	1.494	1.496	0.860	0.858
	Mesitylene	1.494	1.497	0.860	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.494	1.497	0.860	0.870
	Benzene	1.494	1.498	0.860	0.874
Toluene	*t*-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.494	1.495	0.862	0.854
	m-Xylene	1.494	1.495	0.862	0.860
	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.494	1.495	0.862	0.868
	1,4-Diethylbenzene	1.494	1.496	0.862	0.858
	Mesitylene	1.494	1.497	0.862	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.494	1.497	0.862	0.870
	Benzene	1.494	1.498	0.862	0.874
t-Butylethylbenzene (70% *p*, 25% *m*)	*m*-Xylene	1.495	1.495	0.854	0.860
	1,4-Diethylbenzene	1.495	1.496	0.854	0.858
	Mesitylene	1.495	1.497	0.854	0.861
m-Xylene	Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.495	1.495	0.860	0.868
	1,4-Diethylbenzene	1.495	1.496	0.860	0.858
	Mesitylene	1.495	1.497	0.860	0.861
	Benzene	1.495	1.498	0.860	0.874
Hexylethylbenzene (70% *p*, 25% *m*)	1,4-Diethylbenzene	1.495	1.496	0.868	0.858
	Mesitylene	1.495	1.497	0.868	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.495	1.497	0.868	0.870
	Benzene	1.495	1.498	0.868	0.874
1,4-Diethylbenzene	Mesitylene	1.496	1.497	0.858	0.861
	Hexyltoluene (70% *p*, 25% *m*, 5% *o*)	1.496	1.498	0.858	0.870
Mesitylene	Ethylbenzene	1.497	1.497	0.861	0.870
Hexyltoluene (70% *p*, 25% *m*)	Benzene	1.497	1.498	0.870	0.874
	1,2-Diethylbenzene	1.497	1.501	0.870	0.876
Benzene	Mesitylene	1.498	1.498	0.874	0.874
	1,2-Diethylbenzene	1.498	1.501	0.874	0.876
Mesitylene	1,2-Diethylbenzene	1.498	1.501	0.874	0.876
1,2-Diethylbenzene	*o*-Xylene	1.501	1.503	0.867	0.876
β-Picoline	Phenetole	1.504	1.505	0.953	0.961
Phenetole	Pyridine	1.505	1.507	0.961	0.978
Cyclohexylcumene (50% *p*, 20% *m*)	Cyclohexylethylbenzene (60% *p*, 20% *m*, 20% *o*)	1.516	1.520	0.917	0.923
Benzyl acetate	Chloro-*t*-butylbenzene	1.518	1.521	1.051	1.039
Cyclohexylethylbenzene (60% *p*, 20% *m*)	Cyclohexyltoluene	1.520	1.523	0.923	0.923
2-Furfurol	Thiophene	1.524	1.526	1.057	1.059
Benzyl alcohol	*m*-Cresol	1.538	1.542	1.041	1.037
m-Cresol	Benzaldehyde	1.542	1.544	1.037	1.041
m-Toluidine	*o*-Toluidine	1.566	1.570	0.985	0.994

Refractive Indices of Common Solvents

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A. INTRODUCTION

Measurements which depend on the difference in refractive index between the polymer and the solvent will, in general, give greater accuracy as the refractive index increment between polymer and solvent is increased. The magnitude of the increment may be either positive or negative. Systems involving refractive index increments are those of light scattering and ultracentrifugation when either schlieren or interference optics is used. A table of commonly used solvents, arranged according to increasing refractive index, will be useful in practical work with many different polymers. Data at 25°C, D-line.

B. TABLE OF REFRACTIVE INDICES OF COMMON SOLVENTS

Solvents	Refractive index
Trifluoroacetic acid	1.283
Trifluoroethanol	1.290
Octafluoro-1-pentanol	1.316
Dodecafluoro-1-heptanol	1.316
Methanol	1.326
Acetonitrile	1.342
Ethyl ether	1.352
Acetone	1.357
Ethyl formate	1.358
Ethanol	1.359
Methyl acetate	1.360
Propionitrile	1.363
2,2-Dimethylbutane	1.366
Isopropyl ether	1.367
2-Methylpentane	1.369
Formic acid	1.369
Ethyl acetate	1.370
Acetic acid	1.370
Propionaldehyde	1.371
n-Hexane	1.372
2,3-Dimethylbutane	1.372
3-Methylpentane	1.374
2-Propanol	1.375
Isopropyl acetate	1.375
Propyl formate	1.375
2-Chloropropane	1.376
2-Butanone	1.377
Butyraldehyde	1.378
2,4-Dimethylpentane	1.379
Propyl ether	1.379
Acetaldehyde diethyl acetal	1.379
Butyl ethyl ether	1.380
Nitromethane	1.380
Trifluoropropanol	1.381
2-Methylhexane	1.382
Butyronitrile	1.382
Propyl acetate	1.382
Ethyl propionate	1.382
2-Methyl-2-propanol	1.383
Propanol	1.383
Isobutyl formate	1.383
Diethyl carbonate	1.383
Heptane	1.385
t-Butanol	1.385
Propionic acid	1.385
3-Methylhexane	1.386
Propylamine	1.386
3-Methyl-2-butanone	1.386
1-Chloropropane	1.386
2,2,3-Trimethylbutane	1.387
s-Butyl acetate	1.387
Butyl formate	1.387
Isobutyl acetate	1.388
2,2,4-Trimethylpentane	1.389
2,3-Dimethylpentane	1.389
Acetic anhydride	1.389
Diisopropylamine	1.390
2-Butylamine	1.390
2-Pentanone	1.390
3-Pentanone	1.390
Nitroethane	1.390
Methyl *n*-butyrate	1.391
Butyl acetate	1.392
2-Nitropropane	1.392
4-Methyl-2-propanone	1.394
2-Methyl-1-propanol	1.394
Octane	1.395

Solvents	Refractive index
Isobutylamine	1.395
Valeronitrile	1.395
2-Butanol	1.395
2-Hexanone	1.395
2-Chlorobutane	1.395
Butyric acid	1.396
2,2,5-Trimethylhexane	1.397
Dibutyl ether	1.397
Butanol	1.397
Acrolein	1.397
1-Chloro-2-methylpropane	1.397
Methacrylonitrile	1.398
3-Methyl-2-pentanone	1.398
Triethylamine	1.399
Butylamine	1.399
2,4-Dimethyl-3-pentanone	1.399
Isobutyl *n*-butyrate	1.399
1-Nitropropane	1.399
Dodecane	1.400
Amyl acetate	1.400
1-Chlorobutane	1.400
2-Methoxyethanol	1.400
Propionic anhydride	1.400
2,2,3-Trimethylpentane	1.401
1-Chlorobutane	1.401
β-Methoxypropionitrile	1.401
Isovaleric acid	1.402
Nonane	1.403
Dipropylamine	1.403
Isoamyl acetate	1.403
Cyclopentane	1.404
2-Methyl-2-butanol	1.404
3-Methyl-1-butanol	1.404
Tetrahydrofuran	1.404
Capronitrile	1.405
4-Heptanone	1.405
2-Ethoxyethanol	1.405
2-Heptanone	1.406
Valeric acid	1.406
Diisobutylene	1.407
Methylcyclopentane	1.407
Isoamyl ether	1.405
2-Pentanol	1.407
Tributyl borate	1.407
Pentanol	1.408
3-Methyl-2-butanol	1.408
Diethyl oxalate	1.408
Decane	1.409
4-Methyl-2-pentanol	1.409
3-Isopropyl-2-pentanone	1.409
2-Methyl-1-butanol	1.409
Butyric anhydride	1.409
Amyl ether	1.410
Isoamyl isovalerate	1.410
1-Chloropentane	1.410
Allyl alcohol	1.411

Solvents	Refractive index
2,4-Dimethyldioxane	1.412
Ethyl lactate	1.412
Diethyl malonate	1.412
3-Chloropropene	1.413
Ethylene glycol diacetate	1.413
2-Octanone	1.414
3-Octanone	1.414
3-Methyl-2-heptanone	1.415
Caproic acid	1.415
4-Methyldioxane	1.415
1,2-Propylene glycol monobutyl ether	1.415
Ethyl cyanoacetate	1.415
Dibutylamine	1.416
Hexanol	1.416
2-Pentanol	1.416
1,1-Dichloroethane	1.416
Heptachlorodiethyl ether	1.416
3-Methoxypropylamine	1.417
Caprylonitrile	1.418
2-Heptanol	1.418
Allylamine	1.419
1,2-Propylene glycol carbonate	1.419
2-Heptanol	1.420
3-Methyl-2-pentanol	1.420
2-Ethyl-1-butanol	1.420
1-Chloro-2-methyl-1-propene	1.420
1,4-Dioxane	1.420
Methylcyclopropane	1.421
4-Hydroxy-4-methyl-2-pentanone	1.421
Heptanol	1.422
3-Isopropyl-2-heptanone	1.423
Cyclohexane	1.424
2-Bromopropane	1.424
3-Chloro-2-methyl-1-propene	1.425
Caprylic acid	1.426
Ethylene carbonate	1.426
Octanol	1.427
3-Methyl-2-heptanol	1.427
N,*N*-Dimethylformamide	1.427
Sulfuric acid	1.427
1-Chlorooctane	1.428
Triisobutylene	1.429
N-Methylalaninenitrile	1.429
1,2-Ethanediol	1.429
1-Chloro-2-ethylhexane	1.430
Ethylcyclohexane	1.431
1,2-Propanediol	1.431
1-Bromopropane	1.431
2-Methyl-7-ethyl-4-nonanone	1.433
Ethylene glycol monoallyl ether	1.434
Butyrolactone	1.434
2-Methyl-7-ethyl-undecanone	1.435
1,2-Dichloroisobutane	1.435
1,2-Propylene glycol sulfite	1.435
N-Methylmorpholine	1.436
Chloro-*t*-butanol	1.436

Solvents	Refractive index
Epichlorohydrin	1.436
Triethylene glycol monobutyl ether	1.437
2-Methyl-7-ethyl-4-nonanol	1.438
5-Ethyl-2-nonanol	1.438
6-Ethyl-3-octanol	1.438
1,3-Propanediol	1.438
Diethyl maleate	1.438
Butanethiol	1.440
Dibutyl sebacate	1.440
2-Chloroethanol	1.440
6-Ethyl-3-decanol	1.441
1-Chlorododecane	1.441
3-Methyl-2,4-pentanediol	1.441
Dimethyl maleate	1.441
2-Methyl-7-ethyl-4-undecanol	1.442
Ethyl sulfide	1.442
Mesityl oxide	1.442
Butyl stearate	1.442
Cyclohexene	1.443
Lauryl glycidyl ether	1.443
Dibutyl maleate	1.444
1,3-Butylene glycol sulfite	1.444
1,2-Dichloroethane	1.444
Glycol sulfite	1.444
Chloroform	1.444
1-Chlorotetradecane	1.445
Diethylene glycol	1.445
cis-1,2-Dichloroethylene	1.445
2-Butyloctyl-3-aminopropyl ether	1.446
2-Methylmorpholine	1.446
Formamide	1.446
3-Lauryl-1-hydroxypropylamine	1.447
Ethylene glycol diglycidyl ether	1.447
1-Chlorohexanedecane	1.448
Cyclohexanone	1.448
1-Amino-2-propanol	1.448
Diethylene glycol mono-β-hydroxyisopropyl ether	1.448
1-Amino-2-methyl-2-pentanol	1.449
Tetrahydrofurfuryl alcohol	1.450
2-Propylcyclohexanone	1.452
2-Aminoethanol	1.452
1,4-Butanediol glycidyl ether	1.452
4-Chloro-1,3-dioxolane-2-one	1.452
1-Chlorooctadecane	1.453
2-Butylcyclohexanone	1.453
Ethylenediamine	1.454
2-(β-Methyl)propylcyclohexanone	1.454
4-Methylcyclohexanol	1.454
3-Methylcyclohexanol	1.455
Bis(2-chloroethyl) ether	1.455
Cyclohexyl methacrylate	1.456
1,8-Cineole	1.456
2,2′-Dimethyl-2,2′-dipropyl diethanolamine	1.456
1,3-Butanediol glycidyl ether	1.456

Solvents	Refractive index
1-Chloroeicosane	1.459
Oleic acid	1.459
(1,1′,2,2′-Tetramethyl) diethanolamine	1.459
3-Aminopropanol	1.459
Carbon tetrachloride	1.459
3-Methyl-5-ethyl-2,4-heptanediol	1.459
2-(β-Ethyl)butylcyclohexanone	1.461
2-Methylcyclohexanol	1.461
N-(*n*-Butyl)diethanolamine	1.461
4,5-Dichloro-1,3-dioxolane-2-one	1.462
2-Butylcyclohexanol	1.462
N-β-Hydroxypropylmorpholine	1.462
2-(β-Ethyl)hexylcyclohexanone	1.463
2-Ethylcyclohexanol	1.463
Fluorobenzene	1.463
D-α-Pinene	1.464
L-α-Pinene	1.465
Cyclohexanol	1.465
m-Fluorotoluene	1.465
p-Fluorotoluene	1.467
N-(2-Hydroxyethyl)-2-hydroxybutylamine	1.467
4-Chloromethyl-1,3-dioxolane-2-one	1.467
trans-Decahydronaphthalene	1.468
o-Fluorotoluene	1.468
N-(2-Hydroxyethyl)-2-hydroxypropylamine	1.468
3-Allyloxy-2-hydroxypropylamine	1.469
Di(2-hydroxybutyl)ethanolamine	1.469
Di(2-hydroxypropyl)ethanolamine	1.469
D-Limonene	1.471
2-(α-Hydroxybutyl)cyclohexanol	1.473
1,2,3-Trichloroisobutane	1.473
Decahydronaphthalene	1.474
1,2,3-Propanetriol	1.475
Trichloroethylene	1.474
N-(β-Hydroxyethyl)morpholine	1.476
Dimethyl sulfoxide	1.476
cis-Decahydronaphthalene	1.479
2-(α-Hydroxyethyl)cyclohexanol	1.479
1-Methoxy-1-butene-3-yn	1.480
2-Butylidenecyclohexanone	1.481
N-(β-Chloroallyl)morpholine	1.481
n-Dodecyl-4-*t*-butyl phenyl ether	1.482
n-Dodecyl phenyl ether	1.482
n-Dodecyl 4-methylphenyl ether	1.483
N-Hydroxyethyl-1,3-propanediamine	1.483
Morpholyl *N*-(ethylhydroxy)ethylamine	1.485
2-Ethylidenecyclohexanone	1.486
Butylbenzene	1.487
Dioctylbenzene (90% *p*, 10% *m*)	1.487
p-Cymene	1.488
Isopropylbenzene	1.489
Furfuryl alcohol	1.489
t-Butylcumene (80% *p*, 15% *m*, 5% *o*)	1.490
n-Propylbenzene	1.490

Solvents	Refractive index
s-Butylbenzene	1.490
Hexyl-*m*-xylene (mainly 1,3,5)	1.490
t-Butylbenzene	1.490
Dibutyl phthalate	1.490
Isopropylethylbenzene (35% *p*, 60% *m*, 5% *o*)	1.491
t-Butyltoluene (85% *p*, 10% *m*, 5% *o*)	1.491
1-Phenyl-1-hydroxyphenyl ethane	1.491
Hexylcumene (90% *p*, 5% *m*, 5% *o*)	1.492
Octyltoluene (96% *p*, 2% *m*, 2% *o*)	1.492
Octylcumene (90% *p*, 4% *m*, 6% *o*)	1.493
Dihexylbenzene (85% *p*, 10% *m*, 5% *o*)	1.492
p-Xylene	1.493
1,3-Diethylbenzene	1.493
t-Butyl-*m*-xylene (mainly 1,3,5)	1.493
Ethylbenzene	1.493
Octylethylbenzene (80–90% *p*, 10% *m*)	1.493
1,3-Dimorpholyl-2-propanol	1.493
1,1,2,2-Tetrachloroethane	1.493
Isopropyl-*m*-xylene (mainly 1,3,5)	1.494
Toluene	1.494
Benzyl ethyl ether	1.494
t-Butylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.495
m-Xylene	1.495
Hexylethylbenzene (70% *p*, 25% *m*, 5% *o*)	1.495
1,4-Diethylbenzene	1.496
2,3-Dichlorodioxane	1.496
Mesitylene	1.497
Hexyltoluene (90% *p*, 5% *m*, 5% *o*)	1.497
2-Iodopropane	1.497
Benzene	1.498
Propyl benzoate	1.498
α-Picoline	1.499
1,2-Diethylbenzene	1.501
Pentachloroethane	1.501
1-Iodopropane	1.502
o-Xylene	1.503
Ethyl benzoate	1.503
β-Picoline	1.504
Tetrachloroethylene	1.504
Phenetole	1.505
Pyridine	1.507
Iodoethane	1.512
Phenyl methallyl ether	1.514
Anisole	1.515
Methyl benzoate	1.515
Cyclohexylcumene (60% *p*, 25% *m*, 15% *o*)	1.516
Diallyl phthalate	1.517
Benzyl acetate	1.518
Cyclohexylethylbenzene (70% *p*, 20% *m*, 20% *o*)	1.520
2-Methyl-4-*t*-butylphenol	1.521
Phenyl acetonitrile	1.521
(Chloro-*t*-butyl)benzene	1.521
Methyl salicylate	1.522
Cyclohexyltoluene (50% *p*, 20% *m*, 30% *o*)	1.523
Chlorobenzene	1.523
Furfural	1.524
Octachlorodiethyl ether	1.525
Benzonitrile	1.526
Thiophene	1.526
Nonachlorodiethyl ether	1.529
Iodomethane	1.530
4-Phenyldioxane	1.530
3-Phenyl-1-propanol	1.532
Acetophenone	1.532
Benzyl alcohol	1.538
1,2-Dibromoethane	1.538
1,2,3,4-Tetrahydronaphthalene	1.539
m-Cresol	1.542
β,β-Di(butylxanthogenic acid) diethylester	1.543
m-Dichlorobenzene	1.544
Benzaldehyde	1.544
Styrene	1.545
Nitrobenzene	1.550
o-Dichlorobenzene	1.551
Bromobenzene	1.557
o-Nitroanisole	1.560
m-Toluidine	1.566
Benzyl benzoate	1.568
o-Toluidine	1.570
1-Methoxyphenyl-1-phenylethane	1.571
Aniline	1.583
o-Chloroaniline	1.586
Bromoform	1.587
Thiophenol	1.588
2,4-Bis(α-phenylethyl)phenylmethyl ether	1.590
Carbon disulfide	1.628
1,1,2,2-Tetrabromoethane	1.633
Methylene iodide	1.749

Physical Constants of the Most Common Solvents for Polymers

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Values given here have been taken a variety of reference books. Where a range was given, the melting and boiling points given here are average values. Flash point was determined by the closed cup method, using a Setaflash apparatus. The refractive index is reported relative to the sodium D line. Boiling points are given at 760 mmHg unless indicated otherwise by the pressure in mmHg following a slash (/). Viscosity is reported at 20°C unless indicated at another temperature in Celsius following a slash (/).

Compound	CAS Number	Formula wt.	Melting pt. (°C)	Boiling pt. (°C)	Density (at 20°C)	Viscosity at 20°C (cps)	Refractive index (at 20°C)	Flash pt. (°C)
Acetic acid	64-19-7	60.05	17	118	1.049	1.21	1.3716	40
Acetic anhydride	108-24-7	102.09	−73	140	1.082		1.3900	54
Acetone	67-64-1	58.08	−94	56	0.790	0.324	1.3588	−17
Acetonitrile	75-05-8	41.05	−48	82	0.786		1.3440	5
n-Amyl acetate	628-63-7	130.19	−71	148	0.876		1.4026	36
Aniline	62-53-3	93.13	−6	184	1.022		1.5860	70
Anisole	100-66-3	108.14	−37	154	0.995		1.5160	51
Benzene	71-43-2	78.11	6	80	0.874	0.52	1.5010	−11
Benzonitrile	100-47-0	103.12	−13	191	1.010		1.5280	71
Benzyl acetate	140-11-4	150.18	−51	206	1.040		1.5020	102
Benzyl alcohol	100-51-6	108.14	−15	205	1.045	5.8	1.5400	93
Biphenyl	92-52-4	154.21	71	255	0.992			
Bromoform	75-25-2	252.75	8.3	151	2.894		1.5960	
n-Butanol	71-36-3	74.12	−90	118	0.810	2.948	1.3990	35
n-Butyl acetate	123-86-4	116.16	−78	125	0.882	5.8	1.3940	
γ-Butyrolactone	96-48-0	86.09	−45	205	1.120		1.4360	98
Carbon disulfide	75-15-0	76.14	−111	46	1.266		1.6270	−33
Carbon tetrachloride	56-23-5	153.82	−23	77	1.594	0.969	1.4600	
Chlorobenzene	108-90-7	112.56	−45	132	1.107		1.5240	23
2-Chloroethanol	107-07-3	80.51	−89	129	1.201		1.4410	60
Chloroform	67-66-3	119.38	−63	61	1.492	0.568	1.4460	
2-Chlorophenol	95-57-8	128.56	8	176	1.241		1.5580	63
m-Cresol	108-39-4	108.14	9	203	1.034	20.8	1.5400	86
Cyclohexane	110-82-7	84.16	7	81	0.779	0.979	1.4260	−18
Cyclohexanol	108-93-0	100.16	22	161	0.948		1.4650	67
Cyclohexanone	108-94-1	98.15	−47	155	0.947	2.543/15	1.4500	46
Cyclohexyl acetate	622-45-7	142.2		173	0.966		1.4390	57
Cyclopentanol	96-41-3	86.13	−19	140	0.949		1.4530	51
cis-Decahydronaphthalene	493-01-6	138.25	−43	193	0.896		1.4750	57
Diacetone alcohol	123-42-2	116.16		166	0.931	1.4230		61
Di-*n*-amyl ether	693-65-2	158.29	−69	188	0.785		1.4119	57
1,2-Dichlorobenzene	95-50-1	147	−17	180	1.306		1.5510	65
Di-2-chlorodiethylether	111-44-4	143.01	−47	66/15	1.220		1.4560	55
1,2-Dichloroethylene, *cis & trans*	540-59-0	96.94	−57	48–60	1.265		1.4470	6
Dichloromethane/methylene chloride	75-09-2	84.93	−97	40	1.325		1.4240	
1,3,-Dichloro-2,4,6-trifluorobenzene	2368-53-8	200.98		162	1.599	1.4870		76
Diethyl ether	60-29-7	74.12	−116	34.6	0.708	1.26	1.3530	−40

Compound	CAS Number	Formula wt.	Melting pt. (°C)	Boiling pt. (°C)	Density (at 20°C)	Viscosity at 20°C (cps)	Refractive index (at 20°C)	Flash pt. (°C)
Di(ethylene glycol)	111-46-6	106.12	−10	245	1.118		1.4460	143
Di(ethylene glycol mono-*n*-butyl ether	112-34-5	162.23	−68	231	0.967		1.4320	100
Di(ethylene glycol) monoethyl ether	111-90-0	134.18		202	0.999		1.4270	96
Diisobutyl ketone	108-83-8	142.24	−46	169	0.806		1.4113	49
Diisopropyl ketone	565-80-0	114.19	−80	124	0.806		1.4000	15
1,2-Dimethoxyethane/ethylene glycol dimethyl ether	110-71-4	90.12	−58	85	0.867		1.3790	0
N,*N*-Dimethylacetamide	127-19-5	87.12	−20	166	0.937		1.4380	70
N,*N*-Dimethylformamide	68-12-2	73.1	−61	153	0.944		1.4310	57
Dimethylsulfoxide	67-68-5	78.13	18	189	1.101		1.4790	95
1,4-Dioxane	123-91-1	88.11	12	102	1.034	1.26	1.4220	12
Diphenyl ether	101-84-8	170.21	28	259	1.073		1.5790	>110
Di(propylene glycol)	110-98-5	134.18		230	1.023		1.4410	137
Di(propylene glycol) monomethyl ether	34590-94-8	148.2		90/12	0.938		1.4220	74
Ethanol	64-17-5	46.07	−130	78	0.785	1.2	1.3600	8
Ethyl acetate	141-78-6	88.11	−84	77	0.902	0.455	1.3720	−3
Ethylene glycol diacetate	111-55-7	146.14	−41	187	1.128		1.4150	82
Ethylene carbonate	96-49-1	88.06	39	244/740	1.321			160
Ethylene dichloride	107-06-2	98.96	−35	83	1.256		1.4450	15
Ethylene gylcol	107-21-1	62.07	−13	197	1.113		1.4310	>110
Ethylene glycol mono-*n*-butyl ether	111-76-2	118.18	−75	171/743	0.903		1.4190	60
Ethylene glycol diacetate	111-55-7	146.14	−41	187	1.128		1.4150	82
Ethylene glycol diethyl ether	629-14-1	118.18	−74	121	0.842		1.3923	20
Ethylene glycol dimethyl ether	110-71-4	90.12	−58	85	0.867		1.3790	0
Ethylene glycol monoethyl ether	110-80-5	90.12	−90	135	0.930		1.4070	44
Ethylhexyl acetate	103-09-3	172.27		200	0.870			77
(*S*)-(−)-Ethyl lactate	687-47-8	118.13	−26	154	1.042		1.4130	28
Formaldehyde, 37% solution	50-00-0	30.03	−15	97	1.083		1.3765	56
Formamide	75-12-7	45.04	3	210	1.134		1.4470	154
Formic acid, 96%	64-18-6	46.03	8	101	1.220		1.3704	68
Furfurol	98-00-0	98.1	−29	170	1.135		1.4860	65
Glycerine	56-81-5	92.09	20	182/20	1.261		1.4740	160
n-Heptane	142-82-5	100.21	−91	98	0.684		1.3870	−1
Hexachloroethane	67-72-1	236.74	193		2.091			
Hexafluorobenzene	392-56-3	186.05	4	80	1.612		1.3770	10
Hexamethylphosphoric triamide	680-31-9	179.2	7	232/740	1.030		1.4580	105
Hexane	110-54-3	86.18	−95	69	0.659	0.326	1.3750	−23
Isobutanol	78-83-1	74.12	−108	108	0.803		1.3960	27
Isobutyl isobutyrate	97-85-8	144.21	−81	148	0.855		1.3980	37
Isobutyric acid	79-31-2	88.11	−47	154	0.950		1.3930	55
Isophorone	78-59-1	138.21	−8	214	0.923		1.4760	84
Isopropanol	67-63-0	60.1	−89	82	0.785		1.3770	11
Isopropyl acetate	108-21-4	102.13	−73	89	0.872		1.3770	16
Isopropylamine	75-31-0	59.11	−101	34	0.694		1.3746	−32
Mesitylene	108-67-8	120.2	−45	163	0.864		1.4990	44
Mesityl oxide	141-79-7	98.15	−53	129	0.858		1.4450	30
Methanol	67-56-1	32.04	−98	65	0.791		1.3290	11
2-Methoxyethanol	109-86-4	76.1	−85	125	0.965		1.4020	46
Methyl acetate	79-20-9	74.08	−98	58	0.932		1.3610	−9
Methyl ethyl ketone	78-93-3	72.11	−87	80	0.805	0.423/15	1.3970	3
Methyl isoamyl ketone	110-12-3	114.19		145	0.888		1.4070	41
Methyl isobutyl ketone	108-10-1	100.16	−80	118	0.801		1.3960	13
Methyl propyl ketone	107-87-9	86.13	−78	110	0.802		1.3900	7
N-Methylpyrrolidone	872-50-4	99.13	−24	82/10	1.033		1.4700	86
Morpholine	110-91-8	87.12	−7	129	0.999		1.4540	35
Nitrobenzene	98-95-3	123.11	6	211	1.196		1.5510	87
Nitromethane	75-52-5	61.04	−29	101	1.127		1.3820	35
n-Octane	111-65-9	114.23	−57	127	0.703	2.256	1.3980	15
n-Pentanol	71-41-0	88.15	−78	138	0.811		1.4090	48
Perfluoroacetone	684-16-2	166.02	−129	−26				
Perfluorotributylamine	311-89-7	671.1		178	1.883			None
Phenol	108-95-2	94.11	41	182	1.071		1.5425	79
1,4-Phenylenediamine	106-50-3	108.14	144	267				

Compound	CAS Number	Formula wt.	Melting pt. (°C)	Boiling pt. (°C)	Density (at 20°C)	Viscosity at 20°C (cps)	Refractive index (at 20°C)	Flash pt. (°C)
Piperazine	110-85-0	86.14	109	146				109
n-Propanol	71-23-8	60.1	−127	97	0.804	2.256	1.3840	15
Propyl acetate	109-60-4	102.13	−95	102	0.888		1.3840	12
Propylene carbonate	108-32-7	102.09	−55	240	1.189		1.4210	132
Propylene glycol	57-55-6	76.1	−60	187	1.036		1.4320	107
Propylene glycol mono-*n*-butyl ether		132.2		72/20	0.885		1.4160	58
Propylene glycol monomethyl ether	107-98-2	90.12	−97	119	0.922		1.4030	33
Propylene glycol mono-*n*-propyl ether	1569-01-3	118.18		140–160	0.885		1.4110	48
Propylene glycol mono-*t*-butyl ether	57018-52-7	132.20		144	0.874		1.4130	44
Pyridine	110-86-1	79.1	−42	115	0.978		1.5100	20
1,1,2,2-Tetrachloroethane	79-34-5	167.85	−45	147	1.586		1.4935	None
Tetrachloroethylene	127-18-4	165.83	−22	121	1.623		1.5060	None
Tetrahydrofuran	109-99-9	72.11	−108	66	0.889	0.486	1.4070	−17
Tetrahydrofurfuryl alcohol	97-99-4	102.13	−80	178	1.054		1.4520	83
Tetrahydronaphthalene/tetralin	119-64-2	132.21	−35	207	0.973		1.5410	77
Tetramethylene sulfone	126-33-0	120.17	27	285	1.261		1.4840	165
Tetramethylurea	632-22-4	116.16	−1	177	0.971		1.4506	65
Toluene	108-88-3	92.14	−93	111	0.865	0.59	1.4960	4
1,1,1-Trichloroethane	71-55-6	133.41	−35	75	1.338		1.4366	None
Trichloroethylene	79-01-6	131.39	−85	87	1.463		1.4760	None
Tricresyl phosphate	1330-78-5	368.37		265/10	1.143		1.5550	>110
Trifluoroacetic acid	76-05-1	114.02	−15	72	1.480			None
Water	7732-18-5	18.02	0	100	0.998	1.005	1.3329	None
o-Xylene	95-47-6	106.17	−24	144	0.870		1.5050	32
p-Xylene	106-42-3	106.17	13	138	0.866	0.648	1.4950	27

SECTION IV

PHYSICAL DATA OF OLIGOMERS

Physical Data of Oligomers

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A. INTRODUCTION

Oligomers are defined as the low members of the polymeric–homologous series with molecular weights up to about 1000–2000. They are easily obtained by stepwise synthesis from suitably protected monomer derivatives or by separation from the polymers. As monodisperse oligomers are low molecular compounds of defined molecular weights, they represent ideal model substances for the polymers. By physical studies of a complete series of oligomers, substantial knowledge is gained of the relation between chain length and physical properties. In this way certain physical data of the polymers could be explained for the first time. In chemical aspects, oligomers must, on principle, have the same properties as the related polymers; they are, however, much more accessible to all studies, owing to their strictly defined and comparatively simpler structure.

Homologous oligomers differ sufficiently in their physical properties – due to their low molecular weight – so that they can be separated into individual chemicals. They can therefore be used for elucidating the structure of polymers and, in close relation, for the investigation of the mechanisms of polymerization. Oligomers are intermediates in polycondensation and polyaddition reactions and are present in more or less significant amounts in the polymers, due to equilibria between different chains, and between chains and rings. Therefore important conclusions on the structure of the related high polymers can be drawn from isolation and structure determination of oligomers, along with a comparison of the chemical properties of polymers and oligomers. In this way clear evidence can be obtained as regards the type of linkage between the monomer units in the polymer as well as the structure of unknown endgroups. Finally, unequivocal evidence on the mechanism of polymer formation is to be expected from the behavior of oligomers under the conditions of the polyreaction.

The following tables give the physical properties of the most important, linear, cyclic, and branched monodisperse oligomers which are significant for polymer chemistry. Derivatives and co-oligomers (such as co-oligo-peptides, -saccharides, and -nucleotides) as well as organosilicon and inorganic oligomers are not included, nor are any low polymers which have not been proved to be strictly monodisperse. Only those literature references are cited that indicate the best methods of synthesis and the most important physical properties of the oligomers concerned (including spectroscopic and conformational data which could not be mentioned in detail in the tables).

In the last few decades several reviews on linear and cyclic oligomers have appeared. A small list is giving below.

1. H. Zahn, G. B. Gleitsmann, Angew. Chem., **68**, 229 (1956); Angew. Chem., Int. Ed. Engl., **2**, 410 (1963).
2. M. Rothe, Makromol. Chem., **35**, 183 (1960).
3. M. Rothe, I. Rothe, in: J. Brandrup, H. Immergut (Eds.), "Polymer Handbook", 3rd Ed., Wiley, New York, 1989.
4. J. A. Semlyen, "Cyclic Polymers", Elsevier, London, 1986.
5. S. Penczek, S. Slomkowski, Compr. Polym. Sci., **3**, 725 (1989).
6. P. Maravigna, G. Montaudo, Compr. Polym. Sci., **5**, 63 (1989).
7. V. Percec, C. Pugh, O. Nuyken, S. D. Pask, Compr. Polym. Sci., **6**, 281 (1989).
8. C. V. Uglea, I. I. Negulescu, "Synthesis and Characterization of Oligomers", CRC Press, Boca Raton, 1991.
9. M. Rothe, in: E. W. Fischer, R. C. Schulz, H. Sillescu (Eds.), "Chemistry and Physics of Macromolecules", VCH, Weinheim, 1991.
10. K. Hatada, K. Ute, N. Miyatake, Prog. Polym. Sci., **19**, 1067 (1994).
11. J. M. Tour, Trends Polym. Sci., **2**, 332 (1994).

B. OLIGOMERS CONTAINING MAIN CHAIN ACYCLIC CARBON ONLY

TABLE 1. OLIGO(OLEFINS)

1.1. OLIGO(METHYLENES) AND OLIGO(ETHYLENES)

1.1.1. *n*-ALKANES $H[CH_2]_nH$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
1	16.0	−182.6	−161.6/1013	0.4240/−164	21,22,75,84,88,93
2	30.1	−183.3	−88.5/1013	0.5462/89	21,75,84,88,93
3	44.1	−187.1	−42.2/1013	0.5824/−45	21,22,75,84,88,93
4	58.1	−138.4	−0.5/1013	0.6011/0	21,22,75,84,88,112,178,192
5	72.2	−129.7	36.1/1013	0.6263/20	3,7,13,21,22,30,75,84,88,93,112,178,192,195
6	86.2	−94.0	68.7/1013	0.6594/20	3,5,7,21,22,30,75,78,84,88,93,112,178,192,195
7	100.2	−90.5	98.4/1013	0.6838/20	3,5,7,13,21,22,30,75,84,88,93,112,178,192
8	114.2	−56.8	125.5/1013	0.7026/20	3,5,7,13,21,22,30,78,84,87,88,93,112,178,192,195
9	128.3	−53.5	150.8/1013	0.7177/20	1,5,7,13,21,22,30,84,87,88,93,111,112,178,192
10	142.3	−29.7	174.1/1013	0.7301/20	1,3,5,7,18,21,22,30,84,87,88,93,111,112,178,192,195
11	156.3	−25.6	195.9/1013	0.7402/20	1,3,7,13,21,22,30,77,83,84,87,88,93,111,112
12	170.3	−9.7	216.3/1013	0.7487/20	1,7,17,21,22,30,79,84,87,88,93,111,112,178,185,192
13	184.4	−6.0	235.5/1013	0.7563/20	1,7,13,21,30,77,83,84,87,88,93,112,205,206
14	198.4	5.5	253.6/1013	0.7627/20	1,7,17,21,30,84,87,88,93,111,112
15	212.4	10.1	270.7/1013	0.7684/20	1,3,7,21,30,77,83,84,87,88,93,112
16	226.4	18.1	287.1/1013	0.7733/20	1,7,11,15,16,19,21,30,78,84,87,88,93,111,112,192
17	240.5	22.1	302.6/1013	0.7767/22	1,3,7,13,14,21,30,75,77,83,84,88,93,111,112,193
18	254.5	28.2	317.4/1013	0.7768/28	1−3,7,11,14,15,17,19,21,30,75,84,88,93,111
19	268.5	32.0	331.6/1013	0.7776/32	1,3,4,7,14,15,19,21,77,83,84,88,93,97,111,121, 198,221,235
20	282.6	36.7	345.1/1013	0.7550/70	1,3−7,11,14,15,17−19,21,22,79,82,88,91,93,107, 110,111,198
21	296.6	40.4	215/20	0.7583/70	1,3,4,10,14,15,19,21,77,83,86,88,93,111,118, 194,198,232
22	310.6	44.4	222−225/20	0.7631/70	1,3,4,7,11,14,21,78,83,88,91,93,97,111,115,121,198
23	324.6	47.4	234/20	0.7641/70	1,3,4,10,14,21,77,83,86,88,93,121,198
24	338.7	51.1	243/20	0.7657/70	1,3,4,11,14,17,21,83,86,88,91,93,99,182,198,202,204
25	352.7	54	259/20	0.7693/70	1,3−5,14,21,77,83,88,93,118,193,198,232
26	366.7	57.0	262/20	0.7704/70	1−4,7,11,14,21,83,85,88,93,198
27	380.7	60.0	270/20	0.7732/70	1−4,8,14,21,77,83,85,88,93,118,198
28	394.8	61.6	279−281/20	0.7750/70	1,4,11,13−15,17,19,21,83,85,86,88,90,93,99,100, 111,198
29	408.8	64.0	286/20	0.7755/70	1,3,4,11,14,21,77,83,85,88,91,93,118,198
30	422.8	66.0	304/20	0.7795/70	1−4,6,7,11,14,18,21,82,83,85,86,88,93,110,198
31	436.9	67.9	302/20	0.7678/100	1−4,8,77,83,85,86
32	450.9	70.4	310/20	0.7645/100	1,2,4,11,15,16,19,20,83,85,91,95,97,99,109,111, 115,185,198
33	464.9	71.3			1,4,5,77,83,91,109,116,119,198,207
34	478.9	72.9			1−4,7,11,77,83,85,86,198
35	493.0	74.7	331/20	0.7436/135	1,4,7,8,10,15,19,77,79,83,85,86,88,91,95,118
36	507.0	76.1	298.4/4	0.7783/90	1,2,4,7,10,13−15,19,77,78,83,85,86,88,90,91 107,109,111,185,191,193,194,198,207,211,221, 223,232,233
37	531.0	77.2			75,109,198
38	535.0	79.2			75,77,83,198
39	549.1	80.5			12,75,77,83
40	563.1	81.5	150/10^{-4}		6,7,10,17,18,21,75,77,82,83,86,93,99,110,198
41	577.1	81.7			75
42	591.2	82.9			76
43	605.2	85.5	332/4	0.7812/90	10,22,77,83,86
44	619.2	86.4		0.7514/135	3,11,15,19,77,79,80,83,89,91,95,107,110,111,115, 116,118,205
45	633.2	87.2			109
46	647.2	88.0			77,80,83,91,100,198
48	675.3	88.5−89			100,109,202,204
50	703.4	92.1	420−422/20		6,7,11,15,18,21,77,80,82,83,90,93,110,116, 188,193,198
52	731.4	94.0			77,80,83,90

1.1.1. *cont'd*

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
54	759.5	95.0			2,7,77,80,83,90
56	787.6	95.5–96			106,204
60	843.6	99.1	250/10^{-4}		6,7,18,21,77,80,82,83,90,93,104,109,110,187,188, 190,193,198
61	857.6	100.3			109
62	871.7	100.5			2,7,77,80,83,90,116
64	899.7	102.1		0.7654/135	2,7,13,22,77,79,80,83,90
65	913.7	102.1			109
66	927.8	103.6			9,77,80,83,90
67	941.8	104.1			9,77,80,83,90
69	969.9	104.7			109
70	983.9	105.3	300/10^{-4}		6,7,18,21,77,80,82,83,90,93,100,110,116
72	1012.0	103–105			100,202
80	1124.2	108.0			18,82,104,109
82	1152.2	110.1			20,77,80,81,83,89,110,188
94	1320.6	113.8		0.783/115	15,19,20,77,79,80,83,100,103,107,110, 116–118,188,204
96	1348.6	110–112			100,202
98		113–114			229
100	1404.7	115.2			18,22,77,80–83,93,104,110,185
102	1432.8	115.5–116.5			105,108,183,203
104	1460.8	115–115.5			106,203,204
120	1685.3	119.2			82,98,104,200,202
122		116–117			229
140	1965.8	121.0			82
144	2021.9	121			98,200,202
150	2106.1	123–123.5			105,108,183,203,227,248
160	2246.4	122.2			104,108
162		121–121.5			229
168	2358.5	121.9			98,114,177,189,199,200,202,221
192	2695.2	127.5			98,186,189,200–202,225
194		126.9			224,229
198	2779.4	127.0			105,108,183,184,197,203,224–227,229,248
210	2947.7	127.4			224,229
216	3029.8	128.4			98,200,202,225
240	3368.5				189,199
242	3396.6	120–121			229
246	3450.6	128–130			105,108,181,183,184,203,224–227,229,249
258	3621.0	128.8			224,229
288	4039.8	128.8			98,200,202
294	4126.0	130.4			108,183,203,224,225,229,248
384	5386.4	131.3			98,200
390	5470.5	132			105,108,183,203,229,248

Conformation, chain folding: Refs. 108,178 (n = 2–10,12), 179 (n = 9–44), 181,183,184,193,194,202,205–208,211,226,227,248.
Crystallization: Refs. 168,177–179,181,183,184,188–190,194,197,199–202,211,224–227,248.
Thermal behavior, phase transitions, DSC: Refs. 88,98,107,107,109,119,181,184,186–191,193,197–204,217,221,224–226,246.
NMR: Refs. 95,97 (n = 19,22,32), 99 (n = 6,8,10,14,18,24,28), 100,179,187,191,205,206,221,229,246.
IR, Raman: Refs. 98,108,116,119,178,182,192–194,200,202,232,248.
X-ray: Refs. 100,107,109,112,119,183,184,187,190,191,198,206,223,224,226,233.
Electron diffraction: Refs. 187,188.
MS: Ref. 195.
GPC: Ref. 185.

1.1.2. CYCLOALKANES $\text{[}[CH_2]_n\text{]}$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
3	42.1	−127.5	−32.8/1013	0.688/−40	23,24,93,94,178
4	56.1	−90	12.5/1013	0.7038/0	25,93,94,178,232
5	70.1	−93.9	49.3/1013	0.7457/20	26,93,94,178,183,195
6	84.2	6.6	80.7/1013	0.7784/20	27–29,93,94,99,111,178,183,195
7	98.2	−8.0	118.5/1013	0.8098/20	29,31,93,94,178,195
8	112.2	13.5	149/1000	0.8349/20	32–34,93,94,99,178,195,208

1.1.2. *cont'd*

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
9	126.2	9.7	69/19	0.8534/20	29,34,93,94,178,208
10	140.3	9.5	69/16	0.8575/20	29,35,36,93,94,99,178,180,208
11	154.3	−7.3	91/16	0.8591/20	29,37,93,94,99,208
12	168.3	60.6		0.861/20	29,38,39,40,93,94,99,101,111,114,178,180,196, 208,214,215,217,232
13	182.4	24.4	128/27	0.861/20	29,34,38,40,93,94,99,196,208,217
14	196.4	56.2	131/15	0.8259/79	17,29,38,40,41,93,94,99,180,196,208–210,214, 217,218,220
15	210.4	63.4	147/16	0.8240/78	29,38,40,44,93,94,99,196,208,217
16	224.4	60.5	170–171/27	0.819/79	16,38,40–42,93,94,99,111,169,180,196,208,209, 212,214,217,220
17	238.5	65		0.8200/77	40,43,44,93,94
18	252.5	72–73		0.8201/76	17,40–42,45,93,94,99
19	266.5	80			40,93,94,99
20	280.5	61–62			18,40,41,93,94,99,111
21	294.6	64–65			41,94,99,102
22	308.6	46.8	177/0.5	0.8174/75	38,40,43,48,94,96,99,220
23	322.6	53–54	177/0.1	0.8259/64	46,48,94,99
24	336.6	48.8	222–228/0.8		38,40,41,45,48,94–99,100,101,111,114,169,196, 202,210,211,213–217,219,228,235,246
25	350.7	53–54			47,94
26	364.7	44–46	218–219/0.7	0.8120/78	38,40,43,48,94,99,210
27	378.7	47–48			41,94,102
28	392.8	47–48	213–214/0.3	0.8243/58	38,41,48,94,97,99,102,111,219,222
29	406.8	54.5–55.5	215/1.5	0.8232/64	38,94,99
30	420.8	57–58	230/0.3	0.8233/69	18,41,42,45,48,90,93,94,99,102
32	448.9	65.5–66.5		0.8261/70	41,48,94,97,99,102,111,169,222
34	476.9	66–67	230–240/0.4	0.8229/76	48,92,94,113
36	505.0	70.0			41,45,94,97,99,100,101,102,111,114,115,120,196, 211,214,215,217,228,248,256
37	519.0	65–66			99
40	561.1	76–77			18,41,93,97,99,102,111,115,216,219,222,248,256
42	589.2	75–76			42,102
44	617.2	82–82.5			99,100
45	631.2	80–81			41,102
46	645.2	83.5–85			222
48	673.3	89.2			97,98,100,101,111,114,115,120,196,198,200, 202,207,211,213,217,230,235,248,256
50	701.4	87–88			18
54	757.5	90–91			41,102
60	841.6	96.5			101,120,211,247
64	897.7		^{13}C-NMR		97,111,115,234,248,256
72	1009.9	107.0	103.5–104.5		98,100,101,120,196,198,200,202,207,211,213, 216,217,228,230,231,235
80	1122.2		^{13}C-NMR		97,111,115,248,256
84	1178.3	108.8			101
96	1346.6	116.0			96,98,100,120,196,198,202,216,217,230,231,235
100	1402.7				228
104					248,256
120	1683.2				211,228
128	1795.5				219,248,256
144	2019.9	120.8			98,100,198,200,202,230
160	2294.3				219,248,256
168	2356.5				177
192	2693.2	124.3			98
200	2805.4				228,248,256
288	4039.8	129.8			98,100,202,230

Conformation, chain folding: Refs: 94,97,113,178 (n = 3–10,12), 179 (n = 3–96), 180,187,202,207–209,211–213,215–217,219,220,228,230,231,234,235,246,248.
Crystallization: Refs. 178,179,217,230,234,248.
Thermal behavior, phase transition, DSC: Refs. 96,101,179,196,209,216–221,230,246.
NMR: Refs. 95,97,99 (n = 6,8,10–16,18–24,26,28–30,32,36,37,40,44), 100 (n = 24,26,44,48,72,96), 111,114,115,179 (n = 3–96), 180,189,214–219,234,246,248,256.
IR, Raman: Refs. 92,96,178,179,200,202,209,220,235.
X-ray: Refs. 113,120,179,202,210,211,215,216,218,223,230,256.
Electron diffraction: Refs. 230,231.
GPC: Ref. 211.

1.2. OLIGO(PERFLUOROMETHYLENES) AND OLIGO (PERFLUOROETHYLENES)

1.2.1. PERFLUORO-*n*-ALKANES $F[CF_2]_nF$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm³)/°C	Refs.
1	88.0	−183.6	−128/1005	1.619/−129	22,49,63,64,125,130,179
2	138.0	−96.6	−78.3/1013	1.590/−78	49,50,63,64,125,130,154,179
3	188.0	−147.7	−36.7/1013		22,51,52,64,125,130,131,179
4	238.0	−129.2	−2.2/1013		22,52–54,64,125,126,129,132,139,144,179,241
5	288.1	−125.4	29.2/1013	1.620/20	22,52,54–56,124,129
6	338.1	90.7	57.2/1013	1.6995/20	22,52,57,58,124–126,134,144,175,179
7	388.1	93.3	82.2/1013	1.7333/20	22,52,56,59,125,128,134,144,175,179
8	438.1	−19.0	103.3/1013	1.776/25	22,52,59,125,134,144,175,179
9	488.1	−16	125.3/1013	1.799/25	22,52,59,127
10	538.1	36	144.2/1013	1.770/45	22,52,59,127
11	588.1	57	161/1013	1.745/70	22,52,59
12	638.1	75.3	178/1013	1.670/113.5	22,52,59,127,175,179,250,251
14	738.1		IR, Raman spectr.		124,127,133,179,240,249
16	838.2	129.0	232/1013		54,59,127,175
18	938.2		IR, Raman spectr.		124,127,133
20	1038.2	164.7	IR, Raman spectr.		124,127,133,175,179,243–245,249,252
24	1238.2	192.0	IR, Raman spectr.		175,242,243,245

1.2.2. PERFLUORO-CYCLOALKANES $\ulcorner[CH_2]_n\urcorner$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
3	150.0	−80	−31.5/1013	60,61
4	200.0	−38.7	−2/1013	61,63,144
5	250.0	10.3	22.5/1013	51,54–56,61,64
6	300.0	58.2	51 (subl.)	51,56,59,64,65,144
7	350.0	−37	76.2/1013	51,59
8	400.0	−55	101.5/1013	59
9	450.0	−56	124.6/1013	59
10	500.0	−44	146.5/1013	59

1.2.3. α,ω-DISUBSTITUTED PERFLUORO-*n*-ALKANES $X-[CF_2]_n-X$

X	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
H	1	52.0		−51.6/1013	
	2	102.0		−20°/1013	144,145
	3	152.0		10–11/1013	146
	4	202.0		43.5/1013	135–137,144
	5	252.1		66.5/1013	136
	6	302.1		88.5–89.5/1013	135–137,144
	7	352.1		104.2/1013	136,137
	8	402.1		130.6/1013	136,137
	10	502.1	42–43	153–154/1013	135–137
CH_3S	2	194.2		67–70/53	135
	4	294.2		80–83/27	135
	6	394.3		81–84/5	135
	8	494.3		102–106/5	135
	10	594.3		90–94/1	135
	12	694.3		118–122/1	135
CH_3SO_2	2	258.4	125		135
	4	358.4	151		135
	6	458.5	168		135
	8	558.5	171		135
	10	658.5	187		135
CH_3COO	4	318.1		115–116/133	135
	6	418.1		121–123/67	135
	8	518.1		126–127/33	135
	10	618.2		144–146/33	135
$HOCH_2$	2	164.1	81–82		135
	4	264.1	69	135–136/36	135

1.2.3. *cont'd*

X	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
I	2	353.8		113/1013	236
	4	453.8		149/1013	236
	6	553.8		186/1013	236
	8	653.8		222/1013	236
	10	753.8		258/1013	236

1.2.4. α,ω-DISUBSTITUTED PERFLUORO-*n*-ALKANES WITH DIFFERENT ENDGROUPS X-$[CF_2]_n$-Y

X	Y	*n*	Mol. wt.	m.p.	b.p. (°C/mbar)	d_4^{20} (g/cm^3)	Refs.
H	Cl	2	136.5		−10/1013		61
		3	186.5		21/1013	1.556	61
		4	236.5		50/1013	1.607	61
		5	286.5		77/1013	1.661	61
		6	336.5		100/1013	1.719	61
		7	386.5		122.5/1013	1.738	61
		8	436.5		144.5/1013	1.778	61
		9	486.5		162.5/1013		61
		10	536.5	46	162.5/1013		61
		11	586.5	52	177/1013		61
		12	636.5	78	202/1013		61
		13	686.5	87	213/1013		61
		14	736.5	95	227–228/1013		61
H	F	2	120.0		−48.5/1013		240,252
		3	170.0		−16/986		240
		4	220.0		14/986		240
		5	270.0		46/986		240
		7	370.0		94/986		240
		12	620.0	72			250,251
$(CF_3)_2CF$	I	4	495.9		139/1026		237
		6	596.0		189/1008		237

1.2.5. SEMIFLUORINATED *n*-ALKANES F-$[CF_2]_m$-$[CH_2]_n$-H

m	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1	1	84.0		−46.7/1013	155,176
6	12	488.3	−13.0		138,176
7	7	468.3			255
8	8	532.3	27.0		138,176
10		540.3	35		251
10	8	632.4	53.0		138,176
10	10	660.4	64		138,176,251,255
10	12	688.4	65		138,176,251
12	2	648.2	71.0		138,176,250,251,255
12	4	676.2	76.0		138,176,250,251,255
12	6	704.3	84		138,176,250,251,255
12	8	732.4	88		138,176,251
12	10	760.4	92		138,176,251
12	12	788.4	91		138,176,250,251,255
12	14	816.5	93		138,176,251
12	16	844.5	94		138,176,251
12	18	872.6	97		138,176,251
12	20	800.6	100		138,176,250,251,255

1.2.6. ANIONIC OLIGOMERS OF PERFLUORO-*n*-ALKENES (MAJOR OLIGOMERS, *cis/trans* MIXTURES)

n	Compound	Mol. wt.	b.p. (°C/mbar)
$CF_2=CF_2$ (Refs. 238,239,243):			
4	$CF_3CF_2-C(CF_3)=C(CF_3)-CF_2CF_3$	400.1	95–96/1013
5	$(CF_3CF_2)_2(CF_3)C-C(CF_3)=CF(CF_3)$	500.1	135–138/1012

1.2.6. *cont'd*

n	Compound	Mol. wt.	b.p. (°C/mbar)
6	$(CF_3CF_2)_2(CF_3)C-C(=CF_2)-CF(CF_2CF_3)_2$	600.1	160–165/1013
$CF_3-CF=CF_2$ (Refs. 253,254,257):			
2	$(CF_3)_2CF-CF=CF-CF_3$	300.1	46–48.5/1013
	$(CF_3)_2C=CF-CF_2-CF_3$		46–48.5/1013
3	$(CF_3)_2C=C(CF_2-CF_3)-CF(CF_3)_2$	450.1	114.8/1013
	$[(CF_3)_2CF]_2C=CF-CF_3$		110–115/1013
	$(CF_3)_2CF-CF=C(CF_3)-CF_2-CF_2-CF_3$		110–115/1013

1.2.7. PERFLUORO-1-ALKENES $F-[CF_2]_n-CF=CR_2$

n	Mol. wt.	b.p. at ~983 mbar (°C)	n_D^{20}	Refs.
1	150.0	−29		240
2	200.0	1		240
3	250.0	28–29	1.2571	240
5	350.0	81	1.2782	240
7	450.1	123	1.2868	240

Conformation; Crystallization: Refs. 126,179,242,243–245,251.
Phase transition, DSC: Refs. 55,125,138,175,176,179,242,244,245,249,251.
NMR: Refs. 134,179,250,251.
X-ray: Refs. 65,138,175,176,179,244,245.
IR, Raman: Refs. 125–127,133,138,175,176,241,242,245,249,250,255.
MS: Ref. 56.

1.3. OLIGO(ISOBUTENES)

Compound	m.p. (°C)	b.p. (°C/1013 mbar)	d_4^{20} (g/cm³)
MONOMER (mol. wt. 56.1)			
Isobutene $CH_2=C(CH_3)_2$		−6	
DIMERS (mol. wt. 112.2; Refs. 66–70,72,74,122,141,143,147,148,150–152,153)			
2,4,4-Trimethyl-1-pentene $CH_2=C(CH_3)-CH_2C(CH_3)_3$	−93.6	101.4	0.7150
2,4,4-Trimethyl-2-pentene $CH_3-C(CH_3)=CH-C(CH_3)_3$	−106.5	104.9	0.7212
3,4,4-Trimethyl-2-pentene $CH_3-CH=C(CH_3)-C(CH_3)_3$		112.3	0.7392
2,3,4-Trimethyl-2-pentene $(CH_3)_2C=C(CH_3)-CH(CH_3)_2$	−113.4	116.3	0.7434
2,3,4-Trimethyl-1-pentene $CH_2=C(CH_3)-CH(CH_3)_2$		108	0.729
2,3,3-Trimethyl-1-pentene $CH_2=C(CH_3)-C(CH_3)_2-CH_2-CH_3$	−69	108.4	0.7352
1,1,3-Trimethyl-cyclopentane		105.0	0.7481

1.3. *cont'd*

Compound	m.p. (°C)	b.p. (°C/1013 mbar)	d_4^{20} (g/cm^3)
TRIMERS (mol. wt. 168.3, Refs. 71,73,74,123,140,142,143,147–149,151,152)			
1,1-Dineopentylethylene $CH_2=C[CH_2C(CH_3)_2]_2$		178.0/1013	0.7599
2,2,4,6,6-Pentamethyl-3-heptene $(CH_3)_3C-CH=C(CH_3)-CH_2-C(CH_3)_3$		^{13}C-NMR	
TETRAMERS (mol. wt. 224.4; Refs. 74,141–143,147,151,152)			
2,2,6,6,8,8-Hexamethyl-4-methylene-nonane $(CH_3)_3C-CH_2-C(=CH_2)-CH_2-C(CH_3)_2-CH_2-C(CH_3)_3$		106–107/11 ^{13}C-NMR	
2,2,4,6,6,8,8-Heptamethyl-4-nonene $(CH_3)_3C-CH_2-C(CH_3)=CH-C(CH_3)_2-CH_2-C(CH_3)_3$		^{13}C-NMR	

1.4. OLIGO(1-ALKENYLENES)

x	*n*	Configuration	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{25}	Refs.
2	2	*cis/trans*	108.2	−70	150.8/1002		163,170–172
	3	*cis/trans*	162.3		104/15		164,173,174
	4	*cis/trans*	216.3		88–91/0.07		161,164
	5	*cis/trans*	270.4		105–110/0.07		161,164
	6	*cis/trans*	324.5		162/0.1		161
	n = 4–13 *cis/trans* GC Refs. 161,166						
	n = 4–7,10,11 *cis/trans* ^{1}H-, ^{13}C-NMR Ref. 161						
3	2	*cis*	136.2	21–23		1.4972	162
4	2	*cis*	164.3	0	51/0.5 (*cis/trans*)	1.4971	40,157,162
	3	*cis/trans*	246.4		112/0.5		157
	4	*cis*	328.6	25–27			40
5	2	*cis*	192.3	46	48/0.001 (*cis/trans*)		40,157
	3	*cis/trans*	288.5		116/0.001		157
6	2	*cis*	220.4	18.5	89/0.07 (*cis/trans*)		40,157,158
	3	*cis/trans*	330.6		130/0.07		157,158
n = 2–8 (*cis/trans*): GPC, b.p., n_D^{20}, IR, UV, MS Refs. 158–160							
7	2	*cis*	248.4	38.5	122/0.4 (*cis/trans*)		40,157
8	2	*cis*	276.5	38–39	123–124/0.2 (*cis/trans*)		40,157
10	1	*cis/trans*	166.3		100/0.11		156
	2	*cis/trans*	332.6	54–55	150/0.09		156–158,160
		trans/trans		IR: 970 cm^{-1}			156
		trans/cis		970, 725 cm^{-1}			156
		cis/cis		725 cm^{-1}			156
	3	*cis/trans*	498.3		220/0.12		156
	4	*cis/trans*	665.2		255/0.12		156
	n = 2–14: MS Refs. 167,168						
	n = 2–9: GPC Ref. 160						
13	*n* = 2–7: MS Ref. 166						

REFERENCES

1. F. Krafft, Chem. Ber., **15**, 1687, 1711 (1882); **19**, 2218 (1886); **40**, 4479 (1907).
2. A. Gascard, Ann. Chim. (Paris) (9), **15**, 332 (1921).
3. A. Mueller, Proc. Roy. Soc. (London) A, **120**, 437 (1928); **127**, 417 (1930); **138**, 514 (1932).
4. J. H. Hildebrand, A. Wachter, J. Am. Chem. Soc., **51**, 2487 (1929).
5. G. S. Parks, H. M. Huffmann, S. B. Thomas, J. Am. Chem. Soc., **52**, 1032 (1930).
6. W. H. Carothers, J. W. Hill, J. E. Kirby, R. A. Jacobsen, J. Am. Chem. Soc., **52**, 5279 (1930).
7. W. E. Gramer, K. van Bibber, A. M. King, J. Chem. Soc., 1533 (1931).
8. H. Staudinger, F. Staiger, Chem. Ber., **68**, 707 (1935).

9. F. Francis, A. M. King, J. A. V. Willis, J. Chem. Soc., 999 (1937)
10. H. J. Backer, J. Strating, Rec. Trav. Chim. Pays-Bas, **59**, 933 (1940); **55**, 903 (1936).
11. W. F. Seyer, R. F. Patterson, J. L. Keays, J. Am. Chem. Soc., **66**, 179 (1944); W. F. Seyer, Wm. Morris, *ibid.*, **61**, 1114 (1939).
12. E. Stenhagen, B. Taegtstroem, J. Am. Chem. Soc., **66**, 845 (1944).
13. A. K. Doolittle, R. H. Peterson, J. Am. Chem. Soc., **73**, 2145 (1951).
14. A. A. Schaerer, C. J. Busso, A. E. Smith, L. B. Skinner, J. Am. Chem. Soc., **77**, 2017 (1955).
15. P. R. Templin, Ind. Eng. Chem., **48**, 154 (1956).
16. F. Sondheimer, Y. Amiel, J. Am. Chem. Soc., **79**, 5817 (1957).
17. F. Sondheimer, Y. Amiel, R. Wolovsky, J. Am. Chem. Soc., **79**, 6263 (1957).
18. F. Sondheimer, R. Wolovsky, D. A. Ben-Efraim, J. Am. Chem. Soc., **83**, 1686 (1961).
19. A. Odajima, J. A. Sauer, A. E. Woodward, J. Phys. Chem., **66**, 718 (1962).
20. D. W. McCall, D. C. Douglass, E. W. Anderson, Ber. Bunsenges. Phys. Chem., **67**, 336 (1963); D. C. Douglass, D. W. McCall, J. Phys. Chem., **62**, 1102 (1958).
21. J. D. Downer, K. J. Beynon, in: "Rodd's Chemistry of Carbon Compounds", vol. I, Part A, 2nd ed., Elsevier, Amsterdam, 1964, p. 367.
22. R. Simha, A. J. Havlik, J. Am. Chem. Soc., **86**, 197 (1964).
23. M. Trautz, K. Winkler, J. Prakt. Chem., **104**, 37 (1922).
24. R. A. Ruehrwein, T. M. Powell, J. Am. Chem. Soc., **68**, 1063 (1946).
25. G. B. Heisig, J. Am. Chem. Soc., **63**, 1698 (1941).
26. J. G. Aston, H. L. Fink, S. C. Schumann, J. Am. Chem. Soc., **65**, 341 (1943).
27. J. G. Aston, G. J. Szasz, H. L. Fink, J. Am. Chem. Soc., **65**, 1135 (1943).
28. R. A. Raphael, in: E. R. Rodd (Ed.), "Chemistry of Carbon Compounds", vol. II, Part A, 1st ed., Elsevier, Amsterdam, 1953, p. 124.
29. L. Ruzicka, P. A. Plattner, H. Wild, Helv. Chim. Acta, **29**, 1611 (1946).
30. R. M. Deanesley, L. T. Carleton, J. Phys. Chem., **45**, 1104 (1941).
31. R. Willstaetter, T. Kametaka, Chem. Ber., **41**, 1480 (1908).
32. N. D. Zelinsky, M. G. Freimann, Chem. Ber., **63**, 1485 (1930).
33. L. Ruzicka, H. A. Boekenoogen, Helv. Chim. Acta, **14**, 1319 (1931).
34. L. Ruzicka, P. A. Plattner, H. Wild, Helv. Chim. Acta, **28**, 395 (1945).
35. P. A. Plattner, J. Huelstkamp, Helv. Chim. Acta, **27**, 220 (1944).
36. W. Hueckel, A. Gercke, A. Gross, Chem. Ber., **66**, 563 (1933).
37. P. A. Plattner, Helv. Chim. Acta, **27**, 801 (1944).
38. L. Ruzicka, M. Stoll, H. W. Huyser, H. A. Boekenoogen, Helv. Chim. Acta, **13**, 1152 (1930).
39. R. Wolovsky, F. Sondheimer, J. Am. Chem. Soc., **84**, 2844 (1962).
40. J. Dale, A. J. Hubert, G. S. D. King, J. Chem. Soc., 73 (1963).
41. F. Sondheimer, Y. Amiel, R. Wolosky, J. Am. Chem. Soc., **81**, 4600 (1959).
42. L. Ruzicka, W. Bruegger, C. F. Seidel, H. Schinz, Helv. Chim. Acta, **11**, 496 (1928).
43. L. Ruzicka, G. Giacomello, Helv. Chim. Acta, **20**, 548 (1937).
44. L. Ruzicka, W. Bruegger, M. Pfeiffer, H. Schinz, M. Stoll, Helv. Chim. Acta, **9**, 499 (1926).
45. F. Sondheimer, R. Wolovsky, J. Am. Chem. Soc., **84**, 260 (1962).
46. R. Stol, Helv. Chim. Acta, **16**, 493 (1933).
47. A. J. Hubert, J. Dale, J. Chem. Soc., 86 (1963).
48. L. Ruzicka, M. Huerbin, M. Furter, Helv. Chim. Acta, **17**, 78 (1934).
49. O. Ruff, Angew. Chem., **46**, 739 (1933).
50. E. L. Pace, J. G. Aston, J. Am. Chem. Soc., **70**, 566 (1948).
51. J. H. Simons, L. P. Block, J. Am. Chem. Soc., **61**, 2962 (1939).
52. R. N. Haszeldine, J. Chem. Soc., 3761 (1953)
53. J. A. Brown, W. H. Mears, J. Phys. Chem., **62**, 960 (1958).
54. W. B. Burford, R. D. Fowler, J. M. Hamilton, Jr., H. C. Anderson, C. E. Weber, R. G. Sweet, Ind. Eng. Chem., **39**, 319 (1947).
55. L. L. Burger, G. H. Cady, J. Am. Chem. Soc., **73**, 4243 (1951).
56. F. L. Mohler, E. G. Bloom, J. H. Lengel, C. E. Wise, J. Am. Chem. Soc., **71**, 337 (1949).
57. V. E. Stiles, G. H. Cady, J. Am. Chem. Soc., **74**, 3771 (1952).
58. R. D. Dunlap, C. J. Murphy, R. G. Bedford, J. Am. Chem. Soc., **80**, 83 (1958).
59. R. N. Haszeldine, F. Smith, J. Chem. Soc., 3617 (1950); 603 (1951).
60. B. Atkinson, J. Chem. Soc., 2684 (1952).
61. J. D. Park, A. F. Benning, F. Downing, J. F. Laucius, R. C. McHarness, Ind. Eng. Chem., **39**, 354 (1947).
62. J. R. Lacher, G. W. Tompkin, J. D. Park, J. Am. Chem. Soc., **74**, 1693 (1952).
63. G. Bier, R. Schaeff, K.-H. Kahrs, Angew. Chem., **66**, 285 (1954).
64. N. Fukuhara, L. A. Bigelow, J. Am. Chem. Soc., **63**, 2792 (1941).
65. H. J. Christoffers, E. C. Lingafelter, G. H. Cady, J. Am. Chem. Soc., **69**, 2502 (1947).
66. F. C. Whitmore, S. N. Wrenn, J. Am. Chem. Soc., **53**, 3136 (1931).
67. C. O. Tongberg, J. D. Pickens, M. R. Fenske, F. C. Whitmore, J. Am. Chem. Soc., **54**, 3706 (1932).
68. F. C. Whitmore, K. C. Laughlin, J. Am. Chem. Soc., **54**, 4011 (1932).
69. F. C. Whitmore, K. C. Laughlin, J. F. Matuszeski, J. D. Surmatis, J. Am. Chem. Soc., **63**, 756 (1941).

70. J. A. Dixon, N. C. Cook, F. C. Whitmore, J. Am. Chem. Soc., **70**, 3361 (1948).
71. F. C. Whitmore, C. D. Wilson, J. V. Capinjola, C. O. Tongberg, G. H. Fleming, R. V. McGrew, J. N. Cosby, J. Am. Chem. Soc., **63**, 2035 (1941).
72. F. L. Howard, T. W. Mears, A. Fookson, P. Pomerantz, D. B. Brooks, J. Res. Natl. Bur. Std., **38**, 365 (1947).
73. P. D. Bartlett, G. L. Fraser, R. B. Woodward, J. Am. Chem. Soc., **63**, 495 (1941).
74. M. I. Batuev, A. P. Meshcheriakov, A. D. Matveeva, Izv. Akad. Nauk, SSSR, 75 (1958); Bull. Acad. Sci. USSR, 66 (1958).
75. F. D. Rossini, K. S. Pitzer, R. L. Arnett, R. M. Braun, G. C. Pimental, "Selected Values of Physical and Thermodynamic Properties of Hydrocarbons and Related Compounds", Carnegie Press, Pittsburgh, Pa., 1953.
76. G. L. Clark, H. A. Smith, Ind. Eng. Chem., **23**, 700 (1931).
77. P. J. Flory, A. Vrij, J. Am. Chem. Soc., **85**, 3548 (1963).
78. R. A. Orwoll, P. J. Flory, J. Am. Chem. Soc., **89**, 6814 (1967).
79. R. R. Reinhard, J. A. Dixon, J. Org. Chem., **30**, 1450 (1965).
80. M. G. Broadhurst, J. Chem. Phys., **36**, 2578 (1962).
81. G. Staellberg, S. Staellberg-Stenhagen, E. Stenhagen, Acta Chem. Scand., **6**, 313 (1952).
82. W. Heitz, Th. Wirth, R. Peters, G. Strobl, E. W. Fischer, Makromol. Chem., **162**, 63 (1972).
83. M. G. Broadhurst, J. Res. Natl. Bur. Std., **66A**, 241 (1962).
84. M. Kurata, S. Isida, J. Chem. Phys., **23**, 1126 (1955).
85. S. H. Piper, A. C. Chibnall, S. J. Hopkins, A. Pollard, J. A. B. Smith, E. F. Williams, Biochem. J., **25**, 2072 (1931).
86. W. M. Mazee, Rec. Trav. Chim. Pays-Bas, **67**, 197 (1948).
87. H. L. Finke, M. E. Gross, G. Waddington, H. M. Huffmann, J. Am. Chem. Soc., **76**, 333 (1954).
88. A. A. Schaerer, G. G. Baylé, W. M. Mazee, Rec. Trav. Chim. Pays-Bas, **75**, 513 (1956).
89. B. G. Ranby, F. F. Morehead, N. M. Walter, J. Polym. Sci., **44**, 349 (1960).
90. G. Egloff, "Physical Constants of Hydrocarbons", Vols. 1 and 5, A. C. S. Monograph Series, Reinhold, New York, 1953.
91. M. Brini, Bull. Soc. Chim. France, 996 (1955).
92. T. Tasumi, T. Shimanouchi, R. F. Schaufele, Polym. J., **6**, 740 (1971).
93. D. Barton, W. D. Ollis, in: J. F. Stoddart (Ed.), Comprehensive Organic Chemistry, Vol. 1, Pergamon Press, Oxford, 1979, pp. 49–51.
94. J. Dale, J. Chem. Soc., 93 (1963).
95. M. Moeller, H.-J. Cantow, J. K. Krueger, H. Hoecker, Polym. Bull., **5**, 125 (1981).
96. H.-P. Grossmann, Polym. Bull., **5**, 137 (1981).
97. I. Ando, T. Yamanobe, T. Sorita, T. Komoto, H. Sato, K. Deguchi, M. Imanari, Macromolecules, **17**, 1955 (1984).
98. K. S. Lee, G. Wegner, Makromol. Chem., Rapid Commun., **6**, 203 (1985).
99. H. Fritz, E. Logemann, G. Schill, T. Winkler, Chem. Ber., **109**, 1258 (1976).
100. G. Schill, C. Zuercher, H. Fritz, Chem. Ber., **111**, 2901 (1978).
101. H. Hoecker, K. Riebel, Makromol. Chem., **178**, 3101 (1977).
102. F. Sondheimer, Y. Amiel, R. Wolovsky, J. Am. Chem. Soc., **79**, 4247 (1957).
103. J. D. Hoffman, Macromolecules, **18**, 772 (1985).
104. K. Takamizawa, Y. Sasaki, K. Kono, Y. Urabe, Rep. Progr. Polym. Phys. Jpn., **19**, 285 (1976).
105. I. Bidd, M. C. Whiting, J. Chem. Soc., Chem. Commun., 543 (1985).
106. O. I. Paynter, D. J. Simmonds, M. C. Whiting, J. Chem. Soc., Chem. Commun., 1165 (1982).
107. P. K. Sullivan, J. J. Weeks, J. Res. Natl. Bur. Std., **74A**, 203 (1970); R. G. Sullivan, *ibid.*, **78A**, 129 (1974).
108. G. Ungar, I. Stejny, A. Keller, I. Bidd, M. C. Whiting, Science, **229**, 386 (1985).
109. K. Takamizawa, Y. Ogawa, I. Oyama, Polym. J., **14**, 441 (1982).
110. J. N. Hay, J. Polym. Sci., Polym. Chem. Ed., **14**, 2845 (1976).
111. T. Yamanobe, T. Sorita, I. Ando, H. Sato, Makromol. Chem., **186**, 2071 (1985).
112. R. G. Snyder, J. Chem. Phys., **47**, 1316 (1967).
113. B. A. Newman, H. F. Kay, J. Appl. Phys., **38**, 4105 (1967); Acta Cryst. B, **24**, 615 (1968).
114. H.-J. Cantow, D. Emeis, W. Gronski, A. Hasenhindl, D. Lausberg, M. Moeller, Y. Shahab, Makromol. Chem., Suppl., **7**, 63 (1984).
115. T. Sorita, T. Yamanobe, T. Komoto, I. Ando, H. Sato, K. Deguchi, M. Imaneri, Makromol. Chem., Rapid Commun., **5**, 657 (1984).
116. G. R. Strobl, R. Eckel, J. Polym. Sci., Polym. Phys. Ed., **14**, 913 (1976).
117. F. Khoury, J. Appl. Phys., **34**, 73 (1963).
118. G. T. Davis, J. J. Weeks, G. M. Martin, R. K. Eby, J. Appl. Phys., **45**, 4175 (1974).
119. B. Ewen, D. Richter, J. Chem. Phys., **69**, 2954 (1978); G. R. Strobl, B. Ewen, E. W. Fischer, W. Piesczek, J. Chem. Phys., **61**, 5257, 5265 (1974); W. Piesczek, G. R. Strobl, K. Malzahn, Acta Cryst. B, **30**, 1278 (1974); G. R. Strobl, Coll. Polym. Sci., **254**, 170 (1976).
120. T. Trzebiatowski, M. Dräger, G. R. Strobl, Makromol. Chem., **183**, 731 (1982).
121. J. D. Barnes, J. Chem. Phys., **58**, 5193 (1973).
122. F. C. Whitmore, J. M. Church, J. Am. Chem. Soc., **54**, 3710 (1932).
123. R. J. McCubbin, J. Am. Chem. Soc., **53**, 356 (1931); R. J. McCubbin, H. Adkins, *ibid.*, **52**, 2547 (1930).
124. J. L. Koenig, F. J. Brerio, J. Chem. Phys., **50**, 2823 (1909).
125. M. Campos-Vallette, M. Rey-Lafon, R. Lagnier, Chem. Phys. Lett., **89**, 189 (1982).
126. P. Piaggio, P. G. Francese, G. Masetti, G. Dellepiane, J. Mol. Struct., **26**, 421 (1975).
127. J. Rabolt, B. Fanconi, Polymer **18**, 1258 (1977), Macromolecules **11**, 740 (1978).
128. G. D. Oliver, J. W. Grisard, J. Am. Chem. Soc., **73**, 1688 (1951).

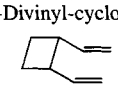
129. J. H. Simons, R. D. Dunlap, J. Chem. Phys., **18**, 335 (1950).
130. J. H. Smith, E. L. Pace, J. Phys. Chem., **73**, 2368, 4232 (1969).
131. E. L. Pace, A. C. Plaush, J. Chem. Phys., **47**, 38 (1967).
132. J. H. Simons, J. W. Mausteller, J. Chem. Phys., **20**, 1516 (1952).
133. G. W. Chantry, E. A. Nicol, R. G. Jones, H. A. Willis, M. E. A. Cudby, Polymer, **18**, 37 (1977).
134. J. R. Lyerla Jr., D. L. VanderHart, J. Am. Chem. Soc., **98**, 1697 (1976).
135. R. B. Ward, J. Org. Chem., **30**, 3009, (1965).
136. K. Matsuo, W. H. Stockmayer, J. Phys. Chem., **85**, 3307 (1981).
137. T. W. Bates, W. H. Stockmayer, Macromolecules, **1**, 12 (1968).
138. J. F. Rabolt, T. P. Russell, R. J. Twieg, Macromolecules, **17**, 2786 (1984).
139. T. W. Bates, Trans. Faraday Soc., **63**, 1825 (1967).
140. T. Higashimura, Y. Miyoshi, H. Hasegawa, J. Appl. Polym. Sci., **28**, 241 (1983).
141. H. Hasegawa, T. Higashimura, J. Appl. Polym. Sci., **27**, 171 (1982).
142. S. A. Francis, E. D. Archer, Anal. Chem., **35**, 1363 (1963).
143. H. Schütz, R. Radeglia, G. Heublein, Acta Polym., **36**, 415 (1985).
144. R. N. Haszeldine, J. Chem. Soc., 4259 (1952); R. N. Haszeldine, R. J. Marklow, J. Chem. Soc. 962 (1956).
145. A. L. Henne, M. W. Renoll, J. Am. Chem. Soc., **58**, 887 (1936).
146. F. B. Downing, A. F. Benning, R. C. McHarness, United States Patent 2.387.247 (1945), 2.413.695 (1947); through Chem. Abstr., 588 (1946), 2427 (1947).
147. J. P. Kennedy, S. Rengachary, Adv. Polym. Sci., **14**, 1 (1974).
148. L. Schmerling, V. N. Ipatieff, Adv. Catal., **2**, 21 (1950).
149. J. Kriz, M. Marek, Makromol. Chem., **163**, 155, 171 (1973).
150. J. B. McKinley, D. R. Stevens, W. E. Baldwin, J. Am. Chem. Soc., **67**, 1455 (1945).
151. T. Higashimura, M. Swamoto, Adv. Polym. Sci., **62**, 50 (1984).
152. W. O. Haag, Chem. Eng. Progr., **63**, 145 (1967).
153. D. H. Rank, E. R. Bordner, J. Chem. Phys., **3**, 248 (1935).
154. E. G. Locke, W. R. Brode, A. L. Henne, J. Am. Chem. Soc., **56**, 1726 (1934).
155. A. L. Henne, M. W. Renoll, J. Am. Chem. Soc., **58**, 889 (1936).
156. R. Wolovsky, Z. Nir, Synthesis, **4**, 134 (1972).
157. H. J. Bestmann, H. Pfüller, Angew. Chem., **84**, 528 (1972).
158. H. Höcker, R. Musch, Makromol. Chem., **157**, 201 1972; **175**, 1395 (1974).
159. E. A. Ofstead, E. Asch, H. Höcker, Macromol. Synth., **6**, 69 (1977).
160. H. Höcker, W. Reimann, K. Riebel, Z. Szentivanyi, Makromol. Chem., **177**, 1707 (1976).
161. H. Sato, Y. Tanaka, T. Taketomi, Makromol. Chem., **178**, 1993 (1977).
162. D. J. Cram, N. L. Allinger, J. Am. Chem. Soc., **78**, 2518 (1956).
163. A. C. Cope, C. L. Stevens, F. A. Hochstein, J. Am. Chem. Soc., **72**, 2510 (1950).
164. K. Saito, T. Yamaguchi, K. Tanabe, T. Ogura, M. Yagi, Bull. Chem. Soc. Japan, **52**, 3192 (1979).
165. H. Kumobayashi, T. Ogura, S. Akutagawa, K. Saito, T. Yamaguchi, K. Tanabe, Chem. Lett., 317 (1976).
166. H. Höcker, W. Reimann, L. Reif, K. Riebel, Makromol. Chem., J. Mol. Catal., **8**, 191 (1980).
167. R. Wolovsky, J. Am. Chem. Soc., **92**, 2132 (1970).
168. D. A. Ben-Efraim, C. Batich, E. Wasserman, J. Am. Chem. Soc., **92**, 2133 (1970).
169. E. Wasserman, D. A. Ben-Efraim, R. Wolovsky, J. Am. Chem. Soc., **90**, 3286 (1968).
170. W. Brenner, P. Heimbach, H. Hey, E. W. Mueller, G. Wilke, Ann. Chem., **727**, 161 (1969).
171. K. Ziegler, H. Wilms, Ann. Chem., **567**, 1 (1950); K. Ziegler, H. Sauer, L. Bruns, H. Froitzheim-Kuehlhorn, J. Schneider, Ann. Chem., **589**, 122 (1954).
172. A. C. Cope, W. J. Bailey, J. Am. Chem. Soc., **70**, 2305 (1948); A. C. Cope, C. L. Stevens, F. A. Hochstein, *ibid.*, **72**, 2510 (1950).
173. G. Wilke, Angew. Chem., **69**, 397 (1957); G. Wilke, E. W. Mueller, M. Kroener, *ibid.*, **73**, 33 (1961); B. Bogdanovic, P. Heimbach, M. Kroener, G. Wilke, E. G. Hoffmann, J. Brandt, Ann. Chem., **727**, 143 (1969).
174. H. Breil, P. Heimbach, M. Kroener, H. Mueller, G. Wilke, Makromol. Chem., **69**, 18 (1963).
175. H. W. Starkweather, Jr., Macromolecules, **19**, 1131 (1986).
176. Th. P. Russell, J. F. Rabolt, R. J. Twieg, R. L. Siemens, Macromolecules, **19**, 1135 (1986).
177. R. G. Alamo, NATO ASI, Ser. C, **405**, 73 (1993); Chem. Abstr., **121**, 110078f (1994).
178. N. L. Allinger, Y. H. Yuh, J.-H. Lii, J. Am. Chem. Soc., **111**, 8551 (1989); J.-H. Lii, N. L. Allinger, *ibid.*, **111**, 8566, 8576 (1989).
179. B. Wunderlich, M. Möller, J. Grebowicz, H. Baur, Adv. Polym. Sci., **87**, 1 (1988).
180. F. A. L. Anet, A. K. Cheng, J. Am. Chem. Soc., **97**, 2420 (1975); F. A. L. Anet, A. K. Cheng, J. Wagner, *ibid.*, **94**, 9250 (1972).
181. J. D. Hoffman, Polymer, **32**, 2828 (1991).
182. K. Ishii, M. Nukaga, Y. Hibino, S. Hagiwara, H. Nakayama, Bull. Chem. Soc. Jpn., **68**, 1323 (1995).
183. A. Keller, G. Ungar, S. I. Organ, Polym. Preprints, **30**, 263 (1989); A. Keller, S. J. Organ, G. Ungar, Makromol. Chem., Macromol. Symp., **48/49**, 93 (1991); J. Polym. Sci., Polym. Phys., **25**, 2409 (1987); J. Polym. Sci., C, Polym. Lett., **26**, 259 (1988); G. Ungar, Integr. Fundament. Polym. Sci. Technol., Ed. P. J. Lemstra, L. A. Kleintjens, Elsevier, London, **2**, 346 (1988).
184. G. Ungar, A. Keller, Polym., **27**, 1835 (1986).
185. D. J. Blundell, A. Keller, I. M. Ward, I. J. Grant, J. Polym. Sci., Part B, Polym. Lett., **4**, 781 (1966).
186. L. Mandelkern, A. Prasad, R. G. Alamo, G. M. Stack, Macromolecules, **23**, 3696 (1990).
187. W. L. Jarrett, L. J. Mathias, R. G. Alamo, L. Mandelkern, D. L. Dorset, Macromolecules, **25**, 3468 (1992).

188. D. L. Dorset, R. G. Alamo, L. Mandelkern, Macromolecules, **25**, 6284 (1992).
189. R. G. Alamo, L. Mandelkern, G. M. Stack, C. Kröhnke, G. Wegner, Macromolecules, **27**, 147 (1994).
190. L. Mandelkern, R. G. Alamo, E. L. Sokolov, Y. Li, B. Chu, Macromolecules, **27**, 2324 (1994).
191. M. J. Stewart, W. L. Jarrett, L. J. Mathias, R. G. Alamo, L. Mandelkern, Polym. Preprints, **35**, 779 (1994).
192. S.-I. Mizushima, T. Simanouti, J. Am. Chem. Soc., **71**, 1320 (1949).
193. Y. Kim, H. L. Strauss, R. G. Snyder, J. Chem. Phys., **93**, 7520 (1989).
194. H. Hagemann, H. L. Strauss, R. G. Snyder, Macromolecules, **20**, 2810 (1987).
195. R. L. Betts, M. A. Park, E. L. Shirey, E. A. Schweikert, Org. Mass Spectr., **29**, 679 (1994).
196. K. Drotloff, M. Möller, Thermochem. Acta, **112**, 57 (1987).
197. S. J. Organ, G. Ungar, A. Keller, Macromolecules, **22**, 1995 (1989).
198. R. D. Heyding, K. E. Russell, T. L. Varty, D. St.-Cyr, Powder Diffraction, **5**, 93 (1990).
199. R. G. Alamo, L. Mandelkern, G. M. Stack, C. Kröhnke, G. Wegner, Macromolecules, **26**, 2743 (1993).
200. G. Wegner, G. Lieser, Polym. Preprints, **30**, 265 (1989).
201. G. M. Stack, L. Mandelkern, C. Kröhnke, G. Wegner, Macromolecules, **22**, 4351 (1989).
202. K.-S. Lee, G. Wegner, S. L. Hsu, Polymer, **28**, 889 (1987).
203. I. Bidd, D. W. Holdup, M. C. Whiting, J. Chem. Soc., Perkin Trans. I, 2455 (1987).
204. E. Igner, O. I. Paynter, D. J. Simmonds, M. C. Whiting, J. Chem. Soc., Perkin Trans. *I*, 2447 (1987).
205. G. D. Smith, D. Y. Yoon, W. Zhu, M. D. Ediger, Macromolecules, **27**, 5563 (1994); G. D. Smith, D. Y. Yoon, R. L. Jaffe, *ibid.*, **28**, 5897 (1995).
206. G. D. Smith, D. Y. Yoon, J. Chem. Phys., **100**, 649 (1994).
207. W. Stocker, G. Bar, M. Kunz, M. Möller, S. N. Magonov, H.-J. Cantow, Polym. Bull., **26**, 215 (1991).
208. J. Dale, Acta Chem. Scand., **27**, 1115, 1130 (1973).
209. G. Borgen, J. Dale, J. Chem. Soc., Chem. Commun., 1340 (1970).
210. P. Groth, Acta Chem. Scand. A, **33**, 199 (1979); **30**, 155 (1979).
211. K. J. Ihn, M. Tsuji, S. Isoda, A. Kawaguchi, K. Katayama, Y. Tanaka, H. Sato, Macromolecules, **23**, 1781, 1788 (1990); K. J. Ihn, M. Tsuji, A. Kawaguchi, K.-I. Katayama, Microscop. Res. Tech., **20**, 205 (1992); K. J. Ihn, M. Tsuji, S. Isoda, A. Kawaguchi, K. Katayama, Y. Tanaka, H. Sato, Makromol. Chem., **190**, 837 (1989).
212. R. Zhang, W. L. Mattice, J. Chem. Phys., **98**, 9888 (1993).
213. A. Wawkuschewski, H.-J. Cantow, S. N. Magonov, M. Möller, W. Liang, M.-H. Whangbo, Adv. Mater., **5**, 821 (1993).
214. M. Möller, Adv. Polym. Sci., **66**, 59 (1985).
215. M. Möller, W. Gronski, H.-J. Cantow, H. Höcker, J. Am. Chem. Soc., **106**, 5093 (1984).
216. M. Möller, G. Kogler, D. Oelfin, H. Drotloff, Solid-State NMR Polymer, Ed. L. Mathias, Plenum Press, NY, 245 (1991); Chem. Abstr, **116**, 236430e (1992).
217. H. Drotloff, D. Emeis, R. F. Waldron, M. Möller, Polymer, **28**, 1200 (1987).
218. H. Drotloff, H. Rotter, D. Emeis, M. Möller, J. Am. Chem. Soc., **109**, 7797 (1987).
219. M. Takenaka, T. Yamanobe, T. Komoto, I. Ando, H. Sato, K. Sato, J. Polym. Sci., Part B, Polym. Phys., **25**, 2165 (1987).
220. V. L. Shannon, H. L. Strauss, R. G. Snyder, C. A. Elliger, W. L. Mattice, J. Am. Chem. Soc., **111**, 1947 (1989).
221. M. Möller, H.-J. Cantow, H. Drotloff, D. Emeis, K.-S. Lee, G. Wegner, Makromol. Chem., **187**, 1237 (1986).
222. G. Schill, N. Schweickert, H. Fritza, W. Vetter, Chem. Ber., **121**, 961 (1988).
223. H. M. M. Shearer, V. Vand, Acta Cryst., **9**, 379 (1956).
224. E. Boda, G. Ungar, G. M. Brooke, S. Burnett, S. Mohammed, D. Proctor, M. C. Whiting, Macromolecules, **30**, 4674 (1977).
225. S. J. Sutton, A. S. Vaughan, D. C. Bassett, Polymer, **25**, 5735 (1996).
226. G. Ungar, A. Keller, Polymer, **28**, 1899 (1987).
227. S. J. Organ, A. Keller, M. Hikosaka, G. Ungar, Polymer, **37**, 2517 (1996).
228. P. R. Sundararajan, T. A. Kavassalis, Macromolecules, **30**, 5172 (1977).
229. G. M. Brooke, S. Burnett, S. Mohammed, D. Proctor, M. C. Whiting, J. Chem. Soc., Perkin Trans. I, 1635 (1996).
230. G. Lieser, K.-S. Lee, G. Wegner, Colloid Polym. Sci., **266**, 419 (1988).
231. D. L. Dorsett, S.-L. Hsu, Polymer, **30**, 1596 (1989).
232. G. Zerbi, P. Roncone, G. Longhi, S. L. Wunder, J. Chem. Phys., **89**, 166 (1988).
233. P. W. Teare, Acta Cryst., **12**, 294 (1959).
234. I. Ando, T. Sorita, T. Yamanobe, T. Komoto, H. Sato, K. Deguchi, M. Imanari, Polymer, **26**, 1864 (1985).
235. P. Grossmann, R. Arnold, K. R. Bürkle, Polymer Bull., **3**, 135 (1980).
236. S. Boneva, St. Kotov, Chromatographia, **25**, 735 (1988).
237. R. D. Chambers, J. Hutchinson, R. H. Mobbs, W. K. R. Musgrove, Tetrahedron, **20**, 497 (1964).
238. D. P. Graham, J. Org. Chem., **31**, 955 (1966).
239. P. L. Coe, S. F. Sellers, J. C. Tatlow, H. C. Fielding, G. Whittaker, J. Fluorine Chem., **18**, 417 (1981).
240. J. D. LaZerte, L. J. Hals, T. S. Reid, G. H. Smith, J. Am. Chem. Soc., **75**, 4525 (1953).
241. B. Albinsson, J. Michl, J. Am. Chem. Soc., **117**, 6378 (1995).
242. R. J. Twieg, J. F. Rabolt, Macromolecules, **21**, 1806 (1988); **16**, 1361 (1985).
243. R. D. Chambers, A. A. Lindley, H. C. Fielding, J. S. Moilliet, G. Whittaker, J. Chem. Soc., Perkin Trans. I, 1064 (1981).
244. T. Albrecht, H. Elber, R. Jaeger, M. Kimmig, W. Petry, C. Ritter, H. Schwickert, R. Steiner, G. Strobl, B. Stühn, Progr. Colloid Polym. Sci., **87**, 46 (1992).
245. S. Tsubakihara, K. Higashi, S. Taki, K. Matsushige, M. Yasuniwa, Polymer J., **24**, 777, 1215 (1992).

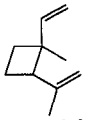
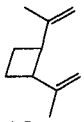
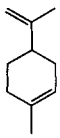
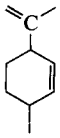
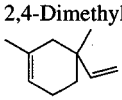
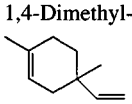
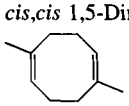
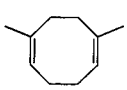
246. J. K. Krüger, J. Albers, M. Möller, H.-J. Cantow, Polym. Bull., **5**, 131 (1981).
247. H. Schwickert, G. Strobl, M. Kimmig, J. Chem. Phys., **95**, 2800 (1991).
248. K. Hatada, K. Ute, N. Miyatake, Progr. Polym. Sci., **19**, 1067 (1994).
249. H. Vanni, J. F. Rabolt, J. Polym. Sci., Polym. Phys., **18**, 587 (1980).
250. R. J. Twieg, J. F. Rabolt, J. Polym. Sci., Polym. Lett., **21**, 901 (1983).
251. J. Höpken, C. Pugh, W. Richtering, M. Möller, Makromol. Chem., **189**, 911 (1988); J. Höpken, M. Möller, Macromolecules, **25**, 2482 (1992).
252. D. S. Young, N. Fukuhara, L. A. Bigelow, J. Am. Chem. Soc., **62**, 1171 (1940).
253. W. Brunskill, W. T. Flowers, R. Gregory, R. N. Haszeldine. J. Chem. Soc., Chem. Commun., 1444 (1970).
254. S. P. V. Halasz, F. Kluge, T. Martini, Chem. Ber., **106**, 2950 (1973).
255. G. Minoni, G. Zerbi, J. Polym. Sci., Polym. Lett., **22**, 533 (1984).
256. T. Takenaka, T. Yamanobe, T. Komoto, I. Ando, H. Sato, Solid State Commun., **61**, 563 (1987).
257. W. Dmowski, W. T. Flowers, R. N. Haszeldine, J. Fluorine Chem., **9**, 94 (1977).

TABLE 2. OLIGO(DIENES)

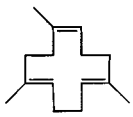
2.1. OLIGOMERS OF 1,3-BUTADIENE

Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
LINEAR DIMERS	2	108.2				
3-Methyl-1-*trans*-4,6-heptatriene CH_2=CH–CH(CH_3)–CH=CH–CH=CH_2				117/1013	1.4657/20	13,14,33,35,36,38
1,3,6-Octatriene CH_2=CH–CH=CH–CH_2–CH=CH–CH_3				129/1013	1.4743/20	33,36
1,3,7-Octatriene CH_2=CH–CH=CH–CH_2–CH_2–CH=CH_2						31,32,35,36,39
trans				124–125/1013	1.4686/20	39
cis				115–117/1013	1.4594/20	39
2,4,6-Octatriene (*cis/trans*) CH_3–CH=CH–CH=CH–CH=CH–CH_3				65–66/53		34,40,41
all-*trans*			52.5	147–148/1019	1.5131/27	34,40,41
$\Delta^{3,3'}$-Octahydro-diphenyl	3	162.3		230–232/1013		42,43
Vinylcyclopolyenes $\lceil[(CH_2CH{=}CHCH_2)_xCH(CH{=}CH_2)CH_2]\rceil$						
4-Vinyl-1-cyclohexene ($x = 1$)	2	108.2		129–130/1013	1.4642/20	1–7,34–37,44,48,49
17-Vinyl-all-*trans*-1,5,9,13-cyclo-octadecatetraene ($x = 4$)	5	270.4		~136/1.3	1.5093/20	37
21-Vinyl-all-*trans*-1,5,9,13,17-cyclodocosapentaene ($x = 5$)	6	324.5		~170/1.3	1.5098/20	37
1,2-Divinyl-cyclobutane	2	108.2				
cis 				38/51	1.4563	1,8,48,49
trans				112–113/1013	1.4451/20	8,44,48,49
CYCLOPOLYENES $\lceil[CH_2{-}CH{=}CH{-}CH_2]_n\rceil$						
1,5-Cyclooctadiene	2	108.2				1,3–10,35–37
cis,cis			−70.1	150.8/1002	1.4936/20	1,4,8,9
trans,trans			−62		1.4933/20	4
1,5,9-Cyclododecatriene	3	162.3				5–7,10–12,35–37
trans,trans,trans			34	237.5/1013	1.5005/20	5–7,12,45,46
cis,trans,trans			−16.8	241.5/1013	1.5078/20	5,6,12,45–47
cis,cis,trans			−9 to −8	244/1013	1.5129/20	6,12,46
All-*trans*-1,5,9,13-cyclohexadecatetraene	4	216.4	19–20	~97/1	1.507/20	37
All-*trans*-1,5,9,13,17-cycloeicosapentaene	5	270.4		~142/1	1.5133/20	37
All-*trans*-1,5,9,13,17,21-cyclotetracosahexaene	6	324.5		~176/1	1.5133/20	37

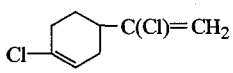
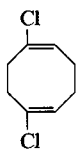
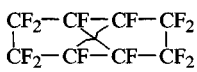
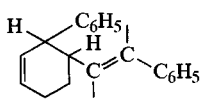
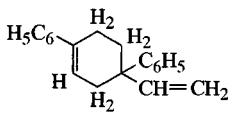
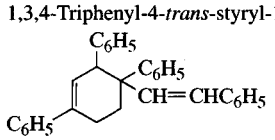
2.2. OLIGOMERS OF ISOPRENE

Oligomers	*n*	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
LINEAR OLIGOMERS					
2,7-Dimethyl-1-*trans*-3,7-octatriene $CH_2=C(CH_3)-CH=CH-CH_2-CH_2-C(CH_3)=CH_2$	2	136.3	176/1013	1.4721/25	50,51
2,6-Dimethyl-1,3,6-Octatriene $CH_2=C(CH_3)-CH=CH-CH_2-C(CH_3)=CH-CH_3$	2	136.3	57–58/16	1.4794	6,30,52,27
2,7-Dimethyl-1-*trans*-3,6-octatriene $CH_2=C(CH_3)-CH=CH-CH_2-CH=C(CH_3)-CH_3$	2	136.3	54/13	1.4786/20	11,29,30
2,6-Dimethyl-1-*trans*-3,7-octatriene $CH_2=C(CH_3)-CH=CH-CH_2-CH(CH_3)-CH=CH_2$	2	136.3	167/1013	1.4704/25	50
3-Methyl-6-methylene-1,7-octadiene $CH_2=CH-C(CH_3)-(CH_2)_2-C(=CH_2)-CH=CH_2$	2	136.3	161/1013	1.4570/25	50
2,6,10-Trimethyl-1-*trans*-3,5,10-dodecatetraene $CH_2=C(CH_3)-CH=CH-CH=C(CH_3)-(CH_2)_3-C(CH_3)=CH-CH_3$	3	204.3	78/1.3	1.4963/20	53
7,11-Dimethyl-3-methylene-1-*trans*-6,11-dodecatriene (β-farnesene) $CH_2=C(CH_3)-(CH_2)_3-C(CH_3)=CH-(CH_2)_2-C(=CH_2)-CH=CH_2$	3	204.3	70/1.3	1.4786/20	53
2,6,10,14-Tetramethyl-1-*trans*-3,6,10,14-hexadecapentaene $CH_2=C(CH_3)CH=CH-[CH_2C(CH_3)=CHCH_2]_3=H$	4	272.5	160/1.3	1.5013/20	53
CYCLIC OLIGOMERS					
trans-1,2-Dimethyl-1,2-divinylcyclobutane 	2	136.3	134.5–135/992	1.4541/27	49
trans-1-Isopropenyl-2-methyl-2-vinylcyclobutane 	2	136.3	152.5–153/992	1.4591/27	49
trans-1,2-Diisopropenylcyclobutane 	2	136.3	155.1–155.4/987	1.4585/29	49
4-Isopropenyl-1-methyl-1-cyclohexene (dipentene) (D,L) 	2	136.3	178/1013	1.4727/18	15–17, 55,62,71
6-Isopropenyl-3-methyl-1-cyclohexene (diprene) (D,L) 	2	136.3	173–175/1000	1.4735/20	15,16, 59,71
2,4-Dimethyl-4-vinyl-1-cyclohexene 	2	136.3	162/1000	1.4654/20	11,29,30, 49,55, 58–60,71
1,4-Dimethyl-4-vinyl-1-cyclohexene 	2	136.3	160.5–161.5/988	1.4658/28	11,17,30, 49,53,55, 57–59,71
cis,cis 1,5-Dimethyl-1,5-cyclooctadiene 	2	136.3	182.4/988	1.4863/28	49,54, 56–59,71
cis,cis 2,5-Dimethyl-1,5-cyclooctadiene 	2	136.3	182.5–184/992	1.4870/27	49,54, 56–59,71

2.2. *cont'd*

Oligomers	*n*	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
1,5,10-Trimethyl-*cis*,*trans*,*trans*-1,5,9-cyclododecatriene 	3	204.3	138/17	1.5120	61

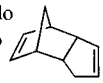
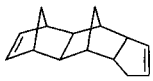
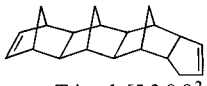
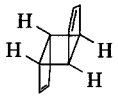
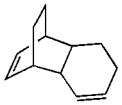
2.3. OLIGOMERS OF 1,3-BUTADIENE DERIVATIVES

Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
2-Chloro-1,3-butadiene	1-Chloro-4-(α-chloro-vinyl)-1-cyclohexene 	2	177.1		53/1.2	1.5135/25	63–66
	1,6-Dichloro-1,5-cyclooctadiene 	2	177.1	13.8	64/0.33	1.5339/25	9,63–66
2,3-Dichloro-1,3-butadiene	3,4,7,8-Tetrachloro-1,5-cyclooctadiene	2	246.0	98–99	140–146/4.8		63
Hexafluoro-1,3-butadiene	Cyclic dimer 	2	324.1	40	80/1013		20,21
2-Cyano-1,3-butadiene	4-Vinyl-1,4-dicyano-1-cyclohexene	2	158.2	55–56	110–112/1.3		69
1-Phenyl-1,3-butadiene	3-Phenyl-4-*trans*-styryl-1-cyclohexene 	2	260.4		189–191/3		67
2-Phenyl-1,3-butadiene	1,4-Diphenyl-4-vinyl-1-cyclohexene 	2	260.4	60	220–222/17		68
1,3-Diphenyl-1,3-butadiene	1,3,4-Triphenyl-4-*trans*-styryl-1-cyclohexene 	2	412.3	137–138		18,19	

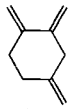
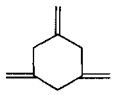
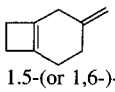
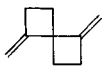
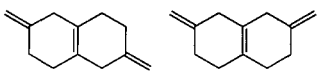
2.4. 1,4-OLIGO(ALKADIENES) $CH_3CH_2-[CH_2-CX=CH-CH_2]_n-CH_2CH_3$

X	*n*	Configuration	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
H	1	*cis*	112.2	121.7/985	1.4136/20	87
		trans		121.4/985	1.4116/20	87
	2	All-*cis*	166.3	90–91/23	1.4455/23	85
		All-*trans*		87–88/23	1.4415/24	85
	3	All-*cis*	220.4	120–125/2	1.4638/19	86
CH_3	1		126.2	138–140/1013	1.4224/20	75
	2		194.3	92/7	1.4540/20	75
	3		262.5	98/0.3	1.4729/20	75
	4		330.6	140/0.2	1.4878/16	75

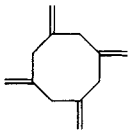
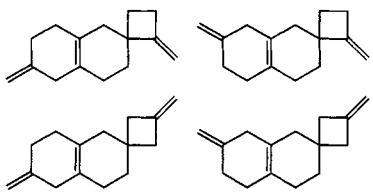
2.5. OLIGOMERS OF CYCLOPENTADIENE AND 1,3-CYCLOHEXADIENE

Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
Dicyclopentadiene	2	132.2				84
endo 			32	163/1021		22,23,49,70,74
exo				170/1017	1.5070/25	49,70,72
Tricyclopentadiene 	3	198.3	68	251/1021		22,23,74,84
Tetracyclopentadiene 	4	264.4	207	344/1021		22,23,74,84
trans-Tricyclo[5,3,0,$0^{2,6}$]-3,9-decadiene 	2	132.2		170–172/1013	1.5080/25	49,70
Dicyclohexadiene 	2	160.3		229/1013		22,73,74

2.6. OLIGOMERS OF ALLENE

Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
DIMERS	2	80.1				
1,2-Dimethylene-cyclobutane				74/1013	1.4690/20	24–27,77,81,82
1,3-Dimethylene-cyclobutane				69/1013	1.4485/20	77–80
TRIMERS	3	120.2				
1,2,4-Trimethylene-cyclohexane 				68–69/80	1.4919/25	76,77
1,3,5-Trimethylene-cyclohexane 			33.5–35	68–69/80	1.4919/25	76
3-Methylenebicyclo-[4,2,0]-octa-1(6)-ene 			UV, ^{1}H-NMR, MS			77
1.5-(or 1,6-)-Dimethylenespiro-[3,3]heptane 			IR, UV, ^{1}H-NMR, MS			77
TETRAMERS	4	160.3				
2,6- + 2,7-Dimethylenebicyclo-[4,4,0]-dec-9,10-ene (β-tetramer) 				107–108/17	1.5248/25	24,25,28,60,83

2.6. *cont'd*

Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
1,3,5,7-Tetramethylene cyclooctane				60/4.7	1.5102/25	76
						
PENTAMERS	5	200.4				
6′-Methylene-3′,4′,7′,8′-tetrahydrospiro[2- and 3-methylene]-cyclobutane-1,2′(1′H,5′H)-naphthalene + 7′-methylene-3′,4′,5′,6′-tetrahydrospiro[2- and 3-methylene]cyclobutane-1,2′-(1′H,8′H)-naphthalene				73/0.03	1.5307/20	24,88
						

REFERENCES

1. W. Brenner, P. Heimbach, H. Hey, E. W. Mueller, G. Wilke, Ann. Chem., **727**, 161 (1969).
2. N. E. Duncan, G. J. Janz, J. Chem. Phys., **20**, 1644 (1952).
3. J. C. Hillyer, J. V. Smith, Jr., Ind. Eng. Chem., **45**, 1133 (1953).
4. K. Ziegler, H. Wilms, Ann. Chem., **567**, 1 (1950); K. Ziegler, H. Sauer, L. Bruns, H. Froitzheim-Kuehlhorn, J. Schneider, Ann. Chem., **589**, 122 (1954).
5. G. Wilke, Angew. Chem., **69**, 397 (1957); G. Wilke, E. W. Mueller, M. Kroener, *ibid.*, **73**, 33 (1961); B. Bogdanovic, P. Heimbach, M. Kroener, G. Wilke, E. G. Hoffmann, J. Brandt, Ann. Chem., **727**, 143 (1969).
6. G. Wilke, Angew. Chem., **75**, 10 (1963).
7. G. Pajaro, D. Fiumani, M. Moro, Gazz. Chim. Ital., **92**, 1452 (1962); C. **1965**, II, 990.
8. E. Vogel, Angew. Chem., **68**, 413 (1956); Ann. Chem., **615**, 1 (1958).
9. A. C. Cope, W. J. Bailey, J. Am. Chem. Soc., **70**, 2305 (1948); A. C. Cope, C. L. Stevens, F. A. Hochstein, *ibid.*, **72**, 2510 (1950).
10. H. W. B. Reed, J. Chem. Soc., 1931 (1954).
11. G. Wilke, J. Polym. Sci., **38**, 45 (1959).
12. H. Breil, P. Heimbach, M. Kroener, H. Mueller, G. Wilke, Makromol. Chem., **69**, 18 (1963).
13. S. Tanaka, K. Mabuchi, N. Shimazaki, J. Org. Chem., **29**, 1626 (1964).
14. S. Otsuka, T. Kikuchi, T. Taketomi, J. Am. Chem. Soc., **85**, 3709 (1963).
15. T. Wagner-Jauregg, Ann. Chem., **488**, 176 (1931).
16. O. Aschan, Ann. Chem., **439**, 221 (1924); Chem. Ber., **57**, 1959 (1924).
17. C. Walling, J. Paisach, J. Am. Chem. Soc., **80**, 5819 (1958).
18. W. Herz, W. Lewis, J. Org. Chem., **23**, 1646 (1958).
19. T. L. Jacobs, M. H. Goodrow, J. Org. Chem., **23**, 1653 (1958).
20. I. L. Karle, J. Karle, T. B. Owen, R. W. Broge, A. H. Fox, J. L. Hoard, J. Am. Chem. Soc., **86**, 2523 (1964).
21. M. Prober, W. T. Miller, Jr., J. Am. Chem. Soc., **71**, 598 (1949).
22. K. Alder, G. Stein, Ann. Chem., **485**, 223 (1931); **496**, 204 (1932); **504**, 216 (1933).
23. H. Staudinger, H. A. Bruson, Ann. Chem., **447**, 97 (1926).
24. S. V. Lebedev, Zh. Fiz. Khim. Obshch., **45**, 1357 (1913); C. A., **9**, 799 (1915).
25. R. N. Meinert, C. D. Hurd, J. Am. Chem. Soc., **52**, 4540 (1930).
26. K. Alder, O. Ackermann, Chem. Ber., **87**, 1567 (1954).
27. A. T. Blomquist, J. A. Verdol, J. Am. Chem. Soc., **78**, 109 (1956).
28. B. Weinstein, A. H. Fenselau, Tetrahedron Lett., 1463, (1963).
29. J. Itakura, H. Tanaka, Makromol. Chem., **123**, 274 (1969).
30. L. I. Zakharkin, Dokl. Akad. Nauk SSSR, **131**, 1069 (1960).
31. E. J. Smutny, J. Am. Chem. Soc., **89**, 6793 (1967).
32. S. Takahashi, T. Shibano, N. Hagihara, Tetrahedron Lett., 2451, (1967).
33. H. Takahasi, S. Tai, M. Yamaguchi, J. Org. Chem., **30**, 1661 (1965).
34. T. Alderson, E. L. Jenner, R. V. Lindsey, Jr., J. Am. Chem. Soc., **87**, 5638 (1965).
35. G. P. Potapov, V. V. Punegov, U. M. Dzhemilev, Izv. Akad. Nauk SSSR, Ser. Khim., 1468 (1985); Bull. Acad. Sci. USSR, 1342 (1985).
36. H. Mueller, D. Wittenberg, H. Seibt, E. Scharf, Angew. Chem., **77**, 318 (1965); D. Wittenberg, Angew. Chem., **75**, 1124 (1963).
37. A. Miyake, H. Kondo, M. Nishino, Angew. Chem., **83**, 851 (1971).
38. T. Saito, Y. Uchida, A. Misono, Bull. Chem. Soc. Japan, **37**, 105 (1964).
39. G. B. Butler, T. W. Brooks, J. Org. Chem., **28**, 2699 (1963).
40. M. Urion, Koll.-Z., **63**, 84 (1933); Compt. Rend. Acad. Sci., **196**, 353 (1933).

41. R. Kuhn, Ch. Grundmann, Chem. Ber., **71**, 442 (1938).
42. K. Alder, H. F. Rickert, Chem. Ber., **71**, 373 (1938).
43. G. B. Kistiakowsky, W. W. Ransom, J. Chem. Phys., **7**, 725 (1939).
44. H. W. B. Reed, J. Chem. Soc., 685, (1951).
45. H. Nozaki, Y. Nisikawa, M. Kawanisi, R. Noyori, Tetrahedron, **23**, 2173 (1967).
46. M. Denecke, H. C. Broecker, Makromol. Chem., **176**, 1471 (1975); V. D. Mochel, J. Polym. Sci. A-1, **10**, 1009 (1972).
47. H. Takahasi, M. Yamaguchi, J. Org. Chem., **28**, 1409 (1963).
48. G. S. Hammond, N. J. Turro, A. Fischer, J. Am. Chem. Soc., **83**, 4674 (1961).
49. G. S. Hammond, N. J. Turro, R. S. H. Liu, J. Org. Chem., **28**, 3297 (1963).
50. K. Takahashi, G. Hata, A. Miyake, Bull. Chem. Soc. Japan, **46**, 600 (1973).
51. A. Josey, J. Org. Chem., **39**, 139 (1974).
52. A. Misono, Y. Uchida, K. Furuhata, S. Yoshida, Bull. Chem. Soc. Japan, **42**, 2303 (1969).
53. U. M. Dzhemilev, G. M. Latypov, G. A. Tolstikov, O. S. Vostrikova, Izv. Akad. Nauk SSSR, Ser. Khim., 553 (1979); Bull. Akad. Sci. USSR, 509 (1979).
54. U. M. Dzhemilev, G. E. Ivanov, G. A. Tolstikov, Zh. Org. Khim, **11**, 1636 (1975); J. Org. Chem. USSR, **11**, 1623 (1975).
55. J. P. Candlin, W. H. Janes, J. Chem. Soc. C, 1856 (1968).
56. A. Misono, V. Uchida, M. Hidai, V. Ohsawa, Bull. Chem. Soc. Japan, **39**, 2425 (1966).
57. S. Watanabe, K. Suga, H. Vikuchi, Austr. J. Chem., **3**, 385 (1970).
58. L. I. Zakharkin, G. G. Zhigareva, Izv. Akad. Nauk SSSR, 168, (1964); Bull. Akad. Sci. USSR, 149 (1964).
59. J. L. Binder, K. C. Eberly, G. E. P. Smith, Jr., J. Polym. Sci., **38**, 229 (1959).
60. S. Lebedev, Zh. Russ. Fiz.-Khim. Obshch., **45**, 1249 (1913); C. **1914** I, 1406; J. W. Lebedew, A. A. Ivanov, *ibid.*, **48**, 997 (1916); C. **1923** I, 1539.
61. Y. Tanaka, H. Sato, Y. Nakafutami, H. Iwasaki, T. Taketomi, Polym. J., **14**, 713 (1982).
62. G. S. Whitby, R. N. Crozier, Can. J. Res., **6**, 203 (1932).
63. R. E. Foster, R. S. Schreiber, J. Am. Chem. Soc., **70**, 2303 (1948).
64. J. G. T. Brown, J. D. Rose, J. L. Simonsen, J. Chem. Soc., 10 (1944).
65. A. C. Cope, W. R. Schmitz, J. Am. Chem. Soc., **72**, 3056 (1950).
66. A. L. Klebanskii, M. M. Denisova, Zh. Obshch. Khim. SSSR, **17**, 703 (1947); Chem. Abstr. **42**, 1215c (1948).
67. K. Alder, J. Haydn, W. Vogt, Chem. Ber., **86**, 1302 (1953); C. Liebermann, C. N. Riiber, Chem. Ber., **35**, 2696 (1902).
68. K. Alder, J. Haydn, Ann. Chem., **570**, 201 (1950).
69. C. S. Marvel, N. O. Brace, J. Am. Chem. Soc., **71**, 37 (1949).
70. N. J. Turro, G. S. Hammond, J. Am. Chem. Soc., **84**, 2841 (1962).
71. I. N. Nazarov, A. I. Kuznetsova, N. V. Kuznetsov, Zh. Obshch. Khim. SSSR, **25**, 291 (1955); J. Gen. Chem., **25**, 307 (1955).
72. P. D. Bartlett, I. S. Goldstern, J. Am. Chem. Soc., **69**, 2553 (1947); P. D. Bartlett, A. Schneider, J. Am. Chem. Soc., **68**, 6 (1946); H. A. Bruson, T. W. Riener, J. Am. Chem. Soc., **67**, 723, 1178 (1945), **68**, 8 (1946).
73. K. Alder, G. Stein, Ann. Chem., **496**, 197 (1932).
74. K. Alder, G. Stein, Chem. Ber., **67**, 613 (1934).
75. C. Pinazzi, D. Reyx, Bull. Soc. Chim. France, 3930 (1972).
76. R. E. Benson, R. V. Lindsey, Jr., J. Am. Chem. Soc., **81**, 4247 (1959).
77. B. Weinstein, A. H. Fenselau, J. Chem. Soc. C, 368 (1967).
78. F. F. Caserio, Jr., S. H. Parker, R. Piccolini, J. D. Roberts, J. Am. Chem. Soc., **80**, 5507 (1958).
79. J. K. Williams, W. H. Sharkey, J. Am. Chem. Soc., **81**, 4269 (1959).
80. Y. M. Slobodin, A. P. Khitrov, Zh. Obshch. Khim., **33**, 153 (1963); J. Gen. Chem. USSR, **33**, 146 (1963).
81. A. T. Blomquist, J. A. Verdol, J. Am. Chem. Soc., **77**, 1806 (1955).
82. Y. M. Slobodin, A. P. Khitrov, Zh. Obshch. Khim., **33**, 2819 (1963); J. Gen. Chem. USSR, **33**, 2745 (1963).
83. B. Weinstein, A. H. Fenselau, J. Org. Chem., **32**, 2278 (1967).
84. H. Staudinger, A. Rheiner, Helv. Chim. Acta, **7**, 23 (1924).
85. C. Pinazzi, D. Reyx, G. Levesque, C. R., Ser. C, **270**, 2120 (1970).
86. C. Pinazzi, D. Reyx, C. R. Ser. C, **274**, 252 (1972).
87. K. N. Campbell, L. T. Eby, J. Am. Chem. Soc., **63**, 2683 (1941).
88. B. Weinstein, A. H. Fenselau, J. Org. Chem., **32**, 2988 (1967).

TABLE 3. OLIGO(ACETYLENES)

3.1. LINEAR OLIGOMERS OF ACETYLENE

	n	Mol. wt.	b.p. (°C/mbar)	*d* (g/cm³)/°C	n_D^{20}	Refs.
Vinylacetylene CH_2=CH–C≡CH	2	52.1	5/1009	0.705/0		1,2
Divinylacetylene CH_2=CH–C≡C–CH=CH_2	3	78.1	84.5–85/1013	0.772/20	1.5055	1–3
1,3-Hexadiene-5-yne *cis/trans* CH_2=CH–CH=CH–C≡CH	3	78.1	81–82/1013	0.780/28	1.4900	1,2,4–6,25
1,3,7-Octatriene-5-yne CH_2=CH–CH=CH–C≡C–CH=CH_2 (E)-: GC; ^{1}H-, ^{13}C-NMR; Ref. 7 (Z)-: GC; ^{1}H-, ^{13}C-NMR; Ref. 7	4	105.1	10/27	0.830/20	1.5834	1,2,7

3.2. POLYENYNE OLIGOMERS $(CH_3)_3C-[C\equiv C-CH=CH]_n-C\equiv C-C(CH_3)_3$

n	Mol. wt.	m.p. (°C)	Refs.
1	188.3	69–72	26
2	238.4	118–120	26
3	288.4	155–156	26
5	388.5	220 d	26
7	488.6	204 d	26

3.3. OLIGOMERIC α,ω-DIYNES $H[C\equiv C-(CH_2)_x-C\equiv C]_nH$

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
1	1	64.1	−20 to −19	63–64/1013		8
	2	126.1	65–66			8,9
	3	188.2	103–104			8
2	1	78.1		87.5–88.5/1011	1.4381/23	10
	2	154.2	99–100			11–14
	4	30.4	167–168 d			11,12
3	1	92.1	−84.8	111.5/1013	1.4423/20	15
	2	182.3	~−5	110–111/0.9		11,12
	4	362.5	60–61			11,12
4	1	106.2		93–95/253	1.4521/18	16
	2	210.3	21–22	119–120/0.1	1.5205/23	11,17
	4	418.6	93–94			11,17
5	1	120.2	−27.3	162/1013	1.4490/20	15
	2	238.4	21–22		1.5119/26	11,12
6	1	134.2		68–71/18		18
	2	266.4	30–31			11,12
	4	530.8	59–61			11,12

3.4. OLIGO(ALKYNES) $H[C\equiv C-(CH_2)_x]_n-C\equiv CH$

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
1	1	64.1	−21 to −19	63–64/1013		8
	2	102.1	21–21.5		1.4833/22	8
	3	140.2	44–45			8
	4	178.2	73–74			8
	6	254.3	112–113.5			8
2	1	78.1		87.5–88.5/1011	1.4381/23	10
	2	130.2	46	61–63/3		19
4	1	106.2		93–95/253	1.4521/18	16
	2	186.2	−3	111/1.3	1.4775/25	9
	3	266.4	19	167/0.4	1.4918/25	9
	7	586.9	36			9
	8	667.0	55			9
5	1	120.2	−27.3	162/1013	1.4490/20	15
	2	214.3	8	113/1.1	1.4791/26	9
	3	308.5	33	170/0.1		9
6	1	134.2		68–71/18		18
	2	242.4	24	131/0.3	1.4806/25	9

3.5. CYCLIC OLIGO(ALKYNES) $\ulcorner[(C\equiv C)_x-(CH_2)_y]_n\urcorner$

x	*y*	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1	4	2	160.2	39	87/0.3	9,20,21
		4	320.5	71	215/0.3	9,20
	5	2	188.3	100	100–110/0.001	9,20
	6	2	216.4	−3.5	110–120/0.001	20
	7	2	244.4	97	160–180/4	20
	8	2	272.5	38–39	145–147/0.01	20
	9	2	300.5	106.5	198–202/0.001	20

3.5. *cont'd*

x	y	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	2	3	228.3	175–180 d		14
		4	304.4	170–180 d		14
		5	380.5	165 d		14
		6	456.6	~180 d		14
		7	532.6	~180 d		14
	3	2	180.2	115–120 (expl.)		11,12
		3	270.4	174–175		13
		4	360.5	213–214		13,22
	4	2	208.3	163		12,13,17,20,22
		3	312.4	173–174		13,22
		4	416.6	154–155		13,22
		5	520.7	155–157		13,22
	5	2	236.3	212		11–13,20,22
		3	354.5	125–126		13,22
		4	472.7	135–136		13,22
		5	590.9	144–145		13,22
		6	709.0	144–145		13,22
	6	2	264.4	82		12,20,22
		3	396.6	135–136		22
		4	528.8	152–154		22
	7	2	292.4	171		20
	8	2	320.5	85		20,22
	9	2	348.5	142		20
	10	1	188.3	18		23
		2	376.6	101–102		23
3	5	2	284.4	>200		20

3.6. **PERICYCLYNES** $\ulcorner[C{\equiv}C{-}C(CH_3)_2]_n\urcorner$

n	Mol. wt.	m.p. (°C)	Refs.
5	330.5	201–202	24
6	396.6	249–250	24
7	462.7	173–174	24
8	528.8	189–190	24

REFERENCES

1. J. A. Nieuwland, W. S. Calcott, F. B. Downing, A. S. Carter, J. Am. Chem. Soc., **53**, 4197 (1931).
2. K. K. Georgieff, W. T. Cave, K. G. Blaikie, J. Am. Chem. Soc., **76**, 5494 (1954).
3. R. Lespieau, Guillemonat, Koll.-Z., **63**, 84 (1933).
4. D. A. Ben-Efraim, F. Sondheimer, Tetrahedron, **25**, 2837 (1969).
5. J. H. Van Boom, L. Brandsma, J. F. Arens, Rec. Trav. Chim. Pay-Bas, **85**, 952 (1966).
6. F. Sondheimer, D. A. Ben-Efraim, Y. Gaoni, J. Am. Chem. Soc., **83**, 1682 (1961).
7. H. Hopf, L. Eisenhuth, V. Lehne, L. Ernst, Chem. Ber., **119**, 1105 (1986).
8. D. A. Ben-Efraim, F. Sondheimer, Tetrahedron, **25**, 2823 (1969).
9. J. H. Wotiz, R. F. Adams, C. G. Parsons, J. Am. Chem. Soc., **83**, 373 (1961).
10. R. A. Raphael, F. Sondheimer, J. Chem. Soc., 120 (1950).
11. Y. Amiel, F. Sondheimer, R. Wolovsky, Proc. Chem. Soc., 1957, p. 22.
12. F. Sondheimer, Y. Amiel, R. Wolovsky, J. Am. Chem. Soc., **79**, 6263 (1957).
13. F. Sondheimer, Y. Amiel, R. Wolovsky, J. Am. Chem. Soc., **79**, 4247 (1957).
14. F. Sondheimer, R. Wolovsky, J. Am. Chem. Soc., **84**, 260 (1962).
15. A. L. Henne, K. W. Greenlee, J. Am. Chem. Soc., **67**, 484 (1945); R. Lespieau, Journaud, Compt. Rend., **188**, 1410 (1929).
16. H. Bader, L. C. Cross, I. M. Heilborn, E. R. H. Jones, J. Chem. Soc., 619 (1949).
17. F. Sondheimer, Y. Amiel, J. Am. Chem. Soc., **79**, 5817 (1957).
18. W. M. Lauer, W. J. Gensler, J. Am. Chem. Soc., **67**, 1171 (1945).
19. F. Sondheimer, Y. Gaoni, J. Am. Chem. Soc., **84**, 3520 (1962).
20. J. Dale, A. J. Hubert, G. S. D., King, J. Chem. Soc., 73, (1963).
21. D. J. Cram, N. L. Allinger, J. Am. Chem. Soc., **78**, 2518 (1956).
22. F. Sondheimer, Y. Amiel, R. Wolovsky, J. Am. Chem. Soc., **81**, 4600 (1959).

23. G. Eglinton, A. R. Galbraith, J. Chem. Soc., 889 (1959); Chem & Ind., 737 (1956).
24. L. T. Scott, G. J. DeCicco, J. L. Hyun, G. Reinhardt, J. Am. Chem. Soc., **105**, 7760 (1983); **107**, 6546 (1985).
25. A. L. Klebanskii, U. A. Dranitzina, I. M. Dobromilskaya, Compt. Rend. Acad. Sci. USSR, **2**, 229 (1935); Chem, Abstr., **29**, 6205 (1935).
26. F. Wudl, S. P. Bitler, J. Am. Chem. Soc., **108**, 4685 (1986).

TABLE 4. OLIGOMERS WITH ALIPHATIC SIDE CHAINS WHICH IN ADDITION CONTAIN HETEROATOMS

4.1. OLIGOMERIC ACRYLIC DERIVATIVES

4.1.1. OLIGO(ACRYLIC ACIDS)

4.1.1.1. $H[CH_2CH(COOH)]_nCH_3$

n	Mol. wt.	Configuration	m.p. (°C)	b.p. (°C/mbar)	Refs.
1	88.1		−46	153/1012	
2	160.2	Isotactic (meso)	128		1–3,11
		Syndiotactic (racemic)	141		1–3,11
3	232.2	Isotactic (meso)	167.5–170.0		3,78
		Heterotactic (racemic)	160.0–163.0		3,78
		Syndiotactic (meso)	153.5–154.5		3,78

4.1.1.2. $(CH_3)_2C(COOH)-[CH_2CH(COOH)]_n-C(COOH)(CH_3)_2$

n	Mol. wt.	m.p. (°C)	Refs.
1	246.3	30–35	4
2	318.3	70–80[a]	4
3	390.4	181–183	4
4	462.5	80–82	5

[a] *p*-Nitrobenzyl ester.

4.1.2. OLIGO (ACRYLATES)

4.1.2.1. $H[CH_2CH(COOH)]_nCH_3$

n	Mol. wt.	Configuration	b.p. (°C/mbar)	n_D/°C	Refs.
1	102.1		91.5/990	1.3838/20	6
2	188.2	Isotactic (meso)	64/0.9	1.4258	3,6,101
		Syndiotactic (racemic)	NMR		3
3	274.3	Isotactic (meso)	163/15	1.4440	3,6,78
		Heterotactic (racemic)	160/15	1.4420	3,6,78
		Syndiotactic (meso)	156/15	1.4407	3,6,78

4.1.2.2. Anionic Oligomers $RO[CH_2CH(COOR)]_nH$

R	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
CH_3	1	118.1	55–56/27		7
CH_3	2	204.2	90/1	1.433	7
C_2H_5	1	146.2	75–77/27		7,42,56
C_2H_5	2	246.3	82/0.15	1.446	7,42,56,62

4.1.2.3. Unsaturated Oligomers

Oligomers	R	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
Dialkyl-Δ^2-dihydromuconate (*trans*)	CH_3	2	172.2	86–88/1.3	1.4497	42,44,58,60,61,63
$ROOC-CH=CH-CH_2CH_2COOR$	C_2H_5	2	200.2	80–84/1.3		58
Dialkyl-Δ^3-dihydromuconate	CH_3	2	172.2	74–75/1.3		42,58,60,63
$ROOC-CH_2-CH=CH-CH_2COOR$	C_2H_5	2	200.2	107–108/1.3		58,63
Dialkyl-α-methyleneglutarate oligomers	CH_3	1	172.2	59/0.3		7,8,42,44,62

4.1.2.3. cont'd

Oligomers	R	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
CH_2=C(COOR)-[CH_2CH(COOR)]$_n$H		2	258.2	120/0.9		8
		3	344.3	166–168/0.4		8
		4	430.4	180/0.03		8
	C_2H_5	1	200.2	52–53/0.2	1.0392	42,52,56,62

4.1.3. ANIONIC OLIGOMERS OF ACRYLAMIDE CH_2=CHCO-[$NHCH_2CH_2CO$]$_n$$NH_2$

n	Mol. wt.	m.p. (°C)	Refs.
1	142.2	149–150	110–112
2	213.2	179–181	112

4.1.4. OLIGO(ACRYLONITRILES)

4.1.4.1. H[CH_2CH(CN)]$_n$H

n	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
1	55.1	97.1/1013	1.3689/15	
2	108.1	116–118/6.5	1.4320/25	9,45
3	161.2	195–196/3.5	1.4609/20	9,10,45

4.1.4.2. H[CH_2CH(CN)]$_n$$CH_3$

n^a	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{64}/°C (g/cm^3)	n_D/°C	Refs.
1	69.1	−71.5	107–108/1013	0.773	1.3720/20	37
2	122.2	8.8[b]	106/3	0.9051	1.4191/60	10,11,38,54,80
		55[c]	94/3	0.8940	1.4155/60	10,11,38,54,80
3	175.2	37–39[b]				10,38,53
		53–54[d]				10,53
		80–81[e]				10,53
4	228.3	130–132[f]				10
		158				10

[a] n = 2–5 (diastereomers): ^{13}C-NMR, Ref. 35.
[b] Isotactic (meso).
[c] Syndiotactic (racemic).
[d] Heterotactic (racemic).
[e] Syndiotactic (meso).
[f] Two stereoisomers.

4.1.4.3. H[CH(CN)CH_2]$_n$$CH_2CN$

n	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
1	94.1	287.4/1013	1.4347/20	
2	147.2	85/0.01	1.4644/25	9,12
3	200.2	95/0.01		9,12

4.1.4.4. 1,2-Dicyanocyclobutanes

n	Configuration	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{25}	Refs.
1	*cis/trans*	106.1				13,42,43,66,81
	cis		67–67.5	190/20	1.4628	13,14,79
	trans		30.5/31.5	160/20		13,14,79

4.1.4.5. Unsaturated Oligomers

Oligomers[a]	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{25}	Refs.
2,3-Dihydromuconitrile $NCCH_2CH=CHCH_2CN$	2	106.1	76			14,15,82
1,4-Dicyano-1-butene $CN-CH=CH-CH_2CH_2CN$	2	106.1				42,46,49,67
cis			−19 to −17	99.5–102.5/0.5	1.4677	49,50,83
trans			−13 to −12	84.5–86.5/0.5	1.4646	49,50
2-Methyleneglutaronitrile $CH_2=C(CN)-CH_2CH_2CN$ (*cis/trans*)	2	106.1	−9.6	65/0.13	1.4550	42,44–48,52
2,4,6-Tricyano-1-hexene $CH_2=C(CN)-CH_2CH(CN)-CH_2CH_2CN$	3	159.2		152–164/2.5	1.4762	42,45
1,1,4,4-Tetracyanoethyl-1,4-dicyano-2-butene $(NCCH_2CH_2)_2C(CN)-CH=CH-C(CN)(CH_2CH_2CN)_2$	6	318.4	240			14,45,46
1,4-Dicyano-1,3-butadiene $NC-CH=CH-CH=CH-CH$	2	104.1				42,51
cis,cis			128–129			83
trans,trans			159–161			97

4.2. OLIGOMERIC METHACRYLIC DERIVATIVES

4.2.1. OLIGO(METHACRYLIC ACIDS)

4.2.1.1. $H[CH_2C(CH_3)(COOH)]_n-C(COOH)(CH_3)_2$

n	Mol. wt.	m.p. (°C)	Refs.
2	260.3	Oil	19
3	346.4	45–48	18,19
4	432.5	56–59	19

4.2.1.2. Anionic Oligo(methacrylic acids)

n	Mol. wt.	m.p. (°C)	Refs.
$CH_3O-[CH_2C(CH_3)(COOH)]_nH$			
2	204.2	82	17
3	290.3	160	17

4.2.2. OLIGO(METHACRYLATES)

4.2.2.1. Anionic Oligo(methacrylates) $RO-[CH_2C(CH_3)(COOOR)]_nH$

R	R′	*n*	Mol. wt.	b.p. (°C/mbar)	d^{20}/(g/cm³) °C	n_D^{20}	Refs.
CH_3O	CH_3O	1	132.2	147/1013	0.7949	1.4053	17,36,41,73
		2	232.3	241/1013	1.0540	1.4378	17,36,41
		3	332.4	116/0.1	1.1045	1.4567	17,36,41
		4	432.5	190/0.1	1.12	1.4681	17
		6	632.8		1.565		17
			n = 2–10: GPC Ref. 36				
$n-C_4H_9O$	$n-C_4H_9O$	1	216.3	59–61/0.13	0.8970	1.4213	74
		2	358.5	128–130/0.13	0.9436	1.4400	74
		3	500.7	186–90/0.13	0.9736	1.4508	74
		4	642.9	246–250/0.4	0.9909	1.4574	74

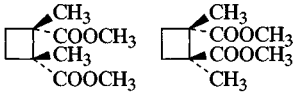
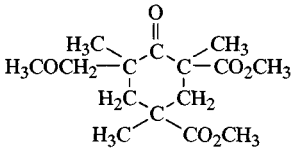
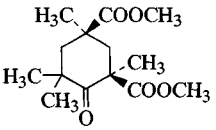
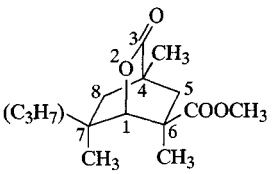
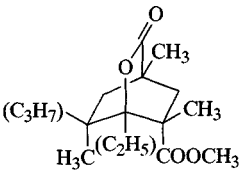
Isotactic and syndiotactic oligomers, $(CH_3)_3C-[CH_2C(CH_3)COOCH_3]_n-H$, n = 2–100 (mol. wt. 158.2–10070.0): HPLC, GPC, SFC; DSC (T_g, T_m), X-ray (n = 2–5), NMR, MS; Refs. 114–124.
Reviews: Refs. 114,115.
Properties of specific oligomers: n (isotactic and syndiotactic, resp.) = 2–8,13,14,16,19,21,22–25,27–29,31,36,41,44,45,50,58,80,100.
m.p. (°C), (isotactic oligomers): n = 28: 98.5, n = 31: 104.5, n = 36: 109.6, n = 41: 117.3, n = 44: 124.3; Ref. 116.

4.2.2.2. Unsaturated Oligomers[a] $CH_2=C(COOCH_3)-[CH_2-X-(COOCH_3)]_n-CH_3$

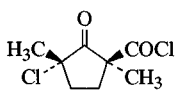
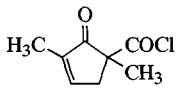
X	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
CH_2CH	1	188.2	107/4	1.4445	16,20,42,59,65,77
	2	276.3	173/4	1.4588	16,20,42,59,65,77
$C(CH_3)$	1	188.2		IR, MS, ^{1}H-NMR	Ref. 65,77 Ref. 75

[a] n = 2–4: 2D-COSY-NMR

4.2.2.3. Cyclic Oligomers

Oligomers	R	*n*	Configuration	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1,2-Dimethylcyclobutane-1,2-dicarboxylates	CH_3	2	*cis*	200.2	IR, MS, ^{1}H-NMR		65
			*trans*		IR, MS, ^{1}H-NMR		65
2,4,6-Trimethyl-6-methoxy-methyl-cyclohexanone-2,4-dicarboxylate	CH_3	3		268.3		122/0.13	17,36
							
1,3,3,5-Tetramethylcyclohexane-2-one-1,5-dicarboxylate	CH_3	3	*cis*	270.1		128–138/3	64,72
			*trans*		52–54	128–138/3	64,72
4,6,7-Trimethyl-3-oxo-7-propyl-2-oxabicyclo-[2,2,2]octane-6-carboxylate	CH_3	3	6-endo	268.3	117–119		64
			6-exo		Oil		64
1-Ethyl-4,6,7-trimethyl-3-oxo-7-propyl-2-oxa-bicyclo[2,2,2]octane-6-carboxylate	CH_3	4	6-exo	296.4	IR, MS, ^{1}H-NMR		64
							

4.2.3. OLIGO(METHACRYLOYL CHLORIDES)

Oligomers	*n*	Mol. wt.	b.p. (°C/mbar)	Refs.
2-Chloro-2,5-dimethylcyclopentanone-5-carboxylic chloride	2	209.1	111–117/23	40
				
2,5-Dimethyl-Δ^2-cyclopentenone-5-carboxylic acid chloride	2	209.1	111–117/23	40
				

4.2.4. ANIONIC OLIGOMERS OF METHACRYLAMIDE $CH_2=C(CH_3)CO-[NHCH_2CH(CH_3)CO]_nNH_2$

n	Mol. wt.	m.p. (°C)	Refs.
1	170.2	118–120	84
2	255.3	155–158	84
3	340.4	195	84
$n = 4–9$: GPC, MS, Ref. 84			

4.2.5. OLIGO(METHACRYLONITRILES)

4.2.5.1. Anionic Oligo(methacrylonitriles) $CH_3O[CH_2C(CH_3)CN]_nH$

n	Mol. wt.	b.p. (°C/mbar)	n_D/°C	Refs.
1	99.1	160.6/1013	1.4038/20	113
2	166.2	92/0.7	1.438	21,34
3	233.3	165/0.7	1.464	21,34
4	300.4	235/0.7	1.478	21,34
5	367.5	300–305/0.7	1.488	21,34

4.2.5.2. 1,2-Dicyano-1,2-dimethyl-cyclobutanes $\overline{\ulcorner[CH_2C(CH_3)(CN)-C(CH_3)(CN)CH_2]_n\urcorner}$

n	Configuration	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	*cis*	134.2	107–108	170/33	16,66,89
2	*trans*	134.2	90.3	120/33	16,66,89

4.2.5.3. Unsaturated Oligomers

Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{25}	Refs.
α-Methylene-δ-methyladiponitrile $CH_3CH(CN)-CH_2CH_2C(CN)=CH_2$	2	134.2		148/33	1.4502	16,42,89
5-Cyano-8-methyl-2-methylene-nonanedinitrile $CH_2=C(CN)-[CH_2CH_2CH(CN)]_2=CH_3$	3	201.3	47–49	194/1.3	1.4720	16

4.3. OLIGOMERS OF β-ALKYL-SUBSTITUTED VINYL DERIVATIVES

4.3.1. OLIGO(ESTERS)

Monomer	R	Oligomers	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
$CH_3CH=CHCOOR$		2-methyl-3-pentene-1,3-dicarboxylate (*cis/trans*) $CH_3CH=C(COOR)-CH(CH_3)CH_2COOR$					
	CH_3		2	200.2	70–80/96		42,55,68,70,71
	C_2H_5		2	228.3	84/0.75	1.4512	42,55,56,69
	CH_3	2-Methyl-4-pentene-1,3-dicarboxylate (*cis/trans*) $CH_2=CH-CH(COOR)-CH(CH_3)CH_2COOR$	2	200.2	90–110/13		42,70,71
n-$C_3H_7CH=CHCOOR$	CH_3	2-*n*-Propyl-3-heptene-1,3-dicarboxylate $C_3H_7-CH=C(COOR)-CH(CH_2COOR)-C_3H_7$	2	252.3	100–102/4		55
i-$C_3H_7CH=CHCOOR$	CH_3	2-Isopropyl-5-methyl-4-hexene-1,3-dicarboxylate $(CH_3)_2C=CH-CH(COOR)-CH(CH_2COOR)-CH(CH_3)_2$	2	252.3	90–93/3		55

4.3.2. OLIGO(CROTONONITRILES)

Oligomers	*n*	Mol. wt.	b.p. (°C/mbar)	n_D^{25}	Refs.
1,3-Dicyano-2-methyl-3-pentene (*cis/trans*) $CH_3CH{=}C(CN){-}CH(CH_3)CH_2CN$	2	134.2	81–83/0.3	1.4730	42,52,55–57
1,3,5-Tricyano-2,4,6-trimethylcyclohexane $\lceil CH(CN){-}CH(CH_3)\rceil_3$	3	201.3	152–155/0.3	1.4950	42,57
1,3,5-Tricyano-2,4-dimethyl-5-heptene $CH_3CH{=}C(CN){-}CH(CH_3)CH(CN){-}CH(CH_3)CH_2CN$	3		150–160/0.13		42,52

4.4. OLIGO(VINYL) DERIVATIVES

4.4.1. OLIGO(VINYL ALCOHOLS) AND OLIGO(VINYL ACETATES) $H[CH_2CHOR]_nCH_3$

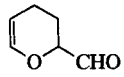
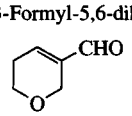
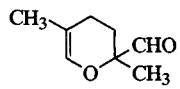
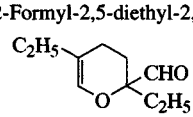
R	*n*	Configuration	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
H	2	Isotactic (meso)	104.1		73/4	1.4327	80,86,87
		Syndiotactic (racemic)		48–49	74/4	1.4378	80,86,87
	3	Isotactic (meso)	148.2	67	120/1.3		84,85
		Heterotactic (racemic)			120/1.3		84,85
		Syndiotactic (meso)		87.5–88			84,85
$COCH_3$	2	Isotactic (meso)	188.2		20/1013	1.4172	86,88,97
		Syndiotactic (racemic)			201/1013	1.4142	86,88,97
		Mixture			103–107/27	1.4170	96
	3	Isotactic	274.3		116–117.5/0.7	1.4342	84,88
		Heterotactic		21–22	115–116/0.7	1.4319	84
		Syndiotactic		32.5–33.5	112–114/0.5	1.4297	84,88
		Mixture			120–122/0.7	1.4320	84,96

4.4.2. OLIGO(VINYL CHLORIDES) $H[CH_2CHCL]_nCH_3$ [a]

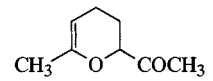
n	Configuration	Mol. wt.	b.p. (°C/mbar)	n_D^{20}	Refs.
2 [b]	Isotactic (meso)	141.0	40/16	1.4409	80,90–95,97
	Syndiotactic (racemic)		36/16	1.4390	80,90–95,97
	Mixture		140–141/1013	1.4409	95
3 [b]	Isotactic	203.5	102.5/20	1.4722	89,94,95
	Heterotactic		103.5/20	1.4704	89,94,95
	Syndiotactic		104.5/20	1.4686	89,94,95
	Mixture		102.5–104.5	1.4699	89

[a] Oligomers $(CH_3)_3C{-}[CH_2CHCl]_n{-}H$, n = 1,2,3 (threo-; erythro-), 4 (ribo-; xylo-; arabino-; lyxo-): HPLC, ^{13}C-NMR, Ref. 39.
[b] 2,4-Dibromopentanes and 2,4,6-tribromoheptanes: Kerr Effekt, Ref. 98.

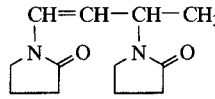
4.4.3. OLIGO(VINYL ALDEHYDES) AND OLIGO(VINYL KETONES)

Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (C/mbar)	Refs.
Acrolein $CH_2{=}CHCHO$	2-Formyl-2,3-dihydropyran 	2	112.1		145–148/1013	22–25
	3-Formyl-5,6-dihydropyran 	2	112.1		77–78/16	26,27
α-Methylacrolein $CH_2{=}C(CH_3)CHO$	2-Formyl-2,5-dimethyl-2,3-dihydropyran 	2	140.2	−75	166/1000	24,28
α-Ethylacrolein $CH_2{=}C(CH_2H_5)CHO$	2-Formyl-2,5-diethyl-2,3-dihydropyran 	2	168.2		195/1013	24

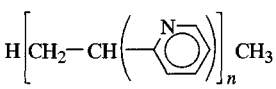
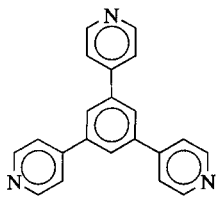
4.4.3. *cont'd*

Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (C/mbar)	Refs.
Methyl vinyl ketone $CH_2=CH-CO-CH_3$	2-Methyl-6-acetyl-5,6-dihydropyran 	2	128.2		68/17	29
	3-Methylene-2,6-heptanedione $CH_3COC(=CH_2)CH_2CH_2COCH_3$	2	140.2	5–7	91–92/7	46
2-Methyl-1-butene-(3)-one $CH_2=C(CH_3)COCH_3$	2,6-Dimethyl-3,7-dioxo-1-octene $CH_3COCH(CH_3)CH_2CH_2COC(CH_3)=CH_2$	2	168.2		83–85/23	30,99,100
Phenyl vinyl ketone $CH_2=CH-CO-C_6H_5$	2,4-Dibenzoylpentane $CH_3CH(COC_6H_5)CH_2CH(COC_6H_5)CH_3$	2	280.4	^{1}H-NMR		99,100
	Isotactic (meso)			64–65	100/1.3	99,100
	Syndiotactic (racemic)			62–63	155/1.3	99,100

4.4.4. OLIGOMERIC *N*-VINYL DERIVATIVES

Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	Ref.
N-Vinyl-pyrrolidone	1,3-Bis-[*N*-pyrrolidone-(2)-yl]-1-butene 	2	212.3	75	31

4.4.5. OLIGO(VINYL PYRIDINES)

Monomer	Oligomers	*n*	Configuration	Mol. wt.	b.p. (°C/mbar)	Refs.
2-Vinylpyridine		2	Isotactic (meso)	226.3	106–108/0.33	102,105–107,109
		3	Isotactic (meso)	331.4	LC, GC, ^{1}H-, ^{13}C-NMR	103,105,107,109
		4	Isotactic (meso)	436.6	LC, GC, ^{1}H-, ^{13}C-NMR	104,103,107,109
		n = 2–7: LC, ^{1}H-, ^{13}C-NMR Refs. 108,109				
4-Vinylpyridine	sym-Tri(4-pyridyl)-cyclohexane 	3		315.4	m.p. (°C): 228.5	32,33

REFERENCES

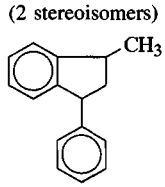
1. K. Auwers, J. Thrope, Ann. Chem., **285**, 337 (1985).
2. E. Moeller, Chem. Ber., **43**, 3250 (1910).
3. H. G. Clark, Makromol. Chem., **86**, 107 (1965).
4. H. Kaemmerer, A. Jung, Makromol. Chem., **101**, 284 (1967).
5. H. Kaemmerer, N. Oender, Makromol. Chem., **111**, 67 (1968).
6. D. Lim, O. Wichterle, J. Polym. Sci., **29**, 579 (1958).
7. B. A. Feit, Eur. Polym. J., **3**, 523 (1967).
8. B. A. Feit, Eur. Polym. J., **8**, 321 (1972).
9. H. Zahn, P. Schaefer, Chem. Ber., **92**, 736 (1959); Makromol. Chem., **30**, 225 (1959).
10. T. Takata, M. Taniyama, Kobunshi Kagaku, **16**, 693 (1955);T. Takata, H. Ishii, Y. Nishiyama, M. Taniyama, *ibid.*, **18**, 235 (1961); C. 729 (1964); Chem. Abstr., **56**, 1591 (1962); T. Takata, I. Hirori, M. Taniyama, J. Polym. Sci. A, **2**, 1567 (1964).
11. H. G. Clark, Makromol. Chem., **63**, 69 (1963); L. E. Alexander, R. Engmann, H. G. Clark, J. Phys. Chem., **70**, 252 (1966).
12. R. C. Houtz, Textile Res. J., **20**, 786 (1950).
13. E. C. Coyner, W. S. Hillman, J. Am. Chem. Soc., **71**, 324 (1949).
14. N. Takashina, C. C. Price, J. Am. Chem. Soc., **84**, 489 (1962).
15. W. Reppe, Ann. Chem., **596**, 133 (1956).
16. C. J. Albisetti, D. C. England, M. J. Hogsed, R. M. Joyce, J. Am. Chem. Soc., **78**, 472 (1956).

17. Th. Voelker, A. Neumann, U. Baumann, Makromol. Chem., **63**, 182 (1963); G. Schreyer, Th. Voelker, Makromol. Chem., **63**, 202 (1963).
18. H. Kaemmerer, J. S. Shukla, G. Scheuermann, Makromol. Chem., **116**, 72 (1968).
19. H. Kaemmerer, J. S. Shukla, Makromol. Chem., **116**, 62 (1968).
20. E. Trommsdorf, Angew. Chem., **68**, 355 (1956).
21. B. A. Feit, J. Wallach, A. Zilkha, J. Polym. Sci. A, **2**, 4743 (1964).
22. K. Alder, A. Rueden, Chem. Ber., **74**, 920 (1941).
23. S. Potnis, K. Shorara, R. C. Schulz, W. Kern, Makromol. Chem., **63**, 78 (1963).
24. H. Schulz, H. Wagner, Angew. Chem., **62**, 105 (1950).
25. S. M. Scherlin, A. J. Berlin, T. A. Sserebrennikowa, F. E. Rabinowitsch, Zh. Obshch. Khim., **8**, 22 (1938); C. **1939** I, 1971.
26. R. Hall, Chem. Ind., 1772 (1955).
27. G. Dumas, P. Rumpff, C. R. Ser. C, **242**, 2574 (1956).
28. G. G. Stoner, J. S. McNulty, J. Am. Chem. Soc., **72**, 1531 (1950).
29. K. Alder, H. Offermanns, E. Rueden, Chem. Ber., **74**, 905 (1941).
30. H. Staudinger, B. Ritzenthaler, Chem. Ber., **67**, 1773 (1934).
31. J. W. Breitenbach, F. Galinovsky, H. Nesvadba, E. Wolf, Naturwissenschaften, **42**, 155, 440 (1955).
32. F. Longo, J. W. Bassi, F. Greco, M. Cambini, Tetrahedron Lett., 995 (1964).
33. A. Segre, Tetrahedron Lett., 1001 (1964).
34. B. A. Feit, E. Heller, A. Zilkha, J. Polym. Sci. A-1, **4**, 1499 (1966); G. Teichmann, A. Zilkha, Eur. Polym. J., **3**, 25 (1967).
35. H. Balard, H. Fritz, J. Meybeck, Makromol. Chem., **178**, 2393 (1977).
36. A. Laguerre, J.-P. Busnel, C.-M. Bruneau, Makromol. Chem., Rapid Commun., **5**, 21 (1984).
37. R. E. Kent, S. M. McElvain, Org. Synth., **25**, 61 (1945).
38. V. Janout, P. Cafelin, Coll. Czech. Chem. Commun., **45**, 1928 (1980).
39. K.-F. Elgert, R. Kosfeld, Polym. Bull., **175**, 244 (1983).
40. W. Fischer, D. Bellus, A. Alder, E. Francotte, A. Roloff, Chimia, **39**, 19 (1985).
41. S. Fujishige, Makromol. Chem., **177**, 375 (1976); **179**, 2251 (1978).
42. V. P. Yurev, G. A. Gailyunas, O. T. Kofanova, Usp. Khim., **53**, 1352 (1984); Russ. Chem. Rev., **53**, 783 (1984).
43. D. M. Gale, J. Org. Chem., **35**, 970 (1970).
44. T. Saegusa, Y. Ito, H. Kinoshita, S. Tomita, Bull. Chem. Soc. Japan, **43**, 877 (1970).
45. M. M. Baizer, J. D. Anderson, J. Org. Chem., **30**, 1357 (1965).
46. J. D. McClure, J. Org. Chem., **35**, 3045 (1970).
47. British Patent 1,003,656; Chem. Abstr., **64**, 605a (1966).
48. French Patent 1,388,444; Chem. Abstr., **63**, 504e (1965).
49. C. M. Langkammer, United States Patent 2,478,285; Chem. Abstr., **44**, 2009c (1950).
50. A. Misono, Y. Uchida, M. Hidai, H. Shinobara, Y. Watanabe, Bull. Chem. Soc. Japan, **41**, 396 (1968).
51. D. J. Milner, R. Whelan, J. Organomet. Chem., **152**, 193 (1978).
52. D. A. White, Synth. React. Inorg. Met.-Org. Chem., **7**, 433 (1977).
53. M. Murano, R. Yamadera, J. Polym. Sci., Polym. Lett. Ed., **5**, 483 (1967).
54. M. Murano, R. Yamadera, J. Polym. Sci. A-1, **5**, 1855 (1967).
55. T. Saegusa, Y. Ito, S. Tomita, H. Kinoshita, J. Org. Chem., **35**, 670 (1970).
56. D. A. White, M. M. Baizer, J. Chem. Soc., Perkin Trans. I, 2230 (1973).
57. J. Shabtai, E. Ney-Igner, H. Pines, J. Org. Chem., **40**, 1158 (1975).
58. G. Oehme, H. Pracejus, J. Prakt. Chem., **322**, 798 (1980).
59. M. Stickler, G. Meyerhoff, Makromol. Chem., **181**, 131 (1980).
60. M. G. Barlow, M. J. Bryant, R. N. Haszeldine, A. G. Mackie, J. Organomet. Chem., **21**, 215 (1970).
61. T. Alderson, E. L. Jenner, R. V. Lindsey, Jr., J. Am. Chem. Soc., **87**, 5638 (1965).
62. M. M. Rauhut, H. Currier, United States Patent 3,074,999 (1963); Chem. Abstr., **58**, 11224b (1963).
63. E. H. Farmer, J. Chem. Soc., **123**, 2531 (1923).
64. K. Yamaguchi, K. Yokota, Y. Takada, Makromol. Chem., **182**, 669 (1981); Makromol. Chem., Rapid Commun., **2**, 645 (1981).
65. J. Lingnau, M. Stickler, G. Meyerhoff, Eur. Polym. J., **16**, 785 (1980).
66. M. Bolte, J. Lemaire, C. R., Ser. C, **288**, 403 (1979).
67. H.-F. Piepenbrink, Ann. Chem., **572**, 83 (1951).
68. T. Saegusa, Y. Ito, S. Kobayashi, S. Tomita, J. Chem. Soc., Chem. Commun., 273 (1968).
69. J. Shabtai, H. Pines, J. Org. Chem., **30**, 3854 (1965).
70. M. Ikeda, T. Hirano, T. Tsuruta, Tetrahedron, **30**, 2217 (1974).
71. M. Ikeda, T. Hirano, T. Tsuruta, Tetrahedron Lett., 4477, (1972).
72. L. Lochmann, M. Rodová, J. Petránek, D. Lim, J. Polym. Sci., Polym. Chem. Ed., **12**, 2295 (1974).
73. D. Dosococilová, J. Stokr, B. Schneider, M. Pradný, S. Sevcik, Coll. Czech. Chem. Commun., **49**, 2275 (1984).
74. J. Trekoval, Coll. Czech. Chem. Commun., **38**, 3769 (1973).
75. P. Cacioli, D. G. Hawthorne, S. R. Johns, D. H. Solomon, E. Rizzardo, R. I. Willing, J. Chem. Soc., Chem. Commun., 1355 (1985).
76. W. von E. Doering, C. A. Guyton, J. Am. Chem. Soc., **100**, 3229 (1978).
77. K. Naruchi, M. Miura, Polymer, **22**, 1716 (1981); K. Naruchi, M. Miura, K. Nagakubo, Nipoon Kagaku Kaishi, 871, (1977); K. Naruchi, O. Yamamoto, M. Miura, K. Nagakubo, *ibid.*, 1974, (1976).
78. D. Lim, B. Obereigner, D. Doskocilová, J. Polym. Sci., Polym. Lett., **3**, 893 (1965).

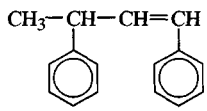
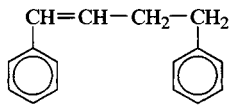
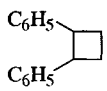
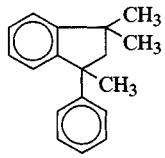
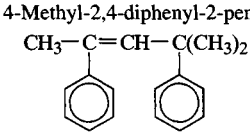
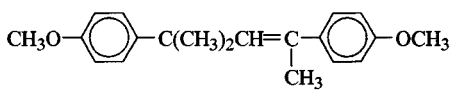
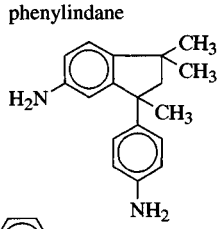
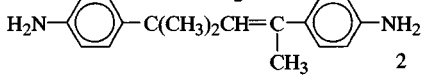
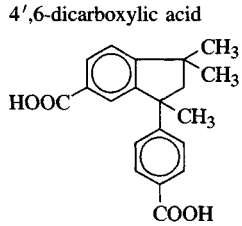
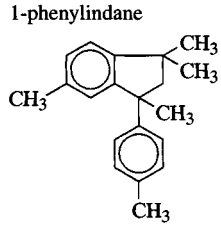
79. J. A. Barltrop, H. A. J. Carless, J. Am. Chem. Soc., **94**, 1951 (1972).
80. P. E. McMohan, W. C. Tincher, J. Mol. Spectr., **15**, 180 (1965).
81. S. Hosaka, S. Wakamatsu, Tetrahedron Lett., 219 (1968).
82. P. Kurtz, Ann. Chem., **572**, 23 (1951).
83. K. Nakagawa, H. Onoue, Tetrahedron Lett., 1433, (1965).
84. D. Lim, E. Votavová, J. Stokr, J. Petránek, J. Polym. Sci., Polym. Lett., **4**, 581 (1966).
85. K. Fujii, J. Polym. Sci., Polym. Lett., **3**, 375 (1965).
86. J. G. Pritchard, R. L. Vollmer, J. Org. Chem., **28**, 1545 (1963).
87. Y. Fujiwara, S. Fujiwara, Bull. Chem. Soc. Japan, **37**, 1010 (1964).
88. Y. Fujiwara, S. Fujiwara, K. Fujii, J. Polym. Sci. A-1, **4**, 257 (1966).
89. D. Lim, M. Kolinsky, J. Stokr, J. Petránek, J. Polym. Sci., Polym. Lett., **4**, 577 (1966).
90. S. Satoh, J. Polym. Sci. A, **2**, 5221 (1964).
91. Y. Inoue, I. Ando, A. Nishioka, Polym. J., **3**, 246 (1972).
92. T. Shimanouchi, M. Tasumi, Spectrochim. Acta, **17**, 755 (1961).
93. T. Shimanouchi, M. Tasumi, Y. Abe, Makromol. Chem., **86**, 43 (1965).
94. Y. Abe, M. Tasumi, T. Shimanouchi, S. Satoh, R. Chujo, J. Polym. Sci. A-1, **4**, 1413 (1966).
95. R. Lukás, J. Jakes, V. Palecková, S. Pokorny, M. Kolinsky, J. Polym. Sci., Polym. Chem. Ed., **21**, 3093 (1983).
96. J. Stokr, B. Schneider, Coll. Czech. Chem. Commun., **28**, 1946 (1963).
97. D. Doskocilová, B. Schneider, Coll. Czech. Chem. Commun., **29**, 2290 (1964).
98. A. E. Tonelli, G. Khanarian, R. E. Cais, Macromolecules, **18**, 2324 (1985).
99. L. Merle-Aubry, Y. Merle, E. Sélégny, C. R., Sér. C., **276**, 249 (1973).
100. Y. Merle, L. Merle-Aubry, E. Sélégny, C. R., Sér. C., **280**, 443 (1975).
101. T. E. Hogen-Esch, C. F. Tien, J. Polym. Sci., Polym. Chem. Ed., **17**, 281 (1979).
102. C. F. Tien, T. E. Hogen-Esch, Macromolecules, **9**, 871 (1976); C. F. Tien, T. E. Hogen-Esch, J. Am. Chem. Soc., **98**, 7109 (1976).
103. C. F. Tien, T. E. Hogen-Esch, J. Polym. Sci., Polym. Lett., **16**, 297 (1978).
104. T. E. Hogen-Esch, C. F. Tien, J. Polym. Sci., Polym. Lett., **17**, 431 (1979).
105. T. E. Hogen-Esch, C. F. Tien, Macromolecules, **13**, 207 (1980).
106. T. E. Hogen-Esch, W. L. Jenkins, Macromolecules, **14**, 510 (1981).
107. S. S. Hung, C. Mathis, T. E. Hogen-Esch, Macromolecules, **14**, 1802 (1981).
108. S. S. Hung, T. E. Hogen-Esch, J. Polym. Sci., Polym. Chem. Ed., **23**, 1203 (1985).
109. W. L. Jenkins, C. F. Tien, T. E. Hogen-Esch, Pure Appl. Chem., **51**, 139 (1979).
110. A. Leoni, S. Franco, Macromolecules, **4**, 355 (1971).
111. D. S. Breslow, G. E. Hulse, A. S. Matlack, J. Am. Chem. Soc., **79**, 3760 (1957).
112. A. Leoni, S. Franco, G. Polla, J. Polym. Sci. A-1, **6**, 3187 (1968).
113. D. S. Tarbell, P. Noble, Jr., J. Am. Chem. Soc., **72**, 2657 (1950).
114. K. Hatada, K. Ute, T. Kitayama, T. Nishiura, N. Miyatake, Macromol. Chem. Macromol. Symp., **85**, 325 (1994).
115. K. Hatada, K. Ute, N. Miyatake, Prog. Polym. Sci., **19**, 1067 (1994).
116. K. Ute, N. Miyatake, K. Hatada, Polymer, **36**, 1415 (1995).
117. T. Kitayama, T. Nishimura, A, Oshima, K. Ute, K. Hatada, Polymer Bull., **32**, 215 (1994).
118. K. Ute, N. Miyatake, Y. Osugi, K. Hatada, Polym. J., **25**, 1153 (1993).
119. K. Ute, N. Miyatake, T. Asada, K. Hatada, Polym. Bull., **28**, 561 (1992).
120. K. Ute, N. Miyatake, K. Hatada, J. Macromol. Sci., Pure Appl. Chem. A, **29**, 599 (1992).
121. K. Hatada, K. Ute, T. Nishimura, M. Kashiyama, T. Saito, M. Takeuchi, Polymer Bull., **23**, 157 (1990).
122. K. Ute, T. Nishimura, K. Hatada, Polym. J., **21**, 1027 (1989).
123. K. Ute, T. Nishimura, Y. Matsuura, K. Hatada, Polym. J., **21**, 231 (1989).
124. K. Hatada, K. Ute, K. Tanaka, T. Kitayama, Polym. J., **19**, 1325 (1987).

TABLE 5. OLIGO(STYRENES)

No.	Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
Styrene		3-Methyl-1-phenylindane	2	208.3	9.5	168–169/21	1.5810	1–5
		(2 stereoisomers)			25.5	157/16	1.5809	



5.1. *cont'd*

No.	Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
		1,3-Diphenyl-1-butene 	2	208.3		181–182/27 134–135/1	1.5930	1–3,6,23
		1,4-Diphenyl-1-butene 	2	208.3		124/1013		6,23
		1,2-Diphenylcyclobutane 	2	208.3			1.5913	7
α-Methylstyrene		1,3,3-Trimethyl-1-phenylindane 	2	236.4	53	158–160/13		8–10
		4-Methyl-2,4-diphenyl-2-pentene 	2	236.4	52	166–167/20	1.5728	8,11,12
α-Methyl-*p*-methoxystyrene			2	296.4		237–240/24	1.5703	9,13
α-Methyl-*p*-aminostyrene		1,3,3-Trimethyl-4′,6-diamino-phenylindane 	2	266.4	93–94			9
			2	266.4	173			9,14
α-Methyl-*p*-carboxystyrene		1,3,3-Trimethyl-1-phenylindane-4′,6-dicarboxylic acid 	2	348.4	297			9,14
α-*p*-Dimethylstyrene		1,3,3,4′,6-Pentamethyl-1-phenylindane 	2	264.4	40	142–144/1.1		9,15,16

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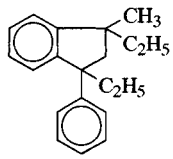
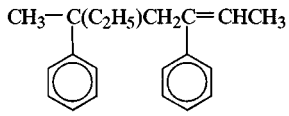
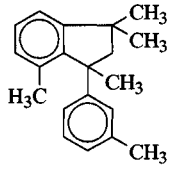
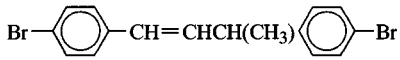
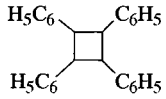
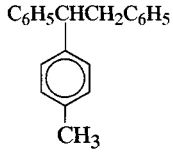
5.1. *cont'd*

No.	Monomer	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
	α-Ethylstyrene	1,3-Diethyl-3-methyl-1-phenylindane 	2	264.4		104–106/0.4	1.5642	17
		3,5-Diphenyl-5-methyl-2-heptene 	2	264.4		133–135/1.6	1.54345/25	17
	α-*m*-Dimethylstyrene	1,3,3,3′,7-Pentamethyl-1-phenylindane 	2	264.4	57			9
	p-Bromostyrene	*cis*-1,3-Di(*p*-bromophenyl)-1-butene 	2	366.1	67–68			18,19
	Stilbene	1,2,3,4-Tetraphenylcyclobutane 	2	360.5	164–165			20–22
		1,2-Diphenyl-1-*p*-tolyethane 	2	272.4	42–43			20

REFERENCES

1. P. E. Spoerri, M. J. Rosen, J. Am. Chem. Soc., **72**, 4918 (1950).
2. M. J. Rosen, J. Org. Chem., **18**, 1701 (1953).
3. B. B. Corson, J. Dorsky, J. E. Nickels, W. M. Kutz, H. I. Thayer, J. Org. Chem., **19**, 17 (1954).
4. M. J. Rosen, Org. Syn. Coll. Vol. IV, 665 (1963).
5. R. Stoemer, H. Kootz, Chem. Ber., **61**, 2330 (1928).
6. R. Fittig, E. Erdmann, Ann. Chem., **216**, 179 (1883); H. Stobbe, G. Posnjak, Ann. Chem., **371**, 287 (1909).
7. I. S. Bengelsdorf, J. Org. Chem., **25**, 1468 (1960).
8. E. Bergmann, H. Taubadel, H. Weiss, Chem. Ber., **64**, 1493 (1931).
9. J. C. Petropoulos, J. J. Fisher, J. Am. Chem. Soc., **80**, 1938 (1958).
10. L. M. Adams, R. J. Lee, F. T. Wadsworth, J. Org. Chem., **24**, 1186 (1959).
11. F. S. Dainton, R. H. Tomlinson, J. Chem. Soc., 151 (1953).
12. J. M. Van der Zanden, Th. R. Rix, Rec. Trav. Chim., **75**, 1343 (1956).
13. J. M. Van der Zanden, Th. R. Rix, Rec. Trav. Chim., **75**, 1166 (1956).
14. J. V. Braun, E. Anton, W. Haensel, G. Werner, Ann. Chem., **472**, 1 (1929).
15. M. Tiffeneau, Ann. Chim. Phys., **10**, p. 145, p. 197 (1907).
16. V. N. Ipatieff, H. Pines, R. C. Oldberg, J. Am. Chem. Soc., **70**, 2123 (1948).
17. C. G. Overberger, E. M. Pearce, D. Thanner, J. Am. Chem. Soc., **80**, 1761 (1958).
18. G. L. Goemer, J. W. Pearce, J. Am. Chem. Soc., **73**, 2304 (1951).
19. J. Hukki, Acta Chem. Scand., **3**, 279 (1949).
20. D. S. Brackman, P. H. Plesch, J. Chem. Soc., 3563 (1958); Chem. Ind., 255 (1955).
21. M. Pailer, U. Mueller, Monatsh. Chem., **79**, 615 (1948).
22. J. D. Fulton, J. D. Dunitz, Nature (London), **160**, 161 (1947).

C. OLIGOMERS CONTAINING HETEROATOMS IN THE MAIN CHAIN

TABLE 6. OLIGOMERS CONTAINING O IN THE MAIN CHAIN

6.1. OLIGO(ETHERS) AND OLIGO(ACETALS)

6.1.1. OLIGO(OXYETHYLENES)

6.1.1.1. Linear Oligo(oxyethylenes) $H[OCH_2CH_2]_nOH$

n[a]	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	(g/cm^3)	d_{20}	n_D^{20}/°C	Refs.
1	62.1	−12.6	197.8/1013	1.1136/20	1.4324		23–26,55,119,149,209
2	106.1	−6	245/1013	1.120/20	1.4479		23–26,55,57,119,129,149,209
3	150.2	−4.3	122–123/0.1	1.1274/15	1.4573		23–25,26,55,57,119,129,149,209
4	194.2	−3	155–156/0.8	1.127/20	1.4611		23–27,55,57,70,119,120,129,149,209
5	238.3	−2 to 0	173–174/0.8		1.4629		25–27,55,57,68,71,119,149,209
6	282.3	9	175–177/0.07	1.127/20	1.4640		23–28,55,68–70,119,120,209,215
7	326.4	17.8–18.8	230–232/0.9		1.4653		25,26,55,57,69,71,119,219
8	370.4	19.7–21.8	206–209/0.02	1.1278/20	1.4660		27,120,215,219
9	414.5	30.0	186–192/0.001				37,50,52–54,56,59,219
10	458.6	25.8–28.3	210/0.1				27,51,54,71,219
12	546.6	36.5–38.5					219
14	634.7	29.5					25
15	678.8	40.0	246–257/0.001				37,50–54,59
16	722.9	49.5–505					219
18	811.0	35.0					25,37,211
20	899.1	44.4					51,54
25	1119.3	47.0					51,53,54,59
27	1207.5	44.6–44.8					37
30	1321.6	46.4					25,54
35	1559.8	50.1					51,53,54,59
36	1604.0	43.0–43.4					37
45	2000.4	53.0					37,54
54	2396.9	54.2					212

[a] n = 1–27: HPLC, Refs. 72–74; n = 1–45: GPC, Ref. 220; n = 6,8,9,10,15,16,18,25,35,54: IR, Raman, NMR, MS, DSC, Ref. 211,212,214,215,219,224.

6.1.1.2. Oligo(oxyethylene) Monoalkyl Ethers $R-[OCH_2CH_2]_n-OH$

R	n[a]	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
CH_3	1	76.1	−85.1	124/1009	1.4024	26,124,125,127–129,177,196
	2	120.1		194.2/1013	1.4260	127
	3	164.2		245.3/1013		180
	5	252.3		110/0.013		178
	6	296.4		125/0.013		178
	7	341.4		150/0.007		178
	8	385.5		165/0.007		178
	9	429.5	13–14	185/0.003		178
	10	473.6	18–18.5	218–220/0.013		178
	11	518.1	22	236–238/0.013		178
	18	825.0	24–25			24
C_2H_5	1	90.1		135/1013	1.4075	24,125–129,176,177,196
	2	134.2		196/1013	1.4273	126,129
	3	178.2		248/1013	1.4380	126
	4	222.3		284/1013	1.4499	129
n-C_3H_7	1	148.2		152–153/1012	1.4133	128,129,175–177,196
	2	192.3		215.8/1023		175
	3	236.3		148–150/23		175
n-C_4H_9	1	118.2		171/1013	1.4196	24,125,129,176,177,194,196
	2	162.2	−68.1	112–114/15		194
	3	206.3		278/1013		206
n-C_6H_{13}	1	146.2	−50.1			102
	2	190.3	−36.9			102
	3	234.3	−30.3			102
	4	278.4	−15.0			102

6.1.1.1. cont'd

n^a	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	(g/cm^3)	d_{20} n_D^{20}/°C	Refs.
	5	322.4	−3.3			102
	6	366.5	1.2			102
	9	498.6	13.0			102
n-C_9H_{19}	4	320.5		180–182/0.5	NMR, MS	225
	8	496.7	22.7			225
n-$C_{12}H_{25}$	1	186.3	20.2	144/0.13	1.4456	70,207
	2	230.4	18.6	164/0.13	1.4485	70,207
	3	274.4	17.0	178/0.13	1.4510	70,207
	4	318.5	20.0	188/0.07	1.4531	70,207
	5	362.5	22.8	212/0.07	1.4548	70,207
	6	406.6	25.4	233/0.13	1.4560	70,207
	7	550.8	28.5	236–240/0.15	1.4576	207
	8	594.8	32.3			207
	9	638.9	33.5			207
	10	683.0	40.0			207
	11	727.0	40.3			207
	12	771.1	42.0			207
n-$C_{16}H_{33}$	9	638.9	40.5			56
	12	771.1	43.5			56
$C_6H_5CH_2$	3	240.3		166–169/0.5	1.503	75,175
	4	284.3			1.501	75
	5	328.4			1.499	75
	6	372.4			1.493	75
	8	460.6			1.488	75
	9	504.6			1.483	75
$(C_6H_5)_3C$	3	392.4			1.540	75
	4	436.5			1.544	75
	5	480.6			1.549	75
	6	524.6			1.561	75
C_6H_5	1	138.2		120.5/12	1.5370	109
	2	182.2		163.0/11	1.5230	109
	3	226.3		190.5/11	1.5144	109
	4	270.3		181/0.3	1.5075	109
	5	314.4		218/0.5	1.5030	109

[a] n = 4,6,8,12,16 with R= n-C_9H_{19} and n-$C_{15}H_{31}$: Raman spectr., Ref. 213.

6.1.1.3. Oligo(oxyethylene) Dialkyl Ethers $H(CH_2)_x$-$[OCH_2CH_2]_n$-$O(CH_2)_xH$

n^a	x	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
1	1	90.1		84/1013	1.3797/20	130,131,133,134,195
	2	118.2		124/1013	1.3922/20	24,131,132,135
	3	145.7		159–160/965		131,175
	4	174.2		204/1013		24,131,132
2	1	134.2		161/1013	1.4183/20	24,26,134,179,195
	2	162.2		189/1013	1.4143/20	8,24,135,179
	3	190.2		219/983	1.4025/25	179
	4	218.3		250/1013	1.4265/20	24,179
3	1	178.8		216/1013	1.4233/20	134
	2	206.3		118/13		135
4	1	222.3		148/19	1.4371/20	8,24,134
	2	250.3		159/15	1.4332/20	24,135
	4	306.4		192/13	1.4384/20	24
5	1	266.3		149–151/3.3		134,136,166
	2	294.4		174/4		135
6	1	310.4		135/0.1	1.4471/20	24,134
	2	338.4		125/0.01	1.4454/20	24,135
	4	394.5		155/0.4	1.4509/20	24
7	1	354.4		205–207/4		134
9	1	442.5	16.0			52,58,59
	2	470.6	21.5			58,59
	3	498.6	12.7			58
	4	526.7	11.7			58
	6	582.8	13.0			58

6.1.1.3. cont'd

n^a	x	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
	8	638.8	19.5			58
	10	695.0	28.5			58,63
	12	751.1	37.0			58,63,64
	16	863.3	49.5			58,63,64
	18	919.4	54.0			58,63
	21	1003.8	62.0			58
	26	1143.8	69.0			58,63
15	1	706.8	28.7			52,59,60
	2	734.9	35.1		Raman	59,60,224
	3	762.9	30.8			60,63
	4	791.0	31.3			60,63
	5	819.0	32.0			60,63
	6	847.1	32.2			60,63
	7	875.2	33.7			60,63
	8	903.2	35.3			60,63
	9	931.2	38.4			60,63
	10	959.3	39.5			60,63
	11	987.4	42.5			60,63
	12	1015.4	45.5			60,63
	13	1043.5	47.3			60,63,64,67
	14	1071.6	50.0			60,63,64,67
	15	1099.6	51.7			60,63,64,67
	16	1127.6	54.4			60,63,64,67
	17	1155.7	57.6			60,63,64,67
	18	1183.7	59.0			60,63,64,67
	21	1267.9	62			60,62,65
	26	1408.2	68			62
	30	1520.4	71			62
25	4	1231.5	43.5			66
	15	1540.1	55.5			66
	30	1960.9	72.7			66
45	1	2028.4	51.3			61
	10	2280.9	52.8			61
	13	2365.0	56.1			61
	15	2449.2	57.1			61
	18	2505.3	61.2			61
	21	2589.4	64.3			61
	30	2841.9	66			66

[a] $n = 2–45$, $x = 1$: GPC, Ref.: 220; α-alkyl (CH_3, C_9H_{19}, $C_{15}H_{31}$), ω-methoxy oligo(oxyethylenes), $n = 4–12,15,16,25$: Ref. 214; $n = 15$, $x = 5,6,7$: micelliation and surface properties, Ref. 221.

6.1.1.4. Amino Derivatives of Oligo(oxyethylenes) α,ω-Diamines $RNH–CH_2CH_2–[OCH_2CH_2]_n–NHR$

R	n	Mol. wt.	b.p. (°C/mbar)	n_D/25°C	Refs.
H	1	94.2	48–50/1.3		76
	2	148.2	77–78/0.3		76
	3	192.3	115/0.3	1.4637	76,77
	4	236.3	133–134/0.2	1.4647	77
	5	280.4	175–177/0.07	1.4668	77
CH_3	3	220.3	97–98/0.2	1.4511	77
	4	264.4	131–133/0.2	1.4549	77
	5	308.4	146–148/0.07	1.4562	77

[a] *N*-Oligoethylene mono- and dialkylamines $R_1R_2N(CH_2CH_2O)_nH$; R_1, $R_2 = C_4H_9$ and $R_1 = n\text{-}C_{10}H_{21}$, $R_2 = H$; $n = 1–6$; GC; Ref. 88.

6.1.1.5. Cyclic Oligo(oxyethylenes) $\square[OCH_2CH_2]_n\square$

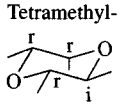
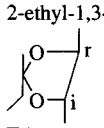
n^a	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm³)/°C	Refs.
1	44.1	−111.3	10.7/1013	1.8922/6	29
2	88.1	11.8	101.3/1013	1.0336/20	30,85–87,92,102,145
3	132.2	0	168–172/1013		78,79,102
4	176.2	17	106/13	1.109/25	31,40,79–81,86,102,145,216
5	220.3	Oil	152/1.7	1.113/20	31,39,79,81,83,102,145,216

6.1.1.5. cont'd

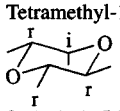
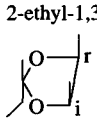
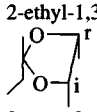
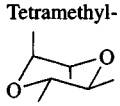
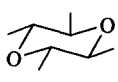
n[a]	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
6	264.3	39.5–40.5	80/0.013		39,41,79,81–83,102,145,216
7	308.4	Oil			39,84,145,216
8	352.4	19			39,84,145
9	396.9	Oil			217
10	441.0	35.5–36.8			217,219
11	485.1	Oil			217
12	529.2	36.5–38.5			217,219
13	573.3	Oil			217
14	617.4	28.5–31			217
15	661.5	38–41.5			217
16	705.6	49.5–50.5			217,219
17	749.7	44–47			217
18	793.8	47–50			217
19	837.9	51–53.5			217
20	882.0	46–50.5			217

[a] n = 2, 4–8; GC, MS; Ref. 145; n = 4–20: IR, NMR, MS, Refs. 216,217,219.

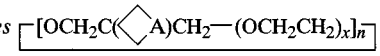
6.1.1.6. Oligomers of Substituted Ethylene Oxides

Epoxide	Oligomers	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
Oxirane (Ethylene oxide)	Crown ethers $\boxed{[OCH_2CH_2]_n}$	1–8		cf. Tab. 1.1.1.5			
						110–112	
	2-Methyl-1,3-dioxolane $\boxed{[OCH(CH_3)-OCH_2CH_2]}$	2	88.1		82–83/1013	1.3970/20	85–87,92, 110–112
	2-Methyl-1,3,6-trioxocane $\boxed{[OCH(CH_3)-(OCH_2CH_2)_2]}$	3	132.2		110/13		70,150
Methyloxirane (Propylene oxide)	Dimers	2	116.2		117–130/1013		80,90,92,93
	cis- + *trans*-2,5-Dimethyl-1,4-dioxane				GC, MS, ^{1}H-NMR, IR		89,114
	trans- $\boxed{[OCH(CH_3)CH_2]_n}$			−4.5	121.5/1000	1.4147/22	112,113
	cis- + *trans*-2,6-Dimethyl-1,4-dioxane $\boxed{[OCH(CH_3)CH_2-OCH_2CH(CH_3)]}$				127.7–129/1001	1.4169/20	89,119
					GC, MS, ^{1}H-NMR, IR		
	cis- + *trans*-2-Ethyl-4-methyl-1,3-dioxolane				GC, MS, ^{1}H-NMR, IR		89
	cis-					1.4029/24	115
	trans- $\boxed{[OCH(C_2H_5)-OCH_2CH(CH_3)]}$					1.4044/24	115
	Higher Oligomers	4	232.3	93	61–65/0.009	1.4453/23	49,89–93
	$\boxed{[OCH(CH_3)CH_2]_n}$	5[a]	290.4		77–83/0.009	1.4453/23	90–92
					n = 4–6: GPC, Ref. 91		
Ethyloxirane (1,2-Butene oxide) 2,3-Dimethyl oxirane (2,3-Butene oxide)	$\boxed{[OCH(CH_2CH_3)CH_2]_n}$	4	288.4		99–101/0.13	1.4503/23	90–92
					n = 4–6: GPC, Ref. 91		
cis-	Monomer	1	72.1		59.9–60.4/996	1.3826/20	116
	Dimers	2	144.2				94,95
	2,*cis*-3,*trans*-5,*cis*-6-Tetramethyl-1,4-dioxane[b]				GC, ^{1}H-NMR		94,95
							
	2,*cis*-4,*trans*-5-Trimethyl-2-ethyl-1,3-dioxane[b]				GC, ^{1}H-NMR		94,95
							
	Trimers	3	216.3		GPC, ^{1}H-NMR		95
	Tetramers	4	288.4		GPC, ^{1}H-NMR		95

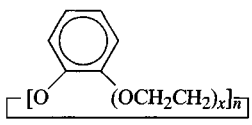
6.1.1.6. cont'd

Epoxide	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
trans-	Monomer	1	72.1		53.6–54.1/996	1.3736/20	116
	Dimers	2	144.2				94,95
	2,*cis*-3,*trans*-5-*cis*-6-Tetramethyl-1,4-dioxane[b]				GC, ^{1}H-NMR		94,95
							
	2,*cis*-4,*cis*-5-Trimethyl-2-ethyl-1,3-dioxolane[b]				GC, ^{1}H-NMR		94,95
							
	2,*trans*-4,*trans*-5-Trimethyl-2-ethyl-1,3-dioxolane[b]				GC, ^{1}H-NMR		94,95
							
cis-, *trans-*	2,*trans*-3,*cis*-5-*trans*-6-Tetramethyl-1,4-dioxane	2			GC, ^{1}H-NMR		94,95,114
							
	2,*trans*-3,*trans*-5,*cis*-6-Tetramethyl-1,4-dioxane		39.5–40.0		GC, ^{1}H-NMR		94,95,114,117
							
2,2-Dimethyloxirane (Isobutene oxide)	Dimers	2	144.2				96
	2,5-Tetramethyl-1,4-dioxane $\boxed{[OC(CH_3)_2CH_2]_{\overline{n}}}$				140/1013		96
	2-Isopropyl-4,4-dimethyl-1,3-dioxolane $\boxed{[OCH\{CH(CH_3)_2\}-OC(CH_3)_2CH_2]}$				135/1013		96
	2,4,6-Triisopropyl-1,3,5-trioxane $\boxed{[OCH\{CH(CH_3)_2\}]_{\overline{n}}}$	3	216.3	64			96
(R)-*tert*, Butyloxirane	$\boxed{[OCH\{C(CH_3)_3\}CH_2]_{\overline{n}}}$	4	400.6	168.0			92,97
	2R, 5R, 8R, 11R-			$[\alpha]_D^{25} = 50{,}8$ ($c = 1.1$, C_6H_{12})			
Methyleneoxirane (Epoxyallene)	$\boxed{[OC(=CH_2)CH_2]_{\overline{n}}}$	4	224.2	120–122	146–147/1007	1.4880/20	98,118
Epichlorohydrin	$\boxed{[OCH(CH_2Cl)CH_2]_{\overline{n}}}$	2	185.0		60/0.2 (subl.)		92,100,103–105
	cis-			65–66	67–75/0.7		99
	trans-			109–110			99
	2R, 5R, 8S, 11S-	4	370.1	136–136.5	190–205/1		90,92,98,100,101, 103,104
				$n = 2$–6: GPC, Refs. 100,103,104			
n-Propoxymethyl-oxirane (*n*-Propyl glycidyl ether)	$\boxed{[OCH(CH_2OC_3H_7)CH_2]_{\overline{n}}}$	2	232.3		88/0.6	1.4330/30	91
				$n = 2$–5: GPC, Ref. 91			
Phenyloxirane (Styrene oxide)	2-Benzyl-4-phenyl-1,3-dioxolane $\boxed{[OCH(CH_2C_6H_5)-OCH(C_6H_5)CH_2]}$	2	240.3	37–40	150–151/1		43,106
	2,5-Diphenyl-1,4-dioxane	2	240.3	152			92,106,107
	cis-			121–122			42,43
	trans-			177			42–44
	$\boxed{[OCH(C_6H_5)CH_2]_{\overline{n}}}$						
	2,4,6-Tribenzyl-1,3,5-trioxane $\boxed{[OCH(CH_2C_6H_5)]_{\overline{n}}}$	3	360.4		GPC, NMR		106
		$3n = 2$–5:			GPC, MS,		106

[a] Stereoisomers.
[b] The letters r and i indicate retention and inversion at the carbon, respectively.

6.1.1.7. Oligo(oxyethylenes) Containing Spirane Residues 

A	*n*	*x*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
O	1	1	144.2		97–99/13	182
		2	188.2	Oil	MS, ^{1}H-NMR	182,183
		3	232.3		97–98/0.008	182
		7	408.3		182–183/0.004	182
	2	1	288.3	163–165		182
		2	376.4	86–87		182–183
		3	464.5	52–53		182
		7	816.6	42–43		182
$OCH(C_6H_5)O$	1	2	294.1	Oil	MS, ^{1}H-NMR	183
	2	2	588.3	99		183
	3	2	882.4	86–87		183

6.1.1.8. Benzo Crown Ethers[a] 

Position of phenolic groups	*x*	*n*	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
o-	1	2	272.3	208.5		102
		3	408.4	190–192		41,188
		4	544.6	150–152		41,188
	2		180.2	67–69		41
		2	360.4	164.0	380–384/1025	41,102,188,189
	3	1	224.2	44–45.0		41
		2	448.5	113–114		41,102,188,189
	4	1	268.3	79–79.5		41
		2	536.3	106.8		41,188
	9	2	977.1	<26		41,188
p	3	2	448.5	95.5–96.5		187,190
	4	2	536.6	93.5–94		187,190
	5	2	624.7	67–68		187,190

[a] Ref. 187.

6.1.2. OLIGO(OXYPOLYMETHYLENES)

6.1.2.1. Linear Oligomers $R-[O(CH_2)_x]_n-R'$

R	R′	*x*	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
H	OH	4	1	90.1	20.1	85/0.01	1.4460/20	137
			2	162.2	3	108/0.01	1.4544/26	108,120,137
			3	234.4	10.1	151/0.01	1.4484/60	33,120,137,222,223
			4	306.4	18	183/0.01		137
			5	378.5	37	226/0.01		137
			6	450.6	31	253/0.01		137
			7	522.7	42			137
			8	594.8	34			137
			9	667.0	41			137
			14	1027.5	38			137
		5	1	104.2	−18	137–138/16	1.4494/20	
			2	190.3	16–18	141–142/0.7	1.4570/25	108
			3	276.4	29.5	188–190/1	1.4495/60	33,222
			7	620.9	36–38		1.4538/60	33
		6	1	118.2	43	132/2		
			3	318.5	56	209–210/1	1.4538/60	33,222
		10	1	174.3	72–75	175–176/19		
			3	486.8	80	290–300/1	1.4518/60	33
				$n = 3$, $x = 3$–6: X-ray, IR, Raman Spectr., DSC				
H	Cl	4	1	108.6		81–82/19		137,138
			2	180.7		87/0.01		137
			3	252.8		132/0.01		137
			4	324.9		160/0.01		137

6.1.2.1. cont'd

R	R′	x	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
C_6H_5NHCO	$OCONHC_6H_5$	4	1	328.4	180			137
			2	400.5	119			137
			3	472.6	74			137
			4	544.7	54			137
			5	616.8	48			137
			6	688.9	47			137
						$n = 2–7$: HPLC, MS; Ref. 146		
CH_3	OCH_3	4				$n = 2–8$: GC,MS; Ref. 139		

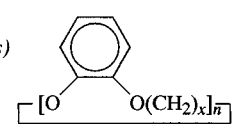
6.1.2.2. Oligo(oxypolymethylene) Ethers $H(CH_2)_x-[O(CH_2)_x]_n-H$

x	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4(g/cm³)/°C	n_D^{20}	Refs.
3	2	160.3		180/1013	0.836/20	1.4090	33
	3	218.3		240–250/1013			33
4	2	202.3		234–236/1013	0.843/20	1.4226	24,32,33
	3	274.4		137/2	0.878/20	1.4357	32,33
	4	346.6		190/1	0.895/20	1.4393	32,33
5	2	244.4		120/03	0.843/20	1.4318	33
	3	330.6		160/0.4	0.870/20	1.4416	33
	4	416.7		200/0.3	0.890/20	1.4471	33
6	2	286.5		130/0.5	0.842/20	1.4379	33
	3	386.7		184/0.8	0.867/20	1.445	33
	4	486.8	20.5	260/2	0.879/20	1.451	33
10	2	454.8	40	233/0.4	0.819/60	1.4370/60	33
	3	611.1	50	310–330/3	0.834/60	1.4416/60	33
	4	767.4	59		0.850/60	1.4457/60	33

6.1.2.3. Cyclic Oligomers

Oligomers	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
$\ulcorner[O(CH_2)_3]_n\urcorner$	1	58.1	47		140
	4	232.3	70		38,92,140,141,143
		$n = 3–8$: GC, MS; Refs. 140,141,143			
$\ulcorner[OCH_2C(CH_3)_2CH_2]_4\urcorner$	1	86.1		79–80/1013	142,143
	4	344.5	157		38,92,140,143
		$n = 3–9$: GLC, MS; Refs. 140,143			
$\ulcorner[OCH_2C(CH_2CH_3)_2CH_2]_n\urcorner$	1	114.2		139–141/1013	142
	4	456.7	138		144
$\ulcorner[OCH_2C(CH_2Cl)_2CH_2]_3\urcorner$	3	465.1	122		34,92
$\ulcorner[O(CH_2)_4]_n\urcorner$	2	144.2	183–185	60/13	147,148
		$n = 2–9$: GC, MS; Refs. 139			

6.1.2.4. Oligomers of Catechol Poly(methylene ethers)



x	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{20}	n_D^{20}	Refs.
1	1	122.1	−18	172–173/1013	1.0640	1.5423	198,199,208–211
	2	244.2	261.6–262.6				198
2	1	136.1		212–214/1009	1.180		208,210
	2	272.3	208–209				41
3	1	150.2	14–15	120/27	1.1343	1.5396	199–203,210
	2	300.3	150–152				41,202
5	1	178.2	Oil	95/3	1.0847	1.5346	199–201
	2	356.4	185.5–187.5				201

6.1.2.4. cont'd

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{20}	n_D^{20}	Refs.
6	1	192.2	38	104/1.3			200,204
	2	384.5	140–141	200–201/0.11			41,204
	3	576.7	112				204
7	1	206.3	17–18	156/13	1.0638		200
	2	412.5	158–159				205
10	2	248.4	Oil	179/13	1.0342		200
	2	496.7	137–138.5				41

6.1.3. OLIGO(FORMALDEHYDE) DERIVATIVES

6.1.3.1. Oligo(oxymethylene) Dihydrates $H[OCH_2]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
4	138.1	95–105 d	1,2
8	258.1	115–120 d	1,2

6.1.3.2. Oligo(oxymethylene) Diacetates $CH_3CO-[OCH_2]_n-OCOCH_3$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	*d* (g/cm³)/°C	n_D/°C	Refs.
1	132.1	−23	39–40/0.13	1.128/24	1.4025/24	2–4
2	162.1	−13	60–62/0.13	1.158/24	1.4124/24	3,4
3	192.2	−3	84/0.13	1.179/24	1.4185/24	3
4	222.2	7	102–104/0.13	1.195/24	1.233/24	2,3
5	252.2	17	124–126/0.13	1.204/24	1.4258/24	3
8	342.3	32–34		1.216/36	1.4297/36	2,3
9	372.3	40–43		1.353/15		3
10	402.4	63.1–66.2				3,218
11	432.4	71.2–73.7				3,218
12	462.4	79.9–82.5				2,3,218
14	522.5	84–86				3
15	552.5	90.5–92		1.364/15		3
16	582.5	93–95				2,3
17	612.6	98.5–99.5		1.370/15		3
19	672.6	107–109		1.390/15		3
20	702.6	111–112				2,23
22	762.7	116–118		1.465		3

6.1.3.3. Oligo(oxymethylene) Dimethyl Ethers $CH_3-[OCH_2]_n-OCH_3$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	*d* (g/cm³)/°C	Refs.
1	76.1	104.8	45.5/1013	0.8538/20	5,8,12,15
2	106.1	−69.7	105.0/1013	0.9597/25	5,13–15,197
3	136.2	−42.5	155.9/1013	1.0242/25	5,197
4	166.2	−9.8	201.8/1013	1.0671/25	5,197
5	196.2	18.3	242.3/1013	1.1003/25	5,197
6	226.3	31–34			3
8	286.4	49–51			3
10	346.4	63.1–66.2			3,218
11	376.4	71.2–73.7			3,218
12	406.5	79.9–82.5			3,218
13	436.5	89–91			3
14	466.5	101–104			3
15	496.5	109–111			3

6.1.3.4. Oligo(oxymethylene) Dipropyl Ethers $C_3H_7-[OCH_2]_n-OC_3H_7$

C_3H_7	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	*d*/°C (g/cm³)	n_D/°C	Refs.
n-	1	132.2		137.5/1013	0.83325/25	1.3913/25	6–9
	2	162.2		67/15	0.89725/25	1.4004/25	6,9
	3	192.2		97/15	0.94325/25	1.4086/23	6,9
	4	222.3	−15 to −13		0.99025/25	1.4137/26	6
	5	252.3	8–8.56		1.01424/25	1.4181/26	6
i-	1	132.2		117–119/1013	0.8242/20	1.3864/20	8,10
	2	162.2		39.5–41/31	0.8897/20	1.3971/20	10
	3	192.2		68.2–68.5/4	0.9348/20	1.4035/20	10
	4	222.3		93.5–94.5/4	0.9751/20	1.4117/20	10
	5	252.3		120–123/4	1.0275/20	1.4235/20	10
	6	282.3	23.4–24.3	159–163/5	1.101/26	1.4467/26	10

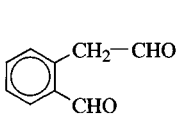
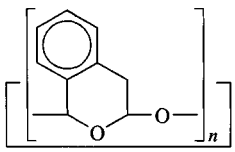
6.1.3.5. Oligo(oxymethylene) Diallyl Ethers $CH_2=CHCH_2-[OCH_2]_n-OCH_2CH-CH_2$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_{25}/(g/cm³)	n_D/°C	Refs.
1	128.2		138–139/1013		1.4226/21	6,11
2	158.2		75–76.5/20	0.946	1.4280/25	6
3	188.2		58.5–64/0.4	0.992	1.4320/25	6
4	218.2	−4.3	82–87/0.4	1.027	1.4350/25	6
5	248.3	15.5	105–107/0.4	1.059	1.4377/25	6
6	278.3	22.5	144–155/0.5	1.079	1.4411/25	6

6.1.3.6. Cyclic Oligo(oxymethylenes) $\ulcorner[OCH_2]_n\urcorner$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	*d* (g/cm³)	Refs.
3	90.1	67–68	114.5/1013	1.39	16–18,31,46–48,121,122
4	120.1	114		1.435	17–19,31,46–48,122,123
5	150.2	61		1.480	31,45–47
6	180.2	72.5		1.440	47
15	450.4	68–70			31

6.1.4. OLIGOMERS OF HIGHER ALDEHYDES

Aldehyde	Oligomers	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_3-CHO	$\ulcorner[OCH(CH_3)]_n\urcorner$	3	132.3	12.6	125/1013	20
		4	176.2	246.2	112–115 (Subl.)	20,21
C_6H_5-CHO	$\ulcorner[OCH(C_6H_5)]_n\urcorner$	3	318.4	175–176	75/1013	35
		3	444.5	284–285		36

6.1.5. HIGHER CYCLIC OLIGO(ACETALS)

6.1.5.1. Oligomers of 1,3-Dioxacycloalkanes $\ulcorner OCH_2-O(CH_2)_x\urcorner_n$

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm³)/°C	n_D/°C	Refs.
2	1	74.1	−95.3	75/1013	1.0595/20	1.4002/20	151
	2	148.2	67				152–155
	5	370.4	28				152

n = 2–5: NMR, Ref. 152
n = 2–9: GC, MS, Refs. 153–155

6.1.5.1. cont'd

x	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	n_D/°C	Refs.
4	1	102.1		117/1009	1.003/20	1.4308/20	22,147,156–158
	2	204.3	90–92				157
5	1	116.2		40–44/15			22,169–171
	2	232.3	56–58				22,152,169
6	2	260.4	71–72				22
8	1	158.2	−63	196/1013	0.985/25	1.4564/25	159,160,162
	2	316.5	82				159,160
9	2	344.5	68–69				22
10	2	372.6	93–94				22
14	2	484.8	103.5–104				22

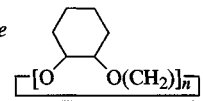
6.1.5.2. Oligomers of Poly(oxacycloalkanes) $\overline{\lceil OCH_2{-}(OCH_2CH_2)_x\rceil_n}$

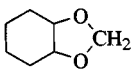
x	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D/°C	Refs.
2	1	118.1		157/1013	1.4421	151,158,168,170
	2	236.3	59.9–60.3			152,168,171
	3	354.4	Oil			152,168
	4	472.5	34.7–35.2			152,168
	5	590.7	Oil			168
	6	708.8	39.7–40.1			168
			n = 1–9: GPC, Ref. 168			
3	1	162.2	27	56/0.5	1.4541/30	22,159,161,164,166–168
	2	324.4	88.0–88.6	85 (subl.)		159,161,164,165,168,214
	3	486.5	27			159,164,165,168,214
	4	648.7	56.5			159,164,165,168,214
	5	810.9	19			159,164,165,168,214
	6	973.1	38.3–39.5			159,164,168,214
	7	1135.3	23			159,164,214
	8	1297.5	28.5			164
			n = 1–8: GPC, ^{1}H-NMR, ^{13}C-NMR, Refs. 163,164,168			
4	1	206.2	23.5	90/0.3	1.4616/25	161,162,166–168
	2	412.5	44.0–44.6			167,168,214
	3	618.7	16			167,168,214
	4	824.9	54			168,214
	5		27			214
	6		30			214
			n = 1–4: GPC, Ref. 168			
5	1	250.3	−6.5	142–144/0.3	1.4642/20	166–168
	2	500.6	79.1–79.5			168,214
	3	750.9	34			168,214
	4	1001.1	63			168,214
	5		32			214
	6		48			214
6	1	294.3		142–144/0.3		166
			n = 1–7: GPC, Ref. 168			

6.1.5.3. Oligomers of Substituted 1,3-Dioxacycloalkanes $\overline{\lceil OCR_2{-}OCH_2{-}X{-}(CH_2]_n\rceil}$

R	X	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	n_D/°C	Refs.
CH_3	$CH_2{-}CH_2$	1	130.2		73–74/139			181
		2	260.4	145.7/146.8				181
H	CH=CH	1	100.1	−55.2	128/979	1.067/20	1.4570/20	173,174
		2	200.2	117				174,184,185
				n = 2–6:	HP-GPC			Ref. 174
H		1	154.2	1	30/0.03		1.4932/24	160,186
		2	308.4	234				186

6.1.5.4. Oligomers of 7,9-Dioxabicyclo[4,3,0]nonane



Monomer	Configuration	*n*	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
	*cis*	1	128.2		52.0/12	191
		2	256.3	142–143		191,192
	trans	1	128.2		57.0/27	191
		2	256.3	107–109		191,192

6.1.5.5. Oligomers of Substituted Poly(oxacycloalkanes)

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	1	132.2		32/1.3	172
	2	264.3	36–37		172
3	1	176.2		60/0.3	172
	2	352.4	61–62		172
4	1	220.3		82/0.2	172
	2	440.5	33–34		172
5	1	264.3		120/0.13	172
	2	528.6	52–53.5		172

REFERENCES

1. H. Staudinger, W. Kern, in "Die hochmolekularen organischen Verbindungen", Springer, Berlin, 1960, p. 224, p. 248, p. 275.
2. W. Kern, H. Cherdron, V. Jaacks, H. Baader, H. Deibig, A. Giefer, L. Höhr, A. Wildenau, Angew. Chem., **73**, 177 (1961).
3. H. Staudinger, H. Johner, R. Signer, G. Mie, J. Hengstenberg, Z. Physik. Chem., **126**, 425 (1927).
4. M. Descudé, Ann. Chim. (Paris), **29**, p. 486, p. 502 (1903).
5. R. H. Boyd, J. Polym. Sci., **50**, 133 (1961).
6. R. F. Webb, A. J. Duke, L. S. A. Smith, J. Chem. Soc., 4307, (1962).
7. M. Ghysels, Bull. Soc. Chim. Belges, **33**, 57 (1924).
8. A. I. Vogel, J. Chem. Soc., 616 (1948).
9. A. Rieche, H. Gross, Chem. Ber., **93**, 259 (1960).
10. E. Klein, J. K. Smith, R. J. Eckert, Jr., J. Appl. Polym. Sci., **7**, 383 (1963).
11. J. A. Trillat, R. Cambier, Bull. Soc. Chim. France, **11**, 752 (1894).
12. E. Fischer, G. Giebe, Chem. Ber., **30**, 3053 (1897).
13. M. Descudé, Compt. Rend., **138**, 1705 (1904).
14. J. Loebering, R. Fleischmann, Chem. Ber., **70**, 1680 (1937).
15. T. Uchida, Y. Kurita, M. Kubo, J. Polym. Sci., **19**, 365 (1956).
16. V. Jaacks, W. Kern, Makromol. Chem., **52**, 37 (1962).
17. J. F. Walker, "Formaldehyde", 3rd ed., Reinhold, New York, 1964.
18. H. Staudinger, M. Luethy, Helv. Chim. Acta, **8**, 65 (1925).
19. K. Hayashi, H. Ochi, M. Nishii, Y. Miyake, S. Okamura, J. Polym. Sci. B, **1**, 427 (1963).
20. A. W. Johnson et al., in: E. H. Rodd (Ed.), "Chemistry of Carbon Compounds", Vol. I, Part A, Elsevier, New York, 1951, p. 476.
21. T. S. Patterson, G. M. Holmes, J. Chem. Soc., 904 (1935).
22. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **57**, 925 (1935).
23. K. J. Rauterkus, H. G. Schimmel, W. Kern, Makromol. Chem., **50**, 166 (1961); K. J. Rauterkus, W. Kern, Chimia, **16**, 114 (1962).
24. P. Rempp, Bull. Soc. Chim. France, 844 (1957).
25. Y. Kuroda, M. Kubo, J. Polym. Sci., **26**, 323 (1957); **36**, 453 (1959); T. Uchida, Y. Kurita, N. Koizumi, M. Kubo, J. Polym. Sci., **21**, 313 (1956).
26. A. F. Gallaugher, H. Hibbert, J. Am. Chem. Soc., **58**, 813 (1936).
27. S. Z. Perry, H. Hibbert, Can. J. Res. B, **14**, 77 (1936).
28. R. Fordyce, E. L. Lovell, H. Hibbert, J. Am. Chem. Soc., **61**, 1905, 1912, 1916 (1939).
29. A. W. Johnson, A. G. Long, C. E. Dalgliesh, in: E. H. Rodd (Ed.), "Chemistry of Carbon Compounds", Vol. I, Part A, Elsevier, New York, 1951, p. 670.
30. G. R. Ramage, E. H. Rodd, J. K. Landquist, in: E. H. Rodd (Ed.), "Chemistry of Carbon Compounds", Vol. IV, Part C, Elsevier, New York, 1951 p. 1529.
31. K. H. Burg, H. D. Hermann, H. Rehling, Makromol. Chem., **111**, 181 (1968).
32. T. P. Hobin, Polymer, **9**, 65 (1968).
33. T. P. Hobin, Polymer, **6**, 403 (1965).
34. Y. Arimatsu, J. Polym. Sci. A-1, **4**, 728 (1966).
35. G. B. Stampa, Macromolecules, **2**, 203 (1969).
36. S. Tagami, T. Kagiyama, C. Aso, Polym. J., **2**, 101 (1971).
37. B. Boemer, H. Heitz, W. Kern, J. Chromatogr., **53**, 51 (1970).
38. J. B. Rose, J. Chem. Soc., 542, (1956).

39. J. Dale, P. O. Kristiansen, Chem. Commun., 670, (1971); Acta Chem. Scand., **26**, 1471 (1972).
40. D. G. Stewart, D. Y. Waddan, E. T. Borrows, B. P., **785**, 229 (1957); Chem. Abstr., 5038th (1958).
41. C. J. Pedersen, J. Am. Chem. Soc., **89**, 7017 (1967).
42. L. A. Bryan, W. M. Smedley, R. K. Summerbell, J. Am. Chem. Soc., **72**, 2206 (1950); W. M. Pasika, J. Polym. Sci. A, **3**, 4287 (1965).
43. R. K. Summerbell, M. J. Kland-Englich, J. Am. Chem. Soc., **77**, 5095 (1955).
44. S. Kondo, L. P. Blanchard, J. Polym. Sci. B, **7**, 621 (1969).
45. M. Nishii, K. Hayashi, S. Okamura, J. Polym. Sci. B, **7**, 891 (1969).
46. Y. Chatani, K. Kitahama, H. Tadokoro, T. Yamauchi, Y. Miyake, J. Macromol. Sci. B: Phys. **4**, 61 (1970).
47. Y. Chatani, T. Ohno, T. Yamauchi, Y. Miyake, J. Polym. Sci. A-2, Polym. Phys. Ed., **11**, 369 (1973).
48. Y. Chatani, T. Uchida, H. Tadokoro, K. Hayashi, M. Nishii, S. Okamura, J. Macromol. Sci. B: Phys. **2**, 567 (1968).
49. J. L. Down, J. Lewis, B. Moore, G. Wilkinson, J. Chem. Soc., 3767, (1959).
50. A. Marshall, R. H. Mobbs, C. Booth, Eur. Polym. J., **16**, 881 (1980).
51. H. H. Teo, R. H. Mobbs, C. Booth, Eur. Polym. J., **18**, 541 (1982).
52. A. Marshall, R. C. Domszy, H. H. Teo, R. H. Mobbs, C. Booth, Eur. Polym. J., **17**, 885 (1981).
53. H. H. Teo, A. Marshall, C. Booth, Makromol. Chem., **183**, 2265 (1982).
54. S. G. Yeates, H. H. Teo, R. H. Mobbs, C. Booth, Makromol. Chem., **185**, 1559 (1984).
55. H. Matsuura, T. Miyazawa, Spectrochim. Acta, **23A**, 2433 (1967).
56. P. H. Elworthy, C. B. Macfarlane, J. Chem. Soc., 537 (1962).
57. V. P. Privalko, A. P. Lobodina, Eur. Polym. J., **10**, 1033 (1974).
58. R. C. Domszy, C. Booth, Makromol. Chem., **183**, 1051 (1982).
59. K. Viras, H. H. Teo, A. Marshall, R. C. Domszy, T. A. King, C. Booth, J. Polym. Sci., Polym. Phys. Ed., **21**, 919 (1983).
60. H. H. Teo, T. G. E. Swales, R. C. Domszy, F. Heatley, C. Booth, Makromol. Chem., **184**, 861 (1983).
61. S. G. Yeates, C. Booth, Makromol. Chem., **186**, 2663 (1985).
62. S. G. Yeates, H. H. Teo. C. Booth, Makromol. Chem., **185**, 2475 (1984).
63. T. G. E. Swales, H. H. Teo, R. C. Domszy, K. Viras, T. A. King, C. Booth, J. Polym. Sci., Polym. Phys. Ed., **21**, 1501 (1983).
64. T. G. E. Swales, R. C. Domszy, R. L. Beddoes, C. Price, C. Booth, J. Polym. Sci., Polym. Phys. Ed., **23**, 1585 (1985).
65. T. G. E. Swales, S. G. Yeates, C. Booth, Br. Polym. J., **17**, 32 (1985).
66. S. G. Yeates, C. Booth, Eur. Polym. J., **21**, 217 (1985).
67. T. G. E. Swales, R. L. Beddoes, C. Price, C. Booth, Eur. Polym. J., **21**, 629 (1985).
68. M. Newcomb, S. S. Moore, D. J. Cram, J. Am. Chem. Soc., **99**, 6405 (1977).
69. J. S. Bradshaw, R. A. Reeder, M. D. Thompson, E. D. Flanders, R. L. Carruth, R. M. Izatt, J. J. Christensen, J. Org. Chem., **41**, 134 (1976).
70. P. Sally, J. Morgós, L. Farkas, I. Rusznák, B. Bartha, Acta Chim. Acad. Sci. Hung., **102**, 85 (1979).
71. A. Warshawsky, N. Shoef, J. Polym. Sci., Polym. Chem. Ed., **22**, 657 (1984).
72. F. Eisenbeiß, S. Ehlerding, Kontakte (Merck), **1**, 22 (1978).
73. Sj. Van der Wal, L. R. Snyder, J. Chromatogr., **255**, 463 (1983).
74. S.-T. Lai, L. Sangermano, D. C. Locke, J. High Resolut. Chromatogr., **7**, 494 (1984).
75. T. Gartiser, C. Selve, L. Mansuy, A. Robert, J.-J. Delpuech, J. Chem. Res. (S), 292 (1984); (M), 2672 (1984).
76. B. Dietrich, J. M. Lehn, J. P. Sauvage, J. Blanzat, Tetrahedron, **29**, 1629 (1973).
77. W. Kern, S. Iwabuchi, H. Sato, V. Böhmer, Makromol. Chem., **180**, 2539 (1979).
78. J. Dale, G. Borgen, K. Daasvatn, Acta Chem. Scand. B, 28, 378 (1974).
79. J. Dale, K. Daasvatn, J. Chem. Soc., Chem. Commun., 295, (1976); Acta Chem. Scand. B, **34**, 327 (1980).
80. F. A. L. Anet, J. Krane, J. Dale, K. Daasvatn, P. O. Kristiansen, Acta Chem. Scand., **27**, 3395 (1973).
81. F. L. Cook, T. C. Caruso, M. P. Byrne, C. W. Bowers, D. H. Speck, C. L. Liotta, Tetrahedron Lett., 4029 (1974).
82. G. W. Gokel, D. J. Cram, C. I. Liotta, H. P. Harris, F. L. Cook, J. Org. Chem., **39**, 2445 (1974); Org. Synth., **57**, 30 (1977).
83. G. Johns, C. J. Ransom, C. B. Reese, Synthesis, 515, (1976).
84. R. N. Greene, Tetrahedron Lett., 1793, (1972).
85. D. J. Worsfold, A. M. Eastham, J. Am. Chem. Soc., **79**, 897, 900 (1957); G. A. Latremouille, G. T. Merrall, A. M. Eastham, J. Am. Chem. Soc., **82**, 120 (1960).
86. S. Kobayashi, K. Morikawa, T. Saegusa, Macromolecules, **8**, 952 (1975); S. Kobayashi, T. Kobayashi, T. Saegusa, Polym. J., **15**, 883 (1983).
87. S. Kobayashi, K. Morikawa, T. Saegusa, Polym. J., **11**, 405 (1979).
88. J. Szymanowski, H. Szewczyk, J. Hetper, J. Beger, J. Chromatogr., **351**, 183 (1986).
89. R. J. Katnik, J. Schaefer, J. Org. Chem., **33**, 384 (1968).
90. R. J. Kern, J. Org. Chem., **33**, 388 (1968).
91. J. H. Hammond, J. F. Hooper, W. G. P. Robertson, J. Polym. Sci. A-1, **9**, 265, 281 (1971).
92. E. J. Goethals, Adv. Polym. Sci., **23**, 103 (1977); Pure Appl. Chem., **48**, 335 (1976).
93. R. O. Colclough, K. Wilkinson, J. Polym. Sci. C, **4**, 311 (1966).
94. Y. Kawakami, A. Ogawa, Y. Yamashita, J. Org. Chem., **44**, 441 (1979).

95. Y. Kawakami, A. Ogawa, Y. Yamashita, J. Polym. Sci., Polym. Chem. Ed., **17**, 3785 (1979).
96. Y. Yamashita, K. Iwao, K. Ito, Polym. Bull., **1**, 73 (1978).
97. A. Sato, T. Hirano, M. Suga, T. Tsuruta, Polym. J., **9**, 209 (1977).
98. M. G. Zelenskaya, F. L. Kolodkin, F. P. Sidelkovskaya, V. A. Ponomarenko, Izv. Akad. Nauk SSSR, Ser. Khim., 1105, (1968); Bull. Acad. Sci. USSR, 1050, (1968).
99. J. D. McClure, U S Patent, **3**, 140, 296 (1963); Chem. Abstr., **61**, 8318b (1964).
100. K. Ito, N. Usami, Y. Yamashita, Polym. J., **11**, 171 (1979).
101. T. Miyasaka, S. Tanaka, Polym. J., **16**, 365 (1984).
102. I. Géczy, Acta Chim. Acad. Sci. Hung., **118**, 131 (1985).
103. Y. I. Estrin, S. G. Entelis, Vysokomol. Soedin. A, **13**, 1654 (1971); Polym. Sci. USSR, **13**, 1862 (1971).
104. S. G. Entelis, G. V. Korovina, Makromol. Chem., **175**, 1253 (1974).
105. K. Weissermel, E. Nölken, Makromol. Chem., **68**, 140 (1963).
106. Y. Yamashita, K. Iwao, K. Ito, J. Polym. Sci., Polym. Lett. Ed., **17**, 1 (1979).
107. R. O. Colclough, G. Geé, W. C. E. Higginson, J. B. Jackson, M. Litt, J. Polym. Sci., **34**, 171 (1959).
108. K. Alexander, L. E. Schniepp, J. Am. Chem. Soc., **70**, 1839 (1948).
109. S. A. Miller, B. Bann, R. D. Thrower, J. Chem. Soc., 3623 (1950).
110. H. J. Dauben, Jr., B. Löken, H. J. Ringold, J. Am. Chem. Soc., **76**, 1359 (1954).
111. W. J. Croxall, F. J. Glavis, H. T. Neher, J. Am. Chem. Soc., **70**, 2805 (1948).
112. A. N. Nesmeyanov, I. F. Lutsenko, Bull. Acad. Sci. USSR, Ser. Khim., 296 (1943); Chem. Abstr. **38**, 5498 (1944).
113. E. Augdahl, Acta Chem. Scand., **9**, 1237 (1955).
114. M. J. Astle, B. E. Jacobson, J. Org. Chem., **24**, 1766 (1959).
115. F. Kametani, Y. Sumi, Chem. Pharm. Bull., **20**, 1479 (1972).
116. C. E. Wilson, H. J. Lucas, J. Am. Chem. Soc., **58**, 2396 (1936).
117. Y. Sumi, F. Kametani, Chem. Pharm. Bull., **21**, 1103 (1973).
118. R. K. Summerbell, G. J. Lestina, J. Am. Chem. Soc., **79**, 6219 (1957).
119. N. Koizumi, T. Hanai, J. Phys. Chem., **60**, 1496 (1956).
120. A. L. Kravchenko, N. A. Lipatnikov, V. T. Burmistrov, T. M. Gritsenko, I. A. Popov, Zh. Prikl. Khim., **45**, 2581 (1972); J. Appl. Chem. USSR, **45**, 2708 (1972).
121. V. Busetti, M. Mammi, G. Carazzolo, Z. Kristallogr., **119**, 310 (1963); G. Carazzolo, S. Leghissa, M. Mammi, Makromol. Chem., **60**, 171 (1963).
122. M. Mansson, E. Morawetz, Y. Nakase, S. Sunner, Acta Chem. Scand., **23**, 56 (1969).
123. J. A. Ladd, J. Mol. Struct., **36**, 329 (1977).
124. M. Gordon, J. G. Miller, A. R. Day, J. Am. Chem. Soc., **71**, 1245 (1949).
125. A. I. Vogel, J. Chem. Soc., 1814 (1948).
126. C. W. Tasker, C. B. Purves, J. Am. Chem. Soc., **71**, 1017 (1949).
127. J. N. Street, H. Adkins, J. Am. Chem. Soc., **50**, 162 (1928).
128. M. H. Palomaa, Chem. Ber., **35**, 3299 (1903); **42**, 3873 (1910).
129. J. G. Davidson, Ind. Eng. Chem., **18**, 669 (1926).
130. M. H. Palomaa, I. Honkanen, Chem. Ber., **70**, 2199 (1937).
131. W. Lippert, Ann. Chem., **276**, 148, 171 (1893).
132. L. Liston, W. M. Dehn, J. Am. Chem. Soc., **60**, 1264 (1938).
133. J. V. Capinjola, J. Am. Chem. Soc., **67**, 1615 (1945).
134. K. Machida, T. Miyazawa, Spectrochim. Acta, **20**, 1865 (1964).
135. A. Kotera, K. Suzuki, K. Matsumura, T. Nakano, T. Oyama, U. Kambayashi, Bull. Chem. Soc. Japan, **35**, 797 (1962).
136. L. L. Chan, K. H. Wong, J. Smid, J. Am. Chem. Soc., **92**, 1955 (1970); U. Takaki, J. Smid, *ibid.*, **96**, 2588 (1974).
137. R. Bill, M. Dröscher, G. Wegner, Makromol. Chem., **179**, 2993 (1978); **182**, 1033 (1981).
138. D. Starr, R. M. Hixon, J. Am. Chem. Soc., **56**, 1595 (1934).
139. G. Pruckmayr, T. K. Wu, Macromolecules, **11**, 265 (1978); J. M. McKenna, T. K. Wu, G. Pruckmayr, Macromolecules, **10**, 877 (1977).
140. M. Bucquoye, E. J. Goethals, Makromol. Chem., **179**, 1681 (1978).
141. P. Dreyfuss, M. P. Dreyfuss, Polym. J., **8**, 81 (1976).
142. L. F. Schmoyer, L. F. Case, Nature, **187**, 592 (1960).
143. M. R. Bucquoye, E. J. Goethals, Makromol. Chem., **182**, 3379 (1981).
144. M. R. Bucquoye, E. J. Goethals, Polym. Bull., **2**, 707 (1980).
145. I. M. Robinson, G. Pruckmayr, Macromolecules, **12**, 1043 (1979).
146. G. D. Andrews, A. Vatvars, G. Pruckmayr, Macromolecules, **15**, 1580 (1982).
147. W. Reppe, Ann. Chem., **596**, 1, 132 (1955).
148. V. Weinmayr, J. Am. Chem. Soc., **81**, 3590 (1959).
149. A. Weissler, J. W. Fitzgerald, I. Resnick, J. Appl. Phys., **18**, 434 (1947).
150. J. Dale, K. Daasvatn, T. Grønneberg, Makromol. Chem., **178**, 873 (1977).
151. M. Astle, J. A. Zaslowsky, P. G. Lafyatis, Ind. Eng. Chem., **46**, 787 (1954); W. Baker, F. B. Field, J. Chem. Soc., 87 (1932).
152. Y. Kawakami, Y. Yamashita, Macromolecules, **10**, 837 (1977).
153. J. A. Semlyen, Adv. Polym. Sci., **21**, 41 (1976).
154. J. A. Semlyen, Pure Appl. Chem., **53**, 1797 (1981).
155. J. M. Andrews, J. A. Semlyen, Polymer, **13**, 142 (1972).
156. P. A. Laurent, P. Tarte, B. Rodrigues, Bull. Soc. Chim. France, 946 (1959).
157. P. H. Plesch, P. H. Westermann, Polymer, **10**, 105 (1969).
158. J. Rübner, R. Gnauck, H. Frommelt, Acta Polym., **32**, 238 (1981).
159. R. C. Schulz, K. Albrecht, Q. V. Tran Thi, J. Nienburg, D. Engel, Polym. J., **12**, 639 (1980).
160. R. C. Schulz, K. Albrecht, W. Hellermann, A. Kane, Q. V. Tran Thi, Pure Appl Chem., **53**, 1763 (1981).

161. K. Albrecht, D. Fleischer, C. Rentsch, H. Yamaguchi, R. C. Schulz, Makromol. Chem., **178**, 3191 (1977).
162. K. Albrecht, D. Fleischer, A. Kane, C. Rentsch, Q. V. Tran Thi, H. Yamaguchi, R. C. Schulz, Makromol. Chem., **178**, 881 (1977).
163. C. Rentsch, R. C. Schulz, Makromol. Chem., **179**, 1403 (1978).
164. C. Rentsch, R. C. Schulz, Makromol. Chem., **178**, 2535 (1977).
165. Y. Yamashita, Y. Kawakami, K. Kitano, J. Polym. Sci., Polym. Lett. Ed., **15**, 213 (1977).
166. Y. Kawakami, T. Sugiura, Y. Yamashita, Bull. Chem. Soc. Japan, **51**, 3053 (1978).
167. Y. Kawakami, J. Suzuki, Y. Yamashita, Polym. J., **9**, 519 (1977).
168. Y. Yamashita, J. Mayumi, Y. Kawakami, K. Ito, Macromolecules, **13**, 1075 (1980).
169. J. Dale, T. Ekeland, Acta Chem. Scand., **27**, 1519 (1973).
170. F. A. L. Anet, P. J. Degen, J. Am. Chem. Soc., **94**, 1390 (1972).
171. J. Dale, T. Ekeland, J. Krane, J. Am. Chem. Soc., **94**, 1389 (1972).
172. V. Gold, C. M. Sghibartz, J. Chem. Soc., Chem. Commun., 507, (1978).
173. K. C. Brannock, G. R. Lappin, J. Org. Chem., **21**, 1366 (1956).
174. W. Hellermann, R. C. Schulz, Makromol. Chem., Rapid Commun., **2**, 585 (1981).
175. R. Riemschneider, W. M. Schneider, Monatsh. Chem., **90**, 510 (1959).
176. L. H. Cretcher, W. H. Pittenger, J. Am. Chem. Soc., **46**, 1503 (1924).
177. R. C. Tallman, J. Am. Chem. Soc., **56**, 126 (1934).
178. M. Matter, A. Kobler, U S Patent 2,798,873 (1957), British Patent 762,761 (1956); Chem. Abstr., **51**, 18012a (1957); U S Patent 2,858,315 (1958); Chem. Abstr., **53**, 6262e (1959); M. Matter, A. Kobler, M. Kuhn, K. Michel, US Patent 2,857,418 (1958); Chem. Abstr. **53**, 10131g (1959); US Patent 2,769,838 (1956); Chem. Abstr., **52**, 1247c (1958).
179. L. H. Cretcher, W. H. Pittenger, J. Am. Chem. Soc., **47**, 163 (1925).
180. M. Lecat, Ann. Chim., **2**, 171 (1947).
181. M. Okada, K. Yagi, H. Sumitomo, Makromol. Chem., **163**, 225 (1973).
182. C. G. Krespan, J. Org. Chem., **39**, 2351 (1974).
183. A. C. Coxon, J. F. Stoddart, J. Chem. Soc., Perkin I, 767 (1977).
184. J. U. R. Nielsen, S. E. Jørgensen, N. Frederiksen, R. B. Jensen, O. Dahl, O. Buchardt, G. Schroll, Acta Chem. Scand. B, **29**, 400 (1975).
185. O. Dahl, J. Dale, N. Frederiksen, H. Henriksen, R. B. Jensen, S. E. Jørgensen, S. Larsen, L. Nørskov, G. Schroll, Acta Chem. Scand. B, **33**, 197 (1979).
186. Ch. Mletzko, R. C. Schulz, Makromol. Chem., Rapid Commun., **4**, 445 (1983).
187. J. S. Bradshaw, P. E. Stott, Tetrahedron, **36**, 461 (1980).
188. C. J. Pedersen, H. K. Frensdorff, Angew. Chem., **84**, 16 (1972).
189. C. J. Pedersen, J. Am. Chem. Soc., **92**, 386 (1970); **89**, 2495 (1967).
190. R. C. Helgeson, T. L. Tarnarski, J. M. Timko, D. J. Cram, J. Am. Chem. Soc., **99**, 6411 (1977).
191. J. Kops, H. Spanggaard, Makromol. Chem., **176**, 299 (1975).
192. S. Brimacombe, A. B. Foster, B. D. Jones, J. J. Willard, J. Chem. Soc. C, 2404, (1967).
193. B. R. Bowsher, A. J. Rest, J. Chem. Soc., Dalton Trans., 1157, (1981).
194. R. Riemschneider, P. Grob, Monatsh. Chem., **90**, 783 (1959).
195. H. Böhme, W. Schürhoff, Chem. Ber., **84**, 28 (1951).
196. T. L. Davis, V. F. Harrington, J. Am. Chem. Soc., **56**, 126 (1934).
197. ICI Ltd., British Patent 603,872 (1948), US Patent 2449,469 (1948); Chem. Abstr. **43**, 1051h (1949).
198. W. J. Gensler, C. M. Samour, J. Org. Chem., **18**, 9 (1953).
199. G. Baddeley, N. H. P. Smith, J. Chem. Soc., 2516, (1961).
200. K. Ziegler, A. Lüttringhaus, K. Wohlgemuth, Ann. Chem., **528**, 162, 179 (1937).
201. G. Illuminati, L. Mandolini, G. Masci, J. Am. Chem. Soc., **96**, 1422 (1974).
202. A. W. Archer, P. A. Claret, Chem. Ind., 171 (1969).
203. R. B. Thompson, E. Symon, J. Am. Oil Chem. Soc., **33**, 414 (1956).
204. A. Lüttringhaus, I. Sichert-Modrow, Makromol. Chem., **18/19**, 511 (1956).
205. G. R. Brown, J. H. P. Tyman, Chem. Ind., 436 (1970).
206. K. F. Ridley, J. Ridley, British Patent 795,866 (1955); Chem. Abstr. **53**, 1154 (1959).
207. S. Schüring, W. Ziegenbein, Tenside, **4**, 161 (1967).
208. B. N. Ghosh, J. Chem. Soc., **107**, 1588 (1915).
209. W. Baker, F. B. Field, J. Chem. Soc., 86 (1931).
210. R. C. Cass, S. E. Fletcher, C. T. Mortimer, P. G. Quincey, H. D. Springall, J. Chem. Soc., 2595 (1958).
211. S. Kinugasa, H. Hayashi, S. Hattori, Polym. J., **22**, 1059 (1990).
212. S. Kinugasa, A. Takatsu, H. Nakanishi, H. Nakahara, S. Hattori, Macromolecules, **25**, 4848 (1992).
213. C. Campbell, K. Viras, C. Booth, J. Polym. Sci., Polym. Phys., **29**, 1613 (1991); J. R. Craven, K. Viras, A. J. Masters, C. Booth, J. Chem. Soc., Faraday Trans, **87**, 3677 (1991).
214. M. Nekomanesk, Ch. V. Nicholas, S. Booth, Makromol. Chem., **191**, 2567 (1990).
215. Y. Nakatsuji, N. Kameda, M. Okahara, Synthesis, 280 (1987).
216. P.-L. Kuo, N. Kawamura, M. Miki, M. Okahara, Bull. Chem. Soc. Japan., **53**, 1689 (1980).
217. R. Chenevert, L. D'Astous, J. Heterocycl. Chem., **23**, 1785 (1986).
218. K. Ute, T. Takahashi, K. Matsui, K. Hatada, Polymer, **25**, 1275 (1993).
219. C. A. Vitali, B. Masci, Tetrahedron, **45**, 2201 (1989).
220. J. R. Craven, H. Tyrer, S. P. L. Li, C. Booth, D. Jackson, J. Chromatogr., **387**, 233 (1987).

221. S. G. Yeates, J. R. Craven, R. H. Mobbs, C. Booth, J. Chem. Soc., Faraday Trans. I, **82**, 1865 (1986).
222. A. D. Bedells, C. Booth, Makromol. Chem., **192**, 2099 (1991).
223. A. D. Bedells, R. H. Mobbs, C. Booth, Makromol. Chem., **192**, 2089 (1991).
224. K. Viras, T. A. King, C. Booth, J. Polym. Sci., Polym. Phys., **23**, 741 (1985).
225. J. Craven, R. H. Mobbs, C. Booth, E. J. Goodwin, D. Jackson, Makromol. Chem., **190**, 1207 (1989).

6.2. OLIGO(CARBONATES)

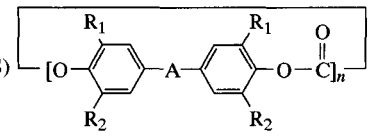
6.2.1. CYCLIC OLIGO(DIOL CARBONATES) $\ulcorner[O(CH_2)_xO{-}CO]_n\urcorner$

x	*n*	Mol. wt.	m.p. (°C)	Refs.
4	2	232.2	175–176	1,2,9
5	2	260.3	117–118	2,8,9
6	2	288.3	128–129	2,8,9
7	2	316.4	97–98	2,9
8	1	172.2	21.5–23	2,9
8	2	344.5	116–117	2,8,9
9	1	186.3	35–35	2,9
9	2	372.5	95–95.5	2,9
10	1	200.3	10–11	2,9
10	2	400.6	105–106	2,8,9
11	1	214.3	40–41	2,9
11	2	428.6	97–97.5	2,9
12	1	228.3	11–12	2,9
12	2	456.7	93–95	2,8,9

6.2.2. CYCLIC OLIGO(CARBONATES) OF DI- AND TRI-ETHYLENE GLYCOL $\ulcorner[O(CH_2CH_2O)_x{-}CO]_n\urcorner$

x	*n*	Mol. wt.	m.p. (°C)	Refs.
2	2	264.2	139–140	8
2	3	396.3	165–167	8
3	1	176.2	39–40	8
3	2	352.4	120–121	8

6.2.3. CYCLIC OLIGO(*p,p'*-BISPHENOL CARBONATES)



A	R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
$C(CH_3)_2$	H	H	3	762.9	335–340	6,7,9,10
	H	H	4	1017.1	375	3,4,9,10
	CH_3	H	4	1129.4	301–303	3,9
	Cl	Cl	4	1568.2	420	3,9
S	H	H	4	977.1	320–322	3,9

6.2.4. CYCLIC OLIGO(*o,o'*-BISPHENOL CARBONATES)

R	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	1	254.3	144–146	7
	2	508.6	276–279	5–7
Cl	2	295.1	365–368	5

REFERENCES

1. W. H. Carothers, F. J. van Natta, J. Am. Chem. Soc., **52**, 314 (1930).
2. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **55**, 5031 (1933).
3. H. Schnell, L. Bottenbruch, Makromol. Chem., **57**, 1 (1962).
4. H. Sotobayashi, S. L. Lie, J. Springer, K. Ueberreiter, Makromol. Chem., **111**, 172 (1968).
5. R. J. Prochaska, Belgian Patent 648.741 (1964); Chem. Abstr., **63**, P 16.262 d (1965).
6. R. J. Prochaska, French Patent 1.385.700 (1965); Chem. Abstr., **62**, 13269 g (1965).
7. L. S. Moody, US Patent 3.155.683 (1964); Chem. Abstr., **62**, 2739c (1965).
8. H. Krimm, H. J. Buysch, Bayer-AG., Ger. Offen. DE 3.103.135, 3.103.137, 3.103.140 (1981); Chem. Abstr., **97**, 216894, 216892, 216893 (1982).
9. J. S. Bradshaw, G. E. Mass, R. M. Izatt, J. J. Christensen, Chem. Rev., **79**, 37 (1979).
10. R. H. Wiley, Macromolecules, **4**, 254 (1971).

6.3. OLIGO(ESTERS)

6.3.1. OLIGO(ESTERS) OF HYDROXY ACIDS

6.3.1.1. Oligo(esters) of Aliphatic Hydroxy Acids

6.3.1.1.1. OLIGO(α-HYDROXY ACID) METHYL ESTERS $H[OCHRCO]_nOCH_3$

R	*n*	Mol. wt.	b.p. (°C/mbar)	Refs.
CH_3	1	104.1	58–59/39	
	2	176.2	79–80/3	53
$CH(CH_3)_2$	1	132.2	81–82/60	51
	2	232.3	106.5–107/1	51
$CH(CH_3)_2$	3	332.4	151–153/1	51
	4	432.5	197–199.5/1	51
	5	532.6	207–212	51

6.3.1.1.2. ACETYL OLIGO(α-HYDROXY ACID) METHYL ESTERS $CH_3CO-[OCHRCO]_nOCH_3$

R	*n*	Mol. wt.	b.p. (°C/mbar)	$[\{\alpha\}]_D^{25}$	Conc.	Refs.
CH_3	1	146.1	67–68/17	−47.3	1.41[a]	48,53
	2	218.2	87–88/0.7	−65.6	1.98[a]	53
	3	290.3	144–146/1	−81.8	1.53[a]	53
$CH(CH_3)_2$	1	174.2	89–89.5	−30.9	1.78[b]	51
$CH(CH_3)_2$	2	274.3	101–101.5	−43.8	1.14[b]	51
	3	374.4	153–154	−49.0	1.21[b]	51
	4	474.5	186–188	−53.6	1.14[b]	51
	5	574.6	230–232	−60.8	0.86[b]	51

[a] In CH_2Cl_2.
[b] In $CHCl_3$.

6.3.1.1.3. CYCLIC OLIGO(ESTERS) OF α-HYDROXY ACIDS $\square[OCHRCO]_n\square$

R	Config.	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	$[\{\alpha\}]_D$	Refs.
H		2	116.1	86–87	169/16		1,2,100,101
		5	290.2	230			102
		6	348.2	215			102
CH_3	L	2	144.1	98.7	150/33	−280[a]	3,4
	DL	2	144.1	128	256		2–7
$CH(CH_3)_2$	L	2	200.2	152–153		−256[b]	103
	DL	2	200.2	134–135			103
$CH_2CH(CH_3)_2$	L	2	228.3	169–170		−226[b]	103
	DL	2	228.3	138–140			103

[a] c = 0.58, C_6H_6, 18°C.
[b] c = 1, $CHCl_3$, 25°C.

6.3.1.1.4. OLIGO((R)-β-HYDROXYBUTYRIC ACIDS) $H[OCH(CH_3)CH_2CO]_nOH$

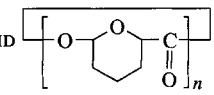
n	Mol. wt.	m.p. (°C)	$[\{\alpha\}]_D^{20}$	(c = 1, CH_2Cl_2); λ (nm)	Method	Refs.
2	190.2		−19.7	(H_2O); 389	IR, NMR, MS	149,150
3	276.3			NMR, MS		149
4	362.4	49	−52.9, 365		NMR, MS	63,64,73,144,149
5	448.5	65			NMR, MS	64,73,144,149
6	534.5	81				64,73
7	620.6	89				64,73
8	706.7	100–101	11.7; 365		IR, NMR, MS	64,73,144,151
12	1051.1	128–129		−7.7; 365	IR, NMR, MS	144
16	1395.5	134.3			NMR	144
24	2084.2	155–156.5			IR, NMR, MS	144
32	2772.9	154.3		3.8; 365	IR, NMR, MS	144
64	5527.9	161.3		7.1; 365		144
128	11037.8				NMR, MS	144

6.3.1.1.5. OLIGO(β-HYDROXYPIVALIC ACIDS) $H[OCH_2C(CH_3)_2CO]_nOH$

n	Mol. wt.	m.p. [°C]	Refs.
1	118.1	123–124	50
2	218.2	71.5–72.5	50
3	318.4	67–68	50
4	418.5	111.5–112.5	50
5	518.6	119.5–120.5	50
6	618.7	138–139	50

6.3.1.1.6. OLIGO(β-HYDROXYPIVALIC ACID) ISOBUTYL ESTERS $H[OCH_2C-(CH_3)_2CO]_nOCH_2CH(CH_3)_2$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	274.4		115–117/0.7	49
3	374.5		175–177/1	49
4	474.6	75.5–76		49
5	574.7	97–98		49
6	674.8	120.5–121.5		49
7	774.9	134–135		49

6.3.1.1.7. Cyclic Oligo(esters) of 6-Hydroxytetrahydropyran-2-Carboxylic Acid 

Config.	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	$[\alpha]_D^{25}$ ($CHCl_3$)	Refs.
RS	1	128.1	18–20	64–65/5		79,82,105
	2	256.2	160			76,79,80,82–84
	4	512.5	155			76,79,80,82,83,85
	5	640.7	175			76,79,80,82,83,86
R	1	128.1			+ 142	81,83,87
	4	512.5	115		− 189	76,78,82–84,87
	5	640.7	125		− 136	76,81–83,86,87

6.3.1.1.8. Cyclic Oligo(esters) of ω-Hydroxy Acids $\overline{[O(CH_2)_xCO]_n}$

x[a]	n[a]	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	1	72.1		51/133	
	3	216.2	56–58		104
	4	288.3	116–120		104
	5	350.3	80–85		104
	6	432.4	133–135		104
	7	504.5	84–86		104
5	1	114.1	5	140–106/13	8–10,91
	2	228.3	112–113	130/0.2	8,9,65,66,69,77,91–95
	3	342.4		250/0.3	9,65,66,72,91–95
6	1	128.2		80–82/15	10
	2	256.3	45	135/0.3	9,69
	3	384.5		202/0.3	9
7	1	142.2		72–73/15	10,91
	2	284.4	93	152/0.2	9,69,91
8	1	156.2	31–31.5	86–87/13	9–11
	2	312.5	57–58	158/0.04	9,10,69,77
	3	468.7	20	240/0.1	9
9	1	170.3	6.4	100/13	9–12,77
	2	340.5	97	192/0.4	9,12–14,69,77
	3	510.8	29	270/0.4	9
10	1	184.3	3	116/13	9–12,77,91
	2	368.6	74		9,12,69,77,91
11	1	198.3	2	130/13	10,11,15,77,91
	2	396.6	104		9,69,77,91
12	1	212.3	27.5	143/13	9–12,16,77,91
	2	424.7	84		9,14,77,91
13	1	226.4	33–33.7	165/20	9,11,16,77
	2	452.7	107		9,12,77
14	1	240.4	37–37.5	169/13	9–11,16,77,91
	2	480.7	90–91		9,15,69,76,77,88,91
15	1	254.4	35.5–36.5	188/20	9,11,16,77
	2	508.8	111		9,69,77
16	1	268.4	42–43	194/20	9,11,16
	2	536.9	97		9
17	1	282.5	37	143/0.3	9
	2	564.9	114		9
22	1	348.6	36	175/0.3	9
	2	697.2	105		9

[a] $x = 10$, $n = 2.7$ (sharp fractions): GPC, NMR, MS, Ref. 154.

6.3.1.1.9. Cyclic Oligoesters of (R)-β-Hydroxybutyric Acid $\overline{[OCH(CH_3)CH_2CO]_n}$

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{20}$ ($c = 1$, $CHCl_3$)	Methods	Refs.
2	172.2	124.5–125	− 91.7	IR, NMR, MS, X-ray	146
3	258.3	110–110.5	− 33.9	IR, NMR, MS, X-ray	144,145,148,151
4	344.4	178–178.5	15.4	IR, NMR, MS, X-ray	144,151
5	430.5	102.5–103.5	6.7	IR, NMR, MS, X-ray	144,151,152
6	516.6	112–113.5	11.1	IR, NMR, MS, X-ray	144,151,152

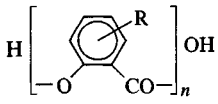
6.3.1.1.9. *cont'd*

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{20}$ ($c = 1$, $CHCl_3$)	Methods	Refs.
7	602.6	118–119.2	−1.0	IR, NMR, MS, X-ray	144,151,152
8	688.7	84–85	3.1	NMR, X-ray	144,151
9	774.8	92.5–93	0.8	IR, NMR, MS	144,151
10	860.9	89.5–90.5	+0.5	IR, NMR, MS	144,151
12	1033.1	98–99	6.5 (CH_2Cl_2)	IR, NMR, MS	144
16	1377.5	122–123	1.3 (CH_2Cl_2)	IR, NMR, MS	144
32	2754.9			NMR, MS	144

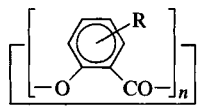
6.3.1.1.10. CYCLIC OLIGESTERS OF (R)-3-HYDROXYPENTANOIC ACID $[OCH(C_2H_5)CH_2CO]_n$

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{20}$ ($c = 1$, $CHCl_3$)	Refs.
3	300.4	98.5–99.0	−19.4	147
4	400.5	36.0–36.6	34.0	147
5	500.6	110.5–111	19.8	147
6	600.7	97.5–98.5	19.6	147
7	700.8	54.0–54.5	13.2	147
8	801.0		15.3	147
9	901.1		14.5	147
10	1001.2		13.9	147
11	1101.3		13.8	147
12	1201.4		13.2	147

6.3.1.2. Oligo(esters) of Aromatic Hydroxy Acids

6.3.1.2.1. LINEAR OLIGO(ESTERS) OF SALICYLIC ACIDS 

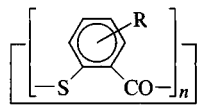
R_3	R_4	R_5	R_6	*n*	Mol. wt.	m.p. (°C)	Refs.
H	H	H	H	2	258.2	148–149	17–19
				3	378.3	150–152	130
				4	498.4	177–183	130
H	Me	H	H	2	286.3	162	20
H	H	Me	H	2	286.3	128–129	20

6.3.1.2.2. CYCLIC OLIGO(ESTERS) OF SALICYLIC ACIDS 

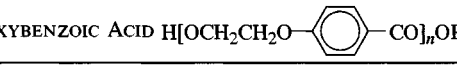
R_3	R_4	R_5	R_6	*n*	Mol. wt.	m.p. (°C)	Refs.
H	H	H	H	2	240.2	234	17,19,132,135
				3	360.3	200	17,21,132,135
				4	480.4	298–300	17,132,135,141
				6	720.7	375	17
Me	H	H	H	2	268.3	240	20–23,135
				3	402.4	264–265	22,135,139
				4	536.5	299–300	22,135,141
H	Me	H	H	2	268.3	255	20,22,23,135
				3	402.4	207–207.5	22,135,139
				4	536.5	305	22
H	H	Me	H	2	268.3	235–235.5	20,22,23,135
				3	402.4	244.5–245	22,135
				4	536.5	347	21,22
Me	H	H	Me	2	296.3	211–212	137
				3	444.5	255	137,138
$CHMe_2$	H	H	Me	2	352.4	209	24,139,142
				3	528.7	217	24,133,134,136,138,142

6.3.1.2.2. *cont'd*

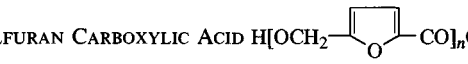
R_3	R_4	R_5	R_6	*n*	Mol. wt.	m.p. (°C)	Refs.
Me	H	H	$CHMe_2$	2	352.4	176–178	137,139
				3	528.7	247	136–138,140
				4	704.8	323	137
CMe_3	H	H	Me	2	380.5	238	137
H	$CHMe_2$	H	Me	2	352.4	136	137
				3	528.7	191	137
Ph	H	H	H	4	784.8	360	137

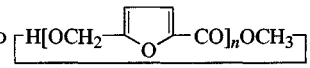
6.3.1.2.3. Cyclic Oligo(esters) of Thiosalicylic Acids 

R_3	R_4	R_5	R_6	*n*	Mol. wt.	m.p. (°C)	Refs.
H	H	H	H	2	272.3	182–183	25,26,131,141,143
				3	408.5	257–258	25,131
				4	544.7	288–290	25
Me	H	H	H	2	300.0	245–250	131,136
				3	450.0	300–302	131,136
				4	600.1	320	131
H	H	Me	H	2	300.0	155–157	131
				3	450.0	250–255	131
H	H	H	Me	2	300.0	155	131,136
				3	450.0	>300	131,136
Me	H	H	Me	2	328.1	235	131
				3	492.1	>300	131
$CHMe_2$	H	H	Me	2	384.1	203	131

6.3.1.2.4. Linear Oligo(esters) of *p*-(β-hydroxy)-Ethoxybenzoic Acid 

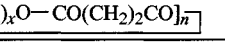
	n	Mol. wt.	m.p. (°C)	Refs.
R = H	1	182.2	177	27
	2	346.3	165	27
	3	510.5	183	27
	4	674.7	192	27
$R = -CH_2CH_2OH$	1	226.2	77	28
	2	390.4	114	28
	3	554.6	136	28
	4	718.7	156	28

6.3.1.2.5. Linear Oligo(esters) of 2,5-Hydroxymethylfuran Carboxylic Acid 
n = 1–5: GPC, HPLC, ^{1}H-NMR; Ref. 67,68.

6.3.1.2.6. Cyclic Oligo(esters) of 2,5-Hydroxymethylfuran Carboxylic Acid 
n = 3–5: GPC, MS, ^{1}H-NMR, ^{13}C-NMR; Ref. 67,68.

6.3.2. OLIGO(ESTERS) OF DIOLS AND DICARBOXYLIC ACIDS

6.3.2.1. Aliphatic Oligo(esters)

6.3.2.1.1. Cyclic Oligo(alkylene succinates) 

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	2	288.3	131		29,30,89,108,116,118
3	1	158.2	82	90–92/3	29,108,118,128
	2	316.3	138		29,108,118,129
4	1	172.2	42	95–96/3	29,108,118,128
	2	344.4	121		29,108,118

6.3.2.1.1. *cont'd*

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
5	1	186.2	19	88–89/1	29,108,118,128
	2	372.4	87		29,108,118
6	1	200.2	−15	108–110/3	29,108,118,128
	2	400.5	110		29,108,118
7	1	214.3	49	116–118/1–3	29,108,118
	2	428.5	86		29,108,118
8	1	228.3	71		29,108,118
	2	456.6	109		29,108,118
10	1	256.3	60	130–132/1	29,108,118,128
	2	512.7	109		15,29,108

6.3.2.1.2. CYCLIC OLIGO(ALKYLENE ADIPATES)

$[OCH_2{-}X{-}CH_2O{-}CO(CH_2)_4CO]_n$

X	*n*	Mol. wt.	m.p. (°C)	Refs.
$C(CH_3)_2$	2	428.5	127–128	47
$(CH_2)_2$	2	400.4	57	106
$(CH_2)_3$	2	428.5	82–84	115
$(CH_2)_8$	n = 1–5 (sharp fractions):		GPC, NMR	156

6.3.2.1.3. CYCLIC OLIGO(ALKYLENE SEBACATES)

$[O(CH_2)_xO{-}CO(CH_2)_8CO]_n$

x	*n*	Mol. wt.	m.p. (°C)	Refs.
2	1	228.3	42	15,29
	2	456.6	81	15,29
3	1	242.3	14	29,31
	2	484.6	113–113.5	15,29,31
8	1	312.5	68	69

6.3.2.1.4. CYCLIC OLIGO(ETHYLENE DICARBOXYLATES)

$[O(CH_2)_2O{-}CO(CH_2)_xCO]_n$

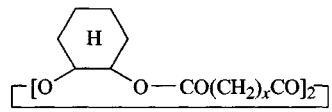
x	*n*	Mol. wt.	m.p. (°C)	Refs.
1	2	260.2	160	109
2	2	288.3	131	29,30,116,118
3	2	316.3	144	106,107
4	2	344.4	56	107
5	2	372.4	147	106,107
6	2	400.5	54	107
7	1	214.3	52	15,29
	2	428.5	147	15,29,106–108
8	1	228.3	42	15,29
	2	456.6	81	15,29,106–108
9	1	242.3	35	15
	2	484.6	143	15
10	1	256.3	18	29
	2	512.7	96	15,29
11	1	270.4	−8	29
	2	540.7	145–146	15,29
12	2	568.8	102–103	15,108

6.3.2.1.5. CYCLIC OLIGO(THIOESTERS) OF DICARBOXYLIC ACIDS

$[S(CH_2)_xS{-}CO(CH_2)_yCO]_n$

x	*y*	*n*	Mol. wt.	m.p. (°C)	Refs.
2	1	2	324.5	144	109
	3	2	380.5	140–145	110,111
	4	2	408.4	97–105	111
	5	1	218.2	75–78	111
	5	2	436.4	125–129	110
3	3	2	408.4	97–102	111
	4	1	218.2	115–117	111
	4	2	436.4	108–110	111

6.3.2.1.6. CYCLIC OLIGO(CYCLOHEXYLENE DICARBOXYLATES)



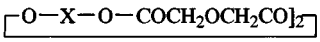
x	Mol. wt.	Config.	m.p. (°C)	Config.	m.p. (°C)	Refs.
1	368.4	*trans*	233–235	*cis*	189–191	112,113
3	424.5		220–222		144–146	112,113
5	480.6		210–212		143–145	112,113
7	536.7		160–162		134–136	112,113
8	564.8		145–148		93–95	112,113

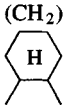
6.3.2.1.7. CYCLIC OLIGO(ESTERS) OF OXALIC ACID

$[OCH_2{-}X{-}CH_2O{-}COCO]_2$

X	Mol. wt.	m.p. (°C)	Refs.
CH_2	260.2	185–187	108,125
CH_2OCH_2	320.2	162.5–164	62,108,115
$(CH_2OCH_2)_2$	408.3	68–72	62,108

6.3.2.1.8. Cyclic Oligo(esters) of Diglycolic Acid

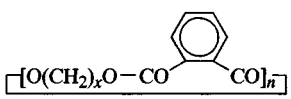


X	Mol. wt.	m.p. (°C)	Refs.
CH_2CH_2	320.3	185–187	114,129
$CH(CH_3)$-$CH(CH_3)$	376.4	128–130	114
$(CH_2)_2O(CH_2)_2$	408.4	125–126	129
 *trans*	428.4	276–278	114
cis	428.4	265–268	114
	416.3	208–209	114

6.3.2.1.9. Cyclic Oligo(tetraethylene glycol succinate)
$n = 1$: X-ray, NMR, MS; $n = 2$–8: (sharp fractions): GPC, NMR, MS, Refs. 160–162.

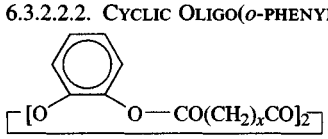
6.3.2.2. Aromatic Oligo(esters)

6.3.2.2.1. Cyclic Oligo(alkylene phthalates)



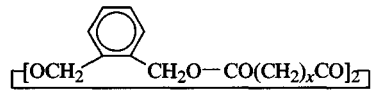
x	n	Mol. wt.	m.p. (°C)	Refs.
2	1	192.2	60–62	118,119
	2	384.3	198	90,108,118–120
3	2	412.4	193	108,120
4	1	220.2	106	108,120
	2	440.5	140	108,120
5	1	234.3	101–102	108,119,120
	2	468.5	139	108,120
6	1	248.3	64–65	108,119,120
	2	496.6	66	108,120
7	2	524.6	163	108,120
8	2	552.7	64–65	108,120
9	2	580.7	140–141	108,120
10	2	608.8	78–79	108,120
11	2	636.8	138–139	108,120
12	2	664.9	84	108,120

6.3.2.2.2. Cyclic Oligo(*o*-phenylene dicarboxylates)



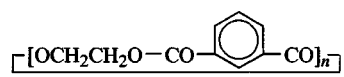
x	Mol. wt.	m.p. (°C)	Refs.
3	412.4	238–240	122
5	468.5	145–147	122
7	524.6	110–112	122
8	552.7	108–110	122

6.3.2.2.3. Cyclic Oligo(*o*-xylylene dicarboxylates)



x	Mol. wt.	m.p. (°C)	Refs.
0	320.3	233–235	117
1	348.4	215–218	117
2	376.5	217	117
3	404.5	173	117
4	432.6	144	117
5	460.6	121	117
6	488.7	134	117
7	516.7	119	117
8	544.8	135	117

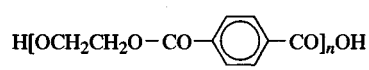
6.3.2.2.4. Cyclic Oligo(ethylene isophthalates)



n	Mol. wt.	m.p. (°C)	Refs.
2	384.4	330	34,90

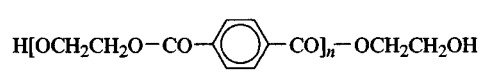
6.3.2.2.5. Oligo(ethylene terephthalates)

6.3.2.2.5.1. Oligomeric Hydroxy Acids



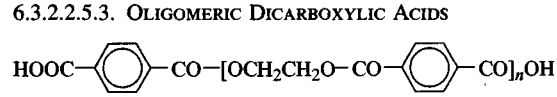
n	Mol. wt.	m.p. (°C)	Refs.
1	210.2	178	35
2	402.4	200–205	36
3	594.6	219–223	36

6.3.2.2.5.2. Oligomeric Diols



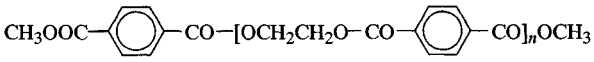
n	Mol. wt.	m.p. (°C)	Refs.
1	254.2	109–110	35,36,59,74,75
2	446.4	173–174	35,36,74,75
3	638.6	200–205	35,36,74,75
4	830.8	213–216	35,36,74
5	1023.0	218–220	36

6.3.2.2.5.3. Oligomeric Dicarboxylic Acids



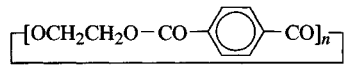
n	Mol. wt.	m.p. (°C)	Refs.
1	358.3	>360	36–38
2	550.5	284–286	33,36–38
3	742.7	274–276	33,37,38
4	934.9	252–255	38
5	1127.1	233–236	38

6.3.2.2.5.4. Oligomeric Dicarboxylates



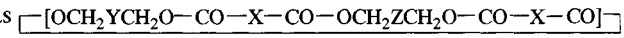
n	Mol. wt.	m.p. (°C)	Refs.
1	386.3	168–170	35,52,74
2	578.5	194–198	35,52,74
3	770.7	215–217	52,74
4	962.8	231–232	52
5	1155.0	242–243	52
6	1347.1	248–249	52
7	1539.3	250–252	52
8	1731.5	251–253	52
9	1923.7	253–254	52
10	2115.8	254–256	52

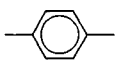
6.3.2.2.5.5. Cyclic Oligo(esters)

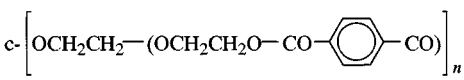


n	Mol. wt.	m.p. (°C)	Refs.
2	384.0	225	32,33,46,57,75,90,159
3	576.5	321	33,39,40,43,44,46,57,58, 60,61,70,75,90,163–165
4	768.7	326	33,40,43,46,57,58,61,75,166
5	960.9	264	33,40,46,57,58,61,75
6	1152.7	306	33,46,58,61,75
7	1344.8	238–240	33,61,75

n = 3–13 (sharp fractions): GPC, MS, NMR, Ref. 158

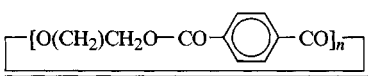
6.3.2.2.6. Cyclic Oligo(esters) of Di- and Tri-Ethylene Glycols 

X	Y	Z	Mol. wt.	m.p. (°C)	Refs.
	–	CH_2OCH_2	428.4	173–179	40,43,57,90
	CH_2OCH_2	CH_2OCH_2	472.4	223	45,90,115
	$(CH_2OCH_2)_2$	$(CH_2OCH_2)_2$	560.5	130	90
	CH_2OCH_2	CH_2OCH_2	472.4	168–170	115
	$(CH_2OCH_2)_2$	$(CH_2OCH_2)_2$	560.5	108–111	115

c- 

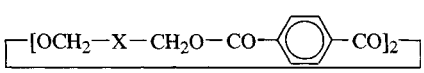
n = 3–7 (sharp fractions): GPC, MS, NMR, Ref. 158

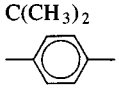
6.3.2.2.7. Cyclic Oligo(esters) of Tetraethylene Glycol



n	Mol. wt.	m.p. (°C)	Refs.
1	324.3	98–99	126
2	648.7	96	90,126,127

6.3.2.2.8. Cyclic Oligo(alkylene terephthalates)

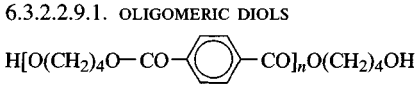


X[a]	n[a]	Mol. wt.	m.p. (°C)	Refs.
(CH_2)		412.4	251	90
$(CH_2)_3$	2	440.4	208	90
$(CH_2)_4$	2	454.5	149	90
$C(CH_3)_2$	2	234.3	284	90
	2	268.3	384	90

[a] X = $(CH_2)_8$, *n* = 2–5 (sharp fractions): GPC, NMR, MS 157.

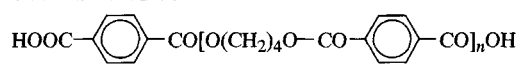
6.3.2.2.9. Oligo(butylene terephthalates)

6.3.2.2.9.1. oligomeric diols



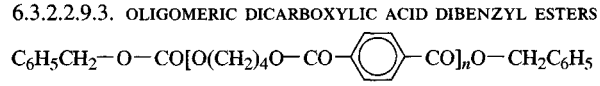
n	Mol. wt.	m.p. (°C)	Refs.
1	310.3	72	96–99
2	530.6	138	43,96–99
3	750.8	175	96–99
5	1191.1	193	96–99
7	1630.5	202	97–99

6.3.2.2.9.2. oligomeric dicarboxylic acids



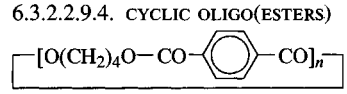
n	Mol. wt.	m.p. (°C)	Refs.
1	386.4	290	43,96–98
2	606.6	266	96–98
3	826.8	254	97,98
4	1047.0	247	97,98
6	1487.5	235	97,98

6.3.2.2.9.3. oligomeric dicarboxylic acid dibenzyl esters



n	Mol. wt.	m.p. (°C)	Refs.
1	566.6	139	96–98
2	786.8	168	96–98
3	1007.1	178	97,98
4	1227.3	187	97,98
6	1667.8	199	97,98
8	2108.2	205	97,98
10	2548.7	208	97,98

6.3.2.2.9.4. cyclic oligo(esters)

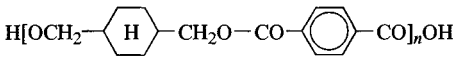


n	Mol. wt.	m.p. (°C)	Refs.
2	440.5	199	54–56,90,96,159
3	660.7	171.5	55,56,96
4	880.9	251	43,55,56,96
5	1101.2	207	96

n = 2–9 (sharp fractions): GPC, MS, X-ray (*n* = 2), Ref. 155

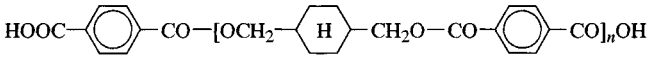
6.3.2.2.10. Oligo(1,4-cyclohexylenedimethylene terephthalates)

6.3.2.2.10.1. oligomeric hydroxy acids



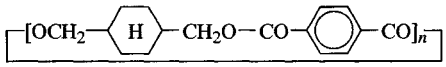
n	Config.	Mol. wt.	m.p. (°C)	Refs.
1	*trans*	292.3	175–178	41

6.3.2.2.10.2. oligomeric dicarboxylic acids



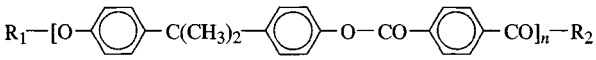
n	Config.	Mol. wt.	m.p. (°C)	Refs.
1	*trans*	440.4	>310	41
1	*cis*	440.4	>300	41

6.3.2.2.10.3. cyclic oligo(esters)

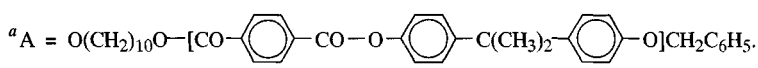


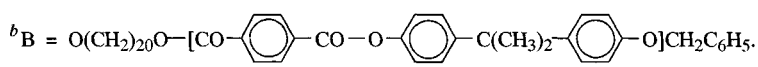
n	Config.	Mol. wt.	m.p. (°C)	Refs.
2	*trans*	548.7	299	90
3	*trans*	823.0	385	41,42,90

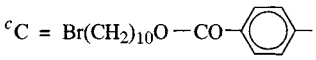
6.1.3.2.2.11. Oligomeric 4,4′-isopropylidenediphenol (bisphenol A) terephthalates



R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
$C_6H_5CH_2$	$O\text{-}C(CH_3)_3$	1	522.6	120	123,124
		2	881.0	197	123,124
		3	1239.4	265	123,124
		4	1597.7	307	123,124
		5	1956.1	–	123,124
H	$O\text{-}C(CH_3)_3$	1	432.5	147	123,124
		2	790.9	160	123,124
		3	1149.2	280	123,124
		4	1513.6	297	123,124
		5	1866.0	314	123,124
$C_6H_5CH_2$	OH	1	466.4	213	123,124
		2	824.9	280	123,124
		3	1183.3	300	123,124
		4	1541.6	305	123,124
		5	1900.0	315	123,124
H	OH	1	376.4	193	123,124
H	OH	2	734.8	275	123,124
		3	913.1	292	123,124
		4	1451.5	305	123,124
$C_6H_5CH_2$	$O(CH_2)_{10}Br$	1	685.7	63	123,124
		2	1044.1	134	123,124
		3	1402.5	232	123,124
	$O(CH_2)_{20}Br$	1	826.0	88	123,124
		2	1184.4	130	123,124
		3	1542.8	236	123,124
	A[a]	1	1071.3	132	123,124
		2	1788.1	170	123,124
	B[b]	1	1211.6	83	123,124
C[c]	$O\text{-}C(CH_3)_3$	1	798.8	117	123,124
		2	1157.5	177	123,124
C[c]	$O(CH_2)_{10}Br$	1	962.8	47	123,124

[a] A = 

[b] B = 

[c] C = 

REFERENCES

1. C. A. Bischoff, P. Walden, Ann. Chem., **279**, 45 (1894); Chem. Ber., **26**, 262 (1893).
2. C. A. Bischoff, P. Walden, Ann. Chem., **279**, 71 (1894); Chem. Ber., **27**, 2939, 2950 (1894).
3. E. Jungfleisch, M. Godchot, Compt. Rend., **141**, 111 (1905); **142**, 637 (1906); **144**, 425 (1908).
4. J. Kleine, K.-H. Kleine, Makromol. Chem., **30**, 23 (1959).
5. R. Dietzel, R. Krug, Chem. Ber., **58**, 1307 (1925).
6. J. Wislicenus, Ann. Chem., **167**, 302, 319 (1873).
7. R. Eder. F. Kutter, Helv. Chim. Acta, **9**, 557 (1926).
8. F. J. van Natta, J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **56**, 455 (1934).
9. M. Stoll, A. Rouvé, Helv. Chim. Acta, **18**, 1087 (1935); **17**, 1283 (1934).
10. R. Huisgen, H. Ott, Tetrahedron, **6**, 253 (1959).
11. H. Hundsdiecker, H. Erlbach, Chem. Ber., **80**, 129 (1947).
12. E. W. Spanagel, W. H. Carothers, J. Am. Chem. Soc., **58**, 654 (1936).
13. W. H. Lycan, R. Adams, J. Am. Chem. Soc., **51**, 3450 (1929).
14. W. H. Carothers, F. J. van Natta, J. Am. Chem. Soc., **55**, 4714 (1933).
15. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **55**, 5031 (1933).
16. L. Ruzicka, M. Stoll, Helv. Chim. Acta, **11**, 1159 (1928).
17. W. Baker, W. D. Ollis, T. S. Zealley, J. Chem. Soc., 201 (1951).
18. A. Einhorn, Chem. Ber., **44**, 431, 437 (1911).
19. G. Schroeter, Chem. Ber., **52**, 2224 (1919).
20. R. Anschuetz, Ann. Chem., **439**, 8 (1924).
21. A. Einhorn, H. Pfeiffer, Chem. Ber., **34**, 2952 (1901); A. Einhorn, C. Mettler, Chem. Ber., **35**, 3644 (1902).
22. W. Baker, B. Gilbert, W. D. Ollis, T. S. Zealley, J. Chem. Soc., 209 (1951).
23. L. Anschuetz, G. Gross, Chem. Ber., **77**, 644 (1944).
24. W. Baker, B. Gilbert, W. D. Ollis, J. Chem. Soc., 1443 (1952).
25. W. Baker, A. S. El-Nawary, W. D. Ollis, J. Chem. Soc., 3163 (1952).
26. R. Anschuetz, E. Rhodius, Chem. Ber., **47**, 2733 (1914).
27. M. Ishibashi, M, Hirai, Chem. High Polym. (Tokyo), **21**, 231 (1964); Chem. Abstr., **62**, 5342f (1965).
28. M. Ishibashi, M. Hirai, Chem. High Polym. (Tokyo), **21**, 235 (1964); Chem. Abstr., **62**, 5342g (1965).
29. E. W. Spanagel, W. H. Carothers, J. Am. Chem. Soc., **57**, 929 (1935).
30. W. H. Carothers, G. L. Dorough, J. Am. Chem. Soc., **52**, 711 (1930).
31. M. Stoll, A. Rouvé, Helv. Chim. Acta, **19**, 253 (1936).
32. H. Repin, E. Papanikolau, J. Polym. Sci. A-1, **7**, 3426 (1969).
33. H. Zahn, H. Repin, Chem. Ber., **103**, 3041 (1970).
34. C. E. Berr, J. Polym. Sci., **40**, 591 (1955).
35. H. Zahn, R. Krzikalla, Makromol. Chem., **23**, 31 (1957).
36. H. Zahn, C. Borstlap, G. Valk, Makromol. Chem., **64**, 18 (1963).
37. H. Zahn, B. Seidel, Makromol. Chem., **29**, 70 (1959).
38. H. Zahn, B. Gleitsmann, Angew. Chem., **75**, 772 (1963).
39. S. D. Ross, E. R. Coburn, W. A. Leach, W. B. Robinson, J. Polym. Sci., **13**, 406 (1954).
40. J. Goodman, B. F. Nesbitt, Polymer, **1**, 384 (1960); J. Polym. Sci., **48**, 423 (1960).
41. H. Zahn, G. Valk, Makromol. Chem., **64**, 37 (1963).
42. H. Zahn, G. Valk, Polym. Lett., **1**, 105 (1963).
43. E. Meraskentis, H. Zahn, J. Polym. Sci. A-1, **4**, 1890 (1966); Chem. Ber., **103**, 3034 (1970).
44. G. L. Binns, J. S. Frost, F. S. Smith, E. C. Yeadon, Polymer, **7**, 583 (1966).
45. T. Morimoto, Y. Fujita, A. Takeda, J. Polym. Sci. A-1, **6**, 1044 (1968).
46. L. H. Peebles, Jr., M. W. Huffman, C. T. Ablett, J. Polym. Sci. A-1, **7**, 479 (1969).
47. I. S. Megna, A. Koroscil, J. Polym. Sci. A-1, **7**, 1371 (1969).
48. M. Goodman, M. D'Alagni, J. Polym. Sci. B, **5**, 515 (1967).
49. Y. Iwakura, K. Hayashi, K. Iwata, S. Matsuo, Makromol. Chem., **108**, 300 (1967); **110**, 90 (1967).
50. Y. Iwakura, K. Hayashi, K. Iwata, S. Matsuo, Makromol. Chem., **123**, 245 (1969).
51. Y. Iwakura, K. Iwata, S. Matsuo, A. Tohara, Makromol. Chem., **146**, 33 (1971).
52. R. Penisson, H. Zahn, Makromol. Chem., **133**, 25 (1970).
53. S. Matsuo, Y. Iwakura, Makromol. Chem., **152**, 203 (1972).
54. K. Burzin, P.-J Frenzel, Angew. Makromol. Chem., **71**, 61 (1978); K. Burzin, W. Holtrup, R. Feinauer, Angew. Makromol. Chem., **74**, 93 (1978).
55. J. Steinke, H. Wald, Melliand Textilber., **58**, 494 (1977).
56. G. C. East, A. M. Girshab, Polymer, **23**, 323 (1982).
57. S. Shiono, J. Polym. Sci., Polym. Chem. Ed., **17**, 4123 (1979).
58. H. Zahn, P. Kusch, Z. Ges. Textilind. **69**, 880 (1967).
59. K. Tomita, Polymer, **14**, 50 (1973).
60. R. Giuffria, J. Polym. Sci., **49**, 427 (1961).
61. D. R. Cooper, J. A. Semlyen, Polymer, **14**, 185 (1973).
62. P. E. Fore, J. S. Bradshaw, S. F. Nielsen, J. Heterocycl. Chem., **15**, 269 (1978).
63. N. Grassie, E. J. Murray, P. A. Holmes, Polym. Deg. Stab., **6**, 47 95, 127 (1984).
64. R. H. Marchessault, S. Coulombe, H. Morikawa, K. Okamura, J. F. Revol, Can. J. Chem., **59**, 38 (1981).
65. K. Ito, Y. Hashizuka, Y. Yamashita, Macromolecules, **10**, 821 (1977).
66. K. Ito, Y. Yamashita, Macromolecules, **11**, 68 (1978).
67. H. Hirai, K. Naito, T. Hamasaki, M. Goto, H. Koinuma, Makromol. Chem., **185**, 2347 (1984).
68. H. Hirai, J. Macromol. Sci.-Chem. A, **21**, 1165 (1984).,
69. J. Dale, J. Chem. Soc., 72 (1965).
70. F. L. Hamb, L. C. Trent, J. Polym. Sci. B, **5**, 1057 (1967).
71. R. H. Wiley, Macromolecules, **4**, 254 (1971).
72. K. Ito, M. Tomida, Y. Yamashita, Polym. Bull., **1**, 569 (1979).
73. B. Hauttecoeur, M. Jolivet, R. Gavard, Compt. Rend., **280**, 2801 (1975).

74. G. Heidemann, P. Kusch, H.-J. Nettelbeck, Z. Anal. Chem., **212**, 401 (1965).
75. W. R. Hudgins, K. Theurer, T. Mariani, J. Appl. Polym. Sci., Appl. Polym. Symp., **34**, 145 (1978).
76. M. Okada, H. Sumitomo, J. Tajima, Macromolecules, **14**, 1180 (1981).
77. W. H. Kruizinga, R. M. Kellogg, J. Chem. Soc., Chem. Commun., 286 (1979).
78. J. Tajima, M. Okada, H. Sumitomo, Makromol. Chem., Rapid Commun., **1**, 197 (1980).
79. M. Okada, H. Sumitomo, J. Tajima, Macromolecules, **10**, 505 (1977).
80. M. Okada, H. Sumitomo, J. Tajima, Polym. Bull., **1**, 41 (1978).
81. M. Okada, H. Sumitomo, J. Tajima, J. Am. Chem. Soc., **101**, 4013 (1979).
82. J. Tajima, M. Okada, H. Sumitomo, J. Am. Chem. Soc., **103**, 4096 (1981).
83. M. Okada, H. Sumitomo, M. Atsumi, Makromol. Chem., Rapid Commun., **4**, 253 (1983).
84. J. Tanaka, J. Tajima, Y. Hayakawa, M. Okada, M. Bitoh, T. Ashida, H. Sumitomo, J. Am. Chem. Soc., **102**, 7873 (1980).
85. M. Imaeda, J. Tanaka, T. Ashida, J. Tajima, M. Okada, H. Sumitomo, Polym. J., **14**, 197 (1982).
86. J. Sakuragi, J. Tanaka, T. Ashida, J. Tajima, M. Okada, H. Sumitomo, J. Am. Chem. Soc., **104**, 6035 (1982).
87. M. Okada, H. Sumitomo, M. Atsumi, J. Am. Chem. Soc., **106**, 2101 (1984).
88. D. A. Jaeger, J. T. Ippoliti, J. Org. Chem., **46**, 4964 (1981).
89. D. Vorländer, Ann. Chem., **280**, 167 (1894).
90. G. Wick, H. Zeitler, Angew. Makromol. Chem., **112**, 59 (1983).
91. E. J. Corey, K. C. Nicolaou, J. Am. Chem. Soc., **96**, 5614 (1974).
92. J. B. Rashkov, J. Gitsov, J. M. Panayotov, J.-P. Pascault, J. Polym. Sci., Polym. Chem. Ed., **21**, 923 (1983).
93. N. E. Manolova, J. Gitsov, R. S. Velichkova, J. B. Rashkov, Polym. Bull., **13**, 285 (1985).
94. S. Sosnowski, S. Slomkowski, S. Penczek, Makromol. Chem., **184**, 2159 (1983).
95. J. S. Megna, A. Koroscil, Polym. Lett., **6**, 653 (1968).
96. F.-J. Müller, P. Kusch, J. Windeln, H. Zahn, Makromol. Chem., **184**, 2487 (1983).
97. H.-W. Hässlin, M. Dröscher, G. Wegner, Makromol. Chem., **179**, 1373 (1978).
98. H.-W. Hässlin, M. Dröscher, Makromol. Chem., **181**, 235 (1980); Polymer Bull., **2**, 769 (1980).
99. P. C. Gillette, S. D. Dirlikov, J. L. Koenig, J. B. Lando, Polymer, **23**, 1759 (1982).
100. F. Andreas, R. Sowada, J. Scholz, J. Prakt. Chem., **18**, 141 (1962).
101. D. K. Gilding, A. M. Reed, Polymer, **20**, 1459 (1979).
102. J. Dale, O. Sevaldsen, K. Tittlestad, Acta Chem. Scand. B, **32**, 306 (1978).
103. Y. Iwakura, K. Iwata, S. Matsuo, A. Tohara, Makromol. Chem., **146**, 21 (1971).
104. A. Shanzer, J. Libman, F. Frolow, J. Am. Chem. Soc., **103**, 7339 (1981).
105. R. R. Wetstone, S. A. Ballard, J. Am. Chem. Soc., **73**, 5280 (1951).
106. A. Shanzer, N. Mayer-Shochet, J. Chem. Soc., Chem. Commun., 176 (1980).
107. A. Shanzer, N. Mayer-Shochet, F. Frolow, D. Rabinovich, J. Org. Chem., **46**, 4662 (1981).
108. J. S. Bradshaw, G. E. Maas, R. M. Izatt, J. J. Christensen, Chem. Rev., **79**, 37 (1979).
109. P. Margaretha, F. P. Schmook, H. Budzikiewicz, O. E. Polansky, Monatsh. Chem., **99**, 2539 (1968).
110. A. Shanzer, E. Schwartz, Tetrahedron Lett., 5019 (1979).
111. A. Shanzer, J. Libmann, Synthesis, 140, (1984).
112. C. Anchisi, L. Corda, A. M. Fadda, A. Maccioni, G. Podda, J. Heterocycl. Chem., **21**, 577 (1984).
113. G. Podda, A. Maccioni, C. Anchisi, S. Daolio, B. Pelli, P. Traldi, J. Heterocycl. Chem., **21**, 491 (1984).
114. G. Podda, C. Anchisi, L. Corda, A. M. Fadda, A. Maccioni, J. Heterocycl. Chem., **21**, 1789 (1984).
115. A. Ninagana, T. Maeda, H. Matsuda, Chem. Lett., 1985 (1984).
116. R. Asay, J. S. Bradshaw, S. F. Nielsen, M. D. Thompson, J. W. Snow, D. R. K. Masilidas, R. M. Izatt, J. J. Christensen, J. Heterocycl. Chem., **14**, 85 (1977).
117. S. E. Drewes, B. G. Riphagen, J. Chem. Soc., Perkin Trans. I, 323 (1974).
118. E. W. Spanagel, United States Patent 2.092.031 (1937), Chem. Abstr., **31**, 7443 (1937); French Patent 796.410 (1936), Chem. Abstr., **30**, 6138 (1936).
119. W. A. Ehrhart, J. Org. Chem., **33**, 2930 (1968).
120. S. E. Drewes, P. C. Coleman, J. Chem. Soc., Perkin Trans. I, 2148, (1972); 2578 (1974).
121. G. R. Brown, J. H. P. Tyman, Chem. Ind., 436 (1970).
122. A. Maccioni, A. Plumitallo, G. Podda, J. Heterocycl. Chem., **20**, 1397 (1983).
123. H. Seliger, M. B. Bitar, H. Nguyen-Trong, A. Marx, R. Roberts, J.-K Krueger, H.-G. Unruh, Makromol. Chem., **185**, 1335 (1984).
124. J.-K. Krueger, A. Marx, R. Roberts, H.-G. Unruh, M. B. Bitar, H. Nguyen-Trong, H. Seliger, Makromol. Chem., **185**, 1469 (1984).
125. W. H. Carothers, J. A. Arvin, G. L. Dorough, J. Am. Chem. Soc., **52**, 3292 (1930).
126. K. Frensch, F. Voegtle, Tetrahedron Lett., 2573 (1977); J. Org. Chem., **44**, 884 (1979).
127. J. D. Bradshaw, M. D. Thompson, J. Org. Chem., **43**, 2456 (1978).
128. A. Bachrach, A. Zilkha, Eur. Polym. J., **18**, 421 (1982).
129. A. V. Bogatskii, N. G. Lukyanenko, V. A. Shapkin, M. S. Salakhov, M. U. Mamina, D. Taubert, Zh. Org. Khim., **16**, 2057 (1980); J. Org. Chem. USSR, **16**, 1750 (1980).
130. J. C. Reepmeyer, J. Pharm. Sci., **72**, 322 (1983).
131. G. B. Guise, W. D. Ollis, J. A. Peacock, J. S. Stephanatou, J. F. Stoddart, Tetrahedron Lett., 4203 (1980) J. Chem. Soc., perkin Trans. I, 1637 (1982).
132. R. Anschütz, Ber. Dtsch. Chem. Ges., **52**, 1875 (1919); R. Anschütz, K. Riepenkröger, Ann. Chem., **439**, 1 (1924).

133. A. P. Downing, W. D. Ollis, I. O. Sutherland, J. Mason, S. F. Mason, J. Chem. Soc., Chem. Commun., 329 (1968); W. D. Ollis, I. O. Sutherland, J. Chem. Soc., Chem. Commun., 402 (1966).
134. A. C. D. Newman, H. M. Powell, J. Chem. Soc., 3747 (1952); D. Lawton, H. M. Powell, J. Chem. Soc., 2339 (1958).
135. P. G. Edgerley, L. E. Sutton, J. Chem. Soc., 1069 (1951).
136. E. Gil, A. Quick, D. J. Williams, Tetrahedron Lett., 4207 (1980); D. J. Williams, D. Lawton, Tetrahedron Lett., 111 (1975).
137. W. Baker, J. B. Harborne, A. J. Price, A. Rutt, J. Chem. Soc., 2042 (1954).
138. A. P. Downing, W. D. Ollis, I. O. Sutherland, J. Chem. Soc. B, 24 (1970).
139. W. D. Ollis, J. S. Stephanatou, J. F. Stoddart, J. Chem. Soc., Perkin Trans. I, 1629 (1982); W. D. Ollis, J. F. Stoddart, J. Chem. Soc., Chem. Commun., 571 (1973).
140. A. P. Downing, W. D. Ollis, I. O. Sutherland, J. Chem. Soc., Chem. Commun., 171 (1967).
141. V. D. Gaitonde, B. D. Hosangadi, Indian J. Chem., **14B**, 97 (1976).
142. C. M. Farias, B. D. Hosangadi, Indian J. Chem., **15B**, 997 (1977).
143. J.-M. Medard, N. Rodier, P. Reynaud, J. D. Brion, N. T. Xuong, Acta Cryst. C, **39**, 1136 (1983).
144. U. D. Lengweiler, M. G. Fritz, D. Seebach, Helv. Chim. Acta, **79**, 670 (1996).
145. D. Seebach, H. M. Bürger, D. A. Plattner, R. Nesper, T. Fässler, Helv. Chim. Acta, **76**, 2581 (1993).
146. D. Seebach, T. Hoffmann, F. N. M. Kühnle, J. N. Kinkel, M. Schulte, Helv. Chim. Acta, **78**, 1525 (1995).
147. D. Seebach, T. Hoffmann, F. N. M. Kühnle, U. D. Lengweiler, Helv. Chim. Acta, **77**, 2007 (1994).
148. D. Seebach, H.-M. Müller, H. M. Bürger, D. A. Platner, Angew. Chem., **104**, 443 (1992); Angew. Chem., Int. Ed. Engl., **31**, 435 (1992).
149. T. Tanio, T. Fukui, Y. Shirakura, T. Saito, K. Tomita, T. Kaiho, S. Masamune, Eur. J. Biochem., **124**, 71 (1982).
150. I. Olsen, J. M. Merrick, I. J. Goldstein, Biochem., **4**, 453 (1965).
151. D. A. Plattner, A. Brunner, M. Dobler, H.-M. Müller, W. Petter, P. Zbinden, D. Seebach, Helv. Chim. Acta, **76**, 2004 (1993).
152. D. Seebach, U. Brändli, P. Schnurrenberger, M. Przybylski, Helv. Chim. Acta, **71**, 155 (1988).
153. B. R. Wood, J. A. Semlyen, P. Hodge, Polymer, **38**, 2287 (1997).
154. B. R. Wood, P. Hodge, J. A. Semlyen, Polymer, **34**, 3052 (1993).
155. J. J. L. Briant, J. A. Semlyen, Polymer, **38**, 4531 (1997).
156. B. R. Wood, J. A. Semlyen, P. Hodge, Polymer, **38**, 191 (1997).
157. S. C. Hamilton, J. A. Semlyen, Polymer, **38**, 1685 (1997).
158. J. J. L. Bryant, J. A. Semlyen, Polymer, **38**, 2475 (1997).
159. Y. Kitano, A. Ishitani, T. Ashida, Polymer J., **23**, 949 (1991).
160. P. Singh, M. Kumar, H. Singh, Indian J. Chem. B, **26**, 64 (1987); J. Chem. Res. (S), 94 (1989) (M), 0675 (1989).
161. T. Ogawa, A Yoshikawa, H. Wada, C. Ogawa, N. Ono, H. Suzuki, J. Chem. Soc., Chem. Commun., **14**, 1407 (1995).
162. J. S. Bradshaw, C. T. Bishop, S. F. Nielson, R. E. Asay, D. R. K. Masihdas, E. D. Flanders, L. D. Hansen, R. M. Izatt, J. J. Christensen, J. Chem. Soc., Perkin Trans. I, 2505, (1976).
163. Y. Kitano, T. Ashida, Polym. J., **24**, 1099 (1992).
164. D. II Yoo, S. W. Seo, W. S. Ha, Polym. Bull., **28**, 593 (1992).
165. J. Hasek, J. Jecny, V. Langer, K. Huml, P. Sedlácek, Acta Cryst. B, **36**, 2698 (1980); **38**, 2710 (1982).
166. T. Yu, H. Bu, Y. Jin, Makromol. Chem., **187**, 2461 (1986).

6.4. OLIGO(URETHANES)

6.4.1. OLIGO(ALKYLENE URETHANES)

6.4.1.1. Oligo(ethylene urethanes) R–[$OCH_2CH_2NHCO]_n$–OCH_3

R	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_3CO	2	248.2	69–70		15
	3	335.3	98–99		15
	4	422.4	138.5–139.5		15
	5	509.5	164–165		15
	6	596.6	174–176		15
	7	683.6	185–186		15
H	1	119.1	~0	116–117/3	15
	2	206.2	62–63		15
	3	293.3	101.5–102.5		15
	4	380.4	129–130		15
	5	467.4	152–153		15
	6	554.5	168.5–169.5		

6.4.1.2. Oligo[(3-methyl)ethylene urethanes] R–[$OCH_2CH(CH_3)NHCO]_n$–OCH_3

R	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	$[\alpha]_D$	*c* (%), EtOH	Refs.
CH_3CO	1	175.2		109–110/4	−23.2	1.39	11
	2	276.3	93–98		−30.2	0.32	11
	3	377.4	113–117		−31.8	0.77	11
	4	478.5	143–147		−30.9	1.14	11
	5	579.6	153–154		−30.6	0.42	11
	6	680.7	167				11
	7	781.8	173				11
H	1	133.2		109–110/3	−15.8	2.8	11,16
	2	234.3	60–64				11
	3	335.4	86–87				11
	4	436.5	120–122				11
	5	537.6	135–137				11
	6	638.7	152–153				11

6.4.1.3. Oligo[(3-isopropyl)ethylene urethanes] $CH_3CO\left[OCH_2CH(CH(CH_3)_2)NHCO\right]_n-OCH_3$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	$[\alpha]_D$	*c* (%), EtOH	Refs.
1	203.2		117.5–118/1	−22.2	0.79	10
2	332.4	111.5–112.5		−33.5	2.0	10
3	461.6	142–143		−28.4	2.1	10
4	590.7	156.5–158		−4.2	1.9	10

6.4.1.4. Oligo[(3-isobutyl)ethylene urethanes] $R\left[OCH_2CH(CH_2CH(CH_3)_2)NHCO\right]_n-OCH_3$

R	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	$[\alpha]_D^{25}$	*c* (%), EtOH	Refs.
CH_3CO	1	217.3		128–129/1			9
	2	360.5	73–74		−41.7	2.0	9,14
	3	503.6	97–98		−42.9	1.0	9,14
	4	646.8	136–138		−41.5	1.4	9,14
	5	790.0	149–150		−38.5	2.0	9,13
	6	933.2	162–164		−35.9	1.0	9,13
	7	1076.4	168–169		−30.5	2.2	9,13
	8	1219.5	180–181				12
	9	1362.7	185.5–186.5				12
H	1	175.3		119–121/1	−34.1	2.6	14
	2			Oil	−47.2	2.3	14
	3			Oil	−45.0	0.99	14

6.4.1.5. Oligo [(3-benzyl)ethylene urethanes] $R\left[OCH_2CH(CH_2C_6H_5)NHCO\right]_n-OCH_3$

R	*n*	Mol. wt.	m.p. (°C)	$[\alpha]_D$	*c* (%), EtOH	Refs.
CH_3CO	2	428.5	104.5–106.5	−17.9	1.2	11,14
	3	605.7	138–139	−18.1	0.71	11,14
	4	782.9	149–150			11
H	1	209.2	51.5–53	−25.9	0.72	11,14
	2	386.4	130–131	−27.8	0.77	11,14
	3	563.6	153–154			11

6.4.1.6. Oligo(trimethylene urethanes)

R	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3CO	2	276.3	57.5–58.5	18
	3	377.4	95–97	18
	4	478.5	118.5–120	18
	5	579.6	133.5–135	18
H	1	133.2	Oil	18
	2	234.3	Oil	18
	3	335.4	76.5–79	18
	4	436.5	106.5–108.5	18
	5	537.6	126–128	18

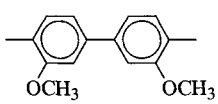
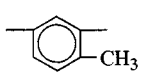
6.4.1.7. Oligo(pentamethylene urethanes)
$R-[O(CH_2)_5NHCO]_n-OCH_3$

R	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_3CO	2	332.4	63.7		1,18–21
	3	461.6	98.7		1,18–21
	4	590.7	109.2		1,18–21
	5	719.9	124.7		1,18–21
	6	849.0	131.2		1,18–21
	7	978.2	138.2		1,18–21
	8	1107.4	143.2		1,18–21
	9	1236.6	150.2		1,18–21
	10	1365.7	155.2		1,18–21
H	1	161.2		131–134/0.3	1,18–20
	2	290.4	75.2		1,18–20
	3	419.5	100.2		1,18–20
	4	548.7	112.2		1,18–20
	5	677.8	116.2		1,18–20
	6	807.0	125.2		1,18–20
	7	936.2	135.2		1,18–20
	8	1065.3	141.2		1,18–20

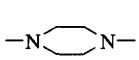
6.4.2. OLIGO(URETHANES) OF DIISOCYANATES AND GLYCOLS

6.4.2.1. Oligo(urethanes) of Diisocyanates and 1,4-Butanediol

6.4.2.1.1. Diol Oligo(urethanes)
$H[O(CH_2)_4O-CO-NH-X-NH-CO]_n-O(CH_2)_4OH$

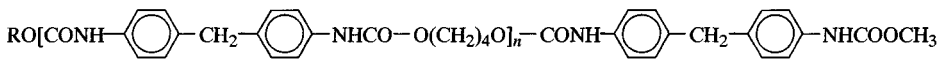
X	*n*	Mol. wt.	m.p. (°C)	Refs.
$(CH_2)_6$	1	348.4	103–105	2,3,5,6
	2	606.8	146	2,3,5
	3	865.1	162–163	2,3,5
	4	1123.4	169–170	2,3
	5	1381.7	171–173	2,3
	6	1640.0	173–174	2,3
	7	1898.3	175–176	2,3
	9	2414.9	177–179	2,3
	1	476.5	133.5	2,3
	2	862.9	127–128	2,3
	3	1249.4	164–166	2,3
	7	2795.0	190–195	2,3
	15	5886.3	210–215	2,3
	1	404.4	115	22
	2	710.7	122	22
	3	1021.0	120	22
	4	1331.3	136	22
	5	1641.6	133	22

6.4.2.1.2. Diamine Oligo(urethanes)
$H[NR-X-NR-CO-O(CH_2)_4O-CO]_n-NR-X-NHR$

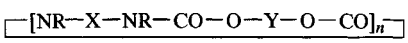
X	R	*n*	Mol. wt.	m.p. (°C)	Refs.
$(CH_2)_6$[a]	H	1	536.4	228	5
		2	794.7	226	5
		3	1053.0	218	5
		4	1311.4	212	5
		1	314.4	97–98	23
		2	542.6	126.5	23
		3	770.9	147–148	23
		4	999.2	169	23

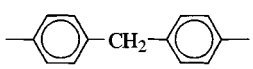
[a] Dihydrobromides.

6.4.2.1.3. Alkoxyacylamine Oligo(urethanes)



R	n	Mol. wt.	m.p. (°C)	Refs.
CH_3	0	314.3	181.7	24
	1	654.7	185.2	24
	2	995.1	201.6	24
	3	1335.5	176.9	24
C_2H_5	0	328.3	125.3	24
	1	668.7	187.6	24
	2	1009.1	208	24
	3	1449.5	182.5	24
C_8H_{17}	0	412.5	114	24
	1	752.9	183.3	24
	2	1093.3	208.2	24
	3	1433.7	185	24
$C_{18}H_{37}$	0	552.7	123.8	24
	1	893.1	166.2	24
	2	1233.5	192.4	24
	3	1573.9	199	24

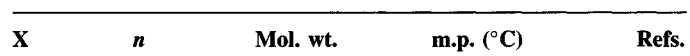
6.4.2.1.4. Cyclic Oligo(urethanes) 

R	X	Y	n	Mol. wt.	m.p. (°C)	Refs.
H	$(CH_2)_6$	$(CH_2)_4$	1	258.3	164	4,6,7
			2	516.6	198	6,8
CH_3		$CH_2C(CH_3)_2CH_2$	2	764.9	254–255	25

6.4.2.2. Oligo(urethanes) of Hexamethylene Diisocyanate and Diglycols

6.4.2.2.1. Diol Oligo(urethanes)
$HO[(CH_2)_2X(CH_2)_2O\text{-}CO\text{-}NH(CH_2)_6NH\text{-}CO\text{-}O]_n\text{-}(CH_2)_2X(CH_2)_2OH$

X	n	Mol. wt.	m.p. (°C)	Refs.
O	1	380.4	68–69	2,3
	2	654.8	103–104	2
	3	929.1	123–125	2,3
	4	1203.4	120–123	2
	5	1478.7	123–124	2,3
	7	2026.3	122–124	2,3
	15	4221.0	119–123	2,3
S	1	412.6	105–106	2
	3	993.4	132–134	2
	7	2155.5	133–135	2

6.4.2.2.2. Cyclic Oligo(urethanes)


X	n	Mol. wt.	m.p. (°C)	Refs.
O	1	274.3	138	4,17
	2	548.6	170	17
	3	722.9	131–133	17
	4	1097.2	151–153	17
	5	1371.5	136	17
	6	1645.8	145–148	17
	7	1920.1	134	17
S	1	290.4	128	4

REFERENCES

1. Y. Iwakura, K. Hayashi, K. Iwata, Makromol. Chem., **89**, 214 (1965).
2. W. Kern, H. Kalsch, K. J. Rauterkus, H. Sutter, Makromol. Chem., **44–46**, 78 (1961).
3. W. Kern, Angew. Chem., **71**, 585 (1959).
4. W. Kern, K. J. Rauterkus, W. Weber, Makromol. Chem., **43**, 98 (1961).
5. H. Zahn, M. Dominik, Makromol. Chem., **44–46**, 290 (1961).
6. H. Zahn, M. Dominik, Chem. Ber., **94**, 125 (1961).
7. O. Bayer, Ann. Chem., **549**, 286 (1941); Angew. Chem., **59A**, 257 (1947).
8. W. Kern, K. J. Rauterkus, W. Weber, W. Heitz, Makromol. Chem., **57**, 241 (1962).
9. K. Iwata, Y. Iwakura, K. Hayashi, Makromol. Chem., **112**, 242 (1968).
10. K. Iwata, Y. Iwakura, K. Hayashi, Makromol. Chem., **116**, 250 (1968).
11. K. Iwata, Y. Iwakura, Makromol. Chem., **135**, 165 (1970).

12. K. Iwata, Y. Iwakura, Makromol. Chem., **134**, 321 (1970).
13. Y. Iwakura, K. Hayashi, K. Iwata, Makromol. Chem., **108**, 296 (1967).
14. Y. Iwakura, K. Hayashi, K. Iwata, Makromol. Chem., **93**, 274 (1966).
15. Y. Iwakura, K. Hayashi, K. Iwata, Makromol. Chem., **95**, 217 (1966).
16. Y. Iwakura, K. Hayashi, K. Iwata, Makromol. Chem., **104**, 46 (1967).
17. W. Heitz, H. Hoecker, W. Kern, H. Ullner, Makromol. Chem., **150**, 73 (1971).
18. J. N. Hay. Makromol. Chem., **178**, 1601 (1977).
19. H. Tagawa, E. Wada, K. Iwata, K. Hayashi, Y. Iwakura, Rep. Prog. Polym. Phys. Japan., **10**, 211 (1967).
20. H. Tagawa, E. Wada, T. Sakurai, Makromol. Chem., **154**, 215 (1972).
21. M. Nagura, H. Tagawa, E. Wada, Makromol. Chem., **165**, 325 (1973).
22. B. Fu, C. Feger, W. J. MacKnight, N. S. Schneider, Polymer, **20**, 889 (1985).
23. L. L. Harrell, Jr., Macromolecules, **2**, 607 (1969).
24. Y. Camberlin, J. P. Pascault, J. M. Letoffe, P. Claudy, J. Polym. Sci., Polym. Chem. Ed., **20**, 383 (1982).
25. S. Foti, P. Maravigna, G. Montaudo, Macromolecules, **15**, 883 (1982).

TABLE 7. OLIGO(SULFIDES) AND OLIGO(SELENIDES)

7.1. CYCLIC OLIGO(THIOMETHYLENES) AND OLIGO(SELENOMETHYLENES) $\sqsubset[XCH_2]_n\sqsupset$

X	*n*	Mol. wt.	m.p. (°C)	Refs.
S	2	92.2	105–106	11
	3	138.3	215–218	1–7,15,16,20
	4	184.4	49–50	1,5,6,8,13,18,20
	5	230.5	123–125	6,9,13,14,19,20
S	6	276.5	227–229	12
Se	3	279.0	226–228	1,17
	4	372.0	80–81	1

7.2. CYCLIC OLIGO(THIOALKYLENES) $\sqsubset[S(CH_2)_x]_n\sqsupset$

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
2	2	120.2	112–113	199–200/1013	43–45
	3	180.4	84		22,24,27
	4	240.5	224–225		23,25,26,30
	5	300.6	97.5–99		25
	6	360.7	91–93		25,28,29
3	2	148.3		86–87/1	25,29,42
	3	222.4	87–88		23,30
	4	296.6	57.5–59		25,29,30,42
	6	444.9	29–30		25
4	2	176.3	94–95.5		25
4	4	352.7	31–32		25
	6	529.0	67–70		25
5	2	204.3	81–82.5		25,26
	4	408.6	33–33.5		25
	6	612.8	36.5–38		25
6	2	232.5	82		21,25
	4	464.9	30–32		25
	6	697.4	56–59.5		25
10	2	344.7	48–49		21,26
16	2	513.0	75–77		26

7.3. SUBSTITUTED CYCLIC OLIGO(THIOETHYLENES) $\sqsubset[S-CR_1R_2-CHR_3]_n\sqsupset$

R_1	R_2	R_3	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	H	H	4	296.6		32,33
			5	370.7	Oil	32,33
C_2H_5			4	352.7		33
$(CH_3)_2CH$			4	408.8		33
C_6H_5			2	272.4		33,34
CH_3	CH_3	H	2	176.3		33
CH_3	H	CH_3	4	352.6	155–180 (NMR)	33,35
H			2	280.5		33

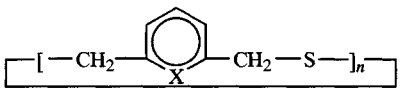
7.4. CYCLIC CO-OLIGOMERS OF FORMALDEHYDE AND THIOFORMALDEHYDE $\sqsubset[OCH_2-(SCH_2)_x]\sqsupset$

x	Mol. wt.	m.p. (°C)	Refs.
4	214.4	106–108	6,10
5	260.5	163–165	6,10

7.5. CYCLIC OLIGO(ETHER SULFIDES) $\square[(SCH_2CH_2)_x-(OCH_2CH_2)_y]_n\square$

x	y	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1	1	1	104.2	−17	147/1007	45,46
	1	2	208.3	22.5–23		26,36
	2	1	148.2		40–48/0.2	30,31
	2	2	296.5	90–91		26,31,36
	3	1	192.3		90–95/1	31,36
	3	2	384.6	Liquid		26,36
	4	1	236.3		123–124/0.1	30,31
	5	1	280.4		164–170/0.1	30,31
2	1	1	164.3		62–64/0.1	30,31
	1	2	328.6	50		28,29
	1	3	492.9	123		28
	2	1	208.3	20–24	134–136/0.1	31
	3	1	252.4	51–52	34–36/0.1	29–31
	4	1	296.4	54–56		30,31
3	1	1	224.4	89–90		30,31
	2	1	268.5	43–44		30,31
	3	1	312.5		200–206/3	30,31
	4	1	356.6	Liquid		30,31
4	1	1	284.5	93–95		30,31
	2	1	328.6	125		31

7.6. THIACYCLOPHANES



X	n	Mol. wt.	m.p. (°C)	Refs.
CH	2	272.4	122–123	30,37,39
	3	408.7	124–125	37,39
	4	544.9	161–162	39
N	2	274.4	220–222	30,37,38,41
	3	411.6	185–188	30,37,40,41

REFERENCES

1. M. Russo, L. Mortillaro, L. Credali, C. De Checchi, J. Polym. Sci. A-1, **4**, 248 (1966).
2. A. W. V. Hofmann, Ann. Chem., **145**, 357 (1868).
3. E. Baumann, Chem. Ber., **23**, 60 (1890).
4. R. W. Bonst, E. W. Constable, Org. Syn. Coll. Vol. II, 610.
5. M. Schmidt, K. Blaettner, Angew. Chem., **71**, 407 (1959).
6. L. Credali, M. Russo, Polymer, **8**, 469 (1967).
7. E. Gipstein, E. Wellisch, O. J. Sweeting, J. Polym. Sci. B, **1**, 237 (1963).
8. M. Russo, L. Mortillaro, C. De Checchi, G. Valle, M. Mammi, J. Polym. Sci. B, **3**, 501 (1965).
9. M. Russo, L. Mortillaro, L. Credali, C. De Checchi, J. Polym. Sci. B, **3**, 455 (1965).
10. M. Russo, L. Mortillaro, L. Credali, C. De Checchi, J. Chem. Soc. C, 428 (1966).
11. E. Block, E. R. Corey, R. E. Penn, T. L. Renken, P. F. Sherwin, J. Am. Chem. Soc., **98**, 5715 (1976).
12. E. Weissflog, Phosph. Sulf., **12**, 141 (1982).
13. M. Schmidt, K. Blaettner, P. Kochendoerfer, H. Ruf, Z. Naturforsch., **21b**, 622 (1966).
14. H. Schmidt, E. Weissflog, Z. Anorg. Allg. Chem., **406**, 271 (1974).
15. J. Lal, J. Org. Chem., **26**, 971 (1961).
16. B. G. Maura, L. Tulli, Ann. Chim., Roma, **55**, 892 (1965).
17. L. Mortillaro, L. Credali, M. Russo, C. De Chjecchi, Polym. Lett., **3**, 581 (1965).
18. G. W. Frank, P. J. Degen, Acta Cryst. B, **29**, 1815 (1973).
19. G. Valle, A. Piazzesi, A. del Pra, Cryst. Struct. Commun., **1**, 289 (1972).
20. M. E. Peach, E. Weissflog, N. Pelz, Org. Mass Spectr., **10**, 781 (1975).
21. H. Stetter, W. Wirth, Ann. Chem., **631**, 144 (1960).
22. D. Sellmann, L. Zapf, Angew. Chem., **96**, 799 (1984).
23. W. Rosen, D. M. Busch, Inorg. Chem., **9**, 262 (1970).
24. D. Gerber, P. Chongsawangvirod, A. K. Leung, L. A. Ochrymowycz, J. Org. Chem., **42**, 2644 (1977).
25. L. A. Ochrymowycz, C.-P. Mak, J. D. Michna, J. Org. Chem., **39**, 2079 (1974).
26. J. Buter, R. M. Kellogg, J. Chem. Soc., Chem. Commun., 466 (1980); J. Buter, R. M. Kellogg, J. Org. Chem., **46**, 4481 (1981).
27. W. N. Setzer, C. A. Ogle, G. S. Wilson, R. S. Glass, Inorg. Chem., **22**, 266 (1983); R. S. Glass, G. S. Wilson, W. N. Setzer, J. Am. Chem. Soc., **102**, 5068 (1980).
28. D. St. C. Black, I. A. McLean, Tetrahedron Lett., 3961 (1969); Austr. J. Chem., **24**, 1401 (1971).
29. J. R. Meadow, E. E. Reid, J. Am. Soc., **56**, 2177 (1934).

Next Page

30. J. S. Bradshaw, J. Y. K. Hui, J. Heterocycl. Chem., **11**, 649 (1974).
31. J. S. Bradshaw, J. Y. K. Hui, B. L. Baymore, R. M. Izatt, J. J. Christensen, J. Heterocycl. Chem., **10**, 1 (1973); J. S. Bradshaw, J. Y. K. Hui, J. Y. Chan, B. L. Haymore, J. J. Christensen, R. M. Izatt, J. Heterocycl. Chem., **11**, 45 (1974).
32. J. L. Lambert, D. van Ooteghem, E. J. Goethals, J. Polym. Sci. A-1, **9**, 3055 (1971).
33. E. J. Goethals, Adv. Polym. Sci., **23**, 103 (1977).
34. A. Noshay, C. C. Price, J. Polym. Sci., **54**, 533 (1961).
35. W. van Craeynest, E. J. Goethals, Eur. Polym. J., **12**, 859.9 (1976).
36. J. R. Dann, P. P. Chiesa, J. W. Gates, J. Org. Chem., **26**, 1991 (1961).
37. F. Vögtle, F. Ley, Chem. Ber., **116**, 3000 (1983).
38. F. Vögtle, L. Schunder, Chem. Ber., **102**, 2677 (1969).
39. R. H. Mitchell, V. Boekelheide, J. Am. Chem. Soc., **96**, 1547 (1974); V. Boekelheide, J. L. Mondt, Tetrahedron Lett., 1203 (1970).
40. K. Galuszko, Rocz. Chem., **49**, 1597 (1975).
41. H. J. J.-B. Martel, M. Rasmussen, Tetrahedron Lett., 3843 (1971).
42. B. V. Gorewit, W. K. Musker, J. Coord. Chem., **5**, 67 (1976).
43. R. G. Gillis, A. B. Lacey, Org. Synth., Coll. Vol. IV, 396 (1963).
44. V. Meyer, Chem. Ber., **19**, 3259 (1886).
45. J. D. A. Johnson, J. Chem. Soc., 1530 (1933).
46. H. T. Clarke, J. Chem. Soc., **101**, 1806 (1912).

TABLE 8. OLIGOMERS CONTAINING N IN THE MAIN CHAIN

8.1. OLIGO(AMIDES)

8.1.1. OLIGO(AMIDES) OF β-AMINO ACIDS (NYLON 3)

8.1.1.1. Linear Oligo(amides) of β-Alanine $H[NH(CH_2)_2CO]_nOH$

n	Mol. wt.	m.p. (d) (°C)	Refs.
1	89.1	206	1
2	160.2	212	2,4,96
3	231.2	>255	3,4,96
4	302.3	>260	4,96
5	373.4	>310	5
6	444.5	>320	5

8.1.1.2. Cyclic Oligo(amides) of β-Alanine $\ulcorner[NH(CH_2)_2CO]_n\urcorner$

n	Mol. wt.	m.p (°C)	Refs.
1	71.1	74–74.5	6,104,113
2	142.2	298–299	4,8,9,72,84,114,116
3	213.2	>360	4,10,84,88,93,101,114,116
4	284.3	353	4,10,84,114,116
5	355.4	327	84,114,116
6	426.5	360	114,116
7	497.6	341	114,116
8	568.8	>360	114,116
9	639.7	349	114,116
10	710.8	>350	116
11	781.8	>350	116

8.1.1.3. Linear Oligo(amides) of C-Methyl-β-Alanines $H[NHCHR_1CHR_2CO]_nOH$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	H	1	103.1	188–189	102
		3	273.3	240–241	92

8.1.1.4. Cyclic Oligo(amides) of C-Methyl-β-Alanines $\ulcorner[NHCHR_1CHR_2CO]_n\urcorner$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_3	H	1	85.1		100/13	104,105
		3	255.3	300		91
H	CH_3	1	85.1		98–99/20	104
		2	170.2	255		84
		3	255.3	>350		84,85,87
		4	340.4	>350		84
		5	425.5	>350		84
		6	510.6	>350		84
		7	595.7	330–335		84
		8	680.8	338–340		84
		9	765.9	327–330		84
		10	851.0	330		84
		11	936.0	325		84

8.1.1.5. Linear Oligo(amides) of C-Dimethyl-β-Alanines $H[NHC(R_1)_2C(R_2)_2CO]_nOH$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
H	CH_3	1	117.1	239	106
		3	315.4	209–212	94
		6	612.8	207.5–210.5	94
CH_3	H	1	117.1	217	103
		3	315.4	182–184	95

8.1.1.6. Cyclic Oligo(amides) of C-Dimethyl-β-Alanines $\ulcorner[NHC(R_1)_2C(R_2)_2CO]_n\urcorner$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
H	CH_3	1	99.1		70–73/1.1	107
		3	297.4	255–259.5		84,91
		4	396.5	270		84
		5	495.7	257		84
		6	594.8	310		84,91
		7	693.9	290		84
		8	793.1	256		84
		9	892.3	315		84
		10	991.3	252		84
		11	1090.5	315		84
		12	1189.6	245		84
CH_3	H	1	99.1	16–17	112/20	104,105
		3	297.4	257–258		91

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36. J. R. Dann, P. P. Chiesa, J. W. Gates, J. Org. Chem., **26**, 1991 (1961).
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40. K. Galuszko, Rocz. Chem., **49**, 1597 (1975).
41. H. J. J.-B. Martel, M. Rasmussen, Tetrahedron Lett., 3843 (1971).
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43. R. G. Gillis, A. B. Lacey, Org. Synth., Coll. Vol. IV, 396 (1963).
44. V. Meyer, Chem. Ber., **19**, 3259 (1886).
45. J. D. A. Johnson, J. Chem. Soc., 1530 (1933).
46. H. T. Clarke, J. Chem. Soc., **101**, 1806 (1912).

TABLE 8. OLIGOMERS CONTAINING N IN THE MAIN CHAIN

8.1. OLIGO(AMIDES)

8.1.1. OLIGO(AMIDES) OF β-AMINO ACIDS (NYLON 3)

8.1.1.1. Linear Oligo(amides) of β-Alanine $H[NH(CH_2)_2CO]_nOH$

n	Mol. wt.	m.p. (d) (°C)	Refs.
1	89.1	206	1
2	160.2	212	2,4,96
3	231.2	>255	3,4,96
4	302.3	>260	4,96
5	373.4	>310	5
6	444.5	>320	5

8.1.1.2. Cyclic Oligo(amides) of β-Alanine $\overline{[NH(CH_2)_2CO]_n}$

n	Mol. wt.	m.p (°C)	Refs.
1	71.1	74–74.5	6,104,113
2	142.2	298–299	4,8,9,72,84,114,116
3	213.2	>360	4,10,84,88,93,101,114,116
4	284.3	353	4,10,84,114,116
5	355.4	327	84,114,116
6	426.5	360	114,116
7	497.6	341	114,116
8	568.8	>360	114,116
9	639.7	349	114,116
10	710.8	>350	116
11	781.8	>350	116

8.1.1.3. Linear Oligo(amides) of C-Methyl-β-Alanines $H[NHCHR_1CHR_2CO]_nOH$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	H	1	103.1	188–189	102
		3	273.3	240–241	92

8.1.1.4. Cyclic Oligo(amides) of C-Methyl-β-Alanines $\overline{[NHCHR_1CHR_2CO]_n}$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_3	H	1	85.1		100/13	104,105
		3	255.3	300		91
H	CH_3	1	85.1		98–99/20	104
		2	170.2	255		84
		3	255.3	>350		84,85,87
		4	340.4	>350		84
		5	425.5	>350		84
		6	510.6	>350		84
		7	595.7	330–335		84
		8	680.8	338–340		84
		9	765.9	327–330		84
		10	851.0	330		84
		11	936.0	325		84

8.1.1.5. Linear Oligo(amides) of C-Dimethyl-β-Alanines $H[NHC(R_1)_2C(R_2)_2CO]_nOH$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
H	CH_3	1	117.1	239	106
		3	315.4	209–212	94
		6	612.8	207.5–210.5	94
CH_3	H	1	117.1	217	103
		3	315.4	182–184	95

8.1.1.6. Cyclic Oligo(amides) of C-Dimethyl-β-Alanines $\overline{[NHC(R_1)_2C(R_2)_2CO]_n}$

R_1	R_2	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
H	CH_3	1	99.1		70–73/1.1	107
		3	297.4	255–259.5		84,91
		4	396.5	270		84
		5	495.7	257		84
		6	594.8	310		84,91
		7	693.9	290		84
		8	793.1	256		84
		9	892.3	315		84
		10	991.3	252		84
		11	1090.5	315		84
		12	1189.6	245		84
CH_3	H	1	99.1	16–17	112/20	104,105
		3	297.4	257–258		91

8.1.2. OLIGO(AMIDES) OF γ-AMINOBUTYRIC ACID (NYLON 4)

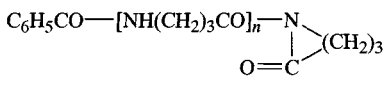
8.1.2.1. Linear Oligo(amides) $H[NH(CH_2)_3CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	103.1	204–205	11
2	188.2	186	4,7,12
3	273.4	198–199	4
4	358.4	202–203	4
6	528.6	196–198	133
8	698.8	206–208	133
10	869.0	199–203	133
12	1039.3	208	133
16	1379.6	207–209	133
20	1719.4	206–208	133

8.1.2.2. Cyclic Oligo(amides) $\ulcorner[NH(CH_2)_3CO]_n\urcorner$

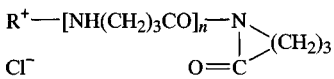
n	Mol. wt.	m.p. (°C)	Refs.
1	85.1	24	13,14
2	170.2	283	4,10,70,71,73
3	255.3	242–243	4,10
4	340.4	255	4,10,70
6	510.6	303	10,135
8	680.9	298	135
10	851.1	312	135
12	1021.3	301	135
14	1191.5	>355	135

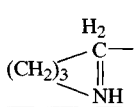
8.1.2.3. Anionic Oligo(amides)



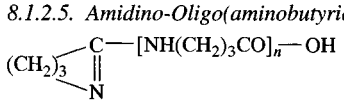
n	Mol. wt.	m.p. (°C)	Refs.
0	189.2	92.5	15
1	274.3	121	16

8.1.2.4. Cationic Oligo(amides)



R	*n*	Mol. wt.	m.p. (°C)	Refs.
H_2	1	206.5	181	71,89,118
	0	188.7	156	89,119
	1	273.5	162	89

8.1.2.5. Amidino-Oligo(aminobutyric Acids)



n	Mol. wt.	m.p. (°C)	Refs.
1	170.2	175–177	89,118,121
2	255.3	175	89
3	340.4	162–163	89

8.1.3. OLIGO(AMIDES) OF-δ-AMINOVALERIC ACID (NYLON 5)

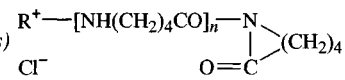
8.1.3.1. Linear Oligo(amides) $H[NH(CH_2)_4CO]_nOH$

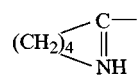
n	Mol. wt.	m.p. (°C)	Refs.
1	117.2	160–162	17
2	216.3	178	18
3	315.4	184–185.5	18
4	414.5	196.5–198.5	18

8.1.3.2. Cyclic Oligo(amides) $\ulcorner[NH(CH_2)_4CO]_n\urcorner$

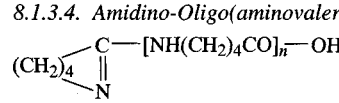
n	Mol. wt.	m.p. (°C)	Refs.
1	98.1	39–40	14,17,19
2	198.3	295–296	8,18
3	297.4	329–331	18
4	396.5	266–267	18

8.1.3.3. Cationic Oligo(amides)



R	*n*	Mol. wt.	m.p. (°C)	Refs.
H_2	1	234.5	114	89
	0	216.5	93–94	89
	1	315.5	138	89

8.1.3.4. Amidino-Oligo(aminovaleric Acids)



n	Mol. wt.	m.p. (°C)	Refs.
1	198.3	166–168	89,118
2	297.4	92–94	89
3	396.5	112	89

8.1.4. OLIGO(AMIDES) OF ε-AMINOCAPROIC ACID (NYLON 6)

8.1.4.1. Linear Oligo(amides) $H[NH(CH_2)_5CO]_nOH$

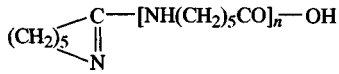
n	Mol. wt.	m.p. (°C)	Refs.
1	131.2	206–208	20,21
2	244.3	204–205	22–24,136
3	357.3	208–209	23–25,69,136
4	470.6	220–221	23,24,36
5	583.8	214–215	23,69,136
6	697.0	212–214	23,24,136
7	810.1	210–211	23,69,136
8	923.3	210–211	23
9	1036.5	208–211	26,69
10	1149.6	212–213	26
11	1262.8	209–212	26,69
12	1375.9	211–213	26
13	1489.1	210–211	69
14	1602.3	207–208	136
15	1715.3	207–208	69
17	1941.5	209–211	69

8.1.4.1. cont'd

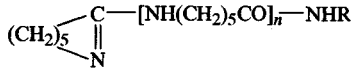
n	Mol. wt.	m.p. (°C)	Refs.
18	2054.9	206–209	136
21	2394.5	212–213	69
22	2507.5	209–211	136
25	2847.0	212–213.5	69
26	2960.2	209–211	136
30	3412.8	210–213	136
34	3865.5	211–214	136
38	4318.1	214–217	136

8.1.4.2. Oligo(aminocaproic Amides)
$H_2^+[NH(CH_2)_5CO]_n$–NHR CL^-

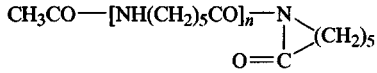
R	*n*	Mol. wt.	m.p. (°C)	Refs.
n-C_4H_9	1	222.8	143	122
	2	335.9	176	122
	3	449.1	189	122
$CH_2C_6H_5$	1	256.8	164.5	122
	2	369.9	179	122
	3	483.1	194–195	122

8.1.4.3. Amidino-Oligo(aminocaproic Acids)


n	Mol. wt.	m.p. (°C)	Refs.
1	226.3	168	120,128
2	339.5	120	120
3	452.6	154	120

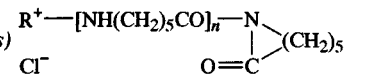
8.1.4.4. Amidino-Oligo(aminocaproic Amides)


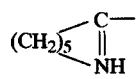
R	*n*	Mol. wt.	m.p. (°C)	Refs.
n-C_4H_9	0	168.3	53	122
	1	281.4	129	122
	2	394.6	148–150	122
$CH_2C_6H_5$	0	202.3	76–78	122
	1	315.5	135	122
	2	428.6	157	122

8.1.4.5. Anionic Oligo(amides)


n	Mol. wt.	b.p. (°C/mbar)	n_D^{20}	R_F^a	Refs.
0	155.2	123/21	1.4898		32
1	268.5		1.4992	0.76	33
2	381.5			0.67	33
3	494.7			0.58	33

[a] PC: solvent, tetrahydrofuran/cyclohexane/water (186 : 14 : 10).

8.1.4.6. Cationic Oligo(amides)


R	*n*	Mol. wt.	m.p. (°C)	Refs.
H_2	1	262.8	112	120
	2	375.9	Oil	120
	3	489.1	Oil	120
	0	244.8	137	89,120
	1	357.9	106–108	120

8.1.4.7. End-group Protected Oligo(amides)
C_2H_5CO–$[NH(CH_2)_5CO]_n$–NHC_3H_7

n	Mol. wt.	m.p. (°C)	Refs.
1	228.3	106	31,112
2	341.5	149	31,112
3	454.7	172	31,112
4	567.8	181	31
5	681.0	187	31,112
6	794.2	191	31,112
7	907.3	197	31,112
8	1020.5	200	31,112
9	1133.6	201	31,112
10	1246.8	201	31,112
11	1360.0	204	31
12	1473.1	202	31,112
13	1586.3	200	31
14	1699.3	205	31
15	1812.6	206	31
16	1925.8	206	31

8.1.4.8. Cyclic Oligo(amides)

			R_F		
n^a	Mol. wt.	m.p. (°C)	TCW[b]	EAW[c]	Refs.
1	113.2	69.5	0.56	0.88	14,17,27,78
2	226.3	348	0.48	0.81	4,24,28,29,45,78,98,110,111,112,148
3	339.5	244	0.34	0.77	4,24,28,29,45,78,98,111,112,147
4	452.6	256–257	0.25	0.69	4,24,28,29,45,78,98,111,112
5	565.8	254	0.16	0.59	4,24,29,78,98,111,112
6	678.9	260	0.10	0.49	4,24,29,60,98,111,112
7	792.1	243	0.08	0.40	29,30,60,98,111
8	905.3	254	0.04		29,30,60,98,111
9	1018.4	245	0.02		29,30,68,138

8.1.4.8. cont'd

n^a	Mol. wt.	m.p. (°C)	R_F TCW[b]	R_F EAW[c]	Refs.
10	1131.8	257			117,137
11	1244.8	249			138
12	1357.9	262			117,137
13	1471.1	249			138
14	1584.3	263			117,137
15	1697.4	256			138
16	1810.6	261			117,137
18	2036.9	263			117,137
20	2263.2	259			117,137
21	2376.4	248			138
22	2489.5	250			117,137
24	2715.9	252			117,137
25	2829.0	249			138
26	2942.2	246			137
27	3055.3	244			138
28	3168.5	247			137
30	3394.8	255			117,137
32	3621.2	248			137
33	3734.3	231			138
34	3847.5	231			137
36	4073.8	228			137
38	4300.1	230			137
40	4526.4	235			137
48	5431.7	234			138
50	5658.1	229			138
55	6223.9	226			138
56	6337.0	229			138
60	6789.7	227			138
64	7242.3	226			138
70	7921.3	225			138
72	8147.6	225			138
80	9052.9	224			138
90	10184.5	222			138
100	11316.1	221			138

[a] n = 2–40: HPLC (trifluoroethanol/water mixtures), Ref. 117; n = 2–18: FD-MS, Ref. 117; n = 2–100: HPLC (trifluoroethanol/water, gradient elution, Ref. 138.
[b] PC, TLC: solvent, tetrahydrofuran/cyclohexane/water (186 : 14 : 10).
[c] PC, TLC: solvent, ethyl acetate/acetone/water (10 : 10 : 2), Refs. 29,134.

8.1.5. OLIGO(AMIDES) OF HIGHER ω-AMINO ACIDS (NYLON 7–12)

8.1.5.1. Linear Oligo(amides) $H[NH(CH_2)_xCO]_nOH$

x	n	Mol. wt.	m.p. (°C)	Refs.
6	1	145.2	195	34,35
	2	272.4	205–208	36
7	1	159.2	188	4,37
	2	300.4	191–192	4,38
8	1	173.3	184	4,34
	2	328.5	184–186	4
10	1	201.3	186–187	34,38,39
	2	384.6	189–190	4,40,140
	3	567.9	183–184	5,40,115,140
	4	751.2	177–179	40,63,115
	5	934.5	177–178	63
	10	1850.8	181–182	63,67
11	1	215.3	186–187	41
	2	412.6	192–193	31

8.1.5.2. Cyclic Oligo(amides) $\overline{}[NH(CH_2)_xCO]_n\overline{}$

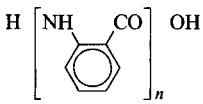
x	n	Mol. wt.	m.p. (°C)	Refs.
6	1	127.2	29–30	14,17,42
	2	254.4	236–237	4,10
	3	381.6	259–261	89
6	4	508.8	215–216	89
	5	636.0	236	89
	6	763.2	244–245	89
	8	1017.5	258–259	89
	10	1271.9	262–263	89
7	1	141.2	72–73	13,14,34,42
	2	282.4	277	4,10,76,77
	3	423.6	217	89
	4	564.8	219	82,89
	5	706.0	215	89
	6	847.2	215	89
8	1	155.3	138–139	14,42
	2	310.5	201	4,10
9	1	169.3	162	14,42
	2	338.4	238	10,144

8.1.5.2. cont'd

x	*n*	Mol. wt.	m.p. (°C)	Refs.
10	1	183.3	155	42,43,75
	2	366.6	188–189	4,10,44,75,98
10	4	741.2	203	140
	5	926.6	204	140
	6	1111.9	217	140
	7	1297.2	205	140
	8	1482.5	221	140
	9	1667.8	201	140
	3	549.9	184	44,60,75,98
11	1	197.3	155	13,14,42,75
	2	394.6	212	60,61,75,98,115
	3	531.9	175	31,75
14	2	478.8	182	139

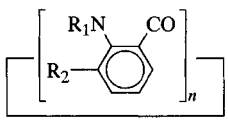
8.1.6. OLIGO(AMIDES) OF AROMATIC AMINO ACIDS

8.1.6.1. Linear Oligo(amides) of o-Aminobenzoic Acid



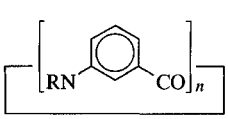
n	Mol. wt.	m.p. (°C)	Refs.
1	137.1	146	46–48
2	256.3	203–204	46,123,124,131
3	375.4	256	46,131
4	494.5	d	46

8.1.6.2. Cyclic Oligo(amides) of o-Aminobenzoic Acids



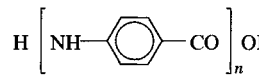
R_1	R_2	*n*	Mol. wt.	m.p. (°C)	Refs.
H	H	2	238.2	335–337	47,125–128,131
CH_3	H	2	266.3	207	47
		3	399.2	>320	131
		4	532.2	>300	130
$C_6H_5CH_2$	H	2	418.2	158–159	131
		3	627.3	260–263	131
	CH_3	3	669.3	260–263	129
CH_3	CH_3	3	441.2	>300	129

8.1.6.3. Cyclic Oligo(amides) of m-Aminobenzoic Acids



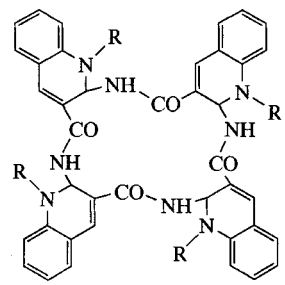
R	*n*	Mol. wt.	m.p. (°C)	Ref.
CH_3	3	399.2	>320	132

8.1.6.4. Linear Oligo(amides) of p-Aminobenzoic Acid



R	*n*	Mol. wt.	m.p. (°C)	Refs.
H	2	256.1	276	86
	3	375.4	>310	86
C_2H_5	2	284.1	176	86
	3	403.2	>266	86
	4	522.2	>330	86

8.1.6.5. Cyclic Oligo(amides) of 4-Amino-2,3-dihydroquinoline-3-Carboxylic Acid



R	*n*	Mol. wt.	m.p. (°C)	Ref.
n-C_3H_7	4	857.0	198	97
$CH_2C_6H_5$	4	1049.2	206	97

8.1.7. CYCLIC DIAMIDES OF ALIPHATIC DICARBOXYLIC ACIDS AND DIAMINES $\urcorner[NH(CH_2)_xNH{-}CO(CH_2)_xCO]_n\urcorner$

x	*n*	Mol. wt.	m.p. (°C)	Refs.
2	1	142.2	201	142
3	1	170.2	238	49,50,74,139
4	1	198.3	304	49,51,139
5	1	226.3	337	49,139
	2	452.6	234	143
	3	679.0	246	143
	4	905.3	244	143
6	1	254.3	238	49,51,139
7	1	282.4	267	49,139
8	1	310.5	209	8,49,139,146
9	1	338.5	228	49,139,147
10	1	366.6	195	8,49,139
11	1	394.6	206	139
12	1	422.7	186	139

8.1.8. OLIGO(AMIDES) OF ADIPIC ACID AND HEXAMETHYLENE DIAMINE (NYLON 6.6)

8.1.8.1. Linear Oligo(amides) $H[NH(CH_2)_6NH{-}CO(CH_2)_4CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	244.3	193	52,53
2	470.7	221–222	52,53,63
3	696.9	246–248	52,53,63
4	923.3	247–249	31,63,65
5	1149.5	247–250	31,63,65
10	2281.1	250–255	63

8.1.8.2. End-Group Protected Oligo(amides)

8.1.8.2.1. Oligomeric Dicarboxylic Acid Dipropylamides

$C_3H_7NH-CO(CH_2)_4CO-[NH(CH_2)_6NH-CO(CH_2)_4CO]_n-NHC_3H_7$

n	Mol. wt.	m.p. (°C)	Refs.
1	454.7	229–230	64
2	681.0	239–242	64
3	907.3	248–250	64
4	1133.6	257–260	64
5	1359.9	258–261	64

8.1.8.2.2. Oligomeric Dipropionyl Diamines

$C_2H_5CO-[NH(CH_2)_6NH-CO(CH_2)_4CO]_n-NH(CH_2)_6NH-COC_2H_5$

n	Mol. wt.	m.p. (°C)	Refs.
1	454.7	202–203	64
2	681.0	226–228	64
3	907.3	229–231	64

8.1.8.3. Oligomeric Diamines

$H_2^+[NH(CH_2)_6NH-CO(CH_2)_4CO]_n-NH(CH_2)_6NH_3^+ \ 2Cl^-$

n	Mol. wt.	m.p. (°C)	Refs.
1	415.4	248–250	65
2	641.7	249–251	65
3	868.0	252–254	65
4	1094.4	251–254	65
5	1320.7	252–255	65

8.1.8.4. Oligomeric Dicarboxylic Acids

$HOOC(CH_2)_4CO-[NH(CH_2)_6NH-CO(CH_2)_4CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	372.4	188–189	56
2	598.7	200–205	56,57
3	825.0	210–211	56
4	1051.3	214–219	31

8.1.8.5. Cyclic Oligo(amides) $\urcorner[NH(CH_2)_6NH-CO(CH_2)_4CO]_n\ulcorner$

n[a]	Mol. wt.	m.p. (°C)	TCW[b]	EAW[c]	Refs.
1	226.3	254	0.47	0.51	51,53–55,62,80,81,90,98,108
2	452.6	243–244	0.28	0.38	53–55,62,80,81,90,98,109
3	678.9	235	0.13	0.23	53,55,62,98
4	905.3	273	0.06	0.16	53,60,62,98
5	1131.6	279			149
6	1357.9	279			149
7	1584.3	291			149
8	1810.6	282			149
9	2036.9	280			149
10	2263.2	280			149
12	2715.9	280			149
14	3168.5	292			149
16	3621.2	284			149
18	4073.8	275			149
20	4526.4	270			149

[a] n = 1–20: HPLC (trifluoroethanol/water, gradient elution), Ref. 149.
[b] PC, TLC: solvent: tetrahydrofuran/cyclohexane/water (186 : 14 : 10).
[c] PC, TLC: solvent: ethyl acetate/acetone/water .(10 : 10 : 2).

8.1.9. OLIGO(AMIDES) OF SEBACIC ACID AND HEXAMETHYLENEDIAMINE (NYLON 6.10)

8.1.9.1. Linear Oligo(amides) $H[NH(CH_2)_6NH-CO(CH_2)_8CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	300.4	187–188	58
2	582.8	196–199	63,67
3	865.3	208–210	63,67

8.1.9.2. Cyclic Oligo(amides) $\urcorner[NH(CH_2)_6NH-CO(CH_2)_8CO]_n\ulcorner$

n	Mol. wt.	m.p. (°C)	Refs.
1	282.4	232	51,98,145
2	564.8	222	58,60,98,145
3	847.3	221	145

8.1.9.3. Oligomeric Diamines

$H[NH(CH_2)_6NH-CO(CH_2)_8CO]_n-NH(CH_2)_6NH_2$

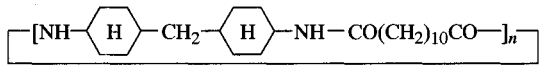
n	Mol. wt.	m.p. (°C)	Refs.
1	398.6	129–131	57,58
2	681.1	177–179	58
3	963.5	203–210	58

8.1.9.4. Oligomeric Dicarboxylic Acids

$HOOC(CH_2)_8CO-[NH(CH_2)_6NH-CO(CH_2)_8CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	494.6	156–159	57,58
2	767.1	182–184	57,58
3	1049.5	181–185	58

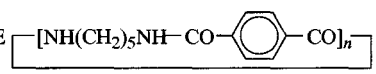
8.1.10. OLIGO(AMIDES) OF DECANE DICARBOXYLIC ACID AND *p,p'*-DIAMINO-DICYCLOHEXYLMETHANE (QIANA TYPE)



n	Mol. wt.	Config.	m.p. (°C)	Refs.
1	404.6	*trans-trans*	310–314	83
		cis-trans	269–272	83
		cis-cis	254–257	83

8.1.11. OLIGO(AMIDES) OF OXALIC ACID AND HEXAMETHYLENEDIAMINE $\left[NH(CH_2)_6NH-COCO\right]_n$

n	Mol. wt.	m.p. (°C)	Refs.
1	170.2	232	66
2	340.4	303	66
3	510.6	345	66

8.1.12. OLIGO(AMIDES) OF TEREPHTHALIC ACID AND PENTAMETHYLENEDIAMINE 

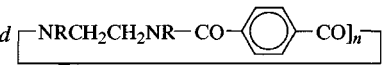
n	Mol. wt.	m.p. (°C)	Refs.
2	464.3	407	59

8.1.13. OLIGO(AMIDES) OF *N,N'*-DIALKYLETHYLENEDIAMINE AND DICARBOXYLIC ACIDS

8.1.13.1. Aliphatic Dicarboxylic Acids

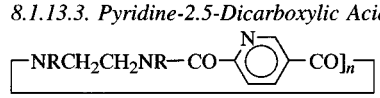
$\left[NRCH_2CH_2NR-CO(CH_2)_xCO\right]_n$

R	*x*	*n*	Mol. wt.	m.p. (°C)	Refs.
$C_6H_5CH_2$	3	2	672.8	231–234	79
	5	2	729.0	199–201	79
	7	2	785.1	199–204	79

8.1.13.2. Terephthalic Acid 

R	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	3	654.8	114–123	99
$C_6H_5CH_2$	3	1111.4	255–257	100
	4	1481.8	143–145	100
	6	2222.8	140–144	100

8.1.13.3. Pyridine-2.5-Dicarboxylic Acid



R	*n*	Mol. wt.	m.p. (°C)	Refs.
CH_3	3	663.8	123–129	99
	4	885.0	132–136	99

REFERENCES

1. J. H. Ford, Org. Syn., Coll. Vol. III, 34; T. W. J. Taylor, J. Chem. Soc., 1897 (1928).
2. H. Th. Hanson, E. L. Smith, J. Biol. Chem., **175**, 833 (1948).
3. E. Adams, N. C. Davis, E. L. Smith, J. Biol. Chem., **199**, 845 (1952).
4. M. Rothe, Habilitationsschrift, Universität Halle, DDR, 1960.
5. M. Rothe, Unpublished data.
6. R. W. Holley, A. D. Holley, J. Am. Chem. Soc., **71**, 2129 (1949).
7. S. Murahashi, H. Sekiguchi, H. Yuki, Compt. Rend., **248**, 1521 (1959).
8. M. Rothe, R. Timler, Chem. Ber., **95**, 783 (1962).
9. H. K. Hall, Jr., J. Am. Chem. Soc., **80**, 6404 (1958).
10. M. Rothe, Angew. Chem., **74**, 725 (1962).
11. C. C. DeWitt, Org. Syn., Coll. vol. II, 25 (1943).
12. R. L. Evans, F. Irreverre, J. Org. Chem., **24**, 863 (1959).
13. K. Dachs, E. Schwarz, Angew. Chem., **74**, 540 (1962).
14. R. Huisgen, H. Brade, H. Walz, I. Glogger, Chem. Ber., **90**, 1437 (1957).
15. S. J. Kanewskaja, Chem. Ber., **69**, 266 (1936).
16. H. Sekiguchi, Bull. Sci. Chim. (France), 1835 (1960).
17. O. Wallach, Ann. Chem., **312**, 171 (1900); **324**, 281 (1902).
18. M. Rothe, R. Hoßbach, Makromol. Chem., **70**, 150 (1964).
19. N. Yoda, A. Miyake, J. Polym. Sci., **43**, 117 (1960).
20. S. Gabriel, A. Maass, Chem. Ber., **32**, 1266 (1899).
21. J. C. Eck, Org. Syn., Coll. vol. II, 28 (1943); C. Y. Meyers, L. E. Miller, Org. Syn., Coll. vol. IV, 39 (1963).
22. G. M. van der Want, A. Staverman, Rec. Trav. Chim., **71**, 379 (1952).
23. H. Zahn, D. Hildebrand, Chem. Ber., **90**, 320 (1957).
24. M. Rothe, F.-W. Kunitz, Ann. Chem., **609**, 88 (1957); Angew. Chem., **68**, 414 (1956).
25. G. M. van der Want, H. Peters, P. Inklaar, Rec. Trav. Chim., **71**, 1221 (1952).
26. H. Zahn, D. Hildebrand, Chem. Ber., **92**, 1963 (1959).
27. C. S. Marvel, J. C. Eck, Org. Syn., Coll. vol. II, 371 (1943).
28. P. H. Hermans, Rec. Trav. Chim., **72**, 798 (1953); Nature, **177**, 126 (1956).
29. M. Rothe, J. Polym. Sci., **30**, 227 (1958).
30. H. Zahn, J. Kunde, Angew. Chem., **70**, 189 (1958).
31. H. Zahn, G. B. Gleitsmann, Angew. Chem., **75**, 772 (1963).
32. H. A. Offe, Z. Naturforsch, **2b**, 182 (1947).
33. K. Gehrke, Faserforsch. Textiltechn., **13**, 556 (1962).
34. D. D. Coffman, N. L. Cox, E. L. Martin, W. E. Mochel, F. J. van Natta, J. Polym. Sci., **3**, 85 (1948).
35. C. F. Horn, B. T. Freure, H. Vineyard, H. J. Decker. Angew. Chem., **74**, 531 (1962).
36. M. Rothe, R. Hoßbach, unpublished data.
37. A. G. Goldsobel, Chem. Ber., **27**, 3121 (1894).

38. T. Gaeumann, H. H. Guenthardt, Helv. Chim. Acta, **35**, 53 (1952).
39. M. Genas, Angew. Chem., **74**, 535 (1962).
40. M. Zahn, H. Roedel, J. Kunde, J. Polym. Sci., **36**, 539 (1959).
41. A. Neuberger, Proc. Roy. Soc. A, **158**, 68, 84 (1937).
42. L. Ruzicka, M. Kobelt, O. Haefliger, V. Prelog, Helv. Chim. Acta, **32**, 544 (1949).
43. W. Ziegenbein, W. Lang, Angew. Chem., **74**, 943 (1962),
44. H. Zahn, J. Kunde, Chem. Ber., **94**, 2470 (1961).
45. I. Rothe, M. Rothe, Chem. Ber., **88**, 284 (1955).
46. H. Meyer, Ann. Chem., **351**, 267 (1907).
47. G. Schroeter, O. Eisleb, Ann. Chem., **367**, 101 (1909); Chem. Ber., **52**, 2224 (1919).
48. E. Mohr, F. Koehler, H. Ulrich, J. Prakt. Chem., **79**, 281 (1909); **80**, 1 (1909).
49. J. Dale, R. Coulon, J. Chem. Soc., 182 (1964).
50. G. I. Glover, H. Rapoport, J. Am.Chem. Soc., **86**, 3397 (1964).
51. H. Stetter, J. Marx, Ann. Chem., **607**, 59 (1957).
52. H. Zahn, F. Schmidt, Makromol. Chem., **36**, 1 (1959).
53. I. Rothe, M. Rothe, Makromol. Chem., **68**, 206 (1963).
54. H. Zahn, F. Schmidt, Chem. Ber., **92**, 1381 (1959); H. Zahn, P. Miro, F. Schmidt, Chem. Ber., **90**, 1411 (1957).
55. M. Rothe, I. Rothe, H. Bruenig, K.-D. Schwenke, Angew. Chem., **71**, 700 (1959).
56. H. Zahn, W. Lauer, Makromol. Chem., **23**, 85 (1957).
57. C. D. Cowell, Chem. Ind., 577 (1954).
58. H. Zahn, G. B. Gleitsmann, Makromol. Chem., **60**, 45 (1963).
59. H. K. Livingston, R. L. Gregory, Polymer, **13**, 297 (1972).
60. P. Kusch, H. Zahn, Angew. Chem., **77**, 720 (1965).
61. H. Zahn, H.-D. Stolper, G. Heidemann, Chem. Ber., **98**, 3251 (1965).
62. H. Zahn., P. Kusch. Chem. Ber., **98**, 2588 (1965).
63. P. Kusch, Kolloid-Z., **208**, 138 (1966); H. Klostermeyer, J. Halstrøm, P. Kusch, J. Foehles, W. Lunkenheimer, Peptides, 113 (1967), North-Holland Publ. Co., Amsterdam.
64. H. Zahn, O. P. Garg, Kolloid-Z., **208**, 132 (1966).
65. H. Zahn, P. Kusch, J. Shah, Kolloid-Z., **216/7**, 298 (1967); H. Zahn, Z. Ges. Textil-Ind., **66**, 928 (1964).
66. O. Vogl, A. C. Knight, Macromolecules, **1**, 311, 315 (1968).
67. H. Zahn, P. Kusch, Z. Ges. Textil-Ind., **69**, 880 (1967).
68. M. Rothe, U. Kress, unpublished data.
69. M. Rothe, W. Dunkel, J. Polym. Sci., Polym. Lett., **5**, 589 (1967).
70. M. Rothe, I. Rothe, Makromol. Chem., **85**, 307 (1965).
71. G. I. Glover, R. B. Smith, H. Rapoport, J. Am. Chem. Soc., **87**, 2003 (1965).
72. D. N. White, J. D. Dunitz, Israel J. Chem., **10**, 249 (1972).
73. T. Winkler, T. Leutert, Helv. Chim. Acta, **65**, 1760 (1982).
74. T. Srikrishnan, J. D. Dunitza, Acta Cryst. B, **31**, 1372 (1975).
75. R. Feldmann, R. Feinauer, Angew. Makromol. Chem., **34**, 9 (1973).
76. J. Marik, J. Mitera, J. Kralicek, J. Stehlicek, Eur. Polym. J., **13**, 961 (1977).
77. I. Falgova, J. Kondelikova, J. Kralicek, Angew. Makromol. Chem., **49**, 75 (1976).
78. R. Okada, T. Fukumura, H. Tanzawa, J. Polym. Sci., Polym. Lett., **4**, 971 (1960).
79. E. Schwartz, A. Shanzer, J. Chem. Soc., Chem. Commun. 634 (1981).
80. P. D. Frayer, J. L. Koenig, J. B. Lando, J. Macromol. Sci.-Phys. B, **3**, 329 (1969).
81. C. J. Brown, A. Hill, R. V. Youle, Nature, **177**, 128 (1956).
82. Y. Okuno, K. Horita, O. Yonemitsu, Chem. Pharm. Bull., **31**, 737 (1983).
83. G. E. Hahn, P. Kusch, V. Rossbach, H. Zahn, Makromol. Chem., **186**, 297 (1985).
84. M. Rothe, B. Benz, unpublished data.
85. M. Guaita, L. F. Thomas, Makromol. Chem., **117**, 171 (1968).
86. H. Bredereck, H. v. Schuh, Chem. Ber., **81**, 715 (1948).
87. L. Trossarelli, M. Guaita, A. Priola, Makromol. Chem., **109**, 253 (1967).
88. G. Camino, L. Costa, L. Trossarelli, J. Polym. Sci., Polym. Chem. Ed., **18**, 377 (1980).
89. M. Rothe, Ch. Seltenreich, R. Hoßbach, H. Emmert, unpublished data.
90. L. E. Alexander, J. Polym. Sci., Polym. Lett. Ed., **10**, 759 (1972).
91. J. Lowbridge, E. Mtetwa, R. J. Ridge, C. N. C. Drey, J. Chem. Soc., Perkin Trans. I, 155, (1986).
92. C. N. C. Drey, E. Mtetwa, J. Chem. Soc., Perkin Trans. I, 1587 (1982).
93. D. N. White, C. Morrow, P. J. Cox, C. N. C. Drey, J. Lowbridge, J. Chem. Soc., Perkin Trans. II, 239 (1982).
94. C. N. C. Drey, R. J. Ridge, J. Chem. Soc., Perkin Trans. I, 2468 (1981).
95. C. N. C. Drey, J. Lowbridge, R. J. Ridge, J. Chem. Soc., Perkin Trans. I, 2001 (1973).
96. W. Ried, K. Marquardt, Ann. Chem., **642**, 141 (1961).
97. H. Berenbold, W.-H. Gündel, Ann. Chem., 532 (1978).
98. G. Heidemann, Encycl. Polym. Sci. Technol., **9**, 485 (1968).
99. F. Voegtle, W. M. Müller, Angew. Chem., **96**, 711 (1984).
100. F. Behm, W. Simon, W. M. Müller, F. Voegtle, Helv. Chim. Acta, **68**, 940 (1985); F. Voegtle, H. Puff, E. Friedrichs, W. M. Müller, J. Chem. Soc., Chem. Commun., 1398 (1982).
101. J. L. Van Winkle, J. D. McClure, P. H. Williams, J. Org. Chem., **31**, 3300 (1966).
102. L. Birkofer, I. Storch, Chem. Ber., **86**, 749 (1953).
103. M. D. Slimmer, Chem. Ber., **35**, 400 (1902).
104. L. Birkofer, J. Schramm, Ann. Chem., 2195 (1975).
105. R. Graf, Ann. Chem., **661**, 111 (1963).
106. D. H. Johnson, J. Chem. Soc. C, 126 (1968).
107. E. Testa, L. Fontanella, Ann. Chem., **625**, 95 (1959).
108. M. G. Northolt, L. E. Alexander, J. Phys. Chem., **72**, 2838 (1968).
109. M. G. Northolt, Acta Cryst. B, **26**, 240 (1970).
110. M. G. Northolt, L. E. Alexander, Acta Cryst. B, **27**, 523 (1971).
111. G. Heidemann, H.-J. Nettelbeck, Faserforsch. Textiltechn., **18**, 183 (1967).

112. H. Halboth, J. Polym. Sci. C, 729 (1969); H. Halboth, G. Rehage, Faserforsch. Textiltechn., **18**, 177 (1967).
113. M. Rothe, A. Häberle, unpublished data.
114. M. Rothe, F. Wehowsky, unpublished data.
115. M. Rothe, W. Fischer, unpublished data.
116. M. Rothe, D. Mühlhausen, unpublished data.
117. M. Rothe, M. Lohmüller, G. Schmidtberg, unpublished data.
118. M. Rothe, W. Helmling, unpublished data.
119. H. Bredereck, K. Bredereck, Chem. Ber., **94**, 2278 (1961).
120. M. Rothe, M. Schlipf, unpublished data.
121. M. Rothe, D. Jacob, unpublished data.
122. M. Rothe, W. Müller, unpublished data.
123. R. P. Staiger, E. B. Miller, J. Org. Chem., **24**, 1214 (1959).
124. S. Petersen, E. Tietze, Ann. Chem., **623**, 166 (1959).
125. N. S. Dokunikhin, L. A. Galva, J. D. Kraft, Dokl. Akad. Nauk SSSR, **81**, 1073 (1951).
126. F. C. Cooper, M. W. Partridge, J. Chem. Soc., 3429 (1954).
127. H. N. Rydon, N. H. P. Smith, D. Williams, J. Chem. Soc., 1900 (1957).
128. D. J. Williams, J. Chem. Soc., Chem. Commun., 170 (1977).
129. S. J. Edge, W. D. Ollis, J. S. Stephanatou, J. F. Stoddart, J. Chem. Soc., Perkin Trans. I, 1701 (1982); S. J. Edge. W. D. Ollis, J. S. Stephanatou, J. F. Stoddart, D. J. Williams, K. A. Woode, Tetrahedron Lett., 2229 (1981).
130. A. Hoorfar, W. D. Ollis, J. F. Stoddart, D. J. Williams, Tetrahedron Lett., 4211 (1980); A. Hoorfar, W. D. Ollis, J. F. Stoddart, J. Chem. Soc., Perkin Trans. I, 1721 (1982).
131. A. Hoorfar, W. D. Oills, J. A. Price, J. S. Stephanatou, J. F. Stoddart, J. Chem. Soc., Perkin Trans. I, 1649 (1982); W. D. Ollis, J. S. Stephanatou, J. F. Stoddart, A. G. Ferrige, Angew. Chem., **88**, 223 (1976); W. D. Ollis, J. A. Price, J. S. Stephanatou, J. F. Stoddart, Angew. Chem., **87**, 169 (1975).
132. F. E. Elhadi, W. D. Ollis, J. F. Stoddart, D. J. Williams, K. A. Woode, Tetrahedron Lett., 4215 (1980); F. E. Elhadi, W. D. Ollis, J. F. Stoddart, J. Chem. Soc., Perkin Trans. I, 1727 (1982).
133. M. Rothe, E. Bigdeli, unpublished data.
134. M. Rothe, Makromol. Chem., **35**, 183 (1960).
135. M. Rothe, G. Ries, unpublished data.
136. M. Rothe, M. Schilling, unpublished data.
137. M. Rothe, M. Lohmüller, U. Breuksch, G. Schmidtberg, Angew. Chem., **106**, 2047 (1994); Angew. Chem., Int. Ed. Engl., **33**, 1960 (1994).
138. M. Rothe, T. Mohr, B. Trnka, R. Berginski, unpublished data.
139. M. Rothe, E. W. Reinold, unpublished data.
140. M. Rothe, H. Ritsche-Thoma, unpublished data.
141. M. Rothe, E. Gatter, unpublished data.
142. M. Rothe, K. Steiner, W. Knobloch, unpublished data.
143. M. Rothe, K. Kitzelmann, unpublished data.
144. M. Rothe, N. Grüner, unpublished data.
145. M. Rothe, M. Gehrmann, unpublished data.
146. H. P. Großmann, J. Schwede, B. Heise, V. Rauschenberger, E. W. Reinold, M. Rothe, Polymer Bull., **32**, 653 (1994).
147. J. Dybal, B. Schneider, D. Doskocilová, J. Baldrian, H. Pavlíková, J. Kvarda, I. Prokopová, Polymer, **38**, 2483 (1997).
148. G. Di Silvestro, P. Sozzani, S. Brückner, L. Malpezzi, C. Guaita, Makromol. Chem., **188**, 2745 (1987).
149. M. Rothe, G. Dangel, G. Ries, unpublished data.

8.2. OLIGO(PEPTIDES)

8.2.1. OLIGO(PEPTIDES) OF GLYCINE

8.2.1.1. Linear Oligo(peptides) $H[NHCH_2CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	75.1	233–236	1
2	132.1	210–215	2,7,75
3	189.2	235	3,4,75
4	246.2	240	4,5,75
5	303.3	270	4,5,75,78
6	360.3	>256 d	6

8.2.1.2. Cyclic Oligo(peptides) $\ulcorner[NHCH_2CO]_n\urcorner$

n	Mol. wt.	m.p. (°C)	Refs.
2	114.1	309	2,7,79,131,174,177,184
4	228.2	340	8,9,77,79,80,191,205,206
5	285.3	>330	9,77–80,136
6	342.3	>360	4,8,10,11,77,79,80,131
7	399.4	>280	77,79,131
8	456.4	>350	77,79,131,205
9	513.5	290	77,79,131
10	570.5	290	77,79,131
11	627.6	300	77,79,131

8.2.2. OLIGO(PEPTIDES) OF SARCOSINE

8.2.2.1. Linear Oligo(peptides) $H[N(CH_3)CH_2CO]_nOH$

n	Mol. wt.	m.p. (°C)	Refs.
1	89.1	211–213	1
2	160.2	190–191	13,17,21

8.2.2.2. Cyclic Oligo(peptides) $\ulcorner[N(CH_3)CH_2CO]_n\urcorner$

n	Mol. wt.	m.p. (°C)	Refs.
2	142.2	147	13,27,51,61,177,187,191,204
3	213.3	221	51,191,192,197
4	284.4	>350	51,166,191–195,198
5	355.5	255	51,191–193,199
6	426.5	315	17,51,163,166,191,193,200
7	497.7	296	51,191,193,201
8	568.7	338	51,191,193,196,202
9	639.8	>320	17,163
10	711.0		193,203
12	853.0	>320	17,163

8.2.3. OLIGO(PEPTIDES) OF L-ALANINE

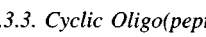
8.2.3.1. Linear Oligo(peptides) H[NHCH(CH_3)CO]$_n$OH

			Specific rotation				
n	Mol. wt.	m.p. (°C)	$[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
1	89.1		+14.5	25	10	6.0 N HCl	1
2	160.2	298	−38.5	27	0.9	0.2 N HCl	6,12,15,73,85,138,139
3	231.3	257–263	−85.7	27	0.6	0.2 N HCl	14,15,73,85,138,139
4	302.3	269–272	−131.0	27	0.6	0.2 N HCl	15,16,73,85,138,139
5	373.4	>320	−150.0	27	0.5	0.2 N HCl	15,16,85,138
6	444.5	>320	−167.0	27	0.3	0.2 N HCl	15,16,85,138
7	515.6	>320					15

8.2.3.2. Endgroup Protected Oligo(peptides) R–[NHCH(CH_3)CO]$_n$–R′

					Specific rotation			
R	R′	*n*	Mol. wt.	m.p. (°C)	$[\alpha]_D^{25}$	*c* (%)	Solvent	Refs.
BOC	OMe	2	274.3	110–111	−71.8[a]		HFIP	86,95,102,103
		3	345.4	193–194	−84.8[a]		HFIP	86,87,95,102,103
		4	416.5	>240 d	−84.0[a]		HFIP	86,87,95,102,103
		5	487.6	>240 d	−83.1[a]		HFIP	86,87,95,102,103
		6	558.6	>240 d	−82.0[a]		HFIP	86,87,102,103
		7	629.7	>240 d	−79.5[a]		HFIP	86,87,102,103
Z[b]	OEt	2	322.4	116	−51.1	0.5	TFE	81–83
		3	393.4	192	−73.0	0.5	TFE	14,81–83
		4	464.5	251–253	−95.1	0.5	TFE	81
		5	535.6	250	−107.1	0.5	TFE	81–83
		6	606.7	>275 d	−119.3	0.25	TFE	81
		7	677.8	>275 d	−125.4	0.25	TFE	81
		10	891.0	>250 d	−136.9	0.2	TFE	82,83
	Mo	2	363.4	135–136	−35.4	0.47	TFE	81
		3	434.5	198–200	−62.5	0.48	TFE	81
		4	505.6	270–271	−78.0	0.46	TFE	81
		5	576.7	>275 d	−92.0	0.45	TFE	81
		6	647.7	>275 d	−102.5	0.45	TFE	81
		7	718.8	>275 d	−108.1	0.45	TFE	81
		8	789.9	>275 d	−1171.1	0.44	TFE	81
		9	861.0	>275 d	−121.6	0.20	TFE	81
MEEA	OEt	3	419.5	138–140	−70.7	0.55	TFE	81
		4	490.6	240–241	−89.2	0.49	TFE	81
		5	561.6	>250 d	−98.5	0.20	TFE	81
		6	632.7	>250 d	−108.2	0.45	TFE	81
		7	703.8	>250 d	−119.0	0.50	TFE	81
	Mo	3	460.5	110–111	−49.9	0.44	TFE	81,84
		5	602.7	250 d	−87.5	0.40	TFE	81,84
		6	673.8	>275 d	−94.5	0.21	TFE	81,84
		7	744.9	>275 d	−102.3	0.23	TFE	81,84
		8	816.0	>275 d	−104.4	0.25	TFE	81,84
		9	887.1	>275 d	−110.2	0.22	TFE	81,84

[a] $t = 22°C$.
[b] Z-, $C_6H_5CH_2$–OCO–; BOC-, $(CH_3)_3C$–OCO–; MEEA-, $CH_3OCH_2CH_2OCH_2CH_2OCH_2CO$–; –OMe, $-CH_3$; –OEt, $-OC_2H_5$; –Mo, ; TFE, trifluorethanol; HFIP, Hexafluoroisopropanol.

8.2.3.3. Cyclic Oligo(peptides) 

			Specific rotation				
n	Mol. wt.	m.p. (°C)	$[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
2	142.2	297	−29.6	20	1.9	H_2O	6,79,174,183–185
4	284.3	250	−68	22	0.5	DMF	134,135,195,206
6	426.5	>310	−210.4	20	0.02	H_2O	136,137

8.2.3.3. cont'd

			Specific rotation				
n	Mol. wt.	m.p. (°C)	$[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
7	497.6	302	−60.5	20	2.1	H_2O	79
8	568.6	331	−62.1	20	1.4	H_2O	79
9	639.7	321	−97.1	20	1	H_2O	79
10	710.8	368	−102.7	20	0.1	H_2O	79
11	781.8	289	−133.0	20	0.002	H_2O	79

8.2.4. OLIGO(PEPTIDES) OF α-AMINOISOBUTYRIC ACID $C_6H_5CH_2OCO-[NHC(CH_3)_2CO]_n-OX$

X	*n*	Mol. wt.	m.p. (°C)	Refs.
$C(CH_3)_3$	1	293.4	60–1	88,90,91,94
	2	378.5	134	90,91,94
	3	463.6	166.5–167.5	88,91–94
	4	558.7	178	88,89,91–94
	5	633.8	236	88,89,91–94
	6	718.9	215–218	88,91
	7	804.0	229–230	91
	8	889.1	247–248	91
	9	974.2	262–263	91
$C(CH_3)_3$	10	1059.3	257–258	91
	11	1144.4	271–272	91
	12	1229.5	>300	91
	1	237.2	88–89	90,151,152
	2	322.4	161–162.5	151
	3	407.5	205–206	88,92
	4	482.5	243–244	88,92
	5	577.7	254–256	88,92
	6	662.8	243–245	88

8.2.5. OLIGO(PEPTIDES) OF HYDROPHOBIC AMINO ACIDS (α-AMINOBUTYRIC ACIDS, NORVALINE, VALINE, NORELEUCINE, LEUCINE, ISOLEUCINE, CYCLOHEXYLALANINE, PHENYLALANINE, TRIYPTOPHAN)

8.2.5.1. Linear Oligo(peptides) $H[NHCHRCO]_nOH$

			Specific rotation				
R	*n*	Mol. wt.	$[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
L-Val	1	117.2	+28.8	20	3.4	6 N HCl	1
$-CH(CH_3)_2$	2	216.3	+10.8	25	2.0	H_2O	18,19
	3	315.4	−41.8	21	2.7	1 N Cl	19
L-Leu	1	131.2	+13.9	25	9.1	4.5 N HCl	1
$-CH_2CH(CH_3)_2$	2	244.3	−13.7	23	1.0	1 N NaOH	18,22–24,74
	3	357.5	−51.4	20	3.1	1 N NaOH	26
	4	470.7	−90.0	20	7.6	1 N NaOH	26
L-Ile	1	131.2	+40.6	20	1	6 N HCl	1
$-CH(CH_3)CH_2CH_3$	2	244.3	+17.1	25	1	H_2O +	20
					1 eq. HCl		

8.2.5.2. Endgroup Protected Oligo(peptides) $(CH_3)_3COCO-[NHCHRCO]_n-OCH_3$

			Specific rotation			
R	*n*	Mol. wt.	m.p. (°C)	$[\alpha]^T$	Solvent	Refs.
L-Abu	2	302.3	108–109	−5.95	HFIP[a]	95
$-CH_2CH_3$	3	387.5	126–127	−10.84	HFIP[a]	95
	4	472.6	228–229	−16.72	HFIP[a]	95
	5	557.7	>250	−20.44	HFIP[a]	95

8.2.5.2. cont'd

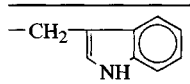
R	*n*	Mol. wt.	Specific rotation			Refs.
			m.p. (°C)	$[\alpha]^T$	Solvent	
	6	642.8	>250	−23.12	HFIP[a]	95
	7	727.8	>250	−27.21	HFIP[a]	95
L-Nva	2	330.4	95–96			98,102,103,105,113,153
$-CH_2CH_2CH_3$						
	3	429.5	119–120			95,97,98,102,103,105,113,153
	4	528.7	207–208			95,97,98,102,103,105,113,153
	5	627.8	220			95,97,98,102,103,153
	6	726.9	220			97–99,102,103,153
	7	826.1	220			97,98,102,103,153
L-Val	2	330.4	166–167	−48.3	TFE[b]	96,102,103,112,113,115
$-CH(CH_3)_2$				−40.8	HFA[b]	
	3	429.5	170–171	−76.7	TFE[b]	96,102,103,112,113,115
				−77.4	HFA[b]	153
	4	528.7	247–248	−91.0	TFE[b]	96,102,103,112,113,115
				−107.5	HFA[b]	153
	5	627.8	>260	−103.6	TFE[b]	96,102,103,112,115,153
				−126.2	HFA[b]	
	6	726.9	>260	−94.3	TFE[b]	96,102,103,153
				193.0	HFA[b]	
	7	826.1	>260	+37.8	TFE[b]	96,102,103,112,153
				−144.5	HFA[b]	
L-Nle	2	358.5	63–64	−3.88	HFIP[c]	95,106
$-CH_2CH_2CH_2CH_3$	3	471.6	146–147	−6.51	HFIP[c]	95,106
	4	584.8	214–215	−6.42	HFIP[c]	95,106
	5	698.0	>240	−7.74	HFIP[c]	95,106
	6	811.1	>240	−7.26	HFIP[c]	106
	7	924.3	>240	−7.51	HFIP[c]	106
L-Leu	2	358.5	141–142	−46.9	HFA[d]	100,102,104,111
$-CH_2CH(CH_3)_2$	3	471.6	158–159	−75.1	HFA[d]	100,102,104,111,153
	4	584.8	208–209	−92.6	HFA[d]	100,102,104,111,153
	5	698.0	>250	−101.0	HFA[d]	100,102,104,111,153
	6	811.1	>250	−111.1	HFA[d]	100,102,104,111,153
	7	924.3	>250	−120.0	HFA[d]	100,102,104,111,153
L-Ile	2	358.5	158–159	−44.0	TFE[e]	56–58,101,102,112,115,116
$-CH(CH_3)CH_2CH_3$	3	471.6	193–194	−67.6	TFE[e]	56–58,101,102,112,115,116
	4	584.8	245	−83.4	TFE[e]	56,57,101,102,112,115,116
	5	698.0	>250 d	−87.8	TFE[e]	56,57,101,102,112,115,116
	6	811.1	>250 d	−93.9	TFE[e]	56,57,101,102,112,115,116
	7	924.3	>240 d	+13.9	TFE[e]	56,101,102
	8	1037.5	>240 d	+24.6	TFE[e]	56,101
D-aIle	2	358.5	133–134	+30.1	TFE[f]	115,116
$-CH(CH_3)CH_2CH_3$				+39.9	HFIP[g]	
	3	471.6	175–176	+48.1	TFE[f]	115,116
				+58.3	HFIP[g]	
	4	584.8	245	+58.3	TFE[f]	115,116
				+73.5	HFIP[g]	
	5	698.0	>250 d	+55.0	TFE[f]	115,116
				+94.8	HFIP[g]	
	6	811.1	>250 d	+55.3	TFE[f]	115,116
				+101.8	HFIP[g]	
	7	924.3	>250 d	+109.4	HFIP[g]	115
L-Cha	2	438.6	117–118	−43.3	HFIP[h]	110,111,115
$-CH_2-$	3	591.8	171–172	−54.1	HFIP[h]	110,111
	4	745.1	203–204	−57.7	HFIP[h]	110,111
	5	898.3	>240	−61.2	HFIP[h]	110,111,114
	6	1051.5	>240	−65.6	HFIP[h]	110,111
	7	1204.7	>240	−66.9	HFIP[h]	110,111
L-Phe	2	426.5	121–122	0.00	TFE[i]	102,107–109,111,112
				−8.7	DMF[i]	
$-CH_2-$	3	573.7	173–174	−15.3	TFE[i]	102,107–109,111,112
				−14.9	DMF[i]	

8.2.5.2. cont'd

R	n	Mol. wt.	Specific rotation			Refs.
			m.p. (°C)	$[\alpha]^T$	Solvent	
	4	720.8	196–197	−22.2	TFE[i]	102,107–109,111,112
				−16.1	DMF[i]	
	5	868.0	213–214	−28.3	TFE[i]	102,107–109,111,112
				−19.2	DMF[i]	
	6	1015.2	223–226	−25.0	TFE[i]	102,108,109,111
				−21.1	DMF[i]	
	7	1162.3	>235	−22.1	DMF[i]	102,108,109,111
	8	1309.5	>235			108,109
	9	1456.7	>235			108,109

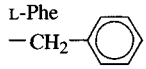
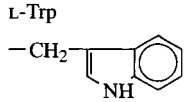
[a] $\lambda = 546$ nm; $T = 20°C$; $c = 5 \times 10^{-4}$ *M*; $\alpha[(\pi/180)$ rad $\times$ cm^2/g]; HFIP = hexafluoroisopropanol.
[b] $\lambda = 589$ nm; $T = 25°C$; $c = 3 \times 10^{-3}$ *M* ($n = 2$–6), 3×10^{-4} *M* ($n = 7$); TFE = trifluoroethanol; HFA = hexafluoroacetone sesquihydrate.
[c] $\lambda = 589$ nm; $T = 22°C$; $c = 10^{-3}$ *M*; HFIP = hexafluoroisopropanol.
[d] $\lambda = 589$ nm; $T = 25°C$; $c = 3 \times 10^{-3}$ *M*; HFA = hexafluoroacetone sesquihydrate.
[e] $\lambda = 589$ nm; $T = 25°C$; $c = 0.025$–0.2%; TFE = trifluoroethanol.
[f] $\lambda = 589$ nm; $T = 25°C$; $c = 0.6\%$; TFE = trifluoroethanol.
[g] $\lambda = 589$ nm; $T = 21°C$; $c = 0.1\%$; hexafluoroacetone sesquihydrate.
[h] $\lambda = 589$ nm; $T = 20°C$; $c = 0.1$–0.2%; HFIP = hexafluoroisopropanol.
[i] $\lambda = 589$ nm; T in TFE = 23°C, in DMF = 18°C; $c = 0.2\%$; TFE = trifluoroethanol.

8.2.5.3. Endgroup Protected Oligo(peptides) of Tryptophan X–[NHCHRCO]$_n$–OCH_3

R	X	n	Mol. wt.	Specific rotation				Refs.
				m.p. (°C)	$[\alpha]_D^{25}$	c (%)	Solvent	
	Z[a]	2	538.5	196	−13	1	MeOH	208,225
		3	721.8	85–95				208
	$(CH_3)_2PS$	1	276.3	108–109	−2.4	1.0	MeOH	140
		2	428.4	153–154	−47.6	0.5	MeOH	140
		3	580.6		−66.8	0.5	MeOH	140
		4	732.7		−65.6	0.5	MeOH	140
		5	884.9	151–154	−65.0	0.5	MeOH	140

[a] Z- = Benzyloxycarbonyl-.

8.2.5.4. Cyclic Oligo(peptides) –[NHCHRCO]$_n$–

R	n	Mol. wt.	Specific rotation					Refs.
			m.p. (°C)	$[\alpha]_D$	(°C)	c (%)	Solvent	
L-Val	2	198.3	269–271	−77.5	18–22	1.0	CH_3COOH	207
$-CH(CH_3)_2$								
L-Leu	2	226.3	277	−46.0	20	0.8	CH_3COOH	22,25,177,184,185, 188,207
$-CH_2CH(CH_3)_2$								
	5	565.8	>300	−71	15	1.0	EtOH	132
L-Ile	2	226.3	308	−60.0	18–22	1.0	CH_3COOH	131,207
$-CH(CH_3)CH_2CH_3$	5	565.8	357					131
L-Phe	2	294.4	315–316	−100	25–28	0.2	CH_3COOH	177,184,185,188
	5	735.9	250–254	−124.3	25	1.0	DMF	78
	10	1471.8	251.5	−142	20	0.4	MeOH	131
L-Trp	2	372.4		−14.7	20		CH_3COOH	230,232
								

8.2.5.5. Endgroup Protected Oligo(Dipeptides) $R'[NHCHR_1CO-NHCHR_2CO]_n-R''$[a]

R′	R″	R_1	R_2	n	Specific rotation: Mol. wt.	m.p. (°C)	$[\alpha]_D^{25}$	c (%)	Solvent	Refs.
Boc[a]	OMe	D-Val	L-Val	1	300.4	110.7	−23.7	1.02	$CHCl_3$	118
		L-$CH(CH_3)_2$	D-$CH(CH_3)_2$	2	528.7	209.6	−17.8	1.00	$CHCl_3$	118
				3	727.0	232.7	−21.0	1.02	$CHCl_3$	118
				4	925.2	261.1	−8.9	1.00	$CHCl_3$	117,118
				6	1321.8	d	−6.2	1.12	$CHCl_3$	118
				8	1718.3	d	−11.4	1.05	$CHCl_3$	118
Nps[a]	OEt	L-Leu	L-Ala	2	567.6	110–111	−66.5	2.0	HFIP/TFE[a] (1:2)	119
		$-CH_2CH(CH_3)_2$	$-CH_3$	3	751.8	203–205	−96.9	2.0	HFIP/TFE (1:2)	119
				4	936.0	257–260 d	−101.3	2.0	HFIP/TFE (1:2)	119
				5	1120.2	>270 d	−84.6	2.0	HFIP/TFE (1:2)	119
				6	1304.4	>270 d	−50.0	2.0	HFIP/TFE (1:2)	119
				8	1672.8	>270 d	−52.6	2.0	HFIP/TFE (1:2)	119

[a] Boc- = *tert.* butyloxycarbonyl-; Nps. = *o*-nitrophenylsulfenyl-; HFIP = hexafluoroisopropanol; TFE = trifluoroethanol.

8.2.6. OLIGO(PEPTIDES) OF L-AND D-PROLINE

8.2.6.1. Linear Oligo(all-L-peptides) $H[N(CH_2)_3CHCO]_nOH$

n	Mol. wt.	m.p. (°C)	E_F-Values[a]				$[\alpha]_D^{20}$[b]	Refs.
1	115.2	220–222	1.00				−87	1
2	212.3	144–145	0.84				−175	52,62,69,70,150,171–173
3	309.4	122	0.72				−220	53,62,69,70,150,171–173
4	406.5	170	0.60				−293	62,69,70,150,171–173
5	503.6	189	0.52	1.00			−338	62,69,70,150,172,173
6	600.7	209		0.91			−376	62,69,70,150,172,173
7	697.8	228		0.77			−394	62,69,70,150
8	794.9	>280		0.72			−412	62,69,70,150
9	892.0	>280		0.66			−435	62,69,150
10	989.1	>300		0.60	1.00		−456	62,69,70,150
11	1086.2	>300			0.92		−466	62,69,70,150
12	1183.3	>300			0.83		−472	62,69,70,150
13	1280.4	>300			0.75	1.00	−480	69,70
14	1377.5	>300			0.70	0.94	−491	69
15	1474.6	>300			0.64	0.90	−496	62,69,70
16	1571.7	>300				0.86	−499	69,70
17	1668.8	>300		1.00		0.81	−505	69
18	1765.9	>300		0.95			−509	69,70
19	1863.0	>300		0.90			−513	69
20	1960.1	>300		0.84			−520	69,70
21	2057.3	>300					−521	70
22	2154.5	>300	100				−523	70
25	2445.7	>300	0.93	1.00			−526	70
28	2737.0	>300		0.92	1.00		−528	70
30	2931.3	>300			0.95		−530	70
31	3028.5	>300			0.93	1.00	−532	70
34	3320.1	>300	1.00			0.93–530	70	
37	3611.4	>300	0.93	1.00			−528	70
40	3902.7	>300		0.94			−532	70

[a] Paper electrophoresis; buffer; $HCOOH/CH_3COOH/H_2O$ (1:1:3), 1000–5000 V, 40–115 mA, 40–240 min.
[b] c (%) = 1, H_2O.

8.2.6.2. Endgroup Protected Oligo(peptides)

8.2.6.2.1. *tert*-BUTYLOXYCARBONYL OLIGO(ALL-L-PROLINES) $(CH_3)_3COCO-[N(CH_2)_3CHCO]_n-OH$

n	Mol. wt.	m.p. (°C)	R_F^a	$[\alpha]_D^{20}$[b]	Refs.
1	215.3	137–138	0.84	−105	62,72,169,179
2	312.4	191–193	0.68	−153.5	62,150,167,179,181
3	409.5	219–220	0.60	−173	62,150,167,179
4	506.6	178	0.53	−194	62,150,167

8.2.6.2.1. *cont'd*

n	Mol. wt.	m.p. (°C)	R_F^a	$[\alpha]_D^{20\,b}$	Refs.
5	603.7	225	0.47	−220	62,150,167
6	700.8	247	0.41	−239	62,150,167
7	797.9	267	0.37	−247	62,150
8	894.0	>280	0.33	−256	62,150
9	991.1	>280	0.29	−265	62,150
10	1088.2	>280	0.25	−272	62,150
11	1185.3	>300	0.23	−269	62,150
12	1282.4	>300	0.22	−277	62,150
15	1573.7	>300	0.22	−291	62,150

[a] TLC: solvent, *n*-butanol/acetone/acetic acid/ammonia, 1:4/water (9:3:2:2:4).
[b] *c* (%) = 1, $CHCl_3$.

8.2.6.2.2. *tert*-AMYLOXYCARBONYL OLIGO(ALL-L-PROLINES) $C_5H_{11}OCO-[N-CHCO]_n-OH$

n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
1	229.3	94–95.5	−47.2	21	1.7	EtOH	63–68,170,179
2	326.4	157–159	−113	25	2.0	EtOH	64–68,179
3	423.5	200–201.5	−170	25	2.2	EtOH	64–68,179,182
4	520.6	207.5–208.5	−227	24	2.1	EtOH	64–68
5	617.7	224.5–226	−248	24	2.3	EtOH	64–68
6	714.8	240–242	−291	24	2.1	EtOH	64–68
8	908.0	280	−318	22	0.9	EtOH	64–68

8.2.6.2.3. *tert*-BUTYLOXYCARBONYL D,L-OLIGO(DIPROLINES)

$(CH_3)COCO-[N-CHCO-N-CHCO]_n-OH$

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{22a}$	Refs.
1	312.4	204	−70.1	158,165
2	506.6	188–189	−38.8	158
3	700.8	220–221	−32.4	158,159

[a] *c* (%) = 1, DMF.

8.2.6.2.4. *tert*-BUTYLOXYCARBONYL OLIGO(D,L-DIPROLINE) BENZYL ESTERS

$(CH_3)_3COCO-[N-CHCO-N-CHCO]_n-OCH_2C_6H_5$

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{22a}$	Refs.
2	612.7	64–66	−24.0	164
3	806.9	157–160	−23.2	164
4	1001.2	146–148	−20.4	164

[a] *c* (%) = 1, DMF.

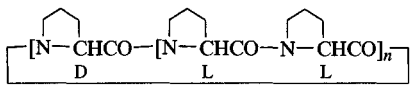
8.2.6.3. Cyclic Oligo(peptides)

8.2.6.3.1. CYCLIC OLIGO(ALL-L-PROLINES) $[N-CHCO]_n$

n	Mol. wt.	m.p. (°C)	$[\alpha]_D$	(°C)	*c* (%)	Solvent	Refs.
2	194.2	146	−151.6	19	0.5	H_2O	54,55,180,185
3	291.3	358	+48.5	25	0.48	CH_3OH	53,154–157,162,163,168,176,189,190

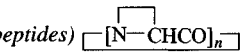
8.2.6.3.2. CYCLIC OLIGO(L,D-DIPROLINES) $[N-CHCO-N-CHCO]_n$ (L, D)

n	Mol. wt.	m.p. (°C)	Refs.
1	194.2	194–199	158,180,185,186
2	388.4	>300 d	158
3	582.6	>310 d	158, 159
4	776.8	>250 d	158
6	1165.2	>280 d	158
9	1747.8	>250 d	158
12	2330.4	>250 d	158

8.2.6.3.3. CYCLIC OLIGO(D,L,L-TRIPROLINES) 

n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D^{22}$	c (%)	Solvent	Refs.
1	291.2	181–184	+70.7	1	H_2O	159–163
2	582.3	304	−60.9	1	H_2O	159–163
3	873.5	230 d	−114.5	1	H_2O	159–163
4	1164.6	250 d	−72.0	1	H_2O	159–163
5	1455.8	250 d	−101.8	0.5	H_2O	159–163

8.2.7. OLIGO(PEPTIDES) OF L-AZETIDINE-2-CARBOXYLIC ACID

8.2.7.1. Cyclic Oligo(peptides) 

n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{578}$ [a]	Refs.
2	166.2	207–212	+8.2	175
3	249.3	334–340	+20.8	175
6	498.5	339–350	−352.0	175

[a] c (%) = 0.5, H_2O.

8.2.8. OLIGO(PEPTIDES) OF L-HYDROXY AND L-MERCAPTO AMINO ACIDS

8.2.8.1. Linear Oligo(peptides) H[NHCHRCO]$_n$OH

	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent [a]	Refs.
L-Ser	1	105.1		+14.5	25	9.3	N HCl	1
$-CH_2OH$	2	192.2		+14.2	25	7	N HCl	28,35,139
L-Tyr	1	181.2		−7.3	25	4	6.1 N HCl	1
$-CH_2C_6H_4OH$	2	344.4	285–288	+30.1	19	4	H_2O + 1 eq. HCl	29–31
	3	507.6	181–182					31
L-Cys	1	121.1		+7.6	26	12	N HCl	1
$-CH_2SH1.5$ HCl	2	224.2		+35	22	1	0.2 N HCl	37
L-Met	1	149.2		+23.4	20	5	3 N HCl	1
$-CH_2CH_2SCH_3$	2	280.4	229–231	+27.0	24	2	H_2O	32–34
	3	411.6	219–220	−70.0	16	1	H_2O	33

8.2.8.2. Endgroup Protected Oligo(peptides) $(CH_3)_3COCO$-[NHCHRCO]$_n$-OCH_3

R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	c (%)	Solvent [a]	Refs.
L-Cys(Me)	2	366.5	94–95				102,121,122,133
$-CH_2SCH_3$	3	483.7	104–105				102,121,122,133
	4	600.8	191–192				102,121,122,133
	5	718.0	211–212				102,121,122,133
	6	835.2	>220				102,121,122,133
	7	952.3	>220				102,121,122,133
L-Met	2	394.5	66–67	−29.6	0.003 M	HFIP [a]	102,123,124,127–129
$-CH_2CH_2SCH_3$	3	525.8	92–96	−37.6	0.003 M	HFIP	102,123,124,127–129
	4	656.9	191–194	−41.4	0.03 M	HFIP	102,123,124,127–129
	5	788.1	236 d	−48.1	0.003 M	HFIP	102,123,124,127–129
	6	919.3	245 d	−50.3	0.03 M	HFIP	102,123,124,126 –129
	7	1050.5	>275	−54.8	0.03 M	HFIP	102,123–129

[a] HFIP = hexafluoroisopropanol.

8.2.8.3. Formyl-Oligo(L-Methionines) HCO–[NHCHRCO]$_n$–OH

R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	c (%)	Solvent[a]	Refs.
L-Met	2	308.4	125–126	−44.8	0.5	HFIP	130
$-CH_2CH_2SCH_3$	3	439.6	176–177	−56.7	0.5	HFIP	130
	4	570.8	195–196	−67.2	0.5	HFIP	130
	5	702.0	216–217	−63.3	0.5	HFIP	130
	6	833.2	230–231 d	−81.8	0.5	HFIP	130

8.2.8.4. Cyclic Oligo(peptides) ⌐[NHCHRCO]$_n$⌐

R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent[a]	Refs.
$-CH_2OH$	2	174.2	247	67.5	25	2.2	H_2O	35,174,183
$-CH_2C_6H_4OH$	2	326.4	277–280	−223.8	20	2.4	NaOH	36,174

8.2.8.5. Oligo(peptides) of Cystine CyS = [−NHCHCO− / CH$_2$S−]

R	n	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	Solvent[a]	Refs.
H−CyS−CyS−OH	222.3	−29	25	1	N HCl	38
H−CyS−CyS−OH H−CyS−CyS−OH	444.6	−58.7	26	1	N HCl	37,38

[a] HFIP = hexafluoroisopropanol.

8.2.9. OLIGO(PEPTIDES) OF ACIDIC AMINO ACIDS

8.2.9.1. α- and ω-Oligo(peptides) of L-Aspartic Acid and L-Glutamic Acid H[NH–X–CO]$_n$OH

X	n	Mol. wt.	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent	Refs.
Asp	2	248.2	+18.5	20	2	0.5 N HCl	76
−CHCH$_2$− COOH	3	363.3	+6.0	20	2	0.5 N HCl	76
	4	478.3	+16.1	20	1	0.5 N HCl	76
	5	593.4	+6.1	20	1	0.5 N HCl	76
	6	708.5	+8.7	20	2	0.5 N HCl	76
Glu	1	147.1	+31.2	22.4	1	6 N HCl	1
−CH− CH$_2$CH$_2$COOH	2	276.3	+18.2	24	1–2	0.5 N HCl	42–44
	3	405.4	−7.2	19	1.4	H_2O	45
Glu	2	276.3	+4.2	24	1–2	0.5 N HCl	43,44,46,47,59,178
−CHCH$_2$CH$_2$− COOH	3	405.4	−7.2	20	2	0.5 N HCl	48,59,178
	4	534.5	−16.0	24	1–2	0.5 N HCl	59,178
	5	663.6	−14.8	20	1–5	0.5 N HCl	59
	6	792.7	−17.2	20	1–5	0.5 N HCl	59
	7	921.8	−19.1	20	1–5	0.5 N HCl	59,71

8.2.9.2. γ-Oligo(peptides) of D-*Glutamic Acid Esters* $H[NH-X-CO]_n-OC(CH_3)_3$

X	n	Mol. wt.	Specific rotation			Solvent	Refs.
			m.p. (°C)	$[\alpha]_D^{25}$	c (%)		
D-Glu	3	629.8	81–84	+17.9	3	MeOH	59
—CHCH$_2$CH$_2$—	4	815.0	125–127.5	+22.3	3	MeOH	59
COOC(CH$_3$)$_3$	5	1000.2	126–131	+27.6	3	MeOH	59
	6	1185.4	>58	+29.4	3	MeOH	59

8.2.9.3. α-Oligo(peptides) of γ-Unsubstituted L-*Glutamic Acid Derivatives* $CH_3CO-[NHCH(CH_2CH_2COOH)CO]_n-NHC_2H_5$

n[a]	Mol. wt.	Specific rotation		Solvent	Ref.
		$[\alpha]_D^{25}$	c (%)		
1	216.2	−23	1	H_2O	226
2	345.4	−36	1	H_2O	226
3	474.3	−49.5	1	H_2O	226

[a] $n = 2–12$: ion-exchange chromatography, IR, CD, MS, pK values, Refs. 227,228.

8.2.9.4. α-Oligo(peptides) of Benzyloxycarbonyl L-*Aspartic Acid and* L-*Glutamic Acid Ethyl Esters* $C_6H_5CH_2OCO-[NHCHRCO]_{n-1}-NHCHR'COOC_2H_5$

R	R′	n	Mol. wt.	m.p. (°C)	Specific rotation			Refs.
					$[\alpha]_D^{25}$	c (%)	Solvent[a]	
Asp		2	452.5	80–81	+17.9	3.1	DCA	49,146
–CH$_2$COOCH$_3$	–CH$_2$COOC$_2$H$_5$	3	581.6	127–128	−1.01	3.1	DCA	49,146
		4	710.7	143–144	−13.1	1.1	DCA	49,146
		5	839.8	161–163	−19.1	0.1	DCA	49,146
		6	968.9	175–178	−26.6	0.7	DCA	49
		8	1227.2	207 d	−35.1	0.5	DCA	49,141,146
		11	1614.5	224 d	−42.9	0.45	DCA	49,146
		14	2001.9	233 d	−46.0	0.3	DCA	49,146
Glu		2	480.5	86	−12.4	2	DCA	50,141,146
–CH$_2$CH$_2$COOCH$_3$	–CH$_2$CH$_2$COOC$_2$H$_5$	3	623.6	124	−18.0	2	DCA	50,146
		4	766.8	139	−21.3	2	DCA	50,141,146
		5	910.0	200	−22.7	2	DCA	50,141,146
		6	1053.1	250	−26.7	2	DCA	50,146
		7	1196.3	259	−28.7	2	DCA	50,141,146
		9	1482.5	d	−32.6	2	DCA	50,146
		11	1768.8	d	−35.6	2	DCA	50,146
Glu		2	494.5	104–105	−17.9		TFE	43,120,142–145
–CH$_2$CH$_2$COOC$_2$H$_5$	–CH$_2$CH$_2$COOC$_2$H$_5$	3	651.7	127–128	−27.1	2	TFE	43,120,142–145
		4	808.9	137–139	−30.9		TFE	142–145
		5	966.0	177–178.5	−26.5		TFE	142–145
		7	1280.4	>235	−20.9	1	TFE	142–144
		8	1437.5	>235	−16.5		TFE	142,143,145
		9	1594.7	>235	−7.6	1	TFE	142,143,145

[a] DCA, dichloroacetic acid; TFE, trifluoroethanol.

8.2.9.5. ω-Oligo(peptides) of Benzyloxycarbonyl L-*Aspartic Acid and* L- *and* D-*Glutamic Acid ω-Alkyl Esters* $C_6H_5CH_2OCO-[NH-X-CO]_n-OR$

X	R	n	Mol. wt.	m.p. (°C)	Specific rotation				Refs.
					$[\alpha]_D$	(°C)	c (%)	Solvent[a]	
L-Asp	C(CH$_3$)$_3$	2	550.7	70–72	−23.0	25	1	DMF	76
—CHCH$_2$—		3	721.9	93–95	−22.1	25	1	DMF	76
COOC(CH$_3$)$_3$		4	893.1	97–99	−19.3	25	1	DMF	76
		5	1064.3	121–123	−18.7	25	1	DMF	76
		6	1235.4	157–162	−20.0	25	1	DMF	76

8.2.9.5. cont'd

X	R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent[a]	Refs.
L-Glu	CH_3	2	452.5	97–98	−25.2	20	5	DMF	59
$-CHCH_2CH_2-$		3	595.5	135–137	−26.1	20	5	DMF	59
$COOCH_3$		4	738.6	135–139	−28.2	20	5	DMF	59
		5	881.7	175–179	−30.1	20	5	DMF	59
		6	1024.8	195–198	−31.5	20	5	DMF	59
		7	1153.9	206–213	−32.8	20	5	DMF	59
D-Glu	$C(CH_3)_3$	2	578.7	88.5–91	+27.4	25	3	MeOH	59
$-CHCH_2CH_2-$		3	763.9	77.5–81.5	+30.6	25	3	MeOH	59
$COOC(CH_3)_3$		4	949.1	95–97	+34.5	25	3	MeOH	59
		5	1134.3	99–102	+36.1	25	3	MeOH	59
		6	1319.6	73–77	+42.4	25	3	MeOH	59
		7	1504.8	74–78	+42.9	25	3	MeOH	59
L-Glu	$C(CH_3)_3$	2	578.7	83–84	−25.4	22	1	MeOH	71
$-CHCH_2CH_2-$		3	763.9	79	−30.2	21	1	MeOH	71
$COOC(CH_3)_3$		4	949.1	87–89	−33.2	22	1	MeOH	71
		5	1134.3	102–103.5	−36.8	23	1	MeOH	71
		6	1319.6	123–125	−39.9	22.5	1	MeOH	71
		7	1504.8	130–132	−43.7	21	1	MeOH	71

8.2.9.6. α-Oligo(peptides) of Acetyl L-Glutamic Acid Esters $CH_3CO-[NHCH(CH_2CH_2COOC_2H_5)CO]_nOC_2H_5$

n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D^{25}$	c (%)	Solvent[a]	Ref.
2	402.4	94–95	−29.1	0.5–1.5	TFE	142
3	559.6	150–151	−31.2	0.5–1.5	TFE	142
4	716.8	171	−39.3	0.5–1.5	TFE	142
5	873.9	>230	−34.9	0.5–1.5	TFE	142
8	1345.5	>230	−17.3	0.5–1.5	TFE	142

[a] TFE, Trifluoroethanol

8.2.9.7. α-Oligo(peptides) of o-Nitrophenylsulfenyl L-Glutamic Acid Amides and Esters $Nps^a-[NHCH(CH_2CH_2COOCH_2C_6H_5)CO]_n-R$

R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c(%)	Solvent	Ref.
NHC_2H_5	1	417.5	92–93					147
	2	636.7	147–148					147
	4	1075.2	180–182					147
	6	1513.6	233–234					148
$OCH_2C_6H_5$	2	699.7	65–67	−40.0	20	1.0	$CHCl_3$	149
	4	1138.1	114–115	−26.2	22	1.0	$CHCl_3$	149
	8	2014.9	198–205	−8.2	25	0.9	$CHCl_3$	149
	12	2891.7	236–238	+7.1	26	0.5	$CHCl_3$	149

[a] Nps = *o*-Nitrophenylsulfenyl.

8.2.9.8. Cyclo-γ-Oligo(peptides) of Glutamic Acid $\ulcorner[NHCHCH_2CH_2CO]_n\urcorner$ (COOR)

R	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent	Ref.
H	2	258.2	189–192	+14.0	22	0.5	0.1 N HCl	60
	3	387.3	235–238	+90.0	22	0.5	0.1 N HCl	60
	4	516.5	232–236	+38.5	22	0.45	0.1 N HCl	60
$C(CH_3)_3$	2	370.4	220–222	+23.0	23	1	DMF	60
	3	555.7	203–206	−52.5	23	1	DMF	60
	4	740.9	230–232	−38.0	23	0.5	DMF	60

8.2.10. OLIGO(PEPTIDES) OF BASIC L-AMINO ACIDS (LYSINE, ORNITHINE, ARGININE, HISTIDINE)

8.2.10.1. Linear Oligo(peptides) H[NHCHRCO]$_n$OH

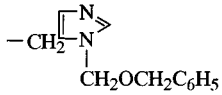
R	n	Mol. wt.	R_F[a]	Specific rotation $[\alpha]_D$	(°C)	c (%)	Solvent	Refs.
Lys	1	146.2	0.24	+25.7	25	1.6	6 N HCl	1
$-(CH_2)_4NH_2$	2	274.4	0.17	+5.6	25	2	6 N HCl	39,40
	3	402.6	0.12	−2.2	24	2	0.5 N HCl	40,41
	4	530.7	0.09					40
	5	658.9	0.06					40
	n = 2–20: Ion-exchange chromatography, Refs. 209–212; n = 2–22; ORD, pK values, Ref. 209; n = 2–5: CD, Ref. 217.							
Lys(Boc)[b]	1	246.3	237–255	+4.7	26	0.88	2 N NH_3	219
$-(CH_2)_4NH-$Boc	2	474.6	188	+16.4	20	1	MeOH	213,214
	3	702.9	185–187	+7.4	20	1	MeOH	214
Arg	2	330.4	224–226[a]	+1780[c,d]		1.4	N HCl	216
$-(CH_2)_3-NHCNH_2$	3	486.6	227–232[b]	+1540[c,d]		3.9	N HCl	126
‖ NH	4	642.7	238–241[b]	+1030[c,d]		1.8	N HCl	216

[a] TLC in *n*-butanol/acetic acid/water/pyridine (30 : 6 : 24 : 20).
[b] Boc = *tert* butyloxycarbonyl-.
[c] As flavianates.
[d] λ = 224 nm

8.2.10.2. Endgroup Protected Oligo(peptides) X-[NHCHRCO]$_n$-Y

R	X	Y	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_\lambda$	λ (nm)	(°C)	c (%)	Solvent	Refs.
Lys(Boc)[b]	Z	OCH_3	1	394.5	Oil	−10.6	589	26	1.9	Acetone	219
$-(CH_2)_4NH-$Boc			2	622.8	97–98	−5.5	589	20	2.1	Acetone	218–220
			4	1079.4	121	−15.3	589	22	2.1	CH_3COOH	220
			8	2192.6	>260	−16.4	589	22	2.1	CH_3COOH	220
	Z	$OC(CH_3)_3$	2	664.8	82–84	−10.4	578	22	1	DMF	213
			4	1121.4	143–145	−15.5	578	20	1	DMF	213
	Z	$NHCH_3$	2	721.8	129	−14.4	589	22		CH_3COOH	220
			4	1078.3	174	−16.6	589	22		CH_3COOH	220
			8	1991.5	>260	−18.0	589	22		CH_3COOH	220
			16	3817.7	>260	−14.6	589	22		CH_3COOH	220
	Z	OH	1	380.4	Oil	−2.4	589	26	2.0	Acetone	219
			2	790.0	112	−3.1	578	22	1	CH_3COOH	213
			3	837.0	107	−12.2	589	20	1	MeOH	214
	Boc	$OC(CH_3)_3$	7	1773.2	268	−19.0	578	22	1	DMF	213
	Boc	OH	3	802.0	114	−5.5	578	22	1	DMF	213
			4	1031.0	92–95	−15.4	589	20	1	MeOH	214
	H	Suco[a]	n = 1,5–8,10,20: TLC, IR, CD								
Lys(Z)[b]	Boc	$OCH_2C_6H_5$	2	732.8	80	−14.0	578		1	DMF	213
$-(CH_2)_4NH-$Z			3	995.1	140	−16.2	578		1	DMF	213
			4	1257.4	156–158	−15.7	578		1	DMF	213
			5	1519.7	170	−16.7	578		1	DMF	213
			7	2044.3	245–248	−13.9	578		1	DMF	213
Lys(Z)	Boc	OH	2	642.7	97–105						215
$-(CH_2)_4NH-$Z			4	1167.3	124–127						215
			8	2216.5	1235–140						215
Orn(Boc)	Z	OCH_3	1	380.4	71	−14.4	589	27	1.08	90% AcOH[b]	221
$-(CH_2)_3NH-$Boc			2	594.7	125	−15.1	589	20	1.34	90% AcOH	221,222
			3	809.0	147	−13.7	589	20	1.1	90% AcOH	222
Orn(Boc)			4	1023.2	152	−17.8	589	20	1	MeOH	221,222
			5	1237.5	165–171	−5.1	589	25	1.05	DMF	222
$Arg(NO_2)$	Z	OCH_3	1	367.4	153–153.5						216
$-(CH_2)_3-NHCNH_2$			3	769.7	133–136						216
‖ $N-NO_2$			4	970.9	139–140						216

8.2.10.2. cont'd

R	X	Y	n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_\lambda$	λ (nm)	(°C)	c (%)	Solvent	Refs.
His(π-Bom)[b]	BOc	OCH_3	3	904.1	67–69	−15.5	589	20	0.49	MeOH	223
			4	1161.4	78–81	−21.2	589	20	0.49	MeOH	223
			5	1418.7	93–97	−22.3	589	20	0.48	MeOH	223
			6	1676.0	117–122	−23.3	589	20	0.52	MeOH	223
			7	1933.3	155–163	−23.2	589	20	0.52	MeOH	223

[a] Suco = *p*-oxymethylbenzylcholestan-3-β-yl succinate.
[b] Z- = Benzyloxycarbonyl-; Boc- = *tert* butyloxycarbonyl-; π-Bom- = *N*(π)-benzyloxymethyl-; AcOH = acetic acid.

REFERENCES

1. J. P. Greenstein, M. Wintz, "Chemistry of the Amino Acids", vol. III, Wiley, New York, 1961.
2. E. Fischer, E. Fourneau, Chem. Ber., **34**, 2868 (1901).
3. E. Fischer, Chem. Ber., **36**, 2982 (1903).
4. M. Rothe, H. Bruenig, G. Eppert, J. Prakt. Chem., **8**, 323 (1959).
5. E. Fischer, Chem. Ber., **37**, 2486 (1904).
6. E. Fischer, Chem. Ber., **39**, 453 (1906).
7. H. F. Schott, J. B. Larkin, L. B. Rockland, M. S. Dunn, J. Org. Chem., **12**, 490 (1947).
8. R. Schwyzer, B. Iselin, W. Rittel, P. Sieber, Helv. Chim. Acta, **39**, 872 (1956).
9. M. Rothe, G. Luedke, unpublished data.
10. J. C. Sheehan, W. L. Richardson, J. Am. Chem. Soc., **76**, 6239 (1954); J. C. Sheehan, M. Goodman, W. L. Richardson, J. Am. Chem. Soc., **77**, 6391 (1955).
11. D. G. Ballard, C. H. Bamford, F. J. Weymouth, Proc. Roy. Soc. A, **227** 155 (1955).
12. B. F. Erlanger, E. Brand, J. Am. Chem. Soc., **73**, 3508 (1951).
13. F. Sigmund, E. Liedl, Hoppe-Seyler's Z. Physiol. Chem., **202**, 268, 275 (1931).
14. E. Brand, B. F. Erlanger, H. Sachs, J. Polatnick, J. Am. Chem. Soc., **73**, 3510 (1951).
15. H. Zahn, A. Meißner, Ann. Chem., **636**, 132 (1960).
16. E. Brand, B. F. Erlanger, H. Sachs, J. Am. Chem. Soc., **74**, 1849 (1952).
17. M. Rothe, R. Theyson, W. Taiber, unpublished data.
18. M. A. Nyman, R. M. Herbst, J. Org. Chem., **15**, 108 (1950).
19. S. Shankman, Y. Schvo, J. Am. Chem. Soc., **80**, 1164 (1958).
20. T. Sugimura, W. K. Paik, unpublished data.
21. S. P. Datta, R. Leberman, B. R. Rabin, Trans. Faraday Soc., **55**, 1982 (1959).
22. E. Fischer, Chem. Ber., **39**, 2893 (1906).
23. F. H. Carpenter, D. T. Gish, J. Am. Chem. Soc., **74**, 3818 (1952).
24. F. C. McKay, N. F. Albertson, J. Am. Soc., **79**, 4686 (1957).
25. M. M. Exner, R. J. Kostelnik, Biopolymers, **16**, 1387 (1977).
26. E. Abderhalden, R. Fleischmann, Fermentforsch., **9**, 524 (1928).
27. Y. Shibata, T. Asahina, Bull. Chem. Soc., Japan, **1**, 71 (1927).
28. J. S. Fruton, J. Biol. Chem., **146**, 463 (1942).
29. M. Bergmann, L. Zervas, L. Salzmann, H. Schleich, Hoppe-Seyler's Z. Physiol. Chem., **224**, 17 (1934).
30. J. L. Bailey, J. Chem. Soc., 3461 (1950).
31. A. E. Barkdoll, W. F. Ross, J. Am. Chem. Soc., **66**, 951 (1944).
32. C. A. Dekker, S. P. Taylor, Jr., J. S. Fruton, J. Biol. Chem., **180**, 155 (1949).
33. M. Brenner, R. W. Pfister, Helv. Chim. Acta, **34**, 2085 (1951).
34. H. B. Milne, C.-H. Peng, J. Am. Chem. Soc., **79**, 645 (1957).
35. E. Fischer, Chem. Ber., **40**, 1501 (1907); E. Fischer, W. A. Jacobs, Chem. Ber., **39**, 2942 (1906).
36. E. Fischer, W. Schrauth, Ann. Chem., **354**, 21 (1907).
37. J. P. Greenstein, J. Biol. Chem., **121**, 9 (1937).
38. N. Izumiya, J. P. Greenstein, Arch. Biochem. Biophys., **52**, 203 (1954); R. Wade, M. Wintz, J. P. Greenstein, J. Am. Chem. Soc., **78**, 373 (1956).
39. B. F. Erlanger, E. Brand, J. Am. Chem. Soc., **73**, 4025 (1951); **72**, 3314 (1950).
40. S. G. Waley, J. Watson, J. Chem. Soc., 475 (1953).
41. E. Brand, B. F. Erlanger, J. Polatnick, H. Sachs, D. Kirschenbaum, J. Am. Chem. Soc., **73**, 4027 (1957).
42. M. Bergmann, L. Zervas, Chem. Ber., **65**, 1192 (1932).
43. S. Goldschmidt, C. Jutz, Chem. Ber., **89**, 518 (1956).
44. H. Sachs, E. Brand, J. Am. Chem. Soc., **75**, 4608 (1953).
45. D. A. Rowlands, G. T. Young, Biochem. J., **65**, 516 (1957).
46. V. Bruckner, M. Szekerke, J. Kovacs, Hoppe-Seyler's Z. Physiol. Chem., **309**, 25 (1957).
47. W. L. Le Quesne, G. T. Young, J. Chem. Soc., 1959, (1950).
48. H. Sachs, E. Brand, J. Am. Chem. Soc., **76**, 1811 (1954).
49. M. Goodman, F. Boardman, J. Listowsky, J. Am. Chem. Soc., **85**, 2483 2491 (1963).
50. M. Goodman, E. E. Schmitt, D. A. Yphantis, J. Am. Chem. Soc., **84**, 1283, 1288, 3770 (1962).
51. J. Dale, K. Titlestad, Chem. Commun., 656 (1969).
52. N. C. Davis, E. L. Smith, J. Biol. Chem., **200**, 373 (1953).
53. M. Rothe, K.-D. Steffen, I. Rothe, Angew. Chem., **77**, 347 (1965).
54. J. Kapfhammer, A. Mathes, Hoppe-Seyler's Z. Physiol. Chem., **223**, 43 (1934).

55. E. Abderhalden, H. Nienburg, Fermentforsch., **13**, 573 (1933).
56. C. Toniolo, Bipolymers, **10**, 1707 (1971).
57. U. Widmer, G. P. Lorenzi, Chimia, **25**, 236 (1971).
58. J. E. Shields, S. T. McDowell, J. Pavlos, G. R. Gray, J. Am. Chem. Soc., **90**, 3549 (1968).
59. M. Kajtár, M. Hollósi, Acta Chim. Acad. Sci. Hung., **65**, 403 (1970); M. Kajtár, V. Bruckner, Tetrahedron Lett., 4813 (1966).
60. M. Hollósi, M. Kajtár, Acta Chim. Acad. Sci. Hung., **73**, 247 (1972); M. Kajtár, M. Hollósi, G. Snatzke, Tetrahedron, **27**, 5659 (1971).
61. S. M. McElvain, P. M. Laughton, J. Am. Chem. Soc., **73**, 448 (1951).
62. M. Rothe, R. Theyson, K.-D. Steffen, Tetrahedron Lett., 4063 (1970).
63. S. Sakakibara, M. Shin, M. Fujino, Y. Shimonishi, S. Inoue, N. Inukai, Bull. Chem. Soc. Japan, **38**, 1522 (1965).
64. M. Miyoshi, T. Kimura, S. Sakakibara, Bull. Chem. Soc. Japan, **43**, 2941 (1970).
65. T. Isemura, H. Okabayashi, S. Sakakibara, Biopolymers, **6**, 307 (1968).
66. H. Okabayashi, T. Isemura, S. Sakakibara, Biopolymers, **6**, 323 (1968).
67. H. Okabayashi, T. Isemura, Bull. Chem. Soc. Japan, **43**, 20 (1970).
68. H. Okabayashi, T. Isemura, Bull. Chem. Soc. Japan, **43**, 359 (1970).
69. M. Rothe, J. Mazánek, Tetrahedron Lett., 3795 (1972).
70. M. Rothe, H. Rott, J. Mazánek, Peptides 1976, Proc. 14th Europ. Peptide Symp., A. Loffet (Ed.), Edition de l'Université de Bruxelles, p. 309.
71. J. Meienhofer, P. M. Jacobs, H. A. Godwin, I. H. Rosenberg, J. Org. Chem., **35**, 4137 (1970).
72. E. Schnabel, Ann. Chem., **702**, 188 (1967).
73. A. K. Mitra, I. Ostashevsky, C. F. Brewer, Int. J. Peptide Protein Res., **22**, 502 (1983).
74. F. C. Carpenter, D. T. Gish, J. Am. Chem. Soc., **74**, 3818 (1952).
75. A. Nakamura, O. Jardetzky, Biochemistry, **7**, 1226 (1968); Proc. Natl. Acad. Sci. USA, **58**, 2212 (1967).
76. M. Kajtár, M. Hollósi, Zs. Riedl, Acta Chim. Acad. Sci. Hung., **88**, 301 (1976).
77. M. Rothe, D. Mühlhausen, Angew. Chem., **88**, 338 (1976); Angew. Chem. Int. Ed. Engl., **15**, 307 (1976).
78. Y. A. Bara, A. Friedrich, W. Hehlein, H. Kessler, P. Kondor, M. Molter, H.-J. Veith, Chem. Ber., **111**, 1029 (1978); Y. A. Bara, A. Friedrich, H. Kessler, M. Molter, *ibid.*, **111**, 1045 (1978).
79. M. Rothe, E. Frank, G. Schmidtberg, in: K. Brunfeldt (Ed.), "Peptides 1980", Proc. 16th Eur. Peptide Symp., Scriptor, Copenhagen 1981, p. 258.
80. M. Rothe, Hj. Schaeffer, unpublished data.
81. M. Goodman, R. Rupp, F. Naider, Bioorg. Chem., **1**, 294, 310 (1971).
82. M. Goodman, M. Langsam, Biopolymers, **4**, 275 (1966).
83. M. Goodman, M. Langsam, I. G. Rosen, Biopolymers, **4**, 305 (1966).
84. M. Goodman, N. Ueyama, F. Naider, Biopolymers, **14**, 901 (1975).
85. I. Schechter, A. Berger, Biochemistry, **5**, 3362 (1966).
86. G. M. Bonora, D. Nisato, C. Toniolo, Makromol. Chem., **176**, 2535 (1975); C. Toniolo, G. M. Bonora, *ibid.*, **176**, 2547 (1975); J. Polym. Sci., Polym. Chem. Ed., **14**, 515 (1976).
87. U. Narayanan, T. A. Keiderling, G. M. Bonora, C. Toniolo, Biopolymers, **24**, 1257 (1985).
88. D. S. Jones, G. W. Kenner, J. Preston, R. C. Sheppard, J. Chem. Soc., 6227 (1965).
89. M. Y. Ali, J. Dale, K. Titlestad, Acta Chem. Scand., **27**, 1509 (1973).
90. D. Leibfritz, E. Haupt, N. Dubischar, H. Lachmann, R. Oekonomopulos, G. Jung, Tetrahedron, **38**, 2165 (1982).
91. C. Toniolo, G. M. Bonora, V. Barone, A. Bavoso, E. Benedetti, B. Di Blasio, P. Grimaldi, F. Lelj, V. Pavone, C. Pedone, Macromolecules, **18**, 895 (1985).
92. E. Benedetti, A. Bavoso, B. Di Blasio, V. Pavone, C. Pedone, M. Crisma, G. M. Bonora, C. Toniolo, J. Am. Chem. Soc., **104**, 2437 (1982).
93. G. M. Bonora, C. Mapelli, C. Toniolo, R. R. Wilkening, E. S. Stevens, Int. J. Biol. Macromol., **6**, 179 (1984).
94. R. R. Wilkening, E. S. Stevens, G. M. Bonora, C. Toniolo, J. Am. Chem. Soc., **105**, 2560 (1983).
95. C. Toniolo, G. M. Bonora, M. Crisma, F. Bertanzon, E. S. Stevens, Makromol. Chem., **182**, 3149 (1981).
96. C. Toniolo, G. M. Bonora, A. Fontana, Int. J. Peptide Protein Res., **6**, 371 (1974).
97. C. Toniolo, G. M. Bonora, A. Fontana, Bull. Soc. Chim. Belg., **84**, 305 (1975).
98. G. M. Bonora, A. Maglione, A. Fontana, C. Toniolo, Bull. Soc. Chim. Belg., **84**, 299 (1975).
99. P. Spadon, A. Del Pra, Int. J. Peptide Protein Res., **15**, 54 (1980).
100. G. M. Bonora, A. Maglione, C. Toniolo, Polymer, **15**, 767 (1974).
101. M. Goodman, F. Naider, C. Toniolo, Biopolymers, **10**, 1719 (1971).
102. M. Palumbo, S. Da Rin, G. M. Bonora, C. Toniolo, Makromol. Chem., **177**, 1477 (1976).
103. J. S. Balcerski, E. S. Pysh, G. M. Bonora, C. Toniolo, J. Am. Chem. Soc., **98**, 3470 (1976).
104. M. M. Kelly, E. S. Pysh, G. M. Bonora, C. Toniolo, J. Am. Chem. Soc., **99**, 3264 (1977).
105. E. S. Pysh, C. Toniolo, J. Am. Chem. Soc., **99**, 6211 (1977).
106. G. M. Bonora, C. Toniolo, Makromol. Chem., **179**, 1453 (1978).
107. C. Toniolo, G. M. Bonora, Gazz. Chim. Ital., **104**, 843 (1974).
108. E. Peggion, M. Palumbo, G. M. Bonora, C. Toniolo, Bioorg. Chem., **3**, 125 (1974).
109. C. Toniolo, G. M. Bonora, Bioorg. Chem., **3**, 114 (1974).
110. G. M. Bonora, C. Toniolo, Gazz. Chim. Ital., **107**, 195 (1977).
111. C. Toniolo, G. M. Bonora, M. Palumbo, E. Peggion, E. S. Stevens, Biopolymers, **17**, 1713 (1978).
112. M. H. Baron, C. De Loze, C. Toniolo, G. D. Fasman, Biopolymers, **17**, 2225 (1978); **18**, 411 (1979).

113. C. Toniolo, G. M. Bonora, G. R. Sullivan, W. H. Bearden, J. D. Roberts, J. Org. Chem., **45**, 288 (1980).
114. D. R. Rueda, P. Spadon, G. Zanotti, Gazz. Chim. Ital., **109**, 389 (1979).
115. C. Toniolo, G. M. Bonora, S. Salardi, Int. J. Biol. Macromol., **3**, 377 (1981).
116. U. Widmer, G. P. Lorenzi, P. Pino, Biopolymers, **18**, 239 (1979).
117. G. P. Lorenzi, L. Tomasic, H. Jaeckle, Makromol. Chem., Rapid Commun., **1**, 729 (1980).
118. L. Tomasic, A. Stefani, G. P. Lorenzi, Helv. Chim. Acta, **63**, 2000 (1980).
119. R. Katakai, J. Chem. Soc., Perkin Trans. I, 905 (1979).
120. J. Semb, J. H. Boothe, R. B. Angier, C. W. Waller, J. H. Mowat, B. L. Hutchings, Y. Subbarow, J. Am. Chem. Soc., **71**, 2310 (1949).
121. G. M. Bonora, A. Maglione, C. Toniolo, Gazz. Chim. Ital., **105**, 1055 (1975).
122. R. T. Coffey, E. S. Stevens, C. Toniolo, G. M. Bonora, Makromol. Chem., **182**, 941 (1981).
123. F. Naider, J. M. Becker, Biopolymers, **13**, 1011, 1747 (1974).
124. G. M. Bonora, C. Toniolo, Biopolymers, **13**, 2179 (1974).
125. M. Goodman, A. A. Ribeiro, F. Naider, Proc. Natl. Acad. Sci. USA, **75**, 4647 (1978).
126. F. Naider, J. M. Becker, A. Ribeiro, M. Goodman, Biopolymers, **17**, 2213 (1978).
127. A. A. Ribeiro, M. Goodman, F. Naider, J. Am. Chem. Soc., **100**, 3903 (1978); Int. J. Peptide Protein Res., **14**, 414 (1979).
128. F. Naider, A. A. Ribeiro, M. Goodman, Biopolymers, **19**, 1799 (1980).
129. D. J. Paskowski, E. S. Stevens, G. M. Bonora, C. Toniolo, Biochim. Biophys. Acta, **535**, 188 (1978).
130. C. Bismara, G. M. Bonora, C. Toniolo, E. L. Becker, R. J. Freer, Int. J. Peptide Protein Res., **26**, 482 (1985).
131. M. Rothe, F. Wehowsky, H. Ott, unpublished data.
132. A. Sakurai, Y. Okumura, Bull. Chem. Soc. Japan, **52**, 540 (1979).
133. C. Toniolo, G. M. Bonora, A. Scatturin, Gazz. Chim. Ital., **105**, 1063 (1975).
134. M. Fridkin, A. Patchornik, E. Katchalski, J. Am. Chem. Soc., **87**, 4646 (1965).
135. B. V. Rozynov, V. M. Burikov, V. V. Shilin, A. A. Kiryushkin, Zh. Obsh. Khim., **38**, 2690 (1968); J. Gen. Chem. USSR, **38**, 2602 (1968).
136. V. T. Ivanov, V. V. Shilin, Y. Bernat, Y. A. Ovchinnikov, Zh. Obsh. Khim., **41**, 2318 (1971); J. Gen. Chem. USSR, **41**, 2341 (1971).
137. S. L. Portnova, T. A. Balashova, V. F. Bystrov, V. V. Shilin, Y. Bernat, V. T. Ivanov, Y. A. Ovchinnikov, Chem. Nat. Compd., **7**, 307 (1971).
138. P. Sutton, J. L. Koenig, Biopolymers, **9**, 615 (1970).
139. J. Beacham, V. T. Ivanov, P. M. Scopes, D. R. Sparrow, J. Chem. Soc. C, 1449 (1966); J. Beacham, V. T. Ivanov, G. W. Kenner, R. C. Sheppard, J. Chem. Soc., Chem. Commun., 386 (1965).
140. M. Ueki, S. Ikeda, T. Handa, Bull. Chem. Soc. Japan, **54**, 3865 (1981).
141. P. A. Temussi, M. Goodman, Proc. Natl. Acad. Sci. USA, **68**, 1767 (1971).
142. M. Goodman, I. G. Rosen, M. Safdy, Biopolymers, **2**, 503 (1964); M. Goodman, I. G. Rosen, Biopolymers, **2**, 537 (1964).
143. M. Goodman, Y. Masuda, A. S. Verdini, Biopolymers, **10**, 1031 (1971).
144. A. Ribeiro, R. P. Saltman, M. Goodman, Biopolymers, **19**, 1771 (1980).
145. M. Goodman, A. S. Verdini, C. Toniolo, W. D. Phillips, F. A. Bovey, Proc. Natl. Acad. Sci. USA, **64**, 444 (1969).
146. M. Goodman, I. Listowsky, Y. Masuda, F. Boardman, Biopolymers, **1**, 33 (1963).
147. K. Okahashi, S. Ikeda, Int. J. Peptide Protein Res., **13**, 462 (1979); K. Okahashi, S. Ikeda, Biopolymers, **18**, 2105 (1979).
148. S. Ikeda, K. Okahashi, Biopolymers, **18**, 2115 (1979); T. Imae, K. Okahashi, S. Ikeda, Biopolymers, **20**, 2553 (1981).
149. T. Ozaki, A. Shoji, M. Furukawa, Makromol. Chem., **183**, 771 (1982).
150. M. Rothe, P. Grauer, unpublished data.
151. M. T. Leplawy, D. S. Jones, G. W. Kenner, R. C. Sheppard, Tetrahedron, **11**, 39 (1960).
152. W. J. McGahren, M. Goodman, Tetrahedron, **23**, 2017 (1963).
153. U. Narayanan, T. A. Keiderling, G. M. Bonora, C. Toniolo, J. Am. Chem. Soc., **108**, 2431 (1986).
154. M. Rothe, R. Theysohn, K.-D. Steffen, M. Kostrzewa, M. Zamani, in: E. Scoffone, (Ed.) "Peptides 1969", Proc. 10th Europ. Peptides Symp., North-Holland, Amsterdam, 1971, p. 179.
155. R. Deslauriers, J. C. P. Smith, M. Rothe, Proc. 4th Amer. Peptide Symp., Ed. R. Walter, J. Meienhofer, Ann Arbor Sci. Publ., 1975, p. 91.
156. G. Kartha, G. Ambady, P. V. Shankar, Nature, **247**, 204 (1974); M. E. Druyan, C. L. Coulter in: R. Walter, J. Meienhofer (Eds.), Proc. 4th Amer. Peptide Symp, Ann Arbor Sci. Publ., 1975, p. 85; M. E. Druyan, C. L. Coulter, R. Walter, G. Kartha, G. K. Ambady, J. Am. Chem. Soc., **98**, 5496 (1976).
157. H. Kessler, A. Friedrich, W. E. Hull, J. Org. Chem., **46**, 3892 (1981).
158. M. Rothe, W. Maestle, unpublished data.
159. M. Rothe, W. Maestle, Angew. Chem., **94**, 223 (1982); Angew. Chem. Int. Ed. Engl., **21**, 220 (1982); Angew. Chem. Suppl., 533 (1982).
160. J. W. Bats, A. Friedrich, H. Fuess, H. Kessler, W. Maestle, M. Rothe, Angew. Chem., **91**, 573 (1979); Angew. Chem. Int. Ed. Engl., **18**, 538 (1979).
161. J. W. Bats, H. Fuess, J. Am. Chem. Soc., **102**, 2065 (1980).
162. H. Kessler, W. Bermel, A. Friedrich, G. Krack, W. E. Hull, J. Am. Chem. Soc., **104**, 6297 (1982); H. Kessler, W. Bermel, H. Förster, Angew. Chem., **94**, 703 (1982); Angew. Chem. Int. Ed. Engl., **21**, 689 (1982).
163. W. Maestle, M. Rothe, in: J. Z. Siemion, G. Kupryszewski (Eds.), "Peptides 1978", Proc. 15th Eur. Peptide Symp., Wroclaw Univ. Press, Wroclaw, p. 179.
164. M. Maestle, T. Weber, M. Rothe, unpublished data.

165. E. Benedetti, B. Di Blasio, V. Pavone, C. Pedone, C. Toniolo, G. M. Bonora, Macromolecules, **13**, 1454 (1980).
166. T. Sugihara, Y. Imanishi, T. Higashimura, Biopolymers, **14**, 723 (1975).
167. C. M. Deber, F. A. Bovey, J. P. Carver, E. R. Blout, J. Am. Chem. Soc., **92**, 6191 (1970).
168. C. M. Deber, D. A. Torchia, E. R. Blout, J. Am. Chem. Soc., **93**, 4893 (1971).
169. E. Benedetti, M. R. Ciajolo, A. Maisto, Acta Cryst. B, **30**, 1783 (1974).
170. E. Benedetti, A. Ciajolo, B. Di Blasio, V. Pavone, C. Pedone, C. Toniolo, G. M. Bonora, Int. J. Peptide Protein Res., **14**, 130 (1979).
171. M. Rothe, H. Rott, Angew. Chem., **88**, 844 (1976); Angew. Chem. Int. Ed. Engl., **15**, 770 (1976).
172. Y.-Y. H. Chao, R. Bersohn, Biopolymers, **17**, 2761 (1978).
173. A. E. Tonelli, J. Am. Chem. Soc., **92**, 6187 (1970).
174. A. Chemizard, S. David, Bull. Soc. Chim. France, 184 (1966).
175. R. Boni, A. S. Verdini, C. M. Deber, E. R. Blout, Biopolymers, **17**, 2385 (1978); J.-P. Meraldi, E. R. Blout, R. Boni, A. S. Verdini, *ibid.*, **17**, 2401 (1978).
176. J. Vicar, P. Malon, A. Trka, J. Smoliková, J. Fric, K. Bláha, Coll. Czech. Chem. Commun., **42**, 2701 (1977).
177. K. Bláha, J. Fric, Coll. Czech. Chem. Commun., **35**, 619 (1970); K. Bláha, J. Smoliková, A. Vitek, *ibid.*, **31**, 4296 (1966); K. Bláha, Z. Samek, *ibid.*, **32**, 3780 (1967).
178. H.-J. Burkhardt, H. K. Mitchell, Arch. Biochem. Biophys., **94**, 32 (1961).
179. E. Benedetti, A. Bavoso, B. Di Blasio, V. Pavone, C. Pedone, C. Toniolo, G. M. Bonora, Biopolymers, **22**, 305 (1983).
180. I. Vicar, J. Smoliková, K. Bláha, Coll. Czech. Chem. Commun., **37**, 4060 (1972); J. Z. Siemion, Ann. Chem., **748**, 88 (1971).
181. M. E. Kamwaya, O. Oster, H. Bradaczek, Acta Cryst. B, **37**, 1564 (1981).
182. G. Kartha, T. Ashida, M. Kakudo, Acta Cryst. B, **30**, 1861 (1974).
183. H. Brockmann, H. Musso, Chem. Ber., **89**, 241 (1956).
184. W. Pickenhagen, P. Dietrich, B. Keil, J. Polonski, F. Nonaille, E. Lederer, Helv. Chim. Acta, **58**, 1078 (1975).
185. C. Eguchi, A. Kakuta, Bull. Chem. Soc. Japan, **47**, 2277 (1974).
186. P. E. Young, V. Madison, E. R. Blout, J. Am. Chem. Soc., **48**, 5365 (1976).
187. T. Sugihara, Y. Imanishi, T. Higashimura, Biopolymers, **12**, 2823 (1973).
188. D. E. Nitecki, B. Halpern, J. W. Westley, J. Org. Chem., **33**, 864 (1968).
189. M. Rothe, I. Rothe, T. Toth, K.-D. Steffen, in: H. C. Beyerman, A. van de Linde, W. Maassen van den Brink (Eds.), "Peptides 1966", Proc. 8th Eur. Peptide Symp., North-Holland Publ. Co., Amsterdam, 1967, p. 8.
190. M. Rothe, R. Theyson, D. Mühlhausen, F. Eisenbeiss, W. Schindler, in: J. Meienhofer (Ed.), "Chemistry and Biology of Peptides", Proc. 3rd Amer. Peptide Symp., Ann Arbor Sci. Publ., Ann Arbor, 1972, p. 51.
191. K. Titlestad, Acta Chem. Scand. B, **29**, 153 (1975).
192. J. Dale, K. Titlestad, Acta Chem. Scand. B, **29**, 353 (1975).
193. J. Dale, P. Groth, K. Titlestad, Acta Chem. Scand. B, **31**, 523 (1977).
194. J. Dale, K. Titlestad, J. Chem. Soc., Chem. Commun., 1403 (1970).
195. J. Dale, K. Titlestad, J. Chem. Soc., Chem. Commun., 255 (1972).
196. K. Titlestad, P. Groth, J. Dale, M. Y. Ali, J. Chem. Soc., Chem. Commun., 346 (1973).
197. P. Groth, Acta. Scand. A, **30**, 838 (1976).
198. P. Groth, Acta. Scand. **24**, 780 (1970).
199. P. Groth, Acta. Scand. **27**, 3419 (1973); K. Titlestad, P. Groth, J. Dale, J. Chem. Soc., Chem. Commun., 646 (1973).
200. P. Groth, Acta. Scand. A, **31**, 232 (1977).
201. P. Groth, Acta. Scand. A, **29**, 38 (1975).
202. P. Groth, Acta. Scand. **27**, 3217 (1973).
203. P. Groth, Acta. Scand. A, **30**, 840 (1976).
204. P. Groth, Acta. Scand. **23**, 3155 (1969).
205. J. Dale, K. Titlestad, Tetrahedron Lett., 379 (1978).
206. K. Titlestad, Acta Chem. Scand. B, **31**, 641 (1977).
207. K. Suzuki, Y. Sasaki, N. Endo, Y. Mihara, Chem. Pharm. Bull., **29**, 233 (1981).
208. H. Eckstein, Ann. Chem., 2145 (1976).
209. A. Yaron, M. C. Otey, H. A. Sober, E. Katchalski, S. Ehrlich-Rogozinski, A. Berger, Biopolymers, **11**, 607 (1972); A. Yaron, E. Katchalski, A. Berger, G. D. Fasman, H. A. Sober, Biopolymers, **10**, 1107 (1971).
210. J. W. Stewart, M. A. Stahmann, J. Chromatogr., **9**, 233 (1962); in: M. A. Stahmann (Ed.), "Polyamino Acids, Polypeptides, and Proteins", The University of Wisconsin Press, Madison, 1962, p. 95; H. A. Sober, in: M. A. Stahmann (Ed.), "Polyamino Acis, Polypeptides, and Proteins", The University of Wisconsin Press, Madison, 1962, p. 105.
211. O. Grahl-Nielsen, G. L. Trisch, Biochemistry, **8**, 187 (1969).
212. N. Yu. Kozhevnikova, G. P. Vlasov, Zh. Obshch, Khim., **49**, 924 (1979); J. Gen. Chem. USSR, **49**, 800 (1979).
213. E. Schnabel, Hoppe-Seyler's Z. Physiol. Chem., **357**, 1365 (1976).
214. G. P. Vlasov, V. R. Glushenkova, T. N. Prokopyuk, Zh. Obshch. Khim., **41**, 430 (1971); J. Gen. Chem. USSR, **41**, 422 (1971).
215. L. B. Nadezhdina, T. A. Komogorova, G. P. Vlasov, Zh. Obshch, Khim., **51**, 214 (1981); J. Gen. Chem. USSR, **51**, 187 (1981).
216. H. Ito, I. Ichikizaki, T. Ando, Int. J. Protein Res., **1**, 59 (1970).
217. W. L. Mattice, W. H. Harrison, III, Biopolymers, **14**, 2025 (1975).
218. H. Yajima, K. Kawasaki, H. Minami, H. Kawatani, N. Mizokami, Y. Kiso, F. Tamura, Chem. Pharm. Bull., **18**, 1394 (1970).
219. R. Schwyzer, W. Rittel, Helv. Chim. Acta, **44**, 159 (1961).
220. I. Géczy, Acta Chim. Acad. Sci. Hung., **115**, 345 (1984); H. A. Van der Schee, G. I. Tesser, unpublished data.
221. G. I. Tesser, J. T. Buis, Rec. Trav. Chim. Pays-Bas, **90**, 444 (1971).

222. F. Marchiori, G. Borin, B. Filippi, V. Moretto, G. M. Bonora, C. Toniolo, Int. J. Peptide Protein Res., **14**, 143 (1979); F. Marchiori, G. Borin, D. Stivanello, V. Moreto, G. Chessa, Hoppe-Seyler's Z. Physiol. Chem., **363**, 1483 (1982).
223. G. M. Bonora, C. Toniolo, J. H. Jones, K. N. Rajasekharan, Int. J. Peptide Protein Res., 27, 1 (1986); K. N. Rajshekharan, J. H. Jones, G. M. Bonora, C. Toniolo, Int. J. Peptide Protein Res., **26**, 605 (1985).
224. C. Toniolo, G. M. Bonora, I. F. Lüscher, C. H. Schneider, Int. J. Peptide Protein Res., **23**, 47 (1984).
225. M. Wilchek, A. Patchornik, J. Org. Chem., **28**, 1874 (1963).
226. M. Rinaudo, A. Domard, G. Spach, Biopolymers, **12**, 2199 (1973); M. Rinaudo, A. Domard, *ibid.*, **12**, 2211 (1973).
227. M. Rinaudo, A. Domard, Biopolymers, **14**, 2035 (1975); J. Am. Chem. Soc., **98**, 6360 (1976).
228. C. Bosso, A. Domard, F. M. Devienne, R. Combarieu, Spectros. Int. J., **3**, 81 (1984).
229. R. Guarnaccia, G. P. Lorenzi, V. Rizzo, P. L. Luisi, Biopolymers, **14**, 2329 (1975); P. L. Luisi, V. Rizzo, G. P. Lorenzi, B. Straub, U. Suter, R. Guarnaccia, *ibid.*, **14**, 2347 (1975).
230. R. Deslauriers, Z. Gronka, K. Schaumburg, T. Shiba, R. Walter, J. Am. Chem. Soc., **97**, 5093 (1975).
231. N. Go, H. A. Scheraga, Macromolecules, **6**, 525 (1973).
232. H. Edelhoch, R. E. Lippoldt, M. Wilchek, J. Biol. Chem., **243**, 4799 (1968); H. Edelhoch, R. S. Bernstein, M. Wilchek, *ibid.*, **243**, 5985 (1968).

8.3. OLIGO(IMINES)

8.3.1. LINEAR OLIGO(IMINES)

8.3.1.1. Linear Oligo(ethylene imines) $H[NHCH_2CH_2]_nNH_2$

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_{20}^{20} (g/cm³)	n_D^{25}	Refs.
1	60.1	11.0	116.5/1013	0.8994	1.4536	1,2
2	103.1		207.1/1013	0.9586	1.4810	1–3
3	146.2	12.0	277.9/1013	0.9839	1.4951	1–4
4	189.2		333/1013	0.9994	1.5015	1–3
7	318.5		109–110/11		1.5132	2,5
9	404.7		199–200/1		1.5161	2,5

8.3.1.2. Linear Oligo(alkylene imines) $R-[NH(CH_2)_6NH(CH_2)_{10}]_n-R'$

R	R′	*n*	Mol. wt.	m.p. (°C)	Ref.
H	OH	1	272.5	64–67	6
		2	526.9	70–72	6
		3	781.4	85–86	6
$(CH_2)_{10}OH$	OH	1	428.8	99–100	6
		2	683.2	90–93	6
		3	937.7	80–84	6
H	$NH(CH_2)_6NH_2$	1	370.7	70–72	6
		2	625.1	86–88	6
		3	879.6	81–85	6

8.3.2. CYCLIC OLIGO(IMINES) $\urcorner[NH(CH_2)_x]_n\ulcorner$

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm³)/°C	Refs.
2	1	43.1		55–56/1013	0.8321/24	2,7,8
	2	86.1	104	145–146/1013		8,9
	4	172.3	35	$110/10^{-4}$		10
3	1	57.1		63/1013	0.8436/20	11
	2	114.2	14–15	186–188/1013		8,11–13
4	1	71.1		88/1013	0.8520/22.5	14
	2	142.2		95/16	0.9020/18	8,13
5	1	85.2	−9	106/1013	0.8606/20	14
	2	170.3		108–110/16	0.9195/13	8
6	1	99.2		138/1013	0.8864/21	8,15,16
	2	198.2	72			13,17,18
	3	297.5	42			17
	4	396.7	59–60			17
	5	495.9	45			17
	6	595.1	67–68			17

8.3.2. *cont'd*

x	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4 (g/cm^3)/°C	Refs.
7	1	113.1	−33	162/1013	0.8895/20	16,19
	2	226.2	26		0.9012/30	19
8	1	127.2		90/32	0.9021/21	16
	2	254.5	55			20
9	1	141.3			0.8982/21	16
	2	282.5	38			19
13	1	197.4	50–51			16
	2	394.7	52			21

REFERENCES

1. A. L. Wilson, Ind. Eng. Chem., **27**, 870 (1935).
2. G. D. Jones, A. Langsjoen, M. M. Ch. Neumann, J. Zomlefer, J. Org. Chem., **9**, 125 (1944).
3. H. B. Jonassen, T. B. Crumpler, T. D. O'Brien, J. Am. Chem. Soc., **67**, 1709 (1945).
4. F. G. Mann, J. Chem. Soc., 461 (1934).
5. G. S. Whitby, N. Wellman, V. W. Floutz, L. H. Stephens, Ind. Eng. Chem., **42**, 445 (1950).
6. H. Zahn, G. B. Gleitsmann, Makromol. Chem., **63**, 129 (1963).
7. C. F. H. Allen, F. W. Spangler, E. R. Webster, Org. Syn., **30**, 38 (1950).
8. J. v. Braun, G. Blessing, F. Zobel, Chem. Ber., **57**, 185 (1924).
9. A. Ladenburg, J. Abel, Chem. Ber., **21**, 758 (1888).
10. H. Stetter, K.-H. Mayer, Chem. Ber., **94**, 1410 (1961).
11. C. C. Howard, W. Marckwald, Chem. Ber., **32**, 2031, 2038 (1899).
12. E. L. Buhle, A. M. Moore, F. Y. Wiselogle, J. Am. Chem. Soc., **65**, 29 (1943).
13. H. Stetter, J. Marx, Ann. Chem., **607**, 59 (1958); H. Stetter, H. Spangenberger, Chem. Ber., **91**, 1982 (1958).
14. T. S. Stevens, In E. R. Rodd (Ed.), "Chemistry of Carbon Compounds", Vol. IV, Part A, 1st ed., Elsevier, Amsterdam, 1953, p. 61; N. Campbell, *ibid.*, p. 570.
15. K. Ziegler, Ph. Orth, Chem. Ber., **66**, 1867 (1933).
16. L. Ruzicka, M. Kobelt, O. Haefliger, V. Prelog, Helv. Chim. Acta, **32**, 544 (1944).
17. H. Zahn, H. Spoor, Chem. Ber., **89**, 1296 (1956); **92**, 1375 (1959).
18. M. A. Th. Rogers, Nature (London), **177**, 128 (1956).
19. A. Mueller, E. Srepel, E. Funder-Fritzsche, F. Dicher, Monatsh. Chem., **83**, 386 (1952).
20. A. Mueller, L. Kindlmann, Chem. Ber., **74**, 416 (1941).
21. A. Mueller, Chem. Ber., **67**, 295 (1934).

8.4. OLIGO(UREAS)

8.4.1. OLIGO(METHYLENE UREAS) H[NHCONHCH$_2$]$_n$NHCONH$_2$

n	Mol. wt.	m.p. (°C)	Refs.
1	132.1	218 d	1,2
2	204.2	227	1
3	276.3	231–233	1,2
5	420.5	236 d	2

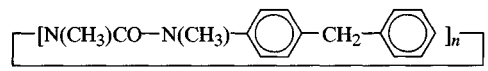
8.4.2. OLIGO(METHYLENE THIOUREAS) H[NHCSNHCH$_2$]$_n$NHCSNH$_2$

zn	Mol. wt.	m.p. (°C)	Refs.
1	164.3	198 d	3,4
2	252.4	210 d	3
3	340.5	215 d	3

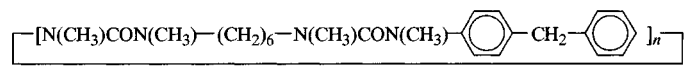
3.4.3. OLIGO(METHYLOL THIOUREAS) R[NHCSNHCH$_2$]$_n$OH

R	*n*	Mol. wt.	m.p. (°C)	Refs.
H	1	106.2	104–105	3,4
	2	194.3	190–192 d	3
CH$_2$OH	1	136.2	92	3
	2	224.3	132	3,4

8.4.4. CYCLIC OLIGO(UREAS)

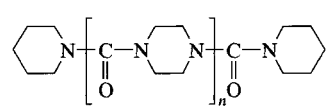


n	Mol. wt.	m.p. (°C)	Refs.
2	504.6	300–301	5



1	422.6	157–158	5
2	845.1	180–181	5

8.4.5. OLIGO(CARBONYLPIPERAZINES)

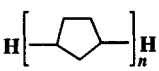


n	Mol. wt.	m.p. (°C)	Refs.
0	196.3	42–43	6,7
1	308.4	177	7,8
2	420.5	267	7
3	532.7	337	7

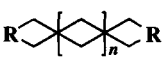
REFERENCES

1. A. A. Wanscheidt, S. K. Naumova, J. P. Melnikowa, Zh. Obshch. Khim., **10**, 1968 (1940); through Chem. Zentralbl., **II**, 184 (1941).
2. H. Kadowaki, Bull. Chem. Soc. Japan, **11**, 248 (1936).
3. H. J. Becher, F. Griffel, Chem. Ber., **91**, 691 (1958).
4. H. Staudinger, K. Wagner, Makromol. Chem., **12**, 168 (1954).
5. S. Foti, P. Maravigna, G. Montaudo, Macromolecules, **15**, 883 (1982).
6. B. Kühn, Chem. Ber., **33**, 2900 (1900).
7. R. Schwalm, W. Heitz, Makromol. Chem., **187**, 1415 (1986).
8. D. E. Rivett, J. F. K. Wilshire, Aust. J. Chem., **19**, 869 (1966).

D. CARBON CHAIN OLIGOMERS CONTAINING MAIN CHAIN CYCLIC UNITS

TABLE 9. OLIGO(CYCLOPENTYLENES) H[]H

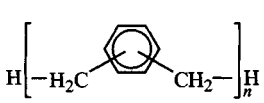
n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{20} (g/cm^3)	Refs.
1	70.1	−93.9	42.3/1013	0.7510	
2	138.3		190/1013	0.8646	1,2
3	206.3		293–294/1013	0.9177	1,2
4	274.5		369–370/1013	0.9564	1,2
6	410.7	143–146	235/0.13		1

TABLE 10. OLIGO(SPIRANES) R  R

R	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
CH_2	5	216.3		78/0.4	3
$(CH_2)_2$	3	244.4		84/0.07	3
	5	324.5	61		3
	7	406.6	107–108		3
$(CH_2)_3$	3	272.5	52.5	120–122/0.07	3
	5	352.6	79.5		3
	7	432.7	103–104		3
	9	512.8	138–140		3

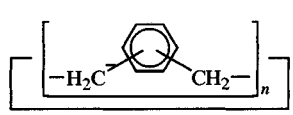
TABLE 11. OLIGO(XYLYLENES)

11.1. LINEAR OLIGO(XYLYLENES)

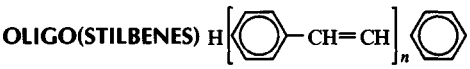


n		Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
o-	1	106.2	−25	144/1013	
	2	210.3	66.5	177–178/127	5–7
m-	1	106.2	−47.4	139/1013	
	2	210.3		296/1013	7,8
p-	1	106.2	13–14	138/1013	
	2	210.3	82	178/24	7,9–12,30
	3	314.5	141–142		10,13

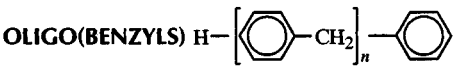
11.2. CYCLIC OLIGO(XYLYLENES)



n		Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
o-	2	208.3	112–112.5		5,6,14
	3	312.5	184.5		5,6
	4	416.6	205		15
m-	2	208.3	132–133	290/1013	10
p-	2	208.3	285–287		8,11–13,19,30
	3	312.5	170–171		10,,13,30
	4	416.6	179–182		13

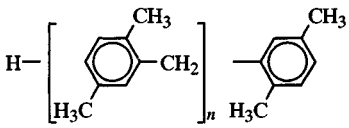
TABLE 12. OLIGO(STILBENES) 

n	Mol. wt.	m.p. (°C)	Refs.
1	180.2	124	17
2	282.4	265	16–18
3	384.5	350–352	17
4	486.5	360–374	16,17
5	588.7	395–400	16,17
6	690.9	415–420	16,17
7	793.0	425	16,17

TABLE 13. OLIGO(BENZYLS) 

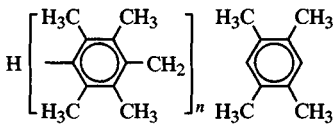
n	Mol. wt.	m.p. (°C)	Refs.
1	168.2	25.1	
2	258.3	82	20,23
3	348.4	90	21,23
4	438.6	120	23
5	528.7	153–156	22,23

TABLE 14. OLIGO-(2,5-DIMETHYL-BENZYLS)



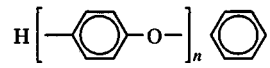
n	Mol. wt.	m.p. (°C)	Refs.
1	224.3	60–61	22,24
2	342.5	153–154	22
4	578.8	234–236	22
6	815.1	262–266	22

TABLE 15. OLIGO(2,3,5,6-TETRAMETHYL-BENZYLS)



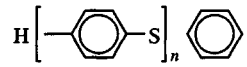
n	Mol. wt.	m.p. (°C)	Refs.
1	280.5	155–156	22,25
2	426.7	263–264	22,26
3	572.9	305–307	22
4	719.1	335–337	22

TABLE 16. OLIGO(*p*-PHENYLENE OXIDES)



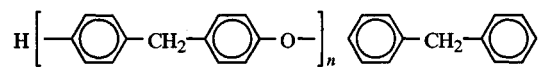
n	Mol. wt.	m.p. (°C)	Refs.
1	170.2	28	
2	262.3	73	23,27
3	354.4	105	23,27
4	446.5	142	23,27

TABLE 17. OLIGO(*p*-PHENYLENE SULFIDES)



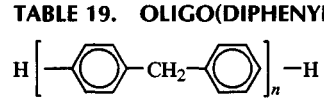
n	Mol. wt.	m.p. (°C)	Refs.
1	186.4	−21.5	
2	294.5	80	23,28
3	402.6	110	23
4	510.7	150	23

TABLE 18. OLIGO(*p*-PHENOXYPHENYLMETHANES)



n	Mol. wt.	m.p. (°C)	Refs.
1	350.4	59	23
2	532.7	101	23

TABLE 19. OLIGO(DIPHENYLMETHANES)



n	Mol. wt.	m.p. (°C)	Refs.
1	168.2	25.1	
2	334.5	115	4,29
3	500.7	180	4
4	666.9	240	4

REFERENCES

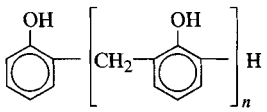
1. J. V. Braun, J. Reitz-Kopp, Chem. Ber., **74**, 1105 (1941).
2. G. E. Goheen, J. Am. Chem. Soc., **63**, 744 (1941).
3. E. Buchta, K. Geibel, Ann. Chem., **678**, 53 (1964).
4. M. Busch, W. Weber, J. Prakt. Chem., **146**, 1 (1936).
5. W. Baker, R. Banks, D. R. Lyon, F. G. Mann, J. Chem. Soc., 27 (1945).
6. A. C. Cope, S. W. Fenton, J. Am. Chem. Soc., **73**, 1668 (1951).
7. G. H. Coleman, W. H. Holst, R. D. Maxwell, J. Am. Chem. Soc., **58**, 2310 (1936).
8. M. Swarcz, J. Chem. Phys., **16**, 128 (1948).
9. T. Reichstein, R. Oppenauer, Helv. Chim. Acta, **16**, 1373 (1933).
10. W. Baker, J. F. W. McOmie, J. M. Norman, J. Chem. Soc., 1114 (1951).

11. C. J. Brown, A. C. Farthing, Nature (London), **164**, 915 (1949); J. Chem. Soc., 3261, 3265, 3270, (1953).
12. D. J. Cram, H. Steinberg, J. Am. Chem. Soc., **73**, 5691 (1951).
13. L. A. Errede, R. S. Gregorian, J. M. Hoyt, J. Am. Chem. Soc., **82**, 5218 (1960).
14. L. A. Errede, J. Am. Chem. Soc., **83**, 949 (1961).
15. E. D. Bergmann, Z. P elchowicz, J. Am. Chem. Soc., **75**, 4281 (1953).
16. G. Drefahl, G. Ploetner, Chem. Ber., **94**, 907 (1961); **91**, 1274 (1958).
17. G. Drefahl, R. Kuehmstedt, H. Oswald, H.-H. Hoerhold, Makromol. Chem., **131**, 89 (1970).
18. T. W. Campbell, R. N. McDonald, J. Org. Chem., **24**, 1246 (1959).
19. H. E. Winberg, F. S. Fawcett, W. E. Mochel, C. W. Theobald, J. Am. Chem. Soc., **82**, 1428 (1960).
20. A. B. Galun, A. Kaluszymer, E. D. Bergmann, J. Org. Chem., **27**, 1426 (1961).
21. E. Connerade, Bull. Soc. Chim. Belges, **44**, 411 (1935).
22. G. Montaudo, F. Bottino, S. Caccamese, P. Finocchiaro, G. Bruno, J. Polym. Sci. A-1, **8**, 2453 (1970).
23. G. Montaudo, G. Bruno, P. Maravigna, P. Finocchiaro, G. Centineo, J. Polym. Sci., **11**, 65 (1973).
24. R. C. Huston, D. T. Ewing, J. Am. Chem. Soc., **37**, 2394 (1915).
25. C. M. Welch, H. A. Smith, J. Am. Chem. Soc., **73**, 4391 (1951).
26. H. Kaemmerer, M. Harris, Makromol. Chem., **66**, 215 (1963).
27. H. Staudinger, F. Staiger, Ann. Chem., **517**, 67 (1935).
28. E. Bourgeois, A. Fonassin, Bull. Chim. France, **9**, 941 (1911).
29. N. Wolf, Chem. Ber., **14**, 2031 (1881).
30. J. R. Schaefgen, J. Polym. Sci., **15**, 203 (1955).

TABLE 20. PHENOL–FORMALDEHYDE AND RELATED OLIGOMERS

20.1. LINEAR PHENOL–FORMALDEHYDE OLIGOMERS

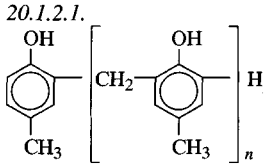
20.1.1. PHENOL–FORMALDEHYDE OLIGOMERS



n	Mol. wt.	m.p. (°C)	Refs.
1	200.2	119–120	1–4
2	306.4	158–159	2–6
3	412.5	161–162	3,4
4	518.6	148–150	4,7
5	624.7	203–204	4
6	730.8	213–214	4

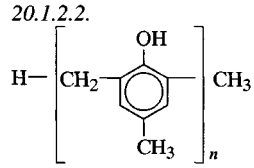
20.1.2. *p*-CRESOL–FORMALDEHYDE OLIGOMERS

20.1.2.1.



n	Mol. wt.	m.p. (°C)	Refs.
	228.3	126	8–10,56–58
2	348.4	214–215	8–11,56–58
3	468.6	173	8,9,56,57
4	588.7	130	12,56,57
5	708.9	215–217	12
6	829.1	225–230	12
7	949.2	167–170	12
8	1069.4	205–210	12
9	1189.5	175–180	12
11	1429.8	245	12

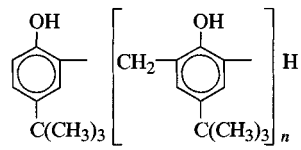
20.1.2.2.



n	Mol. wt.	m.p. (°C)	Refs.
1	136.1	70–72	61
2	256.3	148	11,13–16
3	376.5	183–184	11
4	496.7	204–205	11
5	616.9	152–154	61

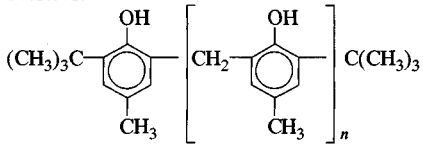
20.1.3. *p-tert* BUTYLPHENOL–FORMALDEHYDE OLIGOMERS

20.1.3.1.



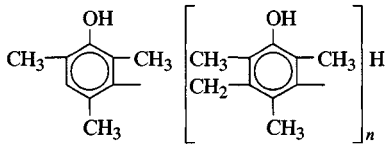
n	Mol. wt.	m.p. (°C)	Refs.
1	312.5	156	17,18,62
2	474.7	218–220	17,18,62
3	636.9	211	17,18,62
4	799.2	216–217	17,18,62
5	961.4	250	17,18,62
6	1123.6	249–250	17,18
7	1285.9	253–256	18
8	1448.1	224–226	17
10	1772.6	140	17

20.1.3.2.



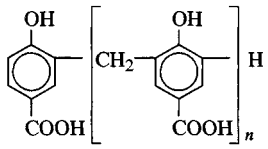
n	Mol. wt.	m.p. (°C)	Refs.
0	220.4	70–71	61
1	340.5	125–130	61
2	460.7	82–84	61
4	701.0	188–190	61

20.1.4. 2,4,6-MESITOL–FORMALDEHYDE OLIGOMERS



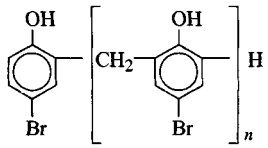
n	Mol. wt.	m.p. (°C)	Refs.
1	284.4	188	19
2	432.6	257	19

20.1.5. 4-CARBOXY-PHENOL–FORMALDEHYDE OLIGOMERS



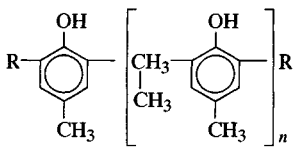
n	Mol. wt.	m.p. (°C)	Refs.
1	288.3	305–307	59
2	438.4	310	59

20.1.6. 4-BROMO-PHENOL–FORMALDEHYDE OLIGOMERS



n	Mol. wt.	m.p. (°C)	Refs.
1	358.1	183–184	1,60
2	543.1	237–238	60

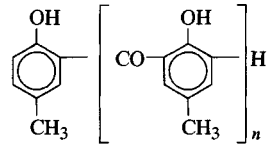
20.1.7. *p*-CRESOL–ACETALDEHYDE OLIGOMERS



R	*n*	Mol. wt.	m.p. (°C)	Refs.
H	1	241.3	141	20,21
H	2	373.5	214–215	21
CH_3	1	270.4	135–135.5	22
CH_3	2	404.6	154–155	63
CH_3	3	538.7	204–205	63

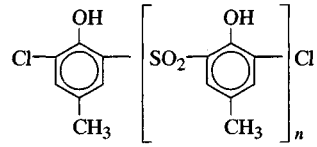
20.1.8. OLIGOMERS WITH CARBONYL AND SULFONYL BRIDGES

20.1.8.1.



n	Mol. wt.	m.p. (°C)	Refs.
1	242.3	106–107	23,24,56–58
2	376.4	122–123	24,56,57

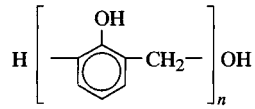
20.1.8.2.



n	Mol. wt.	m.p. (°C)	Refs.
1	347.2	179	25
2	517.4	238–240	25
3	687.6	246–252	25

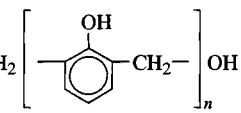
20.2. OLIGOMERIC PHENOL ALCOHOLS

20.2.1. OLIGOMERIC PHENOL ALCOHOLS



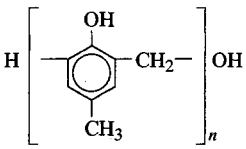
n	Mol. wt.	m.p. (°C)	Refs.
1	124.1	86–87	3,26
2	230.3	121.5–123	3,27,28

20.2.2. PHENOL DIALCOHOLS $HOCH_2$

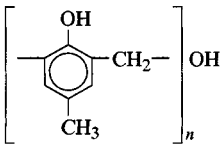


n	Mol. wt.	m.p. (°C)	Refs.
1	154.2	101	29–32
2	260.3	126–127	28

20.2.3. *p*-CRESOL MONOALCOHOLS H

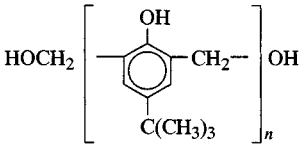


n	Mol. wt.	m.p. (°C)	Refs.
1	138.2	107	8,33,34
2	258.3	148	8,9,35
4	498.6		36

20.2.4. *p*-CRESOL DIALCOHOLS $HOCH_2$  OH

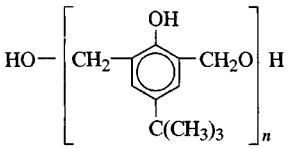
n	Mol. wt.	m.p. (°C)	Refs.
1	168.2	133–134	13,37
2	288.3	151.5	9,35,38,39
3	408.5	203	9,39,40

20.2.5. *p-tert* BUTYLPHENOL DIALCOHOLS



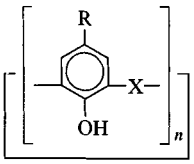
n	Mol. wt.	m.p. (°C)	Refs.
1	210.3	74–75	41
2	372.5	117–118	18

20.2.6. *o*-HYDROXYDIBENZYL ETHERS

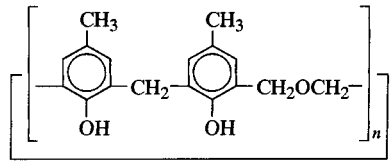


n	Mol. wt.	m.p. (°C)	Refs.
2	318.4	85	42

20.3. CYCLIC PHENOL–FORMALDEHYDE OLIGOMERS

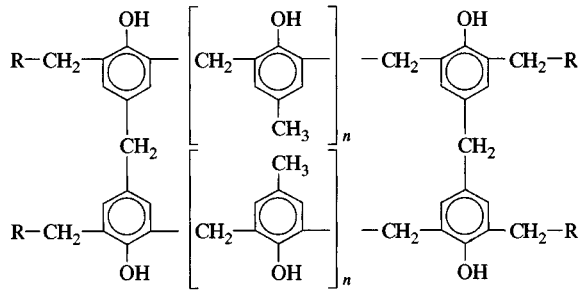
20.3.1. 

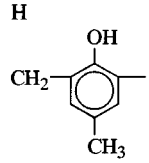
X	R	*n*	Mol. wt.	m.p. (°C)	Refs.
$-CH_2-$	$-CH_3$	4	480.6	300	36,43
	$-C(CH_3)_3$	4	648.9	344–346	18,44, 65,66
		5	811.1	310	68
		6	973.5	380–381	65,66,68
		8	1298.0	411–412	65,66,68
	$-C_6H_5$	4	728.9	407–409	18,66,67
		6	1093.4	408–410	66,67
		8	1457.8	421–423	66,67
	$-C_6H_{11}$	4	753.1	330	18
	$-CH_2C_6H_5$	4	785.0	330	18
	$-C(CH_3)_2CH_2CH_3$	4	705.0	280	18
	$-C(CH_3)_2CH_2C(CH_3)_2$	4	813.2	333	18
$-CH_2OCH_2-$	$-CH_3$	4	600.7	264–266	42
	$-C(CH_3)_3$	3	576–8	245	45

20.3.2. 

n	Mol. wt.	m.p. (°C)	Ref.
2	540.7	325–330	46

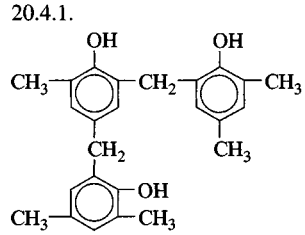
20.3.3.



R	*n*	Mol. wt.	Refs.
H	2	916.2	47
	1	1201.5	43,47

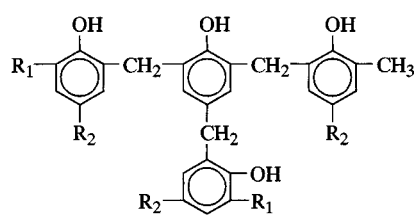
20.4. BRANCHED PHENOL–FORMALDEHYDE OLIGOMERS

20.4.1.



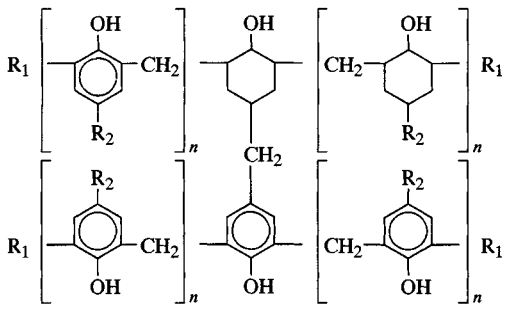
Mol. wt.	m.p. (°C)	Ref.
376.5	158	48

20.4.2.



R_1	R_2	Mol. wt.	m.p. (°C)	Refs.
H	H	411.5		49
H	CH_3	453.6	184–187	30,50
CH_3	CH_3	495.6	185–186	51

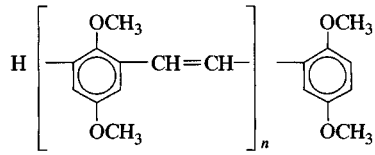
20.4.3.



R_1	R_2	*n*	Mol. w.t	m.p. (°C)	Refs.
H	H	1	624.7	185–187	52,53
CH_3	CH_3	1	737.0	207–208	6,53,54
CH_3	CH_3	2	1217.5	190–191	54

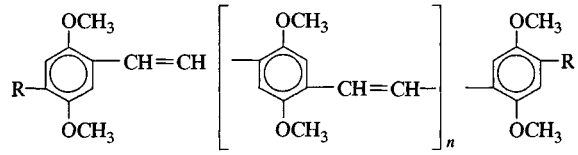
20.5. HYDROQUINONE OLIGOMERS

20.5.1.



n	Mol. wt.	m.p. (°C)	Refs.
1	300.4	97–98	64
2	462.5	146–147	64

20.5.2.



R	*n*	Mol. wt.	m.p. (°C)	Refs.
H	0	300.4	97–98	64
	1	462.5	192–193	64
CH_3	0	328.4	162–163	64
	1	490.6	242–243	64
	2	652.8	283–284	64
	3	815.0	306–307	64
	4	977.2	326–328	64

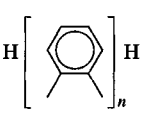
REFERENCES

1. C. A. Buehler, D. E. Cooper, E. O. Scrudder, J. Org. Chem., **8**l, 316 (1943).
2. S. R. Finn, J. W. James, C. J. S. Standen, J. Appl. Chem., **4**, 497 (1954); S. R. Finn, G. Lewis, N. J. L. Megson, J. Soc. Chem. Ind., **69**, 551 (1950); S. R. Finn, G. Lewis, J. Appl. Chem., **1**, 524 (1954).
3. D. A. Fraser, R. W. Hall, P. A. Jenkins, A. L. J. Raum, J. Appl. Chem., **7**, 689 (1957).
4. H. Kaemmerer, H. Lenz, Makromol. Chem., **27**, 162 (1958).
5. H. L. Bender, A. G. Franham, J. W. Guyer, F. N. Apel, T. B. Gibb, Jr., Ind. Eng. Chem., **44**, 1619 (1952).
6. A. C. Davis, B. T. Hayes, R. F. Hunter, J. Appl. Chem., **3**, 312 (1923).
7. S. Seto, H. Horiuchi, A. Takahashi, J. Chem. Soc. Japan, Ind. Chem. Sect., **58**, 378 (1955).
8. H. Kaemmerer, W. Rausch, Makromol. Chem., **24**, 152 (1957).
9. M. Koebner, Z. Angew. Chem., **46**, 251 (1953).
10. N. J. L. Megson, A. A. Drummond, J. Soc. Chem. Ind., **49**, 251 T (1930).
11. E. Ziegler, I. Hontschik, L. Milowiz, Montash. Chem., **78**, 334 (1948).
12. H. Kaemmerer, W. Rausch, H. Schweikert, Makromol. Chem., **56**, 123 (1962).
13. K. Auwers, Chem. Ber., **40**, 2524 (1907).
14. K. Fries, K. Kann, Ann. Chem., **353**, 335 (1907).
15. K. Hultzsch, Chem. Ber., **74**, 989 (1941).
16. A. Zinke, E. Ziegler, Chem. Ber., **74**, 541 (1941).
17. H. Kaemmerer, K. Haberer, Monatsh. Chem., **95**, 1589 (1964).
18. A. Zinke, R. Kretz, E. Leggewie, K. Hoessinger, Monatsh. Chem., **83**, 1213 (1952).
19. S. R. Finn, J. W. G. Musty, J. Soc. Chem. Ind., **69** (Suppl. No. 1), S 3 (1950).
20. E. Adler, H. V. Euler, G. J. Gie, Arkiv Kemi, **16A**, No. 12 (1943).
21. L. M. Debing, Trans. Electrochem. Soc., **90**, 277 (1946).
22. H. V. Euler, E. Adler, J. O. Cedwall, Arkiv Kemi, **15A**, No. 19 (1942).
23. A. V. Bayer, V. Frewsen, Ann. Chem., **212**, 344 (1882).
24. H. Kaemmerer, G. Buesing, H.-G. Haub, Makromol. Chem., **66**, 82 (1963).
25. H. Kaemmerer, M. Harris, Makromol. Chem., **62**, 18 (1963); J. Polym. Sci. A, **2**, 4003 (1964).
26. L. Lederer, J. Prakt. Chem., **50**, 223 (1894).
27. S. R. Finn, J. W. James, C. J. S. Standen, Chem. Ind., 188, (1954); J. Appl. Chem., **4**, 296 (1954).
28. A. T. Carpenter, R. F. Hunter, J. Appl. Chem., **3**, 486 (1953).
29. S. R. Finn, J. W. G. Musty, J. Appl. Chem., **2**, 88 (1952).
30. J. H. Freeman, J. Am. Chem. Soc., **74**, 6257 (1952).
31. H. Kaemmerer, M. Grossmann, Chem. Ber., **86**, 1492 (1953).
32. J. Reese, Angew. Chem., **64**, 399 (1952).
33. H. V. Euler, E. Adler, G. Eklund, O. Toerngren, Arkiv Kemi, **15B**, No. 9 (1942).

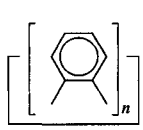
34. O. Manasse, Chem. Ber., **27**, 2409 (1894).
35. P. Maitland, D. C. Pepper, J. Soc. Chem. Ind., **61**, 66 (1942).
36. B. T. Hayes, R. F. Hunter, J. Appl. Chem., **8**, 743 (1958).
37. F. Ullmann, K. Brittner, Chem. Ber., **42**, 2539 (1909).
38. F. Adler, Arkiv Kemi, **14B**, No. 23 (1941).
39. S. R. Finn, G. J. Lewis, J. Soc. Chem. Ind., **69**, 132 (1950).
40. S. Kyrning, Arkiv Kemi, **15A**, No. 2 (1941).
41. F. Hanus, E. Fuchs, J. Prakt. Chem., **153**, 327 (1939).
42. H. Kaemmerer, M. Dahm, Kunstst. Plast., **6**, 1 (1959).
43. B. T. Hayes, R. F. Hunter, Chem. Ind., 193, (1956).
44. A. Zinke, E. Ziegler, Chem. Ber., **77**, 264 (1944).
45. K. Hultzsch, Kunststoffe, **52**, 19 (1962).
46. H. V. Euler, E. Adler, B. Bergstroem, Arkiv Kemi, **14B**, No. 30 (1941).
47. R. F. Hunter, C. Turner, Chem. Ind., 72, (1957).
48. E. Ziegler, Monatsh. Chem., **79**, 142 (1948).
49. E. Ziegler, Oester, Chem. Ztg., **49**, 92 (1948).
50. A. T. Carpenter, R. F. Hunter, J. Appl. Chem., **1**, 217 (1951).
51. A. T. Carpenter, R. F. Hunter, J. Chem. Soc., 2731, (1954).
52. H. Kaemmerer, H. Lenz, Kunststoffe, **51**, 26 (1961).
53. A. C. Davis, B. T. Hayes, R. F. Hunter, J. Appl. Chem., **7**, 521 (1957).
54. R. F. Hunter, C. Turner, J. Appl. Chem., **7**, 528 (1957).
55. H. v. Euler, E. Adler, S. V. Kispeczy, A. M. Fagerlund, Arkiv Kemi, **14A**, No. 10 (1940).
56. H.-J. Eichhoff, H. Kaemmerer, D. Weller, Makromol. Chem., **129**, 109 (1969).
57. H.-J. Eichhoff, H. Kaemmerer, D. Weller, Makromol. Chem., **132**, 163 (1970).
58. H. Kaemmerer, G. Gros, H. Schweikert, Makromol. Chem., **143**, 135 (1971).
59. H. Kaemmerer, W. Lotz, Makromol. Chem., **145**, 1 (1971).
60. H. Kaemmerer, G. Gros, Makromol. Chem., **149**, 85 (1971).
61. M. B. Huglin, G. J. Knight, W. W. Wright, Makromol. Chem., **152**, 67, 83 (1972).
62. T. Carins, G. Eglinton, Nature (London), **196**, 535 (1962).
63. H. Kaemmerer, A. Kiegel, unpublished results.
64. G. Manecke, D. Zeipner, Makromol. Chem., **129**, 183 (1969).
65. C. D. Gutsche, B. Dhawan, K. H. No, R. Muthukrishnan, J. Am. Chem. Soc., **103**, 3782 (1981).
66. C. D. Gutsche, Top. Curr. Chem., 123, (1984); Acc. Chem. Res., **16**, 161 (1983).
67. C. D. Gutcshe, K. H. No, J. Org. Chem., **47**, 2708 (1982); C. D. Gutsche, P. F. Pagoria, *ibid.*, **50**, 5795 (1985).
68. A. Ninagawa, H. Matsuda, Makromol. Chem., Rapid Commun., **3**, 65 (1982).

TABLE 21. OLIGO(PHENYLENES)

21.1. *o*-OLIGO(PHENYLENES)

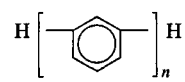
21.1.1. LINEAR *o*-OLIGO(PHENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
3	230.3	59	332	1,2
4	306.4	119	420	1,3–6
6	458.6	217		3,5
8	610.8	320		5

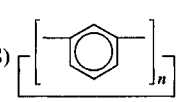
21.1.2. CYCLIC *o*-OLIGO(PHENYLENES) 

n	Mol. wt.	m.p. (°C)	Refs.
2	152.2	111	4,5,7–9
3	228.3	196.5	10,11
4	304.4	233	4,5
6	456.6	432	5
8	608.8	425	5

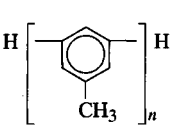
21.2. *m*-OLIGO(PHENYLENES)

21.2.1. LINEAR *m*-OLIGO(PHENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
3	230.3	89	365	12–14
4	306.4	86.5–87.5	419	3,6,15–17
5	382.5	117–117.5		12,15,16
6	458.6	148		3,15,17,18
8	610.8	129–131		15,17
9	689.9	195–200		18

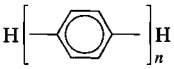
21.2.2. CYCLIC *m*-OLIGO(PHENYLENES) 

n	Mol. wt.	m.p. (°C)	Refs.
6	456.6	509.5–511	19
8	608.8	449–451	19

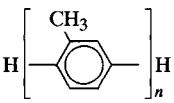
21.2.3. OLIGO(3-METHYL-*m*-PHENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
2	182.3	9–9.5	280	20–23
3	272.4	65		12

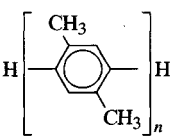
21.3. *p*-OLIGO(PHENYLENES)

21.3.1. *p*-OLIGO(PHENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C/mm)	Solubility (g/L toluene)	Refs.
1	78.1	5.5	80.1	∞	
2	154.2	71	156	430	24
3	230.3	215	376	7.4	3,12,14,25,26
4	306.4	322	428/10	0.12	3,12,16,21,25,26
5	382.5	395		<0.005	12,16,21,25,27,28
6	458.6	465			3,29–31

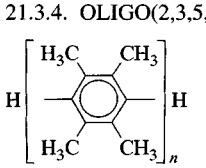
21.3.2. OLIGO(3-METHYL-*p*-PHENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
1	92.1	−95	110.6	
2	182.3		273–274	22,23,32
3	272.4	43		33

21.3.3. OLIGO(2,5-DIMETHYL-*p*-PHENYLENES) 

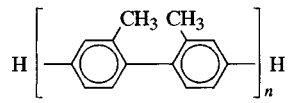
n	Mol. wt.	m.p. (°C)	b.p. (°C)	Solubility (g/L toluene)	Refs.
1	106.2	13.3	138.4	∞	
2	210.3	53–54		700	20,34
3	314.5	182–183		28	34
4	418.6	264–266		1.1	34
5	522.8	307–309		0.24	34

21.3.4. OLIGO(2,3,5,6-TETRAMETHYL-*p*-PHENYLENES)



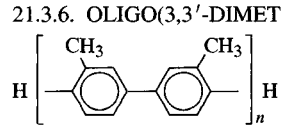
n	Mol. wt.	m.p. (°C)	b.p. (°C)	Solubility (g/L toluene)	Refs.
1	134.2	79.2	196–198	∞	
2	266.4	136–137		365	34,35
3	398.6	270–272		24	34
4	530.8	270–272		3.4	34

21.3.5. OLIGO(2,2′-DIMETHYL-BIPHENYLENES)



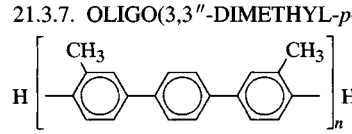
n	Mol. wt.	m.p. (°C)	Solubility (g/L toluene)	Refs.
1	182.3	24		20,23,36
2	362.5	96	490	37

21.3.6. OLIGO(3,3′-DIMETHYL-BIPHENYLENES)



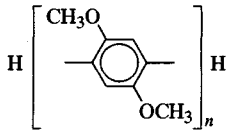
n	Mol. wt.	m.p. (°C)	b.p. (°C)	Solubility (g/L toluene)	Refs.
1	182.3	9–9.5	280	∞	20–23,38
2	362.5	76.5		512	27,37
3	542.8	142		87	39
4	723.0	273		<6.5	39,40
5	903.3	285		~2	39,40
6	1083.5	298		<0.8	39,40

21.3.7. OLIGO(3,3″-DIMETHYL-*p*-TERPHENYLENES)



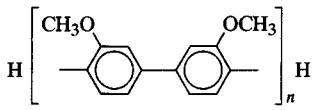
n	Mol. wt.	m.p. (°C)	Solubility (g/L toluene)	Refs.
1	258.4	140–141	4.9	27,37
2	514.7	258	0.4	37

21.3.8. OLIGO(2,5-DIMETHOXY-*p*-PHENYLENES)



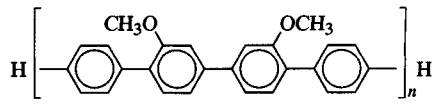
n	Mol. wt.	m.p. (°C)	b.p. (°C)	Refs.
1	138.2	56	212.6	
2	274.3	105		20,41
3	410.5	189		41
4	546.6	246		41

21.3.9. OLIGO(3,3′-DIMETHOXY-BIPHENYLENES)

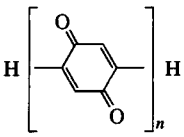


n	Mol. wt.	m.p. (°C)	b.p. (°C)	Solubility (g/L toluene)	Refs.
1	214.3	36	328		42
2	426.5	158		22	38,43

21.3.10. OLIGO(2′,3′′-DIMETHOXY-*p*-QUATERPHENYLENES)



n	Mol. wt.	m.p. (°C)	Solubility (g/L toluene)	Refs.
1	366.5	183–184	13	43
2	730.9	276–277	0.2	43

21.4. OLIGO(*p*-QUINONES) 

n	Mol. wt.	m.p. (°C)	Refs.
1	108.1	116.5	
2	214.2	194	41
3	320.3	>230	41
4	426.3	230–270	41

REFERENCES

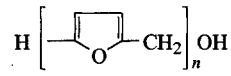
1. W. Bachmann, H. T. Clarke, J. Am. Chem. Soc., **49**, 2089 (1927).
2. C. F. H. Allen, F. P. Pingert, J. Am. Chem. Soc., **64**, 1365 (1942).
3. J. A. Cade, A. Pilbeam, Tetrahedron, **20**, 519 (1964); J. Chem. Soc., 114 (1964).
4. W. S. Rapson, R. G. Shuttleworth, J. N. van Niekerk, J. Chem. Soc., 326 (1943).
5. G. Wittig, G. Lehmann, Chem. Ber., **90**, 875 (1957); G. Wittig, G. Klar, Ann. Chem., **704**, 91 (1967).
6. S. T. Bowden, J. Chem. Soc., 1111 (1931).
7. W. Baker, M. P. V. Boarland, J. F. W. McOmie, J. Chem. Soc., 1476 (1954).
8. W. C. Lothrop, J. Am. Chem. Soc., **63**, 1187 (1941).
9. G. Wittig, W. Herwig, Chem. Ber., **87**, 1511 (1954).
10. C. Mannich, Chem. Ber., **40**, 159 (1907).
11. P. G. Copeland, R. E. Dean, D. McNeil, J. Chem. Soc., 1689 (1960).
12. M. Busch, W. Weber, J. Prakt. Chem., **146**, 1 (1936).
13. A. E. Gillam, D. H. Hey, J. Chem. Soc., 1170 (1939); H. France, I. M. Heilbron, D. H. Hey, J. Chem. Soc., 1288 (1939); 1364 (1938).
14. G. F. Woods, J. W. Tucker, J. Am. Chem. Soc., **70**, 2174 (1948).
15. R. L. Alexander, Jr., J. Org. Chem., **21**, 1464 (1956).
16. G. F. Woods, F. T. Reed, J. Am. Chem. Soc., **71**, 1348 (1949).
17. M. Bennett, N. B. Sunshine, G. F. Woods, J. Org. Chem., **28**, 2514 (1963); W. Davey, D. H. Maass, J. Chem. Soc., 4386 (1963).
18. L. Silverman, W. Houk, Anal. Chem., **27**, 1956 (1955).
19. H. A. Staab, F. Binnig, Tetrahedron Lett., 319 (1964); Chem. Ber., **100**, 293 (1967); H. Bräunling, F. Binnig, H. A. Staab, *ibid.*, **100**, 880 (1967).
20. F. Ullmann, G. M. Meyer, O. Loewenthal, E. Gilli, Ann. Chem., **332**, 38 (1904).
21. E. Mueller, T. Toepel, Chem. Ber., **72**, 273 (1939).
22. G. F. Woods, A. L. Van Artsdale, F. T. Reed, J. Am. Chem. Soc., **72**, 3221 (1950).
23. E. A. Johnson, J. Chem. Soc., 4155 (1957).
24. E. Clar, "Polycyclic Hydrocarbons", Academic, New York, Springer, Berlin, 1964.
25. O. Gerngross, M. Dunkel, Chem. Ber., **57**, 739 (1924); O. Gerngross, C. Schachnow, R. Jonas, Chem. Ber., **57**, 747 (1954).
26. H. O. Wirth, K. H. Goenner, W. Kern, Makromol. Chem., **63**, 53 (1963).
27. H. O. Wirth, K. H. Goenner, R. Stueck, W. Kern, Makromol. Chem., **63**, 30 (1963).
28. T. W. Campbell, R. N. McDonald, J. Org. Chem., **24**, 730 (1959).
29. P. Kovacic, R. M. Lange, J. Org. Chem., **29**, 2416 (1964).
30. T. Nozaki, M. Tamura, Y. Harada, K. Saito, Bull. Chem. Soc. Japan, **33**, 1329 (1960).
31. R. Pummerer, K. Bittner, Chem. Ber., **57**, 84 (1924); R. Pummerer, L. Seligsberger, Chem. Ber., **64**, 2477 (1931).
32. F. Mayer, K. Freitag, Chem. Ber., **54**, 347 (1921).
33. H. O. Wirth, H. Hefner, W. Kern, unpublished results.
34. H. O. Wirth, F. U. Herrmann, W. Kern, Makromol. Chem., **80**, 120 (1964).

35. E. Marcus, W. M. Lauer, R. T. Arnold, J. Am. Chem. Soc., **80**, 3742 (1958).
36. D. M. Hall, M. S. Lesslie, E. E. Turner, J. Chem. Soc., 711 (1950).
37. W. Kern, M. Seibel, H. O. Wirth, Makromol. Chem., **29**, 164 (1959).
38. W. Schlenk, M. Brauns, Chem. Ber., **48**, 661 (1914).
39. W. Kern, W. Gruber, H. O. Wirth, Makromol. Chem., **37**, 198 (1960).
40. W. Heitz, R. Ullrich, W. Kern, Makromol. Chem., **98**, 29 (1966).
41. H. Erdtmann, M. Granath, G. Schultz, Acta Chem. Scand., **8**, 1442 (1954).
42. N. Kornblum, Org. Syn. Coll. Vol. III, 295 (1962).
43. W. Kern, H. W. Ebersbach, I. Ziegler, Makromol. Chem., **31**, 154 (1959).

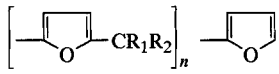
E. OLIGOMERS CONTAINING HETEROCYCLIC RINGS IN THE MAIN CHAIN

TABLE 22. HETEROCYCLIC OLIGOMERS

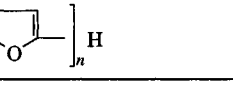
22.1. OLIGO(FURAN)DERIVATIVES

22.1.1. OLIGO(FURFURYL) ALCOHOLS 

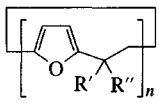
n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
1	98.1		171/1000	1.4868	10,11
2	178.2	−2.5	141–143/12	1.5290	10–12,14
3	258.3	62–62.5	192.5–295/8		10–12,14

22.1.2. OLIGO(FURFURYL) FURANS 

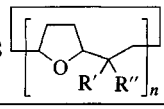
R_1	R_2	*n*	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{20} (g/cm³)	n_D^{20}	Refs.
H	H	1	148.2		78/16	1.102	1.5048	10,12,14–15,39
		2	228.2		141–146/11		1.5324	10,12
		3	308.3	74–75				10,12,14
		4	388.4	99–101				10,14
H	CH_3	1	162.2		91/28	1.073	1.4993	16,17
		2	256.3		114–116/0.13			17
		3	350.4		165–168/0.04			16,17
CH_3	CH_3	1	176.2	−12	73–76/7–8	1.043	1.4966	17,18
		2	284.3	47.0/47.5	120–123/1.3			17,18
		3	392.5		163–167/1.3	1.030	1.5172	17,18
		4	500.6	Oil				17
		5	608.7	83–85				17
CH_3	C_2H_5	1	190.2		68–70/3	1.033	1.4970	18
		2	312.4		143–145/1.3	1.045	1.5094	19
		3	434.6		205–210/1.3	1.046	1.5173	19
C_2H_5	C_2H_5	1	204.3	−9 to −8	76–77/3	1.023	1.4978	20
		2	340.4		162–163/2	1.028	1.5089	20
		3	476.6		212–215/2	1.034	1.5199	20
		1	216.3	13–17	94–95/2			21
		2	364.5	86.8–87.6	200–205/2	1.018	1.5221	21

22.1.3. 2,5-OLIGO(FURYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	d_4^{20} (g/cm³)	n_D^{20}	Refs.
1	68.1	−85.7	31.4/1013	0.9514/20	1.4214/20	
2	134.1		63–64/15		1.5601/22	39–41,42
3	200.2	62–63				32

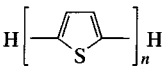
22.1.4. FURANOPHANES 

R′	R″	n	Mol. wt.	m.p. (°C)	Refs.
H	CH_3	4	376.4	140–142	16,17,22
	r-2,c-7,c-12,c-17-			191.0–192.0	22
H	C_2H_5	4	432.5	83–88	22
	r-2,c-7,c-12,c-17-			152.0–153.0	22
CH_3	CH_3	4	432.5	243	18,23,24
CH_3	CH_3	5	540.7	Oil	17
		6	648.8	182	17,18,24
CH_3	C_2H_5	4	488.5	174	18,19,22,24
C_2H_5	C_2H_5	4	544.7	249–250	18,20,24
H		4	592.8	268–269	21,24

22.1.5. TETRAHYDROFURANOPHANES 

n	Mol. wt.	m.p. (°C)	Refs.
4	448.7	204–209	17,18,20,21,23
	r-(1S, 3S, 6R, 8S, 11R, 13R, 16S, 18R)-	218	25
	r-(1S, 3R, 6S, 8R, 11S, 13R, 16S, 18R)-	221	25
5	560.8	Oil	17
6	673.0	75–80	17

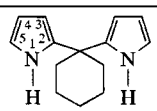
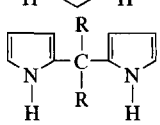
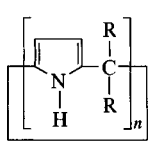
22.2. OLIGO(THIOPHENE) DERIVATIVES

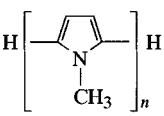
22.2.1. OLIGO(2.5-THIENYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
1	84.1	−38.4	84.2/1013	1.5289	
2	166.3	33–34	125/16		1,27,28,34,41,42,44,49
3	248.4	96–97			1,3,26,28,30–33,42,44,47,49
4	330.5	215–216			1,3,27–29,41,42,44,46,49
5	412.5	257–258			1,3,28,29,42,48
6	494.8	304–305			3,27,28,42,44,46
7	576.9	327–328			3,29,48
8	659.0	364			42,46

22.3. OLIGO(PYRROLE) DERIVATIVES

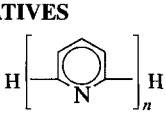
22.3.1. OLIGO(PYRROLE) DERIVATIVES

No.	Oligomers	R	n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1			2	214.3	105–106		21,43
2		CH_3	2	174.2	59	125–126/4	35
3		CH_3	4	428.6	291		36,37
			4	588.8	272.0–272.5		21

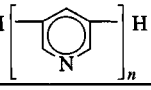
22.3.2. OLIGO[2,5-(*N*-METHYL)PYRROLENES] 

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	Refs.
1	81.1		114–115/99.6	
2	160.2	14–15	77–78/0.5	13,42
4	318.4	134		42
5	397.5	145		42
6	476.6	160		42
8	634.8	178–182		42
16	1267.6	240–256		42

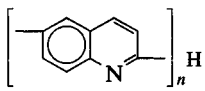
22.4. OLIGO(PYRIDINE) DERIVATIVES

22.4.1. OLIGO(2,6-PYRIDYLENES) 

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
1	79.1	−42	115.5/1013	1.5095	
2	156.2	70.1	273–275/1013		4,5
3	233.3	89–90	370/1013		2,4–6
4	310.4	219–220			5
5	387.5	265			5
6	464.5	350			5

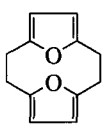
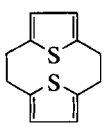
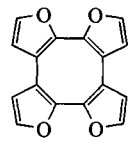
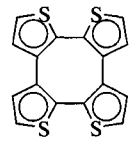
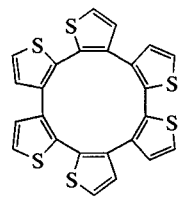
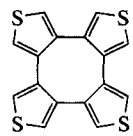
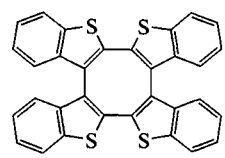
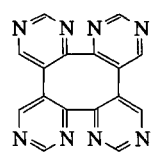
22.4.2. OLIGO(3,5-PYRIDYLENES) H  H

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
1	79.1	−42	115.5	1.5095	
2	156.2	68	291–292/981		4,9
3	233.3	249–251			9

22.4.3. OLIGO(2,6-QUINOLYLENES) H  H

n	Mol. wt.	m.p. (°C)	b.p. (°C/mbar)	n_D^{20}	Refs.
1	129.2	−15.6	238.1/1013	1.6268	
2	256.3	144			7,8
3	383.5	267–269			8
4	510.6	348–350			8

22.5. CYCLIC OLIGO(HETEROCYCLICS)

No.		Mol. wt.	m.p. (°C)	Refs.
1		188.2	189–190	38
2		220.3	194.5–196	38
3		264.2	233	42,45
4		328.5	255–256	42,45
5		492.7	420	42,45
6		328.5	300–301	42,45
7		528.7	410	45
8		312.3	355–360	45

REFERENCES

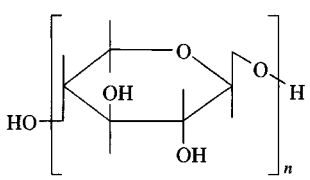
1. J. W. Sease, L. Zechmeister, J. Am. Chem. Soc., **69**, 270 (1947).
2. Th. Kauffmann, J. König, A. Woltermann, Chem. Ber., **109**, 3864 (1976).
3. W. Steinkopff, R. Leitsmann, K. H. Hoffmann, Ann. Chem., **546**, 180 (1941).
4. G. T. Morgan, F. H. Burstall, J. Chem. Soc., 20 (1932).
5. F. H. Burstall, J. Chem. Soc., 1662 (1938).
6. F. H. Case, W. H. Butte, J. Org. Chem., **26**, 4415 (1961).
7. H. Weidel, Monatsh. Chem., **8**, 120, 141 (1887).
8. S. G. Waley, J. Chem. Soc., 2008 (1948).
9. M. Busch, W. Weber, J. Prakt. Chem., **146**, 1 (1936).
10. J. T. Barr, S. B. Wallon, J. Appl. Polym. Sci., **15**, 1079 (1971).
11. A. P. Dunlop, F. N. Peters, Ind. Eng. Chem., **34**, 814 (1942).
12. A. H. Fawcett, W. Dadamba, Makromol. Chem., **183**, 2799 (1982).
13. J. Gjøs, S. Gronowitz, Acta Chem. Scand., **25**, 2596 (1971).
14. K. Takano, Nippon Kagaku Zasski, **79**, 955 (1958); Chem. Abstr., **54**, 4530e (1960); *ibid.*, **80**, 313 (1959); Chem. Abstr., **55**, 5453g (1961).
15. H. Gilman, G. F. Wright, J. Am. Chem. Soc., **55**, 3302 (1933).
16. W. H. Brown, H. Sanatzky, Can. J. Chem., **34**, 1147 (1956).
17. Y. Kobuke, K. Hanji, K. Horiguchi, M. Asada, Y. Nakayama, J. Furukawa, J. Am. Chem. Soc., **98**, 7414 (1976).
18. R. G. Ackman, W. H. Brown, G. F. Wright, J. Org. Chem., **20**, 1147 (1955).
19. W. H. Brown, W. N. French, Can. J. Chem., **36**, 537 (1958).
20. R. E. Beals, W. H. Brown, J. Org. Chem., **21**, 447 (1956).
21. W. H. Brown, B. J. Hutchinson, M. H. MacKinnon, Can. J. Chem., **49**, 4017 (1971).
22. A. G. S. Högberg, M. Weber, Acta Chem. Scand. B, **37**, 55 (1983).
23. M. Chastrette, F. Chastrette, J. Sabadie, Org. Synth., **57**, 74 (1977).
24. M. De Sousa Healy, A. J. Rest, J. Chem. Soc., Chem. Commun., 149 (1981); J. Chem. Soc., Perkin Trans. I, 973 (1985).
25. M. Van Beylen, B. Roland, G. S. D. King, J. Aerts, J. Chem. Res. (S), 388 (1985); (M), 4201 (1985).
26. J. Kagan, S. K. Arora, Heterocycles, **20**, 1941 (1983).
27. J. Kagan, S. K. Arora, Heterocycles, **20**, 1937 (1983).
28. J. Kagan, S. K. Arora, Tetrahedron Lett., **24**, 4043 (1983).

29. J. Kagan, S. K. Arora, J. Org. Chem., **48**, 4317 (1983).
30. K. E. Schulte, J. Reisch, L. Hörner, Chem. Ber., **95**, 1943 (1962).
31. H. J. Kooreman, H. Wynberg, Rec. Trav. Chim. Pays-Bas, **86**, 37 (1967).
32. T. Asano, S. Ito, N. Saito, K. Hatakeda, Heterocycles, **6**, 317 (1977).
33. J.-P. Beny, S. N. Dhawan, J. Kagan, S. Sundless, J. Org. Chem., **47**, 2201 (1982).
34. J.-P. Morizur, Bull. Soc. Chim. France, 1331 (1964); J.-P. Morizur, C. R., Sér. C, **254**, 1093 (1962).
35. W. H. Brown, W. N. French, Can. J. Chem., **36**, 371 (1958).
36. A. V. Baeyer, Chem. Ber., **19**, 2184 (1886).
37. V. V. Chelintzev, B. V. Tronov, S. G. Karmanov, J. Russ. Phys. Chem., **48**, 1210 (1916); Chem. Abstr. **11**, 1418 (1917).
38. H. E. Winberg, F. S. Fawcett, W. E. Mochel, C. W. Theobald, J. Am. Chem. Soc., **82**, 1428 (1960).
39. T. Reichstein, A. Grüssner, H. Zschokke, Helv. Chim. Acta, **15**, 1066 (1932).
40. R. Grigg, J. A. Knight, M. V. Sargent, J. Chem. Soc. C, 976 (1966).
41. R. E. Atkinson, R. F. Curtis, G. T. Phillips, J. Chem. Soc. C, 2011 (1967).
42. Th. Kauffmann, Angew. Chem., **91**, 1 (1979). Angew. Chem., Int. Ed. Engl., **18**, 1 (1979).
43. D. Dolphin, R. Grigg, M. V. Sargent, D. H. Williams, J. A. Knight, Tetrahedron, **21**, 3441 (1965).
44. D. D. Cunningham, L. Laguren-Davidson, H. B. Mark Jr., Ch. V. Pham, H. Zimmer, J. Chem. Soc., Chem. Commun., 1021 (1987).
45. Th. Kauffmann, B. Greving, R. Kriegesmann, A. Mitschker, A. Woltermann, Chem. Ber., **111**, 1330 (1978).
46. J. Nakayama, T. Konishi, S. Murabayashi, M. Hoshino, Heterocycles, **26**, 1793 (1987).
47. J. Nakayama, Y. Nakamura, T. Tajiri, M. Hoshino, Heterocycles, **24**, 637 (1986).
48. J. Nakayama, Y. Nakamura, S. Murabayashi, M. Hoshino, Heterocycles, **26**, 939 (1987).
49. K. Tamao, S. Kodama, I. Nakajima, M. Kumada, A. Minato, K. Suzuki, Tetrahedron, **38**, 3347 (1982).

TABLE 23. OLIGO(SACCHARIDES)

23.1. OLIGOMERIC PENTOSES

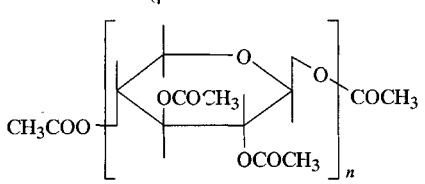
23.1.1. OLIGO(β-XYLOPYRANOSES)[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{25}$ in H_2O
1	150.1	153	19.2
2	282.2	186–187	−25.6
3	414.4	215–216	−48.1
4	546.5	224–226	−61.9
5	678.6	240–242	−72.9
6	810.7	237–242	−78.5
7	942.8	240–242	−74.0

[a] Refs. 1–3.

23.1.2. OLIGO(β-XYLOPYRANOSE ACETATES)[a]

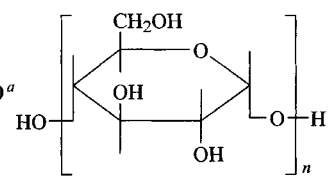


n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{25}$	*c* in $CHCl_3$
2	534.5	155.5–156	−74.5	0.9
3	750.7	109–110	−84.3	0.6
4	966.9	201–202	−93.7	0.8
5	1183.1	249–250	−97.5	1.1
6	1399.3	260–261	−102.0	1.5

[a] Refs. 1–3.

23.2. OLIGOMERIC HEXOSES

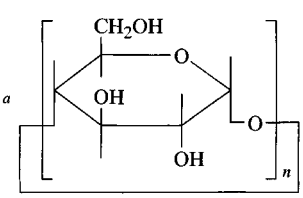
23.2.1. MALTO-OLIGO[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{15}$ in H_2O
1	180.2	146	52.6
2	342.3	160–165	136.0
3	504.4		160.0
4	666.6		177.0
5	828.7		180.3
6	990.9		184.7
7	1153.0		186.4

[a] Refs. 4–6.

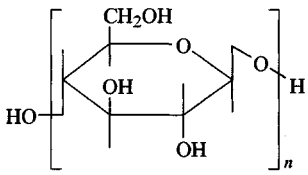
23.2.2. CYCLODEXTRINS[a]



n	Mol. wt.	$[\alpha]_D$ ($c = 1$, H_2O)
6	972.8	149.0
7	1135.0	158.8
8	1297.1	170.0

[a] Refs. 4–6.

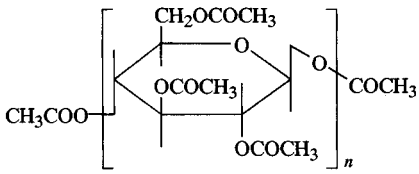
23.2.3. CELLO-OLIGOSES[a]



n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c in H_2O
1	180.2	150	52.5	20	4
2	342.3	225 d	34.6	20	8
3	504.4	238 d	21.6	26	4
4	666.6	253 d	16.5	23	3.4
5	828.7	267 d	11.0	30	4.1
6	990.9	278 d	10.0	30	1.2
7	1153.0	286 d	7.3	30	0.1

[a] Refs. 7–14.

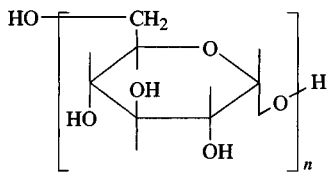
23.2.4. CELLO-OLIGOSE ACETATES[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D^{20-25}$ (c = 5, in $CHCl_3$)
1	390.4	113	101.6
2	678.6	229.5	41.0
3	966.9	223–224	22.6
4	1255.1	230–234	13.4
5	1543.4	240–241	4.2
6	1831.6	252–255	−0.2
7	2119.8	263–266	−4.4

[a] Refs. 7–14.

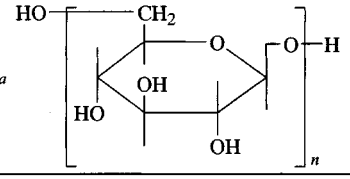
23.2.5. ISOMALTO-OLIGOSES[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D$ in H_2O
1	180.2	146	52.6
2	342.3	225 d	
3	504.4		
4	666.6		153
5	828.7		160
6	990.9		163

[a] Ref. 15.

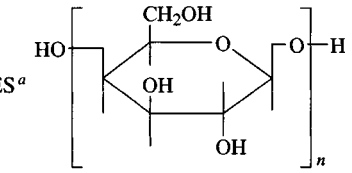
23.2.6. GENTIO-OLIGOSES[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D$ in H_2O
1	180.2	150	52.5
2	342.3	190–195	9.6
3	504.4		−10.5
4	666.6		−19.5

[a] Ref. 15.

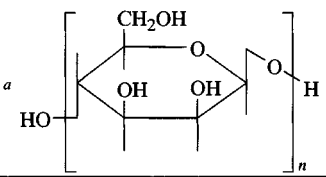
23.2.7. GALACTO-OLIGOSES[a]



n	Mol. wt.	m.p. (°C)	$[\alpha]_D$ in H_2O
1	180.2	167	−52.5
2	342.3	210–211	173
3	504.4	115–120	58
4	666.6		53

[a] Ref. 15.

23.2.8. MANNO-OLIGOSES[a]

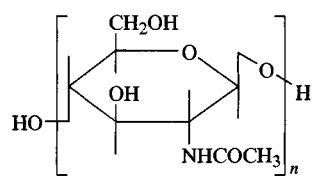


n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	(°C)	c, H_2O
1	180.2	132			
2	342.3	193–194	−7.7	25	0.9
3	504.4	137–137.5	−23.3		1.3
4	666.6	232–234	−31		1.6

[a] Refs. 16,17.

23.3. OLIGOMERIC AMINO SUGARS

23.3.1. *N*-ACETYL CHITO-OLIGOSES[a]



n	Mol. wt.	m.p. (°C)	Specific rotation $[\alpha]_D$	c in H_2O
1	221.2			
2	424.4	260–262 d	17.2	0.5
3	627.6	290–311 d	2.2	0.9
4	830.8	290–300 d	−4.1	1.0
5	1034.0	285–295 d	−9.1	1.0
6	1237.2		−11.4	0.8
7	1440.4		−12.6	0.3

[a] Refs. 18,19.

REFERENCES

1. R. L. Whistler, C.-C. Tu, J. Am. Chem. Soc., **74**, 4334 (1952).
2. R. H. Marchessault, T. E. Timell, J. Polym. Sci. C, **2**, 49 (1963).
3. C. T. Bishop, Can. J. Chem., **33**, 1073 (1955).
4. W. J. Whelan, J. M. Bailey, P. J. P. Roberts, J. Chem. Soc., 1293 (1953).
5. J. M. Bailey, W. J. Whelan, S. Peart, J. Chem. Soc., 3692 (1950).
6. K. Freudenberg, F. Cramer, Chem. Ber., **83**, 296 (1950).
7. E. E. Dickey, M. L. Wolfrom, J. Am. Chem. Soc., **71**, 825 (1949).
8. M. L. Wolfrom, J. C. Dacons, J. Am. Chem. Soc., **74**, 5331 (1952).
9. L. Zechmeister, G. Tóth, Chem. Ber., **64**, 854 (1931).
10. K. Hess, K. Dziengel, Chem. Ber., **68**, 1594 (1935).
11. C. S. Hudson, J. M. Johnson, J. Am. Chem. Soc., **37**, 1276 (1915).
12. R. Willstaetter, L. Zechmeister, Chem. Ber., **46**, 2401 (1913); **62**, 722 (1929).
13. K. Freudenberg, G. Blomquist, Chem. Ber., **68**, 2070 (1935).
14. H. Staudinger, E. V. Leopold, Chem. Ber., **67**, 479 (1934).
15. W. Walter, in: H. M. Rauen (Ed.), "Biochemisches Taschenbuch", part 1, 2nd ed., Springer, Berlin, 1954, p. 98.
16. R. L. Whistler, C. G. Smith, J. Am. Chem. Soc., **74**, 3795 (1952).
17. R. L. Whistler, J. Z. Stein, J. Am. Chem. Soc., **73**, 4187 (1951).
18. S. A. Barker, A. B. Foster, M. Stacey, J. M. Webber, J. Chem. Soc., 2218 (1958).
19. H. P. Lenk, M. Wenzel, E. Schuette, Hoppe-Seyler's Z. Physiol. Chem., **326**, 116 (1961).

SECTION V

PHYSICAL CONSTANTS OF SOME IMPORTANT POLYMERS

Physical Constants of Rubbery Polymers

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A. Introduction V-1
B. Tables V-1
Table 1. 1,4-*cis*(96–98%)Poly(butadiene) V-1
Table 2. Poly(butadiene-*co*-acrylonitrile) V-2
Table 3. Poly(butadiene-*co*-styrene) V-3
Table 4. Poly(chloroprene)(CR Neoprene) V-3
Table 5. Poly(isobutene)-*co*-isoprene)Butyl Rubber (IIR) V-4
Table 6. Polyisoprene, Natural Rubber V-5
Table 7. Ethylene–Propylene–Diene–Terpolymer (EPDM) V-6
C. References V-6

A. INTRODUCTION

Where a range is given, there are available several observations which differ. In most cases the differences are thought to be real, arising from differences in the rubber rather than from errors of observation. Where a single value is given, it is either because no other observations are available or because there seems to be no significant disagreement among values within the errors of observation. Where values are not given, data have not been found. Where dashes are shown, either the physical measurement is impossible or the constant in question is not adequately defined under the given conditions. The values shown refer to specific vulcanizates cited in the corresponding references. Other vulcanizates may yield a broader range of values.

The values are expressed in the International System of Units (SI), the modern metric system, described in National Bureau of Standards Special Publication 330, 1981 edition, and in American Society for Testing and Materials Metric Practice Guide E380–85.

Values are given for constants at a temperature of 25°C (273.15 K) and a pressure of 1 normal atmosphere = 101.325 kPa.

B. TABLES

TABLE 1. 1,4-*cis*(96–98%)POLY(BUTADIENE)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3			1.01	1
Coefficient of expansion	K^{-1} ($\times 10^6$)	669	2		
Glass transition temperature	°C			−110 to −95	3
Specific heat	J/kg/K ($\times 10^3$)	1.96	4	1.960–1.970	1
Thermal conductivity	W/m/K ($\times 10^{-3}$)			200	5
Thermal diffusivity	m^2/s ($\times 10^{-9}$)				
Solubility parameter	$MPa^{1/2}$	17.0	6		
Polymer–solvent interaction parameter (25–30°C)					
	Solvent				
	n-Heptane	0.51	7		
	n-Hexane	0.53	7		
	Cyclohexane	0.45	8		
	Benzene	0.44	7		
	Toluene	0.36	7		

TABLE 1. *cont'd*

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Equilibrium melting temperature	K (°C)	1.0	10		
		0	11		
Temperature of most rapid crystallization	K (°C)	−52 to −57	3		
Heat of fusion of crystal	J/kg ($\times 10^3$)	47	11		
		169	10		
Refractive index	N_D	1.526	12		
Molar polarizability	cm^3 ($\times 10^{-25}$)	71.4	12		
Dielectric constant (50 Hz)		2.3	3		
Dielectric loss factor (50 Hz)		0.8	3		
Compressibility	Pa^{-1} ($\times 10^{-12}$)	500	2		
Young's modulus	Pa ($\times 10^6$)			1.3	9
Shear modulus	Pa ($\times 10^6$)			0.96	9
Storage modulus (1 Hz)	log Pa			5.98	9
Loss modulus	log Pa			4.79	9
Loss tangent				0.065	9

TABLE 2. POLY(BUTADIENE-*co*-ACRYLONITRILE)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3			0.95[a] (20%)	3
				1.02 (45%)	3
Glass transition temperature	°C			−85 + A(20–40%) (A = AN%)	3
Specific heat	J/kg/K ($\times 10^3$)			1.970(40%)	3
Thermal conductivity	W/m/K ($\times 10^{-3}$)			250 (18–35%)	3
Thermal diffusivity	m^2s ($\times 10^{-9}$)				
Solubility parameter	$MPa^{1/2}$	18.5–21.0 (18–39%)	6		
Polymer–solvent interaction parameter (25–30°C)					

	AN%			
Solvent	**18**	**30**	**39**	
n-Heptane	0.96	1.88	3.6	9
n-Hexane	0.99		2.8	9
Dichloromethane	0.39	0.31	0.32	9
Cyclohexane	0.70	1.42	2.6	9
Benzene	0.39	0.31	0.32	9
Toluene	0.43	0.8	0.6	8

Refractive index	N_D			1.519 (20%)	3
				1.521 (45%)	3
Dielectric constant (10^6 Hz)				5.5 (27%)	3
				4.8 (40%)	3
Dielectric loss factor (10^6 Hz)				35 (27%)	3
				42 (40%)	3

[a] Bound acrylonitrile content (= AN%)).

TABLE 3. POLY(BUTADIENE-*co*-STYRENE) (23.5–25% BOUND STYRENE CONTENT)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3	933	43	980	17
		(932.5–933.5)	43	940–1000	17
Coefficient of expansion	K^{-1} ($\times 10^6$)	660	43	660	17,18,44
Glass transition temperature	°C	−64 to −59	43	−52	21
Specific heat	J/kg/K ($\times 10^3$)	1.89	45	1.83	24
Thermal conductivity	W/m/K ($\times 10^{-3}$)			190–250	46,47
Thermal diffusivity	m^2/s ($\times 10^{-9}$)			90	48
Solubility parameter	$MPa^{1/2}$	17.0	6		
Polymer–solvent interaction parameter (25–30°C)					
	Solvent				
	n-Heptane	0.59	9		
	n-Hexane	0.66	9		
	Dichloromethane	0.47	9		
	Cyclohexane	0.48	9		
	Benzene	0.40	9		
	Toluene	0.31	13		
Heat of combustion	J/kg ($\times 10^6$)	56.5	43		
Refractive index	N_D	1.5345	43		
		(1.534–1.535)	43		
Molar polarizability	cm^3 ($\times 10^{-25}$)				
Dielectric constant (1 kHz)		2.5	33	2.66	33
Dielectric loss factor		0.0009	33	0.0009	33
Electric conductivity	Sm^{-1} ($\times 10^{-15}$)				
Compressibility	Pa^{-1} ($\times 10^{-12}$)	530	49	510	50
Bulk modulus (isothermal)	Pa ($\times 10^6$)	1890	49	1960	49
Young's modulus	Pa ($\times 10^6$)			1.6	21,35
				1.0–2.0	35
Shear modulus	Pa ($\times 10^6$)			0.53	21
				0.3–0.7	35
Shear compliance	Pa^{-1} ($\times 10^{-6}$)			7	21,35
				3–10	21,35
Storage modulus	log Pa	5.82	40	5.88	64
		(5.82–5.85)	40	5.64–6.20	64
Loss modulus	log Pa	4.94	40	4.92	64
		(4.56–4.94)	40	4.73–5.04	64
Loss tangent		0.13	40	0.11	64
		(0.05–0.13)	40		
Tensile strength	MPa			1.4-3.0	19,42
Ultimate elongation	%			400–600	19,42

TABLE 4. POLY(CHLOROPRENE) (CR NEOPRENE)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3	1230	43,51,58	1320	17
Coefficient of expansion	K^{-1} ($\times 10^6$)	600	46,58	610–720	17,58,59
Glass transition temperature	°C	−45	60	−45	21,59,60,61
Specific heat	J/kg/K ($\times 10^3$)	2.2–2.2	46	2.1–2.2	46
Thermal conductivity	W/m/K ($\times 10^{-3}$)	192	58	192	58
Thermal diffusivity	m^2/s	10^{-9}			
Solubility parameter	$MPa^{1/2}$	18.5	6		

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TABLE 4. *cont'd*

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Polymer–solvent interaction parameter (25–30°C)					
	Solvent				
	n-Heptane	0.85	9		
	n-Hexane	0.89	9		
	Dicyclomethane	0.53	9		
	Cyclohexane	0.69	9		
	Benzene	0.26	9		
	Toluene	0.70	13		
Equilibrium melting temperature	K (°C)	328–351 (55–78)	62,63		
Temperature of most rapid crystallization	K (°C)	268 (−5)	65	261 (−12)	65,66
Heat of fusion of crystal	J/kg ($\times 10^3$)	95	63		
Refractive index	N_D	1.558	20		
Dielectric constant				6.5–8.1	33
Dielectric loss factor				0.03/0.86	33
Electric conductivity	Sm^{-1} ($\times 10^{-15}$)			3–1400	33
Compressibility	Pa^{-1} ($\times 10^{-12}$)	480	49,50	440	49,50
Bulk modulus (isothermal)	Pa ($\times 10^6$)	2080	50	2270	50
Young's modulus	Pa ($\times 10^6$)			1.6	21,35
Shear modulus	Pa ($\times 10^6$)			0.52	21
Shear compliance	Pa^{-1} ($\times 10^{-6}$)			2.0	19,21,35
Poisson's ratio				0.49974	38
Storage modulus	log Pa			5.81	57,59
Loss modulus	log Pa			5.04	57,59
Loss tangent				0.17	57,59
Tensile strength	MPa			25–38	19,42
Ultimate elongation	%			800–1000	19,42

TABLE 5. POLY(ISOBUTENE-*co*-ISOPRENE) BUTYL RUBBER (IIR)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^{-3}	917	51	933	52
				930–970	17
Coefficient of expansion	K^{-1} ($\times 10^6$)	750	53	560	52,54
Glass transition temperature	°C	−71	53	−63	21
Specific heat	J/kg/K ($\times 10^3$)	1.95	23,24,29	1.85	24
Thermal conductivity	W/m/K ($\times 10^{-3}$)			130	47
Thermal diffusivity	m^2/s ($\times 10^{-9}$)			70	55
Solubility parameter	$MPa^{1/2}$	16.0	6		
Polymer–solvent interaction parameter					
	Solvent				
	n-Heptane	0.48	14		
	n-Hexane	0.52	14		
	Dicyclomethane	0.58	14		
	Cyclohexane	0.44	14		
	Benzene	0.66	14		
	Toluene	0.56	14		
Equilibrium melting temperature	K (°C)	275 (1.5)	53		
Temperature of most rapid crystallization	K (°C)	239 (−34)	53		
Refractive index	N_D	1.5081	43		
Dielectric constant (1 kHz)		2.38	33	2.42	33
Dielectric loss factor		0.003	33	0.0054	33

TABLE 5. *cont'd*

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Compressibility	Pa^{-1} ($\times 10^{12}$)			508	54
Bulk modulus (isothermal)	Pa ($\times 10^6$)			1970	54
Young's modulus	Pa ($\times 10^6$)			1.0	21,35
Shear modulus	Pa ($\times 10^6$)			0.33	21,35
				0.2–0.5	19,21,35
Shear compliance	Pa^{-1} ($\times 10^{-6}$)			3.1	21
Strage modulus	log Pa	6.50	56	5.64	57
Loss modulus	log Pa	5.98	56	5.48	57
Loss tangent		0.3	56	0.7	57
Tensile strength	MPa			18–21	19,42
Ultimate elongation	%			750–950	19,42

TABLE 6. POLYISOPRENE, NATURAL RUBBER

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3	913	16	970	17,18
		906–916	16	(920–1000)	19
Coefficient of expansion	K^{-1} ($\times 10^6$)	670	16	660	16
Glass transition temperature	°C	−72	20	−63	21
		(−74 to −69)	20	(−72 to −61)	21
Specific heat	J/kg/K ($\times 10^3$)	1.905	22,23	1.828	24
Thermal conductivity	W/m/K ($\times 10^{-3}$)	134	16	153	25,26
Thermal diffusivity	m^2/s ($\times 10$)			70	27
Solubility parameter	$MPa^{1/2}$	1.7	6		
Polymer–Solvent interaction parameter (25–30°C)					
	Solvent				
	n-Heptane	0.43	9		
	n-Hexane	0.47	9		
	Dichloromethane	0.40	9		
	Cyclohexane	0.53	9		
	Benzene	0.44	9		
	Toluene	0.39	9		
Equilibrium melting temperature	K (°C)	308.6 (35.5)	28	313(40)	29
Temperature of most rapid crystallization	K (°C)	248 (−25)	30		
Heat of fusion of crystal	J/kg ($\times 10^3$)	67.3	31		
Heat of combustion	J/kg ($\times 10^6$)	45.2	16	44.4	16
Refractive index	$N_D K^{-1}$	1.5191(25°C)	32	1.5264	16
Molar polarizability	cm^3 ($\times 10^{-25}$)				
Dielectric constant (1 kHz)		2.37–2.45	16,33	2.68	33
Dielectric loss factor (1 kHz)		0.001–0.003	33	0.002–0.04	33
Electric conductivity	Sm^{-1} ($\times 10^{-15}$)	2.57	16,33	2–100	16,33
Compressibility	Pa^{-1} ($\times 10^{-12}$)	515	34	514	34
Bulk modulus (isothermal)	Pa ($\times 10^6$)	1940	34	1950	34
Young's modulus	Pa ($\times 10^6$)			1.3	35,36
				(1.0–2.0)	19,35
Shear modulus	Pa ($\times 10^6$)			0.43	21,36
				0.3–0.7	21,37
Shear compliance	Pa^{-1} ($\times 10^{-6}$)			2.3	21,36
				(1.5–3.5)	21,37
Poisson's ratio				0.49989	34,38,39
Storage modulus	log Pa	5.61	40	5.61	41
		5.53–5.75		5.49–5.78	
Loss modulus	log Pa	4.46	40	3.80	41
		(4.43–4.65)		3.72–4.48	

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TABLE 6. *cont'd*

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Loss tangent		0.09	40	0.016	41
		0.07–0.13		0.01–0.05	
Tensile strength	MPa			17–25	19,42
Ultimate elongation	%			750–850	19,42

TABLE 7. ETHYLENE–PROPYLENE–DIENE–TERPOLYMER (EPDM)

Property	Units	Unvulcanized	Refs.	Pure-gum vulcanizate	Refs.
Density	kg/m^3			0.85	3
Glass transition temperature	°C			−60 to −69	1
Specific heat	J/kg/K ($\times 10^3$)			2340	3
Thermal conductivity	W/m/K ($\times 10^{-3}$)				
Thermal diffusivity	m^2/s ($\times 10^{-9}$)				
Solubility parameter	$MPa^{1/2}$	16.0–16.5	6		
Polymer–Solvent interaction parameter (25–30°C)					
	Solvent				
	n-Heptane	0.44	15		
	n-Hexane	0.49	16		
	Dichloromethane	0.32	16		
	Cyclohexane	0.35	15		
	Benzene	0.58	15		
	Toluene	0.49	15		
Refractive index	N_D			1.48	3
Young's modulus	Pa ($\times 10^6$)			2.0	13
Shear modulus	Pa ($\times 10^6$)			1.59	13
Storage modulus (1 Hz)	MPa			1.59	13
Loss modulus (1 Hz)	MPa			0.12	13
Loss tangent (1 Hz)				0.077	13

C. REFERENCES

1. J. Brandrup, E. H. Immergut, (eds.) "Polymer Handbook", 3rd Edition, 1989.
2. J. W. Barlow, Polym. Eng. Sci., **18** (3), 238 (1978).
3. W. J. Roff, J. R. Scott, Fibers, Films, Plastics and Rubbers (A Handbook of Common Polymer).
4. J. Grebowicz, W. Aycock, B. Wunderlich, Polymer, **27** (4), 575 (1986).
5. "Hikinzokuzairyou Data Book", Nihon Kensa Kikaku Kyoukai.
6. "Kobunshi Data Handbook, Ouyouhen", Kobunnsi Gakkai ed. Baihukan.
7. A. F. M. Barton, CRC Handbook of Polymer–Liquid Interaction Parameter and Solubility Parameters, CRC Press, Inc, Boca Raton, FL, 1983.
8. A. G. Shvart, Kolloid Zh., **18**, 755 (1953).
9. C. Shibuya, K. Ogino, T. Nakagawa, Chemical composition dependence of a polymer-solvent interaction parameter.
10. G. Natta, G. Moraglio, Makromol. Chem., **66**, 218 (1963).
11. W. S. Bahary, D. I. Sapper, J. H. Lane, Rubber Chem. Technol., **40**, 1529 (1967).
12. J. Furukawa, S. Yamasita, T. Kotani, M. Kawasima, J. Appl. Polym. Sci., **13**, 2527 (1969).
13. "Gomu Kougyou Binran", Nihon Gomu Kyoukai, (1993).
14. G. M. Bristow, W. F. Watson, Trans. Faraday Soc., **59**, 1731 (1958).
15. A. G. Shvart, E. K. Chefranova, L. A. Itokovskaya, Solubility parameters of resins based on dimethylvinylethyl-p- hydroxyphenylmethane, Kolloidn Zh., **32**, 603 (1970); Colloid J. USSR, **32**, 506 (1970).
16. L. A. Wood, "Values of the Physical Constants of Rubber", Proceedings of the Rubber Technology Conf. (Institution of the Rubber Industry, London), 1938, p. 933; Rubber Chem. Technol., **12**, 130 (1939).
17. N. Bekkedahl, F. L. Roth, National Bureau of Standards, unpublished observations of density and expansivity, 1948.
18. A. J. Wildschut, "Technological and Physical Investigations on Natural and Synthetic Rubbers", Elsevier, New York, 1946.

19. B. B. S. T. Boonstra,"Properties of Elastomers", Chapter 4 of Vol. III in R. Houwink (Ed.), "Elastomers. Their Chemistry, Physics, and Technology", Elsevier, New York, 1948.
20. L. A. Wood, "Synthetic Rubbers: A Review of Their Compositions, Properties, and Uses," Natl. Bur. Std. Circ., C427 (1940); Rubber Chem. Tech., **13**, 861 (1940); India Rubber World, 102, No. 433 (1940).
21. L. A. Wood, F. L. Roth, Proc. 4th Rubber Technology Conference, London, 1962, p. 328, Institution of the Rubber Industry, London, 1963; Rubber Chem. Technol., **36**, 611 (1963).
22. S. S. Chang, A. B. Bestul, J. Res. Nat. Bur. Std., **75A**, 113 (1971).
23. L. A. Wood, N. Bekkedahl, J. Polym. Sci., Part B, Polym. Lett., **5**, 169 (1967).
24. W. H. Hamill, B. Mrowca, R. L. Anthony, Ind. Eng. Chem., **38**, 106 (1946); Rubber Chem. Technol., **19**, 622 (1946).
25. L. C. Carwile, H. J. Hoge, "Thermal Conductivity of Soft Vulcanized Natural Rubber, Selected Values", in: "Advances in Thermophysical Properties at Extreme Temperatures and Pressures", American Society of Mechanical Engineers, New York, 1965; Rubber Chem. Technol., **39**, 126 (1966).
26. M. N. Pilsworth, Jr., H. J. Hoge, H. E. Robinson; J. Mater., **7** (4), 550 (1972).
27. D. Hands, Rubber Chem. Technol. (Rubber Rev.), **50**, 480 (1977).
28. E. N. Dalai, K. D. Taylor, P. J. Phillips, Polymer, **24**, 1623 (1983).
29. G. T. Furukawa, M. L. Reilly, J. Res. Nat. Bur. Std. **56**, 285 (1956); RP 2676.
30. L. A. Wood, N. Bekkedahl, J. Res. Natl. Bur. Std., **36**, 489 (1946); RP 1718; J. Appl. Phys., **17**, 362 (1946); Rubber Chem. Technol., **19**, 1145 (1946).
31. H. G. Kim, L. Mandelkern, J. Polym. Sci. A-2, **10**, 1125 (1972).
32. L. A. Wood, L. W. Tilton, Proc. 2nd Rubber Technology Conference, London; 1948, p. 142, Institution of the Rubber Industry, London; J. Res. Natl. Bur. Std., **43**, 57 (1949), RP 2004.
33. A. T. McPherson, Rubber Chem. Technol. (Rubber Rev.), **36**, 1230 (1963).
34. L. A. Wood, G. M. Martin, J. Res. Natl. Bur. Std., **68A**, 259 (1964); Rubber Chem. Technol., **37**, 850 (1964).
35. G. M. Martin, F. L. Roth, R. D. Stieler, Trans. Inst, Rubber Ind., **32**, 189 (1956); Rubber Chem. Technol., **30**, 876 (1957).
36. F. L. Roth, G. W. Bullman, L. A. Wood, J. Res. Natl. Bur. Std., **29A**, 347 (1965); Rubber Chem. Technol., **39**, 397 (1966).
37. W. Philipoff, J. Appl. Phys., **24**, 685 (1953).
38. B. P. Holownia, J. Inst. Rubber Ind. **8**, 157 (1974); Rubber Chem. Technol., **48**, 246 (1975).
39. G. K. Rightmire, Am. Soc. Mech. End. Trans. Series F, J. Lubrication Technol., 381. July 1970.
40. L. J. Zapas, S. L. Shufler, T. W. deWitt, J. Polym. Sci., **18**, 245 (1955); Rubber Chem. Technol., **29**, 725 (1956).
41. J. D. Ferry, R. G. Mancke, E. Maekawa, Y. Oyanagi, R. A. Dickie; J. Phy. Chem., **68**, 3414 (1964).
42. J. M. Ball, G. C. Maassen, Symp. Applications of Synthetic Rubbers, American Society for Testing Materials, March 2, 1944, p. 27.
43. L. A. Wood, "Physical Chemistry of Synthetic Rubbers", Chapter 10 in: G. S. Whitby (Ed.), "Synthetic Rubbers", J Wiley, New York 1954.
44. I. B. Prettyman, "Physical Properties of Natural and Synthetic Rubber Stocks", Handbook of Chemistry and Physics, Chemical Rubber Pub. Co., Cleveland, in: 1962, p. 1564.
45. R. D. Rands, Jr., W. J. Ferguson, G. Allen, Polymer, **10**, 495 (1964).
46. A. R. Payne, J. R. Scott, "Engineering Design with Rubber". Interscience, New York, 1960.
47. H. Shilling, Kautschuk Gummi, **16**, 84 (1963).
48. J. Rehner, Jr., J. Polym. Sci., **2**, 263 (1947); Rubber Chem. Technol., **21**, 82 (1948).
49. W. H. S. Naunton et al., "Rubber in Engineering," Ministry of Supply, London, 1945, or Chemical Publishing Co., Brooklyn, NY., p. 30.
50. W. S. Cramer, I. Silver, NSVORD Report 1778, Feb. 1951, US Naval Ordnance Lab., White Oak, MD.
51. L. A. Wood, N. Bekkedahl, F. L. Roth, J. Res. Natl. Bur. Std., **29**, 391 (1942) RP 1507; Ind. Eng. Chem., **34**, 1291 (1942); Rubber Chem. Technol., **16**, 244 (1943).
52. N. Bekkdahl, J. Res. Natl. Bur. Std., **43**, 145 (1949).
53. R. M. Kell, B. Bennett, P. B. Stickney; Rubber Chem. Technol., **31**, 499 (1958).
54. C. Price, J. Padget, M. C. Kirkham, G. Allen; Polymer, **10**, 495 (1969).
55. D. R. MacRae, R. L. Zapp, Rubber Age, **82**, 831 (1958).
56. L. Schmieder, K. Wolk, Kolloid-Z., **134**, 149 (1953).
57. J. H. Dillon, I. B. Prettyman, G. L. Hall, J. Appl. Phys., **15**, 309 (1944); Rubber Chem. Technol., **17**, 597 (1944).
58. N. L. Catton,"The Neoprenes", Rubber Chemicals Division, E. I. DuPont de Numours and Company, Wilmington, DE, 1953.
59. T. P. Yin, R. Pariser, J. Appl. Polym. Sci., **7**, 667 (1963).
60. R. M. Kell, B. Bennett, P. B. Stickney, J. Appl. Polym. Sci., **2**, 8 (1959).
61. A. N. Gent, J. Polym. Sci., **A3**, 3787 (1965).
62. W. R. Krigbaum, J. V. Dawkins, G. H. Via, Y. I. Balta; J. Polym. Sci. A-2, **4**, 475 (1966).
63. J. T. Maynard, W. E. Mochel, J. Polym. Sci., **13**, 235 (1954).
64. W. P. Fletcher, A. N. Gent, Brit. J. Appl. Phys., **8**, 194 (1957).
65. R. M. Murray, J. D. Detenber, Rubber Chem. Technol., **34**, 668 (1961).
66. P. R. Johnson, Rubber Chem. Technol., (Rubber Rev.), **49**, 650 (1976).

Physical Constants of Poly(ethylene)*

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A. CRYSTALLOGRAPHIC DATA AND CRYSTALLOGRAPHIC MODIFICATIONS[a]

Crystal system	Space group	Lattice constants (nm) a	b	c	α, β, or γ (deg.)	Number of chains per unit cell	Molecular conformation	Crystal density (g/cm^3)	Refs.
Orthorhombic stable form	Pnam-D_{2h}	0.7417	0.4945	0.2547		2	Planar Zigzag (2/1)	1.00	3–7
Monoclinic metastable form	C2/m-C_{2h}	0.809	f.a. = 0.253[b]	0.479	$\beta = 107.9$	2	Planar Zigzag (2/1)	0.998	4,8
Hexagonal high-pressure form		0.842	–	f.a.[c]			–	–	9–11

[a] Refs. 1,2.
[b] f.a. indicates fiber axis.
[c] Not determined.

Temperature Dependence of Crystallographic Data for Orthorhombic Poly(ethylene)[a]

Temp. (K)	Lattice constants (nm) a	b	c	Specific volume (cm^3/g)	Refs.
4[b]	0.712	0.485	0.2548	0.945	12
10	0.716	0.486	0.2534	0.947	13
77	0.718	0.488	0.2534	0.953	13
77	0.715	0.490	0.2547	0.959	14
90[b]	0.716	0.487	0.2546	0.953	12
195	0.727	0.491	0.2534	0.971	13
293[c]	0.740	0.495	0.2543	0.999	5
	(0.743)	(0.494)	(0.2543)	(1.003)	5
297	0.742	0.496	0.2534	1.001	13
297	0.740	0.493	0.2534	0.993	3
297	0.736	0.492	0.2534	0.985	15
303	0.741	0.494	0.2547	1.002	14

[a] Ref. 1. Data obtained by X-ray diffraction studies unless otherwise indicated.
[b] Data obtained by neutron diffraction.
[c] The data of nonoriented samples are given without parentheses and those of oriented samples are given with parentheses, both for high-density PE.

* Based on a table in the third edition, by R. P. Quirk and M. A. A. Alsamarraie, The Maurice Morton Institute of Polymer Science, University of Akron, Akron, Ohio.

B. MOLECULAR PARAMETERS AND SOLUTION PROPERTIES

Bond Angle and Bond Length

	Wide angle X-ray[a] (Refs. 3,4,16)	Neutron scattering (Ref. 12)	Temp. (K)
BOND ANGLE (in degrees)			
C–C–C	112.0	107.7	4
		108.1	90
H–C–H	107.0	109.0	4
		110.0	90
BOND LENGTH (nm)			
C–C	0.153	0.158	4
		0.157	90
C–H	0.107	0.106	4
		0.107	90

[a] Measurements taken at room temperature.

Dimensions of Linear Polyethylene Molecules[a]

Temperature (°C)	PEH Matrix $\overline{M}_w$ ($\times 10^3$)	$\overline{M}_w$ ($\times 10^3$)	PED[b] $\langle S^2\rangle_z^{1/2}$† (nm)	PED[b] $\langle S^2\rangle_w^{1/2}/\overline{M}_w^{1/2}$‡ [nm/(g/mole)$^{1/2}$]	State	Refs.
23 (296.2)[c]	78	60	13.3	0.046	Crystalline	17
150 (423.2)	100	140	21.5	0.046	Molten	17
150 (423.2)	80	10	4.38	0.044	Molten	18

[a] Effect of temperature and state, neutron scattering measurements.
[b] Deuterated polyethylene.
[c] Values in parentheses indicate temperature in kelvin.
† $\langle S^2\rangle_z^{1/2}$ is the z-average value of the radius of gyration of the molecule.
‡ $\langle S^2\rangle_w^{1/2}$ is the weight-average value of the radius of gyration of the molecule.

Molecular Dimensions of Linear Poly(ethylene) (Unperturbed)[a]

$\overline{M}_w$ ($\times 10^5$)	$(\langle r_0^2\rangle_w/\overline{M}_w)^{1/2}$ [nm/(g/mole)$^{1/2}$]	$\langle r_0^2\rangle_w/Dr_{max}$
1.25	0.192	7.80
2.69	0.215	9.86
4.65	0.190	7,73

[a] In tetralin at 105°C (378.2 K). $\langle r_0^2\rangle_w$ is the weight-average mean-square end-to-end distance in nm; $\overline{M}_w$ is the weight-average molecular weight; D is the diameter of a spherical segment of the lattice model chain; and r_{max} is the length of the fully extended chain (19); temperature dependence of $\langle r_0^2\rangle$ (in long chain paraffinic hydrocarbon solvents) is given by $-d\ln\langle r_0^2\rangle/dT = 1.2\times10^{-3}$ (20).

Effect of Molecular Weight on Paraclustering[a]

Molecular weight of tagged molecules	$\frac{\sum_0}{\overline{M}_w} = \frac{\text{Molecular weight (neutrons)}}{\text{Molecular weight (g.p.c.)}}$	$\frac{\langle S^2\rangle_w^{1/2}}{\overline{M}_w^{1/2}}$†
3000	73 ± 7	0.41
12000	7 ± 1	0.18
60000	0.91 ± 0.1	0.049
140000	0.94 ± 0.1	0.042

[a] For deuterated poly(ethylene) in a protonated poly(ethylene) matrix with molecular weight ca. 100000 by neutron scattering measurements (21). All sample solution were blended in ortho-dichlorobenzene, quenched rapidly from the melt and measured at 25°C (298.2 K) (concentration of tagged molecules $\leq 5\%$).
† $\langle S^2\rangle_w^{1/2}$ the weight-average value of the radius of gyration of the molecule; values are in [nm/(g/mol)$^{1/2}$].

C. CRYSTALLINITY, CRYSTAL SIZE AND CRYSTALLIZATION KINETICS

Crystallinity[a,b]

	Average molecular weight	Density ρ (g/cm^3)	Melting point T (°C)	Degree of branching (CH_3/100C)	Crystallinity (α_ρ)	Crystal thickness l (nm)
Marlex 6015	15.0×10^4	0.980	131 (404.2)[c]	0.05	0.94	23.9
Hostalen GC	5.0	0.969	129 (402.2)	0.10	0.83	19.8
Hostalen GF	12.0	0.968	128 (401.2)	0.64	0.85	18.4
Hoechst PA-190	0.9	0.954	124 (397.2)	0.84	0.76	14.1
Alkathene HD	24.5	0.952	121 (394.2)	0.91	0.76	12.0
Epolene C-11	1.0	0.947	119 (392.2)	1.08	0.74	11.0
Lupolen KR 1051	5.1	0.943	120 (393.2)	1.30	0.70	11.4
BASF a Wachs	1.0	0.933	111 (384.2)	1.84	0.67	8.1
Lupolen KR 1032	5.3	0.931	115 (388.2)	2.50	0.60	9.2
Epolene C-13	1.0	0.919	104 (377.2)	2.80	0.54	6.5
Lupolene 1810 H	5.4	0.928	106 (379.2)	3.00	0.59	6.9
Hoechst PA-130	3.0×10^3	0.957	115 (388.2)	1.58	0.85	9.3
Epolene N-10	2.5	0.925	105 (378.2)	2.25	0.61	6.8
Epolene N-12	1.5	0.941	108 (381.2)	2.27	0.75	7.4
BASF a Wachs	5.6	0.936	106 (379.2)	2.32	0.68	6.9
Epolene N-11	1.5	0.934	104 (377.2)	2.48	0.68	6.6
Hoechst PA-560	6.0	0.914	101 (374.2)	4.10	0.51	6.1
Epolene C-10	7.0	0.919	99 (372.2)	4.19	0.60	5.8
Epolene C-101	3.2	0.936	104 (377.2)	4.42	0.70	6.6
Hoechst PA-520	2.0	0.917	102 (375.2)	4.90	0.54	6.2
Epolene C-12	3.7	0.906	83 (356.2)	6.94	0.50	5.1

[a] Depends upon chain branching; data given are representative values of typical commercial poly(ethylenes) (22).
[b] Crystallization from melt by rapid cooling at ambient temperature; crystallinity estimated from density.
[c] Values in parentheses indicate temperature in kelvin.

Single Crystal Lamella Thickness[a]

Solvent	Cryst. temp. (°C)	Conc. (%)	Long period (nm)	Refs.
HIGH DENSITY, POLY(ETHYLENE) (PHILLIPS MARLEX 50)				
Tetrachloroethylene	50 (323.2)[b]	0.31	11.2	24
	60 (333.2)	0.25	10.8	25
p-Xylene	72 (245.2)	0.46	11.0	26
	78 (351.2)	0.58	14.0	24
Xylene	50 (323.2)	–	9.25	23,26
	60 (333.2)	–	10.2	
	70 (343.2)	–	11.15	
	80 (353.2)		12.05	
	90 (363.2)		15.0	
	75 (348.2)	0.1	11.5	27
	80 (353.2)	0.1	12.5	
	85 (358.2)	0.1	13.3	
Melt	120 (393.2)	100	19.0	28
	125 (398.2)	100	22.3	
	130 (403.2)	100	35.5	
n-Butyl acetate	105 (378.2)	0.45	14.7	25
	110 (383.2)	0.57	16.2	24
Diphenyl ether	120 (393.2)	0.47	20.2	24
	125 (398.2)	0.37	17.3	25
LOW DENSITY POLY(ETHYLENE) (DUPONT ALATHON)				
Butyl stearate	111 (384.2)	0.25	12.6,15.5	23
	113.5 (386.7)	0.25	16.6	
Squalene	155.5 (428.7)	0.25	17.6	
Glycol dipalmitate	155.4 (428.6)	0.25	15.1	
	118 (391.2)	0.25	16.4	
	121 (394.2)	0.25	20.6	
Tripalmitin	121 (394.2)	0.25	18.8	

[a] Long periods from small angle X-ray scattering, effect of crystallization temperature and solvents (23).
[b] Values in parenthesis indicate temperature in kelvin.

Crystallization Kinetic Parameters See table "Rate of Crystallization of Polymer" in this Handbook, and Refs. 29–31.

D. EQUILIBRIUM THERMODYNAMIC PROPERTIES

Entropy of Fusion at Equilibrium Melting Temperature (J/mol/K)

	Value	Refs.
ΔS_u	9.81	32
	9.60	33
	9.91	34
At constant volume $(\Delta S_u)_v$	7.42	33
	7.72	35

Glass Transition Temperature[a,b] (°C)

Value	Refs.
-30 ± 15 (243.2 ± 15)[c]	37,38
-80 ± 10 (193.2 ± 10)	39
-128 ± 5 (145.2 ± 5)	40,41

[a] Ref. 36. See also Transition and Relaxation Temperatures in Section E.
[b] Considerable disagreement exists between different authors on the exact value of the transition which can be identified as the glass transition temperature. See Ref. 36 for a detailed discussion.
[c] Values in parentheses indicate temperature in kelvin.

Glass Transition Activation Energy 46–75 kJ/mol (42).

Heat Capacity, C_p (J/mol/K)[a]

Temperature (K)	Polymer	C_p	Refs.
298.15	Crystalline	21.70	
298.15	Amorphous	30.81	
300	Branched	32.55	43
		32.68	44
		32.86	45
300	Linear	23.93	46
		24.14	47
		24.24	46
400	Molten	34.48	48
		35.22	44,49
		35.36	50
		36.04	48

[a] Ref. 2. For linear, amorphous, crystalline and molten poly(ethylene), (0–600 K), see Table "Heat Capacity of High Polymers" in this Handbook, and also Refs. 2,34.

Heat Capacity in Liquid (2) For molten poly(ethylene) (390–600 K)

$C_p = 4.325 \times 10^{-2} T + 17.919$ [$\pm 1.2\%$ (RMS deviation)]

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Heat Capacity in Amorphous Glass (2) The heat capacity of amorphous poly(ethylene) obtained from crystallinity extrapolations was given in J/mol/K.

From 0.5 to 20 K:

$$C_p = \exp[-0.135328(\ln T)^3 + 0.393949(\ln T)^2 + 2.85597(\ln T) - 7.84553]$$
$$[\pm 2.2\% \text{ (RMS deviation)}]$$

From 10 to 260 K:

$$C_p = \sum_{n=0}^{7} A_n T^n \; [\pm 0.7\% \text{ (RMS deviation)}]$$

$A_0 = 1.0966333$ $A_1 = -0.11209575$
$A_2 = 9.3092851 \times 10^{-3}$ $A_3 = -1.7095906 \times 10^{-4}$
$A_4 = 1.5885817 \times 10^{-6}$ $A_5 = -7.931924 \times 10^{-9}$
$A_6 = 2.0248831 \times 10^{-11}$ $A_7 = -2.0616876 \times 10^{-14}$

Heat Capacity in Crystalline Solid (2) The heat capacity of crystalline poly(ethylene) obtained from extrapolated values was given in J/mol/K.

From 0.4 to 20 K:

$$C_p = \exp[-1.07155 \times 10^{-2}(\ln T)^3 + 4.62622 \times (\ln T)^2 + 2.89948(\ln T) - 9.12864]$$
$$[\pm 5.8\% \text{ (RMS deviation)}]$$

From 10 to 410 K:

$$C_p = \sum_{n=0}^{9} A_n T^n \; [\pm 1.2\% \text{ (RMS deviation)}]$$

$A_0 = 1.1504009$ $A_1 = -0.19842302$
$A_2 = 1.2786634 \times 10^{-2}$ $A_3 = -2.4474207 \times 10^{-4}$
$A_4 = 2.593176 \times 10^{-6}$ $A_5 = 1.6647891 \times 10^{-8}$
$A_6 = 6.5926585 \times 10^{-11}$ $A_7 = -1.5679761 \times 10^{-13}$
$A_8 = 2.0474114 \times 10^{-16}$ $A_9 = -1.1252521 \times 10^{-19}$

Heat Capacity Change at Glass Transition Temperatures 10.5 J/mol/K at 237 K (2)

Heat of Fusion at Equilibrium Melting Temperature 4.1 ± 0.2 kJ/mol (2,51)

Residual Entropy of the Glassy State at 0 K $S_0 = 3.0$ J/mol/K (2).

E. OTHER GENERAL PHYSICAL PROPERTIES

Coefficient of Thermal Expansion[a]

Temp (°C)	Coefficient of expansion ($\times 10^5$)		Specific volume ratio (V_E/V_{25})[b]
	Linear	Cubical	
−35 (238.2)[c]	10.0	30	0.969
−20 (253.2)	13.7	41	0.975
0 (273.2)	18.3	55	0.986
20 (293.2)	23.7	71	0.997
25 (298.2)	24.8	74	1.000
40 (313.2)	29.0	87	1.012
60 (333.2)	33.7	101	1.031
80 (353.2)	40.3	121	1.055
100 (373.2)	46.6	140	1.094
110 (383.2)	51.0	153	1.130
115 (388.2)	25.0	75	1.140
115–150 (388.2–423.2)	25.0	75	–
150 (423.2)	25.0	75	1.168

[a] Branched polyethylene (52,53); for data on amorphous linear poly(ethylene), see Ref. 40.
[b] V_E is relative excess volume, V_{25} is the volume at 25°C (273.2 K).
[c] Values in parentheses indicate temperature in kelvin.

Density[a] (Mg/m^3) ≡ (g/cm^3)

	Value	Refs.
Amorphous (from extrapolation of data above the melting point)	0.855	54
Commercial high-pressure poly(ethylene)	0.915–0.935	55
Experimental high-pressure poly(ethylene)	0.940–0.970	55
Ziegler process (Refs. 56,57) poly(ethylene)	0.940–0.965	55
Phillips process (Ref. 58) poly(ethylene)	0.960–0.970	55
Crystal density (theoretical)	See "Crystallographic Data and Crystallographic Modifications" in Section A	

[a] Unless otherwise stated, the values given are for 25°C (293.2 K); see also Sections G and H.

Dielectric Constant At 100 kHz and 23°C (296.2 K): 2.3 (59)

Effect of Density on Dielectric Constant[a]

Density (g/cm^3)	Dielectric constant (ASTM D 150)
0.920	2.28
0.925	2.29
0.930	2.30
0.935	2.31
0.940	2.32

[a] Ref. 60. The relationship shown in this table is for pure poly(ethylene). Therefore, considerable divergence from this relationship is caused by the presence of impurities as well as addition of additives such as carbon black or other fillers.

Dielectric Loss tan δ (1×10^{-4}–1×10^{-3}) (61,62).
The values of tan δ depend on temperature and structure of poly(ethylene) (61).

Dielectric Strength[a]

Temperature (°C)	Value ($\times 10^6$) (V/cm)[b,c]
−200 to 0 (73.2–273.2)[d]	7
50 (323.2)	5.3
100 (373.2)	1.8

[a] Ref. 63.
[b] Short-time test value (ASTM D149) is 16000–24000 V/cm, and depends upon sample configuration and impurities (64).
[c] Intrinsic dielectric strength of LDPE is of the order of 8.0×10^5 V/cm, if all chain end, surface and impurity effects are eliminated (64).
[d] Values in parentheses indicate temperature in kelvin.

Electrical Properties See also Section H (65).

Flash Ignition Temperature ASTM Method E136-58T: 340°C (613.2 K) (66).

Frictional Properties[a]

Coefficient of friction (μ)	Steel sliding on polymer		Polymer sliding on steel		Polymer sliding on polymer	
	Polished	Abraded	Polished	Abraded	Polished	Abraded
Static (M_S)	0.60	0.33	0.60	0.33	0.60	0.33
Kinetic (M_K)	0.60	0.33	0.60	0.33	0.60	0.33
	0.50	0.25				

[a] Ref. 67.

Heat of Combustion[a]

Density (Mg/m^3) ≡ (g/cm^3)	Methyl groups per 1000 C atoms	Heat of combustion ΔE (kJ/kg)
0.9391	8.3	−46,412
0.9220	24.7	−46,492
0.9053	46.2	−46,542

[a] Ref. 68.

Heat of Fusion and Melting Temperature From Differential Thermal Analysis Data[a]

Poly(ethylene) type	Melting point (°C)	Heat of fusion (kJ/kg)	Refs.
Marlex 50 (Phillips, linear)	135 (408.2)[a]	245.3	69
Super-Dylan (Ziesler, linear)	130 (403.2)	218.6	69
DYNH (Union Carbide, branched)	112 (385.2)	140.6	69
Linear poly(ethylene)			
from dilatometric measurements		280.5	33
from calorimetric measurements		277.1	34,70

[a] Values in parentheses indicate temperature in kelvin.

Ignition Limiting Oxygen Indices (ILOI)[a]

650°C (923.2 K)	600°C (873.2 K)	550°C (823.2 K)
2.8 ± 0.1	3.5 ± 0.1	5.7 ± 0.1

[a] Ref. 71. The method used (ASTM D2863-70) is defined as the minimum volume fraction of oxygen required for ignition to occur.

Infrared Absorption Bands (72–79).
Far Infrared (80).

Intrinsic Viscosity See Section G.

Mechanical Properties See Section H.
Ultra-drawn polyethylene. See Refs. 81–90.

Elastic Compliance See Section G.

Elongation at Break (%) See Section H.

Hardness, Shore D See Section H.

Impact Strength, Izod See Section H.

Low Temperature Brittleness See Section H.

Tensile Modulus See Section H. Also see Ref. 91 for Young's, shear and bulk moduli for more than 1,100 poly(ethylene) samples reported (within density range from 0.90 to 0.98 g/cm^3).

Tensile Strength See Section H.

Melt Index See Section G.

Melt Viscosity See Section G.

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Melting Temperature[a,b] (°C)

	Value	Refs.
Poly(methylene)	136.5 ± 0.5 (409.7 ± 0.5)[c]	92
Linear poly(ethylene)	137.5 (410.7)	33
Linear poly(ethylene), high molecular weight fraction	138.5 (411.7)	93
From extrapolation of MP of *n*-Paraffins	141 ± 2.4 (414.2 ± 2.4)	94
T_m^0 (∞ MW)[b]	141.4–145.5 ± 1 (414.6–418.7 ± 1)	30

[a] See also Section G and under Section F (Effect of chain branching on melting temperature).
[b] The estimated melting point of the infinite poly(ethylene) crystal is in dispute (see Refs. 30,95).
[c] Values in parentheses indicate temperature in kelvin.

Molecular Properties of Typical Poly(ethylenes)[a]

Polymer	$\overline{M}_w$ (×10⁻³)	Density (g/cm³)	Groups per 100 carbon atoms				
			CH_3-	$-CH_2CH_3$	$-CH=CH_2$	$-CH=CH-$	$=C=CH_2$
Ziegler	50–60	0.960	3.6	0.5	0.09	<0.02	0.06
Phillips	50–60	0.965	3.1	0.8	1.58	<0.02	0.08
Solution polymerization	50–60	0.964	3.2	0.5	0.42	<0.02	0.03
Phillips catalyst gas phase	350	0.954	2.8	<0.2	0.65	<0.02	0.04
Low-density PE		0.918–0.928	20–33	6–9	0.08–0.25	<0.02–0.06	0.17–0.33

[a] Ref. 74.

Neutron Scattering Spectra See Refs. 96–98.

Nuclear Magnetic Resonance See Refs. 99–110. Solid State. See Ref. 111.

Permeability and Diffusion Constants See corresponding table in this Handbook, and also Refs. 112,113.

Permeabilities at 30°C (293.2 K) of Poly(ethylene)[a]

	Permeability $\left[\frac{(cm^3\ STP)\ mm \times 10^{10}}{cm^2\ s\ (cm\ Hg)}\right]$			
Crystallinity (%)	N_2	O_2	CO_2	H_2O[b]
60	1.90×10^{-9}	5.5×10^{-9}	35.2×10^{-9}	80×10^{-9}
90	0.66	2.1	7.4	–
78	0.33	1.1	4.3	–
81	0.27	1.06	3.5	13

[a] Ref. 114.
[b] 90% e.h. at 25°C.

Raman Spectra See Refs. 115–123.

Refractive Index n_D^{25}

	Value	Refs.
LDPE	1.51	124
MDPE	1.52	124
HDPE (melt index above 0.1)	1.54	124
Amorphous, n_{5461}^{25}	1.49	125
Crystal, $\alpha \approx \beta$[a]	1.520	125
γ[a]	1.582	122

[a] α, β and γ are refractive indices along the *a*, *b* and *c* crystallographic directions of the crystal, respectively.

Dependence of Refractive Index on Chain Branching, Crystallinity and Density See Section F (Effect of Chain Branching on Physical Properties).

Specific Refractivity[a]

Temp. (°C)	*v* (cm³/g)	*n*	*r*
LOW DENSITY POLY(ETHYLENE) (ALATHON-10)			
90 (363.2)[b]	1.159	1.4801	0.3293
100 (373.2)	1.178	1.4693	0.3283
108 (381.2)	1.209	1.4575	0.3297
113 (386.2)	1.239	1.4432	0.3286
118 (391.2)	1.250	1.4392	0.3289
124.4 (397.6)	1.256	1.4368	0.3288
			Av. 0.3290
HIGH DENSITY POLY(ETHYLENE) (MARLEX 50)			
130 (403.2)	1.261	1.4327	0.3273
139.9 (413.1)	1.270	1.4297	0.3297
150.6 (423.8)	1.281	1.4261	0.3283

[a] $r = v(n^2 - 1)/(n^2 + 1)$, where v and n are specific volume and refractive index respectively. Ref. 52.
[b] Values in parentheses indicate temperature in kelvin.

Refractive Index Increment (dn/dc) See corresponding table in this Handbook, and also Refs. 126–133.

Softening Temperature, Vicat See Section H.

Solvent–Nonsolvent Systems for Fractionation

Solvent	Nonsolvent	Refs.
Xylene	Triethylene glycol	134,135
p-Xylene	Ethylene glycol monoethylene ether	136
Tetralin	2-Butoxyethanol	137
	Poly(ethylene oxide) [mol. wt. 200]	138,139

Transition and Relaxation Temperatures[a]

Designation	Temperature range (°C)	Approximate activation energy (kJ/mol)	Refs.
α	60–80 (333.2–353.2)[b]	>420	
β	−20 to −30 (253.2–243.2)	160–200	141
γ	−80 to −90 (193.2–183.2)[c]	46–75	
	−120 to −130 (153.2 −143.2)	32	

[a] There is considerable disagreement in the literature on the phenomena associated with the various transition and relaxation temperatures observed. Transition temperatures and temperatures associated with peaks in dynamic loss are collected together under the above combined heading. The transition and relaxation temperatures associated with amorphous regions of branched and linear poly(ethylenes) are designated as α, β, γ, etc. in the descending temperature order (140).
[b] Values in parentheses indicate temperature in kelvin.
[c] These frequently merge depending upon the crystallinity and frequency of the test method.

Frequency Dependence of Relaxation Temperatures in Dynamic Mechanical Loss Measurements[a]

α		β		γ	
Hz	***T* (K)**	**Hz**	***T* (K)**	**Hz**	***T* (K)**
HIGH PRESSURE, BRANCHED POLY(ETHYLENES)					
0.3	340	0.3	268	1.25	140
1.2	327	4.1	268	8.6	166
39	333	150	253	324	158
150	355	540	265	1.2×10^3	165
200	360	520	280	1.15×10^3	165
600	385	6000	320	1.9×10^4	≤ 200
4×10^4	≥ 360	4×10^4	275	4×10^4	180
1×10^5	≥ 320	1×10^5	283	1×10^5	≤ 190
–	–	1×10^5	285	1×10^5	200
–	–	5×10^5	285	5×10^5	205
2×10^6	360	2×10^6	295	2×10^6	210
–	–	2×10^6	300	2×10^6	210
LOW PRESSURE, LINEAR POLY(ETHYLENES)					
0.3	373	–		1.25	153
0.2	368	8	273	10	173
≤ 460	≥ 380	1.1×10^3	295	840 and 1.57×10^3	173
3000	420	–		–	–

[a] Ref. 142.

Viscosity–Molecular Weight Relationship See corresponding table in this Handbook, and Ref. 52.

F. EFFECT OF CHAIN BRANCHING (SHORT) ON PHYSICAL PROPERTIES

Effect on Density and Refractive Index[a]

Methyl groups per 1000 C atoms	Density (Mg/m^3) ≡ (g/cm^3)	Refractive index n_D^{25}
83	0.91	1.5060
48	0.917	1.5168
46	0.925	1.5152
26	0.929	1.5227
16	0.926	1.5260

[a] Refs. 143, 52.

Effect on Expansion Coefficient (Mean) and Specific Volume of Crystalline Phase[a]

Methyl groups per 1000 C atoms	$1/V_{20}$ (−150 to 100°C)	$(\Delta V/\Delta T)(\times 10^{-4})$ (0–100°C)	V_{20}
0.3	2.47	3.13	1.001
2	2.59	2.95	0.998
17.5	2.61	2.98	1.010
23	2.84	3.42	1.009
37	2.96	3.70	1.017

[a] Ref. 52,144.

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Effect on Long Period Spacings and Crystallinity of Completely Annealed Samples[a]

Poly(ethylene)	Methyl groups per 1000 C atoms	Long period spacing (nm)	% crystallinity	Refs.
Branched (high pressure)	60	22.0	21	146
	45	20.0	48	
	35	21.0	50	
	28	22.0	53	
	20	23.0	56	
	15	25.0	59	
	10	26.0	62	
Linear	7	32.0	77	143,144
(Ziegler type catalysts)	5	36.0	79	
Linear	2	42.0	88	58
(Phillips petroleum process)				

[a] Ref. 52,145.

Effect on Melting Point Observed[a]

Methyl groups per 1000 C atoms	Melting point (°C)
87	105 (378.2)[b]
28	113 (386.2)
28	108 (381.2)
8	123 (396.2)
0	132 (405.2)

[a] Both the amount and randomness of branching affect the melting point. Experimental conditions used for the following data were not adequate for equilibrium crystallinity and accuracy of melting points (52).
[b] Values in parentheses indicate temperature in kelvin.

G. PROPERTIES OF A SERIES OF SELECTED POLY(ETHYLENE) SAMPLES

Sample No.	Optical melting point (°C)	Density[a] (g/cm^3)	Infrared functional groups per 100 C methyl	Infrared functional groups per 200°C: Vinyl	Trans-unsaturation	Vinylidene	Carbonyl
PE 1	104.2 (377.4)[f]	0.9142	3.68	0.18	0.15	0.79	0.13
PE 2	112.4 (385.6)	0.9225	2.59	0.32	0.11	0.32	0.02
PE 3	112.2 (385.4)	0.9218	2.48	0.10	0.06	0.29	0.05
PE 4	113.7 (386.9)	0.9232	2.55	0.11	0.06	0.29	n.d.
PE 5	114.0 (387.2)	0.9219	2.46	0.11	0.05	0.30	0.01
PE 6	114.5 (387.7)	0.9228	2.31	0.06	0.05	0.26	n.d.
PE 7	113.5 (386.7)	0.9207	2.59	0.11	0.06	0.33	0.02
PE 8	112.0 (385.2)	0.9188	2.54	0.10	0.06	0.31	n.d.
PE 9	121.5 (394.7)	0.9334	1.40	0.04	0.02	0.11	0.87
PE 10	135.8 (409.0)	0.9549	0.1	1.82	0.04	0.15	<0.005
PE 11	–	0.9554	0.165	0.86	–	0.17	–

Sample No.	Molecular weight: Weight Av.	Molecular weight: Number Av.	Intrinsic viscosity[b] (ml/g)	Melt index[c]	Melt viscosity[d] (Pa s)	Elastic compliance[e] (MPa^{-1})
PE 1	510000[g]	10700[h]	79.5	1.80	64	63×10^{-6}
PE 2	300000[g]	13300[h]	75.7	1.95	38	46
PE 3	550000[g]	19100[h]	96.1	0.16	620	46
PE 4	225000[i]	(16000)[j]	62	19.9	24	22
PE 5	500000[i]	(18000)[j]	75	3.30	185	36
PE 6	500000[i]	(22000)[j]	82	1.06	73	54
PE 7	300000[i]	(45000)[j]	77	2.94	23	31
PE 8	800000[i]	(55000)[j]	97	0.21	310	54
PE 9	300000[i]	(27000)[j]	73	3.75	153	39×10^{-6}
PE 10	144000[k]	11500[h]	116	2.92	–	–
PE 11	–	–	–	–	–	–

[a] Samples annealed for 1 hour at about 100°C (373.2 K).
[b] α-Chloronaphthalene, 125°C (398.2 K).
[c] ASTM D-1238-57T.
[d] Newtonian melt viscosity at 150°C (423.2 K) and 400 Pa.
[e] Steady-state elastic compliance from creep recovery at 150°C (423.2 K) and 400 Pa.
[f] Values in parentheses indicate temperature in kelvin.
[g] Light scattering after optical clarification by high temperature ultracentrifugation.
[h] Cryoscopy.
[i] Preliminary value, subject to revision.
[j] "Best guess", subject to drastic revision.
[k] Measured by osmometry.

H. PROPERTIES OF TYPICAL POLY(ETHYLENES)[a]

Property	Low density ASTM type I[b]	Medium density ASTM type II[b]	High density ASTM type III[b]
Abrasion resistance, Taber (mg/1000 cycles)	10–15	6–10	2–5
Brittleness, low temperature (°C)	<– 118 (155.2)[c]	<– 118 (155.2)	<– 118 to – 73 (155.2–200.2)
Coefficient of thermal expansion ($\times 10^{-5}$) [D 696][d] (K^{-1})	10	–	13
Density [D 792] (g/cm^3)	0.910–0.925	0.926–0.940	0.941–0.965
Dielectric constant at 1 kHz [D 150]	2.28	–	2.32
Elongation at break [D 638] (%)	150–600	100–150	12–700
Hardness, Shore D [D 1706]	44–48	45–60	55–70
Impact strength, Izod [D 256] (ft/lb/in notch)	> 16	>16	0.8–14
Power factor at 1 kHz [D 150]	< 0.0001	–	< 0.0001
Heat capacity (kJ/kg/K)	1.916	1.916	1.916
Tensile modulus [D 638] (MPa)≡(N/mm^2)	55.1–172	172–379	413–1034
Tensile strength [D 638] (MPa)≡(N/mm^2)	15.2–78.6	12.4–19.3	17.9–33.1
Vicat, softening temp. [D 1525] (°C)	88–100 (361.2–373.2)	99–124 (372.2–39.2)	112–132 (385.2–405.2)
Volume resistivity [D 257] (cm)	6×10^{15}	–	6×10^5

[a] Ref. 52, 147, 148.
[b] ASTM designation D1248-72, "Standard Specification for Polyethylene Plastics Molding and Extrusion Materials."
[c] Values in parentheses indicate temperature in kelvin.
[d] The numbers in square brackets refer to the ASTM Standards, American Society for Testing Materials.

I. REFERENCES

1. H. Tadokoro, "Structure of Crystalline Polymers," Wiley, New York, 1979, p. 375.
2. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data., **10** (1), 119 (1981).
3. C. W. Bunn, Trans. Faraday Soc., **35**, 482 (1939).
4. S. Kavesh, J. M. Schultz, J. Polym. Sci. A-2, **8**, 243 (1970).
5. Y. Chatani, Y. Ueda, H. Tadokoro, Annual Meeting of the Society of Polymer Science, Japan, Tokyo, Preprint, p. 1326 (1977).
6. P. R. Swan, J. Polym. Sci., **56**, 409 (1962).
7. A. Turner-Jones, A. J. Cobbold, J. Polym. Sci., B, **6**, 539 (1968).
8. T. Seto, T. Hara, K. Tanaka, Jpn. J. Appl. Phys., **7**, 31 (1968).
9. D. C. Bassett, S. Block, G. J. Piermarini, J. Appl. Phys., **45**, 4146 (1974).
10. M. Yasuniwa, R. Enoshita, T. Takemura. Jpn. J. Appl. Phys., **15**, 1421 (1976).
11. T. Yamamoto, H. Miyaji, K. Asai, Rep. Prog. Polym. Phys. Jpn., **19**, 191 (1976).
12. G. Avitabile, R. Napolitano, B. Pirozzi, K. D. Rouse, M. W. Thomas, B. T. M. Willis, J. Polym. Sci. Polym. Lett. Ed., **13**, 351 (1975).
13. M. Shen, W. N. Hansen, P. C. Romo, J. Chem. Phys., **51**, 425 (1969).
14. P. R. Swan, J. Polym. Sci., **56**, 403 (1962).
15. E. R. Walter, F. P. Reding, J. Polym. Sci., **21**, 561 (1956).
16. H. M. M. Shearer, V. Vand, Acta Cryst., **9**, 379 (1956); H. M. M. Shearer and P. W. Teare, *ibid.*, **12**, 294 (1959).
17. J. Schelten, D. G. H. Ballard, G. D. Wignall, G. Longman, W. Schmatz, Polymer, **17**, 751 (1976).
18. G. Lieser, E. W. Fischer, K. Ibel, J. Polym. Sci. Polym. Lett. Ed., **13**, 39 (1975).
19. Q. A. Trementozzi, J. Polym. Sci., **36**, 113 (1959).
20. P. J. Flory, A. Citerri, R. Chiang, J. Am. Chem. Soc., **83**, 1023 (1961).
21. J. Schelten, G. D. Wignall, D. G. H. Ballard, G. W. Longman, Polymer, **18**, 1111 (1977).
22. F. J. Balta Calleja and D. R. Rueda, Polym. J., **6** (3), 216 (1974).
23. P. H. Geil, "Polymer Single Crystals," Wiley Interscience, New York, 1963, p. 8611.
24. B. G. Ranby, H. Brumberger, Polymer, **1**, 399 (1960).
25. B. G. Ranby, F. F. Morehead, N. M. Walter, J. Polym. Sci., **44**, 349 (1960).
26. F. P. Price, J. Chem. Phys., **35**, 1884 (1961).
27. D. C. Bassett, A. Keller, Phil. Mag., **7**, 1553 (1962).
28. L. Mandelkern, A. S. Posner, A. F. Diorio, D. E. Roberts, J. Appl. Phys., **32**, 1509 (1961).
29. W. M. Leung, R. St. John Manley, A. R. Panaras, Macromolecules, **18**, 746, 753, and 760 (1985).
30. J. D. Hoffman, Polymer, **24**, 3 (1983); *ibid.*, **23**, 656 (1982).
31. J. D. Hoffman, L. J. Frolen, G. S. Ross, J. I. Lauritzen, Jr. J. Res. Natl. Bur. Stand. **79A**, 671 (1975).
32. H. W. Starkweather, Jr., R. H. Boyd, J. Phys. Chem., **64**, 410 (1960).
33. F. A. Quinn, L. Mandelkern, J. Am. Chem. Soc., **80**, 3178 (1958).
34. B. Wunderlich, "Macromolecular Physics," Vol. 3, Crystal Melting, Academic, New York, 1980.
35. L. Mandelkern, "Crystallization of Polymers," McGraw-Hill, New York, 1964, p. 130.
36. R. F. Boyer, in: H. F. Mark and N. M. Bikales (Eds.), "Encyclopedia of Polymer Science and Technology," Supplement No. 2, 1977, p. 745.
37. G. T. Davis, R. K. Eby, J. Appl. Phys., **44**, 4274 (1973).

38. S. S. Chang, J. Polym. Sci. Polym. Symp., **43**, 43 (1973).
39. K. H. Illers, Kolloid-Z. Z. Polym., **231**, 622 (1969); *ibid.*, **252**, 1 (1974); *ibid.*, **251**, 394 (1973); *ibid.*, **190**, 16 (1963).
40. F. C. Stehling, L. Mandelkern, Macromolecules, **3** (2), 242 (1970).
41. P. J. Hendra, H. P. Jobic, K. Holland-Moritz, J. Polym. Sci., Polym. Lett. Ed., **13**, 365 (1975).
42. R. F. Boyer, Rubber Chem. Tech. (Rubber Reviews), **36** (5), 1303 (1963).
43. S. S. Chang, E. F. Westrum, H. G. Carlson, J. Res. Natl. Bur. Stand, Sect. A, **79**, 437 (1975).
44. M. Dole, W. P. Hettinger, N. R. Larson, J. A. Wethington, Jr., J. Chem. Phys., **20**, 718 (1952).
45. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 277 (1962).
46. F. Sakaguchi, L. Mandelkern, J. Maxfield, J. Polym. Sci., Polym. Phys. Ed., **14**, 2137 (1976).
47. S. S. Chang, J. Res. Natl. Bur. Stand., Sect. A, **78**, 387 (1974).
48. K. H. Hellwege, W. Knappe, W. Wetzel, Kolloid-Z., **180**, 126 (1962).
49. M. J. Richardson, Trans. Faraday Soc., **61**, 1876 (1965).
50. A. P. Gray and N. Brenner, Polym. Preprint, Am. Chem. Soc., Polym. Div., **6** (2), 956 (1965).
51. B. Wunderlich, G. Czornyj, Macromolecules, **10**, 906 (1977).
52. J. Brandrup, E. H. Immergut, (Eds.), "Polymer Handbook," 3rd. ed., Wiley, 1989.
53. F. C. Hahn, M. L. Macht, D. A. Fletcher, Ind. Eng. Chem., **37**, 526 (1945).
54. G. Allen, G. Gee, G. J. Wilson, Polymer, **1**, 456 (1960).
55. K. W. Doak, A. Schrage, "Polymerization and Copolymerization Processes," in R. A. V. Raff and K. W. Doak (Eds.), "Crystalline Olefin Polymers," Part 1, Interscience, New York, 1965, p. 301 ff.
56. K. Ziegler, Angew. Chem., **64**, 323 (1952).
57. K. Ziegler, E. Holzkamp, H. Breil, H. Martin, Angew. Chem., **67**, 426, 541 (1955).
58. J. P. Hogan, R. L. Banks, US Patent 2,825,721 assigned to Phillips Petroleum Company, March 4, 1958.
59. V. L. Lanza, B. D. Herrmann, J. Polym. Sci., **28**, 622 (1958).
60. H. C. Doepken, K. D. Kiss, D. Mangaraj, "Dielectric Strength of Crosslinked Polyethylene Insulation," Preprints, Organic Coatings and Plastics Chemistry, Vol. 38, American Chemical Society, 1978.
61. G. P. Mikhailov, S. P. Kabin, T. A. Krylova, Zh. Tekh. Fiz., **27**, 2050 (1957).
62. H. D. Anspon, et al., "Polyethylene" in "Manufacture of Plastics," W. M. Smith (Ed.), Reinhold, New York, 1964, p. 150 ff.
63. K. H. Stark, C. G. Garton, Nature, **176**, 1225 (1955).
64. P. Fisher, "The Short-Time Electrical Breakdown Behaviour of Polyethylene", Annual Report, Conference on Electrical Insulation and Dielectric Phenomenon, National Academy of Sciences Publication, 1974.
65. A. J. Bur, Polymer, **26**, 963 (1985).
66. G. A. Patten, Mod. Plastics, **38** (11), 119 (1961).
67. R. A. V. Raff, J. B. Allison, "Polyethylene," Interscience, 1956, p. 353.
68. Reference 42, p. 160.
69. B. Ke, J. Polym. Sci., **42**, 15 (1960).
70. B. Wunderlich, M. Dole, J. Polym. Sci., **24**, 201 (1957).
71. T. Morimoto, T. Mori, S. Enomoto, J. Appl. Polym. Sci., **22**, 1911 (1978).
72. A. R. Wedgewood, J. C. Seferis, Pure Appl. Chem., **55** (5), 873 (1983).
73. M. K. Tse, C. S. P. Sung, Org. Coat. Plast. Chem., **42**, 734 (1980).
74. O. Fuchs, L. Bohn, M. Fleissner, G. Gann, H. H. Suhr, H. D. R. Schüddemage, "Ethylene and Propylene Polymers" in F. D. Snell and L. S. Ettre (Eds.), "Encyclopedia of Industrial Chemical Analysis," Vol. 12, Wiley, New York, 1971, pp. 350–358.
75. S. Krimm. Fortschr. Hochpolymer Forschung, **2**, 51 (1960).
76. J. R. Nielsen, R. F. Holland, J. Mol. Spectrosc., **4**, 488 (1960); *ibid.*, **6**, 394 (1961).
77. M. P. Groenewege, J. Schuyer, J. Smidt, C. A. F. Tuijnman, "Absorption and Relaxation Spectra of Polyolefins," in: R. A. V. Raff and K. W. Doak (Eds.), "Crystalline Olefin Polymers," Part I, Interscience, New York, 1965, p. 762.
78. R. G. Snyder, J. Chem. Phys., **47**, 1316 (1967).
79. B. E. Read, R. S. Stein, Macromolecules, **1**, 116 (1968).
80. G. W. Chantry, "Submillimeter Spectroscopy," Academic, New York, 1971.
81. I. M. Ward (Ed.), "Developments in Oriented Polymers," Elsevier Applied Science, England, 1982.
82. I. M. Ward (Ed.), "Structure and Properties of Oriented Polymers," Elsevier Applied Science, England, 1975.
83. A. Ciferri, I. M. Ward, (Eds.), "Ultra-High Modulus Polymers," Elsevier Applied Science, England, 1979.
84. F. -P. Wolf, V. H. Karl, Colloid Polym. Sci., **259**, 29 (1981).
85. N. Capiati, S. Kojima, W. Perkins, R. S. Porter, J. Mater. Sci., **12**, 334 (1977).
86. J. L. Kardos, S. Piccardo, J. C. Halpin, Polym. Eng. Sci., **18**, 505 (1978).
87. A. J. Pennings, C. J. H. Schouteten, A. M. Kiel, J. Polym. Sci. C, **38**, 167 (1972).
88. W. Wu, P. G. Simpson, W. B. Black, J. Polym. Sci. Polym. Phys. Ed., **18**, 751 (1980).
89. I. M. Ward, "Mechanical Properties of Solid Polymers," 2nd Ed., Wiley, New York, 1983.
90. P. Smith, P. J. Lemstra, J. P. L. Pijpers, J. Polym. Sci. Polym. Phys. Ed., **20**, 2229 (1982).
91. J. Janzen, Polym. Eng. Sci., **32** (17), 1242 (1992).
92. L. Mandelkern, M. Hellman, D. W. Brown, D. E. Roberts, F. A. Quinn, Jr., J. Am. Chem. Soc., **75**, 4093 (1953).
93. R. F. Chiang, P. J. Flory, J. Am. Chem. Soc., **83**, 2857 (1961).
94. M. G. Broadhurst, J. Chem. Phys., **36**, 2578 (1962).
95. See "A General Discussion on Organization of Macromolecules in the Condensed Phase", Faraday Disc. Chem. Soc., **68** (1979).

96. M. Stamm, J. Polym. Sci., Polym. Phys. Ed., **20**, 235 (1982).
97. G. D. Wignall, L. Mandelkem, C. Edwards, M. Glotin, J. Polym. Sci. Polym. Phys. Ed., **20**, 245 (1982).
98. K. Swaminathan, S. P. Tewari, Nucl. Sci. Eng., **91** (1), 95 (1985).
99. F. A. Bovey, "High Resolution NMR of Macromolecules," Academic, New York, 1972.
100. R. Kitamaru, F. Horii, S. H. Hyon, J. Polym. Sci. Polym. Phys. Ed., **15**, 821 (1977).
101. H. Fumitaka, R. Kitamaru, S. Maeda. A. Saika, T. Terao, Polym. Bull., **13**, 179 (1985).
102. K. J. Packer, J. M. Pope, R. R. Yeung, M. E. A. Cudby, J. Polym. Sci. Polym. Phys. Ed., **22**, 589 (1984).
103. D. E. Axelson, G. C. Levy, L. Mandelkern, Macromolecules, **12** (1), 41 (1979).
104. R. J. Havens, D. L. Vanderhart, J. Magn. Res., **61** (2), 389 (1985).
105. I. Kamel, A. Charlesby, J. Polym. Sci., Polym. Phys. Ed., **19** (5), 803 (1981).
106. B. Schroeter, A. Posern, Macromol. Chem. Phys., **182** (2), 675 (1981).
107. H. Schneider, G. Hempel, Wiss. Z. Tech. Hochsch. "Carl Schorlemmer" Leuna-Merseburg, **22** (3), 407 (1980).
108. H. Pranadi, A. J. Manuel, Polymer, **21** (3), 303 (1980).
109. D. L. Vanderhart, Macromolecules, **12** (6), 1232 (1979).
110. F. A. Bovey, F. C. Schilling, F. L. McCrackin, H. L. Wagner, Macromolecules, **9** (1), 76 (1976).
111. I. Ando, T. Sorita, T. Yamanobe, T. Komoto, H. Sato, K. Degochi, M. Imanari, Polymer, **26**, (12), 1864 (1985).
112. S. Steingiser, S. P. Nemphos, M. Salame, in KirkOthmer, "Encyclopedia of Chemical Technology", 3rd ed., Vol. 3, Wiley-Interscience, New York, 1978.
113. W. M. Lee, Polym. Eng. Sci., **20** (1), 65 (1980).
114. J. Comyn (Ed.), "Polymer Permeability", Elsevier, London, 1985.
115. L. B. Shih, R. G. Priest, Appl. Spectroscopy, **38**, (5), 687 (1984).
116. D. L. Gerrard, W. F. Maddams, K. P. J. Williams. Polym. Commun., **25** (6), 182 (1984).
117. M. Glotin, L. Mandelkern, J. Polym. Sci. Polym. Phys. Ed., **21**, 29 (1983).
118. M. Glotin, L. Mandelkern, J. Polym. Sci. Polym. Lett. Ed., **21**, 807 (1983).
119. Y. K. Wang, D. A. Waldman, R. S. Stein, S. L. Hsu, J. Appl. Phys., **53** (10), 6591 (1982).
120. I. S. Butler, A. Neppel, Spectroscopy Lett., **16** (6), 419 (1983).
121. J. E. Mark, A. Eisenberg, W. W. Graessley, L. Mandelkern, J. L. Koenig, "Physical Properties of Polymers," Am. Chem. Soc., Washington, DC., 1984; 2nd ed., 1993, p. 280..
122. M. Glotin, L. Mandelkern, Colloid Polym. Sci., **260**, 182 (1982).
123. R. C. Domszy, M. Glotin, L. Mandelkern, J. Polym. Sci. Polym. Symp., **71**, 151 (1984).
124. R. A. V. Raff, K. W. Doak, (Eds.), "Crystalline Olefin Polymers," No. 20 in High Polymer Series, Parts I and II, Wiley, New York, 1964 and 1965.
125. W. M. D. Bryant, J. Polym. Sci., **2**, 556 (1947).
126. N. Ahmad, S. Ali, Colloid Polym. Sci., **256** (11), 1085 (1978).
127. J. Horska, J. Stejskal, P. Kratochvil, J. Appl. Polym. Sci., **24** (8), 1845 (1979); *ibid.*, **28**, 3873 (1983).
128. C. C. Han, P. H. Verdier, H. L. Wagner, J. Res. Natl. Bur. Stand., Sect. A., **83**, 185 (1978).
129. L. F. Shalaeva, V. V. Arefera, Vysokomol. Soedin., Ser. B, **16**, 802 (1974).
130. H. L. Wagner, J. Res. Natl. Bur. Stand., Sect. A, **76**, 151 (1972).
131. A. Peyrouset, R. Prechner, R. Panaris, H. Benoit, J. Appl. Polym. Sci., **19**, 1363 (1975).
132. E. Brauer, H. Wiegleb, M. Helmstedt, W. Eifert, Plaste Kautsch., **24**, 463 (1977).
133. H. L. Wagner, C. A. J. Hoeve, J. Polym. Sci. A-2, **9**, 1763 (1971).
134. L. H. Tung, J. Polym. Sci., **20**, 495 (1956).
135. L. H. Tung, J. Polym. Sci., **24**, 333 (1957).
136. P. S. Francis, R. C. Cooke, Jr., J. H. Elliot, J. Polym. Sci., **31**, 453 (1958).
137. J. E. Guillet, R. L. Combs, D. F. Slonaker, H. W. Coover, J. Polym. Sci., **47**, 307 (1960).
138. I. V. Mussa, J. Polym. Sci., **28**, 587 (1958).
139. L. Nicolas, Compt. Rend., **242**, 2720 (1956).
140. R. F. Boyer, "Transitions and Relaxations in Polymers," J. Polym. Sci. C, **14**, 3 (1966).
141. Reference 45, p. 1409 ff.
142. A. E. Woodward, J. A. Sauer, Fortschr. Hochpolymer Forsch., **1**, 136 (1958). Data summarized from various sources.
143. M. Baccaredda, G. Schiavinato, J. Polym. Sci., **12**, 155 (1954).
144. E. A. Cole, D. R. Holmes, J. Polym. Sci., **46**, 245 (1960).
145. C. Sella, Compt. Rend., **248**, 1819 (1959).
146. W. D. Niegisch, P. R. Swan, J. Appl. Phys., **31**, 1906 (1960).
147. P. D. Ritchie, Ed., "Vinyl and Allied Polymers," Vol. 1, Iliffe, London, 1968.
148. Trade literature published by Phillips Petroleum Company, Bartlesville, Oklahoma.

Physical Constants of Poly(propylene)*

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A. CRYSTALLOGRAPHIC DATA AND MODIFICATIONS OF ISOTACTIC POLYPROPYLENES

Stereoisomeric form	Crystal system	Space group	Lattice dimensions (nm)			α, β, or γ	Number of chains per unit cell	Molecular conformation	Crystal density (g/cm^3)	Refs.
			a	*b*	*c*					
Isotactic	Monoclinic α	C2/C	0.665	2.096	0.650	$\beta = 99.33°$	4	Helix(3/1)	0.936	1
	Monoclinic α_1	C2/C	0.666	2.078	0.6495	$\beta = 99.62°$	4		0.946	2–4
	Monoclinic α_2	$P2_1/C$	0.666	2.078	0.6495	$\beta = 99.62°$	4	$(TG)_3$	0.946	2–4
	Hexagonal β		1.274		0.635	$\gamma = 120°$	9	Helix(3/1)	0.921	5,6
	Hexagonal β_1		0.636		0.635	$\gamma = 120°$	9	$(TG)_3$	0.921	7
	Hexagonal β_2		1.908		0.649	$\gamma = 120°$	9		0.922	2,8
	Triclinc γ		0.654	2.14	0.650	$\alpha = 89°$ $\beta = 99.6°$ $\gamma = 99°$	2	Helix(3/1) $(TG)_3$	0.954	9
	Quenched "smectic"							Helix(3/1) $(TG)_3$		1,10–13

B. CRYSTALLOGRAPHIC DATA AND MODIFICATIONS OF SYNDIOTACTIC POLYPROPYLENES

Stereoisomeric form	Crystal system	Space group	Lattice dimensions (nm)			α, β, or γ	Number of chains per unit cell	Molecular conformation	Crystal density (g/cm^3)	Refs.
			a	*b*	*c*					
Syndiotactic	Orthorhombic observed in fibers	$C222_1$	1.450	0.580 or 0.560	0.740		2	Helix(4/1) $(T_2G_2)_2$	0.90	14–16
	Orthorhombic high temp. form	Ibca	1.450	1.120	0.740		2	Helix(4/1) $(T_2G_2)_2$	0.90	17–19
	Orthorhombic low temp. form		0.522	1.117	0.506		2	Planar Zigzag	0.945	20–22
	Triclinic	P1	0.572	0.764	1.16	$\alpha = 73.1°$ $\beta = 88.8°$ $\gamma = 112.0°$	1	$T_6G_2T_2G_2$	0.939	23

*Based on a table in the third edition, by R. P. Quirk and M. A. A. Alsamarraie, The Maurice Morton Institute of Polymer Science, University of Akron, Akron, Ohio.

C. DIMENSIONS OF POLY(PROPYLENE) MOLECULES*

Effect of Molecular Weight on Radius of Gyration[a]

Method of Crystallization	PPH matrix $\bar{M}_w$ ($\times 10^3$)	PPD matrix[b] $\bar{M}_w$ ($\times 10^3$)	$\bar{M}_w/\bar{M}_n$	$\langle S^2\rangle_z^{1/2}$† (nm)	$\langle S^2\rangle_w^{1/2}$† (nm)	$\langle S^2\rangle_w^{1/2}/\bar{M}_w^{1/2}$‡
Rapidly quenched	46	34	2.52	18.0	11.3	0.061
	46	140	1.68	18.0	13.9	0.037
	56	340	2.02	25.5	17.9	0.031
	105	575	1.56	32.0	25.6	0.034
	114	1540	1.48	50.3	41.3	0.039
Rapidly quenched from	46	34	2.52	18.0	11.3	0.061
melt and subsequently	46	140	1.68	19.0	14.7	0.039
annealed at	56	340	2.02	26.5	18.6	0.032
137°C (410.2 K)	105	575	1.56	34.7	27.8	0.037
	114	1540	1.48	51.4	42.3	0.040
Isothermally crystallized	46	34	2.52	23.5	14.8	0.080
at 139°C (412.2 K)	46	140	1.68	23.5	18.1	0.048
	56	340	2.02	29.0	21.8	0.037
	105	575	1.56	36.8	29.6	0.039
	114	1540	1.48	58.0	47.7	0.038

[a] Measured by neutron scattering. Polymer >97% isotactic; Ref. 26.
[b] Deutrated poly(propylene).
† $\langle S^2\rangle_z^{1/2}$ and $\langle S^2\rangle_w^{1/2}$ are the z-average and weight-average values, respectively, of the radius of gyration of the molecule.
‡ In nm/(g/mol)$^{1/2}$.

Neutron Scattering Measurements

Method of crystallization	$\langle S^2\rangle_w^{1/2}/\bar{M}_w^{1/2}$† Melt	Crystalline
Rapidly quenched	0.035	0.034
Isothermally crystallized at 139°C (412.2 K)	0.035	0.038
Rapidly quenched from melt and subsequently annealed at 137°C (410.2 K)	0.035	0.036

† $\langle S^2\rangle_w^{1/2}$ is the weight-average value of the radius of gyration of the molecule; values are in nm/(g/mol)$^{1/2}$. For further details and analysis, see Refs. 27–30.

Neutron Scattering See Refs. 24–30.

D. CRYSTALLINITY AND CRYSTALLIZATION KINETICS

Crystallinity and Melting Point of Samples from Multiple Solvent Fractionation[a]

Solubility of fractions in boiling solvents			
Soluble in	Insoluble in	T_m (°C)	Wide angle X-ray crystallinity (%)
–	Trichloroethylene	176 (449.2)[b]	75–85
–	*n*-Octane	174–175 (447.2–448.2)	65–85
n-Octane	2-Ethylhexane	174–175 (447.2–448.2)	60–66
2-Ethylhexane	*n*-Heptane	168–170 (441.2–443.2)	52–64
n-Heptane	*n*-Hexane	147–159 (420.2–432.2)	41–54
n-Hexane	*n*-Pentane	110–135 (383.2–408.2)	25–37
n-Pentane	Diethyl ether	106–114 (379.2–387.2)	15–27

[a] Refs. 31,32.
[b] Values in parenthesis indicate temperature in kelvin.

* See also corresponding table in this Handbook, and Refs. 24,25.

Crystallinities and Melting Temperatures of Mixtures of Amorphous[a] and Isotactic Poly(propylene)[b,c]

Isotactic fraction (%)	Crystallinity		Melting temperature (°C)
	Calculated	Measured	
100	67.2	67.7	174 (447.2)[d]
87.5	59.3	60.3	173 (446.2)
75.8	51.4	49.8	172–173 (445.3–446.2)
60	40.6	40.2	171–172 (444.2–445.2)

[a] Ether extract.
[b] 2-Ethylhexane extraction residues.
[c] Refs. 33,34.
[d] Values in parenthesis indicate temperature in kelvin.

Crystallinity in Isotactic Poly(propylenes)[a]

Sample	Description and condition	Crystalline weight fraction
1.	Heptane extract of crude polypropylene, "amorphous", highly atactic	0.14
2.	Isotactic, water quenched	0.31
3.	Same as 2, followed by annealing at 105°C for 1 hr	0.43
4.	Same as 2, followed by annealing at 160°C for 1/2 hr	0.65

[a] Ref. 35,36.

Effect of Isotacticity on Crystallinity[a]

Sample	Description and condition	Crystalline weight fraction
1.	$\bar{M}_w = 202000$, $\bar{M}_w/\bar{M}_n = 2.6$, Isotacticity = 0.988	0.42–0.75
2.	$\bar{M}_w = 159000$, $\bar{M}_w/\bar{M}_n = 2.3$, Isotacticity = 0.978	0.40–0.75
3.	$\bar{M}_w = 189000$, $\bar{M}_w/\bar{M}_n = 3.0$, Isotacticity = 0.953	0.38–0.60
4.	$\bar{M}_w = 209000$, $\bar{M}_w/\bar{M}_n = 1.8$, Isotacticity = 0.882	0.30–0.45
5.	$\bar{M}_w = 190000$, $\bar{M}_w/\bar{M}_n = 1.6$, Isotacticity = 0.787	0.20–0.25

[a] Ref. 37.

Crystallization Kinetics See Table "Rate of Crystallization of Polymers" and Refs. 38–49 for isotactic poly(propylene), and Refs. 50–52 for syndiotactic poly(propylene).

E. EQUILIBRIUM THERMODYNAMIC PROPERTIES

Equilibrium Melting Temperature, T_m

Isotactic poly(propylene) 187.5°C (460.7 K) for the crystal with an infinite size [reported data ranging from 183°C–220°C (456–493 K); for a detailed discussion, see Ref. 34]. It is also critically dependent upon the isotacticity in crystalline samples.

Effect of Isotacticity on Equilibrium Melting Temperature[a]

Sample	Description and condition	Equilibrium melting temperature (°C)
1.	$\bar{M}_w = 202000$, $\bar{M}_w/\bar{M}_n = 2.6$, Isotacticity = 0.988	183.8 (457.0)[b]
2.	$\bar{M}_w = 159000$, $\bar{M}_w/\bar{M}_n = 2.3$, Isotacticity = 0.978	182.8 (456.0)
3.	$\bar{M}_w = 189000$, $\bar{M}_w/\bar{M}_n = 3.0$, Isotacticity = 0.953	180.2 (453.4)
4.	$\bar{M}_w = 209000$, $\bar{M}_w/\bar{M}_n = 1.8$, Isotacticity = 0.882	173.0 (446.2)
5.	$\bar{M}_w = 190000$, $\bar{M}_w/\bar{M}_n = 1.6$, Isotacticity = 0.787	163.0 (436.2)

[a] Ref. 37.
[b] Values in parenthesis indicate temperature in kelvin.

Syndiotactic poly(propylene) Equilibrium melting temperature is critically dependent upon the tacticity of the samples. For a sample with 95% of [r], 92% of [rr], and 86% of [rrr], and molecular weight higher than 40000, for example, the equilibrium melting temperature is 160°C (433.2 K) (53,54). For a 99% dyad sample, an actual melting temperature of 163°C (436.2 K) can be observed (55).

Equilibrium Enthalpy of Fusion, ΔH_f

Isotactic poly(propylene) 8.7 ± 1.6 kJ/mol (56) [reported data ranging from 2.65 kJ/mol to 10.94 kJ/mol; for a detailed discussion see Ref. 56].

Syndiotactic poly(propylene) Equilibrium enthalpy of fusion is determined for samples with 95% of [r], 92% of [rr], and 86% of [rrr], and molecular weight higher than 40000, and is 8.0 kJ/mol (53).

Equilibrium Entropy of Fusion, ΔS_f

Isotactic poly(propylene) 18.9 ± 3.5 J/K/mol (56).

Syndiotactic poly(propylene) No data is available.

Glass Transition Temperature, T_g

Isotactic poly(propylene) -3.2°C (270 K) (56). The agreement of T_g values from various experimental methods [*e.g.*, dilatometry, dynamic mechanical measurements (at low frequency), heat capacity and NMR (narrowing of line width)] is not very good, but ranges from about -30 to $+20$°C. The glass transition temperature depends on thermal history (may also be tacticity) of the sample (57–61). Activation energy, 117–152 kJ/mol (57).

Syndiotactic poly(propylene) May be close to the glass transition temperature of isotactic poly(propylene).

Heat Capacity, C_p (J/K/mol) See "Thermodynamic Properties" in this Handbook, and Ref. 62.

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Isotactic poly(propylene) in the liquid The amorphous liquid heat capacities above T_g are given in J/K/mol [±2.5% (RMS)] $C_p = 0.151291\ T + 42.956$

Isotactic poly(propylene) in the solid The heat capacities in the solid state are separately fitted to 100% crystalline in various temperature ranges, all in J/K/mol:

10–100 K [±0.4% (RMS)]:
$C_p = \exp[0.241028(\ln T)^3 - 3.01364(\ln T)^2 + 13.5529(\ln T) - 18.7621]$
80–250 K [±0.8% (RMS)]:
$C_p = \exp[0.121683(\ln T)^3 - 1.90162(\ln T)^2 + 10.727(\ln T) - 17.6875]$
230–350 K [±1.7% (RMS)]:
$C_p = 1.5912 \times 10^6\ T^2 + 0.3837\ T - 64.551$

The 0% crystalline amorphous state heat capacities in various temperature ranges, all in J/K/mol.

10–60 K [±1.0% (RMS)]:
$C_p = \exp[0.327068(\ln T)^3 - 3.688(\ln T)^2 + 14.7469(\ln T) - 18.4281]$
50–180 K [±0.7% (RMS)]:
$C_p = \exp[0.00742669(\ln T)^3 - 0.189318(\ln T)^2 + 2.10843(\ln T) - 3.0699]$
160–260 K [±1.1% (RMS)]:
$C_p = \exp[0.0727139(\ln T)^3 - 0.711627(\ln T)^2 + 2.31907(\ln T) + 0.77926]$

Syndiotactic poly(propylene) No data is available.

Heat Capacity Data in the Crystalline and Amorphous Isotactic Poly(propylene)

	C_p (J/K/mol)		
Temperature (K)	Crystalline	Amorphous	Refs.
50	13.00	15.22	53
100	26.51	28.21	53
150	37.68	39.53	53
200	47.47	50.53	53
250	56.53	61.95	53
300	70.39	84.11	53
350	82.12	109.2	53

Heat Capacity Changes at Glass Transition Temperatures (J/K/mol)

Isotactic poly(propylene) 19.2 J/K/mol at −3.2°C (270 K) (63).

Syndiotactic poly(propylene) May be close to that of isotactic poly(propylene).

Residual Entropy of the Glassy State at 0 K S_0 (J/K/mol)

Isotactic poly(propylene) 5.2 J/K/mol (58).

Syndiotactic poly(propylene) No data is available.

F. OTHER GENERAL PROPERTIES

Coefficients of Thermal Expansion (ASTM D 696) (K^{-1}) (Refs. 64–66)

From −30°C (243.2 K) to 0°C (273.2 K)	6.5×10^{-5}
From 0°C (273.2 K) to 30°C (303.2 K)	10.5×10^{-5}
From 30°C (303.2 K) to 60°C (333.2 K)	14.5×10^{-3}

Density at 25°C (298.2 K) (Mg/m^3) = (g/cm^3). See also Section G.

Isotactic	Crystalline	0.932–0.943
	Amorphous (from extrapolation of data above melting point)	0.850–0.854
	"Smectic" form	0.916
Syndiotactic	Crystalline	0.989–0.91
	Amorphous (from extrapolation of data above melting point)	0.856

Dielectric Constant See Section G.

Electrical Properties: Conductivity[a]

Electric field (kV/cm)	Crystallinity (%)	Conductivity at 72°C (mhos/cm)	Refs.
8.5	–	1×10^{-18}	67
10.5	75	3.2×10^{-16}	68
	45	7.4×10^{-17}	68
	78	2.3×10^{-18}	69
	50.5	1×10^{-18}	69
24	81.2	2×10^{-16}	70
	39.7	1×10^{-16}	70
	78	2.2×10^{-18}	69
	50.5	1×10^{-18}	69
48	–	3.2×10^{-18}	71
	78	2.3×10^{-18}	69
144	81.2	1×10^{-14}	70
	39.7	1×10^{-14}	70
	78	2.8×10^{-18}	69
	50.5	2×10^{-18}	69

[a] See also Section G.

G Values for Radiation Crosslinking G(c.l.) and Chain Scission G(Breaks)
Maximum and Minimum Values[a]

	G(breaks)		*G*(c.l.)	
	Maximum	Minimum	Maximum	Minimum
Atactic	0.24	0.10	0.27	0.115
Isotactic film	0.21	0.10	0.14	0.069
Isotactic flake	0.27	0.10	0.18	0.068

[a] Ref. 72.

At Room Temperature and in Vacuum[a]

	G(c.l.)		G(breaks)
Measured by	Solubility	Elasticity	Solubility or viscosity
Atactic	0.12–0.27	0.6–1.3	0.10–0.24
Isotactic	0.07–0.25		0.10–0.24
	0.6		0.9
			5.0

[a] Ref. 72.

G-Values in Terms of Radiolytic Gas Yields[a]

	Gas	Irradiation T (°C)	G (gas)
Atactic	H_2	25	2.34
	CH_4	25	0.095
Isotactic	H_2	−196	2.55
	CH_4	−196	0.058
	H_2	25	2.78
	CH_4	25	0.072

[a] Ref. 72.

Ignition Limiting Indices[a]

650°C (923.2 K)	600°C (873.2 K)	550°C (823.3 K)
2.8 ± 0.1	3.5 ± 0.1	5.7 ± 0.1

[a] This method (ASTM D2803-70) is defined as the minimum volume of oxygen required for ignition to occur; Ref. 73.

Infrared Absorption Bands (36,74–85)

Assignment in Infrared Spectrum Used for Crystallinity Measurements[a,b]

Form	Crystalline phase wave numbers (cm^{-1})	Amorphous phase wave numbers (cm^{-1})
Isotactic	809	790
	842	1158
	894	
	997	
Syndiotactic	866	1131
	977	1199
		1230

[a] Used for tacticity measurements: ratio of absorption at 997 cm^{-1} to that at 975 cm^{-1} (80,84,86).
[b] Ref. 36.

Sequence-Sensitive Vibration Band of Propylene Units[a]

Position of band (cm^{-1})	Type of regularity	Classification	n[b]
998	Isotactic	Helix band	10–12
973	Isotactic	Regularity band	3–4
841	Isotactic	Helix band	12
977	Syndiotactic	Helix band	
962	Syndiotactic	Regularity band	
867	Syndiotactic	Helix band	
936	–	Band of isolated propylene unit	

[a] Refs. 74,87.
[b] Number of the units in sequence for the bands to appear.

Far Infrared. See Ref. 88.

Mechanical Properties: Elongation at Break See Sections G and H

Hardness, Shore D See Sections G and H.

Impact Strength, Izod See Sections G and H.

Low Temperature Brittleness See Sections G and H.

Melt Index (ASTM Method D 1238-57T)

Molecular weight ($\times 10^3$)	Melt index (230°C, 2.16 kg)
142	22.8
180	7.3
220	3.5
292	1.2
358	0.39

Molecular Parameters of Typical Poly(propylenes) (64)

Number of double bonds per 1000 C atoms	<1
Type of unsaturation	Vinylidene
$\bar{M}_w$	220000–700000
$\bar{M}_n$	38000–60000
$\bar{M}_w/\bar{M}_n$	5–12

Nuclear Magnetic Resonance See Refs. 80,89–94.

Permeability and Diffusion Coefficients See corresponding table in this Handbook and Refs. 95–97.

Permeability of Polypropylene Films to Various Gases[a]

	Permeability[b] (cm^3/24 hr/100 $in.^2$/mil/atm)		
Film	O_2	CO_2	N_2
Unoriented	250	800	40
Oriented	105–150	300–500	10–25

[a] Ref. 98.
[b] At 25°C (298.2 K) adn 50% rh (ASTM D 1434–63).

Properties of Typical Poly(propylenes) See Sections G and H.

Refractive Index $n_D^{2.5} = 1.49$ (66,99).

Refractive Index Increment (dn/dc) See corresponding table in this Handbook, and Refs. 30,100.

Softening Temperature, Vicat See Sections G and H.

Sound Velocity (m/s) (101–103).

Unoriented:	at 25°C (298.2 K)	2.5×10^3
	at 125°C (398.2 K)	125×10^3

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Specific Heat See Heat Capacity.

Specific Volume (cm^3/g)[a]

Form	v_a (Amorphous)	v_b (Crystalline)
Isotactic	1.176–1.172	10.73–1.060
Smectic		1.092
Syndiotactic	1.165	1.114–1.10

[a] Ref. 64.

Tensile Modulus See Sections G and H.

Tensile Strength See Sections G and H.

Transition and Relaxation Temperatures Associated with Peaks in Dynamic Loss[a]

Transition	T (°C)	Assignment and remarks
α	30–80	Difficult to resolve
β (at 0.2 Hz)	0 (273.2)[b]	Insensitive to changes in crystallinity; long chain motion in amorphous portion
γ	−80 (193.2)	In atactic poly(propylene); hindered movement of $C-CH_3$ units
δ	<−200 (<73.2)	Hindered rotation of CH_3 groups

[a] Designated as α, β, γ, etc. in order of descending temperatures; Ref. 104.
[b] Values in parentheses indicate temperatures in kelvin.

Ultra-Drawn Polypropylene See Refs. 105–108.

Viscosity–Molecular Weight Relationship See corresponding table in this Handbook.

G. PROPERTIES OF TYPICAL MAINLY ISOTACTIC POLY(PROPYLENES)[a,b]

Property	Value
Brittleness [D 746] (°C (K))	25 (298.2)
Deflection temperatures [D 648] (°C (K))	
at 66 lb/in^2 (4.64 kg/cm^2)	96–110 (369.2–383.2)
at 264 lb/in^2 (18.6 kg/cm^2)	57–63 (330.2–336.2)
Density [D 792] (Mg/m^3) = (g/cm^3)	0.90–0.91
Dielectric constant at 1 kHz [D 150]	2.2–2.3
Dielectric strength [D 149] (V/mil)	610 [430 at 120°C (393.2 K)]
(V/cm) × 10^{-3}	240 [170 at 120°C (393.2 K)]
Dissipation factor (60 Hz–100 MHz) [D 510]	$3 \times 10^{-4} - 1 \times 10^{-3}$
Elongation at break [D 638] (%)	500–900 [30 at −40°C (233.2 K)]
Environment stress cracking or F_{50} time [D 1693]	Does not stress crack
Flexural modulus [D 790] (MPa) = (N/mm^2)	1172
Hardness, shore D [D 1706]	70–80
Heat capacity (kJ/kg/K)	1.926 (see also Table I)
Izod impact strength [D 256] (ft lb/in of notch)	0.4–6.0 [0.1–0.7 at −20°C (253.2 K)]
(cm kg/cm notch)	2.2–12 [0.55–3.9 at −20°C (253.2 K)]
Power factor at 1 kHz	<0.002–0.001
Solvent resistance to hydrocarbons and chlorinated hydrocarbons	Resistant below about 80°C (353.2 K)
Stiffness in flexure [D 747] (MPa) = (N/mm^2)	1172
Tensile modulus [D 638] (MPa) = (N/mm^2)	1032–1720
Tensile strength [D 638] (MPa) = (N/mm^2)	29.3–38.6
Tensile yield elongation [D 638] (%)	11–15 [11% at −40°C (233.2 K)]
Thermal conductivity [C 177] (W/m/K)	11.7
Vicat softening temperature [D 1525] (°C (K))	138–155 (411.2–428.2)
Volume resistivity [D 257] (Ω cm)	10^{16}–10^{17}

[a] Properties at ambient room temperature, unless denoted otherwise. The numbers in square brackets in the first column refer to the ASTM (American Society for Testing Materials) Standards.
[b] Refs. 64,66,99.

H. PROPERTIES OF SOME COMMERCIAL POLY(PROPYLENE) GRADES[a]

		Grade			
Property[b]	Method[c]	Pipes and sheets	Films	Injection molding	Cast and tubular film
Melt flow rate (g/10 min)	D 1238L	0.25–0.35	1.5–2.0	5–7	7–10
Intrinsic viscosity at 230°C (dl/g)	ME 15071	2.6–3.0	2.0–2.4	1.8–2.4	1.5–1.9
Density (g/cm^3)	D 1505	0.900–0.905	0.900–0.905		0.900–0.905
Crystallinity by X-ray (%)	ME 149/67	58–59	59–60	61–62	62.5–64.0

Property[b]	Method[c]	Grade: Pipes and sheets	Films	Injection molding	Cast and tubular film
Crystalline mp by dsc[d] (°C (K))	ME 150702	161–163 (434–436)	161–163 (434–436)	162–164 (435–437)	163–165 (435–437)
Isotactic index	ISO 873	>96.0	>96.0	>95.0	>94.0
$\bar{M}_w/\bar{M}_n$ by gpc[e]	ME 15627	8–10	7–9	6–8	5–7
Spiral flow at 230°C, 60 kg/cm (mm)	ME 17544	390–450	525–550	700–750	800–850
Melt tension test					
swelling per 24 s	ME 17115	1.57–1.60	1.60–1.63	1.63–1.64	1.63–1.64
drawing ratio	ME 17115	500–700	1100–1300	1500–1700	≥2000
Linear mold shrinkage (%)		≤1.9	≤1.9	≤1.9	≤1.9
Tensile yield strength at 5 cm/mm (MPa)[f]	D 638	29–31	32–34	33–35	33–35
Tensile yield elongation (%)	D 638	10–12	10–12	8–10	8–10
Flexural modulus (GPa)[g]	D 790	1.1–1.3	1.3–1.4	1.5–1.6	1.6–1.7
Impact, Izod at 23°C (J/m)[h]	D 256	1.5–2.0	0.5–0.7	0.3–0.5	0.2–0.3
Hardness, Rockwell, C scale	D 785B	55–60	55–60	60–65	60–65
Deflection temperature (°C (K))					
at 0.45 MPa[f]	D 648	88–90 (361–363)	90–92 (363–365)	92–94 (365–367)	>94 (>367)
at 1.8 MPa[f]	D 648	50–55 (323–328)	50–55 (323–328)	55–60 (328–333)	55–60 (328–333)
Vicat point at 9.807 MPa[f] (°C (K))	D 1525	≥150 (≥423)	≥150 (≥423)	≥150 (≥423)	≥150 (≥423)
Yellow index on pellets (%)	D 1925	8–15	4–8	3–6	3–6
Gloss on plate (%)	D 523	48–50	48–50	49–51	49–51
Luminosity on plate	D 1635	68–70	≥70	≥70	≥70
Coefficient of linear thermal expansion ($\times 10^{-5}$)					
from −30°C (243.2 K) to 0°C (273.2 K)		6.5	6.5	6.5	6.5
from 0°C (273.2 K) to 30°C (303.2 K)		10.5	10.5	10.5	10.5
from 30°C (303.2 K) to 60°C (333.2 K)		14.0	14.0	14.0	14.0
Water absorption at 24 h immersion (%)	D 570	0.03	0.03	0.03	0.03
Environmental stress cracking[i]	D 1693	Does not break	Does not break	Does not break	Does not break
Dielectric constant	D 1531				
at 10 kHz		2.2–2.6	2.2–2.6	2.2–2.6	2.2–2.6
at 1 MHz		2.2–2.6	2.2–2.6	2.2–2.6	2.2–2.6
at 1 GHz		2.2–2.6	2.2–2.6	2.2–2.6	2.2–2.6
Dissipation factor, *h*					
at 10 kHz		<0.0005	0.0005	0.0005	0.0005
at 1 MHz		<0.0005	0.0005	0.0005	0.0005
at 1 GHz		<0.0005–0.0018	0.0005–0.0018	0.0005–0.0018	0.0005–0.0018
Arc resistance, *s*	D 495	136–185	136–185	136–185	136–185

[a] Ref. 65.
[b] Determined on Montedison Moplen poly(propylene).
[c] D = ASTM; ME = Montedison.
[d] Differential scanning calorimetry.
[e] Gel-permeation chromatography.
[f] To convert MPa to psi, multiply by 145.
[g] To convert GPa to psi, multiply by 145,000.
[h] To convert J/m to ft-lb/in., divide by 53.38.
[i] Igepal = 100%.

I. MECHANICAL PROPERTIES OF POLY(PROPYLENE) HOMOPOLYMERS[a]

Type	Melt flow rates [ASTM: D 1238] (g/10 min)	Flexural modulus [D 790] (MPa)	Izod impact [D 256] (J/m)	Heat deflection temp. 66 psi [D 648] (°C)	Hardness [D 785] (Rockwell R)
Compr. mold, pipe extrusion	0.5	1585 (230)[b]	160 (3.0)[c]	93 (366.2)[d]	95
Sheet, strapping extrusion	1.5	1480 (215)	70 (1.3)	87 (360.2)	–
High clarity, stiffness	2	1790 (260)	140 (2.6)	110 (383.2)	100
General purpose injection molding	4	1720 (250)	43 (0.8)	97 (370.2)	99
Rapid cycle injection molding	12	1655 (240)	27 (0.5)	99 (372.2)	97
Rapid cycle, high stiffness	12	1895 (295)	32 (0.6)	118 (391.2)	105
Thin wall injection molding	20	1720 (250)	32 (0.6)	100 (373.2)	104
Thin wall, high stiffness	20	2000 (290)	32 (0.6)	124 (397.2)	104
Multi-cavity molding	35	1310 (190)	32 (0.6)	90 (363.2)	98
Extrusion coating	40	1895 (275)	32 (0.6)	121 (394.2)	100
Coextrusion	70	1515 (220)		90 (363.2)	
Fibers	100	1585 (230)			

[a] Ref. 109.
[b] Values in parenthesis in this column indicate Flexural modulus in K psi.
[c] Values in parenthesis in this column indicate Izod impact in ft-lb/in.
[d] Values in parenthesis in this column indicate temperature in kelvin.

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J. MECHANICAL PROPERTIES OF POLY(PROPYLENE) RANDOM COPOLYMERS[a]

Type	Melt flow rates [ASTM: D 1238] (g/10 min)	Flexural modulus [D 790] (MPa)	Izod impact [D 256] (J/m)	Heat deflection temp. 66 psi [D 648] (°C)	Hardness [D 785] (Rockwell R)
Sheet extrusion	1	965 (140)[b]	80 (1.5)[c]	90 (363.2)[d]	80
Sheet extrusion	2.5	690 (100)	85 (1.6)	70 (343.2)	65
Sheet extrusion	3	1515 (220)	37 (7)	102 (375.2)	95
Film extrusion	3	260 (38)			
Inj. molding, film, coextrusion	6	825 (120)	48 (0.9)	85 (358.2)	80
Film extrusion	6	495 (72)			
Inj. molding, film, coextrusion	7	585 (85)	80 (1.5)	65 (338.2)	65
Inj. molding, film, coextrusion	10	860 (125)	37 (0.7)	85 (358.2)	80
Inj. molding, film, coextrusion	10	1135 (165)	54 (1.0)	91 (364.2)	88
Rapid injection molding	25	895 (130)	37 (0.7)	85 (358.2)	80
Rapid injection molding	25	1135 (165)		98 (371.2)	94
Rapid injection molding	35	1135 (165)	43 (0.8)	87 (360.2)	80

[a] Ref. 109.
[b] Values in parenthesis in this column indicate Flexural modulus in K psi.
[c] Values in parenthesis in this column indicate Izod impact in ft-lb/in.
[d] Values in parenthesis in this column indicate temperature in kelvin.

K. MECHANICAL PROPERTIES OF POLY(PROPYLENE) IMPACT COPOLYMERS[a]

Type	Melt flow rates [ASTM: D 1238] (g/10 min)	Flexural modulus [D 790] (MPa)	Izod impact [D 256] (J/m)	Heat deflection temp. 66 psi [D 648] (°C)	Hardness [D 785] (Rockwell R)
Compr. Mold, pipe extrusion	0.5	1240 (180)[b]	540 (10)[c]	90 (363.2)[d]	75
Sheet extrusion	1	860 (125)	650 (12)	79 (352.2)	55
Sheet extrusion	2	965 (140)	650 (12)	72 (345.2)	61
General purpose inj. molding	4	1310 (190)	135 (2.5)	81 (354.2)	82
General purpose inj. molding	4	1035 (150)	540 (10)	75 (348.2)	65
General purpose inj. molding	6	930 (135)	650 (12)	90 (363.2)	70
Blush resist., inj. molding	11	1000 (145)	135 (2.5)	77 (350.2)	72
Blush resist., inj. molding	16	860 (125)	650 (12)	80 (353.2)	40
Blush resist., rapid inj. molding	20	965 (140)	185 (3.4)	77 (350.2)	67
Blush resist., rapid inj. molding	20	790 (115)	540 (10)	85 (358.2)	65
Blush resist., rapid inj. molding	30	965 (140)	55 (1.0)	80 (353.2)	80
Blush resist., rapid inj. molding	35	1310 (190)	70 (1.3)	90 (363.2)	85
Blush resist., rapid inj. molding	50	1170 (170)	55 (1.0)	117 (390.2)	90

[a] Ref. 109.
[b] Values in parenthesis in this column indicate Flexural modulus in K psi.
[c] Values in parenthesis in this column indicate Izod impact in ft-lb/in.
[d] Values in parenthesis in this column indicate temperature in kelvin.

L. REFERENCES

1. G. Natta, P. Corradini, Nuovo Cimento. Suppl., **15**, 40 (1960).
2. A. Turner-Jones, J. M. Aizlewood, D. R. Beckett, Makromol. Chem., **75**, 134 (1964).
3. D. C. Bassett, S. Block, G. J. Piermarini, J. Appl. Phys., **45**, 4146 (1974).
4. Y. Chatani, Y. Ueda, H. Tadokoro, Annual Meeting of the Society of Polymer Science, Japan, Tokyo, Preprint, 1977, p. 1326.
5. H. D. Keith, F. J. Padden, Jr., N. M. Walker, H. W. Wyckoff, J. Appl. Phys., **30**, 1485 (1959).
6. R. J. Samuels, R. Y. Yee, J. Polym. Sci. A-2, **10**, 385 (1972).
7. E. J. Addink, J. Bientema, Polymer, **2**, 185 (1961).
8. A. Turner-Jones, A. J. Cobbold, J. Polym. Sci. B, **6**, 539 (1968).
9. D. R. Morrow, B. A. Newman, J. Appl. Phys., **39**, 4944 (1968).
10. R. L. Miller, Polymer, **1**, 135 (1960).
11. J. A. Gailey, R. H. Ralston, SPE Transactions, **4**, 29 (1964).
12. D. M. Gezovich, P. H. Geil, Polym. Eng. Sci., **8**, 202 (1968).
13. P. Corradini, V. Petraccone, C. De Rosa, G. Guerra, Macromolecules, **19**, 2699 (1986).
14. G. Natta, I. Pasquon, P. Corradini, et al., Atti dell'Accademia Nazionale dei Lincei. Rendiconti, **28**, 539 (1960).

15. P. Corradini, G. Natta, P. Ganis, P. A. Temussi, J. Polym. Sci. C, **16**, 2477 (1967).
16. T. Seto, T. Hara, K. Tanaki, Jpn. J. Appl, Phys., **7**, 31 (1968).
17. B. Lotz, A. J. Lovinger, R. E. Cais, Macromolecules, **21**, 2375 (1988).
18. A. J. Lovinger, B. Lotz, D. Davis, Polymer, **31**, 2253 (1990).
19. A. J. Lovinger, D. Davis, B. Lotz, Macromolecules, **24**, 552 (1991).
20. G. Natta, M. Peraldo, G. Allegra, Makromol. Chem., **75**, 215 (1964).
21. H. Tadokaro, M. Kobayashi, S. Kobayashi, K. Yasufuku, K. Mori, Rep. Prog. Polym. Phys. Japan, **9**, 181 (1966).
22. Y. Chatani, H. Maruyama, K. Noguchi, et al., J. Polym. Sci. Polym. Lett., **28**, 393 (1990).
23. Y. Chatani, H. Maruyama, T. Asanuma, T. Shiomura, J. Polym. Sci., Polym. Phys. Ed., **29**, 1649 (1991).
24. G. D. Wignall, L. Mandelkern, C. Edwards, M. Glotin, J. Polym. Sci., Polym. Phys. Ed., **20**, 245 (1982).
25. J. E. Mark, A. Eisenberg, W. W. Graessley, L. Mandelkern, J. L. Koenig, "Physical Properties of Polymers", American Chemical Society, Washington DC., 1984.
26. D. G. H. Ballard, P. Cheshire, G. W. Longman, J. Schelten, Polymer, **19**, 379 (1978).
27. J. Schelton, D. G. H. Ballard, G. D. Wignall, G. W. Longman, W. Schmatz, Polymer, **17**, 751 (1976).
28. D. Y. Yoon, P. J. Flory, Polymer, **18**, 509 (1977).
29. M. Stamm, E. W. Fischer, M. Dettenmaier, Disc. Faraday Soc., **68**, 263 (1979).
30. P. Parrini, F. Sebastiano, G. Messina, Makromol. Chem., **38**, 27 (1960).
31. G. Natta, J. Polym. Sci., **34**, 531 (1959).
32. Y. V. Kissin, "Isospecific Polymerization of Olefins with Heterogeneous Ziegler-Natta Catalysts", Springer, New York, 1985.
33. G. Natta, G. Mazzanti, O. Grespi, G. Noraglio, Chim. Ind., **39**, 275 (1957).
34. B. Wunderlich, "Macromolecular Physics", Vol. 3: Crystal Melting, Academic Press, New York, 1980.
35. W. Ruland, Acta Cryst., **14**, 1180 (1961).
36. B. Wunderlich, "Macromolecular Physics", Vol. 1: Crystal Structure, Morphology, Defects, Academic Press, New York, 1973.
37. S. Z. D. Cheng, J. J. Janimak, A. Zhang, E. T. Hsieh, Polymer, **32**, 648 (1991).
38. B. von Falkai, H. A. Stuart, Kolloid Z., **162**, 138 (1959).
39. B. von Falkai, Makromol. Chem., **41**, 86 (1960).
40. F. L. Binsbergen, B. G. M. De Lange, Polymer, **11**, 309 (1970).
41. A. Wlochowicz, M. Eder, Polymer, **22**, 1285 (1981).
42. H. D. Keith, F. J. Padden, Jr., J. Appl. Phys., **35**, 1286 (1964).
43. A. J. Lovinger, J. O. Chua, C. C. Gryte, J. Polym. Sci., Polym. Phys. Ed., **15**, 641 (1977).
44. L. Goldfarb, Makromol. Chem., **179**, 2297 (1978).
45. E. Martuscelli, C. Silvestre, G. Abate, Polymer, **23**, 229 (1982).
46. E. J. Clark, J. D. Hoffman, Macromolecules, **17**, 878 (1984).
47. S. Z. D. Cheng, J. J. Janimak, A. Zhang, H. N. Cheng, Macromolecules, **23**, 298 (1990).
48. J. J. Janimak, S. Z. D. Cheng, P. A. Giusti, E. T. Hsieh, Macromolecules, **24**, 2253 (1991).
49. J. J. Janimak, S. Z. D. Cheng, J. Polym. Eng., **10**, 21 (1991).
50. R. L. Miller, E. G. Seeley, J. Polym. Sci., Polym. Phys. Ed., **20**, 2297 (1982).
51. G. Balbontin, D. Dainelli, M. Galimberti, G. Paganetto, Makromol. Chem., **193**, 693 (1993).
52. J. Rodriguez-Arnold, Z. Bu, S. Z. D. Cheng, E. T. Hsieh, T. W. Johnson, R. G. Geerts, S. J. Palackal, G. R. Hawley, M. B. Welch, Polymer, **36**, 5194 (1994).
53. J. Rodriguez, S. Z. D. Cheng, A. J. Lovinger, E. T. Hsieh, P. Chu, T. W. Johnson, K. G. Honnell, R. G. Geerts, S. J. Palackal, G. R. Hawley, M. B. Welch, Polymer, **35**, 1884 (1994).
54. J. Rodriguez, S. Z. D. Cheng, Z. Bu, J. Macromol. Sci., C: Rev. Macromol. Chem. Phys., **35**, 117 (1995).
55. A. J. Lovinger, B. Lotz, D. Davis, F. J. Padden, Jr., Macromolecules, **26**, 3494 (1993).
56. H. Bu, S. Z. D. Cheng, B. Wunderlich, Makromol. Chem. Rapid Commun., **9**, 75 (1988).
57. R. F. Boyer, in: H. F. Mark, N. M. Bikales, (Eds.), "Encyclopedia of Polymer Science and Technology", Supplement Number 2, Wiley, New York, 1977, p. 745.
58. D. L. Beck, A. A. Hiltz, J. R. Knox, SPE Trans., **3**, 279 (1963).
59. W. P. Slichter, E. R. Mandell, J. Appl. Phys., **29**, 1438 (1958).
60. R. F. Boyer, Plast. Polym., Part I, **41**, 15 (1973); Parts II and III, p. 71; Parts I–IV published as Seventh Swinburne Award Address by the Plastics Institute, 11 Hobart Place, London.
61. S. C. Sharma, L. Mandelkern, F. C. Stehling, J. Polym. Sci., Polym. Lett., **10**, 345 (1972).
62. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, 1051 (1981).
63. J. Grebowicz, S.-F. Lau, B. Wunderlich, J. Polym. Sci., Polym. Symp., **71**, 19 (1984).
64. J. Brandrup, E. H. Immergut, (Eds.). "Polymer Handbook", 2nd ed., Wiley, New York, 1975.
65. G. Crespi and L. Luciana, in: Kirk-Othmer, (Ed.), "Encyclopedia of Chemical Technology", vol. 16, 3rd ed., Wiley, New York, 1981, p. 453.
66. H. P. Frank, "Polypropylene", Gordon and Breach, New York, 1968.
67. M. Matsui, N. Murasaki, "Electrects, Charge Storage and Transport in Dielectrics", M. M. Perlman (Ed.), Electrochem. Soc. Inc., Princeton, 1972, p. 141.
68. R. A. Foss, W. Dannhauser, J. Appl. Polym. Sci., **7**, 1015 (1963).
69. K. Ikezaki, T. Kaneko, T. Sakakibara, Jpn. J. Appl. Phys., **20**(3), 609 (1981).
70. S. Ide, Kobunshi Kagaku, **23**, 865 (1966).
71. D. K. Das Gupta, K. Joyner, J. Phys. D, **9**, 2041 (1976).
72. M. Dole, "Mechanism of Chemical Effects in Irradiated Polymers", in: R. A. V. Raff, K. W. Doak, (Eds.), "Crystalline Olefin Polymers", Interscience, New York, 1965, p. 907.

73. T. Morimoto, T. Mori, S. Enomoto, J. Appl. Polym. Sci., **22**, 1911 (1978).
74. J. L. Koening, "Chemical Microstructure of Polymer Chains", Wiley, New York, 1980.
75. D. W. Van Krevelen, "Properties of Polymers, Correlations with Chemical Structure", P. J. Hoftyzer, Collaborator, Elsevier, Amsterdam, 1972.
76. H. Tadokoro, T. Kitazawa, S. Nozakura, S. Murahashi, Bull. Chem. Soc. Japan, **34**, 1209 (1961).
77. C. E. Schildknecht (with I. Skeist), "Polymerization Processes", vol. XXIX, Wiley, New York, 1977.
78. G. Gramsberg, Kolloid-Z., **175**, 119 (1961).
79. C. Y. Liang, M. R. Lytton, C. J. Boone, J. Polym. Sci., **47**, 139 (1960); *ibid.*, **54**, 523 (1961).
80. J. Boor, Jr., "Ziegler-Natta Catalysts and Polymerization", Academic Press, New York, 1979.
81. M. C. Tobin, J. Phys. Chem., **64**, 216 (1960).
82. M. Peraldo, M. Farina, Chim. Ind. (Milan)., **42**, 1349 (1960).
83. M. P. MacDonald, I. M. Ward, Polymer, **2**, 341 (1961).
84. J. P. Luongo, J. Appl. Polym. Sci., **3**, 302 (1960).
85. E. A. Youngman, J. Boor, Jr., Macromol. Rev., **2**, 33 (1967).
86. R. H. Hughers, Polym. Preprints (Am. Chem. Soc. Polym. Div.), **4** (2), 697 (1963).
87. Y. V. Kissen, Adv. Polym. Sci., **15**, 92 (1974).
88. G. W. Chantry, "Submillimetre Spectroscopy", Academic Press, London, 1971.
89. M. Moller, W. Ritter, H. J. Cantow, Polym. Bull., **4**, 609 (1981).
90. T. Asakura, K. Omaki, S. N. Zhu, R. Chujo, Polym. J., **16** (9), 717 (1984).
91. K. J. Packer, J. M. Pope, R. R. Young, M. E. A. Cudby, J. Polym. Sci., Polym. Phys. Ed., **22**, 589 (1984).
92. F. W. Wehrli, T. Wirthlin, "Interpretation of Carbon-13 NMR Spectra", Heyden and Son Ltd., 1980.
93. I. C. Randall, "Polymer Sequence Determination, Carbon-13 NMR Method", Academic Press, New York, 1977.
94. R. Pino, R. Mulhaupt, in: R. P. Quirk, (Ed.), "Transition Metal Catalyzed Polymerizations, Alkenes and Dienes", vol. 4, Part A, Harwood, New York, Published for MMI Press, 1983.
95. S. Steingiser, S. P. Nemphos, M. Salame, in: Kirk-Othmer, (Ed.), "Encyclopedia of Chemical Technology", vol. 3, 3rd ed., Wiley, New York, 1978.
96. D. Jeschke, H. A. Stuart, Z. Naturforsch, **16a**, 37 (1961).
97. V. Stannett, H. Yasuda, in: K. W. Doak (Ed.), "Crystalline Olefin Polymers", Part II, Wiley, New York, 1964, p. 139, p. 144.
98. Modern Plastics Encyclopedia, McGraw-Hill Book Co., Inc., New York, 1966, p. 530.
99. P. D. Ritchie Ed., "Vinyl and Allied Polymers", vol. 1, Illife, London, 1968, p. 143.
100. J. Horska, J. Stejskal, P. Kratochvil, J. Appl. Polym. Sci., **28**, 3873 (1983).
101. N. M. Bikales (Ed.), "Encyclopedia of Polymer Science and Technology", vol. 12, Wiley, New York, 1985, p. 702.
102. T. H. Matim, Iron Age, **197**, 88 (1966).
103. H. L. Price, SPE J., **24**, 54 (1968).
104. M. P. Groenwege, J. Schnuzer, J. Smidt, C. A. F. Tuijnman, in: R. A. V. Raff, K. W. Doak (Eds.). "Crystalline Olefin Polymers", Part II, Wiley, New York. 1965, p. 798.
105. I. M. Ward, "Mechanical Properties of Solid Polymers", 2nd ed., Wiley, New York, 1983.
106. I. M. Ward (Ed.), "Developments in Oriented Polymers", Elsevier, England, 1982.
107. A. Citerri, I. M. Ward (Eds.), "Ultra-High Modulus Polymers", Elsevier, England, 1979.
108. I. M. Ward (Ed.), "Structure and Properties of Oriented Polymers", Elsevier, England, 1975.
109. International Plastics Selector, Plastics Digest, DATA Business, Englewood, CA 1994.

Physical Constants of Fluoropolymers

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A. INTRODUCTION

Fluoropolymers are divided into two sub-classes: (a) Fluorocarbon polymers, i.e., those that are made from perfluoromonomers, and (b) All the others that are made from monomers or comonomer systems that contain hydrogen, chlorine, or both, along with enough fluorine so that it contributes significantly to the properties and performance of the polymer. At times, the term fluorocarbon or fluorocarbon polymers, are used improperly to include all of the fluorine-containing polymers.

The commercially important fluorocarbon polymers are poly(tetrafluoroethylene) (PTFE), poly(tetrafluoroethylene-*co*-hexafluoropropylene) (FEP), poly[tetrafluoroethylene-*co*-(perfluoroalkylvinyl ether)] (PFA), and amorphous fluoropolymer (AF), typically copolymers of tetrafluoroethylene and fluorinated dioxole. The second group of fluoropolymers includes modified poly(tetrafluoroethylene-*co*-ethylene) (ETFE), poly(vinylidene fluoride) (PVDF) (sometimes referred to as PVF2), and copolymers of vinylidene fluoride, poly(chlorotrifluoroethylene) (PCTFE), poly(chlorotrifluorethylene-*co*-ethylene) (E/CTFE) and poly(vinyl fluoride) PVF.

The wide range of fluorine content in the various fluoropolymers (from 76 down to 39 weight percent fluorine), details of crystallinity, conformation, molecular weight, and other aspects of structure result in ranges of properties so wide that a single table listing all the materials and their properties is very cumbersome to use. Tables on the various materials are separated, therefore, with a short general comment on each material or class, at the beginning of each table. Each table has its own notes and list of references.

B. PHYSICAL CONSTANTS OF POLY(TETRAFLUOROETHYLENE) $-(CF_2CF_2)-$

Polytetrafluoroethylene (PTFE) was the first fluorocarbon polymer to be made and still retains the major part of the total business. The extremely high melt viscosity of PTFE having a molecular weight high enough for most applications precludes use of conventional methods of processing that require molten polymer. As a result, innovative procedures, quite different but not difficult, were devised to prepare useful objects. Some properties of PTFE reflect the unusual methods of processing.

Property*	Units	ASTM or other methods	Value	Remarks	Refs.
Abrasion Resistance			See Wear Factor		
ASTM Standards				See Note 2	
Material					
Granular products		D4894			
Coagulated dispersion products		D4895			
Dispersion		D4441			
Film and sheeting		D2686, D3293, D3308, D3369			
Rods, tubes, and pipe lining		D1675, D1710, D2902, D3295			
Molded basic shapes		D3294,D3297			
ISO Standards					
Plastics – Fluoropolymer dispersions and molding and extrusion materials Part 1: Designation system and basis for specifications		ISO 12086–1			
Plastics – Polytetrafluoroethylene (PTFE) Semi-finished products Part 1: Requirements and designations		ISO 13000–1			
Bond Angle, C–C	Deg.		See Conformation		
F–C–F			108		70
Bond Energies	kJ/mol				
C–F			482		70
C–C			290–335		16
Bond Length, C–F	Å		1.3–1.32		70
Bulk Density of Molding Powder	g/l	D1457	350–900	See Note 3	
Chemical Resistance			See Solvents		
Coefficient of Friction		D1894	$0.224\,W^{0.163}$	W is the load in grams, calculated from Ref. 2	
Coefficient of Thermal Expansion		D696			
Linear expansion	K^{-1}		99×10^{-6}		77
Average values for temperature range indicated					
25 to −190°C			86		38
25 to −150°C			96		
25 to −100°C			112		
25 to −50°C			135		
25–0°C			200		
25–50°C			124		
25–100°C			124		
25–150°C			135		
25–200°C			151		
25–250°C			174		
25–300°C			218		
Cohesive Energy Density			12.5 (est.); see also Solvent Absorption		79
Compressibility (calculated)	MPa^{-1}		28.8×10^{-17}		8
Compressive Strength Properties					
Stress at 1% deformation	MPa	D695	4.2		30
	MPa		$47094 - 231295(\%C) + 0.040(\%C)^2 - 0.0391(\%C)^3$		77
Stress at 1% offset	MPa	D695	7		30
Yield strength	MPa		190.4–0.862 (K)	From 4 to 180 K	
At −251°C	MPa	D695	224		52
At −196°C	MPa	D695	145		52
At 23°C	MPa	D695	11.7		52
At 55°C	MPa	D695	9		52
At 100°C	MPa	D695	4.8		52

* See note 1.

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Conformation					
Energy barrier to rotation					
Gauche–*trans* energy difference	kJ/mol		5.9 ± 1.7	C–C bond angle is at its minimum energy at 162°C	6
trans–trans Energy difference	kJ/mol		4.6 ± 2.9		
Contact Angle					
Advancing	Deg.		116		34
Receding	Deg.		92		
Continuous Use Temperature			See Service Temperature		
Creep					
Compressive creep modulus	Mpa	D695	186	After 100 h at 0.6895 MPa at 23°C	42,59
Flexural creep modulus	MPa		$2814 - 158.5(\%C) + 2.919(\%C)^2 - 0.1638(\%C)^3$	6.895 MPa for 100 h %C from 45 to 90	76
Tensile creep modulus				See Note 1	59
	MPa		61	After 100 h at 6.895 MPa and 23°C	
Critical Shear Rate for Melt Fracture	s^{-1}		10^{-5}	At 380°C	97
Critical Surface Tension	mN/m		18.6		29
Crystal Structure Change			See Figure 1 and Transitions		
Crystalline Content	wt.% C		762.25 – (1524.5/Density)	Calc from Ref.	77
Crystallographic Data					
Phase I			See Figure 1	Hexagonal structure similar to Phase IV, but little or no lateral congruence	
Separation of chain axis	Å		$S = 3.68 \times 10^{-2}\ T + 5888$	T from 40 to 220°C	
Phase II: Triclinic structure	Å		16.9		13,25,61,100
Repeat distance for 13 CF_2 groups					
Separation of chain axis	Å		5.62		13
Phase III: High pressure form					
Planar zig-zag structure					27
Monoclinic space group			$B2/m$		
Density	g/cm^3		2.74 at 12 bar		
	Å		$a = 9.59$		
	Å		$b = 5.05$		
	Å		$c = 2.62$		
	Deg.		$\gamma = 105.5$		
Phase IV: Hexagonal structure					16,61
Repeat distance for 15 CF_2 groups	Å		19.5		
Separation of chain axis	Å		5.55		
Deformation Under Load	% Def.	D621	$12.93 - 0.3303\ (\%C) - 0.002372\ (\%C)^3$	1000 psi for 24 h at 23°C; %C from 50 to 90	76
Density	g/cm^3		Depends on crystalline and void contents	See Note 1	
Completely amorphous (never observed)			2.0 at 23°C	See Note 5	77
Crystalline (from X-ray and IR data)					
Phase II (triclinic at < 19°C)			2.344		100
Phase IV (hexagonal at 20–40°C)			2.302 at 23°C		67
As polymerized			2.280–2.290 at 23°C		77
After melting and recrystallization (void free polymer)			1524.5/(762.25 – %C)	See Note 1	77
Melt (380°C)			1.46		49

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Property	Units	ASTM or other methods	Value	Remarks	Refs.
Depolymerization Rate			See Thermal Degradation		
Diffusion			See Permeability		
Electrical Properties					20
Dielectric constant					
At 60 Hz	ε	D150	$(1+0.238\,D)/(1-0.119\,D)$, D is density		24
At 60 Hz to 2 GHz			2.1		30
Dielectric Strength	kV/mm	D149	The exact value depends on the thickness of the test specimen as well as other test variables		
	MV/m		15.7–19.7		77
	V/mil		400–500		
Dissipation factor		D150		See Note 6	
60 Hz to 2 GHz			$<3\times10^{-4}$		30
Surface arc resistance	s	D495	700	See Note 7	77
Surface resistivity, 100% RH	ohms/sq	D257	3.6×10^{6}		77
Volume resistivity, dry at 50% RH	ohmcm	D257	1019		77
Entanglement MW	g/mol		3.7×10^{3}		49
End Groups				See Note 8	9
Entropy of Fusion			See Melting		
Equation of State			$(P+a)\,(V-b)=B\,(T-c)$ $a=405.2$ bar; $b=0.500\,\mathrm{cm^3/g}$ $c=114$°C; $B=0.310\,\mathrm{cm^3/g/K}$		48,87
Flammability		UL94	VE-O	See Note 9	
Also see limiting oxygen index					
Flexural Modulus		D790			76
At 23°C	MPa		29(°C) – 1127	°C from 40 to 90	
At −251°C	GPa		5.2		52
At −196°C	GPa		5		52
At −101°C	GPa		3.2		52
At 23°C	GPa		0.62		52
At 55°C,	GPa		0.4		52
At 100°C	GPa		0.19		52
Flexural Strength		D790	No break in flexure		
Fuel Value			See Heat of Combustion		
Glass Transition Temperature	°C	DSC	−73		45
Glass-I transition			See Transitions, Alpha		
Glass-II transition			See Transitions, Gamma		
Hardness					
Durometer	Scale value	D2240	$D=42+0.2$(°C)		76
Rockwell	Scale value	D785	$J=114.6-0.433$(°C)		
Scleroscope			$S=133-1.3$(°C)		
Heat Capacity	kJ/kg/K		$0.9324+1.05\times10^{-3}\,T$		69,103
		DSC	Crystalline Amorphous	See Note 10	44
At 10.0°K	J/mol/K		1.228 (1.228)		
100.0 K			19.37 19.37		
200.0 K			34.47 (35.59)		
300.0 K			45.09 51.42		
400.0 K			53.81 59.24		
500.0 K			61.62 66.05		
600.0 K			67.58 69.37		
605.0 K (T_m)			67.88 69.54		
700.0 K			73.300 72.69		
Heat Deflection Temperature	°C	D648			77
At 0.455 MPa (66 psi)			132		
At 1.82 MPa (264 psi)			60		

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Heat of Combustion	MJ/kg		5.1		35
Heat of Crystallization			See Melting – latent heat of fusion		
Heat of Formation	kJ/mol		813		10,22
Heat of Fusion			See Melting – latent heat of fusion		
Ignition Temperature	°C		494		49
Impact Strength			See Toughness		
Intrinsic Viscosity	ml/g		300–2000	Also see Solvents	71,74
Infrared Spectra			See Table 1	See Note 1	17,51,56 91,105
Lamellar Thickness	μm		0.02(%C) – 0.9	Calc. from	26,57,80
Lattice Energy (per recurrent unit; calculated)	kJ/mol		8.5 ± 2.5		8
Limiting Oxygen Index	% Oxygen	D2863	> 95		20
Mechanical Loss		D2236		See Note 11	87
Melt Viscosity					
By capillary rheometer		D2116		See Note 12	
	Pa s		1.62×10^{-14} $M^{3.4}$		75,78
By melt creep,	Pa s		4 to 20×10^{10} at 360°C		1,58
Energy of activation for melt flow	kJ/mol		63–84		1,74
Melting					
Melting peak temperature					
Initial (irreversible)	°C		335–345	See Note 13	52
Second and subsequent (reversible)	°C		327		64
Equilibrium melting temperature	°C		331.7		45,85
For porous, expanded PTFE	°C		385		32,33
Entropy of melting	J/K/mol		477		88
Corrected to constant volume	J/K/mol		33.18		
Volume expansion during melting			29%		
Increase of melting point with pressure	K cm^2/kg		0.095		89
Heat of fusion	kJ/kg	D4591	82	See Note 14	44
Molecular Weight				See Note 15	
			$\log M_w = 27.5345 - 12.1405D$	*D* is Density	18,40,77
			$\log M_w = 10.3068 - 5.1318 \log(4.184H)$	*H* is Heat of Crystallization in J/g	94
Particle Size (Average calculated diameter) μm					
Dispersion polymer		D1457	0.22		47
Coagulated dispersion polymer		D1457	350–650		
Granular polymer		D1457			
ASTM Type I			600–800		
ASTM Type IV			< 100		
Permeability to Gases	m^3(STP) m/s/m^2/Pa ($\times 10^{15}$)				104
Helium			$\log P_{He} = 3.505 - 0.037(\%C)$		76
Carbon dioxide			$\log P_{CO_2} = 2.605 - 0.037(\%C)$		76
Nitrogen			$\log P_{N_2} = 2.44203 - 0.037(\%C)$		76
Hydrogen			7.35		60
Energy of activation for diffusion	kJ/mol		27.6		60
Energy of activation for permeability	kJ/mol		19.7		60
Phase Diagram			See Fig. 1		27,77,102
Processing Temperature	°C		350–400		
PV Limit	Pa m/s		50–250		46
Radiation, Effects of degradation in air	Radicals/10^6 C atoms		$3.7 \times 10^{-5} D\,0.89$	Derived from Ref.; *D* is dose in rads	37
Recovery after deformation	%	D1147	% Recovery = 111 – 0.87(%C); % C from 40 to 90	1/4 inch penetrator; 5 lb preload; 250 lb major load	96

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Property	Units	ASTM or other methods	Value	Remarks	Refs.
Refractive Index	n_D^{25}	D542	1.376		7
Resistivity			See Electrical Properties		
Service Temperature			−269 to 260°C	See Note 16	21,95,96
Solid Viscosity	Pa s		$10^4 - 10^6$	At 300°C, shear rate $0.1-10\,s^{-1}$	54,81
Solution Viscosity			See Intrinsic Viscosity		
Solvent Absorption				See Note 17	
			% wt. gain = 24.85 − 2.456 HSP	HSP is Hildebrand solubility parameter	66
Perchloroethylene	% wt. increase		2	1000 h in boiling liquid, 60% cryst.	
Methylene chloride	% wt. increase		0.7	1000 h in boiling liquid, 60% cryst.	
Solvents				See Note 18	
Specific Gravity			See Density		
Specific Heat			See Heat Capacity		
Speed of Sound		1 MHz, 25°C			
Longitudinal	m/sec		1410		49
Shear	m/sec		730		49
Standard Specific Gravity (Relative molecular weight)		D1457	Also see Molecular Weight	See Note 15	18
Surface Tension at 293 K	mN/m		25.6		49
Tensile Strength Properties		D638			
Modulus					
At −251°C	GPa		4.1		52
At −196°C	GPa		3.4		52
At −129°C	GPa		2.5		52
At −73°C	GPa		1.8		52
At 23°C	MPa		340		52
At 100°C	MPa		69		52
Yield Stress,					
At −251°C	MPa		131		52
At −196°C	MPa		110		52
At −129°C	MPa		79		52
At −73°C	MPa		53		52
At −55°C	MPa		26		52
At 0°C	MPa		12		52
At 23°C	MPa		10		52
At 158°C	MPa		5.5		52
At 250°C	MPa		3.4		52
Tensile strength					
At 23°C	MPa		$270 - 0.39(\%C) - 99.3\,D$	D from 2.170 to 2.280 Mg/m^3; %C from 30 to 90	
At 23°C,	MPa		17–34	See Note 5	52
Elongation at break					
At −251°C	%		2		52
At −196°C	%		6		52
At −129°C	%		90		52
At −73°C	%		160		52
At 23°C	%		200–400	See Note 5	52
Plateau modulus, 380°C	MPa		1.7		49
Thermal Conductivity	W/m/K	C177	$K = 4.86 \times 10^{-4}\,T + 0.253$		69
25°C	W/m/K		0.25		49
72°C			0.34		49
Thermal Degradation					
Enthalpy of depolymerization			$\Delta H = 156000 - 27.9\,(T-360)$ J/mol		69
Rate of depolymerization			See Ref. for details		73
Kinetic chain length (CL)			$\log CL = 8320/T - 9.958$	T in kelvin	28,98

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Toughness					
Izod impact strength, notched	J/m	D256			21
−57°C			107		52
23°C			187		52
77°C			320		52
Tensile impact strength	kPa/m^2	D1822	TIS = 3 379 − 45.1 (%C)	See Note 1	75
Transitions	°C			See Note 19	23
Alpha (Glass I)			126	At 1 Hz	53
Beta I (Crystalline transition, crystal disordering relaxation)			19		
Change with pressure	K/bar		0.0128		
Latent heat	kJ/kg		13.4		
Volume expansion	l/kg		0.0058		
Kinetics of the transition				See Note 20	
Beta II			30		57
Beta I + Beta II					
Entropy change, overall (19 + 30)	kJ/kg/K		0.0452		
Due to volume change (corrected to constant volume)	kJ/kg/K		0.028		
Gamma (Glass II)			−80 at 1 Hz		53
Change of frequency (in Hz) of loss peak with temp.					23
Alpha (Glass I)			$\log F = 48.3759 - 368$ kJ/(2.303RT)		
Beta (crystalline relaxation)			$\log F = 24.97 - 142$ kJ/(2.303RT)		
Gamma (glass II)			$\log F = 22.37 - 75$ kJ/(2.303 RT)		
Effect of temperature on transition pressure					50
Transition between phases IV and I	Pa		$P = (-91 - 57.8\,T + 1.99\,T^2) \times 10^5$		
Transition between phases II and IV	Pa		$P = (-820 + 26.9\,T + 0.68\,T^2) \times 10^5$		
Triple Point			See Figure 1, Crystallographic Data, and Transitions		
Unit Cell			See Crystallographic Data		
Viscosity			See Melt Viscosity See Solid Viscosity See Solution Viscosity		
Water Absorption	%	D570	0		77
Wear Factor	Pa^{-1}		3000×10^{-17}	See Note 21	46
Weatherability			Excellent	See Note 22	

NOTES

1. Many of the properties of PTFE are not constant but change systematically with crytalline content, molecular weight, void content, temperature, pressure, or some other independent variable. Where available, these inter-relationships are shown even though they are usually empirical rather than fundamental. Throughout the table, %C indicates percent crystallinity by weight.
2. American Society for Testing and Materials (ASTM) standards on the fluoropolymers provide a convenient source for information on details of the nature of a particular fluoropolymer, typical properties, methods of testing, and the like. For this reason, pertinent Standard numbers are included in the tables for each polymer. The fluoropolymer standards are the responsibility of the Fluoropolymers Section of the Materials Subcommittee of the Technical Committee on Plastics. The numerical designation is D-20.15.12. More recently, the International Organization for Standardization (ISO) has issued materials standards which incorporate various aspects of the ASTM standards.
3. The exact value depends on the particular type or grade of polymer that is being used.
4. Equations based on time–temperature superposition are given in Ref. 42 for PTFE with a crystalline content of approximately 58%. These relationships permit calculation of tensile creep modulus at any stress, time, or temperature in the temperature range of 35–100°C and stress levels below the elastic limit. Ref. 41 provides similar data over a wider range of conditions.

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5. Reference 83 gives a value of 2.060 for the amorphous density of PTFE. Internal inconsistencies in this paper, however, need clarification before this writer can accept the value given.
6. Precise measurements of dielectric loss (Ref. 71) show peaks that correspond to the gamma transition. The measured values of tan delta, however, were all less than 300×10^{-6} with some as low as 20×10^{-6}.
7. No carbon track is formed.
8. Infrared methods show presence of carboxylic acid monomer and dimer, perfluorovinyl, and acid fluoride end groups in low molecular weight products (Ref. 9). The methods currently available are not sufficiently sensitive for quantitative work in polymers of commercially useful molecular weights, but the acid monomer and dimer ends have been found in commercial materials using FTIR procedures (Ref. 93).
9. This numerical rating for flame spread is not intended to reflect hazards presented by this or any other material under actual fire conditions.
10. The values given are selected from extensive tables and plots of data in Ref. 44.
11. The mechanical loss factor is related to the transition behavior. The value depends on temperature and frequency of the dynamic mechanical stress, values of the logarithmic decrement up to 0.55 have been observed (see Ref. 53).
12. This relationship is applicable only to PTFE, with molecular weights well below that required for the applications of plastics (see Ref. 78).
13. The temperature observed for the first (irreversible) melting point decreases with rate of heating. It also differs with the type of polymer. Evidence in Refs. 39 and 72 suggests that details of DSC scan for a first melting reflect aspects of molecular weight and molecular weight distribution.
14. Inconsistencies in the results from various laboratories for the heat of fusion of PTFE and FEP have not been resolved. Some aspects are discussed in Refs. 86, 89, and 90.
15. Standard Specific Gravity (SSG), described in ASTM D4894, D4895 and ISO 12086-1 (see Ref. 18 and 77), is a useful inverse measure of relative molecular weight for homopolymers of TFE. Many commercial polymers today, however, contain very small amounts of modifiers that affect the rate and extent of crystallization. With these materials, therefore, SSG is meaningful as a measure of relative molecular weight only within a given family of PTFE resins. Estimating actual values of the molecular weight of the highest molecular weight commercial polymers has, to date, involved assumptions and extrapolation. These relationships, therefore, are, at best, a first approximation. In addition, some references, such as Refs. 11 and 40, give relationships for molecular weight as a function of SSG with little or no documented basis for the equation.
16. The recommended maximum service temperature is 260°C (Ref. 21). There are, however, examples of satisfactory use at higher temperatures. Gaskets in totally enclosed systems, for instance, are used repeatedly at 500°C and 15 000 psi (103 MPa) (see Ref. 19). PTFE also performs well at low temperatures. In compressive strength tests at 4 K, for example, the polymer has been shown to have ductile rather than brittle failure (Ref. 95).
17. Weight gain is noted with some chlorinated and fluorinated liquids that contain no hydrogen (Ref. 82). The weight increase decreases with increasing crystallinity (Ref. 78).
18. PTFE is inert to essentially all common chemicals (see Refs. 64 and 65). It is, however, attacked by molten alkali metals, elemental fluorine, and pure oxygen at elevated temperatures, it can also react under some conditions with alkaline earth oxides and finely divided metals such as aluminum or magnesium at elevated temperature. Sorption of solvents is mentioned in Note 13. Solutions containing 0.1–2% PTFE were obtained in perfluorokerosene $C_{21}F_{44}$(FCX-412) in the temperature range 290–310°C (see Refs. 71 and 74).
19. Clear evidence for a crystalline transition at about 90°C is presented in a series of papers by Araki, Refs. 3–5. He also presents evidence for doublets in the transitions Ref. 4, and points out that the alpha and gamma transitions follow the so called 2/3 (Trouton) rule when compared with the first order crystalline transitions at 19 and 327°C, Ref. 3. See also evidence for a doublet in the freezing-point plot shown in Ref. 65.
20. Data on the kinetics of the transitions not amenable to presentation in tabular form are available in Ref. 57.
21. Incorporation of finely divided fillers decreases wear factor by as much as 500 to 1000 fold (Ref. 46).
22. No change in appearance or properties has been observed after more than twenty years exposure in Florida (see Ref. 21).

INFRARED ABSORPTION SPECTRUM OF POLY(TETRAFLUOROETHYLENE)

(cm^{-1})	(μm)	Intensity[a]
3670	2.72	VW
3570	2.80	VW
3450	2.90	VW
3090	3.24	VW
2925	3.42	VW
2620	3.82	VW
2590	3.86	VW

(cm^{-1})	(μm)	Intensity[a]
2530	3.95	VW
2450	4.08	VW
2367	4.22	M
2300	4.35	VW
1974	5.07	VW
1935	5.17	W
1883	5.31	W
1859	5.38	W
1792	5.58	M
1703	5.87	W
1545	6.47	M
1451	6.90	M
1420	7.04	M
1242	8.05	VS
1213	8.24	VS
1152	8.68	VS
932	10.73	M
850	11.76	Amorphous
778	12.85	Amorphous
738	13.55	Amorphous
718	13.98	Amorphous
703	14.22	Amorphous
638	15.67	S
553	18.08	S
553	18.08	S
516	19.38	VS
384	26.04	Amorphous

[a] VS, very strong; S, strong; M, medium; W, weak; VW, very weak.

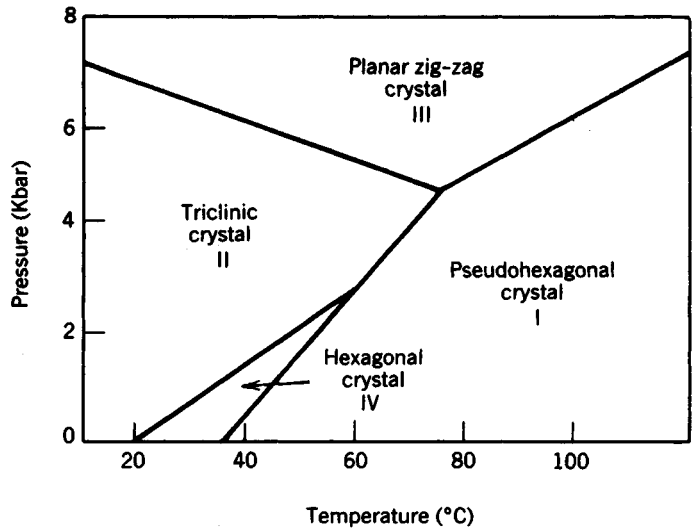

Figure 1. Phase diagram for poly(tetrafluoroethylene).

REFERENCES

1. G. Ajroldi, C. Garbuglio, M. Ragazzini, J. Appl. Polym. Sci., **14**, 79 (1970).
2. A. J. G. Allan, Lubrication Eng., **14**, 211 (1958).
3. Y. Araki, J. Appl. Polym. Sci., **9**, 421 (1965).
4. Y. Araki, J. Appl. Polym. Sci., **9**, 3575 (1965).
5. Y. Araki, J. Appl. Polym. Sci., **11**, 953 (1967).
6. T. W. Bates, W. H. Stockmayer, Macromolecules, **1**, 17 (1968).
7. F. W. Billmeyer, Jr., J. Appl. Phys., **18**, 431 (1947).
8. W. Brandt, J. Chem. Phys., **26**, 262 (1957).
9. M. I. Bro, C. A. Sperati, J. Polym. Sci., **38**, 289 (1959).
10. W. M. D. Bryant, J. Polym. Sci., **56**, 277 (1962).
11. A. J. Cardinal, W. L. Edens, and J. W. Van Dyk, US Patent 3, 142, 665, July 28, 1964 to Du Pont.
12. E. S. Clark, L. T. Muus, Z. Krist., **117** (2/3), 108 (1962).
13. E. S. Clark, L. T. Muus, Z. Krist., **117** (2/3), 119 (1962).
14. E. S. Clark, L. T. Muus, J. Macromol. Sci. B, **1**, 795 (1967).
15. E. S. Clark, Results cited previously in "Polymer Handbook", 2nd (ed.), page V-36.
16. T. L. Cottrell, "The Strengths of Chemical Bonds", 2nd ed., Butterworths, Washington, DC, 1958.
17. D. P. Cutler, P. J. Hendra, R. R. Rahalkar, M. E. A. Cudby, Polymer, **22**, 726–30 (1981).
18. R. C. Doban, A. C. Knight, J. H. Peterson, C. A. Sperati, Meeting Am. Chem. Soc., Atlantic City, Sept. 1956.
19. R. C. Doban, C. A. Sperati, B. W. Sandt, Soc. Plastics Eng. J., **11**, 17 (1955).
20. DuPont Polymer Products Department,
21. DuPont Polymer Products Department, Technical Services Laboratory, Wilmington, DE, Personal Communication.
22. H. C. Duus, Ind. Eng. Chem., **47**, 1445 (1955).
23. R. K. Eby, K. M. Sinnott, J. Appl. Phys., **32**, 1765 (1961).
24. P. Ehrlich, J. Res. Natl. Bur. Std., **51**, 185 (1953).
25. B. L. Farmer, R. K. Eby, Polymer, **22**, 1487 (1981).
26. S. Fischer, N. Brown, J. Appl. Phys., **44**, 4322 (1973).
27. H. D. Flack, J. Polym. Sci. A-2, **10**, 1799 (1972).
28. R. E. Florin, M. S. Parker, L. A. Wall, J. Res. Natl. Bur. Std. A, **70** (2), 115 (1966).
29. H. W. Fox, W. A. Zisman, J. Colloid Sci., **5**, 514 (1950).
30. S. V. Gangal, in Kirk-Othmer: (with 104 references) "Encyclopedia of Chemical Technology", vol. no. 11, 3rd. ed., (1980) p. 24, Wiley, New York, 1980.

31. R. J. Good, J. A. Kvikstad, W. D. Bailey, J. Colloid Interface Sci., **35**, 314 (1971).
32. R. W. Gore, US Patent 3, 953, 566, 4/27/76; US Patent 3, 962, 153, 6/8/76; and US Patent 4, 187, 390, 2/5/80 to W. L. Gore and Associates.
33. R. W. Gore, File history for prosecution of US Patent 4, 187, 390, US Patent office.
34. Imperial Chemical Industries, Ltd.
35. R. L. Johnson, in: "Encyclopedia of Polymer Science and Technology", Supp. no. 1 (1976) p. 269.
36. E. D. Jones, G. P. Koo, J. L. O'Toole, Mod. Plastics, **44** (11), 137 (1967).
37. H. S. Judekis, S. Siegel, IEEE Trans. Nuclear Sci., **14**, 237 (1967).
38. R. E. Kirby, J. Res. Natl. Bur. Std., **57**, 91 (1956).
39. S. Koizumi, S. Ichiba, T. Kadowaki, and K. Yamamoto, US Patent 4, 159, 370, June 26, 1979; RE 31, 341, Aug. 9, 1983 to Daikin Kogyo Co.
40. Y. Kometani, S. Koizumi, S. Fumoto, T. Nakajima, US Patent 3, 462, 401 (August 19, 1969).
41. G. P. Koo, "Cold Drawing Behavior of Polytetrafluoroethylene" Thesis from Stevens Institute of Technology, Sc. D., University Microfilms Inc., Ann Arbor Michigan, 69–21, 294 (1969).
42. G. P. Koo, E. D. Jones, J. M. Riddell, J. L. O'Toole, Soc. Plastics Eng. J., **21**, 1100 (1965).
43. G. P. Koo, J. M. Riddell, J. L. O'Toole, Polym. Eng. Sci., **7**, 182 (1967).
44. S-F. Lau, H. Suzuki, and B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **22**, 379 (1984).
45. S-F. Lau, J. P. Wesson, B. Wunderlich, Macromolecules, **17**, 1102 (1984).
46. R. B. Lewis, J. Eng. Ind., **1**, 1 (1966).
47. J. F. Lontz, W. B. Happoldt, Jr., Ind. Eng. Chem., **44**, 1800 (1952).
48. J. M. Lupton, Meeting Am. Chem. Soc., Chicago, Sept., 1958.
49. J. E. Mark (Ed), "Physical Properties of Polymers Handbook", AIP Press, Woodbury, NY, 1996.
50. C. M. Martin, R. K. Eby, J. Res. Natl. Bur. Std. A, **72**, 467 (1968).
51. G. Masetti, F. Cabassi, G. Morelli, G. Zerbi, Macromolecules, **6**, 700 (1973).
52. D. I. McCane (with 118 references), "Encyclopedia of Polymer Science and Technology", vol. no. 13, Wiley, New York, 1970, p. 623.
53. N. G. McCrum, B. E. Read, G. Williams, "Anelastic and Dielectric Effects in Polymeric Solids", Wiley, New York, 1967.
54. R. L. McGee, J. R. Collier, Polym. Eng. and Sci., **26**, No. 3, 239 (1986).
55. K. Morokuma, J. Chem. Phys., **54**, 962 (1971).
56. R. E. Moynihan, J. Am. Chem. Soc., **81**, 1045 (1959).
57. R. Natarajan, T. Davidson, J. Polym. Sci., Polym. Phys. Ed., **10**, 2209 (1972).
58. A. Nishioka, M. Watanabe, J. Polym. Sci., **24**, 298 (1957).
59. J. L. O'Toole, "Modern Plastics Encyclopedia" **45**, 48 (1968).
60. R. A. Pasternak, M. V. Christensen, J. Heller, Macromolecules, **3**, 366 (1970).
61. R. H. H. Pierce, E. S. Clark, J. F. Whitney, W. M. D. Bryant, Meeting Am. Chem. Soc., Atlantic City, Sept., 1956.
62. C. M. Pooley, D. Tabor, Proc. Roy. Soxc. (London), Ser. A, **329**, 251 (1972).
63. F. J. Rahl, M. A. Evanco, R. J. Fredericks, and A. C. Reimschuessel, J. Polym. Sci. A-2, **10**, 1337 (1972).
64. J. C. Reed, E. J. McMahon, J. R. Perkins, Insulation (Libertyville), **10** (5), 35 (1964).
65. M. M. Renfrew, E. E. Lewis, Ind. Eng. Chem., **38**, 870 (1946).
66. E. Rydel, Thesis Univ. of Strassbourg, March 22, 1965.
67. A. L. Ryland, J. Chem. Educ., **35**, 80 (1958).
68. T. Seguchi. T. Suwa, N. Tamura, and M. Takehisa, J. Polym. Sci., Polym. Phys. Ed., **12**, 2567 (1974).
69. P. H. Settlage, J. C. Siegle, Phys. Chem. Aerody. Space Flight, Proc. Conf., Philadelphia, 1959, **73** (1961), Pergamon, New York.
70. W. A. Sheppard, C. M. Sharts, "Organic Fluorine Chemistry", W. A. Benjamin, New York, 1969.
71. S. Sherratt, Kirk-Othmer, (with 99 references) "Encyclopedia of Chemical Technology", 2nd ed., **9**, 805 (1966).
72. T. Shimizu, Private Communication, Paper Presented at the Symposium on New Fluorine-Containing Materials, Tokyo Institute of Technology, 1985.
73. J. C. Siegle, L. T. Muus, Tung Po Lin, H. A. Larsen, J. Polym. Sci. A, **2**, 391 (1964).
74. C. A. Sperati, Gordon Research Polymers Conference, July 7, (1955).
75. C. A. Sperati, Results cited previously in "Polymer Handbook", 2nd Edition, page V–36.
76. C. A. Sperati, J. L. McPherson, Meeting Am. Chem. Soc., Atlantic City, Sept. 1956.
77. C. A. Sperati, H. W. Starkweather, Jr. (with 114 references), Adv. Polym. Sci., **2**, 465 (1961).
78. C. A. Sperati, "Polytetrafluoroethylene: History of its Development and Some Recent Advances" (with 67 references) in: Seymour and Kirshenbaum (Eds.), "History of High Performance Polymers", Marcel Dekker, New York, 1986.
79. H. W. Starkweather, Jr., Results cited previously in "Polymer Handbook", 2nd (ed.), page V–36.
80. H. W. Starkweather, Jr., Soc. Plastics Eng. Trans., **3**, 1 (1963).
81. H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **17**, 73 (1979).
82. H. W. Starkweather, Jr., Macromolecules, **10**, 1161 (1977).
83. H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **20**, 2159 (1982).
84. H. W. Starkweather, Jr., Macromolecules, **17**, 1178 (1984).
85. H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **23**, 1177 (1985).
86. H. W. Starkweather, Jr., Macromolecules, **19**, 1131 (1986).
87. H. W. Starkweather, Jr., Macromolecules, **19**, 2541 (1986).

88. H. W. Starkweather, Jr., R. H. Boyd, J. Phys. Chem., **64**, 410 (1960).
89. H. W. Starkweather, Jr., P. Zoller, G. A. Jones, A. J. Vega, J. Polym. Sci. Poly. Phys. Ed., **20**, 751 (1982).
90. H. W. Starkweather, Jr., P. Zoller, G. A. Jones, J. Polym. Sci., Poly. Phys. Ed., **22**, 1431 (1984).
91. H. W. Starkweather, R. C. Ferguson, D. B. Chase, J. M. Minor, Macromolecules, **18**, 1684 (1985).
92. H. W. Starkweather, Jr., P. Zoller, G. A. Jones, J. Polym. Sci., Poly. Phys. Ed., **22**, 1431 (1984).
93. J. N.-P. Sun, Report for MS Degree, Ohio University, 1951. See also J. N.-P. Sun, P. R. Griffiths, and C. A. Sperati, Spectrochemica Acta., **39A**, 587 (1983).
94. T. Suwa, M. Takehisa, S. Machi, J. Polym. Sci., **17**, 3253 (1973).
95. C. A. Swenson, Rev. Sci., Instr., **25**, 834 (1954).
96. P. E. Thomas, J. F. Lontz, C. A. Sperati, J. L. McPherson, Soc. Plastics Eng. J., **11**, 17 (1955).
97. J. P. Tordella, Trans. Soc. Rheo., **7**, 231 (1963).
98. L. A. Wall (Ed.), "Fluoropolymers", Wiley Interscience, 1972.
99. J. J. Weeks, I. C. Sanchez, R. K. Eby, C. I. Poser, Polymer, **21**, 325 (1980).
100. J. J. Weeks, E. S. Clark, R. K. Eby, Polymer, **22**, 1480 (1981).
101. J. J. Weeks, R. K. Eby, E. S. Clark, Polymer, **22**, 1496 (1981).
102. C. E. Weir, J. Res. Natl. Bur. Std., **50**, 95 (1953).
103. B. Wunderlich, H. Baur, Adv. Polym. Sci., **7**, 331 (1970).
104. N. Yi-Yan, R. N. Felder, W. J. Koros, J. of Appl. Polym. Sci., **25**, 1755 (1980).
105. G. Zerbi, M. Sacci, Macromolecules, **6**, 692 (1973).

C. PHYSICAL CONSTANTS OF MELT PROCESSIBLE FLUOROCARBON POLYMERS PFA AND FEP

PFA: $-\{(CF_2CF_2)_n(CF_2CF)\}_m-$ (OR_F)

FEP: $-\{(CF_2CF_2)_n(CF_2CF)\}_m-$ (CF_3)

Unlike poly(tetrafluoroethylene), copolymers of tetrafluoroethylene with perfluoro comonomers are made as relatively low molecular weight (i.e., low melt viscosity) products that can be processed by conventional melt procedures and still have properties in the solid state that are similar or superior to those of poly(tetrafluoroethylene). Both PFA, poly[tetrafluoroethylene-*co*-perfluoro(alkoxy vinyl ether)], and FEP, poly(tetrafluoroethylene-*co*-hexafluoropropylene), are included in this table. Various perfluoro(alkoxy vinyl ethers) can be used for PFA. It has been reported that one often used is perfluoro(propylvinyl ether) (14).

Data available for PFA and FEP are not as extensive as for PTFE. In many respects, however, the properties of solid FEP and, especially, PFA, are very similar to those of PTFE. The values given in the table for PTFE, therefore, can often provide useful guidance for the properties of nonmolten PFA or FEP for which no values are included in this table.

Property	Units	ASTM or other methods	PFA Value	PFA Remarks	PFA Refs.	FEP Value	FEP Remarks	FEP Refs.
Abrasion Resistance						See Taber Abrasion		
ASTM or ISO Standards				See Note 1			See Note 1	
Material								
Molding and extrusion powders		D3307, ISO12086-1				D2116		
Film and sheeting						D3368		
Rods, tubes, and pipe lining		F781				D3296,F546		
Bond Energies								
C–C adjoining pendant group	kJ/mol		335	See Note 2		167	See Note 2	
Bulk Density of molding powder	g/l		1250	See Note 3		1230–1250	See Note 3	7
Chemical Resistance			Excellent	See Note 4	15	Excellent	See Note 4	15
Coefficient of Friction		D1894	0.25		7	0.27–0.67		5,7,15
Coefficient of Thermal Expansion		D696						
Linear expansion	K^{-1}							
Average values for temperature indicated								
<23						5.7×10^{-5}		9
>23						9.3×10^{-5}		9
20–100			12×10^{-5}		11			
100–150			17×10^{-5}		11			
150–210			20×10^{-5}		11			
Cohesive Energy Density						36		24
Comonomer Content	wt.%	IR				10–15	See Note 4	11,10
Compressive Strength Properties								
Modulus	MPa		23.7–25.8	See Note 3	7			
Strength	MPa	D695						

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Property	Units	ASTM or other methods	PFA Value	PFA Remarks	PFA Refs.	FEP Value	FEP Remarks	FEP Refs.
−251°C						251		9
−196°C			414		11			
−160°C						207		9
23°C						15		9
55°C						11		9
100°C						3.4		9
Continuous Use Temperature			See Service Temperature					
Creep								
Tensile creep modulus	MPa							
25°C			270		11			
250°C			41		11			
Critical Shear Rate for Melt Fracture	s^{-1}		3–50	See Note 3	6	2–20	See Note 3	6
Critical Surface Tension						17.8–18.6	See Note 3	15
Deformation Under Load	% Def.	D621	0.36–0.41	See Note 2	7	0.5		8
Density	$g/cm^3 = Mg/m^3$		2.12–2.17		11	2.12–2.17		9
Depolymerization Rate			See Thermal Degradation					
Electrical Properties								
Dielectric constant	ε	D150						
At 60 Hz to 2 GHz						2.1		9
At 1 kHz to 1 GHz			2.06		11			
At 240 GHz			2.049		11			
Dielectric strength	kV/mm	D149	78.8 (on 0.25 mm film)		11	13–100	See Note 6	15
	V/mil		2000		11	500–4000		
Dissipation factor		D150						
60 Hz to 1 MHz						<0.0001		
At 1 kHz			0.000027		11	<0.0001		
At 100 kHz						0.00025		15
At 1 MHz			0.00008		11			
At 10 MHz						0.0005		15
At 1 GHz			0.00115		11	0.0011		
At 3 GHz						0.0012		15
At 10 GHz						0.0007		15
At 240 GHz			0.00124		11			
Surface arc resistance	s	D495	>300	See Note 7		>300	See Note 7	9
Surface resistivity, 100% RH	ohms/sq	D257	$>10^{17}$		10	10^{16}		9
Volume resistivity, dry								
At 50% RH	ohm cm	D257	10^{14}		10	$>10^{18}$		9
Flammability								
Also see limiting oxygen index		UL94	VE-O	See Note 8	11	VE-O	See Note 8	6
Flexural Modulus		D790						
At −268°C	GPa					5.31		15
At −196°C	GPa					4.69		
At −101°C	GPa					3.24		
At 23°C	MPa		655–690		11	655		
At 55°C	MPa					345		
At 100°C	MPa					110		
At 200°C	MPa		55–69		11	41.4		
Flex Life	Flexes	MIT	50000–500000	See Note 9	11	5000–30000	See Note 9	2
Flexural Strength		D790	No break			No break		
Fuel Value			See Heat of Combustion					
Hardness								
Durometer	Scale value	D2240	D63–65		10	D55–66		10,9
Rockwell	Scale value	D785	J60			R25		
Heat Capacity			See Specific Heat					

Property	Units	ASTM or other methods	PFA Value	PFA Remarks	PFA Refs.	FEP Value	FEP Remarks	FEP Refs.
Heat Deflection Temperature		D 648						
At 0.455 MPa (66 psi)	°C		63–80	See Note 3	7	70–77		9,7
At 1.82 MPa (264 psi)	°C		48–50	See Note 3	7	51–57		9,7
Heat of Combustion	MJ/kg		5.4		11			
Heat of Fusion				See Melting				
Impact Strength				See Toughness				
Lamellar Thickness	nm					33–53	See Note 3	3
Limiting Oxygen Index	% Oxygen	D2863	>95		11	95		
Melt Viscosity								
By capillary rheometer	kPa s	D2116				$1.62 \times 10^{-13} M^{-3.4}$		18
			4–27	See Note 3	6	8–50	See Note 3	6
Melt flow rate	dg/min	D3307	1–18	See Note 3	11			
		D2116				0.8–18	See Note 3	6,1,12
Energy of activation for melt flow	kJ/mol		43.1	See Note 10	6	46	See Note 10	6
Melting								
Melting point							See Note 11	3
	°C		302–306	See Note 3	11	253–282	See Note 3	9
Volume expansion during melting	%					8		9
Increase of melting point with pressure	K/bar					0.725 to 1.74°C per MPa	See Note 12	9
Latent heat of fusion	kJ/kg	DSC	21–26		7	24.3	See Note 13	20
Molecular Weight			Also see Melt Viscosity					
			200000–450000		17	250000–600000	See Note 3	17
Particle Size (average diameter)	µm							
Dispersion Polymer			0.1			0.1		
Permeability to Gases	ng cm/m^2-s							
Carbon dioxide						273		9
Oxygen						637		9
Nitrogen						91		9
Phase Diagram								22
Processing Temperature	°C		370–395		6	320–380		6
PV Limit	Pam-s							
	psi ft/min		5000		11			
Radiation, Effects of degradation in air			Maintains Tensile properties better than does PTFE		10	10 to 100 times better than PTFE	See Note 14	15
Refractive Index	n_D^{25}	D542	1.35		11	1.344		11
Resistivity			See Electrical Properties					
Service Temperature	°C		−200 to 260		11,7	−250 to 206		9,7
Solvents			None	See Note 4	10	None	See Note 4	15
Specific Gravity			See Density					
Specific Heat	kJ/kg/K	DSC						
20°C						1.09		9
25°C								
100°C						1.17		9
150°C								
260°C						1.3		9
Taber Abrasion	g/MHz	CS-17 wheel; 100 g load				7.5		9
Tensile Strength Properties		D638						
Modulus	MPa							
At 23°C						4	See Note 3	9
Yield stress	MPa							
At 23°C			13–15	See Note 3	11	12	See Note 3	9
At 250°C			3.5–4	See Note 3	11			

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Property	Units	ASTM or other methods	PFA Value	PFA Remarks	PFA Refs.	FEP Value	FEP Remarks	FEP Refs.
Tensile strength	MPa							
At −196°C			129		11			
At 23°C			28–31	See Note 3	11	14.5–21	See Note 3	9
At 250°C			12–14	See Note 15	11			
Elongation at break	%							
At −196°C			8		11			
At 23°C			260–300		11	240–300	See Note 3	9
At 250°C			480–500		11			
Thermal Conductivity	W/m/K	Cenco Fitch						
−253°C						1.4		9
−129 to 182°C						2.4		9
23°C		C177				0.209		7
Thermal Degradation								
Enthalpy of depolymerization			Similar to PTFE					
Rate of depolymerization	%/hr					$\log(\%/h) = 10.08 - 131\,kJ/(2.303RT)$		
							See Note 16	9
Toughness								
Izod impact strength, notched		D256						
−196°C	J/m		64		11			
−54°C						2.9		15
23°C			No break		11	No break		9
Transitions	°C							
Alpha (Glass I)						70–126, 0–15 mol% HFP		
							See Note 3	16
Beta						−70 to −10	See Note 3	16
Gamma (Glass II)						−90		16
Activation energy for change of frequency (in Hz) of loss peak with temp								
Alpha (Glass I)						80		8
Beta (crystalline relaxation)								
Gamma (Glass II)						18		8
Effect of temperature on the frequency of peak loss								
Gamma (Glass II)						$\log F = 22.43 - 76\,kJ/(2.303\,RT)$		
							calc from data in Ref.	8
Unit Cell						Phase II is similar to PTFE		23
Viscosity			See Melt Viscosity					
Water Absorption	% incr. in wt.	D570	0.004		7	<0.1		9
Wear Factor	$in^3 min/ft/lb/hr$		700–1800		10			
Weatherability			Excellent		10	Excellent		9

NOTES

1. Comments on ASTM and ISO standards for fluoropolymers are in Note 2 for the table on PTFE.
2. The values are estimated based on high-spot work on degradation of the polymers.
3. The exact value depends on the type or grade of polymer that is being used.
4. PFA and FEP are insoluble in all common solvents and are highly resistant to essentially all common chemicals (see Ref. 1, 2, and 15). They are however, attacked by molten alkali metals, elemental fluorine, and pure oxygen at elevated temperatures. Also, they can react, under some conditions, with alkaline earth oxides and finely divided metals such as aluminum or magnesium at elevated temperature. Solutions containing 0.1 to 2% FEP were obtained in perfluorokerosene

$C_{21}F_{44}$ (FCX-412) in the temperature range 290–310°C.

5. Determined by 3.0 × IR Ratio. The IR ratio is the absorbance at 980 cm^{-1} (10.2 μm)/absorbance at 2350 cm^{-1} (4.23 μm) (see Ref. 1, 2).
6. Thickness is from 0.025 to 2.5 mm. The value decreases with increasing thickness.
7. No carbon track is formed (see Ref. 7).
8. This numerical rating for flame spread is not intended to reflect hazards presented by this or any other material under actual fire conditions.
9. The MIT flex test is run with a thickness of 0.254 mm.
10. Calculated from data in Ref. 6. The melt flow rate was determined using standard conditions at 372°C.
11. Ref. 3 gives melting temperature, T_m, as a function of wt.% comonomer, X, and lamellar thickness, L. The equation is

$$1/T_m = 1.622 \times 10^{-3} + 2.52 \times 10^{-3}X(1 + X) + (1.92 \times 10^{-10}/L)$$

12. The rate of increase of the melting point decreases with increasing pressure.
13. Inconsistencies in the results from various laboratories on heat of fusion of PTFE and FEP have not yet been resolved. This value is for a sample of FEP with a density of 2.148 Mg/m^3.
14. Moderate crosslinking can occur to offset degradation (see Ref. 15).
15. Mechanical properties of PFA at elevated temperature but below its melting point generally are superior to those of PTFE at the same temperature.
16. Calculated from data in Ref. 9.

REFERENCES

1. Y. Akatsuka, M. Noshiro, Y. Jitsugiri, Reports Res. Lab. Ashai Glass Co., Ltd., **26** [2] 111 (1976); also, Paper Po-8 at the 8th International Symposium on Fluorine Chemistry, Kyoto, Japan, Aug. 26, 1976.
2. M. I. Bro, B. W. Sandt, US Patent 2, 946, 763.
3. J. P. Colson, R. K. Eby, J. Applied Phys., **37**, 3511, (1966).
4. DuPont Polymer Products Department, Technical Information on PFA, 7/19/85.
5. DuPont Polymer Products Department, Teflon® PFA Product Guide, Bulletin E-33272-2.
6. DuPont Polymer Products Department, Extrusion Guide for Melt Processible Copolymers and its Supplement, Bulletins E-41337-1 and E-41338-1.
7. DuPont Polymer Products Department, Technical Services Laboratory, Personal Communication.
8. R. K. Eby, F. C. Wilson, J. Applied Phys., **33**, 2951–55 (1962).
9. S. V. Gangal, (with 68 references), in: Kirk-Othmer (Ed.), Encyclopedia Chem. Tech., vol. 11, 3rd ed., Wiley, New York, 1980, p. 24.
10. R. L. Johnson (with 19 references), "Tetrafluoroethylene Copolymers with Perfluoroalkoxy Pendant Groups, in Encyclopedia of Polymer Science and Technology", Supp. No. 1 (1976).
11. R. L. Johnson (with 20 references) in: Kirk-Othmer (Ed.), Encyclopedia Chem. Tech., vol. 11, 3rd ed., Wiley, New York, 1980, p. 42.
12. W. J. Koros, J. Wang, R. M. Felder, J. Appl. Polym. Sci., **26**, 2805 (1981).
13. H. A. Larsen, G. R. Dehoff, N. W. Todd, "Modern Plastics", **36** (8), 89 August, 1959.
14. V. J. Leslie, and L. S. J. Shipp, "Structure Property Relationships of Fluoropolymers", IX Int. Symp. of Fluorine Chem., Avignon, Sept. 3–7, 1979.
15. D. I. McCane, (with 72 references), "Encyclopedia of Polymer Science and Technology", vol. 13, Wiley, New York, 1990, p. 654.
16. N. G. McCrum, B. E. Read, G. Williams, "Anelastic and Dielectric Effects in Polymeric Solids", Wiley, New York, 1967.
17. R. E. Putnam, (with 12 references), in: R. B. Seymour, G. S. Kirshenbaum (Eds.), "High Performance Polymers: Their Origin and Development", Elvesier, 1986, p. 279.
18. C. A. Sperati, Results cited previously in "Polymer Handbook", 2nd (Ed.), page V–36.
19. H. W. Starkweather, Jr., Macromolecules, **10**, 1161 (1977).
20. Starkweather, H. W., Jr., Zoller, P. and Jones, G. A., J. Polym. Sci., Poly. Phys. Ed., **22**, 1431 (1984).
21. H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys., Ed., **20**, 2159 (1982).
22. J. J. Weeks, I. C.. Sanchez, R. K. Eby, C. I. Poser, Polymer, **21**, 325 (1980).
23. J. J. Weeks, R. K. Eby, E. S. Clark, Polymer, **22**, 1496 (1981).

D. PHYSICAL CONSTANTS OF MODIFIED POLY(ETHYLENE-*co*-TETRAFLUOROETHYLENE) –[CH_2–CH_2–CF_2–CF_2]–

Commercially useful copolymers of ethylene and tetrafluoroethylene became feasible when it was discovered that the use of a small amount of a modifier, often a termonomer, resulted in polymers with properties suitable for long term use (Ref. 7). One key requirement, for example, is the ability to withstand exposure at high temperature without rapid embrittlement. The amount of modifier is very low and the exact amount

and nature of a modifier has been kept proprietary by each manufacturer, and may be different for different grades within a product line.

The commercial modified poly (ETFE) materials are essentially alternating copolymers (Ref. 10). In molecular formula, they are isomeric with poly(vinylidene fluoride)(PVDF), and can be considered equivalent to a PVDF that has a head to head, tail to tail structure. Many important physical properties of modified ETFE copolymers, however, are very different from and superior to those of available PVDF materials. (These properties do not include the remarkable piezoelectric and pyroelectric performance of PVDF.)

Property	Units	ASTM or other methods	Value	Remarks	Refs.
ASTM or ISO Standards					
Molding and extrusion powders		D3159, ISO12086-1		See Note 1	
Chemical Resistance			Excellent	See Note 2	1,2,4
Coefficient of Friction		D1894	0.4		5
Coefficient of Thermal Expansion		D696			
Linear Expansion (average values for temperature indicated)	K^{-1}				
20–30°C			9×10^{-5}		5
50–90°C			9.3×10^{-5}		5
140–180°C			14×10^{-5}		5
Compressive Strength Properties					
Modulus	MPa		860		
Strength	MPa	D695	50		5
Stress at 1% deformation	MPa		10.3		1
Yield strength	MPa		19.3		1
Continuous Use Temperature			See Service Temperature		
Creep					
Flexural creep modulus	GPa	D674	5.5 after 100 h		1
Critical Shear Rate For melt fracture	s^{-1}		200–10000	See Note 3	3
Deformation Under Load	%	D621	4.11	See Note 4	5
Density	g/cm^3				
Crystalline (from X-ray and IR data)			1.9		5
After melting and recrystallization			1.7		6
Melt at processing temp.			1.3		3
Dielectric Constant			See Electrical Properties		
Dielectric Strength			See Electrical Properties		
Dissipation Factor			See Electrical Properties		
Electrical Properties					
Dielectric constant	ε	D150			
At 60 Hz to 1 MHz			2.5		3
At 1 kHz			2.6		5
Dielectric strength		D149			
	kv/mm		59 at 0.25 mm thickness		3
	kv/mm		16 to 80	See Note 5	4
	v/mil		400 to >2000	See Note 5	4
Dissipation factor		D150			
At 1 kHz			0.0008		5
Surface arc resistance	s	D495	75–110	See Note 3	1
Surface resistivity, 100% RH	ohms/sq	D257	$>10^{14}$		4
Volume resistivity, dry					
At 50% RH	ohm cm	D257	$>10^{16}$		5

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Flame Rating					
Also see Limiting Oxygen Index		UL94	V–O	See Note 6	3
Flex Life	flexes	MIT	5500–12000	See Note 7	4
Flexural Modulus		D790			
At 23°C	MPa		827–1380		1,3
Hardness					
Durometer	Scale value	D2240	D70		3
Rockwell	Scale value	D785	R20		5
Heat Deflection Temperature		D 648			
At 0.455 MPa (66 psi)	°C		104		5
At 1.82 MPa (264 psi)	°C		74		5
Heat of Combustion	MJ/kg		13.7		5
Heat of Fusion			See Melting		
Impact Strength			See Toughness		
Limiting Oxygen Index	% Oxygen	D2863	30		5
Low Temperature Embrittlement	°C	D746	< – 100		4
Melt Viscosity					
By capillary rheometer	Pa s	D3159	$(0.7–10) \times 10^3$	See Note 3	3
Melt flow rate	dg/min	D3159	3.7–16	See Note 3	3
Melting					
Melting point	°C	DSC	225–270		6
Heat of fusion	kJ/kg		46.1		2
Molecular Weight			$(5–12) \times 10^5$	See Note 8	10
Permeability to Gases	m^3(STP) m/s/m^2/Pa ($\times 10^{15}$)				
Helium			405		5
Carbon dioxide			113		5
Oxygen			45		5
Nitrogen			14		5
Gas transmission rate	ng cm/m^2/s				
Water vapor		E96	752		5
Processing Temperature	°C		325–345		3
PV Limit	Pa m/s		< 70000		1
Radiation, Effects of				See Note 9	6
Refractive Index	n_D^{25}	D542	1.403		2
Resistivity		See Electrical Properties			
Service Temperature	°C		< – 100 to 150		6,5,8
Specific Gravity		See Density			
Specific Heat	kJ/kg/K		1.9–2.0		5
Tensile Strength Properties		D638			
Modulus					
At 23°C	MPa		830		5
Yield stress					
At 23°C	MPa		21.5		
Tensile strength					
At 23°C	MPa		38–48	See Note 10	3,6
Elongation at break	%				
At 23°C			100–350	See Note 10	3,6
Thermal Conductivity	W/m/K	C177	0.238		6
Toughness					
Izod impact strength (notched)	N	D256			
– 54°C			106–190	See Note 10	6
23°C	J/m		No break		6

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Property	Units	ASTM or other methods	Value	Remarks	Refs.
Transitions					
Alpha	°C		145 at 100 Hz		10
Gamma	°C		− 120 at 1 Hz		10
Unit Cell					
a	A		9.6	See Note 11	5
b	A		9.25		
c	A		5.0		
γ	Degrees		96		
Viscosity		See Melt Viscosity			
Water Absorption	% increase in wt.	D 570	< 0.02		3
Wear Factor	Pa^{-1}				
	in^3 min/ft/lb/hr		6000×10^{-10}	See Note 12	5
Weatherability			Excellent		1

NOTES

1. See additional comments in Note 2 for the table on PTFE.
2. Extensive tables with data on the resistance to a wide variety of chemicals are given in the references.
3. The exact value depends on the particular type or grade of polymer that is being used.
4. Tested at 13.8 MPa and 50°C.
5. The value decreases with increasing thickness.
6. This numerical rating for flame spread is not intended to reflect hazards presented by this or any other material under actual fire conditions.
7. The film tested is 0.25 mm thick.
8. In dibutyl adipate at 250°C.
9. Poly(ETFE) can be crosslinked by ionizing radiation.
10. The exact value depends on the particular type or grade of resin used as well as the method used to prepare the test specimen.
11. The unit cell is believed to be either orthorhombic or orthoclinic.
12. Wear factor is decreased to 16×10^{10} by use of glass fiber filled polymer.

REFERENCES

1. DuPont Polymer Products Department, Tefzel® "Design Handbook", Bulletin E-31301-1.
2. DuPont Polymer Products Department, Tefzel® "Chemical Use Temperature Guide", Bulletin E-18663-1.
3. DuPont Polymer Products Department, "Extrusion Guide for Melt Processible Copolymers and its Supplement", Bulletins E-41337-1 and E-41338-1.
4. DuPont Polymer Products Department, Tefzel® Product Bulletin, E-07229.
5. R. L. Johnson with 24 references, in "Encyclopedia of Polymer Science and Technology", Supp. no. 1 (1976) p. 269.
6. R. L. Johnson with 24 references, in: Kirk-Othmer (Ed.), "Ency. Chem. Tech." vol. no. 11, 3rd ed., Wiley, New York, 1980, p. 35.
7. R. E. Putnam "Development of Thermoplastic Fluoropolymers", in: (Eds.) R. B. Seymour, G. S. Kirshenbaum, "High Performance Polymers: Their Origin and Development", Elsevier 1986, p. 279.
8. J. C. Reed, J. R. Perkins, Presented at 21st International Wire and Cable Symposium, Atlantic City, NJ Dec. 6 1972.
9. H. W. Starkweather, J. Polym. Sci., Polym. Phys. Ed., **11**, 587 (1973).
10. C. Wu, W. Buck, B. Chu, Macromolecules, **20**, 98 (1987).

E. PHYSICAL CONSTANTS OF POLY(VINYLIDENE FLUORIDE) $-(CH_2CF_2)-$

Poly(vinylidene fluoride), PVDF, has been used since the early 1960s because of its chemical resistance combined with very good mechanical properties. In addition, it is now known that PVDF has very unusual and valuable piezoelectric and pyroelectric properties (see Ref. 12).

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Abrasion Resistance					
Sand	μm^{-1}	D968	4.0		9
Tabor	mg/1000 cycles		6–12; 7.0–9.0		8,9
		CS-17 wheel			
ASTM Standards					
Material					
Molding and extrusion powders		D3222		See Note 1	
Rods, tubes, and pipe lining		F491			
Chemical Resistance			Generally excellent	See Note 2	
Coefficient of Friction		D1894	0.48; 0.14–0.17		6,9
Coefficient of Thermal Expansion		D696			
Linear expansion	K^{-1}				
25 to 50			$(7.9–14.1) \times 10^{-5}$		9
Compressive Strength Properties					
Modulus	MPa		1000–1400		12
Strength	MPa		55–100		12
Conformation				See Note 3	12
α-form (also termed Type II)			*Trans*-gauche-*trans*-gauche		
β-form (also termed Type I)			All *trans*, i.e. "planar zig-zag"		
γ-form (also termed Type III)					
δ-form					
Continuous Use Temperature	°C		See Service Temperature		
Crystal Forms			See Conformation		
Density	$g/cm^3 = Mg/m^3$		1.75–1.78		12
Dielectric Constant			See Electrical Properties		
Dielectric Strength			See Electrical Properties		
Dissipation Factor			See Electrical Properties		
Electrical Properties					
Dielectric constant	ϵ	D150			
At 60 Hz			8.4–13.5		12
At 1 kHz			7.46–13.2		12
At 1 MHz			6.0–7.6		12
At 1 GHz			2.9		5
Dielectric strength	kV/mm	D149	260–950	See Note 4	9
Dissipation factor		D150			
60 Hz			0.049, 0.012–0.019		9,12
At 1 kHz			0.013–0.019		12
At 1 MHz			0.15–0.22		12
At 1 GHz			0.110		9
Surface arc resistance	s	D495	50–60		9
Volume resistivity, dry					
At 50% RH	ohm cm	D257	2×10^{14}		9
Flammability		UL94	VO	See Note 5	1
Also see Limiting Oxygen Index					
Flexural Modulus		D790			
At 23°C	MPa		1200–1800		9
Flexural Strength		D790	59–75	See Note 6	9
Fuel Value			See Heat of Combustion		
Hardness					
Durometer	Scale value	D2240	70–80		9
Rockwell	Scale value	D785			

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Property	Units	ASTM or other methods	Value	Remarks	Refs.
Heat Capacity	J/kg/K				13
100 K			443.5		
150 K			618.5		
230 K			891.8		
250 K			785.6		
Heat Deflection Temperature		D648			
At 0.455 MPa	°C		112–140, 150		9,12
At 66 psi	°C		112–140, 150		
At 1.82 MPa	°C		80–90, 100		9,12
At 264 psi	°C		80–90, 100		
Heat of Combustion	MJ/kg		14.9		7
	Btu/lb		6400		7
Heat of Fusion			See Melting		
Impact Strength			See Toughness		
Infrared Spectra				See Note 7	
Limiting Oxygen Index	% oxygen	D2863	44		9,7
Melt Viscosity				See Note 8	
Melt flow rate	dg/min	D1238	5–180 at 265°C		12
Melting					
Melting point	°C	D3418	154–184		5
Piezoelectric Coefficient	pC/N		$d(31) = 21-28$ $d(32) = 2.3$ $d(33) = -32$		12
Polymorphic Phases			Four documented, others suspected		12
See also Conformation					
Processing Temperature	°C		200–300		12
Pyroelectric Coefficient	$\mu C/K/m^2$		27–40		5,12
Radiation, Effects of			Exceptionally resistant to degradation	See Note 9	5,12
Refractive Index	n_D^{25}	D542	1.42		2
Resistivity			See Electrical Properties		
Service Temperature	°C		150		2
Solution Viscosity					
Intrinsic viscosity	dl/g		0.9–1.6 in DMF at 125°C		12
			3.03×10^{-5} $MW^{0.70}$, calc. from Ref.	See Note 10	7
			1.93×10^{-4} $MW^{0.68}$, in dimethyl acetamide		5
Solvent Absorption				See Notes 8 and 11	
Solvents			Soluble in selected solvents; swells in acetone	See Note 8	2,15
Specific Gravity		D792	1.75–1.80		
Specific Heat	kJ/kg/K	DSC	1.26–1.42		5
10 K			0.85	See Note 12	11
100 K			28.4		
200 K			50.3		
300 K			80.6		
340 K			93.1		
Speed of Sound	m/s				
Longitudinal			1930		13
Shear			775		13

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Tensile Strength Properties		D638			
Modulus	MPa				
At 23°C			1340–2000	See Note 6	5
Yield stress					
At 23°C	MPa		36–59	See Note 6	5
At 100°C	MPa		19–23	See Note 6	5
Tensile strength					
At 23°C	MPa		36–56	See Note 6	5
At 100°C	MPa		19–23	See Note 6	
Elongation at break	%				
At 23°C			25–500	See Note 6	5
At 100°C			400–600	See Note 6	9
Thermal Conductivity	W/m/K	C177	100–0.126		9
	W/m/K		0.1–0.13		5
Toughness					
Izod impact strength (notched)		D256			
23°C	J/m		160–530		8
Izod impact strength (unnotched)	N/m		1700–3100		5
Transitions					
α	°C		130 at 100 kHz		14
β	°C		0 at 100 kHz		14
γ (T_g)	°C	Dilatometry	−40		5
			−30 at 100 kHz		14
Viscosity			See Melt Viscosity and Solution Viscosity		
Water Absorption	% increase in wt.	D570	0.03–0.06		12
Weatherability			Superb		12

NOTES

1. Comments on ASTM and ISO standards for fluoropolymers are in Note 2 for the table on PTFE.
2. PVDF is resistant to most inorganic and organic chemicals at room temperature. It is, however, especially sensitive to attack by amines (Ref. 3). References (6), (14), and (15) have many pages of tables with information on the specific effect of different chemicals.
3. Many of the useful properties of PVDF stem from its existence in various conformations. The topic is not amenable for detailed presentation in tabular form, but it is discussed in depth in Ref. (12).
4. The value decreases with increasing thickness of the specimen being tested.
5. This numerical rating for flame spread is not intended to reflect hazards presented by this or any other material under actual fire conditions.
6. The exact value depends on the particular type or grade of resin used as well as the details used to make the test specimen.
7. Each conformation has its characteristic infrared spectra. See Ref. (9) for a general spectra and Ref. (12) for a discussion and references to the extensive literature.
8. Good solvents are dimethyl acetamide, dimethyl formamide, hexamethyl phosphoramide, and dimethyl sulfoxide. Some highly polar solvents such as ketones and ethers, are absorbed.
9. PVDF can be crosslinked by ionizing radiation resulting in an increase in high temperature modulus and very little degradation.
10. Calculated from data in Ref. (12) at page 200.
11. Reference (14) has 23 pages of Tables with information on the effect of many different chemicals.
12. Reference (11) has extensive tables for PVDF at temperatures from 10 to 340 K.

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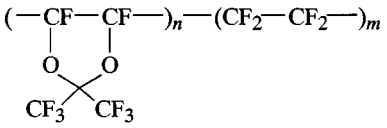
REFERENCES

1. Machine Design, April 19 (1981) p. 128.
2. W. S. Barnhart, and N. T. Hall, Poly(vinylidene fluoride) with 18 references, in: Kirk-Othmer (Ed.) "Encyclopedia of Chemical Technology", Vol. 9, 2nd ed., Wiley, New York, 1966, p. 840.
3. M. I. Bro, J. Appl. Polym. Sci., **1**, 319 (1959).
4. J. E. Dohany, A. A. Dukert, S. S. Preston III, Vinylidene Fluoride Polymers (with 109 references), in: "Encyclopedia of Polymer Science and Technology", 14, (1965) p. 600.
5. J. E. Dohany, and L. E. Robb, Poly(vinylidene fluoride) (with 147 references), in: Kirk-Othmer (Ed.) "Encyclopedia of Chemical Technology", vol. no. 11, 3rd ed., Wiley, New York, 1980, p. 64.
6. Foraflon® Polyvinylidene Fluoride, Product Bulletin from PCUK (with 68 references classified as to properties covered), Produits Chimiques Ugine Kuhlmann, now Atochem, Inc.
7. R. L. Johnson, in: "Encyclopedia of Polymer Science and Technology", Supp. No. 1 (1976) p. 269.
8. KF Polymer® Polyvinylidene Fluoride, Product Bulletin from Kureha Chemical Co., Ltd.
9. Kynar® Poly(vinylidene fluoride), Product Bulletin from Pennwalt Corporation.
10. Kynar® Piezo Film, Technical Manual, 10-M-11-83-M, Pennwalt Corporation.
11. W. K. Lee, C. L. Choy, J. Polym. Sci., Polym. Phys. Ed., **13**, 619 (1975).
12. A. J. Lovinger, Poly(vinylidene fluoride) (with 368 references), Chapter 5, in: Bassett, D. C. (Ed.), "Developments in Crystalline Polymers-1"; Applied Science Publishers, London, 1982.
13. J. E. Mark (Ed)., "Physical Properties of Polymers Handbook", AIP Press, Woodbury, NY, 1996.
14. Solef® Solvay PVDF, Product Bulletin from Solvay & Cie.
15. Tables of Chemical Resistance of Solef® PVDF, Bulletin from Solvay & Cie.

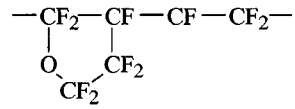
F. PHYSICAL CONSTANTS OF AMORPHOUS FLUOROPOLYMERS

Two major families of amorphous fluoropolymers have been developed. Both rely on in-chain rings to reduce crystallinity in the homopolymer and in copolymers with α-olefins such as tetrafluoroethylene and chlorotrifluoroethylene. One incorporates 2,2-bis-trifluoromethyl-4,5-difluoro-1,3-dioxole (PDD) in copolymers with tetrafluoroethylene or chlorotrifluoroethylene (TEFLON® AF) and the other is based on monomers such as 5-oxaperfluoro-1,6-heptadiene, which when polymerized, produce the structure below (CYTOP®). Amorphous fluoropolymers are soluble in a variety of fluorinated solvents and can be cast into thin films as well as conventionally melt processed. Optical properties are a significant market utility. Cost is a major consideration, however, compared to crystalline fluoropolymers.

TEFLON® AF



CYTOP®



Property	Units	ASTM or other methods	TEFLON® AF Value 1600 grade	TEFLON® AF Value 2400 grade	Remarks	Refs.	CYTOP® Value	CYTOP® Refs.
Melt viscosity	Pa s	D3835				10		
250°C			2650					
350°C				540				
Chemical Properties								
Water absorptivity	%	D570	<0.01	<0.01		4	<0.01	5
Chemical resistance					See Note 1	4		
Solubility					See Note 2	11		
Contact angle, water	Deg		104	105		3,10	110	4
Critical surface energy	dynes/cm		15.6–15.7	15.6–15.7		10		
Critical surface tension						3		
γ_c	mN/M		12.1	11.9				
γ_{GGF}	mN/M		13.9	13.8				
Density	g/cm^3	D792	1.78	1.67		1	2.03	2
Electrical Properties								
Dielectric strength	kv/(0.1 mm)	D149	2.1	1.9		10		
	kv/mm						19	2
Dielectric constant			1.93	1.89		4,9	2.1–2.2	2
1 MHz		D150	1.934	1.904		1		
1 GHz			1.93	1.897		1		
13.6 GHz			1.927	1.89		1		

Property	Units	ASTM or other methods	TEFLON® AF Value 1600 grade	TEFLON® AF Value 2400 grade	Remarks	Refs.	CYTOP® Value	CYTOP® Refs.
Dissipation factor			$(8-20) \times 10^{-5}$	$(8-30) \times 10^{-5}$		4	7×10^{-4}	
1 MHz		D150	0.00012	0.00012		1		
1 GHz			0.00018	0.00024		1		
13.6 GHz			0.0002	0.00035		1		
Surface arc resistivity	s	D495					180	2
Volume resistivity	ohm cm	D257					$>10^{17}$	2
Flammability, limiting oxygen index	$\%O_2$		95	95		4		
Fractional free volume				0.327		8		
Hardness								
Rockwell		D785	103	97.5		10		
Durometer		D1706				10		
23°C			77	75				
150°C			70					
220°C				65				
Optical Properties								
Refractive index	n_D^{25}	D542	1.31	1.29		4	1.34	2
dn/dt	$°C^{-1}$ ($\times 10^{-6}$)					1		
Below T_g			−77	−78				
Above T_g			−329	−378				
Weatherability			Excellent	Excellent			Excellent	
Optical transmission	%	D1003	>95	>95	See Note 3	4,6		
Stress optical coefficient	B		4.5		See Note 4	4		
Infrared spectra					See Note 5	7		
Abbe number			92	113	See Note 6	4	90	4
Permeability coefficients	cB				See Note 7	4,8	34000	4
Oxygen			34000	99000				
Nitrogen			130000	49000				
Carbon dioxide				280000				
Water vapor			117000	410000				
Helium				270000				
Hydrogen				220000				
Methane				340000				
Ethane				180000				
Ethylene				350000				
Tensile Properties								
Tensile strength, at break	MPa	D638				10	35	2
23°C			27	26				
150°C			8					
220°C				4				
Elongation, at break	%	D638				10		
23°C			17	8				
150°C			89					
220°C				8				
Tensile Modulus	MPa	D638	1550	1540		4	1200	2
Compressive strength	MPa						70	2
Compressive modulus	MPa						2000	2
Tensile strength at yield	psi		3900	3600		16		
Flexural modulus	MPa	D790				10		
23°C			1800	1600				
200°C			1000					
220°C				700				
Thermal Properties								
Transition temperatures	°C							
T_g γ			160	240	See Note 8	1	108	2
β		At 10^2 Hz	88			9		
δ			−203			9		
Thermal stability	°C			360		4	>400	2

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Property	Units	ASTM or other methods	TEFLON® AF Value 1600 grade	TEFLON® AF Value 2400 grade	TEFLON® AF Remarks	TEFLON® AF Refs.	CYTOP® Value	CYTOP® Refs.
Upper use temperature	°C			285				
Thermal expansion								
Linear	$°C^{-1}$ ($\times 10^{-6}$)		74	81		4		
Volume	$°C^{-1}$ ($\times 10^{-6}$)		260	301		10		
Weight loss in air								
after 1 hour at temp. 260°C	%			None				
360°C	%			0.29				
380°C	%			0.53				
400°C	%			1.94				
420°C	%			8.83				
Thermal conductivity	W/m/K	295 K	0.116			1		
Deflection temperature		D648				10		
66 psi			156	200				
264 psi			154	174				
Specific volume						1		
Below T_g	cm^3/g		$0.548+(1.22\times10^{-4})T$	$0.570+(1.37\times10^{-4})T$				
Above T_g	cm^3/g		$0.476+(5.64\times10^{-4})T$	$0.428+(7.75\times10^{-4})T$				
Radiation effects					See Note 9	5		
Charge storage effects					See Note 10			

NOTES

1. See Refs. (2,3) for data on chemical resistance and physical property performance in various solvents.
2. See Ref. (15) for types of solvents and degree of solubility of amorphous fluoropolymers.
3. See Ref. (8) for optical transparency data.
4. Brewster $= 10^{-13}\,cm^2/dyn$.
5. See Ref. (10) for infrared spectral interpretation.
6. Abbe Number = (refractive index at sodium D line minus 1)/(refractive index at hydrogen F line minus refractive index at hydrogen C line).
7. centiBarrier $= 10^{-8} \times (cm^3(STP)\ cm/s/cm\ Hg/cm^2)$.
8. T_g of 2,2-bis(trifluoromethyl)-4,5-difluoro-1,3-dioxole homopolymer is 330°C. Glass transition temperature is a strong function of comonomer content.
9. See Ref. (5) for a discussion of radiation effects on optical and physical properties.
10. See Ref. (12) for charge storage data of amorphous fluoropolymers.

REFERENCES

1. W. H. Buck, P. R. Resnick, Teflon Af Technical Information Bulletin H-52454-1, DuPont; Wilmington, DE, 1993.
2. CYTOP Technical Bulletin, Asahi Glass: Yokohama, Japan, 1988.
3. M. H. Hung, P. R. Resnick, B.E. Smart, W. H. Buck, in: (ed); J. C. Salamone, "Polymeric Material Encyclopedia", CRC Press, NY, 1996.
4. I. Pinnau, L. G. Toy, J. Membr. Sci., **109**, No. 1, 125 (1996).
5. M. S. Jahan, D. R. Ermer, D. W. Cooke, Rad. Phys. Chem., **41** (1–2), 77 (1993).
6. H. W. Starkweather, Jr. P. Avakian, R. R. Matheson, Jr. J. J. Fontanella, M. C. Wintersgill, Macromolecules, **24** (13), 3853 (1991).
7. C. J. Drummond, G. Georgaklis, D. Y. C. Chan, Langmuir, **12** (11), 2617 (1996).
8. J. H. Lowry, J. S. Mendlowitz, N. S. Subramanion, Opt. Eng., 1992, **31** (9), p. 1962.
9. S. P. Andersson, O. Andersson, G. Backstrom, Int. J. Thermophysics, **18** (1), 209 (1997).
10. T. C. Nason, T.-M. Lu, Thin Solid Films, **239** (1), 27 (1994).
11. P. Gunther, H. Ding, R. Gerhard-Multhaupt, Proceedings of the 1993 IEEE Conference on Electrical Insulation and Dielectric Phenomena, 197–202 (1993).
12. H. Ding, Proceedings of the 8th International Symposium on Electrets (ISE 8), 89–94 (1994).
13. A. N. Hammoud, E. D. Baumann, E. Overton, I. T. Myers, J. L. Suthar, W. Khachen, J. R. Laghari, IEEE Conference on Electrical Insulation and Dielectric Phenomena, 549–554 (1992).
14. Teflon AF Amorphous Fluoropolymer Product Information product brochure 2041038, Dupont Company, 10/92.
15. Teflon AF Amorphous Fluoropolymer Product Information product brochure 231577B, Dupont Company, 10/92.
16. CAS Subscription Database PLASPEC, accession nos. 400-11002 and 400-11003, (accessed Apr. 1977).

G. PHYSICAL CONSTANTS OF POLY(CHLOROTRIFLUOROETHYLENE)

Poly(chlorotrifluoroethylene) (PCTFE) was the first of the fluoropolymers to be made (Ref. 3). It was not until the work of W. T. Miller and his colleagues during the research for the Manhattan (atom bomb) project (Ref. 8), however, that materials with molecular weight suitable for plastics applications became available.

$$-[CFCl-CF_2]-$$

Property	Units	ASTM or other methods	Value	Remarks	Refs.
ASTM and ISO Standards					
Material					
Molding and extrusion powders		D1430,12086 [-1]		See Note 1	
Film and sheeting		D3595			
Chemical Resistance				See Note 2	
Coefficient of Thermal Expansion					
Linear expansion	K^{-1}	D696			
(average values for temperature indicated)					
− 80 to 20°C			4.5×10^{-5}		4
20–50°C			7×10^{-5}		
Compressive Strength		D695			
Properties					
Modulus	MPa		1 350		6
Strength	MPa		380		6
Stress at 1% deformation	MPa		13.8	See Note 3	
Continuous Use Temperature	°C		See Service Temperature		
Creep			Very low Creep		10
Deformation Under Load	% Def.	D621	6.86		8
Density	g/cm^3	D792			
			2.10 Amorphous		10
			2.12 Crystalline		10
			2.15		16
Electrical Properties					
Dielectric constant	ε	D150			
At 60 Hz, − 40°C			2.2		4
At 60 Hz, 25°C			2.6		4
At 60 Hz, 170°C			3.2		4
At 100 kHz			2.9		4
At 100 MHz			2.5		4
Dielectric strength	kV/mm	D149	15–200	See Note 4	4
Dissipation factor		D150			
100 Hz			0.02–0.06	See Note 8	7
At 1 kHz			0.025–0.027	See Note 8	7
Surface arc resistance	s	D 495	360		6
Surface resistivity, 100% RH [megohms],	ohms/sq	D 257	10^{15}		6
Volume resistivity, dry	ohmcm	D 257			
At 50% RH			1.2×10^{18}		6
Flammability	UL94		VEO	See Note 5	6
Also see Limiting Oxygen Index					
Flexural Modulus		D790			
At 23°C	MPa		1250		
At 23°C	psi		180000		6
At 100°C	MPa		14420		

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Property	Units	ASTM or other methods	Value	Remarks	Refs.
Flexural Strength	MPa	D790	73.8		1
	psi		10700		
Hardness					
Durometer	Scale value	D2240	D 75–80		6
Rockwell	Scale value	D785	S 85		1
Heat Capacity	J/kg/K	DSC	900		6
80 K			278.7		8
200			625.7		8
300			859.5		8
320			906.7		8
Heat Deflection Temperature		D648			
At 0.455 MPa (66 psi)	°C		130		6
At 1.82 MPa (264 psi)			70		6
Heat of Fusion			See Melting		
Impact Strength			See Toughness		
Intrinsic Viscosity	ml/g		0.9–1.7	See Note 11	4
Limiting Oxygen Index	% Oxygen	D2863	100		6
Melt Viscosity					
By capillary rheometer	Pa s		1–10 at 230°C	See Note 6	4
Melting					
Melting point	°C	DTA	210–215		6
Latent heat of fusion	kJ/kg	DSC	1.2		6
	cal/g		3.6–8.4	See Note 7	4
Molecular Weight					
See also Zero Strength Time			50000–100000	See Note 8	5
			150000–190000		7
Permeability to Gases	m^3(STP) m/s/m^2/Pa × 10^{15}			See Note 9	6
Helium			21.7		6
Carbon dioxide			1.4		
Oxygen			0.4		
Nitrogen			0.05		
Hydrogen			9.8		
Processing Temperature	°C		280–305		10
Radiation, Effects of			Little effect	See Note 10	6
Refractive Index	η_D^{25}	D542	1.435		6
Resistivity				See Electrical Properties	
Service Temperature	°C		−200 to 200		6,10
Solution Viscosity			See Intrinsic Viscosity		
Solvent Absorption			Generally low	See Note 2	6,7,10
Solvents			None at room temperature	See Note 11	4
Specific Gravity			See Density		
Specific Heat					
0°C	kJ/kg/K		0.84		4
211°C	kJ/kg/K		1.7–4.6	See Note 7	4
240°C	kJ/kg/K		4.2		4

Property	Units	ASTM or other methods	Value	Remarks	Refs.
Tensile Strength Properties		D638			
Modulus					
At 25°C	MPa		14000		6
Tensile strength					
At 23°C	MPa		40		
At 125°C	MPa		4		
Elongation at break					
At 23°C	%		150		
At 125°C	%		400		
Thermal Conductivity	W/m/K	C177			
At 3.5°C			3.09×10^{-4}		4
At 60.8°C			3.23×10^{-4}		4
Thermal Degradation			Degrades at temperatures above 260°C	See Note 12	4
Toughness					
Izod impact strength, notched					
24°C	J/m	D256	267		6
Transitions	°C				
α			150 at 1 Hz	See Note 13	1
β			90 at 1 Hz	See Note 13	1
γ			−37 at 1 Hz	See Note 13	1
Transmittance Properties					
Luminous transmittance	%	D1003	>90	See Note 14	6
Haze	%	D1003	<4		
Viscosity				See Melt Viscosity and Solution Viscosity	
Water Absorption	% Increase in wt.	D570	0.00		2
Weatherability			Excellent		
Zero Strength Time (ZST)	s	D1430	200–750	See Note 6 and Note 8	6,10

NOTES

1. Comments on ASTM and ISO standards for fluoropolymers are in Note 2 for the table on PTFE.
2. PCTFE is resistant to most inorganic and organic chemicals at room temperature and to most inorganic chemicals such as strong oxidizing agents over a wide range of temperature. It is especially useful in liquid oxygen service. References (4), (6), and (10) have many pages of tables with information on the specific effect of different chemicals.
3. The value is the stress at 0.3% deformation after 24 h.
4. The value decreases with increasing thickness of the test specimen.
5. This numerical rating for flame spread is not intended to reflect hazards presented by this or any other material under actual fire conditions.
6. The exact value depends on the particular type or grade of resin used.
7. The value increases with increasing crystallinity.
8. ZST is a measure of relative molecular weight of PCTFE.
9. PCTFE is noteworthy for its very low permeability.
10. PCTFE generally maintains properties when exposed up to about 18 Mrad of ionizing radiation. Further improvement in resistance can be obtained by incorporating small concentrations of vinylidene fluoride during polymerization.
11. Soluble at 100°C in 2,5-dichlorobenzotrifluoride. Another solvent is 1,2,3,4-tetrafluorohexachlorobutane.
12. The degradation involves random scission rather than "unzippering".
13. Specimens were 80% crystalline.
14. For amorphous polymers.

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REFERENCES

1. A. J. Bur in: (Ed) L. A. Wall, "Fluoropolymers", Wiley, Interscience, New York, 1972.
2. S. Chandrasekaran, Chlorotrifluoroethylene Homopolymer (with 91 plus 2 general references), in: "Encyclopedia of Polymer Science and Engineering", Vol. 3, Wiley, New York, 1987, p. 463.
3. I. G. Farbenindustrie, Brit. Pat. 465, 520, May 3 (1987).
4. F. J. Honn (with 32 references), in: Kirk-Othmer (Ed.) "Encyclopedia of Chemical Technology", 1st ed., Wiley, New York, p. 691.
5. H. S. Haufman, M. S. Muthana, J. Polym. Sci., **6**, 252 (1951).
6. Kel-F®, Engineering Manual, 3M Company.
7. Kel-F®, Technical Information Manual, 3M Company.
8. J. E. Mark (Ed)., "Physical Properties of Polymers Handbook", AIP Press, Woodbury, NY, 1996.
9. W. T. Miller, US Patent 2, 564, 024, Aug. 14, 1951.
10. P. C. U. K. Bulletin on Voltalef® [PCTFE] Molding Powders.
11. A. C. West (with 57 references), in: Kirk-Othmer (Ed.) "Encyclopedia of Chemical Technology", vol. no. 11, 3rd ed., Wiley, New York, 1980, p. 49.

Physical Constants of Poly(acrylonitrile)*

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A. Tables of Physical Constants V-59
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A. TABLES OF PHYSICAL CONSTANTS

TABLE 1. CRYSTALLINITY/CRYSTALLIZATION BEHAVIOR

Property	Value	Remarks	Refs.
Crystallographic data Unit cell dimensions[a]	See table See also corresponding chapter of this Handbook	With molecular modelling calculations of PAN References are made to unit cell parameter, crystal system, density, melting point, heat of fusion	1

	Axis				
Tacticity	*a* (Å)	*b* (Å)	*c* (Å)	System	Refs.
Syndio	5.99	5.99	–	Hexagonal	2
Syndio	21.18	11.60	5.1	Orthorhombic	3
Syndio	10.6	11.60	5.04	Orthorhombic	4
Syndio	10.2	6.10	5.10	Orthorhombic	5
Syndio	10.55	5.8	5.08	Orthorhombic	6
Syndio	21.0	11.9	5.04	Orthorhombic	7
Syndio	10.7	12.1	5.1	Orthorhombic	8
Iso	4.74	4.74	2.55	Tetragonal	9

Crystallinity (%)	See table	Samples: gel spun PAN-fibers. The role of macromolecular entanglements is discussed	
Crystal size L_{100} (Å)	See table		10

Molecular weight $\bar{M}_w$(g/mol)	Draw ratio (fiber)	Crystallinity (%)	Crystal size L_{100} (Å)
6×10^4	2.0–7.0	18.5–32.0	43.5–66.0
12×10^6	2.0–6.0	27.5–39.5	45.5–78.0

See Ref.	Degree of PAN-crystallinity as a function of crystallite thickness and polymer tacticity	11

*Based on a similar table in the third edition, by W. Fester, Hoechst AG, FR Germany.

Property	Value	Remarks	Refs.
Crystallization temperature T_c[b] (°C)	95–100	Determined in propylene carbonate	12,13
	153.6	Crystallization from PAN/H_2O – melt under pressure	14
Density (g/cm^3)	1.15–1.18	Sample: flakes and films	12,16,17
	1.17–1.19	Sample: fiber	18
Orientation factor	See Ref.	Sample: stretched films X-ray diffraction studies Chain-orientation factors were measured by IR-dichroism.	15

[a]The reported unit cell dimensions, especially the *c*-dimension along the chain axis, can only be regarded as estimated because of the diffuse meridian and polar reflections in the X-ray diffraction studies.
[b]The dissolution and crystallization temperatures given here are obtained from a free radical poly(acrylonitrile). They are sensitive to chain irregularities in the polymer. Samples of poly(acrylonitrile) obtained from different sources show marked differences in the dissolution and crystallization temperatures, although they have similar IR-spectra, X-ray diffraction patterns and densities.

TABLE 2. ELECTRIC AND ELECTRONIC PROPERTIES

Property	Value	Remarks	Refs.
Dielectric Constant			
ε	4.2–6.5 (60–10^6 Hz)	Sample: film	19,20
ε_{RT}	5.68 ± 0.84 (293 K)	Sample: film	
ε_{He}	3.29 ± 0.17 (3.8 K)	Temperature dependence: Arrhenius type behavior	21
ε ($T > 373$ K)	See Ref.	Sample: discs from powder Variation of ε with temperature at various frequencies	22,23
Dissipation factor			
$\tan\delta = \frac{\varepsilon''}{\varepsilon'}$			
	0.033–0.113 (60–10^6 Hz)	Sample: film (ε'', ε' loss and storage dielectric constants)	19,20
	See Ref.	Comparison of the mechanical and dielectric values of tan δ as function of temperature (10^2 Hz)	24
tanδ ($T > 373$ K)	See Ref.	Sample: discs from powder Variation of tan δ with temperature at various frequencies	22
Piezoelectric constants			
Driver constant d_{31} (C/N)	1.5×10^{-12}		
Generator constant g_{31} (V m/N)	30.8×10^{-3}		17,25
McGinnies parameter χ	0.963		
Conductivity (S/cm)	4.8×10^{-14} (293 K)	Sample: film (sandwich)	
	8.4×10^{-12} (373 K)	Data from the current–voltage characteristics at various temperatures	26,31
Radiation induced conductivity	See Ref.	Sample: film (40 µm) sandwich	27
Dipole moments in solution	See corresponding chapter of this Handbook		
Magnetic susceptibility (e m n/g)	See Ref.	Sample: PAN-foam	28
Photoelectric properties	See Ref.	Sample: thin film (plasma-polymerized)	29
Photocurrent (A/cm^2)			
Electronic properties			
Ionization potential (eV)	8.2	Sample: thin film (plasma-polymerized)	30
Electron affinity (eV)	3.9		
Surface work function (eV)	5.8		

TABLE 3. FIBER PROPERTIES

Property	Conventional acrylic fibers (≥ 85% AN)[a] (Refs. 18, 32–36)	PAN-fiber Dralon T (100% AN) (Ref. 35)	High strength acrylic fibers[b] (Refs. 38,39)	Acrylic fibers from isotactic PAN[c] (Ref. 40)
Fiber fineness (dtex)	0.6–19.0	3.3–17.0	1.0–4.0	
Density (g/cm^3)	1.14–1.19	1.17–1.19		
Tenacity [21°C/65% RH] (cN/dtex)	1.8–4.5	3.5–6.0	10–20	8.0–20.0
Tenacity [wet/dry ratio] (%)	75–95	80–100		
Elongation ε [21°C/65% RH] (%)	30–60	25–40	7–10	
Elongation [wet/dry ratio] (%)	100–120	≈ 100		
Initial modulus [(Elongation $\varepsilon \rightarrow 0$)] (cN/dtex)	30–100	95–160	140–270	
Modulus in hot water [90°C] (cN/dtex)	1.0–5.5 (Ref. 37)			15.0–21.0
Relative knot tenacity (%)	70–90	≈ 70	4.5–6.5 (cN/dtex)	
Relative loop tenacity (%)	30–80	≈ 60		

[a] The properties of acrylic fibers manufactured by conventional processes of wet or dry spinning are dependent on spinning conditions and the monomer content in the polymer itself. Some trade names of acrylic fibers: Acrilan, Cashmilon, Courtelle, Dolan, Dralon, Euracryl, Leacryl.
[b] Polyacrylonitrile fibers with high tensile strength are prepared under special conditions: Use of polyacrylonitriles with high molecular weight ($\bar{M}_w > 5.0 \times 10^5$ g/mol), wet or dryjet/wet spinning and forming a fiber with a gel structure, afterwards stretching to high degrees (draw ratios 15–30).
[c] Fibers are made from polyacrylonitriles with highly isotactic content (mm > 0.40). They are prepared by anionic polymerization with a special catalyst.

TABLE 4. FURTHER PROPERTIES OF ACRYLIC FIBERS

Property	Value[a]	Remarks	Refs.
Elastic recovery [(1-$\varepsilon_r/\varepsilon$) × 100] (%)		*Effects of acids and alkalis*	36,42
ε = 2.0%	90–95	Good to excellent resistance to mineral acids,	
ε = 5.0%	50–90	fair to good resistance to weak alkali, and moderate resistance to strong cold solutions of alkali	
Torsion modulus (cN/dtex)	10–17		
Fiber shrinkage [in water, 95°C] (%)		*Effects of bleaches and solvents*	
Drawn fiber	14–22	Good resistance to strong bleaches and common solvents;	
Thermoset fiber	≈ 1.0	Unaffected by dry cleaning solvents;	
Water absorption [(21°C/65% RH)] (%)	1.0–1.5	Can be bleached with sodium chlorite	
Water retention (%)	4.0–12.0[b]	*Resistance to mildew, aging, sunlight, abrasion*	41
Glass transition temperature (°C)		Not attacked by mildew;	
Dry	85–95	Good resistance to aging, sunlight and abrasion	
Wet	50–60		
Melting/decomposition temperature (°C)	250–320		
Heat resistance in air (°C)	140		
Fire limiting oxygen index (LOI) (°C)	18		

[a] Refs. 18, 21–36. [b] Ref. 41.

TABLE 5. OPTICAL PROPERTIES

Property	Value	Remarks	Refs.
Birefringence[a]			
$\Delta_n = n_\parallel - n_\perp$	−0.005	Sample: PAN-fiber	32
	−0.0017 (skin)	Sample: PAN-fiber (kidney-shaped)	44
	−0.0047 (core)		
Refractive index			
n_d^{25}	1.158		45
$n_\parallel$	1.50–1.53	Sample: PAN-fiber	19,32,44
$n_\perp$	1.51–1.53		
	See also corresponding chapter of this Handbook		
Refractive index increments	See Ref.	Measured and calculated for different solvents	46
	See also corresponding chapter of this Handbook	(λ = 546 nm)	
Polarizability			
$P_\parallel$	0.0735		
$P_\perp$	0.074		
Optical anisotropy in solution	See corresponding chapter of this Handbook		44

[a] $n_\perp$ and $n_\parallel$ are refractive indices measured with incident light having the vibration vector perpendicular and parallel to the fiber axis, respectively.

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TABLE 6. POLYMERIZATION: KINETIC AND THERMODYNAMIC DATA

Property	Value	Remarks	Ref.
Heat of polymerization (KJ/mol)	-72.4 ± 2.2		43
Rate constants of free radical polymerization (propagation, termination and transfer constants)	See corresponding chapter of this Handbook		
Heats and entropies of polymerization	See corresponding chapter of this Handbook		
Activation energies of polymerization	See corresponding chapter of this Handbook		
Activation enthalpies and entropies of stereo-control in free radical polymerization	See corresponding chapter of this Handbook		
Stereoregularity[a]	See table		

Polymerization	$[\eta]_{DMF}$ (dl/g)	$\bar{M}_{V(DMSO)}$ (g/mol)	Triad tacticity (%)[†] Iso	Hetero	Syndio	Refs.
Radical	1.97–6.87		25–29	47–51	22–27	47
Anionic	2.17–2.26		30–31	43–46	23–27	47,48
Anionic		5.3×10^4	26.7	48.8	24,524	40
Anionic		0.2×10^4–5.1×10^6	47–72	21–36	10–20	49,50,51
Urea clathrate						
UV-irradiation	0.22–1.56		56–71	22–32	7–12	52
γ-irradiation (post)	0.79–3.05		69–87	10–23	3–8	48
γ-irradiation (in source)	1.81–4.96		48–65	25–36	9–16	53

[†] Tacticity of PAN was determined by ^{1}H-NMR, ^{2}H-NMR and ^{13}C-NMR, computing the spectra, and by decoupling techniques.

Spectral data[a]			
Nuclear magnetic resonance spectrum			
^{1}H-NMR, ^{2}H-NMR	See Refs.	Configuration of PAN	54–58
^{13}C-NMR	See Refs.	Stereoregularity of PAN	48,53,59–61
Solid-state NMR	See Refs.	Chain conformation and phase structure of PAN	62–64
Infrared spectrum	See Refs.		65–72

[a] Stereoregularity and spectral data were properly provided with separate generic terms..

TABLE 7. SOLUBILITY/SOLUTION PROPERTIES

Property	Value	Remarks	Refs.
Solvents	Dimethylformamide, dimethyl sulfoxide, dimethylacetamide, ethylene carbonate, propylene carbonate, malononitrile, succinonitrile, adiponitrile, γ-butyrolactone, conc. sulfuric and nitric acid, conc. salt solutions: LiBr, NaCNS, $ZnCl_2$;	See also corresponding chapter of this Handbook Solvents and Nonsolvents	73,74
Solubility			
Hildebrand parameter δ_2 $[J/cm^3]^{1/2}$	31.5 (exp.); ~26.0 (estim.) See also corresponding chapter of this Handbook	Estimated values from empirical formulae of Hildebrand/Scott and Askadskii	75
Dissolution of highly isotactic PAN	See Ref.	Solution temperature as function of isotacticity and molecular weight	76
Intrinsic viscosity (η)			77,78
Solvent dependence	See Ref.	Data and factors which convert [η]-values from one solvent into another	79

Property	Value	Remarks	Ref.
Temperature dependence	See table		

Solvent	T (°C)	Huggins' coefficients	$-(d \ln[\eta]/dT)$
N,N-Dimethyl-	25	34	0.14–0.19
formamide	35	33	
N,N-Dimethylacetamide			0.27
Dimethyl sulfoxide			0.08
60% HNO_3			0.05
γ-Butyrolactone			0.14
			0.13–0.17
Hydroxyacetonitrile			0.07
acetonitrile			

Viscosity–molecular weight relationship	See corresponding chapter in this Handbook	Mark–Houwink–Sakurada equation	
Dilute solution properties			
Unperturbed dimensions of linear chain molecules	See corresponding chapter in this Handbook		
Partial specific volume	See corresponding chapter in this Handbook		
Huggins and Schulz–Blaschke coefficients		See corresponding chapter in this Handbook	
Sedimentation and diffusion coefficients	See corresponding chapter in this Handbook		
Parameters of isotactic PAN	See Ref.	Effects of stereoregularity on Mark–Houwink–Sakurada equation for different solvents, further on the radius of gyration $\langle S^2 \rangle$, the second virial coefficient A_2 and the conformation parameter	80
Parameters of ultrahigh molecular weight PAN	See Ref.	Dependence of molecular weight on radius of gyration $\langle S^2 \rangle$, second virial coefficient A_2 and intrinsic viscosity $[\eta]$	81
Viscosity and related parameters	See Ref.	η-values, activation parameters of viscous flow, voluminosity and shape factor at different temperatures (Solvent: DMF)	82
Flexibility parameter (λ) (Theta conditions)	7.20–7.68	Solvent: DMF λ is defined by Kratky–Porod for random-coiled polymers	83
Dissolution temperature[a] (°C)	125–130	In propylene carbonate	12,13

[a] The dissolution and crystallization temperatures given here are obtained from a free radical poly(acrylonitrile). They are sensitive to chain irregularities in the polymer. Samples of poly(acrylonitrile) obtained from different sources show marked differences in the dissolution and crystallization temperatures, although they have similar IR-spectra, X-ray diffraction patterns and densities.

TABLE 8. SPECIAL SOLID STATE PROPERTIES

Property	Value	Remarks	Refs.
Gas permeability (P) $\left[\frac{g \times cm}{cm^2 \times s \times bar}\right]$			
O_2 (film, 25°C)	2.15×10^{-15}	Relationship between published and CGS permeability units	84,85,86
(film, 25°C/65% RH)	$150–195 \times 10^{-15}$		
CO_2 (film, 25°C)	11.8×10^{-15}	$1 \frac{g \times cm}{s \times cm^2 \times bar} = \frac{298.82}{M} \frac{cm^3(STP) \times cm}{s \times cm^2 \times cmHg}$	
H_2O (film, 25°C)	18.4×10^{-10}	Where M is the molecular weight of the penetrant gas	
Polymer surface energy δ (m N/m)	58.8	Calculated from cohesion parameters and	87
	49.9/54.1	refractometric data	88
Critical surface tension γ_c (dyn/cm)	See corresponding chapter of this Handbook		
Vickers microhardness H_V (kg/mm^2)	11–24	H_V is a function of load L ($L = 20–60$ g) Sample: film	89

TABLE 9. THERMAL AND THERMODYNAMIC DATA

Property	Value	Remarks	Refs.
Heat capacity C_P	See table	C_P-data based on measurements in the solid state	
Enthalpy function $H_T - H_O$	See also corresponding	Enthalpy and entropy functions	90
Entropy function $S_T - S_O$	chapter of this Handbook	are calculated	

T (K)	C_P (J/mol/K)	$H_T - H_O$ (J/mol)	$S_T - S_O$ (J/mol/K)
50	13.77	244.7	7.546
100	30.23	1388	22.73
150	40.44	3167	37.02
200	49.77	5410	49.87
250	58.48	8101	61.84
300	68.84	11277	73.40
350	80.83	15012	84.89
370	86.18	16681	89.53

Heat capacity C_P ($T > T_g$)	See Ref.	Calculated heat capacity data for states above the Glass transition temperature	91
Specific heat of combustion Δh_c (KJ/g)	30.6 (expt.) 31.5 (calc.)	Δh_c is related to other parameters, such as oxygen index and char residue	92
Thermal decomposition temperature[a] (°C)	250–310		33
Initial decomposition temperature (°C)	248		
Exotherm decomposition range (°C)	238–299	Data from DSC and thermogravimetry analysis	93
Heat of oligomerization ΔH (J/g)	530.9		
	See Ref.	Ranges of decomposition and activation energies for the thermal degradation in air and nitrogen	94
Thermal decomposition activation energies and products	See also corresponding chapter of this Handbook		
Glass transition temperature T_g (°C)	85–104	Various data cited in the 3/e of Polymer Handbook	95
	110	Sample: film Method: fluorescence probe technique	96
	65/105	Existence of two transition temperatures in thermomechanical analysis	97
	See Ref.	Effects of solvents and thermal treatment on T_g	97
	See Ref.	Chain-length dependence of T_g	98
	See also corresponding chapter of this Handbook		
Melting point T_m (°C)	~320	Normally PAN decomposes before melting	99,100
T_m (H_2O) (°C)	184.7	Melting temperature in the wet state under self generated pressure	14
	See also corresponding chapter of this Handbook		
Thermal conductivity κ (mW/cm/K)	0.022 (5 K) 0.440 (20 K) 1.600 (100 K)	Sample: discs from powder	101
Coefficient of expansion (K^{-1})			
Volume $(1/V) \times (dV/dT)_P$	2.8×10^{-4}–3.8×10^{-4} (above T_g) 1.4×10^{-4}–1.6×10^{-4} (below T_g)		45,102,103
Linear $(1/V) \times (dV/dT)_P$	1.6×10^{-4}–2.0×10^{-4} (above T_g) 1.0×10^{-4} (below T_g)		104,105

[a]The thermal decomposition temperature determined by thermogravimetric analysis ranges from 250°C for a PAN-sample prepared with an ionic catalyst, to 310°C for a commercial fiber. Pyrolysis of poly(acrylonitrile) carried out in the absence of oxygen at 500–800°C yields HCN and low molecular weight nitriles such as monomer, dimer and methacrylonitrile leaving a residue with a condensed ring structure.

B. REFERENCES

1. R. J. Hobson, A. H. Windle, Polymer, **34**, 3582 (1993).
2. G. Natta, G. Mazzanti, P. Corradini, Atti. Accad. Nazl. Lince., Cl. Sci. Fis., Mat. Nat. Rend., **25**, 3 (1958).
3. J. J. Klement, P. H. Geil, J. Polym. Sci. A-2, **6**, 1381 (1968).
4. G. Hinrichsen, H. Orth, Kolloid Z. Z. Polym., **247**, 844 (1971).
5. R. Stefani, M. Cherreton, M. Garnier, C. Eyroud, Compt. Rend., **251**, 2174 (1960).

6. V. Holland, S. Mitchell, W. Hunter, P. Lindenmeyer, J. Polym. Sci., **62**, 145 (1962).
7. F. Kumamaru, T. Kajiymama, M. Takayanagi, J. Cryst. Growth, **48**, 202 (1980).
8. H. Yamazaki, S. Kajita, K. Kamide, Polym. J., **19**, 995 (1987).
9. B. G. Colvin, P. Storr, Eur. Polym. J., **10**, 377 (1974).
10. B. Qian, W. Lin et al., J. Polym. Eng., **15**, 327 (1995).
11. R. J. Hobson, A. H. Windle, Macromolecules, **26**, 6905 (1993).
12. R. Chiang, J. Polym. Sci. A, **3**, 2109 (1965).
13. R. Chiang, J. H. Rhodes, V. F. Holland, J. Polym. Sci. A, **3**, 479 (1965).
14. H. S. Kim, J. Polym. Sci. Part B; Polym. Phys., **34**, 1181 (1996).
15. Z. Bashir, A. R. Tipping, S. P. Church, Polym. Intern., **33**, 9 (1994).
16. R. Chiang, J. Polym. Sci. A, **1**, 2765 (1963).
17. V. Mc Ginniss, Polymer, **36**, 1127 (1995).
18. Ullmann's Encyclopedia of Industrial Chemistry, 5th ed., Vol. A10 (Fibers), VCH-Verlag, Weinheim – Germany, 1987, pp. 463–474.
19. M. Harris, "Handbook of Textile Fibers", Harris Research Laboratories 1246 Washington, DC, 1954.
20. H. Kita, K. Okamoto, J. Appl. Polym. Sci., **31**, 1383 (1986).
21. R. B. Altmann, I. Renge, L. Kador, D. Haarer, J. Chem. Phys., **97**, 5316 (1992).
22. A. K. Gupta, R. P. Singhal et al., J. Appl. Polym. Sci., **26**, 3599 (1981); **27**, 4101 (1982).
23. A. K. Gupta, R. P. Singhal et al., J. Appl. Polym. Sci., **28**, 1167 (1983); **28**, 2745 (1983).
24. D. Porter, Polymer, **28**, 1652 (1987).
25. H. Ueda, S. H. Carr; Polymer J., **16**, 661 (1984).
26. M. D. Migahed, A. Tawansi, N. A. Bakr, Eur. Polym. J., **18**, 975 (1982).
27. A. P. Tyutnev et al., Phys. Status Solidi, **85** (2), 591 (1984).
28. V. M. Browning et al., Cryogenics, **36**, 391 (1996).
29. A. H. Bhuijan, S. V. Bhoraskar. Indian J. Phys., **62A**, 960 (1988).
30. A. H. Bhuijan et al., Thin Solid Films, **161**, 187 (1988).
31. H. Grünewald, H. S. Munro. Mat. Sci. Eng. A-139, 356 (1991).
32. P. A. Koch, Ed., Faserstoff–Tabellen, 1969. DK 677 494 745 32.
33. J. Luenenschloss, E. Hummel, Fasertafel Textilpraxis, 1967.
34. T. Benton Sevison (Jr.), Man-Made Fiber Chart, Textile World, New York 1964.
35. B. von Falkai, Synthesefasern, Verlag Chemie,Weinheim–Germany, 1981.
36. "Handbook of Fiber Science and Technology", Vol. IV, Marcel Dekker, New York/Basel, 1985.
37. F. Schulze - Gebhardt, Bayer AG, Chemiefasern/Texilindustrie, **43**, 432 (1993).
38. J. Nishihara, J. Furutani, M. Toramaru. T. Jasunaga, Mitsubishi Rayon Co. LTD, European Patent 0255109 A2 (03.02.1988).
39. R. M. A. M. Schellekens, P. J. Lemstra, Stamicarbon B. V., European Patent 0144983 A2 (19.06.1985).
40. K. Hisatani, H. Yamazaki, Asahi Kasei, European Patent 0397394 A2 (14.11.1990).
41. H. Hähne, Chemiefasern, **33**, 839 (1983).
42. H. Hähne, U. Schuster, Melliand. Textilberichte, **66**, June 1985.
43. K. K. J. Tong, W. C. Kenyon, J. Am. Chem. Soc., **69**, 2245 (1947).
44. I. M. Fonda et al., J. Text Inst., **5**, 378 (1987).
45. R. B. Beevers, J. Polym. Sci. A, **2**, 5257 (1964).
46. F. Francuskiewics, G. Glöckner. P. Kratochvil, Angew., Makrom. Chem., **206**, 121 (1993).
47. M. Minagawa, K. Miyano, T. Morita, F. Yoshii, Macromolecules, **22**, 2054 (1989).
48. M. Minagawa, H. Yamada, K. Yamaguchi F. Yoshii, Macromolecules, **25**, 503 (1992).
49. Y. Nakano, K. Hisatani, K. Kamide. Polym. Int., **35**, 249 (1994).
50. Y. Nakano, K. Hisatani, K. Kamide. Polym. Int., **35**, 207 (1994).
51. Y. Nakano, K. Hisatani, K. Kamide. Polym. Int., **36**, 87 (1995).
52. M. Minagawa, K. Nouchi, M. Tozuka, R. Chujo, F. Yoshii, J. Polym. Sci, Part A: Polym. Chem., **33**, 665 (1995).
53. M. Minagawa, T. Takasu, S. Shinozaki, F. Yoshii, N. Morishita, Polymer, **36**, 2343 (1995).
54. R. Yamadera, M. Murano, J. Polym. Sci. A-1, **5**, 1059 (1967); J. Polym. Sci., Polym. Lett., **5**, 333 (1967).
55. M. Murano, R. Yamadera, J. Polym. Sci. A-1, **6**, 843 (1968).
56. K. Matsuzaki, M. Okada, T. Uryu, J. Polym. Sci., Polym. Chem. Ed., **9**, 1701 (1971).
57. H. Balard, Dissertation, Universität Strasbourg, 1975.
58. M. Minagawa et al., Macromolecules, **27**, 3669 (1994).
59. K. Kamide, H. Yamazaki, K. Okayima, K. Hikichi, Polm. J., **17**, 1233 (1985).
60. K. Kamide, H. Yamazaki, K. Okayima, K. Hikichi, Polm. J., **17**, 1291 (1985).
61. K. Kamide, H. Yamazaki, Y. Miyazaki, Polm. J., **18**, 819 (1986).
62. L. J. Mathias, R. F. Coletti, Macromolecules, **24**, 5515 (1991).
63. J. Grobelung, P. Tekely, E. Turska, M. Sokól, Polymer, **22**, 1649 (1981); Polymer, **25**, 1415 (1984).
64. T. Thomsen, H. G. Zachmann, S. Korte, Macromolecules, **25**, 6934 (1992).
65. R. Yamadera, H. Tadokoro, S. Murakashi, J. Chem. Phys., **41**, 1233 (1964).
66. C. Y. Liang, G. Pearson, R. H. Marchessault, Spectrochim. Acta, **17**, 568 (1961); C. Y. Liang, "Newer Methods of Polymer Characterization, Interscience", New York, 1964, p. 47.
67. D. O. Hummel, F. Scholl, "Atlas der Kunststoff-Analyse", Carl Hanser Verlag, München, Verlag Chemie GmbH, 1968.
68. N. Grassie, J. N. Hay, J. Polym. Sci., **56**, 189 (1962).
69. C. A. Levine, G. H. Harris, J. Polym. Sci., **62**, 100 (1962).

70. A. Bernas, R. Bensasson, I. Rossi, P. Barchewitz, J. Chem. Phys., **59**, 1442 (1962).
71. D. Mathien, M. Defranceschi, G. Lécayon, J. Delhalle, Chem. Phys., **171**, 133 (1993); **188**, 183 (1994).
72. M. Minagawa, F. Yoshii et al., Macromolecules, **21**, 2387 (1988).
73. Houben - Weyl, Methoden der Organischen Chemie Makromolekulare Stoffe, Vol. E 20/2, Georg Thieme Verlag, Stuttgart, 1987, p. 1192.
74. C. E. Schildknecht, "Vinyl and Related Polymers", Wiley, New York, 1952, p. 270.
75. A. V. Rajulu, P. M. Sab, A. A. Askadskii, Ind. J. Chem., **33B**, 1105 (1994).
76. M. Minagawa, K. Miyano, T. Morita, F. Yoshii, Macromolecules, **22**, 2054 (1989).
77. M. L. Miller, J. Polym. Sci., **56**, 203 (1962).
78. H. Kobayashi, J. Polym. Sci. B, **1**, 299 (1963).
79. H. Kobayashi, Y. Fujisaki, Chem. High Polm., (Japan) **19**, 81 (1962).
80. K. Kamide., H. Yamazaki, Y. Miyazaki, Polym. J., **18**, 819 (1986).
81. M. Bercea, S. Ioan, B. C. Simionescu, C. I. Simionescu, Polym. Bull, **27**, 571 (1992).
82. R. Joseph et al., Polym. Int., **26**, 89 (1991).
83. C. Gonzales, F. Zamora, G. M. Guzmán, J. Macromol. Sci. - Phys. B, **26**, 257 (1987).
84. M. Salame, J. Polymer Sci., Symp., **45**, 1 (1973).
85. H. Takahashi, T. Sugimori, K. Aoki, H. Itoh, Mitsubishi Rayon Co. LTD, European Patent 0363960 (18.04.1990).
86. S. A. Stern, J. Polym. Sci. A-2, **6**, 1933 (1968).
87. S. M. Yagnyatinskaja et al., USSR. J. Phys. Chem., **44**, 1445 (1970).
88. L. M. Pritykin, J. Colloid Interf. Sci., **112**, 539 (1986).
89. P. Parashar et al., Polymer Testing, **11**, 185 (1992).
90. U. Gaur, S. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1065 (1982).
91. H. S. Bu, W. Aycock, B. Wunderlich, Polymer, **28**, 1165 (1987).
92. S. C. Ng, K. K. Chee, Polymer, **34**, 3870 (1993).
93. P. Bajaj, M. Padmanaban, Eur. Polym. J., **20**, 513 (1984).
94. M. W. Sabaa et al., Polym. Degrad. Stab., **23**, 257 (1989).
95. J. Brandrup, E. H. Immergut, "Polymer Handbook", 3rd. ed., Wiley, New York, 1989.
96. J. Appl. Polym. Sci., **54**, 827 (1994).
97. M. R. Padhye, A. V. Karandikar, Appl. Polym. Sci., **33**, 1675 (1987).
98. K. O'Driskoll, R. A. Sanayei, Macromolecules, **24**, 4479 (1991).
99. W. R. Krigbaum, N. Tokita, J. Polym. Sci., **43**, 467 (1960).
100. G. Hinrichsen, Angew. Makrom. Chem., **20**, 121 (1974).
101. N. S. Batty, D. Greig et al., Polymer, **24**, 258 (1983).
102. H. J. Kolb, E. F. Izard, J. Appl. Phys., **20**, 564 (1949).
103. G. P. Lanzl, quoted in Ref. 1.
104. W. H. Howard, J. Appl. Polm. Sci., **5**, 303 (1961).
105. C. E. Black, quoted in Ref. 1.

Physical Constants of Poly(vinyl chloride)*

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Birefringence See Stress Optical Coefficient.

Branching

Branching Content (Branches per 1000 Monomer Units (C_2)) as a Function of Polymerization Temperature

T_{polym} (°C)	-CH- | CH_2Cl	-CCl- | CH_2CH_2Cl	-CCl- | $CH_2CHClCH_2CH_2Cl$	H/Cl | -C-C-C- | C	Refs.
45	3.9	< 0.1	0.5	< 0.1	1[a]
55	4.2	0.2	0.6	0.2	1[a]
65	4.6	0.2	0.8	0.3	1[a]
80	4.9	0.3	1.3	0.3	1[a]
100	5.0	1.4	2.1	0.8	2[b]

All data from ^{13}C-NMR.
[a]Extrapolated to 0 subsaturation, reduced with Bu_3Sn as per Ref. 3.
[b]Reduced with Bu_3Sn as per Ref. 4.

Brittle to Ductile Transition Pressure (kbar) Value of 0.2 kbar, measured under tensile deformation of 10% per minute at 25°C. (5)

Coefficient of Thermal Expansion (K^{-1})
$< T_g$ before annealing, $6.6–7.3 \times 10^{-5}$ (6)
after annealing, 6.9×10^{-5}
$> T_g$ $17.0–17.5 \times 10^{-5}$
With plasticizer. (7)

Coefficient of Friction

PVC on Steel (8)

Plasticizer[a] (%)	Static	Dynamic
25.9	0.350	0.719
31	0.495	0.787
35.5	0.645	0.857
39.4	0.797	0.925

[a]Di-2-ethylhexyl phthalate (DOP), di-iso-decyl phthalate (DIDP), *n*-octyl-*n*-decyl phthalate (DNODP). A value of 0.45–60 for unplasticized PVC with steel has been reported (9). For further data see Ref. (10).

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Compressibility (MPa^{-1}) ($\times 10^6$) (11)

Uniaxially Stretched PVC

Elongation (%)	$\gamma_{\parallel}$	$\gamma_{\perp}$
85	4.88	8.0
125	4.25	8.45
160	3.57	8.75

See also Refs. 12,13, and 36.

Creep (14,15)

Crystallinity (%)
From Density Measurements (16)

T_{polym} (°C)	Crystallinity (%)
90	11.3[a]
55–60	11.3
50	13.2
20	15.0
−15	57.3
−75	84.2

[a]Assumes amorphous density equals 1.385 g/cm^3 and crystalline density equals 1.44 g/cm^3 (19). Since the crystalline density for highly crystalline (e.g., single crystal) PVC is considerably greater than 1.44 g/cm^3; as shown in the table on crystallographic data, the above % crystallinity values for the low temperature polymerized PVCs are greatly overestimated. As pointed out by Kostyuchenko and co-workers (20), if a crystalline density of 1.497 g/cm^3 or greater is used, the calculated percent crystallinity values agree better with X-ray diffraction measurements.

Crystallinity cont'd

From X-Ray Diffraction Measurements

T_{polym} (°C)	Crystallinity[a] (%) (17)	T_{polym}	Crystallinity[b] (%) (18,40)
90 frac. 1	10.4	50	13[c]
frac. 3	5.4	−20	17
53 frac. 1	10.4	−40	20.5,20
A	6.5	−60	23,25
B	8.2		
25	12.2	polym. in propionaldehyde	33[c]
−60	23.7		
	34.7[c]	polym. in *n*-butyraldehyde	

[a] Values obtained using the X-ray diffraction method of Hermans and Weidinger (21) and a double hump amorphous curve.
[b] X-ray diffraction method with Lorentz-polarization and atomic scattering factor corrections.
[c] Using a single hump amorphous X-ray curve and the method of Hermans and Weidinger (21), Lebedev and co-workers (22) (see also Ref. 17) report several commercial PVC crystallinites in the range 20–27%. In general, their method gives higher values than that of D'Amato and Strella (17).

From IR Measurements

Crystallinity from IR measurements deemed (by the authors) to be less accurate than NMR are in references 42, 44–46 and 56.

From Calorimetric Measurements (Unfractionated Polymers) (23)

M_n	T_{polym} (°C)	Crystallinity (%)
23,200	75	18.4
38,700	65	15.5
53,500	52	15.3
66,700	52	14.4
136,000	25	11.9
155,000	25	11.8

Crystallographic Data See under Unit Cell.

Density (g/cm^3)

Function of Polymerization Temperatures (16)

T_{polym} (°C)	M_n	Density (20°C)
90	23,750	1.391
55–60	75,000	1.391
50	91,250	1.392
20	172,250	1.393
−15	106,300	1.416
−75	105,300	1.431

Dependence on Thermal History (24)

History	Density (20°C)
Original state	1.3743
Quenched from 200°C in ice water	1.3656
Quenched from 130 to −70°C	1.3716
Quenched from 130 to −70°C and kept at RT for 6.5 h	1.3743
Annealed at 65°C for 136 h	1.3745
Annealed at 90°C for 136 h and slowly cooled	1.3834
Annealed at 90°C for 136 h and quenched to −20°C	1.3817

Effect of Temperature on Density (25,38)

T (°C)	Density
20	1.392
51	1.383
82	1.368
90.6	1.362
97	1.357

Dielectric Properties (26)

Dielectric Constant (ε')

	T (°C)								
	25	40	60	80	90	100	110	120	140
60 Hz	3.50	3.51	3.70	4.25	6.30	10.30	11.89	12.05	11.76
1 kHz	3.39	3.40	3.61	4.09	5.05	7.77	10.21	11.30	11.27
10 kHz	3.29	3.34	3.45	3.89	4.45	5.77	8.50	9.96	10.94

Dielectric Loss Factor (ε'')

	T (°C)								
	25	40	60	80	90	100	110	120	140
60 Hz	0.110	0.116	0.125	0.172	0.410	1.20	0.675	0.481	1.65
1 kHz	0.081	0.081	0.080	0.120	0.500	1.415	1.645	0.630	0.319
10 kHz	0.058	0.058	0.050	0.110	0.920	1.37	1.35	1.22	0.490

T_{polym} = 50°C; $[\eta]$ = 1.17; M_n = 66,700; M_w = 162,000. See also Ref. 27.

Dynamic Viscosity (122,134)

Elongation at Break (%) (28)

T_{polym} (°C)	Test temp. (°C)	$[\eta]^a$	Strain rate (s^{-1}) 0.0026	0.020	0.20	2.0	20	200
70	25	68	171.5	18.5	13.5	16.5	17.0	13.3
65	25	75	194.5	20.0	14.2	21.0	19.5	16.0
56	25	91.3	196.7	21.0	18.0	21.0	19.5	16.0
50	25	116.9	210.7	23.0	16.5	17.0	18.5	17.0
70	60	68		71.6	20.0	20.0	16.5	24.0
65	60	75.0		160	23.8	22.5	22.5	27.5
56	60	91.3		207	26.0	25.0	22.0	30.0
50	60	116.9		243	27.0	26.5	22.5	35.0
70	80	68			168	82	60	41
70	100	68				240	168	157

[a] Intrinsic viscosity (ml/g) in cyclohexanone, 30°C.

Enthalpy Entropy as a function of pressure and temperature. (29)

Fatigue (135,136)

First Normal Stress Difference (118,119,145)

Flory–Huggins Parameter See Polymer–Solvent Interaction Parameter.

Flow Activation Energy ΔE_γ (kJ/mol) (30,31,32,33,34)

Dependence on Temperature and Molecular Weight

T_{polym} (°C)	LVN[a]	ΔE_γ (kJ/mol) $10\,s^{-1}$	$100\,s^{-1}$	$1000\,s^{-1}$
5	0.131	–	150.7	134.9
5	0.693	92.1	119.7	60.2
5	0.909	165.5	103.1	54.2
5	1.750	43.4	30.5	30.4
40	0.640	178.4	102.8	54.0
40	0.682	157.4	105.2	45.0
40	1.225	108.6	57.7	34.6
70	0.495	147.7	97.0	52.4
70	0.680	126.9	84.3	43.6

[a] Limiting viscosity number in cyclohexanone, 30°C.

Dependence on Shear Rate

Shear rate (s^{-1})	>220°C	190–220°C	<190°C
0.3	95.6	363.7	152.4
3	–	244.8	145.7
30	–	159.6	121.7

Glass Transition Temperature See also corresponding chapter in this Handbook.

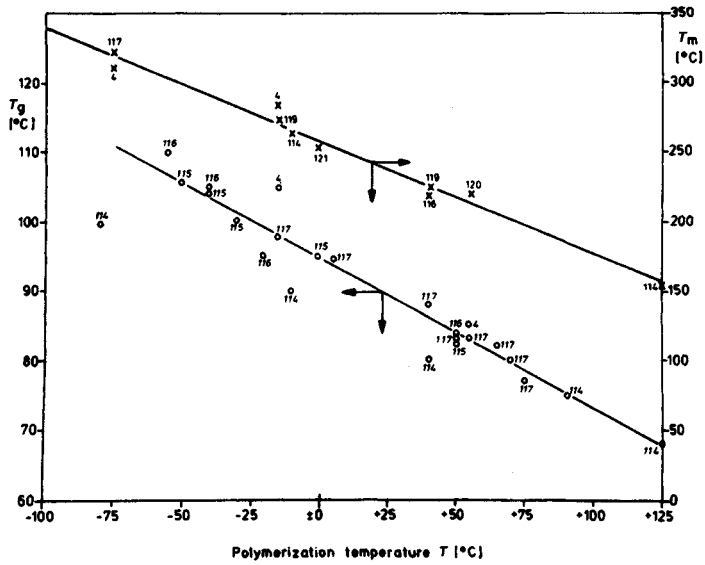
Effect of Pressure: dT_g/dP (°C/atm) 0.013 (35–38)

Effect on Molecular Weight and Polymerization Temperature (39)

T_{polym} (°C)	Intrinsic viscosity $[\eta]^a$ 50	75	100	125	150
70	78.5	82	83	83.5[b]	84.5[b]
50	80	84	85	86	86.5
40	81	85	87	88.5	89
30	84	87	89	89	90
20	85	88	90	90.5	91
5	86	91	93	94.5	95.5
−15	92	87	100	101.5	102[b]

[a] In cyclohexanone, 30°C (ml/g).
[b] Extrapolated from experimental data.

Effect of Polymerization Temperature on T_g and T_m (numbers give Refs.).



The lines through the data points represent the best least squares fit of the collective authors' data.

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Glass Transition Temperature *cont'd*

Dependence on Polymerization Temperature (47)

		T_g (°C)	
T_{polym} (°C)	$[\eta]$ (ml/g)[a]	Initial	Final[b]
90		80	78
50	80	85	83
0	108	97	88
−20	103	100	90
−30	125	100	91
−40		101	90
−50		105	91
−60	90	107	95

[a] $[\eta]$ measured at 25°C in cyclohexanone.
[b] After thermal cycling to 230°C. See also Ref. 48.

Heat Capacity See corresponding table in this Handbook and Refs. 49–52.

Effect of Thermal Cycle on C_P (47)

		ΔC_P (×10²) (J/g/°C)	
T_{polym} (°C)	$[\eta]$ (ml/g)[a]	Initial	Final[b]
90		28.9	29.3
50	80	27.2	27.6
0	108	24.3	25.1
−20	103	15.1	22.2
−30	125	10.9	20.9
−40		10.5	20.1
−50			18.0
−60	90	6.3	16.3

[a] $[\eta]$ measured at 25°C in cyclohexanone. [b] After heating to 140°C.

Heat of Combustion (kJ/kg) −19000. (53)

Heat of Fusion (kJ/mol)

11.3		(41)
12.65		(54)
2.76		(55)
3.28	(most probable value according to Ref. 56)	(16)
3.56		(57)
3.91		(58)

Heat of Dilution 30°C PVC-tetrahydrofuran (23)
PVC-cyclohexanone

Heat of Polymerization (kJ/mol) −96 to −109 (59–61)

Huggins Coefficient (62)

Intrinsic Viscosity – Molecular Weight Relationship See corresponding chapter in this Handbook and Refs. 63–89.

Mark–Houwink Parameters for PVC as a Function of Temperature (115)

Solvent	Temp. range (°C)	*a*	*K* (ml/g)
Cyclohexanone	20–60	0.803	$1.847 \times 10^{-2} - 4.85 \times 10^{-5}\ T$
Cyclopentanone	20–60	0.861	$9.086 \times 10^{-5} - 1.55 \times 10^{-5}\ T$
THF	20–50	0.851	$1.087 \times 10^{-2} - 1.67 \times 10^{-5}\ T$

k-values can be calculated for any temperature *T* (°C) within the temperature range given.

Melt Viscosity See Appendix 1 and Refs. 31–34,118, 119,144.

Permeability See chapter "Permeability Coefficients, Diffusion Constants, and Solubility Coefficients of Polymers" in this Handbook, and Refs. 24,92,93–99.

Diffusion Constants in PVC

Gas	$\log_{10}D$ (cm^2/s)	E_D (kJ/mol)
(A) GASES, AT 25°C (92)		
He	−5.5	20.7
H_2	−6.3	34.5
Ne	−6.6	31.5
N_2	−8.4	62.0
Ar	−8.9	51.5
O_2	−7.9	54.4
CO_2	−8.6	64.5
CH_4	−8.9	70.3
H_2O	−7.6	41.8
Kr	−9.4	62.8
(B) ORGANIC VAPORS, AT 30°C, LOW ACTIVITY (90)		
n-C_4H_{10}	−13.6	
n-C_5H_{12}	−13.9	81.6
n-C_6H_{14}	−14.9	
C_6H_6	−13.7	
CH_3OH	−10.4	
C_2H_5OH	−12.4	
n-C_3H_7OH	−13.2	
n-C_4H_9OH	−13.9	
CH_3Cl	−11.3	59.9
CCl_4	−17.1	108.4
C_2H_3Cl	−11.7	71.2
$(CH_3)_2CO$	−12.8	
SF_6	−16.1	100.0

Poisson Ratio 0.38. (100)

Polymer–Solvent Interaction Parameter See corresponding chapter in this Handbook and Refs. 68,101–114.

Refractive Index PVC ($\rho = 1.384\ g/cm^3$) (14)

Wave length (nm)	Refractive index
486.1	1.54806
589.3	1.54151
656.3	1.53843

Reciprocal dispersive power V_d 59.3
Critical angle ($\lambda = 589.3$ nm) 56.23
Temperature coefficient of refractive index 0.0001142°/°C.

Specific Heat Capacity See chapter "Heat Capacity" in this Handbook, and Refs. 49,50.

Second Virial Coefficient See corresponding chapter in this Handbook and Refs. 43,77,84,116,117.

Shear Modulus (120–122,134)

Dynamic Test Results for Rigid PVC Formulation at Frequencies = 0.06–600 (rad/s) and at Temperatures 140–220°C[a]

	220°C		210°C		200°C		190°C	
ω (rad/s)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)
5.9749 E−2	3.669 E3	3.442 E3	1.256 E4	9.933 E3	3.205 E4	1.684 E4	6.882 E4	2.711 E4
9.4700 E−2	4.775 E3	4.472 E3	1.493 E4	1.096 E4	3.586 E4	1.827 E4	7.470 E4	2.782 E4
1.5008 E−1	6.028 E3	5.736 E3	1.813 E4	1.285 E4	4.153 E4	1.998 E4	7.971 E4	2.941 E4
2.3787 E−1	7.644 E3	7.047 E3	2.201 E4	1.468 E4	4.727 E4	2.245 E4	8.955 E4	3.137 E4
3.7700 E−1	1.012 E4	8.847 E3	2.732 E4	1.706 E4	5.594 E4	2.530 E4	1.011 E5	3.389 E4
5.9749 E−1	1.272 E4	1.096 E4	3.177 E4	1.999 E4	6.340 E4	2.771 E4	1.111 E4	3.598 E4
9.4700 E−1	1.606 E4	1.302 E4	3.859 E4	2.283 E4	7.154 E4	3.031 E4	1.238 E5	3.772 E4
1.5008 E0	1.976 E4	1.684 E4	4.488 E4	2.758 E4	7.935 E4	3.448 E4	1.304 E5	4.199 E4
2.3787 E0	2.506 E4	2.060 E4	5.404 E4	3.205 E4	9.033 E4	3.805 E4	1.440 E5	4.517 E4
3.7700 E0	3.109E4	2.477 E4	6.340 E4	3.645 E4	9.976 E4	4.147 E4	1.510 E5	4.817 E4
5.9749 E0	3.859 E4	3.031 E4	7.374 E4	4.309 E4	1.102 E5	4.859 E4	1.625 E5	5.432 E4
9.4700 E0	4.652 E4	3.645 E4	8.515 E4	4.838 E4	1.218 E5	5.204 E4	1.751 E5	5.526 E4
1.5008 E1	5.650 E4	4.001 E4	9.692 E4	5.166 E4	1.369 E5	5.391 E4	1.876 E5	5.885 E4
2.3787 E1	6.862 E4	4.549 E4	1.103 E5	5.638 E4	1.512 E5	5.885 E4	2.050 E5	6.013 E4
3.7700 E1	8.120 E4	5.219 E4	1.261 E5	6.142 E4	1.663 E5	6.413 E4	2.173 E5	6.552 E4
5.9749 E1	9.904 E4	5.885 E4	1.473 E5	6.810 E4	1.893 E5	7.108 E4	2.431 E5	7.293 E4
9.4700 E1	1.177 E5	6.552 E4	1.684 E5	7.139 E4	2.136 E5	7.618 E4	2.616 E5	7.957 E4
1.5008 E2	1.461 E5	7.293 E5	1.934 E5	7.947 E4	2.400 E5	8.587 E4	2.852 E5	8.369 E4
2.3787 E2	1.699 E5	8.475 E4	2.249 E5	9.039 E4	2.673 E5	9.435 E4	3.136 E5	9.848 E4
3.7700 E2	2.136 E5	9.848 E4	2.560 E5	1.028 E5	3.109 E5	1.073 E5	3.539 E5	1.120 E5
5.9749 E2	2.673 E5	1.189 E5	3.109 E5	1.236 E5	3.696 E5	1.444 E5	4.081 E5	1.594 E5

	178°C		170°C		160°C		140°C	
ω (rad/s)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)	G′ (Pa)	G″ (Pa)
5.9749 E−2	1.055 E5	3.137 E4	1.484 E5	3.724 E4	1.802 E5	3.971 E4	2.613 E5	4.236 E4
9.4700 E−2	1.150 E5	3.360 E4	1.625 E5	3.756 E4	1.850 E5	4.006 E4	2.658 E5	4.272 E4
1.5008 E−1	1.201 E5	3.507 E4	1.697 E5	3.821 E4	1.949 E5	4.058 E4	2.693 E5	4.291 E4
2.3787 E−1	1.352 E5	3.645 E4	1.834 E5	3.904 E4	2.124 E5	4.128 E4	2.751 E5	4.349 E4
3.7700 E−1	1.471 E5	3.821 E4	1.965 E5	4.075 E4	2.256 E5	4.291 E4	2.873 E5	4.537 E4
5.9749 E−1	1.570 E5	4.058 E4	2.078 E5	4.327 E4	2.386 E5	4.517 E4	3.000 E5	4.716 E4
9.4700 E−1	1.675 E5	4.199 E4	2.198 E5	4.365 E4	2.449 E5	4.615 E4	3.039 E5	4.903 E4
1.5008 E0	1.842 E5	4.615 E4	2.295 E5	4.817 E4	2.524 E5	5.051 E4	3.118 E5	5.283 E4
2.3787 E0	1.948 E5	4.817 E4	2.417 E5	5.029 E4	2.751 E5	5.386 E4	3.242 E5	5.621 E4
3.7700 E0	2.025 E5	5.226 E4	2.524 E5	5.456 E4	2.873 E5	5.744 E4	3.356 E5	5.940 E4
5.9749 E0	2.198 E5	5.769 E4	2.635 E5	6.021 E4	2.987 E5	6.205 E4	3.459 E5	6.357 E4
9.4700 E0	2.328 E5	5.844 E4	2.711 E5	6.506 E4	3.073 E5	6.703 E4	3.468 E5	6.829 E4
1.5008 E1	2.452 E5	6.217 E4	2.816 E5	6.523 E4	3.177 E5	6.865 E4	3.601 E5	6.939 E4
2.3787 E1	2.662 E5	6.552 E4	3.030 E5	6.987 E4	3.361 E5	7.169 E4	3.809 E5	7.293 E4
3.7700 E1	2.791 E5	7.139 E4	3.177 E5	7.452 E4	3.434 E5	7.712 E4	3.943 E5	7.959 E4
5.9749 E1	3.069 E5	7.613 E4	3.434 E5	7.947 E4	3.776 E5	8.293 E4	4.170 E5	8.587 E4
9.4700 E1	3.177 E5	8.296 E4	3.617 E5	8.585 E4	4.029 E5	9.049 E4	4.488 E5	9.354 E4
1.5008 E2	3.464 E5	8.848 E4	3.977 E5	9.039 E4	4.298 E5	9.517 E4	4.893 E5	9.899 E4
2.3787 E2	3.744 E5	1.015 E5	4.261 E5	1.028 E5	4.606 E5	1.079 E5	5.321 E5	1.111 E5
3.7700 E2	4.117 E5	1.144 E5	4.586 E5	1.199 E5	–	–	–	–
5.9749 E2	–	–	–	–	–	–	–	–

[a] Storage, G', and Loss, G'', are in Pascal. Data from L. A. Utracki, SPE Techn. Papers **31**, 1024 (1985); J. Vinyl. Technol. **7** (4), 150 (1985). M_w of PVC tested is 90.700 (GPC in THF), LVN 30°C in 93 ml/g cyclohexanone.

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Specific Refractive Index Increment See corresponding chapter in this Handbook.

Specific Volume

As a function of temperature and pressure. (29)

Spectral Data

a. Infrared absorption bands. (123)
b. Nuclear magnetic resonance (high resolution spectra). (124–126)
c. Carbon-13 magnetic resonance (chemical shift assignment) (127–129)

Stress Optical Coefficient (soc)

−200 to +50 −6.5 (132,133)

The value of the SOC becomes positive at T_g, but there is some arbitrariness associated with it because above T_g the plots of birefringence change with stress and have an S shape.

Tacticity (Fraction of syndiotactic dyads) **Dependence on Polymerization Temperature**[a]

T_{polym} (°C)	Tacticity[b]	Refs.
55	0.55	130
50	0.55[c]	129
25	0.57	130
0	0.60	130
−30	0.64	130
−50	0.66	130
−76	0.68	130

[a]Data based on ^{13}C NMR measurements.
[b]High tacticity data may be low due to insolubility during measurement as discussed in Ref. 131.
[c]Bernoullian order.

Tensile Modulus (MPa) = (N/mm^2)

T (°C)	Value	Refs.
−196°C	7584	137[a]
−120°C	5171	137
−75°C	3861	137
20°C	2964	137
30°C	3000	14[b]
40°C	2930	14
50°C	2427	14
60°C	1551	14
70°C	276	14

[a]Stress–Strain measurements at strain rate of 0.00250 s^{-1}.
[b]Measured in creep (100 s, 0.2% strain).

Thermal Conductivity (W/m/K) (138,140)

T (°C)	Thermal conductivity
−170	0.129
−150	0.134
−125	0.139
−100	0.144
−75	0.148
−50	0.152
−25	0.155
0	0.158
20	0.160
30	0.161
40	0.162
50	0.163
60	0.164
70	0.164
80	0.165
90	0.165
100	0.165

Thermal conductivity of system PVC/di-2-ethylhexyl phthalte (see Ref. 139).

Molecular weight, sample polymerization temperature and syndiotactivity do not influence the thermal conductivity of polymers appreciably, except in the case where tacticity leads to crystallization.

Thermal Diffusivity (140)

T (K)	Density (g/cm^3)	Thermal diffusivity ($\times 10^{-4}$) (cm^2/s)
200	1.417	14.40
220	1.415	13.95
240	1.413	13.46
260	1.409	12.96
273	1.407	12.60
280	1.405	12.42
293	1.402	11.92
300	1.400	11.83
320	1.393	11.13
340	1.385	10.14
352	1.379	9.22
360	1.374	8.41
380	1.334	7.65
400	1.306	7.30
420	–	7.13
440	–	7.04
460	–	6.99
480	–	7.06

Unit Cell

PVC sample type	Crystal system	Space group	Unit cell parameters (Å)			Monomers per unit cell	Calc. density (g/cm^3)	Refs.
			a	*b*	*c*			
Commercial, polymerized at 50–60°C	Orthorhombic	Pacm	10.6	5.40	5.10	2	1.44	19
Solution blended high molecular weight, low crystallinity commercial polymer and low molecular weight, high crystallinity polymer	Orthorhombic	Pacm	10.4	5.30	5.10	2	1.48	141
Single crystals, polymerized at −75°C	Orthorhombic		10.32[a]	5.32[a]	–	–	(1.49)[b]	142
Single crystals, low molecular weight, polymer made in *n*-butyraldehyde	Orthorhombic	Pacm	10.24	5.24	5.08	2	1.53	143

[a]Calculated from published *d*-spacings of major diffraction peaks.
[b]Calculated density assuming $c = 5.10$ (Å) and 2 monomers per unit cell.

Unperturbed Dimensions See corresponding chapter in this Handbook.

Viscosity–Molecular Weight Relationship See Intrinsic Viscosity.

Zero Shear Viscosity (34)

APPENDIX 1: APPARENT MELT VISCOSITY OF UNMODIFIED PVC[a] PREPARED AT VARIOUS POLYMERIZATION TEMPERATURES[b] (PAS) ($\times 10^4$)

T_{polym} (°C)	L.V.N.[c] (ml/g)	Melt temp. (°C)	Shear rate (s^{-1})									
			2.95	7.37	14.7	29.5	73.7	147	295	737	1470	2950
110	30.5	190	–	–	–	1.18	0.72	0.82	0.77	0.61	0.52	0.40
110	30.5	205	–	–	–	0.63	0.39	0.48	0.30	0.22	0.21	0.20
70[d]	40.1	140	1128.3	573.1	340.7	202	97.85	54.6	30.0	13.3	6.41	3.08
70	40.1	150	639.83	332.2	200.6	120.8	60.5	35.6	20.44	9.45	4.91	2.47
70	40.1	160	281.4	153.9	97.1	62.4	34.4	21.8	13.41	6.56	3.74	20.4
70	40.1	170	100.9	62.08	42.2	28.7	16.9	11.35	7.65	4.37	2.72	1.62
70	40.1	180	32.12	21.2	15.5	11.42	7.85	5.78	4.38	2.80	1.94	1.27
70	40.1	190	11.95	7.22	5.94	5.15	4.03	3.24	2.65	1.87	1.35	0.95
70	40.1	200	9.03	3.82	2.02	1.86	2.02	1.92	1.4	1.21	0.95	0.68
70	40.1	210	–	5.31	2.71	1.38	0.89	1.02	0.935	0.775	0.65	0.504
70[e]	49.5	140	2336.4	1209.8	721.6	423.5	191.6	102.1	51.11	20.2	9.29	4.24
70	49.5	150	1433.6	711.03	424.5	223.0	119.92	64.7	34.5	15.2	7.43	3.50
70	49.5	160	839.9	431.9	258.9	155.3	76.6	43.8	24.2	10.72	5.47	2.71
70	49.5	170	398.4	231.5	142.9	89.6	47.8	29.1	17.2	8.14	4.43	2.32
70	49.5	180	146.6	91.9	58.7	43.2	26.1	17.3	11.2	5.92	3.49	1.98
70	49.5	190	55.34	34.3	25.5	19.4	13.4	9.75	6.58	3.99	2.56	1.52
70	49.5	200	22.1	13.7	11.3	9.5	7.36	5.87	4.48	2.95	2.02	1.33
70	49.5	210	12.7	7.53	5.54	4.54	3.99	3.43	2.82	2.06	1.5	1.04
70	54.8	210	16.23	8.66	8.12	7.31	6.71	5.74	4.71	2.94	20.3	1.31
70	54.8	220	–	–	–	5.26	4.29	3.82	3.24	2.39	1.72	1.18
70	68	160	2464.2	1172.2	705.5	389.9	173.66	89.55	45.76	17.69	8.47	4.01
70	68	170	1196.9	599.14	359.27	218.53	113.21	64.04	34.39	14.44	7.16	3.49
70	68	180	562.8	320.5	205.9	125.6	64.7	37.7	21.8	10.3	5.52	2.92
70	68	190	244.2	144.9	100.8	69.8	40.1	25.5	15.5	7.75	4.40	2.44
70	68	200	94.2	62.1	46.6	35.0	23.7	16.6	11.0	5.94	3.55	2.07
70	68	210	48.1	29.5	23.5	19.6	14.6	11.1	8.02	4.68	2.92	1.73
70	68	220	19.4	15.5	13.2	11.7	9.34	7.51	5.71	3.62	2.51	1.57
65	75.4	180	976.7	522.7	318.4	195.6	103.5	58.7	31.26	13.3	6.87	3.46
65	75.4	190	545.1	254.7	158.9	101	53.6	31.9	19.0	9.25	5.01	2.63
65	75.4	200	271.1	128.5	85.3	59.8	36.3	23.1	14.2	7.2	4.15	2.32
65	75.4	210	95.6	56.7	43.7	33.1	22.1	15.7	10.5	5.53	3.36	2.02
65	75.4	220	52.6	33.2	25.9	21.4	14.9	10.96	7.75	4.49	2.73	1.64
65	86	180	930	448.5	273.3	198	84	45.1	23.22	9.77	4.92	2.46
65	86	190	686.2	381	243.6	148	82	48.1	25.5	11.1	5.7	2.93

T_{polym} (°C)	L.V.N.[c] (ml/g)	Melt temp. (°C)	Shear rate (s^{-1}) 2.95	7.37	14.7	29.5	73.7	147	295	737	1470	2950
65	86	200	229.5	138.7	100.6	67.75	38.75	34.02	13.67	6.77	3.77	2.01
65	86	210	163.9	94.96	71.05	53.3	33.1	21.4	13.5	7.23	4.37	2.47
65	86	220	–	–	–	26.25	17.13	11.8	7.88	4.22	2.56	1.56
65	91.3	180	1898.2	902.1	573.1	292.0	124.2	65.3	35.7	14.86	7.43	3.66
56	91.3	190	1139.9	584.7	373.8	250.4	113.5	60.1	31.5	13.5	6.87	3.53
56	91.3	200	857.7	350	207.1	130	74	43.3	23.4	10.2	5.45	2.85
56	91.3	210	286.9	155.0	112.0	79.2	45.7	28.5	18.7	9.08	4.81	2.54
56	91.3	220	154.9	90.8	55.9	49.5	31.3	18.5	13.1	7.21	4.27	2.37
56	91.3	230	92.7	60.5	48.7	37.7	24.6	17.0	11.1	6.0	3.98	2.48
56	91.3	240	82.3	11.9	54.7	44.6	31.0	24.1	20.1	10.0	6.367	3.611
50	116.9	180	1884	933.9	562.5	304	126.3	67.12	36.11	14.86	7.32	3.39
50	116.9	185	1452.2	730.1	464.3	284.1	126.8	66.1	34.25	14.22	7.0	3.4
50	116.9	190	1062	633.6	398	251.7	116.2	59.96	30.8	13.16	6.58	3.24
50	116.9	200	783.2	410.7	257.4	177.9	87.02	45.1	24.03	10.61	5.57	2.92
50	116.9	205	584.1	318.4	209.6	131.4	78.9	43.5	22.83	9.87	5.25	2.79
50	116.9	210	424.8	226.0	156.0	105.1	59.43	39.3	21.64	9.29	4.93	2.60
50	116.9	220	265.5	147.5	106.12	75.4	44.4	28.02	18.8	8.49	4.62	2.51
50	116.9	230	–	132.9	79.7	58.1	36.0	23.2	14.7	8.34	4.67	2.57
42	147.2	190	1878.6	897.0	537.1	276.7	113.5	58.4	31.8	13.3	6.36	3.04
42	147.2	200	1626.9	703.2	420.8	252.1	108.5	55.9	28.2	12.3	6.15	3.07
42	147.2	210	1352.7	567.6	319.7	206.3	93.3	48.9	25.3	10.7	5.52	2.77
42	147.2	220	968.4	415.3	231.5	142.5	73.5	39.9	21.2	9.19	4.93	2.85
42	147.2	230	–	299	164.5	106.2	60.2	35.2	19.0	8.44	4.55	2.48
42	147.2	240	–	243.6	199.3	157.7	89.7	53.2	33.5	–	–	–
40	122.5	190	2041.3	867.8	511.0	285.6	117.9	58.5	29.2	12.7	6.45	3.14
40	122.5	200	1162	–	323	–	80.8	41.5	21.0	9.09	–	–
40	122.5	205	445.4	244.5	170.5	111.5	64.4	36.6	19.4	8.35	4.37	2.25
40	122.5	210	–	–	–	98.4	53.5	28.7	14.76	6.28	3.28	1.72
40	122.5	215	338.8	185.6	133.4	94.0	53	35.7	20.2	8.73	4.56	2.39
28	208	190	2050.0	916.5	543.5	284.9	117.8	63.7	33.55	14.92	7.25	3.35
28	208	200	1658.7	740.6	452.3	239.6	102.8	52.2	27.4	11.94	6.07	2.87
28	208	210	1104.9	493.4	282.4	167.7	85.5	46.9	25.03	10.44	5.33	2.66
28	208	220	665.6	332.5	214.7	138.4	69.3	39.5	22.1	9.96	5.33	2.73
25	240.5	200	1513.3	647.4	387.4	223.0	94.98	49.08	26.55	11.67	5.84	2.81
25	240.5	210	1503	603	343	200	90.2	46.6	23.5	10.4	5.4	2.7
25	240.5	215	1340.7	562.5	311.5	180	80.44	41.76	21.4	9.49	5.2	2.6
25	240.5	220	1355.7	572.6	332.2	190.9	83.3	43.9	22.5	9.63	5.07	2.57
25	240.5	225	958.4	421.3	250.9	159.3	75.35	40.33	21.03	9.23	4.92	2.60
25	240.5	230	–	409.8	222.0	136.1	67.7	37.5	19.7	8.86	4.62	2.46
25	240.5	240	–	426.4	223.2	132.3	64.2	36.5	20.2	8.53	4.52	2.53

[a]Ref. original data, E. A. Collins, unpublished.
[b]Melt viscosity data obtained using a capillary having a 90° entrance angle, length 1.0 in and diameter 0.05 in. No correction applied to the data. All samples contained 2.5 parts dibutyltin dioctylthioglycolate stabilizer per 100 parts resin.
[c]Limiting viscosity number, determined in cyclohexanone 30°C.
[d]Modified with 5 parts trichloroethylene/100 parts vinyl chloride monomer.
[e]Modified with 3 parts trichloroethylene/100 parts vinyl chloride monomer.

REFERENCES

1. T. Hjertberg, E. M. Sorvik, ACS Symposium Series, vol. 280, Polymer Stabilization, 1985.
2. W. H. Starnes, F. C. Schilling, I. M. Plitz, R. E. Cais, D. J. Freed, R. L. Hartless, F. A. Bovey, Macromolecules, **16**, 790 (1983).
3. T. Hjertberg, A. Wendel, Polymer, **23**, 1641 (1982).
4. W. H. Starnes, R. L. Hartless, F. C. Schilling, F. A. Bovey. Polym. Preprints, 18 (1), 499 (1977); Adv. Chem. Ser., **169**, 234 (1978).
5. J. Yuan, A. Hiltner, E. Baer, J. Mater. Sci., **18**, 3063 (1983).
6. M. L. Dannis, Mod. Plastics, **31** (7), 120 (1954).
7. L. Dunlap, J. Polym. Sci. C, **30**, 561 (1970).
8. J. B. DeCoste, SPE Antec, **15**, 232 (1969).
9. A. B. Glanvill, "The Plastics Engineer's Data Book", The Machinery Publishing Co., Ltd., Brighton, 1971, p. 181.
10. I. Klein, Plastics World, May 1980, p. 110.
11. J. Hennig, Kolloid-Z., **202** (2), 127 (1965).
12. Y. Yamaguchi, Y. O. Yanagi, Kobunshi Kagaku, **28** (315), 623 (1971).
13. S. K. Bhateja, K. D. Pae, J. Macromol. Sci.: Revs. Macromol. Chem., C, **13** (1), 77 (1975).
14. R. M. Ogorkiewicz, "Engineering Properties of Thermoplastics", Wiley, New York, 1970, p. 251.
15. S. Turner, Brit. Plastics, **37**, 682 (1964).

16. A. Nakajima, H. Hamada, S. Hayashi, Makromol. Chem., **95**, 40 (1966).
17. R. J. D'Amato, S. Strella, Appl. Polym. Symp. **8**, 275 (1969).
18. E. Repko, M. Tureckova, J. Halamek, Petrochemia, **14** (1), 25 (1974).
19. G. Natta, P. Corradini, J. Polym. Sci. **20**, 251 (1956).
20. A. N. Kostyuchenko, N. A. Okladnov, V. P. Lebedev, J. Polym. Sci. USSR, **10** (11), 3025 (1968).
21. P. H. Hermans, A. Weidinger, Makromol. Chem., **44–46**, 24 (1961).
22. V. P. Lebedev, N. A. Okladnov, K. S. Minsker, B. P. Shtarkman, Vysokomol. Soedin., **7** (4), 655 (1965).
23. S. H. Maron, F. E. Filisko, J. Macromol. Sci. B, **6** (2), 413 (1972).
24. K. H. Illers, Makromol. Chem., **127**, 1 (1969).
25. J. Malac, Sb. Prednasek, Makrotest, 4th Celostatni Konf. vol. 2, 345 (1976).
26. H. J. Dietrick, M. L. Dannis, B. F. Goodrich Research Center, Brecksville, Ohio, unpublished data.
27. L. A. Utracki, J. A. Jukes, J. Vinyl Tech., **6** (2), 85 (1984).
28. H. H. Bowerman, Proceedings of Fourth Annual Plastics Conference, Eastern Michigan University, Sept. 24, 1969. H. H. Bowerman, unpublished data.
29. R. G. Griskey, C. A. Gellner, M. W. Din, Mod. Plastics, **43**, 119 (1966).
30. E. A. Collins, Pure Appl. Chem. **49**, 581 (1977).
31. E. A. Collins, C. A. Krier, Trans. Soc. Rheol. **11** (2), 225 (1967).
32. E. A. Collins, A. P. Metzger, Polym. Eng. Sci., **10** (2), 57 (1970).
33. L. A. Utracki, J. Polym. Sci. Polym. Phys. Ed., **12**, 563 (1974).
34. J. Lyngaae-Jorgensen, J. Appl. Polym. Sci., **20**, 2497 (1976).
35. L. A. Chandler, E. A. Collins, B. F. Goodrich Chemical Company, Technical Center, Avon Lake, Ohio, unpublished data.
36. P. Heydemann, H. D. Guicking, Kolloid-Z., **193**, 16 (1963).
37. U. Bianchi, A. Turturro, G. Basile, J. Phys. Chem., **71**, 3555 (1967).
38. K. H. Hellwege, W. Knappe, P. Lehmann, Kolloid-Z., **183** (2), 110 (1962).
39. E. A. Collins, L. A. Chandler, P. R. Kumler, B. F. Goodrich Chemical Company, Technical Center, Avon Lake, Ohio, unpublished data (DTA 1°C/min, T_g taken at inflection point, polymers unmodified).
40. G. A. Garbuglio, A. Rodella, G. C. Borsivi, E. Gallinella, Chem. Ind. (Milan), **46**, 166 (1964).
41. D. Kockott, Kolloid-Z., Z. Polym., **198**, 17 (1964).
42. G. Pezzin, Plastics Polym., **37** (130), 295 (1969).
43. A. J. de Vries, C. Bonnebat, M. Carrega, Pure Appl. Chem., **25**, 209 (1971).
44. A. Michel, A. Guyot, J. Polym. Sci. C **33**, 75 (1971).
45. D. E. Witenhafer, J. Macromol. Sci. B, **4** (4), 915 (1970).
46. F. P. Reding, E. R. Walter, F. J. Welch, J. Polym. Sci., **56**, 225 (1962).
47. G. Ceccorulli, M. Pizzoli, G. Pezzin, J. Macromol. Sci.-Phys. B, **14** (4), 499 (1977).
48. C. A. Daniels, E. A. Collins, Polym. Eng. Sci., **19** (8), 585 (1979).
49. R. Hoffman, W. Knappe, Kolloid Z. -Z. Polym., **240**, 784 (1970).
50. B. Wunderlich, W. Baur, Adv. Polym. Sci., **7**, 151 (1970).
51. U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem., Ref. Data., **12** (1), 29 (1983).
52. S. Chang, J. Res. Natl. Bur. Std. (US) **82**, 9 (1977).
53. E. A. Collins, unpublished data.
54. L. Nielson, "Mechanical Properties of Polymers", Reinhold Publishing Co., New York, 1962.
55. C. E. Anagnastopoulos, A. Y. Coran, H. R. Gamrath, J. Polym. Sci. **4**, 181 (1960).
56. H. Wilski, Kolloid-Z. -Z. Polym., **238**, 426 (1970).
57. J. A. Juijn, Thesis, Technische Hochschule, Delft (1972).
58. R. S. Colborne, J. Appl. Polym. Sci., **14**, 127 (1970).
59. G. C. Sinke, D. R. Stull, J. Phys. Chem., **62**, 397 (1958).
60. R. M. Joshi, Indian J. Chem., **2**, 125 (1964).
61. R. M. Joshi & B. J. Zwolinski, J. Polym. Sci. B, **3**, 779 (1965).
62. L. A. Utracki, Polym. J. (Japan), 3 **(5)** 551 (1972).
63. M. Fournier, Compt. Rend., **222**, 1437 (1946).
64. H. Staudinger, J. Schneiders, Ann., **541**, 151 (1939).
65. J. Breitenbach, E. Forster, A. Renner, Kolloid-Z., **127**, 1 (1952).
66. T. Krasovec, Reports J. Stefan Inst., **3**, 203 (1956).
67. Z. Mencik, Chem. Listy, **49**, 1598 (1955).
68. W. R. Moore, R. J. Hutchinson, Nature, **200**, 1095 (1963).
69. G. Bier, H. Kramer, Makromol. Chem., **18**, 151 (1956).
70. J. Hengstenberg, Angew. Chem., **62**, 26 (1950).
71. F. Danusso, G. Moraglio, S. Gazzera, Chem. Ind. (Milan), **36**, 883 (1954).
72. A. Oth, Ind. Chim. Belge, **20**, 423 (1955).
73. D. Harmon, B. F. Goodrich Research Center, unpublished results.
74. G. M. Guzman, J. M. Fatou, Anales Real. Soc. Espan. Fiz. Quim. (Madrid), **54B**, 609 (1958).
75. G. Ciapa, H. Schwindt, Makromol. Chem., **21**, 169 (1956).
76. G. Pezzin, N. Gligo, J. Appl. Polym. Sci., **10**, 1 (1966).
77. M. Bohdanecky, K. Solc, P. Kratochvil, M. Kolinsky, M. Ryska, D. Lim, J. Polym. Sci. A-2, **5**, 343 (1967).
78. M. Kolinsky, M. Fyska, M. Bohdanecky, P. Kratochvil, A. Soc, D. Lim, J. Polym. Sci. C, **16**, 485 (1967).
79. R. M. Fuoss, D. J. Mead, J. Phys. Chem., **47**, 59 (1943).
80. J. Vaura, J. Lapcek, J. Sabados, J. Polym. Sci. A-2, **5** (6), 1305 (1967).
81. R. Endo, Kobunshi Kagaku, **18**, 143 (1961).
82. H. Batzer, A. Nisch, Makromol. Chem., **22**, 131 (1957).
83. J. Lyngaae-Jorgensen, J. Polym. Sci. C, **33**, 23 (1971).
84. T. Kobayashi, Bull. Chem. Soc. (Japan), **35**, 726 (1962).
85. A. Takahashi, Kogyo Kagaku Zasshi, **66**, 960 (1963).
86. M. Freeman, P. Manning, J. Polym. Sci. A, **2**, 2017 (1964).
87. G. Pezzin, G. Talamini, N. Gligo, Chim. Ind. (Milan), **46**, 648 (1964).
88. C. H. Lu, Theses, Virginia Polytech. Inst. (1962).

89. A. Nakazawa, P. Matsuo, H. Inagaki, Bull. Inst. Chem. Res., Koyoto Univ., **44** (4), 354 (1966).
90. A. R. Berens, H. S. Hopfenberg, J. Membrane Sci., **10**, 283 (1982).
91. A. Lebovis, Mod. Plastics, **43**, 139 (1966).
92. B. P. Takhomirov, H. B. Hopfenberg, V. Stannett, J. L. Williams, Macromol. Chem., **118**, 177 (1968).
93. N. H. Reinkiny, A. E. Barnabeo, W. F. Hale, J. Appl. Polym. Sci., **7**, 2135 (1963).
94. W. S. Penn, "PVC Technology", Wiley-Interscience, New York, 1971.
95. A. W. Myers, V. Tammela, V. Stannett, M. Szwarc, Mod. Plastics, **37** (10), 139 (1960).
96. P. M. Doty, W. H. Aiken, H. Mark, Ind. Eng. Chem. Int. Ed., **38**, 788 (1946).
97. L. C. Grotz, J. Appl. Polym. Sci., **9**, 207 (1965).
98. R. M. Barrer, R. Mallinder, P. S. L. Wong, Polymer, **8** (6), 321 (1967).
99. K. D. Laudenslogel, S. F. Roe, Mod. Packaging, **40**, 139 (1967).
100. R. Raghova, R. M. Caddell, G. S. Y. Yeh, J. Mater. Sci., **8**, 225 (1973).
101. J. L. Williams, H. B. Hopfenberg, V. Stannett, J. Macromol. Sci. B, **3** (4), 711 (1969).
102. A. Nakajima, Kobunshi Kagaku, **7**, 309 (1950).
103. F. Mencik, Collect. Czech. Chem. Commun., **24**, 3291 (1959).
104. R. Gautron, C. Wippler, J. Chim. Phys., **58**, 754 (1961).
105. E. Schroder, Plaste Kautschuk, **15**, 2 (1968).
106. E. J. Mouton, Thesis, Univ. Leiden, 1961.
107. J. Breitenbach, E. Forster, A. Renner, Koloid-Z., **127**, 1 (1952).
108. P. Doty, E. Mishuck, J. Am. Chem. Soc., **69**, 1631 (1947).
109. P. Doty, H. Zable, J. Polym. Sci., **1**, 90 (1946).
110. W. R. Moore, R. J. Hutchinson, J. Appl. Polym. Sci., **8**, 2619 (1964).
111. R. L. Adelman, J. M. Klein, J. Polym. Sci., **31**, 77 (1958).
112. A. Oth, Compt. Pend., 27 Congress Intern. Chem. Ind., Bruxelles, Sept. 1954.
113. J. B. Haehn, B. F. Goodrich Chemical Company, Technological Center, Avon Lake, Ohio, unpublished results.
114. P. Kratochvil, Collect. Czech. Chem. Commun., **29**, 2767 (1964).
115. S. H. Maron, M. S. Lee, J. Macromol. Sci. B, **7** (1), 29 (1973); ibid. **7** (1), 47 (1973); ibid. **7** (1), 61 (1973).
116. C. A. Daniels, E. A. Collins, J. Macromol. Sci., Phys. B, **10** (2), 287 (1974).
117. A. Rudin, J. Beuschop, J. Appl. Polym. Sci., **45**, 2881 (1971).
118. L. A. Utracki, SPE Tech. Papers, **19**, 567 (1973).
119. L. A. Utracki, Polym. Eng. Sci., **14**, 308 (1974).
120. L. A. Utracki, SPE Tech. Papers, **31**, 1024 (1985).
121. L. A. Utracki, J. Vinyl Tech., **7** (4), 150 (1985).
122. C. Bonnebat, A. J. Devries, Polym. Eng. Sci., **18** (10) 824 (1978).
123. S. Krimm, V. L. Folt, J. J. Shipman, A. R. Berens, J. Polym. Sci. A, **1**, 2621 (1963).
124. F. Heatley, F. A. Bovey, Macromolecules, **2** (3), 241 (1969).
125. T. Yoshino, J. Komiyama, J. Polym. Sci. B, **3**, 311 (1965).
126. U. Johnsen, K. Kolbe, Kolloid-Z., **221**, 64 (1967).
127. C. J. Carman, A. R. Tarpley, Jr., J. H. Goldstein, J. Am. Chem. Soc., **93**, 2864 (1971).
128. C. J. Carman, A. R. Tarpley, J. H. Goldstein, Macromolecules, **4**, 445, (1971).
129. C. J. Carman, Macromolecules, **6** (5), 725 (1973).
130. Q. T. Pham, J. Millan, E. L. Madruga, Makromol. Chem., **175**, 945 (1974).
131. C. E. Wilkes, Macromolecules, **4**, 443 (1971).
132. R. D. Andrews, V. Chatre, J. Appl. Phys., **40**, 4266 (1969).
133. R. D. Andrews, Y. Kazama, J. Appl. Phys., **39**, 4891 (1968).
134. C. Rangel-Nafaile, A. Gargia Rejon, A. Gargia Leon, J. Vinyl Tech., **6** (2), 63 (1984).
135. M. D. Skibo, J. A. Manson, S. M. Webler, R. W. Hertzberg, E. A. Collins, in: R. K. Ely (Ed.), ACS Symposium Series 95 (1979), pp. 311–329, Ch. 22.
136. K. V. Gotham, M. J. Hitch, Brit. Polym. J., **10**, 47 (1978).
137. J. Dyment, H. Ziebland, J. Appl. Chem., **8**, 203 (1958).
138. R. P. Sheldon, Sister K. Lane, Polymer, **6**, 77 (1965).
139. K. Eierman, Kunststoffe, **51**, 512 (1961).
140. C. Y. Ho, P. D. Desai, K. Y. Wu, T. N. Havill, T. Y. Lee, NBS Report No. NBS-GCR-77-83 (1977).
141. G. Natta, I. W. Bassi, P. Corradini, Rend. Accad. Naz. Lincei, **31** (1,2), 1 (1961).
142. A. Nakajima, S. Hayashi, Kolloid-Z. -Z. Polym., **229** (1), 12 (1969).
143. C. E. Wilkes, V. L. Folt, S. Krimm, Macromolecules, **6** (2), 235 (1973).
144. J. Brandrup, E. H. Immergut (Eds.), "Polymer Handbook" 3rd ed., Wiley, New York, 1975, p. V/61.
145. L. A. Utracki, Z. Bakerdjian, M. R. Kamal, Trans. Soc. Rheol., **19**, 173 (1975).

Physical Constants of Poly(vinyl acetate)

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Property	Value	Remarks	Refs.
Absorption of Water (%)	Approx. 3–6	At 20°C for 24 to 144 h	26
	4	At 23°C	65,66
	6	At 70°C	
Adhesive Strength			
Lap shear (Mpa)	3.9	Aluminum	130
Peel strength (N/cm)	1.9	Aluminum	
Chain Dimensions and Entanglement Spacings	See Refs.		144–146
Cohesive Energy (J/mol)	26000–36000		52
Cohesive Energy Density (J/cm^3)	3.69×10^2		58
Compressibility (cm^3/g/atm)	1.8×10^{-5}	In the glassy state	1
Isothermal (bar^{-1}) ($\times 10^{-5}$)			
At 80°C	6.2		113
At 100°C	6.6		
At 120°C	7.1		
Decomposition Temperature (°C)	150		15
Degradation, Thermal	$T^{1/2} = 269$°C; $E_{act} = 71$ kJ/mol; monomer yield: none	Temperature at which the polymer loses 50% of its weight, if heated in vacuum for 30 min	52
Density (Mg/m^3) ≡ (g/cm^3)			
At 20°C	1.191		16,24,26
At 25°C	1.19		
At 50°C	1.17		
At 120°C	1.11		
At 200°C	1.05		
At the melting point	1.28		58
At $T = 35$–100°C	$1.2124 - 8.62 \times 10^{-4}\ T + 0.223 \times 10^{-6}\ T^2$		113
At $T = -30$ to $+20$°C	$1.196 - 3.37 \times 10^{-4}\ T$		113
As a function of pressure	See Refs.		113,114,115
Dielectric Constant	3.5	At 50°C and 2×10^3 kHz	34
	8.3	At 150°C and 2×10^3 kHz	
Of oligo PVAC	See Ref.		59
As a function of MW	See Ref.		96
Dielectric Dissipation Factor (tan δ)	1.5×10^2	At 50°C and 2×10^3 kHz	34
	26×10^2	At 120°C and 2×10^3 kHz	
Dielectric Loss Versus Frequency	See Ref.	At various temperatures	95
Dielectric Strength (V/cm) ($\times 10^{-3}$)	394	At 30°C	16
	307	At 60°C	

Property	Value	Remarks	Refs.
Dielectric Relaxation Parameters	See Refs.		121–129
Dipole Moment (ESU (per monomer unit))	2.3×10^{-18}	At 20°C	22,23
	1.77×10^{-18}	At 150°C	
(debye units)	4.8 D	For oligo PVAC, $n = 5$	59
Dynamic Mechanical Loss Peak (°C)	70	At 100 Hz	33
Elongation At Break (%)	10–20	At 20°C, %RH	27
Gas Diffusivity	See Ref.	For N_2, O_2, CO_2, H_2	52
Gas Permeability	See Refs.	Diffusion and sorption of CO_2 (below and above T_g)	53,94
N_2 for PVAC below T_g(cm^2/s/bar)	6×10^{-10}		52
N_2 for PVAC above T_g	5×10^{-10}		
O_2, 73°dry film	2.7×10^{-9}		100
Gas Solubility (cm^3(STP)/cm^3 bar)	N_2 0.02	At 298 K	52
	O_2 0.04		
	H_2 0.023		
Gel Permeation Chromatography	See Refs.	Universal calibration	90,91,92
Gel Properties			
Swelling pressure, elastic modulus and shear modulus parameters of PVAC networks in toluene and acetone	See Ref.		166
Glass Transition Temperature (°C)	28–31		25
	25.8	Isotactic PVAC $M_n \sim 10^5$	63
	31.4	Atactic PVAC $M_n \sim 1.66 \times 10^5$	
	23.6	Atactic PVAC $M_n \sim 3922$	
	34–39	Dry	65
	<30	Wet	
	11	PVAC containing 4.20% water	66
Pressure dependence (K/bar)	0.022		18
As a function of MW	See Refs.		62,64
Activation energy for the glass transition (J/mol)	$\Delta H_a = 1.8 \times 10^5$		60
Hardness (Shore units)	L 80–85	At 20°C	27
Heat Capacity (kJ/kg)	1.465	At 30°C	21
	$C_{p(liquid)} = 1.77$	At $T_g = 304$ K	67
	$C_{p(glass)} = 1.27$		
	0.116	$\Delta C_p = C_{p(liquid)} - C_{p(glass)}$	68
	$C_{v(liquid)} = 1.51$	At $T_g = 304$ K	
	$C_{v(glass)} = 1.20$		
At $T = 320$ K	$C_{p(liquid)} = 1.8409$ (kJ/kg); 158.48 (J/mol/K)		14,147
At $T = 370$ K	$C_{p(liquid)} = 1.898$ (kJ/kg); 163.37 (J/mol/K)		
At $T = 80$ K	$C_{p(glass)} = 0.323$ (kJ/kg); 27.81 (J/mol/K); 53.7 (ΔC_p, J/mol/K)		
At $T = 300$ K	$C_{p(glass)} = 1.183$ (kJ/kg); 101.86(J/mol/K)		
Heat Conductivity (J/s/m/K)	0.159		52
Heat Distortion Point (°C)	50		16
Heat of Polymerization (kJ/mol)	87.5		31,32
Index of Refraction (n_D)			28
At 20°C	1.5013	At 6563 Å	104,105
At 20°C	1.5100	At 4861 Å	
At 20°C	1.5174		
At 20°C, medium acetate type	1.4903		
At 20°C, low acetate type	1.523–1.57		

Property	Value	Remarks	Refs.
At 20°C, molding compound	1.4685		
At 20.7°C	1.4669		
At 30.8°C	1.4657		
At 52.1°C	1.4600		
At 52.1°C	1.4600		
At 80°C	1.4480		
At 142°C	1.4317		
Spec. refraction index increment in various solvents	See Ref.	(d_n/d_c)	52
Infrared Spectrum	Important bands (cm^{-1}): 606, 950, 1025, 1250, 1740, 2850, 2865, 2923, 2932, 2950		39,102,131,132 133
Temperature dependence of IR bands at 100–20°C	See Ref.		69
At very low temperature	See Ref.		70
Interfacial Tension (mN/m)	$\gamma_c = 37$		120
	14.5	At 20°C, with polyethylene	9
	8.4	At 20°C, with poly(dimethylsiloxane)	11
	9.9	At 20°C, with polyisobutene	9
	4.2	At 20°C, with polystyrene	11
	10.5	With polyethylene	101
	11.0	Linear PVAC with polyethylene	101
	9.8	Branched PVAC with polyethylene	101
Internal Pressure (J/m)			
At 0 °C	2.554×10^8		6,29,30
At 20 °C	2.847×10^8		
At 28 °C	3.978×10^8		
At 40 °C	4.313×10^8		
At 60 °C	4.187×10^8		
Intrinsic Viscosity – Molecular Weight Relationship	See corresponding table in this Handbook and also Refs.		40–51,77 136–139
From moderately conc. solutions	See Ref.		73
In 6-methyl-3-heptanone, at 66°C	$[\eta] = 0.078 \times M^{1/2}$		74
In 3-heptanone, at 26.8°C	$[\eta] = 0.082 \times M^{1/2}$		
At Θ temperature, methanol, 6°C	$[\eta] = 0.101 \times M^{1/2}$		134,135
Mechanical Properties			
Modulus of elasticity (MPa) $\equiv$ (N/mm^2)	1274–2255	ASTM-D256	27
Rubbery shear modulus (N/mm^2)	13		52
Notched impact strength (cm kg/cm^2)	>100		27
Molar Volume At 25°C (cm^3/mol)	74.25		99
Nuclear Magnetic Resonance Spectra	See Refs.		36,81,82
Shift with Eu and Pr reagents, improved resolution	See Ref.		78
Solvent and temperature effects	See Ref.		79
Conformational information	See Ref.	Syndiotactic PVAC	80
Optical Configuration Parameters ($\Delta\alpha$) ($Å^3$)			
Acetone	−20	Wavelength of 6328 Å	106
Benzene	4.0–5.9		107
Bromobenzene	9.4		108
Bromoform	−20		106
Carbon tetrachloride	−16		107
Carbon tetrachloride	−26		109
Chlorobenzene	14		107
Chloroform	−34.9		109
Chloroform	−24		107
Cyclohexane	−23		107
Dichloroethane	−36		107
Dichloroethane	−39		107
Tetrabromoethane	−25		107

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Property	Value	Remarks	Refs.
Tetrabromoethane	−33		106
Toluene	10		107
Toluene	13.5		112
Toluene	19		109
o-Xylene	2.0		107
Chloroform	81		107
Chloroform	−2.68	Temp. 50°C; $v^2 = 1.0^a$	110,111
1,3,5-Triethylbenzene	−1.90	Temp. 50°C ; $v^2 = 0.6^a$	110,111
λ-Radiation Effect			
Yields of crosslinking	0.3	Oxygen atmosphere	164
	0.15	Nitrogen atmosphere	
Yields of scission	0.07	Oxygen atmosphere	165
	0.06	Nitrogen atmosphere	
Raman Spectra	See Ref.		37
Softening Temperature (°C)	35–50		7
Solubility	See Ref.	Solvent–polymer activity in ethyl acetate, acetone, benzene	93
	See Refs.	In many solvents	54,57
Solubility Parameter (δ(MPa)$^{1/2}$)	18.6–19.9		2,3,4,5
At 25°C	21.07		56, 140
At 50°C	19.4		55
At 125°C	17.9		57
δ_d	19.0	Dispersion forces contribution	52,141
δ_p	10.2	Polar forces contribution	
δ_h	8.2	Hydrogen bonding contribution	
δ	23.1	Total $\delta^2 = \delta_d^2 + \delta_p^2 + \delta_h^2$	
Solution Properties	See Ref.		35
Unperturbed radius of gyration (Å^2 mol/g)	$Q = 0.107$	Linear PVAC	83
Viscosity	Log $\eta_0 = \log K + \alpha \log c = \beta \log M$, where $\log K = -28.518$, $\alpha = 5.599$, and $\beta = 3.411$	Diethyl phthalate solution	84
η_{spec} in various solvents	See Ref.	At $c = 5$–$40g$/100 cc	89
Huggins constant, K_H	Acetone 0.37		52
	Chlorobenzene 0.41		
	Chloroform 0.34		
	Methanol 0.47		
	Toluene 0.50		
	Benzene 0.37		
	Dioxane 0.34		

Solvent Interaction Parameters

Solvent	Temp. (°C)	Volume fraction, ϕ_2	Interaction parameter, χ	Refs.
Acetaldehyde	125–140	1	0.35–0.32	150
Acetone	25–29	0	0.40	151–153
	30–40	0.8	0.34	154
	30–50	1	0.31–0.39	154
	100–140	1	0.32–0.21	148,150
Acetonitrile	125–140	1	0.54–0.49	150
Allyl chloride	40	1	0.27	154
Benzene	5	0.2	0.46	155
	20	0	0.42	162
	30	0.4–0.8	0.45–0.29	156
	35–62	0	0.51–0.42	157
	30–50	1	0.30–0.26	154
	80–140	1	0.44–0.25	148,150,158
	125–145	1	0.37–0.32	149,159,160
n-Butane	100	1	1.97	148
2-Butanol	135	1	0.31	159
2-Butanone	10–45	0	0.43	151
	100–140	1	0.34–0.20	148,150

Solvent	Temp. (°C)	Volume fraction, ϕ_2	Interaction parameter, χ	Refs.
Butyl acetate	100	1	0.51	148
Butylbenzene	125–145	1	0.95–0.88	160
Butylcyclohexane	125–145	1	1.90–1.75	160
Carbon tetrachloride	90–135	1	0.85–0.63	148,149,158
Chlorobenzene	100–135	1	0.28–0.33	148,149
1-Chlorobutane	100–135	1	0.73–0.66	148,149
Chloroform	80–135	1	20.17–20.09	148,149,158
Chloromethane	100	1	0.25	148
1-Chloropentane	100	1	0.82	148
1-Chloropropane	40	1	0.75	154
Cycloheptane	100	1	1.63	148
Cyclohexane	100–140	1	1.65–1.16	148,149,150,158,160
Cyclohexene	100	1	1.18	148
Cyclohexanol	135	1	0.44	149,159
Cyclooctane	100	1	1.67	148
Cyclopentane	100	1	1.53	148
cis-Decahydronaphthalene	125–145	1	1.65–1.50	160
n-Decane	100–145	1	2.5–2.01	148,149,159,160
1-Decanol	135	1	0.81	159
1,1-Dichloroethane	100	1	0.19	148
1,2-Dichloroethane	100–140	1	20.04–0.00	148,150
Dichloromethane	100	1	20.14	148
Dimethylphthalate	25	0	0.400	151
1,4-Dioxane	25	0	0.407	151
	100–140	1	0.17–0.03	148,150
n-Dodecane	125–145	1	2.48–2.27	159,160
Ethanol	50	0	0.47	157
	100	1	0.80	148
Ethyl acetate	20	0	0.415	162
	100	1	0.36	148
Ethylbenzene	100–135	1	0.66–0.58	148,159
n-Heptane	100–120	1	2.14–1.63	148,150,158
1-Heptanol	135	1	0.55	159
n-Hexadecane	135	1	2.99	159
n-Hexane	100–120	1	2.06–1.71	148,158
1-Hexanol	135	1	0.49	159
Isopropylamine	40	1	0.66	154
Methanol	125–140	1	0.77–0.73	150
Methyl acetate	100	1	0.30	148
2-Methyl-2-propanol	135	1	0.30	159
Nitroethane	125–140	1	0.14–0.19	150
n-Nonane	100–145	1	2.38–1.88	148,149,159,160
n-Octane	90–120	1	2.3–1.94	148,158
1-Octanol	135	1	0.65	159
1-Octene	135	1	1.55	149
n-Pentane	100	1	2.06	148
1-Pentanol	100–135	1	0.59–0.41	148,159
2-Pentanone	135	1	0.38	149
Propane	100	1	3.2	148
1-Propanol	30–50	1	1.3–1.0	154
	100–135	1	0.64–0.38	148,159
2-Propanol	125–140	1	0.44–0.35	150,159
Propyl acetate	100	1	0.42	148
Propylamine	40	1	0.61	154
n-Tetradecane	135	1	2.70	159
Tetrahydrofuran	100–140	1	0.30–0.14	148,150
1,2,3,4-Tetrahydronaphthalene	125–145	1	0.83–0.77	160
3,3,4,4-Tetramethylhexane	125–145	1	1.72–1.56	160
Toluene	80–140	1	0.56–0.40	148,150,158,159
1,1,1-Trichloroethane	100	1	0.49	148
Trichloroethylene	100	1	0.40	148
1,2,3-Trichloropropane	15–50	0	0.38	151
2,2,4-Trimethylpentane	100–120	1	2.17–1.86	158
n-Undecane	100–145	1	2.7–2.14	148,159,160
Vinyl acetate	30	0.4–0.8	0.41–0.22	156
Water	40	1	2.5	161

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Property	Value	Remarks	Refs.
Specific Heat	See also Heat Capacity		
Conformational (dE_{cont}/dT)	1.3	Syndiotactic	85
	1.8	Isotactic	
Softening Point			
As a function of MW	See Refs.		116,117,118
Specific Volume (l/kg)	$0.823 + 6.4 \times 10^{-4} \times T$	$T = 100–200°C$	9,13
	0.84	At T_g	19,61
	1.073	Monomer	77
	0.841	Polymer	
(l/mol)	0.08774	One mole repeating unit	103
Stress Relaxation Curves	See Ref.	22–61.8°C	20,97
Surface Resistance (Ω/cm)	5×10^{-11}		15
Surface Tension (mN/m)			9,10,11,12,86
At 20°C	36.5		
At 140°C	28.6		
At 180°C	25.9		
At 150°C	27.9		58,87,101
	$-d_\gamma/dT = 0.066$ (mN/m)		58,119
	$\gamma_{ds} = 27.4 \pm 0.1$	Dispersive	85
	$\gamma_{ds} = 15.4 \pm 0.7$	Polar	
	$\gamma_{solid} = 42.85 \pm 0.8$	Total	
Tensile Strength (MPa) $\equiv$ (N/mm^2)	29.4–49.0	At 20°C	16
	See Table in Ref.		96
Thermal Conductivity (W/m/K)	0.159		16
Pressure and temperature dependence	See Ref.		88
	0.159		116
Thermal Expansion Coefficient (K^{-1})			
At 0 °C	2.8×10^{-4}		113
At 20 °C	2.8×10^{-4}		
At 40 °C	7.13×10^{-4}		
At 60 °C	7.17×10^{-4}		
At 80 °C	7.20×10^{-4}		
At 100 °C	7.23×10^{-4}		
Cubic	6.7×10^{-4}		15,25
Linear above T_g	22×10^{-5}		17,27
Below T_g	7×10^{-5}		

Theta Temperatures

Polymer	M_w ($\times 10^{-4}$)	Dispersity	Solvent (M_w/M_n)	Θ Temp. (°C)	K_Θ ($\times 10^4$) [dl/g (g mol wt)$^{-1/2}$]	Refs.
Poly(vinyl acetate)	87–346		Methyl isopropyl ketone/*n*-heptane (73.2/26.8)	25		151
Linear	2.7–126.8	2.0	*n*-heptane/methyl isopropyl ketone (27.3/72.7)	30	9.2	152
	4.1–83		Ethyl *n*-butyl ketone	29	9.29	153
			Ethyl isoamyl ketone	66	8.20	
	0.35–150	1.02–1.05	Methanol	6	10.1	154
	3.0–32		Ethyl *n*-butyl ketone	29	9.55	155
			Carbon tetrachloride	46.4		
			Ethanol	19		163
			3-Heptanone	29		
	0.8–130		Cetyl alcohol	123	5.37	156
Branched			Ethanol	12–15		163
			3-Heptanone	26		

Property	Value	Remarks	Refs.
Ultraviolet Radiation λ(nm)			
Absorption, cut-off	240		142
Quantum yields (mol/Einstein)			
(in air) at $\lambda = 254$			143
Acetic acid formation	0.01		
Carbon dioxide formation	0.0065		
Carbon monoxide formation	0.0069		
Methane formation	0.0038		
Chain scission	0.05 (in air), 0.066 (in nitrogen)		
Crosslinking	0.0025 (in air), 0.047 (in nitrogen)		
Ultraviolet Spectra	See Ref.		30,38
Young's Modulus (MPa) ≡ (N/mm^2)	600	At 25°C and 50% RH	8
Viscoelastic Response	See Refs.		167–171
X-Ray Diffraction	See Ref.	Wide angle	98

[a] Polymer volume fraction.

REFERENCES

1. M. M. Martynyuk, V. K. Semenchenko, Kolloid Zh., **26**, 83 (1964).
2. H. Daoust, M. Rinfret, J. Colloid Sci., **7**, 11 (1952).
3. D. Managaraj, S. Patra, P. C. Roy, S. K. Bhatnagar, Makromol. Chem., **84**, 225 (1965).
4. P. A. Small, J. Appl. Chem., **3**, 71 (1953).
5. G. Allen, Polymer, **2**, 375 (1961).
6. D. R. Paul, A. T. DiBenedetto, J. Polym. Sci. C, **16**, 1269 (1967).
7. K. Schmieder, K. Wolf, Kolloid Z., **134**, 149 (1953).
8. J. Galperin, W. Arnheim, J. Appl. Polym. Sci., **11**, 1259 (1967).
9. S. Wu, J. Colloid Interface Sci., **31**, 153 (1969).
10. S. Wu, J. Phys. Chem., **74**, 632 (1970).
11. S. Wu, J. Polym. Sci. C, **34**, 19 (1971).
12. S. Wu, J. Phys. Chem., **72**, 3332 (1968).
13. J. R. Collier, Ind. Eng. Chem., **61** (9), 50 (1969).
14. R. F. Boyer, J. Macromol. Sci. Phys. B, **7**, 487 (1973).
15. Mowilith, Polyvinylacetat, Farbwerke Hoechst AG, Frankfurt, 1969, p. 214.
16. T. P. G. Shaw, "Encyclopedia of Chemical Technology", Vol. 14, Interscience, New York, 1955, p. 692.
17. R. F. Clash, L. M. Rynkiewicz, Ind. Eng. Chem., **36**, 279 (1944).
18. J. M. O'Reilly, J. Polym. Sci., **57**, 429 (1962).
19. A. A. Miller, Macromolecules, **4**, 757 (1971).
20. G. Adam, Kolloid Z., **195**, 1 (1964).
21. H. Wilski, Kolloid Z., **210**, 37 (1966).
22. D. J. Meed, R. M. Fuoss, J. Am. Chem. Soc., **63**, 2839 (1941).
23. O. Broens, F. H. Mueller, Kolloid Z., **141**, 20 (1955).
24. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 618.
25. S. Saito, Kolloid Z. -Z. Polym., **189**, 116 (1963).
26. C. E. Schildknecht, "Vinyl and Related Polymers", Wiley, New York, 1952, p. 336.
27. H. R. Hacobi, "Ullmans Encyklopaedie der Technischen Chemie, vol. 11, 3rd ed., Verlag Chemie, Weinheim, Munich, 1960, p. 34.
28. O. Broens, F. H, Mueller, Kolloid Z., **140**, 121 (1955).
29. S. Fujimoto, Mem. Defense Acad. Math. Phys. Chem. Eng. (Yokosuka, Japan), **7**, 1247 (1967).
30. G. Allen, D. Sims, G. J. Wilson, Polymer, **2**, 375 (1961).
31. R. M. Joshi, Makromol. Chem., **66**, 114 (1963).
32. R. M. Joshi, J. Polym. Sci., **56**, 313 (1962).
33. H. Gramberg, Adhesion, **10**, (3), 97 (1966).
34. H. Thurn, K. Wolf, Kolloid Z., **148**, 16 (1956).
35. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 633.
36. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 551.
37. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 548.
38. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 550.
39. M. K. Lindemann, "Encyclopedia of Polymer Science and Technology", vol. 15, Interscience, New York, 1971, p. 547.
40. P. C. Scherer, A. Tannenbaum, D. W. Levi, J. Polym. Sci., **43**, 531 (1960).
41. W. R. Moore, M. Murphy, J. Polym. Sci., **56**, Sl9 (1962).
42. H.-G. Elias, Kunstoffe Plastics, **8**, 441 (1961).
43. S. N. Chinai, P. C. Schere, D. W. Levi, J. Polym. Sci., **17**, 117 (1955).
44. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960).
45. R. H. Wagner, J. Polym. Sci., **2**, 21 (1947).
46. A. Nakajima, Kobunshi Kagaku, **11**, 142 (1954).
47. M. R. Rao, V. Kalpagam, J. Polym. Sci., **49**, S-14 (1961).
48. H.-G. Elias, F. Patat, Makromol. Chem., **25**, 13 (1957).

49. E. Patrone, U. Bianchi, Makromol. Chem., **94**, 52 (1966).
50. S. Gundiah, N. V. Viswanathan, S. L. Kapur, Indian J. Chem., **4**, 344 (1966).
51. D. Goedhart, A. Opsehoor, J. Polym. Sci. A-2, **8**, 1227 (1970).
52. D. W. Van Krevelen, "Properties of Polymers", Elsevier., New York, 1976.
53. K. Toi, Y. Maeda, T. Tokuda, J. Membr. Sci., **13**, 15 (1983).
54. M. J. Fernandez-Berridi, G. M. Guzman, J. M. Elorza, L. Garijo, Eur. Polym. J., **19**, 445 (1983).
55. M. T. Raetzsch, H. Freydank, G. Goebel, Faserforsch. Textiltech., **27**, 425 (1976).
56. G. DiPaola-Baranyi, J. E. Guillet, Macromolecules, **11**, 228 (1978).
57. W. Merk, R. N. Lichtenthaler, J. M. Prausnitz, J. Phys. Chem., **84**, 1694 (1980).
58. H. G. Cho, S. C. Yoon, M. S. Jhon, J. Polym. Sci., Polym. Chem. Ed., **20**, 1247 (1982).
59. E. Ikada, T. Sugimura, T. Aoyama, T. Watanabe, Polym., **16**, 101 (1975).
60. S. M. Aharoni, J. Appl. Polym. Sci., **16**, 3275 (1972).
61. J. E. McKinney, R. Simha, Macromolecules, **7**, 894 (1974).
62. J. Janacek, J. Polym. Sci., Polym. Lett. Ed., **13**, 409 (1975).
63. W. R. Brown, G. S. Park, Am. Chem. Soc., Div. Org. Coatings Plast. Chem., Pap., **34** (1), 505 (1974).
64. R. P. D'Amelia, H. Jacin, Am. Chem. Soc., Div. Org. Coatings Plast. Chem., Pap., **35** (1), 245 (1975).
65. Z. Miyagi, K. Tanaka, Colloid Polym. Sci., **257**, 259 (1979).
66. G. E. Johnson, H. E. Bair, S. Matsuoka, J. E. Scott. ACS Symp. Ser., 127 (Water Polym.), 451 (1980).
67. J. E. McKinney, R. Simha, Macromolecules, **9**, 430 (1976).
68. R. F. Boyer, J. Macromol. Sci. B: Phys., **7**, 487 (1973).
69. F. Jagodic, Spectrochim. Acta, Part, A, **29**, 1556 (1973).
70. J. K. Haken, R. L. Werner, Brit. Polym. J., **5**, 451 (1973).
71. W. S. Park, W. W. Graessley, J. Polym. Sci., Polym. Phys. Ed., **15**, 71 (1977).
72. F. Cane, Eur. Polym. J., **14**, 185 (1978).
73. K. K. Chee, J. Macromol. Sci. B: Phys., **19**, 257 (1981).
74. K. K. Chee, J. Appl. Polym. Sci., **27**, 1675 (1982).
75. D. Lecacheux, J. Lesec, C. Quivoron, R. Prechner, R. Panaras, J. Appl. Polym. Sci., **29**, 1569 (1984).
76. L. Beltzung, C. Strazielle, Makromol. Chem., **185**, 1155 (1984).
77. A. A. Tager, A. A. Askadskii, M. V. Tsilipotkina, Vysokomol. Soedin. A, **17**, 1346 (1975).
78. A. R. Katritzky, A. Smith, Brit. Polym. J., **4**, 199 (1972).
79. T. K. Wu, D. W. Ovenall, Macromolecules, **7**, 776 (1974).
80. P. R. Sundararajan, Macromolecules, **11**, 256 (1978).
81. H. N. Sung, J. H. Noggle, J. Polym. Sci., Polym. Phys. Ed., **19**, 1593 (1981).
82. F. D. Blum, J. E. Dickson, W. G. Miller, J. Polym. Sci., Polym. Phys. Ed., **22**, 211 (1984).
83. R. K. S. Chan, C. Worman, Polym. Eng. Sci., **12**, 437 (1972).
84. N. M. Chaurasia, Polym., **17**, 570 (1976).
85. R.-J. Roe, A. Tonelli, Macromolecules, **11**, 114 (1978).
86. E. Sacher, J. Colloid Interface Sci., **92**, 275 (1983).
87. K. M. Hong, J. Noolandi, Macromolecules, **14**, 1229 (1981).
88. O. Sandberg, G. Baeckstroem, J. Polym. Sci., Polym. Phys. Ed., **18**, 2123 (1980).
89. E. Bapda, DEFAZET Dtsch. Farben-Z., **32**, 372 (1978).
90. C. M. L. Atkinson, R. Dietz, Eur. Polym. J., **15**, 21 (1979).
91. D. K. Gilding, Polymer, **22**, 505 (1981).
92. J. Bouwma, R. Dietz, L. J. Maisey, B. Wright, Eur. Polym. J., **20**, 471 (1984).
93. T. Oishi, J. M. Prausnitz, Ind. Eng. Chem. Process Des. Dev., **17**, 333 (1978).
94. K. Toi, Y. Maeda, T. Tokuda, J. Appl. Polym. Sci., **28**, 3589 (1983).
95. H. Sasabe, C. T. Moynihan, J. Polym. Sci., Polym. Phys. Ed., **16**, 1447 (1978).
96. S. Negami, R. J. Ruch, R. R. Myers, J. Colloid Interface Sci., **90**, 117 (1982).
97. Kubat, L. A. Nisson, Mater, Sci. Eng., **52**, 223 (1982).
98. A. T. Riga, Polym. Eng. Sci., **18**, 1144 (1978).
99. H. Ahmad, M. Yassen, Polym. Eng. Sci., **19**, 858 (1979).
100. M. Salame, Am. Chem. Soc., Div. Org. Coatings Plast. Chem., Pap., **36** (1), 488 (1976).
101. G. L. Gaines Jr., Polym. Eng. Sci., **12**, 1 (1972).
102. N. K. Roy, Indian J. Phys., **51B**, 65 (1977).
103. Z. Roszkowski, Makromol. Chem., **183**, 669 (1982).
104. H. F. Mark, N. G. Gaylord, N. M. Bikales (Eds.), "Encyclopedia of Polymer Science and Technology: Plastics, Resins, Rubbers, Fibers", Interscience, New York, 1964.
105. G. M. Kline, Analytical Chemistry of Polymers, Part III, Interscience, New York, 1962.
106. E. V. Frisman, A. B. Chzu, Vysokomol. Soedin., **4**, 1564 (1962).
107. E. V. Frisman, A. K. Dadivanyan, J. Polym. Sci. C, **16**, 1001 (1967).
108. V. N. Tsvetkov, A. E. Grishchenko, L. E. De-Millo, E. N. Rostovskii, Vysokomol. Soedin., 6, 384 (1964).
109. E. V. Frisman, V. A. Andreichenko, Vysokomol. Soedin., **4**, 1559 (1962).
110. E. Riande, E. Saiz, "Dipole Moments and Birefringence of Polymers" Prentice Hall, New Jersey, 1992, p. 288.
111. E. Riande, E. Saiz, and J. E. Mark, J. Polym. Sci. Polym. Phys. Ed., **22**, 863 (1984).
112. V. N. Tsvetkov, N. N. Boitsova, M. G. Vitovskaja, Vysokomol. Soedin., **6**, 297 (1964).
113. J. E. McKinney, M. Goldstein, J. Res. Nat. Bur. Stand., **78A**, 331 (1974).
114. S. Beret, J. M. Prausnitz, Macromolecules, **8**, 536 (1975).
115. J. E. McKinney, R. Simha, Macromolecules, **7**, 894 (1974).
116. H. F. Mark, D. F. Othmer, C. G. Overberger, G. T. Seaborg, (Eds.) "Encyclopedia of Chemical Technology", vol. 23, 3rd ed., Wiley-Interscience, New York, 1978. p. 830.
117. H. Fikentscher, Cell. Chem., **13**, 71 (1932).
118. E. O. Kraemer, Ind. Eng. Chem., **30**, 1200 1938).
119. S. Wu, J. Colloid Interface Sci., **31**, 153 (1969).
120. L.-H. Lee, in "Interaction of Liquids at Solid Substrates", Advances in Chemistry Series No. 87, American Chemical Society, Washington, DC, 1968, p. 106.

121. Y. Ishida, M. Matsuo, K. Yamafuje, Kolloid Z. Z. Polym. Sci., **180**, 108 (1962).
122. M. Matsuo, Y. Ishida, K. Yamafugi, M. Takayanagi, F. Irie, Kolloid Z., **201**, 7 (1965).
123. S. Mashima, R. Nozaki, S. Yagihara, S. Takeishi, J. Chem. Phys., **77**, 6262 (1982).
124. S. Havriliak, Jr. D. G. Watts, Polymer, **27**, 1509 (1986).
125. S. Negami, "Dielectric Properties of Polymers", Ph.D. Thesis, Kent University, 1969.
126. X. Von Broens, F. H. Muller, Kolloid Z., **140**, 121 (1955).
127. W. Heinrich, B. Stoll, Colloid Polym. Sci., **263**, 874 (1985).
128. D. E. Beuerger, R. H. Boyd, Macromolecules, **22**, 2699 (1989).
129. X. Von Broens, F. H. Muller, Kolloid Z., **140**, 121 (1955). (1941).
130. Y. Hatano, B. Tomita, H. Mizumachi, Holzforschung, **40**, 255 (1986).
131. N. B. Colthup, L. H. Daly, S. E. Wiberley, "Introduction to Infrared and Raman Spectroscopy", 2nd ed., Academic, New York, 1975.
132. The Infrared Spectra Atlas of Monomers and Polymers Sadtler Res. Labs., Philadelphia, 1980.
133. D. O. Hummel, "Atlas of Polymer and Plastics Analysis", 2nd ed., Verlag Chemie, New York, 1978.
134. M. Ueda, K. Kajitani, Makromol. Chem., **108**, 138 (1967).
135. R. Naito, Kobunshi Kagaku, Chem. High Polym. (Tokyo), **16**, 7 (1959).
136. G. S. Misra, V. P. Gupta, Makromol. Chem., **71**, 110 (1964).
137. M. R. Rao, V. Kalpagam, J. Polym. Sci., **49**, S-14 (1961).
138. A. Nakajima, K. Kagaku, Chem. High Polym. Tokyo, **11**, 142 (1954).
139. V. V. Varadiah, J. Polym. Sci., **19**, 477 (1956).
140. A. T. DiBenedetto, J. Polym. Sci. A, **1**, 3459 (1963).
141. C. M. Hansen, Skand. Tidskr. Farg Lack, **17**, 69 (1971).
142. W. Schnabel, in: "Polymer Degradation – Principles and Practical Applications", Hanser Publishers, Munich, 1981, p. 98.
143. G. Geuskens, M. Borsu, C. David, Europ. Polym. J., **8**, 883 (1972); ibid. 1347 (1972).
144. S. Wu, J. Polym. Sci. Polym. Phys. Ed., **27**, 723 (1989).
145. J. Brandrup, E. H. Immergut (Eds.) "Polymer Handbook", 3rd ed., Wiley, New York, 1989, Sec. VII, p. 37.
146. S. Onogi, T. Masuda, T. Ibaragi, Kolloid Z. Z. Polym., **222**, 110 (1968).
147. U. Gaur, S. F. Lau, B. B. Wunderlich, et al., J. Phys. Chem. Ref. Data, **12**, 29 (1983).
148. P. Munk, P. Hattam, Q. Du, et al., J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 289 (1990).
149. G. DiPaola-Baranyi, J. E. Guillet, H.-E. Jeberien, J. Klein, Makromol. Chem., **181**, 215 (1980).
150. W. Merk, R. N. Lichtenthaler, J. M. Prausnitz, J. Phys. Chem., **84**, 1694 (1980).
151. G. V. Browning, J. D. Ferry, J. Chem. Phys., **17**, 1107 (1949).
152. R. E. Robertson, R. McIntosh, W. E. Grummitt, Can. J. Res., **324**, 150 (1956).
153. R. E. Wagner, J. Polym. Sci., **2**, 27(1947).
154. R. J. Kokes, A. R. DiPietro, F. A. Long, J. Am. Chem. Soc., **75**, 6319 (1953).
155. T. Kawai, J. Polym. Sci., **32**, 425 (1958).
156. A. Nakajima, H. Yamakawa, I. Sakurada, J. Polym. Sci., **35**, 489 (1959).
157. G. R. Cotton, A. F. Sirianni, I. E. Puddington, J. Polym. Sci., **32**, 115 (1958).
158. D. D. Deshpande, O. S. Tyagi, Macromolecules, **11**, 746 (1978).
159. R. C. Castells, G. D. Mazza, J. Appl. Polym. Sci., **32**, 5917 (1986).
160. G. DiPaola-Baranyi, J. E. Guillet, J. Klein, et al., J. Chromatography, **166**, 349 (1978).
161. L. J. Thompson, F. A. Long, J. Am. Chem. Soc., **76**, 5886 (1954).
162. C. Masson, H. W. Melville, J. Polym. Sci., **4**, 337 (1949).
163. F. Candau, C. Strazielle, H. Benoit, Makromol. Chem., **170**, 165 (1973).
164. A. Charlesby, S. H. Pinner, Proc. Roy. Soc. A, **249**, 367 (1959).
165. S. Miller, M. W. Spindler, R. L. Vale, Polym. Sci., Part A, **1**, 2537 (1963).
166. F. Horkay, A. M. Hecht, E. Geissler, J. Chem. Phys., **91**, 2706 (1989).
167. R. Nozaki, S. Mashimo, J. Chem. Phys., **84**, 3575 (1986).
168. D. J. Plazek, Polym. J., **12**, 43 (1980).
169. D. J. Plazek, J. Polym. Sci., Polym. Phys. Ed., **20**, 729 (1982).
170. J. D. Ferry R. F. Landel, Kolloid-Z., **148**,1 (1956).
171. M. L. Williams, J. D. Ferry, J. Colloid Sci., **9**, 479 (1954).

Physical Constants of Poly(methyl methacrylate)*

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Property	Value	Remarks	Refs.
Birefringence $(\alpha_1 - \alpha_2)$ (cm^3)			
Segmental anisotropy	2×10^{-25}	Benzene	1
	20×10^{-25}	Benzene, isotactic	
Ceiling Temperature (K)	478		47
Coefficient of Thermal Expansion (K^{-1})			
Linear	7×10^{-5}	0–50°C	2
Volume	2.60×10^{-4}	$<T_g$	3
	2.55×10^{-4}	$<T_g$	4
	2.72×10^{-4}	$<T_g$	5
	2.25×10^{-4}	$<T_g$	6
	5.80×10^{-4}	$>T_g$	3
	5.60×10^{-4}	$>T_g$	4
	5.80×10^{-4}	$>T_g$	5
	5.75×10^{-4}	$>T_g$	6
Compressibility (MPa^{-1}) $(\times10^6)$			
	245	$T = 20$°C	7
	290	60°C	7
	355	100°C	7
	390	109.3°C	7
	500	119.8°C	7
	530	129.7°C	7
	561	150°C	8
	644	200°C	8
Density $(mg/m^3) \equiv (g/cm^3)$	1.195	0°C	9
	1.190	20°C	10
	1.188	25°C	11
	1.150	T_g	12
Diffusion O_2 in PMMA (cm^2/s) $(\times 10^8)$	0.5	0.21 bar, 25°C	13
	1.0	0.4 bar, 25°C	13
	1.4	1 bar, 25°C	13
	3.8	1 bar, 20°C	14
H_2O in PMMA $(g\,cm/cm^2/h)$	5.2	Permeability	15
Dielectric Constant	3.6	50 Hz, 25°C	16
	3.0	1 KHz, 25°C	16
	2.6	1 MHz, 25°C	16
	2.57	30 GHz, 25°C	17,18
	2.59	138 GHz, 25°C	18
Dissipation Factor $(\tan\delta)$	0.062	50 Hz, 25°C	16
	0.055	1 KHz, 25°C	16
	0.014	1 MHz, 25°C	16
	0.007	30 GHz, 25°C	17,18
	0.010	138 GHz, 25°C	18
Elongation at Break (%)	4	$10^5 \leq M_w \leq 2\times10^5$	48
	5.5	$2\times10^6 \leq M_w$	48

*Data are valid for normal radical polymerized PMMA if no tacticities are given.

Property	Value	Remarks	Refs.
Entanglement Molecular Weight	7000		49
Entropy of Polymerization (kJ/mol/K)*	− 117	25 °C to solid polymer	19
	− 40.3	− 63 °C	19,20
Extinction Modulus (mm^{-1}) ($\times 10^4$)	5.5	647 nm	21
	5	633 nm	2
	1.9	514 nm	21
	1	400 nm	2
Flexural Strength (MPa) ≡ (N/mm^2)	145	DIN 53452	48
Glass Transition Temperature T_g (K)	378	Dilatometric	6,12
	377		5

Dependence on Tacticity

	Tacticity (triad analysis)			
T_g (°C)	Iso	Hetero	Syn	
41.5	0.95	0.05	0.00	
54.3	0.73	0.16	0.11	22
61.6	0.62	0.20	0.18	22
104.0	0.06	0.37	0.56	5
114.2	0.10	0.31	0.59	22
119.0	0.04	0.37	0.59	49
120.0	0.10	0.20	0.70	5
125.6	0.09	0.36	0.64	22
134.0	0.01	0.18	0.81	49

Hardness (MPa) ≡ (N/mm^2)	195	DIN 53456	48
Heat Capacity (kJ/kg/K)	0.585	− 173 °C	23
	0.878	− 100 °C	23
	1.255	0 °C	23
	1.42	25 °C	23
	1.72	100 °C	23
	2.05	120 °C	23
	2.38	180 °C	23
	2.35	240 °C	24
	2.50	300 °C	24
Heat Conductivity (W/m/K)	0.19		50
Heat of Combustion (kJ/kg)	− 26.2		25
Heat of Polymerization (kJ/mol)	− 57.8	To solid polymer	19
Impact Strength, Sharpy (kJ/m^2)	20	ISO 179/1 fU	51
Modulus of Elasticity (MPa) ≡ (N/mm^2)	3300	25 °C	2
Modulus of Shear (MPa) ≡ (N/mm^2)		Dynamic	
	1700	25 °C, 10 Hz	2
Mechanical Loss Factor (tan δ)	0.08	25 °C, 10 Hz	26
Optical Absorption (dB/km)	84	514.5 nm	52
Partial Specific Volume (cm^3/g)	0.812	20 °C in monomer	10
	0.819	20 °C in monomer	27
	0.833	70 °C in monomer	10
Rate Constants of Polymerization			
Contraction constant ε	$-\varepsilon = 0.2256 + 4.81 \times 10^{-4} t + 4.1 \times 10^{-2} t^2$	$0 \leq T \leq 100$ °C	28
Monomer density d (g/cm^3)	$d = 0.9659 - 1.2129 \times 10^{-3} t$		
	$+1.6813 \times 10^{-6} t^2 - 1.0164 \times 10^{-8} t^3$	$0 = T = 130$ °C	31
Chain propagation k_p (l/mol/s)	$\ln k_p = 17.68 - 3762/T$	$313 \leq T \leq 353$ K	29
Chain termination k_t (l/mol/s)	$\ln k_t = 24.91 - 2536/T$	$313 \leq T \leq 353$ K	29
k_p^2/k_t value	$\ln k_p^2/k_t = 10.45 - 4988/T$		29
Termination mode			
Relative contribution of combination π_c	$\ln \pi_c = -4.1 + 1160/T$	$313 \leq T \leq 353$ K	34
Thermal initiation rate $R_{i,th}$ (mol/l/s)	$\ln R_{i,th} = 1.7 - 13100/T$		31
Transfer to monomer C_M	$\ln C_M = -2.60 - 2855/T$	$313 \leq T \leq 353$ K	31
Transfer to polymer C_P	$1.5 - 2.5 \times 10^{-4}$	$313 \leq T \leq 353$ K	30

*mol: monomer unit.

Property	Value	Remarks	Refs.
Pressure dependence of polymerization rate activation volume ΔV_{pol} (cm^3/mol)	−23.6 −19.0	20°C 30°C	32 33
Refractive Index			
n_D	1.492	= 589 nm	2
n_e	1.494	= 546 nm	2,10
n_g	1.502	= 436 nm	22,10
$-dn_D/dT$ (K^{-1})	1.2×10^{-4}	$< T_g$	53
	3.1×10^{-4}	$> T_g$	53
Abbé's number (r_n)	57.8		50
Resistivity (ohm cm)	$>10^{15}$		2
Solvents		See chapter on Solvents–Nonsolvents in this Handbook	
Thermal Conductivity (W/m/K)	0.160	−20°C	35
	0.193	0–50°C	36,37
	0.250	100°C	38
Theta Solvents and Temperatures			
Acetonitrile	$\theta = 30.0$°C		39
Heptanone-4	$\theta = 40.4$°C		40
Isoamyl acetate	$\theta = 57.4$°C		40
n-Butyl chloride	$\theta = 35.0$°C	Syndiotactic	40,41
	$\theta = 26.5$°C	Isotactic	41
Heptanone-3	$\theta = 30.0$°C	Syndiotactic	42
	$\theta = 40.0$°C	Isotactic	42
n-Propanol	$\theta = 75.9$°C	Syndiotactic	42
	$\theta = 84.4$°C	Isotactic	42
Unit Cell		See corresponding chapter in this Handbook	
Unperturbed Dimensions			
$K_\theta = r_\theta^2/M$ (cm^2mol/g)	5.2×10^{-18}	See also corresponding chapter in this Handbook	43,44
Velocity of Sound (m/s)	2700	2 MHz, 25°C	45
	2850	5 MHz, −40°C	46
Vicat Temperature (K)	385	$M_w = 1.2 \times 10^5$	50
	393	$M_w \geq 5 \times 10^6$	50
Viscosity–Molecular Weight Relationship* $[\eta] = KM^a$ (cm^3/g)		See also chapter "Viscosity–Molecular Weight Relationship" in this Handbook	

Solvent	T (°C)	K ($\times 10^3$)	a	
Acetone	20	10	0.68	43
Benzene	30	5.2	0.76	54
THF	25	10.4	0.70	54
n-Butyl chloride	$35 = \theta$	50	0.50	43
Chloroform	25	4.8	0.80	54

Water Absorption (%)	2.0	Saturation $T = 23$°C	55

REFERENCES

1. V. N. Tsetkov, N. N. Boitzkova, Vysokomol. Soedin, **3**, 1176 (1960).
2. G. Schreyer, "Konstruieren mit Kunststoffen", Carl Hanser, München, 1972.
3. P. Heydemann, D. H. Guicking, Kolloid-Z., **193**, 16 (1963).
4. J. Hennig, Diplomarbeit TH, Darmstadt, 1961.
5. J. C. Wittmann, A. J. Kovacs, J. Polym. Sci. C. **16**, 4443 (1969).
6. S. Loshaek, J. Polym. Sci., **15**, 391 (1955).
7. K. H. Hellwege, W. Knappe, P. Lehmann, Kolloid-Z., **183**, 110 (1962).
8. B. P. Shtarkman, J. M. Monich, S. A. Arzhakov, N. Yu Averbakh, Vysokomol. Soedin A, **18**, 1047 (1976).
9. H. J. Kolb, I.F. Izard, J. Appl. Phys., **20**, 564 (1949).
10. D. Panke, W. Wunderlich, Röhm GmbH, Darmstadt, unpublished results.
11. W. G. Gall, N. G. McCrum, J. Polym. Sci., **50**, 489 (1961).
12. S. S. Rogers, L. Mandelkern, J. Phys. Chem., **61**, 945 (1957).
13. R. E. Barker, Jr., J. Polym. Sci., **58**, 553 (1962).
14. J. R. MacCallum, A. L. Rudkin, Eur. Polym. J., **14**, 655 (1978).

* Molecular weights were determined by means of light scattering and sedimentation.

15. F. Bueche, J. Polym. Sci., **14**, 414 (1954).
16. G. Schreyer, Kunststoffe, **55**, 771 (1965).
17. E. Amrhein, Kolloid-Z., **216-7**, 38 (1967).
18. W. Zeil, R. Sistig, W. Frank, V. Hoffman, Ber. Bunsenges. Phys. Chem., **74**, 883 (1970).
19. H. Sawade, J. Macromol. Sci. C, **3**, 313 (1969).
20. R. W. Warfield, M. C. Petree, J. Polym. Sci. A, **1**, 1701 (1963).
21. B. Crist, M. E. Marhic, G. Ravio, M. Epstein, J. Appl. Phys., **51**, 1160 (1980).
22. E. V. Thompson, J. Polym. Sci. A-2, **4**, 199 (1966).
23. R. Hoffmann, W. Knappe, Kolloid-Z., **247**, 763 (1971).
24. V. Bares, B. Wunderlich, J. Polym. Sci. A-2, **11**, 861 (1973).
25. K. Krekeler, P. M. Klinke, Kunststoffe, **55**, 758 (1965).
26. J. Heiboer, Kolloid-Z., **148**, 36 (1956).
27. G. V. Schulz, M. Hoffmann, Makromol. Chem., **22**, 220 (1957).
28. D. Panke, M. Stickler, W. Wunderlich, Makromol. Chem., **184**, 175 (1983).
29. M. Stickler, Internal Report Röhm GmbH, Darmstadt, 1978.
30. T. Corner, Adv. Polym. Sci., **62**, 97 (1984).
31. M. Stickler, G. Meyerhoff, Makromol. Chem., **179**, 2729 (1978).
32. T. Sasuga, M. Takehisa, J. Makromol. Sci.–Chem. A, **12**, 1307 (1978).
33. M. Yokawa, Y. Ogo T. Imoto, Makromol. Chem., **175**, 179 (1974).
34. G. V. Schulz, G. Henrici-Olivé, S. Olivé, Z. Phys. Chem. (Frankfurt), **27**, 1 (1960).
35. R. E. Barker, R. Y. S. Chen, R. S. Frost, J. Polym. Sci., Polym. Phys. Ed., **15**, 1199 (1977).
36. E. Calvet, J. P. Bros, H. Pinelle, Compt. Rend, **260**, 1164 (1965).
37. K. Eiermann, Kolloid-Z., **198**, 5 (1964).
38. A. B. Baschirow, J. W. Selenew, S. K. Achundow, Plaste Kautschuk, **23**, 351 (1976).
39. E. Cohn-Gibsberg, T. G. Fox, H. Maron, Polymer, **3**, 97 (1962).
40. R. G. Kirste, G. V. Schulz, Z. Phys. Chem. (Frankfurt), **27**, 301 (1961).
41. G. V. Schulz, W. Wunderlich, R. G. Kirste, Makromol. Chem., **75**, 22 (1964).
42. J. Sakurada, A. Nakajima, O. Yoshizaki, K. Nahamae, Kolloid-Z., **186**, 41 (1962).
43. G. V. Schulz, R. G. Kirste, Z. Phys. Chem. (Frankfurt), **30**, 171 (1961).
44. W. Wunderlich, Makromol. Chem., **182**, 2465 (1981).
45. F. Pluemer, Kolloid-Z., **176**, 108 (1961).
46. A. M. North, R. A. Pethrick, D. W. Phillips, Polymer, **18**, 324 (1977).
47. H. Sawada, "Thermodynamics of Polymerization", Marcel Dekker, New York, 1976.
48. M. Buck, in: Kunststoff-Handbuch, vol. 9, Carl Hauser, München, 1975.
49. K. Fuchs, Chr. Friedrich, J. Weese, Macro-molecules, **29**, 5893 (1996).
50. G. Schreyer, J. Lehmann, in: Kunststoff-Hand-buch, vol. 1, Carl Hauser, München, 1990.
51. Th. Arndt, Internal Report Röhm GmbH, Darmstadt, 1996.
52. P. Avakian, W. Y. Hsu, P. Meakin, H. L. Snyder, J. Polym. Sci., Polym. Phys. Ed., **21**, 647 (1983).
53. J. M. Cariou, J. Dugas, L. Martin, P. Michel, Appl. Optics, **25**, 334 (1986).
54. H. L. Wagner, J. Phys. Chem. Ref. Data, **16**, 165 (1987).
55. J. Lehmann, Materialien in ihrer Umwelt, DWS, Karlsruhe (1996).

Physical Constants of Poly(styrene)

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Property	Value	Remarks	Refs.
Birefringence Dispersion			
$\frac{\Delta n(\lambda)}{\Delta n\ (546\,\text{nm})}$	$A + B/\lambda^2 + C/\lambda^4$	$A = 0.8905$ $B = 0.275 \times 10^{-9}\,\text{cm}^2$ $C = 0.153 \times 10^{-18}\,\text{cm}^4$	21
Coefficient of Thermal Expansion (K^{-1})			
Linear	$6–8 \times 10^{-5}$	$< T_g$ (unoriented)	3
Volume	$1.7–2.1 \times 10^{-4}$	$< T_g$	7a
	$5.1–6.0 \times 10^{-4}$	$> T_g$	7a
Coefficient of Friction	0.515	20–80°C	7b,7c
	0.744	100°C	7b
	>2	120°C	7b
	0.25	100°C	7c
	2.65	110–120°C	7c
Cohesive Energy Density	See solubility parameter		
Common Solvents	Toluene, perchloroethylene, carbon tetrachloride, carbon disulfide, tetralin, dioxane, MEK, pyridine, cyclohexanone, ethyl acetate		34
Compressibility (MPa^{-1})	220×10^{-6}		3
Compressive Modulus (MPa) ≡ (N/mm^2)	3000	Unoriented	18a
Creep/Stress Relaxation	See Ref.		35

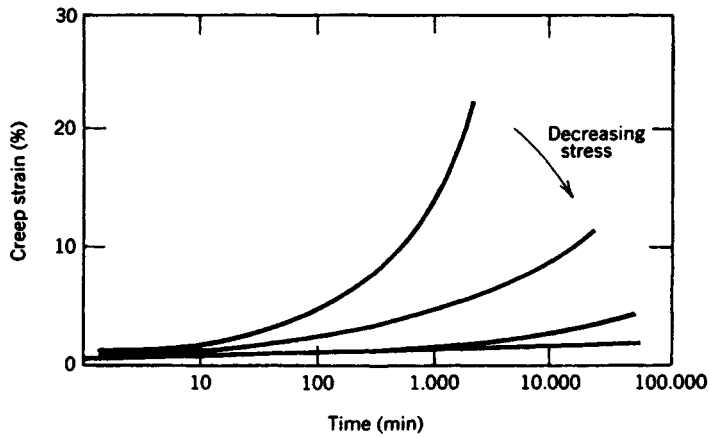


Figure 1. Typical creep behavior for rubber-modified styrene polymers.

Crystallographic Data	See under Unit Cell		
Density (mg/m^3)			
Amorphous	1.04–1.065		3,4
Crystalline	1.111		2
	1.12		1,5
d_ρ/dT ($g/cm^3/K$)	-2.65×10^{-4}	$< T_g$	6
d_ρ/dT ($g/cm^3/K$)	-6.05×10^{-4}	$> T_g$	6

Property	Value	Remarks	Refs.
Dielectric Constant (ε)			
Amorphous	2.49–2.55	at 1 kHz (curve flat to 1 GHz)	3
Crystalline	2.61	at 1 kHz (curve flat to 1 GHz)	16a
Dissipation Factor			
Amorphous	15×10^{-4}	at 1 kHz (curve flat to 1 GHz)	3
Crystalline	3×10^{-4}	at 1 kHz (curve flat to 1 GHz)	16a
Dynamic Mechanical Loss Peaks			
1 Hz	$T_\beta = 300$ K		16b
	$T_\gamma = 138$ K		16b
7 kHz	$T_\delta^\gamma = 48$ K		16c
Entanglement Molecular Weight	19100		43
Glass Transition Temperature, T_g (°C)	80		6
	90	T_g (K) $= 373 - (1.0 \times 10^5 / M_v)$	8
	(100)		9
Heat Capacity, C_P (kJ/kg/K)	1.185 (1.139)	0°C	3,12
	1.256 (1.394)	50°C	3,12
	1.838 (1.821)	100°C	3,12
dC_P/dT (kJ/(kg K^2))	4.04×10^{-3}	50°C	3
Heat Capacity vs. Temperature	See Fig. 2		36–37

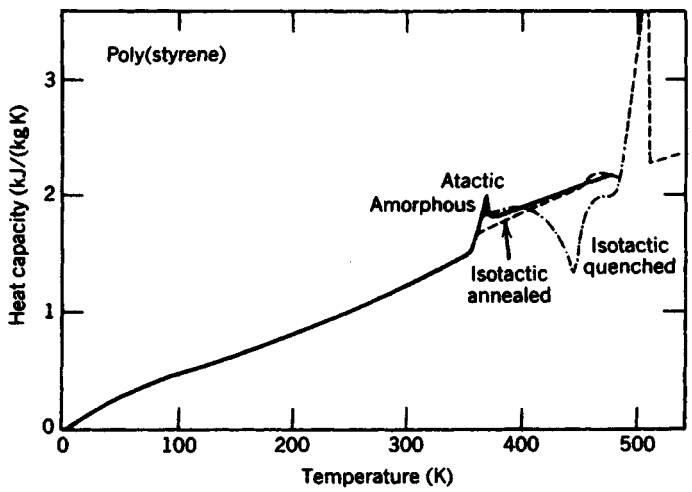


Figure 2. Heat capacity vs. temperature curves for amorphous, atactic, and isotactic poly(styrenes).

Heat of Combustion (kJ/mol)[a]	-4.33×10^3		13a,14
Heat of Fusion (kJ/mol)[a]	8.37 ± 0.08		10
Crystalline	9.00		2
Heat of Polymerization (kJ/mol)[a]	−67.4	To solid polymer	15
	−69.9	To solid polymer	13a
Heat of Solution (kJ/mol)[a]	−3.60	In monomer	13a
Infrared Spectrum		See Ref.	13b
Intrinsic Viscosity vs. MW	Various solvents	See Ref.	38
Limiting Oxygen Index (%)	18.1		44
Melting Point T_M (°C)	240		10
	(250)		11
Melt Viscosity – Molecular Weight Relationship	$\log \eta_r = 3.4 \log M - k$		

	T (°C)	M range	k	
Atactic	217	$\geq$ 38,000	13.04	31
Isotactic	281	100000–600000	14.42	32a
				39

[a]mol: Monomer unit.

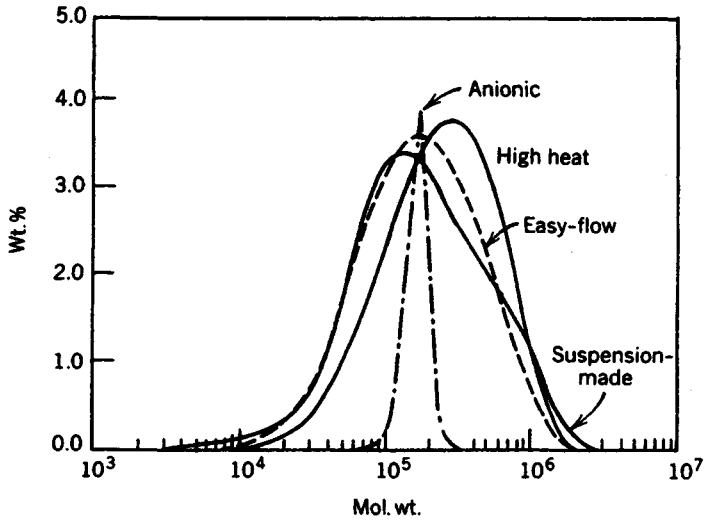
Property	Value	Remarks	Refs.
MW Distribution Curves (For representative poly(styrenes))			32b

Figure 3. Molecular weight distribution curves for representative poly(styrenes).

Nuclear Magnetic Resonance — **Line width** (gauss)

T (°C)	Value
0	6.7
50	6.5
100	5.0
150	1.0

	peaks (ppm) at 3.0, 3.5 (aromatic), 8.4 (CH_2)	Referred to tetramethylsilane as + 10.0	32c
Optical Dispersion, $\eta_F - \eta_C$	1.92×10^{-2}	$\lambda = 486.1$ nm	3
		$\lambda = 656.3$ nm	3
Poisson's Ratio	0.325		3
	0.33		20
Refractive Index, n_D	1.59–1.60	$\lambda = 589.3$ nm	3
dn_D/dT (K^{-1})	-1.42×10^{-4}		3
Resistivity (Ohm cm)	10^{20}–10^{22}		17a
Solubility Parameter $(J/m^3)^{1/2} \times 10^{-3}$	16.6–20.2 (8.1–9.9 $(cal/cm^3)^{1/2}$)		40
Sonic velocity (m/s)	2100		3
Specific Gravity	1.04–1.06		41
Stress–Optical Coefficient (brewsters)	10.1	Monofilament	22
	9.5	Extruded sheet	22
	8.3,8.7	Compression molded	22
Stress–Strain Curves	See Figs. 4–6 (next page)		42
Surface tension, 293 K, mN/m ($\equiv$ dyn/cm) (Ps/air)	40.7		45
syndiotactic			47
Tensile Modulus, E (MPa) $\equiv$ (N/mm^2)	3200	Unoriented	19
	3400		18a
	4200	Oriented monofilament	18b,18c
dE/dT ($N/mm^2/K$) $\equiv$ (MPa/K)	−4.48		19
Thermal Conductivity (W/m/K)	0.105	0°C	3
	0.116	50°C	3
	0.128	100°C	3

Thermogravimetric Analysis In vacuo (10°C min^{-1}) 17b

Wt. loss (%)	*T* (°C)	Wt. loss (%)	*T* (°C)
0	275	40	390
5	330	60	408
10	343	80	422
20	360	100	450

References page V-95

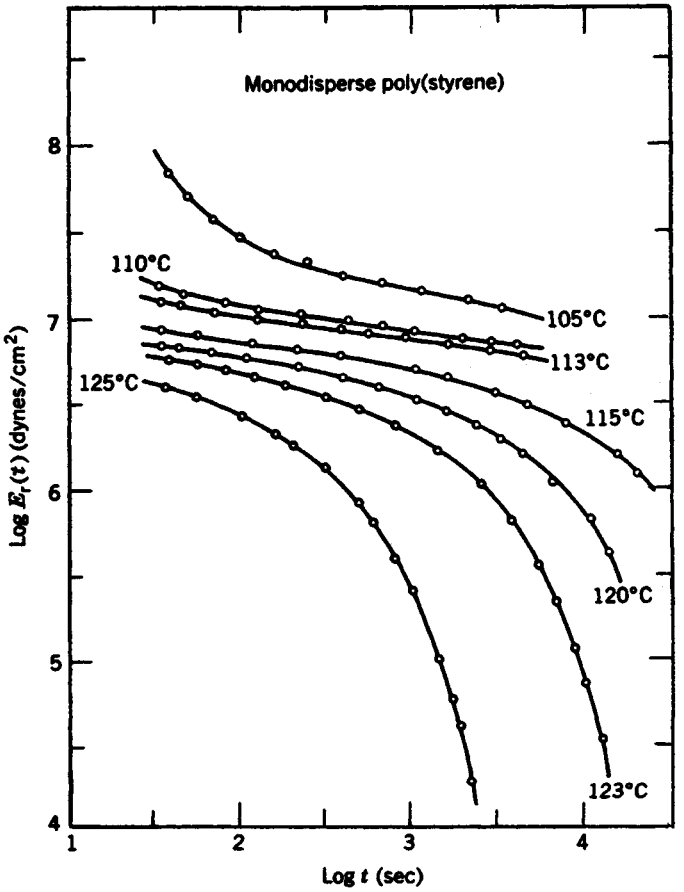


Figure 4. Stress relaxation curves for monodisperse poly(styrene).

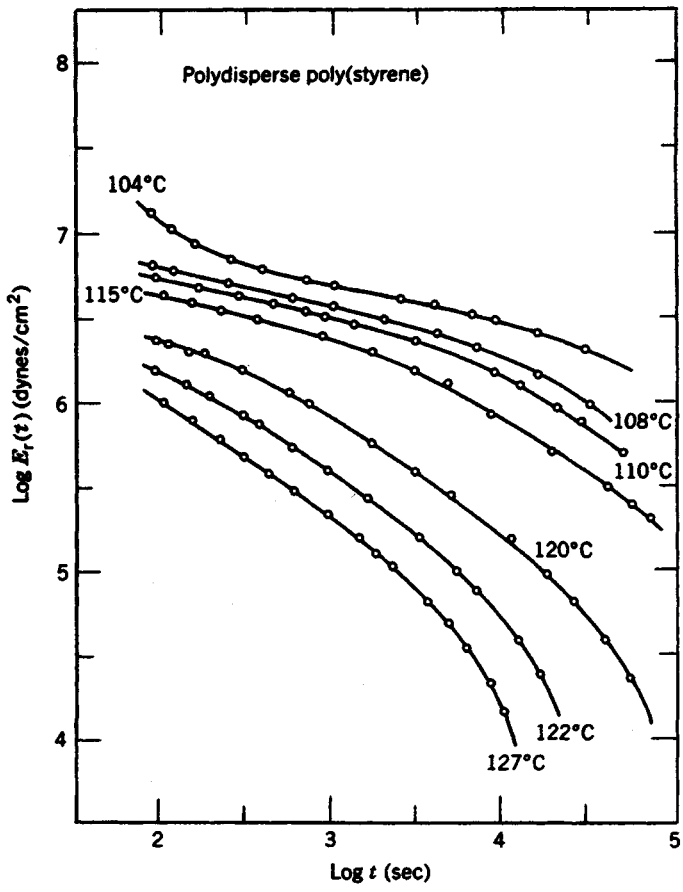


Figure 5. Stress relaxation curves for polydisperse poly(styrene).

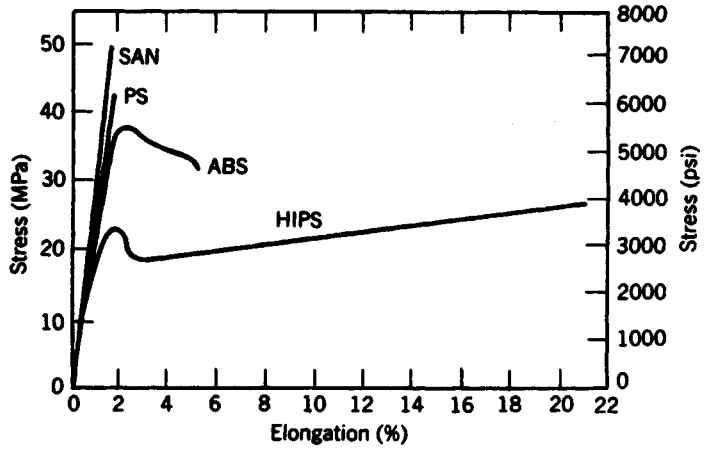


Figure 6. Stress–strain curves for styrene-based polymers.

Property	Value	Remarks	Refs.
Unit Cell (Å)		Rhombohedral system; 18 monomer units in cell	
a_0	21.9		1
	22.08		2
b_0	21.9		1
	22.08		2
c_0	6.63		1
	6.626		2

Property	Value	Remarks	Refs.
Ultraviolet Spectrum (in vacuo)	Absorption bands at 260, 194 and 80 nm		
Viscosity – Molecular Weight Relationship	See also corresponding chapter in this Handbook		33
Water Absorption at Equilibrium (23°C, 50% RH)	0.05%		46

$[\eta]$		M				
Solvent	T (°C)	**Solvent**	**Method**	K ($\times 10^3$)	a	**Refs.**
ATACTIC						
Benzene	30	Benzene	Osmotic	9.7	0.74	23
Benzene	25	Toluene	Osmotic	11.3	0.73	24
Toluene	25	Butanone	Osmotic	13.4	0.71	
Toluene	25	Butanone	Light scattering	17	0.69	25
Toluene	30	Butanone–isopropyl Alcohol	Light scattering	9.23	0.72	26
Benzene	30	Toluene	Osmotic	10.6	0.735	27
Toluene	30	Toluene	Osmotic	11.0	0.725	
Toluene	25	Cyclohexanone	Sedimentation velocity	9.77	0.73	28
ISOTACTIC						
Benzene	30	Toluene	Osmotic	10.6	0.735	27
Toluene	30	Toluene	Osmotic	11.0	0.725	
Benzene	RT	Toluene	Osmotic	9.5	0.77	29
Benzene	25	Toluene	Osmotic	9.79	0.744	30
Toluene	25	Butanone	Light scattering	17	0.69	25

Constants determined by the method and solvent given in these columns; $[\eta] = KM^a$, in ml/g.

REFERENCES

1. G. Natta, Makromol. Chem., **35**, 94 (1960).
2. R. L. Miller, L. E. Nielsen, J. Polym. Sci., **55**, 643 (1961).
3. R. H. Boundy, R. F. Boyer (Eds.), "Styrene, Its Polymers, Copolymers and Derivatives", Reinhold, New York, 1952.
4. G. Natta, J. Polym. Sci., **16**, 143 (1955).
5. G. Natta, P. Pino, P. Corradini, F. Danusso, E. Mantica, G. Mazzanti, G. Moraglio, J. Am. Chem. Soc., **77**, 1708 (1955).
6. W. Patnode, W. J. Scheiber, J. Am. Chem. Soc., **61**, 3449 (1939).

7a. R. S. Spencer, G. D. Gilmore, J. Appl. Phys., **20**, 502 (1949).

7b. H. J. Karam, W. C. Sager, K. S. Hyun, The Dow Chemical Company, unpublished data.

7c. M. Lund, The Dow Chemical Company, unpublished data.

8. F. P. Reding, J. A. Faucher, R. D. Whitman, J. Polym. Sci., **57**, 483 (1962).
9. T. G. Fox, P. J. Flory, J. Appl. Phys., **21**, 581 (1950). (The effect of residual monomer, low-molecular-weight polymer, ethylbenzene, and plasticizers being to depress T_g, the value given by these authors represents the limiting glasstransition temperature at high molecular weights.)
10. R. Dedeurwaerder, J. F. M. Oth, J. Chim. Phys., **56**, 940 (1959).
11. T. W. Campbell, A. C. Haven, Jr., J. Appl. Polym. Sci., **1**, 73 (1959).
12. K. Ueberreiter, E. Otto-Laupenmuhlen, Z. Naturforsch, **8A**, 664 (1953).

13a. D. E. Roberts, W. W. Walton, R. S. Jessup, J. Polym. Sci., **2**, 420 (1947).

13b. C. Y. Liang, S. Krimm, J. Polym. Sci., **27**, 241 (1958).

14. J. W. Breitenbach, J. Derkosch, Monatsh., **81**, 698 (1950).
15. L. K. J. Tong, W. O. Kenyon, J. Am. Chem. Soc., **69**, 1402 (1947).

16a. F. L. Saunders, R. C. Mildner, The Dow Chemical Company, unpublished data.

16b. K. -H. Illers, Z. Electrochem., **65**, 679 (1961).

16c. J. M. Crissman, A. E. Woodward, J. A. Sauer, J. Polym. Sci. A-3, **3**, 2693 (1965).

17a. P. Woodland, The Dow Chemical Company, unpublished data.

17b. J. G. Cobler, The Dow Chemical Company, unpublished data.

18a. "Strength and Stiffness", in: Plastics Design Data, Dow Technical Chemical Publication, 1965 revision.

18b. L. Kin, The Dow Chemical Company, unpublished data.

18c. R. D. Andrews, J. F. Rudd, J. Appl. Phys., **28**, 1091 (1957).

19. J. F. Rudd, Gurnee, J. Appl. Phys., **28**, 1096 (1957).
20. L. E. Nielsen, "Mechanical Properties of Polymers", Reinhold, New York, 1962.
21. E. F. Gurnee, J. Polym. Sci. A-2, **5**, 817 (1967).
22. J. F. Rudd, R. D. Andrews, J. Appl. Phys., **31**, 818 (1960).

23. R. H. Ewart, H. C. Tingey, Abstracts 111th Am. Chem. Soc. Meet., April 1947; quoted in Fox and Flory, J. Am. Chem. Soc., **73**, 1915 (1951).
24. C. E. H. Bawn, R. F. J. Freeman, A. R. Kamaliddin, Trans. Faraday Soc., **46**, 1107 (1950).
25. P. Outer, C. I. Carr, B. H. Zimm, J. Chem. Phys., **18**, 830 (1950).
26. S. N. Chinai, P. C. Scherer, C. W. Bondurant, D. W. Levi, J. Polym. Sci., **22**, 527 (1956).
27. F. Danusso, G. Moraglio, J. Polym. Sci., **24**, 161 (1957).
28. H. W. McCormick, J. Polym. Sci., **36**, 341 (1959).
29. F. Ang, J. Polym. Sci., **25**, 126 (1957).
30. W. R. Krigbaum, D. K. Carpenter, S. Newman, J. Phys. Chem., **62**, 1586 (1958).
31. T. G. Fox, P. J. Flory, J. Polym. Sci., **14**, 315 (1954).

32a. J. Boon, G. Challa, P. H. Hermans, Makromol. Chem., **74**, 129 (1964).

32b. R. Kosfeld, Kolloid-Z., **172**, 182 (1960).

32c. F. A. Bovey, G. V. D. Tiers, G. Filipovich, J. Polym. Sci., **38**, 73 (1959).

33. R. H. Partridge, J. Chem. Phys., **47**, 4223 (1967).
34. Roff, Scott, Pacitti (Eds.), "Handbook of Common Polymers", CRC Press, Florida, 1971, p. 46.
35. Kirk-Othmer, "Encyclopedia of Chemical Technology", vol. 21, 3rd ed., 1978, p. 806.

36–37. "Polymer Handbook", 2nd ed., Wiley, New York, 1974, p. III/217; see also this edition.

38. "Polymer Handbook", 2nd ed., Wiley, New York, 1974, p. IV/17; see also this edition.
39. Kirk-Othmer, "Encyclopedia of Chemical Technology", vol. 21, 3rd ed., 1978, p. 808.
40. Polymer Handbook, 2nd ed., 1974, p. IV/352; see also this edition.
41. Roff, Scott, Pacitti (Eds.), "Handbook of Common Polymers", CRC Press, Florida, 1971, p. 47.
42. Kirk-Othmer, "Encyclopedia of Chemical Technology", vol. 21, 3rd ed., 1978, p. 805.
43. J. T. Seitz, "Measurement of Entanglement length in Broad Molecular Weight Systems", 50th jubilee, Rheol. Soc., Boston, 1979.
44. H. Ohe, K. Matsura, Text. Res. J., **45**, (11), 778 (1975).
45. S. Wu, "Polymer Interface and Adhesion", Marcel Dekker, New York, 1982, p. 88.
46. B. Mauer, Dow Chemical Co., Midland, 1988, unpublished water absorption data.
47. A. Sciutto, F. Ga-bassi, L. Meda, R. Po, L. Pozzi, Appl. surface Sci., **125** (3–4), 287 (1998).

Physical Constants of Poly(oxymethylene)

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Property	Value	Refs.
Absorption		
Homopolymer (density 1.418–1.419 g/cm^3)		
Equilibrium concentration, 60°C		
Aniline (wt.%)	9.60	1
m-Cresol (wt.%)	16.48	1
Dimethylformamide (wt.%)	5.73	1
Water (DIN 53 472, after 96 h), (mg)	20 –30	3
(wt.%)	~0.14–0.21	
Gain in volume, 60°C (vol.%)		
Aniline	13.7	1
m-Cresol	23.0	1
Dimethylformamide	8.6	1
Average sorption coefficient, 25°C		
Water	See Ref.	2
Activation Energy of Polymerization (kJ/mol)		
Of radiation induced trioxane		
In source polymerization	~21; 121	72
Post polymerization >50°C	167	72
<50°C	287	72
44–55°C	126	72
45–55°C	165; 174	72
Of cationic polymerization of trioxane		
In ethylene dichloride	79.5	81
In the solid state	36	81
Anisotropy of Solid State Extruded POM	See Ref.	127
Arc Resistance (s) (ASTM D495)		
Homopolymer	129 (burns)	73
Copolymer	240 (burns)	73
DIN 53484, VDE 0303, part 5		
Copolymer	L4	
Ceiling Temperature (°C)		
Polymerization of formaldehyde (gas) → poly(oxymethylene) (solid)		
At 1 bar	119	66
	127	25
	116	9
	120	22
	137	90
	126	94
	133	95
At different pressures calculated from Ref. 66		67

p (bar)	t_c (°C)
0.36	100
0.62	110
1.06	120
1.75	130
2.81	140
4.44	150
6.85	160
10.36	170

Polymerization of trioxane (gas) → poly(oxymethylene) (solid)	36	94

Property	Value		Refs.
Chemical Resistance	See Refs.		73,79
Coefficient of Linear Thermal Expansion (K^{-1})			
20–100°C (VDE 0304)	$1.0–1.4 \times 10^{-4}$		3
Amorphous > T_g (ASTM D 696)	7.2×10^{-4}		6
Copolymer	0.85×10^{-4}		73
Homopolymer DIN 52328, ASTM 696	0.81×10^{-4}		73
Copolymer	1.1×10^{-4}		
Cohesive Energy, 25°C (kJ/mol)	9.88		4
Cohesive Energy Density $(J/m^3)^{1/2}$ ($\times 10^{-3}$)			
25°C	22.8		1
Liquid, 25°C, theroretical value	20.9		4
Comonomer Distribution by Pyrolysis/GC			107
Comparative Figure of Tracking	600		138
Crystallinity (%)			
Homopolymer			
20°C, from density-data	64–69		5
25°C, from X-ray-data	77		6
135°C, from X-ray-data	75		6
140°C, from X-ray-data	73.0		70
145°C, from X-ray-data	73.1		70
150°C, from X-ray-data	75.4		70
155°C, from X-ray-data	76.9		70
157°C, from X-ray-data	67		6
160°C, from X-ray-data	80.0		70
177°C, from X-ray-data	95.5		70
Delrin 150, 20°C, from density data	64.0		5
Delrin 550, 20°C, from density data	68.7		5
Copolymer			
20°C	56–59		5
Hostaform C2520, M_w 80000			
20°C from density data	56.6		5
Hostaform C9020, M_w 58000			
20°C from density data	58.7		5
Homo- and copolymers, from density data	See Ref.		56
Trioxane/dioxolane copolymers with different comonomer content	See Ref.		71
Dependence on irradiation	See Ref.		98
Crystallographic Data			
Poly(oxymethylene)			
Crystal system	Trig	Orth	61
Space group	P3 or P32	P222	
Mol. helix	2* 9/5	2* 2/1	
Unit cell axes (Å)	4.471	4.767	
	4.471	7.660	
	17.39	3.563	
No. of units	9	4	
Pc ($Mg\,m^{-3}$)	1.491	1.533	
Packing density	0.65		99
Phase transition			
Transition energy	See Ref.		115
Orth → Trig (kJ/mol) ($-OCH_2$)	+0.6		100
Orth, two forms			101
Cryst. packing density	0.78	0.81	60,61
Pa (Mg/m^3)	1.25	–	
Amorphous packing density	0.65	–	
Heat of fusion (kJ/mol)	9.79	–	61
Glass transition temperature (K)	191	–	

Property		Value	Refs.
For trioxane-1,3-dioxolane copolymers		See Ref.	71
Crystal structure of radiation induced polytrioxane and tetroxane		See Ref.	105
Deformation Under Load (%)			
(ASTM D621, 13.8 Mpa at 50°C, 24 hr)			
Homopolymer		0.5	73
Copolymer		1.0	73
Entanglement in Solid State		See Refs.	128,129
Density (10^{-3} kg/m^3) ≡ (g/cm^3)			
Homopolymer			
Delrin 500		1.420 ± 0.005	7
After annealing		1.434 ± 0.004	7
Delrin 150, 20°C		1.427	5
Delrin 550, 20°C		1.435	5
Copolymer, 20°C			
Hostaform C2520, M_w 80000		1.412	5
Hostaform C9020, M_w 58000		1.416	5
Copolymers, 20°C; after annealing; mol fraction of dioxolane comonomer			71
0.075		1.412	
0.15		1.388	
0.18		1.380	
0.25		1.363	
0.30		1.358	
0.34		1.346 (average values)	
Poly(oxymethylene)			
Trigonal, amorphous		1.25	61
Crystalline		1.491	61
Orthorhombic, crystalline		1.533	61
Degradation (initial) Under N_2			121
Homopolymer (°C)		318	
Copolymer (°C)		324	
Depolymerization, Energy of Activation (kJ/mol)			
Poly(oxymethylene)diol thermal depolymerization	90–150°C	42	53
	100–120°C	113	9
	130–150°C	82.9 ± 12.6	8
	135–190°C	83.8	11
	170–285°C	109	10
	172–188°C	71 ± 10	88
Cationic depolymerization	−30 to +33°C	80.0 ± 9.6	12
	130–150°C	87.5 ± 12.6	8
Anionic depolymerization	130–150°C	27.2 ± 6.3	8
	135–170°C	62.0 ± 7.5	13
Poly(oxymethylene)diacetate and dimethyl ether			
Thermal degradation,			
	240–340°C	234	11
	280–350°C	134	10
	199–331°C	117	53
Autoxidative degradation, 160°C		117,121	52
Thermooxidative degradation, 145–160°C		61	53
Depolymerization Rate (min^{-1})		See Ref.	51
Poly(oxymethylene)diol			
$V_{depol} = K_{therm}$ (POM) $+ k_{bas}$ (POM) (Amine)			

T (°C)	K_{therm} ($\times 10^3$)	k_{bas} ($\times 10^3$)
135	1.6 ± 0.2	4.1 ± 0.2
150	3.5 ± 0.1	7.7 ± 0.5
160	5.0 ± 0.5	11.6 ± 1.0
170	7.4 ± 1.0	17.0 ± 2.0

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Property	Value	Refs.
Dielectric Absorption (Homopolymer)		
α-Absorption	Near 10 Hz at about 140°C	49
β-Absorption	1 kHz–1 MHz at 60–100°C	
γ-Absorption	1 GHz	
Dielectric Constant (10^3 Hz)		
DIN 53 483, ASTM D 150–59 T	3.6–4.0	3,73
Dielectric Loss, $\tan\delta$ (10^3 Hz)	$(15–50) \times 10^{-4}$; See also Ref.	3
DIN 53 493, ASTM D 150–59 T		7,14,15,16,138
Dielectric Strength (kV/mm)		
DIN 53 481 (50 Hz, 0.5 kV/s, film 0.2 mm)	50–70	3
Homopolymer	33	138
Copolymer	35	138
Diffusion (films)		
Water		
Activation energy, apparent (kJ/mol)		
Copolymer 0–80°C	41.5	18
Copolymer, branched, 50°C	44.4	2
Diffusion coefficient, average, 25°C (cm^2/s)		
Copolymer	2.7×10^{-8}	17
Copolymer, branched	7.0×10^{-8}	17
Copolymer, linear and branched	See Ref.	2
Carbon dioxide		
Activation energy (kJ/mol)		
Copolymer	48.9	17
Copolymer, branched	38.5	17
Diffusion coefficient (cm^2/s)		
Copolymer	1.4×10^{-9}	17
Copolymer, branched	4.4×10^{-9}	17
Formaldehyde		
Diffusion coefficient (dc) (cm^2/s)		135

T (°C)	dc (cm^2/s)
60	2.14×10^{-9}
40	1.34×10^{-9}
21	1.0×10^{-9}

Dissipation Factor		
At 10^3 Hz (ASTM D150)		
Homopolymer	3×10^{-3}	73
Copolymer	4×10^{-3}	73
Elastic Modulus (N/mm^2)		
DIN 53457 (1 mm/min)	2600–3400	63
Elongation at Break (%)		
DIN 53455 (50 mm/min)		
Low molecular mass	20–40	63
High molecular mass	35–70	63
ASTM D638 (5 mm/min)		
Homopolymer	15–75	73
Copolymer	50–75	73
Elongation at Yield (%)		
Homopolymer	12	138
Copolymer	7	138

Property		Value	Refs.
Enthalpy (kJ/kg)			
Homopolymer	0 K	0	19
	50 K	6.909	19
	100 K	29.69	19
	200 K	102.6	19
	300 K	222.5	19
Copolymer	0 K	0	19
	50 K	6.948	19
	100 K	29.40	19
	200 K	100.9	19
	300 K	211.4	19
Entropy (kJ/kg/K)			
Homopolymer	0 K	0	19
	50 K	0.2097	19
	100 K	0.5167	19
	200 K	1.007	19
	300 K	1.487	19
Copolymer	0 K	0	19
	50 K	0.2123	19
	100 K	0.5153	19
	200 K	0.9974	19
	300 K	1.439	19
Entropy of Fusion (kJ/mol/K)			
Of homopolymer at constant pressure		8.21×10^{-3}	20
Of homopolymer at constant volume		4.98×10^{-3}	20
Poly(oxymethylene), equilibrium value		10.70×10^{-3}	57
Entropy of Polymerization (J/mol/K)			
Formaldehyde (gas, 1 bar, 298 K) → poly(oxymethylene) (solid)			
From heat capacity data		-175 ± 0.8	19
From equilibrium pressure data		-169.5 ± 2.9	66
		-128.4 ± 0.9	25
		−175	9
		−129	19
		−183	21
		-136 ± 17	90
Formaldehyde (liquid, 298 K) → poly(oxymethylene) (solid)		86.7	96,94
Trioxane → poly(oxymethylene) at 298 K			
Gas → solid		−156	22
		−155	96,94
Gas → liquid		−106	96,94
Liquid → liquid		0	96,94
Liquid → solid		−49.0	96,94
Solid → solid		−6.7	96,94
		$+18 \pm 16$	66
		See Ref.	65
At 323 K in dichloroethane		−41	92
Tetraoxane (solid) → poly(oxymethylene) (solid) at 298 K		3.41 ± 0.06	66
Entropy of Trimerization (J/K/mol trioxane)			
Formaldehyde (gas) → trioxane (cryst.) at 298 K		-526 ± 13	91
Formaldehyde (gas) → trioxane (gas) at 298 K		-375 ± 12	91
Fatigue Behavior			
Of acetal homopolymer		See Ref.	54
Flammability			126

Radiant heat flux (kW/m²)	Ignition time (s)	T_{Ig} (°C)
21	248	277
25	184	278
28	138	282
34	79	288

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Property	Value	Refs.
Flammability UL 94		138
(1.5 mm sheet)		
Yellow card	2060	
Class	HB	
Flexural Modulus (N/mm^2)		
ASTM D790 at 23°C		
Homopolymer	2830	73
Copolymer	2590	73
Flexural Strength at Yield (N/mm^2)		
ASTM D790		
Homopolymer	97	73
Copolymer	90	73
Flexural Stress at Conventional Deflection		
MPa ≡ (N/mm^2)		
DIN 53452	105–120	3
Flow Temperature (°C)		
ASTM 569		
Homopolymer	184	73
Copolymer	174	73
Formaldehyde		
Melting point (°C)	−118	73,76
Boiling point (°C)	−19	73,76
Heat of vaporization at bp (kJ/mol)	700	73,76
Heat of combustion at 23°C (gas) (kJ/mol)	−562	73,76
	-570.77 ± 0.42	83
Density of liquid (g/cm^3) at −20°C	0.815	73,76
at −80°C	0.917	73,76
Explosive limits, % by vol. in air	7–72	73
Standard enthalpy		
Formation of formaldehyde (kJ/mol)	-104.9 ± 1.4	66
	−116	82
	−108.57	83
	−97.1	93
Free Energy of Polymerization (kJ/mol)		
Formaldehyde (gas, 298 K) → poly(oxymethylene) (solid)	−2.9	19,94
	−3.9	68
	−17	93
Trioxane (298 K) → poly(oxymethylene)	−1.8	82,65
Gas → solid	−17.6	96,94
Gas → liquid	−6.07	96,94
Liquid → liquid	−0.4	96,94
Liquid → solid	−3.8	96,94
Solid → solid	−6.3	96,94
	−9.9	66
	See Ref.	65
In dichloroethane at 323 K	−5.4	92,94
Tetraoxane (solid, 298 K) → poly(oxymethylene) (solid)	−4.4	66
Friction Coefficient		
Against rotating steel shaft, dry (roughness height ~2 µm, loading pressure $p = 0.05\ N/mm^2$, sliding speed 100 mm/min)		
Dynamic	0.15–0.25	63
Static	0.20–0.30	63
GPC of Macrocyclic POM	See Ref.	114
Glass Transition Temperature (K)	See Second-Order Transition Point	
Hardness Ball Indentation (MPa) ≡ (N/mm^2)		
DIN 53456 E/ISO 2039 after 30 s	130–160	3

Property	Value	Refs.
Heat Capacity (J/mol/K)		
Homopolymer 0 K	0	19,37
50 K	9.94	
100 K	16.69	
200 K	28.82	
300 K	42.79	
Copolymer 0 K	0	
50 K	9.97	
100 K	16.40	
200 K	26.56	
300 K	41.11	
Molten poly(oxymethylene)		
190.0 K (T_g)	53.58	58
200.0 K	53.95	
273.15 K	56.67	
298.15 K	57.60	
300.0 K	57.67	
400.0 K	61.39	
457.0 K (T_m)	63.51	
500.0 K	65.11	
600.0 K	68.83	
Crystalline poly(oxymethylene)		
0 K	0	61
50 K	10.10	
100 K	16.68	
200 K	27.15	
273.15 K	34.94	
298.15 K	38.26	
300 K	38.52	
390 K	54.00	
For the temperature range $T = 25-140$°C		85
At mole fractions of oxyethylene units $X_2 < 0.16$	$Cp(X_2, T) = 36.75 + 0.1425\ (T - 25) + 80X_2$	
For detailed information, see Ref. and corresponding table in this Handbook		55,56,68 61,71
For the temperature range $T = 205$ K to melting point	See Refs.	102,103,104
Heat Deflection Temperature Under Load (°C)		
Martens method (DIN 53 458)	65–70	63
Heat Distortion		
A (ISO R 75)	120–130	63
B (DIN 53 461)	160–170	63
(ASTM D648)		
Homopolymer	124	73
Copolymer	110	73
Heat of Combustion (kJ/mol)		
Poly(oxymethylene)	-508.78 ± 10	68
	-511.6 ± 1.7	25
Heat of Degradation (kJ/mol)		
Needle-like poly(oxymethylene) crystals	55.05	62
Heat of Formation (kJ/mol)		
Homopolymer, 25°C (standard enthalpy)	−171.4	68
Heat of Fusion (kJ/kg)		
Poly(oxymethylene), theoretical value for 100% crystallinity	181–192	15
Homopolymer (acetyl end groups)	236	70
	390	119
Copolymer (trioxane/2% ethylene oxide)	315–336	15
Poly(oxymethylene), at equilibrium	166	57
Poly(oxymethylene), needle like crystals	315	62
Poly(oxymethylene), extended chain crystals	310	87
Poly(oxymethylene), fibrillar products	252	62

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Property	Value	Refs.
From solid state polymerization	264	84
	234	16
Heat of Polymerization (kJ/mol)		
Formaldehyde (gas, 1 bar, 298 K) → poly(oxymethylene) (solid)		
Values obtained from equilibrium data	−66.4 ± 1.0	59,66
	−51.3 ± 0.3	25
	−54.4	25
	−59.8	26
	69.9	59
	−71.1	21
	−72.0	22
	−55.7 ± 6.3	90
353 K	−68.2	27
Value from combustion data	−55.4	68
Value from calorimetric data	−66.2	80
Trioxane → poly(oxymethylene) at 298 K		
Gas → solid	−63.6	94
	−48.1	65
Gas → liquid	−41.5	59,94
Liquid → liquid	−0.4	59,94
Liquid → solid	−22.6	59,94
Solid → solid	−8.4	59,94
	−4.6 ± 3.9	66
	See Ref.	65
Tetraoxane → poly(oxymethylene)		
Solid → solid	−3.32 ± 0.16	86
In dichloroethane		
At 329–333 K	−26.25 ± 0.25	80
At 323 K	−17.6	92
Heat of Solution (kJ/mol)		
Water in branched copolymer, 50°C	19.3	2
Heat of Sublimation	See Lattice Energy	
Heat of Trimerization (kJ/(mol trioxane))		
Formaldehyde (gas) → trioxane (gas), at 298 K	−139.2 ± 2.1	91
Formaldehyde (gas) → trioxane (solid), at 298 K	−194.0 ± 2.5	91
	−188	69
	−187	80
Huggins Coefficient	See Viscosity – Molecular Weight Relationship	
Impact Strength (charpy) (kJ/m^2)		
+23°C	180	138
−30°C	160	138
Impact Strength (charpy) (notched) (kJ/m^2)		
Homopolymer +23°C	9	138
Homopolymer −30°C	7	138
Copolymer +23°C	6.5	138
Copolymer −30°C	6.5	138
Infrared Spectra	See Refs.	28,51,101, 110,136
IR Spectra	937 cm^{-1}	97
Macrocyclic POM	See Ref.	110
Interchain Distance (Å)		
Poly(oxymethylene)	4.47	71
Poly(1,3-dioxolane)	4.54	71
Lattice Energy (kJ/mol)		
Heat of sublimation of poly(oxymethylene) at 0 K (theoretical value)	15.9 ± 1.7	4
Mechanical Properties		
Impact modified poly(oxymethylene)		
Homopolymer	See Ref.	50
Copolymer	See Ref.	77

Property	Value	Refs.
Melt Index (MI) – Molecular Weight (M_w) Relationship (dg/min)		
$M_w = 6200–129000$; 190°C	$MI = 1.30 \times 10^{18} \times M_w^{-3.55}$	29
Melting Data		
At equilibrium	See Ref.	57
Melting Point (°C)		
Homopolymer		
Delrin 150	181	5
Extended chain fibers	182.5 ± 0.5	30
Folded chain spherulites	174	30
As a function of crystallization temperature	See Ref.	31
From solution polymerization of trioxane, ionically initiated		
Fibrous product	186–187	30
Powdery product	178–180	30
From vacuum sublimation of trioxane	195–200	30
Poly(oxymethylene) dimethylethers		
Monomer	−105.0	4
Dimer	−69.7	4
Trimer	−42.5	4
Tetramer	−9.8	4
Pentamer	18.3	4
Copolymer		
Hostaform C2520, M_w = 80000	167	5
Hostaform C9020, M_w = 58000	170	5
Various kinds and contents of comonomers	See Ref.	32
Copolymers with 0/2/4/6/7 mol% dioxolane	See Ref.	32
Trioxane-1,3-dioxolane copolymers (average values)		71
Mole fraction of 1,3-Dioxolane		
0.075	166	
0.15	156	
0.18	154	
0.25	147.5	
0.30	141.5	
0.34	134	
Poly(oxymethylene), homopolymer		
Equilibrium melting temperature	184 ± 2	57
	186.4	70
	189.2	70
Modulus, Tensile (1 mm/min) (MPa) ≡ (N/mm^2)		
DIN 53 457	2600–3400	3
Morphology	See Ref.	78
Mould Shrinkage (cm/cm)		
ASTM D 95		
Homopolymer	0.021	73
Copolymer	0.020	73
	See Refs.	79
NMR-Spectra	See Refs.	33,34,114
2D Dipolar ^{13}C	See Refs.	106,112,116, 122
Crystalline ^{13}C	See Refs.	111,113,117,120
Oligomers R–$(OCH_2)_n$–R		
Boiling points R = $-OCH_3$ (°C)		
$n = 2$	105.0	4
$n = 3$	155.9	4
$n = 4$	201.9	4
$n = 5$	242.3	4
Density R = $-OOC-CH_3$, 25°C (g/cm^3)		
$n = 1$	1.1283	35
$n = 2$	1.1610	35
$n = 3$	1.1823	35
Poly(oxymethylene), liquid, 25°C, theoretical value	1.328	4

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Property	Value	Refs.
Heat of vaporization	See Ref.	4
Infrared data R = $-OOCCH_3$		35

	Wave number (cm^{-1})	Absorption ($cm^{-1} \times mol^{-1}$)
$n = 1$	1760	395
$n = 2$	1755	330
$n = 3$	1755	330

Melting points		
Oligo (oxymethylene) dimethyl ether $n = 2–5$	See Ref.	4
Vapor pressure		
Oligo (oxymethylene) dimethyl ether $n = 2–5$	See Ref.	4
Permeability	See Refs. and corresponding chapter in this Handbook	73,79
Permeability, Apparent Activation Energy (kJ/mol)		
Water		
Homopolymer, melt index 2.3; 50°C	23.0	2
Copolymer, linear, melt index 2.5; 50°C	29.7	2
Copolymer, branched, melt index 3.9; 50°C	25.1	2
Copolymer, 25°C	13.0	17
Copolymer, branched, 25°C	6.3	17
Carbon dioxide		
Copolymer, 25°C	31.4	17
Copolymer, branched	25.1	17
Photoradical Ageing	See Ref.	134
Radiolysis (Gamma)	See Ref.	133
Raman Spectra	See Refs.	110,136
Powder/single crystal (cm^{-1})	909	97
Refractive Index (n_D)		
Extended chain crystals (fibers), different samples		
n	1.545–1.553	30
n_1	1.489	30
Poly(oxymethylene) at 25°C DIN 53491	1.48	63
Rockwell Hardness		
Homopolymer	M94, R120	73
Copolymer	M80	73
Rupture Behavior (Creep) for Medical Application	See Ref.	137
Schulz–Blaschke Coefficient	See Viscosity – Molecular Weight Relationship	
Second-Order Transition Point	See Ref.	55
Glass Transition Temperature (K)	191	61
Homopolymer	198	6
	188	23
	190	23
	199	24
α-Relaxation, activation energy (kJ/mol)		
For POM single crystal		
$T < 40°C$	155	36
$40°C < T < 90°C$	105	36
$T > 90°C$	159	36

Property	Value	Refs.
Shear Strength (N/mm^2) ASTM D732		
Homopolymer	66	73
Copolymer	53	73
Shear Modulus (GPa)		118
5 K	5.0	
77 K	4.7	
293 K	1.5	
Shrinkage of Fibers	See Ref.	109
Solvents		

Solvent	Dissolving temp. (°C)	Gel temp. (°C)	Refs.
Phenol	109	58	1
o-Chlorophenol	–	70	1
Aniline	130	102	1
Cyclopentanone	–	112	1
Ethylene carbonate	145	117	1
Cyclohexanone	–	119	1
Benzyl alcohol	132	119	1
Acetic anhydride	–	120	1
Formamide	150	135	1

Specific Volumes (cm^3/g)		
Homopolymer		
Delrin 150, at 20°C	0.7008	5
Delrin 550, at 20°C	0.6968	5
Crystalline phase		
Varying temperatures, t (°C)	$V_c = 0.669 + 7.85 \times 10^{-5}\ t + 5.93 \times 10^{-7}\ t^{-2}$	38
	$V_c = 6.64 \times 10^{-1} + 2.16 \times 10^{-4}\ t$	71
Amorphous phase	$V_a = 0.792 + 3.07 \times 10^{-4}\ t$	38
Varying temperatures, t (°C)	$V_a = 0.7553 + 5.78 \times 10^{-4}\ (t - 20)$	5
Copolymer		
Hostaform C2520	0.7082	5
Hostaform C9020	0.7064	5
Amorphous, phase (varying amount (X)) of copolymer, wt.% dioxolane, 20°C	$V_a^{20} = 0.7553 + 0.08674\ X$	5
Amorphous phase (varying temperatures, t (°C))	$V_a = 0.7578 + 5.78 \times 10^{-4}\ (t - 20)$	5
	$V_a = 0.7437 + 5.78 \times 10^{-4}\ t$	57
Stress–Strain Curves	See Ref.	39
Surface Resistivity (ohm)		
Homopolymer (ASTM D257)	2×10^{13}	73
Surface Tension, max (dyn/cm)	46.5	130
Taber Abrasion		
ASTM D1044, 1000 g load, CS-17 wheel, mg per 1000 cycles		
Homopolymer	20	73
Copolymer	14	73
Tensile Modulus (MPa)		
DIN 53457	2600–3400	63
Homopolymer	3200	138
Copolymer	2900	138
Ultra high tensile moduli tubes from a tensile drawing process under dielectrical heating	63000	49
Dependence on molecular weight	See Ref.	123
Tensile Creep Modulus (MPa)		
Homopolymer, 1 h	2800	138
Homopolymer, 1000 h	1500	138

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Property	Value	Refs.
Copolymer 1 h (MPa)	2500	139
Copolymer 1000 h (MPa)	1300	139
Tensile Strength at Break (N/mm^2)		
ASTM D638		
Homopolymer	69	73
Copolymer	59	73
Fibers	See Ref.	108
Dependence on molecular weight	See Ref.	131
Thermal Conductivity at 20°C (W/m/K)		
VDE 0304	0.292	3,63
Thermal Expansion	See Ref.	98
Dependent on irradiation		
Thermal Expansion	See Refs.	131,132
Dependent on structure		
Thermo-Acoustic Properties		
At 298 K		124,125
α ($10^{1-3}K^{-1}$)	0.243	
S_o^*	1.114	
B*	1.020	
A*	1.034	
Thermodynamic Data		
Poly(oxymethylene), recommended values	See Refs.	58,61
Tracking Resistance		
DIN 53480	KA 3 b	63
VDE 303	KC > 500	63
Trioxane		
Boiling point (°C)	114.5	64
Boiling point of azeotrope, 30% water, 70% trioxane (°C)	91.3	64
Density, 65°C (g/cm^3)	1.17	40
Dielectric constant, 20°C, 1.6 MHz	3.2–3.4	40
Dipole moment	2.18×10^{18}	64
Enthalpy (kJ/kg)		
0 K	0	22
50 K	7.97	
100 K	32.29	
200 K	104.9	
273.16 K	178.2	
298.16 K	208.0	
Entropy (kJ/kg/K)		
0 K	0	22
50 K	0.244	
100 K	0.573	
200 K	1.064	
273.16 K	1.373	
298.16 K	1.477	
298.16 K (gas, 1 bar)	3.165 ± 0.038	22
	3.207 ± 0.001	65
Explosive mixtures with air (vol.%)	3.5–28.7	64
Flash point (°C)	45	64
Heat of combustion, 23°C (kJ/mol)	1492	73,89
Heat of sublimation (kJ/mol)	56.6	73
	55.39	65
Heat of solution in dichloroethane at 50°C: trioxane (cryst.) → trioxane (soln.) (kJ/mol)	15.0 ± 0.1	80
Heat of vaporization at b.p. (kJ/mol)	41.5	73
	40.6	65
Melting point (°C)	62–64	38

* Acoustic parameters.

Property	Value	Refs.
Molecular weight	90.08	40
Solubility in different solvents	See Ref.	75
Solvents	Water, alcohols and ethers	
Specific heat (kJ/kg/K)		
0 K	0	22
50 K	0.372	
100 K	0.5767	
200 K	0.8783	
273.16 K	1.138	
298.16 K	1.237	
Vapor pressure at 25°C (k Pa)	1.7	73
Standard enthalpy, formation of trioxane (kJ/mol)	-509 ± 5	66
	-521 ± 4	81
Thermodynamic data, others	See Ref.	65
Unit Cell Spacings	See Crystallographic Data	
Vicat Softening Point (°C)		
DIN 53 460/ASTM D1525-65 T	160–170	3
ASTM D1525		
Homopolymer	157	138
Copolymer	151	138
	162	73
Viscosity – Molecular Weight Relationships		
$[\eta] = K \times M^{\alpha}$ (ml/g)	See also corresponding chapter in this Handbook	

Solvent	t (°C)	K ($\times 10^3$)	α	No. of samples FR[a]	No. of samples WP[b]	Molecular weight range ($\times 10^{-3}$)	Method[c]	Refs.
p-Chlorophenol, 2% α-pinene	130	54.3	0.66	?	?	?	OS cited in Ref.	41
	60	41.3[d]	0.724	–	4	62–129	LS	29
1H, 1H, 5H-octafluoropentanol-1	110	13.3	0.81	–	4	62–129	LS	29
Phenol–tetrachloroethane (25–75 wt.), 2% α-pinene	90	12.16	0.64	8	–	1.1–92	EG	42
Phenol	90	11.3	0.76	?	–	?	OS cited in Ref.	43
Dimethylformamide	150	44.0	0.66	–	4	89–285	LS	74
	130	22.4	0.71	7	–	1.5–15	EG	45
Hexafluoroacetone sesquihydrate + triethylamine	25	46.0	0.74	7	–	1.5–15	EG	45

[a] Fractionated.
[b] Polymer unfractionated.
[c] OS – Osmotic measurement; LS – Light scattering; EG – Viscosity measurement.
[d] $[\eta]_{inh}$.

Huggins coefficient k_H, $\eta_{sp/c} = \eta + k_H \eta^2 c$		
Hexafluoroacetone hydrate 25°C	0.41	44
Hexafluoroacetone–water mixtures	See. Ref.	44
Hexafluoroacetone–sesquihydrate + triethylamine, 25°C	0.31	45
Hexafluoroacetone–water (1 : 1.7 mol) + triethylamine, 25°C	0.30 ± 0.02	46
p-Chlorophenol/2% α-pinene (60°C)	0.40	47
Schulz–Blaschke coefficient K_{SB}, $\eta_{sp/c} = K_{SB}[\eta] \times \eta_{hsp} + [\eta]$		
Hexafluoroacetone hydrate, 25°C	0.29	44
Hexafluoroacetone–water mixtures, 25°C	See Ref.	44
Phenol–tetrachloroethane, 1 : 3 (per weight), 90°C	0.23	43
	0.21	48
Volume Resistivity (ohm cm)		
Film 0.2 mm, DIN 53 482/ASTM D257-61	$>10^{15}$	3
Water Absorption		
DIN 53472 (mg)		
After 96 h immersion	20–30	63

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Property	Value	Refs.
ASTM D570 (%)		
After 24 h immersion		
Homopolymer	0.25	73
Copolymer	0.22	73
Equilibrium, (50% r.h.)		
Homopolymer	0.2	73
Copolymer	0.16	73
Yield Stress (MPa)		
Homopolymer	71	138
Copolymer	66	138

REFERENCES

1. R. G. Alsup, J. O. Punderson, G. F. Leverett, J. Appl. Polym. Sci., **1**, 185 (1959).
2. G. F. Hardy, J. Polym. Sci. A-2, **5**, 671 (1967).
3. VDI, Richtlinien 2477, "Polyacetale für die Feinwerktechnik", VDI-Verlag, Düsseldorf, Gruendruck Oktober (1973).
4. R. H. Boyd, J. Polym. Sci., **50**, 133 (1961).
5. H. Wilski, Makromol. Chem., **150**, 209 (1971).
6. Y. Aoki, A. Nobuta, A. Chiba, M. Kaneko, Polym. J., **2**, 502 (1971).
7. B. E. Read, G. Williams, Polymer, **2**, 239 (1961).
8. M. Raetzsch, G. Eckhardt, Plaste Kautschuk, **20**, 424 (1973).
9. Y. Iwasa, T. Imoto, Nippon Kagaku Zasshi, **84**, 31 (1963).
10. L. A. Dudina, N. S. Enikolopyan, Polym. Sci. USSR, **5**, 36 (1964).
11. N. Grassie, R. S. Roche, unpublished.
12. J. Mejzlik, Makromol. Chem., **59**, 184 (1963).
13. H. Pennewiss, V. Jaacks, W. Kern, Makromol. Chem., **103**, 285 (1967).
14. Y. Ishida, Kolloid-Z., **171**, 149 (1960).
15. H. Wilski, Kolloid-Z., **248**, 867 (1971).
16. A. Tanaka, S. Uemura, Y. Ishida, J. Polym. Sci. (Phys.), **10**, 2093 (1972).
17. J. L. Williams, V. Stannett, J. Appl. Polym. Sci., **14**, 1949 (1970).
18. M. Braden, J. Polym. Sci. A-1, **6**, 1227 (1968).
19. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 263 (1962).
20. H. W. Starkweather, Jr., R. H. Boyd, J. Phys. Chem., **64**, 410 (1960).
21. J. B. Thompson (see J. F. Walker, "Formaldehyde", 3rd Ed. p. 180, Reinhold, New York 1964).
22. G. A. Clegg, T. P. Melia, A. Tyson, Polymer, **9**, 75 (1968).
23. F. S. Stehling, L. Mandelkern, J. Polym. Sci. B, **7**, 255 (1969).
24. K. Miki, S. Yamane, M. Kaneko, "Proc. Fifth Intern. Congress Rheology", vol. 3, University of Tokyo Press, p. 335 (1970).
25. F. S. Dainton, K. J. Ivin, D. A. G. Walmsley, Trans. Faraday Soc., **55**, 61 (1959).
26. J. F. Walker, "Formaldehyde", 3rd Ed., p. 180, Reinhold, New York, 1964.
27. Y. Iwasa, T. Imoto, J. Chem. Soc. Japan, Pure Chem. Sect., **84**, 29 (1963).
28. A. Novak, E. Whalley, Trans. Faraday Soc., **55**, 1484 (1959).
29. H. L. Wagner, K. F. Wissbrun, Makromol. Chem., **81**, 14 (1965).
30. M. Jaffe, B. Wunderlich, Kolloid-Z., **216/217**, 203 (1967).
31. K. F. Wissbrun, J. Polym. Sci. A-2, **4**, 827 (1966).
32. M. Inoue, J. Appl. Polym. Sci., **8**, 2225 (1964).
33. D. Fleischer, R. C. Schulz, Makromol. Chem., **166**, 103 (1972).
34. Y. Yamashita, T. Asakura, M. Okada, K. Ito, Makromol. Chem., **129**, 1 (1969).
35. J. Majer, Makromol. Chem., **82**, 169 (1965).
36. K. Miki, Polym. J., **1**, 432 (1970).
37. F. J. Boerio, D. D. Cornell, J. Polym. Sci. (Phys.), **11**, 391 (1973).
38. J. Majer, O. Hainová, Kolloid-Z., **201**, 23 (1965).
39. W. H. Linton, H. H. Goodman, J. Appl. Polym. Sci., **1**, 179 (1959).
40. J. F. Walker, "Formaldehyde", 3rd Ed., p. 192, Reinhold, New York, 1964.
41. A. Tanaka, S. Uemura, Y. Ishida, J. Polym. Sci. A-2, **8**, 1585 (1970).
42. K. Doerffel, H. Friedrich, H. Grohn, D. Wimmers, Plaste-Kautschuk, **12**, 524 (1965).
43. W. Thuemmler, Plaste-Kautschuk, **12**, 582 (1965).
44. N. Grassie, R. S. Roche, J. Polym. Sci. C, **16**, 4207 (1968).
45. L. Höhr. V. Jaacks, H. Cherdron, S. Iwabuchi, W. Kern, Makromol. Chem., **103**, 279 (1967).
46. W. H. Stockmayer, L. L. Chan, J. Polym. Sci. A-2, **4**, 437 (1966).
47. E. Kobayashi, S. Okamura, R. Signer, J. Appl. Polym. Sci., **12**, 1661 (1968).
48. H. Grohn, H. Friedrich, J. Polym. Sci. C, **16**, 3737 (1968).
49. K. Nakagawa, T. Konaka, S. Yamakawa Polymer, **26**, 84 (1985).
50. E. A. Flexman, Jr. Modern Plastics, February 1985, 72.
51. M. Mucha, Colloid Polym. Sci., **262**, 841 (1984).
52. M. Mucha, Colloid Polym. Sci., **262**, 851 (1984).
53. W. Kern, H. Cherdron, Makromol. Chem., **40**, 101 (1960).

54. R. Connolly, R. Gauvin, J. P. Chalifoux, Polym. Eng. Sci., **25**, 548 (1985).
55. H. Suzuki, J. Grebowicz, B. Wunderlich, Br. Polym. J., **17**, 1 (1985).
56. H. Suzuki, J. Grebowicz, B. Wunderlich, Makromol. Chem., **186**, 1109 (1985).
57. B. Wunderlich, Macromolecular Physics, vol. 3, "Crystal Melting", Academic Press, New York, 1980.
58. H. Suzuki, B. Wunderlich, J. Polym. Sci. Polym. Phys. Ed., **23**, 1671, (1985).
59. B. Wunderlich, Macromolecular Physics, vol. 2 "Crystal Nucleation, Growth, Annealing", Academic Press, New York, 1976.
60. B. Wunderlich, Macromolecular Physics, vol. 1, "Crystal Structure, Morphology, Defects", Academic Press, New York 1973.
61. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, 1001 (1981).
62. M. Iguchi, Macromol. Chem., **177**, 549 (1976).
63. Polyacetal-Formstoffe, VDI/VDE Richtlinie 2477, Nov. 1974, Beuth Vlg., Berlin.
64. K. C. Frisch, "Cyclic Monomers", Wiley-Interscience, New York.
65. T. P. Melia, D, Bailey, A. Tyson, J. Appl. Chem., **17**, 15 (1967).
66. W. K. Busfield, D. Merigold, Makromol. Chem., **138**, 65 (1970).
67. G. Kirsch, Hoechst AG, unpublished.
68. G. S. Parks, H. P. Mosher, J. Polym. Sci. A, **1**, 1979 (1963).
69. J. F. Walker, "Formaldehyde", 3rd ed., Reinhold, New York, (1964).
70. F. Salaris, A. Turturro, U. Bianchi, E. Martuscelli, Polymer, **19**, 1163 (1978).
71. P. J. Holdsworth, E. W. Fischer, Makromol. Chem., **175**, 2635 (1974).
72. M. L. Sagu, K. M. Sharan, J. Swarup, K. K. Bhattacharyya, J. Polym. Sci. (Chem), **14**, 1815 (1976).
73. "Encyclopedia of Polymer Science and Technology", vol. 1, Wiley, New York, 1964.
74. I. M. Belgovskii, N. S. Enikolopyan, L. S. Sakhonenko, Vysokomol. Soedin, **4**, 1179 (1962); Polym. Sci. (USSR), **4**, 367 (1963).
75. M. L. Sagu, J. Swarup, K. M. Sharan, K. K. Bhattacharyya, J. Chem. Eng. Data, **28**, 81 (1983).
76. J. F. Walker, "Formaldehyde", 2nd ed., Reinhold, New York, 1953.
77. F. Kloos, E. Wolters, Kunststoffe, **75**, 735 (1985).
78. O. Vogl, "Polyaldehydes", E. Arnold Ltd., London, 1967, p. 125.
79. R. B. Atkin, "Acetal Resins", Reinhold, New York (1962).
80. L. Leese, M. V. Baumber, Polymer, **6**, 269 (1965).
81. S. Okamura, E. Kobayashi, M. Takeda, K. Tomikawa, T. Higashimura, J. Polym. Sci., **4**, Pt. C, 827 (1963).
82. M. Delépine, M. Badoche, C. r. hebd. Séances, Acad. Sci., **214**, 777 (1942).
83. R. A. Fletscher, G. Pilcher, Trans. Faraday. Soc., **66**, 794 (1970).
84. A. Munoz-Escalona, E. W. Fischer, G. Wegner, International Symp. Macromolecules, Aug.–Sept. 1970, Leiden, reprint III, p. 56.
85. M. Dröscher, Makromol. Chem., **178**, 1195 (1977).
86. K. Nakatsuka, H. Suga, S. Seki, Polym. Lett., **7**, 361 (1969).
87. M. Dröscher, K. Hertwig, M. Rodriquez, G. Wegner, Makromol. Chem., **177**, 2793 (1976).
88. G. Opitz, G. Bürger, P. Steinmann, Plaste Kautschuk, **27**, 124 (1980).
89. G. F. Palfrey, E. I. du Pont De Nemours Co., Inc., unpublished communication, cit. in ref. 76, p. 149.
90. F. S. Dainton, K. J. Ivin, Trans. Faraday Soc., **46**, 331 (1959).
91. W. K. Busfield, D. Merigold, J. Chem. Soc. A, 2975 (1969).
92. T. Miki, T. Higashimura, S. Okamura, J. Polym. Sci., **8**, 157 (1970).
93. T. Miki, T. Higashimura, S. Okamura, J. Polym. Sci. A-1, **5**, 2997 (1967).
94. H. Sawada, "Thermodynamics of Polymerization", Marcel Dekker, New York, 1976.
95. T. P. Melia, Polymer, **7**, 640 (1966).
96. A. A. Berlin, S. A. Volfson, E. F. Oleinik, N. S. Yenikolopyan, Vysokolmol. Soedin. A, **12**, 443 (1970).
97. M. Kobayashi, H. Morishita, T. Ishioka, M. Igushi, M. Shimomura, T. Ikeda, J. Mol. Struct., **146**, 155 (1986).
98. H. N. Subramanyam, S. V. Subramanyam, Euro. Polym. J., **23**, 207 (1987).
99. F. Schultze-Gebhardt, Acta Polym., **38**, 36 (1987).
100. M. Kobayashi, H. Morishita, M. Shimomura, M. Iguchi, Macromolecules, **20**, 2453 (1987).
101. P. Schmidt, B. Schneider, J. Baldrian, L. Terlemezyan, M. Mihailov, B. Ivanova, Polymer, **28**, 217 (1987).
102. H. Suzuki, B. Wunderlich, J. Therm. Anal., **29**, 1369 (1984).
103. H. Suzuki, J. Grebowicz, B. Wunderlich, Makromol. Chem., **186**, 1109 (1985).
104. J. Grebowicz, H. Suzuki, B. Wunderlich, Polymer, **26**, 561 (1985).
105. Y. Nakase, I. Kuriyama, A. Odajima, Adv. Polym. Sci., **65**, 79 (1984).
106. W. E. J. R. Maas, A. P. M. Kentgens, W. S. Veeman, J. Chem. Phys., **87**, 6854 (1987).
107. Y. Ishida, H.Oktani, K. Abe, S. Tsuge, K. Yamamoto, K. Kato, Macromolecules, **28**, 6528 (1995).
108. T. Komatsu, S. Enoki, A. Aoshima, Polymer, **33**, 2123 (1992).
109. T. Komatsu, S. Enoki, A. Aoshima, Polymer, **32**, 2992 (1991).
110. M. Kobayashi, M. Sakashita, M. Hasegawa, Macromolekules, **24**, 4796 (1991).
111. P. Tekely, D. Canet, J. J. Depuech, J. Violet, Magn. Reson. Chem., **28**, S10 (1990).
112. A. Hagemeyer, D. Schäfer, K. Schmidt-Rohr, B. Blümich, H. W. Spiess, Mol. Cryst. Liq. Cryst., **187**, 223 (1990).
113. M. Kobayashi, M. Murano, A. Kaji, Macromolecules, **23**, 3051 (1990).
114. M. Hasegawa, K. Yamamoto, T. Shiwaku, T. Hashimoto, Macromolecules, **23**, 2629 (1990).

115. M. Kobayashi, H. Morishita, M. Shimomura, Macromolecules, **22**, 3726 (1989).
116. Y. Yang, A. Hagemeyer, H. W. Spiess, Macromolecules, **22**, 1004 (1989).
117. H. Kurosu, T. Komoto, I. Ando, J. Mol. Struct., **176**, 279 (1988).
118. K. Ahlborn, Cryogenics, **28**, 234 (1988).
119. H. W. Starkweather, G. A. Jones, P. Zoller, J. Polym. Sci. Part B, **26**, 257 (1988).
120. H. Kurosu, T. Yamanobe, I. Ando, J. Chem. Phys., **89**, 5216 (1988).
121. Y.-Y. Fahn, W-Y Chiang, Engng. Plast, **3**, 186 (1990).
122. H. W. Spiess, Bull. Magn. Reson., **11**, 303 (1989).
123. C. J. G. Plummer, Ph. Béguelin, H.H. Kausch ESIS Publ., **19** 265 (1995).
124. B. K. Sharma, J. Polymer Mater., **1**, 193 (1984).
125. B. K. Sharma, R. R. Reddy, J. Polymer Mater., **1**, 178 (1984).
126. H. E. Thomson, D. P. Drysdale, Fire and Materials, **11**, 163 (1987).
127. A. M. Zihlif, Mater. Chem. Phys., **13**, 21 (1985).
128. C. J. G. Plummer, N. Cudre-Mauroux, H. H. Kausch, Polym. Eng. Sci., **34**, 318 (1994).
129. K. Tashiro, M. Kobayashi, H. Tadokoro, Polym. Eng. Sci., **34**, 308 (1994).
130. T. Hata, Y. Kitazahi, T. Santo, J. Adhesion, **21**, 177 (1987).
131. R. R. Zahran, A. A. Abolmagd, Poly. Eng. Sci., **34**, 153 (1994).
132. R. R. Zahran, S. S. Shanouda, Y. A. El-Tawil, A. El-Kayar, Poly. Eng. Sci., **36**, 1377 (1996).
133. A. Faucitano, A. Buttafava, F. Martinotti, P. Ferloni, A. Magistis, Radiat. Phys. Chem., **40**, 347 (1992).
134. M. Y. Mel'Nikov, E. N. Seropegina, Radiat Phys. Chem., **33**, 151 (1989).
135. P. Hennebert, Biomaterials, **9**, 162 (1988).
136. M. Shimomura, Busshitsu Kogaku, **2**, 451 (1994).
137. S. H. Teoh, ASTM Spez. Techn. Publ., **STP 1173**, 77 (1994).
138. Campus 4.0 (1996) CWF GmbH Frankfurt (contains also values of different grades and different sales products of different manufacturers)

Physical Constants of Poly(oxyethylene-oxyterephthaloyl) (Poly(ethylene terephthalate))

Mark Rule
1 Coca Cola Plaza NW, TEC 147, Atlanta, GA, USA

Properties of films are given for Mylar® Type A (DuPont) biaxially oriented and crystalline films. Properties of filaments are given for various products.

Property	Value	Refs.
Birefringence of Filaments (sodium light)*		1
Draw Ratio		
2.0	0.040	
3.0	0.092	
4.0	0.167	
5.0	0.193	
Bursting Strength (g/cm^2)		
1 mil film, 23°C	46×10^2	2,3
Brillouin Light Scattering	See Ref. 4	
Coefficient of Friction (film)		
Kinetic, film to film	0.45	2,3
film to steel	0.14	6
Coefficient of Thermal Conductivity (film) (W/m/s/K)		
(See also Thermal Conductivity)		
100°C	37.5×10^{-3}	2,3
Coefficient of Thermal Expansion (film) (K^{-1})		
20°C–50°C	1.7×10^{-5}	2,3
Melt	6.55×10^{-4}	5
Coefficient of Volume Expansion (K^{-1})		7
Crystalline, below T_g	1.7×10^{-4}	
Crystalline, above T_g	3.94×10^{-4}	
Compressibility (melt) (MPa) ($\times 10^6$)	6.99	5
Conductivity for Direct Current	See Fig. 1 overleaf	
Crystallographic Data	See Unit cell	
Density (Mg/m^3) ≡ (g/cm^3)		
0% Crystalline	1.333	81
100% Crystalline	1.455	82
Amorphous, non-oriented	1.335	9
Partly cryst., non-oriented	1.385	9
Partly cryst., oriented	1.390	9
Highly cryst., non-oriented	1.420	10
Calculated, crystal	1.515	11
	(1.501)	12
Dielectric Strength (film) (V/cm)		2,3
23°C, 60 Hz	2.95×10^6	
150°C, 60 Hz	2.75×10^6	
Dielectric Constant (film)		2,3
23°C, 60 Hz	3.30	
23°C, 1 kHz	3.25	
23°C, 1 MHz	3.0	
23°C, 1 GHz	2.8	
150°C, 60 Hz	3.7	
Dissipation Factor (film)		2,3
23°C, 60 Hz	0.0025	
23°C, 1 kHz	0.0050	
23°C, 1 MHz	0.016	
23°C, 1 GHz	0.003	
150°C, 60 Hz	0.00040	
Elastic Constants (filaments) (Pa^{-1})		13
Oriented		
S_{11} transverse	16×10^{-10}	
S_{33} longitudinal	0.71×10^{-10}	
S_{44} torsional	14×10^{-10}	
S_{12}	-5.8×10^{-10}	
S_{13}	-0.31×10^{-10}	
Unoriented		
S_{11} or S_{33} extensional	4.4×10^{-10}	
S_{44} torsional	11×10^{-10}	
Enthalpy of Fusion (kJ/mol)	2.69	10
Entropy of Fusion (J/mol/K)	48.6	10
Folding Endurance (film) (cycles)		
23°C	300 000	2,3

*Value sensitive to semicrystalline morphology.

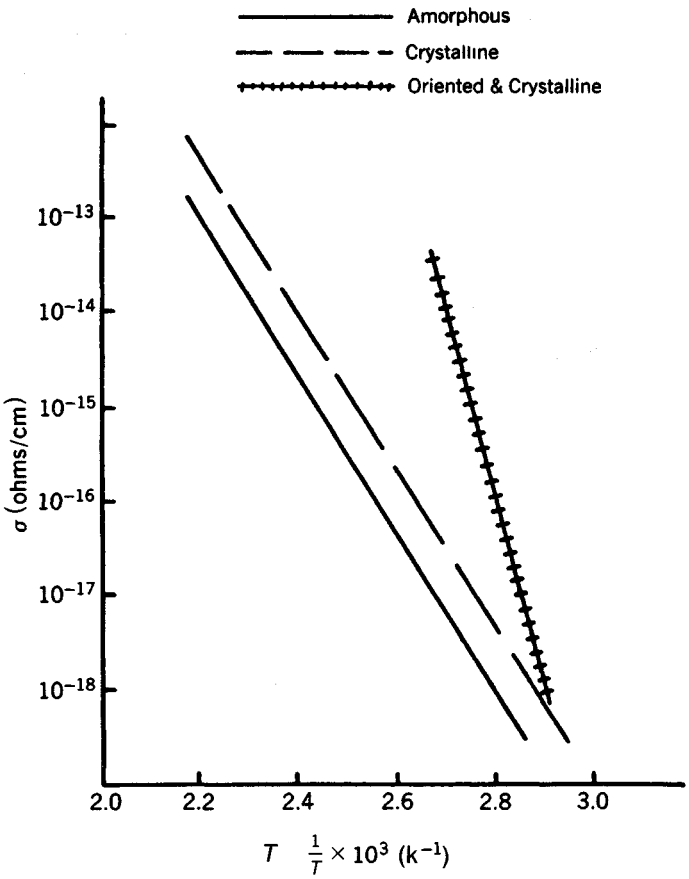

Figure 1. Direct current conductivity at various temperatures and degrees of orientation and crystallinity (8).

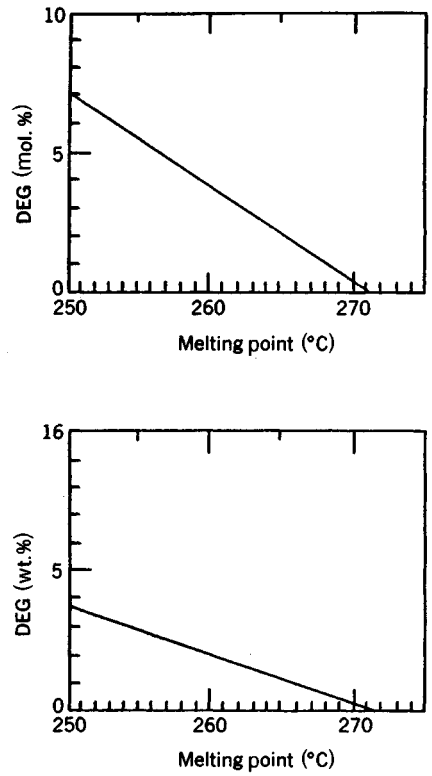

Figure 2. Effect of diethylene glycol (DEG) content on melting point. Melting point (°C) = 271–273 (2,29).

Property	Value	Refs.
Glass Transition Temperature* (°C)		
Amorphous	67	7,14
Crystalline	81	14
Crystalline and oriented	125	15
Heat capacity See Specific heat		
Heat of Combustion (kJ/kg)	-2.16×10^4	16
Heat of Sorption (film) (kJ/kg/mol)		
Carbon dioxide	-3.1×10^4	17
Methane	-2.3×10^4	
Solubility of Carbon Dioxide		
25–115°C, 0–20 atm		83
Hygroscopic Coefficient of Expansion (film) (cm/cm%/RH)		
20–92% RH,	1.1×10^{-5}	2,3
Impact Strength (film) (J/m)		
23°C	2.4×10^2	2,3
Infrared Spectra See Refs. 18–25		
Insulation Resistance (film) (Mohm mfds)		
100°C	5000	26,27,28
130°C	400	
150°C	100	

*Value sensitive to semicrystalline morphology.

Property	Value	Refs.
Melting Point (°C)		
Equilibrium	280 (310)	11(12)
Commercial PET (metastable crystallites)	250–265	2,3
Effect of diethylene glycol content	See Fig. 2	29

Melt Viscosity vs. Intrinsic Viscosity

Melt viscosity (Pa s) at 280°C[a]	Intrinsic viscosity (ml/g), 30°C in *s*-tetrachloroethane/phenol (40/60)
0.45	10
4.5	20
25.0	30
95.0	40
115.0	50
145.0	60
800.0	70
1180.0	80
20000.0	90

[a] Taken from Fig. 14 of Ref. 2.

Moisture Absorption (%)		
Immersion in water at 25°C for 1 w	0.8	27,28

Nuclear Magnetic Resonance Spectrum See Refs. 13,30,31

Oligomers – acyclic

Structure	Molecular weight	Melting point (°C)	Refs.
$H(GA)_1OH$[a]	210.2	178	32
$H(GA)_2OH$	402.4	200–205	32
$H(GA)_3OH$	594.6	219–223	32
		179–183	33
		186	34
$H(GA)_1$–G–H	254.2	109–110	32
$H(GA)_2$–G–H	446.4	173–174	32
$H(GA)_3$–G–H	638.6	200–205	32
$H(GA)_4$–G–H	830.8	213–216	32
$H(GA)_5$–G–H	1023.0	218–220	32
HO–A–$(GA)_1$OH	358.3	>360	32
HO–A–$(GA)_2$OH	550.5	280–281	32
HO–A–$(GA)_3$OH	742.7	268–270	32
HO–A–$(GA)_4$OH	934.9	252–255	32
HO–A–$(GA)_5$OH	1127.1	233–236	32

[a] G = $-OCH_2CH_2O-$; A = $-OC-C_6H_4-CO-$

Property	Value	Refs.
Oligomers – cyclic		
CYCLIC DIMER		
Melting point (°C)	175, 224	33
	229	35
Unit cell (nm)	$a = 0.858$	35
	$b = 1.275$	
	$c = 0.801$	
	$\beta = 90.7°$	
CYCLIC TRIMER (B-TYPE CRYSTAL)		
Melting point (°C)	319	33
	317 –320	36
	321	37
Crystalline transition (°C)	199	37
(A-type → B-type)	195	38
CYCLIC TETRAMER		
Melting point (°C)	326	33
CYCLIC PENTAMER		
Melting point (°C)	256	33
Oligomers – Isolation See Ref. 39		
Optical Haze* (ASTM D1003–61) See Ref.		40
Permeability (film) See corresponding chapter in this Handbook.		
Water (cm^2/s) $(2.11–9.97) \times 10^{-9}$		84
Photoacoustic Spectroscopy See Ref. 41		
Poisson's Ratio (Oriented filament)		
Extensional	0.44	13
Transverse	0.37	
Raman Spectra		21,42–46
Frequency (cm^{-1})	1730	
	1618	
	1096	
	857	
	632	
	278	

*Value sensitive to semicrystalline morphology.

Property	Value	Refs.
Refractive Index (film), (Na Light)		2
Amorphous, 25°C	1.5760	
Crystalline and biaxially oriented, 23°C	1.64	
80,90,100°C		85
Refractive Index Increment (specific) (cm^3/g) (in hexafluoroisopropanol solution)	0.257 ± 0.004	47
Resistivity		2,3
Surface (ohm/cm^2)		
23°C, 30% RH	2×10^{15}	
23°C, 80% RH	2×10^{11}	
Volume (ohm cm)		
23°C	10^{18}	
150°C	10^{14}	
Rheological Spectrum See Refs. 48,49		
Service Temperature (°C)	–60 to +150	2,3
Shrinkage (film) (%) 150°C, 30 min.	2–3	3
Solubility Constants (film) (cm^3/STP cm^3/Pa) ($\times 10^7$)		17
3 mil, 25°C Nitrogen	4.3	
Oxygen	7.5	
Methane	19.7	
Argon	0.8	
Solvents See chapter "Solvents–Nonsolvents" in this Handbook		

Solvent–Nonsolvent for Fractionation

Solvent	Nonsolvent	
o-Chlorophenol	Hexane	50
Phenol/tetrachloroethane (1 : 1)	Gasoline	51
Phenol/chlorobenzene (1 : 1)	Gasoline	52
Phenol/dichlorobenzene (2 : 3)	Benzene	53
Phenol	Cyclohexane	54
Phenol	Ethanol	55

Sonic Velocity (filaments) (m/s)		56
10 kHz unoriented	1400	
Highly oriented	5900	

Specific Heat (kJ/kg/K) $C_p = 4.184\ (A + (B \times T))$ 57

Condition of polymer	*A*	*B* ($\times 10^4$)	Effective temperature *T* (°C)
Molten polymer	0.3243	5.65	270 to 290
Flake	0.2502	9.40	–20 to 60
Yarn (undrawn)	0.2469	10.07	–5 to 60
Yarn (drawn)	0.2482	9.89	–10 to 55
Yarn (drawn + annealed)	0.2431	9.23	–10 to 80
	0.2502	9.31	100 to 200

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Property	Value	Refs.
1.2–40 K	See Ref.	58
Below 1 K	See Ref.	59
Stick Point Temperature (°C)	230–240	4
Stress–Strain Curves for Filaments See Refs.		60,61
Surface Tension (mN/m) ≡ (dyn/cm)		
Solid/liquids, 25°C	39.5	62
	42.1	63
Molten, 290°C	27 ± 3	64
Tear Strength (film (g/m)		
Initial, 23°C	23.6×10^6	2,3
Propagating, 23°C	0.59×10^6	
Tensile Strength (film) (Mpa) ≡ (N/mm^2)		
23°C,	172	2,3
Thermal Conductivity (film) (W/m/K)		
33°C	0.147	65
1.2–40 K	See Ref.	58
Below 1 K	See Ref.	59
Thermal Diffusivity (film) (cm^2 s)		
33°C	9.29×10^{-4}	65
Torsional Modulus (MPa) ≡ (N/mm^2)		
Oriented filament	720	13

Property	Value	Refs.
Transition Temperatures (°C)		
From second moment (NMR) measurements on fibers		
Unoriented	120 ± 5	13,31
Oriented	140 ± 10	
Unit Cell (nm)		66,11,12

	Previously accepted (Ref. 66)	Corrected (Ref. 11)	Corrected (Ref. 12)
X-RAY DIFFRACTION (291 K)			
a	0.456	0.448	0.450
b	0.594	0.585	0.590
c	1.075	1.075	1.076
α	98.5°	99.5°	100.3°
β	118°	118.4°	118.6°
γ	112°	111.2°	110.8°
ELECTRON DIFFRACTION (Ref. 67)			
a	0.452		67
b	0.598		
c	1.077		
α	101°		
β	118°		
γ	111°		

Viscocity–Molecular Weight Relationship See Sections I, II			
Young's Modulus (MPa) ≡ (N/mm^2)			
Oriented Filament – extensional		1.41×10^4	13
– transverse		0.063×10^4	
Zero Strength Temperature (film) (°C)		248	68

Intrinsic Viscosity – Molecular Weight Relationships (Unfractionated)[a]

Solvent	*T* (°C)	*K* ($\times 10^4$)	*a*	Molecular weight range ($\times 10^{-3}$)	Method	Refs.
Trifluoroacetic acid	30	4.33	0.68	26–118	Light scattering	69
Tetrachloroethane/phenol (5 : 3)	30	2.29	0.73	26–118	Light scattering	69
o-Chlorophenol	25	6.56	0.73	12–25	Osmometry	70
o-Chlorophenol	25	3.0	0.77	13–28	End group	71
Tetrachloroethane/phenol (1 : 1)	25	2.1	0.82	5–25	End group	72
Tetrachloroethane/phenol (1 : 1)	20	1.27	0.86	5–21	End group	73
Tetrachloroethane/phenol (1 : 1)	20	7.55	0.685	3–30	End group	54
Phenol/2,4,6-trichlorophenol (10 : 7)	29.8	2.10	0.80	1–8	End group	74
o-Chlorophenol	25	6.31	0.658	6–70	Size exclusion chromatography	
					Light scattering	75
Phenol/Tetrachloroethane (3 : 2)	25	7.44	0.648	6–70	Light scattering	75
Hexafluoroisopropanol	25	5.20	0.695	6–70	Light scattering	75
Pentafluorophenol	25	3.85	0.723	6–70	Light scattering	75
Hexafluoroisopropanol/pentafluorophenol (1 : 1)	25	4.50	0.705	6–70	Light scattering	75
Methylene chloride/hexafluoroisopropanol (7 : 3)	–	4.03	0.691	–	Size exclusion chromatography	76
o-Chlorophenol/chloroform (1 : 3)	25	1.49	0.56	9–100	Chromatography	77

[a] $(\eta) = K \times M^a$.

Intrinsic Viscosity – Molecular Weight Relationships (Fractionated)[a]

Solvent	T (°C)	K ($\times 10^4$)	a	Molecular weight range ($\times 10^{-3}$)	Method	Refs.
o-Chlorophenol	25	1.9	0.81	15–38	End group	78
Dichloroacetic acid	25	67	0.47	15–38	End group	78
Trifluoroacetic acid	25	14	0.64	15–38	End group	78
Phenol/tetrachloroethane (2 : 3)	25	14	0.64	15–38	End group	78
o-Chlorophenol	25	4.25	0.69	20–200	Sedimentation diffusion	79
Tetrachloroethane/phenol (1 : 1)	20	0.9	0.87	5–21	End group	73
Phenol/dichloroethane (2 : 3)	20	0.92	0.85	8–30	End group	53
o-Chlorophenol/chloroform (1 : 9)	25	0.584	0.91	4–30	Size exclusion chromatography	80

[a] $(\eta) = K \times M^a$.

REFERENCES

1. G. Farrow, J. Bagley, Textile Res. J., **32**, 587 (1962).
2. C. J. Heffelfinger, K. L. Knox, in: O. J. Sweeting (Ed.), "Polyester Films", "The Science and Technology of Polymer Films", vol. 11, Wiley-Interscience, New York, 1971, p. 587.
3. J. M. Hawthorne, C. J. Heffelfinger, K. L. Knox, in: H. F. Mark and N. G. Gaylord (Eds.), "Polyester Films", in Encyclopedia of Polymer Science, vol. 11, Interscience, New York, 1969, p. 42.
4. D. B. Cavanaugh and C. H. Wang, J. Polym. Sci., Polym. Phys. Ed., **19**, 1911 (1981).
5. H. W. Starkweather Jr., Paul Zoller, G. A. Jones, J. Polym. Sci., Polym. Phys. Ed., **21**, 295 (1983).
6. Y. Yamada, K. Tanaka, Proc. ACS Div. Polym. Mater. Sci. Eng., **50**, 462 (1984).
7. H. J. Kolb, E. F. Izard, J. Appl. Phys., **20**, 564 (1949).
8. L. E. Amborski, J. Polym. Sci., **62**, 331 (1962).
9. A. B. Thompson, D. W. Woods, Nature, **176**, 78 (1955).
10. A. Mehta, U. Gaur, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **16**, 289 (1978).
11. S. Fakirov, E. W. Fisher, G. H. Schmidt, Makromol. Chem., **176**, 2459 (1975).
12. Y. Kinoshita, R. Nakamura, Y. Kitano, T. Ashida, Polymer Preprints, **20**, 454 (1979).
13. I. M. Ward, J. Macromol. Sci. B, **1**, 667 (1967).
14. O. B. Edgar, R. Hill, J. Polym. Sci., **8**, 1 (1952).
15. D. W. Woods, Nature, **174**, 753 (1954).
16. R. B. LeBlanc, J. Am. Assoc. Textile Chem. Color., **2**, 123 (1970).
17. W. R. Vieth, H. H. Alcalay, A. J. Frabetti, J. Appl. Polym. Sci., **8**, 2125 (1964).
18. S. Krimm, Fortschr. Hochpolymer Forsch., **2**, 51 (1960).
19. J. L. Koenig, M. J. Hannon, J. Macromol. Sci. B, **1**, 119 (1967).
20. T. R. Manley, D. A. Williams, J. Polym. Sci., C, **22**, 1009 (1969).
21. F. J. Boerio, S. K. Bahl, G. E. McGraw, J. Polym. Sci., Polym. Phys. Ed., **14**, 1029 (1976).
22. W. Frank, H. Fiedler, W. Strohmeier, J. Appl. Polym. Sci., Appl. Polym. Symp., **34**, 75 (1978).
23. F. J. Boerio, J. L. Koenig, J. Polym. Sci. A-2, **7**, 1517 (1971).
24. J. L. Koenig, D. Kormos, in: M. Shen (Ed.), "Fourier Transform Infrared Spectroscopy of the Polymeric Amorphous Phase", Contemporary Topics in Polymer Science, vol. 3, Plenum, New York, 1979.
25. W. F. X. Frank, W. Strohmeier, M. L. Hallensleben, Polymer, **22**, 615 (1981).
26. DuPont Technical Bulletin No. 1–2–53.
27. DuPont Technical Bulletin No. M-1A.
28. D. D. Lanning, Prod. Eng., July, 1987 (1956).
29. R. Jannssen, H. Ruysschaert, R. Vroom, Makromol. Chem., **77**, 153 (1964).
30. D. L. Vanderhart, G. G. A. Bohm, V. D. Mochel, Polym. Preprints, **22**, 261 (1981).
31. S. Nohara, Chem. High Polym. (Japan), **14**, 318 (1957).
32. H. Zahn, G. B. Gleitsmann, Angew. Chem., **75**, 772 (1963).
33. L. H. Peebles Jr., M. W. Huffman, C. T. Ablett, J. Polym. Sci. A-1, **7**, 479 (1969).
34. H. Binder, US. Pat. 2,855,432.
35. H. Repin, E. Papanikolau, J. Polym. Sci. A-1, **7**, 3426 (1969).
36. F. L. Hamb, L. C. Trent, J. Polym. Sci. B, **5**, 1057 (1967).
37. E. Ito, S. Okajima, J. Polym. Sci. B, **7**, 583 (1966).
38. G. L. Binns, Polymer, **7**, 583 (1966).
39. W. R. Hudgins, K. Theurer, T. Mariani, J. Appl. Polym. Sci.: Appl. Polym. Symposium, **34**, 145 (1978).
40. S. A. Jabarin, Polym. Engineer. Sci., **22**, 815 (1982).
41. E. Balizer, H. Talaat, J. Phys., **44**, C6 (1983).
42. A. J. Melveger, J. Polym. Sci. A-2, **10**, 317 (1972).
43. G. E. McGraw, Polym. Preprints, **11**, 1122 (1970).
44. F. J. Beblase, M. L. McKenlvy, M. Lewin, B. J. Bulkin, J. Polym. Sci.: Polym. Lett. Ed., **23**, 109 (1985).
45. J. Purvis, D. I. Bower, J. Polym. Sci.: Polym. Phys. Ed., **14**, 1461 (1976).
46. J. Derouault, P. J. Hendra, M. E. A. Cudby, G. Fraser, J. Walker, H. A. W. Willis, in: J. P. Mathieu (Ed.), Proceeding Third International Conference on Raman Spectroscopy, vol. I, Heydensons, 1973, p. 277.
47. S. A. Berkowitz, J. Liquid Chromatog., **6**, 1359 (1983).
48. Y. H. Park, M. S. Rhim, J. Korean Soc. Textile Eng. Chem., 35 (1983).
49. D. R. Gregory, J. Appl. Polym. Sci., **16**, 1479 (1972).
50. W. R. Moore, R. P. Sheldon, J. Textile Inst., **50**, T294 (1959).

51. A. Gordienko, Faserforsch. Textiltechn., **4**, 199 (1953).
52. A. A. Geller, A. A. Konkin, V.A. Myagkov, Khim. Volokna, **3**, 10 (1960).
53. Ye. V. Kuznetsova, A. O. Vizel, I. M. Shermergorn, S. S. Tyulenev, Vysokomolekul. Soedin., **2**, 205 (1960).
54. M. M. Koepp, H. Wemer, Makromol. Chem., **32**, 79 (1959).
55. E. Turska-Kusmierz, T. Skwarski, Prace Inst. Wlokiennictwa, **2**, 49 (1953).
56. W. W. Moseley, Jr., J. Appl. Polym. Sci., **3**, 266 (1960).
57. C. W. Smith, M. Dole, J. Polym. Sci., **20**, 37 (1956).
58. A. Assfalg, J. Phys. Chem. Solids, **36**, 1389 (1975).
59. D. Grieg, M. S. Sahota, J Phys. C: Solid State Phys., **16**, L1051 (1983).
60. A. B. Thompson, J. Polym. Sci., **34**, 741 (1959).
61. Ludewig, "Polyester Fibers, Chemistry and Technology", Wiley-Interscience, New York 1964.
62. D. H. Kaelble, J. Adhesion, **2**, 66 (1970).
63. S. Wu, Polym. Preprints, **11**, 1291 (1970).
64. H. T. Patterson, K. H. Hu, T. H. Grindstaff, Polym. Preprints, **11**, 1299 (1970).
65. R. C. Steere, J. Appl. Phys., **37**, 3338 (1966).
66. R. De P. Daubeny, C. W. Bunn, C. J. Brown, Proc. Roy. Soc. (London) A, **226**, 531 (1954).
67. Y. Y. Tomashpol'skii, G. S. Morkova, Polym. Sci. USSR, **6**, 316 (1964).
68. B. V. Petukhov, "The Technology of Polyester Fibers", Macmillan, New York, 1963, p. 388.
69. M. L. Wallach, Makromol. Chem., **103**, 19 (1967).
70. J. Marshall, A. Todd, Trans. Faraday Soc., **49**, 67 (1953).
71. I. M. Ward, Nature, **180**, 141 (1957).
72. A. Conix, Makromol. Chem., **26**, 226 (1958).
73. W. Griehl, S. Neve, Faserforsch. Textiletechn., **5**, 423 (1954).
74. N. G. Gaylord, S. Rosenbaum, J. Polym. Sci., **39**, 545 (1959).
75. S. Berkowitz, J. Appl. Polym. Sci., **29**, 4353 (1984).
76. J. R. Overton, H. L. Browning, Org. Coatings Appl. Polym. Sci. Proc., **48**, 940 (1983).
77. S. A. Jabarin, D. C. Balduff, J. Liquid Chromatogr., **5**, 1825 (1982).
78. W. R. Moore, D. Sanderson, Polymer, **9**, 153 (1968).
79. Von G. Meyerhoff, S. Shimotsuma, Makromol. Chem., **135**, 195 (1970).
80. M. Sang, N. Jin, J. Liquid Chromatogr., **5**, 1665 (1982).
81. C. W. Bunn, R. P. Daubeny, Proc. Roy. Soc., **226**, 531 (1954).
82. E. W. Fischer, S. Fakirov, J. Mater. Soc., **11**, 104 (1976).
83. W. J. Koros, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **16**, 147 (1978).
84. M. J. Kloppers, F. Bellucci, R. M. Latanision, J. E. Breuuan, J. Appl. Polym. Sci., **48**(12), 2197 (1993).
85. M. Cakmak, J. L. White, J. E. Spruiell Polym. Sci. Eng., **29**(21), 1534 (1989).

Physical Constants of Poly(oxytetramethylene-oxyterephthaloyl) and Copolymers with Tetramethylene Oxide Thermoplastic Elastomers*

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The literature gives different chemical abbreviations for this segmented polyester/polyether copolymer, such as PBT/PTMO, PBT/PTHF (where PTHF abbreviates polytetrahydrofuran), and PBT/POTM (where POTM abbreviates polyoxy-tetramethylene). Occasionally, components of the abbreviation are reversed (as in PTHF/PBT). All these abbreviations represent the same copolymer, which is available with different weight percentages of polyether being present.

Property	Weight percentage polyether in copolymer[a]					
	0%[b]	10%	25%	35%	50%	60%
Hardness, shore D	80	74	63	55	45	38
Density (mg/m^3)	1.30	1.27	1.23	1.20	1.16	1.12
Coefficient of expansion (vol/K)						
@ − 20°C	210×10^{-6}	240×10^{-6}	360×10^{-6}	300×10^{-6}		
20°C	210×10^{-6}	330×10^{-6}	420×10^{-6}	450×10^{-6}	480×10^{-6}	660×10^{-6}
80°C		510×10^{-6}	540×10^{-6}	600×10^{-6}		
THERMAL PROPERTIES						
Glass transition temp. (°C)	55 /48[c]	44 /24[c]	2 /− 28[c]	− 23 /− 46[c]	− 50	− 60
Heat capacity, C_p (kJ kg/K) @ 100°C	1.35	1.61	1.72	1.78	1.95	2.01
Melting temperature (°C)	223	221	210	202	183	192
Heat of fusion (kJ/kg)	54	60	41	38	29	25
ELECTRICAL PROPERTIES						
Dielectric constant (1 kHz)	3.0	3.35	4.1	4.3	4.5	4.5
Dissipation factor (1 kHz)	0.002	0.037	0.017	0.012	0.006	0.005
Volume resistivity (ohm m)	5.10×10^{14}	8.8×10^{13}	1.2×10^{11}	2.5×10^{10}		
MECHANICAL PROPERTIES						
Ultimate elongation (%) (ASTM D 412)	200	360	540	570	700	650
Tensile strength (MPa) (DIN 53504)	52	45	38	32	21	17
Initial slope of stress–strain curve, Young's modulus (MPa)	2600	830	300	170	90	53
Storage modulus G′ (log MPa)	2.74	2.70	2.17	1.94	1.60	1.25
Loss modulus G″ (log MPa)	0.78	1.58	0.96	0.48	0.08	− 0.27
Loss tangent G″/G′	0.011	0.076	0.062	0.035	0.03	0.03
Resilience (rebound) (%)			40	53	60	65

[a] $\bar{M}_n = 1000$ for the polyether except for the 60 wt.% copolymer, in which $\bar{M}_n = 2000$.
[b] 0% refers to the properties for pure poly(oxytetramethylene-oxyterephthaloyl) (poly(butylene terephthalate)).
[c] Two different measurement values are given, top tan δ/top G″, 1 Hz.
[d] Top of highest peak.

* All data are unpublished results from the Akzo Fibers and Polymers Research Laboratories, Arnhem, The Netherlands.

Physical Constants of Various Polyamides: Poly[imino(1-oxohexamethylene)], (Polyamide 6) Poly(iminoadipoyl-iminohexamethylene), (Polyamide 66) Poly(iminohexamethylene-iminosebacoyl), (Polyamide 610) Poly[imino(1-oxododecamethylene)], (Polyamide 12)*

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Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
Abrasion	See Wear Resistance				13
Absorption (%)					
Water, moldings 20–90°C, saturation	9.5 ± 0.5	8.5 ± 0.5	3.3 ± 0.3	1.6 ± 0.2	
Ethanol, moldings 20°C, saturation	9–17	9–12	8–13	9	
Butanol, moldings 20°C, saturation	5–9	4–8	8–12	–	
Glycol, moldings 20°C, saturation	6–13	2–10	2–4	–	
Methanol, moldings 20°C, saturation	12–16	9–14	16	8,5	
Propanol, moldings 20°C, saturation	9–13	9–12	10	–	
Adhesive Bond Strength, tensile (MPa) ≡ (N/mm^2)					
PA-aluminium		68			39
PA-steel		70			39
PA-copper		76			39
Birefringence, (Δ_n)					51
$n_{\parallel}$	1.580	1.582			51
$n_{\perp}$	1.530	1.519			55

* This table includes data compiled by R. Pflüger for the third edition of Polymer Handbook.

Brittleness Temperature, ASTM D746 (°C) 57–59

PA-6		PA-66		PA-610		PA-12	
DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
–	–	−80	−65 [$M_n = 18000$]	−90	−62	–	–
–	–	−100	−85 [$M_n = 34000$]	–	–	–	–

Bulk Modulus B (N/mm^2) (Crystalline rods, dry)	4400	3300	2300	4000

Chemical Resistance of Nylons at 23°C 57,59

Acetic acid (10%)	3[a]	3	–	2[b]
Acetaldehyde (40%)	2	2	–	1[c]
Acetone (100%)	1	1	–	1
Butanol (100%)	1	1	–	1
Carbon tetrachloride (100%)	1	1	–	2
Diesel oil (100%)	1	1	–	1
Ethanol (96%)	1	1	–	1
Formic acid (3%)	3	3	–	1
(10%)	4[d]	4	–	–
Gasoline unleaded (100%)	1	1	–	1
Heptane (100%)	1	1	–	1
Hydrogen peroxide (2%)	4	4	–	2
Dichloromethane (100%)	2	2	–	3
Perchloroethylene (100%)	1	1	–	2
Phenol (75%)	3	3	–	3
Potassium hydroxide (10%)	1	1	–	1
Sulfuric acid (10%)	4	4	–	2
Toluene (100%)	1	1	–	1
Transformer oil (100%)	1	1		1

[a] Considerable absorption and/or attack; limited product life.
[b] Limited resistance, absorption causing dimensional changes and slight reduction in properties.
[c] Resistant, little or no absorption.
[d] Material is soluble or decomposes in short time.

Coefficient of Friction (dry)

(a) Average depth of roughness R_z in (μm)*

0.1	0.35	0.5		0.55
0.5	0.33	0.43		0.45
1.0	0.32	0.38		0.38
2.0	0.32	0.35		0.35
4.0	0.36	0.38		0.37
6.0	0.43	0.46		0.40

(b) Average surface pressure in (MPa) ≡ (N/mm^2)

R_z Optimal	R_z 1.5–3 μm	R_z 1.5–3 μm		R_z 0.5–1 μm
0.02	0.40	0.40		0.40
0.05	0.36	0.36		0.36
0.1	0.35	0.35	0.4	0.35
0.15	0.38	0.38		0.38
0.2			0.32	
0.3			0.29	
0.4			0.26	
0.5			0.24	
0.6			0.23	
0.7			0.22	
0.8			0.21	
1.0	0.43	0.44	0.19	0.44
1.2			0.18	
5.0	0.48	0.50		0.50
15.0	0.48	0.51		0.51
$R_z < 0.3$ μm	Stick-slip-motion	Stick-slip-motion		Stick-slip-motion

13
13
13

*Average surface pressure 0.1 MPa; surface temperature < 40°C.

Coefficient of Thermal Expansion (K^{-1}) (× 10^4)

Coefficient of Linear Thermal Expansion, ASTM D696 (10^{-5} mm/mm°C) 57,59

PA-6		PA-66		PA-610		PA-12		
DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH	
–	–	8.1	–	9	14.9	–	–	
(a) Linear crystalline 20°C			7–10		7–10	8–10	11	
100°C			10–14		10–14		20–23	
Monoclinic α a			6.2					52
α c			23.4					
α b			−4.5					
Triclinic α a					2.1			52
α c					22.0			
(b) Volume, 20°C, cryst. (K^{-1}) ($\times 10^4$)			2.7		2.8	3.8	2.9	55

Cohesive Energy See Solubility Parameter 51

Compressibility (MPa)

(a) Dependence on temperature and pressure (calc. from *PVT*)

20 (°C) 50 (MPa)		62×10^{-6} (50)	200×10^{-6}
100		58	
150		54	
300		50	
120 50		125	320×10^{-6}
100		115	
150		95	
300		75	
200 >100		−>300 (17)	

(b) Dependence on water content

<1.5%		137×10^{-6} (48)	200×10^{-6}
>−2%		226×10^{-6} (48)	
melt	335×10^{-6}	280×10^{-6}	650×10^{-6}

Compressive Strength (MPa) ≡ (N/mm^2)

Compressive Stress, 1% Deformation, ASTM D695 (MPa) 57–59

		20	34	21	13
Temperature 20°C	strain: 1%		14		
(molded; 2.5% H_2O)	strain: 2%		28		
	strain: 4%		56		
	strain: 6%		70		

Compressive Modulus, ASTM D695 (MPa) 58

2410	2830	2070	–

Compressive Yield Stress, 1% Offset, ASTM D695 (MPa) 58

83	91	73	–

Crystallographic Data See also Unit Cell Dimension

Effect of Crystallinity on Solvent Absorption (% Weight Gain) 59

	PA-6		PA-66		PA-610		PA-12	
Agent	b	c	b	c	b	c	b	c
Water	11	9	10	7.5	4	3	–	–
Methanol	19	3	14	9	16	9	–	–
Ethanol	17	3	12	3	13	8	–	–
Benzene	1	0	1	0	1	1	–	–
Toluene	1	0	1	1	3	1	–	–
Acetone	4	3	2	1	5	1	–	–
Benzoyl alcohol	55	11	38	3	40	11	–	–
Dichloromethane	20	8	16	10	24	13	–	–
Chloroform	34	19	27	5	40	21	–	–

b: low crystalline material.
c: high crystalline material.

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Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
Crystallization					
Crystalline Density, X-ray (g/cm^3)	1.23	1.24	1.16	–	65
Max. linear growth rate of crystals v_{max} (μm/s)	3.3	20			51
Activation energy E_a(kJ/mol)	56.5	64.5	53.6		51
Density (Mg/m^3) ≡ (g/cm^3)					
Crystalline α, monoclinic	1.24 (53)				
γ, hexagonal	1.13				
γ, monoclinic	1.17			1.034 (9)	
γ, pseudohexagonal	1.155 (42)				
α, triclinic		1.220 (47)	1.156 (8)		
α, triclinic		1.24			
β, triclinic		1.248			
Melt 270°C, 1 bar	0.960	0.989	0.913	0.840	
Moldings, amorphous	1.09	1.09	1.05	1.01	36
crystalline	1.12–1.14	1.13–1.145	1.09	1.04 (15)	36
Correlation with crystallinity (moldings)	$\alpha = 950 - \frac{1040}{\text{density}}$	$830 - \frac{900}{\text{density}}$	$990 - \frac{1030}{\text{density}}$		36
Deformation Under Load, ASTM D621 (%)					58,59
14 MPa	1.6	1.4	–	–	
28 MPa	1.8	1.5	–	–	

Dielectric Constant ε, Dielectric Loss, tan δ ($\times 10^4$)[a]

Frequency (s^{-1})		10^2		10^3		10^4		10^5		10^6		10^7		10^8		10^9	
PA	**(°C)**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**	ε_r	**tan δ**
6	−30	3.1	140	3.1	100	3.0	100	3.0	160	3.0	200	3.0	180	3.0	120	3.0	55
	0	3.2	120	3.2	120	3.2	140	3.2	170	3.1	220	3.1	230	3.1	160	3.0	100
	30	3.5	65	3.5	100	3.4	140	3.4	190	3.3	240	3.2	300	3.2	300	3.1	200
	60	5.2	840	4.6	600	4.2	520	3.8	540	3.6	550	3.4	550	3.3	480	3.2	260
	90	15.9	2900	11.5	1650	8.8	1500	7.1	1900	5.7	2100	4.4	1900	3.7	1250	3.4	550
	23°/50% RH[b]	10.9	1450	7.3	1400	5.6	1450	4.5	1300	3.8	920	3.5	550	3.3	320	3.2	190
66	−30	3.1	120	3.1	105	3.1	105	3.0	130	3.0	165	3.0	160	3.0	100	3.0	49
	0	3.3	110	3.3	120	3.2	135	3.2	160	3.1	200	3.0	200	3.0	160	3.0	81
	30	3.6	85	3.5	125	3.4	180	3.4	215	3.2	250	3.1	255	3.1	220	3.0	135
	60	5.0	810	4.6	590	4.3	460	4.0	390	3.7	370	3.5	360	3.3	320	3.1	240
	90	10	2000	8.9	1450	7.6	1300	6.2	1450	5.0	1600	4.0	1300	3.4	810	3.2	440
	23°/50%RH[b]	7.5	1100	5.9	1020	4.8	1000	4.1	900	3.7	700	3.4	450	3.3	280	3.2	170
610	−30	3.0	120	3.0	110	3.0	100	3.0	110	3.0	150	3.0	130	3.0	80	3.0	55
	0	3.0	130	3.0	130	3.0	130	3.0	140	3.0	170	3.0	190	3.0	160	3.0	100
	+30	3.2	100	3.2	150	3.1	170	3.0	200	3.0	210	3.0	210	3.0	170	3.0	130
	+60	4.6	900	4.2	650	3.9	570	3.5	540	3.4	540	3.3	500	3.1	360	3.0	250
	+90	13	2500	10.5	1700	8.5	1700	6.5	1900	5.2	1900	4.0	1500	3.5	800	3.1	350
	20°/65%RH[b]	6.5	2000	5.4	1500	4.5	1300	4.0	1000	3.5	800	3.2	550	3.1	350	3.0	200
12	−30	30	170	3.0	160	2.9	100	2.9	50								
	0	3.1	200	3.1	200	3.0	190	3.0	130								
	30	4.0	700	3.5	500	3.3	450	3.1	350								
	60	11.5	2300	8.5	2000	6	2300	4.5	1300								
	90	>20	>5000	15	2000	13	1600	10	2600								

[a] Ref. 5.
[b] Saturated

Dielectric Strength (V/cm) ($\times 10^{-4}$) (VDE 0303, part 2; IEC-243, electrode K20/P50)				
Dry	100	120	100	90
Dry 100°C	40	40	–	25
Moist ISO-1110	60	80	60	–

Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
Diffusion See also table "Permeability Constants" in this Handbook					
Coefficient D H_2O, 20°C	0.4	0.2	0.1	0.3 (16)	
(cm^2/s) $\times 10^8$ 60°C	5.5	3.5	1.5	7 (16)	
100°C	55	35	30	85 (16)	
Activation energy H_2O E_D(kJ/mol)	46	58	65	71 (16)	
Elongation (ISO R527) ultimate (%)					
Molded ($s = 3$ mm), dry	>50	40	>50	>50	
Molded ($s = 3$ mm), moist	>50	>50	>50	>50	
ISO-1110					
Molded at yield	10	6		8	
Dry film, NK 23/50, $s = 25$ µm					
Longitudinal	300				
Transverse	330				
Biaxially oriented					
Longitudinal	70				
Transverse	50				
Enthalpy (kJ/kg)					
Temperature T (°C) (Ref. to 20°C)					
60	60	85	80	80	
100	135	170	160	165	
200	380	420	400	500	
250	585	590	580	650	
300	750	850	700	800	
Pressure (MPa) $\equiv$ (N/mm^2)					
($T = 25$°C)					
100		70			17
200		140			17
($T = 150$°C)					
0		250			17
100		340			17
200		420			17
Entropy of Fusion (J/mol/K)					
Crystalline	44–47.5	83–86	110–114		51
Equation of State* (melt)					
$(p + \pi)(v - \omega) = RT/M_u \pi$ (bar)	2040 (27), 2890 (18)	4590 (18)	540 (51)	3640 (18)	51
ω (cm^3/g)	0.817 (27), 0.778 (18)	0.735 (18)	0.77 (51)	0.837 (18)	
M_u	105 (27), 59 (18)	35.4 (18)		35.3 (18)	

Flexural Strength, ASTM D790 (MPa) 57,59

	PA-6		PA-66		PA-610		PA-12	
Temp. (°C)	DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
−40	145	–	–	–	–	–	–	–
−30	–	–	–	–	–	–	86	–
23	–	–	117	42	–	–	55–57	37–44
24	113	40	–	–	–	–	17	–
80	23–20	–	–	–	–	–	–	–

*Spencer and Gilmore; $R = 82.06$ (cm^3bar/mol/K); v = specific volume (cm^3/g) (51).

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Glass Transition Temperature, T_g (°C) 57–59,61–63,67–69

	DAM				50%RH		100%RH	
	A	B	C	D	C	D	C	D
PA-6	56	65,67	59,65,75	41,60	20	3	−22	−32
PA-66	82	65,67	66,78,80	46,48	35	16	−15	−37
PA-610	56	46,50	65,67,70	42,51	40	–	10	–
PA-12	29	25,32	54,62,–	42,–	–	–	42	–

A: Estimated from melting temperature: $T_g = 2/3 \times T_m$; 9,10.
B: Calculated from group contribution, 9,10.
C: Dynamic measurements with torsion pendulum; 9,10,11,12.
D: Static Measurement: DSC inflexion point of mechanical property versus temperature or method not identified; 9–18.

Grüneisen Parameter γ_G^{**} 0.58 0.49 0.45 0.64 51,55

Hardness
Rockwell Hardness, ASTM D785 57–59

PA-6		PA-66		PA-610		PA-12	
DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
R116–119	R97–101	R118–121	R108	R110–111	–	R106–110	–

Dry, 23°C	*M* 100	*M* 105	*R* 75	*R* 110
Moist ISO-1110, 23°C	*M* 70	*M* 95	*M* 60	
Ball Indentation Hardness H358/30 (MPa) ≡ (N/mm²) (DIN 53 456), (ISO-2039)				
Dry, 23°C	150	160	120	98
Dry, nucleated, 23°C	160	170		
Moist ISO-1110, 23°C	70	100–110	80	

Heat Capacity See also corresponding table in this Handbook

20°C, crystalline (J/mol/K)	204	374	502	354	55
Heat of Combustion (kJ/kg)	−31.400	−31.400			51
Heat of Crystallization (kJ/kg)	−46.5	−54			25

Heat Distortion Temperature
ASTM D648 (°C) 57,59

Load	PA-6		PA-66		PA-610		PA-12	
MPa	DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
0.5	–	–	235	–	175	160	120	–
1.8	55–75	–	90	–	66	60	50	250

(ISO-R 75) (°C)				
Molded, dry, 0.45(N/mm²) ≡ (MPa) method B	>160	>200	>160	150
Molded, dry, 1.8(N/mm²) method A	55–90	100–110	65–85	55

Heat of Formation See Enthalpy

Heat of Fusion (kJ/kg)				
Crystalline from ΔH_m	190 (25)	196 (25)	215	95 (15)
Crystalline from ΔH_m for α-structure	230–278			
	240–260 (24)			
Amorphous, annealed 8 h at 50°C	45			45 (15)

Heat of Sorption (J/mol)

H_2O < 150°C	-45.6×10^3 (43)		
Melt	-76.8×10^3 (14)		-58.5×10^3 (37)

** $\gamma_G = (\alpha B V_m / C_p$ (55) where α is the coefficient of expansion (K^{-1}), B is the bulk modulus (N/mm^2), V_m is the molar volume (cm^3/mol), C_p is the specific heat capacity (mJ/K/mol).

Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
Ignition Temperatures (ASTM D1929–1977) (°C)					
Flash-ignition temperature	400	390	415		
Self-ignition temperature	440				

Impact Strength, Izod, notched, ASTM D256 (J/m) 57–59,62–64

Temp. (°C)	PA-6		PA-66		PA-610		PA-12	
	DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
−40	48	–	43	–	43	–	55	–
23	53	267	53	107	53	85	60	64

Tensile Impact Strength, ASTM D1822 (KJ/m^2) 57–59

	PA-6		PA-66		PA-610		PA-12	
	DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
Short	147–250	–	158 [$M_n = 18000$]	231	168–200	–	–	–
Long	–	Not broken	504 [$M_n = 18000$]	1470	–	–	–	–
			535 [$M_n = 34000$]	Not broken				

Notched, Charpy, DIN 53453 (MPa)					57–59
DAM (20°C)	3–6	2–3	4–10	–	
Conditioned 4 months, 20°C and 65% RH	Not broken	12–30	13–15	–	
Izod (ISO-180, method 1A) (J/m)					
Dry, 23°C	50–100	40–60	50	50–200	
Dry, −40°C	30–50	30–50	35	–	
Moist, ISO-1110, 23°C	120	110	75		
Infrared Spectrum See Ref. 21 and DIN 53746 (draft)					
Magnetic Susceptibility (cm^3/g) ($\times 10^6$)		0.76/0.63			51
Mechanical Loss, $\tan\delta$ (ISO 537)					
($\gamma = 0.1$–10 Hz) dry 20°	0.016	0.016	0.016	0.01	
dry 40°	0.048	0.026	0.10	0.048	
dry 60°	0.175	0.083	0.17	0.16	
dry 80°	0.127	0.127	0.10	0.08	
dry 100°	0.064	0.064	0.055	0.048	
($< 10^4$ Hz) dry 23°		<0.02 (68)			
(10^3 Hz) 0.72% H_2O		0.12 (68)			
(10^3 Hz) 120°		0.2 (68)			
Melting Point (ISO 3146, method C) (moldings) (°C)					
	220	262	220	178	57,59
Equilibrium melting point	231 (32)			187 (15)	
α-Crystalline	260	270			
Pressure dependence	See Ref. (27)	See Ref. (45)			
Melt Viscosity					
Flow activation energy* E_0 (kJ/mol)	60	60	60	60	31
Coefficient K of zero-shear-rate-melt viscosity** (Pa s)	9.5×10^{-14} (250°C)	7.2×10^{-14} (280°C)			31

$^*\eta = \eta_0(T_0)\ \exp\left[-\frac{E_0}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right)\right]$

$^{**}\eta_0(\bar{M}_w) = K\left(\frac{\bar{M}_w}{g\,mol}\right)^{3.4}$

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Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
Modulus (MPa) ≡ (N/mm)					
Bulk modulus – See Compressibility and Bulk Modulus *B*					
Loss modulus *G″* See Mechanical Loss					
Elasticity modulus (Young) (ISO-R527)					
23°, dry	3000	3300	2400	1400	
23°, moist ISO-1110	1500	1700	1500	1200	
100°, dry	500	600	400	200	

Flexural modulus, ASTM D790 (MPa) 57–59

Temp. (°C)	PA-6 DAM	PA-6 50% RH	PA-6 100% RH	PA-66 DAM	PA-66 50% RH	PA-66 100% RH	PA-610 DAM	PA-610 50% RH	PA-610 100% RH	PA-12 DAM	PA-12 50% RH	PA-12 100% RH
−40	3010	3650	3650	3240	3450	–	2240	2520	–	1790	–	–
23	2720	970	410	2830	1210	480	1970	1100	690	1410	1030	–
77	520	340	240	690	570	–	480	–	–	–	–	–

Shear modulus *G* (ISO 537)					
Dry, 23°C	1100	1300	800	500	
Dry, nucleated	1500	1700			
Dry, 100°C	200	300	120	100	
Dry, 200°C	80	150			
Apparent creep modulus (ISO-899)					
Ec/1000/20 (23°C, 50% RH)	230	400	400	300	
Nucleated	440	450			
Mold Shrinkage, (ASTM-D955) (%) (mold temp. 60°C; 80 × 10 × 4 mm)	1.4	1.6	1.2	1.1	
Moisture Absorption See Absorption					
Molecular Weight, (repeat unit)	113.2	226.3	282.4	197.3	
Oxygen Index (ASTM-D2863) (%)					
Dry	25–27	28–29	24	26	
Permeability See table "Permeability Coefficients" in this Handbook					
Poisson Ratio					
(a) Moldings, 20°C	0.33	0.3–0.4	0.3–0.4	0.3–0.4	
(b) Extruded rod; room temp.					
compressive axial strain 0.1		0.1–0.25			29
0.2		0.3–0.36			29
0.4		0.40–0.44			29
0.6		0.43–0.45			29
		0.38			51
(c) Power factor at 10^3 Hz, dry	0.02–0.06	0.04	0.02	–	64
(d) vs. temperature: 100°C	0.46	0.44 (51)	0.47		
Melt	0.50	0.50	0.50	0.50	
(e) Limiting pressure velocity of nylon against steel (kPa × 0.5 m/s)	70	85	70	–	60
Refractive Index, η_D					
Single crystals					
α, calc.		1.475 (8)	1.475		
β, calc.		1.565 (8)	1.525		
γ, obs.		1.58 (8)	1.565		
Moldings	1.53	1.53	1.53		
Resistance to Tracking (IEC-Publ. 112)					
Method CTI	CTI 600	CTI 600	CTI 600	CTI 600	
CTI-M	CTI-M 600	CTI-M 600	CTI-M 600		

Shear Strength, ASTM D732 (MPa) 57–59

PA-6		PA-66		PA-610		PA-12	
DAM	50%RH	DAM	50%RH	DAM	50%RH	DAM	50%RH
59	–	66	57	58	–	34	–

Shrinkage See Mold Shrinkage

Solubility Parameter $(J/cm^3)^{1/2}$

Amorphous, 25°C	27.8	27.8	26	51

Solvents See table "Solvents–Nonsolvents" in this Handbook

Sonic Velocity, 20°C (m/s) 1400–2300 (fibers)(26) 2770 (moulding) (40)

Specific Heat (J/g-K)	1.7	1.7	1.7	1.26	57,59
Effect of temperature (°C)					58,59
0	1.38	1.30	1.38	–	
40	1.67	1.55	1.76	–	
80	1.97	1.97	2.13	–	
120	2.30	2.30	2.22	–	
160	2.68	2.68	2.47	–	
200	–	2.72	2.89	–	
220	–	2.76	8.16	–	
240	–	3.35	3.10	–	
260	–	11.51	2.64	–	
280	–	3.14	2.68	–	

Specific Volume, at varying pressures (MPa) and temperatures (cm^3/g) 61

At 25°C				
Atm	0.875	0.880	1.005	0.985
50	0.865	0.875	0.985	0.975
100	0.860	0.865	0.970	0.960
200	0.850	0.855	0.950	0.945
At 100°C				
Atm	0.895	0.905	1.045	1.020
50	0.885	0.890	1.020	1.000
100	0.875	0.880	1.000	0.985
200	0.860	0.865	0.970	0.960
At 200°C				
Atm	0.945	0.950	1.125	1.130
50	0.925	0.935	1.085	1.095
100	0.910	0.920	1.060	1.070
200	0.890	0.895	1.020	1.030
At 300°C				
Atm	1.055	1.050	1.210	1.210
50	1.020	1.020	1.150	1.150
100	0.995	0.995	1.110	1.110
200	0.960	0.955	1.060	1.070

Surface Tension, critical (mN/m)					
23°C	40–47; 43	40; 44		31	38,51
Melt, 265°C	36	36	37	25	22

Taber Abrasion, D1044 (mg loss/1000 cycles); CS-17 wheel, 1000 g load 57,59

5	7 [$M_n = 18000$]	5–6	–
–	4 [$M_n = 34000$]		

Temperature Index/Thermal Endurance Profile: Typical Values (°C)					57,59
Not stabilized					
5000 h	85	85	–	–	
20000 h	70	70	–	–	
Heat stabilized					
5000	130	130	–	–	
20000	120	120	–	100	

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Tensile Strength 57–59

Temp. (°C)	PA-6 DAM	PA-6 50%RH	PA-66 DAM	PA-66 50%RH	PA-610 DAM	PA-610 50%RH	PA-12 DAM	PA-12 50%RH
ASTM D638 (MPa)								
−40	120	–	108	103	83	83	–	–
23	81	69	83	77	59	49	55	52
77	68	59	62	41	37	37	–	–
Tensile Strength at Yield, ASTM D638 (MPa)								
−40	120	119	108	103	83	83	79	–
23	81	50	83	59	59	49	52	41
77	30	28	62	41	29	–	24	–
Tensile Elongation at Yield, ASTM D638 (%)								
−40	–	–	4	6	10	13	–	–
23	9	–	5	20	10	30	10	20
77	–	–	30	30	30	–	–	–
Tensile Elongation at Break ASTM D638 (%)								
−40	8	4	20	20	20	30	–	–
23	150	300	60	>200	100	220	250	250
77	310	325	340	350	300	–	–	–

(ISO-R 527) (MPa) ≡ (N/mm^2)					
Dry, 23°C	80, 90 (nucleated)	90	70	50	
Dry, 100°C	30	40	20	17	
Moist, 23°C (% water)	50 (2%)	60 (2%)	50 (1%)	30 (0.9%)	
Moist, 23°C (saturated in H_2O)	22	35			
Monofilaments 23°C, 50% RH					
draw ratio 4.3 : 1	475–550	575–600			
5.0 : 1	675–775	650–850			
Thermal Conductivity (W/m/K)					
Moldings	0.23	0.23	0.23	0.25	
Crystalline (moist) 30°C	0.43	0.43			19
Amorphous (moist) 30°C	0.36	0.36	0.35		19
Melt 250°C	0.21 (33)	0.15			12
25°C, dependence on pressure	1.89	1.83	1.90	2.03	2
$\frac{\lambda\,(25\,\text{kbar})}{\lambda_0}$					
Relative thermal index for 3 mm thickness					57–59
Not stabilized					
Electrical	125	125	–	65	
Mechanical					
with impact	75	75	–	65	
without impact	85	85	–	65	
Heat stabilized					
Electrical	130	130	–	110	
Mechanical					
with impact	105	105	–	75	
without impact	110	110	–	95	
Thermal Diffusivity (cm^2/s)		1.8×10^{-3}		1.4×10^{-3}	12
100°C		1.4×10^{-3}			
Melt 250°C		0.65×10^{-3}			

Unit Cell Dimensions

PA	Crystal system	Space group	Unit cell parameters (Å) a	b	c	Angles α	β	γ	Units in cell	Density (Mg/m^3) ≡ (g/cm^3)	Refs.
6	α-Monoclinic	C_2^2	9.56	8.01	17.24			67.5	8	1.23	20
	α-Monoclinic	C_2^2	4.81	7.61	17.1			79.5	4	1.21	
	α-Monoclinic		9.65	8.11	17.2			66.3			
	β-Monoclinic		4.8	3.6–4.1							41

PA	Crystal system	Space group	Unit cell parameters (Å) a	b	c	Angles α	β	γ	Units in cell	Density (Mg/m³) ≡ (g/cm³)	Refs.
	γ-Monoclinic		9.14	4.84	16.68			121	4	1.188	24
66	α-Triclinic	C_1^1	4.9	5.4	17.2	48.5	77	63.5	1	1.24	8,10
	β-Triclinic	C_1^1	4.9	8.0	17.2	90	77	67.5	2	1.248	8
	Pseudohexagonal							60	(>160°C)		7,44
610	α-Triclinic		4.95	5.4	22.4	49	76.5	63.5	1	1.157	8
	β-Triclinic		4.9	8.0	22.4	90	77	67.5	2		8
12	Monoclinic (Pseudohexagonal)	$P2_1/c$	4.79	31.9 (fiber axis)	9.58	120		4	1.034	9	

Melt Viscosity Coefficient K of Zero Shear Rate Melt-Viscosity (Pa s) $\eta_0(M_w) = K(M_w/\text{g mol})^{3.4}$ 70,71

9.5×10^{-4} (250°C) 7.2×10^{-4} (280°C) – –

Viscosity – Molecular-Weight Relationships 71–75

Nylon	Solvent	K ($\times 10^3$) (mol/g)	a	M ($\times 10^4$)	Refs.
PA-6	HCOOH (88% vol.)	22.6	0.820	0.6–12.0	1
	H_2SO_4 (40% vol.)	59.2	0.690	0.3–1.3	1
	Trifluoroethanol	53.6	0.750	1.3–10.0	1
PA-6,6	*m*-cresol	240.0	0.610	1.4–5.0	0
	HCOOH (90% vol.)	35.3	0.786	0.6–6.5	0
	H_2SO_4 (96% vol.)	115.0	0.670	1.4–5.0	0
	o-Chlorophenol	168.0	0.620	1.4–5.0	0
PA-6,10	*m*-cresol	13.5	0.960	0.8–2.4	3
PA-12	*m*-cresol	46.3	0.750	1.3–13.0	2
	H_2SO_4 (96% vol.)	69.4	0.640	1.0–13.0	2

Polyamide viscosity: $\eta = K \times M^a$; see Melt Viscosity, and corresponding table in this Handbook.

Volume, Specific (cm^3/g)					
20°C, Amorphous	0.917	0.917	0.95	0.99	36
270°C, Melt	1.032	1.018	1.095		
20°C	0.813 α-monoclinic	0.82 (47) α-triclinic	0.865 (8)	0.966 (9) γ-monoclinic	
Change on melting	0.02–0.03	0.02–0.03	0.02–0.03	0.03	
Melt, 270°C, 1 bar	1.040	1.010		1.19	
Volume, Molar (cm^3/mol)					
20°C, Crystalline (rods)	100	198	260	195	55
20°C, Amorphous	103	207.5	271	199.5	36
20°C, Amorphous, calcd. from group contribution	104.2	208.3			
Volume Resistivity (Ohm cm)					
Relative humidity 0% (dry)	1×10^{15}	1×10^{15}	1×10^{15}	10^{15}	
Saturated at 50% RH, 20°C	2×10^{11}	3×10^{11}	2×10^{12}	10^{14}	
20°C (saturated) 100% RH	4×10^{8}	1×10^{9}	3×10^{10}	10^{13}	
Temperature 60°C, dry	8×10^{11}	6×10^{11}	5×10^{11}	10^{17}	
Dry, 100°C	3×10^{9}	3×10^{9}	5×10^{8}	10^{9}	
Dry, 150°C		1079			
At 50% RH, 100°C		4×10^{7}			
Wear Resistance (μm/km)					
R_z (μm) average roughness height[a]					
0.1	0.30	0.28		0.20	13
0.5	0.32	0.30		0.25	
1.0	0.35	0.35		0.60	
2.0	0.57	0.48		1.00	
4.0	0.80	0.52		2.50	
6.0	1.00	0.55		3.90	
P (N/mm^2) average surface pressure (R_z optimum, see also Coefficient of Friction)					
0.02	0.11	0.10		0.12	13
0.05	0.29	0.24		0.30	
0.10	0.57	0.48		0.60	
0.25	1.40	1.20		1.50	

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Property	Polyamide 6	Polyamide 66	Polyamide 610	Polyamide 12	Ref.
1.00	5.70	4.80		6.00	
5.00	28.00	24.00		30.00	
15.00	86.00	72.00		90.00	

[a] Average surface pressure: 0.1 N/mm^2; surface temperature: <40°C

REFERENCES

1. As long as no other specific reference is cited, the values for varying polyamides were selected on the basis of the information from the technical manuals of the corresponding suppliers:
   PA66: ®Maranyl A100 (ICI), ®Zytel 101 (DuPont). ®Ultramid A3K (BASF). Designation: ISO-1874-PA66, MR(H), 14-030.
   PA6: ®Capron 8200 (Allied Chemical), ®Durethan B31F (Bayer), ®Ultramid B3K (BASF). Designation: ISO-1874-PA6. MRH, 14-030.
   PA610: ®Ultramid S3K (BASF). Designation: ISO-1874-PA610, MRH, 14-020.
   PA12: ®Grilamid L20 (Ems-Chemie). ®Vestamid L1901 (Hüls). Designation: ISO-1874-PA12. G, 18-010.
   Preparation of test specimens and determination of properties according to ISO-1874-2 (1987) and ISO-DP294. Further property values were taken from the VDI/VDE-Werkstoffblatt, No. 2479, Blatt 1 (1978). VDI-Verlag, D-4000 Düsseldorf (F. R. Germany) "Polyamid-Formstoffe, unverstärkt", a paper which was produced in cooperation with the above named suppliers. This leaflet contains also further approved values for other homopolyamides such as PA11, PA612 and PA6-3-T.
2. P. Anderson, Makromol. Chem., **177**, 271 (1976).
3. G. J. van Amerongen, J. Polym. Sci., **6**, 471 (1951).
4. Bennewitz, R. Faserforschg., Textiltechn., **5** , 155 (1954).
5. K. Bergmann, in G. Schreyer "Konstruieren mit Kunststoffen", C. Hanser-Verl., München 1972 and VDI/VDE-Werkstoffblatt 2479. Teil 1 1978.
6. E. M. Bradbury, L. Brown, A. Elliot, D. A. D. Parry, Polymer, **6**, 465 (1965).
7. R. Brill, J. Prakt. Chem., **161**, (1943), 49; Makromol. Chem., **18/19**, 294 (1956).
8. C. W. Bunn, E. V. Garner, Proc. Roy. Soc. (London) A, **189**, 39 (1947).
9. G. Cojazzi, A. Fichera, C. Garbuglio, V. Malta, R. Zannetti, Makromol. Chemie, **168**, 289 (1973).
10. M. L. Colclough, R. Baker, J. Materials Sci., **73**, 2531 (1978).
11. M. Dole, B. Wunderlich, Makromol. Chem., **34**, 29 (1959).
12. W. Dietz. Colloid Polym. Sci., **255**, 755 (1977).
13. G. Erhard, E. Strickle, Kunststoffe, **62**, 2–9, 232, 283 (1972), and unpubl. measurements (BASF 1985).
14. O. Fukumoto, J. Polym. Sci., **22**, S. 263 (1956).
15. S. Gogolewski, K. Gzerniawska, M. Gasiorek, Colloid & Polymer Sci., **258**, 1130 (1980).
16. G. Goldbach, Angew. Makromol. Chem., **32**, 37 (1973).
17. R. G. Griskey, J. K. P. Shou, Modern Plastics, **45**, 148 (1968).
18. F. Hafner, Th. Mietzner (BASF, Abt. Informatik; 1986), calc. from PVT-functions for ®Ultramid B4, A3K and ®Grilamid L 20.
19. K.-H. Hellwege, R. Hoffmann, W. Knappe, Kolloid-Z. Polymer, **226** (2), S109 (1968).
20. D. R. Holmes, C. W. Bunna, C. J. Smith, J. Polym. Sci., **17**, 159 (1955).
21. Hummel/Scholl, "Atlas der Kunststoff -und Polymeranalyse", C. Hanser-Verl., München/Verl. Chem. GmbH, Weinheim 1984 (2nd ed.).
22. F. Hybart, J. Appl. Polym. Sci., **3**, 118 (1960).
23. K.-H. Illers, H. Haberkorn, Makromol. Chem., **146**, 267 (1971).
24. K.-H. Illers, H. Haberkorn and P. Simák, Makromol. Chem., **158**, 285 (1972); **179**, 497 (1978).
25. M. Inoune, J. Polym. Sci. A, **1**, 2697 (1963).
26. M. Jambrich, I. Diavik, I. Mitterpach, Faserf. Textiltechn., **23** (1), 28 (1972).
27. Y. Katayama, K. Yoneda, Rev. Electrical Comm. Lab. **20** (9/10), 921 (1972).
28. J. Kirshenbaum, J. Polym. Sci. A-3, 1869 (1965).
29. I. Krause, A. J. Segreto, H. Przirembel, L. Mach Materials Sci. Eng., **1**, 239 (1966).
30. H. Kunze, Dissertation TH Aachen 1958.
31. H. M. Laun, Rheol. Acta, **18**, 478 (1979).
32. F. N. Liberti, B. Wunderlich, J. Polym. Sci. A-2, **6**, 833 (1968).
33. P. Lohe, Kolloid-Z., **203** (2), S115 (1965).
34. N. G. McCrum, B. E. Read, G. Williams, "Unelastic and Dielectric Effects in Polymer Solids", Wiley, London 1967.
35. R. W. Moncrieff, "Artificial Fibres", Wiley, New York, 1950.
36. A. Müller, R. Pflüger, Kunststoffe, **50** (4), S203 (1960).
37. N. Ogata, Makromol. Chem., **42**, 52 (1960).
38. D. K. Owens, R. C. Nendt, J. Appl. Polym. Sci., **13**, 1741 (1969).
39. J. Pellon, W. G. Carpenter, J. Polym. Sci. A, **1**, 863 (1963).
40. F. Plümer, Kolloid-Z, **176** (2), 108 (1961).
41. A. Reichle, A. Prietzschk, Angew. Chem., **74**, 562 (1962).
42. L. G. Roldan, H. S. Kaufman, J. Polym. Sci. B, **1**, 603 (1963).
43. E.-O. Schmalz, Faserforschg., Textiltechnik, **20** (11), 533 (1969).
44. G. F. Schmidt, H. A. Stuart, Z. Naturforschg., **13a**, 222 (1958).
45. J. E. Stamhuis, A. J. Pennings, Polymer **18**, 667 (1977).

46. H. W. Starkweather, G. E. Moore, J. E. Hansen, Th. M. Roder, R. E. Brooks, J. Polym. Sci., **21**, 189 (1956).
47. H. W. Starkweather, R. E. Moynihan, J. Polym. Sci., **22**, 363 (1956).
48. H. W. Starkweather, J. Appl. Polym. Sci., **2** (5), 129 (1959).
49. H. W. Starkweather, J. F. Whitney, D. R. Johnson, J. Polym. Sci. A, **1**, 715 (1963).
50. H. Tautz, L. Strobel, Koll. Z., Z. f. Polym., **202** (1), S.33 (1965).
51. Van Krevelen, D. W., P. J. Hoftyzer: "Properties of Polymers-Correlation with Chemical Structure", (2nd ed.) Elsevier Publ. Comp., Amsterdam 1976.
52. J. H. Wakelin, A. Sutherland, L. R. Beck, J. Polym. Sci., **42** (139), 278 (1960).
53. L. G. Wallner, Makromol. Chem., **79**, 279 (1948).
54. R. C. Wilhoit, M. Dole, J. Phys. Chem., **57**, 14 (1953).
55. R. W. Warfield, E. G. Kayser, B. Hartmann, Makromol. Chem., **184**, 1927 (1983).
56. A. Zosel, Colloid Polym. Sci., **263**, 541 (1985).
57. Trade Literature
    (a) "Design Handbook for DuPont Engineering Plastics", DuPont, 1997, Module II, 232409E
    (b) "Ultramid® PA, Product Line, Properties, Processing", BASF Plastics, B568e, (9110) 9.91.
    (c) "Ultramid® T PA, Product Line, Properties, Processing", BASF Plastics, B605e, 8.90.
    (d) "Ultramid® PA, Range Chart", BASF Plastics, F568/1e, 5.90.
    (e) "Product Sheet Nylon Resin 6,10", Monsanto Co.
    (f) "Rilsan®, Design Guide to a Versatile Engineering Plastic", Atochem Polymers, Inc., 4/84.
    (g) "Engineering Thermoplastics, Versamid®" Huls America, Inc.
    (h) "A Guide to Properties and Uses of Nylon 12", Huls America, Inc., Datasheets.
    (i) "Product Selection Chart, Grilamid®, Nylon 12", EMS Industries, GG5-805.
    (j) "Capron® Nylon Homopolymers for Molding and Extrusion", Allied Signal, ERG-02-6/91.
    (k) "Vydyne® Nylon", Monsanto Company, 1994, MPP-5-210.
    (l) "Technical Information, Rilsan®", Elf Atochem, North America, Inc.,Technical Polymers, 1998.
58. R. M. Bonner, M. I. Kohan, E. M. Lacey, P. N. Richardson, T. M. Roder, L. T. Sherwood, in: M. I. Kohan (Ed.), "Nylon Plastics", Wiley, New York, 1973, p. 327.
59. J. C. L. Williams, S. J. Watson, P. Boydell, in: M. I. Kohan, (Ed.), "Nylon Plastics Handbook", Hanser/Gardner Publications, Cincinnati, OH, USA, 1995, p. 291.
60. D. M. Bigg, D. J. Walsh and Roder, T. M., in: M. I. Kohan (Ed.), "Nylon Plastics Handbook", Hanser/Gardner Publications, Inc., Cincinnati, OH, USA, 1995, p. 151.
61. A. Dharia and S. J. Grossman, Antec 1987, 590.
62. D. P. Garner and P. D. Pasulo, Polym. Mater. Sci. Eng., **57**, 467 (1987).
63. S. W. Shalaby, P. Moy, in "Engineering Materials Handbook", ASTM International: Metals Park, OH, Vol. 2, 1988, p. 447.
64. J. A. Brydson, "Plastics Materials", Butterworth-Heinemann Ltd., Oxford, Great Britain, 1989.
65. J. Zimmerman, in "Encyclopedia of Polymer Science and Engineering", Vol. 2, IInd Ed., Wiley, New York, 1988.
66. "Recognized Component Directory", Underwriters Laboratories, Northbrook, IL, 1993: Vol. 2, p. 1786.
67. M. I. Kohan, in: "Ullmann's Encyclopedia of Industrial Chemistry," Vol. A2, VCH: Weinheim, 1992, p. 195.
68. E. S. Ong, Y. Kim, H. L. Williams, J. Appl. Polym. Sci., **31**, 367 (1986).
69. E. Roderdink, J. M. M. Warnier, Polymer, **26**, 1582 (1985).
70. H. M. Laun, Rheol. Acta., **18**, 478 (1979).
71. M. Eystatiev, in: O. Olabisi (Ed.), "Handbook of Thermoplastics", Marcel Dekker, New York, 1997.
72. J. J. Burke, T. A. Orafino, J. Polym. Sci., **7**, 1 (1969).
73. A. Mattiussi, G. B. Gechele, R. Francesconi, J. Polym. Sci., **7**, 411 (1969).
74. Z. Tuzar, M. Bohdanecky, R. Puffr, J. Sebenda, Eur. Polym. J., **11**, 851 (1975).
75. P. W. Morgan, S. L. Kwolek, J. Polym. Sci. A, **1**, 1147 (1963).

Physical Constants of Cellulose*

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A. INTRODUCTION

For the present edition of *Polymer Handbook* the former chapter, "Properties of Cellulosic Materials" has been considerably shortened and restructured. Besides leaving out the infrared spectrum and some properties not considered genuinely physical, the main alteration is the omitting of the vast field of cellulose derivatives altogether. This is to be justified in view of the space given to the other polymers dealt with in the section, "Physical Constants of Some Important Polymers". Each of the cellulose derivatives (or at least each class), we think, would have required an entry of its own. Accordingly, we restricted ourselves to materials, the main component of which is native or regenerated cellulose. Therefore, we explicitly refer to the previous edition (1) for its wealth of data on cellulose derivatives, liquid crystals therefrom, nonglucose hydrocarbons and noncellulosic substances in cellulosic materials, and first order rate constants and energies of activation of homogeneous hydrolysis. Moreover, much useful and recent information can be found in the Proceedings of the Cellucon Conferences (2), the pertinent ACS volumes (3) and a number of monographs (4–9), whereas cellulose research before 1955 is well documented in Refs. (10,11).

* Based on the table in the third edition, by A. Gröbe, Fraunhofer-Institut für Grenzflächen-und Bioverfahrenstechnik, Stuttgart, Germany.

B. SOLID STATE PROPERTIES

TABLE 1. UNIT CELL DIMENSIONS[a]

Modification	Space group	a (Å)	b (Å)	c (Å)	α (°)	β (°)	γ (°)	Remarks	Refs.
I_α	P1	6.74	5.93	10.36	117	113	81	Natural occurrence (algae, bacteria)	12,13[b,c]
I_β	$P2_1$	7.85	8.14	10.34	90	90	96.6	Natural occurrence (higher plants)	14[c],15[b],16[b]
II	$P2_1$	8.01	9.04	10.36	90	90	117.1	Obtained by alkali treatment	17[c],18[b]
III		10.25	7.78	10.34	90	90	122.4	Obtained by liquid ammonia treatment	19
IV		8.01	8.12	10.34	90	90	90	Obtained by heat treatment of cellulose III	20

[a] Due to the very limited number of available X-ray reflections, atomic coordinates are difficult to determine. The question of chain orientation (parallel *vs.* anti-parallel packing) cannot be considered ultimately settled, although a parallel arrangement is widely accepted for Cellulose I, whereas Cellulose II is thought to crystallize in an anti-parallel packing mode. In each of the five cells given in the table, the *c* axis is parallel to the chain axes. The I_α cell contains a single chain with two glucose units, while all the other cells consist of two chains with two glucose units each.
[b] References deal with the same modification.
[c] Atomic coordinates given.

TABLE 2. DENSITY

Type of cellulose	Density (g/cm^3)	Refs.
Cellulose I	1.582–1.630[a]	21,22,23,24,25,26,27
Cellulose I_α	1.58[a]	12
Cellulose I_β	1.62–1.64[a]	14,16
Cellulose II	1.583–1.62[a]	22,24,28,26
Cellulose IV	1.61[a]	22
Cotton	1.545–1.585	29,30,31,32,33,34
Ramie	*ca.* 1.55	26,35
Flax	1.541	36
Hemp	1.541	36
Jute	1.532	36
Wood pulps	1.535–1.547	26,31,37,25,38,39,33
Cuprammonium fibers	1.519–1.531	31,32,34
Polynosics	1.489–1.528	40,41
Viscose fibers (and films) (rayon and staple)	1.508–1.548	31,32,38,42,43,44
High tenacity viscose fibers	1.498–1.524	42,45

[a] Calculated from X-ray or electron diffraction data.

TABLE 3. AVERAGE ORDERED FRACTION OF COTTON AND LINTERS MEASURED BY VARIOUS TECHNIQUES[a,b]

Technique	Average crystallinity value for cotton and linters (%)	Number of publications involved in the average crystallinity value	Approximate average crystallinity value	
			Wood pulp	Regenerated cellulose
PHYSICAL				
X-ray diffraction	73	25	60	35
Density	64	18	50	35
ABSORPTION AND "SWELLING" CHEMICAL				
Deuteration	58	5	45	25
Moisture regain	58		50	25
Hallwood–Horrobin	67	3	55	35
Nonfreezing water	*ca.* 85	1		50
Acid hydrolysis	90	26	85	70
Alcoholysis	*ca.* 90	4		
Periodate oxidation	92	4	92	80
Nitrogen tetroxide oxidation	*ca.* 70	1		50
Formylation	79	7		35
Iodine sorption	87	9	85	60
"NONSWELLING" CHEMICAL				
Chromic acid oxidation	99.7	1		
Thallation	99.6	2		

[a] Ref. 46.
[b] Most of these values are general averages of several published values: the number of publications involved in each average is given in the third column. The term "crystallinity value" does not imply that any particular method measures crystallinity or order in any strictly defined sense, or that the various methods measure precisely the same type or level of order/disorder; with the chemical methods for instance, the value given is merely the fraction of the material that is not readily accessible to, and thus not able to, react with the particular reagent.

TABLE 4. AVERAGE ORDERED FRACTION MEASURED BY VARIOUS TECHNIQUES[a]

Technique	Cotton	Mercerized cotton	Wood pulp	Regenerized cellulose
X-ray diffraction	0.73	0.51	0.60	0.35
Density	0.64	0.36	0.50	0.35
Deuteration or moisture regain	0.58	0.41	0.45	0.25
Acid hydrolysis	0.90	0.80	0.85	0.70
Periodate oxidation	0.92	0.90	0.92	0.80
Iodine sorption	0.87	0.68	0.85	0.60
Formylation	0.79	0.65	0.75	0.35

[a] Ref. 46.

TABLE 5. DEGREE OF CRYSTALLINITY[a]

Type of cellulose	DC[b](%)	Refs.	x_c^c(%)	k (10^{-2}) (nm^2)	Refs.	Cr[d](%)	Refs.
Natural Fibers							
Ramie	60–71	48,49,50,51	44–47		52		
			57	4	53		
	54–74	54					
Ramie, mercerized	49	55	45		52		
			37–46	2.5–3	53		
Flax						67	56
Linen						66 ± 3	57
Hemp						60 ± 3	57
Manila hemp						42 ± 4	57
Sisal						38 ± 2	58
Jute, *Corchorus olitorius*	58–66	59					
Jute, *Corchorus capsularis*	47–67	59				65 ± 3	58
Flax, bleached	Average 67, range 64–69	60					
Cotton, different origins	69–71	49	56–59	2.4–3.9	61	70	62
	71	63	47		52	79	64
	Average 70, range 66–74	60					
Cotton, mercerized			57–60	3.4–4.2	61	51	62
			42		52		
Cotton, linters	70	49	64–65	3.1–3.2	67	57	66
	73	63	56–63		65	68	56
Cotton, linters, mercerized	Average 48, range 41–55	60	43–49	2.1–2.6	65		
Algal Cellulose							
From *Valonia ventricosa*	Average 68, range 65–70	48					
Bacterial Cellulose							
From *Acetobacter xylinum*	40	48	73–79	3.3–3.9	68		
Cellulose from Woody Plants							
Cellulose from different plant stems						49–58	56
Cellulose from different woods						62–64	56
Cellulose from *Pinus radiata* (5, 10, and 15 annual ring)	73–55	69					
Bagasse						58	56
Pulps							
Different wood pulps	62–70	48,49,60,51				56–68	70
						62–65	56
						49–77	71
Different dissolving pulps	64	63	43–56		67	52–66	72
Different wood pulps, mercerized	49	55					

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TABLE 5. *cont'd*

Type of cellulose	DC[b](%)	Refs.	x_c^c(%)	k (10^{-2}) (nm^2)	Refs.	Cr[d](%)	Refs.
Man-made Cellulosic Fibers							
Viscose rayon, normal grade	38–40	48,49 50,75,76	27	1.58	67	35	74
	Average 27, range 13–36	60					
Viscose staple fibers, normal grade	30–39	48,49,76,78				33–36	77
						34	79
	40–41	63					
	Average 27, range 30–52	60					
Viscose fibers, mercerized	40–52	50,55,75					
High tenacity viscose rayon	23–56	48,49,50,75,76,80					
High wet modulus (HWM) viscose fibers	27–44	48,49,75,76,78				38	81
						42	79
Polynosic-fiber			35.6		52	48	81
						48	79
Cuprammonium rayon	40–44	47,48,50					
	41	63					
Cuprammonium rayon, mercerized	49	55					
Saponified acetate fibers	Average 38	49					
Fortisan	39–54	49,76,78,82					
	43	83					
Regenerated carbamate fibers						50	79
Lyocell-type fiber			42	1.75	67	57	79
Cellulosic Films							
Cellophane	30–40	48,49,78	45	2.7	84		
Cuprophane			46	2.9	84		
Blown films from NMMO solution			38–41	2.6–2.8	84		
Others							
"Amorphous" powder, obtained by dry-grinding viscose rayon	8	85,10					
"Amorphous" powder, obtained by dry grinding viscose rayon, recrystallized	35–40	10					
Different cellulose samples						36–44	73
Regenerated cellulose precipitated from viscose as flake	45–53	82					
Precipitated β-cellulose	50–62	83					

[a] According to Hermans (49a) relative and absolute X-ray methods can be distinguished. DC and x_c values are absolute ones as well as some of the Cr values. The Ruland–Vonk-method provides additionally a lattice disorder factor k representing the amount of lattice distortions. The so-called crystallinity index or ratio values derived from relative methods as peak height ratios are omitted here. For these, the reader is referred to the third edition (1).
[b] According to Hermans and Weidinger (47,48).
[c] According to the method of Ruland (86) and Vonk (87).
[d] According to other methods.

TABLE 6. CRYSTALLITE SIZES

Type of cellulose	Lateral dimensions (Å)	Width and thickness (Å)	Range of length (Å)	Refs.
Cotton	50–60			88
	50		500	89
	155–230		128	90
	100			56
	40			91
	58 ± 3			92
	60			93
	64			94
	50–100			95
	100			96
	146			97
	150			98
		62×49		99
		64×50		100

TABLE 6. *cont'd*

Type of cellulose	Lateral dimensions (Å)	Width and thickness (Å)	Range of length (Å)	Refs.
		93 × 32		101
		110 × 25		102
		110 × (69–72)		52
		130 × 25		103
			400–2550	46
Cotton linters	51			104
	47–60			105
	65			56
Ramie	16–26			106
	50–100			107
	59			94,99
	68			97
	70			108
	<400			109
		61 × 167		52
			600–2500	46
	48–55			72
	46–55			110
	47–66			111
	62			112
	46–50			105
	44			91
Hemp	44			97
	49 ± 3		189	58
Flax	51.5			97
	55			56
Linen	52 ± 3		195	58
Jute	55			97
	49 × 4		174	58
Bamboo	<70			88
Manila hemp	38 ± 2		140	58
Sisal	38 ± 3		130	58
Algal cellulose (*Valonia*)		148 × 53		113
	89–124			110
	89–101			105
	112			91
Bacterial cellulose (*Acetobacter xylinum*)		84 × 70		99
	58–83			110
	53–65			105
	55			91
Chaetomorpha melagonium		169 × 114		99
Tunicin	84 × 34			99
Different wood samples	37–40		65–70	56
Cellulose from different plant stems	35–36			56
Bagasse	35			56
Different wood pulps	58–70			56
Different dissolving pulps	32–60			72
		1400–2200*	75–105	67
	41–47			105
Normal viscose fibers	50–70			77
	75–95		110–156	90
	39–42			104
	54		114	67
HWM viscose fibers	76			77
	70–85		154	90
		(80–100) × (30–40)	120–140	115
		(77–90) × (29–49)		79
	42–51			104
High tenacity rayon			120	115
Normal viscose rayon			140	115
	20–31			116
Polynosic fibers	93			77
	80–100		189	90
			160	115

* Mean cross-sectional area ($Å^2$).

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TABLE 6. *cont'd*

Type of cellulose	Lateral dimensions (Å)	Width and thickness (Å)	Range of length (Å)	Refs.
Tufcel-fibers	35–47		112	110,117
Isotropic "Hermans" viscose fibers	50–200			118
Fortisan	100–160		160	90
	43–57			110
	49–51			111
			160	115
Regenerated carbamate fibers		(67–88) × (26–47)		79
Lyocell-type fibers		70 × (25–46)		79
	42		143	67
Cellophane	36–53			84
Cuprophane	33–53			84
Blown films from NMMO-solution	30–41			84

TABLE 7. X-RAY ORIENTATION OF CELLULOSE FIBERS[a]

Material	Φ (°)	Refs.	f_x	Refs.
Cotton	14.25–16.20	61	0.62–0.70	61
	13.25–17.39	127		
Cotton, mercerized	9.7–13.7	61	0.64–0.72	61
Ramie	7.5	128	0.97	10
Ramie, mercerized without tension			0.90	10
Ramie, mercerized and reoriented			0.98	10
Hemp	6.2	128		
Flax	10	128		
Jute	8.0	128	0.96–0.97	119
Sisal	20.0	128		
Coir	45	128		
Viscose rayon, normal grade			0.31–0.91	76,120,10,121
Viscose staple fibers, normal grade			0.30–0.67	76,78
Lilienfeld rayon			0.70–0.94	76,10,121
High-tenacity viscose cord rayon			0.39–0.998	50,76,122,121
Viscose tire cord			0.94–0.97	123
High-wet modulus (HWM) viscose fibers			0.45–0.993	76,122,120,121,78
Modal fibers cond.			0.57	124
Modal fibers wet			0.30	124
Polynosic fibers (Tufcel)			0.94	123
Isotropic viscose model fibers, without stretching			0.06–0.15	125
Isotropic viscose model fibers, after intense stretching			Up to 0.90	125
Cuprammonium rayon			0.86	10
Fortisan			0.7–0.99	76,122,120,78,121
Lyocell-type fibers cond.			0.62–0.63	124
			0.67–0.94	126
			>0.9	67
Tencel			0.94	123
Lyocell-type fibers wet			0.33–0.35	124

[a] For native fibers the spiral angle Φ between cellulose microfibrils (crystallites) and fiber axis is filed. The degree of uniaxial orientation of crystallites is given for regenerated fibers and some of the native by the Hermans orientation factor, f_x, according to Ref. (10). As existing data for regenerated fibers largely depend on spinning conditions, ranges of f_x are given.

TABLE 8. HEAT CAPACITY[a]

Type of cellulose	T_0 (K)	$C_p(T_0)$ (J/g/K)	a ($\times 10^3$) (J/g/K^2)	T_m (K)	Remarks	Refs.
Amorphous	350	1.414	8.08	435		129
Wood	350	1.364	5.06	435		129
Jute	350	1.339	4.64	435		129
Cotton	350	1.318	4.27	435		129
100% crystalline	350	1.230	1.42	435	Extrapolated	129
Wood	10	0.0205	4.20	300	Independent of crystallinity	130

TABLE 8. *cont'd*

Type of cellulose	T_0 (K)	$C_p(T_0)$ (J/g/K)	a ($\times 10^3$) (J/g/K^2)	T_m (K)	Remarks	Refs.
Amorphous	300	1.266	6.0	400	Ball milled	130
Wood (40% cryst.)	300	1.240	5.5	400	Ball milled	130
Wood (65% cryst.)	300	1.226	4.9	400	Ball milled	130
100% crystalline	300	1.238	4.2	400	Extrapolated	130
Ramie	273	1.775	0	373	Average	131
Flax	273	1.344–1.348	0	373	Average	132
Hemp	273	1.327–1.352	0	373	Average	132
Jute	273	1.357	0	373	Average	132
Viscose rayon	273	1.357	0	373	Average	132
	273	1.415–1.595	0	373	Average	131
Cuprammonium fibers	273	1.357	0	373	Average	132
Paper	273	1.17–1.32	0	373	Average	133

[a] The constant pressure specific heat capacity $C_p(T)$ as a function of temperature T is expressed in linear approximation as $C_p(T) = C_p(T_0) + a\,(T - T_0)$, $T_0 < T_m$, a being the slope and T_m marking the interval of validity.

TABLE 9. THERMAL EXPANSION COEFFICIENT

Type of cellulose	T (°C)	α^a ($\times 10^5$) (K^{-1})	α_3^b ($\times 10^5$) (cm^3/g/K)	Remarks	Refs.
Hydrogel	15–70		6.3–25		134
Cellulose I	RT[c]–150	13		X-ray (101) reflexion	135
	150–200	23		(101)	135
	RT–150	5.1		(10-1)	135
	150–200	1.6		(10-1)	135
	RT–150	11		(002)	135
	150–200	16		(002)	135
	RT–150	−0.64		(040)	135
	150–200	1.6		(040)	135
Cellulose II	RT–100	−31		(101)	135
	100–150	68		(101)	135
	RT–100	5.1		(10-1)	135
	100–150	16		(10-1)	135
	RT–100	7.9		(002)	135
	100–150	19		(002)	135
Amorphous sulphate cellulose	−45, −5		$7/\rho$	ρ = Cellulose density in g/cm^3	136
	55, 80		$17/\rho$	ρ = Cellulose density in g/cm^3	136
Cotton (Cellulose I)	−150 to 225	7.1–36.7		X-ray (002) reflexion	137
High strength cord a	105–150	0.7			138
High strength polynosic	105–150	0.4			138
High strength cord b	105–150	1.8			138
Rayon	−4 to 25		3.59		139
Wood pulp	−4 to 25		5.76		139
Cotton linters	−4 to 25		5.52		139
Different papers		0.2–0.75		Machine direction	140
		0.79–1.62		Cross machine direction	140
Cotton, rayon		8			141
Various celluloses			51–60		142
Linters	−30 to 25		40		143

[a] Linear expansion coefficient.
[b] Cubic expansion coefficient $d\Phi/dT$; specific volume.
[c] Room temperature.

TABLE 10. OTHER THERMAL PROPERTIES

Property	Type of cellulose	Value	Remarks	Refs.
Decomposition point (°C)	Cellulose	200–270		144
	Cotton	150		145
	Viscose rayon	180		145
Start of thermal degradation (°C)	Linters	225		146
	Bleached sulfite pulp	225		146
	Kraft pulp	240		146
	Filter paper	220	Under nitrogen	147

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TABLE 10. *cont'd*

Property	Type of cellulose	Value	Remarks	Refs.
Fast endothermal degradation (°C)	Linters	≈300		146
	Bleached sulfite pulp	≈330		146
Flammability of fibers				
Ignition temperature (°C)	Cotton	390, 400		148,149
	Viscose rayon	420		148
Self ignition temperature (°C)	Cotton	400		150
External ignition temperature (°C)	Cotton	350		150
Maximum flame temperature (°C)	Cotton	880		150
		850–1050	19–25% O_2	151
	Cotton fabric	812–945	Heating rates 169–225 K/s	152
Fast endothermal degradation (°C)	Linters	≈300	Under nitrogen	146
	Bleached sulfite pulp	≈330		146
	Cotton	290	Under nitrogen	150
	Cellulose powder	300–500	Under flowing nitrogen, char yields 89.4–8.7 wt.%	153,154
Heat of combustion (kJ/kg)	Cotton fabric	15090	Measured in air	155
	Cotton fleece	15540	Bomb-calorimetric technique	155
	Cotton	18855		150
	Cotton fabric	15452	Measured in air, independent of fabric weight	156
	Regenerated cellulose	14732		156
	Fabric	54.2 (J/cm^2)		157
	Cellulose diacetate	17179		156
	Cellulose triacetate	17598		156
Rate of heat combustion (J/s/cm)	Cotton fabric	331	In air, bomb calorimetric technique	158
Limiting oxygen index	Cotton	0.184	Fabric	149
	Viscose rayon	0.197	Fabric	149
	Cotton	0.183	Twill weave fabric, 241 gm^{-2}	159
		0.185	Bleached and mercerized	152
	Cellulose	1.34		160
	Cotton	1.22		161
		1.15–1.18	0–34 °C	133
		1.32	0–100 °C	133
		1.327–1.251		162
		1.273–1.35		132
	Mercerized cotton	1.235		161
Heat of crystallization (kJ/kg)	Cellulose, extrapolated to 100% crystallinity	105.9 ± 5.0		163
	Cellulose I	121.8		164
	Cellulose II	134.8		164
Heat of recrystallization (kJ/kg)	Amorphous cellulose→ Cellulose II	41.9	By wetting	85
Heat of transition (kJ/kg)	Cellulose I → cellulose II	38.1		165
Heat of formation (kJ/kg)	Cellulose	5949.7		166
Heat of solution (kJ/kg) of dry material	Cotton in cuane	108.0		163
	Cotton in Et_3PhNOH	142.5		167
	Rayon in cuane	95.5		163
	Rayon in Et_3PhNOH	138.3–153		168
	Ball milled cellulose in Et_3PhNOH	243		167
	Cellulose II in Et_3PhNOH	182.7		167
	Cotton	1235, 1194–1327	At zero regain	169,170,171
Differential heat of sorption of liquid water (kJ/kg)	Cotton	1090	At zero regain, from sorption isotherm at zero regain, 27°C	172
		1332		173
		972–1040	At zero regain	174,26
		964–1006		174,26
		1047	At zero regain, 25°C	175
	Egyptian cotton	1214	At zero regain, 26°C	176
	Cotton, dried by solvent exchange	1047	At zero regain, 26°C	176
	Cotton, mercerized	1173	At zero regain	171
	Linters	1143	At zero regain	177
	Linters mercerized	1327	At zero regain	177

TABLE 10. *cont'd*

Property	Type of cellulose	Value	Remarks	Refs.
	Flax	1248,1206	At zero regain	169,174
	Wood pulps	1173–1256, 1098	At zero regain	177,178
	Fortisan	1235, 1169	At zero regain	171,174
	Cuprammonium fibers	1214, 1115	At zero regain	171,174
	Normal viscose fibers	1173–1215	At zero regain	171,174
		1050, 984,		171,174
		917–1022, 1277		171,174
	Viscose fiber	1549	At zero regain, 26°C	176
	Viscose fiber, dried by Solvent exchange	1172	At zero regain, 26°C	176
	Cotton	293	At 60% RH	169
		201–281	At 65% RH	145
	Cotton, mercerized	230	At 60% RH	171
	Viscose rayon	318	At 60% RH	169
		268–297	At 65% RH	169
Heat evolved by 1.0 g of fibers going from 40% RH to 70% RH in Joule	Cotton	84		145
	Viscose rayon	168		145
Integral heat of wetting of dry material (kJ/kg) wetting in water at zero regain	Cotton	33.9–56.2	At 20–25°C	169,171,26,175 177,179,180,181
		40.6	At 20°C	182
	Cotton, average value	46.5	At 20–25°C	46
	Cotton, bleached	40.0, 39.0		182,183
	Cotton, mercerized	62.0–78.8, 62.6	At 20–25°C	171,177
		62.6		181,184,182
	Cotton, mercerized, average value	75.2	At 20–25°C	46
	Cotton, ball milled	90.5	At 20–25°C	185
	Linters	44.8–57.8	At 20–25°C	163,177,185,186
	Ramie	34.4–48.6	At 20–25°C	26,181,180
	Flax	39.4–54.4	At 20–25°C	169,180
	Wood pulps	49.4–71.2	At 20–25°C	177,179,181,180
	Regenerated cellulose	83.8–106.4	At 20–25°C	177
	Fortisan	76.2	At 20–25°C	171
	Cuprammonium rayon	93.8	At 20–25°C	169
	Viscose fibers	68.2–105.5	At 20–25°C	163,169,171,177, 181,180,187,188
		68.2–82.5	At 20–25°C	183
	Polynosic fibers	83.1–87.8	At 20°C	189
Wetting in methanol	Cotton	7.3	At 20–25 °C	46
Wetting in bases, acids and salt solution, $\Delta H_{salt} - \Delta H_{H_2O}$ (kJ/kg)	Cotton	10.2	5% NaOH	190
		62.2	17.5% NaOH	190
		62.0	18% NaOH	191
		83.8	18% NaOH	192
	Cotton, mercerized	56.4	18% NaOH	190
	Ramie	67	20% NaOH	193
	Wood pulp	11.2	5% NaOH	190
		78.3	17.5% NaOH	190
	Wood pulp, different degrees of refining	67.0–81.0	20% NaOH, 25°C	194
	Dissolving pulp	≈68	18% NaOH, 20°C	195
	Different viscose fibers	25–47	5% NaOH	183
		31.8–36.4	5% NaOH	196
		62.8–67.8	17.5% NaOH	196
	Cotton	4.6	0.1 M H_2SO_4	182
		3.5	3 M NaCl	197
		3.1	3 M KCl	197
		5.4	3 M $MgCl_2$	197
		3.8	3 M $CaCl_2$	197
Thermal conductivity (W/m/K)	Cotton	0.071	Density 0.5 g/ml	198
	Rayon	0.054–0.07		149
	Sulfite pulp, wet	0.8		199
	Sulfite pulp, dry	0.067		199
	Laminated kraft paper	0.13		200

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TABLE 10. *cont'd*

Property	Type of cellulose	Value	Remarks	Refs.
	Different papers	0.029–0.17		201
	Alkali cellulose	0.0581	Density 0.2 g/ml	202
		0.0465		203
		0.0674		204
Transition points (°C)				
Glass transition	Cellulose	230		205
		220–245		206
Secondary transitions	Cellulose	19–23		142
		25		207
	Paper	$-20, +15, \approx 35, \approx 60$		208,140,205
	Cellulose	≈ 200		205
Endothermal process	Cotton	134	$H = 134$ J/g	209
	Bleached linters	131	$H = 131$ J/g	209
	Sulfite pulp	137	$H = 145$ J/g	209
	Sulfate pulp	147	$H = 124$ J/g	209
	Microcrystalline powder	109, 131	$H = 120–143$ J/g	209
		114	$H = 118$ J/g	209

See also Table 17, Typical Non-Aqueous Cellulose Solvents.

TABLE 11. REFRACTIVE INDEX AND BIREFRINGENCE

Type of cellulose	$n_{\parallel}$	$n_{\perp}$	$n_{\parallel} - n_{\perp}$	Refs.[a]
Ideally oriented native cellulose fiber (Cellulose I)	1.618	1.544	0.074	26
Ramie	1.595–1.601	1.525–1.534	0.061–0.071	26,210,211,212,213,214,215
Flax	1.571–1.595	1.517–1.532	0.054–0.067	212,214,216
Hemp	1.585–1.591	1.526–1.530	0.055–0.065	216,217
Cotton	1.573–1.595	1.527–1.534	0.040–0.062	210,212,214,218,45,219,215
Jute	1.577	1.536	0.041	214
Ideally oriented regenerated cellulose fiber (Cellulose II)	1.578	1.523	0.055[b]	220,(221)
			0.043[c]	26
Fortisan			0.045	222
High tenacity rayon	1.542–1.553	1.509–1.513	0.029–0.043	45,(212,223,224)
Polynosics	1.556–1.570	1.518–1.531	0.036–0.043	45,40,(221)
Polynosics cord			0.0410–0.0415	225
Normal viscose fibers	1.529–1.552	1.512–1.520	0.013–0.040	226,(212,45,223,224,40,227),215
Viscose cord			0.0350–0.0370	225
Cuprammonium fibers	1.548–1.571	1.519–1.534	0.018–0.037	26,212,218,222,(212,214)
Cellophane			0.011, 0.0145, 0.0044	228

[a] References in parenthesis give additional data.
[b] Bone dry.
[c] Conditioned.

TABLE 12. RESISTIVITY

Material	ρ^a (Ω cm)	R_s^b (Ω/gcm^2)	Remarks	Refs.
Pure cellulose	10^{18}			229,230
Cotton	2×10^7		52% RH[c], 30°C, moisture dependence investigated	231
		7×10^6	65% RH, 20°C	231
	2.4×10^9		35% RH, 0.18 GPa static pressure	232
Raw cotton	8×10^5		75% RH, 25°C	233,234[d]
	5×10^5			233,234[d]

TABLE 12. *cont'd*

Material	ρ^a(Ω cm)	R_s^b(Ω gcm^{-2})	Remarks	Refs.
Cotton, washed	$12 \times 10^6 - 37 \times 10^6$		$M^e = 10\%$, 20°C, moisture dependence investigated	234
Cotton, purified	9.3×10^5		$M = 10\%$, 20°C, moisture dependence investigated	234
Ramie	$6 \times 10^4 - 1.8 \times 10^6$		$M = 10\%$, 20°C, moisture dependence investigated	234
Ramie, purified	5.7×10^6		$M = 10\%$, 20°C, moisture dependence investigated	234
Hemp	1×10^7		$M = 10\%$, 20°C, moisture dependence investigated	234
Jute, purified	2.5×10^8		$M = 10\%$, 20°C, moisture dependence investigated	234
Flax, purified	6.6×10^6		$M = 10\%$, 20°C, moisture dependence investigated	234
Viscose rayon	$1 \times 10^6 - 3 \times 10^6$		≈80% RH, 30°C, moisture dependence investigated	234
		$10^8 - 10^9$	$M = 10\%$, 20°C, moisture dependence investigated	234
	6.75×10^{10}		35% RH, 0.18 GPa static pressure	232
Cuprammonium rayon		2.6×10^8	$M = 10\%$, 20°C, moisture dependence investigated	234
Fortisan		3.5×10^7	$M = 10\%$, 20°C, moisture dependence investigated	234
Mercerized cotton		6.6×10^6	$M = 10\%$, 20°C, moisture dependence investigated	234
Regenerated cellulose	10^{17}			229
Cellophane cuprophane	$10^{10} - 3 \times 10^{12}$		65% RH, 20°C	235
Insulating paper	$\approx 2 \times 10^{14}$		Air dry	236
	5×10^{17}		$M < 1\%$	237
	10^{12}		$M = 12\%$	237
	10^{14}		20°C, air dry	238
	10^{11}		100°C, air dry	238
Insulating kraft paper	4×10^{17}		20°C	239
	1.2×10^{17}		100°C	239

[a] Specific resistance (resistance of unit cross section per unit length).
[b] Mass specific resistance (product of ρ and density).
[c] Relative humidity.
[d] These references give additional data.
[e] Moisture content (wt.%).

TABLE 13. DIELECTRIC CONSTANT AND LOSS FACTOR AT 1 kHz[a]

Material	ε^b	tan δ^c	Range of lg^d (ν)^e	Remarks	Refs.
Pure cellulose (from spruce)	7.2	0.02	2–8	Bone dry, 20°C	240
Cotton	3.2	<0.02		Bone dry, 21°C	241
	6.1			25°C	242
	2.76		3–4	Unbeaten, density 1.0 g/cm^3	243
	2.85–3.05		3–4	Beaten, density 1.0 g/cm^3	243
	4.6	0.03	2–7	Dry, 0°C, temperatures up to 60°C investigated	244
Jute	105	1.4	2–5	Along fiber	245
	16	1.55	2–5	Transverse to fiber	245
Hemp	50	1.2	2–5	Along fiber	245
	11.2	1.28	2–5	Transverse to fiber	245
Ramie	45	1.15	2–5	Along fiber	245
	7.9	0.85	2–5	Transverse to fiber	245
Alpha wood pulp	6.15			30°C, vacuum dried at 30°C	192
Bleached sulphite pulp	2.9		3–4	Unbeaten, density 1.0 g/cm^3	243
	3.03		3–4	Beaten, density 1.0 g/cm^3	243
Paper	1.2–4				299
Viscose rayon	3.0	<0.02	1.7–5.3	Bone dry, 21°C	241
	6.7	0.18	2–7	Dry, 0°C, temperatures up to 60°C investigated	244
Regenerated cellulose film	6.7–7.5			20–70°C	246
Cellophane	7.6	0.01	1.8–6	22°C (−40 to 90°C investigated)	247

[a] More data for the frequency range indicated in the table and for other temperatures, humidities etc. are found in the references given.
[b] Real part of dielectric constant.
[c] Quotient of imaginary and real part of dielectric constant (loss factor).
[d] Decadic logarithm.
[e] Measuring frequency.

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TABLE 14. OTHER ELECTRICAL PROPERTIES

Property	Material	Value	Remarks	Refs.
Dielectric strength (break down voltage) (kV/mm)	Native cellulose fiber	50	Dry	248
	Insulating paper	7.7–9.2	20°C, 50 Hz, thickness 0.11–0.12 mm, 0.71–0.83 g/ml, RH = 65%	249
	Condenser paper	4.7–12.1	Oven dry, DC	250
	Cellophane	30–50	Thickness 1 mm, 50 Hz	235
Isoelectric point (pH)	Wood pulp	2.75		251
	Cotton	2.5	0.02 M HCl-KCl-buffer	252
Zeta-potential (mV)	Cellulose	21.0	Fines from filter paper, Whatman No. 1	253,254[a]
		30		255
	Cotton	21.1	Fines	253,254[a]
		28.5/26	In water	253,254
		21.1	In water, fines	253,254
		9.3/126	In 10^{-2} N KCl	256[b],257[b]
	Cotton linters	−23.7	Poly (1,1-dimethylpiperidinium-3,5-diallymethylene chloride), $M_n = 50000$, monolayer, in 10^{-4} N KCl	258
	Viscose fiber	22/27	In water, fibers washed with distilled water	256[b],257[b]
		17.3	In 10^{-4} N NaCl	259[b]
	Acetate grade dissolving pulp	17.7	In 10^{-5} N KCl	259[b]
	Cross-Bevan cellulose	22.9	Fines	253,254[a]
	Chlorite holocellulose	32.0	Fines from *Eucalyptus reagnans*	253,254[a]
	Chemical pulps, bleached	17–33	Fines	253,254[a]
	Chemical pulps, unbleached	23.5–33.1	Fines	253,254[a]
	Mechanical and semi-mechanical pulps	17.2–46.1	Fines	253,254[a]
	Beech kraft pulp	20.6	Oven dried at 130°C	260
	Unbleached wood sulfite pulp	9	Distilled water, 20°C	261
	Unbleached spruces sulfite pulp	23	Distilled water	262
	Unbleached sulfate pulp	18	Distilled water	262
	Bleached birch sulfate pulp	21	Distilled water	262
	Viscose rayon	26	$NaBH_4$-reduced sample, distilled water	263
	Spruce sulfite bleached	14.2	In water	264
		16.2	In 10^{-4} N KCl	264
	Spruce sulfite unbleached	14.9	In water	264
		16.0	In 10^{-4} N KCl	264
	Beech sulfite bleached	18.6	In water	264
		21.7	In 10^{-4} N KCl	264
	Birch sulfite unbleached	15.1	In water	264
		19.0	In 10^{-4} N KCl	264

[a] These references give additional data.
[b] More concentrations and/or media investigated.

TABLE 15. CRYSTAL ELASTIC MODULUS E_l IN CHAIN DIRECTION

Modification	E_e (GPa)	Refs.
I	138	265
	120–135	266
II	88	265
	106–112	266
III_I	87	265
III_{II}	58	265
IV_I	75	265

TABLE 16. FIBER STRENGTH, ELONGATION, AND MODULUS

Type of cellulose	Breaking strength (cN/tex)[a]		Breaking elongation (%)		Modulus (cN/tex)[a]		Refs.
	Dry	Wet	Dry	Wet	Dry	Wet	
Cotton	26–43	26–47	8–10	11	370–720		1[b]
Flax	40–80	41–85	2.5–3	2.5–3	1760		1[b]
Jute	~30	30–31	1.7–1.9	1.7–1.9	1720		1[b]
Hemp	47		2.2		2170		270
Ramie	59		3.7		1460		270
Sisal	36–43		3.6–5.1		1170–1520		271
Banana	53–58		2.4–3.6		2000–2440		271
Coir	9–15		17–47		260–520		271
Palmyrah	16–20		7–15		404–560		272
Talipot	17–30		2.7–5.2		1045–1494		272
Viscose rayon (normal)	14–22	2–5	25–30	20–40	290–680		1[b]
Viscose rayon (high tenacity)	35–54	23–43	9–20	14–26	609–883		1[b]
Viscose staple (high wet modulus, normal)	16–26	7–17	14–30	22–30	424–600		1[b]
Viscose staple (high wet modulus, polynosics)	26–53	18–42	7–15	8–20	265–795		1[b]
Lyocell type	61	58	8	9	1950	490	273
$ZnCl_2$ type	15–25	7–10	15–25	18–27	450–1100		274
LiCl-DMAc type	20–29	8–13	6	10–12			275
NaOH type	16–17	4–6	15–21	16–23	971–1290	25–63	276

[a] To convert to MPa multiply value by ten times the density value when expressed in g/cm^3.
[b] Data taken from references 76,267–269.

C. SOLUTION PROPERTIES

TABLE 17. TYPICAL NON-AQUEOUS CELLULOSE SOLVENTS[a,b,c]

Number of components	Substance group	Examples
Unicomponent	*N*-Alkylpyridinium halogenides	Ethylpyridinium chloride
	Oxides of tertiary amines	*N*-Methylpiperidine-*N*-oxide
		N-Methylmorpholine-*N*-oxide
		N,*N*-Dimethylcyclohexylamine-*N*-oxide
Bicomponent	Dimethyl sulfoxide (DMSO) containing solvents	DMSO/methyl amine
		DMSO/$CaCl_2$
	Liquid ammonia/sodium or ammonium salts	NH_3/NaI (NH_4I)
		NH_3/NaSCN (NH_4SCN)
		$NH_3/NaNO_3$
		$NH_3/N(C_2H_5)_4Br$
	Dipolar aprotic solvents/LiCl	*N*,*N*-Dimethyl acetamide (DMA)/LiCl
		N-Methylpyrrolidone/LiCl
		N-Methylcaprolactam/LiCl
		N-Hexamethyl phosphoric acid triamide/LiCl
		Dimethyl urea/LiCl
	Pyridine or quinoline containing systems	Pyridine/resorcinol
		Quinoline/$Ca(SCN)_2$
	Liquid SO_3/secondary or tertiary amines	SO_3/ triethyl amine
Tricomponent	NH_3 or amine/salt/polar solvent	NH_3/NaCl/morpholine
		NH_3/NaCl/DMSO
		Ethylene diamine/NaI/*N*,*N*-dimethyl formamide
	NH_3 or amine/SO_2 or $SOCl_2$ polar solvent	Diethylamine/SO_2/DMSO

[a] A very extensive list including sketches of phase diagrams, is given in Ref. (1).
[b] In most cases a preactivation of the cellulose is required.
[c] Ref. 277.

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TABLE 18. OTHER SOLVENTS[a]

Substance group	Examples	Refs.
Mineral acids[b]	HCl, HBr	279
	H_2SO_4	280,281
	HNO_3	282,283,284
	H_3PO_4	280,285,281
	Trifluoroacetic acid	286,287,288,289,290
Inorganic salts in aqueous solution	LiCl	291,292,293
	$ZnCl_2$	281,294,295
	$Ca(SCN)_2$	296,297,298,299,300
Inorganic salts in concentrated solutions or melts, single and/or multicomponent systems	$Ca(SCN)_2$	298,301
	LiCl, LiJ, LiSCN, $CaBr_2$, $Ca(SCN)_2$	302
	$Ca(SCN)_2/CH_2O$	303
	$LiSCN \cdot 2.5H_2O$	304,305
	$KSCN + NaSCN + Ca(SCN)_2 \cdot 3H_2O$	304,305
	KSCN + NaSCN + DMSO	304,305
	$Ca(SCN)_2 \cdot 3H_2O + CH_2O$	304
	LiSCN, NaSCN, LiJ, NaJ, KJ	306,46
Strong bases in aqueous solution	LiOH	307
	NaOH	307,81
	NaOH + ZnO	307,308,309
Strong bases in aqueous solution	NaOH + BeO	307,308
Tetraalkyl bases (molecular weight >150)	Tetraethylammoniumhydroxide and some higher homologs	310,311,312
	Trimethylbenzylammoniumhydroxide ("Triton B")	313,311
	Dimethyldibenzylammoniumhydroxide ("Triton F")	313,311
	Tetraalkylphosphonium hydroxides	314
	Tetraalkylarsonium hydroxides	144
	Trialkylsulfonium hydroxides	315
	Trialkylselonium hydroxides	292
Metal complex solutions	$[Cu(NH_3)_4](OH)_2$ ("Cuoxam", "Cuam")	316,317
	Modified cuoxam	170
	$[Cu(en)_2](OH)_2$ ("Cuprieethylenediamine", "Cuene", "CED")	318,319
	Cu : biuret : alkali	320
	$[Co(en)_3(OH)_2$ ("Cooxene")	321,322
	$[Ni(NH_3)_6](OH)_2$ ("Nioxam")	323,324
	$[Ni(en)_3](OH)_2$ ("Nioxene")	323,325
	$[Zn(en)_3]$ $(OH)_2$ ("Zincoxene")	323,325
	$[Cd(en)_3](OH)_2$ ("Cadoxene")	323,326,327
	Complex solution of the Na-salt of Fe-tartaric acid in diluted NaOH solution ("FeTNa")	323,328

[a] Excerpt from the table in Ref. (1); cf. also (278) as a modern reference.
[b] Dissolution usually accompanied by strong degradation of the cellulose chains.

TABLE 19. VISCOSITY – MOLECULAR WEIGHT RELATIONSHIPS[a]

Solvent[k]	K_m ($\times 10^2$) (ml/g)	a	Viscosity range, η ($\times 10^{-2}$) (ml/g)	G (s^{-1})	T (°C)	Nature of sample	Method of calibration	Refs.
Cuoxam	0.308	1.0	0.9–9		20		OS[b] (acet.)	329
	11.3	0.657	0.2–4		25	F[c]	Comp[d]	330
	10.1	0.661	0.2–4		25	F	Comp.	330
	56.4	0.523	4.7–16.8		20	U[e]	SD[f]	331
	0.85	0.81	4.7–16.8		20	U	SD	332
	1.70	0.77	4.7–16.8		20	U	SD	10
	0.268	1.0	1.75–19.7	500		U	SD (comp.)	333
	0.70	0.9	4.8–28			U	SD (CN)[g]	334
	0.545	0.9	1–28			U	SD (CN)	335
	0.33	1.15	4.7–16.8			U	SD	336
	0.099	10				F, U	SD, LS[h] (CN)	337
	3.3	0.76	>4.25			F, U	SD (CN)	338
	0.18	1.0	<4.25			F, U	SD (CN)	338
	0.237	1.0	1.2–5.05			U	S[i]	339
	1.30	0.81	0.3–6		25	F	LS (CN)	240

TABLE 19. *cont'd*

Solvent[k]	K_m ($\times 10^2$) (ml/g)	a	Viscosity range, η ($\times 10^{-2}$) (ml/g)	G (s^{-1})	T (°C)	Nature of sample	Method of calibration	Refs.
Cuene	1.06	0.905	1–21.4	$G \rightarrow 60$	25	F, U	Comp.	341,342
	0.498	1.0	1–2.4	$G \rightarrow 60$	25	F, U	Comp.	341,342
	0.395	1.0	2.4–21.4	$G \rightarrow 60$	25	F, U	Comp.	341,342
	0.334	1.0	<10			F	OS (CN)	335
	0.412	1.0	<10				Comp.	343
	0.364	1.0	2.3–27.2	500		U	SD (CN)	344
	1.01	0.9	1.0–28	$G \rightarrow 60$		F, U	SD (CN)	334
	0.85	0.9	1.0–28	$G \rightarrow 60$		F, U	SD (CN)	335
	0.358	1.0	6–18			U	SD	345
	0.153	1.0				U	SD (CN)	337
	5.10	0.76	>600		20	F, U	SD (CN)	338
	0.277	1.0	<600		20	F, U	SD (CN)	338
	0.762	0.936	1.9–21.3			U	SD, LS	346
	2.9	0.8	1.7–3.9		20		K[j] (OS)	347
Cadoxene	0.268	1.0	0.5–7.5		25		Comp.	348
	Mean: 0.575	0.96	1.9–6.5		20	F, U	Comp.	349,350,351
	5.47	0.75	<18	500	25	U	SD	352
	0.315	1.01	2–11		20	U	SD	351
	0.235	1.0	2–11		20	U	SD	353
	0.124	1.0				F, U	SD, LS	337
	4.13	0.76	>5.3			F, U	SD (CN)	338
	0.225	1.0	<5.3			F, U	SD (CN)	338
	0.427	0.94			20	U	SD (CN)	338
	0.593	0.94	<10		20	F	SD (CN)	338
	9.33	0.72	>10		20	F	SD (CN)	338
	3.56 (3.38)	0.77	<18	500	25	U	LS	352
	3.85	0.76	0.4–1.2	500	25	F	SD, LS	354
FeTNa	0.563	1.0			20	U	Comp.	355
	0.599	1.0			20	U	Comp.	355
	1.13	0.9	2.5–22		20	F, U	Comp.	Calculated from 356
	3.57	0.805	2.5–22		20	F, U	Comp.	Calculated from 356
	0.519	1.0	3–16	$G \rightarrow 60$	20	U	SD	357
	0.571	1.0	3–16	$G \rightarrow 60$	20	U	SD	357
	0.502	1.0	3–16	$G \rightarrow 60$	20	U	SD	357
	0.407	1.01	6.6–20	$G \rightarrow 60$	20	U	SD	345,357
	0.355	1.0	<13		20	U	SD	353
	0.303	1.0	<16		20	U	SD	353
	0.201	1.0	<8		20	U	SD	353
	0.39–0.46	1.0					Comp.	358
	2.8	0.84	2.4–21.2					346
	5.31	0.78[b]	4.9–31.5				LS	359
	5.31	0.775	1.68–15.15	>400	25	U	LS	360
LiCl/DMAc	1.278	1.19	1.48–11.64		30	U	LS	361
Zincoxene	0.585	0.936			20	U	OS	362
Tetraethylammonium hydroxide (2.3 M)	0.26	1.0			20	U	OS (via CN)	363
	0.58	1.0	2.4–12.2		20		OS, comp.	364
LiOH solution	≈0.34	1.0			20	U	OS	363
$Ca(SCN)_2$ solution	≈0.5	1.0			20	U	OS	363
H_3PO_4	≈1.3	1.0			20	U	OS	363
	1.71	0.94	>2.25	(49, 72)	25	U	K	365
H_2SO_4 (50%) solution	1.64	0.92	>1.78		20	U	K	366
10% NaOH solution	0.43	1.0			20	U	OS	363

[a] The constants K_m and a given in the table are defined by the equation $[\eta] = K_m M^a$, $[\eta]$ and M being intrinsic viscosity and molecular weight, respectively. G is the shear rate.
[b] Osmotic measurements.
[c] Fractionated.
[d] Viscosimetric comparison.
[e] Unfractionated.
[f] Measurement by sedimentation–diffusion.
[g] Cellulose nitrate used for calibration.
[h] Measurement by light scattering.
[i] Measurement by sedimentation.
[j] From kinetic data.
[k] See also Table 17, Typical Non-Aqueous Cellulose Solvents.

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TABLE 20. SECOND VIRIAL COEFFICIENT

The second virial coefficient listed in the table below is defined by the equation

$$\pi/c = RT[1/M + A_2c + A_3c^2 + \cdots]$$

where π is the osmotic pressure, C the concentrations R the gas constant, T the temperature, M the molar mass, and A_3 the third virial coefficient.

Solvent[c]	T (°C)	DP[a] range	Range of A_2 ($\times 10^3$) (cm^3mol/g^2)	Remarks	Refs.
Cuoxam	20	195–12500	0.8–3.0	No trend, $V^b = 12$	367
Cadoxene	25	175–5820	1.1–1.8	No trend, $V = 9$	352
FeTNa	25	230–3950	3.0–4.4	No trend, $V = 10$	360
LiCl/DMAc	30	770–4320	3.5–5.3	Monotonous decrease (one outlier), $V = 9$	361

[a] Degree of polymerization.
[b] Number of single values given.
[c] See also Table 17, Typical Non-Aqueous Cellulose Solvents.

TABLE 21. SEDIMENTATION AND DIFFUSION COEFFICIENTS

The sedimentation and diffusion coefficients for non-vanishing concentrations s and D, respectively, are calculated from $s^{-1} = s_0^{-1}(1 + k_s c)$ and $D = D_0(1 + k_D c)$, respectively, with s_0, D_0, d_s and k_D listed in the table, where c is the cellulose concentration.

Solvent[a]	T (°C)	M^b ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_d (cm^3/g)	Refs.
Cuoxam	20	19			2.85	9.7	367
	20	21			2.75	10.6	367
	20	27	1.6	50	1.97	50	173
	20	35			1.87	109	367
	20	44	1.9	20	1.54	45	173
	20	45			1.57	113	367
	20.7	53	4.6		5.4		368
	20	75	2.6	50	1.04	0	173
	20	115			0.97	149	367
	20	137			0.72	144	367
	20	150	3.8	110	0.84	30	173
	20	160			0.63	201	367
	20	185			0.59	149	367
	20	230	4.1	170	0.54	75	173
	20	320	6.3		0.66	165	173
	20	340	7.2	390	0.72	145	173
	20	354			0.50	143	367
	20	400	6.5	390	0.54	105	173
	20	490	7.2	520	0.43	130	173
	20	500	6.3	380	0.45	115	173
	20	535			0.37	264	367
	20	569			0.37	218	367
	20	570	6.3		0.37	190	173
	20	680	6.5	320	0.32	295	173
	20	1200	9.2	690	0.26	250	173
	20	1275			0.29	854	367
	20	1500	10.3	600	0.23	320	173
	20	1750	10.4	900	0.2	350	173
	20	1900	14.0	1000	0.25	320	173
	20	2000	10.8	860	0.18	250	173
	20	5900	17.5	1700	0.1	1200	173
Cuene	25	9.5	8.3		0.95		369
	25	175	5.5		1.2		369
Cadoxene	12	10.1	0.74				354
	12	18.8	1.04				354
	12	24.5	1.13				354
	12	33.6	1.25				354
	25	43	1.80	153	1.77		352
	25	115	2.87	367	1.06		352
	25	155	2.75	370	0.75		352
	25	210	3.17	465	0.64		352
	25	275	3.80	7100	0.58		352
	25	435	4.53	1000	0.44		352
	25	745	5.49	1410	0.31		352

TABLE 21. *cont'd*

Solvent	T (°C)	M^a ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_d (cm^3/g)	Refs.
FeTNa	20	111	1.75		0.89		345
	20	201	2.17		0.60		345
	20	364	2.98		0.46		345
	20	539	3.35		0.35		345
LiCl/DMAc	30	125			0.318		361
	30	152			0.293		361
	30	162			0.287		361
	30	178			0.248		361
	30	182			0.250		361
	30	320			0.175		361
	30	354			0.145		361
	30	463			0.124		361
	30	700			0.092		361

[a] See also Table 17, Typical Non-Aqueous Cellulose Solvents.
[b] Cellulose molecular weight.

TABLE 22. PARTIAL SPECIFIC VOLUMES, v_2

Solvent[a]	v_2 (cm^3/g)	T (°C)	Refs.
Cuoxam	0.664	20	173
	0.642	19.8	368
Cuene	0.65	25	352
Cadoxene	0.50	25	352
FeTNa	0.654	20	370

[a] See also Table 17, Typical Non-Aqueous Cellulose Solvents.

TABLE 23. GEOMETRICAL CHAIN CHARACTERISTICS

Listed properties:

Characteristic ratio	$C_\infty = \langle R^2\rangle_0/(nl^2)$
Unperturbed chain dimension	$A = \{\langle R^2\rangle_0/M\}^{1/2}$
Conformation parameter	$\sigma = \langle R^2\rangle_0/\langle R^2\rangle_{0,f}$
Persistence length	$q = \lim\{\langle Rl_1\rangle\}/l$ with $nl \to \infty$
Linear expansion coefficient	$\alpha_s = \{\langle s^2\rangle/\langle s^2\rangle_0\}^{1/2}$

where

$\langle R^2\rangle_0$ – Averaged squared chain end-to-end vector in unperturbed state
n – Degree of polymerization
l – Monomer unit length (5.18 Å)
M – Molar mass
$\langle R^2\rangle_{0,f}$ – Same as $\langle R^2\rangle_0$, but for freely rotating chain
$\langle Rl_1\rangle$ – Averaged projection of chain end-to-end vector onto first bond-vector
$\langle s^2\rangle$ – Averaged squared radius of gyration
$\langle s^2\rangle_0$ – Same as $\langle s^2\rangle$ for the unperturbed state.

Property	Solvent[a]	T (°C)	Value	Remarks	Refs.
C_∞	Cuoxam	25	15.8		361
	Cadoxene	25	45.2		361
	FeTNa	30	91.9		361
	LiCl/DMAc	20	48–60		367
A (Å)	Cadoxene		1.18–1.92		371
	LiCl/DMAc	30	2.74–4.12	Monotonous M-dependence	361
σ	Cadoxene	25	2.78		361
	FeTNa	25	4.70		361
	LiCl/DMAc	30	6.71		361
q (Å)	Cadoxene	25	42	Gaussian coil limit	361
	FeTNa	25	123	Gaussian coil limit	361
	LiCl/DMAc	30	252	Gaussian coil limit	361
	LiCl/DMAc	25	110–160	Various methods	372
α_s	Cuoxam	20	≈ 1		367
	LiCl/DMAc	30	1.03–1.07		361

[a] See also Table 17, Typical Non-Aqueous Cellulose Solvents.

TABLE 24. POLYMER–SOLVENT INTERACTION PARAMETERS

This parameter χ (Flory–Huggins parameter) figures in the equation for the reduced partial molar Gibbs energy of mixing of the solvent $\Delta G_1/RT$ (R – gas constant, T – temperature) in the following way:

$$\Delta G_1/RT = \ln(1-\varphi_2) + \varphi_2 + \chi\varphi_2^2$$

φ_2 being the volume fraction of the polymer (cf. the corresponding chapter in this Handbook).

Solvent	Concentration (wt.% cellulose)	χ	Remarks	Refs.
Water	Saturated with water	0.124	25 °C, cotton fiber	373
		0.435–0.44	25°C, viscose rayon	373
		0.528	25°C, cellophane	373
NMMO·1H_2O[a]	5–15	−3	Melting point depression	374
		0.5		
NMMO		0.35–0.40		375
NMMO with 13–15 wt.% water	3–15	<− 20	DP ~450, concept not applicable	376

[a] NMMO-*N*-methylmorpholine-*N*-oxide.

D. REFERENCES

1. J. Brandrup, E. H. Immergut (Eds.), " Polymer Handbook", 3rd edition, V/117, Wiley, New York, 1989.
2. J. F. Kenedy, G. O. Phillips, P. A. Williams (Eds.), "Cellulose – structural and functional aspects", Ellis Horwood Ser. in Polym. Sci. Technol., Ellis Horwood, Chichester 1989, "Cellulosics–Chemical, Biochemical and Material Aspects", and "Cellulosics–pulp, fiber and envrionmental aspects", Ellis Horwood Ser. in Polym. Sci. Tehnol., Ellis Horwood, Chichester 1993; "Cellulose and Cellulose Derivatives: Physico-Chemical Aspects and Industrial Applications", Woodhead, Cambridge, 1995.
3. R. H. Atalla (Ed.), "The Structures of Cellulose", ACS Symp. Ser. 340, Washington DC 1987; Th. Heinze, W. G. Glasser (Eds.), "Cellulose Derivatives: Modification, Characterization, and Nanostructure", ASC Symp. Ser. 688, 1998.
4. D. Klemm, B. Philipp, Th. Heinze, U. Heinze, W. Wagenknecht, "Comprehensive Cellulose Chemistry", Wiley-VCH, Weinheim 1998.
5. R. D. Gilbert, (Ed.), "Cellulosic Polymers, Blends and Composites", Hanser, Munich, 1994.
6. D. Fengel, G. Wegener, "Wood: Chemistry, Ultrastructure, Reactions", De Gruyter, Berlin, 1989.
7. H. A. Krässig, "Cellulose – Structure, Accessibility and Reactivity", Polymer Monographs, **11**, Gordon and Breach, London, 1993.
8. I. A. Tarchevsky, G. N. Marchenko, "Cellulose Biosynthesis and Structure", Springer, Berlin, 1991.
9. R. T. O. Connor, "Instrumental Analysis of Cotton Cellulose and Modified Cotton Cellulose", Marcel Dekker, New York, 1972.
10. P. H. Hermans, "Physics and Chemistry and Cellulose Fibers", Elsevier, New York, 1949.
11. E. Ott, H. M. Spurlin, M. W. Grafflin (Eds.) "Cellulose and Cellulose Derivatives", "High Polymers", vol. V, Part I, II (1954), Part III (1955), Interscience, New York.
12. J. Sugiyama, R. Vuong, H. Chanzy, Macromolecules, **24**, 4168 (1991).
13. A. Aabloo, A. D. French, Macromol. Theory Simul. **3**, 185 (1994).
14. A. Sarko, R. Muggli, Macromolecules, **7**, 486 (1974).
15. K. H. Garnder, J. Blackwell, Biopolymers, **13**, 1975 (1974).
16. C. Woodcock, A. Sarko, Macromolecules, **13**, 1183 (1980).
17. F. J. Kolpak, J. Blackwell, Macromolecules, **9**, 273 (1976).
18. A. J. Stipanovic, A. Sarko, Macromolecules, **9**, 851 (1976).
19. A. Sarko, J. Southwick, J. Hayashi, Macromolecules, **9**, 857 (1976).
20. A. Sarko, Tappi, **61** (2), 59 (1978).
21. W. Kast, R. Schwarz, Z. Electrochem., **56**, 288 (1952).
22. E. Treiber, "Die Chemie der Pflanzenzellwand", Springer, Berlin, 1957.
23. Ch. Legrand, Thesis, Paris, 1953.
24. A. Burgeni, O. Kratky, Z. Phys. Chem. (Leipzig) B, **4**, 401 (1929).
25. N. V. Klenkova, O. M. Kulakova, I. A. Volkova, Zh. Prikl. Khim., **36**, 166 (1963).
26. P. H. Hermans, "Contribution to the Physics of Cellulose Fibers", Elsevier, New York, 1946.
27. W. J. Lyons, J. Chem. Phys., **9**, 377 (1941).
28. W. Kast, Z. Elektrochem., **57**, 525 (1953).
29. A. J. Stamm, L. A. Hansen, J. Phys. Chem., **41**, 1007 (1937).
30. H. Wakeham, Textile Res. J., **19**, 595 (1949).
31. W. Moll, Z. Ver. Deut. Chem. Beih. Nr., **47**, 105 (1973).
32. G. F. Davidson, J. Textile Inst., **18**, T175 (1927).
33. B. Alince, L. Kuniak, Papri Celulosa, **19**, 87 (1964).
34. P. H. Hermans, J. J. Hermans, D. Vermaas, Kolliod-Z., **109**, 9 (1944); J. Polym. Sci., **1**, 149, 156, 162 (1946).
35. J. M. Preston, M. V. Nimkar, J. Textile Inst., **41**, T446 (1950).
36. V. I. Sharkov, V. P. Levanova, Vyskomol. Soedin., **1**, 1027 (1959).
37. J. A. Howsmon, Textile Res. J., **19**, 152 (1949).
38. K. Lauer, U. Westerman, Kollid-Z., **107**, 89 (1944).
39. F. C. Brenner, V. Frilette, H. Mark, J. Am. Chem. Soc., **70**, 877 (1948).
40. E. Klein, K. Bosarge, J. Polym. Sci. C, **2**, 515 (1963).
41. Centre Recherches Ind. Texitle de Roven, Bull. Inst. Textile France, 945, (1962).

42. N. V. Mikhailov, E. Z. Fainberg, M. Kozler, Vysokomol. Soedin., **2**, 1044 (1960).
43. N. V. Mikhailov, E. Z. Zav'yalova, V. O. Gorbacheva, Khim. Volokna, **1**, 19, (1960).
44. J. Juilfs, Forschungsber. Wirtsch. Verkehrsminist. Nordrhein-Westfalen, Nr. 381, **38**, 44 (1957).
45. J. Jacquemart, Bull. Inst. Textile, France, 963 (1962).
46. J. Q. Warwicker, R. Jeffries, R. L. Colbram, R. N. Robinson, "A Review of the Literature on the Effect of Caustic Soda and Other Swelling Agents on the Fine Structure of Cotton", Shirely Inst. Pamphlet No. 93, Altrincham, 1966.
47. P. H. Hermans, A. Weidinger, Kolloid-Z., **115**, 103 (1949); **120**, 3 (1951).
48. P. H. Hermans, A. Weidinger, J. Polym. Sci., **4**, 135 (1949).
49. P. H. Hermans, Makromol. Chem., **6**, 25 (1951); P. H. Hermans, Separatum Experientia, **19**, 553 (1963).
50. P. H. Hermans, A. Weidinger, Textile Res. J., **31**, 558, 571 (1961).
51. G. Jayme, H. Knolle, Papier, **18**, 249 (1964).
52. A. Viswanathan, V. Venkatakrishnan, J. Appl. Polym. Sci., **13**, 571, 785 (1969).
53. D. Chang, K. Miyasaka, K. Ishikawa, Sen-i Gakkaishi, **30**, 63 (1974).
54. P. K. Ray, Textile Res. J., **37**, 434 (1967).
55. P. H. Hermans, A. Weidinger, J. Polym. Sci., **6**, 533 (1951).
56. M. Ioelovitch, Acta Polymerica, **43**, 110 (1992).
57. H. J. Wellard, J. Polym. Sci., **13**, 471 (1954).
58. A. Wlochowicz, Przegl. Wlok., **22**, 471 (1968).
59. P. K. Ray, J. Appl. Polym. Sci., **13**, 2593 (1969).
60. O. Ant-Wuorinen, Paperi ja Puu, **37**, 335 (1955).
61. S. G. Shenouda, A. Viswanathan, J. Appl. Polym. Sci., **15**, 2259 (1971).
62. J. H. Wakelin, H. S. Virgin, E. Crystal, J. Appl. Phys., **30**, 1654 (1959).
63. O. Kratky, A. Krauß, Kolloid-Z., **151**, 14 (1957).
64. L. Segal, J. J. Creely, A. E. Martin, C. M. Conrad, Textile Res. J., **29**, 786 (1959).
65. H.-P. Fink, D. Fanter, B. Philipp, Acta Polymerica, **36**, 1 (1985).
66. O. Ellefsen, K. Kringstad, B. A. Tønnesen, Norsk. Skogind, **18**, 419 (1964).
67. H.-P. Fink, E. Walenta, Das Papier, **48**, 739 (1994).
68. D. C. Dan, H.-P. Fink, H. J. Purz, B. Philipp, Acta Polymerica, **33**, 742 (1982).
69. R. D. Preston, P. H. Hermans, A. Weidinger, J. Exptl. Botany, **1**, 344 (1950).
70. G. L. Clark, H. C. Terford, Anal. Chem., **27**, 888 (1952).
71. M. Ahtee, T. Hattula, J. Maugs, T. Paakkari, Paperi ja Puu-Papper och Trä, **65**, 475 (1983).
72. J. Schurz, A. Janospi, P. Zipper, Das Papier, **41**, 673 (1987).
73. H. Ruck, H. Krässig, Norelco Reptr., **7**, 71 (1960).
74. L. S. Bolontnikova, T. L. Samsonova, Zh. Prikl. Khim., **38**, 2299 (1965).
75. A. Lude, Ann. Sci. Textiles Belges, 2, 36 (1961).
76. E. Treiber, Chemiefasern, **14**, 25 (1964).
77. R. Segula, Melliand Textilber., **51**, 743 (1970).
78. J. K. Smith, W. J. Kitchen, D. B. Mutton, J. Polym. Sci. C, **2**, 499 (1963).
79. J. Lenz, J. Schurz, Cellulose Chem. Technol., **24**, 3 (1990).
80. A. Bartovics, H. Mark, J. Am. Chem. Soc., **65**, 1901 (1943); H. J. Philipp, C. F. Bjoerk, J. Polym. Sci., **6**, 383, 549 (1951).
81. U. Roesner, Deut. Textiltech., **16**, 304 (1966).
82. P. H. Hermans, A. Weidinger, J. Polym. Sci, **5**, 565 (1950).
83. E. Treiber, W. Berndt, M. Ruck, H. Toplak, Chem. Ing. Technik, **26**, 687 (1954).
84. H.-P. Fink, P. Weigel, A. Bohn, Lenzinger Berichte, **76**, 115 (1997).
85. P. H. Hermans, A. Weidinger, J. Am. Chem. Sci., **68**, 2547 (1946).
86. W. Ruland, Acta Cryst., **14**, 1180 (1961).
87. C. G. Vonk, J. Appl. Cryst., **6**, 148 (1973).
88. A. Frey-Wyssling, Protoplasma, **27**, 372 (1937).
89. H. H. Dolmetsch, Melliand Textilber., **48**, 1449 (1967).
90. H. Krässig, Das Papier, **38**, 571 (1984).
91. E. K. Boylston, J. J. Hebert, J. Appl. Polym. Sci., **25**, 2105 (1980).
92. S. G. Shenouda, A. Viswanathan, J. Appl. Polym. Sci., **16**, 395 (1972).
93. E. Husemann, A. Carmap, J. Makromol. Chem., **1**, 16 (1943); J. Gjonnes, N. Norman, Acta Chem. Scand., **12**, 2028 (1958).
94. C. Sterling, B. J. Spit, Am. J. Botany, **44**, 851 (1957).
95. B. G. Rånby, Makromol. Chem., **13**, 40 (1954); B. G. Rånby, E. Ribi, Experientia, **6**, 12 (1950).
96. G. Porod, loc. cit. E. Treiber, "Die Chemie der Pflanzenzellwand", Springer, Berlin, 1957.
97. A. N. J. Heyn, Textile Res. J., **19**, 163 (1949); J. Am. Chem. Soc., **70**, 3138 (1948).
98. C. W. Hock, J. Polym. Sci., **8**, 425 (1952).
99. L. Nieduszynski, R. D. Preston, Nature, **225**, 273 (1970).
100. F. F. Morehead, Textile. Res. J., **20**, 549 (1950).
101. O. Kratky, H. Sembach, Angew. Chem., **67**, 603 (1955).
102. S. M. Betrabet, E. Daruwalla, H. T. Lokhande, Textile Res. J., **36**, 684 (1966).
103. S. M. Mukherjee, J. Sikorski, H. J. Woods, J. Textile Inst., **43**, T196 (1952).
104. D. Hofmann, H.-P. Fink, B. Philipp, Polymer, **30**, 237 (1989).
105. H.-P. Fink, D. Hofmann, B. Philipp, Cellulose, **2**, 51 (1995).
106. M. Oberlin, J. Mering, Compt. Rend. (Paris), **238**, 1046 (1954).
107. J. Hengstenberg, H. Mark, Z. Krist., **69**, 271 (1928).
108. O. Kratky, Kolloid Z., **120**, 24 (1951).
109. R. Hosemann, Z. Physik, **114**, 133 (1939).
110. J. Haase, R. Hosemann, B. Renwanz, Colloid & Polymer Sci., **252**, 712 (1974).
111. A. M. Hindeleh, D. J. Johnson, Polymer, **13**, 423 (1972).
112. H. Nishimura, A. Sarko, J. Appl. Polym. Sci., **33**, 855 (1987).
113. V. Balashov, R. D. Preston, Nature, **176**, 64 (1955); R. D. Preston, A. B. Wardop, E. Nokolai, Nature, **162**, 957 (1948).

114. J. Lenz, J. Schurz, E. Wrentschur, W. Geymayer, Angew. Macromol. Chem., **138**, 1 (1986).
115. S. Watanabe, J. Hayashi, T. Akahori, J. Polym. Sci., **12**, 1065 (1974).
116. A. M. Hindeleh, D. J. Johnson, Polymer, **15**, 697 (1974).
117. J. Haase, B. Renwanz, Colliod & Polymer Sci., **250**, 503 (1972).
118. E. Treiber, Protoplasma, **40**, 166 (1951).
119. P. K. Ray, Text. Res. J., **48**, 573 (1978).
120. H. Krässig, W. Kitchen, J. Polym. Sci., **51**, 123 (1961).
121. H. Vosters, Svensk Papperstid., **65**, 65 (1962).
122. D. S. Jackson, A. Sandig, Textile Res. J., **31**, 421 (1961).
123. H. Coulsey, Proceedings of the 34th Int. Man-made Fibers Congress, Dornbirn, Austria, Sept. 20, 1995.
124. J. Lenz, J. Schurz, E. Wrentschur, Colloid & Polymer Sci., **271**, 460 (1993).
125. P. H. Hermans, J. J. Hermans, D. Vermaas, A. Weidinger, J. Polym. Sci., **1**, 393 (1946); **2**, 632 (1947).
126. S. A. Mortimer, A. A. Peguy, R. C. Ball, Cellulose Chem. Technol., **30**, 251 (1996).
127. A. R. Kalyanaraman, J. Appl. Polym. Sci., **25**, 2523 (1980).
128. P. S. Mukherjee, K. G. Satyanarayana, J. Materials Sci., **21**, 4162 (1986).
129. T. Hatakeyama, K. Nakamura, H. Hatekeyama, Polymer, **23**, 1801 (1982).
130. A. N. Mochalov, T. B. Khlynstova, M. Ya. Ioelovich, I. Kaimins, Khim. Drev., **4**, 66, (1982).
131. N. V. Mikhailov, E. Z. Fainberg, Vysokomol. Soed., **4**, 230 (1962).
132. W. Goetze, F. Winkler, Faserforsch. Textiltechn. **18**, 119, (1967).
133. K. Melzer, Thesis, TH Darmstadt, 1928.
134. L. E. Bromberg, B. S. El'tsefon, Kolloidn. Zh., **50**, 813 (1988).
135. M. Takahashi, H. Takenaka, Polym. J., **14**, 675 (1982).
136. I. F. Kaimins, Vysokomol. Soedin., Ser. B, **21**, 331 (1979).
137. S. Seitsonen, I. Mikkonen, Polymer J., **5**, 263 (1973).
138. L. M. Granschuk, E. Z. Fainberg, N. V. Mikhailov, Khim. Volokna, **15**, 31 (1973).
139. J. L. Neal, D. A. I. Goring, J. Polym. Sci. C, **28**, 103 (1969).
140. J. Kubat, S. Martin-Loef, Ch. Soeremark, Svensk. Papperstidn., **72**, 763 (1969).
141. E. Wuenneberg, W. Fischer, A. Sapper, Z. Phys. Chem. A, **151**, 1 (1930).
142. M. V. Ramiah, D. A. Goring, J. Polym. Sci. C, **11**, 27 (1965).
143. M. V. Tsilipotkina, A. A. Tager, R. S. Petrov, G. Pustobaeva, Vysokomol. Soed., **4**, 1844 (1962).
144. Th. Lieser, "Kurzes Lehrbuch der Cellulosechemie", Borntraeger, Berlin, 1953.
145. W. E. Morton, J. W. S. Hearle, "Physical Properties of Textile Fibers", Butterworths, London, 1962.
146. R. Schneider, O. Zoeppel, Papier, **25**, 849 (1971).
147. A. Broido, Pyrodynamics, **4**, 243 (1966).
148. DuPont de Nemours Bull., X-45 "The Flammability of Textile Fibres", Wilmington, 1955.
149. H. Goerlach, Chemiefasern, **22**, 611 (1972).
150. U. Einsele, Melliand Textilber., **52**, 1395 (1972).
151. B. Miller, J. R. Martin, Ch. H. Meiser, M. Garguillo, Textile Res. J., **46**, 531 (1976).
152. Y. L. Hsieh, J. R. Hardin, Textile Res. J., **54**, 171 (1984).
153. Y. Sekiguchi, F. Shafizadeh, J. Appl. Polym. Sci., **29**, 1267 (1984).
154. F. Shafizadeh, Y. Sekiguchi, Carbon, **21**, 511 (1983).
155. U. Einsele, H. Koch, H. Herlinger, Melliand Textilber, **65**, 200 (1984).
156. M. M. Birky, K. H. Yeh, J. Appl Polym. Sci., **17**, 239 (1973).
157. B. V. Hettich, Textile Res. J., **84**, 389 (1984).
158. K. N. Yeh, M. M. Birky, C. Huggett, J. Appl. Polym. Sci., **17**, 255 (1973).
159. M. A. Hammons, W. A. Reeves, Textile Chem. Colourists, **14**, 26/210 (1982).
160. "International Critical Tables", vol. II, p. 237.
161. F. C. Magne, H. J. Portas, H. Wakeham, J. Am. Chem. Soc., **69**, 1986 (1947).
162. H. Sommer, F. Winkler, in: H. Sommer (Ed.), "Handbuch der Werkstoffprüfung", vol. V, Springer, Berlin, 1960.
163. E. Calvet, P. H. Hermans, J. Polym. Sci., **6**, 33 (1951).
164. H. A. Stuart, " Die Physik der Hochpolymeren", vol. II., Springer, Berlin, 1955.
165. K. Lauer, Kolloid-Z., **121**, 139 (1951).
166. R. S. Jessup, E. I. Proser, J. Res. Natl. Bur. Std., **44**, 385 (1950).
167. S. M. Lipatov, D. V. Zharkovskii, J. M. Zagraevskaya, Kolloidn. Zh., **21**, 526 (1959).
168. N. V. Mikhailov, E. Z. Fainberg, Dokl. Akad. Nauk. SSSR, **109**, 1160 (1956); J. Polym. Sci., **30**, 259 (1958).
169. J. C. Guthrie, J. Textile Inst., **40**, T489 (1949).
170. S. M. Neale, W. A. Stringfellow, Trans. Faraday Soc., **37**, 525 (1941).
171. W. H. Rees, J. Textile Inst., **39**, T351 (1948).
172. J. B. Tailor, J. Textile Inst., **43**, T489 (1952).
173. M. Gralén, Thesis, Uppsala (1944).
174. D. K. Ashpole, Nature, **169**, 37 (1952).
175. D. H. Argue, O. Maass, Can. J. Res. B, **12**, 564 (1935).
176. S. R. Sivaraja Iyer, N. T. Baddi, Cell. Chem. Techn., **3**, 561 (1969).
177. E. Balcerzyk, K. Hempel, Przegl. Papiern., **20**, 309 (1964).
178. W. Morrow, Tappi, **42**, 167 (1959).
179. M. M. Chilikin, Zh. Prikl. Khim., **3**, 221 (1930); W. B. Champbell, Ind. Eng. Chem., **26**, 218 (1934); W. G. MacMillan, R. R. Mukherjee, M. K. Sen, J. Textile Inst., **37**, T 13 (1946); S. M. Lipatov, D. V. Zharkovskii, J. M. Zagraevskaya, Kolloidn. Zh., **21**, 526 (1959); K. P. Mishchenko, S. L. Talmud, V. L. Yakimova, Vysokomol. Soedin., **1**, 662 (1959); D. V. Zharkovskii, Dokl. Akad. Nauk. Belorussk. SSR, **3**, 429 (1959); through Chem. Zentr. (1961), 17966; Kh. U. Usmanov, I. Kh. Khakimov, Uzbeksk. Khim. Zh., 1959 (5), 30; Kh. Usmanov, A. A. Yul'chibaev, Sh. Nadzhimutdinov, Vysokomol. Soedin. **3**, 1217 (1961).
180. A. V. Dumanskii, E. F. Nekryach, Kolloidn. Zh., **17**, 171 (1955).

181. K. Lauer, R. Doederlein, C. Jackel, O. Wilde, J. Makromol. Chem., **1**, 76 (1943).
182. V. Marćokova, M. Pašteka, Faserforsch. Textiltech., **19**, 164 (1968).
183. H. Ruck, Papier, **23**, 872 (1969).
184. J. L. Morrison, W. R. Chambell, O. Maass, Can. J. Res. B, **18**, 168 (1940).
185. M. Wabba, J. Colloid. Chem., **52**, 1197 (1948).
186. E. F. Nekryach, F. Semchenko, Ukr. Khim. Zh., **26**, 700 (1960).
187. M. Wabba, J. Phys. Colloid. Chem., **54**, 1148 (1950).
188. Ya. V. Pak, Kh. U. Usmanov, Kolloidn. Zh., **18**, 233 (1956).
189. J. Chabert, Bull. Inst. Textile France, **20**, 335 (1966).
190. L. Kuniak, B. Alince, Papier, **21**, 248 (1967).
191. T. Barratt, J. W. Lewis, J. Textile Inst., **13**, T113 (1922).
192. N. E. Karasev, N. P. Dymarchuk, K. P. Mishchenko, Zh. Prikl. Khim., **39**, 2301 (1966).
193. I. Okamura, Naturwissenschaften, **21**, 393 (1923).
194. K. Schwabe, U. Trobisch, Zellstoff Papier, **5**, 55 (1956).
195. L. Waengberg, E. Treiber, loc. cit. I. Czajlik, E. Treiber, Holzforschung, **23**, 133 (1969).
196. V. Jancarik, L. Kuniak, Faserforsch. Textiltech., **20**, 491 (1969).
197. N. E. Karasev, N. P. Dymarchuk, K. P. Mishchenko, J. Appl. Chem. USSR, **40 (5)**, 1032 (1967).
198. E. Talpay, R. Talpay, "Der Baumwolltest", 2 ed. Bachmann, Bremen.
199. S. T. Han, T. Ulmanen, Tappi, **41**, 185 (1958).
200. T. Terada, N. Ito, Y. Goto, KAMI Pa Gikyoshi, **23**, 191 (1969).
201. K. Terasaki, K. Matsuura, Japan Tappi, **26**, 511 (1972).
202. E. Treiber, Mittlg. Fachausschuesse, **20**, 15 (1972).
203. F. Endres, Chem. Ing. Techn., **23**, 265 (1959).
204. A. J. Barry, F. C. Peterson, A. King, J. Am. Chem. Soc., **58**, 333 (1936).
205. E. L. Back, E. I. E. Didriksson, Svensk. Papperstidn., **72**, 687 (1969).
206. D. A. I. Goring, Pulp Paper Mag. Can., **64**, T517 (1963); V. A. Kargin, P. V. Koslov, Want Nai-Chang, Dokl. Akad. Nauk SSSR, **130**, 356 (1960).
207. M. Wabba, S. Nasbed, K. Axix, J. Textile Inst., **49**, T519 (1958).
208. J. Kubat, C. Pattyranie, Nature, **215**, 390 (1967).
209. A. Horbach, Cellulose Chem. Technol., **22**, 561 (1988).
210. A. Frey-Wyssling, Helv. Chem. Acta, **19**, 911, 981 (1936).
211. K. Kanamaru, Helv. Chem. Acta, **17**, 1037, 1425 (1934).
212. J. M. Preston, Trans. Faraday Soc., **29**, 65 (1933).
213. A. Atsuki, S. Okajima, J. Soc. Chem. Ind. (Japan), **40**, 360B (1937).
214. A. Frey-Wyssling, Kolloidchem. Beihefte, **23**, 40 (1927).
215. K. R. Krishna Iyer, P. Neelakanthan, T. Radhakrishnan, J. Polym. Sci., A-2, **7**, 983 (1969).
216. A. Herzog, Textile Forsch., **4**, 58 (1922).
217. A. Frey-Wyssling, Jahrb. Wiss. Botanik, **65**, 201 (1926).
218. A. Herzog "Die Unterscheidung von natürlichen und künstlichen Seiden", Th. Steinkopf, Dresden (1920), p. 59; A. Herzog "Die mikroskopische Untersuchung der Seide und Kunstseide", Springer, Berlin (1924), p. 66.
219. R. Meredith, J. Textile Inst., **37**, T205 (1946).
220. P. H. Hermans, A. Weidinger, J. Am. Chem. Soc., **68**, 198 (1946).
221. R. E. Bingham, Makromol. Chem., **77**, 139 (1964).
222. A. N. J. Heyn, Textile Res. J., **22**, 513 (1952).
223. E. Treiber, Lecture at 2nd Intern. Man-Made Fiber Conference, Dornbirn, July 15, 1963.
224. V. A. Berestnev, N. I. Grechushkina, M. B. Lytkina, V. A. Kargin, Khim. Volokna **3**, 45 (1962).
225. S. A. Novikova, R. S. Ivanova, Khim. Volokna, **5**, 32 (1970).
226. L. Rose, J. D. Griffiths, J. Textile Inst., **39**, 265 (1948).
227. N. Barakat, A. M. Hindeleh,Textilte Res. J., **34**, 581 (1964).
228. K. R. Krishna Iyer, S. Sreenivasan, N. B. Patil, Textile Res. J., **53**, 442 (1983).
229. G. T. Kohmann, Ind. Eng. Chem., **31**, 807 (1939).
230. E. J. Murphy, Can. J. Phys., **41**, 1022 (1963).
231. S. P. Hersh, D. J. Montgomery, Textile Res. J., **22**, 805 (1952).
232. B. E. Davidov, A. V. Kuchmenko, B. P. Morin, I. Yu. Tsarevskaya, Khim. Volokna, **14**, 71 (1972).
233. A. C. Walker, M. H. Quell, J. Textile Inst., **24**, T123 (1933).
234. J. W. S. Hearle, J. Textile Inst., **44** T117 (1953).
235. W. Claussnitzer, in: Landolt-Boernstein (Ed.), "Zahlenwerte und Funktionen", 6th ed., vol. IV (Technik), part 3, Springer, Berlin, 1957.
236. H. F. Chruch, J. Soc. Chem. Ind. (London), **66**, 221 (1947).
237. H. Keith, Frequenz, **3**, 165, 216 (1949).
238. H. Liander, E. Bjoerkman, Svensk. Paperstidn., **55**, 597 (1952).
239. A. Walraff, Elekrotech. Z., **63**, 539 (1942).
240. W. Trapp, L. Pungs, Holzforsch., **10**, 65 (1956).
241. J. W. S. Hearle, Textile Res. J., **24**, 307 (1954).
242. H. A. de Luca, W. B. Campbell, O. Maas, Can. J. Res. B, **16**, 273 (1938).
243. A. Wenkateswaran, J. Appl. Polym. Sci., **9**, 1127 (1965).
244. M. M. A. Moteleb, M. M. Naoum, H. G. Shinouda, H. A. Rizk, J. Polym. Sci., Polym. Chem. Ed., **20**, 765 (1982).
245. P. S. Mukherjee, Polymer/Polymer Commun., **25**, 382 (1984).
246. A. Campbell, Proc. Roy. Soc. (London) A, **78**, 196 (1907).
247. W. N. Stoops, J. Am. Chem. Soc., **56**, 1480 (1934).
248. K. Meyer, H. Mark, "Makromolekulare Chemie", Akad. Verlag, Leipzig, 1950.
249. H. Hoering, in E. v. Rzika "Starkstromtechnik", "Taschenbuch für Elektrotechniker." Part I, Berlin, 1951.
250. K. Meien "Elektrische Eigenschaften von Starkstromkondensatoren", SEV-Bull. 49, 1958.
251. G. Sarret, Thesis, University of Grenoble, France (1967).
252. M. Harris, A. Sookne, in: Matthews-Mauersberger (Eds)., "Textile Fibers", 5, Interscience, New York, 1947.
253. A. W. McKenzie, Appita, **22**, 82 (1968).
254. J. Melzer, Papier, **26**, 305 (1972).
255. G. Rabinov, E. Heymann, J. Phys. Chem., **47**, 655 (1943).

256. H. Jacobasch "Oberflächenchemie Faserbildender Polymere", Akademie Verlag, Berlin (1984).
257. M. V. Stackelberg. W. Kling, W. Benzel, F. Wilke, Kolloid-Z, **135**, 67 (1954).
258. F. Onabe, J. Appl. Polym. Sci., **22**, 3495 (1978); **23**, 2909, 2999 (1979).
259. H. J. Flath, H. Salek, Acta Polymer., **31**, 510 (1980).
260. G. Jacquelin, H. Bourlas, Tech. Rech. Papet, **3**, 49 (1964).
261. T. E. Khripunova, V. I. Yurev, S. S. Pozin, Probl. Fiz. Khim. Mekh. Volok., 473 (1967).
262. C. Heinegrad, private communication.
263. C. Heinegrad, E. Treiben, private communication.
264. A. Schausberger, J. Schruz, Das Papier, **33**, 148 (1979).
265. T. Nishino, K. Takano, K. Nakamae, J. Polym. Sci. B: Polym. Phys., **33**, 1647 (1995).
266. M. Matsuo, C. Sawatari, Y. Iwai, F. Ozaki, Macromolcules, **23**, 3266 (1990).
267. R. S. Govil, Man-Made Textiles, **41** (9), 45 (1964).
268. N. S. Wonding, L. W. S. Hearle, (Ed.) in: "Fiber Structure", Butterworth, London, 1963, p. 460, p. 466.
269. W. Wegner, H. Sommer (Ed.), in: "Handbuch der Werkstoffprüfung", vol. V, Springer, Berlin 1960, p. 394, p. 428.
270. W. E. Morton, J. W. S. Hearle, "Physical Properties of Textile Fibers", Butterworths, London, 1962, p. 288.
271. P. S. Mukherjee, K. G. Satyanarayana, J. Mat. Sci., **19**, 3925 (1984).
272. K. G. Satyanarayana, K. K. Ravikumar, K. Sukumaran, P. S. Mukherjee, S. G. K. Pillai, A. G. Kulkarni, J. Mat. Sci., **21**, 57 (1986).
273. H. Coulsey, S. Smith, "Formation and Structure of a New Cellulosic Fiber", 34th International Man-Made Fibers Conference, Dornbirn, Austria, September 20–22 (1995).
274. H. Schleicher, H.-P. Fink Lenzinger Berichte **74**, 5 (1994).
275. W. Berger, Lenzinger Berichte **74**, 11 (1994).
276. Ch. Yamane, M. Mori, M. Satio, K. Okajima, Polymer J., **28**, 1039 (1996).
277. Th. Heinze, T. Liebert, Cell. Chem. Technol. (1997), (in press).
278. B. Philipp, Polymer News, **15**, 170 (1990).
279. R. Willstaetter, I. Zechmeister, Chem. Ber., **46**, 2401 (1913); A. af Ekenstam, "Über die Celluloseösungen in Mineralsäuren", Lund (1936); L. P. Vyrodova, V. I. Sharkov, Zh. Prikl. Khim. **39** (3), 682 (1966).
280. A. af Ekenstam, Chem. Ber., **69**, p. 549, p. 553 (1936); R. Haller, Textilrundschau, **2**, 39 (1947).
281. D. M. MacDonald, ACS-Symposium Series, **58**, 25 (1977).
282. A. F. Turbak, R. B. Hammer, R. E. Davies, H. L. Hergert, Chemtech. Jan., **51** (1980).
283. S. M. Hudson, J. A. Cuculo, J Macromol. Sci. - Rev. Macromol. Chem. C, **18** (1), 1 (1980).
284. C. Trogus, Cellulos. Chem., **15**, 104 (1934).
285. A. af Ekenstam, Svenk. Papperstidn., **45**, 81 (1942).
286. M. Valdemar, R. Dunlap, Am. Chem. Soc. Meeting, Atlantic City NJ, 14-19. Sept, 1952.
287. A. K. Khripunov, et al., Vysokomol. Soed. Ser, B, **17**, (8), 600 (1975); Ref. from Chem. Abstr., **83**, 194208f.
288. A. K. Khripunov, Tezisy Dokl. Vses, Konk. Khim. Fiz. Tsyllyul, 1st, **2**, 183 (1975); Ref. from Chem. Abstr, **85**, 179236.
289. A. L. Geddes, J. Polym. Sci., **22**, 31 (1956).
290. D. L. Patel. R. D. Gilbert, J. Polym. Sci., Polym. Phys. Ed., **19**, 1231 (1981).
291. Th. Tailor, Brit. Patent, 787 (1859).
292. K. Letters, Kolloid-Z., **58**, 229 (1932); N. M. Bikales, L. Segal "Cellulose and Cellulose Derivatives", vol. V, Part IV, Wiley Interscience, New York, 1971.
293. Th. Taylor, Brit. Patent 787 (1859); (from Ref. 295).
294. Brit. Patent 1.6805 (1884), (from Ref. 295).
295. K. Letters, Kolloid-Z., **58** (2), 229 (1932).
296. H. Erbing, H. Geinitz , Kolloid-Z., **84**, 25 (1938).
297. H. Rath, H. Agster, Klepzigs Textil-Zeitschrift, **40**, 617 (1937).
298. F. Beck, DRP 353662, 22 May 1922.
299. P. P. von Weimarn, Kolloid-Z., **11**, 41 (1912).
300. W. Hönsch, Melliand-Textilber., **13**, 84 (1932).
301. S. J. Kuga, Colloid Interface Sci., **77**, 2, 413 (1980).
302. P. P. von Weimarn, DRP 275882, 30.6. 1914.
303. The Koppers Company, DRP 590326, 14. 12. 1933.
304. B. Lukanoff, H. Schleicher, B. Philipp, Cell. Chem. Technol., **17**, 593 (1983).
305. DWP (GDR) 156608, 08.09.82.
306. I. R. Katz. I. Seiberlich Paper Trade J., **110** (7), 37 (1940).
307. G. F. Davidson, J. Textile Inst., **25**, T174 (1934); **27**, T112 (1936); **28**, T27 (1937).
308. E. Treiber, B. Abrahamson, Papier., **13**, 253 (1959).
309. T. N. Kleinert, V. Moessmer, W. Wincor, Monatsh. Chem., **87**, 82 (1956); T. N. Kleinert, O. Dioszegi, R. M. Dlouhy, Svensk Papperstidn., **62**, 834 (1959).
310. A. Dehnert, W. Koenig, Cellulose Chem., **6**, 1 (1925).
311. Th. Lieser, E. Leckzyck. Ann. Chem., **522**, 56 (1936).
312. Th. Lieser, Ann. Chem., **528**, 276 (1937).
313. T. Brownsett, D. A. Clibbens, J. Textile Inst., **32**, T57 (1941).
314. C. T. Liang, R. H. Marchessault, J. Polym. Sci., **35**, 529 (1959); **37**, 385 (1959).
315. R. S. Stutt, US Patent 2, 371, 389.
316. E. Schweizer, J. Prakt. Chem., **72**, 109 (1857).
317. O. Ellefsen, Norelco Repr., **7**, 104 (1960).
318. W. Traube, Chem. Ber. **44**, 3319 (1911); *ibid.* **54**, 3220 (1921); *ibid.* **55**, 1899 (1922).
319. K. Hess, W. Weltzien, E. Messmer, Ann. Chem. **435**, 1 (1924); F. L. Straus, R. M. Levy, Paper Trade J., **114**, 23 (1942).
320. G. Jayme, F. Lang, Kolloid-Z., **150**, 5 (1957).
321. G. Jayme, Papier, **5**, 244 (1951).
322. N. Iso, D. Yamamoto, Bull. Chem. Soc. (Japan), **41**, 1064 (1968).
323. G. Jayme, N. M. Bikales, L. Segal, in: "Cellulose and Cellulose Derivatives", Wiley Interscience, New York, 1971.
324. G. Jayme, K. Neuschaeffer, Papier, **9**, 563 (1955).
325. G. Jayme, K. Neuschaeffer, Papier, **11**, 47 (1957).

326. G. Jayme, K. Neuschaeffer, Makromol. Chem., **23**, 71 (1957); Naturwissenschaften, **44**, 62 (1957).
327. D. Henley, Svensk Papperstidn., **63**, 143 (1960); A. Donetxhuber, Svensk Papperstidn **63**, 447 (1960); H. Reimers, Papier, **16**, 566 (1962).
328. G. Jayme, V. Verburg, Rayon, Zellwolle, Chemiefasern, **32**, 193, 257 (1954); L. Valtasaari, Paperi Puu, **39**, 243, 250, 252 (1957); G. Jayme, G. El-Kodski, Papier, **22**, 120 (1968); G. Jayme, Zellcheming, Mitt. Fachaussch, **19**, 17 (1971).
329. H. Staudinger, G. Daumiller, Ann. Chem., **529**, 219 (1937).
330. R. J. E. Cumberbirch, W. G. Harland, J. Textile Inst., **49**, T679 (1958); W. G. Harland, J. Textile Inst., **49**, T478 (1958).
331. N Gralén, J. Linderot, Svensk Papperstidn., **59**, 14 (1956).
332. W. Badgley, B. L. Frilette, H. Mark, Ind. Eng. Chem., **37**, 227 (1945).
333. S. Newman, L. Loeb, C. M. Conrad, J. Polym. Sci., **54**, 343 (1961).
334. M. Marx, Makromol. Chem., **16**, 157 (1955).
335. M. Marx, Papier, **10**, 135 (1956).
336. M. I. Arkhipov, Izv. Vysskykh Uchebn. Zavedenii Khim. I Khim. Tekhnol., **3**, 1109 (1960).
337. G. Meyerhoff, Fortschr. Hochpolym. Forsch., **3**, 59 (1961).
338. M. Marx-Figini, Papier, **16**, 551 (1962).
339. E. O. Kraemer, Ind. Eng. Chem., **30**, 1200 (1938).
340. R. J. C. Mitchie, Polymer, **2**, 446 (1961).
341. E. H. Immergut, J. Schurz, H. Mark, Monatsh. Chem., **84**, 219 (1953).
342. E. H. Immergut, B. G. Ranby, H. F. Mark, Ind. Eng. Chem., **45**, 2483 (1953).
343. G. Prati, Tinctoria, **59**, 233, 279, 329 (1962).
344. S. Newman, L. Loeb, C. M. Conrad, J. Polym. Sci., **10**, 463 (1953).
345. S. Claesson, W. Bergmann, G. Jayme, Svensk Papperstidn., **62**, 141 (1959).
346. M. Rinaudo, Papetrie, **90**, 479 (1968).
347. H. Vink, Arkiv Kemi, **14**, 195 (1959).
348. D. K. Smith, R. F. Bampton, W. J. Alexander, Ind. Eng. Chem. Process Design Div., **2**, 57 (1963).
349. H. Nadziakiewicz, H. Jedlinska, Polimery, **7**, 131 (1962).
350. H. Sihtola, B. Kyrklund, L. Laamanen, I. Palenius, Paperi Puu, **45**, 225 (1963).
351. G. Jayme, P. Kleppe, Papier, **15**, 6 (1961).
352. D. Henley, Arkiv Kemi, **18**, 327 (1961).
353. G. Jayme, P. Kleppe, Papier, **15**, 492 (1961).
354. W. Brown, R. Wikstroem, Eur. Polym. J., **1**, 1 (1965).
355. G. Jayme, J. Troeften, Melliand Textilber., **47**, 1432 (1966).
356. G. Jayme, G. El-Kodski, Papier, **34**, 410, 501 (1970).
357. W. Bergmann, Thesis, TH Darmstadt, 1958.
358. W. B. Achwal, T. V. Narayan, U. M. Paro, Tappi, **50** (6), 90A (1967).
359. L. Valtasaari, Tappi, **48**, 627 (1965).
360. L. Valtasaari, Makromol. Chem., **151**, 127 (1971).
361. C. L. McCormick, P. A. Callais, B. H. Hutchinson, Jr., Macromolecules, **18**, 2394 (1985).
362. V. P. Saxena, H. L. Bhatnagar, A. B. Biswas, J. Appl. Polym. Sci., **7**, 181 (1963).
363. H. Staudinger, G. Daumiller, Chem. Ber., **70**, 2508 (1937).
364. H. Krässig, E. Seifert, Makromol. Chem., **14**, 1 (1954).
365. H. Vink, Svensk Papperstidn., **70**, 734 (1967).
366. H. Vink, Makromol. Chem., **94**, 1 (1966).
367. B. Seger, W. Burchard, Macromol. Symp., **83**, 291 (1994).
368. J. Stamm, J. Am. Chem. Soc., **52**, 3047 (1930).
369. H. Vink, Arkiv Kemi, **14**, 29 (1957).
370. Ch. Legrand, Compt. Rend., **227**, 529 (1948).
371. K. Kamide, M. Saito, Adv. Polym. Sci., **83**, 1 (1987).
372. E. Bianchi, A. Ciferri, G. Conio, A. Cosani, M. Terbojevich, Macromolecules, **18**, 646 (1985).
373. T. Kawai, J. Polym. Sci., **37**, 181 (1959).
374. H. Chanzy, S. Nawrot, A. Peguy, P. Smith, J. Chevalier, J. Polymer Sci., Polym. Phys. Ed., **20**, 1909 (1982).
375. A. M. Bochek, L. M. Kalyuzhnaya, G. A. Petropavlovskii, S. Ya. Frenkel, Zh. Prikl. Khim. (S.-Petersburg), **65**, 2764 (1992).
376. M. M. Jovleva, V. N. Smirnova, Yu. Ya. Belousov, S. P. Papkov, Vysokomol. Soedin., A, **28**, 749 (1986).

Physical and Mechanical Properties of Some Important Polymers

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A. INTRODUCTION

This section of *Polymer Handbook* focuses on critical polymers in the commercial world. The polymers were chosen because they already have a large economic impact or are likely to have an economic impact during the lifetime of this Handbook. Hundreds of polymers are the subjects of study and application in the world. Many of these have commercial status. However, including all of them would compromise the utility of this chapter. Properties for all the polymers of this section plus many others are included with more details in various chapters of this Handbook.

Because of their cost and properties, polymers in this section often get first consideration for applications, or are the benchmarks against which alternative materials are compared. The data in the tables of this section can be used for rapid comparison against performance requirements.

Brief narrative descriptions of the polymers are given in the next section. More complete descriptions can be found in other places; the Handbook of Plastic Materials and Technology (I. I. Rubin (Ed.), Wiley Interscience, 1990) is an excellent resource.

A common set of properties are given in Section C. In many cases, the data are not available or are not relevant. In some cases, we have chosen to highlight a specific example of a family of polymers. However, in some cases, selecting a single member of a family did not represent the range of applications and properties available, so we included the ranges or subclassifications. Often the sources of data on polymer properties disagree. We have assumed that the disagreements were honest results of subtle differences in the polymer or the test. In some cases we report averages, in some cases we report the "best" value, and in other cases we report ranges. The reader is advised to refer to the specific product literature and to consult the technical service representatives of the suppliers. Standard or typical abbreviations for each polymer are also included.

B. NARRATIVE DESCRIPTIONS

1. Multipurpose Polymers

Poly(ethylene terephthalate) (PET) This polymer is usually made by a two-step condensation polymerization of the dimethyl ester of terephthalic acid and ethylene glycol. Articles made with this polymer can be either semi-crystalline or amorphous. Applications include fibers for carpeting and clothing, bottles (especially for soft drinks), and films for food packaging and magnetic tapes.

Poly(ethylene terephthalate, 1,4-cyclohexane dimethanol) (PETG) This polymer substitutes enough 1,4-cyclohexane dimethanol for ethylene glycol in the backbone (to prohibit crystallization). Applications include food packaging and injection molded parts.

Poly(vinyl chloride) (PVC) Applications for this polymer fall into two broad categories depending upon formulation. The name for this polymer is often freely abbreviated as "vinyl".

Flexible poly(vinyl chloride) This type typically contains 30–40 wt.% additives, especially plasticizers, to lower the glass-transition temperature. Applications include hoses, films, flooring, and automotive interiors.

Rigid poly(vinyl chloride) This glassy polymer is used for molding and profile extrusions with less additives. Typical applications include pipe, house siding, window frames, and bottles.

Ethylene–methacrylic acid copolymers, partially neutralized with sodium or zinc (Ionomer) The most important applications are for food packaging as films or sheets. They adhere to nylon in coextrusion, and they heat seal well.

Ethylene–acrylic acid copolymers (EAA) These adhesive copolymers typically have between 3 and 20 wt.% acrylic acid. They are often used in extrusion coating to provide a heat seal layer for packaging, especially when good flavor/aroma retention is needed.

Ethylene–vinyl acetate copolymers (EVA) The range of compositions can be from 2–50 wt.% vinyl acetate. Tough, clear films are made with copolymers having low levels of vinyl acetate (2–7 wt.%). Hot melt adhesives are an application for higher levels of vinyl acetate (18–40 wt.%).

Polytetrafluoroethylene (PTFE) This homopolymer of tetrafluoroethene has a wide range of valuable properties. Applications often rely on the very low surface energy (useful for non-stick coatings), the chemical resistance to many liquids (useful for gaskets and liners), and the wide range of service temperatures.

Ethylene–styrene interpolymers (ESI) These copolymers exhibit a very wide range of properties depending on composition. At low levels of styrene, the copolymers are crystalline, flexible, thermoplastics. At intermediate levels of styrene, the copolymers are amorphous and highly elastomeric. At high levels of styrene, the copolymers have a mixture of glassy and rubbery properties with a glass-transition temperature ranging to above room temperature. Applications vary as widely as the properties, including new areas for thermoplastics.

Poly(lactic acid) (PLA) This compostible polymer is synthesized from renewable resources. The properties vary according to the composition (D- and L- conformations). Applications include food packaging as film or molded cups.

2. Polyolefins and Barrier Polymers

Polyethylene family The total usage of all members of this family is the greatest of all thermoplastics. Most members of the family are easy to fabricate and have low cost. The properties vary widely across the family. Food packaging is a frequent application.

High density polyethylene (HDPE): This set of highly crystalline polymers is used in molding and film applications, with milk jugs being a prime example. These polymers are less flexible than other polyethylenes.

Low density polyethylene (LDPE): This set of branched polymers is less crystalline than HDPE. Members are flexible and clear, with moderate toughness.

Linear low density polyethylene (LLDPE): These are copolymers of ethylene with modest amounts of butene, hexene, or octene linear α-olefins. Grades are available for films and for molding. Films typically show good toughness. Some grades are suitable for fibers.

Very low density polyethylene or ultra-low density polyethylene (VLDPE or ULDPE): These are also copolymers of ethylene and linear α-olefins; however, they have densities less than 0.915 g/cm^3. They have improved toughness and sealability compared to LLDPE.

Polyolefin plastomer (POP): New catalyst technologies allow synthesis of these materials which include these with lower densities. Copolymers over a wide range of composition yield a wide range of properties for films, fibers, and molding applications.

Polypropylene (PP) Most applications use the isotactic form. Oriented (OPP) and unoriented films are used often for food packaging, sometimes in combination with vinylidene chloride copolymers. Polypropylene fibers are used for webbing in outdoor furniture, carpets, and both woven and non-woven fabrics. Polypropylene is also an important engineering thermoplastic.

Vinylidene chloride copolymers (PVDC) This set of polymers includes resins for extrusion and molding, resins for solution coating, and latexes for coating. A variety of comonomers are used to achieve different properties. The barrier properties are important for food packaging. Typically, these polymers are used with other polymers such as polyethylene or polypropylene in multi-layer structures.

Ethylene–vinyl alcohol copolymers (EVOH) These polymers are actually hydrolyzed copolymers of ethylene and vinyl acetate. They are excellent barriers to mass transport, especially when dry. They find applications in both flexible and rigid packaging for food, typically as part of a multi-layer structure.

3. Styrenics and Engineering Thermoplastics

Polystyrene family These inexpensive polymers span the range of engineering thermoplastics. They are easy to fabricate by either extrusion or molding. Typically, small amounts of additives are used, including extrusion aids, mold releases, FR agents, UV stabilizers, and plasticizers.

General purpose polystyrene (GPPS): This homopolymer is transparent and glossy. It quickly sets after molding to give stiff parts. Polystyrene can be made into foam articles such as insulating board or cups, using extrusion techniques or by expanding beads. Other applications include food packaging and molded parts in appliances.

High impact polystyrene (HIPS): This rubber-modified composite typically contains 1–10 wt.% rubber, generally polybutadiene, to give added toughness. It is also used in many applications including food packaging and molded parts.

Styrene–Acrylonitrile copolymers (SAN): A typical copolymer has 70 wt.% styrene and 30 wt.% acrylonitrile. These clear polymers have excellent dimensional stability and long-term toughness. Furthermore they are resistant to a wide range of solvents.

Rubber-modified styrene–acrylonitrile copolymers (ABS): This material has a continuous SAN phase with about 15 wt.% rubber modification, generally polybutadiene. In most applications, impact resistance is the reason for the selection of ABS. Applications include housings and covers for appliances and tools, plus a variety of automotive moldings.

Syndiotactic polystyrene (SPS): This semi-crystalline polymer can be used at higher temperatures. Applica-

tions include molded parts that must maintain close tolerances.

Polycarbonate (PC) This polymer is usually made in large scale by the condensation of phosgene and bisphenol A. It has great impact strength and excellent optical properties; hence, it is used for lenses, compact discs, glazing, instrument panels, and high-volume food services wares. It can be modified with reinforcing fibers, FR agents, and stabilizers. Development of new applications is intense.

Nylon family These are polyamides resulting from the condensation reaction of an amine function and an acid function. As a family, they are tough and hard. They are resistant to many liquids and have low coefficients of thermal expansion. They can be reinforced with glass fibers, carbon, and minerals. Applications include molded parts for electrical power transmission, molded parts for a wide range of automotive functions, pulleys, bearings and items which need good abrasion resistance and toughness.

Nylon 6: This homopolymer of caprolactam is in the middle of the range of nylons for toughness, hardness, and heat distortion.

Nylon 6,6: This copolymer of hexamethylene diamine and adipic acid has the greatest hardness and highest resistance to heat deformation and lowest toughness of the nylon family.

Poly(methyl methacrylate) (PMMA) This important member of the acrylate family is often freely abbreviated "acrylate". It has moderate toughness and excellent optical properties. It is hard. Hence, applications include glazing, fibers for transporting light, instrument panels, and packaging.

Polyoxymethylene (Acetal) Acetal resins are derived primarily from formaldehyde although copolymers are common. Endcapping is done to reduce the rate of degradation. These polymers are strong, tough, and stiff, and they have a low coefficient of friction against many materials. They are creep resistant and are extremely resistant to fatigue. Applications are often based upon dimensional stability and solvent resistance.

4. Elastomers

Poly(organosiloxanes) (silicone polymers) Most silicone polymers are either homopolymers or copolymers of dimethylsiloxane. They have high flexibility over a wide range of temperatures. They "repel" water strongly. Membranes are an important application for elastomeric grades. Other applications include stopcock grease, car polishes, and anti-stick formulations.

Chlorinated polyethylene (CPE) After a starting polyethylene has been chlorinated, the resulting material is typically 25–50 wt.% chlorine. The properties range from thermoplastic to elastomeric. Applications for CPE include automotive hoses, tubing, and wire jacketing plus extensive membranes for roofing. It can be an impact modifier for PVC.

Polyolefin elastomers (POE) These copolymers of ethylene and an α-olefin are made with single-site catalysts. They have very low densities. Materials can be designed for a wide range of applications.

Polyurethane elastomers (some TUP, thermoplastic polyurethane) These terpolymers can be used uncured for most applications; however, selected grades can be cross-linked to enhance properties. As a family these elastomers are tough and wear resistant. They have excellent chemical resistance and flexibility. They can have excellent clearity. Applications include cable jackets, Automotive fascia, connectors, fittings, and many, many more.

Ethylene-Propylene-Diene Terpolymers (EPDM) These terpolymers typically contain a few percent of a diene such as hexadiene. This material is easy to fabricate and then is usually heat-treated at 125°C to 175°C to induce crosslinking. The physical properties are much different after crosslinking. Commonly, they are heavily formulated for applications which include wire jacketing systems, automotive hoses, and roof sheeting.

C. PROPERTIES OF COMMERCIAL POLYMERS

TABLE 1. MULTIPURPOSE THERMOPLASTICS

Property	Unit	Test method	PET[a]	PETG	Rigid PVC	Flexible PVC[b]
MECHANICAL						
Density	g/cm^3	D792	1.4	1.27	1.39	1.18–1.70
Tensile strength	MPa	D638	Hoop-172 axial-69	50	55	5.5–26.2
Tensile modulus	MPa	D638	Hoop-4275 axial-2206	1724	2800	4.8–12.4
Flexural modulus	MPa	D638		2000		
Elongation to break	%	D638		180	40–80	150–450
Notched Izod at room temp.	J/m	D256		90		Varies
Hardness		D785		R105	Shore D 65–85	
THERMAL						
Deflection T @ 264 psi	°C	D648		63		
Deflection T @ 66 psi	°C	D648		71	66–77	
Vicat softening point	°C	D1525	79	180	63–82	Varies
UL temp. Index	°C	UL746B				Varies
UL flammability code rating		UL94		HB	V-0	Varies
Linear coefficient of thermal expansion	mm/mm/°C	D696		9.1×10^{-5}	$9–18 \times 10^{-6}$	$13–45 \times 10^{-6}$
ENVIRONMENTAL						
Water absorption 24 h	%	D570	<0.1	0.5	<0.5	Varies
Clarity	% Transmission	D1003	88	90	Transparent Opaque	Clear, translucent,
Outdoor weathering	%	D1435	Not suggested	Not suggested		Varies
FDA approval			Yes	Yes	Special formula available	Possible
CHEMICAL RESISTANCE TO:						
Weak acid		D543	Good	Fair	Not attacked	Not attacked
Strong acid		D543	Fair	Poor	Not to slightly attacked	Not attacked
Weak alkali		D543		Fair	Not attacked	Not attacked
Strong alkali		D543	Fair	Poor	Not to slightly attacked	Not attacked
Low molecular weight solvents		D543	Fair	Fair		Ketones, chlorinated hydrocarbons attack
Alcohols		D543	Good	Fair	Resists	Minimally attacked
ELECTRICAL						
Dielectric strength	kV/mm	D149	26	1/8″–15.7 1/16″–22.1	10–50	9.9–15.8
Dielectric constant		D150	3.3	3.2	32–40	3.3–4.5
Power factor		D150	0.03		0.007–0.020	0.04–0.14
OTHERS						
Melting point	°C	D3418	245–265	None	175	Varies
Glass transition temp.	°C	D3418	80	81	81	Varies
Special						

[a] Bottle grade.
[b] Range includes plasticizers and fillers.
[e] 9.5% acrylic acid, extrusion coating resin.
[f] High styrene content.

Polymer name					
Ethylene–methacrylic acid copolymer[c]	EVA[d]	EAA[e]	PTFE	ESI[f]	PLA[g]
0.94	0.93–0.957	0.938	2.2	0.950–1.050	1.25
33	6.2–28	17.6	21	12–41	53
			410	3–12	3450
220	6.9–110		410		
400	650–950	600	40	200–500	4.1
608	No break		No break	Shore A 60–100	16
Shore D 60	A: 73–98 D: 24–47		D55–D70		
			55.6		
44			121		137
73	36–82	76			
			VE-0		
$18–30 \times 10^{-5}$			9.9		
	0.1		<0.05		<0.5
6	Transparent to translucent		Opaque		
			Superb		
Yes	Yes	Yes	Yes		Yes
Not attacked	Not attacked		Not attacked	Good	
Slow attacked	Attacked by sulfuric acid		Not attacked	Good	Not resistant
Not attacked	Not attacked		Not attacked	Good	
Not attacked	Not attacked		Not attacked	Good	
Slight swell	Soluble		Not attacked	Varies	Varies
Slight swell	Not attacked		Not attacked	Good	
0.04			25.4		
2.4			2	2.2–2.6	
0.003			<0.0002		
93			327	0.001–0.008	178
			−100	−20 to +35	63
		Neck in is 6.6 cm at 25μm; minimum coating weight is 10 g/m^2	Surface energy at 20°C ~ 20 erg/cm^2	Has "Dead Fold"	Combustible; amorphous grades are transparent

[c] Ionomer; partially neutralized with sodium.
[d] Properties range for family.
[g] Sheet extrusion grade, low D isomer, semi-cyrstalline.

TABLE 2. POLYOLEFINS AND BARRIER POLYMERS

Property	Unit	Test method	HDPE[a]	LDPE[b]	LDPE[c]	LLDPE[d]
MECHANICAL						
Density	g/cm^3	D792	0.941–0.967	0.921	0.923	0.92
Tensile strength	MPa	D638	19–30	10.3	9.6	MD 56.5 CD 37.0
Tensile modulus	MPa	D638	800–1400	165.5	248	MD 183 CD 169
Flexural modulus	MPa	D638	700–1700			
Elongation to break	%	D638	10–1000	620	270	MD 600 CD 750
Notched izod at room temp.	J/m	D256	27–160	42.7 at −50°C	48 at −50°C	
Hardness		D785				
THERMAL						
Deflection T @ 264 psi	°C	D648				
Deflection T @ 66 psi	°C	D648	65–95	41		66
Vicat softening point	°C	D1525	120–130	91	93	106
UL temp. index	°C	UL7468				
UL flammability code rating		UL94				
Linear coefficient of thermal expansion	mm/mm/°C	D696	$100–200 \times 10^{-6}$	$\sim 250 \times 10^{-6}$	$\sim 250 \times 10^{-6}$	$160–200 \times 10^{-6}$
ENVIRONMENTAL						
Water absorption 24 h	%	D570	<0.01	<0.01	<0.01	
Clarity	%Transmission	D1003		Translucent		
Outdoor weathering	%	D1435				
FDA approval			Yes	Yes	Yes	Yes
CHEMICAL RESISTANCE TO:						
Weak acid		D543	Not attacked	Not attacked	Not attacked	Not attacked
Strong acid		D543	Minimally attacked	Minimally attacked	Minimally attacked	Minimally attacked
Weak alkali		D543	Not attacked	Not attacked	Not attacked	Not attacked
Strong alkali		D543	Not attacked	Not attacked	Not attacked	Not attacked
Low molecular weight solvents		D543	Minimally attacked	Minimally attacked	Minimally attacked	Not soluble
Alcohols		D543	Not attacked	Not attacked	Not attacked	Not soluble
ELECTRICAL						
Dielectric strength	KV/mm	D149	16–24			
Dielectric constant		D150	2.2–3.0	2.2	2.2	2.3
Power factor		D150	0.00005–0.003	0.0003	0.0003	<0.0005
OTHERS						
Melting point	°C	D3418	130–137			125
Glass transition temp.	°C	D3418	Not measurable	−35 inferred	−35 inferred	−35 inferred
Special					Minimum heat seal; temperature: 132°C	

[a] Properties of range.
[b] General purpose, for film & molding.
[e] For injection molding.
[f] Blown film grade, film properties reported.
[i] 38 mol% ethylene

Polymer name					
LLDPE[e]	VLDPE (or ULDPE)[f]	POP[g]	PP	PVDC[h]	EVOH[i]
0.917	0.912	0.896	0.903	1.7	1.17
15.9	MD 57.9 CD 49.6	MD 49 CD 43	35.5	34.5	60
221	MD 122 CD 135		1380	516.8	
283			1690		
850	MD 575 CD 700	MD 600 CD 710	Depends on specimen molding, history	25	240
40 at −50°C			37	37.4	
			100	60	
			55	49	
			101		
96		80	154	135	
			115		
			HB	V-0	
160–200 × 10^{-6}	~250 × 10^{-6}		90 × 10^{-6}	34 × 10^{-5}	
<0.01	<0.01	<0.01	<0.03	0.1	>1.0
				85	
				moderate discoloration	
Yes	Yes		21 CFR 177.1520 (c) 1.1	Yes	Yes
Not attacked	Not attacked	Not attacked	Excellent	Not attacked	
Minimally attacked	Minimally attacked	Minimally attacked	Varies with acid	Not attacked	
Not attacked	Not attacked	Not attacked	Excellent	Minimally to badly attacked	
Not attacked	Not attacked	Not attacked	Good	Minimally to badly attacked	
Not soluble	Minimally attacked	Minimally attacked	Nonpolar swells; polar excellent	Not attacked	
Not soluble	Not attacked	Not attacked	Excellent	Not attacked	
			24–28		
2.3	~2.2		2.25		
<0.0005	~0.0003		0.0003		
			164	160	175
−35 inferred	−38 inferred Seal initiation temperature: 102°C; Oxygen permeability at 23°C: 1900 nmol/m.s.GPa; Toughness (D882) −MD 97 J/cm^3 −CD 127 J/cm^3	−45 Oxygen permeability at 23°C: 3200 nmol/m.s. GPa	−20	−4 Oxygen permeability at 23°C: 0.2 nmol/m.s. GPa; WVTR (37.8°C, 90% RH): 0.06 g mil/100 in.^{2} day	62 (Dynamic) Oxygen permeability at 23°C 65% RH = 0.08 nmol/m.s. GPa; WVTR at 40°C 90%RH = 1.4 g.mil/100 in.^{2} day

[c] For extrusion coating.
[d] For blown film, mechanical properties for blown film with blow up ratio 2:3.
[g] Homogeneous catalyst, blown film grade, film properties reported.
[h] Extrusion grade.

TABLE 3. STYRENICS AND ENGINEERING THERMOPLASTICS

Property	Unit	Test method	GPPS[a]	HIPS[b]	SAN[c]	ABS[d]
MECHANICAL						
Density	g/cm^3	D792	1.04	1.04	1.08	1.04
Tensile strength	MPa	D638	46	24	82	40
Tensile modulus	MPa	D638	2890	1650	3860	2140
Flexural modulus	MPa	D638	3180	1910	4070	2580
Elongation to break	%	D638	1.5	52	3	25
Notched izod at room temp.	J/m	D256	93	112	27	72
Hardness		D785	M60–M84	M29	M80	R104
THERMAL						
Deflection T @ 264 psi	°C	D648	86	74	103	103
Deflection T @ 66 psi	°C	D648	95	87	108	~107
Vicat softening point	°C	D1525	107	101	111	111
UL temp index	°C	UL746B	50	50	60	~60
UL flammability code rating		UL94	NA to V-O	NA to V-O	HB	HB
Linear coefficient of thermal expansion	mm/mm/°C	D696	9.0×10^{-5}	9.0×10^{5}	6.7×10^{-5}	6.7×10^{-6}
ENVIRONMENTAL						
Water absorption 24 h	%	D570	0.05–0.07	0.05–0.07	0.05–0.40	0.20–0.45
Clarity	%Transmission	D1003			87–89	Opaque
Outdoor weathering	%	D1435			Yellows	Fair
FDA approval			Yes	Yes	No	Possible
CHEMICAL RESISTANCE TO:						
Weak acid		D543			Not attacked	Not attacked
Strong acid		D543			Minimally attacked	Minimally attacked
Weak alkali		D543			Not attacked	Not attacked
Strong alkali		D543			Minimally attacked	Minimally attacked
Low molecular weight solvents		D543	Attacked by Hydrocarbons, Aromatic Amines, Ketones, Aldehydes	Attacked by Hydrocarbons, Aromatic Amines, Ketones, Aldehydes	Minimally attacked	Minimally attacked
Alcohols		D543			Minimally attacked	Minimally attacked
ELECTRICAL						
Dielectric strength	kV/mm	D149	19.7	11.8–19.7	~14	16–31
Dielectric constant		D150	2.54	2.50–2.52	~2.8	~3.1
Power factor		D150	2×10^{-4}	6×10^{-4}	~0.009	0.007–0.015
OTHERS						
Melting point	°C	D3418				
Glass transition temp.	°C	D3418	74–110	93–105	115	~110
Special						Other grades include high impact, high gloss, and heat-resistant

[a] General purpose.
[b] Rubber modified, extrusion grade.
[e] 30% Glass-filled, ignition resistant.
[f] General purpose grade.
[i] General purpose grade.
[j] General purpose grade.

Polymer name					
SPS[e]	PC[f]	Nylon 6[g]	Nylon 6.6[h]	PMMA[i]	ACETAL[j]
1.44	1.2	1.13	1.14	1.17–1.20	1.41
95.1	72	80	90	55–76	61
		50	65		
9600	2480	3000	3400	2400–3100	2829
		1500	1700		
9600	2410	2414	2896	2800–3400	2588
		966	2206		
1.8	150	50–100	20	2–5	40–75
		200	80		
70	961	53	37	16–32	53–80
		96	64		
	R118/M74	M-86	M-90	M80–100	M78–M80
221	132	55–75	104	71–102	110
263	138	>160	>200	74–113	157
263	158			95–110	
					105
V-0	V-2	V-2	V-2	HB	94HB
30.5×10^{-6}	6.84×10^{-5}	81×10^{-6}	81×10^{-6}	$5–9 \times 10^{-5}$	8.5×10^{-5}
0.01	0.15	3 ± 0.4	2.8 ± 0.3	0.2–0.4	0.22
	87–91			91	Opaque
				Very slight yellowing at 3 years	
	Possible			Yes; for less than 8% alcohol	Yes
Good		Not resistant	Not resistant	Not attacked	Consult supplier
Good	Attacked	Not resistant	Not resistant	Minimally attacked to soluble	Badly attacked
Good	Attacked	Good resistance	Resistant	Not attacked	Not attacked
Good	Attaked	Good resistance	Resistant	Not attacked soluble to	Not attacked
Varies	Varies	Good resistance	Resistant	Not attacked	Not attacked
Good		Good resistance	Resistant	Not attacked to badly attacked	Not attacked
23.6	17	100	120	~20	20
		60	80		
3.1	2.93	3.5	3.2	2.8	3.7
		7.0	5.0		
0.001	0.005	0.023	0.025	0.02–0.03	0.006
		0.3	0.2		
270	(230)	220	255		165
	150			105	
	Index of refraction: 1.586				

[c] Injection molding grade.
[d] Solution, low gloss.
[g] For injection molding, heat stabilized.
[h] General purpose.

TABLE 4. ELASTOMERS

Property	Unit	Test method	Silicone[a]
MECHANICAL			
Density	g/cm^3	D792	0.97, greater when filled
Tensile strength	MPa	D638	1.0–5.5
Tensile modulus	MPa	D638	
Flexural modulus	MPa	D638	
Elongation to break	%	D638	200–850
Notched izod at room temp.	J/m	D256	
Hardness		D785	Shore A 20–60
THERMAL			
Deflection *T* @ 264 psi	°C	D648	
Deflection *T* @ 66 psi	°C	D648	
Vicat softening point	°C	D1525	
UL temp. index	°C	UL746B	
UL flammability code rating		UL94	
Linear coefficient thermal expansion	mm/mm/°C	D696	$10–19 \times 10^{-6}$
ENVIRONMENTAL			
Water absorption 24 h	%	D570	0.1
Clarity	% Transmission	D1003	
Outdoor weathering	%	D1435	
FDA approval			
CHEMICAL RESISTANCE TO:			
Weak acid		D543	
Strong acid		D543	Fair
Weak alkali		D543	
Strong alkali		D543	Fair
Low molecular weight solvents		D543	Varies
Alcohols		D543	
ELECTRICAL			
Dielectric strength	kV/mm	D149	16–22
Dielectric constant		D150	3.0–3.5
Power factor		D150	0.001–0.01
OTHERS			
Melting point	°C	D3418	
Glass transition temp.	°C	D3418	
Special			

[a] Flexible casting resins, e.g RTV.
[b] Range for elastomeric grades, 25–36 wt.% chlorine.
[c] Homogeneous catalyst, molding grade.
[d] Polyester Polycaprolactone Resin.
[e] Fully formulated.

Polymer name			
CPE[b]	POE[c]	TPu[d]	EPDM[e]
1.10–1.16	0.87	1.20	0.86, Polymer
21-Sep		42	6–16
At 200% elongation 1.2–11.7	18	8	1.2–5.6 at 50% elongation
	22	83	–
300–800	>1000	425	100
No break			–
Shore A>60	Shore A 72	Shore A 93A	Shore A 57–93
Flexible		Varies	
Flexible			
Flexible	41		
		Medium	Varies
	<0.01	0.3	
			Opaque
			Excellent
	Varies	Possible	Some
Not attacked	Not attacked	Stable	Excellent
Not attacked to badly attacked	Minimally attacked	Medium	Excellent, Varies
Not attacked to badly attacked	Not attacked	Stable	Excellent
Not attacked to badly attacked	Not attacked	Medium	Excellent, Varies
Not attacked to badly attacked	Minimally attacked	No effect to attacked	Fair, Varies
Not attacked to minimally attacked	Not attacked	No effect	Good
14.8–19.7			900
4.3–5.1			3.0–3.5
0.1			0.004–0.008
	60	Varies	Most Grades Amorpnous
–20	–55	–15	
		Taber Abrasion resistance 10 mg	

SECTION VI

SOLID STATE PROPERTIES

Crystallographic Data and Melting Points for Various Polymers

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A. INTRODUCTION

These tables contain data for approximately 2700 polymers. In general, the format and organization of the tables follows that used in the 2nd edition of the "Polymer Handbook" (Section B: Crystallographic Data; Section C: Melting Points; Section D: Formula Index and Section E: References). The Formula Index is included as an aid to the searcher: it permits one to determine the polymer name used in these tables from the structural line formula for the polymer of interest.

To avoid the duplication of data for polymers appearing in both the Crystallographic Data and Melting Point tables, the entry in the Melting Point table has been flagged with an asterisk (*).

1. Nomenclature

Contrary to the rest of the *Polymer Handbook,* the ACS-IUPAC nomenclature proposals (1,2) for naming regular single-strand organic polymers have not been adopted for these tables. Basically, the nomenclature used in this portion of the *Polymer Handbook* is that of the 1st and 2nd editions and of the 1967 recommendations of the IUPAC Nomenclature Committee (3). The majority of polymers were named as the bivalent radical(s) comprising the constitutional repeating unit (CRU), with priority in naming given to (a) incorporation of important structural features (amide, ester, oxide, etc.) and (b) emphasis on symmetry of the CRU (when present). As in the past, many common polymers (poly(styrene), poly(vinyl chloride), etc.) were named according to source names.

In the majority of cases, the polymer was named according to its structure, particularly when the CRU was composed of two or more subunits following each other in a regular fashion. An example is poly(ethylene terephthalate). With more complicated structures, the CRU was divided into its possible subunits, each of which was treated as a bivalent radical and so named. One of the subunits was treated as the parent compound, for example, the "terephthalic acid" of the above example. For poly(amides) and poly(esters), the acid was chosen as the parent compound; for poly(urethanes) it was the isocyanate.

Whenever the CRU (or a portion thereof) exhibited a center of symmetry, naming began with the central subunit using Chemical Abstracts/IUPAC names for the constituent bivalent radicals (see under examples). Conjunctive names were used as needed in constructing names. For symmetrical units, this process of naming could be continued indefinitely.

Unsymmetrical bivalent units were named with replacement nomenclature, or from end to end, depending upon the complexity of the bivalent radical and the ease of naming by the alternative choices. Replacement nomenclature was used, for example, to name poly(hexamethylene 6-oxaheptadecanediamide). The polymer from the symmetrical oxyacid would be named poly(hexamethylene oxydicaprylamide).

More complicated structures without symmetry were named end to end. This method was particularly appropriate for the poly(oxides) of the form $-(R-O-R'-O)_n-$ (and for the analogous sulfides, sulfones, ureas, and anhydrides) and for the poly(ω-amino acids) and for the poly (ω-hydroxy acids). The constitutional hydrocarbon subunits were ordered alphabetically in naming the polymer as, for example, in poly(methyleneoxy pentamethylene oxide).

For silicon-containing polymers, bivalent radical names replaced the source-based silane/siloxane names. For example, the source-named polymer, poly[(tetramethylsilphenylene) siloxane], appears in these tables as poly(*p*-phenylene tetramethyldisiloxanylene).

For the regular polymers of multipeptides, a cyclic permutation of the individual peptide residues in the constitutional base unit was performed to permit alphabetical ordering without loss of directional or sequential sense. Thus, poly(alanylglycyl-proline) is the name given to polymers prepared form (ala-gly-pro), (gly-pro-ala), and (pro-ala-gly), but not from (ala-pro-gly) or from either of its cyclic permutations.

2. Examples of Polymer Names

2.1. Polymer Names Based on Source

Olefin, Vinyl, and Acyclic Polymers

$[-CH_2-CH_2-]_n$	Poly(ethylene)
$\left[-CH(CH_2-CH_3)-CH_2-\right]_n$	Poly(1-butene)
$\left[-CH(CN)-CH_2-\right]_n$	Poly(acrylonitrile)
$\left[-CH(Cl)-CH_2-\right]_n$	Poly(vinyl chloride)
$\left[-CH(COOCH_3)-CH_2-\right]_n$	Poly(methyl acrylate)

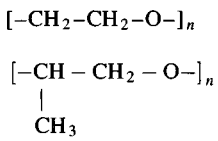
Poly(amides)[1]

$[-NH-CH_2-COO-]_n$	Poly(glycine)
$[-NH-(CH_2)_2-COO-]_n$	Poly(3-aminopropionic acid) (poly(β-alanine))
$[-NH-(CH_2)_3-COO-]_n$	Poly(4-aminobuyric acid)
$[-NH-(CH_2)_4-COO-]_n$	Poly(5-aminovaleric acid)
$[-NH-(CH_2)_5-COO-]_n$	Poly(6-aminocaproic acid)
$[-NH-(CH_2)_6-COO-]_n$	Poly(7-aminoenanthic acid)
$[-NH-(CH_2)_7-COO-]_n$	Poly(8-aminocaprylic acid)
$[-NH-(CH_2)_8-COO-]_n$	Poly(9-aminopelargonic acid)
$[-NH-(CH_2)_9-COO-]_n$	Poly(10-aminocapric acid)
$[-NH-(CH_2)_{10}-COO-]_n$	Poly(11-aminoundecanoic acid)
$[-NH-(CH_2)_{11}-COO-]_n$	Poly(12-aminolauric acid)

Poly(oxides)

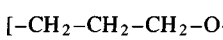
$[-CH_2-O-]_n$	Poly(oxymethylene)
$\left[-CH(CH_3)-O-\right]_n$	Poly(acetaldehyde)

[1] Poly(esters) are treated similarly.



Poly(ethylene oxide)

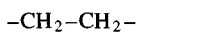
Poly(propylene oxide)



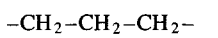
Poly(trimethylene oxide)

2.2. Polymer Names Based on Structure

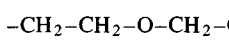
(a) Linking Radicals



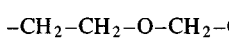
Ethylene



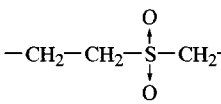
Trimethylene



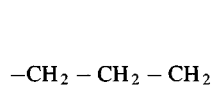
Oxydiethylene



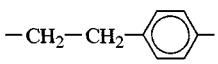
Ethylenedioxy-diethylene



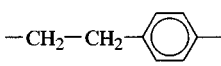
Sulfonyldiethylene



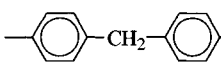
methylphosphinidene-ditrimethylene



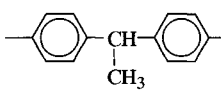
p-Phenylenediethylene



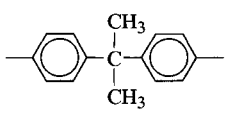
4,4′-Biphenylenediethylene



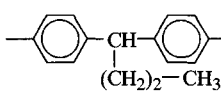
4,4′-Methylenediphenylene



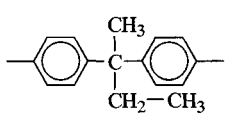
4,4′-Ethylidenediphenylene



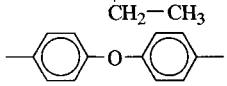
4,4′-Isopropylidenediphenylene



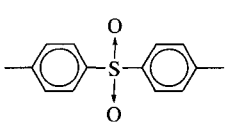
4,4′-Butylidenediphenylene



4,4′-(2,2-Butylidene)diphenylene

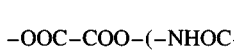


4,4′-Oxydiphenylene

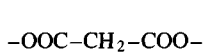


4,4′-Sulfonyldiphenylene

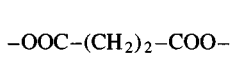
(b) Parent Compounds



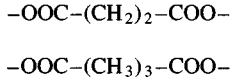
Oxalate (oxamide)



Malonate (malonamide)

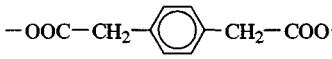
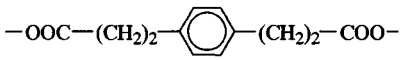
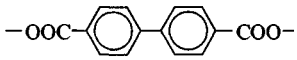
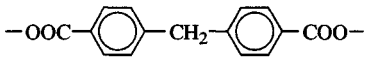
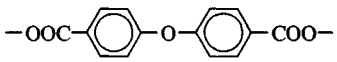
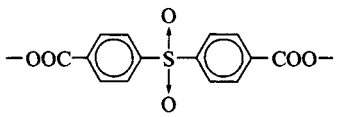
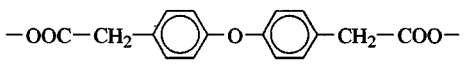
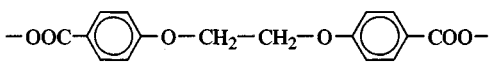
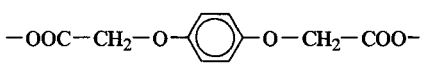
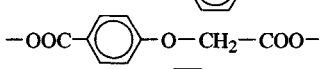


Succinate (succinamide)



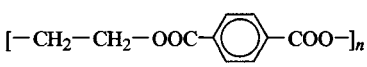
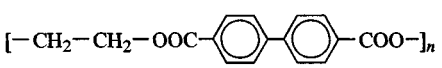
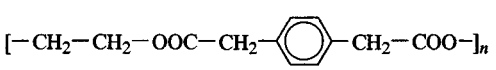
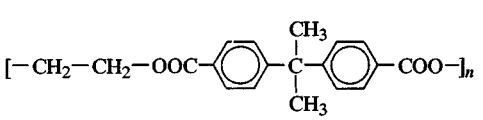
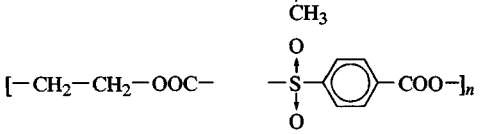
Glutarate (glutaramide)

References page VI-159

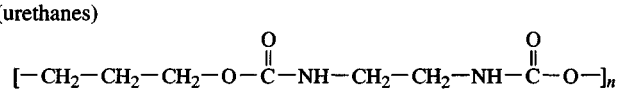
$-OOC-(CH_2)_4-COO-$	Adipate (adipamide)
$-OOC-(CH_2)_5-COO-$	Pimelate (pimelamide)
$-OOC-(CH_2)_6-COO-$	Suberate (suberamide)
$-OOC-(CH_2)_7-COO-$	Azelaate (azelaamide)
$-OOC-(CH_2)_8-COO-$	Sebacate (sebacamide)
$-OOC-(CH_2)_9-COO-$	Nonanedioate (nonanediamide)
$-OOC-CH_2-CH_2-O-CH_2-CH_2-COO-$	Oxydipropionate
	*p*-Phenylene diacetate
	*p*-Phenylenedipropionate
	4,4′-Dibenzoate
	4,4′-Methylenedibenzoate
	4,4′-Oxydibenzoate
	4,4′-Sulfonyldibenzoate
	4,4′-Oxydiphenylenediacetate
	4,4′-(Ethylenedioxy)dibenzoate
	*p*-(Phenylenedioxy)diacetate
	(*p*-Carboxyphenoxy)acetate
	1,4-Piperazinediacetate

(c) Polymer Names

Poly(esters)[2]

	Poly(ethylene terephthalate)
	Poly(ethylene 4,4′-dibenzoate)
	Poly(ethylene *p*-phenylenediacetate)
	Poly(ethylene 4,4′-isopropylidenedibenzoate)
	Poly(ethylene 4,4′-sulfonyldibenzoate)

Poly(urethanes)

	Poly(trimethylene ethylene-urethane)

[2] Poly(amides) are treated similarly.

Poly(oxides), Poly(sulfides), Poly(amines)

$[-(CH_2)_2-O-(CH_2)_3-]_n$	Poly(ethylene trimethylene oxide)
$[-(CH_2)_2-S-(CH_2)_3-]_n$	Poly(ethylene trimethylene sulfide)
$[-(CH_2)_2-NH-(CH_2)_3-]_n$	Poly(ethylene trimethylene amine)

REFERENCES FOR INTRODUCTION

1. ACS Structure-based Nomenclature for Linear Polymers, Macromolecules, **1**, 193 (1968).
2. IUPAC Appendices on Tentative Nomenclature, Symbols, Units, and Standards – Number 29, Nomenclature of Regular Single-strand Organic Polymers, November 1972 – Macromolecules, **6**, 149 (1973).
3. 1967 IUPAC Report on Nomenclature, J. Polym. Sci. B, **6**, 257 (1968).

B. CRYSTALLOGRAPHIC DATA FOR VARIOUS POLYMERS

The following table presents crystallographic data for about 750 polymers (and for their many polymorphs) grouped, according to the generic structure of the constitutional base unit, into poly(olefins), poly(vinyls) and poly(vinylidenes), poly(aromatics) and poly(imides), poly(dienes) and poly(diynes), poly(peptides), poly(amides), poly(esters), poly(urethanes) and poly(ureas), poly(ethers), poly(oxides), poly(sulfides) and poly(sulfones), poly(saccharides), and other polymers. Where a polymer might be included in more than one group the following group definitions were used:

1. *Poly(olefins):* those olefin polymers not containing aromatic rings; all substituted derivatives of these, excluding the vinyl and vinylidene polymers. For example, poly(ethylene), poly(tetrafluoroethylene), and poly(cyclopentene), but not poly(styrene) or poly(vinyl chloride).
2. *Poly(vinyls) and Poly(vinylidenes):* those vinyl and vinylidene polymers containing atoms other than carbon and hydrogen, excepting the poly(ethers). Thus, poly(vinyl chloride) but not poly(methylvinyl ether).
3. *Poly(aromatics) and Poly(imides):* those hydrocarbon polymers containing aromatic rings, including heterocyclic ones. Thus, poly(styrene), poly(*p*-xylylene), and poly(*p*-phenylene pyromellitimide).
4. *Poly(peptides):* those polymers whose basic structure, ignoring substituents, is poly(2-aminoacetic acid), that is, poly(glycine). This group contains mainly the synthetic poly(α-amino acids).
5. *Poly(amides):* all amide polymers except the poly(α-amino acids), which are separately grouped under poly(peptides).
6. The remaining categories contain polymers according to functional groups as, for example, the poly(esters). Poly(ethers) and poly(oxides) are differentiated according to whether the ether linkage is in the side-group or in the chain backbone, respectively. For example, poly(1-butoxy-2-chloroethylene) is a poly(ether) and poly(oxymethylene) is a poly(oxide).
7. Those polymers not otherwise categorized, such as the poly(phosphazenes) and the poly(siloxanes), are grouped in the "other" category.

Within each group, polymers are listed alphabetically according to the basic structure, ignoring substituents (excepting the polypeptides, all of which have the same base structure). Substituted polymers are listed alphabetically according to the substituent under the entry for the unsubstituted polymer. Thus, poly(tetrafluoroethylene) and poly(4-methyl-1-pentene) appear, respectively, under poly(ethylene) and poly(1-pentene).

Included as part of the polymer name column is the molecular weight of the chemical repeat unit of the chain, that is, of the constitutional base unit. This is given in parentheses just below the polymer name. The values reported are in grams per mole.

Unless otherwise indicated, the reference cited in the last column applies to all of the data in that line of the tables. Where an entry in a line has been taken from a source different from that listed in the reference column, a *solidus* (/) separates the value of the entry from the reference citation. For example, one value of the melting point of poly(1-butene) is given as "126/12" which is to be read as "126°C according to Ref. 12".

The crystal system, where known, is given in column 2 of the table, using the abbreviations given later. The next column contains the space group symbol (the Schoenflies notation is used because of the limitations of common printers at mainframe installations) with the subscript portion of the symbol preceding the superscript portion. Thus, according to Ref. 14, the poly(ethylene) unit cell is orthorhombic (Ortho) with space group D_{2h}^{16} (D2h-16). For a fuller discussion of space group symbols, see "International Tables for X-Ray Crystallography", Vol. I, Kynoch Press, Birmingham, England, 1952.

The dimensions of the unit cell (*a*, *b*, *c*) are given next, in Angstroms (1 Å = 0.1 nm). *c* is the fiber axis. Unless otherwise indicated, the unit cell angles are all 90°. Where only one angle differs from 90°, its value is given. When all three angles differ from 90°, their values are listed in the order α, β, γ. The next column contains the number of constitutional base units in the unit cell described.

Columns 9 and 10 contain the densities of the completely crystalline and of the completely amorphous polymer, respectively. The values given, normally appro-

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priate to room temperature, are in grams per cubic centimeter ($1\ g/cm^3 = 10^3\ kg/m^3$). For uniformity crystallographic densities (column 9) have been recalculated on the basis of the 1969 IUPAC atomic weight values. Melting points in degrees Celsius and heats of fusion in kilojoules per mole are listed in the next two columns. The heats of fusion correspond to the amount of substance represented by the molecular weight associated with each polymer name.

Column 13 of the tables indicate briefly the conformation of the polymer chain in the crystal (in a helical notation) either as reported in the reference cited or as inferred from the value of the fiber axis. The designation $n*p/q$ specifies the number (n) of skeletal atoms in the asymmetric unit of the chain and the number of such asymmetric units (p) per q turns of the helix in the crystallographic repeat. Thus, poly(ethylene), as listed, has two carbon atoms in the backbone with one such unit per turn in the repeat – it is designated as a $2*1/1$ helix. Alternatively, poly(ethylene) considered a poly(methylene) would be designated as a $1*2/1$ helix – an entirely equivalent description of the conformation. Note that n may differ from the number of chain atoms in the constitutional base unit. On the one hand, isotactic poly(propene) has two skeletal atoms in the asymmetric unit and three units per turn ($2*3/1$). On the other hand, syndiotactic poly(propene) with the same constitutional base unit as isotactic poly(propene) has four skeletal atoms in the asymmetric unit and two units per turn ($4*2/1$). In this case, n signifies the number of skeletal atoms in the stereobase unit. A fuller discussion of this helical notation is given in Hughes and Lauer, J. Chem. Phys., **30**, 1165 (1959) and in Nagai and Kobayashi, J. Chem. Phys., **36**, 1268 (1961).

This list of polymers and data is not considered to be exhaustive. The compiler expands it as new information is unearthed and he is quite receptive to notification of omissions and/or corrections and to submission of unpublished data. All of the data in this table cannot be considered to have the same validity – the number of polymers for which detailed, accurate, and reliable crystal structure analyses have been conducted is still quite limited. The reader may find it desirable to refer to the original literature cited so as to gain a fuller appreciation of the reasons for apparent discrepancies within the tables.

The following abbrevaitions have been used in the tables.

Tri	Triclinic
Mono	Monoclinic
Ortho	Orthorhombic
Tet	Tetragonal
Rho	Rhombohedral (trigonal)
Hex	Hexagonal
P	Pseudo
Phex	Pseudohexagonal

TABLE 1. POLY(OLEFINS) (–HCH–CHR–)

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm^3)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	b	c	Angles		Crystal.	Amorph.[a]				
Poly(acetylene)													
trans-													
I.	Phex		4.2	4.2	2.43		1	1.16				$2*1/1$	69
(26.04)	Mono		4.26	7.33	2.46	$\beta = 91.4$	2	1.126				$2*1/1$	982
	Ortho	D2h-16	7.32	4.24	2.46		2	1.33				$2*1/1$	981
	Ortho		7.37	4.065	2.45		2	1.178				$2*1/1$	979
	Ortho		5.62	4.92	2.592		2	1.207				$2*1/1$	980
	Mono	C2h-5	4.24	7.32	2.46	$\beta = 91.5$	2	1.133				$2*1/1$	1213
	Port	C2h-5	7.41	4.08	2.48		2	1.153				$2*1/1$	1228
	Ortho		3.99	7.29	2.51		2	1.184				$2*1/1$	1229
	Ortho		7.20	4.15	2.44		2	1.186				$2*1/1$	1281
	Ortho		7.31	4.06	2.45		2	1.189				$2*1/1$	1282
	Ortho		7.26	4.24	2.47		2	1.137				$2*1/1$	1283
			7.46	4.08	2.46		2	1.155				$2*1/1$	1323
			7.38	4.09	2.457		2	1.166				$2*1/1$	1324
	Ortho	D2h-16	7.330	4.090	2.457		2	1.174				$2*1/1$	1369
II.	Mono		3.73	3.73	2.44	$\gamma = 98$	1	1.286				$2*1/1$	980
cis-transoid													
	Ortho	D2h-16	7.61	4.39	4.47[b]		4	1.158				$2*2/1$	945
	Ortho		7.74	4.32	4.47		4	1.157				$2*2/1$	976
	Ortho	D2h-16	7.68	4.46	4.38		4	1.153				$2*2/1$	978
	Ortho		7.45	4.30	4.40[b]		4	1.227				$2*2/1$	1229
	Ortho		7.62	4.46	4.38		4	1.162				$2*2/1$	1282
			7.61	4.39	4.36[b]		4	1.187				$2*2/1$	1323
			7.66	4.42	4.384		4	1.165				$2*2/1$	1324
	Orhto	D2h-16	7.62	4.44	4.384		4	1.166				$2*2/1$	1369
cis-cisoid													
	Hex		5.12	5.12	4.84		3	1.180				$2*3/1$	977

TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, difluoro-													
trans-													
(62.02)	Ortho		7.58	4.29	2.48		2	2.554				2*1/1	1385
cis-													
	Ortho		8.03	4.47	4.64		4	2.473				2*2/1	1385
–, complex with:													
alkali metals													
Cs													
I.	Tet		9.09	9.09									1230
K													
I.	Tet		8.46	8.46									1230
II.	Ortho		5.94	16.37									1231
Na													
I.	Tet		8.5	8.5									1230
Rb													
I.	Tet		8.75	8.75									1230
II.	Ortho		6.16	16.68									1231
SbF_6													
	Mono		8.66	7.36	2.4	$\gamma = 101$							1261
and	Hex		9.8	9.8	5.4								1261
Poly(allene)													
I. (40.06)	Ortho	D2h-6	8.20	7.81	3.88		4	1.071		122/536		2*2/1	497
II.	Mono	C2-2 or C2h-2	6.37	5.12	3.88[b]	$\gamma = 96.6$	2	1.058				2*2/1	497
III.					3.88							2*2/1	497
–, tetrafluoro-													
(112.03)	Tet	C4-2 or D4-3	6.88	6.88	15.4		8	2.042		126		2*8/1	38
Poly(allylbenzene), see Poly(propene) 3-phenyl-													
Poly(1-butene)													
isotactic													
I.	Rho	D3d-6	17.7	17.7	6.50		18	0.951		126/12	13.9/82	2*3/1	35
(56.11)	Hex		17.3	17.3	6.7		18	0.96	0.87	132/250	6.07/277	2*3/1	45
	Rho		17.70	17.70	6.51		18	0.950		133	7.57	2*3/1	1284
								0.950	0.868	136/277	8.0/1285		252
								0.95		139	7.01		983
								0.95/12	0.860	135/320	6.92/320		391
											138/380		
											142/984		
											142/313		
II.	Tet	S4-1	15.42	15.42	21.05		44	0.819		130/380	4.06/320	2*11/3	866
	Tet	S4-1	14.89	14.89	20.87		44	0.886		124/277	6.28/277	2*11/3	207
	Tet		14.85	18.85	20.6		44	0.902		126/313	6.9/1285	2*11/3	350
	Tet		14.91	14.91	20.76		44	0.888		120	8.16	2*11/3	1284
	Tet	D2d-7	14.5	14.5	21.1		44			122/320		*40//11	914
									0.868				1195
III.	Ortho		12.49	8.96	(7.6)		8	(0.876)		106/277	6.49/277	2*4/1	207
	Ortho	D2-4	12.38	8.88	7.56		8	0.897		110/984		2*4/1	876
			12.38	8.92	7.45		8	0.906				2*4/1	531
					7.6								461
syndiotactic													
I.	Ortho	D2-5	16.9	6.05	7.78		8	0.937		45		4*2/1	1435
	Ortho	D2-5	16.81	6.06	7.73		8	0.946				4*2/1	1449
II.	Mono	C2-2	9.94	13.9	20.4		22	0.964?				4*5/3	1435
					20.0								1449
–, 3-methyl-													
(70.14)	Mono		9.55	8.54	6.84	$\gamma = 116.5$	4	0.933		300/48	17.3/82	2*4/1	9
	Mono	C2h-5	9.55	17.08	6.84	$\gamma = 116.5$	8	0.933		306/632		2*4/1	549
	Mono		19.1	17.8	6.85	$\gamma = 116$	16	0.890		310/90		2*4/1	231
	Port		19.25	17.20	6.85	$\gamma = 116.5$	16	0.918		300/282		2*4/1	355
	Tet	C4v-12	34.3	34.3	6.85		64	0.925				2*4/1	431
	Mono		19.25	17.20	6.63	$\gamma = 116.5$	16	0.948				2*4/1	280

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TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 3-methyl- [1,3-via hydride shift]													
(70.14)			5.4		7.8					55			332
										68/319			
–, 4-phenyl-													
(132.21)	Port	CS-2	10.4	18.0	6.61		6	1.064	0.962	158	4.6	2*3/1	483
					6.55					160/90		2*3/1	67
									1.041	159/282	4.36		1195
										168/187			
–, 4-trimethylsilyl-													
(128.29)	Ortho		10.97	20.02	6.46		6	0.901		206		2*3/1	772
Poly(2-butene-*alt*-ethylene)													
(84.16)	Mono	C2h-5	10.92	7.73	9.10	$\alpha = 130$	4	0.950	0.87	135		4*2/1	324
Poly(chlorotrifluoroethylene-*alt*-ethylene)													
(144.52)	Hex		9.86	9.86	5.02		3	1.703		262	18.8	4*1/1	635
					5					264	13.30		637
Poly(1,3-cyclobutylene)													
–, 1-cyano-													
(79.10)					25.3					370		3*8/	847
Poly(1,2-cycloheptylene-*alt*-ethylene)													
(124.23)					9.0					74		9*1/1	259
Poly(1,2-cyclopentylene-*alt*-ethylene)													
(96.17)	Ortho		8.76	7.83	9.02		4	1.032				4*2/1	247
	Ortho	D2h-17 or D2h-14	8.76	8.05	9.00		4	1.006		185		4*2/1	412
Poly(1,3-cyclopentylene-*alt*-methylene)													
(82.15)	Ortho		7.69	6.21	4.80		2	1.190		146		4*1/1	267
	Ortho		13.30	15.52	4.80		8	1.101		128		4*1/1	340
Poly(decenamer)													
trans-													
III. (138.25)	Tri	Ci-1	4.40	5.39	12.30	66,104,118	1	0.985				10*1/1	477
IV.	Mono	C2h-5	7.42	5.00	12.40	$\beta = 94.2$	2	1.001		80/562	32.9/562	10*1/1	530
Poly(1-decene)													
I. (140.27)					13.2					34/250		2*	150
										40/537			
Poly(1-docosene)													
I. (308.59)	Ortho		7.5	88	6.7		8	0.93		90		2*4/1	949
										91/872			
II.	Mono		5.3	108	7.7	$\beta = 94$	8	0.93		83		2*4/1	949
III.										75/949		2*	
Poly(dodecenamer)													
trans-													
III. (166.31)	Tri	Ci-1	4.40	5.39	14.78	66,104,118	1	0.986				12*1/1	477
IV.	Mono	C2h-5	7.43	5.00	14.85	$\beta = 93.5$	2	1.003		80/439	41.2/562	12*1/1	530
										84/562			
Poly(1-dodecene)													
I. (168.32)					13.2					45/48		2*	150
										49/872			
										49/250			
Poly(ethylene)													
I. (28.05)	Ortho	D2h-16	7.40	4.93	2.534		2	1.008		137/85	8.03/85	2*1/1	14
	Ortho		7.428	4.934	2.532		2	1.004		142/315	8.12/262	2*1/1	892
	Ortho		7.36	4.94	2.534		2	1.011		141/206	7.70/86	2*1/1	15
	Ortho		7.418	4.946	2.546		2	0.997		145/351	8.67/700	2*1/1	535
	Ortho		7.406	4.939	2.547		2	1.000		141/286	7.79/156	2*1/1	311
	Ortho	D2h-16	7.407	4.949	2.551		2	0.9961				2*1/1	1421
									0.855	146/362	7.62/343		138
								0.999	0.887		8.37/260		363
								0.991	0.811				406
								1.003	0.852	141	8.2		939
single crystal	Tri		7.84	5.56	120	63,71,82	96	1.004				2*1/1	72
II.	Pmono		4.05	4.85	2.54	$\gamma = 105$	1	0.966				2*1/1	58

TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Pmono		7.11	5.46	2.534	$\gamma = 96.5$	2	0.953				2*1/1	557
	Mono	C2h-3	8.09	4.79	2.53[b]	$\gamma = 107.9$	2	0.998				2*1/1	253
	Tri		4.285	4.820	2.54	90,110,108	1	1.002				2*1/1	254
III.	Ortho		8.46	4.88	2.45		2	0.920				2*1/1	792
(hi press.)	Tet		4.26	4.26	8.52		4	1.20				2*4/1	946
–, chlorotrifluoro-													
(116.47)	Hex		6.34	6.34	35		14	2.222	2.08/49	220/49	5.02/88	2*14/1	208
	Hex		6.5	6.5	35		14	2.11		210/46		2*14/1	21
	Hex		6.438	6.438	41.5	Approx.	17	2.18		222/341			777
	Hex		6.385	6.385	42		17	2.217				2*17/	296
	Hex		6.4	6.4	42.5		17	2.18				2*17/1	629
					43		16					2*16/1	205
								2.19/49	1.925				139
								2.192	2.032	215			116
–, ethylsilyl-													
isotactic	Hex	C3I-2	21.60	21.60	6.50		18	0.981				2*3/1	489
(86.21)													
–, tetrafluoro-													
II. (50.01)	Phex		5.54	5.54	16.8	$\gamma = 119.5$	13	2.406	2.0/1195	327/46	3.4/1195	1*13/1	11
	Tri		4.882	4.875	5.105	90,87,87	4	2.744				1*4/	266
	Tri		5.59	5.59	16.88	$\gamma' = 119.3$	13	2.347				1*13/6	209
	Mono	C2-2	5.59	9.76	16.88	$\beta = 90$	26	2.344				1*13/	602
			9.649	5.648	1.3**	γ	2	2.344				Irreg	987
	Phex		5.61	5.61		$\gamma = 119.3$						1*13/6	1286
	Tri		9.52	5.59	17.06	88,90,92	26	2.381				1*13/1	1321
I and IV.	Hex		5.61	5.61	16.8		13	2.358		330	2.87/91	1*13/1	11
(>20°C)	Hex		5.66	5.66	19.50		15	2.302		330/101	4.10/989	1*15/7	209
	Hex		5.65	5.65	19.54		15	2.306	2.060	346	4.64	1*15/	988
	Hex		5.67	5.67						332/89		1*15/7	1286
								2.304		334/1357			66
III.	Mono	C2h-3	9.50	5.05	2.62	$\gamma = 105.5$	4	2.742				1*2/1	681
(hi press.)	Ortho	D2h-16	8.73	5.69	2.62		4	2.552				1*2/1	857
	Ortho	D2h-16	8.53	5.72	2.62		4	2.598				1*2/1	986
–, trifluoro-													
(82.02)	Hex		5.59	5.59	2.50		1	2.013		222/726	5.4/726	2*1/1	389
	Hex		5.63	5.63	6.75		3	2.205		213/985	5.44/985	2*3/1	854
	Hex		5.61	5.61									1192
Poly(ethylene), complexes with:													
–, perhydrotriphenylene													
	Hex	C6-6 or C6h-2	14.34	14.34	4.78		2	1.071		181		4*1/1	1124
–, urea													
	Hex		8.22	8.22	11.02								992
(at 83 K)	Ortho		8.30	13.81	11.00								993
Poly(ethylidene)													
(28.05)	Ortho		12.38	6.28	2.5		4	0.958		100/617		1*2/1	178
										195/619			
										200/618			
Poly(heptenamer)													
trans-													
I.	Ortho	D2h-16	7.40	5.00	17.10		4	1.010		51/439		7*2/1	97
(96.17)		or C2v-9											
Poly(1-heptene)													
I. (98.19)					6.45					17/250		2*3/1	150
												2*3/1	359
–, 5-methyl-													
(112.22)					6.40					52/129		2*3/1	67
(S)	Mono	C2-2	18.40	10.62	6.36	$\beta = 90$	6	0.900				2*3/1	685
(S), (R)	Tet	S4-1	20.00	20.00	38.76		76	0.913				2*19/6	685
Poly(1-hexadecene)													
II. (224.43)	Ortho		7.5	63.2	6.6		8	0.95		68/250		2*4/1	150
Poly(hexafluoroisobutylene-*alt*-vinylidene fluoride)													
(228.08)	Ortho	C2v-21	10.64	18.37	7.83		8	1.980		327		4*2/1	947

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TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(1-hexene)													
(84.16)	Mono		22.2	8.89	13.7	$\gamma = 94.5$	14	0.726		−55/48		2*7/2	150
	Ortho		11.7	26.9	13.7		28	0.908				2*7/2	150
–, 4-methyl-													
(98.19)	Tet		19.64	19.64	14.00		28	0.845		200/9		2*7/2	67
	Tet	C1-1	19.85	19.85	13.50		28	0.858		190/632		2*7/1	569
–, 5-methyl-													
(98.19)	Hex		10.2	10.2	6.50		3	0.835		130		2*3/1	9
	Mono	C2-2	17.62	10.17	6.33	$\beta = 90$	6	0.862		110/282		2*3/1	685
Poly(isobutene)													
(56.11)	Ortho	D2-4	6.94	11.96	18.63		16	0.964	0.915	44/66	12.0/82	2*8/5	34
	Ortho	D2-4	6.88	11.91	18.60		16	0.978	0.912/53			2*8/3	801
									0.842				139
Poly(norbornene), see Poly(1-vinylene-3-methylene)													
Poly(1-octadecene)													
II. (252.49)	Ortho		7.5	70.4	6.6		8	0.96		80/6		2*4/1	150
										100/90			
III.	Hex		4.24	4.24								2*	493
Poly(octamethylene)													
–, 1-methyl-													
(126.24)										−5	10.9	8*	639
Poly(octenamer)													
trans-													
III.					9.78					67/439	20.1/284		126
(110.20)	Tri	Ci-1	4.34	5.41	9.78	64,105,119	1	1.008		73/284		8*1/1	476
									0.867	76	25.3		1377
IV.	Mono	C2h-5	7.43	5.00	9.90	$\beta = 95.2$	2	0.999		77/562	23.8/562	8*1/1	530
										62/439			
cis-	Mono	C2h-6	4.58	9.50	17.11	$\beta = 98$	4	0.993				8*2/1	905
										38	21.0		1376
Poly(1-octene)													
(at 90 K)	Mono		5.6	38	7.6	$\beta = 97$	8	0.93		5		2*4/1	1209
(112.22)										10– 38/48			
										10/872			
Poly(pentenamer)													
trans-													
I.	Ortho	C2v-9	7.28	4.97	11.90		4	1.051		23	12.0/682	5*2/1	147
(68.12)		or D2h-16								34/682			
Poly(1-pentene)													
isotactic													
Ia.	Mono		11.35	20.85	6.49	$\beta = 99.6$	12	0.923		130/276	6.28/501	2*3/1	355
(70.13)			21.15	11.20	6.49	$\beta = 99.6$	12	0.922		130/380		2*3/1	355
	Mono		22.4	21.2	6.49[b]	$\gamma = 91$	24	0.907		111/501		2*3/1	931
								0.96	0.85				408
Ib.					6.5							2*3/1	355
	Mono		24.3	19.3	6.50[b]	$\gamma = 96$	24	0.922				2*3/1	931
IIa.					6.60					75/48	4.00/501		152
	Mono		35.65	20.2	7.13[b]	$\gamma = 91$	40	0.907		78/282		2*4/1	818
	Port		19.30	16.90	7.08	$\gamma = 116$	16	0.898		80/9		2*4/1	355
										79/501			
IIb.	Port		19.60	16.75	7.08	$\gamma = 115.3$	16	0.887				2*4/1	355
III.	Ortho	D2-4	21.20	11.48	14.39		28	0.931				2*7/2	834
syndiotactic													
										42	2.94		1452
–, 5-amino-													
–, –, *N,N*-diisobutyl-													
(197.37)					6.85					112		2*	191
–, –, *N,N*-diisopropyl-													
(169.31)					7.32					130		2*	191
–, 4,4-dimethyl-													
(98.19)	Tet	S4-1	20.3	20.3	13.8		28	0.803		231/282		2*7/2	431
	Tet		20.35	20.35	7.01		16	0.899		350/90		2*4/1	442
										380/441			

TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 5-hydroxy-													
(86.13)					6.5							2*	191
–, 3-methyl-													
(84.16)	Tet	C4-6	13.35	13.35	6.80		8	0.922		362/632 273/129 200/441		2*4/1	622
–, 4-methyl-													
isotactic													
I.	Tet	S4-1	18.66	18.66	13.80		28	0.814		235/48	19.7/82	2*7/2	55
(84.16)	Tet		18.70	18.70	13.54		28	0.826	>0.835	228/282	11.9/370	2*7/2	936
	Tet		18.53	18.53	13.76		28	0.828		238/632	9.9/1195	2*7/2	894
	Tet		18.54	18.54	13.84		28	0.822		250/94	5.2/1288	2*7/2	452
	Tet		18.50	18.50	13.76		28	0.831			5.5/1331	Irreg.	333
	Tet		18.60	18.60	13.85		28	0.817				2*7/2	67
	Tet	S4-1	18.63	18.63	13.85		28	0.814				2*7/2	689
	Tet	D2d-7	18.70	18.70	13.68		28	0.818				2*7/2	1177
	Tet		18.66	18.66	13.65		28	0.823				2*7/2	1198
								0.828	0.838				94
II.			19.16	19.16	7.12		16	0.855		125		2*4/1	8
III.			19.36	19.36	7.05		16	0.846		75		2*4/1	8
	Tet		19.38	19.38	6.98		16	0.853				2*4/1	1197
	Tet	C4-6	19.46	19.46	7.022		16	0.8408				2*4/1	1453
IV.					6.5							2*	845
(hi press)	Hex		16.88	16.88	6.31		18	0.808				2*3/1	1454
V.	Hex		22.17	22.17	6.69		18	0.883?				2*3/1?	990
syndiotactic													
	Tet	S4-1	18.03	18.03	46.91		48	0.880		210		4*12/7	1455
–, 5-trimethylsilyl-													
(142.32)					6.55					133/71		2*3/1	67
Poly(propene)													
isotactic													
I.	Mono	C2h-6	6.65	20.96	6.50	$\beta = 99.3$	12	0.938	0.85/45	176/10	9.92/82	2*3/1	127
(42.08)	Mono		6.66	20.78	6.495	$\beta = 99.62$	12	0.946		189/392	7.96/478	2*3/1	308
	Tri	Ci-1	13.36	11.00	6.502[b]	93,81,108	12	0.935		186/394	8.8/394	2*3/1	170
	Mono	C2h-5	6.65	20.73	6.50	$\beta = 99.7$	12	0.947		208/676	5.80/676	2*3/1	519
	Mono		6.63	20.78	6.504	$\beta = 99.5$	12	0.949		165/282	10.5/474	2*3/1	626
	Mono		6.64	20.88	6.51	$\beta = 98.7$	12	0.940		178/349	10.0/358	2*3/1	131
	Mono	C2h-5	6.69	20.98	6.504	$\beta = 99.5$	12	0.931	0.854	183/380	6.16/182	2*3/1	136
	Mono	C2h-3								180/250	7.91/251	2*3/1	330
	Mono	C2h-5	6.65	20.80	6.50	$\beta = 99.33$	12	0.945		220/974 220/975 186/1003	10.9/83 6.9/1195 9.3/1212	2*3/1	991
II.	Hex		12.74	12.74	6.35		12	0.939	0.907/112	147/789	4.22/789	2*3/1	166
	Rho		19.08	19.08	6.49		27	0.922		200/975	8.2/1212	2*3/1	485
	Rho	D3-4	6.38	6.38	6.33		3	0.939		170/974	4.0/1404	2*3/1	196
	Ortho		19.08	11.01	6.490		18	0.922		192/1212		2*3/1	308
	Hex		22.03	22.03	6.490		36	0.922		183/1404		2*3/1	308
	Rho	C3-2	19.08	19.08	(6.49)		27	0.922				2*3/1	1456
	Ortho		11.03	19.08	6.49		18	0.921				2*3/1	1457
	Rho		11	11	6.5		9	0.92				3*3/1	1642
III.	Tri		6.47	10.71		$\gamma' = 99.07$	6					2*3/1	308
	Tri		6.54	21.40	6.50	89,100,99	12	0.946				2*3/1	592
	Ortho	D2h-24	8.54	9.93	42.41		48	0.935				cf ref	1408
quenched	Tet		5.97	5.97	6			0.88/167				2*3/1	943
syndiotactic													
I.	Ortho	D2-2	14.5	5.8	7.4		8	0.898	0.858/393	161/415	1.88/545	4*2/1	169
	Ortho	D2-5	14.50	5.60	7.40		8	0.930		138/545	2.1/1195	4*2/1	430
	Ortho	D2h-27	14.50	11.20	7.40		16	0.930			8.3/1458	4*2/1	1330
								0.898	0.858	160/1459	8.0/1459		409
II.	Phex				5.05							4*1/1	306
	Ortho	C2v-9	5.22	11.17	5.06		4	0.947				4*/1	1430
III.					11.8							4*	845

TABLE 1. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm^3)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	*b*	*c*	Angles		Crystal.	Amorph.[a]				
(hi press.)													
IV. (252.48)	Tri	C1-1	5.72	7.64	11.60	73,89,112	1	0.942				12*1/1	1460
–, 3-cyclohexyl-													
I. (124.23)	Tet	C4-5 or S4-2	21.06	21.06	20.09		40	0.926		230/90 215/282 214/328		2*10/3	431
II.	Rho	C4-3 or C3i-2	19.12	19.12	6.33		9	0.926				2*3/1	431
–, 3-cyclopentyl-													
(110.20)	Tet	C4-5 or S4-2	20.34	20.34	47.49		96	0.894		210/328 225/90		2*24/7	431
	Tet		20.30	20.30	47.40		96	0.899				2*24/7	625
–, hexafluoro-													
(150.02)										160		2*4/1	268
–, 3-phenyl- [allylbenzene]													
(118.18)					6.40					230/90 208/187 185/282		2*3/1	67
–, 3-silyl-													
(72.18)					6.45					128/71		2*3/1	67
–, trimethylsilyl-													
(114.26)					6.50					360/71		2*3/1	67
Poly(propene-alt-ethylene) syndiotactic													
(70.14)	Mono		11.19	11.62	9.00	$\gamma = 67.03$				55			1450
Poly(1-tetradecene)													
II. (196.38)	Ortho		7.5	56.0	6.6		8	0.94		57/250 57/872		2*4/1	150
Poly(tetrafluoroethylene-*alt*-ethylene)													
(128.07)	Mono		9.6	9.25	5.0	$\gamma = 96$	4	1.93				4*1/1	722
	Ortho		8.57	11.20	5.04		4	1.758	1.684	267		4*1/1	1287
			9.46	5.81									1321
Poly(vinylcyclobutane)													
(82.15)	Tet		34.12	34.12	6.6		64	1.14		228/437		2*4/1	431
Poly(vinylcycloheptane)													
(124.23)	Tet	C4h-6	23.4	23.4	6.5		16	0.927		>300/437		2*4/1	431
Poly(vinylcyclohexane)													
I. (110.20)	Tet	C4h-6	21.99	21.99	6.43		16	0.942		305/90		2*4/1	431
	Tet		21.76	21.76	6.50		16	0.951		300/282 372/441 383/328		2*4/1	67
II.	Tet	C4-5 or S4-2	20.48	20.48	44.58		96	0.940				2*24/7	431
Poly(vinylcyclopentane)													
I. (96.17)	Tet	C4h-6	20.14	20.14	6.50		16	0.969		270/441 292/328		2*4/7	431
II.	Tet	C4-5 or S4-2	20.14	20.14	19.5		48	0.969				2*12/3	431
III.	Tet	D4h-1 or D4h-7	37.3	37.3	19.8			0.927				2*10/3	431
Poly(vinylcyclopropane)													
I. (68.12)					6.5					230			175
	Rho	C3-2 or C3-3	13.6	13.6	6.48		9	0.981				2*3/1	526
II.	Tet	S4-2	15.21	15.21	20.85		40	0.938				2*10/3	526
Poly(1-vinylene-3-cyclopentylene) [norbornene]													
trans-													
(94.16)					11.8					202		5*2/	620
	Mono		5.13	4.78	11.56	$\gamma = 73.5$	2	1.150				5*2/1	1451
cis-	Mono		4.64	4.22	9.84	$\gamma = 68.1$	2	1.749				5*2/1	1451

[a] The numbers after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 2. POLY(VINYLS) AND POLY(VINYLIDENES) (–HCH–CHR– and –HCH–RCR–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(acrylamide)													
–, *N,N*-dibutyl-													
(183.30)	Hex		26.3	26.3	6.3		12	0.97 (1.06)				2*3/1	93
–, *N*-isopropyl-													
(113.16)								1.118	1.070	200		2*	102
Poly(acrylic acid)													
–, allyl ester													
(112.13)					6.5					90		2*3/1	278
–, biphenyloxyhexamethylene ester													
(324.42)	Phex		23.5	23.5	6.48		6	1.043				2*3/1	1122
–, *sec*-butyl ester													
(128.17)	Ortho		17.92	10.34	6.49		6	1.062		130		2*3/1	401
–, *tert*-butyl ester													
(128.17)					6.45					193/16		2*3/1	67
	Ortho		17.92	10.50	6.48		6	1.047		200		2*3/1	401
–, *p*-carboxyphenyl ester													
(192.17)	Mono		21.50	7.42	6.68[b]	$\beta = 124$	4	1.445				2*	865
–, isobutyl ester													
(128.17)	Ortho		17.92	17.92	6.42		12	1.239		81		2*3/1	401
–, isopropyl ester													
isotactic													
(114.14)					6.5					162/120	5.9/541	2*3/1	67
					6.32			1.08		162			149
										178/541			
syndiotactic													
					5.18			1.18		115		4*1/1	113
Poly(acrylonitrile)													
syndiotactic													
(53.06)	Hex		5.99	5.99						317/77	4.86/77–		76
	Ortho		21.18	11.60	(5.1)		16	1.125		341/499	5.23/77	4*1/1	486
	Ortho		10.6	11.6	5.04		8	1.137				4*1/1	575
	Ortho	C2v-16	10.20	6.10	5.10		4	1.111				4*1/1	133
	Ortho		18.1	6.12	5.00		8	1.273				4*1/1	162
	Ortho		10.55	5.80	5.08		4	1.134				4*1/1	210
	Phex		6.1	6.11	(5.1)	$\gamma = 120$	2	1.07				4*1/1	322
	Ortho		21.0	11.9	5.04		16	1.119				4*1/1	1001
	Ortho	C2v-9	10.7	12.1	(5.1)		8	1.07				4*1/1	1332
			9.03	5.24									1631
II?	Ortho		21.48	11.55	7.096		24	1.201				2*4/	804
			21.18	11.71									1631
isotactic													
	Tet		4.74	4.74	2.55		1	1.538				2*1/1	133
(gel)	Tet		11.4	11.4	8.2		16	1.32				2*4/1	1461
Poly(isopropenylmethyl ketone)													
(84.12)	Tet		15.08	15.08	8.54		16	1.151		240		2*4/1	347
										240/255			
										200/379			
Poly(methacrylic acid)													
–, methyl ester													
isotactic													
(100.12)	Port		21.08	12.17	10.55		20	1.228	1.22/31	160/31	5.02/871	2*5/2	30
	Ortho		21.08	12.17	10.50		20	1.234		220/855		2*5/1	520
	Ortho		20.98	12.06	10.40		20	1.264				vi ref	868
	Tri		20.57	12.49	10.56	93,88,91	20	1.228				*10/1	1002
	Ortho	D2h-24	41.96	24.34	10.50		80	1.240				*10/1	1462
syndiotactic									1.19/31	200/31		4*5/2	30
it-st stereocomplex (1 : 2)													
	Mono		21.5	16.5	18.4	$\gamma = 112$						*9/1	1397
					73.6					260/1006		*30/4	1005
–, –, chloroacetone complex													
syndiotactic													
	Mono		43.5	43.5	35.4	$\gamma = 107$						4*37/4	1004

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TABLE 2. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm^3)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	b	c	Angles		Crystal.	Amorph.[a]				
–, octadecyl ester syndiotactic													
(338.58)	Hex		4.73	4.73	30.2		1	0.961		36			500
Poly(methacrylontirile)													
I. (67.09)	Phex		9.03	9.03	6.87		4	0.918		250/436		2*	427
			7.87	8.97	6.87		4	0.918					729
II.	Mono		13.5	7.71	7.62	$\beta = 97.7$	8	1.134				2*4/1	427
Poly(thiolacrylic acid)													
–, *sec*-butyl ester													
(144.23)	Rho	C3-2			6.35							2*3/1	401
–, isobutyl ester													
(144.23)	Mono	Cs-2			6.42							2*	401
–, isopropyl ester													
(130.20)	Rho	C3-2			6.42							2*	401
–, propyl ester													
(130.20)	Mono	Cs-2			6.4							2*3/1	401
Poly(vinyl acetate)													
–, trifluoro-													
(140.06)	Ortho		9.54	12.44	4.8		4	1.633		175/821		2*	638
Poly(vinyl alcohol)													
(44.05)	Mono	C2h-2	7.81	5.51	2.52[b]	$\gamma = 91.7$	2	1.350	1.291/1	232/310	6.87/433	2*1/1	29
	Mono		7.81	5.43		$\gamma = 91.5$	2	1.34/410	1.27/410	243/480	6.99/480	2*1/1	316
	Mono		7.805	5.485	2.533[b]	$\gamma = 92.2$	2	1.350	1.26/53	265/323	7.1/1195	2*1/1	132
								1.345	1.269	228/433			184
	Mono	C2h-2	7.81	5.50	2.52[b]	$\gamma = 92$	2	1.352		267/450		2*1/1	261
	Mono		7.84	5.52	2.52[b]	$\gamma = 93$	2	1.343				2*1/1	1441
quenched	Ortho	C2v-20	7.42	5.25	2.52[b]		2	1.490				2*1/1	261
single crystal	Hex		5.45	5.45	2.51		1	1.133				2*1/1	151
I2 complex	Mono		7.87	5.53	2.525[b]	$\gamma = 93.3$	2					2*1/1	1424
Poly(vinyl bromide)													
(106.95)	Ortho	D2h-11	11.0	5.6	5.1		4	2.26		117		4*1/1	772
Poly(vinyl chloride)													
(62.50)	Ortho	D2h-11	10.6	5.4	5.1		4	1.42	1.41/413	273/372	11/372	4*1/1	7
	Ortho	D2h-11	10.40	5.30	5.10		4	1.477		310/423	4.9/835	4*1/1	235
	Ortho		10.11	5.27	5.12		4	1.522		212/143	2.8/145	4*1/1	64
	Mono		10.65	5.20	5.15[b]	$\gamma = 90$	4	1.456			3.3/423	4*1/1	140
single crystal	Ortho	D2h-11	10.24	5.24	5.08		4	1.523				4*1/1	718
Poly(vinyl fluoride)													
(46.04)	Hex		4.93	4.93	2.53		1	1.436		200/79	7.54/79	2*1/1	62
	Ortho	C2v-14	8.57	4.95	2.52		2	1.430		230/434		2*1/1	236
	Mono		4.938	4.938	2.518	$\gamma = 120$	1	1.4377				2*1/1	1463
Poly(vinyl formate)													
isotactic													
(72.06)	Rho	C3v-6-D3d-6	15.9	15.9	6.55		18	1.502				2*3/1	232
syndiotactic													
					5.0							4*1/1	232
Poly(vinylidene bromide)													
(185.85)	Mono		25.88	13.87	4.77[b]	$\gamma = 109.8$	16	3.065					33
Poly(vinylidene chloride)													
(96.94)	Mono	C2h-2	6.71	12.51	4.68[b]	$\gamma = 123$	4	1.954	1.66/43	200/721		4*1/1	915
	Mono		13.69	6.296	4.67[b]	$\gamma = 124.8$	4	1.948		190/46		4*1/1	32
	Mono		22.54	12.53	4.68[b]	$\gamma = 95.8$	16	1.958					33
	Mono	C2-2	6.73	12.54	4.68[b]	$\gamma = 123.6$	4	1.957		195	5.68		245
												2*2/1	231
									1.7754				462
Poly(vinylidene fluoride)													
α (II)	Mono		5.02	25.4	4.62	$\beta = 107$	10	1.888	1.68/776	185/434	6.7/823		171
(64.04)	Ortho	C2v-4	4.96	9.64	4.62		4	1.925	1.67/770	170/995	6.2/886	4*1/1	994
	Mono	C2-2	17.72	11.68	4.57[b]	$\gamma = 92.7$	16	1.801		220/443	5.96/726	4*1/1	417
	Ortho	C2v-9	9.66	4.96	4.64		4	1.913		171/373		2*2/1	418
	Mono	C2h-5	4.96	9.64	4.62	$\beta = 90$	4	1.925		178		4*1/1	604
	Mono	C2-2,1	9.64	5.02	4.64[b]	$\gamma = 91.1$	4	1.895		178/726		4*1/1	168

TABLE 2. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Port	C2-2	5.02	9.63	4.62		4	1.904		188/846		2*2/1	563
	Ortho		9.60	5.00	4.655		4	1.904		213/1464			597
			9.80	5.05			4						750
β (I)	Ortho	C2v-14	8.45	4.88	2.55		2	2.022		212/443		2*1/1	417
	Ortho	C2v-14	8.58	4.91	2.56		2	1.972		191		2*1/1	604
	Ortho	C2v-14	8.47	4.90	2.56		2	2.002		207/942		2*1/1	418
	Ortho		8.60	4.97	2.57		2	1.936					597
γ (III)	Mono	Cs-4	4.96	9.58	9.23	$\beta = 92.9$	8	1.942		218/846		8*1/1	996
	Mono	C2-3	8.66	4.93	2.58	$\beta = 97$	2	1.945		197		2*1/1	604
	Ortho	C2v-16	4.97	9.66	9.18		8	1.930		184/995		8*1/1	997
	Mono		4.96	9.67	9.20	$\beta = 93$	8	1.930		231/1464		8*1/1	998
δ (IV)			4.96	9.64								4*1/1	999
(polar alpha)	Ortho	C2v-9	4.96	9.64	4.62		4	1.925				4*1/1	1000
Poly(vinylmethyl ketone)													
(70.09)	Tet	S4-1	14.52	14.52	14.40		28	1.073				2*7/2	164
	Tet		14.56	14.56	14.10		28	1.090		170		2*7/2	379

[a] The numbers after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 3. POLY(AROMATICS) AND POLY(IMIDES)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(4,4′-benzobisthiazole-diphenylene pyromellitimide)													
(556.57)	Ortho	D2-3	5.70	8.11	24.8		2	1.612				22*1/1	1433
Poly(2,6-benzothiazole)													
(133.17)	Port		6.044	3.417	12.19		2	1.756				5*2/1	1235
Poly(2,5(6)-benzoxazole)													
(117.11)	Port		6.061	3.384	11.58		2	1.638				5*2/1	1235
Poly(4,4′-biphenylene 3,3′,4,4′-biphenylenetetracarboximide)													
(442.43)								1.480					1335
–, 2,2′-bis(trifluoromethyl)-													
(578.43)	Mono		15.40	9.90	20.25	$\gamma = 56$	4	1.501				18*1/1	1494
	Mono		15.40	9.918	20.21	$\gamma = 56.2$	4	1.4977				18*1/1	1495
–, 2,2′-dimethyl-													
(470.48)	Tri		20.49	15.30	40.0	62,52,80	16	1.443	1.341			18*2/1	1584
–, 3,3′-dimethyl-													
(470.48)	Ortho		16.7	13.0	20.5		8	1.404				18*1/1	1289
Poly(4,4′-biphenylene 4,4′-carbonyl(3,3′-dicarboxy)-dibenzamide)													
(506.47)					21.2							21*1/1	1178
Poly(4,4′-biphenylene 3,3′,4,4′-carbonyldiphenylenetetracarboximide)													
(470.44)					21.6							19*1/1	1178
Poly(4,4′-biphenylenedioxy-3,3′-biphenylene pyromellitimide)													
(550.53)	Ortho		7.89	6.29	25.11		2	1.467				23*1/1	1475
	Ortho		7.85	6.37	25.31		2	1.444		388		34*1/1	1498
Poly[2,2′-(4,4′-biphenylene) 6,6′-diquinoline]													
–, di-*p*-phenyl-													
(558.63)	Ortho		16.2	5.31	21.4		2	1.008				20*1/1	1500
Poly(4,4′-biphenylene 4,4′-oxy(3,3′-dicarboxy)-dibenzamide)													
(494.46)					21.0							21*1/1	1178
Poly(4,4′-biphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(458.43)					21.3					900/1018		19*1/1	1178
Poly(4,4′-biphenylene 1,4-pyromellitamide)													
(402.36)					16.1							16*1/1	1178
Poly(4,4′-biphenylene pyromellitimide)													
(366.33)					16.6								908
			8.58	5.48	16.6		2	1.559		827/1018		15*1/1	928
	Ortho	D2h-9	8.57	5.51	16.78		2	1.535				15*1/1	1476

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TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(2,2′-bithiophene)													
–, 3,3′didecyl-													
(444.78)	Ortho		17.6	4.5	15.6		2	1.20				6*2/1	1485
–, 3,3′-dihexyl-													
(332.56)	Ortho		11.8	4.5	15.6[b]		2	1.33				6*2/1	1485
–, 3,3′-dioctyl-													
(388.67)	Mono	C2h-4	14.6	4.42	14.73	$\beta = 90$	2	1.36				6*2/1	1484
	Ortho		14.5	4.5	15.6[b]		2	1.27				6*2/1	1485
	Ortho		14.6	4.42	15.70		2	1.27				6*2/1	1486
Poly(4,4′-carbonyldiphenylene pyromellitimide)													
(394.34)					31.0[b]								908
	Ortho		7.9	6.65	32.8		4	1.520				16*2/1	1013
Poly(decamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(446.50)					23.5							21*1/1	896
Poly(decamethylene pyromellitimide)													
(354.40)	Mono		19.8	4.61	19.2	$\gamma = 96$	4	1.350		318		17*1/1	1395
										323/1267			
Poly(dodecamethylene 1,4,5,8-naphthalenetetracarboximide)													
(432.52)	Tri		6.23	4.68	19.1	90,111,100	1	1.409		308		20*1/1	1015
Poly(dodecamethylene pyromellitimide)													
α	Mono		5.58	4.89	18.9	$\gamma = 108.4$	1	1.289				19*1/1	1395
(382.46)					18.96								1371
β	Mono		18.95	4.54	21.5	$\gamma = 101.3$	4	1.401		297		19*1/1	1015
					21.43					305/1267			1371
γ					22.05								1371
δ					20.95								1371
Poly(3,4′-ethylenedioxydiethylenedioxydiphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(606.59)										268	88.0		1493
Poly(3,4′-ethylenedioxydiphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(518.48)										340	72.5		1493
Poly(2,7-fluorenylene 1,4-pyromellitamide)													
(414.37)					15.9							15*1/1	1178
Poly(2,7-fluoenylene pyromellitimide)													
(378.34)					16.3							14*1/1	1178
Poly(hexamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(390.40)					19.5								896
	Tri		8.34	9.18	19.6	90,134,117	2	(1.462)		238		17*1/1	1015
Poly(hexamethylene pyromellitimide)													
(298.30)					24.88					430			1395
Poly[4,4′-(isophthaloyldiamino)-diphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide]													
(620.58)	Ortho		9.84	4.95	28.5		2	1.485		320		26*1/1	1190
Poly[4,4′-(isophthaloyldiamino)-diphenylene *p*-phenyleneditrimellitatediimide]													
(768.69)	Ortho		6.03	7.58	73.6		4	1.518				33*2/1	1191
Poly[4,4′-(isophthaloyldiamino)-diphenylene pyromellitimide]													
(528.48)					24.2							22*1/1	1191
										1.382			1335
Poly[isophthaloyl-4,4′-diphenylenedioxy-4,4′-diphenylene 3,3′,4,4′-carbonyldiphenylenetetracarboximide]													
(786.75)	Ortho		8.07	6.135	37.50		2	1.07		350		34*1/1	1498
Poly(4,4′-methylenediphenylene pyromellitimide)													
(380.36)					29.1								908
	Ortho		8.20	6.55	32.4		4	1.452				16*2/1	1013
					14.1								1014
Poly(methylene-*p*-phenylene)													
(90.12)	Ortho		8.16	6.32	9.90		4	1.172		170		5*2/1	745
										142/744			
–, α-methyl-													
(104.15)	Ortho		10.42	5.97	9.62		4	1.156		219/1007		5*2/1	848
Poly(methylene-3,5-toluene)													
–, 2,6-diamino-													
(134.18)	Ortho		12.82	13.83	9.69		8	1.037				4*2/1	1351
Poly(2,6-naphthalene 2,4-cyclobutylenedicarboxylic acid)													
–, bis(2,4-dichlorophenyl) ester													
(558.24)	Tri	Ci-1	6.00	8.93	13.48	70,94,101	1	1.396				9*1/1	950

TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(4,4′-neopentylenedioxydiphenylene 3,3′,4,4′-carbonyldiphenylenetetracarboximide)													
(572.57)	Mono		9.60	5.82	24.6	$\gamma = 81.1$	2	1.400				24*1/1	1499
Poly(nonamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(432.48)					21.3							20*1/1	1015
Poly(nonamethylene pyromellitimide)													
(340.38)					32.0					303			1395
										>350			1267
Poly(octamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)												19*1/1	
(418.45)	Tri		8.37	4.95	21.5	90,38,104	1	1.372	1.30				896
	Tri		8.37	4.95	21.5	90,38,104	1	1.396		202		19*1/1	1015
Poly(octamethylene pyromellitimide)													
(326.35)	Mono		20.8	4.76	16.9	$\gamma = 95.5$	4	1.301		362		15*1/1	1395
										383/1267			
Poly(3,4′-oxydiethylenedioxdiphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(562.34)										304	80.2		1493
Poly(3,4′-oxydiphenylene 3,3′,4,4′-biphenylenetetracarboximide)													
(458.43)	Mono		14.8	8.20	20.6	$\gamma = 56$	4	1.469				18*1/1	1289
Poly[4,4‴-((4′,4″-oxydiphenylene)-dioxy)-diphenylene *p*-phenylenditrimellitatediimide]													
(806.74)			5.68	8.09	78.6		4	1.484				37*2/1	963
Poly[4,4‴-((4′,4″-oxydiphenylene)-dioxy)-diphenylene pyromellitimide]													
(566.53)			8.44	5.66	53.4		4	1.475		547/1018		26*2/1	963
					53.4								908
Poly(4,4′-oxydiphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)													
(474.43)	Mono		5.17	10.85	38.2	$\gamma = 103.7$	4	1.514		427/1018		20*2/1	684
Poly[4,4′-oxydiphenylene 3,3′-(*p*-phenylene)-ditrimellitatediamide]													
(658.58)					26.3							26*2/1	1178
Poly(4,4′-oxydiphenylene *p*-phenylenditrimellitatediimide)													
A. (622.54)	Mono		5.56	7.94	58.0	$\gamma = 86$	4	1.619				27*2/1	658
			5.56	7.94	59.0	$\beta = 86$	4	1.591				27*2/1	963
					28.7								1178
									1.366				1335
B.			13.12	7.84	57.0		8	1.410				27*2/1	963
Poly(4,4′-oxydiphenylene 1,4-pyromellitamide)													
(418.36)					15.2							17*1/1	1178
									1.410				1335
Poly(3,4′-oxydiphenylene pyromellitimide)													
(382.33)	Ortho		8.48	5.62	33.65		4	1.583				15*2/1	1477
	Ortho	D2-9	5.75	8.48	33.2		4	1.569				15*2/1	1478
Poly(4,4′-oxydiphenylene pyromellitimide)													
I. (382.33)	Ortho		6.35	4.05	32.6		2	1.514		597/1018		16*2/1	642
	Ortho		6.31	3.97	32		2	1.58				16*2/1	684
	Ortho	C2-2	4.66	5.96	32.9[b]	$\gamma = 100$	2	1.411					877
	Ortho		8.36	5.63	33.03		4	1.633				16*2/1	1477
	Mono	C2-2	6.45	4.01	32.58	$\gamma = 90$	2	1.507				16*2/1	1478
					15.9								1178
II.	Ortho		5.93	4.70	33.00		2	1.380				16*2/1	1477
Poly(*m*-phenylene)													
(76.10)										790/964	6.90/964	3*	
Poly(*p*-phenylene)													
(76.10)	Ortho		7.81	5.53	4.20		2	1.393		<550/735	15.0/964	4*1/1	475
	Ortho		7.80	5.56	4.20		2	1.388		1400/964		4*1/1	1290
	Mono		8.06	5.55	4.30	$\beta = 100$	2	1.334				4*1/1	1334
	Ortho		7.82	5.60	4.20		2	1.374				4*1/1	1448
	Ortho	pgg	7.79	5.51	(4.20)		2	1.402				4*1/1	1466
												4*2/1	508
–, complex with:													
K (1/2 K/unit)	Ortho	D2h-9	6.04	15.6	4.35		4					4*1/1	1231
$SbCl_5$ (1/2 Sb/unit)	Ortho		15.6	6.2	4.2		2					4*1/1	1393
SbF_5 (1/3 Sb/unit)	Hex		12.0	12.0	4.2		3					4*1/1	1393

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TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(*p*-phenylene benzobisoxazole) [PBO]													
(234.21)	Mono		5.64	3.58	12.07	$\gamma = 100.4$	1	1.622				11*1/1	1263
	Mono		11.20	3.540	12.050	$\gamma = 101.3$	2	1.660				11*1/1	1488
	Mono	Cs-2	11.20	3.54	12.05	$\gamma = 101.3$	2	1.660				11*1/1	1489
	Port		11.8	3.54	12.1		2	1.55				11*1/1	1490
–, complex with poly(phosphoric acid)													
I.	Mono		12.6	11.6	12.2	$\gamma = 98$							1492
II.	Mono		7.0	5.8	12.0	$\gamma = 99$							1492
Poly(*p*-phenylene benzobisthiazole) [PBT]													
(266.34)	Mono		5.83	3.54	12.35	$\gamma = 96$	1	1.745				11*1/1	1016
	Mono		11.96	3.555	12.35	$\gamma = 100.9$	2	1.716				11*1/1	1017
	Mono		11.79	3.539	12.514	$\gamma = 94.0$	2	1.698				11*1/1	1488
–, complex with poly(phosphoric acid)													
	Ortho		15.9	20.6	36.9					250		11*3/1	1316
	Mono		8.15	24.49	37.2	$\gamma = 94.5$		2.15					1491
Poly(*p*-phenylenecyclobutylene)													
–, 2,4-dipyridyl-													
(284.36)	Ortho	D2h-15	19.16	10.69	7.45		4	1.238				7*1/1	202
	Ortho	D2h-15	18.9	10.5	7.53		4	1.26		340		7*1/1	723
Poly(*p*-phenylene 2,4-cyclobutylenedicarboxylic acid)													
–, diethyl ester													
(274.32)	Mono	C2h-5	8.62	9.98	8.16[b]	$\beta = 102$	2	1.327		347/723		7*1/1	893
–, dimethyl ester													
(246.26)	Tri	Ci-1	6.04	7.42	7.82[b]	101,106,72	1	1.284		415/723		7*1/1	893
–, diphenyl ester													
(370.40)	Mono	C2h-5	7.50	17.3	7.50[b]	$\beta = 102$	2	1.292		420/723		7*1/1	893
–, dipropyl ester													
–, –, 2,4-dicyano-													
(352.39)	Mono	C2h-2	6.19	20.1	7.58	$\beta = 96$	2	1.248		335/723		7*1/1	893
Poly[4,4‴-((4′,4″-(*p*-phenylenedioxy)-diphenylene)-dioxy)-diphenylene *p*-phenyleneditrimellitatediimide]													
(898.84)			5.66	8.14	43.5		2	1.489				42*1/1	963
Poly[4,4‴-((4′,4″-(*p*-phenylenedioxy)-diphenylene)-dioxy)-diphenylene pyromellitimide]													
(658.62)			8.44	5.66	31.3		2	1.463				31*1/1	963
					31.3								908
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene 4,4′-oxy(3,3′-dicarboxy)-dibenzamide]													
(602.56)					24.6							27*1/1	1178
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide]													
(566.52)	Tri		10.52	9.45	26.1	90,95,98	4	1.472				25*1/1	684
					26.0								1178
									1.341				1335
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene *p*-phenyleneditrimellitatediimide]													
(714.64)	Ortho		5.64	8.05	34.5		2	1.515				32*1/1	658
Poly[4,4′-(*m*-phenylenedioxy)-diphenylene 1,4-pyromellitamide]													
(510.46)					19.8							21*1/1	1178
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene 1,4-pyromellitamide]													
(510.46)					19.6							22*1/1	1178
Poly[4,4′-(*m*-phenylenedioxy)-diphenylene pyromellitimide]													
(474.43)			8.76	5.47	43.4		4	1.515				20*2/1	963
					43.4								908
					21.7								1178
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene pyromellitimide]													
(474.43)	Mono		5.64	8.30	21.8	$\beta = 98.7$	2	1.562				21*1/1	684
	Ortho		8.23	5.58	22.09		2	1.553				21*1/1	1477
					21.0								1178
Poly(*p*-phenylene-1,3,4-oxadizaole)													
α	Ortho	D2-4	12.35	6.55	14.0		8	1.69				7*2/1	1496
(144.13)													
Poly[*p*-phenylene 4,4′-oxy(3,3′-dicarboxy)-dibenzamide]													
(418.36)					16.6							17*1/1	1178
Poly(*p*-phenylene 3,3′, 4,4′-oxydiphenylenetetracarboximide)													
(382.33)					16.9							15*1/1	1178
Poly(*p*-phenylene 1,4-pyromellitamide)													
(326.26)					12.2							12*1/1	1178

TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(*m*-phenylene pyromellitimide)													
(290.23)	Ortho		7.72	7.18	21.5		4	1.618				10*2/1	1013
Poly(*p*-phenylene pyromellitimide)													
(290.23)					12.3								908
			8.58	5.48	12.3		2	1.667				11*1/1	928
Poly(*p*-phenylenevinylene)													
(102.14)	Mono		7.90	6.05	6.58	$\alpha = 123$	2	1.286				6*1/1	1236
			8.1	6.1	6.6	$\alpha = 123$	2	1.24				6*1/1	1465
–, complex with:													
AsF_5	Ortho		10.10	6.60	6.58		2					6*1/1	1399
ClO_4^-	Ortho		9.40	6.60	6.58		2					6*1/1	1399
$FeCl_3$	Ortho		10.80	6.70	6.6		2					6*1/1	1465
H_2SO_4	Ortho		10.45	6.60	6.58		2					6*1/1	1399
SbF_5	Ortho		10.32	6.60	6.58		2					6*1/1	1399
Poly(2,5-pyrazinecyclobutylene)													
–, 2,4-diphenyl-													
(284.36)	Ortho		10.8	18.4	7.4		4	1.28		321		7*2/1	558
	Ortho	D2h-15	18.36	10.88	7.52		4	1.257		343/723		7*2/1	564
	Ortho	C2v-5	18.36	10.88	7.52		4	1.257				7*2/1	950
	Ortho		18.40	10.89	73.45		4	1.265				7*2/1	1238
Poly(styrene)													
isotactic													
(104.15)	Rho	D3d-6	22.08	22.08	6.626		18	1.113	1.04–	240/10	9.00/89	2*3/1	6
	Rho	D3d-6	21.9	21.9	6.65		18	1.127	1.065/17	250/90	8.37/174	2*3/1	128
	Rho	D3d-6	22.08	22.08	6.594		18	1.118		242/1415	8.5/1415	2*3/1	1388
									1.024				139
								1.114	1.052	235/282	10/1121		411
gel					30.6							4*6/1	1008
	Ortho		21.0	16.4	30.6		48					4*6/1	1009
syndiotactic													
α	Mono	C2h-5	19.84	12.90	5.06	$\gamma = 95$	8	1.072		266/1372	8.6/1407	4*1/1	1418
	Hex	D3h-4	26.26	26.26	5.045		18	1.033		269/1407		4*1/1	1388
	Hex		26.3	26.3	5.1		18	1.02				4*1/1	1470
					5.06					270		4*1/1	1291
					5.08					286/1417		4*1/1	1370
β	Ortho	D2h-16	8.81	28.82	5.06		8	1.077				4*1/1	1418
γ					7.5							4*2/1	1401
					7.7							4*2/1	1419
–, complex with:													
CH_2Cl_2	Ortho	D2h-10	21.4	17.1	7.5		16	1.01				4*2/1	1370
					7.7							4*2/1	1419
C_7H_8	Mono	C2h-5	17.48	13.27	7.71	$\gamma = 122$	8	1.114				4*2/1	1436
	Mono	C2h-5	17.58	13.26	7.71	$\gamma = 121.2$	8	1.100				4*2/1	1474
I_2	Mono	C2h-5	17.29	12.85	7.79	$\gamma = 120.3$	8	2.054				4*2/1	1473
–, *p-tert*-butyl-													
–, –, isotactic													
(160.26)									0.950	300			198
										308/1372			
–, *o*-fluoro-													
–, –, isotactic													
(122.14)	Rho	C3v-6	22.15	22.15	6.63		18	1.296		270/75		2*3/1	123
–, *p*-fluoro-													
–, –, isotactic													
(122.14)					8.30					265/75		2*4/1	67
	Port	D2-4	17.6	12.1	8.25		8	0.924				2*4/1	791
–, α-methyl-													
–, –, isotactic													
(118.18)	Rho				6.6							2*3/1	357
–, –, (*p*-methyl-													
–, –, –, isotactic													
(132.21)										224/888	10.0/888		

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TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, *m*-methyl-													
–, –, isotactic													
(118.18)	Tet	S4-1	19.81	19.81	21.74		44	1.012		215		2*11/3	80
					57.1					215/74		2*29/8	163
	Tet	C4-5	19.9	19.9	78.9			1.005				2*40/	431
		or S4-2			57.0							11	103
–, *o*-methyl-													
–, –, isotactic													
(118.18)	Tet	C4v-12	19.01	19.01	8.10		16	1.073		360/74		2*4/1	125
–, –, *p*-fluoro-													
–, –, –, isotactic													
(136.17)					8.05					360/75		2*4/1	67
–, *p*-methyl-													
–, –, isotactic													
(118.18)					12.9				1.04/67	205/1402		2*	103
–, –, syndiotactic													
I.					~7.8					180/1472			1471
										173/1372			
II.					~7.8					200/1472			1471
III.	Ortho	D2h-16	13.36	23.21	5.12		8	0.977				4*1/1	1471
IV.					~5.1								1471
(mesomorphic)													
V.					~5.1								1471
–, trimethylsilyl-													
–, –, isotactic													
(176.34)					60.4					284			103
Poly[4,4′-(terephthaloyldiamino)-diphenylene pyromellitimide]													
(528.48)	Mono	C2h-5	8.64	5.18	25.38	$\beta = 90$	2	1.545				23*2/1	1191
Poly[4,4′-(terephthaloyldiamino)-diphenylene *p*-phenyleneditrimellitatediimide]													
(768.69)					37.63							34*2/1	1191
Poly(*p*-terphenylene pyromellitimide)													
(442.43)					20.9								908
			8.58	5.48	20.9		2	1.495				19*1/1	928
Poly(2,2′:5′,2″-terthiophene)													
–, 3,3″-dioctyl-													
(470.79)	Ortho		14.7	4.33	23.55		2	1.04				9*2/1	1486
Poly(4,4′-thiodiphenylene *p*-phenyleneditrimellitatediimide)													
(638.61)	Ortho		6.61	8.02	53.6		4	1.493				27*2/1	808
Poly(4,4′-thiodiphenylene pyromellitimide)													
(398.39)					29.4								908
	Ortho		8.57	6.37	61.5		8	1.576				16*4/1	1013
Poly(2,5-thiophene)													
(82.12)	Ortho		7.80	5.55	8.03		4	1.569				3*2/1	1276
	Mono		7.83	5.55	8.20	$\beta = 96$	4	1.539				3*2/1	1276
			7.79	5.53	7.7		4	1.64				3*2/1	1336
	Ortho		7.80	5.55	8.03		4	1.569				3*2/1	1448
–, 3-decyl-													
α	Mono		22.32	16.76	7.77	$\gamma = 95.5$	8	1.021				3*2/1	1630
(222.39)													
(at 120°C)	Mono		22.08	16.72	7.76	$\gamma = 95.6$	8	1.036				3*2/1	1630
β	Mono	C2-2	22.36	17.40	7.77	$\gamma = 90.5$	8	0.977				3*2/1	1447
	Ortho		23.9	3.8	15.6		4	1.042				3*4/1	1482
	Mono		51.9	7.74	7.8[b]	$\gamma = 86$	9	0.94				3*2/1	1487
(at 173 K)	Mono	C2-2	22.54	17.17	7.749	$\gamma = 90.3$	8	0.985				3*2/1	1447
(at 120°C)	Mono		21.54	17.17	7.750	$\gamma = 90.3$	8	1.0310				3*2/1	1630
–, 3-dodecyl-													
I. (250.44)	Ortho		25.83	7.75	7.77		4	1.069				3*2/1	1479
	Ortho		51.9	7.74	8.0		8	1.04				3*2/1	1481
	Ortho		21.71	3.97	7.8		2	1.24				3*2/1	1483
II.	Mono		19.8	9.37	8.07	$\gamma = 77$	4	1/140				3*2/1	1623
–, 3-hexadecyl-													
(306.55)	Ortho		24.74	4.21	7.8		2	1.25				3*2/1	1483

TABLE 3. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 3-hexyl-													
(166.28)	Ortho		16.63	7.75	7.77		4	1.069				3*2/1	1479
	Ortho		33.6	7.66	7.7		8	1.11				3*2/1	1481
–, –, iodine doped													
	Ortho		37.25	23.15	7.77		24	1.932				3*2/1	1480
–, 3-methyl- (with 50% CF3S03-)													
(96.15)	Hex		19	19	12.2		22	1.63				6*11/1	1259
–, 3-octyl-													
I. (194.34)	Ortho		41.3	7.63	7.7		8	1.06				3*2/1	1481
	Ortho		17.40	3.61	7.8		2	1.32				3*2/1	1483
	Ortho		20.3	3.81	7.85		2	1.063				3*2/1	1486
	Mono		42.0	7.67	7.8	$\gamma = 87$	8	1.03				3*2/1	1487
II.	Mono		15.3	9.43	8.07	$\gamma = 72$	4	1.166				3*2/1	1623
–, 3-phenyl-													
–, –, 4′-octyl-													
(270.43)	Ortho		28.4	5.06	7.44		2	0.840				3*2/1	1637
–, 3-tetradecyl-													
(278.50)	Ortho		24.73	3.82	7.8		2	1.26				3*2/1	1483
Poly(tridecamethylene pyromellitimide)													
(396.49)					39.2					263			1395
Poly(undecamethylene pyromellitimide)													
(368.43)					32.0					292			1395
Poly(*N*-vinyl carbazole)													
(193.25)	Hex		12.30	12.30	7.44		3	0.988		>320/228		2*3/1	126
	Hex		12.00	12.00	6.47		3	1.193				2*3/1	825
	Ortho		21.6	12.5	7.44		6	0.958				2*3/1	1010
	Phex		12.3	12.3			3					2*3/1	607
	Rho		12.00	12.00	6.39		3	1.208				2*3/1	813
Poly(1-vinyl naphthalene)													
(154.21)	Tet	C4v-12	21.20	21.20	8.10		16	1.125		360/75		2*4/1	122
Poly(2-vinylpyridine)													
(105.14)					6.7					212			185
	Rho	C3-2	15.49	15.49	6.56		9	1.153				2*3/1	861
Poly(*p*-xylylene)													
α	Mono		11.68	6.10	9.16	$\beta = 102.5$	4	1.086		425/142	30.2/82	6*2/1	223
(104.15)	Ortho		21.3	33.6	6.58		31	1.138	1.05	420/219		6*1/1	426
	Mono		5.92	10.64	6.55	$\beta = 134.7$	2	1.179		375/82		6*1/1	595
	Mono		5.9	10.64	6.55	$\beta = 135$	2	1.19		435/1187		6*1/1	1181
										400/395			
										440/546			
β	Mono		8.10	5.25	6.53	$\beta = 95$	2	1.250				6*1/1	223
	Rho	C3-1	20.52	20.52	6.55		16	1.158				6*1/1	1011
	Hex		20.52	20.52	6.58		16	1.153		420		6*1/1	426
					6.55					412			224
–, chloro-													
(138.60)	Tri		5.92	12.8	6.55	90,90,135	2	1.312		290/395		6*1/1	1012
	Mono		5.9	12.8	6.55	$\beta = 135$	2	1.32				6*1/1	1181
–, ethyl-													
(132.21)	Mono		5.9	13.4	6.55	$\beta = 135$	2	1.20		170/395		6*1/1	1181
–, isopropyl-													
(146.23)	Mono		5.9	14.1	6.55	$\beta = 135$	2	1.26				6*1/1	1181

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 4. POLY(DIENES) AND POLY(DIYNES)

Polymer	Cryst syst.	Space group	Unit cell parameters					Density (g/cm^3)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	*b*	*c*	Angles	*z*	Crystal.	Amorph.[a]				
1,4-Poly(2,3-azabutadiene)													
–, 1,4-diethyl-													
trans-													
(112.18)					4.72							4*1/1	1504
1,2-Poly(1,3-butadiene)													
syndiotactic													
(54.09)	Ortho	D2h-11	10.98	6.60	5.14		4	0.964		154		4*1/1	7
isotactic													
	Rho	D3d-6	17.3	17.3	6.5		18	0.96		120		2*3/1	12
1,4-Poly(1,3-butadiene)													
trans-													
I.	Phex		4.54	4.54	4.9		1	1.03		100/352	10.0/352	4*1/1	37
(54.09)	Mono	C2h-5	8.63	9.11	4.83	$\beta = 114$	4	1.036		96/451	13.8/451	4*1/1	496
									0.891	55/624	6.31/624		1195
II. (>65°C)	Phex		4.88	4.88	4.68		1	0.930		141/352	4.61/352	4*1/1	154
	Hex		4.95	4.95	4.66		1	0.908		148/44	5.99/342	4*1/1	550
										145/451	4.61/451		
										142/624	3.62/624		
cis-													
	Mono	C2h-5	8.53	8.16	12.66	$\beta = 83.33$	8	0.821		6.3/335	9.2/353		60
	Mono	Cs-4	4.60	9.50	8.60	$\beta = 109$	4	1.011		1		8*1/1	124
	or	C2h-6							0.902	12/1121			1195
–, 2-*tert*-butyl-													
cis-													
(110.20)	Tri		13.95	20.78	15.3		22	0.908		106		4*11/3	367
–, chlorinated													
cis-threo-disyndiotactic													
(125.00)	Mono	C2h-4	7.37	5.30	10.10	$\beta = 134$	2	1.463				4*2/1	768
trans-erythro-diisotactic													
	Mono	C2h-5	7.05	8.05	5.10	$\beta = 100$	2	1.456				4*1/1	768
–, 2-chloro- [chloroprene]													
trans-													
(88.54)	Ortho	D2-4	8.84	10.24	4.79		4	1.356		115/229	8.37/81	4*1/1	40
	Ortho		9.0	8.23	4.79		4	1.658		80/81		4*1/1	109
cis-													
										70/1157			
–, 1-cyano													
trans-													
(79.10)					4.8							4*1/1	275
–, 2,3-dichloro-													
trans-													
I.					4.86								221
(122.98)	Mono	C2h-5	5.34	9.95	4.80	$\beta = 93.5$	2	1.604				4*1/1	565
II.	Ortho	C2v-9	8.81	12.34	4.80		4	1.565				4*1/1	699
–, –, thiourea complex													
	Mono	C2Hh5	9.87	15.83	12.53	$\beta = 114.1$		1.44					565
	Mono	C2h-5	9.91	15.85	12.5	$\beta = 114.1$		1.44					803
–, 2,3-dimethyl-													
trans-													
(82.15)					4.35					260			104
	Ortho	C2v-9	9.13	13.00	4.70		4	0.978		272/221		4*1/1	699
cis-													
					7.0					198			220
–, –, thiourea complex													
	Mono	C2h-5	10.40	15.47	12.5	$\beta = 114.4$		1.22					803
–, 2-methyl- [isoprene]													
trans-													
α					8.70					80/1121	13/1121	4*2/1	288
(68.12)					8.76						10/1128		502
	Mono	C2h-5	7.98	6.29	8.77	$\beta = 102$	4	1.051					715
									0.905				1127
β (LM)	Ortho	D2-4	7.78	11.78	4.72		4	1.046		65/46	8.0/1128	4*1/1	40
	Ortho		7.85	11.9	4.8[b]		4	1.009		64/81	10.6/287	4*1/1	288

TABLE 4. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Ortho		7.83	11.87	4.75		4	1.025		82/287		4*1/1	502
									0.906	78/630			1195
										78/568			
										80/1170			
δ (HM)	Mono		5.9	7.9	9.2	γ = 94	4	1.06		74/81	12.7/81	4*2/1	502
			7.84	5.99			4			80/287			492
										87/568			
										89/1170			
ε			7.80	6.29									492
cis-													
	Mono	C2h-5	12.46	8.89	8.10	β = 92	8	1.009	0.906/53	28/81	4.40/81	8*1/1	40
			25.2	8.97	8.20[b]		16	0.976	0.910/47	36/50	4.6/1195	8*1/1	288
	Ortho	D2h-15								14/287		8*1/1	124
	Mono		26.3	8.9	8.15[b]	γ = 109.5	16	1.01		36/1169			1019
	Ortho		12.4	8.9	8.15[b]		8	1.101					1019
–, –, 3-chloro-													
trans-													
(102.56)					4.9							4*1/1	165
–, –, hydrochlorinated													
(104.58)	Port	C2h-5	5.83	10.38	8.95	β = 90	4	1.282		115		4*2/1	229
	Ortho		11.9	10.4	9.1[b]		8	1.23		110		4*2/1	70
–, 2-methylacetoxy-													
trans-													
(126.15)	Ortho		16.2	9.3	4.75		4	1.17		135		4*1/1	165
–, perhydrotriphenylene complex													
trans-													
2 phtp/monomer	Hex	C6h-2	14.26	14.26	4.78		1	1.079		183/1124			798
cis-													
–, 2-propyl-													
trans-													
(96.17)			10.95	6.65	9.2		4	0.95		42		8*1/1	368
										42/368			
–, urea complex													
	Hex	D6-2	8.22	8.22	11.01								827
1,4-Poly(1,3-butadiyne)													
–, 1,4-(4,4′-diphenyleneglutarate)-													
(330.34)	Mono	C2h-6	23.12	7.87	9.69	β = 111.29	4	1.336				4*2/1	948
	Mono	C2h-6	20.8	8.0	9.7	β = 106	4	1.41				4*2/1	913
Poly(chloroprene), see Poly(butadiene), 2-chloro-													
1,12-Poly(1,11-dodecadiyne)													
macromonomer	Mono	C2h-5	13.25	7.63	14.15[b]	γ = 118.5	4	0.847				12*1/1	1022
(160.26)													
cross-polymer	Mono	C2h-5	9.17	9.92	12.2	γ = 123.5	4	1.146					1022
5,8-Poly(5,7-dodecadiynediol)													
–, 1,12-bis(*N*-butoxycarbonyl)methylurethane-													
(508.61)			5.33	54.36	4.88		2	1.195				4*1/1	1319
	Ortho	D2-3	53.86	10.98	4.880		4	1.170				4*1/1	1329
–, 1,12-bis(ethylurethane)-													
(336.43)	Mono	C2h-5	18.30	10.78	4.89[b]	γ = 95	2	1.163				4*1/1	1318
	Mono		18.13	10.81	4.89[b]	γ = 94	2	1.169				4*1/1	1320
(hi temp)	Mono	C2h-5	19.93	9.58	4.83[b]	γ = 91	2	1.212				4*1/1	1318
	Mono		18.9	10.89	4.83[b]	γ = 91	2	1.124				4*1/1	1320
–, 1,12-bis(phenylurethane)-													
(432.52)	Mono	C2h-5	6.229	39.03	4.909	β = 106.85	2	1.258				4*1/1	95
	Mono	C2h-5	6.205	39.16	4.90[b]	β = 106.42	2	1.259				4*1/1	1353
–, 1,12-bis(*p*-toluylsulfonyl)-													
(502.64)	Tri	Ci-1	20.13	6.11	4.91	95,94,89	1	1.391				4*1/1	381
(110 K)	Tri	Ci-1	20.01	6.02	4.91	95,94,89	1	1.420				4*1/1	381
10,13-Poly(10,12-heptacosadiynoic acid)													
(as Li salt)			65										1268
(400.65)													

TABLE 4. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
1,4-Poly(1,3-heptadiene)													
trans-													
isotactic													
(96.17)	Mono	C2-2	8.62	7.95	4.85	$\beta = 99$	2	0.973		85/371		4*1/1	375
–, 6-methyl-													
trans-													
isotactic													
(110.20)					4.85					119		4*1/1	371
1,4-Poly(1,3-hexadiene)													
trans-													
isotactic													
(82.15)	Ortho	D2-4	14.02	8.02	4.85		4	1.000		82/371		4*1/1	274
--, 5-methyl-													
trans-													
isotactic													
(96.17)					4.85					88		4*1/1	371
2,5-Poly(2,4-hexadiene)													
trans-erythro-diisotactic													
(82.15)					2.3					84		2*1/1	525
–, 2,5-dimethyl-													
trans-													
(110.20)					4.8					265 265/376		4*1/1	183
2,5-Poly(2,4-hexadienedioic acid)													
trans-erythro-diisotactic													
–, diisopropyl ester													
(226.27)	Ortho		14.16	10.28	9.70		4	1.064		230/420		8*1/1	420
2,5-Poly(2,4-hexadienoic acid)													
trans-erythro-diisotactic													
–, 6-amino-													
0.5 $CdCl_4$/unit	Tri	Ci-1	9.67*	10.70	17.79	93,90,90	8	1.838		347		4*2/1	1123
(254.25)	Tri	Ci-1	7.214	7.245	18.59	104,97,96	4	1.823				8*1/1	1188
–, –, hydrochloride													
(163.60)										334			1123
–, butyl ester													
(168.24)	Ortho		11.36	9.70	4.80		2	1.056				4*1/1	384
–, ethyl ester													
(140.18)					4.80							4*1/1	384
–, isoamyl ester													
(182.26)					4.80							4*1/1	384
–, isobutyl ester													
(168.24)					4.80							4*1/1	384
–, isopropyl ester													
(154.21)					4.80							4*1/1	384
–, methyl ester													
(126.16)					4.80							4*1/1	384
					4.8							4*1/1	1208
trans-threo-diisotactic													
–methyl ester													
(126.16)					9.4							4*2/1	1208
2,5-Poly(2,4-hexadiyne)													
–, 1,6-di(*N*-carbazoyl)-													
(408.50)	Mono	C2h-5	13.03	17.55	4.92[b]	$\gamma = 108.7$	2	1.273				4*1/1	951
	Mono	C2h-5	12.82	17.33	4.886[b]	$\gamma = 108.3$	2	1.316				4*1/1	611
	Mono	C2h-5	12.87	17.40	4.907[b]	$\gamma = 108.3$	2	1.300				4*1/1	830
–, –, hexabromo-													
	Mono		14.21	20.67	4.75[b]	$\gamma = 106.8$						4*1/1	1503
2,5-Poly(2,4-hexadiynediol)													
(50 mrad)	Mono	C2h-5	4.176	15.78	4.703	$\beta = 110.34$	2	1.258				4*1/1	903
(" annld)	Mono	C2h-5	4.115	16.05	4.819	$\beta = 106.08$	2	1.196				4*1/1	903
(110.11)	Mono		4.109	16.09	4.793	$\beta = 106.4$	2	1.203				4*1/1	904
–, 1,6-bis(ethylurethane)-													
(252.27)	Mono	C2h-5	15.14	17.96	4.88	$\gamma = 98.42$	4	1.277				4*1/1	1240

TABLE 4. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 1,6-bis(phenylsulfonyl)-													
–, –, di-*p*-fluoro-													
(426.41)	Mono	C2h-5	13.93	14.06	4.90[b]	$\gamma = 113.3$	2	1.606				4*1/1	1378
	Mono	C2h-5	13.89	14.10	4.91[b]	$\gamma = 113.2$	2	1.602				4*1/1	1379
	Mono	C2h-5	13.96	14.10	4.914[b]	$\gamma = 113.56$	2	1.597				4*1/1	1380
–, –, di-*p*-methoxy-													
(450.48)	Tri	Ci-1	12.72	8.543	4.902[b]	110,103,79	1	1.542				4*1/1	1021
–, 1,6-bis(phenylurethane)-													
I. (348.36)	Tri	Ci-1	16.78	12.53	4.89[b]	84,83,69	2	1.340				4*1/1	716
(5 dioxane/unit)					4.9								749
II.					4.9							4*1/1	749
III.					4.9							4*1/1	749
–, 1,6-bis-(*p*-toluylsulfonyl)-													
I. (418.48)	Mono	C2h-5	14.49	14.94	4.910[b]	$\gamma = 118.14$	2	1.482		102/1020	35/1020	4*1/1	898
	Mono	C2h-5	14.53	14.96	4.912[b]	$\gamma = 118.21$	2	1.477				4*1/1	1277
	Mono	C2h-5	14.91	14.48	4.93	$\gamma = 118.0$	2	1.479				4*1/1	1381
II. (120 K)	Mono	C2h-5	14.93	25.56	4.910[b]	$\gamma = 92$	4	1.484				4*1/1	899
–, 1-(*p*-toluylsulfonyl)-6-(*p*-fluorophenylsulfonyl)-													
I. (422.44)	Mono		14.22	14.52	4.909[b]	$\gamma = 116.1$	2	1.541				4*1/1	1337
II. (143 K)	Mono		14.10	14.43	4.906[b]	$\gamma = 116.3$	2	1.566				4*1/1	1337
Poly(isoprene), see Poly(butadiene), 2-methyl-													
10, 13-Poly(10,12-nonacosadiynoic acid)													
(as Li salt) (428.70)	Ortho		70	8.11	4.89		4	1.04				4*1/1	1268
1,9-Poly(1,8-nonadiyne)													
macromonomer (118.18)	Mono	C2-2	7.88	5.67	21.00[b]	$\gamma = 90.8$	4	0.837				9*2/1	1023
cross-polymer	Mono	C2h-5	9.30	17.5	4.85	$\beta = 101.71$	4	1.016				9*2/1	1023
1,4-Poly(1,3-octadiene)													
trans-													
isotactic													
(110.20)					4.85					87		4*1/1	371
2,5-Poly(2,4,6-octatriynediol)													
(134.14)	Mono	C2h-5	4.11	19.59	4.80	$\beta = 109$	2	1.219				4*1/1	780
–, 1,8-bis(phenylurethane)-													
(372.38)	Mono	C2h-5	24.65	30.74	4.89	$\beta = 92.2$	8	1.336				4*1/1	780
10,13-Poly(10,12-pentacosadiynoic acid)													
(as Li salt) (372.59)			60										1268
1,2-Poly(1,3-pentadiene)													
trans-													
syndiotactic													
(68.12)					5.1					10		4*1/1	230
–, 4-methyl-													
isotactic													
II. (82.15)	Tet	D2d-10	17.80	17.80	36.50		72	0.849		167/707		2*18/5	488
syndiotactic													
I.					11.73							12*1/1	1502
II.					5.05							4*1/1	1502
1,4-Poly(1,3-pentadiene)													
trans-													
isotactic													
(68.12)	Ortho		19.73	4.85	4.8		4	0.98		95		4*1/1	189
	Ortho	D2-4	19.80	4.86	4.85		4	0.969		104/1126		4*1/1	200
cis-													
isotactic													
	Ortho	D2-4	9.47	5.97	8.15		4	0.982				4*2/1	844
					8.15					44		4*2/1	263
syndiotactic													
	Mono	C2h-5	6.34	9.12	8.50	$\beta = 105.9$	4	0.957				8*1/1	1431
					8.50					53		8*1/1	234

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TABLE 4. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 2-methyl-													
trans-													
isotactic													
(RT)	Mono	C2h-5	12.89	9.186	4.820[b]	$\beta=93.5$	4	0.958		195/1365		4*1/1	1317
(82.15)													
(153 K)	Mono	C2h-5	12.78	9.084	4.824[b]	$\beta=93.6$	4	0.977				4*1/1	1317
cis-													
isotactic													
α (−130)	Ortho	D2h-15	10.74	13.04	7.87		8	0.990		175		4*2/1	1364
β (RT)	Ortho	D2-4	9.30	7.73	7.90		4	0.961		165		4*2/1	1364
										165/1365			
–, 3-methyl-													
cis-													
isotactic													
(82.15) I.					8.02							*2/1	1497
II.					6.97							*3/1	1497
syndiotactic													
II.					8.6							8*1/1	1501
1,4-Poly(1,3-pentadiynol)													
–, 1-*N*-carbazoly-													
(245.28)	Mono	C2h-5	16.49	19.51	4.873[b]	$\gamma=124.27$	4	1.257				4*1/1	1354
2,5-Poly(5-phenyl-2,4-pentadienoic acid)													
trans-erythro-diisotactic													
–, butyl ester													
(230.31)					4.80							4*1/1	384
–methyl ester													
(188.23)					4.80							4*1/1	384
1,24-Poly(1,11,13,23-tetracostetrayne)													
cross-polymer	Mono	C2h-4	26.8*	8.254	4.91*	$\gamma=119.6$	2	1.136					1264
(322.54)													

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 5. POLY(PEPTIDES) -(NH–CHR–CO)-

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(L-alanine)													
α helix	hex		8.55	8.55	70.3		47	1.246				3*47/13	507
(71.08)													
β	Ortho		4.79	10.7	6.88		4	1.339				3*2/1	507
	Ortho		4.734	10.54	6.89		4	1.373				3*2/1	509
–, phenyl-													
α helix	Hex		11.55	11.55								3*18/5	1025
(147.18)													
ω helix	Tet		11.28	11.28	5.91		4	1.300				3*4/1	1024
	Tet		12.86	12.86	5.59		4	1.057				3*4/1	1025
Poly(β-alanine), see Poly(3-aminopropionic acid)													
Poly(L-alanyl-D-alanyl-L-alanyl-D-valine)													
(312.37)	Ortho		9.5	31.0	11.0		8	1.281				12*1/1	1046
Poly(L-alanyl-L-alanylglycine)													
(200.22)	Ortho		9.44	9.96	20.94[b]		8	1.351				9*2/1	753
–phenyl- [2nd alanyl]													
(275.31)					15							9*	774
Poly(L-alanyl-L-alanylglycyl-L-prolyl-L-prolylglycine)													
(450.49)	Ortho		10.5		29.5							18*	755
Poly(L-alanyl-D-alanyl-L-valyl-D-valine)													
(340.42)	Ortho		9.5	34.1	11.0		8	1.269				12*1/1	1046

TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
Poly(L-alanyl-4-aminobutyric acid)													
(156.18)	Ortho		9.4	9.9	9.2		4	1.21				8*1/1	969
Poly(L-alanyl-6-aminocaproic acid)													
(184.24)	Ortho		9.4	9.8	11.5[b]		4	1.16				10*1/1	969
Poly(L-alanyl-3-aminopropionic acid)													
(142.16)	Ortho		9.4	9.8	16.1[b]		8	1.27				7*2/1	969
Poly(L-alanyl-11-aminoundecanoic acid)													
(254.37)	Ortho	D2-5	4.79	10.35	29.8		4	1.144				15*2/1	1619
Poly(L-alanyl-5-aminovaleric acid)													
(170.21)	Ortho		9.4	10.7	20.7[b]		8	1.09				9*2/1	969
Poly(L-alanyl-L-glutamylglycine)													
–, γ-ethyl-													
(285.30)	Mono		9.54	8.54	21.36[b]	$\gamma=106$	4	1.133				9*2/1	756
Poly(L-alanylglycine)													
I. (128.13)	Ortho		9.42	8.87	6.95[b]		4	1.466				6*1/1	591
	Ortho		9.44	8.96	6.94[b]		4	1.450				6*1/1	754
II.	Ortho		4.72	14.4	9.6		4	1.30				6*2/1	741
Poly(L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-glutamylglycine)													
(990.98)	Ortho		9.64	9.62	48.65		4	1.459				42*1/1	1648
Poly(L-alanylglycyl-L-alanylglyclyl-L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-glutamylglycine)													
(850.84)	Ortho		9.57	9.70	41.70		4	1.460				36*1/1	1648
Poly(L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-glutamylglycine)													
(710.70)	Ortho		9.48	10.28	34.75		4	1.394				30*1/1	1648
Poly(L-alanylglycyl-L-alanylglycyl-L-alanylglycyl-L-glutamylglycine)													
I. (570.56)	Ortho		9.48	17.34	18.8		4	1.226				24*1/1	1633
II.	Ortho		9.48	10.60	27.80		4	1.356				24*1/1	1648
Poly(L-alanylglycyl-L-alanylglycyl-L-serylglycine)													
(400.39)	Ortho		9.39	9.05	20.55[b]		4	1.522				18*	754
Poly(L-alanylglycyl-L-alanyl-L-prolyglycyl-L-proline)													
(450.49)	Ortho		10.5		29.5							18*	755
Poly(L-alanylglycylglycine)													
II. (185.18)	Mono		8.86	22.0	9.42	$\beta=90$	9	1.507				9*1/1	596
(monohydrate)	Mono	C2-2	11.0	4.8	9.45	$\beta=90$	2	1.35				9*1/1	740
Poly(L-alanylglycylglycylglycine)													
II. (242.24)	Hex		4.89	4.89	36.60		3	1.592				12*3/1	740
Poly(L-alanylglycyl-L-proline)													
dry	Hex		11.9	11.9	28.8		10	1.06				9*10/3	614
wet	Hex		11.4	11.4	9.3		3	1.07				9*3/1	614
(225.25)													
Poly(L-alanylglycyl-L-prolyl-L-prolyglycyl-L-proline)													
(476.53)	Ortho		10.6		28.7							18*	755
Poly(L-alanyl-L-prolyglycine)													
with H_2O	Mono	C2-2	8.6	7.2	9.4	$\beta=90$	2	1.4				9*1/1	616
(225.25)													
Poly(L-alanyl-L-prolylglycyl-L-prolyl-L-prolylglycine)													
(476.53)	Hex		11.9	11.9	29.5		5	1.093				18*	755
Poly(L-alanyl-D-valine)													
(170.21)	Port		9.7	14.7	6.2		4	1.28				6*1/1	1176
	Ortho		9.5	29.4	11.1		16	1.459?				6*2/1	1046
Poly(2-aminoisobutyric acid)													
(85.11)	Hex		8.7	8.7	5.96		3	1.08				3*3/1	967
–, dichloroacetic acid complex													
(106.60)	Hex	C3v-6	21.8	21.8	5.95		18	1.301				3*3/1	1053
(1 DCA/6 residues)													
Poly(L-arginine)													
–, hydrochloride													
α helix	Hex		14.7	14.7	27.0		18	1.406				3*18/	641
(2.5 H_2O/residue)													
(192.65)													
β	Mono		9.26	22.05	6.76	$\gamma=108.9$	4	1.438				3*2/1	641
(>5 H_2O/residue)													

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TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm³)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	*b*	*c*	Angles		Crystal.	Amorph.[a]				
Poly(L-aspartic acid)													
–, β-benzyl-													
ω helix	Tet		13.85	13.85	5.30		4	1.341				3*4/1	582
(205.21)	Tet		13.80	13.80	5.42		4	1.320				3*4/1	703
	Tet		13.88	13.88	5.3		4	1.34				3*4/1	837
racemic	Ortho		21.4	4.28	5.68		2	1.310				3*2/1	1026
–, –, *p*-chloro-													
α helix	Hex	D6-3	14.9	14.9	27.0		18	1.380				3*18/5	606
(239.66)	Hex		14.8	14.8	27		4	1.40				3*18/5	1505
ω helix	Tet	D4-4	23.3	23.3	5.20		8	1.128				3*4/1	606
	Mono		16.4	13.4	5.2	$\gamma=99$	18	1.41				3*4/1	1505
–, –, *p*-methyl-													
α helix	Hex		14.9	14.9	27		4	1.26				3*18/5	1505
(219.24)													
ω helix	Mono		16.6	13.4	5.28	$\gamma=108$	4	1.304				3*4/1	1505
–, α-*n*-butyl-													
I. (171.20)	Hex		13.5	13.5	19.9		13	1.177				4*13/4	1507
	Phex	C2-2	13.45	13.45	20.6	$\gamma=120$	13	1.145				4*13/4	1508
II.	Tet	C4-2	14.15	14.15	4.90		4	1.338				4*4/1	1508
–, α-ethyl-													
–, –, 2-methoxy-													
(173.17)	Phex	C2-2	12.40	12.40	20.74	$\gamma=120$	13	1.354				4*13/4	1508
–, β-ethyl-													
–, –, 2-phenyl-													
α helix	Tet		13.2	13.2	27.0		18	1.393				3*18/5	1027
(219.24)													
β	Ortho		17.7	4.76	6.83		2	1.265				3*2/1	1027
π helix	Ortho		20.3	11.7	19.89		17	1.310				3*17/4	1028
>200°C	Hex		17.1	17.1	19.89		17	1.229				3*17/4	1028
?	Hex		23.4	23.4									1027
–, α-isobutyl-													
I. (171.20)	Ortho		13.5	23.4	19.9		26	1.176		286/1310		4*13/4	1312
	Hex		13.5	13.5	19.9		13	1.177				4*13/4	1506
II.	Tet		14.0	14.0	4.95		4	1.172				4*4/1	1312
	Tet		13.98	13.98	4.94		4	1.178				4*4/1	1506
–, β-*n*-propyl													
β	Mono	C2-2	9.57	25.08	6.79[b]	$\gamma=96$	8	1.288				3*2/1	579
(157.17)													
–, –, 3-phenyl-													
(232.26)	Hex		15.6	15.6	10.5		7	1.220				3*7/2	1628
(at 140°C)	Ortho		28.0	16.1	10.5		14	1.141				3*7/2	1628
Poly(L-cysteine)													
–, s-benzyloxycarbonyl-													
β													
I.	Ortho		4.76	32.4	6.95[b]		4	1.470				3*2/1	578
(237.27)													
II.	Ortho		4.89	32.8	6.89[b]		4	1.426				3*2/1	578
–, *s*-benzylthio-													
ω helix	Tet	C4-2	14.28	14.28	5.55		4	1.322				3*4/1	1029
(225.32)													
Poly(2,4-diaminobutyric acid)													
–, *N*-benzyloxycarbonyl-													
(234.26)	Rho		27.5	27.5	27.0		56	1.232				3*18/5	720
Poly(D-glutamic acid)													
–, γ-benzyl-													
–, –, chloro-													
meta	Phex		15.30	15.30	26.99	$\gamma=60.23$	18	1.382				3*18/5	932
(253.68)													
ortho	Phex		15.07	15.07	27.15	$\gamma=60.03$	18	1.420				3*18/5	932
para	Ortho		15.24	26.40	10.68		14	1.372				3*7/2	932
	Hex		15.24	15.24	10.67		7	1.374				3*7/2	1033

TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, γ-*n*-hexyl-													
α helix	Hex		15.89	15.89	27		18	1.08				3*18/5	859
(213.27)													
–, γ-methyl-													
α helix	Hex		12.0	12.0								3*18/5	1162
(143.14)													
β	Ortho		9.49	19.09	7.07		8	1.484				3*2/1	954
			9.44	23.37	6.81	$\gamma=90$	8	1.266					812
Poly(L-glutamic acid)													
β													
I.	Mono		9.66	9.10	6.98[b]	$\gamma=105$	4	1.447				3*2/1	585
(129.12)	Mono		9.79	9.35	6.82[b]	$\gamma=105$	4	1.422				3*2/1	966
II.	Mono		9.75	8.06	6.84[b]	$\gamma=104$	4	1.644				3*2/1	966
	Ortho		4.95	15.6	6.90		4	1.610				3*2/1	836
–, γ-*n*-amyl-													
(199.25)	Hex		15.1	15.1	(27)		18	1.12				3*18/5	1040
	Hex		15	15	27		18	1.13				3*18/5	1279
–, γ-benzyl-													
α helix													
A.	Mono		15.3	14.6		$\gamma=121.7$						3*18/5	1030
(219.24)	Hex		30	30			72					3*18/5	952
B.	Tri		15.25	15.25	27.0	84,84,122	18	1.257				3*18/5	587
	Mono		25.2	15.0	26.8	$\gamma=122.5$	30	1.278				3*18/5	702
	Mono		15.88	13.00	27.1	$\gamma=113.7$	18	1.279				3*18/5	952
(dry)	Ortho		14.76	25.40	64.11		86	1.303				*43/12	757
	Mono		29.06	13.20	27.27	$\beta=96$	36	1.260				3*18/5	1030
C.	Hex		14.95	14.95	27.1	$\gamma=56$	18	1.305				3*18/5	584
(racemic)	Hex		14.95	14.95	106		71	1.260				*71/20	584
	Mono		15.05	14.78	27	$\gamma=119.4$	18	1.252				3*18/5	1030
	Hex		14.8	14.8	10.50		7	1.279				3*7/2	1030
D.	Hex		39.05	39.05	10.58		49	1.277				3*7/	953
E.	Hex		64.36	64.36	10.5			1.285				3*7/2	1030
single crystal racemic mixture	Mono		5.52	5.11	22.25	$\beta=97.8$	2	1.171					926
I.	Ortho		24.4	15.4									1031
II.	Mono		25.6	15.6		$\gamma=107.7$							1031
–, –, chloro-													
meta	Hex		15.50	15.50	27.10		18	1.345				3*18/5	1032
(racemic)	Hex		15.49	15.49	26.84		18	1.360				3*18/5	1035
(253.68)													
ortho	Ortho		26.50	30.60	27.02		72	1.384				3*18/5	1032
(racemic)	Hex		15.24	15.24	27.23		18	1.384				3*18/5	1034
para	Ortho		15.24	26.40	10.67		14	1.374				3*7/2	1032
(racemic)	Ortho		15.12	26.09	10.72		14	1.394				3*7/2	932
–, γ-*n*-butyl-													
(185.22)	Phex		14.1	14.1	(27)	$\gamma=114$	18	1.13				3*18/5	1040
(at 100°C)	Ortho		25.4	14.66	27		36	1.101				3*18/5	1040
–, γ-calcium													
β (wet)	Mono	C2-2	9.40	12.82	6.83[b]	$\gamma=100.3$							585
–, γ-*n*-dodecyl-													
(297.44)	Hex		27.6	27.6									1292
–, γ-ethyl-													
α helix													
I.	Phex		12.5	12.5	27.0	$\gamma=110$	18	1.18				3*18/5	826
(157.17)	Phex		12.16	12.16	108.9	$\gamma=109.5$	73	1.255				*73/20	955
					16.4							3*11/3	1036
T.	Tet		11.72	11.72	108.9		73	1.274				3*73/20	955
III.	Phex		12.9	12.9	108.9	$\gamma=123$	73	1.25				3*73/20	955
H.	Ortho		22.95	13.25	108.9			1.151				3*73/20	955
–, –, 2-chloro-													
(191.61)	Phex		11.7	11.7	16.56	$\gamma=118.7$	11	1.760				3*11/3	1036
–, –, 2-phenyl-													
(233.27)	Hex		15.60	15.60	(27)			1.225				3*18/5	1038
	Ortho		15.38	26.65	54.4		72	1.251				3*18/5	1039

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TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, γ-*n*-hexyl-													
(213.28)	Hex		15.8	15.8	27		18	1.09				3*18/5	1040
	Hex		15.5	15.5	27		18	1.14				3*18/5	1279
–, γ-methyl-													
α helix	Hex		11.96	11.96	27.5		18	1.256				3*18/5	588
(143.14)	Hex	11.95	11.95	27.0			18	1.281				3*18/5	576
	Hex		11.79	11.79	26.9		18	1.321				3*18/5	858
					97.26		65	(1.284)				*65/18	583
	Ortho		11.95	20.70	43.2		58	1.290				3*29/8	589
	Hex		11.82	11.82	103.0		69	1.316				*69/19	1204
	Hex		11.58	11.58	26.33		18	1.399				3*18/5	1650
precursor to α (3 chloroform/18 mers)													
	Hex		12.75	12.75	27.0		18	1.282				3*18/5	1293
	Hex		12.75	12.75	26.9		18	1.287				3*18/5	1338
(at 170°C)	Rho		21.27	21.27	176.0			1.210				117/32	1204
β			4.725		6.83							3*2/1	702
	Ortho		4.70	21	6.83		4	1.410				3*2/1	1037
–, –, dimethyl phthalate complex													
25%	Hex		27.6	27.6			72					3*18/5	1041
extracted	Phex		23.9	24.9	27	γ = 121.4	72	1.248				3*18/5	1041
–, γ-*n*-octyl-													
(241.33)	Hex		17.1	17.1	27		18	1.06				3*18/5	1279
–, γ-*n*-propyl-													
(171.20)	Phex		14.4	14.4	(27)	γ = 130	18	1.19				3*18/5	1040
(at 150°C)	Ortho		24.6	14.2	60.0		80	1.085				*40/11	1040
	Rho		24.3	24.3	60.0		120	1.112				*40/11	1533
–, –, 3-phenyl-													
(247.29)	Hex		16.30	16.30	(27)		18	1.190				3*18/5	1038
–, γ-sodium													
α helix (hydrated)	Hex		15.9	15.9									794
β (wet)	Port		14.4	7.0*	4.7								794
Poly(L-glutamine)													
–, *N*5-(4-hydroxybutyl)-													
(200.24)	Ortho		14.8	12.4	60.4		40	1.200				3*40/11	935
–, *N*5-(2-hydroxyethyl)-													
(172.18)	Ptet		12.1	12.1	27		18	1.30				3*18/5	968
–, *N*5-(3-hydroxypropyl)-													
(186.21)	Ortho		14.6	12.3	43.2		29	1.156				3*29/8	924
Poly(D-glutamyl-L-glutamic acid)													
–, γ-benzyl-													
α helix	Hex		15.2	15.2	28		19	1.235				3*19/5	885
(219.24)	Hex		14.9	14.9	26.6		18	1.281				6*9/5	1042
	Hex		26.0	26.0	32.4		66	1.267				6*11/6	1272
β	Hex		22.9	22.9	9.46		16	1.356				6*1/1	1042
	Ortho		9.52	22.5									1215
ω helix			27	15.8									885
	Ortho		26.92	15.54	5.40		8	1.289				6*2/1	1042
	Ortho		27.1	15.5	5.40		8	1.284				6*2/1	1272
π helix	Hex		17.0	17.0	25.6		22	1.250				6*11/5	885
	Hex		17.0	17.0	25.6		22	1.250				6*11/5	1272
π–π 5.6	Hex		18.2	18.2	56.3		56	1.262				6*14/5	1214
π–π 7.2	Hex		22.6	22.6	53.1		72	1.116				6*18/5	1214
π–π 9.0	Mono		25.3	24.4	20.2	γ = 117	36	1.180				6*9/2	1214
π–π 10.8	Mono		26.4	25.6	50.0	γ = 118		1.318				6*27/5	1214
–, γ-ethyl-													
–, –, 2-phenyl-													
α helix	Hex		15.9	15.9	26.5		18	1.204				6*9/5	1042
(233.27)	Hex		27.5	27.5	32.4		66	1.205				6*11/6	1272
β	Hex		24.2	24.2	4.75		8	1.286				6*1/1	1042
π helix	Hex		17.6	17.6	25.7		22	1.236				6*11/5	1272
Poly(glycine)													
I. (57.05)			4.77	4.77	7.0	γ = 66	2	1.30		365/795		3*2/1	581
	Mono	C2h-5	9.54	3.67	7.044[b]	γ = 113	4	1.669				3*2/1	1239

TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
II.	Hex	C3-2	4.8	4.8	9.3		3	1.53				3*3/1	273
Poly(glycyl-6-aminocaproic acid)													
I. (170.21)	Hex		4.79	4.79	35.1		3	1.216		276/1524		10*3/1	1311
II.	Mono		4.76		3.96	$\beta=113$							1311
Poly(glycyl-12-aminododecanoic acid)													
II. (254.37)	Hex		4.79	4.79	112.8		6	1.131		230/1524		6*16/1	1512
Poly(glycyl-3-aminopropionic acid)													
I. (128.13)	Ortho		9.51	15.20	16.95[b]		16	1.389				7*2/1	1294
II.	Hex		4.79	4.79	22.2		3	1.447				7*3/1	1294
Poly(glycyl-3-aminopropionic acid-3-aminopropionic acid)													
(199.21)	Hex		4.79	4.79	35.2		3	1.419				11*3/4	1313
Poly(glycyl-11-aminoundecanoic acid)													
I. (240.35)	Ortho		9.60		37.5[b]							15*2/1	1391
II.	Hex		4.79	4.79								15*	1391
Poly(glycylglycyl-L-proline)													
with H_2O	Mono	C2-2	12.2	4.9	9.3	$\beta=90$	2	1.4				9*1/1	615
w/formic acid (211.22)	Mono	C2-2	13.5	4.9	9.3	$\beta=94$	2					9*1/1	615
Poly(glycylprolylhydroxyproline)													
(267.28)	Hex		13.7	13.7	28.2		10	0.97				9*10/	612
Poly(glycyl-L-prolyl-L-proline)													
dry (251.29)	Hex		12.5	12.5	28.7		10	1.07				9*10/	613
–, α-phenyl-(glycyl)													
(327.38)	Ortho	D2-4	14.3	13.5	9.4		4	1.198				9*1/1	965
Poly(L-histidine)													
–, 1-benzyl-													
(227.27)	Phex		17.4	17.4	27.0	$\gamma=130$	18	1.085				3*18/5	719
–, hydrochloride													
(1 H_2O/residue) (190.61)	Mono		14.95	14.95	45.0	$\gamma=116.5$	45	1.582				3*15/4	1295
Poly(L-isoleucine)													
β	Ortho		4.80	22.36	6.88		4	1.018				3*2/1	806
(113.16)	Ortho		4.8	23	6.6		4	1.032				3*2/1	1025
Poly(L-leucine)													
α helix (113.16)	Hex		13.22	13.22								3*	806
β	Ortho		4.80	23.56	6.88		4	0.966				3*2/1	806
Poly(L-lysine)													
α helix (128.18)	Hex		19.55	19.55								3*	586
β	Ortho		9.44	17.16	6.8[b]		6	1.16				3*2/1	586
–, *E*-benzyloxycarbonyl-													
α helix (262.31)	Hex		16.69	16.69	26.90		18	1.208				3*18/5	577
–, hydrobromide													
(1.4 H_2O/residue) (234.31)	Ortho	D2-3	9.52	16.44	6.80		4	1.46				3*2/1	923
–, hydrochloride													
α helix >84% RH (164.64)	Hex		16.80	16.80									666
β dry	Ortho		4.62	15.20	6.66		4	2.338					666
65% RH	Ortho		4.71	16.67	6.66		4						666
Poly(L-lysyl-L-lysyl-L-valine)													
α helix (355.48)	Hex		15.87	15.87								9*	1047
β (with 2HCl)	Ortho		9.35	15.04	20.88		4	0.969				9*2/1	1047
–, di-benzyloxycarbonyl-													
β (623.75)	Ortho		9.38	28.24	20.22		8	1.547				9*2/1	1048

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TABLE 5. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm³)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	*b*	*c*	Angles		Crystal.	Amorph.[a]				
Poly(L-methionine)													
α helix (131.19)	Hex		11.49	11.49	70.5		47	1.270				3*47/13	867
Poly(D-methionyl-L-methionine)													
β (131.19)	Mono		18.05	4.71	14.66[b]	$\beta = 110.4$	8	1.492				6*2/1	850
Poly(L-ornithine)													
–, *N*-benzoyl-													
α helix (218.26)	Hex		15.0	15.0	26.5		18	1.263				3*18/5	1373
–, –, *p*-bromo-													
ω helix 8297.15*	Ortho		D2-4	24.6	18.4	5.68	8	1.535				3*4/1	1373
–, –, *p*-chloro-													
ω helix (252.70)	Ortho	D2-4	24.1	18.5	5.68		8	1.326				3*4/1	1373
–, –, *p*-fluoro-													
ω helix (236.25)	Ortho	D2-4	23.8	18.7	5.68		8	1.241				3*4/1	1373
–, *N*-benzyloxycarbonyl-													
I. (248.28)	Mono		23.3	22.7	16.2	$\gamma = 119.2$	22	1.213				3*11/3	720
	Phex		23.0	22.6	(16)	$\gamma = 119.5$	22	1.238				3*11/3	1043
	Hex		22.8	22.8	16.4		22	1.228				3*11/3	1296
II.	Ortho		29.40	14.70	27.0		36	1.272				3*18/5	1043
	Tet		20.8	20.8	60.00		80	1.270				3*40/1	1296
III.	Ortho		22.09	19.95	60.00		80	1.247				3*40/1	1296
–, hydrobromide													
α helix (>86% RH)	Hex		15.0										784
β (wet) (to 85% RH)	Ortho		4.60	30.2	6.64		8					3*2/1	784
Poly(L-proline)													
I. (97.12)	Hex	C2-2	9.05	9.05	19.0		10	1.197				3*10/3	590
II.	Rho	C3-3	6.62	6.62	9.36		3	1.362				3*3/1	510
	Rho	C3-3	6.62	6.62	9.31		3	1.369				3*3/1	512
	Rho	C3-3	6.68	6.68	9.36		3	1.338				3*3/1	511
–, acid complexes													
–, acetic	Tet		9.13	9.13	19.0		10					3*10/3	701
–, formic	Tet		8.92	8.92	19.0		10					3*10/3	701
–, propionic	Tet		9.13	9.13	19.0		10					3*10/3	701
highly solvated	Ortho		9.00	25.1	19.0								701
–, hydroxy-													
A. (113.12)	Hex	C3-3	12.3	12.3	9.15		9	1.410				3*	580
B. >66% RH	Phex		15.75	15.75	22.88	$\gamma = 120.47$						3*8/3	1044
–, –, *O*-acetyl-													
II. (155.15)	Hex		11.4	11.4	9.3		6	1.48				3*3/1	737
Poly(L-serine)													
–, *O*-acetyl-													
β (129.12)	Ortho		9.71	6.48	9.6[b]		3	1.065				3*3/1	706
Poly(L-tyrosine)													
–, *O*-benzyloxycarbonyl- (297.31)	Mono		18.0	18.0	16.5	$\gamma = 122$	11	1.20				3*11/3	849
Poly(L-valine)													
α helix (99.13)	Hex		11.43	11.43	(27)		18	0.970				3*18/5	1025
β	Ortho		4.79	19.00	6.88		4	1.052				3*2/1	806
	Ortho	D2-4	4.80	19.14	6.59		4	1.088				3*2/1	1045

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 6. POLY(AMIDES) (–NH–X–NH–CO–Y–CO– or –NH–X–CO–) see also POLY(PEPTIDES)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(2-aminoacetic acid), see Polyglycine													
Poly(*p*-aminobenzoic acid)													
(119.12)	Ortho	D2-4	7.71	5.14	12.9		4	1.548		550/735		6*2/1	811
	Ortho	D2-4	7.71	5.14	12.8		4	1.560				6*2/1	902
	Ortho	D2-4	7.75	5.30	12.87		4	1.497				6*2/1	1434
Poly(3-amino-2-butenoic acid)													
irrad. in vacuum													
(83.09)	Mono		19.11	8.54	10.665	$\beta=97.6$						4*	1530
irrad. in air													
	Mono		21.90	10.13	12.461	$\beta=96.8$						4*	1530
Poly(3-aminobutyric acid)													
(85.11)	Ortho		10.9	9.6	4.6		4	1.17		330		4*1/1	503
Poly(4-aminobutyric acid)													
α	Mono		9.44	8.22	12.1[b]	$\gamma=116$	8	1.340		260/337		5*2/1	272
(85.11)	Mono	C2-2	9.29	7.97	12.24[b]	$\gamma=114.5$	8	1.375		260/244		5*2/1	457
	Mono		9.79	8.31	12.25	$\gamma=117$	8	1.273				5*2/1	1509
β	Mono				12.24[b]							5*2/1	458
δ	Hex		4.65	4.65									458
–, 4-benzyloxycarbonyl-(L)													
I. (219.24)	Mono		9.62	19.30	12.30	$\beta=71.1$	8	1.35				5*2/1	1359
II.	Hex		13.14	13.14	10.0		5	1.22				5*5/2	1359
–, 4-methyloxycarbonyl-(L)													
(143.14)	Hex		10.90	10.90	10.0		5	1.16				5*5/2	1359
Poly(10-aminocapric acid)													
α	Tri		9.80	5.12	27.54	54,90,110	4	1.110				11*2/1	1237
(169.27)													
γ	Phex	C2h-5	4.78	9.56	26.9	$\gamma=120$	4	1.056		188/337		11*2/1	971
	Hex		4.9	4.9	26.5		2	1.02		177/146		11*2/1	61
Poly(6-aminocaproic acid) [caprolactam]													
α	Mono	C2-2	9.56	8.01	17.24[b]	$\gamma=67.5$	8	1.232	1.09/839	215	20.8/155	7*2/1	3
(113.16)	Mono	C2-2	4.81	7.61	17.10[b]	$\gamma=79.5$	4	1.221		223/153	18.1/216	7*2/1	26
	Mono		9.65	8.11	17.2[b]	$\gamma=66.3$	8	1.220		223/244	23.0/405	7*2/1	135
	Mono	C2-2	9.45	8.02	17.08	$\gamma=68$	8	1.252		226/293	21.6/343	7*2/1	212
	Mono		9.66	8.32	17.0	$\gamma=65$	8	1.214		214/304	26/1121	7*2/1	334
	Mono		4.77	4.06	17.25	$\gamma=66.5$	2	1.224		272/839	19.4/839	7*2/1	891
	Mono	C2-2	9.71	8.19	17.40	$\gamma=115$	8	1.199		260/1051	27/1050	7*2/1	1049
									0.917	228/346	27.3		1050
								1.250	1.114	250/532	24.1/697		449
								1.23	1.10		18.5/534		213
β	Hex		4.8	4.8	8.6		1	1.10				7*1/1	222
	Hex		4.85	4.85	8.4		1	1.10				7*1/1	1052
γ	Mono	C2h-5	9.35	4.81	16.60[b]	$\gamma=120$	4	1.162	1.09/839	272/839	15.6/839	7*2/1	336
	Mono	C2h-5	9.33	4.78	16.88[b]	$\gamma=121$	4	1.165			27/1050	7*2/1	246
	Hex		4.79	4.79	16.7		2	1.132				7*2/1	272
	Ortho		4.82	7.82	16.70		4	1.194				7*2/1	279
	Mono		9.14	4.84	16.68	$\gamma=121$	4	1.188				7*2/1	697
	Ortho		4.83	7.83	16.68		4	1.191				7*2/1	697
>150°C	Rho		4.90	16.28	8.22		4	1.146				7*1/1	26
	Mono		4.86	4.65	16.87	$\gamma=60$	2	1.138				7*2/1	891
–, complex with I2													
	Mono	C2h-5	9.13	17.9	16.2[b]	$\gamma=59.7$							1510
	or	C2-2											
–, D-(-)-3-methyl-													
(127.19)	Mono	C2-2	9.15	4.84	16.96[b]	$\gamma=90$	4	1.125		225/19		7*2/1	440
										226/36			
Poly(6-aminocaproic acid-*alt*-11-aminoundecanoic acid)													
(296.46)					22.5					184		19*1/1	714
Poly(6-aminocaproyl-5-aminovaleryl-3-aminopropionic acid)													
(297.40)	Tri		4.9	10.7	21.6	51,77,62	2	1.28				18*1/1	1437
Poly(8-aminocaprylic acid)													
α	Mono	C2-2	9.8	8.3	22.4[b]	$\gamma=115$	8	1.14		185/146	30/1195	9*2/1	272
(141.21)								1.18		202			153
										200/244			
										218/838			

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TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
β	Mono												121
γ	Hex		4.79	4.79	21.7		2	1.088	1.04/495		18/1121	9*2/1	272
	Hex		4.9	4.9	21.7		2	1.04				9*2/1	61
	Phex	C2h-5	4.77	9.54	21.9	γ=120	4	1.087				9*2/1	971
Poly(2-aminocyclopentylenecarboxy acid)													
(111.14)			8.25	4.54	7.8		2	1.26				4*2/1	1059
Poly(7-aminoenanthic acid)													
(127.19)	Tri	C1-1	9.8	10.0	9.8	56,90,69	4	1.18		233/215		8*1/1	918
	Tri	Ci-1	4.9	5.4	9.85	49,77,63	1	1.21		225/146		8*1/1	61
										217			153
								1.20			223/291		
–, (R)-3-methyl-													
(141.21)					9.24					179			479
–, (S)-4-methyl-													
(141.21)					9.43					176			479
–, (R)-5-methyl-													
(141.21)					9.11					182			479
–, (R)-6-methyl-													
(141.21)					9.57					188			479
Poly(aminoformic acid)													
–, *N*-butyl-													
(99.13)	Phex		13.3	13.3	15.4		16	1.12		175/528		2*8/3	522
Poly(2-aminoisobutyric acid), see Poly(peptide) Section													
Poly(12-aminolauric acid)													
γ	Hex		4.80	4.80	32.1		2	1.023	0.972	208/1172		13*2/1	895
(197.32)	Hex		4.70	4.70	31		2	1.10		179/177		13*2/1	660
	Phex	C2h-5	4.79	9.58	31.9[b]	γ=120	4	1.034				13*2/1	680
	Mono		4.87	9.38	32.2[b]	γ=121.5	4	1.045				13*2/1	759
	Mono		4.90	4.67	32.1	γ=121.7	2	1.048	0.99			13*2/1	1054
	Mono		9.28	4.85	31.35	γ=124.4	4	1.126				13*2/1	1511
formed during planar extrusion													
	Tri		9.28	5.29	31.35	59,90,60	4	1.223				13*2/1	1511
Poly(9-aminopelargonic acid)													
(155.24)	Tri	Ci-1	4.9	5.4	12.5	49,77,64	1	1.15		210		10*1/1	421
	Tri	C1-1	9.7	9.7	12.6	64,90,67	4	1.07		194/146		10*1/1	918
										198/292			
										209/177			
Poly(2-aminopropionic acid), see Poly(alanine)													
Poly(3-aminopropionic acid) [β alanine]													
α	Mono	C2-2	9.33	8.73	4.78[b]	γ=120	4	1.400		340		4*1/1	491
(71.08)	Mono		9.60	8.96	4.78[b]	γ=122.5	4	1.361		330/337		4*1/1	1265
										357/1403			
β	Ortho		9.56	7.56	4.78[a]		4	1.366				4*1/1	1265
–, 2-butyl-2-methyl-													
(141.21)					8.3					72		4*2/1	1093
–, 2,2-dimethyl-													
I. (99.13)										189/73		4*	
II.					8.4					273	13		73
	Ortho		11.57	10.95	8.39		8	1.239		268		4*2/1	1093
										270/402			
–, 2-ethyl-2-methyl-													
(113.16)					8.5					76		4*2/1	1093
–, 3-methyl-, see Poly(3-aminobutyric acid)													
–, 2-methyl-2-propyl-													
(127.19)					8.4					74		4*2/1	1093
Poly(3-aminopropyl-5-aminovaleryl-6-aminocaproic acid)													
(297.40)	Tri		4.9	10.7	21.6	51,77,62	2	1.28				18*1/1	1437
Poly(18-aminostearic acid)													
γ	Mono		4.76	9.52	46.9	γ=120	4	1.016				19*2/1	1055
(281.48)													
Poly(2-aminotetrahydropyran-6-carboxcylic acid)													
(127.14)	Mono	C2-1	11.79	14.40	19.43	γ=98	20	1.292	Double	305		5*10/3	1333

TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(11-aminoundecanoic acid)													
α	Tri	C1-1	9.5	10.0	15.0	60,90,67	4	1.10		183/292	27/1161	12*1/1	918
(183.30)	Tri		4.78	4.13	13.1	90,75,66	1	1.343	1.01	220	41/343	12*1/1	365
	Tri	Ci-1	4.9	5.4	14.9	49,77,63	1	1.15		182/146		12*1/1	61
	Tri		9.6	4.2	15.0	72,90,64	2	1.19		194/101		12*1/1	59
	Tri		4.78	4.13	14.9	82,75,66	1	1.174		186		12*1/1	263
			4.78		14.1	α = 63.5				188/444 226/1161		12*1/1	1057
II.	Mono		9.75	8.02	15.0[b]	γ = 115	4	1.145				12*1/1	529
γ	Mono		9.48	4.5	29.4[b]	γ = 118.5	4	1.10				12*1/1	529
				7.68	14.78								1056
Poly(5-aminovaleric acid)													
α	Tri	C1-1	9.5	5.6	7.5	48,90,67	2	1.30		258/402		6*1/1	918
(99.13)													
Poly(4,4′-benzanilidylene terephthalamide)													
(357.37)					18.63							18*1/1	1425
Poly(1,3-cyclohexylenedimethylene adipamide)													
cis-													
(252.36)	Tri		4.8	12.6	29.6	26,85,75	2	1.19		253/1201		13*2/1	1058
Poly(1,3-cyclohexylenedimethylene dodecanediamide)													
cis-													
(336.52)	Tri		4.8	19.0	44.6	16,90,83	2	1.12				19*2/1	1058
Poly(1,3-cyclohexylenedimethylene suberamide)													
cis-													
(280.41)	Tri		4.8	14.5	34.8	22,85,76	2	1.15				15*2/1	1058
Poly(decamethylene adipamide) [10.6]													
(282.43)					20.0					230 236/244 240/544		18*1/1	110
Poly(decamethylene azelamide) [10.9]													
(324.51)										214/137	68.2/405 36.7/159	21*	
γ	Phex					2							65
Poly(4,4′-decamethylenedipiperazine sebacamide)													
(476.75)										130/159 129/402	69.1/159	28*	
Poly(decamethylene 4-octenediamide)													
trans-													
(308.47)				24.1						243		20*1/1	657
Poly(decamethylene phosphinylidenedipropionamide)													
–, methyl-(diacid)													
(344.44)				20						173		19*1/1	831
Poly(decamethylene sebacamide) [10.10]													
(338.54)	Tri		4.9	5.4	27.8	49,76,63	1	1.14		203/244	51.1/118	22*1/1	1412
					25.6					197	34.7/159	*1/1	110
										198/454	72.0/405		
										216/137	32.7/160		
Poly(decamethylene succinamide) [10.4]													
(254.37)	Tri		4.9	5.5	19.8	49,77,63	1	1.18		242		16*1/1	1626
Poly(dodecamethylene oxamide) [12.2]													
(254.37)					19.5					230		16*1/1	743
Poly(dodecamethylene phosphinylidenedipropionamide)													
–, methyl-(diacid)													
(372.49)					21.55					180			831
Poly(dodecamethylene malonamide) [12.3]													
(268.40)	Mono		8.48	4.71	41.30	β = 101	4	1.101				17*2/1	1523
Poly(dodecamethylene sebacamide) [12.10]													
α	Tri		5.0	5.25	29.2	51,75,63	1	1.15		173/339		24*1/1	1513
(366.59)													
β	Tri		5.0	8.16	29.2	90,75,68	1	1.15				24*1/1	1513
Poly(dodecamethylene succinamide) [12.4]													
(282.43)	Tri		4.9	5.5	22.3	49,77,63	1	1.17		237		18*1/1	1626

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TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly[4,4′-(ethylenedioxydiphenylene)-ditrimethylene adipamide]													
(438.57)					26.3							25*1/1	516
Poly(ethylene-L-tartaramide) –, di-*O*-methyl-													
(202.21)	Tri		4.95	6.87	9.10	59,90,106	1	1.337		312		8*1/1	1629
Poly(heptamethylene adipamide) [7.6]													
γ	Phex	C2-2					2			209/291			65
(240.35)										226/339			
										250/244			
Poly(heptamethylene azelamide) [7.9]													
γ	Phex						1			201/244			65
(282.43)													
Poly(heptamethylene pimelamide) [7.7]													
(254.37)	Phex	Cs-1	4.82	18.95[b]		γ=120	1	1.105		214/244		16*1/1	63
										205/2			
										196/339			
										228/1625			
γ	Phex				18.95		1					16*1/1	65
Poly(heptamethylene sebacamide) [7.10]													
γ	Phex	C2-2					2			187/339			65
(296.46)										208/244			
Poly(heptamethylene suberamide) [7.8]													
γ	Phex	C2-2					2			230/244			65
(268.40)													
Poly(hexamethylene adipamide) [6.6]													
α													
I.	Tri	Ci-1	4.9	5.4	17.2	48,77,63	1	1.24	1.09/52	265/2	46.5/82	14*1/1	25
(226.32)	Mono		15.7	10.5	17.3	β=73	9	1.240	1.12/495	270/289	40/155	14*1/1	402
	Tri		5.00	4.17	17.3	81,76,63	1	1.204		301/1166	36.8/216	14*1/1	407
	Tri		4.87	5.26	17.15	50,76,64	1	1.241		280/1121	68/1121	14*1/1	891
	Tri		4.97	5.47	17.29	48,77,62	1	1.214			58/1166	14*1/1	1180
	Tri		4.96	5.52	17.41	48,76,62	1	1.208			46.9/405	14*1/1	1409
								1.225	1.095		53.2		1179
								1.220	1.069				54
II.	Tri		4.95	5.45	17.12	52,80,63	1	1.152		269.5	43.4	14*1/1	1180
								1.165	1.095		41.9		1179
β	Tri	Ci-1	4.9	8.0	17.2	90,77,67	2	1.25	1.09/495			14*1/1	25
high temp.	Tri		5	5.9	16.23	57,80,60	1	1.10				14*1/1	1060
(170°C)	Tri		4.91	5.87	16.50	56,81,60	1	1.119				14*1/1	1409
Poly(hexamethylene azelamide) [6.9]													
(268.40)	Mono		7.8	5.3	40.15[b]	γ=87	4	1.08		226/244		17*2/1	402
										185/339			
Poly(hexamethylene glutaramide) [6.5]													
(212.29)	Mono	C2h-6	4.60	8.62	30.95[b]	β=114	4	1.258		241/402		13*2/1	1514
(at 190°C)	Mono		5.23	8.50	30.55	β=110	4	1.105				13*2/1	1514
Poly(hexamethylene malonamide) [6.3]													
(184.24)	Mono		8.60	4.62	25.70	β=100.4	4	1.218		241/1524		11*2/1	1523
Poly(hexamethylene 4-octenediamide) *trans*-													
(252.36)					19.1					259		16*1/1	657
Poly[hexamethylene oxamide) [6.2]													
(170.21)	Tri	Ci-1	5.15	7.54	12.39	32,74,62	1	1.280		320/402		10*1/1	956
Poly(hexamethylene *m*-phenylenedisulfonamide)													
(318.41)	Mono	C2h-3	7.70	7.76	14.1	β=117	2	1.409		170		13*1/1	1384
Poly(hexamethylene sebacamide) [6.10]													
α	Tri	Ci-1	4.95	5.4	22.4	49,76,63	1	1.16		228/51	30.6/160	18*1/1	25
(282.43)	Tri		4.86	5.05	22.35	55,76,62	1	1.189		215/291	56.5/137	18*1/1	891
								1.152	1.041	233/244	58.6/405		54
								1.189		225/454			52
								1.17		216			153
β	Tri	Ci-1	4.9	8.0	22.4	90,77,67	2	1.20				18*1/1	25

TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(hexamethylene suberamide) [6.8]													
(254.37)	Tri						1			232/244			65
	Mono		9.60	8.26	19.7	$\gamma = 115$	4	1.193		235/291 241/1202		16*1/1	1515
quenched	Phex		9.7	9.7		$\gamma = 120$							1515
Poly(hexamethylene succinamide) [6.4]													
(198.26)	Tri		4.9	5.3	14.8	51,77,62	1	1.25		275		12*1/1	1626
									212/402				
Poly(hexamethylene-L-tartaramide) –, di-*O*-methyl-													
(252.32)	Tri		5.00	6.84	13.20	62,90,112	1	1.191		230		12*1/1	1629
Poly(hexamethylene terephthalamide) [6.T]													
(246.31)					15.6					371 371/829 D350/291		14*1/1	828
Poly(methylene adipamide) [1.6]													
(156.18)	Mono	C2h-6	8.04	4.79	20.2[b]	$\gamma = 90$	4	1.333		285/1524 306/402		9*2/1	1522
Poly(4,4′-methylenedicyclohexylene azelamide) *trans-, trans-*													
(362.56)	Ortho		9.4	6.1	39.5		4	1.06				20*2/1	1315
Poly(4,4′-methylenedicyclohexylene dodecanediamide) *trans-, trans-*													
I.	Ortho	C2v-5	9.30	6.06	45.0		4	1.060				23*2/1	1315
(404.64)	Ortho		9.4	6.1	45.5		4	1.03				23*2/1	1315
II.	Mono		9.50	5.22	52.6	$\gamma = 96.6$	4	1.037				23*2/1	1315
Poly(4,4′-methylenedicyclohexylene pimelamide) *trans-, trans-*													
I.	Ortho		9.4	6.1	35.8		4	1.08				18*2/1	1315
(334.50)													
II.	Mono		9.57	5.32	40.5	$\gamma = 99.6$	4	1.093				18*2/1	1315
Poly(4,4′-methylenedicyclohexylene sebacamide) *trans-, trans-*													
(376.58)	Ortho		9.4	6.1	41.4		4	1.05				21*2/1	1315
Poly(4,4′-methylenedicyclohexylene suberamide) *trans-, trans-*													
(348.53)	Ortho		9.4	6.1	37.6		4	1.07				19*2/1	1315
Poly(4,4′-methylenedicyclohexylene tetradecanediamide) *trans-, trans-*													
(432.60)	Ortho		9.4	6.1	49.4		4	1.02				25*2/1	1315
Poly(4,4′-methylenedicyclohexylene tridecanediamide) *trans-, trans-*													
(418.66)	Ortho		9.4	6.1	47.4		4	1.02				24*2/1	1315
Poly(4,4′-methylenedicyclohexylene undecandediamide) *trans-, trans-*													
(390.61)	Ortho		9.4	6.1	43.5		4	1.04				22*2/1	1315
Poly(4,4′-methylenediphenylene dodecanediamide)													
I. (392.54)	Ortho		11.10	4.96	44.7		4	1.059		290 256/1443		23*2/1	1440
II.	Ortho		10.52	4.96	44.7		4	1.118				23*2/1	1440
III.	Ortho		9.64	4.96	44.7		4	1.220				23*2/1	1440
Poly(methylene dodecanediamide) [1.12]													
(240.35)	Mono	C2h-6	8.12	4.79	35.2[b]	$\gamma = 90$	3	1.166		259/1524		15*2/1	1522
Poly(methylene glutaramide) [1.5]													
(142.16)	Rho	D3-5	4.79	4.79	26.1		3	1.366				8*3/1	1522
Poly(methylene malonamide) [1.3]													
(114.10)	Hex		4.79	4.79	18.0		3	1.589				6*3/2	1313
Poly(methylene pimelamide) [1.7]													
(170.21)	Rho	D3-5	4.79	4.79	34.5		3	1.237				10*3/1	1522
Poly(methylene sebacamide) [1.10]													
(212.29)	Mono	C2h-6	8.10	4.79	30.0[b]	$\gamma = 90$	4	1.211		260/4024 266/1524 268/1537		13*2/1	1522

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TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(methylene suberamide) [1.8]													
(184.24)	Mono	C2h-6	8.08	4.79	25.2[b]	$\gamma=90$	4	1.255		276/1524		11*2/1	1522
Poly(nonamethylene azelamide) [9.9]													
(310.48)					24.0					177 189/2 165/339		20*1/1	110
γ	Phex						1						65
Poly(nonamethylene malonamide) [9.3]													
(226.32)	Ortho		8.32	4.71	32.70		4	1.173				14*2/1	1523
Poly(octamethylene azelamide) [8.9]													
γ	Phex						2			206/244			65
(296.46)													
Poly(octamethylene malonamide) [8.3]													
(212.29)	Mono		8.50	4.71	30.70	$\beta=101.7$	4	1.172		233/1524		13*2/1	1523
Poly(octamethylene suberamide) [8.8]													
(282.43)	Tri						1			216/291 225/244			65
Poly(octamethylene succinamide) [8.4]													
(226.32)	Tri		4.9	5.4	17.3	51,77,62	1	1.20		254		14*1/1	1626
Poly(octamethylene-L-tartaramide)													
–, di-*O*-methyl-													
(286.37)	Tri		5.00	6.82	15.7	57,90,111	1	1.175		208		14*1/1	1629
Poly(octamethylene terephthalamide) [8.T]													
(274.36)					17.9					335		16*1/1	828
Poly(pentamethylene azelamide) [5.9]													
(254.37)					19.5					179/291		16*1/1	2
Poly(pentamethylene glutaramide) [5.5]													
(198.26)	Mono	C2-3	8.30	4.79	13.8	$\gamma=90$	2	1.200		198/291 247/1625		12*1/1	1624
Poly(pentamethylene malonamide) [5.3]													
(170.21)	Ortho		8.47	4.62	22.40		4	1.290		195/402 248/1524		10*2/1	1523
Poly(pentamethylene pimelamide) [5.7]													
γ	Mono	Cs-2	4.83	9.35	16.62	$\gamma=58.9$	2	1.169		228		14*1/1	1210
(226.32)										183/291			
Poly(pentamethylene terephthalamide) [5.T]													
(232.28)					14.1					354 353/829		13*1/1	828
–, 2-methyl-(diamine)													
(246.31)	Ortho		4.61	36.7	26.5		16	1.460				13*2/1	1529
Poly(3,8-phenanthridinonediyl terephthalamide)													
(355.35)	Mono		8.1	6.5	17.2	$\gamma=60$	2	1.50				16*1/1	1063
Poly(*m*-phenylene adipamide)													
(218.26)					11.8					344 296/402		11*1/1	828
Poly(*p*-phenylene dodecanediamide)													
(302.42)					21.2							19*1/1	828
Poly(*m*-phenylene isophthalamide)													
(238.25)	Ortho		6.7	4.71	11.0		1	1.14		375/735		10*1/1	735
	Tri	C1-1	5.27	5.25	11.3	112,111,88	1	1.470		390/748		10*1/1	856
	Tri	C1-1	5.36	5.36	11.3	113,113,88	1	1.443		1.33		10*1/1	878
	Ortho		5.1	5	23.2		2	1.34				10*2/1	1062
–, hexamethylphosphorictriamide complex													
(596.66)	Mono		9.43	16.94	11.3	$\beta=123.4$	2	1.315					957
(2 HMPT/monomer)													
Poly(*p*-phenylene phthalamide)													
(238.25)	Ortho	D2h-14	22.8	5.5	8.1		4	1.56				10*1/1	396
Poly(*p*-phenylene pimelamide)													
(232.28)					14.7					372		14*1/1	828
Poly(*m*-phenylene sebacamide)													
(274.36)					14.9					258		17*1/1	828
Poly(*p*-phenylene sebacamide)													
(274.36)					18.5					334		18*1/1	828

TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(*m*-phenylene suberamide)													
(246.31)					13.0					302		15*1/1	828
Poly(*p*-phenylene suberamide)													
(246.31)					16.3					374		16*1/1	828
Poly(*m*-phenylene terephthalamide)													
(238.25)										500/748			
–, 4-methyl-(diamine)													
(252.27)	Mono		8.57	7.54	22.11	$\beta=116.3$	4	1.308				11*2/1	1398
Poly(*p*-phenylene terephthalamide)													
I. (238.25)	Port	Cs-1	7.87	5.18	12.9	$\gamma=90$	2	1.504		500/748		12*1/1	761
	Mono		7.728	5.184	12.81	$\gamma=90.04$	2	1.542		600/735		12*1/1	1061
	Mono	C2h-5	7.80	5.19	12.9	$\gamma=90$	2	1.515				12*1/1	902
	Ortho		7.78	5.28	12.9		2	1.493				12*1/1	1525
	Mono		7.79	5.18	12.89	$\gamma=92.2$	2	1.522				12*1/1	1527
	Port	Cs-2	7.88	5.22	12.9		2	1.491				12*1/1	1620
	Ortho		7.60	5.04	12.78		2	1.616				12*1/1	1638
				4.8	12.7	$\alpha=88$							1526
II.			8.0	5.1	12.9		2	1.50				12*1/1	958
				4.8	12.7	$\alpha=88$							1526
single crystals & spherulites													
					11.5								1297
–, 2,2′-difluoro-													
(274.23)	Port		8.6	5.0	13.0		2	1.63				12*1/1	1528
Poly(*p*-phenylene 2,5-thiophenediamide)													
(272.46)	Mono		4.29	5.09	25.2	$\gamma=98$	2	1.489				11*2/1	1532
Poly(piperazine adipamide)													
(196.25)					9.2					355		10*1/1	386
Poly(1,4-piperazine 1,3-cyclobutylenedicarboximide)													
–, 2,4-diphenyl-(diacid)													
trans-													
(346.43)	Ortho		8.74	5.93	9.22		1	1.204				9*1/1	934
Poly(1,4-piperazine 4-octenediamide)													
trans-													
(222.29)	Mono	C2h-5	4.54	10.00	13.50	$\beta=110.2$	2	1.283		252/657		12*1/1	873
Poly(piperazine sebacamide)													
(252.36)										180/81	26.0/81	16*	
Poly(2,5-pyrimidine terephthalamide)													
(240.22)	Ortho		7.3	5.1	12.8		2	1.67				12*1/1	1531
Poly(4,4′-sulfonyldiphenylene terephthalamide)													
(378.40)					15.45								1203
Poly(terephthaloyldihydrazo-di(*p*-aminobenzoyl) terephthalamide)													
(562.54)	Port		8.5	4.9	29.69		2	1.51				28*1/1	760
Poly(tetramethylene adipamide) [4.6]													
(198.26)	Tri	C1-1	4.95	5.47	14.66	48,78,64	1	1.245		295/291		12*1/1	1516
	Mono		9.6	8.26	14.7	$\gamma=115$	4	1.246		308/51		12*1/1	1517
	Tri		4.9	5.5	14.8	46,78,64	1	1.29		350	41.6	12*1/1	1655
										319/1518			
Poly(tetramethylene azelamide) [4.9]													
γ	Phex						2			223/339			65
(240.35)										253/402			
Poly(tetramethylene malonamide) [4.3]													
156.18*	Mono		8.65	4.62	20.40	$\beta=102.5$	4	1.303		277/1524		9*2/1	1523
Poly(tetramethylene 4-octenediamide)													
trans-													
(224.30)					17.0					294		14*1/1	657
Poly(tetramethylene succinamide) [4.4]													
(170.21)	Tri		4.9	5.5	12.3	49,77,62	1	1.29		d		10*1/1	1626
										287/402			
Poly(tetramethylene-L-tartaramide)													
–, di-*O*-methyl-													
(230.26)	Tri		5.00	7.06	10.80	59,90,110	1	1.270		266		10*1/1	1629

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TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(tridecamethylene tridecanediamide) [13.13]													
γ	Mono		9.22	4.94	34.47	$\beta=121.1$	2	1.044	1.01	>183	97.2	28*1/1	1519
(422.70)	Mono		4.88	4.73	34.0	$\gamma=121$	1	1.043		172/1521		28*1/1	1520
Poly(*m*-xylylene adipamide)													
(246.31)	Mono		5.10	4.70	15.2	$\gamma'=69.6$	1	1.198		246/326		13*1/1	84
	Tri	Ci-1	12.01	4.83	29.8	75,26,65	2	1.250		244/544		13*2/1	459
Poly(*p*-xylylene sebacamide)													
(302.42)	Tri		5.74	4.87	20.6	76,55,65	1	1.168		300/290 281/385 291/454 268/291		18*1/1	204

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 7. POLY(ESTERS) (–O–X–O–CO–Y–CO– or –O–X–CO–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(4,4′-bicyclohexylene sebacate)													
–, *trans*-, *trans*-													
(364.53)	Mono		11.16	5.27	19.48	$\beta=49.0$	2	1.400				20*1/1	1636
smectic	Mono		11.98	5.48	19.76	$\beta=52.5$	2	1.176				20*1/1	1636
Poly(4,4′-bicyclohexylene undecanedioate)													
–, *trans*-, *trans*-													
(378.55)	Mono		12.4	5.21	23.0	$\beta=43.5$	2	1.229				21*1/1	1636
smectic	Mono		12.3	5.55	20.6	$\beta=51.4$	2	1.144				21*1/1	1636
Poly(4,4′-biphenylene 4,4′-biphenylenedicarboxylate)													
(392.11)	Ortho		7.83	5.50	20.66		2	1.464				20*1/1	1622
Poly(4,4′-biphenylene terephthalate)													
(316.31)	Ortho		7.20	5.23								16*	1622
Poly(3,4′-carbonyldiphenylene terephthalate)													
(344.32)	Ortho	C2v-17	12.51	7.561	67.66		16	1.4292				16*4/1	1562
Poly(1,3-cyclobutylene carbonate)													
–, 2,2,4,4-tetramethyl-													
trans-													
(170.21)	Tri		9.25	8.28	(6.9)	$\gamma=96.5$		1.08		360		6*1/1	364
cis-													
	Ortho		9.16	8.22	12.9		4	1.164		253		6*2/1	364
Poly(1,4-cyclohexylene adipate)													
trans-													
(226.27)					13.5					225/762		12*1/1	763
Poly(1,4-cyclohexylenedimethylene adipate)													
trans-													
I. (254.33)	Mono	C2h-5	6.938	9.761	15.97	$\beta=41.4$	2	1.181		124/547		14*1/1	1262
cis-													
										55/547			
Poly(1,4-cyclohexylenedimethylene succinate)													
trans-													
(226.27)	Mono	C2h-5	6.486	9.482	13.51	$\beta=45.9$	2	1.259		147/547		12*1/1	1193
cis-													
										62/547			
Poly(1,4-cyclohexylenedimethylene terephthalate)													
trans-													
(274.32)	Tri		6.37	6.63	14.2	89,47,114	1	1.266		318/547		14*1/1	199
	Tri	Ci-1	6.46	6.65	14.2	89,47,115	1	1.260				14*1/1	1077
cis-													
	Tri		6.02	6.01	13.7	89,53,112	1	1.319		256/547		14*1/1	199

TABLE 6. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(tridecamethylene tridecanediamide) [13.13]													
γ	Mono		9.22	4.94	34.47	$\beta=121.1$	2	1.044	1.01	>183	97.2	28*1/1	1519
(422.70)	Mono		4.88	4.73	34.0	$\gamma=121$	1	1.043		172/1521		28*1/1	1520
Poly(*m*-xylylene adipamide)													
(246.31)	Mono		5.10	4.70	15.2	$\gamma'=69.6$	1	1.198		246/326		13*1/1	84
	Tri	Ci-1	12.01	4.83	29.8	75,26,65	2	1.250		244/544		13*2/1	459
Poly(*p*-xylylene sebacamide)													
(302.42)	Tri		5.74	4.87	20.6	76,55,65	1	1.168		300/290 281/385 291/454 268/291		18*1/1	204

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 7. POLY(ESTERS) (–O–X–O–CO–Y–CO– or –O–X–CO–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(4,4′-bicyclohexylene sebacate)													
–, *trans-*, *trans-*													
(364.53)	Mono		11.16	5.27	19.48	$\beta=49.0$	2	1.400				20*1/1	1636
smectic	Mono		11.98	5.48	19.76	$\beta=52.5$	2	1.176				20*1/1	1636
Poly(4,4′-bicyclohexylene undecanedioate)													
–, *trans-*, *trans-*													
(378.55)	Mono		12.4	5.21	23.0	$\beta=43.5$	2	1.229				21*1/1	1636
smectic	Mono		12.3	5.55	20.6	$\beta=51.4$	2	1.144				21*1/1	1636
Poly(4,4′-biphenylene 4,4′-biphenylenedicarboxylate)													
(392.11)	Ortho		7.83	5.50	20.66		2	1.464				20*1/1	1622
Poly(4,4′-biphenylene terephthalate)													
(316.31)	Ortho		7.20	5.23								16*	1622
Poly(3,4′-carbonyldiphenylene terephthalate)													
(344.32)	Ortho	C2v-17	12.51	7.561	67.66		16	1.4292				16*4/1	1562
Poly(1,3-cyclobutylene carbonate)													
–, 2,2,4,4-tetramethyl-													
trans-													
(170.21)	Tri		9.25	8.28	(6.9)	$\gamma=96.5$		1.08		360		6*1/1	364
cis-													
	Ortho		9.16	8.22	12.9		4	1.164		253		6*2/1	364
Poly(1,4-cyclohexylene adipate)													
trans-													
(226.27)					13.5					225/762		12*1/1	763
Poly(1,4-cyclohexylenedimethylene adipate)													
trans-													
I. (254.33)	Mono	C2h-5	6.938	9.761	15.97	$\beta=41.4$	2	1.181		124/547		14*1/1	1262
cis-													
										55/547			
Poly(1,4-cyclohexylenedimethylene succinate)													
trans-													
(226.27)	Mono	C2h-5	6.486	9.482	13.51	$\beta=45.9$	2	1.259		147/547		12*1/1	1193
cis-													
										62/547			
Poly(1,4-cyclohexylenedimethylene terephthalate)													
trans-													
(274.32)	Tri		6.37	6.63	14.2	89,47,114	1	1.266		318/547		14*1/1	199
	Tri	Ci-1	6.46	6.65	14.2	89,47,115	1	1.260				14*1/1	1077
cis-													
	Tri		6.02	6.01	13.7	89,53,112	1	1.319		256/547		14*1/1	199

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(1,4-cyclohexylene sebacate) *trans-*													
(282.38)					18.5					151/762		16*1/1	763
Poly(decamethylene adipate) [10.6]													
(284.40)	Mono		5.0	7.4	22.1		2	1.16		80/81	42.7/81	18*1/1	106
			5.11	7.43						74	44		1079
										77/300	45.6/405		
Poly(decamethylene azelate) [10.9]													
(326.48)	Mono		5.0	7.4	51.7		4	1.13		69/137	41.9/81	21*2/1	106
											50.7/405		
Poly(decamethylene glutarate) [10.5]													
(270.37)	Mono		5.0	7.4	41.6		4	1.17		58/764		17*2/1	106
Poly(decamethylene 3-hexenedioate)													
(282.38)			4.94	7.65						66	36	18*	1079
Poly(decamethylene octadecanedioate) [10.18]													
(452.72)	Mono		5.47	7.38	37.5	$\beta = 115$	2	1.096		93		30*1/1	521
Poly(decamethylene 4-octenedioate)													
(310.44)			4.74	4.05		$\gamma' = 109$				53	44	20*	1079
Poly(decamethylene oxalate) [10.2]													
(228.29)	Mono		5.28	7.00	17.0		2	1.207		79/305		14*1/1	106
	Mono		6.75	7.05	17.0	$\beta = 129$	2	1.206				14*1/1	1171
Poly(decamethylene sebacate) [10.10]													
(340.50)	Mono		5.0	7.4	27.1		2	1.13		80/137	50.2/81	22*1/1	106
										77/767	56.5/405		
										73/291	30.2/301		
											51.5/160		
Poly(decamethylene suberate) [10.8]													
(312.45)	Mono		5.0	7.4	24.6		2	1.14		75/764		20*1/1	106
			5.07	7.50						77	57		1079
Poly(decamethylene succinate) [10.4]													
(256.34)	Mono		5.0	7.4	19.6		2	1.17		68/290		16*1/1	106
										73/764			
Poly(decamethylene terephthalate) [10.T]													
(304.39)	Tri		4.62	6.30	20.10	107,96,113	1	1.022		138/81	46.1/81	18*1/1	68
										129/99	44.0/405		
										131/453			
Poly(decamethylene 4,4′-terephthaloyldioxydibenzoate)													
(544.60)	Ortho		53.7	39.9	57.7			1.287				30*2/1	1352
Poly(dodecamethylene adipate) [12.6]													
			5.04	7.42						78	52	20*	1079
Poly(dodecamethylene *N,N′*-dodecamethylenediterephthalamate)													
(662.91)	Tri		5.01	6.93	46.77	130,55,130	1	1.213		197/782	140/782	40*1/1	874
									1.124				1195
Poly(dodecamethylene 3-hexenedioate)													
(310.44)			4.94	7.64						77	46	20*	1079
Poly(dodecamethylene 4-octenedioate)													
(338.49)			4.74	4.07		$\gamma = 109$				57	55	22*	1079
Poly(dodecamethylene suberate) [12.8]													
(340.50)			5.03	7.52						80	68	22*	1079
Poly(ethylene adipate) [2.6]													
(172.18)	Mono		25.7	30.7	11.71	$\beta = 103.8$	40	1.275		52/27	15.9/265	10*1/1	105
	Mono	C2h-5	5.47	7.23	11.72	$\beta = 113.5$	2	1.345		54/291	21.0/571	10*1/1	203
	Mono		7.26	5.40	10.85	$\alpha = 67.7$	2	1.453		47/265		10*1/1	211
	Mono		5.47	7.24	11.55	$\beta = 113.5$	2	1.363		50/46		10*1/1	523
								1.302	1.204	55/543	21.7		1074
									1.219	65/571			1195
Poly(ethylene *N-N′*-azelaoyldi(*p*-aminobenzoate))													
(452.52)					26.3								1386
Poly(ethylene azelate) [2.9]													
(214.26)	Ortho		7.45	4.97	31.5		4	1.220		46/543		13*2/1	108
	Mono		25.7	30.7	31.2	$\beta = 103.8$	80	1.190				13*2/1	105
	Ortho		5.0	7.4	31.2		4	1.23				13*2/1	109
								1.233	1.119	47	43.7		1074

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TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(ethylene 4,4′-dibenzoate)													
(268.27)	Tri	Ci-1	5.73	3.77	14.73	90,90,79	1	1.426		346/291		14*1/1	1551
										333/1552			
Poly(ethylene *N*,*N*′-dodecamethylenediterephthalamate)													
(522.64)										244	102.6	30*	782
									1.210				1195
Poly(ethylene 4,4′-(ethylenedioxy)-dibenzoate)													
α	Mono		7.83	10.33	18.68	β=83.1	4	1.454		288	26.2	18*1/1	1078
(328.32)										240/290			
										240/291			
β	Ortho		7.28	5.65	18.64		2	1.422				18*1/1	
Poly(ethylene 2,5-furandicarboxylate)													
(182.13)	Tri		5.75	5.35	20.1	134,90,112	2	1.575				9*2/1	742
Poly(ethylene *N*,*N*′-hexadecanedioyldi(*p*-aminobenzoate))													
(550.70)					34.6								1386
Poly(ethylene isophthalate) [2.1]													
I. (192.17)	Tri	Ci-1	5.20	7.08	14.8	109,136,96	2	2.034				9*2/1	
					14.8					143			398
								1.358	1.346	240			111
II.					21.0								398
	Tri	Ci-1	5.41	6.35	21.2	116,136,84	2	1.478				9*2/1	572
Poly(ethylene 2,6-naphthalate)													
(242.23)	Tri	Ci-1	6.51	5.75	13.2	81,114,100	1	1.406		260/291		12*1/1	920
			3.825	5.72	13.2	γ′=96	1	1.400		337/1342	25/1342	12*1/1	1555
Poly(ethylene oxalate) [2.2]													
(116.07)	Ortho	D2h-14	6.44	6.22	11.93		4	1.613		172/678		6*2/1	675
Poly(ethylene *p*-oxybenzoate)													
α	Ortho	D2-4	10.52	4.75	15.68		4	1.392		203/327		9*2/1	494
(164.16)	Ortho	D2-4	10.49	4.75	15.60		4	1.403		224/345		9*2/1	901
										220/354			356
										215/1174			
β			11.2	14.7	18.99		16	1.395				9*2/1	494
	Mono	C2h-5	8.19	11.07	19.05	β=114.8	8	1.391				9*2/1	919
Poly(ethylene pimelate) [2.7]													
(186.21)								4.278	1.162	29/764	29.8	11*	1074
Poly(ethylene 1,4-piperazinedicarboxylate)													
(200.19)					10.4					245		10*1/1	386
Poly(ethylene sebacate) [2.10]													
I. (228.29)	Mono		5.5	15	16.9	β=65	4	1.20		72/265	13.8/265	14*1/1	100
	Mono	C2h-5	5.58	7.31	16.76	β=115.5	2	1.229		79/46	35/405	14*1/1	521
	Mono		5.52	7.30	16.65	β=115.0	2	1.247		73/539	32.0/571	14*1/1	523
	Mono		5.52	7.4	16.9	β=65	2	1.21		76/137	29.1/137	14*1/1	109
	Mono		25.7	30.7	16.67	β=103.8	40	1.187		78/302	25.6/158	14*1/1	105
									1.086	83/571			1195
II. Tri		Ci-1	5.39	7.60	16.76	105,112,72	2	1.264				14*1/1	1072
Poly(ethylene suberate) [2.8]													
(200.23)	Mono	C2h-5	5.51	7.25	14.28	β=114.5	2	1.281		75/571	26.5/571	12*1/1	203
	Mono		5.0	7.4	14.1		2	1.27		55/27		12*1/1	108
	Mono		5.50	7.25	14.10	β=114.5	2	1.300				12*1/1	523
								1.249	1.140	63	26.2		1074
										1.147			1195
Poly(ethylene succinate) [2.4]													
α	Mono		5.0	7.4	8.32		2	1.55		108/290		8*1/1	108
(144.13)	Mono		9.05	11.09	8.32	β=102.8	4	1.176		103/302		8*1/1	105
	Ortho	D2h-10	7.60	10.75	8.33		4	1.407				8*1/1	675
β					9.50							8*1/1	1539
Poly(ethylene terephthalate) [2.T]													
(192.17)	Tri	Ci-1	4.56	5.94	10.75	98,118,112	1	1.457	1.335	265	24.1/87	10*1/1	27
	Tri		5.54	4.14	10.86	107,112,92	1	1.472	1.337	284/264	22.6/157	10*1/1	195
	Tri		4.52	5.98	10.77	101,118,111	1	1.477		267/265	9:2/265	10*1/1	400
	Tri		4.48	5.85	10.75	100,118,111	1	1.515		270/374	25/405	10*1/1	853
	Tri		4.48	5.80	10.71	100,118,111	1	1.530		265/290	16.7/155	10*1/1	1061
	Tri		4.50	5.90	10.76	100,119,111	1	1.503		310	32/1071	10*1/1	1075

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Tri		4.44	5.91	10.67	100,117,112	1	1.515		280/960	26.9/960	10*1/1	1158
	Tri		4.52	5.92	10.70	100,118,111	1	1.480		245/472	27.8/472	10*1/1	1159
	Tri		4.53	5.87	10.73	100,119,111	1	1.501		278/724	23.3/887	10*1/1	1543
			4.01	5.83	10.75	$\gamma' = 120$	1	1.466				10*1/1	1555
	Tri		4.62	5.92	10.68	100,128,105	1	1.58				10*1/1	1644
Poly(glycolic acid), see Poly(2-hydroxyacetic acid)													
Poly(heptamethylene sebacate) [7.10]													
(298.42)											25/301	19*	
Poly(heptamethylene terephthalate) [7.T]													
(262.31)					15.0					98/453		13*1/1	1174
Poly[heptamethylene (4,4′-terephthaloyldioxy)-4,4′-dibenzoate]													
(502.52)					49.2							27*2/1	1561
Poly(hexadecamethylene adipate) [16.6]													
(368.56)			5.05	7.46						86	63	24*	1079
Poly(hexadecamethylene 3-hexenedioate)													
(366.54)			4.93	7.59						87	56	24*	1079
Poly(hexadecamethylene 4-octenedioate)													
trans-													
(394.60)			4.87	4.03		$\gamma = 109$				70	54	26*	1079
										85/879	54.5/879		
Poly(hexadecamethylene suberate) [16.8]													
(396.61)	Mono		6.47	7.38	37	$\beta = 115$	2	0.823?		109	66.4	26*1/1	879
			5.00	7.40						87	66	26*	1079
Poly(hexamethylene adipate) [6.6]													
(228.29)	Ortho		10.08	7.32	16.83		4	1.221	1.096	56/290	38.1	14*1/1	1119
										58/1079	32/1079		
										57/539			
										58/764			
										61/807			
Poly(hexamethylene 4,4′-dibenzoate)													
α	Mono	C2-2	10.98	11.47	19.62	$\beta = 89.7$	6	1.308	1.22	214/291		18*1/1	1556
(324.38)										200/1552			
										230/1566			
β	Mono		13.39	12.84	39.19	$\beta = 84.7$	16	1.284				18*2/1	1556
γ	Mono	C2-2	9.70	9.20	19.39	$\beta = 83.0$	4	1.254				18*1/1	1556
Poly(hexamethylenedi(6-hydroxycaproamide) adipate)													
(454.61)					37.50					142		26*	1105
Poly(hexamethylene *N*-*N*′-dodecamethylenediterephthalamate)													
(578.75)	Tri		5.02	6.87	39.27	128,54,131	1	1.252		214/782	116/782	34*1/1	874
									1.167				1195
Poly(hexamethylene *N*,*N*′-hexamethylenediterephthalamate)													
(494.59)	Tri	Ci-1	5.07	6.87	31.78	127,53,132	1	1.314		253/782	92.9/782	28*1/1	874
									1.236	265/781			1195
Poly(hexamethylene 3-hexenedioate)													
(226.27)										61	21	14*	1079
Poly(hexamethylene 4-octenedioate)													
(254.33)										20	22	16*	1079
Poly(hexamethylene oxalate) [6.2]													
(172.18)	Mono		6.75	6.85	11.95	$\beta = 129$	2	1.332		66/305		10*1/1	1171
										72/764			
Poly(hexamethylene sebacate) [6.10]													
(284.40)	Mono		5.52	7.40	22.15	$\beta = 115$	2	1.152		78	38/809	18*1/1	521
										67/290	22/301		
Poly(hexamethylene suberate) [6.8]													
(256.34)										67	34	16*	1079
										58/291			
										65/764			
Poly(hexamethylene terephthalate) [6.T]													
(248.28)	Tri		4.57	6.10	15.40	105,98,114	1	1.146		160/81	34.8/81	14*1/1	68
	Tri	Ci-1	9.98	9.52	15.40	120,98,95	4	1.133	1.225	161/405	33.5/405	14*1/1	814
										154/99	35.3/159		
α	Mono		9.1	17.2	15.5	$\alpha = 127.3$	6	1.28				14*1/1	1070
	Mono		9.100	17.56	15.74	$\alpha = 127.8$	6	1.245				14*1/1	1184

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
β													
I.	Tri		4.8	5.7	15.7	104,161,108	1	1.25				14*1/1	1070
	Tri		4.91	5.68	15.44	103,118,109	1	1.285				14*1/1	1160
	Tri	Ci-1	5.217	5.284	15.74	129,98,96	1	1.277		174		14*1/1	1184
	Tri	Ci-1	4.68	11.57	15.51	105,114,109	2	1.279				14*1/1	1544
(at 135°C)	Tri	Ci-1	4.84	5.74	15.53	104,116,106	1	1.227				14*1/1	1544
II.	Tri		5.217	10.57	15.74	129,98,96	2	1.277				14*1/1	1184
γ	Tri		5.3	13.9	15.5	124,130,88	2	1.30				14*1/1	1070
(from soln.)													
Poly(2-hydroxyacetic acid) [glycolic acid]													
(58.04)	Ortho	D2-4	6.36	5.13	7.04		4	1.678		223/305	11/1121	3*2/1	312
	Ortho	D2h-16	5.22	6.19	7.02		4	1.700		233	12/464	3*2/1	490
					7		2	1.69				3*2/1	605
								1.707	1.50	230			465
Poly(*p*-hydroxybenzoic acid)													
I. (120.11)	Hex	D3-7	17.8	17.8	18.4		36	1.422		>320/881		vi ref	880
	Ortho	D2-4	7.52	5.70	12.49		4	1.490		>350/882		6*2/1	1065
	Ortho	D2-4	7.62	5.70	12.56		4	1.462				6*2/1	1182
	Ortho		7.47	5.67	12.55		4	1.501				6*2/1	1340
	Ortho		7.5	5.7	12.6		4	1.48				6*2/1	1545
II.					12.6							6*2/1	1065
	Ortho		11.1	3.7	12.6		4	1.54				6*2/1	1545
III.	Ortho		9.2	5.3	12.4		4	1.32				6*2/1	1065
(high temp.)	Ortho		9.24	5.28	12.50		4	1.308				6*2/1	1340
			9.11	5.26									1545
–, 3-*n*-decyl-													
(260.38)	Ortho		26.80	9.6	12.6		8	1.07				6*2/1	1646
–, 3-*n*-hexyl-													
(204.27)	Ortho		19.80	9.6	12.6		8	1.13				6*2/1	1646
–, 3-*n*-pentyl-													
(190.24)	Ortho		17.80	9.6	12.6		8	1.17				6*2/1	1646
–, 3-*n*-propyl-													
Ia.	Tet		16.95	16.95	12.36		16	1.213		294		6*2/1	1646
(162.19)													
Ib.	Tet		16.95	16.95	12.86		16	1.166				6*2/1	1646
II.	Ortho		14.20	9.60	12.48		8	1.266				6*2/1	1646
Poly(4,4′-hydroxybibenzoic acid)													
I. (196.20)	Ortho		7.8	5.55	10.8		2	1.39				10*1/1	1546
	Ortho		7.44	5.79	10.90		2	1.388		585		10*1/1	1550
	Ortho		7.32	5.35	10.44		2	1.594				10*1/1	1554
	Ortho		7.5	5.7	10.9		2	1.40				10*1/1	1622
(at 280°C)	Ortho		9.18	5.30	10.90		2	1.229				10*1/1	1550
II.			15.6	3.6									1546
III.	Ortho		9.0	5.2	10.8		2	1.29				10*1/1	1546
	Ortho		8.86	5.12	10.90		2	1.318				10*1/1	1550
–, 3-bromo-													
(275.10)	Tri	Ci-1	12.71	3.85	10.50	90,90,90	2	1.778				10*1/1	1639
Poly(3-hydroxy-3-butenoic acid)													
(82.07)	Mono		5.50	9.06	7.78[b]	γ = 92	4	1.441		115		4*2/1	217
	Ortho	D2-3	5.43	8.94	7.75		4	1.484		115/190		4*2/1	862
Poly(3-hydroxybutyric acid)													
I. (86.09)	Ortho	D2-4	5.76	13.20	5.96		4	1.262		176/422	13/1216	4*2/1	419
	Ortho	D2-4	5.76	13.20	5.96		4	1.262		168/570		4*2/1	771
	Ortho		5.71	13.06	5.89		4	1.302		170/900		4*2/1	698
	Ortho	D2-4	5.73	13.14	5.93		4	1.281				4*2/1	1382
	Ortho		5.73	13.15	5.96		4	1.273		203/1535		4*2/1	1534
									1.177	197			1216
II.					4.70								771
Poly(10-hydroxycapric acid)													
(170.25)	Ortho		7.45	4.96	27.1		4	1.129		80/309		11*2/1	108
Poly(6-hydroxycaproic acid)													
(114.14)	Ortho	D2-4	7.496	4.974	17.30		4	1.175	1.09/712	55/560	15.4/623	7*2/1	559
	Ortho	D2-4	7.47	4.98	17.05		4	1.195		64/623		7*2/1	561

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Ortho	D2-4	7.48	4.98	17.26		4	1.179		69/1368		7*2/1	1420
											16.4		1368
Poly(5-hydroxy-2-(1,3-dioxane)-caprylic acid)													
trans-													
(228.29)	Ortho		7.487	5.117	13.56		2	1.459	1.093	150/797		13*1/1	796
cis-													
	Mono		7.842	5.476	12.34	$\gamma=96.45$	2	1.440	1.093	120/797		13*1/1	796
Poly(4-hydroxymethylene-2-(1,3-dioxolane)-caprylic acid)													
cis-/trans-													
(228.29)	Ortho		7.367	5.186	13.52		2	1.468	1.093	53/797		13*1/1	796
Poly(2,6-hydroxynaphthoic acid)													
I.	Mono		7.8	12.3	17.4	$\beta=97$	8	1.36				8*2/1	1547
(170.17)	Ortho	C2v-5	7.66	5.98	17.12		4	1.441				8*2/1	1548
(at 370°C)	Ortho	C2v-21	9.28	5.64	17.04		4	1.267				8*2/1	1548
II.			11.6	3.85									1547
–, 5-bromo-													
(249.06)	Mono	C2h-5	13.02	4.06	17.15	$\beta=101$	4	1.859				8*2/1	1639
Poly(2-hydroxypropionic acid) [lactic acid]													
α	Port		10.7	6.45	27.8		20	1.247				3*10/3	663
(72.06)	Ortho		10.34	5.97						215			1064
								1.290	1.248		6.69		1175
	Ortho		10.37	5.98	28.0		20	1.378				3*10/3	1557
										184	14.6		1367
β					8.8							3*3/1	1557
D,L-stereocomplex													
	Tri	C1-1	9.16	9.16	8.70	109,109,110	6	1.274				3*3/1	1557
Poly(3-hydroxypropionic acid) [propiolactone]													
α					7.02					122/317	9.1/1121	4*2/1	314
(72.06)									1.360	84/623	8.6/623		1195
β					4.82							4*1/1	314
			7.76	4.50	4.76		2	1.440				4*1/1	689
			7.73	4.48	4.77	$\gamma'=90$	2	1.449		120		4*1/1	840
–, 3-chloromethyl-													
isotactic					6.19					146			900
(120.54)										275/1195			
–, 3-dichloromethyl-													
isotactic					7.89							4*1/1	1090
(154.98)					5.77					146			900
syndiotactic					11.55							8*1/1	1090
(319.96)													
– 2,2-diethyl-													
α	Ortho		8.3	10.2	4.5		2	1.12		258/1088	21/1088	4*1/1	1087
(128.17)													
β					4.63							4*1/1	1087
–, 2,2-dimethyl-[pivalolactone]													
α													
(100.12)	Mono	C2h-5	9.02	11.64	6.02	$\gamma=121.5$	4	1.234	1.10/551	240/551	14.9/551	4*2/1	321
	Mono	C2h-5	9.05	11.58	6.03	$\gamma=121.5$	4	1.234		268/1085		4*2/1	683
	Mono	C2h-5	9.03	11.62	6.01[b]	$\gamma=121.5$	4	1.237		269/1411		4*2/1	23
β					4.76					234		4*1/1	816
γ	Ortho	D2-4	8.23	11.27	6.04		4	1.187				4*2/1	1092
	Ortho	D2-4	8.23	11.28	6.02		4	1.190				4*2/1	1390
–, 3-ethyl-[3-hydroxyvaleric acid]													
isotactic													
(100.12)	Ortho	D2-4	9.52	10.08	5.56		4	1.246		110/900		4*2/1	1183
(racemic)	Ortho		9.32	10.02	5.56		4	1.281		108		4*2/1	810
										130/1538			
–, 2-ethyl-2-methyl-													
isotactic													
β					4.75					120		4*1/1	1089
(114.14)	Port	C2-1	9.10	7.44	4.75	$\beta=90$	2	1.179		164/1211		4*1/1	1536
–, D,L-2-ethyl-2-methyl-													
	Ortho		11.8	34.2	7.1		16	1.06				4*2/1	1536

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TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 3-isopropyl-isotactic													
(114.14)	Ortho		10.63	18.13	6.49		8	1.212		89 79/900		4*2/1	698
–, 2-methyl-													
opt. active										95	5.2		1341
racemic										100	4.8		1341
(86.09)													
–, 3-methyl-, see Poly(3-hydroxybutyric acid)													
–, 2-methyl-2-propyl-isotactic													
(128.17)	Ortho	D2-3	10.6	11.1	6.7		4	1.08				4*2/1	1536
		or D2-4								152	14.6		1651
–, D,L-2-methyl-2-propyl-													
I. (128.17)					5.9					110		4*2/1	785
	Ortho		17.9	13.5	12.0		16	1.174		97/959		8*2/1	1086
II.	Ortho		7.48	10.1	4.74		2	1.19				4*1/1	785
–, 3-trichloromethyl-isotactic	Mono		11.79	10.99	11.24	$\gamma = 108.0$	8	1.817		275/1199		4*2/1	1091
(189.42)					11.27								1090
Poly(3-hydroxyvaleric acid), see Poly(3-hydroxypropionic acid), 3-ethyl-													
Poly(4,4′-isopropylidenediphenylene adipate)													
–, 3,3′, 5,5′-tetrachloro-													
(476.18)										283	33.9	17*	713
Poly(4,4′-isopropylidenediphenylene carbonate)													
(254.28)	Ortho	D2-2	11.9	10.1	21.5		8	1.307		267/57	27.9/566	12*2/1	5
	Mono		12.3	10.1	20.8	$\gamma = 84$	8	1.314	1.20	230	36.8/299	12*2/1	435
	Ortho		12.1	10.1	22.0		8	1.256		263/370		12*2/1	1549
								1.30	1.20	317/930 230/438 260/736 295/1121 302/1233 335/1066			283
Poly(lactic acid), see Poly(2-hydroxypropionic acid)													
Poly(4,4′-methylenediphenylene carbonate)													
(226.23)	Ortho	C2v-9	5.0	10.5	22.0		4	1.301	1.240	240 300/438 278/736		12*2/1	435
Poly[nonamethylene *N,N′*-adipolydi(*p*-aminobenzoate)]													
(508.62)					31.0								1386
Poly[nonamethylene *N,N′*-azelaoyldi(*p*-aminobenzoate)]													
(550.70)					33.3								1386
Poly(nonamethylene azelate) [9.9]													
(312.45)										65/405	43.1/159 49.0/405	20*	
Poly[nonamethylene *N,N′*-glutaryldi(*p*-aminobenzoate)]													
(494.59)					29.8								1386
Poly[nonamethylene *N,N′*-hexadecanedioyldi(*p*-aminobenzoate)]													
(648.88)					41.0								1386
Poly[nonamethylene *N,N′*-sebacoyldi(*p*-aminobenzoate)]													
(564.72)					34.3								1386
Poly(nonamethylene terephthalate) [9.T]													
(290.36)					17.6					85/99 90/453		17*1/1	1174
Poly[nonamethylene *N,N′*-tetradecanedioyldi(*p*-aminobenzoate)]													
(620.83)					38.6								1386
Poly[nonamethylene *N,N′*-tridecanedioyldi(*p*-aminobenzoate)]													
(606.80)					37.5								1386
Poly(octamethylene adipate) [8.6]													
(256.34)			5.10	7.42						67	41	16*	1079
Poly(octamethylene 3-hexenedioate)													
(254.33)										66	36	16*	1079

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(octamethylene 4-octenedioate)													
(282.38)			4.81	4.07		$\gamma = 109$				39	34	18*	1079
Poly(octamethylene suberate) [8.8]													
(284.40)			5.05	7.44						70	42	18*	1079
Poly(octamethylene terephthalate) [8.T]													
(276.33)					18.1					132/99		16*1/1	1174
Poly(oxydiethylene sebacate)													
(272.34)	Tet		17.6	17.6	38.0		32	1.229		44/325 47/767 46/765		17*2/1	105
Poly(oxydiethylene terephthalate)													
(236.22)										100 139/1654	44.4	13*	1653
Poly[pentamethylene *N,N'*-hexadecanedioyldi(*p*-aminobenzoate)]													
(592.78)					35.6								1386
Poly(pentamethylene pimelate) [5.7]													
(228.29)	Mono		6.14	7.37	17.16	$\beta = 54.6$	2	1.198		43/1120		14*1/1	1119
Poly(pentamethylene terephthalate) [5.T]													
α	Tri	Ci-1	6.21	4.56	12.50	89,114,116	1	1.354		116/291		13*1/1	814
(234.25)	Tri		4.69	5.79	24.66	112,94,105	2	1.318		134/99		13*2/1	1069
	Tri		4.68	5.77	24.42	114,87,106	2	1.341		140/815 142/807		13*2/1	1160
β	Tri		4.96	5.82	28.16	126,74,120	2	1.358				13*2/1	1069
Poly[pentamethylene *N,N'*-tridecanedioyldi(*p*-aminobenzoate)]													
(550.70)					32.0								1386
Poly(*p*-phenylene adipate)													
(220.22)	Tri		14.37	13.55	10.24	125,44,111	4	1.283			20.8	12*1/1	869
Poly(*p*-phenylene 4,4'-dibenzoxyterephthalate)													
(480.43)	Mono		7.83	5.64	26.37	$\beta = 97.9$	2	1.383				24*1/1	1621
Poly(*p*-phenylene *m*-carboranedicarboxylate)													
(306.32)	Ortho		11.4	7.7	18.0		4	1.29	1.172	280			832
Poly(*p*-phenylene isophthalate)													
I. (240.22)	Mono		6.97	6.92	24.32	$\alpha = 95$	4	1.365		425		11*2/1	1553
II.	Ortho		5.58	3.92	24.32		2	1.500				11*2/1	1553
Poly(*m*-phenylene terephthalate)													
(240.22)	Ortho		5.61	3.96	23.90		2	1.502				11*2/1	1553
Poly(*p*-phenylene terephthalate)													
(240.22)	Mono		7.98	5.33	12.64	$\beta = 98.98$	2	1.502		600/1375		12*1/1	1406
	Mono		7.85	5.48	25.38	$\beta = 98.6$	4	1.478				12*2/1	1621
(at 475°C)	Ortho		9.25	5.40	12.56		2	1.272				12*1/1	1406
–, 2,5-di-*n*-hexadecyl-(diacid)													
(721.08)	Mono		26.2	7.3	12.6	$\alpha = 130$	2	1.30				12*1/1	1641
	Ortho		25.6	7.2	12.734[b]		2	1.02				12*1/1	
–, phenyl- (diol)													
(316.31)	Mono	C2-2	12.77	10.08	12.58	$\gamma = 90$	4	1.297		340/1375		12*1/1	1374
	Mono		28.0	4.89	12.48	$\gamma = 114.8$	4	1.354				12*1/1	1558
–, chloro- (diol)													
(274.66)	Ortho		7.88	5.27	12.62		2	1.740				12*1/1	1429
Poly[(phenyl-*p*-phenylene terephthalate)-co-(1-phenylethyl-*p*-phenylene terephthalate)]													
(330.34)	Ortho	C2-2	12.77	10.18	12.58		4	1.342				12*1/1	1389
as spun	Mono		13.04	9.514	12.24	$\gamma = 80.87$	4	1.463				12*1/1	1426
annealed	Mono		13.31	9.732	12.15	$\gamma = 81.01$	4	1.411				12*1/1	1426
Poly(*p*-phenylene thioterephthalate)													
–, phenyl- (dithio acid)													
(348.43)	Mono		29.9	4.81	12.57	$\gamma = 110.7$	4	1.368				12*1/1	1558
Poly(pivalolactone, see Poly(3-hydroxypropionic acid), 2,2-dimethyl-													
Poly(propiolactone), see Poly(3-hydroxypropionic acid)													
Poly(terephthalic anhydride)													
(148.12)	Ortho		6.04	3.97			2			410/1564		7*2/1	1563
	Mono		6.02	3.76	13.49	$\alpha = 90$	2	1.611		400/1565		7*2/1	1645

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TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(tetradecamethylene adipate) [14.6]													
(340.50)			5.00	7.40						80	59	22*	1079
Poly(tetradecamethylene 3-hexenedioate)													
(338.49)			4.94	7.64						80	49	22*	1079
Poly(tetradecamethylene 4-octenedioate)													
(366.54)			4.91	4.05		$\gamma' = 109$				63	55	24*	1079
Poly(tetradecamethylene suberate) [14.8]													
(368.56)			5.03	7.46						82	66	24*	1079
Poly(tetramethylene adipate) [4.6]													
α	Mono		6.73	7.94	14.20	$\beta = 45.5$	2	1.229		48/325		12*1/1	1073
(200.23)										54/539			
										60/766			
										45/696			
β	Ortho		5.062	7.325	14.67		2	1.222				12*1/1	1073
Poly[tetramethylene *N,N'*-azelaoyldi (*p*-aminobenzoate)]													
(480.56)					26.6								1386
Poly[tetramethylene *N,N'*-hexadecanedioyldi(*p*-aminobenzoate)]													
(578.75)					35.2								1386
Poly(tetramethylene isophthalate) [4.1]													
(220.22)								1.309	1.268	152.5	42.3	11*2/1	111
					26.0					150			807
Poly(tetramethylene sebacate) [4.10]													
(256.34)										60//325	17/301	16*	
										62/425	2.08/425		
										67/766			
Poly(tetramethylene succinate) [4.4]													
(172.18)	Mono	C2h-5	5.21	9.14	10.94	$\beta = 124$	2	1.324		34/696		10*1/1	689
	Mono		5.23	9.08	10.79	$\beta = 123.9$	2	1.344		114/766		10*1/1	1540
					10.90								1541
β					11.90							10*1/1	1541
Poly(tetramethylene terephthalate) [4.T]													
α	Tri		4.830	5.953	11.67	100,64,69	1	1.395	1.28/824	234/807	31/405	12*1/1	786
(220.22)	Tri	Ci-1	5.96	4.87	11.71	116,100,110	1	1.394		232/99	32/111	12*1/1	814
	Tri		4.87	5.99	11.67	100,116,111	1	1.392		221/453		12*1/1	933
	Tri	Ci-1	5.96	4.83	11.62	115,100,111	1	1.406		245/1362		12*1/1	851
	Tri		4.83	6.05	11.45	100,117,111	1	1.440				12*1/1	860
	Tri	Ci-1	4.83	5.94	11.59	100,115,111	1	1.403		224		12*1/1	864
	Tri		4.89	5.95	11.67	98,116,110	1	1.36				12*1/1	883
	Tri		4.82	5.93	11.74	100,116,111	1	1.404				12*1/1	1158
								1.08/111	1.256		30.8		1071
β	Tri		4.703	6.738	13.12	55,58,56	1	1.355				14*1/1	786
	Tri	Ci-1	4.95	5.67	12.95	102,122,100	1	1.283				14*1/1	864
	Tri		4.7	5.8	13.0	102,121,105	1	1.37				14*1/1	883
	Tri		4.73	5.83	12.90	102,119,105	1	1.332				14*1/1	933
	Tri		4.73	5.75	13.11	104,121,101	1	1.330				14*1/1	1160
	Tri		4.73	5.88	13.06	103,120,104	1	1.320				14*1/1	1339
Poly[tetramethylene (4,4'-terephthaloyldioxy)-4,4'-dibenzoate]													
(460.44)	Mono	C2-2	8.42	5.79	24.83	$\beta = 101.8$	2	1.290				24*1/1	1559
Poly(4,4'-thiodiphenylene carbonate)													
(244.27)	Ortho	C2v-9	5.6	8.7	22.2		4	1.50	1.35	220		12*2/1	435
										240/540			
Poly(trimethylene adipate) [3.6]													
(186.21)	Mono		5.0	7.4	21.5		4	1.55		38		11*2/1	107
										45/300			
										46/609			
Poly[trimethylene *N,N'*-adipolyldi(*p*-aminobenzoate)]													
(424.45)					22.6								1386
Poly[trimethylene *N,N'*-azelaoyldi(*p*-aminobenzoate)]													
(466.53)					26.4								
Poly(trimethylene azelate) [3.9]													
(228.29)	Mono		5.0	7.4	27.7		4	1.48		50		14*2/1	107
										60/609			

TABLE 7. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(trimethylene dodecanediote) [3.12]													
(270.37)	Mono		5.0	7.4	35.8		4	1.36		61		17*2/1	107
Poly(trimethylene glutarate) [3.5]													
(172.18)	Mono		5.0	7.4	15.4		2	1.00		39 53/609		10*1/1	107
Poly[trimethylene *N,N'*-glutaryldi(*p*-aminobenzoate)]													
(410.43)				21.3									
Poly[trimethylene *N,N'*-hexadecandioyldi(*p*-aminobenzoate)]													
(564.72)					35.0								1386
Poly[trimethylene *N,N'*-malonyldi(*p*-aminobenzoate)]													
(382.37)					18.5								1386
Poly(trimethylene octadecanedioate) [3.18]													
(354.53)	Mono		5.0	7.4	51.6		4	1.23		76 76/307		23*2/1	107
Poly(trimethylene pimelate) [3.7]													
(200.23)	Mono		5.0	7.4	23.6		4	1.52		37 51/609		12*2/1	107
Poly(trimethylene sebacate) [3.10]													
(242.32)	Mono		5.0	7.4	31.3		4	1.39		53		15*2/1	107
	Ptet		31.2	31.2	33.5	$\gamma=90$	96	1.184		56/305		15*2/1	105
	Ortho	D2-4	5.032	7.532	31.33		4	1.3554		58/609 49/696		15*2/1	1542
Poly[trimethylene *N,N'*-sebacoyldi(*p*-aminobenzoate)]													
(480.56)					27.8								1386
Poly(trimethylene suberate) [3.8]													
(214.26)	Mono		5.0	7.4	26.1		4	1.47		41 52/609		13*2/1	107
Poly(trimethylene succinate) [3.4]													
(158.15)	Mono		5.0	7.4	15.2		4	1.87		47 52/300 52/609		9*2/1	107
Poly(trimethylene terephthalate) [3.T]													
(206.20)	Tri	Ci-1	4.58	6.22	18.12	97,89,111	2	1.432		233/99		11*2/1	917
	Tri	Ci-1	4.637	6.266	18.64	98,93,111	2	1.379		227/453		11*2/1	
	Tri		4.59	6.21	18.31	98,90,112	2	1.428		221/291		11*2/1	1068
	Tri		4.64	6.27	18.64	98,93,111	2	1.377				11*2/1	1080
Poly[trimethylene *N,N'*-tridecanedioyldi(*p*-aminobenzoate)]													
(522.64)					31.7								1386
Poly(trimethylene undecanedioate) [3.11]													
(256.34)	Ortho		5.0	7.4	32.4		4	1.42		59		16*2/1	107
Poly(4,4'-vinylenediphenylene dodecanedioate)													
–, methyl-													
1S (420.55)	Mono		11.1	7.9	27.8	$\beta=84$	4	1.15				24*1/1	1076
1M	Mono		13.4	4.94	27.8	$\beta=40$	2	1.81				24*1/1	1076
2 (at 151°C)	Mono		14.5	5.17	27.5	$\beta=40$	2	1.054				24*1/1	1076
3 (at 172°C)	Mono		14.6	5.20	27.5	$\beta=40$	2	1.041				24*1/1	1076
	Mono	C2h-5	13.83	5.28	27.3	$\beta=138.6$	2	1.059				24*1/1	1560

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 8. POLY(URETHANES) AND POLY(UREAS) (–O–X–O–CO–NH–Y–NH–CO– or –O–X–NH–CO) and (–NH–X–NH–CO–NH–Y–NH–CO– or –NH–X–NH–CO–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(decamethylene hexamethylenediurethane)													
(342.48)										154 148.634 161/473	56.5	22*	1653

TABLE 8. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(decamethylene 4,4′-methylenediphenylenediurethane)													
(424.54)										192 166/634 194/473	69.0	25*	1653
Poly(ethylene decamethylenediurethane)													
α					21.8					174		18*1/1	414
(286.37)													
β					18.9							18*1/1	414
Poly(ethylene 4,4′-ethylenediphenylenediurethane)													
(326.35)	Tri				19.7					312		18*1/1	414
Poly(ethylene hexamethylenediurethane)													
α	Tri	Ci-1	4.93	4.58	16.8	113,103,109	1	1.266		170	42/1653	14*1/1	627
(230.26)										184/402 166/473			
γ	Tri	Ci-1	4.59	5.14	13.9	γ = 119	1	1.333		170		14*1/1	627
Poly(ethylene 4,4′-methylenediphenylenediurethane)													
(312.32)	Hex				15.7					239			414
					15.0					237/1653	48/1653	17*1/1	1125
Poly(ethylene nonamethylenediurethane)													
(272.34)					36.2					168		17*2/1	414
Poly(heptamethylene hexamethylenediurethane)													
(300.40)										146 151/473	48.5	19*	1653
Poly(heptamethylene 4,4′-methylenediphenylenediurethane)													
(382.46)					20.7					198/473		22*1/1	1125
										191	50.6		1653
Poly(heptamethyleneurea)													
(156.23)											10.6/404	10*	
Poly(hexamethylene hexamethylenediurethane)													
(286.36)	Tri	Ci-1	5.05	4.54	21.9	112,108,108	1	1.226		165 171/473 164/634 150/291	52/1653	18*1/1	627
Poly(hexamethylene 4,4′-methylenediphenylenediurethane)													
I. (368.43)	Tri		4.99		41.5	114,114,84	2			179/634	52/1653	21*2/1	1125
	Tri		5.1	5.1	41.6	116,116,85	2	1.41		200/473		20*2/1	1627
II.					19.2								1637
Poly(hexamethylene trimethylenediurethane)													
(244.29)	Tri	Ci-1	5.04	5.04	34.65	111,111,111	2	1.275				15*2/1	1081
Poly(hexamethyleneurea)													
(142.20)										300/402 >300/291 268/634	13.9/404	9*	
Poly(4,4′-methylenedicyclohexylenethiourea) *trans-, trans-*													
(252.42)	Ortho		9.67	6.17	24.6		4	1.142				12*2/1	1315
Poly(4,4′-methylenedicyclohexyleneurea) *trans-, trans-*													
(236.36)	Ortho		9.20	6.22	23.8		4	1.153				12*2/1	1315
Poly(4,4′-methylenediphenyleneurea)													
(224.26)	Mono	Cs-2	4.72	11.33	11.64	γ = 116.5	2	1.337				12*1/1	1084
Poly(nonamethylene hexamethylenediurethane)													
(328.45)										147	523	21*	1653
Poly(nonamethylene 4,4′-methylenediphenylenediurethane)													
(410.51)										190 194/473	58.6	24*	1653
Poly(octamethylene hexamethylenediurethane)													
(314.43)										157 152/634 162/473	55.2	20*	1653
Poly(octamethylene 4,4′-methylenediphenylenediurethane)													
(396.49)					22.8					172/634 201/473	59/1653	23*1/1	1125

TABLE 8. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(oxydiethylenedioxydiethylenedioxydiethylene hexamethylenediurethane)													
(534.69)					28.1							29*1/1	875
Poly(pentamethylene hexamethylenediurethane)													
(272.34)	Mono	C2h-3	4.70	8.36	39.0	$\gamma = 115$	4	1.302		158 157/473 235/402 151/633	42/1653	17*2/1	627
Poly(pentamethylene 4,4′-methylenediphenylenediurethane)													
(354.41)					18.6					192/473	51/1653	20*1/1	1125
Poly(pentamethyleneurethane)													
(129.16)										189/944 150/291 155/444	33.7/944	8*	
Poly(tetramethylene hexamethylenediurethane)													
(258.32)	Tri		4.95	8.65	19.17	90,104,60	2	1.248		180/291	48/1653	16*1/1	662
	Tri		9.05	8.38	19.1	90,115,63	4	1.510		173/101		16*1/1	334
	Tri	Ci-1	4.98	4.71	19.4	116,105,109	1	1.258		184 182/473 184/402			627
Poly(tetramethylene 4,4′-methylenedicyclohexylenediurethane)													
trans-, trans-													
I. (352.48)	Tri		5.1	10.2	37.5	115,85,94	4	1.33				19*2/1	1643
II.					41.3							19*2/1	1643
Poly(tetramethylene 4,4′-methylenediphenylenediphenylenediurethane)													
I. (340.38)	Tri	Ci-1	5.31	5.26	38.61	114,115,96	2	1.438				19*2/1	1567
II.	Tri	Ci-1	5.06	5.06	36.97 17.0	115,108,95	2	1.340				19*2/1	1567
III.	Tri		5.05	4.67	37.9	116,116,84	2	1.523		194/634		19*2/1	1082
	Tri		4.92	5.66	38.35 19.2	124,104,86	2	1.324		248/473 232/1653	53/1653	19*2/1	1083 1273
Poly(tetramethylene trimethylenediurethane)													
(216.24)	Tri	Ci-1	5.06	5.04	30.10	112,113,110	2	1.337				13*2/1	1081
Poly(trimethylene hexamethylenediurethane)													
(244.29)	Mono	C2h-3	4.70	8.36	33.9	$\gamma = 115$	4	1.344		168 167/633 163/473	42/1653	15*2/1	627
Poly(trimethylene 4,4′-methylenediphenylenediurethane)													
I. (326.35)					17.6					227/1653	47/1653	18*1/1	1627
II.					16.2					241/473		18*1/1	1125

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 9. POLY(ETHERS) (–HCH–CHOR–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(benzylvinyl ether)													
(134.18)					6.30					162/114		2*3/1	253
Poly(butylvinyl ether)													
(100.16)	Rho	C3i-2	23.7	23.7	6.50		18	0.947	0.92	64/114		2*3/1	360
–, 2-methyl-													
(114.19)					6.50					140 128/608		2*3/1	382
Poly(*sec*-butylvinyl ether), see Poly(propylvinyl ether), 1-methyl-													
Poly(*tert*-butylvinyl ether)													
(100.16)	Tet	C4h-6	18.84	18.84	7.65		16	0.980		160 260/114 238/281		2*4/1	269

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TABLE 9. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(ethylene)													
–, 1-butoxy-2-chloro-													
trans-													
(134.61)					6.5							2*3/1	242
cis-													
					8.6							2*4/1	242
–, 1-chloro-2-isobutoxy-													
trans-													
(134.61)					20.8							2*10/3	242
–, 1-isobutoxy-2-methyl-													
trans-													
(114.19)					13.77							2*7/2	67
					13.8					226		2*7/2	141
Poly(isobutylvinyl ether)													
(100.16)	Ortho		16.8	9.70	6.50		6	0.942		117		2*3/1	161
								0.94		117			269
										165/114			
										170/281		2*3/1	152
										115/46			
Poly(isopropylvinyl ether)													
(86.13)	Tet		17.2	17.2	35.5		68	0.926		191/281		2*17/5	161
								0.93		98			269
										190/114			
Poly(methylvinyl ether)													
(58.08)	Rho	D3d-6	16.20	16.20	6.50		18	1.175		144/114		2*3/1	176
	Rho	D3d-6	16.25	16.25	6.50		18	1.168				2*3/1	429
Poly(α-methylvinyl methyl ether)													
syndiotactic	Tet		15.2	15.2	16.4		32	1.011				2*8/3	383
	(72.11)	Phex	9.02	9.02	16.0		10	1.062				4*5/2	514
	Hex		9.02	9.02	16.6		10	1.024				4*5/2	554
Poly(neopentylvinyl ether)													
(114.19)	Ortho		18.2	10.51	6.50		6	0.915		216/281		2*3/1	161
								0.91		155			269
										216/114			
Poly(propylvinyl ether)													
–, 1-methyl-													
(100.16)	Tet		18.25	35.5			68	0.956		177/608		2*17/5	424
					35.3					170		2*17/5	382

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 10. POLY(OXIDES) (–X–O–Y–O– or –R–O–)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(acetaldehyde)													
(44.05)	Tet	C4h-6	14.63	14.63	4.79		16	1.142		165/329		2*4/1	92
–, 2-chloro-													
(78.50)	Tet	C4h-6			4.80							2*4/1	387
– 2,2-dichloro-													
(112.94)	Tet	C4h-6			5.22							2*4/1	387
–, 2,2,2-trichloro-													
(147.39)	Tet	C4h-6			6.45					>220/338		2*4/1	387
Poly(2-butene oxide)													
trans-													
(72.11)	Orhto	D2-4	13.72	4.60	6.90		4	41.100		114/399		3*2/1	397
	Ortho	D2-4	13.79	4.65	6.96		4	1.073		100/527		3*2/1	800
cis-													
	Ortho		11.20	10.44	7.01		8	1.169		162/399		3*2/1	397

TABLE 10. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(butyraldehyde)													
(72.11)	Tet	C4h-6	20.01	20.01	4.78		16	1.001		225/329		2*4/1	92
Poly(1,3-cyclobutyleneoxymethylene oxide)													
–, 2,2,4,4-tetramethyl-													
trans-													
α					11.5					260			361
(156.23)										260/369			
β					5.75							6*	361
cis-													
α					11.5					285			361
										285/369			
β					5.75							6*	361
Poly(decamethylene oxide)													
(156.27)	Ortho		7.40	4.94	27.49		4	1.033		79		11*2/1	482
	Ortho	D2h-16	7.40	4.93	27.29		4	1.042		79/181		11*2/1	689
Poly(1,3-dioxolane), see Poly(ethyleneoxymethylene oxide)										72/180			
Poly(1,3-dioxolane), see Poly(methyleneoxytetramethylene oxide)													
Poly(dodecamethylene oxide)													
(184.32)	Ortho		7.40	4.94	32.53		4	1.030				13*2/1	482
Poly(epichlorohydrin), see Poly(ethylene oxide), chloromethyl-													
Poly(4,4′-ethylenediphenyleneoxyheptamethylene oxide)													
–, 3-methyl-													
(324.46)	Mono		11.4	10.4	18.9	$\gamma = 84.1$	6	1.450				19*1/1	1574
Poly(4,4′-ethylenediphenyleneoxynonamethylene oxide)													
–, 3-methyl-													
(352.52)	Mono		20.1	28.3	22.3	$\gamma = 88.6$		vi. ref.				21*1/1	1574
Poly(4,4′-ethylenediphenyleneoxypentamethylene oxide)													
–, 3-methyl-													
(296.41)	Tri		12.7	9.61	47.3	44,96,100	12	1.495				17*3/1	1574
Poly(4,4′-ethylenediphenyleneoxyundecamethylene oxide)													
–, 3-methyl-													
(380.57)	Mono		20.8	27.4	25.5	$\gamma = 84.1$		vi. ref.				23*1/1	1574
Poly(ethylene oxide)													
I. (44.05)	Mono		9.5	12.0	19.5[b]	$\gamma = 101$	36	1.207	1.13/366	66/81	8.29/81	3*7/2	109
	Mono	C2h-5	8.05	13.04	19.48	$\beta = 125.4$	28	1.229		70/538	9.5/665	3*7/2	303
	Mono	Cs-2	8.03	13.09	19.52	$\beta = 125.1$	28	1.220		62/180	8.04/498	3*7/2	194
	Mono		7.95	13.11	19.39	$\beta = 124.6$	28	1.231		72/318	11.7/466	3*7/2	188
	Mono		8.02	13.4	19.25	$\beta = 126.9$	28	1.238		69/728	9.41/728	3*7/2	227
	Mono		8.16	12.99	19.30	$\beta = 126.1$	28	1.239		75/964	7.86/964	3*7/2	460
	Mono		7.51	13.35	19.90	$\beta = 118.6$	28	1.169		76/567	8.7/1195	3*7/2	1164
								1.227	1.123	75/665			190
								1.235	1.124	70/1189	7.33		390
II.	Tri	Ci-1	4.71	4.44	7.12	63,93,111	2	1.197				3*2/1	790
–, *tert*-butyl-													
isotactic													
(100.16)	Tet	D2d-8	15.42	15.42	24.65		36	1.022		135/691		3*9/4	717
										140/692			
										150/693			
										152/527			
syndiotactic													
										63/691			
–, chloromethyl- [epichlorohydrin]													
(92.52)					7.05					117			119
	Ortho	D2-4	12.14	4.90	7.07		4	1.461		121		3*2/1	194
		or C2v-9								135/318			
	Ortho	C2v-9	12.16	4.90	7.03		4	1.467				3*2/1	555
	Ortho		12.24	4.92	6.96		4	1.466				3*2/1	601
	ortho	D2-4	12.15	4.86	7.07		4	1.472				3*2/1	800
										116	10.6		1280
–, isopropyl-													
(86.13)	Ortho	D2-4	12.85	7.52	5.55		4	1.067		53		3*2/1	802

TABLE 10. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(ethylene oxide), complexes with:													
–, *p*-bromochlorobenzene (3 per 10 residues)													
	Ortho	C2v-12	16.58	9.58	28.10		40	1.510		91/1232		3*10/3	1278
–, *p*-bromofluorobenzene (3 per 10 residues)													
	Ortho	C2v-12	16.38	9.46	28.56		40	1.449				3*10/3	1278
–, *p*-chloroiodobenzene (3 per 10 residues)													
	Ortho	C2v-12	16.78	9.73	28.24		40	1.665		82/1232		3*10/3	1278
–, *p*-chlorotoluene (3 per 10 residues)													
	Ortho		8.63	9.27	28.41		20	1.199				3*10/3	1232
–, *p*-dibromobenzene (3 per 10 residues)													
	Ortho		8.37	4.84	27.98		10	1.682		96.5		3*10/3	1207
	Ortho	C2v-12	16.74	9.68	27.98		40	1.682		98/1232		3*10/3	1278
–, *p*-dichlorobenzene (3 per 10 residues)													
	Ortho	C2v-12	16.48	9.51	27.86		40	1.341		83/1232		3*10/3	1278
–, mercuric compounds													
–, –, $HgBr_2$													
I.	Ortho		13.73	8.66	11.80		16	2.540					732
(1 Hg/4 residues)													
–, –, HgCl2													
I.	Ortho		13.5	17.1	11.6		32	2.22					731
(1 Hg/4 residues)	Ortho		13.55	8.58	11.75		16	2.177				6*2/1	732
II.	Ortho		7.75	12.09	5.88		4	3.804				6*1/1	733
(1 Hg/residue)													
–, –, HgI_2					13.52								732
–, *p*-nitrophenol													
	Tri	Ci-1	11.72	5.55	15.57	91,87,104	6	1.389				9*2/1	1570
–, perhydrotriphenylene													
					6.87							3*2/1	1173
–, resorcinol													
α	Mono	C2h-5	16.05	14.25	9.84	$\beta = 112$	16	1.262				3*4/1	1099
(1 R/2 residues)	Ortho	C2v-9	10.50	10.18	9.776		8	1.260		93		3*4/1	1568
β										71/1568			
–, sodium iodide													
(1 NaI/3 EO)	Mono	C2h-5	18.15	8.41	7.98	$\gamma = 122.3$	12	1.819				9*2/1	1343
–, sodium perchlorate													
(1 Na/3 EO)	Mono	C2h-5	18.32	8.578	8.346[b]	$\gamma = 123.1$	6	1.540				9*2/1	1571
–, thiocyanates													
–, –, potassium													
1 salt:4 EO	Tri		11.37	8.18	8.10	80,66,89	8	1.343				3*4/1	1100
(dble. helix)													
–, –, sodium													
1 salt:1 EO	Mono	C2h-5	7.55	12.10	5.83	$\beta = 97.5$	4	1.574				6*1/1	1432
1 salt:3 EO	Mono	C2h-5	16.83	10.64	7.19	$\gamma = 125.5$	12	1.351				9*2/1	1432
1 salt:4 EO	Tri		10.25	8.43	7.23	91,61,104	8	1.616				3*4/1	1100
(dble. helix)													
–, trifluoromethane sulfonate													
(1 salt/3 EO)	Mono	C2h-5	16.77	10.07	8.613[b]	$\gamma = 121.0$	6	1.5355				9*2/1	1572
–, urea													
(9 urea/4 R)	Rho		10.43	10.43	9.12					143/734			730
	Rho	D3-4	10.52	10.52	9.259		4	1.3408				3*4/1	1569
Poly(ethyleneoxymethylene oxide) [1,3-dioxolane]													
I. (74.08)	Tri	C1-1	12.32	4.66	24.7	$\gamma = 100.9$	15	1.325		55/506	17.1/973	5*5/1	690
					36.6								505
										93	15.5		1163
II.	Ortho	D2H-15	9.07	7.79	9.85		8	1.414		55/937		5*2/1	688
III.	Hex		8.07	8.07	29.5		18	1.331		60/937		5*6/1	690
Poly(formaldehyde), see Poly(oxymethylene)													
Poly(heptaldehyde)													
(114.19)	Tet	C4H-6	25.9	25.9	4.52		16	1.000	0.880	147		2*4/1	970
										150/329			

TABLE 10. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(hexamethylene oxide)													
(100.16)	Mono	C2h-6	5.65	9.01	17.28	$\beta = 134.5$	4	1.060		58		7*2/1	482
	Mono	C2h-6	5.64	8.98	17.32	$\beta = 134.5$	4	1.063		58/181		7*2/1	689
									0.932	73.3	23.8		941
										62/542			
Poly(hexamethyleneoxymethylene oxide)													
(130.19)	Ortho	C2v-9	8.4	4.85	18.8		4	1.13		38/695		9*2/1	783
Poly(isobutylene oxide)													
(72.11)	Ortho	D2-4	10.76	5.76	7.00		4	1.104		158/593		3*2/1	594
	Ortho	D2-4	10.80	5.88	7.05		4	1.070		175		3*2/1	775
										177/548			
										160/664			
										155/318			
Poly(isobutyraldehyde)													
(72.11)	Tet				5.2					>260/329		2*4/1	96
Poly(4,4′-isopropylidenediphenyleneoxytrimethylene oxide)													
–, 2-hydroxy- (alkyl)													
(284.36)	Ortho	D2h-24	16.31	19.34	7.90		8	1.515				14*1/1	1118
Poly(isopropylidene oxide)													
(58.08)	Tet	S4-1	14.65	14.65	10.22		28	1.231		60		2*7/2	214
Poly(isovaleraldehyde)													
(86.13)	Tet		20.6	20.6	5.2		16	1.04				2*4/1	96
Poly(4,4′-methylenediphenylene oxide)													
(182.22)	Ortho		8.10	5.60	10.00		2	1.334		140		10*1/1	745
Poly(methyleneoxypentamethylene oxide) [1,3-dioxocane]													
(116.16)	tri	Ci-1	8.36	4.84	8.15	90,90,90	2	1.70		39/695		8*1/1	783
										46	7.74		1410
Poly(methyleneoxytetramethylene oxide) [1,3-dioxepane]													
(102.13)	Ortho	C2v-9	8.50	4.79	13.50		4	1.234				14*1/1	686
										30	14.5		1165
Poly(nonaldehyde)													
(142.24)	Tet		30.56	30.56	4.52		16	0.895				2*4/1	1098
Poly(nonamethylene oxide)													
(142.24)	Ortho		7.36	4.94	12.45		2	1.044		73		10*1/1	482
Poly(octamethylene oxide)													
I. (128.22)	Mono	C2h-6	5.67	9.04	22.45	$\beta = 134.5$	4	1.038		67		9*2/1	482
										74/727	29.8/727		
										74/799	29.3/799		
										83/1167	32/1167		
II.	Ortho		7.36	4.93	22.43		4	1.046				9*2/1	482
Poly(oxacyclobutane) [trimethylene oxide]													
I. (hydrate)	Mono	C2h-3	12.3	7.27	4.80	$\beta = 91$	4	1.178		36/180		4*1/1	446
(58.08)										34/181			
										35/447			
II.	Rho	C3v-6	14.13	14.13	8.41		18	1.194				8*1/1	446
III.	Ortho	D2-5	9.23	4.82	7.21		4	1.203		50/1328	8.8/1328	8*1/1	446
IV.					4.79							4*1/1	1096
–, 3,3-bisazidomethyl-													
(152.16)										128	53.6		1269
–, 3,3-bischloromethyl-													
α	Ortho		17.85	8.16	4.8		4	1.47		190/173	32.2/271	4*1/1	344
(155.02)	Ortho	D2h	8.16	17.85	4.82		4	1.466		180/148	23.0/343	4*1/1	378
	Ortho	D2h-16	17.85	8.16	4.67		4	1.514				4*1/1	689
	Ortho	C2v-9	17.85	8.15	4.78		4	1.481				4*1/1	1094
									1.371	203	19.5		1195
								1.47	1.39				343
β	Mono	Cs-1 or Cs-2	6.85	11.42	4.75	$\beta = 109.8$	2	1.472				4*1/1	172
	Mono	Cs-3	11.42	7.06	4.82	$\gamma = 114.5$	2	1.456				4*1/1	378
	Ortho	C2v-12	13.01	11.71	4.67		4	1.447				4*1/1	689

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TABLE 10. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, 3,3-bisethoxymethyl-													
(174.24)	Hex		5.24	5.24	6.46		6?	1.25?		125	9.41	4*2/1?	1209
										83/1271			
–, 3,3-bishydroxymethyl-													
(86.13)	Mono		11.85	10.39	4.79	$\gamma = 90$	4	0.970				4*1/1	1385
–, 2,2-bistrifluoromethyl-													
(194.08)	Ortho	D2-4	10.63	8.00	8.01		4	1.892				4*2/1	921
–, 3-*tert*-butyl-													
(114.19)										77	5.0		1358
–, 3,3-diethyl-													
I. (114.19)	Mono	C2-2	13.33	5.77	4.74	$\gamma = 91.1$	2	1.040		47/1271		4*1/1	1097
	Mono									100	10.46	4*1/1	1298
II.	Ortho	D2-5	20.36	5.58	6.67		4	1.001				4*2/1	1299
					6.7							4*2/1	1097
	Ortho									80	10.04	4*2/1	1298
–, 3,3-dimethyl-													
I. (86.13)					4.83						47/447	4*1/1	1095
II.	Mono	C2h-5	8.93	7.48	8.35	$\beta = 97.9$	4	1.036		64		8*1/1	1095
										75/1328	9.2/1328		
										80/1394			
III.	Ortho	D2-5	15.60	5.74	6.51		4	0.981		66		4*2/1	1095
										56/1328	7.4/1328		
–, 3-ethyl-3-methyl-													
(100.16)	Ortho	D2-2	18.46	5.66	6.57		4	0.969				4*2/1	1299
					6.5				0.927	60	6.28	4*2/1	1300
–, 3-methyl-3-hydroxymethyl-													
(86.13)	Mono		11.69	10.25	4.70	$\gamma = 89.4$	4	1.016				4*1/1	1385
Poly(oxymethylene) [formaldehyde]													
I. (30.03)	Rho	C3-2	4.46	4.46	17.30		9	1.506	1.25	181	7.45/91	2*9/5	42
		or C3-3							1.32/640	198/217	6.66/186	2*9/5	134
	Rho		4.43	4.43	17.25		9	1.531		178/329	7.37/343	2*9/5	258
	Rho	D6h-1	4.471	4.471	17.39		9	1.491		215/964	10.0/628	2*9/5	513
	Rho		4.470	4.470	56.00		29	1.492		200/455	9.8/1121	*29/16	270
	Ortho		7.75	4.46	17.30		18	1.501			5.65/964	2*9/	249
									1.215				1195
											11.67		1344
II.	Ortho	D2-4	4.767	7.660	3.563		4	1.533				2*2/1	270
III.	Ortho		4.57	7.41	3.49		4	1.688				2*2/1	817
(at 14 kbar)													
Poly(*p*-phenylene oxide)													
(92.10)	Ortho	D2h-14	8.07	5.54	9.72		4	1.408	1.27	298	7.83/773	5*2/1	487
										262/773			
–, 2,6-dimethoxy-													
(152.15)										287	3.2	5*	793
–, 2,6-dimethyl-													
(120.15)			8.45	6.02		$\gamma = 91$				261	5.86/524	5*	226
	Tet		11.92	11.92	17.10		16	1.314		272/471	3.8/471	5*4/1	621
										307	5.95		1200
										262/484	5.0/484		
											5.73/972		
											5.08/574		
											5.38/940		
											5.40/598		
–, 2,6-diphenyl-													
I. (244.29)	Tet	D4-4	12.51	12.51	17.08		8	1.214		484/201	12.2/201	5*4/1	57
Poly(*m*-phenyleneoxy-2-cyano-1,3-phenylene oxide)													
(209.20)	Mono	C2h-5	8.62	13.07	9.03	$\beta = 99$	4	1.383		340		8*1/1	1427
Poly(propionaldehyde)													
(58.08)	Tet	C4h-6	17.52	17.52	4.78		16	1.052		185/329		2*4/1	92
–, 3-methoxycarbonyl-													
(116.12)					4.56					150		2*	704
–, 3-cyano-													
(83.09)	Tri		9.44	5.32	4.95	$\gamma = 102$	2	1.135		176		2*2/1	240

TABLE 10. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(propylene oxide)													
(58.08)	Ortho	C2v-9 or D2-4	10.52	4.67	7.16		4	1.097	0.998/139	75/18 75/377	8.4/498 8.4/377	3*2/1	13
	Ortho	D2-4	10.51	4.69	7.09		4	1.104		72/664		3*2/1	800
	Ortho	D2-4	10.52	4.68	7.10		4	1.104		73/285		3*2/1	41
	Ortho	D2-4	10.40	4.64	6.92		4	1.155		80/1649	8.4/1649	3*2/1	78
	Ortho	D2-4	10.46	4.66	7.03		4	1.126				3*2/1	448
–, 3-phenoxy-													
(150.18)	Ortho		17.0	8.2	5.48		4	1.30	1.27	215 210/297 208/318 203/833 203/527		3*2/1	238
–, –, *o*-chloro-													
(184.62)	Ortho		12.6	9.90	6.93		4	1.418		200		3*2/1	445
–, 3,3,3-trifluoro-													
(112.05)	Ortho	D2-4	11.42	6.26	6.26		4	1.663				3*2/1	
Poly(tetrahydrofuran) [tetramethylene oxide]													
(72.11)	Mono	C2h-6	5.48	8.73	12.07	$\beta = 34.2$	4	1.157		35	12.6/425	5*2/1	348
	Mono	C2h-6	5.59	8.90	12.07	$\beta = 134.2$	4	1.112		43	12.4/600	5*2/1	460
				8.89	12.15			1.116	0.982	60/366	14.4	5*2/1	599
	Ortho	D2-4	7.22	8.75	12.25[b]		8	1.238		37/180		5*2/1	533
	Mono	C2h-6	5.61	8.92	12.25	$\beta = 134.5$	4	1.095				5*2/1	403
–, complex with:													
–, –, urea (6 urea:1.8 EO)													
	Hex	C6-1	8.199	8.199	11.032			1.267					1573
Poly(4,4′-thiodiphenylene oxide)													
(200.26)	Ortho		8.16	5.55	10.20		2	1.440		340		10*1/1	745
Poly(trimethylene oxide), see Poly(oxacyclobutane)													

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 11. POLY(SULFIDES) AND POLY(SULFONES)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Polymer of selenium													
(78.96)	Rho	D3-6	4.355	4.355	4.949		3	4.839		219/752	5.20/752	1*3/1	751
									4.289	221/1121	6.2/1121		1195
Polymer of sulfur													
(32.06)	Mono	C2h-2	26.4	12.32	9.26[b]	$\gamma = 100.75$		2.30				1*8	288
	Ortho		8.11	9.20								Indet.	469
	Mono	C2-1	17.6	9.25	13.8	$\beta = 113$	80	2.059				1*10/3	192
									1.92	250			1195
Poly(ethylene disulfide)													
(92.17)					8.8					130/294			134
					8.8					113/295 145/725		4*3/1	468
Poly(ethylene sulfide)													
(60.11)	Hex		4.92	4.92	6.74		2	1.413		210		3*2/1	241
	Ortho	D2h-6	8.50	4.95	6.70		4	1.416		190/181		3*2/1	504
	Ortho	D2-4	8.508	4.938	6.686		4	1.421		216/1104	14/1104	3*2/1	922
–, *tert*-butyl-													
opt. active (116.22)	Rho	C3-2	16.91	16.91	6.50		9	1.079		162		3*3/1	906
inactive	Port	C2h-5	16.67	19.27	6.52	$\beta = 90$	12	1.106		210 205/694		3*3/1	906

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TABLE 11. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, isopropyl- isotactic													
(102.20)	Ortho	C2v-21	17.92	17.92	7.20		16	1.174				3*2/1	
Poly(ethylene tetrasulfide)													
(156.29)	Ortho		8.57	5.0	4.27		1	1.42				4*1/1	256
	Mono		8.68	5.03	4.32	$\gamma=87$	1	1.378				4*1/1	257
					4.32							4*1/1	134
Poly(hexamethylene sulfide)													
(116.22)										90.3	19.8		1322
										79/181			
										68/295			
Poly(hexamethylene sulfone)													
(148.22)	Mono		9.88	9.26	18.24	$\beta=121.7$	8	1.387		220		7*2/1	39
Poly(hexamethylenesulfonylpentamethylene sulfone)													
(282.42)	Mono		9.88	9.26	34.00	$\beta=121.7$	8	1.418		223		13*2/1	39
Poly(hexamethylenesulfonyltetramethylene sulfone)													
(268.39)	Mono		9.88	9.26	15.68	$\beta=121.7$	4	1.460		246		12*1/1	39
Poly(methylene disulfide)													
(78.15)					4.18							3*2/1	467
Poly(methylene selenide)													
I. (92.99)	Hex		5.22	5.22	46.25		21	2.971		190		2*21/11	416
II.	Ortho	D2-4	5.37	9.03	4.27		4	2.983		170		2*2/1	463
Poly(methylene sulfide)													
(46.09)	Ortho		12.7	12.0	5.10		16	1.575		260		2*2/1	237
	Hex		5.07	5.07	36.52		17	1.600		245/181		2*17/9	331
										260/298			
Poly(pentamethylene sulfide)													
(102.20)	Mono	C2h-5	9.61	9.78	7.84	$\beta=131$	4	1.221		65/295		6*1/1	687
Poly(pentamethylene sulfone)													
(134.19)	Mono		9.88	9.26	7.76	$\beta=121.7$	4	1.476		243		6*1/1	39
Poly(pentamethylenesulfonyltetramethylene sulfone)													
(254.36)	Mono		9.88	9.26	28.33	$\beta=121.7$	8	1.532		247		11*2/1	39
Poly(*p*-phenylene sulfide)													
(108.16)	Ortho	D2h-14	8.67	5.61	10.26		4	1.440		295		5*2/1	677
	Ortho		8.68	5.66	10.26		4	1.425		290/679		5*2/1	1345
	Ortho		8.57	5.59	10.33		4	1.452	1.319	315/1234	12.1	5*2/1	1575
										320	8.65		1362
Poly(propylene sulfide) isotactic													
(74.14)	Ortho	D2-4	9.95	4.89	8.20		4	1.234				3*2/1	515
Poly(tetradecamethylene sulfide)													
I. (228.44)	Tri	Ci-1	4.81	9.61	38.8	123,89,106	4	1.058				15*2/1	1422
Poly(tetramethylene sulfide)													
I. (88.17)	Mono	C2h-6	5.73	9.15	13.26	$\beta=135.7$	4	1.206		67/181		5*5*2/1	961
Poly(thiazyl)													
I. (46.07)	Mono	C2h-5	4.153	7.637	4.439[b]	$\gamma=109.7$	4	2.308				2*2/1	910
	Mono	C2h-5	4.12	7.64	4.43	$\gamma=109.5$	4	2.328				2*2/1	912
	Mono	C2h-5	4.15	7.64	4.44	$\gamma=110$	4	2.313				2*2/1	1102
I′.	Mono	C2h-5	4.04	7.86	4.43[b]	$\gamma=109$	4	2.300				2*2/1	911
	Mono	C2h-5	4.040	7.849	4.429[b]	$\gamma=109.3$	4	2.308				2*2/1	925
II.	Ortho	D2-4	6.251	4.807	4.429[b]		4	2.299				2*2/1	925
	Ortho	D2-4	4.81	6.25	4.43		4	2.298				2*2/1	1102
III.	Ortho	D2-4	9.20	10.72	4.93		16	2.517				2*	1102
IV.			6.80		5.24								1103
V.			13.92		5.81								1103
Poly(trimethylene sulfide)													
(74.14)	Mono	Cs-2	5.16	10.33	4.06[b]	$\gamma=120.5$	2	1.320		100/81		4*1/1	916
										90	10.4		1346
–, 3,3-dimethyl-													
(102.20)										19	5.23		1363

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 12. POLY(SACCHARIDES)

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(2,1-β-D-fructofuran)													
(hemihydrate)	Ortho	D2-4	16.70	9.65	14.4		12	1.470				3*6/1	1634
(162.14)													
(monohydrate)	Ortho	D2-4	16.70	9.80	14.7		12	1.492				3*6/1	1634
Poly(1,4-β-D-galacto-1,4-β-D-mannose) (1 : 2) [guar galactomannan]													
dry			13.5	8.66	10.3[b]		2	1.34				10*	643
(486.42)													
16.5% H2O	Mono	C2-2	15.45	8.65	10.3[b]	$\gamma=90$	6						643
Poly(1,4-α-D-galactosamine)													
anhydride	Ortho		5.2	30.8	8.7		8	1.54				5*2/1	1327
(161.16)													
Poly(1,3-β-D-galactosamine-*alt*-1,4-β-D-glucuronic acid)													
–, *N*-acetyl- [chondroitin]													
–, –, 4-sulfate													
free acid	Ortho				19.6							9*2/1	819
(459.38)													
–, –, –, as Ca salt													
7 H_2O/unit	Ortho	D2-3	7.45	17.81	19.64		4					9*2/1	1242
(497.44)	Hex		12.8	12.8	27.4		6					9*3/1	1258
–, –, –, as Na salt													
I. (@ 93% RH)	Hex		14.5	14.5	28.8		6					9*3/1	819
10 H_2O	Hex	D3-6	14.52	14.52	28.32		6					9*3/1	1241
(503.34)					28.5							9*3/1	1254
II.	Ortho		16.0	26.0	24.1[b]		18					9*3/1?	1139
–, –, –, as K salt													
(535.57)	Hex	D3-4	13.85	13.85	27.76		6					9*3/1	1223
–, –, 6-sulfate													
free acid					18.6[b]							9*2/1	705
(459.38)													
–, –, –, as Na salt													
I. (503.34)	Ortho		12.1	14.4	28.5[b]		9?					9*3/1	705
	or Mono		12.1	9.3	28.5[b]	$\gamma=93$	6?					9*3/1	705
	Rho		14.3	14.3	28.7		9?					9*3/1	769
II.	Tet		13.8	13.8	78.2		32					9*8/1	769
Poly(1,4-β-D-glucosamine) [chitosan]													
I. (161.16)	Ortho		8.9	17.0	10.25[b]		8	1.38				5*2/1	1135
	Ortho		7.76	10.91	10.30		4	1.228				5*2/1	1137
	Ortho	D2-4	8.28	8.62	10.43		4	1.44				5*2/1	1579
(anhydrous)	Ortho	D2-4	8.07	8.44	10.34		4	1.520				5*2/1	1580
II.			4.4	10.0	10.3		2	1.18				5*2/1	1137
III.	Ortho		8.24	16.48	10.39		8	1.517				5*2/1	1138
IV.	Mono		13.8	16.3	40.7	$\alpha=96.5$	48?	1.412				5*8/5	1581
–, *N*-acetyl- [chitin]													
α	Ortho	D2-4	4.76	18.85	10.28[b]		4	1.463				5*2/1	652
(203.19)	Ortho	D2-4	4.69	19.13	10.43		4	1.442				5*2/1	669
	Ortho	D2-3	9.25	19.25	10.46[b]		8	1.449				5*2/1	1135
	Ortho	D2-4	4.74	18.86	10.32		4	1.463				5*2/1	1222
β													
α-phase													
(anhydrous)	Mono	C2-2	4.85	9.26	10.38	$\gamma=97.5$	2	1.460				5*2/1	778
	Ortho		4.82	36.7	10.33		8	1.477				5*2/1	1136
	Mono	C2-2	4.80	9.83	10.32[b]	$\gamma=112$	2	1.495				5*2/1	1253
(hemihydrate)	Mono	C2-2	4.7	10.5	10.3	$\beta=90$	2	1.39				5*2/1	653
(monohydrate)	Ortho		9.32	22.15	10.17[b]		8	1.400				5*2/1	670
	Mono	C2-2	4.8	10.5	10.4[b]	$\gamma=97$	2	1.41				5*2/1	778
(dihydrate)	Mono	C2-2	4.8	11.1	10.4[b]	$\gamma=97$	2	1.44				5*2/1	778
β-phase (hydrated)													
I.	Mono		9.67	21.1	10.4[b]	$\gamma=99.5$	8					5*2/1	778
II.	Mono		4.78	23.3	10.1[b]	$\gamma=94$	4					5*2/1	778
γ	Mono	C2-2	4.7	28.4	10.3[b]	$\gamma=90$	6	1.47				5*2/1	779
–, –, nitrate													
	Ortho		9.2	23	10.3[b]		8					5*2/1	1135

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TABLE 12. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(1,3-β-D-glucosamine-*alt*-1,4-β-D-glucuronic acid)													
–, *N*-acetyl- [hyaluronic acid]													
(379.32)													
II.													
–, –, as Na salt													
(401.30)	Ortho		11.4	9.8	33.7		8	1.42				9*4/1	710
	Tet		9.9	9.9	33.9		8	1.60				9*4/1	769
dry	Tet	D4-8	9.89	9.89	33.94		8	1.61				9*4/1	1227
wet	Ortho	D2-4	11.53	9.89	33.86		8	1.50				18*2/1	1227
–, –, as K salt													
(417.41)	Ortho		11.0	9.9	33.0		8	1.54				9*4/1	711
	Tet	D4-10	17.14	17.14	32.80	Double	16	1.55				9*4/1	1225
	Ortho	D2-4	11.73	9.25	35.42		8	1.50				18*2/1	1226
III.													
K salt	Ortho		10.4	9.0	37.1		8	1.60				9*4/1	711
K salt	Tet	D4-8	9.96	9.96	37.87		8	1.54				9*4/1	1224
IV.													
Na salt	Hex	D3-6	11.7	11.7	28.5		6	1.50				9*3/1	659
Na salt	Hex	D3-6	11.70	11.70	28.50		6	1.40				9*3/1	1256
Ca salt	Hex		15.4	15.4	28.5		9	1.43				9*3/1	659
Ca salt	Hex		20.4	20.4	28.5		9	1.53				9*3/1	659
V.													
Na salt	Hex		18.7	18.7	28.5		18	1.58				9*3/1	659
dry Ca salt	Hex	D3-3	18.32	18.32	28.47		18	1.57				9*3/1	1257
wet Ca salt	Hex	D3-3	20.93	20.93	28.30		18	1.46				9*3/1	1257
VI.													
Na salt	Ortho		34.4	11.7	28.5		24	1.58				9*3/1	659
VII.													
free acid	Mono				19.6							9*2/1	659
Poly(1,3-α-D-glucose)													
I. (native)	Mono		5.81	10.0	8.35	$\gamma = 96$	2	1.116				4*2/1	1130
(162.14)													
II. (dihydrate)	Ortho		5.02	9.63	8.35		2	1.630				4*2/1	1130
III. (dry)	Ortho		4.57	8.65	8.35		2	1.631				4*2/1	1130
IV.	Ortho	D2-4	16.46	9.55	8.44		8	1.623				4*2/1	1131
Poly(1,3-β-D-glucose) [curdlan]													
(anhydrous)	Hex	C6-6	14.41	14.41	5.87	Triple	6	1.530				4*6/1	1132
(162.14)	Hex	C6-6	14.38	14.38	5.82	Triple	6	1.550				4*6/1	1168
(hydrate)	Ortho		26.4	16.4	22.65		14					4*7/1	1133
	Hex		17.01	17.01	22.70	Triple	28	1.473				4*7/1	1168
(dihydrate)	Hex	C3-1	15.56	15.56	18.78	Triple	18	1.476				4*6/1	1132
–, triacetate [pachyman]													
I. (288.25)	Ortho	D2-4	11.0	19.0	22.38		12	1.228				4*6/1	1140
	Hex	C6-2	10.99	10.99	22.91		6	1.198				4*6/1	1582
II.	Ortho		11.49	20.13	18.60		12	1.335				4*6/1	1140
Poly(1,4-α-D-glucose) [amylose]													
A. (162.14)	Ortho		11.90	17.70	10.52	Double	12	1.580				5*6/1	1146
(3 H_2/4 g.)	Mono		21.24	11.72	10.68	$\gamma = 123.48$	12	1.457				5*6/1	1347
B.	Ortho	D2-1	16.0	9.2	10.6[b]		8	1.38				5*2/1	644
(anhydrous)	Mono		12.0	16.25	10.48	$\gamma = 96.5$	12	1.591					1145
(monohydrate)	Ortho		15.6	9.0	10.6[b]		8	1.61				5*4/1	645
(trihydrate?)	Ortho		18.33	15.87	10.41		12	1.422				5*6/1	1143
(trihydrate)	Hex		18.50	18.50	10.40	Double	12	1.397				5*6/1	1144
(tetrahydrate?)	Ortho		15.9	18.2	10.4		12	1.55				5*6/1	1219
V-A. "anhydrous" (monohydrate)													
(dehydrate)	Ortho	D2-4	12.92	22.40	7.95		12	1.560				5*6/1	843
	Ortho	D2-4	12.97	22.46	7.91		12	1.558				5*6/1	1243
	Ortho	D2-4	13.0	22.5	7.90		12	1.554				5*6/1	20
			13.2	23.3								5*6/1	647
V-H. "hydrate" (dihydrate?)													
	Ortho	D2-4	13.61	23.60	8.01		12	1.535				5*6/1	1244
	Ortho	D2-4	13.65	23.70	8.05		12	1.516				5*6/1	20

TABLE 12. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
	Ortho	D2-4	13.7	23.8	8.05		12	1.504				5*6/1	646
	Phex		14.94	14.94	7.93		7	1.503				5*7/1	668
–, triacetate													
I. (288.25)	Ortho		10.87	18.83	52.33		28	1.251				5*14/3	648
II.	Ortho		29.04	20.97	34.00		54	1.248				5*9/2	1301
–, tributyrate													
(372.42)	Ortho		22.04	38.18	18.45		30	1.195				5*5/1	1301
–, triethyl-													
1. (246.30)	Ortho	D2-4			15.48							5*4/1	1142
	Ortho	D2-4	16.13	11.66	15.48		8	1.124				5*4/1	1221
3.	Ortho	D2-4	15.36	12.18	15.48		8	1.130				5*4/1	1220
–, –, complexes with:													
chloroform													
(1 : 2)	Ortho	D2-4	16.76	14.28	16.02		8	1.267				5*4/1	1142
(1 : 1)	Ptet		14.70	14.70	15.48		8	1.215				5*4/1	1142
dichloromethane													
(1 : 2)	Ortho	D2-4	16.52	13.95	16.02		8	1.192				5*4/1	1142
(1 : 1)	Ptet		14.70	14.70	15.48		8	1.099				5*4/1	1142
nitromethane													
(1 : 1)	Ptet		14.70	14.70	15.48		8	1.147				5*4/1	1142
–, triisobutyrate													
(372.42)	Ortho		32.38	32.38	16.10		32	1.172				5*4/1	1301
–, trimethyl-													
(204.22)	Ortho	D2-4	17.24	8.704	15.64		8	1.156				5*4/1	820
–, –, complexes with													
chloroform													
(1 : 1)	Ortho	D2-4	14.57	13.28	15.96		8	1.392				5*4/1	1302
dicloromethane													
(1 : 1)	Ptet	D2-4	13.92	13.92	15.80		8	1.255				5*4/1	1302
–, tripropionate													
(330.34)	Ortho		20.26	35.10	18.46		30	1.254				5*5/1	1301
–, trivalerate													
(414.50)	Ortho		23.44	27.85	18.80		20	1.122				5*5/1	1301
Poly(1,4-α-D-glucose), complexes with:													
–, alcohols													
butanol	Ortho	D2-4	27.0	26.4			24					5*6/1	1156
(V-A)	Ortho	D2-4	13.20	27.00	7.92		12					5*6/1	1246
(V-H)	Ortho	D2-4	13.70	25.80	8.10		12					5*6/1	1246
1-naphthol	Tet	D4-4	22.9	22.9	7.8		16					5*8/1	1247
–, DMSO													
(8 DMSO/cell)	Tet	D2-4	19.17	19.17	24.39		36					5*18/	649
	Ptet	D2-4	19.21	19.21	8.12		12					5*6/1	667
	Ortho		30.23	28.18	7.91		14					5*7/1	1245
–, ethylenediamine													
(1 : 2)	Ptet	D2-4	18.87	18.87	7.99		12					5*6/1	1248
–, fatty acids	Ortho	D2-4	13.0	23.0	8.05		12					5*6/1	646
–, hydroxides													
$C(NH_2)_3OH$			13.1	22.6	9.0		12						650
LiOH			12.1	22.6	8.8		12						650
NaOH			12.3	22.6	8.9		12						650
KOH	Ortho	D2-4	12.7	22.6	9.0		12						650
NH_4OH			12.7	22.6	9.0		12						650
CSOH			12.4	22.6	8.9		12						650
–, iodine	Hex		12.97	12.97	7.91		6	1.77					651
(204.44)													
(1 I/3 residues)													
–, potassium salts													
dry bromide	Tet	D4-8	10.2	10.2	16.4		8					5*4/1	661
formate	Tet	D4-8	10.2	10.2	16.4		8					5*4/1	661
iodide	Tet	D4-8	10.5	10.5	15.9		8					5*4/1	661
wet (10% H_2O)													
acetate	Tet	D4-8	10.8	10.8	16.1		8					5*4/1	661
bicarbonate	Tet	D4-8	10.8	10.8	15.8		8					5*4/1	661

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TABLE 12. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
bromide	Tet	D4-8	10.7	10.7	16.1		8					5*4/1	661
formate	Tet	D4-8	10.8	10.8	16.1		8					5*4/1	661
iodide	Tet	D4-8	10.7	10.7	16.1		8					5*4/1	661
atm. dried													
acetate	Ortho		11.0	18.1	17.9		16						661
propionate	Ortho		11.4	18.0	17.6		16						661
Poly(1,4-β-D-glucose) [cellulose]													
I. (162.14)	Mono		8.35	7.9	10.3[b]	$\gamma = 96$	4	1.59				5*2/1	22
	Mono		8.20	7.90	10.3[b]	$\gamma = 96.7$	4	1.625				5*2/1	98
	Mono		16.78	15.88	10.58[b]	$\gamma = 98$	16	1.543				5*2/1	1249
	Mono		8.171	7.846	10.34[b]	$\gamma = 96.4$	4	1.635				5*2/1	28
	Mono		10.85	12.08	10.3[b]	$\gamma = 1.59$	8	1.598				5*2/1	709
	Mono		16.78	15.88	10.3[b]	$\gamma = 98$	16	1.585				5*2/1	805
	Tri		9.41	8.15	10.34	90,58,96	4	1.624				5*2/1	841
	Mono	C2-2	16.34	15.72	10.38	$\beta = 97$	16	1.628				5*2/1	852
	Mono	C2-2	7.78	8.20	10.34[b]	$\gamma = 96.5$	4	1.643				5*2/1	1149
	Mono		16.43	15.70	10.33	$\gamma = 97.0$	16	1.629				5*2/1	1250
	Mono	C2-2	8.34	7.71	10.37	$\gamma = 97.6$	4	1.629				5*2/1	1576
(α)	Tri	C1-1	6.74	5.93	10.36	117,113,81	2	1.586				5*2/1	1577
(β)	Mono	C2-2	8.01	8.17	10.36	$\gamma = 97.3$	4	1.601				5*2/1	1577
II.	Mono		8.14	9.14	10.3[b]	$\gamma = 118$	4	1.592				5*2/1	22
	Mono		8.02	9.03	10.3[b]	$\gamma = 117.2$	4	1.623				5*2/1	98
	Mono		7.917	9.083	10.34[b]	$\gamma = 117.3$	4	1.630				5*2/1	28
	Tri		8.97	7.31	10.34[b]	$\gamma = 99.4$	4	1.610				5*2/1	708
	Mono	C2-2	8.02	8.99	10.36[b]	$\gamma = 116.6$	4	1.612				5*2/1	842
	Mono		15.7	18.4	10.3[b]	$\gamma = 117$	16	1.625				5*2/1	884
	Mono	C2-2	9.09	7.96	10.31	$\gamma = 117.3$	4	1.624				5*2/1	889
	Mono	C2-2	4.46	7.25	10.34[b]	$\gamma = 98.6$	2	1.629				5*2/1	1150
	Mono		7.94	9.12	10.30[b]	$\gamma = 116.33$	4	1.611				5*2/1	1151
(monohydrate)	Mono	C2-2	9.04	9.63	10.34	$\gamma = 116$	4	1.479				5*2/1	1152
	Mono		15.92	18.22	10.31	$\gamma = 117$	16	1.617				5*2/1	1303
	Mono	C2-2	8.00	9.05	10.38	$\gamma = 116.8$	4	1.605				5*2/1	1578
III.	Mono		7.74	9.9	10.3[b]	$\gamma = 122$	4	1.61				5*2/1	28
	Mono	C2-2	10.25	7.78	10.34	$\gamma = 122.4$	4	1.547				5*2/1	890
	Mono		7.78	9.92	10.30[b]	$\gamma = 122$	4	1.597				5*2/1	1151
IV.	Mono		8.12	7.99	10.3[b]	$\gamma = 90$	4	1.612				5*2/1	98
			7.9		10.3[b]							5*2/1	239
	Mono		8.068	7.946	10.3[b]	$\gamma = 90$	4	1.63				5*2/1	28
	Ortho		8.01	8.12	10.34		4	1.601				5*2/1	1153
X.	Mono		8.10	8.16	10.3[b]	$\gamma = 101.7$	4	1.615				5*2/1	98
	Mono		8.12	7.99	10.3[b]	$\gamma = 90$	4	1.612				5*2/1	218
–, dinitrate													
(252.14)			13.9		8.0								24
–, sodium													
I. (1:1:2) (238.17)	Port	C2-2	8.83	25.28	10.29		8	1.377				5*2/1	1445
IV. (2:2:1) (171.15)	Mono		9.57	8.72	10.35	$\gamma = 122.0$	4	1.552				5*2/1	1446
–, triacetate													
I. (288.25)	Ortho		44.3	13.45	10.47		16	1.228		306/114		5*2/1	1148
	Ortho	C2-2	23.63	6.27	10.43		4	1.239				5*2/1	938
II.	Ortho	D2-4	24.68	11.52	10.54		8	1.278				5*2/1	962
	Port	C2-2	24.5	11.56	10.43		8	1.296				5*2/1	4
dry			24.6	11.4									1251
–, –, nitromethane complex													
	Tet		21.15	21.15	41.36							1252	
–, tributyrate													
(372.41)	Ortho		31.3	25.6	10.36		16	1.192		207/118 206/46	12.6/118	5*2/1	1148
–, tricaprylate													
(540.74)										116/117	13.0/117	5*	
–, tricarbanilate													
(519.51)	Ortho		28.6	16.7	15.25		6	1.105				5*3/1	1148

TABLE 12. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
(4 MEK/residue)													
–, triethyl-													
(246.30)	Ortho		15.64	27.09	15.0		18	1.158				5*3/1	1148
–, trimethyl-													
I. (204.22)	Ortho		4.64	43.2	10.43		8	1.298				5*2/1	1136
	Port	C2-2	4.64	43.2	10.42		8	1.299				5*2/1	1148
II.												5*3/1	1148
–, trinitrate													
(297.13)	Ortho		12.25	9.0	25.4[b]		10	1.762		697/77	3.8-	5*5/1	24
	Ortho		9.0	14.6	25.4		10	1.48		700/81	6.3/77	5*5/2	1147
–, tripropionate													
(330.34)	Ortho		31.6	22.1	15.0		24	1.257		234/144		5*3/1	1148
			11.0	15.4									1348
–, 2.44-nitrate										617/81	5.65/81		
Poly(1,4-β-D-glucose), complexes with:													
–, dimethyl-1,3-propanediamine													
(1 : 2)	Mono		33.64	30.40	10.26[b]	γ = 32.74	16	1.00				5*2/1	1255
–, ethylenediamine													
I. (1 : 1)	Mono	C2-2	12.87	9.52	10.35	γ = 118.8	4	1.328				5*2/1	1154
(1 : 2)			12.2	12.3	10.3[b]	γ = 43.23	4	1.394				5*2/1	1270
–, heptamethylenediamine													
(1 : 2)			18.4	17.4	(10)[b]	γ = 29.23	4	1.20				5*2/1	1270
–, hexamethylenediamine													
(1 : 2)			17.2	15.9	10.3[b]	γ = 32.35	4	1.226				5*2/1	1270
–, hydrazine													
II. (1 : 1)	Mono	C2-2	9.37	19.88	10.39	γ = 120	8	1.539				5*2/1	1154
(1 : 2)			9.68	9.96	(10)[b]	γ = 54.80	4	1.59				5*2/1	1270
–, octamethylenediamine													
(1 : 2)			19.1	17.9	(10)[b]	γ = 28.52	4	1.21				5*2/1	1270
–, pentamethylenediamine													
(1 : 2)			16.6	15.3	10.1[b]	γ = 32.92	4	1.259				5*2/1	1270
–, $NaOH/H_2O$													
II. (1 : 1)	Hex		10.0	10.0	15.1		6	1.677				5*3/1	1155
–, trimethylenediamine													
(1 : 2)					10.2[b]								1270
Poly(1,6-α-D-glucose) [dextran]													
(162.14)	Mono	C2-2	9.22	9.22	7.78	β = 91.3	4	1.631				5*2/1	1129
(high temp.)													
Poly(1,4-α-L-guluronic acid)													
(176.12)	Ortho	D2-4	7.75	10.6	8.7[b]		4	1.64				5*2/1	674
(monohydrate)	Ortho	D2-4	8.6	10.74	8.72[b]		4	1.60				5*2/1	673
Poly(1,3-α-D-mannose)													
(dihydrate)	Mono	C2-2	11.29	18.12	8.40		8	1.560				4*2/1	1356
(162.14)													
Poly(1,4-β-D-mannose)													
I. (162.14)			7.21	8.82	10.27[b]		4	1.65				5*2/1	636
	Ortho	D2-4	8.92	7.21	10.21		4	1.640				5*2/1	1141
	Ortho		8.87	7.29	10.19		4	1.634				5*2/1	1355
II.	Mono		18.8	18.7	10.2[b]	γ = 122.5	16	1.43				5*2/1	636
(4 H_2O/cell)	Ortho	D2-8	8.91	16.46	10.3		8	1.505				5*2/1	1583
–, triacetate													
(288.25)				15.24								5*3/1	672
–, trimethyl-													
	Ortho		4.75	40.1	10.30		8	1.383				5*2/1	1304
Poly(1,4-β-D-mannuronic acid)													
(176.12)	Ortho	D2-4	7.58	8.58	10.35[b]		4	1.738				5*2/1	673
Poly(1,3-β-D-xylose) [xylan]													
(132.12)	Hex		13.7	13.7	5.85[b]		6	1.384				4*	654
(monohydrate)	Hex	C6-4	15.4	15.4	6.12		6	1.190					788
Poly(1,4-β-D-xylose) [hardwood xylan]													
I. dry	Hex		8.8	8.8	14.85[b]		6	1.32				5*3/1	671
(132.12)													

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TABLE 12. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
(monohydrate)	Hex	D3db	9.16	9.16	14.85		6	1.386				5*3/1	671
(150.13)	Phex	C1-1	9.16	9.16	14.84	$\gamma = 120$	6	1.387				5*3/1	655
(dihydrate)	Hex		9.64	9.64	14.95		6	1.392				5*3/1	671
(168.15)													
II.	Ortho		11.5	14.4									1134
–, diacetate													
(216.19)	Mono	C2-2	7.64	12.44	10.31	$\gamma = 85$	4	1.471				5*2/1	656

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

TABLE 13. OTHER POLYMERS

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Polyaniline, see Poly(*p*-phenylene amine)													
Poly(barium sebacate)													
(337.58)	Tet		8.04	8.04	15.24		4	2.276				13*1/1	1186
Poly(barium suberate)													
(309.52)	Tet		7.88	7.88	11.53		4	2.871				11*1/1	1186
Poly(4,4′-biphenylenedioxy 4,4′-diphenylene ketone) [PEDEK]													
(364.40)	Ortho		7.72	5.94	37.5		4	1.407		425		19*2/1	1596
Poly[4,4′-biphenylenedioxy)-diphenylene-*m*-phenylene diketone]													
(468.51)	Ortho		7.85	6.05	47.7		4	1.374				23*2/1	1594
Poly(cadmium sebacate)													
(312.64)	Tet		10.16	10.16	15.59		4	1.290				13*1/1	1186
Poly(cadmium suberate)													
(284.58)	Tet		8.32	8.32	12.16		4	2.246				11*1/1	1186
Poly(calcium sebacate)													
(240.32)	Tet		8.20	8.20	15.59		4	1.523				13*1/1	1186
Poly(calcium suberate)													
(212.26)	Tet		7.72	7.72	12.16		4	1.945				11*1/1	1186
Poly(carbon monoxide-*alt*-ethylene) [Poly(propanone-1)]													
α	Ortho	D26-16	6.91	5.12	7.60		4	1.383		278		3*2/1	1586
(56.06)											12.6/1587		
β	Ortho	D2h-16	7.97	4.76	7.57		4	1.296		185		3*2/1	193
(at 140°C)	Ortho	D2h-16	8.24	4.74	7.58		4	1.256				3*2/1	1586
										244/929	7.49/929		
Poly[4,4′-(carbonyldiphenylenedioxy)-diphenylene *p*-phenylene diketone]													
2/5 repeat	Ortho		7.74	6.04	10.05		2	1.404		370		10*1/1	1439
(198.61)													
Poly(decamethylene amine)													
–, *N,N*-dimethyl-													
–, –, salt with ("Ionene")													
$FeCl_4$	Hex		9.71	9.71	11.04		2	1.548		274			1405
SbF_6	Hex		9.71	9.71	11.04		2	1.548		274			1405
Poly(dimethyl ketene)													
I. (ketone)													
α	Ortho	C2v-9	12.85	6.53	8.80		8	1.261		255/130		8*1/1	432
(70.09)					8.8					250			179
β					4.40							2*2/1	456
II. (ester)										170/179			
(140.18)										180/248			
Poly(ethylene amine)													
anhydrate	Ortho	D2h-24	29.8	17.2	4.79	Double	40	1.165		58/738		3*5/1	1107
(43.07)													
hemihydrate	Mono	C2h-6	10.89	9.52	7.31	$\beta = 127.6$	8	1.152				3*2/1	1108
(52.08)													
sesquihydrate	Mono	C2h-6	11.55	9.93	7.36	$\beta = 104.5$	8	1.139				3*2/1	1106
(70.09)													

TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
dihydrate (79.10)	Mono	C2h-6	13.26	4.61	7.36	$\beta = 101.0$	4	1.190				3*2/1	1106
–, *N*-benzoyl- (147.18)										210/747		3*	
–, –, *p*-chloro- (181.62)	Tri		4.74	14.8	6.55	88,86,99	2	1.334		285/747		3*2/1	746
–, *N*-butyryl- (113.16)										155/747		3*	
–, –, 4-(4-methylthiophenoxy)- (251.34)	Tri		4.35	24.0	12.7	90,90,90	4	1.259		105		3*4/1	739
–, *N*-heptanoyl- (155.24)	Tri		5.0	17.7	6.4	85,62,98	2	1.06		175/747		3*2/1	746
–, *N*-hexanoyl- (141.21)	Tri		4.9	15.9	6.4	87,64,95	2	1.06		175/747		3*2/1	746
–, *N*-isobutyryl- (113.16)					6.4					210/747		3*	746
–, *N*-isovaleryl- (127.19)	Tri		4.7	13.0	6.4	80,82,88	2	1.11		210/747		3*2/1	746
–, *N*-octanoyl- (169.27)	Tri		5.03	19.9	6.3	91,60,97	2	1.04		165/747		3*2/1	746
–, *N*-*p*-toluoyl- (161.20)	Tri		4.84	14.8	6.61	91,78,99	2	1.172		260/747		3*2/1	746
–, *N*-valeryl- (127.19)	Tri		4.7	14.4	6.4	86,73,97	2	1.03		172/747		3*2/1	746
Poly(ethylene amine), complexes with:													
–, acetic acid (1 : 1) (103.12)	Mono	C2h-2	5.76	6.20	7.39	$\gamma = 95$	2	1.302		135		3*2/1	1392
–, HCl (1 : 1) (79.53)	Ptet		5.06	5.06	7.57		2	1.362		265/1423		3*2/1	1366
–, oxalic acid and water (2 : 1 : 2) (105.09)	Ortho	D2h-7	6.65	10.05	7.32		4	1.427		240		3*2/1	1428
Poly(ethylisocyanide)													
–, α-phenyl- (131.18)	Phex		14.92	14.92	10.33	93,90,118	10	1.078				1*10/	863
Poly(germylene)													
–, di-*n*-hexyl- (242.93)	Ortho		15.65	21.46	4.06		4	1.183				1*2/1	1635
Poly(germoxane)													
–, phthalocyaninato- (601.12)	Tet	C4h-1	13.27	13.27	3.53		1	1.606				2*1/1	1205
–, –, iodine complex (1 : 1.12)	Tet	D4h-2	13.96	13.96	6.96		2	1.820				2*2/1	1206
Poly(hexamethylene amine)													
–, *N*,*N*-diethyl-													
–, –, salt with ("Ionene")													
$ZnBr_4$	Tet		9.81	9.81	13.88		4	1.734					1405
–, *N*,*N*-dimethyl-													
–, –, salt with ("Ionene")													
BF_4	Hex		7.54	7.54	11.95		2	1.214		276			1405
CdI_4	Tet		9.78	9.78	13.75		4	2.213		261			1405
CF_3O_3	Hex		7.90	7.90	13.06		2	1.305		207			1405
$FeCl_4$	Hex		8.04	8.04	13.10		2	1.476		220			1405
HgI_4	Tet		9.82	9.82	13.54		4	2.454		242			1405
PF_6	Hex		7.75	7.75	11.96		2	1.458		233			1405
SbF_6	Hex		7.90	7.90	12.58		2	1.807		235			1405
$ZnBr_4$	Tet		9.52	9.52	13.46		4	1.746					1405
$Zncl_4$	Tet		9.36	9.36	13.07		4	1.345					1405
ZnI_4	Tet		9.64	9.64	13.89		4	2.134		233			1405
Poly(hydroxymethylene) (30.03)					2.5							1*2/1	338
Poly(1-hydroxytrimethylene) (58.08)					6.89							3*	758

TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
Poly(lead sebacate)													
(407.44)	Tet		7.73	7.73	17.42		4	2.600				13*1/1	1186
Poly(lead suberate)													
(379.38)	Tet		7.64	7.64	13.66		4	3.160				11*1/1	1186
Poly(magnesium sebacate)													
(224.54)	Tet		8.26	8.26	13.11		4	1.667				13*1/1	1186
Poly(magnesium suberate)													
(196.49)	Tet		8.76	8.76	11.23		4	1.514				11*1/1	1186
Poly(manganese sebacate)													
(255.17)	Tet		7.92	7.92	14.56		4	1.856				13*1/1	1186
Poly(manganese suberate)													
(227.12)	Tet		7.52	7.52	11.83		4	2.255				11*1/1	1186
Poly(6-mercaptocaproic acid)													
(130.21)					17.8					106		7*2/1	428
Poly(octamethylene-5,5′-dibenzimidazole)													
(344.46)					21							18*1/1	470
Poly(2,5-octamethylene-1,3,4-triazole)													
–, 1-amino- (triazole)													
(194.28)					25.5					258/610		11*2/1	610
Poly(oxybisdimethylsilylene)													
II. (102.24)	Ortho		8.66	11.94	3.9		2	0.80		52		3*1/1	1618
III.	Ortho		7.92	13.72	3.9		2	0.84				3*1/1	1618
Poly(4,4′-oxydiphenylene 4,4′-biphenylene diketone) [PEKDK]													
(376.41)	Ortho		7.64	6.09	38.00		4	1.413				19*2/1	1595
Poly(4,4′-oxydiphenylene ketone) [PEK]													
(196.20)	Ortho		7.63	5.96	10.0		2	1.433	1.272	367		10*1/1	1113
	Ortho	D2h-4	7.65	5.97	10.09		2	1.414		367/1117		10*1/1	1274
	Ortho		7.76	6.00	10.01		2	1.398		365		10*1/1	1439
Poly(4,4′-oxydiphenylene *m*-phenylene diketone) [PEKK(I)]													
(300.31)	Ortho		7.66	6.11	15.76		2	1.352				14*1/1	1592
Poly(4,4′-oxydiphenylene *p*-phenylene diketone) [PEKK]													
I. 2/3 repeat	Ortho		7.69	6.06	10.16		2	1.404				10*1/1	1590
(200.21)	Ortho		7.67	6.06	10.08		2	1.419				10*1/1	1591
II.	Ortho		4.17	11.34	10.08		2	1.395		385		10*1/1	1439
	Ortho		3.93	5.75	10.16		1	1.448				10*1/1	1590
	Ortho		4.2	11.3	10.1		2	1.39				10*1/1	1591
Poly[(4,4′-oxydiphenylene *p*-phenylene diketone)-*alt*-(4,4′-oxydiphenylene *m*-phenylene diketone)]													
I. (600.62)	Ortho		7.78	6.10	31.13		2	1.350				29*1/1	1597
II.	Ortho		4.17	11.08	31.13		2	1.387				29*1/1	1597
III.	Ortho		6.41	6.10	35.04		2	1.456				29*1/1	1597
Poly(*p*-phenylene amine) [aniline]													
–, emeraldine base													
II. (181.22)	Ortho		7.80	5.75	10.05		2	1.335				10*1/1	1612
	Ortho		11.34	5.75	7.56		2	1.221				10*1/1	1613
–, –, complex with doecylbenzenesulfonate													
	Ortho	D2h-13	11.78	17.91	7.16		2	1.114				10*1/1	1613
–, emeraldine salt (HCl)													
I. (217.68)	Port		4.3	5.9	9.6		1	1.48				10*1/1	1612
II.	Ortho		7.0	8.6	10.4		2	1.15				10*1/1	1612
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene diketone]													
2/3 repeat (210.88)	Ortho	D2h-14	9.96	4.78	10.7		2	1.375				10*1/1	1444
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene ketone] [PEEK]													
2/3 repeat	Ortho		7.75	5.86	10.0		2	1.405	1.265	335		10*1/1	1113
(192.20)	Ortho	D2h-14	7.75	5.89	9.883		2	1.415		384/1117		10*1/1	1114
	Ortho	D2h-14	7.781	5.922	10.06		2	1.378	1.264			10*1/1	1115
	Ortho	D2h-14	7.83	5.94	9.86		2	1.392				10*1/1	1305
	Mono		7.767	5.911	9.878		2	1.407				10*1/1	1438
	Ortho		7.76	5.89	9.95		2	1.404		335		10*1/1	1439
full repeat	Ortho	D2h-14	7.88	5.94	30.50		4	1.341		338/1383	46/1383	15*2/1	1194
(288.30)	Ortho	D2h-14	7.83	5.94	29.58		4	1.392			351589	15*2/1	1305
								1.401	1.263	395	37.5		1116
(at 291°C)	Ortho		8.048	5.975	9.856		2	1.347				10*1/1	1438

TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
–, methyl- (dioxy)													
(302.33)								1.348			22.3		1588
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene 4,4′-oxydiphenylene diketone] [PEEKEK]													
2/5 repeat	Ortho		7.79	5.94	9.96		2	1.396		340		10*1/1	1439
(193.80)													
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene *m*-phenylene diketone] [PEEKmK]													
(392.41)	Ortho		7.71	6.05	39.9		4	1.400				19*2/1	1593
	Ortho	D2h-14	7.747	6.086	19.87		2	1.3911				19*1/1	1640
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene *p*-phenylene diketone] [PEEKK]													
1/2 repeat	Ortho		7.80	6.01	10.01		2	1.388		360		10*1/1	1439
(196.20)	Ortho		7.79	6.025	10.08		2	1.378				10*1/1	1640
Poly(*p*-phenylene disiloxyanylene)													
–, tetramethyl-													
(208.41)	Tet		9.08	9.08	15.38		4	1.092	0.985/787	148/225	18.2/225	7*2/1	115
	Tet	D4-8	9.02	9.02	15.43		4	1.103		160/787	11.3/787	7*2/1	897
Poly(phosphazene)													
–, bis(β-naphthoxy)-													
(331.31)			22.4	13.9		γ = 93							1387
–, bis(4-*n*-butylphenyl)azophenoxyethoxy-													
(639.74)	Ortho		35.7	17.85	9.85		8	1.354				2*4/21	1442
–, bis(2,2,3,3,4,4,4-heptafluorobutoxy)-													
(443.08)	Ortho		27	12.85	9.48		8	1.79				4*2/1	1611
–, bis(2,2,3,3,3-pentafluoropropoxy)-													
I. (343.06)	Mono		20.75	15.22	4.74	γ = 44.6	4	2.168				2*2/1	1610
II.	Ortho		23.96	10.77	4.64		4	1.903				2*2/1	1610
–, bis(phenoxy)-													
α					4.9					390/909			552
(231.19)	Mono		16.6	13.8	9.70	γ = 83	8	1.392				2*4/1	1314
γ			19.2	11.5	9.70		8	1.434				2*4/1	1314
δ	Phex		12.9	12.9									1314
–, –, di-*m*-bromo-													
α	Mono		23.7	10.0	4.85	γ = 86	4	2.253		320		2*2/1	1607
(388.98)													
γ	Ortho		23.7	11.0	4.85		4	2.043				2*2/1	1607
–, –, di-*p*-bromo													
γ	Ortho		21.7	12.9	4.90		4	1.884		420		2*2/1	1607
(388.98)													
–, –, di-*m*-chloro-													
α	Mono		23.8	10.3	4.85	γ = 86	4	1.808		290		2*2/1	1607
(at 180°C)	Phex		14.2	14.2								2*	870
(300.08)													
–, –, di-*p*-chloro-													
γ	Ortho	D2-4	13.08	20.23	4.90		4	1.537		405/553		2*2/1	822
(300.08)	Ortho		20.5	13.1	4.80		4	1.546		390		2*2/1	1607
										360/1387			
(at 180°C)	Phex		14.2	14.2								2*	870
–, –, di-2,4-dichloro-													
(368.97)	Mono		21.6	16.5	4.86	γ = 94	4	1.418		210/553		2*2/1	1185
–, –, di-3,4-dimethyl-													
(287.30)	Port	C2-2	15.85	19.43	9.85		8	1.258				2*4/1	1111
	Ortho	D2h-14	20.5	14.9	9.98		8	1.252				4*2/1	1600
–, –, di-*p*-ethyl-													
(287.30)	Mono	C2-2	30.1	22.1	9.88	γ = 111.5	16	1.248				4*2/1	1604
–, –, di-*m*-fluoro-													
α	Mono		26.2	20.1	4.85	γ = 86	8	1.393		260		2*2/1	1607
(267.17)													
γ	Ortho		25.7	20.2	4.85		8	1.410				2*2/1	1607
–, –, di-*p*-fluoro-													
α	Mono		26.4	19.2	4.91	γ = 86	8	1.430		365/1387		2*2/1	1360
(267.17)													
β	Mono		18.9	13.2	4.90	γ = 77	4	1.490				2*2/1	1306

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TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
γ	Ortho		27.3	19.5	4.75		8	1.404				2*2/1	1387
	Hex		13.5	13.5								2*	1306
	Ortho		26.4	19.2	4.85		8	1.444		370		2*2/1	1607
–, –, di-*p*-isopropyl-													
α	Mono	C2-2	24.55	19.56	5.56	γ = 101.0	6	1.199				2*3/1	1307
(315.35)													
β	Mono		33.24	22.69	9.80	γ = 109.2	16	1.200				2*4/1	1307
	Ortho	D2-4	31.4	11.4	9.92		8	1.180		120		4*2/1	1606
–, –, di-*p*-methoxy-													
α	Ortho		23.9	17.9	9.83		12	1.380				4*2/1	1605
(291.24)													
β	Mono		24.7	19.8	9.83	γ = 102	12	1.234				4*2/1	1605
γ	Ortho		22.1	20.1	9.63		12	1.357				4*2/1	1605
δ			14.4	14.4		γ = 120							1605
–, –, di-*p*-methyl-													
(259.24)	Ortho		14.52	18.62	4.87		4	1.308		417/1387		2*2/1	1110
	Ortho	D2h-15	28.9	18.6	9.95		16	1.288				4*2/1	1598
–, –, di-*p*-phenyl-													
(383.39)	Ortho	C2v-9	41.8	18.3	9.57		16	1.391		205		2*4/1	1349
										398/553			
–, bis(2,2,3,3-tetrafluoropropoxy)-													
I. (307.08)	Mono		24.4	9.96	4.96	γ = 123	4	2.018				2*2/1	1609
–, bis(2,2,2-trifluoroethoxy)-													
α	Ortho		10.16	9.35	4.86		2	1.748		240/553		2*2/1	1185
(243.04)					4.8								552
β	Mono		10.03	9.37	4.86	γ = 91	2	1.767				2*2/1	1112
	Mono		10.17	9.26	4.86	γ = 91	2	1.764				2*2/1	1608
γ	Ortho		20.60	9.40	4.86		4	1.715				2*2/1	1217
	Ortho		20.64	9.29	4.86		4	1.732				2*2/1	1608
δ	Hex		11.90	11.90	4.86		3?	2.031?				2*2/1	1217
–, dichloro-													
(115.89)	Ortho	C2v-9	12.72	11.07	4.92[b]		8	2.222	1.91			2*2/1	56
	Ortho		12.99	11.11	4.92		8	2.168				2*2/1	1109
	Ortho		13.01	11.09	4.92		8	2.169				2*2/1	1109
	Mono	C2h-5	5.98	12.99	4.92	β = 111.7	4	2.168				2*2/1	1326
–, difluoro-													
α	Ortho		21.43	5.83	6.49		12	2.041				2*3/1	1218
(83.04)	Ortho		11.75	11.25	6.49		12	1.929				2*3/1	1218
β	Ortho	C2v-12	8.69	5.38	4.86		4	2.427				2*2/1	1218
–, dimethyl-													
α	Ortho		13.9	5.98	4.9		4	1.22		146/1387		2*2/1	1349
(75.05)	Ortho		13.90	5.98	4.73		8	1.268		143/1350		2*2/1	1599
	Mono	C2h-5	6.345	13.80	4.887	β = 110.4	4	1.2427		146		2*2/1	1603
β					5.85					76			
–, diphenyl-													
α	Ortho		9.9	12.5	9.8		4	1.09				4*2/1	1601
(199.19)	Mono		10.1	10.1	9.95	γ = 101	4	1.328				4*2/1	1602
β					12.5								1602
γ	Ortho		19.9	10.5	9.95		8	1.273				4*2/1	1602
δ	Hex		12.8	12.8									1602
Poly(1,4-piperidinediyl-trimethylene)													
–, 2,6-dioxo-													
(153.18)	Tri		9.64	11.32	15.80	98,98,114	8	1.326		281		7*2/1	481
Poly(siloxane)													
–, diethyl-													
α-1	Mono		14.15	8.75	4.72	γ = 29.8	2	1.169	0.99			2*2/1	1266
(at 193 K)													
(102.21)													
α-2	Mono		14.59	8.90	4.75	γ = 29.66	2	1.112		7		2*2/1	1266
(at 230 K)										7	1.74		1196
β-1	Tet		7.83	7.83	4.72		2	1.173				2*2/1	1266

TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters a	b	c	Angles	z	Density (g/cm³) Crystal.	Amorph.[a]	Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. (n*p/q)	Refs.
(at 193 K)													
β-2	Tet		7.90	7.90	4.72		2	1.152		17/518	2.64/518	2*2/1	1266
(at 233 K)										17	2.15		1196
mesomorphic	Mono		14.75	8.89	4.88	$\gamma = 31.24$	2	1.023		57		2*2/1	1266
(at 290 K)													
–, dimethyl-													
(74.16)	Mono		13.0	7.75	8.3[b]	$\gamma = 120$	6	1.02	0.98	−40/517	2.72/517	2*3/1	197
										−40/631			
–, diphenyl-													
I. (198.30)	Tet		10.2	10.2	9.9		4	1.279				2*4/1	1656
II.	Ortho		20.10	10.51	10.24		8	1.218		230/907	8.0/907	2*4/1	1275
	Mono	C2-1	21.33	9.91	20.3	$\gamma = 108.9$		16		1.300		4*2/1	
–, dipropyl-													
(130.26)	Tet	C4-2	9.52	9.52	9.40		4	1.016		74	3.03	2*4/1	518
										148	5.8		1195
										206/1260			
–, pentaphenyl-*p*-toluyltri- (averaged)													
(202.77)	Ortho		21.06	10.53	10.36		8	1.172				2*4/1	1275
–, phenyl-*p*-toluyl-													
(212.32)	Ortho		21.04	10.86	9.97		8	1.238				2*4/1	1275
–, phthalocyaninato-													
(556.62)	Ortho	D2h-26	13.80	27.59	6.66		4	1.458				2*2/1	1205
–, –, bromine complex (1 : 1.12)													
	Tet	D4h-2	13.97	13.97	6.60		2	1.666				2*2/1	1206
–, –, iodine complex (1 : 1.12)													
	Tet	D4h-2	13.97	13.97	6.60		2	1.802				2*2/1	1206
Poly(silylene)													
–, *n*-butyl-*n*-pentyl-													
I. (156.34)	Ortho		13.23	22.92	13.88		14	0.864				1*7/3	1614
II. (at 100°C)	Hex		13.4	13.4									1614
–, di-*n*-butyl-													
I. (142.32)	Ortho		12.84	22.24	13.9		14	0.828				1*7/3	1396
II.	Phex		13.3	13.3									1617
–, di-*n*-decyl-													
I. (310.64)	Ortho		21.4	24.9	7.8		8	0.99				4*1/1	1617
II.	Phex		19.1	19.1									1617
–, di-*n*-dodecyl-													
I. (366.75)	Ortho		22.0	25.8	7.8		8	1.10				4*1/1	1617
II.	Phex		20.8	20.8									1617
–, diethyl-													
(86.21)	Ortho		11.08	12.10	3.99		4	1.070				1*2/1	1400
(at −20°C)	Ortho		10.9	11.6	3.86		4	1.173				1*2/1	1632
(at 60°C)	Ortho		10.7	12.4	3.87		4	1.115				1*2/1	1632
–, di-*n*-heptyl-													
I. (226.48)	Ortho		14.68	27.38	4.02		4	0.931				1*2/1	1414
II.	Phex		16.2	16.2									1617
–, di-*n*-hexyl-													
I. (RT)	Mono		13.75	21.82	4.07	$\gamma = 88$	4	1.080				1*2/1	1309
(198.43)	Ortho	C2v-9	13.76	23.86	3.99		4	1.006				1*2/1	1615
	Mono				4.00								1308
II.	Phex		15.5	15.5									1617
–, dimethyl-													
(58.16)	Mono		12.18	8.00	3.88	$\gamma = 91$	4	1.022		387/1416		1*2/1	1413
(at 160°C)	Hex		7.79	7.79	3.88		2	0.947				1*2/1	1413
–, di-*n*-nonyl-													
I. (282.59)	Ortho		20.5	25.1	7.8		8	0.94				4*1/1	1617
II.	Phex		18.4	18.4									1617
–, di-*n*-octyl-													
I. (254.53)	Ortho		16.4	26.4	4.0		4	0.98				1*2/1	1617
II.	Ortho		20.1	21.7	7.8		8	0.99				4*1/1	1617
III.	Phex		17.9	17.9									1617

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TABLE 13. *cont'd*

Polymer	Cryst syst.	Space group	Unit cell parameters				z	Density (g/cm³)		Melting point (°C)[a]	Heat of fusion kJ/mol[a]	Chain conform. ($n*p/q$)	Refs.
			a	*b*	*c*	Angles		Crystal.	Amorph.[a]				
–, di-*n*-pentyl-													
I. (170.37)	Phex		13.76	23.83	13.8	$\gamma=120$	14	1.011				1*7/3	1414
	Mono		13.9	24.5	13.9	$\gamma=120$	14	0.966				1*7/3	1616
					13.8							1*7/3	1325
II.	Hex		14.3	14.3									1616
III.	Ortho		13.7	21.4	4.0		4	0.96				1*2/1	1616
–, diphenyl-													
(182.30)	Ptet		22.96	22.96	7.02		16	1.309		>221		1*4/1	927
–, di-*n*-propyl-													
(114.26)	Tet		9.80	9.80	3.99		2	0.990				1*2/1	1400
–, di-*n*-tetradecyl-													
(422.86)	Ortho		29.78	21.37	7.84		8	1.126				4*1/1	1414
	Ortho		21.5	29.5	7.9		8	1.12				4*1/1	1617
–, di-*n*-undecyl-													
I. (338.79)	Ortho		21.6	25.0	7.8		8	1.07				4*1/1	1617
II.	Phex		20.5	20.5									1617
–, *n*-hexyl-*n*-heptyl-													
I. (212.45)	Ortho		14.1	25.0	4.0		4	1.00				4*1/1	1617
II.	Phex		15.3	15.3									1617
–, methyl-*n*-propyl-													
(86.21)	Ortho		9.6	13.1	4.0		4	1.14				1*2/1	1617
–, *n*-pentyl-*n*-hexyl-													
I. (184.40)	Mono		14.7	25.4	4.0	$\gamma=120$	4	0.95				1*2/1	1617
II.	Phex		14.9	14.9									1617
–, *n*-propyl-*n*-pentyl-													
(142.32)	Hex		12.2	12.2									1614
Poly(stannous sebacate)													
(318.92)	Tet		8.92	8.92	13.62		4	1.955				13*1/1	1186
Poly(stannous suberate)													
(290.87)	Tet		7.78	7.78	11.53		4	2.768				11*1/1	1186
Poly(stannyloxane)													
–, phthalocyaninato-													
(647.22)	Tet	C4h-1	12.81	12.81	3.82		1	1.714				2*1/1	1205
Poly(styrene-*alt*-carbon monoxide)													
isotactic													
(132.16)	Mono	C2-2	8.367	5.47	7.574[b]	$\gamma=110$	2	1.347				3*2/1	1469
syndiotactic													
	Mono	C2h-5	15.5	6.15	7.56	$\beta=105$	4	1.261				3*2/1	1467
	Mono	C2h-5	15.20	6.166	7.59	$\beta=102.5$	4	1.264	1.200			3*2/1	1468
–, *p*-fluoro-													
–, –, syndiotactic													
(150.15)	Mono	C2h-5	15.59	6.168	7.551	$\beta=98.8$	4	1.3900	1.306			3*2/1	1468
–, *m*-methyl-													
–, –, syndiotactic													
(146.19)	Mono	C2h-5	16.14	6.816	7.580	$\beta=100.3$	4	1.1835	1.150			3*2/1	1468
–, *p*-methyl-													
–, –, syndiotactic													
(146.19)	Mono	C2h-5	35.54	6.262	7.536	$\beta=96.0$	8	1.1642	1.133			3*2/1	1468
Poly(zinc sebacate)													
(265.60)	Tet		7.92	7.92	13.25		4	2.122				13*1/1	1186
Poly(zinc suberate)													
(237.55)	Tet		8.68	8.68	10.69		4	1.959				11*1/1	1186

[a] The number after the solidus (/) are reference citations.
[b] The unit cell parameter assignment has been changed from that published so as to make *c* the fiber axis.

C. MELTING POINTS OF POLYMERS

In the following table of melting points, entries are alphabetical according to the basic structure of the polymer, ignoring substituents. Substituted polymers are listed alphabetically (according to the substituent) under the entry for the unsubstituted polymer. The molecular weight is that of the chemical repeat (the constitutional base unit) in the polymer. Melting points (in °C) are taken from the reference cited. Melting points are presented in this table only for those polymers without (as yet) available crystallographic data: an asterisk (*) in the reference column signifies that the relevant melting point data is contained within the "Crystallographic Data" table of Section B.

Polymer	Molecular weight	Melting point	Refs.
Cellulose, see Poly (1,4-β-D-glucose)			*
Poly(acetaldehyde)			*
–, 2,2,2-trichloro-			*
Poly(acrylamide)	71.08		
–, *N*-docosyl-	379.67	68	1906
–, *N*-dodecyl-	239.40	– 8	1906
–, *N*-hexadecyl-	295.51	45	1906
–, *N*-isopropyl-			*
–, *N*-octadecyl-	323.56	68	1906
–, *N*-tetradecyl-	267.46	18	1906
Poly(acrylic acid)	72.06		
–, allyl ester			*
–, butyl ester	128.17	47	401
–, sec-butyl ester			*
–, *tert*-butyl ester			*
–, decyl ester	212.33		
–, – 1,1-dihydroperfluoro-	554.15	100	1898
–, docosyl ester	380.66	72	1906
–, dodecyl ester	240.39	12	1906
–, hexadecyl ester	296.50	43	1906
–, isobutyl ester			*
–, isopropyl ester			*
–, octadecyl ester	324.55	56	1906
–, octyl ester	184.28		
–, –, 1,1-dihydroperfluoro-	454.14	35	1898
–, tetradecyl ester	268.44	32	1906
Poly(acrylonitrile)			*
–, α-ethyl-	81.12	200	436
–, α-isopropyl-	95.14	310	436
–, α-propyl-	95.14	210	436
Poly(2,2′-adipamidodibenzoic isophthalic anhydride)	514.49	225	1871
Poly(4,4′-adipamidodibenzoic isophthalic anhydride)	514.49	273	1871
Poly(2,2′-adipamidodibenzoic terephthalic anhydride)	514.49	235	1871
Poly(4,4′-adipamidodibenzoic terephthalic anhydride)	514.49	285	1871
Poly(adipic anhydride)	128.13	85	1840
		98	1841
		77	1564
Poly(adipoyl dithionisophthaloyldihydrazide)	336.43	280	1889
Poly(adipoyl ethylenediurea)	256.26	241	402
Poly(adipoyl octamethylenediurea)	340.42	290	402
Poly(adipoyl pentamethylenediurea)	298.34	222	402
Poly(β-alanine), see poly(3-aminopropionic acid)			*
Poly(allene)			*
–, tetrafluoro-			*
Poly(allylbenzene), see Poly(propene), 3-phenyl-			*
Poly(2-aminoacetic acid), see Poly(glycine)			*
Poly(*m*-aminobenzoic acid)	119.12	425	735
Poly(*p*-aminobenzoic acid)			*
Poly(3-aminobutyric acid)			*
Poly(4-aminobutyric acid)			*
Poly(10-aminocapric acid)			*
Poly(6-aminocaproic acid) [caprolactam]			*
–, D-(-)-3-methyl-			*
–, 6-methyl-	127.19	185	402

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Polymer	Molecular weight	Melting point	Refs.
Poly(6-aminocaproic acid-*alt*-11-aminoundecanoic acid)			*
Poly(8-aminocaprylic acid)			*
Poly(3-aminocyclobutylene)-propionic acid)	125.17		
–, 2,2-dimethyl-	153.22	358	1816
Poly(4-aminocyclohexyleneacetic acid)	139.20	400	402
Poly(aminodihexamethylene suberamide)	353.55	160	402
Poly(aminoditrimethylene oxamide	185.23		
–, *N*-methyl	199.25	202	291
Poly(22-aminodocosanoic acid)	337.59	145	177
Poly(7-aminoenanthic acid)			*
–, (*R*)-3-methyl-			*
–, (*S*)-4-methyl-			*
–, (*R*)-5-methyl-			*
–, (*R*)-6-methyl-			*
–, *N*-methyl-	141.21	65	1828
Poly[*p*-(aminoethylene)-phenyleneacetic acid]	161.20	283	402
Poly[*p*-(aminoethylene)-phenylenebutyric acid]	189.26	224	402
Poly[*p*-(aminoethylene)-phenylenepropionic acid]	175.23	382	402
Poly[*p*-(aminoethylene)-phenylenevaleric acid]	203.28	275	402
Poly(2-aminoethylenesulfonic acid)	107.13	275	1938
Poly(17-aminoheptadecanoic acid)	267.46	150	292
Poly(aminoformic acid)	43.02		
–, *N*- butyl-			*
–, –, 2-methyl-	113.16	220	1909
–, *N*-phenyl-	119.12	275	528
Poly(*p*-aminohydrocinnamic acid)	147.18	310	402
		360	1911
Poly(12-aminolauric acid)			*
–, *N*-methyl-	211.35	52	1929
Poly[(3-aminomethylene)-cyclohexyleneacetic acid]	153.22		
–, 1,3-dimethyl-	181.28	297	1816
Poly[*p*-(aminomethylene)-phenyleneacetic acid]	147.18	355	402
Poly[*p*-(aminomethylene)-phenylenebutyric acid]	175.23	267	402
Poly[*p*-(aminomethylene)-phenylenepropionic acid]	161.20	300	402
Poly[*p*-(aminomethylene)-phenylenevaleric acid]	189.26	233	402
Poly(6-amino-4-oxacaproic acid)	115.13		
–, 3,5-dimethyl-	143.19	210	1902
Poly(5-amino-3-oxavaleric acid)	101.10	148	402
Poly(9-aminopelargonic acid)			*
Poly(3-aminopropionic acid) [β-alanine]			*
–, 2-butyl-2-methyl-			*
–, 2,2-dimethyl			*
–, 2,3-dimethyl-	99.13		
erythro		405	503
threo		355	503
–, 3,3-dimethyl-	99.13	296	503
–, 2-ethyl-2-methyl-			*
–, *N*-isopropyl-	113.16	130	1869
–, *N*-methyl-	85.11	225	1863
		202	1869
–, 3-methyl-, see Poly(3-aminobutyric acid)			*
–, 2-methyl-2-propyl-			*
–, *N*-phenyl-	147.18	205	1869
Poly(18-aminostearic acid)			*
Poly(2-aminotetrahydropyran-6-carboxylic acid)			*
Poly(13-amino-11-thiatridecanoic acid)	229.38	150	402
Poly(13-amino-12-thiatridecanoic acid)	229.38	150	402
Poly(6-aminothiocaproic acid)	129.22	120	402
Poly(7-aminothioenanthic acid)	143.25	235	402
Poly(α-aminotoluic acid)	133.15	300	444
Poly(13-aminotridecanoic acid)	211.35	183	177
Poly(3-aminotrimethylenesulfonic acid)	107.15	260	1932
Poly(11-aminoundecanoic acid)			*
–, *N*-allyl-	223.36	350	402
–, *N*-ethyl-	212.36	– 30	402

Polymer	Molecular weight	Melting point	Refs.
–, 2-methyl-	197.32	130	402
–, *N*-methyl-	197.32	80	402
–, *N*-phenyl-	259.39	−30	402
–, *N*-piperazinyl-	267.42	142	402
Poly(5-aminovaleric acid)			*
Poly(L-aspartic acid)	115.09		
–, α-isobutyl ester			*
Poly(azelaic anhydride)	170.21	54	1823
Poly(azelaoyl oxalodihydrazide)	270.29	260	402
Poly(benzaldehyde-*alt*-dimethylketene)	176.22	290	1817
–, *p*-chloro-	210.66	260	1817
–, *p*-methoxy-	206.24	240	1817
–, *m*-nitro-	221.21	240	1817
Poly(benzylidenethiodecamethylene sulfide)	294.52	135	1923
Poly(benzylidenethiohexamethylene sulfide)	238.42	115	1923
–, *p*-methoxy-	268.44	130	1923
Poly(benzylvinyl ether)			*
Poly(4,4′-biphenyldicarboxaldehyde)	210.23	250	1801
Poly(4,4′-biphenylene adipamide)	294.35	400	402
Poly(4,4′-biphenylene adipate)	296.32	300	1901
Poly(4,4′-biphenylene carbodiimine)	192.22	>300	1907
–, 3,3′-dimethoxy-	220.27	190	1907
Poly(4,4′-biphenylene-4″,4‴-dibenzylidenimino adipate)	502.57		
–, 3″,3‴-dimethoxy-	562.62	220	1893
Poly(4,4′-biphenylene-4″,4‴-dibenzylidenimino sebacate)	558.68		
–, 3″,3‴-dimethoxy	618.73	230	1893
Poly[4,4′-(4,4′-biphenylenedioxy)-diphenylene carbonate]	396.40	250	1915
Poly(4,4′-biphenylenedioxy 4,4′-diphenylene ketone)			*
Poly(4,4′-biphenylene disiloxanylenedipropionamide)	370.56		
–, tetramethyl- (Si)	426.66	300	402
Poly[4,4′-(2,2′-2,2′-biphenylenedithiazole)-oxy-*p*-phenylene]	486.61	240	1810
Poly[4,4′-(2,2′-2,2′-biphenylenedithiazole)-*p*-phenylene]	394.51	250	1810
Poly(4,4′-biphenylene dithiolisophthalate)	348.44	340	1922
Poly(4,4′-biphenylene dithiolterephthalate)	348.44	>460	1922
Poly(4,4′-biphenylenegermylene)	224.79	295	1777
Poly(biphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide)			*
Poly(4,4′-biphenylene pyromellitimide)			*
Poly(4,4′-biphenylene sebacamide)	350.46	435	402
Poly(4,4′-biphenylene terephthalamide)	314.34	500	402
–, 3,3′-dimethyl-	342.40	440	402
–, –, *N,N*-diethyl-	398.51	254	1870
–, –, *N,N′*-diethyl-	370.45	316	1870
Poly(4,4′-biphenylene terephthalate)			*
–, 2,2′-dipropyl-	400.47	350	1901
Poly[4,4′-biphenylene 4″,4‴-(thiodiphenylene)-dioxy-3,3′,4,4′-diphenylenetetracarboximide]	658.68	483	1771
Poly(1,3-butadiene), 1,2-			*
Poly(1,3-butadiene), 1,4-			*
–, 2-*tert*-butyl			*
–, 2-chloro- [chloroprene]			*
–, 2,3-dimethyl-			*
–, 1-methoxy-	84.12		
trans-		118	1803
–, 2-methyl- [isoprene]			*
–, –, hydrochlorinated			*
–, 2-methylacetoxy-			*
–, 2-propyl-			*
Poly(1,3-butadiene(1,4-)-*alt*-methacrylonitrile]	121.18		
–, 1,1,4,4-tetradeutero-	125.22	75	1825
Poly[1,3-butadiene(1,4-)-*alt*-methyl methacrylate]	154.21	56	1849
Poly(butadiene oxide)	70.09	74	1809
Poly(butenamer)	54.09		
I.		210	1864
II.		150	1864
Poly(1-butene)			*
–, 4-cyclohexyl-	138.25	170	282
		138	328

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Polymer	Molecular weight	Melting point	Refs.
–, 4-*n*-*n*-diisopropylamino-	155.28	315	191
–, 3,3-dimethyl-	84.16	260	282
–, 3-methyl-			*
–, 3-methyl- (1,3-via hydride shift)			*
–, 3-phenyl-	132.21	360	90
–, 4-phenyl-			*
–, 4-*o*-tolyl-	146.23	239	187
–, 4-*p*-tolyl-	146.23	196	187
–, 4,4,4-trifluoro-	110.08	263	1813
–, 3-trifluoromethyl-	124.11	300	1813
–, 4-trimethylsilyl-			*
Poly(2-butene-*alt*-ethylene)			*
Poly(2-butene 4-octenediamide)	222.29		
trans-, *trans*-		301	657
Poly(2-butene oxide)			*
–, 2-methyl-	86.13	196	1879
Poly(2-butene sulfide)	88.17		
cis-		156	548
trans-		192	548
Poly(2-butenylene hexamethylenediurethane)	256.30		
cis-		136	402
trans-		177	402
Poly(4,4′-butylidenediphenylene carbonate)	268.31	170	438
Poly[4,4′-(2,2-butylidene)-diphenylene carbonate]	268.31	222	438
Poly(butylvinyl ether)			*
–, 2-methyl-			*
Poly(*sec*-butylvinyl ether), see Poly(propylvinyl ether), methyl-			*
Poly(*tert*-butylvinyl ether)			*
Poly(butyraldehyde)			*
Poly(caprolactam), see Poly(6-aminocaproic acid)			*
Poly(caprylaldehyde)	128.22	35	329
Poly(carbon monoxide-*alt*-ethylene) [ketone]			*
Poly[*m*-(carboxyphenoxy)-acetic anhydride]	178.14	134	1565
Poly[*p*-(carboxyphenoxy)-acetic anhydride]	178.14		
–, 3-bromo-	257.04	179	1565
Poly(chloroprene), see Poly(1,3-butadiene), chloro-			*
Poly(chlorotrifluoroethylene-*alt*-ethylene)			*
Poly(1,3-cyclobutylene)	54.09		
–, 1-cyano-			*
Poly(1,3-cyclobutylene carbonate)	114.10		
–, 2,2,4,4-tetramethyl-			*
Poly(1,3-cyclobutyleneoxymethylene oxide)	110.12		
–, 2,2,4,4-tetramethyl-			*
Poly(1,2-cycloheptylene-*alt*-ethylene)			*
Poly(1,4-poly-1,3-cyclohexadiene)	80.13		
trans-		380	221
Poly(cyclohexene oxide)	98.14	126	527
Poly(1,3-cyclohexylene adipamide)	224.30		
cis-		170	402
trans-		300	402
Poly(1,4-cyclohexylene adipamide)	224.30	400	402
Poly(1,4-cyclohexylene adipate)			*
Poly(1,3-cyclohexylene azelamide)	266.38		
cis-		125	402
trans-		300	402
Poly(1,4-cyclohexylene 3,3′-dibenzamide)	320.39	390	402
Poly(1,3-cyclohexylenedimethylene adipamide)			*
Poly(1,4-cyclohexylenedimethylene adipamide)	252.36		
cis-		246	385
		225	1878
trans-		345	385
		342	1806
		347	1878
Poly(1,4-cyclohexylenedimethylene adipate)			*

Polymer	Molecular weight	Melting point	Refs.
Poly(1,4-cyclohexylenedimethylene azelamide)	294.44		
cis-		195	385
trans-		275	385
Poly(1,4-cyclohexylenedimethylene azelate)	296.41		
cis-		41	547
trans-		50	547
Poly(1,4-cyclohexylenedimethylene 1,4-cyclohexylenedicarboxylate)	280.36		
cis-, trans-		205	547
trans-, trans-		246	547
Poly(1,4-cyclohexylenedimethylene dithiol-1,4-cyclohexylenedicarboxylate)	312.49		
trans-, trans-		310	1922
Poly(1,4-cyclohexylenedimethylene dodecanediamide)	336.52		
cis-		215	385
		190	1878
trans-		278	385
		280	1878
Poly(1,4-cyclohexylenedimethylene dodecanedioate)	338.49		
cis-		46	547
trans-		85	547
Poly[1,4-cyclohexylenedimethylene (ethylenedithio)-diethylenediurethane]	376.53		
trans-		121	1897
Poly(1,4-cyclohexylenedimethylene glutaramide)	238.33		
cis-		167	385
trans-		290	385
Poly(1,4-cyclohexylenedimethylene glutarate)	240.30		
trans-		50	547
Poly[1,4-cyclohexylenedimethylene (hexamethylenedithio)-diethylenediurethane]	432.64		
trans-		124	1897
Poly(1,4-cyclohexylenedimethylene isophthalamide)	272.35		
trans-		310	385
Poly(1,4-cyclohexylenedimethylene isophthalate)	274.32		
trans-		197	547
Poly(1,4-cyclohexylenedimethylene 2,6-naphthalate)	324.38		
cis-		287	547
trans-		341	547
Poly(1,4-cyclohexylenedimethylene octamethylenediurethane)	340.46		
trans-		160	402
		160	291
Poly(1,4-cyclohexylenedimethylene oxalate)	198.22		
trans-		215	547
Poly(1,4-cyclohexylenedimethylene)-oxymethylene oxide)	156.22		
trans-		78	369
		86	1945
Poly[1,4-cylohexylenedimethylene (pentamethylenedithio)-diethylenediurethane]	418.61		
trans-		102	1897
Poly(1,4-cyclohexylenedimethylene *p*-phenylenediacetate)	302.37		
cis-		55	547
Poly(1,4-cyclohexylenedimethylene pimelamide)	266.38		
cis-		191	385
trans-		293	385
Poly(1,4-cyclohexylenedimethylene pimelate)	268.35		
trans-		42	547
Poly(1,4-cyclohexylenedimethylene sebacamide)	308.47		
cis-		208	385
		200	1878
trans-		300	385
		296	1878
		292	1769
Poly(1,4-cyclohexylenedimethylene sebacate)	310.43		
cis-		50	547
trans-		78	547
Poly(1,4-cyclohexylenedimethylene suberamide)	280.41		
cis-		215	385
trans-		311	385

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Polymer	Molecular weight	Melting point	Refs.
Poly(1,4-cyclohexylenedimethylene suberate)	282.38		
cis-		50	547
trans-		96	547
Poly(1,4-cyclohexylenedimethylene succinate)			*
Poly(1,4-cyclohexylenedimethylene terephthalate)			*
Poly[1,4-cyclohexylenedimethylene (tetramethylenedithio)-diethylenediurethane]	404.58		
trans-		129	1897
Poly[1,4-cyclohexylenedimethylene (trimethylenedithio)-diethylenediurethane]	390.56		
trans-		105	1897
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene adipamide]	340.46	196	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene 4,4′-(ethylenedioxy)-dibenzamide]	496.60	250	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene 4,4′-(hexamethylenedioxy)-dibenzamide]	552.71	215	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene oxamide]	284.36	246	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene 4,4′-(oxydiethylenedioxy)-dibenzamide]	540.66	125	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene (*p*-phenylenedioxy)-diacetamide]	420.51	160	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene terephthalamide	360.45	384	402
Poly[(1,4-cyclohexylenedioxy)-ditrimethylene 4,4′-(tetramethylenedioxy)-dibenzamide]	524.66	224	402
Poly(1,4-cyclohexylene dithiolsuccinate)	230.35		
trans-		302	1922
Poly[1,4-cyclohexylene (ethylenedithio)-diethylenediurethane]	348.48		
trans-		214	1897
Poly[1,4-cyclohexylene (hexamethylenedithio)-diethylene-diurethane]	404.58		
trans-		204	1897
Poly(1,4-cyclohexylene 3,3′-methylenedibenzamide)	334.42	174	402
Poly(1,4-cyclohexylene octamethylenediurethane)	312.41		
trans-		255	633
		255	402
		221	291
Poly(1,4-cyclohexyleneoxymethylene oxide)	128.17		
trans-		209	369
Poly[1,4-cyclohexylene (pentamethylenedithio)-diethylene-diurethane]	390.56		
trans-		189	1897
Poly(1,3-cyclohexylene sebacamide)	280.41		
cis-		120	402
trans-		290	402
Poly(1,4-cyclohexylene sebacate)			*
Poly(1,4-cyclohexylene suberate)	254.33		
trans-		181	1174
Poly(1,2-cyclohexylene sulfide)	114.21	130	1917
Poly(1,3-cyclohexylene sulfide)	114.21	139	1917
Poly(1,2-cyclohexylene sulfone)	146.21	284	1917
Poly(1,3-cyclohexylene sulfone)	146.21	309	1917
Poly[1,4-cyclohexylene (tetramethylenedithio)-diethylene-diurethane]	376.53		
trans-		214	1897
Poly[1,4-cyclohexylene (trimethylenedithio)-diethylene-diurethane]	362.50		
trans-		202	1897
Poly(1,4-cylclohexylene urethane)	141.17	>355	444
Poly(cyclohexylidenedimethylene oxide)	126.20	152	1935
–, 3,4-dihydro-	124.18	102	1935
–, –, 4-methyl-	138.21	155	1935
–, 4-methyl-	140.23	165	1935
Poly(4,4′cyclohexylidenediphenylene carbonate)	294.35	260	438
–, 2,2′-dimethyl-	322.40	200	1915
–, 3,3′,5,5′-tetrachloro-	427.09	270	540
Poly(cyclohexylidenethiohexamethylene sulfide)	230.44	75	1923
Poly(1,2-cyclopentylene-*alt*-ethylene)			*
Poly(1,3-cyclopentylene-*alt*-methylene) [1,5-hexadiene]			*
Poly(4,4′-cyclopentylidenediphenylene carbonate)	280.32	250	438
Poly(cyclopropylene cyclopropylenedicarboxamide)	166.18	285	1865
Poly[cyclopropylene (cyclopropylenedicarboxoyl)-diurethane]	254.20		
trans-, trans-		245	1865
Poly(cyclopropylenedimethylene cyclopropylenedicarboxamide)	194.23	220	1865
Poly(cyclopropylenedimethylene cyclopropylenediurethane)	226.23		
cis-		175	1865
trans-		210	1865

Polymer	Molecular weight	Melting point	Refs.
Poly(cyclopropylenedimethylene hexamethylenediurethane)	270.33		
cis-		120	1860
trans-		165	1860
Poly(cyclopropylenedimethylene isophthalamide)	230.27	220	1865
Poly(cyclopropylenedimethylene isophthalate)	232.24	100	1860
Poly[cyclopropylenedimethylene 4,4′-(methylenediphenylene)-diurethane]	352.39		
cis-		230	1860
trans-		290	1860
Poly(cyclopropylenedimethylene piperazinediurethane)	240.26		
cis-		70	1860
trans-		100	1860
Poly(cyclopropylenedimethylene sebacamide)	266.38	223	1865
Poly(cyclopropylenedimethylene terephthalate)	232.24	130	1860
Poly(cyclopropylenedimethylene toluylenediurethane)	276.29		
cis-		170	1860
trans-		200	1860
Poly(cyclopropylene hexamethylenediurea)	240.31	180	1865
Poly(cyclopropylene isophthalamide)	202.21	250	1865
Poly(cyclopropylene piperazinediurea)	210.24	260	1865
Poly(cyclopropylene sebacamide)	238.33	220	1865
Poly(cyclopropylidenedimethylene oxide)	84.12	45	1935
Poly(decamethylene)	140.27		
–, 1,2-dichloro-	209.16		
trans-		47	1925
Poly(decamethylene adipamide)			*
Poly(decamethylene adipate)			*
–, 3-methyl- (diacid)	298.42	−2	1796
Poly[decamethylene *N*,*N*-adipolyldi(6-aminocaproate)]	510.72	137	1792
Poly(decamethylene adipoyldiurethane)	370.45	164	402
Poly(decamethyleneaminohexamethylene amine)	254.46	102	1829
Poly(decamethylene azelamide)			*
Poly(decamethylene azelate)			*
Poly(decamethylene 4,4′-biphenylenediurethane)	410.51		
–, 3,3′-dimethyl- (diisocyanate)	438.57	219	633
Poly(decamethylene carbonate)	200.28	55	1820
Poly(decamethylene *m*-carboxycarbanilate)	319.40	127	1891
Poly(decamethylene *p*-carboxycarbanilate)	319.40	158	1891
Poly[decamethylene (decamethylenedisulfonyl)-dicaproamide]	634.98	207	402
Poly(decamethylene decamethylenediurethane)	398.59	145	291
		142	634
Poly(decamethylene 2,2′-dibenzoate)	380.48	−3	1552
Poly(decamethylene 3,3′-dibenzoate)	380.48	90	1552
Poly(decamethylene 4,4′-dibenzoate)	380.48	132	1552
Poly[(decamethylenedioxy)-dihexamethylene oxide]	356.59	68	542
Poly[4,4‴-(4′,4″-(decamethylenedioxy)-diphenylenedioxy)-diphenylene terephthalate]	672.78	242	1773
Poly[4,4′-(decamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	460.62	211	1789
Poly[3,3′-(decamethylenedioxy)-diphenylene terephthalate]	488.58	108	1773
Poly[4,4′-(decamethylenedioxy)-diphenylene terephthalate]	488.58	259	1773
–, dichloro- (diol)	557.47	160	1773
Poly(4,4′-decamethylenediphenylene carbonate)	352.47	110	1915
		150	1174
Poly(4,4′-decamethylenedipiperazine sebacamide)			*
Poly(decamethylene disiloxanylenedipropionamide)	358.63		
–, tetramethyl- (silicone)	414.74	50	402
Poly(decamethylene disulfide)	204.39	45	1868
		65	725
Poly(decamethylenedithioethylene disulfide)	296.56	80	1837
Poly(decamethylenedithiohexamethylene disulfide)	352.67	46	1784
Poly(decamethylene dithioladipate)	316.53	75	1922
Poly(decamethylene dithiolsebacate)	372.63	103	1922
Poly(decamethylene dithiolterephthalate)	336.52	201	1922
Poly(decamethylenedithiotetramethylene disulfide)	324.62	56	1837
		58	1784
Poly(decamethylene docosanediamide)	506.86	169	454
Poly(decamethylene dodecamethylenediurea)	424.67	190	634

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Polymer	Molecular weight	Melting point	Refs.
Poly(decamethylene phosphinylidenedipropionamide)	330.41		
–, methyl- (*P*)			*
Poly(decamethylene phthalamide)	302.42	115	1868
Poly(decamethylene pimelate)	298.42	66	764
Poly(decamethylene dodecanediamide)	366.59	191	244
		192	454
Poly[decamethylene *N,N'*-dodecanedioyldi(*p*-aminobenzoate)]	606.80	171	1792
Poly(decamethylene eicosanediamide)	478.80	171	244
Poly[decamethylene 4,4'-(ethylenedioxy)-dibenzoate]	440.54	135	290
Poly[decamethylene 4,4'-(ethylenediphenylene)-dioxydiacetamide]	466.62	220	402
Poly(decamethylene *N,N'*-ethylenediterephthalamate)	494.59	277	781
Poly(decamethylene 4,4'-ethylidenedibenzamide)	406.57	150	402
Poly(decamethylene fumaramide)	252.36	50	402
–, 2,3-dimethyl- (diacid)	280.41		
trans-		267	1931
–, methyl- (diacid)	266.38	166	1931
Poly(decamethylene 2,5-furandicarboxamide)	292.38	130	1905
Poly(decamethylene 2,5-furandipropionamide)	348.49	140	402
Poly(decamethylene glutaramide)	268.40		
–, 3-carboxyl- (diacid)	312.41	70	402
Poly(decamethylene glutarate)			*
Poly[decamethylene (hexamethylenedisulfonyl)-dicapromide]	578.87	218	402
Poly(decamethylene *N,N'*-hexamethylenediterephthalamate)	550.70	235	781
Poly(decamethylene hexamethylenediurea)	340.51	210	402
		216	634
Poly(decamethylene hexamethylenediurethane)			*
Poly(decamethylene 3-hexenedioate)			*
Poly(decamethylene isophthalamide)	302.42	186	1868
Poly(decamethylene isophthalate)	304.39	36	1794
Poly[decamethylene 4,4'-(isopropylidenediphenylene)-dioxy-diacetamide]	480.65	105	402
Poly(decamethylene malonamide)	240.35		
–, dodecyl- (diacid)	408.67	123	1774
Poly(decamethylene malonate)	242.32	34	764
Poly(decamethylene 3,3'-methylenedibenzamide)	392.54	65	402
Poly(decamethylene 4,4'-methylenedibenzamide)	392.54	100	402
Poly(decamethylene 4,4'-methylenediphenylenediurea)	422.57	246	634
Poly(decamethylene 4,4'-methylenediphenylenediurethane)			*
Poly[decamethylene (methylene-2,5-furan)-dicarboxamide]	306.41	111	402
Poly[decamethylene (methylene-2,5-tetrahydrofuran)-dicarboxamide]	310.44	85	402
Poly[decamethylene octadecamethylenediurea)	508.84	172	634
Poly(decamethylene ocatadecanediamide)	450.75	170	454
Poly[decamethylene octadecanedioate)			*
Poly[decamethylene octamethylenediurea)	368.57	201	634
Poly[decamethylene octamethylenediurethane)	370.53	146	402
Poly(decamethylene 4-octenediamide)			*
Poly(decamethylene 4-octenedioate)			*
Poly(2,5-decamethylene-1,3,4-oxadiazole)	208.30	100	1921
Poly(decamethylene oxalate)			*
Poly(decamethylene oxamide)	226.32	229	291
		252	743
		290	402
Poly(decamethylene oxamide-*alt*-decamethylene sebacamide)	564.86	231	1781
Poly(decamethylene oxide)			*
Poly(decamethylene oxydiacetate)	272.34	64	1798
Poly(decamethyleneoxymethylene oxide)	186.30	57	695
Poly(decamethyleneoxy-*p*-phenylene oxide)	248.37	< 100	1868
Poly[decamethylene (pentamethylenedisulfonyl)-dicaproamide]	564.84	223	402
Poly(decamethylene *N,N'*-pentamethylenediterephthalamate)	536.67	205	781
Poly(decamethylene *p*-phenylenediacetamide)	330.47	242	291
		265	454
Poly[decamethylene (*p*-phenylenedioxy)-diacetamide]	362.47	188	402
Poly(decamethylene *p*-phenylenedipropionamide)	358.53	265	402
		271	454
Poly(decamethylene 3,3'-phosphinylidenedibenzamide)	426.50		
–, methyl- (*P*)	440.52	155	402

Polymer	Molecular weight	Melting point	Refs.
Poly(decamethylene 4,4′-phosphinylidenedibenzamide)	426.50		
–, methyl- (*P*)	440.52	186	402
–, phenyl- (*P*)	502.59	192	402
Poly(decamethylene 2,4-pyridinedicarboxamide)	303.41	190	1905
Poly(decamethylene 2,5-pyridinedicarboxamide)	303.41	232	1905
Poly(decamethylene 2,6-pyridinedicarboxamide)	303.41	200	1905
–, 1,4-dihydro-4-*oxo*-	319.41	137	1905
Poly(decamethylene 3,5-pyridinedicarboxamide)	303.41	210	1905
Poly(decamethylene pyromelltimide)			*
–, 3-bromo- (imide)	433.30	291	1267
Poly(decamethylene sebacamide)			*
Poly(decamethylene sebacate)			*
Poly(decamethylene sebacoyldiurethane)	426.55	153	402
Poly(decamethylene suberamide)	310.48	217	244
		208	291
Poly(decamethylene suberate)			*
Poly(decamethylene succinamide)			*
Poly(decamethylene succinate)			*
–, D-1,2-dihydroxy- (diacid)	288.34	66	609
Poly(decamethylene sulfide)	172.33	78	1819
		91	181
Poly(decamethylene sulfonyldivalerate)	404.56	101	1797
Poly(decamethylene terephthalamide)	302.42	276	1868
Poly(decamethylene terephthalate)			*
Poly[decamethylene *N,N′*-terephthaloyldi(6-aminocaproate)]	530.71	166	1792
Poly(decamethylene tetradecamethylenediurea)	452.73	180	634
Poly(decamethylene tetradecanediamide)	394.64	189	2
Poly(decamethylene 2,5-tetrahydrofurandipropionamide)	352.52	178	402
Poly[decamethylene (tetramethylenedisulfonyl)-dicaproamide]	550.81	236	402
Poly(decamethylene *N,N′*-tetramethyleneditерephthalamate)	522.64	261	781
Poly(decamethylene tetramethylenediurethane)	314.43	171	633
Poly(decamethylene thiodivalerate)	372.56	57	767
Poly(decamethylenethiohexamethylene sulfide)	288.55	78	1819
Poly(decamethylene 2,5-thiophenedicarboxamide)	308.44	250	1905
Poly(2,5-decamethylene-1,3,4-triazole)	207.32		
–, 1-amino- (triazole)	222.34	242	610
Poly(decamethylenetridecanediamide)	380.62	175	339
Poly(decamethylene *N,N′*-trimethyleneditерephthalamate)	508.61	222	781
Poly(decamethylene urea)	198.31	200	402
		197	634
		210	291
Poly(decamethylene *p*-xylylenediurea)	360.50	260	634
Poly(decamethylene *p*-xylylenediurethane)	362.47	147	634
Poly(decenamer)			*
Poly(1-decene)			*
Poly(decylvinyl ether)	184.32	7	1900
Poly[2,7-(1,6-diazanthrylene)-4,4′-biphenylene]	330.39		
–, 4,9-diphenyl-	482.58	498	1778
Poly[2,8-(1,9-diazanthrylene)-4,4′-biphenylene]	330.39		
– 4,6-diphenyl	482.52	513	1778
Poly[2,7-(1,6-diazanthrylene)-4,4′-oxydiphenylene]	346.39		
–, 4,9-diphenyl-	498.58	490	1778
Poly[2,8-(1,9-diazanthrylene)-4,4′-oxydiphenylene]	346.39		
–, 4,6-diphenyl-	498.58	502	1778
Poly(dimethyl ketene)			*
Poly(di-1,4-naphthyleneoxyethylene oxide)	312.37	291	1787
Poly(di-1,4-naphthyleneoxyheptamethylene oxide)	382.50	239	1787
Poly(di-1,4-naphthyleneoxyhexamethylene oxide)	368.48	280	1787
Poly(di-1,4-naphthyleneoxypentamethylene oxide)	354.45	196	1787
Poly(di-1,4-naphthyleneoxytetramethylene oxide)	340.42	324	1787
Poly(di-1,4-naphthyleneoxytrimethylene oxide)	326.40	278	1787
Poly(1,4-dioxanylene-2,5-dimethylene adipamide)	256.30		
trans-		291	1769
Poly(1,4-dioxanylene-2,5-dimethylene sebacamide)	312.41		
trans-		262	1769

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Polymer	Molecular weight	Melting point	Refs.
Poly(1, 3-dioxolane), see Poly(ethyleneoxymethylene oxide)			*
Poly(1,3-dioxopane), see Poly(methyleneoxytetramethylene oxide)			*
Poly(dithiodiethylene sebacamide)	318.49	155	402
Poly(4,4′-dithiodiphenylene hexamethylenediurea)	416.56	262	402
Poly(divinylbenzal)	176.22	100	1830
–, 2-methyl-	190.24	95	1830
–, 4-methyl-	190.24	115	1830
Poly(divinylfurfural)	166.18	145	1830
Poly(1-docosene)			*
Poly(dodecamethylene)	168.32		
–, 1,2-dichloro-	237.21		
trans-		40	1925
–, 1-methyl-	182.35	30	1926
Poly(dodecamethylene adipamide)	310.48	210	339
		230	544
Poly(dodecamethylene adipate)			*
Poly[dodecamethylene *N,N′*-adipoyldi(6-aminocaproate)]	538.77	134	1792
Poly(dodecamethylene 4,4′-biphenylenediurethane)	438.57		
–, 3,3′-dimethyl- (diisocyanate)	466.62	219	402
Poly(dodecamethylene decamethylenediurethane)	426.64	133	634
Poly[4,4′-(dodecamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	488.67	207	1789
Poly(dodecamethylene docosanediamide)	534.92	164	454
Poly(dodecamethylene *N,N′*-dodecamethylenediterephthalamate)			*
Poly(dodecamethylene dodecamethylenediurethane)	454.70	128	633
Poly(dodecamethylene dodecanediamide)	394.64	183	454
Poly[dodecamethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)]	634.86	167	1792
Poly(dodecamethylene *N,N′*-ethylenediterephthalamate)	522.64	264	781
Poly(dodecamethylene *N,N′*-hexamethylenediterephthalamate)	578.75	231	781
Poly(dodecamethylene hexamethylenediurea)	368.57	205	634
		238	1892
Poly(dodecamethylene hexamethylenediurethane)	370.53	139	634
Poly(dodecamethylene 3-hexenedioate)			*
Poly(dodecamethylene 4,4′-methylenediphenylenediurea)	450.63	240	634
Poly(dodecamethylene 4,4′-methylenediphenylenediurethane)	452.59	164	634
Poly(dodecamethylene 1,4,5,8-naphthalenetetracarboximide)			*
Poly(dodecamethylene octadecanediamide)	478.80	167	454
Poly(dodecamethylene 4-octenedioate)			*
Poly(dodecamethylene oxamide)			*
Poly(dodecamethylene oxamide-*alt*-dodecamethylene sebacamide)	620.96	217	1781
Poly(dodecamethylene *N,N′*-pentamethylenediterephthalamate)	564.72	206	781
Poly(dodecamethylene *p*-phenylenediacetamide)	358.53	256	454
Poly(dodecamethylene *p*-phenylenedipropionamide)	386.58	263	454
Poly(dodecamethylene phosphinylidenedipropionamide)	358.46		
–, methyl- (*P*)			*
Poly(dodecamethylene pyromellitimide)			*
–, 3-bromo- (imide)	461.36	238	1267
Poly(dodecamethylene sebacamide)			*
Poly(dodecamethylene suberate)			*
Poly(dodecamethylene succinamide)			*
Poly(dodecamethylene terephthalamide)	330.47	296	454
		301	544
Poly[dodecamethylene *N,N′*-terephthaloyldi(6-aminocaproate)]	558.76	163	1792
Poly(dodecamethylene *N,N′*-tetramethylenediterephthalamate)	550.70	258	781
Poly(dodecamethylene *N,N′*-trimethylenediterephthalamate)	536.67	216	781
Poly(dodecamethylene *p*-xylylenediurea)	388.56	258	634
Poly(dodecamethylene *p*-xylylenediurethane)	390.52	178	634
Poly(dodecanedioic anhydride)	212.29	87	1823
Poly(dodecenamer)			*
Poly(1-dodecene)			*
Poly(dodecylvinyl ether)	212.38	30	1900
Poly(eicosamethylene adipate)	424.67	87	1790
–, 3-methyl- (diacid)	438.69	60	1796
Poly(eicosamethylene azelate)	466.75	87	1790
Poly(eicosamethylene 3,3′-dibenzoate)	520.76	91	1552
Poly(eicosamethylene 4,4′-dibenzoate)	520.76	115	1552

Polymer	Molecular weight	Melting point	Refs.
Poly(eicosamethylene glutarate)	410.64	80	1790
–, 3-(*p*-nitrophenyl)- (diacid)	531.74	46	1796
Poly(eicosamethylene isophthalate)	444.66	49	1794
Poly(eicosamethylene malonate)	382.58	69	1868
		69	1790
Poly(eicosamethylene oxalate)	368.56	90	1790
Poly(eicosamethylene oxydiacetate)	412.61	89	1798
Poly(eicosamethylene phthalate)	444.66	52	1794
Poly(eicosamethylene pimelate)	438.69	84	1790
Poly(eicosamethylene sebacate)	480.78	92	1790
Poly(eicosamethylene suberate)	452.72	88	1790
Poly(eicosamethylene succinate)	396.61	89	1790
Poly(eicosamethylene sulfonyldivalerate)	544.83	117	1797
Poly(eicosamethylene terephthalate)	444.66	113	1794
Poly(eicosamethylene thiodivalerate)	512.84	79	767
Poly(enanthaldehyde, see Poly(heptaldehyde)			*
Poly(epichlorohydrin), see Poly(ethylene oxide), chloromethyl-			*
Poly(ethylene)			*
–, chlorotrifluoro-			*
–, 1-butoxy-2-methyl-	114.19		
cis- (erythro-di-isotactic)		100	1880
trans- (threo-di-isotactic)		100	1880
–, 1-*tert*-butoxy-2-methyl-	114.19		
cis- (erythro-di-isotactic)		>250	1880
–, 1-ethoxy-2-methoxy-	102.13		
trans-		217	141
–, 1-ethoxy-2-methyl-	86.13		
cis- (erythro-di-isotactic)		191	1880
trans- (threo-di-isotactic)		207	1880
		230	1804
		243	141
–, isobutoxy-2-methyl-			*
–, 1-isopropoxy-2-methyl-	100.16		
cis- (erythro-di-isotactic)		204	1880
trans- (threo-di-isotactic)		211	1880
–, 1-methoxy-2-methyl-	72.11		
cis- or *trans*- (threo-di-isotactic)		230	1880
		287	1804
		210	141
–, 1-methyl-2-propoxy-	100.16		
trans-		168	1804
–, tetrafluoro-			*
–, trifluoro-			*
Poly(ethylene adipamide)	170.21	310	402
Poly(ethylene adipamide-*alt*-ethylene oxamide)	284.32	D359	1781
Poly(ethylene adipate)			*
–, methyl- (diol)	186.21	−20	765
Poly[ethylene *N,N'*-adipoyldi(6-aminocaproate)]	398.50	163	1792
Poly(ethylene adipoyldiurethane)	258.23	210	402
Poly(ethylene amine)			*
–, *N*-acetyl-	85.11	200	747
–, *N*-benzoyl-			*
–, –, *p*-chloro-			*
–, *N*-butyryl-			*
–, –, 4-(4-methylthiophenyoxy)-			*
–, *N*-cyclohexanecarbonyl-	153.22	285	747
–, *N*-dodecanoyl-	225.38	155	747
–, *N*-heptanoyl-			*
–, *N*-hexanoyl-			*
–, *N*-isobutyryl-			*
–, *N*-isovaleryl-			*
–, *N*-2-naphthoyl-	197.24	258	747
–, *N*-octadecanoyl-	309.54	145	747
–, *N*-octanoyl-			*
–, *N*-perfluorooctanoyl-	439.12	245	747

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Polymer	Molecular weight	Melting point	Refs.
–, *N*-perfluoropropionyl-	189.08	216	747
–, *N*-pivaloyl-	127.19	320	747
–, *N*-*P*-toluoyl-			*
–, *N*-valeryl-			*
Poly(ethylene azelamide-*alt*-ethylene oxamide)	326.40	343	1781
Poly(ethylene azelate)			*
–, methyl- (diol)	228.29	–38	765
Poly(4,4′-ethylenebiphenylene)	180.25	>550	1187
Poly[ethylene *p*-(carboxyphenoxy)-acetate]	222.20	140	290
Poly[ethylene *p*-(carboxyphenoxy)-butyrate]	250.25	85	290
Poly[ethylene *p*-(carboxyphenoxy)-caproate]	278.30	45	290
Poly[ethylene *p*-(carboxyphenoxy)-heptanoate]	292.33	55	290
Poly[ethylene *p*-(carboxyphenoxy)-undecanoate]	348.44	65	290
Poly[ethylene *p*-(carboxyphenoxy)-valerate]	264.28	55	290
Poly[ethylene *p*-(carboxyphenylene)-acetamide]	204.23	221	1904
Poly(ethylene 1,4-cyclohexylenedicarboxylate)	198.22		
cis-		<30	1912
trans-		120	1912
Poly(ethylene cyclopropylenedicarboxamide)	154.17		
trans-		350	1860
Poly(ethylene decamethylenediurethane)			*
Poly(ethylene 2,2′-dibenzoate)	268.27	130	1654
–, methyl- (diol)	282.30	55	1552
Poly(ethylene 3,3′-dibenzoate)	268.27	122	1552
–, methyl- (diol)	282.30	97	1552
Poly(ethylene 4,4′-dibenzoate)			*
–, methyl- (diol)	282.30	140	1552
Poly(4,4′-ethylenedibenzoic anhydride)	252.27	340	1854
Poly(4,4′-ethylenedicyclohexylene dodecanediamide)	418.66	316	1926
Poly(4,4′-ethylenedicyclohexylene sebacamide)	390.61	333	1926
Poly[3,3′-(ethylenedioxy)-dibenzoic anhydride]	284.27	237	1854
Poly[4,4′-(ethylenedioxy)-dibenzoic anhydride]	284.27	208	1854
		215	1565
		222	1872
–, 3,3′-dimethoxy-	344.32	220	1855
Poly[4,4′-(ethylenedioxy)-dibenzoic isophthalic anhydride]	432.38	140	1871
Poly[4,4′-(ethylenedioxy)-dibenzoic phthalic anhydride]	432.38	185	1872
Poly[4,4′-(ethylenedioxy)-dibenzoic terephthalic anhydride]	432.38	209	1871
Poly[(ethylenedioxy)-diethylene adipamide]	258.32	160	291
		190	402
Poly[(ethylenedioxy)-diethylene adipate]	260.29	–30	765
Poly[(ethylenedioxy)-diethylene azelate]	302.37	–40	765
Poly[(ethylenedioxy)-diethylene 2,2′-dibenzoate]	356.37	41	1552
Poly[(ethylenedioxy)-diethylene 3,3′-dibenzoate]	356.37	46	1552
Poly[(ethylenedioxy)-diethylene 4,4′-dibenzoate]	356.37	93	1552
Poly[4,4′-(ethylenedioxydiethylenedioxy)-diphenylene 3,3′,4,4′-(carbonyldiphenylene)tetracarboximide]	616.58	405	1808
Poly[3,4′-(ethylenedioxydiethylenedioxy)-diphenylene 3,3′,4,4′-(oxydiphenylene)tetracarboximide]			*
Poly[(ethylenedioxy)-diethylene glutarate]	246.26	–28	765
Poly[(ethylenedioxy)-diethylene isophthalate]	280.28	65	1794
Poly[(ethylenedioxy)-diethylene malonate]	218.21	–28	765
Poly[(ethylenedioxy)-diethylene oxalate]	204.18	–10	765
Poly[(ethylenedioxy)-diethylene pimelate]	274.32	–37	765
Poly[(ethylenedioxy)-diethylene 1,4-piperazinediacetamide]	314.39	115	291
Poly[(ethylenedioxy)-diethylene sebacate]	316.39	29	696
		35	767
Poly[(ethylenedioxy)-diethylene suberate]	288.34	–36	765
Poly[(ethylenedioxy)-diethylene succinate]	232.23	–18	765
Poly[(ethylenedioxy)-diethylene sulfonyldivalerate	348.39	27	1797
Poly[(ethylenedioxy)-diethylene terephthalate]	280.28	65	1794
Poly[(ethylenedioxy)-diethylene thiodivalerate]	348.46	–37	767
Poly[(ethylenedioxy)-diethylene undecanediamide]	328.45		
–, 6-hydroxy- (diacid)	344.45	156	402
Poly[4,4′-(ethylenedioxy)-diphenylene 3,3′,4,4′-(carbonyl-diphenylene)-tetracarboximide]	528.48	470	1808
Poly[3,4′-(ethylenedioxy)-diphenylene 3,3′,4,4′-(oxy-diphenylene)-tetracarboximide]			*
Poly[(ethylenedioxy)-ditrimethylene adipamide]	286.37	180	1791

Polymer	Molecular weight	Melting point	Refs.
Poly[(ethylenedioxy)-ditrimethylene azelamide]	328.45	130	1791
Poly[(ethylenedioxy)-ditrimethylene sebacamide]	342.48	160	1791
Poly[(ethylenedioxy)-ditrimethylene terephthalamide]	306.36	240	1791
Poly(4,4′-ethylenediphenylene adipamide)	322.41	400	402
Poly(4,4′-ethylenediphenylene carbonate)	240.26	300	438
Poly(4,4′-(ethylenediphenylene)-methylene]	194.28	255	1187
Poly(4,4′-ethylenediphenylene sebacamide)	378.52	360	402
Poly(ethylene disiloxanylenedipropionamide)	246.42		
–, tetramethyl- (silicone)	302.52	55	402
Poly(ethylene disulfide)			*
Poly[(ethylenedisulfonyl)-diacetic anhydride]	256.24	185	1856
Poly[(ethylenedisulfonyl)-dipropionic anhydride]	284.30	255	1856
Poly[4,4′-(2,2′-ethylenedithiazole)-*p*-phenylene]	270.37	265	1810
Poly[(ethylenedithio)-diacetic anhydride]	192.25	83	1565
Poly[(ethylenedithio)-dipropionic anhydride]	220.30	75	1565
Poly(ethylenedithiohexamethylene disulfide)	240.56	102	1837
		107	1784
Poly(ethylene dithioladipate)	204.31	125	1922
Poly(ethylene dithiolisophthalate)	224.30	185	1922
Poly(ethylene dithiol-2,5-pyridinedicarboxylate)	252.29	280	1922
Poly(ethylene dithiolterephthalate)	224.29	340	1922
Poly(ethylene dithionisophthalate)	224.29	80	1889
Poly(ethylene dithiotetramethylene disulfide)	212.40	96	1837
		88	1784
Poly(ethylene *N,N′*-dodecamethylenediterephthalamate)			*
Poly(ethylene dodecanediamide)	254.37	261	244
Poly[ethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)]	494.59	242	1792
Poly[ethylene (ethylenediamino)-dipropionamide]	228.30		
–, *N,N′*-diisopropyl-, *N,N′*-dimethyl-	340.51	102	1875
Poly(ethylene 4,4′-ethylenedibenzoate)	296.32	212	1566
		>320	1654
Poly[ethylene (ethylenedioxy)-diacetamide]	202.21	168	1894
Poly[ethylene 3,3′-(ethylenedioxy)-dibenzoate]	328.32	141	111
Poly[ethylene 4,4′-(ethylenedioxy)-dibenzoate]			*
–, 3,3′-dimethoxy-	388.37	210	1914
Poly[ethylene 4,4′-(ethylenedioxydimethylene)-dibenzoate]	356.38	135	1174
Poly[ethylene 4,4′-(ethylenediphenylene)-diurethane]			*
Poly[ethylene 4,4′-(ethylenedithio)-dibenzoate]	360.44	190	291
Poly[ethylene (ethylenedithio)-diethylenediurethane]	294.38	168	1897
Poly(ethylene fumaramide)	140.14	50	402
Poly(ethylene glutarate)	158.15		
–, methyl- (diol)	172.18	−18	765
–, 3-(*p*-nitro)-	279.25	76	1796
Poly(ethylene 2,4-hexadienediamide)	166.18	61	402
Poly(ethylene 4,4′-hexamethylenedibenzoate)	352.43	160	1174
Poly[ethylene (hexamethylenedithio)-diethylene-diurethane]	350.49	130	1897
Poly(ethylene hexamethylenediurea)	228.30	293	1892
		240	1768
Poly(ethylene hexamethylenediurethane)			*
Poly(ethylene isophthalate)			*
–, methyl- (diol)	206.20	87	1794
Poly(ethylene *N,N′*-lysyldiurea)	258.28		
–, methyl ester	272.30	160	1768
Poly(ethylene 4,4′-methylenedibenzoate)	282.30	220	291
		>320	1654
Poly[ethylene 4,4′-(methylenediphenylene)-diurethane]			*
Poly(ethylene 1,5-naphthalate)	242.23	230	291
Poly(ethylene 2,6-naphthalate)			*
Poly(ethylene 2,7-naphthalate)	242.23	270	1911
Poly(ethylene nonamethylenediurethane)			*
Poly(ethylene oxalate)			*
Poly(ethylene oxamide-*alt*-ethylene sebacamide)	340.42	323	1781
Poly(ethylene oxamide-*alt*-ethylene suberamide)	312.37	348	1781
Poly(ethylene oxide)			*
–, 1,1-bischloromethyl-	141.00	180	1816

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Polymer	Molecular weight	Melting point	Refs.
–, bromomethyl- [epibromohydrin]	136.98	112	1809
–, *tert*-butyl-			*
–, chloromethyl- [epichlorohydrin]			*
–, 1,2-dichloromethyl-	141.00		
cis-		235	1779
trans-		145	1779
–, fluoromethyl- [epifluorohydrin]	76.07	68	119
–, isopropyl-			*
–, neopentyl-	114.19	82	527
–, *p*-nitrophenol complex			*
–, resorcinol complex			*
–, tetrafluoro-	116.02	42	1858
–, tetramethyl-	100.16	300	664
–, urea complex			*
Poly(ethylene-*p*-oxybenzoate)			*
–, *m*-chloro-	198.61	285	1174
–, *o*-chloro-	198.61	153	1174
–, *m*-mehthyl-	178.19	292	1174
–, *o*-methyl-	178.19	157	1174
Poly(ethylene oxydiacetate)	160.13	20	1798
–, methyl- (diol)	174.15	21	1798
Poly(ethylene 4,4′-oxydibenzoate)	284.27	152	291
		134	1939
Poly[ethylene 4,4′-(oxydiethylene)-dioxydibenzoate]	372.37		
–, 3,3′-dimethoxy-	432.42	118	1914
Poly[ethylene 4,4′-oxydimethylene)-di-2-(1,3-dioxolane)-caprylate]	500.63	19	797
Poly(ethyleneoxymethylene oxide) [1,3-dioxolane]			*
Poly(ethylene *p*-oxyphenyleneacetate)	178.19	172	1841
Poly(ethyleneoxy-*p*-phenylene oxide)	136.15	270	1868
Poly[ethylene 4,4′-(pentamethylenedioxy)-dibenzoate]	370.40	150	1912
Poly[ethylene (pentamethylenedithio)-diethylenediurethane]	336.46	126	1897
Poly(ethylene *p*-phenylenediacetate)	220.22	107	325
		137	341
Poly(ethylene *m*-phenylenedisulfonamide)	262.30	182	1384
Poly(ethylene phthalamide)	190.20	250	1833
Poly(ethylene phthalate)	192.17	104	1654
–, methyl- (diol)	206.20	50	1794
Poly(ethylene pimelate)			*
–, methyl- (diol)	200.24	−30	765
Poly(ethylene 1,4-piperazinedicarboxylate)			*
Poly(ethylene 1,4-piperazinedipropionamide)	254.33	D231	1875
–, *N,N′*-diisopropyl-	338.50	210	1875
Poly(ethylene 1,4-piperazinedipropionate)	256.30	93	1875
Poly(ethylene sebacamide)	226.32	276	244
		254	291
		280	402
Poly(ethylene sebacate)			*
–, methyl (diol)	242.32	−26	765
Poly(ethylene sebacoyldiurea)	312.37	228	402
Poly(ethylene sebacoyldiurethane)	314.34	198	402
Poly(ethylene suberate)			*
–, methyl- (diol)	214.26	−34	765
Poly(ethylene succinate)			*
–, methyl- (diol)	158.15	0	765
Poly(ethylene sulfide)			*
–, *tert*-butyl-			*
Poly(ethylene 4,4′-sulfonyldibenzamide)	330.36	380	1835
Poly(ethylene sulfonyldivalerate)	292.35	86	1797
–, methyl- (diol)	306.38	68	1797
Poly(ethylene L-tartaramide)	174.16		
–, di-*O*-methyl-			*
Poly(ethylene terephthalamide)	190.20	455	829
–, *N,N′*-dimethyl-	218.26	379	829
Poly(ethylene terephthalate)			*
–, chloro- (diacid)	226.62	92	1913

Polymer	Molecular weight	Melting point	Refs.
–, 2,5-dichloro-	261.06	165	1913
–, 2,5-dimethyl-	220.22	180	1913
–, methyl- (diacid)	206.20	70	1913
–, methyl- (diol)	206.20	111	1794
Poly[ethylene *N,N'*-terephthaloyldi(6-aminocaproate)]	418.49	204	1792
Poly(ethylene 2,5-tetrahydrofurandipropionamide)	240.30	218	402
Poly(ethylene 4,4′-tetramethylenedibenzoate)	324.38	170	291
Poly[ethylene 3,3′-(tetramethylenedioxy)-dibenzoate]	356.37	100	111
Poly[ethylene 4,4′-(tetramethylenedioxy)-dibenzoate]	356.37	252	1841
–, 3,3′-dimethoxy-	416.43	117	1914
Poly[ethylene 4,4′-(tetramethylenediphenylene)-diurethane]	354.41	274	414
Poly[ethylene (tetramethylenedithio)-diethylene-diurethane]	322.44	126	1897
Poly[ethylene 2,4,8,10-tetraoxa-3,9-spiro(5.5)hendecane-dicaprylate]	470.60	85	797
Poly(ethylene 4,4′-thiodibenzoate)	300.33	200	291
Poly(ethylene thiodienanthamide)	314.49	210	402
Poly(ethylene thiodivaleramide)	258.38	220	402
Poly(ethylene thiodivalerate)	260.35	28	1797
–, methyl- (diol)	274.38	– 34	767
Poly(ethylenethiohexamethylene sulfide)	176.34	86	1819
Poly(ethylenethiotetramethylene sulfide)	148.29	89	181
Poly(2,5-ethylene-1,3,4-triazole)	95.10		
–, 1-amino- (triazole)	110.12	360	610
Poly[ethylene 4,4′-(trimethylenedioxy)-dibenzoate]	342.35	190	1912
–, 3,3′-dimethoxy-	403.00	105	1914
Poly[ethylene 4,4′-(trimethylenediphenylene)-diurethane]	340.38	207	414
Poly(ethylene *N,N'*-trimethylenediterephthalamate)	396.40	205	781
Poly(ethylene urea)	86.09	400	402
Poly(ethylidene)			*
Poly(4,4′-ethylidenediphenylene carbonate)	240.26	195	438
–, phenyl-	316.36	230	438
Poly(ethylidene *p*-phenylene)	104.15	205	1876
Poly(ethylvinyl ether)	72.11	86	114
–, 2-chloro-	106.55	150	114
–, 2-methoxy-	102.13	73	114
–, 2,2,2-trifluoro-	126.08	128	114
Poly(formaldehyde), see Poly(oxymethylene)			*
Poly(2,5-furandipropionic anhydride)	194.19	67	1565
Poly(furfural-*alt*-dimethylketene)	166.18	180	1817
Poly(1,4-β-D-glucose) [cellulose]	162.14		
–, triacetate			*
–, tributyrate			*
–, tricaprate	624.90	88	144
–, tricaproate	456.58	94	144
–, tricaprylate			*
–, triheptylate	498.66	88	144
–, trilaurate	709.06	91	144
–, trimyristate	793.22	106	144
–, trinitrate			*
–, tripalmitate	877.38	105	144
–, tripropionate			*
–, trivalerate	414.50	122	144
–, 2.44-nitrate			*
Poly(glycine)			*
Poly(glycolic acid), see Poly(2-hydroxyacetic acid)			*
Poly(glycyl-12-aminododecanoic acid)			*
Poly(1,3-heptadiene), 1,4-			*
–, 5-methyl-	110.20		
trans-			
isotactic		180	1896
–, 6-methyl-			*
Poly(heptaldehyde)			*
Poly(heptamethylene)	98.19		
–, 1,2-dichloro-	167.08		
trans-		53	1925
Poly(heptamethylene adipamide)			*

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Polymer	Molecular weight	Melting point	Refs.
Poly(heptamethylene azelamide)			*
Poly[4,4′-(heptamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	418.54	207	1789
Poly(heptamethylene disulfide)	162.31	130	1868
Poly(heptamethylene *N,N′*-ethylenediterephthalamate)	452.51	275	781
Poly(heptamethylene *N,N′*-hexamethylenediterephthalamate)	508.61	227	781
Poly(heptamethylene hexamethylenediurea)	298.43	243	1892
Poly(heptamethylene malonamide)	198.26		
–, dodecyl- (diacid)	366.59	147	1774
Poly[heptamethylene 4,4′-(methylenediphenylene)diurethane]			*
Poly(heptamethylene 4-octenediamide)	266.38		
trans- I.		261	657
II.		249	657
Poly(heptamethylene oxamide-*alt*-heptamethylene sebacamide)	480.69	248	1781
Poly(heptamethylene *N,N′*-pentamethylenediterephthalamate)	494.59	199	781
Poly(heptamethylene *p*-phenylenediacetamide)	288.39	234	402
Poly(heptamethylene pimelamide)			*
Poly(heptamethylene sebacamide)			*
Poly(heptamethylene sebacate)			*
Poly(heptamethylene suberamide)			*
Poly(heptamethylene 4,4′-sulfonyldibenzamide)	400.49		
–, 4,4-dimethyl-	428.55	268	1835
Poly(heptamethylene terephthalamide)	260.34	328	1775
		330	828
		341	829
Poly(heptamethylene terephthalate)			*
Poly(heptamethylene 2,5-tetrahydrofurandipropionamide)	310.44	148	402
Poly(heptamethylene *N,N′*-tetramethylenediterephthalamate)	480.56	269	781
Poly(2,5-heptamethylene-1,3,4-triazole)	165.24		
–, 1-amino-	180.26	237	610
Poly(heptamethylene *N,N′*-trimethylenediterephthalamate)	466.53	222	781
Poly(heptametylene undecanediamide)	310.48	195	2
Poly(heptenamer)			*
Poly(1-heptene)			*
–, 6,6-dimethyl-	126.24	104	282
–, 5-methyl-			*
Poly[4,4′-heptylidene)-diphenylene carbonate]	310.39	200	438
Poly(hexadecamethylene adipate)			*
Poly(hexadecamethylene decamethylenediurethane)	482.75	128	634
Poly(hexadecamethylene hexamethylenediurethane)	426.64	134	634
Poly(hexadecamethylene 3-hexenedioate)			*
Poly(hexadecamethylene 4,4′-(methylenediphenylene)-diurethane]	508.70	152	634
Poly(hexadecamethylene 4-octenedioate)			*
Poly(hexadecamethylene suberate)			*
Poly(hexadecamethylene *p*-xylylenediurethane)	446.63	168	634
Poly(1-hexadecene)			*
Poly(1,5-hexadiene), see Poly(cyclopentylene-*alt*-methylene)			*
Poly(1,3-hexadiene), 1,4-			*
–, 5-methyl-			*
Poly(2,4-hexadiene), 2,5-			*
–, 2,5-dimethyl-			*
Poly(2,4-hexadienedioic acid), 2,5-	142.11		
–, diisopropyl ester			*
Poly(2,4-hexadienoic acid), 2,5-	112.13		
–, 6-amino-			*
Poly(2,4-hexadiyne adipate)	220.22	130	1782
Poly(2,4-hexadiynediol), 2,5-	110.11		
–, 1,6-bis(*p*-methoxyphenylsulfonyl)-			*
Poly(2,4-hexadiyne hexamethylenediurethane)	278.31	149	1782
Poly[2,4-hexadiyne (4,4′-methylenediphenylene)-diurethane]	360.37	150	1782
Poly(hexafluoroisobutylene-*alt*-vinylidene fluoride)			*
Poly(hexamethylene adipamide)			*
–, 2,5-dihydroxy- (diacid)	258.32	163	402
–, *N,N′*-dimethyl-	254.37	75	402
–, 3-methyl- (diacid)	240.35	216	291
		230	1859

Polymer	Molecular weight	Melting point	Refs.
–, 3-methyl- (diamine)	240.35	180	291
–, *n*-methyl-	240.35	145	291
Poly(hexamethylene adipamide-*alt*-hexamethylene oxamide)	396.53	268	1781
Poly(hexamethylene adipate)			*
Poly[hexamethylene *N,N'*-adipoyldi(6-aminocaproate)]	454.61	147	1792
Poly(hexamethylene adipoyldiurethane)	314.34	206	402
Poly(hexamethylene azelamide)			*
Poly(hexamethylene azelamide-*alt*-hexamethylene oxamide)	438.61	254	1781
Poly(hexamethylene azelate)	270.37	53	539
		55	764
Poly(hexamethylene carbonate)	144.17	60	1820
Poly(hexamethylene *m*-carboxycarbanilate)	263.29	96	1891
Poly[hexamethylene *p*-(carboxyphenoxy)-acetate]	278.30	50	290
Poly[hexamethylene *p*-(carboxyphenoxy)-caproate]	334.41	60	290
Poly[hexamethylene *p*-(carboxyphenoxy)-undecanoate]	404.55	72	290
Poly[hexamethylene *p*-(carboxyphenoxy)-valerate]	320.38	60	290
Poly[hexamethylene *p*-(carboxyphenylene)-acetamide]	260.34	227	1904
Poly(hexamethylene 1,2-cyclohexylenediacetamide)	280.41	255	402
Poly(hexamethylene 1,2-cyclohexylenedicarboxamide)	252.36		
trans-		242	402
Poly(hexamethylene 1,3-cyclohexylenedicarboxamide)	252.36		
trans-		312	402
Poly(hexamethylene 1,4-cyclohexylenedicarboxamide)	252.36		
trans-		360	402
Poly(hexamethylene 1,4-cyclohexylenedicarboxylate)	254.33		
trans-		120	1174
Poly(hexamethylene cyclopropylenedicarboxamide)	210.28		
cis-		180	1860
–, –, *N,N'*-dibutyl	432.45	20	402
–, –, *N,N'*-diethyl-	376.34	5	402
–, –, *N,N'*-diisopropyl-	404.40	20	402
–, –, *N,N'*-dimethyl-	348.29	30	402
Poly(hexamethylene glutarate)	214.26	34	764
Poly[hexamethylene (hexamethylenedioxy)-dipropionamide]	342.48	105	402
Poly[hexamethylene 4,4'-(hexamethylenedisulfonyl)-dibenzoate]	536.66	186	1174
Poly[hexamethylene (hexamethylenedisulfonyl)-dicaproamide]	522.76	222	402
Poly(hexamethylene *N,N'*-hexamethyleneditерephthalamate)			*
Poly[hexamethylene (hexamethylenedithio)-diethylene-diurethane]	406.60	106	1897
Poly(hexamethylene hexamethylenediurethane)			*
Poly(hexamethylene 3-hexenedioate)			*
Poly(hexamethylene isophthalamide)	246.31	220	402
		200	828
–, 5-*tert*-butyl- (diacid)	302.42	210	402
Poly(hexamethylene isophthalate)	248.28	140	111
		80	1794
Poly(hexamethylene 4,4'-isopropylidenedibenzamide)	364.49	180	402
Poly(hexamethylene 4,4'-ketodibenzamide)	350.42	350	402
Poly(hexamethylene ketodipropionamide)	254.33	180	402
Poly(hexamethylene *N,N'*-lysyldiurea)	314.38		
–, methyl ester	328.41	180	1768
Poly(hexamethylene malonamide)			*
–, butyl- (diacid)	240.35	161	1783
		160	1774
–, decyl- (diacid)	324.51	133	1783
		134	1774
–, dimethyl- (diacid)	212.19	117	402
–, dodecyl- (diacid)	352.56	134	1783
		132	1774
–, heptadecyl- (diacid)	422.70	166	1774
–, heptyl- (diacid)	282.43	135	1774
–, hexadecyl- (diacid)	408.67	164	1783
		160	1774
–, hexyl- (diacid)	268.40	145	1783
		144	1774
–, nonyl- (diacid)	310.48	137	1774

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Polymer	Molecular weight	Melting point	Refs.
–, octadecyl- (diacid)	436.72	167	1783
		163	1774
–, octyl- (diacid)	296.45	133	1783
		131	1774
–, pentadecyl- (diacid)	394.64	164	1774
–, pentyl- (diacid)	254.37	151	1774
–, propyl- (diacid)	226.32	163	1774
–, tetradecyl- (diacid)	380.62	152	1783
		148	1774
–, tridecyl- (diacid)	366.59	133	1774
–, undecyl- (diacid)	338.54	133	1774
Poly(hexamethylene 3,3′-methylenedibenzamide)	336.44	113	402
Poly(hexamethylene 4,4′-methylenediphenylene)	336.44	132	402
Poly[hexamethylene 4,4′-(methylenediphenylene)-dioxy-diacetamide]	396.49	174	402
Poly[hexamethylene 4,4′-(methylenediphenylene)-diurea]	366.46	250	402
–, 3,3′-dimethyl-	394.52	290	402
–, –, *N,N′*-diethyl-	450.63	120	402
–, –, *N,N′*-dipropyl-	478.68	90	402
Poly[hexamethylene 4,4′-(methylenediphenylene)-diurethane]			*
Poly[hexamethylene (methylene-2,5-furan)-dicarboxamide]	250.30	115	402
Poly[hexamethylene (*p*-methylenephenoxy)-diacetamide]	304.39	220	402
		226	1202
Poly[hexamethylene (*p*-methylenephenylene)-diacetamide]	283.39	204	1202
Poly[hexamethylene (methylene-2,5-tetrahydrofuran)-dicarboxamide]	254.33	78	402
Poly(hexamethylene naphthalenedicarboxamide)	298.39	200	402
Poly(hexamethylene nonamethylenediurea)	326.48	243	1892
Poly(hexamethylene octadecamethylenediurea)	452.73	192	634
Poly(hexamethylene octadecanediamide)	394.64	192	454
–, 9,10-dihydroxy-	426.64	172	402
–, *N,N′*-dimethyl-	422.70	52	1877
Poly(hexamethylene octamethylenedithiourea)	344.58	160	402
Poly(hexamethylene octamethylenediurea)	312.46	239	634
		253	1892
		255	402
Poly(hexamethylene octamethylenediurethane)	342.48	153	633
Poly(hexamethylene 4-octenediamide)			*
Poly(hexamethylene 4-octenedioate)			*
Poly(hexamethylene 5-oxadodecanediamide)	312.45	149	1847
Poly(hexamethylene 6-oxaheptadecanediamide)	382.59	160	1847
Poly(hexamethylene 5-oxahexadecanediamide)	368.56	159	1847
Poly(hexamethylene oxalate)			*
Poly(hexamethylene oxamide)			*
–, 3-oxypropyl-	228.29	240	402
Poly(hexamethylene oxamide-*alt*-hexamethylene pimelamide)	410.56	259	1781
Poly(hexamethylene oxamide-*alt*-hexamethylene sebacamide)	452.64	257	1781
Poly(hexamethylene oxamide-*alt*-hexamethylene suberamide)	424.58	260	1781
Poly(hexamethylene 6-oxapentadecanediamide)	354.53	154	1847
Poly(hexamethylene 5-oxasebacamide)	284.40	160	1847
Poly(hexamethylene 5-oxatetradecanediamide)	340.51	155	1847
Poly(hexamethylene 5-oxaundecanediamide)	298.43	152	1847
Poly(hexamethylene oxide)			*
Poly(hexamethylene oxydiacetamide)	214.26	143	402
		172	1894
Poly(hexamethylene oxydiacetate)	216.23	51	1798
		78	1791
Poly(hexamethylene 4,4′-oxydibenzoate)	340.38	74	1939
Poly(hexamethylene oxydibutyramide)	270.37	187	1847
Poly(hexamethylene oxydicaproamide)	326.48	175	1847
Poly(hexamethylene oxydienanthamide)	354.53	170	1847
Poly[hexamethylene 4,4′-(oxydimethylene)-di-2-(1,3-dioxolane)-caprylamide]	554.77	145	797
Poly(hexamethylene oxydipelarqonamide)	410.64	158	1847
Poly[hexamethylene 4,4′-(oxydiphenylene)-dioxydiacetamide]	398.46	220	402
Poly(hexamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)			*
Poly(hexamethyelene oxydivaleramide)	298.43	180	1847
Poly(hexamethyleneoxymethylene oxide)			*

Polymer	Molecular weight	Melting point	Refs.
Poly(hexamethyleneoxy-*p*-phenylene oxide)	192.26	170	1868
Poly[hexamethylene (pentamethylenedisulfonyl)-dicaproamide]	508.73	226	402
Poly(hexamethylene *N,N'*-pentamethylenediterephthalamate)	480.56	238	781
Poly[hexamethylene (pentamethylenedithio)-diethylene-diurethane]	392.57	98	1897
Poly(hexamethylene pentamethylenediurea)	270.38	251	1892
Poly[hexamethylene (*p*-phenoxy)-diacetamide]	290.36	86	402
		180	1202
Poly(hexamethylene *m*-phenylenediacetamide)	274.36	182	402
Poly(hexamethylene *p*-phenylenediacetamide)	274.36	300	454
Poly(hexamethylene 3,3'-*p*-phenylenedibutyramide)	330.47	196	1895
–, 3,3'-dimethyl- (diacid)	358.52	193	1895
Poly[hexamethylene (*p*-phenylenedioxy)-diacetamide]	306.36	237	402
		239	1202
Poly(hexamethylene *p*-phenylenedipropionamide)	302.42	290	291
		295	402
		298	1202
		300	1895
–, 3-methyl- (diacid)	316.44	227	1895
Poly(hexamethylene *m*-phenylenedisulfonamide)			*
Poly(hexamethylene *m*-phenylenediurea)	276.34		
–, 4-methyl- (diisocyanate)	290.37	235	402
Poly(hexamethylene 3,3'-phosphinylidenedibenzamide)	370.39		
–, methyl- (*P*)	384.42	172	402
Poly(hexamethylene 4,4'-phosphinylidenedibenzamide)	370.39		
–, hydroxy- (*P*)	386.39	245	402
–, methyl- (*P*)	384.42	213	402
–, phenyl- (*P*)	445.48	231	402
Poly(hexamethylene phthalamide)	246.31	150	1833
Poly(hexamethylene phthalate)	248.28	2	1794
Poly(hexamethylene pimelamide)	240.35	228	244
		202	339
Poly(hexamethylene pimelate)	242.32	50	539
		55	764
Poly(hexamethylene 1,4-piperazinediacetamide)	282.39	168	291
Poly(hexamethylene *N,N'*-piperazinediterephthalamate)	464.52	255	1174
Poly[hexamethylene (1,4-piperazinedithio)-dicarboxylate]	322.44	144	1934
Poly(hexamethylene piperazinediurea)	254.33	245	402
		265	1838
Poly(hexamethylene propylenediurea) racemic	270.33	230	1768
Poly(hexamethylene 4(H)-pyran-2,6-dicarboxamide)	250.30		
–, 4-oxo- (diacid)	264.28	77	1905
Poly(hexamethylene 2,3-pyrazoledicarboxamide)	238.29	140	1905
Poly(hexamethylene 2,4-pyrazoledicarboxamide)	238.29	271	1905
Poly(hexamethylene 2,3-pyridinedicarboxamide)	247.30	103	1905
Poly(hexamethylene 2,4-pyridinedicarboxamide)	247.30	220	1905
Poly(hexamethylene 2,5-pyridinedicarboxamide)	247.30	272	1905
Poly(hexamethylene 2,6-pyridinedicarboxamide)	247.30	220	1905
–, 1,4-dihydro-4-oxo-	263.30	178	1905
Poly(hexamethylene 3,4-pyridinedicarboxamide)	247.30	101	1905
Poly(hexamethylene 3,5-pyridinedicarboxamide)	247.30	257	1905
–, 2,6-dimethyl- (diacid)	275.35	140	1905
Poly(hexamethylene pyromellitimide)			*
Poly(hexamethylene 2,5-pyrroledipropionamide)	291.40		
–, 1-methyl- (diacid)	305.42	180	402
Poly(hexamethylene 2,5-pyrrolidinedipropionamide)	295.43		
–, 1-methyl- (diacid)	309.45	200	402
Poly(hexamethylene 5-pyrrolidonylidenedipropionamide)	295.40	250	402
Poly(hexamethylene sebacamide)			*
–, 3-methyl- (diamine)	296.46	153	402
Poly(hexamethylene sebacate)			*
Poly(hexamethylene sebacoyldiurethane)	370.45	158	402
Poly(hexamethylene 4,4'-silylenedibenzamide)	352.51		
–, dimethyl- (Si)	380.56	190	402
Poly(hexamethylene suberamide)			*

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Polymer	Molecular weight	Melting point	Refs.
Poly(hexamethylene suberate)			*
Poly[hexamethylene *N,N'*-suberoyldi(*p*-aminobenzoate)]	494.59	250	1174
Poly(hexamethylene succinamide)			*
Poly(hexamethylene succinate)	200.23	57	305
Poly(hexamethylene sulfide)			*
Poly(hexamethylene sulfone)			*
Poly(hexamethylene 4,4'-sulfonyldibenzamide)	386.46	310	1835
Poly(hexamethylene 4,4'-sulfonyldibenzoate)	388.44	280	1174
Poly[hexamethylene 4,4'-(sulfonyldimethylene)-dibenzoate]	416.49	252	1174
Poly(hexamethylene sulfonyldivaleramide)	346.49	215	402
Poly(hexamethylene sulfonyldivalerate)	348.46	92	1797
Poly(hexamethylenesulfonylpentamethylene sulfone)			*
Poly(hexamethylenesulfonyltetramethylene sulfone)			*
Poly(hexamethylene L-tartaramide)	230.26		
–, di-*O*-methyl-			*
Poly(hexamethylene terephthalamide)		*	
–, 2,5-dihydroxy- (diacid)	278.31	334	402
–, 2,5-dimethyl- (diacid)	274.36	143	402
–, *N,N'*-dimethyl-	274.36	260	829
–, methyl- (diacid)	260.34	248	402
Poly(hexamethylene terephthalate)			*
Poly[hexamethylene *N,N'*-terephthaloyldi(6-aminocaproate)]	474.60	177	1792
Poly(hexamethylene tetradecamethylenediurea)	396.62	198	634
Poly(hexamethylene tetradecanediamide)	338.54	209	2
Poly(hexamethylene 2,5-tetrahydrofurandipropionamide)	296.41	182	402
Poly[hexamethylene (tetramethylenedioxy)-dipropionamide]	314.43	110	402
Poly[hexamethylene (tetramethylenedisulfonyl)-dicaproamide]	494.70	241	402
Poly(hexamethylene *N,N'*-tetramethylenediterephthalamate)	466.53	290	781
Poly[hexamethylene (tetramethylenedithio)-diethylene-diurethane]	378.55	120	1897
Poly(hexamethylene tetramethylenediurea)	256.35	283	1892
Poly[hexamethylene 2,4,8,10-tetraoxa-3,9-spiro(5.5)hendecane-dicaprylamide]	524.74	186	797
Poly(hexamethylene 4,4'-tetrasiloxanylenedibenzamide)	490.81		
–, tetramethyl-tetraphenyl- (Si)	847.28	140	402
Poly(hexamethylene thiodibutyramide)	286.43	200	291
Poly(hexamethylene thiodienanthamide)	370.60	170	402
Poly(hexamethylene thiodipropionamide)	258.38	216	1859
		219	402
Poly(hexamethylene thiodivaleramide)	314.49	180	402
		185	1853
Poly(hexamethylene thiodivalerate)	316.46	46	767
Poly[hexamethylenethio-4,4'-(methylenediphenylene)-diethylene sulfide]	370.61	88	1819
Poly(hexamethylenethiopentamethylene sulfide)	218.42	65	1819
Poly(hexamethylene 2,5-thiophenediacetamide)	280.39	230	1905
Poly(hexamethylene 2,5-thiophenedicarboxamide)	252.33	315	1905
Poly(hexamethylene 2,5-thiophenedipropionamide)	308.44	232	402
		210	1905
Poly(hexamethylenethiotetramethylene sulfide)	204.39	67	1819
		75	181
Poly(hexamethylenethioundecamethylene sulfide)	302.59	78	1923
Poly(hexamethylene thiourea)	158.26	160	402
Poly(2,5-hexamethylene-1,3,4-triazole)	151.21		
–, 1-amino- (triazole)	166.23	275	610
Poly[hexamethylene 4,4'-(trimethylenediphenylene)-dioxy-diacetamide]	424.54	80	402
Poly[hexamethylene(trimethylenedisulfonyl)-diimino-dicaproamide]	446.59	185	402
Poly(hexamethylene *N,N'*-trimethylenediterephthalamate)	452.51	256	781
Poly(hexamethylene trimethylenediurea)	242.32	266	1892
Poly(hexamethylene undecanediamide)	296.46		
–, 6-hydroxy- (diacid)	312.45	165	402
Poly(hexamethylene urea)			*
Poly(hexamethylene 4,4'-vinylenedibenzamide)	348.44		
–, methyl- (diacid)	362.47	160	402
Poly[hexamethylene 4,4'-(*p*-xylylenedisulfonyl)-dibutyrate]	488.61	196	1174
Poly[hexamethylene *N,N'-p*-xylylenediterephthalamate)	514.58	285	1174
Poly(hexamethylene *p*-xylylenediurea)	304.39	306	634
Poly(hexamethylene *p*-xylylenediurethane)	306.36	209	634

Polymer	Molecular weight	Melting point	Refs.
Poly(1,3,5-hexatriene), 1,6-	80.13	250	1848
Poly(hexenamer)	82.15	61	439
Poly(1-hexene)			*
–, 4,4-dimethyl-	112.22	350	90
–, 4-ethyl-	112.22	234	632
–, 3-methyl-	98.19	285	441
–, 4-methyl			*
–, 5-methyl-			*
Poly[4,4′-(2,2-hexylidene)-diphenylene carbonate]	296.36	200	438
Poly(hydrazo adipamide)	142.16	320	402
Poly(hydrazo azelamide)	184.24	237	402
Poly(hydrazo docosanediamide)	366.59	210	402
Poly(hydrazo dodecanediamide)	226.32	242	402
Poly(hydrazo glutaramide)	128.13	140	402
Poly(hydrazo pimelamide)	156.18	260	402
Poly(hydrazo sebacamide)	198.27	260	402
Poly(hydrazo suberamide)	170.21	275	402
Poly(hydrazo succinamide)	114.10	360	402
Poly(hydrazo thiodivaleramide)	230.33	180	402
Poly(2-hydroxyacetic acid) [glycolic acid]			*
–, diethyl-	114.14	200	1928
–, dimethyl-	86.09	190	1174
–, isopropyl-	100.12	205	1930
Poly(*m*-hydroxybenzoic acid)	120.11	176	1799
Poly(*p*-hydroxybenzoic acid)			*
–, 3,5-di-*tert*-butyl-	232.32	480	1814
–, 3-*n*-propyl-			*
Poly(4,4′-hydroxybibenzoic acid)			*
Poly(3-hydroxy-3-butenoic acid)			*
Poly(3-hydroxybutyric acid)			*
Poly(10-hydroxycapric acid)			*
Poly(6-hydroxycaproic acid)			*
Poly[5-hydroxy-2-(1,3-dioxane)-caprylic acid]			*
Poly(7-hydroxyenanthic acid)	128.17		
–, 4-methyl- (*R* +)	142.20	36	1836
Poly[4-hydroxymethylene-2-(1,3-dioxolane)-caprylic acid]			*
Poly[5-hydroxymethylene-5-methyl-2-(1,3-dioxane)-caprylic acid]	256.34	81	797
Poly(5-hydroxy-3-oxavaleric acid)	102.09	89	305
Poly[(*p*-hydroxyphenylene)-acetic acid]	134.13	280	1174
Poly[3-(*p*-hydroxyphenylene)-propenoic acid]	146.14		
trans-		313	1943
Poly(2-hydroxypropionic acid) [lactic acid]			*
Poly(3-hydroxypropionic acid) [propiolactone]			*
–, 2,2-bischloromethyl-	169.01	302	1174
–, 3-chloromethyl-			*
–, 2,2-dibutyl-	184.28	245	1174
–, 3-dichloromethyl-			*
–, 2,2-diethyl-			*
–, 2,2-dimethyl- [pivalolactone]			*
–, 3-ethyl- [3-hydroxyvaleric acid]			*
–, –, 1,1-dichloro-	169.01	235	1199
–, 2-ethyl-2-methyl-			*
–, 3-isopropyl-			*
–, 2-methyl-			*
–, 3-methyl-, see Poly(3-hydroxybutyric acid)			*
–, 2-methyl-2-propyl-			*
Poly[4-hydroxytetramethylene-2-(1,3-dioxolane)-caprylic acid]	270.37		
cis-		23	797
trans-		23	797
Poly(3-hydroxyvaleric acid), see poly(hydroxypropionic acid), 3-ethyl-			*
Poly(5-hydroxyvaleric acid)	100.12	55	1844
		53	305
–, 2,3,4-trimethoxy-	190.20	138	305
Poly(2-imino-1,3-dithiapentamethylene)	119.20		
–, *N*-ethyl-	147.25	68	1927

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Polymer	Molecular weight	Melting point	Refs.
–, *N*-methyl-	133.23	89	1927
–, *N*-phenyl-	195.30	128	1927
–, –209.32	89	1927	
Poly(isobutene)			*
Poly(isobutylene oxide)			*
Poly(isobutylene sulfide)	88.17	187	548
Poly(4,4′-isobutylidenediphenylene carbonate)	268.31	180	438
Poly(isobutylvinyl ether)			*
Poly(isobutyraldehyde)			*
Poly(isophthalaldehyde)	134.13	80	1801
Poly(isophthalic anhydride)	148.12	259	1565
Poly[isophthalic 2,2′-(isophthaloyldiamino)-dibenzoic anhydride]	534.48	259	1871
Poly[isophthalic (4,4′-isopropylidenediphenylenedioxy)-diacetic anhydride]	474.46	65	1871
Poly[isophthalic 4,4′-(methylenedioxy)-dibenzoic anhydride]	418.36		
–, 3,3′-dimethoxy-	478.41	228	1871
Poly[isophthalic (*p*-phenylenedioxy)-diacetic anhydride]	356.29	297	1871
Poly[isophthalic 2,2′-(terephthaloyldiamino)-dibenzoic anhydride]	534.48	291	1871
Poly[isophthalic 4,4′-(tetramethylenedioxy)-dibenzoic anhydride]	460.44	100	1871
Poly[isophthalic 4,4′-(trimethylenedioxy)-dibenzoic anhydride]	446.41	165	1871
–, 3,3′-dimethoxy-	506.46	216	1871
Poly[isophthaloyl (*m*-carboxyphenoxyacetyl)-dihydrazide]	354.32	125	1833
Poly[isophthaloyl (*m*-carboxyphenoxybutyryl)-dihydrazide]	382.38	265	1833
Poly[isophthaloyl (*m*-carboxyphenoxycapryl)-dihydrazide]	410.43	249	1833
Poly[isophthaloyl (*m*-carboxyphenoxypropionyl)-dihydrazide]	368.35	238	1833
Poly[2,2′-(isophthaloyldiamino)-dibenzoic terephthalic anhydride]	534.48	264	1871
Poly(isophthaloyl dithionisophthaloyldihydrazide)	356.42	324	1889
Poly(isoprene), see Poly(1,3-butadiene), 2-methyl-			*
Poly(isopropenylmethyl ketone)			*
Poly(4,4′-isopropylidenedibenzoic anhydride)	266.30	235	1854
		240	1855
Poly[isopropylidenedimethylene, 4,4′-(methylenediphenylene)-diurethane]	354.41	190	402
Poly[isopropylidenedimethylene (1,4-piperazinedithio)-dicarboxylate]	306.40		
–, 2,5-dimethyl-	334.45	135	1934
Poly(4,4′-isopropylidenediphenylene adipate)	338.40		
–, 3,3′,5,5′-tetrachloro-			*
Poly(4,4′-isopropylidenediphenylene carbonate)			*
–, 3,3′-dichloro-	323.18	210	540
		230	736
–, 3,3′-diisopropyl-	338.45	110	1915
–, 3,3′-dimethyl-	282.34	170	540
–, 3,3′,5,5′-tetrabromo-	569.87	260	540
		267	736
–, 3,3′,5,5′-tetrachloro-	392.06	260	540
		275	736
Poly(4,4′-isopropylidenediphenylene cyclopropylene-dicarboxylate)	322.36		
cis-		130	1860
trans-		180	1860
–, 1-methyl- (diacid)	336.39		
trans-		90	1860
–, 3-methyl- (diacid)	336.39		
trans-		130	1860
Poly[(4,4′-isopropylidenediphenylenedioxy)-diacetic anhydride]	326.35	202	1855
Poly[(4,4′-isoproylidenediphenylenedioxy)-diacetic *p*-phenylenedioxydiacetic anhydride]	502.52	230	1872
Poly[(4,4′-isopropylidenediphenylenedioxy)-diacetic 4-pyridinedicarboxylic anhydride]	475.45	120	1872
Poly[(4,4′-isopropylidenediphenylenedioxy)-diacetic terephthalic anhydride]	474.47	270	1871
Poly(4,4′-isopropylidenediphenylene dithioladipate)	370.53	198	1922
Poly(4,4′-isopropylidenediphenylene dithiol-1,3-cyclohexylenedicarboxylate)	396.57	220	1922
Poly(4,4′-isopropylidenediphenylene dithionisophthalate)	390.52	152	1889
Poly(4,4′-isopropylidenediphenylene hexamethylenediurethane)	396.49	130	402
Poly(4,4′-isopropylidenediphenylene isophthalamide)	356.42	424	1886
Poly(4,4′-isopropylidenediphenylene malonate)	296.32	96	1868
Poly(4,4′-isopropylidenediphenylene 4,4′-(methylenediphenylene)-diurethane)	478.55	193	402
Poly(4,4′-isopropylidenediphenylene 1,4-piperazinedipropionate)	422.52	145	1875
Poly(isopropylidene oxide)			*
Poly(isopropylidene sulfide)	74.14	125	1882
		124	1800

Polymer	Molecular weight	Melting point	Refs.
Poly(isopropylvinyl ether)			*
Poly(ketone), see Poly(carbon monoxide-*alt*-ethylene)			*
Poly(lactic acid), see Poly(2-hydroxypropionic acid)			*
Poly(*N*,*N*′-lysyl nonamethylenediurea)	356.47		
–, methyl ester	370.49	185	1768
Poly(*N*,*N*′-lysyl pentamethylenediurea)	300.36		
–, methyl ester	314.38	175	1768
Poly(*N*,*N*′-lysyl propylenediurea)	272.30		
–, methyl ester	286.33		
racemic		175	1768
Poly(*N*,*N*′-lysyl tetramethylenediurea)	286.33		
–, methyl ester	300.36	185	1768
Poly(*N*,*N*′-lysyl trimethylenediurea)	272.30		
–, methyl ester	286.33	205	1768
Poly(*N*,*N*′-lysylurea)	172.18		
–, methyl ester	186.21	175	1768
Poly(2-mercaptoacetic acid)	74.10	169	1922
–, 2-methyl-	88.13	152	1922
Poly(6-mercaptocaproic acid)			*
–, 4-methyl-	144.24	62	1922
Poly(3-mercaptopropionic acid)	88.13	145	1922
–, 2-phthalimido-	233.24	260	1922
–, 2-(*p*-toluenesulfonamido)-	257.33	250	1922
Poly(methacrylic acid)	86.09		
–, *tert*-butyl ester	142.12		
isotactic		104	1851
syndiotactic		165	1851
–, hexadecyl ester	310.52	22	1900
–, methyl ester			*
–, octadecyl ester			*
Poly(methacrylonitrile)			*
Poly(methylene) (see also Poly(ethylene))	14.03		
–, diphenyl-	166.22	220	1831
–, di-*p*-tolyl-	194.28	180	1831
Poly(methylene adipamide)			*
Poly(4,4′-methylenedibenzoic anhydride)	238.24	332	1854
Poly[4,4′-(methylenedioxy)-dibenzoic anhydride]	270.24	220	1854
Poly[4,4′-(methylenedioxy)-dibenzoic terephthalic anhydride]	418.36		
–, 3,3′-dimethoxy-	478.41	267	1871
Poly(4,4′-methylenediphenylene adipamide)	308.38	356	402
–, 3,3′-dimethyl- (diamine)	336.44	326	402
–, –, *N*,*N*′-dibutyl-	448.65	57	402
–, –, *N*,*N*′-diethyl-	392.54	79	402
–, –, *N*,*N*′-dihexyl-	504.76	45	402
–, –, *N*,*N*′-diisoamyl-	476.70	45	402
–, –, *N*,*N*′-diisopropyl-	420.60	61	402
–, *N*,*N*′-diethyl-	364.49	62	402
–, *N*,*N*′-dimethyl-	336.44	120	402
Poly(4,4′-methylenediphenylene adipate)	310.35	140	1901
Poly(4,4′-methylenediphenylene azelamide)	350.46	275	402
		268	1443
		290	1946
–, *N*,*N*′-diethyl-	406.57	41	402
–, *N*,*N*′-dimethyl-	378.52	58	402
Poly(4,4′-methylenediphenylene carbonate)			*
–, 3,3′-dimethyl-	254.29	250	736
–, diphenyl-	378.43	230	540
–, phenyl-	302.33	215	438
Poly[4,4′-(methylenediphenylene)-diethylene sulfide]	254.39	108	1819
Poly(4,4′-methylenediphenylene disiloxanylenedipropionamide)	384.58		
–, tetramethyl- (Si)	440.69	90	402
–, –, 3,3′-dimethyl- (diamine)	454.72	265	402
Poly[3,3′-(4,4′-methylenediphenylene-*N*,*N*′-disuccinimidediyl)-imino-4,4′- methylenediphenylene amine]	556.62	295	1876
Poly(4,4′-methylenediphenylene dodecanediamide)			*
Poly(4,4′-methylenediphenylene isophthalamide)	328.37	416	1886

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Polymer	Molecular weight	Melting point	Refs.
Poly(4,4′-methylenediphenylene octadecamethylenediurea)	534.79	225	634
Poly(4,4′-methylenediphenylene octamethylenediurea)	394.52	263	634
Poly(4,4′-methylenediphenylene oxide)			*
Poly(4,4′-methylenediphenylene *m*-phenylenediurea)	358.40		
–, 3,3′-dimethyl-, 4-methyl-	400.48	307	402
Poly(4,4′-methylenediphenylene pyromellitimide)			*
Poly(4,4′-methylenediphenylene sebacamide)	364.49	280	402
		270	1443
–, *N*,*N*′-diethyl	420.60	32	402
–, 3,3′-dimethyl- (diamine)	392.54	227	402
–, *N*,*N*′-dimethyl-	392.54	55	402
Poly(4,4′-methylenediphenylene terephthalamide)	328.37	420	402
–, *N*,*N*′-dibutyl-	440.59	195	1870
–, *N*,*N*′-diethyl-	384.48	182	1870
– 3,3′-dimethyl- (diamine)	356.42	380	402
–, –, *N*,*N*′-dibutyl-	468.64	159	1870
–, –, *N*,*N*′-diethyl-	412.53	178	1870
–, –, *N*,*N*′-dimethyl-	384.48	229	1870
–, –, *N*,*N*′-dipropyl-	440.59	190	1870
–, *N*,*N*′-dimethyl-	356.42	264	1870
–, *N*,*N*′-dipropyl-	412.53	156	1870
Poly(4,4′-methylenediphenylene tetradecamethylenediurea)	478.68	229	634
Poly(*N*,*N*′-methylenediterephthalamic anhydride)	324.29	330	1565
Poly(methyleneoxynonamethylene oxide)	172.27	55	695
Poly(methyleneoxyoctadecamethylene oxide)	298.51	72	695
Poly(methylene oxypentamethylene oxide)			*
Poly(methyleneoxytetradecamethylene oxide)	242.40	69	695
Poly(methyleneoxytetramethylene oxide) [1,3-dioxopane]			*
Poly(methylene *p*-phenylene)			*
–, 2,5-dimethyl-	118.18	300	744
–, α-methyl-			*
Poly(methylene sebacamide)			*
Poly(methylene selenide)			*
Poly(methylene sulfide)			*
–, difluoro-	82.07	35	1842
Poly(methylenethiotetramethylene sulfide)	134.26	73	181
Poly(methylvinyl ether)			*
Poly(1,5-naphthalene-4′,4″-dibenzylidenimino isophthalate)	496.52		
–, 3′,3″-dimethoxy-	556.58	220	1893
Poly(1,5-naphthalene-4′,4″-dibenzylidenimino sebacate)	532.64		
–, 3′,3″-dimethoxy-	592.69	180	1893
Poly(1,5-naphthalene-4′,4″-dibenzylidenimino terephthalate)	496.52		
–, 3′,3″-dimethoxy-	556.58	230	1893
Poly(neopentylvinyl ether)			*
Poly(nonamethylene adipamide)	268.40	205	339
	242	1818	
Poly(nonamethylene azelamide)			*
Poly(nonamethylene azelate)			*
Poly[4,4′-(nonamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	446.59	200	1789
Poly(nonamethylene disiloxanylenedipropionamide)	344.60		
–, diethyl-dimethyl- (Si)	428.76	10	402
–, tetraethyl- (Si)	456.82	10	402
Poly(nonamethylene disulfide)	190.36	55	1868
		60	725
Poly(nonamethylenedithiotetramethylene disulfide)	310.59	48	1784
Poly(nonamethylene *N*,*N*′-ethylenediterephthalamate)	480.56	267	781
Poly(nonamethylene *N*,*N*′-hexamethylene diterephthalamate)	536.67	219	781
Poly(nonamethylene hexamethylenediurethane)			*
Poly(nonamethylene malonamide)			*
–, dodecyl- (diacid)	394.64	127	1774
Poly[nonamethylene 4,4′-methylenediphenylenediurethane]			*
Poly(nonamethylene 4-octenediamide)	294.44		
trans- II.		248	657
Poly(2,5-nonamethlene-1,3,4-oxadiazole)	194.28	82	1921
Poly(nonamethylene oxamide-*alt*-nonamethylene sebacamide)	536.80	234	1781

Polymer	Molecular weight	Melting point	Refs.
Poly(nonamethylene oxide)			*
Poly(nonamethylene *N,N'*-pentamethylenediterephthalamate)	522.64	194	781
Poly(nonamethylene 3,3'-phosphinylidenedibenzamide)	412.47		
–, methyl- (*P*)	426.50	141	402
Poly(nonamethylene 4,4'-phosphinylidenedibenzamide)	412.47		
–, methyl- (*P*)	426.50	167	402
–, phenyl- (*P*)	488.57	179	402
Poly(nonamethylene pimelamide)	282.43	196	2
Poly(nonmethylene pyromellitimide)			*
–, 3-bromo- (imide)	419.27	287	1267
Poly(nonamethylene sebacamide)	324.51	176	339
		202	402
Poly(nonamethylene terephthalate)			*
Poly(nonamethylene 2,5-tetrahydrofurandipropionamide)	338.49	149	402
Poly(nonamethylene *N,N'*-tetramethylenediterephthalamate)	508.61	259	781
Poly(nonamethylene tridecanediamide)	366.59	183	2
Poly(nonamethylene *N,N'*-trimethylenediterephthalamate)	494.59	211	781
Poly(nonamethylene undecanediamide)	338.54	196	2
Poly(nonamethylene urea)	184.28	220	1776
Poly(1-nonene)	126.24	19	250
		22	872
Poly[4,4'-(2,2-nonylidene)-diphenylene carbonate]	338.47	190	438
Poly(norbornene), see Poly(1-vinylene-3-cyclopentylene)			*
Poly(octadecamethylene docosanediamide)	619.08	146	454
Poly(octadecamethylene dodecanediamide)	478.80	170	454
Poly(octadecamethylene 4,4'-(ethylenedixy)-dibenzoate]	552.77	122	290
Poly(octadecamethylene octadecanediamide)	562.97	152	454
Poly(octadecamethylene *p*-phenylenediacetamide)	442.69	225	454
Poly(octadecamethylene *p*-phenylenedipropionamide)	470.74	232	454
Poly(octadecamethylene sebacamide)	450.75	171	454
Poly(octadecamethylene terephthalamide)	414.63	255	454
Poly(octadecamethylene terephthalate)	416.60	116	290
Poly(octadecamethylene *p*-xylylenediurea)	472.72	226	634
Poly(octadecamethylene anhydride)	296.45	95	1823
Poly(1-octadecene)			*
Poly(1,3-octadiene), 1,4-			*
Poly(octamethylene)	112.22		
–, 1-methyl-			*
Poly(octamethylene adipamide)	254.37	250	244
		235	291
		254	544
Poly(octamethylene adipate)			*
Poly[octamethylene *N,N'*-adipoyldi(6-aminocaproate)]	482.66	140	1792
Poly(octamethylene azelamide)			*
Poly(octamethylene 4,4'-biphenylenedicarboxylate)	352.43	197	1174
Poly(octamethylene decamethylenediurethane)	370.53	137	634
Poly[4,4'-(octamethylenedicarbonyl)-diphenylene carbonate]	380.44	175	1174
Poly[4,4'-(octamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	432.56	220	1789
Poly(octamethylene disiloxanylenedipropionamide)	330.58		
–,tetramethyl- (Si)	386.68	95	402
Poly(octamethylene disulfide)	176.34	58	725
Poly[4,4'-(2,2'-octamethylenedithiazole)-*p*-phenylene]	354.53	164	1810
Poly(octamethylenedithiotetramethylene disulfide)	296.56	45	1784
Poly(octamethylene docosanediamide)	478.80	175	454
Poly(octamethylene dodecanediamide)	338.54	202	244
		200	454
		194	291
Poly(octamethylene dodecanedioate)	340.50	73	291
Poly[octamethylene *N,N'*-dodecanedioyldi(*p*-aminobenzoate)]	578.75	188	1792
Poly(octamethylene eicosanediamide)	450.75	179	244
Poly(octamethylene *N,N'*-ethylenediterephthalamate)	466.53	282	781
Poly(octamethylene 2,5-furandicarboxamide)	264.32	125	1905
Poly(octamethylene *N,N'*-hexamethylenediterephthalamate)	522.64	238	781
Poly(octamethylene hexamethylenediurethane)			*
Poly(octamethylene 3-hexenedioate)			*

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Polymer	Molecular weight	Melting point	Refs.
Poly(octamethylene isophthalamide)	274.36	186	1868
Poly(octamethylene malonamide)			*
–, dodecyl- (diacid)	380.62	125	1774
Poly[octamethylene 4,4′-(methylenediphenylene)-diurethane]			*
Poly(octamethylene octadecanediamide)	422.70	179	454
Poly(octamethylene octamethylenedithiourea)	372.63	190	402
Poly(octamethylene octamethylenediurethane)	342.48	144	402
Poly(octamethylene 4-octenediamide)	280.41		
trans-		256	657
Poly(octamethylene 4-octenedioate)			*
Poly(2,5-octamethylene-1,3,4-oxadiazole)	180.25	110	1921
		100	1862
Poly(octamethylene oxamide)	198.27	276	743
Poly(octamethylene oxamide-*alt*-octamethylene sebacamide)	508.75	244	1781
Poly(octamethylene oxide)			*
Poly(octamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)			*
Poly(octamethylene *N,N,*′-pentamethylenediterephthalamate)	508.61	207	781
Poly(octamethylene *p*-phenylenediacetamide)	302.42	280	454
Poly(octamethylene *p*-phenylenedipropionamide)	330.47	289	454
Poly(octamethylene 3,3′-phosphinylidenedibenzamide)	398.44		
–, methyl- (*P*)	412.47	158	402
Poly(octamethylene 4,4′-phosphinylidenedibenzamide)	398.44		
–, methyl- (*P*)	412.47	172	402
– phenyl- (*P*)	474.54	202	402
Poly(octamethylene phthalamide)	274.36	123	1868
Poly(octamethylene 2,4-pyridinedicarboxamide)	275.35	190	1905
Poly(octamethylene 2,5-pyridinedicarboxamide)	275.35	232	1905
Poly(octamethylene 2,6-pyridinedicarboxamide)	275.35	192	1905
–, 1,4-dihydro-4-oxo-	291.35	86	1905
Poly(octamethylene 3,5-pyridinedicarboxiamide)	275.35	214	1905
Poly(octamethylene pyromellitimide)			*
–, 3-bromo- (imide)	405.25	320	1267
Poly(octamethylene sebacamide)	310.48	207	153
		206	454
		210	244
Poly(octamethylene sebacoyldiurea)	396.53	212	402
Poly(octamethylene suberamide)			*
Poly(octamethylene suberate)			*
Poly(octamethylene succinamide)			*
Poly(octamethylene L-tartaramide)	258.32		
–, di-*O*-methyl-			*
Poly(octamethylene terephthalamide)			*
Poly(octamethylene terephthalate)			*
Poly[octamethylene *N,N*′-terephthaloyldi(6-aminocaproate)]	502.65	167	1792
Poly(octamethylene tetradecanediamide)	366.59	196	2
Poly(octamethylene 2,5-tetrahydrofurandipropionamide)	324.46	180	402
Poly(octamethylene *N,N*′-tetramethylenediterephthalamate)	494.59	276	781
Poly(octamethylene tetramethylenedithiourea)	316.53	160	402
Poly(octamethylene 2,5-thiophenedicarboxamide)	280.39	304	1905
Poly(2,5-octamethylene-1,3,4-triazole)	179.27		
–, 1-amino- (triazole)	194.28	258	610
Poly(octamethylene *N,N*′-trimethylenediterephthalamate)	480.56	231	781
Poly(octamethylene urea)	170.26	260	291
Poly(octamethylene *p*-xylylenediurea)	332.45	278	634
Poly(octamethylene *p*-xylylenediurethane)	334.42	196	634
Poly(octenamer)			*
Poly(1-octene)			*
Poly(oxacyclobutane) [trimethylene oxide]			*
–, 3,3-bisazidomethyl-			*
–, 3,3-bisbromomethyl-	243.93	220	1807
–, 3,3-bischloromethyl-			*
–, 3,3-bisethoxymethyl-			*
–, 3,3-bisfluoromethyl-	122.12	135	1805
–, 3,3-bishydroxymethyl-	118.13	D280	1271
–, 3,3-bisiodomethyl-	337.93	290	1807

Polymer	Molecular weight	Melting point	Refs.
–, 3-*tert*-butyl-			*
–, 3,3-diethyl-			*
–, 3,3-dimethyl-			*
–, 3-hydroxy-	74.08	166	1944
Poly(oxybisdimethylsilylene)			*
Poly(4,4′-oxydibenzoic anhydride)	240.21	296	1854
Poly(oxydiethylene adipate)	216.23	– 24	765
Poly(oxydiethylene azelate)	258.31	0	696
Poly(oxydiethylene 2,2′-dibenzoate)	312.32	95	1654
Poly(oxydiethylene 3,3′-dibenzoate)	312.32	75	1552
Poly(oxydiethylene 4,4′-dibenzoate)	312.32	170	1654
Poly[(oxydiethylene)-dioxydibenzoic anhydride]	328.32	190	1854
Poly[4,4′-(oxydiethylenedioxydiethylenedioxy)-diphenylene 3,3′,4,4′-(carbonyldiphenylene)-tetracarboximide]	660.64	360	1808
Poly[4,4′-(oxydiethylenedioxy)-diphenylene 3,3′,4,4′-(carbonyldiphenylene)-tetracarboximide]	572.53	402	1808
Poly[3,4′-(oxydiethylenedioxy)-diphenylene] 3,3′,4,4′-(oxydiphenylene)-tetracarboximide]			*
Poly(oxydiethylene 4,4′-ethylenedibenzoate)	340.38	105	1654
Poly(oxydiethylene glutarate)	202.21	– 22	765
Poly(oxydiethylene hexamethylenediurethane)	274.32	120	633
Poly(oxydiethylene isophthalate)	236.22	90	1654
Poly(oxydiethylene malonate)	174.15	– 11	765
Poly(oxydiethylene 4,4′-methylenedibenzoate)	326.35	145	1654
Poly(oxydiethylene oxalate)	160.13	10	765
Poly(oxydiethylene oxydiacetate)	204.18	6	1798
Poly[(oxydiethylene)-oxy-*p*-phenylene oxide]	180.20	136	1868
Poly(oxydiethylene phthalate)	236.22	11	1794
Poly(oxydiethylene pimelate)	230.26	– 26	765
Poly(oxydiethylene sebacate)			*
Poly(oxydiethylene suberate)	244.29	35	765
Poly(oxydiethylene succinate)	188.18	– 8	765
Poly(oxydiethylene sulfonyldivalerate)	304.34	66	1797
Poly(oxydiethylene terephthalate)			*
Poly(oxydiethylene thiodivalerate)	304.40	– 35	767
Poly[(oxydimethylenedioxy)-dibenzoic anhydride]	300.27	192	1854
Poly(oxydipentamethylene 5-oxadodecanediamide)	384.56	102	1847
Poly(oxydipentamethylene 6-oxadodecanediamide)	384.56	100	1847
Poly(oxydipentamethylene 6-oxaheptadecanediamide)	454.70	128	1847
Poly(oxydipentamethylene 5-oxahexadecanediamide)	440.67	122	1847
Poly(oxydipentamethylene 6-oxapentadecanediamide)	426.64	117	1847
Poly(oxydipentamethylene 5-oxasebacamide)	356.51	127	1847
Poly(oxydipentamethylene 5-oxatetradecanediamide)	412.61	118	1847
Poly(oxydipentamethylene 6-oxatridecanediamide)	398.59	106	1847
Poly(oxydipentamethylene 5-oxaundecanediamide)	370.53	108	1847
Poly(oxydipentamethylene oxydibutyramide)	342.48	138	1847
Poly(oxydipentamethylene oxydicaproamide)	398.59	128	1847
Poly(oxydipentamethylene oxydienanthamide)	426.64	129	1847
Poly(oxydipentamethylene oxydipelargonamide)	482.75	129	1847
Poly(oxydipentamethylene oxydivaleramide)	370.53	134	1847
Poly(3,4′-oxydiphenylene adipamide)	310.35	225	1832
Poly(4,4′-oxydiphenylene adipamide)	310.35	360	1832
Poly(3,4′-oxydiphenylene azelamide)	352.43	150	1832
Poly(4,4′-oxydiphenylene azelamide)	352.43	300	1832
Poly(4,4′-oxydiphenylene carbonate)	228.20	235	540
Poly(4,4′-oxydiphenylene-4″,4‴-dibenzylidenimino adipate)	518.57		
–, 3″,3‴-dimethoxy-	578.62	135	1893
Poly(4,4′-oxydiphenylene-4″,4‴-dibenzylidenimino isophthalate)	538.56		
–, 3″,3‴-dimethoxy-	598.61	185	1893
Poly(4,4′-oxydiphenylene-4″,4‴-dibenzylidenimino sebacate)	574.68		
–, 3″,3‴-dimethoxy-	634.73	140	1893
Poly(4,4‴((4′,4″-oxydiphenylene)-dioxy-diphenylene pyromellitimide)			*
Poly(2,2′-(4,4′-oxydiphenylene-4,4′-diquinoline)-4,4′-biphenylene)	574.68	475	1778
–, 3,3′-diphenyl-	726.88	500	1778
Poly(4,4′-oxydiphenylenedithiodecamethylene disulfide)	436.70	80	1837
Poly(4,4′-oxydiphenylenedithioethylene disulfide)	324.49	158	1837
Poly(4,4′-oxydiphenylenedithiotetramethylene disulfide)	352.54	94	1837

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Polymer	Molecular weight	Melting point	Refs.
Poly(3,4′-oxydiphenylene dodecanediamide)	394.52	150	1832
Poly(4,4′-oxydiphenylene dodecanediamide)	394.52	290	1832
Poly(4,4′-oxydiphenylene ketone)			*
Poly(4,4′-oxydiphenylene 3,3′,4,4′-oxydiphenylene-tetracarboximide)			*
Poly(4,4′-oxydiphenylene *m*-phenylenediketone)	300.31		
–, 2,2′-diethyl-	356.42	242	1767
–, 2,6-diisopropyl-	384.48	185	1767
–, 2-ethyl-2′-isopropyl-	370.45	228	1767
–, 2-isopropyl-2′-methyl-	356.42	220	1767
Poly(4,4′-oxydiphenylene *p*-phenylenediketone)	300.31		
–, 2-isopropyl-2′-methyl-	356.42	262	1767
Poly(3,4′-oxydiphenylene pimelamide)	324.38	170	1832
Poly(4,4′-oxydiphenylene pimelamide)	324.38	315	1832
Poly(4,4′-oxydiphenylene pyromellitimide)			*
Poly(3,4′-oxydiphenylene sebacamide)	366.46	165	1832
Poly(4,4′-oxydiphenylene sebacamide)	366.46	310	1832
Poly(3,4′-oxydiphenylene suberamide)	338.41	175	1832
Poly(4,4′-oxydiphenylene suberamide)	338.41	310	1832
Poly(3,4′-oxydiphenylene undecanediamide)	380.49	140	1832
Poly(4,4′-oxydiphenylene undecanediamide)	380.49	270	1832
Poly[2,2′-(6,6′-oxydiquinoline)-4,4′-biphenylene]	422.49		
–, 4,4′-diphenyl-	574.68	480	1778
–, 3,3′,4,4′-tetraphenyl-	726.88	480	1778
Poly[2,2′-(6,6′-oxydiquinoline)-4,4′-oxydiphenylene]	438.49		
–, 4,4′-diphenyl-	590.68	448	1778
–, 3,3′,4,4′-tetraphenyl-	742.88	476	1778
Poly[2,2′-(6,6′-oxydiquinoline)-*p*-phenylene]	346.39		
–, 4,4′-diphenyl-	498.58	455	1778
–, 3,3′,4,4′-tetraphenyl-	650.78	475	1778
Poly(oxydisilyleneditrimethylene pyromellitimide)	374.50		
–, tetramethyl- (Si)	430.61	210	1361
Poly(oxyditetramethylene hexamethylenediurethane)	330.42	124	633
Poly(oxyditrimethylene adipamide)	242.32	190	291
		213	1775
Poly(oxyditrimethylene terephthalamide)	262.31	281	1775
Poly(oxymethylene) [formaldehyde]			*
Poly(1-pentadecene)	210.40	54	872
Poly(1,3-pentadiene), 1,2-			*
–, 4-methyl-			*
Poly(1,3-pentadiene), 1,4-			*
–, 2-methyl-			*
Poly(pentamethylene)	70.14		
–, 1,2-dichloro-	139.02		
trans-		75	1925
Poly(pnetamethylene adipamide)	212.29	223	339
		225	402
		251	1852
		258	1818
–, (L)-1-carboxy- (diamine)	256.30	113	1883
–, (L-1-carboxy- (diamine)	512.60	102	1883
–, 2,2,3,3,4,4-hexafluoro- (diamine)	320.24		
–, –, *N,N′*-dibutyl-	432.45	15	402
–, –, *N,N′*-diethyl-	376.34	20	402
–, –, *N,N′*-diisopropyl-	404.40	35	402
–, –, *N,N′*-dimethyl-	348.29	30	402
Poly(pentamethylene adipamide-*alt*-pentamethylene oxamide)	368.48	281	1781
Poly(pentamethylene adipate)	214.26	40	1120
–, 2,2,3,3,4,4-hexafluoro- (diamine)	322.20	34	1933
Poly(pentamethylene azelamide)			*
Poly(pentamethylene azelamide-*alt*-pentamethylene oxamide)	410.56	268	1781
Poly(pentamethylene azelate)	256.34	41	696
		50	1120
Poly(pentamethylene carbonate)	130.14	46	1820
Poly(pentamethylene *m*-carboxycarbanilate)	249.27	80	1891
Poly(pentamethylene *p*-carboxycarbanilate)	249.27	160	1891

Polymer	Molecular weight	Melting point	Refs.
Poly[pentamethylene *p*-(carboxyphenylene)-acetamide]	246.31	216	1904
Poly[pentamethylene (cyclopropylenedicarboxoyl)-diurethane]	284.27		
trans-		250	1865
Poly(pentamethylene cyclopropylenediurethane)	228.25	170	1865
Poly[pentamethylene (decamethylenedisulfonyl)-dicaproamide]	564.84	202	402
Poly(pentamethylene 2,2′-dibenzoate)	310.35	13	1552
Poly(pentamethylene 3,3′-dibenzoate)	310.35	60	1552
Poly(pentamethylene 4,4′-dibenzoate)	310.35	170	1552
Poly(4,4′-pentamethylenedibenzoic anhydride)	294.35	118	1854
Poly[3,3′-(pentamethylenedioxy)-dibenzoic anhydride]	326.35	176	1854
Poly[4,4′-(pentamethylenedioxy)-dibenzoic anhydride]	326.35	188	1854
Poly[(pentamethylenedioxy)-didecamethylene oxide]	398.67	72	542
Poly[(pentamethylenedioxy)-dihexamethylene oxide]	286.46	46	542
Poly(pentamethylene disulfide)	134.26	44	295
Poly(pentamethylene dithiolisophthalate)	266.38	175	1922
Poly(pentamethylene dithiolterephthalate)	266.38	232	1922
Poly(pentamethylene dodecanedioate)	298.42	60	1792
Poly(pentamethylene *N,N′*-ethylenbediterephthalamate)	424.45	312	781
Poly[pentamethylene (ethylenedithio)-diethylenediuethane]	336.46	130	1897
Poly(pentamethylene glutaramide)			*
Poly(pentamethylene glutarate)	200.23	25	1120
Poly[pentamethylene (hexamethylenedisulfonyl)-dicaproamide]	508.73	210	402
Poly(pentamethylene *N,N′*-hexamethyleneditерephthalamate)	480.56	255	781
Poly[pentamethylene (hexamethylenedithio)-diethylene-diurethane]	392.57	94	1897
Poly(pentamethylene hexamethylenediurethane)			*
Poly(pentamethylene isophthalate)	234.25	82	1794
Poly(pentamethylene malonamide)			*
–, dodecyl- (diacid)	338.54	208	1774
Poly[pentamethylene 4,4′-(methylenediphenylene)-diurethane]			*
Poly(pentamethylene octadecanediamide)	380.62	167	339
Poly(pentamethylene 4-octenediamide)	238.33		
trans- II.		258	657
Poly(pentamethylene oxalate)	158.15	54	1120
Poly(pentamethylene oxamide-*alt*-pentamethylene pimelamide)	382.50	273	1781
Poly(pentamethylene oxamide-*alt*-pentamethylene sebacamide)	424.58	258	1781
Poly(pentamethylene oxamide-*alt*-pentamethylene suberamide)	396.53	273	1781
Poly(pentamethylene oxydiacetamide)	200.24	130	291
Poly(pentamethylene oxydiacetate)	202.21	33	1798
Poly(pentamethylene 4,4′-oxydibenzoate)	326.35	70	1939
Poly(pentamethyleneoxy-*p*-phenylene oxide)	178.23	164	1868
Poly[pentamethylene (pentamethylenedisulfonyl)-dicaproamide]	494.70	212	402
Poly(pentamethylene *N,N′*-pentamethyleneditерephthalamate)	466.53	198	781
Poly[pentamethylene (pentamethylenedithio)-diethylene-diurethane]	378.55	87	1897
Poly(pentamethylene *m*-phenylenedisulfonamide)	304.38	160	1384
Poly(pentamethylene phthalate)	234.25	9	1794
Poly(pentamethylene pimelamide)			*
Poly(pentamethylene pimelate)			*
Poly(pentamethylene sebacamide)	268.40	195	339
Poly(pentamethylene sebacate)	270.37	51	696
		57	1120
Poly(pentamethylene sebacoyldiurea)	354.45	205	402
Poly(pentamethylene suberamide)	240.35	202	339
Poly(pentamethylene suberate)	242.32	46	1120
Poly(pentamethylene succinate)	186.21	35	1120
Poly(pentamethylene sulfide)			*
Poly(pentamethylene sulfone)			*
Poly(pentamethylene sulfonyldivalerate)	334.43	86	1797
Poly(pentamethylenesulfonyltetramethylene sulfone)			*
Poly(pentamethylene terephthalamide)			*
Poly(pentamethylene terephthalate)			*
Poly(pentamethylene tetradecanediamide)	324.51	178	339
Poly(pentamethylene 2,5-tetrahydrofurandipropionamide)	282.38	153	402
Poly(pentamethylene *N,N′*-tetramethyleneditерephthalamate)	452.51	284	781
Poly(pentamethylene (tetramethylenedithio)-diethylene-diurethane]	364.52	95	1897
Poly(pentamethylene thiodivalerate)	302.43	42	767

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Polymer	Molecular weight	Melting point	Refs.
Poly(pentamethylenethiotetramethylene sulfide)	190.37	67	181
Poly(2,5-pentamethylene-1,3,4-triazole)	137.19		
–, 1-amino-	152.20	260	610
Poly(pentamethylene tridecanediamide)	310.48	176	339
Poly(pentamethylene *N,N'*-trimethylenediterephthalamate)	438.48	235	781
Poly(pentamethylene undecamethylenediurethane)	342.48	123	633
Poly(pentamethylene undecanediamide)	282.43	176	291
Poly(pentamethylene urethane)			*
Poly(pentenamer)			*
Poly(1-pentene)			*
–, 5-cyclohexyl-	152.28	123	328
–, 5-diisobutylamino-			*
–, 5-diisopropylamino-			*
–, 4,4-dimethyl-			*
–, 3-ethyl-	98.19	425	632
–, 3-methyl-			*
–, 4-methyl-			*
–, 5,5,5-trifluoro-	124.11	225	1813
–, 4-trifluoromethyl-	138.13	255	1813
–, 5-trimethylsilyl-			*
Poly[4,4'-(2,2-pentylidene)-diphenylene carbonate]	282.34	220	438
–, 4-methyl-	296.37	220	438
Poly[4,4'-(3,3-pentylidene)-diphenylene carbonate]	282.34	195	1915
–, 2,4-dimethyl-	310.39	220	1915
Poly[4,4'-perfluorodiphenyleneoxyperfluoropentamethylene oxide)	578.16		
–, 1,1,5,5-tetrahydro- (pentamethylene)	506.19	145	1908
Poly(*m*-phenylene)			*
Poly(*p*-phenylene)			*
Poly(*m*-phenylene adipamide)	218.26	296	402
		344	828
–, 3-methyl- (diamine)	232.28	225	402
Poly(*o*-phenylene adipamide)	218.26	179	1868
Poly(*m*-phenylene adipamide)			*
Poly(*p*-phenylene azelamide)	260.34	345	828
Poly(*p*-phenylene azelate)	262.31	135	1174
Poly(*p*-phenylene *m*-carboranedicarboxylate)			*
Poly(*p*-phenylene cyclobutylene)	130.19		
–, dipyridyl-			*
Poly(*p*-phenylene cyclobutylenedicarboxamide)	216.24	405	723
Poly(*p*-phenylene cyclobutylenedicarboxylic acid)	218.21		
–, dibutyl ester	330.42		
–, –, dicyano-	380.44	330	723
–, diethyl ester			*
–, –, dicyano-	324.34	340	723
–, diisopropyl ester	302.37	320	723
–, –, dicyano-	352.39	320	723
–, dimethyl ester			*
–, –, dicyano-	296.28	290	723
–, diphenyl ester			*
–, dipropyl ester	302.37		
–, –, dicyano-			*
Poly(*m*-phenylene cyclopropylenedi(carboxylate)	204.18		
cis-		65	1860
trans-		105	1860
Poly(*p*-phenylene cyclopropylenedicarboxylate)	204.18		
cis-		160	1860
trans-		280	1860
Poly(*p*-phenylenediacetic anhydride)	176.17	92	1565
		152	1855
Poly(*m*-phenylene-3,3'-dibenzamido isophthalamide)	476.49	410	1788
Poly(*m*-phenylene-4,4'-dibenzamido isophthalamide)	476.49	450	1788
Poly(*p*-phenylene-3,3'-dibenzamido isophthalamide)	476.49	475	1788
Poly(*p*-phenylene-4,4'-dibenzamido isophthalamide)	476.49	490	1788
Poly(*m*-phenylene-3,3'-dibenzamido terephthalamide)	476.49	460	1788
Poly(*m*-phenylene-4,4'-dibenzamido terephthalamide)	476.49	467	1788

Polymer	Molecular weight	Melting point	Refs.
Poly(*p*-phenylene-3,3′-dibenzamido terephthalamide)	476.49	480	1788
Poly(*p*-phenylene-4,4′-dibenzamido terephthalamide)	476.49	555	1788
Poly(*p*-phenylene-4′,4″-dibenzylidenimino adipate)	426.47		
–, 3′,3″-dimethoxy-	486.52	135	1893
Poly(*p*-phenylene-4′,4″-dibenzylidenimino isophthalate)	446.46		
–, 3′,3″-dimethoxy-	506.52	230	1893
Poly(*p*-phenylene-4′,4″-dibenzylidenimino sebacate)	482.58		
–, 3′,3″-dimethoxy-	542.63	120	1893
Poly(*p*-phenylenediethylene adipamide)	274.36	310	402
Poly(*p*-phenylenediethylene azelamide)	316.44	250	402
		290	454
Poly(*p*-phenylenediethylene docosanediamide)	498.80	230	454
Poly(*p*-phenylenediethylene dodecanediamide)	358.53	280	454
Poly(*p*-phenylenediethylene heptadecanediamide)	428.66	249	454
Poly(*p*-phenylenediethylene hexadecanediamide)	414.63	258	454
Poly(*p*-phenylenediethylene octadecanediamide)	442.69	248	454
Poly(*p*-phenylenediethylene octamethylenediurethane)	362.47	212	402
		212	291
Poly(*p*-phenylenediethylene pentadecanediamide)	400.61	248	454
Poly(*p*-phenylenediethylene *m*-phenylenediacetamide)	322.41	222	402
Poly[*p*-phenylenediethylene 3,3′-(*p*-phenylene)-dipropionate]	352.43	219	1174
Poly(*p*-phenylenediethylene sebacamide)	330.47	285	1841
		285	402
		300	454
Poly(*p*-phenylenediethylene terephthalate)	296.32	330	1881
Poly(*p*-phenylenediethylene tetradecanediamide)	386.58	267	454
Poly(*p*-phenylenediethylene thiodicaproamide)	390.58	251	1824
Poly(*p*-phenylenediethylene thiodienanthamide)	418.64	234	1824
Poly(*p*-phenylenediethylene thiodipelargonamide)	474.75	222	1824
Poly(*p*-phenylenediethylene thiodiundecanoamide)	530.86	207	1824
Poly(*p*-phenylenediethylene thiodivaleramide)	362.53	252	1824
Poly(*p*-phenylenediethylene tridecanediamide)	372.55	262	454
Poly(*p*-phenylenediethylene undecandiamide)	344.50	275	454
Poly[(*m*-phenylenedioxy)-diacetic anhydride]	208.17	130	1565
Poly[(*p*-phenylenedioxy)-diacetic anhydride]	208.17	158	1855
		152	1565
		160	1872
Poly[(*p*-phenylenedioxy)-diacetic 3,4-pyridinedicarboxylic anhydride]	357.27	312	1872
Poly[(*p*-phenylenedioxy)-diacetic terephthalic anhydride]	356.29	324	1871
Poly[(*p*-phenylenedioxy)-diacetic 4,4′-(trimethylenedioxy)-dibenzoic anhydride]	506.46	260	1872
Poly[(*p*-phenylenedioxy)-diethylene octamethylenediurethane]	394.47	212	633
		212	402
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene carbonate]	320.30	215	1915
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene ketone]			*
Poly[4,4′-(*p*-phenylenedioxy)-diphenylene 3,3′,4,4′-(*p*-phenylenedioxy)-diphenylenetetracarboximide]	658.62	660	1018
Poly(*p*-phenylenedipentamethylene terephthalate)	380.48	116	1881
Poly(*p*-phenylenedipropionic anhydride)	204.22	92	1565
Poly(*p*-phenylene disiloxanylene)	152.30		
–, tetramethyl- (Si)			*
Poly(*m*-phenylene disiloxanylenedipropionamide)	294.46		
–, 4-methyl- (diamine)	308.49		
–, –, diethyl-dimethyl- (Si)	392.65	50	402
–, –, tetraethyl- (Si)	420.70	90	402
–, –, tetramethyl- (Si)	364.59	70	402
–, tetraethyl- (Si)	406.67	70	402
–, tetramethyl- (Si)	350.57	90	402
Poly(*o*-phenylene disiloxanylenedipropionamide)	294.46		
–, diethyl-diemthyl- (Si)	378.62	40	402
–, tetraethyl- (Si)	406.67	40	402
–, tetramethyl- (Si)	350.57	40	402
Poly(*p*-phenylene disiloxanylenedipropionamide)	294.46		
–, diethyl-dimethyl- (Si)	378.62	300	402
–, tetraethyl- (Si)	406.67	290	402
–, tetramethyl- (Si)	350.57	300	402

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Polymer	Molecular weight	Melting point	Refs.
Poly(*p*-phenyleneditetramethylene terephthalate)	352.43	217	1881
		220	1174
Poly[4,4′-(2,2′-*m*-phenylenedithiazole)-oxy-*p*-diphenylene]	410.51	340	1810
Poly[(*p*-phenylenedithio)-dipropionic anhydride]	268.34	55	1565
		50	1857
Poly(*p*-phenyleneditrimethylene hexamethylenediurethane)	362.47	158	633
		240	291
Poly(*p*-phenyleneditrimethylene terephthalate)	324.38	162	1881
Poly(*p*-phenyleneethylene adipamide)	246.31		
–, 1,1-dimethyl- (diamine)	274.36	235	1802
Poly[*p*-phenylene 4,4′-(ethylenedioxy)-dibenzoate]	376.37		
–, methyl- (diol)	390.39	316	1375
Poly(*p*-phenyleneethylene sebacamide)	302.42		
–, 1,1-dimethyl- (diamine)	330.47	205	1802
Poly(*p*-phenyleneethylene succinamide)	218.26		
–, 1,1-dimethyl- (diamine)	246.31	250	1802
Poly(*p*-phenyleneethylene terephthalamide)	266.30		
–, 1,1-dimethyl- (diamine)	294.35	360	1802
Poly(*p*-phenylene glutaramide)	204.23	398	828
Poly(*m*-phenylene hexamethylenediurethane)	278.31	150	402
Poly(*p*-phenylene hexamethylenediurethane)	278.31	150	402
Poly(*m*-phenylene isophthalamide)			*
Poly(*p*-phenylene isophthalamide)	238.25	500	748
Poly(*m*-phenylene isophthalate)	240.21	245	1837
Poly(*p*-phenylene isophthalate)			*
Poly(*p*-phenylene malonate)	178.14	233	1868
Poly(*p*-phenylene oxide)			*
–, 2,6-dimethoxy-			*
–, 2,6-dimethyl-			*
–, 2,6-diphenyl-			*
Poly(*p*-phenyleneoxytrimethylene oxide)	150.18	166	1868
Poly(*m*-phenylene 3,3′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	260	402
–, –, 4-methyl- (diamine)	390.38	244	402
Poly(*m*-phenylene 4,4′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	340	402
–, –, 4-methyl- (diamine)	390.38	252	402
–, phenyl- (*P*)	438.42	340	402
–, –, 4-methyl- (diamine)	452.45	280	402
Poly(*o*-phenylene 3,3′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	181	402
Poly(*o*-phenylene 4,4′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	193	402
–, phenyl- (*P*)	438.42	207	402
Poly(*p*-phenylene 3,3′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	264	402
Poly(*p*-phenylene 4,4′-phosphinylidenedibenzamide)	362.32		
–, methyl- (*P*)	376.35	340	402
–, phenyl- (*P*)	438.42	340	402
Poly(*p*-phenylene pimelamide)			*
Poly(*p*-phenylene 1,4-piperazinedipropionate)	304.35	189	1875
–, methyl- (diacid)	318.37	164	1875
Poly(*m*-phenylene sebacamide)			*
–, 3-methyl- (diamine)	288.39	200	402
Poly(*o*-phenylene sebacamide)	274.36	125	1868
Poly(*p*-phenylene sebacamide)			*
Poly(*m*-phenylene suberamide)			*
Poly(*o*-phenylene suberamide)	246.31	150	1868
Poly(*p*-phenylene suberamide)			*
Poly(*m*-phenylene succinate)	192.17	260	1174
Poly(*p*-phenylene succinate)	192.17	300	1841
Poly(*p*-phenylene sulfide)			*
–, 2-methyl-	122.18	100	679
Poly(*p*-phenylene sulfonate)	156.16	276	1785
Poly(*m*-phenylene terephthalamide)			*

Polymer	Molecular weight	Melting point	Refs.
Poly(*p*-phenylene terephthalamide)			*
Poly(*p*-phenylene terephthalate)			*
–, *tert*-butyl- (diol)	296.32	444	1375
–, phenyl- (diol)			*
Poly[*p*-phenylene 4,4′-(tetramethylene dioxy)-dibenzoate]	404.42		
–, methyl- (diol)	418.45	212	1375
Poly[*p*-phenylene 4″,4‴-(thiodiphenylene)-dioxy-3,3′,4,4′-diphenylenetetracarboximide]	582.59	426	1771
Poly(*m*-phenylene urea)	134.14		
–, 4-methyl-	148.16	284	402
Poly(phosphazene)	47.00		
–, bis-β-naphthoxy-	331.31	>350	553
–, bis-phenoxy-			*
–, –, di-*m*-bromo-			*
–, –, di-*p*-bromo-			*
–, –, di-*m*-chloro-			*
–, –, di-*p*-chloro-			*
–, –, di-2,4-dichloro-			*
–, –, di-*m*-fluoro-			*
–, –, di-*p*-fluoro-			*
–, –, di-isopropyl-			*
–, –, di-*p*-phenyl-			*
–, –, di-*m*-trifluoromethyl-	367.19	330	553
–, ,bis-2,2,2-trifluoroethoxy-			*
–, dimethyl-			*
Poly(phosphinideneditrimethylene adipamide)	258.30		
–, octyl-	370.52	135	1859
Poly(pimelic anhydride)	142.15	55	1823
Poly(piperazine adipamide)		*	
–, 3-methyl- (diacid)	210.28	300	402
Poly(piperazine azelamide)	238.33	148	402
Poly(piperazine cyclopropylenedicarboxamide)	180.21		
trans-		330	1860
–, 1-methyl- (diacid)	194.23		
trans-		130	1860
–, 3-methyl- (diacid)	194.23		
trans-		280	1860
Poly[(piperazinedicarbamoyl)-dithiohexamethylene disulfide]	352.54	196	1793
Poly[(piperazinedicarbamoyl)-dithiotetramethylene disulfide]	324.49	204	1793
Poly(1,4-piperazinediethylene hexamethylenediurethane)	342.44	165	291
Poly(1,4-piperazinediethylene sulfone)	204.29	200	1875
Poly[piperazine (ethylenediamino)-dipropionamide]	254.33		
–, *N,N′*-diisopropyl-	338.50	97	1875
–, *N,N′*-diemthyl-	282.39	113	1875
Poly[piperazine (ethylenedixoy)-diacetamide]	228.25	165	1894
Poly(piperazine isophthalamide)	216.24	340	1833
–, 2,5-dimethyl- (diamine)	244.28	315	1833
–, methyl- (diamine)	230.27	280	1833
Poly(piperazine 4-octenediamide)			*
Poly(piperazine oxydiacetamide)	184.20	258	1894
Poly(piperazine phthalamide)	216.24	325	1833
–, 2,5-dimethyl- (diamine)	244.29	>350	1833
–, methyl- (diamine)	230.27	350	1833
Poly(piperazine 1,4-piperazinedipropionamide)	280.37	D270	1875
–, methyl- (diacid)	294.40	218	1875
–, methyl- (diamine)	294.40	217	1875
Poly(piperazine sebacamide)			*
Poly(piperazine suberamide)	224.30	300	402
Poly(piperazine terephthalamide)	216.24	350	1833
		380	402
–, 2,5-dimethyl- (diamine)	244.29	350	1833
–, methyl- (diamine)	230.27	350	1833
Poly[piperazine (trimethylenedithio)-dipropionamide]	302.45	100	402
Poly(1,4-piperidinediyltrimethylene)	125.22		
–, 2,6-dioxo-			*

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Polymer	Molecular weight	Melting point	Refs.
Poly(1,4-piperidene urethane)	127.14	270	444
Poly(pivalolactone), see Poly(3-hydroxypropionic acid), 2,2-dimethyl-			*
Poly(propene)			*
–, 3-cyclohexyl-			*
–, 3-cyclopentyl-			*
–, hexafluoro-			*
–, 3-phenyl- [allylbenzene]			*
–, –, 2,5-dimethyl-	146.23	338	187
–, –, 3,4-dimethyl-	146.23	275	187
–, –, 3,5-dimethyl-	146.23	252	187
–, 3-silyl-			*
–, 3-*m*-tolyl-	132.21	180	187
–, 3-*o*-tolyl-	132.21	290	187
–, 3-*p*-tolyl-	132.21	240	187
–, 3-trimethylsilyl-			*
Poly(propiolactone), see Poly(3-hydroxypropionic acid)			*
Poly(propionaldehyde)			*
–, 3-cyano-			*
–, 3-methoxycarbonyl-			*
Poly(propylene oxalate)	130.10	180	305
Poly(propylene oxide)			*
–, 2-chloromethyl-	106.55	126	1816
		126	593
–, 3-(1-naphthoxy)-	200.24	235	445
–, 3-(2-naphthoxy)-	200.24	297	445
–, 3-phenoxy-			*
–, –, *o*-chloro-			*
–, –, *p*-chloro-	184.62	176	445
		192	833
–, –, dimethyl-	178.23	125	833
–, –, *o*-isopropyl-	192.26	128	445
–, –, *p*-methoxy-	180.20	200	833
–, –, *m*-methyl-	164.20	169	445
–, –, *o*-methyl-	164.20	191	445
–, –, *p*-methyl-	164.20	212	445
		205	833
–, –, *o*-phenyl-	226.28	293	445
–, –, 2,4,6-trichloro-	253.51	130	445
Poly(propylene 4,4′-sulfonyldibenzamide)	344.38	335	1835
Poly(propylidene)	42.08	90	617
Poly(propylvinyl ether)	86.13		
–, 1-methyl-			*
Poly(2,5-pyrazinecyclobutylene)	132.16		
–, 2,4-diphenyl-			*
Poly(2,5-pyrrolidipropionic anhydride)	193.20		
–, *N*-methyl-	207.223	188	1565
Poly(2,6-quinoline)	1127.2		
–, 4-phenyl-	203.24	552	1778
Poly(sebacic anhydride)	184.24	80	1564
		82	1822
Poly(sebacoyl dithionisophthaloyldihydrazide)	392.54	294	1889
Poly(selenium)			*
Poly(siloxane)	46.10		
–, diethyl-			*
–, dimethyl-			*
–, diphenyl-			*
–, dipropyl-			*
Poly(silylene)	28.09		
–, dimethyl-			*
–, diphenyl-			*
Poly[4,4′-(silylenediphenylene)-dimethylene 4,4′-disiloxanylenedibenzamide]	524.80		
–, dimethyl- (diamine), tetramethyl- (Si)	608.96	165	402
Poly[4,4′-(silylenediphenylene)-dimethylene 4,4′-silylenedibenzamide]	478.70		
–, dimethyl- (diamine), dimethyl- (Si)	550.81	215	402

Polymer	Molecular weight	Melting point	Refs.
Poly[4,4′-(silylenediphenylene)-dimethylene 4,4′-tetrasiloxanylenedibenzamide]	617.00		
–, dimethyl- (diamine)	645.06		
–, –, tetramethyl-tetraphenyl- (Si)	1001.5	150	402
Poly(4,4′-stilbenediyloxyheptamethylene oxide)	308.42		
–, α-methyl-	322.45	93	1772
Poly(4,4′-stilbenediyloxyhexamethylene oxide)	294.40		
–, α-methyl-	308.42	166	1772
Poly(4,4′-stilbenediyloxymethylene oxide)	224.46		
–, α-methyl-	238.29	254	1772
Poly(4,4′-stilbenediyloxyoctamethylene oxide)	322.45		
–, α-methyl-	336.48	114	1772
Poly(4,4′-stilbenediyloxypentamethylene oxide)	280.37		
–, α-methyl-	294.40	116	1772
Poly(4,4′-stilbenediyloxytetramethylene oxide)	266.34		
–, α-methyl-	280.37	182	1772
Poly(4,4′-stilbenediyltrimethylene oxide)	252.31		
–, α-methyl-	266.34	128	1772
Poly(styrene)			*
–, *p-tert*-butyl-			
–, –, isotactic			*
–, –, syndiotactic	160.26	310	1372
–, *m*-chloro-			
–, –, isotactic	138.60	124	1372
–, –, syndiotactic	138.60	192	1372
–, *p*-chloro-			
–, –, isotactic	138.60	165	1372
–, –, syndiotactic	138.60	299	1372
–, 2,4-dimethyl-			
–, –, isotactic	132.21	310	75
		350	74
–, 2,5-dimethyl-			
–, –, isotactic	132.21	330	75
		340	74
–, 3,4-dimethyl-			
–, isotactic	132.21	240	75
–, 3,5-dimethyl-			
–, –, isotactic	132.21	290	75
–, *o*-fluoro-			
–, –, isotactic			*
–, *m*-fluoro-			
–, –, isotactic	122.14	250	1910
–, –, syndiotactic	122.14	275	1372
–, *p*-fluoro-			
–, –, isotactic			*
–, –, syndiotactic	122.14	322	1372
–, α-methyl-			*
–, –, *p*-isopropyl-			
–, –, –, isotactic	160.26	203	888
–, –, *p*-methyl-			
–, –, –, isotactic			*
–, *o*-methyl-			
–, –, isotactic			*
–, –, *p*-fluoro-			
–, –, –, isotactic			*
–, *m*-methyl-			
–, –, isotactic			*
–, –, syndiotactic	118.18	206	1372
–, *p*-methyl-			
–, –, isotactic			*
–, –, syndiotactic			*
–, β-nitro-			
–, –, isotactic	149.15	285	1937
–, *p*-trifluoroacetyl-			
–, –, isotactic	200.16	178	1834

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Polymer	Molecular weight	Melting point	Refs.
–, *p*-trimethylsilyl-			
–, –, isotactic			*
Poly(styrene-*alt*-carbon monoxide)			*
–, *o*-hydroxy-	148.16	195	1884
Poly(styrene oxide)	120.15	149	1809
		140	318
		162	693
Poly(styrene sulfone)	168.21	215	1916
Poly(styryl pyridine-*alt*-vinyl pyridine)	286.38	350	1920
Poly(suberic anhydride)	156.18	66	1823
Poly(4,4′-sulfinyldiphenylene carbonate)	260.26	250	540
Poly(4,4′-sulfonyldiphenylene carbonate)	276.26	210	540
Poly(sulfonyldipropionic anhydride)	192.18	237	1856
Poly(*p*-sulfonylphenylene sebacamide)	338.42	117	1267
Poly(sulfur trioxide)	80.06	32	288
Poly(terephthalaldehyde)	134.13	120	1801
		240	1174
–, 2,5-dimethyl-	162.19	140	1801
Poly(terephthalic anhydride)			*
Poly[terephthalic 2,2′-(terephthaloyldiamino)-dibenzoic anhydride]	534.48	312	1871
Poly[terephthalic 4,4′-(tetramethylenedioxy)-dibenzoic anhydride]	460.44	215	1871
Poly[terephthalic 4,4′-(trimethylenedioxy)-dibenzoic anhydride]	446.41	285	1871
–, 3,3′-dimethoxy-	506.46	255	1871
Poly[terephthaloyl (*m*-carboxyphenoxyacetyl)-dihydrazide]	354.32	218	1833
Poly(terephthaloyl dithionisophthaloyldihydrazide)	356.42	309	1889
Poly(tetradecamethylene adipate)			*
Poly(tetradecamethylene docosanediamide)	562.97	153	454
Poly(tetradecamethylene dodecanediamide)	422.70	175	454
Poly(tetradecamethylene *N,N′*-ethylenediterephthalamate)	550.70	249	781
Poly(tetradecamethylene *N,N′*-hexamethylenediterephthalamate)	606.80	222	781
Poly(tetradecamethylene 3-hexenedioate)			*
Poly(tetradecamethylene octadecanediamide)	506.36	158	454
Poly(tetradecamethylene 4-octenedioate)			*
Poly(tetradecamethylene *N,N′*-pentamethylenediterephthalamate)	592.78	200	781
Poly(tetradecamethylene *p*-phenylenediacetamide)	386.58	235	454
Poly(tetradecamethylene *p*-phenylenedipropionamide)	414.63	246	454
Poly(tetradecamethylene sebacamide)	394.64	175	454
Poly(tetradecamethylene suberate)			*
Poly(tetradecamethylene terephthalamide)	358.53	265	454
Poly(tetradecamethylene *N,N′*-tetramethylenediterephthalamate)	578.75	248	781
Poly(tetradecamethylene *N,N′*-trimethylenediterephthalamate)	564.72	208	781
Poly(tetradecamethylene *p*-xylylenediurea)	416.61	248	634
Poly(tetradecanedioic anhydride)	240.34	91	1823
Poly(1-tetradecene)			*
Poly(tetrahydrofuran) [tetramethylene oxide]			*
Poly(2,5-tetrahydrofurandipropionic anhydride)	198.22	135	1565
Poly(2,5-tetrahydropyrroledipropionic anhydride)	197.23		
–, *N*-methyl-	211.26	103	1565
Poly(tetramethylene adipamide)			*
Poly(tetramethylene adipamide-*alt*-tetramethylene oxamide)	340.42	325	1781
Poly(tetramethylene adipate)			*
Poly[tetramethylene *N,N′*-adipyldi(6-aminocaproate)]	426.55	156	1792
Poly(tetramethylene azelamide)			*
Poly(tetramethylene azelamide-*alt*-tetramethylene oxamide)	382.50	300	1781
Poly(tetramethylene azelate)	242.32	37	325
		39	696
		51	766
Poly(tetramethylene carbonate)	116.12	59	1820
Poly(tetramethylene *m*-carboxycarbanilate)	235.24	60	1891
Poly(tetramethylene *p*-carboxycarbanilate)	235.24	186	1891
Poly[tetramethylene *p*-(carboxyphenylene)-acetamide]	232.28	221	1904
Poly(tetramethylene 1,4-cyclohexylenediurethane)	256.30	260	633
Poly[tetramethylene (cyclopropylenedicarboxoyl)-diurethane]	270.24	226	1865
Poly(tetramethylene cyclopropylenediurethane)	214.22	180	1865
Poly[tetramethylene (decamethylenedisulfonyl)-dicapromide]	550.81	219	402

Polymer	Molecular weight	Melting point	Refs.
Poly(tetramethylene 2,2′-dibenzoate)	296.32	36	1552
Poly(tetramethylene 3,3′-dibenzoate)	296.32	66	1552
Poly(tetramethylene 4,4′-dibenzoate)	296.32	280	1566
		260	1552
		328	1174
Poly(4,4′-tetramethylenedibenzoic anhydride)	280.32	263	1854
Poly[4,4′-(tetramethylenedicarbonyl)-diphenylene succinate]	380.40	242	1174
Poly[3,3′-(tetramethylenedioxy)-dibenzoic anhydride]	312.32	199	1854
Poly[4,4′-(tetramethylenedioxy)-dibenzoic anhydride]	312.32	204	1854
–, 3,3′-dimethoxy-	372.37	172	1855
Poly[(tetramethylenedioxy)-didecamethylene oxide]	384.64	73	542
Poly[(tetramethylenedioxy)-dihexamethylene oxide]	272.43	50	542
Poly[(tetramethyleneoxy)-ditrimethylene oxamide]	258.32	160	291
		167	402
Poly(tetramethylene disiloxanylenedipropionamide)	274.47		
–, tetramethyl- (Si)	330.58		
–, –, *N,N′*-dimethyl-	358.63	120	402
Poly(tetramethylene disulfide)	120.23	39	295
		44	725
Poly[4,4′-(2,2′-tetramethylenedithiazole)-*p*-phenylene]	298.42	250	1810
Poly[4,4′-(tetramethylenedithio)-dibenzoic anhydride]	344.44	206	1854
Poly(tetramethylene dithiodicarboxylate)	208.25	115	1936
Poly(tetramethylene dithioladipate)	232.36	128	1922
Poly(tetramethylene dithiolisophthalate)	252.35	174	1922
Poly(tetramethylene dithiolsebacate)	288.47	98	1922
Poly(tetramethylene dithiolterephthalate)	252.35	310	1922
Poly(tetramethylene dithionisophthalate)	252.35	118	1889
Poly(tetramethylenedithiotrimethylene disulfide)	226.43	53	1784
Poly(tetramethylene dodecanediamide)	282.43	245	153
Poly[tetramethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)]	522.64	192	1792
Poly[tetramethylene 4,4′-(ethylenedioxy)dibenzoate]	356.37	180	290
Poly[tetramethylene (ethylenedithio)-diethylenediurethane]	322.44	177	1897
Poly(tetramethylene glutarate)	186.21	38	766
Poly[tetramethylene (hexamethylenedisulfonyl)-dicaproamide]	494.70	228	402
Poly(tetramethylene *N,N′*-hexamethylenediterephthalamate)	466.53	300	781
Poly[tetramethylene (hexamethylenedithio)-diethylene-diurethane]	378.55	117	1897
Poly(tetramethylene hexamethylenediurethane)			*
–, 1,4-dimethyl- (diol)	286.37	104	633
Poly(tetramethylene isophthalamide)	218.26	262	828
Poly(tetramethylene isophthalate)			*
Poly(tetramethylene malonamide)			*
–, dodecyl- (diacid)	324.51	201	1774
Poly(tetramethylene malonate)	158.15	−24	766
Poly[tetramethylene 4,4′-(methylenediphenylene)-diamino-dithiodicarboxylate]	404.50	104	1934
Poly[tetramethylene 4,4′-(methylenediphenylene)-diurethane]			*
Poly(tetramethylene nonamethylenediurethane)	300.40	140	633
Poly(tetramethylene octamethylenediurethane)	286.37	160	633
Poly(tetramethylene 4-octenediamide)			*
Poly(tetramethylene oxalate)	144.13	105	766
Poly(tetramethylene oxamide-*alt*-tetramethylene pimelamide)	354.45	301	1781
Poly(tetramethylene oxamide-*alt*-tetramethylene sebacamide)	396.53	290	1781
Poly(tetramethylene oxamide-*alt*-tetramethylene suberamide)	368.48	310	1781
Poly(tetramethylene oxide), see Poly(tetrahydrofuran)			*
Poly(tetramethylene oxydiacetate)	188.18	70	1798
Poly(tetramethylene 4,4′-oxydibenzoate)	312.32	881	939
Poly(tetramethylene 3,3′,4,4′-oxydiphenylenetetracarboximide)	362.34	352	1015
Poly[tetramethylene (pentamethylenedisulfonyl)-dicaproamide]	480.68	232	402
Poly[tetramethylene (pentamethylenedithio)-diethylene-diurethane]	364.52	113	1897
Poly(tetramethylene pentamethylenediurethane)	244.29	159	633
Poly(tetramethylene *p*-phenylenediacetate)	248.28	63	325
Poly(tetramethylene *m*-phenylenedisulfonamide)	290.35	215	1384
Poly(tetramethylene 4,4′-phosphinylidenedibenzamide)	342.34		
–, methyl- (*P*)	356.36	208	402
–, phenyl- (*P*)	418.43	230	402
Poly(tetramethylene phthalate)	220.22	18	11794

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Polymer	Molecular weight	Melting point	Refs.
Poly(tetramethylene pimelamide)	212.29	233	339
		238	402
Poly(tetramethylene pimelate)	214.26	41	766
Poly[tetramethylene (1,4-piperazinedithio)-dicarboxylate]	294.38	188	1934
–, 2,5-dimethyl-	322.44	113	1934
Poly(tetramethylene 4,4′-propylenedibenzamide)	336.44	180	402
Poly[tetramethylene 4,4′-(propylenedioxy)-dibenzamide]	368.43	140	402
Poly(tetramethylene sebacamide)	254.33	254	153
		239	291
Poly(tetramethylene sebacate)			*
Poly(tetramethylene suberamide)	226.32	250	291
Poly(tetramethylene suberate)	228.29	56	291
Poly(tetramethylene succinamide)			*
–, *N,N′*-dimethyl-	198.27	123	402
Poly(tetramethylene succinate)			*
–, D-1,2-dimethoxy- (diacid)	232.23	92	609
Poly(tetramethylene sulfide)			*
Poly(tetramethylene sulfone)	120.17	100	1845
		>271	39
Poly(tetramethylene 4,4′-sulfonyldibenzamide)	358.41	358	1835
Poly(tetramethylene sulfonyldivalerate)	320.40	88	1797
Poly(tetramethylene L-tartaramide)	202.21		
–, di-*O*-methyl-			*
Poly(tetramethylene terephthalamide)	218.26	436	829
		475	1940
–, *N,N′*-dimethyl-	246.31	272	829
Poly(tetramethylene terephthalate)			*
Poly[tetramethylene *N,N′*-terephthaloyldi(6-aminocaproate)]	446.54	190	1792
Poly(tetramethylene 2,5-tetrahydrofurandipropionamide)	268.36	210	402
Poly[tetramethylene (tetramethylenedithio)-diethylene-diurethane]	350.49	119	1897
Poly(tetramethylene tetramethylenediurethane)	230.26	193	291
		193	633
Poly(tetramethylene tetrathiodicarboxylate}	272.37	92	1936
Poly(tetramethylene thiodicarboxylate)	176.19	90	1936
Poly(tetramethylene thioditrimethylenediurethane)	290.38	133	633
Poly(tetramethylene thiodivalerate)	288.40	38	767
Poly(tetramethylenethiotrimethylene sulfide)	162.32	64	181
Poly(tetramethylene thiourea)	130.21	213	402
Poly(2,5-tetramethylene-1,3,4-triazole)	123.16		
–, 1-amino-	138.17	300	610
Poly[tetramethylene 1,1′-(4,4′-trimethylenedipiperidine)-dithiodicarboxylate]	418.61	80	1934
Poly(tetramethylene *N,N′*-trimethyleneditерephthalamate)	424.45	193	781
Poly(tetramethylene trithiodicarboxylate)	240.31	66	1936
Poly(tetramethylene undecamethylenediurethane)	328.45	146	633
Poly(tetramethylene undecanediamide)	268.40	208	339
		208	402
Poly(tetramethylene urea)	114.15	400	402
Poly(tetramethylene *p*-xylylenediurethane)	278.31	227	634
Poly(tetramethyl-*p*-silphenylene siloxane), see Poly(*p*-phenylene disiloxanylene), tetramethyl-			*
Poly(3,6,13,16-tetraoxa-8,10-octadecadiyne hexamethylenediurethane)	454.52	138	1782
Poly(3,6,13,16-tetraoxa-8,10-octadecadiyne (4,4′-methylenediphenylene)-diurethane)	536.58	130	1782
Poly(thiodiethylene dithioladipate)	264.43	70	1922
Poly(thiodiethylene dithiolterephthalate)	284.42	146	1922
Poly(thiodiethylene hexamethylenediurethane)	290.38	134	633
Poly(thiodiphenylene carbonate)			*
Poly(4,4′-thiodiphenylene oxide)			*
Poly(thiodipropionic anhydride)	160.19	55	1857
Poly(thioditetramethylene hexamethylenediurethane)	346.49	125	633
Poly(2,5-thiophenedipropionic anhydride)	210.25	78	1565
Poly(tridecamethylene *N,N′*-ethyleneditерephthalamate)	536.67	257	781
Poly(tridecamethylene *N,N′*-hexamethyleneditерephthalamate)	592.78	222	781
Poly(tridecamethylene *N,N′*-pentamethyleneditерephthalamate)	578.75	194	781
Poly(tridecamethylene pyromellitimide)			*
Poly(tridecamethylene *N,N′*-tetramethyleneditерephthalamate)	564.72	248	781
Poly(tridecamethylene tridecanediamide)			*

Polymer	Molecular weight	Melting point	Refs.
Poly(tridecamethylene *N,N'*-trimethylenediterephthalamate)	550.70	206	781
Poly(tridecanedioic anhydride)	226.32	78	1823
Poly(1-tridecene)	182.35	48	872
Poly(trimethylene adipamide-*alt*-trimethylene oxamide)	312.37	303	1781
Poly(trimethylene adipate)			*
–, 2,2-bisbromomethyl- (diol)	372.05	120	1874
–, 2,2-bischloromethyl- (diol)	283.15	108	1874
–, 2-bromomethyl-2-chloromethyl- (diol)	327.60	111	1874
–, 2,2-diemthyl- (diol)	214.26	37	609
Poly(trimethylene adipoyldiurethane)	272.26		
–, 2-hydroxy- (diol)	288.26	208	402
Poly(trimethylene amine)	57.10	126	1786
Poly(trimethylene azelamide-*alt*-trimethylene oxamide)	354.45	290	1781
Poly(trimethylene azelate)			*
–, 2,2-dimethyl- (diol)	256.34	0	609
Poly(poly-trimethylenebiphenylene)	194.28		
–, α,α,α',α'',α'',α''-hexafluoro-	302.20	155	1780
Poly[trimethylene *p*-(carboxyphenylene)-acetamide]	218.26	281	1904
Poly(trimethylene 1,4-cyclohexylenedicarboxylate)	212.24		
trans-		110	1918
Poly(trimethylene cyclopropylenedicarboxamide)	168.20		
trans-		310	1860
Poly(trimethylene cyclopropylenediurethane)	200.19	170	1865
Poly(trimethylene 2,2'-dibenzoate)	282.30	78	1552
–, 1-methyl- (diol)	296.32	38	1552
Poly(trimethylene 3,3'-dibenzoate)	282.30	78	1552
–, 1-methyl- (diol)	296.32	87	1552
Poly(trimethylene 4,4'-dibenzoate)	282.30	249	1552
–, 1-methyl- (diol)	296.32	135	1552
Poly(4,4'-trimethylenedibenzoic anhydride)	266.30	215	1854
Poly[3,3'-(trimethylenedioxy)-dibenzoic anhydride]	298.29	197	1854
Poly[4,4'-(trimethylenedioxy)-dibenzoic anhydride]	298.29	267	1854
		263	1872
–, 3,3'-dimethoxy-	358.35	175	1855
Poly[(4,4'-trimethylenedipiperidinediyl)-dicarbamoyldithio-tetramethylene disulfide]	448.72	150	1793
Poly(trimethylene disulfide)	106.20	67	295
		88	725
Poly(trimethylene dodecanedioate)			*
Poly[trimethylene (ethylenedithio)-diethylenediurethane]	308.41	157	1897
Poly(trimethylene glutarate)			*
Poly[trimethylene (hexamethylenedithio)-diethylenediurethane]	364.52	115	1897
Poly(trimethylene hexamethylenediurethane)			*
–, 2-butyl-2-ethyl- (diol)	314.43	60	402
–, 2,2-dimethyl- (diol)	272.35	120	402
–, 2-methyl- (diol)	258.32	50	402
Poly(trimethylene isophthalate)	206.20	132	111
–, 1-methyl- (diol)	220.22	55	1794
Poly(trimethylene malonate)	144.13	33	609
–, 2,2-dimethyl- (diol)	172.18	67	609
Poly[trimethylene 4,4'-(methylenediphenylene)-diurethane]			*
–, 2-butyl-2-ethyl- (diol)	410.51	120	402
–, 2,2-diethyl- (diol)	382.46	150	402
–, 1-ethyl-2-propyl- (diol)	396.49	140	402
Poly(trimethylene octadecanedioate)			*
Poly(trimethylene octamethylenediurethane)	272.35		
–, 1-methyl- (diol)	286.37	82	633
		82	402
Poly(trimethylene 4-octenediamide)	210.28		
trans-		271	657
Poly(trimethylene oxalate)	130.10	86	678
		88	305
		89	609
–, 2,2-bischloromethyl-	227.04	122	1874
–, 2,2-dimethyl-	158.15	111	609
Poly(trimethylene oxamide-*alt*-trimethylene sebacamide)	368.48	276	1781

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Polymer	Molecular weight	Melting point	Refs.
Poly(trimethylene oxamide-*alt*-trimethylene suberamide)	340.42	295	1781
Poly(trimethylene oxide), see Poly(oxacyclobutane)			*
Poly(trimethylene-*p*-oxybenzoate)	178.19	185	1841
		211	327
Poly(trimethylene oxydiacetate)	174.15	32	1798
–, 1-methyl- (diol)	188.18	8	1798
Poly(trimethylene 4,4′-oxydibenzoate)	298.30	90	1939
Poly[trimethylene (pentamethylenedithio)-diethylene-diurethane]	350.49	112	1897
Poly(trimethylene *p*-phenylenediacetate)	234.25	58	1918
Poly(trimethylene *m*-phenylenedisulfonamide)	276.33	157	1384
Poly(trimethylene phthalate)	206.20		
–, 1-methyl- (diol)	220.22	0	1794
Poly(trimethylene pimelate)			*
Poly(trimethylene sebacate)			*
–, 2,2-dimethyl- (diol)	270.37	26	609
Poly(trimethylene sebacoyldiurethane)	328.36		
–, 2-hydroxy- (diol)	344.36	189	402
Poly(trimethylene suberate)			*
–, 2,2-dimethyl- (diol)	242.32	17	609
Poly(trimethylene succinate)			*
–, 2,2-bischloromethyl- (diol)	255.10	74	1874
–, 2,2-dimethyl- (diol)	186.21	86	609
Poly(trimethylene sulfide)			*
–, 3,3-diethyl-	130.25	85	1770
–, 3,3-dimethyl-			*
Poly(trimethylene sulfone)	106.14	300	1890
–, 3,3-diethyl-	162.25	364	1890
–, 3,3-dimethyl-	134.19	266	1890
		303	1917
–, 3,3-dipentyl-	246.41	385	1890
Poly(trimethylene 4,4′-sulfonyldibenzamide)	344.38	298	1835
–, 2,2-dimethyl- (diamine)	372.44	284	1835
–, 1-methyl- (diamine)	358.41	272	1835
Poly(trimethylene sulfonyldivalerate)	306.38		
–, 1-methyl- (diol)	320.40	35	1797
Poly(trimethylene terephthalamide)	204.23	399	829
–, *N,N′*-dimethyl- (diamine)	232.28	220	829
Poly(trimethylene terephthalate)			*
–, 2,2-dimethyl-	234.25	140	291
–, 1-methyl- (diol)	220.22	87	1794
Poly(trimethylene tetradecanedioate)	298.42	66	1792
Poly[trimethylene (tetramethylenedithio)-diethylene-diurethane]	336.46	113	1897
Poly(trimethylene thiodivalerate)	274.38	−36	767
–, methyl- (diol)	288.40	−36	767
Poly(trimethylene *N,N′*-trimethyleneditерephthalamate)	410.43	235	781
Poly(trimethylene undecanedioate)			*
Poly(undecamethylene adipamide)	296.46	218	544
Poly[4,4′-(undecamethylenedioxy)-diphenyleneoxy-*p*-xylylene oxide]	474.64	193	1789
Poly(undecamethylene *N,N′*-ethyleneditерephthalamate)	508.61	251	781
Poly(undecamethylene *N,N′*-hexamethyleneditерephthalamate)	564.72	217	781
Poly(undecamethylene oxamide-*alt*-undecamethylene sebacamide)	592.91	221	1781
Poly(undecamethylene *N,N′*-pentamethyleneditерephthalamate)	550.70	199	781
Poly(undecamethylene pyromellitimide)			*
Poly(undecamethylene sebacamide)	352.56	169	339
Poly(undecamethylene terephthalamide)	316.44	292	544
Poly(undecamethylene *N,N′*-tetramethyleneditерephthalamate)	536.67	248	781
Poly(undecamethylene *N,N′*-trimethyleneditерephthalamate)	522.64	198	781
Poly(undecanedioic anhydride)	198.26	70	1823
Poly(1-undecene)	154.30	36	872
Poly(valeraldehyde)	86.13	155	329
Poly(vinyl acetate)	86.09		
–, trifluoro-			*
Poly(vinyl alcohol)			*
–, 2-methyl-	58.08	140	1885

Polymer	Molecular weight	Melting point	Refs.
Poly(vinyl benzoate)	148.16		
–, *p-tert*-butyl-	204.27	327	1821
Poly(vinyl bromide)			*
Poly(vinyl-*t*-butyl ketone)	112.17	150	1812
		>240	1843
Poly(*N*-vinyl carbazole)			*
Poly(vinyl chloride)			*
Poly(vinylcyclobutane)			*
Poly(vinylcycloheptane)			*
Poly(vinylcyclohexane)			*
–, *o*-methoxy-	140.23	195	1815
–, 3-methyl-	124.23	355	1867
–, 4-methyl-	124.23	250	1867
Poly(vinylcyclohexene)	108.18	418	1850
Poly(vinylcyclohexyl ketone)	138.21	>240	1843
Poly(vinylcyclopentane)			*
Poly(vinylcyclopropane)			*
Poly(*N*-vinyldiphenylamine)	195.26	320	1811
Poly(1-vinylene-3-cyclopentylene) [norbornene]			*
Poly(vinyl fluoride)			*
Poly(vinylidene chloride)			*
Poly(vinylidene fluoride)			*
Poly(vinylisopropyl ketone)	98.14	220	1843
Poly(vinyl laurate)	226.36	16	1906
Poly(vinylmethyl ketone)			*
Poly(vinyl myristate)	254.41	28	1888
Poly(1-vinyl naphthalene)			*
Poly(vinyl palmitate)	282.47	41	1888
		46	1906
Poly(3-vinyl pyrene)	228.39	214	1873
Poly(2-vinylpyridine)			*
Poly(vinyl stearate)	310.52	54	1846
		52	1888
		58	1906
Poly(*m*-xylylene)	104.15	80	546
–, 4,6-dimethyl-	132.21	135	1187
Poly(*p*-xylylene)			*
–, bromo-	183.05	270	395
–, chloro-			*
–, cyano-	129.16	270	395
–, 2,5-dichloro-	173.04	183	1831
		>300	395
–, 2,5-dimethyl-	132.21	350	1187
–, ethyl-			*
–, methyl-	118.18	210	395
		230	142
–, α,α,α′,α′-tetrafluoro-	176.11	500	1794
Poly(*m*-xylylene adipamide)			*
Poly(*p*-xylylene adipamide)	246.31	333	385
		340	402
Poly(*p*-xylylene adipate)	248.28	70	325
		80	547
		81	1919
Poly(*m*-xylylene azelamide)	288.39	172	402
Poly(*p*-xylylene azelamide)	288.39	263	385
		282	454
Poly(*p*-xylylene azelate)	290.36	59	325
		74	547
		79	1919
Poly(*p*-xylylene carbonate)	164.16	185	1820
Poly(*p*-xylylene 1,4-cyclohexylenedicarboxylate)	274.32		
trans-		106	547
Poly(*p*-xylylene disulfide)	168.27	175	1861
Poly[(*p*-xylylenedisulfonyl)-diacetic anhydride]	332.34	170	1856
Poly[(*p*-xylylenedisulfonyl)-dipropionic anhydride]	360.40	260	1856

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Polymer	Molecular weight	Melting point	Refs.
Poly[(*p*-xylylenedithio)-diacetic anhydride]	268.34	88	1565
		87	1857
Poly[(*p*-xylylenedithio)-dipropionic anhydride]	296.40	91	1565
		55	1857
Poly(*p*-xylylene dithioladipate)	280.41	170	1922
–, 2,3,5,6-tetramethyl-	336.52	255	1922
Poly(*p*-xylylene dithiolisophthalate)	300.40	210	1922
Poly(*p*-xylylene dithiol-4,4′-methylenedibenzoate)	390.52	274	1922
Poly(*p*-xylylene dithiol-*p*-phenylenediacetate)	328.45		
–, 2,3,5,6-tetramethyl- (diol)	384.56	262	1922
Poly(*p*-xylylene dithiolsebacate)	336.52		
–, 2,3,5,6-tetramethyl- (diol)	392.62	180	1922
Poly(*p*-xylylene docosanediamide)	470.74	240	402
		225	454
Poly(*p*-xylylene dodecanediamide)	330.47	272	385
		270	454
Poly(*p*-xylylene dodecanedioate)	332.44	94	547
Poly(*p*-xylylene glutaramide)	232.28	280	402
Poly(*p*-xylylene glutarate)	206.20	58	1919
Poly(*p*-xylylene heptadecanediamide)	400.61	239	454
Poly(*p*-xylylene hexadecanediamide)	386.58	248	454
–, *N,N′*-dimethyl-	414.63	85	1877
Poly(*m*-xylylene isophthalamide)	266.30	225	402
Poly(*p*-xylylene isophthalamide)	266.30	290	385
Poly(*p*-xylylene isophthalate)	268.27	100	547
Poly(*p*-xylylene malonamide)	204.23	110	402
Poly(*p*-xylylene octadecanediamide)	414.63	235	402
		242	454
–, *N,N′*-diethyl-	470.74	50	1877
–, *N,N′*-dimethyl-	442.69	80	1877
Poly(*p*-xylyleneoctamethylenediurethane)	334.42	168	402
Poly(*p*-xylylene 5-oxadodecanediamide)	332.44	200	1847
Poly(*p*-xylylene 6-oxadodecanediamide)	332.44	206	1847
Poly(*p*-xylylene 6-oxaheptadecanediamide)	402.58	223	1847
Poly(*p*-xylylene 5-oxahexadecanediamide)	388.55	220	1847
Poly(*p*-xylylene oxalate)	164.12	214	1919
Poly(*p*-xylylene 6-oxapentadecanediamide)	374.52	217	1847
Poly(*p*-xylylene 5-oxasebacamide)	304.39	234	1847
Poly(*p*-xylylene 5-oxatetradecanediamide)	360.50	215	1847
Poly(*p*-xylylene 6-oxatridecanediamide)	346.47	207	1847
Poly(*p*-xylylene 5-oxaundecanediamide)	318.42	190	1847
Poly(*p*-xylylene oxydibutyramide)	290.36	241	1847
		243	1853
Poly(*p*-xylylene oxydicaproamide)	346.47	234	1847
Poly(*p*-xylylene oxydienanthamide)	374.52	229	1847
Poly(*p*-xylylene oxydipelargonamide)	430.63	215	1847
Poly(*p*-xylylene oxydivaleramide)	318.42	243	1847
Poly(*p*-xylylene pentadecanediamide)	372.55	241	454
Poly(*p*-xylylene *p*-phenylenediacetate)	296.32	146	291
Poly(*p*-xylylene phthalamide)	266.30	230	402
Poly(*m*-xylylene pimelamide)	260.34	192	402
Poly(*p*-xylylene pimelamide)	260.34	284	385
Poly(*p*-xylylene pimelate)	262.30	67	547
		66	1919
Poly(*m*-xylylene sebacamide)	302.42	193	402
Poly(*p*-xylylene sebacamide)			*
Poly(*p*-xylylene sebacate)	304.39	84	325
		88	1919
		93	547
Poly(*m*-xylylene suberamide)	274.36	213	402
Poly(*p*-xylylene suberamide)	274.36	305	385
Poly(*p*-xylylene suberate)	276.33	82	547
Poly(*p*-xylylene succinamide)	218.26	360	402
Poly(*p*-xylylene succinate)	220.22	115	547
Poly(*m*-xylylene terephthalamide)	266.30	300	402

Polymer	Molecular weight	Melting point	Refs.
Poly(*p*-xylylene terephthalamide)	266.30	350	402
Poly(*p*-xylylene terephthalate)	268.27	242	547
		267	1881
Poly(*m*-xylylene tetradecanediamide)	358.53	192	402
Poly(*p*-xylylene tetradecanediamide)	358.53	257	454
Poly(*p*-xylylene thiodicaproamide)	362.53	236	1853
Poly(*p*-xylylene thiodienanthamide)	390.58	228	1853
Poly(*p*-xylylene thiodipelargonamide)	446.69	214	1824
Poly(*p*-xylylene thiodiundecanoamide)	502.80	200	1824
Poly(*p*-xylylene thiodivaleramide)	334.48	242	1853
Poly(*p*-xylylene tridecanediamide)	344.50	247	454
Poly(*p*-xylylene undecanediamide)	316.44	264	454

D. APPENDIX: FORMULA INDEX TO THE TABLES

As the number of crystallizable polymers and their complexity increased, it became necessary to devise a simple, reasonably unambiguous scheme to relate polymer structure to the specific name given to that polymer.

Some time ago, the compiler designed a shorthand notation to represent the structure of linear polymers which has proved to be useful. This scheme requires one to write the line formula for the constitutional base unit (without concern for priority or direction rules) and to count the groups/atoms comprising the unit in a specified order. In a sense, this scheme is analogous to the Chemical Abstracts Formula Index, but emphasizes groups commonly occuring in linear organic polymers rather than atoms exclusively. This scheme has the additional advantage of ensuring that data for a given polymer are not entered in the tables under differing polymer names.

The rules of this scheme are simple – count the number of groups/atoms in the constitutional base unit in the following order:

1. $-CH_2-$ (methylene as such, but not the $-CH_2-$ portion of a $-CH_3$ group).
2. $-CO-$ (carbonyls).
3. $-C_6H_4-$ (disubstituted benzenes; phenylenes; monosubstituted benzenes with the fifth proton counted under point 5 below).
4. Carbon atoms not already counted (as in $-CH_3$, $-CH=$, $-C_6H_3=$, etc.).
5. Protons (–H) not already counted (as in $-CH_3$, $-NH-$, etc.).
6. Oxygen atoms (exclusive of those in carbonyls).
7. Nitrogen atoms.
8. Sulfur atoms.
9. Other atoms (F, Cl, Br, Si, etc.).

The string of nine integers thus generated, arranged in the above order, is the required structure code. By looking up this code in the Formula Index Table one can find the name under which the polymer in question appears in the data tables or, alternatively, by its absence, that no data has been found for the polymer in question. The Formula Index Table is arranged in ascending order of the structural code in columns 2–10 with the assigned polymer name in column 1. Note that some very long, complicated names have been truncated. This should cause little difficulty in the use of this table. Examples of the application of this scheme are shown below.

Experience has shown that this scheme generates unique codes for the vast majority of linear polymers capable of crystallizing. Where two or more polymers do have the same structural code, the choice of correct name is usually obvious from the structure. Of course, certain classes of polymers do tend to generate many members all having a single code – for example, all of the condensation poly(amides) having a given total number of methylene groups and differing only in the proportioning of these between the diol and diacid constituents. Again, no ambiguities arise.

An advantage of this structural coding scheme is its adaptability to electronic data processing equipment. The logical step beyond this scheme could be a strict sequential notation detailing each group and its position (similar to the ACS–IUPAC Nomenclature Recommendation) in a short-hand notation for conciseness and for computer compatibility.

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Polymer	Molecular weight	Melting point	Refs.
Poly(*p*-xylylene terephthalamide)	266.30	350	402
Poly(*p*-xylylene terephthalate)	268.27	242	547
		267	1881
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Poly(*p*-xylylene tetradecanediamide)	358.53	257	454
Poly(*p*-xylylene thiodicaproamide)	362.53	236	1853
Poly(*p*-xylylene thiodienanthamide)	390.58	228	1853
Poly(*p*-xylylene thiodipelargonamide)	446.69	214	1824
Poly(*p*-xylylene thiodiundecanoamide)	502.80	200	1824
Poly(*p*-xylylene thiodivaleramide)	334.48	242	1853
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4. Carbon atoms not already counted (as in $-CH_3$, $-CH=$, $-C_6H_3=$, etc.).
5. Protons (–H) not already counted (as in $-CH_3$, $-NH-$, etc.).
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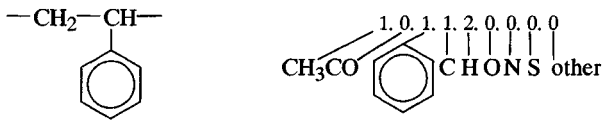
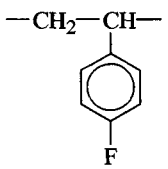
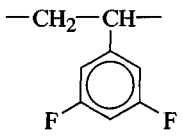
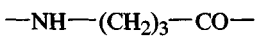
Examples*

Structure		
		Poly(styrene)
	1.0.1.1.1.0.0.0.1	Poly(*p*-fluorostyrene)
	1.0.0.7.4.0.0.0.2	Poly(3,5-difluorostyrene)
	5.1.0.0.1.0.1.0.0	Poly(6-aminocaproic acid)

In the first example, there is one methylene, one phenylene, one carbon atom not otherwise counted (in –CH=), and two protons not otherwise counted (one in –CH= and one in $-C_6H_4H-$). In the second example, the fluorine atom replaces a proton from the substituted benzene ring – no other change in code occurs. In the third example, the trisubstituted benzene ring is removed from category 3 ($-C_6H_4-$) and is placed in category 4 (carbon atoms not otherwise counted) as 6 carbon atoms and in category 5 (protons) as 3 protons. In the last example, there are five methylene groups, one carbonyl, no other carbons, one proton (from –NH–), no uncounted oxygens, and one nitrogen. The approximately 2700 entries in the Formula Index Table were generated in a similar fashion.

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
0	0	0	0	0	0	0	0	0	0
Selenium	0	0	0	0	0	0	0	0	1
Silylene	0	0	0	0	0	0	0	0	1
Sulfur	0	0	0	0	0	0	0	1	0
Dichlorophosphazene	0	0	0	0	0	0	1	0	3
Thiazyl	0	0	0	0	0	0	1	1	0
Difluorophosphazene	0	0	0	0	0	1	0	0	3
Sulfur trioxide	0	0	0	0	0	3	0	1	0
Phosphazene	0	0	0	0	2	0	1	0	1
Germoxane	0	0	0	0	2	1	0	0	1
Siloxane	0	0	0	0	2	1	0	0	1
Stannyloxane	0	0	0	0	2	1	0	0	1
Difluoromethylene sulfide	0	0	0	1	0	0	0	1	2
Hydroxymethylene	0	0	0	1	2	1	0	0	0
Difluoroacetylene	0	0	0	2	0	0	0	0	2
Chlorotrifluoroethylene	0	0	0	2	0	0	0	0	4
Tetrafluoroethylene	0	0	0	2	0	0	0	0	4
Tetrafluoroethylene oxide	0	0	0	2	0	1	0	0	4
Trifluoroethylene	0	0	0	2	1	0	0	0	3
2,2,2-Trichloroacetaldehyde	0	0	0	2	1	1	0	0	3
Acetylene	0	0	0	2	2	0	0	0	0
2,2-Dichloroacetaldehyde	0	0	0	2	2	1	0	0	2
Ethylidene	0	0	0	2	4	0	0	0	0
Acetaldehyde	0	0	0	2	4	1	0	0	0
Dimethylsilylene	0	0	0	2	6	0	0	0	1
Dimethylphosphazene	0	0	0	2	6	0	1	0	1
Dimethylsiloxane	0	0	0	2	6	1	0	0	1
Oxybisdimethylsilylene	0	0	0	2	6	1	0	0	2

* Periods (·) are used to delineate the structural code integers generated.

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Tetrafluoroallene	0	0	0	3	0	0	0	0	4
Hexafluoropropene	0	0	0	3	0	0	0	0	6
Isopropylidene sulfide	0	0	0	3	6	0	0	1	0
Isopropylidene oxide	0	0	0	3	6	1	0	0	0
2-Methylvinylalcohol	0	0	0	3	6	1	0	0	0
1,3-Butadiyne, 1,4-	0	0	0	4	2	0	0	0	0
2,5-Thiophene	0	0	0	4	2	0	0	1	0
2-Butene sulfide	0	0	0	4	8	0	0	1	0
2-Butene oxide	0	0	0	4	8	1	0	0	0
Isobutyraldehyde	0	0	0	4	8	1	0	0	0
1-Methoxy-2-methylethylene	0	0	0	4	8	1	0	0	0
3-Methyl-2,5-thiophene	0	0	0	5	4	0	0	1	0
2-Methyl-2-butene oxide	0	0	0	5	10	1	0	0	0
2,4-Hexadiyne, 2,5-	0	0	0	6	6	0	0	0	0
2,4-Hexadiene, 2,5-	0	0	0	6	10	0	0	0	0
1-Isopropoxy-2-methyethylene	0	0	0	6	12	1	0	0	0
Tetramethylethylene oxide	0	0	0	6	12	1	0	0	0
2,6-Benzothiazole	0	0	0	7	3	0	1	1	0
2,5-Benzoxazole	0	0	0	7	3	1	1	0	0
2-Methyl-*p*-phenylene sulfide	0	0	0	7	6	0	0	1	0
1-*tert*-Butoxy-2-methylethylene	0	0	0	7	14	1	0	0	0
Tetrafluoro-*p*-xylylene	0	0	0	8	4	0	0	0	4
2,2′-Bithiophene	0	0	0	8	4	0	0	2	0
2,6-Dimethyl-1,4-phenylene oxide	0	0	0	8	8	1	0	0	0
2,6-Dimethoxy-1,4-phenylene oxide	0	0	0	8	8	3	0	0	0
2,5-Dimethyl-2,4-hexadiene, 2,5-	0	0	0	8	14	0	0	0	0
2,6-Quinoline	0	0	0	9	5	0	1	0	0
Terthiophene	0	0	0	12	6	0	0	3	0
Bis(2,4-dichlorophenoxy)phosphazene	0	0	0	12	6	2	1	0	5
3,3′-Dimethoxy-4,4′-biphenylene carbodiimine	0	0	0	15	12	0	2	0	0
Bis(3,4-dimethylphenoxy)phosphazene	0	0	0	16	18	2	1	0	1
4,4′-Perfluorodiphenyleneoxyperfluoropentamethylene oxide	0	0	0	17	0	2	0	0	8
Bis-β-naphthoxyphosphazene	0	0	0	20	14	2	1	0	1
m-Phenylene	0	0	1	0	0	0	0	0	0
p-Phenylene	0	0	1	0	0	0	0	0	0
p-Phenylene sulfide	0	0	1	0	0	0	0	1	0
p-Phenylene oxide	0	0	1	0	0	1	0	0	0
p-Phenylene sulfonate	0	0	1	0	0	3	0	1	0
Phenylenedisiloxanylene	0	0	1	0	4	1	0	0	2
p-Phenylene-1,3,4-oxadiazole	0	0	1	2	0	1	2	0	0
p-Phenylenevinylene	0	0	1	2	2	0	0	0	0
β-Nitrostyrene	0	0	1	2	3	2	1	0	0
Ethylidene *p*-phenylene	0	0	1	2	4	0	0	0	0
α-Methylmethylene-*p*-phenylene	0	0	1	2	4	0	0	0	0
α-Phenylethylisocyanide	0	0	1	3	5	0	1	0	0
p-Phenylene tetramethyldisiloxanylene	0	0	1	4	12	1	0	0	2
m-Phenyleneoxy-2-cyano-1,3-phenylene oxide	0	0	1	7	3	2	1	0	0
p-Phenylenebenzobisthiazole	0	0	1	8	2	0	2	2	0
p-Phenylenebenzobisoxazole	0	0	1	8	2	2	2	0	0
4-Phenyl-2,6-quinoline	0	0	1	9	5	0	1	0	0
(*p*-Phenylenedipyridyl)cyclobutylene	0	0	1	14	12	0	2	0	0
6,6′-Oxydiquinolene-*p*-phenylene	0	0	1	18	10	1	2	0	0
4,4′-Biphenylenegermylene	0	0	2	0	0	0	0	0	1
4,4′-Thiodiphenylene oxide	0	0	2	0	0	1	0	1	0
Bis(*m*-bromophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
Bis(*p*-bromophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
Bis(*m*-chlorophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
Bis(*p*-chlorophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
Bis(*m*-fluorophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
Bis(*p*-fluorophenoxy)phosphazene	0	0	2	0	0	2	1	0	3
p-Phenylene amine, emeraldine base	0	0	2	0	1	0	2	0	0
Diphenylsilylene	0	0	2	0	2	0	0	0	1
Diphenylphosphazene	0	0	2	0	2	0	1	0	1
Diphenylsiloxane	0	0	2	0	2	1	0	0	1
Bisphenoxyphosphazene	0	0	2	0	2	2	1	0	1

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4,4′-Biphenylene carbodiimine		0	0	2	1	0	0	2	0	0
Diphenylmethylene		0	0	2	1	2	0	0	0	0
Phenyl-*p*-toluylsiloxane		0	0	2	1	4	1	0	0	1
Bis(*m*-trifluoromethylphenoxy)phosphazene		0	0	2	2	0	2	1	0	7
Bis(*p*-methylphenoxy)phosphazene		0	0	2	2	6	2	1	0	1
Bis(*p*-methoxyphenoxy)phosphazene		0	0	2	2	6	4	1	0	1
Hexafluorotrimethylenebiphenylene		0	0	2	3	0	0	0	0	6
Di-*p*-tolylmethylene		0	0	2	3	6	0	0	0	0
2,6-Diphenyl-1,4-phenylene oxide		0	0	2	6	4	1	0	0	0
Bis(*p*-isopropylphenoxy)phosphazene		0	0	2	6	14	2	1	0	1
2,5-Pyrazine-2,4-diphenylcyclobutylene		0	0	2	8	8	0	2	0	0
1,6-Diazanthrylene-4,4′-biphenylene		0	0	2	12	6	0	2	0	0
1,9-Diazanthrylene-4,4′-biphenylene		0	0	2	12	6	0	2	0	0
1,6-Diazanthrylene-4,4′-oxydiphenylene		0	0	2	12	6	1	2	0	0
1,9-Diazanthrylene-4,4′-oxydiphenylene		0	0	2	12	6	1	2	0	0
6,6′-Oxydiquinoline-4,4′-biphenylene		0	0	2	18	10	1	2	0	0
6,6′-Oxydiquinoline-4,4′-oxydiphenylene		0	0	2	18	10	2	2	0	0
4,4′-Isopropylidenediphenylene dithionisophthalate		0	0	3	5	6	2	0	2	0
2,2′-Biphenylenedithiazole-*p*-phenylene		0	0	3	6	2	0	2	2	0
m-Phenylenedithiazole-oxy-*p*-diphenylene		0	0	3	6	2	1	2	2	0
4,4′-Diphenyl-2,2′-(6,6′-oxydiquinolene)-*p*-phenylene		0	0	3	18	10	1	2	0	0
Bis(*p*-phenylphenoxy)phosphazene		0	0	4	0	2	2	1	0	1
2,2′-Biphenylenedithiazole-oxy-*p*-diphenylene		0	0	4	6	2	1	2	2	0
Phthalocyaninatogermoxane		0	0	4	8	0	1	8	0	1
Phthalocyaninatosiloxane		0	0	4	8	0	1	8	0	1
Phthalocyaninatostannyloxane		0	0	4	8	0	1	8	0	1
4,6-Diphenyl-2,8-(1,9-diazanthrylene)-4,4′-biphenylene		0	0	4	12	6	0	2	0	0
4,9-Diphenyl-2,7-(1,6-diazanthrylene)-4,4′-biphenylene		0	0	4	12	6	0	2	0	0
4,6-Diphenyl-2,8-(1,9-diazanthrylene)-4,4′-oxydiphenylene		0	0	4	12	6	1	2	0	0
4,9-Diphenyl-2,7-(1,6-diazanthrylene)-4,4′-oxydiphenylene		0	0	4	12	6	1	2	0	0
2,2′-Biphenylene 6,6′-bis(4-phenylquinoline)		0	0	4	18	10	0	2	0	0
4,4′-Diphenyl-2,2′-(6,6′-oxydiquinoline)-4,4′-biphenylene		0	0	4	18	10	1	2	0	0
4,4′-Oxydiphenylene-4,4′-diquinoline-4,4′-biphenylene		0	0	4	18	10	1	2	0	0
4,4′-Diphenyl-2,2′-(6,6′-oxydiquinoline)-4,4′-oxydiphenylene		0	0	4	18	10	2	2	0	0
Tetraphenyl-2,2′-(6,6′-oxydiquinolene)-*p*-phenylene		0	0	5	18	10	1	2	0	0
Pentaphenyl-*p*-toluyltrisiloxane		0	0	6	1	8	3	0	0	3
3,3′-Diphenyl-2,2′-(4,4′-oxydiphenylene-4,4′-diquinoline)-biphenylene		0	0	6	18	10	1	2	0	0
Tetraphenyl-2,2′-(6,6′-oxydiquinoline)-4,4′-biphenylene		0	0	6	18	10	1	2	0	0
Tetraphenyl-2,2′-(6,6′-oxydiquinoline)-4,4′-oxydiphenylene		0	0	6	18	10	2	2	0	0
Aminoformic acid		0	1	0	0	1	0	1	0	0
2-Mercapto-2-methylacetic acid		0	1	0	2	4	0	0	1	0
2-Hydroxypropionic acid (lactic acid)		0	1	0	2	4	1	0	0	0
Alanine		0	1	0	2	5	0	1	0	0
3-Amino-2-butenoic acid		0	1	0	3	5	0	1	0	0
Dimethyl ketene (ketone)	I.	0	1	0	3	6	0	0	0	0
Dimethyl-2-hydroxyacetic acid		0	1	0	3	6	1	0	0	0
2-Aminoisobutyric acid		0	1	0	3	7	0	1	0	0
Isopropylhydroxyacetic acid		0	1	0	4	8	1	0	0	0
2,3-Dimethyl-3-aminopropionic acid		0	1	0	4	9	0	1	0	0
Valine		0	1	0	4	9	0	1	0	0
2,4-Hexadienoic acid, 2,5-		0	1	0	5	8	1	0	0	0
α-L-Guluronic acid, 1,4-		0	1	0	5	8	5	0	0	0
β-D-Mannuronic acid, 1,4-		0	1	0	5	8	5	0	0	0
Methyl-2,4-hexadienoate, 2,5-		0	1	0	6	10	1	0	0	0
4-Methyl-1,3-phenyleneurea		0	1	0	7	8	0	2	0	0
Dimethyl ketene (ester)	II.	0	1	0	7	12	1	0	0	0
Furfural-*alt*-dimethylketene		0	1	0	8	10	2	0	0	0
Isopropyl-2,4-hexadienoate, 2,5-		0	1	0	8	14	1	0	0	0
Tetramethylcyclobutylene carbonate		0	1	0	8	14	2	0	0	0
5-Bromo-6-hydroxy-2-naphthoic acid		0	1	0	10	5	1	0	0	1
2,6-Hydroxynaphthoic acid		0	1	0	10	6	1	0	0	0
Di-*tert*-butyl-4-hydroxybenzoic acid		0	1	0	14	20	1	0	0	0
4,4′-Isopropylidene-3,3′,5,5′-tetrabromodiphenylene carbonate		0	1	0	15	10	2	0	0	4
4,4′-Isopropylidene-3,3′,5,5′-tetrachlorodiphenylene carbonate		0	1	0	15	10	2	0	0	4
4,4′-Isopropylidene-3,3′-dichlorodiphenylene carbonate		0	1	0	15	12	2	0	0	2

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
4,4′-Isopropylidene-3,3′-dimethyldiphenylene carbonate	0	1	0	17	18	2	0	0	0
4,4′-Cyclohexylidene-3,3′,5,5-tetrachlorodiphenylene carbonate	0	1	0	18	9	2	0	0	4
4,4′-Isopropylidene-3,3′-diisopropyldiphenylene carbonate	0	1	0	21	26	2	0	0	0
m-Hydroxybenzoic acid	0	1	1	0	0	1	0	0	0
p-Hydroxybenzoic acid	0	1	1	0	0	1	0	0	0
m-Aminobenzoic acid	0	1	1	0	1	0	1	0	0
p-Aminobenzoic acid	0	1	1	0	1	0	1	0	0
N-Phenylaminoformic acid	0	1	1	0	1	0	1	0	0
m-Phenylene urea	0	1	1	0	2	0	2	0	0
3-(*p*-Hydroxyphenylene)-propenoic acid	0	1	1	2	2	1	0	0	0
5-Phenyl-2,4-pentadienoic acid, 2,5-	0	1	1	4	6	1	0	0	0
p-Chlorobenzaldehyde-*alt*-dimethylketene	0	1	1	4	7	1	0	0	1
m-Nitrobenzaldehyde-*alt*-dimethylketene	0	1	1	4	7	3	1	0	0
Benzaldehyde-*alt*-dimethylketene	0	1	1	4	8	1	0	0	0
2,5-Methyl-5-phenyl-2,4-pentadienoate	0	1	1	5	8	1	0	0	0
p-Methoxybenzaldehyde-*co*-dimethylketene	0	1	1	5	10	2	0	0	0
3-Bromo-4,4′-hydroxybibenzoic acid	0	1	1	6	3	1	0	0	1
4,4′-Hydroxybibenzoic acid	0	1	2	0	0	1	0	0	0
4,4′-Oxydiphenylene ketone	0	1	2	0	0	1	0	0	0
4,4′-Thiodiphenylene carbonate	0	1	2	0	0	2	0	1	0
4,4′-Oxydiphenylene carbonate	0	1	2	0	0	3	0	0	0
4,4′-Sulfinyldiphenylene carbonate	0	1	2	0	0	3	0	1	0
4,4′-Sulfonyldiphenylene carbonate	0	1	2	0	0	4	0	1	0
4,4′-Ethylidenediphenylene carbonate	0	1	2	2	4	2	0	0	0
4,4′-Isopropylidenediphenylene carbonate	0	1	2	3	6	2	0	0	0
4,4′-Isobutylidenediphenylene carbonate	0	1	2	4	8	2	0	0	0
4,4′-(Methyl-*p*-phenylenedioxy)-diphenylene ketone	0	1	2	7	6	2	0	0	0
4,4′-(2,4-Dimethyl-3,3-pentylidene)diphenylene carbonate	0	1	2	7	14	2	0	0	0
(*p*-Phenylenedioxy)diphenylene ketone	0	1	3	0	0	2	0	0	0
(*p*-Phenylenedioxy)diphenylene carbonate	0	1	3	0	0	4	0	0	0
4,4′-(Phenylmethylene)diphenylene carbonate	0	1	3	1	2	2	0	0	0
4,4′-Phenylethylidenediphenylene carbonate	0	1	3	2	4	2	0	0	0
4,4′-Biphenylenedioxy 4,4′-phenylene ketone	0	1	4	0	0	2	0	0	0
4,4′-Biphenylenedioxy-diphenylene carbonate	0	1	4	0	0	4	0	0	0
4,4′-(Diphenylmethylene)diphenylene carbonate	0	1	4	1	2	2	0	0	0
2,4-Hexadienedioic acid, 2,5-	0	2	0	4	6	2	0	0	0
Alanylvaline	0	2	0	6	14	0	2	0	0
2,5-Diisopropyl-2,4-hexadienedioate	0	2	0	10	18	2	0	0	0
2-Fluoro-*p*-phenylene 2-fluoroterephthalamide	0	2	0	12	8	0	2	0	2
2,6-Naphthalene-2,4-cyclobutylenedicarboxylic acid	0	2	0	14	12	2	0	0	0
2,6-Naphthalene-2,4-bisdichlorophenylcyclobutylenedicarboxylate	0	2	0	26	16	2	0	0	4
Isophthalic anhydride	0	2	1	0	0	1	0	0	0
Terephthalic anhydride	0	2	1	0	0	1	0	0	0
p-Xylylene oxalate	0	2	1	0	0	2	0	0	0
p-Phenylene *m*-carboranedicarboxylate	0	2	1	2	10	2	0	0	10
p-Phenylene 2,5-thiopheneamide	0	2	1	4	4	0	2	1	0
2,5-Pyrimidine terephthalamide	0	2	1	4	4	0	4	0	0
p-Phenylene cyclobutylenedicarboxylic acid	0	2	1	4	6	2	0	0	0
p-Phenylene cyclobutylenedicarboxamide	0	2	1	4	8	0	2	0	0
Chloro-*p*-phenylene Terephthalate	0	2	1	6	3	2	0	0	1
p-Phenylene dimethylcyclobutylenedicarboxylate	0	2	1	6	10	2	0	0	0
4-Methyl-*m*-phenylene terephthalamide	0	2	1	7	8	0	2	0	0
p-Phenylene dicyanodimethylcyclobutylenedicarboxylate	0	2	1	8	8	2	2	0	0
tert-butyl-*p*-phenylene terephthalate	0	2	1	10	12	2	0	0	0
p-Phenylene diisopropylcyclobutylenedicarboxylate	0	2	1	10	18	2	0	0	0
p-Phenylene dicyanodiisopropylcyclobutylenedicarboxylate	0	2	1	12	16	2	2	0	0
3,3′-Dimethyl-4,4′-biphenylene terephthalamide	0	2	1	14	14	0	2	0	0
2′-Isopropyl-2-methyl-4,4′-oxydiphenylene *m*-phenylenediketone	0	2	1	16	16	1	0	0	0
2′-Isopropyl-2-methyl-4,4′-oxydiphenylene *p*-phenylenediketone	0	2	1	16	16	1	0	0	0
1,5-Naphthalene-4′,4″-(dimethoxy)dibenzylidenimino isophthalate	0	2	1	26	20	4	2	0	0
1,5-Naphthalene-4′,4″-(dimethoxy)dibenzylidenimino terephthalate	0	2	1	26	20	4	2	0	0
4,4′-Oxydibenzoic anhydride	0	2	2	0	0	2	0	0	0
m-Phenylene terephthalate	0	2	2	0	0	2	0	0	0
m-Phenylene isophthalate	0	2	2	0	0	2	0	0	0
p-Phenylene isophthalate	0	2	2	0	0	2	0	0	0

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m-Phenylene terephthalate	0	2	2	0	0	2	0	0	0
p-Phenylene terephthalate	0	2	2	0	0	2	0	0	0
m-Phenylene isophthalamide	0	2	2	0	2	0	2	0	0
p-Phenylene isophthalamide	0	2	2	0	2	0	2	0	0
p-Phenylene phthalamide	0	2	2	0	2	0	2	0	0
m-Phenylene terephthalamide	0	2	2	0	2	0	2	0	0
p-Phenylene terephthalamide	0	2	2	0	2	0	2	0	0
Isophthaloyl (dithionisophthaloyl)dihydrazide	0	2	2	2	4	0	4	2	0
Terephthaloyl dithionisophthaloyldihydrazide	0	2	2	2	4	0	4	2	0
4,4′-Isopropylidenedibenzoic anhydride	0	2	2	3	6	1	0	0	0
p-Phenylene phenylthioterephthalate	0	2	2	6	4	0	0	2	0
Phenyl-*p*-phenylene terephthalate	0	2	2	6	4	2	0	0	0
4,4′-Isopropylidenediphenylene 3-methylcyclopropylenedicarboxylate	0	2	2	7	12	2	0	0	0
4-Methyl-1,3-phenylene 3,3′-methylphosphinylidenedibenzamide	0	2	2	8	11	1	2	0	1
4-Methyl-1,3-phenylene 4,4′-methylphosphinylidenedibenzamide	0	2	2	8	11	1	2	0	1
2,6-Diisopropyl-4,4′-oxydiphenylene *m*-phenylenediketone	0	2	2	12	16	1	0	0	0
p-Phenylene-4′,4″-(dimethoxy)dibenzylidenimino isophthalate	0	2	2	16	14	4	2	0	0
4,4′-Biphenylene dithiolisophthalate	0	2	3	0	0	0	0	2	0
4,4′-Biphenylene dithiolterephthalate	0	2	3	0	0	0	0	2	0
4,4′-Oxydiphenylene *m*-phenylene diketone	0	2	3	0	0	1	0	0	0
4,4′-Oxydiphenylene *m*-phenylene ketone	0	2	3	0	0	1	0	0	0
4,4′-Oxydiphenylene *p*-phenylene diketone	0	2	3	0	0	1	0	0	0
4,4′-Biphenylene terephthalate	0	2	3	0	0	2	0	0	0
(*p*-Phenylenedioxy)diphenylene diketone	0	2	3	0	0	2	0	0	0
4,4′-Biphenylene terephthalamide	0	2	3	0	2	0	2	0	0
4,4′-Sulfonyldiphenylene terephthalamide	0	2	3	0	2	2	1	1	0
m-Phenylene 3,3′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
o-Phenylene 3,3′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
p-Phenylene 3,3′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
m-Phenylene 4,4′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
o-Phenylene 4,4′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
p-Phenylene 4,4′-phosphinylidenedibenzamide	0	2	3	0	3	1	2	0	1
m-Phenylene 3,3′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
o-Phenylene 3,3′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
p-Phenylene 3,3′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
m-Phenylene 4,4′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
o-Phenylene 4,4′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
p-Phenylene 4,4′-(methylphosphinylidene)dibenzamide	0	2	3	1	5	1	2	0	1
4,4′-Isopropylidenediphenylene isophthalamide	0	2	3	3	8	0	2	0	0
p-Phenylene diphenylcyclobutylenedicarboxylate	0	2	3	4	6	2	0	0	0
4-Methyl-1,3-phenylene 4,4′-phenylphosphinylidenedibenzamide	0	2	3	7	9	1	2	0	1
1,5-Naphthalene-4′,4″-dibenzylidenimino isophthalate	0	2	3	12	8	2	2	0	0
1,5-Naphthalene-4′,4″-dibenzylidenimino terephthalate	0	2	3	12	8	2	2	0	0
4,4′-Oxydiphenylene-4″,4‴-(dimethoxy)dibenzylidenimino isophthalate	0	2	3	16	14	5	2	0	0
4,4′-Oxydiphenylene 4,4-biphenylene diketone	0	2	4	0	0	1	0	0	0
4,4′-Biphenylene 4,4′-biphenylenedicarboxylate	0	2	4	0	0	2	0	0	0
(4,4′-Phenylenedioxy)diphenylene *m*-phenylene diketone	0	2	4	0	0	2	0	0	0
(4,4′-Phenylenedioxy)diphenylene *p*-phenylene diketone	0	2	4	0	0	2	0	0	0
m-Phenylene 4,4′-(phenylphosphinylidene)dibenzamide	0	2	4	0	3	1	2	0	1
o-Phenylene 4,4′-(phenylphosphinylidene)dibenzamide	0	2	4	0	3	1	2	0	1
p-Phenylene 4,4′-(phenylphosphinylidene)dibenzamide	0	2	4	0	3	1	2	0	1
p-Phenylene-4′,4″-dibenzylidenimino isophthalate	0	2	4	2	2	2	2	0	0
4,4′-Biphenylenedioxy-diphenylene *m*-phenylene diketone	0	2	5	0	0	2	0	0	0
(*p*-Phenylenedioxy)diphenylene oxydiphenylenediketone	0	2	5	0	0	3	0	0	0
4,4′-Oxydiphenylene-4″,4‴-dibenzylidenimino isophthalate	0	2	5	2	2	3	2	0	0
3,8-Phenanthridinonediyl terephthalamide	0	3	1	12	9	0	3	0	0
3,4′-Carbonyldiphenylene terephthalate	0	3	3	0	0	2	0	0	0
4,4′-Benzanilidylene terephthalamide	0	3	3	0	3	0	3	0	0
4,4′-(Carbonyldiphenylenedioxy)-diphenylene-*p*-phenylene diketone	0	3	5	0	0	2	0	0	0
Alanylalanylalanylvaline	0	4	0	10	24	0	4	0	0
Alanylalanylvalylvaline	0	4	0	12	28	0	4	0	0
3,3′-Bis(trifluoromethyl)biphenylene 3,3′,4,4′-biphenylenetetracarboximide	0	4	0	26	12	0	2	0	6
3,3′-Dimethyl-4,4′-biphenylene 3,3′,4,4′-biphenylenetetracarboximide	0	4	0	26	18	0	2	0	0
m-Phenylene pyromellitimide	0	4	1	6	2	0	2	0	0
p-Phenylene pyromellitimide	0	4	1	6	2	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
p-Phenylene 1,4-pyromellitamide	0	4	1	6	6	2	2	0	0
p-Phenylene 3,3′,4,4′-oxydiphenylenetetracarboximide	0	4	1	12	6	1	2	0	0
p-Phenylene 4,4′-oxy(3,3′-dicarboxy)dibenzamide	0	4	1	12	10	3	2	0	0
4,4′-Biphenylene pyromellitimide	0	4	2	6	2	0	2	0	0
4,4′-Thiodiphenylene pyromellitimide	0	4	2	6	2	0	2	1	0
3,4′-Oxydiphenylene pyromellitimide	0	4	2	6	2	1	2	0	0
4,4′-Oxydiphenylene pyromellitimide	0	4	2	6	2	1	2	0	0
4,4′-Biphenylene 1,4-pyromellitimide	0	4	2	6	6	2	2	0	0
4,4′-Oxydiphenylene 1,4-pyromellitimide	0	4	2	6	6	3	2	0	0
4,4′-Biphenylene 3,3′,4,4′-biphenylenetetracarboximide	0	4	2	12	6	0	2	0	0
4,4′-Biphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide	0	4	2	12	6	1	2	0	0
3,4′-Oxydiphenylene 3,3′,4,4′-biphenylenetetracarboximide	0	4	2	12	6	1	2	0	0
4,4′-Oxydiphenylene 3,3′,4,4′-oxydiphenylenetetracarboximide	0	4	2	12	6	2	2	0	0
4,4′-Biphenylene 4,4′-oxy(3,3′-dicarboxy)dibenzamide	0	4	2	12	10	3	2	0	0
4,4′-Benzobisthiazole-diphenylene pyromellitimide	0	4	2	14	4	0	4	2	0
p-Terphenylene pyromellitimide	0	4	3	6	2	0	2	0	0
(*m*-Phenylenedioxy)diphenylene pyromellitimide	0	4	3	6	2	2	2	0	0
(*p*-Phenylenedioxy)diphenylene pyromellitimide	0	4	3	6	2	2	2	0	0
(*m*-Phenylenedioxy)diphenylene 1,4-pyromellitamide	0	4	3	6	6	4	2	0	0
(*p*-Phenylenedioxy)diphenylene 1,4-pyromellitamide	0	4	3	6	6	4	2	0	0
p-Phenylene thiodiphenylenedioxy-diphenylenetetracarboximide	0	4	3	12	6	2	2	1	0
(*p*-Phenylenedioxy)diphenylene oxydiphenylenetetracarboximide	0	4	3	12	6	3	2	0	0
(*p*-Phenylenedioxy)diphenylene 4,4′-oxy(3,3′-dicarboxy)dibenzamide	0	4	3	12	10	5	2	0	0
p-Phenylene 4,4′-dibenzoxyterephthalate	0	4	4	0	0	4	0	0	0
m-Phenylene-3,3′-dibenzamido isophthalamide	0	4	4	0	4	0	4	0	0
p-Phenylene-3,3′-dibenzamido isophthalamide	0	4	4	0	4	0	4	0	0
m-Phenylene-4,4′-dibenzamido isophthalamide	0	4	4	0	4	0	4	0	0
p-Phenylene-4,4′-dibenzamido isophthalamide	0	4	4	0	4	0	4	0	0
m-Phenylene-3,3′-dibenzamido terephthalamide	0	4	4	0	4	0	4	0	0
p-Phenylene-3,3′-dibenzamido terephthalamide	0	4	4	0	4	0	4	0	0
m-Phenylene-4,4′-dibenzamido terephthalamide	0	4	4	0	4	0	4	0	0
p-Phenylene-4,4′-dibenzamido terephthalamide	0	4	4	0	4	0	4	0	0
4,4′-Biphenylenedioxy-3,3′-diphenylene pyromellitimide	0	4	4	6	2	2	2	0	0
(Oxydiphenylenedioxy)diphenylene pyromellitimide	0	4	4	6	2	3	2	0	0
4,4′-Biphenylene thiodiphenylenedioxydiphenylenetetracarboximide	0	4	4	12	6	2	2	1	0
(*p*-Phenylenedioxy)diphenylene *p*-phenylenedioxydiphenylenetetracarboximide	0	4	4	12	6	4	2	0	0
(*p*-Phenylenedioxy)diphenylenedioxydiphenylene pyromellitimide	0	4	5	6	2	4	2	0	0
4,4′-Carbonyldiphenylene pyromellitimide	0	5	2	6	2	0	2	0	0
4,4′-Biphenylene 3,3′,4,4′-carbonyldiphenylenetetracarboximide	0	5	2	12	6	0	2	0	0
4,4′-Biphenylene 4,4′-carbonyl(3,3′-dicarboxy)dibenzamide	0	5	2	12	10	2	2	0	0
4,4′-(Isophthaloyldiamino)diphenylene pyromellitimide	0	6	3	6	4	0	4	0	0
4,4′-(Terephthaloyldiamino)diphenylene pyromellitimide	0	6	3	6	4	0	4	0	0
4,4′-Thiodiphenylene *p*-phenyleneditrimellitatediimide	0	6	3	12	6	2	2	1	0
4,4′-Oxydiphenylene *p*-phenyleneditrimellitatediimide	0	6	3	12	6	3	2	0	0
4,4′-(Isophthaloyldiamino)diphenylene oxydiphenylenetetracarboximide	0	6	3	12	8	1	4	0	0
4,4′-Oxydiphenylene 5,5′-(*p*-phenylene)ditrimellitatediamide	0	6	3	12	10	5	2	0	0
Isophthalic 2,2′-(isophthaloyldiamino)dibenzoic anhydride	0	6	4	0	2	2	2	0	0
Isophthalic 2,2′-(terephthaloyldiamino)dibenzoic anhydride	0	6	4	0	2	2	2	0	0
2,2′-(Isophthaloyldiamino)dibenzoic terephthalic anhydride	0	6	4	0	2	2	2	0	0
Terephthalic 2,2′-(terephthaloyldiamino)dibenzoic anhydride	0	6	4	0	2	2	2	0	0
Terephthaloyldihydrazodi (*p*-aminobenzoyl) terephthalamide	0	6	4	0	6	0	6	0	0
(*p*-Phenylenedioxy)diphenylene *p*-phenyleneditrimellitatediimide	0	6	4	12	6	4	2	0	0
(Oxydiphenylenedioxy)diphenylene *p*-phenyleneditrimellitatediimide	0	6	5	12	6	5	2	0	0
(*p*-Phenylenedioxy)diphenylenedioxydiphenylene *p*-phenyleneditrimellitamide	0	6	6	12	6	6	2	0	0
Isophthaloyldiphenylenedioxdiphenylene carbonyldiphenylenetetracarboximide	0	7	5	12	6	2	2	0	0
4,4′-(Isophthaloyldiamino)diphenylene *p*-phenyleneditrimellitatediimide	0	8	4	12	8	2	4	0	0
4,4′-(Terephthaloyldiamino)diphenylene *p*-phenyleneditrimellitate	0	8	4	12	8	2	4	0	0
1	1	0	0	0	0	0	0	0	0
Methylene	1	0	0	0	0	0	0	0	0
Methylene selenide	1	0	0	0	0	0	0	0	1
Methylene sulfide	1	0	0	0	0	0	0	1	0
Methylene disulfide	1	0	0	0	0	0	0	2	0
Oxymethylene	1	0	0	0	0	1	0	0	0
Vinylidene bromide	1	0	0	1	0	0	0	0	2
Vinylidene chloride	1	0	0	1	0	0	0	0	2

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Vinylidene fluoride	1	0	0	1	0	0	0	0	2
Vinyl bromide	1	0	0	1	1	0	0	0	1
Vinyl chloride	1	0	0	1	1	0	0	0	1
Vinyl fluoride	1	0	0	1	1	0	0	0	1
2-Chloroacetaldehyde	1	0	0	1	1	1	0	0	1
Vinyl alcohol	1	0	0	1	2	1	0	0	0
Acrylonitrile	1	0	0	2	1	0	1	0	0
3,3,3-Trifluoropropylene oxide	1	0	0	2	1	1	0	0	3
Ethyl isocyanide	1	0	0	2	3	0	1	0	0
Propene	1	0	0	2	4	0	0	0	0
Propylidene	1	0	0	2	4	0	0	0	0
Propylene sulfide	1	0	0	2	4	0	0	1	0
Methyl vinyl ether	1	0	0	2	4	1	0	0	0
Propionaldehyde	1	0	0	2	4	1	0	0	0
Propylene oxide	1	0	0	2	4	1	0	0	0
Methacrylonitrile	1	0	0	3	3	0	1	0	0
Isobutene	1	0	0	3	6	0	0	0	0
Isobutylene sulfide	1	0	0	3	6	0	0	1	0
Isobutylene oxide	1	0	0	3	6	1	0	0	0
α-Methylvinylmethyl ether	1	0	0	3	6	1	0	0	0
1-Cyano-1,3-butadiene, 1,4-	1	0	0	4	3	0	1	0	0
3-Trifluoromethyl-1-butene	1	0	0	4	5	0	0	0	3
1,3-Pentadiene, 1,2-	1	0	0	4	6	0	0	0	0
1,3-Pentadiene, 1,4-	1	0	0	4	56	0	0	0	0
1-Methoxy-1,3-butadiene, 1,4-	1	0	0	4	56	1	0	0	0
β-D-Xylose, 1,3-(xylan)	1	0	0	4	6	4	0	0	0
β-D-Xylose, 1,4-(hardwood xylan)	1	0	0	4	6	4	0	0	0
3-Methyl-1-butene	1	0	0	4	8	0	0	0	0
Isopropylethylene sulfide	1	0	0	4	8	0	0	1	0
1-Ethoxy-2-methylethylene	1	0	0	4	8	1	0	0	0
Isopropylethylene oxide	1	0	0	4	8	1	0	0	0
Isopropylvinyl ether	1	0	0	4	8	1	0	0	0
Isovaleraldehyde	1	0	0	4	8	1	0	0	0
1-Ethoxy-2-methoxyethylene	1	0	0	4	8	2	0	0	0
β-D-Glucotrinitrate, 1,4-	1	0	0	5	5	1	3	0	0
β-D-Glucodinitrate, 1,4-	1	0	0	5	6	9	2	0	0
α-Isopropylacrylonitrile	1	0	0	5	7	0	1	0	0
4-Methyl-1,3-pentadiene, 1,2-	1	0	0	5	8	0	0	0	0
2-Methyl-1,3-pentadiene, 1,4-	1	0	0	5	8	0	0	0	0
3-Methyl-1,3-pentadiene, 1,4-	1	0	0	5	8	0	0	0	0
α-D-Glucose, 1,3-	1	0	0	5	8	5	0	0	0
β-D-Glucose, 1,3- (curdlan)	1	0	0	5	8	5	0	0	0
α-D-Glucose, 1,4- (amylose)	1	0	0	5	8	5	0	0	0
β-D-Glucose, 1,4- (cellulose)	1	0	0	5	8	5	0	0	0
α-D-Glucose, 1,6- (dextran)	1	0	0	5	8	5	0	0	0
α-D-Mannose, 1,3-	1	0	0	5	8	5	0	0	0
β-D-Mannose, 1,4-	1	0	0	5	8	5	0	0	0
1-Chloro-2-isobutoxyethylene	1	0	0	5	9	1	0	0	1
α-D-Galactosamine, 1,4-	1	0	0	5	9	4	1	0	0
β-D-Glucosamine, 1,4- (chitosan)	1	0	0	5	9	4	1	0	0
3,3-Dimethyl-1-butene	1	0	0	5	10	0	0	0	0
tert-Butylethylene sulfide	1	0	0	5	10	0	0	1	0
tert-Butylethylene oxide	1	0	0	5	10	1	0	0	0
tert-Butylvinyl ether	1	0	0	5	10	1	0	0	0
2-Vinyl pyridine	1	0	0	6	5	0	1	0	0
5-Methyl-1,3-hexadiene, 1,4-	1	0	0	6	10	0	0	0	0
1-Isobutoxy-2-methylethylene	1	0	0	6	12	1	0	0	0
1,4-Tetradeuterobutadiene-*alt*-methacrylonitrile	1	0	0	7	5	0	1	0	4
Methylene 3,5-toluene	1	0	0	7	6	0	0	0	0
Methylene 2,6-diamino-3,5-toluene	1	0	0	7	8	0	2	0	0
p-Fluoro-*o*-methylstyrene	1	0	0	8	7	0	0	0	1
Methylene 2,5-dimethyl-*p*-phenylene	1	0	0	8	8	0	0	0	0
Tetramethylcyclobutyleneoxymethylene oxide	1	0	0	8	14	2	0	0	0
α-D-Trimethylglucose, 1,4-	1	0	0	8	14	5	0	0	0
β-D-Trimethylglucose, 1,4-	1	0	0	8	14	5	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
β-D-Trimethylmannose, 1,4-	1	0	0	8	14	5	0	0	0
2,4-Dimethylstyrene	1	0	0	9	10	0	0	0	0
2,5-Dimethylstyrene	1	0	0	9	10	0	0	0	0
3,4-Dimethylstyrene	1	0	0	9	10	0	0	0	0
3,5-Dimethylstyrene	1	0	0	9	10	0	0	0	0
1-Vinyl naphthalene	1	0	0	11	8	0	0	0	0
3-Vinyl pyrene	1	0	0	17	10	0	0	0	0
Methyl-*alt*-*p*-phenylene	1	0	1	0	0	0	0	0	0
m-Chlorostyrene	1	0	1	1	1	0	0	0	1
p-Chlorostyrene	1	0	1	1	1	0	0	0	1
m-Fluorostyrene	1	0	1	1	1	0	0	0	1
o-Fluorostyrene	1	0	1	1	1	0	0	0	1
p-Fluorostyrene	1	0	1	1	1	0	0	0	1
Styrene	1	0	1	1	2	0	0	0	0
Styrene oxide	1	0	1	1	2	1	0	0	0
Styrene sulfone	1	0	1	1	2	2	0	1	0
α-Methylstyrene	1	0	1	2	4	0	0	0	0
m-Methylstyrene	1	0	1	2	4	0	0	0	0
o-Methylstyrene	1	0	1	2	4	0	0	0	0
p-Methylstyrene	1	0	1	2	4	0	0	0	0
N-Phenyl-2-imino-1,3-dithia-4-methylpentamethylene	1	0	1	3	5	0	1	2	0
p-Methyl-α-methylstyrene	1	0	1	3	6	0	0	0	0
3-Phenyl-1-butene	1	0	1	3	6	0	0	0	0
p-Trimethylsilylstyrene	1	0	1	4	10	0	0	0	1
tert-Butylstyrene	1	0	1	5	10	0	0	0	0
p-Isopropyl-α-methylstyrene	1	0	1	5	10	0	0	0	0
Styryl pyridine-*alt*-vinyl pyridine	1	0	1	13	12	0	2	0	0
4,4′-Methylenediphenylene oxide	1	0	2	0	0	1	0	0	0
N-Vinyl carbazole	1	0	2	1	1	0	1	0	0
N-Vinyl diphenylamine	1	0	2	1	3	0	1	0	0
4,4′-Stilbenediyloxymethylene oxide	1	0	2	2	2	2	0	0	0
α-methyl-4,4′-stilbenediyloxymethylene oxide	1	0	2	3	4	2	0	0	0
1-*N*-Carbazoyl-1,3-pentadiynol	1	0	2	4	1	1	01	0	0
2-Mercaptoacetic acid	1	1	0	0	0	0	0	1	0
2-Hydroxyacetic acid (glycolic acid)	1	1	0	0	0	1	0	0	0
Glycine	1	1	0	0	1	0	1	0	0
Thiolacrylic acid	1	1	0	1	2	0	0	1	0
Acrylic acid	1	1	0	1	2	1	0	0	0
Vinyl formate	1	1	0	1	2	1	0	0	0
Acrylamide	1	1	0	1	3	0	1	0	0
Cysteine	1	1	0	1	3	0	1	1	0
Serine	1	1	0	1	3	1	1	0	0
3-Trichloromethyl-3-hydroxypropionic acid	1	1	0	2	1	1	0	0	3
Vinyl trifluoroacetate	1	1	0	2	1	1	0	0	3
3-Dichloromethyl-3-hydroxypropionic acid	1	1	0	2	2	1	0	0	2
Vinylmethyl ketone	1	1	0	2	4	0	0	0	0
3-Hydroxybutyric acid	1	1	0	2	4	1	0	0	0
Methacrylic acid	1	1	0	2	4	1	0	0	0
2-Methyl-3-hydroxypropionic acid	1	1	0	2	4	1	0	0	0
Vinyl acetate	1	1	0	2	4	1	0	0	0
3-Aminobutyric acid	1	1	0	2	5	0	1	0	0
2,4-Pentadienoic acid, 2,5-	1	1	0	3	4	1	0	0	0
3-Dichloroethyl-3-hydroxypropionic acid	1	1	0	3	4	1	0	0	2
Isopropenylmethyl ketone	1	1	0	3	6	0	0	0	0
2,2-Dimethyl-3-hydroxypropionic acid	1	1	0	3	6	1	0	0	0
Methylmethacrylate	1	1	0	3	6	1	0	0	0
2,2-Dimethyl-3-aminopropionic acid	1	1	0	3	7	0	1	0	0
3,3-Dimethyl-3-aminopropionic acid	1	1	0	3	7	0	1	0	0
Histidine	1	1	0	4	5	0	3	0	0
Histidine hydrochloride	1	1	0	4	6	0	3	0	1
6-Amino-2,4-hexadienoic acid, 2,5-	1	1	0	4	7	1	1	0	0
Vinyl isopropyl ketone	1	1	0	4	8	0	0	0	0
Isopropyl thiolacrylate	1	1	0	4	8	0	0	1	0
Isopropyl acrylate	1	1	0	4	8	1	0	0	0
3-Isopropyl-3-hydroxypropionic acid	1	1	0	4	8	1	0	0	0

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6-Amino-2,4-hexadienoic acid hydrochloride, 2,5-	1	1	0	4	8	1	1	0	1
Isoleucine	1	1	0	4	9	0	1	0	0
N-Isopropyl acrylamide	1	1	0	4	9	0	1	0	0
Leucine	1	1	0	4	9	0	1	0	0
Vinyl-*tert*-butyl ketone	1	1	0	5	10	0	0	0	0
tert-Butyl acrylate	1	1	0	5	10	1	0	0	0
Ethyl-2,4-hexadienoate, 2,5-	1	1	0	6	10	1	0	0	0
N-β-D-Acetylglucosamine, 1,4- (chitin)	1	1	0	6	11	4	1	0	0
tert-Butylmethacrylate	1	1	0	6	12	1	0	0	0
2,3,4-trimethoxy-5-hydroxyvaleric acid	1	1	0	6	12	4	0	0	0
2,5-Dimethylterephthalaldehyde	1	1	0	8	8	1	0	0	0
Isobutyl-2,4-hexadienoate, 2,5-	1	1	0	8	14	1	0	0	0
β-D-Galactosamine-*alt*-1,4-β-D-glucuronic acid, 1,3-	1	1	0	10	17	9	1	0	0
β-D-Glucosamine-*alt*-1,4-β-D-glucuronic acid, 1,3-	1	1	0	10	17	9	1	0	0
3,3′-Dimethyl-4,4′-methylenediphenylene carbonate	1	1	0	14	12	2	0	0	0
(*p*-Hydroxyphenylene)acetic acid	1	1	1	0	0	1	0	0	0
Isophthalaldehyde	1	1	1	0	0	1	0	0	0
Terephthalaldehyde	1	1	1	0	0	1	0	0	0
α-Amino-*p*-toluic acid	1	1	1	0	1	0	1	0	0
p-Fluorostyrene-*alt*-carbon monoxide	1	1	1	1	1	0	0	0	1
Styrene-*alt*-carbon monoxide	1	1	1	1	2	0	0	0	0
o-Hydroxystyrene-*alt*-carbon monoxide	1	1	1	1	2	1	0	0	0
Vinyl benzoate	1	1	1	1	2	1	0	0	0
Phenylalanine	1	1	1	1	3	0	1	0	0
Tyrosine	1	1	1	1	3	1	1	0	0
p-Trifluoroacetylstyrene	1	1	1	2	1	0	0	0	3
m-Methylstyrene-*alt*-carbon monoxide	1	1	1	2	4	0	0	0	0
p-Methylstyrene-*alt*-carbon monoxide	1	1	1	2	4	0	0	0	0
3-Mercapto-2-(*p*-toluenesulfonamide)propionic acid	1	1	1	2	5	2	1	2	0
Vinyl-*p*-*tert*-butylbenzoate	1	1	1	5	10	1	0	0	0
4,4′-Biphenyldicarboxaldehyde	1	1	2	0	0	1	0	0	0
4,4′-Methylenediphenyl carbonate	1	1	2	0	0	2	0	0	0
4,4′-Methylenediphenylene urea	1	1	2	0	2	0	2	0	0
4,4′-Butylidenediphenylene carbonate	1	1	2	3	6	2	0	0	0
4,4′-(Methyl-2,2-pentylidene)diphenylene carbonate	1	1	2	5	10	2	0	0	0
Aspartic acid	1	2	0	1	3	1	1	0	0
Propylene oxalate	1	2	0	2	4	2	0	0	0
O-Acetylserine	1	2	0	2	5	1	1	0	0
Alanylglycine	1	2	0	2	6	0	2	0	0
(3-Bromo-4-carboxyphenoxy)acetic anhydride	1	2	0	6	3	2	0	0	1
β-D-Xylodiacetate, 1,4-	1	2	0	6	10	4	0	0	0
N-β-D-Acetylgalactosamine-*alt*-1,4-β-D-glucuronic acid, 1,3-	1	2	0	11	19	9	1	0	0
N-β-D-Acetylglucosamine-*alt*-1,4-β-D-glucuronic acid, 1,3-	1	2	0	11	19	9	1	0	0
N-β-D-Acetylgalactosamine-4-sulfate-*alt*-1,4-β-D-glucuronic acid	1	2	0	11	19	12	1	1	0
N-β-D-Acetylgalactosamine-6-sulfate-*alt*-1,4-β-D-glucuronic acid	1	2	0	11	19	12	1	1	0
3,3′-Dimethyl-4,4′-methylenediphenylene methyl-1,3-phenylenediurea	1	2	0	21	22	0	4	0	0
(*m*-Carboxyphenoxy)acetic anhydride	1	2	1	0	0	2	0	0	0
(*p*-Carboxyphenoxy)acetic anhydride	1	2	1	0	0	2	0	0	0
p-Phenylene malonate	1	2	1	0	0	2	0	0	0
p-Carboxyphenylacrylate	1	2	1	1	2	2	0	0	0
m-Phenylene cyclopropylenedicarboxylate	1	2	1	2	2	2	0	0	0
p-Phenylene cyclopropylenedicarboxylate	1	2	1	2	2	2	0	0	0
Cyclopropylene isophthalamide	1	2	1	2	4	0	2	0	0
Methylethylene isophthalate	1	2	1	2	4	2	0	0	0
Methylethylene phthalate	1	2	1	2	4	2	0	0	0
Methylethylene terephthalate	1	2	1	2	4	2	0	0	0
3,3′-Dimethyl-4,4′-methylenediphenylene terephthalamide	1	2	1	14	14	0	2	0	0
2-ethyl-2′-Isopropyl-4,4′-oxydiphenylene *m*-phenylenediketone	1	2	1	16	16	1	0	0	0
N,N′-Dimethyl-4,4′-(dimethylmethylene)diphenylene terephthalamide	1	2	1	16	18	0	2	0	0
4,4′-Methylenedibenzoic anhydride	1	2	2	0	0	1	0	0	0
4,4′-(Methylenedioxy)dibenzoic anhydride	1	2	2	0	0	3	0	0	0
Methylethylene 2,2′-dibenzoate	1	2	2	2	4	2	0	0	0
Methylethylene 3,3′-dibenzoate	1	2	2	2	4	2	0	0	0
Methylethylene 4,4′-dibenzoate	1	2	2	2	4	2	0	0	0
Propylene 4,4′-sulfonyldibenzamide	1	2	2	2	6	2	2	1	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
4,4′-Isopropylidenediphenylene malonate	1	2	2	3	6	2	0	0	0
p-Phenylene(1,1-dimethyl)ethylene terephthalamide	1	2	2	3	8	0	2	0	0
4,4′-Isopropylidenediphenylene cyclopropylenedicarboxylate	1	2	2	5	8	2	0	0	0
4,4′-Isopropylidenediphenylene 1-methylcyclopropylenedicarboxylate	1	2	2	6	10	2	0	0	0
4,4′-Methylenediphenylene isophthalamide	1	2	3	0	2	0	2	0	0
4,4′-Methylenediphenylene terephthalamide	1	2	3	0	2	0	2	0	0
4,4′-Methylenediphenylene *m*-phenylenediurea	1	2	3	0	4	0	4	0	0
N,N′-Dimethyl-4,4′-methylenediphenylene terephthalamide	1	2	3	2	6	0	2	0	0
4,4′-Isopropylidenediphenylene (methylenediphenylene)diurethane	1	2	4	3	8	2	2	0	0
Alanylalanylglycine	1	3	0	4	12	0	3	0	0
β-D-Glucotriacetate, 1,3-	1	3	0	8	14	5	0	0	0
α-D-Glucotriacetate, 1,4-	1	3	0	8	14	5	0	0	0
β-D-Glucotriacetate, 1,4-	1	3	0	8	14	5	0	0	0
β-D-Mannotriacetate, 1,4-	1	3	0	8	14	5	0	0	0
α-D-Glucotriisobutyrate, 1,4-	1	3	0	14	26	5	0	0	0
3-Mercapto-2-phthalimidopropionic acid	1	3	1	1	1	0	1	1	0
β-D-Glucotricarbanilate, 1,4-	1	3	3	5	11	5	3	0	0
2,7-Fluorenylene pyromellitimide	1	4	0	18	8	0	2	0	0
2,7-Fluorenylene 1,4-pyromellitamide	1	4	0	18	12	2	2	0	0
Isophthalic 4,4′-methylenedioxy-3,3′-dimethoxydibenzoic anhydride	1	4	1	14	12	6	0	0	0
4,4′-(Methylenedioxy)-3,3′-dimethoxydibenzoic terephthalic anhydride	1	4	1	14	12	6	0	0	0
N,N′-Methylenediterephthalamic anhydride	1	4	2	0	2	1	2	0	0
Isophthaloyl (*m*-carboxyphenoxyacetyl)dihydrazide	1	4	2	0	4	1	4	0	0
Terephthaloyl (*m*-carboxyphenoxyacetyl)dihydrazide	1	4	2	0	4	1	4	0	0
4,4′-Methylenediphenylene pyromellitimide	1	4	2	6	2	0	2	0	0
Isophthalic 4,4′-(methylenedioxy)dibenzoic anhydride	1	4	3	0	0	4	0	0	0
4,4′-(Methylenedioxy)dibenzoic terephthalic anhydride	1	4	3	0	0	4	0	0	0
2	2	0	0	0	0	0	0	0	0
Ethylene	2	0	0	0	0	0	0	0	0
Ethylene sulfide	2	0	0	0	0	0	0	1	0
Ethylene disulfide	2	0	0	0	0	0	0	2	0
Ethylene tetrasulfide	2	0	0	0	0	0	0	4	0
Ethylene oxide	2	0	0	0	0	1	0	0	0
Ethylene amine	2	0	0	0	1	0	1	0	0
2-Aminoethylenesulfonic acid	2	0	0	0	1	2	1	1	0
Ethyleneamine-HCl complex	2	0	0	0	2	0	1	0	1
Allene	2	0	0	1	0	0	0	0	0
2-Imino-1,3-dithiapentamethylene	2	0	0	1	1	0	1	2	0
Bromomethylethylene oxide	2	0	0	1	1	1	0	0	1
Chloromethylethylene oxide (epichlorohydrin)	2	0	0	1	1	1	0	0	1
Fluoromethylethylene oxide	2	0	0	1	1	1	0	0	1
1-Hydroxytrimethylene	2	0	0	1	2	1	0	0	0
3-Hydroxyoxacyclobutane	2	0	0	1	2	2	0	0	0
3-Silylpropene	2	0	0	1	4	0	0	0	1
2,3-Dichloro-1,3-butadiene, 1,4-	2	0	0	2	0	0	0	0	2
Chlorotrifluoroethylene-*alt*-ethylene	2	0	0	2	0	0	0	0	4
Tetrafluoroethylene-*alt*-ethylene	2	0	0	2	0	0	0	0	4
Bis(2,2,2-trifluoroethoxy)phosphazene	2	0	0	2	0	2	1	0	7
2-Chloro-1,3-butadiene, 1,4- (chloroprene)	2	0	0	2	1	0	0	0	1
4,4,4-Trifluoro-1-butene	2	0	0	2	1	0	0	0	3
2,5-Ethylene-1,3,4-triazole	2	0	0	2	1	0	3	0	0
2,2,2-Trifluoroethylvinyl ether	2	0	0	2	1	1	0	0	3
3-Cyanopropionaldehyde	2	0	0	2	1	1	1	0	0
1,3-Butadiene, 1,2-	2	0	0	2	2	0	0	0	0
1,3-Butadiene, 1,4-	2	0	0	2	2	0	0	0	0
1,3-Cyclobutylene	2	0	0	2	2	0	0	0	0
Chlorinated 1,3-butadiene, 1,4-	2	0	0	2	2	0	0	0	2
2,5-Ethylene-1-amino-1,3,4-triazole	2	0	0	2	2	0	4	0	0
Butadiene oxide	2	0	0	2	2	1	0	0	0
1,2-Dichloromethylethylene oxide	2	0	0	2	2	1	0	0	2
N-Methyl-2-imino-1,3-dithiapentamethylene	2	0	0	2	3	0	1	2	0
2-Chloromethylpropylene oxide	2	0	0	2	3	1	0	0	1
1-Butene	2	0	0	2	4	0	0	0	0
Butyraldehyde	2	0	0	2	4	1	0	0	0
Ethylvinyl ether	2	0	0	2	4	1	0	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Diethylsilylene	2	0	0	2	6	0	0	0	1
Ethylsilylethylene	2	0	0	2	6	0	0	0	1
Methyl-*n*-propylsilylene	2	0	0	2	6	0	0	0	1
Diethylsiloxane	2	0	0	2	6	1	0	0	1
2,2-Bistrifluoromethyloxacyclobutane	2	0	0	3	0	1	0	0	6
1-Cyano-1,3-cyclobutylene	2	0	0	3	1	0	1	0	0
3-Chloro-2-methyl-1,3-butadiene, 1,4-	2	0	0	3	3	0	0	0	1
α-Ethylacrylonitrile	2	0	0	3	3	0	1	0	0
2-Methyl-1,3-butadiene, 1,4- (isoprene)	2	0	0	3	4	0	0	0	0
3-Methyl-1-butene, 1,3-	2	0	0	3	6	0	0	0	0
3,3-Dimethyltrimethylene sulfide	2	0	0	3	6	0	0	1	0
3,3-Dimethyloxacyclobutane	2	0	0	3	6	1	0	0	0
3,3-Dimethyltrimethylene sulfone	2	0	0	3	6	2	0	1	0
Hexafluoroisobutylene-*alt*-vinylidene fluoride	2	0	0	4	0	0	0	0	8
Bis(2,2,3,3,3-pentafluoropropoxy)phosphazene	2	0	0	4	0	2	1	0	11
2,4-Hexadiyne-1,6-diol, 2,5-	2	0	0	4	2	2	0	0	0
Bis(2,2,3,3-tetrafluoropropoxy)phosphazene	2	0	0	4	2	2	1	0	9
1,3-Cyclohexadiene, 1,4-	2	0	0	4	4	0	0	0	0
1,3,5-Hexatriene, 1,6-	2	0	0	4	4	0	0	0	0
4-Trifluoromethyl-1-pentene	2	0	0	4	5	0	0	0	3
2,3-Dimethyl-1,3-butadiene, 1,4-	2	0	0	4	6	0	0	0	0
1,3-Hexadiene, 1,4-	2	0	0	4	6	0	0	0	0
β-D-Fructofuran	2	0	0	4	6	5	0	0	0
2-Butene-*alt*-ethylene	2	0	0	4	8	0	0	0	0
3-Methyl-1-pentene	2	0	0	4	8	0	0	0	0
4-Methyl-1-pentene	2	0	0	4	8	0	0	0	0
1,4-Diethyl-2,3-azabutadiene, 1,4-	2	0	0	4	8	0	2	0	0
Isobutylvinyl ether	2	0	0	4	8	1	0	0	0
1-Methyl-2-propoxyethylene	2	0	0	4	8	1	0	0	0
1-Methylpropylvinyl ether	2	0	0	4	8	1	0	0	0
3-Trimethylsilylpropene	2	0	0	4	10	0	0	0	1
4,4-Dimethyl-1-pentene	2	0	0	5	10	0	0	0	0
3-*tert*-Butyloxacylobutane	2	0	0	5	10	1	0	0	0
Neopentylethylene oxide	2	0	0	5	10	1	0	0	0
Neopentylvinyl ether	2	0	0	5	10	1	0	0	0
Bis(2,2,3,3,4,4,4-heptafluorobutoxy)phosphazene	2	0	0	6	0	2	1	0	15
2,5-Dichloro-*p*-xylyene	2	0	0	6	2	0	0	0	2
2,4,6-Octatriynediol, 2,5-	2	0	0	6	2	2	0	0	0
Bromo-*p*-xylylene	2	0	0	6	3	0	0	0	1
Chloro-*p*-xylylene	2	0	0	6	3	0	0	0	1
2,5-Pyrazinecyclobutylene	2	0	0	6	4	0	2	0	0
2-*tert*-Butyl-1,3-butadiene, 1,4-	2	0	0	6	10	0	0	0	0
5-Methyl-1,3-heptadiene, 1,4-	2	0	0	6	10	0	0	0	0
6-Methyl-1,3-heptadiene, 1,4-	2	0	0	6	10	0	0	0	0
Cyano-*p*-xylylene	2	0	0	7	3	0	1	0	0
2,4,6-Trichloro-3-phenoxypropylene oxide	2	0	0	7	3	2	0	0	3
Methyl-*p*-xylylene	2	0	0	7	6	0	0	0	0
Divinyl furfural	2	0	0	7	6	3	0	0	0
4,6-Dimethyl-*m*-xylyene	2	0	0	8	8	0	0	0	0
2,5-Dimethyl-*p*-xylyene	2	0	0	8	8	0	0	0	0
2,5-Dimethyl-3-phenylpropene	2	0	0	9	10	0	0	0	0
3,4-Dimethyl-3-phenylpropene	2	0	0	9	10	0	0	0	0
3,5-Dimethyl-3-phenylpropene	2	0	0	9	10	0	0	0	0
Isopropyl-*p*-xylylene	2	0	0	9	10	0	0	0	0
2,6-Dimethyl-3-phenoxypropylene oxide	2	0	0	9	10	2	0	0	0
3-(1-Naphthoxy)propylene oxide	2	0	0	11	8	2	0	0	0
3-(2-Naphthoxy)propylene oxide	2	0	0	11	8	2	0	0	0
4,4′-perfluorodiphenyleneoxytetrahydroperfluoropentamethylene oxide	2	0	0	15	0	2	0	0	4
Di-1,4-naphthaleneoxyethylene oxide	2	0	0	20	12	2	0	0	0
m-Xylylene	2	0	1	0	0	0	0	0	0
p-Xylylene	2	0	1	0	0	0	0	0	0
p-Xylylene disulfide	2	0	1	0	0	0	0	2	0
Ethyleneoxy-*p*-phenylene oxide	2	0	1	0	0	2	0	0	0
Ethylene *m*-phenylenedisulfonamide	2	0	1	0	2	4	2	2	0
N-Phenyl-2-imino-1,3-dithiapentamethylene	2	0	1	1	1	0	1	2	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
o-Chloro-3-phenoxypropylene oxide	2	0	1	1	1	2	0	0	1
p-Chloro-3-phenoxypropylene oxide	2	0	1	1	1	2	0	0	1
3-Phenylpropene (allylbenzene)	2	0	1	1	2	0	0	0	0
Benzylvinyl ether	2	0	1	1	2	1	0	0	0
3-Phenoxypropylene oxide	2	0	1	1	2	2	0	0	0
Ethylene dithionisophthalate	2	0	1	2	0	2	0	2	0
p-Phenylenecyclobutylene	2	0	1	2	2	0	0	0	0
3-*m*-Tolylpropene	2	0	1	2	4	0	0	0	0
3-*o*-Tolylpropene	2	0	1	2	4	0	0	0	0
3-*p*-Tolylpropene	2	0	1	2	4	0	0	0	0
m-Methyl-3-phenoxypropylene oxide	2	0	1	2	4	2	0	0	0
o-Methyl-3-phenoxypropylene oxide	2	0	1	2	4	2	0	0	0
p-Methyl-3-phenoxypropylene oxide	2	0	1	2	4	2	0	0	0
p-Methoxy-3-phenoxypropylene oxide	2	0	1	2	4	3	0	0	0
Divinylbenzal	2	0	1	3	4	2	0	0	0
2-Methyldivinylbenzal	2	0	1	4	6	2	0	0	0
4-Methyldivinylbenzal	2	0	1	4	6	2	0	0	0
o-Isopropyl-3-phenoxypropylene oxide	2	0	1	4	8	2	0	0	0
2,2′-Ethylenedithiazole-*p*-phenylene	2	0	1	6	2	0	2	2	0
4,4′-Ethylenebiphenylene	2	0	2	0	0	0	0	0	0
4,4′-(Oxydiphenylene)dithioethylene disulfide	2	0	2	0	0	1	0	4	0
o-Phenyl-3-phenoxypropylene oxide	2	0	2	1	2	2	0	0	0
Bis(*p*-ethylphenoxy)phosphazene	2	0	2	2	6	2	1	0	1
1,6-Bis(*p*-fluorophenylsulfonyl)-2,4-hexadiynediol, 2,5-	2	0	2	4	0	6	0	2	2
4,4′-Isopropylidenediphenyleneoxy(2-hydroxy)trimethylene oxide	2	0	2	4	8	3	0	0	0
1-(*p*-Toluylsulfonyl)-6-(*p*-fluorobenzylsulfonyl)-hexadiynediol	2	0	2	5	3	6	0	2	1
1,6-Bis(*p*-toluylsulfonyl)-2,4-hexadiynediol, 2,5-	2	0	2	6	6	6	0	2	0
1,6-Bis(*p*-methoxyphenylsufonyl)-2,4-hexadiynediol, 2,5-	2	0	2	6	6	8	0	2	0
1,6-Di(*N*-carbazoyl)-2,4-hexadiyne, 2,5-	2	0	4	4	0	0	2	0	0
Carbon monoxide-*alt*-ethylene	2	1	0	0	0	0	0	0	0
3-Mercaptopropionic acid	2	1	0	0	0	0	0	1	0
3-Hydroxypropionic acid (propiolactone)	2	1	0	0	0	1	0	0	0
3-Aminopropionic acid (β-alanine)	2	1	0	0	1	0	1	0	0
Ethylene urea	2	1	0	0	2	0	2	0	0
Ethyleneamine-oxalic acid-water complex (2 : 1 : 2)	2	1	0	0	3	2	1	0	0
3-Hydroxy-3-butenoic acid	2	1	0	1	0	1	0	0	0
3-chloromethyl-3-hydroxypropionic acid	2	1	0	1	1	1	0	0	1
N-Acetylethylene amine	2	1	0	1	3	0	1	0	0
N-Methyl-3-aminopropionic acid	2	1	0	1	3	0	1	0	0
2,4-Diaminobutyric acid	2	1	0	1	4	0	2	0	0
Ethyleneamine-acetic acid complex (1 : 1)	2	1	0	1	5	1	1	0	0
N-(Perfluoropropionyl)ethylene amine	2	1	0	2	0	0	1	0	5
1,3-Cyclobutylene carbonate	2	1	0	2	2	2	0	0	0
Hydroxyproline	2	1	0	2	3	1	1	0	0
3-Ethyl-3-hydroxypropionic acid	2	1	0	2	4	1	0	0	0
3-Methoxycarbonylpropionaldehyde	2	1	0	2	4	2	0	0	0
Methionine	2	1	0	2	5	0	1	1	0
Diethyl-2-hydroxyacetic acid	2	1	0	3	6	1	0	0	0
2-Ethyl-2-methyl-3-hydroxypropionic acid	2	1	0	3	6	1	0	0	0
2-Ethyl-2-methyl-3-aminopropionic acid	2	1	0	3	7	0	1	0	0
N-Isobutyrylethylene amine	2	1	0	3	7	0	1	0	0
N-Isopropyl-3-aminopropionic acid	2	1	0	3	7	0	1	0	0
2-Methyl-*N*-butylaminoformic acid	2	1	0	3	7	0	1	0	0
sec-butylthiolacrylate	2	1	0	4	8	0	0	1	0
Isobutylthiolacrylate	2	1	0	4	8	0	0	1	0
sec-butyl acrylate	2	1	0	4	8	1	0	0	0
Isobutyl acrylate	2	1	0	4	8	1	0	0	0
N-Pivaloylethylene amine	2	1	0	4	9	0	1	0	0
3,5-Dimethyl-6-amino-4-oxacaproic acid	2	1	0	4	9	1	1	0	0
Ethylene *m*-chloro-*p*-oxybenzoate	2	1	0	6	3	2	0	0	1
Ethylene-*o*-chloro-*p*-oxybenzoate	2	1	0	6	3	2	0	0	1
N-(Perfluorooctanoyl)ethylene amine	2	1	0	7	0	0	1	0	5
3-*n*-Propylhydroxybenzoic acid	2	1	0	7	6	1	0	0	0
Ethylene *m*-methyl-*p*-oxybenzoate	2	1	0	7	6	2	0	0	0
Ethylene *o*-methyl-*p*-oxybenzoate	2	1	0	7	6	2	0	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
1,1-Dihydroperfluorooctyl acrylate	2	1	0	8	1	1	0	0	5
Isoamyl-2,4-hexadienoate, 2,5-	2	1	0	8	14	1	0	0	0
1,1-Dihydroperfluorodecyl acrylate	2	1	0	10	1	1	0	0	9
N-2-Naphthoylethylene amine	2	1	0	10	7	0	1	0	0
N-(*p*-Chlorobenzoyl)ethylene amine	2	1	1	0	0	0	1	0	1
Ethylene-*p*-oxybenzoate	2	1	1	0	0	2	0	0	0
p-Xylylene carbonate	2	1	1	0	0	2	0	0	0
p-Aminohydrocinnamic acid	2	1	1	0	1	0	1	0	0
p-(Aminomethylene)phenyleneacetic acid	2	1	1	0	1	0	1	0	0
N-Benzoylethylene amine	2	1	1	0	1	0	1	0	0
N-Phenyl-3-aminopropionic acid	2	1	1	0	1	0	1	0	0
N-(*p*-Toluoyl)ethylene amine	2	1	1	1	3	0	1	0	0
S-Benzylthiocysteine	2	1	1	1	3	0	1	2	0
1-Benzylhistidine	2	1	1	4	5	0	3	0	0
4,4′-Ethylenediphenylene carbonate	2	1	2	0	0	2	0	0	0
4,4′-Butylidenediphenylene carbonate	2	1	2	2	4	2	0	0	0
4,4′-Pentylidenediphenylene carbonate	2	1	2	3	6	2	0	0	0
Ethylene oxalate	2	2	0	0	0	2	0	0	0
Hydrazo succinamide	2	2	0	0	2	0	2	0	0
Methylene malonamide	2	2	0	0	2	0	2	0	0
Glutamic acid	2	2	0	1	3	1	1	0	0
Glutamine	2	2	0	1	4	0	2	0	0
Ethylene fumaramide	2	2	0	2	4	0	2	0	0
Ethylasparate	2	2	0	2	5	1	1	0	0
Methyl glutamate	2	2	0	2	5	1	1	0	0
4-Methyloxycarbonyl-4-aminobutyric acid	2	2	0	2	5	1	1	0	0
Alanyl-3-aminopropionic acid	2	2	0	2	6	0	2	0	0
Ethylene -L-tartaramide	2	2	0	2	6	2	2	0	0
O-Acetylhydroxyproline	2	2	0	3	5	1	1	0	0
2,2-dimethyltrimethylene oxalate	2	2	0	3	6	2	0	0	0
Ethylene 2,5-furandicarboxylate	2	2	0	4	2	3	0	0	0
Cyclopropylene cyclopropylenedicarboxamide	2	2	0	4	6	0	2	0	0
Ethylene 2,4-hexadienediamide	2	2	0	4	6	0	2	0	0
Isobutyl aspartate	2	2	0	4	9	1	1	0	0
Ethylene di-*O*-methyl-L-tartaramide	2	2	0	4	10	2	2	0	0
Ethylene dithiol-2,5-pyridinedicarboxylate	2	2	0	5	3	0	1	2	0
Ethylene 2,5-dichloroterephthalate	2	2	0	6	2	2	0	0	2
Ethylene chloroterephthalate	2	2	0	6	3	2	0	0	1
Ethylene methylterephthalate	2	2	0	7	6	2	0	0	0
Ethylene 2,5-dimethylterephthalate	2	2	0	8	8	2	0	0	0
Ethylene 1,5-naphthalate	2	2	0	10	6	2	0	0	0
Ethylene 2,6-naphthalate	2	2	0	10	6	2	0	0	0
Ethylene 2,7-naphthalate	2	2	0	10	6	2	0	0	0
4,4′-Ethylenedioxy-3,3′-dimethoxydibenzoic anhydride	2	2	0	14	12	5	0	0	0
2,5-Dimethylterephthalaldehyde	2	2	0	16	16	2	0	0	0
Ethylene dithiolisophthalate	2	2	1	0	0	0	0	2	0
Ethylene dithiolterephthalate	2	2	1	0	0	0	0	2	0
p-Phenylenediacetic anhydride	2	2	1	0	0	1	0	0	0
Ethylene isophthalate	2	2	1	0	0	2	0	0	0
Ethylene phthalate	2	2	1	0	0	2	0	0	0
Ethylene terephthalate	2	2	1	0	0	2	0	0	0
m-Phenylene succinate	2	2	1	0	0	2	0	0	0
p-Phenylene succinate	2	2	1	0	0	2	0	0	0
(*m*-Phenylenedioxy)diacetic anhydride	2	2	1	0	0	3	0	0	0
(*p*-Phenylenedioxy)diacetic anhydride	2	2	1	0	0	3	0	0	0
Ethylene phthalamide	2	2	1	0	2	0	2	0	0
Ethylene terephthalamide	2	2	1	0	2	0	2	0	0
p-Chlorobenzylasparate	2	2	1	1	2	1	1	0	1
Benzyl aspartate	2	2	1	1	3	1	1	0	0
S-Benzyloxycarbonylcysteine	2	2	1	1	3	1	1	1	0
1-Methyltrimethylene isophthalate	2	2	1	2	4	2	0	0	0
1-Methyltrimethylene phthalate	2	2	1	2	4	2	0	0	0
1-Methyltrimethylene terephthalate	2	2	1	2	4	2	0	0	0
Methylbenzyl aspartate	2	2	1	2	5	1	1	0	0
N,N′-Dimethylethylene terephthalamide	2	2	1	2	6	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
2,2-Dimethyltrimethylene terephthalate	2	2	1	3	6	2	0	0	0
2,5-Dimethylpiperazine isophthalamide	2	2	1	4	8	0	2	0	0
2,5-Dimethylpiperazine phthalamide	2	2	1	4	8	0	2	0	0
2,5-Dimethylpiperazine terephthalamide	2	2	1	4	8	0	2	0	0
p-Phenylene diethylcyclobutylenedicarboxylate	2	2	1	6	10	2	0	0	0
p-Phenylene dicyanodiethylcyclobutylenedicarboxylate	2	2	1	8	8	2	2	0	0
2,2′-Diethyl-4,4′-oxydiphenylene *m*-phenylenediketone	2	2	1	14	12	1	0	0	0
N,N′-Diethyl-3,3′-dimethylbiphenylene terephthalamide	2	2	1	16	18	0	2	0	0
p-Xylylene dithiolisophthalate	2	2	2	0	0	0	0	2	0
4,4′-Ethylenedibenzoic anhydride	2	2	2	0	0	1	0	0	0
Ethylene 2,2′-dibenzoate	2	2	2	0	0	2	0	0	0
Ethylene 3,3′-dibenzoate	2	2	2	0	0	2	0	0	0
Ethylene 4,4′-dibenzoate	2	2	2	0	0	2	0	0	0
Isophthalaldehyde	2	2	2	0	0	2	0	0	0
Terephthalaldehyde	2	2	2	0	0	2	0	0	0
p-Xylylene isophthalate	2	2	2	0	0	2	0	0	0
p-Xylylene terephthalate	2	2	2	0	0	2	0	0	0
Ethylene 4,4′-thiodibenzoate	2	2	2	0	0	2	0	1	0
3,3′-(Ethylenedioxy)dibenzoic anhydride	2	2	2	0	0	3	0	0	0
4,4′-(Ethylenedioxy)dibenzoic anhydride	2	2	2	0	0	3	0	0	0
Ethylene 4,4′-oxydibenzoate	2	2	2	0	0	3	0	0	0
4,4′-(Oxydimethylene)dioxydibenzoic anhydride	2	2	2	0	0	4	0	0	0
p-Phenyleneethylene terephthalamide	2	2	2	0	2	0	2	0	0
m-Xylylene isophthalamide	2	2	2	0	2	0	2	0	0
p-Xylylene isophthalamide	2	2	2	0	2	0	2	0	0
p-Xylylene phthalamide	2	2	2	0	2	0	2	0	0
m-Xylylene terephthalamide	2	2	2	0	2	0	2	0	0
p-Xylylene terephthalamide	2	2	2	0	2	0	2	0	0
Ethylene 4,4′-sulfonyldibenzamide	2	2	2	0	2	2	2	1	0
O-Benzyloxycarbonyltyrosine	2	2	2	1	3	2	1	0	0
1-Methyltrimethylene 2,2′-dibenzoate	2	2	2	2	4	2	0	0	0
1-Methyltrimethylene 3,3′-dibenzoate	2	2	2	2	4	2	0	0	0
1-Methyltrimethylene 4,4′-dibenzoate	2	2	2	2	4	2	0	0	0
1-Methyltrimethylene 4,4′-sulfonyldibenzamide	2	2	2	2	6	2	2	1	0
(4,4′-Isopropylidenediphenylene)dioxydiacetic anhydride	2	2	2	3	6	3	0	0	0
2,2-Dimethyltrimethylene sulfonyldibenzamide	2	2	2	3	8	2	2	1	0
1,6-Bis(phenylurethane)-2,4-hexadiynediol, 2,5-	2	2	2	4	4	2	2	0	0
1,8-Bis(phenylurethane)-2,4,6-octatriynediol, 2,5-	2	2	2	6	4	2	2	0	0
Methyl-*p*-phenylene 4,4′-ethylenedioxydibenzoate	2	2	2	7	6	4	0	0	0
p-Phenylene 4,4′-(ethylenedioxy)dibenzoate	2	2	3	0	0	4	0	0	0
N,N′-Diethylbiphenylene terephthalamide	2	2	3	2	6	0	2	0	0
4,4′-Biphenyldicarboxaldehyde	2	2	4	0	0	2	0	0	0
4,4′-(Silylenediphenylene)dimethylene 4,4′-silylenedibenzamide	2	2	4	0	6	0	2	0	2
4,4′-(Silylenediphenylene)dimethylene 4,4′-disiloxanylenedibenzamide	2	2	4	0	8	1	2	0	3
4,4′-(Silylenediphenylene)dimethylene 4,4′-tetrasiloxanylenedibenzamide	2	2	4	0	12	3	2	0	5
4,4′-Dimethylsilylenediphenylenedimethylene tetrasiloxanylenedibenzamide	2	2	4	2	16	3	2	0	5
4,4′-Dimethylsilylenediphenylenedimethylene dimethylsilylenedibenzamide	2	2	4	4	14	1	2	0	2
4,4′-Dimethylsilylenediphenylenedimethylene tetramethyldisiloxanylenedi	2	2	4	6	20	1	2	0	3
4,4′-Dimethylsilylenediphenylenedimethylene tetramethyltetraphenyltetra	2	2	8	6	20	3	2	0	5
Alanylglycylglycine	2	3	0	2	7	0	3	0	0
Alanyl(phenylalanyl)glycine)	2	3	1	3	9	0	3	0	0
Cyclopropylene (cyclopropylenedicarboxoyl)diurethane	2	4	0	4	6	2	2	0	0
(*p*-Phenylenedioxy)diacetic 3,4-pyridinedicarboxylic anhydride	2	4	1	5	3	4	1	0	0
Isophthalic (*p*-phenylenedioxy)diacetic anhydride	2	4	2	0	0	4	0	0	0
(*p*-Phenylenedioxy)diacetic terephthalic anhydride	2	4	2	0	0	4	0	0	0
Isophthaloyl (*m*-carboxyphenoxypropionyl)dihydrazide	2	4	2	0	4	1	4	0	0
4,4′-(Isopropylidenediphenylene)dioxydiacetic pyridinedicarboxylic anhydride	2	4	2	8	9	4	1	0	0
3,4′-(Ethylenedioxy)diphenylene 3,3′,4,4′-Oxydiphenylenetetracarboximide	2	4	2	12	6	3	2	0	0
4,4′-(Ethylenedioxy)dibenzoic isophthalic anhydride	2	4	3	0	0	4	0	0	0
4,4′-(Ethylenedioxy)dibenzoic phthalic anhydride	2	4	3	0	0	4	0	0	0
4,4′-(Ethylenedioxy)dibenzoic terephthalic anhydride	2	4	3	0	0	4	0	0	0
Isophthalic (isopropylidenediphenylene)diacetic anhydride	2	4	3	3	6	4	0	0	0
4,4′-(Isopropylidenediphenylene(dioxydiacetic terephthalic anhydride	2	4	3	3	6	4	0	0	0
4,4′-(Ethylenedioxy)diphenylene carbonyldiphenylenetetracarboximide	2	5	2	12	4	2	2	0	0
Neopentylene-4,4′-diphenylene 3,3′,4,4′-carbonyldiphenylenetetracarbox	2	5	2	15	12	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
3	3	0	0	0	0	0	0	0	0
Trimethylene sulfide	3	0	0	0	0	0	0	1	0
Trimethylene disulfide	3	0	0	0	0	0	0	2	0
Oxacyclobutane (trimethylene oxide)	3	0	0	0	0	1	0	0	0
Ethyleneoxymethylene oxide (dioxolane)	3	0	0	0	0	2	0	0	0
Trimethylene sulfone	3	0	0	0	0	2	0	1	0
Trimethylene amine	3	0	0	0	1	0	1	0	0
3-Aminotrimethylenesulfonic acid	3	0	0	0	1	2	0	1	0
1,1-Bischloromethylethylene oxide	3	0	0	1	0	1	0	0	2
2-Chloroethylvinylether	3	0	0	1	1	1	0	0	1
5,5,5-Trifluoro-1-pentene	3	0	0	2	1	0	0	0	3
Pentenamer	3	0	0	2	2	0	0	0	0
Vinyl cyclopropane	3	0	0	2	2	0	0	0	0
1,2-Dichloropentamethylene	3	0	0	2	2	0	0	0	2
1,3-Cyclobutyleneoxymethylene oxide	3	0	0	2	2	2	0	0	0
Hydrochlorinated 2-methyl-1,3-butadiene, 1,4-	3	0	0	2	3	0	0	0	1
N-Ethyl-2-imino-1,3-dithiapentamethylene	3	0	0	2	3	0	1	2	0
1-Pentene	3	0	0	2	4	0	0	0	0
Propene-*alt*-ethylene	3	0	0	2	4	0	0	0	0
3-Methyl-3-hydroxymethyloxacyclobutane	3	0	0	2	4	1	0	0	0
Propylvinyl ether	3	0	0	2	4	1	0	0	0
Valeraldehyde	3	0	0	2	4	1	0	0	0
2-Methoxyethylvinyl ether	3	0	0	2	4	2	0	0	0
α-Propylacrylonitrile	3	0	0	3	3	0	1	0	0
1-Butoxy-2-chloroethylene	3	0	0	3	5	1	0	0	1
3-Ethyl-3-methyloxacyclobutane	3	0	0	3	6	1	0	0	0
1-Vinylene-3-cyclopentylene	3	0	0	4	4	0	0	0	0
1,3-Heptadiene, 1,4-	3	0	0	4	6	0	0	0	0
3-Ethyl-1-pentene	3	0	0	4	8	0	0	0	0
3-Methyl-1-hexene	3	0	0	4	8	0	0	0	0
4-Methyl-1-hexene	3	0	0	4	8	0	0	0	0
5-Methyl-1-hexene	3	0	0	4	8	0	0	0	0
1-Butoxy-2-methylethylene	3	0	0	4	8	1	0	0	0
2-Methylbutylvinyl ether	3	0	0	4	8	1	0	0	0
4-Trimethylsilylbutene	3	0	0	4	10	0	0	0	1
1,4-Butadiene-*alt*-methacrylonitrile	3	0	0	5	5	0	1	0	0
4,4-Dimethyl-1-hexene	3	0	0	5	10	0	0	0	0
Ethyl-*p*-xylylene	3	0	0	7	6	0	0	0	0
4-Diisopropylamino-1-butene	3	0	0	7	15	0	1	0	0
β-D-Galacto-1,4-β-D-mannose, 1,4- (1 : 2) (guar gallactomannan)	3	0	0	15	24	5	0	0	0
Di-1,4-naphthaleneoxytrimethylene oxide	3	0	0	20	12	2	0	0	0
p-Phenyleneoxytrimethylene oxide	3	0	1	0	0	2	0	0	0
Trimethylene *m*-phenylenedisulfonamide	3	0	1	0	2	4	2	2	0
4-Phenyl-1-butene	3	0	1	1	2	0	0	0	0
4-*o*-Tolyl-1-butene	3	0	1	2	4	0	0	0	0
4-*p*-Tolyl-1-butene	3	0	1	2	4	0	0	0	0
4,4′-(Ethylenediphenylene)methylene	3	0	2	0	0	0	0	0	0
Trimethylene biphenylene	3	0	2	0	0	0	0	0	0
4,4′-Stilbenediyloxytrimethylene oxide	3	0	2	2	2	2	0	0	0
α-Methyl-4,4′-stilbenediyloxytrimethylene oxide	3	0	2	3	4	2	0	0	0
4,4′-Isopropylidenediphenyleneoxytrimethylene oxide	3	0	2	3	6	2	0	0	0
5-Hydroxy-3-oxavaleric acid	3	1	0	0	0	2	0	0	0
4-Aminobutyric acid	3	1	0	0	1	0	1	0	0
5-Amino-3-oxavaleric acid	3	1	0	0	1	1	1	0	0
2,2-Bischloromethyl-3-hydroxypropionic acid	3	1	0	1	0	1	0	0	2
Proline	3	1	0	1	1	0	1	0	0
N-Butylaminoformic acid	3	1	0	1	3	0	1	0	0
Ornithine	3	1	0	1	4	0	2	0	0
Ornithine hydrobromide	3	1	0	1	5	0	2	0	1
Allyl acrylate	3	1	0	2	2	1	0	0	0
2-Aminocyclopentylenecarboxy acid	3	1	0	2	3	0	1	0	0
2,6-Aminotetrahydropyran carboxylic acid	3	1	0	2	3	1	1	0	0
Propylthiolacrylate	3	1	0	2	4	0	0	1	0
Arginine	3	1	0	2	6	0	4	0	0
Arginine hydrochloride	3	1	0	2	7	0	4	0	1

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
2-Methylacetoxy-1,3-butadiene	3	1	0	3	4	1	0	0	0
2,2-Diethyl-3-hydroxypropionic acid	3	1	0	3	6	1	0	0	0
2-Methyl-2-propyl-3-hydroxpropionic acid	3	1	0	3	6	1	0	0	0
N-Isovalerylethylene amine	3	1	0	3	7	0	1	0	0
2-Methyl-2-propyl-3-aminopropionic acid	3	1	0	3	7	0	1	0	0
1,4-Butadiene-*alt*-methyl methacrylate	3	1	0	5	8	1	0	0	0
(2,2-Dimethyl-3-aminocyclobutylene)propionic acid	3	1	0	5	9	0	1	0	0
Butyl-2,4-hexadienoate, 2,5-	3	1	0	6	10	1	0	0	0
Ethylene *p*-oxyphenyleneacetate	3	1	1	0	0	2	0	0	0
Trimethylene *p*-oxybenzoate	3	1	1	0	0	2	0	0	0
p-(Aminoethylene)phenyleneacetic acid	3	1	1	0	1	0	1	0	0
p-(Aminomethylene)phenylenepropionic acid	3	1	1	0	1	0	1	0	0
2,5-Butyl-5-phenyl-2,4-pentadienoate	3	1	1	5	8	1	0	0	0
4,4′-Hexylidenediphenylene carbonate	3	1	2	3	6	2	0	0	0
Trimethylene oxalate	3	2	0	0	0	2	0	0	0
Glycyl-3-aminopropionic acid	3	2	0	0	2	0	2	0	0
Hydrazo glutaramide	3	2	0	0	2	0	2	0	0
Ethylene cyclopropylenedicarboxamide	3	2	0	2	4	0	2	0	0
Methylethylene succinate	3	2	0	2	4	2	0	0	0
Methylethylene oxydiacetate	3	2	0	2	4	3	0	0	0
Ethyl glutamate	3	2	0	2	5	1	1	0	0
n-Propyl aspartate	3	2	0	2	5	1	1	0	0
2-Methoxyethyl aspartate	3	2	0	2	5	2	1	0	0
Alanyl-4-aminobutyric acid	3	2	0	2	6	0	2	0	0
2,2-Dimethyltrimethylene malonate	3	2	0	3	6	2	0	0	0
Cyclopropylenedimethylene toluylenediurethane	3	2	0	9	10	2	2	0	0
4,4′-Trimethylenedioxy-3,3′-dimethoxydibenzoic anhydride	3	2	0	14	12	5	0	0	0
Trimethylene isophthalate	3	2	1	0	0	2	0	0	0
Trimethylene phthalate	3	2	1	0	0	2	0	0	0
Trimethylene terephthalate	3	2	1	0	0	2	0	0	0
p-Xylylene glutarate	3	2	1	0	0	2	0	0	0
Ethylene *p*-(carboxyphenoxy)acetate	3	2	1	0	0	3	0	0	0
Ethylene-*p*-(carboxyphenylene)acetamide	3	2	1	0	2	0	2	0	0
p-Phenylene glutaramide	3	2	1	0	2	0	2	0	0
Trimethylene terephthalamide	3	2	1	0	2	0	2	0	0
p-Xylylene malonamide	3	2	1	0	2	0	2	0	0
m-Chlorobenzylglutamate	3	2	1	1	2	1	1	0	1
o-Chlorobenzylglutamate	3	2	1	1	2	1	1	0	1
p-Chlorobenzylglutamate	3	2	1	1	2	1	1	0	1
N-(*p*-Bromobenzoyl)-L-ornithine	3	2	1	1	3	0	2	0	1
N-(*p*-Chlorobenzoyl)-L-ornithine	3	2	1	1	3	0	2	0	1
N-(*p*-Fluorobenzoyl)-L-ornithine	3	2	1	1	3	0	2	0	1
Benzyl glutamate	3	2	1	1	3	1	1	0	0
4-Benzyloxycarbonyl-4-aminobutyric acid	3	2	1	1	3	1	1	0	0
Phenethylasparate	3	2	1	1	3	1	1	0	0
N-Benzoyl-L-ornithine	3	2	1	1	4	0	2	0	0
N-Benzyloxycarbonyl-2,4-diaminobutyric acid	3	2	1	1	4	1	2	0	0
Cyclopropylenedimethylene isophthalate	3	2	1	2	2	2	0	0	0
Cyclopropylenedimethylene terephthalate	3	2	1	2	2	2	0	0	0
Cyclopropylenedimethylene isophthalamide	3	2	1	2	4	0	2	0	0
Methylpiperazine isophthalamide	3	2	1	2	4	0	2	0	0
Methylpiperazine phthalamide	3	2	1	2	4	0	2	0	0
Methylpiperazine terephthalamide	3	2	1	2	4	0	2	0	0
N,N′-Dimethyltrimethylene terephthalamide	3	2	1	2	6	0	2	0	0
p-Phenylene(1,1-dimethyl)ethylene succinamide	3	2	1	3	8	0	2	0	0
N,N′-Diethyl-3,3′-dimethyl(methylenediphenylene) terephthalamide	3	2	1	16	18	0	2	0	0
4,4′-Trimethylenedibenzoic anhydride	3	2	2	0	0	1	0	0	0
Ethylene 4,4′-methylenedibenzoate	3	2	2	0	0	2	0	0	0
Trimethylene 2,2′-dibenzoate	3	2	2	0	0	2	0	0	0
Trimethylene 3,3′-dibenzoate	3	2	2	0	0	2	0	0	0
Trimethylene 4,4′-dibenzoate	3	2	2	0	0	2	0	0	0
3,3′-(Trimethylenedioxy)dibenzoic anhydride	3	2	2	0	0	3	0	0	0
4,4′-(Trimethylenedioxy)dibenzoic anhydride	3	2	2	0	0	3	0	0	0
Trimethylene 4,4′-oxydibenzoate	3	2	2	0	0	3	0	0	0
Ethylene 4,4′-(methylenediphenylene)diurethane	3	2	2	0	2	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Trimethylene 4,4′-sulfonyldibenzamide	3	2	2	0	2	2	2	1	0
Isopropylidenedimethylene methylenediphenylenediurethane	3	2	2	3	8	2	2	0	0
4,4′-Diphenyleneglutarate-butadiyne, 1,4-	3	2	2	4	0	2	0	0	0
2,4-Hexadiyne (4,4′-methylenediphenylene)diurethane	3	2	2	4	2	2	2	0	0
p-Xylylene dithiol-4,4′-methylenedibenzoate	3	2	3	0	0	0	0	2	0
N,N′-Diethyl(methylenediphenylene) terephthalamide	3	2	3	2	6	0	2	0	0
Alanylglycylglycylglycine	3	4	0	2	8	0	4	0	0
Alanylglutamylglycine	3	4	0	3	9	1	3	0	0
Isophthalic 4,4′-trimethylenedioxy-3,3′-dimethoxydibenzoic anhydride	3	4	1	14	12	6	0	0	0
Terephthalic 4,4′-trimethylenedioxy-3,3′-dimethoxydibenzoic anhydride	3	4	1	14	12	6	0	0	0
Isophthaloyl (*m*-carboxyphenoxybutyryl)dihydrazide	3	4	2	0	4	1	4	0	0
Isophthalic 4,4′-(trimethylenedioxy)dibenzoic anhydride	3	4	3	0	0	4	0	0	0
Terephthalic 4,4′-(trimethylenedioxy)dibenzoic anhydride	3	4	3	0	0	4	0	0	0
4	4	0	0	0	0	0	0	0	0
Tetramethylene sulfide	4	0	0	0	0	0	0	1	0
Tetramethylene disulfide	4	0	0	0	0	0	0	2	0
Tetrahydrofuran (tetramethylene oxide)	4	0	0	0	0	1	0	0	0
Tetramethylene sulfone	4	0	0	0	0	2	0	1	0
3,3-Bisazidomethyloxacylobutane	4	0	0	1	0	0	6	0	0
cyclopropylidenedimethylene oxide	4	0	0	1	0	1	0	0	0
3,3-Bisbromomethyloxacyclobutane	4	0	0	1	0	1	0	0	2
3,3-Bischloromethyloxacyclobutane	4	0	0	1	0	1	0	0	2
3,3-Bisfluoromethyloxacyclobutane	4	0	0	1	0	1	0	0	2
3,3-Bisiodomethyloxacyclobutane	4	0	0	1	0	1	0	0	2
Tetramethylene thiourea	4	0	0	1	2	0	2	1	0
3,3-Bishydroxyoxacyclobutane	4	0	0	1	2	1	0	0	0
5-Hydroxy-1-pentene	4	0	0	1	2	1	0	0	0
3,3-Bishydroxymethyloxacyclobutane	4	0	0	1	2	3	0	0	0
5-Amino-1-pentene	4	0	0	1	3	0	1	0	0
2,5-Tetramethylene-1,3,4-triazole	4	0	0	2	1	0	3	0	0
1,3-Cyclopenylene-*alt*-methylene	4	0	0	2	2	0	0	0	0
Vinyl cyclobutane	4	0	0	2	2	0	0	0	0
1,2-Cyclohexylene sulfide	4	0	0	2	2	0	0	1	0
1,3-Cyclohexylene sulfide	4	0	0	2	2	0	0	1	0
2,5-Tetramethylene-1-amino-1,3,4-triazole	4	0	0	2	2	0	4	0	0
1,2-Cyclohexene oxide	4	0	0	2	2	1	0	0	0
1,2-Cyclohexylene sulfone	4	0	0	2	2	2	0	1	0
1,3-Cyclohexylene sulfone	4	0	0	2	2	2	0	1	0
1-Hexene	4	0	0	2	4	0	0	0	0
n-Butylvinyl ether	4	0	0	2	4	1	0	0	0
Di-*n*-propylsilylene	4	0	0	2	6	0	0	0	1
Di-*n*-propylsiloxane	4	0	0	2	6	1	0	0	1
2-Propyl-1,3-butadiene, 1,4-	4	0	0	3	4	0	0	0	0
3,3-Diethyltrimethylene sulfide	4	0	0	3	6	0	0	1	0
3,3-Diethyloxacyclobutane	4	0	0	3	6	1	0	0	0
3,3-Diethyltrimethylene sulfone	4	0	0	3	6	2	0	1	0
1,3-Octadiene, 1,4-	4	0	0	4	6	0	0	0	0
4-Ethyl-1-hexene	4	0	0	4	8	0	0	0	0
5-Methyl-1-heptene	4	0	0	4	8	0	0	0	0
5-Trimethylsilyl-1-pentene	4	0	0	4	10	0	0	0	1
6,6-Dimethyl-1-heptene	4	0	0	5	10	0	0	0	0
5-Diisopropylamino-1-pentene	4	0	0	7	15	0	1	0	0
α-D-Triethylglucose, 1,4-	4	0	0	8	14	5	0	0	0
β-D-Triethylglucose, 1,4-	4	0	0	8	14	5	0	0	0
Di-1,4-naphthaleneoxytetramethylene oxide	4	0	0	20	12	2	0	0	0
Oxydiethyleneoxy-*p*-phenylene oxide	4	0	1	0	0	3	0	0	0
Tetramethylene *m*-phenylenedisulfonamide	4	0	1	0	2	4	2	2	0
Tetramethylene dithionisophthalate	4	0	1	2	0	2	0	2	0
2,2′-Tetramethylenedithiazole-*p*-phenylene	4	0	1	6	2	0	2	2	0
4,4′-(Oxydiphenylene)dithiotetramethylene disulfide	4	0	2	0	0	1	0	4	0
4,4′-Stilbenediyloxytetramethylene oxide	4	0	2	2	2	2	0	0	0
α-Methyl-4,4′-stilbenediyloxytetramethylene oxide	4	0	2	3	4	2	0	0	0
5-Hydroxyvaleric acid	4	1	0	0	0	1	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Tetramethylene carbonate	4	1	0	0	0	2	0	0	0
5-Aminovaleric acid	4	1	0	0	1	0	1	0	0
6-Amino-4-oxacaproic acid	4	1	0	0	1	1	1	0	0
Tetramethylene urea	4	1	0	0	2	0	2	0	0
1,4-Piperidine urethane	4	1	0	1	1	1	1	0	0
N-Butyrylethylene amine	4	1	0	1	3	0	1	0	0
Lysine	4	1	0	1	4	0	2	0	0
Lysine hydrobromide	4	1	0	1	5	0	2	0	1
Lysine hydrochloride	4	1	0	1	5	0	2	0	1
3-Aminocyclobutylenepropionic acid	4	1	0	2	3	0	1	0	0
1,4-Cyclohexylene urethane	4	1	0	2	3	1	1	0	0
6-Mercapto-4-methylcaproic acid	4	1	0	2	4	0	0	1	0
Butyl acrylate	4	1	0	2	4	1	0	0	0
3-Methyl-6-aminocaproic acid	4	1	0	2	5	0	1	0	0
6-Methyl-6-aminocaproic acid	4	1	0	2	5	0	1	0	0
2-*n*-Butyl-2-methyl-3-aminopropionic acid	4	1	0	3	7	0	1	0	0
3-*n*-Pentylhydroxybenzoic acid	4	1	0	7	6	1	0	0	0
p-(Aminoethylene)phenylenepropionic acid	4	1	1	0	1	0	1	0	0
p-(Aminomethylene)phenylenebutyric acid	4	1	1	0	1	0	1	0	0
4,4′-Cyclopentylidenediphenylene carbonate	4	1	2	1	0	2	0	0	0
4,4′-Heptylidenediphenylene carbonate	4	1	2	3	6	2	0	0	0
Adipic anhydride	4	2	0	0	0	1	0	0	0
Thiodipropionic anhydride	4	2	0	0	0	1	0	1	0
(Ethylenedithio)diacetic anhydride	4	2	0	0	0	1	0	2	0
Ethylene succinate	4	2	0	0	0	2	0	0	0
Tetramethylene oxalate	4	2	0	0	0	2	0	0	0
Trimethylene malonate	4	2	0	0	0	2	0	0	0
Tetramethylene thiodicarboxylate	4	2	0	0	0	2	0	1	0
Tetramethylene dithiodicarboxylate	4	2	0	0	0	2	0	2	0
Tetramethylene Trithiodicarboxylate	4	2	0	0	0	2	0	3	0
Tetramethylene tetrathiodicarboxylate	4	2	0	0	0	2	0	4	0
Ethylene oxydiacetate	4	2	0	0	0	3	0	0	0
Oxydiethylene oxalate	4	2	0	0	0	3	0	0	0
Sulfonyldipropionic anhydride	4	2	0	0	0	3	0	1	0
(Ethylenedisulfonyl)diacetic anhydride	4	2	0	0	0	5	0	2	0
Hydrazo adipamide	4	2	0	0	2	0	2	0	0
Methylene glutaramide	4	2	0	0	2	0	2	0	0
2,2-Bischloromethyltrimethylene oxalate	4	2	0	1	0	2	0	0	2
2-Chloroethylglutamate	4	2	0	1	2	1	1	0	1
*N*5-(2-Hydroxyethyl)glutamine	4	2	0	1	4	1	2	0	0
N,N-Lysylurea	4	2	0	1	4	1	2	0	0
Trimethylene cyclopropylenedicarboxamide	4	2	0	2	4	0	2	0	0
Methylethylene glutarate	4	2	0	2	4	2	0	0	0
Trimethylene cyclopropylenediurethane	4	2	0	2	4	2	2	0	0
1-Methyltrimethylene oxydiacetate	4	2	0	2	4	3	0	0	0
n-Butyl aspartate	4	2	0	2	5	1	1	0	0
n-Propylglutamate	4	2	0	2	5	1	1	0	0
Alanyl-5-aminovaleric acid	4	2	0	2	6	0	2	0	0
Methyl-*N,N*′-lysylurea	4	2	0	2	6	1	2	0	0
Tetramethylene-L-tartaramide	4	2	0	2	6	2	2	0	0
2,2-Dimethyltrimethylene succinate	4	2	0	3	6	2	0	0	0
2,5-Thiophenedipropionic anhydride	4	2	0	4	2	1	0	1	0
2,5-Furandipropionic anhydride	4	2	0	4	2	2	0	0	0
2,5-Pyrroledipropionic anhydride	4	2	0	4	3	1	1	0	0
Cyclopropylenedimethylene cyclopropylenedicarboxamide	4	2	0	4	6	0	2	0	0
Piperazine (3-methylcyclopropylene)dicarboxamide	4	2	0	4	6	0	2	0	0
Cyclopropylenedimethylene cyclopropylenediurethane	4	2	0	4	6	2	2	0	0
Tetramethylene 1,2-dimethoxysuccinate	4	2	0	4	8	4	0	0	0
D-Methionyl-L-methionine	4	2	0	4	10	0	2	2	0
Tetramethylene di-*O*-methyl-L-tartaramide	4	2	0	4	10	2	2	0	0
N-Methyl-2,5-pyrroledipropionic anhydride	4	2	0	5	5	1	1	0	0
1,6-Bis(ethylurethane)-2,4-hexadiynediol, 2,5-	4	2	0	6	8	2	2	0	0
3-Methyl-*m*-phenylene adipamide	4	2	0	7	8	0	2	0	0
4-Methyl-*m*-phenylene disiloxanylenedipropionamide	4	2	0	7	12	1	2	0	2
Isopropylidenedimethylene dimethylpiperazinedithiodicarboxylate	4	2	0	7	14	2	2	2	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
4-Methyl-1,3-phenylene tetramethyldisiloxanylenedipropionamide	4	2	0	11	20	1	2	0	2
4,4′-Tetramethylenedioxy-3,3′-dimethoxydibenzoic anhydride	4	2	0	14	12	5	0	0	0
Ethylene 3,3′-dimethoxy-4,4′-(ethylenedioxy)dibenzoate	4	2	0	14	12	6	0	0	0
4,4′-Tetrachloroisopropylidenediphenylene adipate	4	2	0	15	10	2	0	0	4
3,3′-Dimethylmethylenediphenylene tetramethyldisiloxanylenedipropionamide	4	2	0	18	26	1	2	0	2
Tetramethylene dithiolisophthalate	4	2	1	0	0	0	0	2	0
Tetramethylene dithiolterephthalate	4	2	1	0	0	0	0	2	0
Thiodiethylene dithiolterephthalate	4	2	1	0	0	0	0	3	0
Piperazine isophthalamide	4	2	1	0	0	0	2	0	0
Piperazine phthalamide	4	2	1	0	0	0	2	0	0
Piperazine terephthalamide	4	2	1	0	0	0	2	0	0
p-Phenylenedipropionic anhydride	4	2	1	0	0	1	0	0	0
(*p*-Phenylenedithio)dipropionic anhydride	4	2	1	0	0	1	0	2	0
(*p*-Xylylenedithio)diacetic anhydride	4	2	1	0	0	1	0	2	0
Ethylene *p*-phenylenediacetate	4	2	1	0	0	2	0	0	0
p-Phenylene adipate	4	2	1	0	0	2	0	0	0
Tetramethylene isophthalate	4	2	1	0	0	2	0	0	0
Tetramethylene phthalate	4	2	1	0	0	2	0	0	0
Tetramethylene terephthalate	4	2	1	0	0	2	0	0	0
p-Xylylene succinate	4	2	1	0	0	2	0	0	0
Oxydiethylene isophthalate	4	2	1	0	0	3	0	0	0
Oxydiethylene phthalate	4	2	1	0	0	3	0	0	0
Oxydiethylene terephthalate	4	2	1	0	0	3	0	0	0
(*p*-Xylylenedisulfonyl)diacetic anhydride	4	2	1	0	0	5	0	2	0
Tetramethylene *m*-carboxycarbanilate	4	2	1	0	1	2	1	0	0
Tetramethylene *p*-carboxycarbanilate	4	2	1	0	1	2	1	0	0
m-Phenylene adipamide	4	2	1	0	2	0	2	0	0
o-Phenylene adipamide	4	2	1	0	2	0	2	0	0
p-Phenylene adipamide	4	2	1	0	2	0	2	0	0
p-Phenyleneethylene succinamide	4	2	1	0	2	0	2	0	0
Tetramethylene isophthalamide	4	2	1	0	2	0	2	0	0
Tetramethylene terephthalamide	4	2	1	0	2	0	2	0	0
Trimethylene *p*-(carboxyphenylene)acetamide	4	2	1	0	2	0	2	0	0
p-Xylylene succinamide	4	2	1	0	2	0	2	0	0
m-Phenylene disiloxanylenedipropionamide	4	2	1	0	6	1	2	0	2
o-Phenylene disiloxanylenedipropionamide	4	2	1	0	6	1	2	0	2
p-Phenylene disiloxanylenedipropionamide	4	2	1	0	6	1	2	0	2
Ethylene 3-(*p*-nitro)glutarate	4	2	1	1	1	4	1	0	0
Phenylpropyl-L-aspartate	4	2	1	1	2	1	1	0	0
Phenethylglutamate	4	2	1	1	3	1	1	0	0
N-benzyloxycarbonylornithine	4	2	1	1	4	1	2	0	0
Adipoyl dithionisophthaloyldihydrazide	4	2	1	2	4	0	4	2	0
N,N′-Dimethyltetramethylene terephthalamide	4	2	1	2	6	0	2	0	0
2-Methylpentamethylene terephthalamide	4	2	1	2	6	0	2	0	0
m-Phenylene (tetramethyldisiloxanylene)dipropionamide	4	2	1	4	14	1	2	0	2
o-Phenylene (tetramethyldisiloxanylene)dipropionamide	4	2	1	4	14	1	2	0	2
p-Phenylene (tetramethyldisiloxanylene)dipropionamide	4	2	1	4	14	1	2	0	2
p-Phenylene dipropylcyclobutylenedicarboxylate	4	2	1	6	10	2	0	0	0
p-Phenylene dicyanodipropylcyclobutylenedicarboxylate	4	2	1	8	8	2	2	0	0
Tetramethyl-*p*-xylylene dithiol-*p*-phenylenediacetate	4	2	1	10	12	0	0	2	0
2,2′-Dipropyl-4,4′-biphenylene terephthalate	4	2	1	14	12	2	0	0	0
p-Phenylene-4′,4″-(3′,3″-dimethoxy)dibenzylidenimino adipate	4	2	1	16	14	4	2	0	0
p-Xylylene dithiol-*p*-phenylenediacetate	4	2	2	0	0	0	0	2	0
4,4′-Tetramethylenedibenzoic anhydride	4	2	2	0	0	1	0	0	0
4,4′-(Tetramethylenedithio)dibenzoic anhydride	4	2	2	0	0	1	0	2	0
4,4′-Biphenylene adipate	4	2	2	0	0	2	0	0	0
Ethylene 4,4′-ethylenedibenzoate	4	2	2	0	0	2	0	0	0
p-Phenylenediethylene terephthalate	4	2	2	0	0	2	0	0	0
Tetramethylene 2,2′-dibenzoate	4	2	2	0	0	2	0	0	0
Tetramethylene 3,3′-dibenzoate	4	2	2	0	0	2	0	0	0
Tetramethylene 4,4′-dibenzoate	4	2	2	0	0	2	0	0	0
p-Xylylene *p*-phenylenediacetate	4	2	2	0	0	2	0	0	0
Ethylene 4,4′-(ethylenedithio)dibenzoate	4	2	2	0	0	2	0	2	0
Oxydiethylene 2,2′-dibenzoate	4	2	2	0	0	3	0	0	0
Oxydiethylene 3,3′-dibenzoate	4	2	2	0	0	3	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Oxydiethylene 4,4′-dibenzoate	4	2	2	0	0	3	0	0	0
3,3′-(Tetramethylenedioxy)dibenzoic anhydride	4	2	2	0	0	3	0	0	0
4,4′-(Tetramethylenedioxy)dibenzoic anhydride	4	2	2	0	0	3	0	0	0
Tetramethylene 4,4′-oxydibenzoate	4	2	2	0	0	3	0	0	0
Ethylene 3,3′-(ethylenedioxy)dibenzoate	4	2	2	0	0	4	0	0	0
Ethylene 4,4′-(ethylenedioxy)dibenzoate	4	2	2	0	0	4	0	0	0
(Oxydiethylenedioxy)dibenzoic anhydride	4	2	2	0	0	4	0	0	0
4,4′-Biphenylene adipamide	4	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene adipamide	4	2	2	0	2	1	2	0	0
4,4′-Oxydiphenylene adipamide	4	2	2	0	2	1	2	0	0
Ethylene 4,4′-(ethylenediphenylene)diurethan	4	2	2	0	2	2	2	0	0
Trimethylene 4,4′-(methylenediphenylene)diurethan	4	2	2	0	2	2	2	0	0
Tetramethylene 4,4′-sulfonyldibenzamide	4	2	2	0	2	2	2	1	0
Tetramethylene 4,4′-phosphinylidenedibenzamide	4	2	2	0	3	1	2	0	1
4,4′-Biphenylene disiloxanylenedipropionamide	4	2	2	0	6	1	2	0	2
Tetramethylene 4,4′-(methylphosphinylidene)dibenzamide	4	2	2	1	5	1	2	0	1
1,4-Cyclohexylene 3,3′-dibenzamide	4	2	2	2	4	0	2	0	0
Cyclopropylenedimethylene 4,4′-methylenediphenylenediurethane	4	2	2	2	4	2	2	0	0
4,4′-Isopropylidenediphenylene dithioladipate	4	2	2	3	6	0	0	2	0
4,4′-Isopropylidenediphenylene adipate	4	2	2	3	6	2	0	0	0
Piperazine 2,4-diphenyl-1,3-cyclobutylenedicarboxamide	4	2	2	4	6	0	2	0	0
4,4′-Biphenylene (tetramethyldisiloxanylene)dipropionamide	4	2	2	4	14	1	2	0	2
4,4′-Isopropylidenediphenylene dithiol-1,3-cyclohexylenedicarboxylate	4	2	2	5	8	0	0	2	0
Methyl *p*-phenylene 4,4′-tetramethylenedioxydibenzoate	4	2	2	7	6	4	0	0	0
4,4′-Biphenylene-4″,4‴-(dimethoxy)dibenzylidenimino adipate	4	2	2	16	14	4	2	0	0
4,4′-Oxydiphenylene-4″,4‴-(dimethoxy)dibenzylidenimino adipate	4	2	2	16	14	5	2	0	0
p-Phenylene 4,4′-tetramethylenedioxydibenzoate	4	2	3	0	0	4	0	0	0
Tetramethylene 4,4′-(phenylphosphinylidene)dibenzamide	4	2	3	0	3	1	2	0	1
p-Phenylene-4,4′-dibenzylidenimino adipate	4	2	3	2	2	2	2	0	0
4,4′-Biphenylene-4″,4‴-dibenzylidenimino adipate	4	2	4	2	2	2	2	0	0
4,4′-Oxydiphenylene-4″,4‴-dibenzylidenimino adipate	4	2	4	2	2	3	2	0	0
Alanylglycylproline	4	3	0	3	7	0	3	0	0
Alanylprolylglycine	4	3	0	3	7	0	3	0	0
α-D-Glucotripropionate, 1,4-	4	3	0	8	14	5	0	0	0
β-D-Glucotripropionate, 1,4-	4	3	0	8	14	5	0	0	0
Alanyl(ethylglutamyl)glycine	4	4	0	4	11	1	3	0	0
Tetramethylene 3,3′,4,4′-oxydiphenylenetetracarboximide	4	4	0	12	6	1	2	0	0
Trimethylene *N*,*N*′-malonyldi(*p*-aminobenzoate)	4	4	2	0	2	2	2	0	0
3,4′-Oxydiethylenedioxydiphenylene 3,3′,4,4′-oxydiphenylenetetra-carboximide	4	4	2	12	6	4	2	0	0
Isophthalic 4,4′-(tetramethylenedioxy)dibenzoic anhydride	4	4	3	0	0	4	0	0	0
Terephthalic 4,4′-(tetramethylenedioxy)dibenzoic anhydride	4	4	3	0	0	4	0	0	0
Tetramethylene (4,4′-terephthaloyldioxy)-dibenzoate	4	4	3	0	0	4	0	0	0
4,4′-(Isopropylidenediphenylene)dioxydiacetic *p*-phenylenedioxydiacetic an	4	4	3	3	6	4	0	0	0
Methylenediphenylenedisuccinimidediyliminomethylenediphenyleneamine	4	4	4	2	4	0	4	0	0
4,4′-(Oxydiphenylenedioxy)diphenylene carbonyldiphenylenetetra-carboximide	4	6	2	12	4	3	2	0	0
Alanylglycylalanylglycylserylglycine	4	6	0	5	16	1	6	0	0
2,2′-Adipamidodibenzoic isophthalic anhydride	4	6	3	0	2	2	2	0	0
4,4′-Adipamidodibenzoic isophthalic anhydride	4	6	3	0	2	2	2	0	0
2,2′-Adipamidodibenzoic terephthalic anhydride	4	6	3	0	2	2	2	0	0
4,4′-Adipamidodibenzoic terephthalic anhydride	4	6	3	0	2	2	2	0	0
5	5	0	0	0	0	0	0	0	0
Pentamethylene	5	0	0	0	0	0	0	0	0
Pentamethylene sulfide	5	0	0	0	0	0	0	1	0
Methylenethiotetramethylene sulfide	5	0	0	0	0	0	0	2	0
Pentamethylene disulfide	5	0	0	0	0	0	0	2	0
Methyleneoxytetramethylene oxide	5	0	0	0	0	2	0	0	0
Pentamethylene sulfone	5	0	0	0	0	2	0	1	0
6-Aminothiocaproic acid	5	0	0	1	1	0	1	1	0
2,5-Pentamethylene-1,3,4-triazole	5	0	0	2	1	0	3	0	0
1,2-Cyclopentylene-*alt*-ethylene	5	0	0	2	2	0	0	0	0
Heptenamer	5	0	0	2	2	0	0	0	0
Vinyl cyclopentane	5	0	0	2	2	0	0	0	0
1,2-Dichloroheptamethylene	5	0	0	2	2	0	0	0	2

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Oxydiethylene 4,4′-dibenzoate	4	2	2	0	0	3	0	0	0
3,3′-(Tetramethylenedioxy)dibenzoic anhydride	4	2	2	0	0	3	0	0	0
4,4′-(Tetramethylenedioxy)dibenzoic anhydride	4	2	2	0	0	3	0	0	0
Tetramethylene 4,4′-oxydibenzoate	4	2	2	0	0	3	0	0	0
Ethylene 3,3′-(ethylenedioxy)dibenzoate	4	2	2	0	0	4	0	0	0
Ethylene 4,4′-(ethylenedioxy)dibenzoate	4	2	2	0	0	4	0	0	0
(Oxydiethylenedioxy)dibenzoic anhydride	4	2	2	0	0	4	0	0	0
4,4′-Biphenylene adipamide	4	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene adipamide	4	2	2	0	2	1	2	0	0
4,4′-Oxydiphenylene adipamide	4	2	2	0	2	1	2	0	0
Ethylene 4,4′-(ethylenediphenylene)diurethan	4	2	2	0	2	2	2	0	0
Trimethylene 4,4′-(methylenediphenylene)diurethan	4	2	2	0	2	2	2	0	0
Tetramethylene 4,4′-sulfonyldibenzamide	4	2	2	0	2	2	2	1	0
Tetramethylene 4,4′-phosphinylidenedibenzamide	4	2	2	0	3	1	2	0	1
4,4′-Biphenylene disiloxanylenedipropionamide	4	2	2	0	6	1	2	0	2
Tetramethylene 4,4′-(methylphosphinylidene)dibenzamide	4	2	2	1	5	1	2	0	1
1,4-Cyclohexylene 3,3′-dibenzamide	4	2	2	2	4	0	2	0	0
Cyclopropylenedimethylene 4,4′-methylenediphenylenediurethane	4	2	2	2	4	2	2	0	0
4,4′-Isopropylidenediphenylene dithioladipate	4	2	2	3	6	0	0	2	0
4,4′-Isopropylidenediphenylene adipate	4	2	2	3	6	2	0	0	0
Piperazine 2,4-diphenyl-1,3-cyclobutylenedicarboxamide	4	2	2	4	6	0	2	0	0
4,4′-Biphenylene (tetramethyldisiloxanylene)dipropionamide	4	2	2	4	14	1	2	0	2
4,4′-Isopropylidenediphenylene dithiol-1,3-cyclohexylenedicarboxylate	4	2	2	5	8	0	0	2	0
Methyl *p*-phenylene 4,4′-tetramethylenedioxydibenzoate	4	2	2	7	6	4	0	0	0
4,4′-Biphenylene-4″,4‴-(dimethoxy)dibenzylidenimino adipate	4	2	2	16	14	4	2	0	0
4,4′-Oxydiphenylene-4″,4‴-(dimethoxy)dibenzylidenimino adipate	4	2	2	16	14	5	2	0	0
p-Phenylene 4,4′-tetramethylenedioxydibenzoate	4	2	3	0	0	4	0	0	0
Tetramethylene 4,4′-(phenylphosphinylidene)dibenzamide	4	2	3	0	3	1	2	0	1
p-Phenylene-4,4′-dibenzylidenimino adipate	4	2	3	2	2	2	2	0	0
4,4′-Biphenylene-4″,4‴-dibenzylidenimino adipate	4	2	4	2	2	2	2	0	0
4,4′-Oxydiphenylene-4″,4‴-dibenzylidenimino adipate	4	2	4	2	2	3	2	0	0
Alanylglycylproline	4	3	0	3	7	0	3	0	0
Alanylprolylglycine	4	3	0	3	7	0	3	0	0
α-D-Glucotripropionate, 1,4-	4	3	0	8	14	5	0	0	0
β-D-Glucotripropionate, 1,4-	4	3	0	8	14	5	0	0	0
Alanyl(ethylglutamyl)glycine	4	4	0	4	11	1	3	0	0
Tetramethylene 3,3′,4,4′-oxydiphenylenetetracarboximide	4	4	0	12	6	1	2	0	0
Trimethylene *N*,*N*′-malonyldi(*p*-aminobenzoate)	4	4	2	0	2	2	2	0	0
3,4′-Oxydiethylenedioxydiphenylene 3,3′,4,4′-oxydiphenylenetetra-carboximide	4	4	2	12	6	4	2	0	0
Isophthalic 4,4′-(tetramethylenedioxy)dibenzoic anhydride	4	4	3	0	0	4	0	0	0
Terephthalic 4,4′-(tetramethylenedioxy)dibenzoic anhydride	4	4	3	0	0	4	0	0	0
Tetramethylene (4,4′-terephthaloyldioxy)-dibenzoate	4	4	3	0	0	4	0	0	0
4,4′-(Isopropylidenediphenylene)dioxydiacetic *p*-phenylenedioxydiacetic an	4	4	3	3	6	4	0	0	0
Methylenediphenylenedisuccinimidediyliminomethylenediphenyleneamine	4	4	4	2	4	0	4	0	0
4,4′-(Oxydiphenylenedioxy)diphenylene carbonyldiphenylenetetra-carboximide	4	6	2	12	4	3	2	0	0
Alanylglycylalanylglycylserylglycine	4	6	0	5	16	1	6	0	0
2,2′-Adipamidodibenzoic isophthalic anhydride	4	6	3	0	2	2	2	0	0
4,4′-Adipamidodibenzoic isophthalic anhydride	4	6	3	0	2	2	2	0	0
2,2′-Adipamidodibenzoic terephthalic anhydride	4	6	3	0	2	2	2	0	0
4,4′-Adipamidodibenzoic terephthalic anhydride	4	6	3	0	2	2	2	0	0
5	5	0	0	0	0	0	0	0	0
Pentamethylene	5	0	0	0	0	0	0	0	0
Pentamethylene sulfide	5	0	0	0	0	0	0	1	0
Methylenethiotetramethylene sulfide	5	0	0	0	0	0	0	2	0
Pentamethylene disulfide	5	0	0	0	0	0	0	2	0
Methyleneoxytetramethylene oxide	5	0	0	0	0	2	0	0	0
Pentamethylene sulfone	5	0	0	0	0	2	0	1	0
6-Aminothiocaproic acid	5	0	0	1	1	0	1	1	0
2,5-Pentamethylene-1,3,4-triazole	5	0	0	2	1	0	3	0	0
1,2-Cyclopentylene-*alt*-ethylene	5	0	0	2	2	0	0	0	0
Heptenamer	5	0	0	2	2	0	0	0	0
Vinyl cyclopentane	5	0	0	2	2	0	0	0	0
1,2-Dichloroheptamethylene	5	0	0	2	2	0	0	0	2

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
2,5-Pentamethylene-1-amino-1,3,4-triazole	5	0	0	2	2	0	4	0	0
1,4-Cyclohexyleneoxymethylene oxide	5	0	0	2	2	2	0	0	0
1-Heptene	5	0	0	2	4	0	0	0	0
n-Heptaldehyde	5	0	0	2	4	1	0	0	0
Vinyl cyclohexene	5	0	0	3	2	0	0	0	0
3,4-Dihydrocyclohexylidenedimethylene oxide	5	0	0	3	2	1	0	0	0
1,8-Nonadiyne, 1,9-	5	0	0	4	0	0	0	0	0
3,4-Dihydro-4-methylcyclohexylidenedimethylene oxide	5	0	0	4	4	1	0	0	0
3-Methylvinyl cyclohexane	5	0	0	4	6	0	0	0	0
4-Methylvinyl cyclohexane	5	0	0	4	6	0	0	0	0
2-Methoxyvinyl cyclohexane	5	0	0	4	6	1	0	0	0
3-Hexyl-2,5-thiophene	5	0	0	5	4	0	0	1	0
Di-1,4-naphthaleneoxypentamethylene oxide	5	0	0	20	12	2	0	0	0
Pentamethyleneoxy-*p*-phenylene oxide	5	0	1	0	0	2	0	0	0
Pentamethylene *m*-phenylenedisulfonamide	5	0	1	0	2	4	2	2	0
4,4′-(Methylenediphenylene)diethylene sulfide	5	0	2	0	0	0	0	1	0
4,4′-Stilbenediyloxypentamethylene oxide	5	0	2	2	2	2	0	0	0
α-methyl-4,4′-stilbenediyloxypentamethylene oxide	5	0	2	3	4	2	0	0	0
6-Mercaptocaproic acid	5	1	0	0	0	0	0	1	0
6-Hydroxycaproic acid	5	1	0	0	0	1	0	0	0
Pentamethylene carbonate	5	1	0	0	0	2	0	0	0
6-Aminocaproic acid (caprolactam)	5	1	0	0	1	0	1	0	0
Pentamethylene urethane	5	1	0	0	1	1	1	0	0
N-Valerylethylene amine	5	1	0	1	3	0	1	0	0
4-Aminocyclohexyleneacetic acid	5	1	0	2	3	0	1	0	0
4-Methyl-7-hydroxyenanthic acid	5	1	0	2	4	1	0	0	0
3-Methyl-7-aminoenanthic acid	5	1	0	2	5	0	1	0	0
4-Methyl-7-aminoenanthic acid	5	1	0	2	5	0	1	0	0
5-Methyl-7-aminoenanthic acid	5	1	0	2	5	0	1	0	0
6-Methyl-7-aminoenanthic acid	5	1	0	2	5	0	1	0	0
3-*n*-Hexylhydroxybenzoic acid	5	1	0	7	6	1	0	0	0
4,4′-Cyclohexylidene-2,2′-dimethyldiphenylene carbonate	5	1	0	15	12	2	0	0	0
p-(Aminoethylene)phenylenebutyric acid	5	1	1	0	1	0	1	0	0
p-(Aminomethylene)phenylenevaleric acid	5	1	1	0	1	0	1	0	0
N-4-(Methylthiophenoxy)butyrylethylene amine	5	1	1	1	3	1	1	1	0
4,4′-Cyclohexylidenediphenylene carbonate	5	1	2	1	0	2	0	0	0
Pimelic anhydride	5	2	0	0	0	1	0	0	0
Ethylene glutarate	5	2	0	0	0	2	0	0	0
Pentamethylene oxalate	5	2	0	0	0	2	0	0	0
Tetramethylene malonate	5	2	0	0	0	2	0	0	0
Trimethylene succinate	5	2	0	0	0	2	0	0	0
Oxydiethylene malonate	5	2	0	0	0	3	0	0	0
Trimethylene oxydiacetate	5	2	0	0	0	3	0	0	0
Hydrazo pimelamide	5	2	0	0	2	0	2	0	0
Methylene adipamide	5	2	0	0	2	0	2	0	0
Tetramethylene malonamide	5	2	0	0	2	0	2	0	0
(2,6-Dioxo-1,4-piperidinediyl)trimethylene	5	2	0	1	1	0	1	0	0
*N*5-(3-Hydroxypropyl)glutamine	5	2	0	1	4	1	2	0	0
Piperazine cyclopropylenedicarboxamide	5	2	0	2	2	0	2	0	0
Cyclopropylene piperazinediurea	5	2	0	2	4	0	4	0	0
Methylethylene adipate	5	2	0	2	4	2	0	0	0
Tetramethylene cyclopropylenediurethane	5	2	0	2	4	2	2	0	0
n-Butylglutamate	5	2	0	2	5	1	1	0	0
Alanyl-6-Aminocaproic acid	5	2	0	2	6	0	2	0	0
Piperazine (1-methylcyclopropylene)dicarboxamide	5	2	0	3	4	0	2	0	0
Ethylene 3,3′-dimethoxy-4,4′-(trimethylenedioxy)dibenzoate	5	2	0	14	12	6	0	0	0
3,3′-Dimethyl-4,4′-methylenediphenylene adipamide	5	2	0	14	14	0	2	0	0
N,N′-Diisopropyl-3,3′-dimethyl-methylenediphenylene adipamide	5	2	0	20	26	0	2	0	0
Pentamethylene dithiolisophthalate	5	2	1	0	0	0	0	2	0
Pentamethylene dithiolterephthalate	5	2	1	0	0	0	0	2	0
Pentamethylene isophthalate	5	2	1	0	0	2	0	0	0
Pentamethylene phthalate	5	2	1	0	0	2	0	0	0
Pentamethylene terephthalate	5	2	1	0	0	2	0	0	0
Trimethylene *p*-phenylenediacetate	5	2	1	0	0	2	0	0	0
Ethylene *p*-(carboxyphenoxy)butyrate	5	2	1	0	0	3	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Pentamethylene *m*-carboxycarbanilate	5	2	1	0	1	2	1	0	0
Pentamethylene *p*-carboxycarbanilate	5	2	1	0	1	2	1	0	0
Pentamethylene terephthalamide	5	2	1	0	2	0	2	0	0
p-Phenylene pimelamide	5	2	1	0	2	0	2	0	0
Tetramethylene *p*-(carboxyphenylene)acetamide	5	2	1	0	2	0	2	0	0
p-Xylylene glutaramide	5	2	1	0	2	0	2	0	0
3-Phenylpropylglutamate	5	2	1	1	3	1	1	0	0
N-Benzyloxycarbonyllysine	5	2	1	1	4	1	2	0	0
p-Phenylene(1,1-dimethyl)ethylene adipamide	5	2	1	3	8	0	2	0	0
N,N'-Dipropylmethylene-4,4'-dimethyldiphenylene terephthalamide	5	2	1	16	18	0	2	0	0
4,4'-Pentamethylenedibenzoic anhydride	5	2	2	0	0	1	0	0	0
4,4'-Methylenediphenylene adipate	5	2	2	0	0	2	0	0	0
Pentamethylene 2,2'-dibenzoate	5	2	2	0	0	2	0	0	0
Pentamethylene 3,3'-dibenzoate	5	2	2	0	0	2	0	0	0
Pentamethylene 4,4'-dibenzoate	5	2	2	0	0	2	0	0	0
Oxydiethylene 4,4'-methylenedibenzoate	5	2	2	0	0	3	0	0	0
3,3'-(Pentamethylenedioxy)dibenzoic anhydride	5	2	2	0	0	3	0	0	0
4,4'-(Pentamethylenedioxy)dibenzoic anhydride	5	2	2	0	0	3	0	0	0
Pentamethylene 4,4'-oxydibenzoate	5	2	2	0	0	3	0	0	0
Ethylene 4,4'-(trimethylenedioxy)dibenzoate	5	2	2	0	0	4	0	0	0
4,4'-Methylenediphenylene adipamide	5	2	2	0	2	0	2	0	0
3,4'-Oxydiphenylene pimelamide	5	2	2	0	2	1	2	0	0
4,4'-Oxydiphenylene pimelamide	5	2	2	0	2	1	2	0	0
Ethylene 4,4'-(trimethylenediphenylene)diurethane	5	2	2	0	2	2	2	0	0
Tetramethylene 4,4'-(methylenediphenylene)diurethane	5	2	2	0	2	2	2	0	0
Tetramethylene methylenediphenylenediaminodithiodicarboxylate	5	2	2	0	2	2	2	2	0
4,4'-Methylenediphenylene disiloxanylenedipropionamide	5	2	2	0	6	1	2	0	2
1,4-Cyclohexylene 3,3'-methylenedibenzamide	5	2	2	2	4	0	2	0	0
N,N'-Dimethyl-4,4'-methylenediphenylene adipamide	5	2	2	2	6	0	2	0	0
Tetramethylene 4,4'-propylenedibenzamide	5	2	2	2	6	0	2	0	0
Tetramethylene 4,4'-(propylenedioxy)dibenzamide	5	2	2	2	6	2	2	0	0
2,2-Diethyltrimethylene 4,4'-(methylenediphenylene)diurethane	5	2	2	3	8	2	2	0	0
1-Ethyl-2-propyltrimethylene (methylenediphenylene)diurethane	5	2	2	4	10	2	2	0	0
4,4'-Methylenediphenylene tetramethyldisiloxanylenedipropionamide	5	2	2	4	14	1	2	0	2
N,N'-Dipropyl-4,4'-methylenediphenylene terephthalamide	5	2	3	2	6	0	2	0	0
Glycyl-3-aminopropionic acid-3-aminopropionic acid	5	3	0	0	3	0	3	0	0
Glycylglycylproline	5	3	0	1	3	0	3	0	0
N,N'-Lysyl propylenediurea	5	3	0	3	10	1	4	0	0
Methyl-*N,N'*-lysyl propylenediurea	5	3	0	4	12	1	4	0	0
Tetramethylene (cyclopropylenedicarboxoyl)diurethane	5	4	0	2	4	2	2	0	0
Ethylene *N,N'*-trimethyleneditерephthalamate	5	4	2	0	2	2	2	0	0
Isophthaloyl (*m*-carboxyphenoxycapryl)dihydrazide	5	4	2	0	4	1	4	0	0
(*p*-Phenylenedioxy)diacetic trimethylenedioxydibenzoic anhydride	5	4	3	0	0	6	0	0	0
6	6	0	0	0	0	0	0	0	0
Hexamethylene sulfide	6	0	0	0	0	0	0	1	0
Ethylenethiotetramethylene sulfide	6	0	0	0	0	0	0	2	0
Hexamethylene disulfide	6	0	0	0	0	0	0	2	0
(Ethylenedithio)tetramethylene disulfide	6	0	0	0	0	0	0	4	0
hexamethylene oxide	6	0	0	0	0	1	0	0	0
Methyleneoxypentamethylene oxide	6	0	0	0	0	2	0	0	0
Hexamethylene sulfone	6	0	0	0	0	2	0	1	0
Hexamethylene amine	6	0	0	0	1	0	1	0	0
7-Aminothioenanthic acid	6	0	0	1	1	0	1	1	0
Hexamethylene thiourea	6	0	0	1	2	0	2	1	0
2,5-Hexamethylene-1,3,4-triazole	6	0	0	2	1	0	3	0	0
3-Cyclopentylpropene	6	0	0	2	2	0	0	0	0
Octenamer	6	0	0	2	2	0	0	0	0
Vinyl cyclohexane	6	0	0	2	2	0	0	0	0
2,5-Hexamethylene-1-amino-1,3,4-triazole	6	0	0	2	2	0	4	0	0
1-Octene	6	0	0	2	4	0	0	0	0
Caprylaldehyde	6	0	0	2	4	1	0	0	0
Di-*n*-butylsilylene	6	0	0	2	6	0	0	0	1
n-Propyl-*n*-pentylsilylene	6	0	0	2	6	0	0	0	1
4-Methylcyclohexylidenedimethylene oxide	6	0	0	3	4	1	0	0	0
3,3-Bisethoxymethyloxacyclobutane	6	0	0	3	6	3	0	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
5-Diisobutylamino-1-pentene	6	0	0	7	15	0	1	0	0
Di-1,4-naphthaleneoxyhexamethylene oxide	6	0	0	20	12	2	0	0	0
Hexamethyleneoxy-*p*-phenylene oxide	6	0	1	0	0	2	0	0	0
Hexamethylene *m*-phenylenedisulfonamide	6	0	1	0	2	4	2	2	0
Benzylidenethiohexamethylene sulfide	6	0	1	1	2	0	0	2	0
Hexamethylene dithioterephthalamide	6	0	1	2	2	0	2	2	0
p-Methoxybenzylidenethiohexamethylene sulfide	6	0	1	2	4	1	0	2	0
4,4′-Stilbenediyloxyhexamethylene oxide	6	0	2	2	2	2	0	0	0
α-Methyl-4,4′-stilbenediyloxyhexamethylene oxide	6	0	2	3	4	2	0	0	0
7-Hydroxyenanthic acid	6	1	0	0	0	1	0	0	0
Hexamethylene carbonate	6	1	0	0	0	2	0	0	0
7-Aminoenanthic acid	6	1	0	0	1	0	1	0	0
Hexamethylene urea	6	1	0	0	2	0	2	0	0
N-Hexanoylethylene amine	6	1	0	1	3	0	1	0	0
N-Methyl-7-aminoenanthic acid	6	1	0	1	3	0	1	0	0
Vinylcyclohexyl ketone	6	1	0	2	2	0	0	0	0
(3-Aminomethylene)cyclohexyleneacetic acid	6	1	0	2	3	0	1	0	0
(1,3-Dimethyl-3-aminomethylene)cyclohexyleneacetic acid	6	1	0	4	7	0	1	0	0
p-(Aminoethylene)phenylenevaleric acid	6	1	1	0	1	0	1	0	0
4,4′-Nonylidenediphenylene carbonate	6	1	2	3	6	2	0	0	0
Ethylene dithioladipate	6	2	0	0	0	0	0	2	0
Suberic anhydride	6	2	0	0	0	1	0	0	0
(Ethylenedithio)dipropionic anhydride	6	2	0	0	0	1	0	2	0
Piperazine oxydiacetamide	6	2	0	0	0	1	2	0	0
Ethylene adipate	6	2	0	0	0	2	0	0	0
Hexamethylene oxalate	6	2	0	0	0	2	0	0	0
Tetramethylene succinate	6	2	0	0	0	2	0	0	0
Trimethylene glutarate	6	2	0	0	0	2	0	0	0
Barium suberate	6	2	0	0	0	2	0	0	1
Cadmium suberate	6	2	0	0	0	2	0	0	1
Calcium suberate	6	2	0	0	0	2	0	0	1
Lead suberate	6	2	0	0	0	2	0	0	1
Magnesium suberate	6	2	0	0	0	2	0	0	1
Manganese suberate	6	2	0	0	0	2	0	0	1
Stannous suberate	6	2	0	0	0	2	0	0	1
Zinc suberate	6	2	0	0	0	2	0	0	1
Hexamethylene dithiodicarboxylate	6	2	0	0	0	2	0	2	0
Ethylene 1,4-piperazinedicarboxylate	6	2	0	0	0	2	2	0	0
Oxydiethylene succinate	6	2	0	0	0	3	0	0	0
Tetramethylene oxydiacetate	6	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene oxalate	6	2	0	0	0	4	0	0	0
Oxydiethylene oxydiacetate	6	2	0	0	0	4	0	0	0
(Ethylenedisulfonyl)dipropionic anhydride	6	2	0	0	0	5	0	2	0
Ethylene adipamide	6	2	0	0	2	0	2	0	0
Glycyl-6-aminocaproic acid	6	2	0	0	2	0	2	0	0
Hexamethylene oxamide	6	2	0	0	2	0	2	0	0
Hydrazo suberamide	6	2	0	0	2	0	2	0	0
Methylene pimelamide	6	2	0	0	2	0	2	0	0
Pentamethylene malonamide	6	2	0	0	2	0	2	0	0
Tetramethylene succinamide	6	2	0	0	2	0	2	0	0
Ethylene (ethylenedioxy)diacetamide	6	2	0	0	2	2	2	0	0
Aminoditrimethylene oxamide	6	2	0	0	3	0	3	0	0
Ethylene disiloxanylenedipropionamide	6	2	0	0	6	1	2	0	2
2,2-Bischloromethyltrimethylene succinate	6	2	0	1	0	2	0	0	2
*N*5-(4-Hydroxybutyl)glutamine	6	2	0	1	4	1	2	0	0
(Methylamino)ditrimethylene oxamide	6	2	0	1	5	0	3	0	0
1,4-Cyclohexylene dithiolsuccinate	6	2	0	2	2	0	0	2	0
Piperazine 1,3-cyclobutylenedicarboxamide	6	2	0	2	2	0	2	0	0
1,4-Cyclohexylenedimethylene oxalate	6	2	0	2	2	2	0	0	0
Ethylene 1,4-cyclohexylenedicarboxylate	6	2	0	2	2	2	0	0	0
Hexamethylene fumarate	6	2	0	2	2	2	0	0	0
2,5-Tetrahydrofurandipropionic anhydride	6	2	0	2	2	2	0	0	0
2,5-Tetrahydropyrroledipropionic anhydride	6	2	0	2	3	1	1	0	0
Methylethylene pimelate	6	2	0	2	4	2	0	0	0
Pentamethylene cyclopropylenediurethane	6	2	0	2	4	2	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
n-Amylglutamate	6	2	0	2	5	1	1	0	0
N,N'-Dimethyltetramethylene succinamide	6	2	0	2	6	0	2	0	0
Hexamethylene L-tartaramide	6	2	0	2	6	2	2	0	0
Hexafluoropentamethylene adipate	6	2	0	3	0	2	0	0	6
Hexafluoropentamethylene adipamide	6	2	0	3	2	0	2	0	6
Hexamethylene perfluoroglutaramide	6	2	0	3	2	0	2	0	6
N-Methyl-2,5-tetrahydropyrroledipropionic anhydride	6	2	0	3	5	1	1	0	0
2,2-Dimethyltrimethylene adipate	6	2	0	3	6	2	0	0	0
Isopropylidenedimethylene piperazinedithiodicarboxylate	6	2	0	3	6	2	2	2	0
Hexamethylene 2,2-dimethylmalonamide	6	2	0	3	8	0	2	0	0
2,4-Hexadiyne adipate	6	2	0	4	0	2	0	0	0
Hexamethylene 2,5-thiophenedicarboxamide	6	2	0	4	4	0	2	1	0
Hexamethylene 2,5-furandicarboxamide	6	2	0	4	4	1	2	0	0
2-Butene 4-octenediamide	6	2	0	4	6	0	2	0	0
Hexamethylene (3-methylcyclopropylene)dicarboxamide	6	2	0	4	8	0	2	0	0
Hexamethylene di-*O*-methyl-L-tartaramide	6	2	0	4	10	2	2	0	0
Tetramethylene (2,5-dimethylpiperazine)dithiodicarboxylate	6	2	0	4	10	2	2	2	0
Ethylene (Tetramethyldisiloxanylene)dipropionamide	6	2	0	4	14	1	2	0	2
Hexamethylene 2,3-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
Hexamethylene 2,4-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
Hexamethylene 2,5-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
Hexamethylene 2,6-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
Hexamethylene 3,4-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
Hexamethylene 3,5-pyridinedicarboxamide	6	2	0	5	5	0	3	0	0
N,N'-Dimethyl(hexafluoropentamethylene) adipamide	6	2	0	5	6	0	2	0	6
N,N'-Dimethylhexamethylene perfluoroglutaramide	6	2	0	5	6	0	2	0	6
Hexamethylene 2,5-dihydroxyterephthalamide	6	2	0	6	6	2	2	0	0
Hexamethylene methylterephthalamide	6	2	0	7	8	0	2	0	0
Hexamethylene 2,6-dimethyl-3,5-pyridinedicarboxamide	6	2	0	7	9	0	3	0	0
Hexamethylene 4-methyl-1,3-phenylenediurea	6	2	0	7	10	0	4	0	0
Hexamethylene 2,5-dimethylterephthalamide	6	2	0	8	10	0	2	0	0
N,N'-Diisopropyl(hexafluoropentamethylene) adipamide	6	2	0	9	14	0	2	0	6
N,N'-Diisopropylhexamethylene perfluoroglutaramide	6	2	0	9	14	0	2	0	6
Tetramethyl-*p*-xylylene dithioladipate	6	2	0	10	12	0	0	2	0
Hexamethylene 5-tert-butylisophthalamide	6	2	0	10	14	0	2	0	0
4-Methylphenylene diethyldimethyldisiloxanylenedipropionamide	6	2	0	11	20	1	2	0	2
1,4-Cyclohexylenedimethylene 2,6-naphthalate	6	2	0	12	8	2	0	0	0
Ethylene 3,3'-dimethoxy-4,4'-(tetramethylenedioxy)dibenzoate	6	2	0	14	12	6	0	0	0
Ethylene 3,3'-dimethoxy-4,4'-(oxydiethylene)dioxydibenzoate	6	2	0	14	12	7	0	0	0
Hexamethylene dithiolisophthalate	6	2	1	0	0	0	0	2	0
Hexamethylene dithiolterephthalate	6	2	1	0	0	0	0	2	0
p-Xylylene dithioladipate	6	2	1	0	0	0	0	2	0
(*p*-Xylylenedithio)dipropionic anhydride	6	2	1	0	0	1	0	2	0
Hexamethylene isophthalate	6	2	1	0	0	2	0	0	0
Hexamethylene phthalate	6	2	1	0	0	2	0	0	0
Hexamethylene terephthalate	6	2	1	0	0	2	0	0	0
Tetramethylene *p*-phenylenediacetate	6	2	1	0	0	2	0	0	0
p-Xylylene adipate	6	2	1	0	0	2	0	0	0
Ethylene *p*-(carboxyphenoxy)valerate	6	2	1	0	0	3	0	0	0
(Ethylenedioxy)diethylene isophthalate	6	2	1	0	0	4	0	0	0
(Ethylenedioxy)diethylene terephthalate	6	2	1	0	0	4	0	0	0
(*p*-Xylylenedisulfonyl)dipropionic anhydride	6	2	1	0	0	5	0	2	0
Hexamethylene *m*-carboxycarbanilate	6	2	1	0	1	2	1	0	0
Hexamethylene isophthalamide	6	2	1	0	2	0	2	0	0
Hexamethylene phthalamide	6	2	1	0	2	0	2	0	0
Hexamethylene terephthalamide	6	2	1	0	2	0	2	0	0
Pentamethylene *p*-(carboxyphenylene)acetamide	6	2	1	0	2	0	2	0	0
p-Phenyleneethylene adipamide	6	2	1	0	2	0	2	0	0
m-Phenylene suberamide	6	2	1	0	2	0	2	0	0
o-Phenylene suberamide	6	2	1	0	2	0	2	0	0
p-Phenylene suberamide	6	2	1	0	2	0	2	0	0
m-Xylylene adipamide	6	2	1	0	2	0	2	0	0
p-Xylylene adipamide	6	2	1	0	2	0	2	0	0
Oxyditrimethylene terephthalamide	6	2	1	0	2	1	2	0	0
m-Phenylene hexamethylenediurethane	6	2	1	0	2	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
p-Phenylene hexamethylenediurethane	6	2	1	0	2	2	2	0	0
Tetramethylene *p*-xylylenediurethane	6	2	1	0	2	2	2	0	0
Hexamethylene *m*-phenylenediurea	6	2	1	0	4	0	4	0	0
1,4-Cyclohexylenedimethylene isophthalate	6	2	1	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene terephthalate	6	2	1	2	2	2	0	0	0
p-Xylylene 1,4-cyclohexylenedicarboxylate	6	2	1	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene isophthalamide	6	2	1	2	4	0	2	0	0
N,N'-Dimethylhexamethylene terephthalamide	6	2	1	2	6	0	2	0	0
Hexamethylene naphthalenedicarboxamide	6	2	1	4	6	0	2	0	0
o-Phenylene (diethyldimethyldisiloxanylene)dipropionamide	6	2	1	4	14	1	2	0	2
p-Phenylene (diethyldimethyldisiloxanylene)dipropionamide	6	2	1	4	14	1	2	0	2
p-Phenylene dibutylcyclobutylenedicarboxylate	6	2	1	6	10	2	0	0	0
p-Phenylene dicyanodibutylcyclobutylenedicarboxylate	6	2	1	8	8	2	2	0	0
4,4'-Hexamethylenedibenzoic anhydride	6	2	2	0	0	1	0	0	0
Ethylene 4,4'-tetramethylenedibenzoate	6	2	2	0	0	2	0	0	0
Hexamethylene 2,2'-dibenzoate	6	2	2	0	0	2	0	0	0
Hexamethylene 3,3'-dibenzoate	6	2	2	0	0	2	0	0	0
Hexamethylene 4,4'-dibenzoate	6	2	2	0	0	2	0	0	0
p-Phenyleneditrimethylene terephthalate	6	2	2	0	0	2	0	0	0
Hexamethylene dithiol-4,4'-sulfonyldibenzoate	6	2	2	0	0	2	0	3	0
3,3'-(Hexamethylenedioxy)dibenzoic anhydride	6	2	2	0	0	3	0	0	0
4,4'-(Hexamethylenedioxy)dibenzoic anhydride	6	2	2	0	0	3	0	0	0
Hexamethylene 4,4'-oxydibenzoate	6	2	2	0	0	3	0	0	0
Oxydiethylene 4,4'-ethylenedibenzoate	6	2	2	0	0	3	0	0	0
(Ethylenedioxy)diethylene 2,2'-dibenzoate	6	2	2	0	0	4	0	0	0
(Ethylenedioxy)diethylene 3,3'-dibenzoate	6	2	2	0	0	4	0	0	0
(Ethylenedioxy)diethylene 4,4'-dibenzoate	6	2	2	0	0	4	0	0	0
Ethylene 4,4'-(ethylenedioxydimethylene)dibenzoate	6	2	2	0	0	4	0	0	0
Ethylene 3,3'-(tetramethylenedioxy)dibenzoate	6	2	2	0	0	4	0	0	0
Ethylene 4,4'-(tetramethylenedioxy)dibenzoate	6	2	2	0	0	4	0	0	0
Tetramethylene 4,4'-(ethylenedioxy)dibenzoate	6	2	2	0	0	4	0	0	0
Hexamethylene 4,4'-sulfonyldibenzoate	6	2	2	0	0	4	0	1	0
Ethylene 4,4'-(oxydiethylene)dioxydibenzoate	6	2	2	0	0	5	0	0	0
4,4'-Ethylenediphenylene adipamide	6	2	2	0	2	0	2	0	0
Hexamethylene 3,3'-dibenzamide	6	2	2	0	2	0	2	0	0
Hexamethylene 3,4'-dibenzamide	6	2	2	0	2	0	2	0	0
Hexamethylene 4,4'-dibenzamide	6	2	2	0	2	0	2	0	0
Hexamethylene 2,2'-dibenzamide	6	2	2	0	2	0	2	0	0
p-Phenylenediethylene *m*-phenylenediacetamide	6	2	2	0	2	0	2	0	0
3,4'-Oxydiphenylene suberamide	6	2	2	0	2	1	2	0	0
4,4'-Oxydiphenylene suberamide	6	2	2	0	2	1	2	0	0
Ethylene 4,4'-(tetramethylenediphenylene)diurethane	6	2	2	0	2	2	2	0	0
Pentamethylene 4,4'-(methylenediphenylene)diurethane	6	2	2	0	2	2	2	0	0
Hexamethylene 4,4'-sulfonyldibenzamide	6	2	2	0	2	2	2	1	0
Hexamethylene 3,3'-phosphinylidenedibenzamide	6	2	2	0	3	1	2	0	1
Hexamethylene 4,4'-phosphinylidenedibenzamide	6	2	2	0	3	1	2	0	1
Hexamethylene 4,4'-(hydroxyphosphinylidene)dibenzamide	6	2	2	0	3	2	2	0	1
Hexamethylene 4,4'-silylenedibenzamide	6	2	2	0	4	0	2	0	1
4,4'-Dithiodiphenylene hexamethylenediurea	6	2	2	0	4	0	4	2	0
Hexamethylene 4,4'-disiloxanylenedibenzamide	6	2	2	0	6	1	2	0	2
Hexamethylene 4,4'-tetrasiloxanylenedibenzamide	6	2	2	0	10	3	2	0	4
Hexamethylene 3,3'-(methylphosphinylidene)dibenzamide	6	2	2	1	5	1	2	0	1
Hexamethylene 4,4'-(methylphosphinylidene)dibenzamide	6	2	2	1	5	1	2	0	1
Hexamethylene 4,4'-vinylenedibenzamide	6	2	2	2	4	0	2	0	0
2-Ethylhexamethylene dithiol-4,4'-sulfonyldibenzoate	6	2	2	2	4	2	0	3	0
Hexamethylene 4,4'-ethylidenedibenzamide	6	2	2	2	6	0	2	0	0
Hexamethylene 4,4'-(dimethylsilylene)dibenzamide	6	2	2	2	8	0	2	0	1
Hexamethylene 4,4'-(methylvinylene)dibenzamide	6	2	2	3	6	0	2	0	0
Hexamethylene 4,4'-isopropylidenedibenzamide	6	2	2	3	8	0	2	0	0
4,4'-Isopropylidenediphenylene hexamethylenediurethane	6	2	2	3	8	2	2	0	0
4,4'-Dimethylheptamethylene sulfonyldibenzamide	6	2	2	3	8	2	2	1	0
Hexamethylene 4,4'-(tetramethyldisiloxanylene)dibenzamide	6	2	2	4	14	1	2	0	2
Hexamethylene 4,4'-(phenylphosphinylidene)dibenzamide	6	2	3	0	2	1	2	0	1
Hexamethylene tetramethyltetraphenyltetrasiloxanylenedibenzamide	6	2	6	4	14	3	2	0	4
Ethylene *N,N'*-lysyldiurea	6	3	0	1	6	1	4	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Ethylene methyl-*N*,*N*′-lysyldiurea	6	3	0	2	8	1	4	0	0
Glycylprolylhydroxyproline	6	3	0	3	5	1	3	0	0
Hexamethylene 4-oxo-4(H)-pyran-2,6-dicarboxamide	6	3	0	4	4	1	2	0	0
Hexamethylene 1,4-dihydro-4-oxo-2,6-pyridinedicarboxamide	6	3	0	4	5	0	3	0	0
(Phenylglycyl)prolylproline	6	3	1	3	5	0	3	0	0
Hexamethylene 4,4′-ketodibenzamide	6	3	2	0	2	0	2	0	0
Ethylene adipoyldiurethane	6	4	0	0	2	2	2	0	0
Adipoyl ethylenediurea	6	4	0	0	4	0	4	0	0
2-Hydroxytrimethylene adipoyldiurethane	6	4	0	1	4	3	2	0	0
Pentamethylene (cyclopropylenedicarboxoyl)diurethane	6	4	0	2	4	2	2	0	0
Hexamethylene pyromellitimide	6	4	0	6	2	0	2	0	0
Oxydisilyleneditrimethylene pyromellitimide	6	4	0	6	6	1	2	0	2
Tetramethyloxydisilyleneditrimethylene pyromellitimide	6	4	0	10	14	1	2	0	2
Hexamethylene 3,3′,4,4′-(oxydiphenylene)tetracarboximide	6	4	0	12	6	1	2	0	0
4,4′-(Tetramethylenedicarbonyl)diphenylene succinate	6	4	2	0	0	2	0	0	0
Trimethylene *N*,*N*′-glutaryldi(*p*-aminobenzoate)	6	4	2	0	2	2	2	0	0
Trimethylene *N*,*N*′-trimethylenediterephthalamate	6	4	2	0	2	2	2	0	0
D-Benzylglutamyl-L-benzylglutamate	6	4	2	2	6	2	2	0	0
3,4′-Ethylenedioxydiethylenedioxydiphenylene 3,3′4,4′-oxydiphenylenetet	6	4	2	12	6	5	2	0	0
4,4′-(Ethylenedioxydiethylenedioxy)diphenylene carbonyldiphenylenetetrac	6	5	2	12	4	4	2	0	0
Alanylglycylalanylglycylalanylglycylglutamylglycine	6	9	0	7	22	1	8	0	0
7	7	0	0	0	0	0	0	0	0
Heptamethylene	7	0	0	0	0	0	0	0	0
Heptamethylene disulfide	7	0	0	0	0	0	0	2	0
Tetramethylenethiotrimethylene sulfide	7	0	0	0	0	0	0	2	0
(Tetramethylenedithio)trimethylene disulfide	7	0	0	0	0	0	0	4	0
Hexamethyleneoxymethylene oxide	7	0	0	0	0	2	0	0	0
Cyclohexylidenedimethylene oxide	7	0	0	1	0	1	0	0	0
1,4-Piperidinediyltrimethylene	7	0	0	1	1	0	1	0	0
2,5-Heptamethylene-1,3,4-triazole	7	0	0	2	1	0	3	0	0
1,2-Cycloheptylene-*alt*-ethylene	7	0	0	2	2	0	0	0	0
3-Cyclohexylpropene	7	0	0	2	2	0	0	0	0
Vinyl cycloheptane	7	0	0	2	2	0	0	0	0
2,5-Heptamethylene-1-amino-1,3,4-triazole	7	0	0	2	2	0	4	0	0
1,4-Cyclohexylenedimethyleneoxymethylene oxide	7	0	0	2	2	2	0	0	0
1-Methyloctamethylene	7	0	0	2	4	0	0	0	0
1-Nonene	7	0	0	2	4	0	0	0	0
n-Nonaldehyde	7	0	0	2	4	1	0	0	0
n-Butyl-*n*-pentylsilylene	7	0	0	2	6	0	0	0	1
3-Octyl-2,5-thiophene	7	0	0	5	4	0	0	1	0
Di-1,4-naphthaleneoxyheptamethylene oxide	7	0	0	20	12	2	0	0	0
3-(4′-Octylphenyl)thiophene	7	0	1	5	4	0	0	1	0
4,4′-Ethylene-3-methyldiphenyleneoxypentamethylene oxide	7	0	1	7	6	2	0	0	0
4,4′-Stilbenediyloxyheptamethylene oxide	7	0	2	2	2	2	0	0	0
α-Methyl-4,4′-stilbenediyloxyheptamethylene oxide	7	0	2	3	4	2	0	0	0
8-Aminocaprylic acid	7	1	0	0	1	0	1	0	0
Heptamethylene urea	7	1	0	0	2	0	2	0	0
N-Cyclohexanecarbonylethylene amine	7	1	0	1	1	0	1	0	0
N-Heptanoylethylene amine	7	1	0	1	3	0	1	0	0
2,2-Dibutyl-3-hydroxypropionic acid	7	1	0	3	6	1	0	0	0
N,*N*-Dibutylacrylamide	7	1	0	3	7	0	1	0	0
Biphenyloxyhexamethylene acrylate	7	1	2	1	2	2	0	0	0
Azelaic anhydride	7	2	0	0	0	1	0	0	0
Ethylene pimelate	7	2	0	0	0	2	0	0	0
Pentamethylene succinate	7	2	0	0	0	2	0	0	0
Tetramethylene glutarate	7	2	0	0	0	2	0	0	0
Trimethylene adipate	7	2	0	0	0	2	0	0	0
Oxydiethylene glutarate	7	2	0	0	0	3	0	0	0
Pentamethylene oxydiacetate	7	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene malonate	7	2	0	0	0	4	0	0	0
Hexamethylene malonamide	7	2	0	0	2	0	2	0	0
Hydrazo azelamide	7	2	0	0	2	0	2	0	0
Methylene suberamide	7	2	0	0	2	0	2	0	0
Pentamethylene oxydiacetamide	7	2	0	0	2	1	2	0	0
Tetramethylene trimethylenediurethane	7	2	0	0	2	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Trimethylene 1,4-cyclohexylenedicarboxylate	7	2	0	2	2	2	0	0	0
Cyclopropylenedimethylene piperazinediurethane	7	2	0	2	2	2	2	0	0
Hexamethylene cyclopropylenedicarboxamide	7	2	0	2	4	0	2	0	0
Piperazine 3-methyladipamide	7	2	0	2	4	0	2	0	0
Trimethylene 4-octenediamide	7	2	0	2	4	0	2	0	0
Hexamethylene 2,3-pyrazoledicarboxamide	7	2	0	2	4	0	4	0	0
Hexamethylene 2,4-pyrazoledicarboxamide	7	2	0	2	4	0	4	0	0
Methylethylene suberate	7	2	0	2	4	2	0	0	0
n-Hexylglutamate	7	2	0	2	5	1	1	0	0
Cyclopropylene hexamethylenediurea	7	2	0	2	6	0	4	0	0
3-Oxypropylhexamethylene oxamide	7	2	0	2	6	1	2	0	0
Hexamethylene (1-methylcyclopropylene)dicarboxamide	7	2	0	3	6	0	2	0	0
Hexamethylene (methylene-2,5-furan)dicarboxamide	7	2	0	4	4	1	2	0	0
Hexamethylene 4(H)-pyran-2,6-dicarboxamide	7	2	0	4	4	1	2	0	0
Hexamethylene 3,3′-dimethyl-4,4′-methylenediphenylenediurea	7	2	0	14	16	0	4	0	0
N,N′-diethyl-3,3′-dimethyl-4,4′-methylenediphenylene adipamide	7	2	0	16	18	0	2	0	0
Heptamethylene terephthalate	7	2	1	0	0	2	0	0	0
p-Phenylene azelate	7	2	1	0	0	2	0	0	0
p-Xylylene pimelate	7	2	1	0	0	2	0	0	0
Ethylene *p*-(carboxyphenoxy)caproate	7	2	1	0	0	3	0	0	0
Hexamethylene *p*-(carboxyphenoxy)acetate	7	2	1	0	0	3	0	0	0
Heptamethylene terephthalamide	7	2	1	0	2	0	2	0	0
Hexamethylene *p*-(carboxyphenylene)acetamide	7	2	1	0	2	0	2	0	0
p-Phenylene azelamide	7	2	1	0	2	0	2	0	0
m-Xylylene pimelamide	7	2	1	0	2	0	2	0	0
p-Xylylene pimelamide	7	2	1	0	2	0	2	0	0
p-Phenylene (methyl-1,4-piperazine)dipropionate	7	2	1	2	4	2	2	0	0
N,N′-Dibutyl-4,4′-(dimethylmethylene)diphenylene terephthalamide	7	2	1	16	18	0	2	0	0
Ethylene 4,4′-(pentamethylenedioxy)dibenzoate	7	2	2	0	0	4	0	0	0
Hexamethylene 3,3′-methylenedibenzamide	7	2	2	0	2	0	2	0	0
Hexamethylene 4,4′-methylenedibenzamide	7	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene azelamide	7	2	2	0	2	1	2	0	0
4,4′-Oxydiphenylene azelamide	7	2	2	0	2	1	2	0	0
Hexamethylene 4,4′-(methylenediphenylene)diurethane	7	2	2	0	2	2	2	0	0
Heptamethylene 4,4′-sulfonyldibenzamide	7	2	2	0	2	2	2	1	0
Hexamethylene 4,4′-(methylenediphenylene)diurea	7	2	2	0	4	0	4	0	0
N,N′-Diethyl(methylenediphenylene) adipamide	7	2	2	2	6	0	2	0	0
2-Butyl-2-ethyltrimethylene methylenediphenylenediurethane	7	2	2	3	8	2	2	0	0
N,N′-Dibutyl(methylenediphenylene) terephthalamide	7	2	3	2	6	0	2	0	0
N,N′-Lysyl trimethylenediurea	7	3	0	1	6	1	4	0	0
Glycylprolylproline	7	3	0	2	3	0	3	0	0
Hexamethylene propylenediurea	7	3	0	2	8	0	4	0	0
Methyl-*N,N′*-lysyl trimethylenediurea	7	3	0	2	8	1	4	0	0
α-D-Glucotributyrate, 1,4-	7	3	0	8	14	5	0	0	0
β-D-Glucotributyrate, 1,4-	7	3	0	8	14	5	0	0	0
Trimethylene adipoyldiurethane	7	4	0	0	2	2	2	0	0
Azelaoyl oxalodihydrazide	7	4	0	0	4	0	4	0	0
Pentamethylene *N,N′*-ethylenediterephthalamate	7	4	2	0	2	2	2	0	0
Tetramethylene *N,N′*-trimethylenediterephthalamate	7	4	2	0	2	2	2	0	0
Trimethylene *N,N′*-adipoyldi(*p*-benzoate)	7	4	2	0	2	2	2	0	0
Heptamethylene (4,4′-terephthaloyldioxy)dibenzoate	7	4	3	0	0	4	0	0	0
8	8	0	0	0	0	0	0	0	0
Octamethylene	8	0	0	0	0	0	0	0	0
Ethylenethiohexamethylene sulfide	8	0	0	0	0	0	0	2	0
Octamethylene disulfide	8	0	0	0	0	0	0	2	0
(Ethylenedithio)hexamethylene disulfide	8	0	0	0	0	0	0	4	0
Octamethylene oxide	8	0	0	0	0	1	0	0	0
1,4-Piperazinediethylene sulfone	8	0	0	0	0	2	2	1	0
2,5-Octamethylene-1,3,4-oxadiazole	8	0	0	2	0	1	2	0	0
2,5-Octamethylene-1,3,4-triazole	8	0	0	2	1	0	3	0	0
4-Cyclohexyl-1-butene	8	0	0	2	2	0	0	0	0
Decenamer	8	0	0	2	2	0	0	0	0
1,2-Dichlorodecamethylene	8	0	0	2	2	0	0	0	2
2,5-Octamethylene-1-amino-1,3,4-triazole	8	0	0	2	2	0	4	0	0
1-Decene	8	0	0	2	4	0	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Di-*n*-pentylsilylene	8	0	0	2	6	0	0	0	1
1,11-Dodecadiyne, 1,12-	8	0	0	4	0	0	0	0	0
5,7-Dodecadiynediol, 5,8-	8	0	0	4	2	2	0	0	0
Octamethylene 5,5′-dibenzimidazole	8	0	0	14	8	0	4	0	0
2,2′-Octamethylenedithiazole-*p*-phenylene	8	0	1	6	2	0	2	2	0
4,4′-Stilbenediyloxyoctamethylene oxide	8	0	2	2	2	2	0	0	0
α-methyl-4,4′-stilbenediyloxyoctamethylene oxide	8	0	2	3	4	2	0	0	0
1,12-Bis(*p*-toluylsulfonyl)-5,7-dodecadiynediol, 5,8-	8	0	2	6	6	6	0	2	0
9-Aminopelargonic acid	8	1	0	0	1	0	1	0	0
Octamethylene urea	8	1	0	0	2	0	2	0	0
N-Octanoylethylene amine	8	1	0	1	3	0	1	0	0
Octyl acrylate	8	1	0	2	4	1	0	0	0
Tetramethylene dithioladipate	8	2	0	0	0	0	0	2	0
Thiodiethylene dithioladipate	8	2	0	0	0	0	0	3	0
Piperazine adipamide	8	2	0	0	0	0	2	0	0
(Piperazinedicarbamoyl)dithiotetramethylene disulfide	8	2	0	0	0	0	2	4	0
Sebacic anhydride	8	2	0	0	0	1	0	0	0
Ethylene suberate	8	2	0	0	0	2	0	0	0
Hexamethylene succinate	8	2	0	0	0	2	0	0	0
Pentamethylene glutarate	8	2	0	0	0	2	0	0	0
Tetramethylene adipate	8	2	0	0	0	2	0	0	0
Trimethylene pimelate	8	2	0	0	0	2	0	0	0
Barium sebacate	8	2	0	0	0	2	0	0	1
Cadmium sebacate	8	2	0	0	0	2	0	0	1
Calcium sebacate	8	2	0	0	0	2	0	0	1
Lead sebacate	8	2	0	0	0	2	0	0	1
Magnesium sebacate	8	2	0	0	0	2	0	0	1
Manganese sebacate	8	2	0	0	0	2	0	0	1
Stannous sebacate	8	2	0	0	0	2	0	0	1
Zinc sebacate	8	2	0	0	0	2	0	0	1
Piperazine (ethylenedioxy)diacetamide	8	2	0	0	0	2	2	0	0
Hexamethylene oxydiacetate	8	2	0	0	0	3	0	0	0
Oxydiethylene adipate	8	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene succinate	8	2	0	0	0	4	0	0	0
Heptamethylene malonamide	8	2	0	0	2	0	2	0	0
Hexamethylene succinamide	8	2	0	0	2	0	2	0	0
Hydrazo sebacamide	8	2	0	0	2	0	2	0	0
Octamethylene oxamide	8	2	0	0	2	0	2	0	0
Pentamethylene glutaramide	8	2	0	0	2	0	2	0	0
Tetramethylene adipamide	8	2	0	0	2	0	2	0	0
Hydrazo thiodivaleramide	8	2	0	0	2	0	2	1	0
Hexamethylene oxydiacetamide	8	2	0	0	2	1	2	0	0
Ethylene hexamethylenediuirethane	8	2	0	0	2	2	2	0	0
Tetramethylene tetramethylenediurethane	8	2	0	0	2	2	2	0	0
Ethylene (ethylenedithio)diethylenediurethane	8	2	0	0	2	2	2	2	0
Tetramethylene (1,4-piperazinedithio)dicarboxylate	8	2	0	0	2	2	2	2	0
Ethylene (ethylenediamino)dipropionamide	8	2	0	0	4	0	4	0	0
Ethylene hexamethylenediurea	8	2	0	0	4	0	4	0	0
Tetramethylene disiloxanylenedipropionamide	8	2	0	0	6	1	2	0	2
2,2-Bisbromomethyltrimethylene adipate	8	2	0	1	0	2	0	0	2
2,2-Bischloromethyltrimethylene adipate	8	2	0	1	0	2	0	0	2
2,2-(Bromomethylchloromethyl)trimethylene adipate	8	2	0	1	0	2	0	0	2
Piperazine 4-octenediamide	8	2	0	2	2	0	2	0	0
1,4-Cyclohexylene adipate	8	2	0	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene succinate	8	2	0	2	2	2	0	0	0
Hexamethylene 3-hexenedioate	8	2	0	2	2	2	0	0	0
1,3-Cyclohexylene adipamide	8	2	0	2	4	0	2	0	0
1,4-Cyclohexylene adipamide	8	2	0	2	4	0	2	0	0
Tetramethylene 4-octenediamide	8	2	0	2	4	0	2	0	0
Ethylene 2,5-tetrahydrofurandipropionamide	8	2	0	2	4	1	2	0	0
Methylethylene azelate	8	2	0	2	4	2	0	0	0
2-Butenylene hexamethylenediurethane	8	2	0	2	4	2	2	0	0
1,4-Dioxanylene-2,5-dimethylene adipamide	8	2	0	2	4	2	2	0	0
Hexamethylene 1,4-dioxanylene-2,5-dicarboxamide	8	2	0	2	4	2	2	0	0
Tetramethylene 1,4-cyclohexylenediurethane	8	2	0	2	4	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene propylmalonamide	8	2	0	2	6	0	2	0	0
Hexamethylene 2,5-dihydroxyadipamide	8	2	0	2	6	2	2	0	0
2-Methltrimethylene hexamethylenediurethane	8	2	0	2	6	2	2	0	0
Octamethylene L-tartaramide	8	2	0	2	6	2	2	0	0
2,2-Dimethyltrimethylene suberate	8	2	0	3	6	2	0	0	0
2,2-Dimethyltrimethylene hexamethylenediurethane	8	2	0	3	8	2	2	0	0
2,4-Hexadiyne hexamethylene diurethane	8	2	0	4	2	2	2	0	0
Hexamethylene 2,5-thiophenediacetamide	8	2	0	4	4	0	2	1	0
Octamethylene 2,5-thiophenedicarboxamide	8	2	0	4	4	0	2	1	0
Octamethylene 2,5-furandicarboxamide	8	2	0	4	4	1	2	0	0
1,4-Dimethyltetramethylene hexamethylenediurethane	8	2	0	4	10	2	2	0	0
Octamethylene di-*O*-methyl-L-tartaramide	8	2	0	4	10	2	2	0	0
Tetramethylene tetramethyldisiloxanylenedipropionamide	8	2	0	4	14	1	2	0	2
Octamethylene 2,4-pyridinedicarboxamide	8	2	0	5	5	0	3	0	0
Octamethylene 2,5-pyridinedicarboxamide	8	2	0	5	5	0	3	0	0
Octamethylene 2,6-pyridinedicarboxamide	8	2	0	5	5	0	3	0	0
Octamethylene 3,5-pyridinedicarboxamide	8	2	0	5	5	0	3	0	0
N,N'-Diethyl(hexafluoropentamethylene) adipamide	8	2	0	5	6	0	2	0	6
N,N'-Diethylhexamethyelne perfluoroglutaramide	8	2	0	5	6	0	2	0	6
N,N'-Dimethyltetramethylene tetramethyldisiloxanylenedipropionamide	8	2	0	6	18	1	2	0	2
3-Methyl-*m*-phenylene sebacamide	8	2	0	7	8	0	2	0	0
N,N'-Diisopropylethylene dimethyl(ethylenediamino)dipropionamide	8	2	0	8	20	0	4	0	0
4-Methyl-1,3-phenylene tetraethyldisiloxanylenedipropionamide	8	2	0	11	20	1	2	0	2
1,5-Naphthalene-4′,4″-(dimethoxy)dibenzylidenimino sebacate	8	2	0	26	20	4	2	0	0
Hexamethylene dithiol-*p*-phenylenediacetate	8	2	1	0	0	0	0	2	0
Octamethylene terephthalate	8	2	1	0	0	2	0	0	0
p-Xylylene suberate	8	2	1	0	0	2	0	0	0
p-Phenylene 1,4-piperazinedipropionate	8	2	1	0	0	2	2	0	0
Ethylene *p*-(carboxyphenoxy)heptanoate	8	2	1	0	0	3	0	0	0
Hexamethylene *m*-phenylenediacetamide	8	2	1	0	2	0	2	0	0
Hexamethylene *p*-phenylenediacetamide	8	2	1	0	2	0	2	0	0
Octamethylene isophthalamide	8	2	1	0	2	0	2	0	0
Octamethylene phthalamide	8	2	1	0	2	0	2	0	0
Octamethylene terephthalamide	8	2	1	0	2	0	2	0	0
p-Phenylenediethylene adipamide	8	2	1	0	2	0	2	0	0
m-Phenylene sebacamide	8	2	1	0	2	0	2	0	0
o-Phenylene sebacamide	8	2	1	0	2	0	2	0	0
p-Phenylene sebacamide	8	2	1	0	2	0	2	0	0
m-Xylylene suberamide	8	2	1	0	2	0	2	0	0
p-Xylylene suberamide	8	2	1	0	2	0	2	0	0
Hexamethylene *p*-phenoxydiacetamide	8	2	1	0	2	1	2	0	0
p-Xylylene oxydibutyramide	8	2	1	0	2	1	2	0	0
(Ethylenedioxy)ditrimethylene terephthalamide	8	2	1	0	2	2	2	0	0
Hexamethylene (*p*-phenylenedioxy)diacetamide	8	2	1	0	2	2	2	0	0
Hexamethylene *p*-xylylenediurethane	8	2	1	0	2	2	2	0	0
p-Sulfonylphenylene sebacamide	8	2	1	0	2	2	2	1	0
Hexamethylene *p*-xylylenediurea	8	2	1	0	4	0	4	0	0
1,4-Cyclohexylenedimethylene *p*-phenylenediacetate	8	2	1	2	2	2	0	0	0
Sebacoyl dithionisophthaloyldihydrazide	8	2	1	2	4	0	4	2	0
Hexamethylene 3,3′-*p*-phenylenedibutyramide	8	2	1	4	10	0	2	0	0
m-Phenylene (tetraethyldisiloxanylene)dipropionamide	8	2	1	4	14	1	2	0	2
o-Phenylene (tetraethyldisiloxanylene)dipropionamide	8	2	1	4	14	1	2	0	2
p-Phenylene (tetraethyldisiloxanylene)dipropionamide	8	2	1	4	14	1	2	0	2
Hexamethylene 3,3′-dimethyl-3,3′-*p*-phenylenedibutyramide	8	2	1	6	14	0	2	0	0
p-Phenylene-4′,4″-(3′,3″-dimethoxy)dibenzylidenimino sebacate	8	2	1	16	14	4	2	0	0
Ethylene 4,4′-hexamethylenedibenzoate	8	2	2	0	0	2	0	0	0
4,4′-Hexamethylenediphenylenesuccinate	8	2	2	0	0	2	0	0	0
Octamethylene 4,4′-biphenylenedicarboxylate	8	2	2	0	0	2	0	0	0
p-Phenylenediethylene 3,3′-(*p*-phenylene)dipropionate	8	2	2	0	0	2	0	0	0
p-Phenyleneditetramethylene terephthalate	8	2	2	0	0	2	0	0	0
Hexamethylene 4,4′-(ethylenedioxy)dibenzoate	8	2	2	0	0	4	0	0	0
Hexamethylene 4,4′-(sulfonyldimethylene)dibenzoate	8	2	2	0	0	4	0	1	0
4,4′-Biphenylene sebacamide	8	2	2	0	2	0	2	0	0
4,4′-Methylenediphenylene azelamide	8	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene sebacamide	8	2	2	0	2	1	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
4,4′-Oxydiphenylene sebacamide	8	2	2	0	2	1	2	0	0
Heptamethylene 4,4′-(methylenediphenylene)diurethane	8	2	2	0	2	2	2	0	0
Hexamethylene 4,4′-(oxydiphenylene)dioxydiacetamide	8	2	2	0	2	3	2	0	0
Octamethylene 3,3′-phosphinylidenedibenzamide	8	2	2	0	3	1	2	0	1
Octamethylene 4,4′-phosphinylidenedibenzamide	8	2	2	0	3	1	2	0	1
Octamethylene 3,3′-(methylphosphinylidene)dibenzamide	8	2	2	1	5	1	2	0	1
Octamethylene 4,4′-(methylphosphinylidene)dibenzamide	8	2	2	1	5	1	2	0	1
N,N′-Diemthyl-4,4′-methylenediphenylene azelamide	8	2	2	2	6	0	2	0	0
4,4′-Isopropylidenediphenylene 1,4-piperazinedipropionate	8	2	2	3	6	2	2	0	0
1,12-Bis(phenylurethane)-5,7-dodecadiynediol, 5,8-	8	2	2	4	4	2	2	0	0
1,5-Naphthalene-4′,4″-dibenzylidenimino sebacate	8	2	2	12	8	2	2	0	0
4,4′-Biphenylene-4″,4‴-(dimethoxy)dibenzylidenimino sebacate	8	2	2	16	14	4	2	0	0
4,4′-Oxydiphenylene-4″,4‴-(dimethoxy)dibenzylidenimino sebacate	8	2	2	16	14	5	2	0	0
Octamethylene 4,4′-(phenylphosphinylidene)dibenzamide	8	2	3	0	3	1	2	0	1
p-Phenylene-4,4′,4″-dibenzylidenimino sebacate	8	2	3	2	2	2	2	0	0
4,4′-Biphenylene-4″,4‴-dibezylidenimino sebacate	8	2	4	2	2	2	2	0	0
4,4′-Oxydiphenylene-4″,4‴-dibenzylidenimino sebacate	8	2	4	2	2	3	2	0	0
l-1-Carboxypentamethylene adipamide	8	3	0	1	4	1	2	0	0
N,N′-Lysyl tetramethylenediurea	8	3	0	1	6	1	4	0	0
Methyl-*N,N′*-lysyl tetramethylenediurea	8	3	0	2	8	1	4	0	0
Octamethylene 1,4-dihydro-4-*oxo*-2,6-pyridinedicarboxamide	8	3	0	4	5	0	3	0	0
Lysyllysylvaline	8	3	0	6	17	0	5	0	0
4,4′-(Octamethylenedicarbonyl)diphenylene carbonate	8	3	2	0	0	2	0	0	0
Ethylene adipamide-*alt*-ethylene oxamide	8	4	0	0	4	0	4	0	0
Octamethylene bromopyromellitimide	8	4	0	6	1	0	2	0	1
Octamethylene pyromellitimide	8	4	0	6	2	0	2	0	0
Octamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide	8	4	0	12	6	1	2	0	0
Hexamethylene *N,N′*-ethylenediterephthlamate	8	4	2	0	2	2	2	0	0
Pentamethylene *N,N′*-trimethylenediterephthalamate	8	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-(*p*-xylylene)diterephthalamate	8	4	3	0	2	2	2	0	0
4,4′-(Oxydiethyleneoxydiethylenedioxy)diphenylene carbonyldiphenylenet	8	5	2	12	4	5	2	0	0
Alanylalanylglycylprolylprolylglycine	8	6	0	6	14	0	6	0	0
Alanylglycylalanylprolylglycylproline	8	6	0	6	14	0	6	0	0
(Alanylglycyl)4-glutamylglycine	8	?	0	9	26	1	10	0	0
9	9	0	0	0	0	0	0	0	0
Nonamethylene disulfide	9	0	0	0	0	0	0	2	0
Pentamethylenethiotetramethylenesulfide	9	0	0	0	0	0	0	2	0
Nonamethylene oxide	9	0	0	0	0	1	0	0	0
Pentamethylenesulfonyltetramethylene sulfone	9	0	0	0	0	4	0	2	0
2,5-Nonamethylene-1,3,4-oxadiazole	9	0	0	2	0	1	2	0	0
5-Cyclohexyl-1-pentene	9	0	0	2	2	0	0	0	0
1-Undecene	9	0	0	2	4	0	0	0	0
n-Pentyl-*n*-hexylsilylene	9	0	0	2	6	0	0	0	1
3-Decylthiophene	9	0	0	5	4	0	0	1	0
4,4′-Methylenedicyclohexylenethiourea	9	0	0	5	6	0	2	1	0
4,4′-Ethylene-3-methyldiphenyleneoxyheptamethylene oxide	9	0	1	7	6	2	0	0	0
4,4′-Heptamethylenedioxydiphenyleneoxy *p*-xylylene oxide	9	0	3	0	0	4	0	0	0
10-Hydroxycapric acid	9	1	0	0	0	1	0	0	0
10-Aminocapric acid	9	1	0	0	1	0	1	0	0
Nonamethylene urea	9	1	0	0	2	0	2	0	0
5-Hydroxy-2-(1,3-dioxane)caprylic acid	9	1	0	2	2	3	0	0	0
4-Hydroxymethylene-2-(1,3-dioxolane)caprylic acid	9	1	0	2	2	3	0	0	0
2-Methyl-11-aminoundecanoic acid	9	1	0	2	5	0	1	0	0
4,4′-Methylenedicyclohexyleneurea	9	1	0	4	6	0	2	0	0
3-*n*-Decylhydroxybenzoic acid	9	1	0	7	6	1	0	0	0
Undecanedioic anhydride	9	2	0	0	0	1	0	0	0
Ethylene azelate	9	2	0	0	0	2	0	0	0
Hexamethylene glutarate	9	2	0	0	0	2	0	0	0
Pentamethylene adipate	9	2	0	0	0	2	0	0	0
Tetramethylene pimelate	9	2	0	0	0	2	0	0	0
Trimethylene suberate	9	2	0	0	0	2	0	0	0
Oxydiethylene pimelate	9	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene glutarate	9	2	0	0	0	4	0	0	0
hexamethylene glutaramide	9	2	0	0	2	0	2	0	0
Methylene sebacamide	9	2	0	0	2	0	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Octamethylene malonamide	9	2	0	0	2	0	2	0	0
Pentamethylene adipamide	9	2	0	0	2	0	2	0	0
Tetramethylene pimelamide	9	2	0	0	2	0	2	0	0
Hexamethylene trimethylenediurethane	9	2	0	0	2	2	2	0	0
Tetramethylene pentamethylenediurethane	9	2	0	0	2	2	2	0	0
Trimethylene hexamethylenediurethane	9	2	0	0	2	2	2	0	0
Trimethylene (ethylenedithio)diethylenediurethane	9	2	0	0	2	2	2	2	0
Hexamethylene trimethylenediurea	9	2	0	0	4	0	4	0	0
1,4-Cyclohexylenedimethylene glutarate	9	2	0	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene glutaramide	9	2	0	2	4	0	2	0	0
Cyclopropylene sebacamide	9	2	0	2	4	0	2	0	0
Pentamethylene 4-octenediamide	9	2	0	2	4	0	2	0	0
Hexamethylene (methylene-2,5-tetrahydrofuran)dicarboxamide	9	2	0	2	4	1	2	0	0
Methylethylene sebacate	9	2	0	2	4	2	0	0	0
Methylethylene thiodivalerate	9	2	0	2	4	2	0	1	0
Cyclopropylenedimethylene hexamethylenediurethane	9	2	0	2	4	2	2	0	0
Methylethylene sulfonyldivalerate	9	2	0	2	4	4	0	1	0
n-Octylglutamate	9	2	0	2	5	1	1	0	0
Hexamethylene 2-butylmalonamide	9	2	0	2	6	0	2	0	0
Hexamethylene 3-methyladipamide	9	2	0	2	6	0	2	0	0
3-Methylehexamethylene adipamide	9	2	0	2	6	0	2	0	0
2,2-Dimethyltrimethylene azelate	9	2	0	3	6	2	0	0	0
3,3′-Dimethyl-4,4′-methylenediphenylene sebacamide	9	2	0	14	14	0	2	0	0
Hexamethylene (diethyldimethyl)methylenediphenylenediurea	9	2	0	16	20	0	4	0	0
N,N′-Diisoamyl-3,3′-diemthyl-4,4′-methylenediphenylene adipamide	9	2	0	20	26	0	2	0	0
Nonamethylene terephthalate	9	2	1	0	0	2	0	0	0
p-Xylylene azelate	9	2	1	0	0	2	0	0	0
Heptamethylene *p*-Phenylenediacetamide	9	2	1	0	2	0	2	0	0
Hexamethyelene (*p*-methylenephenylene)-diacetamide	9	2	1	0	2	0	2	0	0
m-Xylylene azelamide	9	2	1	0	2	0	2	0	0
p-Xylylene azelamide	9	2	1	0	2	0	2	0	0
Hexamethylene (methylene-*p*-phenoxy)diacetamide	9	2	1	0	2	1	2	0	0
p-Xylylene 5-oxasebacamide	9	2	1	0	2	1	2	0	0
Hexamethylene 3-Methyl-3,3′-*p*-phenylenedipropionamide	9	2	1	2	6	0	2	0	0
p-Phenylene(1,1-dimethyl)ethylene sebacamide	9	2	1	3	8	0	2	0	0
4,4′-Methylenediphenylene sebacamide	9	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene undecanediamide	9	2	2	0	2	1	2	0	0
4,4′-Oxydiphenylene undecanediamide	9	2	2	0	2	1	2	0	0
Hexamethylene 4,4′-(methylenediphenylene)dioxydiacetamide	9	2	2	0	2	2	2	0	0
Octamethylene 4,4′-(methylenediphenylene)diurethane	9	2	2	0	2	2	2	0	0
Nonamethylene 3,3′-phosphinylidenedibenzamide	9	2	2	0	3	1	2	0	1
Nonamethylene 4,4′-phosphinylidenedibenzamide	9	2	2	0	3	1	2	0	1
4,4′-Methylenediphenylene octamethylenediurea	9	2	2	0	4	0	4	0	0
Nonamethylene 3,3′-(methylphosphinylidene)dibenzamide	9	2	2	1	5	1	2	0	1
Nonamethylene 4,4′-(methylphosphinylidene)dibenzamide	9	2	2	1	5	1	2	0	1
N,N′-Dimethyl-4,4′-methylenediphenylene sebacamide	9	2	2	2	6	0	2	0	0
Nonamethylene 4,4′-(phenylphosphinylidene)dibenzamide	9	2	3	0	3	1	2	0	1
N,N′-Lysyl pentamethylenediurea	9	3	0	1	6	1	4	0	0
Methyl-*N,N′*-lysyl pentamethylenediurea	9	3	0	2	8	1	4	0	0
Adipoyl Pentamethylenediurea	9	4	0	0	4	0	4	0	0
Nonamethylene bromopyromellitimide	9	4	0	6	1	0	2	0	1
Nonamethylene pyromellitimide	9	4	0	6	2	0	2	0	0
Nonamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide	9	4	0	12	6	1	2	0	0
Ethylene *N,N′*-azelaoyldi(*p*-aminobenzoate)	9	4	2	0	2	2	2	0	0
Heptamethylene *N,N′*-ethylenediterephthalamate	9	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-trimethylenediterephthalamate	9	4	2	0	2	2	2	0	0
Pentamethylene *N,N′*-tetramethylenediterephthlalamate	9	4	2	0	2	2	2	0	0
10	10	0	0	0	0	0	0	0	0
Decamethylene	10	0	0	0	0	0	0	0	0
Decamethylene sulfide	10	0	0	0	0	0	0	1	0
Decamethylene disulfide	10	0	0	0	0	0	0	2	0
Hexamethylenethiotetramethylene sulfide	10	0	0	0	0	0	0	2	0
(Hexamethylenedithio)tetramethylene disulfide	10	0	0	0	0	0	0	4	0
Decamethylene oxide	10	0	0	0	0	1	0	0	0
Methyleneoxynonamethylene oxide	10	0	0	0	0	2	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylenesulfonyltetramethyloene sulfone	10	0	0	0	0	4	0	2	0
Decamethylene amine	10	0	0	0	1	0	1	0	0
2,5-Decamethylene-1,3,4-oxadiazole	10	0	0	2	0	1	2	0	0
2,5-Decamethylene-1,3,4-triazole	10	0	0	2	1	0	3	0	0
Dodecenamer	10	0	0	2	2	0	0	0	0
1,2-Dichlorododecamethylene	10	0	0	2	2	0	0	0	2
2,5-Decamethylene-1-amino-1,3,4-triazole	10	0	0	2	2	0	4	0	0
1-Dodecene	10	0	0	2	4	0	0	0	0
Decylvinyl ether	10	0	0	2	4	1	0	0	0
Di-*n*-hexylgermylene	10	0	0	2	6	0	0	0	1
Di-*n*-hexylsilyene	10	0	0	2	6	0	0	0	1
3,3-Dipentyltrimethylene sulfone	10	0	0	3	6	2	0	1	0
3,3′-Dihexyl-2,2′-bithiophene	10	0	0	10	8	0	0	2	0
Decamethyleneoxy-*p*-phenylene oxide	10	0	1	0	0	2	0	0	0
Benzylidenethiodecamethylene sulfide	10	0	1	1	2	0	0	2	0
4,4′-(Oxydiphenylene)dithiodecamethylene disulfide	10	0	2	0	0	1	0	4	0
4,4′-Octamethylenedioxydiphenyleneoxy-*p*-xylylene oxide	10	0	3	0	0	4	0	0	0
Bis-(Butylphenyl)azophenoxyethoxy phosphazene	10	0	4	2	6	4	5	0	1
Decamethylene carbonate	10	1	0	0	0	2	0	0	0
11-Aminoundecanoic acid	10	1	0	0	1	0	1	0	0
Decamethylene urea	10	1	0	0	2	0	2	0	0
N-Methyl-11-aminoundecanoic acid	10	1	0	1	3	0	1	0	0
Decyl acrylate	10	1	0	2	4	1	0	0	0
5-Hydroxymethylene-5-methyl-2-(1,3-dioxane)caprylic acid	10	1	0	3	4	3	0	0	0
N-Phenyl-11-aminoundecanoic acid	10	1	1	0	2	0	1	0	0
4,4′-decamethylenediphenylene carbonate	10	1	2	0	0	2	0	0	0
Hexamethylene dithioladipate	10	2	0	0	0	0	0	2	0
Piperazine suberamide	10	2	0	0	0	0	2	0	0
(Piperazinedicarbamoyl)dithiohexamethylene disulfide	10	2	0	0	0	0	2	4	0
Dodecanedioic anhydride	10	2	0	0	0	1	0	0	0
Decamethylene oxalate	10	2	0	0	0	2	0	0	0
Ethylene sebacate	10	2	0	0	0	2	0	0	0
Hexamethylene adipate	10	2	0	0	0	2	0	0	0
Pentamethylene pimelate	10	2	0	0	0	2	0	0	0
Tetramethylene suberate	10	2	0	0	0	2	0	0	0
Trimethylene azelate	10	2	0	0	0	2	0	0	0
Ethylene thiodivalerate	10	2	0	0	0	2	0	1	0
Ethylene 1,4-piperazinedipropionate	10	2	0	0	0	2	2	0	0
Oxydiethylene suberate	10	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene adipate	10	2	0	0	0	4	0	0	0
Ethylene sulfonyldivalerate	10	2	0	0	0	4	0	1	0
Decamethylene oxamide	10	2	0	0	0	0	2	0	0
Ethylene sebacamide	10	2	0	0	2	0	2	0	0
Hexamethylene adipamide	10	2	0	0	2	0	2	0	0
Hydrazo dodecanediamide	10	2	0	0	2	0	2	0	0
Nonamethylene malonamide	10	2	0	0	2	0	2	0	0
Octamethylene succinamide	10	2	0	0	2	0	2	0	0
Pentamethylene pimelamide	10	2	0	0	2	0	2	0	0
Tetramethylene suberamide	10	2	0	0	2	0	2	0	0
Ethylene thiodivaleramide	10	2	0	0	2	0	2	1	0
Hexamethylene thiodipropionamide	10	2	0	0	2	0	2	1	0
(Hexamethylenedicarbamoyl)dithiotetramethylene disulfide	10	2	0	0	2	0	2	4	0
Ethylene 1,4-piperazinedipropionamide	10	2	0	0	2	0	4	0	0
Hexamethylene piperazinediurea	10	2	0	0	2	0	4	0	0
Piperazine (ethylenediamino)dipropionamide	10	2	0	0	2	0	4	0	0
Oxyditrimethylene adipamide	10	2	0	0	2	1	2	0	0
(Ethylenedioxy)diethylene adipamide	10	2	0	0	2	2	2	0	0
Hexamethylene (ethylenedioxy)diacetamide	10	2	0	0	2	2	2	0	0
(Tetramethylenedioxy)ditrimethylene oxamide	10	2	0	0	2	2	2	0	0
Tetramethylene Hexamethylenediurethane	10	2	0	0	2	2	2	0	0
Tetramethylene (thioditrimethylene)diurethane	10	2	0	0	2	2	2	1	0
Thiodiethylene hexamethylenediurethane	10	2	0	0	2	2	2	1	0
Ethylene (tetramethylenedithio)diethylenediurethane	10	2	0	0	2	2	2	2	0
Hexamethylene (1,4-piperazinedithio)dicarboxylate	10	2	0	0	2	2	2	2	0
Tetramethylene (ethylenedithio)diethylenediurethane	10	2	0	0	2	2	2	2	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Oxydiethylene hexamethylenediurethane	10	2	0	0	2	3	2	0	0
Phosphinideneditrimethylene adipamide	10	2	0	0	3	0	2	0	1
Hexamethylene tetramethylenediurea	10	2	0	0	4	0	4	0	0
Hexamethylene disiloxanylenedipropionamide	10	2	0	0	6	1	2	0	2
N-Methylhexamethylene adipamide	10	2	0	1	4	0	2	0	0
Hexamethylene dithiol-1,4-cyclohexylenedicarboxylate	10	2	0	2	2	0	0	2	0
1,4-Cyclohexylenedimethylene adipate	10	2	0	2	2	2	0	0	0
1,4-Cyclohexylene suberate	10	2	0	2	2	2	0	0	0
Hexamethylene 1,4-cyclohexylenedicarboxylate	10	2	0	2	2	2	0	0	0
hexamethylene 4-octenedioate	10	2	0	2	2	2	0	0	0
Octamethylene 3-hexenedioate	10	2	0	2	2	2	0	0	0
1,3-cyclohexylenedimethylene adipamide	10	2	0	2	4	0	2	0	0
Decamethylene fumaramide	10	2	0	2	4	0	2	0	0
Hexamethylene 1,4-cyclohexylenedicarboxamide	10	2	0	2	4	0	2	0	0
Hexamethylene 1,2-cyclohexylenedicarboxamide	10	2	0	2	4	0	2	0	0
Hexamethylene 1,3-cyclohexylenedicarboxamide	10	2	0	2	4	0	2	0	0
Hexamethylene 4-octenediamide	10	2	0	2	4	0	2	0	0
Tetramethylene 2,5-tetrahydrofurandipropionamide	10	2	0	2	4	1	2	0	0
1-Methyltrimethylene thiodivalerate	10	2	0	2	4	2	0	1	0
(1,4-Cyclohexylenedioxy)ditrimethylene oxamide	10	2	0	2	4	2	2	0	0
Hexamethylene 1,4-dioxanylene-2,5-diacetamide	10	2	0	2	4	2	2	0	0
1,4-Cyclohexylene (ethylenedithio)diethylenediurethane	10	2	0	2	4	2	2	2	0
Decamethylene 1,2-dihydroxysuccinate	10	2	0	2	4	4	0	0	0
1-Methyltrimethylene sulfonyldivalerate	10	2	0	2	4	4	0	1	0
Alanyl-11-Aminoundecanoic acid	10	2	0	2	6	0	2	0	0
N-*N*′-Dimethylhexamethylene adipamide	10	2	0	2	6	0	2	0	0
Hexamethylene pentylmalonamide	10	2	0	2	6	0	2	0	0
Piperazine *N*,*N*′-dimethyl(ethylenediamino)dipropionamide	10	2	0	2	6	0	4	0	0
1-Methyltrimethylene octamethylenediurethane	10	2	0	2	6	2	2	0	0
Decamethylene methylfumaramide	10	2	0	3	6	0	2	0	0
2,2-Dimethyltrimethylene sebacate	10	2	0	3	6	2	0	0	0
1,4-Cyclohexylenedimethylene dithiol-1,4-cyclohexylenedicarboxylate	10	2	0	4	4	0	0	2	0
Decamethylene 2,5-thiophenedicarboxamide	10	2	0	4	4	0	2	1	0
Hexamethylene 2,5-thiophenedipropionamide	10	2	0	4	4	0	2	1	0
Decamethylene 2,5-furandicarboxamide	10	2	0	4	4	1	2	0	0
Hexamethylene 2,5-furandipropionamide	10	2	0	4	4	1	2	0	0
1,4-Cyclohexylenedimethylene 1,4-Cyclohexylenedicarboxylate	10	2	0	4	4	2	0	0	0
Hexamethylene 2,5-pyrroledipropionamide	10	2	0	4	5	0	3	0	0
Decamethylene 2,3-dimethylfumaramide	10	2	0	4	8	0	2	0	0
Hexamethylene (tetramethyldisiloxanylene)dipropionamide	10	2	0	4	14	1	2	0	2
Decamethylene 2,4-pyridine dicarboxamide	10	2	0	5	5	0	3	0	0
Decamethylene 2,5-pyridine dicarboxamide	10	2	0	5	5	0	3	0	0
Decamethylene 2,6-pyridine dicarboxamide	10	2	0	5	5	0	3	0	0
Decamethylene 3,5-pyridine dicarboxamide	10	2	0	5	5	0	3	0	0
Hexamethylene (*N*-methyl-2,5-pyrrole)dipropionamide	10	2	0	5	7	0	3	0	0
1,12-Bis(ethylurethane)-5,7-dodecadiynediol, 5,8-	10	2	0	6	8	2	2	0	0
N,*N*′-diisopropylethylene 1,4-piperazinedipropionamide	10	2	0	6	14	0	4	0	0
Piperazine *N*,*N*′-diisopropyl(ethylenediamino)dipropionamide	10	2	0	6	14	0	4	0	0
Tetramethyl-*p*-xylylenedithiol sebacate	10	2	0	10	12	0	0	2	0
Decamethylene 3,3′-dimethyl-4,4′-biphenylenediurethane	10	2	0	14	14	2	2	0	0
Decamethylene dithiolterephthalate	10	2	1	0	0	0	0	2	0
p-Xylylene dithiolsebacate	10	2	1	0	0	0	0	2	0
Decamethylene isophthalate	10	2	1	0	0	2	0	0	0
Decamethylene terephthalate	10	2	1	0	0	2	0	0	0
p-Xylylene sebacate	10	2	1	0	0	2	0	0	0
Hexamethylene *p*-(carboxyphenoxy)valerate	10	2	1	0	0	3	0	0	0
Decamethylene *m*-carboxycarbanilate	10	2	1	0	1	2	1	0	0
Decamethylene *p*-carboxycarbanilate	10	2	1	0	1	2	1	0	0
Decamethylene isophthalamide	10	2	1	0	2	0	2	0	0
Decamethylene phthalamide	10	2	1	0	2	0	2	0	0
Decamethylene terephthalamide	10	2	1	0	2	0	2	0	0
Hexamethylene 3,3′-*p*-phenylenedipropionamide	10	2	1	0	2	0	2	0	0
Octamethylene *p*-phenylenediacetamide	10	2	1	0	2	0	2	0	0
p-Phenylene dodecanediamide	10	2	1	0	2	0	2	0	0
p-Phenyleneethylene sebacamide	10	2	1	0	2	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
m-Xylylene sebacamide	10	2	1	0	2	0	2	0	0
p-Xylylene sebacamide	10	2	1	0	2	0	2	0	0
p-Xylylene thiodivaleramide	10	2	1	0	2	0	2	1	0
p-Xylylene 5-oxaundecanediamide	10	2	1	0	2	1	2	0	0
p-Xylylene oxydivaleramide	10	2	1	0	2	1	2	0	0
Octamethylene *p*-xylylenediurethane	10	2	1	0	2	2	2	0	0
p-Xylylene octamethylenediurethane	10	2	1	0	2	2	2	0	0
Octamethylene *p*-xylylenediurea	10	2	1	0	4	0	4	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene terephthalamide	10	2	1	2	4	2	2	0	0
4,4′-(Decamethylenedioxy)dichlorodiphenylene terephthalate	10	2	1	12	6	4	0	0	2
Decamethylene 2,2′-dibenzoate	10	2	2	0	0	2	0	0	0
Decamethylene 3,3′-debenzoate	10	2	2	0	0	2	0	0	0
Decamethylene 4,4′-debenzoate	10	2	2	0	0	2	0	0	0
p-Phenylenedipentamethylene terephthalate	10	2	2	0	0	2	0	0	0
4,4′-Ethylenediphenylene sebacamide	10	2	2	0	2	0	2	0	0
3,4′-Oxydiphenylene dodecanediamide	10	2	2	0	2	1	2	0	0
4,4′-Oxydiphenylene dodecanediamide	10	2	2	0	2	1	2	0	0
Decamethylene 4,4′-biphenylenediurethane	10	2	2	0	2	2	2	0	0
Hexamethylene 4,4′-(ethylenediphenylene)dioxydiacetamide	10	2	2	0	2	2	2	0	0
Nonamethylene 4,4′-(methylenediphenylene)diurethane	10	2	2	0	2	2	2	0	0
Decamethylene 3,3′-phosphinylidenedibenzamide	10	2	2	0	3	1	2	0	1
Decamethylene 4,4′-phosphinylidenedibenzamide	10	2	2	0	3	1	2	0	1
Decamethylene 3,3′-(methylphosphinylidene)dibenzamide	10	2	2	1	5	1	2	0	1
Decamethylene 4,4′-(methylphosphinylidene)dibenzamide	10	2	2	1	5	1	2	0	1
4,4′-Vinylenediphenylene dodecanedioate	10	2	2	2	2	2	0	0	0
Decamethylene 4,4′-ethylidenedibenzamide	10	2	2	2	6	0	2	0	0
N,N′-Diethyl(methylenediphenylene)azelamide	10	2	2	2	6	0	2	0	0
4,4′-Methylvinylene)diphenylene dodecanedioate	10	2	2	3	4	2	0	0	0
3,3′-(Decamethylenedioxy)diphenylene terephthalate	10	2	3	0	0	4	0	0	0
4,4′-(Decamethylenedioxy)diphenylene terephthalate	10	2	3	0	0	4	0	0	0
Decamethylene 4,4′-(phenylphosphinylidene)dibenzamide	10	2	3	0	3	1	2	0	1
(Decamethylenedioxy)diphenylenedioxydiphenylene terephthalate	10	2	5	0	0	6	0	0	0
Hexamethylene ketodipropionamide	10	3	0	0	2	0	2	0	0
Hexamethylene *N,N′*-lysyldiurea	10	3	0	1	6	1	4	0	0
Hexamethylene methyl-*N,N′*-lysyldiurea	10	3	0	2	8	1	4	0	0
Decamethylene 1,4-dihydro-4-oxo-2,6-pyridinedicarboxamide	10	3	0	4	5	0	3	0	0
α-D-Glucotrivalerate, 1,4-	10	3	0	8	14	5	0	0	0
β-D-Glucotrivalerate, 1,4-	10	3	0	8	14	5	0	0	0
Ethylene sebacoyldiurethane	10	4	0	0	2	2	2	0	0
Hexamethylene adipoyldiurethane	10	4	0	0	2	2	2	0	0
Ethylene oxamide-*alt*-ethylene suberamide	10	4	0	0	4	0	4	0	0
Ethylene sebacoyldiurea	10	4	0	0	4	0	4	0	0
Trimethylene adipamide-*alt*-trimethylene oxamide	10	4	0	0	4	0	4	0	0
2-Hydroxytrimethylene sebacoyldiurethane	10	4	0	1	4	3	2	0	0
Decamethylene bromopyromellitimide	10	4	0	6	1	0	2	0	1
Decamethylene pyromellitimide	10	4	0	6	2	0	2	0	0
Decamethylene 3,3′,4,4′-oxydiphenylenetetracarboximide	10	4	0	12	6	1	2	0	0
Hexamethylene *N,N′*-piperazinediterephthalamate	10	4	2	0	0	2	2	0	0
Heptamethylene *N,N′*-trimethylenediterephthalamate	10	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-tetramethylenediterephthalamate	10	4	2	0	2	2	2	0	0
Octamethylene *N,N′*-ethylenediterephthalamate	10	4	2	0	2	2	2	0	0
Pentamethylene *N,N′*-pentamethylenediterephthalamate	10	4	2	0	2	2	2	0	0
Tetramethylene *N,N′*-hexamethylenediterephthalamate	10	4	2	0	2	2	2	0	0
Trimethylene *N,N′*-azelaoyldi(*p*-aminobenzoate)	10	4	2	0	2	2	2	0	0
Decamethylene 4,4′-terephthaloyldioxydibenzoate	10	4	3	0	0	4	0	0	0
(Benzyloxycarbonyllysyl)(benzyloxycarbonyllysyl)valine	10	5	2	6	17	2	5	0	0
Alanylglycyl)5-glutamylglycine	10	?	0	11	30	1	12	0	0
11	11	0	0	0	0	0	0	0	0
Hexamethylenethiopentamethylene sulfide	11	0	0	0	0	0	0	2	0
Decamethyleneoxymethylene oxide	11	0	0	0	0	2	0	0	0
Hexamethylenesulfonylpentamethylenesulfone	11	0	0	0	0	4	0	2	0
Cyclohexylidenethiohexamethylene sulfide	11	0	0	1	0	0	0	2	0
1-Methyldodecamethylene	11	0	0	2	4	0	0	0	0
1-Tridecene	11	0	0	2	4	0	0	0	0
n-Hexyl-*n*-heptylsilylene	11	0	0	2	6	0	0	0	1

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
3-Dodecyl-2,5-thiophene	11	0	0	5	4	0	0	1	0
4,4′-Ethylene-3-methyldiphenyleneoxynonamethylene oxide	11	0	1	7	6	2	0	0	0
Hexamethylenethio-4,4′-methylenediphenylenediethylenesulfide	11	0	2	0	0	0	0	2	0
4,4′-Nonamethylenedioxydiphenyleneoxy *p*-xylylene oxide	11	0	3	0	0	4	0	0	0
12-Aminolauric acid	11	1	0	0	1	0	1	0	0
13-Amino-11-thiatridecanoic acid	11	1	0	0	1	0	1	1	0
13-Amino-12-thiatridecanoic acid	11	1	0	0	1	0	1	1	0
N-Methyl-12-aminolauric acid	11	1	0	1	3	0	1	0	0
N-Ethyl-11-aminoundecanoic acid	11	1	0	1	4	0	1	0	0
Vinyl laurate	11	1	0	2	4	1	0	0	0
Hexamethylene dithiolpimelate	11	2	0	0	0	0	0	2	0
Piperazine azelamide	11	2	0	0	0	0	2	0	0
Piperazine (trimethylenedithio)dipropionamide	11	2	0	0	0	0	2	2	0
Tridecanedioic anhydride	11	2	0	0	0	1	0	0	0
Decamethylene malonate	11	2	0	0	0	2	0	0	0
Hexamethylene pimelate	11	2	0	0	0	2	0	0	0
Pentamethylene suberate	11	2	0	0	0	2	0	0	0
Tetramethylene azelate	11	2	0	0	0	2	0	0	0
Trimethylene sebacate	11	2	0	0	0	2	0	0	0
Trimethylene thiodivalerate	11	2	0	0	0	2	0	1	0
Oxydiethylene azelate	11	2	0	0	0	3	0	0	0
(Ethylenedioxy)diethylene pimelate	11	2	0	0	0	4	0	0	0
Trimethylene sulfonyldivalerate	11	2	0	0	0	4	0	1	0
Decamethylene malonamide	11	2	0	0	2	0	2	0	0
Glycyl-11-aminoundecanoic acid	11	2	0	0	2	0	2	0	0
Heptamethylene adipamide	11	2	0	0	2	0	2	0	0
Hexamethylene pimelamide	11	2	0	0	2	0	2	0	0
Methylene dodecanediamide	11	2	0	0	2	0	2	0	0
Pentamethylene suberamide	11	2	0	0	2	0	2	0	0
Tetramethylene azelamide	11	2	0	0	2	0	2	0	0
Ethylene nonamethylenediurethane	11	2	0	0	2	2	2	0	0
Pentamethylene hexamethylenediurethane	11	2	0	0	2	2	2	0	0
Trimethylene octamethylenediurethane	11	2	0	0	2	2	2	0	0
Ethylene (pentamethylenedithio)diethylenediurethane	11	2	0	0	2	2	2	2	0
Pentamethylene (ethylenedithio)diethylenediurethane	11	2	0	0	2	2	2	2	0
Trimethylene (tetamethylenedithio)diethylenediurethane	11	2	0	0	2	2	2	2	0
Hexamethylene pentamethylenediurea	11	2	0	0	4	0	4	0	0
1,4-Cyclohexylenedimethylene pimelate	11	2	0	2	2	2	0	0	0
1,3-Cyclohexylene azelamide	11	2	0	2	4	0	2	0	0
1,4-Cyclohexylenedimethylene pimelamide	11	2	0	2	4	0	2	0	0
Cyclopropylenedimethylene sebacamide	11	2	0	2	4	0	2	0	0
Heptamethylene 4-octenediamide	11	2	0	2	4	0	2	0	0
Methylpiperazine 1,4-piperazinedipropionamide	11	2	0	2	4	0	4	0	0
Piperazine (methyl-1,4-piperazine)dipropionamide	11	2	0	2	4	0	4	0	0
Pentamethylene 2,5-tetrahydrofurandipropionamide	11	2	0	2	4	1	2	0	0
1,4-Cyclohexylene (trimethylenedithio)diethylenediurethane	11	2	0	2	4	2	2	2	0
Hexamethylene 2-hexylmalonamide	11	2	0	2	6	0	2	0	0
2-Butyl-2-ethyltrimethylene hexamethylenediurethane	11	2	0	3	8	2	2	0	0
Decamethylene (methylene-2,5-furan)dicarboxamide	11	2	0	4	4	1	2	0	0
N,N′-dibutyl-3,3′-dimethyl-4,4′-methylenediphenylene adipamide	11	2	0	16	18	0	2	0	0
N,N′-dipropylhexamethylene dimethylmethylenediphenylenediurea	11	2	0	16	20	0	4	0	0
Hexamethylene (diemthyldipropyl)methylenediphenylenediurea	11	2	0	16	20	0	4	0	0
Hexamethylene *p*-(carboxyphenoxy)caproate	11	2	1	0	0	3	0	0	0
p-Phenylenediethylene azelamide	11	2	1	0	2	0	2	0	0
Undecamethylene terephthalamide	11	2	1	0	2	0	2	0	0
p-Xylylene undecanediamide	11	2	1	0	2	0	2	0	0
p-Xylylene 5-oxadodecanediamide	11	2	1	0	2	1	2	0	0
p-Xylylene 6-oxadodecanediamide	11	2	1	0	2	1	2	0	0
Decamethylene 3,3′-methylenedibenzamide	11	2	2	0	2	0	2	0	0
Decamethylene 4,4′-methylenedibenzamide	11	2	2	0	2	0	2	0	0
4,4′-Methylenediphenylene dodecanediamide	11	2	2	0	2	0	2	0	0
Decamethylene 4,4′-(methylenediphenylene)diurethane	11	2	2	0	2	2	2	0	0
Hexamethylene 4,4′-(trimethylenediphenylene)dioxydiacetamide	11	2	2	0	2	2	2	0	0
Decamethylene 4,4′-(methylenediphenylene)diurea	11	2	2	0	4	0	4	0	0
N,N′-diethyl(methylenediphenylene) sebacamide	11	2	2	2	6	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Tetraoxa-8,10-octadecadiyne (4,4′-methylenediphenylene)diurethane	11	2	2	4	2	6	2	0	0
Trimethylene sebacoyldiurethane	11	4	0	0	2	2	2	0	0
Ethylene azelamide-*alt*-ethylene oxamide	11	4	0	0	4	0	4	0	0
Undecamethylene pyromellitimide	11	4	0	6	2	0	2	0	0
Heptamethylene *N,N′*-tetramethyleneditерephthalamate	11	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-pentamethyleneditерephthalamate	11	4	2	0	2	2	2	0	0
Nonmethylene *N,N′*-ethyleneditерephthalamate	11	4	2	0	2	2	2	0	0
Octamethylene *N,N′*-trimethyleneditерephthalamate	11	4	2	0	2	2	2	0	0
Pentamethylene *N,N′*-hexamthyleneditерephthalamate	11	4	2	0	2	2	2	0	0
Tetramethylene *N,N′*-azelaoyldi(*p*- aminobenzoate)	11	4	2	0	2	2	2	0	0
Trimethylene *N,N′*-sebacoyldi(*p*-aminobenzoate)	11	4	2	0	2	2	2	0	0
Alanylglycylprolylprolylglycylproline	11	6	0	5	10	0	6	0	0
Alanylprolylglycylprolylprolylglycine	11	6	0	5	10	0	6	0	0
12	12	0	0	0	0	0	0	0	0
Dodecamethylene	12	0	0	0	0	0	0	0	0
(Decamethylenedithio)ethylene disulfide	12	0	0	0	0	0	0	4	0
(Octamethylenedithio)tetramethylene disulfide	12	0	0	0	0	0	0	4	0
Dodecamethylene oxide	12	0	0	0	0	1	0	0	0
1-Tetradecene	12	0	0	2	4	0	0	0	0
Octamethylene tetramethylenedithiourea	12	0	0	2	4	0	4	2	0
Dodecylvinyl ether	12	0	0	2	4	1	0	0	0
Di-*n*-heptylsilylene	12	0	0	2	6	0	0	0	1
4,4′-decamethylenedioxydiphenyleneoxy *p*-xylylene oxide	12	0	3	0	0	4	0	0	0
13-Aminotridecanoic acid	12	1	0	0	1	0	1	0	0
N-Allyl-11-aminoundecanoic acid	12	1	0	1	1	0	1	0	0
N-Dodecanoylethylene amine	12	1	0	1	3	0	1	0	0
4-Hydroxytetramethyelne-2-(1,3-dioxolane)caprylic acid	12	1	0	2	2	3	0	0	0
Dodecyl acrylate	12	1	0	2	4	1	0	0	0
n-Dodecyl acrylamide	12	1	0	2	5	0	1	0	0
Hexamethylene dithiolsuberate	12	2	0	0	0	0	0	2	0
Tetramethylene dithiolsebacate	12	2	0	0	0	0	0	2	0
Piperazine sebacamide	12	2	0	0	0	0	2	0	0
Piperazine 1,4-piperazinedipropionamide	12	2	0	0	0	0	4	0	0
Tetradecanedioic anhydride	12	2	0	0	0	1	0	0	0
Decamethylene succinate	12	2	0	0	0	2	0	0	0
Hexamethylene suberate	12	2	0	0	0	2	0	0	0
Octamethylene adipate	12	2	0	0	0	2	0	0	0
Pentamethylene azelate	12	2	0	0	0	2	0	0	0
Tetramethylene sebacate	12	2	0	0	0	2	0	0	0
Trimethylene undecanedioate	12	2	0	0	0	2	0	0	0
Tetramethylene thiodivalerate	12	2	0	0	0	2	0	1	0
Decamethylene oxydiacetate	12	2	0	0	0	3	0	0	0
Oxydiethylene sebacate	12	2	0	0	0	3	0	0	0
Oxydiethylene thiodivalerate	12	2	0	0	0	3	0	1	0
(Ethylenedioxy)diethylene suberate	12	2	0	0	0	4	0	0	0
Tetramethylene sulfonyldivalerate	12	2	0	0	0	4	0	1	0
Oxydiethylene sulfonyldivalerate	12	2	0	0	0	5	0	0	0
Decamethylene succinamide	12	2	0	0	2	0	2	0	0
Dodecamethylene oxamide	12	2	0	0	2	0	2	0	0
Ethylene doecanediamide	12	2	0	0	2	0	2	0	0
Glycyl-12-aminododecanoic acid	12	2	0	0	2	0	2	0	0
Heptamethylene pimelamide	12	2	0	0	2	0	2	0	0
Hexamethylene suberamide	12	2	0	0	2	0	2	0	0
Octamethylene adipamide	12	2	0	0	2	0	2	0	0
Pentamethylene azelamide	12	2	0	0	2	0	2	0	0
Tetramethylene sebacamide	12	2	0	0	2	0	2	0	0
Hexamethylene thiodibutyramide	12	2	0	0	2	0	2	1	0
Dithiodiethylene sebacamide	12	2	0	0	2	0	2	2	0
(Hexamethylenedicarbamoyl)dithiohexamethylene disulfide	12	2	0	0	2	0	2	4	0
Hexamethylene 1,4-piperazinediacetamide	12	2	0	0	2	0	4	0	0
Hexamethylene oxydibutyramide	12	2	0	0	2	1	2	0	0
Ethylene decamethylenediurethane	12	2	0	0	2	2	2	0	0
(Ethylenedioxy)ditrimethylene adipamide	12	2	0	0	2	2	2	0	0
Hexamethylene hexamethylenediurethane	12	2	0	0	2	2	2	0	0
Tetramethylene octamethylenediurethane	12	2	0	0	2	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Ethylene (hexamethylenedithio)diethylenediurethane	12	2	0	0	2	2	2	2	0
Hexamethylene (ethylenedithio)diethylenediurethane	12	2	0	0	2	2	2	2	0
Tetramethylene (tetramethylenedithio)diethylenediurethane	12	2	0	0	2	2	2	2	0
Triemthylene (pentamethylenedithio)diethylenediurethane	12	2	0	0	2	2	2	2	0
(Ethylenedioxy)diethylene 1,4-piperazinediacetamide	12	2	0	0	2	2	4	0	0
Octamethylene disiloxanylenedipropionamide	12	2	0	0	6	1	2	0	2
1,4-Cyclohexylenedimethylene suberate	12	2	0	2	2	2	0	0	0
1,4-Cyclohexylene sebacate	12	2	0	2	2	2	0	0	0
Decamethylene 3-hexenedioate	12	2	0	2	2	2	0	0	0
Octamethylene 4-octenedioate	12	2	0	2	2	2	0	0	0
1,3-Cyclohexylenedimethylene suberamide	12	2	0	2	4	0	2	0	0
1,4-Cyclohexylenedimethylene suberamide	12	2	0	2	4	0	2	0	0
1,3-Cyclohexylene sebacamide	12	2	0	2	4	0	2	0	0
Hexamethylene 1,2-cyclohexylenediacetamide	12	2	0	2	4	0	2	0	0
Octamethylene 4-octenediamide	12	2	0	2	4	0	2	0	0
Hexamethylene 2,5-tetrahydrofurandipropionamide	12	2	0	2	4	1	2	0	0
1,4-Cyclohexylene octamethylenediurethane	12	2	0	2	4	2	2	0	0
1,4-Dioxanylene-2,5-dimethylene sebacamide	12	2	0	2	4	2	2	0	0
1,4-Cyclohexylenedimethylene ethylenedithiodiethylenediurethane	12	2	0	2	4	2	2	2	0
1,4-Cyclohexylene (tetramethylenedithio)diethylenediurethane	12	2	0	2	4	2	2	2	0
Hexamthylene 2,5-pyrrolidinedipropionamide	12	2	0	2	5	0	3	0	0
Hexamethylene heptylmalonamide	12	2	0	2	6	0	2	0	0
Hexamethylene (*N*-methyl-2,5-pyrrolidine)dipropionamide	12	2	0	3	7	0	3	0	0
Hexamethylene (diethyldimethyldisiloxanylene)dipropionamid	12	2	0	4	14	1	2	0	2
Octamethylene (tetramethyldisiloxanylene)dipropionamide	12	2	0	4	14	1	2	0	2
N,N′-Dibutyl(hexafluoropentamethylene) adipamide	12	2	0	5	6	0	2	0	6
N,N′-Dibutylhexamethylene perfluoroglutaramide	12	2	0	5	6	0	2	0	6
Dodecamethylene 3,3′-dimethyl-4,4′-biphenylenediurethane	12	2	0	14	14	2	2	0	0
p-Xylylene Dodecanedioate	12	2	1	0	0	2	0	0	0
Ethylene *p*-(carboxyphenoxy)undecanoate	12	2	1	0	0	3	0	0	0
Decamethylene *p*-phenylenediacetamide	12	2	1	0	2	0	2	0	0
Dodecamethylene terephthalamide	12	2	1	0	2	0	2	0	0
Ocatmethylene *p*-phenylenedipropionamide	12	2	1	0	2	0	2	0	0
p-Phenylenediethylene sebacamide	12	2	1	0	2	0	2	0	0
p-Xylylene dodecanediamide	12	2	1	0	2	0	2	0	0
p-Phenylenediethylene thiodivaleramide	12	2	1	0	2	0	2	1	0
p-Xylylene thiodicaproamide	12	2	1	0	2	0	2	1	0
p-Xylylene 6-oxatridecanediamide	12	2	1	0	2	1	2	0	0
p-Xylylene oxydicaproamide	12	2	1	0	2	1	2	0	0
Decamethylene (*p*-phenylenedioxy)diacetamide	12	2	1	0	2	2	2	0	0
Decamethylene *p*-xylylenediurethane	12	2	1	0	2	2	2	0	0
p-Phenylenediethylene octamethylenediurethane	12	2	1	0	2	2	2	0	0
p-Phenyleneditrimethylene hexamethylenediurethane	12	2	1	0	2	2	2	0	0
(*p*-Phenylenedioxy)diethylene octamethylenediurethane	12	2	1	0	2	4	2	0	0
Decamethylene *p*-xylylenediurea	12	2	1	0	4	0	4	0	0
(1,4-Cyclohexylenedixoy)ditrimethylene *p*-phenylenedioxydiacetamide	12	2	1	2	4	4	2	0	0
Decamethylene 4,4′-(ethylenedioxy)dibenzoate	12	2	2	0	0	4	0	0	0
Hexamethylene 4,4′-(hexamethylenedisulfonyl)dibenzoate	12	2	2	0	0	6	0	2	0
Dodecamethylene 4,4′-biphenylenediurethane	12	2	2	0	2	2	2	0	0
4,4′-(Ethylenedioxydiphenylene)ditrimethylene adipamide	12	2	2	0	2	2	2	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene ethylenedioxydibenzamide	12	2	2	2	4	4	2	0	0
Decamethylene isopropylidenedi-*p*-phenylenedioxydiacetamide	12	2	2	3	8	2	2	0	0
6-Aminocaproyl-5-aminovaleryl-3-aminopropionic acid	12	3	0	0	3	0	3	0	0
3-Aminopropyl-5-aminovaleryl-6-aminocaproic acid	12	3	0	0	3	0	3	0	0
Hexamethylene 5-pyrrolidonylidenedipropionamide	12	3	0	1	3	0	2	0	0
Decamethylene 3-carboxylglutaramide	12	3	0	1	4	1	2	0	0
Adipoyl octamethylenediurea	12	4	0	0	4	0	4	0	0
Ethylene oxamide-*alt*-ethylene sebacamide	12	4	0	0	4	0	4	0	0
Tetramethylene adipamide-*alt*-tetramethylene oxamide	12	4	0	0	4	0	4	0	0
Trimethylene oxamide-*alt*-trimethylene suberamide	12	4	0	0	4	0	4	0	0
Dodecamethylene bromopyromellitimide	12	4	0	6	1	0	2	0	1
Dodecamethylene pyromellitimide	12	4	0	6	2	0	2	0	0
Dodecamethylene 1,4,5,8-naphthalenetetracarboximide	12	4	0	10	4	0	2	0	0
Ethylene *N,N′*-terephthaloyldi(6-aminocaproate)	12	4	1	0	2	2	2	0	0
Decamethylene *N,N′*-ethylenediterephthalamate	12	4	2	0	2	2	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Ethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)	12	4	2	0	2	2	2	0	0
Heptamethylene *N,N′*-pentamethylenediterephthalamate	12	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-hexamethylenediterephthalamate	12	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-suberoyldi(*p*-aminobenzoate)	12	4	2	0	2	2	2	0	0
Nonamethylene *N,N′*-glutaryldi(*p*-aminobenzoate)	12	4	2	0	2	2	2	0	0
Nonamethylene *N,N′*-trimethylenediterephthalamate	12	4	2	0	2	2	2	0	0
Octamethylene *N,N′*-tetramethylenediterephthalamate	12	4	2	0	2	2	2	0	0
(Alanylglycyl)6-glutamylglycine	12	?	0	13	34	1	14	0	0
13	13	0	0	0	0	0	0	0	0
(Nonamethylenedithio)tetramethylene disulfide	13	0	0	0	0	0	0	4	0
1-Pentadecene	13	0	0	2	4	0	0	0	0
3-Tetradecyl-2,5-thiophene	13	0	0	5	4	0	0	1	0
4,4′-Ethylene-3-methyldiphenyleneoxyundecamethylene oxide	13	0	1	7	6	2	0	0	0
4,4′-Undecamethylenedioxydiphenyleneoxy *p*-xylylene oxide	13	0	3	0	0	4	0	0	0
Vinyl myristate	13	1	0	2	4	1	0	0	0
Hexamethylene dithiolazelate	13	2	0	0	0	0	0	2	0
Decamethylene glutarate	13	2	0	0	0	2	0	0	0
Hexamethylene azelate	13	2	0	0	0	2	0	0	0
Pentamethylene sebacate	13	2	0	0	0	2	0	0	0
Trimethylene dodecandioate	13	2	0	0	0	2	0	0	0
Pentamethylene thiodivalerate	13	2	0	0	0	2	0	1	0
(Ethylenedioxy)diethylene azelate	13	2	0	0	0	4	0	0	0
Pentamethylene sulfonyldivalerate	13	2	0	0	0	4	0	1	0
Deamethylene glutaramide	13	2	0	0	2	0	2	0	0
Dodecamethylene malonamide	13	2	0	0	2	0	2	0	0
Heptamethylene suberamide	13	2	0	0	2	0	2	0	0
Hexamethylene azelamide	13	2	0	0	2	0	2	0	0
Nonamethylene adipamide	13	2	0	0	2	0	2	0	0
Pentamethylene sebacamide	13	2	0	0	2	0	2	0	0
Tetramethylene undecanediamide	13	2	0	0	2	0	2	0	0
Hexamethylene 5-oxasebacamide	13	2	0	0	2	1	2	0	0
Heptamethylene hexamethylenediurethane	13	2	0	0	2	2	2	0	0
Tetramethylene nonamethylenediurethane	13	2	0	0	2	2	2	0	0
Pentamethylene (tetramethylenedithio)diethylenediurethane	13	2	0	0	2	2	2	2	0
Tetramethylene (pentamethylenedithio)diethylenediurethane	13	2	0	0	2	2	2	2	0
Trimethylene (hexamethylenedithio)diethylenediurethane	13	2	0	0	2	2	2	2	0
Heptamethylene hexamethylenediurea	13	2	0	0	4	0	4	0	0
Nonamethylene disiloxanylenedipropionamide	13	2	0	0	6	1	2	0	2
1,4-Cyclohexylenedimethylene azelate	13	2	0	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene azelamide	13	2	0	2	4	0	2	0	0
Nonamethylene 4-octenediamide	13	2	0	2	4	0	2	0	0
Decamethylene (methylene-2,5-tetrahydrofuran)dicarboxamide	13	2	0	2	4	1	2	0	0
Heptamethylene 2,5-tetrahydrofurandipropionamide	13	2	0	2	4	1	2	0	0
Decamethylene 3-methyladipate	13	2	0	2	4	2	0	0	0
1,4-Cyclohexylenedimethylene trimethylenedithiodiethylenediurethane	13	2	0	2	4	2	2	2	0
1,4-Cyclohexylene (pentamethylenedithio)diethylenediurethane	13	2	0	2	4	2	2	2	0
Dodecyl glutamate	13	2	0	2	5	1	1	0	0
Hexamethylene 2-octylmalonamide	13	2	0	2	6	0	2	0	0
3-Methylhexamethylene sebacamide	13	2	0	2	6	0	2	0	0
Tetramethylene 4,4′-methylenedicyclohexylenediurethane	13	2	0	4	6	2	2	0	0
p-Phenylenediethylene undecanediamide	13	2	1	0	2	0	2	0	0
p-Xylylene tridecanediamide	13	2	1	0	2	0	2	0	0
p-Xylylene 5-oxatetradecanediamide	13	2	1	0	2	1	2	0	0
Dodecamethylene 4,4′-(methylenediphenylene)diurethane	13	2	2	0	2	2	2	0	0
Dodecamethylene 4,4′-(methylenediphenylene)diurea	13	2	2	0	4	0	4	0	0
N,N′-Lysyl nonamethylenediurea	13	3	0	1	6	1	4	0	0
Methyl-*N,N′*-lysyl nonamethylenediurea	13	3	0	2	8	1	4	0	0
β-D-Glucotricaproate, 1,4-	13	3	0	8	14	5	0	0	0
Pentamethylene sebacoyldiurea	13	4	0	0	4	0	4	0	0
Tetramethylene oxamide-*alt*-tetramethylene pimelamide	13	4	0	0	4	0	4	0	0
Trimethylene azelamide-*alt*-trimethylene oxamide	13	4	0	0	4	0	4	0	0
Tridecamethylene pyromellitimide	13	4	0	6	2	0	2	0	0
Decamethylene *N,N′*-trimethylenediterephthalamate	13	4	2	0	2	2	2	0	0
Heptamethylene *N,N′*-hexamethylenediterephthalamate	13	4	2	0	2	2	2	0	0
Nonamethylene *N,N′*-adipoyldi(*p*-aminobenzoate)	13	4	2	0	2	2	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Nonamethylene *N,N'*-tetramethylenediterephthalamate	13	4	2	0	2	2	2	0	0
Octamethylene *N,N'*-pentamethylenediterephthalamate	13	4	2	0	2	2	2	0	0
Undecamethylene *N,N'*-ethylenediterephthalamate	13	4	2	0	2	2	2	0	0
14	14	0	0	0	0	0	0	0	0
Tetradecamethylene sulfide	14	0	0	0	0	0	0	1	0
(Decamethylenedithio)tetramethylene disulfide	14	0	0	0	0	0	0	4	0
(Hexamethylenedithio)octamethylene disulfide	14	0	0	0	0	0	0	4	0
1-Hexadecene	14	0	0	2	4	0	0	0	0
Hexamethylene octamethylenedithiourea	14	0	0	2	4	0	4	2	0
Di-*n*-octylsilylene	14	0	0	2	6	0	0	0	1
3,3′-Dioctyl-2,2′-bithiophene	14	0	0	10	8	0	0	2	0
3,3′′-Dioctyl-2,2′:5′,2′′-terthiophene	14	0	0	14	10	0	0	3	0
4,4′-Dodecamethylenedioxydiphenyleneoxy *p*-xylylene oxide	14	0	3	0	0	4	0	0	0
N-Piperazinyl-11-aminoundecanoic acid	14	1	0	0	1	0	3	0	0
Tetradecyl acrylate	14	1	0	2	4	1	0	0	0
n-Tetradecyl acrylamide	14	1	0	2	5	0	1	0	0
Decamethylene dithioladipate	14	2	0	0	0	0	0	2	0
Hexamethylene dithiolsebacate	14	2	0	0	0	0	0	2	0
Decamethylene adipate	14	2	0	0	0	2	0	0	0
Hexamethylene sebacate	14	2	0	0	0	2	0	0	0
Octamethylene suberate	14	2	0	0	0	2	0	0	0
Hexamethylene thiodivalerate	14	2	0	0	0	2	0	1	0
(Ethylenedioxy)diethylene sebacate	14	2	0	0	0	4	0	0	0
(Ethylenedioxy)diethylene thiodivalerate	14	2	0	0	0	4	0	1	0
Hexamethylene sulfonyldivalerate	14	2	0	0	0	4	0	1	0
(Ethylenedioxy)diethylene sulfonyldivalerate	14	2	0	0	0	6	0	0	0
Decamethylene adipamide	14	2	0	0	2	0	2	0	0
Dodecamethylene succinamide	14	2	0	0	2	0	2	0	0
Heptamethylene azelamide	14	2	0	0	2	0	2	0	0
Hexamethylene sebacamide	14	2	0	0	2	0	2	0	0
Nonamethylene pimelamide	14	2	0	0	2	0	2	0	0
Octamethylene suberamide	14	2	0	0	2	0	2	0	0
Pentamethylene undercanediamide	14	2	0	0	2	0	2	0	0
Tetramethylene dodecanediamide	14	2	0	0	2	0	2	0	0
Ethylene thiodienanthamide	14	2	0	0	2	0	2	1	0
Hexamethylene thiodivaleramide	14	2	0	0	2	0	2	1	0
Hexamethylene 5-oxaundecanediamide	14	2	0	0	2	1	2	0	0
Hexamethylene oxydivaleramide	14	2	0	0	2	1	2	0	0
Decamethylene tetramethylenediurethane	14	2	0	0	2	2	2	0	0
Hexamethylene (tetramethylenedioxy)dipropionamide	14	2	0	0	2	2	2	0	0
Octamethylene hexamethylenediurethane	14	2	0	0	2	2	2	0	0
Hexamethylene sulfonyldivaleramide	14	2	0	0	2	2	2	1	0
Thioditetramethylene hexamethylenediurethane	14	2	0	0	2	2	2	1	0
Hexamethylene (tetramethylenedithio)diethylenediurethane	14	2	0	0	2	2	2	2	0
Pentamethylene (pentamethylenedithio)diethylenediurethane	14	2	0	0	2	2	2	2	0
Tetramethylene (hexamethylenedithio)diethylenediurethane	14	2	0	0	2	2	2	2	0
1,4-Piperazinediethylene hexamethylenediurethane	14	2	0	0	2	2	4	0	0
Oxyditetramethylene hexamethylenediurethane	14	2	0	0	2	3	2	0	0
Decamethylene phosphinylidenedipropionamide	14	2	0	0	3	1	2	0	1
Hexamethylene octamethylenediurea	14	2	0	0	4	0	4	0	0
Decamethylene disiloxanylenedipropionamide	14	2	0	0	6	1	2	0	2
Hexamethylene 6-hydroxyundecanediamide	14	2	0	1	4	1	2	0	0
(Ethylenedioxy)diethylene 6-hydroxyundecanediamide	14	2	0	1	4	3	2	0	0
Decamethylene (methylphosphinylidene)dipropionamide	14	2	0	1	5	1	2	0	1
1,4-Cyclohexylenedimethylene sebacate	14	2	0	2	2	2	0	0	0
Decamethylene 4-octenedioate	14	2	0	2	2	2	0	0	0
Dodecamethylene 3-hexenedioate	14	2	0	2	2	2	0	0	0
1,4-Cyclohexylenedimethylene sebacamide	14	2	0	2	4	0	2	0	0
Decamethylene 4-octenediamide	14	2	0	2	4	0	2	0	0
Octamethylene 2,5-tetrahydrofurandipropionamide	14	2	0	2	4	1	2	0	0
1,4-Cyclohexylenedimethylene octamethylenediurethane	14	2	0	2	4	2	2	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene adipamide	14	2	0	2	4	2	2	0	0
1,4-Cyclohexylenedimethylene tetramethylenedithiodiethylenediurethane	14	2	0	2	4	2	2	2	0
1,4-Cyclohexylene (hexamethylenedithio)diethylenediurethane	14	2	0	2	4	2	2	2	0
Hexamethylene nonylmalonamide	14	2	0	2	6	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene octamethylenediurethane	14	2	0	2	6	2	2	0	0
Decamethylene 2,5-Furandipropionamide	14	2	0	4	4	1	2	0	0
4,4′-Methylenedicyclohexylene pimelamide	14	2	0	4	6	0	2	0	0
Decamethylene (tetramethyldisiloxanylene)dipropionamide	14	2	0	4	14	1	2	0	2
Hexamethylene (tetraethyldisiloxanylene)dipropionamide	14	2	0	4	14	1	2	0	2
Hexamethylene 4,4′-(*p*-xylylenedisulfonyl)dibutyrate	14	2	1	0	0	6	0	2	0
Decamethylene *p*-phenylenedipropionamide	14	2	1	0	2	0	2	0	0
Dodecamethylene *p*-phenylenediacetamide	14	2	1	0	2	0	2	0	0
p-Phenylenediethylene dodecanediamide	14	2	1	0	2	0	2	0	0
Tetradecamethylene terephthalamide	14	2	1	0	2	0	2	0	0
m-Xylylene tetradecanediamide	14	2	1	0	2	0	2	0	0
p-Xylylene tetradecanediamide	14	2	1	0	2	0	2	0	0
p-Phenylenediethylene thiodicaproamide	14	2	1	0	2	0	2	1	0
p-Xylylene thiodienanthamide	14	2	1	0	2	0	2	1	0
p-Xylylene 6-oxapentadecanediamide	14	2	1	0	2	1	2	0	0
p-Xylylene oxydienanthamide	14	2	1	0	2	1	2	0	0
Dodecamethylene *p*-xylylenediurethane	14	2	1	0	2	2	2	0	0
Dodecamethylene *p*-xylylenediurea	14	2	1	0	4	0	4	0	0
Decamethylene 4,4′-(ethylenediphenylene)dioxydiacetamide	14	2	2	0	2	2	2	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene tetramethylenedioxydibenzamide	14	2	2	2	4	4	2	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene oxydiethylenedioxydibenzamide	14	2	2	2	4	5	2	0	0
Decamethylene adipoyldiurethan	14	4	0	0	2	2	2	0	0
Hexamethylene sebacoyldiurethane	14	4	0	0	2	2	2	0	0
Pentamethylene adipamide-*alt*-pentamethylene oxamide	14	4	0	0	4	0	4	0	0
Tetramethylene oxamide-*alt*-tetramethylene suberamide	14	4	0	0	4	0	4	0	0
Trimethylene oxamide-*alt*-trimethylene sebacamide	14	4	0	0	4	0	4	0	0
Tetramethylene *N,N′*-terephthaloyldi(6-aminocaproate)	14	4	1	0	2	2	2	0	0
Decamethylene *N,N′*-tetramethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
Dodecamethylene *N,N′*-ethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
Ethylene *N,N′*-dodecamethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
Nonamethylene *N,N′*-pentamethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
Octamethylene *N,N′*-hexamethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
Tetramethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)	14	4	2	0	2	2	2	0	0
Trimethylene *N,N′*-tridecanedioyldi(*p*-aminobenzoate)	14	4	2	0	2	2	2	0	0
Undecamethylene *N,N′*-trimethyleneditерephthalamate	14	4	2	0	2	2	2	0	0
15	15	0	0	0	0	0	0	0	0
(Hexamethylenedithio)nonamethylene disulfide	15	0	0	0	0	0	0	4	0
Methyleneoxytetradecamethylene oxide	15	0	0	0	0	2	0	0	0
3-Hexadecyl-2,5-thiophene	15	0	0	5	4	0	0	1	0
Vinyl palmitate	15	1	0	2	4	1	0	0	0
Decamethylene pimelate	15	2	0	0	0	2	0	0	0
Heptamethylene sebacate	15	2	0	0	0	2	0	0	0
Pentamethylene dodecanedioate	15	2	0	0	0	2	0	0	0
Trimethylene tetradecanedioate	15	2	0	0	0	2	0	0	0
6-Aminocaproic acid-*alt*-11-aminoundecanoic acid	15	2	0	0	2	0	2	0	0
Heptamethylene sebacamide	15	2	0	0	2	0	2	0	0
Hexamethylene undecanediamide	15	2	0	0	2	0	2	0	0
Octamethylene azelamide	15	2	0	0	2	0	2	0	0
Undecamethylene adipamide	15	2	0	0	2	0	2	0	0
Hexamethylene 5-oxadodecanediamide	15	2	0	0	2	1	2	0	0
(Ethylenedioxy)diethylene undecanediamide	15	2	0	0	2	2	2	0	0
(Ethylenedioxy)ditrimethylene azelamide	15	2	0	0	2	2	2	0	0
Nonamethylene hexamethylenediurethane	15	2	0	0	2	2	2	0	0
Nonamethylene hexamethylenediurethane	15	2	0	0	2	2	2	0	0
Tetramethylene undecamethylenediurethane	15	2	0	0	2	2	2	0	0
Hexamethylene (pentamethylenedithio)diethylenediurethane	15	2	0	0	2	2	2	2	0
Pentamethylene (hexamethylenedithio)diethylenediurethane	15	2	0	0	2	2	2	2	0
Hexamethylene nonamethylenediurea	15	2	0	0	4	0	4	0	0
4,4′-(Trimethylenedipiperidenediyl)dicarbamoyldithiotetramethylene disul	15	2	0	2	2	0	2	4	0
Nonamethylene 2,5-tetrahydrofurandipropionamide	15	2	0	2	4	1	2	0	0
1,4-Cyclohexylenedimethylene pentamethylenedithiodiethylenediurethane	15	2	0	2	4	2	2	2	0
Tetramethylene (trimethylenedipiperidene)dithiodicarboxylate	15	2	0	2	4	2	2	2	0
Hexamethylene 2-decylmalonamide	15	2	0	2	6	0	2	0	0
Tetramethylene dodecylmalonamide	15	2	0	2	6	0	2	0	0
4,4′-Methylenedicyclohexylene suberamide	15	2	0	4	6	0	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Nonamethylene (diethyldimethyldisiloxanylene)dipropionamide	15	2	0	4	14	1	2	0	2
N,N'-dihexyl-3,3'-dimethyl-4,4'-methylenediphenylene adipamide	15	2	0	16	18	0	2	0	0
p-Phenylenediethylene tridecanediamide	15	2	1	0	2	0	2	0	0
p-Xylylene pentadecanediamide	15	2	1	0	2	0	2	0	0
p-Xylylene 5-oxahexadecanediamide	15	2	1	0	2	1	2	0	0
4,4'Methylenediphenylene tetradecamethylenediurea	15	2	2	0	4	0	4	0	0
Pentamethylene oxamide-*alt*-Pentamethylene pimelamide	15	4	0	0	4	0	4	0	0
Tetramethylene azelamide-*alt*-tetramethylene oxamide	15	4	0	0	4	0	4	0	0
Decamethylene *N,N'*-pentamethylenediterephthalamate	15	4	2	0	2	2	2	0	0
Dodecamethylene *N,N'*-trimethylenediterephthalamate	15	4	2	0	2	2	2	0	0
Nonamethylene *N,N'*-hexamethylenediterephthalamate	15	4	2	0	2	2	2	0	0
Tridecamethylene *N,N'*-ethylenediterephthalamate	15	4	2	0	2	2	2	0	0
Undecamethylene *N,N'*-tetramethylenediterephthalamate	15	4	2	0	2	2	2	0	0
16	16	0	0	0	0	0	0	0	0
(Decamethylenethio)hexamethylene sulfide	16	0	0	0	0	0	0	2	0
(Decamethylenedithio)hexamethylene disulfide	16	0	0	0	0	0	0	4	0
(Tetramethylenedioxy)dihexamethylene oxide	16	0	0	0	0	3	0	0	0
Decamethyleneaminohexamethylene amine	16	0	0	0	2	0	2	0	0
1-Octadecene	16	0	0	2	4	0	0	0	0
Octamethylene octamethylenedithiourea	16	0	0	2	4	0	4	2	0
Di-*n*-nonylsilylene	16	0	0	2	6	0	0	0	1
Tetracosatetrayne	16	0	0	8	2	0	0	0	0
17-Aminoheptadecanoic acid	16	1	0	0	1	0	1	0	0
Hexadecyl acrylate	16	1	0	2	4	1	0	0	0
n-Hexadecyl acrylamide	16	1	0	2	5	0	1	0	0
Hexadecyl methacrylate	16	1	0	3	6	1	0	0	0
Octadecanedioic anhydride	16	2	0	0	0	1	0	0	0
Decamethylene suberate	16	2	0	0	0	2	0	0	0
Dodecamethylene adipate	16	2	0	0	0	2	0	0	0
Nonamethylene azelate	16	2	0	0	0	2	0	0	0
Decamethylene suberamide	16	2	0	0	2	0	2	0	0
Dodecamethylene adipamide	16	2	0	0	2	0	2	0	0
Heptamethylene undecanediamide	16	2	0	0	2	0	2	0	0
Hexamethylene dodecanediamide	16	2	0	0	2	0	2	0	0
Nonamethylene azelamide	16	2	0	0	2	0	2	0	0
Octamethylene sebacamide	16	2	0	0	2	0	2	0	0
Pentamethylene tridecanediamide	16	2	0	0	2	0	2	0	0
Hexamethylene oxydicaproamide	16	2	0	0	2	1	2	0	0
Decamethylene hexamethylenediurethane	16	2	0	0	2	2	2	0	0
(Ethylenedioxy)ditrimethylene sebacamide	16	2	0	0	2	2	2	0	0
Hexamethylene decamethylenediurethane	16	2	0	0	2	2	2	0	0
Hexamethylene (hexamethylenedioxy)dipropionamide	16	2	0	0	2	2	2	0	0
Octamethylene octamethylenediurethane	16	2	0	0	2	2	2	0	0
Oxydipentamethylene oxydibutyramide	16	2	0	0	2	2	2	0	0
Pentamethylene undecamethylenediurethane	16	2	0	0	2	2	2	0	0
Hexamethylene (hexamethylenedithio)diethylenediurethane	16	2	0	0	2	2	2	2	0
Dodecamethylene phosphinylidenedipropionamide	16	2	0	0	3	1	2	0	1
Decamethylene hexamethylenediurea	16	2	0	0	4	0	4	0	0
Dodecamethylene (methylphosphinylidene)dipropionamide	16	2	0	1	5	1	2	0	1
1,4-Cyclohexylenedimethylene dodecanedioate	16	2	0	2	2	2	0	0	0
Dodecamethylene 4-octenedioate	16	2	0	2	2	2	0	0	0
Tetradecamethylene 3-hexenedioate	16	2	0	2	2	2	0	0	0
1,3-Cyclohexylenedimethylene dodecanediamide	16	2	0	2	4	0	2	0	0
Decamethylene 2,5-tetrahydrofurandipropionamide	16	2	0	2	4	1	2	0	0
1,4-Cyclohexylenedimethylene hexamethylenedithiodiethylenediurethane	16	2	0	2	4	2	2	2	0
Hexamethylene undecylmalonamide	16	2	0	2	6	0	2	0	0
Pentamethylene dodecylmalonamide	16	2	0	2	6	0	2	0	0
Tetraoxa-8,10-octadecadiyne hexamethylenediurethane	16	2	0	4	2	6	2	0	0
4,4'-Bicyclohexylene sebacate	16	2	0	4	4	2	0	0	0
4,4'-Methylenedicyclohexylene azelamide	16	2	0	4	6	0	2	0	0
Hexamethylene *p*-(carboxyphenoxy)undecanoate	16	2	1	0	0	3	0	0	0
Dodecamethylene *p*-phenylenedipropionamide	16	2	1	0	2	0	2	0	0
p-Phenylenediethylene tetradecanediamide	16	2	1	0	2	0	2	0	0
Tetradecamethylene *p*-phenylenediacetamide	16	2	1	0	2	0	2	0	0
p-Xylylene hexadecanediamide	16	2	1	0	2	0	2	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
p-Phenylenediethylene thiodienanthamide	16	2	1	0	2	0	2	1	0
p-Xylylene 6-oxaheptadecanediamide	16	2	1	0	2	1	2	0	0
Tetradecamethylene *p*-xylylenediurea	16	2	1	0	4	0	4	0	0
N,N'-Dimethyl-*p*-xylylene hexadecanediamide	16	2	1	2	6	0	2	0	0
(1,4-Cyclohexylenedioxy)ditrimethylene hexamethylenedioxydibenzamide	16	2	2	2	4	4	2	0	0
β-D-Glucotriheptylate, 1,4-	16	3	0	8	14	5	0	0	0
Ethylene *N,N'*-adipoyldi(6-aminocaproate)	16	4	0	0	2	2	2	0	0
Hexamethylene adipamide-*alt*-hexamethylene oxamide	16	4	0	0	4	0	4	0	0
Octamethylene sebacoyldiurea	16	4	0	0	4	0	4	0	0
Pentamethylene oxamide-*alt*-pentamethylene suberamide	16	4	0	0	4	0	4	0	0
Tetramethylene oxamide-*alt*-tetramethylene sebacamide	16	4	0	0	4	0	4	0	0
1,12-Bis(*n*-Butoxycarbonylmethylurethane)-5,7-dodecadiynediol	16	4	0	6	8	4	2	0	0
Hexamethylene *N,N'*-terephthaloyldi(6-aminocaproate)	16	4	1	0	2	2	2	0	0
Decamethylene *N,N'*-hexamethylenediterephthalamate	16	4	2	0	2	2	2	0	0
Dodecamethylene *N,N'*-tetramethylenediterephthalamate	16	4	2	0	2	2	2	0	0
Ethylene *N,N'*-hexadecanedioyldi(*p*-aminobenzoate)	16	4	2	0	2	2	2	0	0
Hexamethylene *N,N'*-dodecanedioyldi(*p*-aminobenzoate)	16	4	2	0	2	2	2	0	0
Nonamethylene *N,N'*-azelaoyldi(*p*-aminobenzoate)	16	4	2	0	2	2	2	0	0
Pentamethylene *N,N'*-tridecanedioyldi(*p*-aminobenzoate)	16	4	2	0	2	2	2	0	0
Tetradecamethylene *N,N'*-ethylenediterephthalamate	16	4	2	0	2	2	2	0	0
Tridecamethylene *N,N'*-trimethylenediterephthalamate	16	4	2	0	2	2	2	0	0
Undecamethylene *N,N'*-pentamethylenediterephthalamate	16	4	2	0	2	2	2	0	0
L-1-Carboxypentamethylene adipamide	16	6	0	2	8	2	4	0	0
17	17	0	0	0	0	0	0	0	0
Hexamethylenethioundecamethylene sulfide	17	0	0	0	0	0	0	2	0
(Pentamethylenedioxy)dihexamethylene oxide	17	0	0	0	0	3	0	0	0
18-Aminostearic acid	17	1	0	0	1	0	1	0	0
Vinyl stearate	17	1	0	2	4	1	0	0	0
Decamethylene azelate	17	2	0	0	0	2	0	0	0
Decamethylene azelamide	17	2	0	0	2	0	2	0	0
Nonamethylene sebacamide	17	2	0	0	2	0	2	0	0
Pentamethylene tetradecanediamide	17	2	0	0	2	0	2	0	0
Hexamethylene 5-oxatetradecanediamide	17	2	0	0	2	1	2	0	0
Oxydipentamethylene 5-oxasebacamide	17	2	0	0	2	2	2	0	0
(Octylphosphinidene)ditrimethylene adipamide	17	2	0	1	5	0	2	0	1
Hexamethylene 2-dodecylmalonamide	17	2	0	2	6	0	2	0	0
4,4'-Bicyclohexylene undecanedioate	17	2	0	4	4	2	0	0	0
4,4'-Methylenedicyclohexylene sebacamide	17	2	0	4	6	0	2	0	0
Nonamethylene (tetraethyldisiloxanylene)dipropionamide	17	2	0	4	14	1	2	0	2
p-Phenylenediethylene pentadecanediamide	17	2	1	0	2	0	2	0	0
p-Xylylene heptadecanediamide	17	2	1	0	2	0	2	0	0
Hexadecamethylene 4,4'-(methylenediphenylene)diurethane	17	2	2	0	2	2	2	0	0
Hexamethylene oxamide-*alt*-hexamethylene pimelamide	17	4	0	0	4	0	4	0	0
Pentamethylene azelamide-*alt*-pentamethylene oxamide	17	4	0	0	4	0	4	0	0
Dodecamethylene *N,N'*-pentamethylenediterephthalamate	17	4	2	0	2	2	2	0	0
Nonamethylene *N,N'*-sebacoyldi(*p*-aminobenzoate)	17	4	2	0	2	2	2	0	0
Tetradecamethylene *N,N'*-trimethylenediterephthalamate	17	4	2	0	2	2	2	0	0
Tridecamethylene *N,N'*-tetramethylenediterephthalamate	17	4	2	0	2	2	2	0	0
Trimethylene *N,N'*-hexadecanedioyldi(*p*-aminobenzoate)	17	4	2	0	2	2	2	0	0
Undecamethylene *N,N'*-hexamethylenediterephthalamate	17	4	2	0	2	2	2	0	0
18	18	0	0	0	0	0	0	0	0
Di-*n*-decylsilylene	18	0	0	2	6	0	0	0	1
3,3'-Didecyl-2,2'-bithiophene	18	0	0	10	8	0	0	2	0
N-octadecanoylethylene amine	18	1	0	1	3	0	1	0	0
Octadecyl acrylate	18	1	0	2	4	1	0	0	0
Octadecyl acrylamide	18	1	0	2	5	0	1	0	0
Octadecyl methacrylate	18	1	0	3	6	1	0	0	0
Decamethylene dithiolsebacate	18	2	0	0	0	0	0	2	0
Decamethylene sebacate	18	2	0	0	0	2	0	0	0
Dodecamethylen suberate	18	2	0	0	0	2	0	0	0
Octamethylene dodecanedioate	18	2	0	0	0	2	0	0	0
Tetradecamethylene adipate	18	2	0	0	0	2	0	0	0
Decamethylene thiodivalerate	18	2	0	0	0	2	0	1	0
Decamethylene sulfonyldivalerate	18	2	0	0	0	4	0	1	0
Decamethylene sebacamide	18	2	0	0	2	0	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene tetradecanediamide	18	2	0	0	2	0	2	0	0
Nonamethylene undecanediamide	18	2	0	0	2	0	2	0	0
Octamethylene dodecanediamide	18	2	0	0	2	0	2	0	0
Hexamethylene thiodienanthamide	18	2	0	0	2	0	2	1	0
Hexamethylene 6-oxapentadecanediamide	18	2	0	0	2	1	2	0	0
Hexamethylene oxydienanthamide	18	2	0	0	2	1	2	0	0
Decamethylene octamethylenediurethane	18	2	0	0	2	2	2	0	0
Dodecamethylene hexamethylenediurethane	18	2	0	0	2	2	2	0	0
Octamethylene decamethylenediurethane	18	2	0	0	2	2	2	0	0
Oxydipentamethylene 5-oxaundecanediamide	18	2	0	0	2	2	2	0	0
Oxydiepentamethylene oxydivaleramide	18	2	0	0	2	2	2	0	0
Aminodihexamethylene suberamide	18	2	0	0	3	0	3	0	0
Decamethylene octamethylenediurea	18	2	0	0	4	0	4	0	0
Dodecamethylene hexamethylenediurea	18	2	0	0	4	0	4	0	0
Hexadecamethylene 3-hexenedioate	18	2	0	2	2	2	0	0	0
Tetradecamethylene 4-octenedioate	18	2	0	2	2	2	0	0	0
Heptamethylene dodecylmalonamide	18	2	0	2	6	0	2	0	0
Hexamethylene tridecylmalonamide	18	2	0	2	6	0	2	0	0
4,4′-Ethylenedicyclohexylene sebacamide	18	2	0	4	6	0	2	0	0
4,4′-Methylenedicyclohexylene undecanediamide	18	2	0	4	6	0	2	0	0
Octadecamethylene terephthalate	18	2	1	0	0	2	0	0	0
Octadecamethylene terephthalamide	18	2	1	0	2	0	2	0	0
p-Phenylenediethylene hexadecanediamide	18	2	1	0	2	0	2	0	0
Tetradecamethylene *p*-phenylenedipropionamide	18	2	1	0	2	0	2	0	0
p-Xylylene octadecanediamide	18	2	1	0	2	0	2	0	0
p-Xylylene thiodipelargonamide	18	2	1	0	2	0	2	1	0
p-Xylylene oxydipelargonamide	18	2	1	0	2	1	2	0	0
Hexadecamethylene *p*-xylylenediurethane	18	2	1	0	2	2	2	0	0
N,N′-Dimethyl-*p*-xylylene octadecanediamide	18	2	1	2	6	0	2	0	0
Decamethylene sebacoyldiurethane	18	4	0	0	2	2	2	0	0
Tetramethylene *N,N′*-adipoyldi(6-aminocaproate)	18	4	0	0	2	2	2	0	0
Hexamethylene oxamide-*alt*-hexamethylene suberamide	18	4	0	0	4	0	4	0	0
Pentamethylene oxamide-*alt*-pentamethylene sebacamide	18	4	0	0	4	0	4		0
Octamethylene *N,N′*-terephthaloyldi(6-aminocaproate)	18	4	1	0	2	2	2	0	0
Dodecamethylene *N,N′*-hexamethylenediterephthalamate	18	4	2	0	2	2	2	0	0
Hexamethylene *N,N′*-dodecamethylenediterephthalamate	18	4	2	0	2	2	2	0	0
Octamethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)	18	4	2	0	2	2	2	0	0
Tetradecamethylene *N,N′*-tetramethylenediterephthalamate	18	4	2	0	2	2	2	0	0
Tetramethylene *N,N′*-hexadecanedioyldi(*p*-aminobenzoate)	18	4	2	0	2	2	2	0	0
Tridecamethylene *N,N′*-pentamethylenediterephthalamate	18	4	2	0	2	2	2	0	0
19	19	0	0	0	0	0	0	0	0
Methyleneoxyoctadecamethylene oxide	19	0	0	0	0	2	0	0	0
10,12-Pentacosadiynoic acid, 10,13-	19	1	0	5	2	1	0	0	0
Trimethylene octadecanedioate	19	2	0	0	0	2	0	0	0
Undecamethylene sebacamide	19	2	0	0	2	0	2	0	0
Hexamethylene 5-oxahexadecanediamide	19	2	0	0	2	1	2	0	0
Oxydipentamethylene 5-oxadodecanediamide	19	2	0	0	2	2	2	0	0
Oxydipentamethylene 6-oxadodecanediamide	19	2	0	0	2	2	2	0	0
Tetramethylene (pentamethylenedisulfonyl)dicaproamide	19	2	0	0	2	4	2	2	0
Hexamethylene (trimethylenedisulfonyl)diaminodicaproamide	19	2	0	0	4	4	4	0	0
Hexamethylene 2-tetradecylmalonamide	19	2	0	2	6	0	2	0	0
Octamethylene dodecylmalonamide	19	2	0	2	6	0	2	0	0
4,4′-Methylenedicyclohexylene dodecanediamide	19	2	0	4	6	0	2	0	0
p-Phenylenediethylene heptadecanediamide	19	2	1	0	2	0	2	0	0
4,4′-Methylenediphenylene octadecamethylenediurea	19	2	2	0	4	0	4	0	0
β-D-Glucotricaprylate, 1,4-	19	3	0	8	14	5	0	0	0
Hexamethylene azelamide-*alt*-hexamethylene oxamide	19	4	0	0	4	0	4	0	0
Pentamethylene *N,N′*-hexadecanedioyldi(*p*-aminobenzoate)	19	4	2	0	2	2	2	0	0
Tetradecamethylene *N,N′*-pentamethylenediterephthalamate	19	4	2	0	2	2	2	0	0
Tridecamethylene *N,N′*-hexamethylenediterephthalamate	19	4	2	0	2	2	2	0	0
20	20	0	0	0	0	0	0	0	0
1-Docosene	20	0	0	2	4	0	0	0	0
Di-*n*-undecylsilylene	20	0	0	2	6	0	0	0	1
Eicosamethylene oxalate	20	2	0	0	0	2	0	0	0
Hexadecamethylene adipate	20	2	0	0	0	2	0	0	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Tetradecamethylene suberate	20	2	0	0	0	2	0	0	0
Decamethylene dodecandiamide	20	2	0	0	2	0	2	0	0
Dodecamethylene sebacamide	20	2	0	0	2	0	2	0	0
Hydrazo docosanediamide	20	2	0	0	2	0	2	0	0
Nonamethylene tridecanediamide	20	2	0	0	2	0	2	0	0
Octamethylene tetradecanediamide	20	2	0	0	2	0	2	0	0
Hexamethylene 6-oxaheptadecanediamide	20	2	0	0	2	1	2	0	0
Decamethylene decamethylenediurethane	20	2	0	0	2	2	2	0	0
Oxydipentamethylene 6-oxatridecanediamide	20	2	0	0	2	2	2	0	0
Oxydipentamethylene oxydicaproamide	20	2	0	0	2	2	2	0	0
Hexamethylene (tetramethylenedisulfonyl)dicaproamide	20	2	0	0	2	4	2	2	0
Pentamethylene (pentamethylenedisulfonyl)dicaproamide	20	2	0	0	2	4	2	2	0
Tetramethylene (hexamethylenedisulfonyl)dicaproamide	20	2	0	0	2	4	2	2	0
Hexamethylene tetradecamethylenediurea	20	2	0	0	4	0	4	0	0
Hexadecamethylene 4-octenedioate	20	2	0	2	2	2	0	0	0
Hexamethylene pentadecylmalonamide	20	2	0	2	6	0	2	0	0
Nonamethylene dodecylmalonamide	20	2	0	2	6	0	2	0	0
Hexamethylene 9,10-dihydroxyoctadecanediamide	20	2	0	2	6	2	2	0	0
Ethylene tetraoxa-3,9-spiro(5.5)hendecanedicaprylate	20	2	0	3	2	6	0	0	0
Ethylene 4,4′-(oxydimethylene)di-2-(1,3-dioxolane)caprylate	20	2	0	4	4	7	0	0	0
4,4′-Ethylenedicyclohexylene dodecanediamide	20	2	0	4	6	0	2	0	0
4,4′-Methylenedicyclohexylene tridecanediamide	20	2	0	4	6	0	2	0	0
Eicosamethylene isophthalate	20	2	1	0	0	2	0	0	0
Eicosamethylene phthalate	20	2	1	0	0	2	0	0	0
Eicosamethylene terephthalate	20	2	1	0	0	2	0	0	0
Octadecamethylene *p*-phenylenediacetamide	20	2	1	0	2	0	2	0	0
p-Phenylenediethylene octadecanediamide	20	2	1	0	2	0	2	0	0
p-Phenylenediethylene thiodipelargonamide	20	2	1	0	2	0	2	1	0
Octadecamethylene *p*-xylylenediurea	20	2	1	0	4	0	4	0	0
N,N′-Diethyl-*p*-xylylene octadecanediamide	20	2	1	2	6	0	2	0	0
Eicosamethylene 3,3′-dibenzoate	20	2	2	0	0	2	0	0	0
Eicosamethylene 4,4′-dibenzoate	20	2	2	0	0	2	0	0	0
Octadecamethylene 4,4′-(ethylenedioxy)dibenzoate	20	2	2	0	0	4	0	0	0
Hexamethylene *N,N′*-adipoyldi(6-aminocaproate)	20	4	0	0	2	2	2	0	0
Hexamethylenedi(6-hydroxycaproamide) adipate	20	4	0	0	2	2	2	0	0
Hexamethylene oxamide-*alt*-hexamethylene sebacamide	20	4	0	0	4	0	4	0	0
Decamethylene *N,N′*-terephthaloyldi(6-aminocaproate)	20	4	1	0	2	2	2	0	0
Decamethylene *N,N′*-dodecanedioyldi(*p*-aminobenzoate)	20	4	2	0	2	2	2	0	0
Nonamethylene *N,N′*-tridecanedioyldi(*p*-aminobenzoate)	20	4	2	0	2	2	2	0	0
Tetradecamethylene *N,N′*-hexamethylenediterephthalamate	20	4	2	0	2	2	2	0	0
21	21	0	0	0	0	0	0	0	0
22-Aminodocosanoic acid	21	1	0	0	1	0	1	0	0
10,12-Heptacosadiynoic acid, 10,13-	21	1	0	5	2	1	0	0	0
Eicosamethylene malonate	21	2	0	0	0	2	0	0	0
Decamethylene tridecanediamide	21	2	0	0	2	0	2	0	0
Pentamethylene octadecanediamide	21	2	0	0	2	0	2	0	0
Oxydipentamethylene 5-Oxatetradecanediamide	21	2	0	0	2	2	2	0	0
Hexamethylene (pentamethylenedisulfonyl)dicaproamide	21	2	0	0	2	4	2	2	0
Pentamethylene (hexamethylenedisulfonyl)dicaproamide	21	2	0	0	2	4	2	2	0
Decamethylene dodecylmalonamide	21	2	0	2	6	0	2	0	0
Hexamethylene 2-hexadecylmalonamide	21	2	0	2	6	0	2	0	0
4,4′-Methylenedicyclohexylene tetradecanediamide	21	2	0	4	6	0	2	0	0
Nonamethylene *N,N′*-tetradecanedioyldi(*p*-aminobenzoate)	21	4	2	0	2	2	2	0	0
22	22	0	0	0	0	0	0	0	0
(Decamethylenedioxy)dihexamethylene oxide	22	0	0	0	0	3	0	0	0
Di-*n*-dodecylsilylene	22	0	0	2	6	0	0	0	1
Docosyl acrylate	22	1	0	2	4	1	0	0	0
N-Docosyl acrylamide	21	1	0	2	5	0	1	0	0
Eicosamethylene succinate	22	2	0	0	0	2	0	0	0
Hexadecamethylene suberate	22	2	0	0	0	2	0	0	0
Eicosamethylene oxydiacetate	22	2	0	0	0	3	0	0	0
Decamethylene tetradecanediamide	22	2	0	0	2	0	2	0	0
Dodecamethylene dodecanediamide	22	2	0	0	2	0	2	0	0
Hexamethylene octadecanediamide	22	2	0	0	2	0	2	0	0
Tetradecamethylene sebacamide	22	2	0	0	2	0	2	0	0

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Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene oxydipelargonamide	22	2	0	0	2	1	2	0	0
Dodecamethylene decamethylenediurethane	22	2	0	0	2	2	2	0	0
Hexadecamethylene hexamethylenediurethane	22	2	0	0	2	2	2	0	0
Oxydipentamethylene 6-oxapentadecanediamide	22	2	0	0	2	2	2	0	0
Oxydipentamethylene oxydienanthamide	22	2	0	0	2	2	2	0	0
Hexamethylene (hexamethylenedisulfonyl)dicaproamide	22	2	0	0	2	4	2	2	0
Decamethylene dodecamethylenediurea	22	2	0	0	4	0	4	0	0
N,N′-Dimethylhexamethylene octadecanediamide	22	2	0	2	6	0	2	0	0
Hexamethylene heptadecylmalonamide	22	2	0	2	6	0	2	0	0
Octadecamethylene *p*-phenylenedipropionamide	22	2	1	0	2	0	2	0	0
p-Xylylene docosanediamide	22	2	1	0	2	0	2	0	0
p-Xylylene thiodiundecanoamide	22	2	1	0	2	0	2	1	0
Eicosamethylene 3-(*p*-nitrophenyl)glutarate	22	2	1	1	1	4	1	0	0
Octamethylene *N,N′*-adipoyldi(6-aminocaproate)	22	4	0	0	2	2	2	0	0
Heptamethylene oxamide-*alt*-heptamethylene sebacamide	22	4	0	0	4	0	4	0	0
Hexamethylene oxamide-*alt*-hexamethylene sebacamide	22	4	0	0	4	0	4	0	0
Dodecamethylene *N,N′*-terephthaloyldi(6-aminocaproate)	22	4	1	0	2	2	2	0	0
Dodecamethylene *N,N′-dodecanedioyldi*(p-aminobenzoate)	22	4	2	0	2	2	2	0	0
23	23	0	0	0	0	0	0	0	0
10,12-Nonacosadiynoic acid, 10,13-	23	1	0	5	2	1	0	0	0
Eicosamethylene glutarate	23	2	0	0	0	2	0	0	0
Oxydipentamethylene 5-oxahexadecanediamide	23	2	0	0	2	2	2	0	0
Eicosamethylene 3-methyladipate	23	2	0	2	4	2	0	0	0
Hexamethylene 2-octadecylmalonamide	23	2	0	2	6	0	2	0	0
Nonamethylene *N,N′*-hexadecanedioyldi(*p*-aminobenzoate)	23	4	2	0	2	2	2	0	0
24	24	0	0	0	0	0	0	0	0
(Tetramethylenedioxy)didecamethylene oxide	24	0	0	0	0	3	0	0	0
Eicosamethylene adipate	24	2	0	0	0	2	0	0	0
Hexamethylene eicosanediamide	24	2	0	0	2	0	2	0	0
Octamethylene octadecanediamide	24	2	0	0	2	0	2	0	0
Tetradecamethylene dodecanediamide	24	2	0	0	2	0	2	0	0
Tridecamethylene tridecanediamide	24	2	0	0	2	0	2	0	0
Dodecamethylene dodecamethylenediurethane	24	2	0	0	2	2	2	0	0
Oxydipentamethylene 6-oxaheptadecanediamide	24	2	0	0	2	2	2	0	0
Decamethylene (tetramethylenedisulfonyl)dicaproamide	24	2	0	0	2	4	2	2	0
Tetramethylene (decamethylenedisulfonyl)dicaproamide	24	2	0	0	2	4	2	2	0
Oxydiethylenedioxydiethylenedioxydiethylene hexamethylenediurethane	24	2	0	0	2	7	2	0	0
Decamethylene tetradecamethylenediurea	24	2	0	0	4	0	4	0	0
Hexamethylene octadecamethylenediurea	24	2	0	0	4	0	4	0	0
Hexamethylene tetraoxa-3,9-spiro(5.5)hendecanedicaprylamide	24	2	0	3	4	4	2	0	0
Hexamethylene 4,4′-(oxydimethylene)di-dioxolanecaprylamide	24	2	0	4	6	5	2	0	0
p-Phenylenediethylene docosanediamide	24	2	1	0	2	0	2	0	0
p-Phenylenediethylene thiodiundecanoamide	24	2	1	0	2	0	2	1	0
Decamethylene *N,N′*-adipoyldi(6-aminocaproate)	24	4	0	0	2	2	2	0	0
Octamethylene oxamide-*alt*-octamethylene sebacamide	24	4	0	0	4	0	4	0	0
Dodecamethylene *N,N′*-dodecamethylenediterephthalamate	24	4	2	0	2	2	2	0	0
25	25	0	0	0	0	0	0	0	0
(Pentamethylenedioxy)didecamethylene oxide	25	0	0	0	0	3	0	0	0
Eicosamethylene pimelate	25	2	0	0	0	2	0	0	0
Decamethylene (pentamethylenedisulfonyl)dicaproamide	25	2	0	0	2	4	2	2	0
Pentamethylene (decamethylenedisulfonyl)dicaproamide	25	2	0	0	2	4	2	2	0
β-D-Glucotricaprate	25	3	0	8	14	5	0	0	0
26	26	0	0	0	0	0	0	0	0
(Hexamethylenedioxy)didecamethylene oxide	26	0	0	0	0	3	0	0	0
Di-*n*-tetradecylsilylene	26	0	0	2	6	0	0	0	1
4,4′-Decamethylenedipiperazine sebacamide	26	2	0	0	0	0	4	0	0
Decamethylene octadecanedioate	26	2	0	0	0	2	0	0	0
Eicosamethylene suberate	26	2	0	0	0	2	0	0	0
Decamethylene octadecanediamide	26	2	0	0	2	0	2	0	0
Hexamethylene docosanediamide	26	2	0	0	2	0	2	0	0
Octadecamethylene sebacamide	26	2	0	0	2	0	2	0	0
Octamethylene eicosanediamide	26	2	0	0	2	0	2	0	0
Hexadecamethylene decamethylenediurethane	26	2	0	0	2	2	2	0	0
Oxydipentamethylene oxydipelargonamide	26	2	0	0	2	2	2	0	0
Decamethylene (hexamethylenedisulfonyl)dicaproamide	26	2	0	0	2	4	2	2	0

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene (decamethylenedisulfonyl)dicaproamide	26	2	0	0	2	4	2	2	0
Dodecamethylene *N,N'*-adipoyldi(6-aminocaproate)	26	4	0	0	2	2	2	0	0
Nonamethylene oxamide-*alt*-nonamethylene sebacamide	26	4	0	0	4	0	4	0	0
27	27	0	0	0	0	0	0	0	0
Eicosamethylene azelate	27	2	0	0	0	2	0	0	0
28	28	0	0	0	0	0	0	0	0
Eicosamethylene sebacate	28	2	0	0	2	0	0	0	0
Eicosamethylene thiodivalerate	28	2	0	0	0	2	0	1	0
Eicosamethylene sulfonyldivalerate	28	2	0	0	0	4	0	1	0
Decamethylene eicosanediamide	28	2	0	0	2	0	2	0	0
Dodecamethylene octadecanediamide	28	2	0	0	2	0	2	0	0
Octamecamethylene dodecanediamide	28	2	0	0	2	0	2	0	0
Octamethylene docosanediamide	28	2	0	0	2	0	2	0	0
Decamethylene octadecamethylenediurea	28	2	0	0	4	0	4	0	0
Decamethylene oxamide-*alt*-decamethylene sebacamide	28	4	0	0	4	0	4	0	0
30	30	0	0	0	0	0	0	0	0
Decamethylene docosanediamide	30	2	0	0	2	0	2	0	0
Tetradecamethylene octadecanediamide	30	2	0	0	2	0	2	0	0
Decamethylene (decamethylenedisulfonyl)dicaproamide	30	2	0	0	2	4	2	2	0
p-Phenylene 2,5-di-*n*-hexadecyloxyterephthalate	30	2	1	8	8	4	0	0	0
Undecamethylene oxamide-*alt*-undecamethylene sebacamide	30	4	0	0	4	0	4	0	0
β-D-Glucotrilaurate	31	3	0	8	14	5	0	0	0
Dodecamethylene docosanediamide	32	2	0	0	2	0	2	0	0
Dodecamethylene oxamide-*alt*-dodecamethylene sebacamide	32	4	0	0	4	0	4	0	0
Octadecamethylene octadecanediamide	34	2	0	0	2	0	2	0	0
Tetradecamethylene docosanediamide	34	2	0	0	2	0	2	0	0
β-D-Glucotrimyristate	37	3	0	8	14	5	0	0	0
Octadecamethylene docosanediamide	38	2	0	0	2	0	2	0	0
β-D-Glucotripalmitate	43	3	0	8	14	5	0	0	0

E. REFERENCES

1. H. Tadokoro, K. Kozai, S. Seki, I. Nitta, J. Polym. Sci., **26**, 379 (1957).
2. W. P. Slichter, J. Polym. Sci., **35**, 77 (1959).
3. D. R. Holmes, C. W. Bunn, D. J. Smith, J. Polym. Sci., **17**, 159 (1955).
4. W. J. Dulmage, J. Polym. Sci., **26**, 277 (1957).
5. A. Prietzschk, Kolloid Z., **156**, 8 (1958).
6. R. L. Miller, unpublished results.
7. G. Natta, P. Corradini, Atti Accad, Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **19**, 229 (1955); J. Polym. Sci., **20**, 251 (1956).
8. Y. Tanada, K. Imada, M. Takayanagi, Kogyo Kagaku Zasshi, **69**, 1971 (1966).
9. G. Natta, P. Corradini, I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **19**, 404 (1955).
10. G. Natta, SPE J. **15**, 373 (1959).
11. C. W. Bunn, E. R. Howells, Nature (London), **174**, 549 (1954).
12. G. Natta, L. Porri, P. Corradini, D. Morero, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **20**, 560 (1956).
13. G. Natta, P. Corradini, G. Dall'Asta, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **20**, 408 (1956).
14. C. W. Bunn, Trans. Faraday Soc., **35**, 482 (1939).
15. E. R. Walter, F. P. Reding, J. Polym. Sci., **21**, 561 (1956).
16. M. L. Miller, C. E. Rauhut, J. Polym. Sci., **38**, 63 (1959).
17. G. Natta, J. Polym. Sci., **16**, 143 (1955).
18. C. C. Price, M. Osgan, R. E. Hughes, C. Shambelan, J. Am. Chem. Soc., **78**, 690 (1956).
19. C. G. Overberger, H. Jabloner, J. Am. Chem. Soc., **85**, 3431 (1963).
20. H. F. Zobel, A. D. French, M. E. Hinkle, Biopolymers, **5**, 837 (1967).
21. H. S. Kaufman, J. Am. Chem. Soc., **75**, 1477 (1953).
22. P. H. Hermans, "Physics and Chemistry of Cellulose Fibres", Elsevier, New York, 1949.
23. S. Bruckner, S. V. Meille, W. Porzio, Polymer, **29**, 1586 (1988).
24. H. S. Peiser, H. P. Rooksby, A. J. C. Wilson, "X-Ray Diffraction by Poly-Crystalline Materials", Chapman and Hall, London, 1955.
25. C. W. Bunn, E. V. Garner, Proc. R. Soc. London, Ser. A, **189**, 39 (1947).
26. A. Okada, K. Fuchino, Kobunshi Kagaku, **7**, 122 (1950).
27. R. de P. Daubeny, C. W. Bunn, C. J. Brown, Proc. R. Soc. London, Ser. A, **226**, 531 (1954).
28. H. J. Wellard, J. Polym. Sci., **13**, 471 (1954).
29. C. W. Bunn, Nature (London), **161**, 929 (1948).
30. J. D. Stroupe, R. E. Hughes, J. Am. Chem. Soc., **80**, 1768 (1958).
31. T. G. Fox, B. S. Garrett, W. E. Goode, S. Gratch, J. F. Kincaid, A. Spell, J. D. Stroupe, J. Am. Chem. Soc., **80**, 1768 (1958).

Constitutional Base Unit Poly-	CH_2	CO	C_6H_4	C	H	O	N	S	X
Hexamethylene (decamethylenedisulfonyl)dicaproamide	26	2	0	0	2	4	2	2	0
Dodecamethylene *N,N'*-adipoyldi(6-aminocaproate)	26	4	0	0	2	2	2	0	0
Nonamethylene oxamide-*alt*-nonamethylene sebacamide	26	4	0	0	4	0	4	0	0
27	27	0	0	0	0	0	0	0	0
Eicosamethylene azelate	27	2	0	0	0	2	0	0	0
28	28	0	0	0	0	0	0	0	0
Eicosamethylene sebacate	28	2	0	0	2	0	0	0	0
Eicosamethylene thiodivalerate	28	2	0	0	0	2	0	1	0
Eicosamethylene sulfonyldivalerate	28	2	0	0	0	4	0	1	0
Decamethylene eicosanediamide	28	2	0	0	2	0	2	0	0
Dodecamethylene octadecanediamide	28	2	0	0	2	0	2	0	0
Octamecamethylene dodecanediamide	28	2	0	0	2	0	2	0	0
Octamethylene docosanediamide	28	2	0	0	2	0	2	0	0
Decamethylene octadecamethylenediurea	28	2	0	0	4	0	4	0	0
Decamethylene oxamide-*alt*-decamethylene sebacamide	28	4	0	0	4	0	4	0	0
30	30	0	0	0	0	0	0	0	0
Decamethylene docosanediamide	30	2	0	0	2	0	2	0	0
Tetradecamethylene octadecanediamide	30	2	0	0	2	0	2	0	0
Decamethylene (decamethylenedisulfonyl)dicaproamide	30	2	0	0	2	4	2	2	0
p-Phenylene 2,5-di-*n*-hexadecyloxyterephthalate	30	2	1	8	8	4	0	0	0
Undecamethylene oxamide-*alt*-undecamethylene sebacamide	30	4	0	0	4	0	4	0	0
β-D-Glucotrilaurate	31	3	0	8	14	5	0	0	0
Dodecamethylene docosanediamide	32	2	0	0	2	0	2	0	0
Dodecamethylene oxamide-*alt*-dodecamethylene sebacamide	32	4	0	0	4	0	4	0	0
Octadecamethylene octadecanediamide	34	2	0	0	2	0	2	0	0
Tetradecamethylene docosanediamide	34	2	0	0	2	0	2	0	0
β-D-Glucotrimyristate	37	3	0	8	14	5	0	0	0
Octadecamethylene docosanediamide	38	2	0	0	2	0	2	0	0
β-D-Glucotripalmitate	43	3	0	8	14	5	0	0	0

E. REFERENCES

1. H. Tadokoro, K. Kozai, S. Seki, I. Nitta, J. Polym. Sci., **26**, 379 (1957).
2. W. P. Slichter, J. Polym. Sci., **35**, 77 (1959).
3. D. R. Holmes, C. W. Bunn, D. J. Smith, J. Polym. Sci., **17**, 159 (1955).
4. W. J. Dulmage, J. Polym. Sci., **26**, 277 (1957).
5. A. Prietzschk, Kolloid Z., **156**, 8 (1958).
6. R. L. Miller, unpublished results.
7. G. Natta, P. Corradini, Atti Accad, Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **19**, 229 (1955); J. Polym. Sci., **20**, 251 (1956).
8. Y. Tanada, K. Imada, M. Takayanagi, Kogyo Kagaku Zasshi, **69**, 1971 (1966).
9. G. Natta, P. Corradini, I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **19**, 404 (1955).
10. G. Natta, SPE J. **15**, 373 (1959).
11. C. W. Bunn, E. R. Howells, Nature (London), **174**, 549 (1954).
12. G. Natta, L. Porri, P. Corradini, D. Morero, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **20**, 560 (1956).
13. G. Natta, P. Corradini, G. Dall'Asta, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **20**, 408 (1956).
14. C. W. Bunn, Trans. Faraday Soc., **35**, 482 (1939).
15. E. R. Walter, F. P. Reding, J. Polym. Sci., **21**, 561 (1956).
16. M. L. Miller, C. E. Rauhut, J. Polym. Sci., **38**, 63 (1959).
17. G. Natta, J. Polym. Sci., **16**, 143 (1955).
18. C. C. Price, M. Osgan, R. E. Hughes, C. Shambelan, J. Am. Chem. Soc., **78**, 690 (1956).
19. C. G. Overberger, H. Jabloner, J. Am. Chem. Soc., **85**, 3431 (1963).
20. H. F. Zobel, A. D. French, M. E. Hinkle, Biopolymers, **5**, 837 (1967).
21. H. S. Kaufman, J. Am. Chem. Soc., **75**, 1477 (1953).
22. P. H. Hermans, "Physics and Chemistry of Cellulose Fibres", Elsevier, New York, 1949.
23. S. Bruckner, S. V. Meille, W. Porzio, Polymer, **29**, 1586 (1988).
24. H. S. Peiser, H. P. Rooksby, A. J. C. Wilson, "X-Ray Diffraction by Poly-Crystalline Materials", Chapman and Hall, London, 1955.
25. C. W. Bunn, E. V. Garner, Proc. R. Soc. London, Ser. A, **189**, 39 (1947).
26. A. Okada, K. Fuchino, Kobunshi Kagaku, **7**, 122 (1950).
27. R. de P. Daubeny, C. W. Bunn, C. J. Brown, Proc. R. Soc. London, Ser. A, **226**, 531 (1954).
28. H. J. Wellard, J. Polym. Sci., **13**, 471 (1954).
29. C. W. Bunn, Nature (London), **161**, 929 (1948).
30. J. D. Stroupe, R. E. Hughes, J. Am. Chem. Soc., **80**, 1768 (1958).
31. T. G. Fox, B. S. Garrett, W. E. Goode, S. Gratch, J. F. Kincaid, A. Spell, J. D. Stroupe, J. Am. Chem. Soc., **80**, 1768 (1958).

32. R. C. Reinhardt, Ind. Eng. Chem., **35**, 422 (1943).
33. S. Narita, K. Okuda, J. Polym. Sci., **38**, 270 (1959).
34. C. S. Fuller, C. J. Frosch, N. R. Pape, J. Am. Chem. Soc., **62**, 1905 (1940); A. M. Liquori, Acta Crystallogr., **8**, 345 (1955).
35. G. Natta, P. Corradini, I. W. Bassi, Makromol. Chem., **21**, 240 (1956); Nuovo Cimento, Suppl., **15**, 52 (1960).
36. V. M. Coiro, P. de Santis, L. Mazzarella, L. Picozzi, Chim. Ind. (Milan), **47**, 1236 (1965).
37. G. Natta, P. Corradini, L. Porri, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **20**, 728 (1956).
38. J. D. McCullough, R. S. Bauer, T. L. Jacobs, Chem. Ind. (London), 706, (1957).
39. H. D. Noether, J. Polym. Sci., **25**, 217 (1957); Text. Res. J., **28**, 533 (1958).
40. C. W. Bunn, Proc. R. Soc. London, Ser. A, **180**, 40 (1942).
41. C. Shambelan, Ph.D. Thesis, Univ. of Penna., 1959; Diss. Abstr., **20**, 120 (1959); M. Fujisaka, K. Imada, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **11**, 169 (1968).
42. C. F. Hammer, T. A. Koch, J. F. Whitney, J. Appl. Polym. Sci., **1**, 169 (1959).
43. W. Goggin, R. Lowry, Ind. Eng. Chem., **34**, 327 (1942).
44. G. Natta. Chem. Ind. (London), 1520, (1957).
45. G. Natta, P. Pino, P. Corradini, F. Danusso, E. Mantica, G. Mazzanti, G. Moraglio, J. Am. Chem. Soc., **77**, 1708 (1955).
46. R. Boyer, Compt. rend., de la 2e reunion de chimie physique (2–7 Juin 1952, Paris) P. 383.
47. D. E. Roberts, L. Mandelkern, J. Am. Chem. Soc., **80**, 1289 (1958).
48. F. P. Reding, J. Polym. Sci., **21**, 547 (1956).
49. J. D. Hoffman, J. J. Weeks, J. Res. Natl. Bur. Stand., **60**, 465 (1958).
50. L. Wood, N. Bekkedahl, R. E. Gibson, J. Chem. Phys., **13**, 475 (1945).
51. R. Beaman, F. Cramer, J. Polym. Sci., **21**, 223 (1956).
52. H. Starkweather, Jr., G. Moore, J. Hansen, T. Roder, R. Brooks, J., Polym. Sci. **21**, 189 (1956).
53. R. Wiley, Ind. Eng. Chem., **38**, 959 (1946).
54. H. W. Starkweather, Jr., R. E. Moynihan, J. Polym. Sci., **22**, 363 (1956).
55. F. C. Frank, A. Keller, A. O'connor, Philos. Mag., **4**, 200 (1959).
56. K. H. Meyer, W. Lotmar, G. W. Pankow, Helv. Chim. Acta, **19**, 930 (1936); E. Giglio, F. Pompa, A. Ripamonti, J. Polym. Sci., **59**, 293 (1960).
57. J. Boon, E. P. Magre', Makromol. Chem., **136**, 267 (1970).
58. P. W. Teare, D. R. Holmes, J. Polym. Sci., **24**, 496 (1957).
59. K. Little, Br. J. Appl. Phys., **10**, 225 (1959).
60. C. J. B. Clews, "Proc. Rubber Technology Conference", W. Heffer and Sons, Cambridge, p. 955, (1938).
61. W. P. Slichter, J. Polym. Sci., **36**, 259 (1959).
62. R. C. Golike, J. Polym. Sci., **42**, 583 (1960).
63. Y. Kinoshita, Makromol. Chem., **33**, 21 (1959).
64. P. H. Burleigh, J. Am. Chem. Soc., **82**, 749 (1960).
65. Y. Kinoshita, Makromol. Chem., **33**, 1 (1959).
66. R. E. Moynihan, J. Am. Chem. Soc., **81**, 1045 (1959).
67. G. Natta, Makromol. Chem., **35**, 93 (1960).
68. J. Bateman, R. E. Richards, G. Farrow, I. M. Ward, Polymer, **1**, 63 (1960).
69. P. Corradini, Atti Accad, Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **25**, 517 (1958).
70. S. D. Gehman, J. E. Field, R. P. Dinsmore, Proc. Rubber Technology Conference, W. Heffer and Sons, Cambridge, p. 961 (1938).
71. G. Natta, G. Mazzanti, P. Longi, F. Bernardini, Chim. Ind. (Milan), **40**, 813 (1958).
72. W. D. Niegisch, P. R. Swan, J. Appl. Phys., **31**, 1906 (1960).
73. E. Martuscelli, R. Gallo, G. Paiaro, Makromol. Chem., **103**, 295 (1967).
74. D. Sianesi, G. Natta, P. Corradini, Gazz. Chim. Ital., **89**, 775 (1959).
75. G. Natta, F. Danusso, D. Sianesi, Makromol. Chem., **28**, 253 (1958); D. Sianesi, M. Rampichini, F. Danusso, Chim. Ind. (Milan), **41**, 287 (1959).
76. G. Natta, G. Mazzanti, P. Corradini, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **25**, 3 (1958).
77. W. R. Krigbaum, N. Tokita J. Polym. Sci., **43**, 467 (1960).
78. E. Stanley, M. Litt, J. Polym. Sci., **43**, 453 (1960).
79. D. I. Sapper, J. Polym. Sci., **43**, 383 (1960).
80. P. Corradini, P. Ganis, J. Polym. Sci., **43**, 311 (1960).
81. L. Mandelkern, Chem. Rev., **56**, 903 (1956).
82. J. R. Schaefgen, J. Polym. Sci., **38**, 549 (1959).
83. F. Danusso, G. Moraglio, E. Flores, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **25**, 520 (1958).
84. N. Yoda, I. Matsubara, J. Polym. Sci., Part A, **2**, 253 (1964). (Unit cell constants given refer to a projected base).
85. F. A. Quinn, Jr., L. Mandelkern, J. Am. Chem. Soc., **80**, 3178 (1958); Erratum: J. Am. Chem. Soc., **81**, 6533 (1959).
86. F. W. Billmeyer, Jr., J. Appl. Phys., **28**, 1114 (1957).
87. M. Dole, J. Polym. Sci., **19**, 347 (1956).
88. A. M. Bueche, J. Am. Chem. Soc., **74**, 65 (1952).
89. F. Danusso, G. Moraglio, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **27**, 381 (1959).
90. T. W. Campbell, A. C. Haven, Jr., J. Appl. Polym. Sci., **1**, 73 (1959).
91. H. W. Starkweather, Jr., R. H. Boyd, J. Phys. Chem., **64**, 410 (1960).
92. G. Natta, G. mazzanti, P. Corradini, P. Chini, I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **28**, 8 (1960); G. Natta, G. Mazzanti, P. Corradini, I. W. bassi, Makromol. Chem., **37**, 156 (1960); I. W. Bassi, Rend., Ist. Lomb. Accad. Sci. Lett. A, **94**, 579 (1960); G. Natta, P. Corradini, I. W. Bassi, J. Polym. Sci., **51**, 505 (1961).
93. D. V. Badami, Polymer, **1**, 273 (1960).
94. J. H. Griffith, B. G. Ranby, J. Polym. Sci., **44**, 369 (1960).
95. J. B. Lando, D. Day, V. Enkelmann, J. Polym. Sci., Polym. Symp., **65**, 73 (1978); V. Enkelmann, J. B. Lando, Acta Cryst., Sect. B: Struct. Crystallogr. Cryst. Chem., **34**, 2352 (1978).

96. G. Natta, G. Mazzanti, P. Corradini, A. Valvassori, I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **28**, 18 (1960).
97. G. Natta, I. W. Bassi, Eur. Polym. J., **3**, 33 (1967).
98. O. Ellefsen, Norelco Rep., **7**, 104 (1960).
99. G. Farrow, J. McIntosh, I. M. Ward, Makromol. Chem., **38**, 147 (1960).
100. N. G. Esipova, L. Pan-Tun, N. S. Andreeva, P. V. Kozlov, Vysokomol. Soedin., **2**, 1109 (1960).
101. A. G. M. Last, J. Polym. Sci., **39**, 543 (1959).
102. D. J. Shields, H. W. Coover, Jr., J. Polym. Sci., **39**, 532 (1959).
103. S. Murahashi, S. Nozakura, H. Tadokoro, Bull. Chem. Soc. Japan, **32**, 534 (1959).
104. T. F. Yen, J. Polym. Sci., **38**, 272 (1959).
105. C. S. Fuller, C. L. Erickson, J. Am. Chem. Soc., **59**, 344 (1937).
106. C. S. Fuller, C. J. Frosch, J. Am. Chem. Soc., **61**, 2575 (1939). (Except for poly(decamethylene oxalate), unit cell constants given refer to a projected base).
107. C. S. Fuller, C. J. Frosch, N. R. Pape, J. Am. Chem. Soc., **64**, 154 (1942). (Unit cell constants given refer to a projected base).
108. C. S. Fuller, C. J. Frosch, J. Phys. Chem., **43**, 323 (1939).
109. C. S. Fuller, Chem. Rev., **26**, 143 (1940).
110. W. O. Baker, C. S. Fuller, J. Am. Chem. Soc., **64**, 2399 (1942).
111. A. Conix, R. Van Kerpl, J. Polym. Sci., **40**, 521 (1959).
112. J. A. Gailey, R. H. Ralston, SPE Trans. **4**, 29 (1964).
113. H. S. Yani, Quoted in C. F. Ryan, J. J. Gormley, Macromol. Synth. **1**, 30 (1963).
114. E. J. Vandenberg, R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 519 (1960).
115. J. H. Magill, J. Polym. Sci., Part A-2, **5**, 89 (1967).
116. S. Furuya, M. Honda, J. Polym. Sci., **28**, 232 (1958).
117. P. Goodman, J. Polym. Sci., **24**, 307 (1960).
118. L. Mandelkern, P. J. Flory, J. Am. Chem. Soc., **73**, 3206 (1951).
119. S. Ishida, S. Murahashi, J. Polym. Sci., **40**, 571 (1959).
120. B. S. Garrett, W. E. Goode, S. Gratch, J. F. Kincaid, C. L. Levesque, A. Spell, J. D. Stroupe, W. H. Watanabe, J. Am. Chem. Soc., **81**, 1007 (1959).
121. D. C. Vogelsong, E. M. Pearce, J. Polym. Sci. **45**, 546 (1960).
122. P. Corradini, P. Ganis, Nuovo Cimento, Suppl., **15**, 104 (1960).
123. G. Natta, P. Corradini, I. W. Bassi, Nuovo Cimento, Suppl., **15**, 83 (1960).
124. G. Natta, P. Corradini, Angew. Chem., **68**, 615 (1956); Nuovo Cimento, Suppl., **15**, 111 (1960).
125. P. Corradini, P. Ganis, Nuovo Cimento, Suppl., **15**, 96 (1960).
126. R. G. Crystal, Macromolecules **4**, 379 (1971).
127. G. Natta, P. Corradini, Nuovo Cimento, Suppl., **15**, 40 (1960).
128. G. Natta, P. Corradini, I. W. Bassi, Nuovo Cimento, Suppl., **15**, 68 (1960).
129. P. Pino, G. P. Lorenzi, J. Am. Chem. Soc., **82**, 4745 (1960).
130. G. Natta. G. Mazzanti, G. Preglia, M. Binaghi, M. Peraldo, J. Am. Chem. Soc. 82, 4742 (1960).
131. Z. W. Wilchinsky, J. Appl. Phys., **31**, 1969 (1960).
132. T. Mochizuki, Nippon Kagaku Zasshi **81**, 15 (1960).
133. R. Stefani, M. Chevreton, M. Garnier, C. Eyraud, C. R. Hebd. Seances Acad. Sci., **251**, 2174 (1960).
134. M. L. Huggins, J. Chem. Phys., **13**, 37 (1945).
135. C. Ruscher, H. J. Schroder, Faserforsch. Textiltech., **11**, 165 (1960).
136. Z. Mencik, Chem. Prumysl, **10**, 377 (1960).
137. M. Dole, B. Wunderlich, Makromol. Chem., **34**, 29 (1959).
138. G. Allen, G. Gee, G. J. Wilson, Polymer, **1**, 456 (1960).
139. G. Allen, G. Gee, D. Mangaraj, D. Sims, G. J. Wilson, Polymer, **1**, 466 (1960).
140. M. Asahina, K. Okuda, Kobunshi Kagaku **17**, 607 (1960).
141. G. Natta, M. Farina, M. Peraldo, P. Corradini, G. Bressan, P. Ganis, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **28**, 442 (1960).
142. L. A. Auspos, C. W. Burnham, L. Hall, J. K. Hubbard, W. Kirk, J. R. Schaefgen, S. B. Speck, J. Polym. Sci., **15**, 9 (1955).
143. A. T. Walter, J. Polym. Sci., **13**, 207 (1954).
144. C. J. Malm, J. W. Mench, D. L. Kendall, G. D. Hiatt, Ind. Eng. Chem., **43**, 688 (1951).
145. C. E. Anagnostopoulos, A. Y. Coran, H. R. Gamrath, J. Appl. Polym. Sci., **4**, 181 (1960).
146. C. F. Horn, B. T. Freure, H. Vineyard, H. J. Decker, Angew. Chem. **74**, 531 (1962).
147. G. Natta, G. Dall'Asta, G. Mazzanti, Angew. Chem., **76**, 765 (1964) - Angew. Chem., Int. Ed. Engl., **3**, 723 (1964); G. Natta, I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **38**, 315 (1965); G. Natta, I. W. Bassi, J. Polym. Sci., Part C, **16** 2551 (1967).
148. A. C. Farthing, W. J. Reynolds, J. Polym. Sci., **12**, 503 (1954).
149. H. S. Yanai, quoted in W. E. Goode, R. P. Fellman, F. H. Owens, Macromol. Synth., **1**, 25 (1963).
150. A. Turner-Jones, Makromol. Chem., **71**, 1 (1964). (The monoclinic cell is preferred by the author).
151. M. Niinomi, T. Fukuda, M. Takayanagi, private communication.
152. G. Natta, Angew. Chem., **68**, 393 (1956).
153. G. F. Schmidt, H. A. Stuart, Z. Naturforsch. **13A**, 222 (1958).
154. G. Natta, P. Corradini, Nuovo Cimento, Suppl., **15**, 9 (1960).
155. F. Rybnikar, Chem. Listy, **52**, 1042 (1958).
156. B. Wunderlich, M. Dole, J. Polym. Sci., **24**, 201 (1957).
157. C. W. Smith, M. Dole, J. Polym. Sci., **20**, 37 (1956).
158. B. Wunderlich, M. Dole, J. Polym. Sci., **32**, 125 (1958).
159. P. J. Flory, H. D. Bedon, E. H. Keefer, J. Polym. Sci., **28**, 151 (1958).
160. R. D. Evans, M. R. Mighton, P. J. Flory, J. Am. Chem. Soc., **72**, 2018 (1950).

161. G. Dall'Asta, N. Oddo, Chim. Ind. (Milan), **42**, 1234 (1960).
162. Z. Mencik, Vysokomol. Soedin., **2**, 1635 (1960).
163. Y. Chatani, J. Polym. Sci., **47**, 491 (1960).
164. G. Wasai, T. Tsuruta, J. Furukawa, Kogyo Kagaku Zasshi **66**, 1339 (1963).
165. M. Cesari, private communication.
166. H. D. Keith, F. J. Padden, Jr., N. M. Walter, H. W. Wyckoff, J. Appl. Phys., **30**, 1485 (1959).
167. G. Natta, M. Peraldo, P. Corradini, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **26**, 14 (1959).
168. K. Okuda, T. Yoshida, M. Sugita, M. Asahina, J. Polym. Sci., Part B, **5**, 465 (1967).
169. G. Natta, I. Pasquon, P. Corradini, M. Peraldo, M. Pegoraro, A. Zambelli, Atti Acad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **28**, 539 (1960).
170. N. M. Walter, quoted in C. Y. Liang, F. G. Pearson, J. Mol. Spectrosc., **5**, 290 (1960).
171. S. S. Leshchenko, V. L. Karpov, V. A. Kargin, Vysokomol. Soedin., **1**, 1538 (1959); *ibid.*, **5**, 953 (1963) - Polym. Sci. USSR (Engl. Transl.), **5**, 1 (1964).
172. D. J. H. Sandiford, J. Appl. Chem., **8**, 188 (1958).
173. M. Hatano, S. Kambara, Polymer, **2**, 1 (1961).
174. R. Dedeurwaerder, J. F. M. Oth, Bull. Soc. Chim. Belg., **70**, 37 (1961).
175. G. Natta, D. Sianesi, D. Morero, I. W. Bassi, G. Caporiccio, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **28**, 551 (1960).
176. I. W. Bassi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **29**, 193 (1960).
177. G. Champetier, M. Laualov, J. P. Pied, Bull. Soc. Chim. Fr., 708 (1958).
178. A. G. Nasini, L. Trossarelli, G. Saini, Makromol. Chem., **44/46**, 550 (1961).
179. G. Natta, G. Mazzanti, G. F. Pregaglia, M. Binaghi, Makromol. Chem., **44/46**, 537 (1961).
180. J. C. Swallow, Proc. R. Soc. London, Ser. A, **238**, 1 (1956).
181. J. Lal, G. S. Trick, J. Polym. Sci., **50**, 13 (1961).
182. R. J. Wilkinson, M. Dole, J. Polym. Sci., **58**, 1089 (1962).
183. F. B. Moody, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **2**, 285 (1961).
184. I. Sakurada, Y. Nukushina, Y. Sone, Kobunshi Kagaku, **12**, 506 (1955).
185. G. Natta, G. Mazzanti, P. Longi, G. Dall'Asta, F. Bernardini, J. Polym. Sci., **51**, 487 (1961).
186. M. Inoue, J. Polym. Sci., **51**, S18 (1961).
187. J. A. Price, M. R. Lytton, B. G. Ranby, J. Polym. Sci., **51**, 541 (1961).
188. F. P. Price, R. W. Kilb, J. Polym. Sci., **57**, 395 (1962).
189. G. Natta, L. Porri, P. Corradini, G. Zanini, F. Ciampelli, J. Polym. Sci., **51**, 463 (1961).
190. J. P. Arlie, A. Skoulios, C. R. Hebd. Seances Acad. Sci., **258**, 2570 (1964); J. P. Arlie, P. Spegt, A., Skoulios, Makromol. Chem., **104**, 212 (1967).
191. U. Giannini, G. Bruckner, E. Pellino, A. Cassata, J. Polym. Sci., Part B, **5**, 527 (1967).
192. S. Geller, Science **152**, 644 (1966); M. D. Lind, S. Geller, J. Chem. Phys., **51**, 348 (1969).
193. Y. Chatani, T. Takizawa, S. Murahashi, Y. Sakata, Y. Nishimura, J. Polym. Sci., **55**, 811 (1961).
194. J. R. Richards, Ph.D. Thesis, Univ. of Penna., 1961, Diss. Abstr. **22**, 1029 (1961).
195. H. G. Kilian, H. Haboth, E. Jenckel, Kolloid Z., **172** 166 (1960).
196. E. J. Addink, J. Beintema, Polymer **2**, 185 (1961).
197. G. Damaschun, Kolloid Z., **180**, 65 (1962).
198. F. L. Saunders, J. Polym. Sci., Part A-1, **5**, 2187 (1967).
199. C. A. Boye, J. Polym. Sci., **55**, 275 (1961).
200. I. W. Bassi, G. Allegra, R. Scordamaglia, Macromolecules, **4**, 575 (1971).
201. W. Wrasidlo, Macromolecules **4**, 642 (1971).
202. R. H. Baughman, J. Appl. Phys., **42**, 4579 (1971).
203. A. Turner-Jones, C. W. Bunn, Acta Crystallogr., **15**, 105 (1962).
204. D. C. Vogelsong, J. Polym. Sci., **57**, 895 (1962).
205. C. Y. Liang, S. Krimm, J. Chem. Phys., **25**, 563 (1956).
206. M. G. Broadhurst, J. Res. Natl. Bur. Stand., Sect. A, **66**, 241 (1962); J. Chem. Phys., **36**, 2578 (1962).
207. R. L. Miller, V. F. Holland, J. Polym. Sci., Part B, **2**, 519 (1964); V. F. Holland, R. L. Miller, J. Appl. Phys. **35**, 3241 (1964).
208. A. V. Ermolina, G. S. Markova, V. A. Kargin, Kristallografiya, **2**, 623 (1957).
209. E. S. Clark, L. T. Muus, Z. Kristallogr., **117**, 119 (1962). (Unit cell constants for form II refer to a projected base).
210. V. F. Holland, S. B. Mitchell, W. L. Hunter, P. H. Lindenmeyer, J. Polym. Sci., **62**, 145 (1962).
211. J. J. Point, Bull. Cl. Sci. Acad. R. Belg., **30**, 435 (1953).
212. L. G. Wallner, Montash. Chem., **79**, 279 (1948).
213. H. Hendus, K. Schmieder, G. Schnerl, K. A. Wolf. Festschrift Carl Wurster der BASF vom, **2**, 12 1960.
214. J. Furukawa, T. Saegusa, T. Tsuruta, S. Otha, G. Wasai, Makromol. Chem., **52**, 230 (1962).
215. G. Champetier, J. P. Pied, Makromol. Chem., **44/46**, 64 (1961).
216. F. Rybnikar, Collect. Czech. Chem. Commun., **24**, 2861 (1959).
217. S. Okamura, K. Hayashi, Y. Kitanishi, J. Polym. Sci., **58**, 925 (1962).
218. O. Ellefsen, N. Norman, J. Polym. Sci., **58**, 769 (1962).
219. L. A. Errede, R. S. Gregorian, J. Polym. Sci., **60**, 21 (1962).
220. T. F. Yen, J. Polym. Sci., **35**, 533 (1959).
221. J. F. Brown, Jr., D. M. White, J. Am. Chem. Soc., **82**, 5671 (1960).
222. A. Ziabicki, Kolloid Z., **167**, 132 (1959).
223. C. J. Brown, A. C. Farthing, J. Chem. Soc., 3270 (1953).
224. M. H. Kaufman, H. F. Mark, R. R. Mesrobian, J. Polym. Sci., **13**, 3 (1954).
225. R. L. Merker, M. J. Scott, J. Polym. Sci., Part A, **2**, 15 (1964).
226. W. A. Butte, C. C. Price, R. E. Hughes, J. Polym. Sci., **61**, S28 (1962).

227. E. R. Walter, F. P. Reding, 133RD Natl. Am. Chem. Soc. Meeting, San Francisco, April, (1958).
228. O. F. Solomon, M. Dimonie, K. Ambrozh, M. Tomesku, J. Polym. Sci., **52**, 205 (1961).
229. C. W. Bunn, E. V. Garner, J. Chem. Soc., 654 (1942).
230. G. Natta, L. Porri, G. Sovarzi, Eur. Polym. J., **1**, 81 (1965).
231. F. Sakaguchi, R. Kitamaru, W. Tsuji, Bull. Inst. Chem. Res., Kyoto Univ., **44**, 155 (1966).
232. K. Fujii, T. Mochizuki, S. Imoto, J. Ukida, M. Matsumoto, Makromol. Chem., **51**, 225 (1962).
233. K. Tanaka, T. Seto, T. Hara, J. Phys. Soc. Japan., **17**, 873 (1962); T. Seto, T. Hara, K. Tanaka, Jpn. J. Appl. Phys., **7**, 31 (1968).
234. G. Natta, L. Porri, A. Carbonaro, F. Ciampelli, G. Allegra, Makromol. Chem., **51**, 229 (1962).
235. G. Natta, I. W. Bassi, P. Corradini, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **31**, 17 (1961).
236. G. Natta, I. W. Bassi, G. Allegra, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **31**, 350 (1961).
237. J. B. Lando, V. Stannett, J. Polym. Sci., Part B., **2**, 375 (1964); Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **5**, 969 (1964).
238. A. Takahashi, S. Kambara, Makromol. Chem., **72**, 92 (1964).
239. T. Petitpas, J. Mering, C. R. Hebd. Seances Acad. Sci., **244**, 2611 (1962).
240. H. Sumitomo, K. Kobayashi, J. Polym. Sci., Part A-1, **5**, 2247 (1967). (Unit cell constants given refer to a projected base).
241. S. Boileau, J. Coste, J.-M. Raynal, P. Sigwalt, C. R. Hebd. Seances Acad. Sci., **254**, 2774 (1962).
242. G. Natta, M. Peraldo, M. Farina, G. Bressan, Makromol. Chem., **55**, 139 (1962).
243. R. Aelion, Ann. Chim. (Paris), **3**, 5 (1948).
244. K. Dachs, E. Schwartz, Angew. Chem., **74**, 540 (1962); Angew. Chem., Int. Ed. Engl., **1**, 430 (1962).
245. K. Okuda, J. Polym. Sci., Part A, **2**, 1749 (1964).
246. H. Arimoto, J. Polym. Sci., Part A, **2**, 2283 (1964).
247. G. Natta, P. Corradini, P. Ganis, I. W. Bassi, G. Allegra, Chim. Ind. (Milan), **44**, 532 (1962); G. Natta, G. Allegra, I. W. Bassi, P. Corradini, P. Ganis, Makromol. Chem., **58**, 242 (1962).
248. Y. Yamashita, S. Nunomoto, Makromol. Chem., **58**, 244 (1962).
249. L. Becker, Wiss. Z. Karl-Marx-Univ. Leipzig, Math.-Naturwiss. Reihe, **11**, 3 (1962).
250. K. J. Clark, A. Turner-Jones, D. J. H. Sandiford, Chem. Ind. (London), 2010 (1962).
251. E. Passaglia, H. R. Kevorkian, J. Appl. Phys., **34**, 90 (1963).
252. A. Nishioka, K. Yanigisawa, Kobunshi Kagaku, **19**, 667 (1962).
253. S. Murahashi, J. Yuki, T. Sano, U. Yonemura, H. Tadokoro, Y. Chatani, J. Polym. Sci., **62**, S77 (1962).
254. A. Turner-Jones, J. Polym. Sci., **62**, S53 (1962).
255. H. Watanabe, R. Koyama, H. Nagai, A. Nishioka, J. Polym. Sci., **62**, S74 (1962).
256. J.-J. Trillat, R. Tertian, C. R. Hebd. Seances Acad. Sci., **219**, 395 (1944).
257. L. Ulicky, Chem. Zvesti, **16**, 818 (1962).
258. E. Sauter, Z. Phys. Chem., Abt. B, **18**, 417 (1932).
259. G. Natta, G. Dall'Asta, G. Mazzanti, Chim. Ind. (Milan), **44**, 1212 (1962).
260. M. G. Broadhurst, J. Res. Natl. Bur. Stand., Sect. A, **67**, 233 (1963).
261. L. Becker, Plaste Kautsch., **8**, 557 (1961).
262. B. Wunderlich, C. M. Cormier, J. Polym. Sci., Part A-2, **5**, 987 (1967).
263. G. Natta, L. Porri, G. Stoppa, G. Allegra, F. Ciampelli, J. Polym. Sci., Part B, **1**, 67 (1963).
264. G. W. Taylor, Polymer, **3**, 543 (1962).
265. O. B. Edgar, E. Ellery, J. Chem. Soc., 2633 (1952).
266. H.-G. Kilian, Kolloid Z. Z. Polym., **185**, 13 (1962).
267. H. S. Makowski, K. C. Shim, Z. W. Wilchinsky, J. Polym. Sci., Part A, **2**, (1964).
268. D. Sianesi, G. Caporiccio, Makromol. Chem., **60**, 213 (1963).
269. I. W. Bassi, G. Dall'Asta, U. Campigli, E. Strepparola, Makromol. Chem., **60**, 202 (1963).
270. G. Carazzolo, Gazz. Chim. Ital., **92**, 1345 (1962); G. Carazzolo, G. Putti, Chim. Ind. (Milan), **45**, 771 (1963); G. Carazzolo, M. Mammi, J. Polym. Sci., Part A, **1**, 965 (1963); G. A. Carazzolo, J. Polym. Sci., Part A, **1**, 1573 (1963); G. Carazzolo, S. Leghissa, M. Mammi, Makromol. Chem., **63**, 171 (1963).
271. E. Baer, J. R. Collier, D. R. Carter, SPE Trans., **5**, 22 (1965).
272. D. C. Vogelsong, J. Polym. Sci., Part A, **1**, 1055 (1963).
273. F. H. C. Crick, A. Rich, Nature (London) **176**, 780 (1955).
274. G. Perego, I. W. Bassi, Makromol. Chem., **61**, 198 (1963).
275. U. Giannini, M. Cambini, A. Cassata, Makromol. Chem., **61**, 246 (1963).
276. F. Danusso, G. Giannotti, Makromol. Chem., **61**, 164 (1963).
277. F. Danusso, G. Giannotti, Makromol. Chem., **61**, 139 (1963).
278. M. Donati, M. Farina, Makromol. Chem., **60**, 233 (1963).
279. E. M. Bradbury, A. Elliott, Polymer, **4**, 47 (1963).
280. H. Utsunomiya, N. Kawasaki, M. Niinomi, M. Takayanagi, J. Polym. Sci., Part B, **5**, 907 (1967).
281. E. J. Vandenberg, J. Polym. Sci., Part C, **1**, 207 (1963).
282. K. R. Dunham, J. Vandenberghe, J. W. H. Faber, L. E. Contois, J. Polym. Sci., Part A, **1**, 751 (1963).
283. M. Tomika, Kobunshi Kagaku, **20**, 145 (1963).
284. N. Calderon, M. C. Morris, J. Polym. Sci., Part A-2, **5**, 1283 (1967).
285. N. S. Chu, C. C. Price, J. Polym. Sci., Part A, **1**, 1105 (1963).
286. K. H. Meyer, A. van der Wyk, Helv. Chim. Acta., **20**, 1313 (1937).
287. R. D. Flanagan, A. M. Rijke, J. Polym. Sci., Part A-2, **10**, 1207 (1972).
288. K. H. Meyer, Natural and Synthetic High Polymers, Interscience, New York (1950).
289. J. R. Whinfield, Nature (London), **158**, 930 (1946).

290. E. F. Izard, J. Polym. Sci., **8**, 503 (1952).
291. R. Hill, E. E. Walker, J. Polym. Sci., **3**, 609 (1948).
292. D. D. Coffman, M. L. Cox, F. L. Martin, W. E. Mochel, F. J. van Natta, J. Polym. Sci., **3**, 85 (1948).
293. J. R. Schaefgen, P. J. Flory, J. Am. Chem. Soc., **70**, 2709 (1948).
294. J. C. Patrick, Trans. Faraday Soc., **32**, 347 (1936).
295. C. W. Bunn, J. Polym. Sci., **16**, 323 (1955).
296. Ye. L. Gal'Perlin, S. S. Dubov, E. V. Volkova, M. P. Mlenik, Kristallografiya, **9**, 102 (1964); Sov. Phys. Crystallogr. (Engl. Transl.), **9**, 81 (1964).
297. A. Noshay, C. C. Price, J. Polym. Sci., **34**, 165 (1959).
298. E. Gipstein, E. Wellisch, O. J. Sweeting, J. Polym. Sci., Part B, **1**, 237 (1963).
299. L. D. Jones, F. E. Karasz, J. Polym. Sci., Part B, **4**, 803 (1966).
300. W. J. Carothers, J. A. Arvin, J. Am. Chem. Soc., **51**, 2560 (1929).
301. K. Ueberreiter, N. Steiner, Makromol. Chem., **74**, 158 (1964).
302. W. H. Carothers, G. L. Dorough, J. Am. Chem. Soc., **52**, 711 (1930).
303. Y. Takahashi, H. Tadokoro, macromolecules, **6**, 672 (1973).
304. W. H. Carothers, G. J. Berchet, J. Am. Chem. Soc., **52**, 5289 (1930).
305. W. H. Carothers, Chem. Rev., **8**, 353 (1931).
306. G. Natta, M. Peraldo, G. Allegra, Makromol. Chem., **75**, 215 (1964).
307. W. H. Carothers, J. W. Hill, J. Am. Chem. Soc., **54**, 1559 (1932).
308. A. Turner-Jones, J. M. Aizlewood, D. R. Beckett, Makromol. Chem., **75**, 134 (1964). (Unit cell constants given for poly(propene)-form III refer to a projected base).
309. W. H. Carothers, J. Am. Chem. Soc., **55**, 4714 (1933).
310. M. I. Bessonov, A. P. Rudakov, Fiz. Tverd. Tela (Leningrad), **6**, 1333 (1964), through Chem. Abstr., **61**, 5793D (1964).
311. P. R. Swan, J. Polym. Sci., **56**, 403 (1962).
312. K. Hirono, G. Wasai, T. Saegusa, J. Furukawa, Kogyo Kagaku Zasshi, **67**, 604 (1964).
313. J. Boor, Jr., J. C. Mitchell, J. Polym. Sci., Part A, **1**, 59 (1963).
314. G. Wasai, T. Saegusa, J. Furukawa, Kogyo Kaguku Zasshi, **67**, 601 (1964).
315. B. Wunderlich, T. Arakawa, J. Polym. Sci., Part A, **2**, 3697 (1964).
316. K. Tsuboi, T. Mochizuki, J. Polym. Sci., Part B, **1**, 531 (1963); Kobunshi Kagaku, **23**, 645 (1966).
317. K. Hayashi, Y. Kitanishi, M. Nishii, S. Okamura, Makromol. Chem., **47**, 237 (1961).
318. S. Kambara, A. Takahashi, Makromol. Chem., **63**, 89 (1963).
319. J. P. Kennedy, J. J. Elliott, B. Groten, Makromol. Chem., **77**, 26 (1964); J. J. Elliott, J. P. Kennedy, J. Polym. Sci., Polym. Chem. Ed., **11**, 2991 (1973).
320. J. Wilsky, T. Grewer, J. Polym. Sci., Part C, **6**, 33 (1964).
321. G. Carazzolo, Chim. Ind. (Milan), **46**, 525 (1964).
322. G. W. Urbanczyk, Zest. Nauk. Politech. Lodz, Wlok., **9**, 79 (1962), through chem. Abstr., **61**, 5836B (1964).
323. K. Fujii, T. Mochizuki, S. Imoto, J. Ukida, M. Matsumoto, J. Polym. Sci., Part A, **2**, 2327 (1964).
324. G. Natta, G. Dall'Asta, G. Mazzanti, I. Pasquon, A. Valvassori, A. Zambelli, J. Am. Chem. Soc., **83**, 3343 (1961); P. Corradini, P. Ganis, Makromol. Chem., **62**, 97 (1963).
325. E. N. Zilberman, A. E. Kulikova, N. M. Teplyakov, J. Polym. Sci., **56**, 417 (1962).
326. G. Allegra, A. Ponoglio, I. Pasquon, Rend. Ist. Lomb. Accad. Sci. Lett. A, **95**, 335 (1961).
327. M. Ishibashi, J. Polym. Sci., Part A, **2**, 4361 (1964).
328. A. D. Ketley, R. J. Ehrig, J. Polym. Sci., Part A, **2**, 4461 (1964).
329. O. Vogl, J. Polym. Sci., Part A, **2**, 4621 (1964).
330. A. Chiba, H. Futama, J. Furuichi, Rep. Progr. Polym. Phys. Japan, **7**, 51 (1964).
331. G. Carazzolo, M. Mammi, J. Polym. Sci., Part B, **2**, 1057 (1964); G. Carazzolo, G. Valle, Makromol. Chem., **90**, 66 (1966).
332. J. P. Kennedy, R. M. Thomas, Makromol. Chem., **64**, 1 (1963).
333. M. Litt, J. Polym. Sci., Part A, **1**, 2219 (1963).
334. R. Brill, Z. Phys. Chem., Abt. B, **53**, 61 (1943).
335. J. C. Mitchell, J. Polym. Sci., Part B, **1**, 285 (1963).
336. M. Ogawa, T. Ota, O. Yoshizaki, E. Nagai, J. Polym. Sci., Part B, **1**, 57 (1963); T. Ota, O. Yoshizaki, E. Nagai, Kobunshi Kagaku, **20**, 225 (1963).
337. E. Mueller, Melliand Textilber, **44**, 484 (1963).
338. J. R. Schaefgen, R. Zbinden, J. Polym. Sci., Part A, **2**, 4865 (1964).
339. D. D. Coffman, G. J. Berchet, U. R. Peterson, E. W. Spanagel, J. Polym. Sci., **2**, 306 (1947).
340. H. S. MakowskI, B. K. C. Shim, Z. W. Wilchinsky, J. Polym. Sci., Part A, **2**, 4973 (1964).
341. F. Rybnikar, Colllect. Czech. Chem. Commun., **27**, 2864 (1962).
342. L. Mandelkern, M. Tryon, F. A. Quinn, Jr., J. Polym. Sci., **19**, 77 (1956).
343. M. Inoue, J. Polym. Sci., Part A, **1**, 2697 (1963).
344. I. Heber, Kolloid Z. Z. Polym., **189**, 110 (1963).
345. K. Mihara, Angew. Makromol. Chem., **40/41**, 41 (1974).
346. G. B. Gechele, L. Crescentini, J. Appl. Polym. Sci., **7**, 1349 (1963).
347. R. Koyama, Bull. Inst. Chem. Res., Kyoto Univ., **41**, 207 (1963); A. Nishioka, H. Watanabe, R. Koyama, H. Nagai, Rep. Progr. Polym. Phys. Japan, **6**, 311 (1963).
348. H. Tadokoro, Y. Chatani, M. Kobayashi, T. Yoshihara, S. Murahashi, Rep. Progr. Polym. Phys. Japan, **6**, 303 (1963); K. Imada, T. Miyaka, Y. Chatani, H. Tadokoro, S. Murahashi, Makromol. Chem., **83**, 113 (1965).
349. H. W. Wyckoff, J. Polym. Sci., **62**, 83 (1962).
350. A. Turner-Jones, J. Polym. Sci., Part B, **1**, 455 (1963).
351. M. G. Broadhurst, J. Res. Natl. Bur. Stand., Sect. A, **70**, 481 (1966).
352. G. Natta, G. Moraglio, Rubber Plast. Age, **44**, 42 (1962).

353. G. Natta, G. Moraglio, Makromol. Chem., **66**, 218 (1963).
354. M. Ishibashi, Polymer, **5**, 305 (1964).
355. A. Turner-Jones, J. M. Aizlewood, J. Polym. Sci., Part B, **1**, 471 (1963).
356. M. Ishibashi, J. Polym. Sci., Part B, **1**, 629 (1963).
357. Y. Sakurada, M. Matsumoto, K. Imai, A. Nishioda, Y. Kato, J. Polym. Sci., Part B, **1**, 633, (1963).
358. I. Kirshenbaum, Z. W. Wilchinsky, B. Groten, J. Appl. Polym. Sci., **8**, 2723 (1964).
359. N. P. Borisova, T. M. Birshtein, Vysokomol. Soedin., **5**, 279 (1963) - Polym. Sci. (USSR) (Engl. Transl.), **5**, 907 (1963).
360. G. Dall'Asta, I. W. Bassi, Chim. Ind. (Milan), **43**, 999 (1961).
361. C. A. Boye, Bull. Am. Phys. Soc., **8**, 266 (1963).
362. P. J. Flory, A. Vrij, J. Am. Chem. Soc., **85**, 3548 (1963).
363. P. R. Swan, J. Polym. Sci., **42**, 525 (1960).
364. A. Turner-Jones, R. P. Palmer, Polymer **4**, 525 (1963).
365. M. Genas, Angew. Chem. **74**, 535 (1962).
366. N. G. Gaylord, Ed., Polyethers II., Interscience, New York, (1963).
367. M. Cesari, J. Polym. Sci., Part B, **2**, 453 (1964); W. Marconi, A, Mazzei, S. Cucinella, M. Cesari, J. Polym. Sci., Part A, **2**, 4261 (1964). (a and b obtained from equatiorial reflections alone - true cell is probably triclinic).
368. W. Marconi, A. Mazzei, A. Cucinella, M. Cesari, E. Paulezzi, J. Polym. Sci., Part A, **3**, 123 (1965). (Unit cell constants refer to a projected base).
369. W. J. Jackson, Jr., J. R. Caldwell, J. Appl. Polym. Sci., **7**, 1975 (1963).
370. R. B. Isaacson, I. Kirshenbaum, W. C. Feist, J. Appl. Polym. Sci., **8**, 2789 (1964).
371. G. Natta, L. Porri, M. C. Gallazzi, Chim. Ind. (Milan), **46**, 1158 (1964).
372. D. Kockott, Kolloid Z. Z. Polym., **198**, 17 (1964).
373. F. S. Ingraham, D. F. Wooley, Jr., Ind. Eng. Chem., **56**, No. 9, 53 (1964).
374. R. Janssen, H. Ruysschaert, R. Vroom, Makromol. Chem., **77**, 153 (1964).
375. G. Natta, I. W. Bassi, G. Perego, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **36**, 291 (1964).
376. F. B. Moody, Macromol. Synth., **1**, 67 (1963).
377. G. Allen, C. Booth, M. N. Jones, D. J. Marks, W. D. Taylor, Polymer, **5**, 547 (1964).
378. G. Wasai, T. Saegusa, J. Furukawa, Kogyo Kagaku Zasshi, **67**, 1428 (1964).
379. T. Tsuruta, R. Fujio, J. Furukawa, Makromol. Chem., **80**, 172 (1964).
380. F. Danusso, G. Gianotti, Makromol. Chem., **80**, 1 (1964).
381. D. Siegl, H. Sixl, V. Enkelmann, G. Wenz, Chem. Phys., **72**, 201 (1982).
382. G. P. Lorenzi, E. Benedetti, F. Chiellini, Chim. Ind. (Milan), **46**, 1474 (1964).
383. M. Goodman, Y.-L. Fan, J. Am. Chem. Soc., **86**, 4922 (1964).
384. G. Natta, P. Corradini, P. Ganis, J. Polym. Sci., Part A, **3**, 11 (1965).
385. A. Bell, J. G. Smith, C. J. Kibler, J. Polym. Sci., Part A, **3**, 19 (1965).
386. E. L. Wittbecker, W. S. Spliethoff, G. R. Stine, J. Appl. Polym. Sci., **9**, 213 (1965).
387. G. Wasai, T. Iwata, K. Hirono, M. Kuragano, T. Saegusa, J. Furukawa, Kogyo Kagaku Zasshi, **67**, 1920 (1964).
388. D. E. Ilyina, B. A. Krentsel, G. E. Semenido, J. Polym. Sci., Part C, **4**, 999 (1964).
389. Ye. L. Gal'Perlin, Yu. V. Strogalin, Vysokomol. Soedin., **7**, 16 (1965); Polym. Sci. (USSR) (Engl. Transl.), **7**, 15 (1965).
390. F. T. Simon, J. M. Rutherford, Jr., J. Appl. Phys., **35**, 82 (1964).
391. J. Powers, J. D. Hoffman, J. J. Weeks, F. A. Quinn, Jr., J. Res. Natl. Bur. Stand., Sect. A, **69**, 335 (1965).
392. G. Farrow, Polymer, **4**, 191 (1963).
393. F. Danusso, G. Moraglio, W. Chiglia, L. Motto, G. Talamini, Chim. Ind. (Milan), **41**, 748 (1959).
394. W. R. Krigbaum, I. Uematsu, J. Polym. Sci., Part A, **3**, 767 (1965).
395. W. F. Gorham, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **6**, 73 (1965).
396. H. Morawetz, S. Z. Jakabhazy, J. B. Lando, J. Shafer, Proc. Natl. Acad. Sci. U.S.A., **49**, 789 (1963).
397. M. Barlow, J. Polym. Sci., Part A-2, **4**, 121 (1966).
398. R. Yamadera, C. Sonoda, J. Polym. Sci., Part B, **3**, 411 (1965).
399. E. J. Vandenberg, J. Polym. Sci., **47**, 489 (1960).
400. Yu. Ya. Tomashpol'skii, G. S. Markova, Vysokomol. Soedin., **6**, 27 (1964), - Polym. Sci., USSR (Engl. Transl.), **6**, 316 (1964).
401. A. Kawasaki, J. Furukawa, T. Tsuruta, Y. Nakayama, G. Wasai, Makromol. Chem., **49**, 112 (1961); *ibid.*, p. 136; T. Makimoto, T. Tsuruta, J. Furukawa, ibid., **50**, 116 (1961); G. Wasai, J. Furukawa, A. Kawasaki, Kogyo Kagaku Zasshi, **68**, 210 (1965).
402. V. V. Korshak, T. M. Frunze, Synthetic Hetero-Chain Polyamides, translated by N. Kaner, Daniel Davey and Co., New York, 1964, Tables I, II, XXI, XXVIII, AND XXX.
403. M. Cesari, G. Perego, A. Mazzei, Makromol. Chem., **83**, 196 (1965).
404. H. Iiyama, M. Asakura, K. Kimoto, Kogyo Kagaku Zasshi, **68**, 243 (1965).
405. I. Krishenbaum, J. Polym. Sci., Part A, **3**, 1869 (1965).
406. H. Kojima, K. Yamaguchi, Kobunshi Kagaku, **19**, 715 (1962).
407. E. Echochard, J. Chim. Phys. Phys.-Chim. Biol., **43**, 113 (1946), through Ref., 402, p. 363.
408. F. A. Quinn, Jr., J. Powers, J. Polym. Sci., Part B, **1**, 341 (1963).
409. G. Natta, M. Pegoraro, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **34**, 110 (1963).
410. H. Tadokoro, S. Seki, I. Nitta, Bull. Chem. Soc. Japan., **28**, 559 (1955).
411. G. Natta, F. Danusso, G. Moraglio, Makromol. Chem., **28**, 166 (1958).

412. G. Natta, G. Dall'Asta, G. Mazzanti, I. Pasquon, a. Valvassori, A. Zambelli, Makromol. Chem., **54**, 95 (1962); G. Natta, G. Allegra, I. W. Bassi, P. Corradini, P. Ganis, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., **36**, 433 (1964).
413. V. P. Lebedev, N. A. Okladnov, K. S. Minsker, B. P. Shtarkman, Vysokomol. Soedin., **7**, 655 (1965).
414. D. J. Lyman, J. Heller, M. Barlow, Makromol. Chem., **84**, 64 (1965).
415. J. Boor, Jr., E. A. Youngman, J. Polym. Sci., Part B, **3**, 577 (1965).
416. G. Carazzolo, L. Mortillaro, L. Credali, S. Bezzi, J. Polym. Sci., Part B, **2**, 997 (1964); L. Mortillaro, L. credali, M. Russo, C. de Cheechi, J. Polym. Sci., Part B, **3**, 581 (1965); G. Carazzolo, G. Valle, J. Polym. Sci., Part A, **3**, 4013 (1965).
417. Ye. L. Gal'Perin, Yu. V. Strogalin, M. P. Mlenik, Vysokomol. Soedin., **7**, 933 (1965); Polym. Sci. USSR (Engl. Transl.), **7**, 1031 (1965).
418. J. B. Lando, H. G. Olf, A. Peterlin, J. Polym. Sci., Part A-1, **4**, 941 (1966).
419. K. Okamura, R. H. Marchessault, in G. N. Ramachandran, Ed., Conformation of Biopolymers, **V. 2**, Academic, New York, (1967), p. 709; J. Cornibert, R. H. Marchessault, J. Mol. Biol., **71**, 735 (1972); K. Okamura, Kobunshi, **21**, 525 (1972).
420. M. Donati, G. Perego, M. Farina, Makromol. Chem., **85**, 301 (1965).
421. H. Komoto, K. Saotome, Kobunshi Kagaku, **22**, 337 (1965).
422. W. G. C. Forsyth, A. C. Hayward, J. B. Roberts, Nature (London), **182**, 800 (1958).
423. A. Nakajima, H. Hamada, S. Hayashi, Makromol. Chem., **95**, 40 (1966).
424. G. Natta, I. W. Bassi, G. Allegra, Makromol. Chem., **89**, 81 (1965).
425. K. Miki, R. Nakatsuka, Rep. Progr. Polym. Phys. Japan., **8**, 115 (1963).
426. W. D. Niegisch, J. Appl. Phys., **37**, 4041 (1966); J. Polym. Sci., Part B, **4**, 531 (1966).
427. Y. Joh, T. Yoshihara, Y. Kotake, F. Ide, K. Nakatsuka, J. Polym. Sci., Part B, **3**, 933 (1965); T. Yoshihara, Y. Kotake, Y. Joh, *ibid.*, **5**, 459 (1967).
428. C. G. Overberger, J. K. Weise, J. Polym. Sci., Part B, **2**, 329 (1964); J. Am. Chem. Soc., **90**, 3533 (1968).
429. P. Corradini, I. W. Bassi, J. Polym. Sci., Part C, **16**, 3233 (1968).
430. G. Natta, P. Corradini, P. Ganis, P. A. Temussi, J. Polym. Sci., Part C, **16**, 2477 (1967).
431. M. G. Huguet, Makromol. Chem., **94**, 205 (1966); H. D. Noether, J. Polym. Sci., Part C, **16**, 725 (1967).
432. I. W. Bassi, P. Ganis, P. A. Temussi, J. Polym. Sci., Part C, **16**, 2867 (1967).
433. R. K. Tubbs, J. Polym. Sci., Part A, **3**, 4181 (1965).
434. G. Natta, G. Allegra, I. W. Bassi, D. Sianesi, G. Caporiccio, E. Torti, J. Polym. Sci., Part A, **3**, 4263 (1965).
435. R. Bonart, Makromol. Chem., **92**, 149 (1966).
436. G. Natta, G. Dall'Asta, Chim. Ind. (Milan), **46**, 1429 (1964).
437. C. G. Overberger, H. Kaye, G. Walsh, J. Polym. Sci., Part A, **2**, 755 (1964).
438. H. Schnell, Angew. Chem., **68**, 633 (1956).
439. G. Natta, G. Dall'Asta, I. W. Bassi, G. Carella, Makromol. Chem., **91**, 87 (1966).
440. V. M. Coiro, P. de Santis, L. Mazzarella, L. Picozzi, J Polym. Sci., Part A, **3**, 4001 (1965).
441. J. A. Faucher, F. P. Reding, in R. A. V. Raff, K. W. Doak, Eds., Crystalline Olefin Polymers, Part I, Interscience, New York, 1965, p. 677.
442. A. Turner-Jones, Polymer, **7**, 23 (1966).
443. N. I. Makarevich, Zh. Prikl. Spektroskopii, **2**, 341 (1965), through Chem. Abstr., **63**, 8509a (1965).
444. J. R. Schaefgen, F. H. Koontz, R. F. Tietz, J. Polym. Sci., **40**, 377 (1959).
445. T. B. Gibb, Jr., R. A. Clendinning, W. D. Niegisch, J. Polym. Sci., Part A-1, **4**, 917 (1966).
446. H. Tadokoro, Y. Takahashi, Y. Chatani, H. Kakida, Makromol. Chem., **109**, 96 (1967); H. Kakida, D. Makino, Y. Chatani, M. Kobayashi, H. Tadokoro, Macromolecules, **3**, 569 (1970).
447. J. B. Rose, J. Chem. Soc., 542, *ibid.*, 546 (1956).
448. M. Cesari, G. Perego, W. Marconi, Makromol. Chem., **94**, 194 (1966).
449. I. I. Novak, V. I. Vettegren, Vysokomol. Soedin., **7**, 1027 (1965); Polym. Sci. USSR (Engl. Transl.), **7**, 1136 (1965).
450. F. Hamada, A. Nakajima, Kobunshi Kagaku, **23**, 395 (1966).
451. G. Moraglio, G. Polizzotti, F. Danusso, Eur. Polym. J., **1**, 183 (1965).
452. K.-S. Chan, Ph.D. Thesis, State Univ. College of Forestry at Syracuse Univ., Diss. Abstr., **26**, 4260 (1966).
453. J. G. Smith, C. J. Kibler, B. J. Sublett, J. Polym. Sci., Part A-1, **4**, 1851 (1966).
454. K. Saotome, H. Komoto, J. Polym. Sci., Part A-1, **4**, 1463 (1966).
455. K. F. Wissbrun, J. Polym. Sci., Part A-2, **4**, 827 (1966).
456. P. Ganis, P. A. Temussi, Eur. Polym. J., **2**, 401 (1966).
457. R. J. Fredericks, T. H. Doyne, R. S. Spraque, J. Polym. Sci., Part A-2, **4**, 899 (1966).
458. R. J. Fredericks, T. H. Doyne, R. S. Spraque, J. Polym. Sci., Part A-2, **4**, 913 (1966).
459. T. Ota, M. Yamashita, O. Yoshizaki, E. Nagai, J. Polym. Sci., Part A-2, **4**, 959 (1966).
460. H. Tadokoro, J. Polym. Sci., Part C, **15**, 1 (1966).
461. H. Yasuda, Y. Tanada, M. Takayanagi, Kogyo Kagaku Zasshi, **69**, 304 (1966).
462. R. W. Eykamp, A. M. Schneider, E. W. Merrill, J. Polym. Sci., Part A-2, **4**, 1025 (1966).
463. G. Carazzolo, M. Mammi, J. Polym. Sci., Part C, **16**, 1521 (1967); Makromol. Chem., **100**, 28 (1967).
464. K. Chujo, H. Kobayashi, J. Suzuki, S. Tokuhara, M. Tanabe, Makromol. Chem., **100**, 262 (1967).
465. K. Chujo, H. Kobayashi, J. Suzuki, S. Tokuhara, Makromol. Chem., **100**, 267 (1967).
466. W. Braun, K.-H. Hellwege, W. Knappe, Kolloid Z. Z. Polym., **215**, 10 (1967).

467. M. Hayashi, Y. Shiro, H. Murata, Bull. Chem. Soc. Japan, **39**, 1857 (1966).
468. M. Hayashi, Y. Shiro, H. Murata, Bull. Chem. Soc. Japan., **39**, 1861 (1966).
469. F. Tuinstra, Acta Crystallogr., **20**, 341 (1966).
470. B. M. Ginzburg, L. N. Korzhavin, S. Ya. Frenkel', L. A. laius, M. B. Adrova, Vysokomol. Soledin., **8**, 278 (1966); Polym. Sci. USSR (Engl. Transl.), **8**, 302 (1966).
471. F. E. Karasz, J. M. O'Reilly, H. E. Bair, R. A. Kluge, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **9**, 822 (1968).
472. P. E. Slade, T. A. Orofino, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **9**, 825 (1968).
473. W. J. MacKnight, M. Yang, T. Kajima, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **9**, 860 (1968).
474. V. G. Baranov, Bu. Zhu-Chan, T. I. Volkov, S. Ya. Frenkel', Vysokomol. Soedin., Ser. A, **9**, 81 (1967); Polym. Sci. USSR (Engl. Transl.), **9**, 87 (1967).
475. P. Kovacic, M. B. Feldman, J. P. Kovacic, J. B. Lando, J. Appl. Polym. Sci., **12**, 1735 (1968).
476. I. W. Bassi, G. Fagherazzi, Eur. Polym. J., **4**, 123 (1968).
477. G. Fagherazzi, I. W. Bassi, Eur. Polym. J., **4**, 151 (1968).
478. F. Danusso, G. Gianotti, Eur. Polym. J., **4**, 165 (1968).
479. C. G. Overberger, T. Takekoshi, Macromolecules, **1**, 7 (1968).
480. K. Kikukawa, S. Nozakura, S. Murahashi, Kobunshi Kagaku, **25**, 19 (1968).
481. H. K. Reimschuessel, L. G. Roldan, J. P. Sibilia, J. Polym. Sci., Part A-2, **6**, 559 (1968); H. K. Reimschuessel, Trans. N. Y. Acad. Sci., **33**, 219 (1971).
482. S. Kobayashi, H. Tadokoro, Y. Chatani, Makromol. Chem., **112**, 225 (1968).
483. F. J. Golemba, J. E. Guillet, S. C. Nyburg, J. Polym. Sci., Part A-1, **6**, 1341 (1968).
484. F. E. Karasz, H. E. Bair, J. M. O'Reilly, J. Polym. Sci., Part A-2, **6**, 1141 (1968).
485. A. Turner-Jones, A. J. Cobbold, J. Polym. Sci., Part B, **6**, 538 (1968); R. J. Samuels, R. H. Yee, J. Polym. Sci., Part A-2, **10**, 385 (1972).
486. J. J. Klement, P. H. Geil, J. Polym. Sci., Part A-2, **6**, 1381 (1968).
487. H. M. van Dort, C. A. M. Hoefs, E. P. Magre', A. J. Schopf, K. Yntema, Eur. Polym. J., **4**, 275 (1968); J. Boon, E. P. Magre', Makromol. Chem., **126**, 130 (1969).
488. G. Natta, P. Corradini, I. W. Bassi, G. Fagherazzi, Eur. Polym. J., **4**, 297 (1968).
489. A. Carbonaro, A. Greco, I. W. Bassi, Eur. Polym. J., **4**, 445 (1968); I. W. Bassi, G. Chioccola, *ibid.*, **5**, 163 (1969).
490. Y. Chatani, K. Suehiro, Y. Okita, H. Tadokoro, K. Chujo, Makromol. Chem., **113**, 215 (1968).
491. J. Masamoto, K. Sasaguri, C. Ohizumi, H. Kobayashi, J. Polym. Sci., Part A-2, **8**, 1703 (1970); J. Masamoto, H. Kobayashi, Kobunshi Kagaku **27**, 220 (1970).
492. H. Utsunomiya, T. Mori, K. Imada, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **11**, 153 (1968).
493. S. Minami, S. Manabe, M. Takayanagi, Rep. Progr. Polym. Phys. Japan., **11**, 155 (1968); S. Manabe, S. Minami, M. Takayanagi, Kogyo Kagaku Zasshi, **73**, 1577 (1970).
494. S. Takamuku, K. Imada, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **11**, 159 (1968); M. Kuroishi, M. Fujisaki, T. Kajiyama, M. Takayanagi, Nippon Kagaku Kaishi, 1281 (1972).
495. K.-H. Illers, H. Haberkorn, Makromol. Chem., **146**, 267 (1971).
496. S. Iwayanagi, I. Sakurai, T. Sakurai, T. Seto, Rep. Propgr. Polym. Phys. Japan, **10**, 167 (1967); J. Macromol. Sci. B: Phys., **2**, 163 (1968).
497. H. Tadokoro, Y. Takahashi, S. Otsuka, K. Mori, F. Imaisumi, J. Polym. Sci., Part B, **3**, 697 (1965); H. Tadokoro, M. Kobayashi, K. Mori, Y. Takahashi, S. Taniyan, Rep. Progr. Polym. Phys. Japan., **10**, 181 (1967); idem., J. Polym. Sci., Part C, **22**, 1031 (1969).
498. C. Booth, C. J. Devoy, G. Gee, Polymer, **12**, 327 (1971).
499. T. Yubayashi, H. Sato, N. Yamada, Rep. Progr. Polym. Phys. Japan, **10**, 271 (1967).
500. H. Ailhaud, Y. Gallot, A. Skoulios, C. R. Hebd. Seances Acad. Sci., Ser. C, **267**, 139 (1968).
501. G. Giannotti, A. Capizzi, Eur. Polym. J., **4**, 677 (1968).
502. D. Fisher, Proc. Phys. Soc. London, Sect. B, **66**, 7 (1953).
503. Unpublished work of H. Scherer, in H. Bestian, Angew. Chem., Int. Ed. Engl., **7**, 278 (1968).
504. Y. Takahashi, H. Tadokoro, Y. Chatani, J. Macromol. Sci. B: Phys., **2**, 361 (1968).
505. E. F. Oleinik, N. S. Yenikolopyan, Vysokomol. Soedin., Ser. A, **9**, 2609 (1967); Polym. Sci. USSR (Engl. Transl.), **9**, 2951 (1967).
506. M. Okada, Y. Yamashita, Y. Ishii, Makromol. Chem., **80**, 196 (1964).
507. C. H. Bamford, L. Brown, A. Elliott, W. E. Hanby, I. F. Trotter, Nature (London), **173**, 27 (1954); L. Brown, I. F. Trotter, Trans. Faraday Soc., **52**, 537 (1956); A. Elliott, B. R. Malcolm, Proc. R. Soc. London, Ser. A, **249**, 30 (1959).
508. C. S. Marvel, G. E. Hartzell, J. Am. Chem. Sco., **81**, 448 (1959).
509. S. Arnott, S. D. Dover, A. Elliott, J. Mol. Biol., **30**, 201 (1967).
510. P. M. Cowan, S. McGavin, Nature (London), **176**, 501 (1955).
511. V. Sasisekharan, Acta Crystallogr., **12**, 897 (1959).
512. S. Arnott, S. D. Dover, Acta Crystallogr., Sect. B: Struct. Crystallogr. Cryst. Chem., **24**, 599 (1968).
513. T. Uchida, H. Tadokoro, J. Polym. Sci., Part A-2, **5**, 63 (1967).
514. M. Goodman, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **10**, 36 (1969). (Pseudohexagonal cell based on equatiorial reflections).
515. H. Sakakihara, Y. Takahashi, H. Tadokoro, P. Sigwalt, N. Spassky, Macromolecules, **2**, 515 (1969).
516. N. Yasuoka, N. Kasai, M. Kakudo, T. Ando, S. Kuribayashi, Japan Rep. Progr. Polym. Phys. Jpn., **11**, 149 (1968).
517. C. L. Lee, O. K. Johannson, O. L. Flaningam, P. Hahn, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **10**, 1311 (1969).

518. C. L. Lee, O. K. Johannson, O. L. Flaningam, P. Hahn, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **10**, 1319 (1969); D. R. Peterson, D. R. Carter, C. L. Lee, J. Macromol. Sci. B: Phys., **3**, 519 (1969).
519. M. Hikosaka, T. Seto, Rep. Progr. Polym. Phys. Japan, **12**, 153 (1969); Polym. J. (Tokoyo), **5**, 111 (1973).
520. H. Tadokoro, Kobunshi, **19**, 775 (1970); H. Tadokoro, Y. Chatani, H. Kusanagi, M. Yokayama, Macromolecules, **3**, 441 (1970).
521. T. Kanamoto, H. Nagai, K. Tanaka, Rep. Progr. Polym. Phys. Japan, **9**, 135 (1966); T. Kanamoto, K. Tanaka, H. Nagai, J. Polym. Sci., Part A-2, **9**, 2043 (1971); T. Kanamoto, J. Polym. Sci., Polym. Phys. Ed., **12**, 2535 (1974).
522. U. Schmueli, W. Traub, K. Rosenheck, J. Polym. Sci., Part A-2, **7**, 515 (1969).
523. S. Y. Hobbs, F. W. Billmeyer, Jr., J. Polym. Sci., Part A-2, **7**, 1119 (1969).
524. A. R. Schultz, C. R. McCullough, J. Polym. Sci., Part A-2, **7**, 1577 (1969).
525. S. Murahashi, M. Kamachi, N. Wakabayashi, J. Polym. Sci., Part B, **7**, 135 (1969).
526. H. D. Noether, C. G. Overberger, G. Halek, J. Polym. Sci., Part A-1, **7**, 201 (1969).
527. E. J. Vandenberg, J. Polym. Sci., Part A-1, **7**, 525 (1969).
528. G. Natta, J. di Pietro, M. Cambini, Makromol. Chem., **56**, 200 (1962).
529. A. Kawaguchi, T. Ikawa, Y. Fujiwara, K. Monobe, Rep. Progr. Polym. Phys. Japan, **9**, 157 (1966); A. Kawaguchi, T. Ikawa, Y. Fujiwara, M. Tabuchi, K. Monobe, J. Macromol. Sci. B: Phys., **20**, 1 (1981).
530. G. Natta, I. W. Bassi, Eur. Polym. J., **3**, 43 (1967); G. Natta, I. W. Bassi, G. Fagherazzi, *ibid.*, **339**, G. Natta, I. W. Bassi, G. Fagherazzi, ibid., **5**, 239 (1969).
531. E. W. Fischer, F. Kloos, G. Lieser, J. Polym. Sci., Part B, **7**, 845 (1969).
532. K. Ishikawa, K. Miyasaka, T. Okabe, Makromol. Chem., **122**, 123 (1969).
533. E. F. Vainshtein, M. Ya. Kusherev, A. A. Popov, S. G. Entelis, Vysokomol. Soedin., Ser. A, **11**, 1606 (1969); Polym. Sci. USSR (Engl. Transl.), **11**, 1820 (1969).
534. A. Mattiussi, G. B. Gechele, M. A. Paresi, A. Rondanelli, Chim. Ind. (Milan), **51**, 376 (1969).
535. P. Zugenmaier, H.-J. Cantow, Kolloid Z. Z. Polym., **230**, 229 (1969).
536. W. P. Baker, Jr., J. Polym. Sci., Part A, **1**, 655 (1963).
537. A. Turner-Jones, Polymer, **6**, 249 (1965).
538. A. Kovacs, private comminucation.
539. P. W. T. Willmott, F. W. Billmeyer, Jr., Off. Dig. Fed. Soc. Paint Technol. Eng., **35**, 847 (1963).
540. H. Schnell, Ind. Eng. Chem., **51**, 157 (1959).
541. R. A. Wessling, J. E. Mark, R. E. Hughes, J. Phys. Chem., **70**, 1909 (1966).
542. T. P. Hobin, Polymer, **7**, 367 (1966).
543. Y. Iwakura, Y. Taneda, S. Uchida, J. Appl. Polym. Sci., **5**, 108 (1961).
544. A. J. Yu, R. D. Evans, J. Polym. Sci., **62**, 249 (1960).
545. J. Boor, Jr., E. A. Youngman, J. Polym. Sci., Part A-1, **4**, 1861 (1966).
546. H. E. Lunk, E. A. Youngman, J. Polym. Sci., Part A, **3**, 2983 (1965).
547. C. J. Kibler, A. Bell, J. G. Smith, J. Polym. Sci., Part A, **2**, 2115 (1964).
548. E. J. Vandenberg, J. Polym. Sci., Part A-1, **10**, 329 (1972).
549. P. Corradini, P. Ganis, V. Petraccone, Eur. Polym. J., **6**, 281 (1970).
550. K. Suehiro, M. Takayanagi, J. Macromol. Sci. B: Phys., **4**, 39 (1970).
551. C. Borri, S. Bruckner, V. Crescenzi, G. Della Fortuna, A. Mariano, P. Scarazzoto, Eur. Polym. J., **7**, 1515 (1971).
552. H. R. Allcock, R. L. Kugel, K. J. Valan, Inorg. Chem., **5**, 1709 (1966).
553. G. Allen, C. J. Lewis, S. M. Todd, Polymer, **11**, 44 (1970).
554. V. Y. Chen, G. Allegra, P. Corradini, M. Goodman, Macromolecules, **3**, 274 (1970).
555. G. Perego, M. Cesari, Makromol. Chem., **133**, 133 (1970).
556. W. R. Krigbaum, J. H. O'Mara, J. Polym. Sci., Part A-2, **8**, 1011 (1970).
557. C. Gieniewski, R. S. Moore, Macromolecules, **3**, 97 (1970).
558. H. Kanetsuna, M. Hasegawa, S. Mitsuhashi, T. Kurita, K. Sasaki, K. Maeda, H. Obata, T. Hatakeyama, J. Polym. Sci., Part A-2, **8**, 1027 (1970).
559. H. Bittiger, R. H. Marchessault, W. D. Niegisch, Acta Crystallogr., Sect. B: Struct. Crystallogr. Cryst. Chem., **26**, 1923 (1970).
560. F. J. Van Natta, J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **56**, 455 (1934).
561. Y. Chatani, Y. Okita, H. Tadokoro, Y. Yamashita, Polym. J. (Tokyo), **1**, 555 (1970).
562. G. Gianotti, A. Capizzi, Eur. Polym. J., **6**, 743 (1970).
563. W. W. Doll, J. B. Lando, J. Macromol. Sci. B: Phys., **4**, 309 (1970).
564. H. Nakanishi, N. Nakano, M. Hasegawa, J. Polym. Sci., Part B, **8**, 755 (1970); H. Nakanishi, M. Hasegawa, Y. Sasada, J. Polym. Sci., Part A-2, **10**, 1537 (1972); H. Nakanishi, M. Hasegawa, Y. Sasada, J. Polym. Sci., Polym. Lett. Ed., **17**, 459 (1979).
565. Y. Chatani, S. Nakatani, H. Tadokoro, Macromolecules, **3**, 481 (1970).
566. J. P. Mercier, R. Legras, J. Polym. Sci., Part B, **8**, 645 (1970).
567. D. R. Beech, C. Booth, J. Polym. Sci., Part B, **8**, 731 (1970).
568. E. G. Lovering, D. C. Wooden, J. Polym. Sci., Part A-2, **9**, 175 (1971).
569. I. W. Bassi, O. Bonsignori, G. P. Lorenzi, P. Pino, P. Corradini, P. A. Temussi, J. Polym. Sci., Part A-2, **9**, 193 (1971).
570. J. R. Shelton, J. B. Lando, D. E. Agostini, J. Polym. Sci., Part B, **9**, 173 (1971).
571. S. Y. Hobbs, F. W. Billmeyer, Jr., J. Polym. Sci., Part A-2, **8**, 1387 (1970).

572. M. Hachiboshi, T. Fukuda, S. Kobayashi, J. Macromol. Sci. B: Phys., **3**, 525 (1969); S. Kobayashi, M. Hachiboshi, Rep. Progr. Polym. Phys. Japan, **13**, 157 (1970).
573. S. Kobayashi, M. Hachiboshi, Rep. Progr. Polym. Phys. Japan, **13**, 161 (1970).
574. A. R. Shultz, C. R. McCullough, J. Polym. Sci., Part A-2, **10**, 307 (1972).
575. G. Hinrichsen, H. Orth, Kolloid Z. Z. Polym., **247**, 844 (1971).
576. C. H. Bamford, L. Brown, A. Elliott, W. E. Hanby, I. F. Trotter, Nature (London), **169**, 357 (1952).
577. H. L. Yakel, Acta Crystallogr., **6**, 724 (1953).
578. A. Elliott, R. D. B. Fraser, T. P. Macrae, I. W. Stapleton, E. Suzuki, J. Mol. Biol., **9**, 10 (1964).
579. E. M. Bradbury, L. Brown, A. R. Downie, A. Elliott, W. E. Hanby, T. R. R. McDonald, J. Mol. Biol., **2**, 276 (1960).
580. V. Sasisekharan, Acta Crystallogr., **12**, 903 (1959).
581. W. J. Astbury, Nature (London), **163**, 722 (1949).
582. E. M. Bradbury, L. Brown, A. R. Downie, A. Elliott, W. E. Hanby, J. Mol. Biol., **5**, 230 (1962).
583. Y. Mitsui, Acta Crystallogr., **20**, 694 (1966).
584. A. Elliott, R. D. B. Fraser, T. P. Macrae, J. Mol. Biol., **11**, 821 (1965).
585. H. D. Keith, F. J. Padden, Jr., G. Giannoni, J. Mol. Biol., **43**, 423 (1969).
586. F. J. Padden, Jr., H. D. Keith, G. Giannoni, Biopolymers, **7**, 793 (1969).
587. C. H. Bamford, A. Elliott, W. E. Hanby, Synthetic Poly-Peptides. Preparation, Structure, Properties, Academic, New York, p. 272, (1956).
588. L. Pauling, R. B. Corey, Proc. Natl. Acad. Sci. USA, **37**, 241 (1951).
589. C. H. Bamford, L. Brown, A. Elliott, W. E. Hanby, I. F. Trotter, Proc. R. Soc. London, Ser. B, **141**, 49 (1953).
590. W. Traub, U. Shmueli, Nature (London), **198**, 1165 (1963).
591. R. D. B. Fraser, T. P. macrae, F. H. C. Stewart, E. Suzuki, J. Mol. Biol., **11**, 706 (1965).
592. D. R. Morrow, B. A. Newman, J. Appl. Phys., **39**, 4944 (1968).
593. S. Kambara, A. Takahashi, Makromol. Chem., **58**, 226 (1962).
594. K. Kaji, Thesis, Dept. of Polymer Chem., Kyoto Univ., 1970; K. Kaji, I. Sakurada, Makromol. Chem., **148**, 261 (1971); K. Kaji, Kyoto Daigaku Nippon Kagakuseni Kenkyusho Koenshu, **28**, 1 (1971); through Chem. Abstr., **78**, 136889q (1973).
595. S. Kubo, B. Wunderlich, J. Appl. Phys., **42**, 4565 (1971); J. Polym. Sci., Polym. Chem. Ed., **10**, 1949 (1972); Makromol. Chem., **162**, 1 (1972); R. Iwamoto, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **11**, 2403 (1973).
596. J. C. Andries, J. M. Anderson, A. G. Walton, Biopolymers, **10**, 1049 (1971).
597. Ye. L. Gal'Perin, B. P. Kosmynin, Vysokomol. Soedin., Ser. A, **11**, 1432 (1969) - Polym. Sci. USSR (Engl. Transl.), **11**, 1624 (1969).
598. F. E. Karasz, D. Mangaraj, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **12**, 317 (1971).
599. I. Bowman, D. S. Brown, R. E. Wetton, Polymer, **10**, 715 (1969).
600. G. A. Clegg, D. R. Gee, T. P. Melia, A. Tyson, Polymer, **9**, 501 (1968).
601. S. Mitsuhashi, Kogyo Kagaku Zasshi, **72**, 2160 (1969).
602. F. J. Boerio, J. L. Koenig, J. Chem. Phys., **54**, 3667 (1971).
603. I. V. Kumpanenko, K. S. Kazanskii, N. V. Ptitsyna, M. Ya. Kushnerev, Vysokomol. Soedin., Ser. A, **12**, 822 (1970); Polym. Sci., USSR (Engl. Transl.), **12**, 930 (1970).
604. R. Hasegawa, M. Kobayashi, H. Tadokoro, Polym. J. (Tokyo), **3**, 591 (1972); R. Hasegawa, Y. Takahashi, Y. Chatani, H. Tadokoro, *ibid.*, 600, Y. Takahashi, Y. Matsubara, H. Tadokoro, Macromolecules, **16**, 1588 (1983).
605. D. G. Graber, Microscope, **18**, 203 (1970).
606. Y. Takeda, Y. Iitaka, M. Tsuboi, J. Mol. Biol., **51**, 101 (1970).
607. A. Kimura, S. Yoshimoto, Y. Akana, H. Hirata, S. Kusabayashi, H. Mikawa, N. Nasai, J. Polym. Sci., Part A-2, **8**, 643 (1970).
608. P. Pino, G. P. Lorenzi, E. Chiellini, Ric. Sci., **34**, 193 (1964).
609. K. W. Doak, H. N. Campbell, J. Polym. Sci., **18**, 215 (1955).
610. J. W. Fisher, J. Appl. Chem., **4**, 212 (1954).
611. V. Enkelmann, G. Schleier, in V. Enkelmann, Adv. Polym. Sci., **63**, 92 (1984).
612. V. N. Rogulenkova, M. I. Millionova, N. S. Andreeva, J. Mol. Biol., **9**, 253 (1964).
613. W. Traub, A. Yonath, J. Mol. Biol., **16**, 404 (1966); A. Yonath, W. Traub, ibid., **43**, 461 (1969).
614. W. Traub, A. Yonath, J. Mol. Biol., **25**, 351 (1967).
615. W. Traub, J. Mol. Biol., **43**, 479 (1969).
616. D. M. Segal, W. Traub, J. Mol. Biol., **43**, 487 (1969).
617. G. D. Buckley, L. H. Cross, N. H. Ray, J. Chem. Soc., 2714, (1950).
618. G. Saini, E. Campi, S. Parodi, Gazz. Chim. Ital., **87**, 342 (1957).
619. A. Nasini, J. Polym. Sci., **34**, 106 (1959).
620. W. L. Truett, D. R. Johnson, I. M. Robinson, B. A. Montague, J. Am. Chem. Soc., **82**, 2337 (1960).
621. S. Horikiri, J. Polym. Sci., Part A-2, **10**, 1167 (1972).
622. V. Petraccone, P. Ganis, P. Corradini, G. Montagnoli, Eur. Polym. J. **8**, 99 (1972).
623. V. Crescenzi, G. Manzini, G. Calzolari, C. Borri, Eur. Polym. J., **8**, 449 (1972).
624. S. F. Bermudez, J. M. G. Fatou, Eur. Polym. J., **8**, 575 (1972).
625. V. Petraccone, B. Pirozzi, P. Corradini, Eur. Polym. J., **8**, 107 (1972).
626. Z. Mencik, J. Macromol., Sci. B: Phys., **6**, 101 (1972).
627. Y. Saito, S. Nansai, S. Kinoshita, Polym. J. (Tokyo), **3**, 113 (1972).
628. H. Wilski, Kolloid Z. Z. Polym., **248**, 867 (1971).
629. Y. Miyamoto, C. Nakafuku, T. Takemura, Polym. J. (Tokyo), **3**, 122 (1972).
630. E. Martuscelli, Makromol. Chem., **151**, 159 (1972).

631. K. A. Andrianov, G. L. Slonimskii, A. A. Zhdanov, V. Yu. Levin, Yu. K. Godovskii, V. A. Moskalenko, J. Polym. Sci., Part A-1, **10**, 1 (1972).
632. R. Backsai, J. E. Goodrich, J. B. Wilkes, J. Polym. Sci., Part A-1, **10**, 1529 (1972).
633. O. Bayer, Angew. Chem. A, **59**, 257 (1947).
634. K. Saotome, H. Komoto, J. Polym. Sci., Part A-1, **5**, 119 (1967).
635. J. P. Sibilia, L. G. Roldan, S. Chandrasekaran, J. Polym. Sci., Part A-2, **10**, 549 (1972).
636. H. Bittiger, E. Husemann, J. Polym. Sci., Part B, **10**, 367 (1972).
637. M. Ragazzini, C. Garbuglio, D. Carcano, B. Minasso, G. Cevidalli, Eur. Polym. J., **3**, 129 (1967); C. Garbuglio, M. Ragazzini, O. Pilati, D. Carcano, G. Cevidalli, *ibid.*, 137.
638. T. Takizawa, T. Sano, Y. Chatani, I. Taguchi, Preprints of the 6th Polymer Symp., Polymer Phys. Div. (Nagoyan), p. 83, Oct. 1957; Y. Chatani, I. Taguchi, T. Sano, T. Takizawa, Annu. Rept. Inst. Textile Sci. Japan, **13**, 37 (1960), through K. fujii, J. Polym. Sci., Part D, **5**, 431 (1971).
639. G. Gianotti, G. Dall'Asta, A. Valvassori, V. Zamboni, Makromol. Chem., **149**, 117 (1971); G. Dall'Asta, ibid., **154**, 1 (1972).
640. H. Wilski, Makromol. Chem., **150**, 209 (1971).
641. M. Suwalsky, W. Traub, Biopolymers, **11**, 623 (1972).
642. L. G. Kazaryan, Ye. G. Lur'e, L. A. Igonin, Vysokomol. Soedin., Ser. B, **11**, 779 (1969); Ye. G. Lur'e, L. G. Kazaryan, E. L. Uchastkina, V. V. Kovriga, K. N. Vlasova, M. L. Dobrokhotova, L. M. Yemel'Yanova, Vysokomol. Soedin., Ser. A, **13**, 606 (1971); Polym. Sci., USSR (Engl. Transl.), **13**, 685 (1971).
643. K. J. Palmer, M. Ballantyne, J. Am. Chem. Soc., **72**, 736 (1950).
644. R. E. Rundle, L. Daasch, D. French, J. Am. Chem. Soc., **66**, 130 (1944).
645. D. R. Kreger, Biochim. Biophys. Acta, **6**, 406 (1951); L. C. Spark, *ibid.*, **8**, 101 (1952).
646. R. E. Rundle, F. C. Edwards, J. Am. Chem. Soc., **65**, 2200 (1943); F. F. Mikus, R. M. Hixon, R. E. Rundle, *ibid.*, **68**, 1115 (1946).
647. R. S. J. Manley, J. Polym. Sci., Part A-2, **2**, 4503 (1964).
648. A. Sarko, R. H. Marchessault, J. Am. Chem. Soc., **89**, 6454 (1967).
649. W. T. Winter, A. Sarko, Biopolymers, **11**, 849 (1972); *ibid.*, **13**, 1461 (1974).
650. F. R. Senti, L. P. Witnauer, J. Am. Chem. Soc., **70**, 1438 (1948).
651. R. E. Rundle, D. French, J. Am. Chem. Soc., **65**, 1707 (1943).
652. D. Carlstrom, J. Biophys. Biochem. Cytol., **3**, 699 (1957), quoted in R. H. Marchessault, A. Sarko, Adv. Carbohydr. Chem., **22**, 421 (1967).
653. N. E. Dweltz, Biochim. Biophys. Acta, **51**, 283 (1961).
654. I. M. Mackie, E. E. Percival, J. Chem. Soc., 1151 (1959). quoted in R. H. Marchessault, A. Sarko, Adv. Carbohydr. Chem., **22** 421 (1967).
655. R. H. Marchessault, W. Settineri, J. Polym. Sci., Part B, **2**, 1047 (1964); W. Settineri, R. H. Marchessault, J. Polym. Sci., Part C, **11**, 253 (1965).
656. S. M. Gabbay, P. R. Sundararajan, R. H. Marchessault, Biopolymers, **11**, 79 (1972).
657. N. Lanzetta, G. Maglio, C. Marchetta, R. Palumbo, Preprint III-36, IUPAC Symposium on Macromolecules, Helsinkii, July 2–7, 1972; G. Maglio, C. marchetta, R. Palumbo, F. Riva, M. de Simone, Makromol. Chem., **156**, 321 (1972); N. Lanzetta, G. Maglio, C. Marchetta, R. Palumbo, J. Polym. Sci., Polym. Chem. Ed., **11**, 913 (1973).
658. A. I. Koltsov, Yu. G. Baklagina, B. M. Ginzburg, L. N. Korzhavin, M. M. Koton, N. V. Mikhailova, V. N. Nikitin, A. V. Sidorovich, J. Polym. Sci., Polym. Symp., **42**, 867 (1973); N. A. Adrova, A. I. Artyukhov, Yu. G. Baklagina, T. I. Borisova, M. M. Koton, N. V. Mikhailova, V. N. Nikitin, A. V. Sidorovich, Vysokomol. Soedin., Ser. A, **15**, 153 (1973); Polym. Sci., USSR (Engl. Trans.) **15**, 175 (1973); Yu. G. Baklagina, N. V. Mikhailova, V. N. Nikitin, A. V. Sidorovich, L. N. Korzhavin, Vysokomol. Soedin., Ser. A, **15**, 2738 (1973); Polym. Sci., USSR (Engl. Transl.), **15**, 3109 (1973).
659. E. D. T. Atkins, C. F. Phelps, J. K. Sheehan, Biochem. J., **128**, 1255 (1972); E. D. T. Atkins J. K. Sheehan, Nature (London), New Biol., **235**, 253 (1972); J. K. Sheehan, E. D. T. Atkins, I. A. Nieduszynski, J. Mol. Biol., **91**, 153 (1975).
660. M. G. Northolt, B. J. Tabor, J. J. van Aartsen, J. Polym. Sci., Part A-2, **10**, 191 (1972).
661. F. R. Senti, L. P. Witnauer, J. Polym. Sci., **9**, 115 (1952).
662. W. Z. Borchert, Angew. Chem., **63**, 31 (1951).
663. P. de Santis, A. Kovacs, Biopolymers, **6**, 299 (1968).
664. S. Ishida, Bull. Chem. Soc. Japan, **33**, 924 (1960).
665. A. M. Afifi-Effat, J. N. Hay, J. Chem. Soc., Faraday Trans. 2, **68**,656 (1972).
666. U. Shmueli, W. Traub, J. Mol. Biol., **12**, 205 (1965).
667. A. D. French, H. F. Zobel, Biopolymers, **5**, 457 (1967).
668. B. Zaslow, Biopolymers, **1**, 165 (1963).
669. N. E. Dweltz, Biochim. Biophys. Acta, **44**, 416 (1960).
670. W. Lotmar, L. E. R. Picken, Experientia, **6**, 58 (1950).
671. I. A. Nieduszynski, R. H. Marchessault, Nature (London), **232**, (46 (1971); Biopolymers, **11**, 1335 (1972).
672. H. Bittiger, R. H. Marchessault, Carbohydr. Res., **18**, 469 (1971).
673. E. D. T. Atkins, W. Mackie, E. E. Smolko, Nature (London), **225**, 626 (1970); E. D. T. Atkins, W. Mackie, K. D. Parker, E. E. Smolko, J. Polym. Sci., Part B, **9**, 311 (1971); E. D. T. Atkins, I. A. Nieduszynski, W. Mackie, K. D. Parker, E. E. Smolko, Biopolymers, **12**, 1865 (1973).
674. W. T. Astbury, Nature, (London), **155**, 667 (1945).
675. A. S. Ueda, Y. Chatani, H. Tadokoro, Polym. J. (Tokyo), **2** 387 (1971).
676. J. G. Fatou, Eur. Polym. J., **7**, 1057 (1971).
677. B. J. Tabor, E. P. Magre', J. Boon, Eur. Polym. J., **7**, 1127 (1971).
678. W. H. Carothers, J. A. Arvin, G. L. Dorough, J. Am. Chem. Soc., **5**, 3292 (1930).
679. S. Tsunawaki, C. C. Price, J. Polym. Sci., Part A, **2**, 1511 (1964).

680. G. Cojazzi, A. Fichera, C. Garbuglio, V. Malta, R. Zannetti, Chim. Ind. (Milan), **54**, 40 (1972); Makromol. Chem, **168**, 289 (1973).
681. H. D. Flack, J. Polym. Sci., Part A-2, **10**, 1799 (1972).
682. A. Capizzi, G. Gianotti, Makromol. Chem., **157**, 123 (1972).
683. G. Perego, A. Melis, M. Cesari, Makromol. Chem., **157**, 269 (1972).
684. L. G. Kazaryan, D. Ya. Tsvankin, B. M. Ginzburg, S. Taychiev, L. N. Korzhavin, S. Ya. Frenkel', Vysokomol. Soedin., Ser. A, **14**, 1199 (1972) - Polym. Sci. USSR (Engl. Transl.), **14**, 1344 (1972); Yu. G. Baklagina, V. P. Sklzkova, V. V. Kudryavtsev, N. D. Kozhurnikova, V. V Klechkovskaya, L. G. Yanusova, N. D. Stepina, L. A. Feigin, Polym. Sci., Ser. A, **37**, 847 (1995).
685. P. Corradini, E. Martuscelli, G. Montagnoli, V. Petraccone, Eur. Polym. J., **6**, 1201 (1970).
686. S. Sasaki, Y. Takahashi, H. Tadokoro, Polym. J. (Tokyo), **4**, 172 (1973).
687. Y. Gotoh, H. Sakakihara, H. Tadokoro, Polym. J. (Tokyo), **4**, 68 (1973).
688. S. Sasaki, Y. Takahashi, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **10**, 2363 (1972).
689. H. Tadokoro, in G. Allen, Ed. Molecular Structure and Properties (Physical Chemistry, Series One, V.2) (MTP International Review of Science) Butterworths, London, p. 45 (1972).
690. H. Tadokoro, private communication, 30 September 1972.
691. N. Doddi, W. C. Forsman, C. C. Price Macromolecules, **4**, 648 (1971).
692. H. Tani, N. Oguni, J. Polym. Sci., Part B, **7**, 803 (1969).
693. G. Allen, C. Booth, S. J. Hurst, C. Price, F. Vernon, R. F. Warren, Polymer, **8**, 406 (1967).
694. P. Dumas, N. Spassky, P. Sigwalt, Makromol. chem., **156**, 55 (1972).
695. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **57**, 925 (1935).
696. R. Nakatsuka, Bull. Chem. Soc. Japan, **36**, 1294 (1963).
697. H. K. Illers, H. Haberkorn, P. Simak, Makromol. Chem., **158**, 285 (1972).
698. K. Teranishi, T. Araki, H. Tani, Macromolecules, **5**, 660 (1972).
699. Y. Chatani, quoted in Y. Chatani, S. Nakatani, Macromolecules, **5**, 597 (1972).
700. M. J. Richardson, J. Polym. Sci., Part C, **38**, 251 (1972).
701. W. Traub, U. Shmueli, M. Suwalsky, A. Yonath, in: G. N. Ramachandran, (Ed.), "Conformation of Biopolymers", vol. 2, Academic, New York, p. 449 (1967).
702. B. K. Vainshtein, L. I. Tatarinova, in: G. N. Ramachandran (Ed.), "Conformation of Biopolymers", vol. 2, Academic, New York, p. 569 (1967).
703. D. B. Green, F. Happey, B. M. Watson, in: G. N. Ramachandran (Ed.), "Conformation of Biopolymers", vol. 2, Academic, New York, p. 617 (1967) Eur. Polym. J., **6**, (1970).
704. H. Sumitomo, K. Kobayashi, T. Saji, J. Polym. Sci., Polym Chem. Ed., **10**, 3421 (1972).
705. E. D. T. Atkins, R. Gaussen, D. H. Isaac, V. Nandanwar, J. K. Sheehan, J. Polym. Sci., Polym. Lett. Ed., **10**, 863 (1972).
706. I. Yahara, K. Imahori, Y. Iitaka, M. Tsuboi, J. Polym. Sci., Part B, **1**, 47 (1963).
707. L. Porri, M. C. Gallazzi, Eur. Polym. J., **2**, 189 (1966).
708. B. J. Poppleton, A. M. Mathieson, Nature (London), **219**, 1046 (1968).
709. K. C. Ellis, J. O. Warwicker, J. Polym. Sci., **56**, 339 (1962).
710. I. C. M. Dea, R. Moorhouse, D. A. Rees, S. Arnott, J. M. Guss, E. A. Balazs, Science, **179**, 560 (1973).
711. E. D. T. Atkins, J. K. Sheehan, Science, **179**, 562 (1973).
712. R. Perret, A. Skoulios, Makromol. Chem., **156**, 157 (1972).
713. E. Lanza, H. Berghmans, G. Smets, J. Polym. Sci., Polym. Phys. Ed., **11**, 75 (1973).
714. J. Pisanchyn, J. Polym. Sci., Polym. Chem. Ed., **11**, 135 (1973).
715. Y. Takahashi, T. Sato, H. Tadokoro, Y. Tanaka, J. Polym. Sci., Polym. Phys. Ed., **11**, 233 (1973).
716. E. Hadicke, E. C. Mez, C. H. Krauch, G. Wegner, J. Kaiser, Angew. Chem., **83**, 253 (1971); Anew. Chem., Int. Ed. Engl., **10**, 226 (1971).
717. H. Sakakihara, Y. Takahashi, H. Tadokoro, N. Oguni, H. Tani, Macromolecules, **6**, 205 (1973).
718. C. E. Wilkes, V. L. Folt, S. Krimm, Macromolecules, **6**, 235 (1973).
719. A. Del Pra, M. Mammi, E. Peggion, Biopolymers, **12**, 937 (1973).
720. A. Del Pra, P. Spadon, G. Valle, Biopolymers, **12**, 941 (1973).
721. R. A. Wessling, J. H. Oswald, I. R. Harrison, J. Polym. Sci., Polym. Phys. Ed., **11**, 875 (1973).
722. F. C. Wilson, H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **11**, 919 (1973).
723. M. Hasegawa, Y. Suzuki, H. Nakanishi, F. Nakanishi, in: K. Imahori, S. Murahashi (Eds.), "Progress in Polykmer Science", Japan, vol. 5, Halsted (Wiley), New York, p. 143 (1973).
724. A. Wlochowicz, W. Przygocki, J. Appl. Polym. Sci., **17**, 1197 (1973).
725. N. Kobayashi, A. Osawa, T. Fujisawa, J. Polym. Sci., Polym. Lett. Ed., **11**, 225 (1973); J. Polym. Sci., Polym. Chem. Ed., **13**, 2863 (1975).
726. G. J. Welch, R. L. Miller, J. Polym. Sci., Polym. Phys. Ed., **14**, 1683 (1976).
727. M. Ikeda, H. Suga, S. Seki, Rep. Progr. Polym. Phys. Japan, **15**, 255 (1972).
728. M. Ikeda, H. Suga, S. Seki, Rep. Progr. Polym. Phys. Japan, **15**, 257 (1972).
729. S. Minami, K. Murase, T. Yoshihara, Rep. Progr. Polym. Phys. Japan, **15**, 339 (1972).
730. H. Tadokoro, T. Yoshihara, Y. Chatani, S. Murahashi, J. Polym. Sci., Part B, **2**, 363 (1964).
731. A. A. Blumberg, S. S. Pollack, C. A. J. Hoeve, J. Polym. Sci., Part A, **2**, 2499 (1964).
732. R. Iwamoto, Y. Saito, H. Ishihara, H. Tadokoro, J. Polym. Sci., Part A 2, **6**, 1509 (1968).

733. M. Yokoyama, H. Ishihara, R. Iwamoto, H. Tadokoro, Macromolecules, **2**, 184 (1969).

734. F. E. Bailey, Jr., H. G. France, J. Polym. Sci., **49**, 397 (1961).

735. H. Herlinger, H. -P. Horner, F. Druschke, H. Knoell, F. Haiber, Angew. Makromol. Chem., **29/30**, 229 (1973); H. Herlinger, H. Knoell, H. Menzel, J. Schlaefer, Appl. Polym. Symp., **21**, 215 (1973).

736. P. V. Kozlov, A. N. Perepelkin, Vysokomol. Soedin., Ser. A, **9**, 370 (1967); Polym. Sci. USSR (Engl. Transl.), **9**, 414 (1967).

737. J. C. Andries, A. G. Walton, Biopolymers, **8**, 465 (1969).

738. T. Saegusa, H. Ikeda, H. Fujii, Macromolecules, **5**, 108 (1972).

739. J. W. Summers, M. H. Litt, J. Polym. Sci., Polym. chem., Ed., **11**, 1353 (1973).

740. B. Lotz, H. D. Keith, J. Mol. Biol., **61**, 195 (1971). The authors report c = 9.15 A for three peptide residues in a "glycine II" 3/1 helix. Crystallographic identity occurs at four times this.

741. B. Lotz, H. D. Keith, J. Mol. Biol., **61**, 201 (1971).

742. L. G. Kazaryan, F. M. Medvedeva, Vysokomol. Soedin., Ser. B, **10**, 305 (1968).

743. S. W. Shalaby, E. M. Pearce, R. J. Fredericks, E. A. Turi, J. Polym. Sci., Polym. Chem. Ed., **11**, 1 (1973).

744. J. P. Kennedy, R. B. Isaacson, J. Macromol. Chem., **1**, 541 (1966).

745. G. Montaudo, G. Bruno, P. Maravigna, P. Finocchiaro, P. Centineo, J. Polym. Sci., Polym. Chem. Ed., **11**, 65 (1973).

746. M. Litt, F. Rahl, L. G. Roldan, J. Polym. Sci., Part A-2, **7**, 463 (1969).

747. T. G. Bassiri, A. Levy, M. Litt, J. Polym. Sci., Part B, **5**, 871 (1967).

748. H. F. Mark, S. M. Atlas, N. Ogata, J. Polym. Sci., **61**, S49 (1962).

749. J. Kaiser, G. Wegner, E. W. Fischer, Kolloid Z. Z. Polym., **250**, 1158 (1972).

750. K. Sakaoku, A. Peterloin, J. Macromol. Sci., Phys., **1**, 401 (1967).

751. M. Straumanis, Z. Kristallogr., **102**, 432 (1940).

752. R. G. Crystal, J. Polym. Sci., Part A-2, **8**, 1755 (1970).

753. B. B. Doyle, W. Traub, G. P. Lorenzi F. R. Brown III, E. R. Blout, J. Mol. BIol., **51**, 47 (1970). The authors report b = 6.98 A for two peptide residues in an "antiparallel pleated sheet" structure. Crystallographic identitiy occurs at three times this.

754. R. D. B. Fraser, T. P. Macrae, F. H. C. Stewart, J. Mol. Biol., **19**, 580 (1966). The authors report b = 6.85 A for two peptide residues. Crystallographic identity occurs at three times this.

755. D. M. Segal, W. Traub, A. Yonath, J. Mol. Biol., **43**, 519 (1969).

756. J. C. Andries, A. G. Walton, J. Mol. Biol., **56**, 515 (1971).

757. J. M. Squire, A. Elliott, J. Mol. Biol., **65**, 291 (1972).

758. Y. Morishima, T. Takizawa, S. Murahashi, Eur. Polym. J., **9**, 669 (1973).

759. K. Inoue, S. Hoshino, J. Polym. Sci., Polym. Phys. Ed., **11**, 1077 (1973).

760. V. F. Holland, J. Macromol., Chem. A, **7**, 173 (1973); R. L. Miller, J. Macromol. Sci., Chem. A, **7**, 183 (1973). Both papers are also contained in "High-Modulus Wholly Aromatic Fibers", W. B. Black, J. Preston, Eds., Dekker, New York, (1973).

761. M. G. Northolt, J. J. Aartsen, J. Polym. Sci., Polym. Lett. Ed., **11**, 333 (1973); M. G. Northolt, Eur. Polym. J., **10**, 799 (1974).

762. M. Takahaishi, Kobunshi Kagaku, **15**, 273 (1958), through Chem. Abstr., **54**, 2802f (1960).

763. M. Takahaisihi, Sen-I Gakkaishi, **14**, 374 (1958), through Chem. Abstr., **52**, 21217b (1958).

764. V. V. Korshak, S. V. Vinogradova, E. S. Vlasova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk 1089 (1954); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.), 949 (1954).

765. V. V. Korshak, S. V. Vinogradova, E. S. Vlasova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk 1097 (1954); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.), 957, (1954).

766. V. V. Korshak, S. V. Vinogradova, Zh. Obshch. Khim., **26**, 539 (1956); J. Gen. Chem. USSR (Engl. Transl.), **26**, 575 (1956).

767. V. V. Korshak, S. V. Vinogradova, Zh. Obshch. Khim., **26**, 732 (1956); J. Gen. Chem. USSR (Engl. Transl.), **26**, 841 (1956).

768. I. W. Bassi, R. Scordamaglia, Makromol. Chem., **166**, 283 (1973).

769. S. Arnott, J. M. Guss, D. W. L. Hukins, M. B. Mathews, Science, **180**, 743 (1973).

770. K. Nakagawa, Y. Ishida, Kolloid Z. Z. Polym., **251**, 103 (1973).

771. M. Yokouchi, Y. Chatani, H. Tadokoro, K. Teranishi, H. Tani, Polymer, **14**, 267 (1973).

772. Quoted in H. Tadokoro, Kobunshi, **8**, 223 (1959).

773. W. Wrasidlo, J. Polym. Sci., Part A-2, **10**, 1719 (1972).

774. A. Del Pra, A. M. Tamburro, A. Scatturin, M. Mammi, Ric. Sci., **36**, 1330 (1966).

775. S. Mitsuhashi, Kgoyo Kagaku Zasshi, **72**, 1169 (1969).

776. Ye. L. Gal'Perin, B. P. Kosmynin, V. K. Smirnov, Vysokomol. Soedin., Ser. A, **12**, 1880 (1970); Polym. Sci. USSR (Engl. Transl.), **12**, 2133 (1970).

777. Z. Mencik, J. Polym. Sci., Polym. Phys. Ed., **11**, 1585 (1973).

778. J. Blackwell, Biopolymers, **7**, 281 (1969); K. H. Gardner, J. Blackwell, Biopolymers, **14**, 1581 (1975).

779. K. M. Rudall, Adv. Insect Physiol., **1**, 257 (1963), through A. G. Walton, J. Blackwell, Biopolymers, Academic, New York, (1973).

780. J. Kiji, J. Kaiser, G. Wegner, R. C. Schulz, Polymer, **14**, 433 (1973).

781. J. L. R. Williams, T. M. Laakso, L. E. Contois, J. Polym. Sci., **61**, 353 (1962).

782. G. Manzini, V. Crescenzi, A. Ciana, L. Ciceri, G. Della Fortuna, L. Zotteri, Eur. Polym. J., **9**, 941 (1973).

783. S. Sasaki, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **11**, 1985 (1973).

784. M. Suwalsky, L. Hoz, Biopolymers, **12**, 1997 (1973).
785. J. Cornibert, R. H. Marchessault, A. E. Allegrezza, Jr., R. W. Lenz, Macromolecules, **6**, 676 (1973).
786. C. A. Boye, Jr., Bull. Am. Phys. Soc., **19**, 352 (1974).
787. N. Okui, H. M. Li, J. H. Magill, Polymer, **19**, 411 (1978).
788. E. D. T. Atkins, K. D. Parker, R. D. Preston, Proc. R. Soc. London, Ser. B, **173**, 209 (1969); E. D. T. Atkins, K. D. Parker, J. Polym. Sci., Part C, **28**, 69 (1969).
789. K. Kamide, K. Nakamura, Sen-I, Gakkaishi, **25**, 53 (1969).
790. Y. Takahashi, I. Sumita, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **11**, 2113 (1973).
791. B. L. Farmer, J. B. Lando, J. Macromol. Sci. B: Phys., **10**, 403 (1974).
792. D. C. Bassett, B. Turner, Philos. Mag., **29**, 923 (1974); D. C. Bassett, S. Block, G. J. Piermarini, J. Appl. Phys., **45**, 4146 (1974).
793. A. Savolainen, Eur. Polym. J., **10**, 9 (1974).
794. Y. Mitsui, Biopolymers, **12**, 1781 (1973).
795. U. Haberthur, H.-G. Elias, Makromol. Chem., **144**, 183 (1971).
796. R. W. Lenz, B. W. Delf, J. Polym. Sci., Polym. Lett., Ed., **11**, 609 (1973).
797. R. W. Lenz, J. Neloson, P. Guimond, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **9**, 1301 (1968).
798. A. Colombo, G. Allegra, Macromolecules, **4**, 579 (1971).
799. S. Yoshida, H. Suga, S. Seki, Polym. J. (Tokyo), **5**, 33 (1973).
800. R. E. Hughes, R. J. Cella, Jr., Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **15(1)**, 137 (1974).
801. T. Tanaka, Y. Chatani, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **12**, 515 (1974).
802. Y. Takahashi, H. Tadokoro, T. Hirano, A. Sato, T. Tsuruta, J. Polym. Sci., Polym. Phys. Ed., **13**, 285 (1975).
803. Y. Chatani, H. Tadokoro, J. Macromol. Sci. B: Phys., **8**, 203 (1973).
804. B. G. Colvin, P. Storr, Eur. Polym. J., **10**, 377 (1974).
805. M. M. Beg, J. Aslam, Q. H. Khan, N. M. Butt, S. Rolandson, A. U. Ahmed, J. Polym. Sci., Polym. Lett. Ed., **12**, 311 (1974).
806. T. Komoto, K. Y. Kim, M. Oya, T. Kawai, Makromol. Chem., **175**, 283 (1974).
807. M. Gilbert, F. J. Hybart, Polymer, **13**, 327 (1972); *ibid.*, 407.
808. A. V. Sidorovich, M. M. Koton, E. V. Kuvshinsky, V. V. Nikitin, N. A. Adrova, Yu. G. Baklagina, N. V. Mikhailova, J. Polym. Sci., Polym. Chem. Ed., **12**, 375 (1974).
809. J. J. O'Malley, W. J. Staufer, J. Polym. Sci., Polym. Chem. Ed., **12**, 865 (1974).
810. M. Yokouchi, Y. Chatani, H. Tadokoro, H. Tani, Polym. J. (Tokyo), **6**, 248 (1974).
811. R. K. Hasegawa, Y. Chatani, H. Tadokoro, Rep. Progr. Polym. Phys. Japan, **16**, 217 (1973).
812. T. Kajiyama, M. Tanaka, M. Kuroishi, M. Takayanagi, Chem. Lett. 1159 (1973).
813. C. H. Griffiths, M. S. Walker, Madrid IUPAC, Preprint III, 2-3 (1974).
814. A. M. Joly, G. Nemoz, A. Douillard, G. Vallet, Makromol. Chem., **176**, 479 (1975).
815. V. V. Korshak, S. V. Vinogradova, M. Belyakov, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk, 730, (1957); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.), 749 (1957).
816. R. E. Prud'homme, R. H. Marchessault, Makromol. Chem., **175**, 2705 (1974); Macromolecules, **7**, 541 (1974).
817. H. Miyaji, K. Asai, J. Phys. Soc. Japan, **36**, 1497 (1974).
818. J. Y. Decroix, M. Moser, M. Boudeulle, Eur. Polym. J., **11**, 357 (1975). (The true cell is 5 times as long (b = 35.65 A) and contains 200 units).
819. D. H. Isaac, E. D. T. Atkins, Nature (London), New Biol., **244**, 252 (1973)
820. A. Kuppel and E. Husemann, Colloid. Polym. Sci. 252, 1005 (1974); P. Zugenmaier, A. Kuppel, E. Husemann, in: J. C. Arthur (Ed.), "Cellulose Chemistry and Technology" (ACS Symp. Ser. No. 48), Am. Chem. Soc., Washington, 1977, p. 115.
821. C. R. Bohn, J. R. Schaefgen, W. O. Statton, J. Polym. Sci., **55**, 531 (1961).
822. S. M. Bishop, I. H. Hall, Br. Polym. J., **6**, 193 (1974).
823. K. Nakagawa, Y. Ishida, J. Polym. Sci., Polym. Phys. Ed., **11**, 2153 (1973).
824. S. Y. Hobbs, C. F. Pratt, Am. Phys. Soc. Meeting, Denver, CO., March 31, 1975 paper EE10.
825. C. H. Griffiths, J. Polym. Sci., Polym. Phys. Ed., **13**, 1167 (1975).
826. G. Valle, A. Del Pra, P. Spadon M. Mammi, Biopolymers, **14**, 437 (1975).
827. Y. Chatani, S. Kuwata, Macromolecules, **8**, 12 (1975).
828. P. W. Morgan, S. L. Kwolek, Macromolecules, **8**, 104 (1975).
829. V. E. Shashoua, W. M. Eareckson III, J. Polym. Sci., **40**, 343 (1959).
830. P. A. Apgar, K. C. Yee, Acta Crystallogr., Sect. B: Struct. Crystallogr. Cryst. Chem., **34**, 957 (1978).
831. S. W. Shalaby, E. A. Turi, M. H. Riggi, P. J. Harget, J. Polym. Sci., Polym Chem. Ed., **13**, 669 (1975).
832. V. P. Lebedev, T. M. Babchinitser, N. I. Bekasova, L. G. Komarova, D. Ya. Tsvankin, V. V. Korshak, Vysokomol. Soedin., Ser. A, **16**, 987 (1974); Polym. Sci. USSR (Engl. Transl.), **16**, 1140 (1974).
833. C. C. Price, M. K. Akkapeddi, B. T. Debona, B. C. Furie, J. Macromol. Sci. A: Chem., **7**, 1469 (1973).
834. M. Moser, M. Boudeulle, H. Chanzy, J. Polym. Sci., Polym. Phys. Ed., **14**, 1161 (1976).
835. E. V. Gouinlock, J. Polym. Sci., Polym. Phys. Ed., **13**, 1533 (1975).
836. S. S. Zimmerman, J. C. Clark, L. Mandelkern, Biopolymers, **14**, 585 (1975).
837. J. P. Baldwin, E. M. Bradbury, I. F. McLuckie, R. M. Stephens, Macromolecules, **6**, 83 (1973).
838. G. Ceccorulli, F. Manescalchi, M. Pizzoli, Makromol. Chem., **176**, 1163 (1975).
839. T. Itoh, H. Miyaji, K. Asai, Japan. Appl. Phys., **14**, 206 (1975).

840. K. Suehiro, Y. Chatani, H. Tadokoro, Polym. J. (Tokyo), **7**, 352 (1975). The basal dimensions were determined from equatorial reflections alone.
841. A. Sarko, R. Muggli, Macromolecules, **7**, 486 (1974).
842. F. J. Kolpak, J. Blackwell, Macromolecules, **8**, 563 (1975); idem., **9**, 273 (1976); F. J. Kolpak, M. Weih, J. Blackwell, Polymer, **19**, 123 (1978).
843. V. G. Murphy, B. Zaslow, A. D. French, Biopolymers, **14**, 1487 (1975).
844. R. Napolitano, Makromol. Chem., **191**, 355 (1990). Values of the basal unit cell parameters, a and b, were estimated from energy calculations.
845. R. Hasegawa, Y. Tanabe, M. Kobayashi, H. Tadokoro, A. Sawaoka, N. Kawai, J. Polym. Sci., Part A-2, **8**, 1073 (1970).
846. S. Osaki, Y. Ishida, Rep. Progr. Polym. Phys. Japan, **17**, 253 (1974); J. Polym. Sci., Polym. Phys. Ed., **13**, 1071 (1975).
847. H. K. Hall, Jr., Br. Polym. J., **4**, 37 (1972).
848. G. Bruno, G. Montaudo, N. V. Hien, R. H. Marchessault, P. R. Sundararajan, J. E. Chandler, R. W. Lenz, J. Polym. Sci., Polym. Lett. Ed., **13**, 559 (1975).
849. A. Del Pra, G. Gilli, Bipolymers, **14**, 1769 (1975).
850. E. Dellacherie, J. Neel, F. Colonna-Cesari, Biopolymers, **14**, 1447 (1975).
851. Z. Mencik, J. Polym. Sci., Polym. Phys. Ed., **13**, 2173 (1975).
852. K. H. Gardner, J. Blackwell, Biopolymers, **13**, 1975 (1974); K. H. Gardner, J. Blackwell, Biochim. Biophys. Acta, **343**, 232 (1974); W. Claffey, J. Blackwell, Biopolymers, **15**, 1903 (1976).
853. S. Fakirov, E. W. Fischer, G. F. Schmidt, Makromol. Chem., **176**, 2459 (1975).
854. R. R. Kolda, J. B. Lando, J. Macromol. Sci. B: Phys., **11**, 21 (1975).
855. A. de Boer, G. O. R. A. van Ekenstein, G. Challa, Polymer, **16**, 930 (1975).
856. H. Kakida, Y. Chatani, H. Tadokoro, Rep. Progr. Polym. Phys. Japan, **18**, 197 (1975); J. Polym. Sci., Polym. Phys. Ed., **14**, 427 (1976).
857. C. Nakafuku, T. Takemura Jpn. J. Appl. Phys., **14**, 599 (1975).
858. K. Hikichi, N. Sawatari, A. Tsutsumi, M. Kaneko, Rep. Progr. Polym. Phys. Japan, **18**, 523 (1975).
859. K. Ito, H. Tanizaki, T. Kajiyama, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **18**, 583 (1975); K. Ito, T. Kajiyama, M. Takayanagi, Polym. J. (Tokyo), **9**, 355 (1977).
860. U. Alter, R. Bonart, private communication, March 1976.
861. M. Puterman, F. J. Kolpak, J. Blackwell, J. B. Lando, J. Polym. Sci., Polym. Phys. Ed., **15**, 805 (1977).
862. M. Yokouchi, Y. Chatani, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **14**, 81 (1976).
863. F. Millich, Adv. Polym. Sci., **19**, 117 (1975).
864. M. Yokouchi, Y. Sakakibara, Y. Chatani, H. Tadokoro, T. Tanaka, K. Yoda, Macromolecules, **9**, 266 (1976).
865. A. Blumstein, S. B. Clough L. Patel, R. B. Blumstein, E. C. Hsu, Macromolecules, **9**, 243 (1976). (Chain axis is not identified).
866. V. Petraccone, B. Pirozzi, A. Frasci, P. Corradini, Eur. Polym. J., **12**, 323 (1976).
867. F. Colonna-Cesari, S. Premilat, Polymer, **17**, 267 (1976).
868. H. Kusanagi, H. Tadokoro, Y. Chatani, Macromolecules, **9**, 531 (1976). A double-stranded helix is proposed, each strand of which has a 2*10/1 conformation (C = 20.80 A). The fiber repeat is one-half of this.
869. M. Takayanagi, K. Yamaguchi, Kogyo kagaku Zasshi, **59**, 182 (1956), through Chem. Abstr., **51**, 9247c (1957).
870. C. R. Desper, N. S. Schneider, Macromolecules, **9**, 424 (1976).
871. R. P. Kusy, J. Polym. Sci., Polym. Chem. Ed., **14**, 1527 (1976).
872. I. Modric, K. Holland-Moritz, D. O. Hummel, Colloid Polym. Sci., **254**, 342 (1976).
873. P. Corradini, V. Petraccone, F. Riva, G. Maglio, Eur. Polym. J., **12**, 463 (1976).
874. M. Cesari, G. Perego, A. Melis, Eur. Polym. J., **12**, 585 (1976).
875. L. P. Gul'ko, T. M. Gritsenko, Vysokomol. Soedin., Ser. A, **17**, 1309 (1975) - Polym. Sci., USSR (Engl. Transl.), **17**, 1503 (1975).
876. G. Cojazzi, V. Malta, G. Celotti, R. Zannetti, Makromol. Chem., **177**, 915 (1976).
877. G. Conte, L. D'Ilario, N. V. Pavel, E. Giglio, J. Polym. Sci., Polym. Phys. Ed., **14**, 1553 (1976).
878. L. G. Kazaryan, D. Ya. Tsvankin, V. A. Vasil'ev, M. A. Dakhis, Yu. A. Tolkachev, Vysokomol. Soedin., Ser. A, **17**, 1560 (1975); Polym. Sci. USSR (Engl. Transl.), **17**, 1797 (1975).
879. A. di Meo, G. Maglio, E. Martuscelli, R. Palumbo, Polymer, **17**, 802 (1976). The unit cell parameters, a and c, reported are inconsistent with values tabulated for other linear aliphatic polyesters.
880. J. Economy, R. S. Storm, V. I. Matkovich, S. G. Cottis, B. E. Nowak, J. Polym. Sci., Polym. Chem. Ed., **14**, 2207 (1976). A double-stranded helix is proposed with the chains in a reversed head-to-tail order, each strand has a 2*3/1 conformation.
881. J. R. Caldwell, R. Gilkey, 134th National American Chemical Society Meeting, Chicago, September 1958.
882. R. E. Wilfong, J. Polym. Sci., **54**, 385 (1961).
883. I. H. Hall, M. G. Pass, Polymer, **17**, 807 (1976).
884. A. U. Ahmed, N. Ahmed, J. Aslam, N. M. Butt, Q. H. Khan, M. A. Atta, J. Polym. Sci., Polym. Lett. Ed., **14**, 561 (1976).
885. G. Spach, F. Heitz, C. R. Hebd. Seances Acad. Sci., Ser. C, **276**, 1373 (1973); F. Heitz, B. Lotz, G. Spach, J. Mol. Biol., **92**, 1 (1975).
886. T. Nishi, T. T. Wang, Macromolecules, **8**, 909 (1975).
887. R. C. Robert, Polymer, **10**, 113 (1969).
888. R. W. Lenz, J. E. Sutherland, L. C. Westfelt, Makromol. Chem., **177**, 653 (1976).
889. A. J. Stipanovich, A. Sarko, Macromolecules, **9**, 851 (1976).
890. A. Sarko, J. Southwick, J. Hayashi, Macromolecules, **9**, 857 (1976).
891. T. Itoh, Jpn. J. Appl. Phys., **15**, 2295 (1976). Poly (6-aminocaproic acid) values appear to be halved.

892. A. Charlesby, Proc. Phys. Soc. London, **57**, 496 (1945).
893. H. Nakanishi, M. Hasegawa, Y. Sasada, J. Polym. Sci., Polym. Phys. Ed., **15**, 173 (1977).
894. K. Kaji, I. Sakurada, K. Nakamae, T. Shintaku, E. Shikata, Bull. Inst. Chem. Res., Kyoto Univ., **52**, 308 (1974).
895. T. M. Frunze, A. Sh. Cherdabayev, R. B. Shleifman, V. V. Kurashev, D. Ya. Tsvankin, Vysokomol. Soedin., Ser. A, **18**, 696 (1976) – Polym. Sci. USSR (Engl. Transl.), **18**, 793 (1976).
896. V. V. Korshak, D. Ya. Tsvankin, T. M. Babchinitser, L. G. Kazaryan, Ya. V. Genin, Ya. S. Vygodskii, N. A. Churochkina, S. V. Vinogradova, Vysokomol. Soedin., Ser. A, **18**, 40 (1976); Polym. Sci. USSR (Engl. Transl.), **18**, 46 (1976).
897. K. H. Gardner, J. H. Magill, E. D. T. Atkins, Polymer, **19**, 370 (1978).
898. D. Kobelt, F. Paulus, Acta Crystallogr., Sect. B: Struct. Crystallogr. Cryst. Chem., **30**, 231 (1974).
899. V. Enkelmann, G. Wegner, Makromol. Chem., **178**, 635 (1977); V. Enkelmann, Acta Crystallogr., Sect. B: Struct. Crystallogr. Cryst. Chem., **33**, 2842 (1977).
900. M. Iida, T. Araki, K. Teranishi, H. Tani, Macromolecules, **10**, 275 (1977).
901. H. Kusanagi, H. Tadokoro, Y. Chatani, K. Suehiro, Macromolecules, **10**, 405 (1977).
902. K. Tashiro, M. Kobayashi, H. Tadokoro, Macromolecules, **10**, 413 (1977).
903. R. H. Baughman, J. Appl. Phys., **43**, 4362 (1972).
904. D. Bloor, G. C. Stevens, J. Polym. Sci., Polym. Phys. Ed., **15**, 703 (1977).
905. I. W. Bassi, P. Corradini, V. Petraccone, B. Pirozzi, Eur. Polym. J, **13**, 265 (1977).
906. H. Matsubayashi, Y. Chatani, H. Tadokoro, P. Dumas, N. Spassky, P. Sigwalt, Macromolecules, **10**, 996 (1977).
907. J. R. Fallender, G. S. Y. Yeh, D. S. Chiu, J. E. Mark, J. Polym. Sci., Polym. Phys. Ed., **18**, 389 (1980).
908. L. N. Korzhavin, N. R. Prokopchuk, Yu. G. Baklagina, F. S. Florinskii, N. V. Yefanova, A. M. Dubnova, S. Ya. Frenkel, M. M. Koton, Vysokomol. Soedin. Ser. A, **18**, 707 (1976); Polym. Sci. USSR (Engl. Transl.), **18**, 807 (1976).
909. R. E. Singler, N. S. Schneider, G. L. Hagnauer, Polym. Eng. Sci., **15**, 321 (1975).
910. C. M. Mikulski, P. J. Russo, M. S. Saran, A. G. MacDiarmid, A. F. Garito, A. J. Heeger, J. Am. Chem. Soc., **97**, 6358 (1975); M. J. Cohen, A. F. Garito, A. J. Heeger, A. G. MacDiarmid, C. M. Mikulski, M. S. Saran, J. Kleppinger, ibid, **98**, 3844 (1976).
911. M. Boudeulle, Ph. D. Thesis, Univ. of Lyon, 1974 – from R. H. Baughman, R. R. Chance, J. Polym. Sci., Polym. Phys. Ed., **14**, 2019 (1976).
912. M. Boudeulle, Cryst. Struct. Commun., **4**, 9 (1975) – from R. H. Baughman, R. R. Chance, J. Polym. Sci., Polym. Phys. Ed., **14**, 2019 (1976).
913. R. H. Baughman, K. C. Yee, J. Polym. Sci., Polym. Chem. Ed., **12**, 2467 (1974).
914. T. Miyashita, M. Yokouchi, M. Chatani, H. Tadokoro, Annu. Meeting Soc. Polym. Sci., Japan, Tokyo, Preprints (1974) p. 453.
915. T. Takahagi, Y. Chatani, T. Kusumoto, H. Tadokoro, Polym. J., **20**, 883 (1988).
916. H. Sakakihara, Y. Takahashi, H. Tadokoro, Disc. Meeting Soc. Polym. Sci., Japan, Tokyo, Preprints (1969) p. 407.
917. Y. Chatani, N. Higashibata, M. Takase, H. Tadokoro, T. Hirahara, Annu. Meeting Soc. Polym. Sci., Japan, Kyoto, Preprints p. 427 (1977).
918. R. K. Hasegawa, K. Kimoto, Y. Chatani, H. Tadokoro, H. Sekiguchi, Disc. Meeting Soc. Polym. Sci., Japan, Tokyo, Preprints p. 713 (1974).
919. Y. Takahashi, T. Kurumizawa, H. Kusanagi, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **16**, (1978).
920. Z. Mencik, Chem. Prumysl., **17**, 78 (1967), through Chem. Abstr., **67**, 15849r (1967).
921. H. Matsubayashi, Y. Chatani, H. Tadokoro, Y. Tabata, W. Ito, Polym. J. (Tokyo), **9**, 145 (1977).
922. H. Hasegawa, W. Claffey, P. H. Geil, J. Macromol. Sci. B: Phys., **13**, 89 (1977).
923. M. Suwalsky, A. Llanos, Biopolymers, **16**, 403 (1977).
924. D. R. Rueda, A. S. Verdini, A. Del Pra, Macromolecules, **10**, 715 (1977).
925. R. H. Baughman, P. A. Apgar, R. R. Chance, A. G. MacDiarmid, A. F. Garito, J. Chem. Phys., **66**, 401 (1977).
926. A. Shoji, Y. Yamamoto, T. Ozaki, M. Ida, Y. Saito, Macromolecules, **10**, 723 (1977).
927. C. J. Lee, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **15**, 355 (1977).
928. Yu. G. Baklagina, I. S. Milevskaya, N. V. Yefanova, A. V. Sidorovich, V. A. Zubkov, Vysokomol. Soedin., Ser. A, **18**, 1235 (1976) – Polym. Sci. USSR (Engl. Transl.), **18**, 1417 (1976).
929. H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **15**, 247 (1977).
930. R. Legras, J. P. Mercier, J. Polym. Sci., Polym. Phys. Ed., **15**, 1283 (1977).
931. M. Moser, M. Boudeulle, J. Polym. Sci., Polym. Phys. Ed., **16**, 971 (1978).
932. M. Osanai, A. Tsutsumi, K. Hikichi, M. Kaneko, Y. Konishi, M. Hatano, Rep. Progr. Polym. Phys. Japan, **19**, 517 (1976); M. Osanai, A. Tsutsumi, K. Hikichi, Y. Konishi, M. Hatano, ibid., **20**, 589 (1977).
933. I. J. Desborough, Ph. D. Thesis, Victoria Univ. of Manchester (1976), in I. J. Desborough, I. H. Hall, Polymer, **18**, 825 (1977).
934. G. Bruno, S. Caccamese, P. Maravigna, G. Montaudo, A. Recca, J. Polym. Sci., Polym. Lett. Ed., **15**, 481 (1977).
935. D. R. Rueda, P. Spadon, A. S. Verdini, A. Del Pra, Biopolymers, **16**, 2347 (1977).
936. P. Zoller, J. Appl. Polym. Sci., **21**, 3129 (1977).
937. R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **15**, 1619 (1977).
938. A. J. Stipanovic, A. Sarko, Polymer, **19**, 3 (1978).
939. B. Wunderlich, G. Czornyj, Macromolecules, **10**, 906 (1977).
940. H. Janeczek, E. Turska, T. Szekely, M. Lengyel, F. Till, Polymer, **19**, 85 (1978).

941. C. Marco, J. G. Fatou, A. Bello, Polymer, **18**, 1100 (1977); C. Marco, A. Bello, J. G. Fatou, Makromol. Chem., **179**, 1333 (1978).
942. K. Matsushige, T. Takemura, J. Polym. Sci., Polym. Phys. Ed., **18**, 1665 (1980).
943. P. B. McAllister, T. J. Carter, R. M. Hinde, J. Polym. Sci., Polym. Phys. Ed., **16**, 49 (1978).
944. J. N. Hay, Makromol. Chem., **178**, 1601 (1977).
945. R. H. Baughman, S. L. Hsu, G. P. Pez, A. J. Signorelli, J. Chem. Phys., **68**, 5405 (1978).
946. G. T. Fekete, W. A. Basset, C. L. Beatty, Bull. Am. Phys. Soc., **23**, 408 (1978).
947. S. Weinhold, M. H. Litt, J. B. Lando, J. Polym. Sci., Polym. Phys. Ed., **20**, 535 (1982).
948. D. Day, J. B. Lando, J. Polym. Sci., Polym. Phys. Ed., **16**, 1009 (1978).
949. G. Trafara, R. Koch, E. Sausen, Makromol. Chem., **179**, 1837 (1978).
950. W. Meyer, G. Lieser, G. Wegner, J. Polym. Sci., Polym. Phys. Ed., **16**, 1365 (1978).
951. K. C. Yee, R. R. Chance, J. Polym. Sci., Polym. Phys. Ed., **16**, 431 (1978).
952. A. J. McKinnon, A. V. Tobolsky, J. Phys. Chem., **72**, 1157 (1968).
953. S. Sasaki, M. Watanabe, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **20**, 613 (1977).
954. K. Koga, T. Kajiyama, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **20**, 655 (1977).
955. S. Sasaki, M. Miyamoto, T. Nakamura, I. Uematsu, Biopolymers, **17**, 2715 (1978).
956. Y. Chatani, Y. Ueda, H. Tadokoro, H. Deits, O. Vogl, Macromolecules, **11**, 636 (1978).
957. H. Kakida, J. Nakauchi, T. Yoshihara, S. Minami, Rep. Progr. Polym. Phys. Japan, **20**, 183 (1977).
958. K. Haraguchi, T. Kajiyama, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **20**, 191 (1977); J. Appl. Polym. Sci., **23**, 915 (1979).
959. N. Spassky, A. Leborgne, M. Reix, R. E. Prud'homme, E. Bigdeli, R. W. Lenz, Macromolecules, **11**, 716 (1978).
960. A. Mehta, U. Gaur, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **16**, 289 (1978).
961. S. Kobayashi, H. Nakazawa, S. Iwayanagi, Rep. Progr. Polym. Phys. Japan, **20**, 163 (1977).
962. E. Roche, H. Chanzy, M. Boudeulle, R. H. Marchessault, P. Sundararajan, Macromolecules, **11**, 86 (1978).
963. A. V. Sidorovich, Yu. G. Baklagina, A. V. Kenarov, Yu. S. Nadezhin, N. A. Adrova, F. S. Florinsky, J. Polym. Sci., Polym. Symp., **58**, 359 (1977).
964. J. N. Hay, J. Polym. Sci., Polym. Chem. Ed., **14**, 2845 (1976).
965. A. Del Pra, M. Palumbo, M. Goodman, Biopolymers, **15**, 503 (1976).
966. K. Itoh, B. M. Foxman, G. D. Fasman, Biopolymers, **15**, 419 (1976).
967. B. R. Malcolm, Biopolymers, **16**, 2591 (1977).
968. P. Spadon, A. S. Verdini, A. Del Pra, Biopolymers, **17**, 2029 (1978).
969. A. Del Pra, P. Spadon, R. Boni, G. M. Bonara, A. Scatturin, Makromol. Chem., **179**, 2707 (1978).
970. J. S. Wood, I. Negulescu, O. Vogl, J. Macromol. Sci. A: Chem., **11**, 171 (1977).
971. G. Cojazzi, A. Fichera, V. Malta, R. Zannetti, Makromol. Chem., **179**, 509 (1978).
972. P. T. van Emmerick, C. A. Smolders, Eur. Polym. J., **9**, 293 (1973).
973. G. A. Clegg, T. P. Melia, Polymer, **10**, 912 (1969).
974. R. J. Samuels, J. Polym. Sci., Polym. Phys. Ed., **13**, 1417 (1975).
975. A. J. Lovinger, J. O. Chua, C. C. Gryte, J. Polym. Sci., Polym. Phys. Ed., **15**, 641 (1977).
976. G. Lieser, G. Wegner, W. Muller, V. Enkelmann, Makromol. Chem., Rapid Commun., **1**, 621 (1980).
977. F. S. Bates, G. L. Baker, Macromolecules, **16**, 1013 (1983).
978. J. C. W. Chien, F. E. Karasz, K. Shimamura, Macromolecules, **15**, 1012 (1982).
979. T. Akaishi, K. Miyasaka, K. Ishikawa, H. Shirakawa, S. Ikeda, Rep. Progr. Polym. Phys. Japan, **22**, 125 (1979).
980. G. Lieser, G. Wegner, W. Muller, V. Enkelmann, W. H. Meyer, Makromol. Chem., Rapid Commun., **1**, 627 (1980).
981. K. Shimamura, F. E. Karasz, J. A. Hirsch, J. C. W. Chien, Makromol. Chem., Rapid Commun., **2**, 473 (1981).
982. G. Lieser, Polym. Commun., **25**, 201 (1984).
983. J. Haase, R. Hosemann, S. Kohler, Polymer, **19**, 1358 (1978); J. Haase, S. Kohler, R. Hosemann, Z. Naturforsch., **33A**, 1472 (1978).
984. R. M. Gohil, R. D. Patel, Angew. Makromol. Chem., **64**, 43 (1977).
985. T. Yagi, Polym. J. (Tokyo), **12**, 9 (1980).
986. E. S. Clark, G. J. Mauer, R. K. Eby, S. Block, Bull. Am. Phys. Soc., **25**, 316 (1980).
987. E. S. Clark, J. J. Weeks, R. K. Eby, in A. D. French, K. H. Gardner, Eds., Fiber Diffraction Methods (ACS Symp. Ser. No. 141), Am. Chem. Soc., Washington, 1980, p. 183, J. J. Weeks, E. S. Clark, R. K. Eby, Polymer, **22**, 1480 (1981). Irrational helix of 0.1300 nm/CF2 group with helix designated by 1*948/439 or greater.
988. H. W. Starkweather, Jr., P. Zoller, G. A. Jones, A. J. Vega, J. Polym. Sci., Polym. Phys. Ed., **20**, 751 (1982); H. W. Starkweather, Jr., ibid., **20**, 2159 (1982).
989. S. F. Lau, H. Suzuki, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **22**, 379 (1984).
990. G. Charlet, G. Delmas, Polym. Bull. (Berlin), **6**, 367 (1982).
991. P. Corradini, V. Petraccone, B. Pirozzi, Eur. Polym. J., **19**, 299 (1983).
992. K. Monobe, F. Yokoyama, J. Macromol. Sci. B: Phys., **8**, 277 (1973); F. Yokoyama, K. Monobe, Polymer, **21**, 968 (1980).
993. E. Yokoyama, K. Monobe, J. Polym. Sci., Polym. Lett. Ed., **19**, 91 (1981).
994. M. A. Bachmann, J. B. Lando, Macromolecules, **14**, 40 (1981).
995. D. T. Grubb, K. W. Choi, J. Appl. Phys., **52**, 5908 (1981).

996. Y. Takahashi, H. Tadokoro, Macromolecules, **13**, 1317 (1980); Y. Takahashi, M. Kohyama, Y. Matsubara, H. Iwane, H. Tadokoro, Macromolecules, **14**, 1841 (1981).
997. S. Weinhold, M. H. Litt, J. B. Lando, J. Polym. Sci., Polym. Lett. Ed., **17**, 585 (1979); Macromolecules, **13**, 1178 (1980).
998. A. J. Lovinger, Macromolecules, **14**, 322 (1981).
999. D. Naegele, D. Y. Yoon, M. G. Broadhurst, Macromolecules, **11**, 1297 (1978).
1000. M. Bachmann, W. L. Gordon, S. Weinhold, J. B. Lando, J. Appl. Phys., **51**, 5095 (1980).
1001. F. Kumamaru, T. Kajiyama, M. Takayanagi, J. Cryst. Growth, **48**, 202 (1980).
1002. F. Bosscher, G. ten Brinke, A. Eshuis, G. Challa, Macromolecules, **15**, 1364 (1982).
1003. V. Petraccone, G. Guerra, C. De Rosa, A. Tuzi, Macromolecules, **18**, 813 (1985).
1004. H. Kusuyama, M. Takase, Y. Higashihata, H.-T. Tseng, Y. Chatani, H. Tadokoro, Polymer, **23**, 1256 (1982); H. Kusuyama, M. Miyamoto, Y. Chatani, H. Tadokoro, Polym. Commun., **24**, 119 (1983).
1005. F. Bosscher, G. ten Brinke, G. Challa, Macromolecules, **15**, 1442 (1982).
1006. E. J. Vorenkamp, F. Bosscher, G. Challa, Polymer, **20**, 59 (1979).
1007. J. Kuo, R. W. Lenz, J. Polym. Sci., Polym. Chem. Ed., **15**, 119 (1977).
1008. E. D. T. Atkins, D. H. Isaac, A. Keller, K. Miyasaka, J. Polym. Sci., Polym. Phys. Ed., **15**, 211 (1977); E. D. T. Atkins, A. Keller, J. S. Shapiro, P. J. Lemstra, Polymer, **22**, 1161 (1981).
1009. P. R. Sundararajan, N. J. Tyrer, Macromolecules, **15**, 1004 (1982); Polym. Bull. (Berlin), **6**, 359 (1982).
1010. P. R. Sundararajan, J. Zamin, Polymer, **20**, 1567 (1979).
1011. S. Isoda, M. Tsuji, M. Ohara, A. Kawaguchi, K. Katayama, Polymer, **24**, 1155 (1983).
1012. S. Isoda, A. Kawaguchi, K. Katayama, J. Polym. Sci., Polym. Phys. Ed., **22**, 669 (1984).
1013. N. R. Prokopchuk, Yu. G. Baklagina, L. N. Korzhavin, A. V. Sidorovich, M. M. Koton, Vysokomol. Soedin., Ser. A, **19**, 1126 (1977) – Polym. Sci. USSR (Engl. Transl.), **19**, 1297 (1977).
1014. R. N. Goel, A. Hepworth, B. L. Deopura, I. K. Varma, D. S. Varma, J. Appl. Polym. Sci., **23**, 3541 (1979).
1015. V. V. Korshak, T. M. Babchinitser, L. G. Kazaryan, V. A. Vasilyev, Ya. V. Genin, A. Ye. Azriel, Ya. S. Vygodsky, N. A. Churochkina, S. V. Vinogradova, D. Ya. Tsvankin, J. Polym. Sci., Polym. Phys. Ed., **18**, 247 (1980). Errors in author's parameters: values in their table I cannot be replicated.
1016. E. J. Roche, T. Takahashi, E. L. Thomas, in: A. D. French, K. H. Gardner (Eds.), "Fiber Diffraction Methods" (ACS Symp. Ser. No. 141), Am. Chem. Soc., Washington, p. 303 (1980).
1017. J. A. Odell, A. Keller, E. D. T. Atkins, M. J. Miles, J. Mater. Sci., **16**, 3309 (1981).
1018. M. I. Bessonov, N. P. Kuznetsov, M. M. Koton, Vysokomol. Soedin., Ser. A, **20**, 347 (1978); Polym. Sci. USSR (Engl. Transl.), **20**, 391 (1978).
1019. H. A. Morss, Jr., J. Am. Chem. Soc., **60**, 237 (1938).
1020. E. M. Barrall, II, T. C. Clarke, A. R. Gregges, J. Polym. Sci., Polym. Phys. Ed., **16**, 1355 (1978).
1021. R. L. Williams, D. J. Ando, D. Bloor, M. B. Hurshouse, Polymer, **21**, 1269 (1980).
1022. M. Thakur, J. B. Lando, in A. Hiltner, Ed., Structure-Property Relationships of Polymeric Solids (Polymer Science and Technology, V. 22), Plenum, New York, 1983, p. 85; Macromolecules, **16**, 143 (1983). The cross-polymer has two "fiber" axes, *b* and *c*.
1023. M. Thakur, J. B. Lando, J. Appl. Phys., **54**, 5554 (1983). The cross-polymer has two "fiber" axes, *b* and *c*.
1024. T. Komoto, M. Kodaka, T. Kawai, Makromol. Chem., **180**, 819 (1979).
1025. O. Yamashita, T. Yamane, T. Ashida, S. Yamashita, T. Yamashita, Polym. J., **11**, 763 (1979).
1026. J. Watanabe, E. Kawasaki, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **24**, 559 (1981).
1027. S. Sasaki, Y. Yasumoto, J. Watanabe, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **22**, 611 (1979).
1028. S. Sasaki, Y. Yasumoto, I. Uematsu, Macromolecules, **14**, 1797 (1981).
1029. R. D. B. Fraser, T. P. Macrae, I. W. Stapleton, Nature, **193**, 573 (1962).
1030. J. Watanabe, K. Imai, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **21**, 565 (1970); J. Watanabe, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **22**, 603 (1979); J. Watanabe, K. Imai, R. Gehani, I. Uematsu, J. Polym. Sci., Polym. Phys. Ed., **19**, 653 (1981); J. Watanabe, I. Uematsu, Polymer 25, 1711 (1984); J. Watanabe, K. Imai, I. Uematsu, Macromolecules, **19**, 1489 (1986).
1031. J. Watanabe, K. Imai, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **23**, 671 (1980).
1032. M. Osani, K. Hikichi, Polym. J. (Tokyo), **14**, 411 (1982).
1033. M. Osani, K. Hikichi, Biopolymers, **19**, 1099 (1980).
1034. M. Osani, K. Hikichi, Rep. Progr. Polym. Phys. Japan, **21**, 507 (1978).
1035. M. Osani, K. Hikichi, Rep. Progr. Polym. Phys. Japan, **22**, 567 (1979).
1036. K. Ichikawa, K. Hikichi, Rep. Progr. Polym. Phys. Japan, **24**, 535 (1981); Polym. J. (Tokyo), **14**, 753 (1982); Rep. Progr. Polym. Phys. Japan, **25**, 661 (1982).
1037. A. Nakajima, T. Fujiwara, T. Hayashi, K. Kaji, Biopolymers, **12**, 2681 (1973).
1038. M. Nagao, J. Watanabe, I. Uematsu, Rep. Progr. Polym. Phys. Japan, **23**, 675 (1980).
1039. J. A. Lefelar, J. R. Knox, E. T. Samulski, Biopolymers, **22**, 1071 (1983).
1040. S. Sasaki, T. Nakamura, I. Uematsu, J. Polym. Sci., Polym. Phys. Ed., **17**, 825 (1979).
1041. J. Watanabe, Y. Sasanuma, A. Endo, I. Uematsu, Polymer, **25**, 698 (1984).
1042. M. Nagao, T. Suzuki, S. Sasaki, I. Uematsu, Polym. Bull. (Berlin), **11**, 565 (1984).
1043. S. Sasaki, H. Ban, I. Uematsu, J. Polym. Sci., Polym. Phys. Ed., **21**, 413 (1983). For cell I, a value for the *c* axis was calculated from the reported helical parameters and is so indicated by placing it in parentheses.

1044. S. K. Brahmachari, M. Bansal, V. S. Ananthanarayanan, V. Sasisekharan, Macromolecules, **12**, 23 (1979).
1045. O. Yamashita, T. Ashida, Polym. J. (Tokyo), **15**, 899 (1983).
1046. F. Heitz, G. Detriche, F. Vovelle, G. Spach, Macromolecules, **14**, 47 (1981).
1047. R. Ciaschi, M. D'Alagni, G. Mignucci, Makromol. Chem., **180**, 2883 (1979).
1048. R. Ciaschi, M. D'Alagni, Makromol. Chem., **180**, 2893 (1979).
1049. V. Malta, G. Cojazzi, A. Fichera, D. Ajo, R. Zannetti, Eur. Polym. J., **15**, 765 (1979).
1050. K.-H. Illers, Makromol. Chem., **179**, 497 (1978).
1051. S. Gogolewski, M. Gasiorek, K. Czerniawska, A. J. Pennings, Colloid Polym. Sci., **260**, 859 (1982).
1052. N. Avramova, S. Fakirov, Polym. Commun., **25**, 27 (1984).
1053. B. R. Malcolm, Biopolymers, **22**, 319 (1983); B. R. Malcolm, M. D. Walkinshaw, Biopolymers, **25**, 607 (1986).
1054. A. J. Owen, P. Kollross, Polym. Commun., **24**, 303 (1983).
1055. G. Cojazzi, A. M. Drusiani, A. Fichera, V. Malta, F. Pilati, R. Zannetti, Eur. Polym. J., **17**, 1241 (1981).
1056. M. Macchi, A. A. Giorgi, Makromol. Chem., Rapid Commun., **1**, 563 (1980).
1057. B. A. Newman, T. P. Sham, K. D. Pae, J. Appl. Phys., **48**, 4092 (1977).
1058. K. Inoue, S. Hoshino, J. Polym. Sci., Polym. Phys. Ed., **18**, 697 (1980).
1059. E. M. Macchi, F. M. B. Coutinho, A. S. Gomes, J. Polym. Sci., Polym. Lett. Ed., **22**, 209 (1984).
1060. M. L. Colclough, R. Baker, J. Mater. Sci., **13**, 2531 (1978).
1061. M. G. Northolt, H. A. Stuut, J. Polym. Sci., Polym. Phys. Ed., **16**, 939 (1978).
1062. K. Sakaoku, T. Itoh, K. Kitamura, Rep. Progr. Polym. Phys. Japan, **24**, 139 (1981).
1063. T. Kaneda, S. Ishikawa, H. Daimon, T. Katsura, M. Ueda, K. Oda, M. Horio, Makromol. Chem., **183**, 417 (1981).
1064. B. Kalb, A. J. Pennings, Polymer, **21**, 607 (1980).
1065. G. Lieser, J. Polym. Sci., Polym. Phys. Ed., **21**, 1611 (1983).
1066. J. M. Jonza, R. S. Porter, J. Polym. Sci., Part B: Polym. Phys., **24**, 2459 (1986).
1067. S. Poulin-Dandurand, S. Perez, J.-F. Revol, F. Brisse, Polymer, **20**, 419 (1979).
1068. I. J. Desborough, I. H. Hall, J. Z. Neisser, Polymer, **20**, 545 (1979).
1069. I. H. Hall, M. G. Pass, N. N. Rammo, J. Polym. Sci., Polym. Phys. Ed., **16**, 1409 (1978); I. H. Hall, N. N. Rammo, ibid., **16**, 2189 (1978).
1070. I. H. Hall, B. A. Ibrahim, Polymer, **23**, 805 (1982).
1071. K.-H. Illers, Colloid Polym. Sci., **258**, 117 (1980).
1072. H. D. Keith, Macromolecules, **15**, 122 (1982).
1073. R. Minke, J. Blackwell, J. Macromol. Sci. B: Phys., **16**, 407 (1979); *ibid.*, **18**, 233 (1980).
1074. K. Ueberreiter, V.-H. Karl, A. Altmeyer, Eur. Polym. J., **14**, 1045 (1978).
1075. Y. Kinoshita, R. Nakamura, Y. Kitano, T. Ashida, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **20** (1), 454 (1979); Y. Kitano, Y. Kinoshita, T. Ashida, Polymer, **36**, 1947 (1995).
1076. V. Petraccone, A. Roviello, A. Sirigu, A. Tuzi, E. Martuscelli, M. Pracella, Eur. Polym. J., **16**, 261 (1980).
1077. B. Remillard, F. Brisse, Polymer, **23**, 1960 (1982).
1078. J. Shimizu, N. Okui, Y. Imai, S. Nishide, A. Takaku, J. Polym. Sci., Polym. Phys. Ed., **21**, 275 (1983).
1079. G. Maglio, C. Marchetta, A. Botta, R. Palumbo, M. Pracella, Eur. Polym. J., **15**, 695 (1979).
1080. D. L. Dorset, B. Moss, in: C. D. Craver (Ed.), "Polymer Characterization" (Adv. Chem. Series 203), Am. Chem. Soc., Washington, p. 409 (1983).
1081. Y. Saito, K. Hara, S. Kinoshita, Polym. J. (Tokyo), **14**, 19 (1982).
1082. J. Blackwell, K. H. Gardner, Polymer, **20**, 13 (1979); J. Blackwell, M. Ross, J. Polym. Sci., Polym. Lett. Ed., **17**, 447 (1979).
1083. L. Born, J. Crone, H. Hespe, E. H. Muller, K. H. Wolf, J. Polym. Sci., Polym. Phys. Ed., **22**, 163 (1984).
1084. H. Ishihara, I. Kimura, N. Yoshihara, J. Macromol. Sci. B: Phys., **22**, 713 (1983–1984).
1085. J. Noah, R. E. Prud'homme, Eur. Polym. J., **17**, 353 (1981).
1086. R. H. Marchessault, J. St.-Pierre, M. Duval, S. Perez, Macromolecules, **11**, 1281 (1978).
1087. J. Noah, R. E. Prud'homme, Eur. Polym. J., **16**, 1027 (1980).
1088. Y. Normand, M. Aubin, R. E. Prud'homme, Makromol. Chem., **180**, 769 (1979).
1089. D. Duchesne, R. E. Prud'homme, Polymer, **20**, 1199 (1979).
1090. Y. Chatani, M. Yokouchi, H. Tadokoro, Macromolecules, **12**, 822 (1979).
1091. O. Setokuchi, Y. Chatani, H. Tadokoro, Polym. Prepr. Japan, **28**, (1962) (1979), cited in K. Nakamae, F. Yokoyama, T. Nishino, M. Yosikawa, T. Matsumoto, J. Macromol. Sci. B: Phys., **22**, 591 (1983).
1092. S. V. Meille, T. Konishi, P. H. Geil, Polymer, **25**, 773 (1984).
1093. C. D. Eisenbach, R. W. Lenz, M. Duval, R. H. Marchessault, Makromol. Chem., **180**, 429 (1979).
1094. W. Claffey, K. Gardner, J. Blackwell, J. Lando, P. H. Geil, Philos. Mag., **30**, 1223 (1974).
1095. Y. Takahashi, Y. Osaki, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **18**, 1863 (1980).
1096. Y. Takahashi, Y. Osaki, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **19**, 1153 (1981).
1097. E. Perez, M. A. Gomez, A. Bello, J. G. Fatou, Colloid Polym. Sci. 1097B, **261**, 571 (1983); M. A. Gomez, E. D. T. Atkins, C. Upstill, A. Bello, 1097C J. G. Fatou, Polymer, **29**, 224 (1988).
1098. J. Starr, O. Vogl, J. Polym. Sci., Polym. Chem. Ed., **17**, 1923 (1979).
1099. R. M. Myasnikova, E. F. Titova, E. S. Obolonkova, Polymer, **21**, 403 (1980).
1100. J. M. Parker, P. V. Wright, C. C. Lee, Polymer, **22**, 1305 (1981).
1101. H. Tadokoro, Makromol. Chem., Suppl., **4**, 129 (1981).

1102. S. E. Rickert, J. B. Lando, A. J. Hopfinger, E. Baer, Macromolecules, **12**, 1053 (1979).
1103. S. E. Rickert, H. Ishida, J. B. Lando, J. L. Koenig, E. Baer, J. Appl. Phys., **51**, 5194 (1980).
1104. D. Bhaumik, J. E. Mark, Macromolecules, **14**, 162 (1981).
1105. S. Katayama, T. Murakami, Y. Takahashi, H. Serita, Y. Obuchi, T. Ito, J. Appl. Polym. Sci., **20**, 975 (1976).
1106. Y. Chatani, H. Tadokoro, T. Saegusa, H. Ikeda, Macromolecules, **14**, 315 (1981).
1107. Y. Chatani, T. Kobatake, H. Tadokoro, R. Tanaka, Macromolecules, **15**, 170 (1982).
1108. Y. Chatani, T. Kobatake, H. Tadokoro, Macromolecules, **16**, 199 (1983).
1109. H. R. Allcock, R. A. Arcus, E. G. Stroh, Macromolecules, **13**, 919 (1980).
1110. J. J. Beres, N. S. Schneider, C. R. Desper, R. E. Singler, Macromolecules, **12**, 566 (1979).
1111. C. W. Burkhart, P. C. Gillette, J. B. Lando, J. B. Beres, J. Polym. Sci., Polym. Phys. Ed., **21**, 2349 (1983).
1112. M. Kojima, J. H. Magill, Polym. Commun., **24**, 329 (1983).
1113. P. C. Dawson, D. J. Blundell, Polymer, **21**, 577 (1980).
1114. D. R. Rueda, F. Ania, A. Richardson, I. M. Ward, F. J. Balta-Calleja, Polym. Commun., **24**, 258 (1983).
1115. J. N. Hay, D. J. Kemmish, J. I. Langford, A. I. M. Rae, Polym. Commun., **25**, 175 (1984).
1116. D. J. Blundell, B. N. Osborn, Polymer, **24**, 953 (1983).
1117. T. E. Atwood, P. C. Dawson, J. L. Freeman, L. R. J. Hoy, J. B. Rose, P. A. Staniland, Polymer, **22**, 1096 (1981).
1118. S. M. Aharoni, P. A. Apgar, J. H. Dunsmuir, J. Polym. Sci., Polym. Phys. Ed., **15**, 2055 (1977).
1119. P. E. Aylwin, R. H. Boyd, Polymer, **25**, 323 (1984).
1120. V. V. Korshak, S. V. Vinogradova, Zh. Obshch. Khim., **26**, 544 (1956); J. Gen. Chem. USSR (Engl. Transl.), **26**, 581 (1956).
1121. B. Wunderlich, U. Gaur, Pure Appl. Chem., **52**, 445 (1980).
1122. D. Lupinacci, W. T. Winter, J. Polym. Sci., Polym. Phys. Ed., **20**, 1013 (1982).
1123. B. Tieke, J. Polym. Sci., Polym. Chem. Ed., **22**, 391 (1984).
1124. M. Farina, G. Allegra, G. Natta, J. Am. Chem. Soc., **86**, 516 (1964).
1125. J. Blackwell, M. R. Nagarajan, T. B. Hoitink, Polymer, **22**, 1534 (1981); ibid., **23**, 950 (1982).
1126. M. Farina, G. di Silvestro, Makromol. Chem., **183**, 241 (1982).
1127. W. Cooper, G. Vaughan, Polymer, **4**, 329 (1963).
1128. S. Mukherji, A. E. Woodward, J. Polym. Sci., Polym. Phys. Ed., **22**, 793 (1984).
1129. C. Guizard, H. Chanzy, A. Sarko, Macromolecules, **17**, 100 (1984).
1130. J. Jelsma, D. R. Kreger, in: A. D. French, K. H. Gardner (Eds.), "Fiber, Diffraction Methods" (ACS Symp. Ser. No. 141), Am. Chem. Soc., Washington, p. 351 (1980).
1131. K. Ogawa, A. Misaki, S. Oka, K. Okamura, Carbohydr. Res., **75**, C13 (1979); K. Ogawa, K. Okamura, S. Oka, A. Misaki, in: A. D. French, K. H. Gardner (Eds.), "Fiber Diffraction Methods" (ACS Symp. Ser. No. 141), Am. Chem. Soc., Washington, p. 353 (1980).
1132. R. H. Marchessault, Y. Deslandes, K. Ogawa, P. R. Sundararajan, Can. J. Chem., **55**, 300 (1977); Y. Deslandes, R. H. Marchessault, A. Sarko, Macromolecules, **13**, 1466 (1980); C. T. Chuah, A. Sarko, Y. Deslandes, R. H. Marchessault, Macromolecules, **16**, 1375 (1983).
1133. H. Takeda, N. Yasuoka, N. Kasai, T. Harada, Polym. J. (Tokyo), **10**, 365 (1978).
1134. H. Chanzy, M. Dube, R. H. Marchessault, Polymer, **20**, 1037 (1979).
1135. G. L. Clark, A. F. Smith, J. Phys. Chem., **40**, 863 (1937).
1136. P. Zugenmaier, W. Herth, G. Lieser, Colloid Polym. Sci., **259**, 472 (1981).
1137. R. J. Samuels, J. Polym. Sci., Polym. Phys. Ed., **19**, 1081 (1981).
1138. K. Ogawa, S. Hirano, T. Miyanishi, T. Yui, T. Watanabe, Macromolecules, **17**, 975 (1984).
1139. K. Tanaka, N. Katsura, Biopolymers, **18**, 257 (1979).
1140. T. L. Bluhm, A. Sarko, Biopolymers, **14**, 2639 (1975); *ibid.*, **16**, 2067 (1977).
1141. H. Chanzy, M. Dube, R. H. Marchessault, J. F. Revol, Biopolymers, **18**, 887 (1979); H. Chanzy, S. Perez, D. P. Miller, G. Paradossi, W. T. Winter, Macromolecules, **20**, 2407 (1987).
1142. T. L. Bluhm, P. Zugenmaier, Progr. Colloid. Polym. Sci., **64**, 132 (1978); Polymer, **20**, 23 (1979).
1143. P. Zugenmaier, A. Sarko, Biopolymers, **12**, 435 (1973).
1144. H.-C. H. Wu, A. Sarko, Carbohydr. Res., **61**, 7 (1978).
1145. N. C. Schieltz, Quart. Colo. School Mines, **60** (4), 55 (1965), cited in H.-C. Wu, A. Sarko, Carbohydr. Res., **61**, 7 (1978).
1146. H.-C. Wu, A. Sarko, Carbohydr. Res., **54**, C3 (1977); *ibid.*, **61**, 27 (1978).
1147. D. Meader, E. D. T. Atkins, F. Happey, Polymer, **19**, 1371 (1978).
1148. P. Zugenmaier, Appl. Polym. Symp., **37**, 223 (1983).
1149. C. Woodcock, A. Sarko, Macromolecules, **13**, 1183 (1980).
1150. R. E. Hunter, N. E. Dweltz, J. Appl. Polym. Sci., **23**, 249 (1979).
1151. D. M Lee, J. Blackwell, Biopolymers, **20**, 2165 (1981).
1152. K. M. Paralikar, S. M. Batrabet, J. Polym. Sci., Polym. Lett. Ed., **19**, 555 (1981).
1153. E. S. Gardiner, A. Sarko, Appl. Polym. Symp., **37**, 303 (1983).
1154. D. M. Lee, J. Blackwell, M. H. Litt, Biopolymers, **22**, 1383 (1983); J. Blackwell, D. M. Lee, Appl. Polym. Symp., **37**, 283 (1983).
1155. P. M. Whitaker, I. A. Nieduszynski, E. D. T. Atkins, Polymer, **15**, 125 (1974).
1156. F. B. Booy, H. Chanzy, A. Sarko, Biopolymers, **18**, 2261 (1979).
1157. C. A. Aufdermarsh, R. Pariser, J. Polym. Sci., Part A, **2**, 4727 (1964).
1158. E. Bornschlegl, R., Bonart, Colloid Polym. Sci., **258**, 319 (1980).

1159. I. H. Hall, cited in I. H. Hall, Ed., Structure of Crystalline Polymers, Elsevier, New York, p. 58 (1984).
1160. H. Tadokoro, cited in: I. H. Hall (Ed.), "Structure of Crystalline Polymers", Elsevier, New York, p. 59, (1984).
1161. S. Gogelewski, A. J. Pennings, Polymer, **18**, 660 (1977).
1162. F. Kumamaru, T. Kajiyama, M. Takayanagi, Rep. Progr. Polym. Phys. Japan, **19**, 581 (1976).
1163. R. Alamo, J. G. Fatou, J. Guzman, Polymer, **23**, 374 (1982).
1164. E. Bortel, S. Hodorowicz, R. Lamot, Makromol. Chem., **180**, 2491 (1979).
1165. J. Garza, C. Marco, J. G. Fatou, A. Bello, Polymer, **22**, 477 (1981).
1166. J. H. Magill, M. Girolamo, A. Keller, Polymer, **22**, 43 (1981).
1167. C. Marco, J. G. Fatou, A. Bello, A. Blanco, Makromol. Chem., **181**, 1357 (1980).
1168. W. S. Fulton, E. D. T. Atkins, in: A. D. French, K. H. Gardner (Eds.), "Fiber Diffraction Methods" (ACS Symp. Ser. No. 141), Am. Chem. Soc., Washington, p. 385 (1980).
1169. E. N. Dalal, K. D. Taylor, P. J. Phillips, Polymer, **24**, 1623 (1983).
1170. C. K. L. Davies, O. E. Long, J. Mater. Sci., **14**, 2529 (1979).
1171. J. C. Wittamann, B. Lotz, J. Polym. Sci., Polym. Phys. Ed., **19**, 1853 (1981).
1172. M. Kyotani, J. Polym. Sci., Polym. Phys. Ed., **20**, 345 (1982).
1173. M. Farina, G. Natta, G. Allegra, M. Loffelholz, J. Polym. Sci., Part C, **16**, 2517 (1967).
1174. I. Goodman, Angew. Chem., **74**, 606 (1962).
1175. E. W. Fischer, H. J. Sterzel, G. Wegner, Kolloid Z. Z. Polym., **251**, 980 (1973).
1176. F. Ascoli, G. de Angelis, F. del Bianco, P. de Santis, Biopolymers, **14**, 1109 (1975).
1177. H. Kusanagi, M. Takase, Y. Chatani, H. Tadokoro, J. Polym. Sci., Polym. Phys. Ed., **16**, 131 (1978).
1178. B. M. Ginzburg, V. N. Volosatov, Ye. T. Magdalev, Sh. Tuichiyev, Vysokomol. Soedin., Ser. A, **20**, 900 (1978); Polym. Sci. USSR (Engl. Transl.), **20**, 1017 (1978).
1179. H. Haberkorn K. H. Illers, P. Simak, Polym. Bull. (Berlin), **1**, 485 (1979).
1180. H. W. Starkweather, Jr., P. Zoller, G. A. Jones, J. Polym. Sci., Polym. Phys. Ed., **22**, 1615 (1984).
1181. N. S. Murthy, H. Kim, Polymer, **25**, 1093 (1984).
1182. R. Geiss, W. Volksen, J. Tsay, J. Economy, J. Polym. Sci., Polym. Lett. Ed., **22**, 433 (1984).
1183. R. H. Marchessault, H. Morikawa, J. -F. Revol, T. L. Bluhm, Macromolecules, **17**, 1882 (1984).
1184. A. Palmer, S. Poulin-Dandurand, J.-F. Revol, F. Brisse, Eur. Polym. J., **20**, 783 (1984); F. Brisse, A. Palmer, B. Moss, D. Dorset, W. A. Roughead, D. P. Miller, ibid., 791 (1984).
1185. M. Kojima, W. Kluge, J. H. Magill, Macromolecules, **17**, 1421 (1984).
1186. T. A. Ibidapo, D. J. Johnson, Polymer, **24**, 271 (1983).
1187. E. P. Melnikova, A. A. Vansheidt, M. G. Krakovyak, L. V. Kukhareva, Vysokomol. Soedin., **2**, 1817 (1960); Polym. Sci., USSR (Engl. Transl.), **3**, 494 (1962).
1188. B. Tieke, G. Chapuis, J. Polym. Sci., Polym. Chem. Ed., **22**, 2895 (1984).
1189. O. V. Romankevich, S. Ya Frenkel', Vysokomol. Soedin., Ser. A, **22**, 2416 (1980); Polym. Sci. USSR (Engl. Transl.), **22**, 2647 (1980).
1190. A. V. Sidorovich, Yu. G. Baklagina, N. V. Mikhailova, L. K. Prokhorova, L. M. Shcherbakova, M. M. Koton, Vysokomol. Soedin., Ser. A, **25**, 2096 (1983); Polym. Sci. USSR (Engl. Transl.), **25**, 2433 (1983).
1191. Yu. G. Baklagina, I. S. Milevskaya, N. V. Mikhailova, A. V. Sidorovich, L. K. Prokhorova, Vysokomol. Soedin., Ser. A, **23**, 337 (1981); Polym. Sci. USSR (Engl. Transl.), **23**, 376 (1981).
1192. A. J. Lovinger, R. E. Cais, Macromolecules, **17**, 1939 (1984).
1193. F. Brisse, B. Remillard, H. Chanzy, Macromolecules, **17**, 1980 (1984).
1194. N. T. Wakelyn, Polym. Commun., **25**, 306 (1984).
1195. B. Wunderlich, Macromolecular Physics, Academic, New York, vol. 1, Table IV.1 (1973); vol. 3, Table VIII.6 (1980).
1196. V. S. Papkov, Yu. K. Godovsky, V. S. Svistunov, V. M. Litvinov, A. A. Zhdanov, J. Polym. Sci., Polym. Chem. Ed., **22**, 3617 (1984).
1197. G. Charlet, G. Delmas, J. F. Revol, R. S. J. Manley, Polymer, **25**, 1613 (1984).
1198. G. Charlet,, G. Delmas, Polymer, **25**, 1619 (1984).
1199. C. Lavallee, G. Lemay, A. Leborgne, N. Spassky, R. E. Prud'homme, Macromolecules, **17**, 2457 (1984).
1200. D. M. Koenhen, A. Bakker, L. Broens, J. W. A. van den Berg, C. A. Smolders, J. Polym. Sci., Polym. Phys. Ed., **22**, 2145 (1984).
1201. K. Inoue, S. Hoshino, J. Polym. Sci., Polym. Phys. Ed., **22**, 1969 (1984).
1202. M. Scandola, M. Pizzoli, A. Drusiani, G. Garbuglio, Eur. Polym. J., **10**, 101 (1974).
1203. B. M. Ginzburg, Ye. T. Magdalev, V. N. Volosatov, R. G. Fedorova, S. Ya. Frenkel', Vysokomol. Soedin., Ser. A, **22**, 2200 (1980); Polym. Sci. USSR (Engl. Transl.), **22**, 2410 (1980).
1204. S. Sasaki, I. Uematsu, J. Polym. Sci., Polym. Phys. Ed., **23**, 263 (1985).
1205. C. W. Dirk, T. Inabe, K. F. Schoch, Jr., T. J. Marks, J. Am. Chem. Soc., **105**, 1539 (1983); X. Zhou, T. J. Marks, S. H. Carr, J. Polym. Sci., Polym. Phys. Ed., **23**, 305 (1985).
1206. B. N. Diel, T. Inabe, J. W. Lyding, K. F. Schoch, Jr., C. R. Kannewurf, T. J. Marks, J. Am. Chem. Soc., **105**, 1551 (1983).
1207. J. J. Point, C. Coutelier, J. Polym. Sci., Polym. Phys. Ed., **23**, 231 (1985).
1208. P. Corradini, R. Napolitano, V. Petraccone, B. Pirozzi, A. Tuzi, Eur. Polym. J., **21**, 65 (1985).
1209. G. Trafara, J. Polym. Sci., Polym. Chem. Ed., **18**, 321 (1980).
1210. J.-C. Lin, M. H. Litt, G. Froyer, J. Polym. Sci., Polym. Chem. Ed., **19**, 165 (1981).
1211. D. Grenier, R. E. Prud'homme, A. Leborgne, N. Spassky, J. Polym. Sci., Polym. Chem. Ed., **19**, 1781 (1981).

1212. G.-Y. Shi, B. Huang, J.-Y. Zhang, Makromol. Chem., Rapid Commun., **5**, 573 (1984).
1213. C. R. Fincher, Jr., C.-E. Chen, A. J. Heeger, A. G. MacDiarmid, J. B. Hastings, Phys. Rev. Lett., **48**, 100 (1982).
1214. B. Lotz, F. Colonna-Cesari, F. Heitz, G. Spach, J. Mol. Biol., **106**, 915 (1976).
1215. B. Lotz, F. Heitz, G. Spach, C. R. Hebd. Seances Acad. Sci., Ser. C, **276**, 1715 (1973).
1216. P. J. Barham, A. Keller, E. L. Otun, P. A. Holmes, J. Mater. Sci., **19**, 2781 (1984).
1217. M. Kojima, J. H. Magill, Makromol. Chem., **186**, 649 (1985).
1218. H. R. Allcock, R. L. Kugel, E. G. Stroh, Inorg. Chem., **11**, 1120 (1972).
1219. J. Blackwell, A. Sarko, R. H. Marchessault, J. Mol. Biol., **42**, 379 (1969).
1220. T. L. Bluhm, P. Zugenmaier, Carbohydr. Res., **68**, 15 (1979).
1221. T. L. Bluhm, G. Rappenecker, P. Zugenmaier, Carbohydr. Res. **60**, 241 (1978).
1222. R. Minke, J. Blackwell, J. Mol. Biol., **120**, 167 (1978).
1223. R. P. Millane, A. K. Mitra, S. Arnott, J. Mol. Biol., **169**, 903 (1983).
1224. A. K. Mitra, S. Arnott, J. K. Sheehan, J. Mol. Biol., **169**, 813 (1983).
1225. J. K. Sheehan, K. H. Gardner, E. D. T. Atkins, J. Mol. Biol., **117**, 113 (1977); S. Arnott, A. K. Mitra, S. Raghunathan, ibid., **169**, 861 (1983).
1226. A. K. Mitra, S. Raghunathan, J. K. Sheehan, S. Arnott, J. Mol. Biol., **169**, 829 (1983).
1227. J. M. Guss, D. W. L. Hukins, P. J. C. Smith, W. T. Winter, S. Arnott, R. Moorhouse, D. A. Rees, J. Mol. Biol., **95**, 359 (1975).
1228. R. H. Baughman, S. L. Hsu, L. R. Anderson, G. P. Pez, A. J. Singnorelli, in W. E. Hatfield, Ed., Molecular Metals (NATO Conference Series), Plenum, New York, p. 187 (1979).
1229. C. Riekel, H. W. Hasslin, K. Menke, S. Roth, J. Chem. Phys., **77**, 4254 (1982).
1230. R. H. Baughman, N. S. Murthy, G. G. Miller, J. Chem. Phys., **79**, 515 (1983).
1231. R. H. Baughman, L. W. Shacklette, N. S. Murthy, G. G. Miller, R. L. Elsenbaumer, Mol. Cryst. Liq. Cryst., **118**, 253 (1985).
1232. J. J. Point, private communication, 14 March 1985.
1233. J. M. Jonza, R. S. Porter, Bull. Am. Phys. Soc., **30**, 249 (1985).
1234. A. J. Lovinger, D. D. Davis, F. J. Padden, Jr., Polymer, **26**, 1595 (1985).
1235. A. V. Fratini, E. M. Cross, J. F. O'Brien, W. W. Adams, J. Macromol. Sci. B: Phys., **24**, 159 (1985–1986).
1236. T. Granier, E. L. Thomas, D. R. Gagnon, F. E. Karasz, R. W. Lenz, J. Polym. Sci., Polym. Phys. Ed., **24**, 2793 (1986).
1237. G. Cojazzi, A. M. Fichera, V. Malta, R. Zannetti, Eur. Polym. J., **21**, 309 (1985).
1238. H.-G. Braun, G. Wegner, Makromol. Chem., **184**, 1103 (1983).
1239. B. Lotz, J. Mol. Biol., **87**, 169 (1974).
1240. C. Galiotis, R. J. Young, D. J. Ando, D. Bloor, Makromol. Chem., **184**, 1083 (1983).
1241. W. T. Winter, S. Arnott, D. H. Isaac, E. D. T. Atkins, J. Mol. Biol., **125**, 1 (1978).
1242. J. J. Cael, W. T. Winter, S. Arnott, J. Mol. Biol., **125**, 21 (1978).
1243. W. T. Winter, A. Sarko, Biopolymers, **13**, 1447 (1974).
1244. B. Zaslow, V. G. Murphy, A. D. French, Biopolymers, **13**, 779 (1974).
1245. T. D. Simpson, F. R. Dintzis, N. W. Taylor, Biopolymers, **11**, 2591 (1972).
1246. M. E. Hinkle, H. F. Zobel, Biopolymers, **6**, 1119 (1968).
1247. Y. Yamashita, K. Monobe, J. Polym. Sci., Part A-2, **9**, 1471 (1971).
1248. T. D. Simpson, Biopolymers, **9**, 1039 (1970).
1249. G. Honjo, M. Watanabe, Nature (London) 181, 326 (1958).
1250. I. A. Nieduszynski, E. D. T. Atkins, Biochim. Biophys. Acta, **222**, 109 (1970).
1251. S. Sasaki, E. Fukada, J. Polym. Sci., Polym. Phys. Ed., **14**, 565 (1976).
1252. A. Kuppel, H. Bittiger, E. Husemann, Kolloid, Z. Z. Polym., **250**, 623 (1972).
1253. N. E. Dweltz, J. R. Colvin, A. G. McInnes, Can. J. Chem., **46**, 1513 (1968).
1254. E. D. T. Atkins, T. C. Laurent, Biochem. J., **133**, 605 (1973).
1255. J. J. Creely, R. H. Wade, A. D. French, Text. Res. J., **48**, 37 (1978).
1256. W. T. Winter, P. J. C. Smith, S. Arnott, J. Mol. Biol., **99**, 219 (1975).
1257. W. T. Winter, S. Arnott, J. Mol. Biol., **117**, 761 (1977).
1258. K. Tanaka, J. Biochem., **83**, 325 (1978).
1259. F. Garnier, G. Tourillon, J. Y. Barraud, H. Dexpert, J. Mater. Sci., **20**, 2687 (1985).
1260. Yu. K. Godovsky, N. N. Makarova, V. S. Papkov, N. N. Kuzmin, Makromol. Chem., Rapid Commun., **6**, 443 (1985).
1261. G. Wieners, R. Weizenhofer, M. Monkenbusch, M. Stamm, G. Lieser, V. Enkelmann, G. Wegner, Makromol. Chem., Rapid Commun., **6**, 425, (1985). The authors conclude that the crystal structure consists of two mutually incommensurate sublattices of polymer chains and counterions.
1262. B. Remillard, F. Brisse, Polymer, **26**, 1029 (1985).
1263. A. V. Fratini, W. W. Adams, Abstract PD15, Am. Crystallogr. Assoc. Meeting, Stanford Univ., 18–29 August 1985, Ser., **2**, V. 13.
1264. J. B. Lando, M. K. Thakur, Synth. Met., **9**, 317 (1984). The cross-polymer has two "fiber" axes, *a* and *c*.
1265. S. Munoz-Guerra, J. M. Fernandez-Santin, A. Rodriguez-Galan, J. A. Subirana, J. Polym. Sci., Polym. Phys. Ed., **23**, 733 (1985).
1266. D. Ya. Tsvankin, V. S. Papkov, V. P. Zhukov, Yu. K. Godovsky, V. S. Svistunov, A. A. Zhdanov, J. Polym. Sci., Polym. Chem. Ed., **23**, 1043 (1985).
1267. J. R. Evans, R. A. Orwoll, S. S. Tang, J. Polym. Sci., Polym. Chem. Ed., **23**, 971 (1985).

1268. D. Day, J. B. Lando, Macromolecules, **13**, 1483 (1980); D. R. Day, J. B. Lando, J. Polym. Sci., Polym. Phys. Ed., **19**, 165 (1981).

1269. K. E. Hardenstine, G. V. S. Henderson, Jr., L. H. Sperling, C. J. Murphy, G. E. Manser, J. Polym. Sci., Polym. Phys. Ed., **23**, 1597 (1985); G. V. S. Henderson, Jr., K. E. Hardenstine, C. J. Murphy, L. H. Sperling, Polym. Mater. Sci. Eng., **53**, 801 (1985). Structural parameters quoted for Poly (3,3-bis(ethoxymethyl) oxacyclobutane) are incompatible.

1270. J. J. Creely, L. Segal, L. Loeb, J. Polym. Sci., **36**, 205 (1959). Fiber axis taken as 10.3 A where not explicitly reported.

1271. A. C. Farthing, J. Chem. Soc. 1955, 3648.

1272. M. Nagao, T. Suzuki, S. Sasaki, I. Uematsu, Polym. J. (Tokyo), **17**, 787 (1985).

1273. R. M. Briber, E. L. Thomas, J. Polym. Sci., Polym. Phys. Ed., **23**, 1915 (1985).

1274. J. N. Hay, D. J. Kemmish, J. I. Langford, A. I. M. Rae, Polym. Commun., **26**, 283 (1985).

1275. T. M. Babchinitser, L. G. Kazaryan, L. M. Tartakovskaya, N. G. Vasilenko, A. A. Zhdanov, V. V. Korshak, Polymer, **26**, 1527 (1985); I. L. Dubchak, T. M. Babchinitser, L. G. Kazaryan, L. M. Tartakovskaya, N. G. Vasilenko, A. A. Zhdanov, V. V. Korshak, Vysokomol. Soedin., Ser. A, **31**, 65 (1989); Polym. Sci. USSR (Engl. Transl.), **31**, 70 (1989).

1276. Z. Mo, K.-B. Lee, Y. B. Moon, M. Kobayashi, A. J. Heeger, F. Wudl, Macromolecules, **18**, 1972 (1985).

1277. D. Bloor, R. J. Day, D. J. Ando, M. Motevalli, Br. Polym. J., **17**, 287 (1985).

1278. J. J. Point, C. Coutelier, D. Villers, J. Phys. Chem., **90**, 3277 (1986); J. J. Point, P. Damman, Macromolecules, **24**, 2019 (1991).

1279. J. Watanabe, H. Ono, I. Uematsu, A. Abe, Macromolecules, **18**, 2141 (1985).

1280. H. Wu, B. Wang, F. Hu, S. Li, Gaofenzi Tongxun 1985, 46, Chem. Abstr., **103**, 178856s (1985).

1281. G. Wegner, Makromol. Chem., Makromol. Symp., **1**, 151 (1986).

1282. Y. Yamashita, S. Nishimura, K. Shimamura, K. Monobe, Makromol. Chem., **187**, 1757 (1986).

1283. M. M. Sokolowski, E. A. Marseglia, R. H. Friend, Polymer, **27**, 1714 (1986).

1284. H. W. Starkweather, Jr., G. A. Jones, J. Polym. Sci., Polym. Phys. Ed., **24**, 1509 (1986).

1285. U. Leute, W. Dollhopf, Colloid Polym. Sci., **261**, 299 (1983).

1286. T. Yamamoto, T. Hara, Polymer, **27**, 986 (1986).

1287. T. Tanigami, K. Yamaura, S. Matsuzawa, M. Ishikawa, K. Mizoguchi, K. Miyasaka, Polymer, **27**, 999 (1986).

1288. P. Zoller, H. W. Starkweather, Jr., G. A. Jones, J. Polym. Sci., Polym. Phys. Ed., **24**, 1451 (1986).

1289. T.Kaneda, T. Katsura, K. Nakagawa, H. Makino, M. Horio, J. Appl. Polym. Sci., **32**, 3151 (1986).

1290. M. Komakine, T. Namikawa, Y. Yamazaki, Makromol. Chem., Rapid Commun., **7**, 139 (1986).

1291. N. Ishihara, T. Seimiya, M. Kuramoto, M. Uoi, Macromolecules, **19**, 2464 (1986).

1292. J. Watanabe, H. Ono, Macromolecules, **19**, 1079 (1986).

1293. S. Sasaki, J. Polym. Sci., Polym. Phys. Ed., **24**, 2821 (1986).

1294. J. Puiggali, S. Munoz-Guerra, B. Lotz, Macromolecules, **19**, 1119 (1986).

1295. R. Oshima, J. Kumanotani, Macromolecules, **19**, 938 (1986).

1296. S. Sasaki, Polymer, **27**, 849 (1986).

1297. L.-S. Li, J. Macromol. Sci. B: Phys., **25**, 215 (1986).

1298. M. A. Gomez, J. G. Fatou, A. Bello, Eur. Polym. J., **22**, 43 (1986); *ibid.*, 661.

1299. M. A. Gomez, E. Perez, A. Bello, J. G. Fatou, Polym. Commun., **27**, 166 (1986).

1300. A. Bello, E. Perez, J. M. G. Fatou, Macromolecules, **19**, 2497 (1986).

1301. P. Zugenmaier, H. Steinmeier, Polymer, **27**, 1601 (1986). An apparent misprint for amylose triacetate II had b = 29.97 A instead of 20.97 A.

1302. Ch. Buchele, P. Zugenmaier, Colloid Polym. Sci., **258**, 768 (1980).

1303. W. A. Roughead, D. P. Miller, A. D. French, J. Macromol. Sci. B: Phys., **24**, 229 (1985–1986).

1304. P. Zugenmaier, A. Kuppel, Colloid Polym. Sci., **264**, 231 (1986).

1305. A. V. Fratini, E. M. Cross, R. B. Whitaker, W. W. Adams, Polymer, **27**, 861 (1986).

1306. S. Matsuzawa, K. Yamaura, T. Tanigami, M. Higuchi, Colloid Polym. Sci., **263**, 888 (1985).

1307. S. V. Meille, W. Porzio, G. Allegra, G. Audisio, M. Gleria, Makromol. Chem., Rapid Commun., **7**, 217 (1986); S. V. Meille, W. Porzio, A. Bolognesi, M. Gleria, *ibid.*, **8**, 43 (1987).

1308. A. J. Lovinger, F. C. Schilling, F. A. Bovey, J. M. Ziegler, Macromolecules, **19**, 2657 (1986).

1309. H. Kuzmany, J. F. Rabolt, B. L. Farmer, R. D. Miller, J. Chem. Phys., **85**, 7413 (1986).

1310. H. Yuki, Y. Taketani, J. Polym. Sci., Part B, **10**, 373 (1972).

1311. J. Puiggali, S. Munoz-Guerra, J. A. Subirana, Polymer, **28**, 209 (1987).

1312. J. M. Fernandez-Santin, J. Aymami, A. Rodriguez-Galan, S. Munoz-Guerra, J. A. Subirana, Nature, **311**, 53 (1984); J. M. Fernandez-Santin, S. Munoz-Guerra, A. Rodriguez-Galan, J. Aymami, J. Lloveras, J. A. Subirana, E. Giralt, M. Ptak, Macromolecules, **20**, 62 (1987); S. Munoz-Guerra, J. M. Fernandez-Santin, C. Alegre, J. A. Subirana, Macromolecules, **22**, 1540 (1989).

1313. J. Puiggali, S. Munoz-Guerra, J. Polym. Sci., Polym. Phys. Ed., **25**, 513 (1987).

1314. M. Kojima, H. Satake, T. Masuko, J. H. Magill, J. Mater. Sci. Lett., **6**, 775 (1987).

1315. R. Barton, Jr., Bull. Am. Phys. Soc., **32**, 701 (1987). Also, private communication 14 April 1987. The second set of constants for Poly(4,4′-methylenedicyclohexylene dodecanediamide) is included for comparison with the values for homologs.

1316. Y. Cohen, E. L. Thomas, W. W. Adams, Bull. Am. Phys. Soc., **32**, 700 (1987).

1317. S. Bruckner, S. Luzzati, W. Porzio, P. Sozzani, Macromolecules, **20**, 585 (1987).

1318. R. R. Chance, R. H. Baughman, H. Muller, C. J. Eckhardt, J. Chem. Phys., **67**, 3616 (1977).

1319. H. Ohnuma, K. Inoue, K. Se, T. Kotaka, Macromolecules, **17**, 1285 (1984). Unit cell constants refer to a projected base.

1320. M. J. Downey, G. P. Hamill, M. Rubner, D. J. Sandeman, C. S. Velazquez, Makromol. Chem., **189**, 1199 (1988).

1321. K. Scheerer, W. Wilke, Colloid Polym. Sci., **265**, 206 (1987).

1322. A. Sanchez, C. Marco, J. G. Fatou, A. Bello, Makromol. Chem., **188**, 1205 (1987).

1323. P. Robin, J. P. Pouget, R. Comes, H. W. Gibson, A. J. Epstein, J. Phys. (Orsay, Fr.), **44**, Suppl. No. 6, C3-77 (1983).

1324. G. Perego, G. Lugli, U. Pedretti, E. Cernia, J. Phys. (Orsay, Fr.), **44**, Suppl. No. 6, C3-93 (1983).

1325. R. D. Miller, B. L. Farmer, W. Fleming, R. Sooriyakumaran, J. Rabolt, J. Am. Chem. Soc., **109**, 2509 (1987).

1326. Y. Chatani, K. Yatsuyanagi, Macromolecules, **20**, 1042 (1987).

1327. K. Ogawa, F. Tanaka, J. Tamura, K. Kadowaki, K. Okamura, Macromolecules, **20**, 1172 (1987).

1328. E. Perez, A. Bello, J. G. Fatou, Makromol. Chem., **186**, 439 (1985); E. Perez, J. G. Fatou, A. Bello, Eur. Polym. J., **23**, 469 (1987).

1329. C. Brouty, P. Spinat, M. -C. Sichere, A. Whuler, Z. Kristallogr., **176**, 13 (1986).

1330. B. Lotz, A. J. Lovinger, R. E. Cais, Macromolecules, **21**, 2375 (1988); A. J. Lovinger, B. Lotz, D. D. Davis, Macromolecules, **31**, 2253 (1990).

1331. G. Charlet, G. Delmas, J. Polym. Sci., Polym. Phys. Ed., **26**, 1111 (1988).

1332. H. Yamazaki, S. Kajita, K. Kamide, Polym. J. (Tokyo), **19**, 995 (1987).

1333. Z. Zheng, T. Yamane, T. Ashida, K. Hashimoto, H. Sumitomo, Polym. J. (Tokyo), **18**, 973 (1986).

1334. P. Pradere, A. Boudet, J. Mater. Sci., **22**, 4240 (1987).

1335. N. P. Kuznetsov, M. I. Bessonov, Vysokomol. Soedin., Ser. A, **28**, 100 (1986); Polymer Sci. USSR (Engl. Transl.), **28**, 113 (1986).

1336. S. Bruckner, W. Porzio, Makromol. Chem., **189**, 961 (1988).

1337. M. Bertault, L. Toupet, J. Canceill, A. Collet, Makromol. Chem., Rapid Commun., **8**, 443 (1987).

1338. S. Sasaki, S. Takigawa, Polym. J. (Tokyo), **19**, 1081 (1987).

1339. R. P. Grasso, B. C. Perry, J. L. Koenig, J. B. Lando, Macromolecules, **22**, 1267 (1989).

1340. D. Y. Yoon, N. Masciocchi, L. E. Depero, C. Viney, W. Parrish, Macromolecules, **23**, 1793 (1990).

1341. M. Hamamouchi, R. E. Prud'homme, J. Polym. Sci., Polym. Chem. Ed., **26**, 1593 (1988).

1342. S. Z. D. Cheng, B. Wunderlich, Macromolecules, **21**, 789 (1988).

1343. Y. Chatani, S. Okamura, Polymer, **28**, 1815 (1987).

1344. H. W. Starkweather, Jr., G. A. Jones, P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **26**, 257 (1988).

1345. A. J. Lovinger, F. J. Padden, Jr., D. D. Davis, Polymer, **29**, 229 (1988).

1346. A. Sanchez, C. Marco, J. G. Fatou, A. Bello, Eur. Polym. J., **24**, 355 (1988).

1347. A. Imberty, H. Chanzy, S. Perez, A. Buleon, V. Tran, Macromolecules, **20**, 2634 (1987).

1348. Y. Shuto, J. Sugiyama, H. Harada, K. Okamura, Macromolecules, **20**, 2317 (1987).

1349. M. Kojima, J. H. Magill, Sixth Meeting on Inorganic Polymers, 26–27 Nov. 1987, Tokyo, Japan.

1350. U. Rieck, J. H. Magill, Polym. Commun., **28**, 107 (1987).

1351. J. Z. Ruan, M. H. Litt, Macromolecules, **21**, 876 (1988).

1352. Yu. S. Lipatov, A. Sh. Goikhman, L. P. Zhelevskaya, N. V. Dmitrak, V. V. Shilov, A. Yu. Bilibin, S. S. Skorokhodov, Vysokomol. Soedin., Ser. A, **29**, 1136 (1987); Polym. Sci. USSR (Eng. Transl.), **29**, 1256 (1987).

1353. A. Kobayashi, H. Kobayashi, Y. Tokura, T. Kanetake, T. Koda, J. Chem. Phys., **87**, 4962 (1987).

1354. H. Matsuda, H. Nakanishi, T. Hosomi, M. Kato, Macromolecules, **21**, 1238 (1988).

1355. E. D. T. Atkins, S. Farnell, C. Burden, W. Mackie, B. Sheldrick, Biopolymers, **27**, 1097 (1988).

1356. K. Ogawa, T. Miyanishi, T. Yui, C. Hara, T. Kiho, S. Ukai, A. Sarko, Carbohydr. Res., **148**, 115 (1986).

1357. Y. P. Hanna, J. Mater. Sci. Lett., **7**, 817 (1988).

1358. A. Bello, E. Perez, J. G. Fatou, Makromol. Chem., Makromol. Symp., **20/21**, 159 (1988).

1359. J. Puiggali, S. Munoz-Guerra, A. Rodriguez-Galan, C. Alegre, J. A. Subirana, Makromol. Chem., Makromol. Symp., **20/21**, 167 (1988).

1360. M. Kojima, T. Masuko, J. H. Magill, Makromol. Chem., Rapid Commun., **9**, 565 (1988); T. Masuko, M. Hoshi, J. Kitami, K. Yonetake, J. Mater. Sci. Lett., **7**, 1241 (1988).

1361. N. Tsutsumi, A. Tsuji, C. Horie, T. Kiyotsukuri, Eur. Polym. J., **24**, 837 (1988).

1362. S. Z. D. Cheng, R. Pan, H. S. Bu, M. -Y. Cao, B. Wunderlich, Makromol. Chem., **189**, 1579 (1988).

1363. A. Bello, S. Lazcano, C. Marco, J. G. Fatou, J. Polym. Sci., Polym. Chem. Ed., **22**, 1197 (1984); S. Lazcano, C. Marco, J. G. Fatou, A. Bello, Eur. Polym. J., **24**, 991 (1988).

1364. F. Cabassi, W. Porzio, G. Ricci, S. Bruckner, S. V. Meille, L. Porri, Makromol. Chem., **189**, 2135 (1988); S. Bruckner, S. V. Meille, W. Porzio, G. Ricci, ibid., 2145 (1988).

1365. D. Cuzin, Y. Chauvin, G. Lefebvre, Eur. Polym. J., **5**, 283 (1969).

1366. Y. Chatani, T. Irie, Polymer, **29**, 2126 (1988).

1367. K. Jamshidi, S.-H. Hyon, Y. Ikada, Polymer, **29**, 2229 (1988).

1368. B. V. Lebedev, A. A. Yevstropov, N. K. Lebedev, Ye. A. Karpova, Ye. B. Lyudvig, B. G. Belen'kaya, Vysokomol. Soedin., Ser. A, **20**, 1974 (1978); Polym. Sci. USSR (Engl. Transl.), **20**, 2219 (1978).

1369. G. Perego, G. Lugli, U. Pedretti, M. Cesari, Makromol. Chem., **189**, 2657 (1988).

1370. A. Immirzi, F. de Candia, P. Iannelli, A. Zambelli, V. Vittoria, Makromol. Chem., Rapid Commun., **9**, 761 (1988).

1371. Sh. Tuichiev, A. M. Kuznetsova, A. M. Mukhammadieva, Vysokomol. Soedin., Ser. A, **29**, 1756 (1987); Polym. Sci. USSR (Engl. Transl.), **29**, 1930 (1987).

1372. N. Ishihara, M. Kuramoto, M. Uoi, Macromolecules, **21**, 3356 (1988).

1373. S. Sasaki, Y. Iwanami, Macromolecules, **21**, 3389 (1988).

1374. S.-K. Hong, J. Blackwell, Polymer, **30**, 225 (1989).

1375. W. J. Jackson, Jr., Appl. Polym. Symp., **41**, 25 (1985).

1376. G. Gianotti, A. Capizzi, L. Del Giudice, Rubber Chem. Technol., **49**, 170 (1976).

1377. W. A. Schneider, M. F. Muller, Makromol. Chem., **189**, 2823 (1988).

1378. K. C. Yee, J. Org. Chem., **44**, 2571 (1979).

1379. V. Enkelmann, Makromol. Chem., **184**, 1945 (1983).

1380. J. P. Aime, M. Schott, M. Bertault, L. Toupet, Acta Crystallogr., Sect. B: Struct. Sci., **44**, 617 (1988).

1381. D. Bloor, L. Koski, G. C. Stevens, F. H. Preston, D. J. Ando, J. Mater. Sci., **10**, 1678 (1975).

1382. S. Bruckner, S. V. Meille, L. Malpezzi, A. Cesaro, L. Navarini, R. Tombolini, Macromolecules, **21**, 967 (1988).

1383. P. Zoller, T. A. Kehl, H. W. Starkweather, Jr., G. A. Jones, J. Polym. Sci., Part B: Polym. Phys., **27**, 993 (1989).

1384. S. Sasaki, S. Takigawa, J. Polym. Sci., Part B: Polym. Phys., **27**, 1077 (1989).

1385. M. G. Botros, J. B. Lando, S. Rickert, Bull. Am. Phys. Soc., **34**, 981 (1989).

1386. S. M. Aharoni, S. T. Correale, W. B. Hammond, G. R. Hatfield, N. S. Murthy, Macromolecules, **22**, 1137 (1989).

1387. M. Kojima, J. H. Magill, Polymer, **30**, 579 (1989).

1388. O. Greis, T. Asano, J. Xu, J. Petermann, Z. Kristallogr., **182**, 58 (1988); O. Greis, Y. Xu, T. Asano, J. Peterman, Polymer, **30**, 590 (1989).

1389. S.-K. Hong, J. Blackwell, Polymer, **30**, 780 (1989).

1390. S. V. Meille, S. Bruckner, J. B. Lando, Polymer, **30**, 786 (1989).

1391. J. Puiggali, S. Munoz-Guerra, J. Polym. Sci., Part B: Polym. Phys., **27**, 1563 (1989).

1392. Y. Chatani, Y. Yakura, T. Ishioka, Polymer, **31**, 208 (1990).

1393. D. R. Rueda, M. E. Cagiao, F. J. Balta Calleja, Polym. Bull., **21**, 635 (1989).

1394. E. Perez, A. Bello, J. G. Fatou, J. A. Rausell-Colom, Makromol. Chem., **190**, 613 (1989).

1395. L. G. Kazaryan, A. Ye. Azriel', V. A. Vasil'ev, N. G. Annenkova, N. K. Pinayeva, A. G. Chernova, Vysokomol. Soed., Ser. A, **30**, 644 (1988); Polym. Sci. USSR (Engl. Transl.), **30**, 645 (1988).

1396. F. C. Schilling, A. J. Lovinger, J. M. Ziegler, D. D. Davis, F. A. Bovey, Macromolecules, **22**, 3055 (1989).

1397. E. Schomaker, G. Challa, Macromolecules, **22**, 3337 (1989).

1398. K. Okuyama, H. Hidaka, H. Ichige, M. Osawa, Macromolecules, **22**, 3776 (1989).

1399. M. A. Masse, J. B. Schlenoff, F. E. Karasz, E. L. Thomas, J. Polym. Sci., Part B: Polym. Phys., **27**, 2045 (1989).

1400. A. J. Lovinger, D. D. Davis, F. C. Schilling, F. A. Bovey, Polym. Commun., **30**, 356 (1989).

1401. M. Kobayashi, T. Nakaoki, N. Ishihara, Macromolecules, **22**, 4377 (1989).

1402. B. Gunesin, R. J. Gustafson, J. Polym. Sci., Part A: Polym. Chem., **27**, 4439 (1989).

1403. U. Haberthur, H.-G. Elias, Makromol. Chem., **144**, 213 (1971).

1404. Y. Fujiwara, Colloid Polym. Sci., **265**, 1027 (1987).

1405. L. Dominguez, V. Enkelmann, W. H. Meyer, G. Wegner, Polymer, **30**, 2030 (1989).

1406. P. Coulter, S. Hanna, A. H. Windle, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **30**(2), 49 (1989).

1407. G. Gianotti, A. Valvassori, Polymer, **31**, 473 (1990).

1408. S. Bruckner, S. V. Meille, Nature, **340**, 455 (1989); S. V. Meille, S. Bruckner, W. Porzio, Macromolecules, **23**, 4114 (1990). Note that chains lie parallel to both diagonals in the a-b plane.

1409. J. Hirschinger, H. Miura, K. H. Gardner, A. D. English, Macromolecules, **23**, 2153 (1990).

1410. R. G. Alamo, A. Bello, J. G. Fatou, C. Obrador, J. Polym. Sci., Part B: Polym. Phys., **28**, 907 (1990).

1411. H. Marand, J. D. Hoffman, Macromolecules, **23**, 3682 (1990).

1412. M. Zhishen, Z. Hongfang, M. Qingbo, X. Xiaofu, Z. Lihua, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **31** (2), 153 (1990). The authors apparently adopted the unit cell angle values of poly(hexamethylene sebacamide).

1413. A. J. Lovinger, D. D. Davis, F. C. Schilling, F. J. Padden, Jr., F. A. Bovey, J. M. Ziegler, Macromolecules, **24**, 132 (1991).

1414. E. K. Karikari, B. L. Farmer, R. D. Miller, J. F. Rabolt, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **31** (2), 290 (1990).

1415. T. Suzuki, A. J. Kovacs, Polym. J., **1**, 82 (1970).

1416. J. P. Wesson, T. C. Williams, J. Polym. Sci., Polym. Chem. Ed., **17**, 2833 (1979).

1417. J. Arnauts, H. Berghmans, Polym. Commun., **31**, 343 (1990).

1418. Y. Chatani, Y. Fujii, Y. Shimane, J. Ijitsu, Polym. Prepr. Japan, **37**, 1179 (1988); Y. Chatani, Y. Shimane, T. Ijitsu, T. Yukinari, Polymer, **34**, 1625 (1993).

1419. G. Guerra, V. M. Vitagliano, C. De Rosa, V. Petraccone, P. Corradini, Macromolecules, **23**, 1539 (1990).

1420. H. Hu, D. L. Dorset, Macromolecules, **23**, 4604 (1990).

1421. W. R. Busing, Macromolecules, **23**, 4608 (1990).

1422. S. Kobayashi, M. Ode, T. Ikawa, Polym. Prepr. Japan (Engl. Ed.), **37** (1–4), E435 (1988).

1423. T. Irie, Y. Yakura, T. Ishioka, Y. Chatani, Polym. Prepr. Japan (Engl. Ed.), **37** (5–10), E453 (1988).

1424. Y.-S. Choi, Y. Oishi, K. Miyasaka, Polym. J., **22**, 601 (1990).

1425. S. J. Picken, J. Aerts, R. Visser, M. G. Northolt, Macromolecules, **23**, 3849 (1990).

1426. S. Z. D. Cheng, Z. Wu, A. Zhang, R. L. Johnson, H. H. Wu, Polymer, **31**, 1763 (1990).

1427. Y. Ichikawa, Y. Chatani, H. Ichige, Polym. Prepr. Japan (Engl. Ed.), **38** (5–12), E862 (1989); Polymer, **35**, 18 (1994).

1428. T. Yoneda, Y. Chatani, Polym. Prepr. Japan (Engl. Ed.), **38** (5–12), E863 (1989).

1429. D. J. Johnson, I. Karacan, P. E. P. Maj, J. G. Tomka, Polymer, **31**, 1991 (1990).

1430. Y. Chatani, H. Murayama, K. Noguchi, T. Asanuma, T. Shiomura, J. Polym. Sci., Part C: Polym. Lett., **28**, 393 (1990).

1431. R. Napolitano, Makromol. Chem., **191**, 2435 (1990). Values of the unit cell parameters, a and b, were estimated from energy calculations.

1432. Y. Chatani, Y. Fujii, T. Takayanagi, A. Honma, Polymer, **31**, 2238 (1990).

1433. R. Takagi, H. Tanaka, M. Kimura, S. Mukai, S. Nozawa, Polym. Prepr. Japan (Engl. Ed.) **39** (1–4), E497 (1990).

1434. Y. Ozaki, Y. Takahashi, M. Takse, W. R. Krigbaum, Polym. Prepr. Japan (Engl. Ed.), **39** (1–4), E497 (1990); Y. Takahashi, Y. Ozaki, M. Takase, W. R. Krigbaum, J. Polym. Sci., Part B: Polym. Phys., **31**, 1135 (1993).

1435. K. Noguchi, Y. Chatani, Polym. Prepr. Japan (Engl. Ed.), **39** (1–4), E515 (1990).

1436. Y. Shimane, Y. Inoue, T. Inagaki, T. Ishioka, Y. Chatani, T. Yukinari, Polym. Prepr. Japan (Engl. Ed.), **39** (5–11), E1508 (1990). [see also reference 1473a].

1437. A. Prieto, B. Coutin, H. Sekiguchi, J. A. Subirana, S. M. Guerra, Makromol. Chem., **191**, 2587 (1990).

1438. D. J. Blundell, J. D'Mello, Polymer, **32**, 304 (1991).

1439. D. J. Blundell, A. B. Newton, Polymer, **32**, 308 (1991).

1440. L. S. Li, P. H. Geil, Polymer, **32**, 374 (1991).

1441. A. H. Kusanagi, A. Ishimoto, Polym. Prepr. Japan (Engl. Ed.), **39** (5–11), E1509 (1990).

1442. R. E. Singler, R. A. Willingham, C. Noel, C. Friedrich, L. Bosio, E. Atkins, Macromolecules, **24**, 510 (1991).

1443. D. A. Holmer, O. A. Pickett, Jr., J. H. Saunders, J. Polym. Sci., Part A-1, **10**, 1547 (1972).

1444. V. Petraccone, B. Pirozzi, P. Corradini, G. Giunchi, Polymer, **32**, 396 (1991).

1445. H. Nishimura, T. Okano, A. Sarko, Macromolecules, **24**, 759 (1991).

1446. H. Nishimura, A. Sarko, Macromolecules, **24**, 771 (1991).

1447. A. Bolognesi, M. Catellani, S. Destri, W. Porzio, Makromol. Chem., Rapid. Commun., **12**, 9 (1991).

1448. T. Yamamoto, T. Kanbara, C. Mori, Synth. Met., **38**, 399 (1990).

1449. C. De Rosa, V. Venditto, G. Guerra, B. Pirozzi, P. Corradini, Macromolecules, **24**, 5645 (1991); C. De Rosa, V. Venditto, G. Guerra, P. Corradini, Makromol. Chem., **193**, 1351 (1992).

1450. D. McIntyre, H. Chien, R. Kozulla, Bull. Am. Phys. Soc., **40** (1), 350, 1995; Abstr. G31 7.

1451. K. Sakurai, T. Kashiwagi, T. Takahashi, J. Appl. Polym. Sci., **47**, 937 (1993).

1452. M. Galimberti, G. Balbontin, I. Camurati, G. Paganetto, Macromol. Rapid Commun., **15**, 633 (1994).

1453. C. De Rosa, F. Auriemma, A. Borriello, P. Corradini, Polymer, **36**, 4723 (1995).

1454. S. Rastogi, M. Newman, A. Keller, J. Polym. Sci., Part B: Polym. Phys., **31**, 125 (1993).

1455. C. De Rosa, V. Venditto, G. Guerra, P. Corradini Macromolecules, **25**, 6938 (1992); Polymer, **36**, 3619 (1995).

1456. S. Kopp, D. L. Dorset, B. Lotz, Bull. Am. Phys. Soc., **39** (1), 108 (1994); Abstr A'33 3.

1457. S. V. Meille, D. R. Ferro, S. Bruckner, A. J. Lovinger, F. J. Padden, Macromolecules, **27**, 2615 (1994).

1458. S. Haftka, K. Konnecke, J. Macromol. Sci. B: Phys., **30**, 319 (1991).

1459. J. Rodriguez-Arnold, A. Zhang, S. Z. D. Cheng, A. J. Lovinger, E. T. Hsieh, P. Chu, T. W. Johnson, K. G. Honnell, R. G. Geerts, S. J. Palackal, G. R. Hawley, M. B. Welch, Polymer, **35**, 1884 (1994).

1460. H. Maruyama, Y. Chatani, T. Asanuma, T. Shiomura, Polym. Prepr. Japan (Engl. Ed.), **40** (1–4), E476 (1991); Y. Chatani, H. Maruyama, T. Asanuma, T. Shiomura, J. Polym. Sci., Part B: Polym. Phys., **29**, 1649 (1991).

1461. K. Minagawa, T. Tamura, Y. Okada, K. Hatanaka, M. Matsuo, F. Yoshii, K. Makuuchi, Polym. Prepr. Japan. (Engl. Ed.), **42** (5–11), E1138 (1993).

1462. H. Kusanagi, Y. Chatani, H. Tadokoro, Polymer, **35**, 2028 (1994).

1463. J. B. Lando, M. D. Hanes, Macromolecules, **28**, 1142 (1995).

1464. A. K. Nandi, L. Mandelkern, J. Polym. Sci., Part B: Polym. Phys., **29**, 1287 (1991).

1465. X. B. Zhang, G. Van Tendeloo, J. Van Landuyt, D. Van Dijck, Macromolecules, **29**, 1554 (1996).

1466. S. Sasaki, T. Yamamoto, T. Kanbara, A. Morita, T. Yamamoto, J. Polym. Sci., Part B: Polym. Phys., **30**, 293 (1992).

1467. P. Corradini, C. De Rosa, A. Panunzi, G. Petrucci, P. Pino, Chimia, **44**, 52 (1990); C. De Rosa, P. Corradini, Eur. Polym. J., **29**, 163 (1993).

1468. M. Trifuoggi, C. De Rosa, F. Auriemma, P. Corradini, S. Bruckner, Macromolecules, **27**, 3553 (1994).

1469. S. Bruckner, C. De Rosa, P. Corradini, W. Porzio, A. Musco, Macromolecules, **29**, 1535 (1996).

1470. Z. Sun, R. L. Miller, Polymer, **34**, 1963 (1993).

1471. C. De Rosa, V. Petraccone, D. Dal Poggetto, G. Guerra, B. Pirozzi, M. L. De Lorenzo, P. Corradini, Macromolecules, **28**, 5507 (1995).

1472. M. Iuliano, G. Guerra, V. Pettracone, P. Corradini, C. Pellecchia, New Polymeric Mater., **3**, 133 (1992).

1473. Y. Shimane, U. Inoue, T. Inagaki, K. Orihara, T. Ishioka, Y. Chatani, Polym. Prepr. Japan (Engl. Ed.), **40** (5–11), E1478 (1991); Y. Chatani, T. Inagaki, Y. Shimane, H. Shikuma, Polymer, **34**, 4841 (1993).

1474. Y. Chatani, Y. Shimane, T. Inagaki, T. Ijitsu, T. Yukinari, H. Shikuma, Polymer, **34**, 1620 (1993).

1475. K. Sakaitani, K. Okuyama, H. Arikawa, Polym. Prepr. Japan (Engl. Ed.), **40** (1–4), E478 (1991); K. Okuyama, H. Sakaitani, H. Arikawa, Macromolecules, **25**, 7261 (1992).

1476. Y. Obata, L. Okuyama, S. Kurihara, Y. Kitano, T. Jinda, Macromolecules, **28**, 1547 (1995); Y. Kitano, I. Usami, Y. Obata, K. Okuyama, Polymer, **36**, 1123 (1995).

1477. J. Liu, D. Kim, F. W. Harris, S. Z. D. Cheng, J. Polym. Sci., Part B: Polym. Phys., **32**, 2705 (1994); Polymer, **35**, 4048 (1994).

1478. K. H. Gardner, R. S. Irwin, Am. Cryst. Assoc. Annual Meeting, 23–28 May 1993, Albuquerque, NM, Abstr. L003.

1479. K. Tashiro, K. Ono, Y. Minagawa, M. Kobayashi, T. Kawai, K. Yoshino, J. Polym. Sci., Part B: Polym. Phys., **29**, 1223 (1991).

1480. K. Tashiro, Y. Minagawa, M. Kobayashi, S. Morita, T. Kawai, K. Yoshino, Polym. Prepr. Japan. (Engl. Ed.), **41** (5–11), E1648 (1992).

1481. T. J. Prosa, M. J. Winokur, J. Moulton, P. Smith, A. J. Heeger, Macromolecules, **25**, 4364 (1992).

1482. W. Luzny, Acta Cryst., Sect. B: Struct. Sci., **B51**, 255 (1995).

1483. W. -P. Hsu, K. Levon, K. -S. Ho, A. S. Myerson, T. K. Kwei, Macromolecules, **26**, 1318 (1993).

1484. H. J. Fell, E. J. Samuelson, J. Mardalen, E. Bakken, P. H. J. Carlsen, Synth. Met., **69**, 301 (1995).

1485. W. Luzny, P. Rokita, R. Zareba, I. Kulszewicz-Bajer, Synth. Met., **75**, 49 (1995).

1486. J. Mardalen, H. J. Fell, E. J. Samuelson, E. Bakken, P. H. J. Carlsen, M. R. Andersson, Macromol. Chem. Phys., **196**, 553 (1995).

1487. M. J. Winokur, P. Walmsey, J. Moulton, P. Smith, A. J. Heeger, Macromolecules, **24**, 3812 (1991).

1488. A. V. Fratini, P. G. Lenhert, T. J. Resch, W. W. Adams, in: W. W. Adams, R. K. Eby, D. E. McLemore (Eds.), "The Materials Science and Engineering of Rigid-Rod Polymers," Materials Research Society Symposia Proceedings, **134**, p. 431 (1989).

1489. D. C. Martin, E. L. Thomas, Macromolecules, **24**, 2450 (1991).

1490. S. J. Bai, J. Polym. Sci., Part B: Polym. Phys., **32**, 2575 (1994).

1491. Y. Cohen, Y. Suruyama, E. L. Thomas, Macromolecules, **24**, 1161 (1991).

1492. Y. Cohen in "The Materials Science and Engineering of Rigid-Rod Polymers," W. W. Adams, R. K. Eby, D. E. McLemore, Eds., Materials Research Society Symposia Proceedings, **134**, p. 195 (1989) Y. Cohen, W. W. Adams, Polymer, **37**, 2767 (1996).

1493. S. Z. D. Cheng, D. P. Heberer, H.-S. Lien, F. W. Harris, J. Polym. Sci., Part B: Polym. Phys., **28**, 655 (1990).

1494. S. Z. D. Cheng, Z. Wu, M. Eashoo, S. L. C. Hsu, F. W. Harris, Polymer, **32**, 1803 (1991).

1495. S. Z. D. Cheng, F. E. Arnold, Jr., A. Zhang, S. L.-C. Hsu, F. W. Harris, Macromolecules, **24**, 5856 (1991).

1496. D. Hoffmann, R. Leonhardt, P. Weigel, J. Appl. Polym. Sci., **46**, 1025 (1992).

1497. S. V. Meille, S. Capelli, G. Allegra, G. Ricci, Macromol. Rapid Commun., **16**, 329 (1995).

1498. M. V. Brillhart, P. Cebe, J. Polym. Sci., Part B: Polym. Phys., **33**, 927 (1995).

1499. S. Z. D. Cheng, M. L. Mittleman, J. J. Janimak, D. Shen, T. M. Chalmers, H.-S. Lien, C. C. Tso, P. A. Gabori, F. W. Harris, Polym. International, **29**, 201 (1992).

1500. C. W. Burkhart, J. B. Lando, J. K. Stille, in: W. W. Adams, R. K. Eby, D. E. McLemore (Eds.), "The Materials Science and Engineering of Rigid-Rod Polymers", Materials Research Society Symposia Proceedings, **134**, p. 475, (1989).

1501. V. Venditto, C. De Rosa, G. Guerra, R. Napolitano, Polymer, **33**, 3547 (1992).

1502. S. V. Meille, S. Capelli, G. Ricci, Macromol. Rapid Commun., **16**, 891 (1995).

1503. D. J. Sandman, C. S. Velazquez, G. P. Hammill, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **33** (1), 284 (1992).

1504. A. Harada, H. Fujii, M. Kamachi, Macromolecules, **24**, 5504 (1991).

1505. S. Sasaki, H. Ogawa, S. Kimura, Polym. J., **23**, 1325 (1991); S. Sasaki, S. Kimura, Polym. J., **24**, 1251 (1992).

1506. J. Bella, C. Aleman, J. M. Fernandez-Santin, C. Alegre, J. A. Subirana, Macromolecules, **25**, 5225 (1992).

1507. F. Lopez-Carrasquero, M. Garcia-Alvarez, S. Munoz-Guerra, Polymer, **35**, 4502 (1994).

1508. J. J. Navas, C. Aleman, F. Lopez-Carrasquero, S. Munoz-Guerra, Macromolecules, **28**, 4487 (1995).

1509. M. A. Bellinger, A. J. Waddon, E. D. T. Atkins, W. J. MacKnight, Macromolecules, **27**, 2130 (1994).

1510. A. Kawaguchi, Polymer, **33**, 3981 (1992).

1511. M. Dosiere, Polymer, **34**, 3160 (1993).

1512. M. Bermudez, J. Puiggali, S. Munoz-Guerra, Macromolecules, **27**, 6325 (1994).

1513. L. Franco, J. Puiggali, J. Polym. Sci., Part B: Polym. Phys., **33**, 2065 (1995).

1514. E. Navarro, L. Franco, J. A. Subirana, J. Puiggali, Macromolecules, **28**, 8742 (1995).

1515. M. J. Hill, E. D. T. Atkins, Macromolecules, **28**, 604 (1995).

1516. M. Kashima, H. Kusanagi, A. Ishimoto, Polym. Prepr. Japan (Engl. Ed.), **40** (5–11), E1488 (1991).

1517. E. D. T. Atkins, M. Hill, S. K. Hong, A. Keller, S. Organ, Macromolecules, **25**, 917 (1992).

1518. R. J. Gaymans, T. E. C. van Utteren, J. W. A. van den Berge, J. Schuyer, J. Polym. Sci., Polym. Chem. Ed., **15**, 537 (1977).

1519. L.-H. Wang, F. J. Balta Calleja, T. Kanamoto, R. S. Porter, Polymer, **34**, 4688 (1993).

1520. A. Prieto, I. Iribarren, S. Munoz-Guerra, J. Mater. Sci., **28**, 4059 (1993).

1521. J. L. Greene, Jr., E. L. Huffman, R. E. Burks, Jr., W. C. Sheehan, I. A. Wolff, J. Polym. Sci., Part A-1, **5**, 391 (1967).

1522. L. Franco, E. Navarro, J. A. Subirana, J. Puiggali, Macromolecules, **27**, 4284 (1994).

1523. J. E. Aceituno, V. Tereshko, B. Lotz, J. A. Subirana, Macromolecules, **29**, 1886 (1996).

1524. A. Xenopoulos, B. Wunderlich, J. A. Subirana, Eur. Polym. J., **29**, 927 (1993).

1525. L. Penn, F. Larsen, J. Appl. Polym. Sci., **23**, 59 (1979).

1526. D. Snetivy, G. J. Vancso, G. C. Rutledge, Macromolecules, **25**, 7037 (1992).

1527. Y. Fu, B. Wunderlich, Bull. Am. Phys. Soc., **40** (1), 614 (1995), Abstr. M3534.

1528. B. H. Glomm, M. C. Grob, P. Neuenschwander, U. W. Suter, D. Snetivy, G. J. Vancso, Polym., **35**, 878 (1994).

1529. A. Heroux, F. Brisse, Am. Crystallogr. Assoc. Abstr. Ser 2, V. 23, Montreal, 23–28 July 1995, Abstr. M123, p. 125.

1530. A. Usanmaz, O. K. Melad, J. Polym. Sci., Part A: Polym. Chem., **34**, 1087 (1996).

1531. G. C. Rutledge, A. G. Oertli, Makromol. Chem., **192**, 2993 (1991).

1532. K. H. Gardner, B. S. Hsiao, H. Shih, S. Stompel, E. T. Samulski, J. Preston, Am. Crystallogr. Assoc. Abstr. Ser. 2, vol. 23, Montreal, 23–28 July 1995, Abstr. 1a.5.F, p. 50.

1533. S. Sasaki, J. Polym. Sci., Part B: Polym. Phys., **29**, 527 (1991).

1534. R. Pearce, G. R. Brown, R. H. Marchessault, Polymer, **35**, 3984 (1994).

1535. M. Scandola, G. Ceccorulli, M. Pizzoli, M. Gazzano, Macromolecules, **25**, 1405 (1992).

1536. A. M. Ritcey, J. Brisson, R. E. Prud'homme, Macromolecules, **25**, 2705 (1992).

1537. A. Cannepin, G. Champetier, A. Parisot, J. Polym. Sci., **8**, 35 (1952).

1538. R. P. Pearce, R. H. Marchessault, Macromolecules, **27**, 3869 (1994).

1539. Y. Ichikawa, J. Washiyama, Y. Moteki, K. Noguchi, K. Okuyama, Polym. J., **27**, 1264 (1995).

1540. K. J. Ihn, E. S. Yoo, S. S. Im, Macromolecules, **28**, 2460 (1995).

1541. Y. Ichikawa, J. Suzuki, J. Washiyama, Y. Moteki, K. Noguchi, K. Okuyama, Polymer, **35**, 3338 (1994).

1542. N. Jourdan, S. Deguire, F. Brisse, Macromolecules, **28**, 8086 (1995).

1543. T. Thistlethwaite, R. Jakeways, I. M. Ward, Polymer, **29**, 61 (1988).

1544. K. Inomata, K. Yamanaka, S. Sasaki, Polym. Prepr. Japan (Engl. Ed.), **41** (1–4), E609 (1992); K. Inomata, S. Sasaki, *ibid.*, **42** (1–4), E597 (1993); K. Inomata, S. Sasaki, J. Polym. Sci., Part B: Polym. Phys., **34**, 83 (1996).

1545. F. Rynikar, J. Liu, P. H. Geil, Macromol. Chem. Phys., **195**, 81 (1994).

1546. J. Liu, F. Rybnikar, A. J. East, P. H. Geil, J. Polym. Sci., Part B: Polym. Phys., **31**, 1923 (1993).

1547. J. Liu, F. Rybnikar, P. H. Geil, J. Polym. Sci., Part B: Polym. Phys., **30**, 1469 (1992). Note that the cell dimensions listed for form II were deduced from the reciprocal lattice values reported.

1548. P. Iannelli, D. Y. Yoon, W. Parrish, Macromolecules, **27**, 3295 (1994).

1549. S. Radhakrishnan, V. S. Iyer, S. Sivaram, Polymer, **35**, 3790 (1994).

1550. G. Schwartz, H. R. Kricheldorf, Macromolecules, **28**, 3911 (1995).

1551. X. Li, F. Brisse, Macromolecules, **27**, 2276 (1994).

1552. V. V. Korshak, S. V. Vinogradova, V. M. Belyakov, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk 737 (1957); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) 757 (1957).

1553. D. J. Johnson, I. Karacan, J. G. Tomka, Polymer, **34**, 1749 (1993); J. Cao, I. Karacan, J. G. Tomka, Polymer, **36**, 2133 (1995).

1554. K. Kimura, S. Endo, Y. Kato, T. Inaba, Y. Yamashita, Macromolecules, **28**, 255 (1995).

1555. X. Lu, A. H. Windle, Polymer, **36**, 451 (1995). Note that the parameters listed are for the cell projected onto a plane normal to the chain axis.

1556. F. Brisse, X. Li, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **33**, 282 (1992); X. Li, F. Brisse, Macromolecules, **27**, 7725 (1994).

1557. T. Okihara, M. Tsuji, A. Kawaguchi, K. Katayama, J. Macromol. Sci. B: Phys., **30**, 119 (1991).

1558. S. Chvalun, M. Ishaq, J. Blackwell, Bull Am. Phys. Soc., **39** (1), 633 (1994).

1559. W. P. Zhang, S. P. Gido, F. Ignatious, S. W. Kantor, R. W. Lenz, Bull. Am. Phys. Soc., **39** (1), 108 (1944).

1560. M. Carotenuto, P. Iannelli, Macromolecules, **25**, 4373 (1992).

1561. S. D. Hudson, A. J. Lovinger, M. A. Gomez, J. Lorente, C. Marco, J. G. Fatou, Macromolecules, **27**, 3357 (1994).

1562. K. H. Gardner, R. S. Irwin, W. Sweeny, Abstract D07, Am. Crystallogr. Assoc. Meeting, U. of Toledo, OH, 21–26 July 1991.

1563. T. C. Long, J. Liu, B. -L. Yuan, P. H. Geil, Bull. Am. Phys. Soc., **40** (1), 616 (1995), Abstr. M35 42.

1564. N. Yoda, A. Miyake, Bull. Chem. Soc. Japan **32**, 1120 (1959).

1565. N. Yoda, J. Polym. Sci., Part A, **1**, 1323 (1963).

1566. E. F. Izard, J. Polym. Sci., **9**, 35 (1952).

1567. Y. Saito, T. Okano, Y. Yamamoto, Polym. Prepr. Japan. (Engl. Ed.), **42** (1–4), E591 (1993).

1568. E. Delaite, J. -J. Point, P. Damman, M. Dosiere, Macromolecules, **25**, 4768 (1992).

1569. A. Chenite, F. Brisse, Macromolecules, **24**, 2221 (1991).

1570. J. J. Point, P. Damman, Macromolecules, **25**, 1184 (1992); P. Damman, J. J. Point, Polym. International, **36**, 117 (1995); L. Paternostre, P. Damman, M. Dosiere, C. Bourgaux, Macromolecules, **29**, 2046 (1996).

1571. P. Lightfoot, M. A. Mehta, P. G. Bruce, J. Mater. Chem., **2**, 379 (1992).

1572. P. Lightfoot, M. A. Mehta, P. G. Bruce, Science, **262**, 883 (1993).

1573. A. Chenite, F. Brisse, Macromolecules, **25**, 776 (1992).

1574. J. Chen, A. Zhang, M. A. Yandrasites, S. Z. D. Cheng, V. Percec, Makromol. Chem., **194**, 3135 (1993). The value of z for each polymer was obtained through a private communication from Prof. Cheng.

1575. J. S. Chung, J. Bodziuch, P. Cebe, J. Mater. Sci., **27**, 5609 (1992).

1576. Y. Takahashi, H. Matsunaga, Macromolecules, **24**, 3968 (1991).

1577. J. Sugiyama, R. Vuong, H. Chanzi, Macromolecules, **24**, 3968 (1991).

1578. S. Raymond, A. Kvick, H. Chanzy, Macromolecules, **28**, 8422 (1995).

1579. T. Yui, K. Imada, K. Okuyama, Y. Obata, K. Suzuki, K. Ogawa, Macromolecules, **27**, 7601 (1994).

1580. K. Mazeau, W. T. Winter, H. Chanzy, Macromolecules, **27**, 7606 (1994).

1581. P. Cairns, M. J. Miles, V. J. Morris, M. J. Ridout, G. J. Brownsey, W. T. Winter, Carbohydr. Res., **235**, 23 1992).

1582. T. Kusaba, Y. Obata, K. Okuyama, Polym. Prepr. (Engl. Ed.), **42** (1–4), E599 (1993).

1583. T. L. Hendrixson, R. P. Millane, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **33** (1), 329 (1992).

1584. M. Eashoo, Z. Wu, A. Zhang, D. Shen, C. Tse, F. W. Harris, S. Z. D. Cheng, K. H. Gardner, B. S. Hsiao, Macromol. Chem. Phys., **195**, 2207 (1994).

1585. J. M. Parris, R. H. Marchessault, E. J. Vandenberg, J. C. Mullis, J. Polym. Sci., Part B: Polym. Phys., **32**, 749 (1994).

1586. B. J. Lommerts, E. A. Klop, J. Aerts, J. Polym. Sci., Part B: Polym. Phys., **31**, 1319 (1993); E. A. Klop, B. J. Lommerts, J. Veurink, J. Aerts, R. R. Van Puijenbroek, J. Polym. Sci., Part B: Polym. Phys., **33**, 315 (1995); V. Grayer, B. J. Lommerts, P. Smith, B. Lotz, J. C. Wittman, Polymer, **36**, 1915 (1995).

1587. J. M. Machado, J. E. Flood, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **36** (1), 291 (1995).

1588. Y. P. Handa, J. Roovers, F. Wang, Macromolecules, **27**, 5511 (1994).

1589. A. A. Mehmet-Alkan, J. N. Hay, Polymer, **33**, 3527 (1992).

1590. K. H. Gardner, B. S. Hsiao, R. R. Matheson, Jr., B. A. Wood, Polymer, **33**, 2483 (1992).

1591. R.-M. Ho, S. Z. D. Cheng, B. S. Hsiao, K. H. Gardner, Macromolecules, **27**, 2136 (1994).

1592. R.-M. Ho, S. Z. D. Cheng, H. P. Fisher, R. K. Eby, B. S. Hsiao, K. H. Gardner, Macromolecules, **27**, 5787 (1994).

1593. D. J. Blundell, Polymer, **33**, 3773 (1992).

1594. D. J. Blundell, J. J. Liggat, A. Flory, Polymer, **33**, 2475 (1992).

1595. H. M. Colquhoun, C. C. Dudman, D. J. Blundell, A. Bunn, P. D. Mackenzie, P. T. McGrail, E. Nield, J. B. Rose, D. J. Williams, Macromolecules, **26**, 107 (1993).

1596. D. J. Blundell, V. Bayon, Polymer, **34**, 1354 (1993).

1597. R.-M. Ho, S. Z. D. Cheng, R.-M. Ho, B. S. Hsiao, K. H. Gardner, Macromol. Chem. Phys., **197**, 185 (1996).

1598. H. Hozumi, K. Yonetake, T. Masuko, Polym. Prepr. Japan (Engl. Ed.), **40** (1–4), E477 (1991); H. Hozumi, C. Kohama, Y. Yonetake, T. Masuko, J. Mater. Sci. Lett., **10**, 1187 (1991).

1599. M. Kojima, J. H. Magill, Makromol. Chem., **193**, 2731 (1992).

1600. T. Masuko, T. Miyata, H. Hozumi, K. Yonetake, J. Mater. Sci. Lett., **11**, 157 (1992); T. Miyata, K. Yonetake, T. Masuko, J. Mater. Sci., **29**, 2467 (1994).

1601. M. Cypryk, K. Matyjaszewski, M. Kojima, J. H. Magill, Makromol. Chem., Rapid Commun., **13**, 39 (1992).

1602. M. Kojima, J. H. Magill, M. Cypryk, M. L. White, U. Franz, K. Matyjaszewski, Macromol. Chem. Phys., **195**, 1823 (1994).

1603. S. V. Meille, A. R. Poletti, M. C. Gallazzi, M. Gleria, S. Bruckner, Polymer, **33**, 2364 (1992).

1604. T. Anzai, H. Nakamura, T. Miyata, K. Yonetake, T. Masuko, J. Mater. Sci. Lett., **12**, 1803 (1993).

1605. T. Miyata, T. Masuko, M. Kojima, J. H. Magill, Macromol. Chem. Phys., **195**, 253 (1994).

1606. H. Nakamura, T. Miyata, K. Yonetake, T. Masuko, M. Kojima, Polym. Prepr. Japan (Engl. Ed.), **41** (1–4), E622 (1992); Macromol. Rapid Commun., **16**, 189 (1995).

1607. M. Kojima, S. G. Young, J. H. Magill, Polymer, **33**, 4538 (1992).

1608. S. A. Kuptsov, L. K. Golova, A. N. Zadorin, N. P. Kruchinin, N. V. Vasil'eva, G. Ya. Rudinskaya, D. R. Tur, E. M. Antipov, Vysokomol. Soedin., Ser. A, **35**, 541 (1993) - Polym. Sci., **35**, 635 (1993).

1609. D. Ya Tsvankin, M. V. Gerasimov, V. P. Zhukov, I. I. Dubovik, D. R. Tur, V. S. Papkov, J. Polym. Sci., Part B: Polym. Phys., **30**, 851 (1992).

1610. A. N. Zadorin, E. M. Antipov, V. G. Kulichikhin, D. R. Tur, Vysokomol. Soedin., Ser. A, **35**, 676 (1993); Polym. Sci., **35**, 796 (1993).

1611. E. M. Antipov, A. N. Zadorin, S. A. Kuptsov, N. P. Tsukanova, D. R. Tur, V. G. Kulichikhin, Vysokomol. Soedin., Ser. B, **35**, 1540 (1993) - Polym. Sci., **35**, 1286 (1993).

1612. J. P. Pouget, M. E. Jozefowicz, A. J. Epstein, X. Tang, A. G. MacDiarmid, Macromolecules, **24**, 779 (1991).

1613. C. Y. Yang, P. Smith, A. J. Heeger, Y. Cao, J. E. Osterholm, Polymer, **35**, 1142 (1994).

1614. B. Klemann, R. West, J. A. Koustsky, Macromolecules, **26**, 1042 (1993).

1615. S. S. Patnaik, B. L. Farmer, Polymer, **33**, 4443 (1992).

1616. E. E. Karikari, B. L. Farmer, C. L. Hoffmann, J. F. Rabolt, Macromolecules, **27**, 7185 (1994).

1617. E. E. Karikari, A. J. Greso, B. L. Farmer, R. D. Miller, J. F. Rabolt, Macromolecules, **26**, 3937 (1993).

1618. M. Pluta, T. Pakula, M. Kryszewski, J. Kurjata, J. Chojnowski, Macromol. Chem. Phys., **196**, 1607 (1995).

1619. N. Paredes, A. Rodriguez-Galan, J. Puiggali, Polymer, **37**, 4175 (1996).

1620. J. Liu, S. Z. D. Cheng, P. H. Geil, Polymer, **37**, 1413 (1996).

1621. D. Jehnichen, J. Tobisch, P. Friedel, Doris Pospiech, Polymer, **37**, 1463 (1996).

1622. H. R. Kricheldorf, O. Struve, G. Schwarz, Polymer, **37**, 4311 (1996).

1623. T. J. Prosa, M. J. Winokur, R. D. McCullough, Macromolecules, **29**, 3654 (1996).

1624. E. Navarro, C. Aleman, J. A. Subirana, J. Puiggali, Macromolecules, **29**, 5406 (1996).

1625. J. H. Magill, J. Polym. Sci., Part A-2, **9**, 815 (1971).

1626. N. A. Jones, E. D. T. Atkins, M. J. Hill, S. J. Cooper, L. Franco, Macromolecules, **29**, 6011 (1996).

1627. J. Blackwell, C. D. Lee, J. Polym. Sci., Polym. Phys. Ed., **22**, 759 (1984).

1628. J. Watanabe, S. Okamoto, K. Satoh, K. Sakajiri, H. Furuya, A. Abe, Macromolecules, **29**, 7084 (1996).

1629. I. Iribarren, C. Aleman, J. J. Bou, S. Munoz-Guerra, Macromolecules, **29**, 4397 (1996).

1630. A. Bolognesi, W. Porzio, F. Provasoli, T. Ezquerra, Makromol. Chem., **194**, 817 (1993).

1631. Yu. A. Zubov, D. K. Polyakov, V. I. Selikhova, E. A. Pakshver, V. S. Shirets, Vysokomol. Soedin., Ser. A, **38**, 1527 (1996); Polym. Sci., Ser. A, **38**, 1004 (1996).

1632. L. A. Leites, S. S. Bukalov, M. V. Gerasimov, I. I. Dubovik, V. S. Papkov, Vysokomol. Soedin., Ser. B, **38**, 924 (1996); Polym. Sci., Ser. B, **235**, (1996).

1633. M. T. Krejchi, E. D. T. Atkins, M. J. Fournier, T. L. Mason, D. A. Tirrell, J. Macromol. Sci., Pure Appl. Chem., **A33**, 1389 (1996). Note that the c-axis value has been doubled to correspond to a repeat containing one complete octapeptide.

1634. I. Andre, K. Mazeau, I. Tvaroska, J.-L. Putaux, W. T. Winter, F. R. taravel, H. Chanzy, Macromolecules, **29**, 4626 (1996).

1635. S. S. Patnaik, A. J. Greso, B. L. Farmer, Polymer, **33**, 5115 (1992).

1636. A. Coassolo, M. Foa, D. Dainelli, R. Scordamaglia, L. Barino, L. L. Chapoy, F. Rustichelli, B. Yang, G. Torquati, Macromolecules, **24**, 1701 (1991).

1637. H. J. Fell, E. J. Samuelson, J. Mardalen, M. R. Andersson, Synth. Met., **69**, 283 (1995).

1638. A. M. Hindeleh, A. A. A. Obaid, Acta Polym., **47**, 55 (1996).

1639. J. Liu, P. H. Geil, S.-M. Huh, J.-I. Lin, Acta Polym., **47**, 290 (1996).

1640. Y.-C. Ke, Z.-J. Fang, J.-Z. Wang, Z.-W. Wu, J. Appl. Sci., **61**, 1293 (1996).

1641. A. Biswas, J. Blackwell, K. Deutscher, G. Wegner, Acta Polym., **45**, 182 (1994).

1642. B. Lotz, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Sci.), **37** (2), 430 (1996).

1643. M. Nigar, J. Blackwell, S. N. Chvalun, S. D. Seneker, H. G. Schmeltzer, Bull. Am. Phys. Soc., **41** (1), 105 (1996), Abstr. B28 11.

1644. P. H. Geil, J. Liu, Bull. Am. Phys. Soc., **41** (1), 395 (1996), Abstr. I28 9.

1645. F. Rybnikar, J. Liu, P. H. Geil, Bull. Am. Phys. Soc., **41** (1), 444 (1996), Abstr. J'32 1.

1646. R. Stern, M. Ballauff, G. Lieser, G. Wegner, Polymer, **32**, 2096 (1991).

1647. L. Cervinka, M. Ballauff, Colloid Polym. Sci., **270**, 859 (1992).

1648. M. T. Krejchi, E. D. T. Atkins, A. J. Waddon, M. J. Fournier, T. L. Mason, D. A. Tirrell, Science, **265**, 1427 (1994). Values for the c axis have been increased from those published in order to correspond to one chemical repeat along the chain.

1649. C. Booth, C. J. Devoy, D. V. Dodgson, I. H. Hillier, J. Polym. Sci., Part A-2, **8**, 519 (1970).

1650. H. L. Yakel, Jr., L. Pauling, R. B. Corey, Nature, **169**, 920 (1952).

1651. D. Grenier, A. Leborgne, N. Spassky, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **19**, 33 (1981).

1652. J. Guzman, J. G. Fatou, Eur. Polym. J., **14**, 943 (1978).

1653. T. Kajiyama, W. J. MacKnight, Polym. J., **1**, 548 (1970).

1654. V. V. Korshak, T. A. Soboleva, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk 1952, 526 - Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.), 1952, 505.

1655. R. J. Gaymans, D. K. Doeksen, S. Harkema, in: L. A. Kleintjens, P. J. Lemstra (Eds.), "Integration of Fundamental Polymer Science and Technology", Elsevier Appl. Sci., New York, 1986, p. 573.

1656. V. S. Papkov, M. V. Gerasimov, M. I. Buzin, M. N. Il'ina, L. G. Kazaryan, Vysokomol. Soedin., Ser. A, **38**, 1687 (1996); Polym. Sci., Ser. A, 1097 (1996).

1767. J. Lee, C. S. Marvel, J. Polym. Sci., Polym. Phys. Ed., **21**, 2189 (1983).

1768. T. Yasuzawa, H. Yamaguchi, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **17**, 3387 (1979).

1769. M. K. Akkapeddi, J. Polym. Sci., Polym. Chem. Ed., **17**, 2989 (1979).

1770. C. Marco, S. Lazcano, J. G. Fatou, Makromol. Chem., **191**, 1151 (1990).

1771. T. L. St. Clair, A. K. St. Clair, J. Polym. Sci., Polym. Chem. Ed., **15**, 1529 (1977).

1772. V. Percec, T. D. Shaffer, H. Nava, J. Polym. Sci., Polym. Lett. Ed., **22**, 637 (1984).

1773. G. Chen, R. W. Lenz, J. Polym. Sci., Polym. Chem. Ed., **22**, 3189 (1984).

1774. P. Corradini, N. Lanzetta, G. Maglio, R. Plaumbo, F. Riva, J. Polym. Sci., Polym. Chem. Ed., **13**, 1107 (1975).

1775. F. B. Cramer, R. G. Beaman, J. Polym. Sci., **21**, 237 (1956).

1776. N. G. Gaylord, Ed., Macromolecular Syntheses, vol. 3, Wiley, New York, 1969.

1777. E. S. W. Kong, B. Wunderlich, Makromol. Chem., **182**, 81 (1981).

1778. W. Wrasidlo, S. O. Norris, J. F. Wolfe, T. Katto, J. K. Stille, Macromolecules, **9**, 512 (1976).

1779. E. J. Vandenberg, Carl Marvel Symposium, Tucson, Ariz., Jan. 13–14, 1975, J. Polym. Sci., Polym. Chem. Ed., **13**, 2221 (1975).

1780. W. A. Lee, R. A. Rutherford, Polymer Transition Temperature Data Sheets. Fluoropolymers. 1. Carbon-Chain, RAPRA Data Sheet 8:1, May, 1973.

1781. H. J. Chang, O. Vogl, J. Polym. Sci., Polym. Chem. Ed., **15**, 1043 (1977).

1782. G. Wegner, Makromol. Chem., **134**, 219 (1970).

1783. G. Maglio, E. Musco, R. Palumbo, F. Riva, J. Polym. Sci., Polym. Lett. Ed., **12**, 129 (1974).

1784. N. Kobayashi, A. Osawa, T. Fujisawa, J. Polym. Sci., Polym. Lett. Ed., **11**, 679 (1973).

1785. R. W. Campbell, H. W. Hill, Jr., Macromolecules, **6**, 492 (1973).

1786. T. Saegusa, Y. Nagura, S. Kobayashi, Macromolecules, **6**, 495 (1973).

1787. R. G. Feasey, A. Turner-Jones, P. C. Daffurn, J. L. Freeman, Polymer, **14**, 241 (1973).

1788. J. Preston, J. Polym. Sci., Part A-1, **4**, 529 (1966).

1789. H. Jonsson, P. -E. Werner, U. W. Gedde, A. Hult, Macromolecules, **22**, 1683 (1989).

1790. V. V. Korshak, S. V. Vinogradova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk p. 1121 (1953); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) p. 995 (1953).

1791. V. V. Korshak, G. N. Chelnokova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk p. 103 (1956); Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) p. 97 (1956).

1792. F. de Candia, G. Maglio, R. Palumbo, Polym. Bull. (Berlin), **8**, 109 (1982).

1793. N. Kobayashi, A. Osawa, T. Fujisawa, J. Polym. Sci., Polym. Chem. Ed., **11**, 553 (1973).

1794. S. W. Chow, W. E. Loeb, C. E. White, J. Appl. Polym. Sci., **13**, 2325 (1969).

1796. V. V. Korshak, S. V. Vinogradova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk p. 746 (1957); (Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) p. 765 (1957)).

1797. V. V. Korshak, S. V. Vinogradova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk p. 863 (1957); (Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) p. 885 (1957)).

1798. V. V. Korshak, S. V. Vinogradova, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk p. 863 (1957); (Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) p. 889(1957)).

1799. R. Gilkey, J. R. Caldwell, J. Appl. Polym. Sci., **2**, 198 (1959).

1800. V. C. E. Burnop, K. G. Latham, Polymer, **8**, 589 (1967).

1801. Yu. V. Mitikin, Yu. N. Sazanov, G. P. Vlasov, M. M. Koton, Vysokomol. Soedin., **2**, 716 (1960); Polym. Sci. USSR (Engl. Transl.), **2**, 423 (1961).

1802. L. T. C. Lee, J. Polym. Sci., Polym. Chem. Ed., **16**, 2025 (1978).

1803. R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 521 (1960).

1804. R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 520 (1960).

1805. Y. Etienne, Ind. Plast. Mod. (Parts), **9**, 37 (1957).

1806. T. I. Shein, G. I. Kudryavtsew, G. P. Lyubimtseva, Vysokomol. Soedin., **7**, 1447 (1965); Polym. Sci. USSR (Engl. Transl.), **7**, 1603 (1965).

1807. T. W. Campbell, J. Org. Chem., **22**, 1029 (1957).

1808. F. W. Harris, A. J. Karnavas, C. N. Cucuras, S. Das, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **26** (2), 287 (1985).

1809. E. J. Vandenberg, J. Polym. Sci., **47**, 486 (1960).

1810. W. C. Sheehan, T. B. Cole, L. G. Picklesimer, J. Polym. Sci., Part A, **3**, 1443 (1965).

1811. P. Longi, Atti Accad. Nazl. Lincei, Cl. Sci. Fis., Mat. Nat., Rend., Rend., **31**, 273 (1961).

1812. C. G. Overberger, A. M. Schiller, J. Polym. Sci., **54**, S30 (1961).

1813. C. G. Overberger, E. B. Davidson, J. Polym. Sci., **62**, 23 (1962).

1814. D. R. Stevenson, J. E. Mulvaney, J. Polym. Sci., Part A-1, **10**, 2713 (1972).

1815. G. Natta, G. Dall'Asta, G. Mazzanti, A. Casale, Makromol. Chem., **58**, 217 (1962).

1816. H. K. Hall, Jr., J. Org. Chem., **28**, 3213 (1963).

1817. G. Natta, G. Mazzanti, G. F. Pregaglia, G. Pozzi, J. Polym. Sci., **58**, 1201 (1962).

1818. J. H. Magill, J. Polym. Sci., Part A, **3**, 1195 (1965).

1819. C. S. Marvel, R. R. Chambers, J. Am. Chem. Soc., **70**, 993 (1948).

1820. W. H. Carothers, F. J. van Natta, J. Am. Chem. Soc., **52**, 314 (1930).

1821. M. A. Osman, H. -G. Elias, J. Macromol. Sci. A: Chem., **5**, 805 (1971).

1822. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **54**, 1569 (1932).

1823. J. W. Hill, W. H. Carothers, J. Am. Chem. Soc., **55**, 5023 (1933).

1824. H. Komoto, Rev. Phys. Chem. Japan, **37**, 112 (1967).

1825. T. L. Ang, R. C. Chang, E. R. Santee, Jr., H. J. Harwood, J. Polym. Sci., Polym. Lett. Ed., **10**, 791 (1972).

1828. R. S. Muromova, A. A. Strepikheyev, Z. A. Rogovin, Vysokomol. Soedin., **5**, 1096 (1963); Polym. Sci. USSR (Engl. Transl.), **5**, 157 (1964).

1829. H. Zahn, G. B. Gleitsmann, Makromol. Chem., **63**, 129 (1963).

1830. S. G. Matsoyan, L. M. Akopyan, Vysokomol. Soedin., **3**, 1311 (1961); Polym. Sci. USSR (Engl. Transl.), **3**, 915 (1962).

1831. V. V. Korshak, S. L. Sosin, V. P. Alekseyeva, Vysokomol. Soedin., **3**, 1332 (1961) - Polym. Sci. USSR (Engl. Transl.), **3**, 925 (1962).

1832. Y. Imai, M. Kajiyama, S. Ogata, M. Kakimoto, J. Polym. Sci., Polym. Chem. Ed., **22**, 3183 (1984).

1833. N. Ogata, K. Sanui, K. Shiraishi, Kobunshi Kagaku, **30**, 51 (1973).

1834. S.-C. Chu, E. M. Pearce, J. Polym. Sci., Polym. Lett. Ed., **14**, 375 (1976).

1835. C. W. Stephens, J. Polym. Sci., **40**, 359 (1959).

1836. C. G. Overgerger, S. Ozaki, D. M. Braunstein, Makromol. Chem., **93**, 13 (1966).

1837. N. Kobayashi, T. Fujisawa, J. Polym. Sci., Polym. Chem. Ed., **11**, 545 (1973).

1838. D. J. Lyman, S. L. Jung, J. Polym. Sci., **40**, 407 (1959).

1839. R. J. Gaymans, T. E. C. van Utteren, J. W. A. van den Berge, J. Schuyer, J. Polym. Sci., Polym. Chem. Ed., **15**, 537 (1977).

1840. J. W. Hill, J. Am. Chem. Soc., **52**, 4110 (1930).

1841. O. B. Edgar, R. Hill, J. Polym. Sci., **8**, 1 (1952).

1842. W. J. Middleton, H. W. Jacobson, R. E. Putnam, H. C. Walter, D. G. Pye, W. H. Sharkey, J. Polym. Sci., Part A, **3**, 4115 (1965); A. L. Barney, J. M. Bruce, Jr., J. N. Coker, H. W. Jacobson, W. H. Sharkey, ibid., **4**, 2617 (1966).

1843. P. R. Thomas, G. J. Tyler, T. E. Edwards, A. T. Radcliffe, R. C. P. Cubbon, Polymer, **5**, 525 (1964).

1844. W. H. Carothers, G. L. Dorough, F. J. van Natta, J. Am. Chem. Soc., **54**, 761 (1932).

1845. E. Wellisch, E. Gipstein, O. J. Sweeting, J. Polym. Sci., Part B, **2**, 35 (1964).

1846. D. A. Lutz, L. P. Witnauer, J. Polym. Sci., Part B, **2**, 31 (1964).

1847. K. Saotome, K. Sato, J. Polym. Sci., Part A-1, **4**, 1303 (1966).

1848. V. L. Bell, J. Polym. Sci., Part A, **2**, 5291 (1964).

1849. F. Shepherd, H. J. Harwood, J. Polym. Sci., Polym. Lett. Ed., **10**, 799 (1972).

1850. W. Marconi, S. Cesca, G. D. Fortuna, J. Polym. Sci., Part B, **2**, 301 (1964).

1851. K. Matsuzaki, T. Okamoto, A. Ishida, H. Sobue, J. Polym. Sci., Part A, **2**, 1105 (1964).

1852. B. Ke, A. W. Sisko, J. Polym. Sci., **50**, 87 (1961).

1853. K. Saotome, H. Komoto, J. Polym. Sci., Part A-1, **4**, 1475 (1966).

1854. A. Conix, J. Polym. Sci., **29**, 343 (1958).

1855. N. Yoda, Makromol. Chem., **32**, 1 (1959).

1856. N. Yoda, Kobunshi Kagaku, **19**, 495 (1962).

1857. N. Yoda, Kobunshi Kagaku, **19**, 553 (1962).

1858. P. Barnaba, D. Cordischi, M. Lenzi, A. Mele, Chim. Ind. (Milan), **47**, 1060 (1965).

1859. J. Pellon, J. Polym. Sci., Part A, **1**, 3561 (1963).

1860. R. Oda, T. Shono, A. Oku, H. Takao, Makromol. Chem., **67**, 124 (1963).

1861. K. Murayama, S. Morimoto, Chem. Ind. (London) p. 402 (1967).

1862. M. Hasegawa, T. Unishi, J. Polym. Sci., Part B, **2**, 237 (1964).

1863. T. Kagiya, H. Kishimoto, S. Narisawa, K. Fukui, J. Polym. Sci., Part A, **3**, 145 (1965).

1864. G. Natta, G. Dall'Asta, G. Mazzanti, G. Motroni, Makromol. Chem., **69**, 163 (1963).

1865. T. Shono, T. Morikawa, R. -I. Okayama, R. Oda, Makromol. Chem., **81**, 142 (1965).

1866. D. A. Holmer, O. A. Pickett, Jr., J. H. Saunders, J. Polym. Sci., Part A-1, **10**, 1547 (1972).

1867. C. G. Overberger, J. E. Mulvaney, J. Am. Chem. Soc., **81**, 4697 (1959).

1868. P. W. Morgan, Condensation Polymers, by Interfacial and Solution Methods (Polymer Reviews, vol. 10), Interscience, New York, 1965 (pp. 243, 366, 418, 424, and 427).

1869. A. Leoni, M. Guaita, G. Saini, Chim. Ind. (Milan), **47**, 373 (1965).

1870. O. Ya. Fedotova, M. L. Kerber, I. P. Losev, Vysokomol. Soedin., **6**, 452 (1964); Polym. Sci. USSR (Engl. Transl.), **6**, 502 (1964).

1871. H. Sawada, A. Yasue, Kogyo Kagaku Zasshi, **67**, 1444 (1964), through Chem. Abstr., **62**, 10525b (1965).

1872. H. Sawada, A. Yasue, Kogyo Kagaku Zasshi, **67**, 1449 (1964), through Chem. Abstr., **62**, 10525f (1965).

1873. K. Tanikawa, S. Kusabayashi, H. Hirata, H. Mikawa, J. Polym. Sci., Part B, **6**, 275 (1968).

1874. H. Miyake, K. Hayashi, S. Okamura, J. Polym. Sci., Part A, **3**, 2731 (1965).

1875. F. Danusso, P. Ferruti, G. Ferroni, Chim. Ind. (Milan), **49**, 453 (1967); F. Danusso, P. Ferruti, Polymer, **11**, 88 (1970).

1876. J. V. Crivello, J. Polym. Sci., Polym. Chem. Ed., **11**, 1185 (1973).

1877. K. Saotome, H. Komoto, J. Polym. Sci., Part A-1, **5**, 107 (1967).

1878. F. R. Prince, E. A. Turi, E. M. Pearce, J. Polym. Sci., Part A-1, **10**, 465 (1972).

1879. N. D. Field, J. A. Kieras, A. E. Borchert, J. Polym. Sci., Part A-1, **5**, 2179 (1967).

1880. T. Higashimura, S. Kusudo, Y. Ohsumi, A. Mizote, S. Okamura, J. Polym. Sci., Part A-1, **6**, 2511 (1968); A. Mizote, T. Higashimura, S. Okamura, Pure Appl. Chem., **16**, 457 (1968).

1881. A. A. Nishimura, Polymer, **8**, 446 (1967).

1882. V. C. E. Burnop, K. G. Latham, Polymer, **8**, 589 (1967).

1883. K. Saotome, R. C. Schulz, Makromol. Chem., **109**, 239 (1967).

1884. M. M. Koton, I. V. Andreeva, A. I. Turbina, V. G. Sinyauskii, Dokl. Akad. Nauk SSSR, **159**, 602 (1964).

1885. V. B. Luskchik, S. S. Skorokhodov, Vysokomol. Soedin., Ser. B, **9**, 797 (1967).

1886. F. J. Serna, J. de Abajo, J. G. de la Campa, J. Appl. Polym. Sci., **30**, 61 (1985).

1888. W. S. Port, J. E. Hansen, E. F. Jordan, Jr., T. T. Dietz, D. Swern, J. Polym. Sci., **7**, 207 (1951).

1889. Y. Kishimoto, K. Sanui, N. Ogata, Polym. J., **4**, 18 (1973).

1890. V. S. Foldi, W. Sweeney, Makromol. Chem., **72**, 208 (1964).

1891. Y. Iwakura, K. Hayashi, S. Kang, K. Inagaki, Makromol. Chem., **95**, 205 (1966).

1892. H. V. Boenig, N. Walker, E. H. Myers, J. Appl. Polym. Sci., **5**, 384 (1961).

1893. D. Sek, Eur. Polym. J., **20**, 805 (1984).

1894. W. J. Peppel, J. Polym. Sci., **51**, S64 (1961).

1895. F. Manescalchi, M. Pizzoli, A. Drusiani, F. Zanetti, Makromol. Chem., **178**, 863 (1977).

1896. L. Porri, D. Pini, Chim. Ind. (Milan), **55**, 196 (1973).

1897. Y. Iwakura, M. Sakamoto, J. Polym. Sci., **47**, 277 (1960); Y. Iwakura, M. Sakamoto, Y. Awata, *ibid.*, Part A, **2**, 881 (1964).

1898. F. A. Bovey, J. F. Abere, G. B. Rathmann, C. L. Sandberg, J. Polym. Sci., **15**, 520 (1955).

1899. J. L. Greene, Jr., E. L. Huffman, R. E. Burks, Jr., W. C. Sheehan, I. A. Wolff, J. Polym. Sci., Part A-1, **5**, 391 (1967).

1900. J. Lal, G. S. Trick, J. Polym. Sci., Part A, **2**, 4559 (1964).

1901. M. Levine, S. C. Temin, J. Polym. Sci., **28**, 179 (1958).

1902. N. Ogata, T. Asahara, S. Tohyama, J. Polym. Sci., Part A-1, **4**, 1359 (1966).

1904. D. A. Holmer, J. Polym. Sci., Part A-1, **6**, 3177 (1968).

1905. H. Hopff, A. Krieger, Makromol. Chem., **47**, 93 (1961).

1906. E. F. Jordan, Jr., D. W. Feldeisen, A. N. Wrigley, J. Polym. Sci., Part A-1, **9**, 1835 (1971). (Side chain crystallization only).

1907. T. W. Campbell, K. Z. Smeltz, J. Org. Chem., **28**, 2069 (1963).

1908. P. Johncock, M. A. Hewins, A. V. Cunliffe, J. Polym. Sci., Polym. Chem. Ed., **14**, 365 (1976).

1909. M. Goodman, S.-C. Chen, Macromolecules, **5**, 625 (1971).

1910. W. C. Wooten, H. W. Coover, J. Polym. Sci., **37**, 560 (1959).

1911. S. W. Shalaby, E. A. Turi, P. J. Harget, J. Polym. Sci., Polym. Chem. Ed., **14**, 2407 (1976).

1912. A. S. Carpenter, J. Soc. Dyers Colour., **65**, 469 (1949), through S. W. Shalaby et al, J. Polym. Sci., Polym. Chem. Ed., **14**, 2407 (1976).

1913. P. M. Cachia, Ann. Chim. Paris, **4**, 5 (1959), through S. W. Shalaby et al, J. Polym. Sci., Polym. Chem. Ed., **14**, 2407 (1976).

1914. L. H. Bock, J. K. Anderson, J. Polym. Sci., **17**, 553 (1955).

1915. W. F. Christopher, D. W. Fox, Polycarbonates, Reinhold, New York, 1962.

1916. A. Noshay, C. C. Price, J. Polym. Sci., **54**, 533 (1961).

1917. J. K. Stille, J. A. Empen, J. Polym. Sci., Part A-1, **5**, 273 (1967).

1918. V. V. Korshak, S. V. Vinogradova, V. M. Belyakov, Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) 1957, 1029, through C. J. Kibler et al, J. Polym. Sci., Part A, **2**, 2115 (1964).

1919. V. V. Korshak, S. V. Vinogradova, Bull. Acad. Sci. USSR, Div. Chem. Sci. (Engl. Transl.) 1959, 138, through C. J. Kibler et al, J. Polym. Sci., Part A, **2**, 2115 (1964).

1920. G. Natta, P. Longi, V. Nordio, Makromol. Chem., **83**, 161 (1965).

1921. T. Unishi, M. Hasegawa, J. Polym. Sci., Part A, **3**, 3191 (1965).

1922. H. G. Buhrer, H.-G. Elias, in N. A. J. Platzer, Ed., Polymerization Reactions and New Polymers (Adv. Chem. Ser. No. 129), Am. Chem. Soc., Washington, 1973, p. 105.

1923. N. G. Gaylord, Polyethers, Part III, Interscience, New York, 1962.

1925. G. Dall'Asta, P. Meneghini, U. Gennard, Makromol. Chem., **154**, 279 (1972).

1926. H. Komoto, F. Hayano, T. Takami, S. Yamato, J. Polym. Sci., Part A-1, **9**, 2983 (1971).

1927. G. Belonovskaya, Zh. Tchernova, B. Dolgoplosk, Eur. Polym. J., **8**, 35 (1972).

1928. G. P. Blackburn, B. J. tighe, J. Polym. Sci., Part A-1, **8**, 3591 (1970).

1929. S. W. Shalaby, R. J. Fredericks, E. M. Pearce, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **13**, 316 (1971).

1930. Y. Iwakura, K. Iwata, S. Matsuo, A. Tohara, Makromol. Chem., **146**, 21 (1971).

1931. V. Guidotti, M. Russo, L. Mortillaro, Makromol. Chem., **147**, 111 (1971).

1932. A. D. Bliss, W. K. Cline, C. E. Hamilton, O. J. Sweeting, J. Org. Chem., **28**, 3537 (1963).

1933. E. V. Gouinlock, Jr., C. J. Verbanic, G. C. Schweiker, J. Appl. Polym. Sci., **1**, 361 (1959).

1934. N. Kobayashi, T. Fujisawa, J. Polym. Sci., Polym. Chem. Ed., **10**, 3165 (1972).

1935. T. W. Campbell, V. S. Foldi, J. Org. Chem., **26**, 4654 (1961).

1936. N. Kobayashi, T. Fujisawa, J. Polym. Sci., Polym. Chem. Ed., **10**, 3317 (1972).

1937. R. W. H. Berry, R. J. Mazza, Polymer, **14**, 172 (1973).

1938. H. Hirukawa, Y. Imai, J. Polym. Sci., Polym. Lett. Ed., **11**, 271 (1973).

1939. S. Nishizaki, S. Eto, T. Moriwaki, Kogyo Kagaku Zasshi, **71**, 265 (1968).

1940. R. J. Gaymans, J. Polym. Sci., Polym. Chem. Ed., **23**, 1599 (1985).

1942. H. Batzer, Makromol. Chem., **10**, 13 (1953).

1943. H.-G. Elias, J.-H. Tsao, J. A. Palacios, Makromol. Chem., **186**, 893 (1985).

1944. E. J. Vandenberg, J. Polym. Sci., Polym. Chem. Ed., **23**, 915 (1985).

1945. E. Riande, J. Guzman, E. Siaz, Polymer, **22**, 465 (1981).

1946. K. Onder, A. T. Chen, Polym. Eng. Sci., **25**, 942 (1985).

Glass Transition Temperatures of Polymers

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A. INTRODUCTION

Amorphous (noncrystalline) polymeric solids are either glasses or rubbers. A glassy polymer lacks long range order, and is below the temperature at which molecular motions take place on the time scale of the experiment. A rubbery polymer is above the temperature at which molecular motions take place on the time scale of the experiment. The glass transition temperature, T_g, is the critical temperature that separates glassy behavior from rubbery behavior. Many amorphous solids, including polymers, organic liquids, biomaterials, some metals and alloys, and inorganic oxide glasses, exhibit glass transition temperatures.

The dramatic change in the local movement of polymer chains at T_g leads to large changes in a host of physical properties. These properties include density, specific heat, mechanical modulus, mechanical energy absorption, dielectric coefficients, acoustical properties, viscosity, and the rate of gas or liquid diffusion through the polymer, to name a few. Any of these properties can be used, at least in a crude manner, to determine T_g.

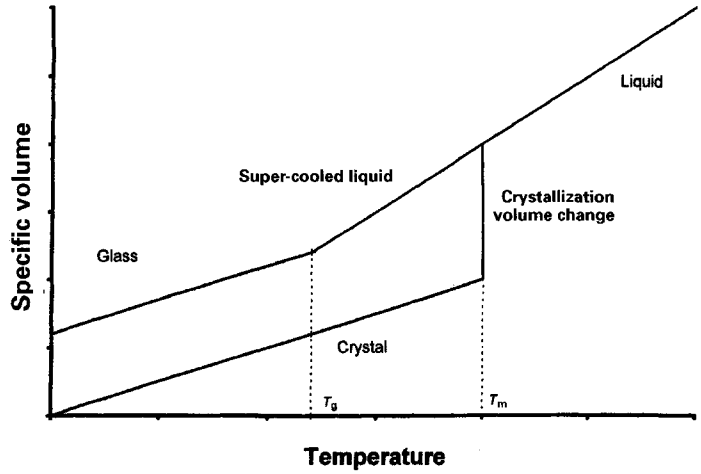

Figure 1.

1. Example of a Property Change at T_g

A classic method for determining T_g is dilatometry, in which the specific volume of the polymer is measured as a function of temperature. Figure 1 compares the specific volume vs. temperature curves for two idealized samples: a 100% amorphous material and a 100% crystalline material. Both materials follow the same path as their liquids are cooled from the melt. At the melting temperature, T_m, the crystallizing material orders into its crystal habit and exhibits a discontinuous decrease in specific volume. Below T_m, this solid has a reduction in specific volume with decreasing temperature, but the slope of this line, the volumetric thermal expansion coefficient, is less than the slope of the liquid line.

The amorphous material does not crystallize as the temperature falls below T_m, but continues to contract with a thermal expansion coefficient similar to the molten liquid: it acts as a supercooled liquid. At T_g, large-scale molecular motion (often considered to be the movement of 20–50 carbon atoms along a chain) becomes greatly reduced, the chain segments no longer rearrange rapidly in experimental time, and further cooling does not result in a similar volume change. Below T_g, the thermal expansion coefficient is similar to that of the crystalline solid. T_g can be defined as the temperature at the intersection of the two line segments. The specific volume is not discontinuous at T_g, as would be the case for the melting point, but the slope dV/dT, is. Therefore, T_g is sometimes referred to as a second-order transition, as compared to the melting point, which is a first-order transition. Both the primary property, specific volume, and its differential, the volumetric thermal expansion coefficient, change at T_g and can be used to identify the event.

Most polymer samples are either completely amorphous or partially crystalline. Figure 2 shows the specific volume vs. temperature curve for a semicrystalline polymer. The crystallizable fraction becomes ordered as it cools to T_m. Below the melting point, the amorphous material continues to contract as though it were a subcooled liquid, while the crystalline portion has a smaller thermal

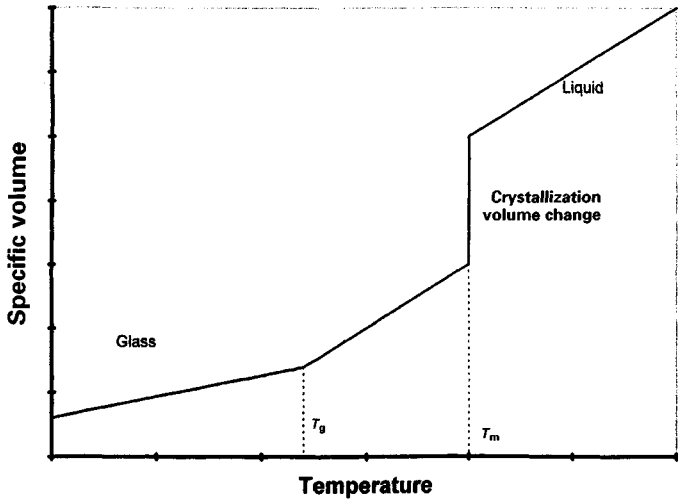

Figure 2.

expansion coefficient and contributes less to the volumetric decrease. Rapid amorphous phase motion stops at the glass transition temperature, and both solid components have similar thermal expansion coefficients below this temperature.

2. T_g: A "Non-Equilibrium" Transition

The glass transition temperature is not a fundamental thermodynamic property like the melting point because it is not thermodynamically stable, it is not defined by state variables, and its measurement is highly dependent on the time scale of the experiment used to determine its value. For example, experiments in which temperature or a deforming force is changed rapidly give higher T_g values than experiments in which conditions are changed slowly. Sample age and history affect the measurements as well.

The thermodynamic transition issue has been studied by a number of researchers. Staverman (1210) and Breuer and Rehage (1211) have concluded that T_g is not a true second-order transition because the glassy state is not completely defined by the normal state variables, p, V and T. Ehrenfest (1212) derived the following relationship that should hold for a true second-order transition:

$$\frac{dT_g}{dp} = \frac{T_g V(T_g)\Delta\alpha}{\Delta C_p} = \frac{\Delta\kappa}{\Delta\alpha}$$

where T_g is the glass transition temperature, p is the pressure, $V(T_g)$ is the molar volume at T_g, $\Delta\alpha$ is the thermal expansion coefficient difference (rubber to glass) at T_g, ΔC_p is the molar heat capacity difference, and $\Delta\kappa$ is the compressibility difference. Available data for a number of polymer systems show large deviations from the above equation for well-studied polymers, including polystyrene (1211), poly(isobutylene), poly(vinyl acetate), poly(vinyl chloride) and poly(methyl methacrylate) (1213,1214). The glass transition temperature is thought to be a kinetically controlled phenomenon (1215–1218).

B. T_g MEASUREMENT METHODS

The classical method of mercury dilatometer requires moderate amounts of sample, and time to permit thermal equilibrium to be achieved. Instruments that are easier to use and require only milligram quantities of sample are the thermal mechanical analyzer (TMA), the dynamic mechanical analyzer (DMA), and the differential scanning calorimeter (DSC). A TMA deforms a sample under a static load as the temperature is changed. At very low loads, it measures the specific volume change of the sample. A DMA measures the response of the material to an oscillating deformation. A DSC measures the change in sample enthalpy with time, dH/dt, for a known temperature program, dT/dt. The ratio of these two quantities gives dH/dT (also the specific heat capacity), which is a derivative measurement for T_g. (Experimental heat capacity curves of amorphous and semicrystalline polymers are given in the Heat Capacity chapter of this Handbook.) DSC is the most commonly used method for measuring the glass transition temperature of non-crosslinked polymers. All of these instruments can be interfaced to computers for data acquisition, test control, data reduction, and analysis.

The interpretation of T_g data can be controversial (1069–1071). The rate at which the sample is cooled or heated will affect the result. If the specific volume experiment of Fig. 2 was repeated for various cooling rates, T_g would occur at lower temperatures as the cooling rate is lowered, the transition region would become sharper, and a denser polymer would be produced. When a sample is allowed to remain below its glass transition temperature over a long period of time (aging), its density will increase. This process is called annealing; the rate of densification increases with annealing temperature.

1. Data Interpretation

The previous history of a sample will affect the measurement of T_g, unless the material's "memory" is erased by heating it well above the transition. The sample is heated well above its expected T_g, cooled to the original temperature, and reheated along a temperature profile to find the glass transition temperature of the material with minimal history. Some information can be inferred from the first heat, including residual stresses in the material, the presence of low-boiling diluents, etc.

The process of heating a polymer solid to find T_g often gives a curve similar to Fig. 1 and will produce a melt with the same density as that of the cooling experiment. However, in heating experiments, chain motions dominate the change in properties, and sudden property changes can occur near the transition temperature. At temperatures much higher than T_g, the property response once again becomes linear with temperature. In DSC measurements, sudden expansions can lead to an endothermic peak past the T_g. This endothermic peak has also been called enthalpy overshoot, and is thought to relate to the difference between the rate of temperature increase and the rate of increase of

chain mobility around T_g. The size of this overshoot depends on the annealing of the sample and the heating rate.

In a direct property measurement experiment (Fig. 1 is an example), it is common practice to ignore the kinetic effects of the transition region and to define T_g as the intersection of the two straight lines. For this case, the same T_g is found both on heating and cooling at a given polymer density. In an experiment that measures the differential of a property (DSC for example), integrating the curve gives the change in enthalpy from before to after the transition region, to which the simple intersection method of Fig. 1 can be applied, giving a T_g that is independent of heating rate (1079–1081). This method is well-known with inorganic glasses.

In DSC experiments, T_g is usually determined either by the onset point, the intersection of the initial straight line and the transition region straight line, or by the midpoint of the transition region (inflection point). The integration method's T_g is the same as the midpoint T_g if there is no endotherm, and is closer to the onset T_g if there is an endotherm. Moreover, it is common to measure T_g upon heating without any mention of previous thermal history or the heating rate. Because of these practices, the T_g values reported in this article should be considered as approximate by the reader. Authors reporting T_g should describe the thermal history of their measurements and also consider the integration method of determining T_g.

The classical methods of T_g measurement discussed previously can be quite insensitive for some polymers, e.g., highly crosslinked polymers. A penetration probe, on the other hand, is very sensitive and can determine T_g even on a thin film. But this method depends on the force of the penetrator and does not measure T_g in a fundamental way. Softening temperature methods are popular and are part of various ASTM standards.

2. Oscillating Load Methods

The measurement of the material response to an oscillating load gives a direct measure of the sample's modulus. The measurement of the lag of the material response (phase shift) to the mechanical forcing function gives a measure of the damping, or loss factor, of the sample. This last quantity is called the tan δ peak value. Dielectric thermal analysis is done by using an oscillating electric field as the forcing function. At T_g, there is often an order of magnitude change in the modulus and a large maximum in tan δ. This α-transition is distinguished from other thermal transitions (β, γ, etc.) by being the most dominant and occurring at the highest temperature. An onset point, a midpoint, or a change in modulus by a certain amount have all been used to mark the glass transition temperature. The tan δ maximum clearly specifies a thermal transition. However, it is necessary to independently determine that it is a glass transition and not another of the many transitions that appear in oscillating deformation methods. A classical T_g measured at a low programmed temperature rate (1 deg/min) may correspond to a tan δ measurement at a low frequency of measurement (1 Hz). The "loading rate" of a DMA experiment is the oscillating frequency, and T_g increases with frequency (analogous to the increase in T_g with heating rate in a DSC experiment). It is possible to measure a tan δ maximum at constant temperature while the frequency is being scanned and there is an inverse relationship between temperature and frequency. Even at a reasonably low frequency (10 Hz), a tan δ T_g can easily be 20–30°C greater than a classical one.

C. OTHER FACTORS AFFECTING T_g

Though the factors that govern T_g have been known for some years, there is still a wide variation in values for particular polymers. Polymer T_g's are sensitive to parameters which may or may not have been evaluated by the authors. Published values should be reviewed considering all the factors which affect T_g. The main factors affecting T_g values are polymer structure, sample crystallinity, diluent types and concentrations, molecular weight distributions, previous thermal history of the sample, and system pressure. More detailed treatments are given in reviews (6,48,49,1241–1249).

1. Structure

Within families of similar polymers, increasing chain stiffness and interchain cohesion increase the glass transition temperature. Copolymers may have one or multiple T_g's, depending on the ordering of monomers along the chain. The glass transition temperatures of copolymers may be higher, lower, or in between those of the homopolymers of their comonomers.

Most polymer structures are idealized and presumed from the characteristics of the reactants rather than proven by chemical and structural analysis. Structural uncertainties can arise both from the multiplicity of chemical reactions occurring during polymerization (especially those taken to high conversions), downstream processing, and from the structure of the reactants themselves. For example, a polymer with an asymmetric in-chain tetravalent atom can produce several stereoregular forms.

Alternatively, a polymer containing a residual double bond in the repeating unit could be in a *cis*, or *trans* conformation. Polymers of 1,3-dienes can have various combinations of *cis*, *trans*, 1,2- or 1,4-structures. Variations in these structural features can greatly effect the T_g values. Despite these uncertainties, the data are often useful if they can be regarded as pertaining to a polymer with reproducible properties.

2. Crystallinity/Crosslinking

Some workers suggest that the presence of crystallinity does not appreciably affect T_g. However, steric constraints imposed by crystalline regions on neighbouring amorphous polymer segments might result in an increase in T_g because

of the reduced mobility of these segments. In samples with crystallinity and moderate fractions of high molecular weight chains, tie molecules (long chains that have segments in more than one crystallite) provide physical crosslinking and increase T_g.

Some polymer families, such as the methacrylates and α-chloroacrylates, show high dependencies of T_g on tacticity. Intermolecular bonding can affect T_g either by increasing the cohesive energy of chain segments, or by decreasing the backbone degrees of freedom of chain segments. This last concept has been used (1240) to aid group contribution models.

As crystallinity increases, polymer T_g's may be little affected (50), may increase (51,52) (at least for isothermal crystallization (53)) or may decrease (51,54). T_g values selected in this work are the highest quoted on the sample with the lowest degree of crystallinity, other factors being equal.

In general, the presence of crosslinks in a sample increases its T_g relative to an uncrosslinked sample. This effect can be independent of the chemical composition of the crosslinking agent, caused by the restricted motion of chain segments near crosslinking sites. However, the crosslinking agent can behave similarly to a second monomer, inducing either an increase or a decrease in T_g due to the copolymer effect.

3. Diluents

Much of the variation in published T_g data is caused by the use of impure samples. Common impurities are unpolymerized monomer, low molecular weight polymer, solvents, and water. Great care should be taken to remove such impurities. Their presence in small concentrations can lead to a shift (48,98–103) in T_g of over 40°C and sometimes the occurrence of "diluent transitions", for example "water peaks". Preferred values are from publications that describe the precautions taken to exclude diluents and the residual levels of these diluents. Few references contain this information and most values should be regarded as only provisional.

4. Molecular Weight

The T_g of a homopolymer generally increases with increasing molecular weight up to a limiting value, known as the limiting or persistent T_g value (48,104,105). The reverse may hold for polymers with particular end-groups (106), or where crystallinity decreases with increasing molecular weight (107). For some polymers, T_g's are independent of molecular weight (108).

Many data, especially on condensation polymers, are for polymers of rather low molecular weight and it seems likely that higher T_g values would be obtained if higher molecular weight samples were tested. Many polymers are not properly characterized with respect to molecular weight and few have reported molecular weight distributions. In many cases, the only measure of molecular weight is a viscosity value, which itself can be very dependent on solvent–polymer interactions and, to some extent on the temperature. Usually, the highest viscosities and the highest molecular weight polymers are associated with the most reliable data.

The classical model for the effect of molecular weight on T_g is (1219–1226)

$$T_g = T_g^{\infty} - \frac{A}{\bar{M}_n}$$

This model suggests that the glass transition temperature reaches a limiting value when the number average molecular weight of the polymer is large. Cowie and Toporowski (1220) have shown that there is no further increase in T_g when the molecular weight is above a critical value, which is similar to the critical molecular weight for viscosity.

5. Thermal History

We have previously discussed in detail how thermal history (cooling rate, annealing time, and temperature), as well as the method of T_g measurement affects the reported T_g.

6. Pressure

Increasing pressure increases T_g in a linear relationship. A simple model is (Refs. 1227–1230)

$$T_g(p) = T_g(0) + sp$$

where $T_g(p)$ is the glass transition temperature as a function of pressure, p is pressure, and s is the linear pressure coefficient. This coefficient is 0.2 K/MPa for flexible aliphatic chains, and 0.55 K/MPa for semirigid aromatic chains (1231–1233). The effect of pressure on T_g can be important in some processing applications, such as injection molding. A different method for modeling the effect of temperature takes into account pVT data near T_g (1235–1237), giving an equation that includes the bulk modulus of the polymer glass.

D. ESTIMATION METHODS FOR THE GLASS TRANSITION TEMPERATURE

Several researchers have developed group contribution methods for correlating polymer properties, including the glass transition temperature (1238–1240). These techniques emphasize quantitative modeling of the various effects of polymer structure on T_g, and are a valuable aid to interpreting experimental data and estimating glass transition temperatures for new materials.

E. CLASSIFICATION, NOMENCLATURE, AND ABBREVIATIONS

Over 10000 papers contain glass transition data (43). This section of *Polymer Handbook* represents a fraction of these

data. Most of the data in the tables are for linear homopolymers. In general, the polymers contain no additives or diluents, and are thought to have low or no branching.

Polymers are subdivided into principle classes by the composition of their repeating chain segment: acyclic carbon polymers, carbocyclic polymers, acyclic heteroatom polymers, heterocyclic polymers, and copolymers. All entries are placed in the most senior class their structure commands (109,110) and appear in only one class. The subclasses and their entries are organized in alphabetical order.

1. Naming Conventions

With the exception of common polymers with accepted trivial names, the polymers are named substantially according to the ACS recommendations for polymer nomenclature (110) in conjunction with IUPAC rules (109); less common polymers are cross-referenced from the trivial to the systematic name. Systematic names are not given for all the polymers in order to save space. Substitutive nomenclature is generally used for simple radicals, but for long combinations of radicals replacement nomenclature has been used to provide a much shorter name (as for some fluorocrylates with ether side chains).

When sequences of radicals have repeated, the repeating sequence has been written once and prefixed "di", "tri", etc. as appropriate, for example, di(oxyethylene) for the sequence $-O-CH_2-CH_2-O-CH_2-CH_2-$. Note that the diradical "di(oxyethylene)" must be distinguished from the diradical "dioxyethylene" which has the structure, $-O-O-CH_2-CH_2-$, and also the diradical "ethylenedioxy" which has the structure, $-O-CH_2-CH_2-O-$ (IUPAC rule C205.2). The principle underlying the last-named diradical has not generally been extended to the naming of polymers in this section, *i.e.*, diradicals of structure $-X-Y-X-$ are not named YdiX, with the exception of alkanedioyl diradicals, because of the difficulty of locating indexed polymer names in which the diradicals are not named from left to right. Many polymers are derivatives of the diradical "propylene" $-CH(CH_3)-CH_2-$; the substituted diradical "propylene" is used in naming polymers instead of "1-methylethylene" which could be preferred.

Polymer names are tabulated in alphabetical order within each subsection, but

1. prefixes like *sec-*, *tert-*, including designated atoms and the numbers showing locations of substituents are ignored except as secondary and tertiary indicators of order. For example, poly(ethylene 2,6-naphthalate) appears before poly(ethylene 1,4-terephthalate); poly(4-*p*-anisoylstyrene) appears before poly(4-benzoylstyrene).
2. multiplying prefixes for various substituents, such as dimethyl or trimethyl, are observed in alphabetical ordering rather than being grouped together as in the 3rd edition of this Handbook.
3. the locations of substituents in otherwise identical polymers are taken as tertiary indicators of order: the numbers are arranged in increasing order at the first point of difference. Thus, 2,3,8- comes before 2,4,1-.

Comments may include information as to the instrument of measurement and, whenever possible, information regarding the method of DSC measurement (e.g., onset), conditions of measurement, thermal history, and whether measurements were made as a function of a particular variable such as molecular weight (*f*(MW)).

2. Abbreviations

HR	Heating rate
CR	Cooling rate
0CR	Zero cooling rate
Xp	Extrapolate
TH	Thermal history
DSC	Differential scanning calorimeter
TMA	Thermal mechanical analyzer
DTA	Differential thermal analysis
DMA	Dynamic mechanical analyzer
MW	Molecular weight
f( )	Function of a variable
Mdpt	Midpoint
Intg	Integration

F. TABLES OF GLASS TRANSITION TEMPERATURES OF POLYMERS

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
1.1. POLY(ACRYLICS) AND POLY(METHACRYLICS)				
1.1.1. POLY(ACRYLIC ACID) AND POLY(ACRYLIC ACID ESTERS)				
Poly(acrylic acid)	9003-01-4	379		720,811–818
		348		1293
Poly(1-adamantyl acrylate)		426		1401,1349
Poly(adamantyl crotonate)		507		1349
Poly(adamantyl sorbate)		388		1349
Poly(benzyl acrylate)		279		746,1447

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(4-biphenylyl acrylate)		~383		819
Poly(4-butoxycarbonylphenyl acrylate)		286		746
Poly(butyl acrylate)	9003-49-0	219	Mechanical method	1, 23,634,775, 802–822
		224		1401
Poly(*sec*-butyl acrylate) conventional		251		823,824
		253		1401
syndiotactic		253		
isotactic		250		
Poly(*tert*-butyl acrylate)		380, 316	Conflicting data	746,824,825,1401
		346		
		304		1432
Poly(2-*tert*-butylphenyl acrylate)		345		826
Poly(4-*tert*-butylphenyl acrylate)		344		826
Poly(cesium acrylate)		447	Extrapolated from DSC data on water plasticized samples	817
Poly[3-chloro-2,2-bis(chloromethyl)propyl acrylate]		319		746
Poly(2-chlorophenyl acrylate)		326		746
Poly(4-chlorophenyl acrylate)		331		826
Poly(2,4-dichlorophenyl acrylate)		333		746
Poly(4-cyanobenzyl acrylate)		317		746
Poly(2-cyanobutyl acrylate)	25154-80-7	384–396	Dilatomer, 10 deg/min HR, DTA, *f*(polymerization)	1088
Poly(2-cyanoisobutyl acrylate)	26809-38-1	324	Dilatomer, 10 deg/min HR	1089
Poly(4-cyanobutyl acrylate)		233–238	No experimental details	827
Poly(2-cyanoethyl acrylate)	25067-30-5	277		746,820
		388	Dilatomer, 10 deg/min HR	1090
Poly(2-cyanoheptyl acrylate)	26936-29-8	389	DTA	1092
Poly(2-cyanohexyl acrylate)	26877-39-4	358	Dilatomer, 10 deg/min HR	1093
Poly(cyanomethyl acrylate)		433	Dilatomer, 10 deg/min HR, DTA, *f*(polymerization)	1087
Poly(2-cyanomethyl acrylate)		296	No experimental details	820
Poly(5-cyano-3-oxapentyl acrylate)		250	No experimental details	820
Poly(4-cyanophenyl acrylate)		363		746
Poly(2-cyanoisopropyl acrylate)	25931-02-6	339	Dilatomer, 10 deg/min HR	1091
Poly(4-cyano-3-thiabutyl acrylate)		249		828
Poly(6-cyano-3-thiahexyl acrylate)		215		828
Poly(6-cyano-4-thiahexyl acrylate)		215		828
Poly(8-cyano-7-thiaoctyl acrylate)		214		828
Poly(5-cyano-3-thiapentyl acrylate)		223		828
Poly(cyclododecyl acrylate)	56710-66-8	310	DSC, onset, HR, 32 deg/min, quenched	1086
Poly(cyclohexyl acrylate) conventional		292		824
syndiotactic		289		
isotactic		285		
Poly(1,2:3,4-di-*O*-isopropylidene-α-D-galactopyranos-6-*O*-yl acrylate)		371		11,842
Poly(3,5-dimethyladamantyl acrylate)		379	DSC heating rate; data corrected (sci)	849,850
		378		1401,1349
Poly(3,5-dimethyladamantyl crotonate)		432		1349
Poly(3-dimethylaminophenyl acrylate)		320		746
Poly(1,3-dimethylbutyl acrylate)		258	Brittle point	823
Poly[2,2-difluoro-2-(2-heptafluorotetrahydrofuranyl) ethyl acrylate]		275	Brittle temperature	830
Poly(dodecyl acrylate)		270	Brittle point	821,829
Poly(2-ethoxycarbonylphenyl acrylate)		303		746
Poly(3-ethoxycarbonylphenyl acrylate)		297		746
Poly(4-ethoxycarbonylphenyl acrylate)		310		746
Poly(2-ethoxyethyl acrylate)		223		830
Poly(3-ethoxypropyl acrylate)		218		830,831
Poly(ethyl acrylate) conventional		249		23,634,775
	9003-32-1	251		1401
syndiotactic		249		820,821,824
isotactic		248		832, 833

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TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(2-ethylbutyl acrylate)		223	Brittle point	823
Poly(2-ethylhexyl acrylate)	9003-77-4	223	Brittle point	821
Poly(ferrocenylethyl acrylate)		430	No experimental details	834
Poly(ferrocenylmethyl acrylate)		470–483	DSC heating rate	835
Poly(3-fluoroalkyl α-fluoroacrylate)		398		1263
Poly(4-fluoroalkyl α-fluoroacrylate)		368		1263
Poly(5-fluoroalkyl α-fluoroacrylate)		374		1263
Poly(8-fluoroalkyl α-fluoroacrylate)		338		1263
Poly(17-fluoroalkyl α-fluoroacrylate)		388		1263
Poly(fluoromethyl acrylate)		288	Estimated T_g	838
Poly(furfuryl acrylate)		321		1279
Poly(1H, 1H-heptafluorobutyl acrylate)		243		155,836,837
Poly(5,5,6,6,7,7,7-heptafluoro-3-oxaheptyl acrylate)		228		830
Poly(2,2,3,3,5,5,5-heptafluoro-4-oxapentyl acrylate)		218		830,837
Poly(heptafluoro-2-propyl acrylate)		278–283	No details on sample or measurement	839
Poly(heptyl acrylate)		213	Brittle point	821,1401
Poly(2-heptyl acrylate)		235	Brittle point	823
Poly(hexadecyl acrylate)		308	Brittle point	23,821,840,841
Poly(1H,1H,3H-hexafluorobutyl acrylate)		251		836
Poly(hexyl acrylate)		216	Brittle point	823
				1401
Poly(3-hydroxyalkanoate)		293		1451
Poly(isobornyl acrylate) conventional		367		824
syndiotactic		369		
isotactic		363		
Poly(isobutyl acrylate)		249	Brittle point	823
		230		1401
Poly(isopropyl acrylate) conventional		267–270		746,823,824
syndiotactic		271–284		
isotactic		262		
Poly(magnesium acrylate)		673	Estimated from copolymer data	843
Poly(3-methoxybutyl acrylate)		217		844
Poly(2-methoxycarbonylphenyl acrylate)		319		746
Poly(3-methoxycarbonylphenyl acrylate)		311		746
Poly(4-methoxycarbonylphenyl acrylate)		340		746
Poly(2-methoxyethyl acrylate)		223		830
Poly(4-methoxyphenyl acrylate)		324		826
Poly(3-methoxypropyl acrylate)		198		830
Poly(methyl acrylate) conventional	9003-21-8	283		18,22,23,81
		284	Dilatomer	1094
		290		1343
		282		1401,1432
head to tail		278		576,720,775–777
head to head		304		821,824,831,841, 845–848
Poly(2-methylbutyl acrylate)		241	Brittle point	823,1401
Poly(3-methylbutyl acrylate)		228	Brittle point	823
Poly(2-methyl-7-ethyl-4-undecyl acrylate)		253	Brittle point	823
Poly(2-methylpentyl acrylate)		235	Brittle point	823
Poly(2-naphthyl acrylate)		358		826
Poly(neopentyl acrylate)		295		746,1401
Poly(1H,1H-nonafluoro-4-oxahexyl acrylate)		224		830
Poly(1H,1H-nonafluoropentyl acrylate)		236		836
Poly(nonyl acrylate)		215	Brittle point	821
		184		1401
Poly(octyl acrylate)	25266-13-1	208	Brittle point	821,841,1401
Poly(2-octyl acrylate)		228	Brittle point	823
Poly(1H,1H,5H-octafluoropentyl acrylate)		238		836
Poly(pentabromobenzyl acrylate)		453		1447
Poly(pentachlorophenyl acrylate)		420		746
Poly(1H,1H-pentadecafluorooctyl acrylate)		256	Crystalline	836
Poly(1H,1H-pentafluoropropyl acrylate)		247		836
Poly(*n*-pentyl acrylate)		216		1401

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(3-pentyl acrylate)		267		746,823
Poly(phenyl ethyl)acrylate		270		746
Poly(phenyl acrylate)		330		746,826
Poly(potassium acrylate)		467	Extrapolated from data on water plasticized samples	817
Poly(propyl acrylate)		236	Brittle point	23,592,832,836, 851, 852
		228		1401
Poly(sodium acrylate)		503	Estimated from copolymer data	817,843
Poly(tertural acrylate)		321		1279
Poly(tetradecyl acrylate)		297	Brittle point; probably 1st order transition	23,821
Poly(7,7,8,8-tetrafluoro-3,6-dioxaoctyl acrylate)		233		830
Poly(4,4,5,5-tetrafluoro-3-oxapentyl acrylate)		251		830
Poly(3-thiabutyl acrylate)		213		831
Poly(4-thiahexyl acrylate)		197		831
Poly(5-thiahexyl acrylate)		203		831
Poly(3-thiapentyl acrylate)		202		831
Poly(4-thiapentyl acrylate)		208		831
Poly(*m*-tolyl acrylate)		298		826
Poly(*o*-tolyl acrylate)		325		826
Poly(*p*-tolyl acrylate)		316		826
Poly(2,2,2-trifluoroethyl acrylate)		263		836
Poly(5,5,5-trifluoro-3-oxapentyl acrylate)		235		830
Poly(3,3,5-trimethylcyclohexyl acrylate)		288		1401
Poly(1H,1H-tridecafluoro-4-oxaoctyl acrylate)		205		830
Poly(1H,1H-undecafluorohexyl acrylate)		234		836
Poly(1H,1H-undecafluoro-4-oxahexyl acrylate)		205		830
Poly(zinc acrylate)		694		1261
1.1.2. POLY(ACRYLAMIDES)				
Poly(acrylamide)	9003-05-8	438	No experimental details	820
Poly(*N*-butylacrylamide)		319	Mechanical method	853
Poly(*N*-*sec*-butylacrylamide)		390	No experimental details	820
Poly(*N*-*tert*-butylacrylamide)		401	No experimental details	820
Poly(*N*,*N*-dibutylacrylamide)		333	Softening point, amorphous	854
Poly(*N*,*N*-diisopropylacrylamide)		~393	Softening point, almost amorphous	854
Poly(*N*,*N*-dimethylacrylamide)	26793-34-0	362		746
Poly(*N*-dodecylacrylamide)		198–320	Conflicting data, 198 K more probable in amorphous sample	820,853
Poly(isodecylacrylamide)		313	No experimental details	820
Poly(isohexylacrylamide)		344	No experimental details	820
Poly(isononylacrylamide)		325	No experimental details	820
Poly(isooctylacrylamide)		339	No experimental details	820
Poly(*N*-isopropylacrylamide)	25189-55-3	358, 403	Conflicting data	820,855
Poly([*N*-(1-methylbutyl)acrylamide)		380	No experimental details	820
Poly(*N*-methyl-*N*-phenylacrylamide)		~453	Softening point, amorphous	854
Poly(morpholylacryamide), [Poly(morpholinocarbonylethylene)]		420		9,856
Poly(*N*-octadecylacrylamide)		162	Mechanical method	853
Poly(*N*-octylacrylamide)		220		853
Poly(piperidylacrylamide)		381		9,856
1.1.3. POLY(METHACRYLIC ACID) AND POLY(METHACRYLIC ACID ESTERS)				
Poly(acrylonitrile methacrylate)		383		1362
Poly(adamantyl methacrylate)		414	DSC heating rate	849
		456		1349
		456/532		1401
Poly(benzyl methacrylate)	25085-83-0	327		746
Poly(2-bromoethyl methacrylate)		325		746
Poly(2-*tert*-butylaminoethyl methacrylate)		306		746,857

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TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(4-*tert*-butylcyclohexyl methacrylate)	34903-89-4	356	DSC, *f*(HR)	1096
		403–451		1286
Poly(butyl methacrylate)	9003-63-8	293	~286 to 308 K	22,69,272,695, 720,775,822,824, 847,858–862, 1359,1367
		300		1272
		307		1254
		328	298 K elsewhere	1342
		297		1401
		294		1255
		299		1413
isotactic		249		824
syndiotactic		328		1342,1555
Poly(*sec*-butyl methacrylate)		333		746,863,1401
Poly(*tert*-butyl methacrylate) atactic		391	Maximum value	746,847,864
		395		1254
		380		1401
syndiotactic		387	Maximum value	
isotactic		280		
Poly(4-*tert*-butylphenyl methacrylate)	29696-27-3	371	DSC, *f*(HR)	1104
Poly(2-chloroethyl methacrylate)		~365	Likely to be slightly high	262
Poly(2-cyanoethyl methacrylate)		364		746,865
Poly(4-cyanomethylphenyl methacrylate)		~401		746
Poly(4-cyanophenyl methacrylate)		428		746
Poly(cyclobutyl methacrylate)		351		1401
Poly(cyclodecyl methacrylate)		331		1401
Poly(cyclododecyl methacrylate)		329		1401
Poly(cyclohexyl methacrylate) atactic		356		263,353,820,824
	25768-50-7	377		1401
		406		1254
isotactic		324		847,862
Poly(cyclooctyl methacrylate)		346		1401
Poly(cyclooctylmethyl methacrylate)		326		1401
Poly(cyclopentyl methacrylate)		348		1401
Poly(2-decanhydronapthyl methacrylate)		418		1401
Poly(decyl methacrylate)		203		23,821,840,846
Poly(diethylaminoethyl methacrylate)		~289–297		857
Poly(1,2:3,4-di-*O*-isopropylidene-α-D-galactopyranos-6-*O*-yl methacrylate)		399		11,842
Poly(3,5-dimethyladamantyl methacrylate)		469	Temperature reported as corrected	849,850
		467		1401,1349
Poly(dimethylaminoethyl methacrylate)		292		746
Poly(3,3-dimethylbutyl methacrylate)		~318	Vicat softening point 332 K	863
		318		1401
Poly(3,3-dimethyl-2-butyl methacrylate)		~381	Vicat softening poing 396 K	863
		381		1401
Poly(1H,1H,7H-dodecafluoroheptyl methacrylate)		286	Mechanical method	876
Poly(dodecanidiol dimethacrylate)		277		1528
Poly(dodecyl methacrylate)		208	Conflicting data	1,821,858,866, 867
Poly(2-ethylhexyl methacrylate)	25719-51-1	263	Brittle point	821
Poly(ethyl methacrylate)	9003-42-3	–	DSC, onset, 16 deg/min HR, quenched, *f*(MW)	1095
atactic		338	Data covers range 320–343 K	22,69,286,352, 353,521,695,821, 824,857,858,860, 862,868–874, 1342,1401,1363, 1545,1432
		339		1315
		352		1552
		362		1254

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
		331		1359
		330		1360
		344		1413
		359	with 25% wt. of chromaphore I-doped polymer	1418
		378	with 25% wt. of chromaphore II-doped polymer	1418
		348		1438
		347	DSC	1484
isotactic		285		824
syndiotactic		339		847
Poly(2-ethylsulfinylethyl methacrylate)		298		746
Poly(ferrocenylethyl methacrylate)		482	No experimental details	834
Poly(ferrocenylmethyl methacrylate)		~458–468	DSC heating rate	835
Poly(3-fluoroalkyl methacrylate)		355		1263
Poly(4-fluoroalkyl methacrylate)		353		1263
Poly(5-fluoroalkyl methacrylate)		350		1263
Poly(8-fluoroalkyl methacrylate)		320		1263
Poly(17-fluoroalkyl methacrylate)		310		1263
Poly(glycidyl methacrylate)		347		1269
		336		1360
Poly(1H,1H-heptafluorobutyl methacrylate) syndiotactic		~330		875
Poly(1H,1H,9H-hexadecafluorononyl methacrylate)		258	Mechanical method	876
Poly(hexadecyl methacrylate)	25986-80-5	288	Brittle point, sample probably crystalline-may be T_m	821,866
Poly(hexyl methacrylate)	25087-17-6	268		695,846, 858,877, 1251,1443
		273		1401
			DSC	1484
		270		1416
		274		1413
Poly(2-hydropropyl methacrylate)	25249-16-5	328, 359	Conflicting data	746,878–880
80% isotactic		311	DSC, dry Xp	1098
58% syndiotactic		393	DSC, dry Xp	1199
Poly(2-hydropropyl methacrylate)		349		846,878
Poly(2-hydroxyethyl methacrylate)		358		1279
Poly(D,L-isobornyl methacrylate)		396/464/443		1401
Poly(isobornyl methacrylate)	64114-51-8	383		824
		423		1508
Poly(isobutyl methacrylate) random		326		746,821,824
isotactic		281		846,881
80% syndiotactic, 20% isotactic		326		
Poly(isobutyl methacrylate)	9011-15-8	326		1401
		337		1438
Poly(isopropyl methacrylate) atactic		354		746,824,862
	26655-94-7	358		1401
isotactic		300		
syndiotactic		358		
Poly(2,3-*O*-isopropylidene-D,L-glyceritol-1-*O*-yl methacrylate)		335	Heating rate: 20 K/min	842
Poly(magnesium methacrylate)		~763	Xp value	843
Poly(methacrylic acid)	54193-36-1	501	Xp data from plasticized samples	882
Poly(methacrylic anhydride) see Section 4.27				
Poly(4-methoxycarbonylphenyl methacrylate)		379		746
Poly(methacrylate)	9003-21-8	273		1255
Poly(methyl methacrylate)	9011-14-7			1102,1112,1101, 1107,1108
			DSC, onset, 16 deg/min HR, quenched, *f*(MW)	1101
			Dilatomer, CR 3 deg/h; creep relaxation, quenched	1109
		378		1432,1315, 1401,1288,1318

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
		379		1320,1300
		400		1343
		387		1365
			Aldrich chemicals	1547
		391		1552
		384	DSC	1277
		373		1359,1508,1531
		390		1363,1545
		367		1413,1457
		382	DSC	1489
		423		1533
		395		1254
atactic		378		1106
				1,17,22,25–27,
				69,78,79,81,190,
				201,263,286,287,
				317,318,352–354,
				400,614,684,698,
				720,775,777–779,
				789,804,821,824,
				846,858,860,862,
				880,883–895
		379		1271
		380		1255
isotactic		311		6,122,720,824,
				847,884,886,890,
				895–901
		323		1255
		325		1271
		319		1432
		373		1254
syndiotactic		378		122,720,824,847,
				890,895,896,898,
				900,901
		414		1256
		403		1254
heterotactic		372	DSC, rapid cooling, 10 deg/min HR, onset point, M_n^∞	1103
plasticized with 5% dibutylphthalate		363		1254
plasticized with 10% dibutylphthalate		355		1254
plasticized with 25% dibutylphthalate		373		1254
with bis(2-ethylhexyl phthalate)		200–398	DS	1557
		252–373	PCS	1557
Poly[(2-nitrosoethyl) methacrylate]		328		903
Poly(octadecyl methacrylate)	25639-21-8	173		1,720,904
Poly(1H,1H,5H-octafluoropentyl methacrylate)		309	Mechanical method	876
Poly(octyl methacrylate)		203, 253	Conflicting data	23,695,821,840,
				846,858
		253		1401
Poly(3-oxabutyl methacrylate)		289		846,857
Poly(3-oxa-5-hydroxypentyl methacrylate)		278–280	Mechanical method	880
Poly(pentyl methacrylate)		268	Brittle point	821
				1401
Poly(neopentyl methacrylate)	34903-87-2	299-312	DSC, f(HR)	1105
		299		1401
Poly(phenethyl methacrylate)		299		746
Poly(phenyl methacrylate)		383		353,746,820,847,
				863,875,1334
		407		1254
Poly(propyl methacrylate)		308	Conflicting data, 308–345 K reported	22,262,272,821,
				847,857,
				858,860,
				1432
		308		1401
		332		1254

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
		319		1413
			DSC	1484
Poly(sodium methacrylate)		~583	Xp value	410,547,616
Poly(3-tetracyclododecyl methacrylate)		477		1401
Poly(tetradecyl methacrylate)		201–264	Conflicting data	843
Poly(1,1,1-trifluoro-2-propyl methacrylate)		354	Vicat softening temperature	863
Poly(3,3,5-trimethylcyclohexyl methacrylate)		398		1401
Poly(3,5,5-trimethylhexyl methacrylate)		274		867
Poly(trimethylsilyl methacrylate) isotactic		341	Heating rate: 15 K/min	902
syndiotactic		400	Weak T_g for syndiotactic polymer	
Poly(2,3-xylenyl methacrylate)		398		1334
Poly(2,6-xylenyl methacrylate)		440		1334
1.1.4. POLY(METHACRYLAMIDES)				
Poly(4-butoxycarbonylphenylmethacrylamide)		401	Softening point	905
Poly(*N-tert*-butylmethacrylamide)		433	No experimental details	820
Poly(4-carboxyphenylmethacrylamide)		473	Softening point	905
Poly(4-ethoxycarbonylphenylmethacrylamide)		441	Softening point	905
Poly(4-methoxycarbonylphenylmethacrylamide)		453	Softening point	905
1.1.5. OTHER α-AND β-SUBSTITUTED POLY(ACRYLICS) AND POLY(METHACRYLICS)				
Poly(butyl butoxycarbonylmethacrylate)		298		906
Poly(butyl chloroacrylate)		330	Vicat softening point	863
Poly(*sec*-butyl chloroacrylate)		347	Vicat softening point	863
Poly(butyl cyanoacrylate)		358		907
Poly(cyclohexyl chloroacrylate)		387	Vicat softening point	863
Poly(dibutyl itaconate) see Poly(butylbutoxy-carbonylmethacrylate)				
Poly(ethyl chloroacrylate)		366	Vicat softening point	832,863, 908,909
10% isotactic		308	Calculated for infinite M_n; heating rate: 20 K/min	
100% syndiotactic		404	Calculated for infinite M_n; heating rate: 20 K/min	
Poly(ethyl ethoxycarbonylmethacrylate)		325	Intrinsic vicosity only 0.24 dL/g	906
Poly(ethyl ethacrylate)		300		746
Poly(ethyl fluoromethacrylate)		316		910
Poly(hexyl hexyloxycarbonylmethacrylate)		269	Intrinsic vicosity only 0.24 dL/g	906
Poly[(1-heptoxycarbonyl-1-heptoxycarbonylmethylene)ethylene]	28451-56-1	188, 250	DSC, onset, 20 deg/min HR	1110
Poly(isobutyl chloroacrylate)		363		832,911
Poly(isopropyl chloroacrylate)		363	Vicat softening point	832,863
Poly[(1-methoxycarbonyl-1-methoxycarbonylmethylene)ethylene]		372	DSC, onset, 20 deg/min HR	1110
Poly(methyl chloroacrylate)		413	Vicat softening point	863
Poly(methyl β-chloroacrylate)		416	No measurement details	820
Poly(methyl fluoroacrylate)		404	No details on samples or measurement	820,912
Poly(methyl fluoromethacrylate)		357	No experimental details	820,910
Poly(methyl phenylacrylate) atactic		391		913
isotactic		397		
Poly(propyl chloroacrylate)		344	Vicat softening point	832,863
1.2. POLY(ALKENES)				
Poly(butene-1) see Poly(ethylethylene)	9003-28-5			
Poly(butylethylene)		223		1,272,574, 629–633
Poly(*tert*-butylethylene)		337	Softening point, highly crystalline sample	631
Poly(cyclohexylethylene) atactic		393		634–639
isotactic		406	Dynamic method	640
Poly(2-cyclohexylethylene)		313	Mechanical method	634

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly[(cyclohexylmethyl)ethylene]		348	Mechanical method	634
Poly(3-cyclohexylpropylethylene)		248	Softening point, comparative data reported	631
Poly(cyclopentylethylene)		348	Dynamic method	634
Poly[(cyclopentylmethyl)ethylene]		333	Mechanical method	634
Poly(decylethylene), poly(1-decene)	25189-70-2	237		629,632,641
Poly(3,3-dimethylbutylethylene)		326	Softening point	631
Poly(1,1-dimethylethylene), poly(isobutane)	9003-27-4	200		1,23,24,58,61, 216,223,695–704
		205	Volumetric measurements	1252
		195		1255
		202		1288
		170–370		1540
		203		1549
		199		1432
Poly(4,4-dimethylpentene-1) see Poly(neopentylethylene)				
Poly(4,4-dimethylpentylethylene)		313	Softening point, crystalline sample	631
Poly(1,1-dimethyltetramethylene)		253	Dynamic method	706
Poly(1,1-dimethyltrimethylene)		263	Dynamic method	706–709
Poly(dodecylethylene), poly(1-dodecene)	25067-08-7	241		629,632,641,1078
Poly(ethylene)	9002-88-4	–		
		195		1255
		275		1254
		190		1287
		148	Conflicting interpretations of data; branch point transition at 252 K	1,6,16,28, 61,80,191, 223,238,261, 262,272,317, 318,344,349, 395,396,469, 521–523,574, 589,592,608,629, 641–676
		About 148	DSC,10/deg/min HR	1080
		222–240	WLF fit of volume relaxation	1076
–C–		163		1432
–C– and –C–C–		248		1432
Poly(ethylene) [Poly(methylene)]	9002-88-4	155		489,897,598,574, 694
Poly(ethyl ethylene)		235–240	Calorimeter	1073
		249	Wide spread in reported values	1,61,67,273,349, 395,397,574,629, 632,641,645,646, 677–680
		522		1541
Poly(1-ethyl-1-methyltetramethylene)		~250	DTA heating rate	681
Poly(ethyl-2-propylene)		268	Dynamic method	682
Poly(furylene ethylene)		293		1477
Poly(heptylethylene), poly(1-heptene)	25511-64-2	226	Dynamic method	629
Poly(1-hexene)	25067-06-5	About 215	Calorimeter	1072
Poly(hexyldecylethylene)		328	Dynamic method, stereoregular sample; may be first-order transition	629,641,683
Poly(hexylethylene)		208–228	Conflicting data	1,574,629,632
Poly(hydroxymethylene)		407	DSC, onset, 6 deg/min HR, dry, *f*(water)	1085
Poly(isobutylene) see Poly(1,1-dimethylethylene)	9003-27-4			
Poly(isobutylethylene)		302	Reported values range from 297 to 333 K	1,51,675,684–692
		313		1330
		300 ± 1		1356
		302		1432
		573		1451

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(isohexylethylene)		239	Softening point	631
Poly(isopentylethylene)		259	Softening point	631
Poly(isopropylethylene) atactic		323	Dynamic method	634,641,685,691, 692
isotactic		367	Dilatometry; suggested transition crystal/crystal type	693
Poly(1-methyloctamethylene)		215	Heating rate: 4–8 K/min	705
Poly(4-methylpentene-1) see Poly(isobutylethylene)	25068-26-2			
Poly(neopentylethylene)		332	Softening point, crystalline sample	631
Poly(nonylethylene)		236		632
Poly(octylethylene), poly(1-octene)	25511-67-5	232		629,632,641
Poly(pentene-1)		238		1356
		511		1541
Poly(pentylethylene)		242	Dynamic method	629
Poly(propylene) atactic	9003-07-0	–		1084
		238		1320
		~260	Conflicting data; most values reported range 258 to 270 K	1,6,15,80,122, 191,223, 261,272, 273,282,318,326, 394–397,469,574, 615,629,632,641, 645,640,659,666, 668,675,677,678, 684,710–726,1255
		266	DSC, onset, *f*(MW)	1082
		253		1271
		267	Light scattering under terminal mode	1452
			Light scattering under segmental mode	1452
		170–370		1540
		243		1432
		263		1499
isotactic	25085-53-4	–		1083
		~265	Most values range 238 to 260 K	
		272	DSC, onset, quenched, *f*(MW)	1081
syndiotactic (c)		~265	Most values range 263 to 267 K	
Poly(propylethylene)		~233	Conflicting data	1,272,574,595, 629,630,632,645, 685,727
Poly(propyl-2-propylene)		300	Dynamic method	682
Poly(tetradecylethylene)		246		629,632,641
Poly(2,5-thienylene ethylene)		298		1477
		353		1477
Poly(1,1,2-trimethyltrimethylene)		310		682

1.3. POLY(DIENES)

EPDM, maleated		236		1455
Natural rubber see Poly(isoprene) *cis*				
Neoprene see Poly(1-chloro-1-butenylene)				
Poly(1-bromo-1-butenylene)		241	DSC, annealed	1077
Poly(butadiene)s see Poly(1-butylene)s and Poly(vinylethylene)				
Poly(1,2-butadiene)		293		1432
		258		1337
ionically-terminated		273		1337
Poly(1,3-butadiene) *cis*		170–370		1540
Poly(1,4-butadiene)	9903-17-2	218		1432
cis		170		1285
Poly(1-butenylene) *cis*		164	DSC, onset, HR 20 deg/min, quenched	1075
		171		26,61,574–597

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
trans		215	Wide spread in published data	580,583,585–587, 589,594,595,598, 599
Poly(1-butyl-1-butenylene)		192	Dynamic method	600
Poly(1-*tert*-butyl-1-butenylene)		293		80,348,601,602
Poly(2-chloro-1,3-butadiene) see Poly(1-chloro-1-butenylene)				
Poly(1-chloro-1-butenylene) *cis*		253		603
trans		233		1,24,604–611
Poly(2-chloro-1,4,4-trifluoro-1-butenylene)		256	No experimental details	587
Poly(chloroprene)	9010-98-4	227		1360
Poly(1-decyl-1-butenylene)		220		1
Poly(endo-dicyclopentadiene)		370		1486
Poly(exo-dicyclopentadiene)		339		1486
Poly(1,2-dimethyl-1-butenylene)		262		576
Poly(1-ethyl-1-butenylene)		197		600,612
Poly(1-heptyl-1-butenylene)		190		80,601
Poly(isoprene) *cis*	9003-31-0, 104389-31-3	200		349,388,394,464, 469,574,591,594, 595,613–623
		203		1255,1271
		204		1432,1309
trans	104389-32-4	266		1255
		206–218	DSC, Mdpt, *f*(HR,TH)	1079
		207	DSC, onset, 20 deg/min HR	1074
		215		514,581,589,594, 608,617,624,1329
protonated		215		1495
Poly(isoprene), gutta percha		311		1323
Poly(1-isopropyl-1-butenylene)		221		600,601,612
Poly(4-methoxy-1-butenylene)		256		591
Poly(4-methoxycarbonyl-3-methyl-1-butenylene)		326	Dynamic method	625
Poly(methyl sorbate) see Poly(4-methoxycarbonyl-3-methyl-1-butenylene)				
Poly(norbornene)	25038-76-0	304		1295
		337–313	With increase in spacer length	1405
Poly(octafluoro-4-methyl-1-butenylene)		270	DSC heating rate	97
Poly(1-pentenylene) *cis*		159	Heating rate: 16 K/min	626
trans		183	DTA heating rate	594
Poly(1-phenyl-1-butenylene)		~283	Low molecular weight	584
Poly(1-propyl-1-butenylene)		196		600
Poly[5-(1,1,3,3-tetramethyl-1,3-disilabutyl)norbornene]		297		1295
Poly(1,4,4-trifluoro-1-butenylene)		238	Slightly crystalline	587
Poly(5-trimethylsilylnorbornene)		386		1295
Poly(vinylethylene) atactic		269	Published values range from 245 to 283 K	585,586,592,597, 627,628
	9003-17-2	304–294	With increase in spacer length	1405
		258	DSC	1354
		249 ± 1		1356
		261	Volumetric measurements	1252
		276		1329
cis	40022-03-5	171		1255
deuterated		186		1495
diol		205		1258
diol (crosslinked)		199		1326
diol 50%, Poly(propylene oxide) diol 50%		211		1258
diol 80%, Poly(propylene oxide) diol 20%		213		1258
high		250	DSC	1559
		258	DSC	1559
		264	DSC	1559
		274	DSC	1354,1559
low		268	DSC	1354,1559
		261	DSC	1559
protonated		182		1495
PVE-0		272		1550

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
PVE-5		276		1550
PVE-11		279		1550
PVE-22		282		1550
PVE-44		285		1550
PVE-66		287		1550
trans	25038-44-2	215		1255

1.4. POLY(STYRENES)

Poly(4-acetylstyrene)		389	Mechanical method	728
Poly(4-*p*-anisoylstyrene)		376	Mechanical method	728
Poly(4-benzoylstyrene)		371	Mechanical method	728
Poly[(2-benzoyloxymethyl)styrene]		345	Mechanical method	729
Poly[3-(4-biphenylyl)styrene]		~471	Softening point	730
Poly[4-(4-biphenylyl)styrene]		593	Softening point	730
Poly(5-bromo-2-butoxystyrene)		320	Mechanical method	731
Poly(5-bromo-2-ethoxystyrene)		353	Mechanical method	731
Poly(5-bromo-2-isopentyloxystyrene)		310	Mechanical method; low viscosity	731
Poly(5-bromo-2-isopropoxystyrene)		308	Mechanical method	731
Poly(5-bromo-2-methoxystyrene)		359	Mechanical method	731
Poly(5-bromo-2-pentyloxystyrene)		322	Mechanical method; low viscosity	731
Poly(5-bromo-2-propoxystyrene)		327	Mechanical method; low viscosity	731
Poly(4-bromostyrene)	24936-50-3	391		732–734
		414–430	DSC, Mdpt, *f*(TH)	1124
Poly(2-butoxycarbonylstyrene)		339	Mechanical method	735
Poly(4-butoxycarbonylstyrene)		349	Mechanical method	728
Poly(-[(2-butoxyethoxy)methyl]styrene)		<235		736
Poly(2-butoxymethylstyrene)		340	Mechanical method	729
Poly(4-butoxymethylstyrene)		<283		736
Poly[4-(*sec*-butoxymethyl)styrene]		313	Dynamic method	736
Poly(4-butoxystyrene)		~320	Mechanical method	736,737
Poly(5-*tert*-butyl-2-methylstyrene)		360		732
Poly(4-butylstyrene)		279		736,738,739
Poly(4-*sec*-butylstyrene)		359	Softening point	739
Poly(*tert*-butylstyrene)		422		1523
Poly(4-*tert*-butylstyrene)	26009-55-2	399–404	DSC, Mdpt, X. 1 deg/min, *f*(CR)	1122
Poly(4-butyrylstyrene)		347	Mechanical method	728
Poly(2-carboxystyrene)		450	Mechanical method	742
Poly(4-carboxystyrene)		386	Mechanical method	728
Poly(4-chloro-3-fluorostyrene)		395		732
Poly(4-chloro-2-methylstyrene)		418		732
Poly(4-chloro-3-methylstyrene)		387		732
Poly(2-chlorostyrene)		392		732
Poly(3-chlorostyrene)		363		732
Poly(4-chlorostyrene)	24991-47-7	383		83,732–734
		388–401	DSC, Mdpt, *f*(TH)	1123
		298		1423
Poly(4-cyanostyrene)		393	Sample thought to be crosslinked	746
Poly(4-decylstyrene)		208		738
Poly(2,4-dichlorostyrene)		406		732,743
Poly(2,5-dichlorostyrene)		379		44,732,744
Poly(2,6-dichlorostyrene)		440		83,569,745
Poly(3,4-dichlorostyrene)		401		732,744
Poly[4-(2-diethylaminoethoxycarbonyl)styrene hydrochloride]		347	Mechanical method	728
Poly(4-diethylcarbamoylstyrene)		375	Mechanical method	728
Poly(2,5-difluorostyrene)		374	Softening point	83
Poly(2,4-diisopropylstyrene)		~435	Softening point	83
Poly(2,5-diisopropylstyrene)		441	Softening point	83

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(2-dimethylaminocarbonylstyrene)		463	Mechanical method; low viscosity	742
Poly(4-dimethylaminocarbonylstyrene)		398	Mechanical method	728
Poly[2-(2-dimethylaminoethoxycarbonyl)styrene]		342	Mechanical method	735
Poly[4-(2-dimethylaminoethoxycarbonyl)styrene]		373	Mechanical method	728
Poly[4-(2-dimethylaminoethoxycarbonyl)styrene hydrochloride]		355	Mechanical method	728
Poly(2,4-dimethylstyrene)		385		122,732,739,749, 768
Poly(2,5-dimethylstyrene)		416	DTA heating rate	732
Poly(3,4-dimethylstyrene)		384	DTA heating rate	732
Poly(3,5-dimethylstyrene)		377		749
Poly(4-dodecylstyrene)		221		738
Poly(2-ethoxycarbonylstyrene)		391	Mechanical method	735
Poly(4-ethoxycarbonylstyrene)		367	Mechanical method	728
Poly[4-(2-ethoxymethyl)styrene]		273	Dynamic method	736
Poly(2-ethoxymethylstyrene)		347	Mechanical method; low viscosity	729
Poly(4-ethoxystyrene)		~359	Mechanical method	737
Poly[4-(1-ethylhexyloxymethyl)styrene]		250		736
Poly(2-ethylstyrene)		376	Softening point	739
Poly(3-ethylstyrene)		~303	Softening point	739
Poly(4-ethylstyrene)		300, <351	Conflicting data	736,738,739
Poly(2-fluoro-5-methylstyrene)		384	Mechanical method	748
Poly(4-fluorostyrene)		368		732–734,749–751
Poly(4-hexadecylstyrene)		278		738
Poly(4-hexanoylstyrene)		339	Mechanical method	728
Poly(2-hexyloxycarbonylstyrene)		318	Mechanical method	735
Poly(4-hexyloxycarbonylstyrene)		339	Mechanical method	728
Poly(4-hexyloxymethylstyrene)		253	Dynamic method	736
Poly(4-hexylstyrene)		246		738
Poly[4-(4-hydroxybutoxymethyl)styrene]		293	Dynamic method	736
Poly[4-(2-hydroxybutoxymethyl)styrene]		319	Dynamic method	736
Poly[4-(1-hydroxyiminoethyl)styrene]		407	Mechanical method	728
Poly(4-[(1-hydroxyimino)-2-phenethyl]styrene)		384	Mechanical method	728
Poly[4-(1-hydroxy-3-dimethylaminopropyl)styrene]		316	Mechanical method; low viscosity	754
Poly[4-(1-hydroxy-1-methylbutyl)styrene]		~403	Softening point	754
Poly[4-(1-hydroxy-1-methylethyl)styrene]		~438	Softening point	754
Poly[4-(1-hydroxy-1-methylhexyl)styrene]		~364	Softening point	754
Poly[4-(1-hydroxy-1-methylpentyl)styrene]		~356	Softening point	754
Poly[4-(1-hydroxy-1-methylpropyl)styrene]		~459	Softening point	754
Poly(2-hydroxymethylstyrene)		433		1
Poly(3-hydroxymethylstyrene)		398	Dynamic method	1
Poly(4-hydroxymethylstyrene)		413		1
Poly[4-(1-hydroxy-3-morpholinopropyl)styrene]		323	Mechanical method; low viscosity	754
Poly[4-(1-hydroxy-3-piperidinopropyl)styrene]		327	Mechanical method; low viscosity	754
Poly(4-hydroxystyrene)		433		1423
Poly(4-iodostyrene)		429		733,734
Poly(2-isobutoxycarbonylstyrene)		400	Mechanical method	735
Poly(4-isobutoxycarbonylstyrene)		363	Mechanical method	728
Poly(2-isopentyloxycarbonylstyrene)		341	Mechanical method	735
Poly(2-isopentyloxymethylstyrene)		351	Mechanical method; low viscosity	729
Poly(4-isopentyloxystyrene)		~330	Mechanical method	737
Poly(2-isopropoxycarbonylstyrene)		419	Mechanical method	735
Poly(4-isopropoxycarbonylstyrene)		368	Mechanical method	728
Poly(2-isopropoxymethylstyrene)		361	Mechanical method; low viscosity	729
Poly(4-isopropylstyrene)		306	Softening point	739
Poly(2-methoxycarbonylstyrene)		403	Mechanical method; low viscosity	742
Poly(4-methoxycarbonylstyrene)		386	Mechanical method	728

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(2-methoxymethylstyrene)		362	Mechanical method; low viscosity	729
Poly(4-methoxymethylstyrene)		350	Dynamic method	736
Poly(4-methoxy-2-methylstyrene)		~358	Softening point	755
Poly(2-methoxystyrene)		~348	Softening point	755
Poly(4-methoxystyrene)		~362	Mechanical method	737,755–757
Poly(2-methylaminocarbonylstyrene)		462	Mechanical method	742
Poly(α-methylstyrene)	25014-31-7	293	DSC, onset, 20 deg/min HR	1130
		371–375	DSC, Mdpt, 20 deg/min HR, *f*(TH)	1135
		453		1549,1432
		443		1324
		455	DSC	1277
Poly(2-methylstyrene)		409		63,732,739,749,755
Poly(3-methylstyrene)		370	DTA heating rate	732,739,749,766
Poly(4-methylstyrene)	24936-41-2	366, 374	Conflicting data	732,739,756,766,767
		382		1423
Poly(4-methoxystyrene)	24936-44-5	386	DSC, Mdpt, 320 deg/min CR, 1 deg/min HR, *f*(TH, HR, M_n)	1140
Poly(4-morpholinocarbonylstyrene)		400	Mechanical method	728
Poly[4-(3-morpholinopropionyl)styrene]		314	Mechanical method	754
Poly(4-nonadecylstyrene)		305		1
Poly(4-nonylstyrene)		220		738
Poly(4-octadecylstyrene)		305		738
Poly(4-octanoylstyrene)		323	Mechanical method	728
Poly[4-(octyloxymethyl)styrene]		231	Dynamic method	736
Poly(2-octyloxystyrene)		286		769
Poly(4-octylstyrene)		228		738
Poly[4-(pentadexafluoroheptyl)styrene]		320	Heating rate 32 K/min, *f*(TH)	747
Poly(2,3,4,5,6-pentafluorostyrene)		378	Heating rate: 5–20 K/min	97
Poly(2-pentyloxycarbonylstyrene)		365	Mechanical method	735
Poly(2-pentyoxymethylstyrene)		320	Mechanical method	729
Poly(perfluorostyrene)		467	Heating rate: 5–20 K/min	97
Poly(2-phenethyloxymethylstyrene)		336	Mechanical method; low viscosity	729
Poly(2-phenoxycarbonylstyrene)		397	Mechanical method; low viscosity	742
Poly(4-phenoxystyrene)		~373	Softening point	83
Poly(4-phenylacetylstyrene)		351	Mechanical method	728
Poly(2-phenylaminocarbonylstyrene)		464	Mechanical method; low viscosity	742
Poly(4-phenylstyrene)		434	Extrapolated to zero rate	763, 770
Poly(4-piperidinocarbonylstyrene)		387	Mechanical method	728
Poly[4-(3-piperidinopropionyl)styrene]		311	Mechanical method	754
Poly(4-propionylstyrene)		375	Mechanical method	728
Poly(2-propoxycarbonylstyrene)		381	Mechanical method	735
Poly(4-propoxycarbonylstyrene)		365	Mechanical method	728
Poly(2-propoxymethylstyrene)		370	Mechanical method; low viscosity	729
Poly(4-propoxymethylstyrene)		295	Dynamic method	736
Poly(4-propoxystyrene)		343	Mechanical method	737
Poly(4-propoxysulfonylstyrene) isotactic		490	DTA heating rate	771
Poly(styrene) alkylated		356		1521
amorphous		373		1506
crosslinked		373		1352
deuterated PS		378		1495
isotactic and atactic	9003-53-6	373		9,17,21,22,25,47,51, 57,63,64,72,78,79,188, 190,191,261,263,317,318, 344,350–352,394,397, 399,469,524,569,576,619, 628,630,635,637,640,646, 647,684,699,731–734, 746,750,766,768,

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
				772–810,1125, 1133,1138,1139,1141, 1254,1276,1288, 1308,1318,1324,1423, 1432,1523,1549,1255
			DSC	1489
			DSC, onset, 16 deg/min HR, quenched, *f*(MW)	1126
			DSC, Mdpt, 20 deg/min HR, *f*(MW)	1129
			DSC, Intg, *f*(HR, CR, MW)	1143
			DSC, Intg, *f*(HR, CR)	1144
			Dilatometer, CR 3 deg/h; creep relaxation, quenched	1148
		368	Dilatometer, 2.5 deg/min CR	1145
		371	DSC, onset, HR 32 deg/min, quenched	1132
		371	DTA, DSC, onset, 1 deg/min HR, *f*(HR)	1147
		371–377	DSC, penetration, onset MP, zero HR	1146
		373	DSC, Intg, OHR, 10 deg/min CR	1142
		376	DSC, Mdpt, 1 deg/min X HR, TH, *f*(MW)	1128
		377	DSC, onset, 20 deg/min HR after similar cool, *f*(MW)	1149
		382	DSC, onset, 20 deg/min HR	1130
		364		1271
		374	GPC	1284
		377	PCS	1309
		378	NR and SE	1319
		380		1433
		374		1437
		400		1530
		401	DSC	1530
		170–370		1540
p-cyanobenzyl		365		1253
p-cyanobiphenyl		370		1253
sulfonated syndiotactic		433–463		1322
Poly(4-tetradecylstyrene)		237		738
Poly(4-*p*-toluoylstyrene)		372	Mechanical method	728
Poly(α, β, β-trifluorostyrene)		475	Heating rate: 5–20 K/min; values up to 513 K reported	83,97,114,752, 753
Poly{3-[bis(trimethylsiloxy)boryl]styrene}		308	Refractive index	1136
Poly{4-[bis(trimethylsiloxy)boryl]styrene}		357	Refractive index	1136
Poly(2,4,5-trimethylstyrene)		~409	Softening point	83
Poly(2,4,6-trimethylstyrene)		~435	Softening point	83
Poly(4-[bis(trimethylstannyl)methyl]styrene)		413		1556,1347
Poly(4-valerylstyrene)		343	Mechanical method	728
1.5. POLY(VINYL ALCOHOL) AND POLY(VINYL KETONES)				
Poly(1-acetyl-1-fluoroethylene)		415		923
Poly(benzoylethylene)		314, 347	Conflicting data	924,925
Poly(4-bromo-3-methoxybenzoylethylene)		317	Mechanical data	924
Poly(4-*tert*-butylbenzoylethylene)		377		925
Poly(4-chlorobenzoylethylene)		310, 362	Conflicting data	924,925
Poly(3,4-dimethylbenzoylethylene)		315	Mechanical data	924
Poly(4-ethylbenzoylethylene)		325		924,925
Poly(4-isopropylbenzoylethylene)		336		925
Poly(4-methoxybenzoylethylene)		319	Mechanical data	924
Poly(phenyl vinyl ketone) see Poly(benzoylethylene)	26742-84-7			
Poly(4-propylbenzoylethylene)		317		925

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(*p*-toluoylethylene)		344		924,925
Poly(vinyl alcohol) [Poly(hydroxyethylene)]	9002-89-5	358	Dynamic method	1,191,202,216, 230,261,573,926, 927
			From hydrolysis of poly(vinyl acetate)	1424
		308		1498
		350		1513
1.6. POLY(VINYL ESTERS)				
Poly[(2-acetoxybenzoyloxy)ethylene]		333		29,925
Poly(4-acetoxybenzoyloxyethylene)		~349		29,925
Poly(acetoxyethylene) [Poly(vinyl acetate)]	9003-20-7	305		1,9,12,19,22,62, 187,216,223,230, 286,352,354,399, 400,573,593,695, 775,777,792,816, 840,845,888,926, 928,932,1016, 1034–1044,1121, 1302,1320,1288
		302		1432
		590		1317
		315		1339,1516
		304		1513
		303		1255
		311		1437,1468
		313	DSC	1489
Poly[(1-acetylindazol-3-ylcarbonyloxy)ethylene]		423	Heating rate: 20 K/min-"onset" value	1045
Poly(4-benzoylbutyryloxyethylene)		318	DTA heating rate	1046
Poly(benzoyloxyethylene)		344		29,354,755,820, 925,1047,1048
Poly(3-bromobenzoyloxyethylene)		331		29,925
Poly(4-bromobenzoyloxyethylene)		365		29,925,1047,1048
Poly[(*tert*-butoxycarbonylamino)ethylene]		393	DTA heating rate	1049
Poly(4-*tert*-butylbenzoyloxyethylene)		374		29,925,1050
Poly(4-butyryloxybenzoyloxyethylene)		334		29,925
Poly(2-chlorobenzoyloxyethylene)		335		29,925,1048
Poly(3-chlorobenzoyloxyethylene)		338		29,925,1048
Poly(4-chlorobenzoyloxyethylene)		357		29,925,1047,1048
Poly(cyclohexanoyloxyethylene)		349	Mechanical method, heating rate: 30 K/h	1043
Poly(cyclohexylacetoxyethylene)		298	Mechanical method, heating rate: 30 K/h	1043
Poly(4-cyclohexylbutyryloxyethylene)		~263	Mechanical method, heating rate: 30 K/h	1043
Poly(cyclopentanoyloxyethylene)		309	Mechanical method, heating rate: 30 K/h	1043
Poly(cyclopentylacetoxyethylene)		270	Mechanical method, heating rate: 30 K/h	1043
Poly(3,3-dimethyl-3-phenylpropionyloxyethylene)		293	Mechanical method, heating rate: 30 K/h	1043
Poly[(2,2-dimethylvaleryloxy)ethylene]		283	Mechanical method	1043
Poly(4-ethoxybenzoyloxyethylene)		343		29,925
Poly(4-ethylbenzoyloxyethylene)		326		29,925
Poly(2-ethyl-2-methylpropylene terephthalate)		340		1532
		328–338		1532
Poly[(2-ethyl-2,3,3-trimethylbutyryloxy)ethylene]		388	Mechanical method	1043
Poly(formyloxyethylene)		310	~60% syndiotactic diads	593
		306	50% syndiotactic diads	
Poly[(heptafluorobutyryloxy)ethylene]		300		1051
Poly(3-hydroxybutyrate)		277		1262
		278 ± 2		1536
		223		1494

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TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(isonicotinoyloxyethylene)		372	From polyvinyl alcohol	29,288,1053
Poly(4-isopropylbenzoyloxyethylene)		342		29,925
Poly[(2-isopropyl-2,3-dimethylbutyryloxy)ethylene]		393	Mechanical method	1043
Poly[(2-methoxybenzoyloxy)ethylene]		338		29,925
Poly[(3-methoxybenzoyloxy)ethylene]		~317		29,925
Poly[(4-methoxybenzoyloxy)ethylene]		360		29,757,925, 1047,1048
Poly[(2-methylbenzoyloxy)ethylene]		321		29,820,925,1048
Poly[(3-methylbenzoyloxy)ethylene]		324		29,925,1048
Poly[(4-methylbenzoyloxy)ethylene]		343		29,925,1047,1048
Poly[(1-methylcyclohexanoyloxy)ethylene]		359	Mechanical method, heating rate: 30 K/h	1043
Poly(nicotinoyloxyethylene)		360	From polyvinyl alcohol	29,1053
Poly(nitratoethylene)		307		903
Poly[(3-nitrobenzoyloxy)ethylene]		366		29,925
Poly[(4-nitrobenzoyloxy)ethylene]		395		29,925
Poly(nonadecafluorodecanoyloxy)ethylene]		253–255		1051
Poly[(nonafluorovaleryloxy)ethylene]		288–293		1051
Poly(octamethylene *p,p'*-dibenzoate)		315		1519
Poly[(pentadecafluorooctanyloxy)ethylene]		258–263		1051
Poly[(pentafluoropropionyloxy)ethylene]		315		1051
Poly[(4-phenylbenzoyloxy)ethylene]		358		29,925
Polyl(pivaloyloxyethylene)		359	Mechanical method	1043
Poly[(4-propionyloxybenzoyloxy)ethylene]		346		29,925
Poly(propionyloxyethylene)		283	Mechanical method	1043
Poly[(2,2,3,3-tetramethylvaleryloxy)ethylene]		363	Mechanical method	1043
Poly[(2,2,3,4-tetramethylvaleryloxy)ethylene]		323	Mechanical method	1043
Poly[(2,2,4,4-tetramethylvaleryloxy)ethylene]		328	Mechanical method	1043
Poy[(4-*p*-toluoylbutyryloxy)ethylene]		313	DTA heating rate	1046
Poly(triethylene glycol *p,p'*-dibenzoate)		293	Annealed	1505
Poly[(trifluoroacetoxy)ethylene]		~319, <348	Conflicting data	1051,1052
Poly[(3-trimethylsilylbenzoyloxy)ethylene]		353	DTA heating rate	1054
Poly[(4-trimethylsilylbenzoyloxy)ethylene]		408	DTA heating rate	1054,1055
Poly[(undecafluorocyclohexylcarbonyloxy)ethylene]		327		1051
Poly[(undecafluorohexanoyloxy)ethylene]		264	Plasticizer may be present	1051
Poly(vinyl acetate) see Poly(acetoxyethylene)	9003-20-7			
Poly(vinyl formate) see Poly(formyloxyethylene)				
Poly(vinyl-4-isopropylbenzoate) see Poly(4-isopropyl benzoyloxyethylene)				
Poly(4-(11-(vinyloxy)undecyloxy)-4-cyanophenylbenzoate)		310		1548
Poly(4-(11-(vinyloxy)undecyloxy)-4-ethoxyphenylbenzoate)		310		1548
1.7. POLY(VINYL ETHERS) AND POLY(VINYL THIOETHERS)				
Poly(butoxyethylene)		218		1,832,866,914–918
		217		1432
Poly(*sec*-butoxyethylene)		253		866,914
Poly(*tert*-butoxyethylene)		361		1,918
Poly(butylthioethylene)		253		866
Poly(butyl vinyl ether) see Poly(butoxyethylene)				
Poly(chlorotrifluoroethylene)	9002-83-9	325		1470
Poly(cyclohexyloxyethylene)		354	Softening point	918
Poly(decyloxyethylene)		~183, 211	Conflicting data, independent DSC data supports higher value	866,914,916,917
Poly(2,2-dimethylbutoxyethylene)		282	Mechanical method	918
Poly(1,1-difluoro-2-trifluoromethoxyethylene)		263–273	Estimated from copolymer data	457
Poly(1,2-difluoro-1-trifluoromethoxyethylene)		263–273	Estimated from copolymer data	457
Poly(dodecafluorobutoxyethylene)		263–273	T_g estimated from copolymer data	457
Poly(ethoxyethylene)		230		1,223,521,573, 832,866,914–920
	25104-37-4	240		1432
Poly[(2-ethylhexyloxy)ethylene]		207		866,915
Poly(ethylthioethylene)		266	Viscosity only 0.2 dl/g	866

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(2-furyloxirane)		288		1419
Poly[(heptafluoro-2-propoxy)ethylene]		~328–338		839
Poly(hexafluoromethoxyethylene)		268	DTA heating rate	457
Poly(hexyloxyethylene)		199		866,914-918,1432
Poly(isobutoxyethylene)		254		1,223,573,832, 866,914,916–918, 920,922
Poly(isopropenyl methyl ether) see Poly(2-methoxy propylene)				
Poly(isopropoxyethylene)		270		832,866,914,918
		261		1432
Poly(methoxyethylene)		242		1,223,805,866, 914–918,920,1255
			NR and SE	1319
		246	GPC	1284
		228		1300
		245		1308
		248		1348
		251		1495,1432
		191		1502
		199		1502
		247		1516
Poly(2-methoxypropylene)		340		866
Poly(methylthioethylene)		272		866
Poly(neopentyloxyethylene)		~424	Softening point	918
Poly(octyloxyethylene)		194		866,914–917
Poly(pentyloxyethylene)		207		866,915
Poly(propoxyethylene)		224		832,915
Poly(2,2,2-trifluoroethoxytrifluoroethylene)		308	DTA heating rate	457,921
Poly[1,1-bis(trifluoromethoxy)difluoroethylene]		213	Poorly defined DTA endotherms	457
Poly(vinyl methyl ether) see Poly(methoxyethylene)	9003-09-2			
Poly(4-vinyl phenol)	24979-70-2	429		1457
Poly(*p*-vinyl phenol)		428		1552

1.8. POLY(VINYL HALIDES) AND POLY(VINYL NITRILES)

Poly(1-acetoxy-1-cyanoethylene)		420	Heating rate: 8 K/min	255
Poly(acrylonitrile) [poly(cyanoethylene)]	25014-41-9	398	Dielectric, 1 Hz, *f*(Hz)	1113
		370		1,7,26,80,191, 202,393,400,521, 525,576,752,772, 779,787,928–941
		355–413		1382
Poly(chlorotrifluoroethylene)	9002-83-9	~325	By "static" methods	66,80,191,261, 349,355
		373	By mechanical methods even at low frequencies	396,401,521,647, 893,945–961
Poly(1,1-dichloroethylene), poly(vinylidene chloride)		255		1,6,62,80,216,521, 524,648,942–944
Poly(1,1-dichloro-2-fluoroethylene)		~320	[η] only 0.035 dl/g	962
Poly(1,2-dichloro-1,2-difluoroethylene)		350	Heating rate: 5–20 K/min, high pressure, radiation synthesis	97
Poly(1,1-difluoroethylene), poly(vinylidene fluoride)	24937-79-9	–		1118
		~233		401,963–970
		223		1321
		235	DSC	1333
Poly(1,2-difluoroethylene)		371	DTA heating rate	114,457,971
	9002-84-0	200		1287
Poly(ethylene), chlorinated	63231-66-3, 110028-31-4	281	48 wt.% Cl	1325
Poly(hexafluoropropylene)		425	Heating rate: 5–20 K/min	97,984,1006
Poly[(heptafluoropropyl)ethylene]		331	Heating rate: 5–20 K/min, high pressure, radiation synthesis	97

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TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(2-iodoethylethylene)		343	Dynamic method	1009
Poly(9-iodononylethylene)		267	Dynamic method	1009
Poly(3-iodopropylethylene)		~303	Dynamic method	1009
Poly(methacrylonitrile)		393		1
Poly[(pentafluoroethyl)ethylene]		314	Heating rate: 5–20 K/min, high pressure, radiation synthesis	97
Poly(tetradecafluoropentylethylene)		503?	Heating rate: 5–20 K/min, high pressure, radiation synthesis, value uncertain	97
Poly(tetrafluoroethylene) [poly(difluoromethylene)]	9002-84-0	390		1117
		160, 400	Much data, some conflicting	1,6,66,190,191, 253,261,355,256, 296,521,522,641, 651,659,891,893, 914,921,952, 972–1005
Poly(2,3,3,3-tetrafluoropropylene)		315	Heating rate: 5–20 K/min, high pressure, radiation synthesis	97
Poly(trifluoroethylene)		304	Quoted value, DTA?	457
Poly(3,3,3-trifluoropropylene)		300		97,914,1007,1008
Poly(vinyl chlorate)		347		1271
Poly(vinyl chloride)	9002-86-2	–		1112,1115,1116
		–	Dilatometer, CR 3 deg/h; creep relaxation, quenched	1119
		354	Increasing syndiotactic content increases T_g to 371 K	1,21,52,78,80, 104,188,191, 230,286,349, 353,354,356, 469,608,696, 720,772,777, 787,798,861, 926,942,944, 972,1010–1031
		366	Dekryptonation, f(MW)	1114
		371	DSC, 20 deg/min HR, M_n infinity, onset, varied annealing	1120
		359		1325
		357		1335
		363		1432
		342		1398
		353		1255
		347		1288
		361		1489
chlorinated	68648-82-8	408	67 wt.% Cl	1325
with Poly(butyl methacrylate) (30/70)		313	Semi-IPNs	1359
with Poly(butyl methacrylate) (50/50)		315	Blend	1359
with Poly(butyl methacrylate) (50/50)		323	Semi-IPNs	1359
with Poly(ethyl methacrylate) (30/70)		343	Semi-IPNs	1359
with Poly(ethyl methacrylate) (50/50)		325	Blend	1359
with Poly(ethyl methacrylate) (50/50)		343	Semi-IPNs	1359
with Poly(methyl methacrylate) (30/70)		370	Semi-IPNs	1359
with Poly(methyl methacrylate) (50/50)		363	Blend	1359
with Poly(methyl methacrylate) (50/50)		366	Semi-IPNs	1359
with Poly(methyl methacrylate) (80/20)		364		1446
H-H Poly(vinyl chloride)		348	DSC	1299
		352	DMA	1299
Poly(vinyl fluoride)	24981-14-4	314	Mechanical method	521,1032,1033
Poly(vinylidene chloride) see Poly(1,1-dichloroethylene)	9002-85-1			

1.9. OTHERS

Poly(1,1-bis[(1-adamantyloxy)carbonyl]-2-vinylcyclopropane)		307		1492
Poly(benzylethylene)		333		631,634,1056

TABLE 1. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(*N*-carbazolylethylene)	25067-59-8	500	DSC, 10 deg/min HR, X to infinite MW	1127
		500		1330
		357, 423, 481	Conflicting values reported	1,272,573,845
Poly[dimethylamino(ethoxy)phosphinylethylene]		305		1057
Poly[dimethylamino(phenoxy)phosphinylethylene]		300		1057
Poly(4,4-dimethyl-oxazolonylethylene)		365	DTA heating rate; low viscosity	1058
Poly(4,4-dimethyl-oxazolonyl-2-propylene)		380, 438	Conflicting data	1058,1059
Poly(2,4-dimethyl-1,3,5-triazinylethylene)		350		820
Poly(diphenylphosphinylethylene)		453		1065
Poly(ferrocenylethylene)		457–467	DSC heating rate	835
Poly(β-hydroxyvalerate)		257		1355
		262		1355
Poly(indazol-2-ylethylene)		331, 298	Two values for different preparations, heating rate: 20 K/min	1045
Poly(isopropyl benzene)		126		1271
Poly(indene)		473		1501
Poly(3-methyl hexane)		90		1271
Poly(3-methyl pentane)		80		1271
Poly(2-methyl pentane)		76		1271
Poly[(2-methyl-5-pyridyl)ethylene]		403	Mechanical method	1060–1062
Poly[(2-methyl-6-pyridyl)ethylene]		365	Heating rate 20 K/min	1061
Poly(1-naphthylethylene)		432	Values range from 323 to 453 K	83,755, 763,1063
Poly(2-naphthylethylene)		424	Extrapolated to zero heating rate	770
Poly(phenethylethylene)		283	Crystalline sample	631,634,1056,1064
Poly(phenethylmethylethylene)		245	Softening point	631
Poly(1,1-bis(phenoxycarbonyl)-2-vinylcyclopropane)		349–357		1535
Poly(phenylacetylene) see Poly(phenylvinylene)				
Poly(phenylvinylene)		393	Crystalline	1066
Poly(phthalimidoethylene)		497	No experimental details	820
Poly(2-pyridylethylene)		377		1061,1062,1067
Poly(4-pyridylethylene)	9017-40-7	415		1061,1062,1272
		433	Molar concentration 5%	1305
		456	Molar concentration 10%	1305
Poly(*N*-pyrrolidinylethylene)	9003-39-8	327		573
		359		1,4,907
		448	DSC, onset, Xp to dry	1150
Poly(*o*-terphenyl)		233		1271
		243		1271
Poly(*m*-tolylmethylethylene)		313	Crystalline sample	1056
Poly(*o*-tolylmethylethylene)		353	Crystalline sample	1056
Poly(*p*-tolylmethylethylene)		338	Crystalline sample	1056
Poly(2,4,6-tribromophenyl acetate)		423		1269
Poly(vinyl carbazole) see Poly(*N*-carbazolylethylene)				
Poly(vinyltrimethylgermanium)		463	Mechanical method	1068
Poly(vinylpyridine) see Poly(pyridylethylene)				
Poly(vinyl pyrrolidine) see Poly(*N*-pyrrolidinylethylene)				

TABLE 2. MAIN-CHAIN CARBOCYCLIC POLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
2.1. POLY(PHENYLENES)				
Poly(3,3′-biphenylylenehexafluorotrimethylene)		324	Heating rate: 32 K/min, η_{inh} only 0.05 dl/g	563
Poly(2-bromo-1,4-phenylene ethylene)		353	Mechanical method	564
Poly(2-chloro-1,4-phenylene ethylene)		343	Mechanical method, heating rate: 1.5 K/min	300,564,565
Poly(2-cyano-1,4-phenylene ethylene)		363	Mechanical method	564
Poly(2,5-dichloro-1,4-phenylene ethylene)		613	Softening point, no measurement details	565
Poly(2,5-dimethyl-1,4-phenylene ethylene)		373	Mechanical method, heating rate: 1.5 K/min	566
Poly(2-ethyl-1,4-phenylene ethylene)		298	Mechanical method	564
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-*p,p,m,p,p*-quinquephenylylenedecamethylene)		453	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-4, 4⁗-*p*-quinquephenylylenedecamethylene)		508	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-*p,p,m,p,p*, quinquephenylylenehexamethylene)		488	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-4, 4⁗-*p*-quinquephenylylenehexamethylene)		523	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-*p,p,m,p,p*-quinquephenylylenetetradecamethylene)		433	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-4,4⁗-*p*-quinquephenylylenetetradecamethylene)		458	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-*p,p,m,p,p*-quinquephenylylenetetramethylene)		453	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-4, 4‴-*p*-quinquephenylylenetetramethylene)		478	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-*p,p,m,p,p*-quinquephenylylenetrimethylene)		513	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2′,3′,6′,2‴,3‴,5‴-hexaphenyl-4,4‴-quinquephenylylenetrimethylene)		563	Heating rate: 20 K/min, structure may contain more *m*-links	568
Poly(2-methyl-1,4-phenylene ethylene)		328	Mechanical method, heating rate: 1.5K/min	564,565
Poly(1,4-phenylenetrichloroethylene)		433	Polymer contains small amount of different structure, mechanical method; heating rate: 1.5K/min	565
Poly(1,4-phenylene ethylene)		~353	Mechanical method, variable data	300,564, 565,567
Poly(1,3-phenylenehexafluorotrimethylene)		303	Heating rate: 32 K/min, η_{inh} only 0.1 dl/g	148
Poly(1,4-phenylene-1-phenylethylene)		428	Softening point, no experimental details	565
Poly(trimethyl-phenylene ether)		483		1432
2.2. OTHERS				
Poly(acenaphthylene)	25036-01-5	487–618	Conflicting data reported	31,83,566, 569–571
		628–649	DSC, onset, 32 deg/min HR after cooling from above T_g	1151
Poly(5-chlorononafluoro-1,3-cyclohexylene-difluoromethylene)		420	Heating rate: 5-20 K/min, high pressure, radiation synthesis, ring structure unproven	97

TABLE 2. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(cyclobutene) erythro di-isotactic		293	Degree of polymerization 150	572
erythro di-syndiotactic		273	Degree of polymerization 15	
Poly(decafluoro-1,3-cyclohexylenedifluoromethylene)		374?	Heating rate: 5–20 K/min	97
Poly(dodecafluoro-1,3-cycloheptylenedifluoromethylene)		471?	Heating rate: 5–20 K/min, high pressure, radiation synthesis, data queried in original paper, ring structure unproven	97
Poly(hexafluoro-1,3-cyclobutylenedifluoromethylene)		395	Heating rate: 5–20 K/min, high pressure, radiation synthesis, ring structure unproven; [η] only 0.03 dl/g	97
Poly(indenylene)		358		573
Poly(octafluoro-1,3-cyclohexylenedifluoromethylene)		390	Heating rate: 5–20 K/min, high pressure, radiation synthesis, ring structure unproven	97
Poly(1,4-naphthyleneethylene)		433	Mechanical method	565

TABLE 3. MAIN-CHAIN ACYCLIC HETEROATOM POLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
3.1. MAIN-CHAIN –C–O–C– POLYMERS				
3.1.1. POLY(ANHYDRIDES)				
Poly(oxycarbonyl-1,4-phenylenehexafluorotrimethylene-1,4-phenylenecarbonyl)		271		433
Poly(oxycarbonyl-1,4-phenyleneisopropylidene-1,4-phenylenecarbonyl)		333		247
Poly(oxycarbonyl-1,4-phenylenemethylene-1,4-phenylenecarbonyl)		395		247
Poly(oxycarbonyl-1,4-phenylenepentamethylene-1,4-phenylenecarbonyl)		321	Amorphous	247
		319	Crystalline	
Poly(oxycarbonyl-1,4-phenylenetetramethylene-1,4-phenylenecarbonyl)		319		247
Poly(oxycarbonyl-1,4-phenylenethiotetramethylenethio-1,4-phenylenecarbonyl)		335		247
Poly(oxyisophthaloyl)		403	Dynamic mechanical method	398,444
3.1.2. POLY(CARBONATES)				
Bisphenol A + 2,2-(fm-biphenyl carboxylate)		421		1432
Bisphenol A + Carboxylate		420		1432
Bisphenol A + Terephthalate + Isophthalate 50/50		468		1432
Bisphenol A of 4,4′-(2,2′-propylidene)-diphenol		313		1431
Bisphenol AP + 2,2-(fm-biphenyl carboxylate)		445		1432
Bisphenol AP + Carboxylate		460		1432
Bisphenol AP + Terephthalate + Isophthalate		516		1432
Bisphenol A-based homopoly(formal)		361	Elsewhere 367 K	1311
Bisphenol AF-based homopoly(formal)		396		1311
Cyclohexane-bisphenol-poly(carbonate)		458		1353
Norbanane-bisphenol-poly(carbonate)		506		1353
Poly(1,4-cyclohexanecarbonate)		411		1393
Poly(2-hydroxypropyl ether Bisphenol A)		370		1516
Polyarylate of Bisphenol A (1 mol), isophthalic acid (0.5 mol) and terephthalic acid (0.5 mol)		472		1254
Polyarylate of Bisphenol A (1 mol), isophthalic acid (0.7 mol) and terephthalic acid (0.3 mol)		466		1254
Polyarylate of Bisphenol A and isophthalic acid		457		1254

TABLE 2. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(cyclobutene) erythro di-isotactic		293	Degree of polymerization 150	572
erythro di-syndiotactic		273	Degree of polymerization 15	
Poly(decafluoro-1,3-cyclohexylenedifluoromethylene)		374?	Heating rate: 5–20 K/min	97
Poly(dodecafluoro-1,3-cycloheptylenedifluoromethylene)		471?	Heating rate: 5–20 K/min, high pressure, radiation synthesis, data queried in original paper, ring structure unproven	97
Poly(hexafluoro-1,3-cyclobutylenedifluoromethylene)		395	Heating rate: 5–20 K/min, high pressure, radiation synthesis, ring structure unproven; [η] only 0.03 dl/g	97
Poly(indenylene)		358		573
Poly(octafluoro-1,3-cyclohexylenedifluoromethylene)		390	Heating rate: 5–20 K/min, high pressure, radiation synthesis, ring structure unproven	97
Poly(1,4-naphthyleneethylene)		433	Mechanical method	565

TABLE 3. MAIN-CHAIN ACYCLIC HETEROATOM POLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
3.1. MAIN-CHAIN –C–O–C– POLYMERS				
3.1.1. POLY(ANHYDRIDES)				
Poly(oxycarbonyl-1,4-phenylenehexafluorotrimethylene-1,4-phenylenecarbonyl)		271		433
Poly(oxycarbonyl-1,4-phenyleneisopropylidene-1,4-phenylenecarbonyl)		333		247
Poly(oxycarbonyl-1,4-phenylenemethylene-1,4-phenylenecarbonyl)		395		247
Poly(oxycarbonyl-1,4-phenylenepentamethylene-1,4-phenylenecarbonyl)		321	Amorphous	247
		319	Crystalline	
Poly(oxycarbonyl-1,4-phenylenetetramethylene-1,4-phenylenecarbonyl)		319		247
Poly(oxycarbonyl-1,4-phenylenethiotetramethylenethio-1,4-phenylenecarbonyl)		335		247
Poly(oxyisophthaloyl)		403	Dynamic mechanical method	398,444
3.1.2. POLY(CARBONATES)				
Bisphenol A + 2,2-(fm-biphenyl carboxylate)		421		1432
Bisphenol A + Carboxylate		420		1432
Bisphenol A + Terephthalate + Isophthalate 50/50		468		1432
Bisphenol A of 4,4′-(2,2′-propylidene)-diphenol		313		1431
Bisphenol AP + 2,2-(fm-biphenyl carboxylate)		445		1432
Bisphenol AP + Carboxylate		460		1432
Bisphenol AP + Terephthalate + Isophthalate		516		1432
Bisphenol A-based homopoly(formal)		361	Elsewhere 367 K	1311
Bisphenol AF-based homopoly(formal)		396		1311
Cyclohexane-bisphenol-poly(carbonate)		458		1353
Norbanane-bisphenol-poly(carbonate)		506		1353
Poly(1,4-cyclohexanecarbonate)		411		1393
Poly(2-hydroxypropyl ether Bisphenol A)		370		1516
Polyarylate of Bisphenol A (1 mol), isophthalic acid (0.5 mol) and terephthalic acid (0.5 mol)		472		1254
Polyarylate of Bisphenol A (1 mol), isophthalic acid (0.7 mol) and terephthalic acid (0.3 mol)		466		1254
Polyarylate of Bisphenol A and isophthalic acid		457		1254

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Polycarbonate		423		1250,1340,1353
		420		1432
		297	In vacuum of 297 K	1442
		418	DSC	1442
		420	DSC	1489
Polycarbonate of Bisphenol A see Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	24936-68-3			
Poly(2-methoxycyanurate) of Bisphenols F and A		433		1546
Poly(oxycarbonyloxy-2,6-dibromo-1,4-phenylene-isopropylidene-3,5-dibromo-1,4-phenylene)		430		333,334
Poly(oxycarbonyloxy-2-chloro-6-methyl-1,4-phenylene-isopropylidene-3-chloro-5-methyl-1,4-phenylene)		427	Mechanical method, heating rate: 1 K/min	335
Poly(oxycarbonyloxy-2-chloro-1,4-phenylene-cyclohexylidene-3-chloro-1,4-phenylene)		443–452	Mechanical method, heating rate: 1 K/min	335
Poly(oxycarbonyloxy-2-chloro-1,4-phenylene-isopropylidene-3-chloro-1,4-phenylene)		420	Mechanical method	334,335
Poly(oxycarbonyloxy-2-cyclohexyl-1,4-phenylene-isopropylidene-1,4-phenylene)		405		338
Poly(oxycarbonyloxy-2,6-dichloro-1,4-phenylene-cyclohexylidene-3,5-dichloro-1,4-phenylene)		436		334
Poly(oxycarbonyloxy-2,6-dichloro-1,4-phenylene isopropylidene-3,5-dichloro-1,4-phenylene)		453, 493, 504	Conflicting data	334,336,337
Poly(oxycarbonyloxyhexamethylene)		230	Low molecular weight sample	339
Poly(oxycarbonyloxy-2-isopropyl-1,4-phenylene-isopropylidene-1,4-phenylene)		385	No experimental details	338
Poly(oxycarbonyloxy-2-methoxy-1,4-phenylene isopropylidene-1,4-phenylene)		418	No experimental details	338
Poly(oxycarbonyloxy-2,2,3,3,4,4,5,5-octafluorohexamethylene)		232	DTA heating rate	339
Poly[oxycarbonyloxy-1,3-(2,2,4,4-tetra-methylcyclobutylene)]		~500, <433, 418	Conflicting data	340–342
Poly(oxycarbonyloxy-2-methyl-1,4-phenylene-cyclohexylidene-3-methyl-1,4-phenylene)		408	Mechanical method	335,343
Poly(oxycarbonyloxy-2-methyl-1,4-phenylene-isopropylidene-3-methyl-1,4-phenylene)		368, 373, 418, 363–383	Conflicting data	334–336,343
Poly(oxycarbonyloxy-2-methyl-1,4-phenyleneisopropylidene-1,4-phenylene)		413		338
Poly(oxycarbonyloxy-2-methyl-1,4-phenylenemethylene-3-methyl-1,4-phenylene)		324	Mechanical method	335
Poly[oxycarbonyloxy-4,6-dimethyl-1,2-phenylene-methylene-3,5-dimethyl-1,2-phenylene]		410	No experimental details	338
Poly(oxycarbonyloxy-3-methyl-1,4-phenylenebenzylidene-2-methyl-1,4-phenylene)		455	Mechanical method	335
Poly(oxycarbonyloxy-1,4-phenylenebenzylidene-1,4-phenylene)		394		335,338
Poly(oxycarbonyloxy-1,4-phenylene-1,1-butylidene-1,4-phenylene)		396		344
Poly(oxycarbonyloxy-1,4-phenylene-2,2-butylidene-1,4-phenylene)		407		335,344
Poly[oxycarbonyloxy-1,4-phenylene(cyano) phenylmethylene-1,4-phenylene]		411		345
Poly(oxycarbonyloxy-1,4-phenylenecyclohexylidene-1,4-phenylene)		448		334,335,344
Poly(oxycarbonyloxy-1,4-phenylenecyclopentylidene-1,4-phenylene)		440		334,346
Poly(oxycarbonyloxy-1,4-phenylene-1,3-dichloro-1,1,3,3-tetrafluoro-2,2-propylidene-1,4-phenylene)		456	DTA heating rate	337
Poly(oxycarbonyloxy-1,4-phenyleneethylidene-1,4-phenylene)		403		344
Poly(oxycarbonyloxy-1,4-phenylenefluoren-9-ylidene-1,4-phenylene)		653	Heating rate: 10 K/min	347
Poly(oxycarbonyloxy-1,4-phenylenehexafluoro-2,2-propylidene-1,4-phenylene)		449	DTA heating rate	337
Poly(oxycarbonyloxy-1,3-phenylenehexafluorotrimethylene-1,3-phenylene)		319		148

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxycarbonyloxy-1,4-phenylene-4,4-heptylidene-1,4-phenylene)		421		334
Poly(oxycarbonyloxy-1,4-phenyleneisobutylidene-1,4-phenylene)		422		344
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,3-phenyleneisopropylidene-1,4-phenylene)		393	No experimental details	338
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	24936-68-3			1162
			Dielectric	1163
		About 417	DSC, onset, HR 10 deg/min	1161
		418		78,102,131,145, 230,233,262,263, 287,317–319,321, 335–337, 344, 348–378
		419–420	DSC, Mdpt, 10 deg/min HR, TSD, 2 deg/min HR, *f*(crystallinity)	1164
		428		1254
		417		1259
		421		1297
		420	DSC	1264
		435	No experimental details	338
Poly(oxycarbonyloxy-1,4-phenylenemethylene-1,4-phenylene)		420	No experimental details	338,367
Poly[oxycarbonyloxy-1,4-phenylene (methyl)phenylmethylene-1,4-phenylene]		449		334,336
Poly(oxycarbonyloxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene)		318		247
Poly(oxycarbonyloxy-1,4-phenylenediphenylmethylene-1,4-phenylene)		394		334
Poly(oxycarbonyloxy-1,4-phenylene-2,2-pentylidene-1,4-phenylene)		410		334
Poly[di(oxyethylene)oxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene]		314		247
Poly(oxymethyleneoxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene)		357		247
Poly[di(oxymethylene)oxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene]		325	Amorphous	247
Poly(oxypentamethyleneoxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene)		326		247
Poly(oxy-1,3-phenylenecarbonyloxycarbonyl-1,3-phenyleneoxypentamethylene)		334		247
Poly(oxy-1,3-phenylenecarbonyloxycarbonyl-1,3-phenyleneoxytetramethylene)		<293		247
3,3,5-trimethyl cyclohexane-1-polycarbonate		508		1353
Poly(oxytetramethyleneoxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene)		348		247
Poly(oxytrimethyleneoxy-1,3-phenylenecarbonyl-oxycarbonyl-1,3-phenylene)		326		247
Poly(oxytrimethyleneoxy-1,4-phenylenecarbonyl-oxycarbonyl-1,4-phenylene)		368		247
3.1.3. POLY(ESTERS)				
Polyarylate of terephthalic acid		471		1254
Poly(bisphenol A terephthalate) see Poly(oxyterephthaloyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)				
Poly(butylene adipate)		223		1331
Poly(butylene isophthalate)		291		1331
Poly(butylene 2,6-naphthalate)		321		1542
Poly(butylene naphthalate)		512		1461

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(butylene 1,4-terephthalate)		499		1461
Poly(butylene terephthalate)		304		1331
Poly(diethyl phthalate)		180		1271
Poly[(1,2-diethoxycarbonyl)ethylene]		286	Dilatomer	1158
Poly[(1,2-dimethoxycarbonyl)ethylene]		367	Dilatomer	1159
Poly(dimethyl phthalate)		192		1271
Poly[di(oxyethylene)oxyadipoyl]		227		
Poly[di(oxyethylene)oxyazelaoyl]		205		
Poly[di(oxyethylene)oxydodecanedioyl])		202		96
Poly[di(oxyethylene)oxyglutary]		226		96
Poly[di(oxyethylene)oxyheptylmalonyl]		215		96
Poly[di(oxyethylene)oxymalonyl]		244		96
Poly[di(oxyethylene)oxymethylmalonyl]		244		96
Poly[di(oxyethylene)oxynonylmalonyl]		214		96
Poly[di(oxyethylene)oxyoctadecanedioyl]		205		96
Poly[di(oxyethylene)oxyoxalyl]		265		96
Poly[di(oxyethylene)oxypentylmalony]		226		96
Poly[di(oxyethylene)oxypimeloyl]		213		96
Poly[di(oxyethylene)oxypropylmalonyl]		235		96
Poly[di(oxyethylene)oxysebacoyl]		199		96
Poly[di(oxyethylene)oxysuberoyl]		212		96
Poly[di(oxyethylene)oxysuccinyl]		244		96
Poly[di(oxy-1,4-phenylene)oxyisophthaloyl]		446	Heating rate: 20 K/min	313
Poly[di(oxy-1,4-phenylene)oxy-5-pentyloxyisophthaloyl]		399	Heating rate: 20 K/min	313
Poly[(1,2-dipropoxycarbonyl)ethylene]		262	Dilatomer	1160
Poly(oxy-2,6-diphenyl-1,4-phenylenemethylene-3,5-diphenyl-1,4-phenyleneoxysebacoyl)		365	Heating rate: 8 K/min	371
Polyesters of 3-phenyl-4,4′-biphenyl dicarboxylic acid, aromatic		398–443		1409
Poly(ethylene adipate) see Poly(oxyethyleneoxyadipoyl)	24938-37-2			
Poly(ethylene naphthalate)		534		1461
Poly(ethylene 2,6-naphthalate)		398		1542
Poly(ethylene 2,6-naphthalene-dicarboxylate)		398		1341,1554
Poly(ethylene 1,4-terephthalate)		526		1461
Poly(ethylene terephthalate) see Poly(oxyethyleneoxyterephthaloyl)	25038-59-9			
Poly(lactic acid)		327		1427
Poly(oxyadipoyloxy-2,6-dichloro-1,4-phenylene-isopropylidene-3,5-dichloro-1,4-phenylene)		383	No measurement details	380
Poly(oxyadipoyloxy-3,3′,5,5′-tetramethyl-4,4′-biphenylylene)		381	Heating rate: 8 K/min	371
Poly(oxyadipoyloxydecamethylene)		217		96,381
Poly(oxyadipoyloxy-2,6-dimethyl-1,4-phenylene isopropylidene-3,5-dimethyl-1,4-phenylene)		366	Heating rate: 8 K/min	371
Poly(oxyadipoyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)		341		382
Poly(oxyadipoyloxy-2,6-diphenyl-1,4-phenylenemethylene-3,5-diphenyl-1,4-phenylene)		388		305,371
Poly(oxy-2-butenyleneoxysebacoyl) *cis*		232		383
trans		233		
Poly(oxy-5-butyl-1,3-phenyleneoxyisophthaloyl)		359		384
Poly(oxy-2-butynyleneoxysebacoyl)		246		383,429
Poly(oxycarbonyl-3,3′-biphenylylenecarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)		460	Heating rate: 8 K/min	386
Poly(oxycarbonyl-1,4-cyclohexylenecarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene) *trans*		423	Heating rate: 20 K/min	42
Poly(oxycarbonyl-1,5-dimethylpentamethylene)		240	Torsion pendulum	1173
Poly(oxycarbonyl-3-methylpentamethylene)		220	Torsion pendulum	1172
Poly(oxycarbonyl-2,6-naphthylene-carbonyloxydecamethylene)		287	No measurement details	387
Poly(oxycarbonylpentamethylene)		213	Torsion pendulum	1170
Poly(oxycarbonyl-1,3-phenyleneoxy-1,3-phenylenecarbonyloxy-2,2,3,3,4,4-hexafluoropentamethylene)		293		148

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxydibutyltinoxyadipoyl)		383	Heating rate: 20 K/min	385
Poly(oxydibutyltinoxyfurnaroyl)		459	Heating rate: 20 K/min	385
Poly(oxydibutyltinoxyterephthaloyl)		588	Heating rate: 20 K/min	385
Poly(oxy-2,6-dimethyl-1,4-phenyleneisopropylidene-3,5-dimethyl-1,4-phenyleneoxysebacoyl)		318	Heating rate: 8 K/min	371
Poly(oxy-2,5-dimethylterephthaloyloxy-1,4-phenylenisopropylidene-1,4-phenylene)		457	Mechanical method	366
Poly(oxyethyleneoxycarbonyl-1,4-cyclohexylenecarbonyl) *trans*		291	Heating rate: 20 K/min	42
Poly(oxyethyleneoxycarbonyl-2,2′-dimethyl-4,4′-biphenylylenecarbonyl)		346		391
Poly(oxyethyleneoxycarbonyl-1,4-naphthylenecarbonyl)		337		393
Poly(oxyethyleneoxycarbonyl-1,5-naphthylenecarbonyl)		344	Amorphous	393
		351	Crystalline	
Poly(oxyethyleneoxycarbonyl-2,6-naphthylenecarbonyl)		386		387,393
Poly(oxyethyleneoxycarbonyl-2,7-naphthylenecarbonyl)		392		393
Poly(oxyethyleneoxycarbonyl-1,4-phenylene-*sec*-butylidene-1,4-phenylenecarbonyl)		~380		165
Poly(oxyethyleneoxycarbonyl-1,1,3-trimethylindan-3,5-ylene-1,4-phenylenecarbonyl)		427	DTA heating rate	392
Poly(oxyethyleneoxyisophthaloyl)		324		393
Poly(oxyethyleneoxyterephthaloyl)	25038-59-9			1166
		333–358	Dielectric, *f*(annealing)	1165
		334	DSC, onset, *X*p zero HR, f(annealing)	1168
		335–350	DSC, onset, *f*(HR, draw ratio)	1167
		342		1,42,61,78,94,183,202,233,261,262,272,319,344,349,352–354,360,362,371,381,388,389,393–418
		359		1169
		355		1254
		355–357	DSC	1338
		337		1432
amorphous		333–349		1383
Poly(oxy-5-ethyl-1,3-phenyleneoxyisophthaloyl)		395		384
Poly(oxyglutaryloxy-2,2,3,3,4,4-hexafluoropentamethylene)		218–223	Brittle point	422
Poly(oxyglutaryloxy-2,2,3,3,4,4,5,5-octafluorohexamethylene)		218–223	Brittle point	422
Poly(oxy-3-heptafluoropropylglutaryloxy-2,2,3,3,4,4-hexafluoropentamethylene)		243–248	Brittle point	422
Poly(oxy-2,2,3,3,4,4-hexafluoropenta-methyleneoxyadipoyl)		216		419–423
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethyleneoxycarbonyl-3,3′-biphenylylenecarbonyl)		318		148
Poly(oxy-2,2,3,3,4,4-hexafluoropenta-methyleneoxycarbonyl-1,3-phenylenedecafluoropentamethylene-1,3-phenylenecarbonyl)		301	Heating rate: 32 K/min; η_{inh} low	148
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethyleneoxycarbonyl-1,3-phenylenehexafluorotrimethylene-1,3-phenylenecarbonyl)		290		148
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethyleneoxy-3,6-dithiaoctanedioyl)		~233	Brittle point	422
Poly(oxy-3-heptafluoropropylglutaryloxy-2,2,3,3,4,4,5,5-octafluorohexamethylene)		~248	Brittle point	422
Poly(oxyhexamethyleneoxycarbonyl-2,6-naphthylenecarbonyl)		317	No measurement details	387
Poly(oxy-5-hexyl-1,3-phenyleneoxyisophthaloyl)		335		384
Poly(3-hydroxybutyrate)		233		1278
Poly(4-hydroxybutyrate)		223	NMR	1494
Poly(oxyisophthaloyloxy-4,4′-biphenylylene)		437, 583	Conflicting data	131,424

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxyisophthaloyloxy-2,2,3,3,4,4-hexafluoropentamethylene)		298	Heating rate: 32 K/min; η_{inh} 0.14 dl/g only	386,425
Poly(oxyisophthaloyloxy-2-methyl-1,4-phenylene isopropylidene-3-methyl-1,4-phenylene)		438	Dynamic method	366
Poly(oxyisophthaloyloxy-2,6-dimethyl-1,4-phenylene isopropylidene-3,5-dimethyl-1,4-phenylene)		498	Heating rate: 8 K/min	371
Poly(oxyisophthaloyloxy-2-methyl-1,4-phenylene methylene-3-methyl-1,4-phenylene)		418	Mechanical method	369
Poly(oxyisophthaloyloxy-2,6-dimethyl-1,4-phenylene methylene-3,5-dimethyl-1,4-phenylene)		461	Mechanical method	369
Poly(oxyisophthaloyloxy-1,4-phenylenebenzylidene-1,4-phenylene)		433		426
Poly(oxyisophthaloyloxy-1,4-phenylenecyclohexylidene-1,4-phenylene)		400		145
Poly(oxyisophthaloyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)		462	Heating rate: 8 K/min	145,369,371,427
Poly(oxyisophthaloyloxy-1,4-phenylenemethylene-1,4-phenylene)		423	Heating rate: 20 K/min	313
Poly(oxymethylene-1,4-cyclohexylenemethyleneoxycarbonyl-*trans*-1,4-cyclohexylenecarbonyl)		325	Heating rate: 20 K/min	42
Poly(oxy-5-methyl-1,3-phenyleneoxyisophthaloyl)		426		384
Poly(oxyneopentyleneoxycarbonyl-1,4-cyclohexylenecarbonyl) *trans*		303	Heating rate: 20 K/min	42
Poly(oxyneopentyleneoxyterephthaloyl)		341	Heating rate: 20 K/min	42
Poly(oxy-5-nonyl-1,3-phenyleneoxy-2-fluoroisophthaloyl)		307		432
Poly(oxy-5-nonyl-1,3-phenyleneoxy-5-fluoroisophthaloyl)		293		255
Poly(oxy-5-nonyl-1,3-phenyleneoxyisophthaloyl)		304		384
Poly(oxy-2,2,3,3,4,4,5,5-octafluorohexamethyleneoxy-3,6-dithiaoctanedioyl)		~235	Brittle point	422
Poly(oxy-5-octyl-1,3-phenyleneoxyisophthaloyl)		314		384
Poly[oxy-5-(pentadecafluoroheptyl)isophthaloyloxy-2,2,3,3,4,4-hexafluoropentamethylene]		300	Heating rate: 32 K/min; η_{inh} only 0.10 dl/g	386
Poly(oxypentamethyleneoxyadipoyl)		204		96
Poly(oxypentamethyleneoxycarbonyl-2,6-naphthylenecarbonyl)		311	No measurement details	387
Poly(oxypentamethyleneoxyterephthaloyl)		283	DTA heating rate	411
Poly(oxy-5-pentyloxyisophthaloyloxy-4,4′-biphenylene)		411	Heating rate: 20 K/min	313
Poly(oxy-2-pentyloxyisophthaloyloxy-2,2,3,3,4,4-hexafluoropentamethylene)		290	DTA heating rate; η_{inh} only 0.12 dl/g	425
Poly(oxy-4-pentyloxyisophthaloyloxy-2,2,3,3,4,4-hexafluoropentamethylene)		282	DTA heating rate; η_{inh} only 0.25 dl/g	425
Poly(oxy-5-pentyloxyisophthaloyloxy-2,2,3,3,4,4-hexafluoropentamethylene)		243	DTA heating rate	425
Poly(oxy-5-pentyloxyisophthaloyloxy-1,4-phenylenemethylene-1,4-phenylene)		383	Heating rate: 20 K/min; η_{inh} only 0.13	313
Poly(oxy-1,4-phenylenefluoren-9-ylidene-1,4-phenyleneoxysebacoyl)		424	Heating rate: 10 K/min	347
Poly(oxy-1,3-phenylenehexafluorotrimethylene-1,3-phenyleneoxycarbonyl-1,3-phenylene-hexafluorotrimethylene-1,3-phenylenecarbonyl)		345		148
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxycarbonyl-1,4-phenylene-*sec*-butylidene-1,4-phenylenecarbonyl)		~480		165
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxycarbonyl-1,3-phenylene-hexafluorotrimethylene-1,3-phenylenecarbonyl)		389		433
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxysebacoyl)		280	Heating rate: 8 K/min	363,371,409
Poly[oxy-1,3-phenyleneoxy-5-(10H-eicosafluorodecyl) isophthaloyl]		356		432
Poly[oxy-1,3-phenyleneoxy-5-(pentadecafluoroheptyl)isophthaloyl]		374		432
Poly(oxy-1,3-phenyleneoxy-2-fluoroisophthaloyl)		393	Heating rate: 32 K/min	255

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly[oxy-1,3-phenyleneoxy-5-(heptafluoropropyl)isophthaloyl]		394	Sample of low η_{inh}	434
Poly(oxy-1,3-phenyleneoxyisophthaloyl)		411	Values range from 411 to 463 K	313,384,427, 435–437
Poly(oxy-1,3-phenyleneoxy-5-pentyloxyisophthaloyl)		269	Heating rate: 20 K/min; η_{inh} only 0.13	313
Poly(oxypimeloyloxy-3,3′,5,5′-tetramethyl-4,4′-biphenylylene)		369	Heating rate: 8 K/min	371
Poly(oxypimeloyloxy-2,6-dimethyl-1,4-phenylene-isopropylidene-3,5-dimethyl-1,4-phenylene)		357	Heating rate: 8 K/min	371
Poly(oxypimeloyloxy-2,6-diphenyl-1,4-phenylene-methylene-3,5-diphenyl-1,4-phenylene)		384	Heating rate: 8 K/min	371
Poly(oxypropyleneoxycarbonyl-2,6-naphthylenecarbonyl)		361	No measurement details	387
Poly(oxypropyleneoxyterephthaloyl)		341		389
Poly(oxy-5-propyl-1,3-phenyleneoxyisophthaloyl)		394		384
Poly(oxyterephthaloyloxy-2-butyl-1,4-phenylene isopropylidene-3-butyl-1,4-phenylene)		433		439
Poly(oxyterephthaloyloxy-2-*sec*-butyl-1,4-phenylene-isopropylidene-3-*sec*-butyl-1,4-phenylene)		373	Mechanical method	439
Poly(oxyterephthloyloxy-2-chloro-1,4-phenylene-isopropylidene-3-chloro-1,4-phenylene)		463	Mechanical method	439
Poly(oxyterephthloyloxy-2-isopropyl-1,4-phenylene-isopropylidene-3-isopropyl-1,4-phenylene)		403		439
Poly(oxyterephthaloyloxydecamethylene)		298, 268	Conflicting data	6,403,411
Poly(oxyterephthaloyloxyheptamethylene)		276	DTA heating rate	411
Poly(oxyterephthaloyloxyhexamethylene)		318, 264	Conflicting data	6,403,411
Poly(oxyterephthaloyloxynonamethylene)		308, 270	Conflicting data	6,403,411
Poly(oxyterephthaloyloxyoctamethylene)		318	Dynamic method	403
Poly(oxyterephthaloyloxypentamethylene)		318	Dynamic method	6,403
Poly(oxyterephthaloyloxymethylene-1,4-cyclohexylenemethylene) 70% *trans*		358	Heating rate: 20 K/min	42,440
Poly(oxyterephthaloyloxy-2-methyl-1,4-phenylene-isopropylidene-3-methyl-1,4-phenylene)		~428	Dilatometric method	366,439,411
		461	Dynamic method	
Poly(oxyterephthaloyloxy-2,6-dimethyl-1,4-phenylene-isopropylidene-3,5-dimethyl-1,4-phenylene)		498	Heating rate: 8 K/min	371
Poly(oxyterephthaloyloxy-1,4-phenylene) 9,9-anthronylidene-1,4-phenylene)		570	Heating rate: 10 K/min	347
Poly(oxyterephthaloyloxy-1,4-phenylenebenzylidene-1,4-phenylene)		473		426
Poly(oxyterephthaloyloxy-1,4-phenylene-cyclohexylmethylene-1,4-phenylene)		543	No measurement details	442
Poly(oxyterephthaloyloxy-1,4-phenylenefluoren-9-ylidene-1,4-phenylene)		654	Heating rate: 10 K/min	347
Poly(oxyterephthaloyloxy-1,4-phenylene-hexafluoroisopropylidene-1,4-phenylene)		534	Softening point	443
Poly(oxyterephthaloyloxy-1,4-phenylene-1,1-indanylidene-1,4-phenylene)		608	Heating rate: 10 K/min	347
Poly(oxyterephthaloyloxy-1,4-phenyleneisoproplyidene-1,4-phenylene)		478	DTA heating rate	363,369,371,382, 427,439,441,443
Poly(oxy-3,3′,5,5′-tetramethyl-4,4′-biphenylyleneoxysebacoyl)		330	Heating rate: 8 K/min	371
Poly(oxy-2,2,4,4-tetramethyl-1,3-cyclobutyleneoxycarbonyl-*trans*-1,4-cyclohexylenecarbonyl)		442	Heating rate: 20 K/min	42
Poly(oxy-2,2,4,4-tetramethyl-1,3-cyclobutylene oxyterephthaloyl) 60–68% *trans*		457	Heating rate: 20 K/min	42,319,428
Poly(oxytetramethyleneoxyadipoyl)		155	Transition at 194–206 K, 1st order (Ref. 243), rate: 10 K/min	80,96,243
Poly(oxytetramethyleneoxycarbonyl-1,4-cyclohexylenecarbonyl)		263	Heating rate: 20 K/min	42
Poly(oxytetramethyleneoxycarbonyl-2,6-naphthylenecarbonyl)		349	No measurement details	387
Poly(oxytrimethyleneoxycarbonyl-1,4-cyclohexylenecarbonyl) *trans*		267	Heating rate: 20 K/min	42

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxytrimethyleneoxycarbonyl-2,6-naphthylenecarbonyl)		346	No measurement details	387
Poly(oxytetramethyleneoxysebacoyl)		216	T_g decreases as intrinsic viscosity increases	1,383,429
Poly(oxytetramethyleneoxyterephthaloyl)		290, 353	Conflicting data	6,42,403,411
Poly(oxy-3,3′,5,5′-tetraphenyl-4,4′-biphenylylenesebacoyl)		371	Heating rate: 8 K/min	371
Poly(oxytrimethyleneoxyadipoyl)		214		96
Poly(oxytrimethyleneoxyterephthaloyl)		368, 308	Conflicting data	42,183,403,411
Poly(oxy-5-tridecyl-1,3-phenyleneoxyisophthaloyl)		291		384
Poly(oxy-5-undecyl-1,3-phenyleneoxyisophthaloyl)		295		384
Poly(resorcinol isophthalate) see Poly(oxy-1,3-phenyleneoxyisophthaloyl)				
Poly(trifluoro-chloroethylene)		352		1432
Poly(1,4,7-trioxy-3,3,5,5-tetrafluoro-heptamethylenecarbonyl-1,3-phenylene-hexafluorotrimethylene-1,3-phenylenecarbonyl)		303	Heating rate: 32 K/min	218
Poly(1,4,7-trioxy-3,3,5,5-tetrafluoro-heptamethyleneisophthaloyl)		301	Heating rate: 32 K/min; η_{inh} 0.23 only	218
Poly(1,4,7-trioxy-3,3,5,5-tetrafluoroheptamethylene-5-pentyloxyisophthaloyl)		287	Heating rate: 32 K/min; [η] only 0.24 dl/g	218
3.1.4. POLY(ETHER KETONES)				
Biphenylene-ether-ketone		454		1375
Methyl poly(aryl ether ether ketone)		418		1551
Poly(aryl ether ether ketone)		423		1291
		429		1291
Poly(aryl ether ketone)		491		1406
		495		1507
Poly(aryl ether ketone) of 1,8-bis(4-fluorobenzoyl) naphthalene and aromatic bisphenols		487–543		1525
Poly(ether ether ketone)		420		1344
		425		1472
		418	DSC with 10% crystallinity	1332
		430	DSC with 28% crystallinity	1332
		405		1260
		431		1432
Poly(ether ketone),fluorinated		428–496		1385
Poly(phenylene ether ether ketone), isopropyl subs		418		1399
Poly(phosphene oxide ether ketone)		473		1368
3.1.5. POLY(OXIDES)				
Poly(acetaldehyde) see Poly(oxyethylidene)				
Poly[3,3-bis(chloromethyl)oxacyclobutane] see Poly[oxy-2,2-bis(chloromethyl)trimethylene]				
Poly(1,2-dihydroxypropane)	25322-69-4	200		1281
Br-Poly(2,6-dimethyl-1,4-phenylene oxide), linear		481		1326
Poly(2,6-dimethyl-1,4-phenylene oxide) [PPO] see Poly(oxy-2,6-dimethyl-1,4-phenylene)	25134-01-4			
Poly[di(oxymethylene)oxy-2,2,3,3,4,4-hexafluoropentamethylene]		218		285
Poly[di(oxy-1,4-phenylene)carbonyl-1,4-phenylene]		433	Mechanical method	116
Poly(epichlorohydrin) see Poly[oxy(chloromethyl)ethylene]	24969-06-0			
Poly(ethylene oxide)s see Poly(oxyethylene)s	25322-68-3			
Poly(ethylene oxide) 1000		209		1266
Poly(ethylene oxide) 400		210		1266
Poly(ethylene oxide) 600		210		1266
Poly(formaldehyde) see Poly(oxymethylene)	9002-81-7			
Poly(methylene oxide) see Poly(oxymethylene)				
Poly(4,4-(2,7-napthalenedioxy) dibenzoic acid)		471		1374
Poly(oxacyclobutane) see Poly(oxytrimethylene)				
Poly(oxetane)s see Poly(oxytrimethylene)s				
Poly(oxy-2-acetoxytrimethyleneoxy-2,6-dichloro-1,4-phenyleneisopropylidene-3,5-dichloro-1,4-phenylene)		373		219
Poly(oxy-2-acetoxytrimethyleneoxy-1,4-pheylene)		322		219

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxy-2-acetoxytrimethyleneoxy-1,4-phenylene-1-ethyl-1,3-cyclohexylene-1,4-phenylene)		380		219
Poly(oxy-2-acetoxytrimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		333		219
Poly[oxy-2-acetoxytrimethyleneoxy-1,4-phenylenemethyl(phenyl)methylene-1,4-phenylene]		383		219
Poly[oxy(allyloxymethyl)ethylene]		195		220
Poly(oxy-2-benzoyloxytrimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		338		219
Poly[oxy-2-benzoyloxytrimethyleneoxy-1,4-phenylenemethyl(phenyl)methylene-1,4-phenylene]		399		219
Poly[oxy-2-(2-biphenylyl)-6-phenyl-1,4-phenylene]		484	DSC heating rate; low viscosity	221
Poly[oxy(bromomethyl)ethylene]		259	Heating rate: 20 K/min	222
Poly(oxybutadiene)		198	Structure not given	223,224
Poly[oxy(butoxymethyl)ethylene]		194	Heating rate: 10 K/min	220
Poly(oxybutylene)		185		1432
Poly(oxybutylethylene)		203	Heating rate: 10 K/min	220
Poly(oxy-*tert*-butylethylene)		308		223,225–227
Poly[oxy-2-(4-*tert*-butylphenyl)-6-phenyl-1,4-phenylene]		513	DSC heating rate	221
Poly[oxy-2-(chloroacetoxy)trimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene]		338		219
Poly[oxy-2-(2-chlorobenzoyloxy)trimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene]		339		219
Poly[oxy(chloromethyl)ethylene]		251	Heating rate: 20 K/min	222
		248		1345
Poly[oxy-2,2-bis(chloromethyl)trimethylene]		265		1,122,230–236
Poly(oxydecylethylene)		232		220,223
Poly(oxy-2,6-dimethoxy-1,4-phenylene)		440		260
Poly(oxy-1,1-dimethylethylene)		264		223
Poly(oxy-1,2,dimethylethylene) *trans*		277		223
Poly[oxy(2,6-dimethyl-1,4-phenylene)]	25134-01-4			1156
		482		102,114,202,262, 286–300
		483–507	DSC,DTA, confused by large endotherm, *f*(TH, filler)	1155
		493		1254
		483		1276
crosslinked		469		1326
Poly(oxydiphenoxymethyleneoxy-4,4′-biphenylene)		385	DSC heating rate	379
Poly(oxydiphenoxymethyleneoxy-3,3′-dimethyl-4,4′-biphenylylene)		398	DSC heating rate	379
Poly(oxydiphenoxymethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		365	DSC heating rate	379
Poly[oxy(ethoxymethyl)ethylene]		212		220
Poly(oxy-2-ethyl-2-chloromethyltrimethylene)		~293	Dynamic method	237
Poly(oxyethylene)	25322-68-3, 9004-74-4	232?	Conflicting data, values range from 158 to 223 K	107,223,234, 238–244
		–		1153
		–		1154
		206		1432,1287
		340		1316
		165		1255
		373		1422
		220		1271,1478
with dipivaloyl methane		340		1316
with neodymium nitrate		222		1316
Poly(oxyethylethylene)		203		220
Poly(oxyethylidene)		243		250,251
		253		1304
Poly[oxy-1-(heptafluoro-2-propoxymethyl)ethylene]		~230	DTA heating rate	258
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethyleneoxy-4,4′-octafluorobiphenylylene)		314	Heating rate: 32 K/min	255

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxy-2,2,3,3,4,4-hexafluoro-pentamethyleneoxymethylene-1,4-phenylenemethylene)		247	Heating rate: 32 K/min	256
Poly(oxyhexylethylene)		206	Heating rate: 10 K/min	220
Poly[oxy(hexyloxymethyl)ethylene]		190		220
Poly(oxy-2-hydroxytrimethyleneoxy-2-chloro-1,4-phenyleneisopropylidene-3-chloro-1,4-phenylene)		358		259
Poly(oxy-2-hydroxytrimethyleneoxy-2,6-dichloro-1,4-phenyleneisopropylidene3,5-dichloro-1,4-phenylene)		388		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene)		333		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene-1-ethyl-1,4-cyclohexylene-1,4-phenylene)		413		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenyleneisobutylidene-1,4-phenylene)		368		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		373		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene-5-isopropyl-2-methyl-1,2-cyclohexylene-1,4-phenylene)		448		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene-1-methyl-1,4-cyclohexyleneisopropylidene-1,4-phenylene)		408		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylenemethylene-1,4-phenylene)		353		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene-3,3-dimethyl-1,6-indanylene)		393		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylene-1,3,3-trimethyltrimethylene-1,4-phenylene)		348		259
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylenemethyl(phenyl)methylene-1,4-phenylene)		388		219
Poly[oxy(methoxymethyl)ethylene]		211	Heating rate: 10 K/min	220
Poly(oxymethylene)	9002-81-7			1152
		191	Values range from 188 to 263 K	1,6,68,191,230, 233,238,239, 261–271
		200		1432
		260		1254
		183		1304
		190		1287
Poly(oxypropylene)	25322-69-4	205		1304
		200		1394
diol		202		1314
diol-based urethane prepolymer		225		1258
Poly(oxymethyleneoxy-2,2,3,3,4,4-hexafluoropentamethylene)		220	Heating rate: 32 K/min	284
Poly(oxymethyleneoxy-2,2,3,3,4,4,5,5-octafluorohexamethylene)		220	Heating rate: 32 K/min	284
Poly(oxymethylene-1,3-phenylenemethyleneoxy-2,2,3,3,4,4-hexafluoropentamethylene)		238		256
Poly(oxymethylene-1,4-phenyleneoxy-1,4-phenylenemethyleneoxy-2,2,3,3,4,4-hexafluoropentamethylene)		279	Heating rate: 32 K/min; high rate effect	256
Poly(oxy-2-methyl-6-phenyl-1,4-phenylene)		428		299
Poly(oxy-1-methyltrimethylene)		223	Dynamic method	281
Poly(oxy-2-methyltrimethylene)		218	Dynamic method	281
Poly[oxy-2-(1-naphthyl)-6-phenyl-1,4-phenylene]		507	DSC heating rate; low viscosity	221
Poly(oxy-3-nitro-1,3-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		423	Mechanical method	116
Poly(oxyoctafluorotetramethylene)		~208	Heating rate: 10 K/min	254,257
Poly(oxy-1,3-phenylene)		~318	DTA heating rate; low viscosity	114,295
Poly(oxy-1,4-phenylene)		358	DTA heating rate	288,305–307
Poly(oxyphenylene)	9041-80-9	363		1432
Poly(oxy-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-4-methyl-1,4-cyclohexylene-1,4-phenylene)		493	Mechanical method	116

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxy-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		428	Mechanical method	116
Poly(oxy-1,4-phenylene-2,2-dicyanotrimethylene-1,4-phenylene)		420	DTA heating rate	310
Poly[oxy-1,4-phenylene-(2-cyano)-2-phenyltrimethylene-1,4-phenylene]		416	Proposed structure; DTA heating rate	310
Poly[oxy-1,4-phenylene-2,2-di(ethoxycarbonyl)trimethylene-1,4-phenylene]		327	DTA heating rate	310
Poly(oxy-1,4-phenylenehexafluoro-2,2-propylidene-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene]		448	Mechanical method	116
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenyleneazo-1,4-phenylene)		448	Mechanical method	116
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenyleneterephthaloyl-1,4-phenylene)		438	Mechanical method	116
Poly(oxy-1,4-phenylenemethylenefluoroen-9-ylidenemethylene-1,4-phenylene)		411	Proposed structure; DTA heating rate	310
Poly(oxyphenylethylene)		313		220,223, 225,324
Poly(oxy-2,6-diphenyl-1,4-phenylene)		493		464,730,477
		498	DSC heating rate; low zero HR	1157
Poly[oxy-6-phenyl-2-(3′-*o*-terphenylyl)-1,4-phenylene]		485	DSC heating rate; low viscosity	221
Poly[oxy-6-phenyl-2-(*m*-tolyl)-1,4-phenylene]		492	DSC heating rate; low viscosity	221
Poly[oxy-6-phenyl-2-(*p*-tolyl)-1,4-phenylene]		491	DSC heating rate; low viscosity	221
Poly(oxy-2-propionyloxytrimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		331		219
Poly(oxypropylene)		198	Conflicting data	220,223,243, 251,269,273, 276,281, 326–332,1432
		201		1271
Poly[oxy-1-(2,2,3,3-tetrafluorocyclobutyl)ethylene]		248	Mechanical method, heating rate: 2 K/min	252
Poly(oxytetrafluoroethylene)		225	Low molecular weight	253,254
Poly(oxytetramethylene)	25190-06-1	189		237,238,243, 272–278,1287
		178		1304
Poly(oxy-2,2,2-trichloroethylethylene)		271	No measurement details	229
Poly(oxytrimethylene)		195		238,272,273, 278–281,1287
Poly(perfluorotrimethylene oxide)		172		1287
Poly(phenylene oxide)s see Poly(oxyphenylene)s	9041-80-9			
Poly(tetrahydrofuran) see Poly(oxytetramethylene)	25190-06-1			
Poly(trihydroxypropane)		347		1271
Poly[tri(oxy-1,4-phenylene)carbonyl-1,4-phenylene]		423	Mechanical method	116
3.1.6. POLY(URETHANES)				
Poly(oxy-2-butenylene oxycarbonyliminohexamethylene iminocarbonyl)		234		383
cis–trans		228		429
Poly(oxycarbonyliminodecamethylene iminocarbonyloxyhexadecamethylene)		307	Heating rate: 1 K/min	445
Poly(oxycarbonyliminohexamethylene iminocarbonyloxydecamethylene)		309	Heating rate: 1 K/min	445
Poly(oxycarbonyliminohexamethylene iminocarbonyloxyhexadecamethylene)		312	Heating rate: 1 K/min	445
Poly(oxycarbonyliminomethylene-1,4-phenylene methyleneiminocarbonyloxydecamethylene)		322	Heating rate: 1 K/min	445
Poly(oxycarbonyliminomethylene-1,4-phenylene methyleneiminocarbonyloxydodecamethylene)		318	Heating rate: 1 K/min	445
Poly(oxycarbonyliminomethylene-1,4-phenylene methyleneiminocarbonyloxyhexadecamethylene)		320	Heating rate: 1 K/min	445
Poly(oxycarbonylimino-4-methyl-1,3-phenylene iminocarbonyloxydecamethylene)		291	Heating rate: 10 K/min	446
Poly(oxycarbonylimino-4-methyl-1,3-phenylene iminocarbonyloxynonamethylene)		335	Heating rate: 10 K/min	446

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TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxycarbonylimino-4-methyl-1,3-phenylene iminocarbonyloxyoctamethylene)		337	Heating rate: 10 K/min	446
Poly(oxycarbonylimino-1,4-phenylenemethylene-1,4-phenylene iminocarbonyloxyhexadecamethylene)		313	Heating rate: 1 K/min	445
Poly(oxydecamethyleneoxycarbonyliminohexamethylene-iminocarbonyl)		328	Heating rate: 10 K/min	446
Poly(oxy-2,2-diethyltrimethyleneoxycarbonylimino-4-methyl-1,3-phenylene iminocarbonyl)		213		448
Poly(oxydodecamethyleneoxycarbonyliminodecamethylene iminocarbonyl)		307	Heating rate: 1 K/min	445
Poly(oxyethyleneoxycarbonyliminodecamethylene iminocarbonyl)		334		447
Poly(oxyethyleneoxycarbonyliminohexamethylene iminocarbonyl)		329	Heating rate: 10 K/min	446
Poly(oxyethyleneoxycarbonyliminononamethylene iminocarbonyl)		317		447
Poly(oxyethyleneoxycarbonylimino-4-methyl-1,3-phenylene iminocarbonyl)		326	Heating rate: 10 K/min	446,448
Poly(oxyethyleneoxycarbonylimino-1,4-phenylene ethylene-1,4-phenyleneiminocarbonyl)		390		447
Poly(oxyethyleneoxycarbonylimino-1,4-phenylene methylene-1,4-phenyleneiminocarbonyl)		412, 366	Conflicting data	446,447
Poly(oxyethyleneoxycarbonylimino-1,4-phenylene tetramethylene-1,4-phenyleneiminocarbonyl)		379		447
Poly(oxyethyleneoxycarbonylimino-1,4-phenylene trimethylene-1,4-phenyleneiminocarbonyl)		347		447
Poly(oxyheptamethyleneoxycarbonyliminohexamethylene iminocarbonyl)		328	Heating rate: 10 K/min	446
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethylene oxycarbonyliminohexamethyleneiminocarbonyl)		289	DTA heating rate	449
Poly(oxy-2,2,3,3,4,4-hexafluoropentamethylene oxycarbonylimino-2,2,3,3,4,4,5,5-octafluoro hexamethyleneiminocarbonyl)		298	DTA heating rate	449
Poly(oxydecamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		321	Heating rate: 1 K/min	445,446
Poly(oxydodecamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		316	Heating rate: 1 K/min	445
Poly(oxyheptamethyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		334	Heating rate: 10 K/min	446
Poly(oxyheptamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		357	Heating rate: 10 K/min	446
Poly(oxyhexamethyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		305	Heating rate: 10 K/min	446
Poly(oxyhexamethyleneoxycarbonliminomethylene-1,4-phenylenemethyleneiminocarbonyl)		329	Heating rate: 1 K/min	445
Poly(oxyhexamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		324	Heating rate: 1 K/min	445,446
Poly(oxymethylene-5-*tert*-butyl-1,3-phenylenemethylene oxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		387	DTA heating rate	453
Poly(oxymethylene-1,3-phenylenemethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		379	Viscosity only 0.20	453
Poly(oxyneopentyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		241	DTA heating rate	448
Poly(oxynonamethyleneoxycarbonyliminohexamethylene iminocarbonyl)		331	Heating rate: 10 K/min	446
Poly(oxynonamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		345	Heating rate: 10 K/min	446
Poly(oxyhexamethyleneoxycarbonyliminohexamethylene iminocarbonyl)		332	Heating rate: 10 K/min	446
Poly(oxyhexamethyleneoxycarbonylimino-2,2,3,3,4,4,5,5-octafluorohexamethyleneiminocarbonyl)		278	DTA heating rate	451,452
Poly(oxy-2,2,3,3,4,4,5,5-octafluorohexamethylene oxycarbonyliminohexamethyleneiminocarbonyl)		271	DTA heating rate	451,452

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxy-2,2,3,3,4,4,5,5-octafluorohexamethylene oxycarbonylimino-2,2,3,3,4,4,5,5-octafluoro-hexamethyleneiminocarbonyl)		293	DTA heating rate	449
Poly(oxyoctamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		352	Heating rate: 10 K/min	446
Poly(oxypentamethyleneoxycarbonyliminohexamethylene iminocarbonyl)		331, 281	Conflicting data	446,449
Poly(oxypentamethyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		325	Heating rate: 10 K/min	446
Poly(oxypentamethyleneoxycarbonylimino-2,2,3,3,4,4,5,5-octafluorohexamethyleneiminocarbonyl)		306	DTA heating rate	449
Poly(oxypentamethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		368	Heating rate: 10 K/min	446
Poly(oxy-2,2,3,3-tetrafluorotetramethylene oxycarbonyliminohexamethyleneiminocarbonyl)		282	DTA heating rate	450
Poly(oxy-2,2,3,3-tetrafluorotetramethylene oxycarbonylimino-2,2,3,3,4,4,5,5-octafluoro-hexamethyleneiminocarbonyl)		311	DTA heating rate	450
Poly(oxytetramethyleneoxycarbonyliminohexamethylene iminocarbonyl)		215, 253, 303, 273, 332	Conflicting data	160,261,344,429, 446
Poly(oxytetramethyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		315	Heating rate: 10 K/min	446,454
Poly(oxytetramethyleneoxycarbonylimino-2,2,3,3,4,4,5,5-octafluorohexamethyleneiminocarbonyl)		306	DTA heating rate	450
Poly(oxytetramethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		382	Heating rate: 10 K/min	446
Poly(oxytrimethyleneoxycarbonyliminohexamethyleneimino carbonyl)		328	Heating rate: 10 K/min	446
Poly(oxytrimethyleneoxycarbonylimino-4-methyl-1,3-phenyleneiminocarbonyl)		345, 213	Conflicting data	446,448
Poly(oxytrimethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		392	Heating rate: 10 K/min	446
3.2. MAIN-CHAIN O-HETEROATOM POLYMERS				
3.2.1. NITROSO-POLYMERS				
Poly(2-bromotetrafluoroethyliminotetrafluoroethylene)		256	Heating rate: 15 K/min	455
Poly(oxy-1,1-difluoroethyliminotetrafluoroethylene)		238	No measurement details	456
Poly(oxytrifluoromethyliminotetrafluoroethylene)		219		423,457–462
Polymer from trifluoronitrosomethane and bromotrifluoroethylene		276	Heating rate: 15 K/min	455,461
Polymer from trifluoronitrosomethane and chlorotrifluoroethylene		248–253		423
Polymer from trifluoronitrosomethane and perfluorobutadiene		266, 230	Heating rate: 15 K/min conflicting data	455,461
Polymer from trifluoronitrosomethane and 1,1,2-trifluorobutadiene		218	No measuremetnt details	456
Polymer from trifluoronitrosomethane and trifluoromethyl trifluorovinyl ether		268	DSC heating rate	457
Polymer from trifluoronitrosomethane and trifluoromethyl trifluorovinyl sulfide		225	No measurement details	461
Polymer from trifluoronitrosomethane and hexafluoropropylene		264	No measurement details	456
3.2.2. POLY(SILOXANES)				
Poly(ethylene) - poly(dimethylsiloxane) 50 : 50, linear		154		1327
Poly(dimethylsiloxane)-1		150		1320
Poly(dimethylsiloxane)-3		150		1320
Poly(dimethylsiloxane)-5		150		1320
Poly(dimethylsiloxane)-7		150		1320
Poly(dimethylsiloxane) see Poly(oxydimethylsilylene)	9016-00-6			
Poly[di(oxydimethylsilylene)-1,4-phenylenedimethylsilylene]		210	DSC heating rate	478

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TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(disiloxane)		269		1313
		331		1313
Poly(methylphenylsilylenemethylene)		301		1417
Poly(oxydimethylsilylene)	9016-00-6	146		1,6,223,243,467, 469–474,1478
		129–150		1476
		150		1255
		143		1441
Poly(oxydimethylsilylene-2,4,5,6-tetrafluorophenylenedimethylsilylene)		245	No measurement details	463
Poly(oxydimethylsilyleneoxypentylene oxyisophthaloyloxyneopentylenedimethylsilylene)		221		430
Poly(oxydimethylsilyleneoxypentylene oxyterephthaloyloxyneopentylenedimethylsilylene)		238		431
Poly(oxydimethylsilyleneoxydimethylsilylene-2,4,5,6-tetrafluorophenylenedimethylsilylene)		212	No measurement details	463
Poly(oxydimethylsilyleneoxy-1,4-phenylene)		363	Mechanical method	468,476
Poly(oxydimethylsilyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		318	Mechanical method	468,476
Poly(oxydimethylsilylene-1,3-phenylenehexafluorotrimethylene-1,3-phenylenedimethylsilylene)		265	Heating rate: 32 K/min	148
Poly(oxydimethylsilylene-1,3-phenylenetetrafluoroethylene-1,3-phenylenedimethylsilylene)		271	Heating rate: 32 K/min; crystalline samples	477
Poly(oxydimethylsilylene-1,4-phenyleneoxy-1,4-phenylenedimethylsilylene)		293	Heating rate: 8 K/min	301
Poly[oxydi(pentafluorophenyl)silylenedi-(oxydimethylsilylene)]		231	No experimental details	463
Poly(oxydiphenylsilyleneoxydimethylsilylene-1,4-phenylenedimethylsilylene)		~273	DSC heating rate	478
Poly(oxydiphenylsilylene-1,3-phenylene)		~331	DTA heating rate	114
Poly[oxymethylchlorotetrafluorophenylsilylenedi-(oxydimethylsilylene)]		198	No experimental details	463
Poly(oxymethylpentafluorophenylsilylene)		248	No measurement details; [η] only 0.2 dl/g	463
Poly[oxymethylpentafluorophenylsilylenedi-(oxydimethylsilylene)]		190	No measurement details	463
Poly(oxymethylpentafluorophenylsilyleneoxydimethylsilylene)		208	No measurement details; [η] only 0.24 dl/g	463
Poly[oxy(methyl)phenylsilylene]		187		467
Poly(oxymethyl-3,3,3-trifluoropropylsilylene)		<193		421,423,464,465, 466
Poly[oxy(methyl)-3,3,3-trifluoropropylsilylene-3,3-difluoropentamethylene(methyl)-3,3,3-trifluoropropylsilylene]		235	DSC heating rate	466
Poly[oxy(methyl)-3,3,3-trifluoropropylsilylene ethyldodecafluorohexamethyleneethylene(methyl)-3,3,3-trifluoropropylsilylene]		249	DSC heating rate	466
Poly[oxy(methyl)-3,3,3-trifluoropropylsilylene ethyleneeicosafluorodecamethyleneethylene(methyl)-3,3,3-trifluoropropylsilylene]		262	DSC heating rate	466
Poly[oxy(methyl)-3,3,3-trifluoropropylsilylene ethylenehexadecafluorooctamethyleneethylene-(methyl)-3,3,3-trifluoropropylsilylene]		245	DSC heating rate	466
Poly[oxy(methyl)-3,3,3-trifluoropropylsilylene ethyleneoctafluorotetramethyleneethylene(methyl)-3,3,3-trifluoropropylsilylene]		248	DSC heating rate	466
Poly[oxy(methyl)-3,3,3-trifluoropropysilylene-3,3,4,4-tetrafluorohexamethylene(methyl)-3,3,3-trifluoropropylsilylene]		246	DSC heating rate	466
Poly[oxy(methyl)phenylsilyleneoxy-1,4-phenylene]		368	Low viscosity	468
Poly[oxy(methyl)phenylsilyleneoxy 1,4-phenyleneisopropylidene-1,4-phenylene]		331	Low viscosity	468
Poly[oxy-1,3-phenyleneoxy-5-(triphenylsiloxydimethylsilyl)isophthaloyl]		376	Heating rate: 32 K/min	438

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly[oxytri(dimethylsilylenoxy)(methyl)phenylsilylene-1,3-phenylene(methyl)phenylsilylene]		231	DTA heating rate	302
Poly[penta(oxydiethylsilylene)-1,4-phenylenedimethylsilylene]		193	DSC heating rate	302
Poly[penta(oxydiethylsilylene)-1,4-phenyleneoxy-1,4-phenylenedimethylsilylene]		208	DTA heating rate	302
Poly[tetra(oxydimethylsilylene)-1,4-phenyleneoxy-1,4-phenylenedimethylsilylene]		221	DTA heating rate	302
Poly[tetra(oxydimethylsilylene)-1,3-phenylenedimethylsilylene]		198	DSC heating rate	302
Poly[tetra(oxydimethylsilylene)-1,4-phenylenedimethylsilylene]		201	DSC heating rate	302
Poly(tetrasiloxane)		257		1313
Poly(1-trimethylsilyl-1-propane)		623		1538
Poly[tri(oxydimethylsilylene)oxy(methyl)-2-phenylethylsilylene]		171		475
Poly[tri(oxydimethylsilylene)oxy(methyl)phenylsilylene]		201		475
Poly[tri(oxydimethylsilylene)oxy(methyl)trimethylsiloxysilylene]		148		475
Poly[tri(oxydimethylsilylene)-1,4-phenylenedimethylsilylene]		211	DSC heating rate	302
Poly[tri(oxydimethylsilylene-1,4-phenyleneoxy)-1,4-phenylenedimethylsilylene]		236	DTA heating rate	302
Poly(trisiloxane)		287		1313
3.2.3. POLY(SULFONATES)				
Poly(4,4′-cyclopentylidene diphenylene toluene-2,4-disulfonate)		407		1407
Poly(oxysulfonyl-1,3-phenylenecarbonyl-1,3-phenylenesulfonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)		395	Mechanical method	361
Poly(oxysulfonyl-1,3-phenylenesulfonyloxy-1,4-phenyleneisoproplyidene-1,4-phenylene)		~385	Not well defined; mechanical method	361
3.3. MAIN-CHAIN –C–(S)$_n$–C– AND –C–S–N– POLYMERS				
3.3.1. POLY(SULFIDES)				
Poly(dithiodecamethylene)		208	DTA heating rate	489
Poly(dithioethylene)		246	Mechanical method	248,249
Poly(dithiohexamethylene)		199		248
Poly(dithiomethylene-1,4-phenylenemethylene)		296	DTA heating rate	249,489
Poly(dithiopentamethylene)		201		248
Poly(oxycarbonyloxy-2-methyl-1,4-phenylenethio-3-methyl-1,4-phenylene)		340	No experimental details	338
Poly(oxycarbonyloxy-1,4-phenylenethio-1,4-phenylene)		~383	DTA heating rate	367
Poly(oxyethylenedithioethylene)		220	Mechanical method	248,249
Poly(oxyethylenetetrathioethylene)		233		248
Poly(oxymethyleneoxyethylenedithioethylene)		214		248,249,283
Poly(oxymethyleneoxytetramethylenedithiotetramethylene)		197		248
Poly(oxytetramethylenedithiotetramethylene)		197		248,249
Poly(phenylene sulfide)		359		1497
		383		1432
10% *m*-phenylene		347		1497
Poly(*p*-phenylene sulfide)		353		1290
		365	Under reactants (diiodobenzene and phenyl ethylene)	1454
		363		1488
Poly(*p*-phenylene sulfide) + HX1000,amorphous LCP		445		1396
Poly(*p*-phenylene sulfide) + HX4000, amorphous LCP		433		1396
Poly(tetrathiodecamethylene)		197	DTA heating rate	489
Poly(tetrathioethylene)		249		248,249
Poly(tetrathiomethylene-1,4-phenylenemethylene)		276	DTA heating rate	489
Poly[thio-1-(allyloxymethyl)ethylene]		213		220
Poly(thiocarbonyl fluoride) see Poly(thiodifluoromethylene)				

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TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(thio-1,2-cyclohexylene)		~256, ~228	Conflicting data	479,480
Poly(thio-1,3-cyclohexylene)		221	DTA heating rate	480
Poly(thioethylene)	24936-67-2	223	DTA, quenched possibly	1174
		~253	Interpretation difficult	481–483
Poly(thiodifluoromethylene)		155		484–486
Poly(thio-2,2-dimethylene) Poly(thioisobutylene)	26373-01-3	259	DTA, quenched	1175
Poly(thio-1-ethylethylene)		218		220
Poly(thio-2-ethyl-2-methyltrimethylene)		~223		160
Poly(thiomethylene)		218	No measurement details	482
Poly(thio-1-methyltrimethylene)		214	DSC heating rate	490
Poly(thioneopentylene)		233	Heating rate: 20 K/min	160,480
Poly(thio-1,4-phenylene)		370	DSC heating rate	305,307, 493,494
Poly(thiopropylene)		226		220,482,483, 495,496
Poly(thiotrimethylene)		~228	Conflicting data	480,487,488
Poly(trithiodecamethylene)		203	DTA heating rate	489
Poly(trithiomethylene-1,4-phenylenemethylene)		291	DTA heating rate	489
Thiokols see Poly(thiomethylene)s				
3.3.2. POLY(SULFONES) AND POLY(SULFONAMIDES)				
Camphorquinone in poly(sulfone)		458		1316
Poly(aryl ether sulfone)		520		1507
		393–433		1378
Poly[di(oxy-1,4-phenylene)sulfonyl-1,4-phenylene]		483	Mechanical method	116
Poly(ether ether sulfone)		483		1270
Poly(ether sulfone)		498		1432,1487
Poly(ether sulfone ketone), aromatic		440–479		1514
Poly(oxy-2-acetoxytrimethyleneoxy-1,4-phenylenesulfonyl-1,4-phenylene)		403		219
Poly(oxy-4,4′-biphenylyleneoxy-1,4-phenylenesulfonyl-1,4-phenylene)		503	Mechanical method	116
Poly[oxycarbonyldi(oxy-1,4-phenylene)sulfonyl-1,4-phenyleneoxy-1,4-phenylene]		~478	Mechanical method	228
Poly(oxycarbonylneopentylenesulfonylneopentylene)		323		160
Poly(oxy-2-hydroxytrimethyleneoxy-1,4-phenylenesulfonyl-1,4-phenylene)		428		219
Poly(oxyneopentylenesulfonylneopentyleneoxycarbonylimino hexamethyleneiminocarbonyl)		303		160
Poly(oxy-5-pentyloxyisophthaloyloxy-1,4-phenylenesulfonyl-1,4-phenylene)		443	Heating rate: 20 K/min; η_{inh} only 0.11	313
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenylene-sulfonylmethyliminotetramethylene-methyliminosulfonyl-1,4-phenylene)		393	Mechanical method	116
Poly(oxy-1,4-phenylenesulfinyl-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)		478	Mechanical method	116
Poly(oxy-1,4-phenylenesulfinyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		438	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-4,4′-biphenylylenesulfonyl-1,4-phenylene)		533		184
Poly(oxy-1,4-phenylenesulfonyl-2,7-naphthylenesulfonyl-1,4-phenylene)		523		184
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenylene)		487	Calculated T_g for infinite molecular weight: 511K (Ref. 316)	116,184, 314–316
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-2-chloro-1,4-phenyleneisopropylidene-3-chloro-1,4-phenylene)		478	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-2,6-dimethyl-1,4-phenyleneisopropylidene-3,5-dimethyl-1,4-phenylene)		508	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)		478	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenecyclohexylidene-1,4-phenylene)		478	Mechanical method	116

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneethylidene-1,3-cyclohexylene-1,4-phenylene)		503	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyl(1,4-phenylene)neoxy-1,4-phenylenehexafluoro-2,2-propylidene-1,4-phenylene)		478	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneisobutylidene-1,4-phenylene)		473	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		449		102,116,262, 287,316–323
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenemethylene-1,4-phenylene)		453	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenemethylphenylmethylene-1,4-phenylene)		473	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylene-2,2-norbornylene-1,4-phenylene)		523	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenediphenylmethylene-1,4-phenylene)		503	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenethio-1,4-phenylene)		448	Mechanical method	116
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxyterephthaloyl)		522	Heating rate: 20 K/min	313
Poly(oxyterephthaloyloxyneopentylene sulfonylneopentylene)		378	? DTA heating rate; η low	160
Poly(oxytetramethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)		413	Mechanical method	116
Poly(sulfone)	25135-51-7	459		1320
		460		1432
Polysulfone of Bisphenol A and diphenylene sulfone		459		1254
Poly(sulfonyl-1,2-cyclohexylene)		401	Heating rate: 20 K/min	480
Poly(sulfonyl-1,3-cyclohexylene)		381	Heating rate: 20 K/min	480
Poly(sulfonylneopentylene)		386	Heating rate: 20 K/min	160,480
Poly(sulfonyl-1,4-phenylenemethylene-1,4-phenylene)		497	Heating rate: 40 K/min calculated T_g at infinite molecular weight; 511 K	316
Poly[tri(oxy-1,4-phenylene)sulfonyl-1,4-phenylene]		453	Mechanical method	116
3.3.3. POLY(THIOESTERS)				
Poly(thio-3-methyl-6-oxohexamethylene)		293	DTA heating rate	497
Poly(thio-1-methyl-3-oxotrimethylene)		285		498
Poly(thio-6-oxohexamethylene)		292	DTA heating rate	499
3.4. MAIN-CHAIN –C–N–C– POLYMERS				
3.4.1. POLY(AMIDES)				
Nylon		323		1320
Nylon 3 [Poly(imino-1-oxotrimethylene)]		384	Crystalline, DTA heating rate	502
Nylon 4,6 [Poly(iminotetramethyleneiminoadipoyl)]	50327-22-5	316	Dynamic method	503
Nylon 5,6 [Poly(iminopentamethyleneiminoadipoyl)]		318	Heating rate: 2 K/min	504,505
Nylon 6 [Poly(imino-1-oxohexamethylene)]	25038-54-4	–		1181,1193
		373	Dilatomer, dry	1187
		313–360	Conflicting data, values ~313–325 K	1,75,101,122, 187,191,261, 262,318,394, 396,399,416, 506–520
		228		1366
27% crys.		326–327	DSC, torsion tester, TH	1176
Nylon 6,2 [Poly(iminooxalyliminohexamethylene)]		427–432	DSC, TMA, DTA, annealed	1188
Nylon 6,6 [Poly(iminoadipoyliminohexamethylene)]	32131-17-2	~323	Conflicting data, most values about 323 K	6,75,78,161, 170,171,191, 234,261,262, 272,317,326, 344,381,388,

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
				394,396,399, 506,510,521, 522–533
27% crys.		330–331	DSC, torsion tester, TH	1177
		341		1254
Nylon 6,7 [Poly(iminohexamethyleneiminopimeloyl)]		331		492,506
Nylon 6,8 [Poly(iminohexamethyleneiminosuberoyl)]		330	Heating rate: 0.5 K/min	506
Nylon 6,9 [Poly(iminohexamethyleneiminoazelaoyl)]	28757-63-3	331		521
Nylon 6,10 [Poly(iminohexamethyleneiminosebacoyl)]	9008-66-6	323	Variable data below 323 K	75,161,167,191, 399,503,505,506, 528,534,535
19% crys.		318	DSC, torsion tester, TH	1178
Nylon 6,12 [Poly(iminohexamethyleneiminododecanedioyl)]	24936-74-1	319	Heating rate: 0.5 K/min	506,535,536
Nylon 7 [Poly(imino-1-oxoheptamethylene)]		–		1183
		325		191,506,537,538
Nylon 7,6 [Poly(iminoadipoyliminoheptamethylene)]		318, 333	Conflicting data	282,492,505,506
Nylon 7,7 [Poly(iminopimeloyliminoheptamethylene)]		328	Heating rate: 0.5 K/min	506
Nylon 8 [Poly(imino-1-oxooctamethylene)]		323		1184,538
Nylon 8,2 Poly[(iminooxalyliminooctamethylene)]46% crys.		373	Penetration, quenched	1190
		359	Penetration, annealed	1189
Nylon 8,6 [Poly(iminoadipoyliminooctamethylene)]		318	Heating rate: 2 K/min	505
Nylon 8,10 [Poly(iminooctamethyleneiminodecanedioyl)]		333	Heating rate: 2 K/min	505
Nylon 8,12 [Poly(iminoctamethyleneiminododecanedioyl)]		323	Dynamic method	535
Nylon 8,22 [Poly(iminooctamethyleneiminodocosanedioyl)]		321	Dynamic method	535
Nylon 9 [Poly(imino-1-oxononamethylene)]	25035-03-4, 25748-72-5	–		1182
		319		506,537,538
Nylon 9,6 [Poly(iminoadipoyliminononamethylene)]		318	Estimated	504
Nylon 10 [Poly(imino-1-oxodecamethylene)]		315		538
Nylon 10,2 54% crys.		330	DSC, TMA, annealed	1191
Nylon 10,6 [Poly(iminoadipoyliminodecamethylene)]		313	Heating rate: 2 K/min	505
Nylon 10,10 [Poly(iminosebacoyliminodecamethylene)]		333, 319	Conflicting data	505,506
Nylon 10,12 [Poly(iminodecamethyleneiminododecanedioyl)]		322	Dynamic method	535
Nylon 11 [Poly(imino-1-oxoundecamethylene)]	25035-04-5			1185
19% crys.		315	DSC, torsion tester, TH	1179
		316, 365	Transitions affected by thermal history and relaxation effects	538
		333–343		1483
Nylon 12 [Poly(imino-1-oxododecamethylene)]	24937-16-4	–		1186
26% crys.		313–316	DSC, torsion tester, TH	1180
		314	Affected by thermal history and relaxation effects	75,520,538
Nylon 12,2 33% crys.		330–333	DSC, TMA	1192
Nylon 12,18 [Poly(iminododecamethyleneiminooctadecanedioyl)]		323	Dynamic method	535
Nylon 13 [Poly(imino-1-oxotridecamethylene)]		314		538
Nylon 14,18 [Poly(iminotetradecamethyleneiminooctadecanedioyl)]		321	Dynamic method	535
Nylon 18,18 [Poly(iminooctadecanedioyliminooctadecamethylene)]		323	Dynamic method	535,536
Polyamide of 1,7-bis(aminophenoxy)napthalene and dicarboxylic acid		412–536		1491
Polyamide of 2,4′-biphenyldiamine		521–557		1511
Polyamide of bis(4-carboxyphenyl)phenylphosphene oxide		498–527		1445
Poly(amide-sulfonamide)		>523		1510
Poly(*N*,*N*′-bis(4′-amino-4-biphenylene isophthalamide))		498–508		1392
Poly(γ-benzyl L-glutamate)		~288	Dynamic method	273
Poly(butyliminohexafluoroglutaryl butyliminohexamethylene)		~293	Brittle point	539
Poly(butylimino-2,2,3,3,4,4-hexafluoropentamethylenebutyliminoadipoyl)		283–288	Brittle point	539
Poly(ε-caprolactam) see Nylon 6	25038-54-4			
Poly(3,4-dibenzyl terephthalamide)		476		1408
		518–600		1429
Poly[di(oxyethylene)oxycarbonyliminohexamethyleneimino carbonyl]		272	Heating rate: 1–2 K/min	245
Poly(ethyliminohexafluoroglutarylethyliminohexamethylene)		~278	Brittle point	539

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(ethylimino-2,2,3,3,4,4-hexafluoropentamethyleneethyliminoadipoyl)		293	Brittle point	539
Poly(hexamethylene adipamide) see Nylon 6,6	32131-17-2			
Poly(hydroxyamide ether)		363–406		1439
Poly(iminoadipoylimino-1,4-cyclohexylenemethylene) 1,4-cyclohexylene) 98% *trans/trans*, 2% *cis/trans*		458	T_g depends on *cis/trans* isomer content	540
Poly(iminoadipoyliminomethylene-2,5-dimethyl-1,4-phenylenemethylene)		343	DTA heating rate	303
Poly(iminoadipoyliminotrimethylenefluoren-9-ylidenetrimethylene)		393	Mechanical method, heating rate: 1 K/min	78,161
Poly(iminoadipoyliminotrimethyleneisobutylphosphinidene trimethylene)		344	DTA heating rate	492
Poly(iminoadipoyliminotrimethylenemethyliminotrimethylene)		278		492
Poly(iminoadipoyliminotrimethylenemethylphosphinylidene-tetramethylene)		332	DTA heating rate	492
Poly(iminoadipoyliminotrimethylenemethylphosphinylidene trimethylene)		332	DTA heating rate	492
Poly(iminoadipoyliminotrimethyleneoctylphosphinylidene trimethylene)		285	DTA heating rate	492
Poly[iminoadipoyliminotrimethylene-(phenylphosphinidene)trimethylene]		322		492
Poly[iminoadipoyliminotrimethylene-(phenylphosphinylidene)trimethylene]		328		492
Poly[imino-5-*tert*-butylisophthaloylimino-(2,5-dimethylhexamethylene)]		422	DTA heating rate	453
Poly[imino-5-*tert*-butylisophthaloylimino-(3,4-dimethylhexamethylene)]		446	DTA heating rate	453
Poly(imino-5-*tert*-butylisophthaloyliminohexamethylene)		436	DTA heating rate	453
Poly(imino-5-*tert*-butylisophthaloyliminomethylene-1,4-cyclohexylenemethylene)		509	DTA heating rate	453
Poly(imino-5-*tert*-butylisophthaloyliminomethylene-1,3-phenylenemethylene)		465	DTA heating rate	453
Poly(imino-5-*tert*-butylisophthaloyliminomethylene-1,4-phenylenemethylene)		477	DTA heating rate	453
Poly(iminocarbonyl-1,3-adamantylenecarbonyliminohexamethylene)		~385	Values from 378 to 393 K obtained	255
Poly(iminocarbonyl-2,2-biphenylylenecarbonyliminotrimethylenefluoroen-9-ylidenetrimethylene)		438	Mechanical method, heating rate: 1 K/min	78,161
Poly(iminocarbonyl-1,4-cyclohexylenemethylene)		466	DTA heating rate	517
Poly(iminocarbonyl-1,4-phenylene-2-oxoethyleneiminohexamethylene)		377	Heating rate: 20 K/min	541
Poly(imino-1,4-cyclohexylenemethylene-1,4-cyclohexyleneiminododecanedioyl) 98% *trans/trans*, 2% *cis/trans*		408	T_g depends on *cis/trans* isomer content	540
Poly(imino-1,4-cyclohexylenemethylene-1,4-cyclohexyleneiminosebacoyl) 98% *trans/trans*, 2% *cis/trans*		417	T_g depends on *cis/trans* isomer content	540
Poly(imino-2,2-dimethylpentamethyleneiminoadipoyl)		350	DTA heating rate	542
Poly(imino-2,2-dimethylpentamethyleneiminoazelaoyl)		336	DTA heating rate	542
Poly(imino-2,2-dimethylpentamethyleneiminopimeloyl)		344	DTA heating rate	542
Poly(imino-2,2-dimethylpentamethyleneiminoterephthaloyl)		430	DTA heating rate	308
Poly(iminoethylene-1,4-phenyleneethyleneimino-1,16-dioxohexadecamethylene)		358	Dynamic method	536
Poly(iminoethylene-1,4-phenyleneethyleneimino-1,18-dioxooctadecamethylene)		348	Dynamic method	536
Poly(iminoethylene-1,4-phenyleneethyleneimino-1,14-dioxotetradecamethylene)		366	Dynamic method	536
Poly(iminoethylene-1,4-phenyleneethyleneimino-1,11-dioxoundecamethylene)		369	Dynamic method	536
Poly(iminoethylene-1,4-phenyleneethyleneiminosebacoyl)		378	Dynamic method	536
Poly(iminoglutarylimino-2,2-dimethylpentamethylene)		355	DTA heating rate	542
Poly(iminohexamethyleneiminocarbonyl-2,2′-biphenylenecarbonyl)		400	Mechanical method, heating rate: 1 K/min	78,161

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TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(iminohexamethyleneiminocarbonyl-1,4-phenylene-2,2-butylidene-1,4-phenylenecarbonyl)		427–437		165
Poly(iminohexamethyleneimino-4-methylpimeloyl)		323		521
Poly(iminohexamethyleneimino-1-oxotrimethylenefluoren-9-ylidene-3-oxotrimethylene)		395	Mechanical method, heating rate: 1 K/min	78,161
Poly(iminohexamethyleneimino-1-oxotrimethylene-phenylphosphinothioylidene-3-oxotrimethylene)		316		492
Poly(iminohexamethyleneimino-1-oxotrimethylenephenylphosphinylidene-3-oxotrimethylene)		302		492
Poly(iminoisophthaloylimino-4,4′-biphenylylene)		558	DTA heating rate	311,312
Poly(iminoisophthaloylimino-3,4-dimethylhexamethylene)		398	DSC heating rate	453
Poly(iminoisophthaloylimino-2,2-dimethylpentamethylene)		426	DTA heating rate	308
Poly(iminoisophthaloyliminohexamethylene)		390		78,161,255,543
Poly(iminoisophthaloyliminomethylene-1,4-cyclohexylenemethylene)		481	DSC heating rate	453
Poly(iminoisophthaloyliminomethylene-1,3-phenylenemethylene)		438	Mechanical method, heating rate: 1 K/min	78,161,453
Poly(iminoisophthaloyliminooctamethylene)		388	Dynamic method	544
Poly(iminoisophthaloylimino-1,4-phenylenemethylene-1,4-phenylene)		<500	Dynamic method	311
Poly(iminoisophthaloyliminotrimethylenefluoren-9-ylidenetrimethylene)		448	Mechanical method	78,161
Poly(iminomesaconoyliminodecamethylene)		347	Heating rate: 16 K/min	545
Poly(iminomesaconoyliminohexamethylene)		367	Heating rate: 16 K/min	545
Poly(iminomesaconoyliminomethylene-1,3-phenylenemethylene)		408	Heating rate: 16 K/min	545
Poly(imino-3-methyladipoyliminohexamethylene)		290	DTA heating rate	492
Poly(iminomethylene-5-*tert*-butyl-1,3-cyclohexylenemethyleneiminoisophthaloyl)		482	DSC heating rate	453
Poly(iminomethylene-5-*tert*-butyl-1,3-phenylenemethyleneiminoadipoyl)		382	DTA heating rate	453
Poly(iminomethylene-5-*tert*-butyl-1,3-phenylenemethyleneiminoisophthaloyl)		461	DSC heating rate	453
Poly(iminomethylene-1,3-cyclohexylenemethyleneimino-5-*tert*-butylisophthaloyl)		473	DTA heating rate	453
Poly(iminomethylene-1,3-cyclohexylenemethyleneiminoisophthaloyl)		453	DSC heating rate	453
Poly[iminomethylene(2,5-dimethyl-1,4-phenylene)methyleneiminosuberoyl]		351	DTA heating rate	303
Poly(iminomethylene-1,3-phenylenemethyleneiminoadipoyl)		346	Mechanical method, heating rate: 1 K/min	78
Poly(iminomethylene-1,3-phenylenemethyleneiminocarbonyl-2,2′-biphenylenecarbonyl)		432	Mechanical method, heating rate: 1 K/min	78
Poly[iminomethylene-1,4-phenylenemethyleneiminocarbonyl (1,3,3-trimethyl-5-oxopentamethyl)]		398	No experimental details	381
Poly(iminomethylene-1,4-phenylenemethyleneiminododecanedioyl)		378		491,536
Poly(iminomethylene-1,4-phenylenemethyleneiminooctadecanedioyl)		348	Dynamic method	536
Poly(iminomethylene-1,3-phenylenemethyleneimino-1-oxotrimethylenefluoren-9-ylidene-3-oxotrimethylene)		423	Mechanical method, heating rate: 1 K/min	78,161
Poly(iminomethylene-1,4-phenylenemethyleneiminopentadecanedioyl)		363	Dynamic method	536
Poly(iminomethylene-1,4-phenylenemethyleneiminosebacoyl)		388	Dynamic method	536
Poly(iminomethylene-1,4-phenylenemethyleneiminotridecanedioyl)		373	Dynamic method	536
Poly(iminomethylene-1,4-phenylenemethyleneiminoundecanedioyl)		380		491,536
Poly(iminomethylene-1,3,3-trimethyl-1,5-cyclohexyleneiminoadipoyl)		433	No experimental details	381
Poly(iminomethylene-1,3,3-trimethyl-cyclohexyleneiminoterephthaloyl)		473	No experimental details	381

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(imino-5-methylisophthaloyliminohexamethylene)		393		255
Poly(imino-1-methyl-3-oxotrimethylene)		369	Crystalline; DTA heating rate	502
Poly(imino-1,5-naphthyleneiminoisophthaloyl)		598	DTA heating rate	312
Poly(imino-1,5-naphthyleneiminoterephthaloyl)		578	Heating rate: 10 K/min	312
Poly(iminooctamethyleneimino-1-oxoethylene-1,4-phenylene-2-oxoethylene)		383	Dynamic method	536
Poly(iminooxalylimino-2,2-dimethylpentamethylene)		382	DTA heating rate	542
Poly(imino-1-oxoethylene-1,4-phenylene-2-oxoethyleneiminooctadecamethylene)		351	Dyamic method	536
Poly(imino-1-oxotrimethylenefluoren-9-ylidene-3-oxotrimethyleneiminotrimethylenefluoren-9-ylidenetrimethylene)		438	Mechanical method, heating rate: 1 K/min	78
Poly(imino-1-oxotrimethylene-1,4-phenylene-3-oxotrimethyleneiminododecamethylene)		358	Dynamic method	536
Poly(imino-1-oxotrimethylene-1,4-phenylene-3-oxotrimethyleneiminooctadecamethylene)		338	Dynamic method	536
Poly(iminopentamethyleneiminocarbonyl-1,4-phenylene-2-oxoethylene)		376	Heating rate: 20 K/min	541
Poly(iminopentamethyleneiminoterephthaloyl)		~500	Dynamic method	518
Poly(imino-1,3-phenyleneiminocarbonyl-1,3-adamantylenecarbonyl)		580	Heating rate: 32 K/min	255
Poly(imino-1,3-phenyleneiminoisophthaloyl)		553	Mechanical method, heating rate: 2 K/min	311,518,546
Poly(imino-1,4-phenyleneiminoisophthaloyl)		<500	Dynamic method	311
Poly(imino-1,3-phenyleneimino-5-methylisophthaloyl)		578	Heating rate: 32 K/min	255
Poly(imino-1,3-phenyleneiminosebacoyl)		383	Dynamic method	544
Poly(imino-1,3-phenyleneiminoterephthaloyl)		<500	Dynamic method	311
Poly(imino-1,4-phenyleneiminoterephthaloyl)		618	Heating rate: 10 K/min	311,312
Poly(iminoterephthaloylimino-4,4′-biphenylylene)		613	DTA heating rate	311,312
Poly(iminoterephthaloylimino-3-ethylhexamethylene)		403	DSC heating rate	381
Poly(iminoterephthaloylimino-3-isopropylhexamethylene)		416	DSC heating rate	381
Poly(iminoterephthaloyliminododecamethylene)		393	Dynamic method	543,547
Poly(iminoterephthaloyliminoheptamethylene)		470	Dynamic method	518
Poly(iminoterephthaloyliminohexamethylene)		413	No experimental details	381
Poly(iminoterephthaloyliminononamethylene)		388	No experimental details	381
Poly(iminoterephthaloyliminomethylene-2,5-dimethyl-1,4-phenylenemethylene)		498	DTA heating rate	303
Poly(iminoterephthaloylimino-4,4-dimethylheptamethylene)		425	No experimental details	381
Poly(iminoterephthaloylimino-1,4,4-trimethylheptamethylene)		423	DTA heating rate	548
Poly(iminoterephthaloylimino-2,2,4-trimethylheptamethylene)		418	DSC heating rate	262,381
Poly(iminoterephthaloylimino-2,4,4-trimethylheptamethylene)		421, 432	Two different samples; DSC heating rate	381
Poly(iminoterephthaloylimino-1,4-phenylenemethylene-1,4-phenylene)		<500	Dynamic method	311
Poly[imino(tetrachloroterephthaloyl)iminohexamethylene]	71164-49-3	569	DSC, HR 16 deg/min, *f*(MW)	1194
Poly(iminotetramethyleneiminocarbonyl-1,4-phenylene-2,2-butylidene-1,4-phenylenecarbonyl)		446–455		165
Poly(iminotetramethyleneiminocarbonyl-1,4-phenylene-2-oxoethylene)		357	Heating rate: 20 K/min	541
Poly(iminotrimethylenefluoren-9-ylidenetrimethyleneiminosebacoyl)		358	Mechanical method, heating rate: 1 K/min	78, 161
Poly(iminotrimethyleneiminocarbonyl-1,4-phenylene-2-oxoethylene)		382	Heating rate: 20 K/min	541
Poly(imino-2,2,4-trimethylpentamethyleneiminoadipoyl)		338	No measurement details	381
Poly(isopropyliminohexafluoroglutarylisopropyliminohexamethylene)		~293	Brittle point	539
Poly(isopropylimino-2,2,3,3,4,4-hexafluoropentamethyleneisopropyliminoadipoyl)		303–308	Brittle point	539
Poly(methyliminohexafluoroglutarylmethyliminohexamethylene)		298– 303	Brittle point	539

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TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(methylimino-2,2,3,3,4,4-hexafluoro-pentamethylenemethyliminoadipoyl)		298–303	Brittle point	539
Poly(2-methylpentamethylene terephthalamide) amorphous		416		435
crystalline		424		1435
drawn fiber		453		1435
Poly(1,3,4-oxidiazole-amide)		498–545		1388
Poly(oxytrimethyleneiminoadipoyliminotrimethylene)		307		282
Poly(oxy-3-oxotrimethyleneiminomethylene-2,5-dimethyl-1,4-phenylenemethyleneimino-1-oxotrimethylene)		353	DTA heating rate	303
Poly(oxy-5-oxopentamethyleneiminomethylene-1,4-phenylenemethyleneimino-1-oxopentamethylene)		343	Estimated from copolymer data	304
Poly(oxy-1,4-phenylenecarbonylimino-2,2-dimethyl-pentamethyleneiminocarbonyl-1,4-phenylene)		428	DTA heating rate	308
Poly(oxy-1,4-phenylenecarbonylimino-1,4-phenylenesulfonyl-1,4-phenyleneiminocarbonyl-1,4-phenylene)		571	Softening point	309
Poly(oxy-1,4-phenyleneiminoisophthaloylimino-1,4-phenylene)		463, 554	Conflicting data	130,311,312
Poly(oxy-1,4-phenyleneiminoterephthaloylimino-1,4-phenylene)		613	Heating rate: 10 K/min	311,312
Poly(*p*-phenylene terephthalamide) Kevlar		508		1432
Poly(*N*-phenyl-2-hydroxytrimethylene amine)		343		1400
Poly(phenylquinoxaline-imide-amide)		557–581		1389
Poly(sulfonyl-1,3-phenyleneiminoadipoylimino-1,3-phenylene)		413	DTA heating rate	500,501
Poly(sulfonyl-1,3-phenyleneiminoadipoylimino-1,4-phenylene)		467	DTA heating rate	500,501
Poly(sulfonyl-1,3-phenyleneiminoazelaoylimino-1,3-phenylene)		398	DTA heating rate	500,501
Poly(sulfonyl-1,4-phenyleneiminoazelaoylimino-1,4-phenylene)		451	DTA heating rate	500,501
Poly(sulfonyl-1,3-phenyleneiminocarbonyl-1,4-naphthylenecarbonylimino-1,3-phenylene)		<573	Softening point; mechanical method	309
Poly(sulfonyl-1,4-phenyleneiminocarbonyl-1,4-naphthylenecarbonylimino-1,4-phenylene)		<573	Softening point; mechanical method	309
Poly(sulfonyl-1,4-phenyleneiminocarbonyl-1,4-phenylenemethylene-1,4-phenylenecarbonylimino-1,4-phenylene)		<573	Softening point	309
Poly(sulfonyl-1,3-phenyleneiminododecanedioylimino-1,3-phenylene)		380	DTA heating point	500,501
Poly(sulfonyl-1,4-phenyleneiminododecanedioylimino-1,4-phenylene)		433	DTA heating point	500,501
Poly(sulfonyl-1,3-phenyleneiminoisophthaloylimino-1,3-phenylene)		573–593	Softening point; mechanical method	309
Poly(sulfonyl-1,4-phenyleneimino-2-methoxy-isophthaloylimino-1,4-phenylene)		568–583	Softening point	309
Poly(sulfonyl-1,4-phenyleneiminopimeloylimino-1,4-phenylene)		436	DTA heating point	500,501
Poly(sulfonyl-1,3-phenyleneiminosebacoylimino-1,3-phenylene)		385	DTA heating point	500,501
Poly(sulfonyl-1,4-phenyleneiminosebacoylimino-1,4-phenylene)		444	DTA heating point	500,501
Poly(sulfonyl-1,3-phenyleneiminosuberoylimino-1,3-phenylene)		398	DTA heating point	500,501
Poly(sulfonyl-1,4-phenyleneiminosuberoylimino-1,4-phenylene)		453	Softening point	500,501
Poly(sulfonyl-1,4-phenyleneiminoterephthaloyl-1,4-phenylenecarbonylimino-1,4-phenylene)		590	Softening point; mechanical method	309
Poly(terephthalamide), pendant ester		451–490		1462
Poly(thio-11-oxoundecamethyleneiminoethylene-1,4-phenyleneethyleneimino-1-oxoundecamethylene)		331	Dynamic method; low viscosity	491
Poly(thio-3-oxotrimethyleneiminohexamethyleneimino-1-oxotrimethylene)		287–300		492

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(thio-7-oxoheptamethyleneiminomethylene-1,4-phenylenemethyleneimino-1-oxoheptamethylene)		338	Dynamic method	491
Poly(thio-5-oxopentamethyleneiminomethylene-1,4-phenylenemethyleneimino-1-oxopentamethylene)		343	Dynamic method	491
Poly(tridecanolactone)		237		1361,1544
Poly[tri(oxyethylene)oxycarbonyliminohexamethyleneimino carbonyl]		260		246
3.4.2. POLY(ANILINES)				
Poly(aniline)		373		1298
Poly(4-(methacryloyloxy)acetanilide)		471		1526
Poly(4-(2-methacryloyloxy)ethyloxy)acetanilide)		585		1526
Poly(4-(6-methacryloyloxy)hexyloxy)acetanilide)		318		1526
Poly(2-methyl-4-nitro-[*N*,*N*-bis(2-hydroxyethyl)]-aniline)		371		1280
Poly(2′-methyl-4′-nitrophenylazo)-[*N*,*N*-bis(2-hydroxyethyl)]-aniline)		410		1280
Poly(4′-nitrophenylazo)-[*N*,*N*-bis(2-hydroxyethyl)]-aniline)		404		1280
Poly([2(4′-nitrophenylethylenyl)]-[*N*,*N*-bis(2-hydroxyethyl)]-aniline)		394		1280
Poly(4-tricyanovinyl-[*N*,*N*-bis(2-hydroxyethyl)]-aniline)		398		1280
3.4.3. POLY(IMIDES)				
Phenylethynyl-terminated imide, cured		522		1458
Phenylethynyl-terminated imide, uncured oligomer (6300 M_w)		491		1458
Poly(imide) of 3,3-bis [4-(4′-aminophenoxy)phenyl]phthalimidime with bis(trimellitimide)		540–595		1493
Poly(ester-amide-imide) of phenyltrimellitimide		373–403		1467
Poly(ester imide)		364		1534
Poly(ether imide)		473		1432
		538		1376
		485		1254
Poly(bis(*p*-ethynylphenyl)pyromellitimide)		>713		1303
Poly(4-(4-fluorobenzoyl)-*N*-(4-hydroxyphenyl)phthalimide)		570		1558
Poly([4-(*N*,*N*-bis(2-hydroxyethyl)amino)-benzylidene] imidazolidine-2-dione)		451		1280
Poly(imide)		495		1432
Poly(imide), di-*t*-butyl subs		515–571		1373
nanofoam		>673		1372
PMR-15		613		1453
Poly(imides) of 4,4′-diaminotriphenylmethane, aromatic		533–593		1466
Poly(imide) of 4,4′-oxidiphthalic anhydride and 2,2′-dimethyl-4-4′-diaminobiphenyl diamine		571		1434
Poly(methacryl-methyl-imide)		>423		1397
Poly[4,4′-oxydiphenylene oxydiphthalimide)		543		1448
3.4.4. POLY(IMINES)				
Poly(acetyliminoethylene)		~353	Mechanical method	549
Poly(acetyliminotrimethylene)		~303	Mechanical method; low viscosity	550
Poly(benzoyliminoethylene)		378	Mechanical method	549
Poly(benzoyliminotrimethylene)		345	Mechanical method	550
Poly(butyryliminoethylene)		~303	Mechanical method	549
Poly(dodecanoyliminoethylene)		~283	Mechanical method-apparent T_g	549
Poly[(pentadecafluorooctanoylimino)trimethylene]		298	Mechanical method	550
Poly(heptanoyliminoethylene)		~283	Mechanical method-apparent T_g	549
Poly(hexanoyliminoethylene)		~283	Mechanical method-apparent T_g	549
Poly(hexanoyliminotrimethylene)		257	Mechanical method	550
Poly(isobutyryliminoethylene)		~303	Mechanical method	549
Poly(isovaleryliminoethylene)		~303	Mechanical method	549
Poly(3-methoxycarbonylpropyionyliminoethylene)		304	Mechanical method	551
Poly(2-naphthyliminoethylene)		403	Mechanical method	549
Poly(octadecanoyliminoethylene)		~283	Mechanical method-apparent T_g	549

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(octanoyliminoethylene)		~283	Mechanical method-apparent T_g	549
Poly(propionyliminoethylene)		~343	Mechanical method	549
Poly(propionyliminotrimethylene)		281	Mechanical method; low viscosity	550
Poly(valeryliminoethylene)		~286	Mechanical method	549
3.4.5. POLY(UREAS)				
Poly(di(aminophenyl)acetylene diurea)		508–583		1421
Poly(urethane), hemicyanine dye		394		1456
PU-30		233	Soft microphase	1268
		364	Hard microphase	1268
PU-40		230	Soft microphase	1268
		356	Hard microphase	1268
PU-50		230	Soft microphase	1268
		353	Hard microphase	1268
PU-60		232	Soft microphase	1268
		351	Hard microphase	1268
PU-70		239	Soft microphase	1268
		346	Hard microphase	1268
Poly(urethane)s, shape-memory		283–323		1496
Poly(ureylenemethylene-1,4-phenylenemethyleneureylenedecamethylene)		328	Heating rate: 1 K/min	445
Poly(ureylenemethylene-1,4-phenylenemethyleneureylenehexamethylene)		345	Heating rate: 1 K/min	445
Poly(ureylenemethylene-1,4-phenylenemethyleneureyleneoctadecamethylene)		325	Heating rate: 1 K/min	445
Poly(ureylenemethylene-1,4-phenylenemethyleneureyleneoctamethylene)		341	Heating rate: 1 K/min	445
Poly(ureylenemethylene-1,4-phenylenemethyleneureylenetetradecamethylene)		327	Heating rate: 1 K/min	445
Poly(ureylenedecamethyleneureylenetetradecamethylene)		313	Heating rate: 1 K/min	445
Poly(ureylenehexamethyleneureylenedodecamethylene)		322	Heating rate: 1 K/min	445
Poly(ureylenehexamethyleneureylene-1,4-phenylenemethylene-1,4-phenylene)		328	Heating rate: 1 K/min	445
Poly(ureylenehexamethyleneureylenetetradecamethylene)		320	Heating rate: 1 K/min	445
Poly(ureyleneoctamethyleneureylene-1,4-phenylenemethylene-1,4-phenylene)		323	Heating rate: 1 K/min	445
Poly(ureylene-1,4-phenylenemethylene-1,4-phenyleneureylenedecamethylene)		319	Heating rate: 1 K/min	445
Poly(ureylene-1,4-phenylenemethylene-1,4-phenyleneureylenedodecamethylene)		324	Heating rate: 1 K/min	445
Poly(ureylene-1,4-phenylenemethylene-1,4-phenyleneureyleneoctadecamethylene)		321	Heating rate: 1 K/min	445
3.5. POLY(PHOSPHAZENES)				
Aminoorganosiloxane poly(phosphazene)		190–251		1441
Poly[(amino)thionylphosphazene]		233–355		1450
Poly[bis(4-benzylphenoxy)phosphazene]		286		1518
Poly[bis(2,3-bis(2-*n*-butoxyethoxy)propoxy)phosphazene]		187		1422
Poly(bis(*t*-butyl-*p*-phenoxyphosphazene))		323		1296
Poly(chloro-2,2,2-trifluoroethoxyphosphazene) (average structure)		213	Heating rate: 20 K/min; probably branched structure	552
Poly(dichlorophosphazene)	26085-02-9	210	DTA heating rate	553
Poly(diethoxyphosphazene)		189	DTA heating rate	553
Poly[bis(dimethylamino)phosphazene]		233	Heating rate: 20 K/min	552
Poly(dimethoxyphosphazene)		197	DTA heating rate	553
Poly[bis(2,3-dimethoxypropoxy)phosphazene]		194		1422
Poly[bis(dimethylamino)phosphazene]		269	Dynamic mechanical method	554
Poly(diphenoxyphosphazene)		265	DTA heating rate	553
Poly[bis(ethylamino)phosphazene]		303	DTA heating rate	554
Poly[bis(4-fluorophenoxy)phosphazene]		259	Heating rate: 20 K/min	552
Poly[bis(1H,1H-pentafluoropropoxy)phosphazene]		198, 218		557
Poly[bis(2-methoxyethoxy)phosphazene]		192		1422
Poly[bis((methoxyethoxy)ethoxy)phosphazene]		189		1449,1441

TABLE 3. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly[bis(methylphenoxy)phosphazene]		269		1441
Poly[bis(2-(2-(2-(2-(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)ethoxy)ethoxy)ethoxy)ethoxy)phosphazene]		198		1422
Poly[bis(2-(2-(2-(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)ethoxy)ethoxy)ethoxy)phosphazene]		197		1422
Poly[bis(2-(2-(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)ethoxy)ethoxy)phosphazene]		194		1422
Poly[bis(2-(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)ethoxy)phosphazene]		193		1422
Poly[bis(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)phosphazene]		193		1422
Poly[bis(2,3-bis(2-(2-(2-methoxyethoxy) ethoxy)ethoxy)propoxy)phosphazene]		191		1422
Poly[bis(2-(2′-methoxyethoxy)ethoxy)phosphazene]		189		1422
Poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene]		189		1422
Poly[bis(2,3-bis(2-(2-methoxyethoxy) ethoxy)propoxy)phosphazene]		192		1422
Poly[bis(2,3-bis(2-methoxyethoxy)propoxy)phosphazene]		192		1422
Poly(organophosphazene), ethyleneoxy subs		203–217		1449
Poly[bis(2,3-bis(2-*sec*-propoxyethoxy) propoxy)phosphazene]		196		1422
Poly[bis(phenylamino)phosphazene]		364	DTA heating rate	554
Poly(phosphazene)s		168		1441
fluorinated		311–367		1512
Poly[bis(piperidino)phosphazene]		292		554
Poly[bis(trifluoroethoxy)phosphazene]		210		1441
Poly[bis(2,2,2-trifluoroethoxy)phosphazene]		203		552,555,556
Poly[bis(3-trifluoromethylphenoxy)phosphazene]		238	Heating rate: 20 K/min	552
3.6. POLY(SILANES) AND POLY(SILAZANES)				
Poly(carbosilazanes)		198–253		1370
Poly(di-*n*-butylsilylenemethylene)		243		1440
Poly(di-*n*-hexylsilylenemethylene)		217		1440
Poly[(4-dimethylaminophenyl)methylsilylenetrimethylene]		267		558
Poly[(4-dimethylaminophenyl)phenylsilylenetrimethylene]		325		558
Poly(1,1-dimethylsilazane)		191		560,561
Poly(dimethylsilylenemethylene)		173	No measurement details	562
		186		1440
Poly(dimethylsilylenetrimethylene)		203		559
Poly(di-*n*-pentylsilylenemethylene)		218		1440
Poly(di-*n*-propylsilylenemethylene)		323		1440
Poly(di-*p*-tolysilylenetrimethylene)		311	Mechanical method	558
Poly[(methyl)phenylsilylenetrimethylene]		243	No measurement details	559
Poly(silethylene)		328		1313
Poly(silethylene glycol)		204		1478
Poly(siloxyethylene glycol)		220–373	Increases with Si content	1478
Poly(silylethylene)		>653		1395
Poly(vinyl trimethyl silane)		378–417		1295

TABLE 4. MAIN-CHAIN HETEROCYCLIC POLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
4.1. CARBOHYDRATES				
Acetoxypropyl cellulose	9004-39-1	257		1358
Amylose hexanoate (2.9)		315	Heating rate: 5–10 K/min	189
Amylose triacetate		440	Heating rate: 5–10 K/min	189
Amylose tributyrate		365	Heating rate: 5–10 K/min	189
Amylose tripropionate		406	Heating rate: 5–10 K/min	189
Amylose valerate: (2.8)		330	Heating rate: 5–10 K/min	189
Cellulose	9004-34-6	243–433	Conflicting data	190–200

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TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Cellulose acetate	9004-35-7	473		1320
Cellulose eicosanoate		407		388,1464
Cellulose laurate		367		1464
Cellulose triacetate	9012-09-3	322–751	Conflicting data; depend on acetate and water content and degree of crystallinity	1,21,93,143,188, 191,197,201–214
Cellulose tributyrate	39320-16-6	388		1,206,215
Cellulose tridecanoate		321		215
Cellulose triheptanoate		320		215
Cellulose trihexanoate		237		215
Cellulose nitrate	9004-70-0	326, 339	Conflicting data	18,21,188,201, 216,217
Cellulose tripropionate		400		215
Cellulose trivalerate		338		215
Cyanoethyl cellulose	9004-41-5	453	Dynamic method	197
Ethyl cellulose	9004-57-3	316		1,201
Methyl cellulose	9004-67-5	423	Dynamic method	197
Poly(1,4-butanediol succinate)		346		1527
Starch acetate, dry		438–458		1469
humid		368–408		1469
Triphenylmethyl cellulose		426	Dynamic method	197
4.2. LIQUID CRYSTALS				
LC Poly(acrylate)		294		1463
LC Poly(butadiene) + *p*-nitrostilbene		287–304		1405
LC Poly(hydroquinone terephthalate), *t*-butyl		458		1480
LC Poly(norborene) + *p*-nitrostilbene		317–337		1405
LC Polylster		333		1499
4.3. NATURAL POLYMERS				
Amorphous silk fibroinchains		453		1364
Collagen		308		1498
Gelatin		343		1498
4.4. POLY(ACETALS)				
Poly(1,3-dioxa-4,6-cyclohexylenemethylene)		378		1,187
Poly(1,3-dioxa-2-propyl-4,6-cyclohexylenemethylene)	63148-65-2	322		1,188
Poly(2-ethyl-1,3-dioxa-4,6-cyclohexylenemethylene)		345		1
Poly(2-isopropyl-1,3-dioxa-4,6-cyclohexylenemethylene)		329		1
Poly(2-methyl-1,3-dioxa-4,6-cyclohexylenemethylene)		355		1
Poly(vinyl acetal) see Poly(2-methyl-1,3-dioxa-4,6-cyclohexylenemethylene)				
Poly(vinyl butyral) see Poly(1,3-dioxa-2-propyl-4,6-cyclohexylenemethylene)	63148-65-2			
Poly(vinyl pentanal acetal)		327		1524
4.5. POLY(ANHYDRIDES)				
Poly(amide imide) of biphenyltetracarboxylic dianhydride		493–563		1404
Poly(3,3′,4,4′-benzophene tetracarboxylic dianhydride)		597		1350
Poly(3,3′,4,4′-biphenyl tetracarboxylic dianhydride)		616		1350
Poly(2,2-bis(3,4-dicarboxyphenyl)hexafluoropropane dianhydride)		597		1350
Poly(methacrylic anhydride)		432	Vicat softening point	186
Poly(3,3′,4,4′-oxydiphthalic anyhydride)		536		1350
Poly(pyromellitic dianhydride) Kapton		635		1350
		693		1320
Poly(N,N,N′,N′-tetraglycidyl-4-4′-diaminodiphenylmethane hexahydrophthalic anhydride)		479		1381
Poly(N,N,N′,N′-tetraglycidyl-4-4′-diaminodiphenylmethane tetrahydrophthalic anhydride)		457		1381

TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
4.6. POLY(BENZIMIDAZOLES)				
Poly(2,6-benzimidazolediyl-6,2-benzimidazolediylocatamethylene)		507	No experimental details	112
Poly(2,5-benzimidazolediyl-6,2-benzimidazolediylpentamethyleneimino-4-diethylamino-1,3,5-triazinediyliminopentamethylene)		463	Heating rate: 2 K/min	142
Poly(2,5-benzimidazolediyl-5,2-benzimidazolediyl-1,3-phenylene)		703	T_g increased after storage at 773 K	143
Poly(2,5-benzimidazolediyl-6,2-benzimidazolediyl-1,4-phenyleneimino-4-diphenylamino-1,3,5-triazinediylimino-1,4-phenylene)		602	Heating rate: 2 K/min; η_{inh} only 0.21 dl/g	142
Poly(2,6-benzimidazolediyl-6,2-benzimidazolediyltrimethylene)		548	No experimental details	112
Poly(2,6-benzimidazole-diylsulfonyl-5,2-benzimidazolediyl-1,3-phenylene)		~560	DTA heating rate	114
Poly(5,5′-bibenzimidazole-2,2′-biphenylylene)		663	Mechanical method	144
4.7. POLY(BENZOTHIAZINOPHENOTHIAZINES)				
Poly(7,7′-[1″,4″-dihydro-(2,3H)benzothiazino(2′,3′-b)hexahydro(1,2,3,4H)phenothiazine]diyl)		483	Dynamic method	117
4.8. POLY(BENZOTHIAZOLES)				
Poly(2,6-benzothiazolediyl-6,2-benzothiazolediyloctamethylene)		511	No method	112
Poly(2,6-benzothiazolediyl-6,2-benzothiazolediyl-1,3-phenylene)		768	Dynamic method	118
4.9. POLY(BENZOXAZINES)				
Polybenzoxazine B-b		408		1410
Polybenzoxazine B-e		453		1410
Polybenzoxazine B-ip		413		1410
Polybenzoxazine B-m		453		1410
Polybenzoxazine B-p		403		1410
4.10. POLY(BENZOXAZOLES)				
Poly(arylene ether benzoxazole)		>573		1539
Poly(benzoxazole)		328–370		1346
Poly(2,6-benzoxazolediyl-6,2-benzoxazolediyloctamethylene)		326	No method	112
Poly(benzoxazole-5,2-diylhexamethylenebenzoxazole-2,5-diyliminosebacoylimino)		359	Heating rate: 20 K/min	113
Poly(benzoxazole-5,2-diylhexamethylenebenzoxazole-2,5-diyliminosuberoylimino)		350	Heating rate: 20 K/min	113
Poly(benzoxazole-5,2-diyloctamethylenebenzoxazole-2,5-diyliminoadipoylimino)		358	Heating rate: 20 K/min	113
Poly(benzoxazole-5,2-diyloctamethylenebenzoxazole-2,5-diyliminosebacoylimino)		366	Heating rate: 20 K/min	113
Poly(benzoxazole-5,2-diyl-1,3-phenylenebenzoxazole-2,5-diyliminosebacoylimino)		393	Heating rate: 20 K/min	113
Poly(*m*-phenylene),*o*,*m*-2-benzoxazole		494–523		1474
Poly(*m*-phenylene),*o*,*m*-5-methyl-2-benzoxazole		494–523		1474
4.11. POLY(CARBORANES)				
Poly($C_2B_5H_5$-carboranylenedi(dimethylsilyleneoxy)dimethylsilylene)		<188	Dynamic method	180
Poly($C_2B_5H_5$-carboranylenedimethylsilyleneoxy-dimethylsilylene)		277, 211	Conflicting data	178,179
Poly($C_2B_{10}H_{10}$-carboranylenedimethylsilyleneoxy-dimethylsilylene)		298	Mechanical method	181
Poly($C_2B_{10}H_{10}$-carboranylenedi(dimethylsilyleneoxy)dimethylsilylene)		239		180–182

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TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly($C_2B_{10}H_{10}$-carboranylenedi(dimethylsilyleneoxy)dimethylsilylene)		213	Mechanical method	180–182
4.12. POLY(DIBENZOFURANS)				
Poly(3,6-dibenzofurandiylsulfonyl)		633		184
Poly(3,6-dibenzofurandiylsulfonyl-1,4-phenyleneoxy-1,4-phenylenesulfonyl)		563		184
4.13. POLY(DIOXOISOINDOLINES)				
Poly(1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylene)		~638	Dynamic method	428,905,437
Poly(1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenedecafluoropentamethylene-1,3-phenylene)		533–543	Softening point; viscosity only 0.11 dl/g	148
Poly(1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenehexafluorotrimethylene-1,3-phenylene)		~393	Sample hygroscopic	149
Poly(1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		<469	Heating rate: 32 K/min	124,125,150
Poly(1,3-dioxoisoindoline-2,5-diylcarbonylimino-1,4-phenyleneiminocarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene)		641	Dynamic method	151
Poly(1,3-dioxoisoindoline-2,5-diyldodecafluorohexamethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		458	Heating rate: 32 K/min	153
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenedecafluoropentamethylene-1,3-phenylene)		371	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenehexafluorotrimethylene-1,3-phenylene)		418		148
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenemethylene-1,4-phenylene)		498	Heating rate: 32 K/min	152
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneoxy-1,3-phenylene)		451	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		483		148
Poly(1,3-dioxoisoindoline-2,5-diylhexadecafluorooctamethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		457	Heating rate: 32 K/min; highly crystalline	152
Poly(1,3-dioxoisoindoline-2,5-diylhexafluorotrimethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene)		533	Heating rate: 32 K/min	154
Poly(1,3-dioxoisoindoline-2,5-diyloctafluorotetramethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenedecafluoropentamethylene-1,3-phenylene)		417	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyloctafluorotetramethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenehexafluorotrimethylene-1,3-phenylene)		432	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyloctafluorotetramethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		485	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyltetradecafluoroheptamethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylene-decafluoropentamethylene-1,3-phenylene)		385	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyltetradecafluoroheptamethylene-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylene-hexafluorotrimethylene-1,3-phenylene)		406	Heating rate: 32 K/min	148

TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(1,3-dioxoisoindoline-2,5-diyltetradeca-fluoroheptamethylene-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		460	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenedecafluoropentamethylene-1,3-phenylene)		393		148
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenylenehexafluorotrimethylene-1,3-phenylene)		457	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneimino-4-dimethylamino-1,3,5-triazinediylimino-1,3-phenylene)		567	Heating rate: 2 K/min	142
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneimino-4-diphenylamino-1,3,5-triazinediylimino-1,3-phenylene)		563	Heating rate: 2 K/min	142
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneimino-4-phenyl-1,3,5-triazinediylimino-1,3-phenylene)		565	Heating rate: 2 K/min	142
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneimino-1,3,5-triazinediylimino-1,3-phenylene)		563	Heating rate: 2 K/min	142
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,3-phenyleneoxy-1,3-phenylene)		480	Heating rate: 32 K/min	148
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		523	Heating rate: 2 K/min	129
Poly(1,3-dioxoisoindoline-2,5-diyloxy-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylene-4-phenyl-1,3,5-triazinediyl-1,4-phenylene)		628	Heating rate: 2 K/min	142
Poly(1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonylimino-4,4′-biphenylyleneiminocarbonyl)		643	Dynamic method	151
Poly(1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonylimino-1,3-phenyleneiminocarbonyl)		584	Dynamic method	151
Poly(1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)		597	Dynamic method	151
Poly(1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonylimino-1,4-phenyleneoxy-1,4-phenylene-iminocarbonyl)		583	Dynamic method	151
Poly(1,3-dioxoisoindoline-5,2-diyl-1,4-phenylenesulfonyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonylimino-1,4-phenylenesulfonyl-1,4-phenyleneiminocarbonyl)		580	Dynamic method	151
4.14. POLY(FLUORESCEINS)				
Poly(fluorescein-3′,6′-diyloxyisophthaloyloxy)		549		183
4.15. POLY(FURAN TETRACARBOXYLIC ACID DIIMIDES)				
Poly(4,5,6,8,9,10-hexahydro-1,3,6,8-tetraoxofuro[3,2-c:4,5-c′] dipyrrole-2,7[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		528	Dynamic method	111
Poly(4,5,6,8,9,10-hexahydro-1,3,6,8-tetraoxofuro[3,2-c:4,5-c′] dipyrrole-2,7[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		583	Dynamic method	111
Poly(4,5,6,8,9,10-hexahydro-1,3,6,8-tetraoxofuro[3,2-c:4,5-c′] dipyrrole-2,7[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		503	Dynamic method	111
4.16. POLY(OXABICYCLONONANES)				
Poly[9-oxabicyclo(3,3,1)nonane-2,6-diyloxycarbonylimino-1,4-cyclohexylenemethylene-1,4-cyclohexyleneiminocarbonyloxy] 53% *trans, trans*		494	DTA heating rate	164
70% *trans, trans*		502		164

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TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly[9-oxabicyclo(3,3,1)nonane-2,6-diyloxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyloxy]		473		164
Poly[9-oxabicyclo(3,3,1)nonane-2,6-diyloxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylenecarbonyloxy]		467	DTA heating rate	164
4.17. POLY(OXADIAZOLES)				
Poly(arylene-1,3,4-oxadiazole)		535–551		1490
Poly(1,3,4-oxadiazolediyl-1,3-phenylene-1,3,4-oxadiazolediyl-1,4-phenylene)		551	DTA heating rate	114
Poly(1,3,4-oxadiazolediyl-1,4-phenyleneoxy-1,4-phenylene)		513	Mechanical method	115
Poly(1,3,4-oxadiazolediyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenylene)		453	Mechanical method	116
Poly(1,3,4-oxadiazolediyl-1,4-phenylene-3,3-phthalidylidene-1,4-phenylene)		653	Mechanical method	115
Starburst oxadiazole		370–415		1371
4.18. POLY(OXINDOLES)				
Poly(5,7-dichloro-3,3-oxindolylidene-1,4-phenyleneoxyisophthaloyl-1,4-phenylene)		543		145
Poly(3,3-oxindolylidene-1,4-phenyleneoxyisophthaloyloxy-1,4-phenylene)		529		145
4.19. POLY(OXOISOINDOLINES)				
Poly(*N*-methyl-3,3-oxoisoindolylidene-1,4-phenyleneoxyisophthaloyloxy-1,4-phenylene)		~558	Transition indistinct	145
Poly(*N*-methyl-3,3-oxoisoindolylidene-1,4-phenyleneoxyterephthaloyloxy-1,4-phenylene)		~555	Transition indistinct	145
Poly(3,3-oxoisoindolylidene-1,4-phenyleneoxyisophthaloyloxy-1,4-phenylene)		598		145
Poly(3,3-oxoisoindolylidene-1,4-phenyleneoxyterephthaloyloxy-1,4-phenylene)		600		145
4.20. POLY(PHTHALAZINE)				
Poly(aryl ether phthalazine)		551–603		1460
Poly(aryl ether phthalazine), 2-napthalenyl		560–607		1460
Poly(aryl ether phthalazinium)		570–636		1479
4.21. POLY(PHTHALIDES)				
Poly(3,3-phthalidylidene-1,4-phenyleneoxy-5-*tert*-butylisophthaloyloxy-1,4-phenylene)		552		145
Poly(3,3-phthalidylidene-1,4-phenyleneoxycarbonyloxy-1,4-phenylene)		513, 538	Conflicting data	145,185
Poly(3,3-phthalidylidene-1,4-phenyleneoxy-5-chloroisophthaloyloxy-1,4-phenylene)		586		145
Poly(3,3-phthalidylidene-1,4-phenyleneoxyisophthaloyloxy-1,4-phenylene)		591		145
4.22. POLY(PIPERAZINES)				
Poly(2,5-dimethyl-1,4-piperazinediylcarbonyloxyneopentylenesulfonyl-neopentyleneoxycarbonyl)		358		160
Poly(2,5-dimethyl-1,4-piperazinediylcarbonyloxyneopentylene-thioneopentyleneoxycarbonyl)		313		160
Poly(1,4-piperazinediyladipoyl)		399	No experimental details	161,162
Poly(1,4-piperazinediylcarbonyl-2,2′-biphenylylenecarbonyl)		466	Mechanical method, heating rate: 1 K/min; low viscosity	78,161
Poly(1,4-piperazinediylcarbonyloxyethyleneoxycarbonyl)		333		163
Poly(1,4-piperazinediylcarbonyloxyneopentyleneoxycarbonyl)		343		160
Poly(1,4-piperazinediylcarbonyloxy-9-oxabicylco-[3,3,1]nonan-2,6-yleneoxycarbonyl)		386	DTA heating rate	164

TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(1,4-piperazinediylcarbonyl-1,4-phenylene-2,2-butylidene-1,4-phenylenecarbonyl)		492–505		165
Poly(1,4-piperazinediylisophthaloyl)		465	Mechanical method, heating rate: 1 K/min	78,161
Poly(1,4-piperazinediyl-1-oxotrimethylenefluoren-9-ylidene-3-oxotrimethylene)		418	Mechanical method, heating rate: 1 K/min	78,161
Poly(1,4-piperazinediylsebacoyl)		~355	Rather ill-defined transition	161,166–168
4.23. POLY(PIPERIDINES)				
Poly(2,6-dioxopiperidine-1,4-diyltrimethylene)		363	Heating rate: 10 K/min	172
4.24. POLY(PYRAZINOQUINOXALINES)				
Poly[3,7-diphenylpyrazino(2,3-g)quinoxaline-2,8-diyl-1,3-phenylene]		638	Heating rate: 20 K/min	119
Poly[3,7-diphenylpyrazino(2,3-g)quinoxaline-2,8-diyl-1,4-phenylene]		668	Heating rate: 20 K/min	119
Poly[pyrazino(2,3-g)quinoxaline-2,8-diyl-1,4-phenylene]		665	Heating rate: 20 K/min	119
Poly[pyrazino(2,3-g)quinoxaline-2,8-diyl-1,4-phenyleneoxy-1,4-phenylene]		626	Heating rate: 20 K/min	119
4.25. POLY(PYRAZOLES)				
Poly(arylene ether pyrazole)		463–566		1390
Poly(1,3-pyrazolediyl-1,3-phenylene-3,1-pyrazolediylhexamethylene)		343	DTA heating rate; low viscosity	175
Poly(1,3-pyrazolediyl-1,4-phenylene-3,1-pyrazolediylhexamethylene)		353	DTA heating rate; high crystallinity	175
Poly(1,3-pyrazolediyl-1,3-phenylene-3,1-pyrazolediyl-1,4-phenylene)		353	DTA heating rate; high crystallinity	175
Poly(1,3-pyrazolediyl-1,4-phenylene-3,1-pyrazolediyl-1,4-phenylene)		373	DTA heating rate; high crystallinity	175
Poly(pyrrole),		438		1328
p-toluene sulfonated		428		1529
4.26. POLY(PYRIDAZINES)				
Poly(3,6-pyridazinediyloxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy)		453	Mechanical method	116
4.27. POLY(PYRIDINES)				
Poly(2,5-pyridinediylcarbonyliminohexamethyleneiminocarbonyl)		322		170
Poly(2,3,5-trifluoropyridinediyloxy-2,2,3,3,4,4-hexafluoropentamethyleneoxy)		260	Polymer structure may contain *o*-linkage	169
4.28. POLY(PYROMELLITIMIDES)				
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-4,4-dimethylheptamethylene)		408	No experimental details	121
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-2,7-fluorenylene)		~623	Heating rate: 3 K/min	120
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]diyl-3-methylheptamethylene)		408	No experimental details	121
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diylnonamethylene)		383	No experimental details	121
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-1,4-phenyleneoxy-1,4-phenylene)		Probably ~773	Conflicting data, transitions from 523 K upwards reported as T_g	119,120,122–131
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6-[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		513	Dynamic method, heating rate 5 K/min	132
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′]dipyrrol-2,6[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		547	Dynamic method, heating rate 5 K/min	132

TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		490	Dynamic method, heating rate 5 K/min	132
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-1,4-phenylene-3-phenylquinoxaline-2,7-diylsulfonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		459	Dynamic method, heating rate 5 K/min	132
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-1,4-phenylenethio-1,4-phenylene)		618	Dynamic method	125
Poly([N,N′(*p,p′*-oxydiphenylene)pyromellitimeide] see Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′] dipyrrole-2,6[1H,3H]-diyl-1,4-phenyleneoxy-1,4-phenylene)				

4.29. POLY(PYRROLIDINES)

Poly(1-butyl-2,5-diazo-3,4-pyrrolidinediyl)		461	DSC, Mdpt, Xp zero HR	1196
Poly(1-decyl-2,5-diazo-3,4-pyrrolidinediyl)		368–370	DSC, Mdpt, Xp zero HR, dilatomer	1196
Poly(2,5-dioxo-1,3-pyrrolidinediylethylene)		408	Heating rate: 20 K/min	176,177
Poly(dodecyl-2,5-diazo-3,4-pyrrolidinediyl)		255	Dilatomer	1196
Poly(1-ethyl-2,5-diazo-3,4-pyrrolidinediyl)		523	DSC, Mdpt, Xp zero HR	1196
Poly(1-hexadecyl-2,5-diazo-3,4-pyrrolidinediyl)		348	Dilatomer	1196
Poly(1-hexyl-2,5-diazo-3,4-pyrrolidinediyl)		422	DSC, Mdpt, Xp zero HR	1196
Poly(1-octadecyl-2,5-diazo-3,4-pyrrolidinediyl)		348	Dilatomer	1196
Poly(1-octyl-2,5-diazo-3,4-pyrrolidinediyl)		393–395	DSC, Mdpt, Xp zero HR, dilatomer	1196
Poly(1-tetradecyl-2,5-diazo-3,4-pyrrolidinediyl)		351	Dilatomer	1196

4.30. POLY(QUINONES)

Poly(2-acrylaminoanthraquinone)		401		1294
Poly(*t*-butylhydroquinone)		450		1312
Poly(methylhydroquinone)		426		1312
Poly(phenylhydroquinone)		430		1312

4.31. POLY(QUINOXALINES)

Poly(3-phenylquinoxaline-2,6-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		583	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		595	Values range from 531 to 651 K	119,134,135
Poly(3-phenylquinoxaline-2,7-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylene)		528	Dynamic method, heating rate: 5 K/min	111
Poly(3-phenylquinoxaline-2,7-diylcarbonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		516, 544	Conflicting data	119,134
Poly(3-phenylquinoxaline-2,6-diyloxy-3-phenylquinoxaline-6,2-diyl-4,4′-biphenylene)		683	Dynamic method	136
Poly(3-phenylquinoxaline-2,6-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		565	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		526, 543	Conflicting data possibly arising from different rate effects	119,134,137
Poly(3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		571	Dynamic method	119,134,137–139
Poly(3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylene)		527	Dynamic method	111
Poly(3-phenylquinoxaline-2,6-diyloxy-3-phenylquinoxaline-6,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		710	Dynamic method	136

TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(3-phenylquinoxaline-2,6-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		516	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		541		119,134,137
Poly(3-phenylquinoxaline-2,6-diyl-3-phenylquinoxaline-6,2-diyl-4,4′-biphenylylene)		677	Dynamic method	136
Poly(3-phenylquinoxaline-2,6-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		639–698	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,3-phenylene)		593	Dynamic method	134,137
Poly(3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		590–638	Conflicting data; dilatometric T_g 638 K	119,134,139
Poly(3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisoindoline-5,2-diyl-1,4-phenylene)		533	Dynamic method, heating rate: 5 K/min	111
Poly(3-phenylquinoxaline-2,6-diyl-3-phenylquinoxaline-6,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		567	Values range from 558 to 693 K	119,134,136,140
Poly(3-phenylquinoxaline-2,6-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		558	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diyl-3-phenylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		561, 533	Conflicting data	119,134
Poly(3-phenylquinoxaline-2,6-diylsulfonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		543	Dynamic method	133
Poly(3-phenylquinoxaline-2,7-diylsulfonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)		485–618	Conflicting data	119,134,135
Poly(3-phenylquinoxaline-2,7-diylsulfonyl-3-phenylquinoxaline-7,2-diyl-1,4-phenylene-1,3-dioxoisoindoline-2,5-diylcarbonyl-1,3-dioxoisiondoline-5,2-diyl-1,4-phenylene)		513	Dynamic method, heating rate: 5 K/min	111
Poly(3-phenylquinoxaline-2,7-diylsulfonyl-3-phenylquinoxaline-7,2-diyl-1,4-pheyleneoxy-1,4-phenylene)		468, 563	Conflicting data	119,134
Poly(quinoxaline-2,7-diylcarbonylquinoxaline-7,2-diyl-1,4-phenylene)		591	Heating rate: 20 K/min	119
Poly(quinoxaline-2,6-diyloxyquinoxaline-6,2-diyl-4,4′-biphenylene)		663	Dynamic method	136
Poly(quinoxaline-2,7-diyloxyquinoxaline-7,2-diyl-1,4-phenylene)		579	Heating rate: 20 K/min	119
Poly(quinoxaline-2,6-diyloxyquinoxaline-6,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		655	Dynamic method	136
Poly(quinoxaline-2,6-diyloxyquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		543		137
Poly(quinoxaline-2,6-diylquinoxaline-6,2-diyl--4-4′-biphenylylene)		659	Dynamic method	136,141
Poly[quinoxaline-2,6-diylquinoxaline-6,2-diyl-di(1,4-phenyleneoxy)-1,4-phenylene]		468	Dynamic method	141
Poly[quinoxaline-2,7-diylquinoxaline-7,2-diyl-di(1,4-phenyleneoxy)-1,4-phenylene]		508	Heating rate: 20 K/min	119
Poly(quinoxaline-2,6-diylquinoxaline-6,2-diyl-1,4-phenylene)		623	Dynamic method	141
Poly(quinoxaline-2,7-diylquinoxaline-7,2-diyl-1,4-phenylene)		649	Heating rate: 20 K/min	119
Poly(quinoxaline-2,6-diylquinoxaline-6,2-diyl-1,4-phenyleneoxy-4,4′-biphenylyleneoxy-1,4-phenylene)		486	Dynamic method	141
Poly(quinoxaline-2,6-diylquinoxaline-6,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		661, 553	Dynamic method, conflicting data	136,137,141
Poly(quinoxaline-2,7-diylquinoxaline-7,2-diyl-1,4-phenyleneoxy-1,4-phenylene)		580		119
Poly(quinoxaline-2,7-diylquinoxaline-7,2-diyl-*p*-terphenyl-4,4′-ylene)		578	Heating rate: 20 K/min	119
Poly[quinoxaline-2,6-diylquinoxaline-6,2-diyl-tri-(1,4-phenyleneoxy)-1,4-phenylene]		406	Dynamic method	141
Poly[quinoxaline-2,7-diylquinoxaline-7,2-diyl-tri-(1,4-phenyleneoxy)-1,4-phenylene]		489	Heating rate: 20 K/min	119
Poly(quinoxaline-2,7-diylsulfonylquinoxaline-7,2-diyl-1,4-phenylene)		615	Heating rate: 20 K/min	119

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TABLE 4. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
4.32. POLY(TRIAZINES)				
Poly(heptafluoropropyl-1,3,5-triazinediyltrimethylene)		253–255	No measurement details	155
Poly[(4-H-octafluorobutyl)-1,3,5-triazinediyltrimethylene]		253–255	No measurement details	155
Poly(pentafluoroethyl-1,3,5-triazinediyltrimethylene)		253–255	No measurement details	155
Poly[perfluoro(propyl-1,3,5-triazinediylhexamethylene)]		261	DTA heating rate	156
Poly(6-phenyl-1,3,5-triazinediylphenylimino-1,3,5-triazinediylphenylimino)		541–552	DTA heating rate	114
Poly(6-phenyl-1,2,4-triazine-5,3-diyl-2,6-pyridinediyl-6-phenyl-1,2,4-triazine-3,5-diyl-1,4-phenylene)		533	Dynamic method; different structural isomers may be present	157
Poly(6-phenyl-1,2,4-triazine-5,3-diyl-2,6-pyridinediyl-6-phenyl-1,2,4-triazine-3,5-diyl-1,4-phenylenemethylene-1,4-phenylene)		478	Dynamic method; different structural isomers may be present	157
Poly(6-phenyl-1,2,4-triazine-5,3-diyl-2,6-pyridinediyl-6-phenyl-1,2,4-triazine-3,5-diyl-1,4-phenyleneoxy-1,4-phenylene)		488	Dynamic method; different structural isomers may be present	157,158
Poly(2-H-tetrafluoroethyl-1,3,5-triazinediyltrimethylene)		253–255	No measurement details	155
Poly(1,2,4-triazine-5,3-diyl-2,6-pyridinediyl-1,2,4-triazine-3,6-diyl-1,4-phenyleneoxy-1,4-phenylene)		496	Heating rate: 20 K/min	159
4.33. POLY(TRIAZOLES)				
Poly(4-phenyl-1,2,4-triazolediyl-1,3-phenyleneimino-terephthaloylimino-1,3-phenylene)		623	DTA heating rate	173
Poly(4-phenyl-1,2,4-triazolediyl-1,3-phenylene-4-phenyl-1,2,4-triazolediyl-1,4-phenylene)		538		174

TABLE 5. COPOLYMERS

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Copolyester (*p*-hydroxybenzoic acid-isophthalic acid-hydroquinone) 40:30:30		398–414		1436
Copolyester, aromatic		403–433		1509
Copolyester, aromatic		377		1269
Poly(alkylene *p,p'*-bibenzoate-b-tetramethylene ether)		205		1485
Poly(aryl ether sulfone-*b*-aryl ether ketone)		>473		1425
Poly(*t*-butyl methacrylate-*co*-styrene-*co*-butadiene-*co*-styrene-*co*-*t*-butyl methacrylate)		423		1377
Poly(ε-caprolactone-b-1,5-dioxepen-2-one-β-ε-caprolactone)		235		1515
Poly(carbonate-*co*-butylene terephthalate) 0.5/0.5		363		1386
Poly(3-carboxyphenyl maleimide-*co*-styrene)		493		1273
Poly(4-carboxyphenyl maleimide-*co*-styrene)		493		1273
Poly(2-chloro-1,3-butadiene-*co*-ethyl methacrylate)		238		1360
Poly(2-chloro-1,3-butadiene-*co*-glycidyl methacrylate)		257		1360
Poly(chloroprene-*co*-ethyl methacrylate)		238		1543,1360
Poly(chloroprene-*co*-glycidyl methacrylate)		256		1543,1360
Poly(chloroprene-g-ethyl methacrylate)		230		1360
Poly(chloroprene-g-glycidyl methacrylate)		243		1360
Poly(*N*-cyclohexylmaleimide-*alt*-isobutene)		465		1465
Poly(ethylene oxide-b-dimethyl siloxane-b-ethylene oxide)		156		1500
Poly(ethylene oxide-*co*-epychloridrine)		233		1415
Poly(ethylene terephthalate-*co*-*p*-hydroxybenzoic acid)		333		1517
Poly(ethylene-*co*-methacrylate)	25103-74-6	237		1307
Poly(ethylene-*co*-methacrylic acid) ionomer	25053-53-6	330	DSC	1265
Poly(ethyl methacryate-*co*-methyl methacrylate, 40:60)		377		1552
Poly(ethyl methacrylate-*co*-methyl methacrylate, 30:70)		365		1552
Poly(furfuryl acrylate-*alt*-2-hydroxyethyl methacrylate) 50:50		326		1279
Poly(3-hydroxybutyrate-*co*-4-hydroxybutyrate)-18		269	NMR	1494
Poly(3-hydroxybutyrate-*co*-4-hydroxybutyrate)-69		237	NMR	1494
Poly(β-hydroxybutyrate-*co*-hydroxyvalerate)		257		1560

TABLE 5. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(*N*-(4-hydroxyphenyl)maleimide-*co*-allytrimethylsilane)		499		1289
Poly(1,4-isoprene-b-styrene)		286		1283
Poly(L-lactic acid-*co*-DL-maleic acid) 0–45 mol%		306–329		1412
Poly(maleimide-*alt*-*p*-trimethylsilyl-a-methylstyrene)		499		1324
Poly(maleimide-*alt*-*p*-trimethylsilylstyrene)		503		1324
Poly(methyl methacrylate-*co*-styrene)		375		1318
Poly(methyl methacrylate-*co*-styrene-*co*-butadiene-*co*-styrene-*co*-methyl methacrylate)		433		1377
Poly(norbornene-*co*-ethylene)		428		1403
Poly(styrene-b-isoprene)		378/213		1472
Poly(styrene-b-methylphenylsiloxane) 58% styrene		303		1282
Poly(styrene-*co*-acrylonotrile-*co*-chromium acrylate)		369		1537
Poly(styrene-*co*-maleic anhydride)		381		1348
Poly[styrene-b-(ethylene-*co*-butylene)-b-styrene]		233		1321
Poly(*N*-vinyl carbazole-*co*-pyrrole)		453–473		1328
Poly(vinyl chloride-*co*-vinyl acetate)	9003-22-9	342		1359
Resorcinol-Bisphenol A copolymer, *o*-adamantyl		373		1369

G. REFERENCES

1. L. E. Nielsen, "Mechanical Properties of Polymers", Reinhold, New York, 1962.
2. C. A. Kumins, J. Roteman, J. Polym. Sci. A, **1**, 527 (1963).
3. K. Kanamaru, M. Sugiura, Kolloid Z. -Z. Polym., **194**, 110 (1964).
4. M. Suguira, E. Fujii, Toyko Kogyo Shikensho Hokoku, **58**, 534 (1963).
5. K. Kanamaru, M. Suguira, J. Chem. Soc. Japan, **65**, 1434 (1962).
6. R. F. Boyer, Rubber Chem. Technol., **63**, 1303 (1963).
7. T. M. A. Hossain, T. Iijima, S. Morita, H. Maeda, J. Appl. Polym. Sci., **13**, 541 (1969).
8. T. R. Manley, Polymer, **10**, 148 (1969).
9. A. J. Kovacs, J. Polym, Sci., **30**, 131 (1958).
10. N. Bekkedahl, J. Res. Nat. Bur. Std., **43**, 145 (1949).
11. J. H. Sewell, Royal Aircraft Establishment, Farnborough, Hants. UK, Unpublished report.
12. T. Holleman, Rheol. Acta, **10**, No. 2, (1971).
13. A. Eisenberg, A. V. Tolbolsky, J. Polym. Sci., **61**, 483 (1962).
14. S. Nakamura, A. V. Tolbolsky, Rept. Progr. Polym. Phys. Japan, **12**, 303 (1969).
15. J. J. Maurer, H. C. Tsein, J. Appl. Polym. Sci., **8**, 1719 (1964).
16. F. C. Stehling, L. Mandelkern, Macromolecules, **3**, 242 (1970).
17. G. M. Martin, S. S. Rogers, L. Mandelkern, J. Polym. Sci., **20**, 579 (1956).
18. R. F. Boyer, R. S. Spencer, "Advances in Colloid Science", vol. II, Interscience, New York, 1946, p.1.
19. R. H. Wiley, J. Polym. Sci., **2**, 10 (1947).
20. L. Turunen, Kunststoffe, **52**, 672 (1962).
21. B. S. Sully, "Science of Surface Coatings", Ernest Benn, 1962, p. 281.
22. R. H. Wiley, G. M. Brauer, J. Polym. Sci., **3**, 455 (1948).
23. R. H. Wiley, G. M. Brauer, J. Polym. Sci., **3**, 647 (1948).
24. R. H. Wiley, G. M. Brauer, A. R. Bennett, J. Polym. Sci., **5**, 609 (1950).
25. R. B. Beevers, Trans. Faraday Soc., **58**, 1465 (1962).
26. R. B. Beevers, E. F. T. White, Trans. Faraday Soc., **56**, 1529 (1960).
27. R. B. Beevers, E. F. T. White, Trans. Faraday Soc., **56**, 744 (1960).
28. K. H. Illers, Kolloid Z. -Z. Polym., **190**, 16 (1963).
29. P. L. Magagnini, Chim. Ind. (Milan), **49**, 1041 (1967).
30. R. Kaneko, Chem. High Polym. (Toyko), **24**, 272 (1967).
31. K. R. Dunham, J. Vandenberghe, J. W. H. Faber, W. F. Fowler, J. Appl. Polym. Sci., **7**, 143 (1963).
32. B. Maxwell, L. F. Rahm, SPE. J., **6**, No. 9, 7 (1950).
33. S. Newman, S. Strella, J. Appl. Polym. Sci., **9**, 2297 (1965).
34. S. L. Rosen, Polym. Eng. Sci., **7**, 115 (1967).
35. R. J. Seward, Rubber Chem. Techno., **43**, 1 (1970).
36. I. Momiyama, H. Iwai, S. Matsuzaki, Shikizai Kyokaishi, **43**, 427 (1970); CA **74**, 4067c (1971).
37. G. Salee, J. Appl. Polym Sci., **15**, 2049 (1971)
38. L. E. Nielsen, R. Buchdahl, R. Levreault, J. Appl. Phys., **21**, 607 (1950).
39. F. S. Conant, J. W. Liska, J. Appl. Phys., **15**, 767 (1944).
40. F. S. Owens, US pat. 3,514,427, 26 May 1970.
41. J. A. Gorbatkina, V. G. Ivanova-Mumjieva, IUPAC Preprints, Intern, Symp, Macromol. Chem. Toronto, Section A9.9 (1968).
42. W. J. Jackson, T. F. Gray, J. R. Caldwell, J. Appl. Polym. Sci., **14**, 685 (1970).
43. "RAPRA Data Handbook Series on Transition Temperatures" (1973) by Rubber and Plastics Research Association of Great Britain, Shawbury, Shrewsbury, SY4 4NR, England.
44. O. G. Lewis, "Physical Constants of Linear Homopolymers", Springer-Verlag, 1968.

TABLE 5. *cont'd*

Polymer	CAS No.	T_g (K)	Remarks	Refs.
Poly(*N*-(4-hydroxyphenyl)maleimide-*co*-allytrimethylsilane)		499		1289
Poly(1,4-isoprene-b-styrene)		286		1283
Poly(L-lactic acid-*co*-DL-maleic acid) 0–45 mol%		306–329		1412
Poly(maleimide-*alt*-*p*-trimethylsilyl-a-methylstyrene)		499		1324
Poly(maleimide-*alt*-*p*-trimethylsilylstyrene)		503		1324
Poly(methyl methacrylate-*co*-styrene)		375		1318
Poly(methyl methacrylate-*co*-styrene-*co*-butadiene-*co*-styrene-*co*-methyl methacrylate)		433		1377
Poly(norbornene-*co*-ethylene)		428		1403
Poly(styrene-b-isoprene)		378/213		1472
Poly(styrene-b-methylphenylsiloxane) 58% styrene		303		1282
Poly(styrene-*co*-acrylonotrile-*co*-chromium acrylate)		369		1537
Poly(styrene-*co*-maleic anhydride)		381		1348
Poly[styrene-b-(ethylene-*co*-butylene)-b-styrene]		233		1321
Poly(*N*-vinyl carbazole-*co*-pyrrole)		453–473		1328
Poly(vinyl chloride-*co*-vinyl acetate)	9003-22-9	342		1359
Resorcinol-Bisphenol A copolymer, *o*-adamantyl		373		1369

G. REFERENCES

1. L. E. Nielsen, "Mechanical Properties of Polymers", Reinhold, New York, 1962.
2. C. A. Kumins, J. Roteman, J. Polym. Sci. A, **1**, 527 (1963).
3. K. Kanamaru, M. Sugiura, Kolloid Z. -Z. Polym., **194**, 110 (1964).
4. M. Suguira, E. Fujii, Toyko Kogyo Shikensho Hokoku, **58**, 534 (1963).
5. K. Kanamaru, M. Suguira, J. Chem. Soc. Japan, **65**, 1434 (1962).
6. R. F. Boyer, Rubber Chem. Technol., **63**, 1303 (1963).
7. T. M. A. Hossain, T. Iijima, S. Morita, H. Maeda, J. Appl. Polym. Sci., **13**, 541 (1969).
8. T. R. Manley, Polymer, **10**, 148 (1969).
9. A. J. Kovacs, J. Polym, Sci., **30**, 131 (1958).
10. N. Bekkedahl, J. Res. Nat. Bur. Std., **43**, 145 (1949).
11. J. H. Sewell, Royal Aircraft Establishment, Farnborough, Hants. UK, Unpublished report.
12. T. Holleman, Rheol. Acta, **10**, No. 2, (1971).
13. A. Eisenberg, A. V. Tolbolsky, J. Polym. Sci., **61**, 483 (1962).
14. S. Nakamura, A. V. Tolbolsky, Rept. Progr. Polym. Phys. Japan, **12**, 303 (1969).
15. J. J. Maurer, H. C. Tsein, J. Appl. Polym. Sci., **8**, 1719 (1964).
16. F. C. Stehling, L. Mandelkern, Macromolecules, **3**, 242 (1970).
17. G. M. Martin, S. S. Rogers, L. Mandelkern, J. Polym. Sci., **20**, 579 (1956).
18. R. F. Boyer, R. S. Spencer, "Advances in Colloid Science", vol. II, Interscience, New York, 1946, p.1.
19. R. H. Wiley, J. Polym. Sci., **2**, 10 (1947).
20. L. Turunen, Kunststoffe, **52**, 672 (1962).
21. B. S. Sully, "Science of Surface Coatings", Ernest Benn, 1962, p. 281.
22. R. H. Wiley, G. M. Brauer, J. Polym. Sci., **3**, 455 (1948).
23. R. H. Wiley, G. M. Brauer, J. Polym. Sci., **3**, 647 (1948).
24. R. H. Wiley, G. M. Brauer, A. R. Bennett, J. Polym. Sci., **5**, 609 (1950).
25. R. B. Beevers, Trans. Faraday Soc., **58**, 1465 (1962).
26. R. B. Beevers, E. F. T. White, Trans. Faraday Soc., **56**, 1529 (1960).
27. R. B. Beevers, E. F. T. White, Trans. Faraday Soc., **56**, 744 (1960).
28. K. H. Illers, Kolloid Z. -Z. Polym., **190**, 16 (1963).
29. P. L. Magagnini, Chim. Ind. (Milan), **49**, 1041 (1967).
30. R. Kaneko, Chem. High Polym. (Toyko), **24**, 272 (1967).
31. K. R. Dunham, J. Vandenberghe, J. W. H. Faber, W. F. Fowler, J. Appl. Polym. Sci., **7**, 143 (1963).
32. B. Maxwell, L. F. Rahm, SPE. J., **6**, No. 9, 7 (1950).
33. S. Newman, S. Strella, J. Appl. Polym. Sci., **9**, 2297 (1965).
34. S. L. Rosen, Polym. Eng. Sci., **7**, 115 (1967).
35. R. J. Seward, Rubber Chem. Techno., **43**, 1 (1970).
36. I. Momiyama, H. Iwai, S. Matsuzaki, Shikizai Kyokaishi, **43**, 427 (1970); CA **74**, 4067c (1971).
37. G. Salee, J. Appl. Polym Sci., **15**, 2049 (1971)
38. L. E. Nielsen, R. Buchdahl, R. Levreault, J. Appl. Phys., **21**, 607 (1950).
39. F. S. Conant, J. W. Liska, J. Appl. Phys., **15**, 767 (1944).
40. F. S. Owens, US pat. 3,514,427, 26 May 1970.
41. J. A. Gorbatkina, V. G. Ivanova-Mumjieva, IUPAC Preprints, Intern, Symp, Macromol. Chem. Toronto, Section A9.9 (1968).
42. W. J. Jackson, T. F. Gray, J. R. Caldwell, J. Appl. Polym. Sci., **14**, 685 (1970).
43. "RAPRA Data Handbook Series on Transition Temperatures" (1973) by Rubber and Plastics Research Association of Great Britain, Shawbury, Shrewsbury, SY4 4NR, England.
44. O. G. Lewis, "Physical Constants of Linear Homopolymers", Springer-Verlag, 1968.

45. W. A. Lee, J. Polym. Sci. A-2, **8**, No. 4, 444 (1970).
46. J. H. Gibbs, in: J. D. Mackenzie (Ed.), "Modern Aspects of the Vitreous State", Butterworths, London, 1960, p.152.
47. S. Krimm, A. V. Tobolsky, J. Polym. Sci., **6**, 667 (1951).
48. M. C. Shen, A. Eisenberg, in: H. Reiss (Ed.), "Progress in Solid State Chemistry", vol. 3, Pergamon Press, NY, 1966, p. 407; reprint in Rubber Chem. Technol., **43**, 95 (1970).
49. A. Eisenberg, M. Shen, Rubber Chem. Technol., **43**, 156 (1970).
50. G. Allen, SCI Plastics Polymer Group Symp., London, Sept. 1962.
51. B. G. Ranby, K. S. Chan, H. Brumberger, J. Polym. Sci., **58**, 545 (1962).
52. S. A. Iobst, J. A. Manson, ACS Polymer Preprints, **11**, 765 (1970).
53. W. Schemann, H. G. Zachmann, Kolloid Z. -Z. Polym., **241**, 921 (1970).
54. V. P. Roshchupkin, V. V. Kochervinskii, Vysokomol. Soedin. B, **13**, 194 (1971).
55. J. M. Barton, Polymer, **10**, 151 (1969).
56. W. Wrasidlo, Macromolecules, **4**, 642, (1971).
57. G. W. Miller, J. Appl. Polym. Sci., **15**, 2335 (1971).
58. M. L. Williams, R. F. Landel, J. D. Ferry, J. Am. Chem. Soc., **77**, 3701 (1955).
59. A. F. Lewis, J. Polym. Sci. B, **1**, 649 (1963).
60. Y. V. Zelenev, V. I. Vysokomol. Soedin. A, **11**, 920 (1969); (transl'n in Polymer Sci. USSR, **11**, 1040 (1969)).
61. R. Zannetti, P. Manaresi, L. Baldi, J. Polym. Sci., **62**, S33 (1962).
62. J. Heller, D. J. Lyman, J. Polym. Sci. B, **1**, 317 (1963).
63. A. Quach, R. Shima, J. Appl. Phys., **42**, 4592 (1971).
64. S. Ichihara, A. Komatsu, Y. Tsujita, T. Noe, T. Hata, Polym. J. (Japan), **2**, 530 (1971).
65. A. J. Kovacs, S. Y. Hobbs, J. Appl. Polym. Sci., **16**, 301, (1972).
66. H. G. Killian, E. Jenckel, in: K. A. Wolf (Ed.), "Struktur und Physikalisches Verhalten der Kunststoffe", Springer-Verlag, 1962, p. 176.
67. E. W. Fischer, F. Kloos, G. Lieser, J. Polym. Sci. B, **7**, 845 (1969).
68. Y. Aoki, A. Nobuta, A. Chiba, M. Kaneko, Polym. J. (Japan), **2**, 502 (1971).
69. A. B. Romberger, D. P. Eastman, J. L. Hunt, J. Chem. Phys., **51**, 3723 (1963).
70. G. A. Clegg, T. P. Melia, Polymer, **11**, 245 (1970).
71. R. Hoffman, W. Knappe, Kolloid -Z. -Z Polym., **247**, 763 (1971).
72. A. Lambert, Polymer, **10**, 319 (1969).
73. J. J. Maurer, Rubber Chem. Technol., **42**, 110 (1969).
74. S. M. Wolpert, A. Weitz, B. Wunderlich, J. Polym. Sci. A-2, **9**, 1887 (1971).
75. G. A. Gordon, J. Polym. Sci. A-2, **9**, 1693 (1971).
76. R. W. Warfield, SPE J., **15**, 625 (1959).
77. A. Rembaum, J. Polym. Sci. C, 157, (1970).
78. S. C. Temin, J. Polym. Sci., **9**, 471 (1965).
79. J. Periad, A. Banderet, G. Reiss, Angew. Makromol. Chem., **15**, 37 (1971).
80. R. F. Boyer, Changements de Phases, Soc. Chim. Phys., Paris, 1952, p. 383.
81. E. H. Riddle, Monomeric Acrylic Esters, Reinhold, New York, 1954, p. 59.
82. A. L. Machek, J. Elastoplastics, **1**, 213 (1969).
83. W. G. Barb, J. Polym. Sci., **37**, 515 (1959).
84. H. Aida, M. Urushizaki, H. Takeuchi, Fukui Daigaku Kogakubu Kenyu Hokoku, **18**, 173 (1970); Chem. Abstr., **74**, 42806a (1971).
85. M. Tsuge, S. Tanimoto, S. Tanaka, Kogyo Kagaku Zasshi, **73**, 440 (1970); Chem. Abstr., **72**, 133340u (1970).
86. A. Charlesby, R. H. Partridge, Proc. Roy. Soc. A, **271**, 170 (1963).
87. M. Mosiesk, Intern. J. Appl. Radiat. Isotopes, **21**, 11 (1970).
88. J. R. Stevens, A. C. Mao, J. Appl. Phys., **41**, 4273 (1970).
89. S. Y. Chuang, S. J. Tao, J. M. Wilkenfeld, J. Appl. Phys., **43**, 737 (1972).
90. J. E. Guillet, E. Dan, R. S. Mitchell, J. P. Valleau, Nature Phys. Sci., **234**, 135 (1971).
91. J. N. Gayles, W. L. Peticolas, Light Scattering Spectr. Solids, Proc. Intern. Conf., 1968, Ed. G.B. Wright, Springer-Verlag, New York, p. 715 (1969); Chem. Abstr., **72**, 22089z (1970).
92. A. Lavoie, J. E. Guillet, Macromolecules, **2**, 443 (1963).
93. S. Nakamura, S. Shindo, K. Matsuzaki, J. Polym. Sci. B, **9**, 591 (1971).
94. E. Sacher, J. Macromol. Sci. B, **4**, 449 (1970).
95. T. Takamatsu, E. Fukada, Rika Gaku Kenkyusho Hokoku, **46**, 47 (1970); CA, **74**, 23281j (1971).
96. B. M. Grieveson, Polymer, **1**, 499 (1960).
97. D. W. Brown, L. A. Wall, J. Polym. Sci. A-2, **7**, 601 (1969).
98. E. Jenckel, R. Heusch, Kolloid -Z., **130**, 89 (1953).
99. H. Jacobs, E. Jenckel, Makromol. Chem., **47**, 72 (1961).
100. E. A. DiMarsio, J. H. Gibbs, J. Polym. Sci. A, **1**, 1417 (1963).
101. J. Kolarik, J. Janacek, J. Polym. Sci. C, **16**, Pt. 1, 441 (1967).
102. G. Allen, J. McAinsh, G.M. Jeffs, Polymer, **12**, 85 (1971).
103. R. D. McCammon, R. G. Saba, R. N. Work, J. Polym. Sci. A-2, **7**, 1721 (1969).
104. G. Pezzin, F. Zilio-Grandi, P. Sanmartin, Euro. Polym. J., **6**, 1053 (1970).
105. F. M. Smekhov, A. I. Nepomnyashchii, A. T. Sanzharovskii, S.V. Yakubovich, Vysokomol. Soedin. A., **13**, 2012 (1971); (transl'n in Polym. Sci., USSR, **13**, 2362 (1971)).
106. K. Ueberreiter, U. Rohde-Liebenau, Makromol. Chem., **49**, 164 (1961).
107. J. A. Faucher, J. V. Koleske, E. R. Santee, J. J Stratta, C. W. Wilson, J. Appl. Phys., **37**, 3962 (1966).
108. J. A. Faucher, J. Polym. Sci. B, **3**, 143 (1965).
109. IUPAC, "Nomenclature of Organic Chemistry", Sections A & B, Third Edit., Sect. C, Second Edit., London, Butterworths, 1969.
110. Macromolecules, **1**, 193, (1968) and IUPAC Information Bulletin Appendix No. 29, 1972.
111. J. M. Augl, J. V. Duffy, J. Polym. Sci. A-1, **9**, 1343 (1971).
112. F. D. Trischler, K. J. Kjoller, H. H. Levine, ACS Org. Coatings Plast. Chem., **27**, 381 (1967).

113. W. De Winter, R. Stein, H. Unwents, C. Masquelier, J. Polym. Sci. A-1, **8**, 1955 (1970).
114. G. F. L. Ehlers, K. R. Fisch, Appl. Polym. Symp., **8**, 171 (1969).
115. V. V. Korshak, V. M. Mamedov, G. E. Golubkov, D. R. Tur, Vysokomol. Soedin. B, **12**, 57 (1970).
116. R. N. Johnson, A. G. Farnham, R. A. Glendinning, W. F. Hale, C. N. Merriam, J. Polym. Sci. A-1, **5**, 2375 (1967).
117. J. M. Augl, W. J. Wrasidlo, J. Polym. Sci. A-1, **8**, 63 (1970).
118. P. M. Hergenrother, H. H. Levine, J. Polym. Sci. A-1, **4**, 2341 (1966).
119. W. Wrasidlo, AD 718834 (1971); in part in ACS Polym. Preprints, **12**, 755 (1971).
120. J. K. Gilham, M. B. Roller, Polym. Eng. Sci., **11**, 295 (1971).
121. W. M. Edwards, I. M. Robinson, US pat. 2,710,853, 14 June 1955, via Ref. 128.
122. K. W. Doak, Modern Plastics Encyclopedia, **45**, 1A, 14 (1967).
123. J. M. Barton, J. P. Critchley, Polymer, **11**, 212 (1970).
124. S. L. Cooper, A. D. Mair, A. V. Tobolsky, Textile Res. J., **35**, 1110 (1965).
125. J. H. Freeman, L. W. Frost, G. M. Bower, E. J. Traynor, Conf. Structural Plastics, Adhesives, Filament Wound Composites, WPAFB, Dayton, Ohio, **1**, 30 (1962).
126. G. A. Bernier, D. E. Kline, J. Appl. Polym. Sci., **12**, 593 (1968).
127. A. D. Mair, M. C. Shen, A. V. Tobolsky, AD 604010 (1964) "High temperature polymers: H-Film and SP-Polymer", ONR Tech. Rept. RLT-83; via Ref. 126.
128. C. E. Sroog, J. Polym. Sci. C., **16**, 1191 (1967); data quoted from W. E. Tatum, L. E. Amborski, C. W. Gerow, J. F. Heacock, R. S. Mallouk, Electrical Insulation Conf., Chicago, 1963.
129. A. V. Sidorovich, Ye. V. Kuvshinskii, Polym. Sci. USSR, **10**, 1627 (1968); (transl'n of Vysokomol. Soedin. A, **10**, 1401 (1968).
130. W. Wrasidlo, J.M. Augl, J. Polym. Sci. A-1, **7**, 321 (1969).
131. G. A. Progany, Polymer, **11**, 66 (1970).
132. J. M. Augl, Polym. Sci. A-1, **8**, 3145 (1970).
133. W. Wrasidlo, ACS Org. Coatings Plast. Chem., **30**, 97 (1970).
134. P. M. Hergenrother, AD 710504 (1970). Includes values of N. J. Johnston and W. J. Wrasidlo.
135. W. Wrasidlo, J. M. Augl, Macromolecules, **3**, 544 (1970).
136. P. M. Hergenrother, H. H. Levine, J. Polym. Sci. A-1, **5**, 1453 (1967).
137. W. Wrasidlo, J. Polym. Sci. A-1, **8**, 1107 (1970).
138. W. Wrasidlo, J. M. Augl, J. Polym. Sci. B, **7**, 281 (1969).
139. W. Wrasidlo, J. M. Augl, J. Polym. Sci. A-1, **7**, 3393 (1969).
140. P. M. Hergenrother, SAMPE Quarterly, **3**, 1 (1971).
141. P. M. Hergenrother, D. E. Kiyohara, Macromolecules, **3**, 387 (1970).
142. R. J. Kray, R. Seltzer, R. A. E. Winter, ACS Org. Coatings Plast. Chem., **31**, 569 (1971).
143. J. K. Gilham, Polym. Eng. Sci., **7**, 225 (1967).
144. J. K. Gilham, Science, **139**, 494 (1963).
145. P. W. Morgan, J. Polym. Sci. A, **2**, 437 (1964).
146. G. M. Bower, J. H. Freeman, E. J. Traynor, L. W. Frost, H. A. Burgman, C. R. Ruffing, J. Polym. Sci. A-1, **6**, 877 (1968).
147. J. H. Freeman, L. W. Frost, G. M. Bower, E. J. Traynor, H. A. Burgman, C. R. Ruffing, Polymer. Eng. Sci., **9**, 56 (1969).
148. J. P. Critchley, V. C. R. McLoughlin, J. Thrower, I. M. White, Brit. Polym. J., **2**, 288 (1970).
149. R. Dine-Hart, W. W. Wright, Royal Aircraft Establishment, Farnborough, Hants., UK, Unpublished results.
150. J. Thrower, W. A. Lee, Royal Aircraft Establishment, Farnsborough, Hants., UK, Unpublished results.
151. W. Wrasidlo, J. M. Augl, J. Polym. Sci. A-1, **7**, 1589 (1969).
152. J. P. Critchely, M. A. White, Royal Aircraft Establishment, Farnsborough, Hants., UK, Unpublished results.
153. J. P. Critchley, M. A. White, Royal Aircraft Establishment, Farnsborough, Hants., UK, J. Polym. Sci. A-1, **10**, 1809 (1972).
154. J. P. Critchley, Royal Aircraft Establishment, Farnsborough, Hants., UK, Unpublished results.
155. I. M. Dolgopol'skii, Kauch. I Rezina, **30**, No. 2, 31 (1971), (transl'n in Soviet Rubber Technol., **30**, No. 2, 28 (1971).
156. E. Dorfman, W. E. Emerson, R. J. Gruber, A. A. Lemper, B. M. Rushton, T. L. Graham, Angew. Makromol. Chem., **16/17**, 75 (1971).
157. P. M. Hergenrother, D. E. Kiyohara, J. Macromol. Sci. A, **5**, 365 (1971).
158. W. H. Wrasidlo, P. M. Hergenrother, Macromolecules, **3**, 548 (1970).
159. W. Wrasidlo, J. Polym. Sci., A-2, **9**, 1603 (1971).
160. G. L. Brode, ACS Polym. Preprints, **6**, 626 (1965).
161. R. G. Beaman, J. Appl. Polym. Sci., **9**, 3949 (1965).
162. W. W. Moseley, R. G. Parrish; private communication to Ref. 161.
163. E. L. Wittbecker, Chem. Eng. News, **41**, 41 (1963).
164. A. H. Frazer, Macromolecules, **1**, 199 (1968).
165. A. Schiller, J. C. Petropoulos, C. S. H. Chen, J. Appl. Polym. Sci., **8**, 1699 (1964).
166. M. E. Baird, G. T. Goldsworthy, C. J. Creasey, J. Polym. Sci. B, **6**, 737 (1968).
167. M. E. Baird, J. Polym. Sci. A-2, **8**, 739 (1970).
168. P. J. Flory, L. Mandelkern, H. K. Hall, J. Am. Chem. Soc., **73**, 2532 (1951).
169. P. Johncock, P. A. Grattan, M. A. H. Hewins, Royal Aircraft Establishment, Farnsborough, Hants., UK -Unpublished report.
170. O. Ishizuka, A. Okada, S. Ueda, T. Muroi, I. Ikoma, Chem. High Polymers (Tokyo), **17**, 143 (1960); Chem. Abstr., **55**, 15997c (1961).
171. F. D. Hartley, F. W. Lord, L. B. Morgan, Intern. Symp. Macromol. Chem., Ric. Sci., **25**, 577 (1955).
172. H. K. Reimschuessel, L. G. Roldan, J. P. Sibilia, J. Polym. Sci. A-2, **6**, 559 (1968).
173. J. Preston, US pat. 3,376,28, 2 April, 1968.

174. M. R. Lilyquist, J. R. Holden, J. Polym. Sci. C, **19**, 77 (1967).
175. J. K. Stille, M. A. Bedford, J. Polym. Sci. A-1, **6**, 2331 (1968).
176. H. K. Reimschuessel, K. P. Klein, G. J. Schmitt, Macromolecules, **2**, 567 (1969).
177. H. K. Reimschuessel, K. P. Klein, J. Polym. Sci. A-1, **9**, 3071 (1971).
178. R. E. Kesting, K. F. Jackson, E. B. Klusmann, F. J. Gerhart, J. Appl. Polym. Sci., **14**, 2525 (1970).
179. R. E. Kesting, K. F. Jackson, J. M. Newman, J. Appl. Polym. Sci., **15**, 1527 (1971).
180. J. M. Augl, AD 718329 (1970).
181. E. J. Zaganiaris, L. H. Sperling, A. V. Tobolsky, J. Macromol. Sci. A-1, **6**, 1111 (1967).
182. L. H. Sperling, S. L. Cooper, A. V. Tobolsky, J. Appl. Polym. Sci., **10**, 1725 (1966).
183. A. B. Thompson, D. W. Woods, Trans. Faraday Soc., **52**, 1383 (1956).
184. H. A. Vogel, J. Polym. Sci. A-1, **8**, 2035 (1970).
185. Z. Dobkowski, B. Krajewski, Z. Weilgosz, Polim. Tworz. Wielk., **15**, 428 (1970).
186. J. C. H. Hwa, W. A. Fleming, L. Miller, J. Polym. Sci. A, **2**, 2385 (1964).
187. K. Kanamura, M. Hirata, Kolloid-Z. -Z. Polym., **230**, 206 (1969).
188. R. F. Clash, L. M. Runkieweiz, Ind. Eng. Chem., **36**, 279 (1944).
189. J. M. G. Cowie, P. M. Toporowski, F. Costaschuk, Makromol. Chem., **121**, 51 (1969).
190. C. Klason, J. Kubat, A. de Ruvo, Rheol. Acta, **6**, 390 (1967).
191. K. E. Perepelkin, Mekh. Polim., **5**, 790 (1971).
192. T. Hatakeyama, H. Kanetsuna, Chem. High Polymer (Tokyo), **26**, 76 (1969).
193. J. Kubat, C. Pattyranie, Nature, **215**, 390 (1967).
194. E. Fukada, M. Date, N. Hirai, J. Polym. Sci. C, **23**, Pt. 2, 509 (1968).
195. V. A. Kargin, P. V. Kozlov, Nai-Ch'ang Wang, Dokl. Akad. Nauk SSSR, **130**, 356 (1960); Chem. Abstr., **56**, 605b (1962).
196. K. Tsuge, Y. Wada, J. Phys. Soc. (Japan), **17**, 156 (1962); via Ref. 199.
197. G. P. Mikhailov, A. I. Artyukhov, V. A. Shevelev, Polym. Sci. USSR, **11**, 553 (1969)(transl'n of Vysokomol. Seodin. A, **11**, 553 (1969)).
198. E. L. Back, I. E. Dioriksson, Svensk Paperstid., **72**, 687 (1969); via Chem. Titles, No. 26 (1969).
199. Y. Ogiwara, H. Kubota, S. Hayashi, N. Mitomo, J. Appl. Polym. Sci., **14**, 303 (1970).
200. Y. Nakamura, J. C. Arthur, M. Negishi, K. Doi, E. Kageyama, K. Kudo, J. Appl. Polym. Sci., **14**, 929 (1970).
201. F. E. Wiley, Ind. Eng. Chem., **34**, 1052 (1942).
202. J. K. Gillham, R. F. Schwenker, Appl. Polym. Symp., **2**, 59 (1966).
203. A. Sharples, F. L. Swinton, J. Polym. Sci., **50**, 53 (1961).
204. J. H. Daane, R. E. Barker, J. Polym. Sci. B, **2**, 343 (1964).
205. K. Nakamura, Chem. High Polym. (Tokyo), **13**, 47 (1956).
206. L. Mandelkern, P. J. Flory, J. Am. Chem. Soc., **73**, 3206 (1951).
207. J. Russell, R. G. Van Kerpel, J. Polym. Sci., **25**, 77 (1957).
208. R. E. Barker, C. R. Thomas, J. Appl. Phys., **35**, 87 (1964).
209. B. S. Sprague, J. Polym. Sci. C, **20**, 159 (1967); via ref. 7.
210. N. N. Druzhinina, M. P. Pen'kova, R. M. Livshits, Z. A. Rogovin, Polym. Sci. USSR, **10**, 3181 (1968) (transl'n of Vysokomol. Soedin. A, **10**, 2743 (1968)).
211. K. S. Patel, K. C. Patel, R. D. Patel, Makromol. Chem., **132**, 7 (1970).
212. J. K. Gillham, Appl. Polym. Symp., **2**, 45 (1966).
213. J. M. G. Cowie, R. J. Ranson, Makromol. Chem., **143**, 105 (1971).
214. K. Ueberreiter, Z. Physik. Chem. B, **48**, 197 (1941); Chem. Abstr., **36**, 1489 (1942).
215. A. F. Klarman, A. V. Galanti, L. H. Sperling, J. Polym. Sci. A-2, **7**, 1513 (1969).
216. R. Houwink, "Fundamentals of Synthetic Polymer Technology", Elsevier, New York, 1949, p. 49.
217. G. A. Sorokin, I. V. Tishunin, E. N. Forminykh, Vysokomol. Soedin. B, **11**, 522 (1969).
218. W. J. Feast, W. K. R. Musgrave, N. Reeves, J. Polym. Sci. A-1, **9**, 2733 (1971).
219. N. H. Reinking, A. E. Bamabeo, W. F. Hale, J. Appl. Polymer. Sci., **7**, 2153 (1963).
220. J. Lal, G. S. Trick, J. Polym. Sci. A-1, **8**, 2339 (1970).
221. A. S. Hay, R. F. Clark, Macromolecules, **3**, 533 (1970).
222. A. R. Blythe, G. M. Jeffs, J. Macromol. Sci. B, **3**, 141 (1969).
223. D. J. Marks, Thesis, Manchester, Univ, 1961.
224. G. Allen, J. Hurst, Unpublished work.
225. G. Allen, C. Booth, S. J. Hunt, C. Price, F. Vernon, R. F. Warren, Polymer, **8**, 406 (1967).
226. P. G. Wapner, (supervisor: W. C. Forsman), PhD. Thesis, Univ. Pennsylvania 1971; via Diss. Abstr. Intern. B, **32**, 2332 (1971).
227. N. Doddi, W. C. Forsman, C. C. Price, Macromolecules, **4**, 648 (1971).
228. H. Schnell, L. Bottenbruch, G. Darsow, K. Weirauch, Brit. 1,400,00, 15. Jan. 1969.
229. P. E. Wei, P. E. Butler, J. Polym. Sci. A-1, **6**, 2461 (1968).
230. V. Frosini, E. Butta, M. Calamia, J. Appl. Polym. Sci., **11**, 527 (1967).
231. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Meila, Polymer, **3**, 271 (1962).
232. D. J. H. Sandiford, J. Appl. Chem., **8**, 188 (1958).
233. K. -H. Helwege, J. Hennig, W. Knappe, Kolloid Z. -Z. Polym., **186**, 29 (1962).
234. K. -H. Helwege, R. Hoffman, W. Knappe, Kolloid Z.-Z. Polym., **226**, 109 (1968).
235. M. Barccaredda, E. Butta, V. Frosini, Eur. Polym. J., **2**, 423 (1966).
236. A. Nagai, T. Ishibashi, M. Takayanagi, Rept. Progr. Polym. Phys. Japan, **10**, 341 (1967).

237. A. I. Marei, Ye. A. Sidorovich, G. Ye. Novikova, E. I. Rodina, Polym. Sci. USSR, **10**, 630 (1968) (transl'n of Vysokomol. Soedin. A, **10**, 542 (1968)).
238. J. A. Faucher, J. V. Koleske, Polymer, **9**, 44 (1968).
239. J. C. Swallow, Proc. Roy. Soc., (London) A, **238**, 1 (1957).
240. L. Mandelkern, N. L. Jain, H. Kim, J. Polym. Sci., A-2, **6**, 165 (1968); from results of Ref. 241.
241. Y. Ishida, M. Matsuo, M. Takayanagi, J. Polym. Sci. B, **3**, 321 (1965).
242. A. Aoki, S. Sawada, Repts. Progr. Polym. Phys. (Japan), **10**, 261,509 (1967).
243. W. G. Miller, J. H. Saunders, J. Appl. Polym. Sci., **13**, 1277 (1969).
244. E. J. Vandenberg, R. H. Ralston, B. J. Kocher, Rubber Age, **102**, 47 (1970).
245. Yu. K. Godovskii, Yu. S. Lipatov, Polym. Sci. USSR, **10**, 34 (1968) (transl'n of Vysokomol. Soedin. A, **10**, 32 (1968)).
246. Yu. K. Govodskii, Yu. S. Lipatov, Vysokomol. Soedin. B, **10**, 323 (1968).
247. A. Conix, J. Polym. Sci., **29**, 343 (1958).
248. R. H. Gobran, M. B. Berenbaum, ACS Meeting 133, April 1958; via E.R. Bertozzi, Rubber Chem. Technol., **41**, 114 (1968).
249. A. V. Tobolsky, W. J. MacKnight, in: H. F. Mark, E. H. Immergut (Eds.), "Polymer Reviews", vol. 13: Polymeric Sulfur and Related Polymer, Interscience, 1965, p. 35.
250. G. Williams, Trans. Faraday Soc., **59**, 1397 (1963).
251. B. E. Read, Polymer, **3**, 529 (1962).
252. V. A. Ponomarenko, N. M. Khomutova, Izv. Akad. Nauk USSR, Ser. Khim., **5**, 1153 (1969).
253. A. Mele, A. Delle Site, C. Bettinali, A. Di Domenico, J. Chem. Phys., **49**, 3297 (1968).
254. J. L. Zollinger, J. R. Throckmorton, S. T. Ting, R. A. Mitsch, D. E. Elrick, J. Macromol. Sci. A, **3**, 1443 (1969).
255. W. A. Lee, Royal Aircraft Establishment, Farnsborough, Hants., UK-Unpublished results.
256. P. Johncock, M. A. H. Hewins, Royal Aircraft Establishment, Farnsborough, Hants., UK-Unpublished report.
257. D. E. Rice, J. Polym. Sci. B, **6**, 335 (1968).
258. A. G. Pittman, D. L. Sharp, J. Polym. Sci. B, **3**, 379 (1965).
259. N. H. Reinking, A. E. Barnabeo, W. F. Hale, J. Appl. Polym. Sci., **7**, 2135 (1963).
260. H. M. Van Dort, C. R. H. I. De Jonge, W. J. Mijs, J. Polym. Sci. C, **22**, 431 (1968).
261. K. Fujimoto, Chem. High Polym. (Japan), **18**, 415 (1961).
262. J. Heijboer, Brit. Polym. J., **1**, 3 (1969).
263. S. G. Turley, ACS Org. Coatings Plast. Chem., **27**, 453 (1967); ACS Polymer Preprints, **8**, 1524 (1967).
264. B. E. Read, G. Williams, Polymer, **2**, 239 (1961).
265. W. H. Linton, H.H. Goodman, J. Appl. Polym. Sci., **1**, 179 (1959).
266. N. G. McCrum, J. Polym. Sci., **54**, 561 (1961).
267. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 263 (1962).
268. G. Williams, Polymer, **4**, 27 (1963).
269. R. E. Wetton, G. Allen, Polymer, **7**, 331 (1966).
270. H. Fischer, W. Langbein, Kolloid Z.-Z. Polym., **216/217**, 329 (1967).
271. L. Bonn, Kolloid Z.-Z. Polym., **201**, 20 (1965).
272. A. H. Willbourn, Trans. Faraday Soc., **54**, 717 (1958).
273. R. G. Saba, J. A. Sauer, A. E. Woodward, J. Polym. Sci. A, **1**, 1483 (1963).
274. D. Simms; private communication: Ministry of Aviation E.R.D.E., Waltham Abbey, Essex, UK.
275. G. A. Clegg, D. R. Gee, T. P. Melia, A. Tyson, Polymer, **9**, 501 (1968).
276. J. A. Faucher, J. Polym. Sci. B, **3**, 143 (1965).
277. R. E. Wetton, G. Williams, Trans. Faraday Soc., **61**, 2132 (1965).
278. R. E. Wetton, G. S. Fielding-Russell, K. U. Fulcher, J. Polym. Sci. C, **30**, 219 (1968).
279. S. Yoshida, M. Sakiyama, S. Seki, Rept. Progr. Polym. Phys. (Japan), **12**, 247 (1969).
280. S. Yoshida, M. Sakiyama, S. Seki, Polym. J. (Japan), **1**, 573 (1970).
281. J. J. Stratta, F. P. Reding, J. A. Faucher, J. Polym. Sci. A, **2**, 5017 (1964).
282. F. B. Cramer, R. G. Beaman, J. Polym. Sci., **21**, 237 (1956).
283. W. W. Schartz, R. D. Lowrey, J. Appl. Polym. Sci., **11**, 553 (1967).
284. P. Johncock, P. A. Grattan, Royal Aircraft Establishment, Farnsborough, Hants., UK-Unpublished report.
285. P. Johncock, J.D. Lee, Royal Aircraft Establishment, Farnsborough, Hants., UK-Unpublished report.
286. O. G. Lewis, L. V. Gallagher, AD 649869 (1966).
287. A. T. DiBenedetto, K. L. Trachte, J. Appl. Polym. Sci., **14**, 2249 (1970).
288. E. Butta, Mater, Plastics Elastomers, **35**, 1411 (1969).
289. S. de Petris, V. Frosini, E. Butta, M. Baccaredda, Makromol. Chem., **109**, 54 (1967).
290. F. E. Karasz, J. M. O'Reilly, J. Polym. Sci. B, **3**, 561 (1965).
291. F. E. Karasz, H. E. Bair, J. M. O'Reilly, J. Polym. Sci. A-2, **6**, 1141 (1968).
292. G. A. Bernier, R. P. Kambour, Macromolecules, **1**, 393 (1968).
293. M. H. Litt, A. V. Tobolsky, J. Macromol. Sci. B1, **3**, 433 (1967).
294. A. J. Chalk, A. S. Hay, J. Polym. Sci. A-1, **7**, 691 (1969).
295. G. F. L. Ehlres, K. R. Fisch, W. R. Powell, J. Polym. Sci. A-1, **7**, 2931 (1969).
296. G. Allen, M. W. Coville, R. M. John, R. F. Warren, Polymer, **11**, 492 (1970).
297. F. E. Karasz, W. J. MacKnight, J. Stoelting, J. Appl. Phys., **41**, 4357 (1970).
298. J. M. Barrales-Rienda, J. M. G. Fatou, Kolloid Z.-Z. Polym., **244**, 317 (1971).
299. A. Eisenberg, B. Cayrol, J. Polym. Sci. C, **35**, 129 (1971).
300. C. I. Chung, J. A. Sauer, J. Polym. Sci. A-2, **9**, 1097 (1971); T_g values mostly from manufacturers' data.
301. P. A. Gratten, V. C. R. McLoughlin, W. A. Lee, Royal Aircraft Establishment, Farnsborough, Hants., UK-Unpublished data.

302. L. W. Breed, R. L. Elliot, M. E. Whitehead, J. Polym. Sci. A-1, **5**, 2745 (1967).
303. L. T. C. Lee, E. M. Pearce, J. Polym. Sci. A-1, **9**, 557 (1971).
304. K. Saotome, H. Komoto, J. Polym. Sci. A-1, **4**, 1475 (1966).
305. H. G. Weyland, P. J. Hoftyzer, D. W. van Krevelen, Polymer, **11**, 79 (1970).
306. H. M. van Dort, C. A. M. Hoefs, E. P. Magre, A. J. Schopf, K. Yntema, Eur. Polym. J., **4**, 275 (1968).
307. B. J. Tabor, E. P. Magre, J. Boon, Eur. Polym. J., **7**, 1127 (1971).
308. J. C. Mileo, B. Sillion, G. De Gaudemaris, Compt. Rend. C, **268**, 1949 (1969).
309. M. E. B. Jones, I. Goodman, Brit. Pat. 1,128,807, 2. Oct. 1968.
310. D. Brown, M. E. B. Jones, W. R. Maltman, J. Polym. Sci. B, **6**, 635 (1968).
311. V. Frosini, M. Pasquini, E. Butta, Chim. Ind., **53**, 140 (1971).
312. E. Butta, S. de Petris, V. Frosini, M. Pasquini, Eur. Polym. J., **7**, 387 (1971).
313. G. F. L. Ehlers, R. C. Evers, K. R. Fisch, J. Polym. Sci. A-1, **7**, 3413 (1969).
314. B. E. Jennings, M. E. B. Jones, J. B. Rose, J. Polym. Sci. C, **16**, 715 (1967).
315. M. E. A. Cudby, R. G. Feasey, B. E. Jennings, M. E. B. Jones, J. B. Rose, Polymer, **6**, 589 (1965).
316. J. E. Kurz, J. C. Woodbrey, M. Ohta, J. Polym. Sci. A-2, **8**, 1169 (1970).
317. G. W. Miller, ACS Polymer Preprints, **8**, 1072 (1967).
318. G. W. Miller, "Thermal Analysis", Academic Press Inc., New York, Vol. I, 1969, p. 435.
319. T. F. Gray, R. L. Combs, D. F. Slonaker, W. C. Wooten, SPE Tech. Papers, **13**, 370 (1967).
320. M. Baccaredda, E. Butta, V. Frosini, S. de Petris, J. Polym. Sci. A-2, **5**, 1296 (1967).
321. L. M. Robeson, J. A. Faucher, J. Polym. Sci. B, **7**, 35 (1969).
322. G. F. L. Ehlers, K. R. Fisch, W. R. Powell, J. Polym. Sci. A-1, **7**, 2955 (1969).
323. N. J. Mills, A. Nevin, J. McAinsh, J. Macromol. Sci. B, **4**, 863 (1970).
324. E. J. Vandenberg, J. Polym. Sci., **47**, 486,489a (1960).
325. A. S. Hay, Macromolecules, **2**, 107 (1969).
326. J. M. Crissman, J. A. Sauer, A. E. Woodward, J. Polym. Sci., A. E. Woodward, J. Polym. Sci. A, **2**, 5075 (1964).
327. W. P. Slichter, Makromol. Chem., **34**, 67 (1959).
328. L. E. St. Pierre, C. C. Price, J. Am. Chem. Soc., **78**, 3432 (1956).
329. T. M. Connor, A. Hartland, Polymer, **9**, 591 (1968).
330. G. Williams, Trans. Faraday Soc., **61**, 1564 (1965); data used by ref. 329.
331. G. Allen, C. Booth, M. N. Jones, D. J. Marks, W. D. Taylor, Polymer, **5**, 547 (1964).
332. I. Kuntz, C. Cozewith, H. T. Oakley, G. Via, H. T. White, Z. W. Wilchinsky, Macromolecules, **4**, 4 (1971).
333. I. P. Losev, O. V. Smimova, Ye. V. Korovina, Vysokomol. Soedin., **5**, 1603 (1963); (transl'n in Polymer Sci. USSR, **5**, 707 (1964)).
334. H. Schnell, Ind. Eng. Chem., **51**, 157 (1959).
335. A. N. Perepelkin, P. V. Kozlov, Polymer Sci. USSR, **8**, 57 (1966); (transl'n of Vysokomol. Soedin., **8**, 56 (1966)).
336. F. P. Reding, J. A. Faucher, R. D. Whitman, J. Polym. Sci., **54**, S56 (1961).
337. L. J. Garfield, J. Polym. Sci. C, **30**, 561 (1970).
338. H. Schnell, "Chemistry and Physics of Polycarbonates", Interscience, New York, 1964, Chap. III.
339. B. F. Malichenko, V. I. Feoktistova, A. E. Nesterov, Vysokomol, Soedin. B, **11**, 543 (1969).
340. E. N. Elam, J. C. Martin, R. Gilkey, US pat. 3,313,777, 11. April 1967 (Brit. 962,913).
341. W. Gawlak, R. P. Palmer, J. B. Rose, D. J. Sandford, A. Turner-Jones, Chem. Ind., **25**, 1148 (1962).
342. A. A. D'Onofrio, Brit. pat. 1,011,283, 24. Nov. 1965.
343. O. V. Smirnova, El Said Ali Kasan, I. P. Losev, G. S. Kolesnikov, Polymer Sci. USSR, **7**, 557 (1965); (transl'n of Vysokomol. Soedin., **7**, 503 (1965)).
344. H. Schnell, Angew. Chem., **68**, 633 (1956).
345. H. Schnell, Plastics Intst. Trans. J., **28**, 143 (1960).
346. O. V. Smirnova, Ye. V. Korovina, G. S. Kolesnikov, A. M. Lipkin, S. I. Kuzina, Polym. Sci. USSR, **8**, 776 (1966); translation of Vysokomol. Soedin., **8**, 708 (1966).
347. US Patent 3,546,165, Dec. 8, 1970, assigned to P. W. Morgan.
348. D. Stefan, H. L. Williams, D. R. Renton, M. M. Pintar, J. Macromol. Sci. B, **4**, 853 (1970).
349. D. W. McCall, D. R. Falcone, Trans. Faraday Soc., **66**, 262 (1970).
350. R. E. Robertson, General Electric Co., R & D Center, Schenectady, N. Y., Gen Chem. Lab Report No. 67 C-353, 1967; also Appl. Polym. Symp., **7**, 201 (1968).
351. W. G. Miller, Appl. Polym. Symp., **10**, 35 (1969).
352. M. S. Ali, R. P. Shelson, J. Appl. Polym. Sci., **14**, 2619, (1970).
353. J. van Turnhout, Polym. J. (Japan), **2**, 173 (1971).
354. S. Matsuoka, Y. Ishida, J. Polym. Sci. C, **14**, 247 (1966).
355. F. Krum, F. H. Muller, Kolloid Z., **164**, 81 (1959).
356. I. I. Perepechko, L. A. Bodrova, Soviet Plastics, **7**, 58 (1967); translation of Plasticheskie Massy, **7**, 56 (1967).
357. J. P. Mercier, J. J. Aklonis, A. V. Tobolsky, AD 410922 (1963), The Viscoelastic Behavior of Polycarbonates of Bisphenol A, ONR Tech. Report RLT 60.
358. J. H. Golden, B. L. Hammant, E. A. Hazell, J. Appl. Polym. Sci., **11**, 1571 (1967).
359. J. P. Mercier, J. J. Aklonis, M. Litt, A. V. Tobolsky, J. Appl. Polym. Sci., **9**, 447 (1965).
360. W. J. Jackson, J. R Caldwell, Ind. Eng. Chem. Prod. Res. Develop., **2**, 246 (1963).
361. J. L. Work, J. E. Herweh, J. Polym. Sci. A-1, **6**, 2022 (1968).
363. G. P. Mikhailov, M. P. Eidel'nant, Vysokomol. Soedin., **2**, 287 (1960).
364. J. M. O'Reilly, F. E. Karasz, H. E. Bair, J. Polym. Sci. C, **6**, 109 (1964).
365. A. Conix, L. Jeurissen, Adv. Chem. Ser., **48**, 172 (1964).

366. J. Bussink, J. Heijboer, J. A. Prins (Ed.), "Physics of Non-Crystalline Solids", Proc. Intern. Conf. Delft, 1964, p. 388, North Holland Publishing Co., Amsterdam, 1965.
367. B. von Falkai, W. Rellensmann, Makromol. Chem., **75**, 112 (1964).
368. T. Sulzberg, R. J. Cotter, Macromolecules, **2**, 146, (1969).
369. E. V. Gouinlock, E. J. Quinn, H. W. Marciniak, R. R. Hindersinn, IUPAC Preprints, Intern. Symp. Macromol. Chem., Toronto, Sect. A13.3 (1968).
370. Du Pont trade literature, A 49783: 940 Thermomechanical Analyzer.
371. H. G. Weyland, C. A. M. Hoefs, K. Yntema, W. J. Mijs, Eur. Polym. J., **6**, 1339 (1970).
372. P. W. Morgan, Macromolecules, **3**, 536 (1970).
373. I. V. Yannas, A. C. Lunn, J. Macromol. Sci. B, **4**, 603 (1970).
374. I. I. Perepechko, L. A. Kvacheva, I. I. Levantovskaya, Vysokomol. Soedin. A, **13**, 702 (1971), translation of Polym. Sci. USSR, **13**, 796 (1971).
375. Y. Yamaoto, S. Tsuge, T. Takeuchi, Bull. Chem. Soc. (Japan), **44**, 1145 (1971).
376. A. Conix, J. Polym. Sci. C, **16**, 3821 (1968).
377. H. van Hoorn, J. Appl. Polym. Sci., **12**, 871, (1968).
378. I. I. Perepechko, L. A. Kvacheva, Soviet Plastics, **8**, (1971), translation of Plasticheskie Massy, **8**, 42 (1971).
379. T. Takekoshi, ACS Polym. Preprints, **10**, 103 (1969).
380. Brit. Pat. 1,052,757, Dec. 30, 1966.
381. G. Bier, Adv. Chem. Ser., **91**, 612 (1969).
382. A. Conix, IUPAC Preprints, Intern. Symp. Macromol. Chem., Montreal, Sect. D39 (1961); via ref. 44.
383. C. S. Marvel, C. H. Young, J. Am. Chem. Soc., **73**, 1066 (1951).
384. W. A. Lee, B. Stag, Royal Aircraft Establishment, Farnborough, Hants, UK, TR 71223 (1971).
385. S. D. Bruck, J. Polym. Sci. B, **4**, 933 (1966).
386. V. C. R. McLoughlin, J. Thrower, M. A. H. Hewins, J. S. Pippett, M. A. White, Royal Aircraft Establishment, Farnborough, Hants, UK, TR 70160 (1970).
387. I. N. Duling, US Pat. 3,436,376, April 1, 1969.
388. R. G. Beaman, J. Polym. Sci., **9**, 470 (1952).
389. O. B. Edgar, J. Chem. Soc., 2638 (1952).
390. O. B. Edgar, R. Hill, J. Polym. Sci., **8**, 1 (1952).
391. E. F. Izard, J. Polym. Sci., **9**, 35 (1952).
392. A. Steitz, J. O. Knobloch, J. Paint Technol., **40**, 384 (1968).
393. H. J. Kolb, E. F. Izard, J. Appl. Phys., **20**, 564 (1949).
394. H. Mark, Kolloid Z.-Z. Polym., **216/217**, 126 (1967).
395. N. G. Gaylord, H. F. Mark, "Linear and Stereoregular Addition Polymers", Interscience, New York, 1959.
396. Y. Wada, J. Phys. Soc. (Japan), **16**, 1226 (1961).
397. R. Zannetti, P. Manaresi, L. Baldi, Shim. Ind., **43**, 1310 (1961).
398. I. M. Ward, Textile Res. J., **13**, 650 (1961).
399. A. Anton, J. Appl. Polym. Sci., **12**, 2117 (1968).
400. S. Saito, T. Nakajim, J. Appl. Polym. Sci., **2**, 93 (1959).
401. L Mandelkern, G. M. Martin, F. A. Quinn, J. Res. Nat. Bur. Std., **58**, 137 (1957).
402. I. M. Ward, Polymer, **5**, 59 (1964).
403. G. Farrow, J. McIntosh, I. M. Ward, Makromol. Chem., **38**, 147 (1960).
404. J. Bateman, R. E. Richards, G. Farrow, I. M. Ward, Polymer, **1**, 63 (1960).
405. B. Ke, J. Appl. Polym. Sci., **6**, 624 (1962).
406. J. H. Dumbleton, T. Murayama, Kolloid Z.-Z., Polym., **220**, 41 (1967).
407. W. R. Vieth, E. X. Matulevicius, S. R. Mitchell, Kolloid Z.-Z. Polym., **220**, 49 (1967).
408. G. S. Y. Yeh, P. H. Geil, J. Macromol. Sci. B, **1**, 235 (1967).
409. G. P. Michailov, V. A. Shevelev, Polym. Sci., USSR **8**, 840 (1966), translation of Vysokomol. Soedin., **8**, 763 (1966).
410. J. H. Dumbleton, J. P. Bell, T. Murayama, J. Appl. Polym. Sci., **12**, 2491 (1968).
411. J. G. Smith, C. J. Kibler, B. J. Sublett, J. Polym. Sci. A-1, **4**, 1851 (1966), values from Ref. 412.
412. R. M. Schulken, R. E. Boy, R. H. Cox, J. Polym. Sci. C, **6**, 17 (1964).
413. I. Ito, S. Okajima, F. Shibata, J. Appl. Polym. Sci., **14**, 551 (1970).
414. J. B. Jackson, G. W. Longman, Polymer, **10**, 873 (1969).
415. J. H. Dumbleton, T. Murayama, J. Appl. Polym. Sci., **14**, 2912 (1970).
416. E. A. Egorov, V. V. Zhizhenkov, Mekh. Polim., **192**, 24 (1971).
417. T. R. Manley, J. Thermal Anal., **2**, 411 (1970).
418. H. Sasabe, K. Sawamura, S. Salto, K. Yoda, Polymer J. (Japan), **2**, 518 (1971).
419. I. M. Dolgopol'skii, Kh. A. Dobina, V. S. Fikhtengol'ts, S. A. Kamysheva, M. I. Sinaiskaya, L. G. Balashova, R. V. Zollotareva, Polymer Sci. USSR, **9**, 1723 (1967), translation of Vysokomol. Soedin. A, **9**, 1536 (1967).
420. E. V. Gouinlick, C. J. Verbanic, G. C. Schweiker, J. Appl. Polym. Sci., **1**, 361 (1959).
421. W. R. Griffin, Rubber World, **136**, 687 (1957).
422. G. C. Schweiker, P. Robitschek, J. Polym. Sci., **24**, 33 (1957).
423. J. H. Sewell, Royal Aircraft Establishment, Farnborough, Hants, UK, unpublished results.
424. I. J. Goldfarb, R. McGuchan, AD 697987 (1987).
425. R. C. Evers, G. F. L. Ehlers, J. Polym. Sci. A-1, **7**, 3020 (1969).
426. A. Conix, Ind. Eng. Chem., **51**, 147 (1959).
427. V. Frosini, G. Vallebona, Chim. Ind., **52**, 499 (1970).
428. N. H. Shearer, SPE Tech. Papers, 12, 22nd ANTEC, Montreal, 1966, Session XIV, paper 1.
429. C. S. Marvel, J. H. Johnson, J. Am. Chem. Soc., **72**, 1674 (1950).
430. S. M. Somerville, I. M. White, Royal Aircraft Establishment, Farnborough, Hants, UK, TR 66345 (1966).
431. I. M. White, M. A. H. Hewins, S. M. Somerville, paper presented at 2nd Intern. Sump. Organosilicon Chem., Univ. Bordeaux, 9-12 July, 1968.
432. W. A. Lee, I. H. Sewell, Royal Aircraft Establishment, Farnborough, Hants, UK, unpublished results.

433. J. H. Sewell, B. Stagg, Royal Aircraft Establishment, Farnborough, Hants, UK, unpublished results.
434. W. A. Lee, F. Pinchin, Royal Aircraft Establishment, Farnborough, Hants, UK, unpublished results.
435. French pat. 1,175,362 (1959); Brit. pat. 863,704 (1961), via Ref. 313.
436. W. M. Eareckson, J. Polym. Sci., **40**, 399 (1959).
437. V. V. Korshak, S. V. Vinogradova, C. L. Slonimskii, Ya. S. Vygodskii, S. N. Salazkin, A. A. Askadskii, A. I. Mzhel'skii, V. P. Sidorova, Polym. Sci. USSR, **10**, 2395 (1968), translation of Vysokomol Soedin. A, **10**, 2058 (1968).
438. W. A. Lee, I. M. White, F. Pinchin, Royal Aircraft Establishment, Farnborough, Hants, UK, unpublished results.
439. K. A. Gol'dgammer, G. G. Pimenov, A. I. Maklakov, V. V. Korshak, S. V. Vinogradova, P. M. Valetskii, A. N. Baskakov, Vysokomol. Soedin. A, **10**, 821 (1968), translation of Polym. Sci. USSR, **10**, 953 (1968).
440. R. L. Combs, R. G Nations, J. Polym. Sci. C, **30**, 407 (1970). unpublished results of T. F. Gray.
441. V. V. Korshak, G. L. Berestneva, S. V. Vinogradova, A. N. Baskakov, P. M. Valetskii, Polymer Sci. USSR, **10**, 2300 (1968), translation of Vysokomol. Soedin. A, **10**, 1984 (1968).
442. W. J. Jackson, J. R. Caldwell, Brit. pat. 956,206, April 22, 1964.
443. S. S. Karapetyan, A. Ya. Yakubovich, I. L. Knunyants, Polym. Sci. USSR, 1718 (1964), translation of Vysokomol. Soedin., **6**, 1550 (1964).
444. J. E. McIntyre, E. C. Pugh, Brit. pat. 838,986, 1960.
445. K. Saotome, H. Komoto, J. Polym. Sci. A-1, **5**, 119 (1967).
446. W. J. MacKnight, M. Yang, T. Kajiyama, ACS Polym. Preprints, **9**, 860 (1968).
447. D. J. Lyman, J. Heller, M. Barlow, Makromol. Chem., **84**, 64 (1965).
448. A. Anton, H. C. Beachel, J. Polym. Sci. B, **7**, 215 (1969).
449. F. Malichenko, E. V. Shelud'ko, Yu. Yu. Kercha, R. L. Savchenko, Vysokomol. Soedin. A, **11**, 377 (1969), translation in Polym. Sci. USSR, **11**, 423 (1969).
450. B. F. Malichenko, E. V. Shelud'ko, Yu. Yu. Kercha, R. L. Savchenko, Vysokomol. Soedin. A, 11 (1969), translation in Polym. Sci. USSR, **11**, 1721 (1969).
451. B. F. Malichenko, E. V. Shelud'ko, Yu. Yu. Kercha, Vysokomol. Soedin. A, **9**, 2482 (1967), translation in Polym. Sci., USSR, **9**, 2808 (1967).
452. Yu. Yu. Kercha, L. I. Ryabokon, B. F. Malichenko, in: Yu. S. Lipatov, (Ed.), "Synthesis and Physico-Chemistry of Polyurethanes", Naukova Dumka, Kiev, 1968, p. 198.
453. M. A. McCall, J. R. Caldwell, H. G. Morre, H. M. Beard, J. Macromol. Sci., A, **3**, 911 (1969).
454. A. Ye. Nesterov, Yu. Yu. Kercha, Yu. S. Lipatov, L. M. Sushko, L. I. Ryabokon, Polym. Sci. USSR, **9**, 2782 (1967), translation of Vysokomol. Soedin. A, **9**, 2459 (1967).
455. C. B Griffis, C. W. Shurtleff, AD 635114 (1966).
456. M. C. Henry, C. B. Griffis, AD 632196 (1966).
457. P. D. Schumann, E. C. Stump, G. Westmoreland, J. Macromol. Sci. B, **1**, 815 (1967).
458. D. A. Barr, R. N. Haszeldine, C. J. Willis, J. Chem. Soc., 1351 (1961).
459. G. H. Crawford, D. E. Rice, Chem. Eng. News, **38**, 107 (1960).
460. A. Hesling, Plastica, **21**, 206 (1968).
461. E. C. Stump, C. D. Padgett, AD 666801 (1967).
462. G. L. Ball, I. O. Salyer, J. V. Pustinger, H. S. Wilson, AD 672523 (1967).
463. Yu. A. Yuzhelevskii, E. G. Kagan, A. L. Klebanskii, I. A. Zevakin, A. V. Kharlamova, Vysokomol. Soedin. B, **11**, 854 (1969).
464. M. S. Paterson, J. Appl. Phys., **35**, 176 (1964).
465. J. B. Alexopoulos, J. A. Barrie, J. C. Tye, M. Fredrickson, Polymer, **9**, 56 (1968).
466. O. R. Pierce, Y. K. Kim, J. Elastoplastics, **3**, 82 (1971).
467. K. E. Polmanteer, M. J. Hunter, J. Appl. Polym. Sci., **1**, 3 (1959).
468. K. A. Andrianov, T. K. Dzhashiashvili, V. V. Astakin, G. N. Shumakavo, Vysokomol. Soedin. B, **10**, 766 (1968).
469. W. P. Gergen, T. L. Keelan, 11th Annual Meeting Proc. Intern. Inst. Synth. Rubber Producers, Inc., 1970, p. 81.
470. C. E. Weir, W. H. Leser, L. A. Wood, J. Res. Nat. Bur. Std., **44**, 367 (1950).
471. L. A. Wood, J. Polym. Sci., **28**, 319 (1958).
472. M. Sh. Yagfarov, V. S. Ionkin, Polym. Sci. USSR, **10**, 1867 (1968), translation of Vysokomol. Soedin. A, **10**, 1613 (1968).
473. A. Yi, L. E. St. Pierre, J. Polym.Sci. B, **7**, 237 (1969).
474. J. D. Helmer, K. E. Polmanteer, J. Appl. Polym. Sci., **13**, 2113 (1969).
475. K. A. Andrianov, S. E. Yakushkina, Vysokomol. Soedin., **4**, 1193 (1962).
476. K. A. Andrianov, T. K. Dzhashiashvili, V. V. Astakin, G. N. Shumakova, Soviet Plastics, **2**, 47 (1968), translations of Plasticheskie Massy, **2**, 44 (1968).
477. V. C. R. McLoughlin, P. A. Grattan, Royal Aircraft Establishment, Farnborough, Hants, UK, TR 71224 (1971).
478. R. E. Burks, J. L. Greene, N68-16509; Southern Res. Inst. Rept. 25 (1967).
479. J. A. Empen, J. K. Stille, in: N. G. Gaylord (Ed.), "Macromolecular Synthesis", 3, Wiley, New York, 1969, p. 56.
481. E. H. Catsiff, J. Appl. Polym. Sci., **15**, 1641 (1971).
482. A. Nicco, J. P. Machon, H. Fremaux, J. Ph. Pied, B. Zindy, M. Thiery, Eur. Polym. J., **6**, 1427 (1970).
483. R. S. Nevin, E. M. Pearce, J. Polym. Sci. B, **3**, 487 (1965).
484. W. J. Middleton, H. W. Jacobson, R. E. Putnam, H. C. Walter, D. G. Pye, W. H. Sharkey, J. Polym. Sci. A, **3**, 4115 (1965).
485. V. A. Engelhardt; report in Chem. Eng. News, **43**, 80 (1965).
486. W. H. Sharkey, W. J. Middleton, H. W. Jacobson, O. S. Acker, H. C. Walter, Chem. Eng. News, **41**, 46 (1963).
487. J. A. Empen, J. K. Stille, in: N. G. Gaylord (Ed.), "Macromolecular Synthesis", 3, Wiley, New York, 1969, p. 53.
488. C. C. Price, E. A. Blair, J. Polym. Sci. A-1, **5**, 171 (1967).
489. R. M. Fitch, D. C. Helgeson, J. Polym. Sci. C, **22**, 1101 (1969).

490. M. Morton, R. F. Kammereck, L. J. Fetters, Brit. Polym. J., **3**, 120 (1971).
491. H. Komoto, Rev. Phys. Chem. Japan, **37**, 112 (1967).
492. J. Pellon, J. Polym. Sci. A, **1**, 3561 (1963).
493. R. L. Fyans; private communication to Ref. 114.
494. M. Russo, Mater. Plastics Elastomers, **34**, 61 (1968); via RAPRA abstr. 1968, abstr. 4571.
495. E. A. Peterson; private communication to Ref. 481.
496. S. Adamek, B. B. J. Wood, R. T. Woodhams, Rubber Plastics Age Intern., **46**, 56 (1965); Rubber Age, **96**, 581 (1965).
497. C. G. Overberger, J. K. Weise, J. Am. Chem. Soc., **90**, 3538 (1968).
498. H. G. Buhrer. H.-G. Elias, Makromol. Chem., **140**, 41 (1970).
499. C. G. Overberger, J. K. Weise, J. Am. Chem. Soc., **90**, 3533 (1968).
500. M. E. B. Jones, SPE Tech. Papers, 15, 27th ANTEC, Chicago 1969, p. 453.
501. G. Jarrett, M. E. B. Jones, Brit. Polym. J., **2**, 229 (1970).
502. H. Wexler, Makromol. Chem., **115**, 262 (1968).
503. R. G. Beaman, F. B. Cramer, J. Polym. Sci., **21**, 223 (1956).
504. J. H. Magill, J. Polym. Sci. A, **3**, 1195 (1965).
505. B. Ke, A. W. Sisko, J. Polym. Sci., **50**, 87 (1961).
506. Y. Nishijima, J. Selki, T. Kawai, Rept. Progr. Polym. Phys. (Japan), **10**, 473 (1967).
507. M. J. Forster, Textile Res. J., **38**, 474 (1968).
508. K. Neki, M. Takayanagi, Rept. Progr. Polym. Phys. (Japan), **8**, 281 (1965).
509. W. H. Charch, W. W. Moseley, Textile Res. J., **29**, 552 (1959); via Ref. 507.
510. F. Rybnikar, J. Polym. Sci., **28**, 633 (1958).
511. M. Takayanagi, Memoirs Faculty Eng. Kyushu Univ., **23**, 2 (1963); via Ref. 507.
512. S. R. Urzendowski, A. H. Guenther, J. R. Asay, ACS Polym. Preprints, **9**, 878 (1968).
513. J. Rubin, R. D. Andrews, Polym. Eng. Sci., **8**, 302 (1968).
514. E. R. Neuhausl, Plastics Polym., **36**, 93 (1968).
515. D. R. Gee, T. P. Melia, Polymer, **11**, 192 (1970).
516. T. Shibusawa, J. Appl. Polym. Sci., **14**, 1553 (1970).
517. F. R. Prince, E. M. Pearce, R. J. Fredericks, J. Polym. Sci. A-1, **8**, 3533 (1970).
518. V. Frosini, E. Butta, J. Polym. Sci. B, **9**, 253 (1971).
519. V. Zilvar, I. Boukal, J. Hell, Plastics Inst. Trans. J., **35**, 402 (1967).
520. D. C. Prevorsek, R. H. Butler, K. H. Reimschuessel, J. Polym. Sci. A-2, **9**, 867 (1971).
521. K. Schmieder, K. Wolf, Kolloid Z., **134**, 149 (1953).
522. K. H. Illers, H. G. Kilian, R. Kosfeld, Ann. Rev. Phys. Chem., **12**, 49 (1961).
523. K. C. Dao, J. H. Percy, Inst. Physics and Physical Soc., Proc. Conf. on the Science of Materials, p. 151, Aug. 1969; New Zealand, DSIR, Information Series 71.
524. R. F. Boyer, R. S. Spencer, J. Appl. Phys., **15**, 398 (1944).
525. R. Meredith, B. Hsu. Polym. Sci., **61**, 271 (1962).
526. G. F. D'Alelio, "Fundamental Principles of Polymerization", Wiley, New York, 1952, p. 124.
527. D. W. McCall, E. W. Anderson, J. Chem. Phys., **32**, 237 (1960).
528. A. E. Woodward, J. M. Crissman, J. A. Sauer, J. Polym. Sci., **44**, 23 (1960).
529. A. M. Thomas, Nature, **179**, 862 (1957).
530. F. P. Chappel, M. F. Culpin, R. G. Gosden, T. C. Tranter, Conference on Adv. Polym. Sci. Technol., London, 1963.
531. J. S. Ridgeway, J. Polym. Sci. A-1, **7**, 2195 (1969).
532. J. S. Ridgeway, J. Polym. Sci. A-1, **8**, 3089 (1970).
533. J. P. Bell, J. Appl. Polym. Sci., **12**, 627 (1968).
534. E. A. Tippetts, J. Zimmerman, J. Appl. Polym. Sci., **8**, 2465 (1964).
535. H. Komoto, Rev. Phys. Chem. (Japan), **37**, 105 (1967).
536. K. Saotome, H. Komoto, J. Polym. Sci. A-1, **4**, 1463 (1966).
537. H. Komoto, K. Saotome, Chem. High Polymers (Japan), **22**, 337 (1965); Chem. Abstr., **63**, 15045f (1965).
538. G. Champetier, J. P. Pied, Makromol. Chem., **44/46**, 64 (1961); Chem. Abstr., **56**, 1584c (1962).
539. B. S. Marks, G. C. Schweiker, J. Polym. Sci., **43**, 229 (1960).
540. F. R. Prince, E. M. Pearce, Macromolecules, **4**, 347 (1971).
541. D. A. Holmer, J. Polym. Sci. A-1, **6**, 3177 (1968).
542. J. C. Mileo, B. Sillion, G. De Gaudemaris, Compt. Rend. C, **268**, 2007 (1969).
543. J. B. Jackson, Polymer, **10**, 159 (1969).
544. L. C. Glover, B. J. Lyons, ACS Polymer Preprints, **9**, 243 (1968).
545. V. Guidotti, M. Russo, L. Mortillaro, Makromol. Chem., **147**, 111 (1971).
546. G. A. Kuznetsov, V. D. Gerasimov, L. N. Fomenko, A. I. Maklakov, G. G. Pimenov, L. B. Sokolov, Polym. Sci. USSR, **7**, 1763 (1965) (transl'n of Vysokomol. Soedin., **7**, 1592 (1965)).
547. G. S. Kolesnikov, O. Ya. Fedotova, V. V. Trezov, V. N. Kuz'micheva, Polym. Sci. USSR, **10**, 2612 (1968), (transl'n of Vysokomol. Soedin. A, **10**, 2248 (1968)).
548. E. Bessler, G. Bier, Makromol. Chem., **122**, 30 (1969).
549. T. G. Bassiri, A. Levy, M. Litt, J. Polym. Sci. B, **5**, 871 (1967).
550. A. Levy, M. Litt, J. Polym. Sci. B, **5**, 881 (1967).
551. A. Levy, M. Litt, J. Polym. Sci. A-1, **6**, 1883 (1968).
552. G. Allen, C. J. Lewis, S. M. Todd, Polymer, **11**, 44 (1970).
553. H. R. Allcock, R. L. Kugel, K. J. Valan, Inorg. Chem., **5**, 1709 (1966).
554. H. R. Allcock, R. L. Kugel, Inorg. Chem., **5**, 1716 (1966).
555. G. Allen, Brit. Polym. J., **1**, 168 (1969).
556. H. R. Allcock, R. L. Kugel, J. Am. Chem. Soc., **87**, 4216 (1965).
557. S. H. Rose, K. A. Reynard, J. R. Cable, AD 704332 (1970).
558. N. S. Nametkin, N. V. Ushakov, V. M. Vdovin, Polym. Sci. USSR, **13**, 31 (1971) (transl'n of Vysokomol. Soedin. A, **13**, 29 (1971)).
559. N. S. Nametkin, V. M. Vdovin, V. I. Zav'yalov, Dokl. Akad. Nauk SSSR, **162**, 824 (1965).

560. F. S. Model, G. Redl, E.-G. Rochow, J. Polym. Sci. A-1, **4**, 639 (1966).
561. F. S. Model, E. G. Rochow, J. Polym. Sci. A-2, **8**, 999 (1970).
562. N. S. Nametkin, V. M. Vdovin, V. I. Zav'yalov, Polym. Sci. USSR, **7**, 836 (1965) (transl'n of Vysokomol. Soedin., **7**, 757 (1965)).
563. V. C. R. McLoughlin, J. Thrower, P. A. Grattan, M. A. H. Hewins, Royal Aircraft Establishment, Farnborough, Hants, UK, TR 71111 (1971).
564. W. F. Gorham, J. Polym. Sci. A4, **4**, 3027 (1966).
565. H. G. Gilch, W. L. Wheelwright, J. Polym. Sci. A-1, **4**, 1337 (1966).
566. I. N. Markevich, S. I. Beilin, M. P. Teterina, G. P. Karpacheva, B. A. Dolgoplosk, Dokl. Akad. SSSR, 191, 362 (1970).
567. L. A. Auspos, C. W. Burnham, L. A. R. Hal'l, J. K. Hubbard, W. Kirk, J. R. Schaefgen, S. B. Speck, J. Polym. Sci., **15**, 19 (1955).
568. J. K. Stille, R. O. Rakutis, K. Mukamal, F. W. Harris, Macromolecules, **1**, 431 (1968).
569. M. Morton, J. Elastoplastics, **3**, 112 (1971).
570. R. J. Schaffhauser, M. G. Shen, A. V. Tobolsky, J. Appl. Polym. Sci., **8**, 2825 (1964).
571. M. Kaufman, A. F. Williams, J. Appl. Chem., **1**, 489 (1951).
572. A. Chierico, G. Del Nero, G. Lanzi, E. R. Mognaschi, Eur. Polym. J., **3**, 245 (1967).
573. E. Jenckel, Kolloid Z., **100**, 163 (1942).
574. M. L. Dannis, J. Appl. Polym. Sci., **1**, 121 (1959).
575. M. L. Dannis, J. Appl. Polym. Sci., **7**, 231 (1963).
576. R. H. Gerke, J. Polym. Sci., **13**, 295 (1954).
577. M. Baccaredda, E. Butta, Chim. Ind. (Milan), **42**, 978 (1960).
578. E. Pedemonte, U. Bianchi, J. Polym. Sci. B, **2**, 1025 (1964).
579. A. W. Meyer, R. R. Hampton, J. A. Davison, J. Am. Chem. Soc., **74**, 2294 (1952).
580. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 297 (1962).
581. R. J. Morgan, L. E. Nielson, R. Buchdahl, J. Appl. Phys., **42**, 4653 (1971).
582. R. N. Kienle, E. S. Dizon, T. J. Brett, C. F. Eckert, Rubber Chem. Technol., **44**, 996 (1971).
583. G. Dall'asta, G. Motroni, Angew. Makromol. Chem., **16/17**, 51 (1971).
584. W. Marconi, A. Mazzel, G. Lugli, M. Bruzzone, J. Polym. Sci. C, **16**, 805 (1967).
585. W. S. Bahary, D. I. Sapper, J. H. Lane, Rubber Chem. Technol., **40**, 1529 (1967).
586. G. Kraus, C. W. Childers, J. T. Gruver, J. Appl. Polym. Sci., **11**, 1581 (1967); reprint in Rubber Chem. Technol., **42**, 520 (1969).
587. D. I. Relyea, H. P. Smith, A. N. Johnson, AD 657675 (1967).
588. K. A. Wolf, Z. Electrochem., **65**, 604 (1961); via R. Havinga, A. Schors, J. Macromol. Sci. A, **2**, 12 (1968) and Sci. Abstr., A, 65, No. 773, abs. 8141 (1962).
589. E. N. Tinyakova, B. A. Dolgoplosk, T. G. Zhuravleva, R. N. Kovalevskaya, T. N. Kuren'gina, J. Polym. Sci., **52**, 159 (1961).
590. M. Sh. Yagfarov, Polym. Sci. USSR, **10**, 1465 (1968); (transl'n of Vysokomol. Soedin. A, **10**, 1264 (1968)).
591. G. S. Trick, J. Appl. Polym. Sci., **3**, 253 (1960).
592. K.-H. Illers, Eur. Polym. J. (Suppl.) 133 (Aug. 1969).
593. W. R. Brown, G. S. Park, J. Paint Technol., **42**, 16 (1970).
594. F. Haas, K. Nutzel, G. Pampus, D. Theisen, Rubber Chem. Technol., **43**, 1116 (1970). See also Kautschuk Gummi Kunststoffe, **23**, 502 (1970).
595. M. Takeda, K. Tanaka, R. Nagao, J. Polym. Sci., **57**, 517 (1962).
596. Shell Chem. Bull., RB/70/14.
597. N. K. Kalfoglou, H. L. Williams, J. Appl. Polym. Sci., **14**, 2481 (1970).
598. M. Baccaredda, E. Butta, J. Polym. Sci., **51**, S39 (1961).
599. R. Nagao, Nippon Gomu Kyokaishi, **41**, 509 (1968).
600. I. A. Livshits, L. M. Korobova, E. A. Sidorovich, Mekh. Polim., **4**, 596 (1967).
601. C. G. Overberger, L. H. Arond, R. H. Wiley, R. R. Garrett, J. Polym. Sci., **7**, 431 (1951).
602. W. Marconi, A. Mazzei, S. Cucinella, M. Cesari, J. Polym. Sci. A, **2**, 4261 (1964).
603. C. A. Aufdermarsh, R. Parsier, J. Polym. Sci. A, **2**, 4727 (1964).
604. R. M. Kell, B. Bennett, P. B. Stickney, J. Appl. Polym. Sci., **2**, 8 (1959).
605. J. B. Campbell, Science, **141**, 329 (1963).
606. B. L. Dolgoplosk, S. P. Erusalimskii, A. V. Merkur'eva, Vysokomol. Soedin., **4**, 1333 (1962); Chem. Abstr., **59**, 4045d (1963).
607. R. R. Garrett, Unpublished results quoted by C. A. Aufdermarsh, R. Parsier, J. Polym. Sci. A, **2**, 4727 (1964)
608. R. Nagao, Polymer, **9**, 517 (1968).
609. R. F. Robbins, R. P. Reed, Adv. Cryogenic Eng., **13**, 252 (1968).
610. U. Eisele, Kautschuk Gummi Kunststoffe, **23**, 486 (1970).
611. R. R. Garrett, C. A. Hargreaves, D. N. Robinson, J. Macromol. Sci. A, **4**, 1679 (1970).
612. I. A. Livshits, L. M. Korobova, Dokl. Akad. Nauk USSR, **121**, 474 (1958).
613. N. V Zakharenko, R. A. Gavrilina, Soviet Rubber Technol., **25**, No. 11, 44 (1966) (transl'n of Kauchuk i Rezina, **25**, No. 11, 47 (1966)).
614. J. P. Mercier, Rev. Gen. Caoutchouc, Edn. Plast., **5**, 25 (1968).
615. J. J. Maurer, ACS Polymer Preprints, **9**, 866 (1968).
616. D. Heinze, K. Schmieder, G. Schnell, K. A. Wolf, Kautschuk Gummi, **14**, WT208 (1961) (transl'n in Rubber Chem. Technol., **35**, 776 (1962)); via R. Havinga, A. Schors, J. Macromol. Sci. A, **2**, 12 (1968), and Ref. 262.
617. H. S. Gutowsky, A. Saika, M. Takeda, D. E. Woessner, J. Chem. Phys., **27**, 534 (1957).
618. W. V. C. van Beek, W. F. Koch, B. F. Steggerda, O. F. K. Bussemaker, Assoc. Francaise Ing. Caoutchouc,

Plastiques Conf. Intern. Caoutchouc, Paris, June 1970, paper 55; also Rev. Gen. Caoutchouc, **48**, 929 (1971).
619. M. Girolamo, J. R. Urwin, Eur. Polym. J., **7**, 225 (1971).
620. D. Reichenbach, W. Eckelmann, Angew. Makromol. Chem., **16/17**, 157 (1971).
621. S. S. Chang, A. B. Bestul, J. Res. Nat. Bur. Std., **75A**, 113 (1971).
622. R. T. Humpidge, A. Maclean, S. H. Morrell, Rubber Chem. Technol., **44**, 479 (1971).
623. Z. R. Glaser, F. R. Eirich, J. Polym. Sci. C, **31**, 275 (1970).
624. K. Ueberreiter, Z. Physik. Chem. B, **45**, 361 (1940).
625. E. R. Mognaschi, A. Chierico, Trans. Faraday Soc., **66**, 86 (1970).
626. G. Dall'asta, P. Scaglione, Rubber Chem. Technol., **42**, 1235 (1969).
627. J. N. Short, G. Kraus, R. P. Zelinski, F. E. Naylor, Rubber Chem. Technol., **23**, 614 (1959).
628. R. M. Ikeda, M. L Wallach, R. J. Angelo, in: S. L. Aggarwal (Ed.), "Block Copolymers", Plenum Press, New York, 1970, p 43; ACS Polym. Preprints, **10**, 1446 (1969).
629. K. J. Clark, A. T. Jones, D. H. Sandford, Chem. Ind., **47**, 2010 (1962).
630. G. Natta, F. Danusso, G. Moraglio, J. Polym. Sci., **25**, 119 (1957).
631. K. R. Dunham, J. Vandenberghe, J. W. H. Faber, L. E. Contois, J. Polym. Sci. A, **1**, 751 (1963).
632. G. Natta, Rend. Accad. Nazi. Lincei. Ser., **8**, 24 (1958); via Ref. 223.
633. S. F. Kurath, E. Passaglia, R. Pariser, J. Appl. Phys., **28**, 499 (1957).
634. F. P. Reding, J. A. Faucher, R. D. Whitman, J. Polym. Sci., **57**, 483 (1962).
635. H.-G. Elias, O. Etter, J. Macromol. Sci. A, **1**, 943 (1967).
636. W. H. McCarty, G. Parravano, J. Polym. Sci. A, **3**, 4029 (1965).
637. G. L. Taylor, S. Davison, J. Polym. Sci. B, **699**, (1968).
638. S. T. Barsamyan, A. S. Apresyan, V. I. Kleiner, L. L. Stotskaya, Soviet Plastics, **3**, 17 (1968) (transl'n of Plasticheskie Massy, **3**, 13 (1968)).
639. R. E. Kelchner, J. J. Aklonis, J. Polym. Sci. A-2, **8**, 799 (1970).
640. A. Abe, T. Hama, J. Polym. Sci. B, **7**, 427 (1969).
641. M. Takayanagi, Pure Appl. Chem., **23**, 151 (1970).
642. P. R. Swan, J. Polym. Sci., **42**, 525 (1960).
643. T. Shibukawa, V. D. Gupta, R. Turner, J. H. Dillon, A. V. Tobolsky, Textile Res. J., **32**, 910 (1962).
644. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 277 (1962).
645. M. Baccaredda, E. Butta, Chim. Ind. (Milan), **44**, 1228 (1962).
646. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 286 (1962).
647. A. V Tobolsky, "Properties and Structure of Polymers", Wiley, New York, 1960.
648. N. L. Zutty, S. J. Whitworth, J. Polym. Sci. B, **2**, 709 (1964).
649. F. Danusso, G. Moraglio, G. Talamini, J. Polym. Sci., **21**, 139 (1956).
650. H. Thurn, Kolloid Z., **173**, 72 (1960).
651. R. Simha, R. F. Boyer, J. Chem. Phys., **37**, 1003 (1962).
652. A. Odajima, J. A. Sauer, A. E. Woodward, J. Phys. Chem., **66**, 718 (1962).
653. W. G. Oakes, D. W. Robinson, J. Polym. Sci., **14**, 505 (1954).
654. O. D. Frampton, J. F. Nobis, Ind. Eng. Chem., **45**, 404 (1953).
655. K. Wolf, K. Schmieder, Intern. Symp. Macromol. Chem., Ric. Sci., **25**, 732 (1955).
656. F. Wuerstlin, Kunststoffe Plastiks, **40**, 158 (1950).
657. A. E. Woodward, J. A. Sauer, Adv. Polym. Sci., **1**, 114 (1958).
658. B. Wunderlich, J. Polym. Sci. C, **1**, 41 (1963).
659. N. G. McCrum, Makromol. Chem., **34**, 50 (1959).
660. P. Manaresi, V. Giannela, J. Appl. Polym. Sci., **4**, 251 (1960).
661. R. Nakane, J. Appl. Polym. Sci., **3**, 125 (1960).
662. E. Hunter, E. W. Oakes, Trans. Faraday Soc., **41**, 49 (1945).
663. F. A. Quinn, L. Mandelkern, J. Am. Chem. Soc., **80**, 3178 (1958).
664. D. E. Kline, J. A. Sauer, A. E. Woodward, J. Polym. Sci., **22**, 455 (1956).
665. G. Kontos, W. P. Slichter, J. Polym. Sci., **61**, 61 (1962).
666. H. A. Floeck, Kolloid Z.-Z. Polym., **180**, 118 (1962).
667. W. Pechhold, S. Blasenbrye, S. Woemer, Kolloid Z. Polym., **189**, 14 (1963).
668. C. A. F. Tuijnman, J. Polym. Sci. C, **16**, 2379 (1967).
669. J. Heijboer, J. Polym. Sci. C, **16**, 3755 (1968).
670. H. Schlein, M. Shen, Rev. Sci. Instrum., **40**, 587 (1969).
671. Perepechko, L. A. Kvacheva, Vysokomol. Soedin. B, **12**, 484 (1970).
672. B. Wunderlich, J. Chem. Phys., **37**, 1203 (1962).
673. E. W. Fischer, F. Kloos, J. Polym. Sci. B, **8**, 685 (1970).
674. S. Matsuoka, J. H. Daane, H. E. Blair, T. K. Kwei, J. Polym. Sci. B, **6**, 87 (1968).
675. R. A. Jackson, S. R. D. Oldland, A. Pajaczkowski, J. Appl. Polym. Sci., **12**, 1297 (1968).
676. T. K. Kwei, T. T. Wang, H. E. Bair, J. Polym. Sci. C, **31**, 87 (1970).
677. G. Natta, Angew. Chem., **68**, 393 (1956).
678. F. P. Reding, J. Polym. Sci., **21**, 547 (1956).
679. R. W. Warfield, R. Brown, J. Polym. Sci. A-2, **5**, 791 (1967).
680. G. Vidotto, A. J. Kovacs, Kolloid Z.-Z. Polym., **220**, 1 (1967).
681. J. P. Kennedy, W. Naegele, J. J. Elliott, J. Polym. Sci. B, **3**, 729 (1965).
682. O. E. Van Lohuizen, K. S. De Vries, J. Polym. Sci. C, **16**, 3943 (1968).
683. D. W. Aubrey, A. Bamatt, J. Polym. Sci. A-2, **6**, 241 (1968).
684. A. E. Woodward, Trans. N. Y. Acad. Sci., **24**, 250 (1962).
685. A. E. Woodward, J. A. Sauer, R. A. Wall, J. Polym. Sci., **50**, 117 (1961).
686. J. H. Griffith, B. G. Ranby, J. Polym. Sci., **44**, 369 (1960).
687. R. W. Penn, Trans. Soc. Rheol., **3**, 416 (1963).
688. W. A. Hewett, F. E. Weir, J. Polym. Sci. A, **1**, 1239 (1963).

689. F. E. Karasz, H. E. Bair, J. M. O'Reilly, Polymer, **8**, 547 (1967).
690. T. P. Melia, A. Tyson, Makromol. Chem., **109**, 87 (1967).
691. I. Kirshenbaum, R. B. Isaacson, M. Druin, J. Polym. Sci. B, **3**, 525 (1965).
692. N. Kawasaki, M. Takayanagi, Rept. Progr. Polym. Phys. (Japan), **10**, 337 (1967).
693. R. G. Quynn, B. S. Sprague, J. Polym. Sci. A-2, **8**, 1971 (1970).
694. A. A. Miller, J. Polym. Sci. A-2, **6**, 249 (1968).
695. J. D. Ferry, Viscoelastic Properties of Polymers, Wiley, New York, 1961.
696. N. Hirai, H. Eyring, J. Polym. Sci., **37**, 51 (1959).
697. J. D. Ferry, G. S. Parks, J. Chem. Phys., **4**, 70 (1936).
698. T. G. Fox, S. Loshaek, J. Polym. Sci., **15**, 371 (1955).
699. L. A. Wood, in: G. S. Wiley (Ed.), "Synthetic Rubber", Wiley, New York, 1954, Chap. 10.
700. G. T. Furukawa, R. E. McCoskey, M. L. Reilly, J. Res. Nat. Bur. Std., **55**, 127 (1955).
701. G. M. Zhidomirov, U. D. Tsvetkov, Y. S. Lebedev, Zh. Strukt. Khim., **2**, 696 (1961).
702. E. Butta, P. Giusti, Ric. Sci. Rend. Sec. A, **2**, 362 (1962).
703. T. L. Smith, J. Polym. Sci. C, **16**, 841 (1967).
704. J. J. Maurer, ACS Division of Rubber Chem., Spring Meeting, May 1967, paper 26 (33).
705. G. Gianotti, G. Dall'asta, A. Valvassori, V. Zamboni, Makromol. Chem., **149**, 117 (1971).
706. A. Turner, F. E. Bailey, J. Polym. Sci. B, **1**, 601 (1963).
707. J. P. Kennedy, R. M. Thomas, Makromol. Chem., **64**, 1 (1963).
708. J. P. Kennedy, R. M. Thomas, Makromol. Chem., **53**, 28 (1962).
709. J. P. Kennedy, J. J. Elliott, B. Groten, Makromol. Chem., **77**, 26 (1964).
710. D. L. Beck, A. A. Hiltz, J. R. Knox, SPE Trans., **3**, 279 (1963).
711. B. Ke, J. Polym. Sci. B, **1**, 167 (1963).
712. L. T. Muus, N. G. McCrum, F. C. McGrew, SPE J., **15**, 368 (1959).
713. N. G. McCrum, J. Polym. Sci. B, **2**, 495 (1964).
714. R. W. Wilkinson, M. Dole, J. Polym. Sci., **58**, 1089 (1962).
715. H. Wilski, Kuntstoffe, **54**, 90 (1964).
716. J. A. Sauer, R. A. Wall, N. Fuschillo, A. E. Woodward, J. Appl. Phys., **29**, 1385 (1958).
717. V. A. Kargin, I. Yu. Marchenko, Polym. Sci. USSR, **2**, 370 (1961); (transl'n of Vysokomol. Soedin., **2**, 549 (1960)).
718. E. Passaglia, G. M. Martin, J. Res. Nat. Bur. Std., **68A**, 273 (1964).
719. S. Nara, H. Kashiwabara, J. Sohma, J. Polym. Sci. A-2, **5**, 929 (1967).
720. J. P. Mercier, Ind. Chim. Belge, **30**, 813 (1965).
721. D. R. Gee, T. P. Melia, Makromol. Chem., **116**, 122 (1968).
722. D. R. Gee, T. P. Melia, Polymer, **10**, 239 (1969).
723. D. R. Gee, T. P. Melia, Makromol. Chem., **132**, 195 (1970).
724. I. Abu-Isa, V. A. Crawford, A. R. Haly, M. Dole, J. Polym. Sci. C, **6**, 149 (1964).
725. E. Passaglia. H. K. Kevorkian, J. Appl. Phys., **34**, 90 (1963).
726. M. S. Akutin, Z. I. Salina, L. Yu. Ziatkevich, B. V. Andrianov, L. E. Vlaskina, Soviet Plastics, No. 2, p. 19 (1971).
727. G. Gianotti, A. Capizzi, Eur. Polym. J., **4**, 677 (1968).
728. G. M. Pogosyan, G. A. Zhamkochyan, S. G. Matsoyan, Arm. Khim. Zh., **22**, 364 (1969).
729. G. M. Pogosyan, L. M. Akopyan, E. V. Vanyan, S. G. Matsoyan, Vysokomol. Soedin. B, **13**, 242 (1971).
730. W. Kern, W. Heitz, M. Jager, K. Pfitzner, H. O. Wirth, Makromol. Chem., **126**, 73 (1969).
731. G. M. Pogosyan, L. M. Akopyan, E. V. Vanyan, S. G. Matsoyan, Vysokomol. Soedin. B, **12**, 142 (1970).
732. K. R. Dunham, J. W. H. Faber, J. Vandenberghe, W. F. Fowler, J. Appl. Polym. Sci., **7**, 897 (1963).
733. R. Kosfeld, Kolloid Z., **172**, 182 (1960).
734. K. H. Illers, Z. Elektrochen, **65**, 679 (1961).
735. G. M. Pogosyan, T. G. Karapetyan, S. G. Matsoyan, Vysokomol. Soedin. B, **13**, 228 (1971).
736. E. C. Chapin, J. G. Abrams, V. L. Lyons, J. Org. Chem., **27**, 2595 (1962).
737. G. S. Kolesnikov, G. M. Pogosyan, Izv. Akad. Nauk SSSR Otd. Khim. Nauk, **2**, 227 (1958).
738. C. G. Overberger, C. Frazier, J. Mandelman, H. F. Smith, J. Am. Chem. Soc., **75**, 3326 (1953).
739. T. E. Davies, British Plastics, **32**, 283 (1959).
740. US pat. 2,723,261, 8. Nov. 1955.
741. L. L. Ferstandig, J. C. Butler, A. E. Straus, J. Am. Chem. Soc., **76**, 5779 (1954).
742. G. M. Pogosyan, T. G. Karapetyan, S. G. Matsoyan, Vysokomol. Soedin. B, **12**, 463 (1970).
743. G. S. Kolesnikov, Izv. Akad. Nauk USSR, Otd. Khim. Nauk, 1333 (1959).
744. K. R. Dunham, J. Vandenberghe, J. W. H. Faber, W. F. Fowler, J. Appl. Polym. Sci., **7**, 1531 (1963).
745. Brit. Pat. 609,482, 1 Oct. 1948.
746. S. Krause, J. J. Gormley, N. Roman, J. A. Shetter, W. H. Watanabe, J. Polym. Sci. A, **3**, 3573 (1965).
747. J. Thrower, M. A. H. Hewins, Royal Aircraft Aircraft Establishment, Farnborough, Hants, UK, TR 70056 (1970).
748. R. A. Abdrashitov, N. M. Bazhenov, M. V. Vol'kenshtein, A. I. Kol'tsov, A. S. Khachaturov, Vysokomol. Soedin., **5**, 405 (1963); (transl'n in Polymer Sci. USSR, **4**, 1066 (1963)); Z. Phys. Chem., **220**, 413 (1962).
749. F. Danusso, G. Polizzotti, Makromol. Chem., **61**, 157 (1963).
750. K. Tanaka, O. Yano, S. Maru, R. Nagao, Rept. Progr. Polym. Phys. (Japan), **12**, 379 (1969).
751. R. A. Abdrashitov, N. M. Bazhenov, M. V. Vol'kenshtein, A. I. Kol'tsov, A. S. Khachaturov, Polym. Sci. USSR, **6**, 2074 (1964); (transl'n of Vysokomol. Soedin., **6**, 1871 (1964)).
752. US Pat. 2,651,627, 8 Sept. 1953.
753. D. I. Livingstone, P. M. Kamath, R. S. Corley, J. Polym. Sci., **20**, 485 (1956).
754. G. M. Pogosyan, G. A. Zhamkochyan, R. A. Stepanyan, S. G. Matsoyan, Arm. Khim. Zh., **22**, 915 (1969).

755. Polaroid Corp., Offic. Sci. Res. Dev. Rept. No. 4417, p. 246, Feb. 1945.
756. V. Frosini, P. L. Magagini, Eur. Polym. J., **2**, 129 (1966)
757. V. Frosini, E. Butta, Mater, Sci. Eng., **6**, 274 (1970).
758. T. Fujimoto, N. Ozaki, M. Nagasawa, J. Polym. Sci. A-2, **6**, 129 (1968).
759. H.-G. Elias, V. S. Kamat, Makromol. Chem., 61 (1968).
760. J. M. G. Cowie, P. M. Toporowski, Eur. Polym. J., **4**, 621 (1968).
761. M. Baccaredda, E. Butta, V Frosini, P. Magagnini, F. Andruzzi, J. Polym. Sci. C, **16**, 3581 (1968).
762. J. M. G. Cowie, P. M. Toporowski, J. Macromol. Sci., **B3**, 81 (1969).
763. E. Rovira, A. Eisenberg, Unpublished results quoted by Ref. 48.
764. H. Endo, T. Fujimoto, M. Nagasawa, J. Polym. Sci. A-2, **7**, 1669 (1969).
765. S. Ichihara, A. Komatsu, T. Hata, Polym. J. (Japan), **2**, 650 (1971).
766. G. T. Kennedy, F. Morton, J. Chem. Soc., 2383 (1949).
767. L. C. Corrado, J. Chem. Phys., **50**, 2260 (1969).
768. W. Hodes, American Cyanamid Co. Unpublished results quoted by Ref. 44.
769. G. S. Kolesnikov, G. M. Pogosyan, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk, 2098 (1962); Chem. Abstr., **58**, 8047a (1963).
770. L. A. Utracki, R. Simha, Makromol. Chem., **117**, 94 (1968).
771. N. N. Aylward, J. Polym. Sci. B, **8**, 337 (1970).
772. J. J. Keavney, E. C. Eberlin, J. Appl. Polym. Sci., **3**, 47 (1960).
773. R. S. Spencer, J. Colloid Sci., **4**, 229 (1949).
774. T. G. Fox, P. J. Flory, J. Appl. Phys., **21**, 581 (1950).
775. L. J. Hughes, G. L. Brown, J. Appl. Polym. Sci., **5**, 580 (1961).
776. E. Jenckel, K. Ueberreiter, Z. Physik. Chem. (Leipzig) A, **182**, 361 (1938).
777. F. Wuerstlin, H. Thurn, in: H. A. Stuart (Ed.), "Die Physik der Hochpolymeren", Springer, Berlin, 1956.
778. H. A. Stuart, (Ed.), "Die Physik der Hochpolymeren", vol. III, Springer, Berlin, 1955.
779. R. B. Beevers, E. F. T. White, J. Polym. Sci. B, **1**, 171 (1963).
780. S. Newman, W. P. Cox, J. Polym. Sci., **46**, 29 (1960).
781. R. S. Spencer, R. F. Boyer, J. Appl. Phys., **17**, 398 (1946).
782. T. G. Fox, P. J. Flory, J. Polym. Sci., **14**, 315 (1954).
783. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **70**, 2384 (1948).
784. T. G. Fox, P. J. Flory, J. Phys. Colloid Chem., **55**, 221 (1951).
785. R. H. Boundy, R. F. Boyer, "Styrene", Reinhold, New York, 1952.
786. W. Patnode, W. J. Scheiber, J. Am. Chem. Soc., **61**, 3449 (1939).
787. W. J. Roff, "Fibres, Plastics and Rubbers", Academic Press, New York, 1956,
788. Yu. V. Zelenev, Vysokomol. Soedin., **4**, 1486 (1962); (transl'n in Polym. Sci. USSR, **4**, 457 (1963))
789. M. S. Parker, V. J. Krasnansky, B. G. Achbarnmer, J. Appl. Poly. Sci., **8**, 1825 (1964).
790. A. D. Pasquino, M. N. Pilsworth, J. Polym. Sci. B, **2**, 253 (1964).
791. A. E. Martin, H. F. Rase, IEC Prod, Res. Dev., **6**, 104 (1967).
792. S. N. Kolesov, Vysokomol. Soedin. A, **9**, 1860 (1967); (transl'n in Polym. Sci. USSR, **9**, 2098 (1967)).
793. G. Rehage, H. Breuer, J. Polym. Sci. C, **16**, 2299 (1967).
794. W. Simpson, Chem. Ind., **5**, 215 (1965).
795. S. G. Turley, Appl. Polym. Symp., **7**, 237 (1968).
796. K. C. Rusch, J. Macromol. Sci. B, **2**, 421 (1968).
797. J. Stoelting, F. E. Karasz, W. J. McKnight, Polym. Eng. Sci., **10**, 133 (1970).
798. K. Thinius, B. Hoesselbarth, Plaste Kautschuk, **17**, 653 (1970).
799. N. Nemoto, Polym. J. (Japan), **1**, 485 (1970).
800. F. E. Karasz, H. E. Bair, J. M. O'Reilly, J. Phys. Chem., **69**, 2657 (1965).
801. I. Abu-Isa, M. Dole, J. Phys. Chem., **69**, 2668 (1965).
802. E. H. Merz, L. E. Nielsen, R. Buchdahl, Ind. Eng. Chem., **43**, 1396 (1951).
803. D. J. Plazek, V. M. O'Rourke, J. Polym. Sci. A-2, **9**, 209 (1971).
804. C. H. Bamford, G. C. Eastmond, D. Whittle, Polymer, **12**, 247 (1971).
805. M. Bank, J. Leffingwell, C. Thies, Macromolecules, **4**, 43 (1971).
806. S. Hozumi, T. Wakabayashi, K. Sugihara, Polym. J. (Japan), **1**, 632 (1970).
807. G. V. Vinogradov, E. A. Dzyura, A. Ya. Malkin, V. A. Grechanovskii, J. Polym. Sci. A-2, **9**, 1153 (1971).
808. R. A. Wallace, J. Polym. Sci. A-2, **9**, 1325 (1971).
809. S. Ichihara, A. Komatsu, T. Hata, Polym. J. (Japan), **2**, 644 (1971).
810. O. Yamamoto, H. Kambe, Polym. J. (Japan), **2**, 623 (1971).
811. E. Jenckel, E. Braucker, Z. Physik. Chem. (Leipzig) A, **185**, 465 (1940).
812. L. J. T. Hughes, D. B. Fordyce, J. Polym. Sci., **22**, 509 (1956).
813. K. H. Illers, Z. Elektrochem., **70**, 353 (1966); via Ref. 815.
814. L. E. Nielsen, ACS Polym. Preprints, **9**, 596 (1968).
815. A. Eisenberg, T. Yokoyama, E. Sambalido, J. Polym. Sci. A-1, **7**, 1717 (1969).
816. A. Chifor, Mater, Plast., **7**, 575 (1970).
817. A. Eisenberg, H. Matsuura, T. Yokoyama, AD 715093 (1970).
818. M. Pegoraro, L. Szilagyi, A. Penati, G. Alessandrini, Eur. Polym. J., **7**, 1709 (1971).
819. M. Baccaredda, P. L. Magagnini, G. Pizzirani, P. Giusti, J. Polym. Sci. B, **9**, 303 (1971).
820. E. C. Eberlin,-Unpublished values quoted by Ref. 44.
821. C. E. Rehberg, C. H. Fisher, Ind. Eng. Chem., **40**, 1429 (1948).
822. J. Hrouz, J. Janacek, J. Macromol. Sci. B, **5**, 245 (1971).
823. C. E. Rehberg, W. A. Faucette, C. H. Fisher, J. Am. Chem. Soc., **66**, 1723 (1944).

824. J. A. Shetter, J. Polym. Sci. B, **1**, 209 (1963).
825. A. R. Monahan, J. Polym. Sci. A-1, **4**, 2381 (1966).
826. G. Pizzirani, P. L. Magagnini, Chim. Ind. (Milan), **50**, 1218 (1968).
827. B. Vollmert, H. Haas, Brit. Pat. 1,116,933, 12. June 1968.
828. J. H. Prager, R. M. McCurdy, C. B. Rathmann, J. Polym. Sci. A, **2**, 1941 (1964).
829. V. P. Shibayev, B. S. Petrukhin, Yu. A. Zubov, N. A. Plate, V A. Kargin, Polym. Sci. USSR, **10**, 258 (1968).
830. F. A. Boyer, J. F. Abere, J. Polym. Sci., **15**, 537 (1955).
831. R. M. McCurdy, J. H. Prager, J. Polym. Sci. A, **2**, 1185 (1964).
832. G. P. Mikhailov, V. A. Shevelev, Polym. Sci. USSR, **9**, 2762 (1967). (transl'n of Vysokomol. Soedin. A, **9**, 2442 (1967)).
833. J. Lawler, D. C. Chalmers, J. Timar, ACS Div. Rubber Chem., Spring Meeting, May 1967, paper 42. (10).
834. C. U. Pittman, R. L. Voges-Unpublished results quoted by Ref. 835.
835. C. U. Pittman, J. C. Lai, D. P. Vanderpool, M. Good, R. Prado, Macromolecules, **3**, 746 (1970).
836. F. A. Bovey, J. F. Abere, G. B. Rathmann, C. L. Sandberg, J. Polym. Sci., **15**, 520 (1955).
837. A. S. Khachaturov, N. M. Bazhenov, M. V. Vol'kenshtein, I. M. Dolgopol'skii, A. I. Kol'tsov, Soviet Rubber Technol., **24**, No. 12, 9 (1965); (transl'n of Kaucbuk i Rezina, **24**, No. 12, 6 (1965)).
838. G. K. Dyvik, Polymer, **2**, 449 (1961).
839. A. G. Pittman, B. A. Ludwig, D. L. Sharp, J. Polym. Sci. A-1, **6**, 1741 (1968).
840. R. H. Wiley, G. M. Brauer, J. Polym. Sci., **4**, 351 (1949).
841. W. R. Sorenson, T. W. Campbell, "Preparative Methods of Polymer Chemistry", 2nd ed., 1968, p. 248.
842. W. A. P. Black, J. A. Colquhoun, E. T. Dewar, Makromol. Chem., **122**, 244 (1969).
843. E. P. Otocka, T. K. Kwei, Macromolecules, **1**, 401 (1968).
844. J. R. Constanza, J. A. Vona, J. Polym. Sci. A-1, **4**, 2659 (1966).
845. F. Wuerstlin, in: H. A. Stuart (Ed.), "Die Physik der Hochpolymeren", Springer, Berlin 1955, Chap. 11.
846. R. A. Haldon, R. Simha, J. Appl. Phys., **39**, 1890 (1968).
847. J. Heijboer, in: J. A. Prins (Ed.)," Physics of Non-Crystalline Solids", Proc. Intern. Conf., Delft 1964, p. 231, North Holland Publishing Co., Amsterdam, 1965.
848. T. Otsu, S. Aoki, R. Nakatani, Makromol. Chem., **134**, 331 (1970).
849. M. E. Hoagland, I. N. Duling, ACS Div. Petrol. Chem. Preprints, **15**, B85 (1970).
850. I. N. Duling, A. Schneider, R. E. Moore, US Pat. 3,518,241, 30. June 1970.
851. C. E. Rehberg, C. H. Fisher, J. Am. Chem. Soc., **66**, 1203 (1944).
852. P. Weitz, Paint Varn. Prod., **57**, 99 (1967).
853. E. F. Jordan, G. R. Riser, B. Artymyshyn, W. E. Parker, J. W. Pensabene, A. N. Wrigley, J. Appl. Polym. Sci., **13**, 1777 (1969).
854. K. Butler, P. R. Thomas, G. J. Tyler, J. Polym. Sci., **48**, 357 (1960).
855. O. Smidsrod, J. E. Guillet, Macromolecules, **2**, 272 (1969).
856. J. Elles, Chim. Mod., **4**, 53 (1959).
857. A. V. Tobolsky, M. C. Shen, J. Phys. Chem., **67**, 1886 (1963).
858. S. S. Rogers, L. Mandelkern, J. Phys. Chem., **61**, 985 (1957).
859. A. F. Wilde, J. J. Ricca, AD 675463, Proc. Army Symp. on Solid Mechanics, p. 39 (1968).
860. T. Hata, T. Nose, J. Polym. Sci. C, **16**, 2019 (1967).
861. N. W. Johnston, ACS Polym. Preprints, **10**, 608 (1969).
862. Z. G. Gardlund, J. J. Laverty, J. Polym. Sci. B, **7**, 719 (1969).
863. E. A. W. Hoff, D. W. Robinson, A. H. Willbourn, J. Polym. Sci., **18**, 161 (1955).
864. Z. A. Azimov, S. P. Mitsengendler, A. A. Korotkov, Vysokomol. Soedin., **7**, 843 (1965); (transl'n in Polym. Sci. USSR, **7**, 929 (1965)).
865. G. P. Mikhailov, A. I. Artyukhov, T. I. Borisova, Vysokomol. Soedin. A, **10**, 1755 (1968); (transl'n in Polym. Sci. USSR, **10**, 2033 (1968)).
866. J. Lal, G. S. Trick, J. Polym. Sci. A, **2**, 4559 (1964).
867. G. Williams, D. C. Watts, Trans. Faraday Soc., **67**, 2793 (1971).
868. K. M. Sinnott, SPE Trans., **2**, 65 (1962).
869. M.O. Samsoen, Ann. Phys., **9**, 35 (1928).
870. E. Baer, Offic. Dig. J. Paint Technol. Eng., **36**, 464 (1964).
871. L. de Brouckere, G. Offergeld, Bull. Soc. Chim. Belges, **67**, 96 (1958).
872. J. D. Ferry, W. C. Child, R. Zard, D. M. Stem, M. L. Williams, R. F. Landel, J. Colloid Sci., **12**, 53 (1957).
873. S. Saito, Kolloid Z.-Z. Polym., **189**, 116 (1963) and Res. Electrotech. Lab. (Toyoko), No. 648 (1964); via Ref. 286.
874. W. L. Peticolas, G. I. A. Stegeman, B. P. Stoicheff, Phys. Rev. Lett., **18**, 1130 (1967).
875. W. M. Lee, (advisers: F. R. Eirich, B. R. McGarvey), Dissertation, Polytechnic. Inst., Brooklyn 1967: via Diss. Abs., B, **28**, 1897 (1967).
876. E. N. Rostovskii, L. D. Rubinovich, Vysokomol. Soedin. Karb. Vys. Soedin. Sb. Statei, p. 140 (1963).
877. M. Ilavsky, G. L. Slonimskii, J. Janacek, J. Polym. Sci. C, **16**, 329 (1967).
878. M. C. Shen, J. D. Strong, F. J. Matusik, J. Macromol. Sci. B, **1**, 15 (1967).
879. J. Janacek, J. D. Ferry, J. Macromol. Sci. B, **5**, 219 (1971).
880. F. Lednicky, J. Janacek, J. Macromol. Sci. B, **5**, 335 (1971).
881. H. Ochiai, H. Shindo, H. Yamamura, J. Polym. Sci. A-2, **9**, 431 (1971).
882. I. N. Razinskaya, N. E. Kharitonova, B. P. Shtarkman, Vysokomol. Soedin. B, **11**, 892 (1969).
883. R. W. Warfield, M. C. Petree, P. Donovan, SPE J., **15**, 1055 (1959).
884. T. G. Fox, B. S. Garrett, W. E. Goode, S. Gratch, J. F. Kincaid, A. Spell, J. D. Stroupe, J. Am. Chem. Soc., **80**, 1768 (1958).
885. S. Loshaek, J. Polym. Sci., **15**, 391 (1955).
886. A. Odajima, A. E. Woodward, J. A. Sauer, J. Polym. Sci., **55**, 181 (1961).

887. W. G. Gall, N. G. McCrum, J. Polym. Sci., **50**, 489 (1961).
888. G. Allen, D. Sims, G. J. Wilson, Polymer, **2**, 375 (1961).
889. B. E. Read, J. Polym. Sci. C, **16**, 1887 (1967).
890. M. F. Margaritova, S. D. Stavrova, S. N. Trubitsyna, S. D. Medvedev, J. Polym. Sci. C, **16**, 2251 (1967).
891. S. R. Urzendowski, A. H. Guenther, J. R. Asay, ACS Polym. Preprints, **9**, 878 (1968).
892. A. A. Berlin, E. F. Samarin, R. V. Kronman, I. G. Sumin, Soviet Plastics, **2**, 11 (1971); (transl'n of Plasticheskie Massy, **2**, 26 (1971)).
893. A. K. Schulz, J. Chim. Phys., **53**, 933 (1956).
894. F. P. Wolf, K. Ueberreiter, Kolloid Z.-A. Polym., **245**, 399 (1971).
895. S. Bywater, P. M. Toporowski, Polymer, **13**, 94 (1972).
896. G. V. Schulz, W. Wunderlich, R. Kirste, Makromol. Chem., **75**, 23 (1964).
897. J. P. Mercier, H. Berghmans, G. Smets, Paper given at VIII FATIPEC Congress, Scheveningen (The Hague), p. 26 (1966).
898. E. V. Thompson, J. Polym. Sci. A-2, **4**, 199 (1966).
899. E. V. Thompson, J. Polym. Sci. A-2, **6**, 433 (1968).
900. S. Havriliak, Polymer, **9**, 289 (1968).
901. H. Shindo, I. Murakami, H. Yamamura, J. Polym. Sci. A-1, **7**, 297 (1969).
902. N. N. Aylward, J. Polym. Sci. A-1, **8**, 319 (1970).
903. T. I. Borisova, Vysokomol. Soedin. A, **12**, 932 (1970); (trans'n in Polymer Sci. USSR, 12, 1060 (1970)).
904. M. J. Bowden, J. H. O'Donnell, J. Polym. Sci. A-1, **7**, 1665 (1969).
905. G. M. Chetyrkina, T. A. Sokolova, M. M. Koton, Vysokomol. Soedin., Vsesoyuz. Khim. Obshchestvo im. D. I. Mendeleeva, **1**, 248 (1959); Chem. Abstr., **53**, 23059i (1959).
906. B. E. Tate, Adv. Polym. Sci., **5**, 214 (1967)-Unpublished results of Charles Pfizer and Co. Inc.
907. J. B. Kinsinger, J. R. Panchak, R. L. Kelso, J. S. Bartlett, R. K. Graham, J. Appl. Polym. Sci., **9**, 429 (1965).
908. B. Wesslen, R. W. Lenz, ACS Polym. Preprints, **11**, 105 (1970).
909. B. Wesslen, R. W. Lenz, W. J. McKnight, F. E. Karasz, Macromolecules, **4**, 24 (1971).
910. J. A. Powell, R. K. Graham, J. Polym. Sci. A, **3**, 3451 (1965).
911. V. A. Kargin, T. I. Sogolova, Zh. Fiz. Khim., **23**, 540 (1949).
912. H. D. Anspon, J. J. Baron, WADC Tech' 1 Report 57-24, Pt. 1, 1957, Pt. 11, 1958; via ref. 910.
913. H. Yuki, K. Hatada, T. Nijnomi, M. Hashimoto, J. Ohshima, Polym. J. (Japan), **2**, 629 (1971).
914. R. A. Haldon, W. J. Schell, R. Simha, J. Macromol. Sci. B, **1**, 759 (1967).
915. D. S. Otto, Vinyl Technol. Newsletter, **5**, 20 (1968).
916. W. J. Schell, R. Simha, J. J. Aklonis, J. Macromol. Sci., **A3**, 1297 (1969).
917. W. J. Schell, Thesis, Univ. S. California 1969; via Diss. I Abstr. Intern. B, **30**, 2120 (1969).
918. L. Fishbein, B. F. Crowe, Makromol. Chem., **48**, 221 (1961).
919. H. Thurn, K. Wolf, Kolloid-Z., **148**, 16 (1956).
920. E. J. Vandenberg, R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 519 (1959).
921. W. S. Durrell, E. C. Stump, P. D. Schuman, J. Polym. Sci. B, **3**, 831 (1965).
922. C. E. Schildknecht, S. T. Gross, H. R. Davidson, J. M. Lambert, A. O. Zoss, Ind. Eng. Chem., **40**, 2104 (1948).
923. J. A. Sedlak, K. Matsuda, J. Polym. Sci. A, **3**, 2329 (1965).
924. G. M. Pogosyan, E. S. Avanesyan, S. G. Matsoyan, Arm. Khim. Zh., **24**, 694 (1971).
925. G. Pizzirani, P. Magagnini, P. Giusti, J. Polym. Sci. A-2, **9**, 1133 (1971).
926. J. B. Clark, Polym. Eng. Sci., **7**, 137 (1967).
927. T. N. Kalinina, G. N. Afanas'eva, L. A. Vol'f, A. I. Meos, E. B. Kremer, S. Ya. Frenkel', S. S. Mnatsakanov, Vysokomol. Soedin. B, **12**, 661 (1970).
928. W. H. Howard, J. Appl. Polym. Sci., **5**, 303 (1961).
929. L. V. Holroyd, R. S. Codrington, B. A. Mrowca, E. Guth, Rubber Chem. Technol., **25**, 767 (1952).
930. W. R. Krigbaum, N. Tokita, J. Polym. Sci., **43**, 467 (1960).
931. T. G. Fox, Bull. Am. Phys. Soc., **1**, 123 (1956).
932. R. J. Kokes, F. A. Long, J. L. Hoard, J. Chem. Phys., **20**, 1711 (1952).
933. Rohm and Haas Co., Philadelphia. Leaflet SP-251 (1/66) of Spec. Prod. Dept.
934. S. Okajima, A. Takeuchi, J. Polym. Sci. A-2, **5**, 1317 (1967).
935. K.-H. Illers, Makromol. Chem., **124**, 278 (1969).
936. R. D. Andrews, R. M. Kimmel, J. Polym. Sci. B, **3**, 167 (1965).
937. S. H. Ronel, D. H. Kohn, J. Polym. Sci. A-1, 2221 (1969).
938. C. R. Bohn, J. R. Schaefgen, W. O. Statton, J. Polym. Sci., **55**, 531 (1961).
939. R. B. Beevers, J. Polym. Sci. A, **2**, 5257 (1964).
940. T. M. A. Hossain, H. Maeda, T. Iijima, Z. Morita, J. Polym. Sci. B, **5**, 1069 (1967).
941. K. Ogura, S. Kawamura, H. Sobue, Macromolecules, **4**, 79 (1971).
942. H. Kakutani, M. Asahina, J. Polym. Sci. A-1, **5**, 1717 (1967).
943. M. Kyrszewski, M. Mucha, J. Appl. Polym. Sci., **15**, 2687 (1971).
944. V. Heidingsfeld, J. Zelinger, V. Kuska, J. Appl. Polym. Sci., **15**, 2447 (1971).
945. J. D. Hoffman, J. Am. Chem. Soc., **74**, 1696 (1952).
946. R. F. Boyer, J. Appl. Phys., **25**, 825 (1954).
947. F. P. Reding; private communication to ref. 946.
948. J. D. Hoffman, J. J. Weeks, J. Polym. Sci., **28**, 472 (1958); J. Res. Nat. Bur. Std., **60**, 465 (1958).
949. T. Nakajima, S. Saito, J. Polym. Sci., **31**, 423 (1958).
950. A. Nishioka, J. Polym. Sci., **37**, 163 (1959).
951. F. Krum, Kolloid-Z., **165**, 77 (1959).
952. K.-H. Illers, E. Jenckel, Kolloid-Z., **165**, 84 (1959).
953. M. Baccaredda, E. Butta, J. Polym. Sci., **44**, 421 (1960).

Previous Page

954. A. W. Meyers, V. Tammela, V. Stannett, M. Szwarc, Mod. Plast., **37**, 139 (1960).
955. N. G. McCrum, J. Polym. Sci., **60**, 53 (1962).
956. A. H. Scott, D. J. Scheiber, A. J. Curtis, J. L. Lauritzen, J. D. Hoffman, J. Res. Nat. Bur. Std., **66A**, 269 (1962).
957. E. Passaglia, J. M. Crissman, R. R. Stromberg, ACS Polym. Preprints, **6**, 590 (1965).
958. J. D. Hoffman, G. Williams, E. Passagilia, J. Polym. Sci. C, **14**, 173 (1966).
959. J. M. Crissman, E. Passaglia, J. Polym. Sci. C, **14**, 237 (1966).
960. I. I. Perepechko, L. A. Bodrova, Vysokomol. Soedin. B, **10**, 148 (1968).
961. I. I. Perepechko, L. A. Kvacheva, L. A. Ushakov, A. Ya. Svetov, V. A. Grechishkin, Soviet Plastics, **8**, 39 (1970); (transl'n of Plasticheskie Massy, **8**, 43 (1970)).
962. G. S. Kolesnikov, M. G. Avetyan, Izv. Akad. Nauk SSSR, Otd. Khim. Nauk, 331 (1959).
963. L. A. Wood, Army Elastomer Conference 1955.
964. R. N. Haszeldine, U. M. I. S. T., Manchester, U.K.; private communication.
965. A. Peterlin, J. D. Holbrook, Kolloid Z.-Z. Polym., **203**, 68 (1965).
966. N. Koizumi, S. Yano, K. Tsunashima, J. Polym. Sci., B, **7**, 59 (1969).
967. H. Sasabe, S. Saito, M. Asahina, H. Kakutani, J. Polym. Sci. A-2, **7**, 1405 (1969).
968. A. Miglierina, G. Ceccato, SRS 4 (4th Intern. Synth. Rubber Symp. London 1969), **2**, 65 (1969).
969. S. Yano, J. Polym. Sci. A-2, **8**, 1057 (1970).
970. H. Kakutani, J. Polym. Sci. A-2, **8**, 1177 (1970).
971. W. S. Durrell, G. Westmoreland, M. G. Moshonas, J. Polym. Sci. A, **3**, 2975 (1965).
972. G. T. Furukawa, R. E. McCoskey, G. J. King, J. Res. Nat. Bur. Std., **49**, 273 (1952).
973. A. J. Warner; private communication to Ref. 80.
974. H. Sack; private communication to Ref. 946.
975. J. A. S. Smith, Diss. Faraday Soc., **19**, 207 (1955).
976. S. Nobara, Chem. High Polym. (Japan), **14**, 318 (1957).
977. S. P. Kabin, Zh. Tekh. Fiz., **26**, 2628 (1956); (transl'n in Soviet Phys.-Tech. Phys., **1**, 2542 (1957)).
978. Y. Maeda, Chem. High Polym. (Japan), **14**, 442 (1957); Chem. Abstr., **52**, 5022f (1958).
979. K. Nagamatsu, T. Yoshitomi, T. Takemoto, J. Colloid. Sci., **13**, 257 (1958).
980. M. Baccaredda, E. Butta, J. Polym. Sci., **31**, 189 (1958).
981. K. H. Illers, E. Jenckel, Kolloid-Z., **160**, 97 (1958).
982. N. G. McCrum, J. Polym. Sci., **27**, 555 (1958).
983. N. G. McCrum, J. Polym. Sci., **34**, 355 (1959).
984. A. V Tobolsky, J. Polym. Sci., **35**, 555 (1959).
985. L. Mandelkern; private communication to Ref. 659.
986. G. W. Becker, Kolloid-Z., **167**, 44 (1959).
987. M. Baccaredda, E. Butta, Ann. Chim. (Rome), **49**, 559 (1959); Chem. Abstr., **53**, 18538a (1959).
988. J. A. Sauer, A. E. Woodward, Rev. Mod. Phys., **32**, 88 (1960).
989. C. A. Sperati, H. W. Starkweather, Adv. Polym. Sci., **2**, 465 (1961).
990. R. K. Eby, K. M. Sinnott, J. Appl. Phys., **32**, 1765 (1961).
991. T. Satokawa, S. Koisumi, Kogyo Kagaku Zasshi, **65**, 1211 (1962); Chem. Abstr., **58**, 1544h (1963).
992. A. V. Tobolsky, D. Katz, M. Takahashi, J. Polym. Sci. A, **1**, 483 (1963).
993. Y. Ohzawa, Y. Wada, Rept. Progr. Polym. Phys. (Japan), **6**, 147 (1963).
994. Y. Ohzawa, Y. Wada, Japan J. Appl. Phys., **3**, 436 (1964).
995. Y. Araki, J. Appl. Polym. Sci., **9**, 421 (1965).
996. Y. Araki, J. Appl. Polym. Sci., **9**, 1515 (1965).
997. K. L. Hsu, D. E. Kline, J. N. Tomlinson, J. Appl. Polym. Sci., **9**, 3567 (1965).
998. Y. Araki, J. Appl. Polym. Sci., **9**, 3575 (1965).
999. Y. Araki, J. Appl. Polym. Sci., **9**, 3585 (1965).
1000. Y. Araki, J. Appl. Polym. Sci., **11**, 953 (1967).
1001. E. G. Howard, P. B. Sargeant, J. Macromol. Sci. A-1, 1011 (1967).
1002. K. Frigge, G. Dube, H. Kriegsman, Plaste Kautschuk, **15**, 470 (1968).
1003. S. Miyakawa, T. Takemura, Japan J. Appl. Phys., **7**, 814 (1968); via ref. 641.
1004. A. W. Neumann, W. Tanner, J. Colloid Interface Sci., **34**, 1 (1970).
1005. D. W. Brown, L. A. Wall, ACS Polym. Preprints, **12**, 302 (1971).
1006. H. S. Eleuterio, E. P. Moore, Chem. Eng. News, **40**, 44 (1962).
1007. D. W. Brown, L. A. Wall, ACS Polym. Preprints, **7**, 1116 (1966).
1008. E. M. Sullivan, E. W. Wise, F. P. Reding, US Pat. 3,110,705, 12. Nov. 1962.
1009. J. A. E. Kail, Polymer, **6**, 535 (1965).
1010. F. P. Reding, E. R. Walter, F. J. Welch, J. Polym. Sci., **56**, 225 (1962).
1011. L. Bohn, Kunststoffe Plastiks, **53**, 93 (1963).
1012. F. E. Bailey, J. P. Henry, R. D. Lundberg, J. M. Whelan, J. Polym. Sci. B, **2**, 447 (1964).
1013. R. M. Barrer, R. Mallinder, P. S. L. Wong, Polymer, **8**, 321 (1967).
1014. G. Pezzin, G. Sanmartin, F. Zilio-Grandi, J. Appl. Polym. Sci., **11**, 1539 (1967).
1015. K. S. Minsker, Yu. A. Sangalov, G. A. Razuwayev, J. Polym. Sci. C, **16**, 1489 (1967).
1016. U. Bianchi, A. Turturro, G. Basile, J. Phys. Chem., **71**, 3555 (1967).
1017. K. Nakamura, T. Nakagawa, Rept. Progr. Polym. Phys. (Japan), **10**, 277 (1967).
1018. A. M. Jendrychowska-Bonamour, Eur. Polym. J., **4**, 627 (1968).
1019. A. M. Sharetskii, S. V. Svetozarskii, I. B. Kotlyar, E. N. Zil'berman, Soviet Plastics, **3**, 10 (1968); (transl'n of Plasticheskie Massy, **3**, 6 (1968)).
1020. I. I. Perepechko, L. I. Trepelkova, L. A. Bodrova, E. K. Kosikova, L. O. Bunina, Soviet Plastics, **5**, 7 (1968); (transl'n of Plasticheskie Massy, **5**, 11 (1968)).

1021. J. Zelinger, J. Polym. Sci. C, **16**, 4259 (1969).
1022. J. V Koleske, R. D. Lundberg, J. Polym. Sci. A-2, 795 (1969).
1023. J. Petersen, B. Ranby, Makromol. Chem., **133**, 251 (1970).
1024. M. Kryszewski, M. Mucha, J. Polym. Sci. B, **5**, 1095 (1967).
1025. D. E. Witenhafer, J. Macromol. Sci. B, **4**, 915 (1970).
1026. G. Pezzin, G. Ajroldi, C. Garbuglio, J. Appl. Polym. Sci., **11**, 2553 (1967).
1027. S. A. Liebman, C. R. Foltz, J. F. Reuwer, R. J. Obremski, Macromolecules, **4**, 134 (1971).
1028. J. Barton, M. Pegoraro, L. Szilagyi, G. Pagani, Makromol. Chem., **144**, 245 (1971).
1029. Kh. U. Usmanov, A. A. Yulchibaev, M. K. Asamov, A. Valiev, J. Polym. Sci. A-1, **9**, 1459 (1971).
1030. J. Furukawa, Pure Appl. Chem., **26**, 153 (1971).
1031. A. M. Sharetskii, S. V. Svetozarskii, E. N. Zil'berman, I. B. Kotlyar, Soviet Plastics, **9**, 1 (1971); (transl'n of Plasticheskie Massy, **9**, 5 (1971)).
1032. E. Sacher, J. Polym. Sci. A-2, **6**, 1813 (1968).
1033. Kh. U. Usmanov, A. A. Yulchibaev, T. Sirlibaev, J. Polym. Sci. A-1, **9**, 1779 (1971).
1034. J. M. O'Reilly, J. Polym. Sci., **57**, 429 (1962).
1035. D. M. Chackraburtty, J. Chem. Phys., **26**, 427 (1957).
1036. E. W. Merrill, D. A. Gibbs, Chem. Eng. News, **41**, 41 (1963).
1037. P. Meares, Trans. Faraday Soc., **53**, 31 (1957).
1038. P. L. Magagnini, F. Sardelli, G. Pizzirani, Ann. Chem. (Rome), **57**, 805 (1967).
1039. S. Saito, Denki Sikense Kenkyu Khakoku, Res. Electrotech. Lab., **1**: 648 (1964); via Ref. 792.
1040. M. Scatena, P. Samnartin, F. Zilio-Grandi, FATIPEC Congr., 9, Section 3, p. 68 (1968), also Ind. Vernice, **23**(2), 17 (1969); Chem. Abstr., 71, 50752h (1969).
1041. R. M. Holsworth, J. Paint Technol., **41**, 167 (1969).
1042. J. M. Carcamo, F. Arranz, Rev. Plast. Mod., **185**, 1683 (1971).
1043. H. van Hoorn, Rheol. Acta, **10**, 208 (1971).
1044. S. Ichihara, A. Komatsu, T. Hata, Polym. J. (Japan), **2**, 640 (1971).
1045. H. Hopff, P. Perlstein, Makromol. Chem., **125**, 247 (1969).
1046. H. Hopff, M. A. Osman, Angew. Makromol. Chem., **6**, 39 (1969).
1047. M. Kinoshita, T. Irie, M. Imoto, Makromol. Chem., **110**, 47 (1967).
1048. P. L. Magagnini, V. Frosini, Eur. Polym. J., **2**, 139 (1966).
1049. A. R. Monahan, Macromolecules, **1**, 408 (1968).
1050. M. A. Osman, H. -G. Elias, J. Macromol. Sci. A, **5**, 805 (1971).
1051. T. S. Reid, D. W. Codding, F. A. Bovey, J. Polym. Sci., **18**, 417 (1955).
1052. H. C. Haas, E. S. Emerson, N. W. Schuler, J. Polym. Sci., **22**, 291 (1956).
1053. G. Pizzirani, P. L. Magagnini, J. Appl. Polym. Sci., **11**, 1173 (1967).
1054. H. Hopff, M. A. Osman, Makromol. Chem., **135**, 175 (1970).
1055. H. Hopff, M. A. Osman, Makromol. Chem., **143**, 289 (1971).
1056. J. A. Price, M. R. Lytton, B. G. Ranby, J. Polym. Sci., **51**, 541 (1961).
1057. G-S. Kolesnikov, E. F. Rodionova, L. S. Fedorova, T. Ya. Medved, M. I. Kabacknik, Vysokomol. Soedin., **4**, 1385 (1962); Chem. Abstr., **58**, 14108e (1963).
1058. L. D. Taylor, C. K. Chiklis, T. E. Platt, J. Polym. Sci. B, **9**, 187 (1971).
1059. K. Hubner, F. Kollinsky, G. Markert, H. Pennewiss, Angew. Makromol. Chem., **11**, 109 (1970).
1060. L. D. Lovyagina, N. V. Meiya, A. F. Nikolaev, Zb. Prikl. Khim., **44**, 2056 (1971).
1061. C. Noel, L. Monnerie, J. Chim. Phys. Phys.-Chim. Biol., **65**, 2089 (1968).
1062. V. Frosini, S. de Petris, Chim. Ind. (Milan), **49**, 1178 (1967).
1063. L. A. Utracki, R. Simha, Makromol. Chem., **117**, 94 (1968).
1064. F. J. Golemba, J. E. Guillet, S. C. Nyburg, J. Polym. Sci. A-1, **6**, 1341 (1968).
1065. B. L. Tsetlin, T. Ya. Medved, Yu. G. Chikishev, Yu. M. Polikarpov, S. R. Rafikov, M. I. Kabuchnik, Vysokomol. Soedin., **3**, 1117 (1961); Chem. Abstr., **56**, 2568h (1962).
1066. P. Ehrlich, R. J. Kern, E. D. Pierron, T. Provder, J. Polym. Sci. B, **5**, 911 (1967).
1067. C. Noel, Compt. Rend., **258**, 3702 (1964).
1068. N. S. Nametkin, S. G. Durgar'yan, L. I. Tikhonova, V. G. Fillippova, Vysokomol. Soedin. A, **13**, 672 (1971); (transl'n in Polym. Sci. USSR, **13**, 75 (1971)).
1069. J. H. Flynn, Thermochim. Acta, **8**, 69 (1974).
1070. P. Peyser, W. D. Bascom, J. Macromol. Sci, B: Phys., **13** (4), 597, (1977).
1071. M. J. Richardson, N. G. Savill, Polymer, **16**, 753 (1975).
1072. J. Bourdariat, R. Isnard, J. Odin, J. Polym. Sci., Polym. Phys. Ed., **11** (9), 1817 (1973).
1073. S. S. Chang, Polym. Prep., ACS, Div. Polym. Chem., **13** (1), 322 (1972).
1074. J. M. G. Cowie, Eur. Polym. J., **11** (4), 297 (1975).
1075. J. M. G. Cowie, I. J. McEwen, Macromolecules, **10** (5), 1124 (1977).
1076. G. T. Davis, R. K. Eby, J. Appl. Phys., **44** (10), 4274 (1973).
1077. K. A. Movsisyan, G. T. Ovanesov, K. A. Gasparyan, Uch. Zap. Erevan. Un-t. Estestv. N., **2**, 140 (1980).
1078. A. N. Paauvve, J. R. Knox, Polym. Eng. Sci., **16** (1), 36 (1976).
1079. J. Prud'homme, P. Goursot, A. Laramee, Makromol. Chem., **178** (5), 1561 (1977).
1080. J. Simon, C. L. Beatty, F. E. Karasaz, J. Therm. Anal., **7**(1), 187 (1975).
1081. J. M. G. Cowie, Eur. Polym. J., **9** (10), 1041 (1973).
1082. Same as Ref. 1081.
1083. M. Jarigeon, B. Chabert, D. Chatain, C. Lacabanne, G. Nenoz, J. Macromol. Sci. B: Phys., **17** (1), 1 (1980).
1084. I. V. Kuleshov, Zh. Fiz. Khim., **49** (3), 803 (1975).
1085. H. K. Reimschuessel, E. A. Turi, M. K. Akkapeddi, J. Polym. Sci., Polym. Chem. Ed., **17** (9), 2769 (1979).
1086. H. Daimon, H. Okitsu, J. Kumanotani, Polym. J., **7** (4), 460 (1975).

1087. R. K. Kulkarni, H. J. Porter, F. Leonard, J. Appl. Polym. Sci., **17** (11), 3509 (1973).
1088. Same as Ref. 1087.
1089. Same as Ref. 1087.
1090. R. K. Kulkarni, H. J. Porter, F. Leonard, J. Appl. Polym. Sci., **17** (11), 3509 (1973).
1091. Same as Ref. 1090.
1092. Same as Ref. 1090.
1093. Same as Ref. 1090.
1094. H. Aida, H. Senda, Fukui Daigaku Kogakubu Kenkyu Hokoku, **28** (1), 95 (1980).
1095. M. S. Ali, R. P. Sheldon, Bangladesh, J. Sci., Ind. Res., **10** (3–4), 197 (1975).
1096. L. Gargallo, M. Russo, Makromol. Chem., **176** (9), 2735 (1975).
1097. Y. S. Lipatov, V A. Vilenskii, Vysokomol. Soedin., Ser. A, **17** (9), 2069 (1975).
1098. Y. K. Sung, D. E. Gregonis, G. A. Russell, J. D. Andrade, Polymer, **19** (11), 1362 (1978).
1099. Same as Ref. 1098.
1100. A. M. Abushihada, F. E. Shunbo, F. Al-Hajjar, Y. Y. Al-Sultan, HRC CC, J. High Resolut. Chromatogr. Chromatogr. Commun., **2** (8), 512 (1979).
1101. M. S. Ali, R. P. Sheldon, Bangladesh J. Sci., Ind. Res., **10** (3–4), 197 (1975).
1102. J. Ballesteros, G. J. Howard, L. Teasdale, J. Macromol. Sci. A, Chem., **11** (1), 39 (1977).
1103. F. Feranadez-Martin, I. Ferdanadez-Pierola, A. Horta, J. Polym. Sci., Polym. Phys. Ed., **19** (9), 1353 (1981).
1104. L. Gargallo, M. Russo, Makromol. Chem., **176** (9), 2735 (1975).
1105. Same as Ref. 1104.
1106. W. Knappe, H. J. Ott, Colloid Polym. Sci., **255** (9), 837 (1977).
1107. R. P. Kusy, W. F. Simmons, A. R. Greenberg, Polymer, **22** (2), 268 (1981).
1108. Y. S. Lipatov, V. A. Vilenskii, Vysokomol. Soedin., Ser. A, **17** (9), 2069 (1975).
1109. F. R. Schwarzl, Rheol., [Proc. Int. Congr], 8th, vol. 1, 243. Edited by Astarita, Giovanni et al., Plenum, NY.
1110. J. M. G. Cowie, Z. Haq, I. J. McEwen, J. Polym. Sci., Polym. Lett. Ed., **17** (12), 771 (1979).
1111. Same as Ref. 1110
1112. A. M. Abushihada, F. E. Shunbo, F. Al-Hajjar, Y. Y. Al-Sultan, HRC CC, J. High Resolut. Chromatogr. Chromatogr. Commun., **2** (8), 512 (1979).
1113. A. K. Gupta, N. Vhand, J. Polym. Sci., Polym. Phys. Ed., **18** (5), 1125 (1980).
1114. P. Lukac, L. Lapcik, Radiochem. Radioanal. Lett., **29** (5), 227 (1977).
1115. W. Mischok, B. Hoesselbarth, Plaste Kautsch., **23** (7), 491 (1976).
1116. Z. Petrik, Sb. Prednasek, 'MAKROTEST 1973', **1**, 231 (1973).
1117. V. P. Privalko, Vysokomol. Soedin., Ser. A, **18** (6), 1213 (1976).
1118. A. A. Remizova, I. V. Kuleshov, N. A. Novikov, Zh. Fiz. Khim., **50** (3), 787 (1976).
1119. F. R. Schwarzi, Rheol. [Proc. Int. Congr], 8th, vol. 1, 243. Edited by Astarita, Giovanni et al., Plenum, NY.
1120. P. Singh, J. Lyngaae-Joergensen, J. Macromol. Sci. B: Phys., **19** (2), 177 (1981).
1121. A. De Ruvo, E. Alfthan, Polymer, **19** (8), 872 (1978).
1122. S. L. Malholtra, P. Lessard, L. P. Blanchard, J. Macromol. Sci. A: Chem., **15** (1), 121 (1981).
1123. S. L. Malhotra, P. Lessard, L. P. Blanchard, J. Macromol. Sci. A: Chem., **15** (2), 279 (1981).
1124. S. L. Malhotra, P. Lessard, L. P. Blanchard, J. Macromol. Sci. A: Chem., **15** (8), 1125,1577 (1981).
1125. A. M. Abusbihada, F. E. Shunbo, F. Al-Hajjar, Y. Y. Al-Sultan, HRC CC, J. High Resolut. Chromatogr. Chromatogr. Commun., **2** (8), 512 (1979).
1126. M. S. Ali, R. P. Sheldon, Bangladesh J. Sci., Ind. Res., **10** (3–4), 197 (1975).
1127. J. A. Bergfjord, R. C. Penwell, M. Stolka, J. Polym. Sci., Polym. Phys. Ed., **17** (4), 711 (1979).
1128. L. P. Blanchard, J. Hesse, S. L. Malhotra, Can. J. Chem., **52** (18), 3170 (1974).
1129. P. R. Couchman, Polym. Eng. Sci., **2** (17), 377 (1981). J. M. G. Cowie, Eur. Polym. J., **11** (4), 297 (1975).
1133. A. De Ruvo, E. Alfthan, Polymer, **19** (8), 872 (1978).
1134. P. D. Garn, O. Menis, J. Macromol. Sci., Phys. B, **13** (4), 611 (1977).
1135. U. Gaur, B. Wunderlich, Macromolecules, **13** (6), 1618 (1980).
1136. M. Hartmann, H. Carlsohn, F. Rudolf, Faserforsch. Textiltech., **28** (2), 59 (1977).
1137. Same as Ref. 1136.
1138. G. Hentze, Thermochim. Acta, **15** (2), 139 (1976).
1139. H. Lee, A. M. Jamieson, R. Simha, Macromolecules, **12** (2), 329 (1979).
1140. S. L. Malhotra, P. Lessard, L. P. Blanchard, J. Macromol. Sci., Chem., **A15** (2), 301 (1981).
1141. W. Mischok, B. Hoesselbarth, Plaste Kautsch., **23** (7), 491 (1976).
1142. P. Peyser, W. D. Bascom, J. Macromol. Sci. B: Phys., **13** (4), 597 (1977).
1143. M. J. Richardson, Plastics and Rubber: Materials and Applications, Sept., 162–167, 1976.
1144. M. J. Richardson, N. G. Savill, Polymer, **16**, 753 (1975).
1145. M. J. Richardson, N. G. Savill, Polymer, **18**, 3 (1977).
1146. J. E. L. Roovers, P. M. Toporowski, J. Appl. Polym. Sci., **18** (6), 1685 (1974).
1147. A. Rudin, D. Burgin, Polymer, **16** (4), 291 (1975).
1148. F. R. Schwarzt, Rheol., [Proc. Int. Congr.], 8th vol. 1, 243. Edited by Astarita, Giovanni, et al., Plenum, NY.
1149. L. A. Wall, Roestamsjah, M. H. Aldridge, J. Res. Natl. Bur. Stand., Sect. A, **78** (4), 447 (1974).
1150. Y. Y. Tan, G. Challa, Polymer, **17** (8), 739 (1976).
1151. J. Ballesteros, G. J. Howard, L. Teasdale, J. Macromol. Sci., Chem. A, **11** (1), 39 (1977).
1152. V. P. Privalko, Vysokomol. Soedin., Ser. A, **18** (6), 1213 (1976).

1153. V. P. Privalko, A. P. Lobodina, Eur. Polym. J., **10** (11), 1033 (1974).
1154. P. Tormala, Eur. Polym. J., **10** (6), 519 (1974).
1155. L. Nicolais, R. F. Landel, P. Giordano-Orsini, Polym. J., **7** (2), 259 (1975).
1156. L. M. Panova, 1. S. Filàtov, B. I. Sazhin, B. I. Yudkin, Plast. Massy, **5**, 75 (1974).
1157. W. Wrasidlo, Macromolecules, **4** (5), 642 (1971).
1158. H. Aida, H. Senda, Fukui Daigaku Kogakubu Kenkyu Hokoku, **28** (1), 95 (1980).
1159. Same as Ref. 1158.
1160. Same as Ref. 1158.
1161. E. Ito, T. Hatakeyama, J. Polym. Sci., Polym. Phys. Ed., **13** (12), 2313 (1975).
1162. Y. S. Lipatov, V. A. Vilenskii, Vysokomol. Soedin., Ser. A, **17** (9), 2069 (1975).
1163. J. M. Pochan, D. F. Pochan, Macromolecules, **13** (6), 1577 (1980).
1164. G. E. Wissler, B. Crist Jr., J. Polym. Sci., Phys. Ed., **18** (6), 1257 (1980).
1165. P. Dobbert, F. Fabry, V. Koos, W. Mischok, Plaste Kautsch., **26** (1), 18 (1979).
1166. G. Hentze, Thermochim. Acta, **15** (2), 139 (1976).
1167. E. Ito, T. Hatakeyarna, J. Polym. Sci., Polym. Phys. Ed., **12** (7), 1477 (1974).
1168. E. Ito, K. Yamamoto, Y. Kobayashi, T. Hatakeyama, Polymer, **19** (1), 39 (1978).
1169. C. Otto, Faserforsch. Textiltech., **28** (8), 347 (1974).
1170. C. G. Seefried Jr., J. V. Koleske, J. Polym. Sci., Polym. Phys. Ed., **13** (4), 851 (1975).
1171. Same as Ref 1170.
1172. Same as Ref. 1170.
1173. Same as Ref. 1170.
1174. E. Sorta, A. De Chirico, Polymer, **17** (4), 348 (1976).
1175. Same as Ref. 1174.
1176. Same as Ref. 1174.
1177. Same as Ref. 1174.
1178. Same as Ref. 1174.
1179. Same as Ref. 1174.
1180. Same as Ref. 1174.
1181. K. H. Illers, Polymer, **18** (6), 551 (1977).
1182. Same as Ref. 1181.
1183. Same as Ref. 1181.
1184. Same as Ref. 1181.
1185. Same as Ref. 1181.
1186. Same as Ref. 1181.
1187. G. J. Kettle, Polymer, **18** (7), 742 (1977).
1188. S. W. Shalaby, E. A. Turi, W. M. Wenner, Therm. Methods Polym. Anal., East. Anal. Symp., **17**, 35, (1978).
1189. Same as Ref. 1188.
1190. Same as Ref. 1188.
1191. Same as Ref. 1188.
1192. Same as Ref. 1188.
1193. A. Siegrmann, Z. Baraam, Makromol. Chem., Rapid Commun., **1** (2), 113 (1980).
1194. R. Larrain, L. H. Tagle, F. R. Diaz, Polym. Bull. (Berlin), **4** (8), 487 (1981).
1195. W. Wrasidlo, J. K. Stille, Macromolecules, **9** (3), 505 (1976).
1196. J. M. Barrales-Rienda, Ramos J. Gonzalez, M. Sanchez.
1197. Same as Ref. 1196.
1198. Same as Ref. 1196.
1199. Same as Ref. 1196.
1200. Same as Ref. 1196.
1201. Same as Ref. 1196.
1202. Same as Ref. 1196.
1203. Same as Ref. 1196.
1204. Same as Ref. 1196.
1210. J. Staverman, Rheologica Acta, **5**, 283 (1966).
1211. H. Breuer and G. Rehage, Kolloid Z., **216/217**, 158 (1967).
1212. P. Ehrenfest, Proc. Kon. Akad. Wetensch, Amsterdam, **36**, 153 (1933).
1213. U. Bianchi, J. Phys. Chem., **69**, 1497 (1965).
1214. A. J. Kovacs, Fortschr. Hochpolym. Forschung, **3**, 394 (1963).
1215. M. V. Volkenstein, O. B. Ptitsyn, Sov. Phys.-Tekhn. Phys., **1**, 2138 (1957).
1216. B. Wunderlich et al., J. Polym. Sci. C, **6**, 173 (1963); J. Appl. Phys., **35**, 95 (1964); Adv. Polym. Sci., **7**, 151 (1970); J. Polymer Sci. A-2, **9**, 1887 (1971); A2, **12**, 2473 (1974).
1217. C. T. Moynihan, J. Phys. Chem., **78**, 2673 (1974); J. Am. Chem. Soc., **59**, 12 (1976).
1218. C. T. Vijayakumar, H. Kothandaraman, Thermochim. Acta, **115**, 159 (1987).
1219. T. G. Fox and P. J. Flory, J. Appl. Phys., **21**, 581 (1950).
1220. J. M. G. Cowie, P. M. Toporowski, Eur. Polym. J., **4**, 621 (1968).
1221. F. E. Karasz, W. J. MacKnight, Macromolecules, **1**, 537 (1968).
1222. G. A. Adam, J. N. Hay, I. W. Parsons, R. N. Haward, Polymer, **17**, 51 (1976).
1223. G. P. Johari, A. Hallbrucker, E. Mayer, J. Polym. Sci., Part B: Polym. Phys., **26**, 1923 (1988).
1224. Styrene Polymers, in: "Encyclopedia of Polymer Science and Engineering", **16**, 1–246 (1989).
1225. S. J. Clarson, J. A. Semlyen, K. Dodgson, Polymer, **32**, 2823 (1991).
1226. K. O'Driscoll, R. A. Sanayei, Macromolecules, **24**, 4479 (1991).
1227. E. A. DiMarzio, J. A. Gibbs, P. D. Flemming, I. C. Sanchez, Macromolecules, **9**, 763 (1976).
1228. G. Gee, Polymer, **7**, 177 (1966).
1229. S. Ichihara, A. Komatsu, Y. Tsujita, T. Nose, T. Hata, Polymer J., **2**, 530 (1971).
1230. G. Rehage, J. Macromol. Sci. B: Phys., **18**, 423 (1980).
1231. P. Zoller, J. Polymer Sci., Polymer Physi. Ed., **16**, 1261 (1978).
1232. P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **20**, 1453 (1982).
1233. H. W. Starkweather, G. A. Jones, P. Zoller, J. Polym. Sci., Part B: Polym. Phys., **26**, 257 (1988).

1234. P. Zoller, T. A. Kehl, H. W. Starkweather, G. A. Jones, J. Polym. Sci., Part B: Polym. Phys., **27**, 993 (1989).
1235. T. Ougizawa, G. D. Dee, D. J. Walsh, Polymer, **30**, 1675 (1989).
1236. J. Tatibouet, L. Pichet, Polymer, **32**, 3147 (1991).
1237. K. H. Hellwege, W. Knappe, P. Lehmann, Kolloidzschr., **183**, 110 (1962).
1238. D. W. van Krevelen, "Properties of Polymers"., 3rd ed., Elsevier, New York, 1990, Chapter 6.
1239. J. Bicerano, "Prediction of Polymer Properties", 2nd ed., Marcel Dekker, Inc., New York, 1996, Chapter 6.
1240. D. Porter, "Group Interaction Modelling of Polymer Properties", Marcel Dekker, Inc., New York, 1995, Part 3.
1241. C. J. Lee, J. Macromol. Sci. C: Rev. Macromol. Chem. Phys., **29**, 431 (1989).
1242. A. Bondi, "Physical Properties of Molecular Crystals, Liquids and Glasses", Wiley, Inc., New York, 1968.
1243. R. Zallen, "The Physics of Amorphous Solids", Wiley, Inc., New York, 1983.
1244. D. Meier, Ed., "Molecular Basis of Transitions and Relaxations", Gordon and Breach, New York, 1978.
1245. G. B. McKenna, in: G. Allen (Ed.), "Glass Formation and Glassy Behavior in Comprehensive Polymer Science", Vol. 2, Pergamon Press, Oxford, 1987.
1246. E. A. Turi, Ed., "Thermal Characterization of Polymeric Materials", Academic Press, New York, 1981.
1247. J. A. Bryson, "The Glass Transition, Melting Point and Structure", in: A. D. Jenkins (Ed.), "Polymer Science", North Holland, Amsterdam, 1972.
1248. M. C. Shen, A. Eisenberg, "Glass Transtions in Polymers", Progress Solid State Chem., 3, 407 (1966).
1249. "Encyclopedia of Polymer Science and Engineering", vol. 17, "Transitions and Relaxations", Wiley, New York, 1989.
1250. I. A. Barbari, R. M. Conforti, J. Polym. Sci., Part B: Polym. Phys., **30**, 1261 (1992).
1251. L. Giebel, G. Meier, G. Fytas, E. W. Fischer, J. Polym. Sci., Part B: Polym. Phys., **30**, 1291 (1992).
1252. K. J. McGrath, K. L. Ngai, C. M. Roland, Macromolecules, **25**, 4911 (1992).
1253. A. A. Mansour, R. Junge, B. Stoll, W. Pechhold, Colloid Polym. Sci., **270**, 325 (1992).
1254. S. Wu, J. Appl. Polym. Sci., **46**, 619 (1992).
1255. R. F. Boyer, Macromolecules, **25**, 5326 (1992).
1256. T. Kitayama, E. Masuda, M. Yamaguchi, T. Nishiura, K. Hatada, Polymer Journal, **24**, 817 (1992).
1257. Y. Oishi, S. Nakata, M. Kajiyama, M. Kakimoto, Y. Imai, J. Polym. Sci., Part A: Polym. Chem., **30**, 2357 (1992).
1258. H. D. Kim, S. G. Kang, C. S. Ha, J. Appl. Polym. Sci., **46**, 1339 (1992).
1259. S. Vleeshouwers, J.-E. Kluin, J. D. McGervey, A. M. Jamieson, R. Simha, J. Polym. Sci., Part B: Polym. Phys., **30**, 1429 (1992).
1260. S. J. Bai, M. Dotrong, R. C. Evers, J. Polym. Sci., Part B: Polym. Phys., **30**, 1515 (1992).
1261. B. P. Agrawal, A. K. Srivastava, J. Macromol. Sci. A: Chem., **29**, 325 (1992).
1262. M. Scandola, G. Ceccorulli, M. Pizzoli, Macromolecules, **25**, 6441 (1992).
1263. S. Koizumi, K. Tadano, Y. Tanaka, T. Shimidzu, S. Kutsumizu, S. Yano, Macromolecules, **25**, 6563 (1992).
1264. T. Inoue, H. Okamoto, K. Osaki, Macromolecules, **25**, 7069 (1992).
1265. S. Kutsumizu, N. Nagao, K. Tadano, H. Tachino, E. Hirasawa, S. Yano, Macromolecules, **25**, 6829 (1992).
1266. L. Lestel, P. Guegan, S. Boileau, H. Cheradame, F. Laupretre, Macromolecules, **25**, 6024 (1992).
1267. J. A. Puertolas, P. J. Alonso, L. Oriol, J. L. Serrano, Macromolecules, **25**, 6018 (1992).
1268. J. T. Koberstein, L. M. Leung, Macromolecules, **25**, 6205 (1992).
1269. T. Narasimhaswamy, S. C. Sumathi, B. S. R. Reddy, J. Polym. Sci., Part A: Polym. Chem., **30**, 2165 (1992).
1270. Y. Oishi, S. Nakata, M.-A. Kakimoto, Y. Imai, J. Polym. Sci., Part A: Polym. Chem., **30**, 2217 (1992).
1271. S. S. N. Murthy, J. Polym. Sci., Part B: Polym. Phys., **31**, 475 (1993).
1272. J. Scheirs, S. W. Bigger, E. T. H. Then, N. C. Billingham, J. Polym. Sci., Part B: Polym. Phys., **31**, 287 (1993).
1273. J.-L. Liang, J. P. Bell, J. O. Iroh, A. Mehta, J. Appl. Polym. Sci., **48**, 465 (1993).
1274. S. Kole, A. Bhattacharya, D. K. Tripathy, A. K. Bhowmick, J. Appl. Polym. Sci., **48**, 529 (1993).
1275. S. L. Simon, J. K. Gillham, J. Appl. Polym. Sci., **47**, 461 (1993).
1276. Y. H. Chin, C. Zhang, P. Wang, P. T. Inglefield, A. A. Jones, R. P. Kambour, J. T. Bendler, D. M. White, Macromolecules, **25**, 3031 (1992).
1277. J. M. G. Cowie, S. Miachon, Macromolecules, **25**, 3295 (1992).
1278. G. Ceccorulli, M. Pizzoli, M. Scandola, Macromolecules, **25**, 3304 (1992).
1279. D. Zaldivar, C. Peniche, A. Bulay, J. San Roman, J. Polym. Sci., Part A: Polym. Chem., **31**, 625 (1993).
1280. P. Kitipichai, R. La Peruta, G. M. Korenowski, G. E. Wnek, J. Polym. Sci., Part A: Polym. Chem., **31**, 1365 (1993).
1281. D. L. Sidebottom, R. Bergman, L. Borjesson, L. M. Torell, Prog. Colloid Polym. Sci., **91**, 43 (1993).
1282. B. Gerharz, S. Vogt, E. W. Fischer, G. Fytas, Prog. Colloid Polym. Sci., **91**, 58 (1993).
1283. A. K. Rizos, G. Fytas, J. E. L. Roovers, K. L. Ngai, Prog. Colloid Polym. Sci., **91**, 72 (1993).
1284. E. Jabbari, N. A. Peppas, Macromolecules, **26**, 2175 (1993).
1285. T. Kanaya, T. Kawaguchi, K. Kaji, Physica B: Condensed Matter, **182**, 403 (1992).
1286. A. Matsumoto, K. Mizuta, T. Otsu, Macromolecules, **26**, 1659 (1993).
1287. G. Marchionni, G. Ajroldi, M. C. Righetti, G. Pezzin, Macromolecules, **26**, 1751 (1993).
1288. P. Goldstein, L. F. del Castillo, L. S. Garcia-Colin, Macromolecules, **26**, 655 (1993).
1289. W.-Y. Chiang, J.-Y. Lu, Eur. Polym. J., **29**, 837 (1993).
1290. E. M. Cavalcante, M. Campos, Synth. Met., **57**, 4918 (1993).

1291. B. S. Hsiao, B. B. Sauer, J. Polym. Sci., Part B: Polym. Phys., **31**, 901 (1993).
1292. B. K. Coltrain, W. T. Ferrar, J. M. Salva, J. Polym. Sci., Part A: Polym. Chem., **31**, 2261 (1993).
1293. C.-H. Chu, B. Berner, J. Appl. Polym. Sci., **47**, 1083 (1993).
1294. N. S. Allen, J. P. Hurley, A. Rahman, G. W. Follows, I. Weddell, Eur. Polym. J., **29**, 1155 (1993).
1295. V. I. Bondar, Yu. M. Kukharskii, Y. P. Yampol'skii, E. Sh. Finkelshtein, K. L. Makovetskii, J. Polym. Sci., Part B: Polym. Phys., **31**, 1273 (1993).
1296. Y. P. Yampol'skij, S. M. Shishatskij, V. I. Bondar', S. A. Kuptsov, E. M. Antipov, V. A. Gubanov, D. Golemme, E. Drioli, N. A. Plate, Vysokomolekularnye Soedineniya. Seriya A, **35**, 1486 (1993).
1297. T. Kyu, C.-C. Ko, D.-S. Lim, S. D. Smith, I. Noda, J. Polym. Sci., Part B: Polym. Phys., **31**, 1641 (1993).
1298. A. J. Milton, A. P. Monkman, J. Phys. D: Appl. Phys., **26**, 1468 (1993).
1299. S. F. Hahn, M. P. Dreyfuss, J. Polym. Sci., Part A: Polym. Chem., **31**, 3039 (1993).
1300. E. Espi, M. Alberdi, J. J. Iruin, Macromolecules, **26**, 4586 (1993).
1301. R. W. Venderbosch, J. G. L. Nelissen, A. A. J. M. Peijs, P. J. Lemstra, 837 (1993).
1302. P. Krishnan, H. Le, S. H. Lee, E. Gelerinter, J. Polym. Sci., Part B: Polym. Phys., **31**, 1885 (1993).
1303. A. P. Melissaris, M. H. Litt, Macromolecules, **26**, 6734 (1993).
1304. K. L. Ngai, C. M. Roland, Macromolecules, **26**, 6824 (1993).
1305. L. A. Belfiore, M. P. McCurdie, E. Ueda, Macromolecules, **26**, 6908 (1993).
1306. H. W. J. Starkweather, P. Avakian, J. J. Fontanella, M. C. Wintersgill, Macromolecules, **26**, 5084 (1993).
1307. S.-I. Chen, C.-C. Yang, Polymer Journal, **25**, 1015 (1993).
1308. A. R. Khokhlov, I. Ya. Erukhimovich, Macromolecules, **26**, 7195 (1993).
1309. A. Hoffmann, T. Koch, B. Stuhn, Macromolecules, **26**, 7288 (1993).
1310. P. J. Burchill, J. Polym. Sci., Part A: Polym. Chem., **32**, 1 (1994).
1311. Y. Saegusa, M. Kuriki, A. Kawai, S. Nakamura, J. Polym. Sci., Part A: Polym. Chem., **32**, 57 (1994).
1312. G. S. Bennett, R. J. Farris, J. Polym. Sci., Part A: Polym. Chem., **32**, 73 (1994).
1313. S. A. Nye, S. A. Swint, J. Polym. Sci., Part A: Polym. Chem., **32**, 131 (1994).
1314. W. Hu, J. T. Koberstein, J. Polym. Sci., Part B: Polym. Phys., **32**, 437 (1994).
1315. P. D. Condo, K. P. Johnston, J. Polym. Sci., Part B: Polym. Phys., **32**, 523 (1994).
1316. C. J. Twomey, S. H. Chen, T. N. Blanton, A. Schmid, K. L. Marshall, J. Polym. Sci., Part B: Polym. Phys., **32**, 551 (1994).
1317. P. J. Scott, A. Penlidis, G. L. Rempel, A. D. Lawrence, J. Polym. Sci., Part A: Polym. Chem., **32**, 539 (1994).
1318. P. D. Condo, D. R. Paul, K. P. Johnston, Macromolecules, **27**, 365 (1994).
1319. B. B. Sauer, D. J. Walsh, Macromolecules, **27**, 432 (1994).
1320. A. Higuchi, T. Nakagawa, J. Polym. Sci., Part B: Polym. Phys., **32**, 149 (1994).
1321. Anon, J. Polym. Sci., Part B: Polym. Phys., **32**, 201 (1994).
1322. Z. Su, X. Li, S. L. Hsu, Macromolecules, **27**, 287 (1994).
1323. I. Arvanitoyannis, I. Kolokuris, J. M. V. Blanshard, C. Robinson, J. Appl. Polym. Sci., **48**, 987 (1993).
1324. W.-Y. Chiang, J.-Y. Lu, J. Appl. Polym. Sci., **49**, 893 (1993).
1325. M. F. Champagne, R. E. Prud'Homme, J. Polym. Sci., Part B: Polym. Phys., **32**, 615 (1994).
1326. Y. Xue, H. L. Frisch, J. Polym. Sci., Part A: Polym. Chem., **32**, 257 (1994).
1327. R. N. Santra, B. K. Samantaray, A. K. Bhowmick, G. B. Nando, J. Appl. Polym. Sci., **49**, 1145 (1993).
1328. M. Biswas, A. Roy, J. Appl. Polym. Sci., **49**, 2189 (1993).
1329. G. C. Chung, J. A. Kornfield, S. D. Smith, Macromolecules, **27**, 964 (1994).
1330. H. Kumazawa, J.-S. Wang, K. Naito, B. Messaoudi, E. Sada, J. Appl. Polym. Sci., **51**, 1015 (1994).
1331. S.-J. Chang, H.-B. Tsai, J. Appl. Polym. Sci., **51**, 999 (1994).
1332. M. Mourgues-Martin, A. Bernes, C. Lacabanne, J. Therm. Anal., **40**, 697 (1993).
1333. G. Teyssedre, A. Bernes, C. Lacabanne, J. Therm. Anal., **40**, 711 (1993).
1334. D. M. Petrovic-Djakov, J. M. Filipovic, Lj. P. Vrhovac, J. S. Velickovic, J. Therm. Anal., **40**, 741 (1993).
1335. J. J. Del Val, J. Colmenero, G. Martinez, J. L. Millan, J. Polym. Sci., Part B: Polym. Phys., **32**, 871 (1994).
1336. P. J. A. In't Veld, Z.-R. Shen, G. A. J. Takens, P. J. Dijkstra, J. Feijen, J. Polym. Sci., Part A: Polym. Chem., **32**, 1063 (1994).
1337. S. A. M. Ali, D. J. Hourston, J. Appl. Polym. Sci., **52**, 1129 (1994).
1338. L. Parravicini, B. Leone, F. Auriemma, G. Guerra, V. Petraccone, G. D. Dino, R. Bianchi, R. Vosa, J. Appl. Polym. Sci., **52**, 875 (1994).
1339. A. B. Dias, N. T. Correia, J. J. Moura-Ramos, A. C. Fernandes, Polym. Int., **33**, 293 (1994).
1340. G. Pollock, S. Nazarenko, A. Hiltner, E. Bear, J. Appl. Polym. Sci., **52**, 163 (1994).
1341. R. A. Huijts, A. J. De Vries, International Journal Of Polymeric Materials, **22**, 231 (1993).
1342. G. D. Patterson, P. K. Jue, D. J. Ramsay, J. R. Stevens, J. Polym. Sci., Part B: Polym. Phys., **32**, 1137 (1994).
1343. M. Fernandez-Garcia, M. M. C. Lopez-Gonzalez, J. M. Barrales-Rienda, E. L. Madruga, C. Arias, J. Polym. Sci., Part B: Polym. Phys., **32**, 1191 (1994).
1344. C. L. Choy, K. W. Kwok, W. P. Leung, F. P. Lau, J. Polym. Sci., Part B: Polym. Phys., **32**, 1389 (1994).
1345. X. Zhang, M. Shimoda, A. Toyoda, J. Polym. Sci., Part B: Polym. Phys., **32**, 1399 (1994).
1346. K. Itoya, M. Arata, M. Kakimoto, Y. Imai, J. Maeda, T. Kurosaki, J. Macromol. Sci. A: Chem., **31**, 817 (1994).
1347. N. Kato, N. Yamazaki, Y. Nagasaki, M. Kato, Polym. Bull. (Berlin), **32**, 55 (1994).
1348. G. Muller, R. Stadler, Macromolecules, **27**, 1555 (1994).

1349. A. Matsumoto, K. Shimizu, K. Mizuta, T. Otsu, J. Polym. Sci., Part A: Polym. Chem., **32**, 1957 (1994).
1350. T. M. Moy, J. E. McGrath, J. Polym. Sci., Part A: Polym. Chem., **32**, 1903 (1994).
1351. M. T. DeMeuse, J. Polym. Sci., Part B: Polym. Phys., **32**, 1749 (1994).
1352. D. Kim, J. M. Caruthers, N. A. Peppas, J. Polym. Sci., Part B: Polym. Phys., **32**, 1593 (1994).
1353. M.-B. Hagg, W. J. Koros, J. C. Schmidhauser, J. Polym. Sci., Part B: Polym. Phys., **32**, 1625 (1994).
1354. P. G. Santangelo, K. L. Ngai, C. M. Roland, Macromolecules, **27**, 3859 (1994).
1355. R. P. Pearce, R. H. Marchessault, Macromolecules, **27**, 3869 (1994).
1356. B. V. Lebedev, L. Ya. Tsvetkova, N. N. Smirnova, J. Therm. Anal., **41**, 1371 (1994).
1357. C. He, A. C. Griffin, A. H. Windle, J. Appl. Polym. Sci., **53**, 561 (1994).
1358. I. Rusig, M. H. Godinho, L. Varichon, P. Sixou, J. Dedier, C. Filliatre, A. F. Martins, J. Polym. Sci., Part B: Polym. Phys., **32**, 1907 (1994).
1359. A. Mathew, P. C. Deb, J. Appl. Polym. Sci., **53**, 1103 (1994).
1360. C. K. Park, C. S. Ha, J. K. Lee, W. J. Cho, J. Appl. Polym. Sci., **53**, 967 (1994).
1361. P. Skoglund, A. Fransson, J. Polym. Sci., Part B: Polym. Phys., **32**, 1999 (1994).
1362. M. D. Migahed, M. Ishra, A. El-Khodary, T. Fahmy, J. Appl. Polym. Sci., **53**, 1315 (1994).
1363. A. S. Kulik, H. W. Beckham, K. Schmidt-Rohr, D. Radloff, U. Pawelzik, C. Boeffel, H. W. Spiess, Macromolecules, **27**, 4746 (1994).
1364. M. Tsukada, Y. Gotoh, M. Nagura, N. Minoura, N. Kasai, G. Freddi, J. Polym. Sci., Part B: Polym. Phys., **32**, 961 (1994).
1365. O. Prucker, S. Christian, H. Bock, C. W. Frank, W. Knoll, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **38**, 918 (1997).
1366. K. Nagapudi, H. W. Beckham, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **38**, 790 (1997).
1367. J. J. Schwab, J. D. Lichtenhan, K. P. Chaffee, J. C. Gordon, Y. A. Otonari, M. J. Carr, A. G. Bolf, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **38**, 518 (1997).
1368. J. W. Fitch, V. S. Reddy, P. W. Youngman, G. A. Wohlfahrt, P. E. Cassidy, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **37**, 290 (1996).
1369. C. M. Lewis, L. J. Mathias, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **37**, 243 (1996).
1370. C. Brochon, E. Duguet, M. Schappacher, A. F. Mingotaud, A. Soum, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **37**, 807 (1996).
1371. J. Bettenhausen, P. Strohriegl, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **37**, 504 (1996).
1372. J. E. McGrath, S. K. Jayaraman, P. Lakshmanan, J. C. Abed, F. Afchar-Taromi, Polymer Preprints, Division of Polymer Chemistry, American Chemical Society, **37**, 136 (1996).
1373. D.-J. Liaw, B.-Y. Liaw, J. Polym. Sci., Part A: Polym. Chem., **35**, 1527 (1997).
1374. S.-H. Hsiao, C.-P. Yang, K.-Y. Chu, J. Polym. Sci., Part A: Polym. Chem., **35**, 1469 (1997).
1375. S. Shirouzu, S. Chino, M. Hayashi, T. Ishii, T. Iga, M. Yoshida, N. Senda, Japanese Journal of Applied Physics, Part 1: Regular Papers & Short Notes & Review Papers, **36**, 260 (1997).
1376. J. E. McGrath, B. Tan, V. Vasudevan, G. W. Meyer, A. C. Loos, T. Bullions, International SAMPE Technical Conference, **28**, 29 (1996).
1377. Y. S. Yu, Ph. Dubois, R. Jerome, Ph. Teyssie, J. Polym. Sci., Part A: Polym. Chem., **34**, 2221 (1996).
1378. V. Vasudevan, S. J. Mecham, S. Liu, S. Srinivasan, S. Claus, A. C. Loos, J. E. McGrath, International SAMPE Symposium and Exhibition (Proceedings), **41**, 1092 (1996).
1379. S. B. Sastri, J. P. Armistead, T. M. Keller, International SAMPE Symposium and Exhibition (Proceedings), **41**, 171 (1996).
1380. W. K. Shu, Proceedings - Electronic Components and Technology Conference, 219 (1996).
1381. M. A. Corcuera, I. Mondragon, C. C. Riccardi, R. J. J. Williams, J. Appl. Polym. Sci., **64**, 157 (1997).
1382. X.-P. Hu, Y.-L. Hsieh, Polymer, **38**, 1491 (1997).
1383. J. S. Zaroulis, M. C. Boyce, Polymer, **38**, 1303 (1997).
1384. J.-O. Choi, R. Kirsten, S. Rojstaczer, D. Riordan, Proceedings of the International Symposium and Exhibition on Advanced Packaging Materials Processes, Properties and Interfaces, 56 (1997).
1385. F. W. Mercer, M. M. Fone, V. N. Reddy, A. A. Goodwin, Polymer, **38**, 1989 (1997).
1386. A. N. Wilkinson, S. B. Tattum, A. J. Ryan, Polymer, **38**, 1923 (1997).
1387. J. P. Habas, J. Peyrelasse, M. F. Grenier-Loustalot, High Performance Polymers, **8**, 515 (1996).
1388. C. Hamciuc, E. Hamciuc, M. Bruma, High Performance Polymers, **8**, 507 (1996).
1389. C. Hamciuc, E. Hamciuc, F. W. Mercer, N. M. Belomoina, M. Bruma, High Performance Polymers, **8**, 445 (1996).
1390. K. R. Srinivasan, R. G. Bass, J. G. Jr. Smith, High Performance Polymers, **8**, 381 (1996).
1391. C. M. Rocha, M. A. Gomez, J. G. Fatou, P. Evans, J. Cronauer, H. G. Zachmann, Polymer, **38**, 1601 (1997).
1392. O. Ozarslan, T. Yilmaz, E. Yildiz, U. Fiedeldei, A. Kuyulu, A. Gungor, J. Polym. Sci., Part A: Polym. Chem., **35**, 1149 (1997).
1393. D. H. Bolton, K. L. Wooley, J. Polym. Sci., Part A: Polym. Chem., **35**, 1133 (1997).
1394. S. W. Smith, B. D. Freeman, C. K. Hall, Macromolecules, **30**, 2052 (1997).
1395. X. Wan, H. S. Kwok, B. Z. Tang, Proceedings of the Asian Symposium on Information Display, 187 (1997).
1396. G. Gabellini, R. E. S. Bretas, Journal De Physique. IV: JP, **6**, 575 (1996).
1397. D. M. Rueck, J. Schulz, W. F. Frank, Proc. SPIE-Int. Soc. Opt. Eng., **2851**, 118 (1996).

1398. J. Reyes Romero, M. D. Chipara, V. Mihai, M. I. Chipara, Conference on Electrical Insulation and Dielectric Phenomena (CEIDP), Annual Report, **2**, 679 (1996).
1399. M. Ueda, T. Nakayama, T. Mitsuhashi, J. Polym. Sci., Part A: Polym. Chem., **35**, 371 (1997).
1400. Q. Guo, S. Zheng, J. Li, Y. Mi, J. Polym. Sci., Part A: Polym. Chem., **35**, 211 (1997).
1401. C. C. Cypcar, P. Camelio, V. Lazzeri, L. J. Mathias, B. Waegell, Macromolecules, **29**, 8954 (1996).
1402. K. Hiltunen, M. Harkonen, J. V. Seppala, T. Vaananen, Macromolecules, **29**, 8677 (1996).
1403. C. H. Bergstrom, J. V. Seppala, J. Appl. Polym. Sci., **63**, 1063 (1997).
1404. A. Shiotani, M. Kohda, J. Appl. Polym. Sci., **63**, 865 (1997).
1405. B. R. Maughon, M. Weck, B. Mohr, R. H. Grubbs, Macromolecules, **30**, 257 (1997).
1406. Z. Wang, T. Chen, J. Xu, J. Appl. Polym. Sci., **63**, 1127 (1997).
1407. K. M. Rajkotia, M. M. Kamani, P. H. Parsania, Polymer, **38**, 715 (1997).
1408. K. Inoue, Y. Imai, Kobunshi Ronbunshu/Japanese Journal of Polymer Science and Technology, **53**, 869 (1996).
1409. T.-G. Choi, Y.-K. Yun, J. Jin, Polymer-Plastics Technology and Engineering, **36**, 135 (1997).
1410. H. Ishida, H. Y. Low, Macromolecules, **30**, 1099 (1997).
1411. S. Srinivas, F. E. Caputo, M. Graham, S. Gardner, R. M. Davis, J. E. McGrath, G. L. Wilkes, Macromolecules, **30**, 1012 (1997).
1412. J. Kylma, M. Harkonen, J. V. Seppala, J. Appl. Polym. Sci., **63**, 1865 (1997).
1413. E. Donth, M. Beiner, S. Reissig, J. Korus, F. Garwe, S. Vieweg, S. Kahle, E. Hempel, K. Schroeter, Macromolecules, **29**, 6589 (1996).
1414. A. Gu, G. Liang, L. Lan, J. Appl. Polym. Sci., **62**, 799 (1996).
1415. A. E. Wolfenson, R. M. Torresi, T. J. Bonagamba, M.-A. De Paoli, H. Panepucci, Solid State Ionics, **85**, 219 (1996).
1416. Y. Hwang, G. D. Patterson, J. R. Stevens, J. Polym. Sci., Part B: Polym. Phys., **34**, 2291 (1996).
1417. F. Koopmann, A. Burgath, R. Knischka, J. Leukel, H. Frey, Acta Polym., **47**, 377 (1996).
1418. M. A. Pauley, C. H. Wang, A. K. Y. Jen, Macromolecules, **29**, 7064 (1996).
1419. H. Amri, M. N. Belgacem, A. Gandini, Polymer, **37**, 4815 (1996).
1420. T. S. Haddad, E. Choe, J. D. Lichtenhan, Materials Research Society Symposium Proceedings, **435**, 25 (1996).
1421. J. A. Mikroyannidis, J. Polym. Sci., Part A: Polym. Chem., **34**, 3389 (1996).
1422. H. R. Allcock, S. J. M. O'Connor, D. L. Olmeijer, M. E. Napierala, C. G. Cameron, Macromolecules, **29**, 7544 (1996).
1423. A. Brunacci, J. M. G. Cowie, R. Ferguson, J. L. G. Ribelles, A. V. Garayo, Macromolecules, **29**, 7976 (1996).
1424. R. M. Hodge, T. J. Bastow, G. H. Edward, G. P. Simon, A. J. Hill, Macromolecules, **29**, 8137 (1996).
1425. Y. Bourgeois, Y. Charlier, J. Devaux, R. Legras, Polymer, **37**, 5503 (1996).
1426. A. H. Tsou, D. L. DelleFave, Polymer, **37**, 5381 (1996).
1427. S. Jacobsen, H. G. Fritz, Polym. Eng. Sci., **36**, 2799 (1996).
1428. Ch. R. Davis, J. Appl. Polym. Sci., **62**, 2237 (1996).
1429. C. Hamciuc, E. Hamciuc, I. Sava, A. Stoleriu, F. W. Mercer, M. Bruma, Polymers for Advanced Technologies, **7**, 847 (1996).
1430. O. Schaefer, A. Greiner, J. Pommerehne, W. Guss, H. Vestweber, H. Y. Tak, H. Baessler, C. Schmidt, G. Luessem, B. Schartel, V. Stuempflen, J. H. Wendorff, S. Spiegel, C. Moeller, H. W. Spiess, Synth. Met., **82**, 1 (1996).
1431. J. S. Grebowicz, J. Therm. Anal., **46**, 1151 (1996).
1432. S. Matsuoka, J. Therm. Anal., **46**, 985 (1996).
1433. J. Rieger, J. Therm. Anal., **46**, 965 (1996).
1434. Y. H. Kim, B. S. Moon, F. W. Harris, S. Z. D. Cheng, J. Therm. Anal., **46**, 921 (1996).
1435. J. D. Menczel, M. Jaffe, C. K. Saw, T. P. Bruno, J. Therm. Anal., **46**, 753 (1996).
1436. J. D. Menczel, M. Jaffe, C. K. Saw, J. Therm. Anal., **46**, 733 (1996).
1437. S. Reissig, M. Beiner, S. Vieweg, K. Schroeter, E. Donth, Macromolecules, **29**, 3996 (1996).
1438. D. D. Deppe, A. Dhinojwala, J. M. Torkelson, Macromolecules, **29**, 3898 (1996).
1439. D. J. Brennan, J. E. White, A. P. Haag, S. L. Kram, M. N. Mang, S. Pikulin, C. N. Brown, Macromolecules, **29**, 3707 (1996).
1440. F. Koopmann, H. Frey, Macromolecules, **29**, 3701 (1996).
1441. H. R. Allcock, S. E. Kuharcik, C. J. Nelson, Macromolecules, **29**, 3686 (1996).
1442. L. M. Weldon, M. J. Pomeroy, S. Hampshire, Key Eng. Mater., **118/119**, 241 (1996).
1443. G. Meier, F. Kremer, G. Fytas, A. Rizos, J. Polym. Sci., Part B: Polym. Phys., **34**, 1391 (1996).
1444. X. Meng, A. Natansohn, P. Rochon, J. Polym. Sci., Part B: Polym. Phys., **34**, 1461 (1996).
1445. Y. Zhang, J. C. Tebby, J. W. Wheeler, J. Polym. Sci., Part A: Polym. Chem., **34**, 1561 (1996).
1446. R. Flores, J. Perez, P. Cassagnau, A. Michel, J. Y. Cavaille, J. Appl. Polym. Sci., **60**, 1439 (1996).
1447. Y. Yuan, A. Siegmann, M. Narkis, J. Appl. Polym. Sci., **60**, 1475 (1996).
1448. N. C. Stoffel, M. Hsieh, E. J. Kramer, W. Volksen, IEEE Transactions on Components, Packaging, and Manufacturing Technology Part B: Advanced Packaging, **19**, 417 (1996).
1449. H. R. Allcock, S. E. Kuharcik, C. S. Reed, M. E. Napierala, Macromolecules, **29**, 3384 (1996).
1450. Y. Ni, P. Park, M. Liang, J. Massey, C. Waddling, I. Manners, Macromolecules, **29**, 3401 (1996).
1451. Y. Kim, Y. H. Rhee, S.-H. Han, G. S. Heo, J. S. Kim, Macromolecules, **29**, 3432 (1996).
1452. P. G. Santangelo, K. L. Ngai, C. M. Roland, Macromolecules, **29**, 3651 (1996).
1453. J. A. Dugger, D. E. Hirt, Polymer Composites, **17**, 492 (1996).
1454. Y. Ding, A. S. Hay, Macromolecules, **29**, 4811 (1996).

1455. S. Datta, A. K. Bhattacharya, S. K. De, E. G. Kontos, J. M. Wefer, Polymer, **37**, 2581 (1996).
1456. K.-J. Moon, H.-K. Shim, K.-S. Lee, J. Zieba, P. N. Prasad, Macromolecules, **29**, 861 (1996).
1457. D. Li, J. Brisson, Macromolecules, **29**, 868 (1996).
1458. R. G. Bryant, B. J. Jensen, P. M. Hergenrother, J. Appl. Polym. Sci., **59**, 1249 (1996).
1459. Y.-H. Kuo, L.-W. Chen, Polym. Int., **40**, 93 (1996).
1460. E. Elce, A. S. Hay, J. Polym. Sci., Part A: Polym. Chem., **34**, 1897 (1996).
1461. Y.-M. Sun, C.-S. Wang, J. Polym. Sci., Part A: Polym. Chem., **34**, 1783 (1996).
1462. I. K. Spiliopoulos, J. A. Mikroyannidis, J. Polym. Sci., Part A: Polym. Chem., **34**, 1703 (1996).
1463. R. Sadamoto, A. Uedono, T. Kawano, S. Tanigawa, Y. Kosaka, T. Uryu, J. Polym. Sci., Part B: Polym. Phys., **34**, 1659 (1996).
1464. J. E. Sealey, G. Samaranayake, J. G. Todd, W. G. Glasser, J. Polym. Sci., Part B: Polym. Phys., **34**, 1613 (1996).
1465. T. Doi, N. Ishikawa, A. Akimoto, J. Appl. Polym. Sci., **61**, 859 (1996).
1466. D. Likhatchev, L. Alexanrova, M. Tlenkopatchev, A. Martinez-Richa, R. Vera-Graziano, J. Appl. Polym. Sci., **61**, 815 (1996).
1467. H. Reinecke, J. G. de la Campa, J. de Abajo, H. R. Kricheldorf, G. Schwarz, Polymer, **37**, 3101 (1996).
1468. M. Beiner, J. Korus, H. Lockwenz, K. Schroeter, E. Donth, Macromolecules, **29**, 5183 (1996).
1469. R. L. Shogren, Carbohydr. Polym., **29**, 57 (1996).
1470. P. Ramachandra, R. Ramani, T. S. G. Ravichandran, G. Ramgopal, S. Gopal, C. Ranganathaiah, N. S. Murthy, Polymer, **37**, 3233 (1996).
1471. Y. Zhang, U. Wiesner, Y. Yang, T. Pakula, H. W. Spiess, Macromolecules, **29**, 5427 (1996).
1472. M. Chen, H. W. Gibson, Macromolecules, **29**, 5502 (1996).
1473. B. L. Weick, B. Bhushan, IEEE Trans. Magn., **32**, 3319 (1996).
1474. L. J. Mathias, G. L. Tullos, Polymer, **37**, 3771 (1996).
1475. K. Nawa, M. Ohkita, Circuit World, **22**, 16 (1996).
1476. C. M. Roland, K. L. Nagi, Macromolecules, **29**, 5747 (1996).
1477. Y.-H. Lu, C.-S. Hsu, C.-H. Chou, T.-H. Chou, Macromolecules, **29**, 5546 (1996).
1478. Y. Nagasaki, F. Matsukura, M. Kato, H. Aoki, T. Tokuda, Macromolecules, **29**, 5859 (1996).
1479. H. Yang, A. S. Hay, J. Polym. Sci., Part A: Polym. Chem., **34**, 2621 (1996).
1480. P. Avakian, J. C. Coburn, M. S. Connolly, B. B. Sauer, Polymer, **37**, 3843 (1996).
1481. D. E. Maurer, E. M. O'Brien, International SAMPE Electronics Conference, **7**, 188 (1994).
1482. M. A. Del Nobile, G. Mensitieri, A. Sommazzi, Polymer, **36**, 4943 (1995).
1483. L. F. Brown, J. I. Scheinbeim, B. A. Newman, Ferroelectrics, **171**, 321 (1995).
1484. F. Garwe, A. Schoenhals, H. Lockwenz, M. Beiner, K. Schroeter, E. Donth, Macromolecules, **29**, 247 (1996).
1485. H.-B. Tsai, D.-K. Lee, J.-L. Liu, Y.-S. Tsao, R.-S. Tsai, J. Appl. Polym. Sci., **59**, 1027 (1996).
1486. J. Kodemura, T. Natsuume, Polymer Journal, **27**, 1167 (1995).
1487. E. J. Kim, T. Takeda, Y. Ohki, Conference on Electrical Insulation and Dielectric Phenomena (CEIDP), Annual Report, 492 (1995).
1488. M. Campos, E. M. Cavalcante, J. Kalinowski, J. Polym. Sci., Part B: Polym. Phys., **34**, 623 (1996).
1489. P. Kaatz, P. Pretre, U. Meier, U. Stalder, C. Bosshard, P. Gunter, B. Zysset, M. Stahelin, M. Ahlheim, F. Lehr, Macromolecules, **29**, 1666 (1996).
1490. I. Sava, B. Schulz, S. Zhu, M. Bruma, High Performance Polymers, **7**, 493 (1995).
1491. C.-P. Yang, J.-H. Lin, J. Polym. Sci., Part A: Polym. Chem., **34**, 341 (1996).
1492. J. Sugiyama, N. Kanamori, S. Shimada, Macromolecules, **29**, 1943 (1996).
1493. J.-H. Lin, C.-P. Yang, J. Polym. Sci., Part A: Polym. Chem., **34**, 747 (1996).
1494. A. Spyros, R. H. Marchessault, Macromolecules, **29**, 2479 (1996).
1495. H. Takeno, S. Koizumi, H. Hasegawa, T. Hashimoto, Macromolecules, **29**, 2440 (1996).
1496. T. Takahashi, N. Hayashi, S. Hayashi, J. Appl. Polym. Sci., **60**, 1061 (1996).
1497. S. S. Wu, D. S. Kalika, R. R. Lamonte, S. Makhija, J. Macromol. Sci. B: Phys., **35**, 157 (1996).
1498. B. Sarti, M. Scandola, Biomaterials, **16**, 785 (1995).
1499. M. Kawagoe, W. Mizuno, J. Qiu, M. Morita, Kobunshi Ronbunshu/Japanese Journal of Polymer Science and Technology, **52**, 335 (1995).
1500. H.-Y. Shih, W. F. Kuo, E. M. Pearce, T. K. Kwei, Polymers for Advanced Technologies, **6**, 413 (1995).
1501. D. Li, R. Faust, Macromolecules, **28**, 4893 (1995).
1502. H. Maeda, Macromolecules, **28**, 5156 (1995).
1503. Y. Zhang, L. Wang, T. Wada, H. Sasabe, Molecular Crystals and Liquid Crystals Science and Technology, Section A: Molecular Crystals and Liquid Crystals, **267**, 65 (1995).
1504. M. S. Ho, A. Natansohn, P. Rochon, Macromolecules, **28**, 6124 (1995).
1505. E. Perez, R. Benavente, A. Bello, J. M. Perena, D. L. VanderHart, Macromolecules, **28**, 6211 (1995).
1506. T. Kawaguchi, T. Kanaya, K. Kaji, Physica B: Condensed Matter, **213/214**, 510 (1995).
1507. Y. Di, A. D'Amore, G. Marino, L. Nicolais, B. Li, J. Appl. Polym. Sci., **57**, 989 (1995).
1508. F. Alvarez, J. Colmenero, C. H. Wang, J. L. Xia, G. Fytas, Macromolecules, **28**, 6488 (1995).
1509. N. Khan, Z. Bashir, D. M. Price, J. Appl. Polym. Sci., **58**, 1509 (1995).
1510. W.-H. Chan, S. Y. Lam-Leung, C.-F. Ng, J. Ding, S. Xi, Polymer, **36**, 4503 (1995).
1511. F. Akutsu, N. Sugiyama, N. Ando, Y. Kasashima, M. Inoki, K. Naruchi, Polymer Journal, **27**, 1025 (1995).
1512. P. Wisian-Neilson, G.-F. Xu, T. Wang, Macromolecules, **28**, 8657 (1995).

1513. D. Dibbern-Brunelli, T. D. Z. Atvars, J. Appl. Polym. Sci., **55**, 889 (1995).
1514. Y.-K. Han, S. D. Chi, Y. H. Kim, B. K. Park, J.-I. Jin, Macromolecules, **28**, 916 (1995).
1515. A. Lofgren, R. Renstad, A.-C. Albertsson, J. Appl. Polym. Sci., **55**, 1589 (1995).
1516. A. Alegria, E. Guerrica-Echevarria, L. Goitiandia, I. Telleria, J. Colmenero, Macromolecules, **28**, 1516 (1995).
1517. A. Uedono, R. Sadamoto, T. Kawano, S. Tanigawa, T. Uryu, J. Polym. Sci., Part B: Polym. Phys., **33**, 891 (1995).
1518. L. Fambri, A. Pegoretti, M. Gleria, F. Minto, Polymer, **35**, 4813 (1994).
1519. E. Perez, Z. Zhen, A. Bello, R. Benavente, Polymer, **35**, 4794 (1994).
1520. G. Consolati, T. Genco, M. Pegoraro, L. Zanderighi, Mater. Sci. Forum, **175/178**, 335 (1995).
1521. M. Coskun, K. Demirelli, E. Ozdemir, Polym. Degrad. Stab., **47**, 251 (1995).
1522. J. M. I. McKillen, International SAMPE Technical Conference, **26**, 423 (1994).
1523. M. Takenaka, K. Linliu, Q. Ying, B. Chu, D. Peiffer, Macromolecules, **28**, 2700 (1995).
1524. S. Ito, K. Kawano, M. Yamamoto, H. Hasegawa, T. Hashimoto, Macromolecules, **28**, 3736 (1995).
1525. Z. Y. Wang, A. Le Guen, Macromolecules, **28**, 3728 (1995).
1526. B. Levenfeld, J. San Roman, Macromolecules, **28**, 3650 (1995).
1527. D. K. Song, Y. K. Sung, J. Appl. Polym. Sci., **56**, 1381 (1995).
1528. J. Huang, Y. Hu, I. E. Ruyter, J. Appl. Polym. Sci., **56**, 1253 (1995).
1529. V.-T. Truong, B. C. Ennis, M. Forsyth, Polymer, **36**, 1933 (1995).
1530. J. Zhao, Y. H. Chin, Y. Liu, A. A. Jones, P. T. Inglefield, R. P. Kambour, D. M. White, Macromolecules, **28**, 3881 (1995).
1531. X. Ma, J. A. Sauer, M. Hara, Macromolecules, **28**, 3953 (1995).
1532. J. S. Kollodge, R. S. Porter, Macromolecules, **28**, 4089 (1995).
1533. G. B. Blanchet, Macromolecules, **28**, 4603 (1995).
1534. R. Pardey, S. S. Wu, J. Chen, F. W. Harris, S. Z. D. Cheng, A. Keller, J. Aducci, J. V. Facinelli, R. W. Lenz, Macromolecules, **27**, 5794 (1994).
1535. J. Sugiyama, K. Ohashi, M. Ueda, Macromolecules, **27**, 5543 (1994).
1536. H. Abe, I. Matsubara, Y. Doi, Y. Hori, A. Yamaguchi, Macromolecules, **27**, 6018 (1994).
1537. P. Shukla, A. K. Srivastava, Polymer, **35**, 4665 (1994).
1538. K. Nagai, A. Higuchi, T. Nakagawa, J. Appl. Polym. Sci., **54**, 1207 (1994).
1539. W. D. Joseph, J. C. Abed, R. Mercier, J. E. McGrath, Polymer, **35**, 5046 (1994).
1540. J. Han, R. H. Gee, R. H. Boyd, Macromolecules, **27**, 7781 (1994).
1541. B. V. Lebedev, L. Ya. Tsvetkova, N. N. Smirnova, J. Therm. Anal., **41**, 1371 (1994).
1542. K. H. Yoon, S. C. Lee, O. O. Park, Polymer Journal, **26**, 816 (1994).
1543. C. K. Park, C. S. Ha, J. K. Lee, W. J. Cho, J. Appl. Polym. Sci., **53**, 967 (1994).
1544. P. Skoglund, A. Fransson, J. Polym. Sci., Part B: Polym. Phys., **32**, 1999 (1994).
1545. A. S. Kulik, H. W. Beckham, K. Schmidt-Rohr, D. Radloff, U. Pawelzik, C. Boeffel, H. W. Spiess, Macromolecules, **27**, 4746 (1994).
1546. S. Kansara, G. M. Patel, N. K. Patel, J. Macromol. Sci. A: Chem., **31**, 525 (1994).
1547. X. Zhang, D. H. Solomon, Macromolecules, **27**, 4919 (1994).
1548. F. Sahlen, A. Hult, U. W. Gedde, F. Ania, J. Martinez-Salazar, Polymer, **35**, 4041 (1994).
1549. K. Osaki, T. Inoue, E.-J. Hwang, H. Okamoto, O. Takiguchi, J. Non-Cryst. Solids, **172/177**, 838 (1994).
1550. C. M. Roland, Macromolecules, **27**, 4242 (1994).
1551. Y. P. Handa, J. Roovers, F. Wang, Macromolecules, **27**, 5511 (1994).
1552. J. A. Pomposo, I. Eguiazabal, E. Calahorra, M. Cortazar, Polymer, **34**, 95 (1993).
1553. R. Storbeck, M. Ballauff, Polymer, **34**, 5003 (1993).
1554. R. A. Huijts, A. J. De Vries, "International Journal of Polymeric Materials", **22**, 231 (1993).
1555. G. D. Patterson, P. K. Jue, D. J. Ramsay, J. R. Stevens, J. Polym. Sci., Part B: Polym. Phys., **32**, 1137 (1994).
1556. N. Kato, N. Yamazaki, Y. Nagasaki, M. Kato, Polym. Bull. (Berlin), **32**, 55 (1994).
1557. G. Floudas, A. Rizos, W. Brown, K. L. Ngai, Macromolecules, **27**, 2719 (1994).
1558. S. Matsuo, K. Mitsuhashi, J. Polym. Sci., Part A: Polym. Chem., **32**, 1969 (1994).
1559. P. G. Santangelo, K. L. Ngai, C. M. Roland, Macromolecules, **27**, 3859 (1994).
1560. R. P. Pearce, R. H. Marchessault, Macromolecules, **27**, 3869 (1994).

Rates of Crystallization of Polymers

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A. INTRODUCTION

1. Background

1.1. General Remarks This chapter is intended to serve as a useful compendium of crystallization data for many types of polymers and polymeric systems. Some information is also provided on the fundamentals of crystallization (nucleation and growth) of these materials, with appropriate references relating to the techniques that are usually employed to study phase transformation processes in polymers. Controversial issues on growth and morphology will be mentioned briefly so that the reader who is new to this field will gain some perspective on the theoretical and experimental developments that have evolved over many decades, although this has little bearing on the numerical data presented here. This presentation serves not only as a guide to the data, but as a timely attempt to alert the vigilant reader to some difficulties, and some intricacies and problems of polymer crystallization that may be encountered in the literature (1–5).

The third edition of this Handbook reported the rates of crystallization for homopolymers (1–5), polymer blends (6,7), and copolymers (8). This edition updates this information, adds polymeric mesophases, and factors other than temperature that effect crystallization behavior. Within the past decade, crystallization of polymer blends have become increasingly important industrially. Besides the property-synergism provided through blending, there is a further economical incentive to use available materials rather than attempt to develop new polymers. Solid-state multiphase systems provide challenging academic research problems because polymer compatibility, *per se,* dominates morphology, whatever the crystallinity content of the system. The phase transformation depends upon the polymer type, monomer size and placement of the monomer units comprising the macromolecules (10). Sequence length and blockiness often feature strongly in determining rates of crystallization and levels of crystallinity. Related monotonic changes sometimes occur in physical parameters such as melting points and glass temperatures, crystallinities, and heats of fusion (11). The usefulness of a material depends upon the magnitude of these parameter changes. Polymer blending can often provide dispersions with overall properties that are superior to those of the respective components in the system, whether they crystallize or not. Blended mixtures frequently phase separate, but others maintain their morphological and physical integrity in practical applications (12). Cocrystallization may occur sometimes if there is lattice matching of the crystallographic chain axis repeat dimension and direction, and packing in the lateral directions within the crystallites is not too disruptive. In these circumstances the property changes may be striking. A fundamental knowledge of the kinetics of crystallization helps one to modify and improve properties (15).

The crystallization and morphology of lyotropic and thermotropic phases have only been developed comparatively recently, though liquid crystal polymers were recognized experimentally (13) and theoretically (14) a long time ago. The guidelines established for small-molecule liquid crystals (15) have been helpful for macromolecules, but they too may be misleading at times, particularly for polydisperse high-molecular-weight polymeric mesophases (16) and where their structural features are submicroscopic. Kinetic information for liquid crystal polymer systems is growing but still sparse (17,18). Synchrotron radiation sources and rapid monitoring techniques (19,20) are now available nationally and internationally for measuring isothermal and nonisothermal changes (21) and these simplify the monitoring of *in situ* transformations. Spectroscopic procedures (22) are now more capable than ever for sensing and tracking rapid and specific motions that have escaped detection by conventional means until the present. In addition to high intensity X-ray and light (laser) sources (23), deuterium NMR methods (24) are being used and developed for studying molecular dynamics, structural, or orientational changes (25). These new classes of nondestructive experiments are essential for improving our understanding of polymer behavior in the condensed state.

1.2. Background Over the years, two parallel themes describing aspects of crystallization of synthetic polymers have been developed. The oldest of these (26,27,30) uses a thermodynamic approach and provides a statistical description of morphology related to polymer crystallization, morphology, and properties. The other approach (28–29) is about half a century old and prefers a kinetic description of the development of lamellar structures that are basic to nucleation, growth, and stability in this context. Each has its merits and shortcomings but neither model is a panacea. Now, there is a consensus over the molecular dimensions that characterize macromolecules in the condensed state (4), but controversy remains over the intricacies of crystallization and its role in determining structure–property relationships. However, with new techniques now available for surface investigations and data processing, the time may not be too far away for an atomic view of matter in real space. For information on this topic, the reader should consult the current literature (15,31).

Whenever a polymer melt (liquid) or solution is sufficiently cooled, it may spontaneously crystallize. For instance, a supersaturated dilute solution produces tiny single[1] crystals (32). Spherulites are formed from concentrated polymer solutions, and these most frequently grow from the supercooled molten state. These growth patterns only occur from "privileged" nuclei, i.e., those that are capable of growing with a decrease in free energy (i.e., enhanced stability) to form ordered structures. If the supercooled melt (liquid) does not undergo this discontinuous transformation, then it is destined to form a glass, a disordered condensed phase. The crystalline texture and

[1] More appropriately described here as solution-grown crystals.

morphology, as well as degree of crystallinity, depends to some degree upon experimental conditions (33,34). "Single" crystals are comprised of thin lamellae with diverse habits. These morphological forms are discussed later in this article. Lamellae are basic crystal structures, although their intricate morphology is still open to debate (15,16). The next steps in the hierarchical level of structure are many different kinds of spherulites that are polycrystalline. The exact morphology of these moieties is still unsettled though they were discovered about a century ago. Several factors may be involved, namely: (a) monomeric chemical repeat unit, (b) crystallization temperature, (c) molecular weight and molecular weight distribution, (d) degree of branching and crosslinking, (e) tacticity, (f) degree of supersaturation, and (g) environmental factors (stereochemical defects, stress, solvent, impurities, etc.) and so on. These items are not placed in order of importance. Morphology is often dictated by other circumstances. Merely by changing the concentration of polymer in solution or its solvent, it is often possible to cover the "morphological spectrum" of crystalline polymeric forms, ranging from the monocrystal (formed in very dilute solution) to spherulites (formed in relatively concentrated solutions or from the polymer melt) under quiescent conditions. Other external factors such as pressure or stress and orientation can also alter the morphology (5).

This article deals mainly with isothermal crystallization conditions, but kinetic results obtained under constrained conditions are also listed. These are in the minority but the literature is growing. Although nonisothermal transformations (35) dominate in industrial operations, the data are frequently proprietary and often peculiar to specific process conditions. For example, molding processes (36) and fiber extrusion (37) operations create complex crystallization patterns, but the literature and even patents are still sparse on kinetic details. Most manufacturing procedures actually originated in industry. Still, some aspects of the topic have been examined theoretically (38).

Again, the influences of nucleants or additives on crystallization behavior have made considerable inroads in the scientific and patent[2] literature (39), but the importance of shear rates (40,41) on crystallization kinetics is beginning to receive more attention in the literature (42). In the field of "single crystal" growth many new results have been added to older kinetic data (43,44) tabulated in previous editions of "Polymer Handbook". The most recent are concerned with the crystallization of oligomers of polyethylene, polyethylene oxide and so on. Many aspects of crystallization are reported in some recent reviews (45,45a,45b) and later in this article.

1.3. Morphological Outline/Growth Features Under quiescent conditions, most often in very dilute solutions, and in self-seeding procedures, crystalline lamellae or "single" crystals are formed. They are seldom more than 85% crystalline even for annealed samples. Within these crystals, the chain macromolecules are disposed with their chain axis (*c* axis) more or less perpendicular to the flat faces of the thin lamellar crystals. However, angles of inclination as large as 48° have been reported in polyamides (5). Lamellar thickness (45–48) is controlled by nucleation/growth conditions. Electron microscopy (48) and other studies (49) have shown that crystallization (growth) proceeds outward from a primary center (a nucleation site) until growth is arrested or slowed to an imperceptible level at very low supersaturations. Screw dislocations (50) that often grow perpendicular to the basal plane of the crystal have an unusually large Burger's vector (equivalent to the lamellar thickness). Growth rates perpendicular on the basal plane of the crystal have not been reported, but a knowledge of this process should provide a better understanding of morphology and mechanistic relationships in high polymers and copolymers as they interrelate to each other and determine properties.

Crystallization from more concentrated solution(s) or from the supercooled molten state produces spherulites that seem to have nucleated similarly, and the high supersaturations encountered lead to polycrystalline spherical arrays. These arrays are comprised of radiating platelets or lamellae (often splayed and branched) in which the polymer chains are also folded to varying degrees, dependent upon polymer molecular weight and experimental conditions. The chains are usually oriented tangentially to their growth direction. Because of this morphology, most spherulites show a negative birefringence since the polarizability along the molecular chain axis is much greater than it is transverse to it, except in polymers with highly polar side groups or in hydrogen bonded polyamides. A variety of textural/optical features are associated with changes in crystallite orientation. The literature (51,52,5a) is replete with details of spherulite morphology, but no unique growth mechanism has been reported. Indeed, one might question its uniqueness in materials science! It is appropriate to mention that many classes of other materials also form spherulites; see for example, sulfur (53) selenium (54), antimony (55), proteins and carbohydrates (56), graphite (57), silicates (58), and inorganic (59) and polar and nonpolar low-molecular-weight organic molecules (60), to cite some examples. Spherulites are almost ubiquitous given suitable growth conditions and their forms are diverse, but it is important to realize that other morphological forms exist.

Polygonized structures are produced when spherulites impinge upon each other, limiting further outward radial growth. Growth rate is extremely temperature-sensitive and it is nucleation-controlled but this has been questioned recently (61). Even so, it is generally agreed that polymer transformations from solutions and melts are nucleation controlled. There has been confusion sometimes over the terms used to describe (a) primary and (b) secondary processes. Nucleation is termed primary for the onset of crystallization; the propagation of lamellae/spherulites (*i.e.*, growth) is named secondary crystallization. Crystallization

[2] Patents are excluded from this article.

that occurs after the radial growth ceases, *i.e.*, "filling-in" or densification behavior in spherulites, is sometimes termed secondary crystallization. However, kineticists often restrict the term primary crystallization to the nucleation step (creation) of the phase transformation process, and secondary is given to that stage in the transformation where significant deviations from Avrami[3] kinetics may take place. Several *in situ* experiments have been made with synchrotron radiation to monitor intricacies of nucleation and growth with time using WAXS and SAXS modes (62). During the secondary stage, densification takes place (63) and improvements in crystallinity have been reported by dilatometry (64). Even so, it must be recognized that the density and perfection of a spherulite decreases as it grows outwards from its center. Changes in crystallite thickness that may occur with annealing are another matter that is still unsettled.

The temperature dependencies of spherulitic and crystal growth rate have been claimed to correspond closely (65) under comparable conditions of undercooling, but there are results to the contrary. In monodisperse poly(oxyethylene) (66) and poly(ethylene)s (67,68), the rate of growth of single crystals in the melt depends on the crystal thickness, determined primarily by the undercooling temperature. Chain ends have been reported to reside sometimes inside (67) but more often outside (68, 68a) the crystalline core of low-molecular-weight poly(ethylene)s below 400 CH_2's per molecule[4].

2. General Principles and Techniques Involved in Crystallization Rate Studies

2.1. Kinetics of Bulk Transformations Whenever a polymer crystallizes, the extent of the phase transformation depends upon the crystallizing species and the experimental conditions. High molecular weight polymers do not crystallize completely (9,69) because of topological constraints that lower crystallinity considerably. The classical isothermal transformation kinetics, initially formulated by Kolmogorov (70) and Göler et al. (71) were extended later by the Avrami theory that was initially formulated for metals and later modified, for example, by Evans (72) and others, for polymers. The Avrami (73) equation

$$\theta = \exp(-kt^n) \qquad \text{(A1)}$$

relates the fraction or amount of uncrystallized material that remains after time t, during an isothermal transformation to its growth rate parameter, k, and nucleation parameter, n. As formulated by Avrami, crystallization was presumed to take place under isovolume conditions.[5] Modifications have been made to equation (A1) to correct for incomplete crystallization (74). The mode of nucleation, n, is an integer that varies from 1 to 4, reflecting the geometry and dimensionality of the growth process, under ideal conditions. The different forms are summarized in the following table.

AVRAMI EXPONENT AND MORPHOLOGY

Exponent	Nucleation Type	Crystallite Shape	
1	Instantaneous	Rod	Constant
2	Instantaneous Homogeneous	Discoid/ Rod	Constant
3	Instantaneous Homogeneous	Sphere/ Discoid	Constant
4	Homogeneous	Sphere	Constant

There are also classical growth conditions where the diffusion limited rate proceeds as $t^{1/2}$ and n exhibits 1/2, $1\frac{1}{2}$ and $2\frac{1}{2}$ values. These examples have been addressed in detail in other texts (31). Still, it is worthwhile pointing out that other terminology has been used to describe the mode of nucleation. For example, $n=3$ may be termed sporadic or predetermined in time where the form of growth is disklike (2D) or spherulitic, respectively. Anyhow, it happens that in polymer phase transformation the growth velocity proceeds linearly with time. Now that mesophase crystallization is receiving more attention (75), rod-like morphologies with $n=1$ resemble those found in well-oriented fibrous homopolymer systems (76). Some non-integral values of n correspond to heterogeneous transformations for $n=2$ and 3. Other values of n determined in kinetic experiments may arise from many factors connected with the morphological complexity of the polymeric system. Computer simulations (77–79) have been conducted to model the bulk phase transformation, but the analysis is seldom free of ambiguity for many reasons, except in the early stages of crystallization. Note that in a recent paper, Levine et al. (80) have pointed out that the Avrami–Kolmogorov isothermal growth equation should not be applied to nonisothermal transformations without proper concern for the microstructural features that are inherent in the sample itself. Again, they claim that measurements employing small samples are subject to error whenever anisotropicity features in this mean field theory. They obtained a modified equation from extensive computer simulations. Polymeric materials are likely to be subjected to these criticisms though they were mainly concerned with metallic and related systems. Polymers exhibit more serious side-effects[6] such as mixed types of nucleation (that causes more than one type of spherulite morphology), nucleation on impurities, severe volume shrinkage upon crystallization and variations in spherulite

[3] The basic concepts in this model are hardly commensurate with the known crystallization behavior of polymers; see for example, Ref. 5, p. 250.

[4] The most recent *in situ* synchrotron studies made with monodisperse *n*-alkanes ($C_{246}H_{494}$) have indicated that once folded chains have considerable ciliated disorder as formed at first, but they undergo a slower improvement in crystallinity as chains equilibrate (adjust) to twice folded lamellae of much higher crystallinity (68a).

[5] In polymer crystallization experiments this situation is never realized, but large single polymer crystals accompanied by small volume changes, have been produced in some solid state polymerizations. So far, kinetic measurements have not been made on these systems.

[6] These are intrinsic factors that are known to influence the Avrami exponent.

density (crystallinity) within and between the spherulite(s) all of which may contribute to nonintegral values of n. In view of these complexities, it is surprising that Avrami n values are taken seriously in practice, except for making comparative rate studies. It is even claimed (81) that the mechanical properties of spherulitic samples may be related to internal morphological defects, such as tie-molecules, and even voids created between confined spherulites during melt crystallization. Laser microbeam and X-ray microdiffraction has also been used to examine the internal morphology of spherulites (82,83). The kinetics of the embryonic nuclei and subsequent stages of spherulitic developments (84,85) from a sheaf-like morphology to a spherical entity (spherulite) have been studied using low-angle light scattering. Of course, electron microscopy (86,87) has recorded more detailed morphological changes at higher resolution, and atomic force microscopy (AFM) investigations have probed even further in this respect (88,89,89a).

In principle, any technique that can follow unambiguously the amorphous → crystalline transformation with time, under isothermal or non-isothermal conditions, can be employed as a bonafide method to detect and monitor the crystallization rate(s) that occur. Many experimental techniques exist, such as calorimetry (90), dilatometry (91), density balance (92) and density gradient (93), light depolarization (or transmission) technique (94,95) and/or small-angle light scattering (96) (for average spherulite growth rate measurements), infrared spectra (97), X-ray (98), nuclear magnetic resonance (99,102), electrical resistivity (100), and small-angle X-ray scattering (101), neutron scattering (103), atomic force microscopy (103a) and more. The Avrami θ may involve any property parameters, P_i's, that are relevant in the transformation time domain. In this respect, specific volume changes are related to θ by the equation

$$\theta = (v_\infty - v_t)/(v_\infty - v_a) \qquad \text{(A2)}$$

Here v_a, v_t, and v_∞ correspond to the initial, intermediate, and final stages of the isothermal crystallization (71,104) and the other P_i's may be similarly incorporated. Wood and Bekkedahl (104) were the first to show that the overall isothermal transformation rate of rubber expressed as $(t_{1/2})^{-1}$ passes through a maximum as its crystallization temperature was lowered. It was later shown experimentally and theoretically that the spherulitic growth rate also proceeded similarly, as related by Eq. (A3). Simply expressed (105), k may be written as in Eq. (A1):

$$k = f(\dot{N}G^\alpha) \qquad \text{(A3)}$$

where α is a geometric parameter that is three for spherulites, and it may take on other values for other geometrical shapes. Another formulation (106) obtained experimentally, at constant crystallization temperature, is

$$k = f[\dot{N}(MW,t)G^\alpha(MW,t)] \qquad \text{(A4)}$$

since it was observed that nucleation rate, $\dot{N}$, and spherulitic growth rate, G, are each dependent on molecular weight, molecular weight distribution and time.[7]

The type of property, P, being monitored (107) depends upon the amorphous contribution, P_a, and crystalline portion, P_c, where each phase can be expressed as a weight fraction or a volume fraction in the simple two-phase model expressed as

$$P = xP_c + (1 - x)P_a \qquad \text{(A5)}$$

If P is specific volume then a weight fraction $x_c(w)$ is obtained, whereas a volume fraction $x_c(v)$ results whenever the sensing is made (for example) by X-ray or dilatometry. The relationship then becomes

$$\frac{1 - x_c(w)}{x_c(w)} = \left(\frac{\rho_a}{\rho_c}\right)\frac{1 - x_c(v)}{x_c(v)} \qquad \text{(A6)}$$

where ρ_a and ρ_c represent the amorphous and crystalline densities, respectively. This model is an oversimplification (109) in that it does not allow for the increasing interdependence of the properties of each phase with changes in crystallinity. Other complications also arise because (i) crystallinity is molecular weight dependent (9,45,64) and (ii) for oriented systems, accurate determinations of crystallinity can be difficult (110,111). It strongly depends upon morphology.

Bulk crystallization rates are of some practical value in a relative sense (112) and depend markedly on the concentration of impurity nuclei (of adventitious origin or from additives or nucleants introduced deliberately for property or process control). No satisfactory way (other than counting primary nuclei, *i.e.*, spherulites) exists to quantify their concentration. Generally, spherulitic growth rate (after nucleation) is independent of how the spherulite originated (94,113) (*i.e.*, whether the nucleation is homogeneous, heterogeneous, or pseudo-homogeneous). On the other hand, it must also be recognized that bulk transformations can be greatly influenced by impurities.

Adequate fusion conditions and sample purification (94,114) must always be established for best results in a given system to suppress or eliminate adventitious heterogeneous nuclei in crystallization. Homogeneous nucleation is seldom encountered at small undercoolings (115); although it may be mimicked in the earlier stages of crystallization (116). Because of these complications, it is always advisable to combine complementary techniques; for example, optical microscope studies with dilatometric or other rate measurement. Reliable nucleation rates may only be deduced from independent measurements of

[7]On this account bulk rate measurements are intrinically controlled by two convoluted independent factors or parameters. Only a unique insight into interfacial growth mechanisms is obtained from spherulitic or lamellar ("single crystal-like") interfacial tranformations *per se* from the polymer melt or solution growth respectively, and not from a composite of G and N contained in (4) above.

(i) spherulitic growth rate and (ii) nucleation counts made over relatively long transformation times.

Investigations of liquid→solid phase transformations with SAXS and WAXS techniques now employ rotating anode X-ray (117) or the more intense synchrotron radiation sources for real time kinetic measurements. Normally, good policy to check samples for radiation damage after each experiment. Even so, it has been reported that the development of two or more crystal modifications may be identified and monitored *in situ* by synchrotron WAXS and SAXS investigations (118,119). Dilatometry, and/or DSC measurements for example, may be usefully combined with SAXS investigations where the absolute integrated scattering intensity Q was expressed (21) as

$$Q = K\phi(1-\phi)(\rho_c - \rho_a)^2 \tag{A7}$$

and later modified to read:

$$Q = Kv_s\phi(1-\phi)(\rho_c - \rho_a)^2 \tag{A8}$$

In this equation K is an instrumental constant, ρ_c and ρ_a being the densities of the crystalline and amorphous regions, respectively, ϕ is the volume fraction crystallinity, and v_s is the spherulitic volume. Note that Q increases with crystallinity, reaching a maximum around $\phi = 0.5$ as Sheldon had shown earlier. Rate profiles for poly(tetramethyl-*p*-silphenylene siloxane) [poly(TMPS)] fractions (21) of high and low molecular weights differ in some respects from measurements made by other techniques. Strangely, changes in Q for relatively long isothermal transformation times show no signs of the secondary crystallization often encountered in isothermal dilatometry for the same high molecular weight samples. Long period thickening becomes invariant with the time of crystallization in some polymers, but not in poly(ethylene) (25,117,121). Interestingly, synchrotron SAXS work on poly(β-hydroxybutyrate) (20) has shown that the long period is unaltered with time during isothermal crystallization from the melt even though crystals of this polymer have been found in another study (122) to thicken isothermally upon annealing. Elsewhere (123) in oriented poly(ethylene) the long periods are found to change reversibly between 30°C (217 Å) and 100°C (252 Å). Upon cooling of poly(TMPS) fractions, following isothermal crystallization, under quiescent conditions and reheating crystallized sample(s), a reversal of the long period,that is dependent upon molecular weight is observed (21). Concomitantly, the crystallinity increases upon cooling specimens from the crystallization temperature to room temperature (125). More detailed measurements have been made recently (62) where it was demonstrated that the increase in the degree of crystallinity by WAXS and SAXS cannot be correlated with the long period. Clearly, more investigations of this kind are needed.

2.2. Thermodynamic Crystallization Models Thermodynamic and statistical theories of morphology (124,126) *do not provide rate data* but only important crystalline/amorphous information, *i.e.*, morphological features consistent with thermodynamic quantities predicated to change with crystallization temperature or undercooling. Kinetically, these models hardly merit a place in this article on rate measurements except that they do provide information on the "equilibrium" crystallite size l_c in relation to its melting temperature T_m of the finite crystal, from the general expression of the form

$$\left(\frac{1}{T_m} - \frac{1}{T_m^0}\right) = \frac{R}{\Delta H_u}\left[\frac{2\sigma_e}{l_e R T_m} - \frac{1}{l_e}\ln\left(\frac{x_c - l_e + 1}{x_c}\right)\right] \tag{A9}$$

that may be simplified for very long chains when the last term vanishes. In Eq. (A9), x_c is the molecular chain length, R, the gas constant, and ΔH_u, the ideal enthalpy of fusion per mole of repeat units comprising the crystallites of end interfacial energy, σ_e. $T_m^0 = \Delta H^0/\Delta S^0$ is the thermodynamic temperature of the infinite crystal with enthalpy and entropy, ΔH^0 and ΔS^0, respectively. Theories of melting of polymers have been the subject of much discussion (127) over the years. For example, the reported thermodynamic melting temperatures of poly(ethylene) range from 414 to 419 K and from poly(oxyethylene) from 342 to 348 K are in dispute based upon theory and experiment. At best, T_m^0 is only reasonably well established for a few well-characterized polymers, yet so much morphological, growth kinetics and structural interpretations are based upon T_m^0. Only a fraction ($\leq 70\%$) of the chains are folded into the end surfaces of the crystallite (126) but the quality and even the extent of the re-entry adjacency is still disputed.

The crystal fold surface of poly(ethylene) crystals and interfaces have been studied more than any other polymer (126–130). This dispute is likely to continue until the "crystal" surface morphology in real space (88,89) can be resolved and the corresponding surface(s) chemically analyzed by the newer experimental techniques of the last decade or more (131,132,133). It should be realized that the thermodynamic model, unlike the kinetic one, does not require a chain-folded nucleus, nor does it provide kinetic information.

2.3. Growth Kinetics of Lamellar Structures: Crystals and Spherulites Growth rate depends primarily upon the degree of undercooling rather than on the crystallization temperature, but the influence of chemical and physical structure and molecular weight and molecular weight distribution are also important factors in linear chain polymers. Without impurities, growth occurs isothermally at a constant rate, and the temperature dependence of these rates are typical of a nucleation controlled process (134), whatever the model employed to describe it.

Theories for isothermal growth rate-temperature, $G(T)$, in polymeric and other less complex systems can be formulated simply by

$$G(T) = f(T)D(T) \tag{A10}$$

which is often approximated by

$$G(T) = \frac{f(T)}{\eta(T)} \qquad \text{(A11)}$$

It is frequently assumed that the temperature dependence of the diffusive term $D(T)$ is reciprocally related via the Stokes–Einstein relationship[8] to the zero shear viscosity $\eta(T)$ which is more readily available than $D(T)$. The $f(T)$ term in equations (A10) or (A11) differentiates between the growth geometries introduced in various growth rate models. Different formalisms (1(a),4,15,61,119-121,134-137) invoking chain folding and diverse nucleus/growth geometries have been used to simulate crystallization. One of the most elaborate and widely used kinetic descriptions[9] is the modified multiparameter model of the original Hoffman–Lauritzen theory (1). Another salient model is that to Point (137), who first predicted the correct crystal thickness temperature up to large undercooling, that is accordance with experimental evidence (138–140). Point and others (137), have disputed some of the basic ideas in Hoffman's model, that pertain to a stable nucleus and to surface energy and other related features connected with growth Regimes that are now postulated to describe interface roughness and interfacial energy assigned to three stages of undercooling (see below in text). Point's test of Regime theory (141) questions its validity, though there is much experimental evidence (142) *pro and con* for "transitional breaks" in growth rate with undercooling. The matter is still unsettled.

The Hoffman–Lauritzen spherulitic growth rate equation (A1) is similar in principle to the Becker–Döring model (except for the complexions that are invoked to incorporate chain folding). The basic equation can be represented formally by

$$G = G_0 \exp\left(-\frac{\Delta E^*}{RT}\right) \exp\left(-\frac{\Delta G^*}{RT}\right) \qquad \text{(A12)}$$

In which G_0 is a constant for a given molecular weight, ΔE^* is the activation barrier to molecular chain transport, and ΔG^* is the free energy barrier to nucleation.[10]

However, the L–H model differs in several important macromolecular details that distinguish it from small molecule morphology. Following the *ad hoc* assumption that the first exponential in the crystallization equation (A12) may be approximated by a Vogel or modified[11] Williams–Landel–Ferry (WLF) type of term,[12], Eq. (A12) becomes

$$G = G_0 \exp\left[-\frac{A}{R(T - T_0)}\right] \exp\left[-\frac{\Delta G^*}{RT}\right] \qquad \text{(A13)}$$

Here T_0 lies at least 30° below the glass transition temperature T_g, at a temperature where growth enabling motions vanish. A is an activation energy for such segmental transport, usually ~ 5 kcal/mol. The nucleation term is often expressed as

$$\exp\left[-\frac{BT_m^0}{T\Delta T}\right] \qquad \text{(A14)}$$

Here the undercooling $\Delta T = (T_m^0 - T)$ and T_m^0 is the melting temperature of the perfect polymer crystal, most often obtained by extrapolation. The term ΔT^{-1} implies two-dimensional surface nucleation (growth) but it has been claimed (143) that a ΔT^{-2} dependence provides an equally valid fit to experimental growth rate data. The geometric parameter B may be expressed as

$$B = nb_0\sigma\sigma_e/f\Delta h_f k \qquad \text{(A15)}$$

where b_0 is the inter planar distance in the growth direction, σ and σ_e are the lateral and end surface (interfacial) free energy of the growing crystal or lamella[13], k is Boltzmann's constant, and Δh_f is the heat of fusion of the crystal. The adjustable parameter $f(0 \leq f \leq 1)$ accounts for the temperature dependence of the heat of fusion away from the melting point. Another fitting parameter ψ, $(0 \leq \psi \leq 1)$ is an "apportionment factor" of the bulk free energy associated with nucleation and growth away from the melting point. This has been troublesome in the past, amounting to serious discrepancies between theoretical and experimental values of lamellar thickness, l, with undercooling and it was referred to (by Hoffman et al.) as the "δl catastrophe" (1). Now, in a recent paper by Snyder et al., (146), claim to have resolved this problem. This integer n, (is not to be confused with the Avrami, n, in equation (1)). The Hoffman parameter, n, is both morphology and growth rate related and is claimed to coincide with nucleation Regime changes at the growth front with undercooling in some polymers

[8] Only valid for spherical molecules, but it has been used indiscriminately.

[9] Questions have been posed (61) that challenge the mechanistic nature of the nucleation processes *per se*. Recent AFM work has shown (68a) that growing polymer lamellae at the melt-crystal interface exhibited sporadic behavior with time, suggesting that stress may be a significant influence. Interestingly, reports on asymmetric aromatic hydrocarbons (60) have cited and demonstrated stress related growth behavior at faceted spherulitic interfaces. In this paper it was stated that this could complicate growth rate measurements unless proper care was exercised. The implications of this commonality in the behavior of polymers and small molecules may lead to further interesting investigations into materials, generally.

[10] Recently it was reported (144a) that the β-relaxation process is responsible for nucleation from the quenched state but this seems to be irrevalent to isothermal crystalliztion that takes place directly from the supercooled molten state well above the glass transition temperature. Crystallization from the supercooled melt and from glassy polymers (see Ref.5, p.258 and elsewhere) and small molecules were investigated (60) in some detail a long time ago and explained along conventional lines by the writer.

[11] The writer (60,144) pointed out decades ago that it differed in magnitude, from the macroscopic viscosity (or fluidity) temperature dependence by fitting experimental data for asymmetric aromatic hydrocarbons.

[12] Just recently (145), further considerations of Eq. (A13) concludes that a better "fit" to transport data is obtained by compromising the C_1 and C_2 parameters in the classical WLF type formulation for transport in the melt.

[13] This geometry is consistent with many polymer morphological habits.

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(1,147,148). It alters for a smooth interface ($n = 1$) to a rough one ($n = 3$); $n = 2$ corresponds to an intermediate morphology. Several "transitions" in growth rates (149–152) are claimed to be consistent with Hoffman's predictions. Before this theory, "breaks" in the growth rate with undercooling were assigned to morphological changes, but now it is accepted by many investigators though the entire issue is still under discussion.

The concept of reptation[14] (7,152(a)) is another process that has been claimed facilitate chain folding[15] of lamellae, and to circumvent chain entanglements at the melt-crystal interface during crystallization. Experimental evidence (see for example 30, 45,69,144,153–155) questions its validity in interfacial growth kinetics, *whenever spherulitic growth over a wide span of molecular weight fractions is investigated.* For example, the dependence of the interfacial surface energy of crystallites on molecular weight is contrary to expectations if reptation was indeed functional. Clearly, it does not override topological constraints and multiple nucleation events that suppress chain folding and seriously diminish sample crystallinity in poly(TMPS) and polyethylene – see for example (Ref. 155). It is well known that low molecular weight crystalline fractions are mostly chain folded and brittle because of the lack of tie-molecules. On the other hand high molecular weight fractions are tough due to the existence of disordered regions as well as the presence of tie molecules. Hikosaka (156) has proposed another kinetic model for crystallization that simultaneously deals with lateral lamellar growth (chain-folding) coupled with crystal thickening for extended chain growth. Hikosaka and co-workers (157) believe this theory accounts for pressure crystallized polymers (158) and chain extended polymeric mesophases (159). Now, an all-encompassing crystallization scenario involving both of these features is claimed to be relevant in all polymer transformation kinetics (157). Although interesting, this model is open to question because other investigations dealing with *in situ* real time monitoring of "solid" phase transformations (1162) points out that melting *per se* precedes chain extension.

Another model termed Molecular Nucleation has been emphasized by Cheng and Wunderlich (160a) as a significant factor in the crystallization of PEO fractions and mixtures but it is not readily amenable to experimental testing yet, although it has received favorable mention from Point (159a). *So far, all theories of polymer crystallization and morphological models formulated, do not have à priori predictive capability whether they invoke chain folding implicitly or not.* The formation of lamellar structures is amply backed by electron and X-ray diffraction evidence (4,47,160) but real surface details are still lacking. Studies of well-defined oligomers of poly (ethylene) (67,68) and poly(oxy-ethylene)'s (66) support chain folding in short-chain polymers crystallized from the melt, yet the latest WAXS and SAXS measurements indicate that there are conflicting details too (68a,160).At this stage, it appears that the evidence is insufficient to take a polarized stance on this problem bearing in mind the balance of existing evidence and not opinion.

[14] A recent review (45b) still insists on this process even though it is unable to facilitate chain folding in high molecular weight polymers, nor is it required to account for crystallization behavior below the entanglement molar mass, M_c.

[15] If folding is a consequence of kinetics, it is an inherent surface defect that compromises the attainment of high degrees of crystallinity even under the most favorable circumstances. Multiple nucleation and topological contraints worsens this flawed situation even more as physical measurements have shown (eg. see situation summary in Ref. 155). Another publication is in press.

3. Other Significant Aspects of Crystallization

In the preceding sections, some of the following items have been mentioned, but the writer believes that they deserve more attention as many of these areas have grown since the publication of the third edition of this Handbook. For example, there have been developments in:

(I) Non-isothermal crystallization
(II) Crystallization of blends
(III) Solvent and pressure induced crystallization
(IV) Mesophase crystallization
(V) Flow induced crystallization
(VI) Epitaxial crystallization
(VII) Orientational crystallization from the melts and solutions
(VIII) Crystallization of copolymers
(IX) Computer simulations

A brief foreword is given by way of introduction in these important areas. Rate measurements in all of these fields except for epitaxial crystallization appears in the tabulated data in this chapter. In some respect, the growth experienced on a given topic is connected with practical developments and manufacturing and that often provides the opportunities for further investigations.

3.1. Nonisothermal Crystallization This important aspect of polymer crystallization is responsible for the morphology and properties of processed polymers that are crystallizable (ie. polymers, blends, mesophases or any combination of these ingredients in a plastic product) to whatever extent. in the developments. There are many complex events responsible for polymer properties. Several approaches have been used to gain a better representation and understanding of non-isothermal crystallization processes, since the quality of manufactured plastics goods depends upon it. Processing induced structures are never homogeneous even when the materials are thin films or fibers. Because of their complexity, years ago fundamental knowledge was gained through experiments, but now computer modeling has increased in popularity. Several approaches to this problem exist. The techniques of Ozawa (162) Nakamura et al. (163) are still popular. In essence, they have been usually modeled as an extension of

Avrami theory

$$\theta(t) = \sum_{j=1}^{k} \theta_j(t) \qquad \text{(A16)}$$

where the phase change over the crystallization range is considered to be comprised of many incremental isothermal parts so that where $j = 1$ to k involves all the types of nuclei (i.e., homogeneous, heterogeneous, pseudo homogeneous, thermal, athermal and so on). The nucleation rates $\dot{N}(t)$ and growth rates $G(t)$ has been patterned as

$$\dot{N}(t) = \dot{N}[f(t)] \quad \text{and} \quad G(t) = G[f(t)] \qquad \text{(A17)}$$

Simplifying assumptions lead to a more general equation. After some manipulations, it becomes:

$$\log\{-\ln(1 - x(t)\} = \log k(T) - n \log \beta \qquad \text{(A18)}$$

where $x(t)$ is the fraction of material transformed at T and β is the cooling rate, $k(T)$ is the cooling rate function and n is the Avrami exponent. From plots of $\log\{-\ln[1 - x(t)]\}$ vs. $\log \beta$ a straight line is obtained if Osawa's method is valid. From this plot, $k(T)$ and n respectively can be obtained.

Harnisch and Muschik (164) derived the Avrami equation from non-isothermal crystallization thermograms where

$$x(t) = 1 - \exp(-k_n t^n) \qquad \text{(A19)}$$

A plot of $\log\{-ln[1 - x(t)]\}$ vs. $\log t$ for each cooling rate is linear, providing the adjustable parameter k_n and n that are needed. Two parameters are different from the isothermal values. Besides this, it is important to realize that the heat of fusion of the 100% crystalline polymer is needed for the data analysis.

Ziabicki (165) developed a procedure where he assumed that crystallization in polymers may be represented by first order kinetics, i.e.,

$$\frac{dx}{dt} = (1 - x)kT \qquad \text{(A20)}$$

wherein overlap of growing moieties is neglected and $k(T)$ is a temperature dependent rate constant. Plots of rate constant vs. T is Gaussian-like; so that the kinetic crystallizability C, may be expressed in terms of D, the width of the crystallization exotherm at half-height and k_{max} equals k at the top of the exotherm

$$C = \int_{T_g}^{T_m} k(T)dT = (\pi/\ln 2)^{1/2} k_{max} D/2 \qquad \text{(A21)}$$

Depending upon the polymer and its crystallizability (fast vs. slow kinetically), C varies more than three or more orders of magnitude (polymer to polymer) but kinetic parameters obtained from Ziabicki's method do not always compare well with other methods. Except for tests on blended systems in recent literature (164) others were reasonable. A generalized theory by Ziabicki (165), containing the thermodynamics and kinetics of phase transitions, while sound in concepts, appears complex and not readily amenable to testing.

Whenever polymer melts or solutions are sheared (166) the nucleation rate can often be drastically enhanced so that the overall transformation rate becomes accelerated, if the conditions are conducive to crystallization (167,168). If the external cooling is sufficiently rapid, the level of crystallinity may be compromised for instance in molded parts that may subsequently crystallize slowly (ie. age in storage) causing serious changes in dimensions that could cause distortion in the workpiece. Moisture conditioning of polyamides, for example, is a well practised industrial procedure that is used to stabilize fibers "on-line" and as processed molded parts. Network systems undergo non-isothermal crystallization or isothermal crystallization when they are oriented uniaxially (169) or biaxially (170). Some of them have been analyzed from the viewpoint of kinetics and/or thermodynamics as further alluded to in the next section. Non-isothermal crystallization at varying cooling rates is representative of actual polymer processing situations. Integral methods have been developed (171) for simulating the material fraction of transformed material under arbitrary cooling rates but the widespread use of Avrami theory has been criticized (see Computer Modeling section).

In industrial processes rapid sample cooling and monitoring has been a problem. Years ago, the writer introduced the light depolarization intensity method (LDI) to follow rapid changes taking place "on the spin line" and in laboratory tests (94). This procedure is now widely used and has just been further developed (95) to cope with average cooling rates as high as 5000°C/min. for the investigation of non isothermal crystallization generally (i.e., with constant and varying cooling rates) as Spruiell and coworkers have demonstrated for polyolefins. Over the years, Janeschitz-Kriegl and coworkers (see for example, (38)) have investigated the theoretical and experimental difficulties associated with transport and transformation processes encountered in polymer crystallization. In recent papers, they have detailed their findings.

3.2. Crystallization of Blends The first polymer blend was made from two polymeric rubbers in 1846, but polymer blend technology and a scientific understanding of the underlying principles controlling the compatibility (or lack of) in polymer mixtures (alloys as they have been named recently) has taken place only in the latter part of the current century. Many blends are non-crystalline but our interest in this document is focused on the kinetics of phase transformations of binary and ternary systems that receives more attention annually. Some of these systems can be very complicated, often comprised of multiple phases that may involve homopolymers, copolymers, mesophases and the like. Polymorphism and even isomorphism may occur

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because of inherent chemical interactions, imposed temperature and pressure conditions and so on. Despite these complexities, blends comprise a significant part of industrial production where synergism among materials is often exploited through various processing techniques.

The lattice theory of Flory–Huggins captures the essential features of mixing and demixing and the magnitudes of their corresponding entropic and enthalpic changes have provided a basis of understanding for some fifty years ago (172). Since then, experimental practice and more subtle theoretical developments, including computer simulations, have advanced our understanding of homopolymer and copolymer blends (see for example, Binder (173)). With mesophases the situation becomes more complex but it is still beneficial in practical ways. On the technological side, Utracki (174,175) has provided informative overviews of the processing science and technology of blends or alloys in detailed accounts of processing-morphology-property relationships.

Crystallization always involves phase changes and often insufficient attention is given to transformation rates that may correlate strongly with structural and morphological details. For polymer mixtures/blends and copolymers with one or more random components, certain generalizations may be made when they are crystallizable. For instance in bulk crystallization, the Avrami or a related theory has often been employed to analyze bulk kinetics in blends to demonstrate the strong effect of molecular weight and composition on transformations rates. There is an analogy here with homopolymers in that crystallization rate is usually suppressed by the non-crystallizable component(s) in the system. Consequently transformation curves are moved to lower temperatures, although their transformation isothermals are seldom superposable (as they are in most homopolymer transformations). Spherulitic growth rates are usually attenuated too and shifted toward lower temperatures as the results compiled in this article show. The crystallization of blends has been modeled by thermodynamic or kinetic type equations employing computer simulations.

3.3. Solvent and Pressure-Induced Crystallization Diluent or solvent in a polymer system, alters its crystallizability and often its crystallinity is enhanced. Solvent-induced morphological changes (176) may occur and mechanical properties and deform ability including aging behavior (177) of systems can be affected so that the topic is of considerable importance technologically. Plasticizers (178), nucleants (179), and chain length (180) may attenuate or sometimes accelerate the phase transformation kinetics. Several types of investigations and analysis have been employed. Mandelkern and others have adopted a thermodynamic approach where the focus is on the overall Gibbs free energy for nucleus formation, while crystallization kinetics has been investigated by using the Avrami theory as a basis for detecting mechanistic changes with time, temperature, and diluent concentration. For spherulitic growth kinetics the effect of diluent on morphology and crystallization kinetics has been formulated (74) according to

$$G = G_0' \exp\left\{\frac{-A'}{R(T - T_0)}\right\} \exp\left(\frac{-\Delta G^* - \Delta G_s^*}{RT}\right) \quad \text{(A22)}$$

where ΔG_s^* is the free energy contribution from the solvent (diluent), the other terms having been defined before. Equation (A22) may be modified to take care of crystal geometry and polymer diluent interaction. These effects are very marked in many polymers, especially where plasticizers that modify physical behavior are involved, as for instance moisture in polyamides. In these circumstances the defining parameters A', G_0', T_0' are altered (usually decreased in magnitude compared with homopolymers). In non-compatible polymer blends (181) this is often a rare event, and changes in morphology and crystallization behavior have not always been documented even when rates have been measured. Copolymer systems are being studied more extensively (181), and spherulitic growth rates (181,182) are receiving some attention as later tabulated results illustrate.

Pressure induced crystallization has been documented for a few systems (183), poly(ethylenes) (31,46,184) in especially being the most extensively investigated materials from a morphological and properties viewpoint. Conformational constraints are responsible for gauche bond formation in polyethylene, for example, lead to the formation of the higher entropy hexagonal phase as a morphological high-pressure intermediary that is essential to the overall thermodynamic picture of its melting temperature-pressure phase diagram. Other kinetics data (185–190) are also listed in this article. Pressure is a significant processing parameter. In the past, experimentation was the "yard stick" rather than theory for optimization of polymer processing conditions. Crystallization kinetics made by dilatometry at small undercoolings (190a) has moved the focus to the process rather than on the product itself for nylon 66, polypropylene and poly(ethylene terephthalate). According to Zoller (190a), the Malkin modification of the Avrami equation best describes the kinetic data in this investigation. Other systems such as polypropylene behave likewise, and PVT modeling studies have been published (190b,190c).

3.4. Crystallization of Mesophases Although polymeric mesophases were discovered more than one century ago (191), their exploitation has only been developed within the recent decades. Synthesis, properties and applications have been the focus of most studies because of the technological potential of these systems. Now that it is known that mesophases form supermolecular structures through self-organization, this discovery has lead researchers to investigate the criteria underlying liquid crystalline formation. The morphology and structure of these polymers have been the subject of theories (14,192,193) and experiments (194). Molecular design *vis a vis* properties (somewhat analogous to activities in drug chemistry/

industry) has become an attractive research topic. The molecular requirements for liquid crystallinity have been featured in papers and reviews (195) that have also covered lyotropic and thermotropic phases. Structure in these types of phases have sometimes been tailored to ease processing and/or to facilitate or even to reduce rates of crystallization. Blending has been tackled through processing and modeling, including phase separation dynamics (196); yet crystallization kinetics, despite its importance, has had a low profile until comparatively recently, kinetic measurements on mesophases are still sparse. Most results known to the writer, are tabulated in this article. Rate data have been analyzed along the conventional lines normally used for homopolymers in Section B. Insufficient research has been devoted to the development of a new theory and to examine the influence of crystallization rate on morphology-processing-property relationships; viscosity and diffusion considerations notwithstanding.

Crystallization of some thermotropic copolymers cannot be explained on the Flory model (192) through homopolymer segregation. However, Windle and coworkers (197), by means of computer modeling have demonstrated that 3D ordered regions, described as non-periodic layers (NPL) are possible. Structurally, NPL crystallites agree with experimental X-ray observations even when random intramolecular statistical sequencing exists. This ongoing project is of interest as the NPL model may also be explained by a paracrystalline lattice (PCL) model, that does not require explicit sequence matching (198).

3.5. Flow-Induced Crystallization This topic covers polymer solutions and melts (199,200) generally and may be further categorized into

(a) flow induced Poiseuille flow (201),
(b) elongational flow (199),
(c) shear induced transformations (202,203),

and there are others.

Polymers, other than homopolymers (i.e., blends, copolymers, mesophases) have been listed according to processing characteristics. Rates of crystallization in these categories are sparse compared with synthesis, structure, morphology, properties, processing and even computer simulation (the latter being a rapidly growing field), with some information on the factors/forces that modify behavior. An exception is the work of McHugh and coworkers (204,205) who have reported the strain induced crystallization rates of swollen poly(ethylene) networks by using dynamometry coupled with *in situ* fiber video-imaging with birefringence measurements in order to obtain one dimensional heterogeneous growth rates. However, the deformation rates encountered in these experiments are much lower than the values encountered in commercial fiber melt spinning and molding processes.Still, modeling is now underway to link transient deformation histories with flow induced shear results (205). These authors have just developed a continuum model for flow induced crystallization that embodies strain induced crystallization by employing an irreversible thermodynamic formalism based upon principles set forth by Beris and Edwards (206) for describing non equilibrium irreversible thermodynamics of flowing systems.

The Hoffman has developed a kinetic model for fibrous crystal growth from solutions (207). It is patterned after the well established pioneering solution spinning "shish-kebab" textures, and properties by Pennings and coworkers (see for example 208,208a) and others (208b) especially for UHMWPE and their blends. In this model, "shish's" comprise a well-oriented central core fibrils ($\sim$15–50 nm) flanked by chain-folded "kebab" overgrowths off the core. The "shish" cores are interconnected by highly strained amorphous regions. They (core regions) contain many defects. Here again, the paucity of experimental rate data hampers testing of the theoretical model.

3.6. Epitaxial Crystallization This aspect of crystallization deserves to be mentioned. Polymer–polymer epitaxy (209–210) involving mutual interactions between different macromolecules where crystallinity, crystal structure, lattice matching, molecular conformation and chemistry have been considered. The kinetics of epitaxial processes have been largely ignored despite their plausible significance in surface bonding that has practical implications.

Recent experiments (211) have shown that the lamellar thickness of poly(ethylene), PE, on oriented isotactic poly(propylene), changes from 16 to 39 nm upon annealing. This thickening is attributed to dimensional matching and secondary nucleation with the requirement (212) that the PE crystal(s) are thinner than the ^{i} PP dimension along the chain-row matching direction. However, no detailed kinetic measurements were reported on the rate of epitaxial growth here, or elsewhere in the literature.

3.7. Orientational Crystallization Strain, shear, and elongational flow under isothermal or nonisothermal conditions promote crystallization. The problem is of long-standing commercial importance in fiber manufacturing, injection molding (213) and other important processes. On the academic side, the problem has been recognized for a long time, but it has been studied only sporadically until now (203). Although published data are still sparse, useful behavioral trends have been established.

Crystallization of *cis*-1,4-polybutadiene between rollers during shearing has been modeled (214) using a novel procedure in which the influences of shear flow on lateral heat conduction is approximately avoided. This technique permits the crystalline layer build-up to be determined as a function of time, temperature, and shear rates in on-line processing. The morphology created under these conditions was studied. This is not an easy "kinetic" problem to handle theoretically.

Crystallization in polymer networks has received some attention (169) since Flory proposed the basic principles many years ago (215). Gaylord (216) modified this theory

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for finite chains while others studied and determined the interrelation between elongation, melting, and crystallinity. Calculated strain dependence of the melting temperatures (max), average crystallite sizes and crystallinity extension ratios, $\lambda's$, as high as four for crosslinked rubber networks have been correlated, but rates of crystallization were not included in this kind of work that is further addressed by Negahban in Section IX.

3.8. Crystallization of Copolymers In this category, the focus has been on morphology and properties though Flory (215) developed an equilibrium thermodynamic description of the semi crystalline state of these materials almost half a century ago. Fusion of copolymers is expected to occur over a wider range of temperature in contrast to homopolymers. Random and block copolymers have been studied and melting theories are widely used to describe them. Whenever one species of a copolymer is crystallizable and is dependent upon its sequence probability, p, and not on composition, its melting behavior has been described[16]. The technological importance of copolymers has developed over the years, but kinetic descriptions of the phase changes observed (often multiple) have been mostly patterned after models for blends that are analogously related to equation (A22) whenever crystallization proceeds according to a spherulitic morphology. The non-crystallizable component(s) affects crystallization parameters in accordance with their degree of compatibility and interfacial transport that often constrains and usually suppresses growth rate. Even so, the morphological features that determine the overall crystallization kinetics and properties do change with molecular weight and co-unit content and rates have been monitored and analyzed mostly according to procedures outlined in Section A. Morphology varies with molecular weight akin to the behavior encountered in homopolymers. Avrami type transformations and spherulitic growth rates tabulated in this article indicate that the time scale of the transformation is strongly influenced here (more so than it is in blends of the same composition) for species with similar compositions. Other copolymers behave similarly even though the range of molecular weights for these random copolymers of ethylene is not extensive (216,217). Other kinetic theories of copolymer crystallization have been formulated (218) and tested. Modeling of copolymers with short crystallizable sequences using a Monte Carlo approach (219) has been aimed to reveal the 3D morphology of A-B type crystallites such as ethylene-propylene with high propylene content and low crystallinity. This work shows promise. Goldbeck-Wood (220,221) has written an informative overview of computer simulations where the focus has been on molecular dynamics and Monte Carlo simulations pertaining to classical lamellar crystallization and 3D models. The reader should consult the original texts for details.

3.9. Computer Simulations On this topic, there have been a number of developments that have a kinetics basis, apparently connected with assumptions used in the Avrami theory that only provides analytic solutions. In simulations, the approach has been used to model bulk transformations for:

(a) volume shrinkage, non-random nucleation and crystallinity variability (222),
(b) spherulite impingement in two dimensions on SALS patterns (223),
(c) the morphology of spherulites (thermally and athermally nucleated) (224) and the implications of various nucleation modes, as well as density changes and size distribution effects,
(d) isothermal kinetics in thin films, including the influence of surface nucleation on morphology (225,226), and
(e) simulation of crystallization in polymeric composites in two dimensions with some representation of the three dimensional situation (227), and for several other criteria too.

Simulation of phase transformations in these examples have some predictive capability. For composite systems, the Avrami model as such, is claimed not to apply because nucleation is constrained by fiber loading and surface sites. For instance, factors such as fiber fraction and diameter, spherulitic growth (normal and transcrystalline) must all figure in the overall composite morphology.

Another different, but recent approach by Negahban and coworkers (228–230) to crystallization kinetics involves modeling:

(a) thermo-mechanical effects that occur during the phase transformation in "simple" homogeneous processes,
(b) simulation of the thermo-mechanical effects of crystallization in non-homogeneous processes and,
(c) developments of models for the prediction and verification of material responses.

There are other criticisms of the Avrami method here. Negahban points out that modeling of crystallization kinetics using Avrami-type equations "is incompatible with the entropy production inequality" (i.e., Clausius Duhem inequality) that is basic to the termination of crystallization in real situations. Much of his work has concentrated on the crystallization of rubber (literature results) with all its ancillary effects/property-wise material functions that are amenable to simulation. In the same manner, the kinetic theory of crystallization, based upon concepts of molecular chain-folding and it's related parameters,at the same time recognizing some of the shortcomings involved. An attempt has been made to keep

[16] Goldbeck-Wood [Polymer **33**, 778 (1992)] has developed a new melting point equation that is comparable with Eq. (A23) (see below) for copolymer melting point depression, but it predicts a much steeper dependence on composition than the Flory model which states

$$\frac{1}{T_m} - \frac{1}{T_m^0} = -\frac{R}{\Delta H_u} \ln p \qquad \text{(A23)}$$

where p is the sequence probalility [P. J. Flory, Trans. Farad Soc., **51**, 848 (1955)].

this section relevant to crystallization kinetics even though modeling/simulation procedures are becoming widely applied now, on account of growing availability of computers. In this regard one of the more recent practical computer simulations that has been developed for injection molding,embodies nonisothermal and stress–induced crystallization kinetics in poly(phenylenesulfide) during processing (232). The authors claim to be able to "predict" the thermomechanical history and crystallinity developments in PPS in reasonable agreement with experiment. Undoubtedly, there will be further developments in this field,because of its technological and economic relevance.

ADDENDUM

This chapter on "Rate of Crystallization of Polymers" is incomplete for several reasons. Sometimes original data were not available. Where the deconvolution of complex graphical kinetic scenarios was unnecessarily tedious and resulted in questionable accuracy, these data were omitted. However, results are listed for rate plots that could be deciphered and digitized satisfactorily. Original data provided directly by authors are included and gratefully acknowledged by the writer.

B. TABLE OF RATES OF CRYSTALLIZATION OF VARIOUS POLYMERS

1. Rates of Crystal Growth

TABLE 1. POLY(ALKENES), POLY(DIENES), POLY(VINYLS)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
1.1. HOMOPOLYMER MELTS					
Poly(ethylene) oligomers C246 (alkane)		118	680	Melt crystallization as thin film; phase contrast hot stage microscopy; $1/t_{0.1}$, transformation rates for C198 follow a similar pattern with T_c	233
		119	638		
		119.6	137		
		120	124		
		120.6	67		
		121.1	138		
		121.5	103		
		122	138		
		122.5	142		
		123.1	159		
		123.5	103		
		124.0	75		
		124.6	67		
		125.1	25		
Poly(ethylene) oligomers C294 (monodisperse)		118	1100	Optical and electron microscopy, melt crystallized banded and and lamellar spherulites	234
		119	840		
		120	960		
		121	530		
		122	260		
		123	170		
		123.5	120		
		124	20		
		124.5	2.1		
		125	2		
		126	4		
		127	4		
		128	2		
		129	2		
Poly(ethylene) oligomers C246		118	4310		234
		119	3000		
		120	2250		
		120.5	2430		
		121	910		
		121.8	30		
		122	60		
		122.5	100		
		123	150		
		124	220		
		125	230		
		126	200		
		127	130		
		128	70		

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TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
C198		118	1330		234
		119	1440		
		120	1330		
		121.8	1140		
		122	1340		
		122.2	1180		
		122.5	970		
		123	1040		
		123.5	1020		
		124	980		
		125	590		
		125.5	500		
Poly(ethylene)		−45	3.2	Crystallization from quenched droplets	235
$M_n = 32000$		−30	13	upon heating; lamellar thickness over	
$M_w = 119000$		−15	20	the same range; $45 \leq l \leq 75$ nm cited;	
		0	800	crystallization measured upon heating	
		10	1000		
		24.5	20000		
		39.5	2000000		
Poly(isoprene) *cis*		−40.2	0.023	Electron microscopy; thin films	238
		−25.1	0.125		
		−18.2	0.138		
		−13.0	0.098		
		0.2	0.019		
		−38.4	0.011	Electron microscopy; thin films	238
		−32.8	0.020		
		−26.3	0.030		
		−15.7	0.024		
		4.2	0.006		
			0.001		
Poly (isoprene) *cis* with 2–2.5% *trans*		−39.2	0.027	Electron microscopy after staining;	186
(α) growth in Hevea		−25.3	0.118	for further details see text	
		−19.4	0.138		
$M_w/M_n = 4.81$		−11.3	0.110		
$M_w/M_n = 16.93$		−0.4	0.027		
		−39.3	0.007		
(α) growth in Natsyn		−34.3	0.014		186
$M_w/M_n = 4.12$		−29.5	0.025		
$M_w/M_n = 10.17$		−25.2	0.034		
		−19.3	0.024		
		−14.5	0.010		
		−9.6	0.014		
		−4.9	0.009		
		0.6	0.003		
(α) growth in Natsyn		−39.2	0.004		
		−34.3	0.007		
		−29.7	0.013		
		−25.3	0.019		
		−19.6	0.016		
		−14.4	0.012		
		−10.1	0.009		
		−5.1	0.005		
cis with 2–2.5% *trans*		−10	0.071	0.0 kbar	186
(α) growth in Hevea			0.206	0.5 kbar	
			0.370	0.7 kbar	
			0.432	1.0 kbar	
			0.233	1.5 kbar	
			0.069	1.7 kbar	
(α) growth in Natsyn		−10	0.017	0.0 kbar	186
			0.021	0.5 kbar	
			0.123	0.7 kbar	
			0.117	1.0 kbar	
			0.051	1.3 kbar	
			0.036	1.5 kbar	

TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
(α) growth in Hevea		0	0.009	0.0 kbar	186
			0.028	0.5 kbar	
			0.156	1.0 kbar	
			0.698	1.5 kbar	
			0.731	1.7 kbar	
			0.124	2.0 kbar	
(α) growth in Natsyn		0	0.003	0.0 kbar	186
			0.006	0.5 kbar	
			0.011	1.0 kbar	
			0.186	1.5 kbar	
			0.141	2.0 kbar	
Poly(isoprene)*cis*		−42	0.021	Electron microscopy of thin films	239
$M_w = 543000$		−36.5	0.049	(~80.0 nm); growth (lamellar)	
		−31	0.068	arrested by OsO_4	
		−27.3	0.113		
		−20.8	0.093		
		−16.1	0.065		
		−10.6	0.036		
		−5.6	0.014		
		−2.5	0.013		
		−0.9	0.011		
		1.4	0.005[5]		
		3.1	0.005[1]		
Poly(isoprene)*cis* (from Hevea rubber)					
$M_w = 313000$		−41.2	0.040	Electron microscopy of thin film	239
		−36.5	0.061	(~80 nm) growth; stained with	
		−33.4	0.074	OsO_4; lamellar growth rates reported	
		−31.3	0.082		
		−27.1	0.145		
		−21.1	0.117		
		−16.4	0.096		
		−15.3	0.091		
		−13.2	0.085		
		−10.3	0.076		
		−8.3	0.070		
		−6.0	0.060		
		−4.1	0.029		
		−1.6	0.0098		
		−0.0	0.0046		
		1.1	0.0038		
		2.7	0.00335		
Poly(isoprene)*cis* GR-F α-lamellar		−36	0.080	Electron microscopy with OsO_4	240
		−27.5	0.104	staining; α and β spherulitic	
		−21.5	0.144	morphologies studied in quayle	
		−16.7	0.114	rubber extracts as (a) fresh	
		−11.8	0.093	extract GR-F and (b) aged	
		−5.0	0.069	specimen GR-AP	
GR-AP α-lamellar		−35.5	0.063		240
		−27.0	0.093		
		−22.0	0.111		
		−16.8	0.094		
		−11.7	0.067		
		−5.0	0.041		
GR-AP β-lamellar					
Poly(isoprene)*trans*-1,4		41.7	4.2	Optical hot stage microscopy	241
(low melting crystals) LMF		44.3	2.1	T_m (LMF) = 64°C	
		46.5	0.95	$\sigma_e = 23$ erg/cm^2	
		47.5	0.50	(negative spherulites)	
		49.0	0.23		
(high melting crystals) HMF		41.7	7.0	T_m(HMF) = 74°C	241
		44.3	4.7	$\sigma_e = 35$ erg/cm^2	
		46.5	2.35	(dendritic spherulites)	
		47.5	1.90		
		49.0	1.21		

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TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(isoprene)*trans*-1,4 (high molecular weight fraction)		−10.2	0.096	Electron microscopy with OsO_4 staining; thin film ~100 nm thick spherulitic growth pressure = 1 bar	187,242
		−5.3	0.154		
		−0.6	0.42		
		4.3	0.80		
		9.9	1.43		
		14.7	2.08		
		47.3	1.47		
		49.3	1.09		
		51.3	0.77		
		54	0.44		
		55.3	0.30		
		60	0.052		
		−0.7	0.11	0.5 kbar	187,242
		4.8	0.22		
		9.4	0.44		
		14.7	0.82		
		19.3	1.38		
		55.6	1.25		
		58.6	0.77		
		60.5	0.55		
		62	0.37[4]		
		64.3	0.14		
		68.0	0.079		
		4.5	0.102	0.7 kbar	187,242
		9.8	0.26		
		14.3	0.49		
		19.5	0.89		
		25.6	1.58		
		57.5	1.29		
		61.5	0.80		
		63.5	0.57		
		66.0	0.31		
		68.4	0.14		
		14.2	0.23	1.0 kbar	187,242
		19.3	0.44		
		24.4	0.79		
		68	0.64		
		68.3	0.46		
		71.6	0.23		
		76	0.11		
		29.3	0.104	2.0 kbar	187,242
		34.5	0.202		
		39.5	0.39		
		44.6	0.69		
		49.2	1.10		
		54	1.46		
		59.5	1.75		
		63.7	1.80		
		70	1.67		
		75	1.17		
		79.3	0.58		
		82	0.35		
		84.5	0.20[6]		
		87	0.10[3]		
		49.3	0.10[4]	3.0 kbar	187,242
		59	0.44		
		64.5	0.66		
		74	1.06		
		79.3	1.11		
		84.7	1.01		
		89	0.63		
		92.5	0.50		
		94.6	0.27		
		100	0.15		
		102.7	0.085		

TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		59.5	0.124	3.5 kbar	187,242
		63.7	0.23		
		69.3	0.38		
		74	0.53		
		79.2	0.73		
		85	0.90		
		89.7	0.83		
		95	0.68[3]		
		100	0.45		
		103	0.25[4]		
		105	0.15		
		107.7	0.095		
Poly(isoprene)*trans*-1,4		−5.3	0.099	Pressure, 1 bar	187,242
(low molecular weight fraction)		−0.6	0.197		
		4.4	0.385		
		9.8	0.630		
		14.7	0.915		
		39.4	1.81		
		41.3	1.33		
		44.7	0.90		
		47.7	0.62		
		49.5	0.41		
		51.3	0.26		
		54.0	0.12		
		55.8	0.063		
		4.4	0.11	0.5 kbar	187,242
		9.5	0.19		
		14.5	0.32		
		19.5	0.59		
		25	0.98		
		50.5	0.96		
		55.3	0.51		
		57.6	0.33		
		59.5	0.20		
		62.3	0.07[3]		
		9.5	0.088	0.7 kbar	187,242
		14.8	0.22		
		20	0.40		
		25	0.63		
		51.5	1.08		
		56	0.78		
		57.6	0.57		
		60	0.42		
		61	0.24		
		64	0.082		
		14.4	0.089	1 kbar	187,242
		19.4	0.180		
		24	0.303		
		29.5	0.57		
		36.6	1.02		
		56	0.98		
		57.7	0.83		
		59.8	0.66		
		63.5	0.39		
		69	0.16		
		35	0.10[4]		
		39.5	0.15[4]		
		45	0.31		
		49.7	0.46		
		54.5	0.64		
		59.8	0.79		
		64	0.79[4]		
		70	0.72		
		75	0.46		
		84.7	0.072		

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TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		34.5	0.096	2 kbar	187,242
		40	0.15		
		45	0.32		
		50	0.48		
		54.5	0.64		
		60	0.77		
		64	0.82		
		70	0.72		
		74.4	0.47		
		85	0.085		
		59	0.08	3 kbar	187,242
		64.5	0.19[5]		
		70	0.33		
		74.5	0.42		
		80	0.48		
		84.7	0.40		
		89.5	0.26		
		92.4	0.14		
		94.7	0.052		
		67	0.091	3.5 kbar	187,242
		70	0.15		
		74.7	0.18		
		79.5	0.29		
		84.5	0.35		
		89.5	0.36		
		94.5	0.29		
		98.2	0.19		
		99.5	0.14		
		102.7	0.06[2]		
Poly(isoprene)/epoxide *cis*					242
100/0, w/w		−30.2	8.5×10^{-3}		
		−20.8	30×10^{-3}		
		−20	10×10^{-3}		
		−10.6	6.3×10^{-3}		
		−5.7	4.5×10^{-3}		
95.8/4.2, w/w		−30.2			292
		−24.8	1.8×10^{-3}		
		−20.0	2.3×10^{-3}		
		−15			
		−5.7	4.5×10^{-3}		
90.3/9.7, w/w		−30.2	2.4×10^{-3}		292
		−24.8	5.5×10^{-3}		
		−20			
		−5.7	0.5×10^{-3}		
87.8/12.2, w/w		−30.2	0.75×10^{-3}		292
		−24.8	0.10×10^{-3}		
		−20	0.53×10^{-3}		
		−15			
		−5.7	0.12×10^{-3}		
1.2. SOLUTIONS					
Poly(ethylene)		70.45	7.67	Electron microscopy; temperature	42
$M_w = 3.1 \times 10^3$		72.93	6.53	control ± 0.01°C; data for [1 1 0]	
$M_n = 2.9 \times 10^3$		76.10	4.41	faces; 0.05 wt.% in xylene;	
$T_s = 365.9$ K		77.00	4.40	crystallized from stock	
$T_d = 367.4$ K		74.05	5.27	solution prepared at 75°C	
		78.25	2.93	seeded growth used	
		81.00	1.25		
		83.00	0.72		
		84.00	0.62		
		85.00	0.36		
		86.25	0.18		

TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_w = 3.1 \times 10^3$		68.90	5.77	Temperature control ±0.01°C;	42
$M_n = 2.9 \times 10^3$		70.45	5.11	data for [1 1 0] faces; 0.01 wt.%	
$T_s = 365.9$ K		72.10	4.83	in xylene	
$T_d = 367.4$ K		74.05	3.98		
		77.00	2.03		
		79.80	0.88		
		81.00	0.71		
		84.00	0.35		
		85.00	0.25		
		86.25	0.13		
		87.70	0.06		
$M_w = 4.05 \times 10^3$		70.85	9.93	Data for [1 1 0] faces; 0.05 wt.%	42
$M_n = 3.90 \times 10^3$		74.85	5.41	in xylene	
$T_s = 367.6$ K		77.00	3.21		
$T_d = 371.5$ K		77.50	2.91		
		78.05	2.18		
		80.00	1.88		
		82.00	1.22		
		83.00	0.72		
		84.10	0.52		
		85.00	0.30		
		86.10	0.24		
		88.00	0.07		
$M_w = 4.05 \times 10^3$		68.90	6.04	Data for [1 1 0] faces; 0.01 wt.%	42
$M_n = 3.90 \times 10^3$		69.90	5.80	in xylene	
$T_s = 367.6$ K		70.85	4.62		
$T_d = 371.5$ K		72.10	4.80		
		74.00	3.01		
		77.00	1.42		
		78.05	0.97		
		80.00	0.79		
		81.60	0.63		
		82.40	0.59		
		85.00	0.26		
		87.70	0.05		
$M_w = 3.1 \times 10^3$		68.90	3.93	Temperature control ± 0.01°C; data for	42
$M_n = 2.9 \times 10^3$		69.90	3.57	[1 1 0] faces 0.005 wt.% in xylene	
$T_s = 365.9$ K		71.20	2.88		
$T_d = 367.4$ K		74.05	1.68		
		74.95	1.27		
		76.10	1.20		
		77.00	1.11		
		78.40	0.96		
		80.30	0.54		
		83.20	0.34		
		84.00	0.21		
		86.25	0.10		
		87.70	0.03		
$M_w = 3.1 \times 10^3$		70.45	1.06	Temperature control ±0.01°C; data	42
$M_n = 2.9 \times 10^3$		71.20	1.01	for [1 1 0] faces 0.001 wt.% in xylene	
$T_s = 365.9$ K		72.93	0.89		
$T_d = 367.4$ K		74.05	0.69		
		77.00	0.48		
		78.25	0.47		
		79.80	0.32		
		81.00	0.20		
		83.00	0.17		
		85.00	0.10		
		86.40	0.05		
$M_w = 4.05 \times 10^3$		68.90	4.43	Data for [1 1 0] faces; 0.005 wt.%	42
$M_w = 3.90 \times 10^3$		69.10	4.29	in xylene	
$T_s = 367.6$ K		70.85	3.63		
$T_d = 371.5$ K		73.06	2.71		
		75.00	1.52		
		76.00	1.17		

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TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		77.00	1.13		
		78.40	0.70		
		78.90	0.67		
		80.30	0.57		
		82.00	0.38		
		83.20	0.28		
		84.10	0.17		
		85.00	0.13		
		86.70	0.06		
		87.70	0.03		
$M_w = 4.05 \times 10^3$		69.10	1.21	Data for [1 1 0] faces; 0.001 wt.%	42
$M_w = 3.90 \times 10^3$		71.00	1.16	in xylene	
$T_s = 367.6$ K		73.40	0.82		
$T_d = 371.5$ K		75.00	0.58		
		77.00	0.30		
		77.60	0.30		
		78.90	0.22		
		80.00	0.21		
		82.00	0.13		
		82.95	0.12		
		85.00	0.05		
		86.70	0.03		
$M_w = 6.75 \times 10^3$		75.50	21.60	Data for [1 1 0] faces; 0.05 wt.%	42
$M_n = 6.10 \times 10^3$		77.80	13.10	in xylene	
$T_s = 367.8$ K		78.95	11.33		
$T_d = 368.4$ K		81.60	3.33		
		83.60	1.08		
		86.00	0.25		
		88.10	0.07		
		75.50	11.25	Data for [1 1 0] faces; 0.01 wt.%	42
		77.80	5.55	in xylene	
		81.60	1.09		
		83.60	0.36		
		86.00	0.13		
		88.10	0.04		
		75.50	3.40	Data for [1 1 0] faces; 0.001 wt.%	
		77.80	1.50	in xylene	
		78.95	0.96		
		81.60	0.35		
		83.60	0.13		
		86.00	0.05		
		88.10	0.02		
$M_w > 10^4$	67	133	232	$\theta = 9.6°$ Growth rates from dilute	236
solution conc. 2.7×10^{-4} wt.%	68		180	$\theta = 8.0°$ tetrachloroethylene;	
	69	75	163	$\theta = 6.9°$ single crystal and [1 1 0]	
	70		127	$\theta = 7.0°$ twins measured, $G \sim C^{1/3}$	
	71	41	125	$\theta = 4.8°$ without [1 0 0] sectors;	
	71	41	102	$\theta = 3.8°$ $G < G_t$, enhanced growth	
	71.8	29	113	rate where θ is the angle w.r.t. growth	
	72.6	21		direction in twinned crystal (see	
	73.3	16	78	text figures)	
	74.1	11			
	75.4	6	33		
	76.7	2.9			
	77.8	1.5	8.5		
	77.8	1.33			
	79.0	0.70			
	79.0	0.58			
2.5×10^{-5} wt.%	69.0	12.0	20	$\theta = 10°$	
4.5×10^{-5} wt.%		20.8	35	$\theta = 8.5°$	
8.0×10^{-5} wt.%		33.3	65	$\theta = 8.5°$	
1.5×10^{-4} wt.%		53.3	87	$\theta = 8.0°$	
2.7×10^{-4} wt.%		75	163	$\theta = 7.5°$	
5.0×10^{-4} wt.%		105	260	$\theta = 7.6°$	
9.0×10^{-4} wt.%		142	488	$\theta = 7.2°$	

TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
2.5×10^{-5} wt.%	73.3	3.5	7.2		
8.0×10^{-5} wt.%		7.8			
2.7×10^{-4} wt.%		15.8	78.3		
9.0×10^{-4} wt.%		31.7	167		
4.5×10^{-3} wt.%		75			
8.0×10^{-5} wt.%	75.4				
2.7×10^{-4} wt.%		32	15.5		
9.0×10^{-4} wt.%		61	32.5		
4.5×10^{-3} wt.%		123	71.7		
		317			
		54.5	43	$\theta = 7.3°$; θ is oblique angle with $\theta = 5.9°$;	
		58.4	37	with respect to [1 1 0] faces; $\theta = 4.8°$;	
		61.7	27	see Fig. 3 of text	
7×10^{-6} wt.%		50.0	28		
1.3×10^{-5} wt.%			48		
2.5×10^{-5} wt.%			105		
		82	450	Growth rates G measured by	246
$M_w = 120000$		83	340	Dynamic Light Scattering (DLS)	
$M_w/M_n = 1.19$		84	208	from self-seeded crystals in	
		85	125	p-xylene. $G \sim M_z^{\alpha}$ where	
		86	58	$0.6 \leq \alpha \leq 0.8$ for the molecular	
		86.5	51	weights studied; surface energies	
		87.2	35	reported as $\sigma\sigma_e$ erg^2/cm^4	
		87.6	27		
		87.6	24		
		88	18		
		88.5	7.3		
		89	5.5		
		89.5	2.9		
$M_w = 32000$					
$M_w/M_n = 1.11$		85	530		
		85.9	463		
		86.1	403		
		86.8	260		
		85.5	530		
		85.9	463		
		86.1	403		
		86.8	260		
		86.9	250		
		87.0	214		
		87.4	178		
		88.2	109		
		88.7	76		
		89.2	58		
		89.6	37		
		90.2	28		
		90.6	20		
		91	13		
					246
$M_w = 13600$		86.8	2710		
$M_w/M_n = 1.19$		87.1	2130		
		87.8	1810		
		88.5	1016		
		89.0	734		
		89.5	570		
		90	420		
		90.3	335		
		91	213		
		91.3	178		
		91.7	110		
		92.1	80		
		92.4	68		
		92.8	40		
		93	33		
		88.3	1186		

TABLE 1. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(1-cyanoethylene); poly(acrylonitrile)		105	5.56×10^{-8}	Crystallized from propylene carbonate; 0.06 wt./vol% solution 0.25 wt./vol% solution	237
$M_w = 668 \times 10^3$		115	7.18×10^{-8}		
		130	9.84×10^{-8}		
		140	2.00×10^{-7}		
		155	3.20×10^{-7}		
		160	2.89×10^{-8}		

TABLE 2. POLY(OXIDES)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(oxyethylene)		13.5	18330	Growth rates made from self-seeded polymer crystal stocks; dynamic light scattering measurements; $T_m^0 = 69°C$; 25.4 erg/cm $\leq \sigma_e \leq$ 81 erg/cm^2; T_s^0 solution temperatures reported for each fraction	243
$M_n = 5.6 \times 10^4$		14.0	14968		
$M_w/M_n = 1.05$		14.4	11045		
		14.9	8730		
		15.3	9344		
		15.7	6022		
		16.4	4920		
		16.6	2865		
		17.4	2680		
		17.7	2422		
		18.3	1915		
		18.6	1193		
		19.6	567		
		20.5	252		
		21.1	174		
		21.6	19.4		
		22.0	77.3		
		22.5	9.24		
		23.6	19.4		
$M_n = 1.05 \times 10^5$		16.3	19060		243
$M_w/M_n = 1.03$		17.1	12722		
		17.6	6692		
		18.2	5658		
		18.8	4782		
		19.1	4000		
		19.6	2980		
		20.2	2912		
		20.7	1625		
		21.2	1013		
		21.7	625		
		22.1	435		
		22.5	221		
		23.0	125		
		23.5	67.8		
		24.0	37.6		
		24.5	16.9		
		24.9	10.9		
$M_n = 1.60 \times 10^5$		14.7	42570		243
$M_w/M_n = 1.05$		15.1	34983		
		15.7	33865		
		16.8	19745		
		17.3	15070		
		17.8	13178		
		18.2	10060		
		18.8	6486		
		19.2	5480		
		19.7	4184		
		20.2	3089		
		20.7	1990		
		21.2	1327		
		22.7	321		

TABLE 2. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		23.0	194		
		23.5	117		
		24.0	53.1		
		24.5	48.4		
		25.0	23		
		25.5	14.3		
$M_n = 3.25 \times 10^5$		16.3	40156		243
$M_w/M_n = 1.06$		17.3	31750		
		17.6	24235		
		18.6	18520		
		19.2	13672		
		19.8	11173		
		20.2	10450		
		20.8	7715		
		21.0	6517		
		22.0	5514		
		22.6	2904		
		23.6	1207		
		24.2	860		
		24.6	752		
		25.2	468		
		25.6	215		
		26.0	159		
		26.6	34.7		
$M_n = 7.70 \times 10^5$		16.3	54469		243
$M_w/M_n = 1.12$		17.8	36396		
		19.3	20538		
		19.7	15674		
		20.2	14813		
		20.7	10463		
		21.2	8262		
		22.1	4500		
		23.2	3682		
		23.8	3220		
		24.1	2025		
		24.7	1639		
		25.2	1056		
		25.6	955		
		26.0	807		
		26.6	520		
		27.2	384		
		27.6	209		
		28.1	134		
		28.6	69.0		
		29.1	19.6		
Poly(oxyethylene)	69 (15 min)	35.0	58900	Interference optical	160(a)
$M_w = 3.5 \times 10^3$		39.3	48900	microscopy for linear	
$M_w/M_n < 1.2$		40.0	46800	growth measurements	
		40.4	43700		
		41.7	38000		
		42.0	36300		
		43.5	26300		
		44	23400		
		44.4	19900		
		45.3	19500		
		46.0	10700		
		46.4	8910		
		47.0	7590		
		48.0	4680		
		49.0	2400		
		50.0	1260		
		51.0	436		
		52.0	141		
		54.0	42.7		
		55.0	8.51		

TABLE 2. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_w = 4.5 \times 10^3$	69 (15 min)	35.0	53700		160(a)
$M_w/M_n < 1.04$		37.0	50100		
		39.0	47900		
		39.8	45700		
		40.4	42600		
		42.8	33100		
		42.7	28800		
		43.4	24800		
		43.8	23400		
		45.0	16600		
		45.8	12900		
		46.8	9550		
		47.8	5010		
		49.8	1120		
		50.8	575		
		51.8	275		
		52.8	107		
		53.8	14.8		
		54.8	4.89		
$M_w = 7.0 \times 10^3$	69 (15 min)	35.0	70800		160(a)
$M_w/M_n < 1.2$		39.8	60200		
		40.0	55000		
		41.2	53700		
		43.4	43600		
		44.9	33100		
		45.0	34700		
		47.0	21900		
		49.0	13500		
		49.4	13800		
		50.0	9120		
		51.0	4890		
		52.0	2820		
		53.0	1580		
		54.0	871		
		55.0	389		
		56.0	145		
		57.0	45.7		
		58.0	12.6		
$M_w = 9.0 \times 10^3$	69 (15 min)	40.1	38900		160(a)
$M_w/M_n < 1.07$		40.8	38900		
		42.8	32600		
		43.4	30200		
		45.8	20400		
		48.4	11200		
		48.8	10200		
		49.6	7940		
		49.6	7410		
		50.8	7410		
		51.8	1900		
		52.8	1200		
		53.8	589		
		55.8	214		
		56.8	91.2		
		57.8	38.0		
		58.8	13.8		
		59.8	49.0		
$M_w = 20.0 \times 10^3$	69 (15 min)	40.0	22400		160(a)
$M_w/M_n < 1.3$		42.0	17400		
		44.4	14500		
		45.0	13800		
		46.0	11700		
		46.3	11000		
		47.0	9120		
		48.0	7760		
		48.4	6920		

TABLE 2. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		49.0	5250		
		50.0	4470		
		51.0	3240		
		51.7	2570		
		52.0	2290		
		53.0	1170		
		53.2	1150		
		54.0	631		
		55.0	219		
		56.0	97.7		
		57.0	32.4		
		58.0	92.1		
$M_w = 100.0 \times 10^3$	69 (15 min)	39.8	26300		160(a)
M_w/M_n is unknown		40.0	24000		
		42.0	19100		
		43.4	19100		
		44.0	19100		
		46.0	17000		
		47.3	15100		
		48.0	10700		
		49.4	7240		
		51.5	3470		
		52.0	2950		
		53.0	2340		
		53.9	1380		
		54.0	1380		
		56.0	724		
		57.0	389		
$M_w = 3.5 \times 10^3$		39.5	35600	Interference optical microscopy	160(a)
$M_w = 100 \times 10^3$		40.0	27700		
Blend with mole fraction of		45.6	17000		
$M_w = 100 \times 10^3$ at 0.279 mol		46.8	12400		
fraction		47.9	10300		
		48.9	6550		
		50.0	3730		
		51.3	2450		
		51.9	1010		
		54.4	221		
		55.2	80.5		
		45.6	14100		
		46.8	7210		
		47.9	4590		
		48.9	2310		
		50.0	3470(?)		
		51.3	384		
		51.9	113		
		54.4	424		
		55.2	125		
Poly(oxyethylene)		34	129628		245
		36	110640		
		38.5	65120		
		40.5	44545		
		42.7	22615		
		45	9183		
		46.7	2952		
		48	1052		
		49	625		
		50.5	189		
		51	83.2		
		51.5	18.8		
		52.5	17.3		
		53.5	17.3		
		54.0	16.0		
		56.0	8.8		
		56.6	6.0[3]		

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TABLE 2. *cont'd*

Polymer	T_t (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		57.7	1.46		
Poly(oxymethyoxyethylene)		34	217920		245
$M_w = 3000$		36.3	200355		
		38.2	137060		
		40	93780		
		42.5	68980		
		45	19280		
		47	68975		
		48.6	2968		
		55.8	23.1		
		55.8	23.1		
		57.2	8.72		
		57.5	4.15		
		52.5	294		
		51.4	130		
		53.7	31.6		
		55.8	23.1		
		57.2	8.72		
		57.5	4.15		
Poly(oxy-1,4-phenoxyethylene)					
$M_w = 3000$		28	46989	*In situ* synchrotron rate studies;	245
		30	40180	optical microscopy and DSC	
		32	27472	runs made	
		34	25292		
		36	13842		
		38	7034		
		39.5	3857		
		40.3	1969		
		41.9	515		
		43	125		
		44	25.9		
		45	9.90		
		46	2.07		
		47	1.14		
Poly(oxy-2,2-bischloromethyl-trimethylene)	139	45	7.000	High resolution NMR; experiment	244
$M_n = 5.0–10.0 \times 10^3$		60	3.830	in *m*-xylene 0.1–0.001 wt.%,	
		78	2.160	nucleation densities listed	
		86	2.160	initial concentration	
		95	1.170	2×10^{-4} g/cm^3	
		105	0.333		
		60	4.500	Initial concentration	244
		78	3.500	5×10^{-4} g/cm^3	
		86	2.000		
		95	1.500		
		105	0.330		
		86	3.670	Initial concentration	244
		95	2.830	2×10^{-3} g/cm^3	
		105	0.670		
		112	0.333		

2. Rates of Radial Spherulitic Growth

TABLE 3. POLY(DIENES), POLY(ALKENES), POLY(VINYLS)

Polymer	T_t (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
3.1. HOMOPOLYMER MELTS					
Poly(1-butene)	160 (10 min)	90.0	338.0	Supercooled liquid, form II	249
		96.0	168.0		
		100.0	60.4		
		105.0	19.6		

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TABLE 2. *cont'd*

Polymer	T_t (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		57.7	1.46		
Poly(oxymethyoxyethylene)		34	217920		245
$M_w = 3000$		36.3	200355		
		38.2	137060		
		40	93780		
		42.5	68980		
		45	19280		
		47	68975		
		48.6	2968		
		55.8	23.1		
		55.8	23.1		
		57.2	8.72		
		57.5	4.15		
		52.5	294		
		51.4	130		
		53.7	31.6		
		55.8	23.1		
		57.2	8.72		
		57.5	4.15		
Poly(oxy-1,4-phenoxyethylene)					
$M_w = 3000$		28	46989	*In situ* synchrotron rate studies;	245
		30	40180	optical microscopy and DSC	
		32	27472	runs made	
		34	25292		
		36	13842		
		38	7034		
		39.5	3857		
		40.3	1969		
		41.9	515		
		43	125		
		44	25.9		
		45	9.90		
		46	2.07		
		47	1.14		
Poly(oxy-2,2-bischloromethyl-trimethylene)	139	45	7.000	High resolution NMR; experiment	244
$M_n = 5.0–10.0 \times 10^3$		60	3.830	in *m*-xylene 0.1–0.001 wt.%,	
		78	2.160	nucleation densities listed	
		86	2.160	initial concentration	
		95	1.170	2×10^{-4} g/cm^3	
		105	0.333		
		60	4.500	Initial concentration	244
		78	3.500	5×10^{-4} g/cm^3	
		86	2.000		
		95	1.500		
		105	0.330		
		86	3.670	Initial concentration	244
		95	2.830	2×10^{-3} g/cm^3	
		105	0.670		
		112	0.333		

2. Rates of Radial Spherulitic Growth

TABLE 3. POLY(DIENES), POLY(ALKENES), POLY(VINYLS)

Polymer	T_t (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
3.1. HOMOPOLYMER MELTS					
Poly(1-butene)	160 (10 min)	90.0	338.0	Supercooled liquid, form II	249
		96.0	168.0		
		100.0	60.4		
		105.0	19.6		

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
	95	100	0	Parallel plate shearing device; shear rate (s^{-1})	200
		100	0.23		
		94	0.46		
		92	0.9		
	100	23	0		
		27	0.23		
		25	0.66		
		23	1.0		
	105	3.2	0		
		3.1	0.23		
		3.3	0.5		
		3.3	1.0		
		4.1	2.2		
94% Isotactic Shell Chem. Co.	180 (5 min)	65	1192	Hot stage optical microscopy; crystallinity values listed	251
		70	1082		
$M_n = 73 \times 10^3$		80	675		
$M_w = 750 \times 10^3$		85	449		
		90	244		
		95	111		
		100	32.6		
Poly(chlorotrifluoroethylene)	305	163	450.1	Hot stage microscopy	121,252
	277	169	440.1		
	305	169	430.1		
	277	172	360.1		
	277	180	195.0		
	267	185	110.0		
	267	191	40.0		
		195	7.25		
		197	2.50		
Kel-F NST 300	270 (10 min)	170	406.7		121,252
		175	310.0		
		180	216.7		
		185	124.2		
		190	36.3		
Poly(1-difluoroethylene)					
Kureha KF-1100		154.7	120	Hot stage microscopy	253
$M_w = 525 \times 10^3$; $M_v = 348 \times 10^3$		159.5	36.2		
(α)-spherulites		165.3	7.2		
		167.2	3.4		
		170.2	0.82		
		174.0	0.07		
mixed spherulites		157.5	13.5		253
(α and γ)		161.4	4.26		
		165.2	1.68		
		167.3	0.86		
		170.1	0.39		
		174.1	0.10		
Kynar 881		151.6	57.2		253
$M_w = 555 \times 10^3$; $M_n = 375 \times 10^3$		155.7	20.3		
(α)-spherulites		159.9	4.51		
		162.2	1.60		
		165.1	0.33		
		168.8	0.03		
mixed spherulites (α and γ)		153.3	6.13		253
		159.3	1.20		
		162.3	0.56		
		165.1	0.26		
		167.4	0.12		
		168.9	0.07		
Poly(ethylene)					
Sinclair 2907		123	300	Spherulitic growth rates by laser light scattering; at atmospheric pressure	254
		122	500		
		121	750		
		120	1100		
		119	1700		

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TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		118	2000		
		132	310		
		130.9	490	Pressure = 0.5 kbar	
		129.5	820		
		128.5	1170		
		127.7	1410		
		148.5	140		
		147.5	200	Pressure = 1.0 kbar	254
		146.5	370		
		145.5	590		
		144.5	820		
		157.5	300		
		156.5	520		
		155.5	810		
		153.5	1820		
		151.5	2420		
		169.6	150		
		168.6	310	Pressure = 2.0 kbar	254
		168.0	540		
		166.6	910		
		166.2	1220		
		165.5	1560		
$M_w = 3.60 \times 10^3$	155	118.3	2265.2	Hot stage microscopy; selected	255
$M_w/M_n = 1.13$		119.6	271.3	values only listed for axialites	
		120.8	24.2		
		122.0	1.44		
$M_w = 4.20 \times 10^3$		118.9	5350.6		255
$M_w/M_n = 1.26$		119.3	3389.2		
		121.4	70.8		
		123.7	0.35		
$M_w = 6.15 \times 10^3$		122.3	3043.7		
$M_w/M_n = 1.09$		123.6	68.9		
		124.2	7.03		
		124.9	0.47		
$M_w = 6.29 \times 10^3$		122.7	2146.8		255
$M_w/M_n = 1.17$		124.0	87.8		
		124.6	23.5		
		125.1	3.5		
$M_w = 8.56 \times 10^3$		123.5	11663		255
$M\sim/M\sim = 1.09$		124.8	575.6		
		125.4	114.8		
		126.1	7.03		
$M_w = 8.59 \times 10^3$		124.6	1641		255
$M_w/M_n = 1.11$		125.1	623.9		
		126.0	52.7		
		126.8	3.90		
$M_w = 17.0 \times 10^3$		125.1	4678		255
$M_w/M_n = 1.08$		125.7	1435		
		126.9	108.8		
		127.5	14.5		
$M_w = 266 \times 10^3$	155	122.7	80.7	Hot stage microscopy;	255
$M_w/M_n = 1.81$		123.8	29.0	irregular shaped spherulites	
		125.4	5.6		
		127.7	0.2		
$M_w = 210 \times 10^3$		122.5	124.3		255
$M_w/M_n = 1.43$		126.1	6.9		
		127.4	1.7		
		128.5	0.5		
		124.9	52.4		
$M_w = 500 \times 10^3$		126.2	18.1		255
$M_w/M_n = 1.60$		127.7	3.0		
		128.9	0.6		
$M_w = 807 \times 10^3$		123.1	233.0		255
$M_w/M_n = 2.9$		125.5	50.4		

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
SRM 1475		127.5	8.1		
		129.3	0.7		
$M_w = 74.4 \times 10^3$	155	125.6	22.95	Hot stage microscopy; axialites	255
$M_w/M_n = 1.12$		126.3	13.8	above 129°C approx.	
		127.6	4.86		
		128.5	2.54		
		128.9	1.53		
		129.6	0.41		
		130.1	0.12		
		130.6	0.04		
$M_w = 18.1 \times 10^3$		123.2	542.1	Axialites above 125°C	255
$M_w/M_n = 1.39$		124.5	199.1	approx.	
		125.2	108.3		
		125.5	98.2		
		125.6	63.7		
		126.7	10.1		
		128.0	0.78		
		128.7	0.14		
$M_w = 30.6 \times 10^3$		124.7	242.33	Axialites above 127°C	255
$M_w/M_n = 1.19$		125.8	90.76	approx.	
		126.7	48.41		
		127.1	32.68		
		127.5	17.78		
		128.1	5.16		
		128.5	1.59		
		130.0	0.05		
$D = 0$ MR*	292	120.7	117	Hot stage optical microscopy	256
	192	122.9	103.4		
	193	125.1	18.2		
	196	127.0	1.08		
	168	122.4	88.4		
	153	122.5	100.4		
$D = 40$ MR*	195	118.9	41.68		256
	195	113.1	291.7		
	195	115.8	65.0		
	195	117.8	2.33		
$D = 100$ MR*	195	100.0	71.7		256
	195	100.0	61.68		
	195	101.8	31.67		
	195	104.0	7.84		
$M_w = 19 \times 10^3$	200 (20 min)	125.0	126.7		257
		126.0	36.67		
		127.0	16.67		
		128.0	3.33		
$M_w = 4.0 \times 10^3$		81.0	510.0	# $CH_3/100\,CH_2 = 5.3$	258
		83.0	256.7		
		85.0	143.4		
		88.0	63.4		
$M_w = 7.0 \times 10^3$		91.0	215.0	# $CH_3/100\,CH_2 = 5$	258
		93.0	86.7		
		95.2	35.0		
		97.0	14.0		
		98.0	7.84		
$M_w = 10.0 \times 10^3$		92.0	175.0	# $CH_3/100\,CH_2 = 4.7$	258
		93.0	101.2		
		95.0	40.01		
		97.0	20.01		
		98.0	11.50		
$M_w = 12.0 \times 10_3$		90.3	316.7	$CH_3/100\,CH_2 = 4.5$	258
		92.0	185.0		
		98.0	12.2		
		100.3	3.33		

* D: Radiation does; MR: megarad.

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(ethylene)					
$M_n = 5.14 \times 10^3$	160 (30 min)	121.39	422.1	Fractionated samples;	259
$M_w = 6.29 \times 10^3$		125.12	1.069	NBS material SRM 1475	
$M_z = 7.62 \times 10^3$					
$M_n = 10.51 \times 10^3$		125.32	93.39	Fractionated samples;	259
$M_w = 11.74 \times 10^3$		127.81	0.433	NBS material SRM 1475	
$M_z = 13.22 \times 10^3$					
$M_n = 56.35 \times 10^3$		234.46	25.10	Fractionated samples;	259
$M_w = 68.57 \times 10^3$		128.65	0.377	NBS material SRM 1475	
$M_z = 83.93 \times 10^3$					
$M_n = 76.76 \times 10^3$		123.07	117.3	Fractionated samples;	259
$M_w = 134.30 \times 10^3$		127.74	1.039	NBS material SRM 1475	
$M_z = 276.00 \times 10^3$					
$M_n = 195.00 \times 10^3$		120.25	196.3	Fractionated samples;	259
$M_w = 323.20 \times 10^3$		126.30	1.062	NBS material SRM 1475	
$M_z = 500.80 \times 10^3$					
Poly(isoprene), unfractionated		−10	210.8×10^{-3}	Optical microscopy	283
		0	27.8×10^{-3}	0.5 kbar	
		10	2.11×10^{-3}		
		−10	392.5×10^{-3}	1.0 kbar	283
		0	156.1×10^{-3}		
		8	11.9×10^{-3}		
		0	782.8×10^{-3}	1.5 kbar	283
		10	98.6×10^{-3}		
		21	4.44×10^{-3}		
		25	3.89×10^{-3}		
		0	124.2×10^{-3}	2.0 kbar	283
		10	554.2×10^{-3}		
		21	43.9×10^{-3}		
		25	$3 5 \times 10^{-3}$		
		0	19.72×10^{-3}	2.5 kbar	283
		10	392.5×10^{-3}		
		25	110.6×10^{-3}		
		25	554.2×10^{-3}	3.0 kbar	283
		29	175.3×10^{-3}		
		35	31.1×10^{-3}		
cis-Poly(isoprene)					
$M_v = 313 \times 10^3$		−40	39.7×10^{-3}	Optical microscopy; rates	239,240
		−26	146.1×10^{-3}	I to II at −8°C; lamellar	
		−15	92.5×10^{-3}	thickness and surface	
		−12	82.2×10^{-3}	energy data are listed	
		−5	55.6×10^{-3}		
		−1	9.17×10^{-3}		
		0.8	4.08×10^{-3}		
		1.8	3.64×10^{-3}		
		3.0	2.92×10^{-3}		
$M_v = 351 \times 10^3$		−40	33.3×10^{-3}	Optical microscopy; rates	239,240
		−32	66.7×10^{-3}	I to II at −8°C; lamellar	
		−26	138.9×10^{-3}	thickness and surface	
		−20	100×10^{-3}	energy data are listed	
		−12	72.5×10^{-3}		
		−7.3	47.5×10^{-3}		
		1.0	10.0×10^{-3}		
$M_v = 543 \times 10^3$		−40	20.83×10^{-3}	Optical microscopy; rates	239,240
		−35	50.83×10^{-3}	I to II at −8°C; lamellar	
		−30	69.4×10^{-3}	thickness and surface	
		−26	111.1×10^{-3}	energy data are listed	
		−20	97.2×10^{-3}		
		−15	64.4×10^{-3}		
		−10	34.7×10^{-3}		
		−5	12.8×10^{-3}		
		−2.2	13.6×10^{-3}		
		−0.6	10.6×10^{-3}		
		1.4	5.39×10^{-3}		
		2.2	5.0×10^{-3}		

TABLE 3. *cont'd*

Polymer	T_t (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_v = 897 \times 10^3$		− 40	38.20×10^{-3}	Optical microscopy; rates	239,240
		− 35	51.7×10^{-3}	I to II at − 8°C; lamellar	
		− 26	104.2×10^{-3}	thickness and surface	
		− 20	83.3×10^{-3}	energy data are listed	
		− 8.0	48.4×10^{-3}		
$M_v = 1421 \times 10^3$		− 40	11.1×10^{-3}	Optical microscopy; rates	239,240
		− 30	47.2×10^{-3}	I to II at − 8°C; lamellar	
		− 26	61.1×10^{-3}	thickness and surface	
		− 20	50.0×10^{-3}	energy data are listed	
		− 5.0	6.94×10^{-3}		
cis-Poly(isoprene)				Electron microscopy with OsO_4 staining	
		0.0	8.37×10^{-3}	0.0 kbar	272
		0.0	15.74×10^{-3}	0.3 kbar	
		0.0	28.02×10^{-3}	0.5 kbar	
		0.0	55.37×10^{-3}	0.8 kbar	
		0.0	179.3×10^{-3}	1.0 kbar	
		0.0	783×10^{-3}	1.5 kbar	
		0.0	118.3×10^{-3}	2.0 kbar	
		0.0	16.38×10^{-3}	2.5 kbar	
		0.0	13.60×10^{-3}	3.0 kbar	
trans-Poly(isoprene)					
$M_v = 7.0 \times 10^3$	100	33.6	72.18	Hot stage optical microscopy	286
		35.6	63.01		
		37.5	48.01		
		37.6	53.18		
		40.4	39.01		
		43.0	22.67		
$M_v = 12.0 \times 10^3$	100	37.6	44.01	Hot stage optical microscopy	286
		39.5	36.01		
		43.0	21.34		
		45.6	14.34		
$M_v = 45.0 \times 10^3$	100	34.0	24.67	Hot stage optical microscopy	286
		38.0	11.67		
		40.0	9.34		
		43.0	5.67		
		46.0	2.83		
$M_v = 91.0 \times 10^3$	100	34.0	16.50	Hot stage optical microscopy	286
		38.0	7.92		
		40.0	5.85		
		43.0	3.52		
		46.0	1.71		
$M_v = 165.0 \times 10^3$	100	38.0	5.83	Hot stage optical microscopy	286
		40.0	4.25		
		43.0	2.10		
		46.0	1.05		
$M_v = 400.0 \times 10^3$	100	34.0	14.00	Hot stage optical microscopy	286
		38.0	6.67		
		40.0	4.38		
		43.0	2.33		
		46.0	0.95		
$M_v = 12.0 \times 10^3$	100	25.8	173.37		287
		29.6	111.69		
		33.6	66.68		
		37.6	34.01		
		43.0	9.17		
$M_v = 45.0 \times 10^3$	100	26.0	51.01		287
		30.0	34.84		
		34.0	16.50		
		38.0	8.17		
		40.0	4.83		
		43.0	1.83		
$M_v = 91.0 \times 10$	100	26.0	36.01		287
		30.0	20.83		
		34.0	11.16		
		38.0	4.33		

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		40.0	2.50		
		43.0	1.17		
$M_v = 30 \times 10^3$		−9.5	0.080	Electron microscopy with	187
High melting form		−4.6	0.125	OsO_4 staining; pressure @1 bar	
		0.4	0.401		
		5.3	0.811		
		9.5	1.390		
		14.6	2.067		
		47.1	1.470		
		49.4	1.123		
		51.3	0.775		
		53.4	0.481		
		55.4	0.330		
		59.3	0.134		
Low melting form		−5.1	0.071	Electron microscopy with	187
		0.0	0.196	OsO_4 staining; pressure @1 bar	
		5.5	0.383		
		9.9	0.650		
		14.3	0.900		
		38.3	1.791		
		42.0	1.337		
		45.0	0.909		
		47.5	0.624		
		49.4	0.446		
		51.7	0.294		
		54.2	0.169		
		55.7	0.107		
$M_n = 30 \times 10^3$					
High melting form		29.8	0.096	Electron microscopy and	187
		34.6	0.213	OsO_4 staining; pressure @ 2 kbar	
		39.9	0.426		
		45.0	0.714		
		49.6	1.086		
		55.1	1.470		
		59.7	1.740		
		64.1	1.800		
		69.3	1.651		
		74.6	1.179		
		79.4	0.589		
		80.8	0.366		
		84.0	0.209		
		86.3	0.114		
Low melting form		34.2	0.092	Electron microscopy and	187
		39.3	0.170	OsO_4 staining; pressure @ 2 kabr	
		44.7	0.323		
		49.4	0.490		
		54.6	0.653		
		59.4	0.785		
		63.8	0.802		
		69.5	0.714		
		73.2	0.465		
		78.7	0.305		
		84.0	0.075		
Poly(propylene)	191–195	125	87.52	Hot stage optical microscopy	260
		130	61.51		
		134	28.67		
		138	8.668		
		140	4.501		
	180 (15 min)	122.0	300.1	Hot stage optical microscopy	261
		125.0	200.1		
		130.0	71.68		
		135.0	26.67		
		140.0	9.835		
		145.0	4.501		

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_v = 178.0 \times 10^3$		120	490.1	Hot stage optical microscopy	262
		125	216.7		
		131	64.68		
		135	27.17		
	220	120.5	480.1	Pro-fax 6513E	263
		129	120.0		
		145	4.501		
		121	408.4	Pro-fax 6501	263
		132	46.67		
		145	2.334		
Hostalen PPN 106OF	164.6	127.5	116.7	Hot stage microscopy;	264
(Hoechst AG) $T_m = 180$°C	164.8	130.0	71.7	rate constants also listed;	
	165.3	132.5	43.3	crystallinity also provided	
	165.8	135.0	26.7		
	166.8	138.0	143.3		
	167.5	140.0	98.3		
	170.1	145.0	45.0		
$T_m = 190$°C	164.8	130.0	65.0		264
	167.5	140.0	83.3		
$T_m = 200$°C	164.8	130.0	63.3		264
	167.5	140.0	78.3		264
$M_w = 289 \times 10^3$		93.2	1733	Optical microscopy; linear	265
$M_n = 51 \times 10^3$		98.6	3363	crystallization under unilateral	
(KS10 from Petrochemica Danubia)		104.0	8531	quench, modeled and	
		108.7	7505	measured; comparisons	
		113.2	1794	made with literature	
		122.9	470.9	spherulitic growth rates	
		131.8	69.9		
		136.1	23.1		
		145.4	3.7		
(heterotactic/isotactic/syndiotactic		97.4	122.0	Hot stage microscopy	149
= 0.145/0.211/0.644; from		101.4	123.7		
^{13}C NMR analysis)		104.4	82.0		
		105.4	71.3		
		109.4	70.3		
		113.4	30.5		
		117.4	26.9		
		121.4	15.6		
		125.4	7.4		
		129.4	4.4		
		133.4	1.7		
		137.3	0.3		
Poly(1-phenylethylene)					
$M_v = 60 \times 10^3$		140	0.333	Hot stage optical microscopy	262
		155	1.500		
		177	4.334		
		190	3.667		
		200	1.667		
		210	0.500		
$M_v = 190 \times 10^3$	270	140	0.83		268
		150	0.333		
		170	1.334		
		180	334		
		190	0.500		
$M_v = 330 \times 10^3$	250	120	0.040		269
$M_n = 185 \times 10^3$		130	0.162		
		140	0.600		
		150	1.43		
		160	2.50		
		170	3.42		
		180	3.42		
		190	2.50		
		200	1.43		
		210	0.45		

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TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_v = 1250 \times 10^3$		155	2.167		270
		170	4.834		
		184	4.501		
		200	2.167		
		210	0.666		
$M_v = 1380 \times 10^3$	270	130	0.333		268
		150	1.834		
		170	4.501		
		180	7.502		
		190	3.167		
$M_v = 2200 \times 10^3$	260	112.01	0.00454		270
		119.62	0.02582		
		129.70	0.1518		
		150.26	1.526		
		159.22	2.532		
		172.15	3.844		
		179.70	3.980		
		195.06	2.529		
		200.52	1.832		
Isotactic	260	113.3	0.005	Hot stage microscopy; average	271
$M_v = 220 \times 10^3$		121.9	0.040	of many literature results	
		133.0	0.158		
		143.8	0.582		
		154.9	1.343		
		174.8	3.063		
		180.0	3.380		
		190.6	3.063		
		203.6	1.872		
		203.4	1.279		
		202.4	0.821		
		208.3	0.582		
		212.2	0.422		
		212.5	0.611		
$M_v = 2100 \times 10^3$	250 (30 min)	111.1	0.004	Electron microscopy; thin films	272
(Dow Chem.)		123.2	0.069	(80–100 nm) solution cast	
		144.7	0.498	and then melt crystallized	
		161.6	2.009		
		180.0	3.711		
		192.2	3.529		
		206.3	1.738		
		215.2	0.404		
		215.2	0.355		
		220.7	0.037		
Poly(methyl methacrylate) isotactic	180 (5 min)	90	0.072	Hot stage microscopy	273
$M = 213 \times 10^3$		100	0.018		
(heterotactic/isotactic/syndiotactic = 5/94/1)		110	0.444		
		120	0.606		
		13	4.80		
Poly(2-vinyl pyridine) isotactic	230 (1 min)	100	100	Hot stage microscopy	274
$M_v = 400 \times 10^3$		110	79		
		120	413		
		130	1524		
		140	4163		
		150	8574		
		160	13058		
		170	13565		
		180	7707		
3.2. BLENDS					
Poly(isoprene)					
Blend of *cis/trans*					
5% trans		40	0.4	Hot stage optical	Ref. 6, p. 49
20			1	microscopy; other results	
40			1.6	in this paper including	
60			2.1	crystallinities	

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
80			3.6		
100			7.7		
Poly(ethylene)/isotactic poly(propylene)		125.4	265.4	Hot stage optical microscopy	275
PE: $M_w = 16.6 \times 10^3$		129.3	114.9	growth rates of PP spherulites	
$M_n = 10.2 \times 10^3$		131.2	71.7	in blends of 0, 10, 20,	
i-PP: $M_w = 307 \times 10^3$		135.1	27.3	and 50 wt.%; DSC rates listed	
$M_n = 15.6 \times 10^3$					
		137.1	19.0		
		125.6	267.9	Hot stage microscopy	276
		127.5	175.4	*average values cited for	
		129.5	116.9	blends of *i*-PP/LDPE:	
		131.4	74.7	100/0	
		133.4	47.9	90/10	
		135.3	29.3	70/30	
		137.3	20.3	50/150	
Poly(propylene)/poly(isobutene)					
Med. *M*w.					
100/0; PP/PIB		121.0	550.0	Hot stage microscopy	6,277
		123.0	366.7		
		125.0	250.0		
		127.0	166.7		
		131.0	83.3		
90/10		121.0	400.0	Hot stage microscopy	6,277
		123.0	300.0		
		125.0	216.7		
		127.0	150.0		
		131.0	66.7		
80/20		121.0	333.3	Hot stage microscopy	6,277
		123.0	266.7		
		125.0	166.7		
		127.0	116.7		
		131.0	33.3		
60/40		121.0	316.7	Hot stage microscopy	6,277
		123.0	200.0		
		125.0	150.0		
		127.0	83.3		
		131.0	50.0		
High *M*w.					
100/0; PP/PiB:		121.0	550.0	Hot stage microscopy	278
		123.0	366.7		
		125.0	250.0		
		127.0	166.7		
		131.0	83.3		
90/10		121.0	250.0	Hot stage microscopy	278
		123.0	166.7		
		125.0	116.7		
		127.0	66.7		
		131.0	33.3		
80/20		121.0	300.0	Hot stage microscopy	278
		123.0	216.7		
		125.0	133.3		
		127.0	100.0		
		131.0	50.0		
60/40		121.0	316.7	Hot stage microscopy	278
		123.0	166.7		
		125.0	150.0		
		127.0	100.0		
		131.0	50.0		
Poly(1-phenylethylene)/poly(1-phenylethylene); isotactic/atactic polystyrene					
i-PS, $M_w = 550 \times 10^3$; *a*-PS, $M_w = (900–1800) \times 10^3$					
i-PS/a-PS; 100/0		140	1.56		279
		160	3.43		
		170	4.27		
		180	4.65		

TABLE 3. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		190	3.42		
		200	1.76		
80/20		140	1.14		279
		160	2.67		
		170	3.50		
		180	3.56		
		190	2.46		
		200	0.92		
60/40		140	0.56		
		160	1.77		279
		170	2.30		
		180	2.50		
		190	1.72		
Poly(1-phenylethylene)(polystyrene)/poly(vinyl methyl ether)					
$M_v = 134 \times 10^3$; $M_v = 971 \times 10^3$					
PS/PVMe; 100/0		150.0	0.8	Hot stage microscopy	277
		180.0	2.5		
90/110		150.0	1.3	Hot stage microscopy	277
		180.0	3.0		
80/20		150.0	1.5	Hot stage microscopy	277
		180.0	3.2		
Poly(propylene)/poly(propylene) isotactic/atactic		121.3	503	Interpolated *i*-PP	280
i-PP: $M_w = 3.07 \times 10^5$; $M_n = 1.56 \times 10^4$		123.4	337		
a-PP: $M_w = 2.4 \times 10^4$		125.4	235		
		127.4	175		
		129.1	113		
		131.4	83.3		
		121.3	433	*i*-PP/*a*-PP = 9 : 1	280
		123.4	308		
		125.4	218		
		127.4	132		
		129.1	97		
		131.4	67		
		121.4	392	*i*-PP/*a*-PP = 8 : 2.	280
		123.4	263		
		125.4	180		
		127.4	130		
		129.1	70		
		131.4	50		
		212.4	358	*i*-PP:*a*-PP = 7 : 3.	280
		123.4	247		281
		125.4	163		
		127.4	117		
		129.1	67		
		131.4	33		
Poly(propylene)/poly(ethylene); isotactic/linear low density poly(ethylene)					
100/0, w/w		126	240	Hot stage microscopy; $t_{0.5}$ values are also listed microscopy	282
90/10, w/w		126	243		
80/20, w/w		126	242		
50/50, w/w		126	248		
		180.0	2.5		
90/110, w/w		150.0	1.3	Hot stage microscopy; $t_{0.5}$ values are also listed microscopy	282
		180.0	3.0		
80/20, w/w		150.0	1.5	Hot stage microscopy; $t_{0.5}$ values are also listed microscopy	282
		180.0	3.2		

TABLE 4. POLY(OXIDES)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
4.1. HOMOPOLYMER MELTS					
Poly[oxy(1-oxo-2,2-dimethyltrimethylene)]		180	7.93×10^3	Spherulitic growth; hot	308
Poly(pivalolactone); PPVL		185	3.60×10^3	stage, photographic method;	
		190	1.37×10^3	regime growth; α phase	
		192.5	7.25×10^2		
		195	3.65×10^2		
		197.5	1.90×10^2		
		200.0	95.0		
		202.5	40.2		
		205.0	23.5		
		207.5	12.2		
		210.0	6.38		
		213.0	3.88		
		215.0	2.18		
		218.0	0.99		
Poly(oxy-2,2-bischloromethyltrimethylene)	210 (25 min)	75	208.4	Hot stage optical microscopy	289
		80	183.4		
		90	150.0		
		100	125.0		
		110	100.0		
		120	66.68		
		130	33.34		
		140	25.01		
Poly(oxy-2,2-dimethyltrimethylene)	80 (15 min)	25.2	2.11	Hot stage microscopy; average of	290
$M_w = (18.5 \times 10^3 - 130 \times 10^3)$		29.1	1.38	several molecular weights cited	
		33.2	0.73		
		36.8	0.35		
		40.5	0.14		
		42.2	0.08		
		44.0	0.05		
		44.3	0.04		
Poly(oxyethylene)		42.0	12769.4	Hot stage optical microscopy	291
$M_v = 5.47 \times 10^3$		44.2	4734.2		
		45.2	2217.1		
		47.2	395.1		
$M_v = 12.4 \times 10^3$		47.2	11752.3	Hot stage optical microscopy	291
		48.8	6034.5		
		51.2	1514.0		
		55.2	55.51		
$M_v = 33.1 \times 10^3$		47.6	1784.0	Hot stage optical microscopy	291
		48.2	1516.9		
		51.0	421.8		
		53.8	94.19		
		56.4	1.60		
$M_v = 45.0 \times 10^3$		156.0	346.7	Hot stage optical microscopy	292–294
		157.0	180.0		
		158.0	99.19		
		159.8	77.02		
		160.8	56.61		
		163.0	20.84		
		163.9	14.25		
		164.5	4.801		
		166.1	2.951		
		169.1	0.475		
Poly(oxyethylene); Polyox WRS-205, Union Carbide					
$M_v = 920 \times 10^3$	120 (45 min)	53.7	352.2	Hot stage microscopy	11
		54.7	194.3		
		55.7	111.4		
		56.6	54.0		
		57.5	25.5		
$M_n = 1.6 \times 10^3$	100 (min)	44.8	36.5×10^3	Hot stage optical microscopy; selected	295
$M_w/M_n = 1.6$		45.4	18.2×10^3	molecular weights only are listed	
$[\eta] = 0.062$		45.9	6.91×10^3		
		46.5	1.82×10^3		

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TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_n = 4.3 \times 10^3$		48.0	30.33×10^3		
$M_w/M_n = 1.07$		50.0	19.33×10^3		
		51.0	6.06×10^3		
		52.0	4.33×10^3		
		53.0	1.53×10^3		
		54.0	0.55×10^3		
		55.0	0.078×10^3		
$M_n = 20 \times 10^3$		53.6	40.0×10^3	Hot stage optical microscopy; selected	295
$M_w/M_n = 1.2$		55.1	8.00×10^3	molecular weights	
$[\eta] = 0.374$		55.9	4.00×10^3	only are listed	
		56.8	1.67×10^3		
		57.8	0.433×10^3		
		58.7	0.145×10^3		
$M_n = 600 \times 10^3$		51.7	31.00×10^3	Hot stage optical microscopy; selected	295
$M_w/M_w = 1.2$		52.5	25.50×10^3	molecular weights only are listed	
$[\eta] = 3.65$		53.6	21.83×10^3		
		54.3	11.67×10^3		
		55.1	7.80×10^3		
		56.8	1.67×10^3		
		57.8	0.233×10^3		
		59.1	0.163×10^3		
Poly(oxymethylene)					
$M_n = 66 \times 10^3$		137.3	19.82×10^4	Optical hot stage microscopy;	
		142.1	63.83×10^3	Regime III crystallization	
		145.2	26.80×10^3		
		147.0	18.37×10^3		
		150.0	81.50×10^2		
		152.0	40.08×10^2		
		153.4	17.58×10^2		
		155.0	863.3		
		156.5	359.0		
Poly(oxymethylene)					
$M_w = 2 \times 10^3$		29.6	84275	Melt crystallized; data interpolated	296
		32.6	51892	from original plots	
		37.2	17195		
		40.1	3908		
		41.8	841		
		44.1	678		
		48.0	231		
		50.4	19		
		51.2	1		
$M_w = 3 \times 10^3$		32.9	86579	Melt crystallized; data interpolated	296
		36.8	50513	from original plots	
		40.4	21913		
		45.2	3324		
		47.8	319		
		48.5	100		
		49.7	17		
		51.8	16		
		54.5	6		
		55.5	1		
$M_w = 4 \times 10^3$		34.6	96427	Melt crystallized; data interpolated	296
		37.7	57796	from original plots	
		42.5	22511		
		45.7	6013		
		48.3	1072		
		50.6	121		
		52.2	52		
		53.3	16		
		54.0	4		
		54.4	2		
		56.9	1		
		36.4	73655	Melt crystallized; data interpolated	296
$M_w = 6 \times 10^3$		41.5	37558	from original plots	
		45.7	14240		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		49.0	3415		
		51.9	407		
		54.1	43		
		55.4	6		
		57.2	2		
$M_w = 10 \times 10^3$		39.3	57796	Melt crystallized; data interpolated from original plots	296
		43.8	29471		
		47.0	15438		
		50.2	5399		
		52.2	1227		
		55.4	109		
		56.9	15		
		58.3	4		
		59.6	1		
$M_w = 150 \times 10^3$		30.2	64372	Melt crystallized; data interpolated from original plots	296
		34.6	54765		
		39.1	355		
		43.6	21913		
		46.8	10877		
		49.8	4471		
		52.9	1404		
		55.2	465		
		57.6	74		
		59.8	7		
		61.1	1		
Poly(oxymethylene oxyethylene)	80	−18	1840	Hot stage microscopy; 2 crystal modifications; form I	297
$M_v = 9.4 \times 10^3$		−9.8	3980		
		3.0	4680		
		8.5	5650		
		14.0	5410		
		16.0	2450		
		17.0	2630		
		19.0	1730		
		22.0	385		
		25.0	255		
		28.0	94.5		
		31.0	29.1		
		34.0	21		
		22.0	370	Form II	297
		25.0	240		
		28.0	90		
		31.0	10		
		34.0	9		
Poly(oxypropylene)	120 (2–5 min)	42.3	325.8	Hot stage optical microscopy	299
		46.3	132.5		
		46.7	39.1		
		51.0	16.7		
		40.5	165.8		
		42.3	104.1		
		44.2	66.7		
		46.3	73.3		
		39.0	8.3		
		40.5	40		
		42.3	33.3		
		44.2	8.3		
	85	45	1.37	Mean isotactic sequence length = 122	300
		50	0.175		
		55	0.14		
		38	1.00	Mean isotactic sequence length = 14	
		43	0.42		
		48	0.15		
$M_n = 10.3 \times 10^3$	120 (5 min)	0	500.1	Optically active hot stage optical microscopy	301
		5	746.8		
		10	723.4		
		15	838.5		

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